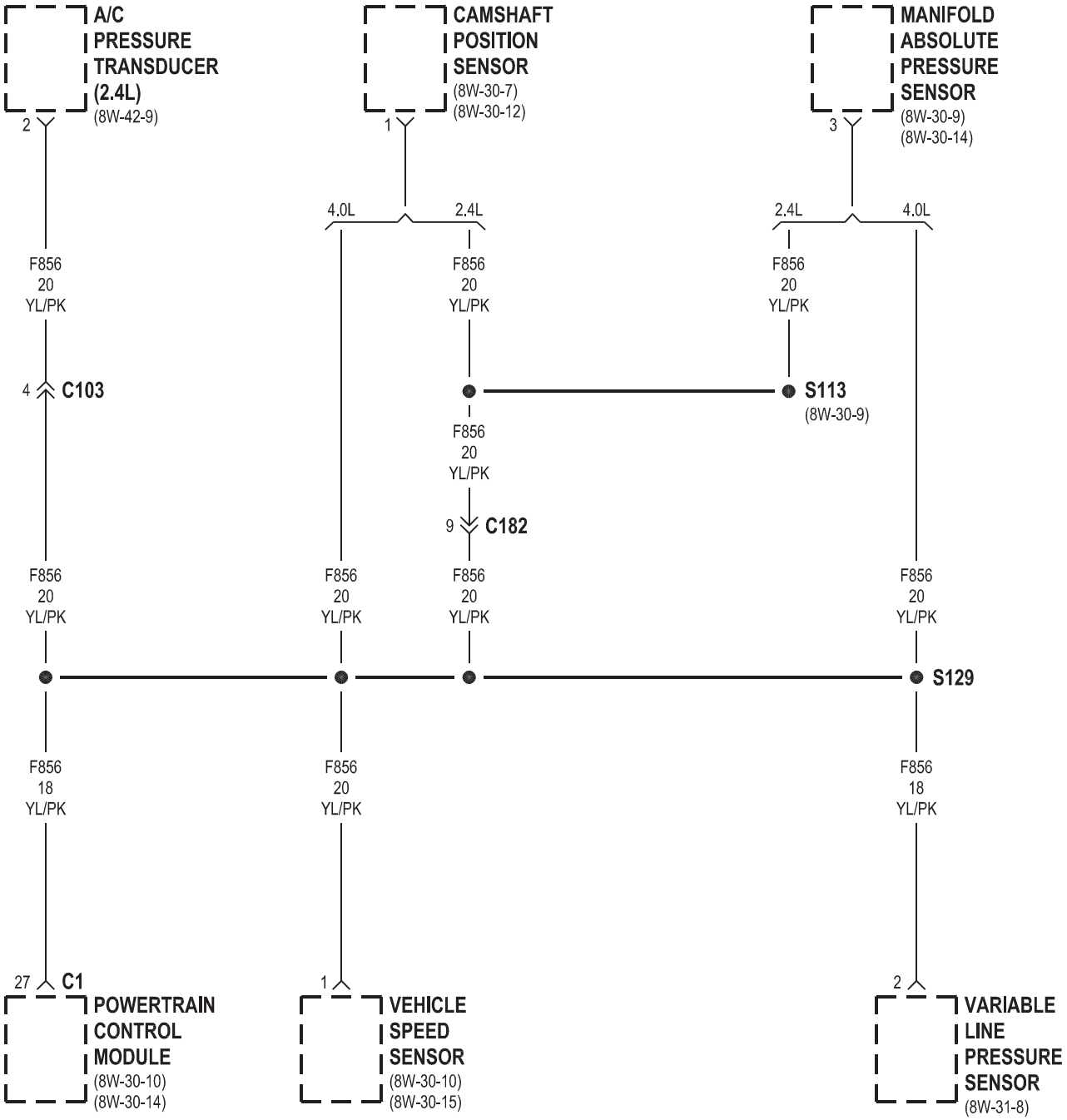
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2.4L
2.4L
4.0L
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F856 20 YL/PK
F856 20 YL/PK
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9 C182
F856 20 YL/PK
F856 20 YL/PK
F856 20 YL/PK
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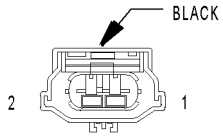
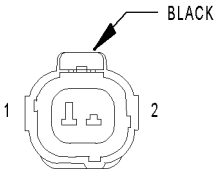


A/C COMPRESSOR CLUTCH

A/C COMPRESSOR CLUTCH - BLACK 2 WAY

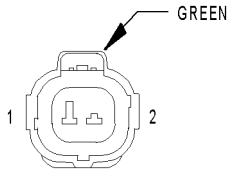
CAV	CIRCUIT	FUNCTION
1	C3 18DB/YL (2.4L)	A/C CLUTCH RELAY OUTPUT
1	C3 20DB/YL (4.0L)	A/C CLUTCH RELAY OUTPUT
2	Z939 18BK (2.4L)	GROUND
2	Z987 20BK (4.0L)	GROUND



A/C HIGH PRESSURE SWITCH (4.0L)

A/C HIGH PRESSURE SWITCH (4.0L) - BLACK 2 WAY

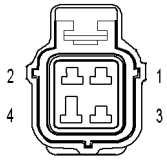
CAV	CIRCUIT	FUNCTION
1	C221 20LB/OR	A/C SELECT SIGNAL
2	C22 20LB/WT	A/C SWITCH SIGNAL



A/C LOW PRESSURE SWITCH

A/C LOW PRESSURE SWITCH - GREEN 2 WAY

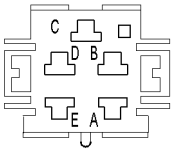
CAV	CIRCUIT	FUNCTION
1	C20 20DB/YL (2.4L)	A/C REQUEST SIGNAL
1	C20 18DB/YL (4.0L)	A/C REQUEST SIGNAL
2	C221 20LB/OR (2.4L)	A/C SELECT SIGNAL
2	C22 20LB/WT (4.0L)	A/C SELECT SIGNAL



A/C PRESSURE TRANSDUCER (2.4L)

A/C PRESSURE TRANSDUCER (2.4L) - 4 WAY

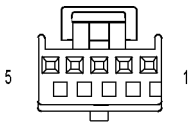
CAV	CIRCUIT	FUNCTION
1	K900 20DB/DG	SENSOR GROUND
2	F856 20YL/PK	5 VOLT SUPPLY
3	C18 20LB/BR	A/C PRESSURE SIGNAL
4	-	-



A/C-HEATER CONTROL C1 (EXCEPT HEATER ONLY)

A/C-HEATER CONTROL C1 (EXCEPT HEATER ONLY) - 5 WAY

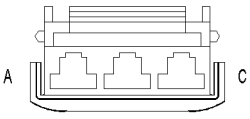
CAV	CIRCUIT	FUNCTION
A	C75 12DB/GY	BLOWER MOTOR HIGH SPEED
B	C73 14DB/VT	BLOWER MOTOR M2 SPEED
C	C72 16DB/OR	BLOWER MOTOR M1 SPEED
D	C71 18DB/BR	BLOWER MOTOR LOW SPEED
E	C805 12DB/WT	BLOWER MOTOR FEED



A/C-HEATER CONTROL C2 (EXCEPT HEATER ONLY)

A/C-HEATER CONTROL C2 (EXCEPT HEATER ONLY) - 5 WAY

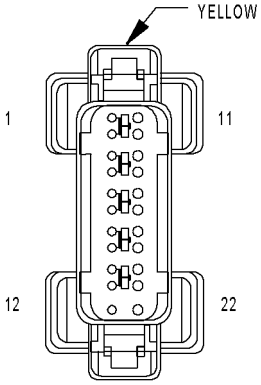
CAV	CIRCUIT	FUNCTION
1	Z134 20BK/LG (LHD)	GROUND
1	F504 20GY/PK (RHD)	FUSED IGNITION SWITCH OUTPUT (RUN)
2	C132 20DB/YL	BLEND DOOR FEEDBACK SIGNAL
3	F504 20GY/PK (LHD)	FUSED IGNITION SWITCH OUTPUT (RUN)
3	Z134 20BK/LG (RHD)	GROUND
4	E2 20OR/BR	PANEL LAMPS FEED
5	Z1 20BK	GROUND



A/C-HEATER CONTROL C3 (EXCEPT HEATER ONLY)

A/C-HEATER CONTROL C3 (EXCEPT HEATER ONLY) - 3 WAY

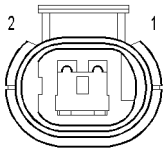
CAV	CIRCUIT	FUNCTION
A	C221 20LB/OR	A/C SELECT SIGNAL
B	C805 12DB/WT	BLOWER MOTOR FEED
C	Z1 12BK	GROUND



AIRBAG CONTROL MODULE

AIRBAG CONTROL MODULE - YELLOW 22 WAY

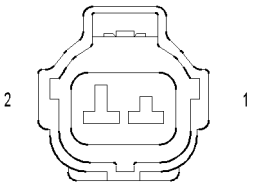
CAV	CIRCUIT	FUNCTION
1	R43 18LG/BR	DRIVER SQUIB 1 LINE 1
2	R45 18LG/OR	DRIVER SQUIB 1 LINE 2
3	-	-
4	-	-
5	R44 18LB/OR	PASSENGER SQUIB 1 LINE 2
6	R42 18LB/BR	PASSENGER SQUIB 1 LINE 1
7	-	-
8	G104 18VT/GY (PAD)	PASSENGER AIRBAG INDICATOR DRIVER
9	-	-
10	Z104 18BK/LG	GROUND
11	R106 18LG/LB (PAD)	PASSENGER AIRBAG MUX SWITCH SENSE
12	-	-
13	-	-
14	-	-
15	-	-
16	-	-
17	F201 18PK/OR	FUSED IGNITION SWITCH OUTPUT (RUN-START)
18	D25 18WT/VT	PCI BUS
19	-	-
20	F100 18PK/VT	FUSED IGNITION SWITCH OUTPUT (RUN)
21	-	-
22	R104 18LB/LG (PAD)	PASSENGER AIRBAG MUX SWITCH RETURN



AMBIENT TEMPERATURE SENSOR (COMPASS/ TEMPERATURE MIRROR)

AMBIENT TEMPERATURE SENSOR (COMPASS/TEMPERATURE MIRROR) - 2 WAY

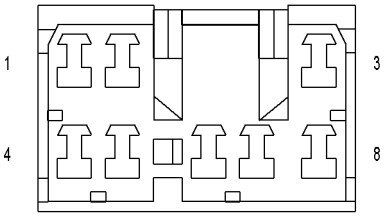
CAV	CIRCUIT	FUNCTION
1	G231 20VT/YL	AAT SIGNAL
2	G32 20VT/LB	SENSOR GROUND



AMBIENT TEMPERATURE SENSOR (ENGINE)

AMBIENT TEMPERATURE SENSOR (ENGINE) - 2 WAY

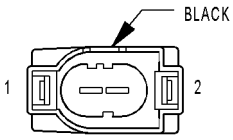
CAV	CIRCUIT	FUNCTION
1	G31 20VT/LG	AAT SIGNAL
2	K900 20DB/DG	SENSOR GROUND



AXLE LOCK SWITCH (OFF-ROAD PACKAGE)

AXLE LOCK SWITCH (OFF-ROAD PACKAGE) - 8 WAY

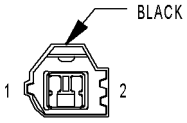
CAV	CIRCUIT	FUNCTION
1	Z1 18BK	GROUND
2	E2 20OR/BR	PANEL LAMPS FEED
3	M21 20YL/BR	FUSED B(+)
4	G302 20VT/OR	LOCKER ENABLE SIGNAL 1
5	G303 20VT/DG	LOCKER ENABLE SIGNAL 2
6	G305 20VT/LG	REAR LOCKER REQUEST
7	G304 20VT/DB	FRONT LOCKER REQUEST
8	-	-



BACK-UP LAMP SWITCH (M/T)

BACK-UP LAMP SWITCH (M/T) - BLACK 2 WAY

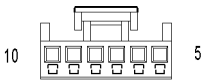
CAV	CIRCUIT	FUNCTION
1	F102 20PK/DB	FUSED IGNITION SWITCH OUTPUT (RUN)
2	L1 20WT/LG	BACK-UP LAMP FEED



BATTERY TEMPERATURE SENSOR

BATTERY TEMPERATURE SENSOR - BLACK 2 WAY

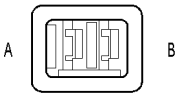
CAV	CIRCUIT	FUNCTION
1	K25 20DB/VT	BATT TEMP SIGNAL
2	K900 20DB/DG	SENSOR GROUND



BLEND DOOR ACTUATOR

BLEND DOOR ACTUATOR - 6 WAY

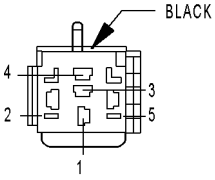
CAV	CIRCUIT	FUNCTION
5	-	-
6	-	-
7	Z134 20DB/WT	GROUND
8	C132 20YL	BLEND DOOR FEEDBACK SIGNAL
9	-	-
10	F504 20OR	FUSED IGNITION SWITCH OUTPUT (RUN)



BLOWER MOTOR

BLOWER MOTOR - 2 WAY

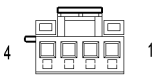
CAV	CIRCUIT	FUNCTION
A	C1 12DG	BLOWER MOTOR RELAY OUTPUT
B	C75 12BK/TN (EXCEPT HEATER ONLY)	BLOWER MOTOR HIGH DRIVER
B	C7 12BK/TN (HEATER ONLY)	BLOWER MOTOR HIGH DRIVER



BLOWER MOTOR RELAY

BLOWER MOTOR RELAY - BLACK 5 WAY

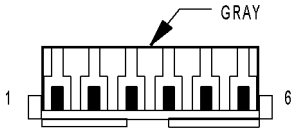
CAV	CIRCUIT	FUNCTION
1	C1 12DG	BLOWER MOTOR RELAY OUTPUT
2	Z1 20BK	GROUND
3	-	-
4	A502 12RD/LG (EXCEPT HEATER ONLY)	FUSED B(+)
4	A111 12RD/LB (HEATER ONLY)	FUSED B(+)
5	F504 20DB/WT (EXCEPT HEATER ONLY)	FUSED IGNITION SWITCH OUTPUT (RUN)
5	F24 20DB/WT (HEATER ONLY)	FUSED IGNITION SWITCH OUTPUT (RUN)



BLOWER MOTOR RESISTOR BLOCK

BLOWER MOTOR RESISTOR BLOCK - 4 WAY

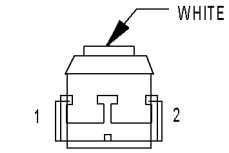
CAV	CIRCUIT	FUNCTION
1	C71 14TN (EXCEPT HEATER ONLY)	BLOWER MOTOR LOW DRIVER
1	C4 14TN (HEATER ONLY)	BLOWER MOTOR LOW DRIVER
2	C72 14LG (EXCEPT HEATER ONLY)	BLOWER MOTOR M1 DRIVER
2	C5 14LG (HEATER ONLY)	BLOWER MOTOR M1 DRIVER
3	C73 14LB (EXCEPT HEATER ONLY)	BLOWER MOTOR M2 DRIVER
3	C6 14LB (HEATER ONLY)	BLOWER MOTOR M2 DRIVER
4	C75 12BK/TN (EXCEPT HEATER ONLY)	BLOWER MOTOR HIGH DRIVER
4	C7 12BK/TN (HEATER ONLY)	BLOWER MOTOR HIGH DRIVER



BRAKE LAMP SWITCH

BRAKE LAMP SWITCH - GRAY 6 WAY

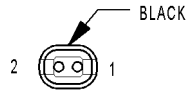
CAV	CIRCUIT	FUNCTION
1	B15 20DG/WT	BRAKE SWITCH NO. 1 SIGNAL
1	B15 20DG/WT	BRAKE SWITCH NO. 1 SIGNAL
2	Z249 20BK/OR	GROUND
3	V32 20VT/YL (EXCEPT POSTAL)	S/C SUPPLY
4	V30 20VT/WT (EXCEPT POSTAL)	S/C BRAKE SWITCH OUTPUT
5	A103 18GY/RD	FUSED B(+)
6	L50 18WT/TN	BRAKE LAMP SWITCH OUTPUT



BRAKE TRANSMISSION SHIFT INTERLOCK SOLENOID

BRAKE TRANSMISSION SHIFT INTERLOCK SOLENOID - WHITE 2 WAY

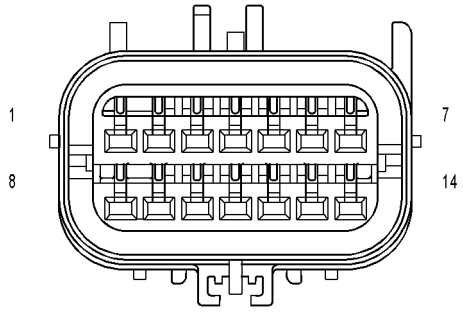
CAV	CIRCUIT	FUNCTION
1	B15 20DG/WT	BRAKE SWITCH NO. 1 SIGNAL
2	F505 20PK/YL	FUSED IGNITION SWITCH OUTPUT (RUN-START)



BRAKE WARNING INDICATOR SWITCH

BRAKE WARNING INDICATOR SWITCH - BLACK 2 WAY

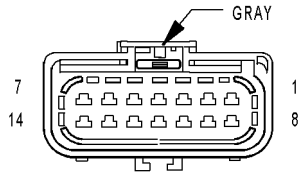
CAV	CIRCUIT	FUNCTION
1	G9 20GY/BK	BRAKE WARNING INDICATOR DRIVER
2	B83 20DG/WT	BRAKE PRESSURE SENSOR SUPPLY



C102

C102 - (ENGINE SIDE)

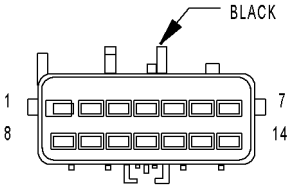
CAV	CIRCUIT
1	A209 18RD
2	F26 16PK/OR
3	B1 18DG/OR (4.0L EXCEPT ABS)
3	G301 20VT/LB (4.0L OFF-ROAD PACKAGE)
4	B2 18DG/LB (4.0L EXCEPT ABS)
4	A850 18RD/WT (4.0L OFF-ROAD PACKAGE)
5	C20 20DB/YL (2.4L A/C)
6	B3 18DG/YL (4.0L EXCEPT ABS)
6	G300 20VT/WT (4.0L OFF-ROAD PACKAGE)
7	B4 18DG/GY (4.0L EXCEPT ABS)
7	A750 18TN/RD (4.0L OFF-ROAD PACKAGE)
8	N4 20DB/WT
9	L1 20WT/LG
10	F102 20PK/DB
11	K334 18BR
12	T750 14YL/GY
13	G93 20VT/LB
14	A109 18OR/RD



C102

C102 - GRAY (HEADLAMP AND DASH SIDE)

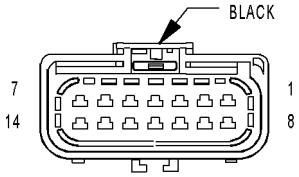
CAV	CIRCUIT
1	A209 18RD/YL
2	F26 16PK/OR
3	B1 18DG/OR (ABS)
3	G301 20VT/LB (OFF-ROAD PACKAGE)
4	B2 18DG/LB (ABS)
4	A850 18RD/WT (OFF-ROAD PACKAGE)
5	C20 20DB/YL (2.4L A/C)
6	B3 18DG/YL (ABS)
6	G300 20VT/WT (OFF-ROAD PACKAGE)
7	B4 18DG/GY (ABS)
7	A750 18TN/RD (OFF-ROAD PACKAGE)
8	N4 20DB/WT
9	L1 20WT/LG
10	F102 20PK/DB
11	K334 18BR
12	T750 14YL/GY
13	G93 20VT/LB
14	A109 18OR/RD



C103

C103 - BLACK (ENGINE SIDE)

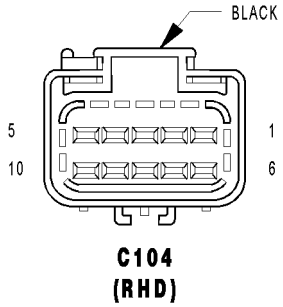
CAV	CIRCUIT
1	D20 18WT/LG
2	G31 18VT/LG
3	C221 20LB/OR (4.0L)
4	F856 20YL/PK (2.4L)
4	K399 18BR/GY (4.0L)
5	K299 18BR/WT
6	D25 18WT/VT
7	D21 18WT/BR
8	K900 20DB/DG
9	F1 18PK/WT
10	Y3 18LG
11	Z11 20BK/LG
12	K75 20DB/OR (DRL)
13	K66 18DB/WT (2.4L)
14	C3 20DB/BK (2.4L A/C)
14	C3 20DB/YL (4.0L A/C)



C103

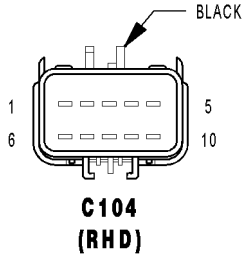
C103 - BLACK (HEADLAMP AND DASH SIDE)

CAV	CIRCUIT
1	D20 20WT/LG
2	G31 20VT/LG
3	C221 20LB/OR (4.0L)
4	F856 20YL/PK (2.4L)
4	K399 20BR/GY (4.0L)
5	K299 20BR/WT
6	D25 18WT/VT
7	D21 20WT/BR
8	K900 20DB/DG
9	F1 20PK/WT
10	Y3 18LG
10	Y3 20LG
11	Z11 20BK/LG
12	K75 20DB/OR (DRL)
13	K66 20DB/WT (2.4L)
14	C3 20DB/YL (A/C)



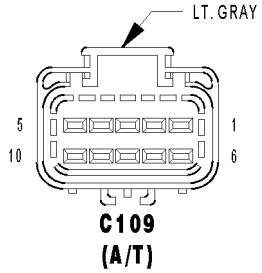
C104 (RHD) - BLACK (HEADLAMP AND DASH SIDE)

CAV	CIRCUIT
1	A209 18RD
2	F26 16PK/OR
3	A109 18OR/RD
4	K334 18BR
5	-
6	T750 14YL/GY
7	G93 20VT/LB
8	N4 20DB/WT
9	L1 20WT/LG
10	F102 20PK/DB



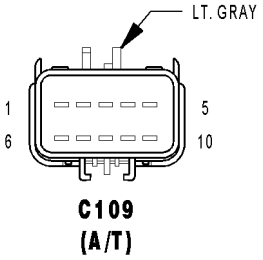
C104 (RHD) - BLACK (ENGINE SIDE)

CAV	CIRCUIT
1	A209 18RD
2	F26 16PK/OR
3	A109 18OR/RD
4	K334 18BR
5	-
6	T750 14YL/GY
7	G93 20VT/LB
8	N4 20DB/WT
9	L1 20WT/LG
10	F102 20PK/DB



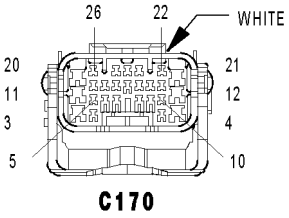
C109 (A/T) - LT. GRAY (HEADLAMP AND DASH SIDE)

CAV	CIRCUIT
1	-
2	T515 20YL/DB
3	T16 18YL/OR
4	D15 20WT/DG
5	D16 20WT/OR
6	-
7	T56 20DG/LB
8	-
9	T6 20DG
10	-



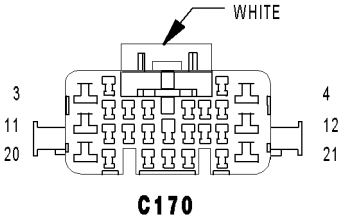
C109 (A/T) - LT. GRAY (ENGINE SIDE)

CAV	CIRCUIT
1	-
2	T515 18YL/DB
3	T16 16YL/OR
4	D15 18WT/DG
5	D16 18WT/OR
6	-
7	T56 18DG/LB
8	-
9	T6 18DG
10	-



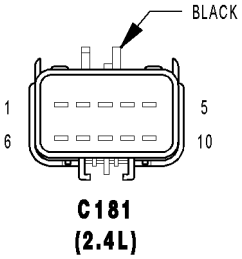
C170 - WHITE (HEADLAMP AND DASH SIDE)

CAV	CIRCUIT
1	L13 20WT/YL (HEADLAMP LEVELING)
2	C115 20DB
3	A502 12RD/LG (EXCEPT HEATER ONLY)
3	A111 12RD/LB (HEATER ONLY)
4	F306 16DB/PK
5	T56 20DG/LB (A/T)
6	C221 20LB/OR (A/C)
7	T6 20DG (A/T)
8	G231 20VT/YL
9	D20 20WT/LG
10	D21 20WT/BR
11	A205 18RD/OR
12	L300 14LG/WT
12	L340 14WT/OR
13	G305 20VT/LG (OFF-ROAD PACKAGE)
14	Z11 20BK/LG
15	D16 20WT/OR
16	G300 20VT/WT (OFF-ROAD PACKAGE)
17	G301 20VT/LB (OFF-ROAD PACKAGE)
18	G32 20VT/LB
19	G4 20VT/LB (ABS)
20	L39 18WT/OR
21	-
22	B83 20DG/WT
23	D15 20WT/DG (A/T)
24	G304 20VT/DB (OFF-ROAD PACKAGE)
25	-
26	-



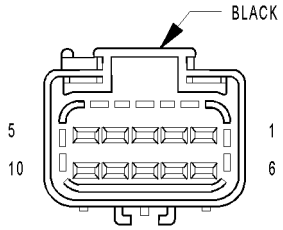
C170 - WHITE (I/P SIDE)

CAV	CIRCUIT
1	L13 20WT/YL (EXPORT)
2	C115 20DB (HARD TOP)
3	A502 12RD/LG (EXCEPT HEATER ONLY)
3	A111 12RD/LB (HEATER ONLY)
4	F306 16DB/PK
5	T56 20DB/LB
6	C221 20LB/OR
7	T6 20DG
8	G231 20VT/YL
9	D20 20WT/LG
10	D21 20WT/BR
11	A205 18RD/OR
12	L300 16LG/WT
13	G305 20VT/LG (OFF-ROAD PACKAGE)
14	Z11 20BK/LG
15	D16 20WT/OR
16	G300 20VT/WT (OFF-ROAD PACKAGE)
17	G301 20VT/LB (OFF-ROAD PACKAGE)
18	G32 20VT/LB
19	G4 20VT/LB
20	L39 16WT/OR
21	-
22	B83 20DG/WT
23	D15 20WT/DG
24	G304 20VT/DB (OFF-ROAD PACKAGE)
25	-
26	-



C181 (2.4L) - BLACK (ENGINE INTAKE SIDE)

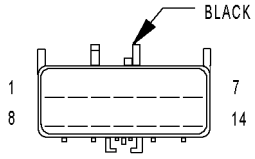
CAV	CIRCUIT
1	K334 18OR/DG
2	C3 18DB/YL
3	Z987 18BK/GR
4	K11 18WT/DB
5	K12 18TN
6	K13 18YL/WT
7	K14 18LB/BR
8	K44 18TN/YL
9	K900 18BK/LB
10	-



C181
(2.4L)

C181 (2.4L) - BLACK (ENGINE SIDE)

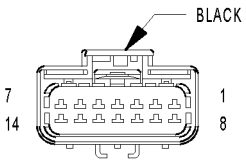
CAV	CIRCUIT
1	K334 18BR
2	C3 20DB/BK
3	Z987 20BK
4	K11 18BR/YL
5	K12 18BR/DB
6	K13 18BR/LB
7	K14 18BR/TN
8	K44 18DB/GY
9	K900 18DB/DG
10	-



C182
(2.4L)

C182 (2.4L) - BLACK (ENGINE INTAKE SIDE)

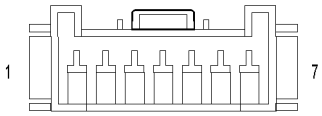
CAV	CIRCUIT
1	-
2	-
3	K61 18YL/BK
4	K961 18GY/RD
5	K21 18BK/RD
6	K2 18TN/BK
7	K1 18DG/RD
8	K22 18OR/DB
9	F856 18YL/PK
10	G6 18GY/YL
11	-
12	F855 18OR
13	-
14	-



C182
(2.4L)

C182 (2.4L) - BLACK (ENGINE SIDE)

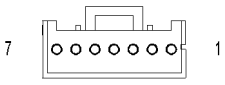
CAV	CIRCUIT
1	-
2	-
3	K61 18VT/GY
4	K961 18BR/VT
5	K21 18DB/LG
6	K2 18VT/OR
7	K1 18VT/BR
8	K22 18BR/OR
9	F856 18YL/PK
10	G6 18VT/GY
11	-
12	F855 18PK/YL
13	-
14	-



C203

C203 - (I/P SIDE)

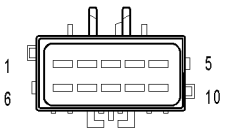
CAV	CIRCUIT
1	F505 20PK/YL
2	Z10 20BK
3	L1 20WT/LG
4	G231 20VT/YL
5	G32 20VT/LB
6	M22 20YL/OR
7	M21 20YL/BR



C203

C203 - (MIRROR SIDE)

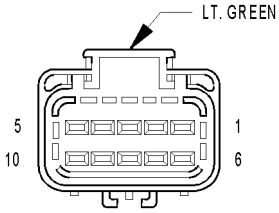
CAV	CIRCUIT
1	F505 20DB/WT
2	Z2 20BK/LG
3	L1 20VT/BK
4	G31 20VT/LG
5	G32 20BK/LB
6	M22 20YL
7	M21 20PK/WT



C205

C205 - (HVAC SIDE)

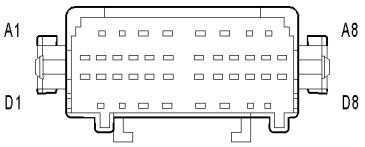
CAV	CIRCUIT
1	Z134 20DB/WT (EXCEPT HEATER ONLY)
1	Z11 20DB/WT (HEATER ONLY)
2	C71 14TN (EXCEPT HEATER ONLY)
2	C4 14TN (HEATER ONLY)
3	C72 14LG (EXCEPT HEATER ONLY)
3	C5 14LG (HEATER ONLY)
4	C73 14LB (EXCEPT HEATER ONLY)
4	C6 14LB (HEATER ONLY)
5	F504 20OR (EXCEPT HEATER ONLY)
5	F24 20OR (HEATER ONLY)
6	C132 20YL (EXCEPT HEATER ONLY)
6	C132 20YL (HEATER ONLY)
7	C75 12BK/TN (EXCEPT HEATER ONLY)
7	C7 12BK/TN (HEATER ONLY)
8	A502 12RD/LG (EXCEPT HEATER ONLY)
8	A111 12RD/LG (HEATER ONLY)
9	F504 20DB/WT (EXCEPT HEATER ONLY)
9	F24 20DB/WT (HEATER ONLY)
10	Z1 20BK



C205

C205 - LT. GREEN (I/P SIDE)

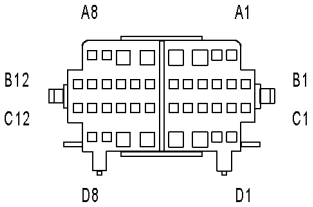
CAV	CIRCUIT
1	Z134 20BK/LG (EXCEPT HEATER ONLY)
1	Z11 20DB/WT (HEATER ONLY)
2	C71 18DB/BR (EXCEPT HEATER ONLY)
2	C4 18TN (HEATER ONLY)
3	C72 16DB/OR (EXCEPT HEATER ONLY)
3	C5 16LG/LB (HEATER ONLY)
4	C73 14DB/VT (EXCEPT HEATER ONLY)
4	C6 14LB (HEATER ONLY)
5	F504 20GY/PK (EXCEPT HEATER ONLY)
5	F504 20GY/PK (EXCEPT HEATER ONLY)
5	F24 20RD/DG (HEATER ONLY)
6	C132 20DB/YL (EXCEPT HEATER ONLY)
6	C36 20DB/RD (HEATER ONLY)
7	C75 12DB/GY (EXCEPT HEATER ONLY)
7	C7 12BR/TN (HEATER ONLY)
8	A502 12RD/LG (EXCEPT HEATER ONLY)
8	A111 12RD/LB (HEATER ONLY)
9	F504 20GY/PK (EXCEPT HEATER ONLY)
9	F24 20RD/DG (HEATER ONLY)
10	Z1 20BK



C300

C300 - (BODY SIDE)

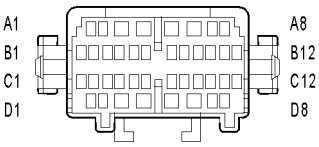
CAV	CIRCUIT
A1	L45 20LG/DG
A2	G75 20VT
A3	X55 18DG/BR
A4	X53 18DG
A5	L7 18WT/YL
A5	L7 18WT/YL
A6	L61 18GY
A6	L61 18GY
A7	L1 20WT/LG
A8	A207 18RD/LG
B1	F201 18PK/OR
B2	F505 20PK/YL
B3	X52 18GY/DB
B4	X58 18GY/OR
B5	L60 18TN
B6	-
B7	G15 20VT/TN
B8	G104 18VT/GY
B9	W20 18BR/YL
B10	G93 20VT/LB
B11	C515 20LB/OR
B12	L36 16WT/DB
C1	X112 16DG/GY
C2	X60 20GY/OR
C3	W13 18BR/LG
C4	G10 20VT/LG
C5	M22 20YL/OR
C6	F302 18GY/PK
C7	E19 20OR/BR
C8	-
C9	R104 18LB/LG
C10	X51 18DG/DB
C11	X57 18DG/OR
C12	D25 20WT/VT
D1	-
D2	G76 20VT/YL
D3	M21 20YL/BR
D3	M21 20YL/BR
D4	F504 20GY/PK (EXCEPT HEATER CONTROL)
D4	F24 20RD/DG (HEATER CONTROL)
D5	X54 18GY
D6	X56 18GY/BR
D7	E2 20OR/BR
D8	R106 18LG/LB



C300

C300 - (I/P SIDE)

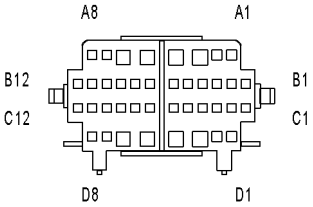
CAV	CIRCUIT
A1	L45 20WT/OR
A2	G75 20VT
A3	X55 18DG/BR
A3	X55 18DG/BR
A4	X53 18DG
A4	X53 18DG
A5	L7 18WT/YL
A6	L61 18WT/LG
A7	L1 20WT/LG
A8	A207 18RD/LG
B1	F201 18PK/OR
B2	F505 20PK/YL
B3	X52 18GY/DB
B4	X58 18GY/OR
B5	L60 18WT/TN
B6	-
B7	G15 20VT/TN
B8	G104 18VT/GY
B9	W20 18BR/YL
B10	G93 20VT/LB
B11	C515 20LB/OR
B12	L36 16WT/DB
C1	X112 16DG/GY
C2	X60 20GY/OR
C3	W13 18BR/LG
C4	G10 20VT/LG
C5	M22 20YL/OR
C6	F302 18GY/PK
C7	E19 20OR/BR
C8	-
C9	R104 18LB/LG
C10	X51 18DG/DB
C11	X57 18DG/OR
C12	D25 20WT/VT
D1	-
D2	G76 20VT/YL
D3	M21 20YL/BR
D4	F504 20GY/PK (EXCEPT HEATER ONLY)
D4	F504 20GY/PK (EXCEPT HEATER ONLY)
D4	F24 20RD/DG (HEATER ONLY)
D4	F24 20RD/DG (HEATER ONLY)
D5	X54 18GY
D5	X54 18GY
D6	X56 18GY/BR
D6	X56 18GY/BR
D7	E2 20OR/BR
D8	R106 18LG/LB



C301

C301 - (BODY SIDE)

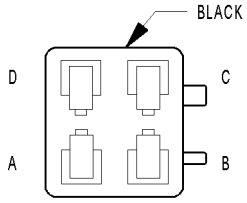
CAV	CIRCUIT
A1	V32 20VT/YL
A2	G117 20VT/LG
A3	L300 14LG/WT
A4	L4 14WT/LG
A5	A333 14WT/RD
A6	A991 12RD
A7	F26 16PK/OR
A8	K900 20DB/DG
B1	V37 20VT
B2	A701 18BR/RD
B3	M21 20YL/BR
B4	B15 20DG/WT
B5	G93 20VT/LB
B6	L60 18TN
B7	A1 16RD
B8	L1 20WT/LG
B9	L61 18GY
B10	L7 18WT/YL
B11	L50 18WT/TN
B12	X3 20RD/YL
C1	W10 16BR
C2	L139 18WT/YL
C3	V30 20VT/WT
C4	K106 20VT/LB
C5	K107 20VT/WT
C6	L45 20LG/DG
C7	F102 20PK/DB
C8	F1 20PK/WT
C9	A39 18RD/LG
C10	F960 20PK/LG
C11	G9 20GY/BK
C12	G119 20VT
D1	T141 18YL/OR
D2	G118 20VT/YL
D3	A906 12RD
D4	A940 12RD
D5	C15 12BK/WT
D6	A8 14RD
D7	D25 20WT/VT
D8	-



C301

C301 - (HEADLAMP AND DASH SIDE)

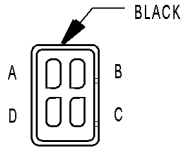
CAV	CIRCUIT
A1	V32 20VT/YL
A2	G117 18VT/LG
A3	L300 14LG/WT
A4	L4 14WT/LG
A5	A333 14WT/RD
A6	A991 12RD
A7	F26 16PK/OR
A8	K900 20DB/DG
B1	V37 20VT
B2	A701 18BR/RD
B3	M21 20YL/BR
B4	B15 20DG/WT
B5	G93 20VT/LB
B6	L60 18WT/TN
B7	A1 16RD
B8	L1 20WT/LG
B9	L61 18WT/LG
B10	L7 18WT/YL
B11	L50 18WT/TN
B12	X3 20RD/YL
C1	W10 16BR
C2	L139 18WT/YL
C3	V30 20VT/WT
C4	K106 20VT/LB
C5	K107 20VT/WT
C6	L45 20WT/OR
C7	F102 20PK/DB
C8	F1 20PK/WT
C9	A39 18RD/DG
C9	A39 16RD/DG (4.0L LHD M/T EXCEPT EXPORT OFF-ROAD PACKAGE)
C10	F960 20PK/LG
C11	G9 20GY/BK
C12	G119 18VT
D1	T141 18YL/OR
D2	G118 18VT/YL
D3	A906 12RD
D4	A940 12RD
D5	C15 12BK/WT
D6	A8 14RD
D7	D25 18WT/VT
D8	-



C324

C324 - BLACK (BODY SIDE)

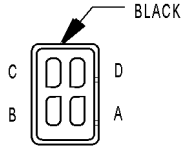
CAV	CIRCUIT
A	M21 20YL/BR
B	M22 20YL/OR
C	X58 18GY/OR
D	X52 18GY/DB



C324

C324 - BLACK (SPEAKER POD JUMPER)

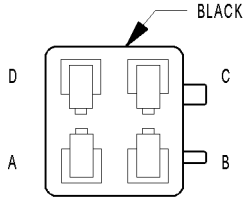
CAV	CIRCUIT
A	M21 20PK/WT
B	M22 20YL
C	X58 18DB/PK
D	X52 18GY/DB



C325

C325 - BLACK (SPEAKER POD JUMPER)

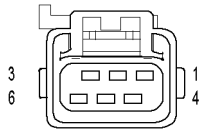
CAV	CIRCUIT
A	M21 20PK/WT
B	M22 20YL
C	X57 18BR/LB
D	X51 18BR/YL



C325

C325 - BLACK (BODY SIDE)

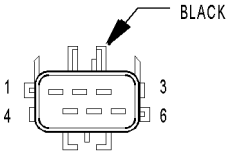
CAV	CIRCUIT
A	M21 20YL/BR
B	M22 20YL/OR
C	X57 18DG/OR
D	X51 18DG/DB



C326
(HARD TOP)

C326 (HARD TOP) - (BODY SIDE)

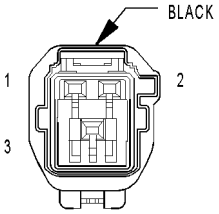
CAV	CIRCUIT
1	C15 12BK/WT
2	-
3	W13 18BR/LG
4	F302 18GY/PK
5	F302 18GY/PK
6	Z913 12BK



C326 (HARD TOP)

C326 (HARD TOP) - BLACK (HARD TOP SIDE)

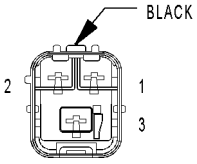
CAV	CIRCUIT
1	C15 12BK/WT
2	-
3	W13 18BR/LG
4	F302 18BR/PK
5	F302 18BR/PK
6	Z913 12BK



C329 (EXPORT)

C329 (EXPORT) - BLACK (BODY SIDE)

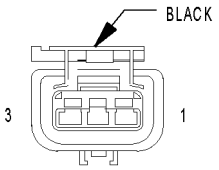
CAV	CIRCUIT
1	L36 16WT/DB
2	L7 18WT/YL
3	Z913 20BK



C329 (EXPORT)

C329 (EXPORT) - BLACK (BUMPER SIDE)

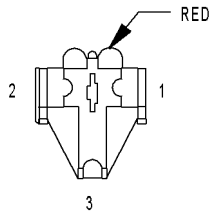
CAV	CIRCUIT
1	L36 18BR/WT
2	L7 18WT/YL
3	Z913 20BK



CAMSHAFT POSITION SENSOR

CAMSHAFT POSITION SENSOR - BLACK 3 WAY

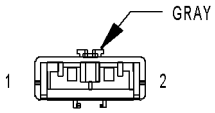
CAV	CIRCUIT	FUNCTION
1	F856 20YL/PK	5 VOLT SUPPLY
2	K900 20DB/DG	SENSOR GROUND
3	K44 20DB/GY (2.4L)	CMP SIGNAL
3	K44 18DB/GY (4.0L)	CMP SIGNAL



CIGAR LIGHTER/ POWER OUTLET

CIGAR LIGHTER/POWER OUTLET - RED 3 WAY

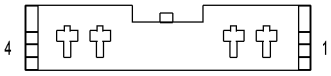
CAV	CIRCUIT	FUNCTION
1	A207 18RD/LG	FUSED CIGAR LIGHTER/ACCESSORY RELAY OUTPUT
2	-	-
3	Z1 18BK	GROUND



CIRCUIT BREAKER

CIRCUIT BREAKER - GRAY 2 WAY

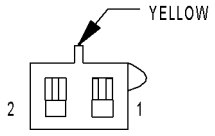
CAV	CIRCUIT	FUNCTION
1	A940 12RD	FUSED B(+)
2	A333 14WT/RD	FUSED B(+)



CLOCKSPRING C1

CLOCKSPRING C1 - 4 WAY

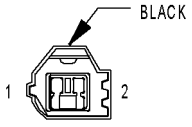
CAV	CIRCUIT	FUNCTION
1	X3 20RD/YL	HORN RELAY CONTROL
2	V37 20VT	S/C SWITCH NO. 1 SIGNAL
3	K900 20DB/DG	SENSOR GROUND
4	-	-



CLOCKSPRING C2

CLOCKSPRING C2 - YELLOW 2 WAY

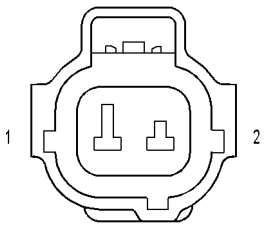
CAV	CIRCUIT	FUNCTION
1	R43 18LG/BR	DRIVER SQUIB 1 LINE 1
2	R45 18LG/OR	DRIVER SQUIB 1 LINE 2



CLUTCH PEDAL POSITION SWITCH (M/T)

CLUTCH PEDAL POSITION SWITCH (M/T) - BLACK 2 WAY

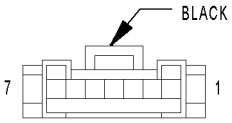
CAV	CIRCUIT	FUNCTION
1	T141 18YL/OR	CLUTCH INTERLOCK SWITCH SENSE
2	Z902 18BK	GROUND



COIL CAPACITOR

COIL CAPACITOR - 2 WAY

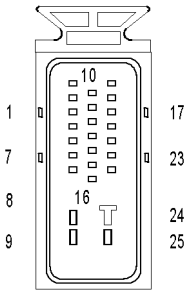
CAV	CIRCUIT	FUNCTION
1	K334 18BR	ASD RELAY OUTPUT
2	-	-



COMPASS/ TEMPERATURE MIRROR

COMPASS/TEMPERATURE MIRROR - BLACK 7 WAY

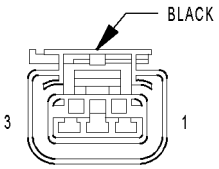
CAV	CIRCUIT	FUNCTION
1	F505 20DB/WT	FUSED IGNITION SWITCH OUTPUT (RUN-START)
2	Z2 20BK/LG	GROUND
3	L1 20VT/BK	BACK-UP LAMP FEED
4	G31 20VT/LG	AAT SIGNAL
5	G32 20BK/LB	SENSOR GROUND
6	M22 20YL	COURTESY LAMPS DRIVER
7	M21 20PK/WT	FUSED B(+)



CONTROLLER ANTILOCK BRAKE

CONTROLLER ANTILOCK BRAKE - 25 WAY

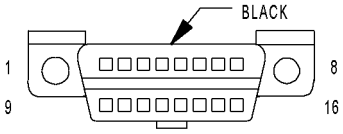
CAV	CIRCUIT	FUNCTION
1	B1 18DG/OR	RIGHT REAR WHEEL SPEED SENSOR (-)
2	B3 18DG/YL	LEFT REAR WHEEL SPEED SENSOR (-)
3	B7 18DG/VT	RIGHT FRONT WHEEL SPEED SENSOR (+)
4	B9 18DG/LG	LEFT FRONT WHEEL SPEED SENSOR (+)
5	-	-
6	G117 18VT/LG	G-SWITCH NO. 1 SENSE
7	G118 18VT/YL	G-SWITCH NO. 2 SENSE
8	Z127 12BK/DG	GROUND
9	A111 12DG/RD	FUSED B(+)
10	B4 18DG/GY	LEFT REAR WHEEL SPEED SENSOR (+)
11	B8 18DG/TN	LEFT FRONT WHEEL SPEED SENSOR (-)
12	L50 18WT/TN	BRAKE LAMP SWITCH OUTPUT
13	G119 18VT	G-SWITCH TEST SIGNAL
14	-	-
15	-	-
16	G83 18VT/WT	ABS RELAY CONTROL
17	B2 18DG/LB	RIGHT REAR WHEEL SPEED SENSOR (+)
18	B6 18DG/WT	RIGHT FRONT WHEEL SPEED SENSOR (-)
19	-	-
20	D21 18WT/BR	SCI TRANSMIT
21	-	-
22	-	-
23	F102 18PK/DB	FUSED IGNITION SWITCH OUTPUT (RUN)
24	Z107 12BK/DG	GROUND
25	A107 12TN/RD	FUSED B(+)



CRANKSHAFT POSITION SENSOR

CRANKSHAFT POSITION SENSOR - BLACK 3 WAY

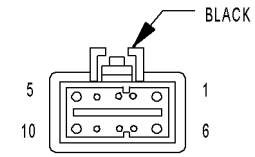
CAV	CIRCUIT	FUNCTION
1	F855 20PK/YL	5 VOLT SUPPLY
2	K900 20DB/DG	SENSOR GROUND
3	K24 18BR/LB	CKP SIGNAL



DATA LINK CONNECTOR

DATA LINK CONNECTOR - BLACK 16 WAY

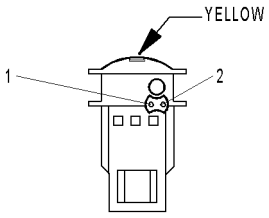
CAV	CIRCUIT	FUNCTION
1	-	-
2	D25 20WT/VT	PCI BUS
3	-	-
4	Z111 20BK/WT	GROUND
5	Z11 20BK/LG	GROUND
6	-	-
7	D21 20WT/BR	SCI TRANSMIT (PCM/CAB)
8	-	-
9	D16 20WT/OR	SCI RECEIVE (TCM)
10	-	-
11	-	-
12	D20 20WT/LG	SCI RECEIVE (PCM)
13	-	-
14	-	-
15	D15 20WT/DG	SCI RECIEVE (TCM)
16	M21 20YL/BR	FUSED B(+)



DAYTIME RUNNING LAMP MODULE (EXCEPT EXPORT)

DAYTIME RUNNING LAMP MODULE (EXCEPT EXPORT) - BLACK 10 WAY

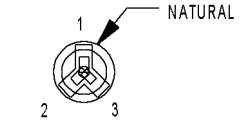
CAV	CIRCUIT	FUNCTION
1	L300 14LG/WT	DIMMER SWITCH HIGH BEAM OUTPUT
2	-	-
3	-	-
4	L340 14WT/OR	HIGH BEAM INDICATOR DRIVER
5	F960 20PK/LG	FUSED IGNITION SWITCH OUTPUT (RUN-START)
6	A333 14WT/RD	FUSED B(+)
7	K75 20DB/OR	VEHICLE SPEED SENSOR SIGNAL
8	Z901 18BK	GROUND
9	-	-
10	L4 14WT/LG	DIMMER SWITCH LOW BEAM OUTPUT



DRIVER AIRBAG SQUIB 1

DRIVER AIRBAG SQUIB 1 - YELLOW 2 WAY

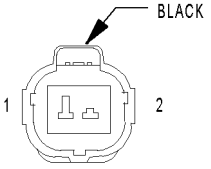
CAV	CIRCUIT	FUNCTION
1	R45 18DG/LB	DRIVER SQUIB 1 LINE 2
2	R43 18BK/LB	DRIVER SQUIB 1 LINE 1



DRIVER DOOR AJAR SWITCH

DRIVER DOOR AJAR SWITCH - NATURAL 3 WAY

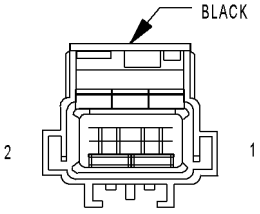
CAV	CIRCUIT	FUNCTION
1	M23 20YL/DG	DOOR AJAR SWITCH OUTPUT
2	-	-
3	G75 20VT	DRIVER DOOR AJAR SWITCH SENSE



ENGINE COOLANT TEMPERATURE SENSOR

ENGINE COOLANT TEMPERATURE SENSOR - BLACK 2 WAY

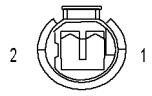
CAV	CIRCUIT	FUNCTION
1	K900 20DB/DG (2.4L)	SENSOR GROUND
1	K2 18VT/OR (4.0L)	ECT SIGNAL
2	K2 20VT/OR (2.4L)	ECT SIGNAL
2	K900 20DB/DG (4.0L)	SENSOR GROUND



ENGINE OIL PRESSURE SWITCH

ENGINE OIL PRESSURE SWITCH - BLACK 2 WAY

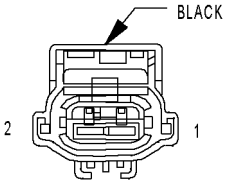
CAV	CIRCUIT	FUNCTION
1	G6 20VT/GY	OIL PRESSURE SIGNAL
2	-	-



EVAP/PURGE SOLENOID

EVAP/PURGE SOLENOID - 2 WAY

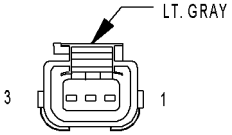
CAV	CIRCUIT	FUNCTION
1	K70 20DB/BR	EVAP/PURGE SOL SIGNAL
2	K52 20DB/WT	EVAP/PURGE SOL CONTROL



FRONT LOCKER INDICATOR SWITCH (OFF-ROAD PACKAGE)

FRONT LOCKER INDICATOR SWITCH (OFF-ROAD PACKAGE) - BLACK 2 WAY

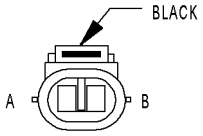
CAV	CIRCUIT	FUNCTION
1	Z987 20BK	GROUND
2	G300 20VT/WT	FRONT LOCKER INDICATOR SWITCH SENSE



FRONT LOCKER PUMP (OFF-ROAD PACKAGE)

FRONT LOCKER PUMP (OFF-ROAD PACKAGE) - LT. GRAY 3 WAY

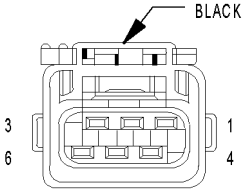
CAV	CIRCUIT	FUNCTION
1	Z987 18BK	GROUND
2	-	-
3	A750 18TN/RD	FRONT LOCKER RELAY OUTPUT



FRONT WASHER PUMP

FRONT WASHER PUMP - BLACK 2 WAY

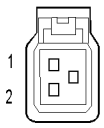
CAV	CIRCUIT	FUNCTION
A	W10 16BR	WASHER PUMP CONTROL SWITCH OUTPUT
B	Z950 18BK	GROUND



FRONT WIPER MOTOR

FRONT WIPER MOTOR - BLACK 6 WAY

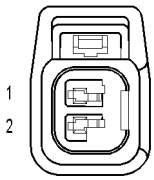
CAV	CIRCUIT	FUNCTION
1	F300 16PK	FUSED IGNITION SWITCH OUTPUT (RUN-ACC)
2	W7 16BR/GY	WIPER PARK SWITCH SENSE
3	-	-
4	Z902 18BK	GROUND
5	W3 16BR/WT	LOW SPEED WIPER SWITCH OUTPUT
6	W4 16BR/OR	WIPER SWITCH HIGH SPEED OUTPUT



FUEL INJECTOR NO. 1 (2.4L)

FUEL INJECTOR NO. 1 (2.4L) - 2 WAY

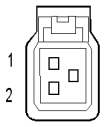
CAV	CIRCUIT	FUNCTION
1	F142 16PK/GY	ASD RELAY OUTPUT
2	K11 16BR/YL	INJECTOR CONTROL NO. 1



FUEL INJECTOR NO. 1 (4.0L)

FUEL INJECTOR NO. 1 (4.0L) - 2 WAY

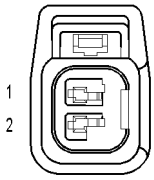
CAV	CIRCUIT	FUNCTION
1	K334 18BR	ASD RELAY OUTPUT
2	K11 18BR/YL	INJECTOR CONTROL NO. 1



FUEL INJECTOR NO. 2 (2.4L)

FUEL INJECTOR NO. 2 (2.4L) - 2 WAY

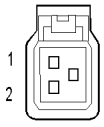
CAV	CIRCUIT	FUNCTION
1	F142 16PK/GY	ASD RELAY OUTPUT
2	K12 16BR/DB	INJECTOR CONTROL NO. 2



FUEL INJECTOR NO. 2 (4.0L)

FUEL INJECTOR NO. 2 (4.0L) - 2 WAY

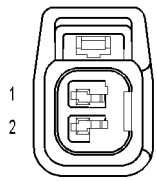
CAV	CIRCUIT	FUNCTION
1	K334 18BR	ASD RELAY OUTPUT
2	K12 18BR/DB	INJECTOR CONTROL NO. 2



FUEL INJECTOR NO. 3 (2.4L)

FUEL INJECTOR NO. 3 (2.4L) - 2 WAY

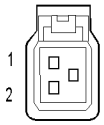
CAV	CIRCUIT	FUNCTION
1	F142 16PK/GY	ASD RELAY OUTPUT
2	K13 16BR/LB	INJECTOR CONTROL NO. 3



FUEL INJECTOR NO. 3 (4.0L)

FUEL INJECTOR NO. 3 (4.0L) - 2 WAY

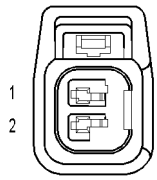
CAV	CIRCUIT	FUNCTION
1	K334 18BR	ASD RELAY OUTPUT
2	K13 18BR/LB	INJECTOR CONTROL NO. 3



FUEL INJECTOR NO. 4 (2.4L)

FUEL INJECTOR NO. 4 (2.4L) - 2 WAY

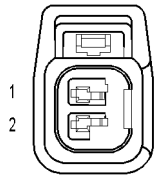
CAV	CIRCUIT	FUNCTION
1	F142 16PK/GY	ASD RELAY OUTPUT
2	K14 16BR/TN	INJECTOR CONTROL NO. 4



FUEL INJECTOR NO. 4 (4.0L)

FUEL INJECTOR NO. 4 (4.0L) - 2 WAY

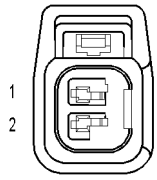
CAV	CIRCUIT	FUNCTION
1	K334 18BR	ASD RELAY OUTPUT
2	K14 18BR/TN	INJECTOR CONTROL NO. 4



FUEL INJECTOR NO. 5 (4.0L)

FUEL INJECTOR NO. 5 (4.0L) - 2 WAY

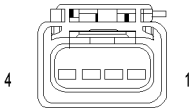
CAV	CIRCUIT	FUNCTION
1	K334 18BR	ASD RELAY OUTPUT
2	K38 18BR/OR	INJECTOR CONTROL NO. 5



FUEL INJECTOR NO. 6 (4.0L)

FUEL INJECTOR NO. 6 (4.0L) - 2 WAY

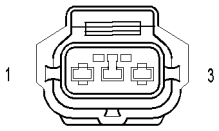
CAV	CIRCUIT	FUNCTION
1	K334 18BR	ASD RELAY OUTPUT
2	K58 18BR/VT	INJECTOR CONTROL NO. 6



FUEL PUMP MODULE

FUEL PUMP MODULE - 4 WAY

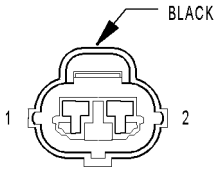
CAV	CIRCUIT	FUNCTION
1	Z987 18BK	GROUND
2	N4 20DB/WT	FUEL LEVEL SENSOR SIGNAL
3	K900 20DB/DG	SENSOR GROUND
4	A109 18OR/RD	FUEL PUMP RELAY OUTPUT



G-SWITCH

G-SWITCH - 3 WAY

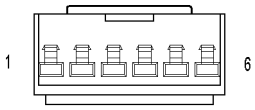
CAV	CIRCUIT	FUNCTION
1	G119 20VT	G-SWITCH TEST SIGNAL
2	G117 20VT/LG	G-SWITCH NO. 1 SENSE
3	G118 20VT/YL	G-SWITCH NO. 2 SENSE



GENERATOR

GENERATOR - BLACK 2 WAY

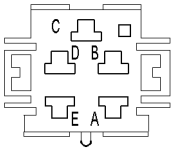
CAV	CIRCUIT	FUNCTION
1	K125 18BR/DG	GEN FIELD CONTROL
2	Z20 18BR/BK	GROUND



HEADLAMP LEVELING SWITCH (EXPORT)

HEADLAMP LEVELING SWITCH (EXPORT) - 6 WAY

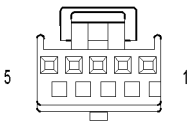
CAV	CIRCUIT	FUNCTION
1	-	-
2	L45 20WT/OR	FUSED HEADLAMP SWITCH OUTPUT
3	-	-
4	Z1 20BK	GROUND
5	L13 20WT/YL	HEADLAMP ADJUST SIGNAL
6	-	-



HEATER CONTROL C1 (HEATER ONLY)

HEATER CONTROL C1 (HEATER ONLY) - 5 WAY

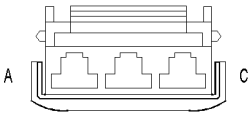
CAV	CIRCUIT	FUNCTION
A	C7 12BR/TN	BLOWER MOTOR HIGH SPEED
B	C6 14LB	BLOWER MOTOR M2 SPEED
C	C5 16LG/LB	BLOWER MOTOR M1 SPEED
D	C4 18TN	BLOWER MOTOR LOW SPEED
E	C1 12DG	BLOWER MOTOR FEED



HEATER CONTROL C2 (HEATER ONLY)

HEATER CONTROL C2 (HEATER ONLY) - 5 WAY

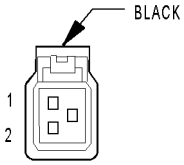
CAV	CIRCUIT	FUNCTION
1	Z11 20BK/WT (LHD)	GROUND
1	F24 20RD/DG (RHD)	FUSED IGNITION SWITCH OUTPUT (RUN)
2	C36 20DB/RD	BLEND DOOR FEEDBACK SIGNAL
3	F24 20RD/DG (LHD)	FUSED IGNITION SWITCH OUTPUT (RUN)
3	Z11 20BK/WT (RHD)	GROUND
4	E2 20OR	PANEL LAMPS FEED
5	Z1 20BK	GROUND



HEATER CONTROL C3 (HEATER ONLY)

HEATER CONTROL C3 (HEATER ONLY) - 3 WAY

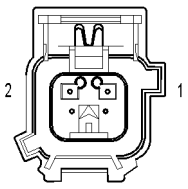
CAV	CIRCUIT	FUNCTION
A	-	-
B	C1 C12DG	BLOWER MOTOR FEED
C	Z1 12BK	GROUND



HORN

HORN - BLACK 2 WAY

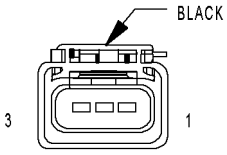
CAV	CIRCUIT	FUNCTION
1	Z950 18BK	GROUND
2	X2 18DG/OR	HORN RELAY OUTPUT



IDLE AIR CONTROL MOTOR

IDLE AIR CONTROL MOTOR - 2 WAY

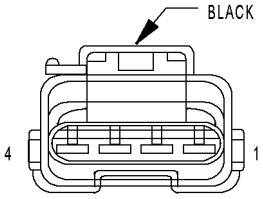
CAV	CIRCUIT	FUNCTION
1	K961 20BR/VT (2.4L)	IAC SIGNAL
1	K961 18BR/VT (4.0L)	IAC SIGNAL
2	K61 20VT/GY (2.4L)	IAC CONTROL
2	K61 18VT/GY (4.0L)	IAC CONTROL



IGNITION COIL PACK (2.4L)

IGNITION COIL PACK (2.4L) - BLACK 3 WAY

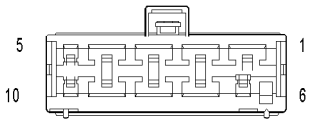
CAV	CIRCUIT	FUNCTION
1	K17 18DB/TN	COIL CONTROL NO. 2
2	K334 18BR	ASD RELAY OUTPUT
3	K19 18DB/DG	COIL CONTROL NO. 1



IGNITION COIL PACK (4.0L)

IGNITION COIL PACK (4.0L) - BLACK 4 WAY

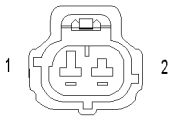
CAV	CIRCUIT	FUNCTION
1	K19 18DB/DG	COIL CONTROL NO. 1
2	K334 18BR	ASD RELAY OUTPUT
3	K17 18DB/TN	COIL CONTROL NO. 2
4	K18 18DB/OR	COIL CONTROL NO. 3



IGNITION SWITCH

IGNITION SWITCH - 10 WAY

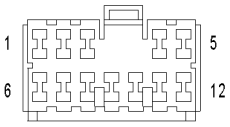
CAV	CIRCUIT	FUNCTION
1	A1 16RD	FUSED B(+)
2	F951 16PK/LG	IGNITION SWITCH OUTPUT (RUN-START)
3	F993 12PK/YL	IGNITION SWITCH OUTPUT (RUN-ACC)
4	A991 12 RD	FUSED B(+)
5	G15 20VT/TN	KEY-IN IGNITION SWITCH SENSE
6	F26 16PK/OR	IGNITION SWITCH OUTPUT (START)
7	F982 16PK/YL	IGNITION SWITCH OUTPUT (RUN-ACC)
8	F921 14PK/YL	IGNITION SWITCH OUTPUT (RUN)
9	A8 14RD	FUSED B(+)
10	Z902 16BK	GROUND



INPUT SPEED SENSOR

INPUT SPEED SENSOR - 2 WAY

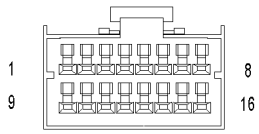
CAV	CIRCUIT	FUNCTION
1	T52 18DG/OR	INPUT SPEED SENSOR SIGNAL
2	T13 18DG/VT	SPEED SENSOR GROUND



INSTRUMENT CLUSTER C1

INSTRUMENT CLUSTER C1 - 12 WAY

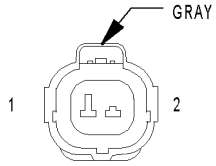
CAV	CIRCUIT	FUNCTION
1	L61 18WT/LG	LEFT TURN SIGNAL
2	L60 18WT/TN	RIGHT TURN SIGNAL
3	L300 16LG/WT	DIMMER SWITCH HIGH BEAM OUTPUT
4	L39 16WT/OR (EXCEPT EXPORT)	FOG LAMP RELAY OUTPUT
4	L36 16WT/DB (EXPORT)	REAR FOG LAMP FEED
5	L7 18WT/YL	HEADLAMP SWITCH OUTPUT
6	G305 20VT/LG (OFF-ROAD PACKAGE)	REAR LOCKER REQUEST
7	G301 20VT/LB (OFF-ROAD PACKAGE)	REAR LOCKER INDICATOR SWITCH SENSE
8	Z18 18BK/LB	GROUND
9	G303 20VT/DG (OFF-ROAD PACKAGE)	LOCKER ENABLE SIGNAL 2
10	D16 20WT/OR	SCI RECEIVE (TCM)
11	F505 20PK/YL	FUSED IGNITION SWITCH OUTPUT (RUN-START)
12	M21 20YL/BR	FUSED B(+)



INSTRUMENT CLUSTER C2

INSTRUMENT CLUSTER C2 - 16 WAY

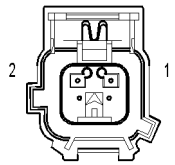
CAV	CIRCUIT	FUNCTION
1	C80 20DB (HARD TOP)	REAR WINDOW DEFOGGER SWITCH SENSE
2	G10 20VT/LG	SEAT BELT SWITCH SENSE
3	G76 20VT/YL	PASSENGER DOOR AJAR SWITCH SENSE
4	G75 20VT	DRIVER DOOR AJAR SWITCH SENSE
5	M22 20YL/OR	COURTESY LAMPS FEED
6	E2 20OR/BR	PANEL LAMPS FEED
7	C115 20DB (HARD TOP)	REAR WINDOW DEFOGGER RELAY CONTROL
8	G4 20VT/LB (ABS)	ABS WARNING INDICATOR DRIVER
9	B83 20DG/WT	BRAKE PRESSURE SENSOR SUPPLY
10	G304 20VT/DB (OFF-ROAD PACKAGE)	FRONT LOCKER REQUEST
11	G93 20VT/LB (OFF-ROAD PACKAGE)	4WD INDICATOR
12	D25 20WT/VT	PCI BUS
13	G15 20VT/TN	KEY-IN IGNITION SWITCH SENSE
14	G302 20VT/OR (OFF-ROAD PACKAGE)	LOCKER ENABLE SIGNAL 1
15	E19 20OR/BR	PANEL LAMPS DIMMER SIGNAL
16	G300 20VT/WT (OFF-ROAD PACKAGE)	FRONT LOCKER INDICATOR SWITCH SENSE



INTAKE AIR TEMPERATURE SENSOR (4.0L)

INTAKE AIR TEMPERATURE SENSOR (4.0L) - GRAY 2 WAY

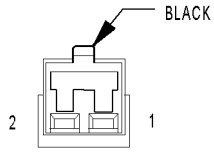
CAV	CIRCUIT	FUNCTION
1	K900 20DB/DG	SENSOR GROUND
2	K21 18BK/RD	IAT SIGNAL



INTAKE AIR TEMPERATURE SENSOR (2.4L)

INTAKE AIR TEMPERATURE SENSOR (2.4L) - 2 WAY

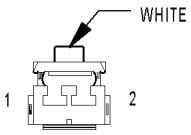
CAV	CIRCUIT	FUNCTION
1	K900 20DB/DG	SENSOR GROUND
2	K21 20BR/WT	IAT SIGNAL



LEFT COURTESY LAMP

LEFT COURTESY LAMP - BLACK 2 WAY

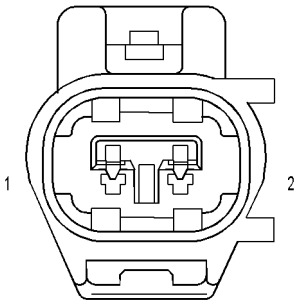
CAV	CIRCUIT	FUNCTION
1	M21 20YL/BR	FUSED B(+)
2	M22 20YL/OR	COURTESY LAMPS DRIVER



LEFT DOME LAMP

LEFT DOME LAMP - WHITE 2 WAY

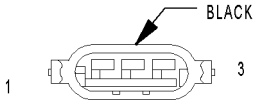
CAV	CIRCUIT	FUNCTION
1	M21 20PK/WT	FUSED B(+)
2	M22 20YL	COURTESY LAMPS DRIVER



LEFT FOG LAMP (EXCEPT EXPORT)

LEFT FOG LAMP (EXCEPT EXPORT) - 2 WAY

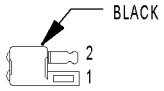
CAV	CIRCUIT	FUNCTION
1	Z950 18BK	GROUND
2	L39 18WT/OR	FOG LAMP RELAY OUTPUT



LEFT FRONT PARK/ TURN SIGNAL LAMP

LEFT FRONT PARK/TURN SIGNAL LAMP - BLACK 3 WAY

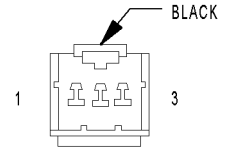
CAV	CIRCUIT	FUNCTION
1	L61 18WT/LG	LEFT TURN SIGNAL
2	L7 18WT/YL (EXCEPT EXPORT)	HEADLAMP SWITCH OUTPUT
3	Z950 18BK	GROUND



LEFT FRONT POSITION LAMP (EXPORT)

LEFT FRONT POSITION LAMP (EXPORT) - BLACK 2 WAY

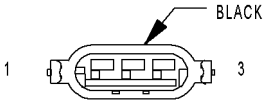
CAV	CIRCUIT	FUNCTION
1	L7 18WT/YL	HEADLAMP SWITCH OUTPUT
2	Z950 20BK	GROUND



LEFT FRONT SPEAKER

LEFT FRONT SPEAKER - BLACK 3 WAY

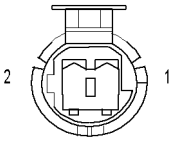
CAV	CIRCUIT	FUNCTION
1	X53 18DG	LEFT FRONT SPEAKER (+)
2	-	-
3	X55 18DG/BR	LEFT FRONT SPEAKER (-)



LEFT FRONT TURN SIGNAL LAMP (EXPORT)

LEFT FRONT TURN SIGNAL LAMP (EXPORT) - BLACK 3 WAY

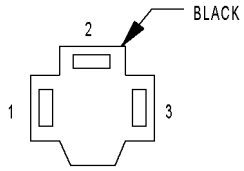
CAV	CIRCUIT	FUNCTION
1	L61 18WT/LG	LEFT TURN SIGNAL
2	-	-
3	Z950 18BK	GROUND



LEFT FRONT WHEEL SPEED SENSOR

LEFT FRONT WHEEL SPEED SENSOR - 2 WAY

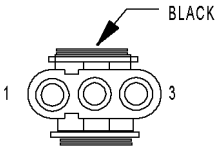
CAV	CIRCUIT	FUNCTION
1	B8 18DG/TN	LEFT FRONT WHEEL SPEED SENSOR (-)
2	B9 18DG/LG	LEFT FRONT WHEEL SPEED SENSOR (+)



LEFT HEADLAMP

LEFT HEADLAMP - BLACK 3 WAY

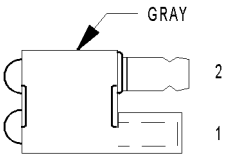
CAV	CIRCUIT	FUNCTION
1	L300 14LG/WT	DIMMER SWITCH HIGH BEAM OUTPUT
1	L300 14LG/WT	DIMMER SWITCH HIGH BEAM OUTPUT
2	L4 14WT/LG	DIMMER SWITCH LOW BEAM OUTPUT
2	L4 14WT/LG	DIMMER SWITCH LOW BEAM OUTPUT
3	Z950 20BK	GROUND



LEFT HEADLAMP LEVELING MOTOR (EXPORT)

LEFT HEADLAMP LEVELING MOTOR (EXPORT) - BLACK 3 WAY

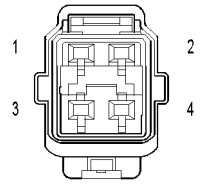
CAV	CIRCUIT	FUNCTION
1	L45 20WT/OR	FUSED HEADLAMP SWITCH OUTPUT
2	L13 20WT/YL	HEADLAMP ADJUST SIGNAL
3	Z950 20BK	GROUND



LEFT LICENSE LAMP (EXPORT)

LEFT LICENSE LAMP (EXPORT) - GRAY 2 WAY

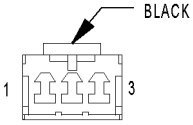
CAV	CIRCUIT	FUNCTION
1	L7 18WT/YL	HEADLAMP SWITCH OUTPUT
2	Z913 20BK	GROUND



LEFT REAR LAMP ASSEMBLY

LEFT REAR LAMP ASSEMBLY - 4 WAY

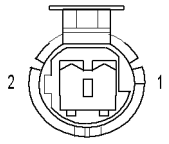
CAV	CIRCUIT	FUNCTION
1	L1 20WT/LG (EXCEPT EXPORT)	BACK-UP LAMP FEED
1	L63 18DG/RD (EXPORT)	LEFT TURN/STOP SIGNAL
2	L63 18DG/RD (EXCEPT EXPORT)	LEFT TURN/STOP SIGNAL
2	L1 20WT/LG (EXPORT)	BACK-UP LAMP FEED
3	L7 18WT/YL (EXCEPT EXPORT)	HEADLAMP SWITCH OUTPUT
3	L50 18WT/TN (EXPORT)	BRAKE LAMP SWITCH OUTPUT
4	L7 18WT/YL (EXPORT)	HEADLAMP SWITCH OUTPUT



LEFT REAR SPEAKER

LEFT REAR SPEAKER - BLACK 3 WAY

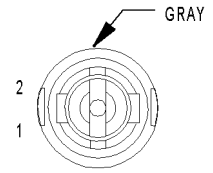
CAV	CIRCUIT	FUNCTION
1	X57 20BR/LB	LEFT REAR SPEAKER (-)
2	-	-
3	X51 18BR/YL	LEFT REAR SPEAKER (+)



LEFT REAR WHEEL SPEED SENSOR

LEFT REAR WHEEL SPEED SENSOR - 2 WAY

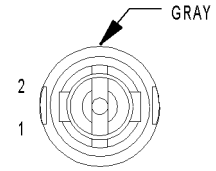
CAV	CIRCUIT	FUNCTION
1	B3 18DG/YL	LEFT REAR WHEEL SPEED SENSOR (-)
2	B4 18DG/GY	LEFT REAR WHEEL SPEED SENSOR (+)



LEFT SIDE MARKER LAMP (EXCEPT EXPORT)

LEFT SIDE MARKER LAMP (EXCEPT EXPORT) - GRAY 2 WAY

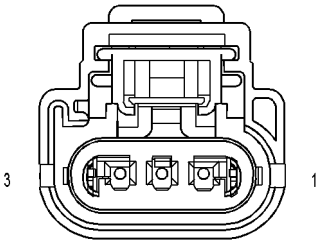
CAV	CIRCUIT	FUNCTION
1	L7 18WT/YL	HEADLAMP SWITCH OUTPUT
2	L61 18WT/LG	LEFT TURN SIGNAL



LEFT SIDE REPEATER LAMP (EXPORT)

LEFT SIDE REPEATER LAMP (EXPORT) - GRAY 2 WAY

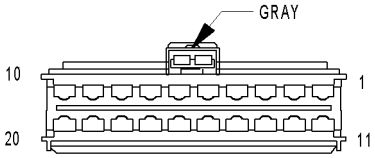
CAV	CIRCUIT	FUNCTION
1	L61 18WT/LG	LEFT TURN SIGNAL
2	Z950 18BK	GROUND



MANIFOLD ABSOLUTE PRESSURE SENSOR

MANIFOLD ABSOLUTE PRESSURE SENSOR - 3 WAY

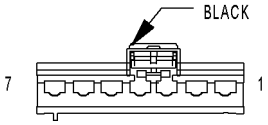
CAV	CIRCUIT	FUNCTION
1	K1 20VT/BR (2.4L)	MAP SIGNAL
1	K1 18VT/BR (4.0L)	MAP SIGNAL
2	K900 20DB/DG	SENSOR GROUND
3	F856 20YL/PK	5 VOLT SUPPLY



MULTI-FUNCTION SWITCH C1

MULTI-FUNCTION SWITCH C1 - GRAY 20 WAY

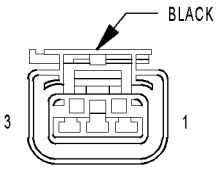
CAV	CIRCUIT	FUNCTION
1	L50 18WT/TN (EXCEPT EXPORT)	BRAKE LAMP SWITCH OUTPUT
2	L61 18GY	LEFT TURN SIGNAL
3	L5 20WT/LB	FUSED IGNITION SWITCH OUTPUT (RUN-ACC)
4	L63 18DG/RD	LEFT TURN/STOP SIGNAL
5	L62 18BR/RD	RIGHT TURN/STOP SIGNAL
6	L60 18TN	RIGHT TURN SIGNAL
7	Z902 18BK	GROUND
8	E19 20OR/BR	PANEL LAMPS DRIVER
9	L7 18WT/YL	HEADLAMP SWITCH OUTPUT
10	-	-
11	A701 18BR/RD	FUSED B(+)
12	L36 16WT/DB (EXPORT)	REAR FOG LAMP FEED
13	A39 18RD/LG	FUSED B(+)
14	L139 18WT/YL (EXCEPT EXPORT)	FOG LAMP SWITCH SIGNAL
15	-	-
16	L4 14WT/LG	DIMMER SWITCH LOW BEAM OUTPUT
17	L300 14LG/WT	DIMMER SWITCH HIGH BEAM OUTPUT
18	A333 14WT/RD	FUSED B(+)
19	A333 14WT/RD	FUSED B(+)
20	L177 18WT/OR	FUSED B(+)



MULTI-FUNCTION SWITCH C2

MULTI-FUNCTION SWITCH C2 - BLACK 7 WAY

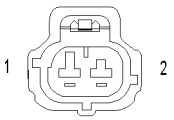
CAV	CIRCUIT	FUNCTION
1	W7 16BR/GY	WIPER PARK SWITCH SENSE
2	Z902 20BK	GROUND
3	W10 16BR	WASHER PUMP CONTROL SWITCH OUTPUT
4	F300 16OR/PK	FUSED IGNITION SWITCH OUTPUT (RUN-ACC)
5	W4 16BR/OR	WIPER SWITCH HIGH SPEED OUTPUT
6	W3 16BR/WT	LOW SPEED WIPER SWITCH OUTPUT
7	-	-



NATURAL VACUUM LEAK DETECTION ASSEMBLY

NATURAL VACUUM LEAK DETECTION ASSEMBLY - BLACK 3 WAY

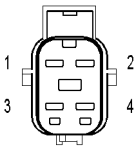
CAV	CIRCUIT	FUNCTION
1	Z913 20BK	GROUND
2	K107 20VT/WT	NVLD SWITCH SIGNAL
3	K106 20VT/LB	NVLD SOL CONTROL



OUTPUT SPEED SENSOR

OUTPUT SPEED SENSOR (A/T) - 2 WAY

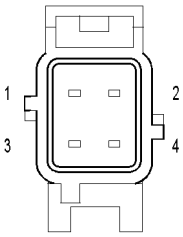
CAV	CIRCUIT	FUNCTION
1	T14 18DG/BR	OUTPUT SPEED SENSOR SIGNAL
2	T13 18DG/VT	SPEED SENSOR GROUND



OXYGEN SENSOR 1/1 UPSTREAM (2.4L)

OXYGEN SENSOR 1/1 UPSTREAM (2.4L) - 4 WAY

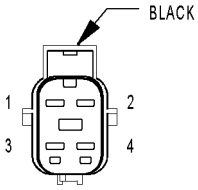
CAV	CIRCUIT	FUNCTION
1	K99 18BR/LG	O2 1/1 HEATER CONTROL
2	Z42 18BK/LG	GROUND
3	K902 18BR/DG	O2 RETURN (UPSTREAM)
4	K41 18DB/LB	O2 1/1 SIGNAL



OXYGEN SENSOR 1/1 UPSTREAM (4.0L)

OXYGEN SENSOR 1/1 UPSTREAM (4.0L) - 4 WAY

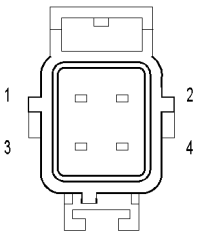
CAV	CIRCUIT	FUNCTION
1	K99 18BR/LG	O2 1/1 HEATER CONTROL
2	Z42 18BK/LG	GROUND
3	K902 18BR/DG	O2 RETURN (UPSTREAM)
4	K41 18DB/LB	O2 1/1 SIGNAL



OXYGEN SENSOR 1/2 DOWNSTREAM

OXYGEN SENSOR 1/2 DOWNSTREAM - BLACK 4 WAY

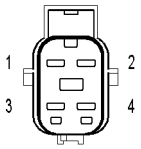
CAV	CIRCUIT	FUNCTION
1	K299 18BR/WT	O2 1/2 HEATER CONTROL
2	Z43 18BK/LB	GROUND
3	K904 18DB/DG	O2 RETURN (DOWNSTREAM)
4	K141 18DB/YL	O2 1/2 SIGNAL



OXYGEN SENSOR 2/1 UPSTREAM (4.0L)

OXYGEN SENSOR 2/1 UPSTREAM (4.0L) - 4 WAY

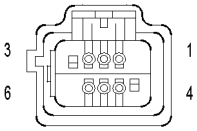
CAV	CIRCUIT	FUNCTION
1	K199 18BR/VT	O2 2/1 HEATER CONTROL
2	Z42 18BK/LG	GROUND
3	K902 18BR/DG	O2 RETURN (UPSTREAM)
4	K43 18DB/LG	O2 2/1 SIGNAL



OXYGEN SENSOR 2/2 DOWNSTREAM (4.0L)

OXYGEN SENSOR 2/2 DOWNSTREAM (4.0L) - 4 WAY

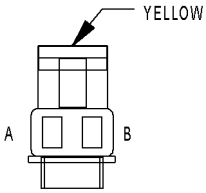
CAV	CIRCUIT	FUNCTION
1	K399 18BR/GY	O2 2/2 HEATER CONTROL
2	Z43 18BK/LB	GROUND
3	K904 18BD/DG	O2 RETURN (DOWNSTREAM)
4	K243 18BR	O2 2/2 SIGNAL



PASSENGER AIRBAG ON-OFF SWITCH (LHD)

PASSENGER AIRBAG ON-OFF SWITCH (LHD) - 6 WAY

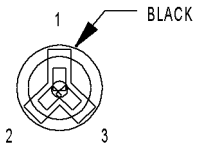
CAV	CIRCUIT	FUNCTION
1	F201 18PK/OR	FUSED IGNITION SWITCH OUTPUT (RUN-START)
2	G104 18VT/GY	PASSENGER AIRBAG INDICATOR DRIVER
3	R106 18LG/LB	PASSENGER AIRBAG MUX SWITCH SENSE
4	-	-
5	-	-
6	R104 18LB/LG	PASSENGER AIRBAG MUX SWITCH RETURN



PASSENGER AIRBAG SQUIB 1

PASSENGER AIRBAG SQUIB 1 - YELLOW 2 WAY

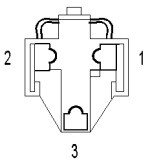
CAV	CIRCUIT	FUNCTION
A	R42 18LB/BR	PASSENGER SQUIB 1 LINE 1
B	R44 18LB/OR	PASSENGER SQUIB 1 LINE 2



PASSENGER DOOR AJAR SWITCH

PASSENGER DOOR AJAR SWITCH - BLACK 3 WAY

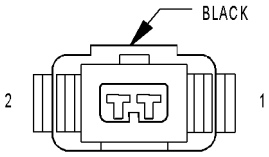
CAV	CIRCUIT	FUNCTION
1	M23 20YL/DG	DOOR AJAR SWITCH OUTPUT
2	-	-
3	G76 20VT/YL	PASSENGER DOOR AJAR SWITCH SENSE



POWER OUTLET

POWER OUTLET - 3 WAY

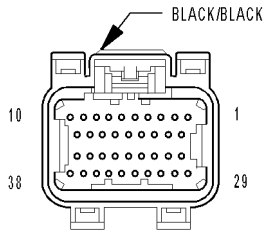
CAV	CIRCUIT	FUNCTION
1	A205 18RD/OR	FUSED B(+)
2	-	-
3	Z1 16BK	GROUND



POWER STEERING PRESSURE SWITCH (2.4L)

POWER STEERING PRESSURE SWITCH (2.4L) - BLACK 2 WAY

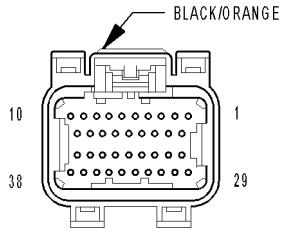
CAV	CIRCUIT	FUNCTION
1	K66 18DB/WT	P/S PRESSURE SWITCH SIGNAL
2	Z987 20BK	GROUND



POWERTRAIN CONTROL MODULE C1

POWERTRAIN CONTROL MODULE C1 - BLACK/BLACK 38 WAY

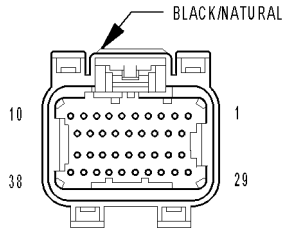
CAV	CIRCUIT	FUNCTION
1	-	-
2	-	-
3	-	-
4	-	-
5	-	-
6	-	-
7	-	-
8	-	-
9	Z130 18BK/BR	GROUND
10	C20 20DB/YL (2.4L)	A/C REQUEST SIGNAL
10	C20 18DB/YL (4.0L)	A/C REQUEST SIGNAL
11	F1 18PK/WT	FUSED IGNITION SWITCH OUTPUT (RUN-START)
12	F1 18PK/WT (A/T)	FUSED IGNITION SWITCH OUTPUT (RUN-START)
13	K75 18DB/OR	VEHICLE SPEED SENSOR SIGNAL
14	-	-
15	-	-
16	-	-
17	-	-
18	Z131 18BK/DG	GROUND
19	-	-
20	G6 18VT/GY	OIL PRESSURE SIGNAL
21	-	-
22	G31 18VT/LG	AAT SIGNAL
23	-	-
24	-	-
25	D20 18WT/LG	SCI RECEIVE (PCM)
26	D16 18WT/OR	SCI RECEIVE (TCM)
27	F856 18YL/PK	5 VOLT SUPPLY
28	T56 18DG/LB	OVERDRIVE OFF SWITCH INDICATOR
29	A209 18RD	FUSED B(+)
30	F26 16PK/OR	IGNITION SWITCH OUTPUT (START)
31	K141 18DB/YL	O2 1/2 SIGNAL
32	K902 18BR/DG	O2 RETURN (UPSTREAM)
33	K243 18BR (4.0L)	O2 2/2 SIGNAL
34	-	-
35	-	-
36	D21 18WT/BR	SCI TRANSMIT (PCM/CAB)
37	D15 18WT/DG	SCI RECEIVE (TCM)
38	D25 18WT/VT	PCI BUS



POWERTRAIN CONTROL MODULE C2

POWERTRAIN CONTROL MODULE C2 - BLACK/ORANGE 38 WAY

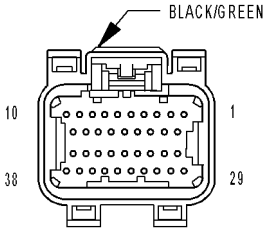
CAV	CIRCUIT	FUNCTION
1	-	-
2	-	-
3	-	-
4	K58 18BR/VT (4.0L)	INJECTOR CONTROL NO. 6
5	K38 18BR/OR (4.0L)	INJECTOR CONTROL NO. 5
6	-	-
7	K18 18DB/OR (4.0L)	COIL CONTROL NO. 3
8	-	-
9	K17 18DB/TN	COIL CONTROL NO. 2
10	K19 18DB/DG	COIL CONTROL NO. 1
11	K14 18BR/TN	INJECTOR CONTROL NO. 4
12	K13 18BR/LB	INJECTOR CONTROL NO. 3
13	K12 18BR/DB	INJECTOR CONTROL NO. 2
14	K11 18BR/YL	INJECTOR CONTROL NO. 1
15	-	-
16	-	-
17	K199 18BR/VT	O2 2/1 HEATER CONTROL
18	K99 18BR/LG	O2 1/1 HEATER CONTROL
19	K125 18BR/DG	GEN FIELD CONTROL
20	K2 18VT/OR	ECT SIGNAL
21	K22 18BR/OR	TP NO. 1 SIGNAL
22	-	-
23	K1 18VT/BR	MAP SIGNAL
24	-	-
25	-	-
26	Y3 18LG (EXCEPT OFF-ROAD PACKAGE)	FUSED SENSOR GROUND
26	K77 18BR/WT (OFF-ROAD PACKAGE)	TRANSFER CASE POSITION SENSOR INPUT
27	K900 18DB/DG	SENSOR GROUND
28	K961 18BR/VT	IAC SIGNAL
29	F855 18PK/YL	5 VOLT SUPPLY
30	K21 18DB/LG (2.4L)	IAT SIGNAL
30	K21 18BK/RD (4.0L)	IAT SIGNAL
31	K41 18DB/LB	O2 1/1 SIGNAL
32	K904 18DB/DG	O2 RETURN (DOWNSTREAM)
33	K43 18DB/LG (4.0L)	O2 2/1 SIGNAL
34	K44 18DB/GY	CMP SIGNAL
35	K24 18BR/LB	CKP SIGNAL
36	-	-
37	-	-
38	K61 18VT/GY	IAC CONTROL



POWERTRAIN CONTROL MODULE C3

POWERTRAIN CONTROL MODULE C3 - BLACK/NATURAL 38 WAY

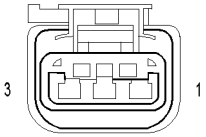
CAV	CIRCUIT	FUNCTION
1	-	-
2	-	-
3	K342 20BR/WT	ASD RELAY CONTROL
4	N112 20DB/OR	RADIATOR FAN RELAY CONTROL
5	V35 20VT/LG (SPEED CONTROL)	S/C VENT SOL CONTROL
6	N201 20DB/LG	LOW SPEED RADIATOR FAN RELAY CONTROL
7	V32 20VT/YL	S/C SUPPLY
8	K106 20VT/LB	NVLD SOL CONTROL
9	K299 20BR/WT	O2 1/2 HEATER CONTROL
10	K399 20BR/GY (4.0L)	O2 2/2 HEATER CONTROL
11	C13 20LB/OR	A/C CLUTCH RELAY CONTROL
12	V36 20VT/YL	S/C VACUUM SOL CONTROL
13	-	-
14	-	-
15	-	-
16	-	-
17	-	-
18	-	-
19	K334 18BR	ASD RELAY OUTPUT
20	K52 20DB/WT	EVAP/PURGE SOL CONTROL
21	T141 18YL/OR	CLUTCH INTERLOCK SWITCH SENSE
22	-	-
23	B15 20DG/WT	BRAKE SWITCH NO. 1 SIGNAL
24	C221 20LB/OR	A/C SELECT SIGNAL
25	-	-
26	-	-
27	-	-
28	K334 18BR	ASD RELAY OUTPUT
29	K70 20DB/BR	EVAP/PURGE SOL SIGNAL
30	K66 20DB/WT	P/S PRESSURE SWITCH SIGNAL
31	C18 20LB/BR	A/C PRESSURE SIGNAL
32	K25 20DB/VT	BATT TEMP SIGNAL
33	N4 20DB/WT	FUEL LEVEL SENSOR SIGNAL
34	V37 20VT	S/C SWITCH NO. 1 SIGNAL
35	K107 20VT/WT	NVLD SWITCH SIGNAL
36	-	-
37	K31 20BR	FUEL PUMP RELAY CONTROL
38	T752 20DG/OR	STARTER MOTOR RELAY CONTROL



POWERTRAIN CONTROL MODULE C4

POWERTRAIN CONTROL MODULE C4 - BLACK/GREEN 38 WAY

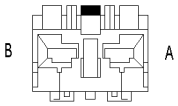
CAV	CIRCUIT	FUNCTION
1	T60 16YL/GY	OD SOLENOID CONTROL
2	T59 16YL/LB	UD SOLENOID CONTROL
3	-	-
4	-	-
5	-	-
6	T19 16YL/DB	2-4 SOLENOID CONTROL
7	-	-
8	-	-
9	-	-
10	T20 16DG/WT	L/R SOLENOID CONTROL
11	T140 18YL/GY	MS SOLENOID CONTROL
12	Z903 16BK	GROUND
13	Z903 16BK	GROUND
14	-	-
15	T1 18DG/LB	TRS T1 SENSE
16	T3 18DG/DB	TRS T3 SENSE
17	T6 18DG	TOW/HAUL OVERDRIVE OFF SWITCH SENSE
18	T515 18YL/DB	TRANSMISSION CONTROL RELAY CONTROL
19	T16 16YL/OR	TRANSMISSION CONTROL RELAY OUTPUT
20	-	-
21	-	-
22	T9 18DG/TN	OD PRESSURE SWITCH SENSE
23	-	-
24	-	-
25	-	-
26	-	-
27	T41 18YL/DB	TRS T41 SENSE (P/N)
28	T16 16YL/OR	TRANSMISSION CONTROL RELAY OUTPUT
29	T50 18YL/TN	L/R PRESSURE SWITCH SENSE
30	T47 18YL/DG	2-4 PRESSURE SWITCH SENSE
31	T38 18YL/BR	LINE PRESSURE SENSOR SIGNAL
32	T14 18DG/BR	OUTPUT SPEED SENSOR SIGNAL
33	T52 18DG/OR	INPUT SPEED SENSOR SIGNAL
34	T13 18DG/VT	SPEED SENSOR GROUND
35	T54 18DG/OR	TRANSMISSION TEMPERATURE SENSOR SIGNAL
36	-	-
37	T42 18DG/YL	TRS T42 SENSE
38	-	-



RADIATOR FAN MOTOR (2.4L)

RADIATOR FAN MOTOR (2.4L) - 3 WAY

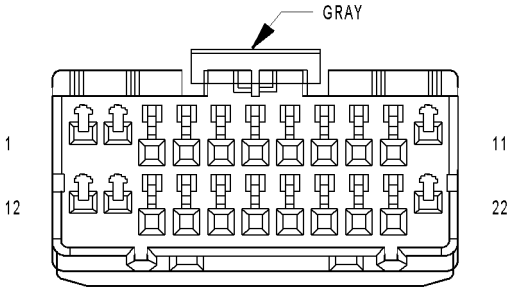
CAV	CIRCUIT	FUNCTION
1	C25 12DG	HIGH SPEED RADIATOR RELAY OUTPUT
2	Z213 12BK	GROUND
3	C23 12DB	LOW SPEED FAN RELAY OUTPUT



RADIATOR FAN MOTOR CIRCUIT BREAKER (2.4L)

RADIATOR FAN MOTOR CIRCUIT BREAKER (2.4L) - 2 WAY

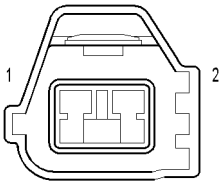
CAV	CIRCUIT	FUNCTION
A	Z213 12BK	GROUND
B	Z823 12BK/DG	GROUND



RADIO

RADIO - GRAY 22 WAY

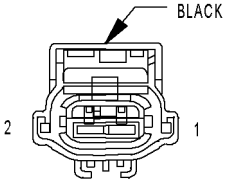
CAV	CIRCUIT	FUNCTION
1	F306 16DB/PK	FUSED B(+)
2	X112 16DG/GY	FUSED IGNITION SWITCH OUTPUT (RUN-ACC)
3	E2 20OR/BR	PANEL LAMPS FEED
4	-	-
5	-	-
6	-	-
7	X54 18GY	RIGHT FRONT SPEAKER (+)
8	X56 18GY/BR	RIGHT FRONT SPEAKER (-)
9	X55 18DG/BR	LEFT FRONT SPEAKER (-)
10	X53 18DG	LEFT FRONT SPEAKER (+)
11	Z514 16BK/LG	GROUND
12	F306 16DB/PK	FUSED B(+)
13	X60 20GY/OR	RADIO 12V OUTPUT
14	D25 20WT/VT	PCI BUS
15	-	-
16	-	-
17	-	-
18	X51 18DG/DB	LEFT REAR SPEAKER (+)
19	X57 18DG/OR	LEFT REAR SPEAKER (-)
20	X58 18GY/OR	RIGHT REAR SPEAKER (-)
21	X52 18GY/DB	RIGHT REAR SPEAKER (+)
22	Z515 16BK	GROUND



REAR FOG LAMP (EXPORT)

REAR FOG LAMP (EXPORT) - 2 WAY

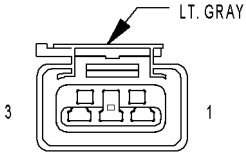
CAV	CIRCUIT	FUNCTION
1	L36 18BR/WT	REAR FOG LAMP FEED
2	Z913 20BK	GROUND



REAR LOCKER INDICATOR SWITCH (OFF-ROAD PACKAGE)

REAR LOCKER INDICATOR SWITCH (OFF-ROAD PACKAGE) - BLACK 2 WAY

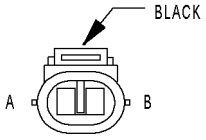
CAV	CIRCUIT	FUNCTION
1	Z987 20BK	GROUND
2	G301 20VT/LB	REAR LOCKER INDICATOR SWITCH SENSE



REAR LOCKER PUMP (OFF-ROAD PACKAGE)

REAR LOCKER PUMP (OFF-ROAD PACKAGE) - LT. GRAY 3 WAY

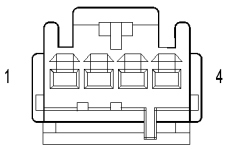
CAV	CIRCUIT	FUNCTION
1	Z987 18BK	GROUND
2	-	-
3	A850 18RD/WT	REAR LOCKER RELAY OUTPUT



REAR WASHER PUMP (HARD TOP)

REAR WASHER PUMP (HARD TOP) - BLACK 2 WAY

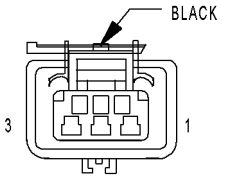
CAV	CIRCUIT	FUNCTION
A	Z913 18BK	GROUND
B	W20 18BR/YL	REAR WASHER MOTOR CONTROL



REAR WINDOW DEFOGGER SWITCH (HARD TOP)

REAR WINDOW DEFOGGER SWITCH (HARD TOP) - 4 WAY

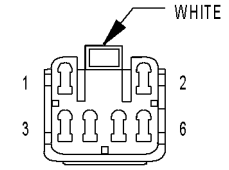
CAV	CIRCUIT	FUNCTION
1	Z1 20BK	GROUND
2	C80 20DB	REAR WINDOW DEFOGGER SWITCH SENSE
3	C515 20LB/OR	FUSED REAR WINDOW DEFOGGER RELAY OUTPUT
4	E2 20OR/BR	PANEL LAMPS FEED



REAR WIPER MOTOR (HARD TOP)

REAR WIPER MOTOR (HARD TOP) - BLACK 3 WAY

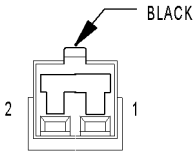
CAV	CIRCUIT	FUNCTION
1	F302 18BR/PK	FUSED IGNITION SWITCH OUTPUT (RUN)
2	Z913 16BK	GROUND
3	W13 18BK/LG	REAR WIPER MOTOR CONTROL



REAR WIPER/ WASHER SWITCH (HARD TOP)

REAR WIPER/WASHER SWITCH (HARD TOP) - WHITE 6 WAY

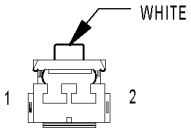
CAV	CIRCUIT	FUNCTION
1	Z1 18BK	GROUND
2	W20 18BR/YL	REAR WASHER MOTOR CONTROL
3	E2 20OR/BR	PANEL LAMPS FEED
4	W13 18BR/LG	REAR WIPER MOTOR CONTROL
5	F302 18GY/PK	FUSED IGNITION SWITCH OUTPUT (RUN)
6	-	-



RIGHT COURTESY LAMP

RIGHT COURTESY LAMP - BLACK 2 WAY

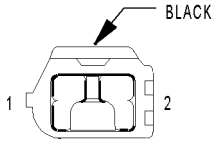
CAV	CIRCUIT	FUNCTION
1	M21 20YL/BR	FUSED B(+)
2	M22 20YL/OR	COURTESY LAMPS DRIVER



RIGHT DOME LAMP

RIGHT DOME LAMP - WHITE 2 WAY

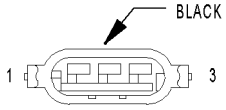
CAV	CIRCUIT	FUNCTION
1	M21 20PK/WT	FUSED B(+)
2	M22 20YL	COURTESY LAMPS DRIVER



RIGHT FOG LAMP (EXCEPT EXPORT)

RIGHT FOG LAMP (EXCEPT EXPORT) - BLACK 2 WAY

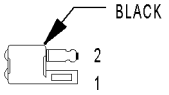
CAV	CIRCUIT	FUNCTION
1	Z962 18BK	GROUND
2	L39 16LB	FOG LAMP RELAY OUTPUT



RIGHT FRONT PARK/ TURN SIGNAL LAMP

RIGHT FRONT PARK/TURN SIGNAL LAMP - BLACK 3 WAY

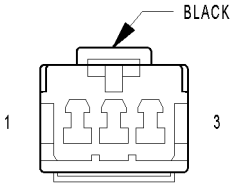
CAV	CIRCUIT	FUNCTION
1	L60 18WT/TN	RIGHT TURN SIGNAL
2	L7 18WT/YL (EXCEPT EXPORT)	HEADLAMP SWITCH OUTPUT
3	Z962 18BK	GROUND



RIGHT FRONT POSITION LAMP (EXPORT)

RIGHT FRONT POSITION LAMP (EXPORT) - BLACK 2 WAY

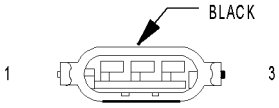
CAV	CIRCUIT	FUNCTION
1	L7 18WT/YL	HEADLAMP SWITCH OUTPUT
2	Z962 20BK	GROUND



RIGHT FRONT SPEAKER

RIGHT FRONT SPEAKER - BLACK 3 WAY

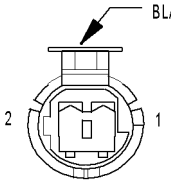
CAV	CIRCUIT	FUNCTION
1	X54 18GY	RIGHT FRONT SPEAKER (+)
2	-	-
3	X56 18GY/BR	RIGHT FRONT SPEAKER (-)



RIGHT FRONT TURN SIGNAL LAMP (EXPORT)

RIGHT FRONT TURN SIGNAL LAMP (EXPORT) - BLACK 3 WAY

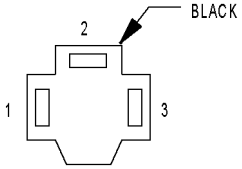
CAV	CIRCUIT	FUNCTION
1	L60 18WT/TN	RIGHT TURN SIGNAL
2	-	-
3	Z962 18BK	GROUND



RIGHT FRONT WHEEL SPEED SENSOR

RIGHT FRONT WHEEL SPEED SENSOR - BLACK 2 WAY

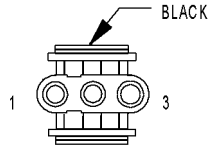
CAV	CIRCUIT	FUNCTION
1	B6 18DG/WT	RIGHT FRONT WHEEL SPEED SENSOR (-)
2	B7 18DG/VT	RIGHT FRONT WHEEL SPEED SENSOR (+)



RIGHT HEADLAMP

RIGHT HEADLAMP - BLACK 3 WAY

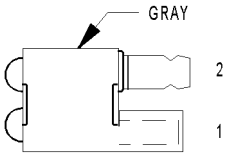
CAV	CIRCUIT	FUNCTION
1	L300 14LG/WT	DIMMER SWITCH HIGH BEAM OUTPUT
2	L4 14WT/LG	DIMMER SWITCH LOW BEAM OUTPUT
3	Z962 18BK	GROUND



RIGHT HEADLAMP LEVELING MOTOR (EXPORT)

RIGHT HEADLAMP LEVELING MOTOR (EXPORT) - BLACK 3 WAY

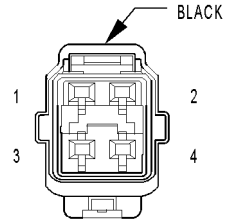
CAV	CIRCUIT	FUNCTION
1	L45 20WT/OR	FUSED HEADLAMP SWITCH OUTPUT
2	L13 20WT/YL	HEADLAMP ADJUST SIGNAL
3	Z962 20BK	GROUND



RIGHT LICENSE LAMP (EXPORT)

RIGHT LICENSE LAMP (EXPORT) - GRAY 2 WAY

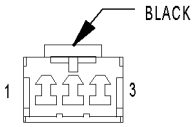
CAV	CIRCUIT	FUNCTION
1	L7 18WT/YL	HEADLAMP SWITCH OUTPUT
2	Z913 20BK	GROUND



RIGHT REAR LAMP ASSEMBLY

RIGHT REAR LAMP ASSEMBLY - BLACK 4 WAY

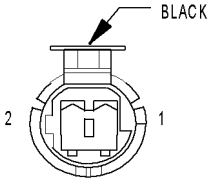
CAV	CIRCUIT	FUNCTION
1	L1 20WT/LG (EXCEPT EXPORT)	BACK-UP LAMP FEED
1	L62 18BR/RD (EXPORT)	RIGHT TURN/STOP SIGNAL
2	L63 18DG/RD (EXCEPT EXPORT)	LEFT TURN/STOP SIGNAL
2	L1 20WT/LG (EXPORT)	BACK-UP LAMP FEED
3	L7 18WT/YL (EXCEPT EXPORT)	HEADLAMP SWITCH OUTPUT
3	L50 18WT/TN (EXPORT)	BRAKE LAMP SWITCH OUTPUT
4	L62 18BR/RD (EXCEPT EXPORT)	RIGHT TURN/STOP SIGNAL
4	L7 18WT/YL (EXPORT)	HEADLAMP SWITCH OUTPUT



RIGHT REAR SPEAKER

RIGHT REAR SPEAKER - BLACK 3 WAY

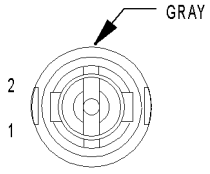
CAV	CIRCUIT	FUNCTION
1	X58 18DB/PK	RIGHT REAR SPEAKER (-)
2	-	-
3	X52 18DB/WT	RIGHT REAR SPEAKER (+)



RIGHT REAR WHEEL SPEED SENSOR

RIGHT REAR WHEEL SPEED SENSOR - BLACK 2 WAY

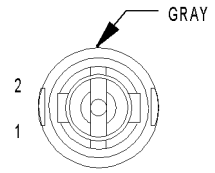
CAV	CIRCUIT	FUNCTION
1	B1 18DG/OR	RIGHT REAR WHEEL SPEED SENSOR (-)
2	B2 18DG/LB	RIGHT REAR WHEEL SPEED SENSOR (+)



RIGHT SIDE MARKER LAMP (EXCEPT EXPORT)

RIGHT SIDE MARKER LAMP (EXCEPT EXPORT) - GRAY 2 WAY

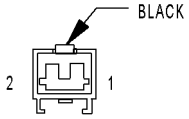
CAV	CIRCUIT	FUNCTION
1	L7 18WT/YL	HEADLAMP SWITCH OUTPUT
2	L60 18WT/TN	RIGHT TURN SIGNAL



RIGHT SIDE REPEATER LAMP (EXPORT)

RIGHT SIDE REPEATER LAMP (EXPORT) - GRAY 2 WAY

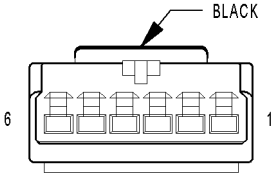
CAV	CIRCUIT	FUNCTION
1	L60 18WT/TN	RIGHT TURN SIGNAL
2	Z962 18BK	GROUND



SEAT BELT SWITCH (EXCEPT LHD EXPORT)

SEAT BELT SWITCH (EXCEPT LHD EXPORT) - BLACK 2 WAY

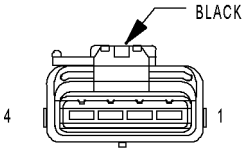
CAV	CIRCUIT	FUNCTION
1	G10 20VT/LG	SEAT BELT SWITCH SENSE
2	Z913 20BK	GROUND



SENTRY KEY IMMOBILIZER MODULE

SENTRY KEY IMMOBILIZER MODULE - BLACK 6 WAY

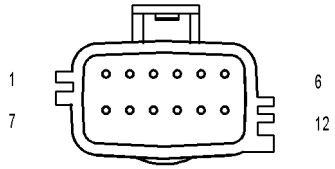
CAV	CIRCUIT	FUNCTION
1	-	-
2	D25 20WT/VT	PCI BUS
3	-	-
4	F1 20PK/WT	FUSED IGNITION SWITCH OUTPUT (RUN-START)
5	Z120 20BK/WT	GROUND
6	L177 20WT/OR	FUSED B(+)



SPEED CONTROL SERVO

SPEED CONTROL SERVO - BLACK 4 WAY

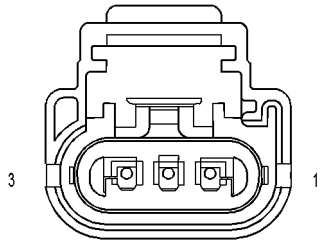
CAV	CIRCUIT	FUNCTION
1	V36 20VT/YL	S/C VACUUM SOL CONTROL
2	V35 20VT/LG	S/C VENT SOL CONTROL
3	V30 20VT/WT	S/C BRAKE SWITCH OUTPUT
4	Z901 18BK	GROUND



SUBWOOFER

SUBWOOFER - 12 WAY

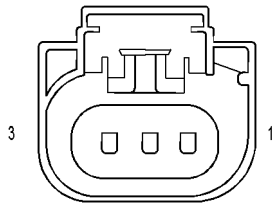
CAV	CIRCUIT	FUNCTION
1	X54 18GY	RIGHT FRONT SPEAKER (+)
2	X56 18GY/BR	RIGHT FRONT SPEAKER (-)
3	X53 18DG	LEFT FRONT SPEAKER (+)
4	X55 18DG/BR	LEFT FRONT SPEAKER (-)
5	X60 20GY/OR	RADIO 12V OUTPUT
6	X13 16BK/RD	FUSED B(+)
7	X52 18GY/DB	RIGHT REAR SPEAKER (+)
8	X58 18GY/OR	RIGHT REAR SPEAKER (-)
9	X51 18DG/DB	LEFT REAR SPEAKER (+)
10	X57 18DG/OR	LEFT REAR SPEAKER (-)
11	-	-
12	Z513 16BK	GROUND



THROTTLE POSITION SENSOR (2.4L)

THROTTLE POSITION SENSOR (2.4L) - 3 WAY

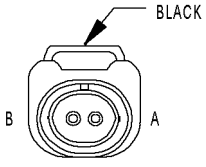
CAV	CIRCUIT	FUNCTION
1	F855 20PK/YL	5 VOLT SUPPLY
2	K900 20DB/DG	SENSOR GROUND
3	K22 20BR/OR	TP NO. 1 SIGNAL



THROTTLE POSITION SENSOR (4.0L)

THROTTLE POSITION SENSOR (4.0L) - 3 WAY

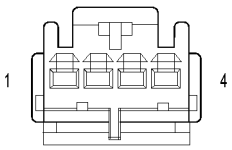
CAV	CIRCUIT	FUNCTION
1	F855 20PK/YL	5 VOLT SUPPLY
2	K900 20DB/DG	SENSOR GROUND
3	K22 18BR/OR	TP NO. 1 SIGNAL



TORQUE CONVERTER CLUTCH SOLENOID

TORQUE CONVERTER CLUTCH SOLENOID - BLACK 2 WAY

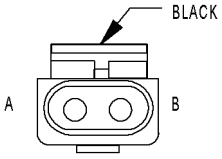
CAV	CIRCUIT	FUNCTION
A	F12 20RD/LG	FUSED IGNITION SWITCH OUTPUT (RUN-START)
B	T23 18OR/LG	TRANSMISSION LOW/OVERDRIVE SOLENOID



TOW/HAUL OVERDRIVE OFF SWITCH

TOW/HAUL OVERDRIVE OFF SWITCH - 4 WAY

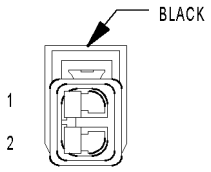
CAV	CIRCUIT	FUNCTION
1	Z1 20BK	GROUND
2	T6 20DG	TOW/HAUL OVERDRIVE OFF SWITCH SENSE
3	T56 20DB/LB	OVERDRIVE OFF SWITCH INDICATOR
4	E2 20OR/BR	PANEL LAMPS FEED



TRANSFER CASE SWITCH (EXCEPT OFF-ROAD PACKAGE)

TRANSFER CASE SWITCH (EXCEPT OFF-ROAD PACKAGE) - BLACK 2 WAY

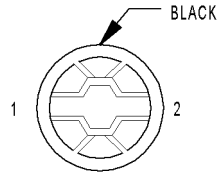
CAV	CIRCUIT	FUNCTION
A	G93 20VT/LB	4WD INDICATOR
B	Z987 18BK	GROUND



TRANSFER CASE SWITCH (OFF-ROAD PACKAGE)

TRANSFER CASE SWITCH (OFF-ROAD PACKAGE) - BLACK 2 WAY

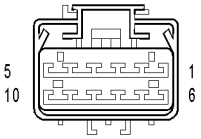
CAV	CIRCUIT	FUNCTION
1	K77 18BR/WT	TRANSFER CASE POSITION SENSOR INPUT
2	K900 20DB/DG	SENSOR GROUND



TRANSMISSION RANGE INDICATOR ILLUMINATION (PRNDL)

TRANSMISSION RANGE INDICATOR ILLUMINATION (PRNDL) - BLACK 2 WAY

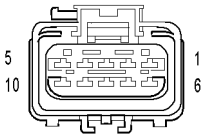
CAV	CIRCUIT	FUNCTION
1	E2 20OR/BR	PANEL LAMPS FEED
2	Z902 20BK	GROUND



TRANSMISSION RANGE SENSOR

TRANSMISSION RANGE SENSOR - 10 WAY

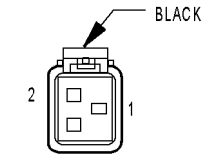
CAV	CIRCUIT	FUNCTION
1	F102 20PK/DB	FUSED IGNITION SWITCH OUTPUT (RUN)
2	-	-
3	T13 18DG/VT	SPEED SENSOR GROUND
4	T54 18DG/OR	TRANSMISSION TEMPERATURE SENSOR SIGNAL
5	-	-
6	L1 20WT/LG	BACK-UP LAMP FEED
7	T1 18DG/LB	TRS T1 SENSE
8	T3 18DG/DB	TRS T3 SENSE
9	T42 18DG/YL	TRS T42 SENSE
10	T41 18YL/DB	TRS T41 SENSE (P/N)



TRANSMISSION SOLENOID/ PRESSURE SWITCH ASSEMBLY

TRANSMISSION SOLENOID/PRESSURE SWITCH ASSEMBLY - 10 WAY

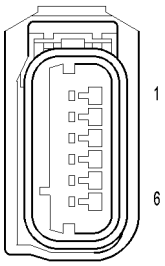
CAV	CIRCUIT	FUNCTION
1	T60 16YL/GY	OD SOLENOID CONTROL
2	T59 16YL/LB	UD SOLENOID CONTROL
3	T16 16YL/OR	TRANSMISSION CONTROL RELAY OUTPUT
4	T19 16YL/DB	2-4 SOLENOID CONTROL
5	T47 18YL/DG	2-4 PRESSURE SWITCH SENSE
6	T9 18DG/TN	OD PRESSURE SWITCH SENSE
7	T20 16DG/WT	L/R SOLENOID CONTROL
8	-	-
9	-	-
10	T50 18YL/TN	L/R PRESSURE SWITCH SENSE



UNDERHOOD LAMP

UNDERHOOD LAMP - BLACK 2 WAY

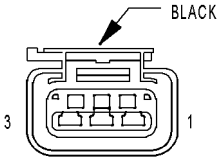
CAV	CIRCUIT	FUNCTION
1	Z901 18BK	GROUND
2	M21 20YL/BR	FUSED B(+)



VARIABLE LINE PRESSURE SENSOR

VARIABLE LINE PRESSURE SENSOR - 6 WAY

CAV	CIRCUIT	FUNCTION
1	-	-
2	F856 18YL/PK	5 VOLT SUPPLY
3	T38 18YL/BR	LINE PRESSURE SENSOR SIGNAL
4	K900 18DB/DG	SENSOR GROUND
5	T16D 18YL/OR	TRANSMISSION CONTROL RELAY OUTPUT
6	T140 18YL/GY	MS SOLENOID CONTROL



VEHICLE SPEED SENSOR

VEHICLE SPEED SENSOR - BLACK 3 WAY

CAV	CIRCUIT	FUNCTION
1	F856 20YL/PK	5 VOLT SUPPLY
2	K900 20DB/DG	SENSOR GROUND
3	K75 18DB/OR	VEHICLE SPEED SENSOR SIGNAL

8W-91 CONNECTOR/GROUND/SPLICE LOCATION

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CONNECTOR/GROUND/SPLICE LOCATION
DESCRIPTION 1

CONNECTOR/GROUND/SPLICE LOCATION

DESCRIPTION

This section provides illustrations identifying connector, ground, and splice locations in the vehicle. Connector, ground, and splice indexes are provided. Use the wiring diagrams in each section for connector, ground, and splice identification. Refer to the index for the proper figure number. For items that are not shown in this section N/S is placed in the Fig. column.

CONNECTORS

Connector Name/Number	Color	Location	Fig.
A/C Compressor Clutch	BK	At A/C Compressor Clutch	5, 6
A/C Heater Control - C1		Center of Instrument Panel	26, 29
A/C Heater Control - C2		Center of Instrument Panel	26, 29
A/C Heater Control - C3		Center of Instrument Panel	26, 29
A/C High Pressure Switch	BK	Left Side of Engine	5
A/C Low Pressure Switch	GN	Near Powertrain Control Module	5, 6
A/C Pressure Transducer (2.4L)	BK	Near Compressor	6
Airbag Control Module	YL	Front of Floor Pan Tunnel	20
Ambient Temperature Sensor (Compass/Temperature Mirror)		Left Front of Engine Compartment	1
Ambient Temperature Sensor (Engine)		Left Front Engine Compartment	2
Axle Lock Switch(Off Road)		Near Cigar Lighter	26
Back-up Lamp Switch (M/T)	BK	Right Side of Transmission	35, 37
Battery		Rear Engine Compartment	11
Battery Temperature Sensor	BK	Under Battery Tray	10, 12
Blend Door Actuator		On HVAC Harness	N/S
Blower Motor		On HVAC Harness	N/S
Blower Motor Relay	BK	On HVAC Harness	N/S
Blower Motor Resistor Block		On HVAC Harness	N/S
Brake Lamp Switch	GY	Top of Brake Pedal Bracket	22, 30
Brake Transmission Shift Interlock Solenoid	WT	Near Steering Column	30
Brake Warning Indicator Switch	BK	On Brake Master Cylinder	12, 14
C102	GY	Rear Engine Compartment	10, 12

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

Connector Name/Number	Color	Location	Fig.
C103	BK	Rear of Engine Compartment	10, 12, 17, 18
C104	BK	Rear of Engine Compartment	17, 18
C109 (A/T)	LT-GY	Rear of Engine Compartment	10, 12, 17, 18
C170 (LHD)	WT	Left Kick Panel	24
C170 (RHD)	WT	Right Kick Panel	27
C180	BK	Below Evap/Purge Solenoid	N/S
C181	BK	Left Rear of Engine Head	4
C182	BK	Left Rear of Engine Head	4
C203 (LHD)	GN	Left Kick Panel	26
C203 (RHD)	GN	Right Kick Panel	29
C205	LT GN	Left Kick Panel	25, 27
C300 (LHD)	GY	Left Kick Panel	22, 24, 38
C300 (RHD	GY	Right Kick Panel	23, 27
C301 (LHD)	YL	Left Kick Panel	22, 38
C301 (RHD)	YL	Right Kick Panel, see LHD Similar	23
C324 (Speaker Pod)	WT/BK	Right Side of Sport Bar	N/S
C325 (Speaker Pod)	WT/BK	Left Side of Sport Bar	38, 39
C326 (Hard Top)	BK	Left Rear Quarter Panel	38, 39
C329 (Export)	BK	Left Rear of Rear Bumper	41
Camshaft Position Sensor (4.0L)	BK	Right Side of Engine	7
Camshaft Position Sensor (2.4L)	BK	Right Front Corner of Engine Head, behind Air Cleaner	N/S
Center High Mounted Stop Lamp	BK	Near Right Rear Lamp Assembly	41, 42
Cigar Lighter/Power Outlet	RD	Rear of Cigar Lighter	26, 28
Circuit Breaker	GY	Near Day Time Running Lamp Module	19, 21
Clockspring - C1	NAT	Rear of Clockspring	30
Clockspring - C2	YL	Rear of Clockspring	30
Clutch Pedal Position Switch (M/T)	BK	Top of Clutch Pedal Bracket	21, 22
Coil Capacitor (2.4L)		Rear Engine	4, 17
Coil Capacitor (4.0L)	BK	Right Rear of Engine Head	7
Compass/Temperature Mirror	BK	At Top Center of Windshield	N/S
Controller Anti-Lock Brake	BK	Rear Left Side of Dash Panel	12, 14
Crankshaft Position Sensor (4.0L)	BK	At Rear of Intake Manifold	7
Crankshaft Position Sensor (2.4L)	BK	Left Bellhousing	4
Data Link Connector	BK	Bottom Driver Side of Instrument Panel	24, 27
Daytime Running Lamp Module (Except Export)	BK	Left Side of Engine Compartment	10
Driver Airbag Squib	YL	Under Horn Pad	31
Driver Door Ajar Switch	NAT	At ″A″ Pillar	22, 23

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

Connector Name/Number	Color	Location	Fig.
Engine Coolant Temperature Sensor	BK	On Thermostat Housing	5, 18
Engine Oil Pressure Switch (4.0L)	BK	Right Side of Engine Block	7
Engine Oil Pressure Switch (2.4L)	BK	Below A/C Compressor	N/S
Engine Starter Motor		Left Rear of Engine	3, 8
Engine Starter Solenoid		At Starter Motor	4, 7
Evap/Purge Solenoid	BK	Left Fender Side Shield	13, 14
Front Locker Indicator Switch (Off Road)	BK	Top of Front Differential	33
Front Locker Pump (Off Road)	LT/GY	Top of Locker Pump Bracket	32
Front Washer Pump	BK	Under Washer Fluid Reservoir	1
Front Wiper Motor	BK	At Motor	15
Fuel Injector NO.1 (4.0L)	BK	At Injector	5
Fuel Injector NO.2 (4.0L)	BK	At Injector	5
Fuel Injector NO.3 (4.0L)	BK	At Injector	5
Fuel Injector NO.4 (4.0L)	BK	At Injector	5
Fuel Injector NO.5 (4.0L)	BK	At Injector	5
Fuel Injector NO.6 (4.0L)	BK	At Injector	5
Fuel Pump Module	BK	Above Fuel Tank	40
G Switch	BK	Near T/O for Controller Anti-Lock Brake	20
Generator	BK	Rear of Generator	3, 7, 8
Headlamp Leveling Switch (Export)	WT	At Switch	24, 27
Horn	BK	Left Front Fender Side Shield	1
Horn Switch		Rear of Steering Wheel Pad	31
Idle Air Control Motor	BK	Side of Throttle Body	4, 5
Ignition Coil Pack (2.4L)	GY	Right Side of Engine	4, 17
Ignition Coil Pack (4.0L)	BK	Right Rear of Engine	7
Ignition Switch		On Steering Column	30
Input Speed Sensor	BK	Left Center of Trans	16, 36
Instrument Cluster - C1		Rear of Cluster	24, 27
Instrument Cluster - C2		Rear of Cluster	24, 27
Intake Air Temperature Sensor	GY	Rear of Intake Manifold	5
Left Courtesy Lamp	BK	Left Side of Instrument Panel	24, 27
Left Dome Lamp	WT	In Left Speaker Pod	N/S
Left Fog Lamp (Except Export)	BK	At Lamp, See Right	2
Left Front Park/Turn Signal Lamp (Except Export)	BK	At Lamp, See Right	2

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

Connector Name/Number	Color	Location	Fig.
Left Front Position Lamp (Export)	BK	At Lamp	1
Left Front Speaker	BK	At Speaker	24, 27
Left Front Turn Signal Lamp (Export)	BK	At Lamp, See Right	2
Left Front Wheel Speed Sensor	BK	Left Side of Engine Compartment Near Hydraulic Control Unit	13, 14
Left Headlamp	BK	Rear of Lamp	1
Left Headlamp Leveling Motor (Export)	BK	Near Headlamp at Motor	1
Left License Lamp (Export)	GY	At Lamp	41, 42
Left Rear Lamp Assembly	BK	At Lamp	41, 42
Left Rear Speaker	BK	In Left Speaker/lamp Pod	N/S
Left Rear Wheel Speed Sensor	BK	Near Vehicle Speed Sensor T/O	16, 40
Left Side Marker Lamp	GY	At Lamp, See Right	2
Left Side Repeater Lamp (Export)	GY	At Lamp, See Right	2
Left Speed Control Switch		Left Steering Wheel	31
Manifold Absolute Pressure Sensor	BK	Side of Throttle Body	4, 5
Multi-Function Switch C1	GY	Steering Column	30
Multi-Function Switch C2	BK	Steering Column	30
Natural Vacuum Leak Detection Assembly	BK	Under Rear Body	41, 42
Output Speed Sensor	BK	Rear Left Side of Trans	16, 36
Oxygen Sensor 1/1 Upstream		On Front Exhaust Pipe	5, 17
Oxygen Sensor 1/2 Downstream	BK	Rear of Catalytic Converter	5, 36, 37
Oxygen Sensor 2/1 Upstream (4.0L California/European III)		Near Idle Air Control Motor T/O	5
Oxygen Sensor 2/2 Downstream (4.0L California/ European III)	GY	T/O near Ignition Coil Pack T/O	5, 35
Park Brake Switch		Left of Seat Belt Switch	20
Passenger Airbag	YL	Rear of Airbag	19, 21
Passenger Airbag On/Off Switch		Lower Center of Instrument Panel	26
Passenger Door Ajar Switch	BK	Near Top Hinge of Passenger Door	19, 21
Power Distribution Center	BK	Engine Compartment Right or Left Fenter	13
Power Distribution Center Eyelet	N/A	Engine Compartment Right or Left Fenter	11
Power Outlet		Center of Instrument Panel	26, 28
Power Steering Pressure Switch (2.4L)	BK	Near Power Steering Pump	4

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

Connector Name/Number	Color	Location	Fig.
Powertrain Control Module - C1 (LHD)	BK/BK	Right Rear of Engine Compartment	9, 17
Powertrain Control Module - C1 (RHD)	BK/BK	Left Rear of Engine Compartment	9
Powertrain Control Module - C2 (LHD)	BK/OR	Right Rear of Engine Compartment	9, 17
Powertrain Control Module - C2 (RHD)	BK/OR	Left Rear of Engine Compartment	9
Powertrain Control Module - C3 (LHD)	BK/NAT	Right Rear of Engine Compartment	10, 17
Powertrain Control Module - C3 (RHD)	BK/NAT	Left Rear of Engine Compartment	12
Powertrain Control Module - C4 (LHD)	BK/GN	Left Rear Engine Compartment	17
Powertrain Control Module - C4 (RHD)	BK/GN	Right Rear Engine Compartment	9
Radiator Fan Motor (2.4L)	BK	On Right Radiator Fan Shroud	6
Radiator Fan Motor Circuit Breaker (2.4L)	BK	On Right Radiator Fan Shroud	6
Radio	GY	Rear of Radio	26, 29
Rear Fog Lamp (Export)	BK	At Lamp	41, 42
Rear Locker Indicator Switch (Off Road)	BK	Top of Rear Differential	34
Rear Locker Pump (Off Road)	LT-GY	Top of Locker Pump Bracket	32
Rear Washer Pump (Hard Top)	BK	Under Washer Fluid Reservoir	N/S
Rear Window Defogger		At Rear Window	38, 39
Rear Window Defogger Switch (Hard Top)	GY	Behind Rear Window Defogger Switch	26, 28
Rear Wiper Motor (Hard Top)	BK	At Rear Wiper Motor	38, 39
Rear Wiper/Washer Switch (Hard Top)	WT	Behind Rear Wiper/Washer Switch	26, 28
Right Courtesy Lamp	BK	Right Side of Instrument Panel	25, 27
Right Dome Lamp	WT	In Right Speaker Pod	
Right Fog Lamp (Except Export)	BK	At Lamp	2
Right Front Park/Turn Signal Lamp (Except Export)	BK	At Lamp	2
Right Front Position Lamp (Export	BK	At Lamp	2
Right Front Speaker	BK	At Speaker	25, 27
Right Front Turn Signal Lamp (Export)	BK	At Lamp	2
Right Front Wheel Speed Sensor	BK	Rear of Engine	10, 12

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

Connector Name/Number	Color	Location	Fig.
Right Headlamp	BK	Rear of Lamp	2
Right Headlamp Leveling Motor (Export)	BK	Near Headlamp at Motor	2
Right License Lamp	GY	At Lamp	41, 42
Right Rear Lamp Assembly	BK	At Lamp	41, 42
Right Rear Speaker	BK	In Right Speaker/lamp Pod	N/S
Right Rear Wheel Speed Sensor	BK	Near Vehicle Speed Sensor T/O	16, 40
Right Side Marker Lamp (Except Export)	GY	At Lamp	2
Right Side Repeater Lamp	GY	At Lamp	2
Right Speed Control Switch	BK	Right Steering Wheel	31
Seat Belt Switch	BK	Near Park Brake Switch	20
Sentry Key Immobilizer Module	BK	At Immobilizer	30
Speed Control Servo	BK	Left Side Engine Compartment	13, 14
Subwoofer	LT-GY	Floor Pan	20
Throttle Position Sensor	WT	Side of Throttle Body	5, 18
Tow/Haul Overdrive Off Switch		Center of Instrument Panel	26, 28
Transfer Case Switch	BK	Left Side of Transfer Case	16, 36, 37
Transmission Range Indicator Illumination (PRNDL)	BK	Between Seats	20
Transmission Range Sensor	GN	Left Center of Transmission	16, 36
Transmission Solensid/ Pressure Switch Assembly	NAT	Right Center of Transmission	7, 36
Underhood Lamp	BK	Under Hood	10, 12, 18
Variable Line Pressure Sensor		Left Side of Transmission	16, 36
Vehicle Speed Sensor	BK	Left Rear of Transfer Case	16, 36, 37

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

GROUNDS

Connector Name/ Number	Location	Fig.
G100	Rear Center of Engine Compartment	10, 12
G101	Rear Center of Engine Compartment	10, 12
G102	Left Radiator Closure Panel	1
G103	Right Radiator Closure Panel	2
G104	Near Generator	N/S
G105	Right Rear of Engine Block (4.0L) Near Left Engine Mount (2.4L)	4, 7
G106	Rear of Engine Compartment	3, 11
G107	Left Front of Engine Block	3, 8, 11
G108	Right Rear Engine Block	8
G110 (2.4L)	Near Left Engine Mount	4
G111	Near Right Rear of Engine	N/S
G190	Near Right Headlamp	2
G191	Near Right Headlamp	N/S
G200 (LHD)	Left Rear of Instrument Panel	24
G200 (RHD)	Right Rear of Instrument Panel	27
G201 (LHD)	Left Rear of Instrument Panel	24
G201 (RHD)	Right Rear of Instrument Panel	27
G202 (LHD)	Right Rear of Instrument Panel	25
G202 (RHD)	Left Rear of Instrument Panel	27
G203 (LHD)	Right Rear of Instrument Panel	25
G203 (RHD)	Left Rear of Instrument Panel	27
G204	Center of Instrument Panel	25, 27
G300 (LHD)	Left Kick Panel	22
G300 (RHD)	Right Kick Panel	23
G301 (LHD)	At Base of Center Stack	20
G301 (RHD)	Left Kick Panel	21
G302 (LHD)	Left Kick Panel	22
G302 (RHD)	Right Kick Panel	23
G303 (LHD)	Near Steering Column Mounting Bracket	30

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

SPLICES

Splice Number	Location	Fig.
S101	Near T/O for Power Distribution Center	10, 12
S102	Near T/O for G100 or Daytime Running Lamp Module	10
S103	Near T/O for G101	10, 12
S105	Left Cowl Panel	22
S106	Near T/O for G100	10
S108	Near T/O for Powertrain Control Module	10, 12
S111	Near T/O for Battery Temperature Sensor or Power Distribution Center	10
S117	Near T/O for Horn	1
S122	Near Grommet for Left Front Turn Signal Lamp, See Right	2
S123 (4.0L)	Near T/O for G105	N/S
S123 (2.4L)	Near T/O for Ignition Coil Pack	N/S
S125 (Export)	Near Grommet for Right Front Turn Signal Lamp	2
S126	Near T/O for C104	7
S127	Rear Engine Compartment	9, 17
S128 (4.0L)	Between Fuel Injector NO. 2 and Fuel Injector NO. 3	5
S128 (2.4L)	Near T/O for C103 on Engine Harness	N/S
S129	Rear of Engine	4, 7, 35
S130 (4.0L)	Left Rear of Engine	5
S130 (2.4L)	Near T/O for C103 on Engine Harness	N/S
S131	Left Rear Engine	7
S132 (4.0L)	Top of Transmission	7, 35
S132 (2.4L)	Near T/O for Engine Starter Solenoid	4
S133 (4.0L)	Trans Harness	7
S133 (2.4L)	Near T/O for Input Speed Sensor	4
S134 (4.0L)	Top Front of Engine	5
S134 (2.4L)	Between Engine Starter Solenoid and C182	N/S
S135 (2.4L)	Rear of Engine	4
S135 (4.0L)	In T/O for PCM	9
S136	In Battery harness near PDC	N/S
S137	Near T/O for Throttle Position Sensor	N/S
S138	Left Engine Compartment	13, 14
S140	Near T/O for C104	N/S
S141	Near T/O for Left Front Turn Signal Lamp, See Right	2
S142	Near T/O for Right Front Turn Signal Lamp	2
S143	Rear of Engine Compartment	10, 12
S144	Near T/O for Right Front Park/Turn Signal Lamp	N/S
S147	Between T/O's for Underhood Lamp and C103, C104	10, 12
S150	Right Headlamp T/O	2
S151	Left Rear Engine Compartment	14
S151 (RHD)	Left Side Engine Compartment	13

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

Splice Number	Location	Fig.
S152	Left Rear Engine Compartment	14
S152 (RHD)	Right Rear Engine Compartment	12
S153	Rear Engine Compartment	7, 17
S155	Left Rear Engine Compartment	10, 12
S156	Near T/O for ABS Pump Motor	10
S158 (2.4L)	Near T/O For Power Steering Pressure Switch	N/S
S159 (2.4L)	In T/O for PCM	4, 9
S160 (2.4L)	Left Rear Engine Compartment	4
S160 (4.0L)	Front of Transmission	5, 7
S161	Rear of Engine Compartment	9, 17
S162 (2.4L)	Near T/O for Oxygen Sensor 1/1 Upstream	N/S
S164	Near T/O For Rer Locker Indicator Switch Inline C116	32
S170	Rear of Engine Compartment	10, 12
S172	Between G100 and C103	10, 12
S173	Near G100	10, 12
S176	In Fuel InjectorHarness	N/S
S177	Rear of Engine Compartment	7, 17
S178	Near T/O for C103 on Engine Harness	N/S
S180 (LHD)	In PDC	10
S180 (RHD)	Left Rear Engine Compartment	12
S184	Left Side of Engine	5
S185	Rear of Engine	7
S202 (LHD)	Center of Instrument Panel	24, 26
S202 (RHD)	Center Rear of Instrument Panel	26, 27
S204	Center Rear of Instrument Panel	26, 27
S206	Center Rear of Instrument Panel	26, 27
S207	Near Cigar Lighter T/O	26, 29
S208	Near Blower Motor	N/S
S209	Near T/O for C202, C203, and C204	24, 26, 27, 28
S210	Near T/O for A/C Heater Control C3	26, 29
S211	Near T/O for C202	N/S
S212	Near T/O to Right Courtesy Lamp	N/S
S213	Center Rear of Instrument Panel	N/S
S214	Center Rear of Instrument Panel	N/S
S215	Center Rear of Instrument Panel	25, 27
S216	Left Side of Instrument Panel	24
S217	Left Side of Instrument Panel	24
S218	Driver Side Instrument Panel	24, 27
S302(LHD)	In T/O to Steering Column	19
S302(RHD)	Between S315 and S331, near T/O to ACM	21
S304 (LHD)	Center Rear of Dash Panel, Near Front Wiper Motor T/O	19
S304 (RHD)	Near T/O for C202 and C107	21, 23

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

Splice Number	Location	Fig.
S309	Front of Left Door Opening	38, 39
S310	Left Rear Quarter Panel, Near Body Grommet	21, 38
S311	Near Left Rear Lamp Assembly Connector	38, 39, 42
S315	Above Steering Column	19, 21
S316	In T/O for Rear Fog Lamp	41,42
S317	Near T/O for Left License Lamp	41, 42
S320 (LHD)	Near T/O for C107	22 38
S320 (RHD)	Near T/O for Fuse Block	21
S331 (LHD)	Near T/O to Steering Column	22, 38
S331 (RHD)	Near T/O for Fuse Block	21
S335 (Except-Export)	Near Left Rear Lamp Assembly Connector	38, 39
S350	Near Fuse Block	38
S351	Near T/O to Sport Bar	21, 38
S352	Forward of S351	21, 38
S353	Near T/O to Body Harness	19, 42
S354	At Top of Right Kick Panel	19, 21, 23
S355 (LHD)	Near T/O to Steering Column	19
S355 (RHD)	Near T/O to Steering Column	23
S356	10″(254 mm) Forward from T/O to Rear Washer	21, 38
S357	20″(508 mm) Forward from T/O to Rear Washer	21, 38
S358	30″(762 mm) Forward from T/O to Rear Washer	21, 38
S359	Near Fuse Block	21, 38
S360	Near T/O for Rear Window Defogger Feed	38, 39
S361	Near T/O for Rear Window Defogger Feed	38, 39

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

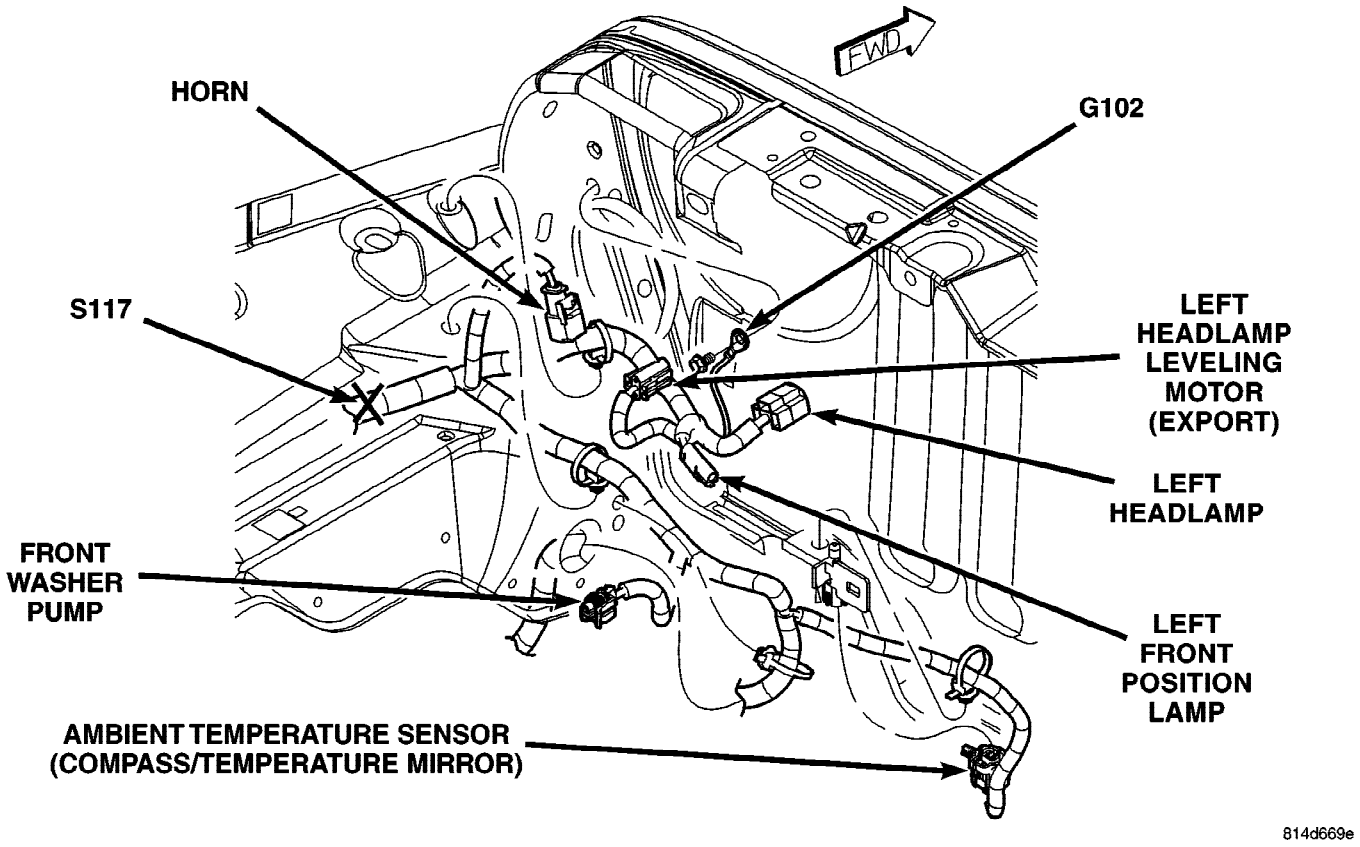


Fig. 1 -LEFT HEADLAMP

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

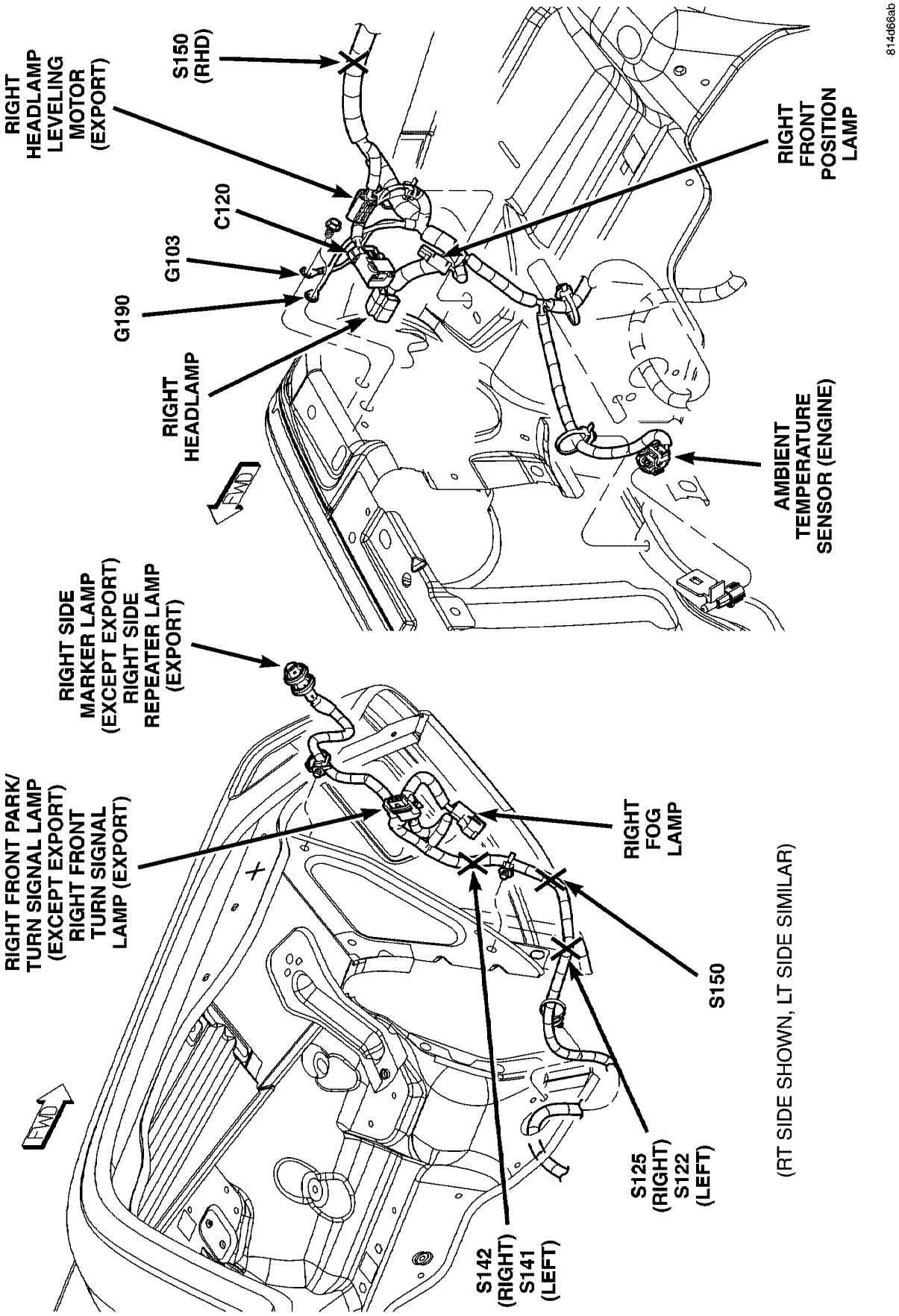


Fig. 2 -RIGHT HEADLAMP

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

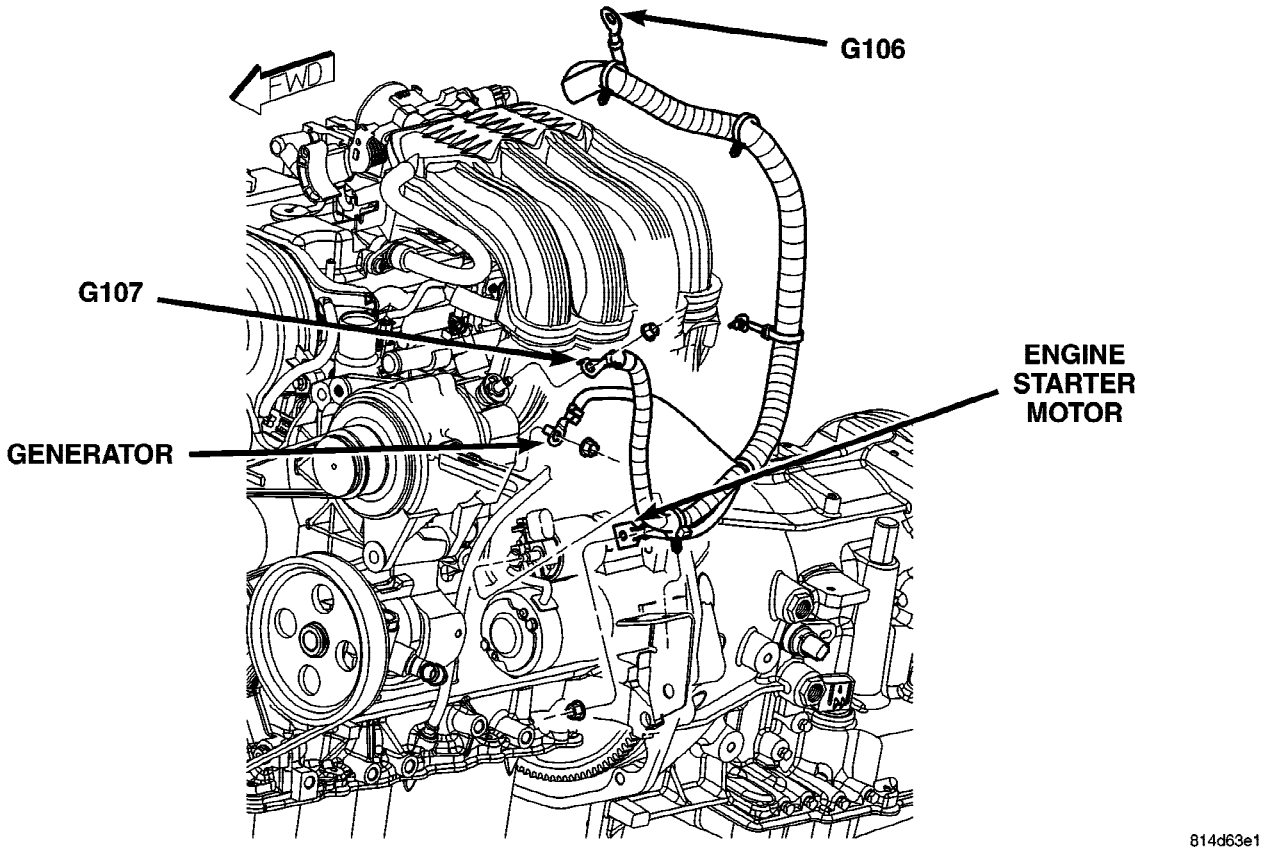


Fig. 3 -2.4L ENGINE LEFT FRONT

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

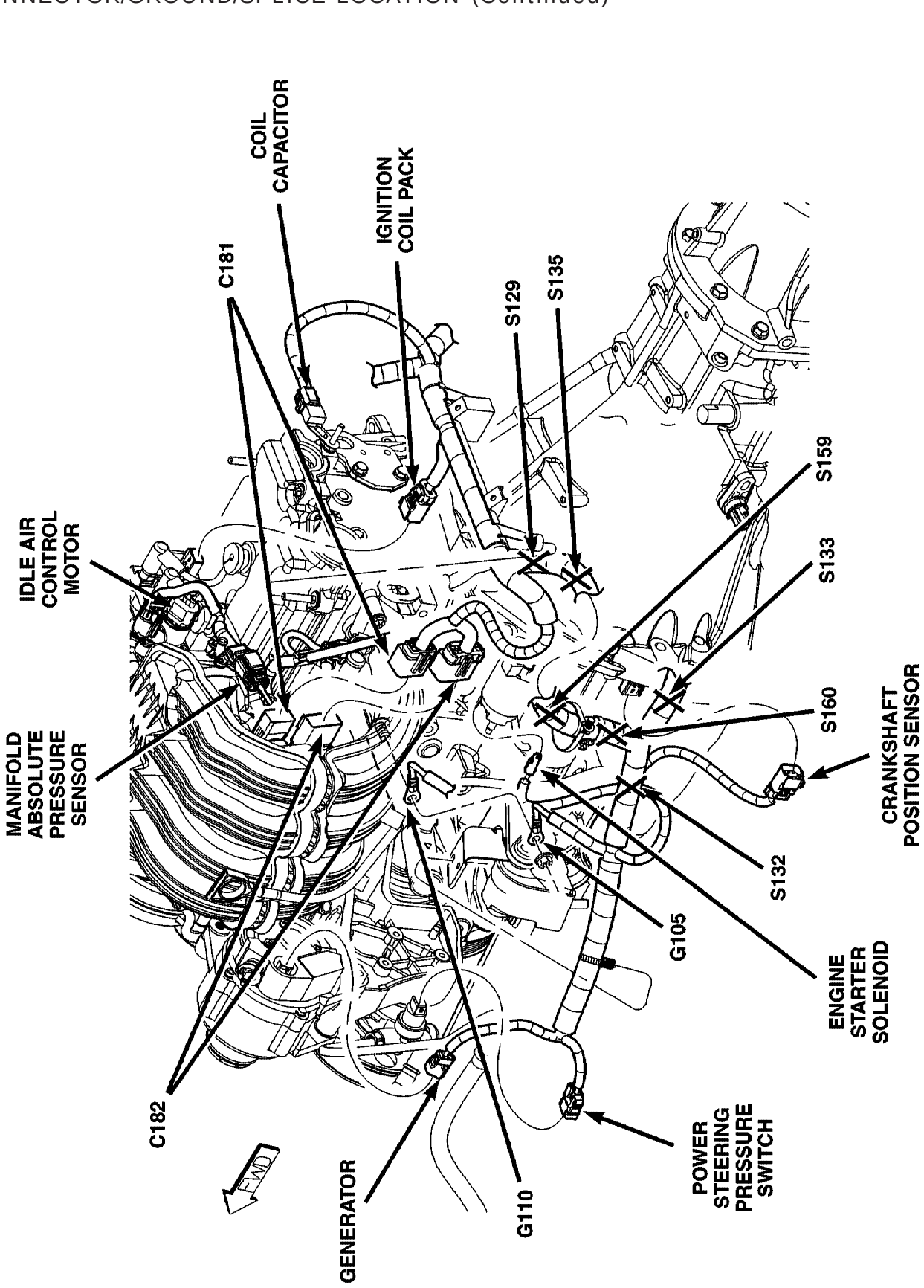


Fig. 4 -2.4 L ENGINE LEFT REAR

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

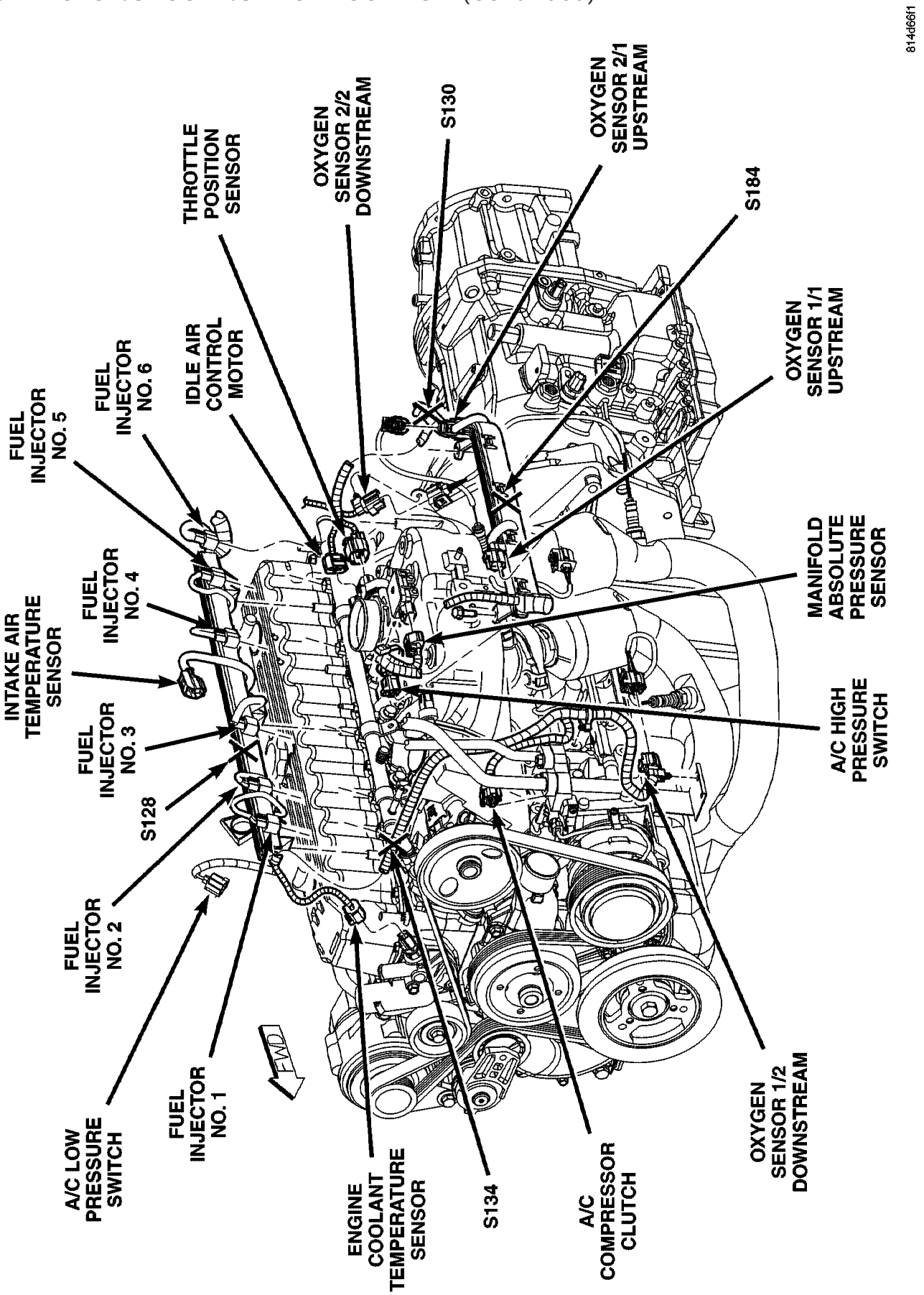


Fig. 5 -4.0 L ENGINE LEFT SIDE

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

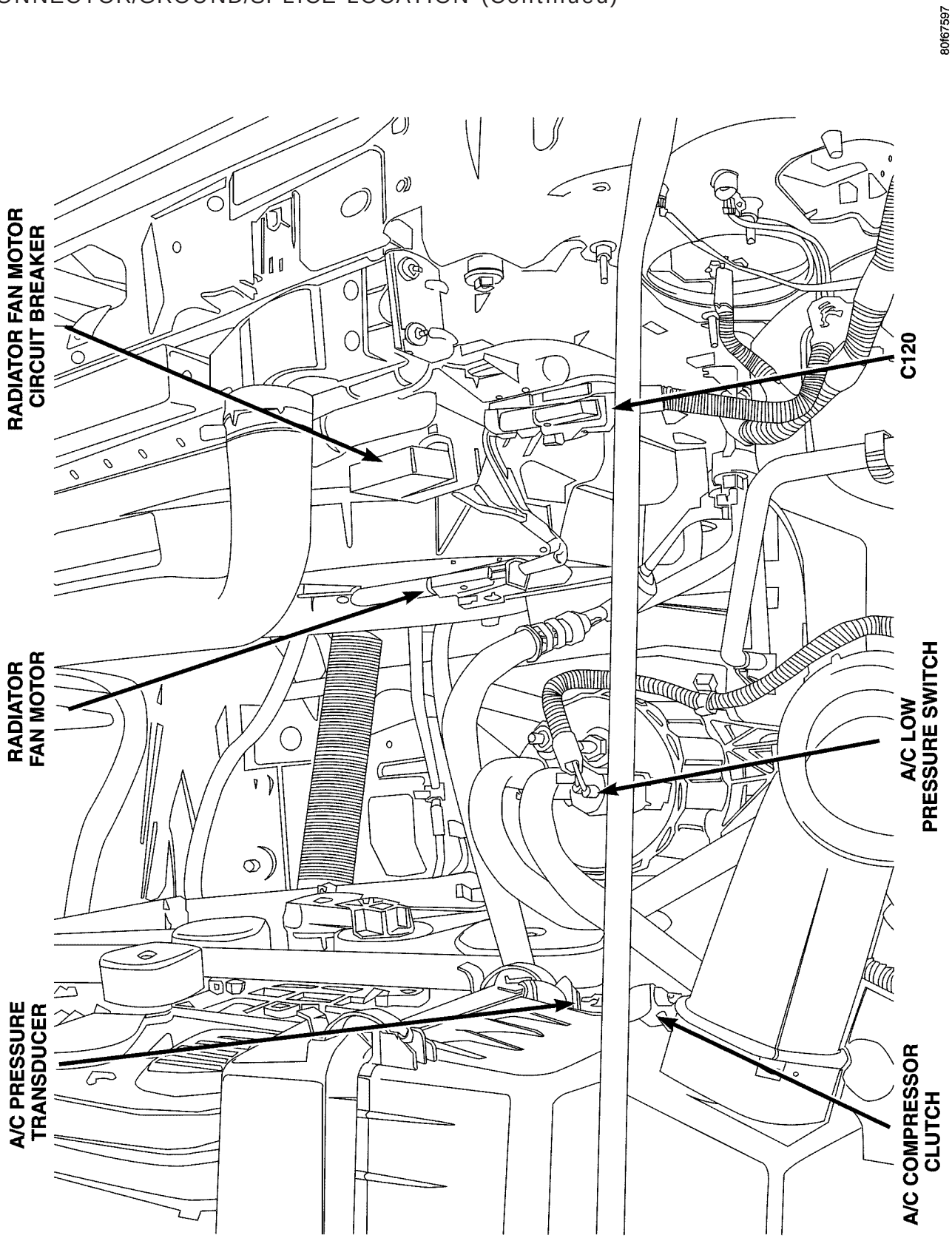


Fig. 6 -2.4 L A/C RIGHT SIDE

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

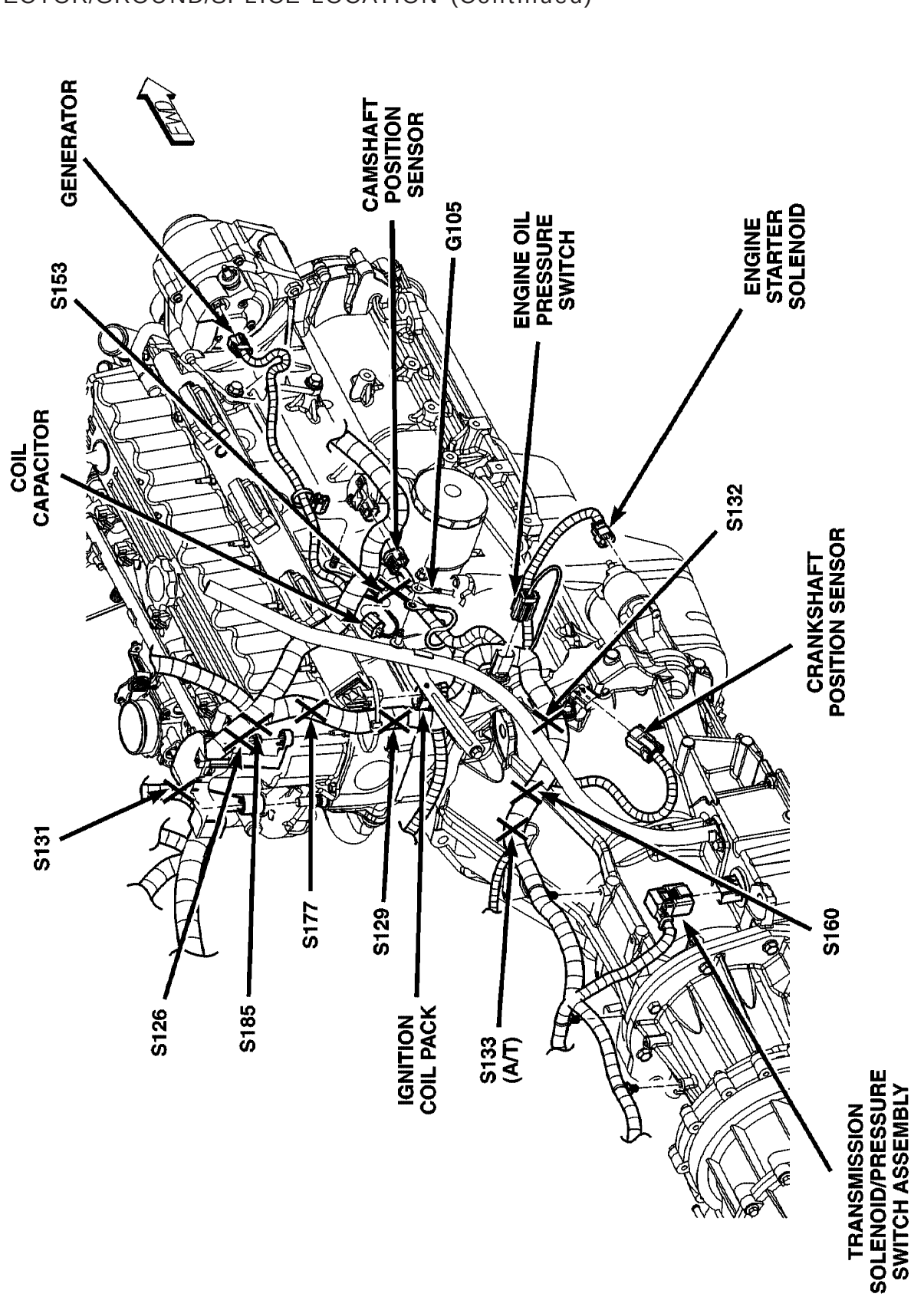


Fig. 7 -4.0 L ENGINE RIGHT SIDE

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

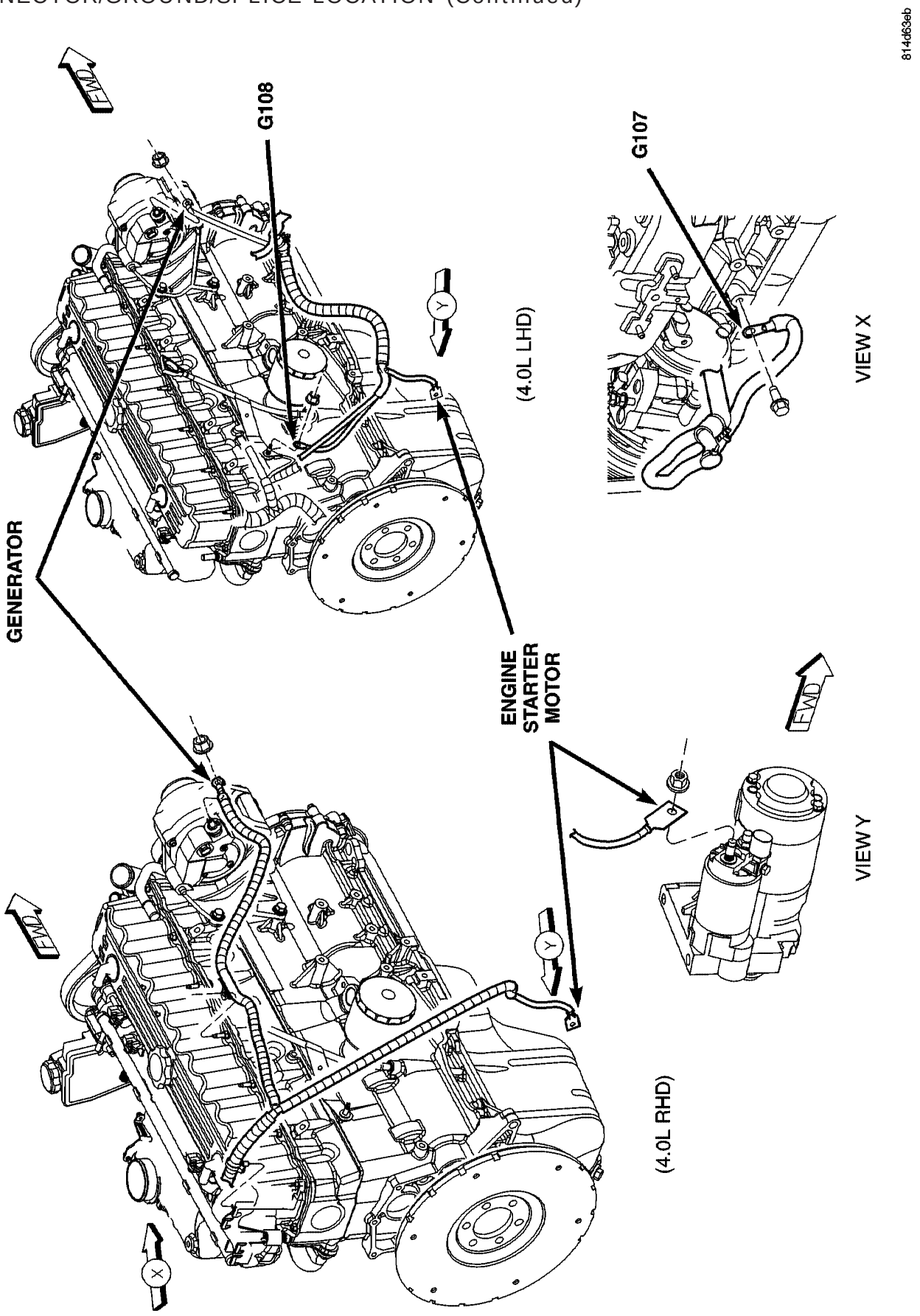


Fig. 8 -4.0 L ENGINE STARTER & GENERATOR

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

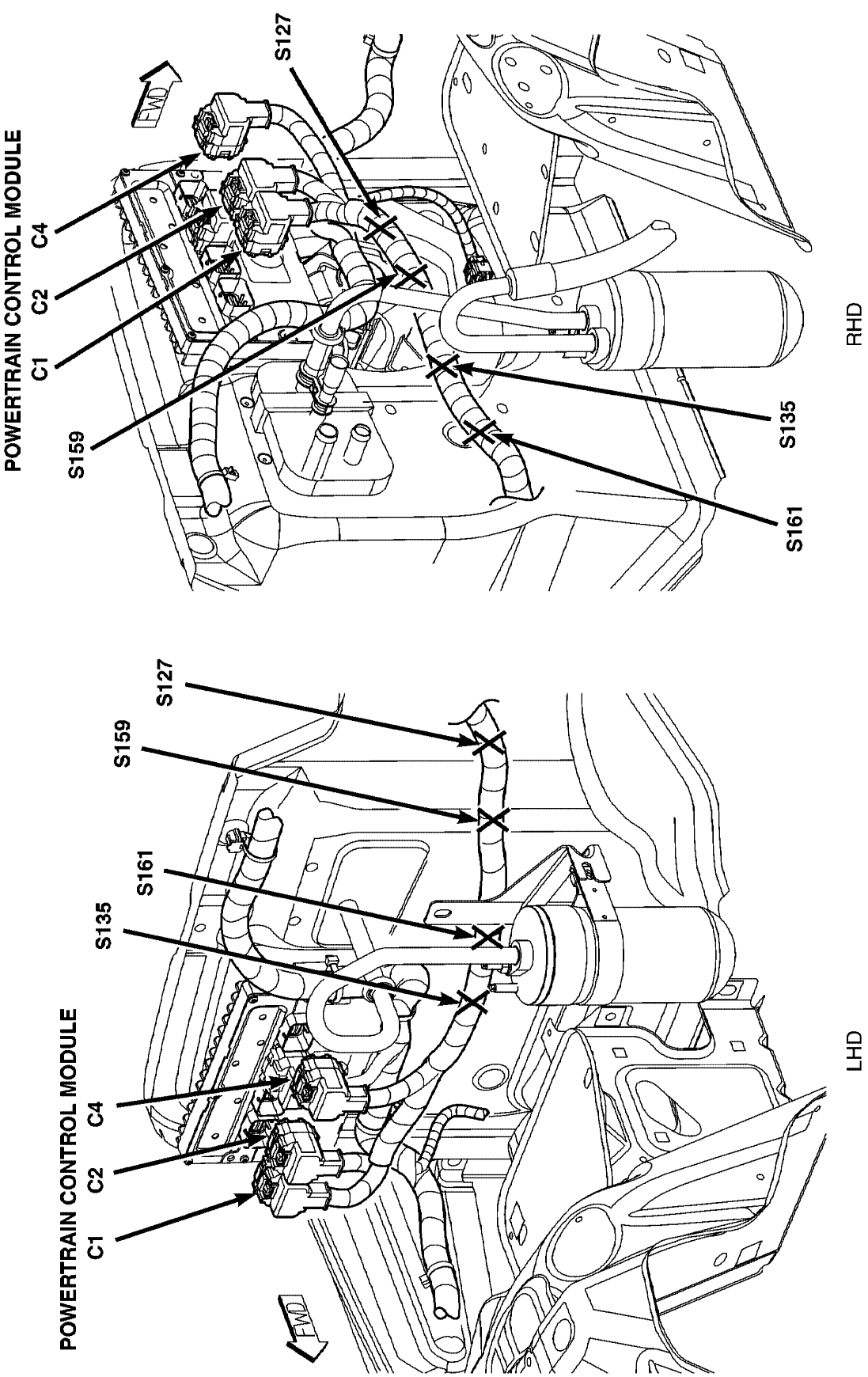


Fig. 9 -ENGINE COMPARTMENT REAR

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

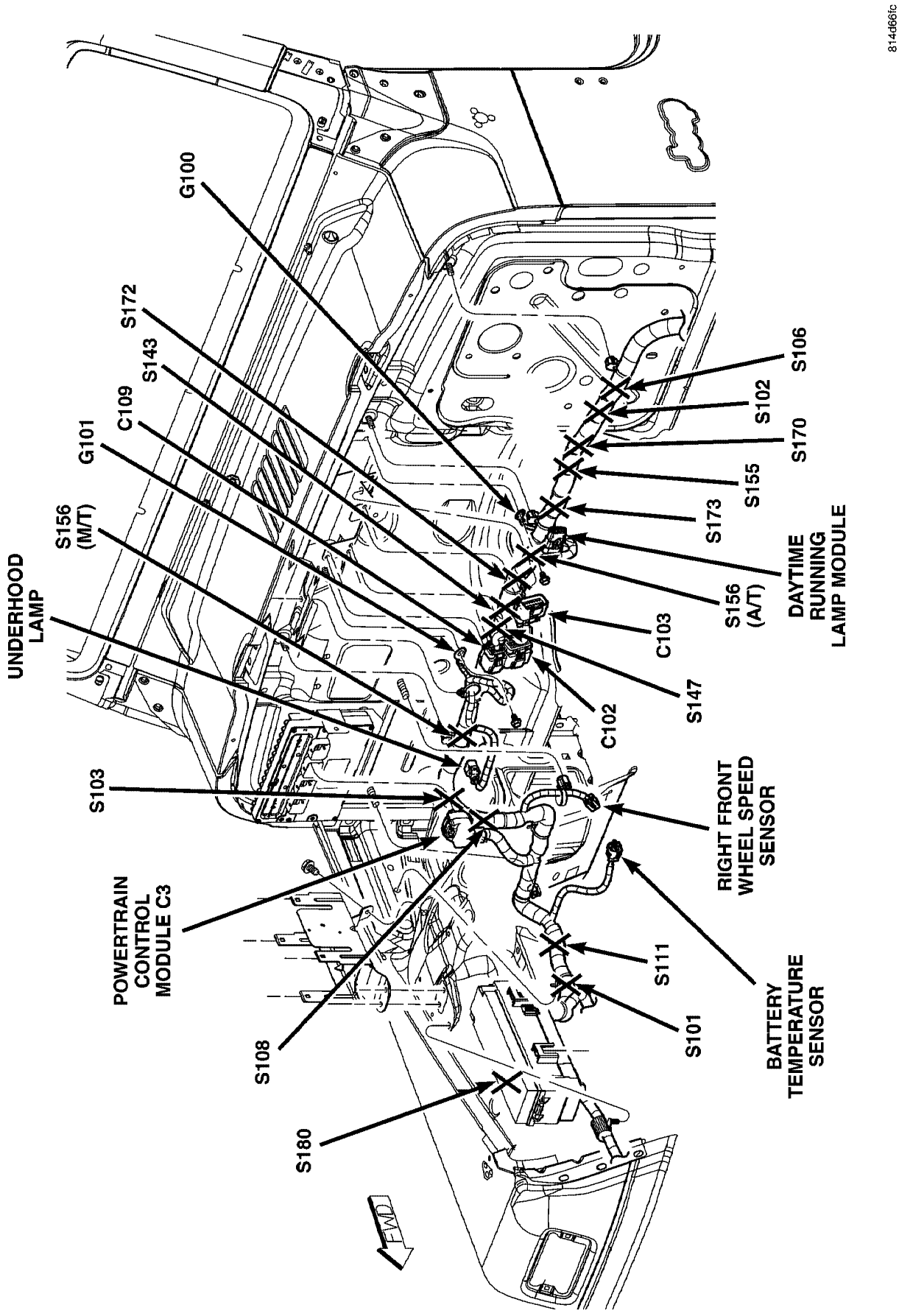


Fig. 10 -ENGINE COMPARTMENT REAR (LHD)

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

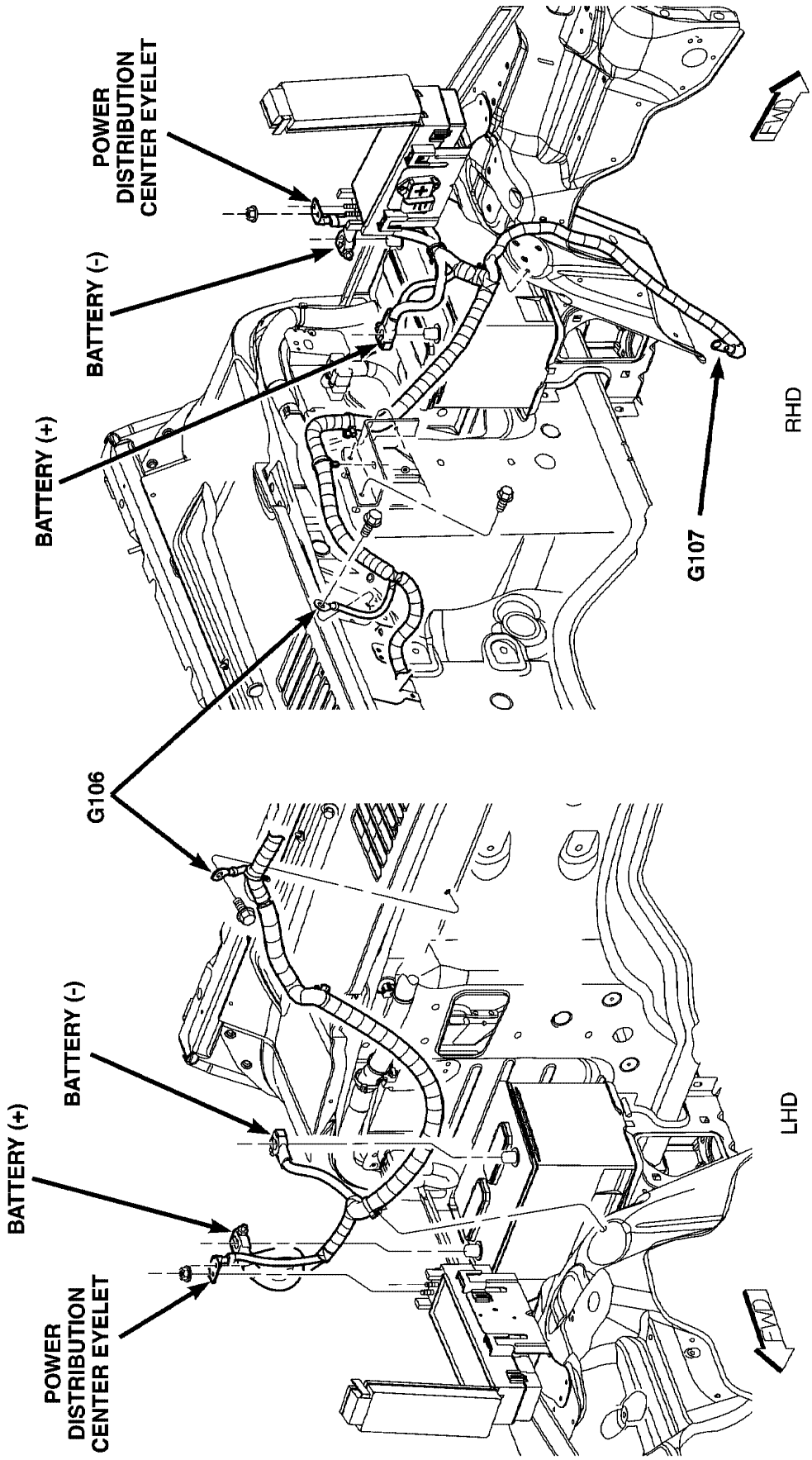


Fig. 11 -ENGINE COMPARTMENT REAR BATT

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

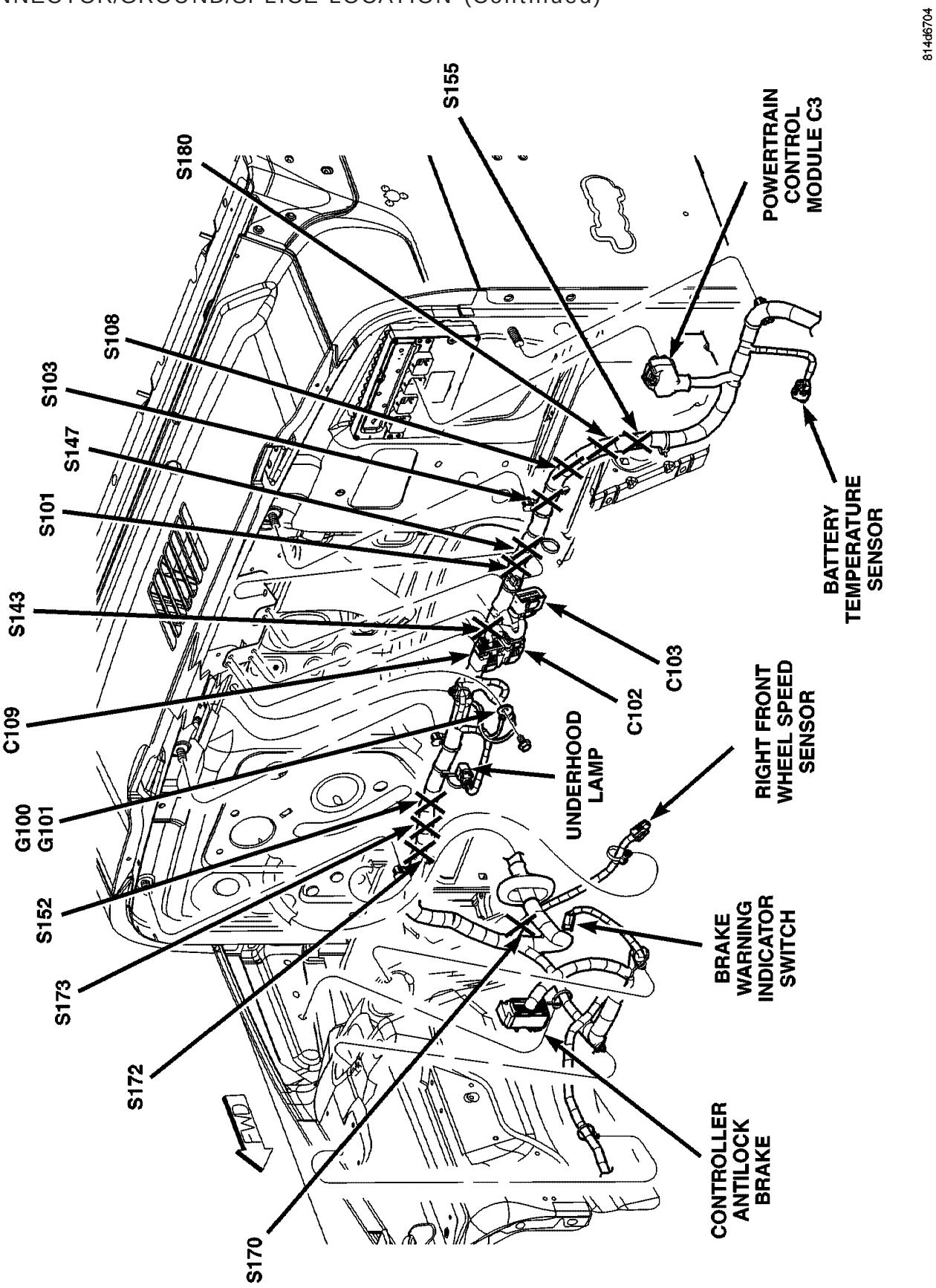


Fig. 12 -ENGINE COMPARTMENT REAR (RHD)

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

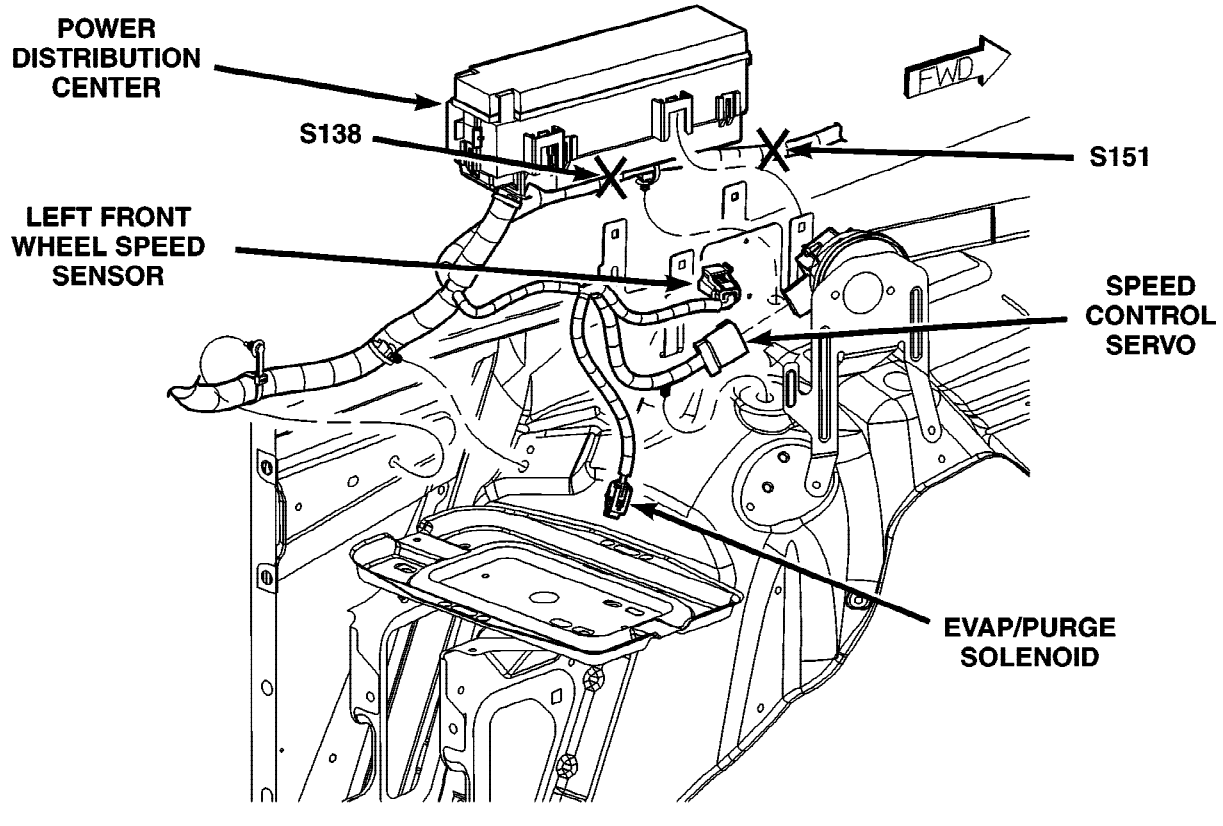


Fig. 13 -ENGINE COMPARTMENT LEFT SIDE (RHD)

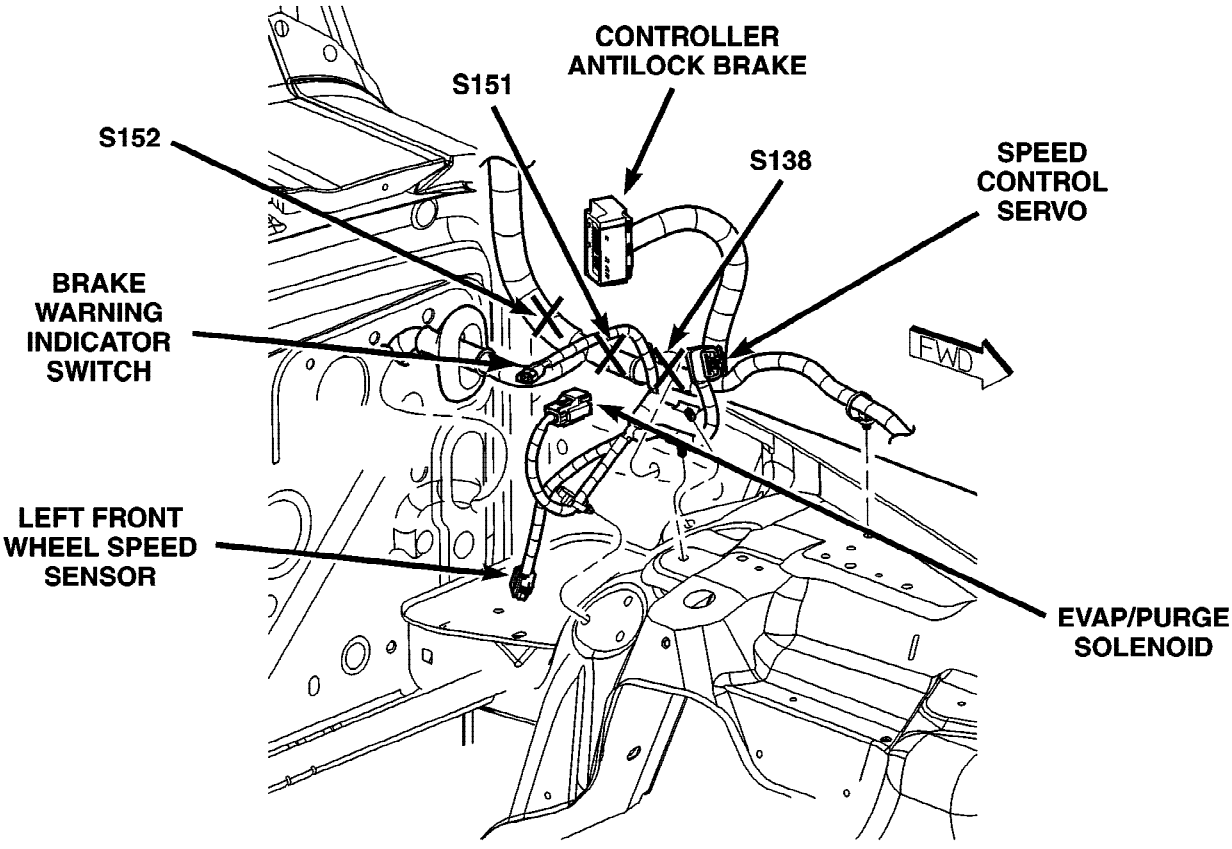


Fig. 14 -LEFT REAR ENGINE COMPARTMENT (LHD)

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

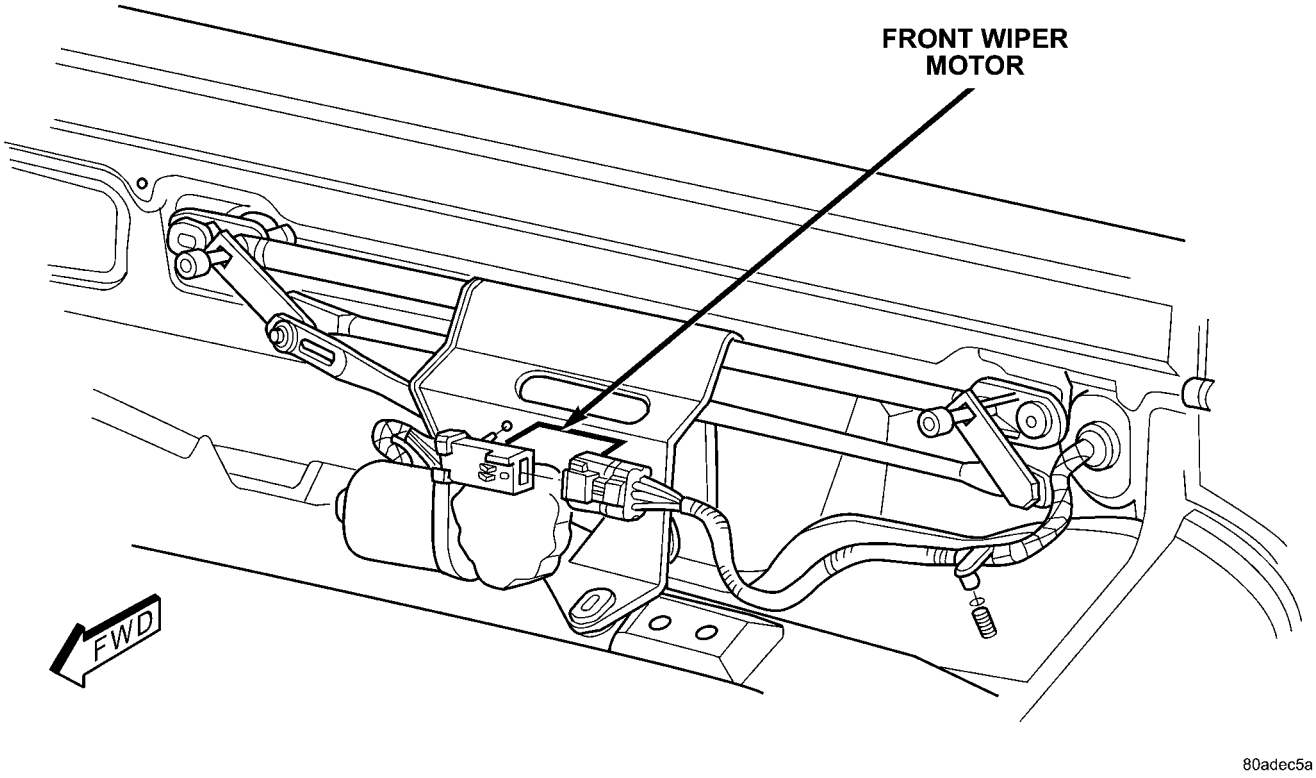


Fig. 15 -FRONT WIPER MOTOR

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

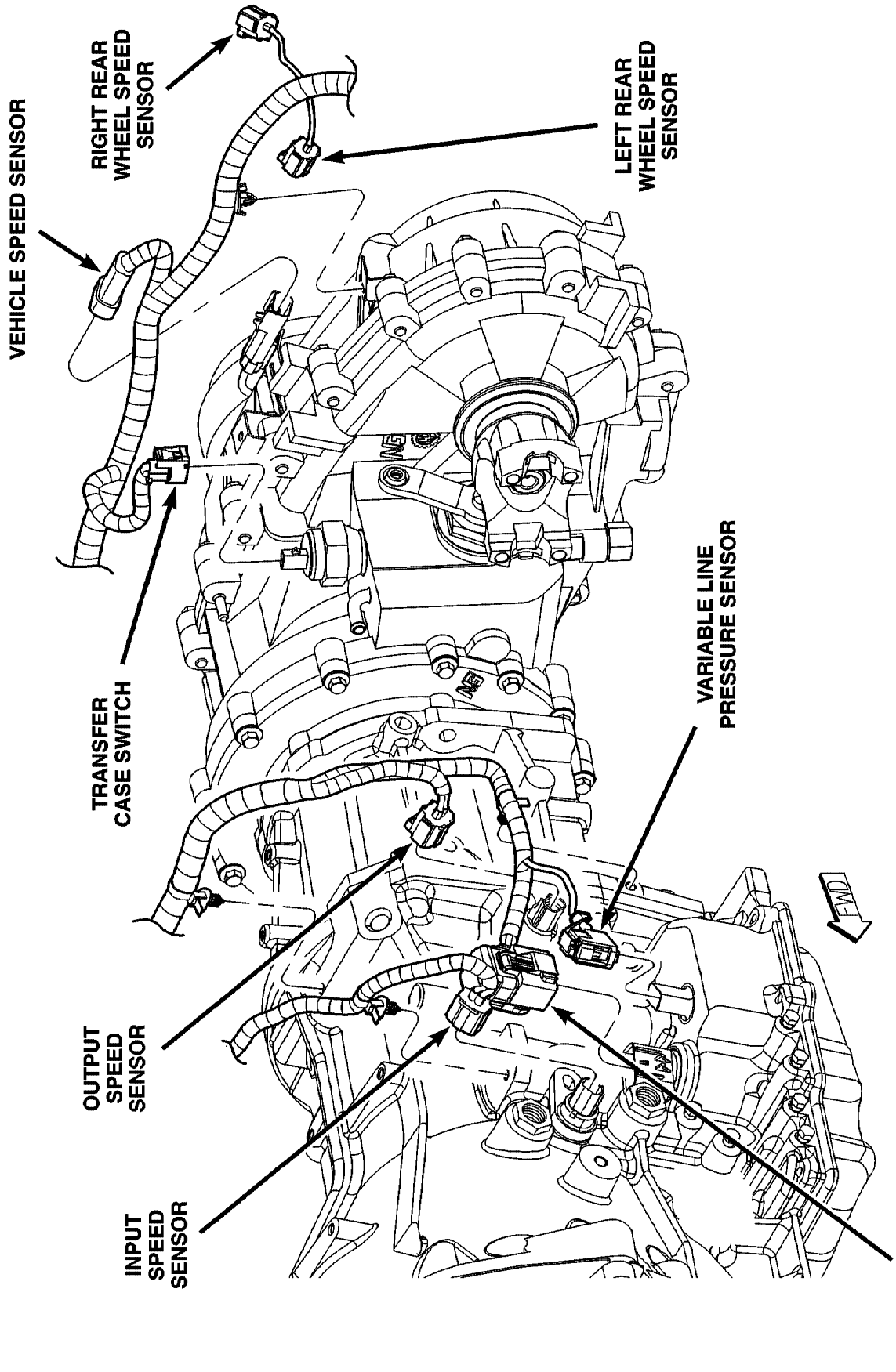


Fig. 16 -4.0 L ENGINE AND TRANS LEFT

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

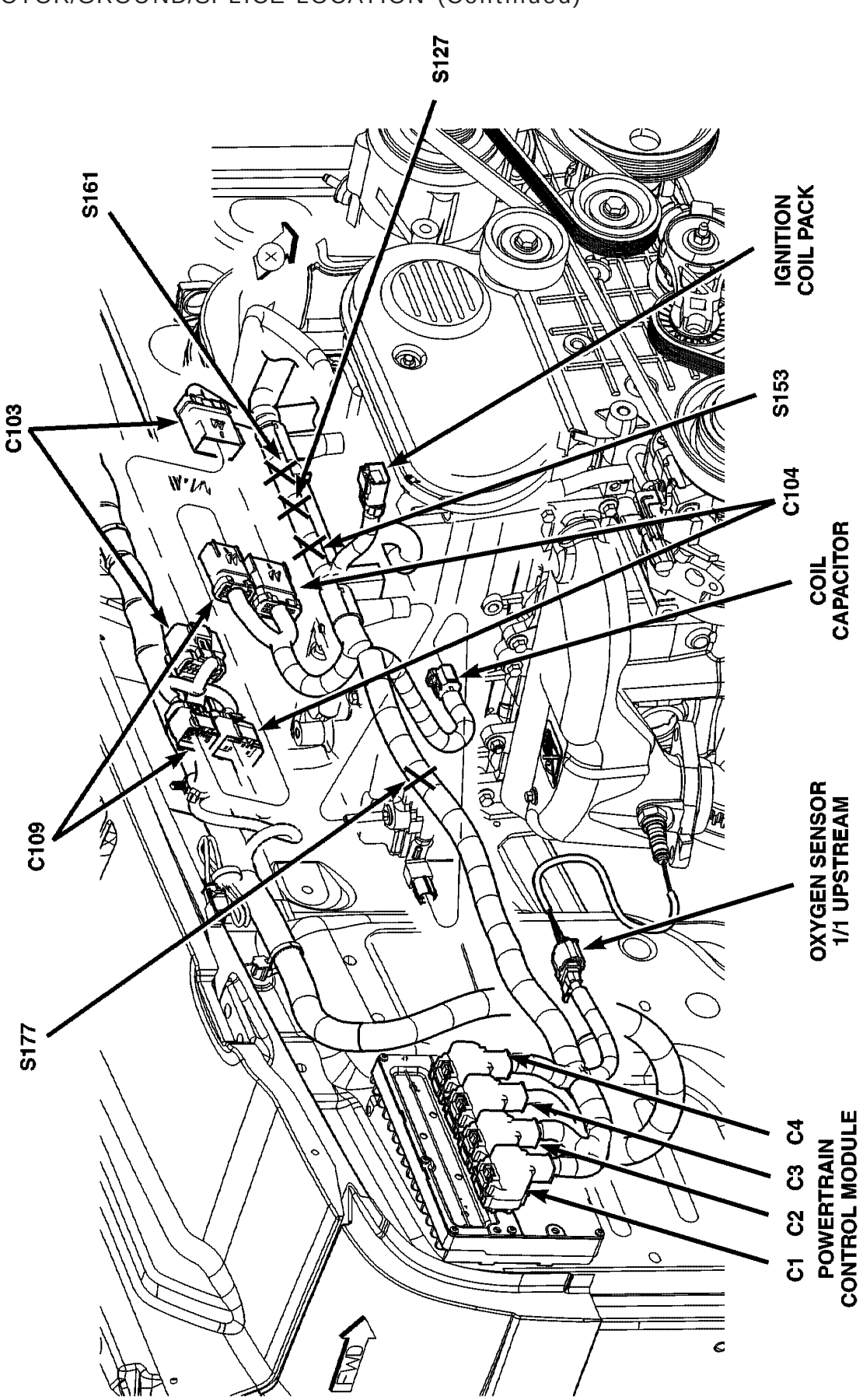


Fig. 17 -2.4 L RIGHT REAR ENGINE COMPARTMENT

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

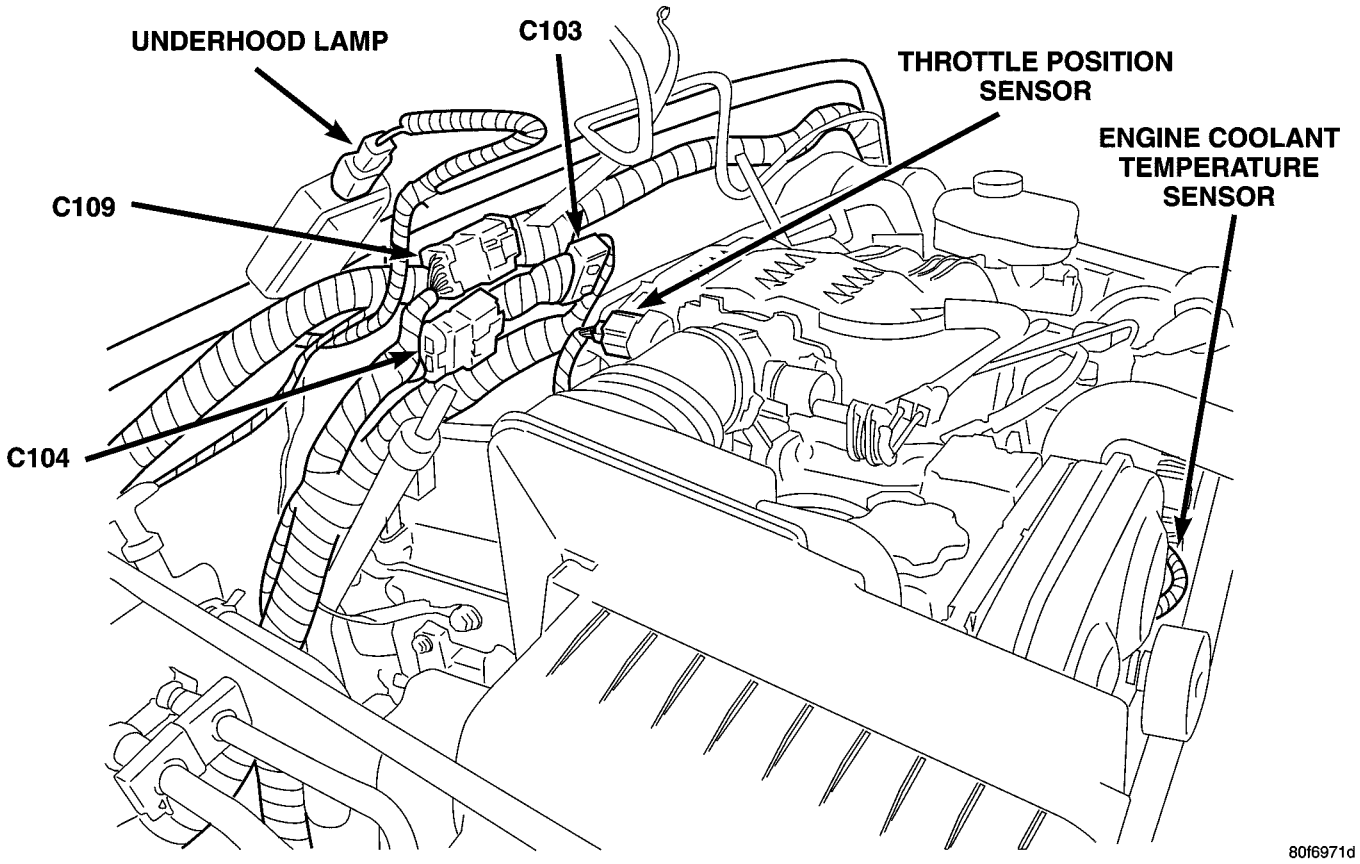


Fig. 18 -2.4 L REAR ENGINE

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

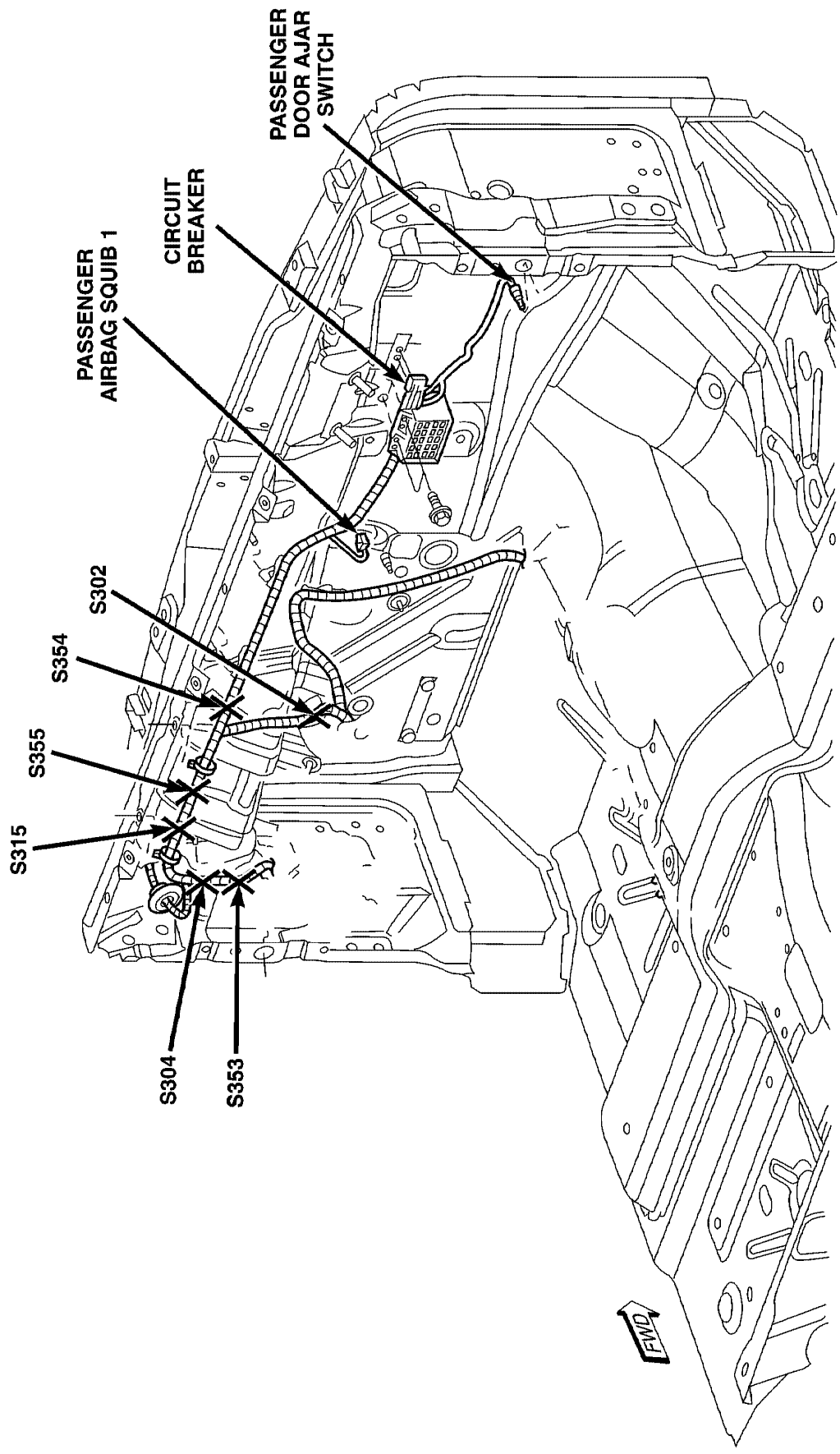


Fig. 19 -DASH PANEL (LHD)

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

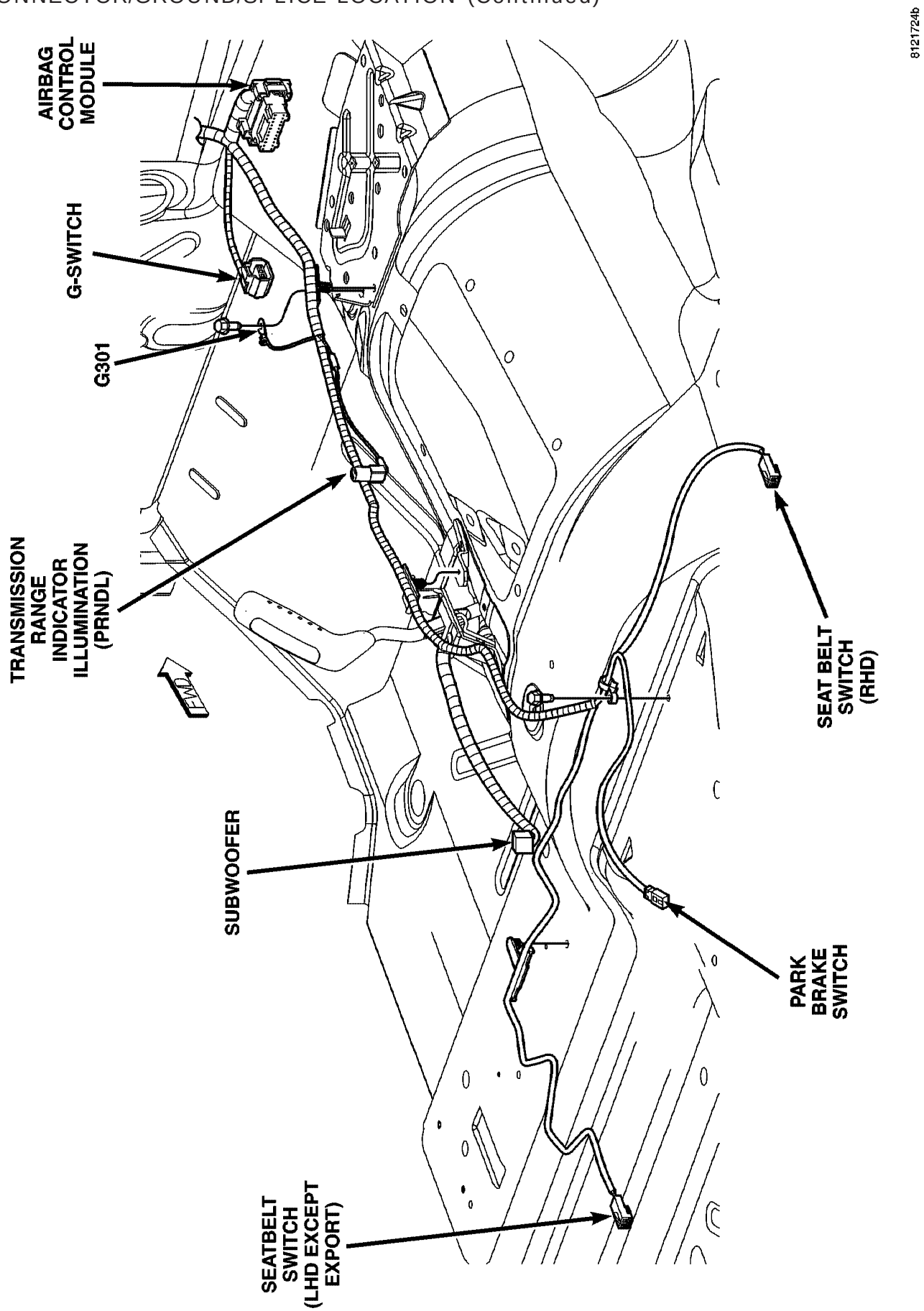


Fig. 20 -BODY HARNESS

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

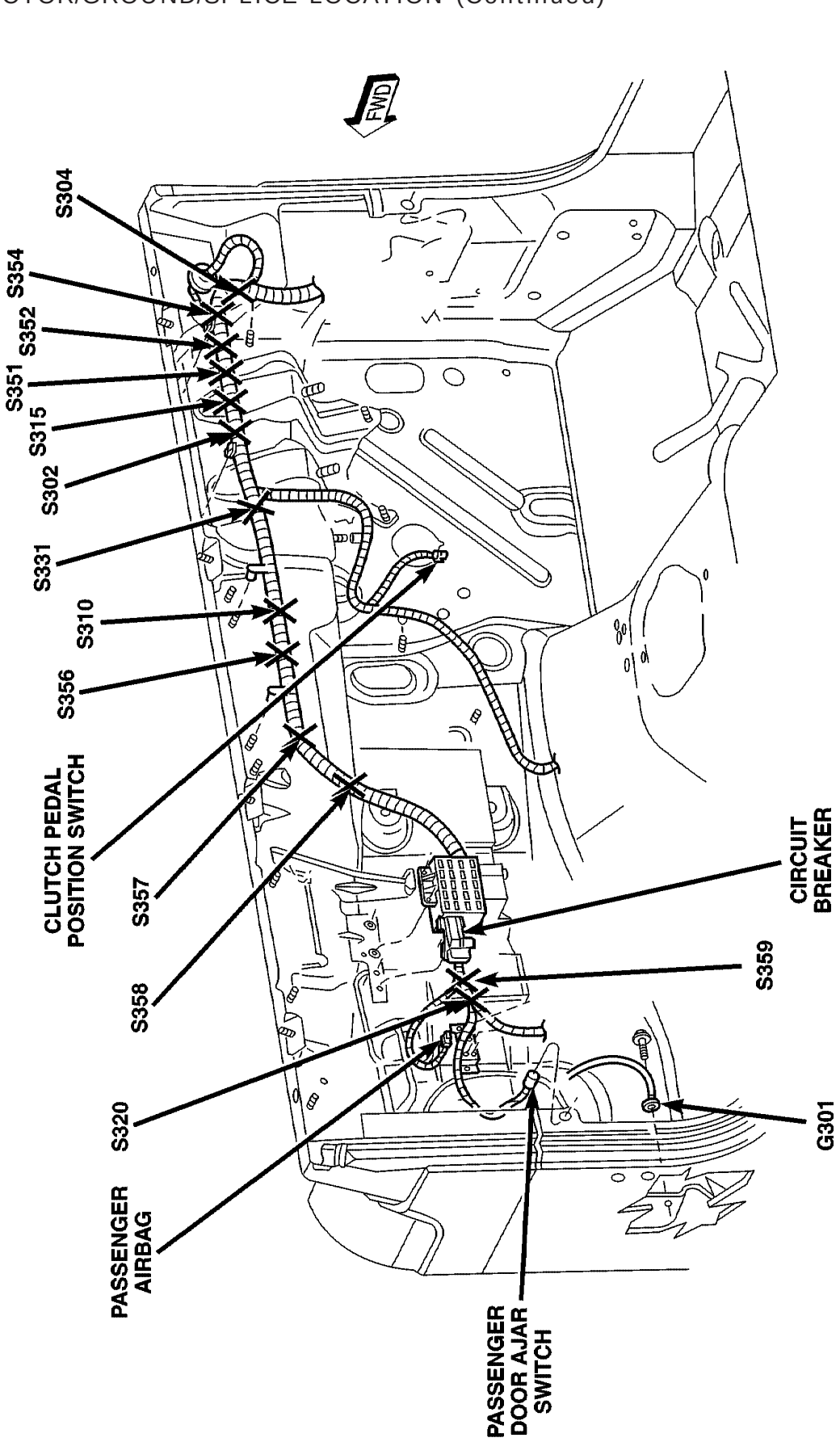


Fig. 21 -DASH PANEL (RHD)

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

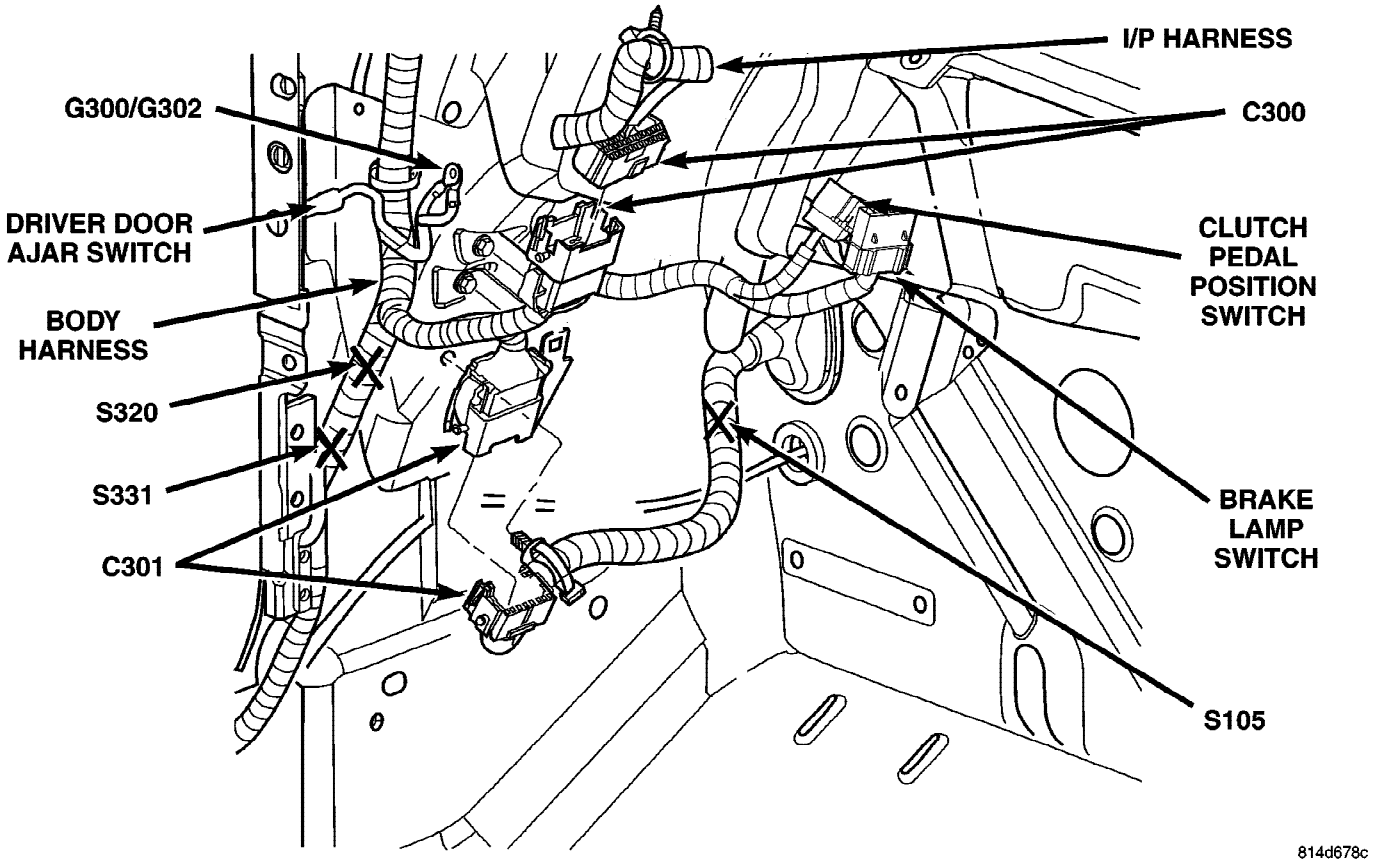


Fig. 22 -LEFT COWL PANEL (LHD)

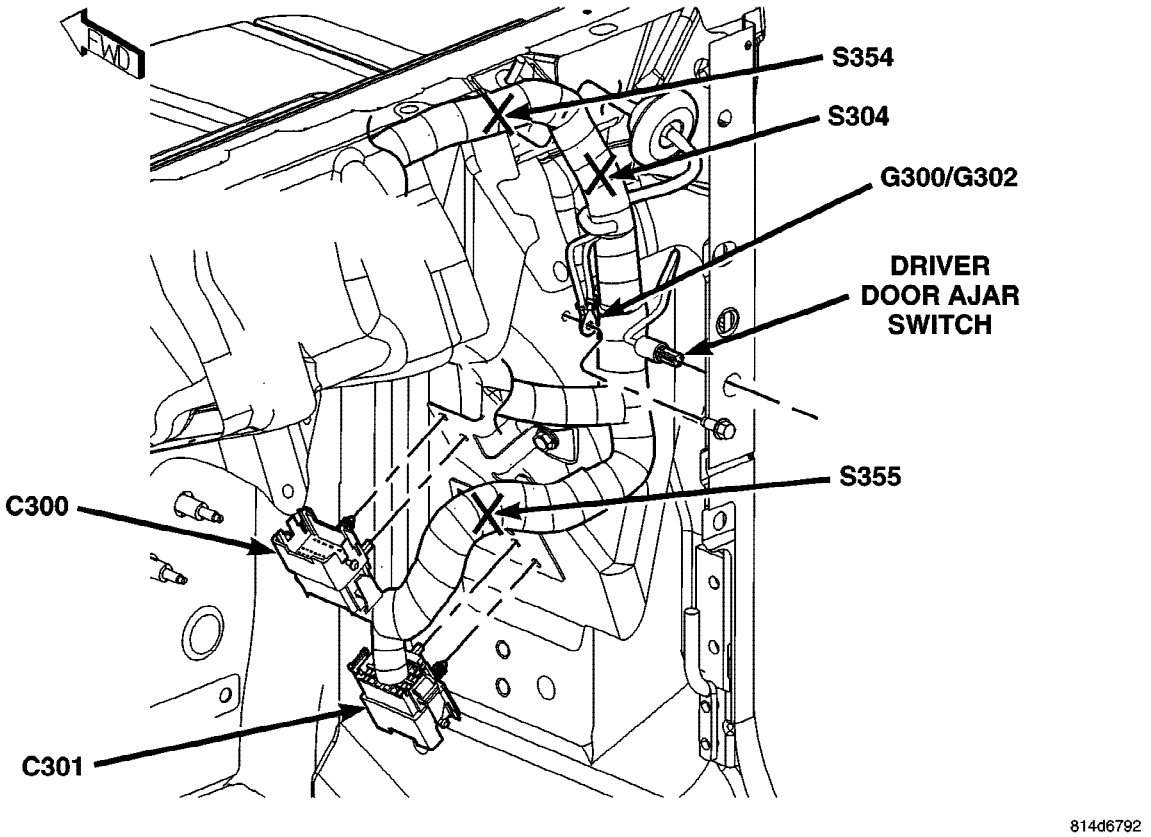
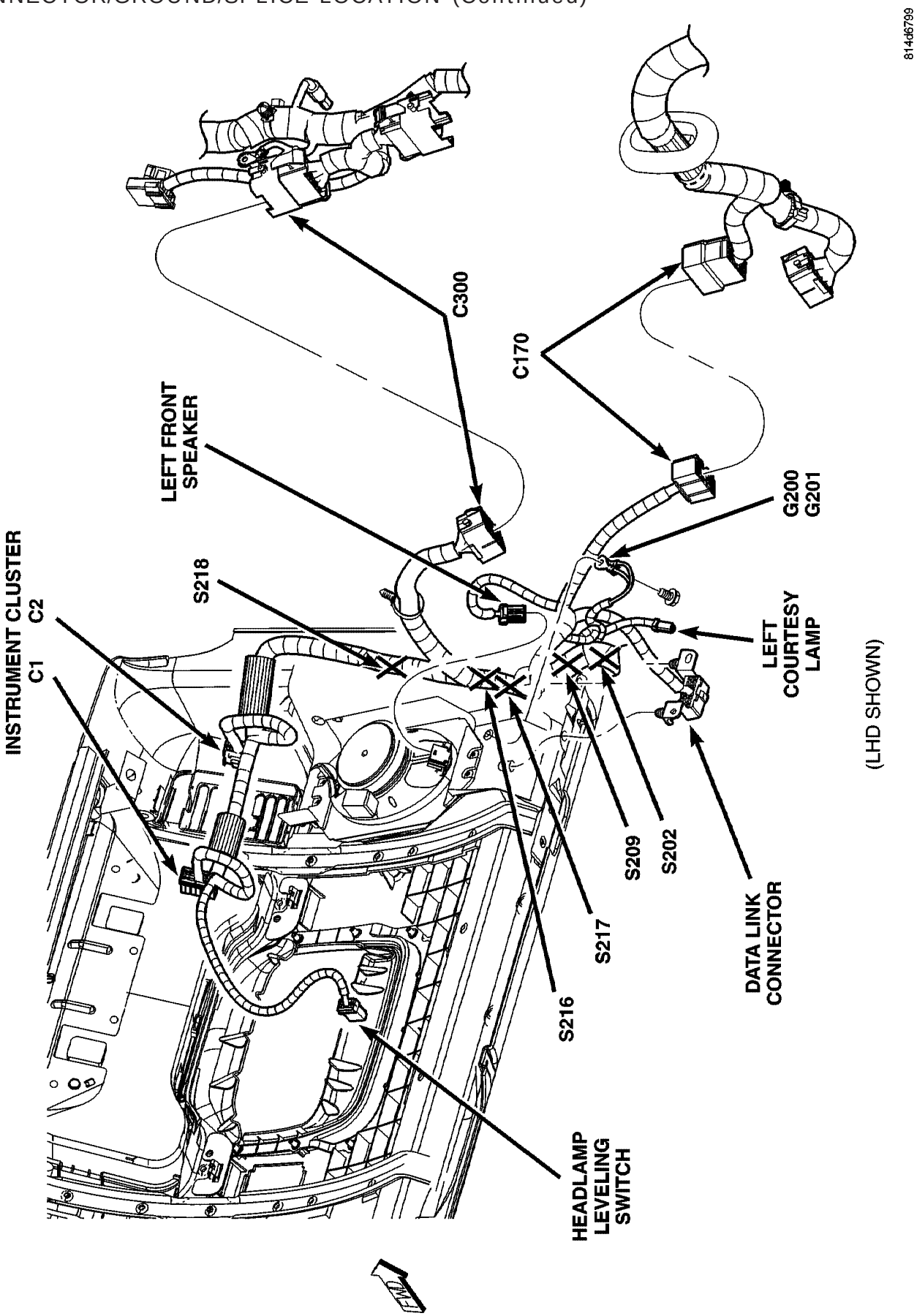


Fig. 23 -RIGHT COWL PANEL (RHD)

CONNECTOR/GROUND/SPLICE LOCATION (Continued)



(LHD SHOWN)

Fig. 24 -LEFT SIDE INSTRUMENT PANEL (LHD)

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

814d679d

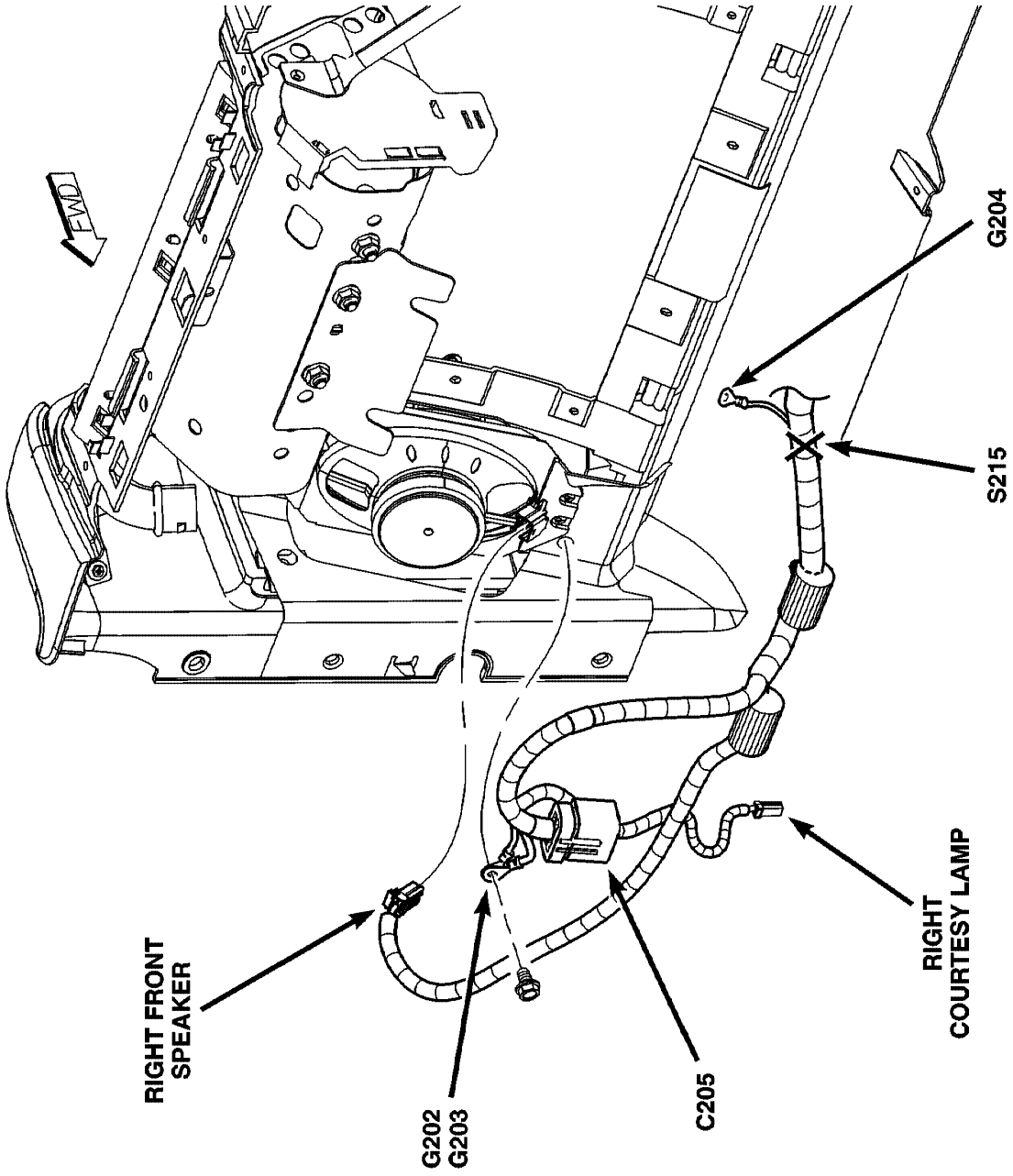


Fig. 25 -RIGHT SIDE INSTRUMENT PANEL (LHD)

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

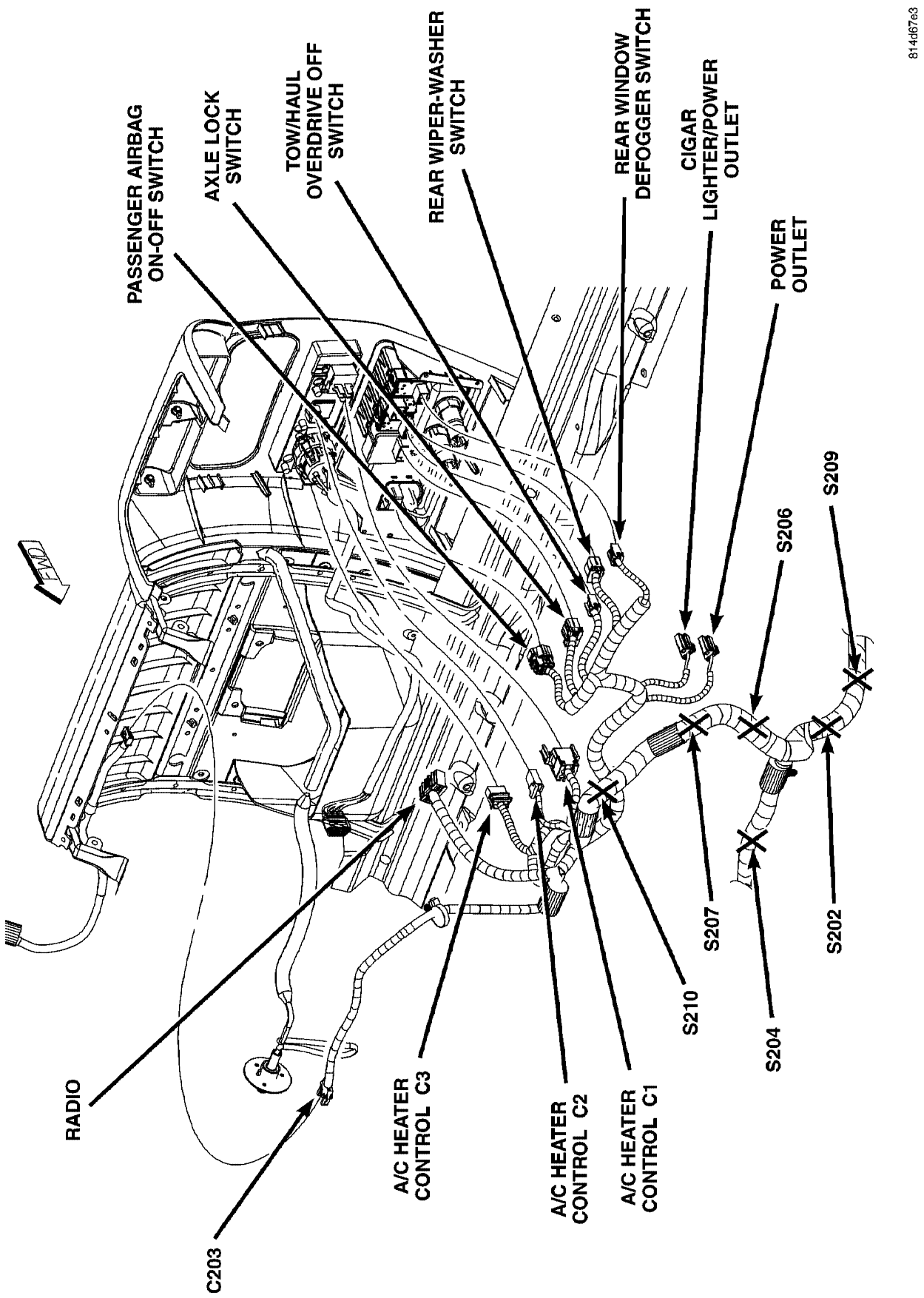


Fig. 26 -CENTER INSTRUMENT PANEL (LHD)

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

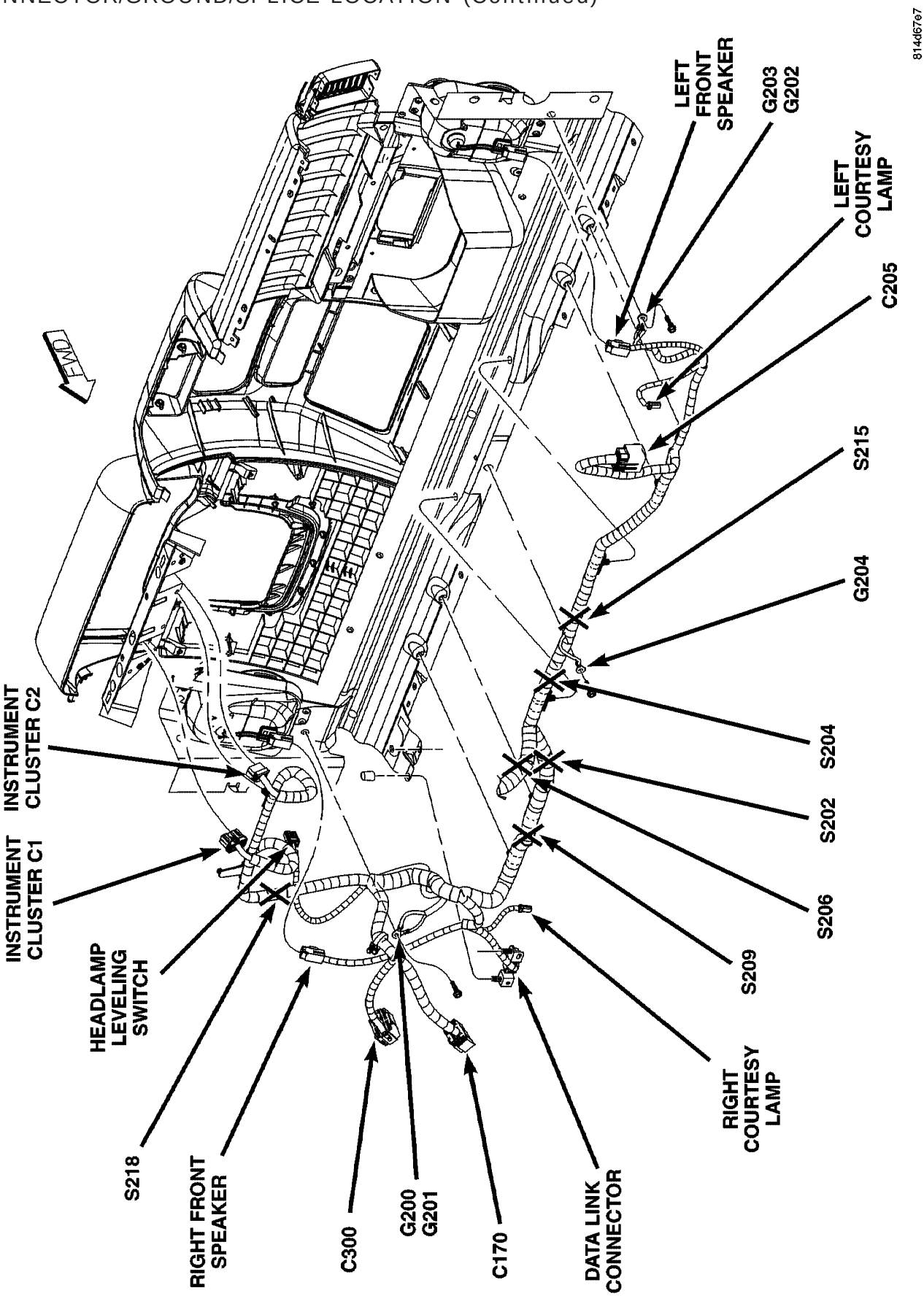


Fig. 27 -INSTRUMENT PANEL (RHD)

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

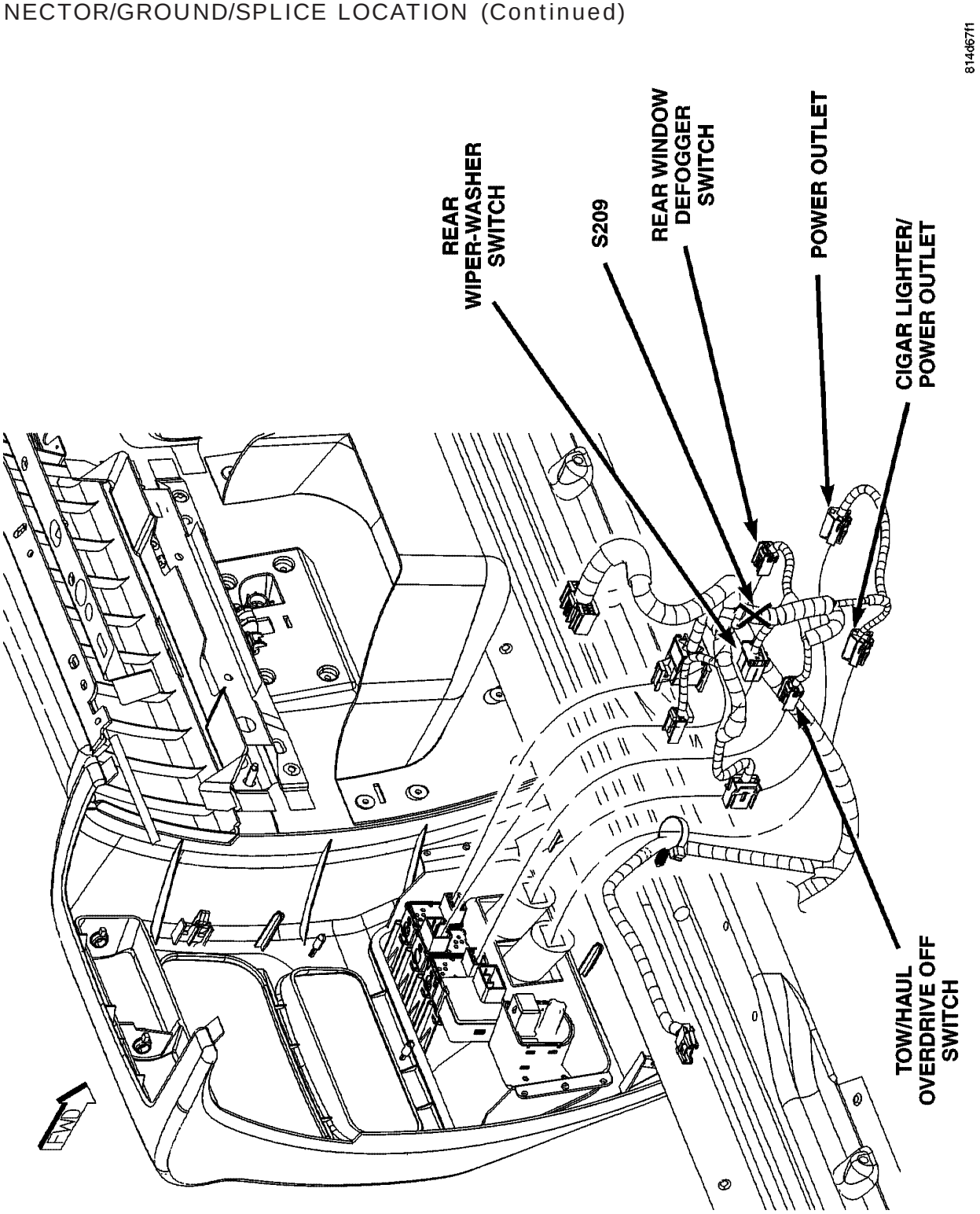


Fig. 28 -CENTER INSTRUMENT PANEL (RHD)

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

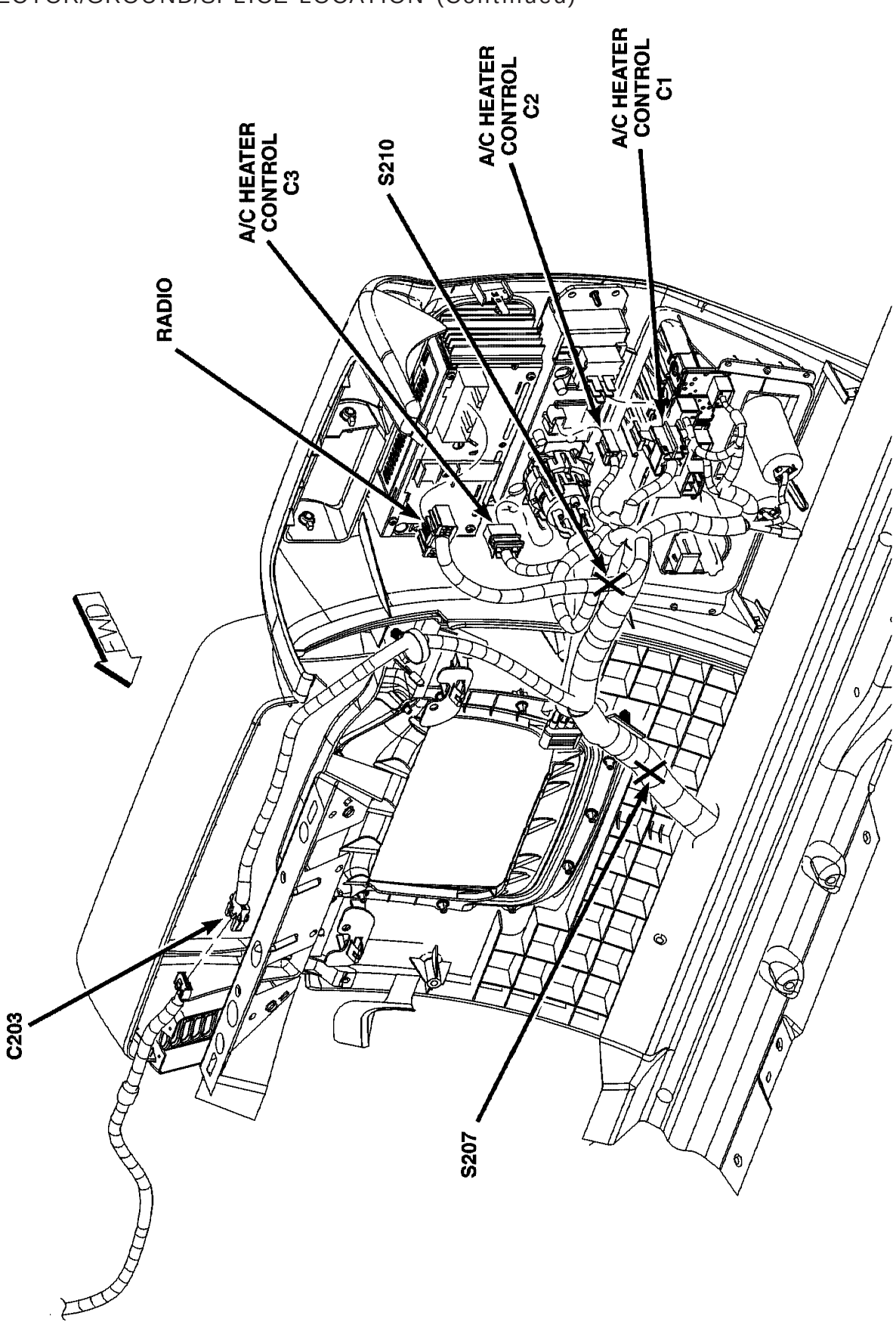


Fig. 29 -RIGHT AND CENTER INSTRUMENT PANEL (RHD)

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

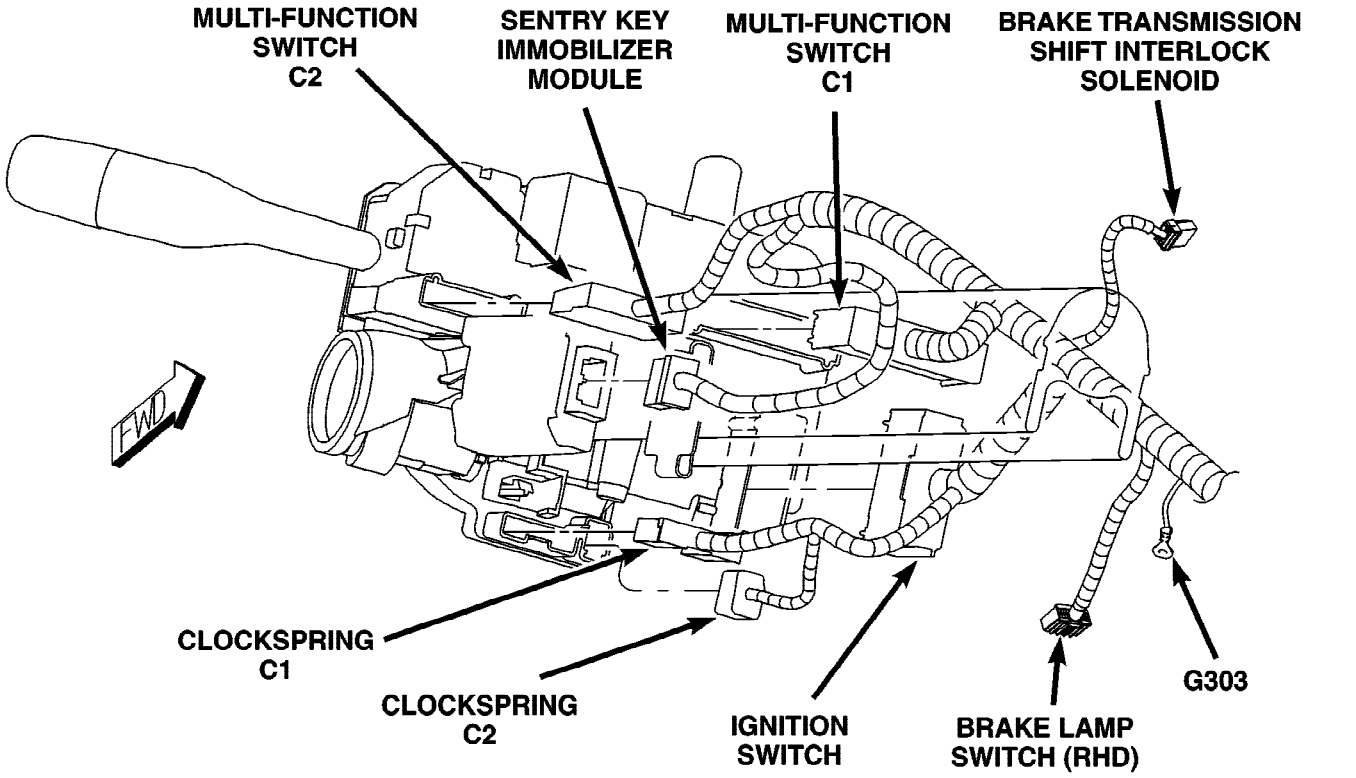


Fig. 30 -STEERING COLUMN CONNECTIONS

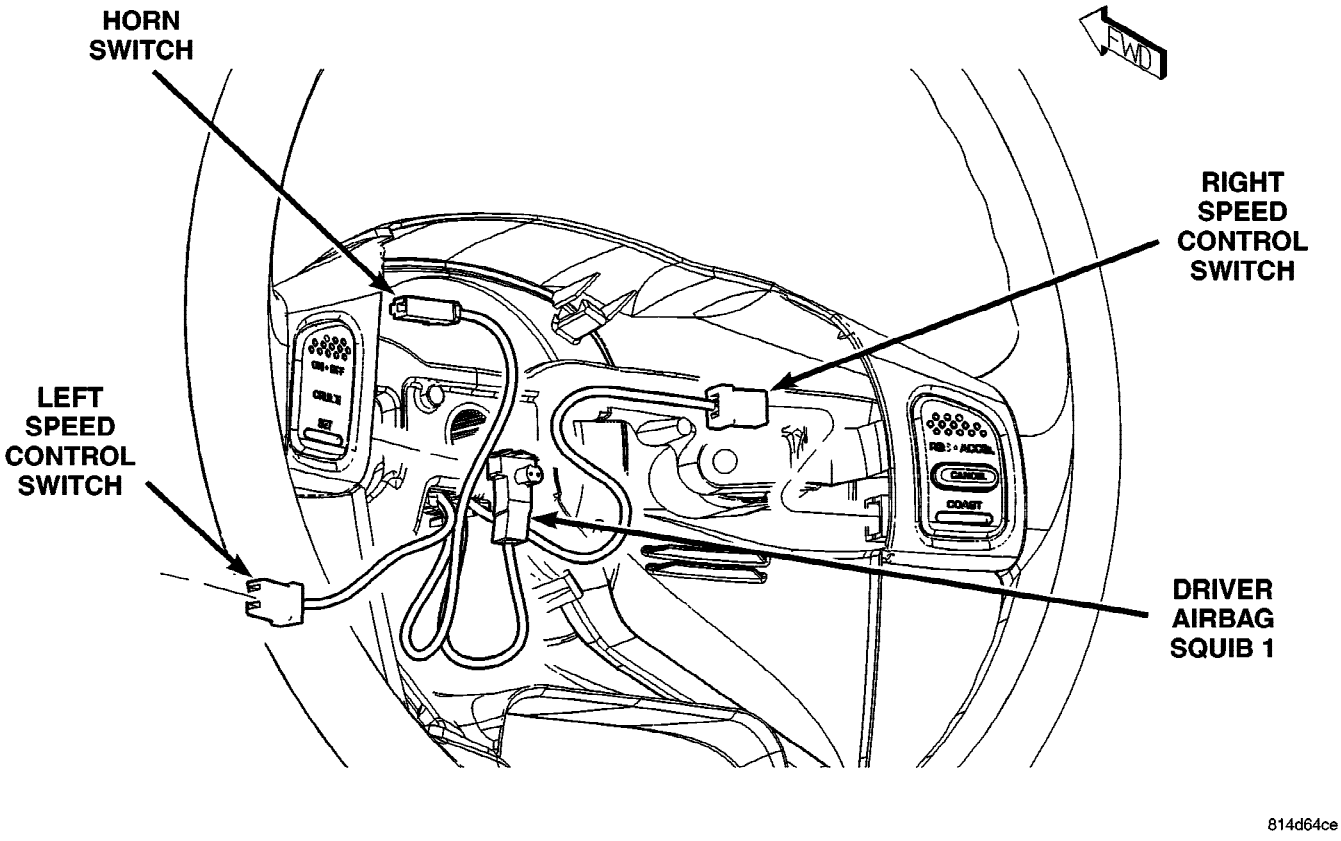


Fig. 31 -STEERING WHEEL CONNECTIONS

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

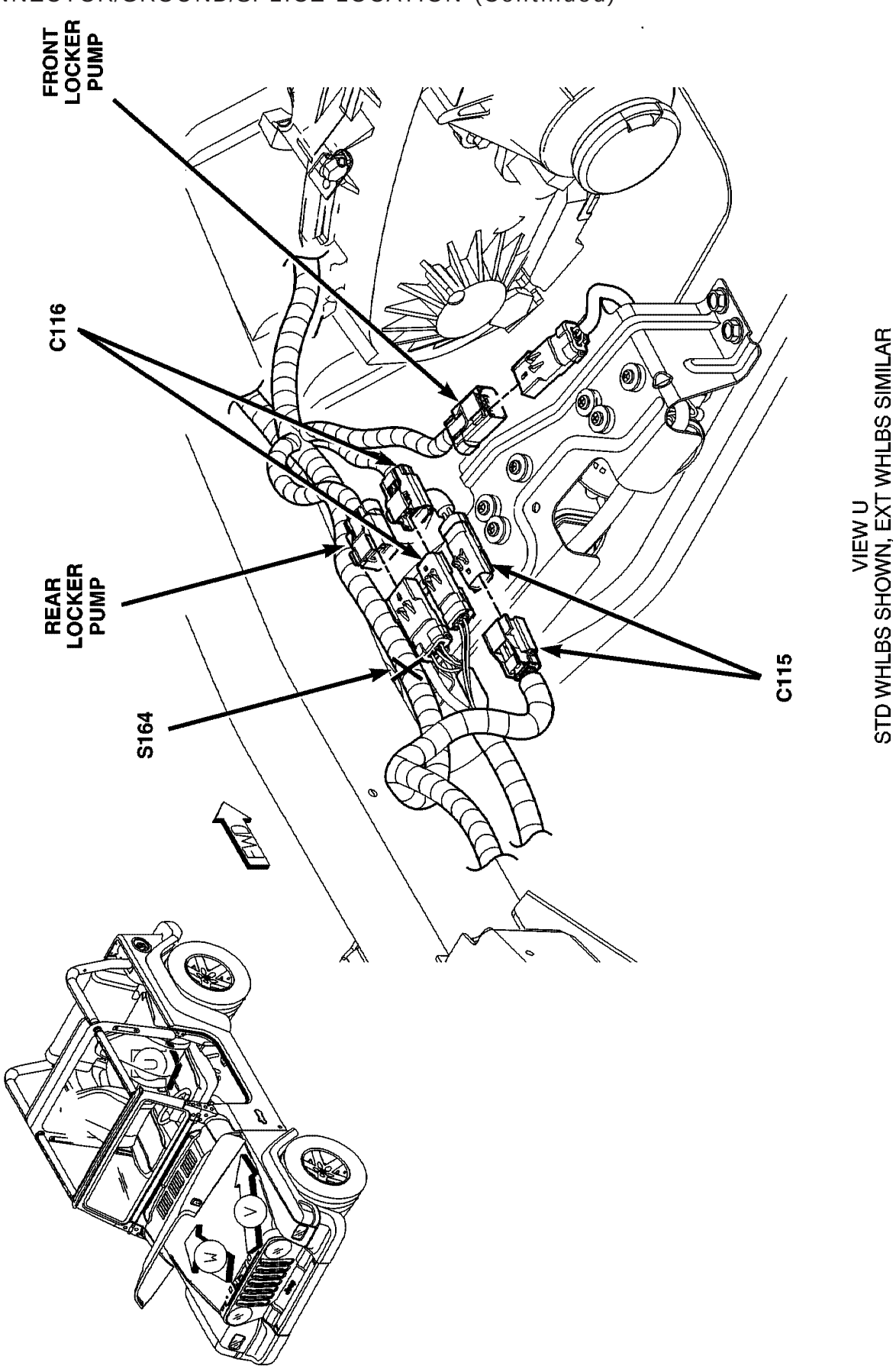


Fig. 32 -TRANSFER CASE

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

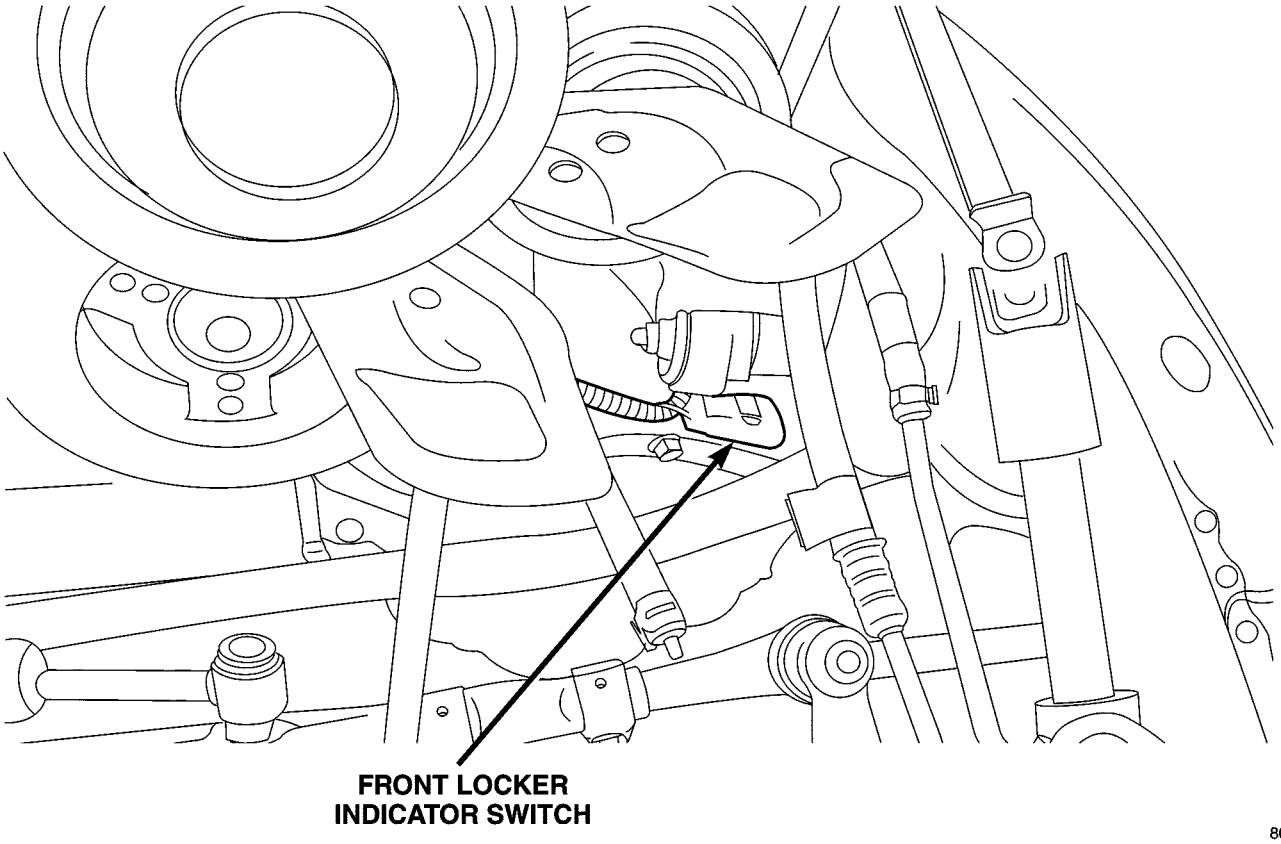


Fig. 33 -FRONT LOCKER INDICATOR SWITCH

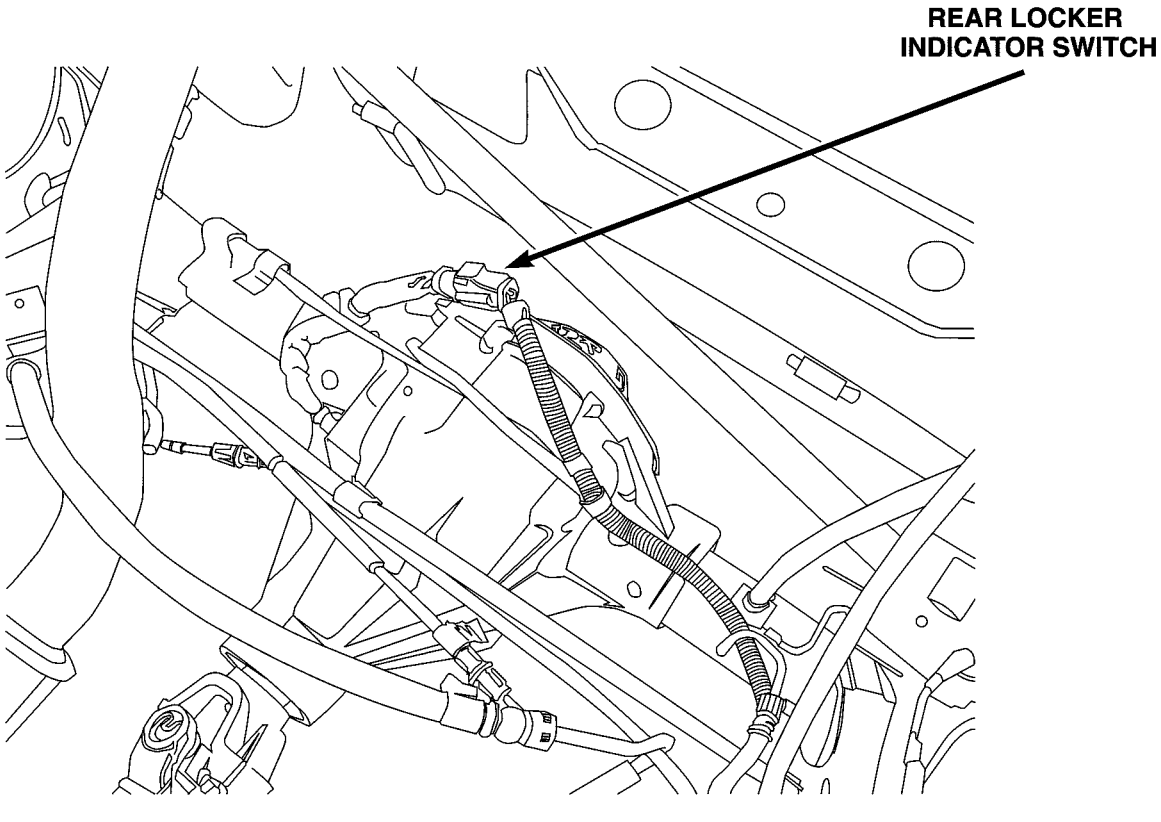


Fig. 34 -REAR LOCKER INDICATOR SWITCH

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

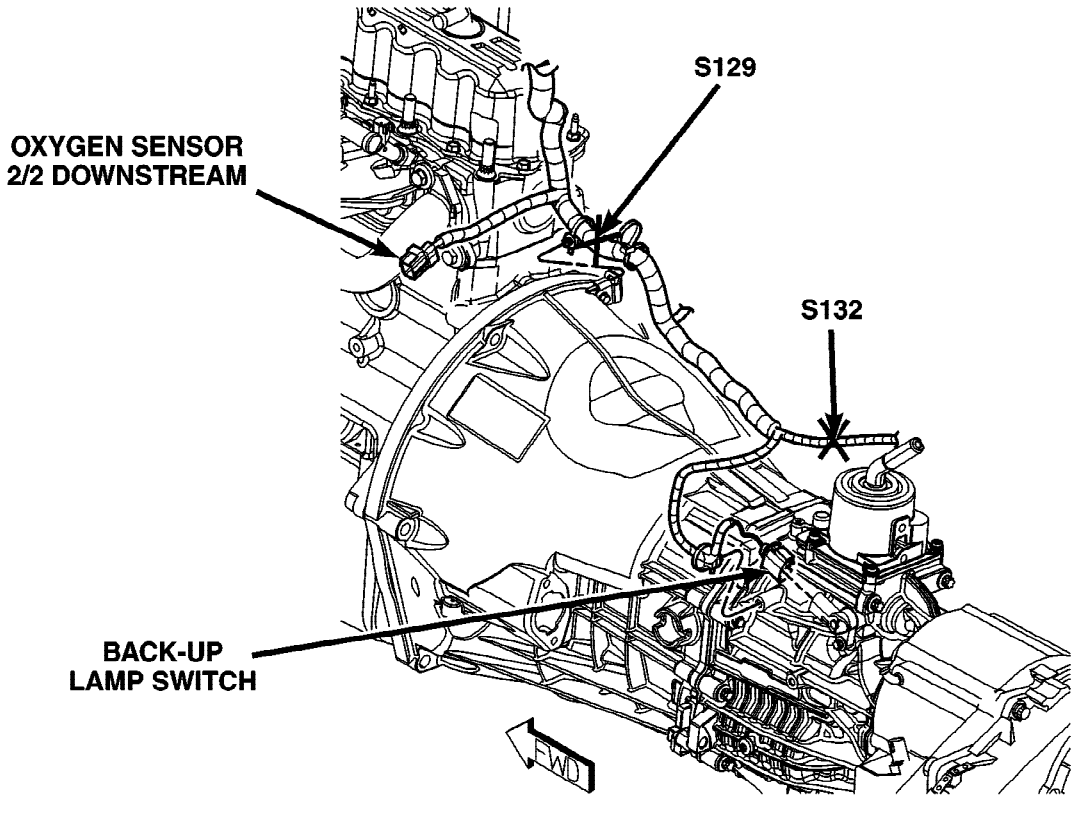


Fig. 35 -4.0 L MANUAL TRANSMISSION

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

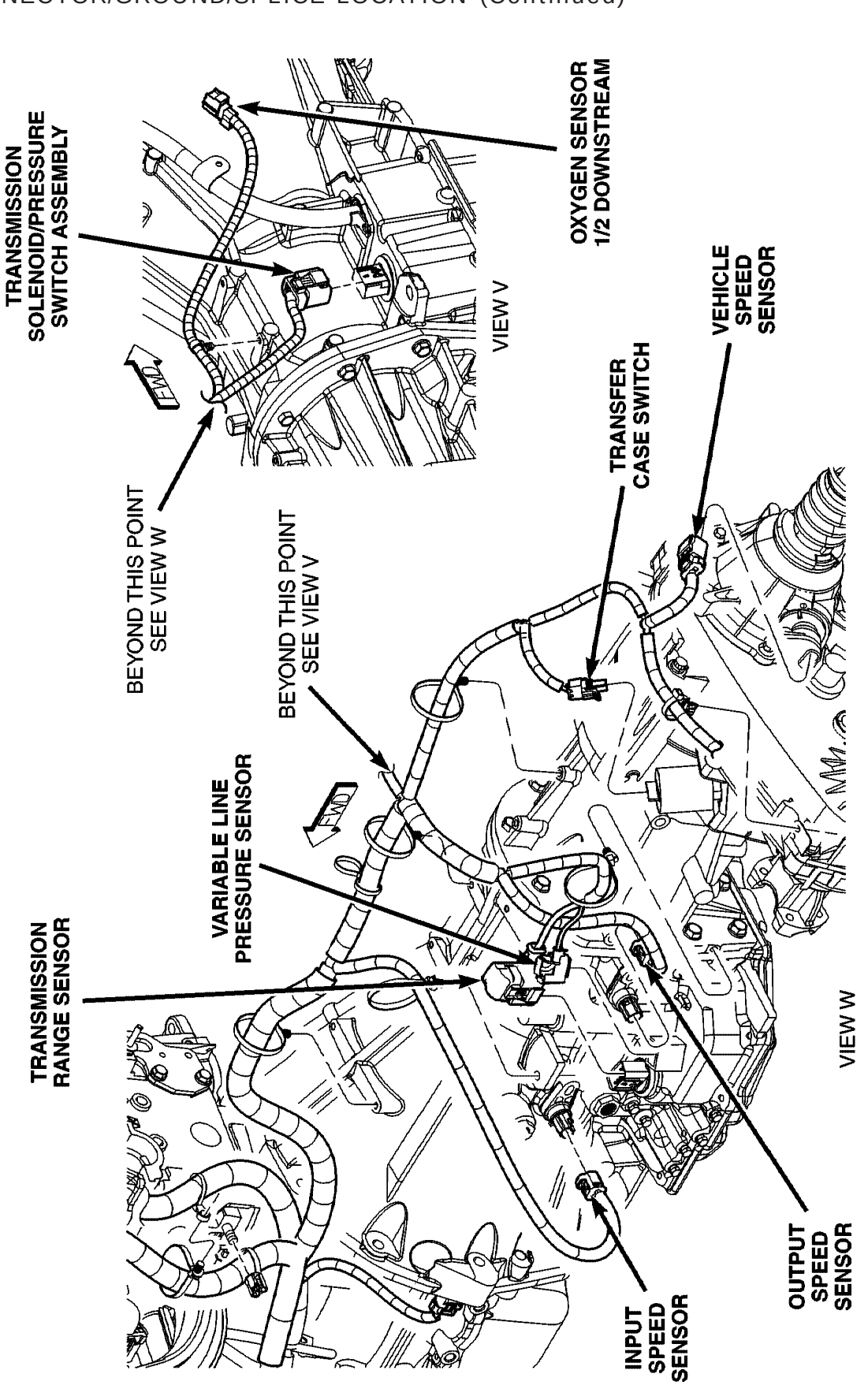


Fig. 36 -2.4 L AUTOMATIC TRANSMISSION

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

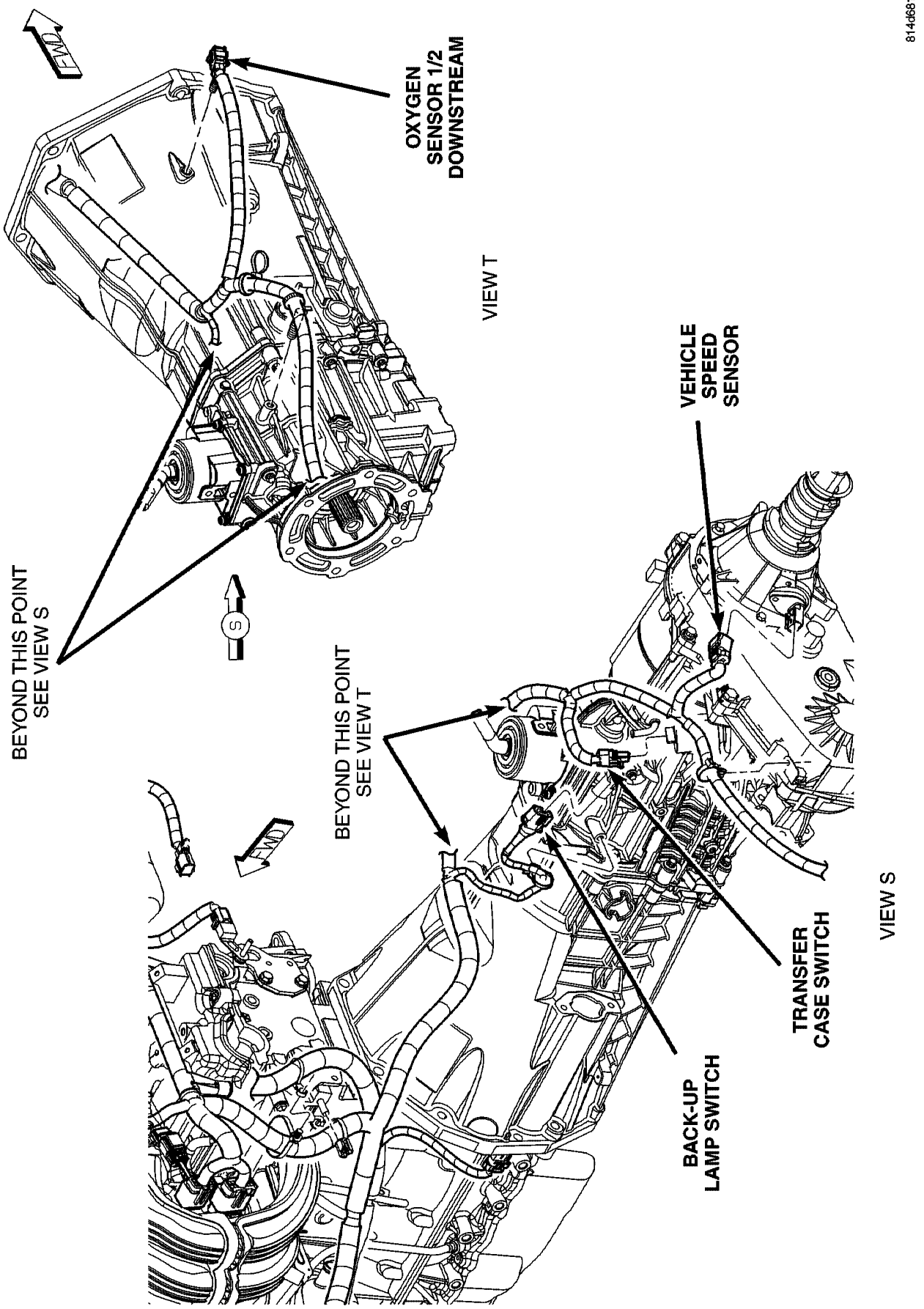


Fig. 37 -2.4 L MANUAL TRANSMISSION

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

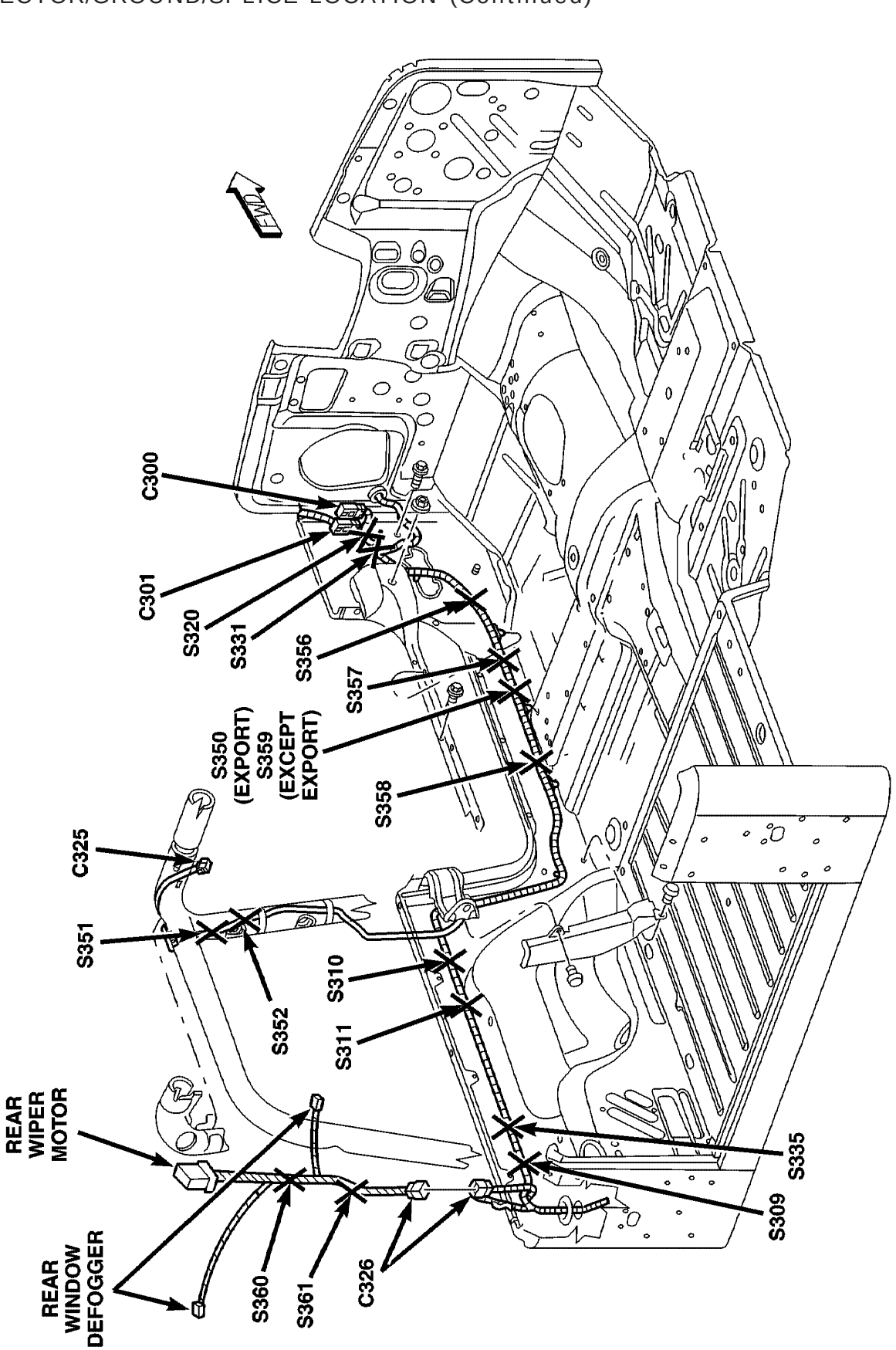


Fig. 38 -BODY CONNECTORS (LHD)

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

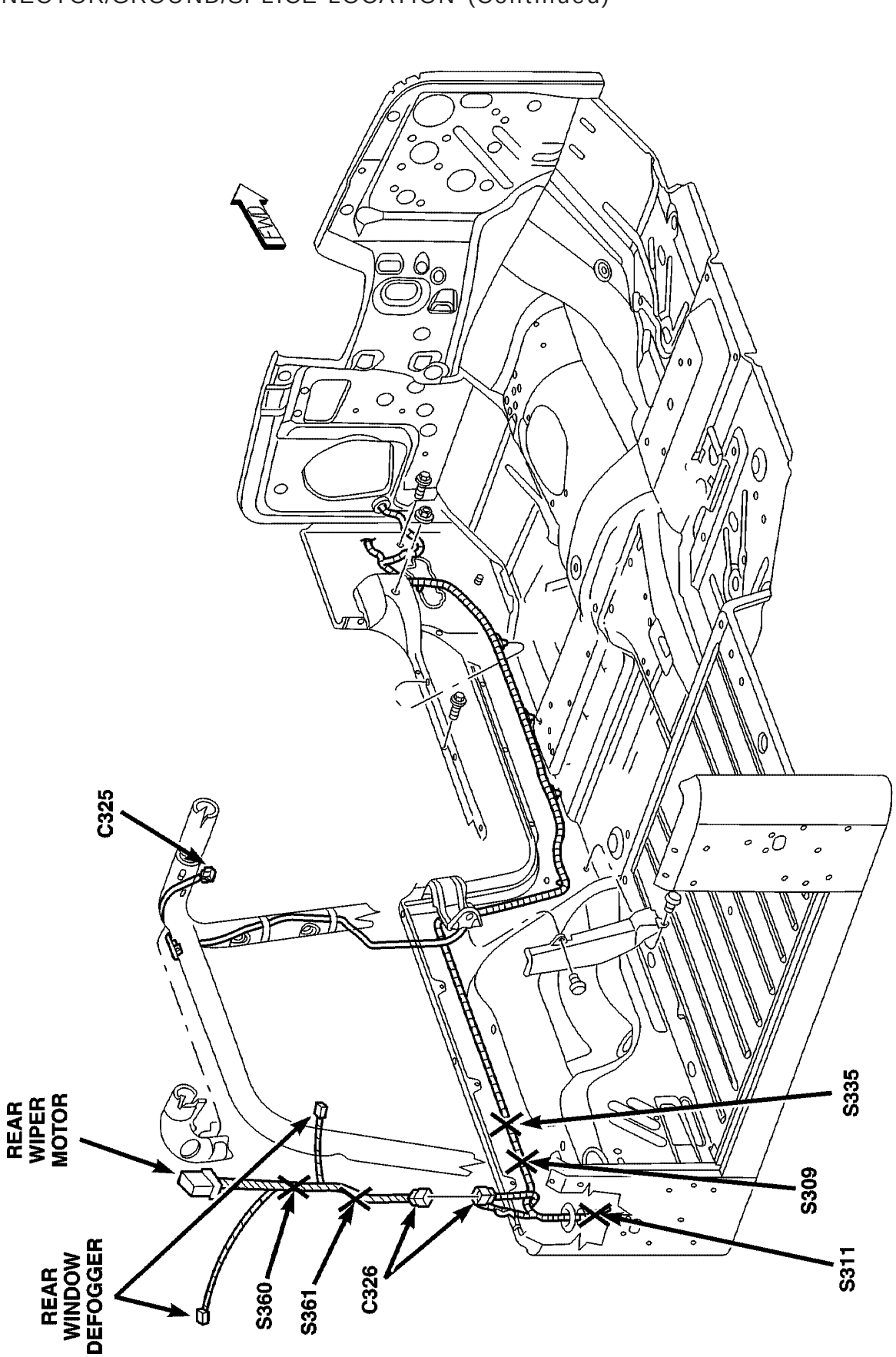


Fig. 39 -BODY CONNECTORS (RHD)

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

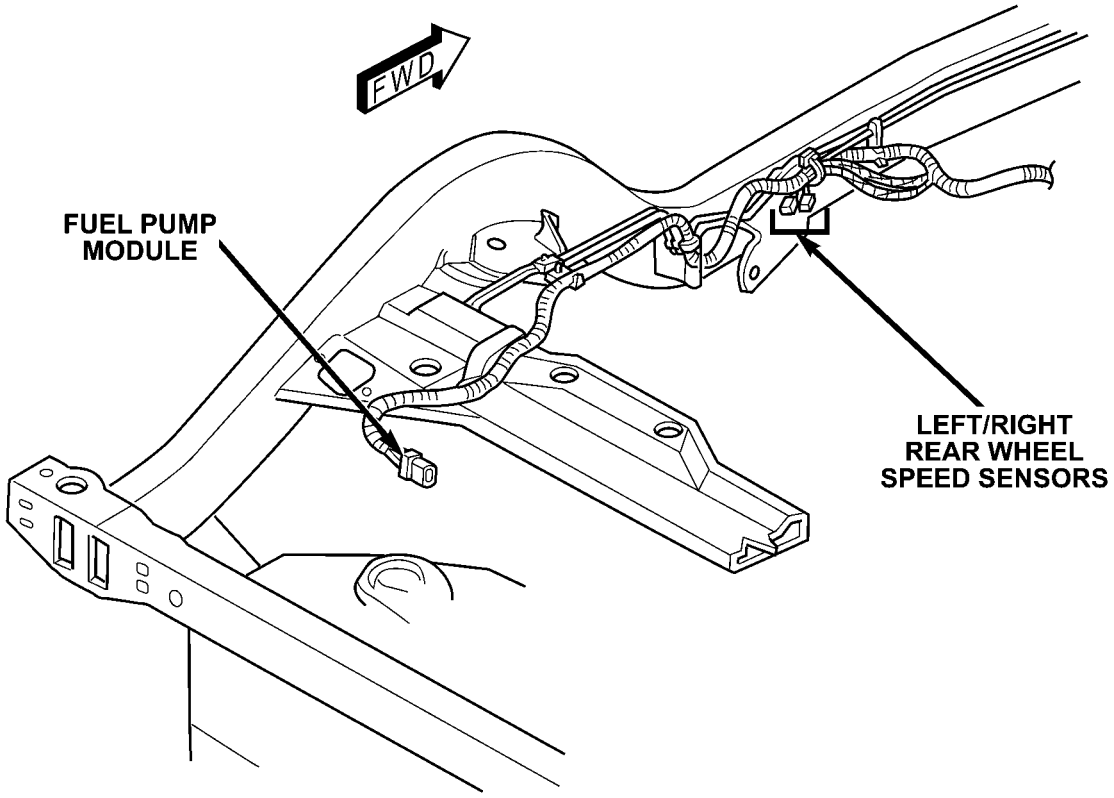


Fig. 40 -FUEL PUMP

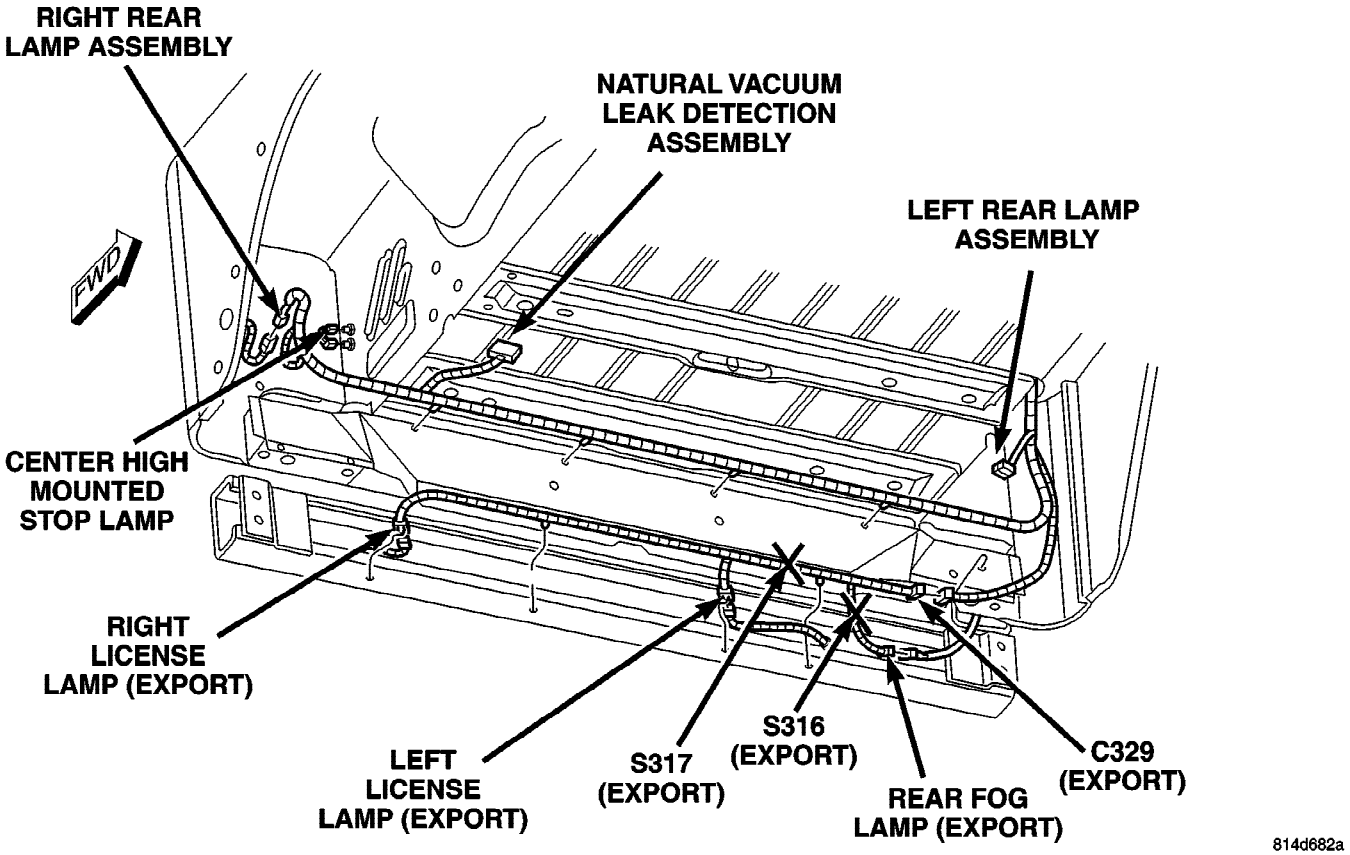


Fig. 41 -REAR LAMPS (LHD)

CONNECTOR/GROUND/SPLICE LOCATION (Continued)

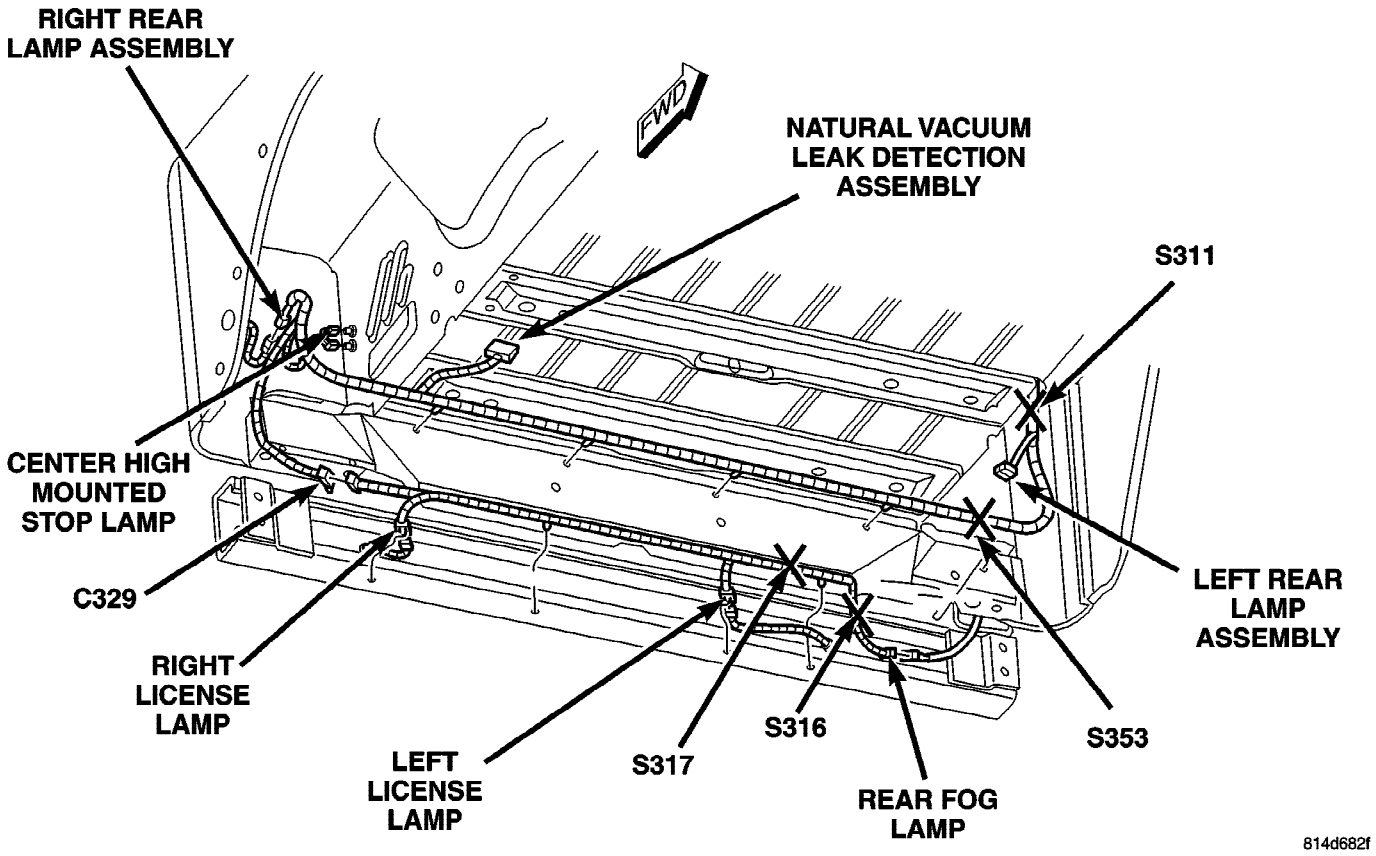


Fig. 42 -REAR LAMPS (RHD)

8W-97 POWER DISTRIBUTION

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POWER DISTRIBUTION

DESCRIPTION

This group covers the various standard and optional power distribution components used on this model. The power distribution system for this vehicle consists of the following components:

- Power Distribution Center (PDC)
- Fuseblock
- Cigar Lighter Outlet
- Power Outlet

The power distribution system also incorporates various types of circuit control and protection features, including:

- Blade-type fuses
- Cartridge fuses
- Relays

Following are general descriptions of the major components in the power distribution system. See the owner's manual in the vehicle glove box for more information on the features and use of all of the power distribution system components. Refer to the index in this service manual for the location of complete circuit diagrams for the various power distribution system components.

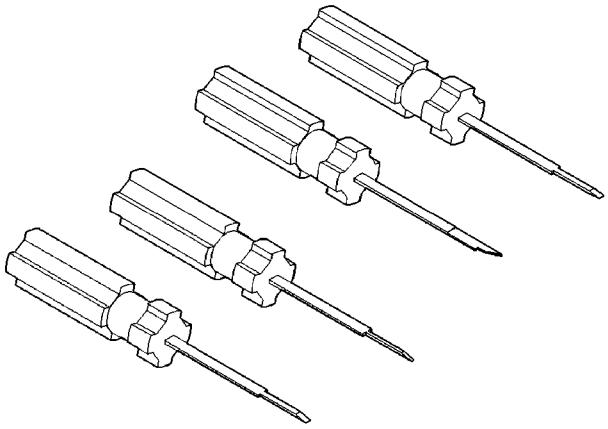
OPERATION

The power distribution system for this vehicle is designed to provide safe, reliable, and centralized distribution points for the electrical current required to operate all of the many standard and optional factory-installed electrical and electronic powertrain, chassis, safety, security, comfort and convenience systems. At the same time, the power distribution system was designed to provide ready access to these electrical distribution points for the vehicle technician to use when conducting diagnosis and repair of faulty circuits. The power distribution system can also prove useful for the sourcing of additional elec-

trical circuits that may be required to provide the electrical current needed to operate many accessories that the vehicle owner may choose to have installed in the aftermarket.

SPECIAL TOOLS

POWER DISTRIBUTION SYSTEMS



Terminal Pick Kit 6680

CIGAR LIGHTER OUTLET

DESCRIPTION

An instrument panel cigar lighter outlet is standard factory-installed equipment on some models. Models equipped with the optional smoker's package will include a cigar lighter knob and heating element. Models without the smoker's package are equipped with a snap fit plastic cap and the outlet is treated as an extra accessory power outlet. The outlet is installed in the instrument panel accessory switch bezel, which is located near the bottom of the instrument panel center bezel area, below the heater and air conditioner controls.

The cigar lighter outlet is serviced only as a part of the accessory switch bezel unit. If the base is faulty or damaged, the accessory switch bezel unit must be replaced. The cigar lighter knob and heating element unit is available for service. This component cannot be repaired and, if faulty or damaged, it must be replaced.

OPERATION

The cigar lighter base or receptacle shell is connected to ground, and an insulated contact in the bottom of the shell is connected to battery current. The cigar lighter receives battery voltage from a fuse (f13) in the Power Distribution Center (PDC) through the ignition switch, only when in the Accessory or On position and through a fuse (f19) in the fuseblock.′

The cigar lighter knob and heating element are encased within a spring-loaded housing, which also features a sliding protective heat shield. When the knob and heating element are inserted in the receptacle shell, the heating element resistor coil is grounded through its housing to the receptacle shell. If the cigar lighter knob is pushed inward, the heat shield slides up toward the knob exposing the heating element, and the heating element extends from the housing toward the insulated contact in the bottom of the receptacle shell.

Two small spring-clip retainers are located on either side of the insulated contact inside the bottom of the receptacle shell. These clips engage and hold the heating element of the cigar lighter against the insulated contact long enough for the resistor coil to heat up. When the heating element is engaged with the contact, battery current can flow through the resistor coil to ground, causing the resistor coil to heat.

When the resistor coil becomes sufficiently heated, excess heat radiates from the heating element causing the spring-clips to expand. Once the spring-clips expand far enough to release the heating element, the spring-loaded housing forces the knob and heating element to pop back outward to their relaxed position. When the cigar lighter knob and element are pulled out of the receptacle shell, the protective heat shield slides downward on the housing so that the heating element is recessed and shielded around its circumference for safety.

DIAGNOSIS AND TESTING - CIGAR LIGHTER OUTLET

For complete circuit diagrams, refer to **Wiring Diagrams**.

WARNING: REFER TO THE RESTRAINTS SECTION OF THE SERVICE MANUAL BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Remove the cigar lighter knob and element from the cigar lighter receptacle shell. Check for continuity between the inside circumference of the cigar lighter receptacle shell and a good ground. there should be continuity. If OK, go to Step 2. If not OK, go to Step 3.

(2) Turn the ignition switch to the On position. Check for battery voltage at the insulated contact located at the back of the cigar lighter receptacle

CIGAR LIGHTER OUTLET (Continued)

shell. If OK, replace the faulty cigar lighter knob and element. If not OK, go to Step 3.

(3) Turn the ignition switch to the Off position. Disconnect and isolate the battery negative cable. Remove the instrument panel accessory switch bezel. Check for continuity between the ground circuit cavity #3 of the cigar lighter wire harness connector and a good ground. There should be continuity. If OK, go to Step 4. If not OK, repair the open ground circuit to ground as required.

(4) Connect the battery negative cable. Turn the ignition switch to the Accessory or On positions. Check for battery voltage at cavity #1 of the cigar lighter wire harness connector. If OK, replace the faulty accessory switch bezel unit. If not OK, check for blown fuse in the fuseblock (f19) or in the PDC (f13). If fuse is blown check for short circuit. Repair the circuit as required and replace blown fuse.

FUSE BLOCK

DESCRIPTION

An electrical fuseblock module is mounted on the dash panel in the passenger compartment of the vehicle (Fig. 1). The fuseblock module serves to distribute electrical current to many of the accessory systems in the vehicle. The fuseblock module houses up to twenty blade-type mini fuses.

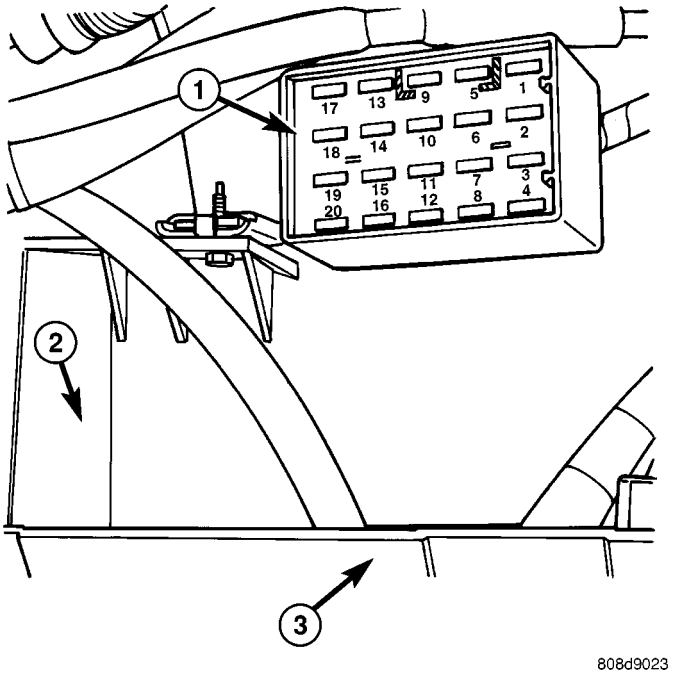


Fig. 1 Fuseblock Module Location

1 - FUSE BLOCK
2 - HEATER CASE
3 - GLOVE BOX OPENING

The molded plastic fuseblock module housing has an integral mounting bracket that is secured with two screws to a bracket welded on the dash panel just above the heater and air conditioner housing. The glove box is rolled down from the instrument panel for service access of the fuseblock module fuses. An adhesive-backed fuse layout map (Fig. 2)is located on the outside of the glove box bin (Fig. 3) to ensure proper fuse identification.

FRT WIPE (25A) 17	TURN SIG (10A) 13	AIRBAG (10A) 9	AIRBAG (10A) 5	PARK LPS SKIM (20A) 1
CIGAR LIGHTER (20A) 18	RADIO (10A) 14	CLUSTER (10A) 10	REAR WIPE (20A) 6	STOP LPS (20A) 2
SPARE FUSE (20A) 19	HBL SW (10A) 15	SOLENOIDS DRL (10A) 11	BACK-UP LP ABS (10A) 7	SUB-WOOFER SYSTEM (20A) 3
TRANS SEE OWNERS MANUAL (20A) 20	HEADLAMP AIM SW (10A) 16	PDC RELAYS SKIM (10A) 12	HEVAC (BLOWER FAN) (10A) 8	DOOR SW DEFEAT (10A) 4
56009391AL				

80d9ff50

Fig. 2 Fuseblock Label

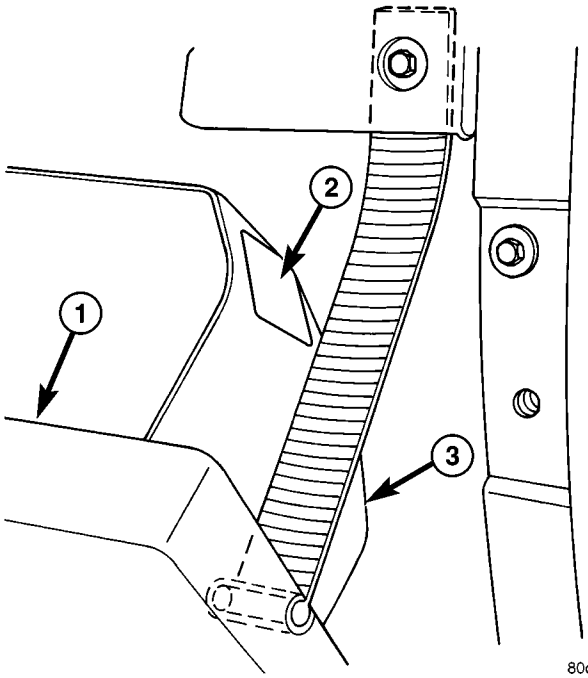


Fig. 3 Fuseblock Label Location

1 - GLOVE BOX DOOR
2 - FUSEBLOCK LABEL
3 - GLOVE BOX BIN

FUSE BLOCK (Continued)

The fuseblock module is integral to the body wire harness. If any internal circuit or the fuseblock module housing is faulty or damaged, the entire fuseblock module and body harness unit must be replaced.

OPERATION

All of the circuits entering and leaving the fuseblock module do so through the body wire harness. Internal connection of all of the fuseblock module circuits is accomplished by an intricate combination of hard wiring and bus bars. Refer to **Wiring Diagrams** for the location of complete fuseblock module circuit diagrams.

REMOVAL

The fuseblock module is serviced as a unit with the body wire harness. If any internal circuit of the fuseblock module or if the fuseblock module housing is faulty or damaged, the entire fuseblock module and the body wire harness unit must be replaced.

(1) Disconnect and isolate the battery negative cable.

(2) Remove the instrument panel assembly from the dash panel (Refer to 23 - BODY/INSTRUMENT PANEL/INSTRUMENT PANEL ASSEMBLY - REMOVAL).

(3) Disconnect each of the body wire harness connectors. Refer to **Connector Locations** in Wiring for the location of the body wire harness connector locations.

(4) Remove all of the fasteners that secure each of the body wire harness ground eyelets to the vehicle body and chassis components. Refer to **Connector Locations** in Wiring for the location of the body wire harness ground locations.

(5) Disengage each of the retainers that secure the body wire harness to the vehicle body and chassis components. Refer to **Connector Locations** in Wiring for the location of the body wire harness retainer locations.

(6) Remove the two screws that secure the fuseblock module to the bracket on the dash panel (Fig. 4).

(7) Remove the fuseblock module and the body wire harness from the dash panel as a unit.

INSTALLATION

The fuseblock module is serviced as a unit with the body wire harness. If any internal circuit of the fuseblock module or if the fuseblock module housing is faulty or damaged, the entire fuseblock module and the body wire harness unit must be replaced.

NOTE: If the fuseblock module is being replaced with a new unit, be certain to transfer each of the fuses from the faulty fuseblock module to the proper cavities of the replacement fuseblock module. Refer to Fuse/Fuse Block in the index of this service manual for the location of complete fuseblock module circuit diagrams and cavity assignments.

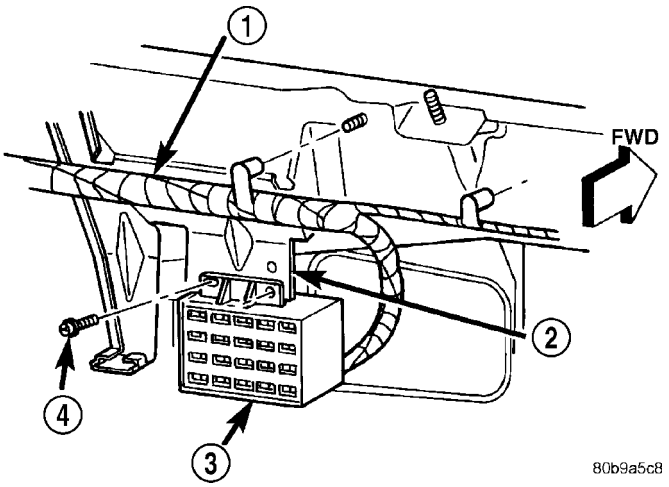


Fig. 4 Fuseblock Module Remove/Install

1 - BODY WIRE HARNESS
2 - BRACKET
3 - FUSEBLOCK MODULE
4 - SCREWS (2)

(1) Position the fuseblock module and the body wire harness onto the dash panel as a unit.

(2) Install and tighten the two screws that secure the fuseblock module to the bracket on the dash panel. Tighten the screws to 3.3 N·m (30 in. lbs.).

(3) Engage each of the retainers that secure the body wire harness to the vehicle body and chassis components. Refer to **Connector Locations** in Wiring for the location of the body wire harness retainer locations.

(4) Install all of the fasteners that secure each of the body wire harness ground eyelets to the vehicle body and chassis components. Refer to **Connector Locations** in Wiring for the location of the body wire harness ground eyelet locations.

(5) Reconnect each of the body wire harness connectors. Refer to **Connector Locations** in Wiring for the location of the body wire harness connector locations.

(6) Install the instrument panel assembly onto the dash panel (Refer to 23 - BODY/INSTRUMENT PANEL/INSTRUMENT PANEL ASSEMBLY - INSTALLATION).

(7) Reconnect the battery negative cable.

IOD FUSE

DESCRIPTION

All vehicles are equipped with an Ignition-Off Draw (IOD) fuse (Fig. 5) that is removed from its cavity in the Power Distribution Center (PDC) when the vehicle is shipped from the factory. Dealer personnel are to remove the IOD fuse from the storage location and install it into PDC fuse cavity 15 as part of the preparation procedures performed just prior to new vehicle delivery.

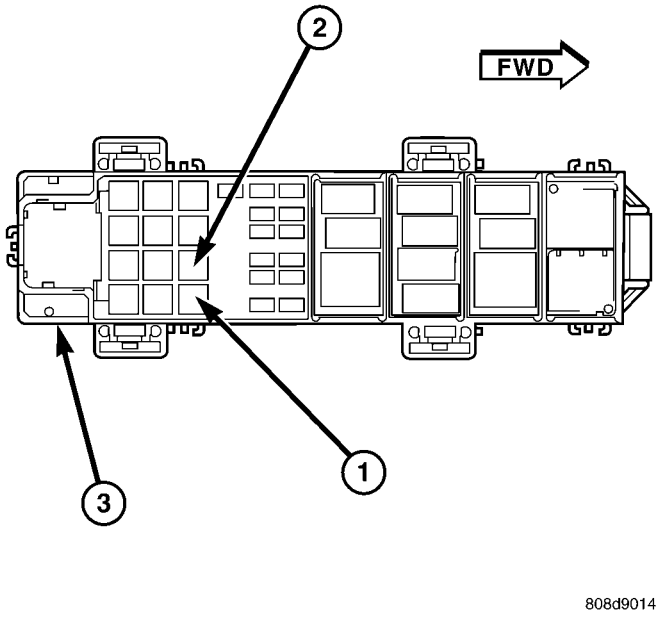


Fig. 5 Ignition-Off Draw Fuse

1 - IGNITION-OFF DRAW FUSE
2 - IGNITION-OFF DRAW FUSE STORAGE LOCATION
3 - POWER DISTRIBUTION CENTER

The PDC has a molded plastic cover that can be unlatched and opened to provide service access to all of the fuses and relays in the PDC. An integral latch and hinge tabs are molded into the PDC cover for easy removal. A fuse layout map is integral to the underside of the PDC cover to ensure proper fuse and relay identification. The IOD fuse is a 50 ampere cartridge-type fuse and, when removed, it is stored in the empty fuse cavity 11 of the PDC.

OPERATION

The term ignition-off draw identifies a normal condition where power is being drained from the battery with the ignition switch in the Off position. The IOD fuse feeds the memory and sleep mode functions for many of the electronic modules in the vehicle as well as various other accessories that require battery current when the ignition switch is in the Off position, including the clock. The only reason the IOD fuse is removed is to reduce the normal IOD of the vehicle electrical system during new vehicle transportation and pre-delivery storage to reduce battery depletion, while still allowing vehicle operation so that the vehicle can be loaded, unloaded and moved as needed by both vehicle transportation company and dealer personnel.

The IOD fuse is removed from PDC fuse cavity #15 when the vehicle is shipped from the assembly plant. Dealer personnel must install the IOD fuse when the vehicle is being prepared for delivery in order to restore full electrical system operation. Once the vehicle is prepared for delivery, the IOD function of this fuse becomes transparent and the fuse that has been assigned the IOD designation becomes only another Fused B(+) circuit fuse. The IOD fuse serves no useful purpose to the dealer technician in the service or diagnosis of any vehicle system or condition, other than the same purpose as that of any other standard circuit protection device.

The IOD fuse can be used by the vehicle owner as a convenient means of reducing battery depletion when a vehicle is to be stored for periods not to exceed about thirty days. However, it must be remembered that removing the IOD fuse will not eliminate IOD, but only reduce this normal condition. If a vehicle will be stored for more than about thirty days, the battery negative cable should be disconnected to eliminate normal IOD; and, the battery should be tested and recharged at regular intervals during the vehicle storage period to prevent the battery from becoming discharged or damaged.

REMOVAL

The Ignition-Off Draw (IOD) fuse is removed from Power Distribution Center (PDC) fuse cavity #15 when the vehicle is shipped from the assembly plant. Dealer personnel must install the IOD fuse when the vehicle is being prepared for delivery in order to restore full electrical system operation.

(1) Turn the ignition switch to the Off position.
(2) Unlatch and open the cover from the PDC.
(3) Remove the IOD fuse from fuse cavity #15 of the PDC.
(4) Store the removed IOD fuse by inserting the terminal blades of the fuse into the empty fuse cavity #11 of the PDC.
(5) Close and latch the cover onto the PDC.

INSTALLATION

The Ignition-Off Draw (IOD) fuse is removed from Power Distribution Center (PDC) fuse cavity #15 when the vehicle is shipped from the assembly plant. Dealer personnel must install the IOD fuse when the

IOD FUSE (Continued)

vehicle is being prepared for delivery in order to restore full electrical system operation.

(1) Turn the ignition switch to the Off position.

(2) Unlatch and open the cover from the PDC.

(3) Remove the stored IOD fuse from fuse cavity #11 of the PDC.

(4) Align the terminal blades of the IOD fuse with the terminal receptacles in fuse cavity #15 of the PDC.

(5) Use a thumb to press the IOD fuse firmly down into PDC fuse cavity #15.

(6) Close and latch the cover onto the PDC.

MICRO-RELAY

DESCRIPTION

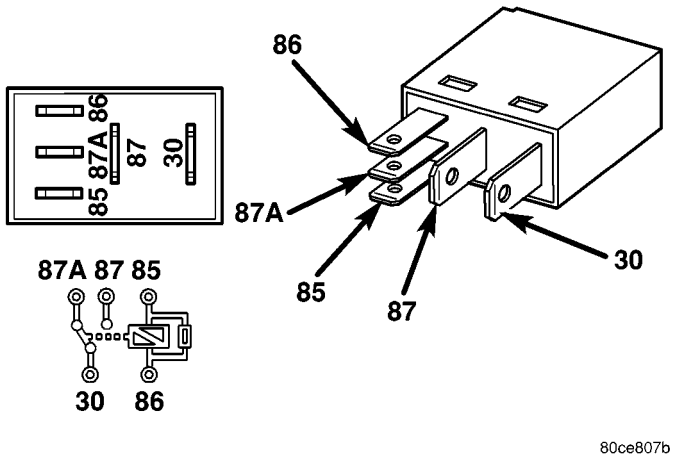


Fig. 6 ISO Micro Relay

30 - COMMON FEED
85 - COIL GROUND
86 - COIL BATTERY
87 - NORMALLY OPEN
87A - NORMALLY CLOSED

A micro-relay is a conventional International Standards Organization (ISO) micro relay (Fig. 6). Relays conforming to the ISO specifications have common physical dimensions, current capacities, terminal patterns, and terminal functions. The relay is contained within a small, rectangular, molded plastic housing and is connected to all of the required inputs and outputs by five integral male spade-type terminals that extend from the bottom of the relay base.

Relays cannot be adjusted or repaired and, if faulty or damaged, the unit must be replaced.

OPERATION

A micro-relay is an electromechanical switch that uses a low current input from one source to control a high current output to another device. The movable common feed contact point is held against the fixed normally closed contact point by spring pressure. When the relay coil is energized, an electromagnetic field is produced by the coil windings. This electromagnetic field draws the movable relay contact point away from the fixed normally closed contact point, and holds it against the fixed normally open contact point. When the relay coil is de-energized, spring pressure returns the movable contact point back against the fixed normally closed contact point. A resistor is connected in parallel with the relay coil in the relay, and helps to dissipate voltage spikes and electromagnetic interference that can be generated as the electromagnetic field of the relay coil collapses.

DIAGNOSIS AND TESTING - MICRO-RELAY

(1) Remove the relay from its mounting location.

(2) A relay in the de-energized position should have continuity between terminals 87A and 30, and no continuity between terminals 87 and 30 (Fig. 6). If OK, go to Step 3. If not OK, replace the faulty relay.

(3) Resistance between terminals 85 and 86 (electromagnet) should be 67.5 - 82.5 ohms. If OK, go to Step 4. If not OK, replace the faulty relay.

(4) Connect a battery to terminals 85 and 86. There should now be continuity between terminals 30 and 87, and no continuity between terminals 87A and 30. If OK, reinstall the relay and use a DRBIII® scan tool to perform further testing. Refer to the appropriate diagnostic information.

Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

REMOVAL

(1) Disconnect and isolate the negative battery cable.

(2) Remove the relay by grasping it firmly and pulling it straight out from its receptacle. A slight back and fourth rocking motion may help the removal process.

INSTALLATION

(1) Align the micro-relay terminals with the terminal cavities in the receptacle.

(2) Push firmly and evenly on the top of the relay until the terminals are fully seated in the terminal cavities in the receptacle.

(3) Connect the battery negative cable.

POWER DISTRIBUTION CENTER

DESCRIPTION

All of the electrical current distributed throughout this vehicle is directed through the standard equipment Power Distribution Center (PDC) (Fig. 7). The molded plastic PDC housing is located on the right side of the engine compartment, forward of the battery on the top of the right front inner fender shield. The PDC houses up to fifteen maxi-type cartridge fuses, which replace all in-line fusible links. The PDC also houses up to thirteen blade-type mini fuses, and up to twelve International Standards Organization (ISO) relays (four standard-type and eight micro-type).

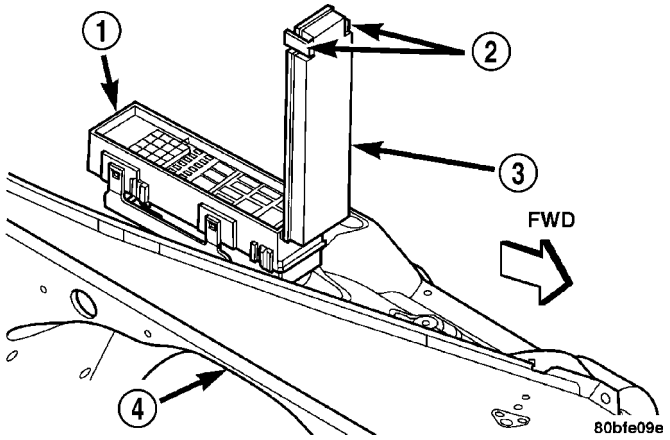


Fig. 7 Power Distribution Center Location

1 - POWER DISTRIBUTION CENTER (PDC)
2 - LATCHES
3 - COVER
4 - RIGHT FRONT FENDER

The PDC housing is secured in the engine compartment at four points. Integral mounts on both sides of the PDC housing engage and latch to stanchions that are integral to the stamped steel PDC bracket. The PDC bracket is secured to the top of the right front inner fender shield with three screws, two at the front of the bracket and one at the rear. The PDC is integral to the dash wire harness, which exits from the bottom of the PDC housing. The PDC housing has a molded plastic cover that includes two integral latches at the rear and pivot hooks at the front that snap over a hinge pin on the front of the PDC housing. The PDC cover is easily opened or removed for service access and has a convenient fuse and relay layout map integral to the inside surface of the cover to ensure proper component identification.

The PDC cover, the PDC housing lower cover, the PDC relay wedges, the PDC relay cassettes and the PDC B(+) terminal stud module are available for service replacement. The PDC main housing unit, the fuse wedges and the bus bars cannot be repaired and are only serviced as a unit with the dash wire harness. If the PDC main housing unit, fuse wedges or the bus bars are faulty or damaged, the dash wire harness unit must be replaced.

OPERATION

All of the current from the battery and the generator output enters the PDC through two cables and a single two-holed eyelet that is secured with nuts to the two PDC B(+) terminal studs near the back of the PDC housing. The molded plastic PDC cover is unlatched and opened to access the battery and generator output connection B(+) terminal studs. Internal connection of all of the PDC circuits is accomplished by an intricate combination of hard wiring and bus bars. Refer to the**Wiring** section of this service manual for wiring diagrams.

REMOVAL

The Power Distribution Center (PDC) main housing unit, the PDC fuse wedges and the PDC bus bars cannot be repaired and are only serviced as a unit with the dash wire harness. If the PDC main housing unit, the fuse wedges or the bus bars are faulty or damaged, the entire PDC and dash wire harness unit must be replaced.

(1) Disconnect and isolate the battery negative cable.

(2) Disconnect each of the dash wire harness connectors. Refer to **Connector Locations** in the Wiring section of this service manual for the location of the dash wire harness connector locations.

(3) Remove all of the fasteners that secure each of the dash wire harness ground eyelets to the vehicle body and chassis components. Refer to **Connector Locations** in the Wiring section of this service manual for the location of the ground eyelet locations.

(4) Disengage each of the retainers that secure the dash wire harness to the vehicle body and chassis components. Refer to **Connector Locations** in the Wiring section of this service manual for the location of the dash wire harness retainer locations.

POWER DISTRIBUTION CENTER (Continued)

(5) Unlatch and open the PDC cover (Fig. 8).

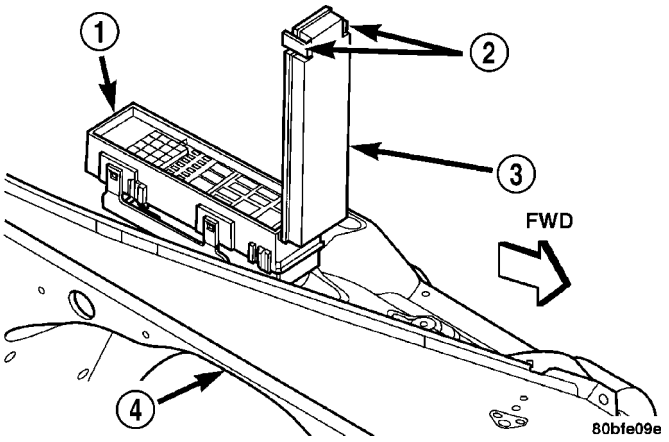


Fig. 8 Power Distribution Center

1 - POWER DISTRIBUTION CENTER (PDC)
2 - LATCHES
3 - COVER
4 - RIGHT FRONT FENDER

(6) Remove the two nuts that secure the eyelet terminal of the battery positive cable take out and the engine wire harness generator output take out to the PDC B(+) terminal studs.

(7) Remove the battery positive cable take out and the engine wire harness generator output take out eyelet terminal from the B(+) terminal studs.

(8) Disengage the latches on the PDC housing mounts from the tabs on the PDC bracket stanchions, and pull the PDC housing upward to disengage the mounts from the stanchions of the bracket.

(9) Remove the PDC and the dash wire harness from the engine compartment as a unit.

DISASSEMBLY

POWER DISTRIBUTION CENTER DISASSEMBLY

The Power Distribution Center (PDC) cover, the PDC housing lower cover, the PDC relay wedges, the PDC relay cassettes and the PDC B(+) terminal stud module are available for service replacement. The PDC cover can be simply unlatched and removed from the PDC housing without the PDC being removed or disassembled. Service of the remaining PDC components requires that the PDC be removed from its mounting and disassembled. Refer to **Wiring Repair** in the index of this service manual for the location of the proper wiring repair procedures.

PDC HOUSING LOWER COVER REMOVAL

(1) Disconnect and isolate the battery negative cable.

(2) Unlatch and remove the cover from the PDC.

(3) Unlatch and remove the B(+) terminal stud cover from the PDC.

(4) Remove the two nuts that secure the eyelet terminal to the two B(+) terminal studs of the PDC.

(5) Remove the eyelet terminal from the PDC B(+) terminal studs.

(6) Disengage the latches on the PDC mounts from the tabs on the PDC bracket stanchions, and pull the PDC housing upward to disengage the mounts from the stanchions of the bracket.

(7) Using a trim stick or another suitable wide flat-bladed tool, gently pry the latches on each side and one end of the PDC housing that secure the housing lower cover to the PDC and remove the housing lower cover (Fig. 9).

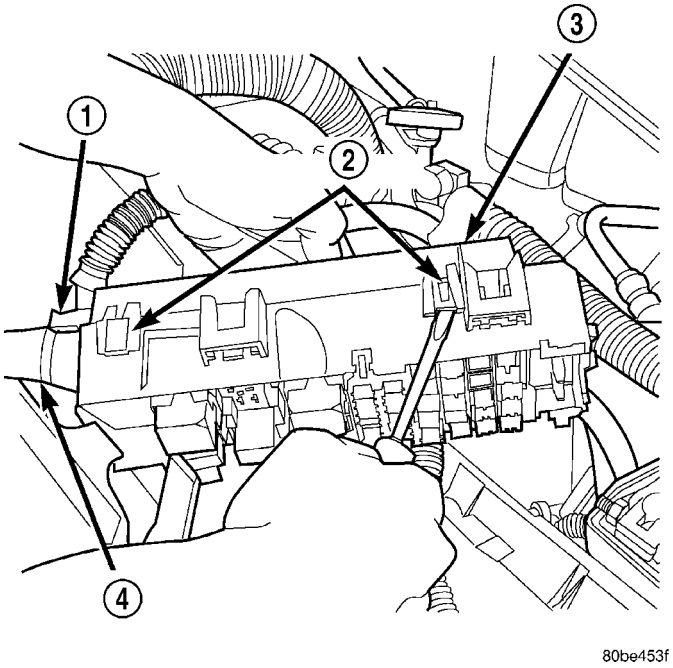


Fig. 9 PDC Housing Lower Cover Remove/Install - Typical

1 - THROUGH FORMATION
2 - LATCHES (5)
3 - PDC HOUSING LOWER COVER
4 - WIRE HARNESS

POWER DISTRIBUTION CENTER (Continued)

PDC B(+) TERMINAL MODULE REMOVAL

(1) Remove the PDC housing lower cover.

(2) From the top of the PDC housing, use a small screwdriver or a terminal pick tool (Special Tool Kit 6680) to release the two latches that secure the B(+) terminal module in the PDC (Fig. 10).

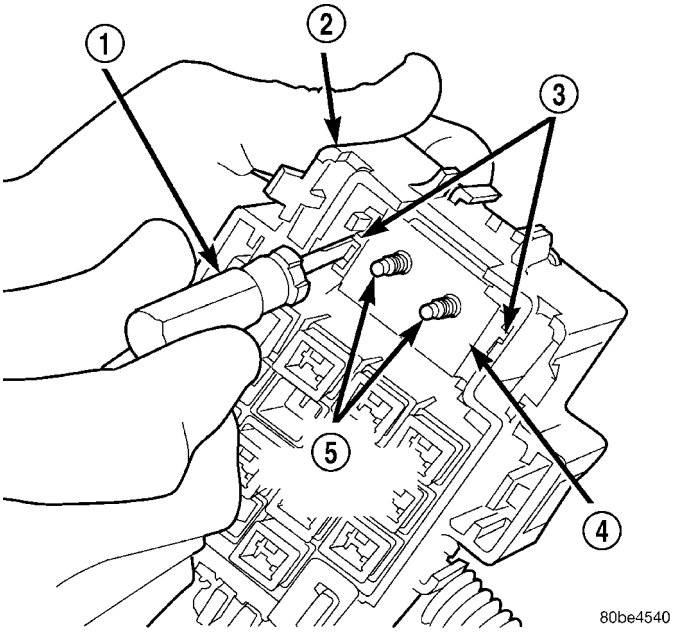


Fig. 10 PDC B(+) Terminal Module Latches

1 - FROM SPECIAL TOOL KIT 6680
2 - PDC HOUSING
3 - LATCHES
4 - BUS BAR
5 - B+ TERMINAL STUDS

(3) Gently and evenly press the two B(+) terminal studs down through the bus bar in the PDC.

(4) From the bottom of the PDC housing, remove the B(+) terminal module from the PDC (Fig. 11).

PDC RELAY WEDGE REMOVAL

(1) Remove the PDC housing lower cover.

(2) Remove each of the relays from the PDC relay wedge to be removed.

(3) From the bottom of the PDC housing, use a small screwdriver or a terminal pick tool (Special Tool Kit 6680) to release the two latches (yellow) that secure the relay wedge to the PDC relay cassette.

(4) From the top of the PDC housing, remove the relay wedge from the PDC relay cassette (Fig. 12).

PDC RELAY CASSETTE REMOVAL

(1) Remove the relay wedge from the PDC relay cassette to be removed.

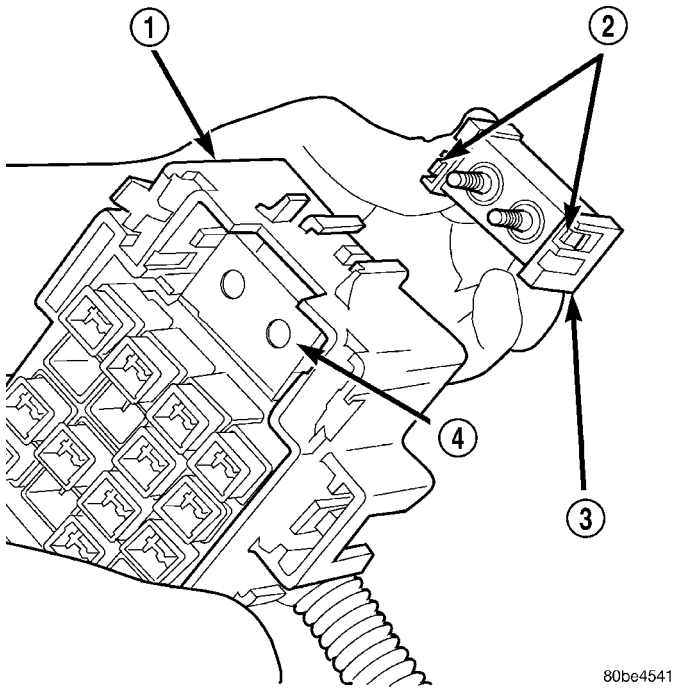


Fig. 11 PDC B(+) Terminal Module Remove/Install

1 - PDC HOUSING
2 - LATCHES
3 - B+ TERMINAL MODULE
4 - BUS BAR

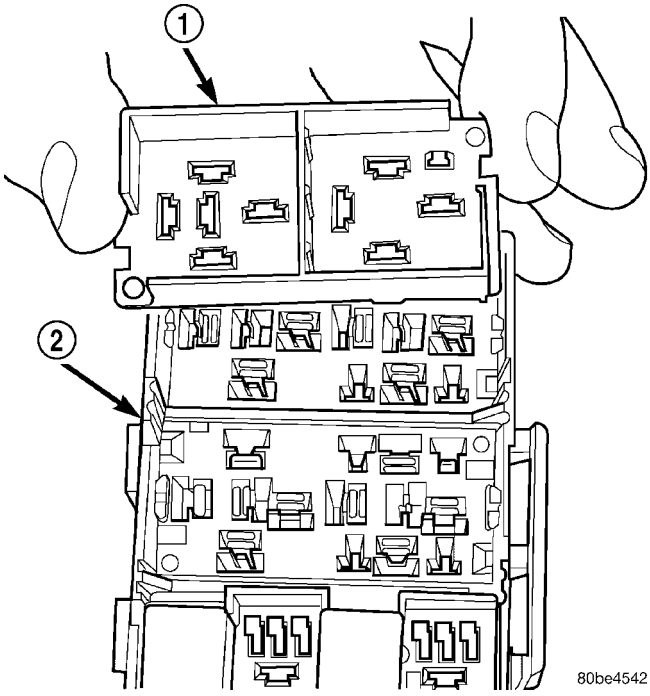


Fig. 12 PDC Relay Wedge Remove/Install - Typical

1 - RELAY WEDGE
2 - PDC HOUSING

POWER DISTRIBUTION CENTER (Continued)

NOTE: It may be necessary to remove relay cassettes that are not being serviced from the PDC housing in order to obtain sufficient clearance to service the faulty relay cassette. The same service procedure is repeated as necessary to remove each of the interfering relay wedges and relay cassettes from the PDC housing.

(2) From the top of the PDC housing, use a small screwdriver or a terminal pick tool (Special Tool Kit 6680) to release the two latches that secure the relay cassette in the PDC (Fig. 13).

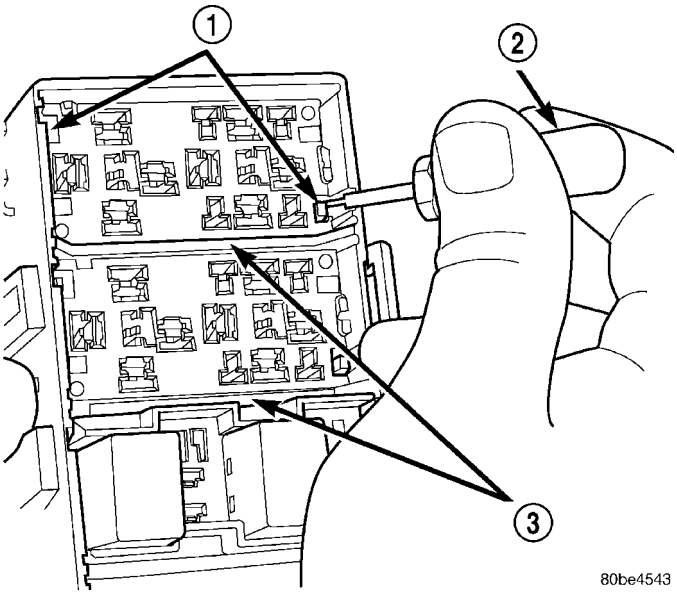


Fig. 13 PDC Relay Cassette Latches - Typical

1 - LATCHES
2 - FROM SPECIAL TOOL KIT 6680
3 - PDC RELAY CASSETTES

(3) Gently and evenly press the relay cassette down through the PDC housing.

(4) From the bottom of the PDC housing, remove the relay cassette from the PDC (Fig. 14).

CAUTION: Do not remove the wiring and terminals from the terminal cavities of the faulty PDC relay cassette at this time. Refer to the Assembly procedure that follows for the proper procedures for transferring the wiring and terminals to the replacement PDC relay cassette.

ASSEMBLY

POWER DISTRIBUTION CENTER ASSEMBLY

PDC HOUSING LOWER COVER INSTALLATION

(1) Align the PDC housing lower cover to the bottom of the PDC.

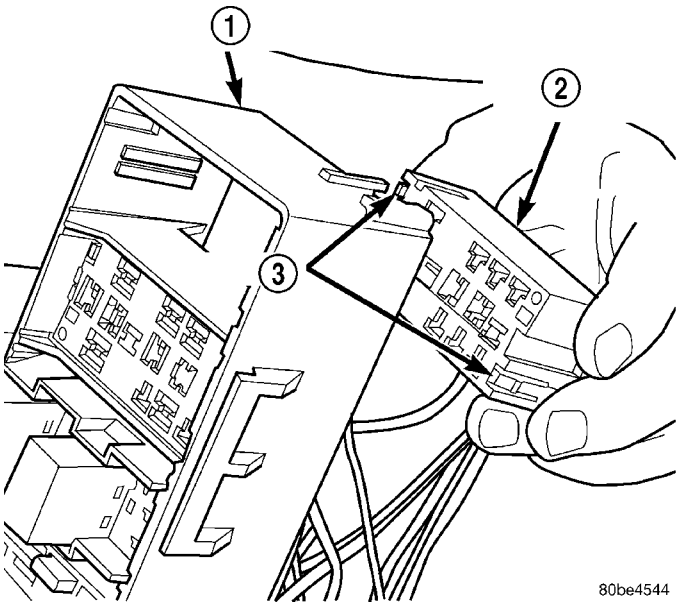


Fig. 14 PDC Relay Cassette Remove/Install - Typical

1 - PDC HOUSING
2 - PDC RELAY CASSETTE
3 - LATCHES

(2) Press the PDC housing lower cover gently and evenly onto the PDC until each of the latches that secure the cover to the PDC is fully engaged.

(3) Engage the mounts on the PDC housing with the stanchions of the PDC bracket and push the unit downward until the mount latches fully engage the mounting tabs on the PDC bracket.

(4) Install the eyelet terminal over the two PDC B(+) terminal studs.

(5) Install and tighten the two nuts that secure the eyelet terminal to the B(+) terminal studs. Tighten the nuts to 11.3 N·m (100 in. lbs.).

(6) Install the B(+) terminal stud cover onto the PDC.

(7) Install the cover onto the PDC.

(8) Reconnect the battery negative cable.

PDC B+ TERMINAL MODULE INSTALLATION

(1) From the bottom of the PDC housing, align and insert the B(+) terminal module into the PDC.

(2) From the bottom of the PDC housing, align and insert the two studs of the PDC B(+) terminal module through the bus bar in the PDC.

(3) From the bottom of the PDC housing, press the B(+) terminal module gently and evenly into the PDC until both of the latches are fully engaged.

(4) Install the PDC housing lower cover.

PDC RELAY WEDGE INSTALLATION

(1) From the top of the PDC housing, align and insert the PDC relay wedge latch arms into the correct cavities in the relay cassette.

POWER DISTRIBUTION CENTER (Continued)

(2) Gently and evenly press the PDC relay wedge down into the relay cassette until both of the latches are fully engaged.

(3) Install each of the removed relays into the proper cavities of the PDC relay wedge.

(4) Install the PDC housing lower cover.

PDC RELAY CASSETTE INSTALLATION

(1) Move the faulty PDC relay cassette with its wiring away from the bottom of the PDC housing far enough to allow the replacement relay cassette to be installed into the PDC.

(2) Using the faulty relay cassette as a guide, be certain that the replacement relay cassette is correctly oriented before installing it into the PDC housing.

(3) From the bottom of the PDC housing, align and insert the replacement relay cassette into the PDC. Press the relay cassette up into the PDC until both of the latches are fully engaged.

CAUTION: Proper care must be taken to be certain that the wiring and terminals from the faulty PDC relay cassette are installed in the correct terminal cavities of the replacement relay cassette. To prevent mistakes it is recommended that the wiring and terminals be removed from the faulty relay cassette one cavity at a time, repaired or spliced as necessary, then installed securely into the correct cavity of the replacement relay cassette. If you are not absolutely certain into which cavity a terminal should be installed, refer to Power Distribution in the index of this service manual for the location of complete PDC wiring diagrams.

(4) While pulling gently on the wire from the bottom of the faulty PDC relay cassette, use a terminal pick tool (Special Tool Kit 6680) from the top of the relay cassette to release the latch that secures the terminal in the relay cassette terminal cavity (Fig. 15).

(5) From the bottom of the faulty PDC relay cassette, remove the wire and terminal from the relay cassette terminal cavity.

(6) Make all necessary repairs and splices to the wire for the removed terminal. Refer to **Wiring Repair** in the index of this service manual for the location of the proper wiring repair procedures.

(7) From the bottom of the PDC housing, align and insert the removed wire and terminal into the correct terminal cavity of the replacement relay cassette. Push the wire and terminal up into the relay cassette terminal cavity until it is fully engaged by the latch.

(8) Repeat Step 4, Step 5, Step 6 and Step 7 one wire and terminal at a time until each of the wires and terminals have been transferred from the faulty PDC relay cassette into the replacement relay cassette.

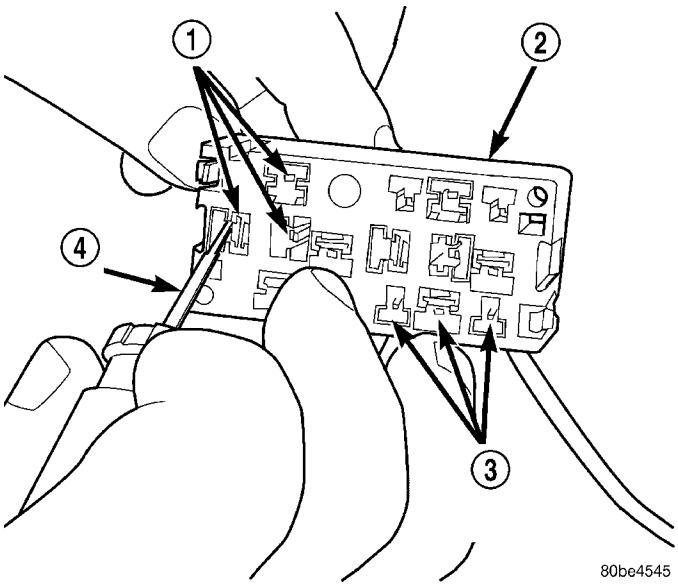


Fig. 15 PDC Relay Cassette Terminal Remove/Install

1 - TERMINAL CAVITIES
2 - PDC RELAY CASSETTE
3 - TERMINAL LATCHES
4 - FROM SPECIAL TOOL KIT 6680

(9) Install the PDC relay wedge into the replacement PDC relay cassette.

INSTALLATION

The Power Distribution Center (PDC) main housing unit, the PDC fuse wedges and the PDC bus bars cannot be repaired and are only serviced as a unit with the dash wire harness. If the PDC main housing unit, the fuse wedges or the bus bars are faulty or damaged, the entire PDC and dash wire harness unit must be replaced.

NOTE: If the PDC is being replaced with a new unit, be certain to transfer each of the fuses and relays that have not been included with the replacement PDC from the faulty PDC to the proper cavities of the replacement unit. Refer to Power Distribution in the index of this service manual for the location of complete PDC wiring diagrams and cavity assignments.

(1) Position the PDC and the dash wire harness unit in the engine compartment.

(2) Engage the mounts on the PDC housing with the stanchions of the PDC bracket and push the unit downward until the mount latches engage the mounting tabs on the PDC bracket.

(3) Install the eyelet terminal of the battery positive cable take out and the engine wire harness gen-

POWER DISTRIBUTION CENTER (Continued)

erator output take out onto the PDC B(+) terminal studs.

(4) Install and tighten the two nuts that secure the eyelet terminal of the battery positive cable take out and the engine wire harness generator output take out to the B(+) terminal studs. Tighten the nuts to 11.3 N·m (100 in. lbs.).

(5) Engage the tabs on the lower edge of the B(+) terminal stud cover in the slots on the back of the PDC housing, then engage the latch on the top of the cover with the latch receptacle on the PDC housing.

(6) Engage each of the retainers that secure the dash wire harness to the vehicle body and chassis components. Refer to **Connector Locations** in the index of this service manual for the location of more information on the dash wire harness retainer locations.

(7) Install all of the fasteners that secure each of the dash wire harness ground eyelets to the vehicle body and chassis components. Refer to **Connector Locations** in the index of this service manual for the location of more information on the ground eyelet locations.

(8) Reconnect each of the dash wire harness connectors. Refer to **Connector Locations** in the index of this service manual for the location of more information on the dash wire harness connector locations.

(9) Reconnect the battery negative cable.

POWER DISTRIBUTION CENTER SUPPORT BRACKET

REMOVAL

(1) Disconnect and isolate the battery negative cable.

(2) Remove the Power Distribution Center (PDC) from the PDC bracket (Refer to 8 - ELECTRICAL/ POWER DISTRIBUTION/POWER DISTRIBUTION CENTER - REMOVAL).

(3) Remove the three screws that secure the PDC bracket to the right front inner fender (Fig. 16).

(4) Remove the PDC bracket from the right front inner fender.

INSTALLATION

(1) Position the PDC bracket onto the right front inner fender.

(2) Install and tighten the three screws that secure the PDC mounting bracket to the right front inner fender. Tighten the screws to 3.9 N·m (35 in. lbs.).

(3) Install the PDC onto the PDC bracket (Refer to 8 - ELECTRICAL/POWER DISTRIBUTION/POWER DISTRIBUTION CENTER - INSTALLATION).

(4) Reconnect the battery negative cable.

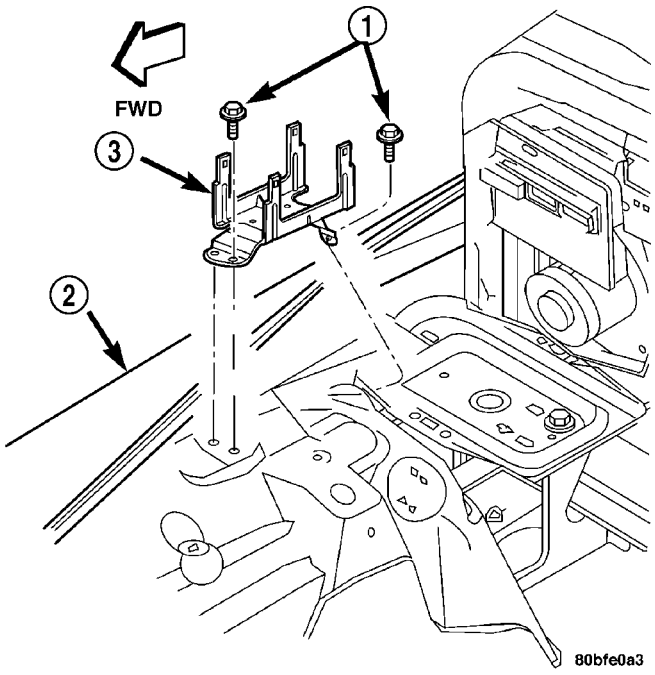


Fig. 16 PDC Bracket

1 - SCREWS (3)
2 - RIGHT FRONT FENDER
3 - PDC BRACKET

POWER OUTLET

DESCRIPTION

An instrument panel power outlet is standard factory installed equipment on this model. The power outlet is installed in the instrument panel accessory switch bezel, which is located near the bottom of the center bezel area below the heater and air conditioning controls. The power outlet base is secured by a snap fit within the instrument panel. A plastic protective cap snaps into the power outlet base when the power outlet is not being used, and hangs from the power outlet base mount by an integral bail strap while the power outlet is in use.

The power outlet receptacle unit and the accessory power outlet protective cap are available for service. The power outlet receptacle cannot be repaired and, if faulty or damaged, it must be replaced.

OPERATION

The power outlet base or receptacle shell is connected to ground, and an insulated contact in the bottom of the shell is connected to battery current. The power outlet receives battery voltage from a fuse in the Power Distribution Center at all times.

While the power outlet is very similar to a cigar lighter base unit, it does not include the two small spring-clip retainers inside the bottom of the receptacle shell that are used to secure the cigar lighter heating element to the insulated contact.

POWER OUTLET (Continued)

DIAGNOSIS AND TESTING - POWER OUTLET

For complete circuit diagrams, refer to **Power Outlet** in Wiring Diagrams.

(1) Check the fused B(+) fuse in the Power Distribution Center. If OK, go to Step 2. If not OK, repair the shorted circuit or component as required and replace the faulty fuse.

(2) Check for battery voltage at the fused B(+) fuse in the Power Distribution Center. If OK, go to Step 3. If not OK, repair the open fused B(+) circuit to the battery as required.

(3) Remove the plastic protective cap from the power outlet receptacle. Check for continuity between the inside circumference of the power outlet receptacle and a good ground. There should be continuity. If OK, go to Step 4. If not OK, go to Step 5.

(4) Check for battery voltage at the insulated contact located at the back of the power outlet receptacle. If not OK, go to Step 5.

(5) Disconnect and isolate the battery negative cable. Remove the power outlet receptacle from the instrument panel. Disconnect the wire harness connector from the power outlet receptacle. Check for continuity between the ground circuit cavity of the power outlet wire harness connector and a good ground. There should be continuity. If OK, go to Step 6. If not OK, repair the open ground circuit to ground as required.

(6) Connect the battery negative cable. Check for battery voltage at the fused B(+) circuit cavity of the power outlet wire harness connector. If OK, replace the faulty power outlet receptacle. If not OK, repair the open fused B(+) circuit to the Power Distribution Center fuse as required.

REMOVAL

(1) Disconnect and isolate the battery negative cable.

(2) Pull the cigar lighter knob and element out of the cigar lighter receptacle base, or unsnap the protective cap from the power outlet receptacle base.

(3) Look inside the cigar lighter or power outlet receptacle base and note the position of the rectangular retaining bosses of the mount that secures the receptacle base to the instrument panel (Fig. 17).

(4) Insert a pair of external snap ring pliers into the cigar lighter or power outlet receptacle base and engage the tips of the pliers with the retaining bosses of the mount.

(5) Squeeze the pliers to disengage the mount retaining bosses from the receptacle base and, using a gentle rocking motion, pull the pliers and the receptacle base out of the mount.

(6) Pull the receptacle base away from the instrument panel far enough to access the instrument panel wire harness connector.

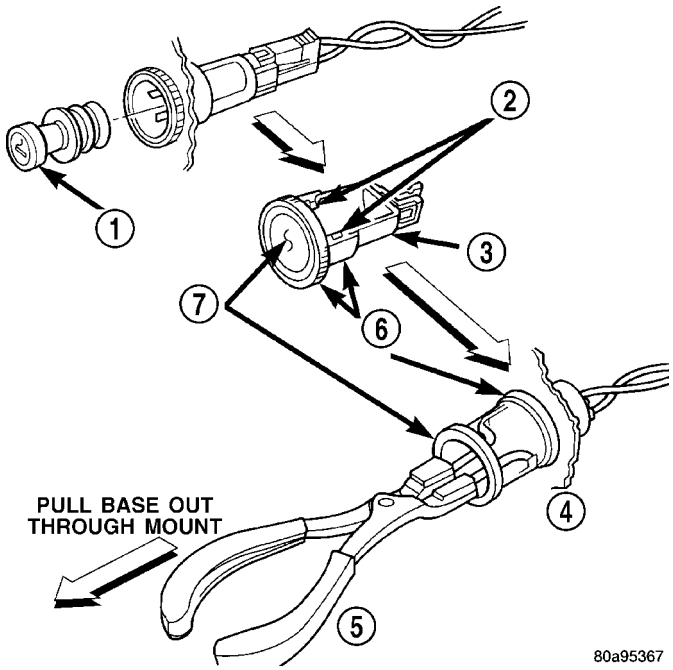


Fig. 17 Cigar Lighter and Power Outlet Remove/ Install

1 - KNOB AND ELEMENT
2 - RETAINING BOSSES-ENGAGE PLIERS HERE
3 - BASE
4 - PARTIALLY REMOVED
5 - EXTERNAL SNAP-RING PLIERS
6 - MOUNT
7 - BASE

(7) Disconnect the instrument panel wire harness connector from the cigar lighter or power outlet receptacle base connector receptacle.

(8) Remove the cigar lighter or power outlet mount from the instrument panel.

INSTALLATION

(1) Reconnect the instrument panel wire harness connector to the cigar lighter or power outlet receptacle base connector receptacle.

(2) Install the cigar lighter or power outlet mount into the instrument panel.

(3) Align the splines on the outside of the cigar lighter or power outlet receptacle base connector receptacle with the grooves on the inside of the mount.

(4) Press firmly on the cigar lighter or power outlet receptacle base until the retaining bosses of the mount are fully engaged in their receptacles.

(5) Install the cigar lighter knob and element into the cigar lighter receptacle base, or the protective cap into the power outlet receptacle base.

(6) Reconnect the battery negative cable.

RELAY

DESCRIPTION

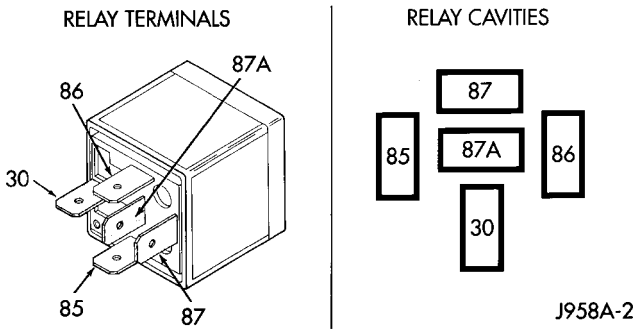


Fig. 18 ISO Relay

30 - COMMON FEED
85 - COIL GROUND
86 - COIL BATTERY
87 - NORMALLY OPEN
87A - NORMALLY CLOSED

A relay is an electromechanical device that switches fused battery current to a electrical component when the ignition switch is turned to the Accessory or Run positions, or when controlled by a electronic module. The relays are located in the junction block or power distribution center (Fig. 18).

The relay is a International Standards Organization (ISO) relay. Relays conforming to the ISO specifications have common physical dimensions, current capacities, terminal patterns, and terminal functions.

A relay cannot be repaired or adjusted and, if faulty or damaged, it must be replaced.

OPERATION

The ISO relay consists of an electromagnetic coil, a resistor and three (two fixed and one movable) electrical contacts. The movable (common feed) relay contact is held against one of the fixed contacts (normally closed) by spring pressure. When the electromagnetic coil is energized, it draws the movable contact away from the normally closed fixed contact, and holds it against the other (normally open) fixed contact.

When the electromagnetic coil is de-energized, spring pressure returns the movable contact to the normally closed position. The resistor is connected in parallel with the electromagnetic coil in the relay, and helps to dissipate voltage spikes that are produced when the coil is de-energized.

DIAGNOSIS AND TESTING - RELAY

The relays are located in the junction block or power distribution center. For complete circuit diagrams, refer to **Wiring Diagrams**.

(1) Remove the relay from its mounting location.

(2) A relay in the de-energized position should have continuity between terminals 87A and 30, and no continuity between terminals 87 and 30. If OK, go to Step 3. If not OK, replace the faulty relay.

(3) Resistance between terminals 85 and 86 (electromagnet) should be 60.7 - 80.3 ohms. If OK, go to Step 4. If not OK, replace the faulty relay.

(4) Connect a battery to terminals 85 and 86. There should now be continuity between terminals 30 and 87, and no continuity between terminals 87A and 30. If OK, perform the Relay Circuit Test that follows. If not OK, replace the faulty relay.

DIAGNOSIS & TESTING - RELAY CIRCUIT TEST

(1) The relay common feed terminal cavity (30) of the junction block or power distribution center is connected to battery voltage and should be hot at all times. Check for battery voltage at the fused B(+) circuit cavity in the junction block receptacle for the relay. If OK, go to Step 2. If not OK, repair the fused B(+) circuit to the Power Distribution Center (PDC) fuse as required.

(2) The relay normally closed terminal (87A) is connected to terminal 30 in the de-energized position, but is not used for this application. Go to Step 3.

(3) The relay normally open terminal (87) is connected to the common feed terminal (30) in the energized position. This terminal supplies battery voltage to the fused B(+) fuse in the junction block that feeds the accessory when the relay is energized by the ignition switch. There should be continuity between the junction block cavity for relay terminal 87 and the fused B(+) fuse in the junction block at all times. If OK, go to Step 4. If not OK, repair the open fused B(+) circuit to the junction block fuse as required.

(4) The coil ground terminal (85) is connected to the electromagnet in the relay. It receives battery feed to energize the relay when the ignition switch is in the Accessory or Run positions. Turn the ignition switch to the On position. Check for battery voltage at the fused ignition switch output (acc/run) circuit cavity for relay terminal 85 in the junction block receptacle for the relay. If OK, go to Step 5. If not OK, repair the open fused ignition switch output (acc/run) circuit to the ignition switch as required.

(5) The coil battery terminal (86) is connected to the electromagnet in the relay. The junction block cavity for this terminal should have continuity to ground at all times. If not OK, repair the open ground circuit to ground as required.

RELAY (Continued)

REMOVAL

(1) Disconnect and isolate the negative battery cable.

(2) Remove the relay by grasping it firmly and pulling it straight out from its receptacle. A slight back and fourth rocking motion may help the removal process.

INSTALLATION

(1) Position the relay to the proper receptacle.

(2) Align the relay terminals with the terminal cavities in the receptacle.

(3) Push firmly and evenly on the top of the relay until the terminals are fully seated in the terminal cavities in the receptacle.

(4) Connect the negative battery cable.

ENGINE

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ENGINE 2.4L

DESCRIPTION

The 2.4 Liter (148 cu. in.) in-line four cylinder engine is a double over head camshaft with hydraulic lifters and four valve per cylinder design. The engine is free-wheeling; meaning it has provisions for piston-to-valve clearance. However valve-to-valve interference can occur, if camshafts are rotated independently.

The cylinders are numbered from front of the engine to the rear. The firing order is 1–3–4–2.

The engine identification number is located on the rear of the cylinder block (Fig. 1).

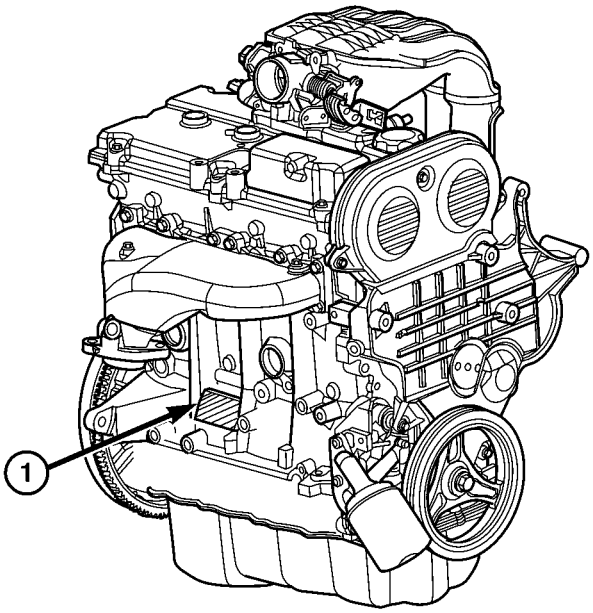


Fig. 1 ENGINE IDENTIFICATION

1 - ENGINE IDENTIFICATION

DIAGNOSIS AND TESTING

CYLINDER COMPRESSION PRESSURE TEST

The results of a cylinder compression pressure test can be utilized to diagnose several engine malfunctions.

Ensure the battery is completely charged and the engine starter motor is in good operating condition. Otherwise the indicated compression pressures may not be valid for diagnosis purposes.

(1) Check engine oil level and add oil if necessary.

(2) Drive the vehicle until engine reaches normal operating temperature. Select a route free from traffic and other forms of congestion, observe all traffic laws, and accelerate through the gears several times briskly.

(3) Remove all spark plugs from engine. As spark plugs are being removed, check electrodes for abnormal firing indicators fouled, hot, oily, etc. Record cylinder number of spark plug for future reference.

(4) Remove the Auto Shutdown (ASD) relay from the PDC.

(5) Be sure throttle blade is fully open during the compression check.

(6) Insert compression gage adaptor Special Tool 8116 or the equivalent, into the #1 spark plug hole in cylinder head. Connect the 0–500 psi (Blue) pressure transducer with cable adaptors to the DRBIII®.

(7) Crank engine until maximum pressure is reached on gage. Record this pressure as #1 cylinder pressure.

(8) Repeat the previous step for all remaining cylinders.

(9) Compression should not be less than 689 kPa (100 psi) and not vary more than 25 percent from cylinder to cylinder.

(10) If one or more cylinders have abnormally low compression pressures, repeat the compression test.

(11) If the same cylinder or cylinders repeat an abnormally low reading on the second compression test, it could indicate the existence of a problem in the cylinder in question. **The recommended compression pressures are to be used only as a guide to diagnosing engine problems. An engine should not be disassembled to determine the cause of low compression unless some malfunction is present.**

ENGINE 2.4L (Continued)

ENGINE DIAGNOSIS - PERFORMANCE

CONDITION	POSSIBLE CAUSE	CORRECTION
ENGINE WILL NOT START	1. Weak battery.	1. Test battery. Charge or replace as necessary. (Refer to 8 - ELECTRICAL/BATTERY SYSTEM - DIAGNOSIS AND TESTING)
	2. Corroded or loose battery connections.	2. Clean and tighten battery connections. Apply a coat of light mineral grease to terminals.
	3. Faulty starter.	3. Test starting system. (Refer to 8 - ELECTRICAL/STARTING - DIAGNOSIS AND TESTING)
	4. Faulty coil(s) or control unit.	4. Test and replace as needed. (Refer to Appropriate Diagnostic Information)
	5. Incorrect spark plug gap.	5. Set gap. (Refer to 8 - ELECTRICAL/IGNITION CONTROL - SPECIFICATIONS)
	6. Contamination in fuel system.	6. Clean system and replace fuel filter.
	7. Faulty fuel pump.	7. Test fuel pump and replace as needed. (Refer to Appropriate Diagnostic Information)
	8. Incorrect engine timing.	8. Check for a skipped timing belt/chain.
ENGINE STALLS OR IDLES ROUGH	1. Idle speed too low.	1. Test minimum air flow. (Refer to Appropriate Diagnostic Information)
	2. Incorrect fuel mixture.	2. (Refer to Appropriate Diagnostic Information)
	3. Intake manifold leakage.	3. Inspect intake manifold, manifold gasket, and vacuum hoses.
	4. Faulty ignition coil(s).	4. Test and replace as necessary. (Refer to Appropriate Diagnostic Information)

ENGINE 2.4L (Continued)

CONDITION	POSSIBLE CAUSE	CORRECTION
ENGINE LOSS OF POWER	1. Dirty or incorrectly gapped plugs.	1. Clean plugs and set gap.
	2. Contamination in fuel system.	2. Clean system and replace fuel filter.
	3. Faulty fuel pump.	3. Test and replace as necessary. (Refer to Appropriate Diagnostic Information)
	4. Incorrect valve timing.	4. Correct valve timing.
	5. Leaking cylinder head gasket.	5. Replace cylinder head gasket.
	6. Low compression.	6. Test compression of each cylinder.
	7. Burned, warped, or pitted valves.	7. Replace valves.
	8. Plugged or restricted exhaust system.	8. Perform exhaust restriction test. (Refer to 11 - EXHAUST SYSTEM - DIAGNOSIS AND TESTING) Install new parts, as necessary.
	9. Faulty ignition coil(s).	9. Test and replace as necessary. (Refer to Appropriate Diagnostic Information)
ENGINE MISSES ON ACCELERATION	1. Dirty or incorrectly gapped spark plugs.	1. Clean spark plugs and set gap.
	2. Contamination in Fuel System.	2. Clean fuel system and replace fuel filter.
	3. Burned, warped, or pitted valves.	3. Replace valves.
	4. Faulty ignition coil(s).	4. Test and replace as necessary. (Refer to Appropriate Diagnostic Information)
ENGINE MISSES AT HIGH SPEED	1. Dirty or incorrect spark plug gap.	1. Clean spark plugs and set gap.
	2. Faulty ignition coil(s).	2. Test and replace as necessary. (Refer to Appropriate Diagnostic Information)
	3. Dirty fuel injector(s).	Test and replace as necessary. (Refer to Appropriate Diagnostic Information)
	4. Contamination in fuel system.	4. Clean system and replace fuel filter.

CYLINDER COMBUSTION PRESSURE LEAKAGE TEST

The combustion pressure leakage test provides an accurate means for determining engine condition.

Combustion pressure leakage testing will detect:

- Exhaust and intake valve leaks (improper seating).
- Leaks between adjacent cylinders or into water jacket.
- Any causes for combustion/compression pressure loss.

WARNING: DO NOT REMOVE THE PRESSURE CAP WITH THE SYSTEM HOT AND UNDER PRESSURE BECAUSE SERIOUS BURNS FROM COOLANT CAN OCCUR.

Check the coolant level and fill as required. DO NOT install the pressure cap.

Start and operate the engine until it attains normal operating temperature, then turn the engine OFF.

Clean spark plug recesses with compressed air.

Remove the spark plugs.

Remove the oil filler cap.

Remove the air cleaner.

Calibrate the tester according to the manufacturer's instructions. The shop air source for testing should maintain 483 kPa (70 psi) minimum, 1,379 kPa (200 psi) maximum, with 552 kPa (80 psi) recommended.

Perform the test procedures on each cylinder according to the tester manufacturer's instructions. While testing, listen for pressurized air escaping through the throttle body, tailpipe and oil filler cap opening. Check for bubbles in the coolant.

All gauge pressure indications should be equal, with no more than 25% leakage per cylinder.

FOR EXAMPLE: At 552 kPa (80 psi) input pressure, a minimum of 414 kPa (60 psi) should be maintained in the cylinder.

ENGINE OIL LEAK INSPECTION

Begin with a thorough visual inspection of the engine, particularly at the area of the suspected leak. If an oil leak source is not readily identifiable, the following steps should be followed:

(1) Do not clean or degrease the engine at this time because some solvents may cause rubber to swell, temporarily stopping the leak.

(2) Add an oil soluble dye (use as recommended by manufacturer). Start the engine and let idle for approximately 15 minutes. Check the oil dipstick to make sure the dye is thoroughly mixed as indicated with a bright yellow color under a black light.

(3) Using a black light, inspect the entire engine for fluorescent dye, particularly at the suspected area of oil leak. If the oil leak is found and identified, repair as necessary.

(4) If dye is not observed, drive the vehicle at various speeds for approximately 24 km (15 miles), and repeat inspection.

(5) **If the oil leak source is not positively identified at this time**, proceed with the air leak detection test method as follows:

- Disconnect the fresh air hose (make-up air) at the cylinder head cover and plug or cap the nipple on the cover.
- Remove the PCV valve hose from the cylinder head cover. Cap or plug the PCV valve nipple on the cover.
- Attach an air hose with pressure gauge and regulator to the dipstick tube.

CAUTION: Do not subject the engine assembly to more than 20.6 kpa (3 PSI) of test pressure.

- Gradually apply air pressure from 1 psi to 2.5 psi maximum while applying soapy water at the suspected source. Adjust the regulator to the suitable test pressure that provides the best bubbles which will pinpoint the leak source. If the oil leak is detected and identified, repair per service manual procedures.
- If the leakage occurs at the crankshaft rear oil seal area, refer to the section, Inspection for Rear Seal Area Leak.

(6) If no leaks are detected, turn off the air supply. Remove the air hose, all plugs, and caps. Install the PCV valve and fresh air hose (make-up air). Proceed to next step.

(7) Clean the oil off the suspect oil leak area using a suitable solvent. Drive the vehicle at various speeds approximately 24 km (15 miles). Inspect the engine for signs of an oil leak by using a black light.

NOTE: If oil leakage is observed at the dipstick tube to block location; remove the tube, clean and reseal using Mopar® Stud & Bearing Mount (press fit tube applications only), and for O-ring style tubes, remove tube and replace the O-ring seal.

INSPECTION FOR REAR SEAL AREA LEAKS

Since it is sometimes difficult to determine the source of an oil leak in the rear seal area of the engine, a more involved inspection is necessary. The following steps should be followed to help pinpoint the source of the leak.

If the leakage occurs at the crankshaft rear oil seal area:

(1) Disconnect the battery.

(2) Raise the vehicle.

(3) Remove torque converter or clutch housing cover and inspect rear of block for evidence of oil. Use a black light to check for the oil leak. If a leak is present in this area, remove transmission for further inspection.

(a) Circular spray pattern generally indicates seal leakage or crankshaft damage.

(b) Where leakage tends to run straight down, possible causes are a porous block, oil gallery cup plug, bedplate to cylinder block mating surfaces and seal bore. See proper repair procedures for these items.

(4) If no leaks are detected, pressurize the crankcase as previously described.

CAUTION: Do not exceed 20.6 kPa (3 psi).

(5) If the leak is not detected, very slowly turn the crankshaft and watch for leakage. If a leak is detected between the crankshaft and seal while slowly turning the crankshaft, it is possible the crankshaft seal surface is damaged. The seal area on the crankshaft could have minor nicks or scratches that can be polished out with emery cloth.

ENGINE 2.4L (Continued)

CAUTION: Use extreme caution when crankshaft polishing is necessary to remove minor nicks and scratches. The crankshaft seal flange is especially machined to complement the function of the rear oil seal.

(6) For bubbles that remain steady with shaft rotation, no further inspection can be done until disassembled.

(7) After the oil leak root cause and appropriate corrective action have been identified, replace component(s) as necessary.

ENGINE DIAGNOSIS

Engine diagnosis is helpful in determining the causes of malfunctions not detected and remedied by routine maintenance.

These malfunctions may be classified as either mechanical (e.g., a strange noise), or performance (e.g., engine idles rough and stalls).

Refer to the Engine Mechanical and the Engine Performance diagnostic charts, for possible causes and corrections of malfunctions (Refer to 9 - ENGINE - DIAGNOSIS AND TESTING - MECHANICAL) (Refer to 9 - ENGINE - DIAGNOSIS AND TESTING - PERFORMANCE).

For fuel system diagnosis, (Refer to 14 - FUEL SYSTEM/FUEL DELIVERY - DIAGNOSIS AND TESTING).

Additional tests and diagnostic procedures may be necessary for specific engine malfunctions that cannot be isolated with the Service Diagnosis charts. Information concerning additional tests and diagnosis is provided within the following:

- Cylinder Compression Pressure Test
- Cylinder Combustion Pressure Leakage Test
- Engine Cylinder Head Gasket Failure Diagnosis
- Intake Manifold Leakage Diagnosis
- Lash Adjuster (Tappet) Noise Diagnosis
- Engine Oil Leak Inspection

ENGINE MECHANICAL

CONDITION	POSSIBLE CAUSES	CORRECTION
VALVETRAIN NOISE	1. High or low oil level in crankcase.	1. Check and correct engine oil level.
	2. Thin or diluted oil.	2. Change oil to correct viscosity.
	3. Thick oil	3. (a) Change engine oil and filter. (b) Run engine to operating temperature. (c) Change engine oil and filter again.
	4. Low oil pressure.	4. Check and correct engine oil level.
	5. Dirt in tappets/lash adjusters.	5. Replace rocker arm/hydraulic lash adjuster assembly.
	6. Worn rocker arms.	6. Inspect oil supply to rocker arms.
	7. Worn tappets/lash adjusters.	7. Install new rocker arm/hydraulic lash adjuster assembly.
	8. Worn valve guides.	8. Replace cylinder head assembly.
	9. Excessive runout of valve seats on valve faces.	9. Grind valve seats and valves.
	10. Missing adjuster pivot.	10. Replace rocker arm/hydraulic lash adjuster assembly.

ENGINE 2.4L (Continued)

CONDITION	POSSIBLE CAUSES	CORRECTION
CONNECTING ROD NOISE	1. Insufficient oil supply.	1. Check engine oil level.
	2. Low oil pressure.	2. Check engine oil level. Inspect oil pump relief valve and spring.
	3. Thin or diluted oil.	3. Change oil to correct viscosity.
	4. Thick oil	4. (a) Change engine oil and filter. (b) Run engine to operating temperature. (c) Change engine oil and filter again.
	5. Excessive bearing clearance.	5. Measure bearings for correct clearance. Repair as necessary.
	6. Connecting rod journal out-of-round.	6. Replace crankshaft or grind surface.
	7. Misaligned connecting rods.	7. Replace bent connecting rods.
MAIN BEARING NOISE	1. Insufficient oil supply.	1. Check engine oil level.
	2. Low oil pressure.	2. Check engine oil level. Inspect oil pump relief valve and spring.
	3. Thin or diluted oil.	3. Change oil to correct viscosity.
	4. Thick oil	4. (a) Change engine oil and filter. (b) Run engine to operating temperature. (c) Change engine oil and filter again.
	5. Excessive bearing clearance.	5. Measure bearings for correct clearance. Repair as necessary.
	6. Excessive end play.	6. Check thrust bearing for wear on flanges.
	7. Crankshaft journal out-of-round or worn.	7. Replace crankshaft or grind journals.
	8. Loose flywheel or torque converter.	8. Tighten to correct torque.

ENGINE 2.4L (Continued)

CONDITION	POSSIBLE CAUSES	CORRECTION
OIL PRESSURE DROP	1. Low oil level.	1. Check engine oil level.
	2. Faulty oil pressure sending unit.	2. Install new sending unit.
	3. Low oil pressure.	3. Check sending unit and main bearing oil clearance.
	4. Clogged oil filter.	4. Install new oil filter.
	5. Worn parts in oil pump.	5. Replace worn parts or pump.
	6. Thin or diluted oil.	6. Change oil to correct viscosity.
	7. Oil pump relief valve stuck.	7. Replace oil pump.
	8. Oil pump suction tube loose.	8. Remove oil pan and install new tube or clean, if necessary.
	9. Oil pump cover warped or cracked.	9. Install new oil pump.
	10. Excessive bearing clearance.	10. Measure bearings for correct clearance.
OIL LEAKS	1. Misaligned or deteriorated gaskets.	1. Replace gasket(s).
	2. Loose fastener, broken or porous metal part.	2. Tighten, repair or replace the part.
	3. Misaligned or deteriorated cup or threaded plug.	3. Replace as necessary.
OIL CONSUMPTION OR SPARK PLUGS FOULED	1. PCV system malfunction.	1. Check system and repair as necessary. (Refer to 25 - EMISSIONS CONTROL/ EVAPORATIVE EMISSIONS/PCV VALVE - DIAGNOSIS AND TESTING)
	2. Worn, scuffed or broken rings.	2. Hone cylinder bores. Install new rings.
	3. Carbon in oil ring slots.	3. Install new rings.
	4. Rings fitted too tightly in grooves.	4. Remove rings and check grooves. If groove is not proper width, replace piston.
	5. Worn valve guide(s).	5. Replace cylinder head assembly.
	6. Valve stem seal(s) worn or damaged.	6. Replace seal(s).

STANDARD PROCEDURE

REPAIR OF DAMAGED OR WORN THREADS

Damaged or worn threads (excluding spark plug and camshaft bearing cap attaching threads) can be repaired. Essentially, this repair consists of drilling out worn or damaged threads, tapping the hole with a special Heli-Coil Tap, (or equivalent) and installing an insert into the tapped hole. This brings the hole back to its original thread size.

CAUTION: Be sure that the tapped holes maintain the original center line.

Heli-Coil tools and inserts are readily available from automotive parts jobbers.

HYDROSTATIC LOCKED ENGINE

When an engine is suspected to be hydrostatically locked, regardless of what caused the problem, the following steps should be used.

ENGINE 2.4L (Continued)

CAUTION: DO NOT use starter motor to rotate the engine, severe damage may occur.

(1) Inspect air cleaner, induction system and intake manifold to insure system is dry and clear of foreign material.
(2) Remove negative battery cable.
(3) Place a shop towel around the spark plugs when removing them from the engine. This will catch any fluid that may possibly be in the cylinder under pressure.
(4) With all spark plugs removed, rotate engine crankshaft using a breaker bar and socket.
(5) Identify the fluid in the cylinder(s) (i.e., coolant, fuel, oil or other).
(6) Make sure all fluid has been removed from the cylinders. Inspect engine for damage (i.e., connecting rods, pistons, valves, etc.)
(7) Repair engine or components as necessary to prevent this problem from re-occurring.

CAUTION: Squirt approximately one teaspoon of oil into the cylinders, rotate engine to lubricate the cylinder walls to prevent damage on restart.

(8) Install new spark plugs.
(9) Drain engine oil and remove oil filter.
(10) Install a new oil filter.
(11) Fill engine with specified amount of approved oil.
(12) Connect negative battery cable.
(13) Start engine and check for any leaks.

ENGINE CORE AND OIL GALLERY PLUGS

Using a blunt tool such as a drift and a hammer, strike the bottom edge of the cup plug. With the cup plug rotated, grasp firmly with pliers or other suitable tool and remove plug (Fig. 2).

CAUTION: Do not drive cup plug into the casting as restricted cooling can result and cause serious engine problems.

Thoroughly clean inside of cup plug hole in cylinder block or head. Be sure to remove old sealer. Lightly coat inside of cup plug hole with Mopar® Stud and Bearing Mount. Make certain the new plug is cleaned of all oil or grease. Using proper drive plug, drive plug into hole so that the sharp edge of the plug is at least 0.5 mm (0.020 in.) inside the lead-in chamfer.

It is not necessary to wait for curing of the sealant. The cooling system can be refilled and the vehicle placed in service immediately.

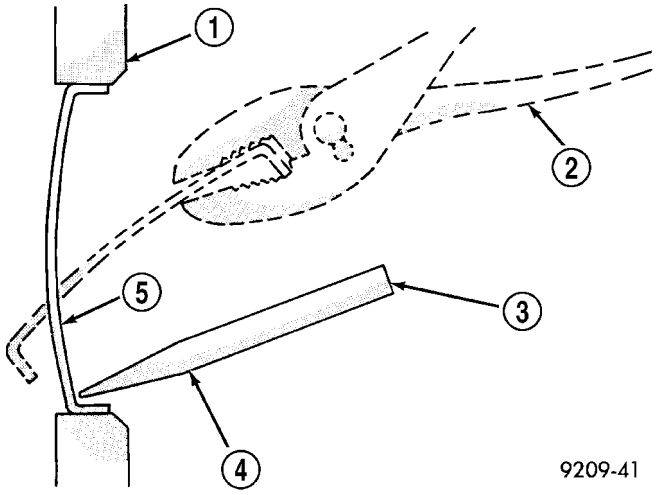


Fig. 2 Core Hole Plug Removal

1 - CYLINDER BLOCK
2 - REMOVE PLUG WITH PLIERS
3 - STRIKE HERE WITH HAMMER
4 - DRIFT PUNCH
5 - CUP PLUG

ENGINE GASKET SURFACE PREPARATION

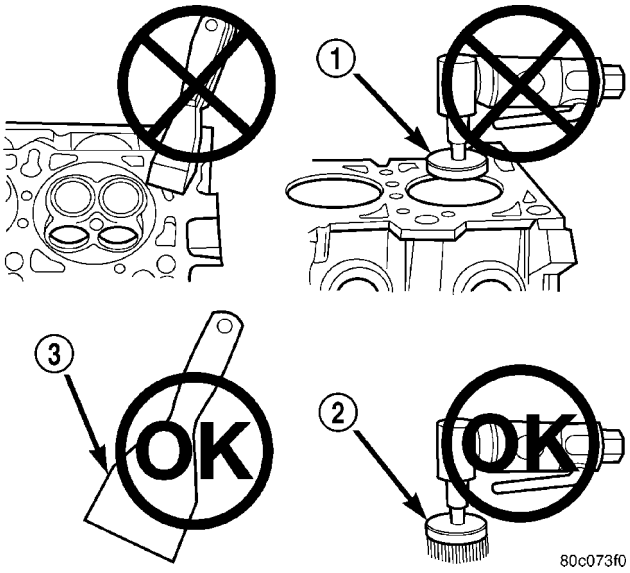


Fig. 3 Proper Tool Usage For Surface Preparation

1 - ABRASIVE PAD
2 - 3M ROLOC™ BRISTLE DISC
3 - PLASTIC/WOOD SCRAPER

To ensure engine gasket sealing, proper surface preparation must be performed, especially with the use of aluminum engine components and multi-layer steel cylinder head gaskets.

Never use the following to clean gasket surfaces: (Fig. 3)

- Metal scraper.
- Abrasive pad or paper to clean cylinder block and head.

- High speed power tool with an abrasive pad or a wire brush.

NOTE: Multi-Layer Steel (MLS) head gaskets require a scratch free sealing surface.

Only use the following for cleaning gasket surfaces: (Fig. 3)

- Solvent or a commercially available gasket remover
- Plastic or wood scraper.
- Drill motor with 3M Roloc™ Bristle Disc (white or yellow).

CAUTION: Excessive pressure or high RPM (beyond the recommended speed), can damage the sealing surfaces. The mild (white, 120 grit) bristle disc is recommended. If necessary, the medium (yellow, 80 grit) bristle disc may be used on cast iron surfaces with care.

MEASURING BEARING CLEARANCE USING PLASTIGAGE

Engine crankshaft bearing clearances can be determined by use of Plastigage or equivalent. The following is the recommended procedure for the use of Plastigage:

(1) Remove oil film from surface to be checked. Plastigage is soluble in oil.

(2) Place a piece of Plastigage across the entire width of the bearing shell in the cap approximately 6.35 mm (1/4 in.) off center and away from the oil holes (Fig. 4). (In addition, suspected areas can be checked by placing the Plastigage in the suspected area). Torque the bearing cap bolts of the bearing being checked to the proper specifications.

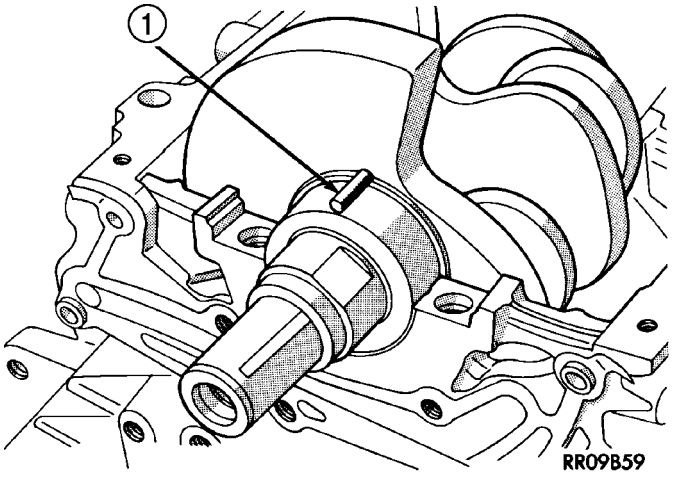


Fig. 4 Plastigage Placed in Lower Shell - Typical

1 - PLASTIGAGE

(3) Remove the bearing cap and compare the width of the flattened Plastigage with the metric scale provided on the package. Locate the band closest to the same width. This band shows the amount of clearance in thousandths of a millimeter. Differences in readings between the ends indicate the amount of taper present. Record all readings taken. Compare clearance measurements to specs found in engine specifications (Refer to 9 - ENGINE - SPECIFICATIONS). **Plastigage generally is accompanied by two scales. One scale is in inches, the other is a metric scale.**

NOTE: Plastigage is available in a variety of clearance ranges. Use the most appropriate range for the specifications you are checking.

(4) Install the proper crankshaft bearings to achieve the specified bearing clearances. (Refer to 9 - ENGINE/ENGINE BLOCK/CRANKSHAFT MAIN BEARINGS - STANDARD PROCEDURE) (Refer to 9 - ENGINE/ENGINE BLOCK/CONNECTING ROD BEARINGS - STANDARD PROCEDURE)

FORM-IN-PLACE GASKETS AND SEALERS

There are numerous places where form-in-place gaskets are used on the engine. Care must be taken when applying form-in-place gaskets to assure obtaining the desired results. **Do not use form-in-place gasket material unless specified.** Bead size, continuity, and location are of great importance. Too thin a bead can result in leakage while too much can result in spill-over which can break off and obstruct fluid feed lines. A continuous bead of the proper width is essential to obtain a leak-free gasket.

There are numerous types of form-in-place gasket materials that are used in the engine area. Mopar® Engine RTV GEN II, Mopar® ATF-RTV, and Mopar® Gasket Maker gasket materials, each have different properties and can not be used in place of the other.

MOPAR® ENGINE RTV GEN II is used to seal components exposed to engine oil. This material is a specially designed black silicone rubber RTV that retains adhesion and sealing properties when exposed to engine oil. Moisture in the air causes the material to cure. This material is available in three ounce tubes and has a shelf life of one year. After one year this material will not properly cure. Always inspect the package for the expiration date before use.

MOPAR® ATF RTV is a specifically designed black silicone rubber RTV that retains adhesion and sealing properties to seal components exposed to automatic transmission fluid, engine coolants, and moisture. This material is available in three ounce tubes and has a shelf life of one year. After one year this material will not properly cure. Always inspect the package for the expiration date before use.

ENGINE 2.4L (Continued)

MOPAR® GASKET MAKER is an anaerobic type gasket material. The material cures in the absence of air when squeezed between two metallic surfaces. It will not cure if left in the uncovered tube. The anaerobic material is for use between two machined surfaces. Do not use on flexible metal flanges.

MOPAR® BED PLATE SEALANT is a unique (green-in-color) anaerobic type gasket material that is specially made to seal the area between the bedplate and cylinder block without disturbing the bearing clearance or alignment of these components. The material cures slowly in the absence of air when torqued between two metallic surfaces, and will rapidly cure when heat is applied.

MOPAR® GASKET SEALANT is a slow drying, permanently soft sealer. This material is recommended for sealing threaded fittings and gaskets against leakage of oil and coolant. Can be used on threaded and machined parts under all temperatures. This material is used on engines with multilayer steel (MLS) cylinder head gaskets. This material also will prevent corrosion. Mopar® Gasket Sealant is available in a 13 oz. aerosol can or 4oz./16 oz. can w/applicator.

SEALER APPLICATION

Mopar® Gasket Maker material should be applied sparingly 1 mm (0.040 in.) diameter or less of sealant to one gasket surface. Be certain the material surrounds each mounting hole. Excess material can easily be wiped off. Components should be torqued in place within 15 minutes. The use of a locating dowel is recommended during assembly to prevent smearing material off the location.

Mopar® Engine RTV GEN II or ATF RTV gasket material should be applied in a continuous bead approximately 3 mm (0.120 in.) in diameter. All mounting holes must be circled. For corner sealing, a 3.17 or 6.35 mm (1/8 or 1/4 in.) drop is placed in the center of the gasket contact area. Uncured sealant may be removed with a shop towel. Components should be torqued in place while the sealant is still wet to the touch (within 10 minutes). The usage of a locating dowel is recommended during assembly to prevent smearing material off the location.

Mopar® Gasket Sealant in an aerosol can should be applied using a thin, even coat sprayed completely over both surfaces to be joined, and both sides of a gasket. Then proceed with assembly. Material in a can w/applicator can be brushed on evenly over the sealing surfaces. Material in an aerosol can should be used on engines with multi-layer steel gaskets.

REMOVAL - ENGINE ASSEMBLY

(1) Disconnect the battery negative cable.
(2) Remove hood. Mark hood hinge location for reinstallation.
(3) Remove air cleaner assembly.
(4) Remove radiator core support bracket.
(5) Remove fan shroud with electric fan assembly.
(6) Remove drive belt.

NOTE: It is NOT necessary to discharge the A/C system to remove the engine.

(7) Remove A/C compressor and secure away from engine with lines attached.
(8) Remove generator and secure away from engine.

NOTE: Do NOT remove the phenolic pulley from the P/S pump. It is not required for P/S pump removal.

(9) Remove power steering pump with lines attached and secure away from engine.
(10) Drain cooling system.
(11) Remove coolant bottle.
(12) Disconnect the heater hoses from the engine.
(13) Disconnect heater hoses from heater core and remove hose assembly.
(14) Disconnect throttle and speed control cables.
(15) Remove upper radiator hose from engine.
(16) Remove lower radiator hose from engine.
(17) Disconnect the engine to body ground straps at the left side of cowl.
(18) Disconnect the engine wiring harness at the following points:

- Intake air temperature (IAT) sensor
- Fuel Injectors
- Throttle Position (TPS) Switch
- Idle Air Control (IAC) Motor
- Engine Oil Pressure Switch
- Engine Coolant Temperature (ECT) Sensor
- Manifold Absolute Pressure MAP) Sensor
- Camshaft Position (CMP) Sensor
- Coil Over Plugs
- Crankshaft Position Sensor

(19) Remove coil over plugs.
(20) Release fuel rail pressure.
(21) Remove fuel rail and secure away from engine.
(22) Remove the PCV hose.
(23) Remove the breather hoses.
(24) Remove the vacuum hose for the power brake booster.
(25) Disconnect knock sensors.
(26) Secure the left and right engine wiring harnesses away from engine.
(27) Raise vehicle.
(28) Disconnect oxygen sensor wiring.

ENGINE 2.4L (Continued)

(29) Disconnect crankshaft position sensor.
(30) Disconnect the engine block heater power cable, if equipped.
(31) Disconnect the front propshaft at the front differential and secure out of way.
(32) Remove the starter.
(33) Remove the ground straps from the engine
(34) Disconnect the exhaust pipes at the manifold.
(35) Remove the structural cover, if equipped.
(36) Remove torque convertor bolts, and mark location for reassembly.
(37) Remove transmission bellhousing to engine bolts.
(38) Loosen left and right engine mount thru bolts.

NOTE: It is not necessary to completely remove engine mount thru bolts, for engine removal.

(39) Lower the vehicle.
(40) Support the transmission with a suitable jack.
(41) Connect a suitable engine hoist to the engine.

CAUTION: The 2.4L engine with manual transmissions, can be removed without removing the manual transmission. Use caution when attempting this procedure as the clearance is tight.

(42) Remove engine from vehicle.

INSTALLATION - ENGINE ASSEMBLY

(1) Position the engine in the vehicle.

CAUTION: Use caution when installing 2.4L engine into vehicle equipped with manual transmission, as clearance is tight.

(2) Install both left and right side engine mounts into the frame mounts.
(3) Raise the vehicle.
(4) Install the transmission bellhousing to engine mounting bolts. Tighten the bolts to 41 N·m (30 ft. lbs.).
(5) Tighten the engine mount thru bolts.
(6) Install the torque convertor bolts.
(7) Connect the ground straps on the left and right side of the engine.
(8) Install the starter.
(9) Connect the crankshaft position sensor.
(10) Install the engine block heater power cable, if equipped.

CAUTION: The structural cover requires a specific torque sequence. Failure to follow this sequence may cause severe damage to the cover.

(11) Install the structural cover.
(12) Install the exhaust pipe.
(13) Connect the oxygen sensors.
(14) Lower vehicle.
(15) Connect the knock sensors.
(16) Connect the engine to body ground straps.
(17) Install the power brake booster vacuum hose.
(18) Install the breather hoses.
(19) Install the PCV hose.
(20) Install the fuel rail.
(21) Install the coil over plugs.
(22) Reconnect the engine wiring harness at the following points:

- Intake air temperature (IAT) sensor
- Fuel Injectors
- Throttle Position (TPS) Switch
- Idle Air Control (IAC) Motor
- Engine Oil Pressure Switch
- Engine Coolant Temperature (ECT) Sensor
- Manifold Absolute Pressure MAP) Sensor
- Camshaft Position (CMP) Sensor
- Coil Over Plugs
- Crankshaft Position Sensor

(23) Connect lower radiator hose.
(24) Connect upper radiator hose.
(25) Connect throttle and speed control cables.
(26) Install the heater hose assembly.
(27) Install coolant recovery bottle.
(28) Install the power steering pump.
(29) Install the generator.
(30) Install the A/C compressor.
(31) Install the drive belt.
(32) Install the fan shroud with the electric fan assembly.
(33) Install the radiator core support bracket.
(34) Install the air cleaner assembly.
(35) Refill the engine cooling system.
(36) Install the hood.
(37) Check and fill engine oil.
(38) Connect the battery negative cable.
(39) Start the engine and check for leaks.

ENGINE 2.4L (Continued)

SPECIFICATIONS

SPECIFICATIONS - 2.4L ENGINE

GENERAL SPECIFICATIONS

DESCRIPTION	SPECIFICATION	
Type	In-Line OHV, DOHC	
Number of Cylinders	4	
Firing Order	1-3-4-2	
Compression Ratio	9.5:1	
Max. Variation Between Cylinders	25%	
	Metric	**Standard**
Displacement	2.4 Liters	148 cu. in.
Bore	87.5 mm	3.445 in.
Stroke	101.0 mm	3.976 in.
Compression Pressure	1172-1551 kPa	170-225 psi

CYLINDER BLOCK

DESCRIPTION	SPECIFICATIONS	
	Metric	**Standard**
Cylinder Bore Diameter	87.4924 - 87.5076 mm	3.4446 - 3.4452 in.
Out-of-Round (Max.)	0.051 mm	0.002 in.
Taper (Max.)	0.051 mm	0.002 in.

PISTONS

DESCRIPTION	SPECIFICATION	
	Metric	**Standard**
Piston Diameter	87.456 - 87.474 mm	3.4431 -3.4439 in.
Clearance @ 24.6 mm (0.551 in.) from bottom of skirt	0.018 - 0.0516 mm	0.0007 - 0.0020 in.
Weight	345 - 355 grams	12.17 - 12.52 oz.
Land Clearance (Diametrical)	0.563 - 0.621 mm	0.022 - 0.024 in.
Piston Length	66.25 mm	2.608 in.
Piston Ring Groove Depth No. 1	3.946 - 4.045mm	0.155 - 0.159 in.
Piston Ring Groove Depth No. 2	4.555 - 4.680 mm	0.179 - 0.184 in.
Piston Ring Groove Depth No. 3	4.108 - 4.220 mm	0.162 - 0.166 in.

PISTON PINS

DESCRIPTION	SPECIFICATION	
	Metric	**Standard**
Clearance in Piston	0.005 - 0.018 mm	0.0002 - 0.0008 in.
Clearance in Connecting Rod	Interference	
Diameter	21.998 - 22.003 mm	0.8660 - 0.8662 in.
End Play	None	
Length	72.75 - 73.25 mm	2.864 - 2.883 in.

ENGINE 2.4L (Continued)

PISTON RINGS

DESCRIPTION	SPECIFICATION	
	Metric	Standard
Ring Gap		
Top Compression Ring	0.25 - 0.51 mm	0.0098 - 0.020 in.
Wear Limit	0.8 mm	0.031 in.
2nd Compression Ring	0.23 - 0.48 mm	0.009 - 0.018 in.
Wear Limit	0.8 mm	0.031 in.
Oil Control Steel Rails	0.25 - 0.64 mm	0.0098 - 0.025 in.
Wear Limit	1.00 mm	0.039 in.
Compression Rings	0.030 - 0.080 mm	0.0011 - 0.0031 in.
Wear Limit	0.10 mm	0.004 in.
Ring Side Clearance - Oil Ring Pack	0.012 - 0.178 mm	0.0004 - 0.0070 in.
Ring Width - Compression Rings	1.47 - 1.50 mm	0.057 - 0.059 in.
Ring Width - Oil Ring Pack	2.72 - 2.88 mm	0.107 - 0.1133 in.

CONNECTING ROD

DESCRIPTION	SPECIFICATION	
	Metric	Standard
Bearing Clearance	0.025 - 0.071 mm	0.0009 - 0.0027 in.
Wear Limit	0.075 mm	0.003 in.
Bore Diameter - Piston Pin	20.96 - 20.98 mm	0.8252 - 0.8260 in.
Bore Diameter - Crankshaft End	53.007 - 52.993 mm	2.0868 - 2.0863 in.
Side Clearance	0.13 - 0.38 mm	0.005 - 0.015 in.
Wear Limit	0.40 mm	0.016 in.
Weight - Total (Less Bearing)	565.8 grams	19.96 oz.

CRANKSHAFT

DESCRIPTION	SPECIFICATION	
	Metric	Standard
Connecting Rod Journal Diameter	49.984 - 50.000 mm	1.968 - 1.9685 in.
Main Bearing Journal Diameter	59.992 - 60.008 mm	2.362 - 2.3625 in.
Journal Out-of-Round (Max.)	0.0035 mm	0.0003 in.
Journal Taper (Max.)	0.007 mm	0.0001 in.
End Play	0.09 - 0.24 mm	0.0035 - 0.0094 in.
Wear Limit	0.38 mm	0.015 in.
Main Bearing Diametrical Clearance	0.018 - 0.062 mm	0.0007 - 0.0024 in.

HYDRAULIC LASH ADJUSTER

DESCRIPTION	SPECIFICATION	
	Metric	Standard
Body Diameter	15.901 - 15.913 mm	0.626 - 0.6264 in.
Plunger Travel Minimum (Dry)	3.0 mm	0.118 in.

CYLINDER HEAD CAMSHAFT BEARING BORE DIAMETER

DSECRIPTION	SPECIFICATION	
	Metric	Standard
Journals No.1 - 6	26.020 - 26.041 mm	1.024 - 1.025 in.

ENGINE 2.4L (Continued)

CAMSHAFT

DESCRIPTION	SPECIFICATION	
	Metric	Standard
Journal Diameter No. 1 - 6	25.976 - 25.995 mm	1.022 - 1.023 in.
Bearing Clearance - Diametrical	0.025 - 0.065 mm	0.0009 - 0.0025 in.
End Play	0.05 - 0.17 mm	0.0019 - 0.0066 in.
Lift (Zero Lash)		
Intake	8.25 mm	0.324 in.
Exhaust	6.60 mm	0.259 in.
Intake Valve Timing*		
Closes (ABDC)	44.3°	
Opens (ATDC)	6.2°	
Duration	218.1°	
Exhaust Valve Timing*		
Closes (ATDC)	0.8°	
Opens (BBDC)	39.9°	
Duration	220.7°	
Valve Overlap	5.4°	

*
All reading in crankshaft degrees at 0.5 mm (0.019 in.) valve lift.

CYLINDER HEAD

DESCRIPTION	SPECIFICATION	
	Metric	Standard
Material	Cast Aluminum	
Gasket Thickness (Compressed)	0.71 mm	0.028 in.

VALVE SEAT

DESCRIPTION	SPECIFICATION	
	Metric	Standard
Angle	44.5 - 45°	
Seat Diameter - Intake	34.37 - 34.63 mm	1.353 - 1.363 in.
Seat Diameter - Exhaust	27.06 - 27.32 mm	1.065 - 1.075 in.
Runout (Max.)	0.05 mm	0.002 in.
Valve Seat Width - Intake and Exhaust	0.9 - 1.3 mm	0.035 - 0.051 in.
Service Limit - Intake	2.0 mm	0.079 in.
Service Limit - Exhaust	2.5 mm	0.098 in.

VALVE GUIDE

DESCRIPTION	SPECIFICATION	
	Metric	Standard
Diameter I.D.	5.975 - 6.000 mm	0.235 - 0.236 in.
Guide Bore Diameter	11.0 - 11.02 mm	0.4330 - 0.4338 in.
Guide Height (spring seat to guide tip)	13.25 - 13.75 mm	0.521 - 0.541 in.

ENGINE 2.4L (Continued)

VALVES

DESCRIPTION	SPECIFICATION	
	Metric	Standard
Face Angle - Intake and Exhaust	44.5 - 45°	
Head Diameter - Intake	34.67 - 34.93 mm	1.364 - 1.375 in.
Head Diameter - Exhaust	28.32 - 28.52 mm	1.114 - 1.122 in.
Valve Length (Overall)		
Intake	112.76 - 113.32 mm	4.439 - 4.461 in.
Exhaust	110.89 - 111.69 mm	4.365 - 4.397 in.
Valve Stem Diameter		
Intake	5.934 - 5.952 mm	0.2337 - 0.2344 in.
Exhaust	5.906 - 5.924 mm	0.2326 - 0.2333 in.

VALVE MARGIN

DESCRIPTION	SPECIFICATION	
	Metric	Standard
Intake	1.2 - 1.7 mm	0.047 - 0.066 in.
Service Limit	0.95 mm	.0037 in.
Exhaust	0.985 - 1.315 mm	0.038 - 0.051 in.
Service Limit	1.05 mm	.039 in.

VALVE STEM TIP HEIGHT

DESCRIPTION	SPECIFICATION	
	Metric	Standard
Intake	48.04 mm	1.891 in.
Exhaust	47.99 mm	1.889 in.

VALVE STEM TO GUIDE CLEARANCE

DESCRIPTION	SPECIFICATION	
	Metric	Standard
Intake	0.048 - 0.066 mm	0.0018 - 0.0025 in.
Max. Allowable	0.076 mm	0.003 in.
Service Limit	0.25 mm	0.010 in.
Exhaust	0.0736 - 0.094 mm	0.0029 - 0.0037 in.
Max. Allowable	0.101 mm	0.004 in.
Service Limit	0.25 mm	0.010 in.

VALVE SPRINGS

DESCRIPTION	SPECIFICATION	
	Metric	Standard
Free Length (Approx.)	49.2 mm	1.937 in.
Nominal Force (Valve Closed)	334 ±17 N @ 38.0 mm	75.08 ±3.8 lbs. @ 1.496 in.
Nominal Force (Valve Open)	598 ±30N @ 29.75 mm	134 ±6.7 lbs. @ 1.171 in.
Installed Height	38.00 mm	1.496 in.
Number of Coils	6.9	
Wire Diameter	3.61 mm	0.142 in

OIL PUMP

DESCRIPTION	SPECIFICATION	
	Metric	Standard
Clearance Over Rotors (Max.)	0.10 mm	0.004 in.
Cover Out-of-Flat (Max.)	0.025 mm	0.001 in.
Inner Rotor Thickness (Min.)	10.699 mm	0.421 in.
Outer Rotor Thickness (Min.)	10.699 mm	0.421 in.
Outer Rotor Clearance (Max.)	0.039 mm	0.015 in.
Outer Rotor Diameter (Min.)	85.924 mm	3.383 in.
Tip Clearance Between Rotors (Max.)	0.20 mm	0.008 in.

ENGINE 2.4L (Continued)

OIL PRESSURE

DESCRIPTION	SPECIFICATION	
	Metric	Standard
At Curb Idle Speed*	25 kPa	4 psi
At 3000 rpm	170 - 550 kPa	25 - 80 psi
CAUTION: *If pressure is ZERO at curb idle, DO NOT run engine at 3000 rpm.		

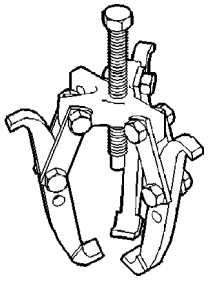
SPECIFICATIONS - TORQUE

DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Bolts, Balance Shaft Carrier to Block	54	40	-
Fastener - Double Ended, Balance Shaft Gear Cover	12	-	105
Bolt, Balance Shaft Sprocket	28	-	250
Bolts, Balance Shaft Chain Tensioner	12	-	105
Bolts, Balance Shaft Carrier Cover	12	-	105
Bolt, Camshaft Sprocket	115	85	-
Bolts, Connecting Rod Cap	27 +¼ turn	20 +¼ turn	-
Bolts, Crankshaft Main Bearing Cap/Bedplate			
M8 Bolts	28	-	250
M11 Bolts	75	55	-
Crankshaft Damper	136	100	-
Bolts, Cylinder Head	(Refer to 9 - ENGINE/ CYLINDER HEAD - INSTALLATION)		
Bolts, Cylinder Head Cover	12	-	105
Flex Plate to Crankshaft	95	70	-
Flywheel Mounting Bolts	81	60	-
Bolts, Engine Mount Bracket Right	61	45	-
Bolts, Engine Mounting	(Refer to 9 ENGINE/ ENGINE MOUNTING)		
Bolts, Exhaust Manifold to Cylinder Head	23	-	200
Bolts, Exhaust Manifold Heat Shield	12	-	105
Bolts, Intake Manifold - Lower	28	-	250
Oil Filter	20	15	-
Bolts, Oil Pan	12	-	105
Plug, Oil Pan Drain	27	20	-
Bolts, Oil Pump to Block	28	-	250
Bolts, Oil Pump Cover Plate	12	-	105
Bolt, Oil Pump Pick-up Tube	28	20	-
Cap - Oil Pump Relief Valve	41	30	-
Spark Plugs	18	13	-
Bolts, Timing Belt Covers			
- Front Covers to Rear Covers	12	-	105
- Rear Cover	12	-	105
Bolts, Timing Belt Tensioner Assembly	61	45	-

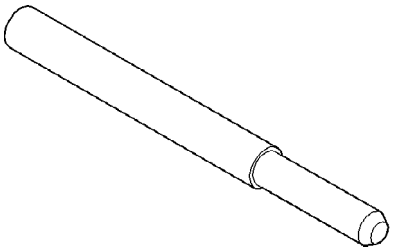
ENGINE 2.4L (Continued)

SPECIAL TOOLS

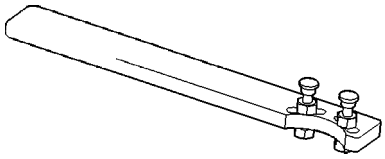
2.4L ENGINE



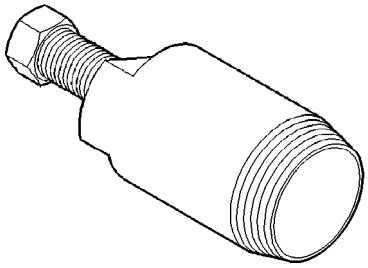
Puller 1026



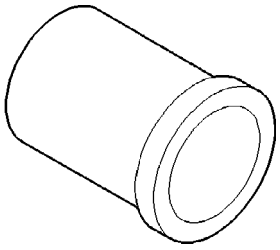
Crankshaft Damper Removal Insert 6827-A



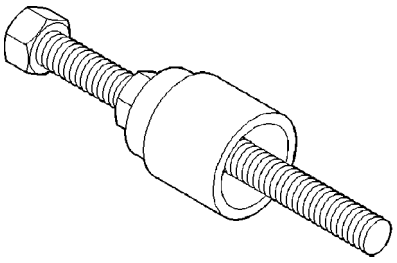
Camshaft Sprocket Holder 6847



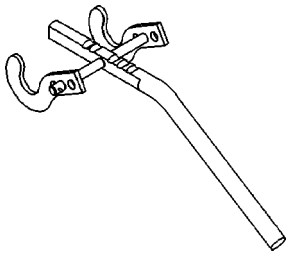
Camshaft Seal Remover C-4679-A



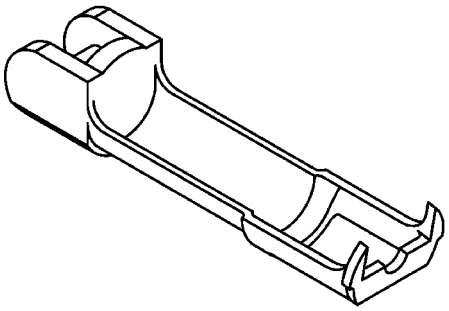
Camshaft Seal Installer MD-998306



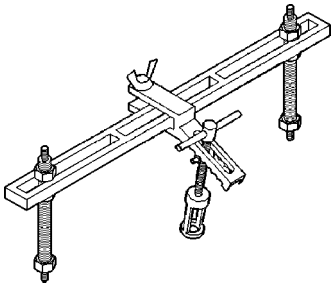
Crankshaft Damper Installer 6792



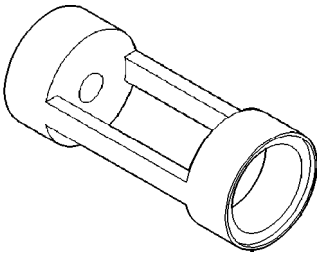
Valve Spring Compressor 8215



Adaptor 8436

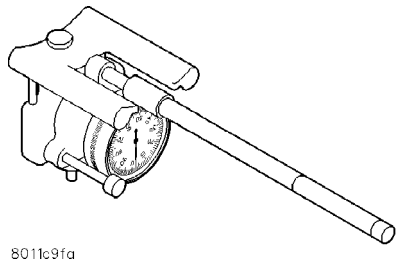


Valve Spring Compressor MD998772A

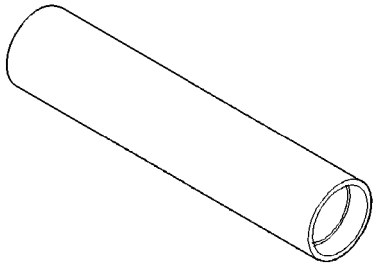


Valve Spring Compressor Adapter 6779

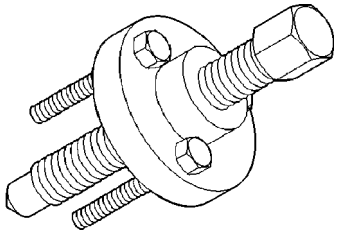
ENGINE 2.4L (Continued)



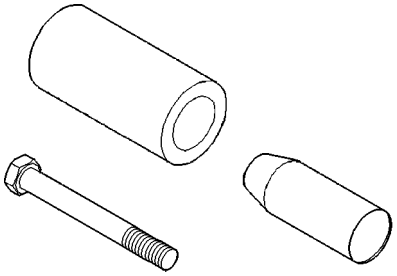
Cylinder Bore Gage C-119



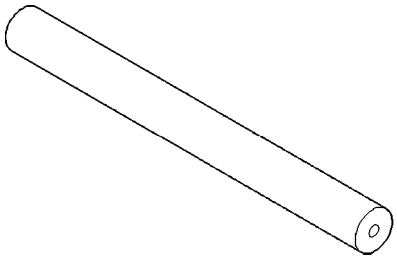
Balance Shaft Sprocket Installer 6052



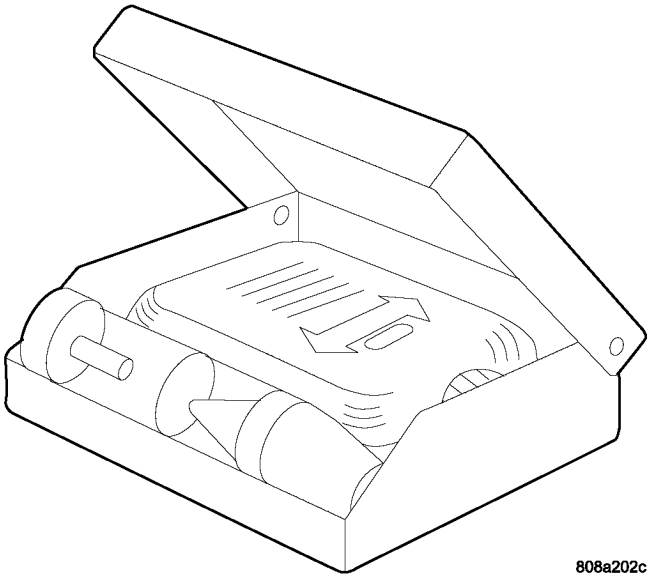
Crankshaft Sprocket Remover 6793



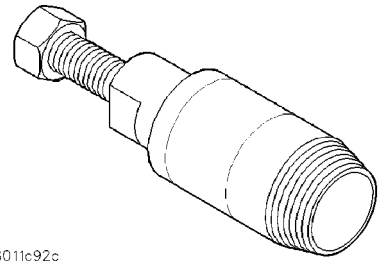
Front Crankshaft Oil Seal Installer 6780



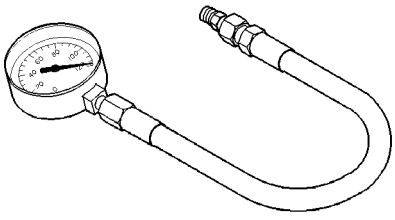
Crankshaft Sprocket Remover Insert C-4685-C2



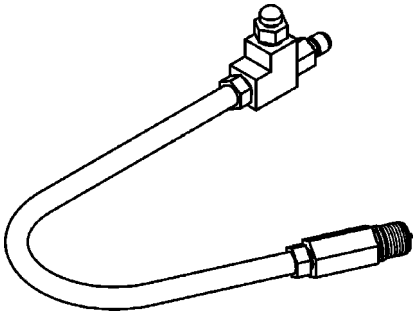
Combustion Leak Tester C-3685-A



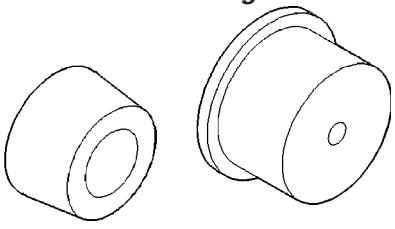
Crankshaft Seal Remover 6771



Oil Pressure Gauge C-3292



Cylinder Compression Pressure Adaptor 8116



Rear Crankshaft Seal Guide and Installer 6926-1 and 6926-2

AIR CLEANER ELEMENT - 2.4L

REMOVAL - 2.4L

Housing removal is not necessary for element (filter) replacement.

(1) Disconnect air intake duct at front of element cover.

(2) Pry up spring clips (Fig. 5) from housing cover (spring clips retain cover to housing).

(3) Release housing cover from locating tabs located on housing, and remove cover.

(4) Remove air cleaner element (filter) from housing.

(5) Clean inside of housing before replacing element.

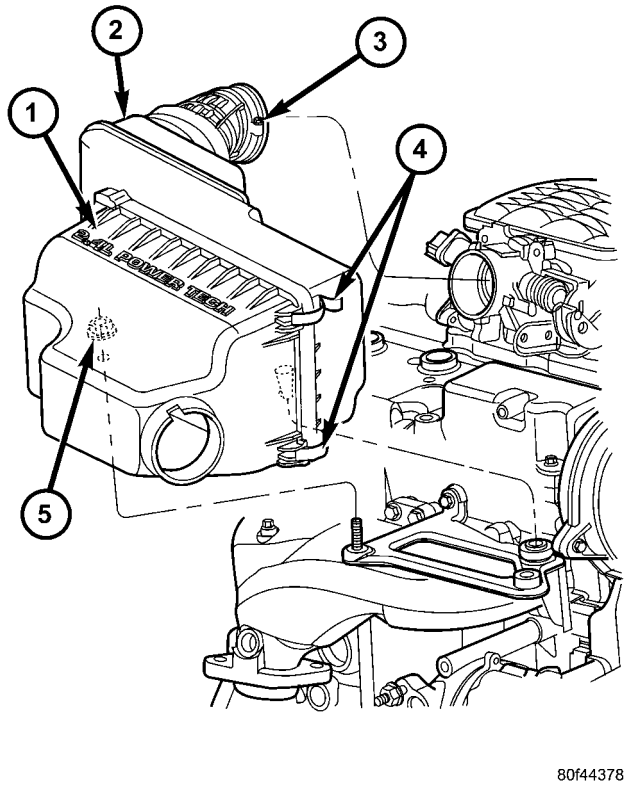


Fig. 5 AIR CLEANER ELEMENT - 2.4L

1 - COVER
2 - HOUSING
3 - CLAMP
4 - SPRING CLIPS
5 - HOUSING MOUNTING NUT

INSTALLATION - 2.4L

(1) Install element into housing.

(2) Position housing cover into housing locating tabs.

(3) Pry up spring clips and lock cover to housing.

(4) Connect air intake duct.

If any air filter, air resonator, air intake tubes or air filter housing clamps had been loosened or removed, tighten them to 5 N·m (40 in. lbs.) torque.

AIR CLEANER HOUSING

REMOVAL

(1)

(2) Disconnect air intake duct at front of element cover.

(3) Pry up spring clips from housing cover (spring clips retain cover to housing).

(4) Release housing cover from locating tabs located on housing, and remove cover. (Fig. 6)

(5) Remove air cleaner element (filter) from housing.

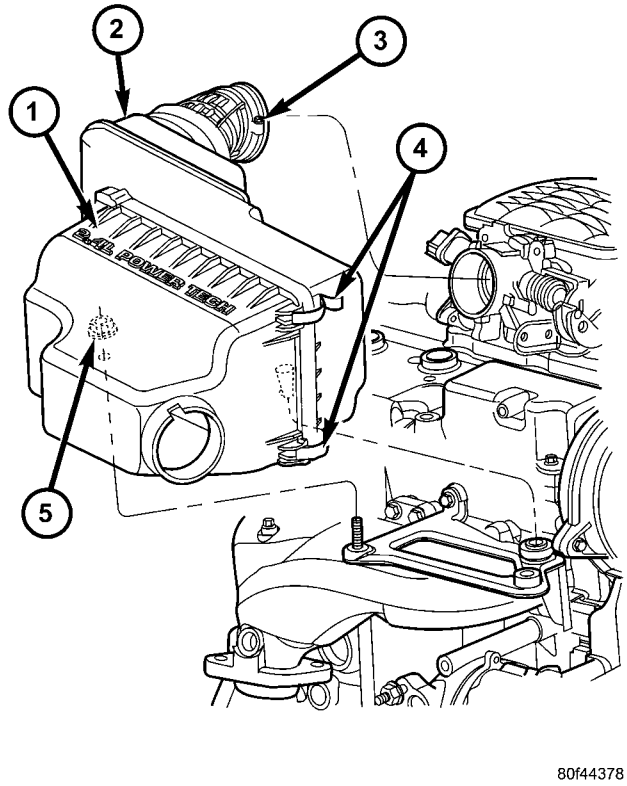


Fig. 6 AIR CLEANER ELEMENT - 2.4L

1 - COVER
2 - HOUSING
3 - CLAMP
4 - SPRING CLIPS
5 - HOUSING MOUNTING NUT

(6) Remove vent hose.

(7) Remove the housing mounting nut.

(8) Remove the air cleaner housing.

AIR CLEANER HOUSING (Continued)

INSTALLATION

(1) Install the air cleaner housing mounting bracket.
(2) Install the air cleaner housing and mounting nut. Tighten nut to 4.5 N·m (40 in. lbs.).
(3) Install vent hose.
(4) Install air cleaner element into the housing.
(5) Position the housing cover into the locating tabs on the housing and install the cover.
(6) Pry up the spring clips and lock the cover in place.
(7) Install the air intake duct. (Fig. 6)
(8) Install the air outlet duct.

CYLINDER HEAD

DESCRIPTION

The cross flow designed, aluminum cylinder head contains dual over-head camshafts with four valves per cylinder (Fig. 7). The valves are arrange in two in-line banks. The intake valves face toward the left side of the vehicle. The exhaust valves face the right side. The cylinder head incorporates powdered metal valve guides and seats. The cylinder head is sealed to the block using a multi-layer steel head gasket and retaining bolts.

Integral oil galleries providing lubrication passages to the hydraulic lash adjusters, camshafts, and valve mechanisms.

OPERATION

The cylinder head closes the combustion chamber, allowing the pistons to compress the fuel/air mixture for ignition. The valves are actuated by the lobe profiles on the camshaft to open and close at specified duration to either allow clean air in the combustion chamber or the exhaust gases out; depending on the stroke of the engine.

DIAGNOSIS AND TESTING

CYLINDER HEAD GASKET

A cylinder head gasket leak can be located between adjacent cylinders or between a cylinder and the adjacent water jacket.

Possible indications of the cylinder head gasket leaking between adjacent cylinders are:

- Loss of engine power
- Engine misfiring
- Poor fuel economy

Possible indications of the cylinder head gasket leaking between a cylinder and an adjacent water jacket are:

- Engine overheating

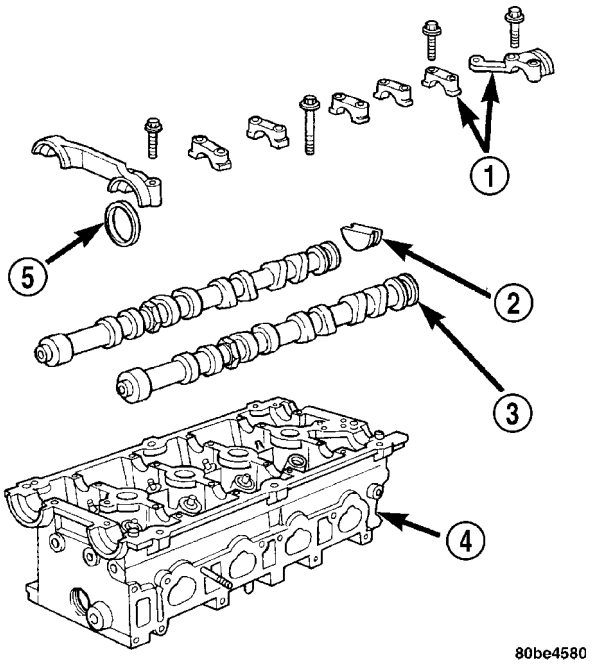


Fig. 7 Cylinder Head and Camshafts

1 - CAMSHAFT BEARING CAPS
2 - PLUG
3 - CAMSHAFT
4 - CYLINDER HEAD
5 - CAMSHAFT OIL SEAL

- Loss of coolant
- Excessive steam (white smoke) emitting from exhaust
- Coolant foaming

CYLINDER-TO-CYLINDER LEAKAGE TEST

To determine if an engine cylinder head gasket is leaking between adjacent cylinders, follow the procedures in Cylinder Compression Pressure Test (Refer to 9 - ENGINE - DIAGNOSIS AND TESTING). An engine cylinder head gasket leaking between adjacent cylinders will result in approximately a 50 - 70% reduction in compression pressure.

CYLINDER-TO-WATER JACKET LEAKAGE TEST

WARNING: USE EXTREME CAUTION WHEN THE ENGINE IS OPERATING WITH COOLANT PRESSURE CAP REMOVED.

VISUAL TEST METHOD

With the engine cool, remove the coolant pressure cap. Start the engine and allow it to warm up until thermostat opens.

If a large combustion/compression pressure leak exists, bubbles will be visible in the coolant.

CYLINDER HEAD (Continued)

COOLING SYSTEM TESTER METHOD

WARNING: WITH COOLING SYSTEM TESTER IN PLACE, PRESSURE WILL BUILD UP FAST. EXCESSIVE PRESSURE BUILT UP, BY CONTINUOUS ENGINE OPERATION, MUST BE RELEASED TO A SAFE PRESSURE POINT. NEVER PERMIT PRESSURE TO EXCEED 138 kPa (20 psi).

Install Cooling System Tester 7700 or equivalent to pressure cap neck. Start the engine and observe the tester's pressure gauge. If gauge pulsates with every power stroke of a cylinder a combustion pressure leak is evident.

CHEMICAL TEST METHOD

Combustion leaks into the cooling system can also be checked by using Bloc-Chek Kit C-3685-A or equivalent. Perform test following the procedures supplied with the tool kit.

REMOVAL - CYLINDER HEAD

(1) Perform fuel system pressure release procedure **before attempting any repairs.** (Refer to 14 - FUEL SYSTEM/FUEL DELIVERY - SPECIFICATIONS)
(2) Disconnect battery negative cable.
(3) Drain cooling system. (Refer to 7 - COOLING - STANDARD PROCEDURE)
(4) Remove air filter housing and inlet tube.
(5) Remove intake manifold.
(6) Remove heater tube support bracket from cylinder head.
(7) Disconnect radiator upper and heater supply hoses from water outlet connections.
(8) Remove accessory drive belts. (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - REMOVAL)
(9) Raise vehicle and remove exhaust pipe from manifold.
(10) Remove power steering pump and set aside. Do not disconnect lines.
(11) Remove accessory drive bracket
(12) Remove ignition coil and wires from engine.
(13) Disconnect cam sensor and fuel injector wiring connectors.
(14) Remove timing belt and camshaft sprockets. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - REMOVAL)
(15) Remove timing belt idler pulley and rear timing belt cover. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT / CHAIN COVER(S) - REMOVAL)
(16) Remove cylinder head cover. (Refer to 9 - ENGINE/CYLINDER HEAD/CYLINDER HEAD COVER(S) - REMOVAL)
(17) Remove camshafts (Refer to 9 - ENGINE/CYLINDER HEAD/CAMSHAFT(S) - REMOVAL).

NOTE: Identify rocker arm position to ensure correct re-installation in original position, if reused.

(18) Remove rocker arms. (Refer to 9 - ENGINE/CYLINDER HEAD/ROCKER ARMS - REMOVAL).
(19) Remove cylinder head bolts in REVERSE sequence of tightening.
(20) Remove cylinder head from engine block.
(21) Inspect and clean cylinder head. (Refer to 9 - ENGINE/CYLINDER HEAD - INSPECTION) (Refer to 9 - ENGINE/CYLINDER HEAD - CLEANING)

CLEANING

To ensure engine gasket sealing, proper surface preparation must be performed, especially with the use of aluminum engine components and multi-layer steel cylinder head gaskets.

NOTE: Multi-Layer Steel (MLS) head gaskets require a scratch free sealing surface.

Remove all gasket material from cylinder head and block (Refer to 9 - ENGINE - STANDARD PROCEDURE). Be careful not to gouge or scratch the aluminum head sealing surface.
Clean all engine oil passages.

INSPECTION

(1) Cylinder head must be flat within 0.1 mm (0.004 in.) (Fig. 8).
(2) Inspect camshaft bearing journals for scoring.

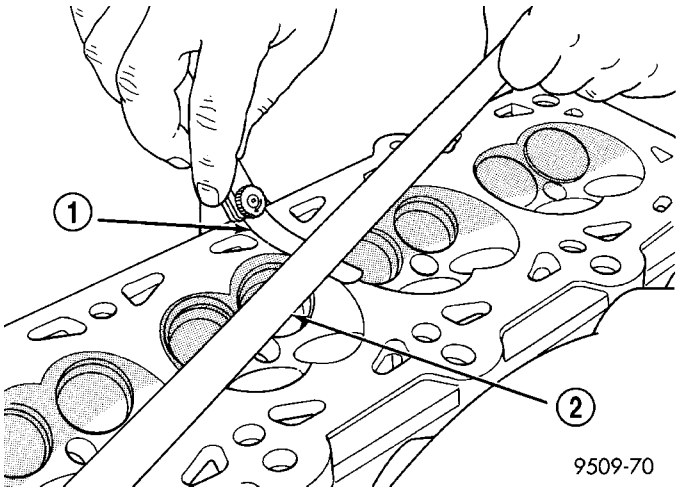


Fig. 8 Checking Cylinder Head Flatness

1 - FEELER GAUGE
2 - STRAIGHT EDGE

(3) Remove carbon and varnish deposits from inside of valve guides with a reliable guide cleaner.

CYLINDER HEAD (Continued)

(4) Using a small hole gauge and a micrometer, measure valve guides in 3 places top, middle and bottom (Fig. 9). (Refer to 9 - ENGINE - SPECIFICATIONS) Replace guides if they are not within specification.

(5) Check valve guide height (Fig. 10).

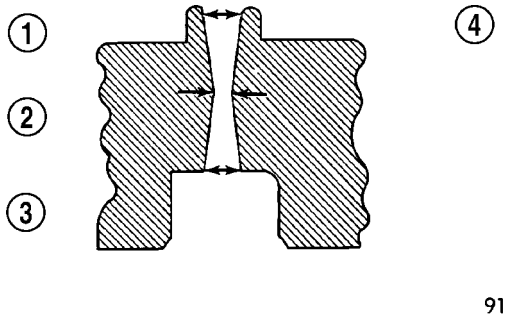


Fig. 9 Checking Wear on Valve Guide - Typical

1 - TOP
2 - MIDDLE
3 - BOTTOM
4 - CUT AWAY VIEW OF VALVE GUIDE MEASUREMENT LOCATIONS

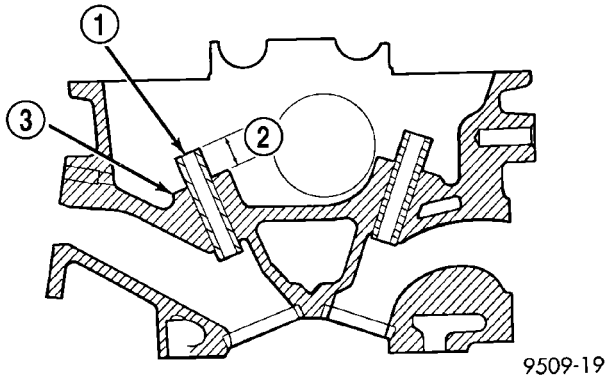


Fig. 10 Valve Guide Height

1 - VALVE GUIDE
2 - 13.25 - 13.75 MM (.521 - .541 IN.)
3 - SPRING SEAT

INSTALLATION - CYLINDER HEAD

NOTE: The Cylinder head bolts should be examined BEFORE reuse. If the threads are necked down, the bolts must be replaced (Fig. 11).

Necking can be checked by holding a scale or straight edge against the threads. If all the threads do not contact the scale, the bolt should be replaced.

(1) Before installing the bolts, the threads should be coated with engine oil.

(2) Position cylinder head gasket on engine block (Fig. 12).

(3) Install cylinder head on engine block.

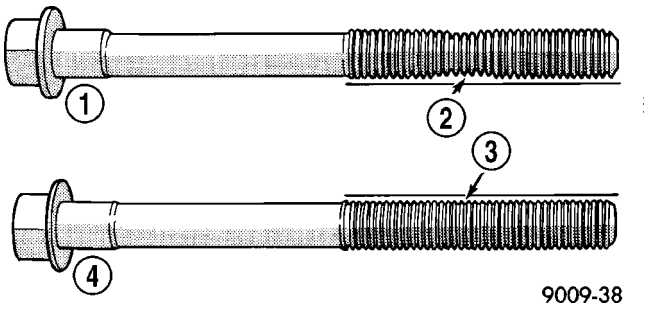


Fig. 11 Checking Bolts for Stretching (Necking)

1 - STRETCHED BOLT
2 - THREADS ARE NOT STRAIGHT ON LINE
3 - THREADS ARE STRAIGHT ON LINE
4 - UNSTRETCHED BOLT

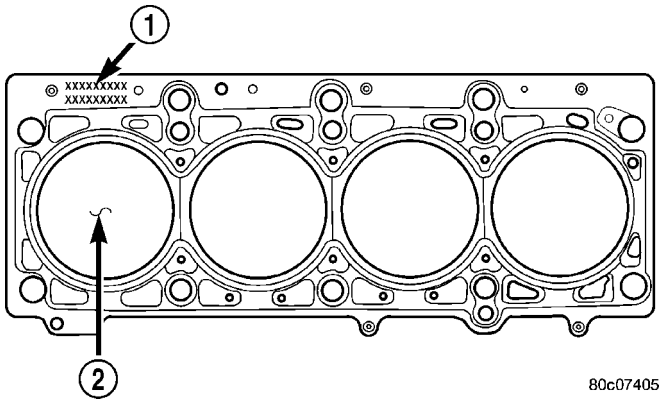


Fig. 12 Cylinder Head Gasket Positioning

1 - PART NUMBER FACES UP
2 - NO. 1 CYLINDER

(4) Tighten the cylinder head bolts in the sequence shown in (Fig. 13). Using the 4 step torque turn method, tighten according to the following values:

- First All to 34 N·m (25 ft. lbs.)
- Second All to 68 N·m (50 ft. lbs.)
- Third All to 68 N·m (50 ft. lbs.)

CAUTION: Do not use a torque wrench for the following step.

- Fourth Turn an additional 1/4 Turn,

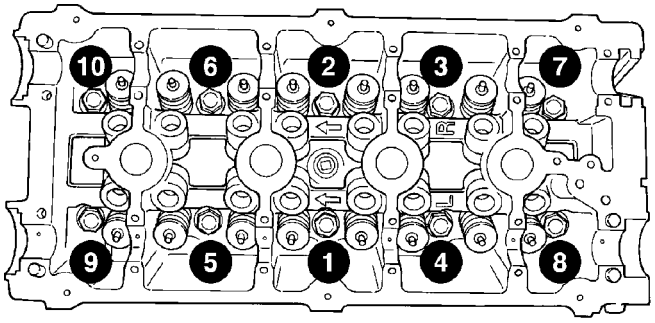


Fig. 13 Cylinder Head Tightening Sequence

CYLINDER HEAD (Continued)

(5) Install rocker arms. (Refer to 9 - ENGINE/CYLINDER HEAD/ROCKER ARMS - INSTALLATION)

(6) Install camshafts. (Refer to 9 - ENGINE/CYLINDER HEAD/CAMSHAFT(S) - INSTALLATION).

(7) Install cylinder head cover. (Refer to 9 - ENGINE/CYLINDER HEAD/CYLINDER HEAD COVER(S) - INSTALLATION)

(8) Install timing belt rear cover and timing belt idler pulley. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT / CHAIN COVER(S) - INSTALLATION)

(9) Install timing belt and camshaft sprockets. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - INSTALLATION)

(10) Connect cam sensor and fuel injectors wiring connectors.

(11) Install ignition coil and wires. Connect ignition coil wiring connector.

(12) Install accessary drive bracket.

(13) Install power steering pump to cylinder head.

(14) Raise vehicle and install the exhaust pipe to the manifold.

(15) Install accessory drive belts. (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - INSTALLATION)

(16) Install heater tube support bracket to cylinder head.

(17) Install intake manifold.

(18) Connect all vacuum lines, electrical wiring, ground straps and fuel line.

(19) Fill cooling system. (Refer to 7 - COOLING - STANDARD PROCEDURE)

(20) Connect battery negative cable.

CAMSHAFT OIL SEAL(S)

REMOVAL

(1) Remove timing belt. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - REMOVAL)

(2) Hold each camshaft sprocket with Special Tool 6847 while removing center bolt (Fig. 14).

(3) Remove camshaft sprockets.

(4) Remove exhaust camshaft target ring.

(5) Remove exhaust camshaft sensor.

CAUTION: Inspect sensor and target ring for excessive wear. Clean sensor face and install new spacer pad.

(6) Remove rear timing belt cover. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT / CHAIN COVER(S) - REMOVAL)

(7) Remove camshaft seal using Special Tool C-4679-A (Fig. 15).

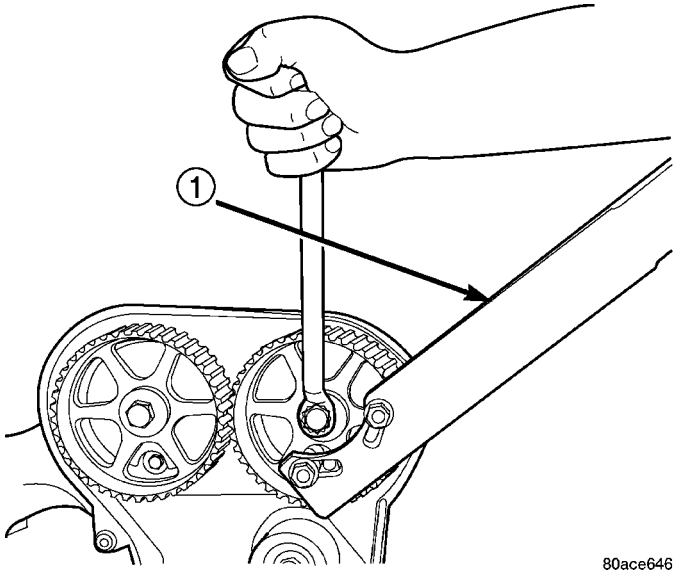


Fig. 14 Camshaft Sprocket - Removal/Installation

1 - SPECIAL TOOL 6847

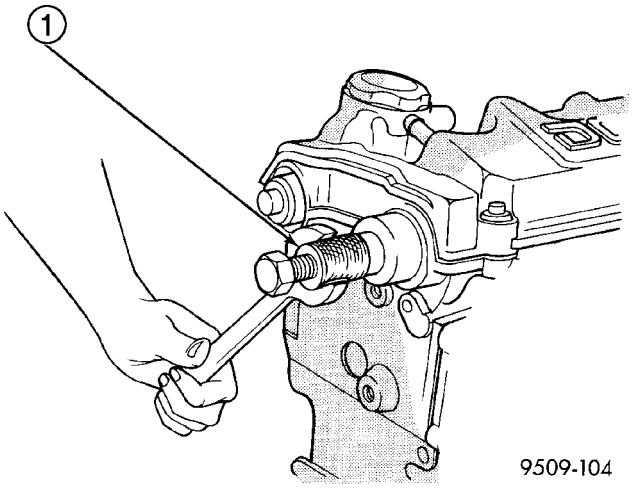


Fig. 15 Camshaft Oil Seal - Removal With C-4679-A

1 - SPECIAL TOOL C-4679

CAUTION: Do not nick shaft seal surface or seal bore.

CAMSHAFT OIL SEAL(S) (Continued)

INSTALLATION

NOTE: Clean and inspect sensor and target ring for excessive wear. Clean sensor face and always install a new spacer pad.

(1) Shaft seal surface must be free of varnish, dirt or nicks. Polish with 400 grit paper if necessary.

(2) Install camshaft seals into cylinder head using Special Tool MD-998306 until flush with head (Fig. 16).

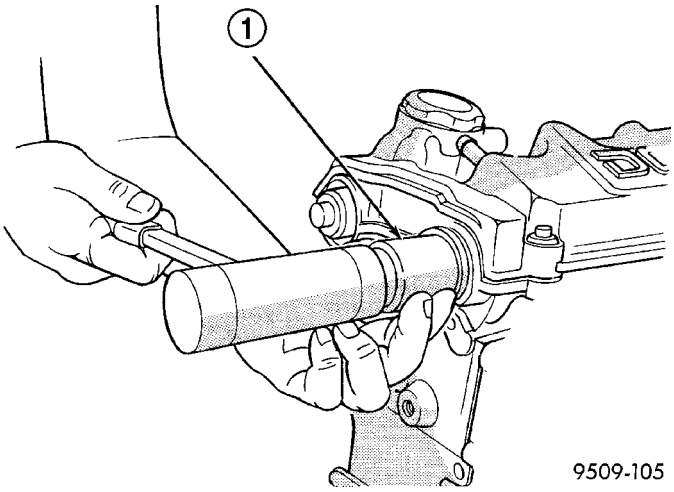


Fig. 16 Camshaft Seal - Installation

1 - SPECIAL TOOL MD 998306

(3) Install timing belt rear cover. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT / CHAIN COVER(S) - INSTALLATION)

NOTE: Target ring tab should provide positive snap-on fit on the camshaft.

(4) Install exhaust camshaft target ring with the word **FRONT** facing forward.

(5) Install exhaust camshaft sensor.

(6) Install camshaft sprockets. Hold each sprocket with Special Tool 6847 and tighten center bolt to 101 N·m (75 ft. lbs.).

(7) Install timing belt and front covers. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - INSTALLATION) (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT / CHAIN COVER(S) - INSTALLATION)

CAMSHAFT(S)

DESCRIPTION

Both nodular iron camshafts have six bearing journal surfaces and two cam lobes per cylinder (Fig. 17). Flanges at the rear journals control camshaft end play. Provision for a cam position sensor is located on the exhaust camshaft on the front of the cylinder head. A hydrodynamic oil seal is used for oil control at the front of the camshaft.

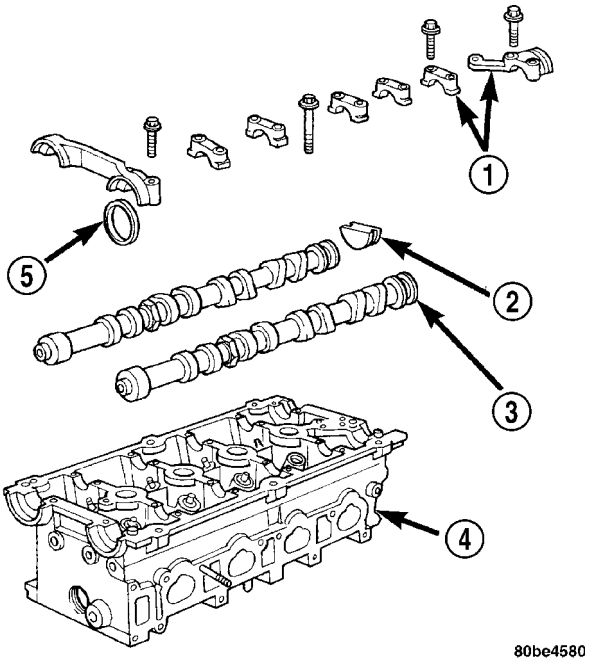


Fig. 17 Camshafts

1 - CAMSHAFT BEARING CAPS
2 - PLUG
3 - CAMSHAFT
4 - CYLINDER HEAD
5 - CAMSHAFT OIL SEAL

OPERATION

The camshaft is driven by the crankshaft via drive sprockets and belt. The camshaft has precisely machined lobes to provide accurate valve timing and duration.

CAMSHAFT(S) (Continued)

STANDARD PROCEDURE - CAMSHAFT END-PLAY

(1) Oil camshaft journals and install camshaft **WITHOUT** cam follower assemblies. Install rear cam caps and tighten screws to specified torque.

(2) Using a suitable tool, move camshaft as far rearward as it will go.

(3) Zero dial indicator (Fig. 18).

(4) Move camshaft as far forward as it will go.

(5) Record reading on dial indicator. For end play specification, (Refer to 9 - ENGINE - SPECIFICATIONS).

(6) If end play is excessive, check cylinder head and camshaft for wear; replace as necessary.

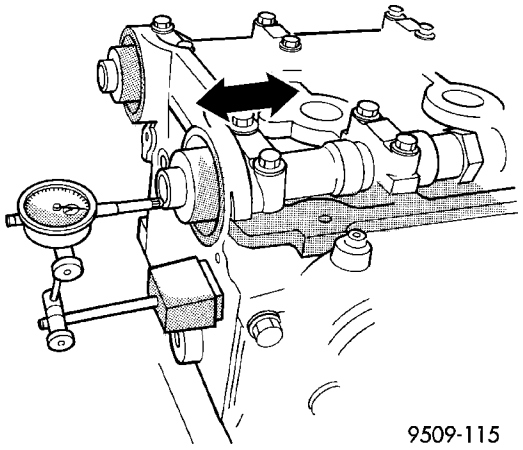


Fig. 18 Camshaft End Play - Typical

REMOVAL

(1) Remove cylinder head cover. (Refer to 9 - ENGINE/CYLINDER HEAD/CYLINDER HEAD COVER(S) - REMOVAL)

(2) Remove camshaft position sensor and camshaft target magnet. (Refer to 8 - ELECTRICAL/IGNITION CONTROL/CAMSHAFT POSITION SENSOR - REMOVAL)

(3) Remove timing belt. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - REMOVAL)

(4) Remove camshaft sprockets and timing belt rear cover. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT / CHAIN COVER(S) - REMOVAL)

(5) Bearing caps are identified for location. Remove the outside bearing caps first (Fig. 19).

(6) Loosen the camshaft bearing cap attaching fasteners in sequence shown (Fig. 20) one camshaft at a time.

CAUTION: Camshafts are not interchangeable. The intake cam number 6 thrust bearing face spacing is wider.

(7) Identify the camshafts before removing from the head. The camshafts are not interchangeable.

(8) Remove camshafts from cylinder head.

NOTE: If removing rocker arms, identify for reinstallation in the original position.

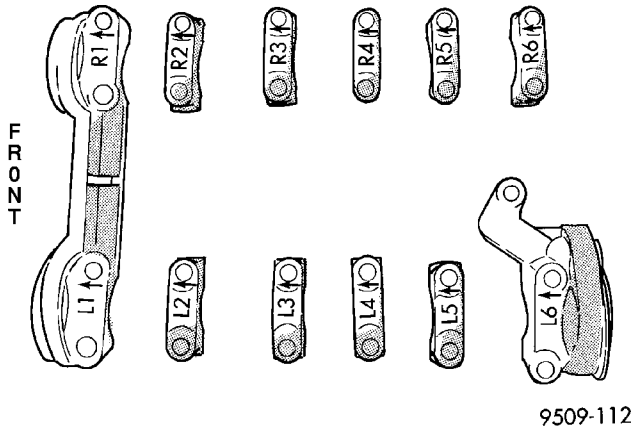


Fig. 19 Camshaft Bearing Cap Identification

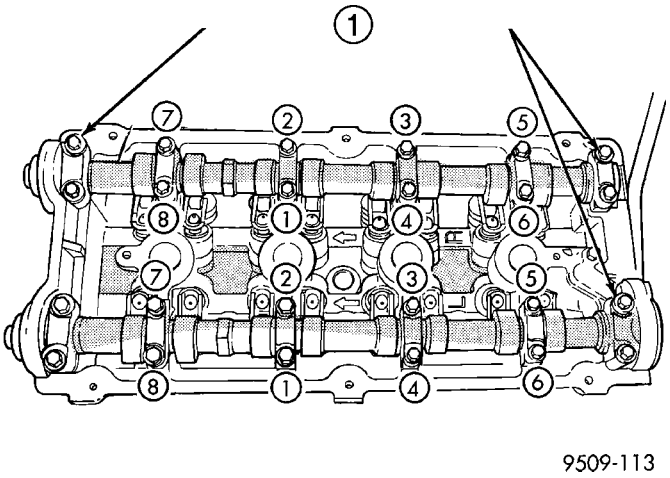


Fig. 20 Camshaft Bearing Cap - Removal

1 - REMOVE OUTSIDE BEARING CAPS FIRST

CLEANING

Clean camshaft with a suitable solvent.

INSPECTION

(1) Inspect camshaft bearing journals for damage and binding (Fig. 21). If journals are binding, check the cylinder head for damage. Also check cylinder head oil holes for clogging.

(2) Check the cam lobe and bearing surfaces for abnormal wear and damage. Replace camshaft if defective.

NOTE: If camshaft is replaced due to lobe wear or damage, always replace the rocker arms.

(3) Measure the lobe actual wear (unworn area - wear zone = actual wear) (Fig. 21) and replace cam-

CAMSHAFT(S) (Continued)

shaft if out of limit. Standard value is 0.0254 mm (0.001 in.), wear **limit** is 0.254 mm (0.010 in.).

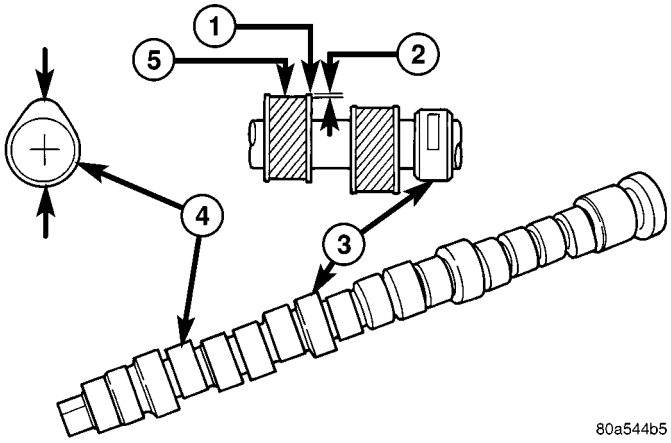


Fig. 21 Checking Camshaft(s) for Wear

1 - UNWORN AREA
2 - ACTUAL WEAR
3 - BEARING JOURNAL
4 - LOBE
5 - WEAR ZONE

INSTALLATION

CAUTION: Ensure that NONE of the pistons are at top dead center when installing the camshafts.

(1) Lubricate all camshaft bearing journals, rocker arms and camshaft lobes.

(2) Install all rocker arms in original positions, if reused.

(3) Position camshafts on cylinder head bearing journals. Install right and left camshaft bearing caps No. 2 – 5 and right No. 6. Tighten M6 fasteners to 12 N·m (105 in. lbs.) in sequence shown in (Fig. 22).

(4) Apply Mopar® Gasket Maker to No. 1 and No. 6 bearing caps (Fig. 23). Install bearing caps and tighten M8 fasteners to 28 N·m (250 in. lbs.).

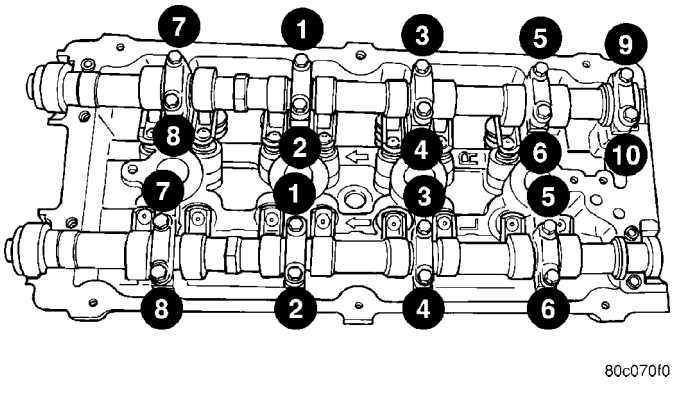


Fig. 22 Camshaft Bearing Cap Tightening Sequence

NOTE: Bearing end caps must be installed before seals can be installed.

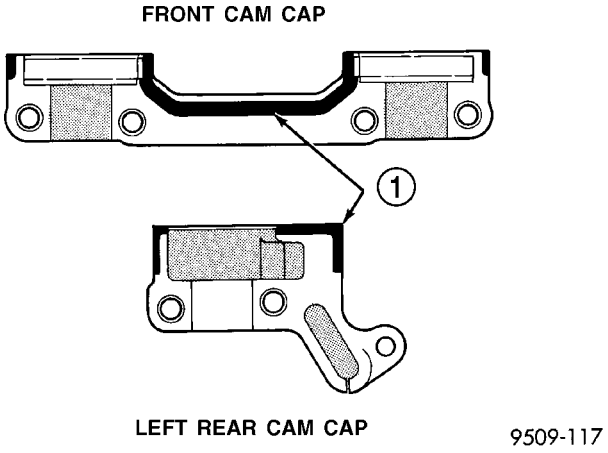


Fig. 23 Camshaft Bearing Cap Sealing

1 - 1.5 mm (.060 in.) DIAMETER BEAD OF MOPAR GASKET MAKER

(5) Install camshaft oil seals. (Refer to 9 - ENGINE/CYLINDER HEAD/CAMSHAFT OIL SEAL(S) - INSTALLATION)

(6) Install camshaft target magnet and camshaft position sensor.

(7) Install cylinder head cover. (Refer to 9 - ENGINE/CYLINDER HEAD/CYLINDER HEAD COVER(S) - INSTALLATION)

(8) Install timing belt rear cover and camshaft sprocket. (Refer to 9 - ENGINE/VALVE TIMING/ TIMING BELT / CHAIN COVER(S) - INSTALLATION)

(9) Install timing belt. (Refer to 9 - ENGINE/ VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - INSTALLATION)

CYLINDER HEAD COVER

REMOVAL

(1) Remove intake manifold. (Refer to 9 - ENGINE/MANIFOLDS/INTAKE MANIFOLD - REMOVAL)

(2) Remove ignition coil and spark plug wires.

(3) Disconnect PCV and make-up air hoses from cylinder head cover.

(4) Remove cylinder head cover bolts.

(5) Remove cylinder head cover from cylinder head.

CLEANING

Clean cylinder head and cover mating surfaces using a suitable solvent.

INSPECTION

Inspect cover rails for flatness.

CYLINDER HEAD COVER (Continued)

INSTALLATION

NOTE: Replace spark plug well seals and bolt assemblies when installing a new cylinder head cover gasket.

(1) Install new cylinder head cover gaskets and spark plug well seals (Fig. 24).

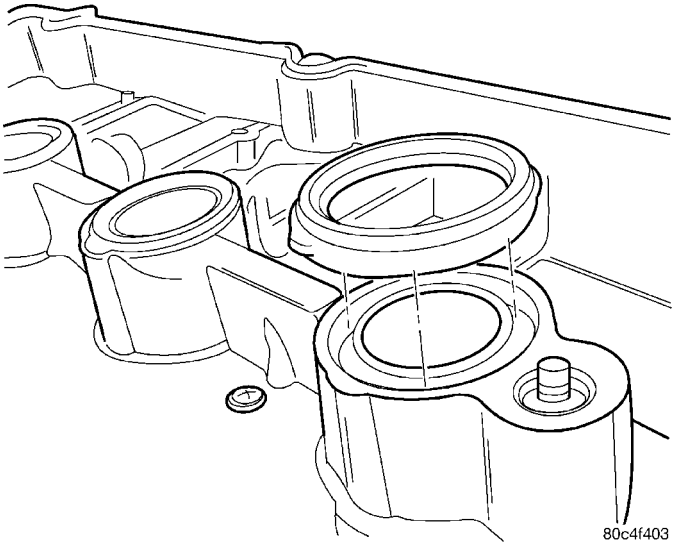


Fig. 24 Spark Plug Well Seals

(2) Replace cylinder head cover bolt assemblies (Fig. 25).

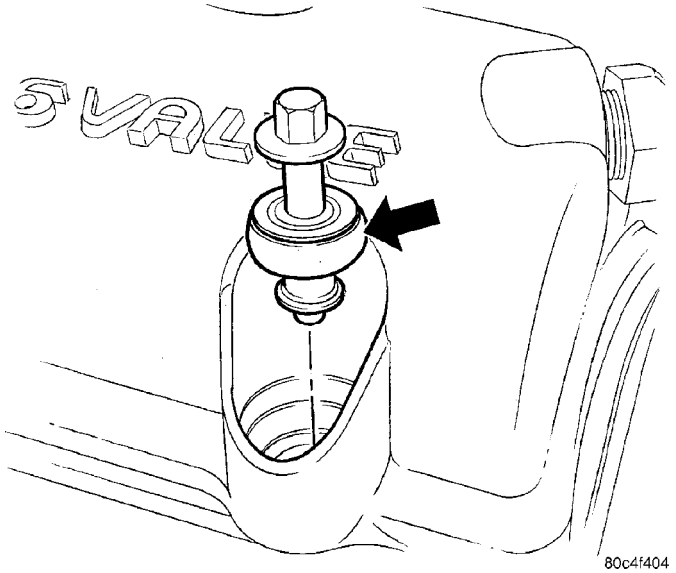


Fig. 25 Cylinder Head Cover Bolt Assembly

CAUTION: Do not allow oil or solvents to contact the timing belt as they can deteriorate the rubber and cause tooth skipping.

(3) Apply Mopar® Engine RTV GEN II at the camshaft cap corners and at the top edges of the 1/2 round seal (Fig. 26).

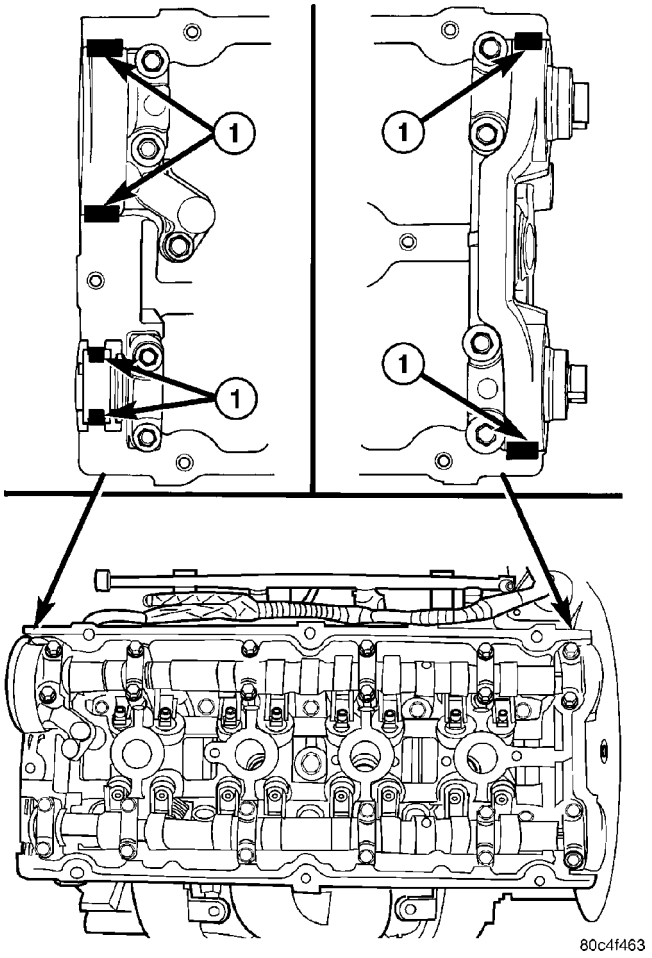


Fig. 26 Sealer Locations - Typical

1 – SEALER LOCATION

CYLINDER HEAD COVER (Continued)

(4) Install cylinder head cover assembly to cylinder head. Install all bolts, ensuring the two (2) bolts containing the sealing washer are located in the center locations of cover. Tighten bolts in sequence shown in (Fig. 27). Using a 3 step torque method as follows:

(a) Tighten all bolts to 4.5 N·m (40 in. lbs.).
(b) Tighten all bolts to 9.0 N·m (80 in. lbs.).
(c) Tighten all bolts to 12 N·m (105 in. lbs.).

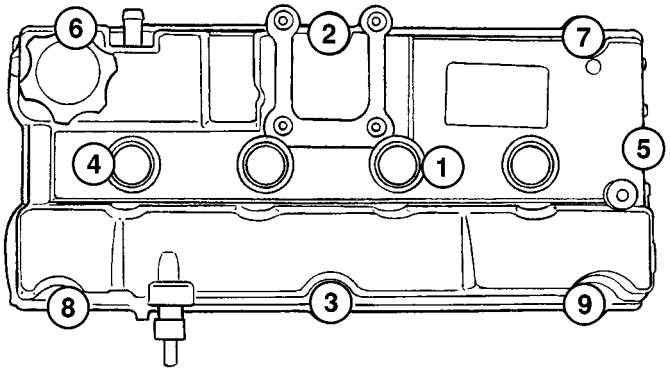


Fig. 27 CYLINDER HEAD TIGHTENING SEQUENCE

(5) Install intake manifold.

(6) Install ignition coil and spark plug wires. Tighten fasteners to 12 N·m (105 in. lbs.).

(7) If the PCV valve was removed, apply Mopar® Thread Sealant with Teflon to threads and install valve to cylinder head cover. Tighten PCV valve to 8 N·m (70 in. lbs.).

(8) Connect PCV and make-up air hoses to cylinder head cover.

INTAKE/EXHAUST VALVES & SEATS

DESCRIPTION

The four valves per cylinder are opened by using roller rocker arms which pivot on hydraulic lash adjusters. The valves have chrome plated valve stems. Viton rubber valve stem seals are integral with the spring seats. They have chrome plated stems to prevent scuffing. Viton rubber valve stem seals are integral with the spring seats. The valves, spring retainers,and locks, are the 3 - bead lock design

CLEANING

(1) Clean all valves thoroughly and discard burned, warped and cracked valves.

VALVE SPRINGS

REMOVAL

REMOVAL - CYLINDER HEAD ON

(1) Remove camshafts.

(2) Rotate crankshaft until piston is at TDC on compression.

(3) With air hose attached to adapter tool installed in spark plug hole, apply 90-120 psi air pressure.

(4) Using Special Tool MD-998772-A with adapter 6779 (Fig. 28), compress valve springs and remove valve locks.

(5) Remove valve spring(s).

(6) Remove valve stem seal(s) by a using valve stem seal tool (Fig. 30).

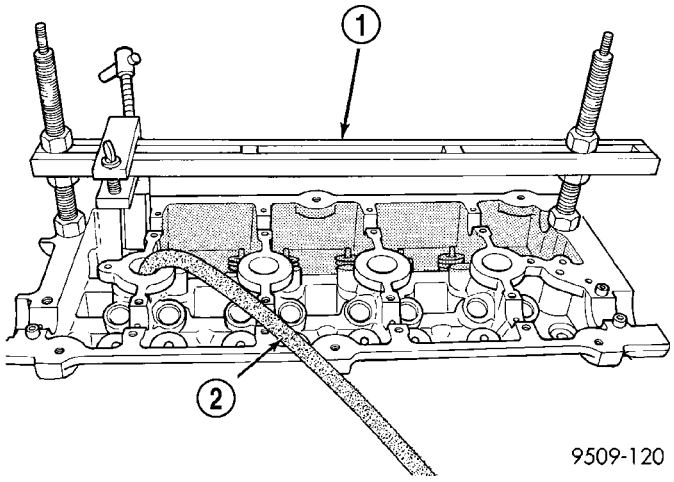


Fig. 28 Valve Spring - Removal/Installation

1 - VALVE SPRING COMPRESSOR MD 998772A
2 - AIR HOSE

REMOVAL - CYLINDER HEAD OFF

(1) With cylinder head removed from cylinder block, compress valve springs using a universal valve spring compressor.

(2) Remove valve retaining locks, valve spring retainers, valve stem seals and valve springs.

(3) Before removing valves, **remove any burrs from valve stem lock grooves to prevent damage to the valve guides.** Identify valves, locks and retainers to insure installation in original location.

(4) Inspect the valves. (Refer to 9 - ENGINE/CYLINDER HEAD/VALVE SPRINGS - INSPECTION)

VALVE SPRINGS (Continued)

INSPECTION

(1) Whenever valves have been removed for inspection, reconditioning or replacement, valve springs should be tested for correct tension. Discard the springs that do not meet specifications. The following specifications apply to both intake and exhaust valves springs:

- Valve Closed Nominal Tension - 75 lbs. @ 38.0 mm (1.50 in.)
- Valve Open Nominal Tension - 134 lbs. @ 29.75 mm (1.17 in.)

(2) Inspect each valve spring for squareness with a steel square and surface plate, test springs from both ends. If the spring is more than 1.5 mm (1/16 inch) out of square, install a new spring.

INSTALLATION

INSTALLATION - CYLINDER HEAD ON

(1) Install valve seal/valve spring seat assembly (Fig. 29). Push the assembly down to seat it onto the valve guide.

(2) Install valve spring and retainer, use Special Tool MD-998772-A with adapter 6779 to compress valve springs only enough to install locks. Correct alignment of tool is necessary to avoid nicking valve stems.

(3) Remove air hose and install spark plugs.

(4) Install camshafts and cylinder head cover .

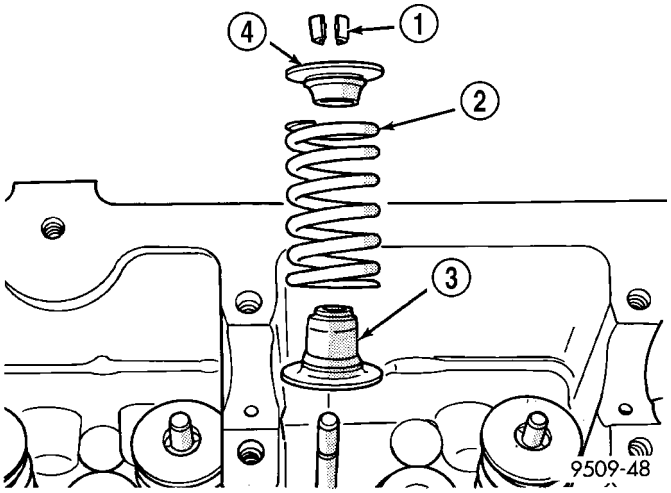


Fig. 29 Valve Stem Seal/Valve Spring Seat - Typical

1 - 3-GROOVE -VALVE RETAINING LOCKS
2 - VALVE SPRING
3 - VALVE SEAL AND VALVE SPRING SEAT ASSEMBLY
4 - VALVE SPRING RETAINER

INSTALLATION - CYLINDER HEAD OFF

(1) Coat valve stems with clean engine oil and insert in cylinder head.

(2) Install new valve stem seals on all valves using a valve stem seal tool (Fig. 30). The valve stem seals should be pushed firmly and squarely over valve guide.

CAUTION: When oversize valves are used, the corresponding oversize valve seal must also be used. Excessive guide wear may result if oversize seals are not used with oversize valves.

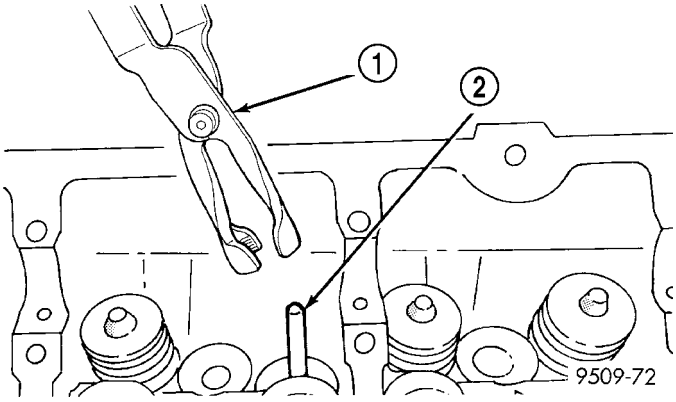


Fig. 30 Valve Stem Oil Seal Tool

1 - VALVE SEAL TOOL
2 - VALVE STEM

(3) Install valve springs and retainers. Compress valve springs only enough to install locks, taking care not to misalign the direction of compression. Nicked valve stems may result from misalignment of the valve spring compressor.

CAUTION: When depressing the valve spring retainers with valve spring compressor the locks can become dislocated. Ensure both locks are in the correct location after removing tool.

(4) Check the valve spring installed height B after refacing the valve and seat (Fig. 31). Make sure measurements are taken from top of spring seat to the bottom surface of spring retainer. If height is greater than 38.75 mm (1.525 in.), install a 0.762 mm (0.030 in.) spacer under the valve spring seat to bring spring height back within specification.

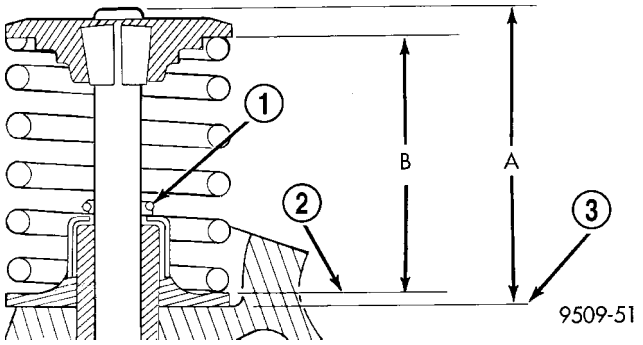


Fig. 31 Checking Spring Installed Height and Valve Tip Height Dimensions

1 - GARTER SPRING
2 - VALVE SPRING SEAT
3 - CYLINDER HEAD SURFACE

HYDRAULIC LIFTERS

DIAGNOSIS AND TESTING

LASH ADJUSTER (TAPPET) NOISE DIAGNOSIS

A tappet-like noise may be produced from several items. Check the following items.

(1) Engine oil level too high or too low. This may cause aerated oil to enter the adjusters and cause them to be spongy.

(2) Insufficient running time after rebuilding cylinder head. Low speed running up to 1 hour may be required.

(3) During this time, turn engine off and let set for a few minutes before restarting. Repeat this several times after engine has reached normal operating temperature.

(4) Low oil pressure.

(5) The oil restrictor (integral to the head gasket) in the vertical oil passage to the cylinder head is plugged with debris.

(6) Air ingested into oil due to broken or cracked oil pump pick up.

(7) Worn valve guides.

(8) Rocker arm ears contacting valve spring retainer.

(9) Rocker arm loose, adjuster stuck or at maximum extension and still leaves lash in the system.

(10) Faulty lash adjuster.

a. Check lash adjusters for sponginess while installed in cylinder head. Depress part of rocker arm over adjuster. Normal adjusters should feel very firm. Spongy adjusters can be bottomed out easily.

b. Remove suspected lash adjusters, and replace as necessary.

REMOVAL

NOTE: This procedure is for in-vehicle service with camshafts installed.

(1) Remove cylinder head cover. (Refer to 9 - ENGINE/CYLINDER HEAD/CYLINDER HEAD COVER(S) - REMOVAL)

(2) Remove the camshafts (Refer to 9 - ENGINE/CYLINDER HEAD/CAMSHAFT(S) - REMOVAL).

(3) Remove rocker arm. (Refer to 9 - ENGINE/CYLINDER HEAD/ROCKER ARMS - REMOVAL)

(4) Remove hydraulic lifter (Fig. 32).

(5) Repeat removal procedure for each hydraulic lifter.

(6) If reusing, mark each hydraulic lifter for reassembly in original position. Lifters are serviced as an assembly.

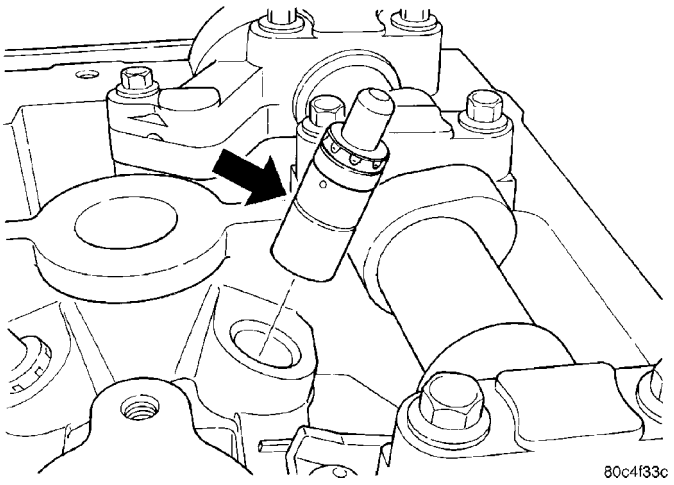


Fig. 32 Hydraulic Lash Adjuster

INSTALLATION

(1) Install hydraulic lifter (Fig. 32). Ensure the lifters are at least partially full of engine oil. This is indicated by little or no plunger travel when the lifter is depressed.

(2) Install rocker arm. (Refer to 9 - ENGINE/CYLINDER HEAD/ROCKER ARMS - INSTALLATION)

(3) Repeat installation procedure for each hydraulic lifter.

(4) Install camshafts (Refer to 9 - ENGINE/CYLINDER HEAD/CAMSHAFT(S) - INSTALLATION).

(5) Install cylinder head cover. (Refer to 9 - ENGINE/CYLINDER HEAD/CYLINDER HEAD COVER(S) - INSTALLATION)

ROCKER ARMS

REMOVAL

NOTE: This procedure is for in-vehicle service with camshafts installed.

(1) Remove cylinder head cover. (Refer to 9 - ENGINE/CYLINDER HEAD/CYLINDER HEAD COVER(S) - REMOVAL)

(2) Remove fuel rail. (Refer to 14 - FUEL SYSTEM/FUEL DELIVERY/FUEL RAIL - REMOVAL)

(3) Remove spark plugs.

(4) Rotate engine until the camshaft lobe, on the follower being removed, is position on its base circle (heel). Also, the piston should be a minimum of 6.3 mm (0.25 in) below TDC position.

CAUTION: If cam follower assemblies are to be reused, always mark position for reassembly in their original positions.

ROCKER ARMS (Continued)

(5) Using Special Tools 8215 and 8436 slowly depress valve assembly until rocker arm can be removed (Fig. 33).

NOTE: It may be necessary to remove additional brackets or components to allow clearance for tool handle movement.

(6) Repeat removal procedure for each rocker arm.

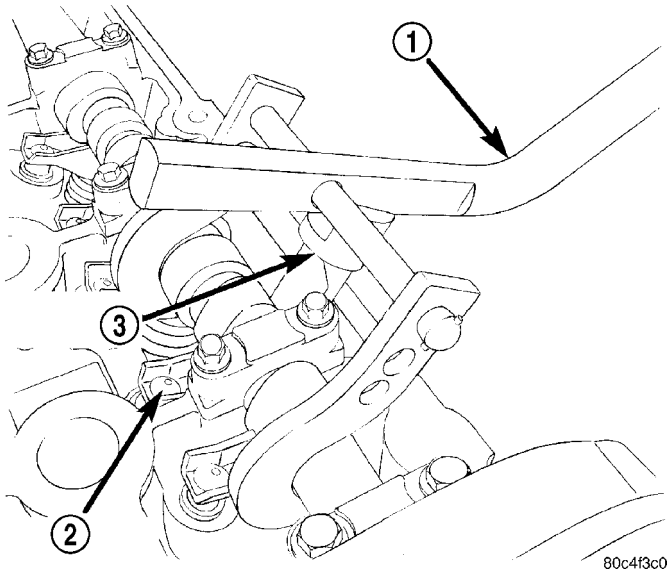


Fig. 33 Rocker Arm - Removal/Installation

1 - SPECIAL TOOL 8215
2 - ROCKER ARM
3 - SPECIAL TOOL 8436

INSPECTION

Inspect the rocker arm for wear or damage (Fig. 34). Replace as necessary.

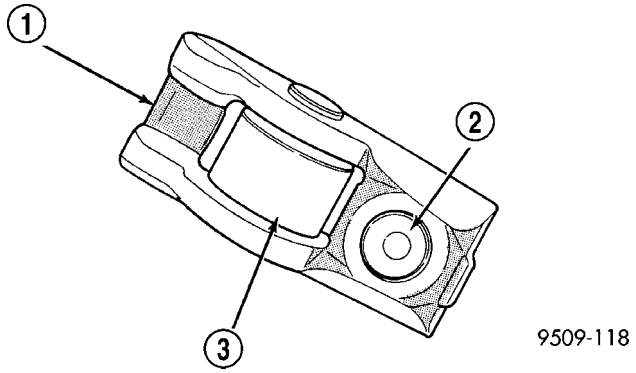


Fig. 34 Rocker Arm - Typical

1 - TIP
2 - LASH ADJUSTER POCKET
3 - ROLLER

INSTALLATION

(1) Lubricate rocker arm with clean engine oil.

(2) Using Special Tools 8215 and 8436 slowly depress valve assembly until rocker arm can be installed on the hydraulic lifter and valve stem.

(3) Repeat installation procedure for each rocker arm.

(4) Install spark plugs.

(5) Install fuel rail. (Refer to 14 - FUEL SYSTEM/ FUEL DELIVERY/FUEL RAIL - INSTALLATION)

(6) Install cylinder head cover. (Refer to 9 - ENGINE/CYLINDER HEAD/CYLINDER HEAD COVER(S) - INSTALLATION)

ENGINE BLOCK

DESCRIPTION

The cast iron cylinder block is a two-piece assembly, consisting of the cylinder block and bedplate (Fig. 35). The bedplate incorporates the main bearing caps and bolts to the cylinder block. This design offers a much stronger lower end and increased cylinder block rigidity. The rear oil seal retainer is integral with the block. The bedplate and block are serviced as an assembly.

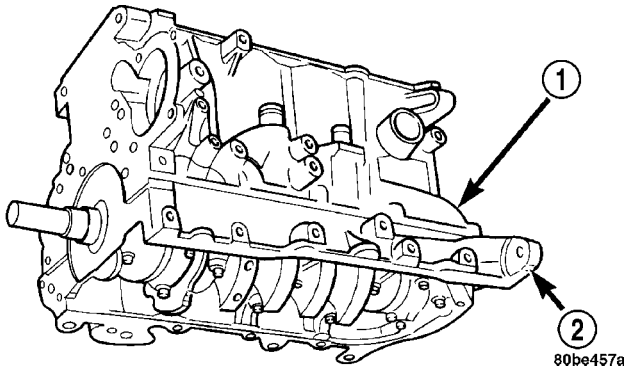


Fig. 35 2.4L Cylinder Block and Bedplate - Typical

1 - CYLINDER BLOCK
2 - BEDPLATE

STANDARD PROCEDURE

PISTON TO CYLINDER BORE FITTING

Piston and cylinder wall must be clean and dry. Piston diameter should be measured 90 degrees to piston pin about 14 mm (9/16 inch.) from the bottom of the skirt as shown in (Fig. 37). Cylinder bores should be measured halfway down the cylinder bore and transverse to the engine crankshaft center line shown in (Fig. 36). Refer to for Engine Specifications (Refer to 9 - ENGINE - SPECIFICATIONS). Correct piston to bore clearance must be established in order to assure quiet and economical operation.

NOTE: Pistons and cylinder bores should be measured at normal room temperature, 21°C (70°F).

ENGINE BLOCK (Continued)

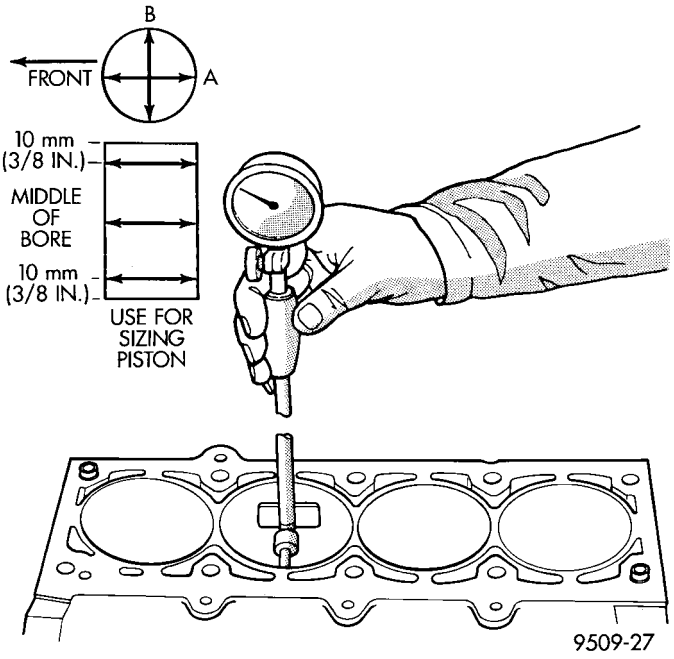


Fig. 36 Checking Cylinder Bore -Typical

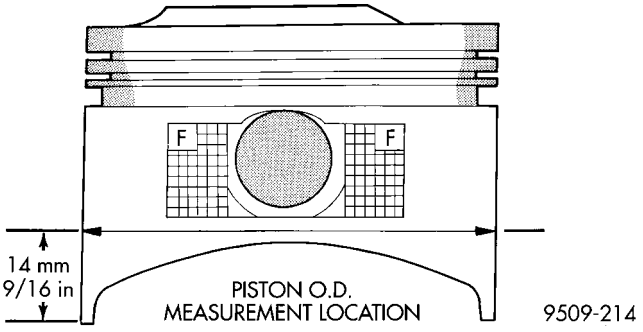


Fig. 37 Piston Measurement - Typical

CYLINDER BORE HONING

(1) Used carefully, the cylinder bore resizing hone, recommended tool C-823 or equivalent, equipped with 220 grit stones, is the best tool for this honing procedure. In addition to deglazing, it will reduce taper and out-of-round as well as removing light scuffing, scoring or scratches. Usually a few strokes will clean up a bore and maintain the required limits.

(2) Deglazing of the cylinder walls may be done using a cylinder surfacing hone, recommended tool C-3501 or equivalent, equipped with 280 grit stones, if the cylinder bore is straight and round. 20–60 strokes depending on the bore condition, will be sufficient to provide a satisfactory surface. Use a light honing oil. **Do not use engine or transmission oil, mineral spirits or kerosene.** Inspect cylinder walls after each 20 strokes.

(3) Honing should be done by moving the hone up and down fast enough to get a cross-hatch pattern. When hone marks **intersect** at 40-60 degrees, the cross hatch angle is most satisfactory for proper seating of rings (Fig. 38).

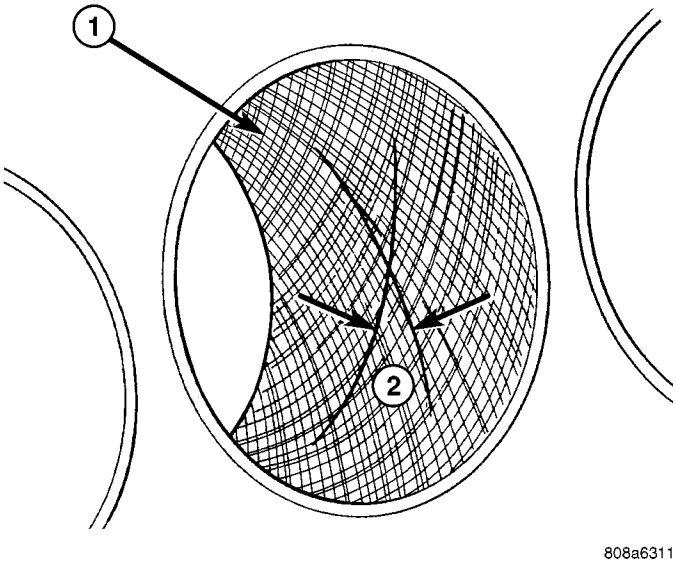


Fig. 38 Cylinder Bore Cross-Hatch Pattern

1 - CROSS-HATCH PATTERN
2 - 40°–60°

(4) A controlled hone motor speed between 200–300 RPM is necessary to obtain the proper cross-hatch angle. The number of up and down strokes per minute can be regulated to get the desired 40–60 degree angle. Faster up and down strokes increase the cross-hatch angle.

(5) After honing, it is necessary that the block be cleaned again to remove all traces of abrasive.

CAUTION: Ensure all abrasives are removed from engine parts after honing. It is recommended that a solution of soap and hot water be used with a brush and the parts then thoroughly dried. The bore can be considered clean when it can be wiped clean with a white cloth and cloth remains clean. Oil the bores after cleaning to prevent rusting.

CLEANING

Clean cylinder block thoroughly using a suitable cleaning solvent.

INSPECTION

ENGINE BLOCK

(1) Clean cylinder block thoroughly and check all core hole plugs for evidence of leaking.

(2) If new core plugs are to be installed, (Refer to 9 - ENGINE - STANDARD PROCEDURE - ENGINE CORE AND OIL GALLERY PLUGS).

ENGINE BLOCK (Continued)

(3) Examine block and cylinder bores for cracks or fractures.

(4) Check block deck surfaces for flatness. Deck surface must be within service limit of 0.1 mm (0.004 in.).

CYLINDER BORE

NOTE: The cylinder bores should be measured at normal room temperature, 21°C (70°F).

The cylinder walls should be checked for out-of-round and taper with Tool C119 or equivalent (Fig. 39) (Refer to 9 - ENGINE - SPECIFICATIONS). If the cylinder walls are badly scuffed or scored, the cylinder block should be replaced, and new pistons and rings fitted.

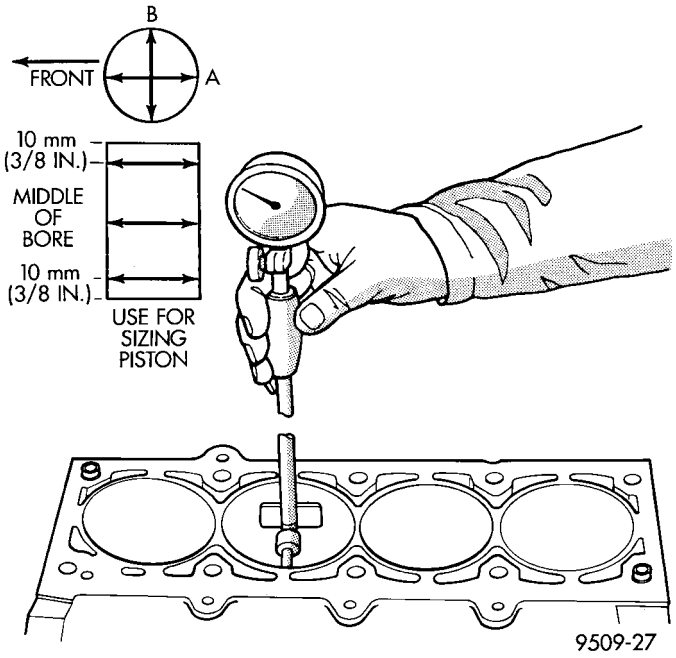


Fig. 39 Checking Cylinder Bore Size

Measure the cylinder bore at three levels in directions A and B (Fig. 39). Top measurement should be 10 mm (3/8 in.) down and bottom measurement should be 10 mm (3/8 in.) up from bottom of bore. (Refer to 9 - ENGINE - SPECIFICATIONS).

CONNECTING ROD BEARINGS

STANDARD PROCEDURE

CONNECTING ROD - FITTING

(1) For measuring connecting rod bearing clearance procedure and use of Plastigage (Refer to 9 - ENGINE - STANDARD PROCEDURE). For bearing clearance refer to Engine Specifications. (Refer to 9 - ENGINE - SPECIFICATIONS)

NOTE: The rod bearing bolts should not be reused.

(2) Before installing the **NEW** bolts the threads should be oiled with clean engine oil.

(3) Install each bolt finger tight then alternately torque each bolt to assemble the cap properly.

(4) Tighten the bolts to 27 N·m PLUS 1/4 turn (20 ft. lbs. PLUS 1/4 turn) **Do not use a torque wrench for last step.**

(5) Using a feeler gauge, check connecting rod side clearance (Fig. 40). Refer to clearance specifications (Refer to 9 - ENGINE - SPECIFICATIONS).

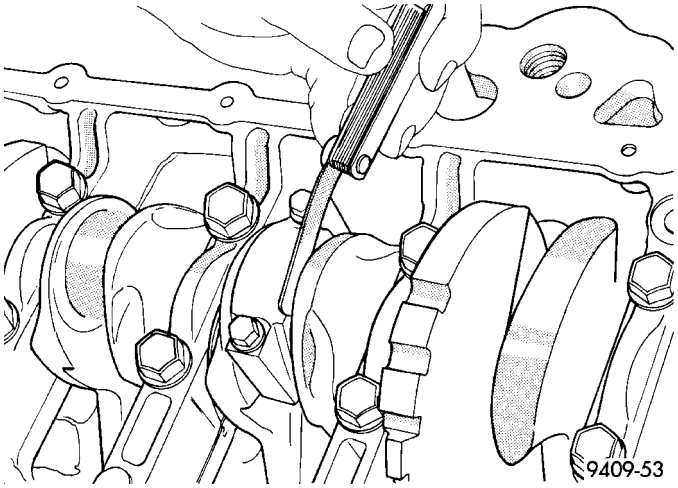


Fig. 40 Connecting Rod Side Clearance

CRANKSHAFT

DESCRIPTION

The crankshaft is made of nodular cast iron and includes five main bearing journals and four connecting rod journals (Fig. 41). The number three journal is the location for the thrust bearing. The mains and connecting rod journals have undercut fillet radiuses that are rolled for added strength. To optimize bearing loading, eight counterweights are used.

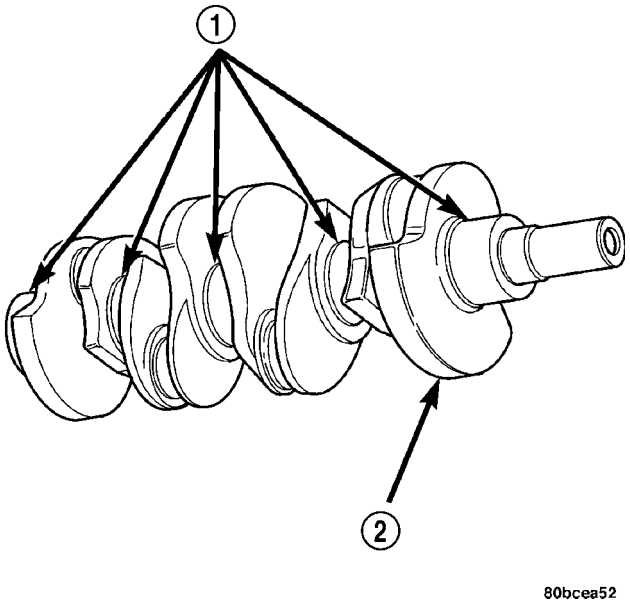


Fig. 41 Crankshaft - Typical

1 - MAIN BEARING JOURNALS
2 - COUNTER BALANCE WEIGHTS

OPERATION

The crankshaft transfers force generated by combustion within the cylinder to the flywheel or flexplate.

STANDARD PROCEDURE - CRANKSHAFT END PLAY

(1) Using Dial Indicator C-3339 and Mounting Post L-4438, attach to front of engine, locating probe perpendicular on nose of crankshaft (Fig. 42).

(2) Move crankshaft all the way to the rear of its travel.

(3) Zero the dial indicator.

(4) Move crankshaft all the way to the front and read the dial indicator. Refer to Engine Specifications.

REMOVAL

NOTE: Crankshaft can not be removed when engine is in vehicle.

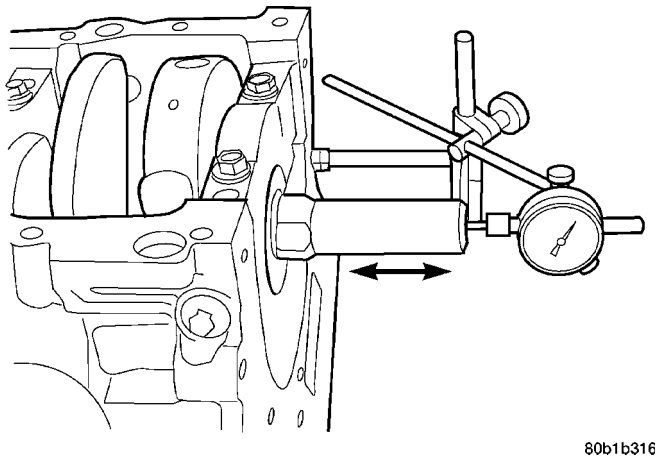


Fig. 42 CHECKING CRANKSHAFT END PLAY

(1) Remove engine assembly from vehicle. (Refer to 9 - ENGINE - REMOVAL)

(2) Remove flex plate and crankshaft rear oil seal.

(3) Mount engine on a repair stand.

(4) Drain engine oil and remove oil filter.

(5) Remove the oil pan. (Refer to 9 - ENGINE/LUBRICATION/OIL PAN - REMOVAL)

(6) Remove the timing belt covers. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT / CHAIN COVER(S) - REMOVAL)

(7) Remove the timing belt. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - REMOVAL)

(8) Remove the oil pump. (Refer to 9 - ENGINE/LUBRICATION/OIL PUMP - REMOVAL)

(9) Remove balance shafts and housing assembly. (Refer to 9 - ENGINE/VALVE TIMING/BALANCE SHAFT - REMOVAL)

(10) Using a permanent ink or paint marker, identify cylinder number on each connecting rod cap (Fig. 43).

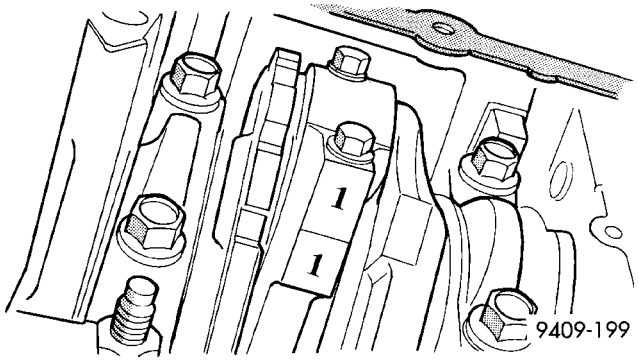


Fig. 43 Identify Connecting Rod to Cylinder-Typical

CAUTION: DO NOT use a number stamp or a punch to mark connecting rods. Damage to connecting rod could occur.

CRANKSHAFT (Continued)

(11) Remove all connecting rod bolts and caps. Care should be taken not to damage the fracture rod and cap surfaces.

NOTE: Do not reuse connecting rod bolts.

(12) Remove all bedplate bolts from the engine block (Fig. 44).

(13) Using a mallet gently tap the bedplate loose from the engine block dowel pins.

CAUTION: Do not pry up on one side of the bedplate. Damage may occur to cylinder block to bedplate alignment and thrust bearing.

(14) Bedplate should be removed evenly from the cylinder block dowel pins to prevent damage to the dowel pins and thrust bearing.

(15) Lift out crankshaft from cylinder block. Do not damage the main bearings or journals when removing the crankshaft.

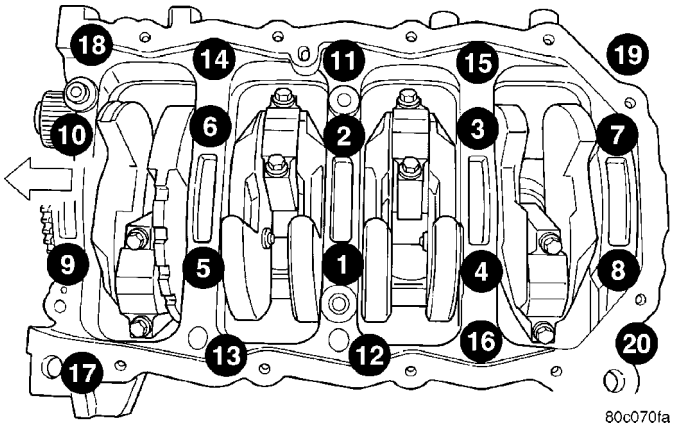


Fig. 44 Bedplate Bolt Tightening Sequence

(16) Remove the target ring mounting screws and discard.

(17) Remove the target ring from the crankshaft.

INSPECTION

The crankshaft journals should be checked for excessive wear, taper and scoring (Fig. 45). Limits of taper or out of round on any crankshaft journals should within specitifications. (Refer to 9 - ENGINE - SPECIFICATIONS) Journal grinding should not exceed 0.305 mm (0.012 in.) under the standard journal diameter. DO NOT grind thrust faces of No. 3 main bearing. DO NOT nick crank pin or bearing fillets. After grinding, remove rough edges from crankshaft oil holes and clean out all passages.

CAUTION: With the nodular cast iron crankshafts, it is important that the final paper or cloth polish be in the same direction as normal rotation in the engine.

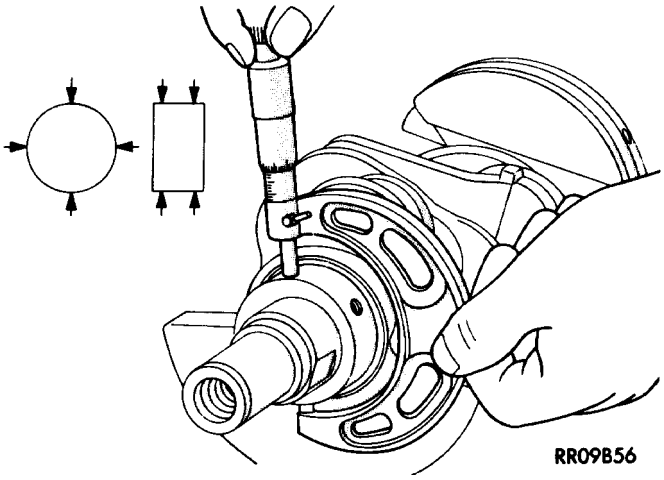


Fig. 45 Crankshaft Journal Measurements

INSTALLATION

(1) Install the main bearing shells with the lubrication groove in the cylinder block (Fig. 46).

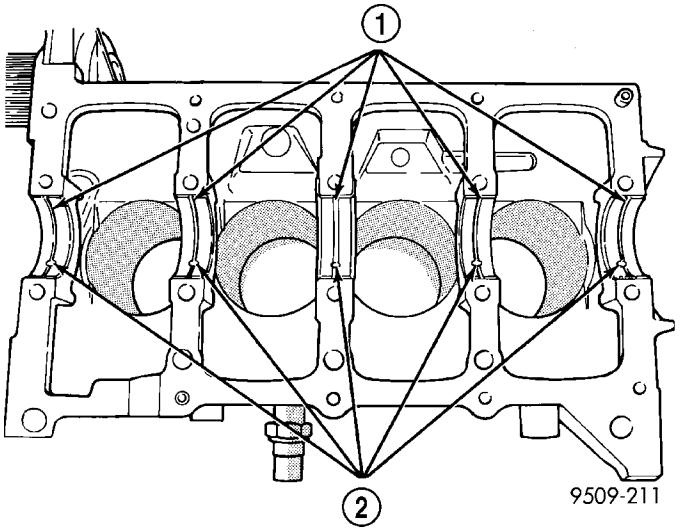


Fig. 46 Installing Main Bearing Upper Shell

1 - LUBRICATION GROOVES
2 - OIL HOLES

(2) Make certain oil holes in block line up with oil hole in bearings and bearing tabs seat in the block tab slots.

CAUTION: Do not get oil on the bedplate mating surface. It will affect the sealer ability to seal the bedplate to cylinder block.

NOTE: If the crankshaft is sent out for machine work, it must be balenced as an assembly with the target ring installed.

(3) Clean crankshaft and target ring with MOPAR® brake parts cleaner and dry with com-

CRANKSHAFT (Continued)

pressed air to ensure that the crankshaft mating surface and target ring mounting holes are free from oil and lock patch debris.

NOTE: Always use NEW mounting screws whether installing original or new target ring.

(4) Install **NEW** mounting screws finger tight starting with the #1 location. (Fig. 47) Make sure engagment occurs with the shoulder of the screw and mounting hole before starting all other screws.

(5) Torque all mounting screws with T30 torx bit to 13 Nm (110 in-lbs) following the torque sequence.

CAUTION: Use extreme care when handling crankshaft. Tone wheel damage can occur if crankshaft is mis-handled.

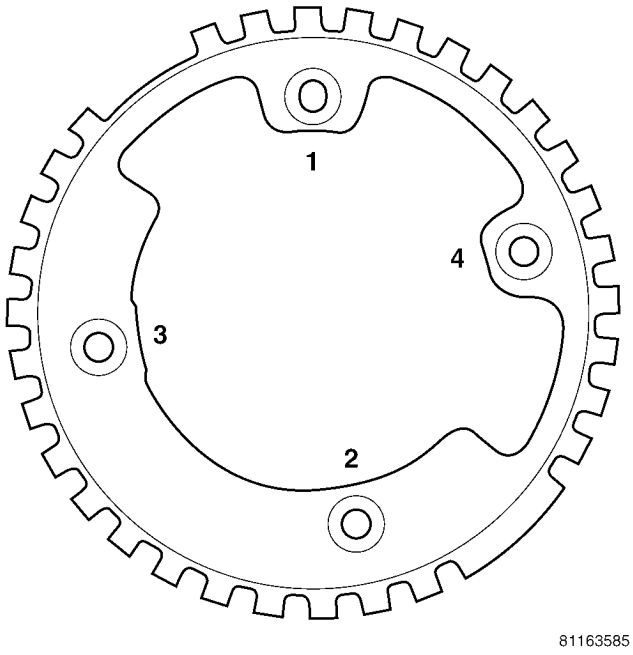


Fig. 47 Target Ring Torque Sequence

(6) Oil the bearings and journals. Install crankshaft.

CAUTION: Use only the specified anaerobic sealer on the bedplate or damage may occur to the engine.

(7) Apply 1.5 to 2.0 mm (0.059 to 0.078 in.) bead of Mopar® Bed Plate Sealant to cylinder block as shown in (Fig. 48).

(8) Install lower main bearings into main bearing cap/bedplate. Make certain the bearing tabs are seated into the bedplate slots. Install the main bearing/bedplate into engine block.

(9) Before installing the bolts the threads should be oiled with clean engine oil, wipe off any excess oil.

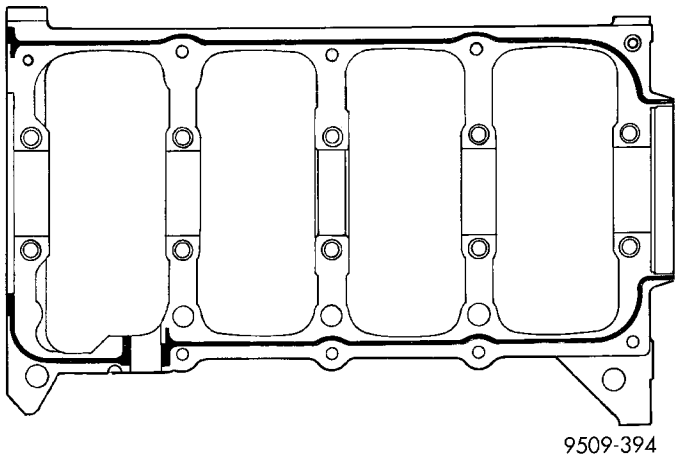


Fig. 48 Bedplate Sealing

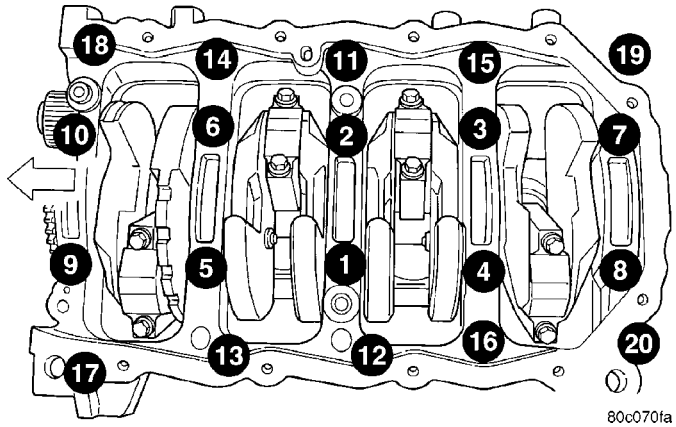


Fig. 49 Bedplate Bolt Torque Sequence

(10) Install main bearing bedplate to engine block bolts 11, 17, and 20 finger tight. Tighten these bolts down together until the bedplate contacts the cylinder block.

(11) To ensure correct thrust bearing alignment, perform the following steps:

- Step 1: Rotate crankshaft until number 4 piston is at TDC.
- Step 2: Move crankshaft rearward to limits of travel.
- Step 3: Then, move crankshaft forward to limits of travel.
- Step 4: Wedge an appropriate tool between the rear of the cylinder block **(NOT BED PLATE)** and the rear crankshaft counterweight. This will hold the crankshaft in it's furthest forward position.
- Step 5: Install and tighten bolts (1–10) in sequence shown in (Fig. 49) to 41 N·m (30 ft. lbs.).
- Step 6: Remove wedge tool used to hold crankshaft.

(12) Tighten bolts (1–10) again to 41 N·m (30 ft. lbs.) +1/4 turn in sequence shown in (Fig. 49).

CRANKSHAFT (Continued)

(13) Install main bearing bedplate to engine block bolts (11–20), and torque each bolt to 28 N·m (20 ft. lbs.) in sequence shown in (Fig. 49).

(14) After the main bearing bedplate is installed, check the crankshaft turning torque. The turning torque should not exceed 5.6 N·m (50 in. lbs.).

(15) Install connecting rod bearings and caps. **Do Not Reuse Connecting Rod Bolts.** Torque connecting rod bolts to 27 N·m (20 ft. lbs.) plus 1/4 turn.

(16) Install balance shafts and housing assembly. (Refer to 9 - ENGINE/VALVE TIMING/BALANCE SHAFT - INSTALLATION)

(17) Install the oil pump and pickup tube. (Refer to 9 - ENGINE/LUBRICATION/OIL PUMP - INSTALLATION)

(18) Install the timing belt rear cover. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT / CHAIN COVER(S) - INSTALLATION)

(19) Install the timing belt. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - INSTALLATION)

(20) Install the timing belt front covers. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT / CHAIN COVER(S) - INSTALLATION)

(21) Install engine support bracket.

(22) Install the oil pan. (Refer to 9 - ENGINE/LUBRICATION/OIL PAN - INSTALLATION)

(23) Install the oil filter.

(24) Install crankshaft rear oil seal. (Refer to 9 - ENGINE/ENGINE BLOCK/CRANKSHAFT OIL SEAL - REAR - INSTALLATION)

(25) Install flex plate. Apply Mopar® Lock & Seal Adhesive to bolt threads and tighten to 95 N·m (70 ft. lbs.).

(26) Install the engine assembly. (Refer to 9 - ENGINE - INSTALLATION)

CRANKSHAFT MAIN BEARINGS

STANDARD PROCEDURE - MAIN BEARING - FITTING

For crankshaft specifications (Refer to 9 - ENGINE - SPECIFICATIONS).

CRANKSHAFT MAIN BEARINGS

The crankshaft is supported in five main bearings. All upper and lower bearing shells in the crankcase have oil grooves. Crankshaft end play is controlled by a flanged bearing on the number three main bearing journal (Fig. 50).

Upper and lower Number 3 bearing halves are flanged to carry the crankshaft thrust loads and are NOT interchangeable with any other bearing halves in the engine (Fig. 50). All bearing cap bolts removed during service procedures are to be cleaned and oiled before installation. Bearing shells are available in standard and the following undersized: 0.025 mm (0.001 in.) and 0.250 mm (0.010 in.). Never install an undersize bearing that will reduce clearance below specifications. Replace or machine the crankshaft as necessary to obtain proper bearing clearances.

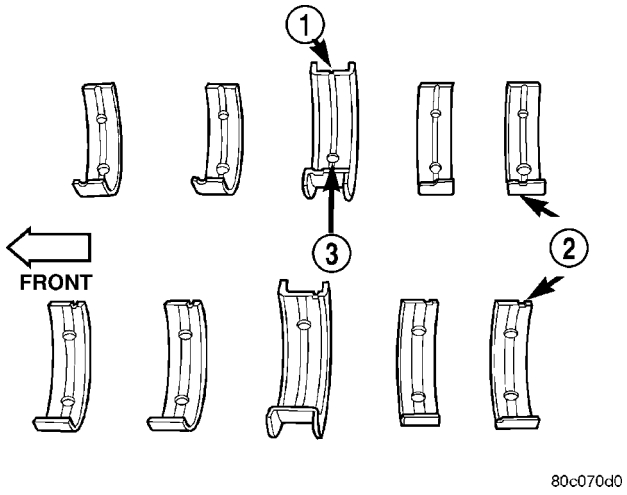


Fig. 50 Main Bearing Identification

1 - OIL GROOVE
2 - MAIN BEARINGS
3 - OIL HOLE

MAIN BEARING INSTALLATION

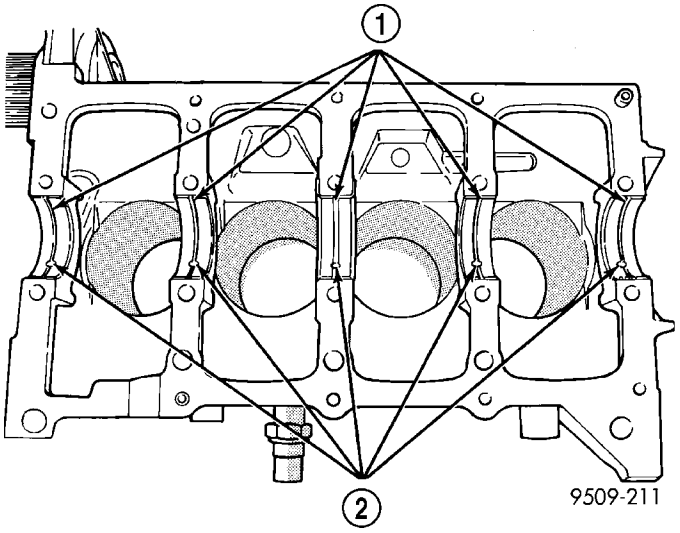


Fig. 51 Installing Main Bearing Upper Shell

1 - LUBRICATION GROOVES
2 - OIL HOLES

(1) Install the main bearing shells with the lubrication groove in the cylinder block (Fig. 51).

(2) Make certain oil holes in block line up with oil holes in bearings. Bearing tabs must seat in the block tab slots.

CRANKSHAFT MAIN BEARINGS (Continued)

CAUTION: Do not get oil on the bedplate mating surface. It will may effect the sealer ability to seal the bedplate to cylinder block.

(3) Oil the bearings and journals and install crankshaft.

CAUTION: Use only the specified anaerobic sealer on the bedplate or damage may occur to the engine. Ensure that both cylinder block and bedplate surfaces are clean.

(4) Apply 1.5 to 2.0 mm (0.059 to 0.078 in.) bead of anaerobic sealer Mopar® Bed Plate Sealant to cylinder block as shown in (Fig. 52).

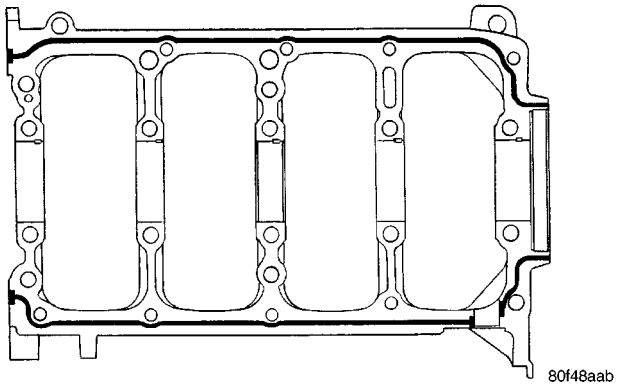


Fig. 52 Bed plate Sealing

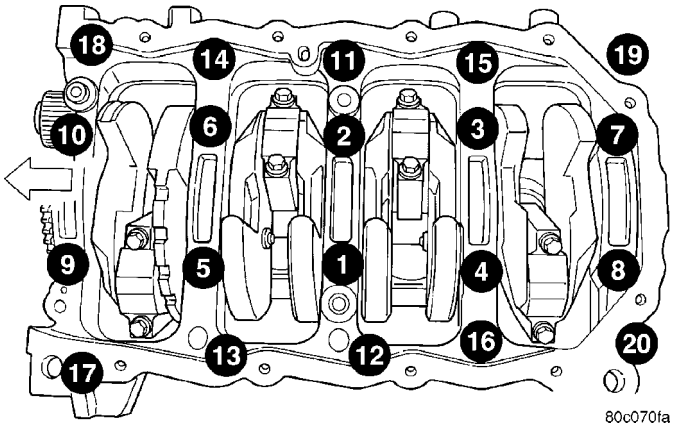


Fig. 53 Main Bearing Caps/Bedplate Tightening Sequence

(5) Install lower main bearings into main bearing cap/bedplate. Make certain the bearing tabs are seated into the bedplate slots.

(6) Position the main bearing/bedplate onto the engine block.

(7) Before installing bolts, lubricate the threads with clean engine oil, wipe off any excess oil.

(8) Install main bearing bedplate to engine block bolts 11, 17 and 20 finger tight. Tighten these bolts down together until the bedplate contacts the cylinder block.

(9) To ensure correct thrust bearing alignment, perform the following steps:

- Step 1: Rotate crankshaft until number 4 piston is at TDC.
- Step 2: Move crankshaft rearward to limits of travel.
- Step 3: Then, move crankshaft forward to limits of travel.
- Step 4: Wedge an appropriate tool between the rear of the cylinder block **(NOT BED PLATE)** and the rear crankshaft counterweight. This will hold the crankshaft in it's furthest forward position.
- Step 5: Install and tighten bolts (1–10) in sequence shown in (Fig. 53) to 41 N·m (30 ft. lbs.).
- Step 6: Remove wedge tool used to hold crankshaft.

(10) Tighten bolts (1–10) again to 41 N·m (30 ft. lbs.) **PLUS** 1/4 turn in sequence shown in (Fig. 53).

(11) Install main bearing bedplate to engine block bolts (11–20), and torque each bolt to 28 N·m (20 ft. lbs.) in sequence shown in (Fig. 53).

(12) After the main bearing bedplate is installed, check the crankshaft turning torque. The turning torque should not exceed 5.6 N·m (50 in. lbs.).

CRANKSHAFT OIL SEAL - FRONT

REMOVAL

(1) Remove the crankshaft vibration damper (Fig. 54). (Refer to 9 - ENGINE/ENGINE BLOCK/VIBRATION DAMPER - REMOVAL)

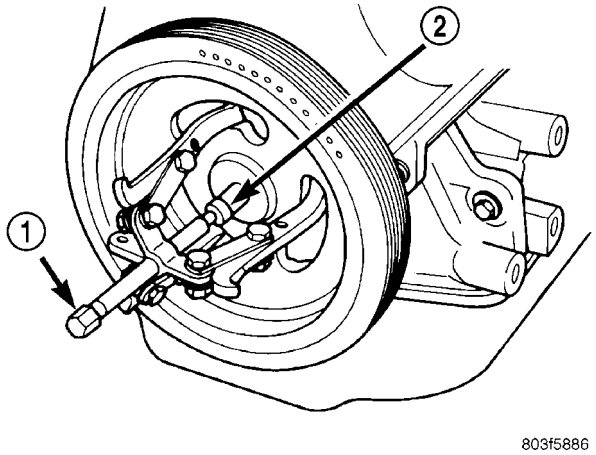


Fig. 54 Crankshaft Vibration Damper - Removal - Typical

1 - SPECIAL TOOL 1026 3-JAW PULLER
2 - SPECIAL TOOL 6827–A INSERT

CRANKSHAFT OIL SEAL - FRONT (Continued)

(2) Remove timing belt. (Refer to 9 - ENGINE/ VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - REMOVAL)

(3) Remove crankshaft sprocket using Special Tool 6793 and insert C-4685-C2 (Fig. 55).

CAUTION: Do not nick shaft seal surface or seal bore.

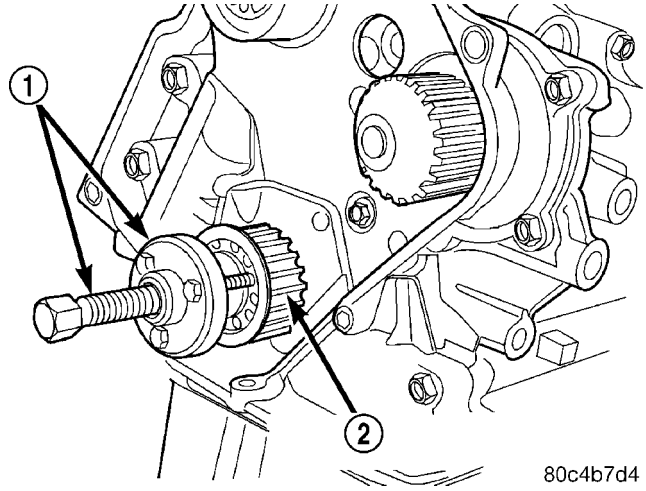


Fig. 55 Crankshaft Sprocket - Removal

1 - SPECIAL TOOL 6793
2 - CRANKSHAFT SPROCKET

(4) Using Tool 6771 to remove front crankshaft oil seal (Fig. 56). Be careful not to damage the seal surface of cover.

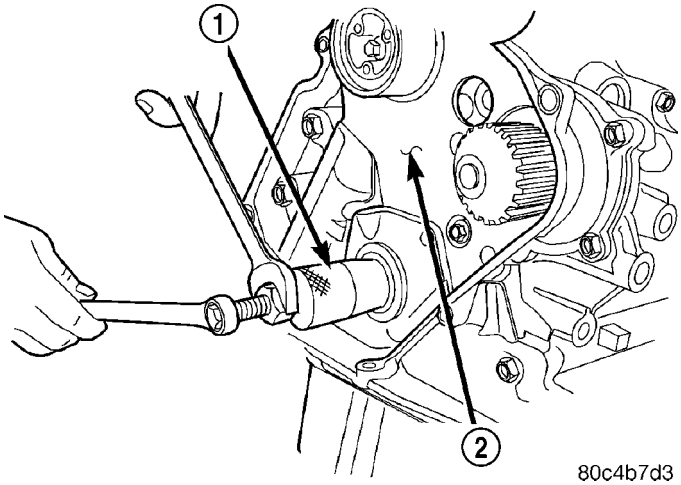


Fig. 56 Front Crankshaft Oil Seal - Removal

1 - SPECIAL TOOL 6771
2 - REAR TIMING BELT COVER

INSTALLATION

(1) Install new seal by using Special Tool 6780 (Fig. 57).

(2) Place seal into opening with seal spring towards the inside of engine. Install seal until flush with cover.

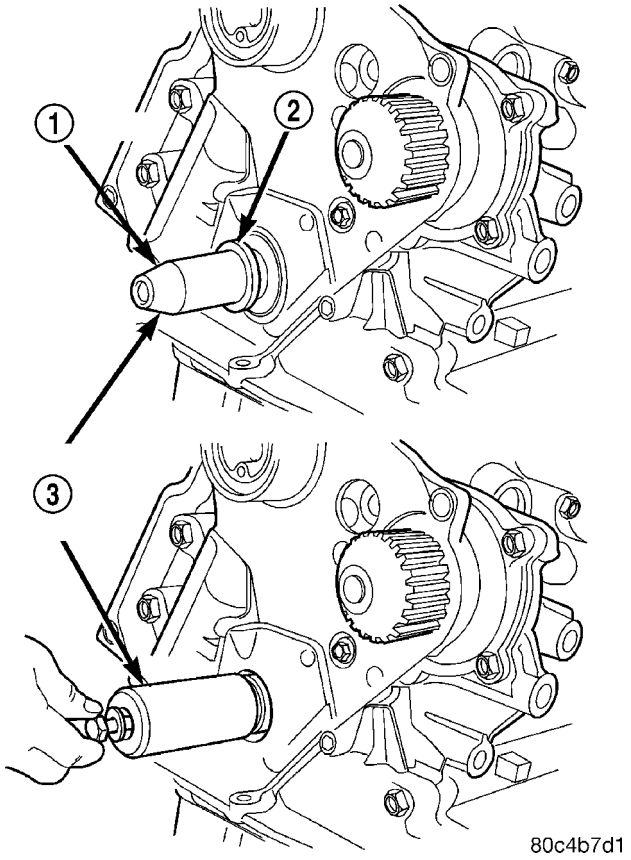


Fig. 57 Crankshaft Front Oil Seal - Installation

1 - PROTECTOR
2 - SEAL
3 - SPECIAL TOOL 6780

CRANKSHAFT OIL SEAL - FRONT (Continued)

(3) Install crankshaft sprocket using Special Tool 6792 (Fig. 58).

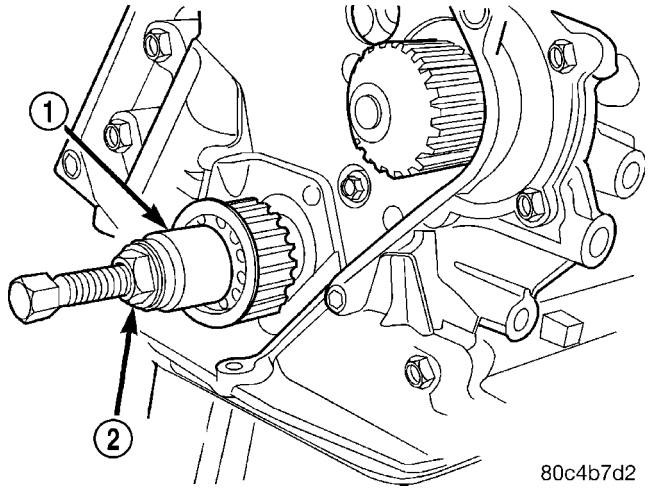


Fig. 58 Crankshaft Sprocket - Installation

1 - SPECIAL TOOL 6792
2 - TIGHTEN NUT TO INSTALL

(4) Install timing belt. (Refer to 9 - ENGINE/ VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - INSTALLATION)

(5) Install crankshaft vibration damper (Fig. 59). (Refer to 9 - ENGINE/ENGINE BLOCK/VIBRATION DAMPER - INSTALLATION)

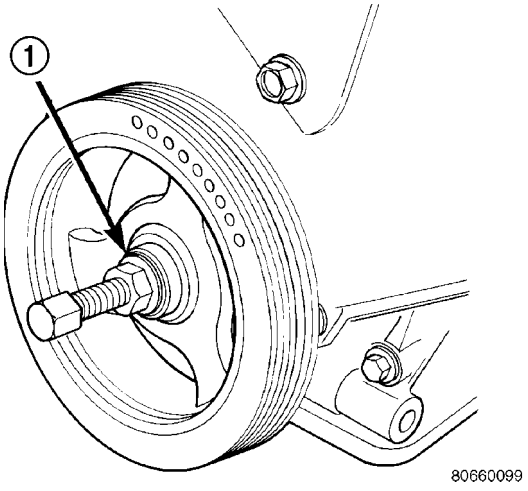


Fig. 59 Crankshaft Vibration Damper - Installation - Typical

1 - M12–1.75 × 150 MM BOLT, WASHER AND THRUST BEARING FROM SPECIAL TOOL 6792

CRANKSHAFT OIL SEAL - REAR

REMOVAL

(1) Remove transmission.
(2) Remove flex plate.
(3) Insert a 3/16 flat bladed screwdriver between the dust lip and the metal case of the crankshaft seal. Angle the screwdriver (Fig. 60)through the dust lip against metal case of the seal. Pry out seal.

CAUTION: Do not permit the screwdriver blade to contact crankshaft seal surface. Contact of the screwdriver blade against crankshaft edge (chamfer) is permitted.

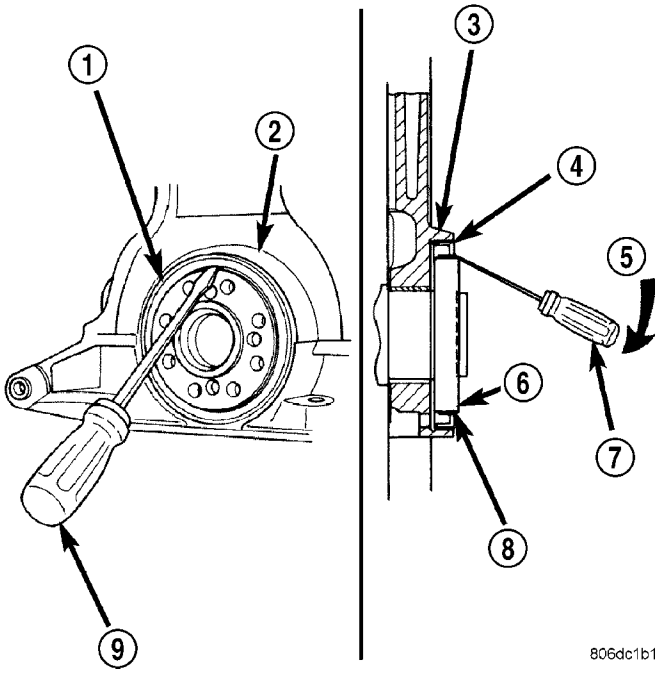


Fig. 60 Rear Crankshaft Oil Seal - Removal

1 - REAR CRANKSHAFT SEAL
2 - ENGINE BLOCK
3 - ENGINE BLOCK
4 - REAR CRANKSHAFT SEAL METAL CASE
5 - PRY IN THIS DIRECTION
6 - CRANKSHAFT
7 - SCREWDRIVER
8 - REAR CRANKSHAFT SEAL DUST LIP
9 - SCREWDRIVER

INSTALLATION

CAUTION: If burr or scratch is present on the crankshaft edge (chamfer), cleanup with 400 grit sand paper to prevent seal damage during installation of new seal.

(1) Lubricate the crankshaft flange with engine oil.
(2) Place Special Tool 6926-1 Seal Guide on crankshaft (Fig. 61).
(3) Position seal over guide tool (Fig. 61). Guide tool should remain on crankshaft during installation of seal. Ensure that the lip of the seal is facing towards the crankcase during installation.

CRANKSHAFT OIL SEAL - REAR (Continued)

CAUTION: If the seal is driven into the block past flush, this may cause an oil leak.

(4) Drive the seal into the block using Special Tool 6926-2 and handle C-4171 (Fig. 62) until the tool bottoms out against the block (Fig. 63).

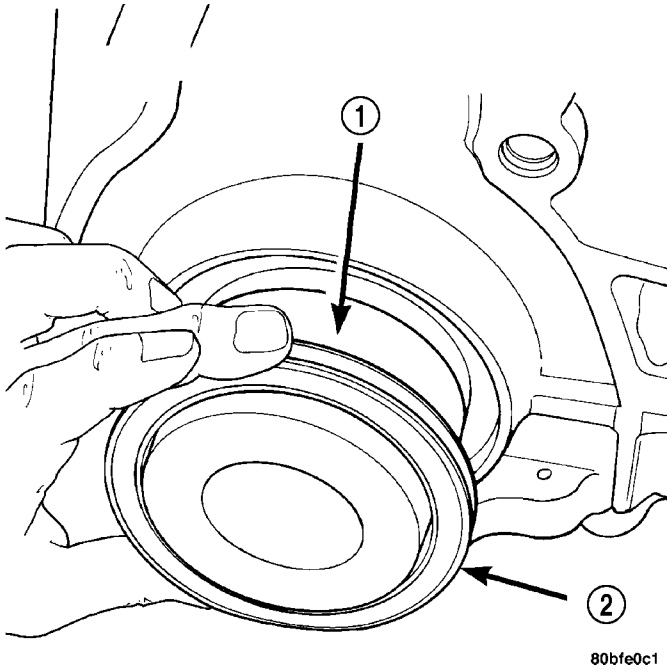


Fig. 61 Rear Crankshaft Seal and Special Tool 6926-1

1 - SPECIAL TOOL 6926–1 PILOT
2 - SEAL

(5) Install flex plate. Apply Mopar® Lock & Seal Adhesive to bolt threads and tighten bolts to 95 N·m (70 ft. lbs.).

(6) Install the transmission.

PISTON & CONNECTING ROD

DESCRIPTION

The pistons are made of a cast aluminum alloy. The pistons have pressed-in pins attached to forged powdered metal connecting rods. The pistons pin is offset 1 mm (0.0394 in.) towards the thrust side of the piston. The connecting rods are a cracked cap design and are not repairable. Hex head cap screws are used to provide alignment and durability in the assembly. The pistons and connecting rods are serviced as an assembly.

OPERATION

The piston and connecting rod is the link between the combustion force to the crankshaft.

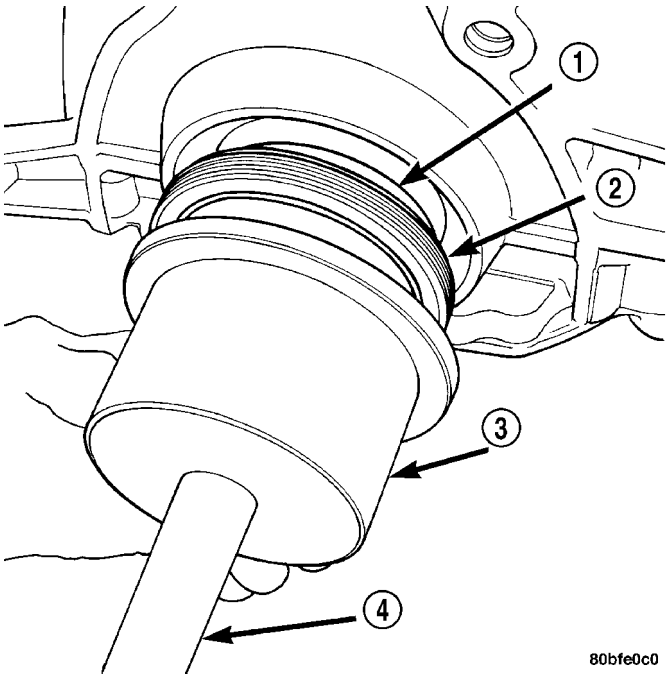


Fig. 62 Crankshaft Seal and Special Tools 6926-2 & C-4171

1 - SPECIAL TOOL 6926–1 PILOT
2 - SEAL
3 - SPECIAL TOOL 6926–2 INSTALLER
4 - SPECIAL TOOL C-4171

REMOVAL

NOTE: Cylinder Head must be removed before Pistons and Rods. Refer to Cylinder Head Removal in this section.

(1) Remove top ridge of cylinder bores with a reliable ridge reamer before removing pistons from cylinder block. **Be sure to keep tops of pistons covered during this operation**. Mark piston with matching cylinder number (Fig. 64).

(2) Remove oil pan. Scribe the cylinder number on the side of the rod and cap (Fig. 65) for identification.

(3) Pistons have a directional stamping in the front half of the piston facing towards the **front** of engine.

(4) Pistons and connecting rods must be removed from top of cylinder block. Rotate crankshaft so that each connecting rod is centered in cylinder bore.

(5) Remove Balance Shaft Assembly. Refer to Balance Shaft Removal in this section.

(6) Remove connecting rod cap bolts. Push each piston and rod assembly out of cylinder bore.

NOTE: Be careful not to nick crankshaft journals.

(7) After removal, install bearing cap on the mating rod.

(8) Piston and Rods are serviced as an assembly.

PISTON & CONNECTING ROD (Continued)

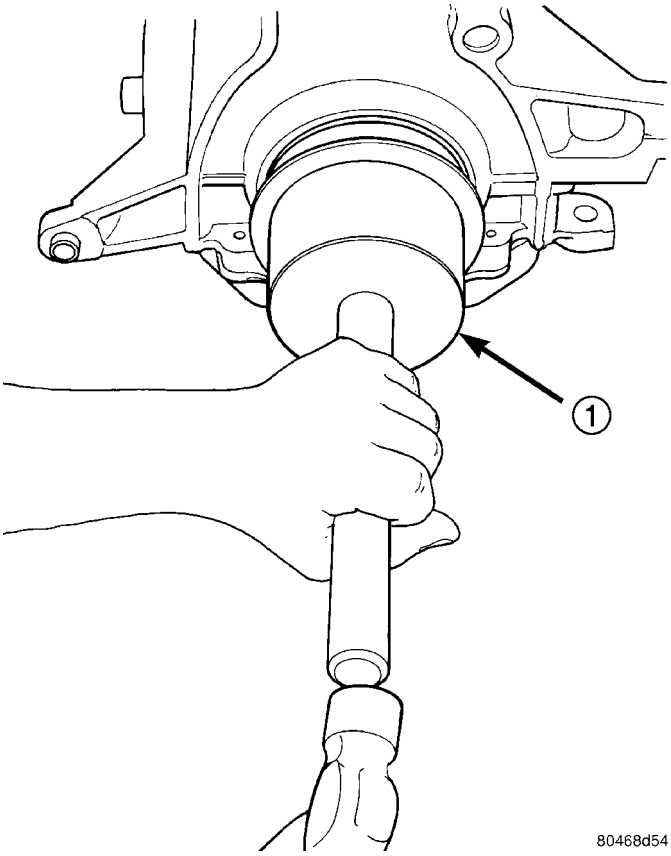


Fig. 63 Rear Crankshaft Seal - Installation

1 - SPECIAL TOOL 6926–2 INSTALLER

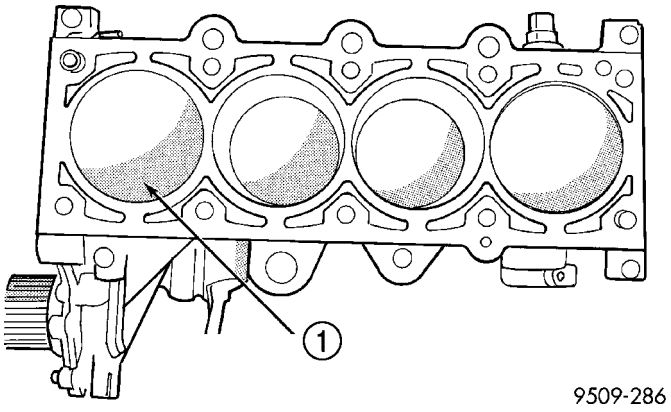


Fig. 64 Piston Markings

1 - DIRECTIONAL ARROW WILL BE IMPRINTED IN THIS AREA

INSTALLATION

(1) Before installing pistons and connecting rod assemblies into the bore, be sure that compression ring gaps are staggered so that neither is in line with oil ring rail gap.

(2) Before installing the ring compressor, make sure the oil ring expander ends are butted and the rail gaps located as shown in (Fig. 66). As viewed from top.

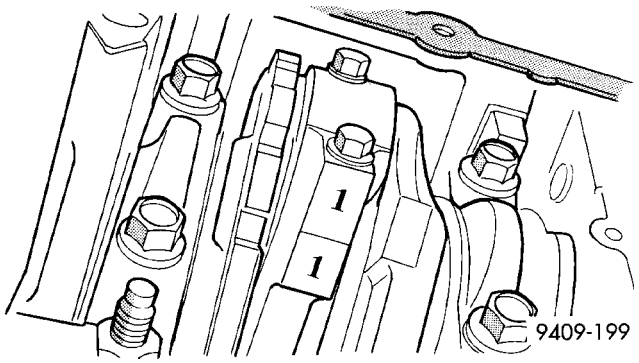


Fig. 65 Identify Connecting Rod to Cylinder

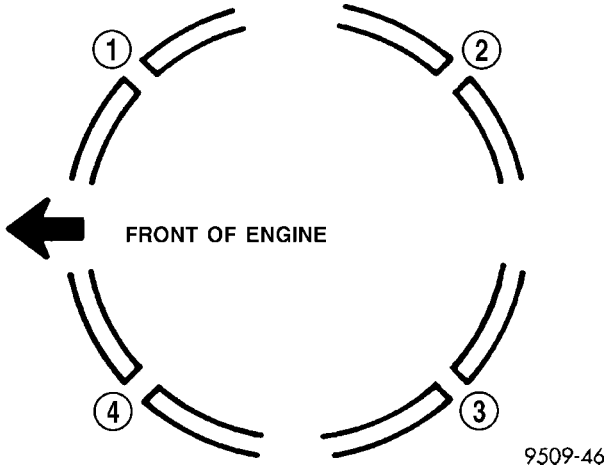


Fig. 66 Piston Ring End Gap Position

1 - GAP OF LOWER SIDE RAIL
2 - NO. 1 RING GAP
3 - GAP OF UPPER SIDE RAIL
4 - NO. 2 RING GAP AND SPACER EXPANDER GAP

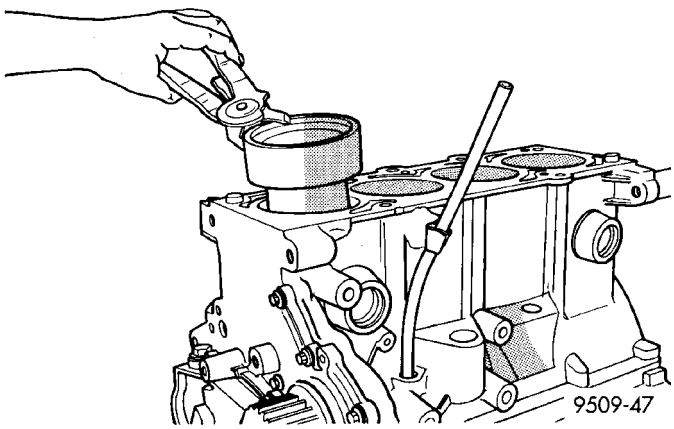


Fig. 67 Piston - Installation

(3) Immerse the piston head and rings in clean engine oil, slide the ring compressor, over the piston (Fig. 67). **Be sure position of rings does not change during this operation** .

(4) The directional stamp on the piston should face toward the front of the engine.

PISTON & CONNECTING ROD (Continued)

(5) Rotate crankshaft so that the connecting rod journal is on the center of the cylinder bore. Insert rod and piston assembly into cylinder bore and guide rod over the crankshaft journal.

(6) Tap the piston down in cylinder bore, using a hammer handle. At the same time, guide connecting rod into position on connecting rod journal.

NOTE: The connecting rod cap bolts should not be reused.

(7) Before installing the **NEW** bolts the threads should be coated with clean engine oil.

(8) Install each bolt finger tight than alternately torque each bolt to assemble the cap properly.

CAUTION: Do not use a torque wrench for second part of last step.

(9) Tighten the bolts to 54 N·m PLUS 1/4 turn (40 ft. lbs. PLUS 1/4 turn).

(10) Using a feeler gauge, check connecting rod side clearance (Fig. 68).

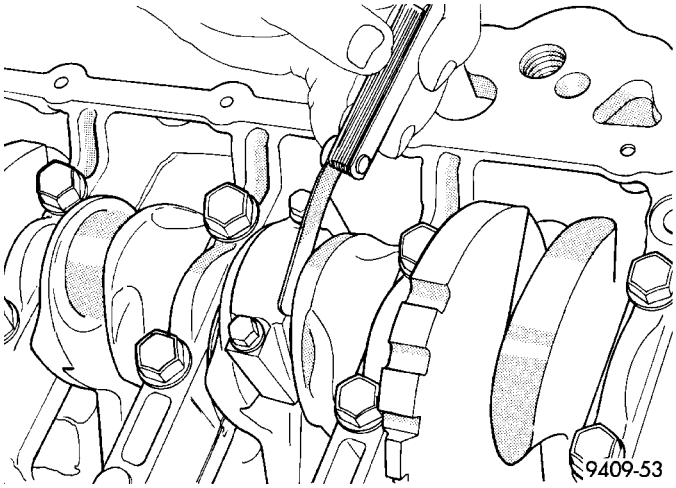


Fig. 68 Checking Connecting Rod Side Clearance

PISTON RINGS

STANDARD PROCEDURE

PISTON RING - FITTING

(1) Wipe cylinder bore clean. Insert ring and push down with piston to ensure it is square in bore. The ring gap measurement must be made with the ring positioning at least 12 mm (0.50 inch) from bottom of cylinder bore. Check gap with feeler gauge (Fig. 69). Refer to Engine Specifications.

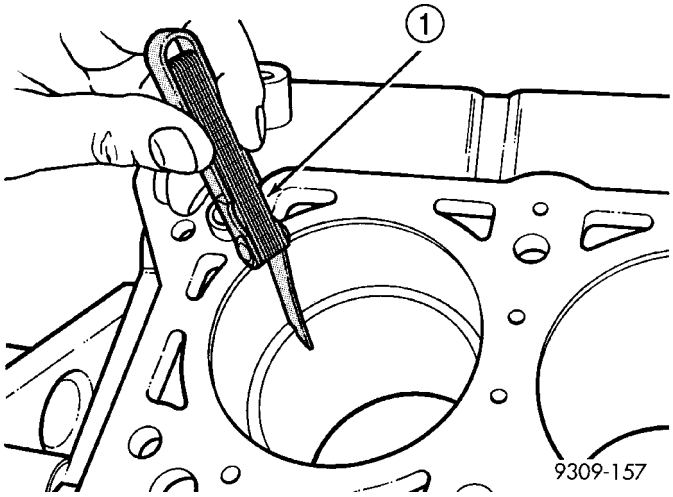


Fig. 69 Piston Ring Gap

1 - FEELER GAUGE

(2) Check piston ring to groove side clearance (Fig. 70). Refer to Engine Specifications.

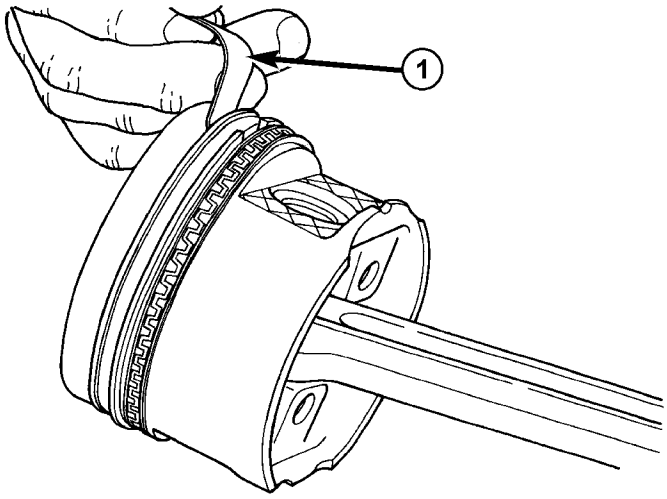


Fig. 70 Piston Ring Side Clearance

1 - FEELER GAUGE

PISTON RINGS (Continued)

PISTON RINGS - INSTALLATION

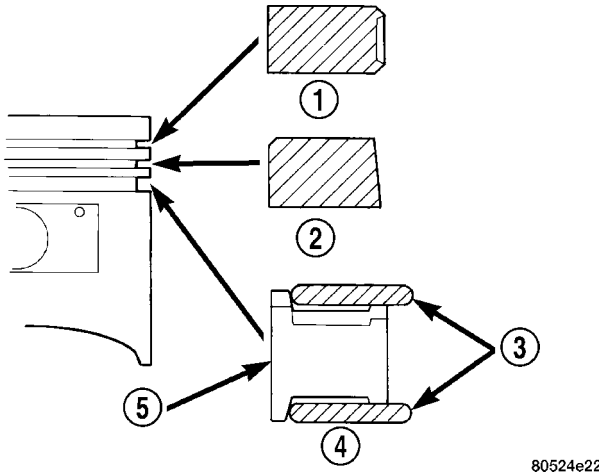


Fig. 71 Piston Ring Installation

1 - NO. 1 PISTON RING
2 - NO. 2 PISTON RING
3 - SIDE RAIL
4 - OIL RING
5 - SPACER EXPANDER

(1) Install rings with manufacturers I.D. mark facing up, to the top of the piston (Fig. 71).

CAUTION: Install piston rings in the following order:

a. Oil ring expander.
b. Upper oil ring side rail.
c. Lower oil ring side rail.
d. No. 2 Intermediate piston ring.
e. No. 1 Upper piston ring.

(2) Install the side rail by placing one end between the piston ring groove and the expander. Hold end firmly and press down the portion to be installed until side rail is in position. **Do not use a piston ring expander** (Fig. 72).

(3) Install upper side rail first and then the lower side rail.

(4) Install No. 2 piston ring and then No. 1 piston ring.

(5) Position piston ring end gaps as shown in (Fig. 73).

(6) Position oil ring expander gap at least 45° from the side rail gaps but **not** on the piston pin center or on the thrust direction. Staggering ring gap is important for oil control.

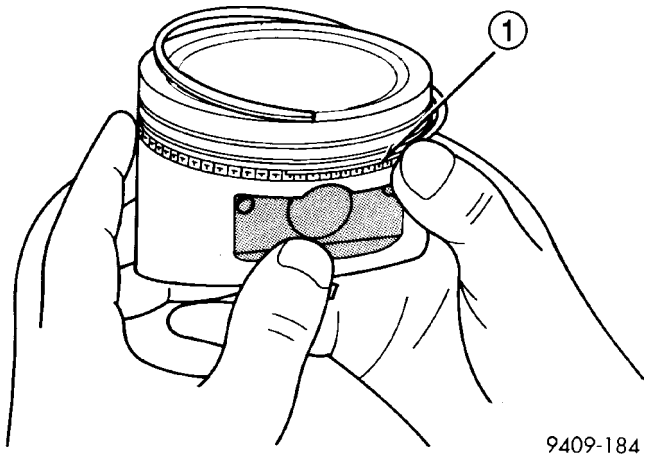


Fig. 72 Installing Side Rail - Typical

1 - SIDE RAIL END

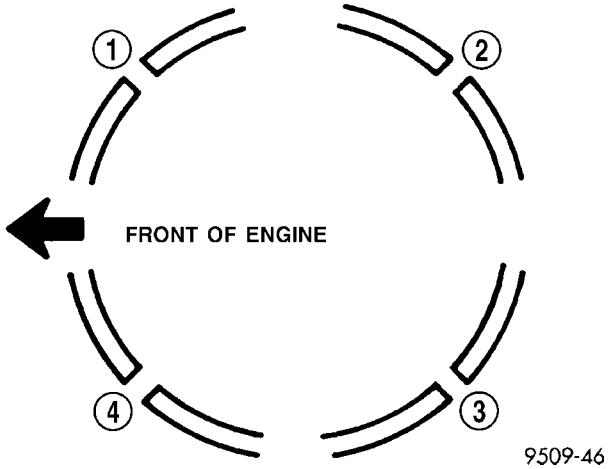


Fig. 73 Piston

1 - GAP OF LOWER SIDE RAIL
2 - NO. 1 RING GAP
3 - GAP OF UPPER SIDE RAIL
4 - NO. 2 RING GAP AND SPACER EXPANDER GAP

VIBRATION DAMPER

REMOVAL

(1) Remove accesory drive belts. (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - REMOVAL)

(2) Remove crankshaft damper bolt.

(3) Remove damper using Special Tool 3-Jaw Puller 1026 and Insert 6827–A (Fig. 74).

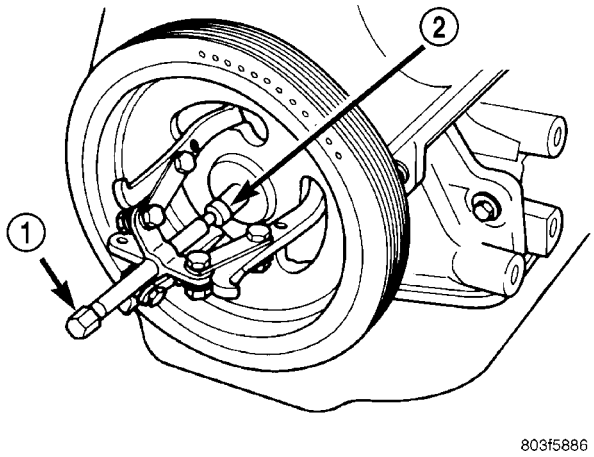


Fig. 74 Crankshaft Vibration Damper - Removal - Typical

1 - SPECIAL TOOL 1026 3-JAW PULLER
2 - SPECIAL TOOL 6827–A INSERT

INSTALLATION

(1) Install crankshaft vibration damper using M12 1.75 x 150 mm bolt, washer, thrust bearing and nut from Special Tool 6792 (Fig. 75).

(2) Install crankshaft vibration damper bolt and tighten to 142 N·m (105 ft. lbs.).

(3) Install accessory drive belts. (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - INSTALLATION)

STRUCTURAL COLLAR

REMOVAL

(1) Raise vehicle on hoist.

(2) Remove structural collar attaching bolts.

(3) Remove collar.

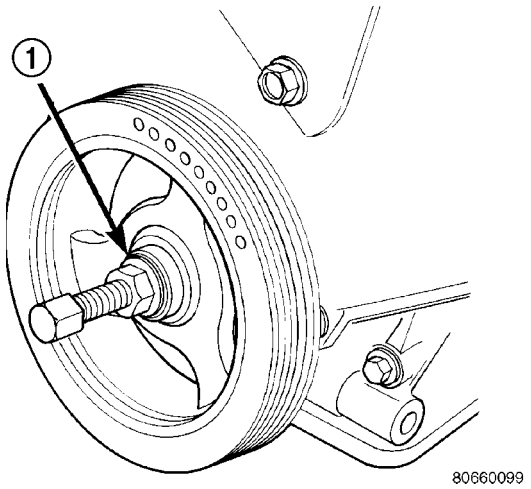


Fig. 75 Crankshaft Vibration Damper - Installation - Typical

1 - M12–1.75 × 150 MM BOLT, WASHER AND THRUST BEARING FROM SPECIAL TOOL 6792

INSTALLATION

CAUTION: Torque procedure for the structural collar must be followed or damage could occur to oil pan and collar.

(1) Perform the following steps for installing structural collar.

- Step 1: Position collar between transmission and oil pan. Install collar to transmission bolts, **hand start only**.
- Step 2: Install collar to oil pan bolts, **hand snug only.**
- Step 3: Tighten collar to transmission bolts.
- Step 4: Tighten collar to oil pan bolts.

(2) Lower vehicle.

ENGINE MOUNTING

DESCRIPTION

The engine mounting system consist of three mounts; right and a left side support the powertrain, and rear mount to control powertrain torque. The mounts are of molded rubber material.

FRONT MOUNT

REMOVAL

(1) Raise vehicle on hoist.
(2) Remove the front engine mount through bolt from the insulator.
(3) Remove the engine front mount bolts and remove the insulator assembly.
(4) Remove the front mounting bracket from engine, if necessary.

INSTALLATION

(1) Install the insulator mount assembly (Fig. 76)and (Fig. 77).

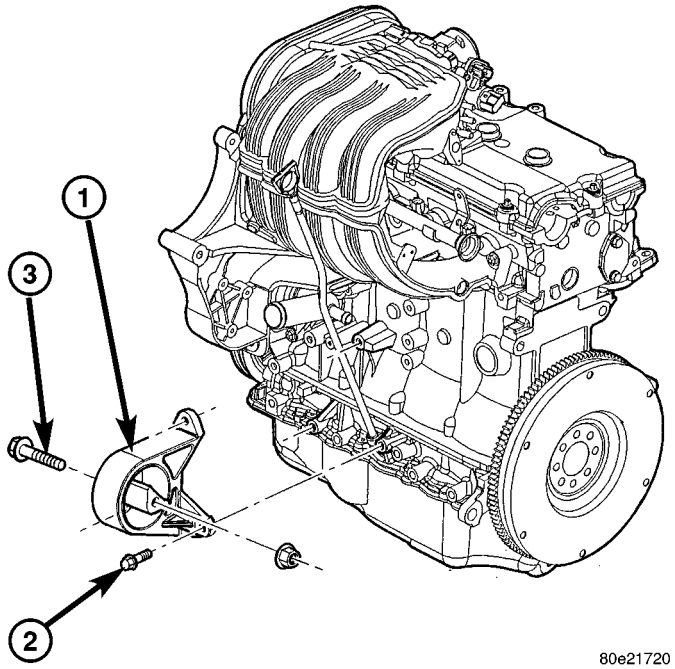


Fig. 76 LH ENGINE MOUNT

1 - ENGINE MOUNT
2 - ENGINE MOUNT BOLT (3)
3 - ENGINE MOUNT THROUGH BOLT

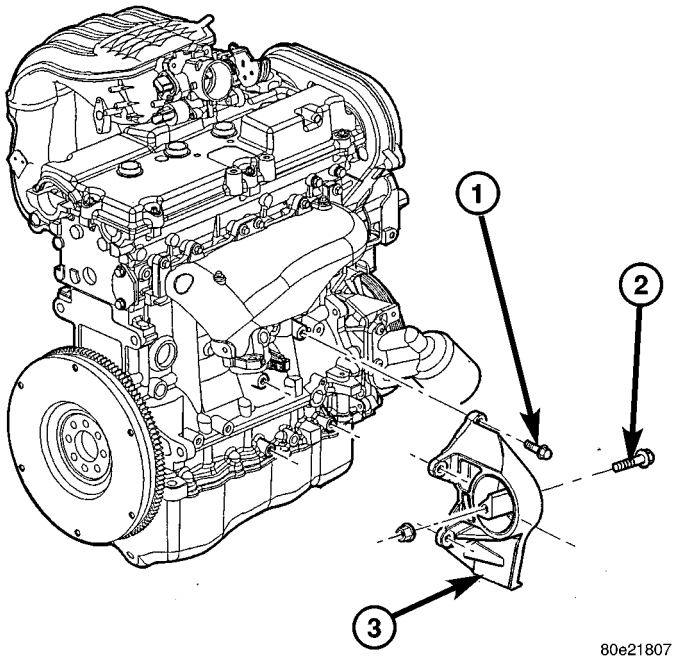


Fig. 77 RH ENGINE MOUNT

1 - ENGINE MOUNT BOLT (4)
2 - ENGINE MOUNT THROUGH BOLT
3 - ENGINE MOUNT

(2) Tighten the mount to engine bolts.
(3) Loosely install the front engine mount through bolt to the insulator.
(4) Lower the engine.
(5) Tighten the through bolt.
(6) Lower the vehicle.

REAR MOUNT

REMOVAL

NOTE: A resilient rubber cushion supports the transmission at the rear between the transmission extension housing and the rear support crossmember or skid plate.

(1) Disconnect negative cable from battery.
(2) Raise the vehicle and support the transmission.
(3) Remove the nuts holding the support cushion to the crossmember. Remove the crossmember.
(4) Remove the support cushion nuts and remove the cushion.
(5) Remove the transmission support bracket bolts and remove the bracket from the transmission.

INSTALLATION

(1) Install the support cushion to the transmission (Fig. 78)or (Fig. 79). Install the bolts and tighten.

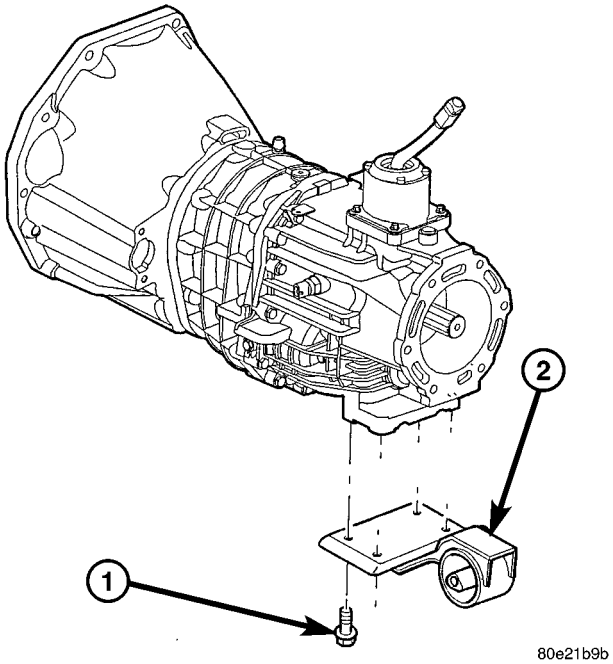


Fig. 78 TRANSMISSION MOUNT - 2.4L MANUAL TRANS

1 - TRANSMISSION MOUNT
2 - MOUNTING BOLT

(2) Position the crossmember in the vehicle. Install the crossmember to mount through bolt and nut.
(3) Install crossmember-to-sill bolts and tighten to 41 N·m (30 ft. lbs.) torque.
(4) Remove the transmission support.
(5) Lower the vehicle.
(6) Connect negative cable to battery.

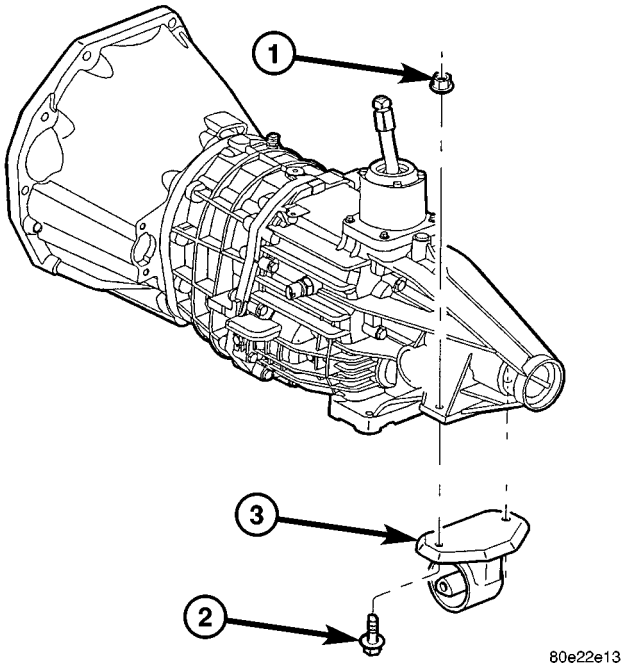


Fig. 79 TRANSMISSION MOUNT - 3.7L MANUAL TRANS 2WD

1 - NUT
2 - BOLT
3 - TRANS MOUNT

LUBRICATION

DESCRIPTION

The lubrication system is a full-flow filtration, pressure feed type. The oil pump is mounted in the front engine cover and driven by the crankshaft.

OPERATION

Engine oil drawn up through the pickup tube and is pressurized by the oil pump and routed through the full-flow filter to the main oil gallery running the length of the cylinder block. A diagonal hole in each bulkhead feeds oil to each main bearing. Drilled passages within the crankshaft route oil from main bearing journals to connecting rod journals. Balance shaft lubrication is provided through an oil passage from the number one main bearing cap through the balance shaft carrier support leg. This passage directly supplies oil to the front bearings and internal machined passages in the shafts that routes oil from front to the rear shaft bearing journals. A vertical hole at the number five bulkhead routes pressurized oil through a restrictor (integral to the cylinder head gasket) up past a cylinder head bolt to an oil gallery running the length of the cylinder head. The camshaft journals are partially slotted to allow a predetermined amount of pressurized oil to pass into the bearing cap cavities. Lubrication of the camshaft

LUBRICATION (Continued)

lobes are provided by small holes in the camshaft bearing caps that are directed towards each lobe. Oil returning to the pan from pressurized components supplies lubrication to the valve stems. Cylinder bores and wrist pins are splash lubricated from directed slots on the connecting rod thrust collars.

DIAGNOSIS AND TESTING

ENGINE OIL PRESSURE CHECKING

(1) Disconnect and remove oil pressure switch. (Refer to 9 - ENGINE/LUBRICATION/OIL PRESSURE SENSOR/SWITCH - REMOVAL)

(2) Install Special Tools C-3292 Gauge with 8406 Adaptor fitting.

(3) Start engine and record oil pressure. Refer to Specifications for correct oil pressure requirements. (Refer to 9 - ENGINE - SPECIFICATIONS)

CAUTION: If oil pressure is 0 at idle, do not perform the 3000 RPM test

(4) If oil pressure is 0 at idle. Shut off engine, check for pressure relief valve stuck open, a clogged oil pick-up screen or a damaged oil pick-up tube O-ring.

(5) After test is complete, remove test gauge and fitting.

(6) Install oil pressure switch and connector. (Refer to 9 - ENGINE/LUBRICATION/OIL PRESSURE SENSOR/SWITCH - INSTALLATION)

OIL

STANDARD PROCEDURE

ENGINE OIL LEVEL CHECK

The best time to check engine oil level is after it has sat overnight, or if the engine has been running, allow the engine to be shut off for at least 5 minutes before checking oil level.

Checking the oil while the vehicle is on level ground will improve the accuracy of the oil level reading. Remove dipstick and observe oil level. Add oil only when the level is at or below the ADD mark (Fig. 80).

ENGINE OIL AND FILTER CHANGE

Change engine oil at mileage and time intervals described in the Maintenance Schedule. (Refer to LUBRICATION & MAINTENANCE/MAINTENANCE SCHEDULES - DESCRIPTION)

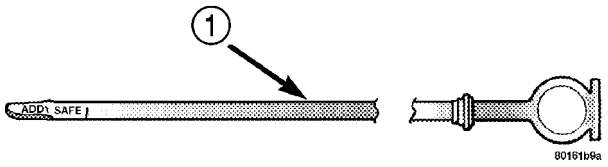


Fig. 80 Oil Level

1 - ENGINE OIL LEVEL DIPSTICK

WARNING: NEW OR USED ENGINE OIL CAN BE IRRITATING TO THE SKIN. AVOID PROLONGED OR REPEATED SKIN CONTACT WITH ENGINE OIL. CONTAMINANTS IN USED ENGINE OIL, CAUSED BY INTERNAL COMBUSTION, CAN BE HAZARDOUS TO YOUR HEALTH. THOROUGHLY WASH EXPOSED SKIN WITH SOAP AND WATER. DO NOT WASH SKIN WITH GASOLINE, DIESEL FUEL, THINNER, OR SOLVENTS, HEALTH PROBLEMS CAN RESULT. DO NOT POLLUTE, DISPOSE OF USED ENGINE OIL PROPERLY. CONTACT YOUR DEALER OR GOVERNMENT AGENCY FOR LOCATION OF COLLECTION CENTER IN YOUR AREA.

Run engine until achieving normal operating temperature.

(1) Position the vehicle on a level surface and turn engine off.

(2) Hoist and support vehicle on safety stands. Refer to Hoisting and Jacking Recommendations. (Refer to LUBRICATION & MAINTENANCE/HOISTING - STANDARD PROCEDURE)

(3) Remove oil fill cap.

(4) Place a suitable drain pan under crankcase drain.

(5) Remove drain plug from crankcase and allow oil to drain into pan. Inspect drain plug threads for stretching or other damage. Replace drain plug and gasket if damaged.

(6) Remove oil filter. (Refer to 9 - ENGINE/LUBRICATION/OIL FILTER - REMOVAL)

(7) Install and tighten drain plug in crankcase.

(8) Install new oil filter. (Refer to 9 - ENGINE/LUBRICATION/OIL FILTER - INSTALLATION)

(9) Lower vehicle and fill crankcase with specified type and amount of engine oil. (Refer to LUBRICATION & MAINTENANCE/FLUID TYPES - DESCRIPTION)

(10) Install oil fill cap.

(11) Start engine and inspect for leaks.

(12) Stop engine and inspect oil level.

OIL (Continued)

NOTE: Care should be exercised when disposing used engine oil after it has been drained from a vehicle engine. Refer to the WARNING listed above.

OIL FILTER

DESCRIPTION

The engine oil filter is a high quality full-flow, disposable type. Replace the oil filter with a Mopar® or the equivalent.

REMOVAL

(1) Raise vehicle on hoist.
(2) Position an oil collecting container under oil filter location.

CAUTION: When servicing the oil filter avoid deforming the filter can by installing the remove/install tool band strap against the can to base lock seam. The lock seam joining the can to the base is reinforced by the base plate.

(3) Using a suitable filter wrench, turn oil filter counterclockwise to remove (Fig. 81).

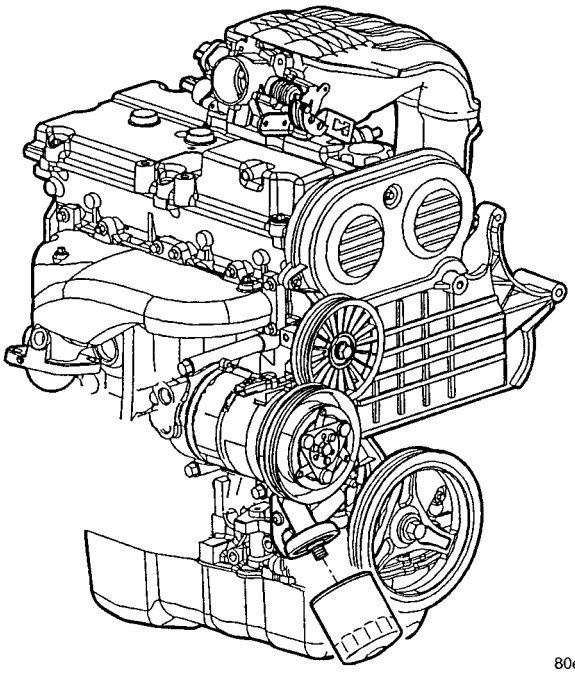


Fig. 81 2.4 OIL FILTER

INSTALLATION

(1) Clean and check filter mounting surface. The surface must be smooth, flat and free of debris or pieces of gasket.
(2) Lubricate new oil filter gasket with clean engine oil.
(3) Screw oil filter on until the gasket contacts base. Tighten to 21 N·m (15 ft. lbs.).

OIL PAN

REMOVAL

(1) Remove air cleaner assembly.
(2) Raise vehicle on hoist and drain engine oil.
(3) Loosen the engine mount thru bolts.
(4) Disconnect exhaust pipe at manifold.
(5) Remove structural collar, if equipped.
(6) Remove front axle mounting bolts, and lower axle as far possible, if equipped.
(7) Position Special Tool 8534 on fender lip and align the slots in the brackets with the fender mounting holes.
(8) Secure brackets to the fender using four M6 X 1.0 X 25 MM flanged cap screws.
(9) Tighten the thumbscrews to secure the sleeves to the support tube.
(10) Secure the support tube in an upright position.
(11) Assemble the flat washer, thrust bearing, hook and T handle.
(12) Using the M10 X 1.5 X 40 mm capscrew supplied with the support fixture, secure the chain to the front cover and the hook.
(13) Support engine as needed.
(14) Remove oil pan attaching bolts.
(15) Remove oil pan.
(16) Clean oil pan and all gasket surfaces.

OIL PAN (Continued)

INSTALLATION

(1) Install the oil pan gasket to the block.

(2) Apply a 3MM (1/8 inch) bead of Mopar® Engine RTV at the oil pump to engine block parting line (Fig. 82).

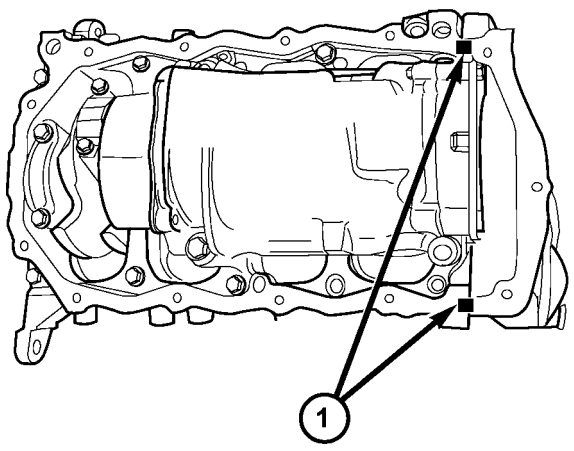


Fig. 82 OIL PAN GASKET INSTALLATION

1 - SEALER LOCATION

(3) Install pan and tighten the screws to 12 N·m (105 in. lbs.).

(4) Lower engine, and remove Special Tool 8534.

(5) Tighten engine mount thru bolts.

(6) Raise the front axle into position, and reinstall front axle mounting bolts. If equipped.

(7) Reconnect exhaust pipe to manifold.

(8) Install structural collar, if equipped.

(9) Lower vehicle.

(10) Fill engine crankcase with proper oil to correct level.

(11) Reinstall air cleaner assembly.

OIL PRESSURE SENSOR/ SWITCH

DESCRIPTION

The 3–wire, electrical/mechanical engine oil pressure sensor (sending unit) is located in an engine oil pressure gallery.

OPERATION

The oil pressure sensor uses three circuits. They are:

- A 5 volt power supply from the Powertrain Control Module (PCM)
- A sensor ground through the PCM's sensor return
- A signal to the PCM relating to engine oil pressure

The oil pressure sensor has a 3 wire electrical function very much like the Manifold Absolute Pressure (MAP) sensor. Meaning different pressures relate to different output voltages.

A 5 volt supply is sent to the sensor from the PCM to power up the sensor. The sensor returns a voltage signal back to the PCM relating to engine oil pressure. This signal is then transferred (bussed) to the instrument panel on either a CCD or PCI bus circuit (depending on vehicle line) to operate the oil pressure gauge and the check gauges lamp. Ground for the sensor is provided by the PCM through a low-noise sensor return.

OIL PUMP

REMOVAL

(1) Disconnect negative cable from battery.

(2) Remove timing belt. (Refer to 9 - ENGINE/ VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - REMOVAL)

(3) Remove timing belt rear cover. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT / CHAIN COVER(S) - REMOVAL)

(4) Remove oil pan. (Refer to 9 - ENGINE/LUBRICATION/OIL PAN - REMOVAL)

(5) Remove crankshaft sprocket using Special Tools 6793 and C-4685-C2 (Fig. 83).

(6) Remove crankshaft key (Fig. 84).

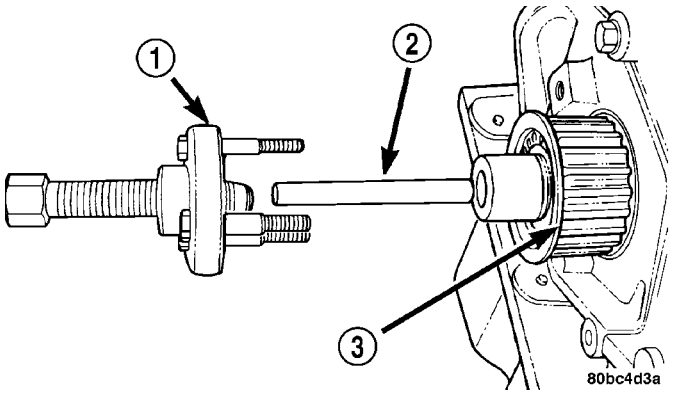


Fig. 83 Crankshaft Sprocket - Removal

1 - SPECIAL TOOL 6793
2 - SPECIAL TOOL C-4685–C2
3 - CRANKSHAFT SPROCKET

(7) Remove oil pick-up tube.

(8) Remove oil pump (Fig. 85) and front crankshaft seal.

OIL PUMP (Continued)

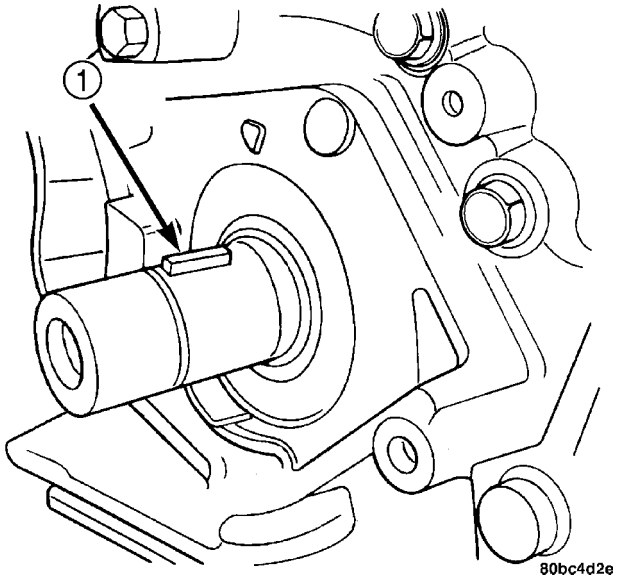


Fig. 84 Crankshaft Key

1 - CRANKSHAFT KEY

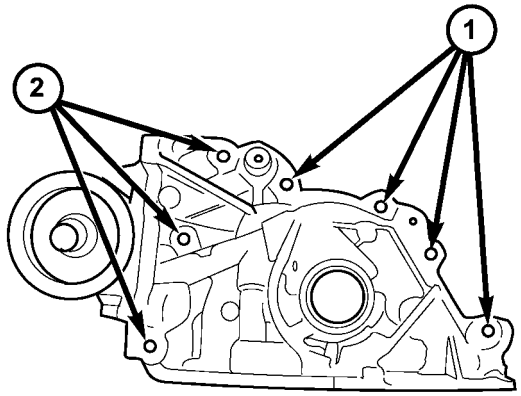


Fig. 85 2.4L OIL PUMP

1 - BOLTS
2 - BOLTS

DISASSEMBLY

(1) To remove the relief valve, proceed as follows:
(a) Remove the threaded plug and gasket from the oil pump.
(b) Remove spring and relief valve.
(2) Remove oil pump cover fasteners, and lift off cover.
(3) Remove pump rotors.
(4) Wash all parts in a suitable solvent and inspect carefully for damage or wear.

CLEANING

(1) Clean all parts thoroughly in a suitable solvent.

INSPECTION

(1) Inspect the mating surface of the oil pump. Surface should be smooth. Replace pump cover if scratched or grooved.
(2) Lay a straightedge across the pump cover surface (Fig. 86). If a 0.025 mm (0.001 in.) feeler gauge can be inserted between cover and straight edge, cover should be replaced.
(3) Measure thickness and diameter of outer rotor. If outer rotor thickness measures 9.40 mm (0.370 in.) or less (Fig. 87), or if the diameter is 79.95 mm (3.148 in.) or less, replace outer rotor.

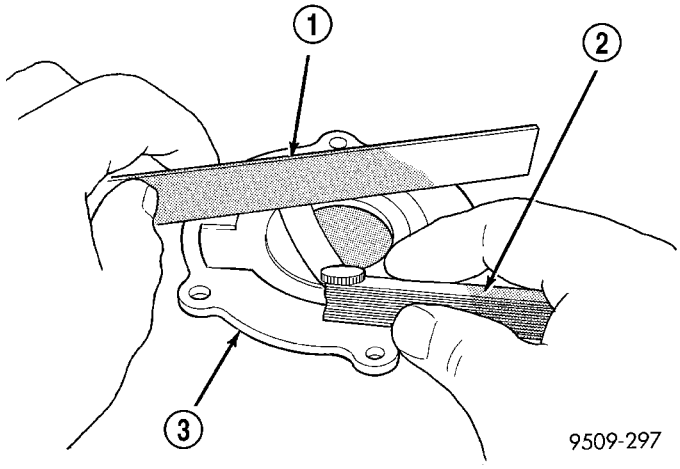


Fig. 86 Checking Oil Pump Cover Flatness

1 - STRAIGHT EDGE
2 - FEELER GAUGE
3 - OIL PUMP COVER

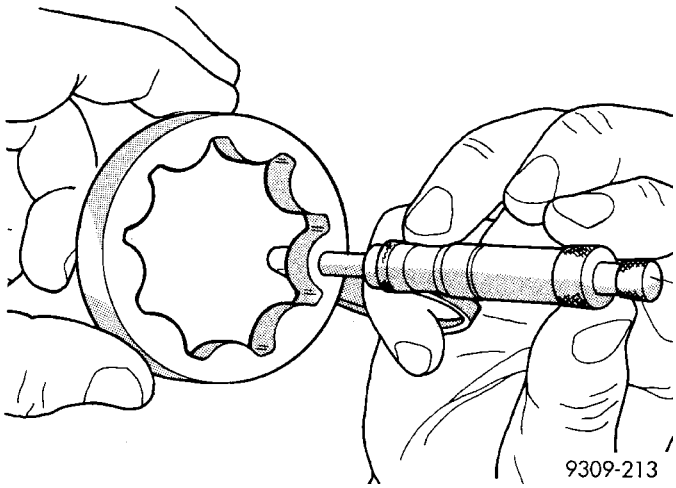


Fig. 87 Measuring Outer Rotor Thickness

OIL PUMP (Continued)

(4) If inner rotor measures 9.40 mm (0.370 in.) or less replace inner rotor (Fig. 88).

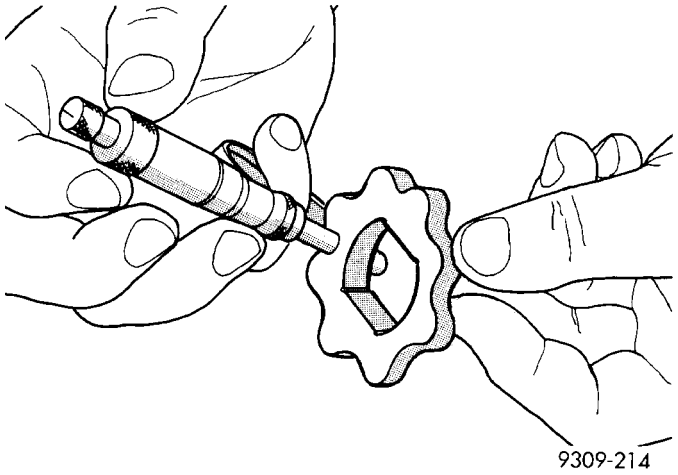


Fig. 88 Measuring Inner Rotor Thickness

ASSEMBLY

(1) Assemble pump, using new parts as required. **Install the inner rotor with chamfer facing the cast iron oil pump cover.**

(2) Prime oil pump before installation by filling rotor cavity with engine oil.

(3) Install cover and tighten fasteners to 12 N·m (105 in. lbs.).

CAUTION: Oil pump pressure relief valve must be installed correctly or serious engine damage may occur.

(4) Install relief valve, spring, gasket and cap. Tighten cap to 41 N·m (30 ft. lbs.).

INSTALLATION

(1) Make sure all surfaces are clean and free of oil and dirt.

(2) Apply Mopar® Gasket Maker to oil pump as shown in (Fig. 89). Install O-ring into oil pump body discharge passage.

(3) Prime oil pump with engine oil before installation.

(4) Align oil pump rotor flats with flats on crankshaft. Install the oil pump to the block.

CAUTION: To align, the front crankshaft seal MUST be out of pump, or damage may result.

(5) Install new front crankshaft seal using Special Tool 6780 (Fig. 90).

(6) Install crankshaft key (Fig. 84).

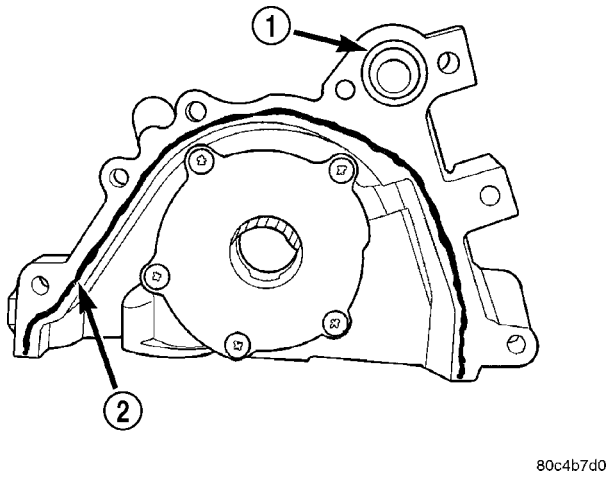


Fig. 89 Oil Pump Sealing - Typical

1 - O-RING
2 - SEALER LOCATION

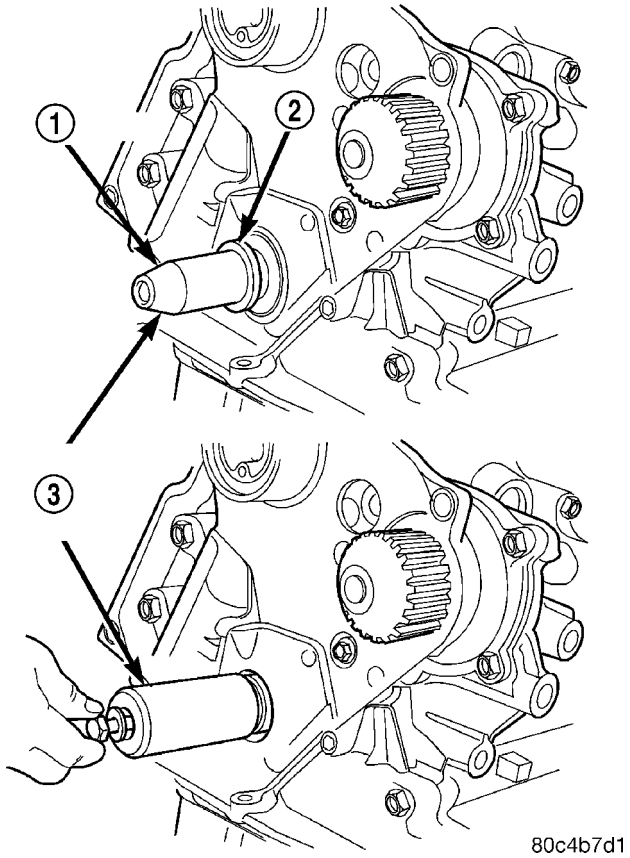


Fig. 90 Front Crankshaft Seal - Installation

1 - PROTECTOR
2 - SEAL
3 - SPECIAL TOOL 6780

OIL PUMP (Continued)

CAUTION: The crankshaft sprocket is set to a predetermined depth from the factory for correct timing belt tracking. If removed, use of Special Tool 6792 is required to set the sprocket to original installation depth. An incorrectly installed sprocket will result in timing belt and engine damage.

(7) Install crankshaft sprocket using Special Tool 6792 (Fig. 91).

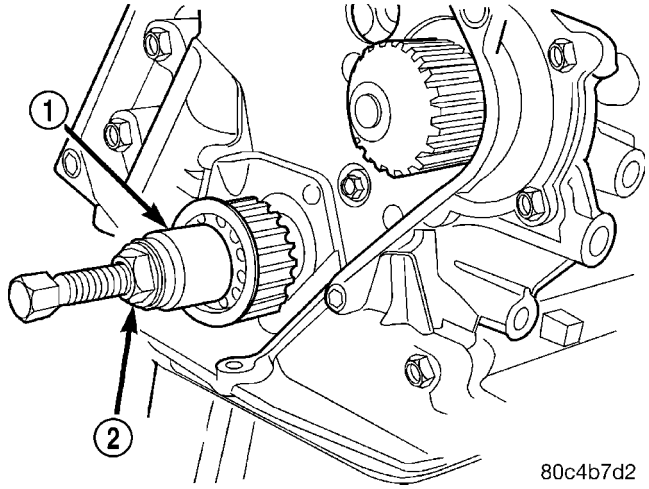


Fig. 91 Crankshaft Sprocket - Installation

1 - SPECIAL TOOL 6792
2 - TIGHTEN NUT TO INSTALL

(8) Install oil pump pick-up tube.
(9) Install oil pan. (Refer to 9 - ENGINE/LUBRICATION/OIL PAN - INSTALLATION)
(10) Install timing belt rear cover. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT / CHAIN COVER(S) - INSTALLATION)
(11) Install timing belt. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - INSTALLATION)

INTAKE MANIFOLD

DESCRIPTION

The intake manifold is a one piece composite module that attaches to the cylinder head with fasteners. The manifold is a long branch design to enhance low and mid-range torque

OPERATION

The intake manifold delivers air to the combustion chambers. This air allows the fuel delivered by the fuel injectors to ignite when the spark plug fire.

DIAGNOSIS AND TESTING

INTAKE MANIFOLD LEAKS

An intake manifold air leak is characterized by lower than normal manifold vacuum. Also, one or more cylinders may not be functioning.

WARNING: USE EXTREME CAUTION WHEN THE ENGINE IS OPERATING. DO NOT STAND IN A DIRECT LINE WITH THE FAN. DO NOT PUT YOUR HANDS NEAR THE PULLEYS, BELTS OR THE FAN. DO NOT WEAR LOOSE CLOTHING.

(1) Start the engine.
(2) Spray a small stream of water (Spray Bottle) at the suspected leak area.
(3) If engine RPM'S change, the area of the suspected leak has been found.
(4) Repair as required.

REMOVAL

(1) Disconnect negative cable from battery.
(2) Disconnect connector from inlet air temperature sensor.
(3) Disconnect air intake tube at throttle body and remove upper air cleaner housing.
(4) Disconnect connector from throttle position sensor (TPS).
(5) Disconnect connector from idle air control (IAC) motor.
(6) Disconnect connector from MAP sensor.
(7) Remove vacuum lines for purge solenoid and PCV valve at intake manifold.
(8) Remove vacuum lines for power brake booster, LDP, EGR transducer, and speed control vacuum reservoir (if equipped) at intake manifold fittings.
(9) Disconnect throttle, speed control (if equipped), and transaxle control (if equipped) and cables from throttle lever and bracket. (Refer to 14 - FUEL SYSTEM/FUEL INJECTION/THROTTLE CONTROL CABLE - REMOVAL)
(10) Perform fuel system pressure release procedure **before attempting any repairs.** (Refer to 14 - FUEL SYSTEM/FUEL DELIVERY - STANDARD PROCEDURE)
(11) Disconnect fuel line. (Refer to 14 - FUEL SYSTEM/FUEL DELIVERY/QUICK CONNECT FITTING - STANDARD PROCEDURE)
(12) Disconnect coolant temperature sensor/fuel injector wire harness connector.
(13) Disconnect fuel injector harness.
(14) Remove intake manifold to cylinder head fasteners.
(15) Remove the manifold from engine.

INTAKE MANIFOLD (Continued)

CAUTION: Cover intake manifold openings to prevent foreign material from entering engine.

(16) Inspect the manifold. (Refer to 9 - ENGINE/MANIFOLDS/INTAKE MANIFOLD - INSPECTION)

INSPECTION

(1) Check manifold surfaces for flatness with straight edge. Surface must be flat within 0.15 mm per 300 mm (0.006 in. per foot) of manifold length.

(2) Inspect manifold for cracks or distortion. Replace manifold if necessary.

INSTALLATION

(1) Clean manifold sealing surfaces.

(2) Install new manifold to cylinder head seals.

(3) Install manifold to head.

(4) Install and tighten intake manifold fasteners to 28 N·m (250 in. lbs.) in the sequence shown (Fig. 92). Repeat procedure until all bolts are at specified torque.

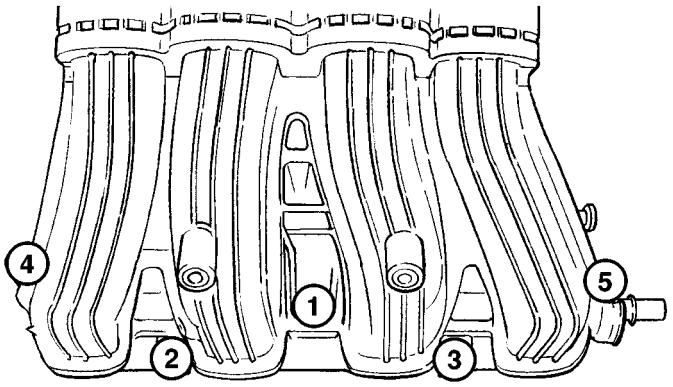


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Fig. 92 LOWER INTAKE MANIFOLD TIGHTENING SEQUENCE

(5) Install throttle cables in bracket.

(6) Connect throttle, speed control, (if equipped), cables to throttle lever.

(7) Connect vacuum lines for power brake booster, LDP, EGR transducer, and speed control vacuum reservoir (if equipped) at upper intake manifold fittings.

(8) Connect vacuum lines for purge solenoid and PCV valve.

(9) Connect electrical connectors for MAP sensor, throttle position sensor (TPS), and idle air control (IAC) motor.

(10) Connect the fuel line. (Refer to 14 - FUEL SYSTEM/FUEL DELIVERY/QUICK CONNECT FITTING - STANDARD PROCEDURE)

(11) Connect coolant temperature sensor/fuel injector wiring harness electrical connector.

(12) Install the air cleaner housing and air intake tube to throttle body.

(13) Connect inlet air temperature sensor connector.

(14) Connect negative cable to battery.

EXHAUST MANIFOLD

DESCRIPTION

The exhaust manifold is made of Hi-Silicone Moly nodular cast iron for strength and high temperatures. The manifold attaches to the cylinder head.

OPERATION

The exhaust manifold collects the exhaust gasses exiting the combustion chambers. Then it channels the exhaust gasses to the exhaust pipe attached to the manifold.

REMOVAL

(1) Raise vehicle and disconnect exhaust pipe from the exhaust manifold.

(2) Lower the vehicle.

(3) Disconnect upstream oxygen sensor connector at the rear of exhaust manifold.

(4) Remove the air cleaner bracket (Fig. 93).

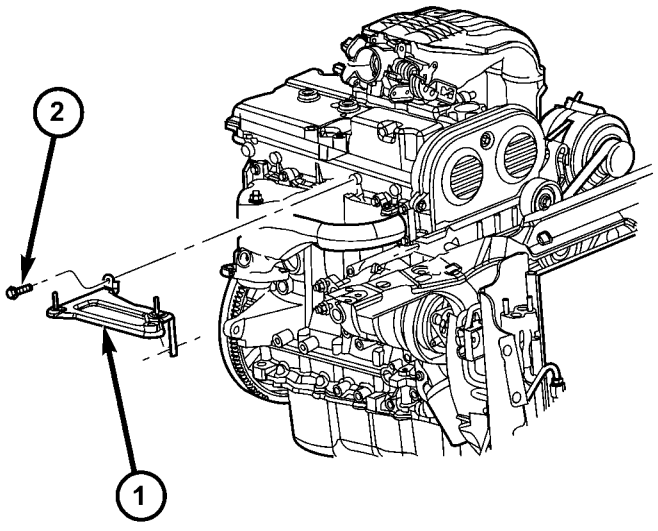

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Fig. 93 AIR CLEANER BRACKET

1 - AIR CLEANER BRACKET
2 - BOLT (2)

EXHAUST MANIFOLD (Continued)

(5) Remove the heat shield.
(6) Remove the bolts attaching the manifold to the cylinder head.
(7) Remove exhaust manifold.
(8) Inspect the manifold. (Refer to 9 - ENGINE/ MANIFOLDS/EXHAUST MANIFOLD - INSPECTION)

CLEANING

(1) Discard gasket (if equipped) and clean all surfaces of manifold and cylinder head.

INSPECTION

(1) Inspect manifold gasket surfaces for flatness with straight edge. Surface must be flat within 0.15 mm per 300 mm (0.006 in. per foot) of manifold length.
(2) Inspect manifolds for cracks or distortion. Replace manifold as necessary.

INSTALLATION

(1) Clean the manifold mating surfaces.
(2) Install exhaust manifold with a new gasket. Tighten attaching nuts to 20 N·m (175 in. lbs.).
(3) Attach exhaust pipe to exhaust manifold and tighten fasteners to 37 N·m (27 ft. lbs.).
(4) Install and connect the oxygen sensor. (Refer to 14 - FUEL SYSTEM/FUEL INJECTION/O2 SENSOR - COMPONENT LOCATION)
(5) Install the heat shield.
(6) Install the air cleaner bracket.

TIMING BELT COVER(S)

REMOVAL

FRONT COVER

(1) Remove crankshaft vibration damper. (Refer to 9 - ENGINE/ENGINE BLOCK/VIBRATION DAMPER - REMOVAL)
(2) Remove generator drive belt tensioner assembly. (Refer to 7 - COOLING/ACCESSORY DRIVE/ BELT TENSIONERS - REMOVAL)
(3) Remove timing belt front cover bolts, and remove covers.

REAR COVER

(1) Remove front covers.
(2) Remove timing belt. (Refer to 9 - ENGINE/ VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - REMOVAL)
(3) Hold camshaft sprocket with Special Tool 6847 while removing center bolt.
(4) Remove timing belt idler pulley.
(5) Remove rear cover fasteners and remove cover from engine.

INSTALLATION

REAR COVER

(1) Install timing belt rear cover and bolts (Fig. 94). Torque bolts to 12 N·m (105 in. lbs).

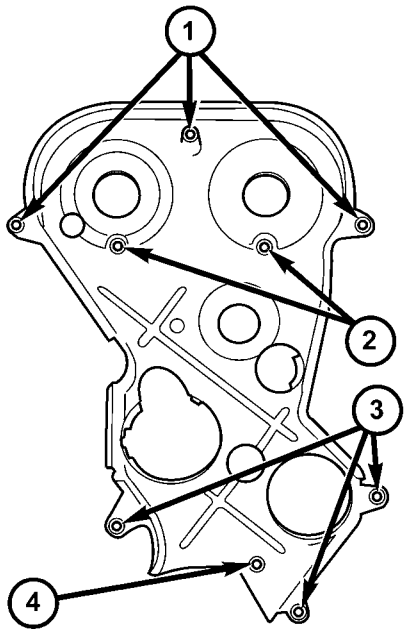

Fig. 94 TIMING BELT REAR COVER FASTENERS

1 - OUTER COVER TO REAR COVER FASTENERS (3)
2 - REAR COVER TO CYLINDER HEAD FASTENERS
3 - OUTER COVER TO REAR COVER FASTENERS (3)
4 - INNER COVER TO BLOCK FASTENERS

CAUTION: Do not use an impact wrench for tightening camshaft sprocket bolt. Damage to the timing locating pin can occur. Hand tighten using a wrench ONLY.

(2) Install camshaft sprockets, and camshaft target ring. Hold sprockets with Special Tool 6848 and tighten center bolt to 101 N·m (75 ft. lbs.).
(3) Install timing belt idler pulley and tighten mounting bolt to 61 N·m (45 ft. lbs.).
(4) Install timing belt. (Refer to 9 - ENGINE/ VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - INSTALLATION)
(5) Install accessory drive bracket (Fig. 95).
(6) Install front covers.

FRONT COVER

(1) Install timing belt front covers (Fig. 96). Tighten fasteners to 7 N·m (60 in. lbs.).

TIMING BELT COVER(S) (Continued)

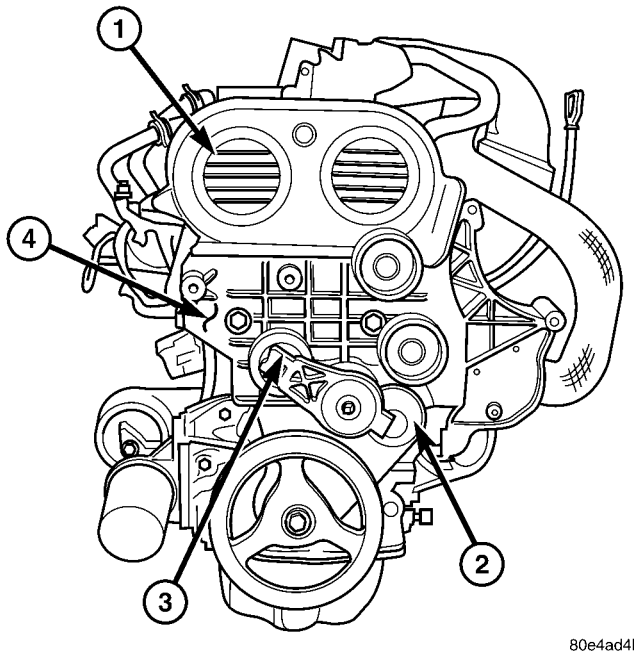

Fig. 95 ACCESSORY DRIVE BRACKET

1- UPPER TIMING BELT COVER
2- LOWER TIMING BELT COVER
3- BELT TENSIONER
4- ACCESSORY DRIVE BRACKET

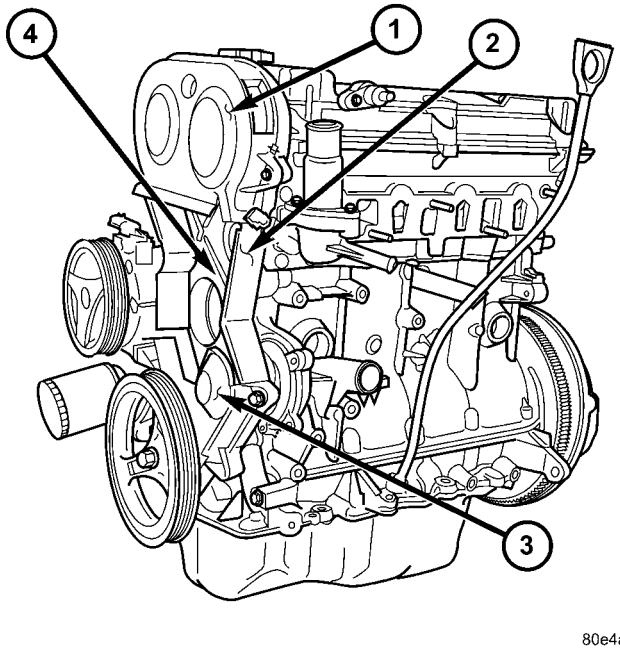

Fig. 96 TIMING BELT COVERS

1- UPPER TIMING BELT COVER
2- REAR TIMING BELT COVER
3- LOWER TIMING BELT COVER
4- ACCESSORY DRIVE BRACKET NOT SHOWN

(2) Install generator drive belt tensioner. (Refer to 7 - COOLING/ACCESSORY DRIVE/BELT TENSIONERS - INSTALLATION)

(3) Install crankshaft vibration damper. (Refer to 9 - ENGINE/ENGINE BLOCK/VIBRATION DAMPER - INSTALLATION)

TIMING BELT AND SPROCKET(S)

REMOVAL

TIMING BELT

(1) Remove air cleaner upper cover, housing, and clean air tube.

(2) Raise vehicle on hoist.

(3) Remove accessory drive belts. (Refer to 7 - COOLING/ACCESSORY DRIVE/BELT TENSIONERS - REMOVAL)

(4) Remove crankshaft vibration damper. (Refer to 9 - ENGINE/ENGINE BLOCK/VIBRATION DAMPER - REMOVAL)

(5) Remove air conditioner/generator belt tensioner and pulley assembly. (Refer to 7 - COOLING/ACCESSORY DRIVE/BELT TENSIONERS - REMOVAL)

(6) Remove timing belt lower front cover bolts and remove cover. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT / CHAIN COVER(S) - REMOVAL)

(7) Lower vehicle.

(8) Remove bolts attaching timing belt upper front cover and remove cover. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT / CHAIN COVER(S) - REMOVAL)

CAUTION: When aligning crankshaft and camshaft timing marks always rotate engine from crankshaft. Camshaft should not be rotated after timing belt is removed. Damage to valve components may occur. Always align timing marks before removing timing belt.

(9) Before the removal of the timing belt, rotate crankshaft until the TDC mark on oil pump housing aligns with the TDC mark on crankshaft sprocket (trailing edge of sprocket tooth) (Fig. 97).

NOTE: The crankshaft sprocket TDC mark is located on the trailing edge of the sprocket tooth. Failure to align trailing edge of sprocket tooth to TDC mark on oil pump housing will cause the camshaft timing marks to be misaligned.

TIMING BELT AND SPROCKET(S) (Continued)

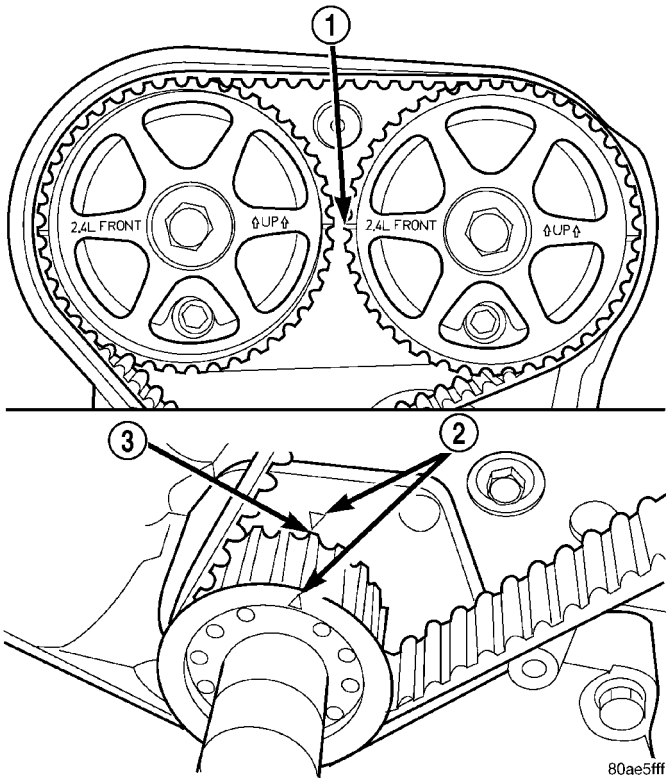

Fig. 97 Crankshaft and Camshaft Timing

1 - CAMSHAFT TIMING MARKS
2 - CRANKSHAFT TDC MARKS
3 - TRAILING EDGE OF SPROCKET TOOTH

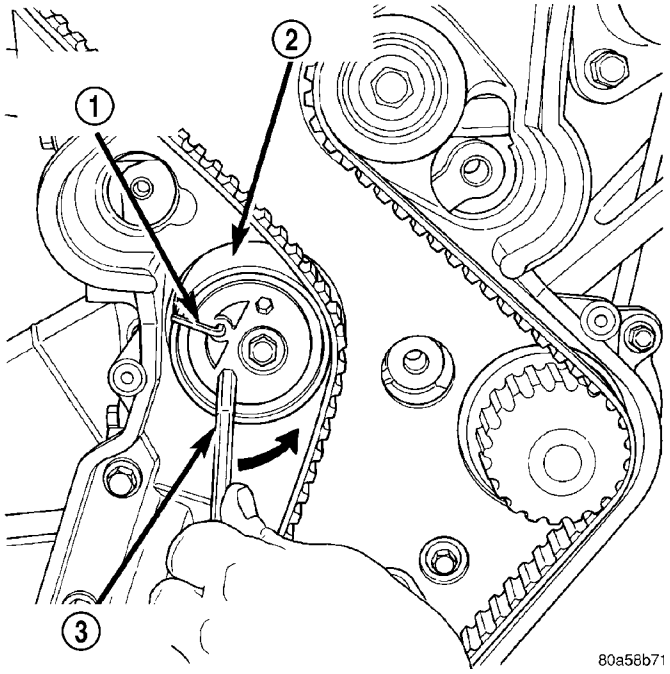

Fig. 98 Locking Timing Tensioner

1 - 1/8 OR 3mm ALLEN WRENCH
2 - BELT TENSIONER
3 - 6mm ALLEN WRENCH

(10) Install 6 mm Allen wrench into belt tensioner. Before rotating the tensioner, insert the long end of a 1/8" or 3 mm Allen wrench into the pin hole on the front of the tensioner (Fig. 98). While rotating the tensioner counterclockwise, push in lightly on the 1/8" or 3 mm Allen wrench, until it slides into the locking hole.

(11) Remove timing belt.

CRANKSHAFT SPROCKET

(1) Disconnect negative battery cable.

(2) Remove timing belt (Refer to 9 - ENGINE/ VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - REMOVAL).

(3) Remove crankshaft sprocket using Special Tools 6793 and insert C-4685-C2 (Fig. 99).

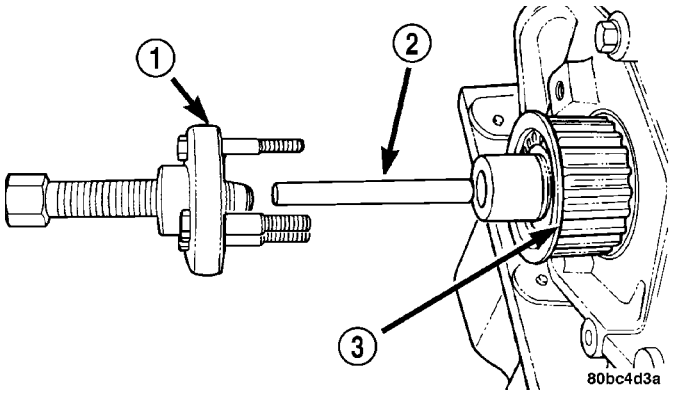

Fig. 99 Crankshaft Sprocket - Removal

1 - SPECIAL TOOL 6793
2 - SPECIAL TOOL C-4685–C2
3 - CRANKSHAFT SPROCKET

CLEANING

Do Not attempt to clean a timing belt. If contamination from oil, grease, or coolants have occurred, the timing belt should be replaced.

Clean all sprockets using a suitable solvent. Clean all sprocket grooves of any debris.

INSTALLATION

CRANKSHAFT SPROCKET

CAUTION: The crankshaft sprocket is set to a predetermined depth from the factory for correct timing belt tracking. If removed, use of Special Tool 6792 is required to set the sprocket to original installation depth. An incorrectly installed sprocket will result in timing belt and engine damage.

TIMING BELT AND SPROCKET(S) (Continued)

(1) Install crankshaft sprocket using Special Tool 6792 (Fig. 100).

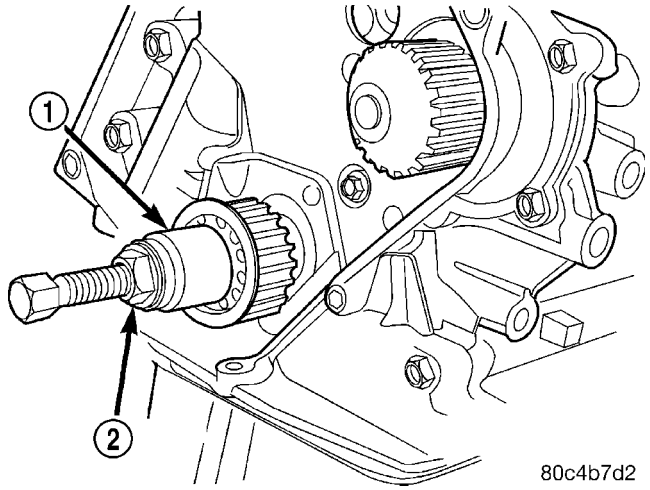

Fig. 100 Crankshaft Sprocket - Installation

1 - SPECIAL TOOL 6792
2 - TIGHTEN NUT TO INSTALL

(2) Install timing belt. (Refer to 9 - ENGINE/ VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - INSTALLATION)

TIMING BELT

CAUTION: The crankshaft sprocket is set to a predetermined depth from the factory for correct timing belt tracking. If removed, use of Special Tool 6792 is required to set the sprocket to original installation depth. An incorrectly installed sprocket will result in timing belt and engine damage.

(1) Set crankshaft sprocket to TDC by aligning the sprocket with the arrow on the oil pump housing.

(2) Set camshafts timing marks so that the exhaust camshaft sprocket is a 1/2 notch below the intake camshaft sprocket (Fig. 101).

CAUTION: Ensure that the arrows on both camshaft sprockets are facing up.

(3) Install timing belt. Starting at the crankshaft, go around the water pump sprocket, idler pulley, camshaft sprockets and then around the tensioner (Fig. 102).

(4) Move the exhaust camshaft sprocket counterclockwise (Fig. 102) to align marks and take up belt slack.

(5) Insert a 6 mm Allen wrench into the hexagon opening located on the top plate of the belt tensioner pulley. Rotate the top plate **COUNTERCLOCKWISE**. The tensioner pulley will move against the belt and the tensioner setting notch will eventually start to move clockwise. Watching the movement of the setting notch, continue rotating the top plate

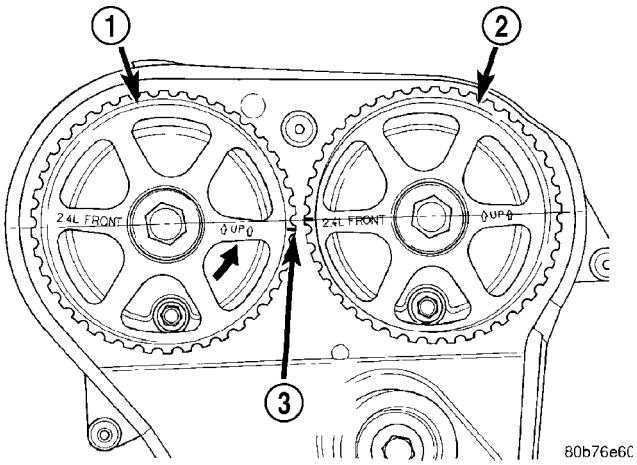

Fig. 101 Camshaft Sprocket Alignment

1 - CAMSHAFT SPROCKET-EXHAUST
2 - CAMSHAFT SPROCKET-INTAKE
3 - 1/2 NOTCH LOCATION

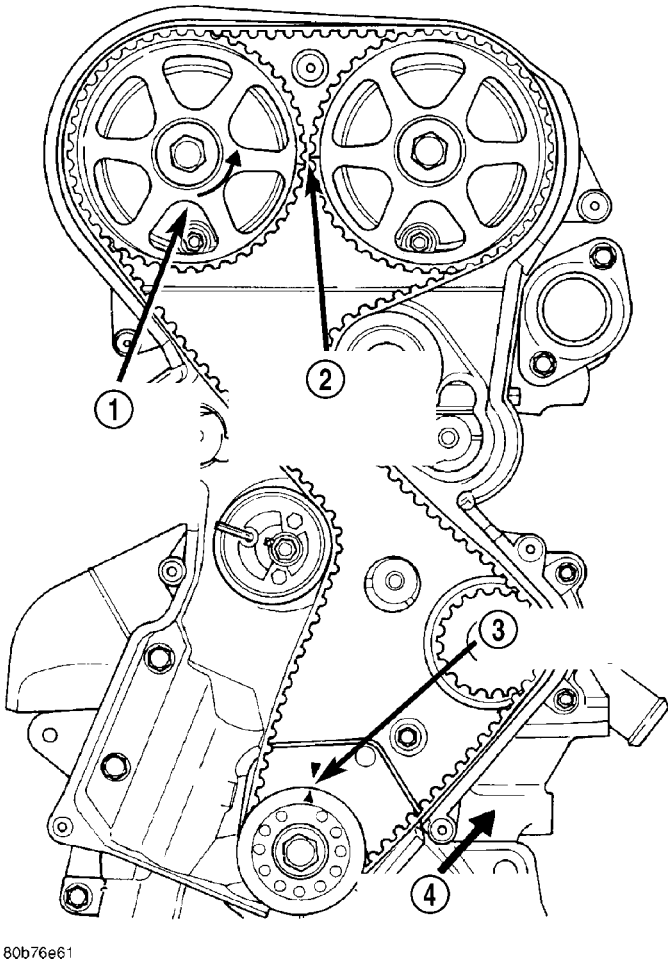

Fig. 102 Timing Belt - Installation - Typical

1 - ROTATE CAMSHAFT SPROCKET TO TAKE UP BELT SLACK
2 - CAMSHAFT TIMING MARKS 1/2 NOTCH LOCATION
3 - CRANKSHAFT AT TDC
4 - INSTALL BELT IN THIS DIRECTION

TIMING BELT AND SPROCKET(S) (Continued)

counterclockwise until the setting notch is aligned with the spring tang (Fig. 103). Using the allen wrench to prevent the top plate from moving, torque the tensioner lock nut to 30 N·m (22 ft. lbs.). Setting notch and spring tang should remain aligned after lock nut is torqued.

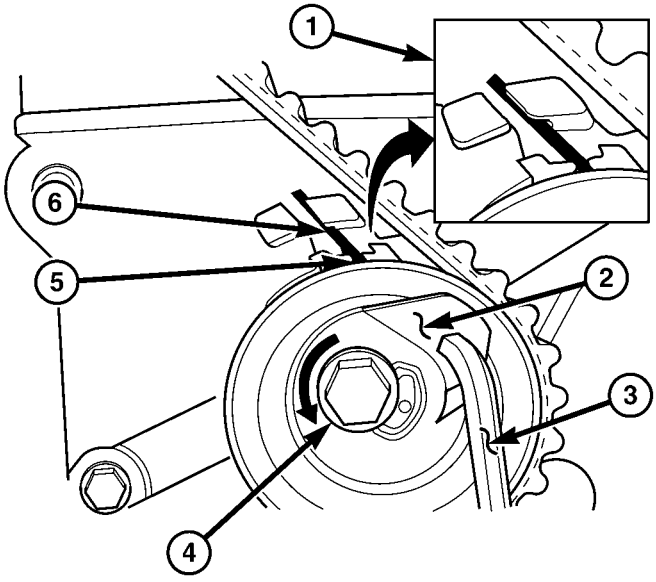

Fig. 103 Timing Belt Tension Adjustment

1 - ALIGN SETTING NOTCH WITH SPRING TANG
2 - TOP PLATE
3 - 6mm ALLEN WRENCH
4 - LOCK NUT
5 - SETTING NOTCH
6 - SPRING TANG

(6) Remove allen wrench and torque wrench.

NOTE: Repositioning the crankshaft to the TDC position must be done only during the CLOCKWISE rotation movement. If TDC is missed, rotate a further two revolutions until TDC is achieved. DO NOT rotate crankshaft counterclockwise as this will make verification of proper tensioner setting impossible.

(7) Once the timing belt has been installed and tensioner adjusted, rotate the crankshaft CLOCKWISE two complete revolutions manually for seating of the belt, until the crankshaft is repositioned at the TDC position. Verify that the camshaft and crankshaft timing marks are in proper position (Fig. 104).

(8) Check if the spring tang is within the tolerance window (Fig. 105). If the spring tang is within the tolerance window, the installation process is complete and nothing further is required. If the spring tang is not within the tolerance window, repeat Steps 5 through 7.

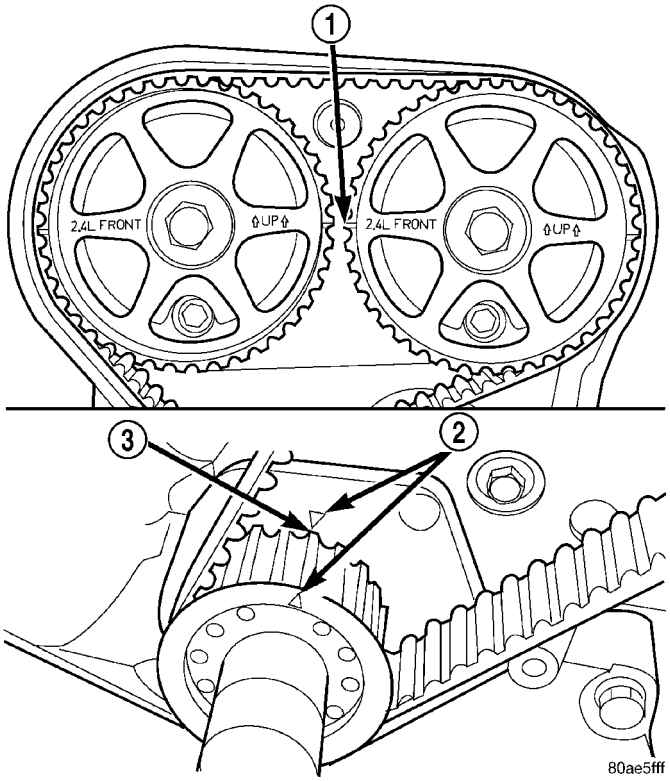

Fig. 104 Crankshaft and Camshaft Timing

1 - CAMSHAFT TIMING MARKS
2 - CRANKSHAFT TDC MARKS
3 - TRAILING EDGE OF SPROCKET TOOTH

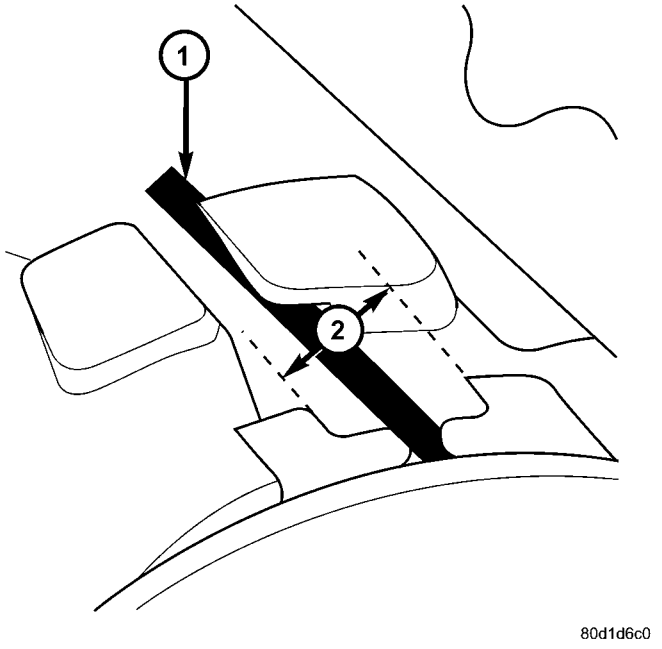

Fig. 105 Timing Belt Tension Verification

1 - SPRING TANG
2 - TOLERANCE WINDOW

TIMING BELT AND SPROCKET(S) (Continued)

(9) Install timing belt front covers and bolts. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT / CHAIN COVER(S) - INSTALLATION)

(10) Install air conditioning/generator belt tensioner and pulley. (Refer to 7 - COOLING/ACCESSORY DRIVE/BELT TENSIONERS - INSTALLATION)

(11) Install crankshaft vibration damper. (Refer to 9 - ENGINE/ENGINE BLOCK/VIBRATION DAMPER - INSTALLATION)

(12) Install accessory drive belts. (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - INSTALLATION)

(13) Install drive belt splash shield.

(14) Install air cleaner housing, upper cover, and clean air tube.

TIMING BELT TENSIONER & PULLEY

REMOVAL

(1) Remove the timing belt. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - REMOVAL)

(2) Remove timing belt idler pulley.

(3) Hold camshaft sprocket with Special Tool 6847 while removing bolt (Fig. 106). Remove both cam sprockets.

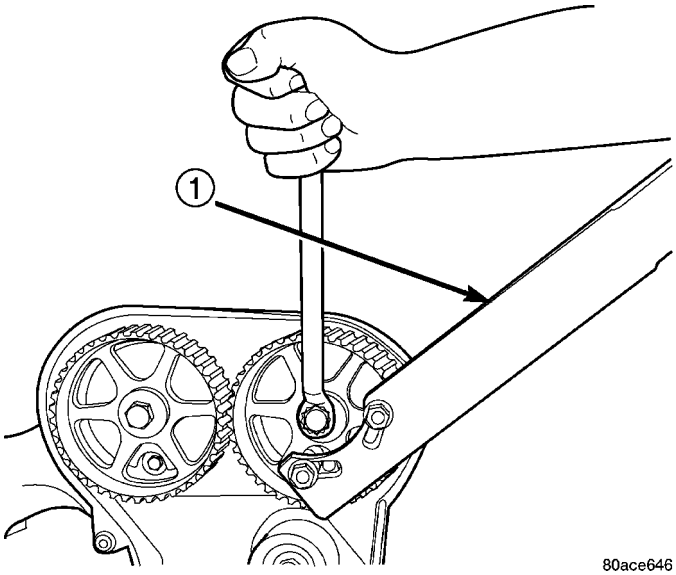

Fig. 106 Camshaft Sprocket - Removal/Installation

1 - SPECIAL TOOL 6847

(4) Remove rear timing belt cover fasteners and remove cover from engine (Fig. 107).

(5) Remove lower bolt attaching timing belt tensioner assembly to engine and remove tensioner **as an assembly**.

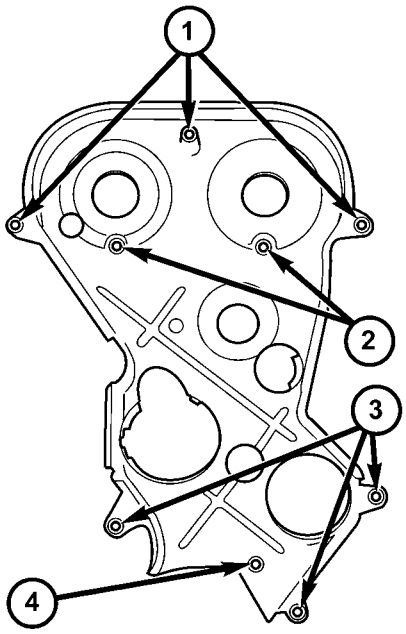

Fig. 107 TIMING BELT REAR COVER FASTENERS

1 - OUTER COVER TO REAR COVER FASTENERS (3)
2 - REAR COVER TO CYLINDER HEAD FASTENERS
3 - OUTER COVER TO REAR COVER FASTENERS (3)
4 - INNER COVER TO BLOCK FASTENERS

INSTALLATION

(1) Align timing belt tensioner assembly to engine and install lower mounting bolt **but do not tighten**. To properly align tensioner assembly—install one of the engine bracket mounting bolts (M10) 5 to 7 turns into the tensioner's upper mounting location.

(2) Torque the tensioner's lower mounting bolt to 61 N·m (45 ft. lbs.). Remove the upper bolt used for tensioner alignment.

(3) Install rear timing belt cover and fasteners.

(4) Install timing belt idler pulley and torque mounting bolt to 61 N·m (45 ft. lbs.).

(5) Install camshaft sprockets. Use Special Tool 6847 to hold sprockets, torque bolts to 101 N·m (75 ft. lbs.).

(6) Install the timing belt. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - INSTALLATION)

BALANCE SHAFTS AND CARRIER ASSEMBLY

DESCRIPTION

The 2.4L engine is equipped with two nodular cast iron balance shafts installed in a cast aluminum carrier attached to the lower cylinder block (Fig. 108).

OPERATION

The balance shafts are driven by the crankshaft via a roller chain and sprockets. The balance shafts are connected by helical gears. The dual counter rotating shafts decrease second order vertical shaking forces caused by component movement.

REMOVAL

BALANCE SHAFTS/CHAIN/SPROCKETS

NOTE: For service procedures requiring only temporary relocation of carrier assembly refer to BALANCE SHAFT CARRIER procedure below.

(1) Drain engine oil.
(2) Remove the oil pan and pick-up tube (Refer to 9 - ENGINE/LUBRICATION/OIL PAN - REMOVAL).
(3) If replacing crankshaft sprocket, remove oil pump (Refer to 9 - ENGINE/LUBRICATION/OIL PUMP - REMOVAL).
(4) Remove chain cover, guide and tensioner. Discard pivot screw and adjuster screw. (Fig. 109).
(5) Remove screw retaining balance shaft drive sprocket (Fig. 110). Remove chain and sprocket.
(6) Using two wide pry bars, work the crankshaft sprocket back and forth until it is off the crankshaftshaft.
(7) Remove gear cover retaining stud (double ended to also retain chain guide). Remove cover and balance shaft gears (Fig. 111).
(8) Remove rear cover and balance shafts (Fig. 112).
(9) Remove four carrier to crankcase attaching bolts to separate carrier from engine bedplate.

BALANCE SHAFT CARRIER

The following components will remain intact during carrier removal: Gear cover, gears, balance shafts and the rear cover (Fig. 108).
(1) Drain engine oil.
(2) Remove the oil pan and pick-up tube (Refer to 9 - ENGINE/LUBRICATION/OIL PAN - INSTALLATION).
(3) Remove chain cover, guide and tensioner (Fig. 109).

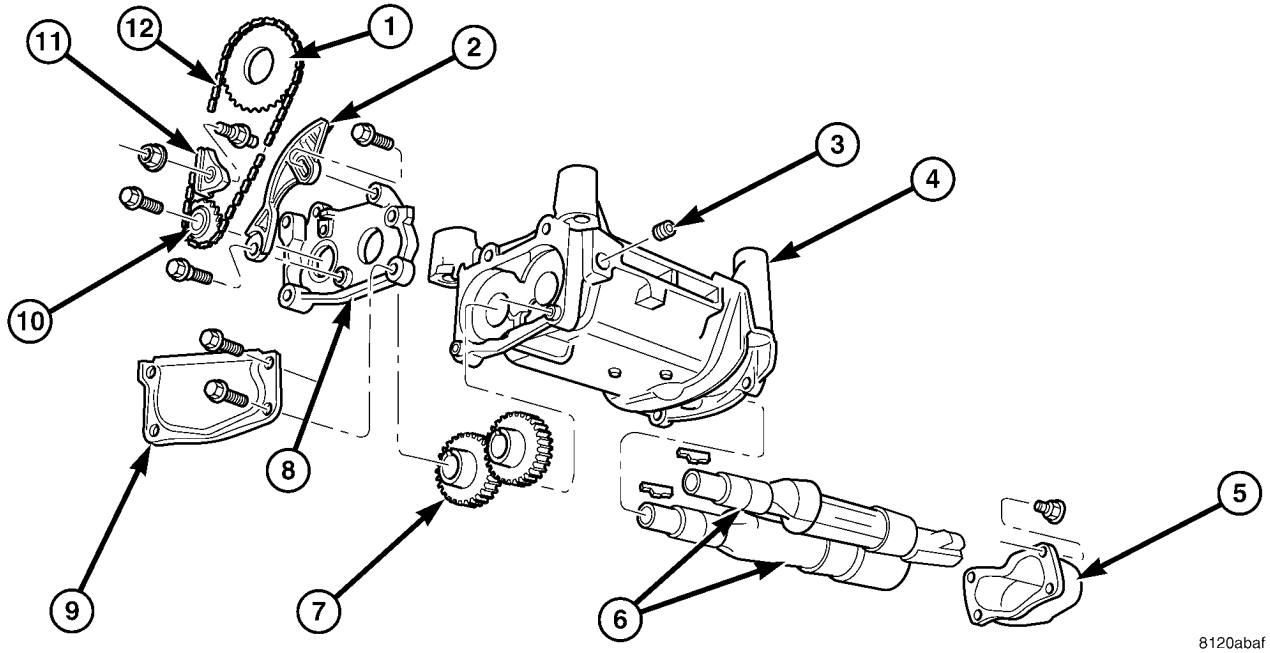

Fig. 108 Balance Shafts and Carrier Assembly

1 - SPROCKET
2 - TENSIONER
3 - PLUG
4 - CARRIER
5 - REAR COVER
6 - BALANCE SHAFTS
7 - GEARS
8 - GEAR COVER
9 - CHAIN COVER
10 - SPROCKET
11 - GUIDE
12 - CHAIN

BALANCE SHAFTS AND CARRIER ASSEMBLY (Continued)

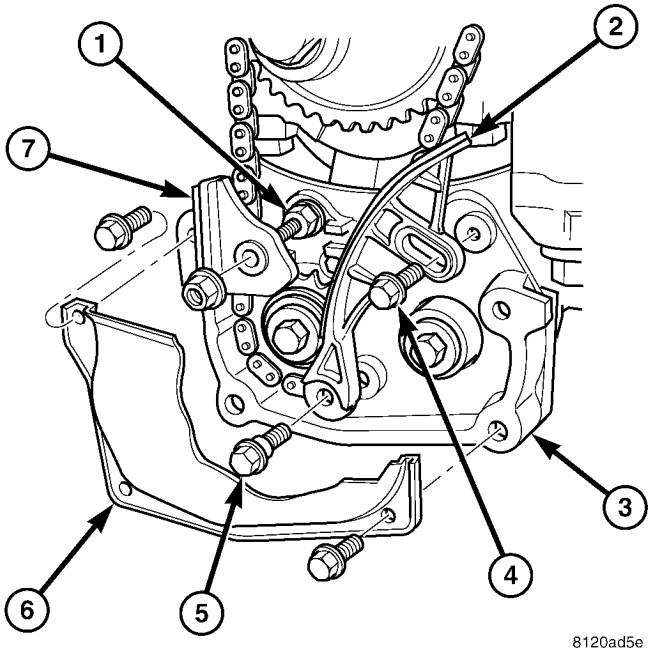

Fig. 109 Chain Cover, Guide and Tensioner

1 - STUD
2 - TENSIONER (ADJUSTER)
3 - GEAR COVER
4 - ADJUSTER SCREW
5 - SHOULDERED PIVOT SCREW
6 - CHAIN COVER (CUTAWAY)
7 - GUIDE

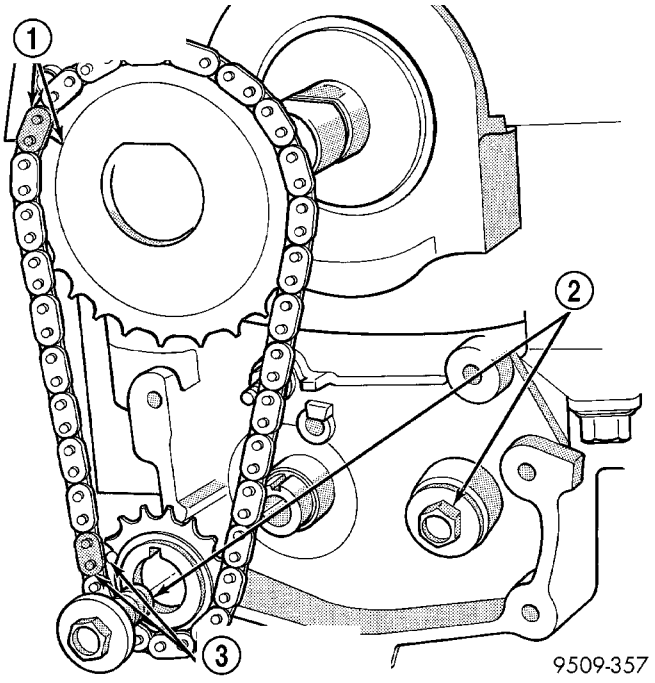

Fig. 110 Drive Chain and Sprockets

1 - NICKEL PLATED LINK AND MARK
2 - GEAR/SPROCKET SCREWS
3 - NICKEL PLATED LINK AND DOT

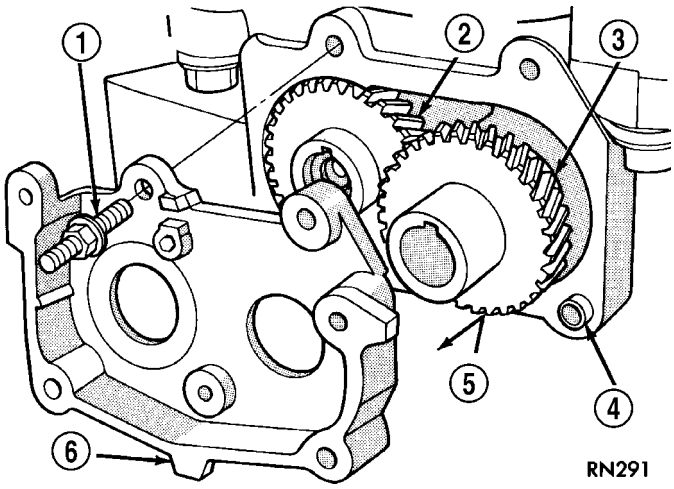

Fig. 111 Gear Cover and Gears

1 - STUD (DOUBLE ENDED)
2 - DRIVE GEAR
3 - DRIVEN GEAR
4 - CARRIER DOWEL
5 - GEAR(S)
6 - GEAR COVER

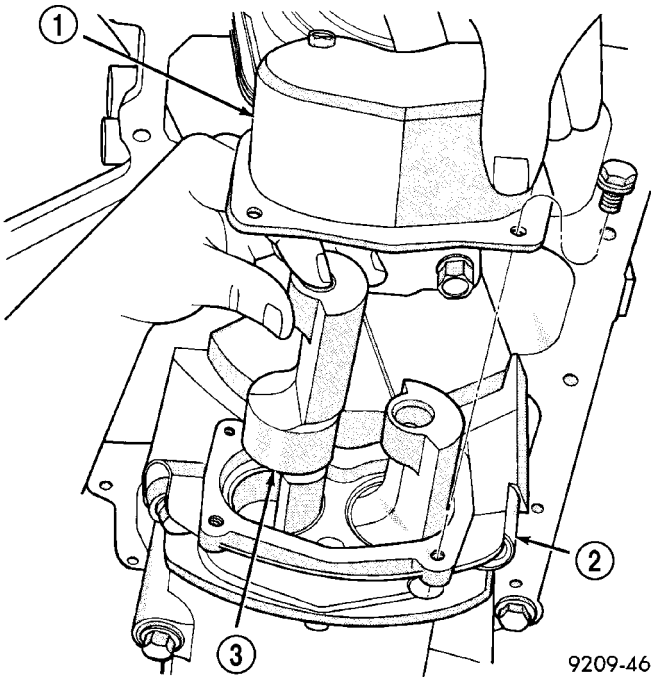

Fig. 112 Balance Shaft(s)—Removal/Installation

1 - REAR COVER
2 - CARRIER
3 - BALANCE SHAFT

(4) Remove screw retaining balance shaft drive sprocket (Fig. 110).

(5) Move balance shaft inboard through drive chain sprocket. Sprocket will hang in lower chain loop.

(6) Remove carrier to crankcase attaching bolts to remove carrier.

BALANCE SHAFTS AND CARRIER ASSEMBLY (Continued)

INSTALLATION

BALANCE SHAFT INSTALLATION/TIMING

Balance shaft and carrier assembly installation is the reverse of the removal procedure. **During installation crankshaft-to-balance shaft timing must be established. Refer to Timing procedure in this section.**

(1) With balance shafts installed in carrier (Fig. 108) position carrier on crankcase and install four attaching bolts and tighten to 54 N·m (40 ft. lbs.).

(2) Turn balance shafts until both shaft key ways are up, parallel to vertical centerline of engine. Install short hub drive gear on sprocket driven shaft and long hub gear on gear driven shaft. After installation gear and balance shaft keyways must be up with gear timing marks meshed as shown in (Fig. 113).

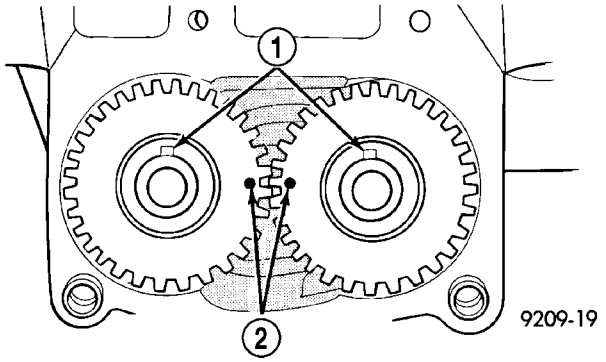

Fig. 113 Gear Timing

1 - KEY WAYS UP
2 - GEAR ALIGNMENT DOTS

(3) Install gear cover and tighten double ended stud/washer fastener to 12 N·m (105 in. lbs.).

(4) Align flat on balance shaft drive sprocket to the flat on crankshaft (Fig. 114).

(5) Install balance shaft drive sprocket on crankshaft using Special Tool 6052 (Fig. 115).

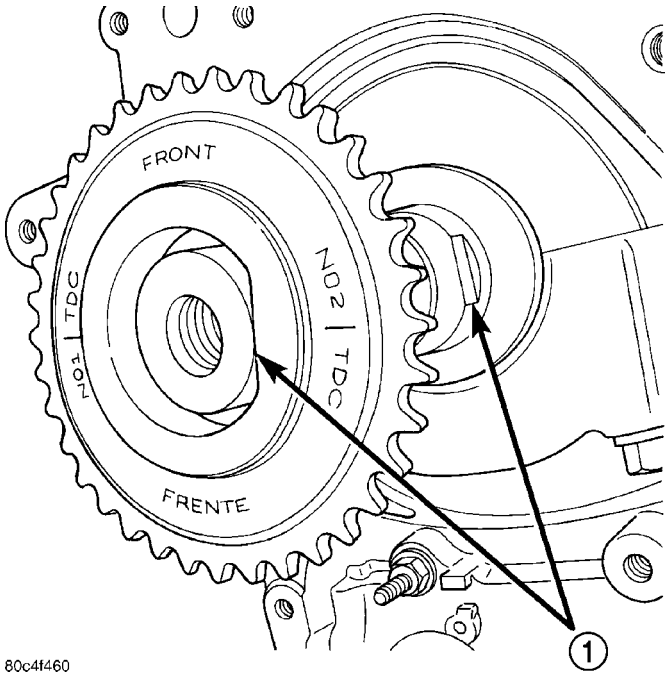

Fig. 114 Balance Shaft Sprocket Alignment to Crankshaft

1 - ALIGN FLATS

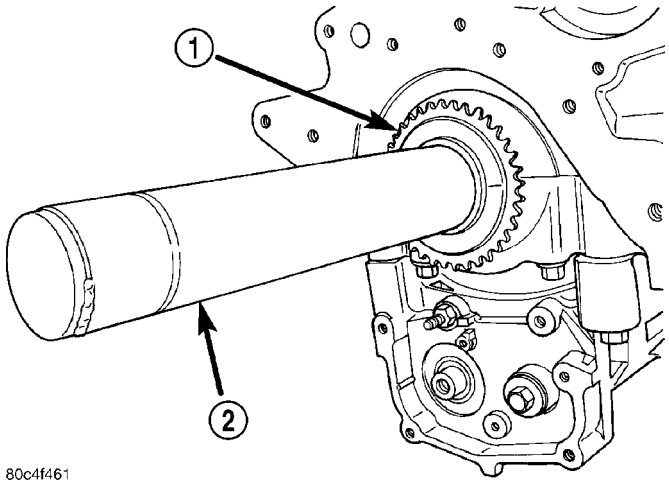

Fig. 115 Balance Shaft Drive Sprocket - Installation

1 - SPROCKET
2 - SPECIAL TOOL 6052

BALANCE SHAFTS AND CARRIER ASSEMBLY (Continued)

(6) Turn crankshaft until number 1 cylinder is at top dead center (TDC). The timing marks on the chain sprocket should line up with the parting line on the left side of number one main bearing cap. (Fig. 116).

(7) Place chain over crankshaft sprocket so that the plated link of the chain is over the number 1 cylinder timing mark on the balance shaft crankshaft sprocket (Fig. 116).

(8) Place balance shaft sprocket into the timing chain (Fig. 116) and align the timing mark on the sprocket (dot) with the (lower) plated link on the chain.

NOTE: The lower plated link is 8 links from the upper link.

(9) With balance shaft keyways pointing up (12 o'clock) slide the balance shaft sprocket onto the nose of the balance shaft. The balance shaft may have to be pushed in slightly to allow for clearance.

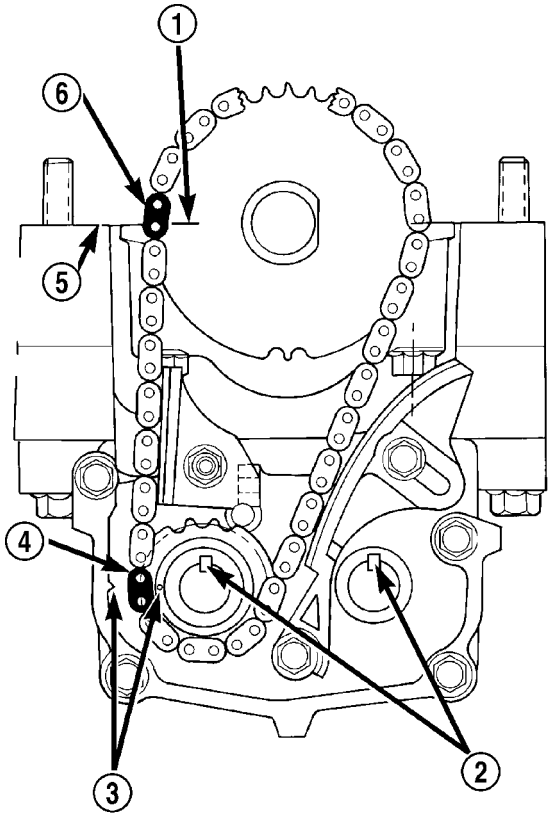

Fig. 116 Balance Shaft Timing

1 - MARK ON SPROCKET
2 - KEYWAYS UP
3 - ALIGN MARKS
4 - PLATED LINK
5 - PARTING LINE (BEDPLATE TO BLOCK)
6 - PLATED LINK

NOTE: THE TIMING MARK ON THE SPROCKET, THE (LOWER) NICKEL PLATED LINK, AND THE ARROW ON THE SIDE OF THE GEAR COVER SHOULD LINE UP WHEN THE BALANCE SHAFTS ARE TIMED CORRECTLY.

(10) If the sprockets are timed correctly, install the balance shaft bolts and tighten to 28 N·m (250 in. lbs.). A wood block placed between crankcase and crankshaft counterbalance will prevent crankshaft and gear rotation.

(11) **CHAIN TENSIONING:**

(a) Install chain tensioner loosely assembled with **new** adjuster screw and shouldered pivot screw.

(b) Position guide on double ended stud making sure tab on the guide fits into slot on the gear cover. Install and tighten nut/washer assembly to 12 N·m (105 in. lbs.).

(c) Place a shim 1 mm (0.039 in.) thick x 70 mm (2.75 in.) long between tensioner and chain. Push tensioner and shim up against the chain. **Apply firm pressure 2.5–3 Kg (5.5–6.6 lbs.) directly behind the adjustment slot to take up all slack.** Chain must have shoe radius contact as shown in (Fig. 117).

(d) With the load applied, tighten top tensioner adjuster bolt first, then bottom shouldered pivot bolt. Tighten bolts to 12 N·m (105 in. lbs.). Remove shim.

(e) Install carrier covers and tighten screws to 12 N·m (105 in. lbs.).

(12) If removed, install oil pump (Refer to 9 - ENGINE/LUBRICATION/OIL PUMP - INSTALLATION).

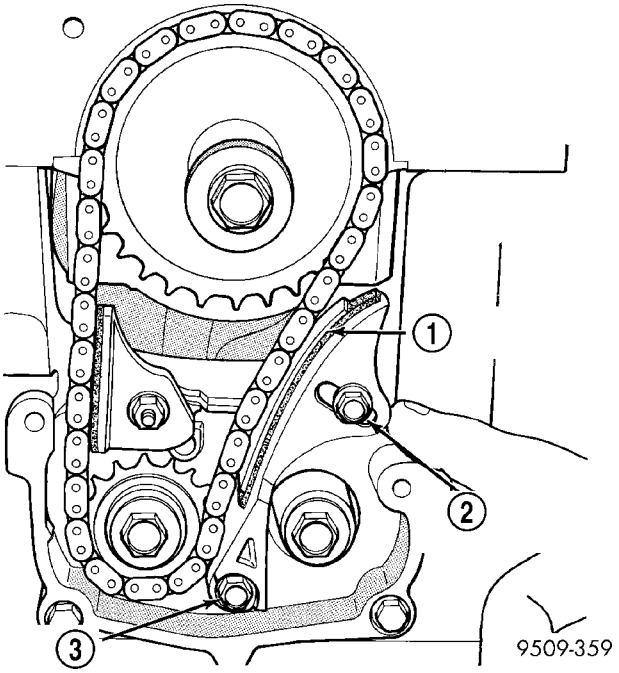

Fig. 117 Chain Tension Adjustment

1 - 1MM (0.039 IN.) SHIM
2 - TENSIONER (ADJUSTER) BOLT
3 - SHOULDERED PIVOT BOLT

BALANCE SHAFTS AND CARRIER ASSEMBLY (Continued)

(13) Install pick-up tube and oil pan (Refer to 9 - ENGINE/LUBRICATION/OIL PAN - INSTALLATION).

(14) Fill engine crankcase with proper oil to correct level.

BALANCE SHAFT CARRIER

DESCRIPTION

The 2.4L engine is equipped with two nodular cast iron balance shafts installed in a cast aluminum carrier attached to the lower cylinder block (Fig. 118).

OPERATION

The balance shafts are driven by the crankshaft via a roller chain and sprockets. The balance shafts are connected by helical gears. The dual counter rotating shafts decrease second order vertical shaking forces caused by component movement.

REMOVAL

(Refer to 9 - ENGINE/VALVE TIMING/BALANCE SHAFT - REMOVAL)

INSTALLATION

(Refer to 9 - ENGINE/VALVE TIMING/BALANCE SHAFT - INSTALLATION)

BALANCE SHAFT CHAIN

REMOVAL

(Refer to 9 - ENGINE/VALVE TIMING/BALANCE SHAFT - REMOVAL)

INSTALLATION

(Refer to 9 - ENGINE/VALVE TIMING/BALANCE SHAFT - INSTALLATION)

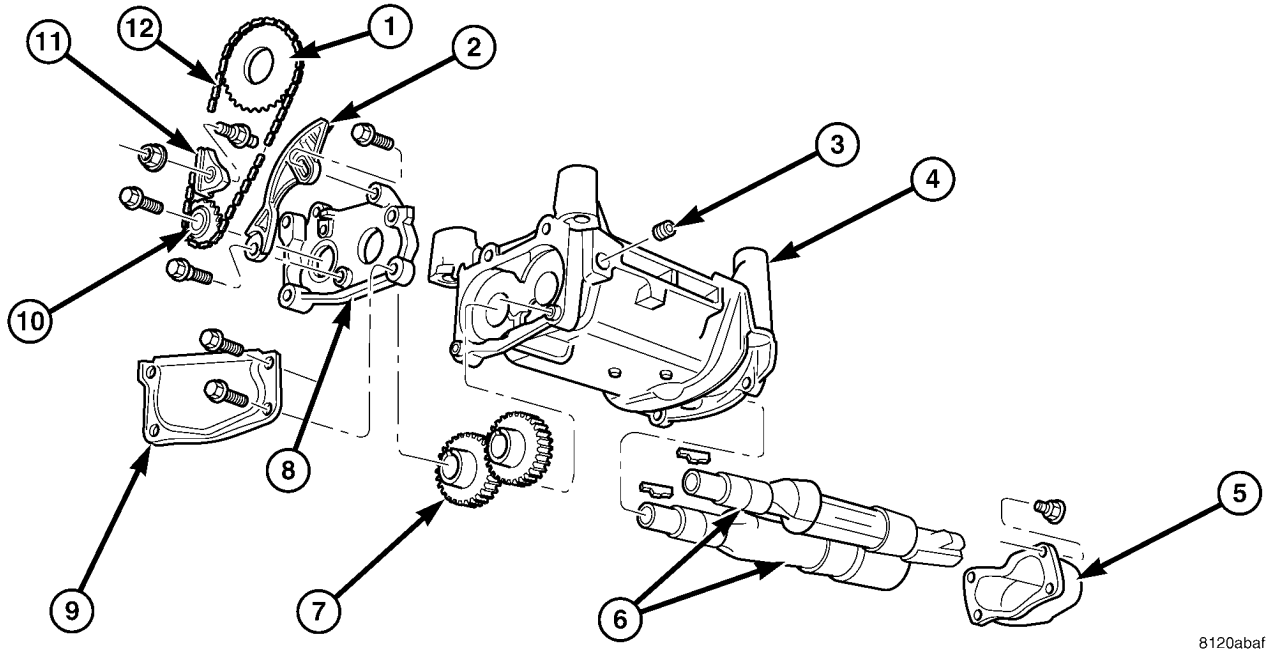

Fig. 118 Balance Shafts and Carrier Assembly

1 - SPROCKET
2 - TENSIONER
3 - PLUG
4 - CARRIER
5 - REAR COVER
6 - BALANCE SHAFTS
7 - GEARS
8 - GEAR COVER
9 - CHAIN COVER
10 - SPROCKET
11 - GUIDE
12 - CHAIN

ENGINE 4.0L

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ENGINE 4.0L

DESCRIPTION

DESCRIPTION - ENGINE BLOCK

The cylinder block is a cast iron inline six cylinder design. The cylinder block is drilled forming galleries for both oil and coolant (Fig. 1).

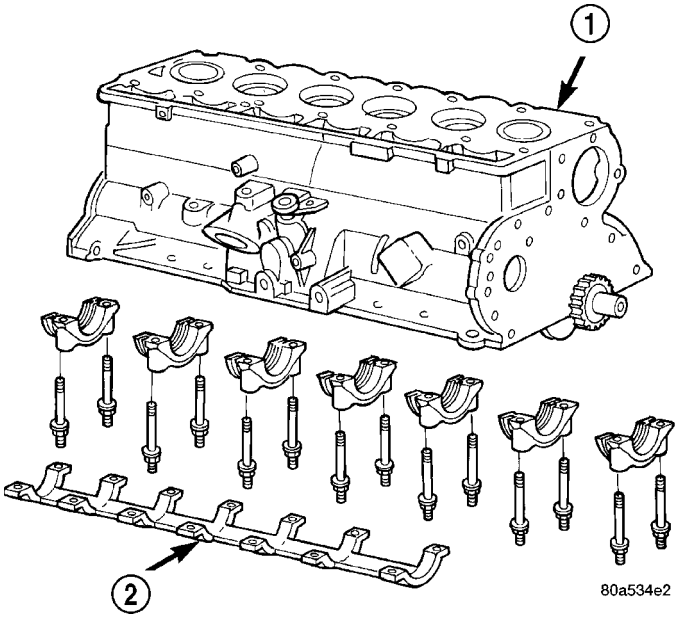

Fig. 1 4.0L Cylinder Block with Main Bearing Caps and Cap Brace

1 - BLOCK
2 - MAIN BEARING CAP BRACE

DESCRIPTION - ENGINE

The 4.0 Liter (242 CID) six-cylinder engine is an In-line, lightweight, overhead valve engine.

This engine is designed for unleaded fuel. The engine cylinder head has dual quench-type combustion chambers that create turbulence and fast burning of the air/fuel mixture. This results in good fuel economy.

The cylinders are numbered 1 through 6 from front to rear. The firing order is 1-5-3-6-2-4 (Fig. 2).

The crankshaft rotation is clockwise, when viewed from the front of the engine. The crankshaft rotates within seven main bearings. The camshaft rotates within four bearings.

BUILD DATE CODE

The engine Build Date Code is located on a machined surface on the right side of the cylinder block between the No.2 and No.3 cylinders (Fig. 3).

The digits of the code identify:

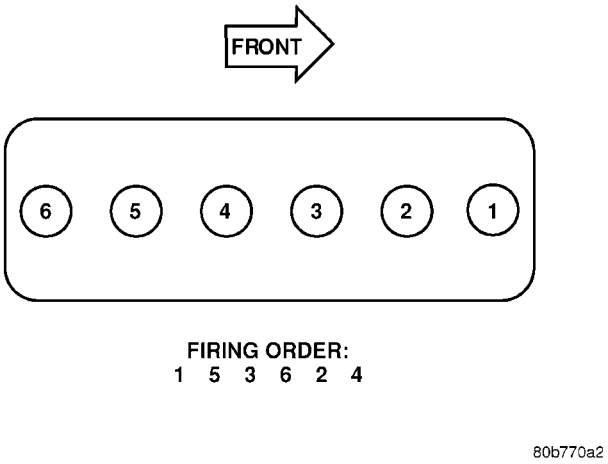

Fig. 2 Engine Firing Order

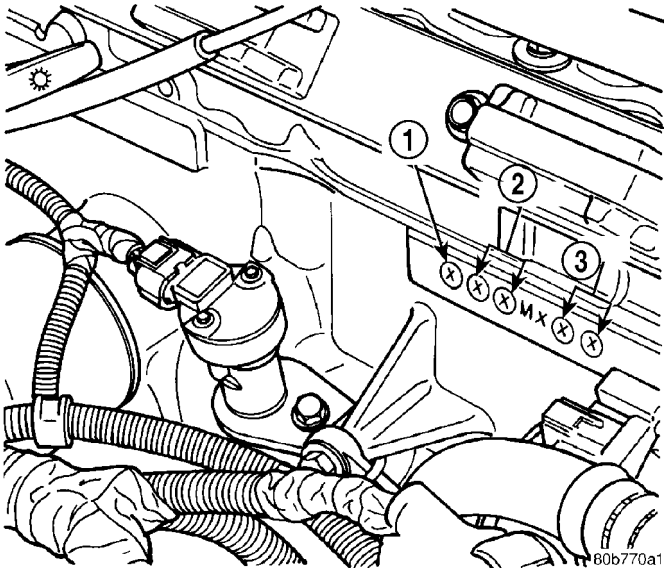

Fig. 3 Build Date Code Location

1 - YEAR
2 - MONTH
3 - DAY

- 1st Digit—The year (0 = 2000).
- 2nd & 3rd Digits—The month (01 - 12).
- 4th & 5th Digits—The engine type/fuel system/ compression ratio (MX = A 4.0 Liter (242 CID) engine with a multi-point fuel injection system).
- 6th & 7th Digits—The day of engine build (01 - 31).

(1) **FOR EXAMPLE:** Code * 001MX12 * identifies a 4.0 Liter (242 CID) engine with a multi-point fuel injection system, and built on January 12, 2000.

ENGINE 4.0L (Continued)

DIAGNOSIS AND TESTING

DIAGNOSIS AND TESTING - ENGINE DIAGNOSIS - INTRODUCTION

Engine diagnosis is helpful in determining the causes of malfunctions not detected and remedied by routine maintenance.

These malfunctions may be classified as either performance (e.g., engine idles rough and stalls) or mechanical (e.g., a strange noise).

(Refer to 9 - ENGINE - DIAGNOSIS AND TESTING) (PERFORMANCE) or (Refer to 9 - ENGINE - DIAGNOSIS AND TESTING) (MECHANICAL) for possible causes and corrections of malfunctions.

Additional tests and diagnostic procedures may be necessary for specific engine malfunctions that can not be isolated with the Service Diagnosis charts. Information concerning additional tests and diagnosis is provided within the following diagnosis:

- (Refer to 9 - ENGINE - DIAGNOSIS AND TESTING) (PERFORMANCE)
- (Refer to 9 - ENGINE - DIAGNOSIS AND TESTING) (MECHANICAL)

DIAGNOSIS AND TESTING - ENGINE DIAGNOSIS - PERFORMANCE

ENGINE PERFORMANCE DIAGNOSIS CHART

CONDITION	POSSIBLE CAUSES	CORRECTION
ENGINE WILL NOT CRANK	1. Weak or dead battery	1. Charge/Replace Battery. (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/BATTERY - STANDARD PROCEDURE), for correct procedures. Check charging system. (Refer to 8 - ELECTRICAL/ CHARGING - DIAGNOSIS AND TESTING), for correct procedures.
	2. Corroded or loose battery connections	2. Clean/tighten suspect battery/ starter connections
	3. Faulty starter or related circuit(s)	3. Check starting system. (Refer to 8 - ELECTRICAL/STARTING - DIAGNOSIS AND TESTING), for correct diagnostics/procedures
	4. Siezed accessory drive component	4. Remove accessory drive belt and attempt to start engine. If engine starts, repair/replace siezed component.
	5. Engine internal mechanical failure or hydro-static lock	5. (Refer to 9 - ENGINE - DIAGNOSIS AND TESTING), for correct diagnostics/procedures

ENGINE 4.0L (Continued)

CONDITION	POSSIBLE CAUSES	CORRECTION
ENGINE CRANKS BUT WILL NOT START	1. No spark	1. Check for spark. (Refer to 8 - ELECTRICAL/IGNITION CONTROL - SPECIFICATIONS), for correct procedures.
	2. No fuel	2. Perform fuel pressure test (Refer to 14 - FUEL SYSTEM/FUEL DELIVERY/FUEL PUMP - DIAGNOSIS AND TESTING), and if necessary, inspect fuel injector(s) and driver circuits. (Refer to 14 - FUEL SYSTEM/FUEL INJECTION/ FUEL INJECTOR - DIAGNOSIS AND TESTING), for correct procedures.
	3. Low or no engine compression	3. Perform cylinder compression pressure test. (Refer to 9 - ENGINE - DIAGNOSIS AND TESTING).
ENGINE LOSS OF POWER	1. Worn or burned distributor rotor	1. Install new distributor rotor
	2. Worn camshaft position sensor shaft	2. Remove and repair camshaft position sensor. (Refer to 8 - ELECTRICAL/IGNITION CONTROL/ CAMSHAFT POSITION SENSOR - REMOVAL).
	3. Worn or incorrect gapped spark plugs	3. Clean plugs and set gap. (Refer to 8 - ELECTRICAL/IGNITION CONTROL/SPARK PLUG - CLEANING).
	4. Dirt or water in fuel system	4. Clean system and replace fuel filter
	5. Faulty fuel pump	5. Install new fuel pump
	6. Incorrect valve timing	6. Correct valve timing
	7. Blown cylinder head gasket	7. Install new cylinder head gasket
	8. Low compression	8. Test cylinder compression. (Refer to 9 - ENGINE - DIAGNOSIS AND TESTING).
	9. Burned, warped, or pitted valves	9. Install/Reface valves as necessary
	10. Plugged or restricted exhaust system	10. Install new parts as necessary
	11. Faulty ignition coil rail	11. Test and replace, as necessary. (Refer to 8 - ELECTRICAL/ IGNITION CONTROL/COIL RAIL - REMOVAL).
ENGINE STALLS OR ROUGH IDLE	1. Carbon build-up on throttle plate	1. Remove throttle body and de-carbon. (Refer to 14 - FUEL SYSTEM/FUEL INJECTION/ THROTTLE BODY - REMOVAL) for correct procedure.

ENGINE 4.0L (Continued)

CONDITION	POSSIBLE CAUSES	CORRECTION
	2. Engine idle speed too low	2. Check Idle Air Control circuit.
	3. Worn or incorrectly gapped spark plugs	3. Replace or clean and re-gap spark plugs. (Refer to 8 - ELECTRICAL/IGNITION CONTROL/ SPARK PLUG - CLEANING)
	4. Faulty coil rail	4. Test and replace, if necessary. (Refer to 8 - ELECTRICAL/ IGNITION CONTROL/COIL RAIL - REMOVAL)
	5. Intake manifold vacuum leak	5. Inspect intake manifold gasket and vacuum hoses. Replace if necessary. (Refer to 9 - ENGINE/ MANIFOLDS/INTAKE MANIFOLD - DIAGNOSIS AND TESTING).
ENGINE MISSES ON ACCELERATION	1. Worn or incorrectly gapped spark plugs	1. Replace spark plugs or clean and set gap.
	2. Spark plug cables defective or crossed	2. Replace spark plug cables.
	3. Dirt in fuel system	3. Clean fuel system
	4. Burned, warped or pitted valves	4. Install new valves
	5. Faulty coil rail	5. Test and replace as necessary. (Refer to 8 - ELECTRICAL/ IGNITION CONTROL/COIL RAIL - REMOVAL)

ENGINE 4.0L (Continued)

DIAGNOSIS AND TESTING - ENGINE DIAGNOSIS - MECHANICAL

ENGINE MECHANICAL DIAGNOSIS CHART

CONDITION	POSSIBLE CAUSES	CORRECTION
NOISY VALVES/LIFTERS	1. High or low oil level in crankcase	1. Check for correct oil level. Adjust oil level by draining or adding as needed
	2. Thin or diluted oil	2. Change oil. (Refer to 9 - ENGINE/LUBRICATION/OIL - STANDARD PROCEDURE)
	3. Low oil pressure	3. Check engine oil level. If ok, Perform oil pressure test. (Refer to 9 - ENGINE/LUBRICATION - DIAGNOSIS AND TESTING) for engine oil pressure test/ specifications
	4. Dirt in tappets/lash adjusters	4. Clean/replace hydraulic tappets/lash adjusters
	5. Bent push rod(s)	5. Install new push rods
	6. Worn rocker arms	6. Inspect oil supply to rocker arms and replace worn arms as needed
	7. Worn tappets/lash adjusters	7. Install new hydraulic tappets/lash adjusters
	8. Worn valve guides	8. Inspect all valve guides and replace as necessary
	9. Excessive runout of valve seats or valve faces	9. Grind valves and seats
CONNECTING ROD NOISE	1. Insufficient oil supply	1. Check engine oil level.
	2. Low oil pressure	2. Check engine oil level. If ok, Perform oil pressure test. (Refer to 9 - ENGINE/LUBRICATION - DIAGNOSIS AND TESTING) engine oil pressure test/specifications
	3. Thin or diluted oil	3. Change oil to correct viscosity. (Refer to 9 - ENGINE/ LUBRICATION/OIL - STANDARD PROCEDURE) for correct procedure/engine oil specifications
	4. Excessive connecting rod bearing clearance	4. Measure bearings for correct clearance with plasti-gage. Repair as necessary
	5. Connecting rod journal out of round	5. Replace crankshaft or grind journals
	6. Misaligned connecting rods	6. Replace bent connecting rods

ENGINE 4.0L (Continued)

CONDITION	POSSIBLE CAUSES	CORRECTION
MAIN BEARING NOISE	1. Insufficient oil supply	1. Check engine oil level.
	2. Low oil pressure	2. Check engine oil level. If ok, Perform oil pressure test. (Refer to 9 - ENGINE/LUBRICATION - DIAGNOSIS AND TESTING)
	3. Thin or diluted oil	3. Change oil to correct viscosity.
	4. Excessive main bearing clearance	4. Measure bearings for correct clearance. Repair as necessary
	5. Excessive end play	5. Check crankshaft thrust bearing for excessive wear on flanges
	6. Crankshaft main journal out of round or worn	6. Grind journals or replace crankshaft
	7. Loose flywheel or torque converter	7. Inspect crankshaft, flexplate/ flywheel and bolts for damage. Tighten to correct torque
LOW OIL PRESSURE	1. Low oil level	1. Check oil level and fill if necessary
	2. Faulty oil pressure sending unit	2. Install new sending unit
	3. Clogged oil filter	3. Install new oil filter
	4. Worn oil pump	4. Replace oil pump assembly.
	5. Thin or diluted oil	5. Change oil to correct viscosity.
	6. Excessive bearing clearance	6. Measure bearings for correct clearance
	7. Oil pump relief valve stuck	7. Remove valve to inspect, clean and reinstall
	8. Oil pickup tube loose, broken, bent or clogged	8. Inspect oil pickup tube and pump, and clean or replace if necessary
	9. Oil pump cover warped or cracked	9. Install new oil pump
OIL LEAKS	1. Misaligned or deteriorated gaskets	1. Replace gasket
	2. Loose fastener, broken or porous metal part	2. Tighten, repair or replace the part
	3. Front or rear crankshaft oil seal leaking	3. Replace seal
	4. Leaking oil gallery plug or cup plug	4. Remove and reseal threaded plug. Replace cup style plug

ENGINE 4.0L (Continued)

CONDITION	POSSIBLE CAUSES	CORRECTION
EXCESSIVE OIL CONSUMPTION OR SPARK PLUGS OIL FOULED	1. CCV System malfunction	1. (Refer to 25 - EMISSIONS CONTROL/EVAPORATIVE EMISSIONS - DESCRIPTION) for correct operation
	2. Defective valve stem seal(s)	2. Repair or replace seal(s)
	3. Worn or broken piston rings	3. Hone cylinder bores. Install new rings
	4. Scuffed pistons/cylinder walls	4. Hone cylinder bores and replace pistons as required
	5. Carbon in oil control ring groove	5. Remove rings and de-carbon piston
	6. Worn valve guides	6. Inspect/replace valve guides as necessary
	7. Piston rings fitted too tightly in grooves	7. Remove rings and check ring end gap and side clearance. Replace if necessary

DIAGNOSIS AND TESTING—CYLINDER COMPRESSION PRESSURE

The results of a cylinder compression pressure test can be utilized to diagnose several engine malfunctions.

Ensure the battery is completely charged and the engine starter motor is in good operating condition. Otherwise, the indicated compression pressures may not be valid for diagnosis purposes.

(1) Clean the spark plug recesses with compressed air.

(2) Remove the spark plugs (Refer to 8 - ELECTRICAL/IGNITION CONTROL/SPARK PLUG - REMOVAL).

(3) Secure the throttle in the wide-open position.

(4) Disconnect the ignition coil.

(5) Insert a compression pressure gauge and rotate the engine with the engine starter motor for three revolutions.

(6) Record the compression pressure on the third revolution. Continue the test for the remaining cylinders.

(Refer to 9 - ENGINE - SPECIFICATIONS) for the correct engine compression pressures.

DIAGNOSIS AND TESTING - CYLINDER COMBUSTION PRESSURE LEAKAGE

The combustion pressure leakage test provides an accurate means for determining engine condition.

Combustion pressure leakage testing will detect:

- Exhaust and intake valve leaks (improper seating).
- Leaks between adjacent cylinders or into water jacket.
- Any causes for combustion/compression pressure loss.

(1) Check the coolant level and fill as required. DO NOT install the radiator cap.

(2) Start and operate the engine until it attains normal operating temperature, then turn the engine OFF.

(3) Remove the spark plugs (Refer to 8 - ELECTRICAL/IGNITION CONTROL/SPARK PLUG - REMOVAL).

(4) Remove the oil filler cap.

(5) Remove the air cleaner.

(6) Calibrate the tester according to the manufacturer's instructions. The shop air source for testing should maintain 483 kPa (70 psi) minimum, 1,379 kPa (200 psi) maximum and 552 kPa (80 psi) recommended.

(7) Perform the test procedures on each cylinder according to the tester manufacturer's instructions. While testing, listen for pressurized air escaping through the throttle body, tailpipe and oil filler cap opening. Check for bubbles in the radiator coolant.

All gauge pressure indications should be equal, with no more than 25% leakage.

FOR EXAMPLE: At 552 kPa (80 psi) input pressure, a minimum of 414 kPa (60 psi) should be maintained in the cylinder.

Refer to CYLINDER COMBUSTION PRESSURE LEAKAGE DIAGNOSIS CHART

ENGINE 4.0L (Continued)

CYLINDER COMBUSTION PRESSURE LEAKAGE DIAGNOSIS CHART

CONDITION	POSSIBLE CAUSE	CORRECTION
AIR ESCAPES THROUGH THROTTLE BODY	Intake valve bent, burnt, or not seated properly	Inspect valve and valve seat. Reface or replace, as necessary
AIR ESCAPES THROUGH TAILPIPE	Exhaust valve bent, burnt, or not seated properly	Inspect valve and valve seat. Reface or replace, as necessary
AIR ESCAPES THROUGH RADIATOR	Head gasket leaking or cracked cylinder head or block	Remove cylinder head and inspect. Replace defective part
MORE THAN 50% LEAKAGE FROM ADJACENT CYLINDERS	Head gasket leaking or crack in cylinder head or block between adjacent cylinders	Remove cylinder head and inspect. Replace gasket, head, or block as necessary
MORE THAN 25% LEAKAGE AND AIR ESCAPES THROUGH OIL FILLER CAP OPENING ONLY	Stuck or broken piston rings; cracked piston; worn rings and/or cylinder wall	Inspect for broken rings or piston. Measure ring gap and cylinder diameter, taper and out-of-round. Replace defective part as necessary

STANDARD PROCEDURE

STANDARD PROCEDURE - FORM-IN-PLACE GASKETS

There are several places where form-in-place gaskets are used on the engine. **DO NOT use form-in-place gasket material unless specified.**Care must be taken when applying form-in-place gaskets. Bead size, continuity and location are of great importance. Too thin a bead can result in leakage while too much can result in spill-over. A continuous bead of the proper width is essential to obtain a leak-free joint.

Two types of form-in-place gasket materials are used in the engine area (Mopar® Silicone Rubber Adhesive Sealant and Mopar® Gasket Maker). Each have different properties and cannot be used interchangeably.

MOPAR® SILICONE RUBBER ADHESIVE SEALANT

Mopar® Silicone Rubber Adhesive Sealant, normally black in color, is available in 3 ounce tubes. Moisture in the air causes the sealant material to cure. This material is normally used on flexible metal flanges. It has a shelf life of a year and will not properly cure if over aged. Always inspect the package for the expiration date before use.

MOPAR® GASKET MAKER

Mopar® Gasket Maker, normally red in color, is available in 6 cc tubes. This anaerobic type gasket material cures in the absence of air when squeezed between smooth machined metallic surfaces. It will not cure if left in the uncovered tube. DO NOT use on flexible metal flanges.

SURFACE PREPARATION

Parts assembled with form-in-place gaskets may be disassembled without unusual effort. In some instances, it may be necessary to lightly tap the part with a mallet or other suitable tool to break the seal between the mating surfaces. A flat gasket scraper may also be lightly tapped into the joint but care must be taken not to damage the mating surfaces.

Scrape or wire brush all gasket surfaces to remove all loose material. Inspect stamped parts to ensure gasket rails are flat. Flatten rails with a hammer on a flat plate, if required. Gasket surfaces must be free of oil and dirt. Make sure the old gasket material is removed from blind attaching holes.

GASKET APPLICATION

Assembling parts using a form-in-place gasket requires care.

Mopar® Silicone Rubber Adhesive Sealant should be applied in a continuous bead approximately 3 mm (0.12 inch) in diameter. All mounting holes must be circled. For corner sealing, a 3 or 6 mm (1/8 or 1/4 inch) drop is placed in the center of the gasket contact area. Uncured sealant may be removed with a shop towel. Components should be torqued in place while the sealant is still wet to the touch (within 10 minutes). The use of a locating dowel is recommended during assembly to prevent smearing the material off location.

Mopar® Gasket Maker should be applied sparingly to one gasket surface. The sealant diameter should be 1.00 mm (0.04 inch) or less. Be certain the material surrounds each mounting hole. Excess material can easily be wiped off. Components should be torqued in place within 15 minutes. The use of a locating dowel is recommended during assembly to prevent smearing the material off location.

ENGINE 4.0L (Continued)

STANDARD PROCEDURE - HYDROSTATIC LOCK

When an engine is suspected of hydrostatic lock (regardless of what caused the problem), follow the steps below.

(1) Perform the Fuel Pressure Release Procedure .

(2) Disconnect the negative cable from the battery.

(3) Inspect air cleaner, induction system and intake manifold to ensure system is dry and clear of foreign material.

(4) Place a shop towel around the spark plugs to catch any fluid that may possibly be under pressure in the cylinder head. Remove the plugs from the engine.

CAUTION: DO NOT use the starter motor to rotate the crankshaft. Severe damage could occur.

(5) With all spark plugs removed, rotate the crankshaft using a breaker bar and socket.

(6) Identify the fluid in the cylinders (i.e. coolant, fuel, oil, etc.).

(7) Make sure all fluid has been removed from the cylinders.

(8) Repair engine or components as necessary to prevent this problem from occurring again.

(9) Squirt engine oil into the cylinders to lubricate the walls. This will prevent damage on restart.

(10) Install new spark plugs. Tighten the spark plugs to 37 N·m (27 ft. lbs.) torque.

(11) Drain engine oil. Remove and discard the oil filter.

(12) Install the drain plug. Tighten the plug to 34 N·m (25 ft. lbs.) torque.

(13) Install a new oil filter.

(14) Fill engine crankcase with the specified amount and grade of oil.

(15) Connect the negative cable to the battery.

(16) Start the engine and check for any leaks.

STANDARD PROCEDURE - SERVICE ENGINE ASSEMBLY (SHORT BLOCK)

A service replacement engine assembly (short block) may be installed whenever the original cylinder block is defective or damaged beyond repair. It consists of the cylinder block, crankshaft, piston and rod assemblies. If needed, the camshaft must be procured separately and installed before the engine is installed in the vehicle.

A short block is identified with the letter “S” stamped on the same machined surface where the build date code is stamped for complete engine assemblies.

Installation includes the transfer of components from the defective or damaged original engine. Follow the appropriate procedures for cleaning, inspection and torque tightening.

STANDARD PROCEDURE - REPAIR DAMAGED OR WORN THREADS

CAUTION: Be sure that the tapped holes maintain the original center line.

Damaged or worn threads can be repaired. Essentially, this repair consists of:

- Drilling out worn or damaged threads.
- Tapping the hole with a special Heli-Coil Tap, or equivalent.
- Installing an insert into the tapped hole to bring the hole back to its original thread size.

STANDARD PROCEDURE - ENGINE PERFORMANCE

It is important that the vehicle is operating to its optimum performance level to maintain fuel economy and the lowest emission levels. If vehicle is not operating to these standards, refer to Engine Diagnosis outlined in this section. The following procedures can assist in achieving the proper engine diagnosis.

(1) Test cranking amperage draw. (Refer to 8 - ELECTRICAL/STARTING - DIAGNOSIS AND TESTING).

(2) Check intake manifold bolt torque (Refer to 9 - ENGINE/MANIFOLDS/INTAKE MANIFOLD - INSTALLATION).

(3) Perform cylinder compression test. (Refer to 9 - ENGINE - DIAGNOSIS AND TESTING).

(4) Clean (Refer to 8 - ELECTRICAL/IGNITION CONTROL/SPARK PLUG - CLEANING) or replace spark plugs (Refer to 8 - ELECTRICAL/IGNITION CONTROL/SPARK PLUG - REMOVAL) as necessary.

(5) Test coil output voltage and primary resistance. (Refer to 8 - ELECTRICAL/IGNITION CONTROL - SPECIFICATIONS) Replace parts as necessary.

(6) Test fuel pump for pressure. (Refer to 14 - FUEL SYSTEM/FUEL DELIVERY/FUEL PUMP - DIAGNOSIS AND TESTING).

(7) The air filter elements should be replaced as specified.

(8) Inspect crankcase ventilation system.

(9) Road test vehicle as a final test.

REMOVAL

(1) Disconnect the battery negative cable.

(2) Mark the hinge locations on the hood panel for alignment reference during installation. Remove the engine compartment lamp. Remove the hood.

(3) Drain engine coolant (Refer to 7 - COOLING - STANDARD PROCEDURE), drain the coolant into a clean container for reuse.

ENGINE 4.0L (Continued)

(4) Remove the upper radiator hose and coolant recovery hose (Fig. 4).

(5) Remove the lower radiator hose.

(6) Remove upper radiator support retaining bolts and remove radiator support.

(7) Remove the fan assembly from the water pump (Refer to 7 - COOLING/ENGINE/FAN DRIVE VISCOUS CLUTCH - REMOVAL).

(8) Remove the fan shroud (Fig. 4).

(9) Disconnect the transmission fluid cooler lines (automatic transmission).

(10) Discharge the A/C system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE).

(11) Remove the service valves and cap the compressor ports.

(12) Remove the radiator (Refer to 7 - COOLING/ENGINE/RADIATOR - REMOVAL) or radiator/condenser (if equipped with A/C).

(13) Disconnect the heater hoses at the engine thermostat housing and water pump (Fig. 4).

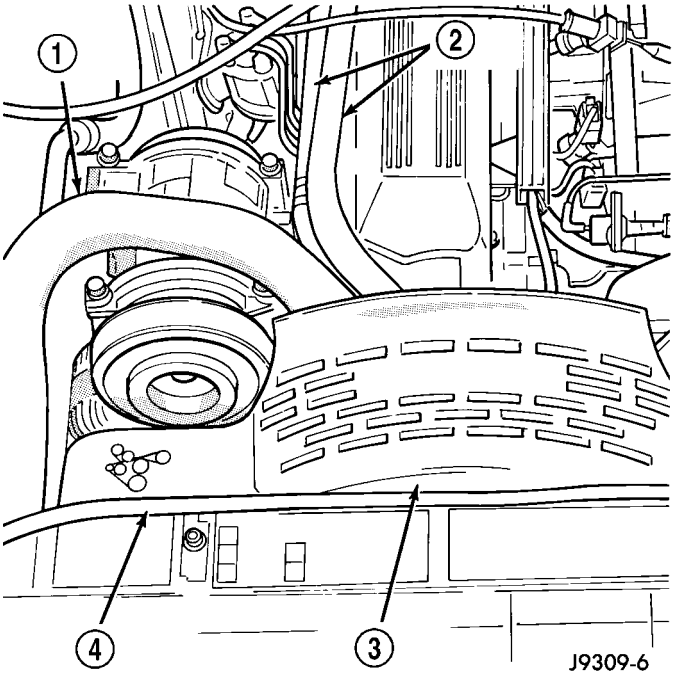

Fig. 4 Upper Radiator Hose, Coolant Recovery Hose, Fan Shroud

1 - UPPER RADIATOR HOSE
2 - HEATER HOSES
3 - FAN SHROUD
4 - COOLANT RECOVERY HOSE

(14) Disconnect the accelerator cable, transmission line pressure cable and speed control cable (if equipped) from the throttle body (Fig. 5).

(15) Remove cables from the bracket and secure out of the way.

(16) Disconnect the body ground at the engine.

(17) Disconnect the following connectors and secure their harness out of the way.

- Power steering pressure switch
- Coolant temperature sensor
- Six (6) fuel injector connectors
- Intake air temperature sensor
- Throttle position sensor
- Map sensor
- Crankshaft position sensor
- Oxygen sensor
- Camshaft position sensor
- Generator connector and B+ terminal wire

(18) Disconnect the coil rail electrical connections and the oil pressure switch connector.

(19) Perform the fuel pressure release procedure (Refer to 14 - FUEL SYSTEM/FUEL DELIVERY - STANDARD PROCEDURE).

(20) Disconnect the fuel supply line at the injector rail (Refer to 14 - FUEL SYSTEM/FUEL DELIVERY/QUICK CONNECT FITTING - STANDARD PROCEDURE).

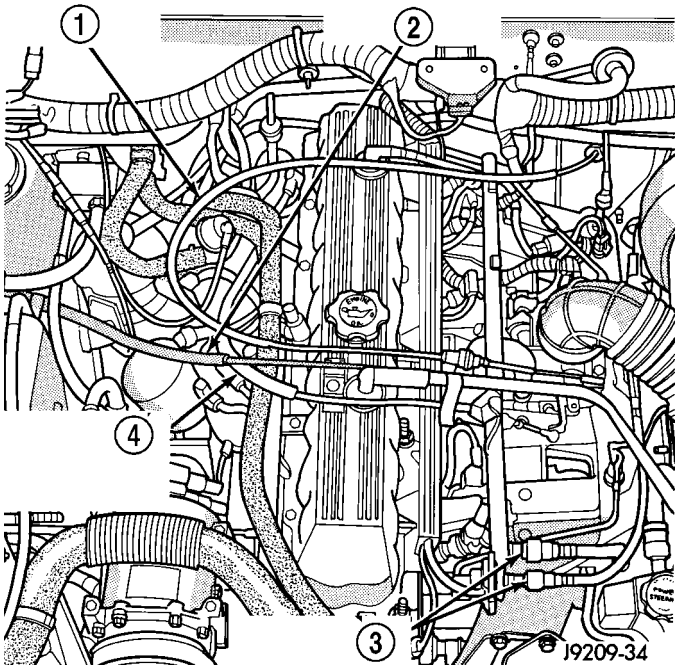

Fig. 5 Accelerator Cable, Vehicle Speed Control Cable, Automatic Transmission

1 - ACCELERATOR CABLE
2 - SPEED CONTROL CABLE
3 - QUICK-CONNECT FUEL LINES
4 - AUTOMATIC TRANSMISSION CONTROL CABLE

(21) Remove the fuel line bracket from the intake manifold.

(22) Remove the air cleaner assembly (Fig. 6).

(23) Disconnect the hoses from the fittings at the steering gear.

(24) Drain the pump reservoir.

ENGINE 4.0L (Continued)

(25) Cap the fittings on the hoses and steering gear to prevent foreign objects from entering the system.

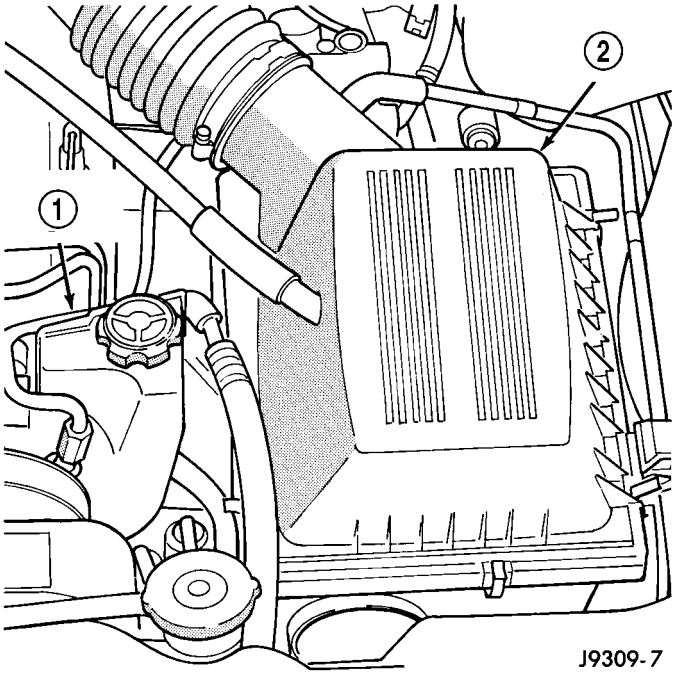

Fig. 6 Air Cleaner Assembly & Power Steering Pump

1 - POWER STEERING PUMP
2 - AIR CLEANER ASSEMBLY

(26) Raise and support the vehicle.

(27) Disconnect the wires from the engine starter motor solenoid.

(28) Remove the engine starter motor (Refer to 8 - ELECTRICAL/STARTING/STARTER MOTOR - REMOVAL).

(29) Disconnect the oxygen sensor from the exhaust pipe.

(30) Disconnect the exhaust pipe from the manifold.

(31) Remove the exhaust pipe support.

(32) Remove the bending brace (Refer to 9 - ENGINE/ENGINE BLOCK/WINDAGE TRAY / STRUCT SUPPORT - REMOVAL).

(33) Remove the engine flywheel/converter housing access cover.

(34) Mark the converter and drive plate location.

(35) Remove the converter-to-drive plate bolts.

(36) Remove the upper engine flywheel/converter housing bolts and loosen the bottom bolts.

(37) Remove the engine mount cushion-to-engine compartment bracket bolts.

(38) Lower the vehicle.

(39) Attach a lifting device to the engine.

(40) Raise the engine off the front supports.

(41) Place a support or floor jack under the converter (or engine flywheel) housing.

(42) Remove the remaining converter (or engine flywheel) housing bolts.

(43) Lift the engine out of the engine compartment.

INSTALLATION

CAUTION: When installing the engine into a vehicle equipped with an automatic transmission, be careful not to damage the trigger wheel on the engine flywheel.

(1) Attach a lifting device to the engine and lower the engine into the engine compartment. For easier installation, it may be necessary to remove the engine mount bracket as an aid in alignment of the engine to the transmission.

(2) Align the transmission torque converter housing with the engine.

(3) Loosely install the converter housing lower bolts and install the next higher bolt and nut on each side.

(4) Tighten all 4 bolts finger tight.

(5) Install the engine mount brackets (if removed).

(6) Lower the engine and engine mount brackets onto the engine compartment cushions. Install the bolts and finger tighten the nuts.

(7) Remove the engine lifting device.

(8) Raise and support the vehicle.

(9) Install the remaining engine flywheel/converter housing bolts. Tighten all bolts to 38 N·m (28 ft. lbs.) torque.

(10) Install the converter-to-drive plate bolts.

(11) Ensure the installation reference marks are aligned.

(12) Install the engine flywheel/converter housing access cover.

(13) Install the exhaust pipe support and tighten the screw.

(14) Install the engine bending brace (Refer to 9 - ENGINE/ENGINE BLOCK/WINDAGE TRAY / STRUCT SUPPORT - INSTALLATION).

(15) Tighten the engine mount-to-bracket bolts.

(16) Connect the vehicle speed sensor wire connections and tighten the screws.

(17) Connect the exhaust pipe to the manifold.

(18) Install the engine starter motor (Refer to 8 - ELECTRICAL/STARTING/STARTER MOTOR - INSTALLATION).

(19) Connect the wires to the engine starter motor solenoid.

(20) Lower the vehicle.

(21) Connect vacuum hoses and wire connectors disconnected during engine removal.

(22) Remove protective caps from the power steering hoses.

ENGINE 4.0L (Continued)

(23) Connect the hoses to the fittings at the steering gear. Tighten the nut to 52 N·m (38 ft. lbs.) torque.

(24) Fill the pump reservoir with fluid (Refer to 19 - STEERING/PUMP - STANDARD PROCEDURE).

(25) Inspect the fuel supply line o-ring(s) and replace if necessary. Connect fuel supply line to injectior rail and verify connection by pulling outward on the line.

(26) Install the fuel line bracket to the intake manifold.

(27) Connect the coil rail electrical connectors and oil pressure switch connector.

(28) Connect the following electrical connectors:

- Power steering pressure switch
- Coolant temperature sensor
- Six (6) fuel injector connectors
- Intake air temperature sensor
- Throttle position sensor
- Map sensor
- Crankshaft position sensor
- Oxygen sensor
- Camshaft position sensor
- Generator connector and B+ terminal wire

(29) Connect all previously removed vacuum hoses.

(30) Connect the body ground strap.

(31) Install the throttle, transmission line pressure, and speed control cables to their mounting bracket and connect them to the throttle body.

(32) Connect the heater hoses at the engine thermostat housing and water pump.

(33) Install the fan assembly to the water pump (Refer to 7 - COOLING/ENGINE/FAN DRIVE VISCOUS CLUTCH - INSTALLATION).

(34) Place the fan shroud in position over the fan.

(35) Install the radiator or radiator/condenser (Refer to 7 - COOLING/ENGINE/RADIATOR - INSTALLATION).

(36) Connect the service valves to the A/C compressor ports, if equipped with A/C.

(37) Charge the air conditioner system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE).

(38) Connect the radiator hoses and automatic transmission fluid cooler pipes, if equipped.

(39) Install the fan shroud to the radiator or radiator/condenser (if equipped with A/C).

(40) Install upper radiator support.

(41) Connect the upper radiator hose and tighten the clamp.

(42) Connect the lower radiator hose and tighten the clamp.

(43) Fill crankcase with engine oil (Refer to LUBRICATION & MAINTENANCE - SPECIFICATIONS).

(44) Fill the cooling system (Refer to 7 - COOLING - STANDARD PROCEDURE).

(45) Align the hood to the scribe marks. Install the hood.

(46) Install the air cleaner assembly.

(47) Install the battery and connect the battery cable.

WARNING: USE EXTREME CAUTION WHEN THE ENGINE IS OPERATING. DO NOT STAND IN A DIRECT LINE WITH THE FAN. DO NOT PUT YOUR HANDS NEAR THE PULLEYS, BELTS OR FAN. DO NOT WEAR LOOSE CLOTHING.

(48) Start the engine, inspect for leaks and correct the fluid levels, as necessary.

SPECIFICATIONS

ENGINE - 4.0L

GENERAL DESCRIPTION

DESCRIPTION	SPECIFICATION
Engine Type	In-Line 6 Cylinder
Displacement	4.0 Liters / 4000 cc 242 (Cubic Inches)
Bore	98.4 mm (3.88 in.)
Stroke	86.69 mm (3.413 in.)
Compression Ratio	8.8:1
Firing Order	1-5-3-6-2-4
Lubrication	Pressure Feed-Full Flow-Filtration
Cooling System	Liquid Cooled-Forced Circulation
Cylinder Block	Cast Iron
Crankshaft	Cast Nodular Iron
Cylinder Head	Cast Iron
Camshaft	Cast Iron
Pistons	Aluminum Alloy
Combustion Chamber	Dual-Quench
Connecting Rods	Cast Malleable Iron

ENGINE 4.0L (Continued)

CYLINDER BLOCK

DESCRIPTION	SPECIFICATION	
	Metric	Standard
CylinderBore Diameter	98.45 - 98.48 mm	3.8759 - 3.8775 in.
Out of Round (MAX)	0.025 mm	0.001 in.
Taper (MAX)	0.025 mm	0.001 in.
Deck Height	240.03 - 240.18 mm	9.450 - 9.456 in.
Deck Clearance (Below Deck)	0.546 mm	0.0215 in.
Tappet Bore Diameter	23.000 - 23.025 mm	0.9055 - 0.9065
Flatness	0.03 mm per 25 mm 0.05 mm per 152 mm	0.001 in. per 1 in. 0.002 in. per 6 in.
Flatness (MAX)	0.20 mm (MAX) for total length	0.008 in. MAX) for total length
Main Bearing Bore Diameter	68.351 - 68.376 mm	2.691 - 2.692 in.

PISTONS

DESCRIPTION	SPECIFICATION	
	Metric	Standard
Weight (Less Pin)	417 - 429 grams	14.7 - 15.1 oz
Piston Pin Bore Diameter	23.650 - 23.658 mm	0.9312 - 0.9315 in.
Piston Pin To Bore (Centerline To Piston Top)	40.61 - 40.72 mm	1.599 - 1.603 in.
Piston To Bore Clearance	0.018 - 0.038 mm	0.0008 - 0.0015 in.
Piston Ring Groove Height		
Compression Rings	1.530 - 1.555 mm	0.0602 - 0.0612 in.
Oil Control Rings	4.035 - 4.060 mm	0.1589 - 0.1598 in.
Piston Ring Groove Diameter		
No. 1 Compression	88.39 - 88.65 mm	3.48 - 3.49 in
No. 2 Compression	87.63 - 87.88 mm	3.45 - 3.46 in.
No. 3 Oil Control	89.66 - 89.92 mm	3.53 - 3.54 in.

PISTON PINS

DESCRIPTION	SPECIFICATION	
	Metric	Standard
Piston To Pin Clearance	0.0102 - 0.0208 mm	0.0005 - 0.0009 in.
Diameter	23.637 - 23.640 mm	0.9306 - 0.9307 in.
Piston-Pin To Connecting Rod (Press Fit)	8.9kN	2000 lbf.

PISTON RINGS

DESCRIPTION	SPECIFICATION	
	Metric	Standard
Ring Gap		
Top Compression Ring	0.229 - 0.610 mm	0.0090 - 0.0240 in.
Second Compression Ring	0.483 - 0.965 mm	0.0190 - 0.0380 in.
Oil Control (Steel Rails)	0.254 - 1.500 mm	0.010 - 0.060 in.
Side Clearance		
Top Compression Ring	.042 - .084 mm	0.0017 - 0.0033 in.
Second Compression Ring	0.042 - 0.084 mm	0.0017 - 0.0033 in.
Oil Ring (Steel Ring)	.06 - .21 mm	.0024 - .0083 in.

ENGINE 4.0L (Continued)

CONNECTING RODS

DESCRIPTION	SPECIFICATION	
	Metric	Standard
Bearing Clearance	0.025 - 0.076 mm	0.001 - 0.003 in.
Preferred	0.044 - 0.050	0.0015 - 0.0020
Side Clearance	0.25 - 0.48 mm	0.010 - 0.019 in.
Piston Pin Bore Diameter	.023.59 - 23.62 mm	0.9288 - 0.9298 in.
Bearing Bore (Less Bearings)	56.08 - 56.09mm	2.2080 - 2.2085 in.
Total Weight (Less Bearing)	663 - 671 grams	23.39 - 23.67 oz.
Length (Center To Center)	155.52 - 155.62 mm	6.123 - 6.127 in.
Twist (MAX.)	0.002 mm per mm	0.002 in. per inch
Bend (MAX.)	0.002 mm per mm	0.002 in. per inch.

CRANKSHAFT

DESCRIPTION	SPECIFICATION	
	Metric	Standard
Main Bearing Journal Diameter		
No. 1-6	63.489 - 63.502 mm	2.499 - 2.500 in.
No. 7	63.449 - 63.487	2.498 - 2.499 in.
Bearing Clearance	0.03 - 0.06 mm	0.001 - 0.0025 in.
Preferred	0.051 mm	0.002 in.
Out of Round (MAX)	0.013 mm	0.0005 in.
Taper (MAX)	0.013 mm	0.0005 in.
End Play	0.038 - 0.165 mm	0.0015 - 0.0065 in.
Connecting Rod Journal Diameter	53.17 - 53.23 mm	2.0934 - 2.0955 in.

CAMSHAFT

DESCRIPTION	SPECIFICATION	
	Metric	Standard
Hydraulic Tappet Clearance	Zero Lash	
Bore Diameter	26.02 - 26.04 mm	1.0245 - 1.0252 in.
Bearing Journal Diameter		
No. 1	51.54 - 51.56 mm	2.029 - 2.030 in.
No. 2	51.28 - 51.31 mm	2.019 - 2.020 in.
No. 3	51.03 - 51.05 mm	2.009 - 2.010 in.
No. 4	50.78 - 50.80 mm	1.999 - 2.000 in.
Base Circle Runout (MAX)	0.033 mm	0.001 in.
Valve Lift		
Intake	10.350 mm	0.4075 in.
Exhaust	10.528 mm	0.4145 in.

VALVE TIMING

DESCRIPTION	SPECIFICATION
Intake	
Opens (BTDC)	12.4°
Closes (ABDC)	60.9°
Duration	253.3°
Exhaust	
Opens (BBDC)	49.8°
Closes (ATDC)	29.2°
Duration	259.0°
Valve Overlap	41.6°

ENGINE 4.0L (Continued)

VALVES

DESCRIPTION	SPECIFICATION
Face Angle	46.5°
Head Diameter	
Intake	48.38 - 48.64 mm (1.905 - 1.915 in.)
Exhaust	37.97 - 38.22 mm (1.495 - 1.505 in.)
Length (Overall)	
Intake	122.47 - 122.86 mm (4.822 - 4.837)
Exhaust	122.86 - 123.24 mm (4.83 - 4.85 in.)
Stem Diameter	7.89 - 7.92 mm (0.311 - 0.312 in.)
Stem - to - Guide Clearance	0.025 - 0.076 mm (0.001 - 0.003 in.)
Tip Refinishing (Max. Allowable)	0.025 mm (0.010 in.)

VALVE SPRING

DESCRIPTION	SPECIFICATION
Free Length (Approx)	47.65 mm (1.876 in.)
Spring Force (Valve Closed)	
Intake and Exhaust	316.0 - 351.0 N @ 41.65 mm (71.0 - 79.0 lbs. @ 1.64 in.)
Spring Force (Valve Open)	
Intake and Exhaust	898.6 - 969.7 N @ 30.89 mm (202.0 - 218.0 lbs. @ 1.216 in.)
Wire Diameter (Inside) Intake and Exhaust	21.0 - 21.51 mm (0.827 - 0.847 in.)
Installed Height (Spring Seat to Bottom of Retainer)	41.65 mm (1.64 in.)

CYLINDER HEAD

DESCRIPTION	SPECIFICATION
Combustion Chamber	55.22 - 58.22 cc (3.37 - 3.55 cu. in.)
Valve Seat Angle	44.5°
Valve Seat Runout (MAX)	0.064 mm (0.0025 in.)
Valve Seat Width	1.02 - 1.52 mm (0.040 - 0.060 in.)
Valve Guide I. D (Integral)	7.95 - 7.97 mm (0.313 - 0.314 in.)
Valve Stem to Guide Clearance	0.025 - 0.076 mm (0.001 - 0.003 in.)
Cylinder Head Warpage (Flatness)	0.03 mm per 25 mm (0.001 in. per 1 in.) 0.05 mm per 152 mm (0.002 in. per 6 in.)
Flatness (MAX)	0.20 mm - max for total length

OIL PUMP

DESCRIPTION	SPECIFICATION
Gear to Body Clearance (Radial)	0.051 - 0.102 mm (0.002 - 0.004 in.)
Gear to Body Clearance (Radial Preferred)	0.051 mm (0.002 in.)
Gear End Clearance (Plastigage)	0.051 - 0.152 mm (0.002 - 0.006 in.)
Gear End Clearance (Plastigage Preferred)	.051 mm (.002 in.)
Gear End Clearance (FeelerGauge)	0.1016 - 0.2032 mm (0.004 - 0.008 in.)
Gear End Clearance (FeelerGauge Preferred)	0.1778 mm (0.007 in.)

ENGINE 4.0L (Continued)

OIL PRESSURE

SPECIFICATION	SPECIFICATION
At Curb Idle Speed*	89.6 kPa (13 psi)
@ 1600 and Higher Rpm	255 - 517 kPa (37 - 75 psi)
Oil Pressure Relief	517 kPa (75 psi.)
*** CAUTION: If pressure is zero at curb idle, DO NOT run engine at 3000 rpm.**	

TORQUE - 4.0L ENGINE

DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
A/C Compressor—Bolts	28	—	250
Block Heater—Nut	2	—	16
Camshaft Sprocket—Bolt	68	50	—
Camshaft Thrust Plate to Cylinder Block—Screws	24	18	—
Clutch Cover to Flywheel—Bolts	54	40	—
Coil Bracket to Block—Bolts	22	—	192
Connecting Rod—Nuts	45	33	—
Cylinder Block—Drain Plugs	34	25	—
Cylinder Head—Bolts	135	100	—
Cylinder Head Cover—Bolts	10	—	85
Distributor Clamp—Bolts	23	—	204
Engine Mounts—Front			
Support Bracket Bolts	61	45	—
Support Cushion Bolts/Nuts	41	30	—
Support Cushion Bracket Bolts	54	40	—
Support Cushion Bracket Stud Nuts	41	30	—
Support Cushion Thru-Bolt	65	48	—
Engine Mounts—Rear Crossmember to Sill Bolts—(Automatic)	41	30	—
Insulator Stud Assembly—Nut	41	30	—
Support Cushion/Crossmember—Nuts	22	—	192
Support Cushion/Bracket—Nuts (Manual)	75	55	—
Transmission Support Bracket —Bolt (Manual)	46	34	—
Transmission Support Bracket/ Cushion—Bolt (4WD Auto)	75	55	—
Transmission Support Adaptor Bracket—Bolts (2WD Auto)	75	55	—
Exhaust Manifold/Pipe—Nuts	27	20	—
Intake/Exhaust Manifold			
Fasteners #1-5	33	24	—
Fasteners #6 and 7	14	—	126
Fasteners #8-11	33	24	—
Flywheel to Converter Housing—Bolts	38	28	—
Flywheel to Crankshaft—Bolts	143	105	—
Front Cover to Block—Bolts			
1/4-20	7	—	60
5/16-18	22	—	192
Fuel Rail—Bolts/Stud	12	—	108
Generator—Bolts	57	42	—
Generator Bracket to Engine—Bolts	47	35	—
Idler Pulley to Cylinder Head—Bolt	47	35	—
Main Bearing Cap—Bolts	108	80	—
Oil Filter	18	—	156
Oil Filter Connector to			
Adaptor	47	35	—
Block	68	50	—
Adaptor Bolts	102	50	—
Oil Galley—Plug	41	30	—
Oil Pan—Bolts			
1/4-20	9.5	—	84
5/16-18	15	—	132
Oil Pan—Drain Plug	34	25	—
Oil Pump			
Mounting Bolts	23	—	204
Cover Bolts	8	—	70
Rocker Arm Assembly to Cylinder Head—Capscrews	30	21	—
Spark Plugs	37	27	—
Starter Motor—Mounting Bolts	45	33	—
Thermostat Housing—Bolts	18	—	156
Throttle Body—Bolts	10	—	90
Vibration Damper—Bolt	108	80	—
Water Pump to Block—Bolts	23	17	—

AIR CLEANER ELEMENT

REMOVAL

(1) Loosen air tube clamp at housing cover (Fig. 7).
(2) Disconnect air tube at cover.
(3) Pry back the clips retaining air cleaner cover to air cleaner housing.
(4) Lift cover up to expose air cleaner element.
(5) Remove air cleaner element.
(6) Clean inside of air cleaner housing and its cover before installing new element.

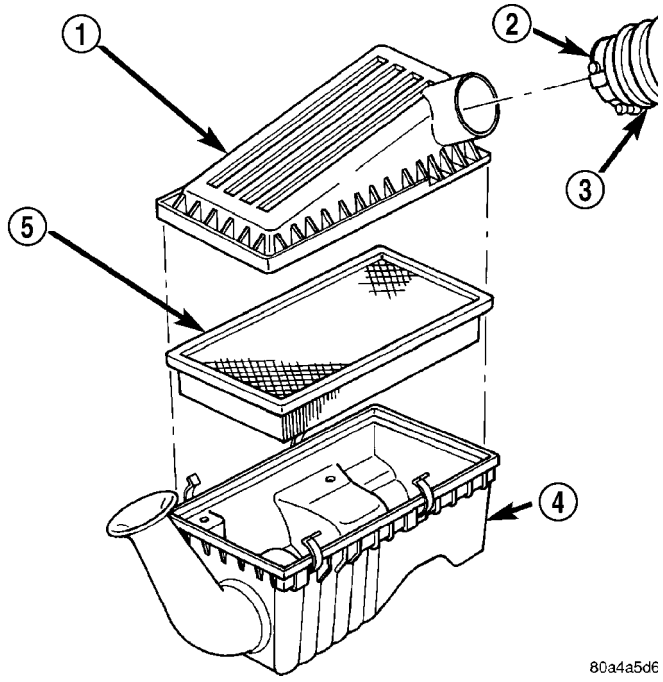

Fig. 7 AIR CLEANER ELEMENT

1 - COVER
2 - CLAMP
3 - AIR TUBE
4 - HOUSING
5 - FILTER

INSTALLATION

(1) Install air cleaner element into housing.
(2) Install housing cover to housing. Be sure cover is properly seated to air cleaner housing.
(3) Connect air tube at cover.

AIR CLEANER HOUSING

REMOVAL

(1) Loosen air tube clamp at housing cover.
(2) Disconnect air tube at cover. (Fig. 8)
(3) Pry back the clips retaining air cleaner cover to air cleaner housing.
(4) Lift cover up to expose air cleaner element.
(5) Remove air cleaner element.

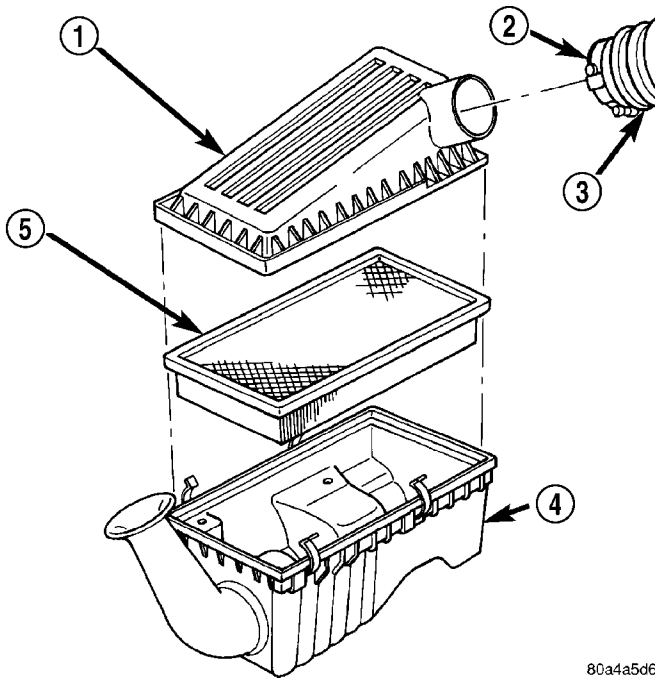

Fig. 8 AIR CLEANER ELEMENT

1 - COVER
2 - CLAMP
3 - AIR TUBE
4 - HOUSING
5 - FILTER

(6) Remove the air inlet duct.
(7) Lift up and remove the air cleaner housing.
(8)

INSTALLATION

(1) Push down on the air cleaner housing to lock in place.
(2) Install the air inlet tube.
(3) Install the air cleaner element.
(4) Pry back the clips retaining air cleaner cover to air cleaner housing. (Fig. 8)
(5) Install the air cleaner housing cover. Be sure cover is properly seated.
(6) Install the air out let tube.

CYLINDER HEAD

DESCRIPTION

The cylinder head is made of cast iron containing twelve valves made of chrome plated heat resistant steel, valve stem seals, springs, retainers and keepers. The cylinder head and valve seats can be resurfaced for service purposes.

The valve guides are integral to the cylinder head, They are not replaceable. However, they are serviceable.

CYLINDER HEAD (Continued)

The cylinder head uses dual quench-type design combustion chambers which cause turbulence in the cylinders allowing faster burning of the air/fuel mixture, resulting in better fuel economy (Fig. 9).

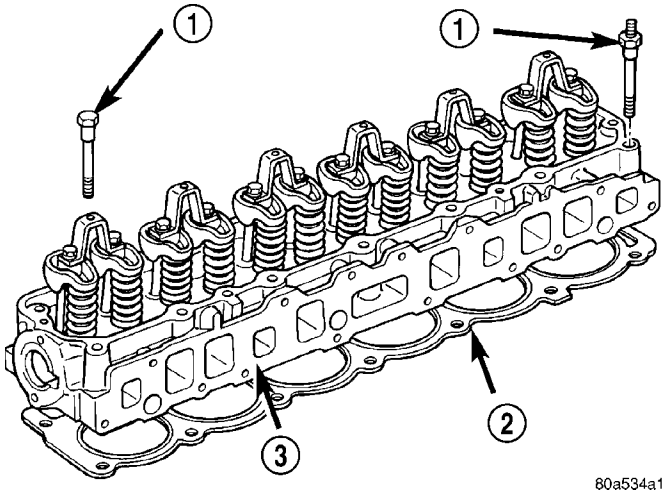

Fig. 9 Cylinder Head 4.0L Engine

1 - CYLINDER HEAD BOLTS
2 - CYLINDER HEAD GASKET
3 - CYLINDER HEAD

DIAGNOSIS AND TESTING - ENGINE CYLINDER HEAD GASKET FAILURE

A leaking engine cylinder head gasket usually results in loss of power, loss of coolant and engine misfiring.

An engine cylinder head gasket leak can be located between adjacent cylinders or between a cylinder and the adjacent water jacket.

- An engine cylinder head gasket leaking between adjacent cylinders is indicated by a loss of power and/or engine misfire.
- An engine cylinder head gasket leaking between a cylinder and an adjacent water jacket is indicated by coolant foaming or overheating and loss of coolant.

CYLINDER-TO-CYLINDER LEAKAGE TEST

To determine if an engine cylinder head gasket is leaking between adjacent cylinders; follow the procedures outlined . An engine cylinder head gasket leaking between adjacent cylinders will result in approximately a 50-70% reduction in compression pressure.

CYLINDER-TO-WATER JACKET LEAKAGE TEST

WARNING: USE EXTREME CAUTION WHEN THE ENGINE IS OPERATING. DO NOT STAND IN A DIRECT LINE WITH THE FAN. DO NOT PUT YOUR HANDS NEAR THE PULLEYS, BELTS OR THE FAN. DO NOT WEAR LOOSE CLOTHING.

Remove the radiator cap.

Start the engine and allow it to warm up until the engine thermostat opens.

If a large combustion/compression pressure leak exists, bubbles will be visible in the coolant.

If bubbles are not visible, install a radiator pressure tester and pressurize the coolant system.

If a cylinder is leaking combustion pressure into the water jacket, the tester pointer will pulsate with every combustion stroke of the cylinder.

REMOVAL

NOTE: This procedure can be done with the engine in or out of the vehicle.

(1) Disconnect the battery negative cable.

WARNING: DO NOT REMOVE THE CYLINDER BLOCK DRAIN PLUGS OR LOOSEN THE RADIATOR DRAIN COCK WITH THE SYSTEM HOT AND PRESSURIZED BECAUSE SERIOUS BURNS FROM THE COOLANT CAN OCCUR.

(2) Drain the coolant (Refer to 7 - COOLING - STANDARD PROCEDURE) and disconnect the hoses at the engine thermostat housing and the water pump inlet. DO NOT waste reusable coolant. If the solution is clean and is being drained only to service the engine or cooling system, drain the coolant into a clean container for reuse.

(3) Remove the air cleaner assembly.

(4) Remove the cylinder head cover (Refer to 9 - ENGINE/CYLINDER HEAD/CYLINDER HEAD COVER(S) - REMOVAL).

(5) Remove the capscrews, bridge and pivot assemblies and rocker arms (Refer to 9 - ENGINE/CYLINDER HEAD/ROCKER ARM / ADJUSTER ASSY - REMOVAL).

(6) Remove the push rods.**Retain the push rods, bridges, pivots and rocker arms in the same order as removed.**

(7) Remove the accessory drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - REMOVAL).

(8) Remove the A/C compressor mounting bolts and secure the compressor to the side.

(9) Remove the power steering pump and bracket from the intake manifold and water pump. Set the pump and bracket aside. DO NOT disconnect the hoses.

(10) Perform the Fuel System Pressure Release procedure (Refer to 14 - FUEL SYSTEM/FUEL DELIVERY - STANDARD PROCEDURE).

(11) Disconnect the fuel supply line at the fuel rail.

CYLINDER HEAD (Continued)

(12) Remove the intake and exhaust manifolds from the engine cylinder head (Refer to 9 - ENGINE/ MANIFOLDS/INTAKE MANIFOLD - REMOVAL).

(13) Disconnect the coil rail electrical connectors and remove the coil rail (Refer to 8 - ELECTRICAL/ IGNITION CONTROL/COIL RAIL - REMOVAL).

(14) Remove spark plugs (Refer to 8 - ELECTRICAL/IGNITION CONTROL/SPARK PLUG - REMOVAL).

(15) Disconnect the temperature sending unit wire connector.

(16) Remove the engine cylinder head bolts. Bolt No.14 cannot be removed until the head is moved forward (Fig. 10). Pull bolt No.14 out as far as it will go and then suspend the bolt in this position (tape around the bolt).

(17) Remove the engine cylinder head and gasket (Fig. 10).

(18) If this was the first time the bolts were removed, put a paint dab on the top of the bolt. If the bolts have a paint dab on the top of the bolt or it isn't known if they were used before, discard the bolts.

(19) Stuff clean lint free shop towels into the cylinder bores.

NOTE: If the valves, springs, or seals are to be inspected/replaced at this time, refer to Valves and Valve Springs in this section for proper inspection procedures.

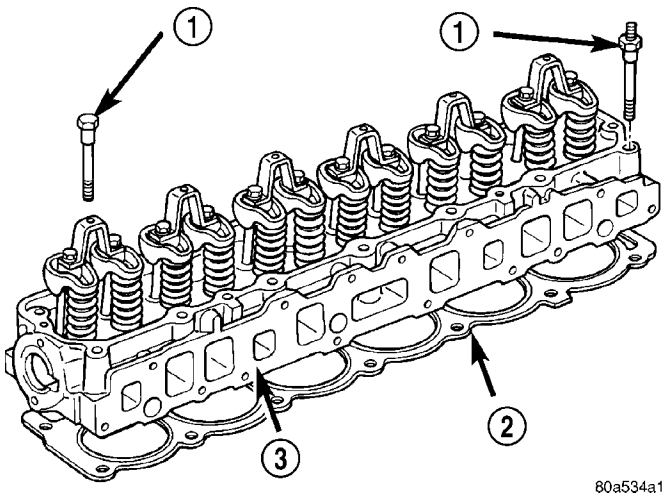

Fig. 10 Engine Cylinder

1 - CYLINDER HEAD BOLTS
2 - CYLINDER HEAD GASKET
3 - CYLINDER HEAD

CLEANING

Thoroughly clean the engine cylinder head and cylinder block mating surfaces. Clean the intake and engine exhaust manifold and engine cylinder head mating surfaces. Remove all gasket material and carbon.

Check to ensure that no coolant or foreign material has fallen into the tappet bore area.

Remove the carbon deposits from the combustion chambers and top of the pistons.

INSPECTION

Use a straightedge and feeler gauge to check the flatness of the engine cylinder head and block mating surfaces.

INSTALLATION

NOTE: This procedure can be done with the engine in or out of the vehicle.

The engine cylinder head gasket is a composition gasket. The gasket is to be installed DRY. **DO NOT use a gasket sealing compound on the gasket.**

If the engine cylinder head is to be replaced and the original valves used, measure the valve stem diameter. Only standard size valves can be used with a service replacement engine cylinder head unless the replacement head valve stem guide bores are reamed to accommodate oversize valve stems. Remove all carbon buildup and reface the valves.

(1) Remove the shop towels from the cylinder bores. Coat the bores with clean engine oil.

(2) Position the engine cylinder head gasket (with the numbers facing up) using the alignment dowels in the cylinder block, to position the gasket.

CAUTION: Engine cylinder head bolts should be reused only once. Replace the head bolts if they were used before or if they have a paint dab on the top of the bolt.

(3) With bolt No.14 held in place (tape around bolt), install the engine cylinder head over the same dowels used to locate the gasket. Remove the tape from bolt No.14.

(4) Coat the threads of stud bolt No.11 with Loctite 592 sealant, or equivalent.

(5) Tighten the engine cylinder head bolts in sequence according to the following procedure (Fig. 11).

CAUTION: During the final tightening sequence, bolt No.11 will be tightened to a lower torque than the rest of the bolts. DO NOT overtighten bolt No.11.

(a) Tighten all bolts in sequence (1 through 14) to 30 N·m (22 ft. lbs.) torque.

(b) Tighten all bolts in sequence (1 through 14) to 61 N·m (45 ft. lbs.) torque.

CYLINDER HEAD (Continued)

(c) Check all bolts to verify they are set to 61 N·m (45 ft. lbs.) torque.

(d) Tighten bolts in sequence:

- Bolts 1 through 10 to 149 N·m (110 ft. lbs.) torque.
- Bolt 11 to 135 N·m (100 ft. lbs.) torque.
- Bolts 12 through 14 to 149 N·m (110 ft. lbs.) torque.

(e) Check all bolts in sequence to verify the correct torque.

(f) If not already done, clean and mark each bolt with a dab of paint after tightening. Should you encounter bolts which were painted in an earlier service operation, replace them.

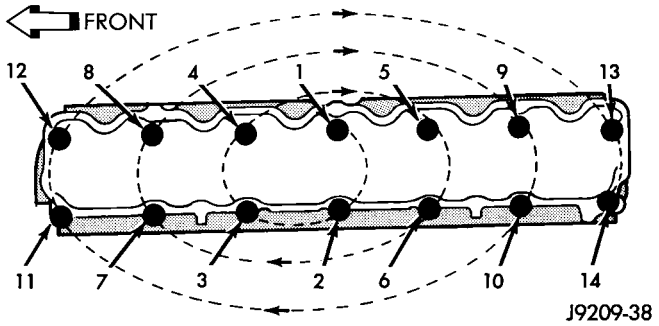

Fig. 11 Engine Cylinder Head Bolt Tightening Sequence

(6) Install the spark plugs (Refer to 8 - ELECTRICAL/IGNITION CONTROL/SPARK PLUG - INSTALLATION).

(7) Connect the temperature sending unit wire connector.

(8) Install the ignition coil rail and coil rail electrical connectors (Refer to 8 - ELECTRICAL/IGNITION CONTROL/COIL RAIL - INSTALLATION).

(9) Install the intake and exhaust manifolds (Refer to 9 - ENGINE/MANIFOLDS/INTAKE MANIFOLD - INSTALLATION).

(10) Install the fuel line and the vacuum advance hose.

(11) Attach the power steering pump and bracket.

(12) Install the push rods, rocker arms, pivots and bridges in the order they were removed (Refer to 9 - ENGINE/CYLINDER HEAD/ROCKER ARM / ADJUSTER ASSY - INSTALLATION).

(13) Install the engine cylinder head cover (Refer to 9 - ENGINE/CYLINDER HEAD/CYLINDER HEAD COVER(S) - INSTALLATION).

(14) Attach the air conditioner compressor mounting bracket to the engine cylinder head and block. Tighten the bolts to 40 N·m (30 ft. lbs.) torque.

(15) Attach the air conditioning compressor to the bracket. Tighten the bolts to 27 N·m (20 ft. lbs.) torque.

CAUTION: The serpentine drive belt must be routed correctly. Incorrect routing can cause the water pump to turn in the opposite direction causing the engine to overheat.

(16) Install the serpentine drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - INSTALLATION).

(17) Install the air cleaner and ducting.

(18) Connect the hoses to the engine thermostat housing and fill the cooling system to the specified level (Refer to 7 - COOLING - STANDARD PROCEDURE).

(19) The automatic transmission throttle linkage and cable must be adjusted after completing the engine cylinder head installation (Refer to 21 - TRANSMISSION/TRANSAXLE/AUTOMATIC - 30RH/THROTTLE VALVE CABLE - ADJUSTMENTS) or (Refer to 21 - TRANSMISSION/TRANSAXLE/AUTOMATIC - 32RH/THROTTLE VALVE CABLE - ADJUSTMENTS).

(20) Install the temperature sending unit and connect the wire connector.

(21) If equipped with air conditioning, install A/C compressor (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/A/C COMPRESSOR - INSTALLATION) and charge A/C system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE).

(22) Connect negative cable to battery.

WARNING: USE EXTREME CAUTION WHEN THE ENGINE IS OPERATING. DO NOT STAND IN DIRECT LINE WITH THE FAN. DO NOT PUT HANDS NEAR THE PULLEYS, BELTS OR FAN. DO NOT WEAR LOOSE CLOTHING.

(23) Operate the engine with the radiator cap off. Inspect for leaks and continue operating the engine until the engine thermostat opens. Add coolant, if required.

CYLINDER HEAD COVER(S)

DESCRIPTION

The cylinder head cover (Fig. 12) is made of stamped steel and incorporates the Crankcase Ventilation (CCV) Hoses and the oil fill opening.

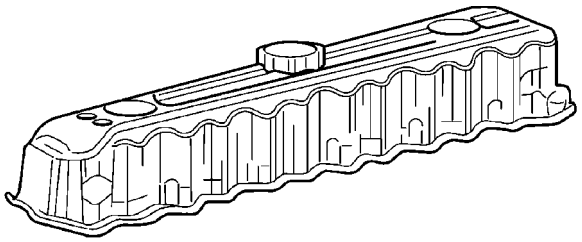

Fig. 12 Cylinder Head Cover

REMOVAL

The cylinder head cover is isolated from the cylinder head via grommets and a reusable molded rubber gasket. The grommet and limiter are retained in the cylinder head cover.

(1) Disconnect negative cable from battery.

(2) Disconnect the Crankcase Ventilation (CCV) vacuum hose from engine cylinder head cover.

(3) Disconnect the fresh air inlet hose from the engine cylinder head cover.

(4) Disconnect the accelerator, transmission, and speed (if equipped) control cables from the throttle body (Fig. 13).

(5) Remove the three bolts that fasten the control cable bracket to the intake manifold.

(6) Remove control cables from cylinder head cover clip.

(7) Position control cables and bracket away from cylinder head cover secure with tie straps.

NOTE: The wiring harness loom on the left hand side of the engine must be disconnected and moved over to the right hand side of the engine to gain clearance for cylinder head cover removal.

(8) Disconnect the left hand side of the wiring harness loom from all components.

(9) Remove fasteners from plastic wiring loom trough on rear of cylinder head cover.

NOTE: Before trying to move the Left side of the harness to the right side, remove the tie strap from the harness on the right side, just below the cover, near the coil pack area, if equipped.

(10) Move the left hand side harness loom and the plastic trough to the right side.

(11) Remove the engine cylinder head cover mounting bolts.

(12) Remove the engine cylinder head cover and gasket.

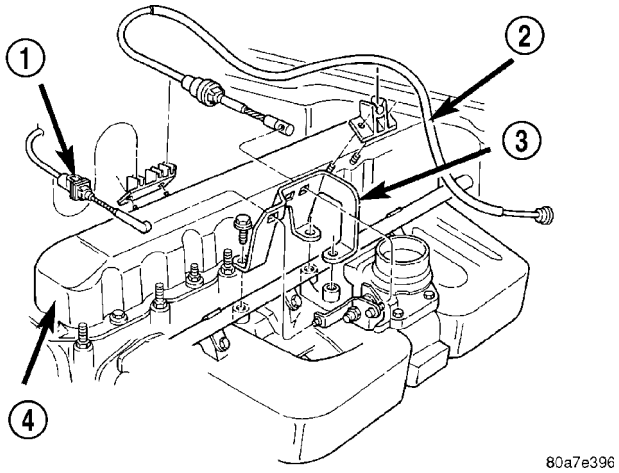

Fig. 13 Engine Cylinder Head Cover

1 - TRANS CONTROL CABLE
2 - ACCELERATOR CABLE
3 - CONTROL CABLE BRACKET
4 - CYLINDER HEAD COVER

CLEANING

Remove any original sealer from the cover sealing surface of the engine cylinder head and clean the surface using a fabric cleaner.

Remove all residue from the sealing surface using a clean, dry cloth.

INSPECTION

Inspect the engine cylinder head cover for cracks. Replace the cover, if cracked.

The original dark grey gasket material should NOT be removed. If sections of the gasket material are missing or are compressed, replace the engine cylinder head cover. However, sections with minor damage such as small cracks, cuts or chips may be repaired with a hand held applicator. The new material must be smoothed over to maintain gasket height. Allow the gasket material to cure prior to engine cylinder head cover installation.

INSTALLATION

The cylinder head cover is isolated from the cylinder head via grommets and a reusable molded rubber gasket. The grommet and limiter are retained in the cylinder head cover.

CYLINDER HEAD COVER(S) (Continued)

(1) If a replacement cover is installed, transfer the CCV valve grommet and oil filler cap from the original cover to the replacement cover.

(2) Install cylinder head cover and gasket (Fig. 14). Tighten the mounting bolts to 10 N·m (85 in. lbs.) torque.

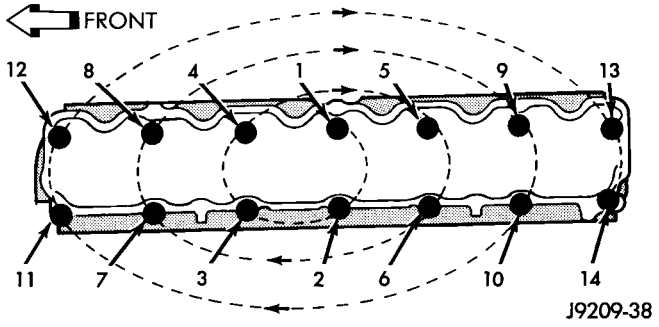

Fig. 14 Cylinder Head Cover Gasket Locator Pins at #8 & #9

(3) Reinstall the plastic wiring loom trough at the rear of cylinder head cover.

(4) Reconnect the left hand side of the wiring harness loom to all components.

(5) Connect the CCV hoses.

(6) Install control cables and bracket on intake manifold and tighten bolts to 8.7 N·m (77 in. lbs.) torque.

(7) Connect control cables to throttle body linkage.

(8) Snap control cables into cylinder head cover clip.

(9) Connect negative cable to battery.

INTAKE/EXHAUST VALVES & SEATS

DESCRIPTION

The valves are made of heat resistant steel and have chrome plated stems to prevent scuffing. All valves use three bead locks to promote valve rotation (Fig. 15).

STANDARD PROCEDURE - VALVE SERVICE

VALVE REFACING

Clean all carbon deposits from the combustion chambers, valve ports, valve stems, valve stem guides and head.

(1) Use a valve refacing machine to reface the intake and exhaust valves to the specified angle.

(2) After refacing, a margin of at least 0.787 mm (0.031 inch) must remain (Fig. 16). If the margin is less than 0.787 mm (0.031 inch), the valve must be replaced.

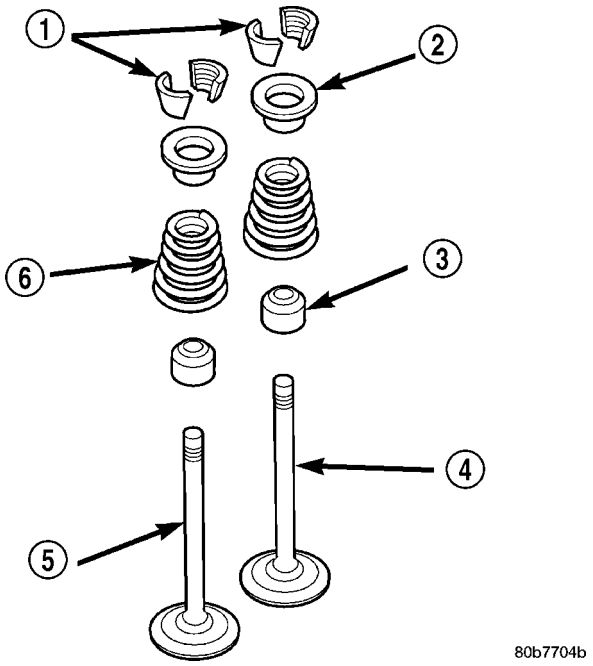

Fig. 15 VALVE AND KEEPER CONFIGURATION 4.0L

1 - VALVE LOCKS (3–BEAD)
2 - RETAINER
3 - VALVE STEM OIL SEAL
4 - INTAKE VALVE
5 - EXHAUST VALVE
6 - VALVE SPRING

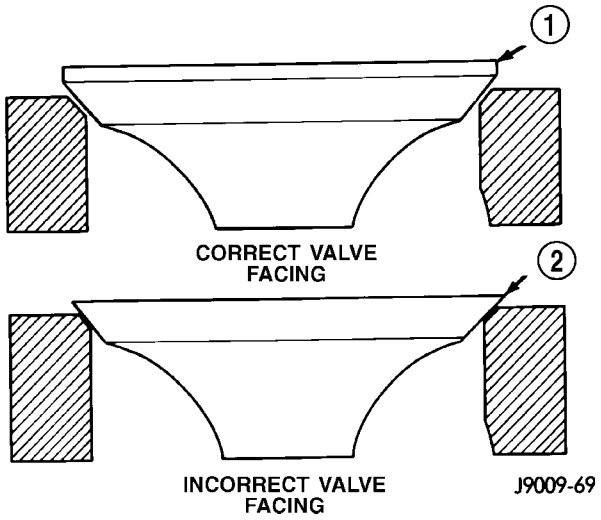

Fig. 16 Valve Facing

1 - VALVE MARGIN
2 - NO MARGIN

VALVE SEAT REFACING

Clean all carbon deposits from the combustion chambers, valve ports, valve stems, valve stem guides and head.

(1) Install a pilot of the correct size in the valve guide bore. Reface the valve seat to the specified

angle with a good dressing stone. Remove only enough metal to provide a smooth finish.

(2) Use tapered stones to obtain the specified seat width when required.

(3) Control valve seat runout to a maximum of 0.0635 mm (0.0025 in.) (Fig. 17).

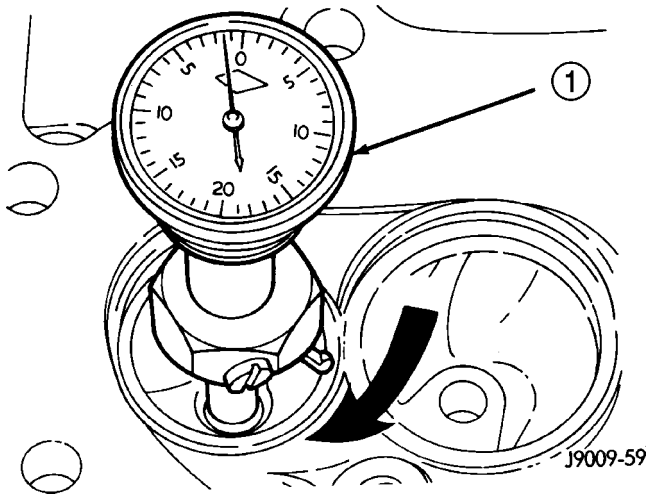

Fig. 17 Measurement of Valve Seat Runout

1 - DIAL INDICATOR

VALVE STEM-TO-GUIDE CLEARANCE MEASUREMENT

PREFERRED METHOD

(1) Remove the valve from the head.

(2) Clean the valve stem guide bore with solvent and a bristle brush.

(3) Insert a telescoping gauge into the valve stem guide bore approximately 9.525 mm (.375 inch) from the valve spring side of the head (Fig. 18).

(4) Remove and measure telescoping gauge with a micrometer.

(5) Repeat the measurement with contacts lengthwise to engine cylinder head.

(6) Compare the crosswise to lengthwise measurements to determine out-of-roundness. If the measurements differ by more than 0.0635 mm (0.0025 in.), ream the guide bore to accommodate an oversize valve stem.

(7) Compare the measured valve guide bore diameter with specifications (7.95-7.97 mm or 0.313-0.314 inch). If the measurement differs from specification by more than 0.076 mm (0.003 inch), ream the guide bore to accommodate an oversize valve stem.

ALTERNATIVE METHOD

(1) Use a dial indicator to measure the lateral movement of the valve stem (stem-to-guide clearance). This must be done with the valve installed in its guide and just off the valve seat (Fig. 19).

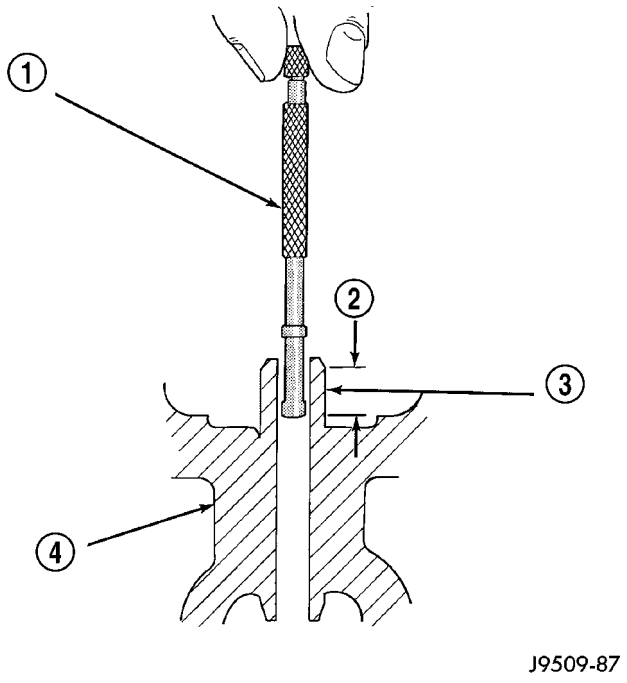

Fig. 18 Measurement of Valve Guide Bore Diameter

1 - GAUGE
2 - 9.525 MM (3/8 INCH)
3 - VALVE STEM GUIDE
4 - CYLINDER HEAD

(2) Correct clearance is 0.025-0.0762 mm (0.001-0.003 inch). If indicated movement exceeds the specification ream the valve guide to accommodate an oversize valve stem.

NOTE: Valve seats must be ground after reaming the valve guides to ensure that the valve seat is concentric to the valve guide.

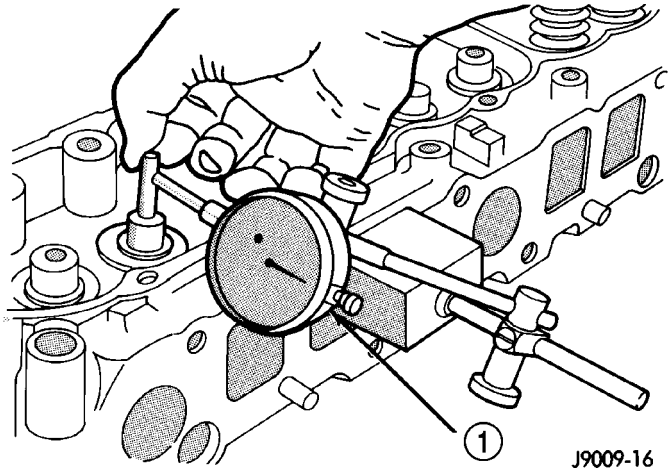

Fig. 19 Measurement of Lateral Movement Of Valve Stem

1 - DIAL INDICATOR

INTAKE/EXHAUST VALVES & SEATS (Continued)

REMOVAL

NOTE: This procedure is done with the engine cylinder head removed from the block.

(1) Remove the engine cylinder head (Refer to 9 - ENGINE/CYLINDER HEAD - REMOVAL) from the cylinder block.

(2) Use Valve Spring Compressor Tool MD-998772A and compress each valve spring.

(3) Remove the valve locks, retainers, springs and valve stem oil seals. Discard the oil seals (Fig. 20).

(4) Use a smooth stone or a jewelers file to remove any burrs on the top of the valve stem, especially around the groove for the locks.

(5) Remove the valves, and place them in a rack in the same order as removed.

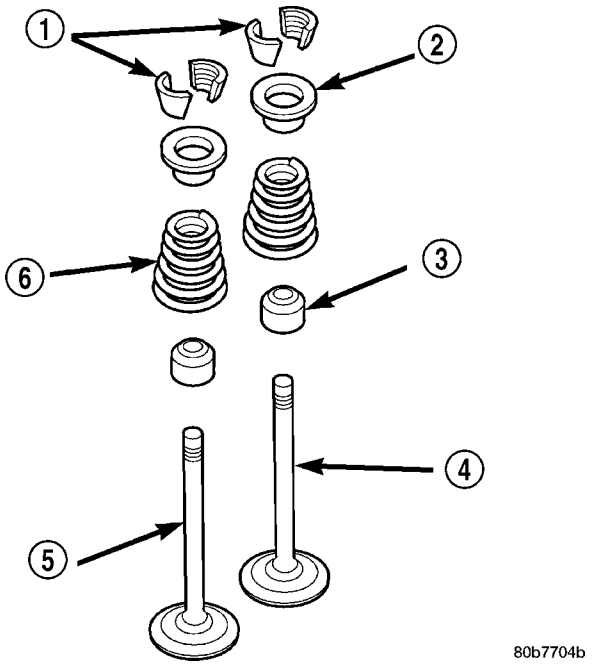

Fig. 20 Valve and Valve Components

1 - VALVE LOCKS (3–BEAD)
2 - RETAINER
3 - VALVE STEM OIL SEAL
4 - INTAKE VALVE
5 - EXHAUST VALVE
6 - VALVE SPRING

INSTALLATION

NOTE: This procedure is done with the engine cylinder head removed from the block.

(1) Thoroughly clean the valve stems and the valve guide bores.

(2) Lightly lubricate the stem.

(3) Install the valve in the original valve guide bore.

(4) Install the replacement valve stem oil seals on the valve stems (Fig. 21). If the 0.381 mm (0.015 inch) oversize valve stems are used, oversize oil seals are required.

(5) Position the valve spring and retainer on the engine cylinder head and compress the valve spring with Valve Spring Compressor Tool MD-998772A.

(6) Install the valve locks and release the tool.

(7) Tap the valve spring from side to side with a hammer to ensure that the spring is properly seated at the engine cylinder head. Also tap the top of the retainer to seat the valve locks.

(8) Install the engine cylinder head (Refer to 9 - ENGINE/CYLINDER HEAD - INSTALLATION).

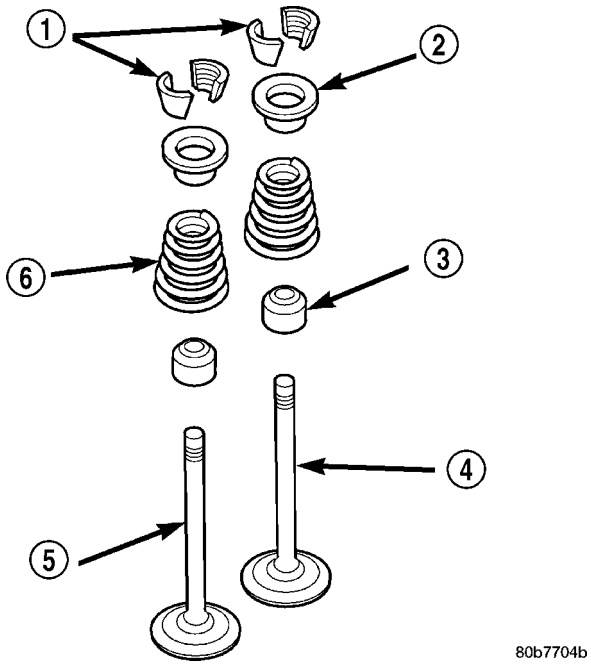

Fig. 21 Valve and Valve Components

1 - VALVE LOCKS (3–BEAD)
2 - RETAINER
3 - VALVE STEM OIL SEAL
4 - INTAKE VALVE
5 - EXHAUST VALVE
6 - VALVE SPRING

ROCKER ARM / ADJUSTER ASSEMBLY

DESCRIPTION

The rocker arms are made of stamped steel and have a operational ratio of 1.6:1 (Fig. 22).

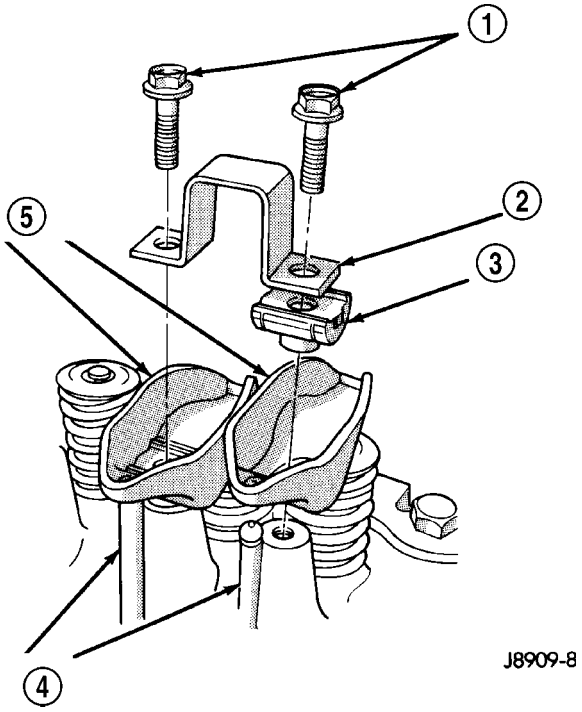

Fig. 22 Rocker Arms—Typical

1 - CAPSCREWS
2 - BRIDGE
3 - PIVOT ASSEMBLY
4 - PUSH RODS
5 - ROCKER ARMS

OPERATION

When the push rods are forced upward by the camshaft lobes the push rod presses upward on the rocker arms, the rocker arms pivot, forcing downward pressure on the valves forcing the valves to move downward and off from their seats.

REMOVAL

NOTE: This procedure can be done with the engine in or out of the vehicle.

(1) Remove the cylinder head cover (Refer to 9 - ENGINE/CYLINDER HEAD/CYLINDER HEAD COVER(S) - REMOVAL).

(2) Check for rocker arm bridges which are causing misalignment of the rocker arm to valve tip area.

(3) Remove the capscrews at each bridge and pivot assembly (Fig. 23). Alternately loosen the capscrews one turn at a time to avoid damaging the bridges.

(4) Remove the bridges, pivots and corresponding pairs of rocker arms (Fig. 23). Place them on a bench in the same order as removed.

(5) Remove the push rods and place them on a bench in the same order as removed.

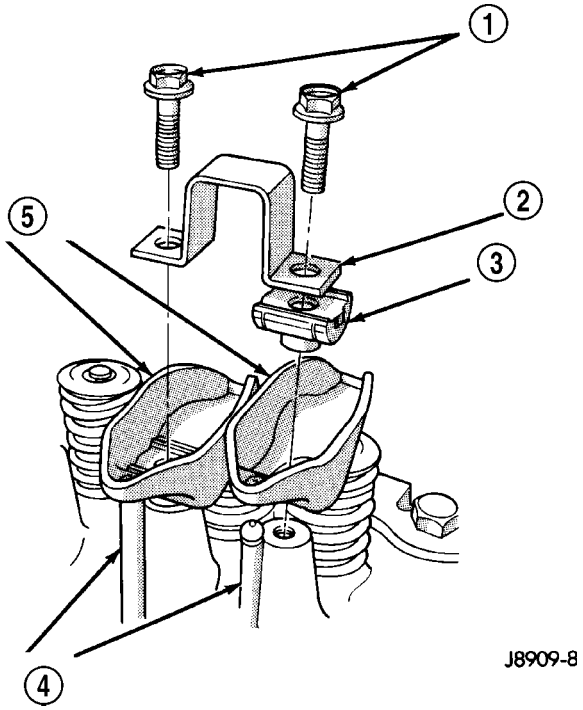

Fig. 23 Rocker Arm

1 - CAPSCREWS
2 - BRIDGE
3 - PIVOT ASSEMBLY
4 - PUSH RODS
5 - ROCKER ARMS

CLEANING

Clean all the components with cleaning solvent.

Use compressed air to blow out the oil passages in the rocker arms and push rods.

INSPECTION

Inspect the pivot surface area of each rocker arm. Replace any that are scuffed, pitted, cracked or excessively worn.

Inspect the valve stem tip contact surface of each rocker arm and replace any rocker arm that is deeply pitted.

Inspect each push rod end for excessive wear and replace as required. If any push rod is excessively worn because of lack of oil, replace it and inspect the corresponding hydraulic tappet for excessive wear.

Inspect the push rods for straightness by rolling them on a flat surface or by shining a light between the push rod and the flat surface.

A wear pattern along the length of the push rod is not normal. Inspect the engine cylinder head for obstruction if this condition exists.

ROCKER ARM / ADJUSTER ASSEMBLY (Continued)

INSTALLATION

NOTE: This procedure can be done with the engine in or out of the vehicle.

(1) Lubricate the ball ends of the push rods with Mopar® Engine Oil Supplement, or equivalent and install push rods in their original locations. Ensure that the bottom end of each push rod is centered in the tappet plunger cap seat.

(2) Using Mopar® Engine Oil Supplement, or equivalent, lubricate the area of the rocker arm that the pivot contacts. Install rocker arms, pivots and bridge above each cylinder in their originally position (Fig. 24).

(3) Loosely install the capscrews through each bridge.

(4) At each bridge, tighten the capscrews alternately, one turn at a time, to avoid damaging the bridge. Tighten the capscrews to 28 N·m (21 ft. lbs.) torque.

(5) Install the engine cylinder head cover (Refer to 9 - ENGINE/CYLINDER HEAD/CYLINDER HEAD COVER(S) - INSTALLATION).

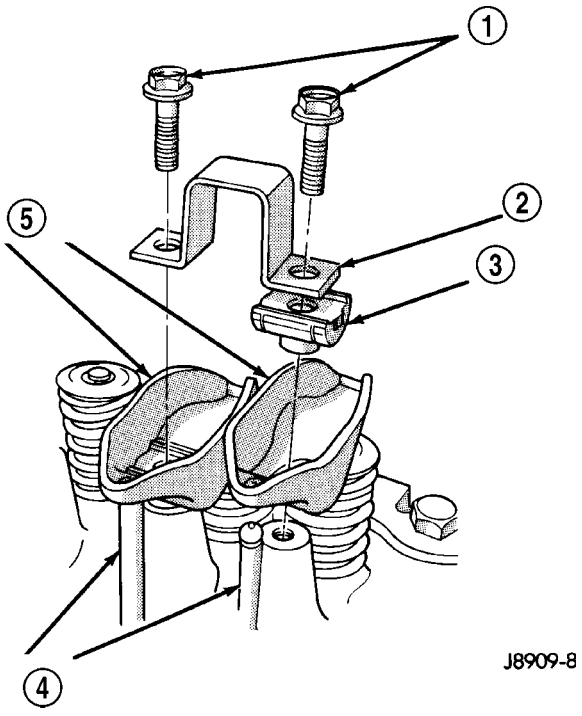

Fig. 24 Rocker Arm

1 - CAPSCREWS
2 - BRIDGE
3 - PIVOT ASSEMBLY
4 - PUSH RODS
5 - ROCKER ARMS

VALVE STEM SEALS

DESCRIPTION

The valve stem seals (Fig. 25) are made of rubber and incorporate a garter spring to maintain consistent lubrication control.

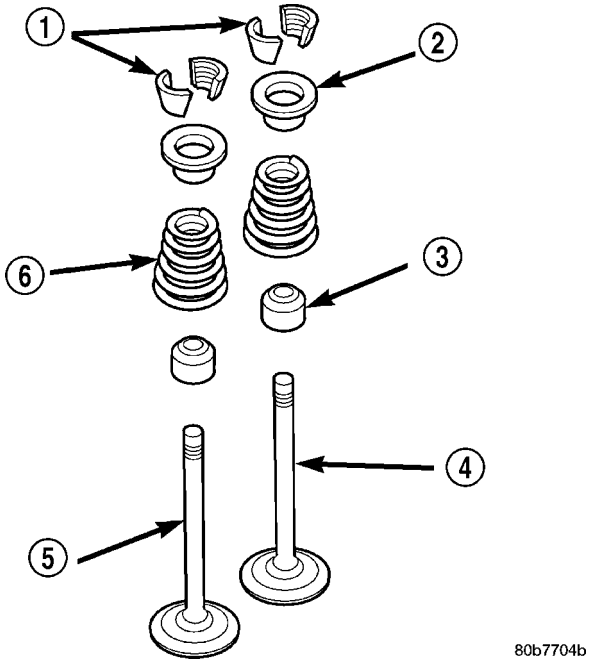

Fig. 25 Valve

1 - VALVE LOCKS (3–BEAD)
2 - RETAINER
3 - VALVE STEM OIL SEAL
4 - INTAKE VALVE
5 - EXHAUST VALVE
6 - VALVE SPRING

VALVE SPRINGS

DESCRIPTION

The valve springs (Fig. 26) are made of high strength silicon chrome spring steel. The springs are common for both intake and exhaust valves.

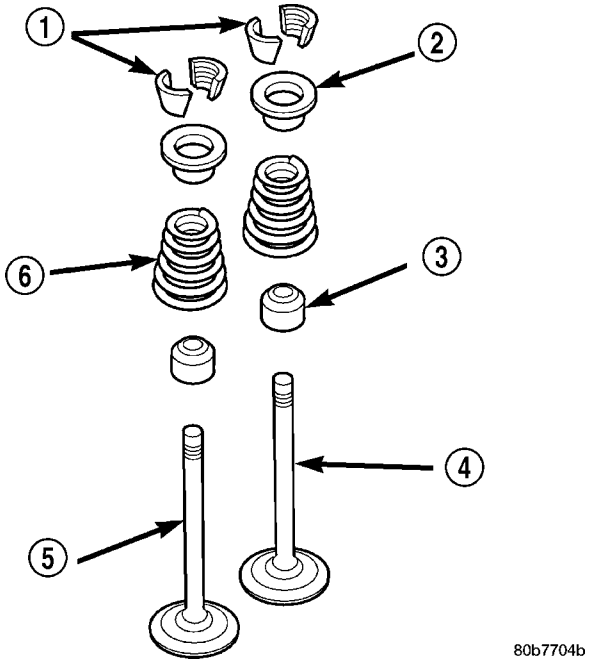

Fig. 26 VALVE AND KEEPER CONFIGURATION 4.0L

1 - VALVE LOCKS (3–BEAD)
2 - RETAINER
3 - VALVE STEM OIL SEAL
4 - INTAKE VALVE
5 - EXHAUST VALVE
6 - VALVE SPRING

STANDARD PROCEDURE - VALVE SPRING TENSION TEST

Clean all carbon deposits from the combustion chambers, valve ports, valve stems, valve stem guides and head.

Use a universal Valve Spring Tester and a torque wrench to test each valve spring for the specified tension value (Fig. 27).

Replace valve springs that are not within specifications.

REMOVAL

NOTE: This procedure can be done with the engine cylinder head installed on the block.

Inspect the valve stems, especially the grooves. An Arkansas smooth stone should be used to remove nicks and high spots.

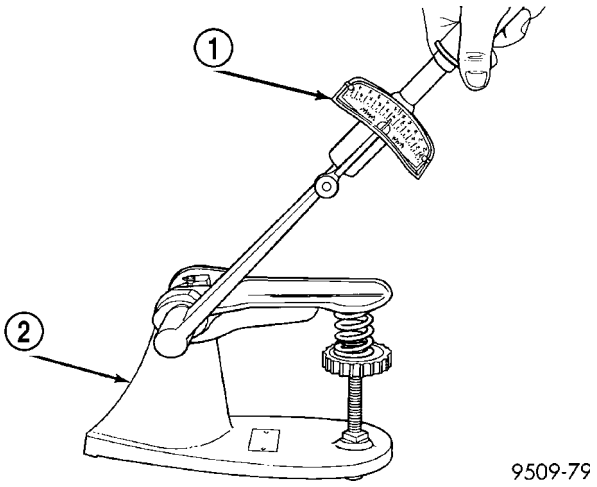

Fig. 27 Valve Spring Tester

1 - TORQUE WRENCH
2 - VALVE SPRING TESTER

Each valve spring is held in place by a retainer and a set of conical valve locks. The locks can be removed only by compressing the valve spring.

(1) Remove the engine cylinder head cover (Refer to 9 - ENGINE/CYLINDER HEAD/CYLINDER HEAD COVER(S) - REMOVAL).

(2) Remove cap screws, bridge and pivot assemblies and rocker arms (Refer to 9 - ENGINE/CYLINDER HEAD/ROCKER ARM / ADJUSTER ASSY - REMOVAL) for access to each valve spring to be removed.

(3) Remove push rods. **Retain the push rods, bridges, pivots and rocker arms in the same order and position as removed.**

(4) Inspect the springs and retainer for cracks and possible signs of weakening.

(5) Remove the spark plug(s) adjacent to the cylinder(s) below the valve springs to be removed.

(6) Connect an air hose to the adapter and apply air pressure slowly. Maintain at least 621 kPa (90 psi) of air pressure in the cylinder to hold the valves against their seats. For vehicles equipped with an air conditioner, use a flexible air adaptor when servicing the No.1 cylinder.

(7) Tap the retainer or tip with a rawhide hammer to loosen the lock from the retainer. Use Valve Spring Compressor Tool MD-998772A to compress the spring and remove the locks (Fig. 28).

(8) Remove valve spring and retainer (Fig. 28).

(9) Remove valve stem oil seals (Fig. 28). Note the valve seals are different for intake and exhaust valves. The top of each seal is marked either INT (intake/black in color) or EXH (exhaust/brown in color). DO NOT mix the seals.

VALVE SPRINGS (Continued)

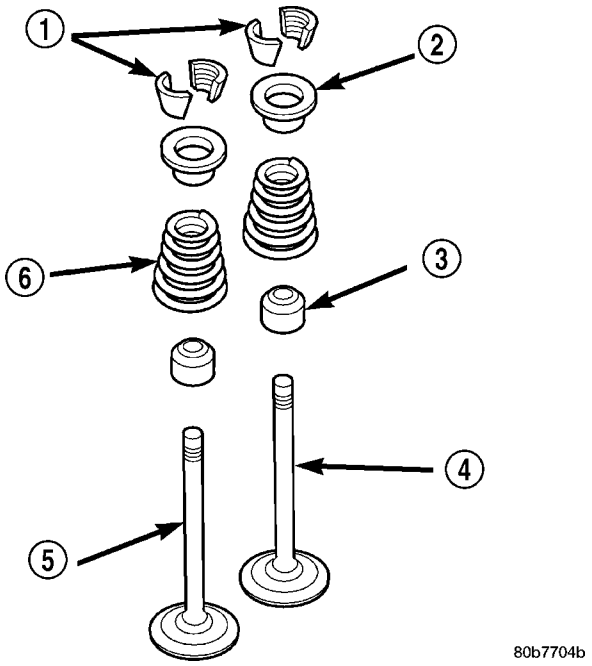

Fig. 28 Valve and Valve Components

1 - VALVE LOCKS (3–BEAD)
2 - RETAINER
3 - VALVE STEM OIL SEAL
4 - INTAKE VALVE
5 - EXHAUST VALVE
6 - VALVE SPRING

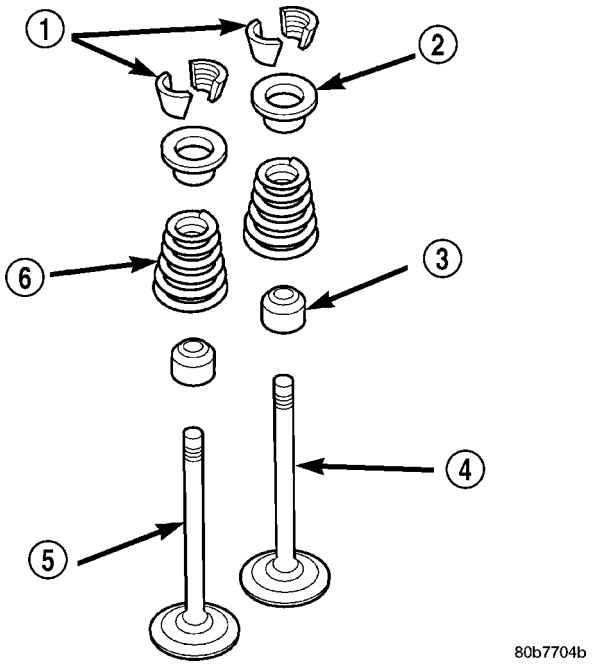

Fig. 29 Valve and Valve

1 - VALVE LOCKS (3–BEAD)
2 - RETAINER
3 - VALVE STEM OIL SEAL
4 - INTAKE VALVE
5 - EXHAUST VALVE
6 - VALVE SPRING

INSTALLATION

NOTE: This procedure can be done with the engine cylinder head installed on the block.

CAUTION: Install oil seals carefully to prevent damage from the sharp edges of the valve spring lock grove.

(1) Lightly push the valve seal over the valve stem and valve guide boss. Be sure the seal is completely seated on the valve guide boss.

(2) Install valve spring and retainer (Fig. 29).

(3) Compress the valve spring with Valve Spring Compressor Tool MD-998772A and insert the valve locks. Release the spring tension and remove the tool. Tap the spring from side-to-side to ensure that the spring is seated properly on the engine cylinder head.

(4) Release air pressure and disconnect the air hose. Remove the adaptor from the spark plug hole and install the spark plug.

(5) Repeat the procedures for each remaining valve spring to be removed.

(6) Install the push rods. Ensure the bottom end of each rod is centered in the plunger cap seat of the hydraulic valve tappet.

(7) Install the rocker arms, pivots and bridge (Refer to 9 - ENGINE/CYLINDER HEAD/ROCKER ARM / ADJUSTER ASSY - INSTALLATION) at their original location.

(8) Install the engine cylinder head cover (Refer to 9 - ENGINE/CYLINDER HEAD/CYLINDER HEAD COVER(S) - INSTALLATION).

ENGINE BLOCK

CLEANING

Thoroughly clean the oil pan and engine block gasket surfaces.

Use compressed air to clean out:

- The galley at the oil filter adaptor hole.
- The front and rear oil galley holes.
- The feed holes for the crankshaft main bearings.

Once the block has been completely cleaned, apply Loctite PST pipe sealant with Teflon 592 to the threads of the front and rear oil galley plugs. Tighten the plugs to 34 N·m (25 ft. lbs.) torque.

ENGINE BLOCK (Continued)

INSPECTION

(1) It is mandatory to use a dial bore gauge to measure each cylinder bore diameter (Fig. 30). To correctly select the proper size piston, a cylinder bore gauge, capable of reading in 0.003 mm (.0001 in.) INCREMENTS is required. If a bore gauge is not available, do not use an inside micrometer.

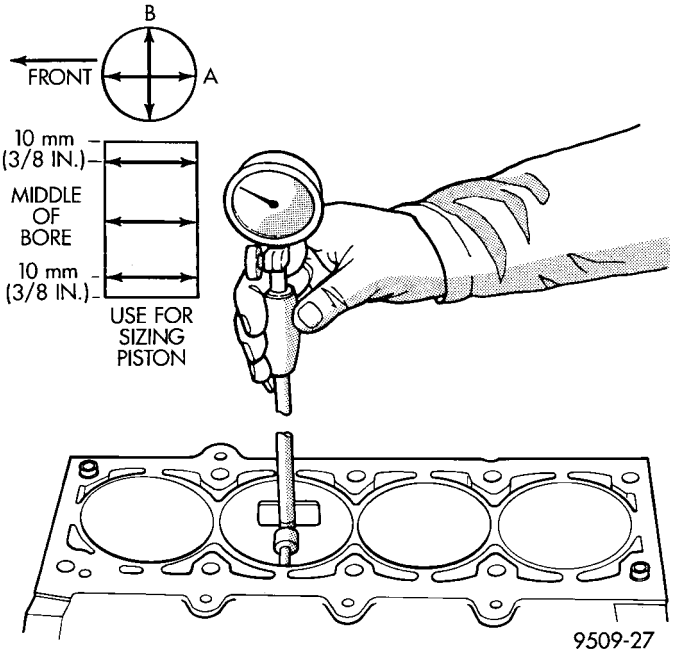

Fig. 30 Cylinder Bore Measurement

(2) Measure the inside diameter of the cylinder bore at three levels below top of bore. Start perpendicular (across or at 90 degrees) to the axis of the crankshaft and then take two additional reading.

(3) Measure the cylinder bore diameter crosswise to the cylinder block near the top of the bore. Repeat the measurement near the middle of the bore, then repeat the measurement near the bottom of the bore.

(4) Determine taper by subtracting the smaller diameter from the larger diameter.

(5) Rotate measuring device 90° and repeat steps above.

(6) Determine out-of-roundness by comparing the difference between each measurement.

(7) If cylinder bore taper does not exceed 0.025 mm (0.001 inch) and out-of-roundness does not exceed 0.025 mm (0.001 inch), the cylinder bore can be honed. If the cylinder bore taper or out- of-round condition exceeds these maximum limits, the cylinder must be bored and then honed to accept an oversize piston. A slight amount of taper always exists in the cylinder bore after the engine has been in use for a period of time.

CAMSHAFT & BEARINGS (IN BLOCK)

DESCRIPTION

The camshaft is made of gray cast iron with twelve machined lobes and four bearing journals (Fig. 31). When the camshaft rotates the lobes actuate the tappets and push rods, forcing upward on the rocker arms which applies downward force on the valves.

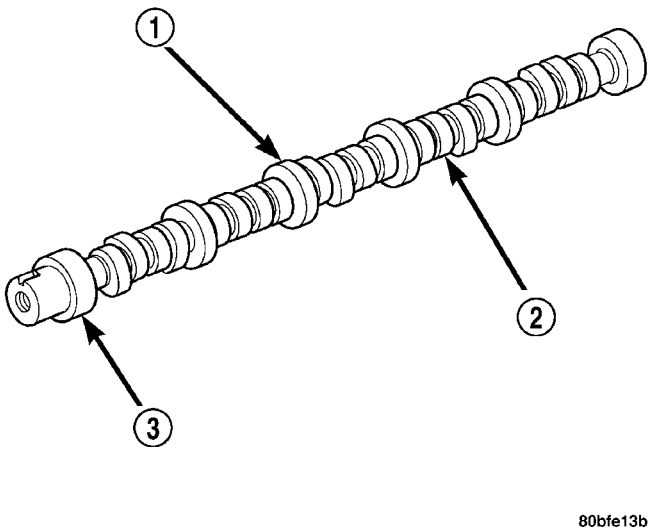

Fig. 31 Camshaft—Typical

1 - CAMSHAFT
2 - LOBES
3 - BEARING JOURNAL

REMOVAL

REMOVAL - CAMSHAFT BEARINGS

The camshaft rotates within four steel-shelled, babbitt-lined bearings that are pressed into the cylinder block and then line reamed. The camshaft bearing bores and bearing diameters are not the same size. They are stepped down in 0.254 mm (0.010 inch) increments from the front bearing (largest) to the rear bearing (smallest). This permits easier removal and installation of the camshaft. The camshaft bearings are pressure lubricated. Camshaft end play is maintained by the thrust plate.

(1) Remove the camshaft (Refer to 9 - ENGINE/ ENGINE BLOCK/CAMSHAFT & BEARINGS (IN BLOCK) - REMOVAL).

CAMSHAFT & BEARINGS (IN BLOCK) (Continued)

NOTE: It is not advisable to attempt to replace camshaft bearings unless special removal and installation tools are available, such as recommended tool 8544 Camshaft Bushing Remover Installer.

(2) Using Special tool 8544 Camshaft Bushing Remover Installer, remove the camshaft bearings.

REMOVAL - CAMSHAFT

WARNING: THE COOLANT IN A RECENTLY OPERATED ENGINE IS HOT AND PRESSURIZED. RELEASE THE PRESSURE BEFORE REMOVING THE DRAIN COCK, CAP AND DRAIN PLUGS.

(1) Disconnect negative cable from battery.
(2) Drain the cooling system (Refer to 7 - COOLING - STANDARD PROCEDURE).
(3) Remove the radiator (Refer to 7 - COOLING/ENGINE/RADIATOR - REMOVAL) and condenser (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/A/C CONDENSER - REMOVAL), if equipped with A/C.
(4) Remove the engine cylinder head cover (Refer to 9 - ENGINE/CYLINDER HEAD/CYLINDER HEAD COVER(S) - REMOVAL).
(5) Remove the rocker arms, bridges and pivots (Refer to 9 - ENGINE/CYLINDER HEAD/ROCKER ARM / ADJUSTER ASSY - REMOVAL).
(6) Remove the push rods.
(7) Remove the engine cylinder head and gasket (Refer to 9 - ENGINE/CYLINDER HEAD - REMOVAL).
(8) Remove the hydraulic valve tappets from the engine cylinder block (Refer to 9 - ENGINE/ENGINE BLOCK/HYDRAULIC LIFTERS (CAM IN BLOCK) - REMOVAL).
(9) Remove the vibration damper (Refer to 9 - ENGINE/ENGINE BLOCK/VIBRATION DAMPER - REMOVAL).
(10) Remove the timing case cover (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT / CHAIN COVER(S) - REMOVAL).
(11) Rotate the crankshaft until the crankshaft sprocket timing mark is aligned on centerline with the camshaft sprocket timing mark (Fig. 33).
(12) Remove the timing chain and sprockets (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - REMOVAL).
(13) Remove the front bumper and/or grille, as required.
(14) Remove the two thrust plate retaining screws, thrust plate and camshaft (Fig. 32).

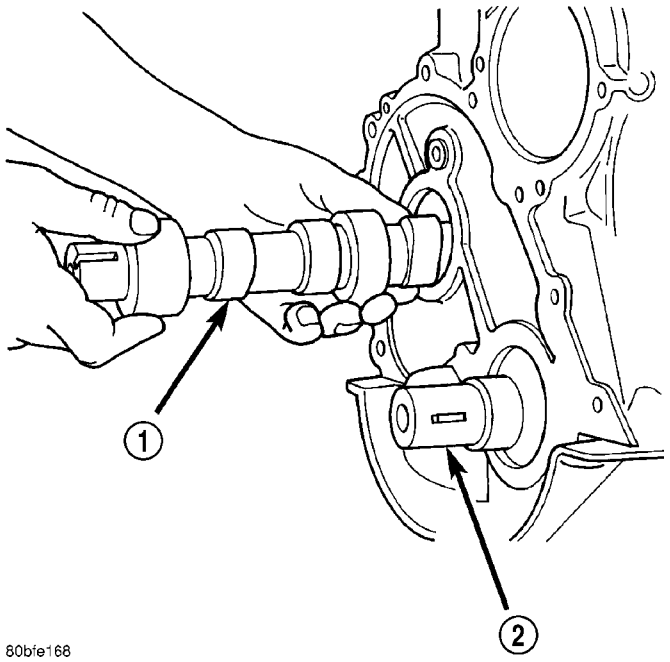

Fig. 32 Camshaft Removal

1 - CAMSHAFT
2 - CRANKSHAFT

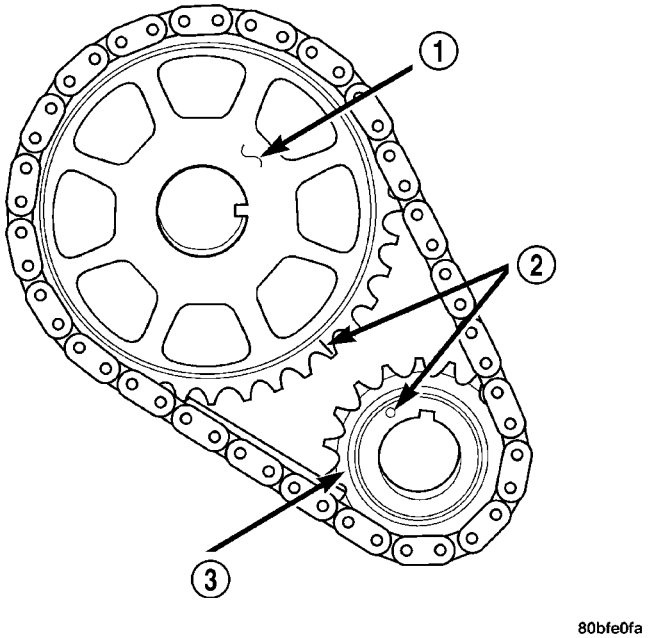

Fig. 33 Crankshaft / Camshaft Sprocket Timing Mark Alignment

1 - CAMSHAFT SPROCKET
2 - TIMING MARKS
3 - CRANKSHAFT SPROCKET

CAMSHAFT & BEARINGS (IN BLOCK) (Continued)

INSTALLATION

INSTALLATION - CAMSHAFT BEARINGS

CAUTION: Make sure outside diameter of number 1 bearing is clean. Make sure that the bearing is properly installed in the engine block, align the oil hole in the bearing with the oil gallery in the bearing bore. Failure to do so will cause inadequate oil supply for the sprockets and timing chain.

(1) Using recommended special tool 8544 Camshaft Bearing Remover/Installer, install new camshaft bearings.

INSTALLATION - CAMSHAFT

(1) Lubricate the camshaft with Mopar® Engine Oil Supplement, or equivalent.

(2) Carefully install the camshaft to prevent damage to the camshaft bearings.

(3) Position thrust plate and install retaining screws. Tighten screws to 24 N·m (18 ft. lbs.).

(4) Lubricate the camshaft with Mopar® engine oil supplement, or equivalent.

(5) Install the camshaft sprocket, crankshaft sprocket and timing chain (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - INSTALLATION).

(6) Tighten the camshaft sprocket bolt and washer to 68 N·m (50 ft. lbs.).

(7) To verify correct installation of the timing chain, turn the crankshaft two full revolutions then position the camshaft sprocket timing mark as shown in (Fig. 34).

(8) Install the timing case cover with a replacement oil seal (Fig. 35). (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT / CHAIN COVER(S) - INSTALLATION).

(9) Install the vibration damper (Fig. 35) (Refer to 9 - ENGINE/ENGINE BLOCK/VIBRATION DAMPER - INSTALLATION).

(10) Install the hydraulic valve tappets (Refer to 9 - ENGINE/ENGINE BLOCK/HYDRAULIC LIFTERS (CAM IN BLOCK) - INSTALLATION).

(11) Install the cylinder head gasket with the numbers facing up.

(12) Install the cylinder head (Refer to 9 - ENGINE/CYLINDER HEAD - INSTALLATION).

(13) Install the push rods.

(14) Install the rocker arms and pivot and bridge assemblies (Refer to 9 - ENGINE/CYLINDER HEAD/ROCKER ARM / ADJUSTER ASSY - INSTALLATION).

(15) Install the engine cylinder head cover (Refer to 9 - ENGINE/CYLINDER HEAD/CYLINDER HEAD COVER(S) - INSTALLATION).

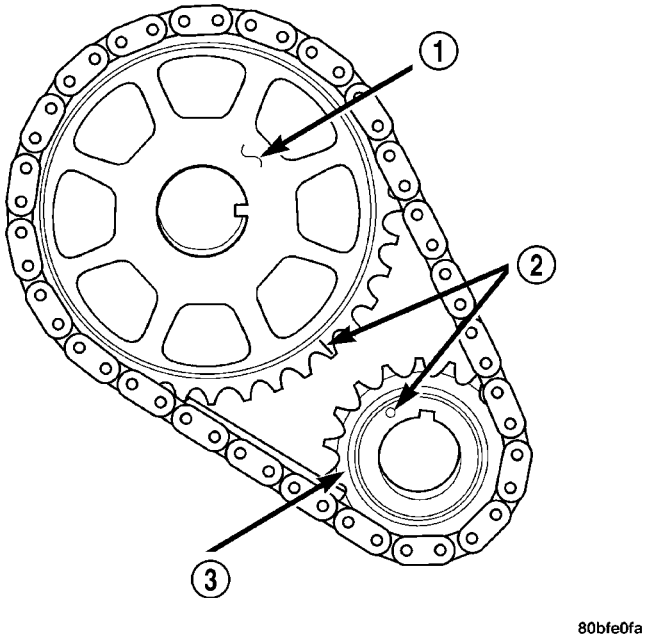

Fig. 34 Crankshaft / Camshaft Chain Drive Installation—Typical

1 - CAMSHAFT SPROCKET
2 - TIMING MARKS
3 - CRANKSHAFT SPROCKET

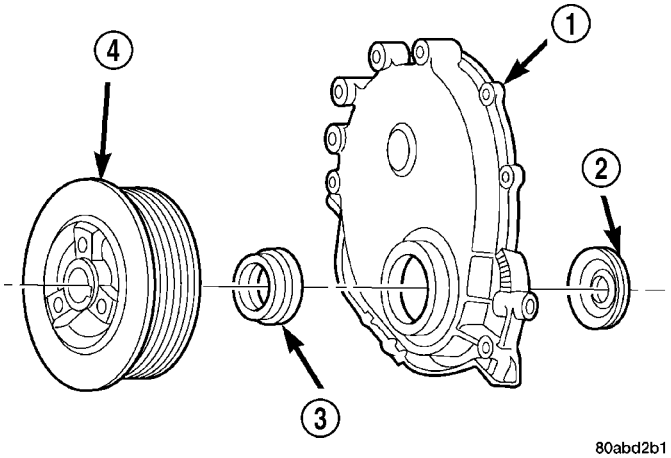

Fig. 35 Timing Case Cover Components

1 - TIMING CASE COVER
2 - OIL SLINGER
3 - CRANKSHAFT OIL SEAL
4 - VIBRATION DAMPER PULLEY

(16) Install the serpentine drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - INSTALLATION).

CAMSHAFT & BEARINGS (IN BLOCK) (Continued)

NOTE: During installation, lubricate the hydraulic valve tappets and all valve components with Mopar® Engine Oil Supplement, or equivalent. The Mopar® Engine Oil Supplement, or equivalent must remain with the engine oil for at least 1609 km (1,000 miles). The oil supplement need not be drained until the next scheduled oil change.

(17) Install the radiator (Refer to 7 - COOLING/ ENGINE/RADIATOR - INSTALLATION).

(18) Check the ignition timing and adjust as necessary.

(19) Install the grille and bumper, if removed.

(20) Connect negative cable to battery.

CONNECTING ROD BEARINGS

STANDARD PROCEDURE - FITTING CONNECTING ROD BEARINGS

Inspect the connecting rod bearings for scoring and bent alignment tabs (Fig. 36) (Fig. 37). Check the bearings for normal wear patterns, scoring, grooving, fatigue and pitting (Fig. 38). Replace any bearing that shows abnormal wear.

Inspect the connecting rod journals for signs of scoring, nicks and burrs.

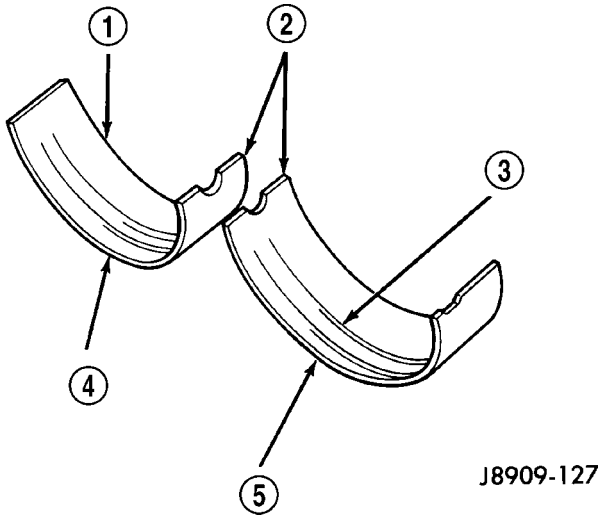

Fig. 36 Connecting Rod Bearing Inspection

1 - UPPER BEARING HALF
2 - MATING EDGES
3 - GROOVES CAUSED BY ROD BOLTS SCRATCHING JOURNAL DURING INSTALLATION
4 - WEAR PATTERN - ALWAYS GREATER ON UPPER BEARING
5 - LOWER BEARING HALF

Misaligned or bent connecting rods can cause abnormal wear on pistons, piston rings, cylinder walls, connecting rod bearings and crankshaft connecting rod journals. If wear patterns or damage to any of these components indicate the probability of a misaligned connecting rod, inspect it for correct rod alignment. Replace misaligned, bent or twisted connecting rods.

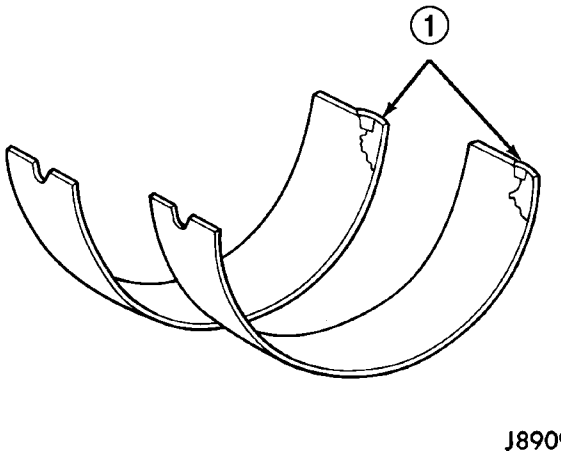

Fig. 37 Locking Tab Inspection

1 - ABNORMAL CONTACT AREA CAUSED BY LOCKING TABS NOT FULLY SEATED OR BEING BENT

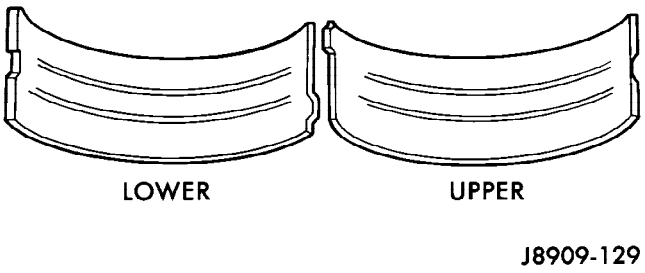

Fig. 38 Scoring Caused by Insufficient Lubrication or Damaged Crankshaft Journal

BEARING-TO-JOURNAL CLEARANCE

(1) Wipe the oil from the connecting rod journal.

(2) Use short rubber hose sections over rod bolts during installation.

(3) Lubricate the upper bearing insert and install in connecting rod.

(4) Use piston ring compressor to install the rod and piston assemblies. The oil squirt holes in the rods must face the camshaft. The arrow on the piston crown should point to the front of the engine (Fig. 39). Verify that the oil squirt holes in the rods face the camshaft and that the arrows on the pistons face the front of the engine.

(5) Install the lower bearing insert in the bearing cap. The lower insert must be dry. Place strip of Plastigage across full width of the lower insert at the center of bearing cap. Plastigage must not crumble in use. If brittle, obtain fresh stock.

(6) Install bearing cap and connecting rod on the journal and tighten nuts to 45 N·m (33 ft. lbs.) torque. DO NOT rotate crankshaft. Plastigage will smear, resulting in inaccurate indication.

CONNECTING ROD BEARINGS (Continued)

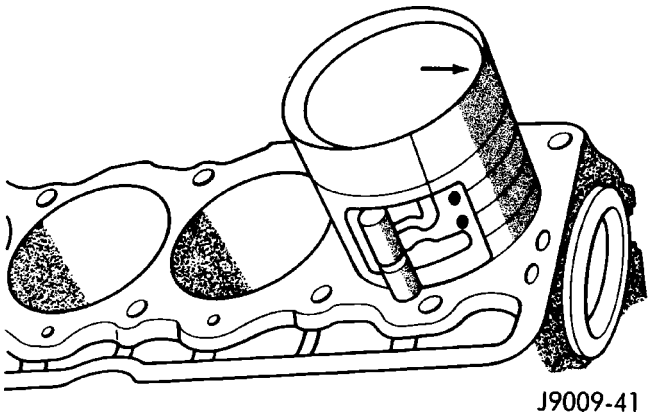

Fig. 39 Rod and Piston Assembly Installation

(7) Remove the bearing cap and determine amount of bearing-to- journal clearance by measuring the width of compressed Plastigage (Fig. 40). **Plastigage should indicate the same clearance across the entire width of the insert. If the clearance varies, it may be caused by either a tapered journal, bent connecting rod or foreign material trapped between the insert and cap or rod.**

(8) If the correct clearance is indicated, replacement of the bearing inserts is not necessary. Remove the Plastigage from crankshaft journal and bearing insert. Proceed with installation.

(9) If bearing-to-journal clearance exceeds the specification, install a pair of 0.0254 mm (0.001 inch) undersize bearing inserts. All the odd size inserts must be on the bottom. The sizes of the service

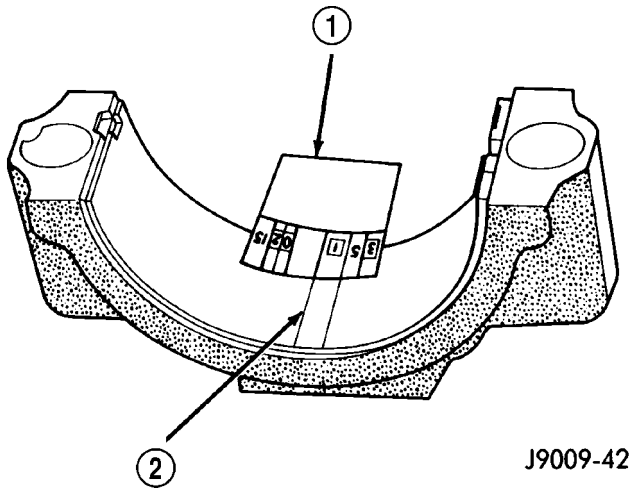

Fig. 40 Measuring Bearing Clearance with Plastigage

1 - PLASTIGAGE SCALE
2 - COMPRESSED PLASTIGAGE

replacement bearing inserts are stamped on the backs of the inserts. Measure the clearance as described in the previous steps.

(10) The clearance is measured with a pair of 0.0254 mm (0.001 inch) undersize bearing inserts installed. This will determine if two 0.0254 mm (0.001 inch) undersize inserts or another combination is needed to provide the correct clearance. Refer to CONNECTING ROD BEARING FITTING CHART.

CONNECTING ROD BEARING FITTING CHART

CRANKSHAFT JOURNAL		CORRESPONDING ROD BEARING INSERT	
Color Code	**Diameter**	**Upper Insert Size**	**Lower Insert Size**
Yellow	53.2257 - 53.2079 mm (2.0955 - 2.0948 in.)	Yellow - Standard	Yellow - Standard
Orange	53.2079 - 53.1901 mm (2.0948 - 2.0941 in.) 0.0178 mm (0.0007 in.) Undersize	Yellow - Standard	Blue - Undersize 0.025 mm (0.001 in.)
Blue	53.1901 - 53.1724 mm (2.0941 - 2.0934 in.) 0.0356 mm (0.0014 in.) Undersize	Blue - Undersize 0.025 mm (0.001 in.)	Blue - Undersize 0.025 mm (0.001 in.)
Red	52.9717 - 52.9539 mm (2.0855 - 2.0848 in.) 0.254 mm (0.010 in.) Undersize	Red - Undersize 0.254 mm (0.010 in.)	Red - Undersize 0.254 mm (0.010 in.)

CONNECTING ROD BEARINGS (Continued)

(11) **FOR EXAMPLE:** If the initial clearance was 0.0762 mm (0.003 inch), 0.025 mm (0.001 inch) undersize inserts would reduce the clearance by 0.025 mm (0.001 inch). The clearance would be 0.002 inch and within specification. A 0.051 mm (0.002 inch) undersize insert would reduce the initial clearance an additional 0.013 mm (0.0005 inch). The clearance would then be 0.038 mm (0.0015 inch).

(12) Repeat the Plastigage measurement to verify your bearing selection prior to final assembly.

(13) Once you have selected the proper insert, install the insert and cap. Tighten the connecting rod bolts to 45 N·m (33 ft. lbs.) torque.

SIDE CLEARANCE MEASUREMENT

Slide snug-fitting feeler gauge between the connecting rod and crankshaft journal flange (Fig. 41). (Refer to 9 - ENGINE - SPECIFICATIONS). Replace the connecting rod if the side clearance is not within specification.

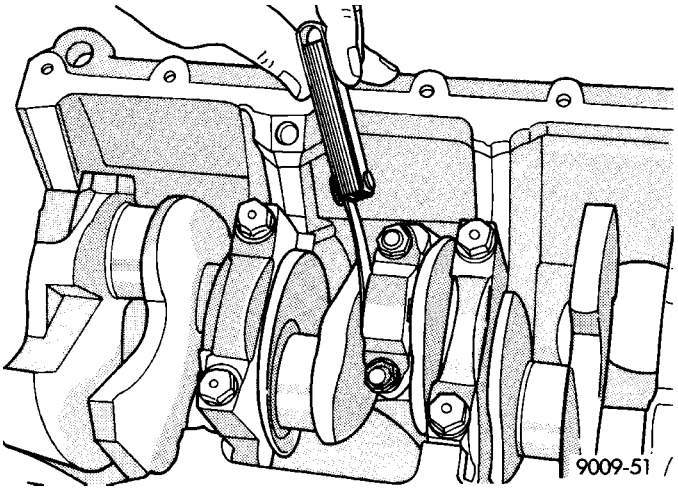

Fig. 41 Checking Connecting Rod Side Clearance - Typical

CRANKSHAFT

DESCRIPTION

The crankshaft is constructed of nodular cast iron. The crankshaft is a crosshaped four throw design with eight counterweights for balancing purposes. The crankshaft is supported by seven select main bearings with the number three serving as the thrust washer location. The main journals of the crankshaft are cross drilled to improve rod bearing lubrication. The select fit main bearing markings are located on the crankshaft counter weights. The crankshaft rear oil seal is a two piece design. The front oil seal is a one piece design retained in the timing chain cover (Fig. 42).

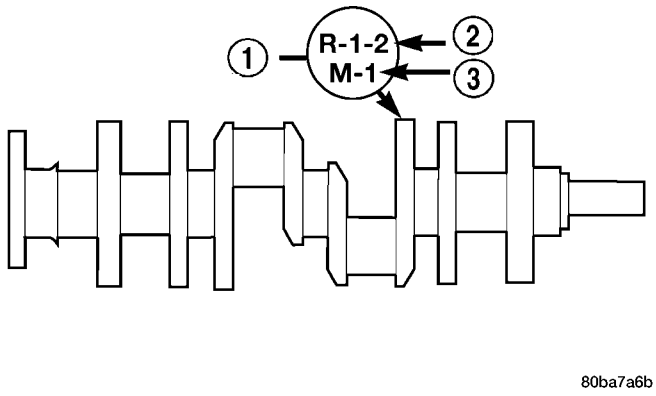

Fig. 42 Crankshaft with Select Fit Marking Location

1 - 1/4" LETTERS
2 - (ROD)
3 - (MAIN)

CRANKSHAFT MAIN BEARINGS

STANDARD PROCEDURE - FITTING CRANKSHAFT MAIN BEARINGS

FITTING BEARINGS (CRANKSHAFT INSTALLED)

The main bearing caps, numbered (front to rear) from 1 through 7 have an arrow to indicate the forward position. The upper main bearing inserts are grooved to provide oil channels while the lower inserts are smooth.

Each bearing insert pair is selectively fitted to its respective journal to obtain the specified operating clearance. In production, the select fit is obtained by using various-sized color-coded bearing insert pairs as listed in the Main Bearing Fitting Chart. The bearing color code appears on the edge of the insert. **The size is not stamped on bearing inserts used for engine production.**

The main bearing journal size (diameter) is identified by a color-coded paint mark (Fig. 43)on the adjacent cheek or counterweight towards the rear of the crankshaft (flange end). The rear main journal, is identified by a color-coded paint mark on the crankshaft rear flange.

When required, upper and lower bearing inserts of different sizes may be used as a pair. A standard size insert is sometimes used in combination with a 0.025 mm (0.001 inch) undersize insert to reduce the clearance by 0.013 mm (0.0005 inch). **Never use a pair of bearing inserts with greater than a 0.025 mm (0.001 inch) difference in size. Refer to the Bearing Insert Pair Chart.**

NOTE: When replacing inserts, the odd size inserts must be either all on the top (in cylinder block) or all on the bottom (in main bearing cap).

CRANKSHAFT MAIN BEARINGS (Continued)

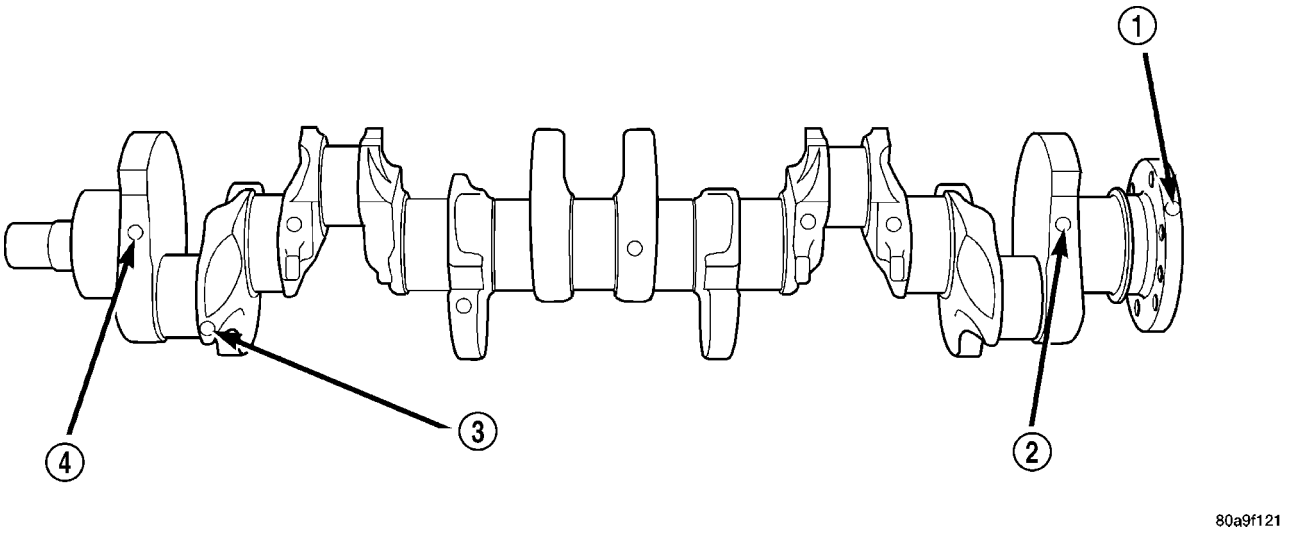

Fig. 43 Crankshaft Journal Size Paint I.D. Location

1 - NO. 7 MAIN JOURNAL SIZE PAINT MARK
2 - NO. 6 CONNECTING ROD JOURNAL SIZE PAINT MARK
3 - NO. 1 CONNECTING ROD JOURNAL SIZE PAINT MARK
4 - NO. 1 MAIN JOURNAL SIZE PAINT MARK

Once the bearings have been properly fitted, (Refer to 9 - ENGINE/ENGINE BLOCK/CRANKSHAFT MAIN BEARINGS - INSTALLATION).

BEARING-TO-JOURNAL CLEARANCE (CRANKSHAFT INSTALLED)

When using Plastigage, check only one bearing clearance at a time.

Install the grooved main bearings into the cylinder block and the non-grooved bearings into the bearing caps.

Install the crankshaft into the upper bearings dry.

Place a strip of Plastigage across full width of the crankshaft journal to be checked.

Install the bearing cap and tighten the bolts to 108 N·m (80 ft. lbs.) torque.

NOTE: DO NOT rotate the crankshaft. This will cause the Plastigage to shift, resulting in an inaccurate reading. Plastigage must not be permitted to crumble. If brittle, obtain fresh stock.

Remove the bearing cap. Determine the amount of clearance by measuring the width of the compressed Plastigage with the scale on the Plastigage envelope (Fig. 44). (Refer to 9 - ENGINE - SPECIFICATIONS) for the proper clearance.

Plastigage should indicate the same clearance across the entire width of the insert. If clearance varies, it may indicate a tapered journal or foreign material trapped behind the insert.

If the specified clearance is indicated and there are no abnormal wear patterns, replacement of the bearing inserts is not necessary. Remove the Plastigage from the crankshaft journal and bearing insert. Proceed to (Refer to 9 - ENGINE/ENGINE BLOCK/CRANKSHAFT MAIN BEARINGS - INSTALLATION).

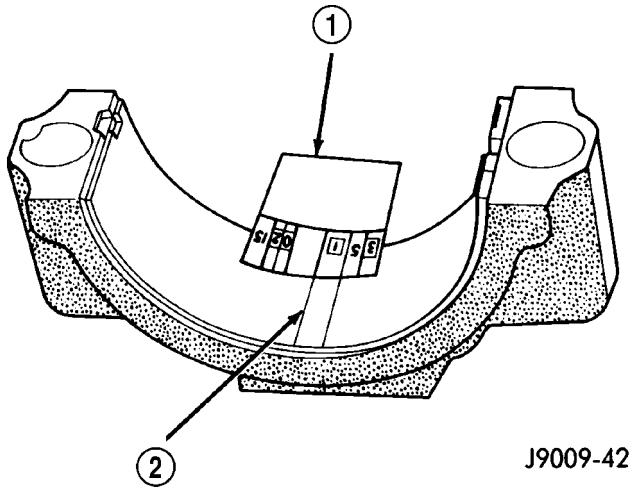

Fig. 44 Measuring Bearing Clearance with Plastigage

1 - PLASTIGAGE SCALE
2 - COMPRESSED PLASTIGAGE

If the clearance exceeds specification, install a pair of 0.025 mm (0.001 inch) undersize bearing inserts and measure the clearance as described in the previous steps.

The clearance indicate with the 0.025 mm (0.001 inch) undersize insert pair installed will determine if this insert size or some other combination will provide the specified clearance. **FOR EXAMPLE:** If the clearance was 0.0762 mm (0.003 inch) originally, a pair of 0.0254 mm (0.001 inch) undersize inserts would reduce the clearance by 0.0254 mm (0.001

inch). The clearance would then be 0.0508 mm (0.002 inch) and within the specification. A 0.051 mm (0.002 inch) undersize bearing insert and a 0.0254 mm (0.001 inch) undersize insert would reduce the original clearance an additional 0.0127 mm (0.0005 inch). The clearance would then be 0.0381 mm (0.0015 inch).

CAUTION: Never use a pair of inserts that differ more than one bearing size as a pair.

FOR EXAMPLE: DO NOT use a standard size upper insert and a 0.051 mm (0.002 inch) undersize lower insert.

If the clearance exceeds specification using a pair of 0.051 mm (0.002 inch) undersize bearing inserts, measure crankshaft journal diameter with a micrometer. If the journal diameter is correct, the crankshaft bore in the cylinder block may be misaligned, which requires cylinder block replacement or machining to true bore.

Replace the crankshaft or grind to accept the appropriate undersize bearing inserts if:

- Journal diameters 1 through 6 are less than 63.4517 mm (2.4981 inches)
- Journal 7 diameter is less than 63.4365 mm (2.4975 inches).

Once the proper clearances have been obtained, proceed to (Refer to 9 - ENGINE/ENGINE BLOCK/ CRANKSHAFT MAIN BEARINGS - INSTALLATION).

JOURNAL DIAMETER (CRANKSHAFT REMOVED)

Remove the crankshaft from the cylinder block.

Clean the oil off the main bearing journal.

Determine the maximum diameter of the journal with a micrometer. Measure at two locations 90° apart at each end of the journal.

The maximum allowable taper and out of round is 0.013 mm (0.0005 inch). Compare the measured diameter with the journal diameter specification MAIN BEARING FITTING CHART. Select inserts required to obtain the specified bearing-to-journal clearance.

Install the crankshaft into the cylinder block.

MAIN BEARING FITTING CHART

Crankshaft Journals #1-6		Corresponding Crankshaft Bearing Insert	
Color Code	**Diameter**	**Upper Insert Size**	**Lower Insert Size**
Yellow	63.5025 -63.4898 mm (2.5001 - 2.4996 in.)	Yellow - Standard	Yellow - Standard
Orange	63.4898 - 63.4771 mm (2.4996 - 2.4991 in.) 0.0127 mm (0.0005 in.) Undersize	Yellow - Standard	Blue - Undersize 0.025 mm (0.001 in.)
Blue	63.4771 - 63.4644 mm (2.4991 - 2.4986 in.) 0.0254 mm (0.001 in.) Undersize	Blue - Undersize 0.025 mm (0.001 in.)	Blue - Undersize 0.025 mm (0.001 in.)
Green	63.4644 - 63.4517 mm (2.4986 - 2.4981 in.) 0.0381 mm (0.0015 in.) Undersize	Blue - Undersize 0.025 mm (0.001 in.)	Green - Undersize 0.051 mm (0.002 in.)
Red	63.2485 - 63.2358 mm (2.4901 - 2.4896 in.) 0.254 mm (0.010 in.) Undersize	Red - Undersize 0.254 mm (0.010 in.)	Red - Undersize 0.254 mm (0.010 in.)

CRANKSHAFT MAIN BEARINGS (Continued)

Crankshaft Journal #7 Only		Corresponding Bearing Insert	
Color Code	Diameter	Upper Insert Size	Lower Insert Size
Yellow	63.4873 - 63.4746 mm (2.4995 - 2.4990 in.)	Yellow - Standard	Yellow - Standard
Orange	63.4746 - 63.4619 mm (2.4990 - 2.4985 in.) 0.0127 mm (0.0005 in.) Undersize	Yellow - Standard	Blue - Undersize 0.025 mm (0.001 in.)
Blue	63.4619 - 63.4492 mm (2.4985 - 2.4980 in.) 0.0254 mm (0.001 in.) Undersize	Blue - Undersize 0.025 mm (0.001 in.)	Blue - Undersize 0.025 mm (0.001 in.)
Green	63.4492 - 63.4365 mm (2.4980 - 2.4975 in.) 0.0381 mm (0.0015 in.) Undersize	Blue - Undersize 0.025 mm (0.001 in.)	Green - Undersize 0.051 mm (0.002 in.)
Red	63.2333 - 63.2206 mm (2.4895 - 2.4890 in.) 0.254 mm (0.010 in.) Undersize	Red - Undersize 0.254 mm (0.010 in.)	Red - Undersize 0.254 mm (0.010 in.)

REMOVAL

(1) Disconnect negative cable from battery.

(2) Remove the spark plugs (Refer to 8 - ELECTRICAL/IGNITION CONTROL/SPARK PLUG - REMOVAL).

(3) Raise the vehicle.

(4) Remove the oil pan (Refer to 9 - ENGINE/LUBRICATION/OIL PAN - REMOVAL) and oil pump (Refer to 9 - ENGINE/LUBRICATION/OIL PUMP - REMOVAL).

(5) Remove main bearing cap brace (Fig. 45).

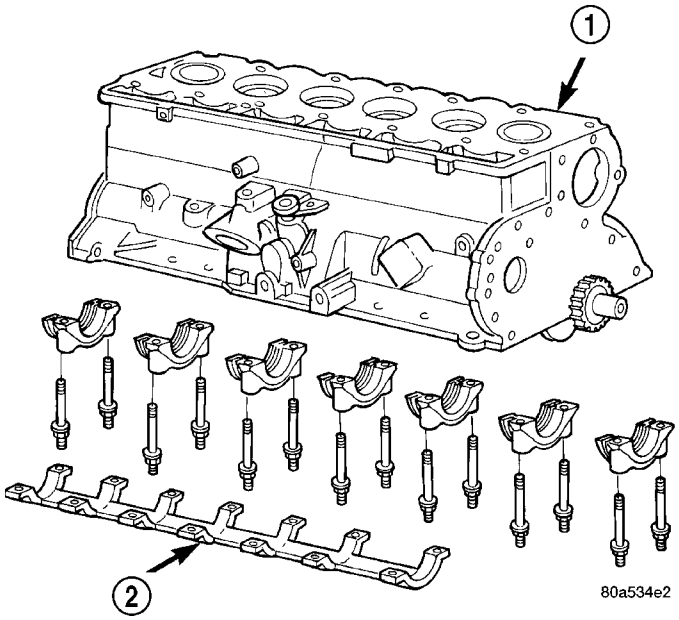

Fig. 45 Main Bearing Caps and Brace.

1 - BLOCK
2 - MAIN BEARING CAP BRACE

(6) Remove only one main bearing cap and lower insert at a time (Fig. 46).

(7) Remove the lower insert from the bearing cap.

(8) Remove the upper insert by LOOSENING (DO NOT REMOVE) all of the other bearing caps. Now insert a small cotter pin tool in the crankshaft journal oil hole. Bend the cotter pin as illustrated to fabricate the tool (Fig. 47). With the cotter pin tool in place, rotate the crankshaft so that the upper bearing insert will rotate in the direction of its locking tab. Because there is no hole in the No.3 main journal, use a tongue depressor or similar soft-faced tool to remove the bearing insert (Fig. 47). After moving the insert approximately 25 mm (1 inch), it can be removed by applying pressure under the tab.

(9) Using the same procedure described above, remove the remaining bearing inserts one at a time for inspection.

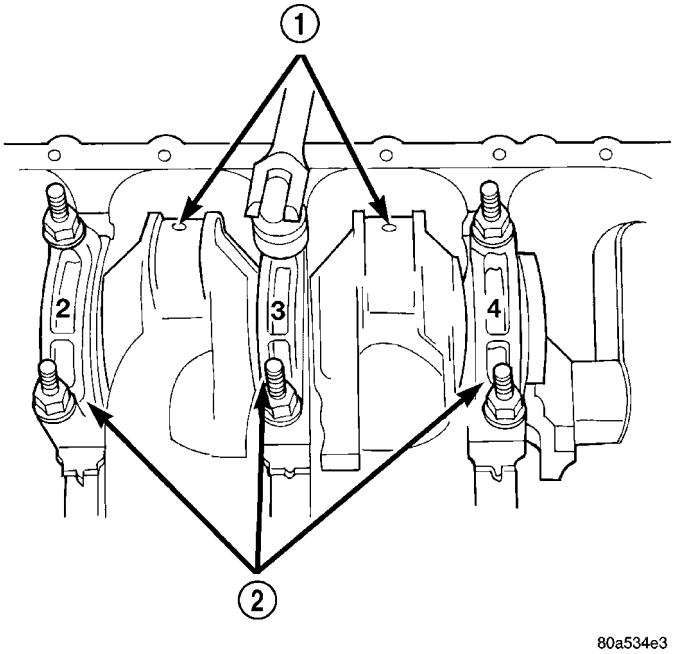

Fig. 46 Removing Main Bearing Caps and Lower Inserts

1 - CONNECTING ROD JOURNAL
2 - MAIN BEARING CAPS

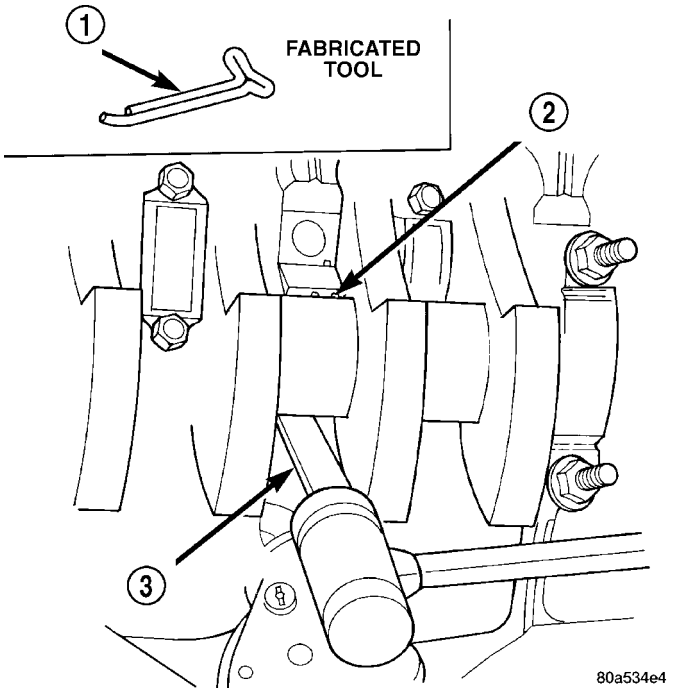

Fig. 47 Removing Upper Inserts

1 - COTTER PIN
2 - BEARING INSERT
3 - TONGUE DEPRESSOR

INSPECTION

Wipe the inserts clean and inspect for abnormal wear patterns and for metal or other foreign material imbedded in the lining. Normal main bearing insert wear patterns are illustrated (Fig. 48). In general the lower bearing half will have a heaver wear pattern.

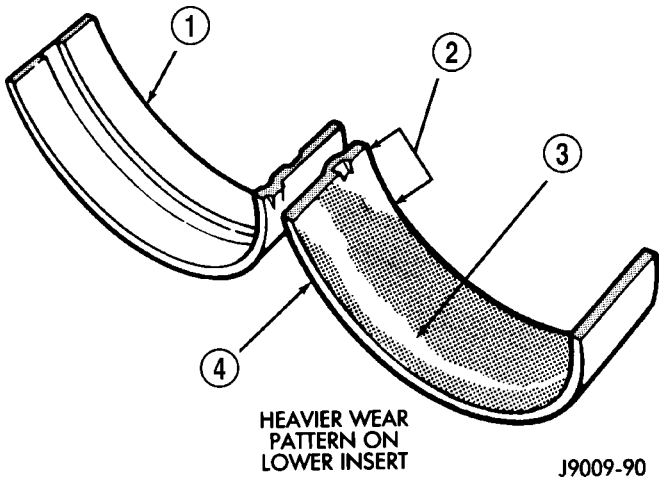

Fig. 48 Main Bearing Wear Patterns

1 - UPPER INSERT
2 - NO WEAR IN THIS AREA
3 - LOW AREA IN BEARING LINING
4 - LOWER INSERT

NOTE: If any of the crankshaft journals are scored, remove the engine for crankshaft repair.

Inspect the back of the inserts for fractures, scrapings or irregular wear patterns.

Inspect the upper insert locking tabs for damage.

Replace all damaged or worn bearing inserts.

INSTALLATION

(1) Lubricate the bearing surface of each insert with engine oil.

(2) Loosen all the main bearing caps. Install the main bearing upper inserts.

(3) Install the lower bearing inserts into the main bearing caps.

(4) On the rear main cap, apply Mopar® Gasket Maker sealer on both sides of cylinder block as shown in (Fig. 49). The dab of sealer should be 3 mm (0.125 in.) in diameter.

(5) Apply Mopar® Gasket Maker on the rear bearing cap. The bead should be 2.3 mm (0.09 in.) in diameter. DO NOT apply sealer to the lip of the seal.

(6) Install the main bearing cap(s) and lower insert(s).

(7) Tighten the bolts of caps 1, 2, 4, 5, 6, and 7 to 54 N·m (40 ft. lbs.) torque. Now tighten these bolts to 95 N·m (70 ft. lbs.) torque. Finally, tighten these bolts to 108 N·m (80 ft. lbs.) torque.

(8) Push the crankshaft forward and backward. Load the crankshaft front or rear and tighten cap bolt No.3 to 54 N·m (40 ft. lbs.) torque. Then tighten

CRANKSHAFT MAIN BEARINGS (Continued)

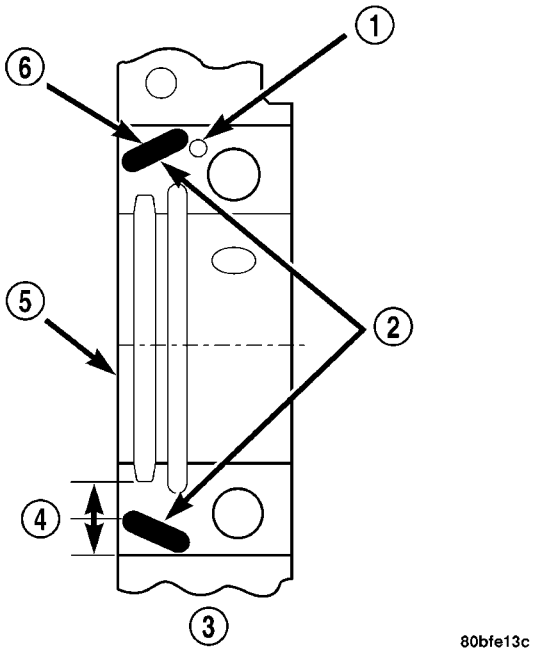

Fig. 49 Location of Sealer

1 - **DOWEL**
2 - **SEALER LOCATIONS**
3 - **CYLINDER BLOCK**
4 - **HALFWAY BETWEEN**
5 - **REAR FACE OF CYLINDER BLOCK**
6 - **3mm (0.125 in.)**

to 95 N·m (70 ft. lbs.) torque and finally tighten to 108 N·m (80 ft. lbs.) torque.

(9) Rotate the crankshaft after tightening each main bearing cap to ensure the crankshaft rotates freely.

(10) Check crankshaft end play. Crankshaft end play is controlled by the thrust bearing which is flange and installed at the No.2 main bearing position.

(a) Attach a magnetic base dial indicator to the cylinder block at either the front or rear of the engine.

(b) Position the dial indicator rod so that it is parallel to the center line of the crankshaft.

(c) Pry the crankshaft forward, position the dial indicator to zero.

(d) Pry the crankshaft forward and backward. Note the dial indicator readings. End play is the difference between the high and low measurements (Fig. 50). Correct end play is 0.038-0.165 mm (0.0015-0.0065 inch). The desired specifications are 0.051-0.064 mm (0.002-0.0025 inch).

(e) If end play is not within specification, inspect crankshaft thrust faces for wear. If no wear is apparent, replace the thrust bearing and measure end play. If end play is still not within specification, replace the crankshaft.

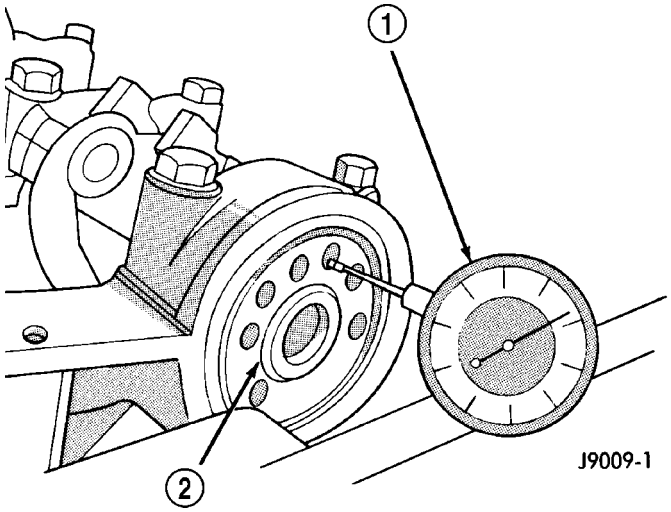

Fig. 50 Crankshaft End Play Measurement

1 - DIAL INDICATOR
2 - CRANKSHAFT

(11) If the crankshaft was removed, install the crankshaft into the cylinder block.

(12) Install main bearing cap brace tighten nuts to 47 N·m (35 ft. lbs.) torque.

(13) Install oil pump assy. and tighten attaching bolts to 23 N·m (17 ft. lbs.)

(14) Install the oil pan (Refer to 9 - ENGINE/LUBRICATION/OIL PAN - INSTALLATION).

(15) Install the drain plug. Tighten the plug to 34 N·m (25 ft. lbs.) torque.

(16) Lower the vehicle.

(17) Install the spark plugs. Tighten the plugs to 37 N·m (27 ft. lbs.) torque.

(18) Fill the oil pan with engine oil to the full mark on the dipstick level.

(19) Connect negative cable to battery.

CRANKSHAFT OIL SEAL - FRONT

REMOVAL

This procedure is done with the timing case cover installed.

(1) Disconnect negative cable from battery.

(2) Remove the serpentine drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - REMOVAL).

(3) Remove the vibration damper (Refer to 9 - ENGINE/ENGINE BLOCK/VIBRATION DAMPER - REMOVAL).

(4) Remove the radiator shroud.

(5) Carefully remove the oil seal. Make sure seal bore is clean.

CRANKSHAFT OIL SEAL - FRONT (Continued)

INSTALLATION

This procedure is done with the timing case cover installed.

(1) Position the replacement oil seal on Timing Case Cover Alignment and Seal Installation Tool 6139 with seal open end facing inward. Apply a light film of Perfect Seal, or equivalent, on the outside diameter of the seal. Lightly coat the crankshaft with engine oil.

(2) Position the tool and seal over the end of the crankshaft and insert a draw screw tool into Seal Installation Tool 6139 (Fig. 51). Tighten the nut against the tool until it contacts the cover.

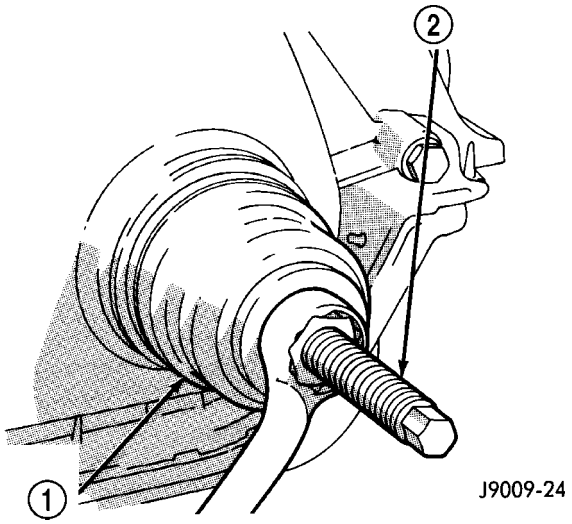

Fig. 51 Timing Case Cover Oil Seal Installation

1 - SEAL INSTALLATION TOOL
2 - DRAW SCREW TOOL

(3) Remove the tools. Apply a light film of engine oil on the vibration damper hub contact surface of the seal.

(4) Apply Mopar® Silicone Rubber Adhesive Sealant to the keyway in the crankshaft and insert the key. With the key inserted in the keyway in the crankshaft, install the vibration damper (Refer to 9 - ENGINE/ENGINE BLOCK/VIBRATION DAMPER - INSTALLATION).

(5) Install the serpentine belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - INSTALLATION).

(6) Install the radiator shroud.

(7) Connect negative cable to battery.

CRANKSHAFT OIL SEAL - REAR

REMOVAL

The crankshaft rear main bearing oil seal consists of two half pieces of viton with a single lip that effectively seals the rear of the crankshaft. Replace the upper and lower seal halves as a unit to ensure leak-free operation.

(1) Remove transmission inspection cover.

(2) Remove oil pan. (Refer to 9 - ENGINE/LUBRICATION/OIL PAN - REMOVAL)

(3) Remove main bearing cap brace.

(4) Remove rear main bearing cap (No.7).

(5) Push upper seal out of the groove. Ensure that the crankshaft and seal groove are not damaged.

(6) Remove lower half of the seal from the bearing cap.

INSTALLATION

The crankshaft rear main bearing oil seal consists of two half pieces of viton with a single lip that effectively seals the rear of the crankshaft. Replace the upper and lower seal halves as a unit to ensure leak-free operation.

(1) Wipe the seal surface area of the crankshaft until it is clean.

(2) Apply a thin coat of engine oil.

(3) Coat lip of the seal with engine oil.

(4) Carefully position the upper seal into the groove in the cylinder block. The lip of the seal faces toward the front of the engine.

(5) Apply Mopar® Gasket Maker sealer on both sides of cylinder block as shown in (Fig. 52). The dab of sealer should be 3 mm (0.125 in.) in diameter.

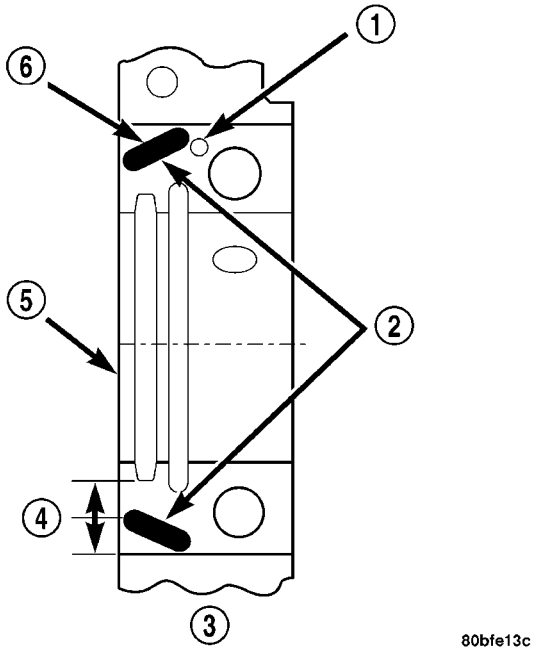

Fig. 52 Location of Sealer

1 - **DOWEL**
2 - **SEALER LOCATIONS**
3 - **CYLINDER BLOCK**
4 - **HALFWAY BETWEEN**
5 - **REAR FACE OF CYLINDER BLOCK**
6 - **3mm (0.125 in.)**

CRANKSHAFT OIL SEAL - REAR (Continued)

(6) Apply Mopar® Gasket Maker on the rear bearing cap (Fig. 52). The bead should be 2.3 mm (0.09 in.) in diameter. DO NOT apply sealer to the lip of the seal.

(7) Position the lower seal into the bearing cap recess and seat it firmly. Be sure the seal is flush with the cylinder block pan rail.

(8) Coat the outer curved surface of the lower seal with soap and the lip of the seal with engine oil.

(9) Install the rear main bearing cap. DO NOT strike the cap more than twice for proper engagement.

(10) Tighten all main bearing bolts to 108 N·m (80 ft. lbs.) torque.

(11) Install the main bearing cap brace. Tighten nuts to 47 N·m (35 ft. lbs.).

(12) Install the oil pan gasket and oil pan (Refer to 9 - ENGINE/LUBRICATION/OIL PAN - INSTALLATION).

(13) Apply Mopar® Silicone Rubber Adhesive Sealant on cylinder block to rear main bearing cap corners and cylinder block to front cover joints (four places) (Fig. 53)

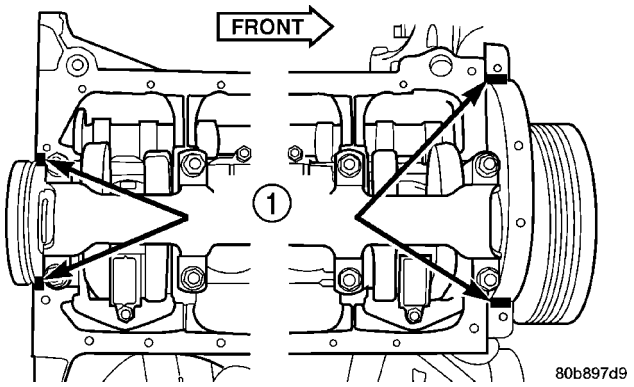

Fig. 53 Oil Pan

1 - SEALER LOCATIONS

(14) Install transmission inspection cover.

HYDRAULIC LIFTERS

DESCRIPTION

Valve lash is controlled by hydraulic tappets located inside the cylinder block, in tappet bores above the camshaft.

REMOVAL

NOTE: Retain all the components in the same order as removed.

(1) Remove the engine cylinder head (Refer to 9 - ENGINE/CYLINDER HEAD - REMOVAL).

(2) Remove the push rods.

(3) Remove the tappets through the push rod openings in the cylinder block with a Hydraulic Valve Tappet Removal/Installation Tool (Fig. 54).

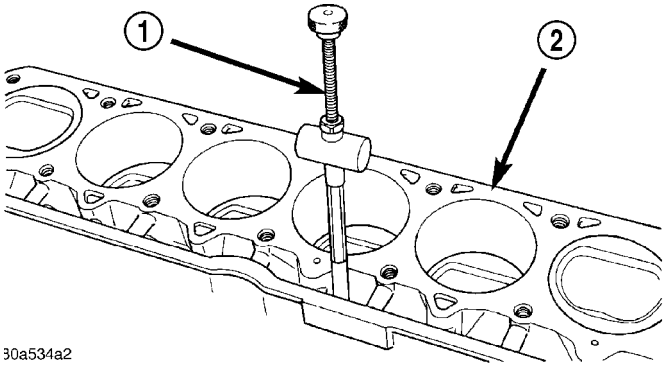

Fig. 54 HYDRAULIC VALVE TAPPET REMOVAL - 4.0L

1 - HYDRAULIC TAPPET REMOVAL TOOL
2 - CYLINDER BLOCK

CLEANING

Clean each tappet assembly in cleaning solvent to remove all varnish, gum and sludge deposits.

INSPECTION

Inspect for indications of scuffing on the side and base of each tappet body.

Inspect each tappet base for concave wear with a straightedge positioned across the base. If the base is concave, the corresponding lobe on the camshaft is also worn. Replace the camshaft and tappets.

After cleaning and inspection, test each tappet for specified leak-down rate tolerance to ensure zero-lash operation (Fig. 55).

Swing the weighted arm of the hydraulic valve tappet tester away from the ram of the Leak-Down Tester.

(1) Place a 7.925-7.950 mm (0.312-0.313 inch) diameter ball bearing on the plunger cap of the tappet.

(2) Lift the ram and position the tappet (with the ball bearing) inside the tester cup.

(3) Lower the ram, then adjust the nose of the ram until it contacts the ball bearing. DO NOT tighten the hex nut on the ram.

(4) Fill the tester cup with hydraulic valve tappet test oil until the tappet is completely submerged.

(5) Swing the weighted arm onto the push rod and pump the tappet plunger up and down to remove air. When the air bubbles cease, swing the weighted arm away and allow the plunger to rise to the normal position.

(6) Adjust the nose of the ram to align the pointer with the SET mark on the scale of the tester and tighten the hex nut.

HYDRAULIC LIFTERS (Continued)

(7) Slowly swing the weighted arm onto the push rod.

(8) Rotate the cup by turning the handle at the base of the tester clockwise one revolution every 2 seconds.

(9) Observe the leak-down time interval from the instant the pointer aligns with the START mark on the scale until the pointer aligns with the 0.125 mark. A normally functioning tappet will require 20-110 seconds to leak-down. Discard tappets with leak-down time interval not within this specification.

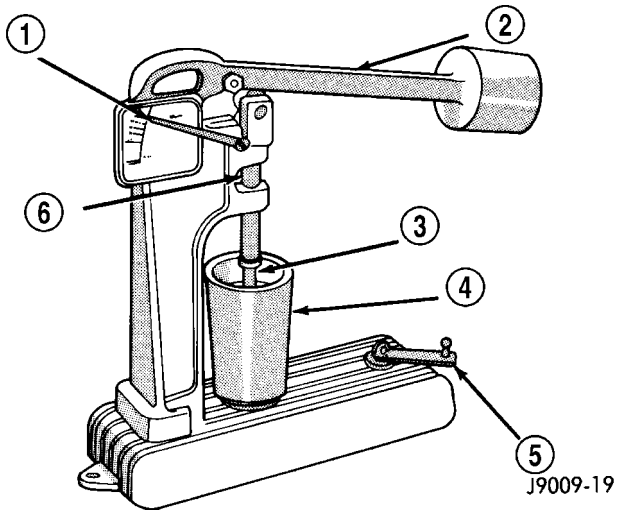

Fig. 55 Leak-Down Tester

1 - POINTER
2 - WEIGHTED ARM
3 - RAM
4 - CUP
5 - HANDLE
6 - PUSH ROD

INSTALLATION

Retain all the components in the same order as removed.

It is not necessary to charge the tappets with engine oil. They will charge themselves within a very short period of engine operation.

(1) Dip each tappet in Mopar® Engine Oil Supplement, or equivalent.

(2) Use Hydraulic Valve Tappet Removal/Installation Tool to install each tappet in the same bore from where it was originally removed.

(3) Install the cylinder head (Refer to 9 - ENGINE/CYLINDER HEAD - INSTALLATION).

(4) Install the push rods in their original locations.

(5) Install the rocker arms and bridge and pivot assemblies at their original locations. Loosely install the capscrews at each bridge.

(6) Tighten the capscrews alternately, one turn at a time, to avoid damaging the bridges. Tighten the capscrews to 28 N·m (21 ft. lbs.) torque.

(7) Pour the remaining Mopar® Engine Oil Supplement, or equivalent over the entire valve actuating assembly. The Mopar® Engine Oil Supplement, or equivalent must remain with the engine oil for at least 1 609 km (1,000 miles). The oil supplement need not be drained until the next scheduled oil change.

(8) Install the cylinder head cover (Refer to 9 - ENGINE/CYLINDER HEAD/CYLINDER HEAD COVER(S) - INSTALLATION).

PISTON & CONNECTING ROD

DESCRIPTION

The pistons (Fig. 56) are made of a high strength aluminum alloy, the piston skirts are coated with a solid lubricant (Molykote) to reduce friction and provide scuff resistance. The connecting rods are made of cast iron.

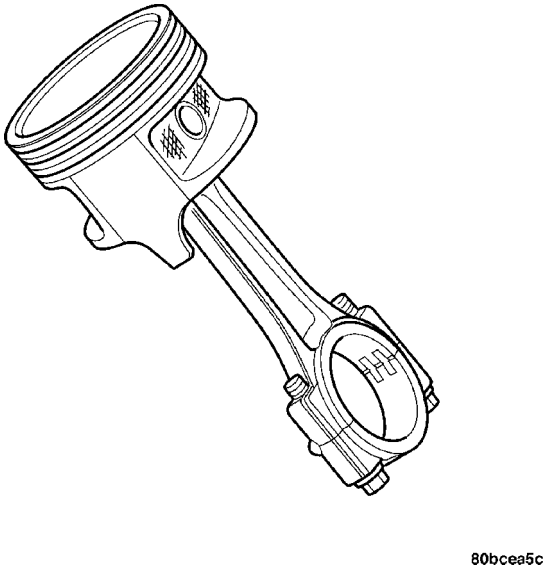

Fig. 56 Piston and Connecting Rod Assembly

STANDARD PROCEDURE - PISTON FITTING

(1) To correctly select the proper size piston, a cylinder bore gauge, capable of reading in 0.003 mm (.0001 in.) INCREMENTS is required. If a bore gauge is not available, do not use an inside micrometer.

(2) Measure the inside diameter of the cylinder bore at a point 49.5 mm (1-15/16 inches) below top of bore. Start perpendicular (across or at 90 degrees) to the axis of the crankshaft at point A and then take an additional bore reading 90 degrees to that at point B (Fig. 58).

(3) The coated pistons will be serviced with the piston pin and connecting rod pre-assembled. **The coated piston connecting rod assembly can be used to service previous built engines and**

PISTON & CONNECTING ROD (Continued)

MUST be replaced as complete sets. Tin coated pistons should not be used as replacements for coated pistons.

(4) The coating material is applied to the piston after the final piston machining process. Measuring the outside diameter of a coated piston will not provide accurate results (Fig. 57). Therefore measuring the inside diameter of the cylinder bore with a dial Bore Gauge is **MANDATORY**. To correctly select the proper size piston, a cylinder bore gauge capable of reading in 0.003 mm (.0001 in.) increments is required.

(5) Piston installation into the cylinder bore requires slightly more pressure than that required for non-coated pistons. The bonded coating on the piston will give the appearance of a line-to-line fit with the cylinder bore.

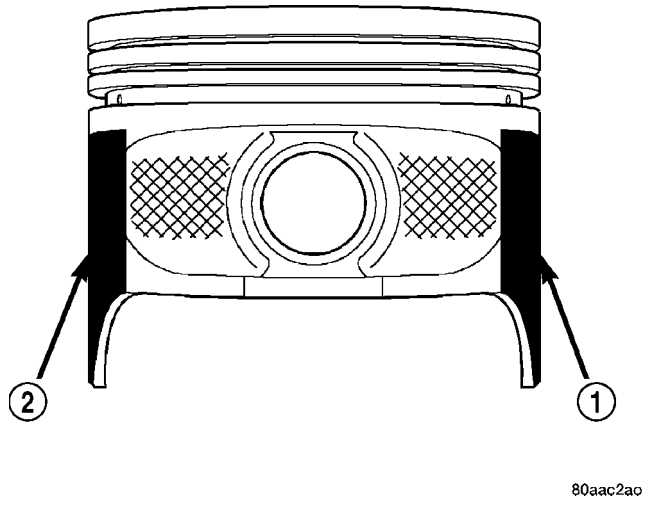

Fig. 57 Moly Coated Piston

1 - MOLY COATED
2 - MOLY COATED

PISTON SIZE CHART

CYLINDER BORE SIZE	PISTON LETTER SIZE
98.438 - 98.448 mm (3.8755 - 3.8759 in.)	A
98.448 - 98.458 mm (3.8759 - 3.8763 in.)	B
98.458 - 98.468 mm (3.8763 - 3.8767 in.)	C
98.468 - 98.478 mm (3.8767 - 3.8771 in.)	D
98.478 - 98.488 mm (3.8771 - 3.8775 in.)	E
98.488 - 98.498 mm (3.8775 - 3.8779 in.)	F

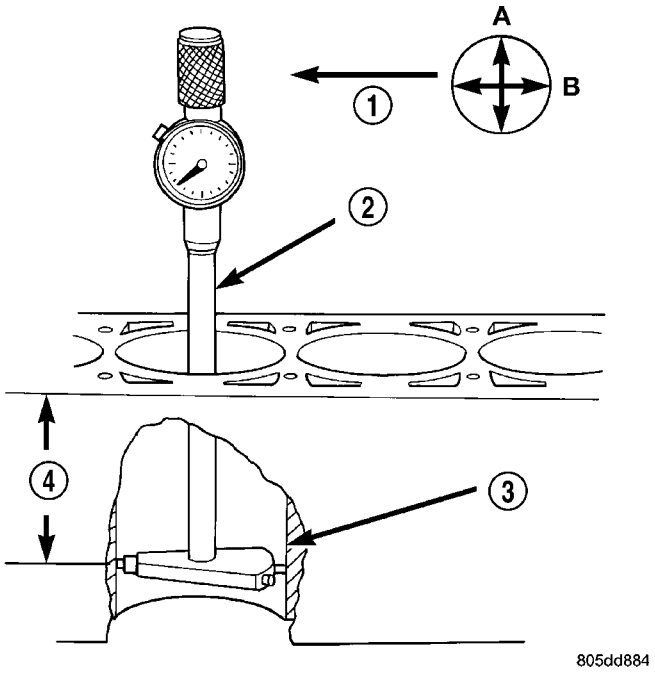

Fig. 58 Bore Gauge

1 - FRONT
2 - BORE GAUGE
3 - CYLINDER BORE
4 - 49.5 MM (1-15/16 in.)

REMOVAL

(1) Remove the engine cylinder head cover. (Refer to 9 - ENGINE/CYLINDER HEAD/CYLINDER HEAD COVER(S) - REMOVAL).

(2) Remove the rocker arms, bridges and pivots.

(3) Remove the push rods.

(4) Remove the engine cylinder head. (Refer to 9 - ENGINE/CYLINDER HEAD - REMOVAL).

(5) Position the pistons one at a time near the bottom of the stroke. Use a ridge reamer to remove the ridge from the top end of the cylinder walls. Use a protective cloth to collect the cuttings.

(6) Raise the vehicle.

(7) Drain the engine oil.

PISTON & CONNECTING ROD (Continued)

(8) Remove the oil pan and gasket. (Refer to 9 - ENGINE/LUBRICATION/OIL PAN - REMOVAL).

(9) Remove main bearing cap brace (Fig. 59).

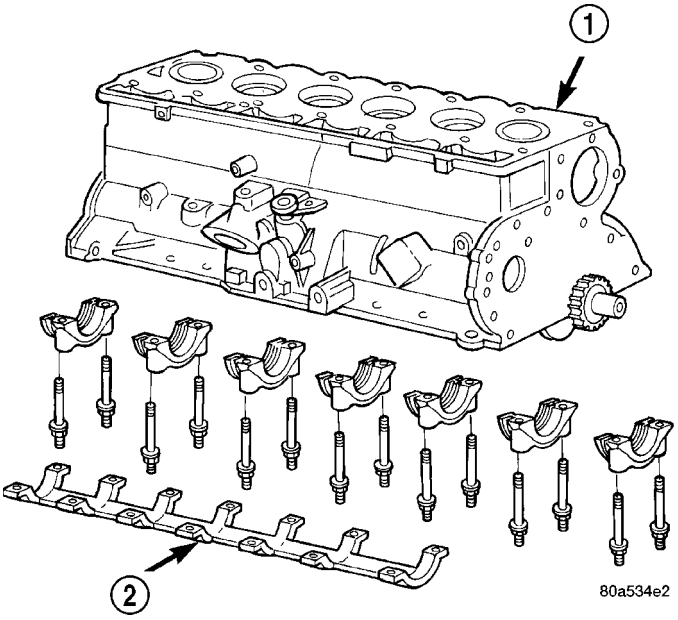

Fig. 59 Main Bearings Caps and Brace

1 - BLOCK
2 - MAIN BEARING CAP BRACE

(10) Remove the connecting rod bearing caps and inserts. Mark the caps and rods with the cylinder bore location. The connecting rods and caps are stamped with a two letter combination (Fig. 60).

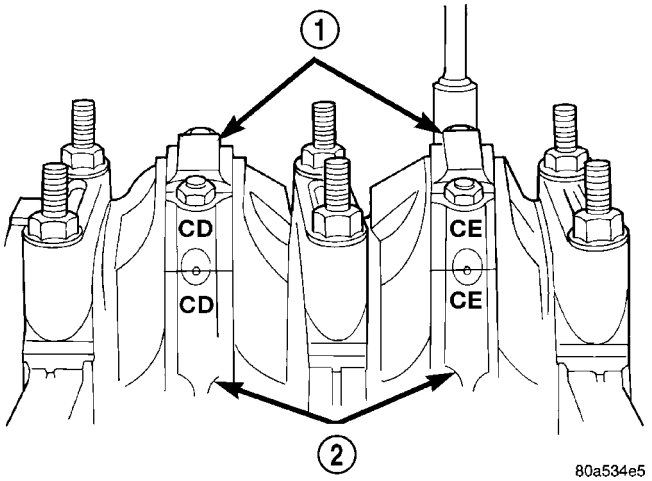

Fig. 60 Stamped Connecting Rods and Caps

1 - CONNECTING ROD CAP
2 - CONNECTING ROD

(11) Lower the vehicle until it is about 2 feet from the floor.

CAUTION: Ensure that the connecting rod bolts DO NOT scratch the crankshaft journals or cylinder walls. Short pieces of rubber hose, slipped over the rod bolts will provide protection during removal.

(12) Have an assistant push the piston and connecting rod assemblies up and through the top of the cylinder bores (Fig. 61).

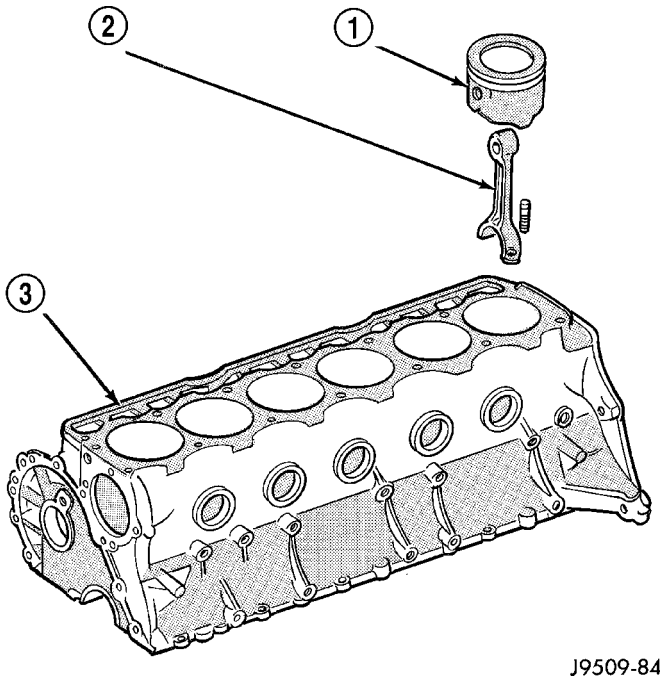

Fig. 61 Removal of Connecting Rod and Piston Assembly

1 - PISTON
2 - CONNECTING ROD
3 - BLOCK

INSTALLATION

(1) Clean the cylinder bores thoroughly. Apply a light film of clean engine oil to the bores with a clean lint-free cloth.

(2) Install the piston rings on the pistons if removed (Refer to 9 - ENGINE/ENGINE BLOCK/ PISTON RINGS - STANDARD PROCEDURE).

(3) Lubricate the piston and rings with clean engine oil.

CAUTION: Ensure that connecting rod bolts DO NOT scratch the crankshaft journals or cylinder walls. Short pieces of rubber hose slipped over the connecting rod bolts will provide protection during installation.

PISTON & CONNECTING ROD (Continued)

(4) Use a piston ring compressor to install the connecting rod and piston assemblies through the top of the cylinder bores (Fig. 62).

(5) Ensure the arrow on the piston top points to the front of the engine (Fig. 62).

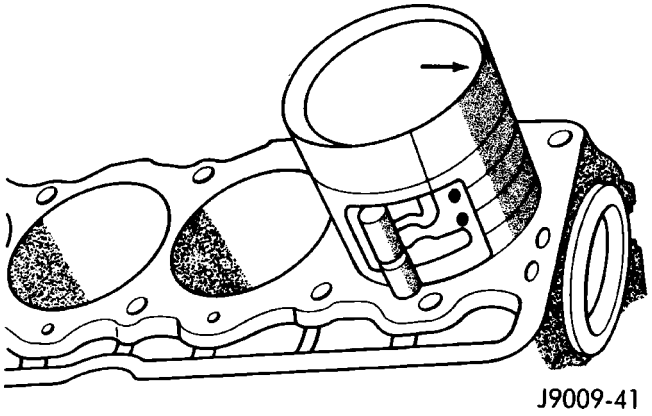

Fig. 62 Rod and Piston Assembly Installation

(6) Raise the vehicle.

(7) Each bearing insert is fitted to its respective journal to obtain the specified clearance between the bearing and the journal. In production, the select fit is obtained by using various-sized, color-coded bearing inserts as listed in the Connecting Rod Bearing Fitting Chart. The color code appears on the edge of the bearing insert. The size is not stamped on inserts used for production of engines.

(8) The rod journal is identified during the engine production by a color-coded paint mark on the adjacent cheek or counterweight toward the flange (rear) end of the crankshaft. The color codes used to indicate journal sizes are listed in the Connecting Rod Bearing Fitting Chart.

(9) When required, upper and lower bearing inserts of different sizes may be used as a pair (refer to Connecting Rod Bearing Fitting Chart). A standard size insert is sometimes used in combination with a 0.025 mm (0.001 inch) undersize insert to reduce clearance 0.013 mm (0.0005 inch).

CAUTION: DO NOT intermix bearing caps. Each connecting rod and bearing cap are stamped with the cylinder number. The stamp is located on a machined surface adjacent to the oil squirt hole that faces the camshaft side of the cylinder block.

(10) Install the connecting rod bearing caps and inserts in the same positions as removed.

CAUTION: Verify that the oil squirt holes in the rods face the camshaft and that the arrows on the pistons face the front of the engine.

(11) Install main bearing cap brace (Fig. 59). Tighten nuts to 47 N·m (35 ft. lbs.).

(12) Install the oil pan and gasket (Refer to 9 - ENGINE/LUBRICATION/OIL PAN - INSTALLATION).

(13) Lower the vehicle.

(14) Install the engine cylinder head (Refer to 9 - ENGINE/CYLINDER HEAD - INSTALLATION), push rods, rocker arms, bridges, pivots and engine cylinder head cover (Refer to 9 - ENGINE/CYLINDER HEAD/CYLINDER HEAD COVER(S) - INSTALLATION).

(15) Fill the crankcase with engine oil.

PISTON RINGS

STANDARD PROCEDURE - PISTON RING FITTING

(1) Carefully clean the carbon from all ring grooves. Oil drain openings in the oil ring groove and pin boss must be clear. DO NOT remove metal from the grooves or lands. This will change ring-to-groove clearances and will damage the ring-to-land seating.

(2) Be sure the piston ring grooves are free of nicks and burrs.

(3) Measure the ring side clearance with a feeler gauge fitted snugly between the ring land and ring (Fig. 63) (Fig. 64). Rotate the ring in the groove. It must move freely around circumference of the groove.

GROOVE HEIGHT

A 1.530-1.555 mm (0.0602-0.0612 in)
B 4.035-4.060 mm (0.1589-0.1598 in)

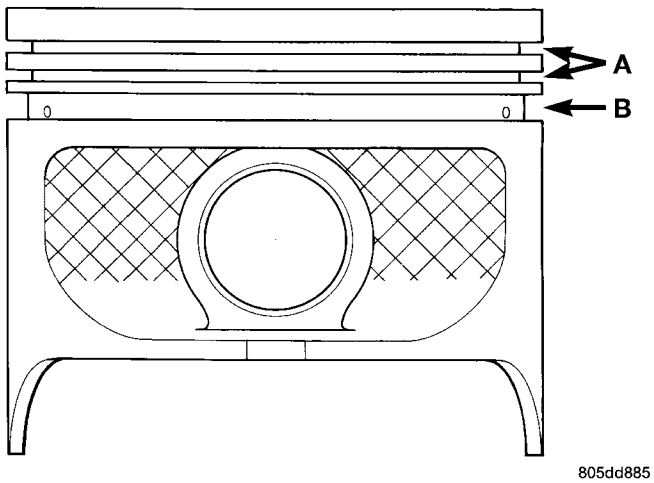

Fig. 63 Piston Dimensions

PISTON RINGS (Continued)

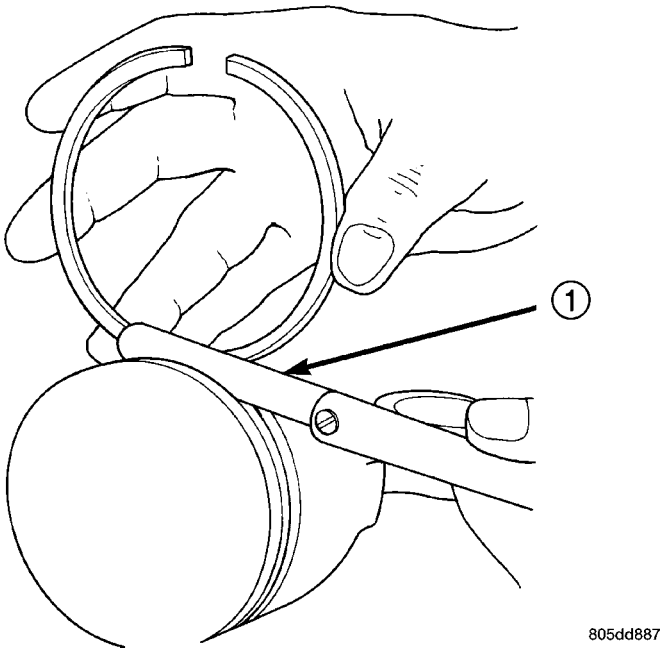

Fig. 64 Ring Side Clearance Measurement

1 - FEELER GAUGE

RING SIDE CLEARANCE CHART

ITEM	SPECIFICATION
Top Compression Ring	0.042 - 0.084 mm (0.0017 - 0.0033 in.)
Second Compression Ring	0.042 - 0.084 mm (0.0017 - 0.0033 in.)
Oil Control Ring	0.06 - 0.21 mm (0.0024 - 0.0083 in.)

(4) Place ring in the cylinder bore and push down with inverted piston to position near lower end of the ring travel. Measure ring gap with a feeler gauge fitting snugly between ring ends (Fig. 65).

RING GAP MEASUREMENT CHART

ITEM	SPECIFICATION
Top Compression Ring	0.229 - 0.610 mm (0.0090 - 0.0240 in.)
Second Compression Ring	0.483 - 0.965 mm (0.0190 - 0.080 in.)
Oil Control Ring	0.254 - 1.500 mm (0.010 - 0.060 in.)

(5) The oil control rings are symmetrical, and can be installed with either side up. It is not necessary to use a tool to install the upper and lower rails. Insert oil rail spacer first, then side rails.

(6) The two compression rings are different and cannot be interchanged. The top compression ring can be identified by the shiny coating on the outer sealing surface and can be installed with either side up. (Fig. 66).

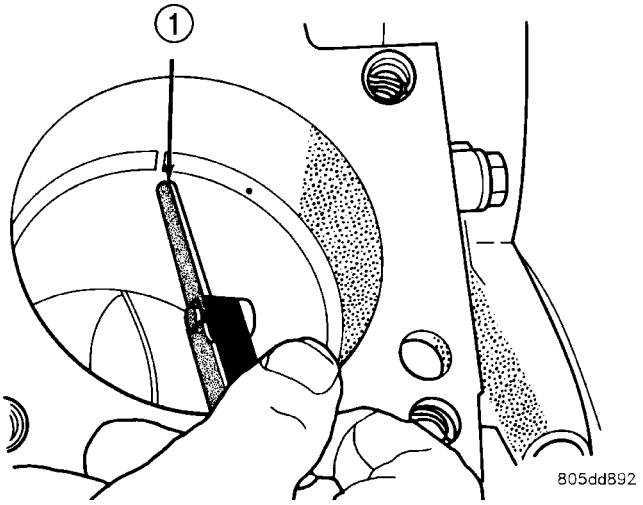

Fig. 65 Gap Measurement

1 - FEELER GAUGE

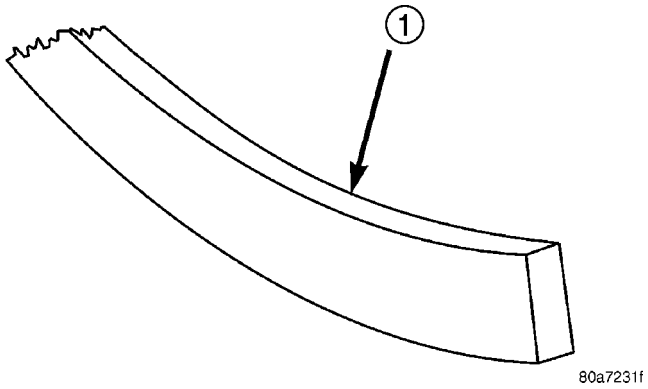

Fig. 66 Top Compression ring identification

1 - TOP COMPRESSION RING

PISTON RINGS (Continued)

(7) The second compression ring has a slight chamfer on the bottom of the inside edge and a dot on the top for correct installation (Fig. 67) and (Fig. 68).

(8) Using a ring installer, install the second compression ring with the dot facing up (Fig. 67) (Fig. 69).

(9) Using a ring installer, install the top compression ring (either side up).

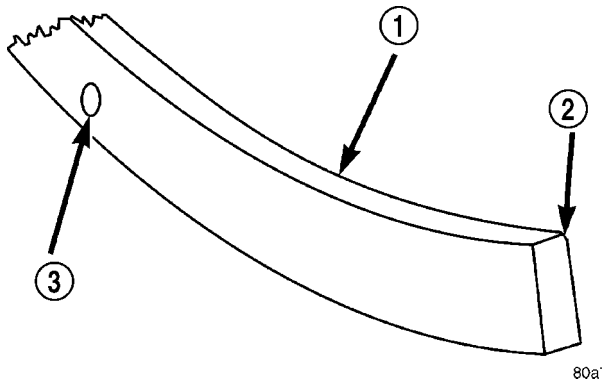

Fig. 67 Second Compression Ring Identification

1 - SECOND COMPRESSION RING
2 - CHAMFER
3 - ONE DOT

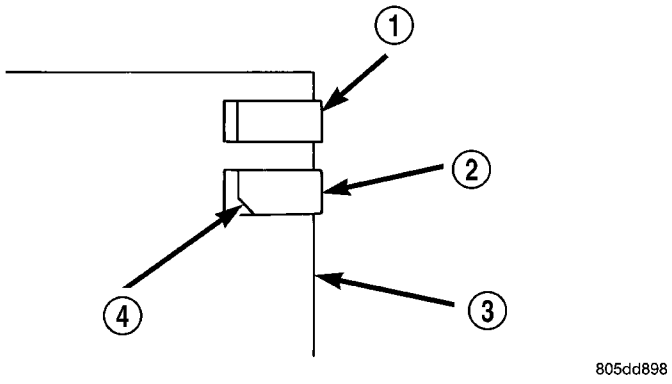

Fig. 68 Compression Ring Chamfer Location

1 - TOP COMPRESSION RING
2 - SECOND COMPRESSION RING
3 - PISTON
4 - CHAMFER

Ring Gap Orientation

- Position the gaps on the piston as shown (Fig. 70).
- Oil spacer - Gap on center line of piston skirt.
- Oil rails - gap 180° apart on centerline of piston pin bore.
- No. 2 Compression ring - Gap 180° from top oil rail gap.
- No. 1 Compression ring - Gap 180° from No. 2 compression ring gap.

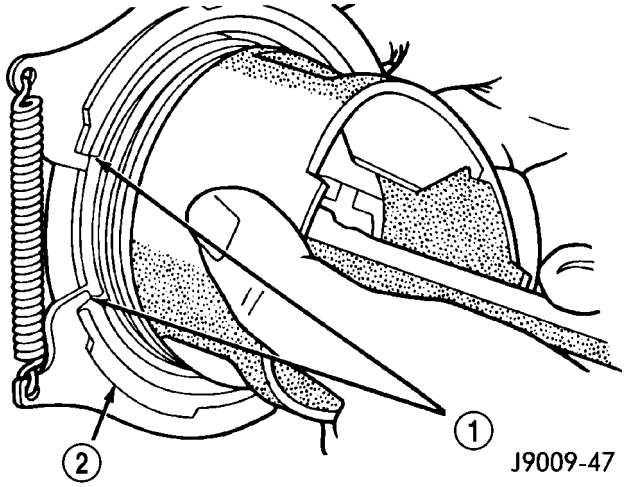

Fig. 69 Compression Ring Installation

1 - COMPRESSION RING
2 - RING EXPANDER RECOMMENDED

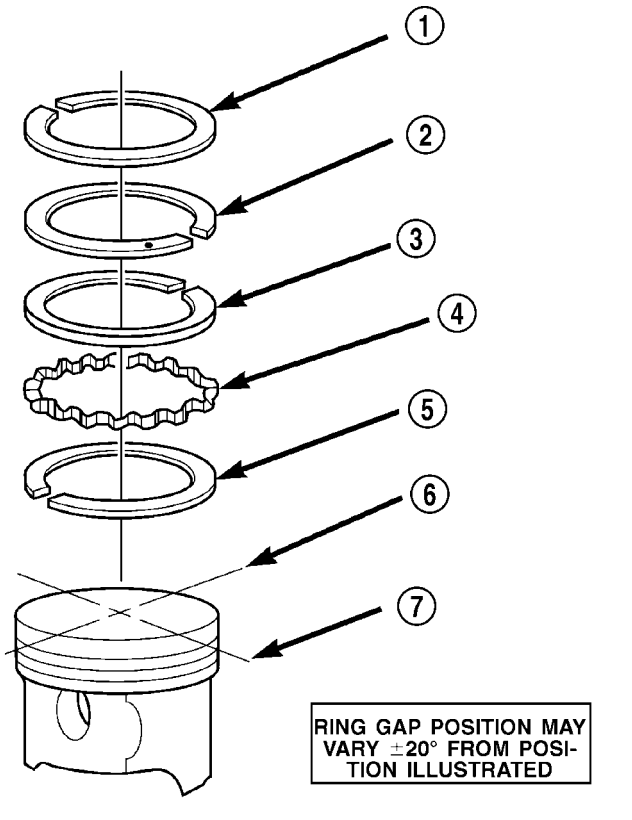

Fig. 70 Ring Gap Orientation

1 - TOP COMPRESSION RING
2 - BOTTOM COMPRESSION RING
3 - TOP OIL CONTROL RAIL
4 - OIL RAIL SPACER
5 - BOTTOM OIL CONTROL RAIL
6 - IMAGINARY LINE PARALLEL TO PISTON PIN
7 - IMAGINARY LINE THROUGH CENTER OF PISTON SKIRT

VIBRATION DAMPER

REMOVAL

(1) Disconnect negative cable from battery.
(2) Remove the serpentine drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - REMOVAL) and fan shroud.
(3) Remove the vibration damper retaining bolt and washer.
(4) Use Vibration Damper Removal Tool 7697 to remove the damper from the crankshaft (Fig. 71).

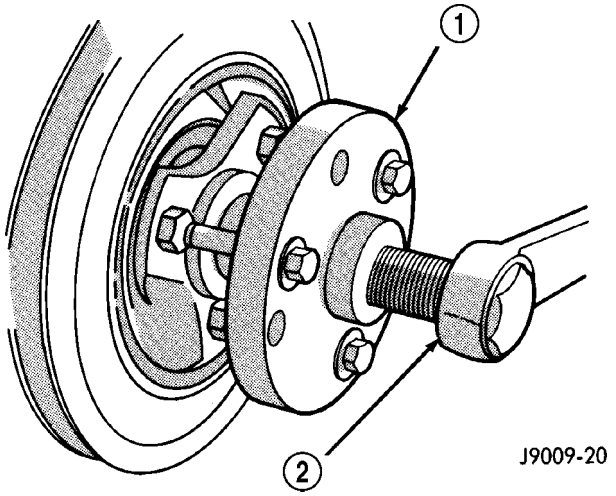

Fig. 71 Vibration Damper Removal Tool 7697

1 - VIBRATION DAMPER REMOVAL TOOL
2 - WRENCH

INSTALLATION

(1) Apply Mopar® Silicone Rubber Adhesive Sealant to the keyway in the crankshaft and insert the key. With the key in position, align the keyway on the vibration damper hub with the crankshaft key and tap the damper onto the crankshaft.
(2) Install the vibration damper retaining bolt and washer.
(3) Tighten the damper retaining bolt to 108 N·m (80 ft. lbs.) torque.
(4) Install the serpentine drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - INSTALLATION) and fan shroud.
(5) Connect negative cable to battery.

STRUCTURAL SUPPORT

REMOVAL

The engine bending braces are used to add strength to the powertrain and to address some minor NVH concerns.

NOTE: Before the engine or the transmission can be removed the engine bending braces must be removed.

(1) Raise and support vehicle.

NOTE: Both left and right side bending braces are removed the sameway. Only the right side is shown.

NOTE: The exhaust does not require removal to preform this procedure.

(2) Remove the exhaust hanger bracket retaining bolt.
(3) Remove locknut and transmission bending brace bar.
(4) Remove engine-to-bending brace retaining bolt, bending brace bar and cross bar.

INSTALLATION

NOTE: DO NOT tighten the retaining hardware until all bending braces are in place.

(1) Position the cross brace into the engine-to-transmission brace, then position the engine-to-transmission brace and install retaining bolt.
(2) Position the transmission bending brace onto through brace and install new locknut.
(3) Position exhaust hanger and transmission brace, install retaining bolt (Fig. 73).
(4) Tighten engine-to-transmission brace retaining bolt (Fig. 72)to 40 N·m (30 ft. lbs.).

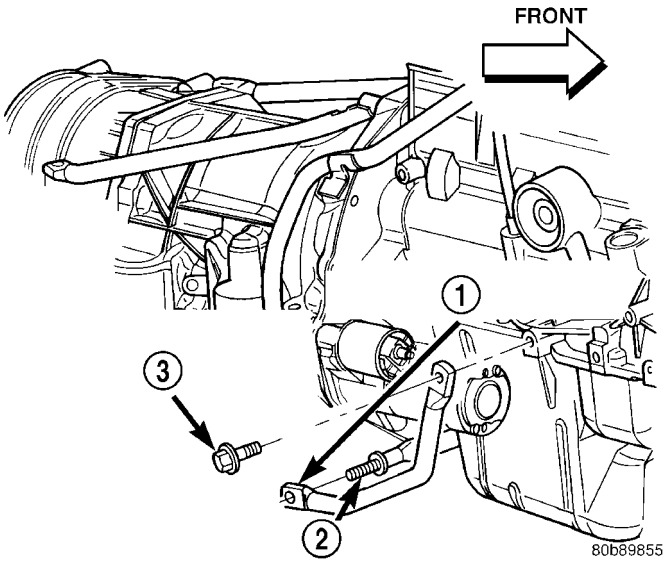

Fig. 72 Engine-to-Transmission Bending Braces

1 - ENGINE-TO-TRANSMISSION BENDING BRACE
2 - CROSS BRACE
3 - ENGINE-TO-TRANSMISSION BENDING BRACE RETAINING BOLT

STRUCTURAL SUPPORT (Continued)

(5) Tighten transmission brace retaining bolts (Fig. 73)to 40 N·m (30 ft. lbs.), then tighten transmission brace retaining lock nuts (Fig. 73) to 108 N·m (80 ft. lbs.).

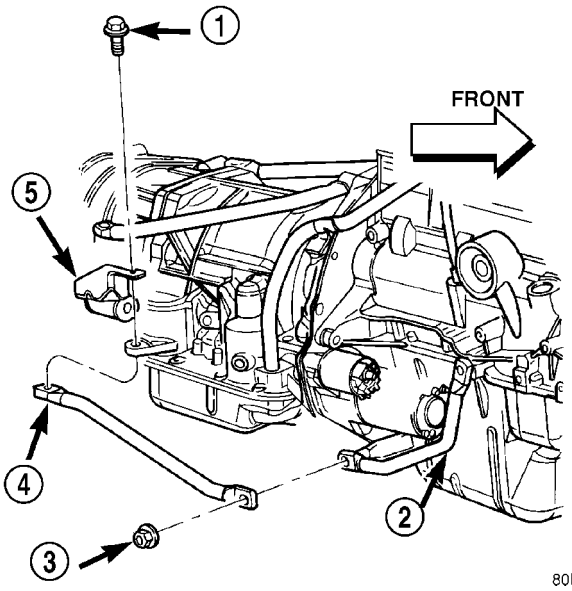

Fig. 73 Transmission Bending Braces and Exhaust Hanger

1 - TRANSMISSION BENDING BRACE RETAINING BOLT
2 - ENGINE-TO-TRANSMISSION BENDING BRACE
3 - LOCKNUT
4 - TRANSMISSION BRACE
5 - EXHAUST HANGER

FRONT MOUNT

REMOVAL

The front mounts support the engine at each side. These supports are made of resilient rubber.

(1) Disconnect negative cable from battery.
(2) Raise the vehicle.
(3) Support the engine.
(4) Remove the nut from the through bolt (Fig. 74) (Fig. 75). DO NOT remove the through bolt.
(5) Remove the retaining bolts and nuts from the insulator.
(6) Remove the through bolt.
(7) Remove the insulator.

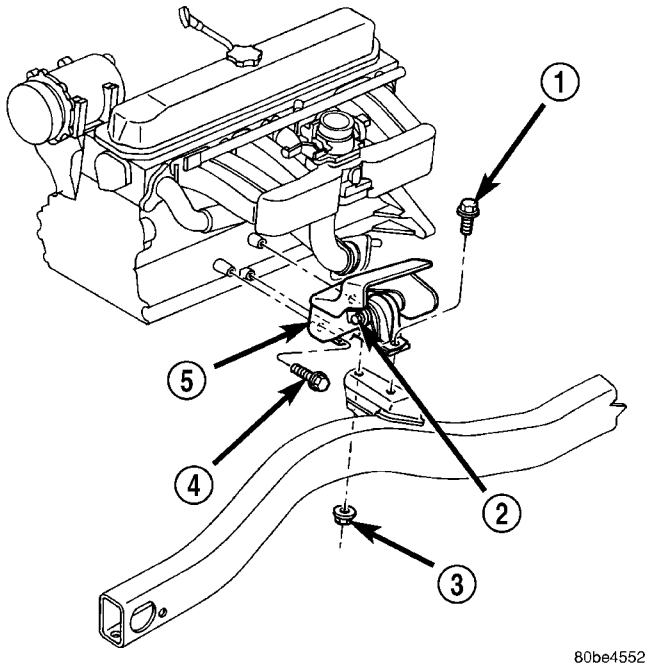

Fig. 74 Left Front Engine Mount

1 - **BOLT (2)**
2 - **THROUGH BOLT**
3 - **NUT (2)**
4 - **BOLT (3)**
5 - **LEFT HAND ENGINE MOUNT ASSEMBLY**

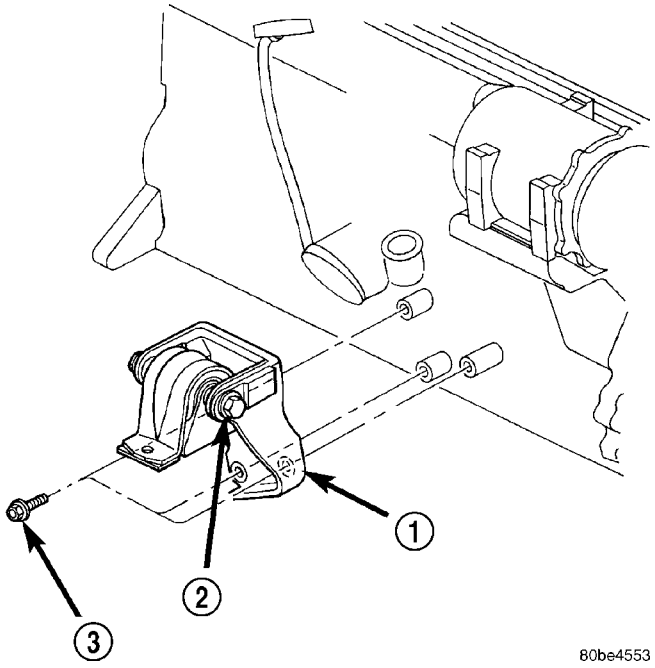

Fig. 75 Right Front Engine Mount

1 - RIGHT HAND ENGINE MOUNT ASSEMBLY
2 - THROUGH BOLT
3 - BOLT (3)

FRONT MOUNT (Continued)

INSTALLATION

The front mounts support the engine at each side. These supports are made of resilient rubber.

(1) If the engine support bracket was removed, position the bracket onto the block and install the attaching bolts (Fig. 74) (Fig. 75). Tighten the bolts to 50 N·m (37 ft. lbs.) torque.

(2) Place the insulator on the support bracket. Install the insulator retaining bolts and nuts. Tighten the bolts and nuts to 40 N·m (30 ft. lbs) torque.

(3) Install the through bolt and the retaining nut. Tighten the through bolt nut to 48 N·m (35 ft. lbs.) torque.

(4) Remove the engine support.

(5) Lower the vehicle.

(6) Connect negative cable to battery.

REAR MOUNT

REMOVAL

A resilient rubber cushion supports the transmission at the rear between the transmission extension housing and the rear support crossmember or skid plate.

ALL TRANSMISSIONS

(1) Disconnect negative cable from battery.

(2) Raise the vehicle and support the transmission.

(3) Remove the nuts holding the support cushion to the skid plate (Fig. 76) (Fig. 77).

MANUAL TRANSMISSIONS

(1) Remove nuts holding support cushion to transmission support bracket.

(2) Remove the support cushion.

(3) Remove bolts holding transmission support bracket to transmission.

(4) Remove the transmission support bracket.

AUTOMATIC TRANSMISSIONS

(1) Remove nuts holding support cushion to transmission support bracket (Fig. 77). Remove the support cushion.

(2) Remove the bolts holding the transmission support bracket to transmission.

(3) Remove the transmission support bracket.

INSTALLATION

MANUAL TRANSMISSION

(1) Position the transmission mount bracket to the transmission and install the bolts (Fig. 76).

(2) Tighten the bolts to 54 N·m (40 ft. lbs.) torque.

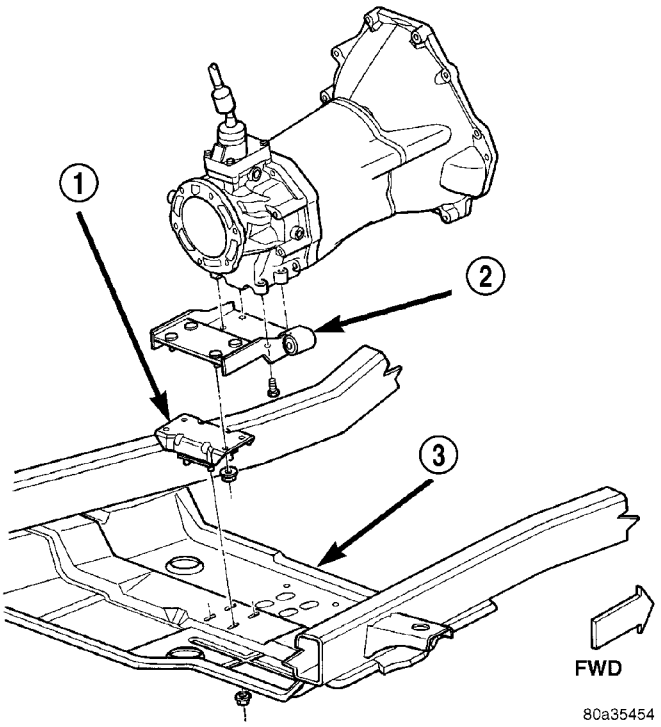

Fig. 76 Rear Mount (Manual Transmission)

1 - CUSHION
2 - BRACKET
3 - SKID PLATE

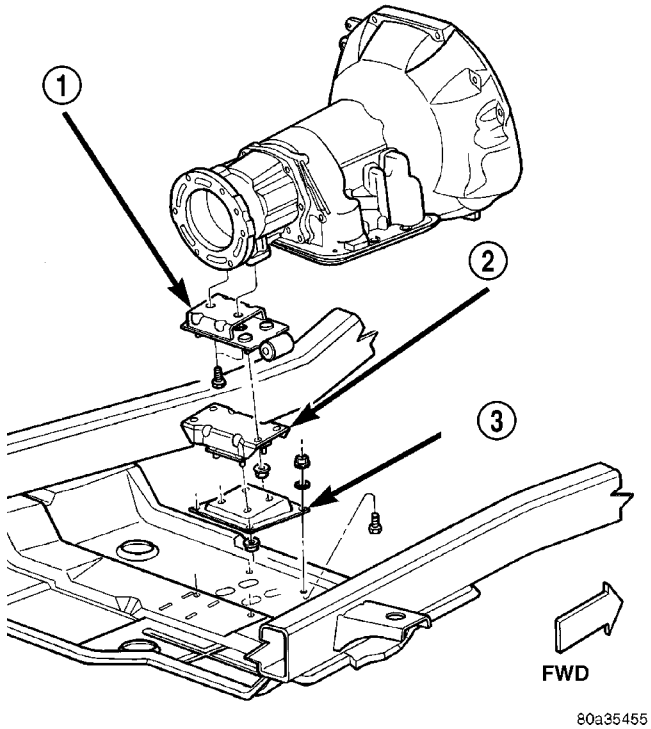

Fig. 77 Rear Mount (AutomaticTransmission)

1 - BRACKET
2 - CUSHION
3 - BRACKET

REAR MOUNT (Continued)

(3) Position the support cushion to the transmission mount bracket and install nuts (Fig. 76).

AUTOMATIC TRANSMISSION

(1) Position the transmission mount bracket to the transmission and install the bolts. Tighten the bolts to 54 N·m (40 ft. lbs.) torque.

(2) Position the support cushion to the transmission mount bracket and install nuts. Tighten the nuts to 41 N·m (30 ft. lbs.) torque (Fig. 77).

(3) If the support cushion bracket was removed from the skid plate, position the bracket on the skid plate and install the nuts and bolts. Tighten the nuts to 28 N·m (21 ft. lbs.) torque.

ALL TRANSMISSIONS

(1) Position the skid plate to the studs of the support cushion and install the nuts (Fig. 76) (Fig. 77). Tighten the nuts to 28 N·m (21 ft. lbs.) torque.

(2) Install the skid plate bolts to the sill and tighten to 75 N·m (55 ft. lbs.) torque.

(3) Remove the transmission support.

(4) Lower the vehicle.

(5) Connect negative cable to battery.

LUBRICATION

DESCRIPTION

A gear—type positive displacement pump is mounted at the underside of the block opposite the No. 4 main bearing.

OPERATION

The pump draws oil through the screen and inlet tube from the sump at the rear of the oil pan. The oil is driven between the drive and idler gears and pump body, then forced through the outlet to the block. An oil gallery in the block channels the oil to the inlet side of the full flow oil filter. After passing through the filter element, the oil passes from the center outlet of the filter through an oil gallery that channels the oil up to the main gallery which extends the entire length of the block.

Galleries extend downward from the main oil gallery to the upper shell of each main bearing. The crankshaft is drilled internally to pass oil from the main bearing journals (except number 4 main bearing journal) to the connecting rod journals. Each connecting rod bearing cap has a small squirt hole, oil passes through the squirt hole and is thrown off as the rod rotates. This oil throwoff lubricates the camshaft lobes, distributor drive gear, cylinder walls, and piston pins.

The hydraulic valve tappets receive oil directly from the main oil gallery. Oil is provided to the camshaft bearing through galleries. The front camshaft bearing journal passes oil through the camshaft sprocket to the timing chain. Oil drains back to the oil pan under the number one main bearing cap.

The oil supply for the rocker arms and bridged pivot assemblies is provided by the hydraulic valve tappets which pass oil through hollow push rods to a hole in the corresponding rocker arm. Oil from the rocker arm lubricates the valve train components, then passes down through the push rod guide holes in the cylinder head past the valve tappet area, and returns to the oil pan (Fig. 78).

LUBRICATION (Continued)

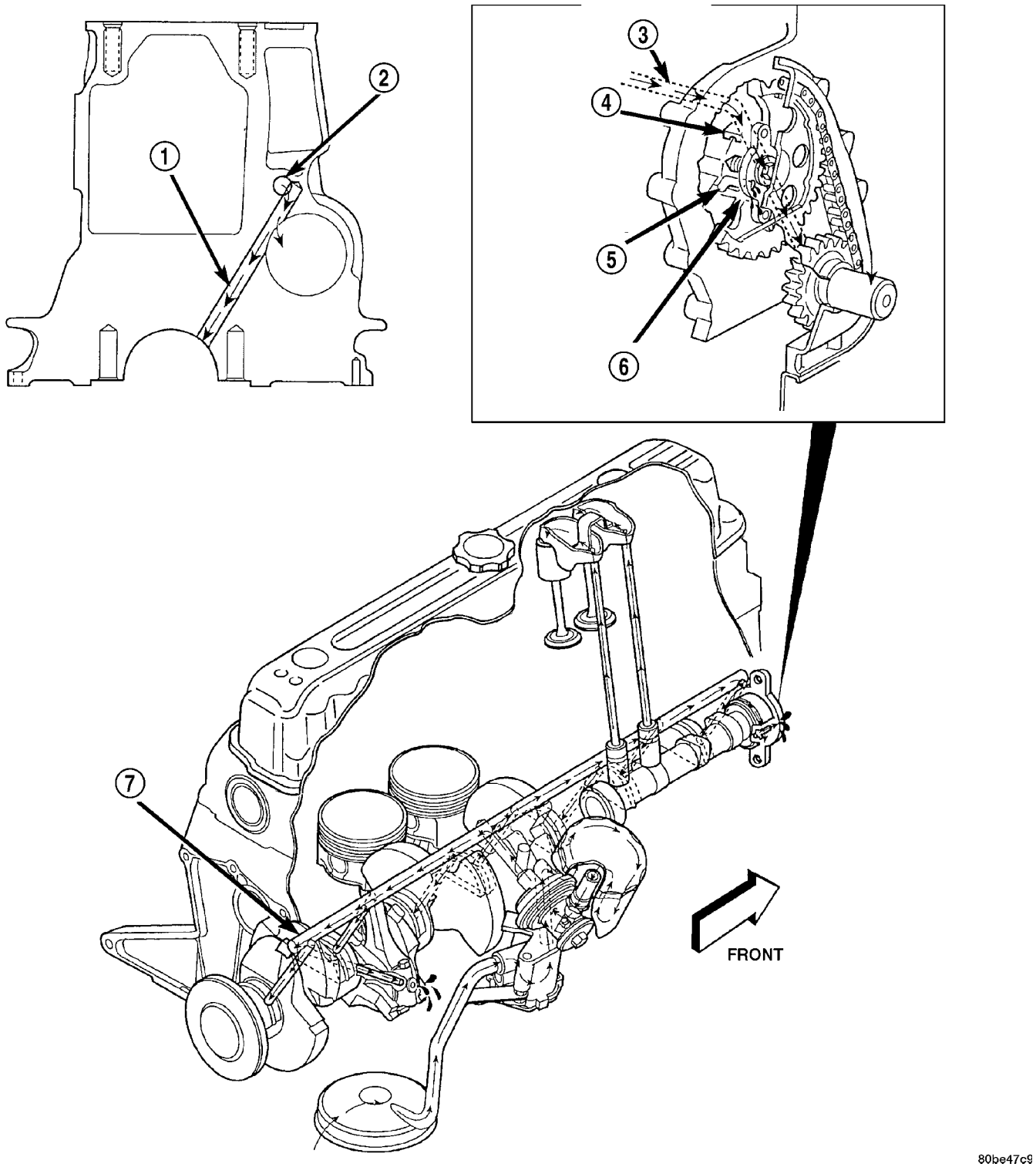

Fig. 78 Oil Lubrication System—4.0L Engine

1 - CAM/CRANK MAIN GALLERY (7)
2 - TAPPET GALLERY
3 - TAPPET GALLERY
4 - CAMSHAFT BEARING
5 - NUMBER 1 CAMSHFT BEARING JOURNAL
6 - CAMSHAFT SPROCKET
7 - TAPPET GALLERY

LUBRICATION (Continued)

DIAGNOSIS AND TESTING

DIAGNOSIS AND TESTING - ENGINE OIL PRESSURE

(1) Disconnect connector and remove oil pressure sending unit.

(2) Install Oil Pressure Line and Gauge Tool C-3292 or equivalent. Start engine and record pressure. (Refer to 9 - ENGINE - SPECIFICATIONS) for the correct pressures.

DIAGNOSIS AND TESTING - ENGINE OIL LEAK

Begin with a thorough visual inspection of the engine, particularly at the area of the suspected leak. If an oil leak source is not readily identifiable, the following steps should be followed:

(1) Do not clean or degrease the engine at this time because some solvents may cause rubber to swell, temporarily stopping the leak.

(2) Add an oil soluble dye (use as recommended by manufacturer). Start the engine and let idle for approximately 15 minutes. Check the oil dipstick to make sure the dye is thoroughly mixed as indicated with a bright yellow color under a black light.

(3) Using a black light, inspect the entire engine for fluorescent dye, particularly at the suspected area of oil leak. If the oil leak is found and identified, repair per service manual instructions.

(4) If dye is not observed, drive the vehicle at various speeds for approximately 24km (15 miles), and repeat inspection.**If the oil leak source is not positively identified at this time**, proceed with the air leak detection test method.

Air Leak Detection Test Method

(1) Disconnect the breather cap to air cleaner hose at the breather cap end. Cap or plug breather cap nipple.

(2) Remove the CCV valve from the cylinder head cover. Cap or plug the CCV valve grommet.

(3) Attach an air hose with pressure gauge and regulator to the dipstick tube.

CAUTION: Do not subject the engine assembly to more than 20.6 kpa (3 PSI) of test pressure.

(4) Gradually apply air pressure from 1 psi to 2.5 psi maximum while applying soapy water at the suspected source. Adjust the regulator to the suitable test pressure that provide the best bubbles which will pinpoint the leak source. If the oil leak is detected and identified, repair per service information procedures.

(5) If the leakage occurs at the rear oil seal area, INSPECTION FOR REAR SEAL AREA LEAKS.

(6) If no leaks are detected, turn off the air supply and remove the air hose and all plugs and caps. Install the CCV valve and breather cap hose.

(7) Clean the oil off the suspect oil leak area using a suitable solvent. Drive the vehicle at various speeds approximately 24 km (15 miles). Inspect the engine for signs of an oil leak by using a black light.

INSPECTION FOR REAR SEAL AREA LEAKS

Since it is sometimes difficult to determine the source of an oil leak in the rear seal area of the engine, a more involved inspection is necessary. The following steps should be followed to help pinpoint the source of the leak.

If the leakage occurs at the crankshaft rear oil seal area:

(1) Disconnect the battery.

(2) Raise the vehicle.

(3) Remove torque converter or clutch housing cover and inspect rear of block for evidence of oil. Use a black light to check for the oil leak:

(a) Circular spray pattern generally indicates seal leakage or crankshaft damage.

(b) Where leakage tends to run straight down, possible causes are a porous block, distributor seal, camshaft bore cup plugs oil galley pipe plugs, oil filter runoff, and main bearing cap to cylinder block mating surfaces.

(4) If no leaks are detected, pressurize the crankcase as outlined in the, Inspection (Engine oil Leaks in general)

CAUTION: Do not exceed 20.6 kPa (3 psi).

(5) If the leak is not detected, very slowly turn the crankshaft and watch for leakage. If a leak is detected between the crankshaft and seal while slowly turning the crankshaft, it is possible the crankshaft seal surface is damaged. The seal area on the crankshaft could have minor nicks or scratches that can be polished out with emery cloth.

CAUTION: Use extreme caution when crankshaft polishing is necessary to remove minor nicks and scratches. The crankshaft seal flange is especially machined to complement the function of the rear oil seal.

(6) For bubbles that remain steady with shaft rotation, no further inspection can be done until disassembled.

OIL

STANDARD PROCEDURE - ENGINE OIL SERVICE

ENGINE OIL CHANGE

WARNING: NEW OR USED ENGINE OIL CAN BE IRRITATING TO THE SKIN. AVOID PROLONGED OR REPEATED SKIN CONTACT WITH ENGINE OIL. CONTAMINANTS IN USED ENGINE OIL, CAUSED BY INTERNAL COMBUSTION, CAN BE HAZARDOUS TO YOUR HEALTH. THOROUGHLY WASH EXPOSED SKIN WITH SOAP AND WATER. DO NOT WASH SKIN WITH GASOLINE, DIESEL FUEL, THINNER, OR SOLVENTS, HEALTH PROBLEMS CAN RESULT. DO NOT POLLUTE, DISPOSE OF USED ENGINE OIL PROPERLY. CONTACT YOUR DEALER OR GOVERNMENT AGENCY FOR LOCATION OF COLLECTION CENTER IN YOUR AREA.

CAUTION: Do not use oil filter with metric threads. The proper oil filter has SAE type 3/4 X 16 threads. An oil filter with metric threads can result in oil leaks and engine failure.

All Jeep engines are equipped with a high quality full-flow, throw-away type oil filter. DaimlerChrysler Corporation recommends a Mopar® or equivalent oil filter be used.

Change engine oil at mileage and time intervals described in Maintenance Schedules.

Run engine until achieving normal operating temperature.

(1) Position the vehicle on a level surface and turn engine off.

(2) Hoist and support vehicle on safety stands.

(3) Remove oil fill cap.

(4) Place a suitable drain pan under crankcase drain.

(5) Remove drain plug from crankcase and allow oil to drain into pan. Inspect drain plug threads for stretching or other damage. Replace drain plug if damaged.

(6) Install drain plug in crankcase.

(7) Position a drain pan under the oil filter.

(8) Using a suitable oil filter wrench loosen filter.

(9) Rotate the oil filter counterclockwise to remove it from the cylinder block oil filter boss or filter adapter housing.

(10) When filter separates from adapter nipple, tip gasket end upward to minimize oil spill. Remove filter from vehicle.

(11) Make sure old gasket comes off with oil filter. With a wiping cloth, clean the gasket sealing surface (Fig. 79) of oil and grime.

(12) Lightly lubricate oil filter gasket with engine oil or chassis grease.

(13) Thread filter onto adapter nipple. When gasket makes contact with sealing surface, (Fig. 79)hand tighten filter one full turn, do not over tighten.

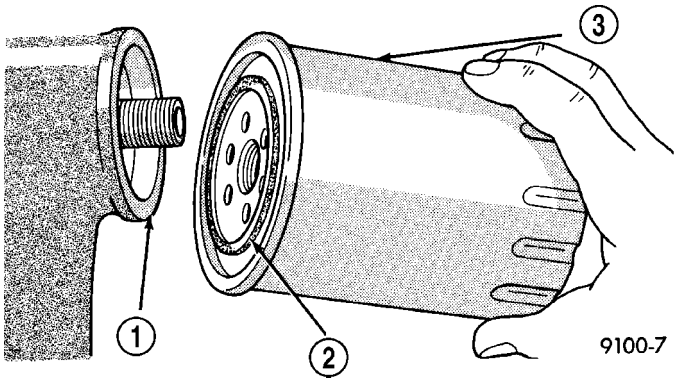

Fig. 79 Oil Filter Sealing Surface—Typical

1 - SEALING SURFACE
2 - RUBBER GASKET
3 - OIL FILTER

(14) Lower vehicle and fill crankcase with specified type and amount of engine oil described in this section.

(15) Install oil fill cap.

(16) Start engine and inspect for leaks.

(17) Stop engine and inspect oil level.

CRANKCASE OIL LEVEL INSPECTION

CAUTION: Do not overfill crankcase with engine oil, oil foaming and oil pressure loss can result.

The engine oil level indicator (Dipstick) is located at the right rear of the 4.0L engine. Inspect engine oil level approximately every 800 kilometers (500 miles). Unless the engine has exhibited loss of oil pressure, run the engine for about five minutes before checking oil level. Checking engine oil level on a cold engine is not accurate.

To ensure proper lubrication of an engine, the engine oil must be maintained at an acceptable level. The acceptable levels are indicated between the ADD and SAFE marks on the engine oil dipstick.

(1) Position vehicle on level surface.

(2) With engine OFF, allow approximately ten minutes for oil to settle to bottom of crankcase, remove engine oil dipstick.

(3) Wipe dipstick clean.

(4) Install dipstick and verify it is seated in the tube.

OIL (Continued)

(5) Remove dipstick, with handle held above the tip, take oil level reading.

(6) Add oil only if level is below the ADD mark on dipstick.

USED ENGINE OIL DISPOSAL

Care should be exercised when disposing used engine oil after it has been drained from a vehicle engine. Refer to the WARNING at beginning of this procedure.

OIL FILTER

REMOVAL

CAUTION: Do not use oil filter with metric threads. The proper oil filter has SAE type 3/4 X 16 threads. An oil filter with metric threads can result in oil leaks and engine failure.

All Jeep engines are equipped with a high quality full-flow, throw-away type oil filter. DaimlerChrysler Corporation recommends a Mopar® or equivalent oil filter be used.

(1) Position a drain pan under the oil filter.

(2) Using a suitable oil filter wrench loosen filter.

(3) Rotate the oil filter counterclockwise to remove it from the cylinder block oil filter boss or filter adapter housing.

(4) When filter separates from adapter nipple, tip gasket end upward to minimize oil spill. Remove filter from vehicle.

(5) Make sure old gasket comes off with oil filter. With a wiping cloth, clean the gasket sealing surface of oil and grime.

INSTALLATION

(1) Lightly lubricate oil filter gasket with engine oil or chassis grease.

(2) Thread filter onto adapter nipple. When gasket makes contact with sealing surface, (Fig. 80) hand tighten filter one full turn, do not over tighten.

(3) Add oil, verify crankcase oil level and start engine. Inspect for oil leaks.

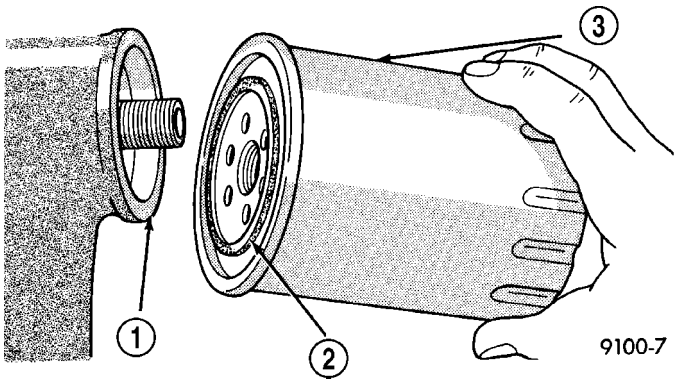

Fig. 80 Oil Filter Sealing Surface—Typical

1 - SEALING SURFACE
2 - RUBBER GASKET
3 - OIL FILTER

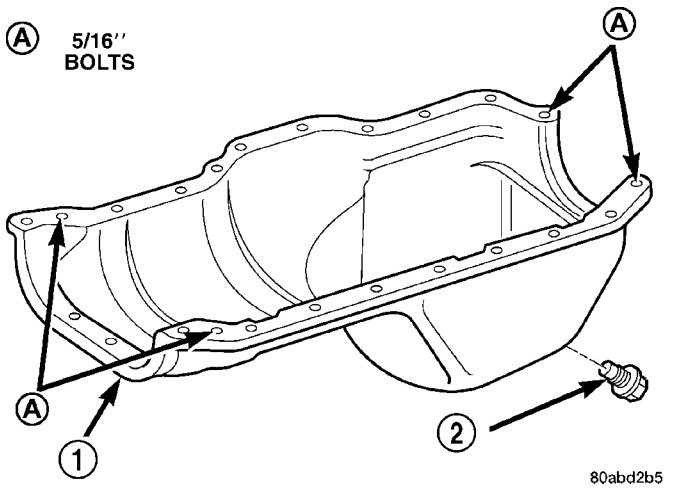

Fig. 81 Oil Pan

1 - OIL PAN
2 - OIL PAN DRAIN PLUG

OIL PAN

DESCRIPTION

The oil pan is made of stamped steel. The oil pan gasket is a one piece steel backbone silicone coated gasket (Fig. 81).

REMOVAL

(1) Disconnect negative cable from battery.

(2) Raise the vehicle.

(3) Remove the oil pan drain plug and drain the engine oil.

(4) Disconnect the exhaust pipe at the exhaust manifold.

(5) Disconnect the exhaust hanger at the catalytic converter and lower the pipe.

(6) Remove the starter motor. (Refer to 8 - ELECTRICAL/STARTING/STARTER MOTOR - REMOVAL).

(7) Remove the engine flywheel and transmission torque converter housing access cover.

(8) If equipped with an oil level sensor, disconnect the sensor.

(9) Position a jack stand directly under the engine vibration damper.

(10) Place a piece of wood (2 x 2) between the jack stand and the engine vibration damper.

(11) Remove the engine mount through bolts.

(12) Using the jack stand, raise the engine until adequate clearance is obtained to remove the oil pan.

OIL PAN (Continued)

(13) Remove transmission oil cooling lines (if equipped) and oxygen sensor wiring supports that are attached to the oil pan studs.

(14) Remove the oil pan bolts and studs. Carefully slide the oil pan and gasket to the rear. If equipped with an oil level sensor, take care not to damage the sensor.

INSTALLATION

(1) Clean the block and pan gasket surfaces.

(2) Fabricate 4 alignment dowels from 1 1/2 x 1/4 inch bolts. Cut the head off the bolts and cut a slot into the top of the dowel. This will allow easier installation and removal with a screwdriver (Fig. 82).

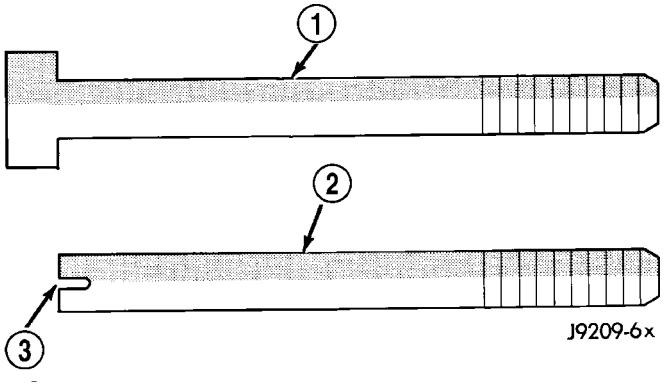

Fig. 82 Fabrication of Alignment Dowels

1 - 1/4'' × 1 1/2'' BOLT
2 - DOWEL
3 - SLOT

(3) Install two dowels in the timing case cover. Install the other two dowels in the cylinder block (Fig. 83).

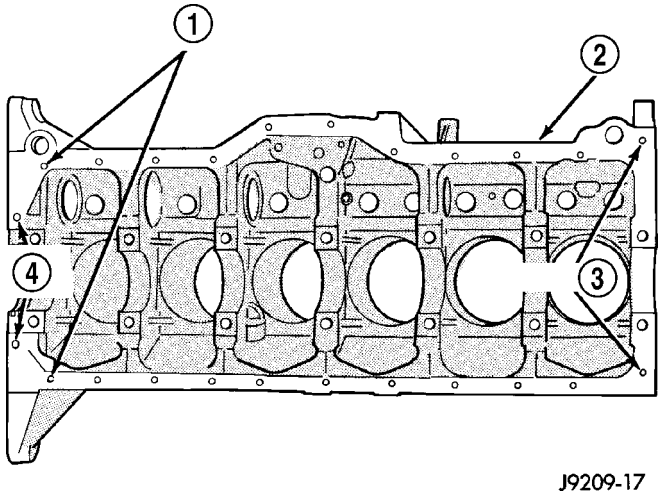

Fig. 83 Position of Dowels in Cylinder Block

1 - DOWEL HOLES
2 - CYLINDER BLOCK
3 - 5/16'' HOLES
4 - 5/16'' HOLES

(4) Apply Mopar® Silicone Rubber Adhesive Sealant on cylinder block to rear main bearing cap corners and cylinder block to front cover joints (four places) (Fig. 84).

(5) Slide the one-piece gasket over the dowels and onto the block and timing case cover.

(6) Position the oil pan over the dowels and onto the gasket. If equipped with an oil level sensor, take care not to damage the sensor.

(7) Install the 1/4 inch oil pan bolts. Tighten these bolts to 9.5 N·m (84 in. lbs.) torque. Install the 5/16 inch oil pan bolts (Fig. 85). Tighten these bolts to 15 N·m (132 in. lbs.) torque.

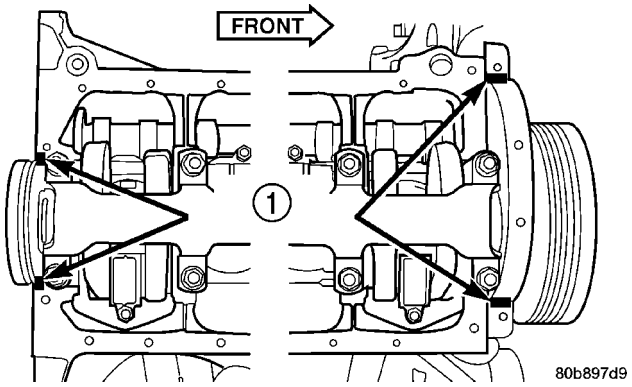

Fig. 84 Oil Pan Sealer Location

1 - SEALER LOCATIONS

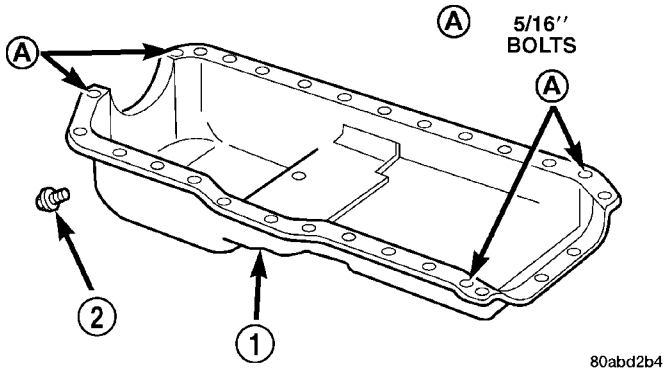

Fig. 85 Position of 5/16 inch Oil Pan Bolts

1 - OIL PAN
2 - OIL PAN DRAIN PLUG

(8) Remove the dowels. Install the remaining 1/4 inch oil pan bolts. Tighten these bolts to 9.5 N·m (84 in. lbs.) torque.

(9) Lower the engine until it is properly located on the engine mounts.

(10) Install the through bolts and tighten the nuts.

(11) Lower the jack stand and remove the piece of wood.

(12) Install the engine flywheel and transmission torque converter housing access cover.

OIL PAN (Continued)

(13) Install the engine starter motor. (Refer to 8 - ELECTRICAL/STARTING/STARTER MOTOR - INSTALLATION).

(14) Connect the exhaust pipe to the hanger and to the engine exhaust manifold.

(15) Install transmission oil cooling lines (if equipped) and oxygen sensor wiring supports that attach to the oil pan studs.

(16) Install the oil pan drain plug (Fig. 85). Tighten the plug to 34 N·m (25 ft. lbs.) torque.

(17) Lower the vehicle.

(18) Connect negative cable to battery.

(19) Fill the oil pan with engine oil to the specified level.

WARNING: USE EXTREME CAUTION WHEN THE ENGINE IS OPERATING. DO NOT STAND IN A DIRECT LINE WITH THE FAN. DO NOT PUT YOUR HANDS NEAR THE PULLEYS, BELTS OR FAN. DO NOT WEAR LOOSE CLOTHING.

(20) Start the engine and inspect for leaks.

ENGINE OIL PRESSURE SENSOR

DESCRIPTION

The 3 wire, solid-state engine oil pressure sensor (sending unit) is located in an engine oil pressure gallery.

OPERATION

The oil pressure sensor uses three circuits. They are:

- A 5 volt power supply from the Powertrain Control Module (PCM)
- A sensor ground through the PCM's sensor return
- A signal to the PCM relating to engine oil pressure

The oil pressure sensor has a 3 wire electrical function very much like the Manifold Absolute Pressure (MAP) sensor. Meaning different pressures relate to different output voltages.

A 5 volt supply is sent to the sensor from the PCM to power up the sensor. The sensor returns a voltage signal back to the PCM relating to engine oil pressure. This signal is then transferred (bussed) to the instrument panel on either a CCD or PCI bus circuit (depending on vehicle line) to operate the oil pressure gauge and the check gauges lamp. Ground for the sensor is provided by the PCM through a low-noise sensor return.

OIL PUMP

REMOVAL

A gear-type oil pump is mounted at the underside of the cylinder block opposite the No.4 main bearing.

(1) Drain the engine oil.

(2) Remove the oil pan (Refer to 9 - ENGINE/LUBRICATION/OIL PAN - REMOVAL).

(3) Remove the pump-to-cylinder block attaching bolts. Remove the pump assembly with gasket (Fig. 86).

CAUTION: If the oil pump is not to be serviced, DO NOT disturb position of oil inlet tube and strainer assembly in pump body. If the tube is moved within the pump body, a replacement tube and strainer assembly must be installed to assure an airtight seal.

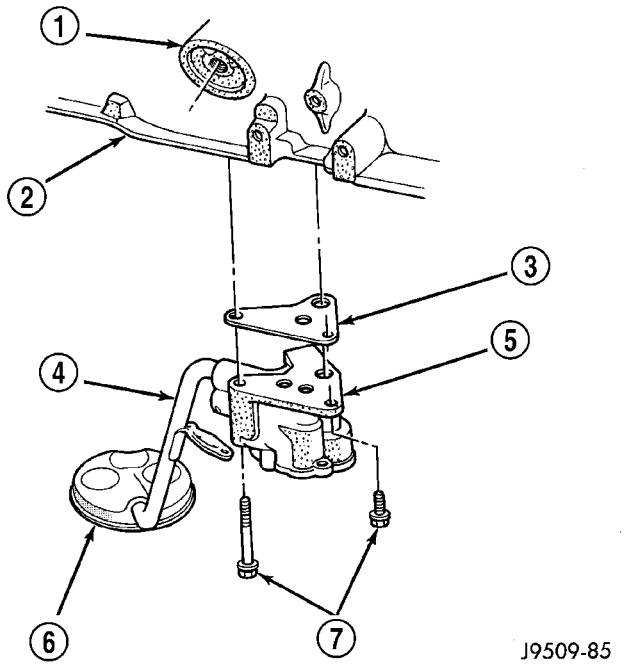

Fig. 86 Oil Pump Assembly

1 - OIL FILTER ADAPTOR
2 - BLOCK
3 - GASKET
4 - OIL INLET TUBE
5 - OIL PUMP
6 - STRAINER ASSEMBLY
7 - ATTACHING BOLTS

INSTALLATION

A gear-type oil pump is mounted at the underside of the cylinder block opposite the No.4 main bearing.

(1) Install the oil pump on the cylinder block using a replacement gasket. Tighten the bolts to 23 N·m (17 ft. lbs.) torque.

(2) Install the oil pan (Refer to 9 - ENGINE/LUBRICATION/OIL PAN - INSTALLATION).

OIL PUMP (Continued)

(3) Fill the oil pan with oil to the specified level.

VALVE TIMING

STANDARD PROCEDURE - VALVE TIMING

(1) Disconnect the spark plug wires and remove the spark plugs.

(2) Remove the engine cylinder head cover .

(3) Remove the capscrews, bridge and pivot assembly, and rocker arms from above the No.1 cylinder.

(4) Alternately loosen each capscrew, one turn at a time, to avoid damaging the bridge.

(5) Rotate the crankshaft until the No.6 piston is at top dead center (TDC) on the compression stroke.

(6) Rotate the crankshaft counterclockwise (viewed from the front of the engine) 90°.

(7) Install a dial indicator on the end of the No.1 cylinder intake valve push rod. Use rubber tubing to secure the indicator stem on the push rod.

(8) Set the dial indicator pointer at zero.

(9) Rotate the crankshaft clockwise (viewed from the front of the engine) until the dial indicator pointer indicates 0.305 mm (0.012 inch) travel distance (lift).

(10) The timing notch index on the vibration damper should be aligned with the TDC mark on the timing degree scale.

(11) If the timing notch is more than 13 mm (1/2 inch) away from the TDC mark in either direction, the valve timing is incorrect.

NOTE: If the valve timing is incorrect, the cause may be a broken camshaft pin. It is not necessary to replace the camshaft because of pin failure. A spring pin is available for service replacement.

TIMING BELT / CHAIN COVER(S)

REMOVAL

(1) Disconnect negative cable from battery.

(2) Remove the vibration damper (Refer to 9 - ENGINE/ENGINE BLOCK/VIBRATION DAMPER - REMOVAL).

(3) Remove the fan, hub assembly and fan shroud (Refer to 7 - COOLING/ENGINE/RADIATOR FAN - REMOVAL).

(4) Remove the accessory drive brackets that are attached to the timing case cover.

(5) Remove the A/C compressor (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/A/C COMPRESSOR - REMOVAL) (if equipped) and generator bracket assembly from the engine cylinder head and move to one side.

(6) Remove the oil pan-to-timing case cover bolts and timing case cover-to-cylinder block bolts.

(7) Remove the timing case cover and gasket from the engine.

(8) Pry the crankshaft oil seal from the front of the timing case cover (Fig. 87).

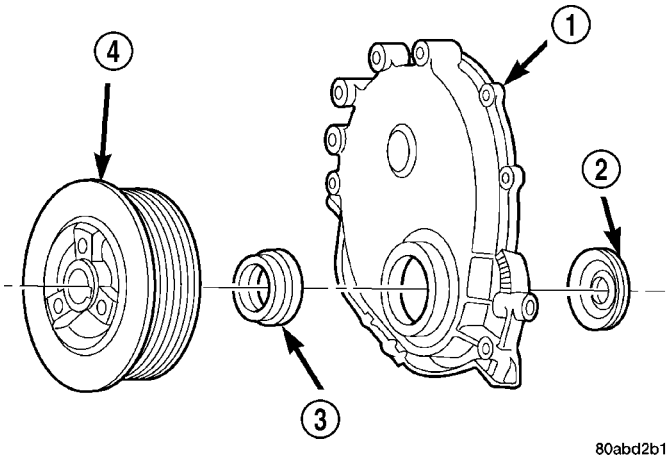

Fig. 87 Timing Case Cover Components

1 - TIMING CASE COVER
2 - OIL SLINGER
3 - CRANKSHAFT OIL SEAL
4 - VIBRATION DAMPER PULLEY

INSTALLATION

Clean the timing case cover, oil pan and cylinder block gasket surfaces.

(1) Install a new crankshaft oil seal in the timing case cover. The open end of the seal should be toward the inside of the cover. Support the cover at the seal area while installing the seal. Force it into position with Seal Installation Tool 6139.

(2) Position the gasket on the cylinder block.

(3) Position the timing case cover on the oil pan gasket and the cylinder block.

(4) Insert Timing Case Cover Alignment and Seal Installation Tool 6139 in the crankshaft opening in the cover (Fig. 88).

(5) Install the timing case cover-to-cylinder block and the oil pan-to-timing case cover bolts.

(6) Tighten the 1/4 inch cover-to-block bolts to 7 N·m (60 in. lbs.) torque. Tighten the 5/16 inch front cover-to-block bolts to 22 N·m (192 in. lbs.) torque. Tighten the oil pan-to-cover 1/4 inch bolts to 9.5 N·m (84 in. lbs.) torque.

(7) Remove the cover alignment tool.

(8) Apply a light film of engine oil on the vibration damper hub contact surface of the seal.

(9) Apply Mopar® Silicone Rubber Adhesive Sealant to the keyway in the crankshaft and insert the key. With the key inserted in the keyway in the

TIMING BELT / CHAIN COVER(S) (Continued)

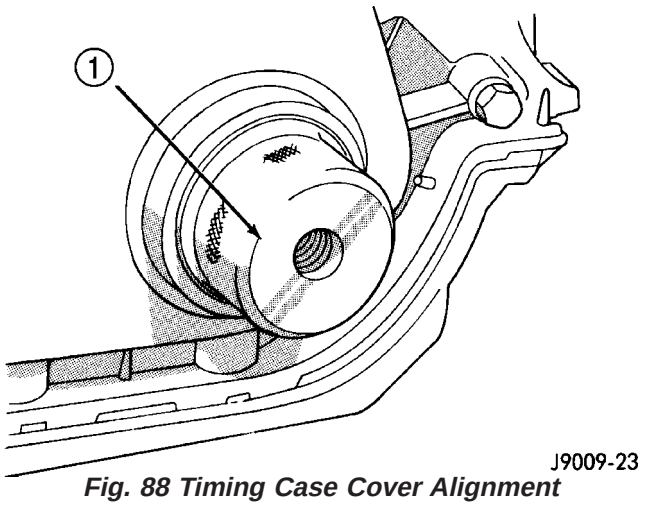

Fig. 88 Timing Case Cover Alignment

1 - TIMING CASE COVER ALIGNMENT AND SEAL INSTALLATION TOOL

crankshaft, install the vibration damper (Refer to 9 - ENGINE/ENGINE BLOCK/VIBRATION DAMPER - INSTALLATION).

(10) Install the A/C compressor (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/A/C COMPRESSOR - INSTALLATION) (if equipped) and generator bracket assembly.

(11) Install the engine fan, hub assembly and shroud (Refer to 7 - COOLING/ENGINE/RADIATOR FAN - INSTALLATION).

(12) Install the serpentine drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - INSTALLATION).

(13) Connect negative cable to battery.

TIMING BELT/CHAIN AND SPROCKETS

REMOVAL

(1) Disconnect negative cable from battery.

(2) Remove the fan and shroud (Refer to 7 - COOLING/ENGINE/RADIATOR FAN - REMOVAL).

(3) Remove the serpentine drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - REMOVAL).

(4) Remove the crankshaft vibration damper (Refer to 9 - ENGINE/ENGINE BLOCK/VIBRATION DAMPER - REMOVAL).

(5) Remove the timing case cover (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT / CHAIN COVER(S) - REMOVAL).

(6) Rotate crankshaft until the “0” timing mark is closest to and on the center line with camshaft sprocket timing mark (Fig. 89).

(7) Remove the oil slinger from the crankshaft.

(8) Remove the camshaft sprocket bolt and washer (Fig. 90).

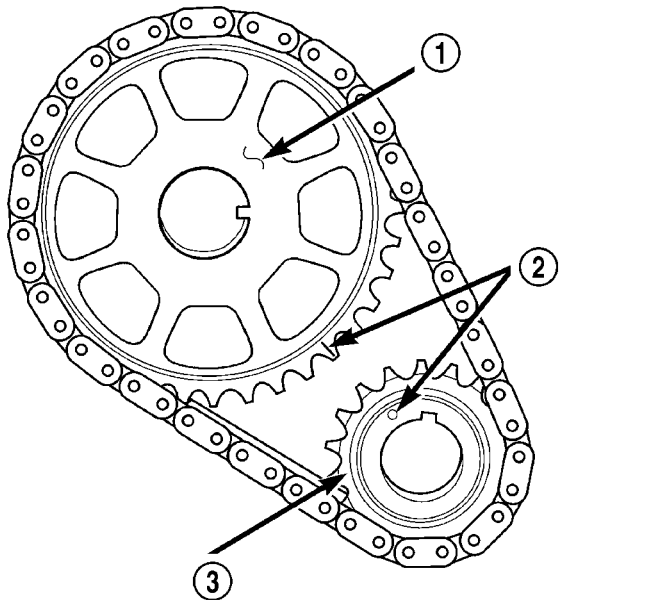

Fig. 89 Crankshaft—Camshaft Alignment

1 - CAMSHAFT SPROCKET
2 - TIMING MARKS
3 - CRANKSHAFT SPROCKET

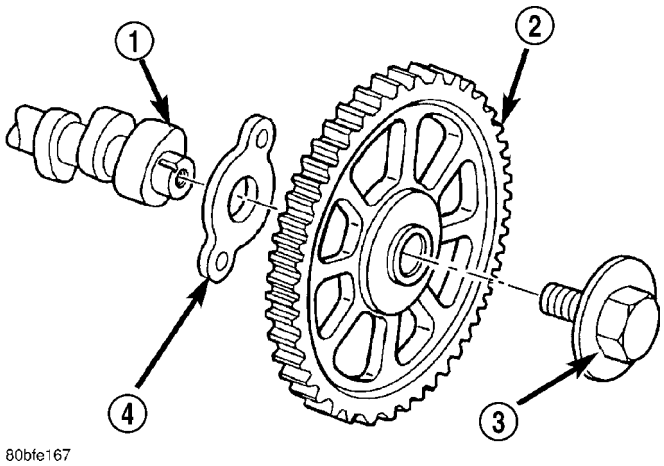

Fig. 90 Camshaft Sprocket and Thrust Plate

1 - CAMSHAFT
2 - CAMSHAFT SPROCKET W/INTEGRAL KEY
3 - BOLT AND WASHER
4 - THRUST PLATE

(9) Remove the crankshaft sprocket, camshaft sprocket and timing chain as an assembly.

(10) Installation of the timing chain with the timing marks on the crankshaft and camshaft sprockets properly aligned ensures correct valve timing. A worn or stretched timing chain will adversely affect valve timing. If the timing chain deflects more than 12.7 mm (1/2 inch) replace it.

INSTALLATION

Assemble the timing chain, crankshaft sprocket and camshaft sprocket with the timing marks aligned (Fig. 89).

(1) Apply Mopar® Silicone Rubber Adhesive Sealant to the keyway in the crankshaft and insert the key. With the key in the keyway on the crankshaft, install the assembly on the crankshaft and camshaft.

(2) Install the camshaft sprocket bolt and washer (Fig. 90). Tighten the bolt to 68 N·m (50 ft. lbs.) torque.

(3) To verify correct installation of the timing chain, rotate the crankshaft 2 revolutions. The camshaft and crankshaft sprocket timing mark should align (Fig. 89).

(4) Install the crankshaft oil slinger.

(5) Replace the oil seal in the timing case cover (Refer to 9 - ENGINE/ENGINE BLOCK/CRANKSHAFT OIL SEAL - FRONT - REMOVAL).

(6) Install the timing case cover and gasket (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT / CHAIN COVER(S) - INSTALLATION).

(7) With the key installed in the crankshaft keyway, install the vibration damper (Refer to 9 - ENGINE/ENGINE BLOCK/VIBRATION DAMPER - INSTALLATION).

(8) Install the serpentine drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - INSTALLATION).

(9) Install the fan, hub assembly and shroud (Refer to 7 - COOLING/ENGINE/RADIATOR FAN - INSTALLATION).

(10) Connect negative cable to battery.

INTAKE MANIFOLD

DESCRIPTION

The intake manifold (Fig. 91) is made of cast aluminum and uses eleven bolts to mount to the cylinder head. This mounting style improves sealing and reduces the chance of leaks.

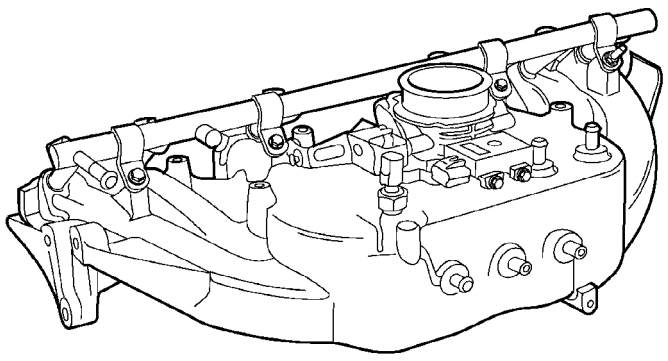

Fig. 91 Intake Manifold 4.0L Engine

DIAGNOSIS AND TESTING - INTAKE MANIFOLD LEAKAGE

An intake manifold air leak is characterized by lower than normal manifold vacuum. Also, one or more cylinders may not be functioning.

WARNING: USE EXTREME CAUTION WHEN THE ENGINE IS OPERATING. DO NOT STAND IN A DIRECT LINE WITH THE FAN. DO NOT PUT YOUR HANDS NEAR THE PULLEYS, BELTS OR THE FAN. DO NOT WEAR LOOSE CLOTHING.

(1) Start the engine.

(2) Spray a small stream of water at the suspected leak area.

(3) If a change in RPM is observed the area of the suspected leak has been found.

(4) Repair as required.

REMOVAL

NOTE: THE ENGINE INTAKE AND EXHAUST MANIFOLD MUST BE REMOVED AND INSTALLED TOGETHER. THE MANIFOLDS USE A COMMON GASKET AT THE CYLINDER HEAD.

(1) Disconnect the battery negative cable.

(2) Remove air cleaner inlet hose from the resonator assembly.

(3) Remove the air cleaner assembly.

(4) Remove the throttle cable, vehicle speed control cable (if equipped) and the transmission line pressure cable (Refer to 21 - TRANSMISSION/TRANSAXLE/AUTOMATIC - AW4/THROTTLE VALVE CABLE - REMOVAL).

(5) Disconnect the following electrical connections and secure their harness out of the way:

- Throttle Position Sensor
- Idle Air Control Motor
- Coolant Temperature Sensor (at thermostat housing)
- Intake Air Temperature Sensor
- Oxygen Sensor
- Crank Position Sensor
- Six (6) Fuel Injector Connectors
- Manifold Absolute Pressure (MAP) Sensor.

(6) Disconnect HVAC, and Brake Booster vacuum supply hoses at the intake manifold.

(7) Perform the fuel pressure release procedure. (Refer to 14 - FUEL SYSTEM/FUEL DELIVERY - STANDARD PROCEDURE).

(8) Disconnect and remove the fuel system supply line from the fuel rail assembly.

(9) Remove the accessory drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - REMOVAL).

INTAKE MANIFOLD (Continued)

(10) Remove the power steering pump from the intake manifold and set aside.
(11) Raise the vehicle.
(12) Disconnect the exhaust pipes from the engine exhaust manifolds.
(13) Lower the vehicle.
(14) Remove the intake manifold and exhaust manifold bolts and manifolds (Fig. 92).

INSTALLATION

If the manifold is being replaced, ensure all the fitting, etc. are transferred to the replacement manifold.
(1) Install a new engine exhaust/intake manifold gasket over the alignment dowels on the cylinder head.
(2) Position the engine exhaust manifolds to the cylinder head. Install fastener Number 3 and finger tighten at this time (Fig. 92).
(3) Install intake manifold on the cylinder head dowels.
(4) Install washer and fastener Numbers 1, 2, 4, 5, 8, 9, 10 and 11 (Fig. 92).
(5) Install washer and fastener Numbers 6 and 7 (Fig. 92).
(6) Tighten the fasteners in sequence and to the specified torque (Fig. 92).
- Fastener Numbers 1 through 5—Tighten to 33 N·m (24 ft. lbs.) torque.
- Fastener Numbers 6 and 7—Tighten to 31 N·m (23 ft. lbs.) torque.
- Fastener Numbers 8 through 11—Tighten to 33 N·m (24 ft. lbs.) torque.

(7) Install the power steering pump to the intake manifold.
(8) Install the accessory drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - INSTALLATION).
(9) Install the fuel system supply line to the fuel rail assembly.

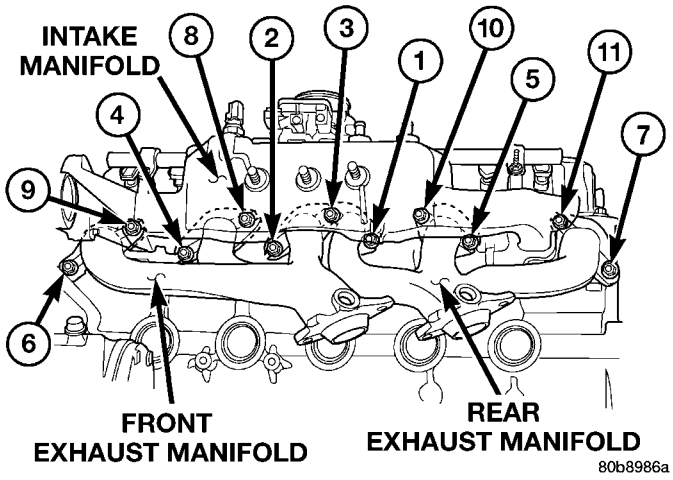

Fig. 92 Intake and Exhaust Manifolds Installation

(10) Connect all electrical connections on the intake manifold.
(11) Connect the vacuum hoses previously removed.
(12) Install throttle cable, vehicle speed control cable (if equipped).
(13) Install the transmission line pressure cable (if equipped) (Refer to 21 - TRANSMISSION/TRANSAXLE/AUTOMATIC - AW4/THROTTLE VALVE CABLE - INSTALLATION).
(14) Install air cleaner assembly.
(15) Connect air inlet hose to the resonator assembly.
(16) Raise the vehicle.
(17) Connect the exhaust pipes to the engine exhaust manifolds. Tighten the bolts to 31 N·m (23 ft. lbs.)
(18) Lower the vehicle.
(19) Connect the battery negative cable.
(20) Start the engine and check for leaks.

EXHAUST MANIFOLD

DESCRIPTION

The two exhaust manifolds (Fig. 93) are log style and are made of high silicon molybdenum cast iron. The exhaust manifolds share a common gasket with the intake manifold. The exhaust manifolds also incorporate ball flange outlets for improved sealing and strain free connections.

REMOVAL

The intake and engine exhaust manifolds on the 4.0L engine must be removed together. The manifolds use a common gasket at the cylinder head.

(Refer to 9 - ENGINE/MANIFOLDS/INTAKE MANIFOLD - REMOVAL).

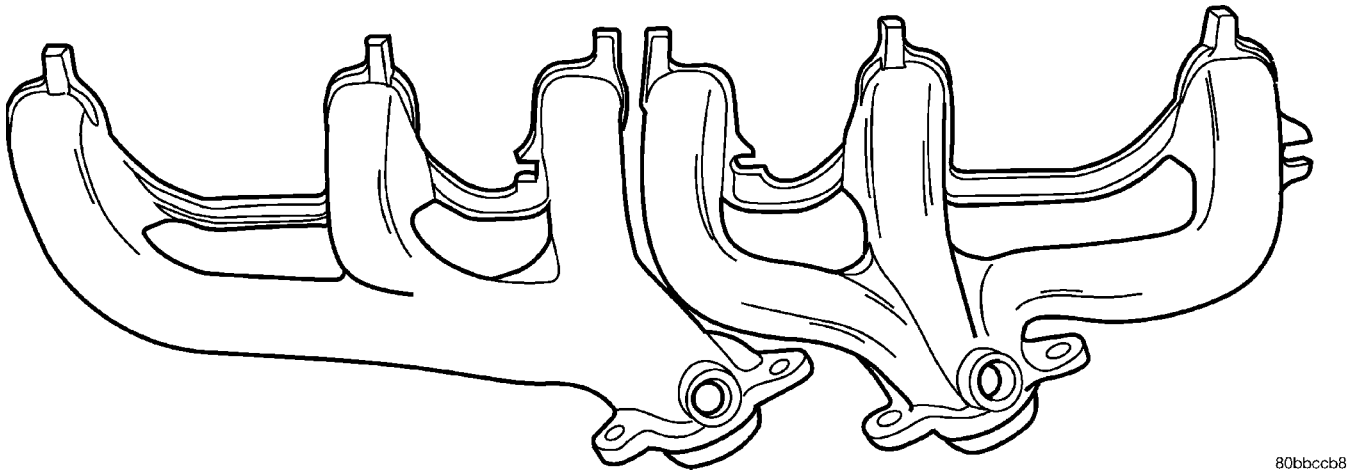

Fig. 93 EXHAUST MANIFOLDS 4.0L ENGINE

EXHAUST SYSTEM

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EXHAUST SYSTEM

DESCRIPTION

WARNING: THE NORMAL OPERATING TEMPERATURE OF THE EXHAUST SYSTEM IS VERY HIGH. THEREFORE, NEVER WORK AROUND OR ATTEMPT TO SERVICE ANY PART OF THE EXHAUST SYSTEM UNTIL IT IS COOLED. SPECIAL CARE SHOULD BE TAKEN WHEN WORKING NEAR THE CATALYTIC CONVERTER. THE TEMPERATURE OF THE CONVERTER RISES TO A HIGH LEVEL AFTER A SHORT PERIOD OF ENGINE OPERATION TIME.

The basic exhaust system consists of exhaust manifold(s), exhaust pipe with oxygen sensors, catalytic converter(s), heat shield(s), muffler and tailpipe (Fig. 1) and (Fig. 2)

EXHAUST SYSTEM (Continued)

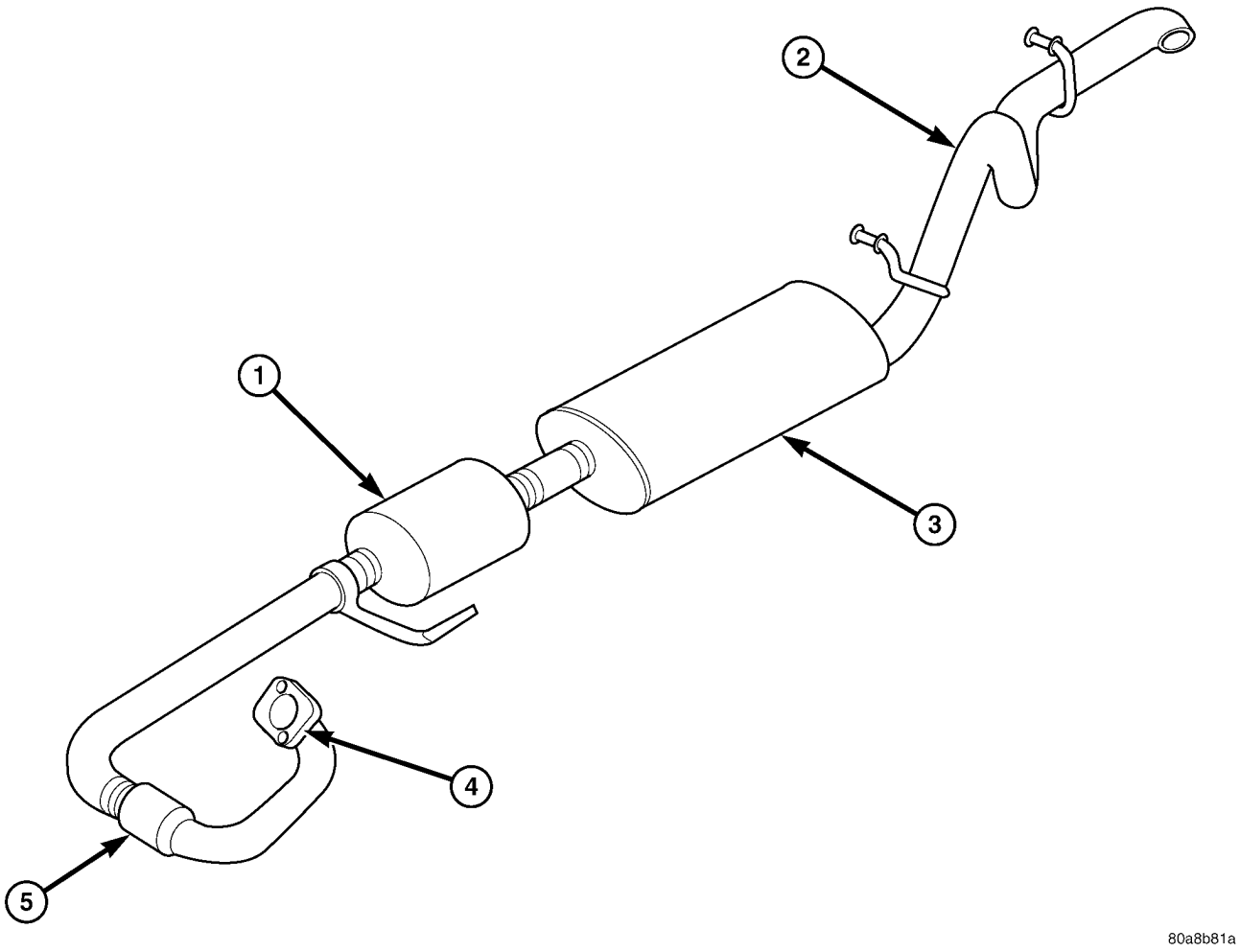

Fig. 1 Exhaust system - 2.4L Engine

1 - CATALYTIC CONVERTER
2 - TAILPIPE
3 - MUFFLER
4 - EXHAUST PIPE TO EXHAUST MANIFOLD FLANGE
5 - MINI CATALYTIC CONVERTER

EXHAUST SYSTEM (Continued)

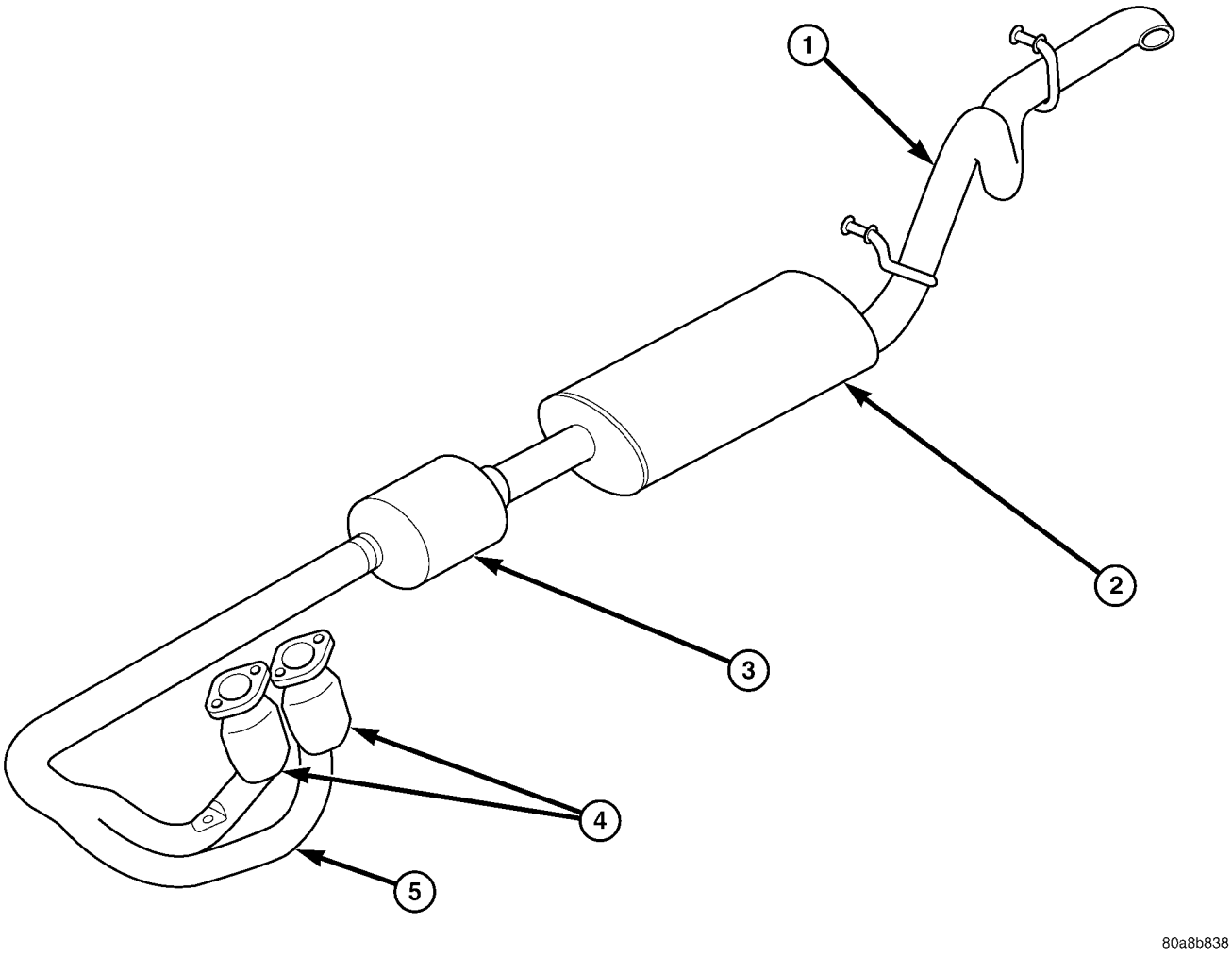

Fig. 2 Exhaust System - 4.0L

1 - TAILPIPE
2 - MUFFLER
3 - CATALYTIC CONVERTER
4 - MINI CATALYTIC CONVERTER
5 - EXHAUST PIPE

EXHAUST SYSTEM (Continued)

DIAGNOSIS AND TESTING - EXHAUST SYSTEM

EXHAUST SYSTEM DIAGNOSIS CHART

CONDITION	POSSIBLE CAUSE	CORRECTION
EXCESSIVE EXHAUST NOISE OR LEAKING EXHAUST GASES	1. Leaks at pipe joints.	1. Tighten clamps/bolts to specified torque at leaking joints.
	2. Rusted or blown out muffler.	2. Replace muffler. Inspect exhaust system.
	3. Broken or rusted out exhaust pipe.	3. Replace exhaust pipe.
	4. Exhaust pipe leaking at manifold flange.	4. Tighten/replace flange attaching nuts/bolts.
	5. Exhaust manifold cracked or broken.	5. Replace exhaust manifold.
	6. Leak between exhaust manifold and cylinder head.	6. Tighten exhaust manifold to cylinder head bolts.
	7. Catalytic converter rusted or blown out.	7. Replace catalytic converter assy.
	8. Restriction in exhaust system.	8. Remove restriction, if possible. Replace restricted part if necessary.
CAUTION: When servicing and replacing exhaust system components, disconnect the oxygen sensor connector(s). Allowing the exhaust to hang by the oxygen sensor wires will damage the harness and/or sensor.		

SPECIFICATIONS

TORQUE

DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Bolts, Crossmember to Sill	42	31	-
Nuts, Crossmember to Transmission Mount	22	16	-
Nuts, Exhaust Pipe to Manifold	31	23	-
Nuts/Bolt, Exhaust Manifold to Engine			
#6&7	31	23	-
#1,2,3,4,5,8,9,10&11	33	24	-
Nuts, Exhaust Pipe to Catalytic Converter Flange	28.5	21	-
Clamp, Tailpipe to Rear Tailpipe Hanger	27	20	-
Oxygen Sensors	27	20	-
Heat Shields	45	33	-

CATALYTIC CONVERTER

DESCRIPTION

WARNING: THE NORMAL OPERATING TEMPERATURE OF THE EXHAUST SYSTEM IS VERY HIGH. THEREFORE, NEVER WORK AROUND OR ATTEMPT TO SERVICE ANY PART OF THE EXHAUST SYSTEM UNTIL IT IS COOLED. SPECIAL CARE SHOULD BE TAKEN WHEN WORKING NEAR THE CATALYTIC CONVERTER. THE TEMPERATURE OF THE CONVERTER RISES TO A HIGH LEVEL AFTER A SHORT PERIOD OF ENGINE OPERATION TIME.

CAUTION: DO NOT remove spark plug wires from plugs or by any other means short out cylinders. Failure of the catalytic converter can occur due to a temperature increase caused by unburned fuel passing through the converter.

The stainless steel catalytic converter body is designed to last the life of the vehicle. Excessive heat can result in bulging or other distortion, but excessive heat will not be the fault of the converter. If unburned fuel enters the converter, overheating may occur. If a converter is heat-damaged, correct the

CATALYTIC CONVERTER (Continued)

cause of the damage at the same time the converter is replaced. Also, inspect all other components of the exhaust system for heat damage.

The catalytic converter and muffler are serviced as an assembly.

Unleaded gasoline must be used to avoid contaminating the catalyst core.

REMOVAL

REMOVAL - 2.4L

WARNING: THE NORMAL OPERATING TEMPERATURE OF THE EXHAUST SYSTEM IS VERY HIGH. THEREFORE, NEVER ATTEMPT TO SERVICE ANY PART OF THE EXHAUST SYSTEM UNTIL IT IS COOLED. SPECIAL CARE SHOULD BE TAKEN WHEN WORKING NEAR THE CATALYTIC CONVERTER. THE TEMPERATURE OF THE CONVERTER RISES TO A HIGH LEVEL AFTER A SHORT PERIOD OF ENGINE OPERATION TIME.

WARNING: IF TORCHES ARE USED WHEN WORKING ON THE EXHAUST SYSTEM, DO NOT ALLOW THE FLAME NEAR THE FUEL LINES.

CAUTION: When servicing exhaust system components, disconnect the oxygen sensor connector(s). Allowing the exhaust system to hang by the oxygen sensor harness will damage the wiring and/or sensor.

(1) Disconnect the oxygen sensors.

(2) Remove the two bolts and flanged nuts at the manifold (Fig. 3).

(3) Remove the two bolts and flanged nuts at the flange.

(4) Remove the mini catalyst assembly from the vehicle.

(5) Slide exhaust pipe forward until exhaust pipe hanger disengages from transmission support. Remove exhaust pipe and catalytic converter from vehicle.

REMOVAL - 4.0L

WARNING: THE NORMAL OPERATING TEMPERATURE OF THE EXHAUST SYSTEM IS VERY HIGH. THEREFORE, NEVER ATTEMPT TO SERVICE ANY PART OF THE EXHAUST SYSTEM UNTIL IT IS COOLED. SPECIAL CARE SHOULD BE TAKEN WHEN WORKING NEAR THE CATALYTIC CONVERTER. THE TEMPERATURE OF THE CONVERTER RISES TO A HIGH LEVEL AFTER A SHORT PERIOD OF ENGINE OPERATION TIME.

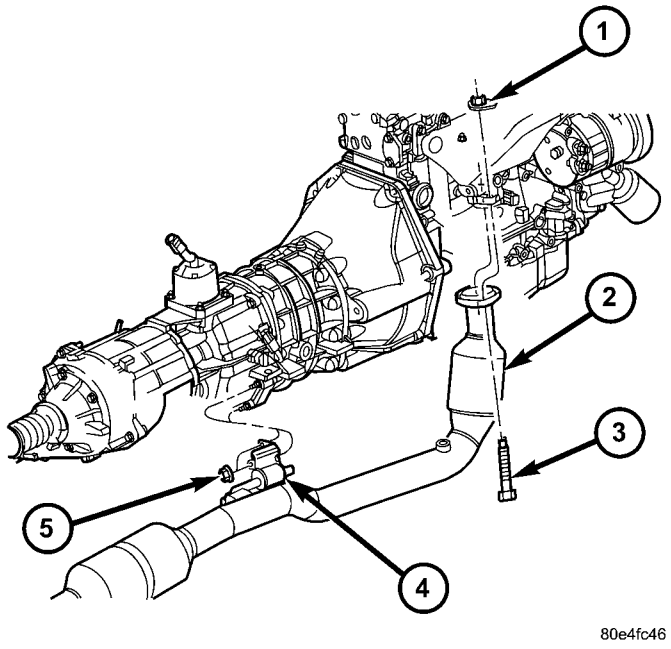

Fig. 3 2.4L Exhaust Pipe and Catalytic Converter - 4x4

1- FLANGED NUT
2- MINI CATALYST ASSEMBLY
3- BOLT
4- HANGER
5- NUT

WARNING: IF TORCHES ARE USED WHEN WORKING ON THE EXHAUST SYSTEM, DO NOT ALLOW THE FLAME NEAR THE FUEL LINES.

CAUTION: When servicing exhaust system components, disconnect the oxygen sensor connector(s). Allowing the exhaust system to hang by the oxygen sensor harness will damage the wiring and/or sensor.

(1) Raise and support the vehicle.

(2) Saturate the studs and nuts with a Mopar® rust penetrant. Allow 5 minutes for penetration.

(3) Remove the oxygen sensors from the exhaust pipe and the catalytic converter.

(4) Disconnect the exhaust pipe from the engine exhaust manifold (Fig. 4).

(5) Remove mini catalytic converter flange retaining nuts (Fig. 4).

(6) Slide exhaust pipe forward until exhaust pipe hanger disengages from transmission support. Remove exhaust pipe and catalytic converter from vehicle.

CATALYTIC CONVERTER (Continued)

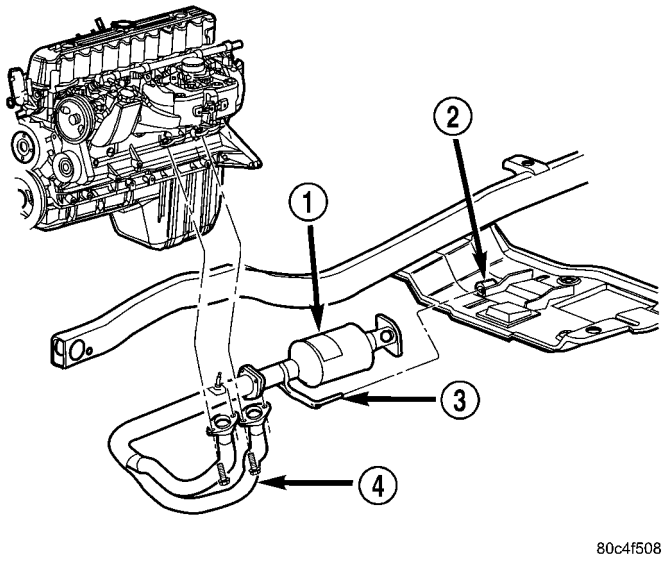

Fig. 4 Exhaust Pipe and Catalytic Converter - 4.0L

1 - CATALYTIC CONVERTER
2 - TRANSMISSION SUPPORT
3 - EXHAUST HANGER
4 - EXHAUST PIPE

INSTALLATION

INSTALLATION - 2.4L

(1) Position the mini catalytic converter assembly onto the exhaust pipe flange and the exhaust manifold. Tighten the nuts to 31 N·m (23 ft.lbs.) torque. Tighten the flange nuts to 28 N·m (21 ft. lbs.).

(2) Connect oxygen sensor wiring.

(3) Lower the vehicle.

(4) Start engine and inspect for leaks. Repair exhaust leaks as necessary.

(5) Check the exhaust system for contact with the body panels. Make the necessary adjustments, if needed.

INSTALLATION - 4.0L

(1) Position exhaust pipe and catalytic converter into vehicle.

(2) Insert exhaust pipe hanger into transmission support (Fig. 4).

(3) Install exhaust pipe onto exhaust manifold **DO NOT** tighten bolts at this time.

(4) Position muffler flange onto catalytic converter flange and install retaining bolts and nuts (Fig. 4). **DO NOT** tighten nuts at this time.

(5) Make sure the exhaust system is aligned and has the proper clearance. The minimum clearance is 25mm (1 inch).

(6) Tighten muffler to catalytic converter flange retaining nuts to 28.5 N·m (21 ft. lbs.).

(7) Tighten exhaust pipe to exhaust manifold mounting bolts to 31 N·m (23 ft. lbs.).

(8) Install the oxygen sensors in the exhaust pipe and catalytic converter.

(9) Lower vehicle.

(10) Start engine and inspect for leaks. Repair exhaust leaks as necessary.

(11) Check the exhaust system for contact with the body panels. Make the necessary adjustments, if needed.

EXHAUST PIPE

REMOVAL

REMOVAL - 2.4L

WARNING: THE NORMAL OPERATING TEMPERATURE OF THE EXHAUST SYSTEM IS VERY HIGH. THEREFORE, NEVER ATTEMPT TO SERVICE ANY PART OF THE EXHAUST SYSTEM UNTIL IT IS COOLED. SPECIAL CARE SHOULD BE TAKEN WHEN WORKING NEAR THE CATALYTIC CONVERTER. THE TEMPERATURE OF THE CONVERTER RISES TO A HIGH LEVEL AFTER A SHORT PERIOD OF ENGINE OPERATION TIME.

WARNING: IF TORCHES ARE USED WHEN WORKING ON THE EXHAUST SYSTEM, DO NOT ALLOW THE FLAME NEAR THE FUEL LINES.

CAUTION: When servicing exhaust system components, disconnect the oxygen sensor connector(s). Allowing the exhaust system to hang by the oxygen sensor harness will damage the wiring and/or sensor.

(1) Raise and support the vehicle.

(2) Saturate the studs and nuts with a Mopar® rust penetrant. Allow 5 minutes for penetration.

(3) Remove the oxygen sensors from the exhaust pipe and the catalytic converter.

(4) Disconnect the exhaust pipe from the engine exhaust manifold (Fig. 5).

(5) Remove catalytic converter to muffler flange retaining nuts (Fig. 5).

(6) Slide exhaust pipe forward until exhaust pipe hanger disengages from transmission support. Remove exhaust pipe and catalytic converter from vehicle.

EXHAUST PIPE (Continued)

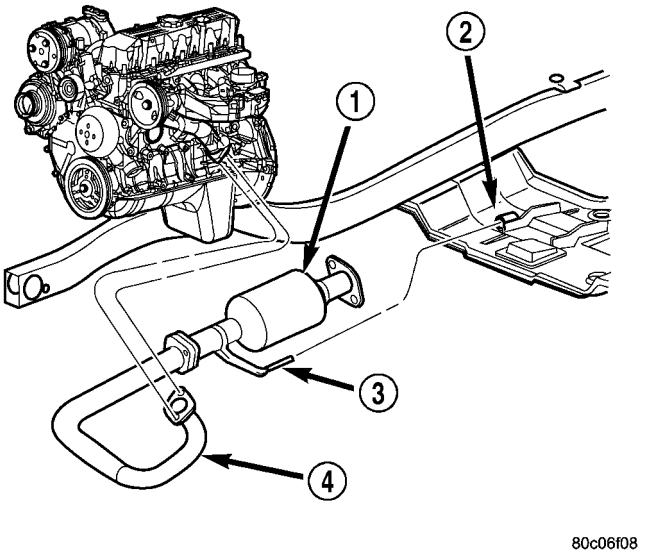

Fig. 5 Exhaust Pipe and Calalytic Converter - 2.4L

1 - CATALYTIC CONVERTER
2 - TRANSMISSION SUPPORT
3 - EXHAUST HANGER
4 - EXHAUST PIPE

REMOVAL - 4.0L

WARNING: THE NORMAL OPERATING TEMPERATURE OF THE EXHAUST SYSTEM IS VERY HIGH. THEREFORE, NEVER ATTEMPT TO SERVICE ANY PART OF THE EXHAUST SYSTEM UNTIL IT IS COOLED. SPECIAL CARE SHOULD BE TAKEN WHEN WORKING NEAR THE CATALYTIC CONVERTER. THE TEMPERATURE OF THE CONVERTER RISES TO A HIGH LEVEL AFTER A SHORT PERIOD OF ENGINE OPERATION TIME.

WARNING: IF TORCHES ARE USED WHEN WORKING ON THE EXHAUST SYSTEM, DO NOT ALLOW THE FLAME NEAR THE FUEL LINES.

CAUTION: When servicing exhaust system components, disconnect the oxygen sensor connector(s). Allowing the exhaust system to hang by the oxygen sensor harness will damage the wiring and/or sensor.

(1) Raise and support the vehicle.

(2) Saturate the studs and nuts with a Mopar® rust penetrant. Allow 5 minutes for penetration.

(3) Remove the oxygen sensors from the exhaust pipe and the catalytic converter.

(4) Disconnect the exhaust pipe from the engine exhaust manifold (Fig. 6).

(5) Remove catalytic converter to muffler flange retaining nuts (Fig. 6).

(6) Slide exhaust pipe foward until exhaust pipe hanger disengages from transmission support. Remove exhaust pipe and catalytic converter from vehicle.

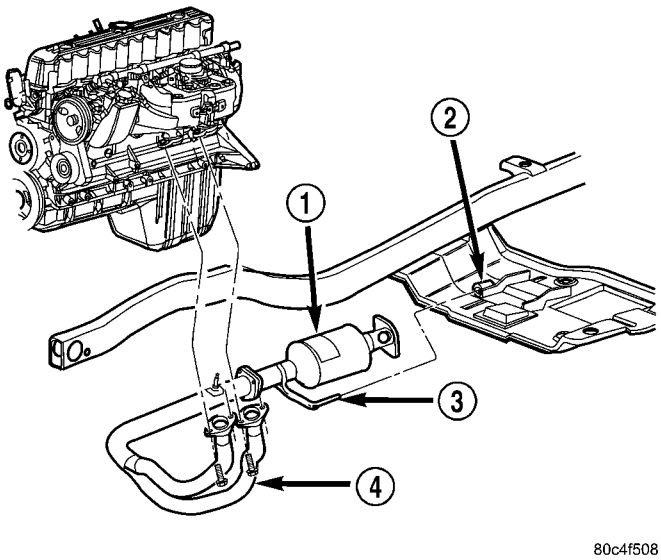

Fig. 6 Exhaust Pipe and Catalytic Converter - 4.0L

1 - CATALYTIC CONVERTER
2 - TRANSMISSION SUPPORT
3 - EXHAUST HANGER
4 - EXHAUST PIPE

INSTALLATION

INSTALLATION - 2.4L

(1) Position exhaust pipe and catalytic converter into vehicle.

(2) Insert exhaust pipe hanger into transmission support (Fig. 5).

(3) Install exhaust pipe onto exhaust manifold **DO NOT** tighten bolts at this time.

(4) Position muffler flange onto catalytic converter flange and install retaining bolts and nuts (Fig. 5). **DO NOT** tighten nuts at this time.

(5) Make sure the exhaust system is aligned and has the proper clearance. The minimum clearance is 25mm (1 inch).

(6) Tighten muffler to catalytic converter flange retaining nuts to 28.5 N·m (21 ft. lbs.).

(7) Tighten exhaust pipe to exhaust manifold mounting bolts to 31 N·m (23 ft. lbs.).

(8) Install the oxygen sensors in the exhaust pipe and catalytic converter.

(9) Lower vehicle.

(10) Start engine and inspect for leaks. Repair exhaust leaks as necessary.

(11) Check the exhaust system for contact with the body panels. Make the necessary adjustments, if needed.

EXHAUST PIPE (Continued)

INSTALLATION - 4.0L

(1) Position exhaust pipe and catalytic converter into vehicle.

(2) Insert exhaust pipe hanger into transmission support (Fig. 6).

(3) Install exhaust pipe onto exhaust manifold **DO NOT** tighten bolts at this time.

(4) Position muffler flange onto catalytic converter flange and install retaining bolts and nuts (Fig. 6). **DO NOT** tighten nuts at this time.

(5) Make sure the exhaust system is aligned and has the proper clearance. The minimum clearance is 25mm (1 inch).

(6) Tighten muffler to catalytic converter flange retaining nuts to 28.5 N·m (21 ft. lbs.).

(7) Tighten exhaust pipe to exhaust manifold mounting bolts to 31 N·m (23 ft. lbs.).

(8) Install the oxygen sensors in the exhaust pipe and catalytic converter.

(9) Lower vehicle.

(10) Start engine and inspect for leaks. Repair exhaust leaks as necessary.

(11) Check the exhaust system for contact with the body panels. Make the necessary adjustments, if needed.

HEAT SHIELDS

DESCRIPTION

Heat shields (Fig. 7) are made of stamped/formed steel, or metal foil.

Exhaust heat shields are needed to protect both the vehicle and the environment from the high temperatures developed by the catalytic converter. The catalytic converter releases additional heat into the exhaust system. Under severe operating conditions, the temperature increases in the area of the converter. Such conditions can exist when the engine misfires or otherwise does not operate at peak efficiency.

REMOVAL

(1) Raise and support the vehicle.

(2) Remove the screws and/or nuts holding the heat shields to the frame and/or floor pan (Fig. 8) and (Fig. 9).

(3) When removing muffler heat shield, the muffler front support bracket must be removed first.

(4) Slide the shields out around the exhaust system.

INSTALLATION

(1) Position the heat shields to the floor pan or the frame and install the screws and/or nuts (Fig. 8) and (Fig. 9).

(2) Tighten the nuts and/or screws to 45 N·m (33 ft. lbs.).

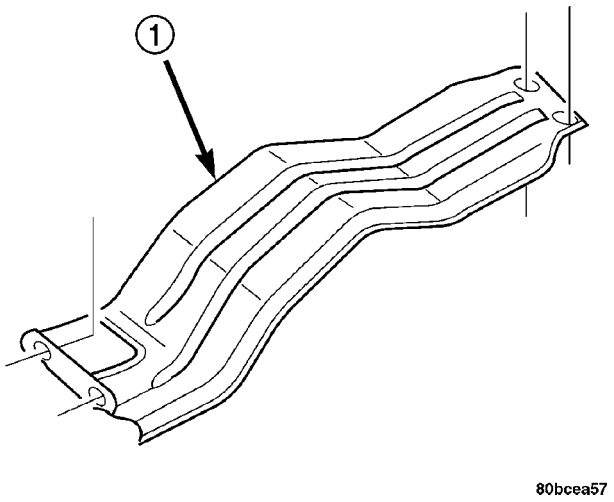

Fig. 7 Exhaust Heat Shield - Typical

1 - HEAT SHIELD

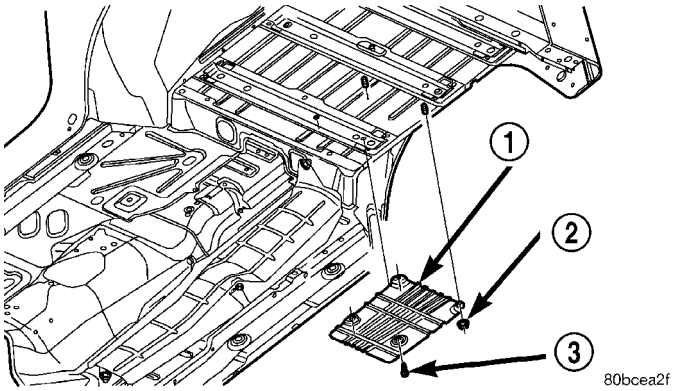

Fig. 8 Muffler Heat Shield - Rear

1 - MUFFLER HEAT SHIELD - REAR
2 - NUTS
3 - SELF TAPPING SCREWS

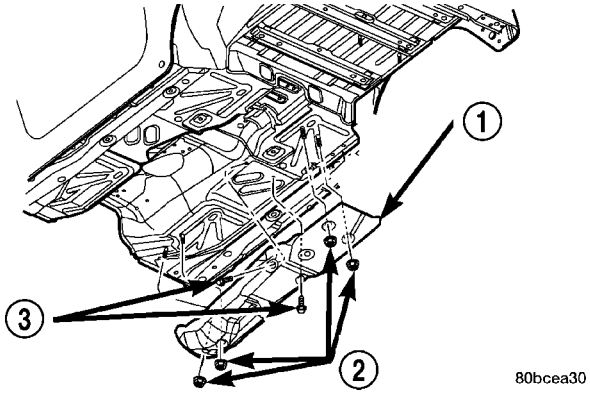

Fig. 9 Muffler Heat Shield - Middle

1 - MUFFLER HEAT SHIELD - MIDDLE
2 - NUTS
3 - SELF TAPPING SCREWS

(3) Lower the vehicle.

MUFFLER

DESCRIPTION

Both the 2.4L and 4.0L engines use a galvanized steel muffler (Fig. 10) to control exhaust noise levels and exhaust back pressure.

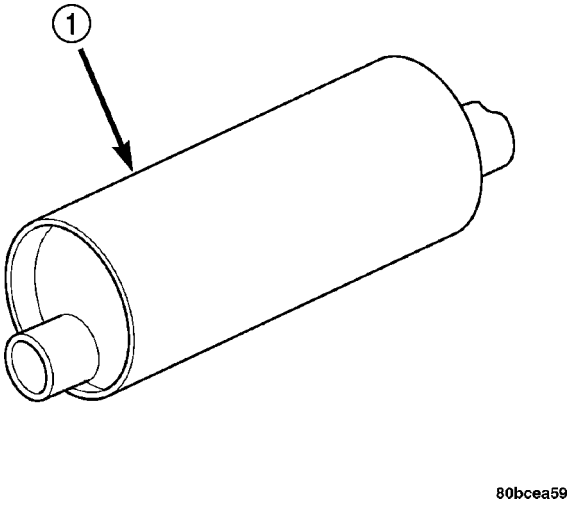

Fig. 10 Muffler

1 - MUFFLER

REMOVAL

WARNING: THE NORMAL OPERATING TEMPERATURE OF THE EXHAUST SYSTEM IS VERY HIGH. THEREFORE, NEVER ATTEMPT TO SERVICE ANY PART OF THE EXHAUST SYSTEM UNTIL IT IS COOLED. SPECIAL CARE SHOULD BE TAKEN WHEN WORKING NEAR THE CATALYTIC CONVERTER. THE TEMPERATURE OF THE CONVERTER RISES TO A HIGH LEVEL AFTER A SHORT PERIOD OF ENGINE OPERATION TIME.

WARNING: IF TORCHES ARE USED WHEN WORKING ON THE EXHAUST SYSTEM, DO NOT ALLOW THE FLAME NEAR THE FUEL LINES.

CAUTION: When servicing exhaust system components, disconnect the oxygen sensor connector(s). Allowing the exhaust system to hang by the oxygen sensor harness will damage the wiring and/or sensor.

All original equipment exhaust systems are manufactured with the exhaust tailpipe welded to the muffler. Service replacement mufflers and exhaust tailpipes are either clamped together or welded together.

(1) Raise the vehicle and support the rear of the vehicle by the side rails and allow the axle to hang free.

(2) Remove the tailpipe hangers from the insulators (Fig. 11).

(3) Remove muffler to catalytic converter flange retaining nuts (Fig. 11).

(4) Remove muffler and tailpipe assembly from vehicle.

MUFFLER (Continued)

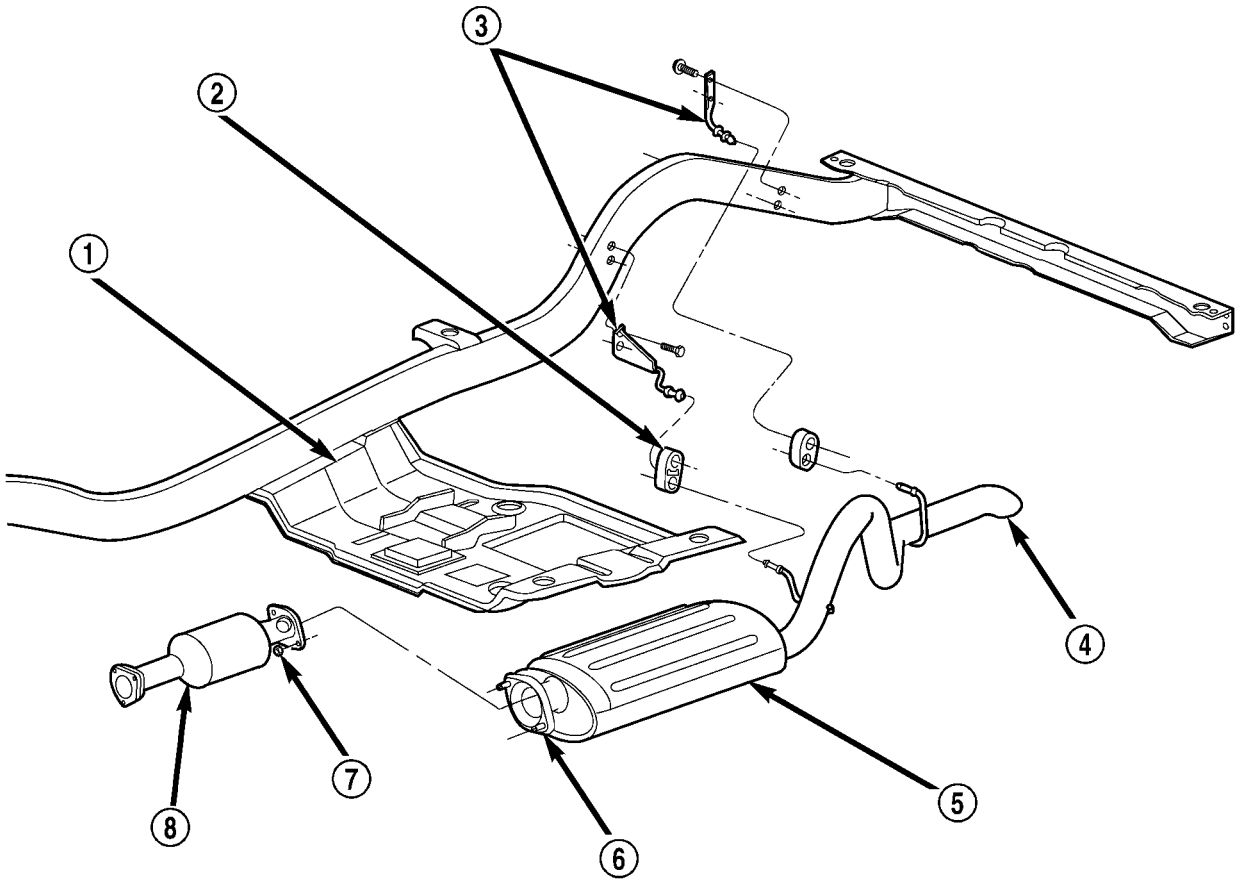

Fig. 11 Muffler and Tailpipe

1 - TRANSMISSION SUPPORT
2 - TAILPIPE HANGER INSULATORS
3 - TAILPIPE HANGERS
4 - TAILPIPE
5 - MUFFLER
6 - MUFFLER TO CATALYTIC CONVERTER FLANGE
7 - NUTS
8 - CATALYTIC CONVERTER

INSTALLATION

(1) Position muffler and tailpipe assembly into vehicle.

(2) Position muffler and catalytic converter flanges together and install nuts (Fig. 11). **DO NOT** tighten nuts at this time.

(3) Install tailpipe hangers into the insulators (Fig. 11).

(4) Make sure the muffler and tailpipe are correctly positioned and the proper alignment. The minimum clearance between components is 25mm (1 inch).

(5) Tighten muffler to catalytic converter flange nuts to 28.5 N·m (21 ft. lbs.).

(6) Lower vehicle.

(7) Start engine and inspect for leaks. Repair exhaust leaks as necessary.

(8) Check the exhaust system for contact with the body panels. Make the necessary adjustments, if needed.

TAILPIPE

DESCRIPTION

TAILPIPE

DESCRIPTION

The tailpipe (Fig. 12) is made of galvanized steel

OPERATION

The tailpipe channels the exhaust out of the muffler and out from under the vehicle to control noise and prevent exhaust gas fumes from entering the passenger compartment.

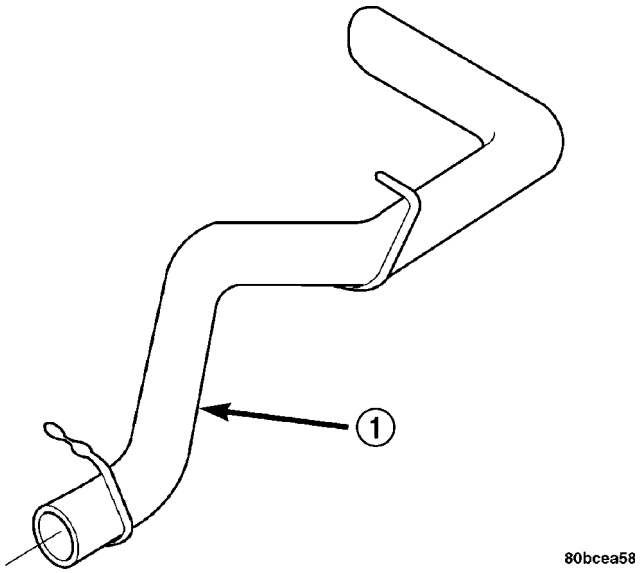

Fig. 12 Tailpipe - Typical

1 - TAILPIPE

FRAME & BUMPERS

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FRAME & BUMPERS

SPECIFICATIONS

TORQUE SPECIFICATIONS

TORQUE SPECIFICATIONS

DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
FRONT BUMPER SCREW	108	80	—
FRONT TOW HOOK SCREW	108	80	—
FUEL TANK SKID PLATE NUTS	16	12	138
FUEL TANK STRAP NUTS	5	—	40
MAIN FLOOR TO HOLD DOWN BOLT	68	50	—
RADIATOR TO FRAME HOLD DOWN BOLT	60	45	—
REAR BUMPER BOLT	77	57	—
REAR FLOOR TO FRAME BOLT	47	35	—
REAR TOE HOOK SCREW	77	57	—
TRANFER CASE SKID PLATE BOLTS	45	33	—
TRANSMISSION MOUNT NUTS	35	26	—
TRANSMISSION SKID PLATE TO FRAME BOLTS	45	33	—
TRANSMISSION SKID PLATE TO TRANSFER CASE SKID PLATE BOLTS	28	21	—

FRONT EXTENSION

REMOVAL

(1) Remove the bolts attaching the bumper extension to the bumper (Fig. 1).

(2) Separate the extension from the bumper.

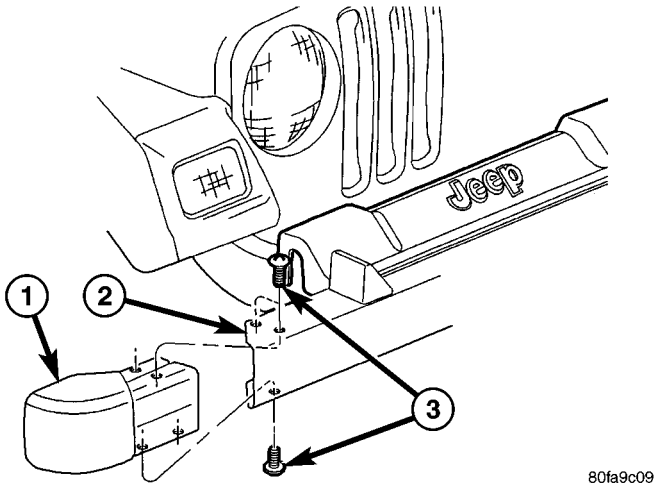

Fig. 1 BUMPER EXTENSION

1 - BUMPER EXTENSION
2 - FRONT BUMPER
3 - BOLTS (2 ON TOP AND 1 ON BOTTOM)

INSTALLATION

(1) Position the extension to the bumper

(2) Install the screws retaining the front bumper extension. (Fig. 1)

FRONT BUMPER

REMOVAL

(1) If equipped, disconnect the fog lamp harness connector.

(2) Remove the screws that attach the bumper to the frame rail (Fig. 2).

(3) If equipped, remove the tow hook. (Refer to 13 - FRAME & BUMPERS/FRAME/FRONT TOW HOOK - REMOVAL)

(4) Separate the bumper from the vehicle.

INSTALLATION

(1) Position the bumper on the vehicle.

(2) If equipped, install the tow hook. (Refer to 13 - FRAME & BUMPERS/FRAME/FRONT TOW HOOK - INSTALLATION)

(3) Install the screws that attach the bumper to the frame rail and tighten to 108 N·m (80 ft. lbs.).

(4) If equipped, Connect the fog lamp harness connector.

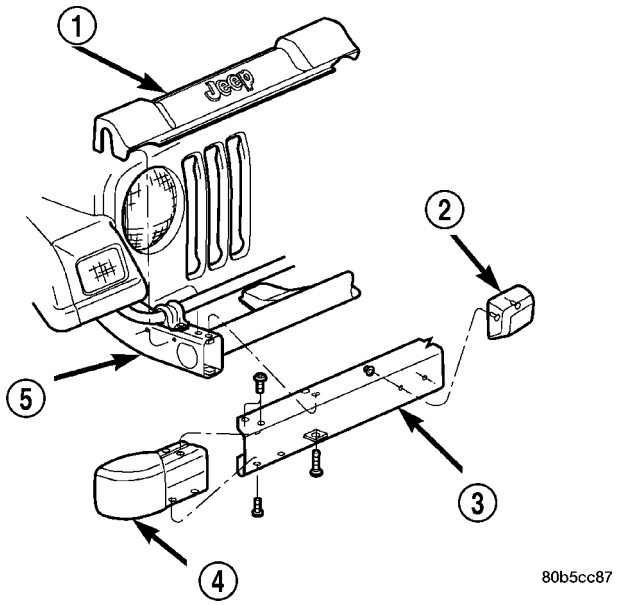

Fig. 2 FRONT BUMPER COMPONENTS

1 - VALANCE
2 - BUMPER GUARD
3 - FRONT BUMPER
4 - BUMPER EXTENSION
5 - FRAME

REAR EXTENSION

REMOVAL

(1) Remove the screws attaching the bumper extension to the bumper (Fig. 3).

(2) Separate the extension from the bumper.

INSTALLATION

(1) Position the extension on the bumper.

(2) Install the screws attaching the bumper extension to the bumper.

REAR BUMPER

REMOVAL

(1) Remove the bolt attaching the bumper to frame rail. (Fig. 3)

(2) If equipped, remove the rear tow eye/hook. (Refer to 13 - FRAME & BUMPERS/FRAME/REAR TOW HOOK - REMOVAL)

(3) Disconnect the rear fog lamp electrical connector, if equipped.

(4) Remove the nuts attaching the bumper to the rear frame crossmember.

(5) Separate the bumper from the vehicle.

REAR BUMPER (Continued)

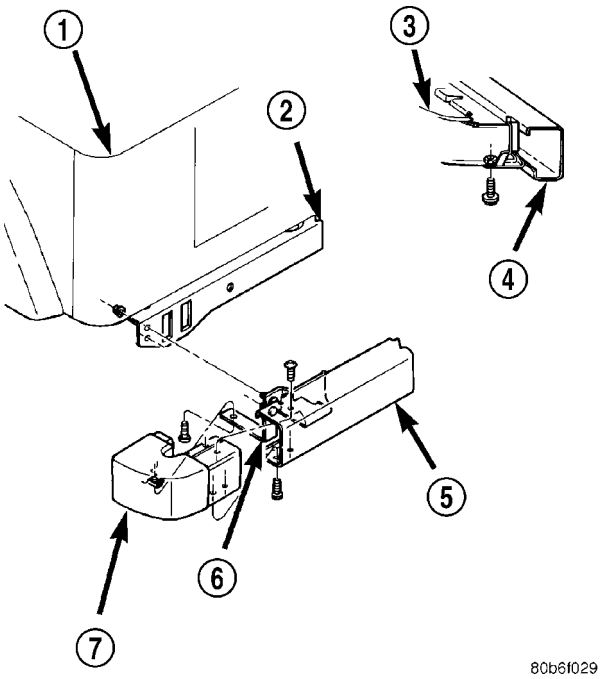

Fig. 3 REAR BUMPER COMPONENTS

1 - BODY
2 - CROSSMEMBER
3 - FRAME
4 - REAR BUMPER
5 - REAR BUMPER
6 - BRACE
7 - REAR BUMPER EXTENSION

INSTALLATION

(1) Position the bumper on the vehicle.

(2) Install the nuts attaching the bumper to the rear frame crossmember and tighten to 77 N·m (57 ft. lbs.).

(3) If equipped, install the rear tow eye/hook. (Refer to 13 - FRAME & BUMPERS/FRAME/REAR TOW HOOK - INSTALLATION)

(4) Install the bolt attaching the bumper to frame rail and tighten to 77 N·m (57 ft. lbs.).

(5) Connect the rear fog lamp electrical connectors, if equipped.

FRAME

STANDARD PROCEDURE - FRAME SERVICE

SAFETY PRECAUTIONS AND WARNINGS

WARNING: USE EYE PROTECTION WHEN GRINDING OR WELDING METAL, SERIOUS EYE INJURY CAN RESULT. BEFORE PROCEEDING WITH FRAME REPAIR INVOLVING GRINDING OR WELDING, VERIFY THAT VEHICLE FUEL SYSTEM IS NOT LEAKING OR IN CONTACT WITH REPAIR AREA, PERSONAL INJURY CAN RESULT. DO NOT ALLOW OPEN FLAME TO CONTACT PLASTIC BODY PANELS. FIRE OR EXPLOSION CAN RESULT. WHEN WELDED FRAME COMPONENTS ARE REPLACED, 100% PENETRATION WELD MUST BE ACHIEVED DURING INSTALLATION. IF NOT, DANGEROUS OPERATING CONDITIONS CAN RESULT. STAND CLEAR OF CABLES OR CHAINS ON PULLING EQUIPMENT DURING FRAME STRAIGHTENING OPERATIONS, PERSONAL INJURY CAN RESULT. DO NOT VENTURE UNDER A HOISTED VEHICLE THAT IS NOT SUPPORTED ON SAFETY STANDS, PERSONAL INJURY CAN RESULT.

CAUTION: Do not reuse damaged fasteners, quality of repair would be suspect. Do not drill holes in top or bottom frame rail flanges, frame rail failure can result. Do Not use softer than Grade 3 bolts to replace production fasteners, loosening or failure can result. When using heat to straighten frame components do not exceed 566°C (1050°F), metal fatigue can result. Welding the joints around riveted cross members and frame side rails can weaken frame.

FRAME STRAIGHTENING

When necessary, a conventional frame that is bent or twisted can be straightened by application of heat. The temperature must not exceed 566°C (1050°F). The metal will have a dull red glow at the desired temperature. Excessive heat will decrease the strength of the metal and result in a weakened frame.

Welding the joints around riveted cross members and frame side rails is not recommended.

A straightening repair process should be limited to frame members that are not severely damaged. The replacement bolts, nuts and rivets that are used to join the frame members should conform to the same specifications as the original bolts, nuts and rivets.

FRAME REPAIRS

DRILLING HOLES

Do not drill holes in the top and bottom of frame rail, metal fatigue can result causing frame failure. Holes drilled in the side of the frame rail must be at least 38 mm (1.5 in.) from the top and bottom flanges.

Additional drill holes should be located away from existing holes.

WELDING

Use MIG, TIG or arc welding equipment to repair welded frame components.

FRAME (Continued)

Frame components that have been damaged should be inspected for cracks before returning the vehicle to use. If cracks are found in accessible frame components perform the following procedures.

(1) Drill a hole at each end of the crack with a 3 mm (O.125 in.) diameter drill bit.

(2) Using a suitable die grinder with 3 inch cut off wheel, V-groove the crack to allow 100% weld penetration.

(3) Weld the crack.

(4) If necessary when a side rail is repaired, grind the weld smooth and install a reinforcement channel (Fig. 4) over the repaired area.

CAUTION: A reinforcement should never be used on the front section of the frame. The frame section forward of the suspension mounts contains energy management holes (Fig. 5). Reinforcing this area may effect energy management.

NOTE: If a reinforcement is required, it should completely cover the repaired area. The reinforcement should also overlap the top and bottom of the frame by more than 50% of its width. Weld as indicated (Fig. 4).

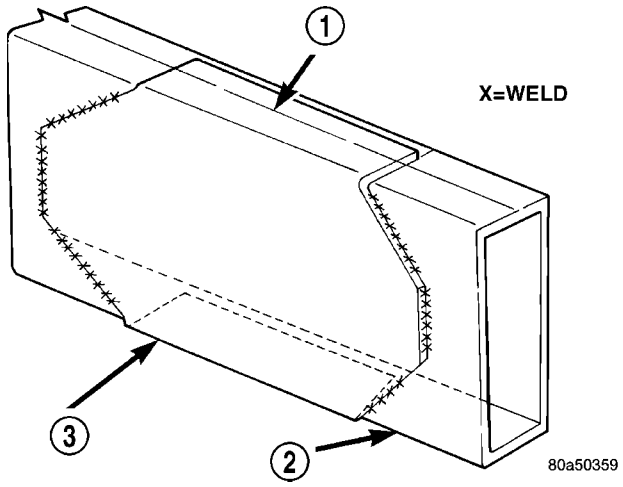

Fig. 4 FRAME REINFORCEMENT

1 - FRAME CENTER LINE
2 - FRAME
3 - FRAME REPAIR REINFORCEMENT

FRAME FASTENERS

Bolts and nuts can be used to repair frames or to install a reinforcement section on the frame.

Conical-type washers are preferred over the split-ring type lock washers. Normally, grade-5 bolts are adequate for frame repair. **Grade-3 bolts or softer should not be used.** Tightening bolts/nuts with the correct torque, refer to the Introduction Group at the front of this manual for tightening information.

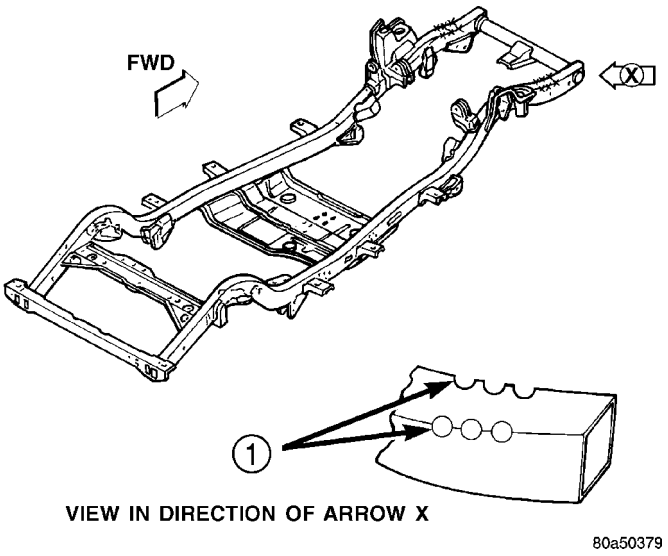

Fig. 5 ENERGY MANAGEMENT HOLES

1 - ENERGY MANAGEMENT HOLES

SPECIFICATIONS

FRAME DIMENSIONS

Frame dimensions are listed in metric scale. All dimensions are from center to center of Principle Locating Point (PLP), or from center to center of PLP and fastener location.

VEHICLE PREPARATION

Position the vehicle on a level work surface. Using screw or bottle jacks, adjust the vehicle PLP heights to the specified dimension above a level work surface. Vertical dimensions can be taken from the work surface to the locations indicated were applicable.

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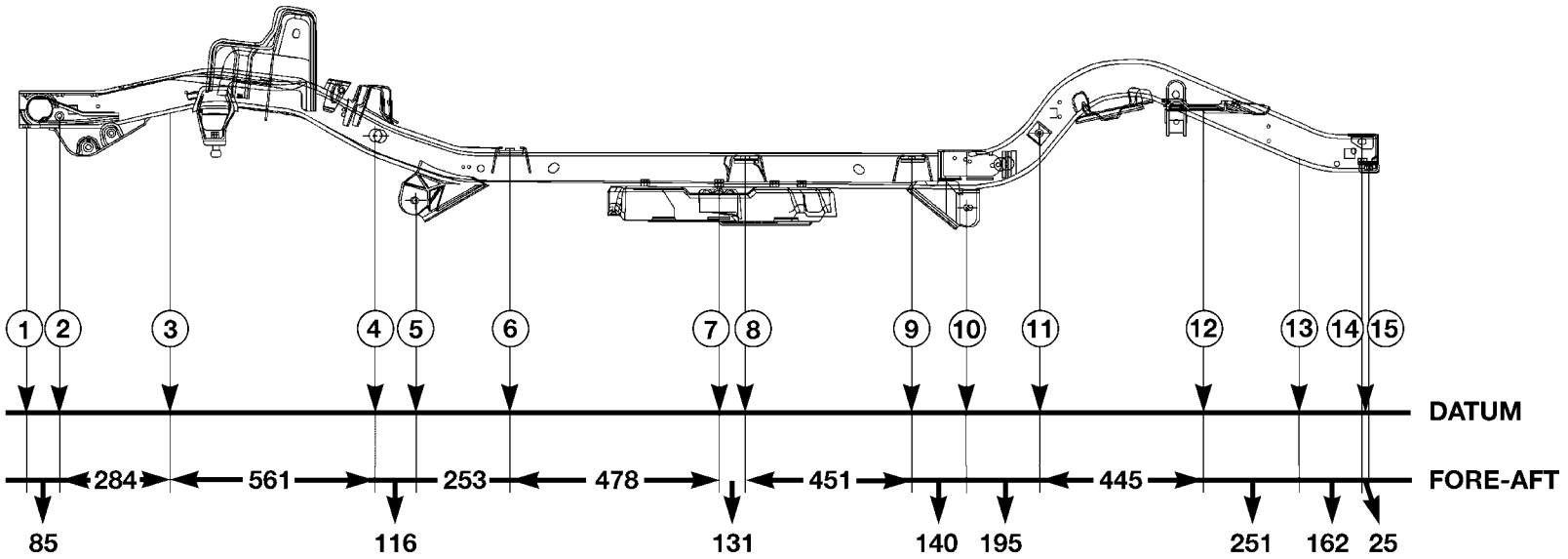
FRAME SIDE VIEW LWB	8
FRAME TOP VIEW LWB	9



POINT #	DISTANCE TO DATUM (mm)	
1.	500	BOTTOM OF SIDERAIL
2.	534	CENTER OF HOLE
3.	543	BOTTOM OF SIDERAIL, CENTER OF SLOT
4.	494	CENTER OF HOLE, OUTBOARD
5.	317	CENTER OF HOLE, OUTBOARD
6.	460	BOTTOM OF BRACKET
7.	359	BOTTOM OF SIDERAIL

POINT #	DISTANCE TO DATUM (mm)	
8.	445	BOTTOM OF BRACKET
9.	445	BOTTOM OF BRACKET
10.	309	CENTER OF HOLE, OUTBOARD
11.	512	CENTER OF HOLE, INBOARD
12.	577	BOTTOM OF CROSSMEMBER
13.	474	CENTER OF HOLE, OUTBOARD
14.	505	BOTTOM OF BRACKET
15.	406	BOTTOM OF FRAME

805fe562

Fig. 6 FRAME SIDE VIEW

FRAME (Continued)

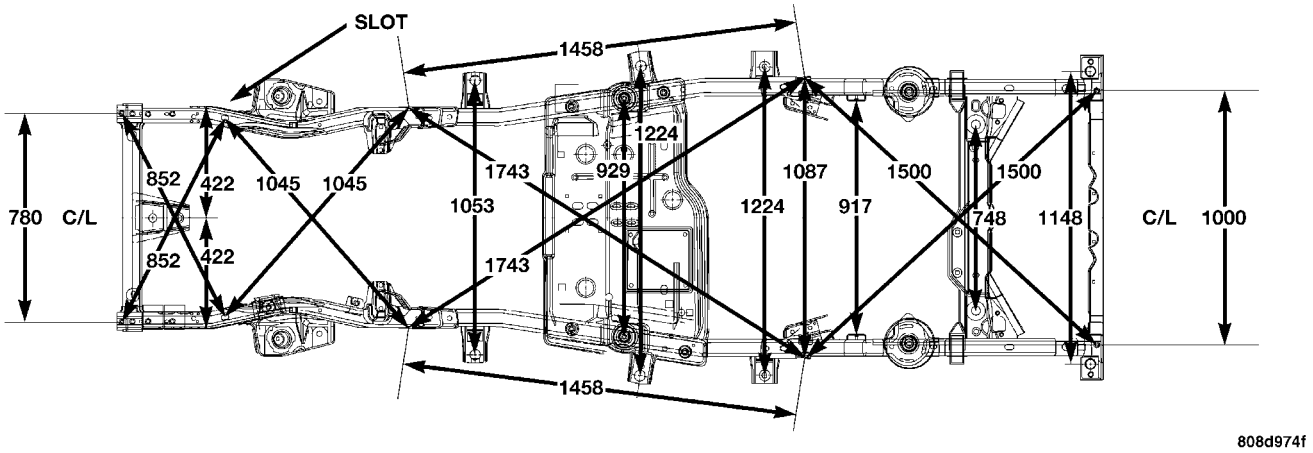


Fig. 7 FRAME TOP VIEW

FRAME (Continued)

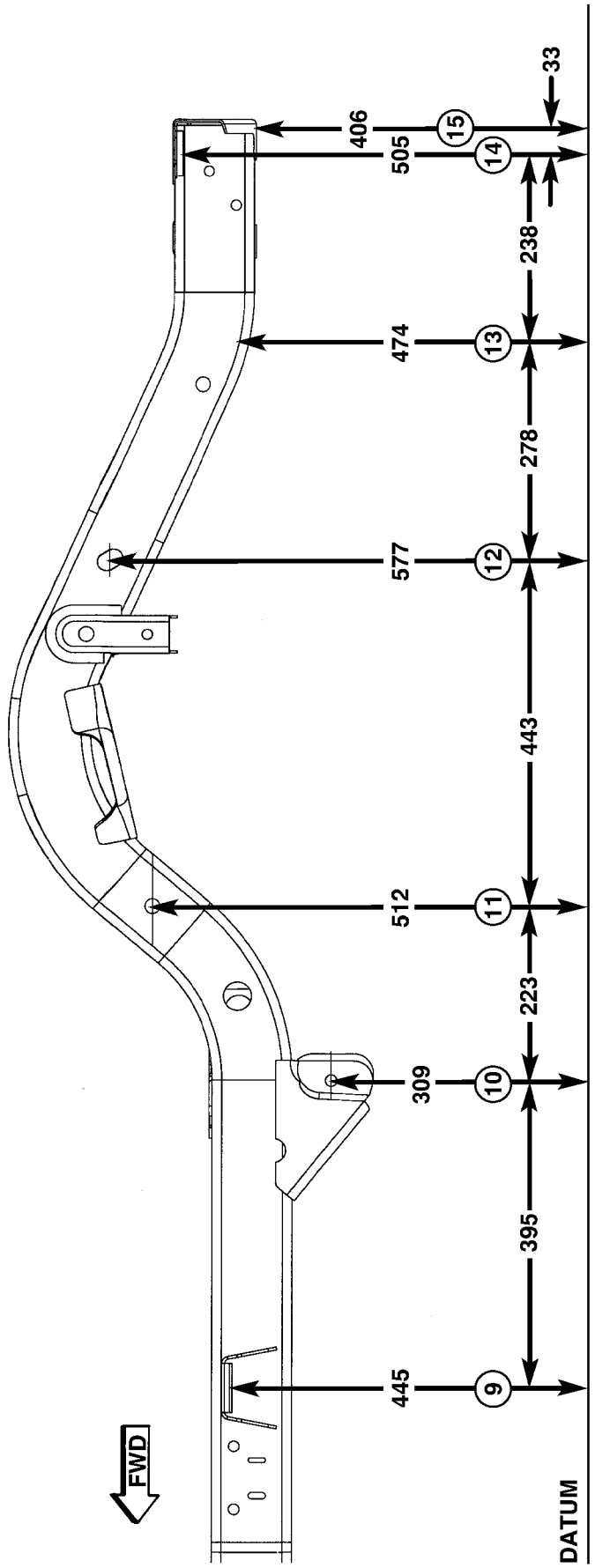


Fig. 8 FRAME SIDE VIEW LWB

FRAME (Continued)

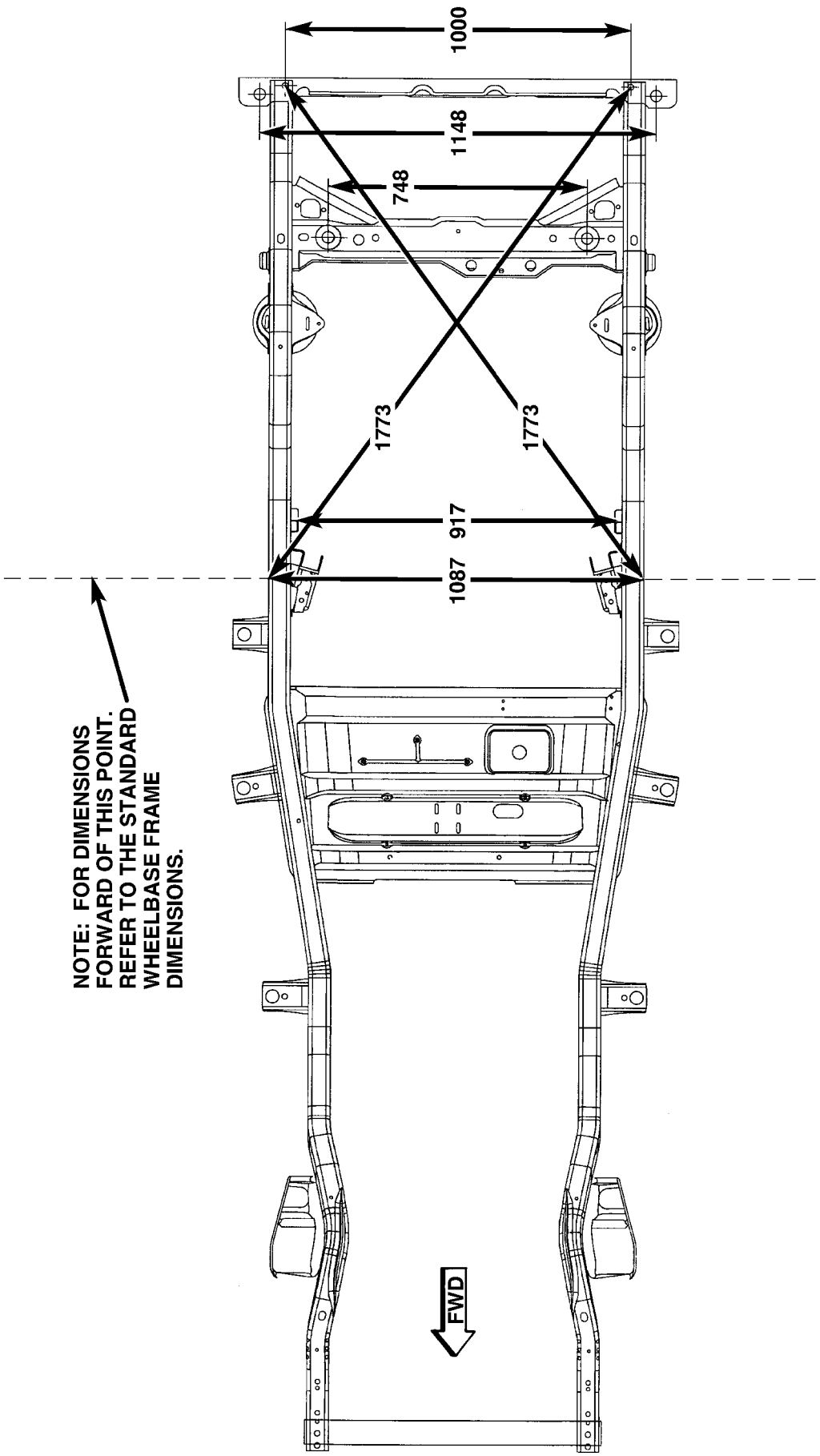


Fig. 9 FRAME TOP VIEW LWB

TRANSFER CASE SKID PLATE

REMOVAL

NOTE: The transmission and transfer case crossmember is integrated with the transfer case skid plate.

WARNING: THE TRANSFER CASE AND TRANSMISSION ARE SUPPORTED BY THE TRANSFER CASE SKID PLATE. BEFORE REMOVING THE TRANSFER CASE SKID PLATE, ENSURE THAT THE TRANSMISSION IS PROPERLY SUPPORTED.

(1) Raise and support the vehicle. (Refer to LUBRICATION & MAINTENANCE/HOISTING - STANDARD PROCEDURE)

(2) Remove the transmission skid plate, if equipped. (Refer to 13 - FRAME & BUMPERS/FRAME/FRONT SKID PLATE - REMOVAL)

(3) Support the transmission with a suitable lifting device.

(4) Remove the nuts attaching the transmission mount to the skid plate. (Fig. 10)

(5) Remove the bolts attaching the skid plate to the frame and remove the skid plate. (Fig. 11)

INSTALLATION

(1) Position the skid plate on the vehicle.

(2) Install the bolts attaching the skid plate to the frame and tighten to 74 N·m (55 ft. lbs.).

(3) Install the nuts attaching the transmission mount to the skid plate and tighten to 35 N·m (26 ft. lbs.).

(4) Install the transmission skid plate. (Refer to 13 - FRAME & BUMPERS/FRAME/FRONT SKID PLATE - INSTALLATION)

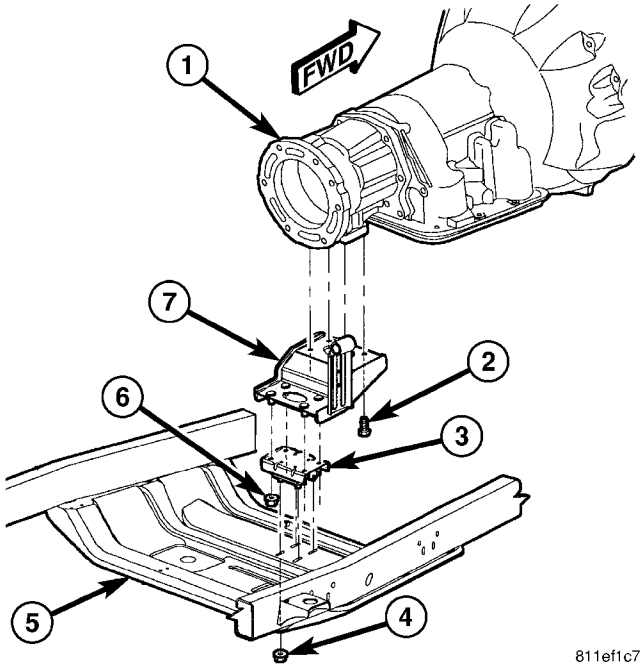


Fig. 10 TRANSMISSION MOUNT (Auto shown, manual similar)

1 - TRANSMISSION ASSEMBLY
2 - SUPPORT BRACKET BOLTS (4)
3 - TRANSMISSION MOUNT
4 - SKID PLATE/CROSSMEMBER NUTS (4)
5 - SKID PLATE/CROSSMEMBER
6 - MOUNT NUTS (4)
7 - SUPPORT BRACKET

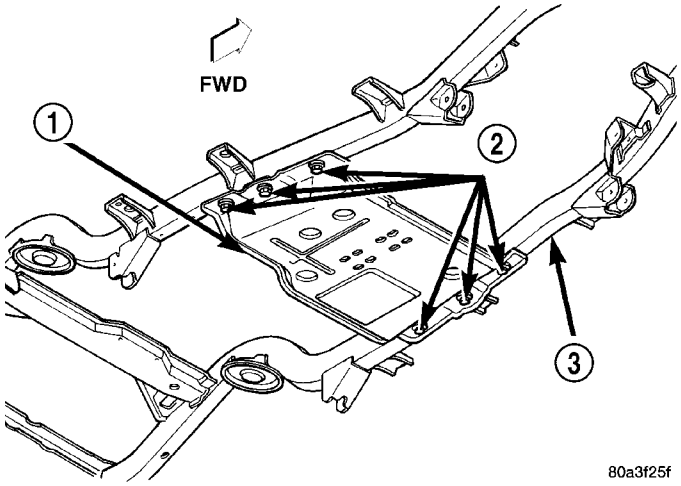


Fig. 11 TRANSFER CASE SKID PLATE

1 - SKID PLATE
2 - BOLTS
3 - FRAME

FUEL TANK SKID PLATE

REMOVAL

(1) Position a support under the fuel tank skid plate.

(2) Remove the protective caps from the end of the strap studs.

(3) Remove the nuts that attach the skid plate to the straps and to the crossmembers (Fig. 12).

(4) Separate the fuel tank strap from the skid plate.

(5) Support the fuel tank and remove the skid plate from the vehicle.

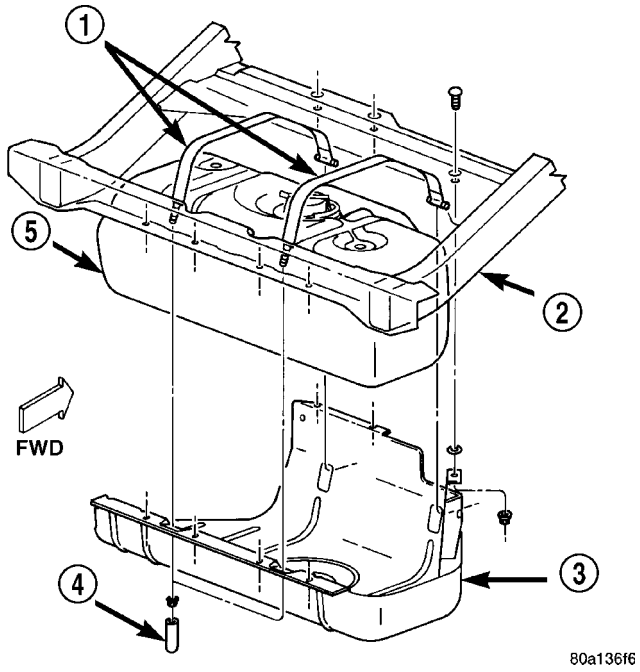


Fig. 12 FUEL TANK SKID PLATE

1 - STRAP
2 - FRAME
3 - SKID PLATE
4 - PROTECTIVE CAP
5 - FUEL TANK

INSTALLATION

(1) Attach the skid plate to the fuel tank strap.

(2) Position and support the skid plate under the fuel tank.

(3) Install the nuts to attach the skid plate to the straps and to the frame crossmembers. Tighten the fuel tank strap nuts to 5 N·m (40 in. lbs.). Tighten the skid plate-to-crossmember nuts with 16 N·m (138 in. lbs.).

(4) Install the protective caps on the end of the strap studs.

(5) Remove the support from under the skid plate.

FRONT TOW HOOK

REMOVAL

(1) Remove the torx bolts that attach the tow hook to the bumper (Fig. 13).

(2) Separate the tow hook from the bumper.

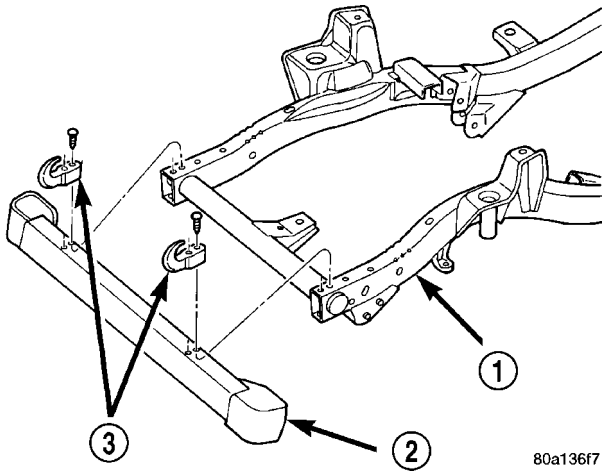


Fig. 13 FRONT TOW HOOK

1 - FRAME
2 - BUMPER
3 - TOW HOOK

INSTALLATION

(1) Position the tow hook on the bumper.

(2) Install the torx bolts that attach the tow hook to the bumper and tighten to 108 N·m (80 ft. lbs.).

REAR TOW HOOK

REMOVAL

(1) Remove the fasteners that attach the rear tow hook to the frame (Fig. 14).

(2) Separate the tow hook from the frame.

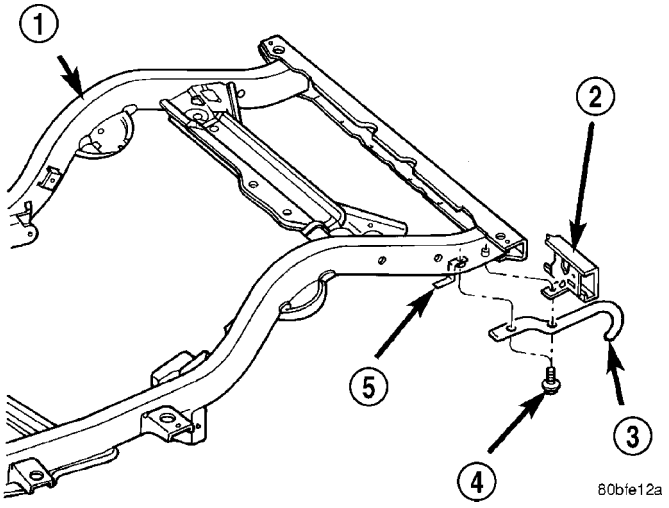


Fig. 14 REAR TOW HOOK

1 - FRAME
2 - REAR BUMPER
3 - TOW HOOK
4 - BOLT
5 - NUT

INSTALLATION

(1) Position the tow hook on the frame.

(2) Install the bolts and tighten to 77 N·m (57 ft. lbs.).

TRANSMISSION SKID PLATE

REMOVAL

(1) Raise and support the vehicle. (Refer to LUBRICATION & MAINTENANCE/HOISTING - STANDARD PROCEDURE)

(2) Support the skid plate.

(3) Remove the bolts and remove the skid plate. (Fig. 15)

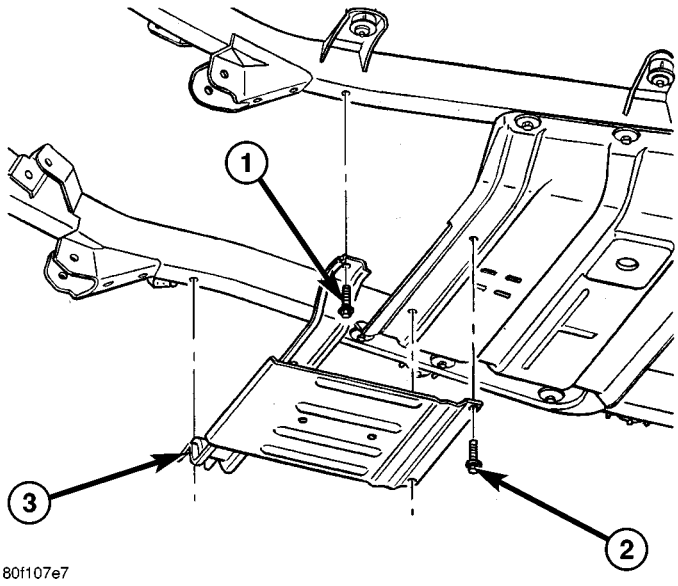


Fig. 15 TRANSMISSION SKID PLATE

1 - FRONT BOLTS (2)
2 - REAR BOLTS (2)
3 - SKID PLATE

INSTALLATION

(1) Position the skid plate in place and support.

(2) Install the bolts and tighten the plate to frame bolts to 45 N·m (33 ft. lbs.).

(3) Tighten the plate to transfer case skid plate bolts to 35 N·m (26 ft. lbs.).

FUEL SYSTEM

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FUEL DELIVERY

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FUEL DELIVERY

DESCRIPTION

The fuel delivery system consists of:

- the fuel pump module containing the electric fuel pump, fuel filter/fuel pressure regulator, fuel gauge sending unit (fuel level sensor) and a separate fuel filter located at bottom of pump module
- fuel tubes/lines/hoses
- quick-connect fittings
- fuel injector rail
- fuel injectors
- fuel tank
- fuel tank filler/vent tube assembly
- fuel tank filler tube cap
- check valve(s)
- accelerator pedal
- throttle cable

OPERATION

Fuel is returned through the fuel pump module and back into the fuel tank through the fuel filter/fuel pressure regulator. A separate fuel return line from the engine to the tank is not used.

The fuel tank assembly consists of: the fuel tank, fuel pump module assembly, fuel pump module locknut/gasket, and fuel tank check valve (refer to Emis-

FUEL DELIVERY (Continued)

sion Control System for fuel tank check valve information).

A fuel filler/vent tube assembly using a pressure/vacuum fuel filler cap is used. The fuel filler tube contains a flap door located below the fuel fill cap.

Also to be considered part of the fuel system is the evaporation control system. This is designed to reduce the emission of fuel vapors into the atmosphere. The description and function of the Evaporative Control System is located in Emission Control Systems.

Both fuel filters (at bottom of fuel pump module and within fuel pressure regulator) are designed for extended service. They do not require normal scheduled maintenance. Filters should only be replaced if a diagnostic procedure indicates to do so.

STANDARD PROCEDURE - FUEL SYSTEM PRESSURE RELEASE

Use following procedure if the fuel injector rail is, or is not equipped with a fuel pressure test port.

(1) Remove fuel fill cap.

(2) Remove fuel pump relay from Power Distribution Center (PDC). For location of relay, refer to label on underside of PDC cover.

(3) Start and run engine until it stalls.

(4) Attempt restarting engine until it will no longer run.

(5) Turn ignition key to OFF position.

CAUTION: Steps 1, 2, 3 and 4 must be performed to relieve high pressure fuel from within fuel rail. Do not attempt to use following steps to relieve this pressure as excessive fuel will be forced into a cylinder chamber.

(6) Unplug connector from any fuel injector.

(7) Attach one end of a jumper wire with alligator clips (18 gauge or smaller) to either injector terminal.

(8) Connect other end of jumper wire to positive side of battery.

(9) Connect one end of a second jumper wire to remaining injector terminal.

CAUTION: Powering an injector for more than a few seconds will permanently damage the injector.

(10) Momentarily touch other end of jumper wire to negative terminal of battery for no more than a few seconds.

(11) Place a rag or towel below fuel line quick-connect fitting at fuel rail.

(12) Disconnect quick-connect fitting at fuel rail. Refer to Quick-Connect Fittings.

(13) Return fuel pump relay to PDC.

(14) One or more Diagnostic Trouble Codes (DTC's) may have been stored in PCM memory due to fuel pump relay removal. The DRB® scan tool must be used to erase a DTC.

SPECIFICATIONS

FUEL SYSTEM PRESSURE

58 psi +/- 2 psi

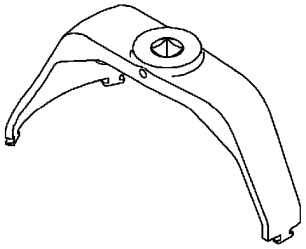
TORQUE - FUEL DELIVERY

DESCRIPTION	N-m	Ft. Lbs.	In. Lbs.
Accelerator Pedal Bracket Mounting Nuts	8.5	-	75
Fuel Hose Clamps	3	-	25
Fuel Rail Mounting Bolts - 2.4L	28	-	250
Fuel Rail Mounting Bolts or Nuts - 4.0L	11	-	100
Fuel Tank Mounting Strap Bolts	Refer to service manual text.	-	-
Fuel Tank Skidplate Bolts	16	-	141

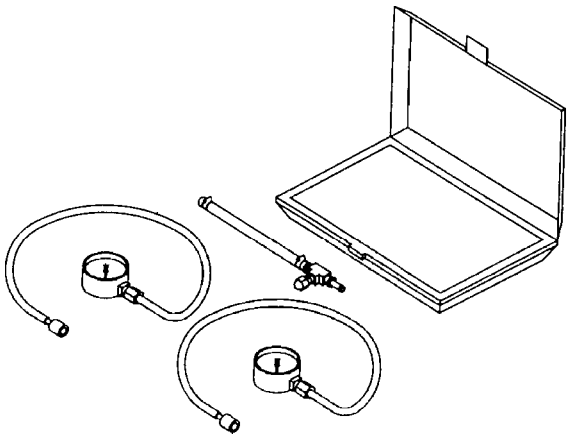
FUEL DELIVERY (Continued)

SPECIAL TOOLS

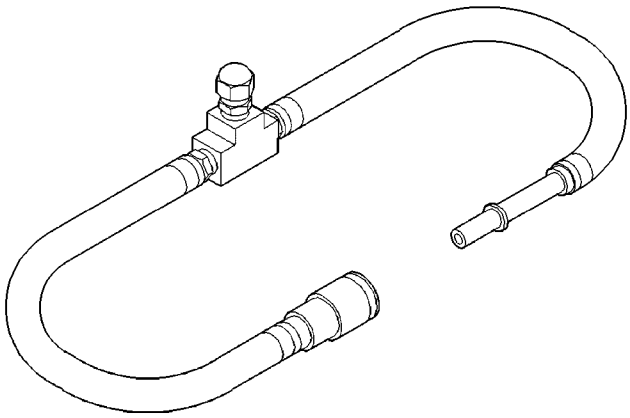
FUEL SYSTEM



LOCKRING REMOVER/INSTALLER #9340



TEST KIT, FUEL PRESSURE #5069



ADAPTERS, FUEL PRESSURE TEST - #6539 AND/OR #6631

FUEL PRESSURE REGULATOR

DESCRIPTION

Refer to Fuel Pump Module for information.

O2S (OXYGEN SENSOR) REMOVER/INSTALLER - #C-4907

FUEL LEVEL SENDING UNIT / SENSOR

DESCRIPTION

Refer to Fuel Pump Module for information.

FUEL LINES

DESCRIPTION

Also refer to Quick-Connect Fittings.

WARNING: THE FUEL SYSTEM IS UNDER A CONSTANT PRESSURE (EVEN WITH THE ENGINE OFF). BEFORE SERVICING ANY FUEL SYSTEM HOSES, FITTINGS OR LINES, THE FUEL SYSTEM PRESSURE MUST BE RELEASED. REFER TO THE FUEL SYSTEM PRESSURE RELEASE PROCEDURE IN THIS GROUP.

The lines/tubes/hoses used on fuel injected vehicles are of a special construction. This is due to the higher fuel pressures and the possibility of contaminated fuel in this system. If it is necessary to replace these lines/tubes/hoses, only those marked EFM/EFI may be used.

If equipped: The hose clamps used to secure rubber hoses on fuel injected vehicles are of a special rolled edge construction. This construction is used to prevent the edge of the clamp from cutting into the hose. Only these rolled edge type clamps may be used in this system. All other types of clamps may cut into the hoses and cause high-pressure fuel leaks.

Use new original equipment type hose clamps.

FUEL PUMP

DESCRIPTION

Refer to Fuel Pump Module for information.

FUEL PUMP MODULE

DESCRIPTION

The fuel pump module assembly (6) (Fig. 1) is located on the top of the fuel tank.

The module assembly (6) contains the following components:

- An internal fuel filter
- A separate fuel pick-up, or inlet filter
- A fuel pressure regulator (4)
- An electric fuel pump
- A lockring to retain pump module to tank
- A soft gasket between tank flange and module
- A fuel gauge sending unit (fuel level sensor)
- Fuel line connection

If the fuel gauge sending unit , electrical fuel pump, primary inlet filter, fuel filter or fuel pressure regulator require service, the fuel pump module must be replaced.

Electric Fuel Pump

The electric fuel pump is located inside of the fuel pump module. A 12 volt, permanent magnet, electric motor powers the fuel pump. The electric fuel pump is not a separate, serviceable component.

Fuel Filters

Two fuel filters are used. One is located at the bottom of the fuel pump module. The other is located inside the module. A separate frame mounted fuel filter is not used with any engine.

Both fuel filters are designed for extended service. They do not require normal scheduled maintenance. Filters should only be replaced if a diagnostic procedure indicates to do so.

Fuel Pressure Regulator

The fuel pressure regulator is located within fuel pump module.

Fuel Gauge Sending Unit (Fuel level sensor)

The fuel gauge sending unit (fuel level sensor) is attached to the side of the fuel pump module. The sending unit consists of a float, an arm, and a variable resistor track (card).

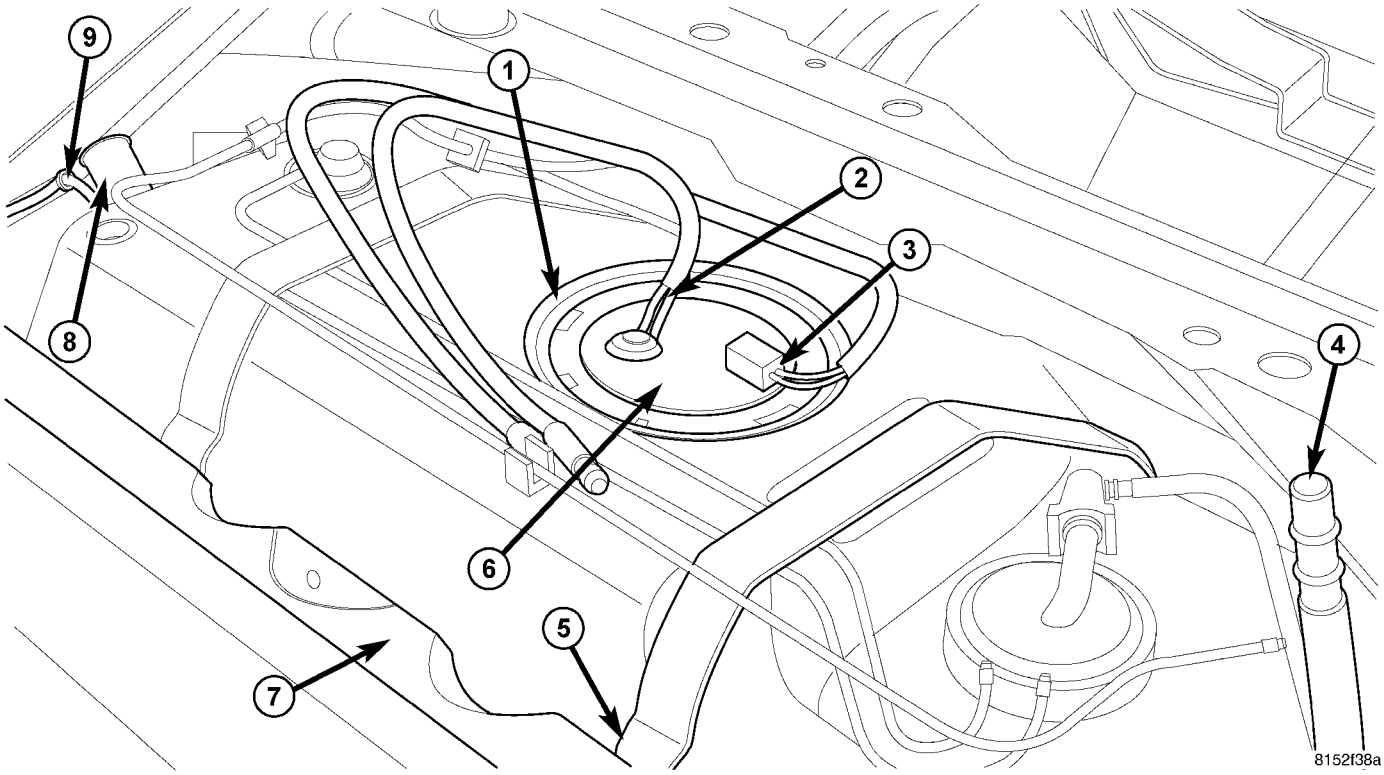


Fig. 1 TOP OF FUEL TANK

1 - FUEL PUMP MODULE LOCKRING
2 - FUEL SUPPLY LINE
3 - ELECTRICAL CONNECTOR
4 - VAPOR LINE
5 - FUEL TANK MOUNTING STRAPS (2)
6 - FUEL PUMP MODULE ASSEMBLY
7 - SKID PLATE
8 - FUEL FILL FITTING
9 - FUEL VENT FITTING

FUEL PUMP MODULE (Continued)

REMOVAL

WARNING: THE FUEL SYSTEM IS UNDER A CONSTANT PRESSURE (EVEN WITH THE ENGINE OFF). BEFORE SERVICING THE FUEL PUMP MODULE, THE FUEL SYSTEM PRESSURE MUST BE RELEASED.

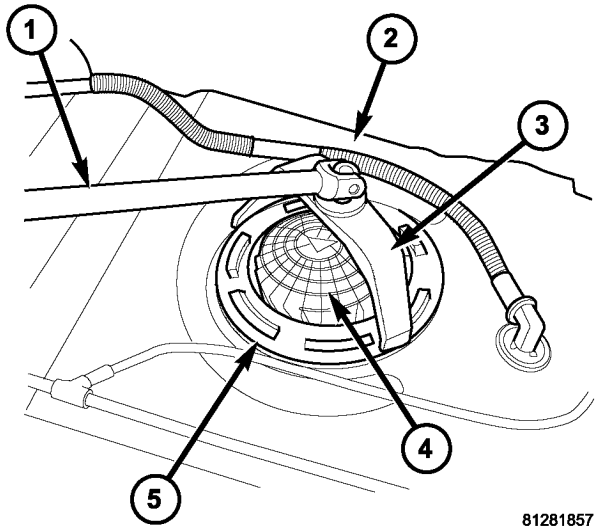


Fig. 2 FUEL PUMP MODULE LOCATION

1 - 1/2″ BREAKER BAR
2 - TOP OF FUEL TANK (TYPICAL)
3 - SPECIAL TOOL #9340
4 - FUEL PUMP MODULE (TYPICAL)
5 - MODULE LOCK RING

(1) Drain and remove fuel tank. Refer to Fuel Tank Removal/Installation.

(2) Note rotational position of module before attempting removal. An indexing arrow is located on top of module for this purpose.

(3) Position Special Tool 9340 (3) (Fig. 2) into notches on outside edge of lockring (5).

(4) Install 1/2 inch drive breaker bar (1) (Fig. 2) to tool 9340 (3).

(5) Rotate breaker bar counter-clockwise to remove lockring.

(6) Remove lockring. The module will spring up slightly when lockring is removed.

(7) Remove module from fuel tank. Be careful not to bend float arm while removing.

INSTALLATION

CAUTION: Whenever the fuel pump module is serviced, the rubber seal (gasket) must be replaced.

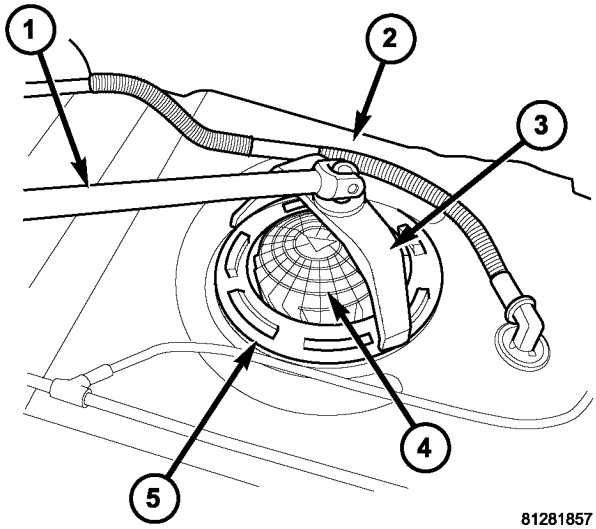


Fig. 3 FUEL PUMP MODULE LOCKRING

1 - 1/2″ BREAKER BAR
2 - TOP OF FUEL TANK (TYPICAL)
3 - SPECIAL TOOL #9340
4 - FUEL PUMP MODULE (TYPICAL)
5 - MODULE LOCK RING

(1) Using a new seal (gasket), position fuel pump module into opening in fuel tank.

(2) Position lockring (5) (Fig. 3) over top of fuel pump module.

(3) Rotate module until embossed alignment arrow points to center alignment mark. This step must be performed to prevent float from contacting side of fuel tank. Also be sure fuel fitting on top of pump module is pointed to front of vehicle.

(4) Install Special Tool 9340 (3) to lockring.

(5) Install 1/2 inch drive breaker (1) into Special Tool 9340 (3).

(6) Tighten lockring (clockwise) until all seven notches have engaged.

(7) Install fuel tank. Refer to Fuel Tank Removal/Installation.

FUEL RAIL/FUEL DAMPER - 2.4L

DESCRIPTION - 2.4L

The fuel injector rail is used to mount the fuel injectors to the engine (Fig. 4). On the 2.4L 4-cylinder engine, a **fuel damper** is located near the front of the fuel rail (Fig. 4).

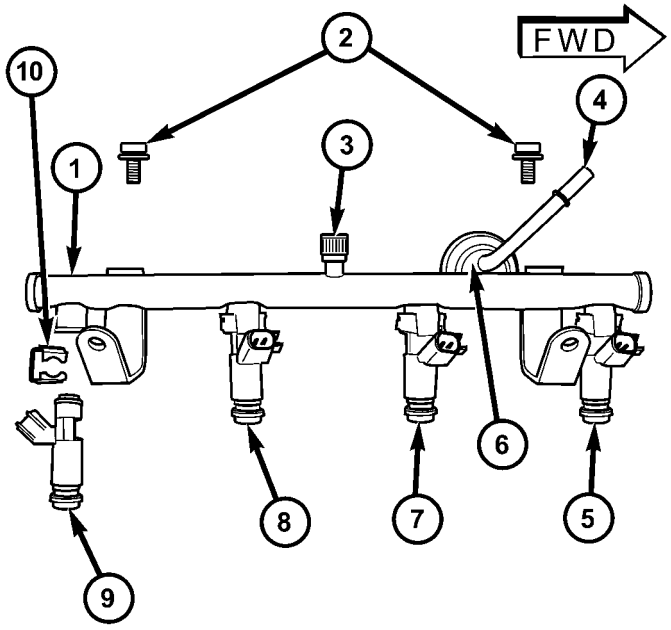


Fig. 4 FUEL RAIL - 2.4L

1 - FUEL RAIL
2 - MOUNTING BOLTS
3 - TEST PORT (IF USED)
4 - QUICK-CONNECT FITTING
5 - INJ. #1
6 - DAMPER
7 - INJ #2
8 - INJ #3
9 - INJ #4
10- INJECTOR RETAINING CLIP

OPERATION - 2.4L

The fuel injector rail supplies the necessary fuel to each individual fuel injector.

The fuel damper is used only to help control fuel pressure pulsations. These pulsations are the result of the firing of the fuel injectors. It is **not used** as a fuel pressure regulator. The fuel pressure regulator is **not mounted** to the fuel rail on any engine. It is located on the fuel tank mounted fuel pump module. Refer to Fuel Pressure Regulator for additional information.

The fuel rail is not repairable.

A quick-connect fitting with a safety latch is used to attach the fuel line to the fuel rail.

REMOVAL - 2.4L

WARNING: THE FUEL SYSTEM IS UNDER CONSTANT PRESSURE EVEN WITH ENGINE OFF. BEFORE SERVICING FUEL RAIL, FUEL SYSTEM PRESSURE MUST BE RELEASED.

The fuel rail can be removed without removing the intake manifold if the following procedures are followed.

(1) Remove fuel tank filler tube cap.
(2) Perform Fuel System Pressure Release Procedure.
(3) Remove negative battery cable at battery.
(4) Remove air duct at throttle body.
(5) Disconnect fuel line latch clip and fuel line at fuel rail. A special tool will be necessary for fuel line disconnection. Refer to Quick-Connect Fittings.
(6) Remove necessary vacuum lines at throttle body.
(7) Drain engine coolant and remove thermostat and thermostat housing.
(8) Remove PCV hose and valve at valve cover.
(9) Remove 3 upper intake manifold mounting bolts (Fig. 6), but only loosen 2 lower bolts about 2 turns.
(10) Disconnect 2 main engine harness connectors at rear of intake manifold (Fig. 5).

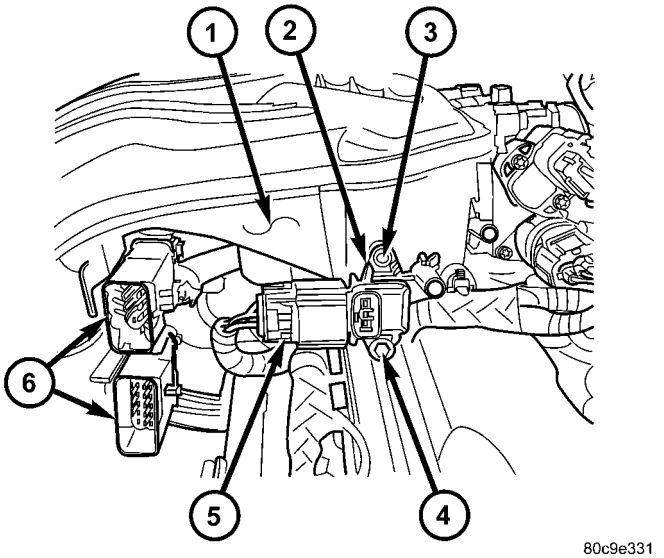


Fig. 5 MAP SENSOR LOCATION - 2.4L

1 - REAR OF INTAKE MANIFOLD
2 - MAP SENSOR
3 - ALIGNMENT PIN
4 - MOUNTING BOLT (TORX)
5 - ELECTRICAL CONNECTOR
6 - MAIN ENGINE HARNESS CONNECTORS

FUEL RAIL/FUEL DAMPER - 2.4L (Continued)

(11) Disconnect 2 injection wiring harness clips at harness mounting bracket (Fig. 6).

(12) Disconnect electrical connectors at all 4 fuel injectors. To remove connector refer to (Fig. 7). Push red colored slider away from injector (1). While pushing slider, depress tab (2) and remove connector (3) from injector. The factory fuel injection wiring harness is numerically tagged (INJ 1, INJ 2, etc.) for injector position identification. If harness is not tagged, note wiring location before removal.

(13) Remove 2 injection rail mounting bolts (Fig. 4).

(14) Gently rock and pull fuel rail until fuel injectors just start to clear machined holes in intake manifold.

(15) Remove fuel rail (with injectors attached) from intake manifold.

(16) If fuel injectors are to be removed, refer to Fuel Injector Removal/Installation.

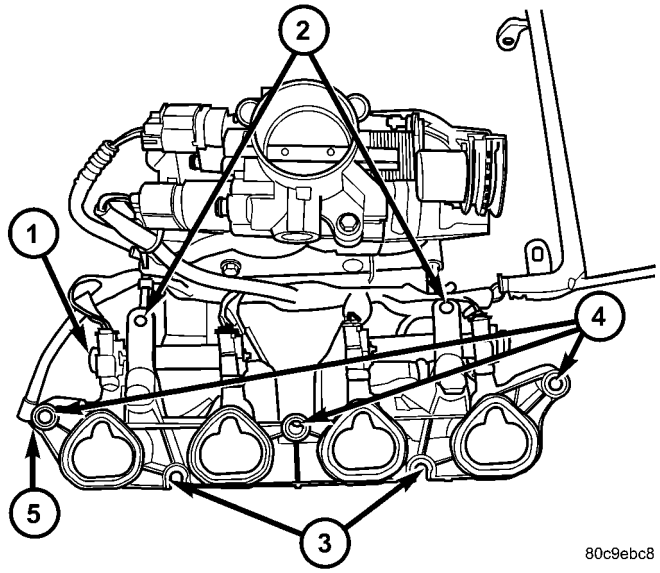


Fig. 6 FUEL RAIL MOUNTING - 2.4L

1 - FUEL RAIL
2 - INJECTION HARNESS CLIPS
3 - LOWER MOUNTING HOLES
4 - UPPER MOUNTING HOLES
5 - INTAKE MANIFOLD

INSTALLATION - 2.4L

(1) If fuel injectors are to be installed, refer to Fuel Injector Removal/Installation.

(2) Clean out fuel injector machined bores in intake manifold.

(3) Apply a small amount of engine oil to each fuel injector o-ring. This will help in fuel rail installation.

(4) Position fuel rail/fuel injector assembly to machined injector openings in intake manifold.

(5) Guide each injector into cylinder head. Be careful not to tear injector o-rings.

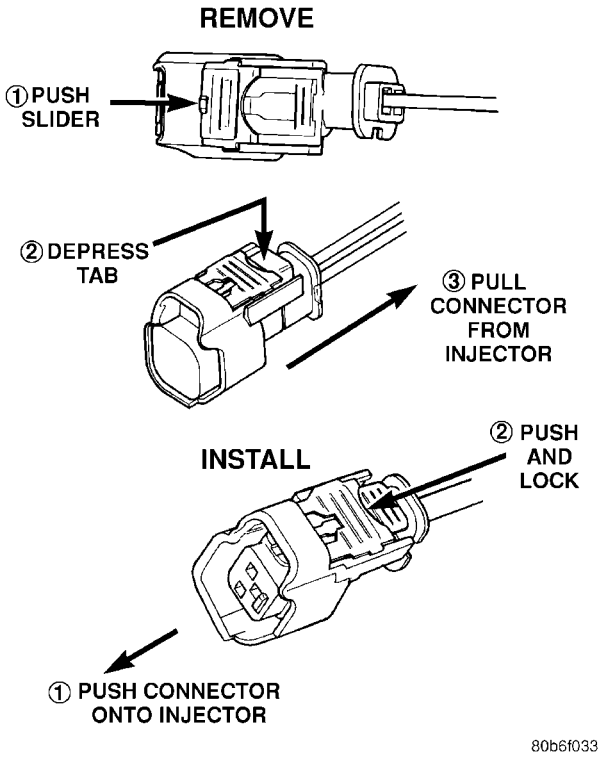


Fig. 7 REMOVE/INSTALL INJECTOR CONNECTOR

(6) Push fuel rail down until fuel injectors have bottomed on shoulders.

(7) Install 2 fuel rail mounting bolts and tighten. Refer to torque specifications.

(8) Connect electrical connectors at all fuel injectors. To install connector, refer to (Fig. 7). Push connector onto injector (1) and then push and lock red colored slider (2). Verify connector is locked to injector by lightly tugging on connector.

(9) Snap 2 injection wiring harness clips (Fig. 6) into brackets.

(10) Connect 2 main engine harness connectors at rear of intake manifold (Fig. 5).

(11) Tighten 5 intake manifold mounting bolts. Refer to Engine Torque Specifications.

(12) Install PCV valve and hose.

(13) Install thermostat and radiator hose. Fill with coolant. Refer to Cooling.

(14) Connect necessary vacuum lines to throttle body.

(15) Connect fuel line latch clip and fuel line to fuel rail. Refer to Quick-Connect Fittings.

(16) Install air duct to throttle body.

(17) Connect battery cable to battery.

(18) Start engine and check for leaks.

FUEL RAIL - 4.0L ENGINE

DESCRIPTION - 4.0L

The fuel rail is mounted to the intake manifold (Fig. 8). It is used to mount the fuel injectors to the engine. On the 4.0L 6–cylinder engine, a **fuel damper** is located near the center of the fuel rail (Fig. 8).

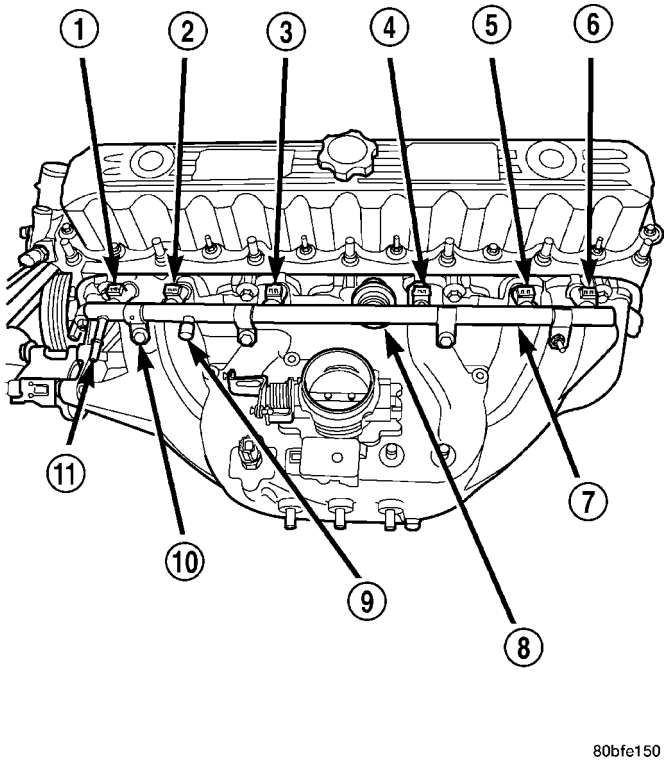


Fig. 8 FUEL RAIL/FUEL DAMPER - 4.0L ENGINE

1 - INJ. #1
2 - INJ. #2
3 - INJ. #3
4 - INJ. #4
5 - INJ. #5
6 - INJ. #6
7 - FUEL INJECTOR RAIL
8 - FUEL DAMPER
9 - PRESSURE TEST PORT CAP
10 - MOUNTING BOLTS (4)
11 - QUICK-CONNECT FITTING

OPERATION - 4.0L

The fuel injector rail supplies the necessary fuel to each individual fuel injector.

The fuel damper is used only to help control fuel pressure pulsations. These pulsations are the result of the firing of the fuel injectors. It is **not used** as a fuel pressure regulator. The fuel pressure regulator is **not mounted** to the fuel rail on any engine. It is located on the fuel tank mounted fuel pump module. Refer to Fuel Filter/Fuel Pressure Regulator for information.

The fuel rail is not repairable.

REMOVAL - 4.0L

The fuel damper is not serviced separately.

WARNING: THE FUEL SYSTEM IS UNDER CONSTANT FUEL PRESSURE EVEN WITH ENGINE OFF. THIS PRESSURE MUST BE RELEASED BEFORE SERVICING FUEL RAIL.

(1) Remove fuel tank filler tube cap.
(2) Perform Fuel System Pressure Release Procedure.
(3) Disconnect negative battery cable from battery.
(4) Remove air tube at top of throttle body. Note: Some engine/vehicles may require removal of air cleaner ducts at throttle body.
(5) Disconnect electrical connectors at all 6 fuel injectors. To remove connector refer to (Fig. 9). Push red colored slider away from injector (1). While pushing slider, depress tab (2) and remove connector (3) from injector. The factory fuel injection wiring harness is numerically tagged (INJ 1, INJ 2, etc.) for injector position identification. If harness is not tagged, note wiring location before removal.
(6) Disconnect fuel supply line latch clip and fuel line at fuel rail. Refer to Quick-Connect Fittings.
(7) Disconnect throttle cable at throttle body. Refer to Throttle Cable Removal/Installation.
(8) Disconnect speed control cable at throttle body (if equipped). Refer to Speed Control Cable in Speed Control System.
(9) Disconnect automatic transmission cable at throttle body (if equipped).
(10) Remove cable routing bracket at intake manifold.
(11) If equipped, remove wiring harnesses at injection rail studs by removing nuts.
(12) Clean dirt/debris from each fuel injector at intake manifold.
(13) Remove fuel rail mounting nuts/bolts (Fig. 10).
(14) Remove fuel rail by gently rocking until all the fuel injectors are out of intake manifold.

INSTALLATION - 4.0L

(1) Clean each injector bore at intake manifold.
(2) Apply a small amount of clean engine oil to each injector o-ring. This will aid in installation.
(3) Position tips of all fuel injectors into the corresponding injector bore in intake manifold. Seat injectors into manifold.
(4) Install and tighten fuel rail mounting bolts to 11 ±3 N·m (100 ±25 in. lbs.) torque.
(5) If equipped, connect wiring harnesses to injection rail studs.
(6) Connect electrical connectors at all fuel injectors. To install connector, refer to (Fig. 9). Push con-

FUEL RAIL - 4.0L ENGINE (Continued)

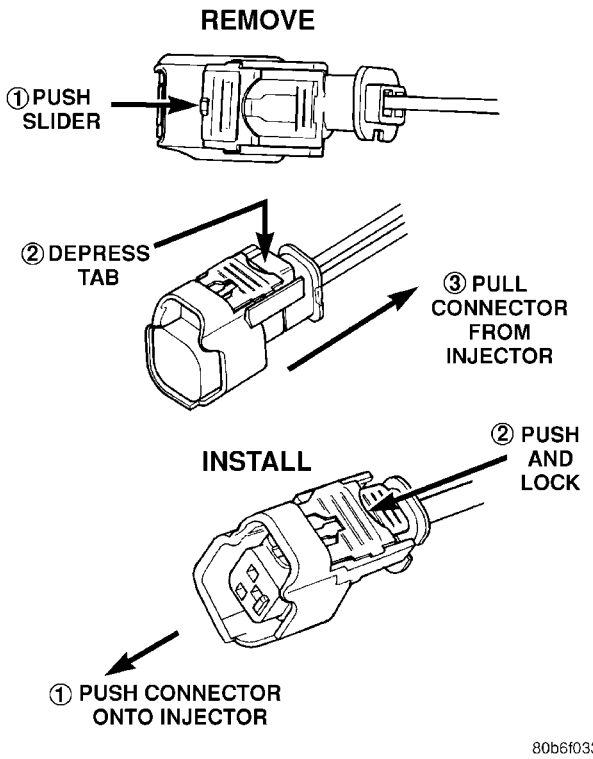


Fig. 9 REMOVE/INSTALL FUEL INJECTOR CONNECTOR

nector onto injector (1) and then push and lock red colored slider (2). Verify connector is locked to injector by lightly tugging on connector.

(7) Connect fuel line and fuel line latch clip to fuel rail. Refer Quick-Connect Fittings.

(8) Install protective cap to pressure test port fitting (if equipped).

(9) Install cable routing bracket to intake manifold.

(10) Connect throttle cable at throttle body.

(11) Connect speed control cable at throttle body (if equipped).

(12) Connect automatic transmission cable at throttle body (if equipped).

(13) Install air tube (or duct) at top of throttle body.

(14) Install fuel tank cap.

(15) Connect negative battery cable to battery.

(16) Start engine and check for fuel leaks.

FUEL TANK

DESCRIPTION

The fuel tank is constructed of a plastic material. Its main functions are for fuel storage and for placement of the fuel pump module.

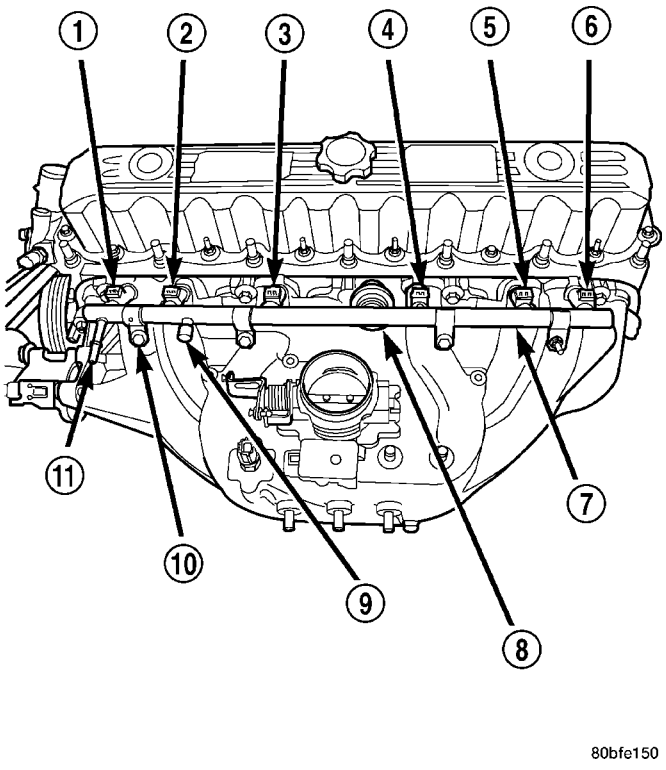


Fig. 10 FUEL RAIL MOUNTING - 4.0L

1 - INJ. #1
2 - INJ. #2
3 - INJ. #3
4 - INJ. #4
5 - INJ. #5
6 - INJ. #6
7 - FUEL INJECTOR RAIL
8 - FUEL DAMPER
9 - PRESSURE TEST PORT CAP
10 - MOUNTING BOLTS (4)
11 - QUICK-CONNECT FITTING

OPERATION

All models pass a full 360 degree rollover test without fuel leakage. To accomplish this, fuel and vapor flow controls are required for all fuel tank connections.

A fuel tank check valve(s) is mounted into the top of the fuel tank (or pump module). Refer to Fuel Tank Check Valve for additional information.

An evaporation control system is connected to the check valve(s) to reduce emissions of fuel vapors into the atmosphere. When fuel evaporates from the fuel tank, vapors pass through vent hoses or tubes to a charcoal canister where they are temporarily held. When the engine is running, the vapors are drawn into the intake manifold. Certain models are also equipped with a self-diagnosing system using either a Leak Detection Pump (LDP), or an NVLD pump. Refer to Emission Control System for additional information.

Refer to ORVR for On-Board Refueling Vapor Recovery system information.

FUEL TANK (Continued)

REMOVAL

This vehicle is equipped with an ORVR (On-Board Refueling Vapor Recovery) system. Because of this, the fuel tank may be drained the conventional way through the filler cap opening.

On this model, the fuel tank is mounted to the vehicle skid plate. The skid plate is mounted to vehicle body. The tank and skid plate are removed as one assembly.

(1) Remove fuel filler cap.

(2) Perform the Fuel System Pressure Release Procedure as described elsewhere in this group.

(3) Disconnect negative battery cable.

(4) Using an approved portable gasoline siphon/storage tank, drain fuel from tank through filler cap opening.

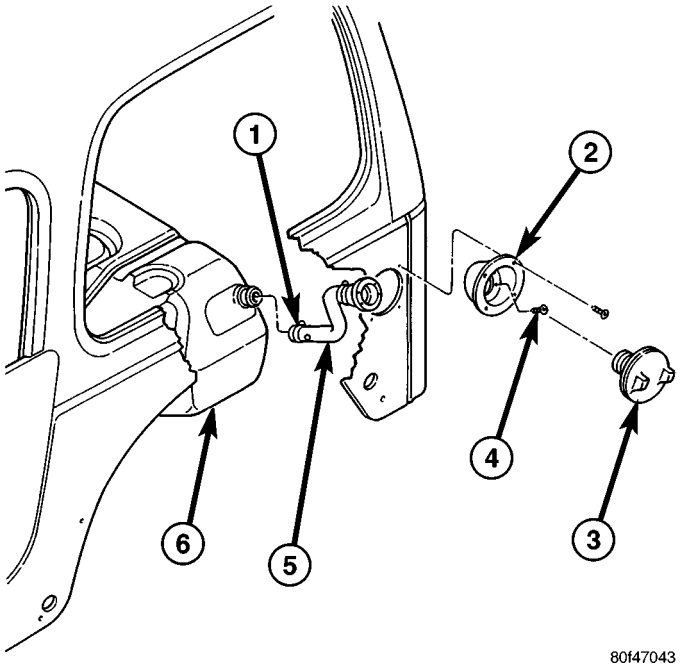


Fig. 11 FUEL FILL HOSE AT BODY

1 - FILLER HOSE CLAMP
2 - BEZEL
3 - FUEL FILLER CAP
4 - BEZEL SCREWS (8)
5 - FUEL FILLER HOSE
6 - FUEL TANK

(5) Remove 8 screws (4) retaining plastic fuel filler bezel to body (Fig. 11). Remove plastic fuel filler bezel (2).

(6) To prevent contaminants from entering tank, temporarily install fuel cap to fill hoses.

(7) Remove right/rear tire/wheel.

(8) Remove wheelhouse liner at right/rear wheel.

(9) Remove vertical support bracket (2) (Fig. 12) to gain access to ORVR vapor line.

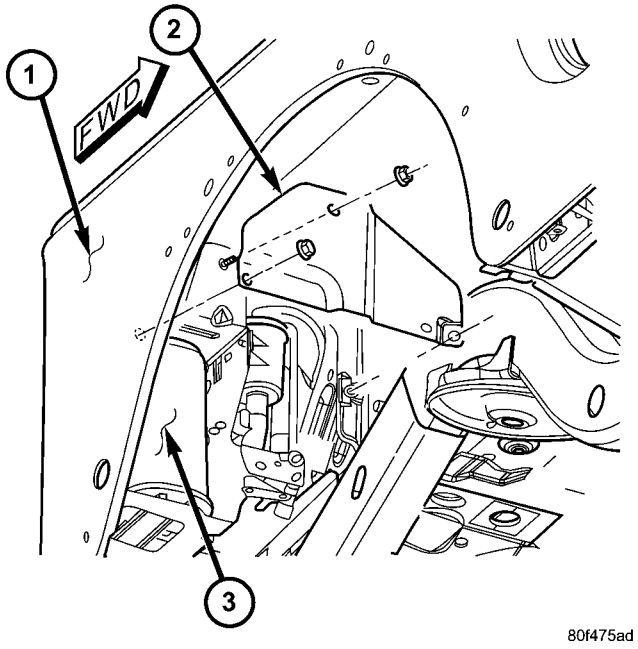


Fig. 12 EVAP CANISTER / NVLD PUMP LOCATION

1 - RIGHT / REAR FENDER (WHEELHOUSE)
2 - BRACKET
3 - EVAP CANISTER

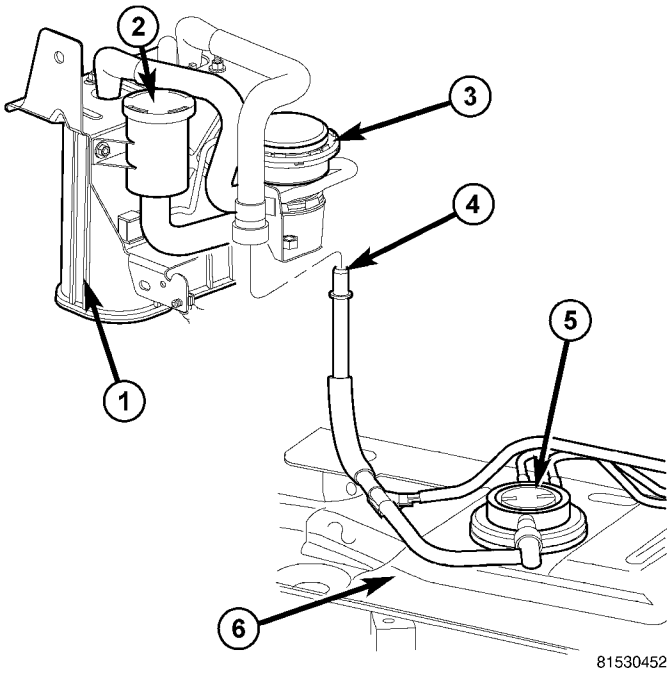


Fig. 13 ORVR VAPOR LINE

1 - EVAP CANISTER
2 - NVLD FILTER
3 - NVLD PUMP
4 - VAPOR LINES
5 - FLOW VALVE
6 - TOP OF FUEL TANK)

FUEL TANK (Continued)

(10) A vapor line (4) connects the fuel tank to the EVAP canister and NVLD pump. This connection is made near the right/rear corner of the fuel tank. Carefully disconnect this vapor line (4) (Fig. 13). Be very careful not to bend or kink the vapor lines. If vapor lines leak, a Diagnostic Trouble Code (DTC) will be set.

(11) Cut plastic tie wrap securing rear axle vent hose to fuel fill hose.

(12) Disconnect fuel tank electrical connector at left/front of fuel tank.

(13) Disconnect 2 vapor lines at left/front of fuel tank.

(14) Disconnect quick-connect fitting from fuel supply line at front of fuel tank (Fig. 14). Refer to Quick-Connect Fittings in this group for procedures.

(15) The fuel tank and skid plate are removed as an assembly. Centrally position a transmission jack (or equivalent lifting device) under skid plate/fuel tank assembly. Secure tank assembly to jack.

(16) Remove three skid plate-to-body nuts (1) at front of tank (Fig. 15). Remove one of the nuts through access hole (3) on skid plate (Fig. 15).

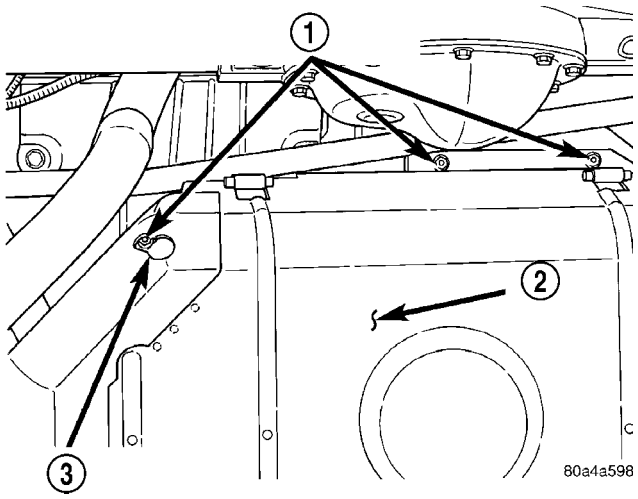


Fig. 15 FUEL TANK MOUNTING NUTS - FRONT

1 - FRONT SKID PLATE NUTS (3)
2 - SKID PLATE
3 - ACCESS HOLE

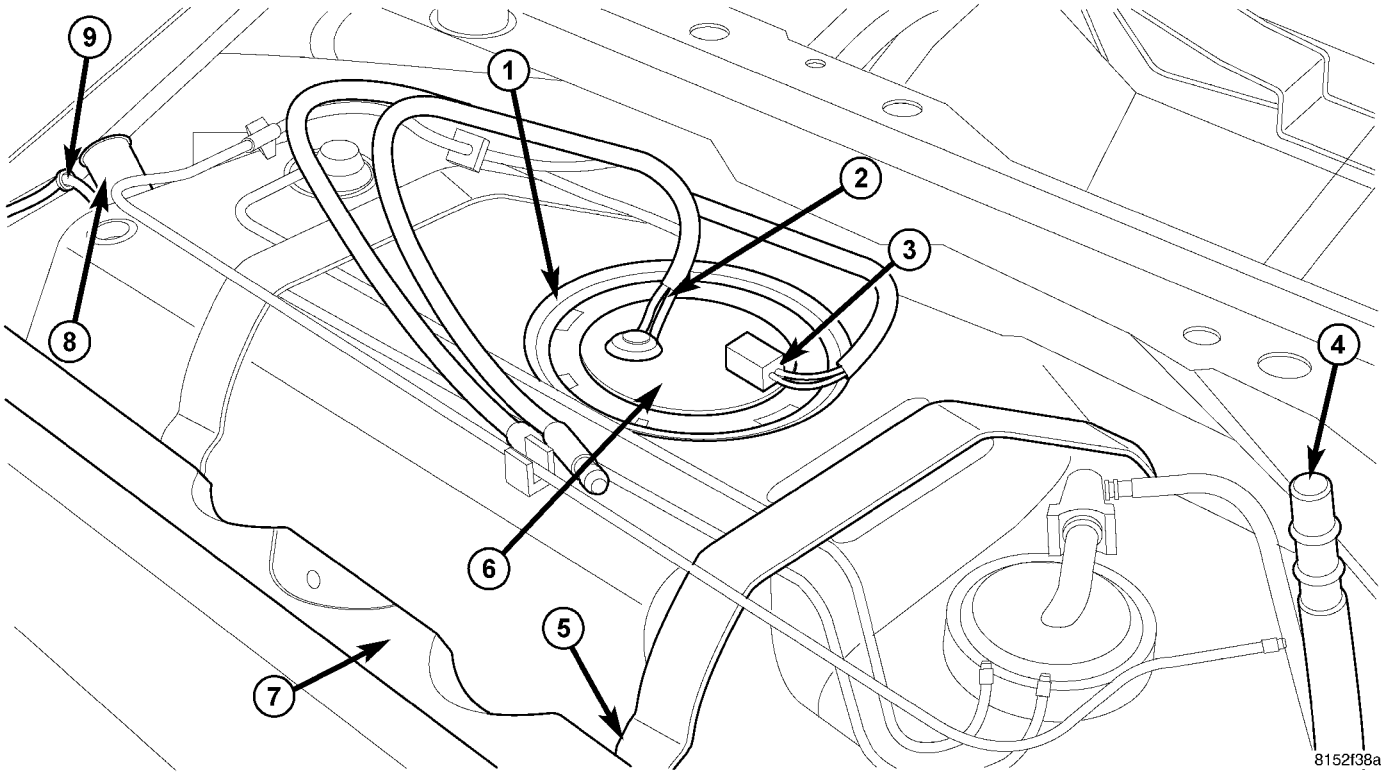


Fig. 14 TOP OF FUEL TANK

1 - FUEL PUMP MODULE LOCKRING
2 - FUEL SUPPLY LINE
3 - ELECTRICAL CONNECTOR
4 - VAPOR LINE
5 - FUEL TANK MOUNTING STRAPS (2)
6 - FUEL PUMP MODULE ASSEMBLY
7 - SKID PLATE
8 - FUEL FILL FITTING
9 - FUEL VENT FITTING

FUEL TANK (Continued)

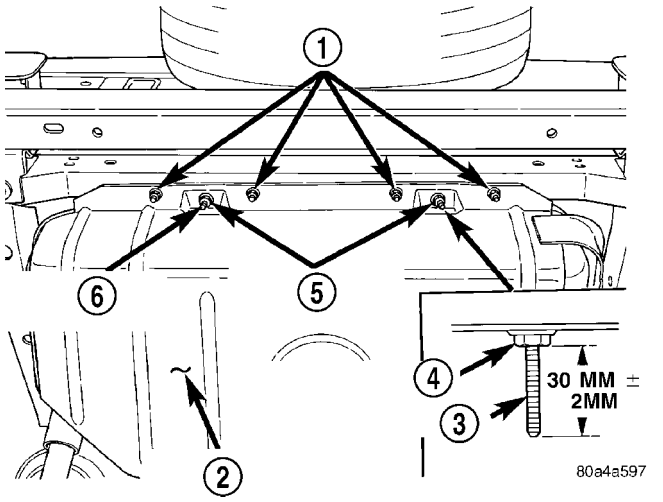


Fig. 16 FUEL TANK MOUNTING NUTS - REAR

1 - REAR SKID PLATE NUTS (4)
2 - SKID PLATE
3 - TANK STRAP STUD
4 - TANK STRAP NUT
5 - TANK STRAP NUTS (2)
6 - PROTECTIVE CAPS (2)

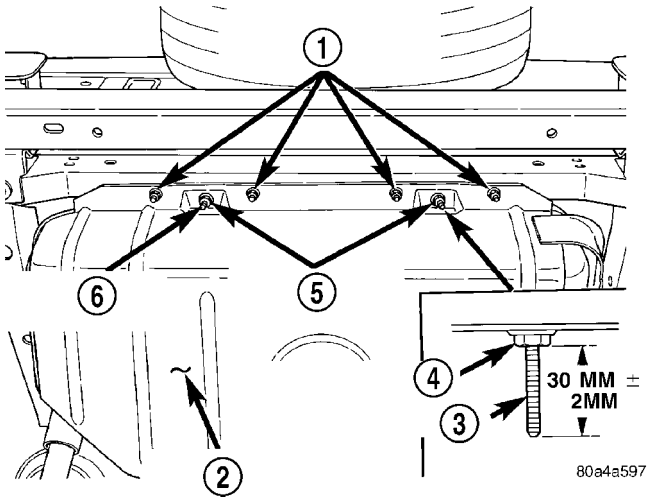


Fig. 17 FUEL TANK MOUNTING NUTS - REAR

1 - REAR SKID PLATE NUTS (4)
2 - SKID PLATE
3 - TANK STRAP STUD
4 - TANK STRAP NUT
5 - TANK STRAP NUTS (2)
6 - PROTECTIVE CAPS (2)

(17) Remove four skid plate-to-body nuts (1) at rear of tank (Fig. 16). **Do not loosen tank strap nuts (Fig. 16).**

(18) Lower the tank assembly.

(19) If fuel pump module is to be removed, refer to Fuel Pump Module Removal/Installation.

(20) Disconnect fuel filler hose at tank. Before disconnecting, mark and note the hose rotational position in relation to tank fitting.

(21) To separate tank from skid plate, remove two protective caps (6) at tank strap studs (Fig. 16) and remove tank strap nuts.

(22) Remove both straps and remove tank from skid plate.

INSTALLATION

(1) If necessary, install fuel pump module to fuel tank. Refer to Fuel Pump Module Removal / Installation.

(2) Place fuel tank into skid plate. Wrap straps around tank with strap studs inserted through holes in skid plate. Tighten strap nuts (5) (Fig. 17) to attain 30 mm (±2 mm) between bottom of nut to end of strap stud. **Do not over tighten nuts.**

(3) Install two protective caps (6) to tank strap studs.

(4) Connect fuel fill hose at tank. Tighten hose clamp.

(5) Raise skid plate/fuel tank assembly into position on body while carefully guiding plastic vapor lines and fill hose.

(6) Install four skid plate mounting nuts (1) at rear of tank (Fig. 17). Tighten to 16 N·m (141 in. lbs.) torque.

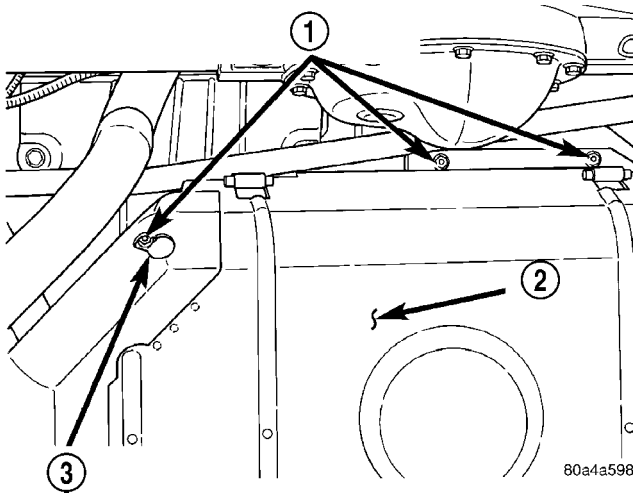


Fig. 18 FUEL TANK MOUNTING NUTS - FRONT

1 - FRONT SKID PLATE NUTS (3)
2 - SKID PLATE
3 - ACCESS HOLE

(7) Install three skid plate mounting nuts (1) at front of tank (Fig. 18). Tighten to 16 N·m (141 in. lbs.) torque.

(8) Remove tank jacking device.

FUEL TANK (Continued)

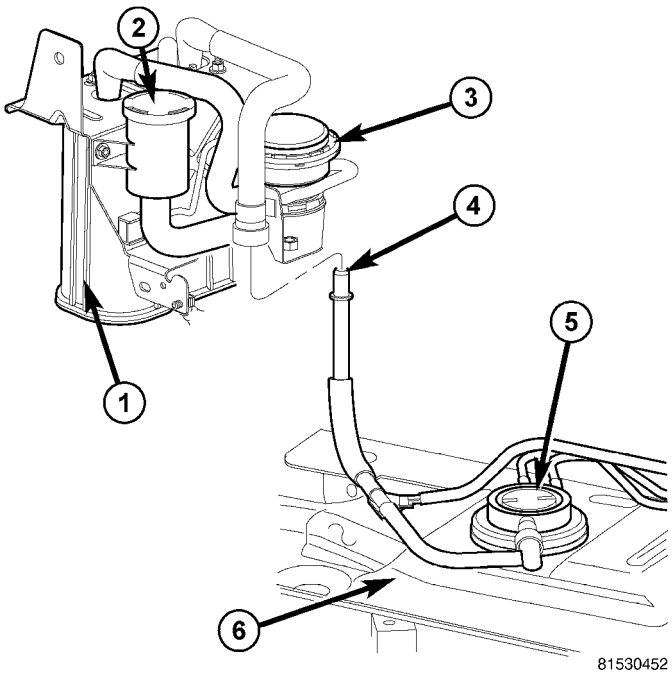


Fig. 19 ORVR VAPOR LINE

1 - EVAP CANISTER
2 - NVLD FILTER
3 - NVLD PUMP
4 - VAPOR LINES
5 - FLOW VALVE
6 - TOP OF FUEL TANK)

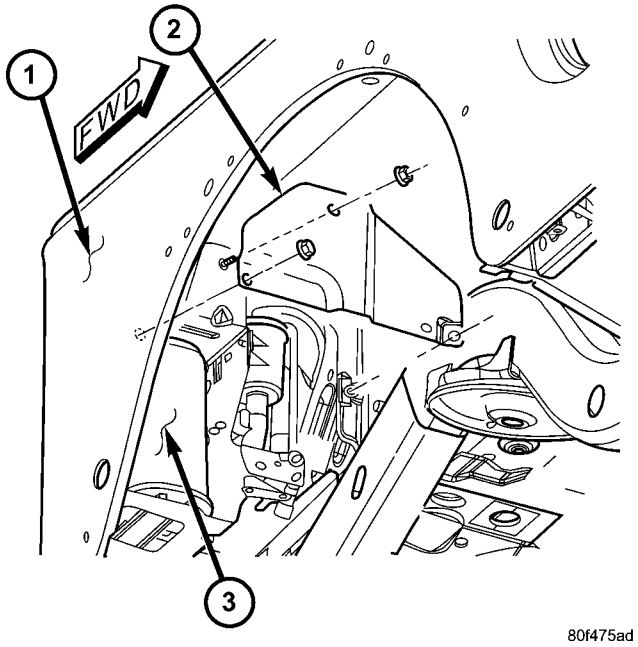


Fig. 20 EVAP CANISTER / NVLD PUMP LOCATION

1 - RIGHT / REAR FENDER (WHEELHOUSE)
2 - BRACKET
3 - EVAP CANISTER

(9) Carefully connect the vapor line (4) (Fig. 19). Be very careful not to bend or kink the vapor lines. If lines leak, a Diagnostic Trouble Code (DTC) will be set.

(10) Install EVAP canister bracket (2) (Fig. 20).

(11) Install wheelhouse liner at right/rear wheel.

(12) Install right/rear tire/wheel.

(13) Connect electrical connector at of fuel tank.

(14) Connect vapor line at left/front of fuel tank.

(15) Connect quick-connect fitting to fuel supply fitting (2) (Fig. 21). Refer to Quick-Connect Fittings in this group for procedures.

(16) Use a new plastic tie wrap to secure rear axle vent hose to fuel fill hose.

FUEL TANK (Continued)

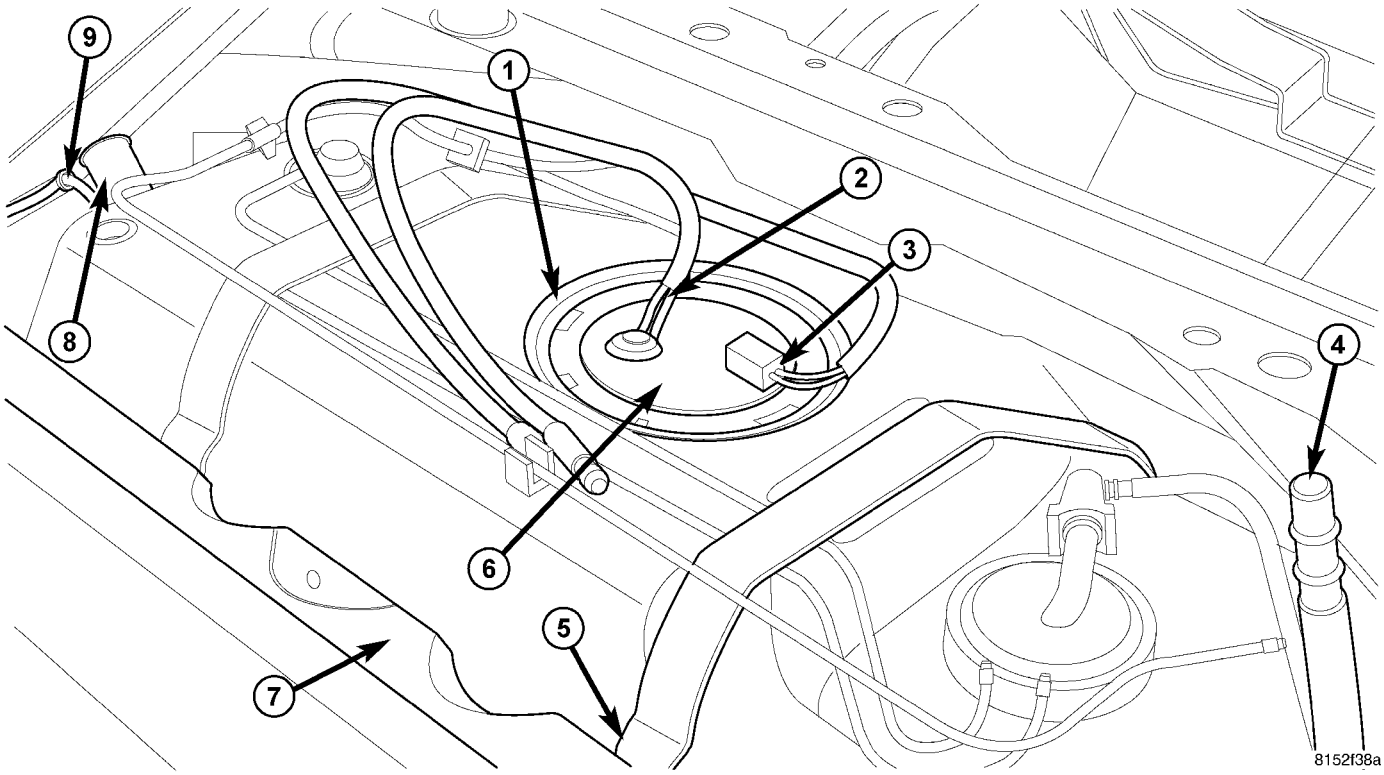


Fig. 21 TOP OF FUEL TANK

1 - FUEL PUMP MODULE LOCKRING
2 - FUEL SUPPLY LINE
3 - ELECTRICAL CONNECTOR
4 - VAPOR LINE
5 - FUEL TANK MOUNTING STRAPS (2)
6 - FUEL PUMP MODULE ASSEMBLY
7 - SKID PLATE
8 - FUEL FILL FITTING
9 - FUEL VENT FITTING

FUEL TANK (Continued)

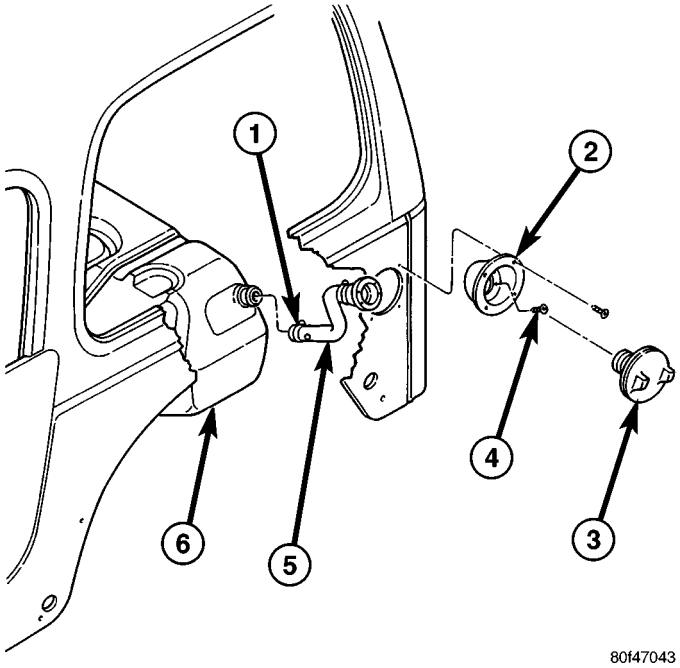


Fig. 22 FUEL FILL HOSE AT BODY

1 - FILLER HOSE CLAMP
2 - BEZEL
3 - FUEL FILLER CAP
4 - BEZEL SCREWS (8)
5 - FUEL FILLER HOSE
6 - FUEL TANK

(17) Position fuel fill bezel to body. Install eight screws (4) (Fig. 22) and tighten.

(18) Fill fuel tank. Install filler cap.

(19) Connect negative battery cable to battery.

(20) Start vehicle and inspect for leaks.

INLET FILTER

REMOVAL

The fuel pump inlet filter (strainer) is located on the bottom of fuel pump module (Fig. 23). The fuel pump module is located on top of fuel tank.

(1) Remove fuel tank. Refer to Fuel Tank Removal/Installation.

(2) Remove fuel pump module. Refer to Fuel Pump Module Removal/Installation.

(3) Remove filter by prying from bottom of module with 2 screwdrivers. Filter is snapped to module.

(4) Clean bottom of pump module.

INSTALLATION

The fuel pump inlet filter (strainer) is located on the bottom of fuel pump module (Fig. 24). The fuel pump module is located on top of fuel tank.

(1) Snap new filter to bottom of module.

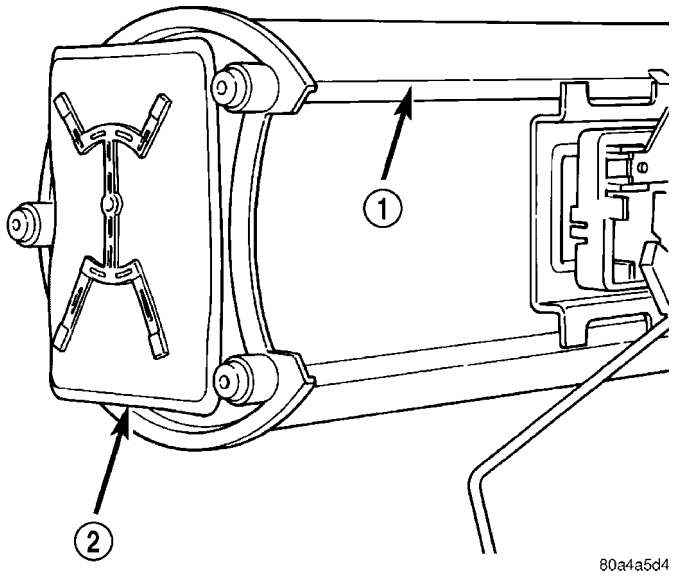


Fig. 23 Fuel Pump Inlet Filter

1 - FUEL PUMP MODULE
2 - FUEL PUMP INLET FILTER

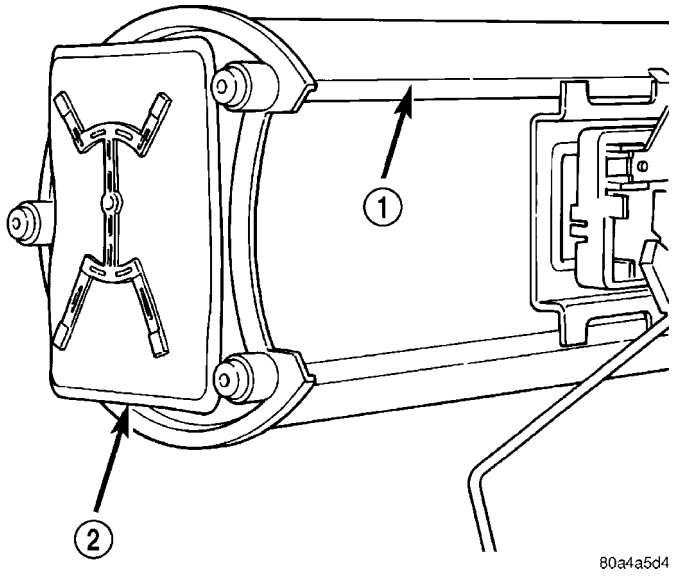


Fig. 24 Fuel Pump Inlet Filter

1 - FUEL PUMP MODULE
2 - FUEL PUMP INLET FILTER

(2) Install fuel pump module. Refer to Fuel Pump Module Removal/Installation.

(3) Install fuel tank. Refer to Fuel Tank Removal/Installation.

QUICK CONNECT FITTING

DESCRIPTION

Different types of quick-connect fittings are used to attach the various fuel system components, lines and tubes. These are: a single-button type, a two-button type, a pinch type, a single-tab type, a two-tab type or a plastic retainer ring type. Some are equipped with safety latch clips. Some may require the use of a special tool for disconnection and removal. Refer to Quick-Connect Fittings Removal/Installation for more information.

CAUTION: Before separating a quick-connect fitting, pay attention to what type of fitting is being used by referring to Quick-Connect Fitting Removal. This will prevent unnecessary fitting or fitting latch breakage.

CAUTION: The interior components (o-rings, clips) of quick-connect fittings are not serviced separately, but new plastic spacers and latches are available for some types. If service parts are not available, do not attempt to repair the damaged fitting or fuel line (tube). If repair is necessary, replace the complete fuel line (tube) assembly.

STANDARD PROCEDURE - QUICK-CONNECT FITTINGS

Different types of quick-connect fittings are used to attach the various fuel system components, lines and tubes. These are: a single-button type, a two-button type, a pinch type, a single-tab type, a two-tab type or a plastic retainer ring type. Some are equipped with safety latch clips. Some may require the use of a special tool for disconnection and removal.

DISCONNECTING

WARNING: THE FUEL SYSTEM IS UNDER A CONSTANT PRESSURE (EVEN WITH ENGINE OFF). BEFORE SERVICING ANY FUEL SYSTEM HOSE, FITTING OR LINE, FUEL SYSTEM PRESSURE MUST BE RELEASED. REFER TO FUEL SYSTEM PRESSURE RELEASE PROCEDURE.

CAUTION: Before separating a quick-connect fitting, pay attention to what type of fitting is being used by referring to Quick-Connect Fitting Removal. This will prevent unnecessary fitting or fitting latch breakage.

CAUTION: The interior components (o-rings, clips) of quick-connect fittings are not serviced separately, but new plastic spacers and latches are available for some types. If service parts are not available, do not attempt to repair the damaged fitting or fuel line (tube). If repair is necessary, replace the complete fuel line (tube) assembly.

(1) Perform fuel pressure release procedure. Refer to Fuel Pressure Release Procedure.

(2) Disconnect negative battery cable from battery.

(3) Clean fitting of any foreign material before disassembly.

(4) **Single–Button Type Fitting:** This type of fitting is equipped with a single push-button located on the quick-connect fitting (Fig. 25). The push-button is attached to two internal latches (Fig. 26). To disconnect, press on button with your thumb and unlatch fitting from fuel line. Special tools are not required for disconnection. **DO NOT ATTEMPT TO PRY OR PULL UP ON PUSH-BUTTON. LATCHES WILL BE BROKEN.**

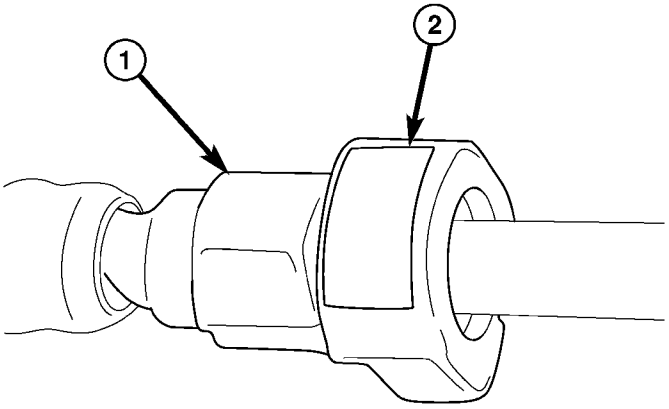


Fig. 25 SINGLE BUTTON FITTING

1 - QUICK-CONNECT FITTING
2 - SINGLE PUSH-BUTTON

QUICK CONNECT FITTING (Continued)

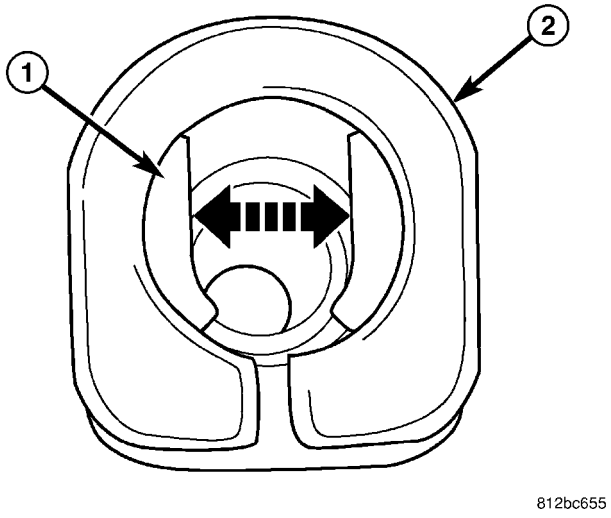


Fig. 26 FITTING LATCHES

1 - LATCHES
2 - QUICK-CONNECT FITTING

(5) **2–Button Type Fitting:** This type of fitting is equipped with a push-button located on each side of quick-connect fitting (Fig. 27). Press on both buttons simultaneously for removal. Special tools are not required for disconnection.

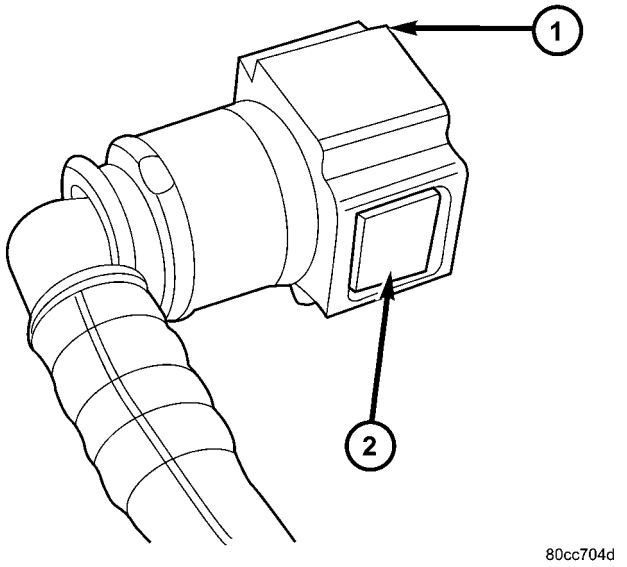


Fig. 27 2-BUTTON TYPE FITTING

1 - QUICK-CONNECT FITTING
2 - PUSH-BUTTONS (2)

(6) **Pinch-Type Fitting:** This fitting is equipped with two finger tabs. Pinch both tabs together while removing fitting (Fig. 28). Special tools are not required for disconnection.

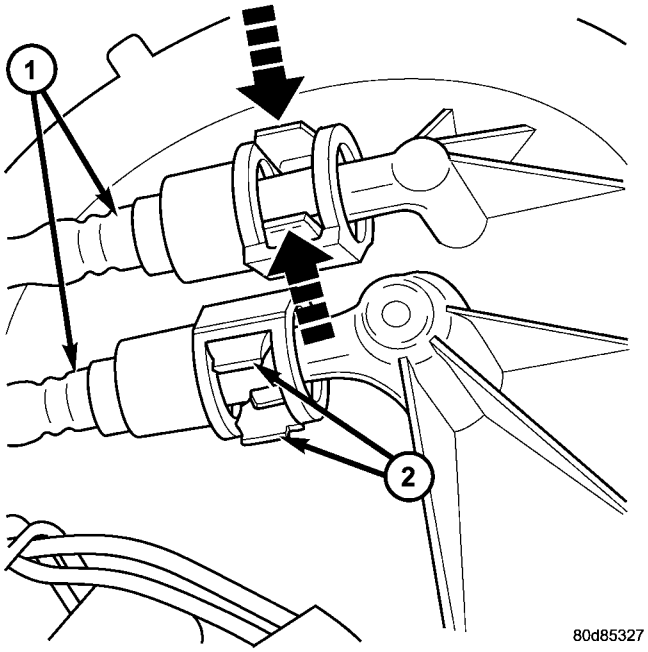


Fig. 28 PINCH TYPE QUICK-CONNECT FITTING

1 - QUICK-CONNECT FITTINGS
2 - PINCH TABS

(7) **Single-Tab Type Fitting:** This type of fitting is equipped with a single pull tab (Fig. 29). The tab is removable. After tab is removed, quick-connect fitting can be separated from fuel system component. Special tools are not required for disconnection.

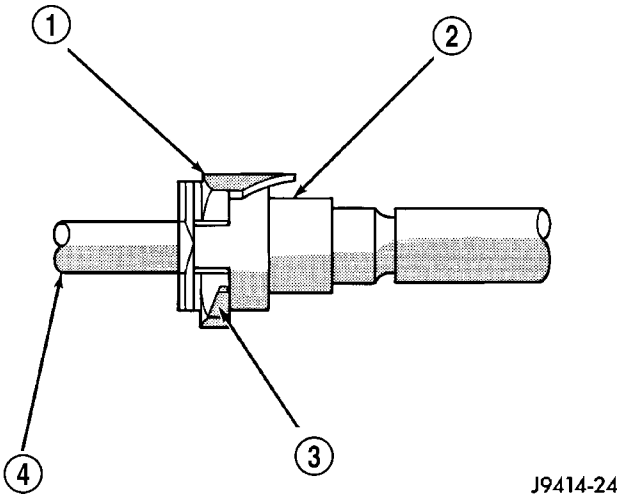


Fig. 29 SINGLE-TAB TYPE FITTING

1 - PULL TAB
2 - QUICK-CONNECT FITTING
3 - PRESS HERE TO REMOVE PULL TAB
4 - INSERTED TUBE END

QUICK CONNECT FITTING (Continued)

(a) Press release tab on side of fitting to release pull tab (Fig. 30). **If release tab is not pressed prior to releasing pull tab, pull tab will be damaged.**

(b) While pressing release tab on side of fitting, use screwdriver to pry up pull tab (Fig. 30).

(c) Raise pull tab until it separates from quick-connect fitting (Fig. 31).

(8) **Two-Tab Type Fitting:** This type of fitting is equipped with tabs located on both sides of fitting (Fig. 32). The tabs are supplied for disconnecting quick-connect fitting from component being serviced.

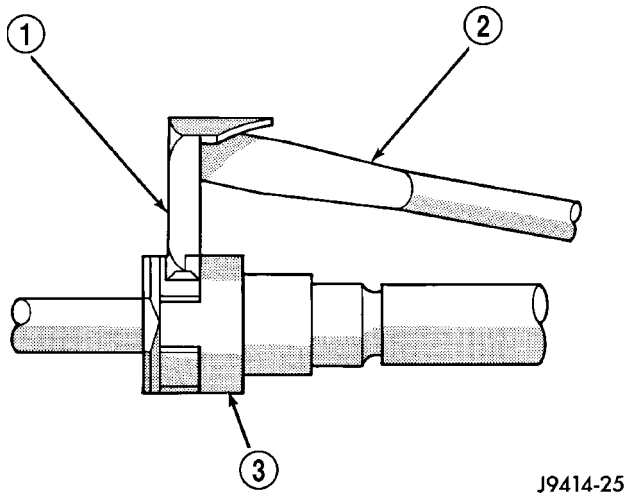


Fig. 30 DISCONNECTING SINGLE-TAB TYPE FITTING

1 - PULL TAB
2 - SCREWDRIVER
3 - QUICK-CONNECT FITTING

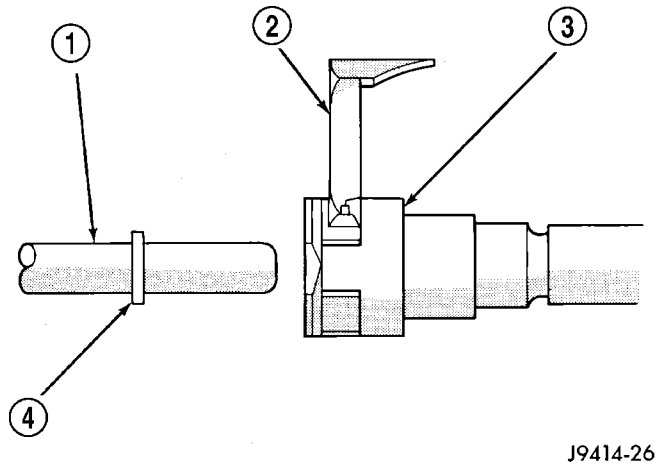


Fig. 31 REMOVING PULL TAB

1 - FUEL TUBE OR FUEL SYSTEM COMPONENT
2 - PULL TAB
3 - QUICK-CONNECT FITTING
4 - FUEL TUBE STOP

(a) To disconnect quick-connect fitting, squeeze plastic retainer tabs (Fig. 32) against sides of quick-connect fitting with your fingers. Tool use is not required for removal and may damage plastic retainer.

(b) Pull fitting from fuel system component being serviced.

(c) The plastic retainer will remain on component being serviced after fitting is disconnected.

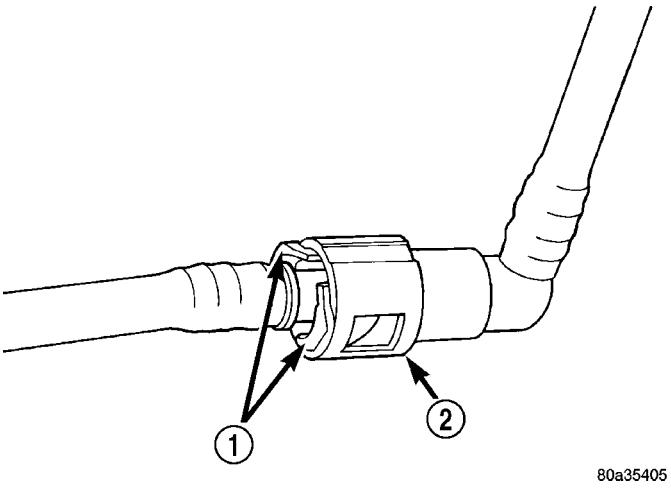
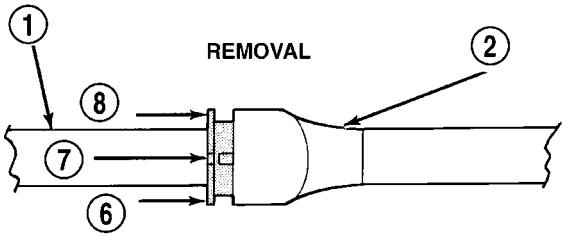


Fig. 32 TYPICAL 2–TAB TYPE FITTING

1 - TAB(S)
2 - QUICK-CONNECT FITTING



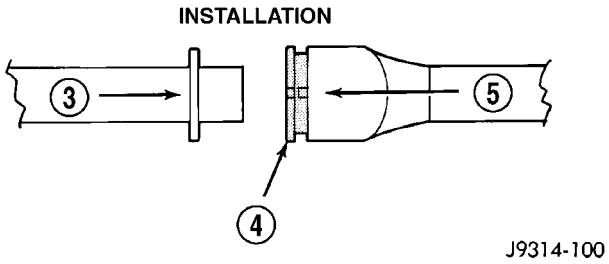


Fig. 33 PLASTIC RETAINER RING TYPE FITTING

1 - FUEL TUBE
2 - QUICK CONNECT FITTING
3 - PUSH
4 - PLASTIC RETAINER
5 - PUSH
6 - PUSH
7 - PUSH
8 - PUSH

QUICK CONNECT FITTING (Continued)

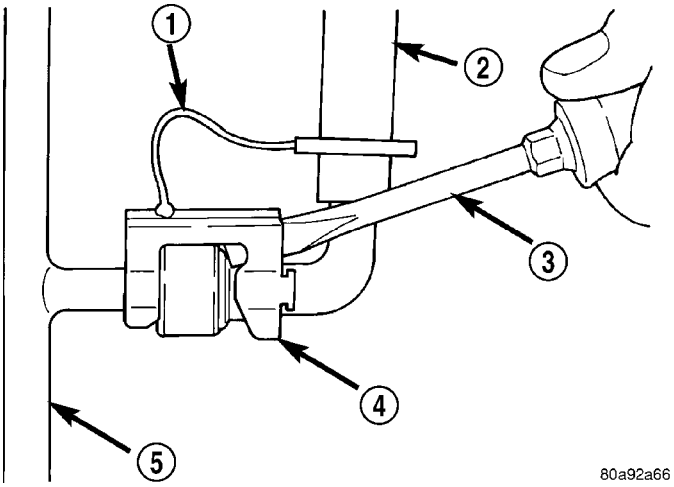


Fig. 34 LATCH CLIP-TYPE 1

1 - TETHER STRAP
2 - FUEL LINE
3 - SCREWDRIVER
4 - LATCH CLIP
5 - FUEL RAIL

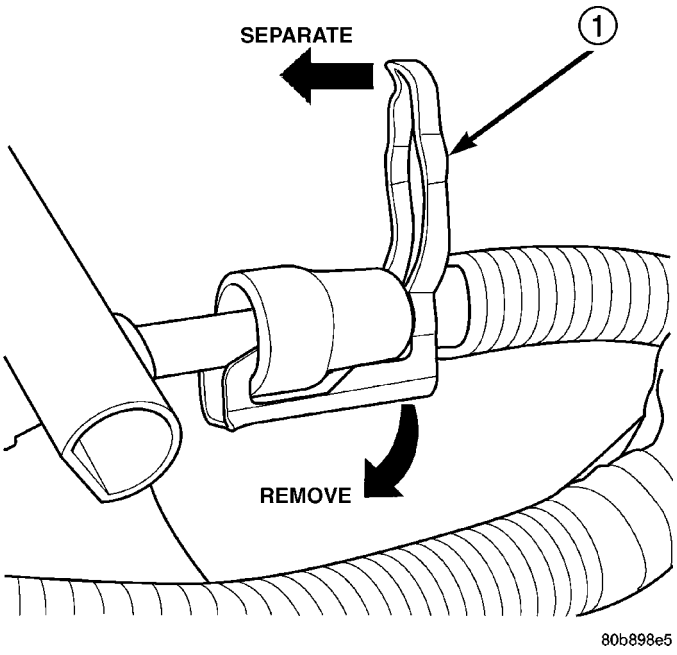


Fig. 35 LATCH CLIP-TYPE 2

1 - LATCH CLIP

The o-rings and spacer will remain in quick-connect fitting connector body.

(9) **Plastic Retainer Ring Type Fitting:** This type of fitting can be identified by the use of a full-round plastic retainer ring (Fig. 33) usually black in color.

(a) To release fuel system component from quick-connect fitting, firmly push fitting towards component being serviced while firmly pushing plastic retainer ring into fitting (Fig. 33). With plastic ring depressed, pull fitting from component. **The plastic retainer ring must be pressed squarely into fitting body. If this retainer is cocked during removal, it may be difficult to disconnect fitting. Use an open-end wrench on shoulder of plastic retainer ring to aid in disconnection.**

(b) After disconnection, plastic retainer ring will remain with quick-connect fitting connector body.

(c) Inspect fitting connector body, plastic retainer ring and fuel system component for damage. Replace as necessary.

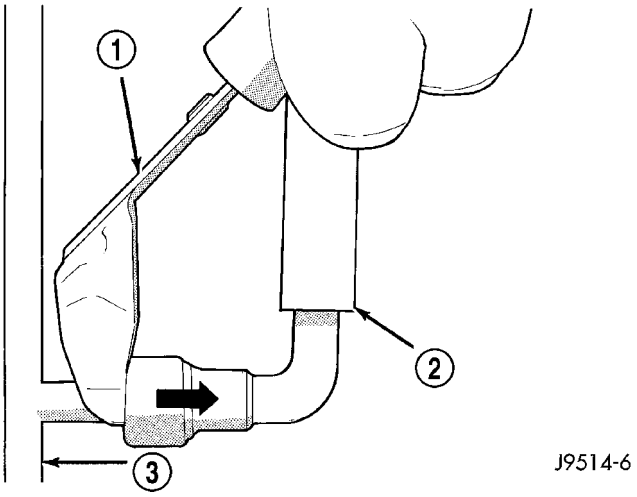


Fig. 36 FUEL LINE DISCONNECTION USING SPECIAL TOOL

1 - SPECIAL FUEL LINE TOOL
2 - FUEL LINE
3 - FUEL RAIL

(10) **Latch Clips:** Depending on vehicle model and engine, 2 different types of safety latch clips are used (Fig. 34) or (Fig. 35). Type-1 is tethered to fuel line and type-2 is not. A special tool will be necessary to disconnect fuel line after latch clip is removed. The latch clip may be used on certain fuel line/fuel rail connection, or to join fuel lines together.

(a) Type 1: Pry up on latch clip with a screwdriver (Fig. 34).

(b) Type 2: Separate and unlatch 2 small arms on end of clip (Fig. 35) and swing away from fuel line.

(c) Slide latch clip toward fuel rail while lifting with screwdriver.

(d) Insert special fuel line removal tool (Snap-On number FIH 9055-1 or equivalent) into fuel line (Fig. 36). Use tool to release locking fingers in end of line.

(e) With special tool still inserted, pull fuel line from fuel rail.

(f) After disconnection, locking fingers will remain within quick-connect fitting at end of fuel line.

QUICK CONNECT FITTING (Continued)

(11) Disconnect quick-connect fitting from fuel system component being serviced.

CONNECTING

(1) Inspect quick-connect fitting body and fuel system component for damage. Replace as necessary.

(2) Prior to connecting quick-connect fitting to component being serviced, check condition of fitting and component. Clean parts with a lint-free cloth. Lubricate with clean engine oil.

(3) Insert quick-connect fitting into fuel tube or fuel system component until built-on stop on fuel tube or component rests against back of fitting.

(4) Continue pushing until a click is felt.

(5) Single-tab type fitting: Push new tab down until it locks into place in quick-connect fitting.

(6) Verify a locked condition by firmly pulling on fuel tube and fitting (15-30 lbs.).

(7) Latch Clip Equipped: Install latch clip (snaps into position). **If safety latch clip will not fit, this indicates fuel line is not properly installed to fuel rail (or other fuel line). Recheck fuel line connection.**

(8) Connect negative cable to battery.

(9) Start engine and check for leaks.

FUEL INJECTION

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FUEL INJECTION

DESCRIPTION

The Powertrain Control Module (PCM) operates the fuel injection system. Refer to Powertrain Control Module in Electronic Control Modules for information.

SPECIFICATIONS

TORQUE - FUEL INJECTION

DESCRIPTION	N-m	Ft. Lbs.	In. Lbs.
Accelerator Pedal Bracket Mounting Nuts	8.5	-	75
Engine Coolant Temperature Sensor - 2.4L/4.0L	11	-	96
IAC Motor-To-Throttle Body Bolts - 2.4L/4.0L	7	-	60
Intake Manifold Air Temp. Sensor - 4.0L	28	20	-
MAP Sensor Mounting Screws - 2.4L/4.0L	3	-	25
Oxygen Sensor - 2.4L/4.0L	30	22	-
PCM Mounting Screws	4	-	35
Throttle Body Mounting Bolts - 2.4L/4.0L	11	-	100
Throttle Position Sensor Mounting Screws - 2.4L/4.0L	7	-	60
Vehicle Speed Sensor Mounting Bolt	2.2	-	20

ACCELERATOR PEDAL

REMOVAL

The accelerator pedal is connected to the throttle body linkage by the throttle cable. The cable is protected by a plastic sheathing and is connected to the throttle body linkage by a ball socket. It is connected to the upper part of the accelerator pedal arm by a plastic retainer (clip) (Fig. 1). This retainer (clip) snaps into the top of the accelerator pedal arm. Retainer tabs (built into the cable sheathing) (Fig. 1) fasten the cable to the dash panel.

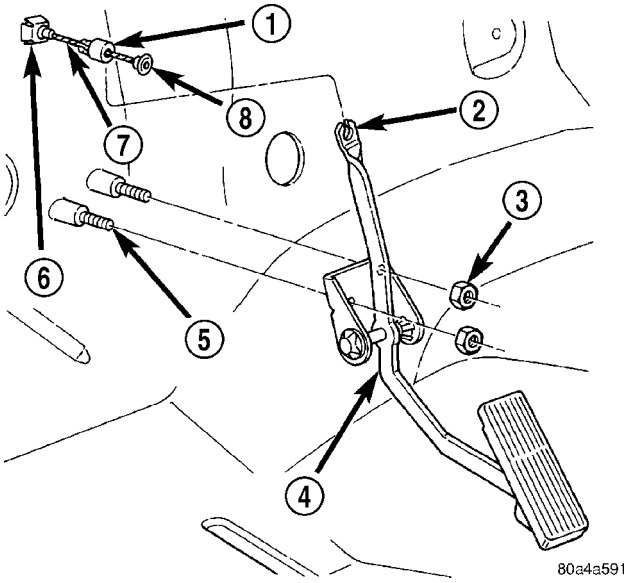

Fig. 1 Accelerator Pedal Mounting

1 - CABLE RETAINER
2 - SLOT
3 - MOUNTING STUDS (2)
4 - PEDAL/BRACKET ASSEMBLY
5 - MOUNTING STUDS
6 - RETAINER TABS
7 - CABLE
8 - CABLE STOP

CAUTION: Be careful not to damage or kink the cable core wire (within the cable sheathing) while servicing the accelerator pedal or throttle cable.

(1) From inside vehicle, hold up accelerator pedal. Remove plastic cable retainer (clip) and throttle cable core wire from upper end of accelerator pedal arm (Fig. 1). Plastic cable retainer (clip) snaps into pedal arm.

(2) Remove accelerator pedal mounting bracket nuts.

(3) Remove accelerator pedal assembly.

INSTALLATION

(1) Place accelerator pedal assembly over mounting studs protruding from floor pan. Tighten mounting nuts to 8.5 N·m (75 in. lbs.) torque.

(2) Slide throttle cable into opening (slot) in top of pedal arm. An index tab is located on pedal arm. Rotate and push plastic cable retainer (clip) into accelerator pedal arm opening until it snaps into place on index tab.

(3) Before starting engine, operate accelerator pedal to check for any binding.

CRANKSHAFT POSITION SENSOR

DESCRIPTION

2.4L

The Crankshaft Position (CKP) sensor is mounted into the right front side of the cylinder block. It is positioned and bolted into a machined hole.

4.0L

The Crankshaft Position (CKP) sensor is located near the outer edge of the flywheel (starter ringear).

OPERATION

2.4L

Engine speed and crankshaft position are provided through the CKP (Crankshaft Position) sensor. The sensor generates pulses that are the input sent to the Powertrain Control Module (PCM). The PCM interprets the sensor input to determine the crankshaft position. The PCM then uses this position, along with other inputs, to determine injector sequence and ignition timing.

The sensor is a hall effect device combined with an internal magnet. It is also sensitive to steel within a certain distance from it.

A tonewheel (targetwheel) is a part of the engine crankshaft (Fig. 2). This tonewheel has sets of notches at its outer edge.

The notches cause a pulse to be generated when they pass under the sensor. The pulses are the input to the PCM.

CRANKSHAFT POSITION SENSOR (Continued)

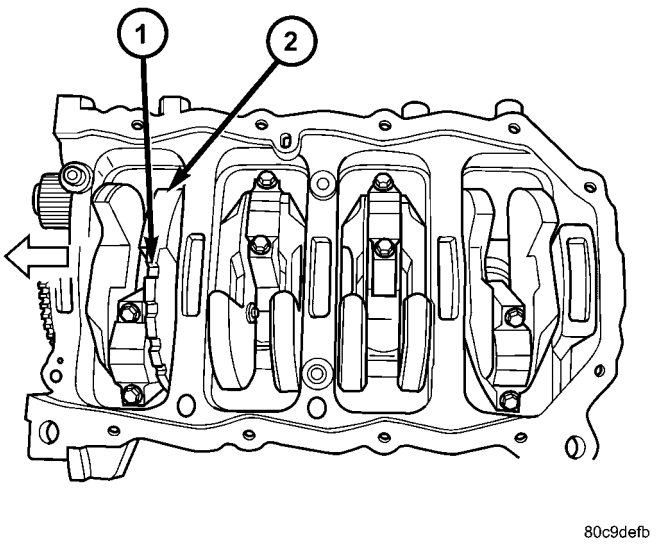

Fig. 2 CKP OPERATION - 2.4L

1 - NOTCHES
2 - CRANKSHAFT

4.0L

Engine speed and crankshaft position are provided through the CKP sensor. The sensor generates pulses that are the input sent to the Powertrain Control Module (PCM). The PCM interprets the sensor input to determine the crankshaft position. The PCM then uses this position, along with other inputs, to determine injector sequence and ignition timing.

The sensor is a hall effect device combined with an internal magnet. It is also sensitive to steel within a certain distance from it.

The flywheel/drive plate has groups of four notches at its outer edge. On 4.0L 6-cylinder engines there are three sets of notches (Fig. 3).

The notches cause a pulse to be generated when they pass under the sensor. The pulses are the input to the PCM. For each engine revolution there are 3 groups of four pulses generated on 4.0L 6-cylinder engines.

The trailing edge of the fourth notch, which causes the pulse, is four degrees before top dead center (TDC) of the corresponding piston.

The engine will not operate if the PCM does not receive a CKP sensor input.

REMOVAL

2.4L

The Crankshaft Position (CKP) sensor is mounted into the right front side of the cylinder block (Fig. 4). It is positioned and bolted into a machined hole.

(1) Disconnect sensor electrical connector.
(2) Remove sensor bolt.

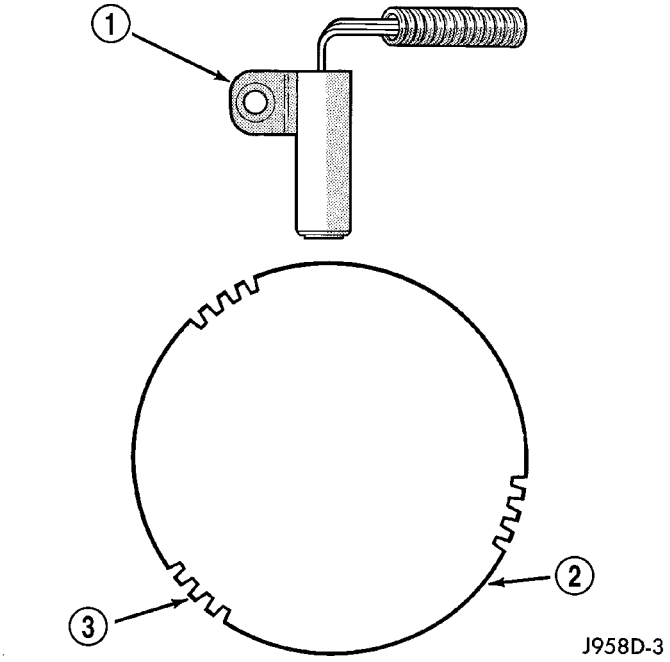

Fig. 3 CKP OPERATION - 4.0L

1 - CRANKSHAFT POSITION SENSOR
2 - FLYWHEEL
3 - FLYWHEEL NOTCHES

(3) Carefully twist sensor from cylinder block.
(4) Check condition of sensor o-ring (Fig. 5).

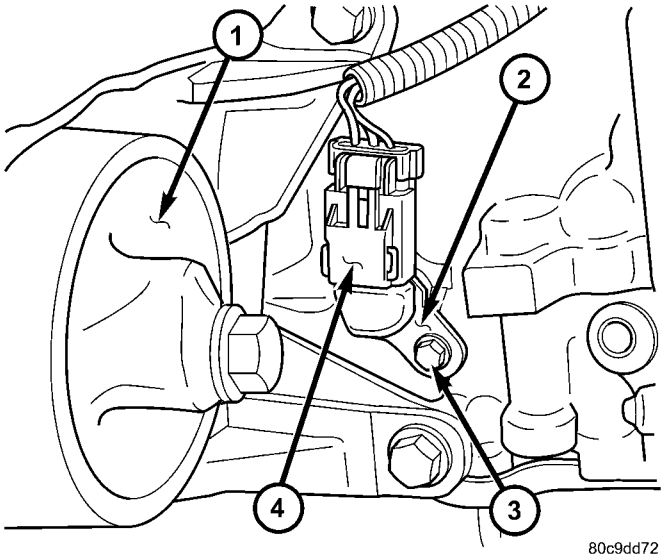

Fig. 4 CKP SENSOR LOCATION - 2.4L

1 - RIGHT FRONT ENGINE MOUNT
2 - CKP SENSOR
3 - MOUNTING BOLT
4 - ELECTRICAL CONNECTOR

CRANKSHAFT POSITION SENSOR (Continued)

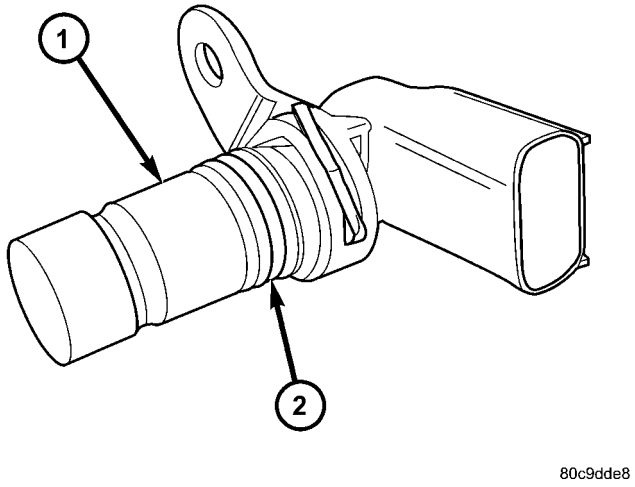

Fig. 5 CKP AND O-RING - 2.4L

1 - CKP SENSOR
2 - O-RING

4.0L

The crankshaft position (CKP) sensor is mounted to the transmission bellhousing near the rear of the engine block.

The sensor may be mounted to the transmission with one of the following four different configurations:

- with one bolt to the right side of the transmission if equipped with a 42RLE automatic transmission (Fig. 6).
- with one bolt to the left side of the transmission (Fig. 7). If sensor is equipped with one mounting bolt, **it is adjustable.**.
- with two nuts to the left side of the transmission (Fig. 8).
- with two bolts to the left side of the transmission (Fig. 9).

(1) Disconnect sensor pigtail harness (electrical connector) from main electrical harness.

(2) Depending upon application, remove either sensor mounting bolt(s) or nuts.

(3) Remove sensor from engine.

INSTALLATION

2.4L

(1) Clean out machined hole in engine block.

(2) Apply a small amount of engine oil to sensor o-ring.

(3) Install sensor into engine block with a slight rocking action. Do not twist sensor into position as damage to o-ring may result.

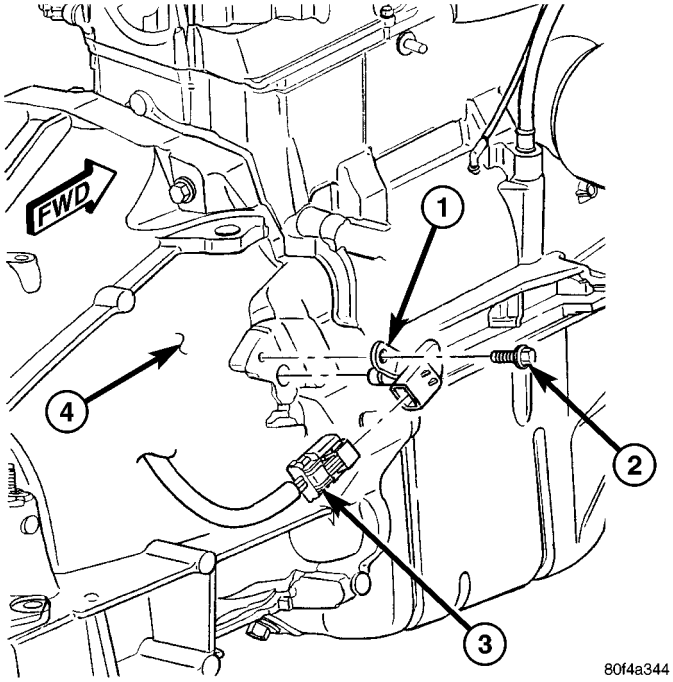

Fig. 6 CKP (WITH 42RLE AUTO. TRANS.)

1 - CKP SENSOR
2 - MOUNTING BOLT
3 - ELEC. CONNECTOR
4 - TRANS. BELLHOUSING

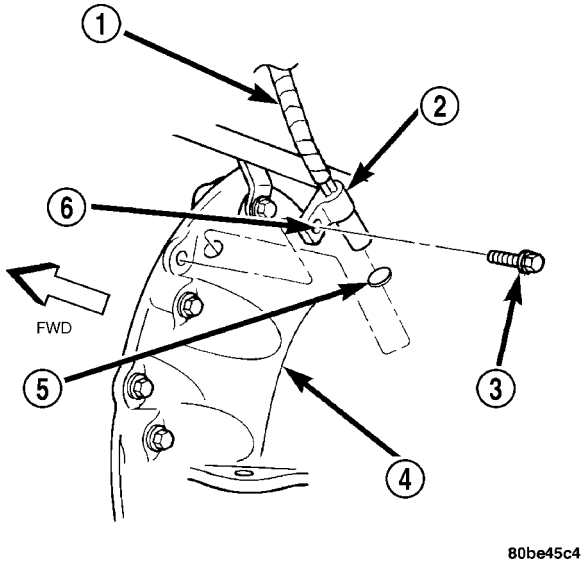

Fig. 7 CKP - ONE-BOLT MOUNTING (EXCEPT 42RLE AUTO. TRANS.)

1 - SENSOR PIGTAIL
2 - CRANKSHAFT POSITION SENSOR
3 - MOUNTING BOLT
4 - TRANSMISSION HOUSING
5 - PAPER SPACER
6 - SLOTTED HOLE

CRANKSHAFT POSITION SENSOR (Continued)

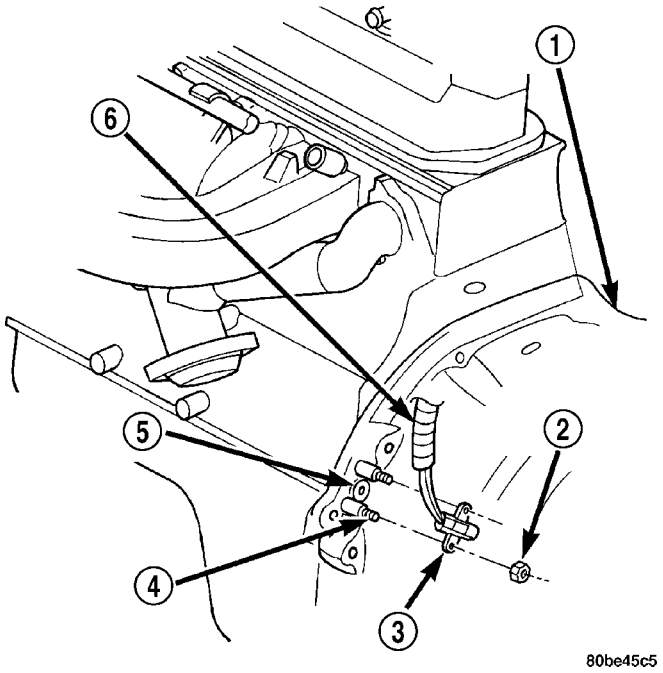

Fig. 8 CKP - TWO-NUT MOUNTING

1 - TRANSMISSION BELLHOUSING
2 - MOUNTING NUTS (2)
3 - CRANKSHAFT POSITION SENSOR
4 - MOUNTING STUDS (2)
5 - RUBBER GROMMET
6 - SENSOR PIGTAIL

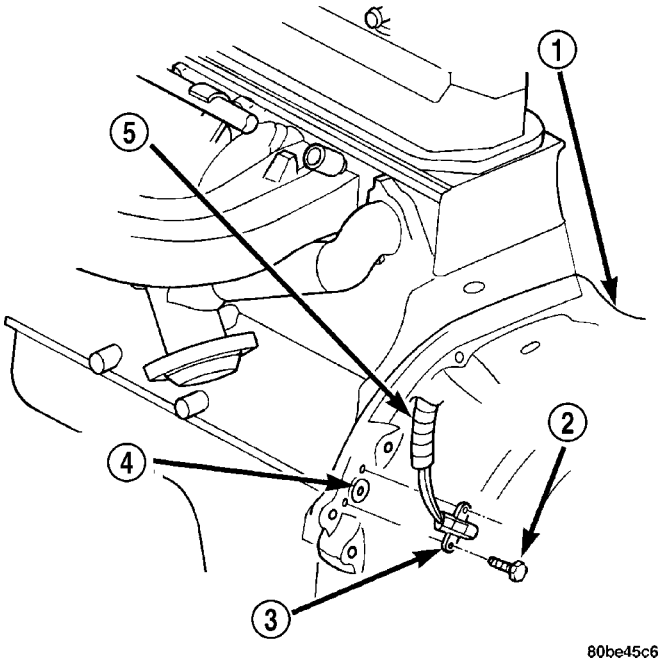

Fig. 9 CKP - TWO-BOLT MOUNTING

1 - TRANSMISSION BELLHOUSING
2 - MOUNTING BOLTS (2)
3 - CRANKSHAFT POSITION SENSOR
4 - RUBBER GROMMET
5 - SENSOR PIGTAIL

CAUTION: Before tightening sensor mounting bolt, be sure sensor is completely flush to cylinder block. If sensor is not flush, damage to sensor mounting tang may result.

(4) Install mounting bolt and tighten to 28 N·m (21 ft. lbs.) torque.
(5) Connect electrical connector to sensor.

4.0L

Sensor With 1–Bolt Mounting (Equipped With 42 RLE Automatic Transmission):
(1) Install sensor flush against opening in transmission housing.
(2) Install and tighten sensor mounting bolt to 12 N·m (9 ft. lbs.) torque.
(3) Connect electrical connector to sensor.

Sensor With 2–Bolt Mounting:
(4) Install sensor flush against opening in transmission housing.
(5) Install and tighten two sensor mounting bolts to 12 N·m (9 ft. lbs.) torque. The two sensor mounting bolts are specially machined to correctly space unit to flywheel. Do not attempt to install any other bolts.
(6) Connect sensor pigtail harness electrical connector to main wiring harness.

Sensor With 2–Nut Mounting:
(7) Install and tighten two sensor mounting nuts to 12 N·m (9 ft. lbs.) torque.
(8) Connect sensor pigtail harness electrical connector to main wiring harness.

Sensor With One-Bolt Mounting (Not Equipped With 42 RLE Automatic Transmission):
New replacement sensors will be equipped with a paper spacer glued to bottom of sensor. If installing (returning) a **used** sensor to vehicle, a new paper spacer must be installed to bottom of sensor. This spacer will be ground off the first time engine is started. If spacer is not used, sensor will be broken the first time engine is started.
(9) New Sensors: Be sure paper spacer is installed to bottom of sensor. If not, obtain spacer PN05252229.
(10) Used Sensors: Clean bottom of sensor and install spacer PN05252229.
(11) Install sensor into transmission bellhousing hole.
(12) Push sensor against flywheel/drive plate. With sensor pushed against flywheel/drive plate, tighten mounting bolt to 7 N·m (60 in. lbs.) torque.
(13) Connect sensor pigtail harness electrical connector to main wiring harness.

FUEL INJECTOR

DESCRIPTION

An individual fuel injector (Fig. 10) is used for each individual cylinder.

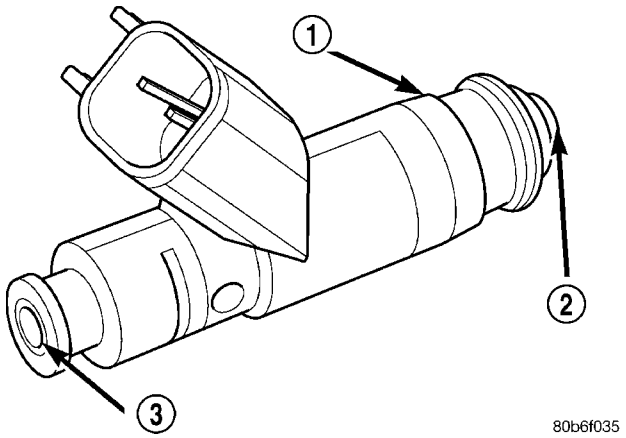

Fig. 10 FUEL INJECTOR - TYPICAL

1 - FUEL INJECTOR
2 - NOZZLE
3 - TOP (FUEL ENTRY)

OPERATION

OPERATION - PCM OUTPUT

The nozzle ends of the injectors are positioned into openings in the intake manifold just above the intake valve ports of the cylinder head. The engine wiring harness connector for each fuel injector is equipped with an attached numerical tag (INJ 1, INJ 2 etc.). This is used to identify each fuel injector with its respective cylinder number.

The injectors are energized individually in a sequential order by the Powertrain Control Module (PCM). The PCM will adjust injector pulse width by switching the ground path to each individual injector on and off. Injector pulse width is the period of time that the injector is energized. The PCM will adjust injector pulse width based on various inputs it receives.

Battery voltage (12 volts +) is supplied to the injectors through the ASD relay. The ASD relay will shutdown the 12 volt power source to the fuel injectors if the PCM senses the ignition is on, but the engine is not running. This occurs after the engine has not been running for approximately 1.8 seconds.

The PCM determines injector on-time (pulse width) based on various inputs.

OPERATION - FUEL INJECTOR

The top (fuel entry) end of the injector (Fig. 10) is attached into an opening on the fuel rail.

The fuel injectors are electrical solenoids. The injector contains a pintle that closes off an orifice at the nozzle end. When electric current is supplied to the injector, the armature and needle move a short distance against a spring, allowing fuel to flow out the orifice. Because the fuel is under high pressure, a fine spray is developed in the shape of a pencil stream. The spraying action atomizes the fuel, adding it to the air entering the combustion chamber.

The nozzle (outlet) ends of the injectors are positioned into openings in the intake manifold just above the intake valve ports of the cylinder head. The engine wiring harness connector for each fuel injector is equipped with an attached numerical tag (INJ 1, INJ 2 etc.). This is used to identify each fuel injector.

The injectors are energized individually in a sequential order by the powertrain control module (PCM). The PCM will adjust injector pulse width by switching the ground path to each individual injector on and off. Injector pulse width is the period of time that the injector is energized. The PCM will adjust injector pulse width based on various inputs it receives.

Battery voltage is supplied to the injectors through the ASD relay.

The PCM determines injector pulse width based on various inputs.

REMOVAL

(1) Remove fuel rail. Refer to Fuel Injector Rail Removal in this section.

(2) Disconnect clip(s) that retain fuel injector(s) to fuel rail (Fig. 11).

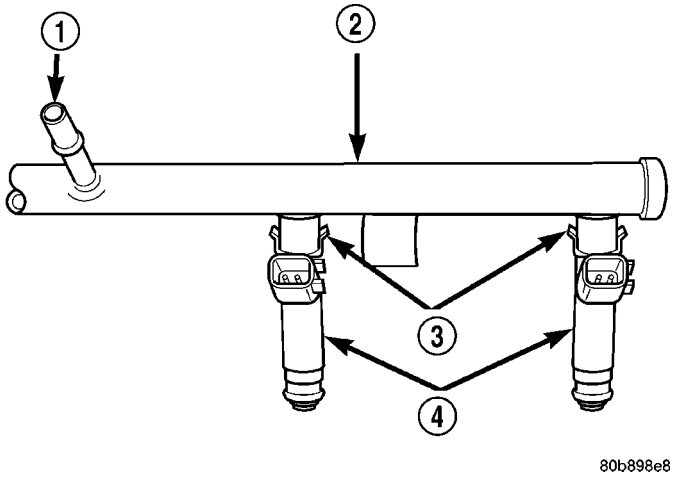

Fig. 11 FUEL INJECTOR MOUNTING

1 - INLET FITTING
2 - FUEL INJECTOR RAIL
3 - CLIP
4 - FUEL INJECTOR

FUEL INJECTOR (Continued)

INSTALLATION

(1) Install fuel injector(s) into fuel rail assembly and install retaining clip(s).

(2) If same injector(s) is being reinstalled, install new o-ring(s). Two different o-rings are being used. These can be easily identified by color. Install black o-ring at intake manifold end of injector. Install red/rust colored o-ring at fuel rail end of injector.

(3) Apply a small amount of clean engine oil to each injector o-ring. This will aid in installation.

(4) Install fuel rail. Refer to Fuel Rail Installation.

(5) Start engine and check for fuel leaks.

FUEL PUMP RELAY

DESCRIPTION

The 5–pin, 12–volt, fuel pump relay is located in the Power Distribution Center (PDC). Refer to the label on the PDC cover for relay location.

OPERATION

The Powertrain Control Module (PCM) energizes the electric fuel pump through the fuel pump relay. The fuel pump relay is energized by first applying battery voltage to it when the ignition key is turned ON, and then applying a ground signal to the relay from the PCM.

Whenever the ignition key is turned ON, the electric fuel pump will operate. But, the PCM will shutdown the ground circuit to the fuel pump relay in approximately 1–3 seconds unless the engine is operating or the starter motor is engaged.

REMOVAL

The fuel pump relay is located in the Power Distribution Center (PDC) (Fig. 12). Refer to label on PDC cover for relay location.

(1) Remove PDC cover.

(2) Remove relay from PDC.

(3) Check condition of relay terminals and PDC connector terminals for damage or corrosion. Repair if necessary before installing relay.

(4) Check for pin height (pin height should be the same for all terminals within the PDC connector). Repair if necessary before installing relay.

INSTALLATION

The fuel pump relay is located in the Power Distribution Center (PDC) (Fig. 12). Refer to label on PDC cover for relay location.

(1) Install relay to PDC.

(2) Install cover to PDC.

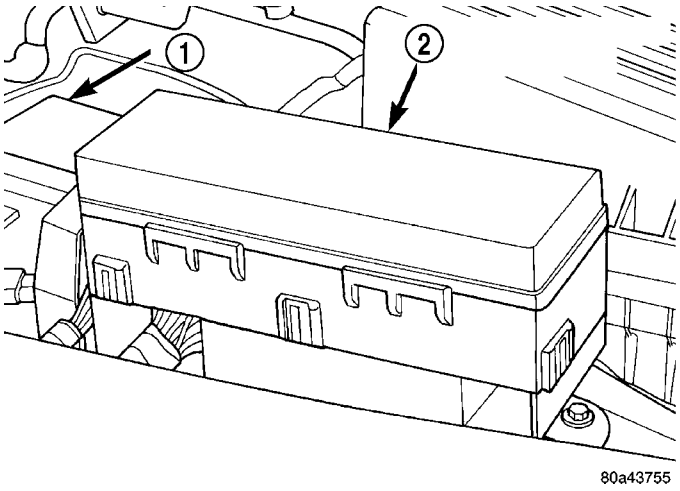

Fig. 12 Power Distribution Center (PDC)

1 - BATTERY
2 - POWER DISTRIBUTION CENTER (PDC)

IDLE AIR CONTROL MOTOR

DESCRIPTION

The IAC stepper motor is mounted to the throttle body, and regulates the amount of air bypassing the control of the throttle plate. As engine loads and ambient temperatures change, engine rpm changes. A pintle on the IAC stepper motor protrudes into a passage in the throttle body, controlling air flow through the passage. The IAC is controlled by the Powertrain Control Module (PCM) to maintain the target engine idle speed.

OPERATION

At idle, engine speed can be increased by retracting the IAC motor pintle and allowing more air to pass through the port, or it can be decreased by restricting the passage with the pintle and diminishing the amount of air bypassing the throttle plate.

The IAC is called a stepper motor because it is moved (rotated) in steps, or increments. Opening the IAC opens an air passage around the throttle blade which increases RPM.

The PCM uses the IAC motor to control idle speed (along with timing) and to reach a desired MAP during decel (keep engine from stalling).

The IAC motor has 4 wires with 4 circuits. Two of the wires are for 12 volts and ground to supply electrical current to the motor windings to operate the stepper motor in one direction. The other 2 wires are also for 12 volts and ground to supply electrical current to operate the stepper motor in the opposite direction.

To make the IAC go in the opposite direction, the PCM just reverses polarity on both windings. If only 1 wire is open, the IAC can only be moved 1 step

IDLE AIR CONTROL MOTOR (Continued)

(increment) in either direction. To keep the IAC motor in position when no movement is needed, the PCM will energize both windings at the same time. This locks the IAC motor in place.

In the IAC motor system, the PCM will count every step that the motor is moved. This allows the PCM to determine the motor pintle position. If the memory is cleared, the PCM no longer knows the position of the pintle. So at the first key ON, the PCM drives the IAC motor closed, regardless of where it was before. This zeros the counter. From this point the PCM will back out the IAC motor and keep track of its position again.

When engine rpm is above idle speed, the IAC is used for the following:

- Off-idle dashpot (throttle blade will close quickly but idle speed will not stop quickly)
- Deceleration air flow control
- A/C compressor load control (also opens the passage slightly before the compressor is engaged so that the engine rpm does not dip down when the compressor engages)
- Power steering load control

The PCM can control polarity of the circuit to control direction of the stepper motor.

IAC Stepper Motor Program: The PCM is also equipped with a memory program that records the number of steps the IAC stepper motor most recently advanced to during a certain set of parameters. For example: The PCM was attempting to maintain a 1000 rpm target during a cold start-up cycle. The last recorded number of steps for that may have been 125. That value would be recorded in the memory cell so that the next time the PCM recognizes the identical conditions, the PCM recalls that 125 steps were required to maintain the target. This program allows for greater customer satisfaction due to greater control of engine idle.

Another function of the memory program, which occurs when the power steering switch (if equipped), or the A/C request circuit, requires that the IAC stepper motor control engine rpm, is the recording of the last targeted steps into the memory cell. The PCM can anticipate A/C compressor loads. This is accomplished by delaying compressor operation for approximately 0.5 seconds until the PCM moves the IAC stepper motor to the recorded steps that were loaded into the memory cell. Using this program helps eliminate idle-quality changes as loads change. Finally, the PCM incorporates a ″No-Load″ engine speed limiter of approximately 1800 - 2000 rpm, when it recognizes that the TPS is indicating an idle signal and IAC motor cannot maintain engine idle.

A (factory adjusted) set screw is used to mechanically limit the position of the throttle body throttle plate. **Never attempt to adjust the engine idle speed using this screw.** All idle speed functions are controlled by the IAC motor through the PCM.

REMOVAL

2.4L

The Idle Air Control (IAC) motor is located on the rear side of the throttle body (Fig. 13).

(1) Disconnect electrical connector from IAC motor.
(2) Remove two mounting bolts (screws).
(3) Remove IAC motor from throttle body.

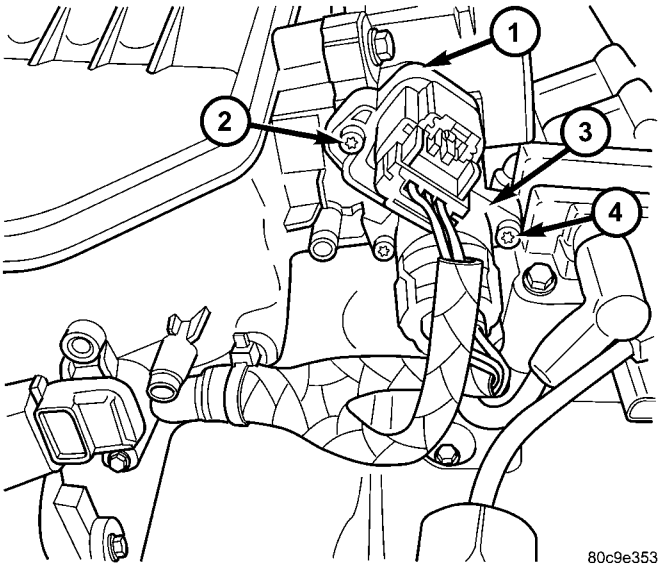

Fig. 13 TPS/IAC MOTOR - 2.4L

1 - THROTTLE POSITION SENSOR (TPS)
2 - MOUNTING SCREWS
3 - IDLE AIR CONTROL MOTOR (IAC)
4 - MOUNTING SCREWS

IDLE AIR CONTROL MOTOR (Continued)

4.0L

The IAC motor is located on the side of the throttle body.

(1) Remove air cleaner tube at throttle body.
(2) Disconnect electrical connector from IAC motor.
(3) Remove two mounting bolts (screws) (Fig. 14).
(4) Remove IAC motor from throttle body.

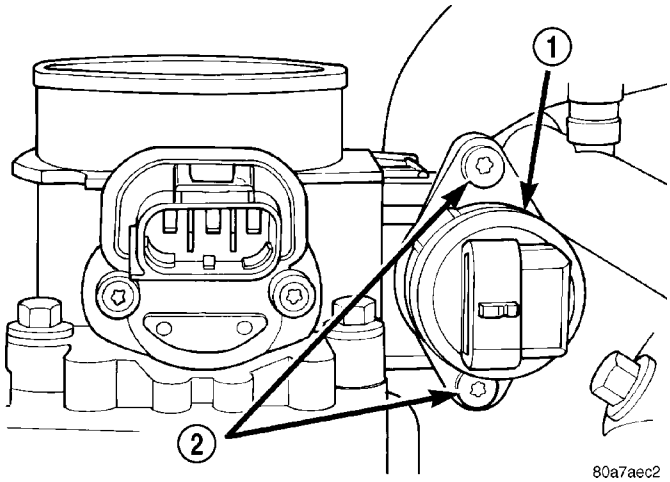

Fig. 14 MOUNTING BOLTS

1 - IDLE AIR CONTROL MOTOR
2 - MOUNTING SCREWS

INSTALLATION

(1) Install IAC motor to throttle body.
(2) Install and tighten two mounting bolts (screws) to 7 N·m (60 in. lbs.) torque.
(3) Install electrical connector.
(4) Install air cleaner tube to throttle body.

INTAKE AIR TEMPERATURE SENSOR

DESCRIPTION

The 2–wire Intake Manifold Air Temperature (IAT) sensor is installed in the intake manifold with the sensor element extending into the air stream.

The IAT sensor is a two-wire Negative Thermal Coefficient (NTC) sensor. Meaning, as intake manifold temperature increases, resistance (voltage) in the sensor decreases. As temperature decreases, resistance (voltage) in the sensor increases.

OPERATION

The IAT sensor provides an input voltage to the Powertrain Control Module (PCM) indicating the density of the air entering the intake manifold based upon intake manifold temperature. At key-on, a 5–volt power circuit is supplied to the sensor from the PCM. The sensor is grounded at the PCM through a low-noise, sensor-return circuit.

The PCM uses this input to calculate the following:

- Injector pulse-width
- Adjustment of spark timing (to help prevent spark knock with high intake manifold air-charge temperatures)

The resistance values of the IAT sensor is the same as for the Engine Coolant Temperature (ECT) sensor.

REMOVAL

2.4L

The intake manifold air temperature (IAT) sensor is installed into the intake manifold plenum at the rear end of the intake manifold (Fig. 15).

(1) Disconnect electrical connector from IAT sensor.
(2) Clean dirt from intake manifold at sensor base.
(3) Gently lift on small plastic release tab (Fig. 15) or (Fig. 16) and rotate sensor about 1/4 turn counterclockwise for removal.
(4) Check condition of sensor o-ring (Fig. 16).

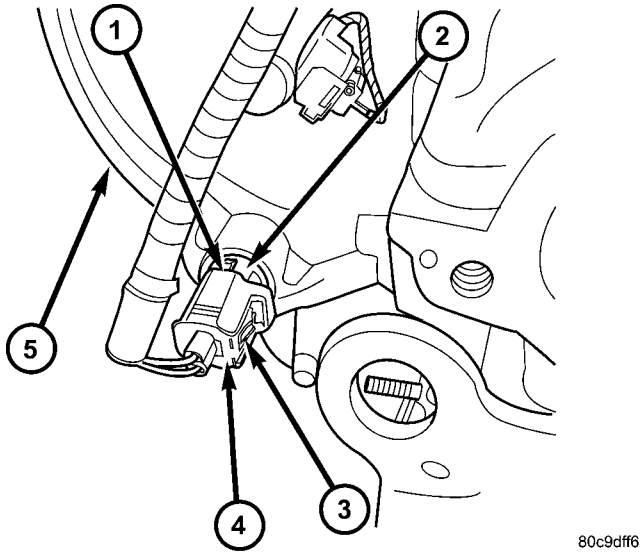

Fig. 15 IAT SENSOR - 2.4L

1 - RELEASE TAB
2 - IAT SENSOR
3 - PRESS HERE FOR REMOVAL
4 - ELECTRICAL CONNECTOR
5 - REAR END OF INTAKE MANIFOLD

INTAKE AIR TEMPERATURE SENSOR (Continued)

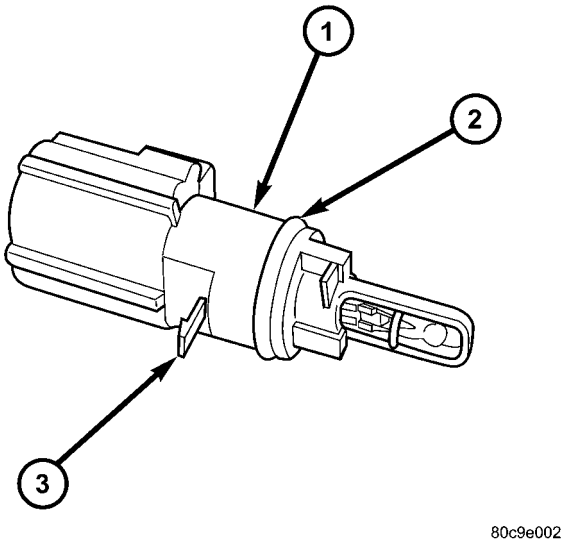

Fig. 16 IAT SENSOR TAB / O-RING - 2.4L

1 - IAT SENSOR
2 - SENSOR O-RING
3 - RELAST TAB

4.0L

The intake manifold air temperature (IAT) sensor is installed into intake manifold plenum near throttle body (Fig. 17).

(1) Disconnect electrical connector from IAT sensor.

(2) Remove sensor from intake manifold.

INSTALLATION

2.4L

(1) Check condition of sensor o-ring.

(2) Clean sensor mounting hole in intake manifold.

(3) Position sensor into intake manifold and rotate clockwise until past release tab.

(4) Install electrical connector.

4.0L

(1) Install IAT sensor into intake manifold. Tighten sensor to 28 N·m (20 ft. lbs.) torque.

(2) Connect electrical connector to sensor.

MAP SENSOR

DESCRIPTION

2.4L

The Manifold Absolute Pressure (MAP) sensor is mounted into the rear of the intake manifold with 1 screw.

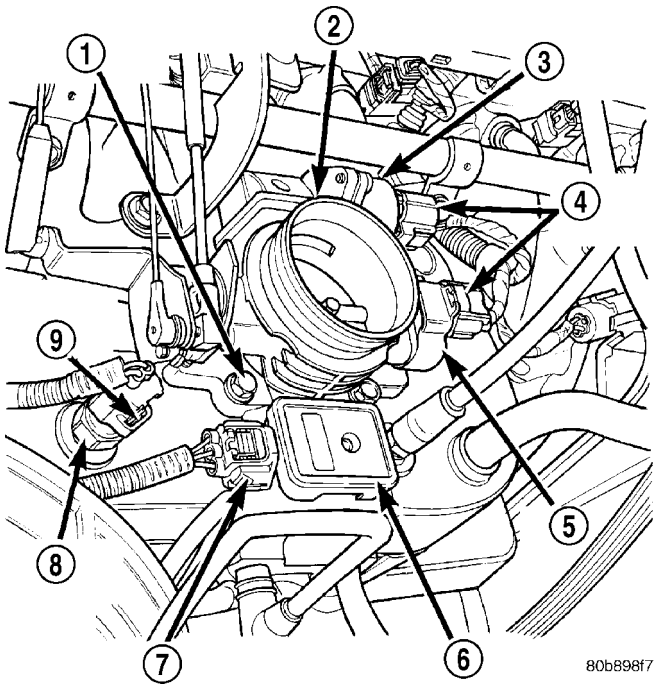

Fig. 17 IAT SENSOR - 4.0L

1 - MOUNTING BOLTS (4)
2 - THROTTLE BODY
3 - IAC MOTOR
4 - ELEC. CONN.
5 - TPS
6 - MAP SENSOR
7 - ELEC. CONN.
8 - IAT SENSOR
9 - ELEC. CONN.

4.0L Early

The Manifold Absolute Pressure (MAP) sensor is attached to the side of the engine throttle body with 2 screws. The sensor is connected to the throttle body with a rubber L-shaped fitting.

4.0L Late

The Manifold Absolute Pressure (MAP) sensor is attached to the side of the engine throttle body with 1 screw. The sensor is connected to the throttle body with a rubber L-shaped fitting.

OPERATION

The MAP sensor is used as an input to the Powertrain Control Module (PCM). It contains a silicon based sensing unit to provide data on the manifold vacuum that draws the air/fuel mixture into the combustion chamber. The PCM requires this information to determine injector pulse width and spark advance. When manifold absolute pressure (MAP) equals Barometric pressure, the pulse width will be at maximum.

A 5 volt reference is supplied from the PCM and returns a voltage signal to the PCM that reflects manifold pressure. The zero pressure reading is 0.5V

MAP SENSOR (Continued)

and full scale is 4.5V. For a pressure swing of 0–15 psi, the voltage changes 4.0V. To operate the sensor, it is supplied a regulated 4.8 to 5.1 volts. Ground is provided through the low-noise, sensor return circuit at the PCM.

The MAP sensor input is the number one contributor to fuel injector pulse width. The most important function of the MAP sensor is to determine barometric pressure. The PCM needs to know if the vehicle is at sea level or at a higher altitude, because the air density changes with altitude. It will also help to correct for varying barometric pressure. Barometric pressure and altitude have a direct inverse correlation; as altitude goes up, barometric goes down. At key-on, the PCM powers up and looks at MAP voltage, and based upon the voltage it sees, it knows the current barometric pressure (relative to altitude). Once the engine starts, the PCM looks at the voltage again, continuously every 12 milliseconds, and compares the current voltage to what it was at key-on. The difference between current voltage and what it was at key-on, is manifold vacuum.

During key-on (engine not running) the sensor reads (updates) barometric pressure. A normal range can be obtained by monitoring a known good sensor.

As the altitude increases, the air becomes thinner (less oxygen). If a vehicle is started and driven to a very different altitude than where it was at key-on, the barometric pressure needs to be updated. Any time the PCM sees Wide Open Throttle (WOT), based upon Throttle Position Sensor (TPS) angle and RPM, it will update barometric pressure in the MAP memory cell. With periodic updates, the PCM can make its calculations more effectively.

The PCM uses the MAP sensor input to aid in calculating the following:

- Manifold pressure
- Barometric pressure
- Engine load
- Injector pulse-width
- Spark-advance programs
- Shift-point strategies (certain automatic transmissions only)
- Idle speed
- Decel fuel shutoff

The MAP sensor signal is provided from a single piezoresistive element located in the center of a diaphragm. The element and diaphragm are both made of silicone. As manifold pressure changes, the diaphragm moves causing the element to deflect, which stresses the silicone. When silicone is exposed to stress, its resistance changes. As manifold vacuum increases, the MAP sensor input voltage decreases proportionally. The sensor also contains electronics that condition the signal and provide temperature compensation.

The PCM recognizes a decrease in manifold pressure by monitoring a decrease in voltage from the reading stored in the barometric pressure memory cell. The MAP sensor is a linear sensor; meaning as pressure changes, voltage changes proportionately. The range of voltage output from the sensor is usually between 4.6 volts at sea level to as low as 0.3 volts at 26 in. of Hg. Barometric pressure is the pressure exerted by the atmosphere upon an object. At sea level on a standard day, no storm, barometric pressure is approximately 29.92 in Hg. For every 100 feet of altitude, barometric pressure drops 0.10 in. Hg. If a storm goes through, it can change barometric pressure from what should be present for that altitude. You should know what the average pressure and corresponding barometric pressure is for your area.

REMOVAL

2.4L

The Manifold Absolute Pressure (MAP) sensor is mounted into the rear of the intake manifold (Fig. 18). An o-ring is used to seal the sensor to the intake manifold (Fig. 19).

(1) Disconnect electrical connector at sensor.
(2) Clean area around MAP sensor.
(3) Remove sensor mounting screw (TORX head).
(4) Remove MAP sensor from intake manifold.
(5) Check condition of sensor o-ring (Fig. 19).

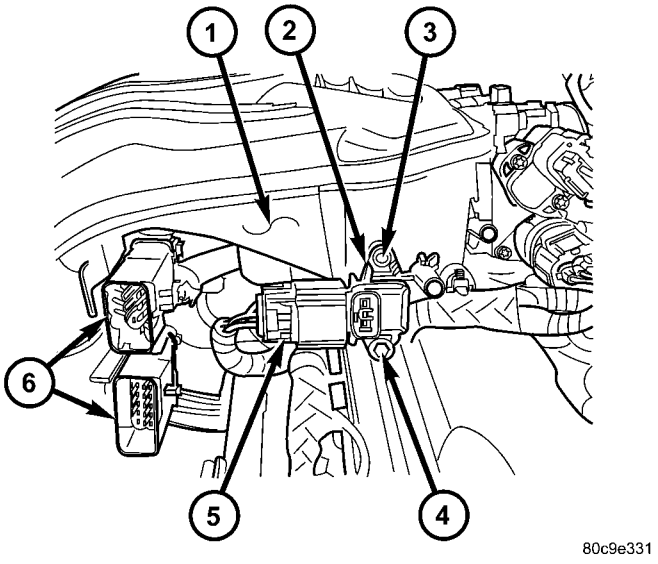

Fig. 18 MAP SENSOR LOCATION - 2.4L

1 - REAR OF INTAKE MANIFOLD
2 - MAP SENSOR
3 - ALIGNMENT PIN
4 - MOUNTING BOLT (TORX)
5 - ELECTRICAL CONNECTOR
6 - MAIN ENGINE HARNESS CONNECTORS

MAP SENSOR (Continued)

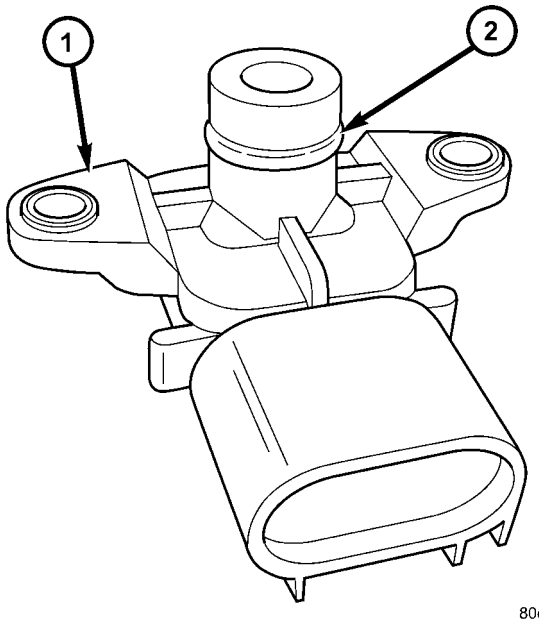

Fig. 19 MAP SENSOR O-RING - 2.4L

1 - MAP SENSOR
2 - O-RING

4.0L Early

An L-shaped rubber fitting is used to connect the MAP sensor to throttle body (Fig. 20).

(1) Remove air cleaner intake tube at throttle body.

(2) Remove two MAP sensor mounting bolts (screws) (Fig. 20).

(3) While removing MAP sensor, slide the rubber L-shaped fitting (Fig. 20) from throttle body.

(4) Remove rubber L-shaped fitting from MAP sensor.

4.0L Late

An L-shaped rubber fitting is used to connect the MAP sensor to throttle body (Fig. 21).

(1) Remove air cleaner intake tube at throttle body.

(2) Remove MAP sensor mounting bolt (screw) (Fig. 21).

(3) While removing MAP sensor, slide the rubber L-shaped fitting (Fig. 21) from throttle body.

(4) Remove rubber L-shaped fitting from MAP sensor.

INSTALLATION

2.4L

(1) Clean MAP sensor mounting hole at intake manifold.

(2) Check MAP sensor o-ring seal for cuts or tears.

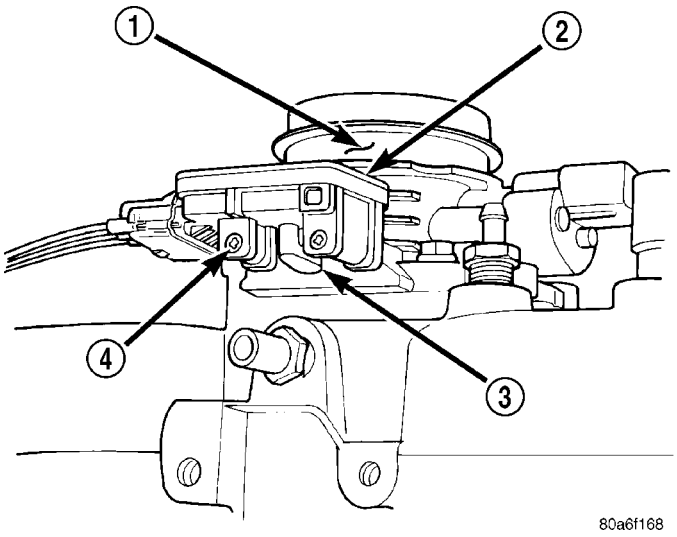

Fig. 20 4.0L MAP SENSOR MOUNTING - EARLY

1 - THROTTLE BODY
2 - MAP SENSOR
3 - RUBBER FITTING
4 - MOUNTING SCREWS (2)

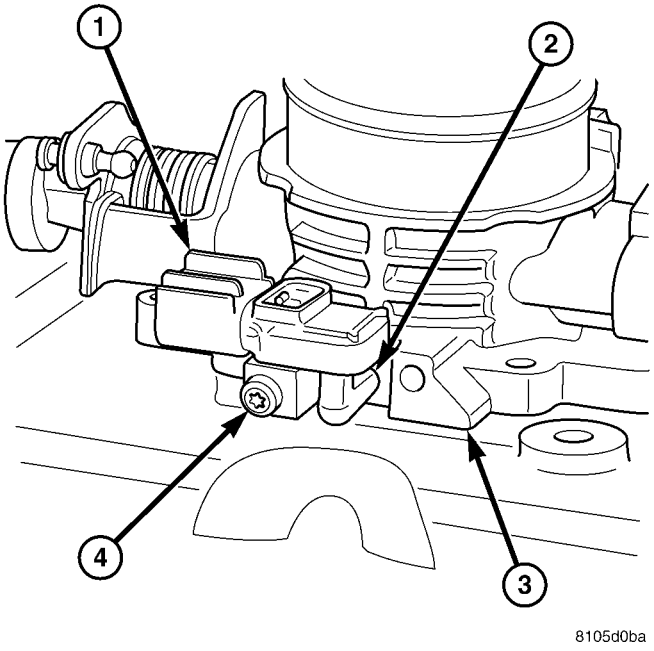

Fig. 21 4.0L MAP SENSOR MOUNTING - LATE

1 - MAP SENSOR
2 - RUBBER FITTING
3 - THROTTLE BODY
4 - MOUNTING SCREW (1)

(3) Position sensor into manifold.

(4) Install MAP sensor mounting screws. Tighten screw to 3 N·m (25 in. lbs.) torque.

(5) Connect electrical connector.

4.0L

(1) Install rubber L-shaped fitting to MAP sensor.

MAP SENSOR (Continued)

(2) Position sensor to throttle body while guiding rubber fitting over throttle body vacuum nipple.
(3) Install MAP sensor mounting bolt(s) (screws). Tighten screw(s) to 3 N·m (25 in. lbs.) torque.
(4) Install air cleaner intake tube.
(5) Connect electrical connector.

O2 HEATER RELAY

DESCRIPTION

The oxygen (O2) sensor heater relay is located in the Powertrain Distribution Center (PDC).

OPERATION

Refer to Oxygen Sensor for oxygen sensor relay information.

REMOVAL

The oxygen sensor heater relay is located in the Power Distribution Center (PDC). Refer to label on PDC cover for relay location.
(1) Remove PDC cover.
(2) Remove relay from PDC.
(3) Check condition of relay terminals and PDC connector terminals for damage or corrosion. Repair if necessary before installing relay.
(4) Check for pin height (pin height should be the same for all terminals within the PDC connector). Repair if necessary before installing relay.

INSTALLATION

The oxygen sensor heater relay is located in the Power Distribution Center (PDC). Refer to label on PDC cover for relay location.
(1) Install relay to PDC.
(2) Install cover to PDC.

O2 SENSOR

DESCRIPTION

The Oxygen Sensors (O2S) are attached to, and protrude into the vehicle exhaust system. Depending on the engine or emission package, the vehicle may use a total of either 2 or 4 sensors.

OPERATION

An O2 sensor is a galvanic battery that provides the PCM with a voltage signal (0-1 volt) inversely proportional to the amount of oxygen in the exhaust. In other words, if the oxygen content is low, the voltage output is high; if the oxygen content is high the output voltage is low. The PCM uses this information to adjust injector pulse-width to achieve the 14.7–to–1 air/fuel ratio necessary for proper engine operation and to control emissions.

The O2 sensor must have a source of oxygen from outside of the exhaust stream for comparison. Current O2 sensors receive their fresh oxygen (outside air) supply through the O2 sensor case housing.

Four wires (circuits) are used on each O2 sensor: a 12–volt feed circuit for the sensor heating element; a ground circuit for the heater element; a low-noise sensor return circuit to the PCM, and an input circuit from the sensor back to the PCM to detect sensor operation.

Oxygen Sensor Heater Relay: If the vehicle is equipped with 4 oxygen sensors, a separate oxygen sensor relay is used to supply voltage to the sensor heating elements. This particular relay is used only for the 1/2 and 2/2 downstream sensors. Voltage for the other 2 sensor heating elements is supplied directly from the ASD relay. Refer to 8, Wiring Diagrams to determine which relay is used.

To avoid the large simultaneous current surge needed to operate all 4 sensors, power is delayed to the 2 downstream heater elements by the PCM for approximately 2 seconds.

Oxygen Sensor Heater Elements:

The O2 sensor uses a Positive Thermal Co-efficient (PTC) heater element. As temperature increases, resistance increases. At ambient temperatures around 70°F, the resistance of the heating element is approximately 4.5 ohms. As the sensor's temperature increases, resistance in the heater element increases. This allows the heater to maintain the optimum operating temperature of approximately 930°-1100°F (500°-600° C). Although the sensors operate the same, there are physical differences, due to the environment that they operate in, that keep them from being interchangeable.

Maintaining correct sensor temperature at all times allows the system to enter into closed loop operation sooner. Also, it allows the system to remain in closed loop operation during periods of extended idle.

In Closed Loop operation, the PCM monitors certain O2 sensor input(s) along with other inputs, and adjusts the injector pulse width accordingly. During Open Loop operation, the PCM ignores the O2 sensor input. The PCM adjusts injector pulse width based on preprogrammed (fixed) values and inputs from other sensors.

Upstream Sensor - Engine Equipped With 2 Sensors: The upstream sensor (1/1) provides an input voltage to the PCM. The input tells the PCM the oxygen content of the exhaust gas. The PCM uses this information to fine tune fuel delivery to maintain the correct oxygen content at the downstream oxygen sensor. The PCM will change the air/fuel

O2 SENSOR (Continued)

ratio until the upstream sensor inputs a voltage that the PCM has determined will make the downstream sensor output (oxygen content) correct.

The upstream oxygen sensor also provides an input to determine catalytic convertor efficiency.

Downstream Sensor - Engine Equipped With 2 Sensors: The downstream oxygen sensor (1/2) is also used to determine the correct air-fuel ratio. As the oxygen content changes at the downstream sensor, the PCM calculates how much air-fuel ratio change is required. The PCM then looks at the upstream oxygen sensor voltage and changes fuel delivery until the upstream sensor voltage changes enough to correct the downstream sensor voltage (oxygen content).

The downstream oxygen sensor also provides an input to determine catalytic convertor efficiency.

Upstream Sensors - Engine Equipped With 4 Sensors: Two upstream sensors are used (1/1 and 2/1). The 1/1 sensor is the first sensor to receive exhaust gases from the #1 cylinder. They provide an input voltage to the PCM. The input tells the PCM the oxygen content of the exhaust gas. The PCM uses this information to fine tune fuel delivery to maintain the correct oxygen content at the downstream oxygen sensors. The PCM will change the air/fuel ratio until the upstream sensors input a voltage that the PCM has determined will make the downstream sensors output (oxygen content) correct.

The upstream oxygen sensors also provide an input to determine mini-catalyst efficiency. Main catalytic convertor efficiency is not calculated with this package.

Downstream Sensors - Engine Equipped With 4 Sensors: Two downstream sensors are used (1/2 and 2/2). The downstream sensors are used to determine the correct air-fuel ratio. As the oxygen content changes at the downstream sensor, the PCM calculates how much air-fuel ratio change is required. The PCM then looks at the upstream oxygen sensor voltage, and changes fuel delivery until the upstream sensor voltage changes enough to correct the downstream sensor voltage (oxygen content).

The downstream oxygen sensors also provide an input to determine mini-catalyst efficiency. Main catalytic convertor efficiency is not calculated with this package.

Engines equipped with either a downstream sensor(s), or a post-catalytic sensor, will monitor catalytic convertor efficiency. If efficiency is below emission standards, the Malfunction Indicator Lamp (MIL) will be illuminated and a Diagnostic Trouble Code (DTC) will be set. Refer to Monitored Systems in Emission Control Systems for additional information.

REMOVAL

2.4L

CAUTION: Never apply any type of grease to the oxygen sensor electrical connector, or attempt any soldering of the sensor wiring harness.

Refer to (Fig. 22) for O2S (oxygen sensor) location.

WARNING: THE EXHAUST MANIFOLD, EXHAUST PIPES AND CATALYTIC CONVERTER BECOME VERY HOT DURING ENGINE OPERATION. ALLOW ENGINE TO COOL BEFORE REMOVING OXYGEN SENSOR.

(1) Raise and support vehicle.
(2) Disconnect wire connector from O2S sensor.

CAUTION: When disconnecting sensor electrical connector, do not pull directly on wire going into sensor.

(3) Remove O2S sensor with an oxygen sensor removal and installation tool.
(4) Clean threads in exhaust pipe using appropriate size tap.

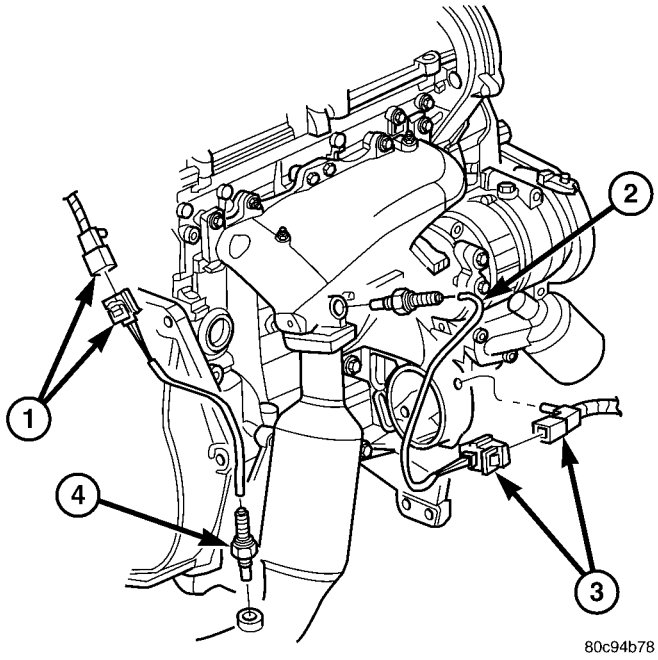

Fig. 22 OXYGEN SENSORS - 2.4L

1 - ELECTRICAL CONNECTORS
2 - UPSTREAM SENSOR (1/1)
3 - ELECTRICAL CONNECTORS
4 - DOWNSTREAM SENSOR (1/2)

O2 SENSOR (Continued)

4.0L

CAUTION: Never apply any type of grease to the oxygen sensor electrical connector, or attempt any soldering of the sensor wiring harness.

Refer to (Fig. 23), (Fig. 24) for O2S (oxygen sensor) location.

WARNING: THE EXHAUST MANIFOLD, EXHAUST PIPES AND CATALYTIC CONVERTER BECOME VERY HOT DURING ENGINE OPERATION. ALLOW ENGINE TO COOL BEFORE REMOVING OXYGEN SENSOR.

(1) Raise and support vehicle.
(2) Disconnect wire connector from O2S sensor.

CAUTION: When disconnecting sensor electrical connector, do not pull directly on wire going into sensor.

(3) Remove O2S sensor with an oxygen sensor removal and installation tool.
(4) Clean threads in exhaust pipe using appropriate size tap.

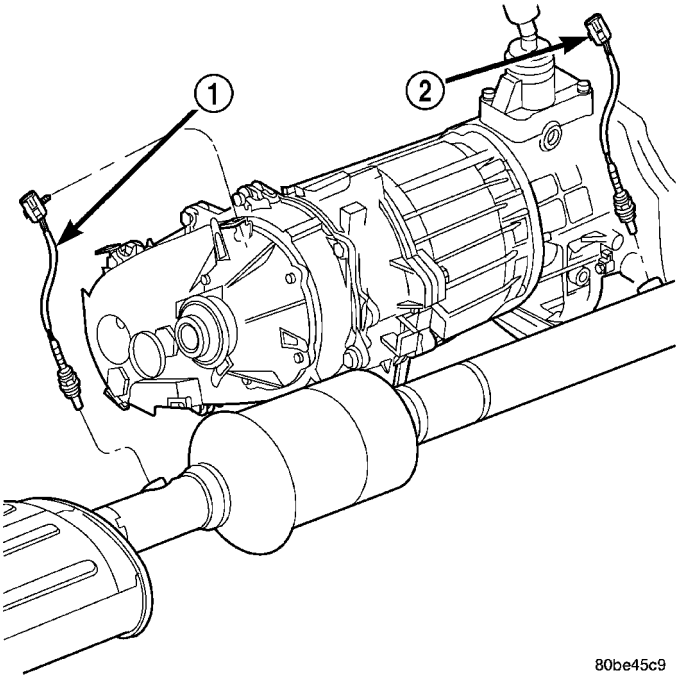

Fig. 23 OXYGEN SENSORS - 4.0L - FEDERAL EMISSIONS

1 - 1/2 O2S
2 - 1/1 02S

INSTALLATION

Threads of new oxygen sensors are factory coated with anti-seize compound to aid in removal. **DO NOT add any additional anti-seize compound to the threads of a new oxygen sensor.**

(1) Install O2S sensor. Tighten to 30 N·m (22 ft. lbs.) torque.
(2) Connect O2S sensor wire connector.
(3) Lower vehicle.

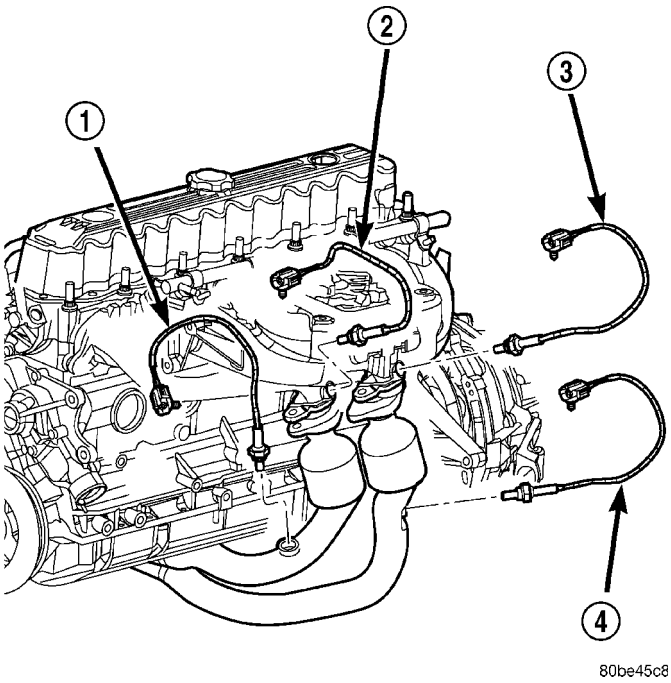

Fig. 24 OXYGEN SENSORS - 4.0L - CALIFORNIA EMISSIONS

1 - 1/2 O2S
2 - 1/1 O2S
3 - 2/1 O2S
4 - 2/2 O2S

THROTTLE BODY

DESCRIPTION

The throttle body is located on the intake manifold. Fuel does not enter the intake manifold through the throttle body. Fuel is sprayed into the manifold by the fuel injectors.

OPERATION

Filtered air from the air cleaner enters the intake manifold through the throttle body. The throttle body contains an air control passage controlled by an Idle Air Control (IAC) motor. The air control passage is used to supply air for idle conditions. A throttle valve (plate) is used to supply air for above idle conditions.

Certain sensors are attached to the throttle body. The accelerator pedal cable, speed control cable and transmission control cable (when equipped) are connected to the throttle body linkage arm.

THROTTLE BODY (Continued)

A (factory adjusted) set screw is used to mechanically limit the position of the throttle body throttle plate. **Never attempt to adjust the engine idle speed using this screw.** All idle speed functions are controlled by the PCM.

REMOVAL

2.4L

A (factory adjusted) set screw is used to mechanically limit the position of the throttle body throttle plate. **Never attempt to adjust the engine idle speed using this screw.** All idle speed functions are controlled by the Powertrain Control Module (PCM).

(1) Remove air cleaner tube at throttle body.

(2) Disconnect throttle body electrical connectors at IAC motor and TPS.

(3) Remove all control cables from throttle body (lever) arm. Refer to the Accelerator Pedal and Throttle Cable section for removal/installation procedures.

(4) Disconnect necessary vacuum lines at throttle body.

(5) Remove 3 throttle body mounting bolts (Fig. 25).

(6) Remove throttle body from intake manifold.

(7) Check condition of old throttle body-to-intake manifold o-ring.

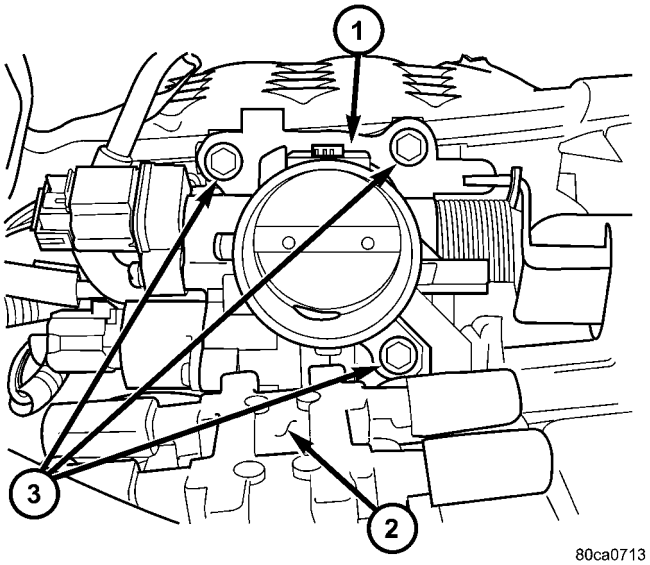

Fig. 25 THROTTLE BODY MOUNTING BOLTS - 2.4L

1 - THROTTLE BODY
2 - IGNITION COIL
3 - MOUNTING BOLTS (3)

4.0L

A (factory adjusted) set screw is used to mechanically limit the position of the throttle body throttle plate. **Never attempt to adjust the engine idle speed using this screw.** All idle speed functions are controlled by the Powertrain Control Module (PCM).

(1) Remove air cleaner tube at throttle body.

(2) Disconnect throttle body electrical connectors at MAP sensor, IAC motor and TPS (Fig. 26).

(3) Remove all control cables from throttle body (lever) arm. Refer to the Accelerator Pedal and Throttle Cable section of this group for additional information.

(4) Remove four throttle body mounting bolts.

(5) Remove throttle body from intake manifold.

(6) Discard old throttle body-to-intake manifold gasket.

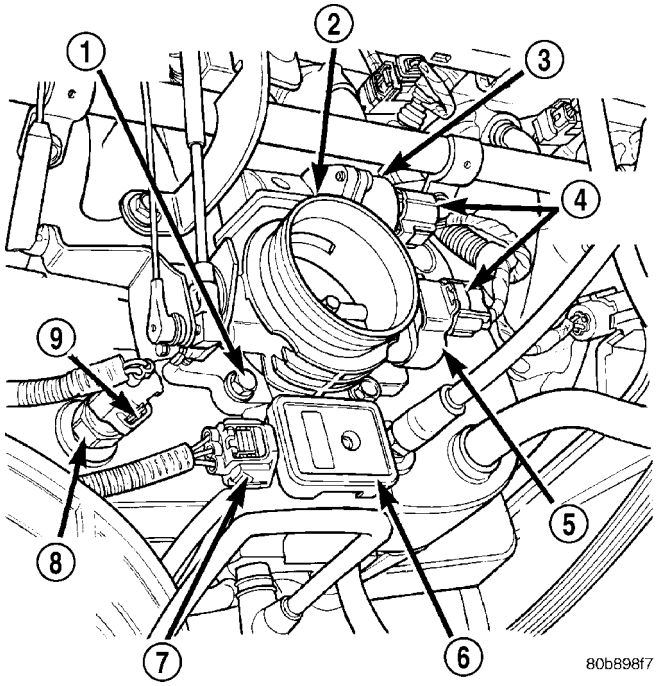

Fig. 26 THROTTLE BODY AND SENSOR LOCATIONS - 4.0L

1 - MOUNTING BOLTS (4)
2 - THROTTLE BODY
3 - IAC MOTOR
4 - ELEC. CONN.
5 - TPS
6 - MAP SENSOR
7 - ELEC. CONN.
8 - IAT SENSOR
9 - ELEC. CONN.

INSTALLATION

2.4L

(1) Check condition of throttle body-to-intake manifold o-ring. Replace as necessary.

(2) Clean mating surfaces of throttle body and intake manifold.

(3) Install throttle body-to-intake manifold o-ring.

(4) Install throttle body to intake manifold.

(5) Install 3 mounting bolts. Tighten bolts to 12 N·m (105 in. lbs.) torque.

THROTTLE BODY (Continued)

(6) Install control cables.
(7) Install electrical connectors.
(8) Install necessary vacuum lines.
(9) Install air cleaner duct at throttle body.

4.0L

A (factory adjusted) set screw is used to mechanically limit the position of the throttle body throttle plate. **Never attempt to adjust the engine idle speed using this screw.** All idle speed functions are controlled by the Powertrain Control Module (PCM).

(1) Clean mating surfaces of throttle body and intake manifold.
(2) Install new throttle body-to-intake manifold gasket.
(3) Install throttle body to intake manifold.
(4) Install four mounting bolts. Tighten bolts to 11 N·m (100 in. lbs.) torque.
(5) Install control cables.
(6) Install electrical connectors.
(7) Install air cleaner at throttle body.

THROTTLE CONTROL CABLE

REMOVAL

2.4L

CAUTION: Be careful not to damage or kink the cable core wire (within the cable sheathing) while servicing accelerator pedal or throttle cable.

(1) From inside vehicle, hold up accelerator pedal. Remove plastic cable retainer (clip) and throttle cable core wire from upper end of pedal arm (Fig. 27). Plastic cable retainer snaps into top of pedal arm.
(2) Remove cable core wire at pedal arm.
(3) From inside vehicle, pinch both sides of cable housing retainer tabs (Fig. 27) at dash panel. Remove cable housing from dash panel and pull into engine compartment.
(4) Remove air box at throttle body.
(5) Unsnap cable from dashpanel routing clip.
(6) Remove cable housing from dash panel and pull into engine compartment.
(7) Hold throttle in wide open position. While held in this position, slide throttle cable pin (Fig. 28) from throttle body bellcrank.
(8) Using a pick or small screwdriver, press release tab (Fig. 28) to release plastic cable mount from bracket. **Press on tab only enough to release cable from bracket. If tab is pressed too much, it will be broken.**To remove throttle cable from throttle body bracket, slide cable towards front of vehicle.
(9) Remove throttle cable from vehicle.

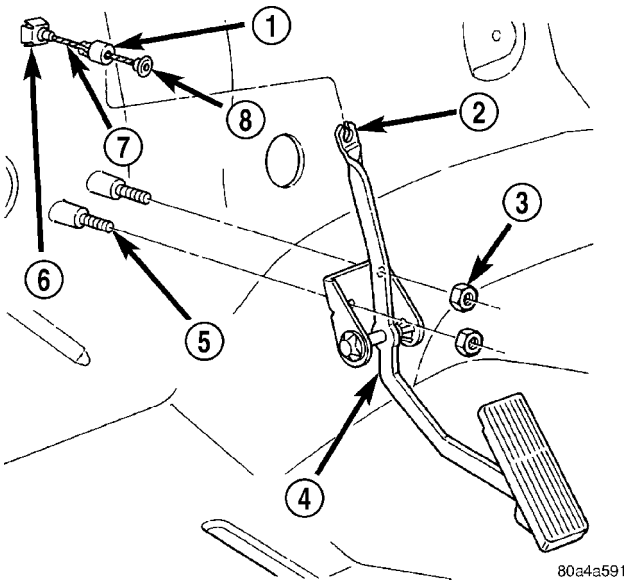

Fig. 27 ACCELERATOR PEDAL MOUNTING

1 - CABLE RETAINER
2 - SLOT
3 - MOUNTING STUDS (2)
4 - PEDAL/BRACKET ASSEMBLY
5 - MOUNTING STUDS
6 - RETAINER TABS
7 - CABLE
8 - CABLE STOP

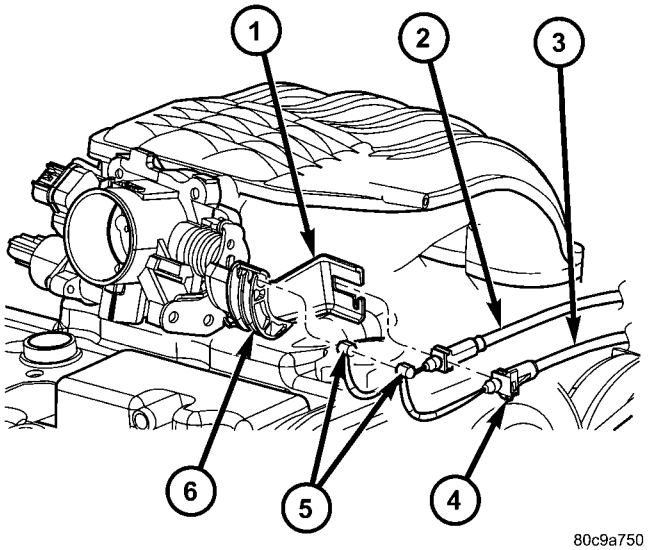

Fig. 28 THROTTLE CABLE, PIN, RELEASE TAB - 2.4L

1 - MOUNTING BRACKET
2 - SPEED CONTROL CABLE
3 - THROTTLE CABLE
4 - RELEASE TAB
5 - CABLE PINS
6 - BELLCRANK

THROTTLE CONTROL CABLE (Continued)

4.0L

CAUTION: Be careful not to damage or kink the cable core wire (within the cable sheathing) while servicing accelerator pedal or throttle cable.

(1) From inside vehicle, hold up accelerator pedal. Remove plastic cable retainer (clip) and throttle cable core wire from upper end of accelerator pedal arm (Fig. 29). Plastic cable retainer (clip) snaps into pedal arm.

(2) Remove cable core wire at pedal arm.

(3) From inside vehicle, pinch both sides of cable housing retainer tabs (Fig. 29) at dash panel. Remove cable housing from dash panel and pull into engine compartment.

(4) Remove cable from clip guides on engine cylinder head (valve) cover (Fig. 30).

(5) Remove throttle cable ball socket at throttle body by pushing ball socket towards rear of vehicle (ball snaps off of throttle body pin) (Fig. 31).

(6) Remove throttle cable from throttle body mounting bracket by compressing release tabs (Fig. 31) and pushing cable through hole in bracket.

(7) Remove throttle cable from vehicle.

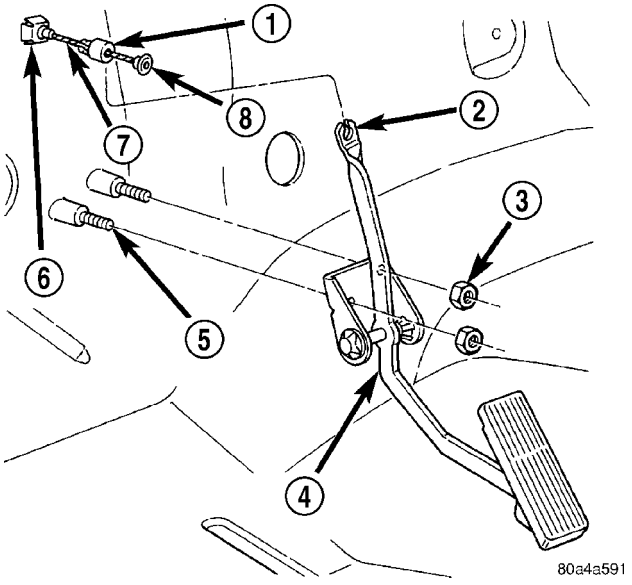

Fig. 29 ACCELERATOR PEDAL MOUNTING

1 - CABLE RETAINER
2 - SLOT
3 - MOUNTING STUDS (2)
4 - PEDAL/BRACKET ASSEMBLY
5 - MOUNTING STUDS
6 - RETAINER TABS
7 - CABLE
8 - CABLE STOP

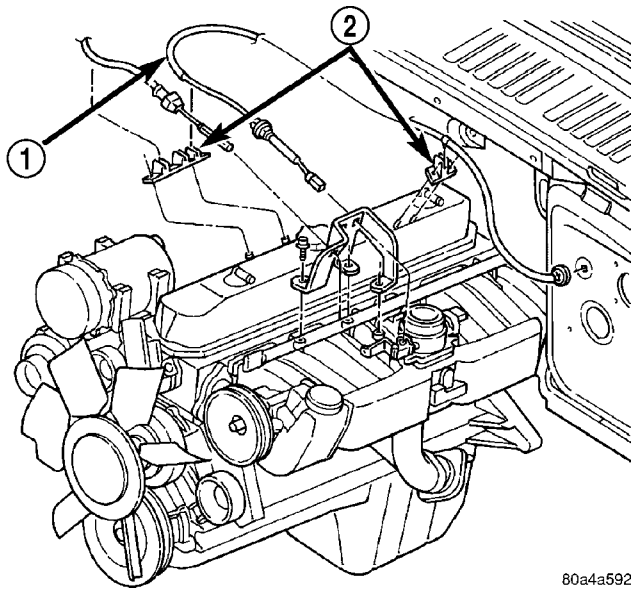

Fig. 30 THROTTLE CABLE ROUTING - 4.0L

1 - THROTTLE CABLE
2 - GUIDE

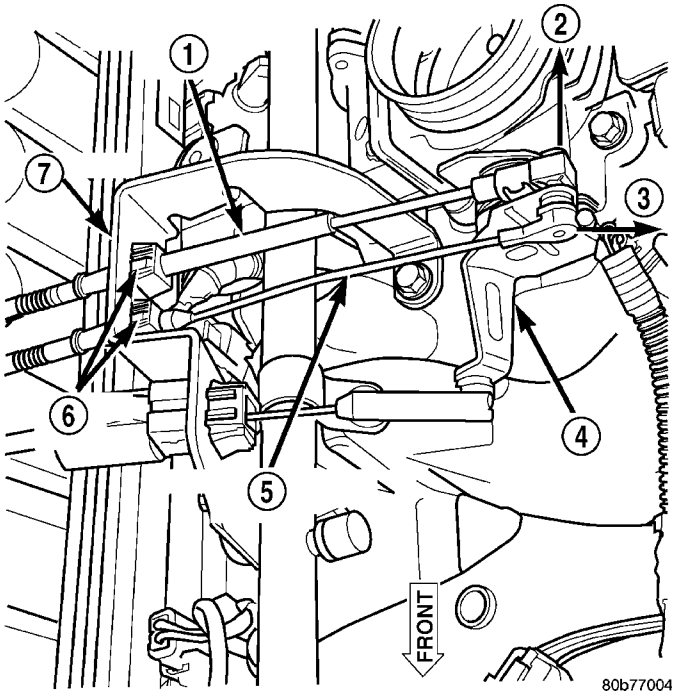

Fig. 31 THROTTLE CABLE AT THROTTLE BODY - 4.0L - TYPICAL

1 - ACCELERATOR CABLE
2 - OFF
3 - OFF
4 - THROTTLE BODY BELLCRANK
5 - SPEED CONTROL CABLE
6 - RELEASE TABS
7 - BRACKET

THROTTLE CONTROL CABLE (Continued)

INSTALLATION

2.4L

(1) Slide accelerator cable plastic mount into throttle body mounting bracket. Continue sliding until release tab is aligned to hole in mounting bracket.

(2) Hold throttle in wide open position. While held in this position, slide throttle cable pin into throttle body bellcrank.

(3) Push cable housing into rubber grommet and through opening in dash panel.

(4) Push other end of cable through opening in dash panel until retaining tabs lock into panel.

(5) From inside vehicle, slide throttle cable core wire into opening (slot) in top of pedal arm.

(6) Push plastic cable retainer (clip) into pedal arm opening until it snaps in place.

(7) Install air box to throttle body.

(8) Before starting engine, operate accelerator pedal to check for any binding.

4.0L

(1) Slide throttle cable through hole in throttle body bracket until retainer tabs lock into bracket. Connect cable ball end to throttle body linkage ball (snaps on).

(2) Snap cable into clip guides on engine cylinder head (valve) cover.

(3) Push other end of cable through opening in dash panel until retaining tabs lock into panel.

(4) From inside drivers compartment, slide throttle cable core wire into opening in top of accelerator pedal arm. An index tab is located on pedal arm. Rotate and push cable retainer (clip) into pedal arm opening until it snaps in place on index tab.

(5) Install air box to throttle body.

(6) Before starting engine, operate accelerator pedal to check for any binding.

THROTTLE POSITION SENSOR

DESCRIPTION

The 3–wire Throttle Position Sensor (TPS) is mounted on the throttle body and is connected to the throttle blade.

OPERATION

The TPS is a 3–wire variable resistor that provides the Powertrain Control Module (PCM) with an input signal (voltage) that represents the throttle blade position of the throttle body. The sensor is connected to the throttle blade shaft. As the position of the throttle blade changes, the resistance (output voltage) of the TPS changes.

The PCM supplies approximately 5 volts to the TPS. The TPS output voltage (input signal to the PCM) represents the throttle blade position. The PCM receives an input signal voltage from the TPS. This will vary in an approximate range of from .26 volts at minimum throttle opening (idle), to 4.49 volts at wide open throttle. Along with inputs from other sensors, the PCM uses the TPS input to determine current engine operating conditions. In response to engine operating conditions, the PCM will adjust fuel injector pulse width and ignition timing.

The PCM needs to identify the actions and position of the throttle blade at all times. This information is needed to assist in performing the following calculations:

- Ignition timing advance
- Fuel injection pulse-width
- Idle (learned value or minimum TPS)
- Off-idle (0.06 volt)
- Wide Open Throttle (WOT) open loop (2.608 volts above learned idle voltage)
- Deceleration fuel lean out
- Fuel cutoff during cranking at WOT (2.608 volts above learned idle voltage)
- A/C WOT cutoff (certain automatic transmissions only)

REMOVAL

2.4L

The Throttle Position Sensor (TPS) is mounted to the throttle body (Fig. 32).

(1) Disconnect TPS electrical connector.

(2) Remove 2 TPS mounting screws.

(3) Remove TPS.

4.0L

The TPS is mounted to the throttle body.

(1) Remove air cleaner tube at throttle body.

(2) Disconnect TPS electrical connector.

(3) Remove TPS mounting screws (Fig. 33).

(4) Remove TPS.

THROTTLE POSITION SENSOR (Continued)

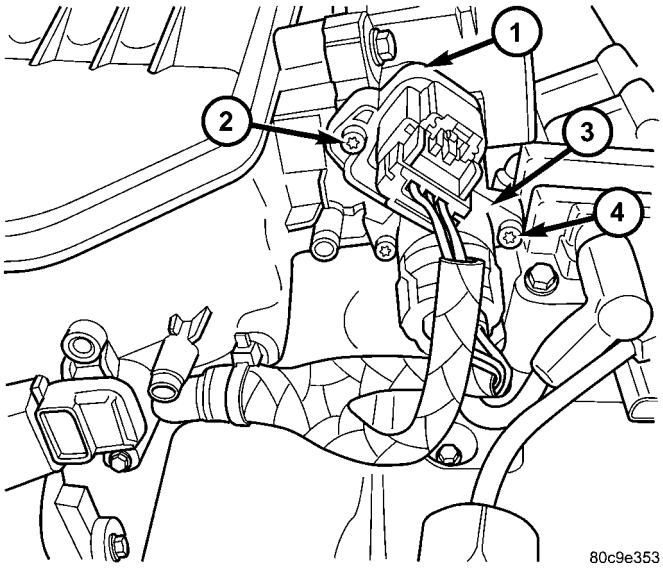

Fig. 32 TPS/IAC MOTOR - 2.4L

1 - THROTTLE POSITION SENSOR (TPS)
2 - MOUNTING SCREWS
3 - IDLE AIR CONTROL MOTOR (IAC)
4 - MOUNTING SCREWS

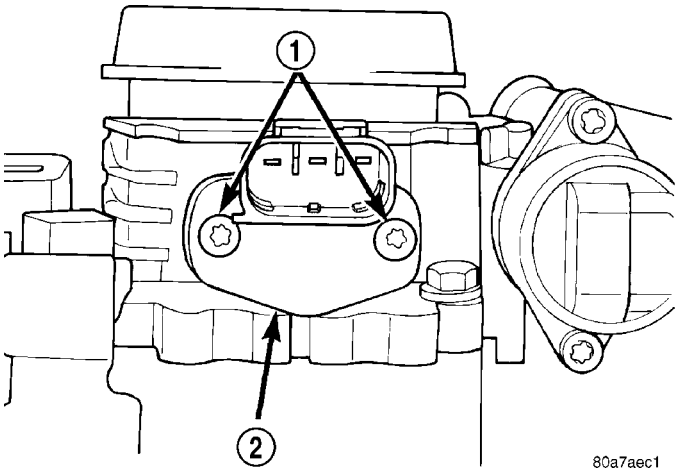

Fig. 33 TPS MOUNTING SCREWS - 4.0L

1 - MOUNTING SCREWS
2 - TPS

INSTALLATION

2.4L

The throttle shaft end of the throttle body slides into a socket in the TPS. The TPS must be installed so that it can be rotated a few degrees. (If sensor will not rotate, install sensor with throttle shaft on other side of socket tangs). The TPS will be under slight tension when rotated.

(1) Install TPS and retaining screws.
(2) Tighten screws to 7 N·m (60 in. lbs.) torque.
(3) Connect TPS electrical connector to TPS.
(4) Manually operate throttle (by hand) to check for any TPS binding before starting engine.
(5) Install air cleaner tube to throttle body.

4.0L

The throttle shaft end of the throttle body slides into a socket in the TPS (Fig. 34). The TPS must be installed so that it can be rotated a few degrees. (If the sensor will not rotate, install the sensor with the throttle shaft on the other side of the socket tangs). The TPS will be under slight tension when rotated.

(1) Install TPS and retaining screws.
(2) Tighten screws to 7 N·m (60 in. lbs.) torque.
(3) Connect TPS electrical connector to TPS.
(4) Manually operate throttle (by hand) to check for any TPS binding before starting engine.
(5) Install air cleaner tube to throttle body.

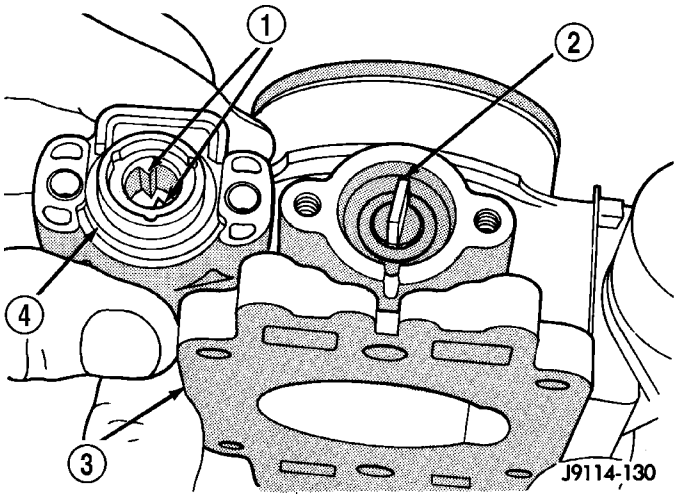

Fig. 34 TPS INSTALLATION - 4.0L

1 - TANGS
2 - THROTTLE SHAFT
3 - THROTTLE BODY
4 - TPS

STEERING

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STEERING

DESCRIPTION

CAUTION: MOPAR® ATF+4 is to be used in the power steering system. No other power steering or automatic transmission fluid is to be used in the system. Damage may result to the power steering pump and system if any other fluid is used, and do not overfill.

The power steering system has a hydraulic pump. The pump is a constant flow rate and displacement vane-type pump. The pump reservoir on the 4.0L engine is mounted to the pump body (Fig. 1)& (Fig. 2). The 2.4L engine has a remote pump reservoir mounted to the fan shroud (Fig. 3).

The steering gear used is a straight ratio recirculating ball type gear. A tilt and non-tilt column provide steering input.

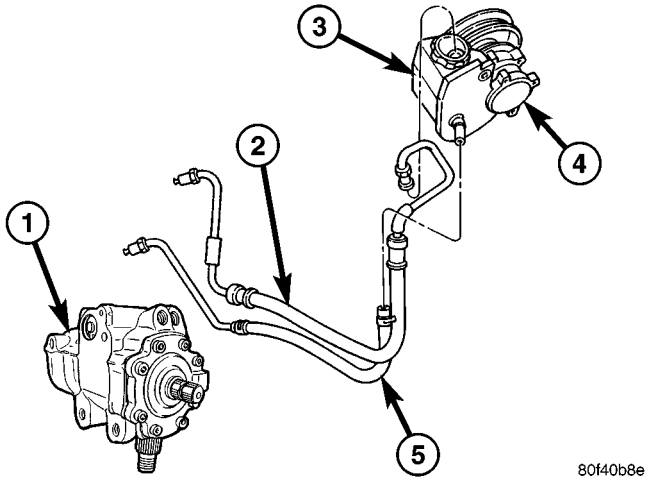

Fig. 1 POWER STEERING GEAR & PUMP - 4.0L

1 - POWER STEERING GEAR
2 - PRESSURE HOSE
3 - POWER STEERING RESERVOIR
4 - POWER STEERING PUMP
5 - RETURN HOSE

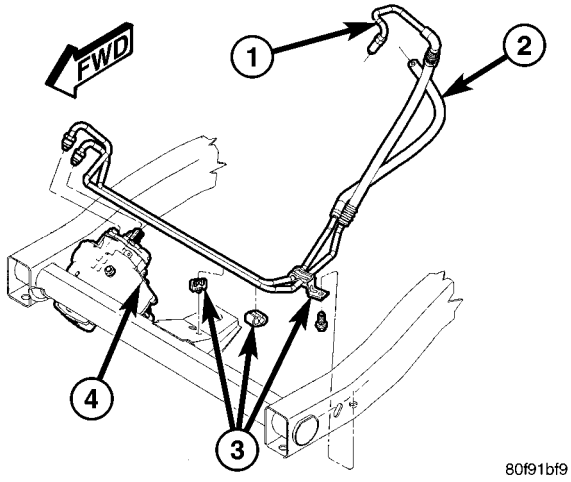

Fig. 2 RHD POWER STEERING HOSES

1 - PRESSURE HOSE
2 - RETURN HOSE
3 - MOUNTING BRACKETS
4 - STEERING GEAR

STEERING (Continued)

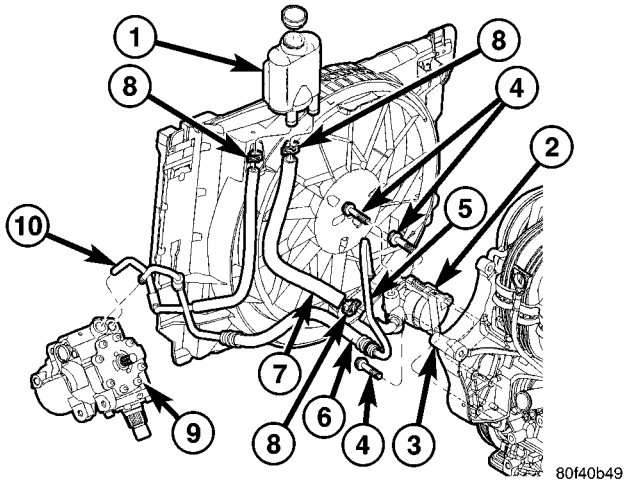

Fig. 3 POWER STEERING GEAR & PUMP - 2.4L

1 - REMOTE RESERVOIR
2 - POWER STEERING PUMP
3 - PUMP MOUNT BRACKET
4 - MOUNTING BOLTS (3)
5 - PRESSURE SWITCH LOCATION
6 - PRESSURE OUTLET HOSE (GEAR TO PUMP)
7 - RETURN HOSE (RESERVOIR TO PUMP)
8 - CLAMPS
9 - STEERING GEAR
10 - INLET HOSE (RESERVOIR TO GEAR)

OPERATION

The gear acts as a rolling thread between the worm shaft and rack piston. The worm shaft is supported by a thrust bearing at the upper end. When the worm shaft is turned from input from the steering column the rack piston moves. The rack piston teeth mesh with the pitman shaft. Turning the worm shaft turns the pitman shaft, which turns the steering linkage.

DIAGNOSIS AND TESTING

DIAGNOSIS AND TESTING - POWER STEERING SYSTEM DIAGNOSIS CHARTS

There is some noise in all power steering systems. One of the most common is a hissing sound evident at a standstill parking. Or when the steering wheel is at the end of it's travel. Hiss is a high frequency noise similar to that of a water tap being closed slowly. The noise is present in all valves that have a high velocity fluid passing through an orifice. There is no relationship between this noise and steering performance.

STEERING NOISE

CONDITION	POSSIBLE CAUSES	CORRECTION
OBJECTIONAL HISS OR WHISTLE	1. Steering intermediate shaft to dash panel seal.	1. Check and repair seal at dash panel.
	2. Noisy valve in power steering gear.	2. Replace steering gear.
RATTLE OR CLUNK	1. Gear mounting bolts loose.	1. Tighten bolts to specification.
	2. Loose or damaged suspension components/track bar.	2. Inspect and repair suspension.
	3. Loose or damaged steering linkage.	3. Inspect and repair steering linkage.
	4. Internal gear noise.	4. Replace gear.
	5. Pressure hose in contact with other components.	5. Reposition hose.

STEERING (Continued)

CONDITION	POSSIBLE CAUSES	CORRECTION
CHIRP OR SQUEAL	1. Loose belt.	1. Adjust or replace.
WHINE OR GROWL	1. Low fluid level. 2. Pressure hose in contact with other components. 3. Internal pump noise. 4. Air in the system.	1. Fill to proper level. 2. Reposition hose. 3. Replace pump. 4. Perform pump initial operation.
SUCKING AIR SOUND	1. Loose return line clamp. 2. O-ring missing or damaged on hose fitting. 3. Low fluid level. 4. Air leak between pump and reservoir.	1. Replace clamp. 2. Replace o-ring. 3. Fill to proper level. 4. Repair as necessary.
SCRUBBING OR KNOCKING	1. Wrong tire size. 2. Wrong gear.	1. Verify tire size. 2. Verify gear.

BINDING AND STICKING

CONDITION	POSSIBLE CAUSE	CORRECTION
DIFFICULT TO TURN WHEEL STICKS OR BINDS	1. Low fluid level. 2. Tire pressure. 3. Steering component. 4. Loose belt. 5. Low pump pressure. 6. Column shaft coupler binding. 7. Steering gear worn or out of adjustment. 8. Ball joints binding.	1. Fill to proper level. 2. Adjust tire pressure. 3. Inspect and lube. 4. Adjust or replace. 5. Pressure test and replace if necessary. 6. Replace coupler. 7. Replace gear. 8. Inspect and repair as necessary.

INSUFFICIENT ASST. OR POOR RETURN TO CENTER

CONDITION	POSSIBLE CAUSE	CORRECTION
HARD TURNING OR MOMENTARY INCREASE IN TURNING EFFORT	1. Tire pressure. 2. Low fluid level. 3. Loose belt. 4. Lack of lubrication. 5. Low pump pressure. 6. Internal gear leak.	1. Adjust tire pressure. 2. Fill to proper level. 3. Adjust or replace. 4. Inspect and lubricate steering and suspension compnents. 5. Pressure test and repair as necessary. 6. Pressure and flow test, and replace as necessary.

STEERING (Continued)

CONDITION	POSSIBLE CAUSE	CORRECTION
STEERING WHEEL DOES NOT WANT TO RETURN TO CENTER POSITION	1. Tire pressure.	1. Adjust tire pressure.
	2. Wheel alignment.	2. Align front end.
	3. Lack of lubrication.	3. Inspect and lubricate steering and suspension compnents.
	4. High friction in steering gear.	4. Test and adjust as necessary.
	5. Ball joints binding.	5. Inspect and repair as necessary.

NOTE:
Some roads will cause a vehicle to drift, due to the crown in the road.

LOOSE STEERING AND VEHICLE LEADS/DRIFTS

CONDITION	POSSIBLE CAUSE	CORRECTION
EXCESSIVE PLAY IN STEERING WHEEL	1. Worn or loose suspension or steering components.	1. Repair as necessary.
	2. Worn or loose wheel bearings.	2. Repair as necessary.
	3. Steering gear mounting.	3. Tighten gear mounting bolts to specification.
	4. Gear out of adjustment.	4. Adjust gear to specification.
	5. Worn or loose steering coupler.	5. Repair as necessary.
VEHICLE PULLS TO ONE SIDE DURING BRAKING	1. Tire Pressure.	1. Adjust tire pressure.
	2. Air in brake hydraulics system.	2. Bleed brake system.
	3. Worn brake components.	3. Repair as necessary.
VEHICLE LEADS OR DRIFTS FROM STRAIGHT AHEAD DIRECTION ON UNCROWNED ROAD.	1. Tire pressure.	1. Adjust tire pressure.
	2. Radial tire lead.	2. Cross front tires.
	3. Brakes dragging.	3. Repair as necessary.
	4. Wheel alignment.	4. Align vehicle.
	5. Weak or broken spring.	5. Replace spring.
	6. Loose or worn steering/ suspension components.	6. Repair as necessary.
	7. Cross caster out of spec.	7. Adjust or replace axle as necessary.

DIAGNOSIS AND TESTING - POWER STEERING FLOW AND PRESSURE

The following procedure is used to test the operation of the power steering system on the vehicle. This test will provide the gallons per minute (GPM) or flow rate of the power steering pump along with the maximum relief pressure. Perform test any time a power steering system problem is present. This test will determine if the power steering pump or power steering gear is not functioning properly. The following pressure and flow test is performed using Power Steering Analyzer Tool 6815 (Fig. 4) and Adapter kit 6893.

FLOW AND PRESSURE TEST

(1) Check the power steering belt to ensure it is in good condition and adjusted properly.

(2) Connect pressure gauge hose from the Power Steering Analyzer to Tube 6865.

(3) Connect Adapter 6826 to Power Steering Analyzer test valve end.

STEERING (Continued)

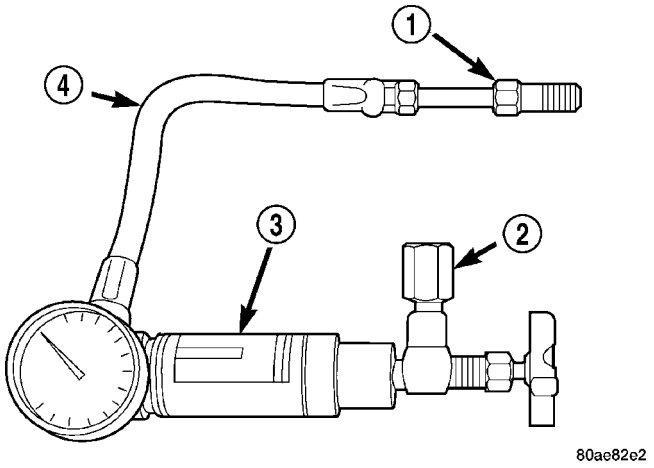

Fig. 4 Power Steering Analyzer

1 - TUBE
2 - ADAPTER FITTINGS
3 - ANALYZER
4 - GAUGE HOSE

(4) Disconnect the high pressure hose from the power steering pump.

(5) Connect Tube 6865 to the pump hose fitting.

(6) Connect the power steering hose from the steering gear to Adapter 6826.

(7) Open the test valve completely.

(8) Start engine and let idle long enough to circulate power steering fluid through flow/pressure test gauge.

(9) Shut off the engine and check the fluid level, add fluid as necessary. Start engine again and let idle.

(10) Gauge should read below 862 kPa (125 psi), if above, inspect the hoses for restrictions and repair as necessary. The initial pressure reading should be in the range of 345-552 kPa (50-80 psi).

(11) Increase the engine speed to 1500 RPM and read the flow meter. The reading should be 2.4 - 2.8 GPM, if the reading is below this specification the pump should be replaced.

CAUTION: This next step involves testing maximum pump pressure output and flow control valve operation. Do not leave test valve closed for more than three seconds as the pump could be damaged.

(12) Close valve fully three times for three seconds and record highest pressure indicated each time. **All three readings must be above pump relief pressure specifications and within 345 kPa (50 psi) of each other.**

- Pressures above specifications but not within 345 kPa (50 psi) of each other, replace pump.
- Pressures within 345 kPa (50 psi) of each other but below specifications, replace pump.

(13) Open the test valve and turn the steering wheel to the extreme left and right positions against the stops. Record the highest pressure reading at each position. Compare readings to pump specifications chart. If pressure readings are not within 50 psi. of each other, the gear is leaking internally and must be repaired.

CAUTION: Do not force the pump to operate against the stops for more than 2 to 4 seconds at a time because, pump damage will result.

PUMP SPECIFICATIONS

ENGINE	RELIEF PRESSURE ± 50	FLOW RATE (GPM)
2.4L	9653 kPa (1400 psi)	1500 RPM 2.4 - 2.8 GPM
4.0L	9653 kPa (1400 psi)	

COLUMN

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COLUMN

DESCRIPTION

NOTE: The steering column on vehicles with an automatic transmission may not be equipped with an internal locking shaft that allows the ignition key cylinder to be locked with the key. Alternative methods of locking the steering wheel for service will have to be used.

The standard non-tilt and tilt steering column has been designed to be serviced as an assembly. The column is connected to the steering gear with an upper and lower shaft. The lower shaft has a support bearing mounted to a bracket. The bracket mounts to the frame rail with two bolts. These shafts and bearing are serviceable. The key cylinder, switches, clock spring, trim shrouds and steering wheel are serviced separately.

OPERATION - SERVICE PRECAUTIONS

Safety goggles should be worn at all times when working on steering columns.

To service the steering wheel, switches or airbag, refer to Electrical - Restraints and follow all WARNINGS and CAUTIONS.

WARNING: THE AIRBAG SYSTEM IS A SENSITIVE, COMPLEX ELECTRO-MECHANICAL UNIT. BEFORE ATTEMPTING TO DIAGNOSE, REMOVE OR INSTALL THE AIRBAG SYSTEM COMPONENTS YOU MUST FIRST DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE. THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE. FAILURE TO DO SO COULD RESULT IN ACCIDENTAL DEPLOYMENT OF THE AIRBAG AND POSSIBLE PERSONAL INJURY. THE FASTENERS, SCREWS, AND BOLTS, ORIGINALLY USED FOR THE AIRBAG COMPONENTS, HAVE SPECIAL COATINGS AND ARE SPECIFICALLY DESIGNED FOR THE AIRBAG SYSTEM. THEY MUST NEVER BE REPLACED WITH ANY SUBSTITUTES. ANYTIME A NEW FASTENER IS NEEDED, REPLACE WITH THE CORRECT FASTENERS PROVIDED IN THE SERVICE PACKAGE OR FASTENERS LISTED IN THE PARTS BOOKS.

REMOVAL

(1) Position front wheels **straight ahead.**

(2) Remove and isolate the negative ground cable from the battery.

(3) Remove the airbag, (Refer to 8 - ELECTRICAL/ RESTRAINTS/DRIVER AIRBAG - REMOVAL).

NOTE: If equipped with cruise control, disconnect clock spring harness from the cruise switch harness on the steering wheel.

(4) Remove the steering wheel with an appropriate puller (Fig. 1).

NOTE: Ensure the puller jaws are seated in the pockets (Fig. 2) of the steering wheel armature.

(5) Turn ignition cylinder to the on position and remove cylinder by pressing release through lower shroud access hole (Fig. 3) (Refer to 19 - STEERING/ COLUMN/LOCK CYLINDER HOUSING - REMOVAL).

COLUMN (Continued)

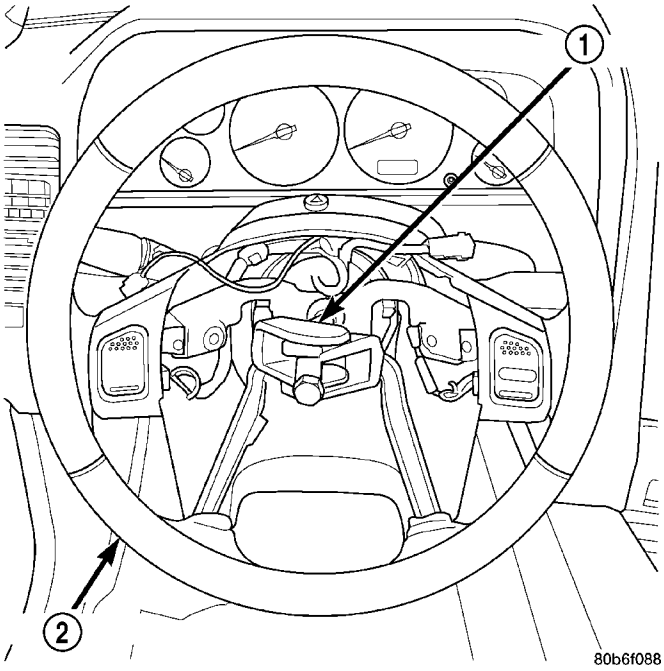

Fig. 1 STEERING WHEEL PULLER

1 - PULLER C-3894-A
2 - STEERING WHEEL

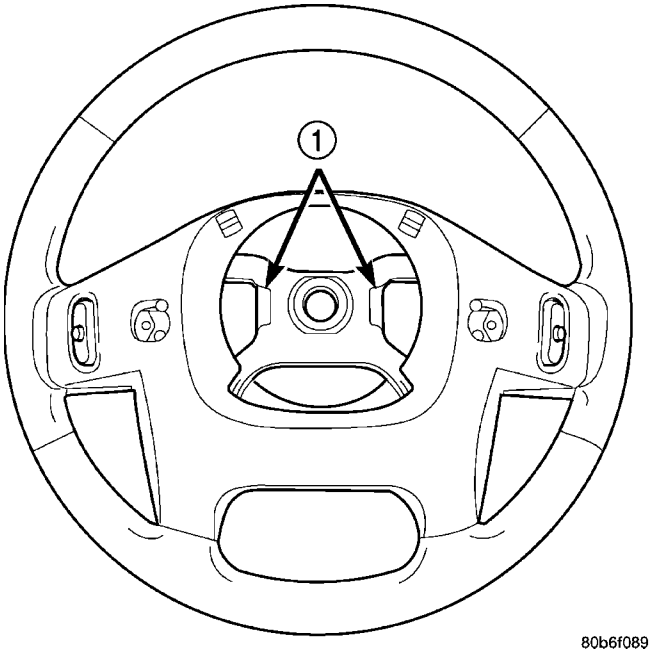

Fig. 2 Steering Wheel Pockets

1 - STEERING WHEEL POCKETS

(6) Remove knee blocker cover and knee blocker, (Refer to 23 - BODY/INSTRUMENT PANEL/KNEE BLOCKER - REMOVAL).

(7) Remove screws from the lower column shroud (Fig. 4) and remove the shroud.

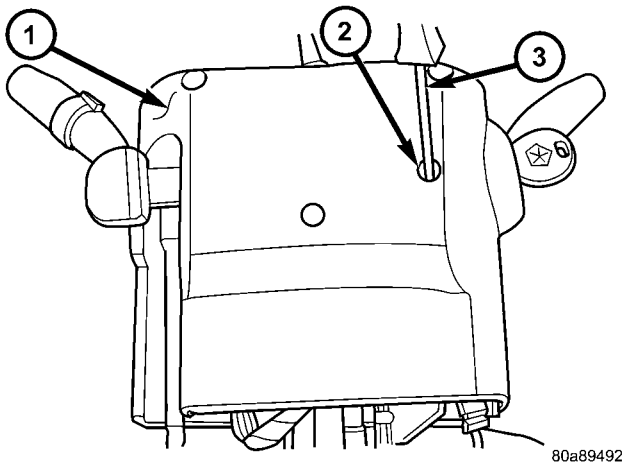

Fig. 3 LOCK CYLINDER RELEASE

1 - Lower Shroud
2 - Lock Cylinder Release Access Hole
3 - Pin Punch

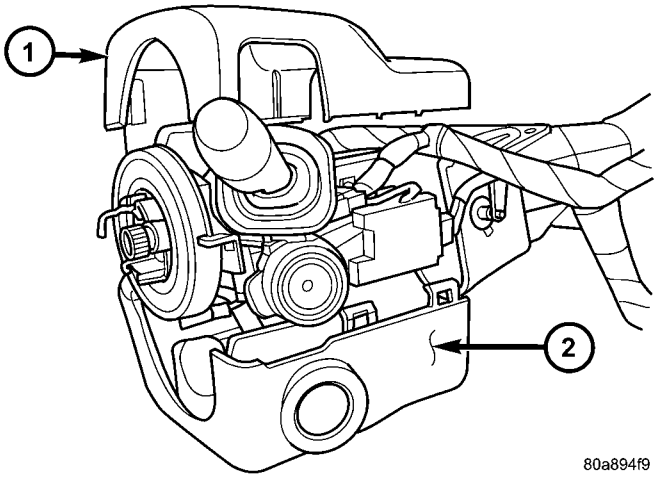

Fig. 4 SHROUD REMOVAL/INSTALL

1 - Upper Shroud
2 - Lower Shroud

(8) Remove the steering coupler bolt and column mounting nuts (Fig. 5) then lower column off the mounting studs.

(9) Remove upper column shroud (Fig. 4).

(10) Disconnect and remove the wiring harness from the column (Fig. 6).

NOTE: If vehicle is equipped with automatic transmission, remove shifter interlock cable from the column. (Refer to 21 - TRANSMISSION/TRANSAXLE/ AUTOMATIC - 30RH/GEAR SHIFT CABLE - REMOVAL).

(11) Remove column.

COLUMN (Continued)

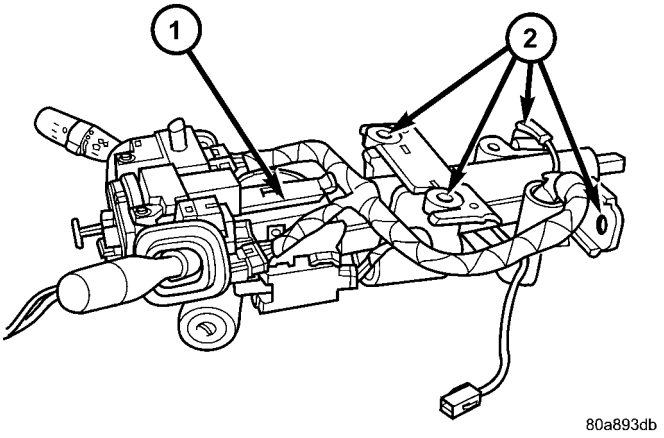

Fig. 5 STEERING COLUMN MOUNTING

1 - Steering Column
2 - Mounting Holes

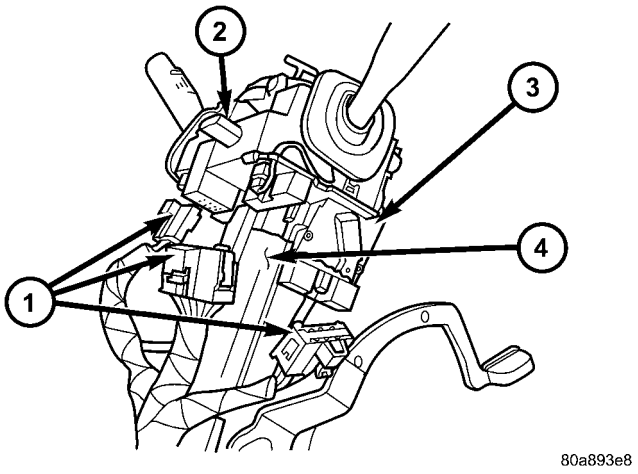

Fig. 6 WIRING HARNESS COLUMN

1 - Column Wiring Harness
2 - Multi-function Switch
3 - Ignition Switch
4 - Steering Column

(12) Remove clock spring (Fig. 7), switches, (SKIM if equipped) (Refer to 8 - ELECTRICAL/RESTRAINTS/CLOCKSPRING - REMOVAL).

INSTALLATION

(1) Align and install column into the steering coupler.

(2) Install column harness and connect harness to switches.

NOTE: If vehicle is equipped with automatic transmission, install shifter interlock cable. (Refer to 21 - TRANSMISSION/TRANSAXLE/AUTOMATIC - 30RH/ GEAR SHIFT CABLE - ADJUSTMENTS).

(3) Install the upper column shroud.

(4) Install the column onto the mounting studs.

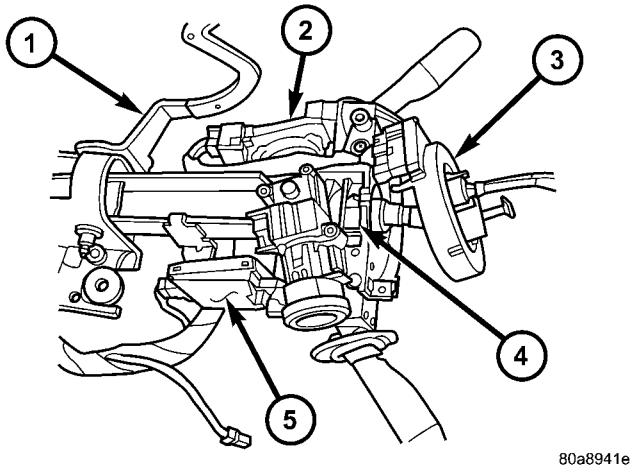

Fig. 7 CLOCK SPRING

1 - Tilt Lever
2 - Ignition Switch
3 - Clockspring
4 - Steering Column
5 - SKIM

CAUTION: Lower nuts must be installed and tightened first then the upper nuts in order to prevent damage to the capsules.

(5) Install the lower mounting nuts and tighten to 17 N·m (150 in. lbs.).

(6) Install the upper mounting nuts and tighten to 17 N·m (150 in. lbs.).

(7) Install the steering column coupler bolt and tighten to 49 N·m (36 ft. lbs.).

(8) Center the clock spring (if necessary) and install it on the column, (Refer to 8 - ELECTRICAL/ RESTRAINTS/CLOCKSPRING - INSTALLATION).

(9) Install the lower column shroud and install mounting screws.

(10) Install the ignition lock cylinder. (Refer to 19 - STEERING/COLUMN/LOCK CYLINDER HOUSING - INSTALLATION).

(11) Install the knee blocker and the knee blocker cover, (Refer to 23 - BODY/INSTRUMENT PANEL/ KNEE BLOCKER - INSTALLATION).

NOTE: Do not reuse the old steering wheel bolt (a new bolt must be used)

NOTE: If equipped with cruise control, connect clock spring harness to cruise switch harness on the steering wheel.

(12) Install the steering wheel and tighten bolt to 54 N·m (40 ft. lbs.).

(13) Install the airbag, (Refer to 8 - ELECTRICAL/ RESTRAINTS/DRIVER AIRBAG - INSTALLATION).

(14) Install the negative battery terminal.

COLUMN (Continued)

SPECIFICATIONS

TORQUE CHART

TORQUE SPECIFICATIONS

DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Tilt Steering Column Steering Wheel Nut	54	40	—
Tilt Steering Column Mounting Nuts	17	—	150
Tilt Steering Column Coupler Bolt	49	36	—
Non-Tilt Steering Column Steering Wheel Nut	54	40	—
Non-Tilt Steering Column Mounting Nuts	17	—	150
Non-Tilt Steering Column Coupler Bolt	49	36	—

IGNITION SWITCH

DESCRIPTION

The electrical ignition switch is located on the steering column. It is used as the main on/off switching device for most electrical components. The mechanical key cylinder is used to engage/disengage the electrical ignition switch.

DIAGNOSIS AND TESTING - IGNITION SWITCH

ELECTRICAL DIAGNOSIS

For ignition switch electrical schematics, Refer to the appropriate section for the component.

MECHANICAL DIAGNOSIS (KEY DIFFICULT TO ROTATE)

Vehicles equipped with an automatic transmission and a floor mounted shifter: a cable is used to connect the interlock device in the steering column assembly, to the transmission floor shift lever. This interlock system is used to lock the transmission shifter in the PARK position when the key cylinder is rotated to any position. If the ignition key is difficult to rotate to or from any position, it may not be the fault of the key cylinder or the steering column components. The brake transmission shift interlock cable may be out of adjustment. (Refer to 21 - TRANSMISSION/TRANSAXLE/AUTOMATIC - 30RH/GEAR SHIFT CABLE - ADJUSTMENTS). The interlock system within the steering column is not serviceable. If repair is necessary, the steering column assembly must be replaced. (Refer to 19 - STEERING/COLUMN - REMOVAL).

Vehicles equipped with a manual transmission and a floor mounted shifter: on certain models, a button is located on the steering column behind the ignition key cylinder. The button must be manually depressed to allow rotation of the ignition key cylinder from the **off** to **lock** positions. If it is difficult to rotate the key to any position, the lever mechanism may be defective. This mechanism is not serviceable. If repair is necessary, the steering column assembly must be replaced.(Refer to 19 - STEERING/COLUMN - REMOVAL).

REMOVAL

The ignition key must be in the key cylinder for cylinder removal. The key cylinder must be removed first before removing ignition switch.

(1) Remove key cylinder. (Refer to 19 - STEERING/COLUMN/LOCK CYLINDER HOUSING - REMOVAL).

(2) Remove lower steering column cover screws and remove cover.

(3) Remove the multi-function switch.

(4) Disconnect the electrical connector at the rear of the ignition switch.

(5) Remove the ignition switch mounting screw (Fig. 8). Use tamper proof torx bit to remove the screw.

(6) Pull the ignition switch straight out to remove from the locking tabs (Fig. 9)

IGNITION SWITCH (Continued)

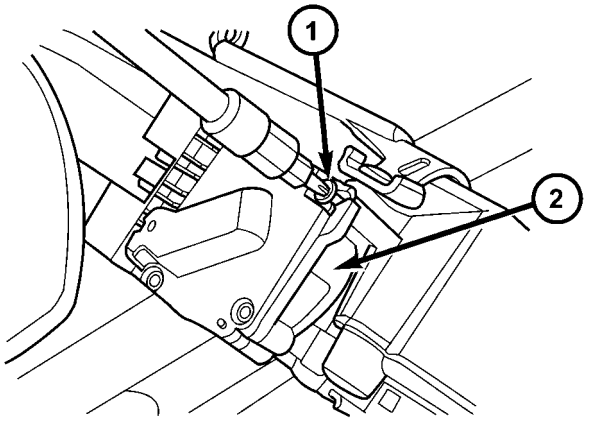

Fig. 8 IGNITION SWITCH MOUNTING SCREW

1 - Tamper Proof Torx Screw
2 - Ignition Switch

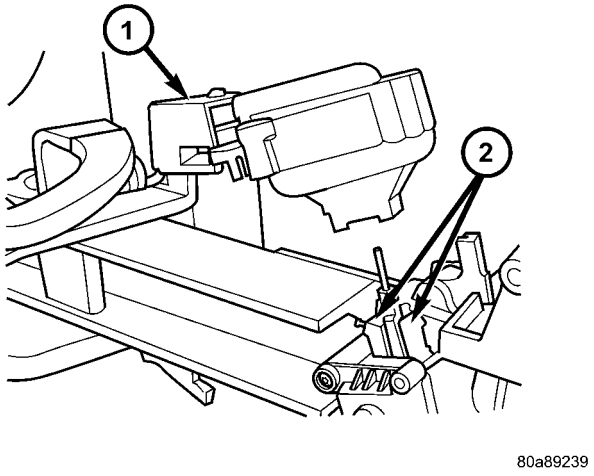

Fig. 9 IGNITION SWITCH TABS

1 - Ignition Switch
2 - Locking Tabs

INSTALLATION

The ignition key must be in the key cylinder for cylinder installation. The key cylinder must be aligned with the ignition switch for installation.

(1) Before installing ignition switch, rotate the slot in the switch to the ON position (Fig. 10).

(2) Connect the electrical connector to rear of ignition switch. Make sure that locking tab is fully seated into wiring connector.

(3) Position the switch to the column and install tamper proof screw. Tighten screw to 3 N·m (26 in. lbs.).

(4) Test the operation of the lock cylinder for smooth rotating.

(5) Install the multi-function switch.

(6) Install steering column lower cover.

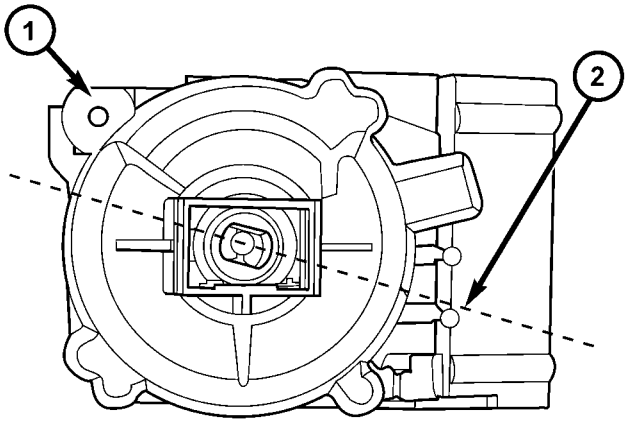

Fig. 10 IGNITION SWITCH ON POSITION

1 - Ignition Switch
2 - Rotate to On Position

KEY-IN IGNITION SWITCH

DESCRIPTION

The key-in ignition switch is integral to the ignition switch, which is mounted on the left side of the steering column, opposite the ignition cylinder. It closes a path to ground for the instrument cluster chime warning circuitry when the ignition key is inserted in the ignition cylinder and the driver door jamb switch is closed (driver door is open). The key-in ignition switch opens the ground path when the key is removed from the ignition cylinder.

The key-in ignition switch cannot be repaired and, if faulty or damaged, the entire ignition switch must be replaced. (Refer to 19 - STEERING/COLUMN/IGNITION SWITCH - REMOVAL).

DIAGNOSIS AND TESTING - KEY-IN IGNITION SWITCH

For circuit descriptions and diagrams, Refer to the appropriate sections on the individual components.

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, REFER TO ELECTRICAL - PASSIVE RESTRAINT SYSTEMS BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Disconnect and isolate the battery negative cable. Remove the steering column shrouds. Unplug the key-in ignition switch wire harness connector from the ignition switch.

KEY-IN IGNITION SWITCH (Continued)

(2) Check for continuity between the key-in switch sense circuit and the left front door jamb switch sense circuit terminals of the key-in ignition switch. There should be continuity with the key in the ignition cylinder, and no continuity with the key removed from the ignition cylinder. If OK, go to Step 3. If not OK, replace the faulty ignition switch assembly.

(3) Check for continuity between the left front door jamb switch sense circuit cavity of the key-in ignition switch wire harness connector and a good ground. There should be continuity with the driver door open, and no continuity with the driver door closed. If OK, see the diagnosis for Instrument Cluster in this group. If not OK, repair the circuit to the driver door jamb switch as required.

KEY CYLINDER

REMOVAL

The ignition key must be in the key cylinder for cylinder removal. The key cylinder must be removed first before removing ignition switch.

(1) If equipped with an automatic transmission, place shifter in PARK position.

(2) Rotate key to ON position.

(3) A release tang is located on bottom of key cylinder (Fig. 11).

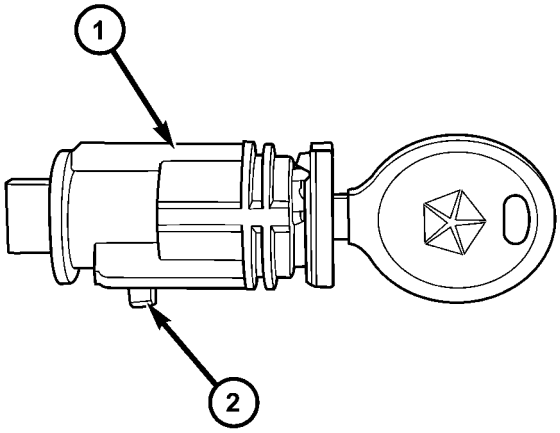

Fig. 11 KEY CYLINDER RELEASE TANG

1 - KEY CYLINDER
2 - RELEASE TANG

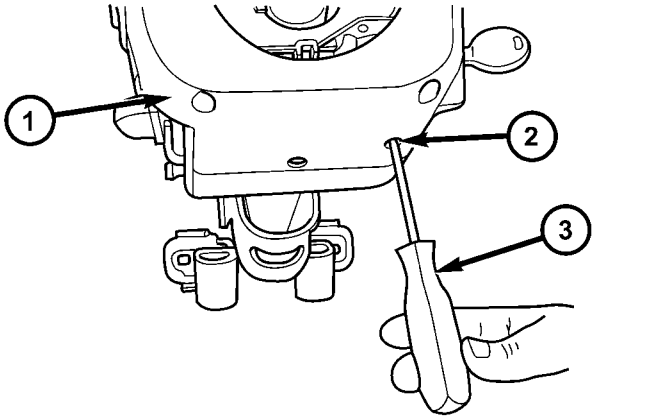

Fig. 12 KEY CYLINDER RELEASE HOLE

1 - Lower Cover
2 - Access Hole
3 - Pin Punch

(4) Position a small screwdriver or pin punch into tang access hole on bottom of steering column lower cover (Fig. 12).

(5) Push the pin punch up while pulling key cylinder from steering column.

INSTALLATION

The ignition key must be in the key cylinder for cylinder installation.

(1) Install the lock cylinder into the housing using care to align the end of the lock cylinder with the ignition switch.

(2) Push the lock cylinder in until it clicks.

(3) Rotate the key to the lock position.

STEERING WHEEL

REMOVAL

(1) Disable and remove the drivers side airbag.(Refer to 8 - ELECTRICAL/RESTRAINTS/DRIVER AIRBAG - REMOVAL).

(2) Partially remove the steering wheel bolt and leave the bolt in the column.

(3) Install puller C-3894–A or equivalent using the top of the bolt to push on. (Fig. 13)

STEERING WHEEL (Continued)

NOTE: Ensure the puller jaws are seated in the pockets (Fig. 14) of the steering wheel armature.

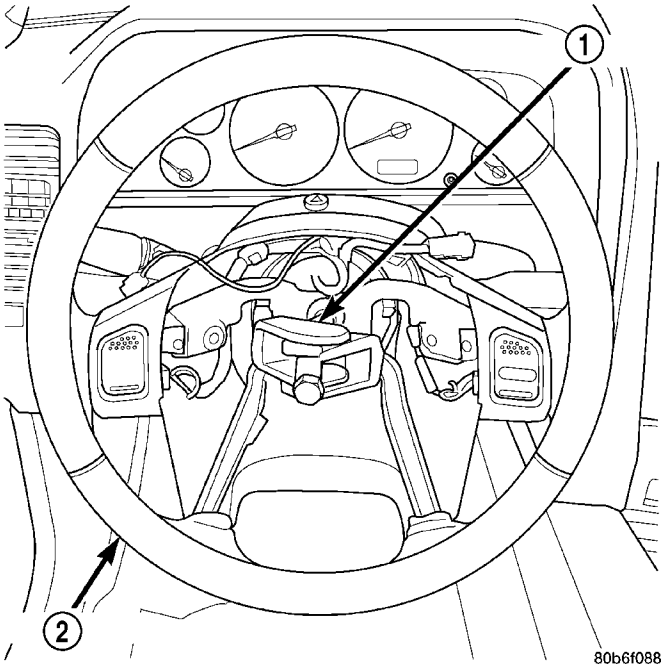

Fig. 13 STEERING WHEEL PULLER

1 - PULLER C-3894-A
2 - STEERING WHEEL

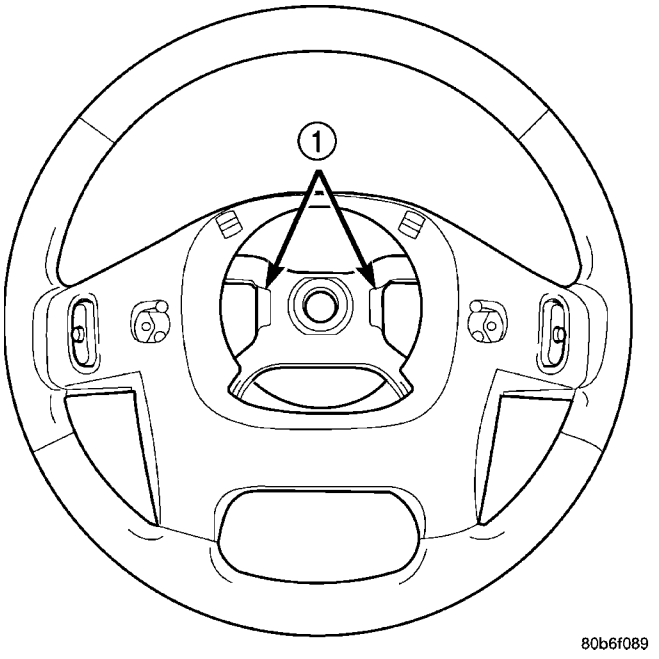

Fig. 14 Steering Wheel Pockets

1 - STEERING WHEEL POCKETS

(4) Remove the steering wheel.

INSTALLATION

NOTE: Do not reuse the old steering wheel bolt (a new bolt must be used)

(1) Install steering wheel to the column

NOTE: Be certain that the steering wheel mounting bolt is tightened to the proper torque specification to ensure proper clockspring operation.

(2) Install the new steering wheel bolt. Tighten the bolt to 54 N·m (40 ft. lbs.).

(3) Install the drivers side air bag. (Refer to 8 - ELECTRICAL/RESTRAINTS/DRIVER AIRBAG - INSTALLATION).

GEAR

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GEAR

DESCRIPTION

CAUTION: MOPAR® ATF+4 is to be used in the power steering system. No other power steering or automatic transmission fluid is to be used in the system. Damage may result to the power steering pump and system if any other fluid is used, and do not overfill.

The power steering system has a hydraulic pump. The pump is a constant flow rate and displacement vane-type pump. The pump reservoir on the 4.0L engine is mounted to the pump body (Fig. 1)& (Fig. 2). The 2.4L engine has a remote pump reservoir mounted to the fan shroud (Fig. 3).

The steering gear used is a straight ratio recirculating ball type gear. A tilt and non-tilt column provide steering input.

OPERATION

The gear acts as a rolling thread between the worm shaft and rack piston. The worm shaft is supported by a thrust bearing at the upper end. When the worm shaft is turned the rack piston moves. The rack piston teeth mesh with the pitman shaft. Turning the worm shaft turns the pitman shaft, which turns the steering linkage.

CAUTION: Components attached with a nut and cotter pin must be torqued to specification. Then if the slot in the nut does not line up with the cotter pin hole, tighten nut until it is aligned. Never loosen the nut to align the cotter pin hole.

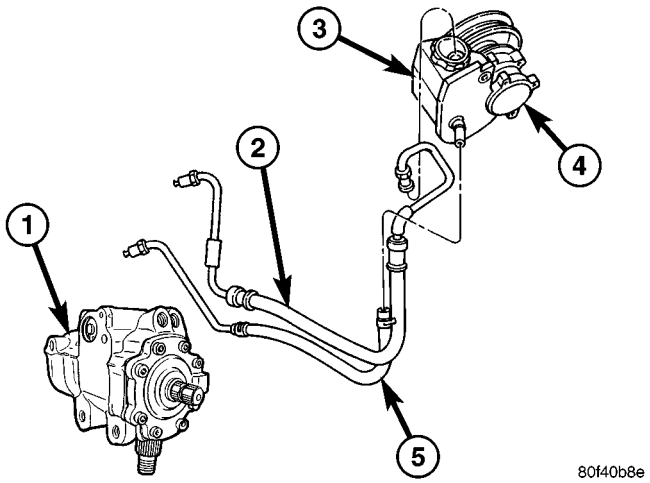


Fig. 1 POWER STEERING GEAR & PUMP - 4.0L

1 - POWER STEERING GEAR
2 - PRESSURE HOSE
3 - POWER STEERING RESERVOIR
4 - POWER STEERING PUMP
5 - RETURN HOSE

DIAGNOSIS AND TESTING - POWER STEERING GEAR LEAKAGE AREAS

NOTE: Do not service the Steering Gear, if the gear is leaking it must be replaced with a new gear assembly.

(1) Possible power steering gear leakage areas. (Fig. 4).

REMOVAL

NOTE: The steering column on vehicles with an automatic transmission may not be equipped with an internal locking shaft that allows the ignition key cylinder to be locked with the key. Alternative methods of locking the steering wheel for service will have to be used.

GEAR (Continued)

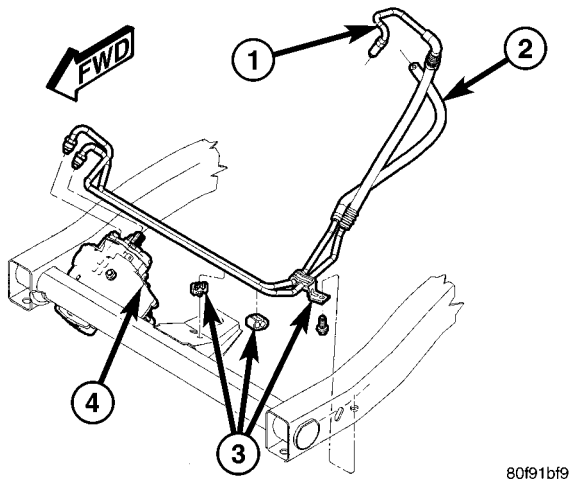


Fig. 2 RHD POWER STEERING HOSES

1 - PRESSURE HOSE
2 - RETURN HOSE
3 - MOUNTING BRACKETS
4 - STEERING GEAR

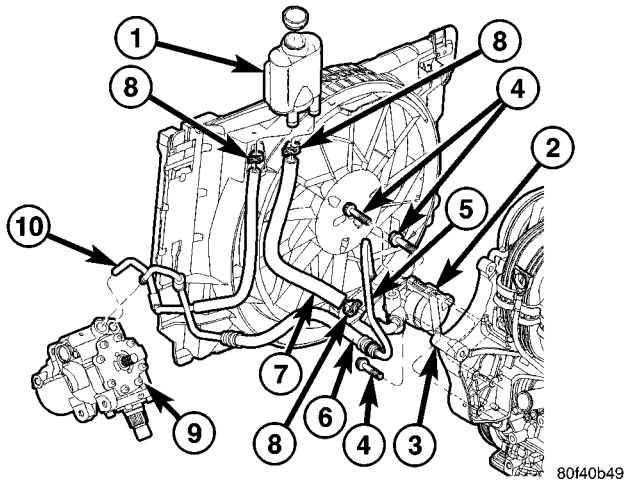


Fig. 3 POWER STEERING GEAR & PUMP - 2.4L

1 - REMOTE RESERVOIR
2 - POWER STEERING PUMP
3 - PUMP MOUNT BRACKET
4 - MOUNTING BOLTS (3)
5 - PRESSURE SWITCH LOCATION
6 - PRESSURE OUTLET HOSE (GEAR TO PUMP)
7 - RETURN HOSE (RESERVOIR TO PUMP)
8 - CLAMPS
9 - STEERING GEAR
10 - INLET HOSE (RESERVOIR TO GEAR)

(1) Place the front wheels in the straight ahead position with the steering wheel centered and locked.

(2) Siphon out as much power steering fluid as possible.

(3) Remove the bumper shield.

(4) Remove power steering hoses/tubes from steering gear.

(5) Remove the column coupler shaft from the gear.

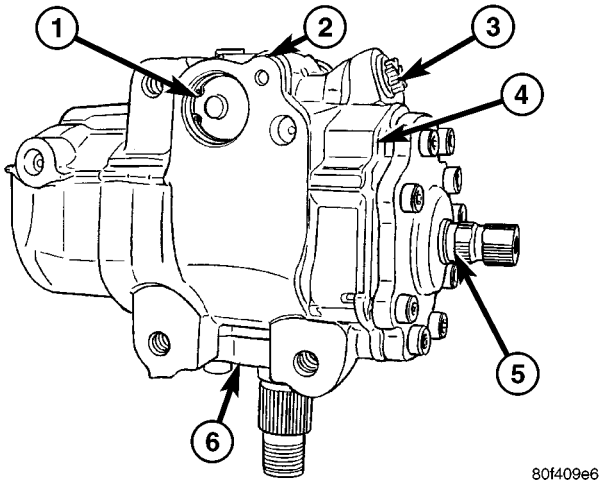


Fig. 4 STEERING GEAR

1 - VALVE HOUSING O-RING
2 - ADJUSTER NUT
3 - POWER STEERING LINE FITTINGS
4 - INPUT SHAFT HOUSING GASKET
5 - INPUT SHAFT SEAL
6 - PITMAN SHAFT SEAL HOUSING / COVER O-RING

(6) Remove pitman arm from gear. (Refer to 19 - STEERING/LINKAGE/PITMAN ARM - REMOVAL).

(7) Remove the steering gear retaining bolts and remove the gear (Fig. 5).

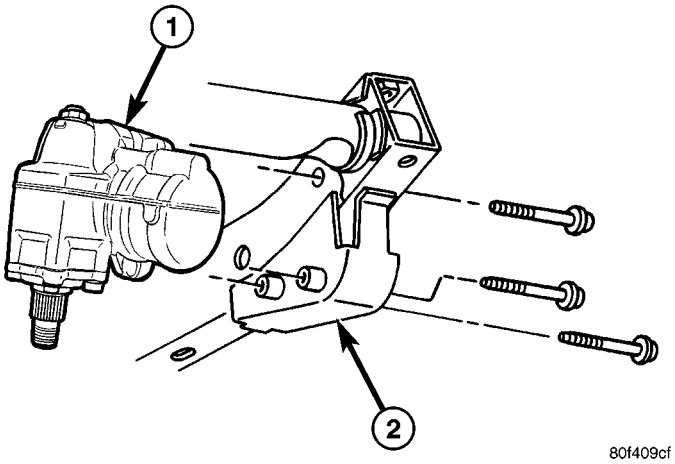


Fig. 5 STEERING GEAR REMOVAL/INSTALL

1 - STEERING GEAR
2 - FRAME MOUNT

INSTALLATION

(1) Install steering gear on the frame rail and tighten bolts to 95 N·m (70 ft. lbs.)

(2) Align and install the pitman arm and tighten nut to 251 N·m (185 ft. lbs.).

(3) Align the column coupler shaft to steering gear. Install a **new** coupler pinch bolt and tighten to 49 N·m (36 ft. lbs.).

GEAR (Continued)

(4) Install power steering hoses/tubes to steering gear and tighten to 28 N·m (21 ft. lbs.).

(5) Install the bumper shield.

(6) Fill power steering system to proper level, (Refer to 19 - STEERING/PUMP - STANDARD PROCEDURE).

ADJUSTMENTS

CAUTION: Steering gear must be adjusted in the proper order. If adjustments are not performed in order, gear damage and improper steering response may result.

NOTE: Adjusting the steering gear in the vehicle is not recommended. Remove gear from the vehicle and drain the fluid. Then mount gear in a vise to perform adjustments.

WORM THRUST BEARING PRELOAD

NOTE: Off center torque (180 to 360 degrees from gear center) = 0.4 to 0.8 Nm (3.5 in-lb to 7.0 in-lb) This torque is set in the assembly plant and can't be adjusted in the field

OVER-CENTER ROTATING TORQUE

(1) Mount the gear carefully into a vise.

CAUTION: Do not overtighten the vise on the gear case. This may affect the adjustment

(2) Rotate the input shaft with a crows foot socket from stop to stop and count the number of turns.

(3) Starting at either stop, turn the input shaft back 1/2 the total number of turns. This is the center of the gear travel.

(4) Loosen the adjuster plug locknut.

(5) Place the torque wrench in the vertical position on the input shaft. Rotate the wrench 50 degrees each side of the center and record the highest rotational torque in this range (Fig. 6). This is the Over-Center Rotating Torque.

NOTE: The input shaft must rotate smoothly without sticking or binding.

(6) The Over-Center Rotating Torque should be 0.53-0.93 N·m (4.5 - 8 in. lbs.) **higher** in addition to the off center torque from above (Fig. 6). than the Preload Rotating Torque.

(7) If an adjustment to the Over-Center Rotating Torque is necessary, first loosen the adjuster lock nut. Then turn the pitman shaft adjuster screw back

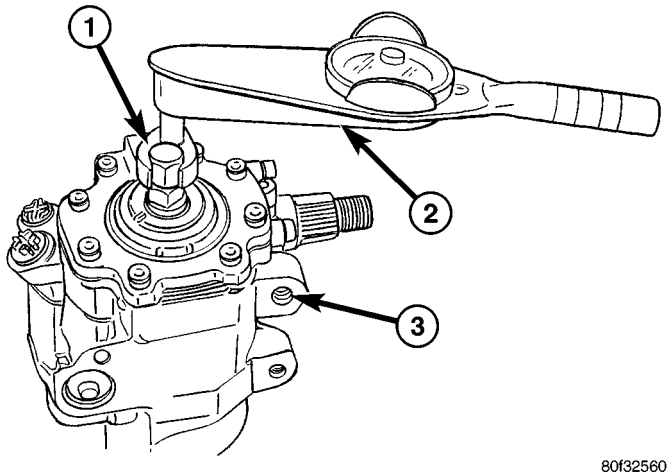


Fig. 6 OVER-CENTER TORQUE

1 - CROWSFOOT WRENCH
2 - INCH POUND TORQUE WRENCH
3 - STEERING GEAR

(COUNTERCLOCKWISE) increases torque, (CLOCKWISE) reduces torque (Fig. 7).

(8) Remeasure Over-Center Rotating Torque. If necessary turn the adjuster screw and repeat measurement until correct Over-Center Rotating Torque is reached (Fig. 7).

NOTE: To increase the Over-Center Rotating Torque turn the screw COUNTERCLOCKWISE.

(9) Prevent the adjuster screw from turning while tightening adjuster lock nut (Fig. 7). Tighten the adjuster lock nut to 65 N·m (48 ft. lbs.).

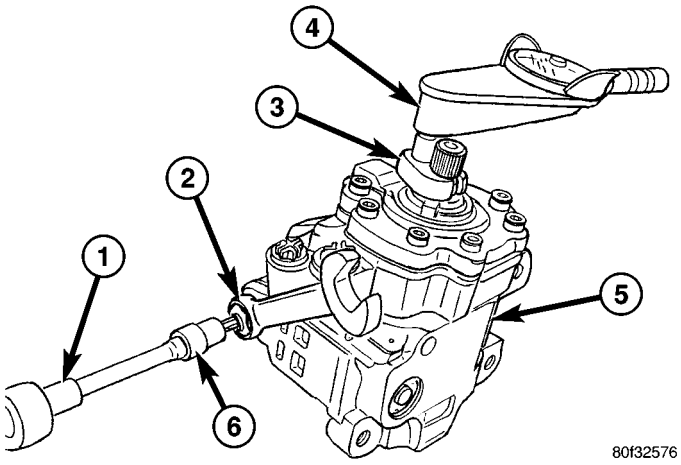


Fig. 7 OVER-CENTER ADJUSTMENT

1 - RATCHET WITH ADAPTER
2 - WRENCH
3 - CROWSFOOT WRENCH
4 - INCH POUND TORQUE WRENCH
5 - STEERING GEAR
6 - ALLEN HEAD SOCKET

GEAR (Continued)

SPECIFICATIONS

POWER STEERING GEAR

SPECIFICATIONS

DESCRIPTION	SPECIFICATION
Steering Gear Type	Recirculating Ball
Steering Gear Gear Ratio	15:24
Worm Shaft Bearing Preload	0.53–.93 N·m (3.5–8 in. lbs.)
Pitman Shaft Over-Center Drag New Gear (under 400 miles)	.133 N·m (1.17 in. lbs.) + Worm Shaft Preload
Pitman Shaft Over-Center Drag Used Gear (over 400 miles)	.133 N·m (1.17 in. lbs.) + Worm Shaft Preload

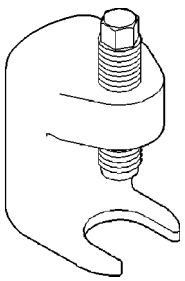
TORQUE CHART

TORQUE SPECIFICATIONS

DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Power Steering Gear Adjustment Screw Locknut	65	48	—
Power Steering Gear Gear to Frame Bolts	95	70	—
Power Steering Gear Pitman Shaft Nut	251	185	—
Power Steering Gear Pressure Line	28	21	—
Power Steering Gear Return Line	28	21	—

SPECIAL TOOLS

POWER STEERING GEAR



Remover, Pitman Arm C-4150A

LINKAGE

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LINKAGE

DESCRIPTION

The steering linkage consists of a pitman arm, drag link, tie rod, and steering dampener (Fig. 1). Adjustment sleeves are used on the tie rod and drag link for toe and steering wheel alignment.

The service procedures and torque specifications are the same for LHD and RHD vehicles.

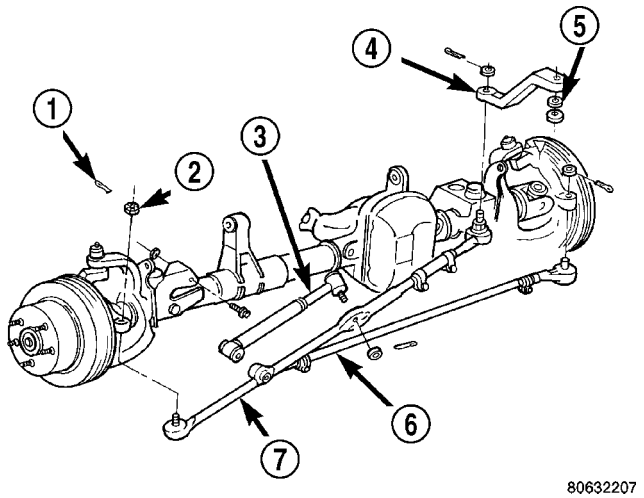


Fig. 1 Steering Linkage

1 - COTTER PIN
2 - NUT
3 - DAMPENER
4 - PITMAN ARM
5 - WASHER
6 - TIE ROD
7 - DRAG LINK

STANDARD PROCEDURE

STANDARD PROCEDURE - LUBRICATION

Periodic lubrication of the steering system components is required. Refer to Lubrication And Maintenance for the recommended maintenance schedule.

The following components must be lubricated:

- Tie rod ends
- Drag link

STANDARD PROCEDURE - STEERING LINKAGE

The tie rod end and ball stud seals should be inspected during all oil changes. If a seal is damaged, it should be replaced. Before installing a new seal, inspect ball stud at the throat opening. Check for lubricant loss, contamination, ball stud wear or corrosion. If these conditions exist, replace the tie rod. A replacement seal can be installed if lubricant is in good condition. Otherwise, a complete replacement ball stud end should be installed.

CAUTION: If any steering components are replaced or serviced an alignment must be performed, to ensure the vehicle meets all alignment specifications.

CAUTION: Components attached with a nut and cotter pin must be torqued to specification. Then if the slot in the nut does not line up with the cotter pin hole, tighten nut until it is aligned. Never loosen the nut to align the cotter pin hole.

LINKAGE (Continued)

SPECIFICATIONS

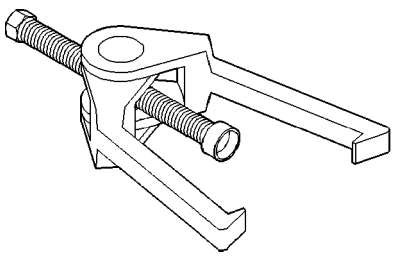
TORQUE CHART

TORQUE SPECIFICATIONS

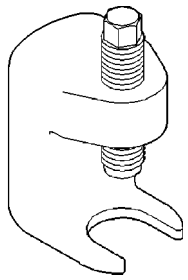
DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Pitman Arm Shaft	251	185	—
Drag Link Ball Studs	74	55	—
Drag Link Clamp	49	36	—
Tie Rod Ends Ball Studs	74	55	—
Tie Rod Ends Clamp	27	20	—
Tie Rod Ball Stud	88	65	—
Steering Damper Frame	74	55	—
Steering Damper Drag Link	74	55	—

SPECIAL TOOLS

STEERING LINKAGE



Puller C-3894–A



Remover Pitman C-4150A

DAMPER

REMOVAL

(1) Place the front wheels in a straight ahead position.

(2) Remove the steering dampener retaining nut and bolt from the axle bracket (Fig. 1).

(3) Remove the cotter pin and nut from the ball stud at the drag link.

(4) Remove the steering dampener ball stud from the drag link using C-3894-A puller.

INSTALLATION

(1) Install the steering dampener to the axle bracket and drag link.

(2) Install the steering dampener bolt in the axle bracket and tighten nut to 74 N·m (55 ft. lbs.).

(3) Install the ball stud nut at the drag link and tighten nut to 74 N·m (55 ft. lbs.). Install a new cotter pin.

DRAG LINK

REMOVAL

(1) Remove the cotter pins and nuts at the steering knuckle and drag link (Fig. 1).

(2) Remove the steering dampener ball stud from the drag link with a puller tool.

(3) Remove the drag link from the steering knuckle with a puller tool. Remove the same for tie rod and pitman arm.

(4) If necessary, loosen the end clamp bolts and remove the tie rod end from the link.

INSTALLATION

(1) Install the drag link adjustment sleeve and tie rod end. Position clamp bolts (Fig. 3).

(2) Position the drag link at the steering linkage. Install the drag link to the steering knuckle nut. Do the same for the tie rod and pitman arm.

(3) Tighten the nut at the steering knuckle to 47 N·m (35 ft. lbs.). Tighten the pitman nut to 81 N·m (60 ft. lbs.) and tie rod ball stud nut to 47 N·m (35 ft. lbs.). Install new cotter pins and bend end 60°.

(4) Install the steering dampener onto the drag link and tighten the nut to 74 N·m (55 ft. lbs.). Install a new cotter pin and bend end 60°.

PITMAN ARM

REMOVAL

(1) Remove the cotter pin and nut from the drag link at the pitman arm.

(2) Remove the drag link ball stud from the pitman arm with a puller.

(3) Remove the nut and washer from the steering gear shaft. Mark the pitman shaft and pitman arm for installation reference. Remove the pitman arm from steering gear with Puller C-4150A (Fig. 2).

INSTALLATION

(1) Align and install the pitman arm on steering gear shaft.

(2) Install the washer and nut on the shaft and tighten the nut to 251 N·m (185 ft. lbs.).

(3) Install drag link ball stud to pitman arm. Install nut and tighten to 81 N·m (60 ft. lbs.). Install a new cotter pin.

TIE ROD END

REMOVAL

(1) Remove the cotter pins and nuts at the steering knuckle and drag link (Fig. 1).

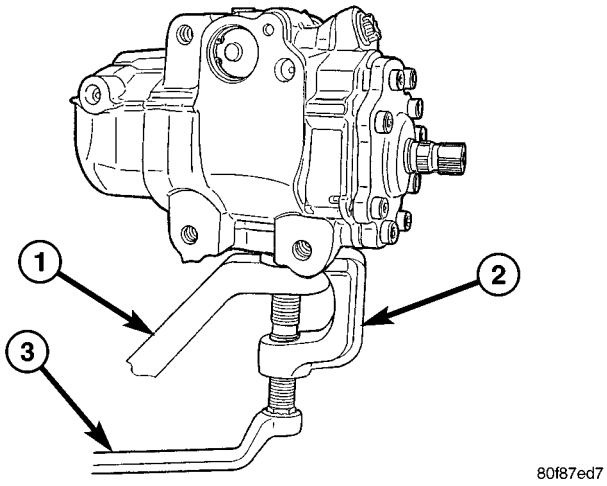


Fig. 2 PITMAN ARM PULLER

1 - PITMAN ARM
2 - SPECIAL TOOL C-4150-A
3 - WRENCH

(2) Remove the ball studs with puller tool C-4150R.

(3) If necessary, loosen the end clamp bolts and remove the tie rod ends from the tube.

INSTALLATION

(1) If necessary, install the tie rod ends in the tube. Position the tie rod clamp (Fig. 3) and tighten to 27 N·m (20 ft. lbs.).

(2) Install the tie rod on the drag link and steering knuckle.

(3) Tighten the ball stud nut on the steering knuckle to 47 N·m (35 ft. lbs.). Tighten the ball stud nut to drag link to 47 N·m (35 ft. lbs.) torque. Install new cotter pins.

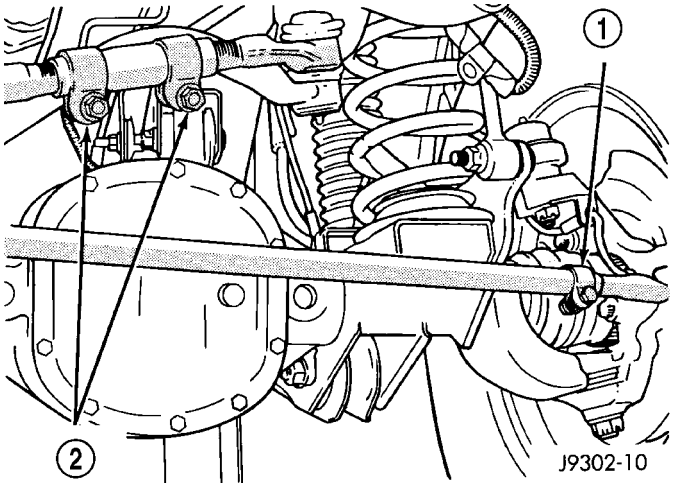


Fig. 3 Tie Rod

1 - TIE ROD CLAMP
2 - DRAG LINK CLAMPS

PUMP

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PUMP

DESCRIPTION

Hydraulic pressure for the power steering system is provided by a belt driven power steering pump (Fig. 1). The pump shaft has a pressed-on high strength plastic drive pulley that is belt driven by the crankshaft pulley. The reservoir is attached to the pump body with spring clips on the 4.0L engine. A remote pump reservoir is used on the 2.4L engine mounted to the fan shroud. The power steering pump is connected to the steering gear by the pressure and return hoses.

OPERATION

The power steering pump is a constant flow rate and displacement, vane-type pump. The pump internal parts operate submerged in fluid. The flow control orifice is part of the high pressure line fitting. The pressure relief valve inside the flow control valve limits the pump pressure.

NOTE: Power steering pumps have different pressure rates and are not interchangeable with other pumps.

DIAGNOSIS AND TESTING - PUMP LEAKAGE

(1) Possible areas of pump leakage (Fig. 2).

STANDARD PROCEDURE - POWER STEERING PUMP - INITIAL OPERATION

WARNING: THE FLUID LEVEL SHOULD BE CHECKED WITH ENGINE OFF TO PREVENT INJURY FROM MOVING COMPONENTS.

CAUTION: MOPAR® ATF+4 is to be used in the power steering system. No other power steering or automatic transmission fluid is to be used in the system. Damage may result to the power steering pump and system if any other fluid is used, and do not overfill.

Wipe filler cap clean, then check the fluid level. The dipstick should indicate **COLD** when the fluid is at normal ambient temperature.

(1) Fill the pump fluid reservoir to the proper level and let the fluid settle for at least two minutes.

(2) Start the engine and let run for a few seconds then turn engine off.

PUMP (Continued)

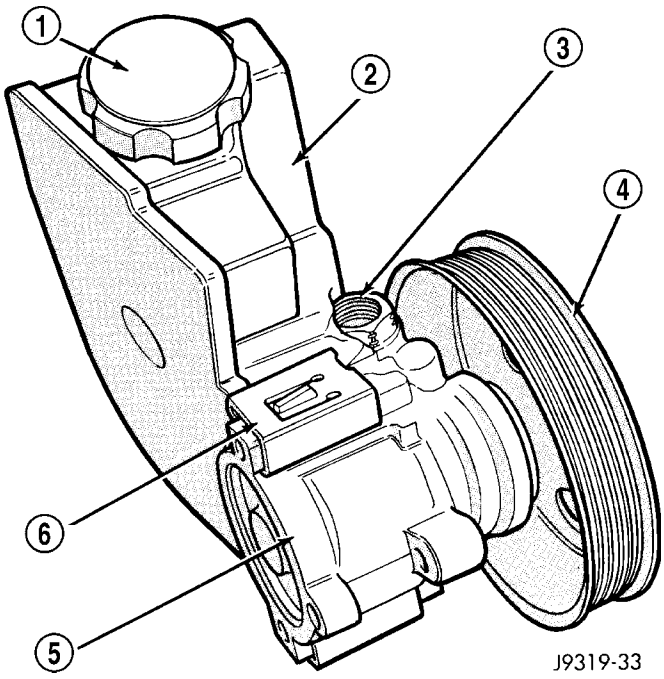
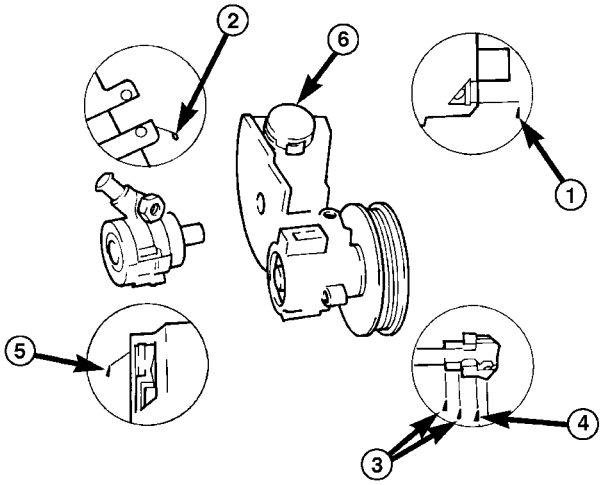


Fig. 1 Pump With Integral Reservoir

1 - CAP
2 - FLUID RESERVOIR (TYPICAL)
3 - HIGH-PRESSURE FITTING
4 - DRIVE PULLEY
5 - PUMP BODY
6 - RESERVOIR CLIP



1. BUSHING (BEARING) WORN, SEAL WORN. REPLACE PUMP.
2. REPLACE RESERVOIR O-RING SEAL.
3. TORQUE HOSE FITTING NUT TO SPECIFICATIONS. IF LEAKAGE PERSISTS, REPLACE O-RING SEAL.
4. TORQUE FITTING TO SPECIFICATIONS. IF LEAKAGE PERSISTS, REPLACE O-RING SEAL.
5. REPLACE PUMP.
6. CHECK OIL LEVEL: IF LEAKAGE PERSISTS WITH THE LEVEL CORRECT AND CAP TIGHT, REPLACE THE CAP.

80a1c3c3

Fig. 2 Power Steering Pump

(3) Add fluid if necessary. Repeat the above procedure until the fluid level remains constant after running the engine.

(4) Raise the front wheels off the ground.

(5) Slowly turn the steering wheel right and left, lightly contacting the wheel stops at least 20 times.

(6) Check the fluid level add if necessary.

(7) Lower the vehicle, start the engine and turn the steering wheel slowly from lock to lock.

(8) Stop the engine and check the fluid level and refill as required.

(9) If the fluid is extremely foamy or milky looking, allow the vehicle to stand a few minutes and repeat the procedure.

CAUTION: Do not run a vehicle with foamy fluid for an extended period. This may cause pump damage.

REMOVAL

REMOVAL - 4.0L

(1) Remove serpentine drive belt, (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - REMOVAL).

(2) Remove pressure and return hoses from pump and drain the pump.

(3) Loosen the pump bracket bolt at the engine block.

(4) Remove 3 pump mounting bolts (Fig. 3) through pulley access holes.

(5) Tilt pump downward and remove from engine.

(6) Remove pulley from pump. (Refer to 19 - STEERING/PUMP/PULLEY - REMOVAL).

REMOVAL - 2.4L

(1) Remove serpentine drive belt, (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - REMOVAL).

(2) Remove pressure and return hoses from pump and drain the pump.

(3) Remove 3 pump mounting bolts (Fig. 4) through pulley access holes.

(4) Loosen the 3 pump bracket bolts.

(5) Tilt pump downward and remove from engine.

(6) Remove pulley from pump. (Refer to 19 - STEERING/PUMP/PULLEY - REMOVAL).

INSTALLATION

INSTALLATION - 4.0L

(1) Install pulley on pump. (Refer to 19 - STEERING/PUMP/PULLEY - INSTALLATION).

(2) Install pump on the engine mounting bracket.

(3) Install 3 pump mounting bolts and tighten to 27 N·m (20 ft. lbs.).

PUMP (Continued)

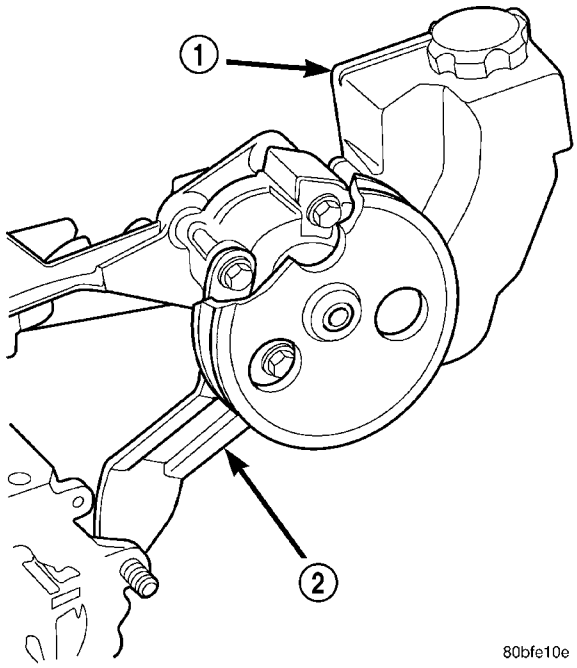


Fig. 3 Pump Mounting – 4.0L

1 - PUMP ASSEMBLY
2 - PUMP BRACKET

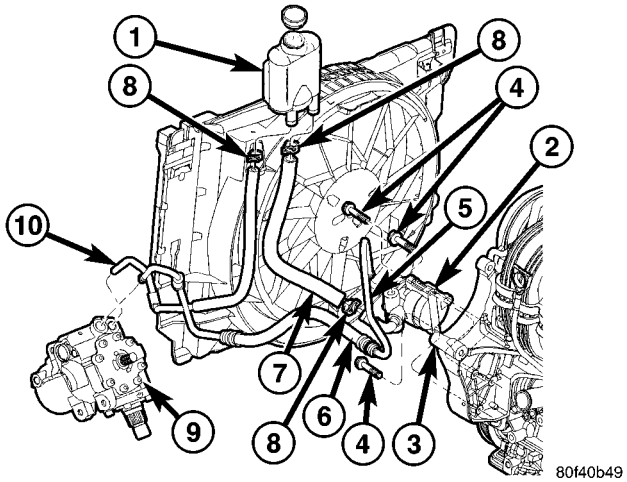


Fig. 4 POWER STEERING GEAR & PUMP - 2.4L

1 - REMOTE RESERVOIR
2 - POWER STEERING PUMP
3 - PUMP MOUNT BRACKET
4 - MOUNTING BOLTS (3)
5 - PRESSURE SWITCH LOCATION
6 - PRESSURE OUTLET HOSE (GEAR TO PUMP)
7 - RETURN HOSE (RESERVOIR TO PUMP)
8 - CLAMPS
9 - STEERING GEAR
10 - INLET HOSE (RESERVOIR TO GEAR)

(4) Tighten pump bracket bolt to 57 N·m (42 ft. lbs.).

(5) Install the pressure line on the pump and tighten to 28 N·m (21 ft. lbs.).

(6) Install return hoses on pump.

(7) Install drive belt, (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - INSTALLATION).

(8) Add power steering fluid, refer to Power Steering Pump Initial Operation. (Refer to 19 - STEERING/PUMP - STANDARD PROCEDURE).

INSTALLATION - 2.4L

(1) Install pulley on pump. (Refer to 19 - STEERING/PUMP/PULLEY - INSTALLATION).

(2) Install pump on the engine mounting bracket.

(3) Tighten pump bracket bolts to 47 N·m (35 ft. lbs.).

(4) Install 3 pump mounting bolts and tighten to 27 N·m (20 ft. lbs.).

(5) Install the pressure line on the pump and tighten to 28 N·m (21 ft. lbs.).

(6) Install return hoses on pump.

(7) Install drive belt, (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - INSTALLATION).

(8) Add power steering fluid, refer to Power Steering Pump Initial Operation. (Refer to 19 - STEERING/PUMP - STANDARD PROCEDURE).

PUMP (Continued)

SPECIFICATIONS

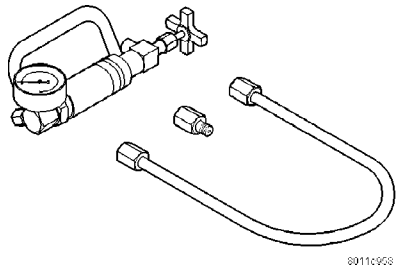
TORQUE CHART

TORQUE SPECIFICATIONS

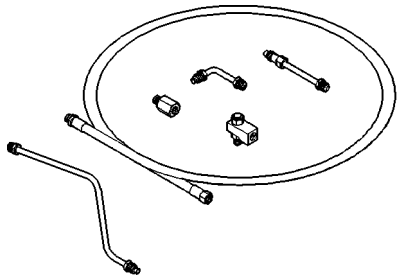
DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Power Steering Pump Bracket to Pump	28	21	—
Power Steering Pump Bracket to 4.0L Engine	57	42	—
Power Steering Pump Bracket to 2.4L Engine	28	21	—
Power Steering Pump Flow Control Valve	75	55	—
Power Steering Pump Pressure Line	28	21	—

SPECIAL TOOLS

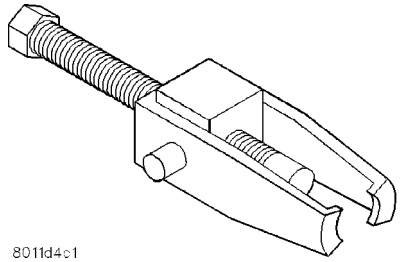
POWER STEERING PUMP



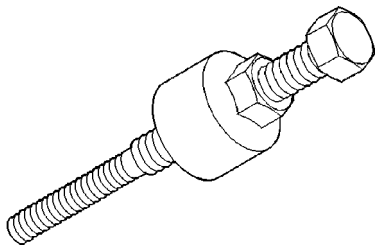
Analyzer Set, Power Steering Flow/Pressure 6815



Adapters, Power Steering Flow/Pressure Tester 6893



Puller C-4333



Installer, Power Steering Pulley C-4063B

HOSES

DESCRIPTION

DESCRIPTION - PRESSURE LINE

The hose consists of two metal ends and rubber center section that contains a tuning cable.

DESCRIPTION - RETURN LINE

Power steering return line is a hose which is clamped at the pump and the gear.

OPERATION

OPERATION - PRESSURE LINE

Power steering pressure line, is used to transfer high pressure power steering fluid, from the power steering pump to the power steering gear.

OPERATION - RETURN LINE

Power steering return line, is used to transfer low pressure power steering fluid, from the power steering gear to the power steering pump.

FLUID

DESCRIPTION

The recommended fluid for the power steering system is Mopar® ATF +4.

Mopar® ATF+4, when new is red in color. The ATF+4 is dyed red so it can be identified from other fluids used in the vehicle such as engine oil or antifreeze. The red color is not permanent and is not an indicator of fluid condition, As the vehicle is driven, the ATF+4 will begin to look darker in color and may eventually become brown. **THIS IS NORMAL.** ATF+4 also has a unique odor that may change with age. Consequently, odor and color cannot be used to indicate the fluid condition or the need for a fluid change.

STANDARD PROCEDURE - POWER STEERING FLUID LEVEL CHECKING

WARNING: FLUID LEVEL SHOULD BE CHECKED WITH THE ENGINE OFF TO PREVENT PERSONAL INJURY FROM MOVING PARTS.

CAUTION: MOPAR® ATF+4 is to be used in the power steering system. No other power steering or automatic transmission fluid is to be used in the system. Damage may result to the power steering pump and system if any other fluid is used, and do not overfill.

The power steering fluid level can be viewed on the dipstick attached to the filler cap. There are two ranges listed on the dipstick, COLD and HOT. Before opening power steering system, wipe the reservoir filler cap free of dirt and debris. Remove the cap and check the fluid level on its dipstick. When the fluid is at normal ambient temperature, approximately 21°C to 27°C (70°F to 80°F), the fluid level should read between the minimum and maximum area of the cold range. When the fluid is hot, fluid level is allowed to read up to the highest end of the HOT range. Only add fluid when the vehicle is cold.

Use only Mopar® ATF+4 Do not overfill the power steering system.

POWER STEERING PRESSURE SWITCH

DESCRIPTION

A pressure sensing switch (Fig. 5) is included in the power steering system (mounted on the high-pressure line). This switch will be used only on vehicles equipped with a 2.4L engine and power steering.

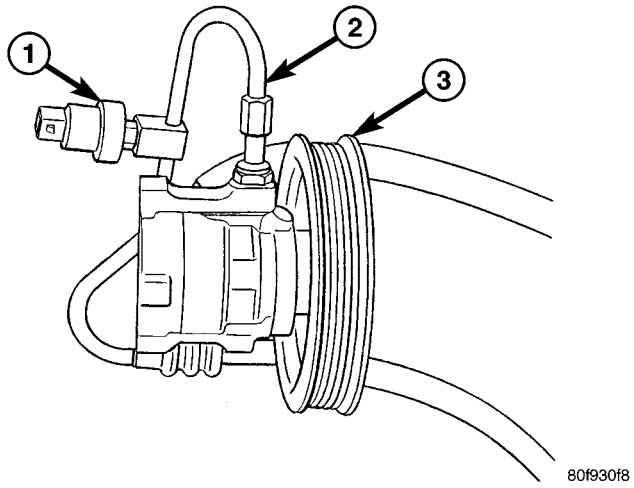


Fig. 5 2.4L POWER STEERING PRESSURE SWITCH

1 - PRESSURE SWITCH
2 - PRESSURE HOSE
3 - POWER STEERING PUMP AND PULLEY ASSEMBLY

OPERATION

The power steering pressure switch provides an input to the Powertrain Control Module (PCM). This input is provided during periods of high pump load and low engine rpm; such as during parking maneuvers. The PCM will then increase the idle speed through the Idle Air Control (IAC) motor. This is done to prevent the engine from stalling under the increased load.

POWER STEERING PRESSURE SWITCH (Continued)

When steering pump pressure exceeds 5860 kPa ± 690 kPa (850 psi ± 100 psi), the normally closed switch will open and the PCM will increase the engine idle speed. This will prevent the engine from stalling.

When pump pressure drops to approximately 1379 kPa (200 psi), the switch circuit will re-close and engine idle speed will return to its previous setting.

REMOVAL

This switch is not used with 4.0L six-cylinder engines.

The power steering pressure switch is installed in the power steering high-pressure hose (Fig. 5).

(1) Disconnect electrical connector from power steering pressure switch.

(2) Place a small container or shop towel beneath switch to collect any excess fluid.

(3) Remove switch. Use back-up wrench on power steering line to prevent line bending.

INSTALLATION

This switch is not used with 4.0L six-cylinder engines.

(1) Install power steering switch into power steering line.

(2) Tighten to 14–22 N·m (124–195 in. lbs.) torque.

(3) Connect electrical connector to switch.

(4) Check power steering fluid and add as necessary.

(5) Start engine and again check power steering fluid. Add fluid if necessary.

PULLEY

REMOVAL

CAUTION: On vehicles equipped with the 4.0L, Do not reuse the old power steering pump pulley it is not intended for reuse. A new pulley must be installed if removed.

(1) Remove pump assembly.

(2) Remove pulley from pump with Puller C-4333 or equivalent puller (Fig. 6).

INSTALLATION

NOTE: The pulley is marked front for installation.

CAUTION: On vehicles equipped with the 4.0L, Do not reuse the old power steering pump pulley it is not intended for reuse. A new pulley must be installed if removed.

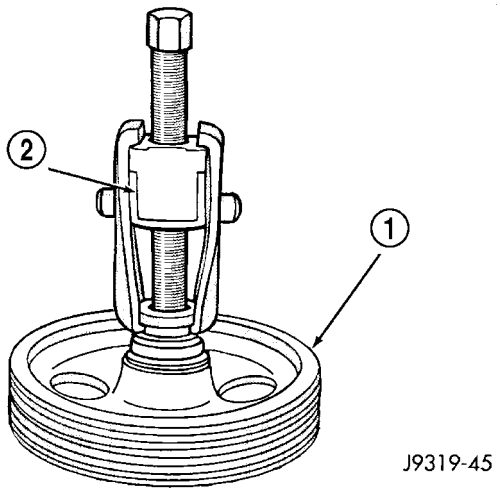


Fig. 6 Pulley Removal

1 - POWER STEERING PUMP DRIVE PULLEY
2 - SPECIAL TOOL C-4333

(1) Replace pulley if bent, cracked, or loose.

(2) Install pulley on pump with Installer C-4063-B or equivalent installer (Fig. 7). The pulley must be flush with the end of the shaft. Ensure the tool and pulley are aligned with the pump shaft.

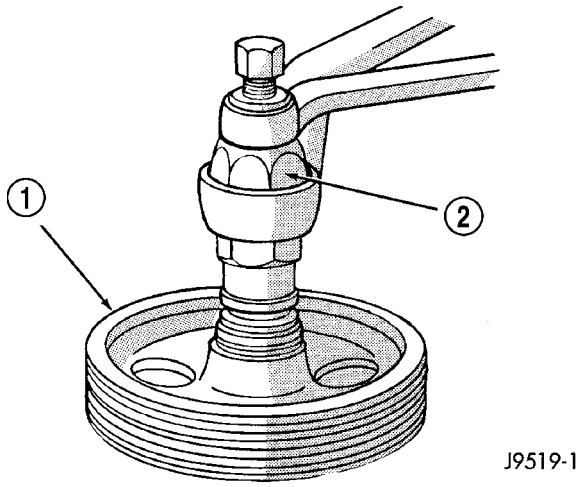


Fig. 7 Pulley Installation

1- POWER STEERING PUMP DRIVE PULLEY
2 - SPECIAL TOOL C-4063-B

(3) Install pump assembly.

(4) With Serpentine Belt, run engine until warm (5 min.) and note any belt chirp. If chirp exists, move pulley outward approximately 0.5 mm (0.020 in.). If noise increases, press on 1.0 mm (0.040 in.). **Be careful that pulley does not contact mounting bolts.**

RESERVOIR

REMOVAL

REMOVAL - 4.0L

(1) Remove power steering pump. (Refer to 19 - STEERING/PUMP - REMOVAL).
(2) Clean exterior of pump.
(3) Clamp the pump body in a soft jaw vice.
(4) Pry up tab and slide the retaining clips off (Fig. 8).

NOTE: Use new retaining clips for installation.

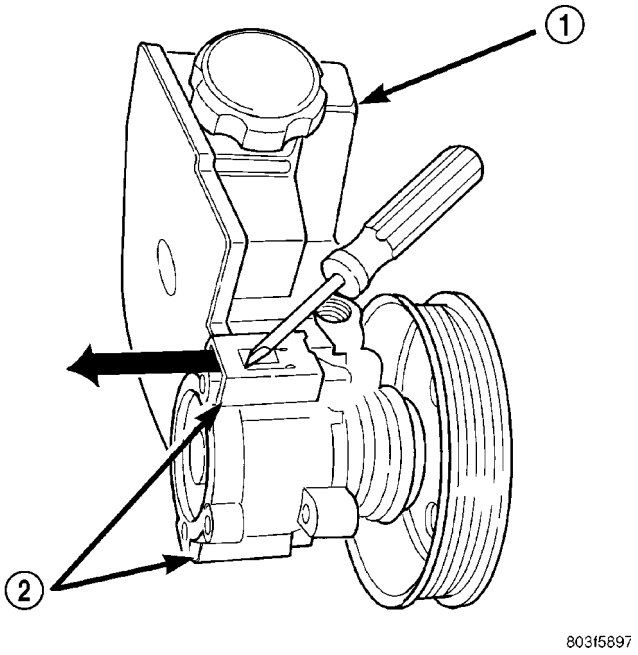


Fig. 8 Pump Reservoir Clips

1 - RESERVOIR
2 - RETAINING CLIPS

(5) Remove fluid reservoir from pump body. Remove and discard O-ring seal.

REMOVAL - 2.4L

(1) Remove the pump return hoses from the reservoir and drain the reservoir.
(2) Remove the push-in fastener from the reservoir (Fig. 9).
(3) Slide the reservoir up out of the fan shroud mount.

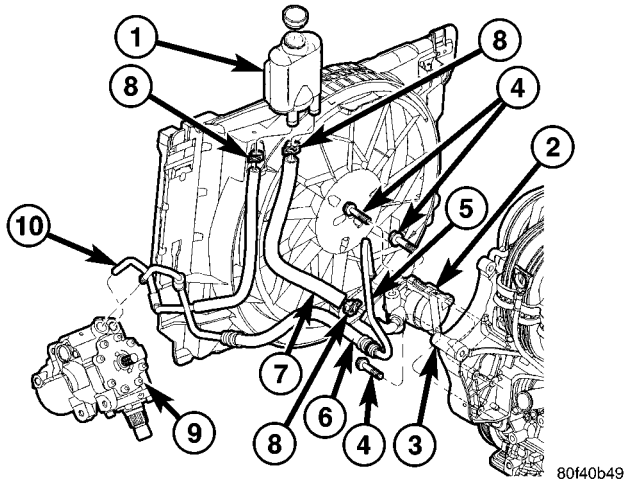


Fig. 9 POWER STEERING GEAR & PUMP - 2.4L

1 - REMOTE RESERVOIR
2 - POWER STEERING PUMP
3 - PUMP MOUNT BRACKET
4 - MOUNTING BOLTS (3)
5 - PRESSURE SWITCH LOCATION
6 - PRESSURE OUTLET HOSE (GEAR TO PUMP)
7 - RETURN HOSE (RESERVOIR TO PUMP)
8 - CLAMPS
9 - STEERING GEAR
10 - INLET HOSE (RESERVOIR TO GEAR)

INSTALLATION

INSTALLATION - 4.0L

(1) Lubricate new O-ring Seal with Mopar Power Steering Fluid or equivalent.
(2) Install O-ring seal in housing.
(3) Install reservoir onto housing.
(4) Slide and tap in **new** reservoir retainer clips until tab locks to housing.
(5) Install power steering pump. (Refer to 19 - STEERING/PUMP - INSTALLATION).
(6) Add power steering fluid, (Refer to 19 - STEERING/PUMP - STANDARD PROCEDURE).

INSTALLATION - 2.4L

(1) Slide reservoir down onto the fan shroud mount until it clicks in place.
(2) Install the push-in fastener.
(3) Install the hoses.
(4) Fill reservoir to proper level, (Refer to 19 - STEERING/PUMP - STANDARD PROCEDURE).

TRANSMISSION AND TRANSFER CASE

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MANUAL TRANSMISSION - NSG370

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MANUAL TRANSMISSION - NSG370

DIAGNOSIS AND TESTING

MANUAL TRANSMISSION - NSG370

LOW LUBRICANT LEVEL

A low transmission lubricant level is generally the result of a leak, inadequate lubricant fill or incorrect lubricant level check.

Rear transmission leaks will be from the oil seals or component mating surfaces.

Front transmission leaks will be from the front input shaft retainer seal. Lubricant may drip from the clutch housing after extended operation. If leak is severe, it may contaminate the clutch disc.

Lubricant level check can only be made when the vehicle is level and allowing the lubricant to settle for a minute before checking. This will ensure an accurate check and avoid an underfill or overfill condition.

HARD SHIFTING

Hard shifting is usually caused by low lubricant level, improper or contaminated lubricants. This will cause noise, excessive wear, internal bind, and hard shifting. Substantial lubricant leaks can result in gear, shift rail, synchro, and bearing damage. The first indications of component damage is usually hard shifting and noise.

Shift component damage, clutch adjustment, worn pressure plate or disc are also causes of increased shift effort. If clutch problem is advanced, gear clash during shifts can result. Worn or damaged synchronizer rings can cause gear clash when shifting into any forward gear. In some new or rebuilt transmissions, new synchro rings may tend to stick slightly causing hard or noisy shifts. In most cases, this condition will decline as the rings wear-in.

TRANSMISSION NOISE

Most manual transmissions make some noise during normal operation. Rotating gears generate a mild whine that is audible, but generally only at extreme speeds.

Severe, highly audible transmission noise is generally the initial indicator of a lubricant problem. Insufficient, improper or contaminated lubricant will promote rapid wear of gears, synchros, shift rails, forks and bearings. The overheating caused by a lubricant problem, can also lead to gear and bearing damage.

MANUAL TRANSMISSION - NSG370 (Continued)

STANDARD PROCEDURE - DRAIN AND FILL

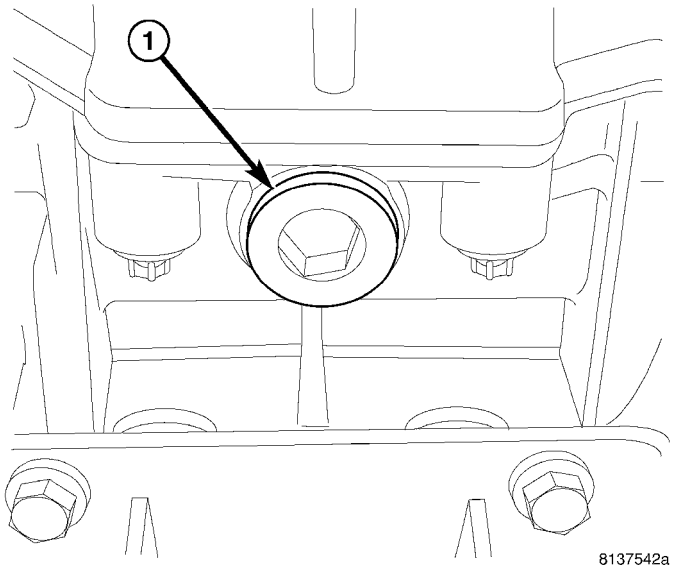


Fig. 1 DRAIN PLUG

1 - DRAIN PLUG

(1) With vehicle in neutral, position vehicle on hoist.
(2) Remove drain plug (1) (Fig. 1) and drain fluid.

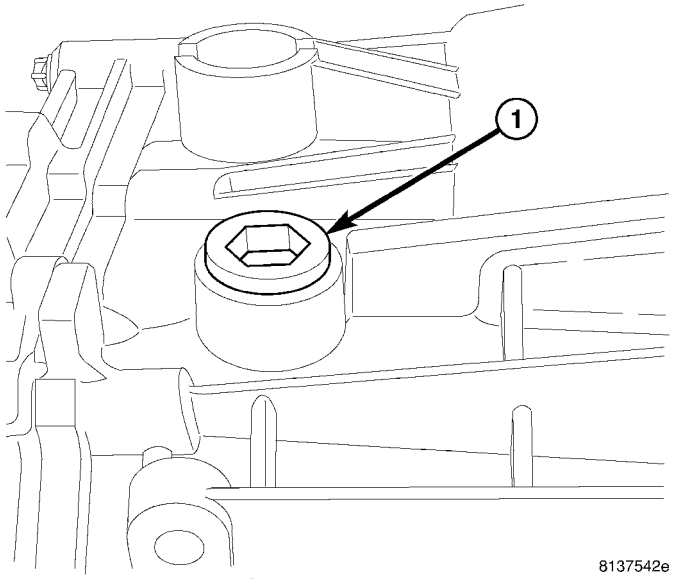


Fig. 2 FILL PLUG

1 - FILL PLUG

(3) Install drain plug and remove fill plug (1) (Fig. 2).
(4) Fill transmission with 1.5L (3.17 pts.) of Mopar® Manual Transmission Lubricant MS-9224 or to the bottom of the fill plug (1) hole.

REMOVAL

(1) Disconnect negative battery cable.
(2) With vehicle in neutral, position vehicle on hoist.
(3) Remove drain plug (1) (Fig. 3) and drain fluid.

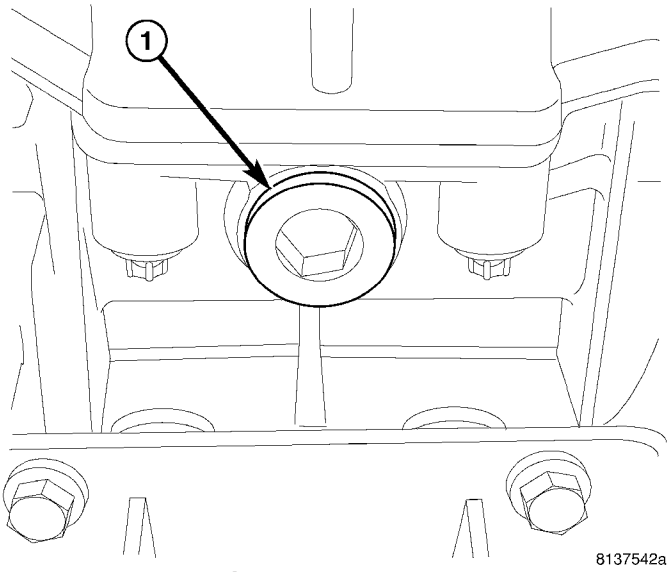


Fig. 3 DRAIN PLUG

1. DRAIN PLUG

(4) Mark installation reference marks on propeller shaft/shafts and remove shafts.
(5) Remove transfer case shift linkage, wiring connector, and vent hose, if equipped.
(6) Remove transfer case, if equipped.
(7) Support transmission with jack.
(8) Remove transmission mount (1) and crossmember (2).

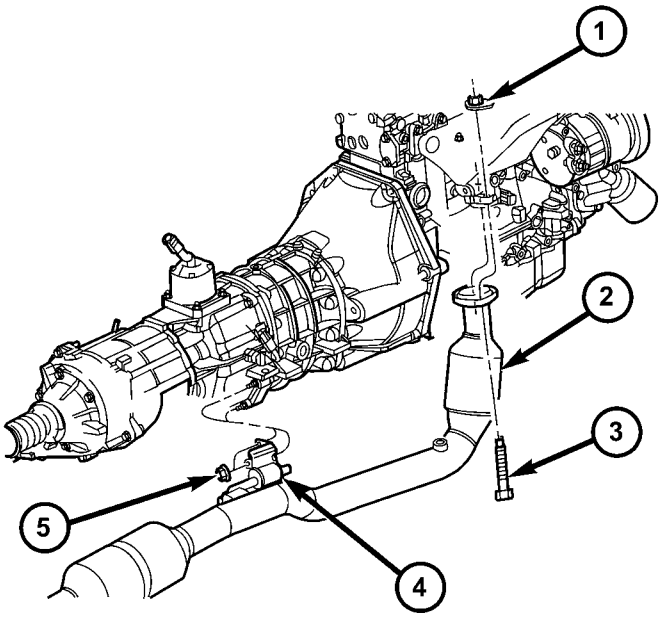


Fig. 4 2.4L CATALYST ASSEMBLY - 4x4

1- FLANGED NUT
2- CATALYST ASSEMBLY
3- BOLT
4- HANGER
5- NUT

MANUAL TRANSMISSION - NSG370 (Continued)

(9) Remove catalyst assembly (2) (Fig. 4).

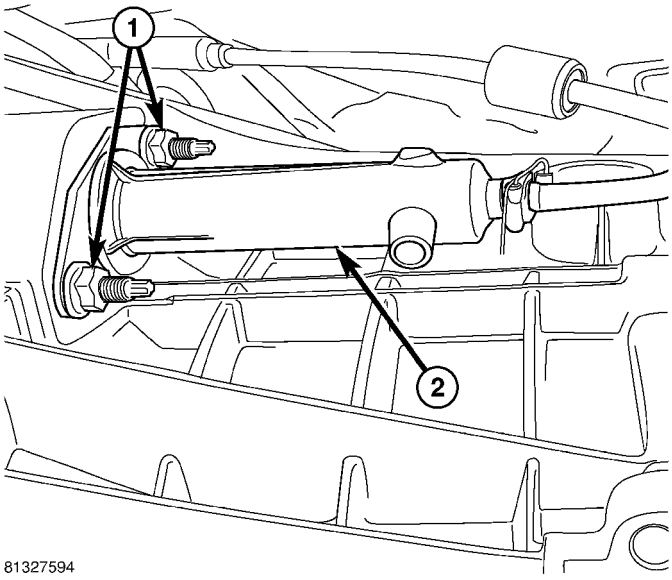


Fig. 5 CLUTCH SLAVE CYLINDER

1. NUTS
2. SLAVE CYLINDER

(10) Remove clutch slave cylinder nuts (1) and remove cylinder (2) (Fig. 5).

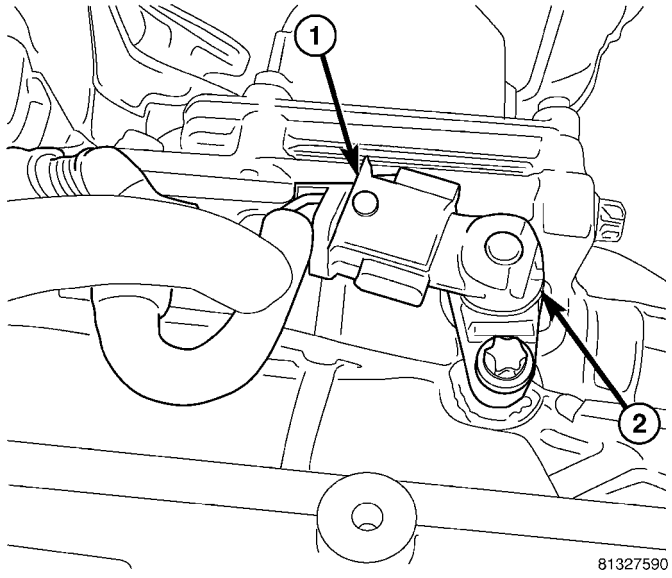


Fig. 6 BACKUP LAMP SWITCH

1. BACKUP LAMP SWITCH
2. SWITCH

(11) Remove backup lamp switch (2) wiring connector (1) (Fig. 6).

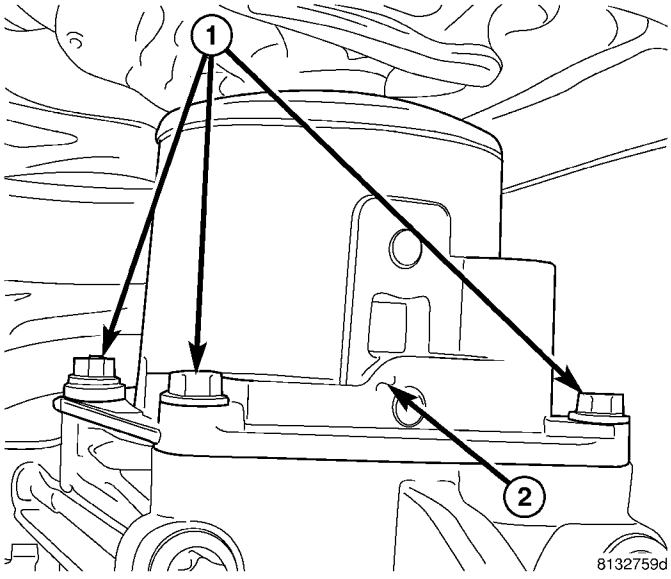


Fig. 7 SHIFT LEVER

1. BOLTS
2. SHIFT TOWER

(12) Remove shift lever tower bolts (1) and remove shift lever tower (2) (Fig. 7).

(13) Remove starter bolts and remove starter.

(14) Remove transmission bolt and remove transmission.

DISASSEMBLY

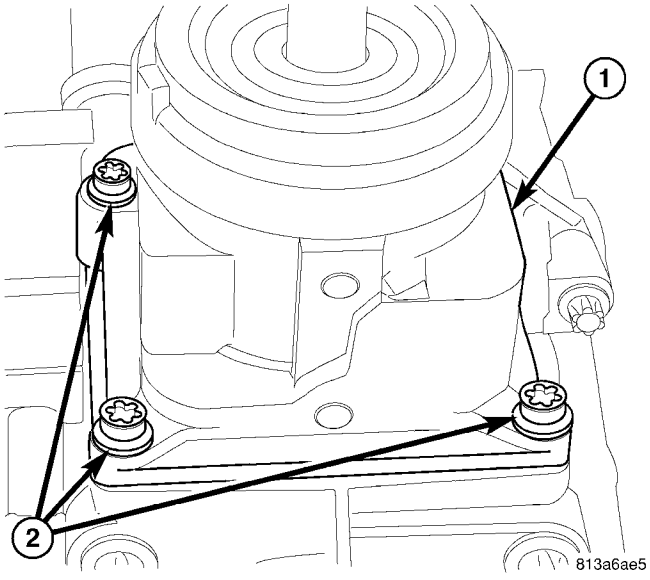


Fig. 8 SHIFT TOWER

1 - SHIFT TOWER
2 - BOLTS

(1) Remove shift tower (1) (Fig. 8) bolts (2) and remove tower.

MANUAL TRANSMISSION - NSG370 (Continued)

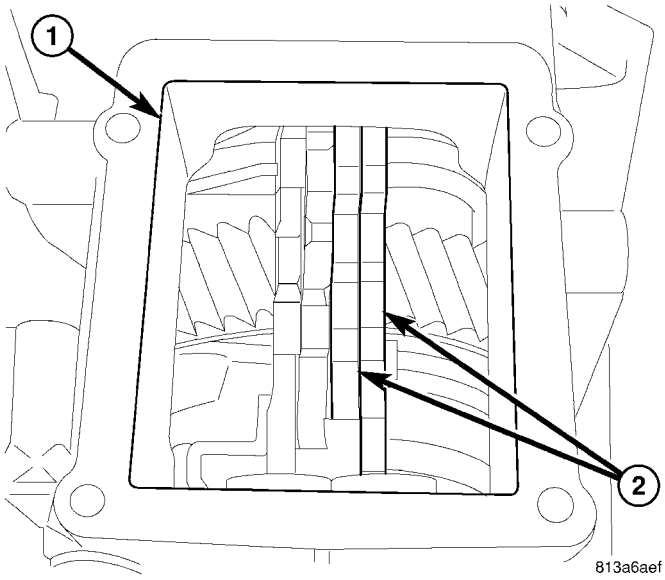


Fig. 9 SHIFT TOWER OPENING

1 - SHIFT TOWER OPENING
2 - SHIFT RAILS

(2) Through shift tower opening (1) (Fig. 9) shift two shift rails (2) forward to engage two gears.

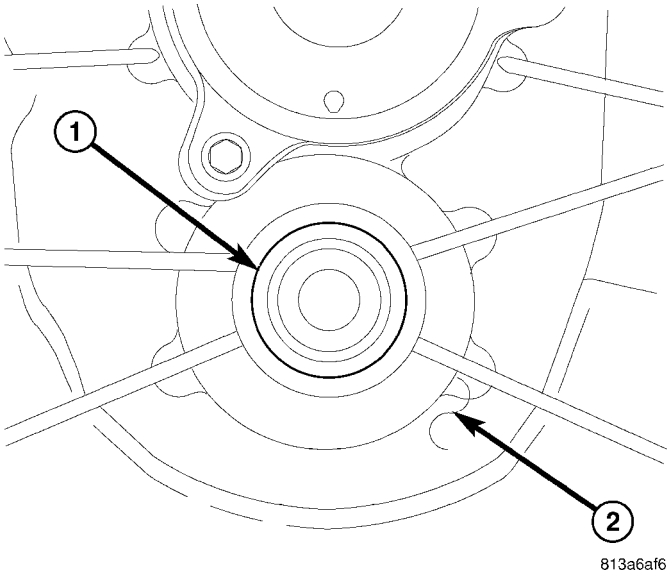


Fig. 10 COUNTERSHAFT PLUG

1 - COUNTERSHAFT PLUG
2 - FRONT HOUSING

(3) Remove countershaft plug (1) (Fig. 10) from the front housing (2) with a seal pick.

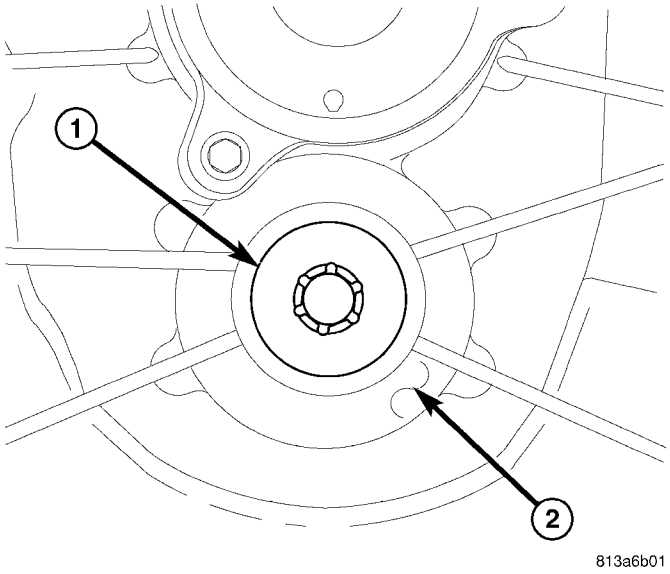


Fig. 11 COUNTERSHAFT BOLT

1 - COUNTERSHAFT BOLT
2 - FRONT HOUSING

(4) Remove countershaft bolt (1) (Fig. 11) from the front housing (2).

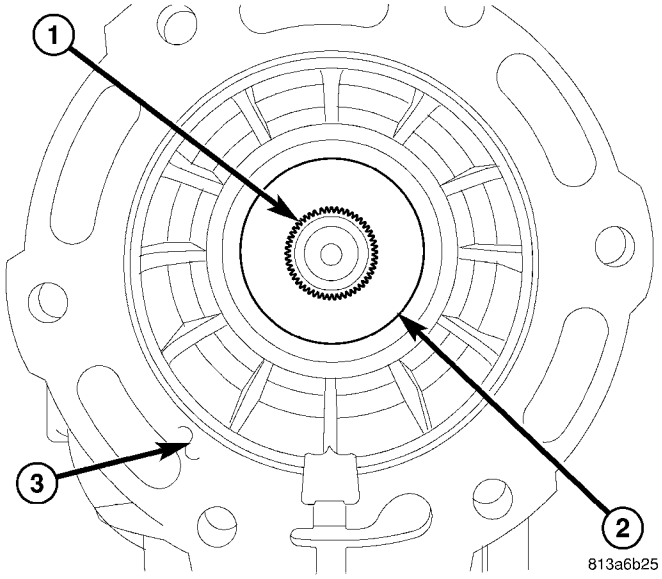


Fig. 12 OUTPUT SHAFT SEAL

1 - OUTPUT SHAFT
2 - SEAL
3 - REAR HOUSING

(5) Remove output shaft (1) (Fig. 12) seal (2) from rear housing (3) with a seal pick.

MANUAL TRANSMISSION - NSG370 (Continued)

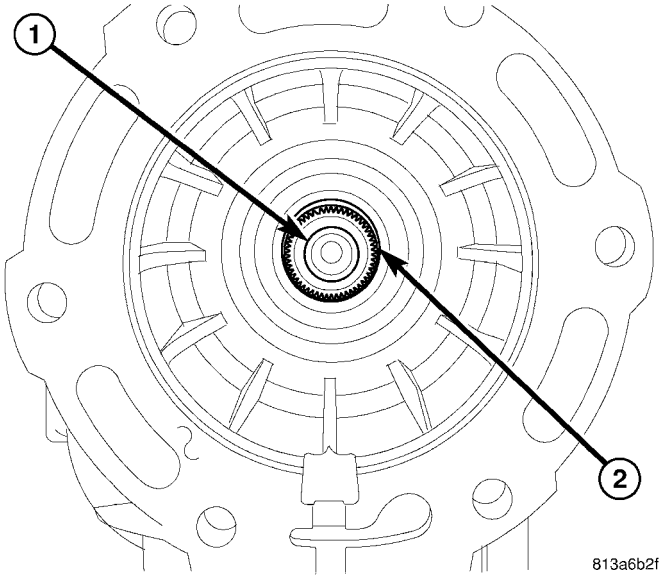


Fig. 13 OUTPUT SHAFT SNAP RING

1 - OUTPUT SHAFT
2 - SNAP RING

(6) Remove output shaft (1) (Fig. 13) snap ring (2) 4x4 only.

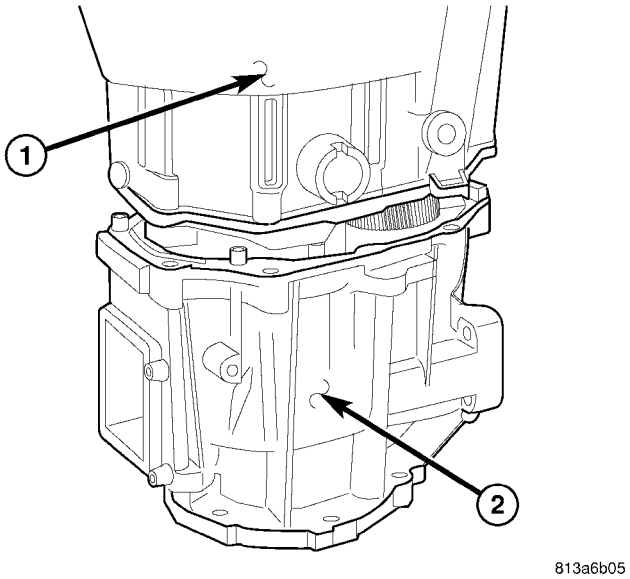


Fig. 14 SPLIT HOUSING

1 - FRONT HOUSING
2 - REAR HOUSING

(7) Stand transmission on the rear housing and remove the housing bolts.

(8) Remove front housing (1) (Fig. 14) from the rear housing (2).

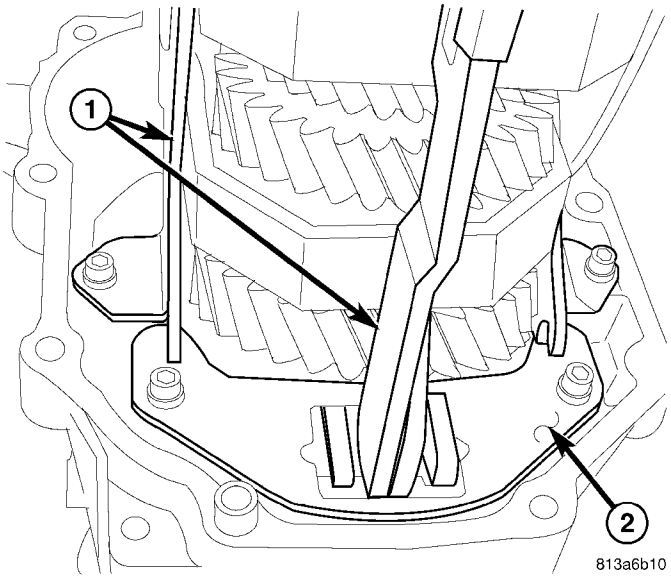


Fig. 15 SHIFT RAIL SUPPORT PLATE

1 - SHIFT RAILS
2 - SUPPORT PLATE

(9) Remove shift rails (1) (Fig. 15) support plate (2) bolts.

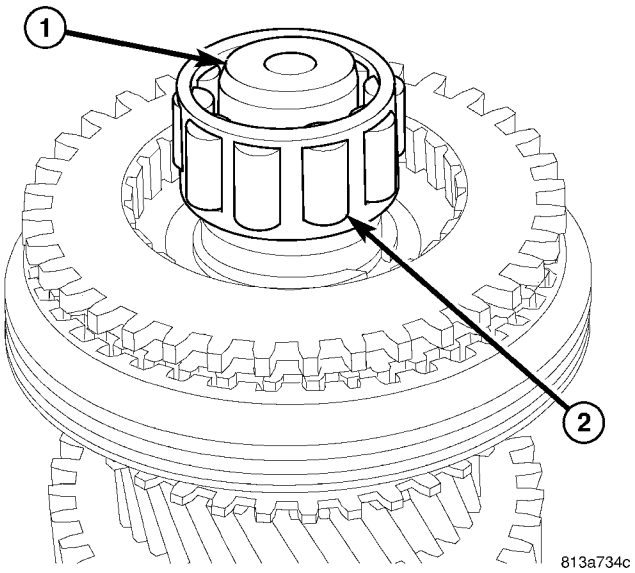


Fig. 16 INPUT SHAFT ROLLER BEARING

1 - INPUT SHAFT
2 - ROLLER BEARING

(10) Remove input shaft (1) (Fig. 16) roller bearing (2).

MANUAL TRANSMISSION - NSG370 (Continued)

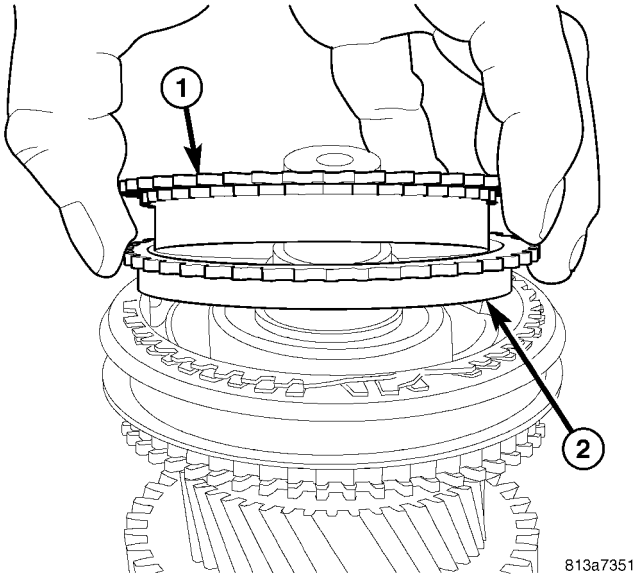


Fig. 17 FIFTH GEAR SYNCHRO

1 - FIFTH GEAR BLOCKER RING
2 - FRICTION RING

(11) Remove fifth gear synchronizer blocker ring (1) (Fig. 17) and friction ring (2).

(12) Set geartrain and shift rails with rear housing into Build Fixture 9633.

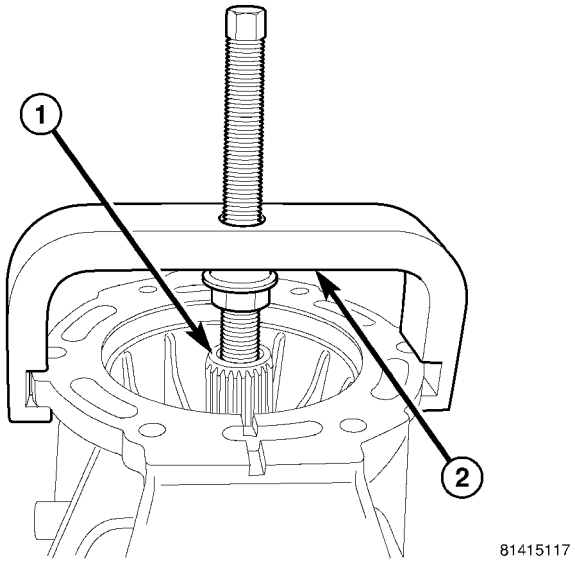


Fig. 18 MAINSHAFT REMOVER

1 - MAINSHAFT
2 - REMOVER/INSTALLER

(13) Remove mainshaft shaft (1) (Fig. 18) from rear housing bearing with Remover 9637 (2).

(14) Remove mainshaft (1), countershaft (2) and shift rails/forks (3) from fixture (Fig. 19).

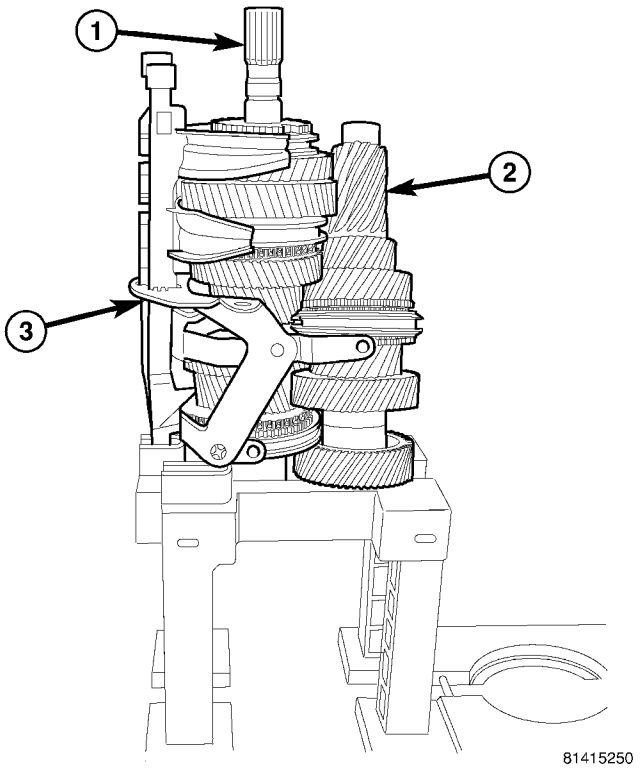


Fig. 19 MAIN/COUNTERSHAFT

1 - MAINSHAFT
2 - COUNTERSHAFT
3 - SHIFT RAILS AND FORKS

MAINSHAFT

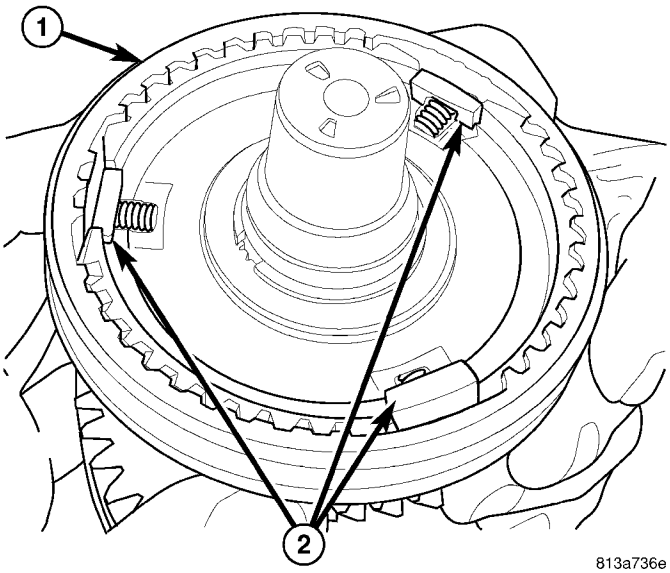


Fig. 20 5-6 SYNCHRO SLEEVE DETENTS

1 - SYNCHRONIZER SLEEVE
2 - DETENTS

(1) Place mainshaft with 5-6 synchronizer facing up in build Fixture 9648 and place in a vise.

MANUAL TRANSMISSION - NSG370 (Continued)

(2) Push 5-6 synchronizer sleeve (1) (Fig. 20) down on hub and remove detents (2) springs and balls.

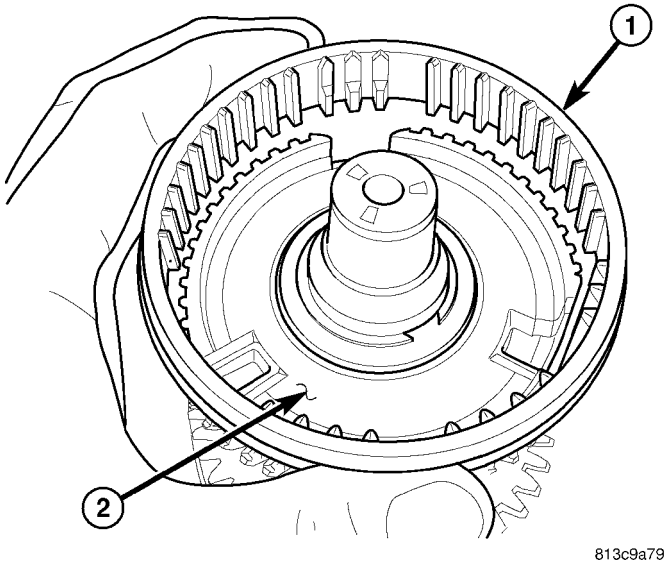


Fig. 21 5-6 SYNCHRONIZER SLEEVE

1 - SYNCHRONIZER SLEEVE
2 - HUB

(3) Remove 5-6 synchronizer sleeve (1) (Fig. 21) from hub (2).

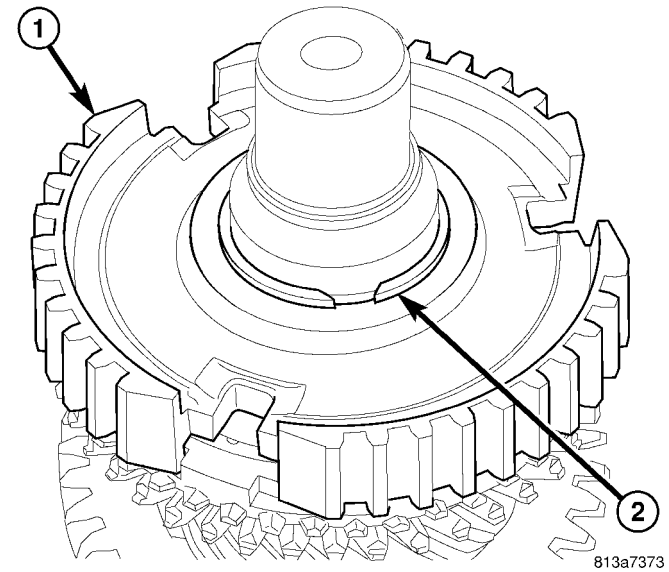


Fig. 22 5-6 SYNCHRO HUB SNAP RING

1 - SYNCHRONIZER HUB
2 - SNAP RING

(4) Remove 5-6 synchronizer hub (1) (Fig. 22) snap ring (2).

NOTE: All snap rings are select fit from the factory.

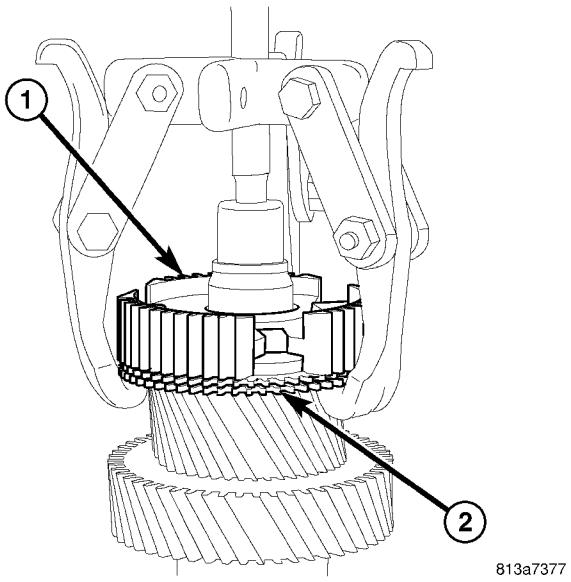


Fig. 23 5-6 SYNCHRO HUB

1 - 5-6 SYNCHRONIZER HUB
2 - SIXTH GEAR SYNCHRONIZER RINGS

(5) Remove 5-6 synchronizer hub (1) (Fig. 23) and sixth gear synchronizer rings (2) with a three jaw puller. Place puller jaws under sixth gear synchronizer rings (2).

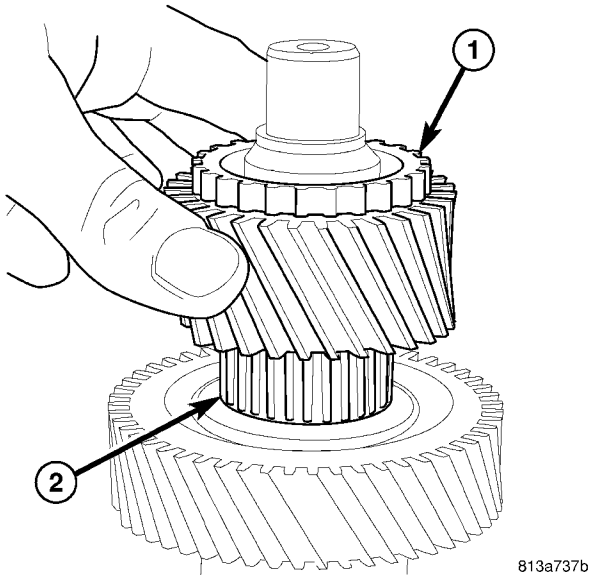


Fig. 24 SIXTH GEAR & BEARING

1 - SIXTH GEAR
2 - BEARING

(6) Remove sixth gear (1) (Fig. 24) and bearing (2).

MANUAL TRANSMISSION - NSG370 (Continued)

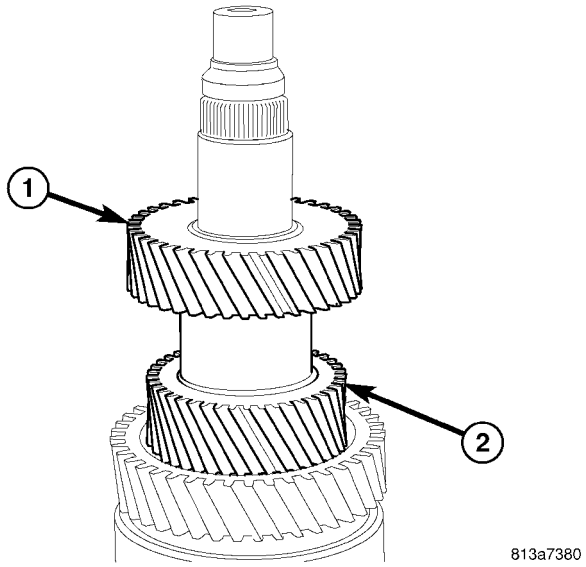


Fig. 25 3-4 GEAR

1 - THIRD GEAR
2 - FOURTH GEAR

(7) Third gear (1) (Fig. 25) and fourth gear (2) are serviced with the mainshaft only.

(8) Remove mainshaft from Fixture 9648. Turn fixture over in vise and set opposite end of mainshaft in the fixture.

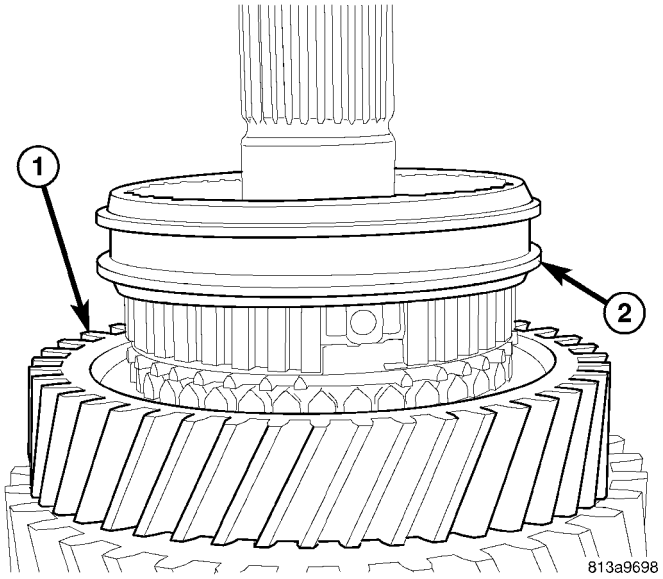


Fig. 26 REVERSE GEAR SYNCHRO SLEEVE

1 - REVERSE GEAR
2 - SYNCHRONIZER SLEEVE

(9) Remove reverse gear (1) (Fig. 26) synchronizer sleeve (2) off synchronizer hub.

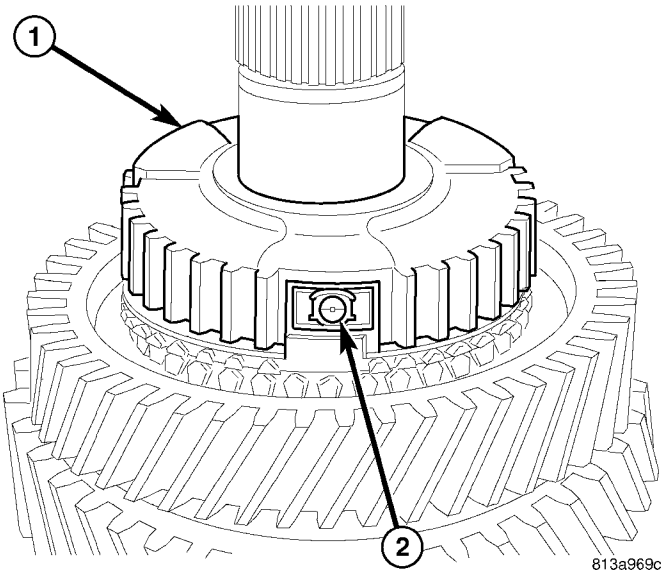


Fig. 27 REVERSE SYNCHRO HUB DETENT

1 - SYNCHRONIZER HUB
2 - DETENTS

(10) Remove reverse synchronizer hub (1) (Fig. 27) detents (2).

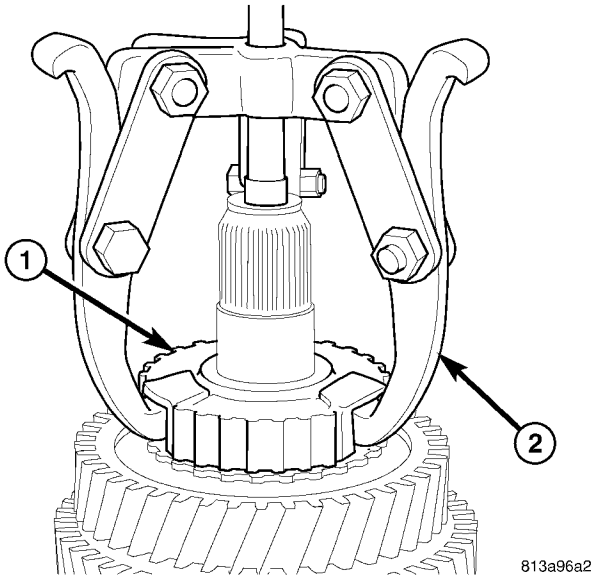


Fig. 28 REVERSE SYNCHRO HUB

1 - SYNCHRONIZER HUB
2 - THREE JAW PULLER

(11) Remove reverse synchronizer hub (1) (Fig. 28) with three jaw puller (2). Place puller jaws in hub detent openings.

MANUAL TRANSMISSION - NSG370 (Continued)

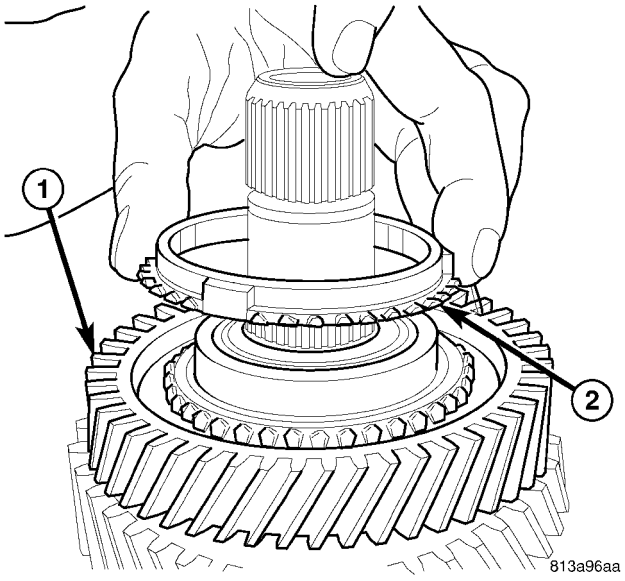


Fig. 29 REVERSE SYNCHRONIZER

1 - REVERSE GEAR
2 - SYNCHRONIZER FRICTION RING

(12) Remove reverse gear (1) (Fig. 29) synchronizer ring (2).

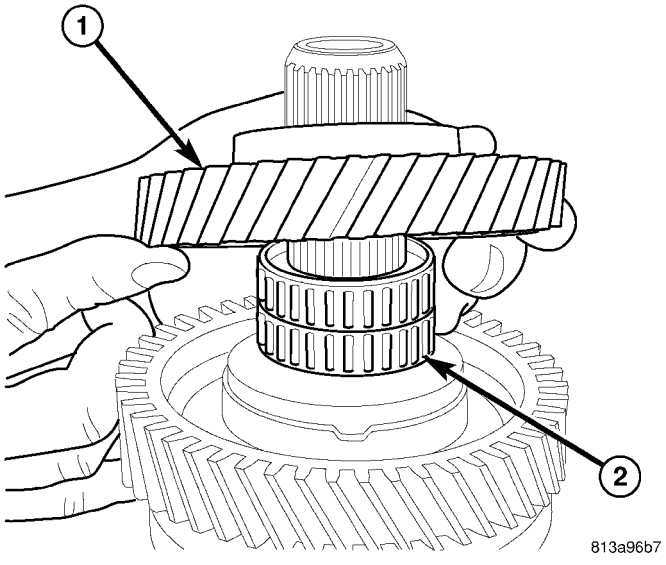


Fig. 30 REVERSE GEAR

1 - REVERSE GEAR
2 - BEARING

(13) Remove reverse gear (1) (Fig. 30) and bearing (2).

(14) Remove reverse gear bearing race (1) (Fig. 31), thrust washer (2) and first gear (3) with puller (4). Place puller jaws under first gear (3).

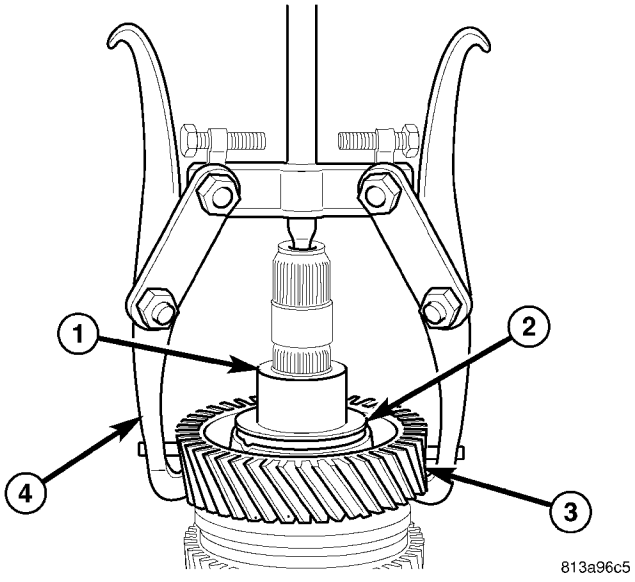


Fig. 31 FIRST GEAR

1 - BEARING RACE
2 - THRUST WASHER
3 - FIRST GEAR
4 - PULLER

(15) Remove first gear bearing.

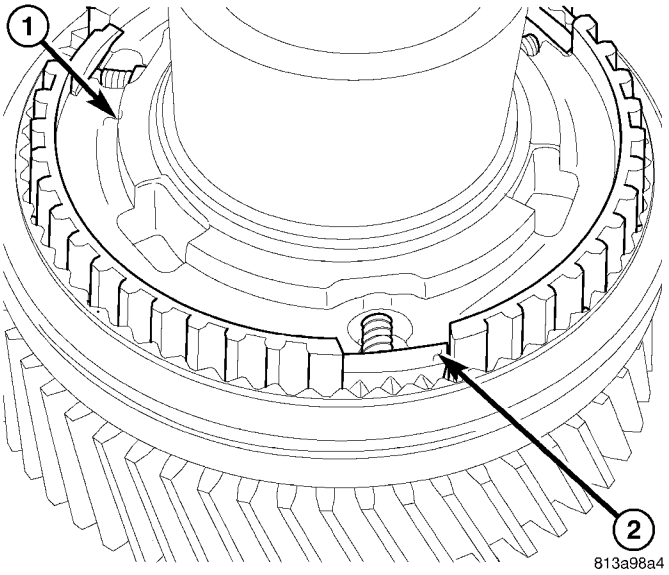


Fig. 32 1-2 SYNCHRO DETENT

1 - 1-2 SYNCHRONIZER HUB
2 - DETENTS

(16) Remove first gear synchronizer rings.

(17) Push 1-2 synchronizer sleeve down on hub. Remove 1-2 synchronizer hub (1) (Fig. 32) detents (2), balls and springs.

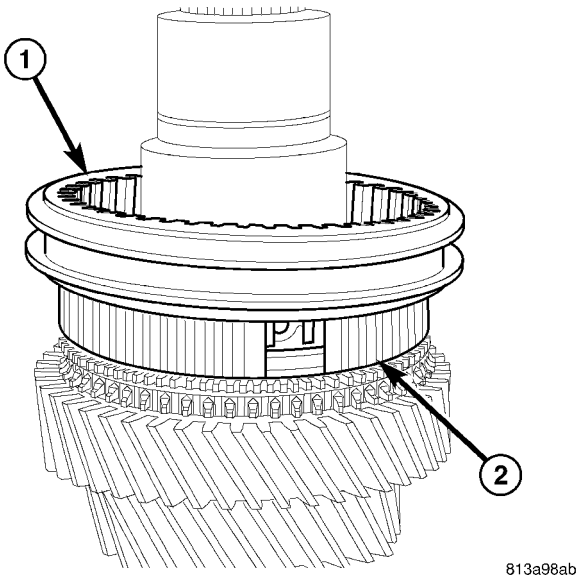


Fig. 33 1-2 SYNCHRO SLEEVE

1 - 1-2 SYNCHRONIZER SLEEVE
2 - SYNCHRONIZER HUB

(18) Remove 1-2 synchronizer sleeve (1) (Fig. 33) from synchronizer hub (2).

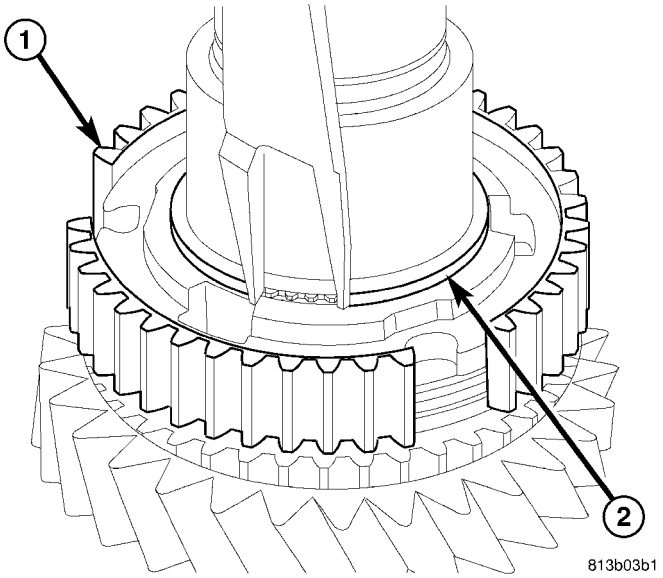


Fig. 34 1-2 SYNCHRO HUB SNAP RING

1 - SYNCHRONIZER HUB
2 - SNAP RING

(19) Remove 1-2 synchronizer hub (1) (Fig. 34) snap ring (2).

(20) Remove 1-2 synchronizer hub (1) (Fig. 35), synchronizer rings and second gear (2) off shaft with a press. Place second gear (2) on press plates and press shaft through 1-2 synchronizer hub (1), synchronizer rings and second gear (2)

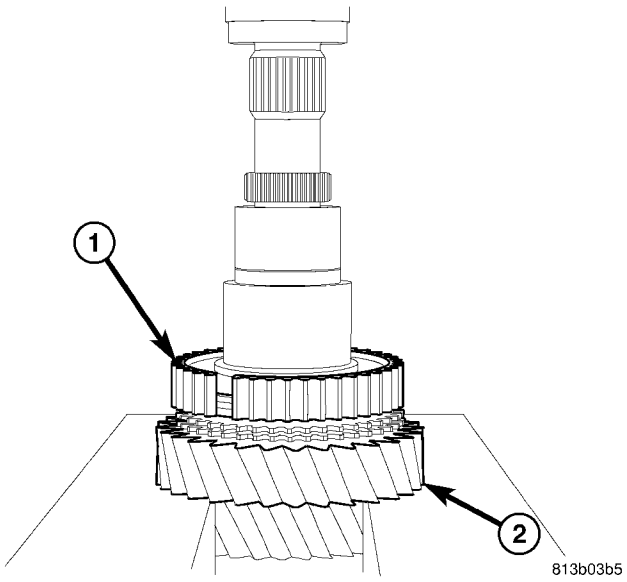


Fig. 35 SECOND GEAR

1 - SYNCHRONIZER HUB
2 - SECOND GEAR

COUNTERSHAFT

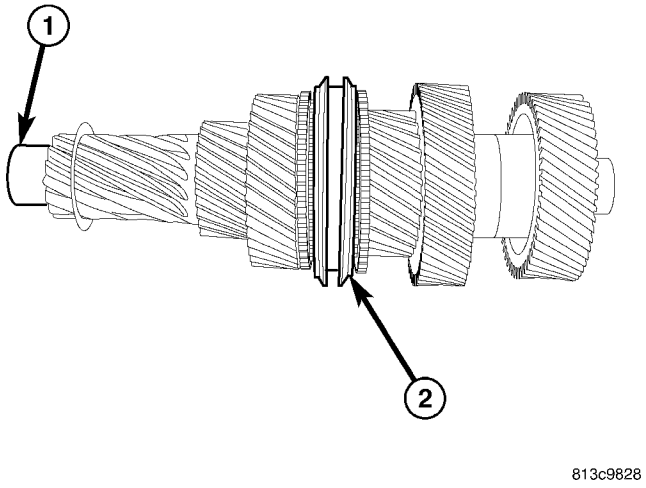


Fig. 36 COUNTERSHAFT

1 - BEARING SURFACES
2 - 3-4 SYNCHRONIZER ASSEMBLY

The countershaft (1) and 3-4 synchronizer (2) are serviced as an assembly only (Fig. 36).

MANUAL TRANSMISSION - NSG370 (Continued)

FRONT HOUSING

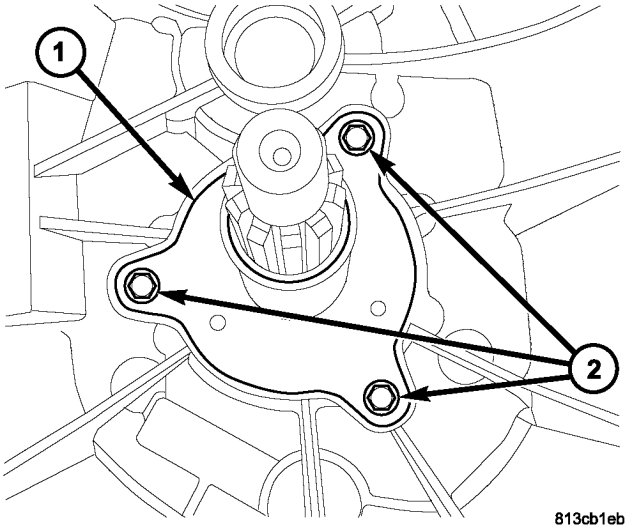


Fig. 37 INPUT SHAFT RETAINER

1 - INPUT SHAFT RETAINER
2 - BOLTS

(1) Remove input shaft retainer (1) (Fig. 37) bolts (2) and remove retainer.

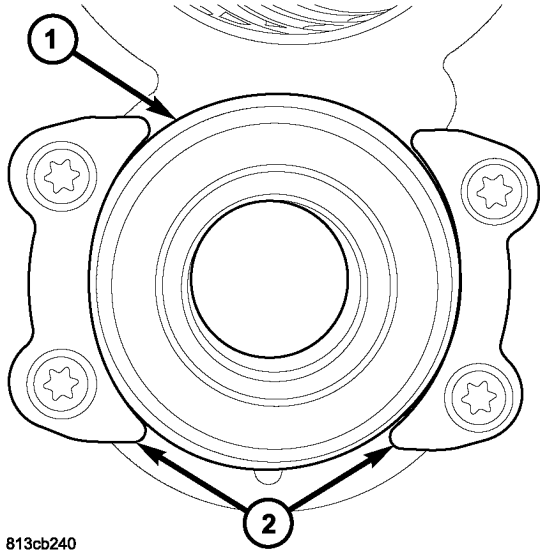


Fig. 38 COUNTERSHAFT BEARING RETAINER

1 - COUNTERSHAFT BEARING
2 - RETAINERS

(2) Remove countershaft bearing (1) (Fig. 38) retainer (2) bolts and remove retainers.

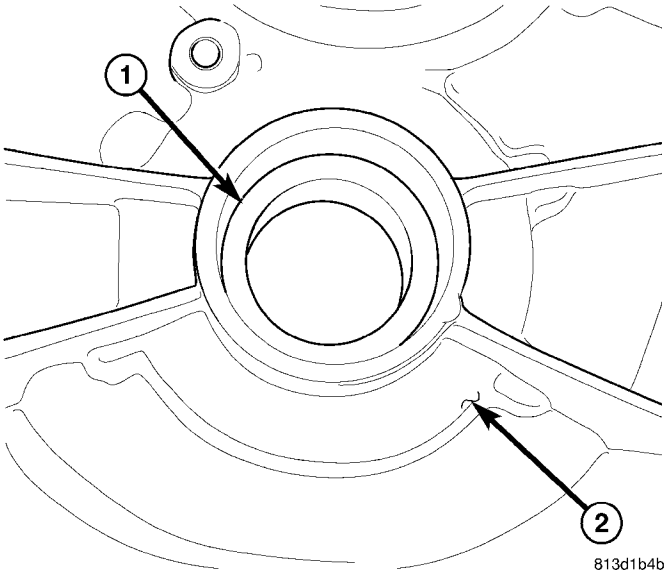


Fig. 39 COUNTERSHAFT BEARING FRONT HOUSING

1 - COUNTERSHAFT BEARING
2 - HOUSING

(3) Remove countershaft bearing (1) (Fig. 39) from housing (2) with a hammer and driver.

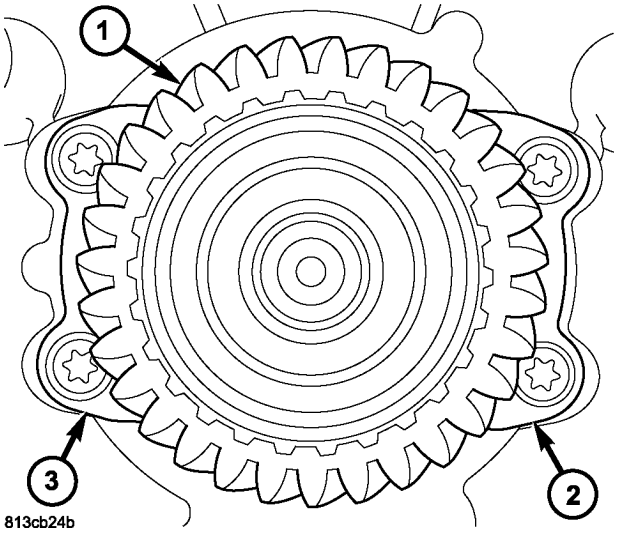


Fig. 40 INPUT SHAFT BEARING RETAINER

1 - INPUT SHAFT
2 - BEARING RETAINER
3 - BEARING RETAINER

(4) Remove input shaft (1) (Fig. 40) bearing retainer (2) bolts and remove retainers.

(5) Remove input shaft and bearing from housing with a dead blow hammer.

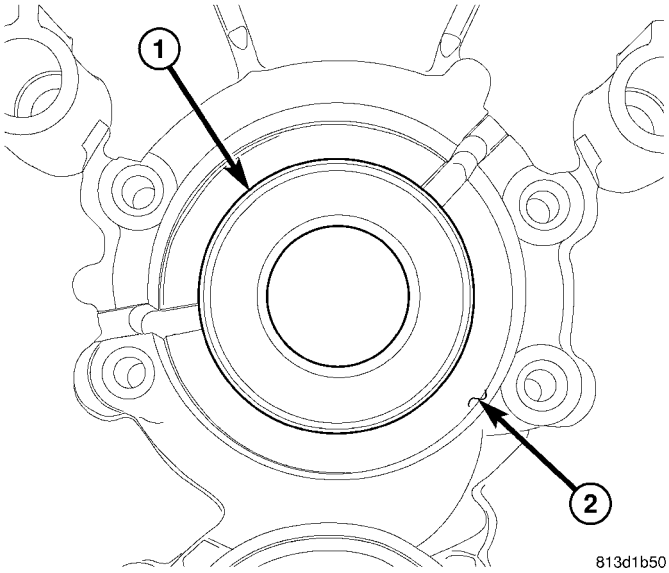


Fig. 41 INPUT SHAFT SEAL

1 - INPUT SHAFT SEAL
2 - HOUSING

(6) Remove input shaft seal (1) (Fig. 41) from housing (2) with a hammer and driver.

INPUT SHAFT

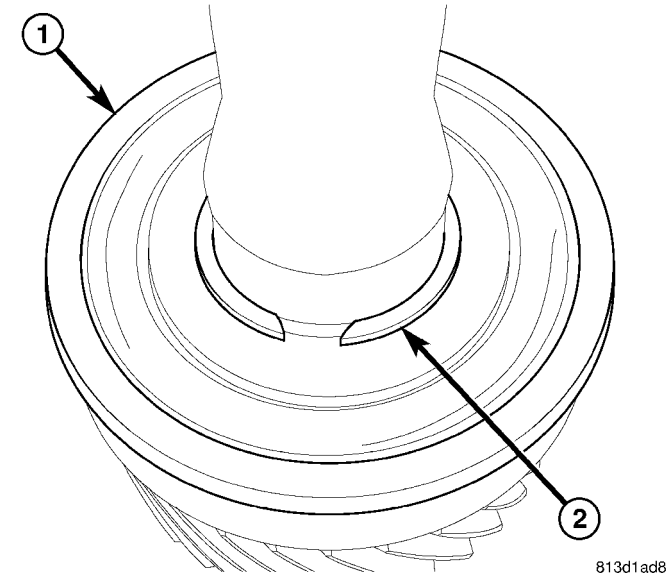


Fig. 42 INPUT SHAFT BEARING SNAP RING

1 - INPUT SHAFT BEARING
2 - SNAP RING

(1) Remove input shaft bearing (1) (Fig. 42) snap ring (2).

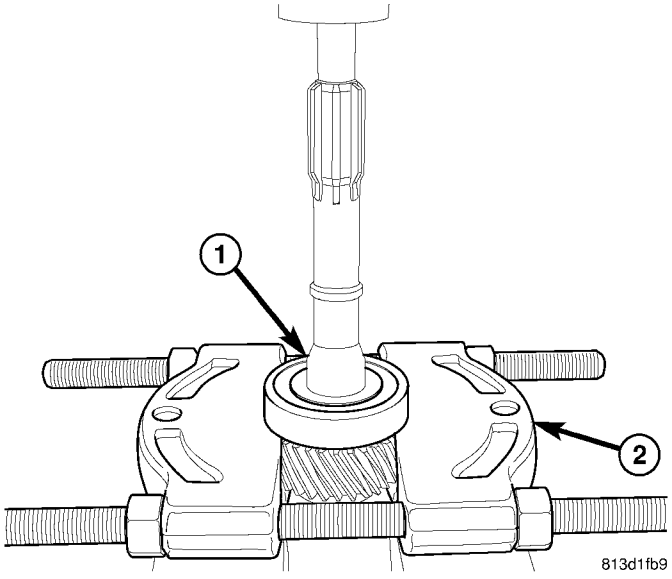


Fig. 43 INPUT SHAFT BEARING REMOVAL

1 - INPUT SHAFT BEARING
2 - BEARING SPLITTER 1130

(2) Remove input shaft bearing (1) (Fig. 43) with Splitter 1130 (2) and press. Place splitter around bearing retainer lip.

REAR HOUSING

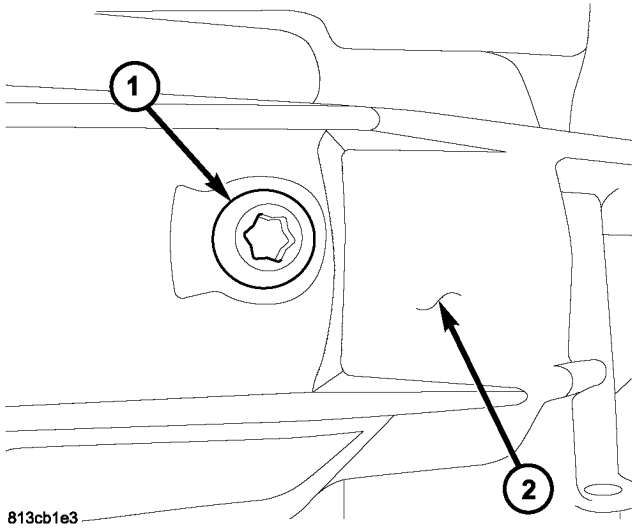


Fig. 44 IDLER SHAFT BOLT

1 - IDLER GEAR SHAFT BOLT
2 - HOUSING

(1) Remove reverse idler gear shaft bolt (1) (Fig. 44) from housing (2).

MANUAL TRANSMISSION - NSG370 (Continued)

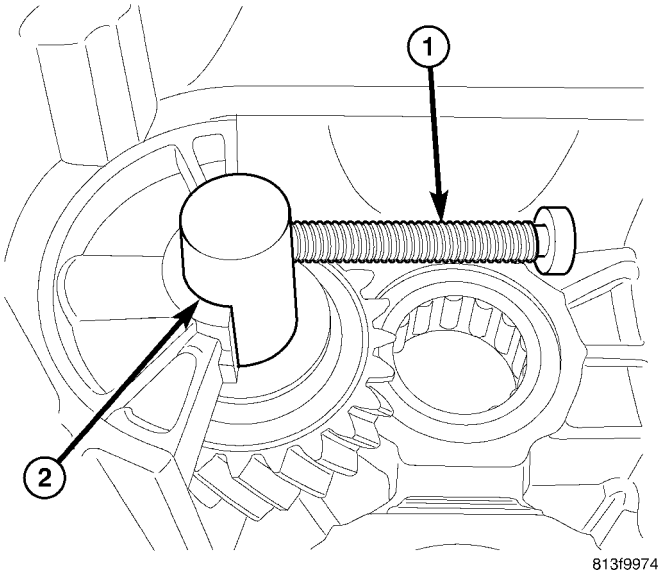


Fig. 45 PULL IDLER SHAFT

1 - IDLER GEAR SHAFT BOLT
2 - IDLER GEAR SHAFT

(2) Thread idler gear shaft bolt (1) (Fig. 45) into the shaft (2). Then work shaft (2) out of the idler gear.

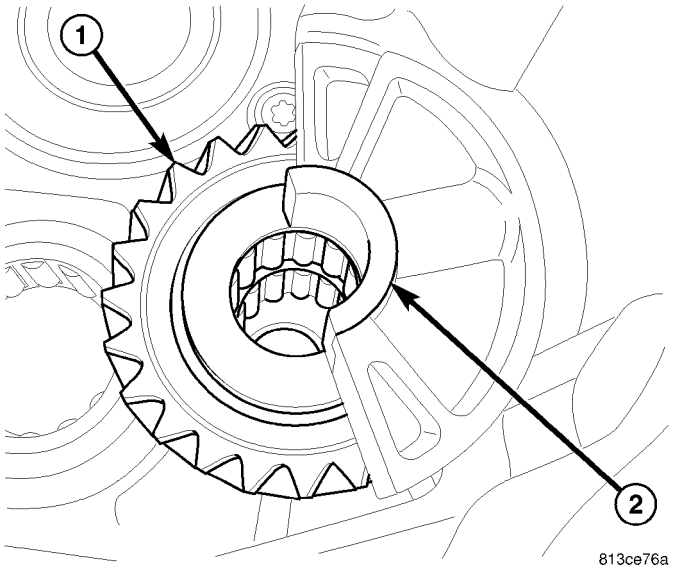


Fig. 46 IDLER SHAFT SUPPORT

1 - REVERSE IDLER GEAR
2 - SHAFT SUPPORT

(3) Remove reverse idler gear (1) (Fig. 46) shaft support (2).

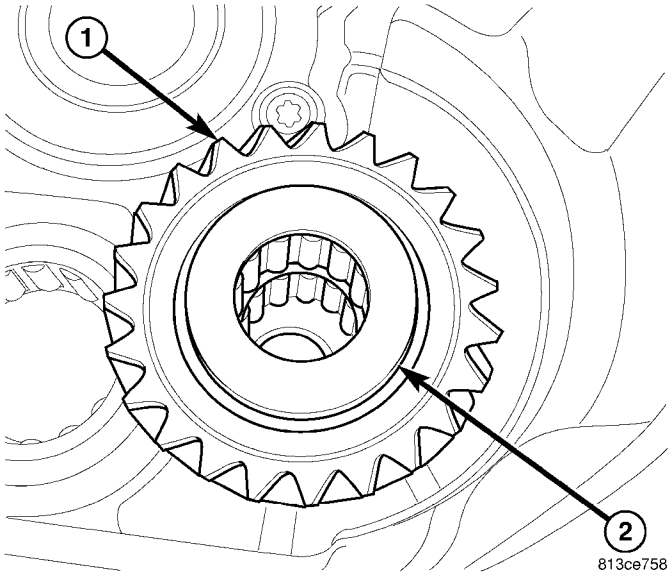


Fig. 47 IDLER GEAR THRUST WASHER

1 - REVERSE IDLER GEAR
2 - THRUST WASHER

(4) Remove reverse idler gear (1) (Fig. 47) thrust washer (2).

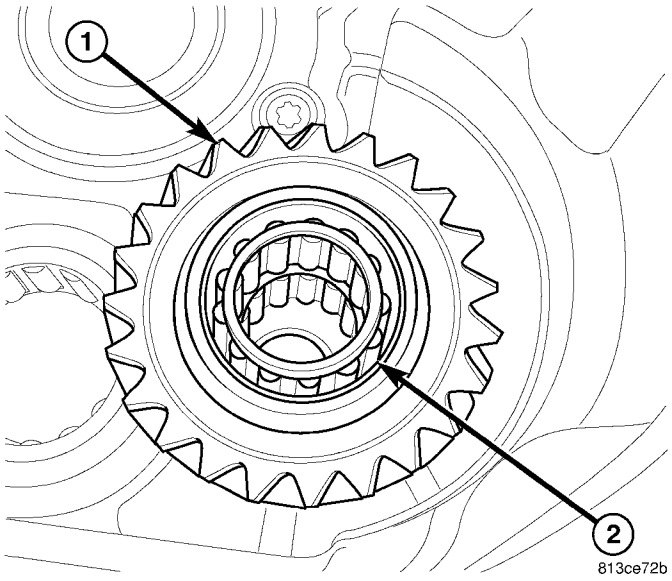


Fig. 48 IDLER GEAR BEARING

1 - REVERSE IDLER GEAR
2 - BEARING

(5) Remove reverse idler gear (1) (Fig. 48) and bearing (2).

MANUAL TRANSMISSION - NSG370 (Continued)

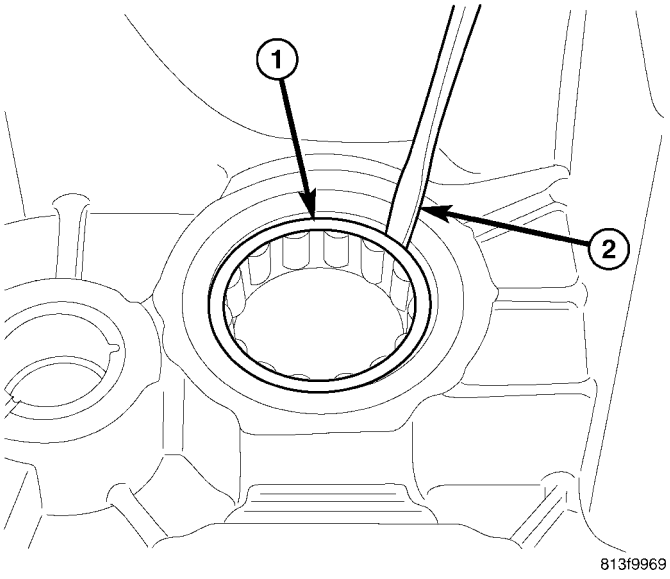


Fig. 49 BEARING CAGE

1 - ROLLER BEARING CAGE
2 - SCREWDRIVER

(6) Break plastic countershaft roller bearing cage (1) (Fig. 49) with a screw driver (2). Then remove cage and roller bearing.

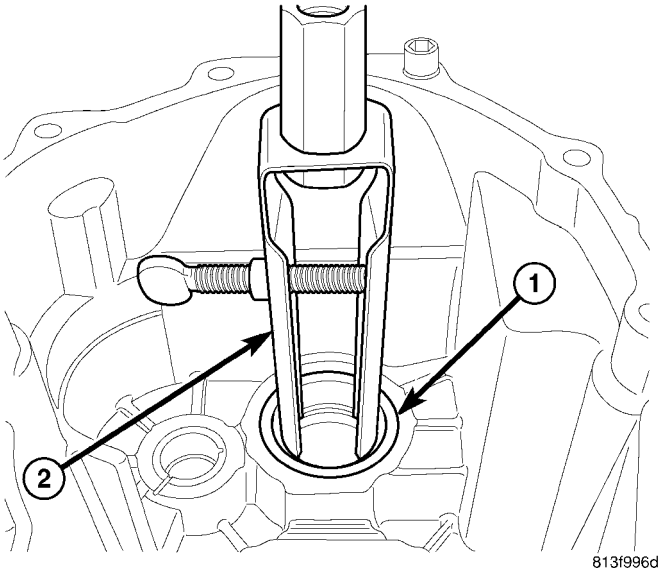


Fig. 50 BEARING SHELL PULLER

1 - ROLLER BEARING CAGE
2 - REMOVER

(7) Remove countershaft roller bearing shell (1) (Fig. 50) from the rear housing with Remover 7794-A (2) and Slide Hammer C-637.

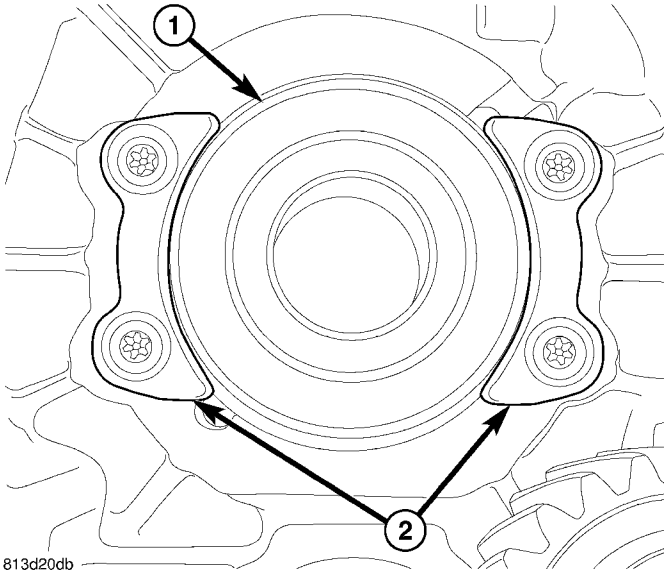


Fig. 51 REAR MAINSHAFT BEARING RETAINER

1 - MAINSHAFT BEARING
2 - RETAINERS

(8) Remove mainshaft bearing (1) (Fig. 51) retainer (2) bolts and remove retainers.

(9) Remove bearing from housing with a hammer and driver.

CLEANING

Clean gears, shafts, shift components and transmission housings with a standard parts cleaning solvent. Do not use acid or corrosive base solvents. Dry all parts except bearings with compressed air.

Clean shaft bearings with a mild solvent such as Mopar degreasing solvent, Gunk or similar solvents. Do not dry the bearings with compressed air. Allow the bearings to either air dry or wipe them dry with clean shop towels.

MANUAL TRANSMISSION - NSG370 (Continued)

INSPECTION

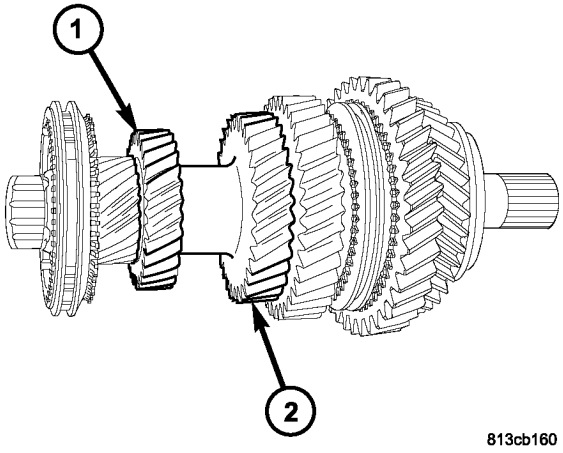


Fig. 52 MAINSHAFT

1 - THIRD GEAR
2 - FOURTH GEAR

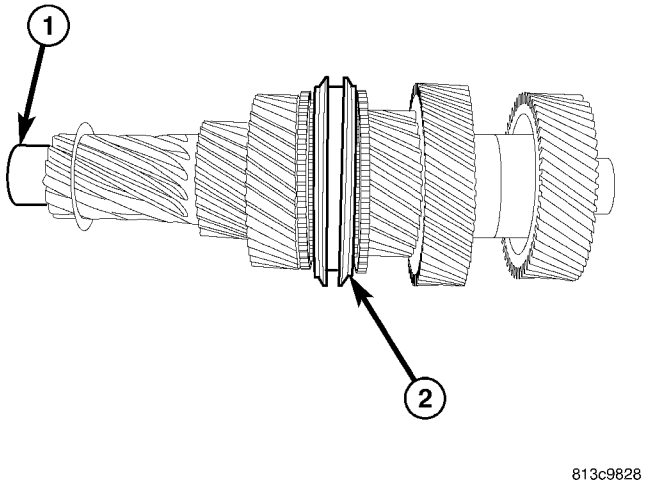


Fig. 53 COUNTERSHAFT

1 - BEARING SURFACES
2 - 3-4 SYNCHRONIZER ASSEMBLY

NOTE: Minor corrosion, nicks, or pitting can be smoothed with 400 grit emery and polished out with crocus cloth.

Bearings: Inspect for worn, cracked, flat-spotted or brinnelled.

Gears: Inspect for worn, chipped or cracked teeth. Inspect bearing surfaces for ware or flat-spotted.

Mainshaft: Inspect for worn splines, snap ring grooves and threads. Inspect bearing surfaces for ware or flat-spotted.**Third gear (1) (Fig. 52) and fourth gear (2) are serviced with mainshaft only.**

Countershaft: Inspect for worn, chipped or cracked teeth. Inspect bearing surfaces (1) (Fig. 53) for wear or flat-spots. Inspect 3-4 synchronizer assembly (2).**The countershaft is serviced as an assembly.**

Input Shaft: Inspect for worn, chipped or cracked teeth (1). Inspect bearing surface (2) for wear or flat-spots. Inspect for worn splines (3) (Fig. 54).

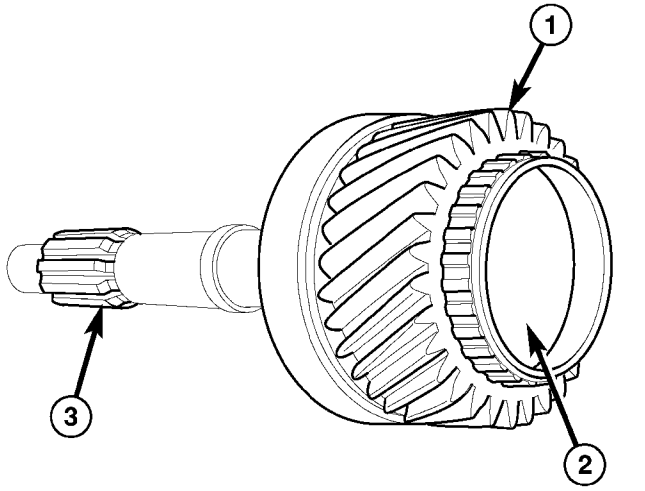


Fig. 54 INPUT SHAFT

1 - TEETH
2 - BEARING SURFACE
3 - SPLINE

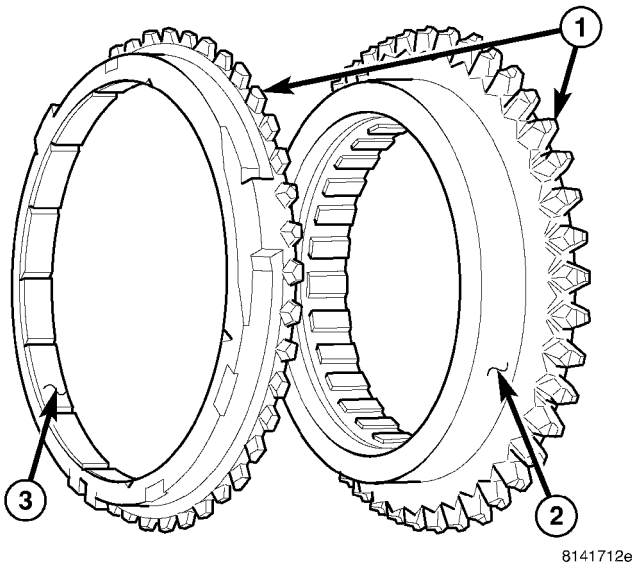


Fig. 55 SYNCHRONIZER RINGS

1 - TEETH
2 - FRICTION SURFACE
3 - FRICTION MATERIAL

Synchronizer components: Inspect for worn, chipped or cracked teeth (1) and burned friction surface (2) or flaking off friction material (3) (Fig. 55). **Synchronizers are serviced as an assembly. 3-4 synchronizer assembly (2) is serviced with countershaft.**

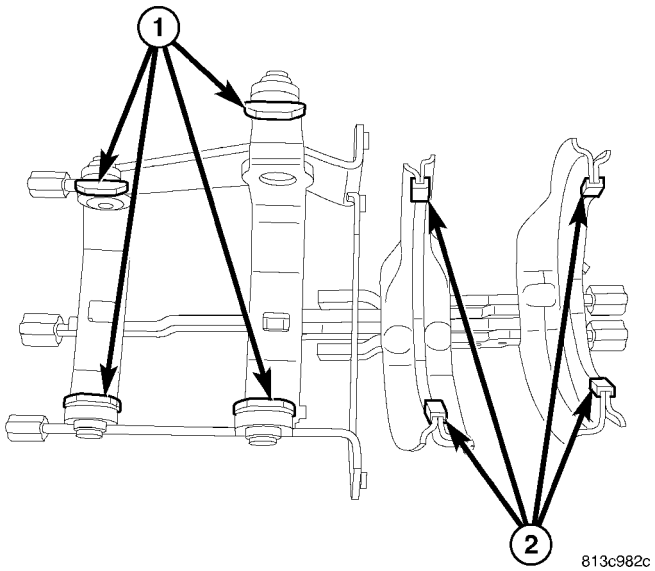


Fig. 56 SHIFT FORKS

1 - SHIFT FORK SHOES
2 - SHIFT FORK PADS

Shift forks: Inspect shift forks and shoes (1) (Fig. 56) for wear and distortion. Check fit of fork shoes in synchronizer sleeve to ensure parts fit and work smoothly, replace if necessary. If shift fork pads (2) are worn, the shift fork must be replaced.

Housing/Tail housing: Inspect sealing and mating surfaces are free of burrs and nicks. Inspect alignment dowels are tight and in good condition.

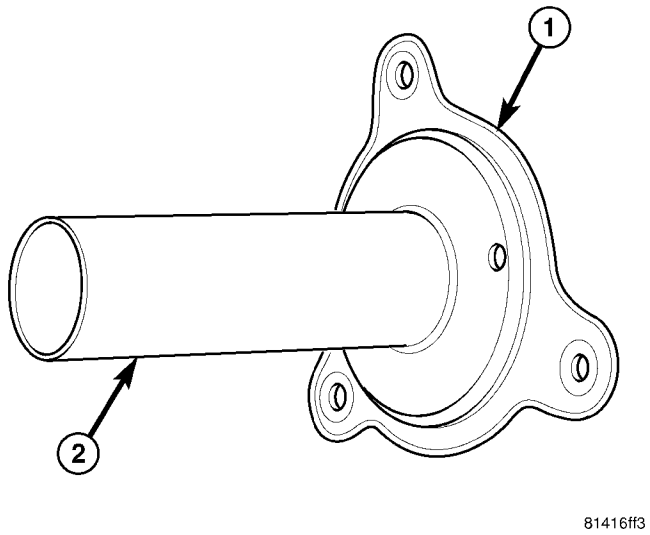


Fig. 57 RETAINER INPUT SHAFT

1 - RETAINER
2 - SLIDE SURFACE

Input Shaft Retainer: Inspect retainer (1) release bearing slide surface (2) (Fig. 57).

ASSEMBLY

WARNING: Use welding gloves when handling heated components. Failure to follow these instructions will result in personal injury.

CAUTION: A bearing heater is used to assembly some components. Use only a bearing heater/hot plate and follow manufactures instructions. Heat components to 100 - 177 Celsius (212° Min - 350° Max Fahrenheit). Never use an open flame to heat components. Never leave components on heater for and extended amount of time. If component is discolored after heating the component has been overheated and must not be used. Failure to follow these instructions will result in component damage.

MAINSHAFT

NOTE: Reuse snap rings in there original locations. Or thickest snap ring that will fit.

(1) Place mainshaft in Fixture 9648 mounted in a vise.

MANUAL TRANSMISSION - NSG370 (Continued)

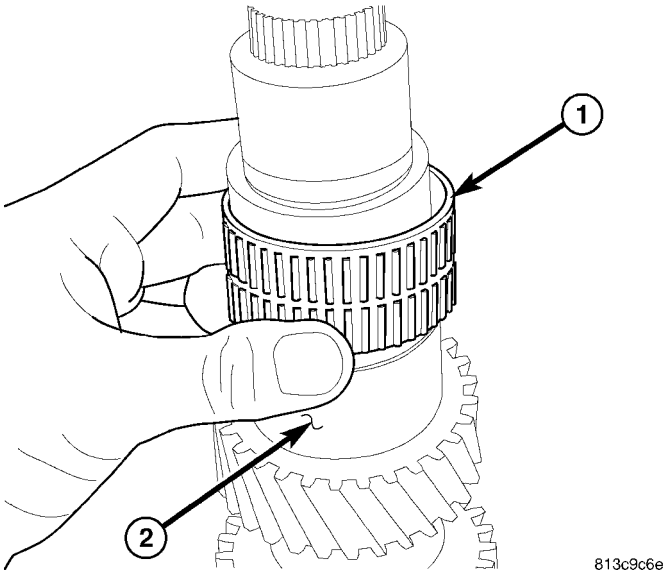


Fig. 58 SECOND GEAR BEARING

1 - SECOND GEAR BEARING
2 - MAINSHAFT

(2) Install second gear bearing (1) (Fig. 58) on mainshaft (2).

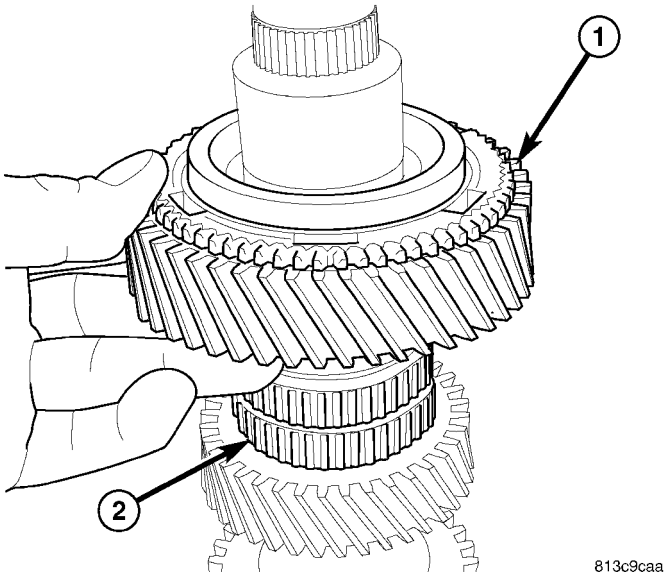


Fig. 59 SECOND GEAR

1 - SECOND GEAR
2 - BEARING

(3) Install second gear (1) (Fig. 59) on bearing (2).

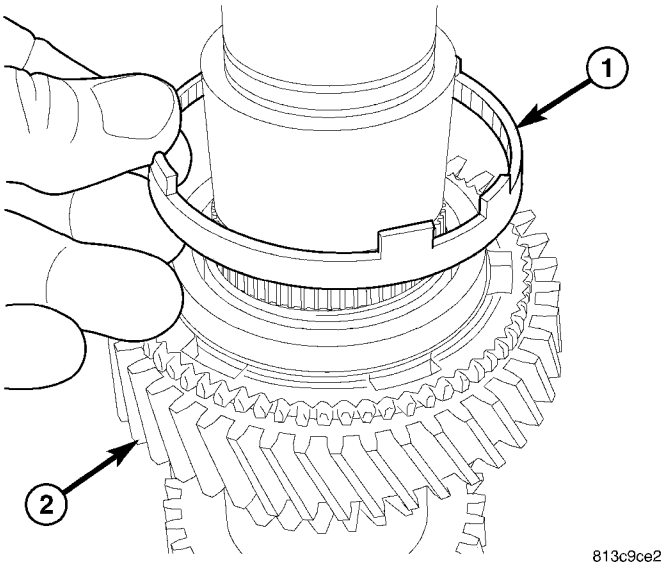


Fig. 60 2ND INNER FRICTION RING

1 - SYNCHRONIZER FRICTION RING
2 - SECOND GEAR

(4) Install inner synchronizer friction ring (1) (Fig. 60) on second gear (2).

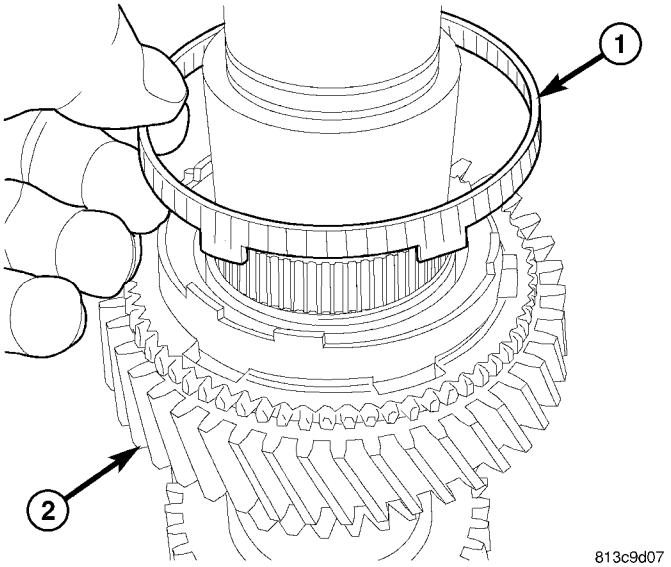


Fig. 61 2ND OUTER FRICTION RING

1 - OUTER SYNCHRONIZER FRICTION RING
2 - SECOND GEAR

(5) Install outer synchronizer friction ring (1) (Fig. 61) on second gear (2).

MANUAL TRANSMISSION - NSG370 (Continued)

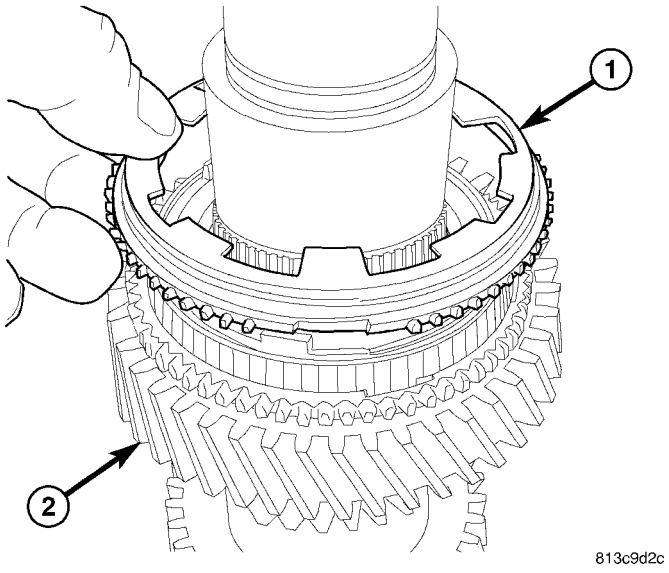


Fig. 62 2ND BLOCKER RING

1 - SYNCHRONIZER BLOCKER RING
2 - SECOND GEAR

(6) Install synchronizer blocker ring (1) (Fig. 62) on second gear (2).

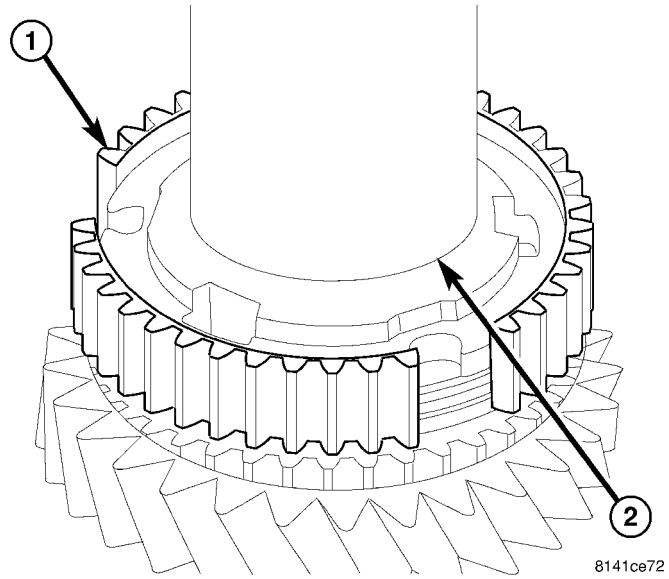


Fig. 63 1-2 SYNCHRO HUB INSTALLER

1 - 1-2 SYNCHRONIZER HUB
2 - INSTALLER

(7) Install 1-2 synchronizer hub (1) (Fig. 63) with Installer 8228 (2) and a press. Align hub detent openings with synchronizer rings.

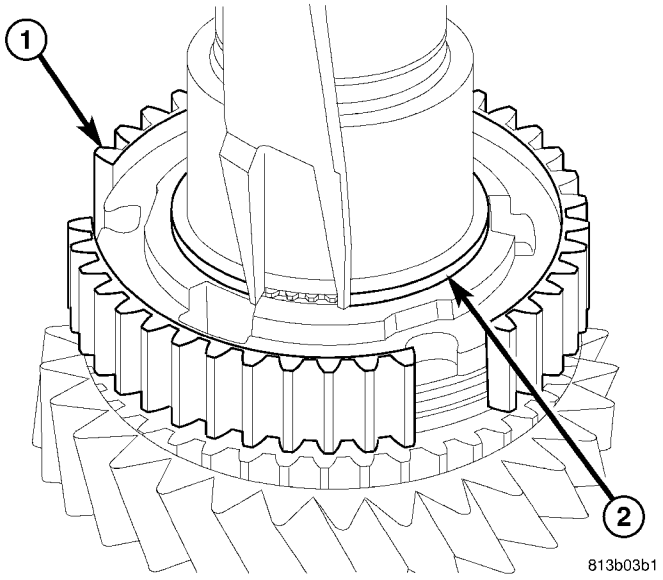


Fig. 64 1-2 SYNCHRO HUB SNAP RING

1 - SYNCHRONIZER HUB
2 - SNAP RING

(8) Install 1-2 synchronizer hub (1) (Fig. 64) snap ring (2).

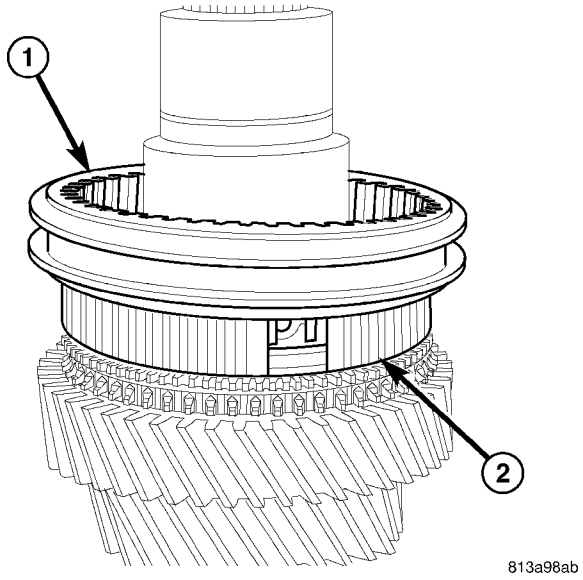


Fig. 65 1-2 SYNCHRO SLEEVE

1 - 1-2 SYNCHRONIZER SLEEVE
2 - SYNCHRONIZER HUB

(9) Install 1-2 synchronizer sleeve (1) (Fig. 65) and push to the bottom of the synchronizer hub (2).

MANUAL TRANSMISSION - NSG370 (Continued)

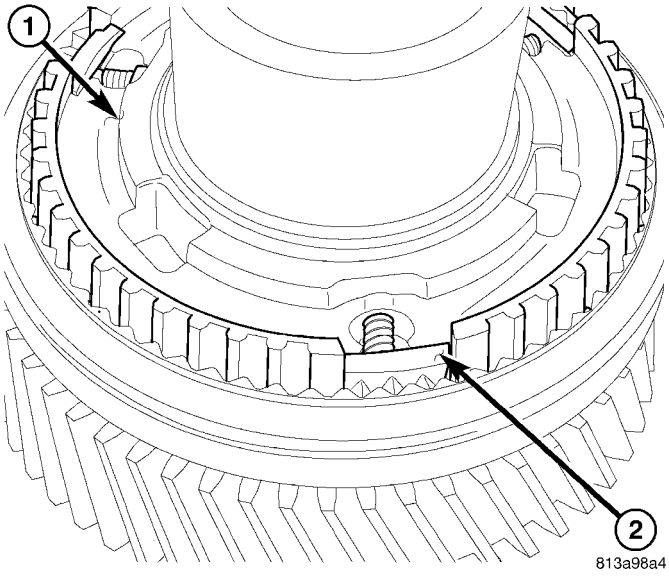


Fig. 66 1-2 SYNCHRO DETENT

1 - 1-2 SYNCHRONIZER HUB
2 - DETENTS

(10) Install 1-2 synchronizer detents, springs, and balls (2) (Fig. 66).

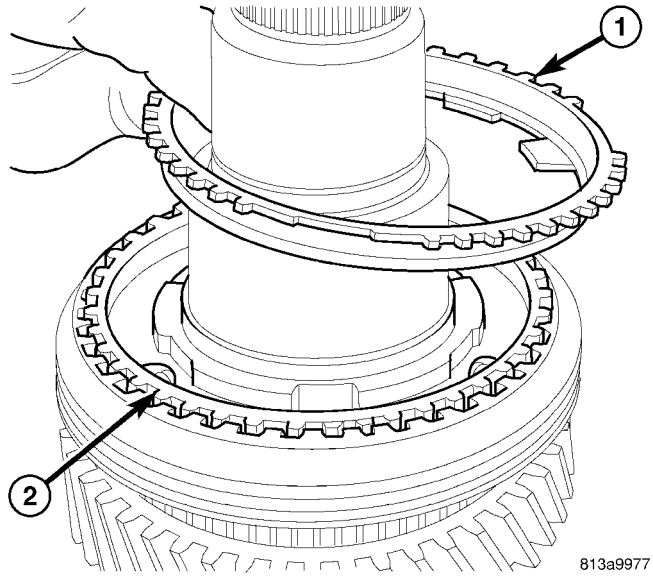


Fig. 67 FIRST GEAR OUTER BLOCKER RING

1 - FIRST GEAR BLOCKER RING
2 - SYNCHRONIZER HUB

(11) Install first gear blocker ring (1) (Fig. 67) into synchronizer hub (2).

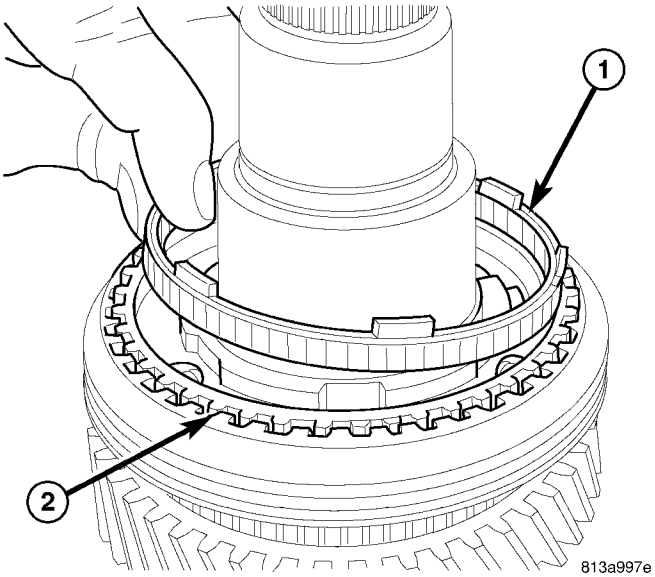


Fig. 68 FIRST GEAR FRICTION RING

1 - FIRST GEAR OUTER FRICTION RING
2 - SYNCHRONIZER HUB

(12) Install first gear outer friction ring (1) (Fig. 68) into synchronizer hub (2).

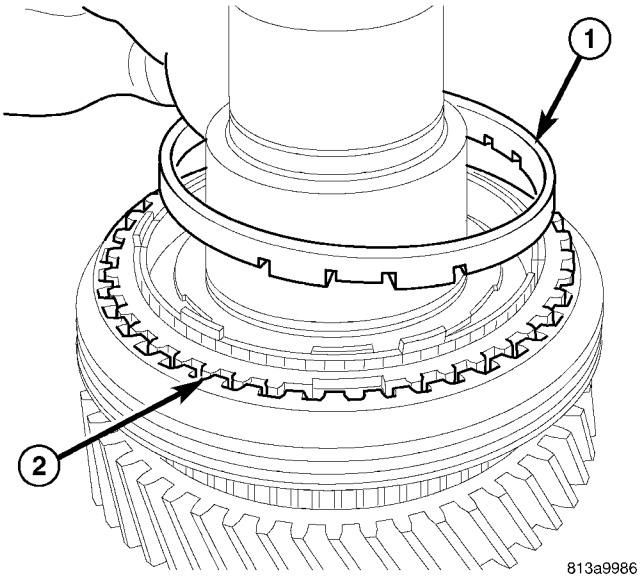


Fig. 69 FIRST GEAR INNER BLOCKER RING

1 - FIRST GEAR INNER BLOCKER RING
2 - SYNCHRONIZER HUB

(13) Install first gear inner blocker ring (1) (Fig. 69) into synchronizer hub (2).

MANUAL TRANSMISSION - NSG370 (Continued)

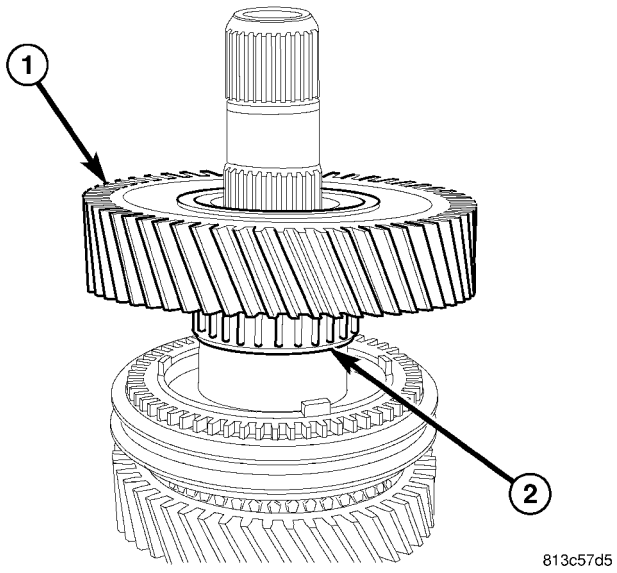


Fig. 70 1ST GEAR AND BEARING

1 - FIRST GEAR
2 - BEARING

(14) Install first gear (1) (Fig. 70) and bearing (2). Then center 1-2 synchronizer sleeve on synchronizer hub.

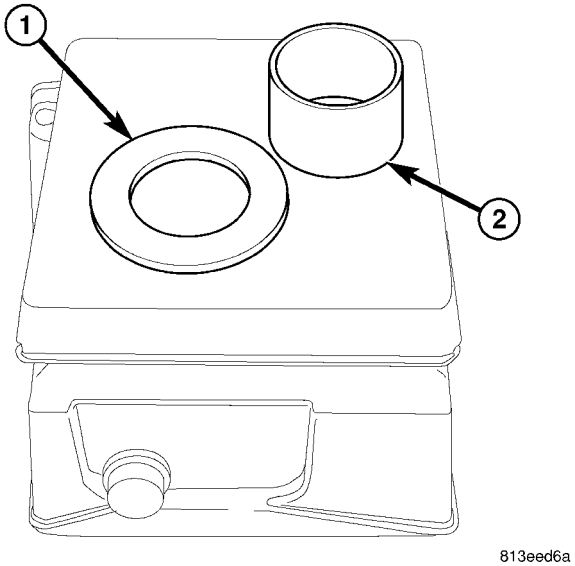


Fig. 71 WASHER AND RACE

1 - FIRST GEAR THRUST WASHER
2 - REVERSE GEAR BEARING RACE

(15) Heat first gear thrust washer (1) (Fig. 71) and reverse gear bearing race (2) with bearing heater.

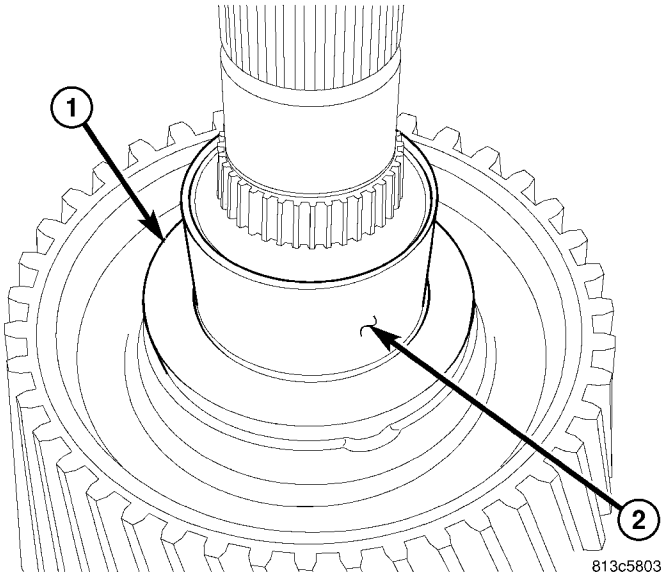


Fig. 72 THRUST WASHER & RACE

1 - FIRST GEAR THRUST WASHER
2 - REVERSE GEAR BEARING RACE

(16) Using welding gloves, install first gear thrust washer (1) (Fig. 72) and reverse gear bearing race (2).

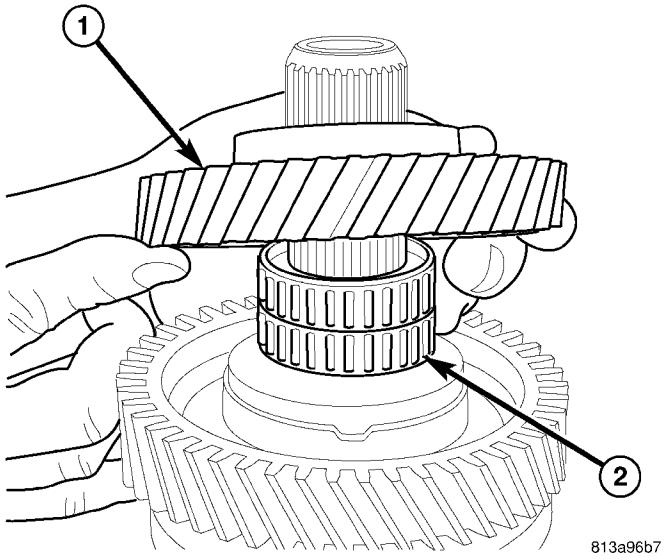


Fig. 73 REVERSE GEAR

1 - REVERSE GEAR
2 - BEARING

(17) Install reverse gear (1) (Fig. 73) and bearing (2).

MANUAL TRANSMISSION - NSG370 (Continued)

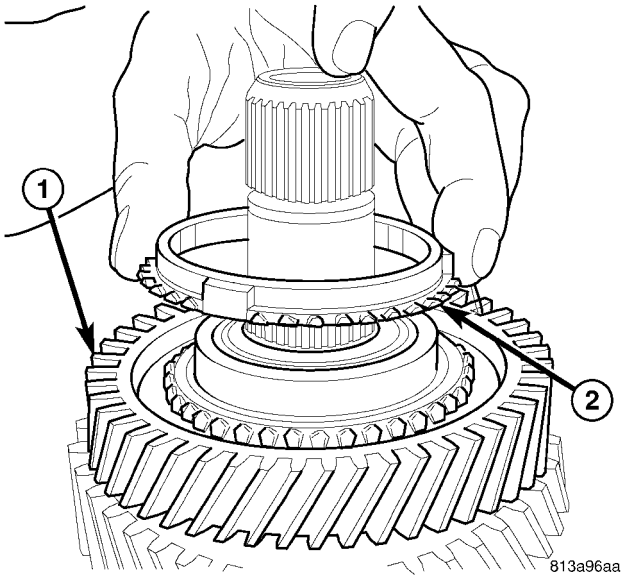


Fig. 74 REVERSE SYNCHRONIZER

1 - REVERSE GEAR
2 - SYNCHRONIZER FRICTION RING

(18) Install reverse gear (1) (Fig. 74) synchronizer friction ring (2).

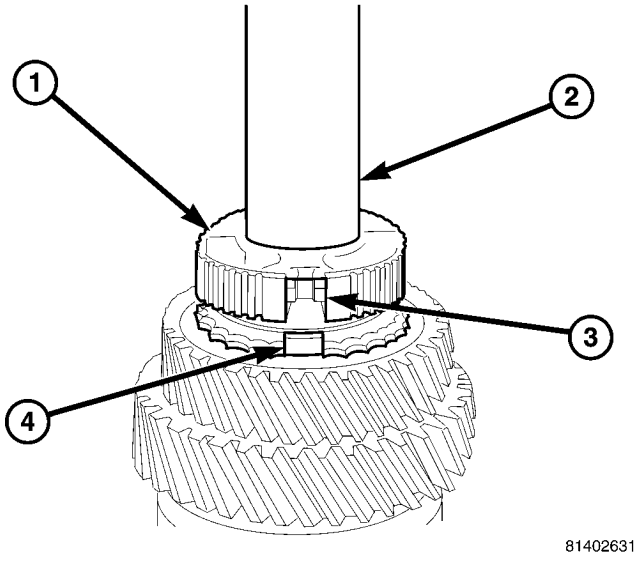


Fig. 75 REVERSE HUB INSTALLER

1 - SYNCHRONIZER HUB
2 - INSTALLER
3 - DETENT OPENING
4 - FRICTION RING TAB

(19) Install reverse gear synchronizer hub (1) on mainshaft with Installer W-262 (2) and a press. Align hub detent opening (3) with friction ring (4) lug (Fig. 75).

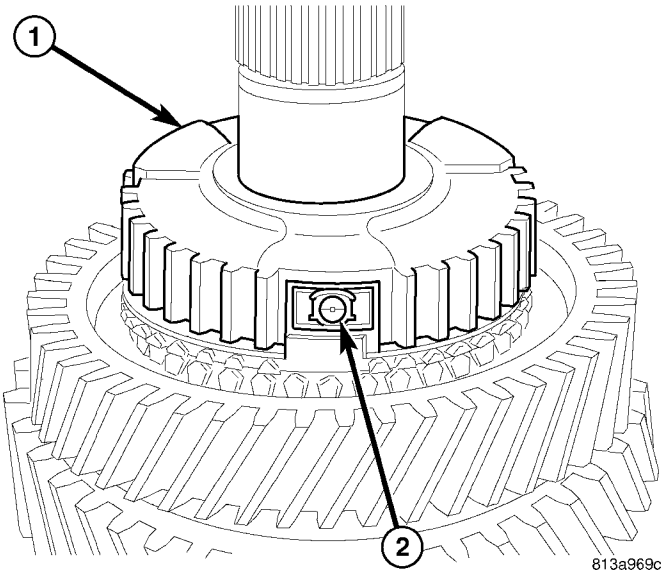


Fig. 76 REVERSE SYNCHRO HUB DETENT

1 - SYNCHRONIZER HUB
2 - DETENTS

(20) Install reverse gear synchronizer hub (1) (Fig. 76) detents (2).

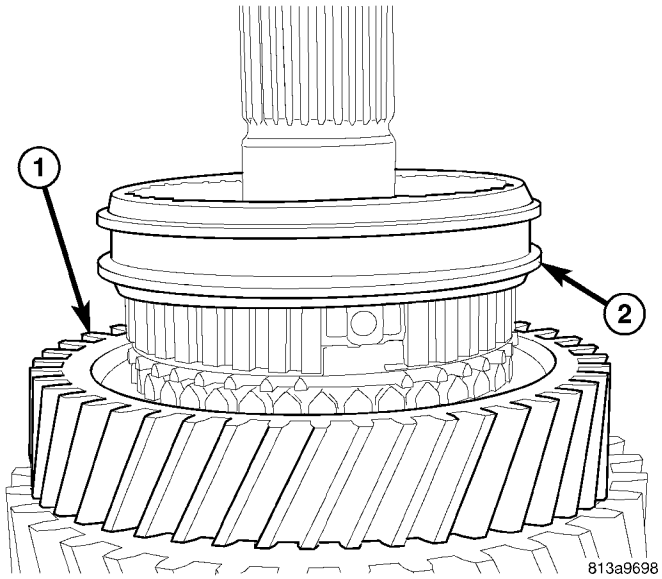


Fig. 77 REVERSE GEAR SYNCHRO SLEEVE

1 - REVERSE GEAR
2 - SYNCHRONIZER SLEEVE

(21) Install reverse gear (1) (Fig. 77) synchronizer sleeve (2) and center sleeve on synchronizer hub.

(22) Remove mainshaft from Fixture 9648. Turn fixture over in the vise, then install opposite end of mainshaft in the fixture.

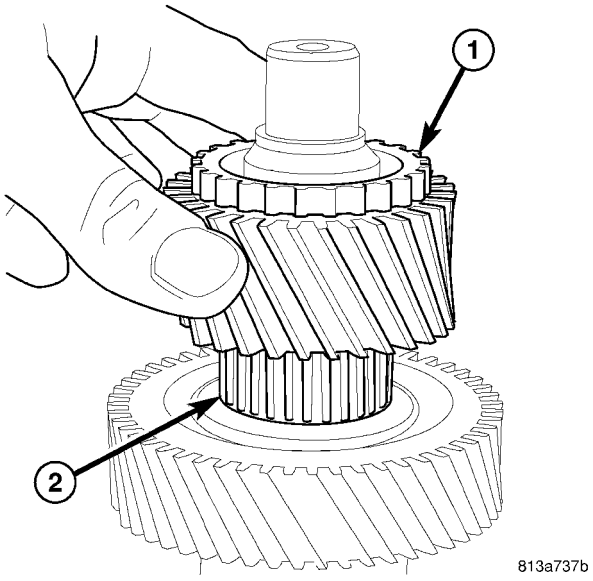


Fig. 78 SIXTH GEAR AND BEARING

1 - SIXTH GEAR
2 - BEARING

(23) Install sixth gear (1) (Fig. 78) and bearing (2).

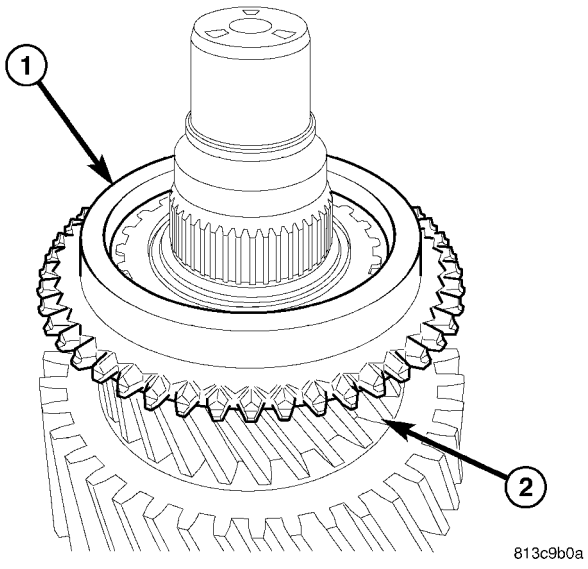


Fig. 79 6TH GEAR BLOCKER RING

1 - SIXTH GEAR BLOCKER RING
2 - SIXTH GEAR

(24) Install sixth gear synchronizer blocker ring (1) (Fig. 79) on sixth gear (2).

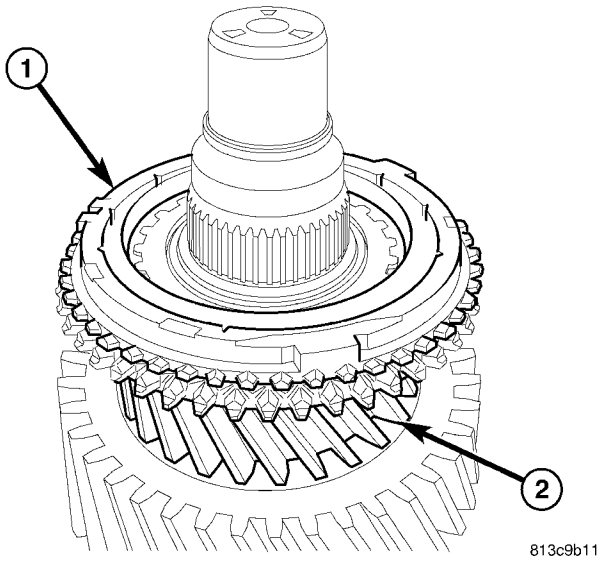


Fig. 80 6TH GEAR FRICTION RING

1 - SIXTH GEAR FRICTION RING
2 - SIXTH GEAR

(25) Install sixth gear synchronizer friction ring (1) (Fig. 80) on sixth gear (2).

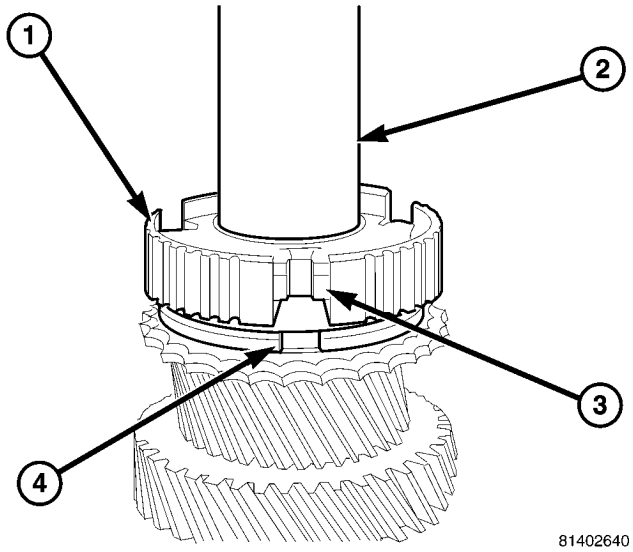


Fig. 81 5-6 HUB INSTALLER

1 - SYNCHRONIZER HUB
2 - INSTALLER
3 - DETENT OPENING
4 - FRICTION RING TAB

(26) Install 5-6 synchronizer hub (1) with Installer W-262 (2). Align hub detent opening (3) with friction ring lugs (4) and press hub on mainshaft (Fig. 81).

NOTE: 5-6 synchronizer hub center, is offset and must be install large offset down.

MANUAL TRANSMISSION - NSG370 (Continued)

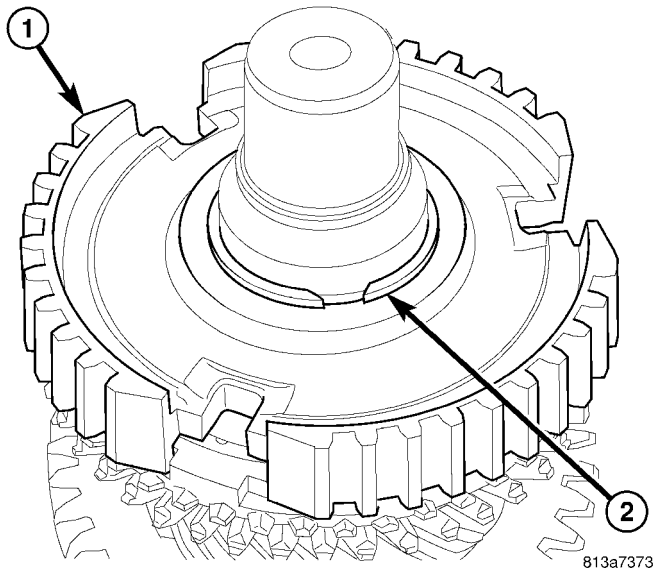


Fig. 82 5-6 SYNCHRO HUB SNAP RING

1 - SYNCHRONIZER HUB
2 - SNAP RING

(27) Install 5-6 synchronizer hub (1) (Fig. 82) snap ring (2).

(28) Install 5-6 synchronizer sleeve on hub (1).

REAR HOUSING

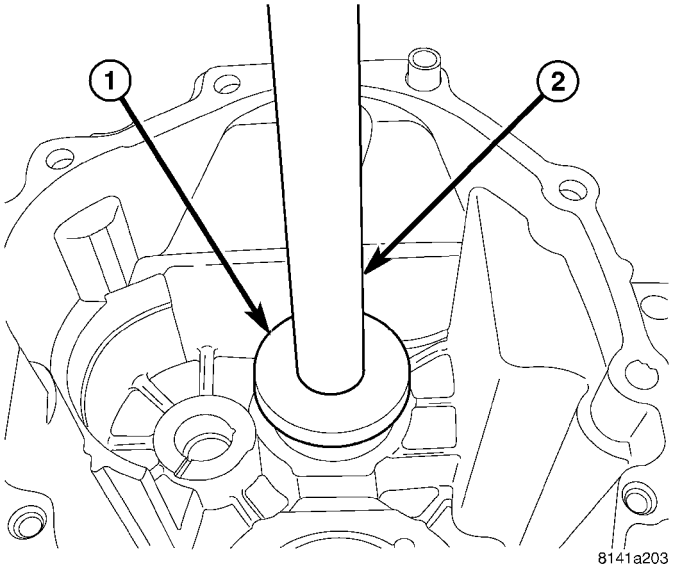


Fig. 83 REAR COUNTERSHAFT BEARING

1 - INSTALLER
2 - HANDLE

(1) Install countershaft bearing (Fig. 83) into rear housing with Installer 9643 (1) and Handle C-4171.

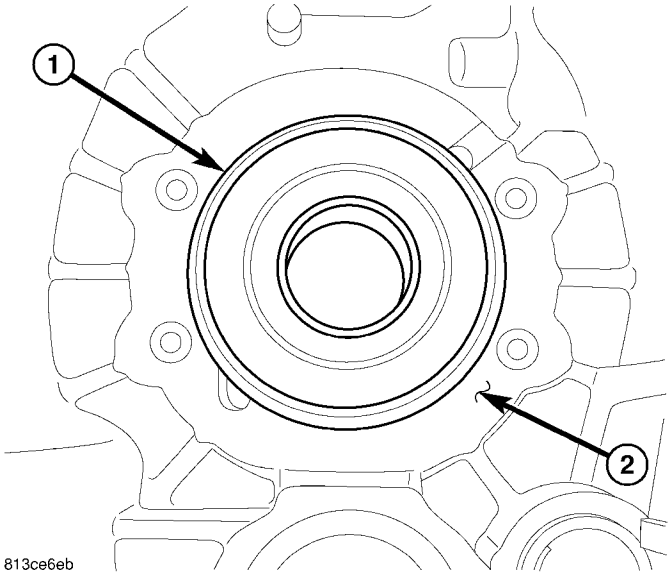


Fig. 84 REAR HOUSING MAINSHAFT BEARING

1 - MAINSHAFT BEARING
2 - HOUSING

(2) Install mainshaft bearing (1) (Fig. 84) into housing (2).

(3) Install bearing retainer and tighten bolts to 10 N·m (7.3 ft. lbs.).

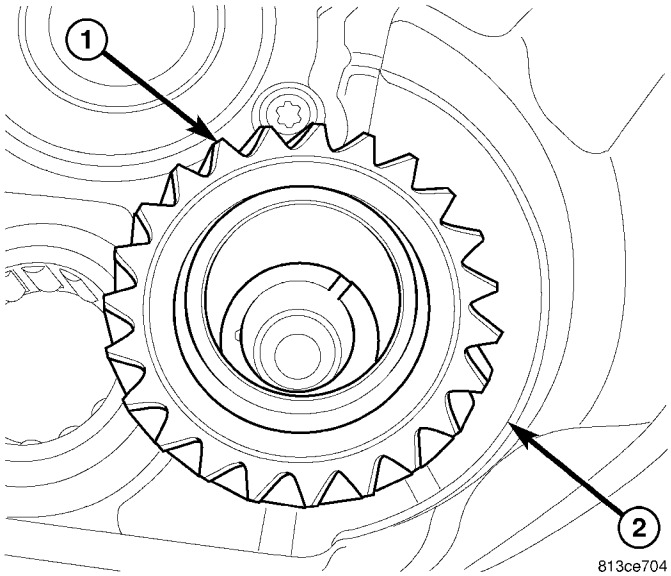


Fig. 85 REVERSE IDLER GEAR

1 - REVERSE IDLER GEAR
2 - HOUSING

(4) Install reverse idler gear (1) (Fig. 85) into housing (2).

MANUAL TRANSMISSION - NSG370 (Continued)

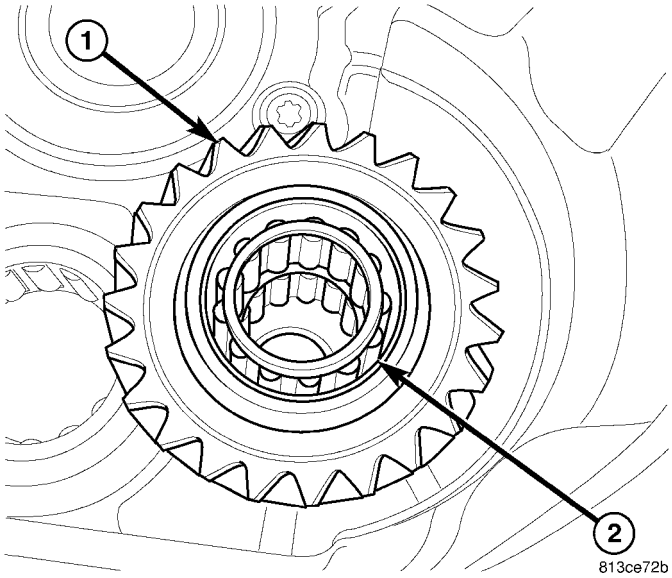


Fig. 86 IDLER GEAR BEARING

1 - REVERSE IDLER GEAR
2 - BEARING

(5) Install idler gear (1) (Fig. 86) bearing (2) into gear.

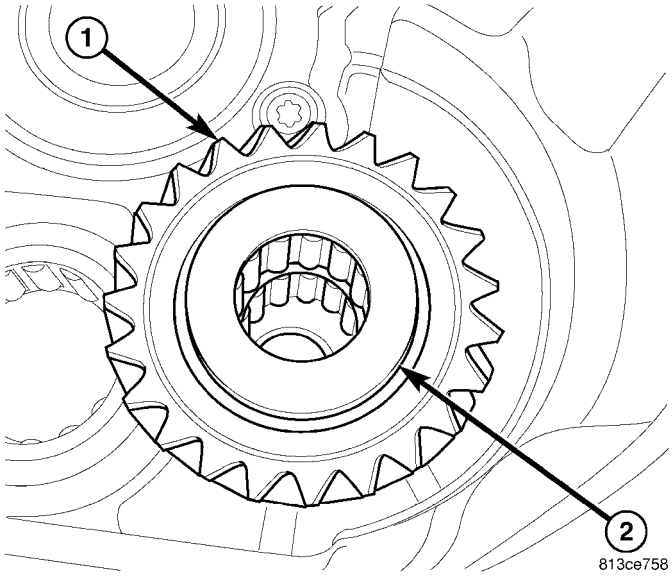


Fig. 87 IDLER GEAR THRUST WASHER

1 - REVERSE IDLER GEAR
2 - THRUST WASHER

(6) Install idler gear (1) (Fig. 87) thrust washer (2) on idler gear.

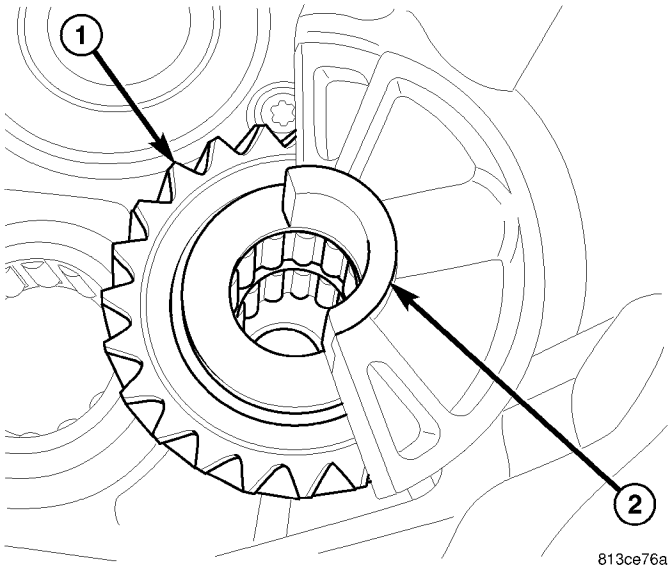


Fig. 88 IDLER SHAFT SUPPORT

1 - REVERSE IDLER GEAR
2 - SHAFT SUPPORT

(7) Install idler gear (1) (Fig. 88) shaft support (2) into housing.

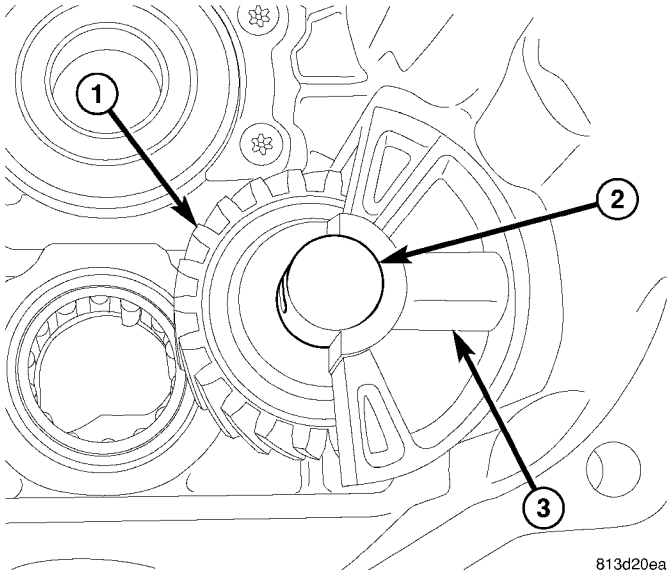


Fig. 89 IDLER GEAR SHAFT

1 - IDLER GEAR
2 - IDLER GEAR SHAFT
3 - SHAFT SUPPORT HOLE

(8) Install idler gear (1) (Fig. 89) shaft (2) with shaft bolt hole alined with shaft support hole (3).

MANUAL TRANSMISSION - NSG370 (Continued)

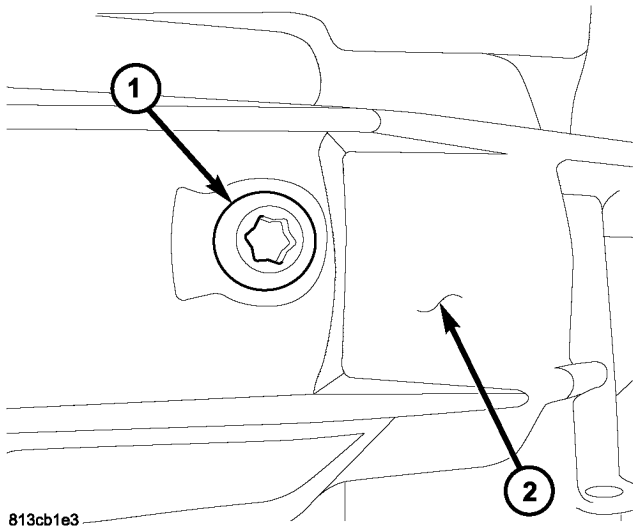


Fig. 90 IDLER SHAFT BOLT

1 - IDLER GEAR SHAFT BOLT
2 - HOUSING

(9) Install idler gear shaft bolt (1) (Fig. 90) into the housing (2) and tighten to 20 N.m (15 ft. lbs.).

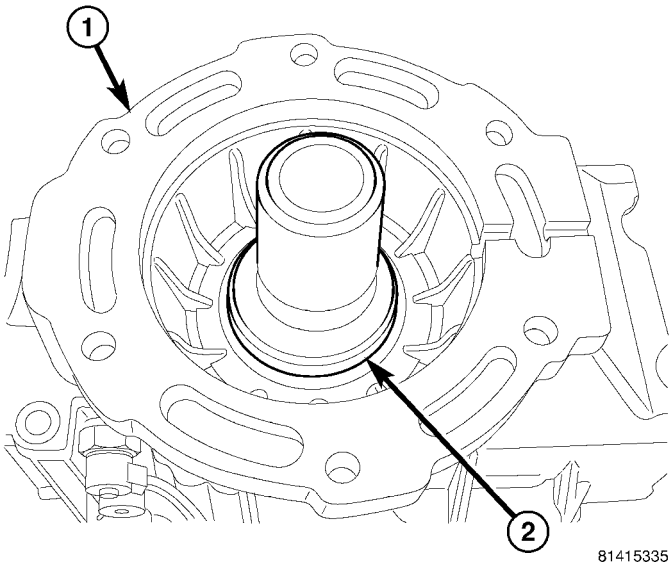


Fig. 91 OUTPUT SHAFT SEAL 4X4

1 - HOUSING
2 - INSTALLER

(10) Install 4x4 output shaft seal into rear housing (1) with Installer 9638 (2) (Fig. 91). Install 4x2 output shaft seal into rear housing with Installer 9635 and Tube 6448A.

INPUT SHAFT

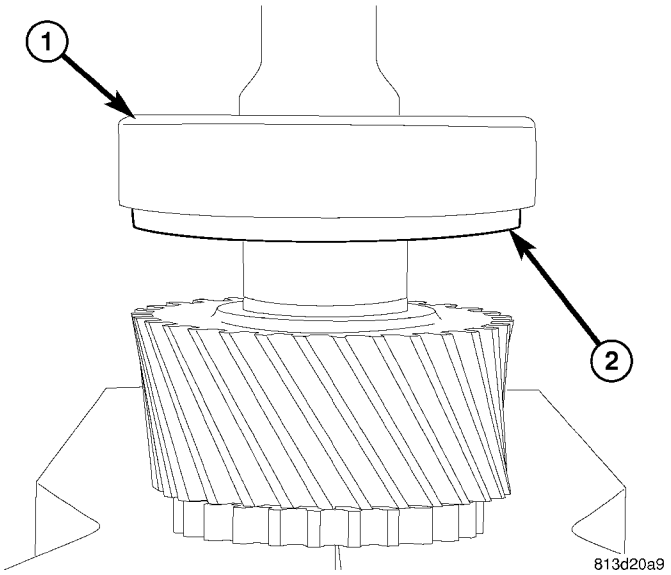


Fig. 92 INPUT SHAFT BEARING LIP

1 - INPUT SHAFT BEARING
2 - BEARING RETAINER LIP

(1) Install input shaft bearing (1) (Fig. 92) on input shaft with the bearing retainer lip (2) facing fifth gear.

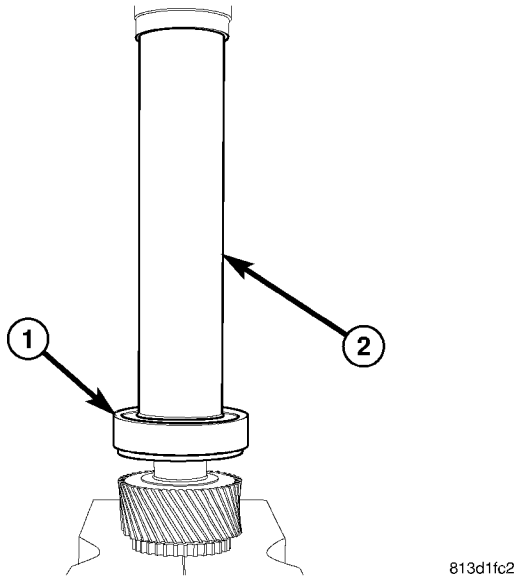


Fig. 93 INPUT SHAFT BEARING INSTALLER

1 - BEARING
2 - INSTALLER

(2) Install input shaft bearing (1) (Fig. 93) with Installer 6448A (2) and a press.

MANUAL TRANSMISSION - NSG370 (Continued)

FRONT HOUSING

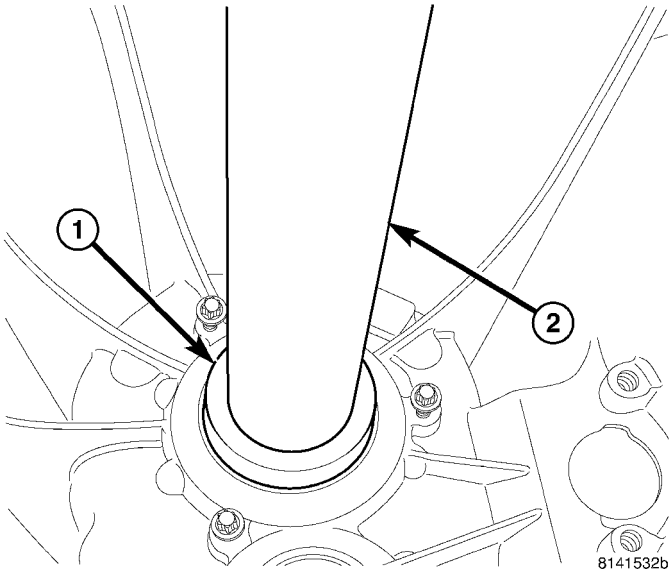


Fig. 94 INPUT SHAFT SEAL INSTALLER

1 - INSTALLER
2 - TUBE

(1) Install input shaft seal (Fig. 94) with Installer 9635 (1) tube 6448A (2).

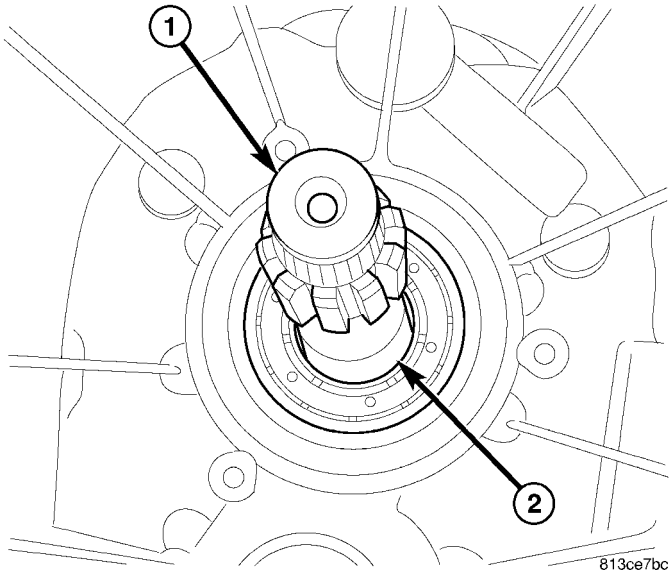


Fig. 95 INPUT SHAFT IN HOUSING

1 - INPUT SHAFT
2 - BEARING

(2) Install input shaft (1) (Fig. 95) with bearing (2) into housing.

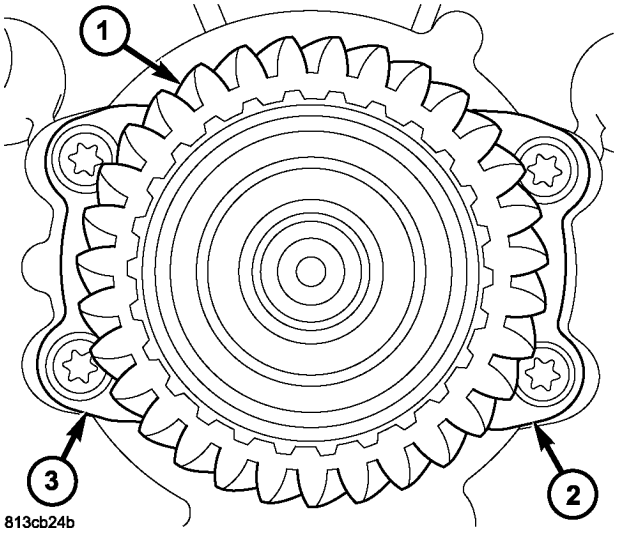


Fig. 96 INPUT SHAFT BEARING RETAINER

1 - INPUT SHAFT
2 - BEARING RETAINER
3 - BEARING RETAINER

(3) Install input shaft (1) (Fig. 96) bearing retainers (2, 3) and tighten bolts to 10 N·m (88 in. lbs.).

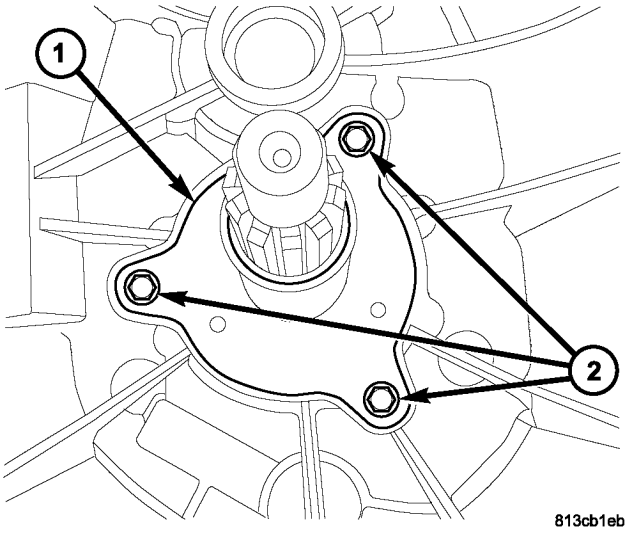


Fig. 97 INPUT SHAFT RETAINER

1 - INPUT SHAFT RETAINER
2 - BOLTS

(4) Install input shaft retainer (1) (Fig. 97) and tighten bolts (2) to 9 N·m (80 in. lbs.).

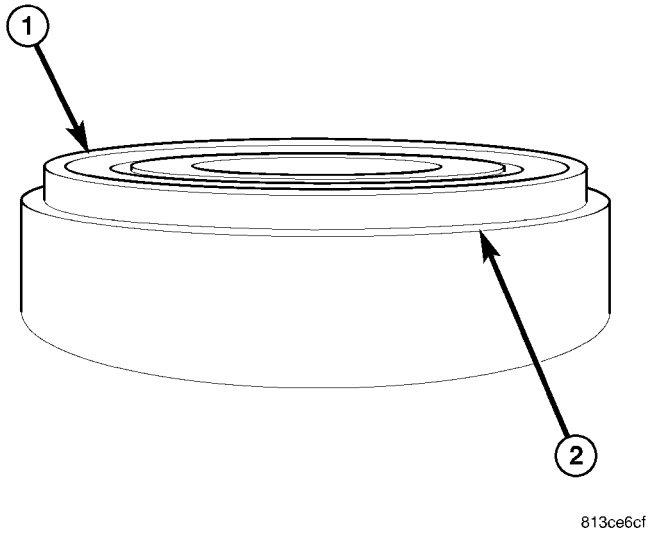


Fig. 98 BEARING RETAINER LIP

1 - COUNTERSHAFT BEARING
2 - BEARING RETAINER LIP

(5) Install countershaft bearing (1) (Fig. 98) into housing with bearing retainer lip (2) facing up.

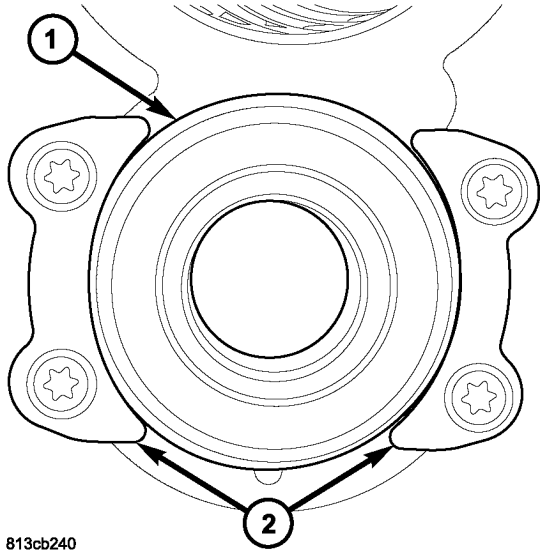


Fig. 99 COUNTERSHAFT BEARING RETAINER

1 - COUNTERSHAFT BEARING
2 - RETAINERS

(6) Install countershaft bearing (1) (Fig. 99) retainers (2) and tighten bolts to 10 N·m (88 in. lbs.).

FINAL ASSEMBLY

(1) Install mainshaft (1) onto Build Fixture 9633 (2) (Fig. 100).

(2) Install countershaft (1) onto Build Fixture 9633 (2) (Fig. 101).

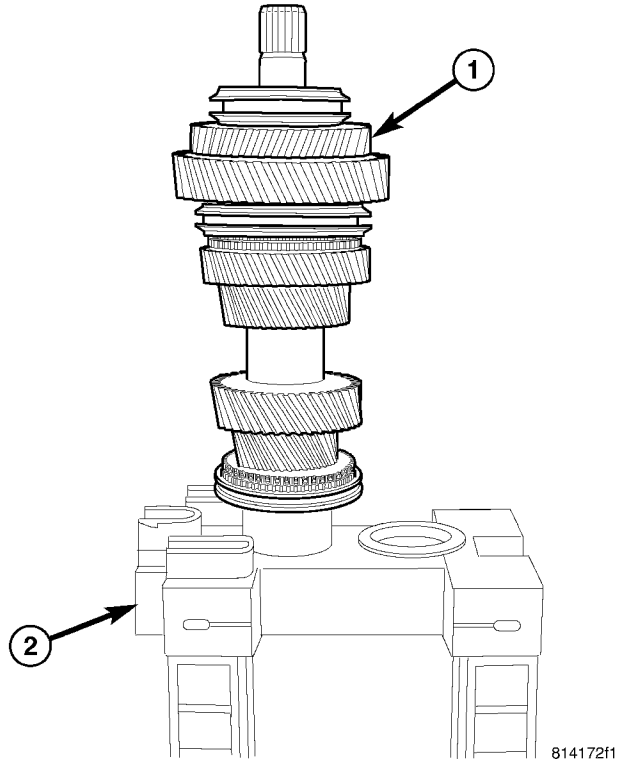


Fig. 100 MAINSHAFT IN FIXTURE

1 - MAINSHAFT
2 - FIXTURE

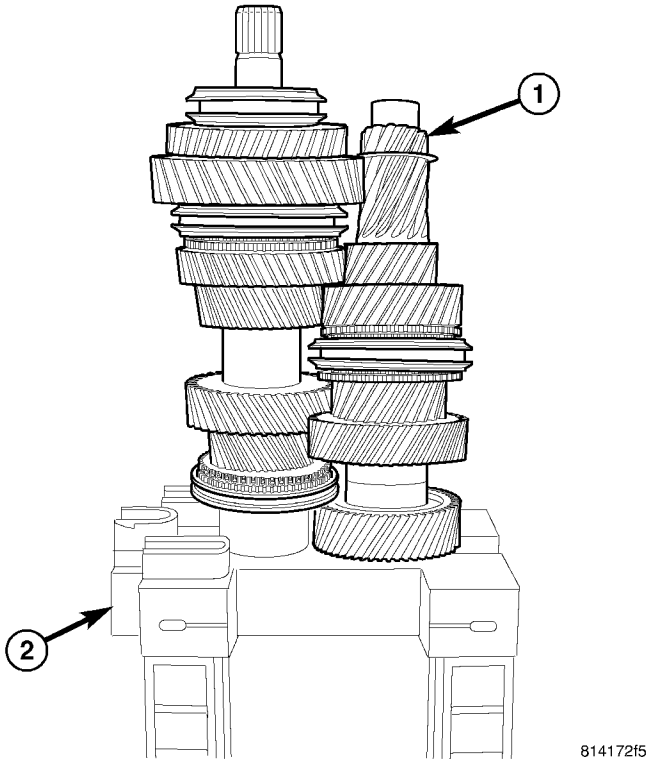


Fig. 101 COUNTERSHAFT IN FIXTURE

1 - COUNTERSHAFT
2 - FIXTURE

MANUAL TRANSMISSION - NSG370 (Continued)

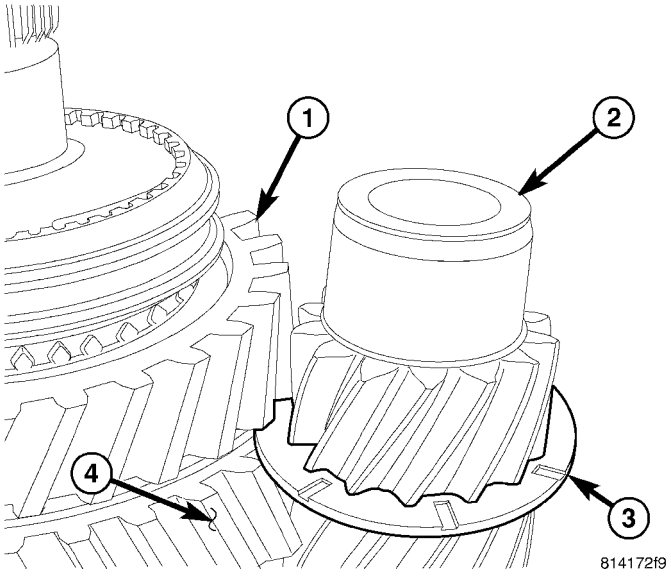


Fig. 102 OIL SILINGER

1 - REVERSE GEAR
2 - COUNTERSHAFT
3 - OIL SLINGER
4 - FIRST GEAR

(3) Install oil slinger (3) on countershaft (2) between reverse (1) and first gear (4) (Fig. 102).

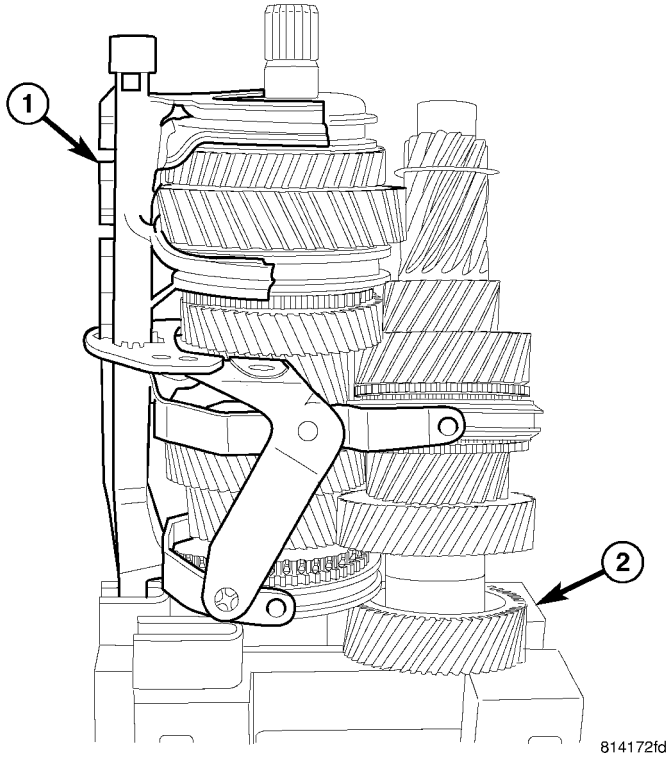


Fig. 103 SHIFT RAILS/FORKS IN FIXTURE

1 - SHIFT RAILS/FORKS
2 - FIXTURE

(4) Install shift rails and forks (1) onto Build Fixture 9633 (2) and countershaft/mainshaft. (Fig. 103).

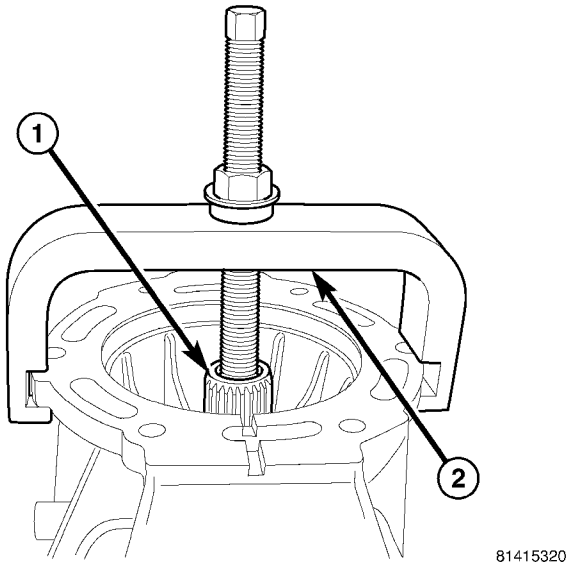


Fig. 104 MAINSHAFT INSTALLER

1 - MAINSHAFT
2 - REMOVER/INSTALLER

(5) Install rear housing on mainshaft (1) (Fig. 104) with Installer 9636 (2).

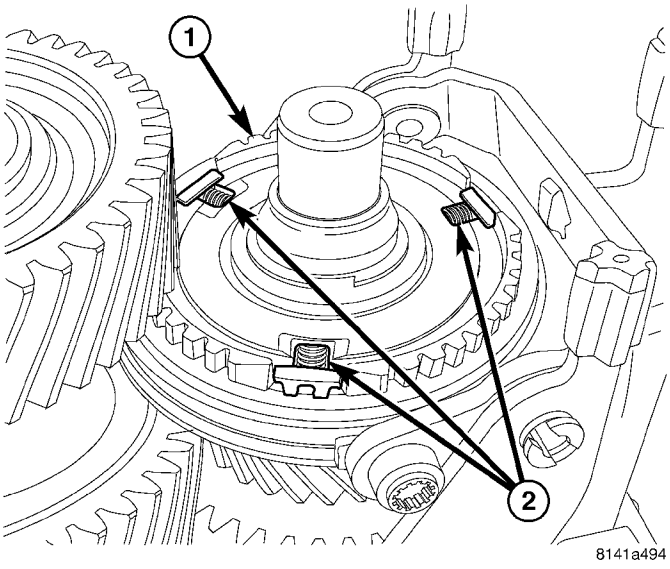


Fig. 105 5-6 SYNCRO DETENTS

1 - HUB
2 - DETENTS

(6) Remove rear housing with mainshaft, countershaft and shift rails from fixture. Set assembly on rear housing with shafts pointing up.

(7) Install 5-6 synchronizer hub (1) (Fig. 105) detents, springs, and balls (2).

MANUAL TRANSMISSION - NSG370 (Continued)

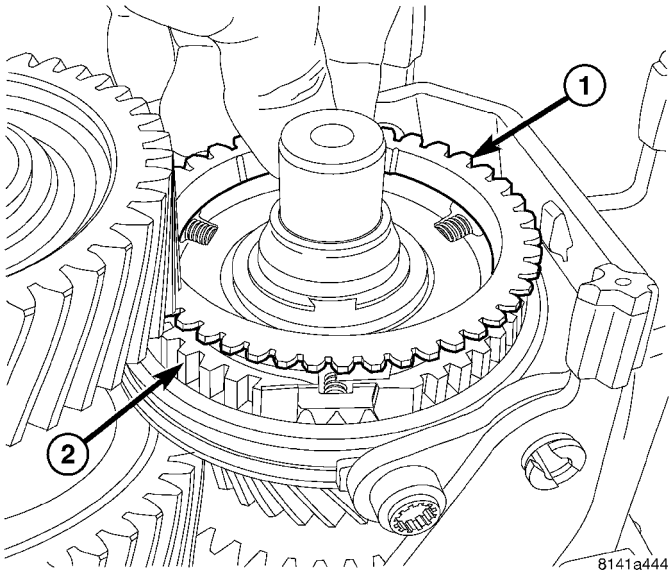


Fig. 106 FIFTH GEAR FRICTION RING

1 - FRICTION RING
2 - HUB

(8) Install fifth gear synchronizer friction ring (1) (Fig. 106) on 5-6 synchronizer hub (2).

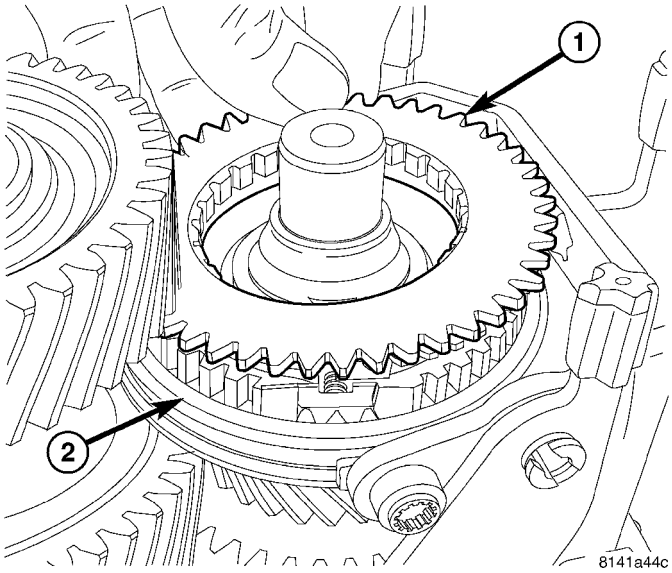


Fig. 107 FIFTH GEAR BLOCKER RING

1 - BLOCKER RING
2 - HUB

(9) Install fifth gear (Fig. 107) blocker ring (1) on 5-6 synchronizer hub (2). Then hold blocker ring and center synchronizer sleeve.

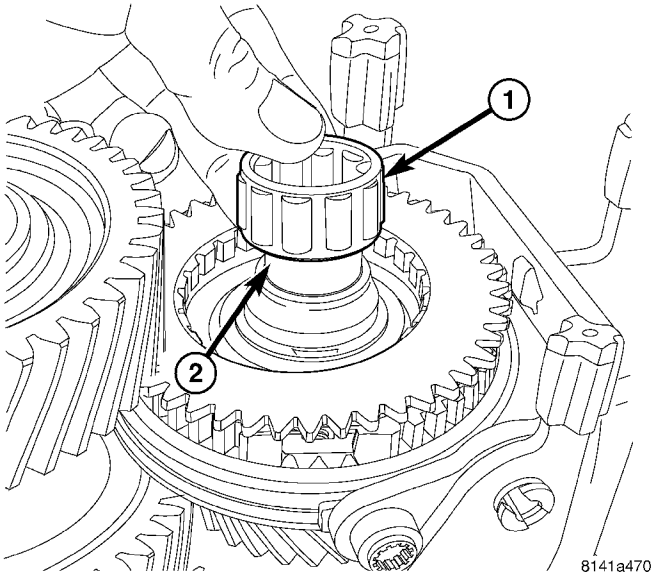


Fig. 108 INPUT SHAFT ROLLER BEARING

1 - ROLLER BEARING
2 - MAINSHAFT

(10) Install input shaft (Fig. 108) roller bearing (1) on mainshaft (2)..

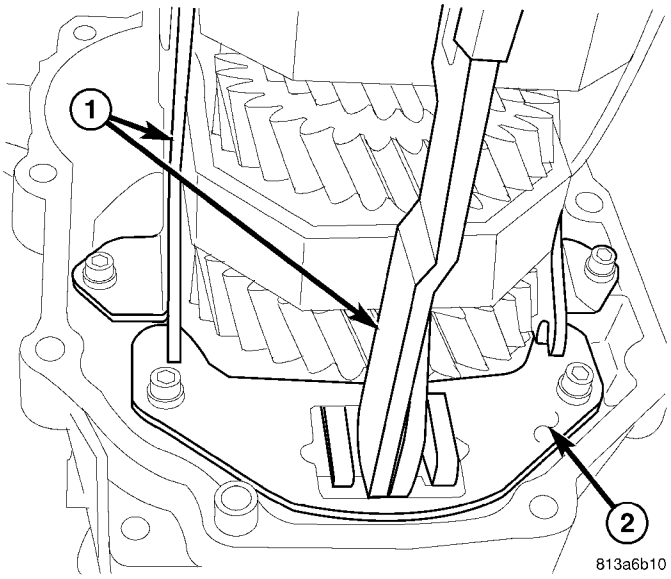


Fig. 109 SHIFT RAIL SUPPORT PLATE

1 - SHIFT RAILS
2 - SUPPORT PLATE

(11) Install shift rail (1) (Fig. 109) support plate (2) bolts and tighten to 8 N·m (71 in. lbs.).

MANUAL TRANSMISSION - NSG370 (Continued)

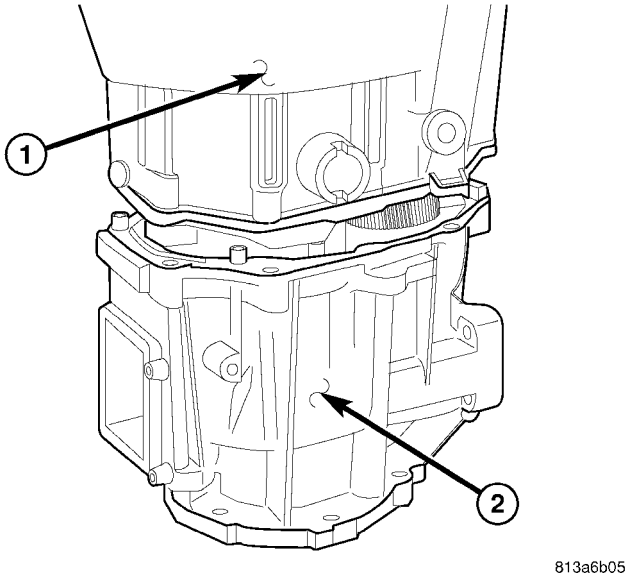


Fig. 110 SPLIT HOUSING

1 - FRONT HOUSING
2 - REAR HOUSING

(12) Apply MOPAR® Gasket Maker to front housing.

(13) Install front housing (1) (Fig. 110) on rear housing (2) and tighten bolts to 28 N·m (21 ft. lbs.).

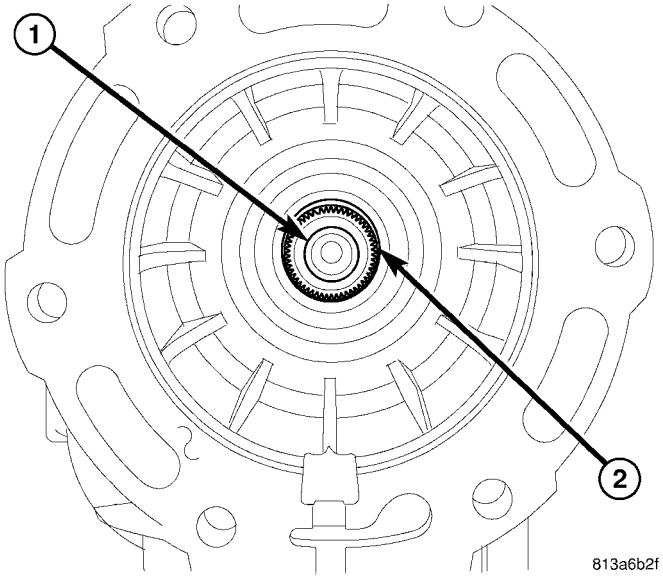


Fig. 111 OUTPUT SHAFT SNAP RING

1 - OUTPUT SHAFT
2 - SNAP RING

(14) Install output shaft (1) (Fig. 111) snap ring (2) 4x4 only.

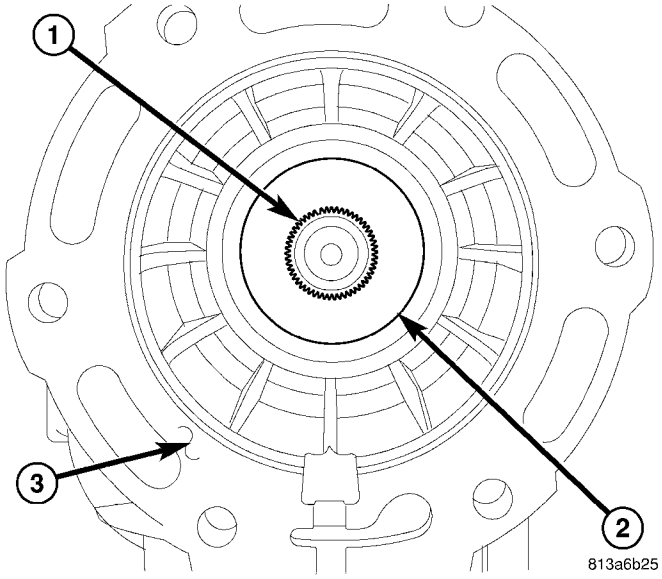


Fig. 112 OUTPUT SHAFT SEAL

1 - OUTPUT SHAFT
2 - SEAL
3 - REAR HOUSING

(15) Install output shaft (1) (Fig. 112) seal (2) in rear housing with Installer 9638.

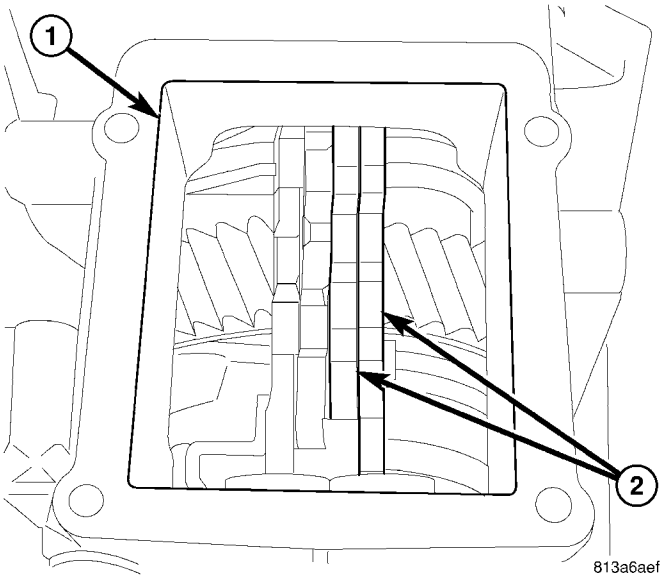


Fig. 113 SHIFT TOWER OPENING

1 - SHIFT TOWER OPENING
2 - SHIFT RAILS

(16) Through the shift tower opening (1) (Fig. 113) move two shift rails (2) forward to lock the transmission in two gear.

MANUAL TRANSMISSION - NSG370 (Continued)

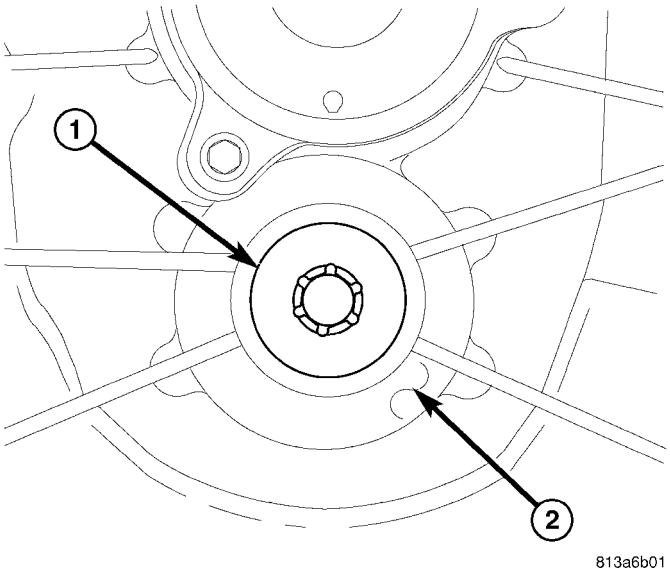


Fig. 114 COUNTERSHAFT BOLT

1 - COUNTERSHAFT BOLT
2 - FRONT HOUSING

(17) Install countershaft bolt (1) (Fig. 114) into the front housing (2) and tighten to 100 N·m (74 ft. lbs.).

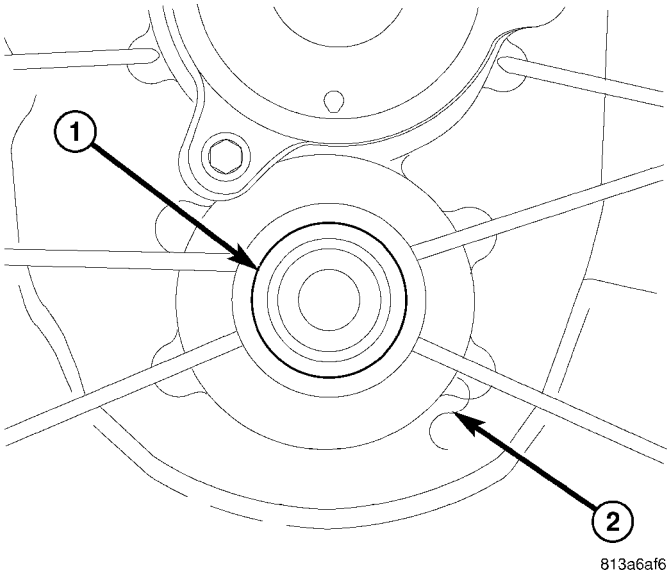


Fig. 115 COUNTERSHAFT PLUG

1 - COUNTERSHAFT PLUG
2 - FRONT HOUSING

(18) Install countershaft plug (1) (Fig. 115) in front housing (2) with Installer 7829-A and Handle C-4171.

(19) Move shift rails to neutral.

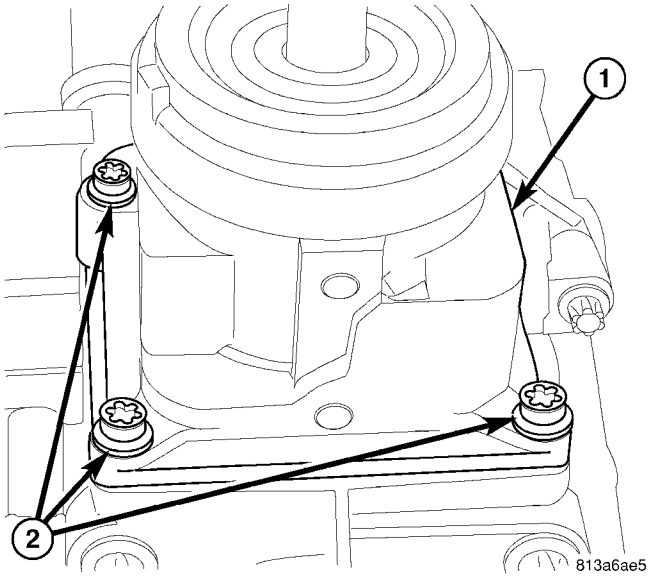


Fig. 116 SHIFT TOWER

1 - SHIFT TOWER
2 - BOLTS

(20) Apply MOPAR® Gasket Maker to shift tower.

(21) Install shift tower (1) (Fig. 116) and tighten bolts (2) to 14 N·m (10 ft. lbs.).

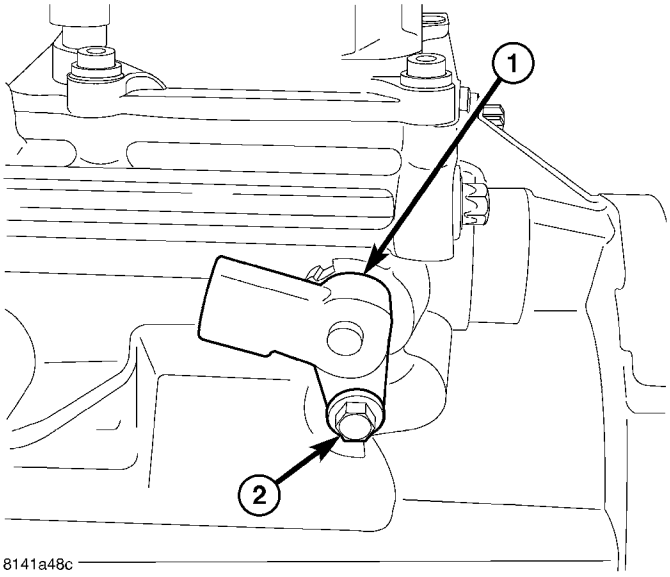


Fig. 117 BACK-UP LAMP SWITCH

1 - SWITCH
2 - BOLTS

(22) Install back up lamp switch (1) and bolt (2) (Fig. 117).

MANUAL TRANSMISSION - NSG370 (Continued)

INSTALLATION

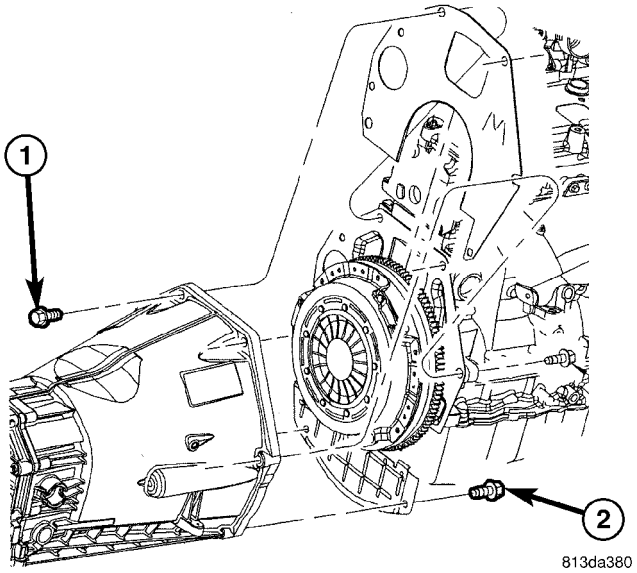


Fig. 118 2.4L ENGINE

1. TRANSMISSION BOLT
2. DUST SHIELD BOLT

(1) Install transmission on engine.

(2) On 2.4L engine tighten bolts (1) to 75 N·m (55 ft. lbs.). Install lower dust shield and tighten bolts (2) to 34 N·m (25 ft. lbs.) (Fig. 118).

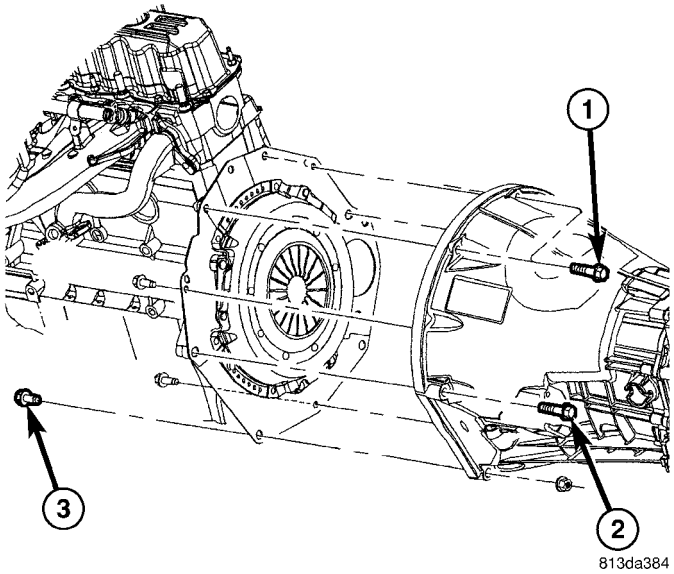


Fig. 119 4.0L ENGINE

1. UPPER TRANSMISSION BOLTS
2. SIDE DUST COVER BOLTS
3. LOWER DUST COVER BOLTS

(3) On 4.0L engine tighten bolts (1) to 37 N·m (27 ft. lbs.). Install side and lower dust shield and tighten bolts (2) (3) to 58 N·m (43 ft. lbs.) (Fig. 119).

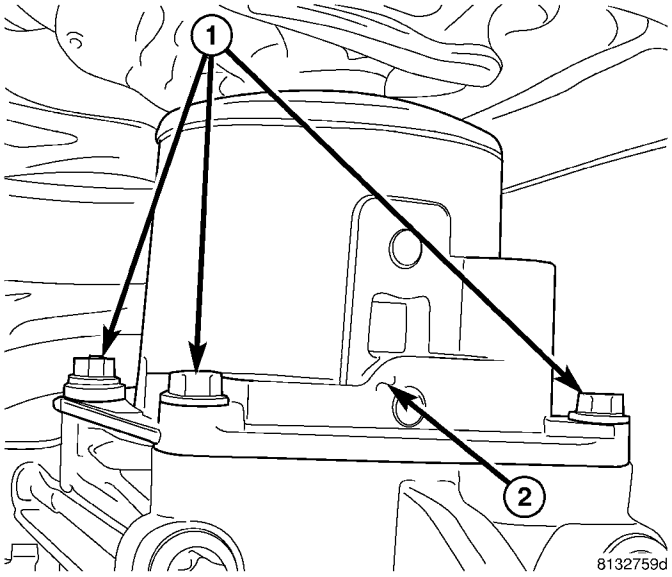


Fig. 120 SHIFT LEVER

1. BOLTS
2. SHIFT TOWER

(4) Clean shift tower and mating surface then apply Mopar® Gasket Maker to shift tower.

(5) Install shift tower (2) and tighten bolts (1) to 10 N·m (7 ft. lbs.) (Fig. 120).

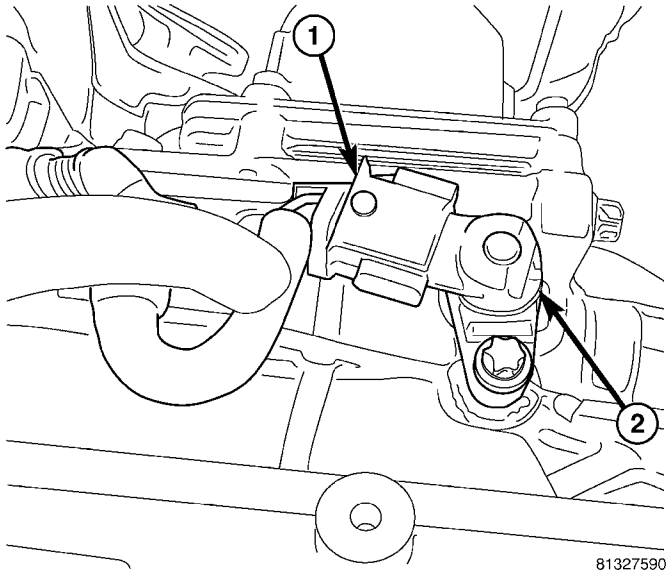


Fig. 121 BACKUP LAMP SWITCH

1. CONNECTOR
2. BACKUP LAMP SWITCH

(6) Install backup lamp (2) wiring connector (1) (Fig. 121).

MANUAL TRANSMISSION - NSG370 (Continued)

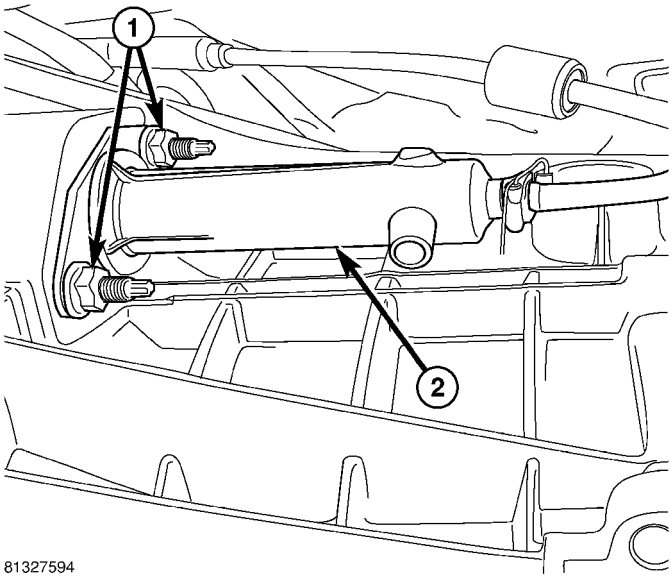


Fig. 122 CLUTCH SLAVE CYLINDER

1. BOLTS
2. SLAVE CYLINDER

(7) Install clutch slave cylinder (2) and mounting nuts (1) (Fig. 122).

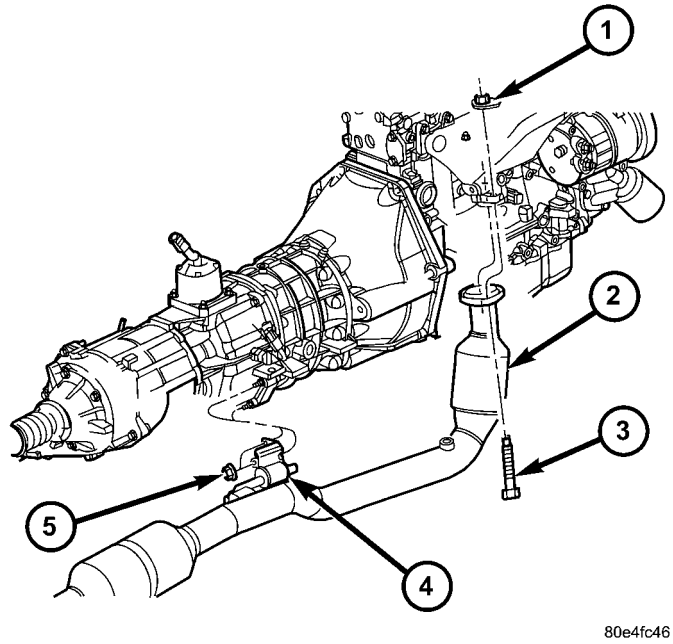


Fig. 123 2.4L CATALYST ASSEMBLY - 4x4

1- FLANGED NUT
2- CATALYST ASSEMBLY
3- BOLT
4- HANGER
5- NUT

(8) Install catalyst assembly (2) on 2.4L engine (Fig. 123).

(9) Install transmission crossmember and tighten bolts to 47 N·m (35 ft. lbs.). Install transmission mount bolts and tighten to 47 N·m (35 ft. lbs.).

(10) Install transfer case, if equipped.

(11) Install transfer case shift linkage.

(12) Install transfer case wiring connector and vent hose.

(13) Install propeller shaft/shafts with reference marks aligned.

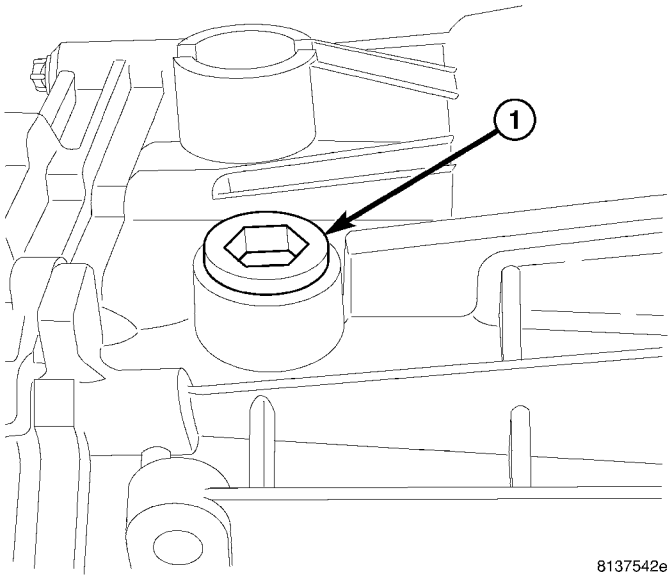


Fig. 124 FILL PLUG

1. FILL PLUG

(14) Remove fill plug (1) and fill transmission withn lubricant (Fig. 124).

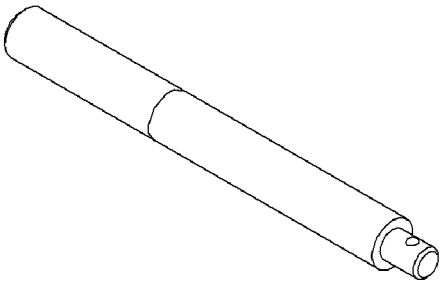
MANUAL TRANSMISSION - NSG370 (Continued)

SPECIFICATIONS - NSG370

TORQUE SPECIFICATIONS

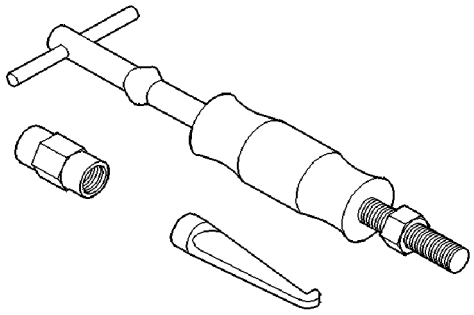
DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Drain Plug	30	22	-
Fill Plug	30	22	-
Shift Tower Bolts	14	10	-
Housing Bolts	28	20	-
Transmission Mounting Bolts	38	28	-
Input Shaft Retainer Bolts	9	-	80
Bearing Retainer Bolts	10	-	88
Shift Rail Support Bolts	8	-	71
Idler Gear Shaft Bolt	20	15	-
Countershaft Bolt	100	74	-

SPECIAL TOOLS



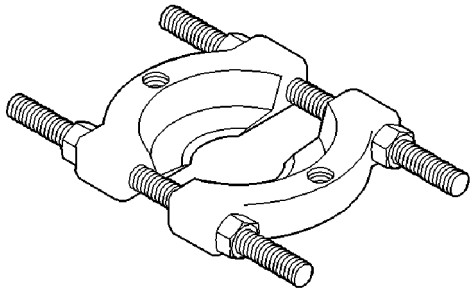
HANDLE C-4171

HUB INSTALLER W-262

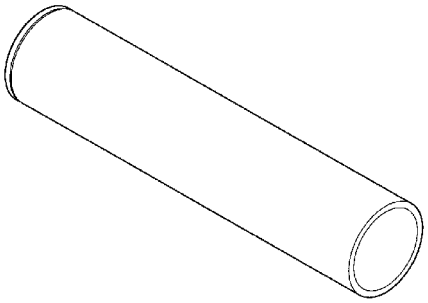


SLIDE HAMMER C-637

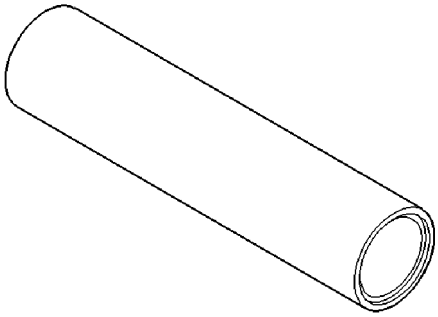
MANUAL TRANSMISSION - NSG370 (Continued)



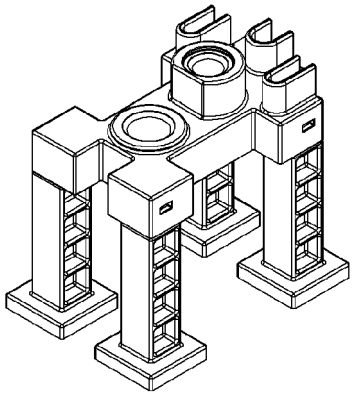
BEARING SPLITTER 1130



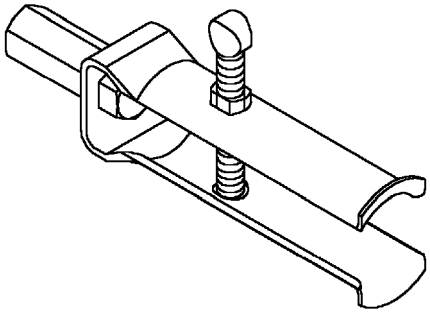
HUB INSTALLER 8228



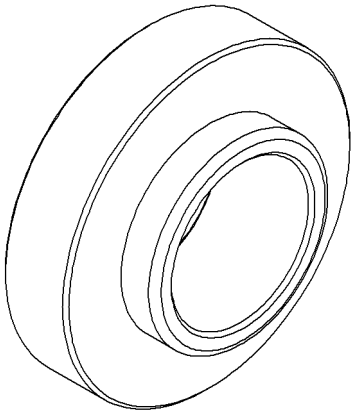
BEARING INSTALLER 6448A



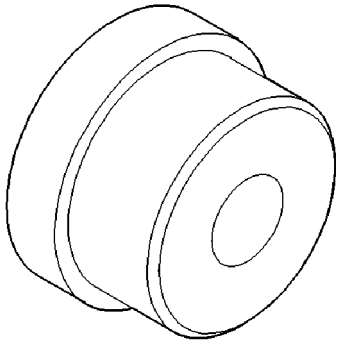
BUILD FIXTURE 9633



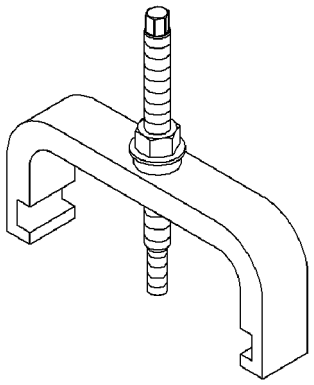
BEARING REMOVER 7794-A



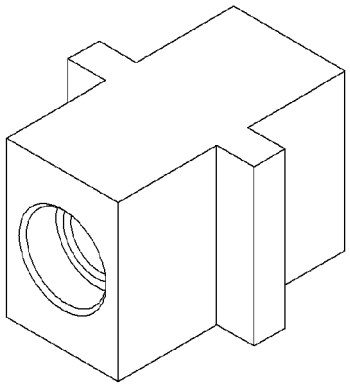
SEAL INSTALLER 9635



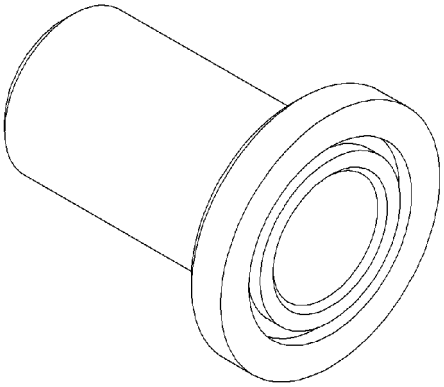
PLUG INSTALLER 7829-A



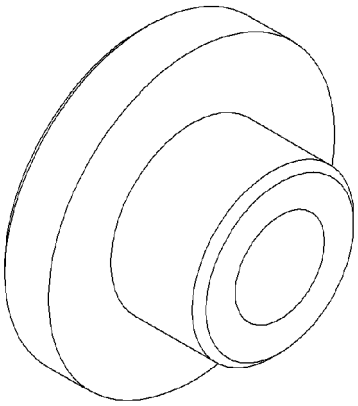
SHAFT REMOVER/INSTALLER 9636



MAINSHAFT BUILD FIXTURE 9648



SEAL INSTALLER 9638



BEARING INSTALLER 9643

AUTOMATIC TRANSMISSION - 42RLE

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AUTOMATIC TRANSMISSION - 42RLE

DESCRIPTION

The 42RLE (Fig. 1) is a four-speed transmission that is a conventional hydraulic/mechanical assembly controlled with adaptive electronic controls and monitors. The hydraulic system of the transmission consists of the transmission fluid, fluid passages, hydraulic valves, and various line pressure control components. An input clutch assembly which houses the underdrive, overdrive, and reverse clutches is used. It also utilizes separate holding clutches: 2nd/4th gear and Low/Reverse. The primary mechanical components of the transmission consist of the following:

- Three multiple disc input clutches
- Two multiple disc holding clutches
- Four hydraulic accumulators
- Two planetary gear sets
- Hydraulic oil pump
- Valve body
- Solenoid/Pressure switch assembly

Control of the transmission is accomplished by fully adaptive electronics. Optimum shift scheduling is accomplished through continuous real-time sensor feedback information provided to the Transmission Control Module (TCM) portion of the Powertrain Control Module (PCM).

The TCM is the heart of the electronic control system and relies on information from various direct and indirect inputs (sensors, switches, etc.) to determine driver demand and vehicle operating conditions. With this information, the TCM can calculate and perform timely and quality shifts through various output or control devices (solenoid pack, transmission control relay, etc.).

The TCM also performs certain self-diagnostic functions and provides comprehensive information (sensor data, DTC's, etc.) which is helpful in proper diagnosis and repair. This information can be viewed with the DRB® scan tool.

AUTOMATIC TRANSMISSION - 42RLE (Continued)

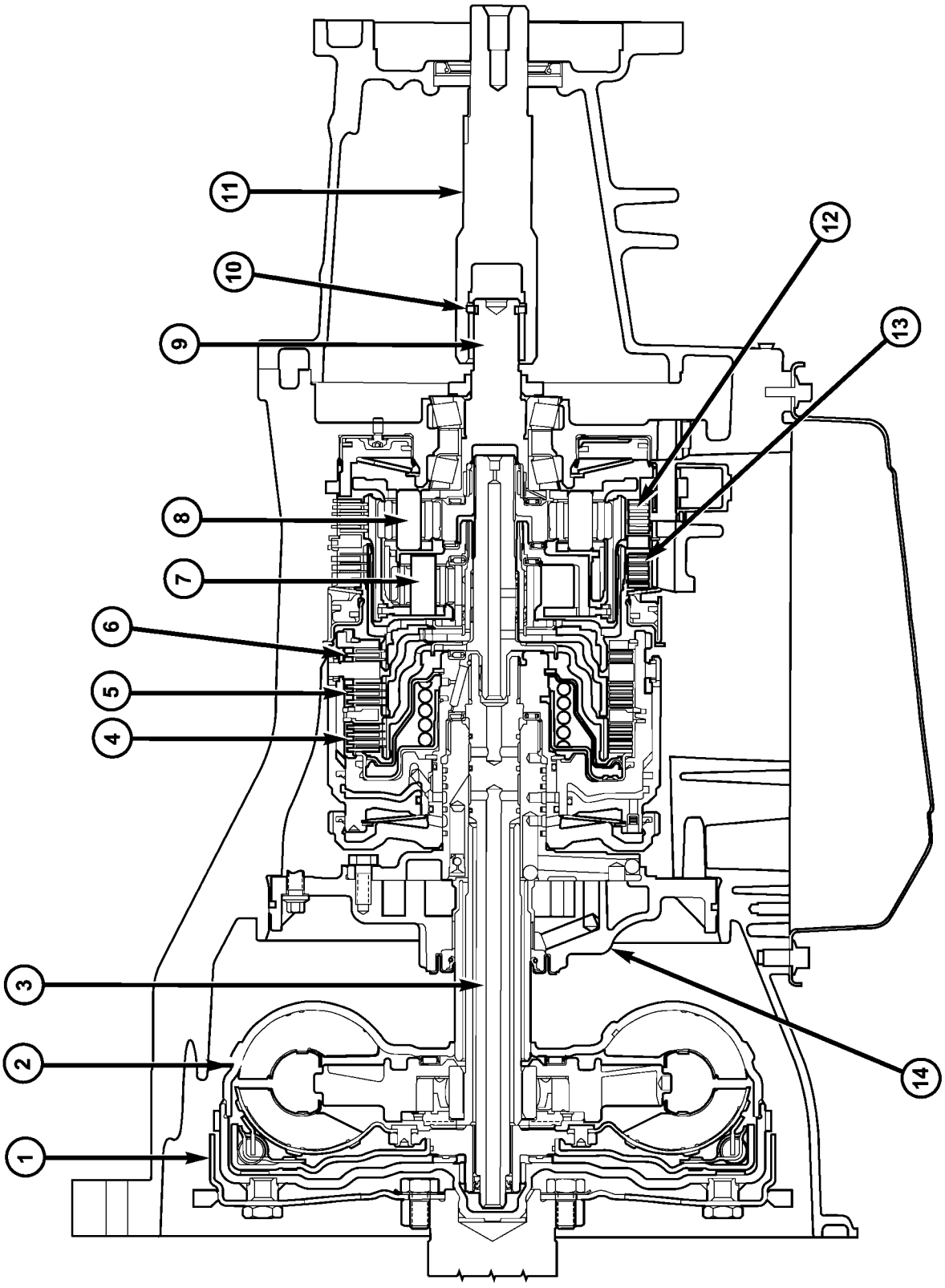

Fig. 1 42RLE Automatic Transmission

1 - DRIVEPLATE
2 - TORQUE CONVERTER
3 - INPUT SHAFT
4 - UNDERDRIVE CLUTCH
5 - OVERDRIVE CLUTCH
6 - REVERSE CLUTCH
7 - FRONT PLANET CARRIER
8 - REAR PLANET CARRIER
9 - OUTPUT SHAFT
10 - SNAP RING
11 - STUB SHAFT
12 - LOW/REVERSE CLUTCH
13 - 2/4 CLUTCH
14 - OIL PUMP

TRANSMISSION IDENTIFICATION

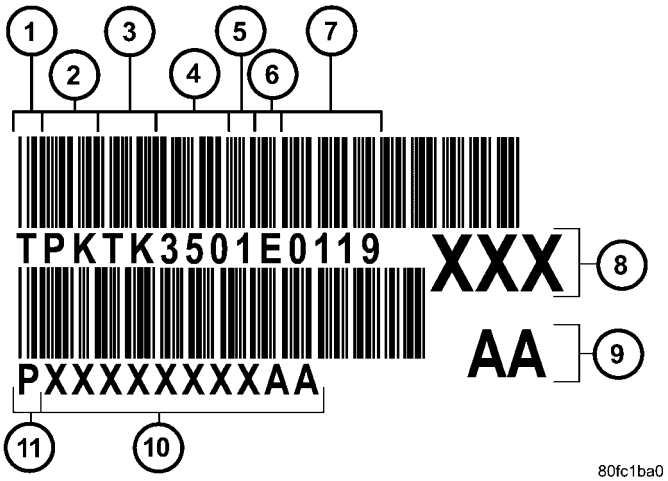

Fig. 2 Identification Label Breakdown

1 - T=TRACEABILITY
2 - SUPPLIER CODE (PK=KOKOMO)
3 - COMPONENT CODE (TK=KOKOMO TRANSMISSION)
4 - BUILD DAY (350=DEC. 15)
5 - BUILD YEAR (1=2001)
6 - ASSEMBLY LINE CODE
7 - BUILD SEQUENCE NUMBER
8 - LAST THREE OF P/N
9 - CHANGE LEVEL
10 - TRANSMISSION PART NUMBER
11 - P=PART NUMBER

The 42RLE transmission can be identified by a barcode label that is affixed to the upper left area of the bellhousing.

The label contains a series of digits that can be translated into useful information such as transmission part number (10), date of manufacture (4, 5), manufacturing origin (2), assembly line identifier (6), build sequence number (7), etc. (Fig. 2).

If the tag is not legible or is missing, the “PK” number, which is stamped into the left rear flange of the transmission case, can be referred to for identification. The entire part number, build code, and sequence number are stamped into the flange.

OPERATION

The 42RLE transmission ratios are:

First	2.84 : 1
Second	1.57 : 1
Third	1.00 : 1
Overdrive	0.69 : 1
Reverse	2.21 : 1

AUTOMATIC TRANSMISSION - 42RLE (Continued)

FIRST GEAR POWERFLOW

In first gear range, torque input is through the underdrive clutch (1) (Fig. 3) to the underdrive hub assembly. The underdrive hub is splined to the rear sun gear. When the underdrive clutch is applied, it rotates the underdrive hub and rear sun gear. The L/R clutch (2) is applied to hold the front carrier/rear annulus assembly. The rear sun gear drives the rear planetary pinion gears. The rear planetary pinion gears are forced to walk around the inside of the stationary rear annulus gear. The pinions are pinned to the rear carrier and cause the rear carrier assembly to rotate as they walk around the annulus gear. This provides the torque output for first gear. The other planetary gearset components are freewheeling. The first gear ratio is 2.84:1.

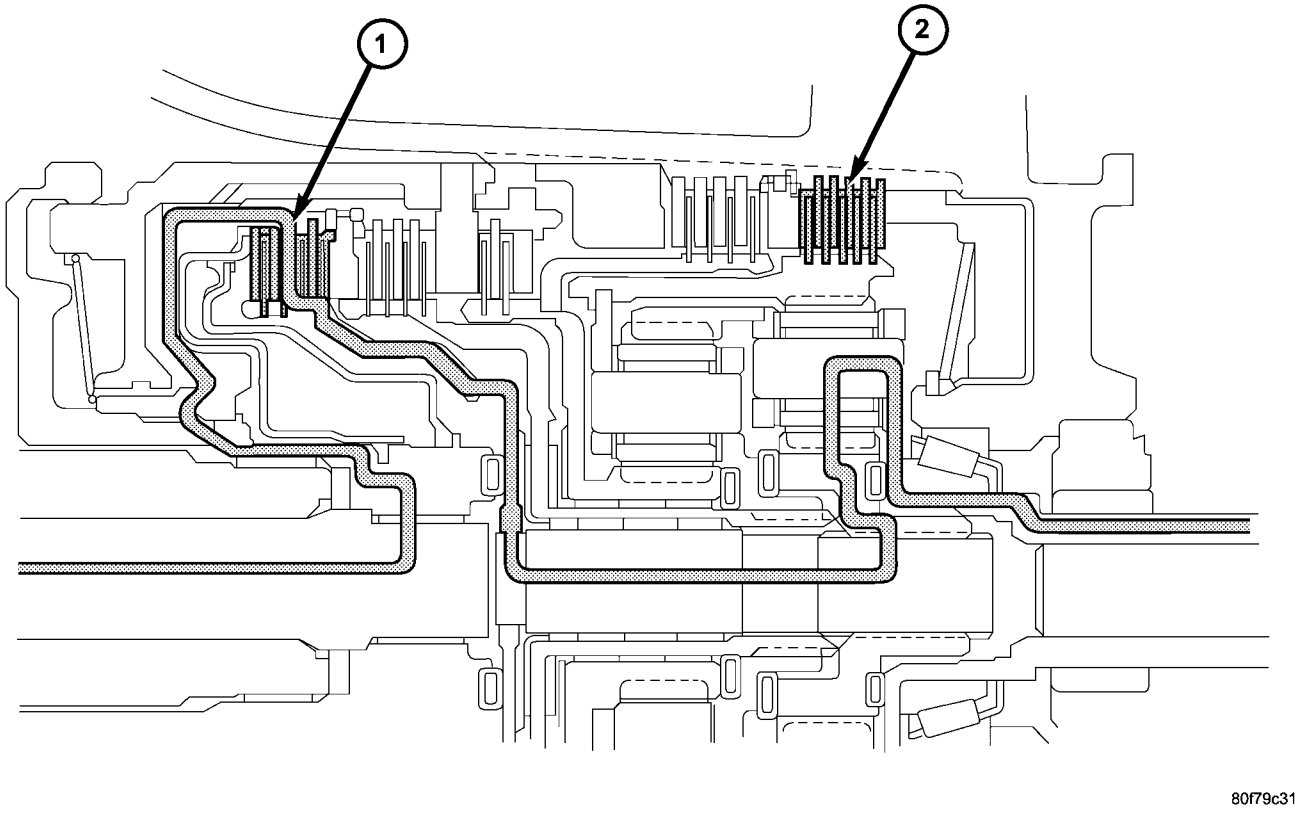

Fig. 3 First Gear Powerflow

1 - UNDERDRIVE CLUTCH APPLIED (Turns Rear Sun)
2 - LOW-REVERSE CLUTCH APPLIED (Holds Rear Annulus/Front Carrier)

AUTOMATIC TRANSMISSION - 42RLE (Continued)

SECOND GEAR POWERFLOW

Second gear is achieved by having both planetary gear sets (Fig. 4) contribute to torque multiplication. As in first gear, torque input is through the underdrive clutch (1) to the rear sun gear. The 2/4 clutch (2) is applied to hold the front sun gear stationary. The rotating rear sun gear turns the rear planetary pinions. The rear pinions rotate the rear annulus/front carrier assembly. The pinions of the front carrier walk around the stationary front sun gear. This transmits torque to the front annulus/rear carrier assembly, which provides output torque and a gear ratio of 1.57:1.

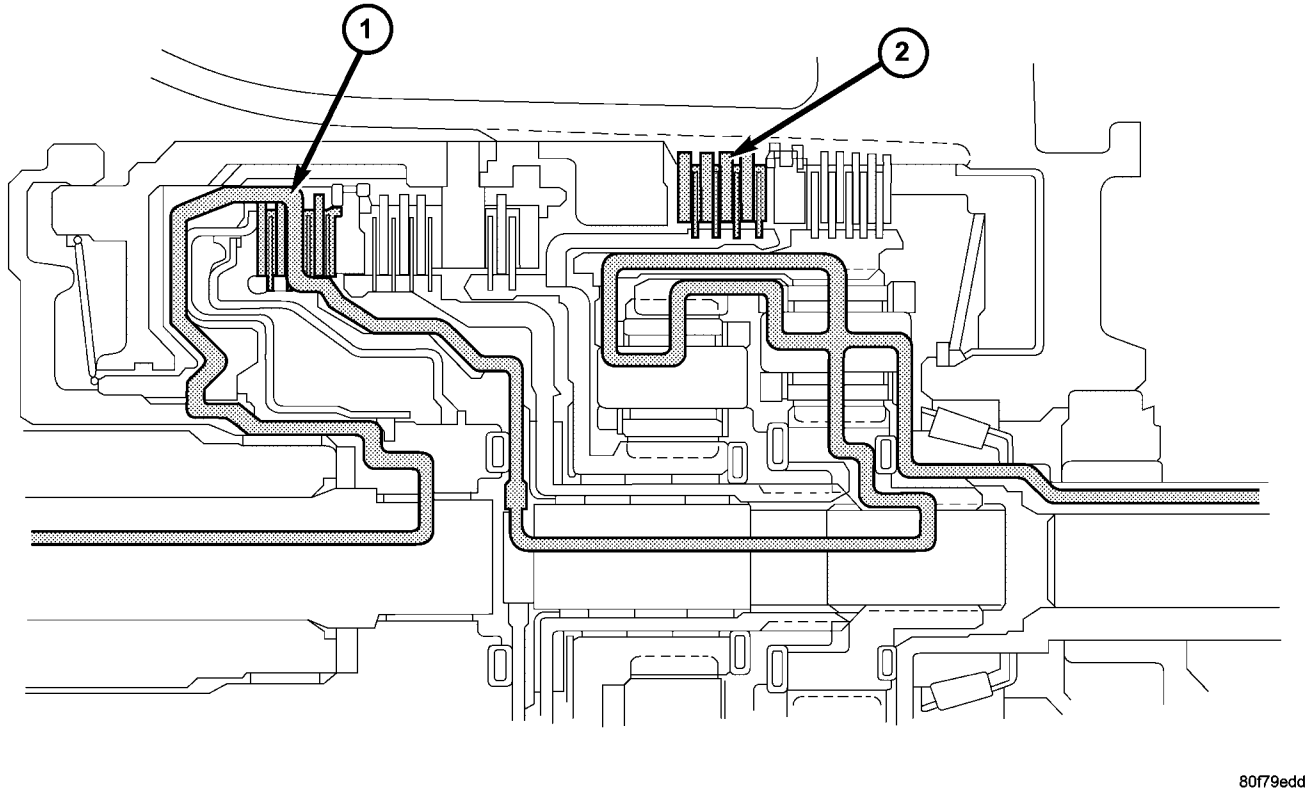

Fig. 4 Second Gear Powerflow

1 - UNDERDRIVE CLUTCH APPLIED (Turns Rear Sun)
2 - 2-4 CLUTCH APPLIED (Holds Front Sun)

AUTOMATIC TRANSMISSION - 42RLE (Continued)

THIRD GEAR POWERFLOW

In third gear, two input clutches are applied to provide torque input: the underdrive clutch (1) (Fig. 5) and overdrive clutch (2). The underdrive clutch rotates the rear sun gear, while the overdrive clutch rotates the front carrier/rear annulus assembly. The result is two components (rear sun gear and rear annulus gear) rotating at the same speed and in the same direction. This effectively locks the entire planetary gearset together and is rotated as one unit. The gear ratio in third is 1:1.

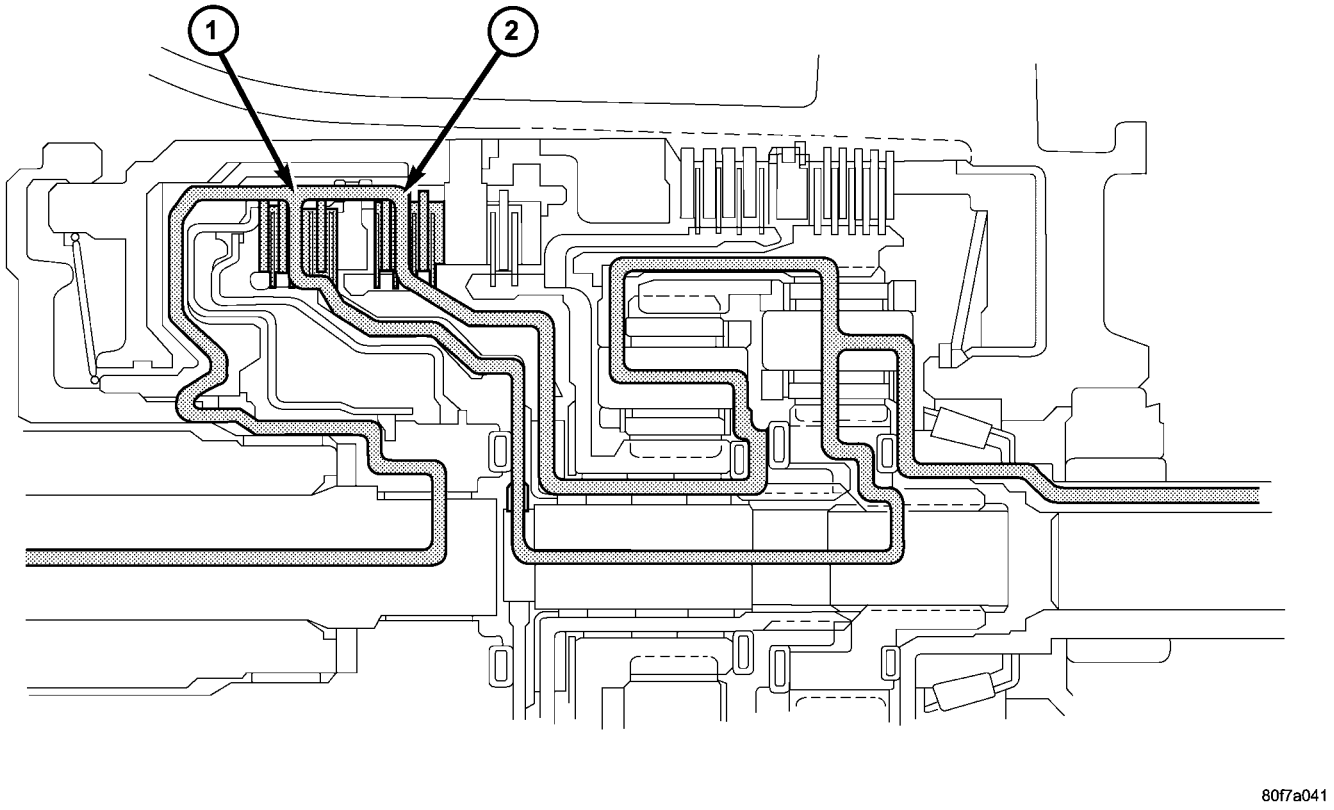

Fig. 5 Third Gear Powerflow

1 - UNDERDRIVE CLUTCH APPLIED (Turns Rear Sun)
2 - OVERDRIVE CLUTCH APPLIED (Turns Front Carrier/Rear Annulus)

FOURTH GEAR POWERFLOW

In fourth gear input torque is through the overdrive clutch (1) (Fig. 6) which drives the front carrier. The 2/4 clutch (2) is applied to hold the front sun gear. As the overdrive clutch rotates the front carrier, it causes the pinions of the front carrier to walk around the stationary front sun gear. This causes the front carrier pinions to turn the front annulus/rear carrier assembly which provides output torque. In fourth gear, transmission output speed is more than engine input speed. This situation is called overdrive and the gear ratio is 0.69:1.

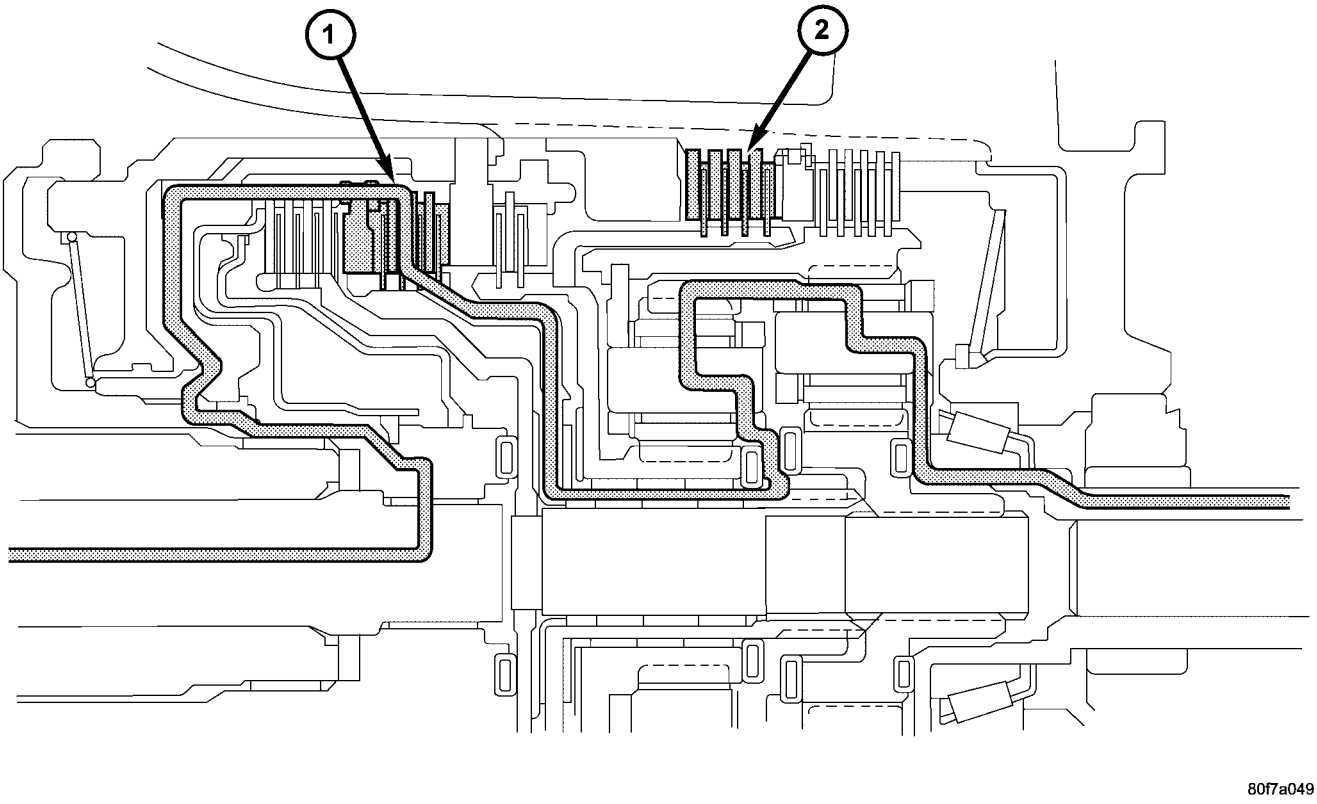

Fig. 6 Fourth Gear Powerflow

1 - OVERDRIVE CLUTCH APPLIED (Turns Rear Sun)
2 - 2-4 CLUTCH APPLIED (Holds Front Sun)

AUTOMATIC TRANSMISSION - 42RLE (Continued)

REVERSE GEAR POWERFLOW

In reverse, input (Fig. 7) power is through the reverse clutch (1). When applied, the reverse clutch drives the front sun gear through the overdrive hub and shaft. The L/R clutch (2) is applied to hold the front carrier/rear annulus assembly stationary. The front carrier is being held by the L/R clutch so the pinions are forced to rotate the front annulus/rear carrier assembly in the reverse direction. Output torque is provided, in reverse, with a gear ratio of 2.21:1.

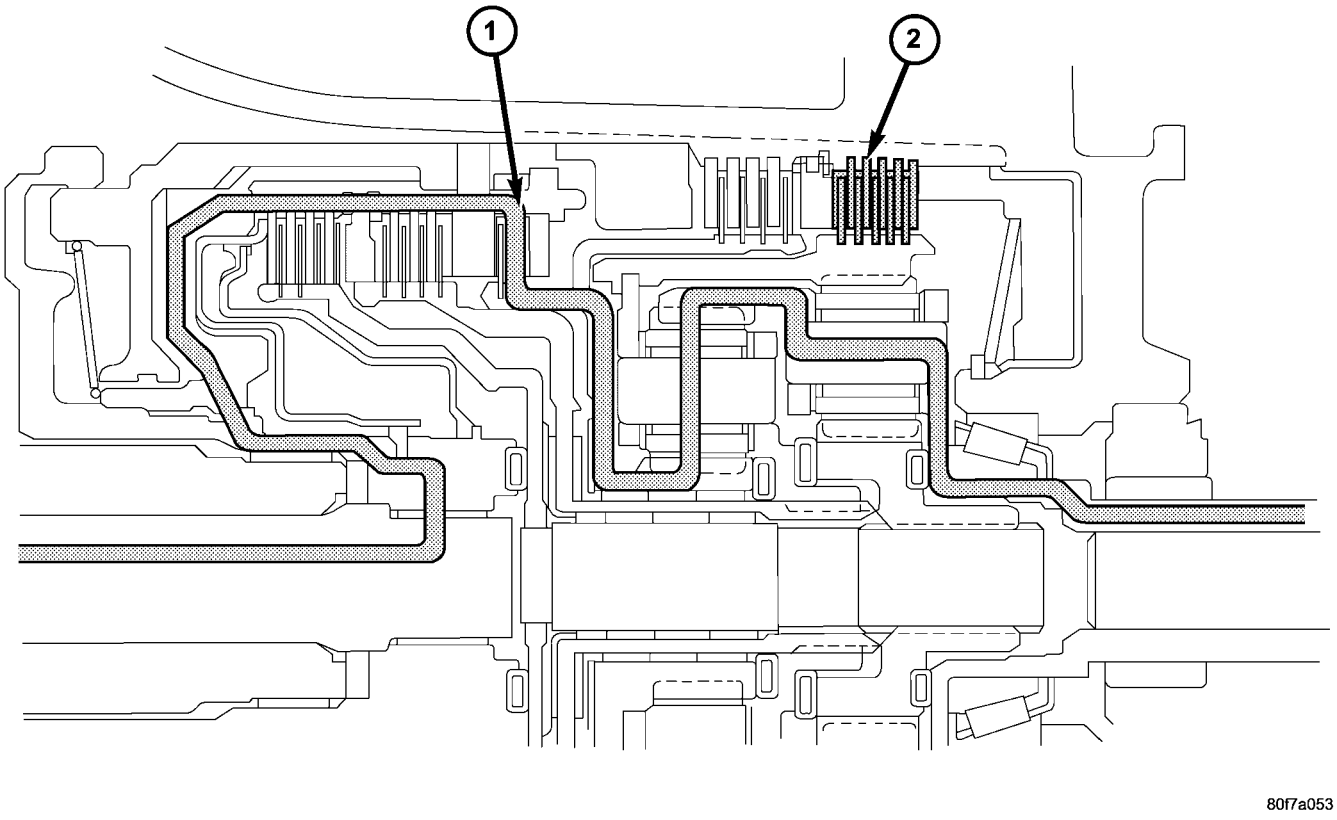

Fig. 7 Reverse Gear Powerflow

1 - LOW-REVERSE CLUTCH APPLIED (Holds Rear Annulus Front Carrier)
2 - REVERSE CLUTCH APPLIED (Turns Front Sun)

DIAGNOSIS AND TESTING

DIAGNOSIS AND TESTING - AUTOMATIC TRANSMISSION

CAUTION: Before attempting any repair on the 42RLE Four Speed Automatic Transmission, always check for proper shift cable adjustment. Also check for diagnostic trouble codes with the DRB® scan tool and the 42RLE Transmission Diagnostic Procedure Manual.

42RLE automatic transmission malfunctions may be caused by these general conditions:

- Poor engine performance
- Improper adjustments
- Hydraulic malfunctions
- Mechanical malfunctions
- Electronic malfunctions

When diagnosing a problem always begin with recording the complaint. The complaint should be defined as specific as possible. Include the following checks:

- Temperature at occurrence (cold, hot, both)
- Dynamic conditions (acceleration, deceleration, upshift, cornering)
- Elements in use when condition occurs (what gear is transmission in during condition)
- Road and weather conditions
- Any other useful diagnostic information.

After noting all conditions, check the easily accessible variables:

- Fluid level and condition
- Shift cable adjustment
- Diagnostic trouble code inspection

Then perform a road test to determine if the problem has been corrected or that more diagnosis is necessary. If the problem exists after the preliminary tests and corrections are completed, hydraulic pressure checks should be performed.

DIAGNOSIS AND TESTING - ROAD TEST

Prior to performing a road test, verify that the fluid level, fluid condition, and linkage adjustment have been approved.

During the road test, the transmission should be operated in each position to check for slipping and any variation in shifting.

If the vehicle operates properly at highway speeds, but has poor acceleration, the converter stator overrunning clutch may be slipping. If acceleration is normal, but high throttle opening is needed to maintain highway speeds, the converter stator clutch may have seized. Both of these stator defects require replacement of the torque converter and thorough transmission cleaning.

Slipping clutches can be isolated by comparing the “Elements in Use” chart with clutch operation encountered on a road test. This chart identifies which clutches are applied at each position of the selector lever.

A slipping clutch may also set a DTC and can be determined by operating the transmission in all selector positions.

AUTOMATIC TRANSMISSION - 42RLE (Continued)

ELEMENTS IN USE AT EACH POSITION OF SELECTOR LEVER

Shift Lever Position	INPUT CLUTCHES			HOLDING CLUTCHES	
	Underdrive	Overdrive	Reverse	2/4	Low/Reverse
P - PARK					X
R - REVERSE			X		X
N - NEUTRAL					X
OD - OVERDRIVE					
First	X				X
Second	X			X	
Direct	X	X			
Overdrive		X		X	
D - DRIVE*					
First	X				X
Second	X			X	
Direct	X	X			
L - LOW*					
First	X				X
Second	X			X	
Direct	X	X			
* Vehicle upshift and downshift speeds are increased when in these selector positions.					

The process of elimination can be used to detect any unit which slips and to confirm proper operation of good units. Road test analysis can diagnose slipping units, but the cause of the malfunction cannot be determined. Practically any condition can be caused by leaking hydraulic circuits or sticking valves.

DIAGNOSIS AND TESTING - HYDRAULIC PRESSURE TESTS

Pressure testing is a very important step in the diagnostic procedure. These tests usually reveal the cause of most transmission problems.

Before performing pressure tests, be certain that fluid level and condition, and shift cable adjustments have been checked and approved. Fluid must be at operating temperature (150 to 200 degrees F.).

Install an engine tachometer, raise vehicle on hoist which allows the wheels to turn, and position tachometer so it can be read.

Using special adapters L-4559, attach 300 psi gauge(s) C-3293SP to the port(s) required for test being conducted.

Test port locations are shown in the Pressure Taps graphic. (Fig. 8)

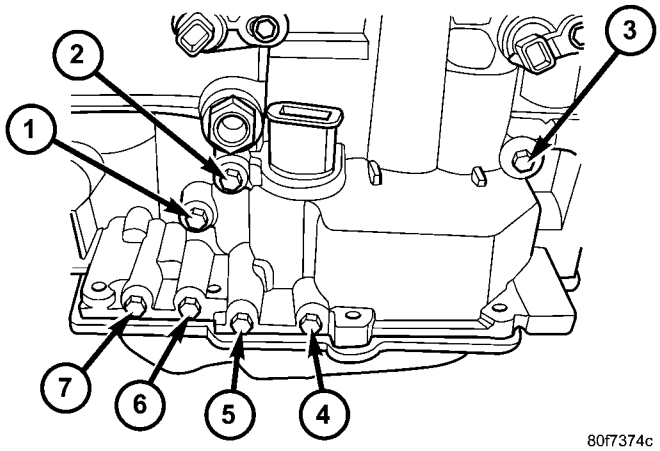

Fig. 8 Pressure Taps

1 - TORQUE CONVERTER CLUTCH OFF
2 - REVERSE
3 - LOW/REVERSE
4 - 2/4
5 - UNDERDRIVE
6 - TORQUE CONVERTER CLUTCH ON
7 - OVERDRIVE

TEST ONE - SELECTOR IN MANUAL 1 (1st Gear)

NOTE: This test checks pump output, pressure regulation and condition of the low/reverse clutch hydraulic circuit and shift schedule.

(1) Attach pressure gauge to the low/reverse clutch tap.
(2) Move selector lever to the MANUAL 1 position.
(3) Allow vehicle wheels to turn and increase throttle opening to achieve an indicated vehicle speed to 20 mph.
(4) Low/reverse clutch pressure should read 115 to 145 psi.

TEST TWO - SELECTOR IN MANUAL 2 (Second Gear)

NOTE: This test checks the underdrive clutch hydraulic circuit as well as the shift schedule.

(1) Attach gauge to the underdrive clutch tap.
(2) Move selector lever to the MANUAL 2 position.
(3) Allow vehicle wheels to turn and increase throttle opening to achieve an indicated vehicle speed of 30 mph.
(4) In second gear the underdrive clutch pressure should read 110 to 145 psi.

TEST TWO A - SELECTOR IN DRIVE (OD ON - Fourth Gear)

NOTE: This test checks the underdrive clutch hydraulic circuit as well as the shift schedule.

(1) Attach gauge to the underdrive clutch tap.
(2) Move selector lever to the DRIVE position. Verfy that the OD switch is ON.
(3) Allow wheels to rotate freely and increase throttle opening to achieve an indicated speed of 40 mph.
(4) Underdrive clutch pressure should read below 5 psi. If not, than either the solenoid assembly or controller is at fault.

TEST THREE - SELECTOR IN DRIVE (OD OFF - Third and Second Gear)

NOTE: This test checks the overdrive clutch hydraulic circuit as well as the shift schedule.

(1) Attach gauge to the overdrive clutch tap.
(2) Move selector lever to the DRIVE position.
(3) Allow vehicle wheels to turn and increase throttle opening to achieve an indicated vehicle speed of 20 mph.
(4) Overdrive clutch pressure should read 74 to 95 psi.
(5) Move selector lever to the DRIVE position and increase indicated vehicle speed to 30 mph.
(6) The vehicle should be in second gear and overdrive clutch pressure should be less than 5 psi.

TEST FOUR - SELECTOR IN DRIVE (OD ON - Fourth Gear)

NOTE: This test checks the 2/4 clutch hydraulic circuit.

(1) Attach gauge to the 2/4 clutch tap.
(2) Move selector lever to the DRIVE position.
(3) Allow vehicle front wheels to turn and increase throttle opening to achieve an indicated vehicle speed of 30 mph. Vehicle should be in fourth gear.
(4) The 2/4 clutch pressure should read 75 to 95 psi.

TEST FIVE-SELECTOR IN DRIVE (OD ON - Fourth Gear, CC on)

NOTE: These tests check the torque converter clutch hydraulic circuit.

(1) Attach gauge to the torque converter clutch off pressure tap.
(2) Move selector lever to the DRIVE position.
(3) Allow vehicle wheels to turn and increase throttle opening to achieve an indicated vehicle speed of 50 mph. Vehicle should be in 4th gear, CC on.

CAUTION: Both wheels must turn at the same speed.

(4) Torque converter clutch off pressure should be less than 5 psi.
(5) Now attach the gauge to the torque converter clutch on pressure tap.
(6) Move selector to the OD position.
(7) Allow vehicle wheels to turn and increase throttle opening to achieve an indicated vehicle speed of 50 mph.
(8) Verify the torque converter clutch is applied mode using the RPM display of the DRB scan tool.
(9) Torque converter clutch on pressure should be 60-90 psi.

TEST SIX-SELECTOR IN REVERSE

NOTE: This test checks the reverse clutch hydraulic circuit.

(1) Attach gauge to the reverse and low/reverse clutch tap.

AUTOMATIC TRANSMISSION - 42RLE (Continued)

(2) Move selector lever to the REVERSE position.
(3) Read reverse clutch pressure with output stationary (foot on brake) and throttle opened to achieve 1500 rpm.
(4) Reverse and low/reverse clutch pressure should read 165 to 235 psi.

TEST RESULT INDICATIONS

(1) If proper line pressure is found in any one test, the pump and pressure regulator are working properly.
(2) Low pressure in all positions indicates a defective pump, a clogged filter, or a stuck pressure regulator valve.
(3) Clutch circuit leaks are indicated if pressures do not fall within the specified pressure range.
(4) If the overdrive clutch pressure is greater than 5 psi in Step 6 of Test Three, a worn reaction shaft seal ring or a defective solenoid assembly is indicated.
(5) If the underdrive clutch pressure is greater than 5 psi in Step 4 of Test Two-A, a defective solenoid/pressure switch assembly or controller is the cause.

ALL PRESSURE SPECIFICATIONS ARE PSI (ON HOIST, WITH WHEELS FREE TO TURN)

Gear Selector Position	Actual Gear	PRESSURE TAPS						
		Under-drive Clutch	Over-drive Clutch	Reverse Clutch	Torque Converter Clutch Off	Torque Converter Clutch On	2/4 Clutch	Low/ Reverse Clutch
PARK - 0 mph *	PARK	0-2	0-5	0-2	60-110	45-100	0-2	115-145
REVERSE - 0 mph *	REVERSE	0-2	0-7	165-235	50-100	35-85	0-2	165-235
NEUTRAL - 0 mph *	NEUTRAL	0-2	0-5	0-2	60-110	45-100	0-2	115-145
Low - 20 mph #	FIRST	110-145	0-5	0-2	60-110	45-100	0-2	115-145
Third - 30 mph #	SECOND	110-145	0-5	0-2	60-110	45-100	115-145	0-2
Third - 45 mph #	DIRECT	75-95	75-95	0-2	60-90	45-80	0-2	0-2
OD - 30 mph #	OVERDRIVE	0-2	75-95	0-2	60-90	45-80	75-95	0-2
OD - 50 mph #	OVERDRIVE WITH TCC	0-2	75-95	0-2	0-5	60-95	75-95	0-2

*** Engine Speed at 1500 rpm**

 # CAUTION: Both wheels must be turning at same speed.

DIAGNOSIS AND TESTING - CLUTCH AIR PRESSURE TESTS

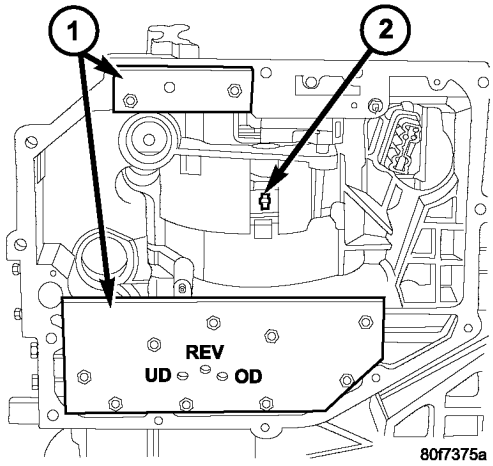

Fig. 9 Air Pressure Test Plate

1 - AIR PRESSURE TEST PLATES
2 - 2/4 CLUTCH RETAINER HOLE

Inoperative clutches can be located by substituting air pressure for fluid pressure. The clutches may be tested by applying air pressure to their respective passages after the valve body has been removed. Use Special Tool 6599-1 (1) and 6599-2 (1) to perform test (Fig. 9).

To make air pressure tests, proceed as follows:

NOTE: The compressed air supply must be free of all dirt and moisture. Use a pressure of 30 psi.

(1) Remove oil pan and valve body. (Refer to 21 - TRANSMISSION/AUTOMATIC - 42RLE/VALVE BODY - REMOVAL)

(2) Apply air pressure to the holes in the special tool (1), one at a time.

(3) Listen for the clutch to apply. It will give a slight thud sound. If a large amount of air is heard escaping, the transmission must be removed from vehicle, disassembled and all seals inspected.

2/4 CLUTCH

Apply air pressure to the feed hole located on the 2/4 clutch retainer (2). Look in the area where the 2/4 piston contacts the first separator plate and watch carefully for the 2/4 piston to move rearward. The piston should return to its original position after the air pressure is removed.

OVERDRIVE CLUTCH

Apply air pressure to the overdrive clutch apply passage and watch for the push/pull piston to move forward. The piston should return to its starting position when the air pressure is removed.

REVERSE CLUTCH

Apply air pressure to the reverse clutch apply passage and watch for the push/pull piston to move rearward. The piston should return to its starting position when the air pressure is removed.

LOW/REVERSE CLUTCH

Apply air pressure to the low/reverse clutch feed hole passage. Look in the area where the low/reverse piston contacts the first separator plate. Watch carefully for the piston to move forward. The piston should return to its original position after the air pressure is removed.

UNDERDRIVE CLUTCH

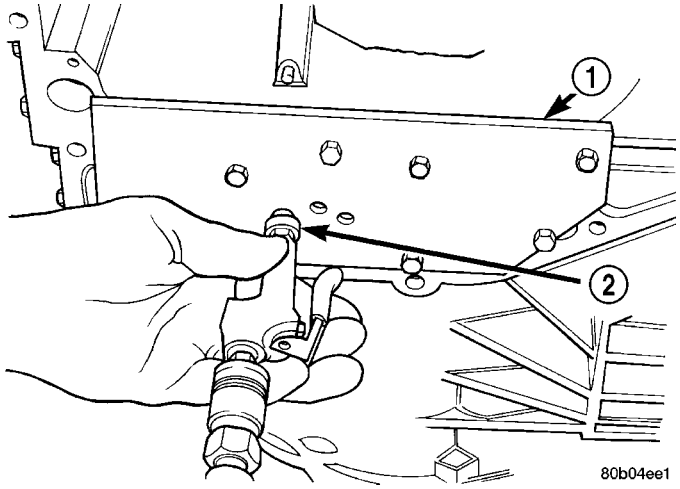

Fig. 10 Testing Underdrive Clutch

1 - AIR PRESSURE TEST PLATE 6599-1
2 - AIR NOZZLE

Because this clutch piston cannot be seen, its operation is checked by function. Air pressure is applied to the low/reverse or the 2/4 clutches. This locks the output shaft. Use a piece of rubber hose wrapped around the input shaft and a pair of clamp-on pliers to turn the input shaft. Next apply air pressure (Fig. 10) to the underdrive clutch. The input shaft should not rotate with hand torque. Release the air pressure and confirm that the input shaft will rotate.

DIAGNOSIS AND TESTING - FLUID LEAKAGE

FLUID LEAKAGE - TORQUE CONVERTER HOUSING AREA

When diagnosing converter housing fluid leaks, three actions must be taken before repair:

(1) Verify proper transmission fluid level.

(2) Verify that the leak originates from the converter housing area and is transmission fluid.

(3) Determine the true source of the leak.

AUTOMATIC TRANSMISSION - 42RLE (Continued)

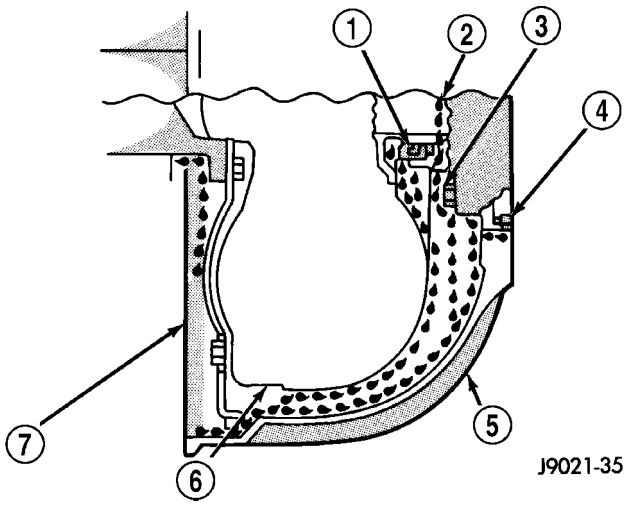

Fig. 11 Converter Housing Leak Paths

1 - PUMP SEAL
2 - PUMP VENT
3 - PUMP BOLT
4 - PUMP GASKET
5 - CONVERTER HOUSING
6 - CONVERTER
7 - REAR MAIN SEAL LEAK

Fluid leakage at or around the torque converter area (Fig. 11) may originate from an engine oil leak. The area should be examined closely. Factory fill fluid is red and, therefore, can be distinguished from engine oil.

Some suspected converter housing fluid leaks may not be leaks at all. They may only be the result of residual fluid in the converter housing, or excess fluid spilled during factory fill, or fill after repair. Converter housing leaks have several potential sources. Through careful observation, a leak source can be identified before removing the transmission for repair.

Pump seal leaks tend to move along the drive hub (Fig. 11) and onto the rear of the converter. Pump o-ring or pump body leaks follow the same path as a seal leak. Pump attaching bolt leaks are generally deposited on the inside of the converter housing and not on the converter itself. Pump seal or gasket leaks (Fig. 11)usually travel down the inside of the converter housing.

TORQUE CONVERTER LEAKAGE

Possible sources of torque converter leakage are:

- Torque converter weld leaks at the outside diameter weld. (Fig. 12)
- Torque converter hub weld. (Fig. 12)

STANDARD PROCEDURE - ALUMINUM THREAD REPAIR

Damaged or worn threads in the aluminum transmission case and valve body can be repaired by the use of Heli-Coils®, or equivalent. This repair consists of drilling out the worn-out damaged threads. Then tap the hole with a special Heli-Coil® tap, or equivalent, and installing a Heli-Coil® insert, or equivalent, into the hole. This brings the hole back to its original thread size.

Heli-Coil®, or equivalent, tools, and inserts are readily available from most automotive parts suppliers.

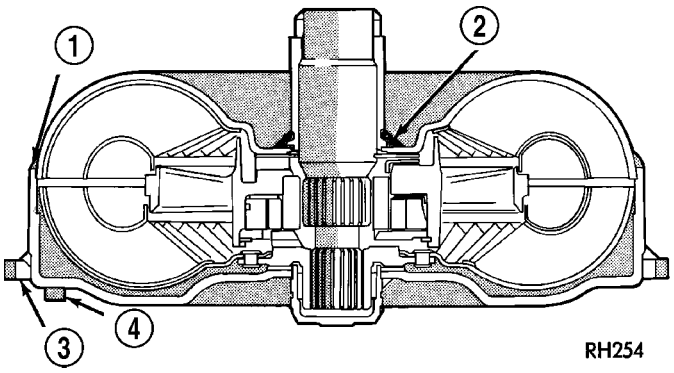

Fig. 12 Converter Leak Points - Typical

1 - OUTSIDE DIAMETER WELD
2 - TORQUE CONVERTER HUB WELD
3 - STARTER RING GEAR
4 - LUG

REMOVAL

(1) Disconnect battery negative cable.

(2) Raise and support vehicle.

(3) Disconnect and lower or remove necessary exhaust components.

(4) Remove engine-to-transmission bending braces or engine collar.

(5) Remove starter motor. (Refer to 8 - ELECTRICAL/STARTING/STARTER MOTOR - REMOVAL)

(6) On 4.0L engine equipped vehicles, disconnect and remove crankshaft position sensor (Fig. 13). Retain sensor attaching bolt.

CAUTION: The crankshaft position sensor can be damaged during transmission removal (or installation) if the sensor is left in place. To avoid damage, remove the sensor before removing the transmission.

(7) If transmission is being removed for overhaul, remove transmission oil pan, drain fluid and reinstall pan. (Refer to 21 - TRANSMISSION/AUTOMATIC - 42RLE/FLUID - STANDARD PROCEDURE)

(8) Remove torque converter access cover.

(9) Rotate crankshaft in clockwise direction until converter bolts are accessible. Then remove bolts one at a time. Rotate crankshaft with socket wrench on dampener bolt.

AUTOMATIC TRANSMISSION - 42RLE (Continued)

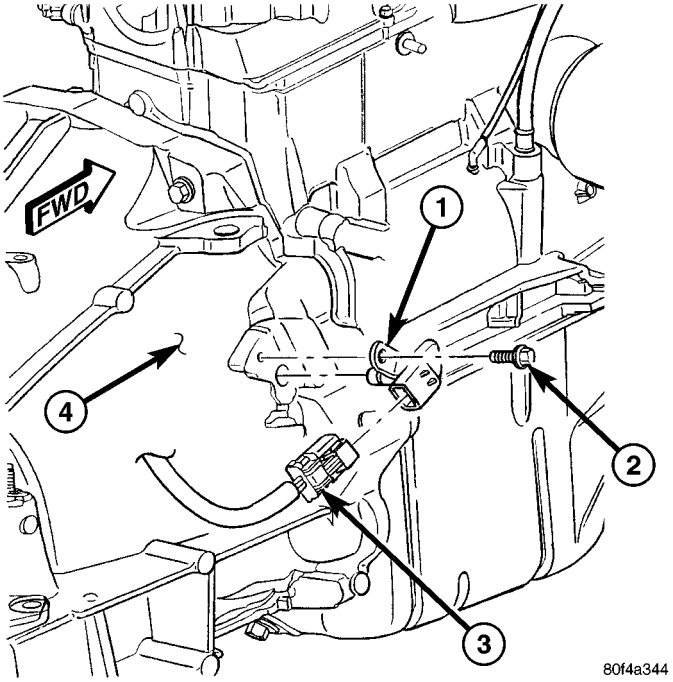

Fig. 13 Crankshaft Position Sensor

1 - CRANKSHAFT POSITION SENSOR
2 - MOUNTING BOLT
3 - ELECTRICAL CONNECTOR
4 - TRANSMISSION BELLHOUSING

(10) Mark propeller shaft and axle yokes for assembly alignment. Then disconnect and remove propeller shafts. (Refer to 3 - DIFFERENTIAL & DRIVELINE/PROPELLER SHAFT/PROPELLER SHAFT - REMOVAL)

(11) Disconnect wires from the input and output speed sensors (Fig. 14).

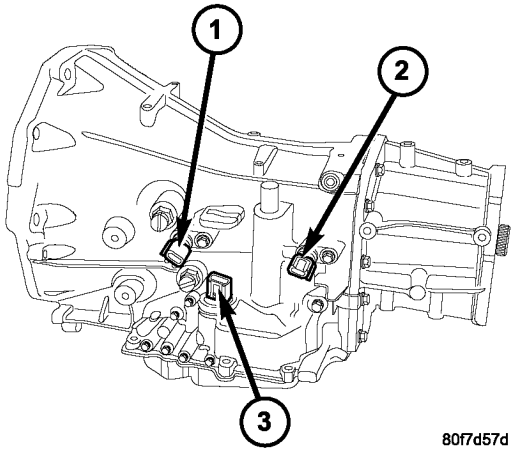

Fig. 14 Input and Output Speed Sensors and Transmission Range Sensor

1 - INPUT SPEED SENSOR
2 - OUTPUT SPEED SENSOR
3 - TRANSMISSION RANGE SENSOR

(12) Disconnect wires from the transmission range sensor (Fig. 14) and the solenoid/pressure switch assembly (Fig. 15).

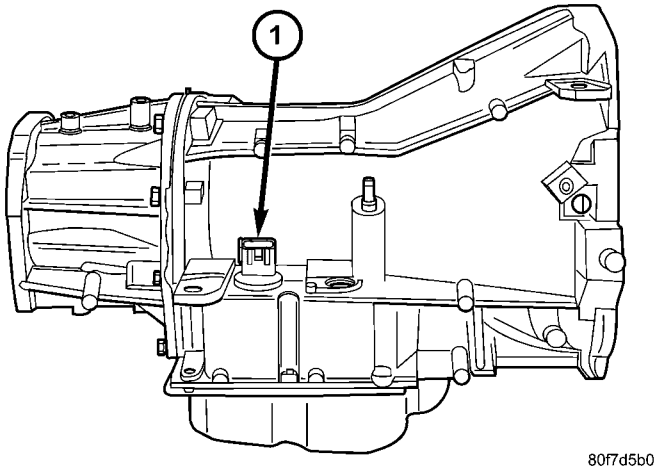

Fig. 15 Solenoid/Pressure Switch Assembly

1 - SOLENOID/PRESSURE SWITCH ASSEMBLY CONNECTOR

(13) Disconnect gearshift cable from transmission manual valve lever.

(14) Disconnect shift rod from transfer case shift lever or remove shift lever from transfer case.

(15) Support rear of engine with safety stand or jack.

(16) Raise transmission slightly with service jack to relieve load on skid plate and transmission support.

(17) Remove bolts securing rear support (Fig. 16) and cushion to transmission and skid plate. Raise transmission slightly, slide exhaust hanger arm from bracket and remove rear support.

(18) Remove bolts attaching skid plate (Fig. 16) to frame and remove skid plate. (Refer to 13 - FRAME

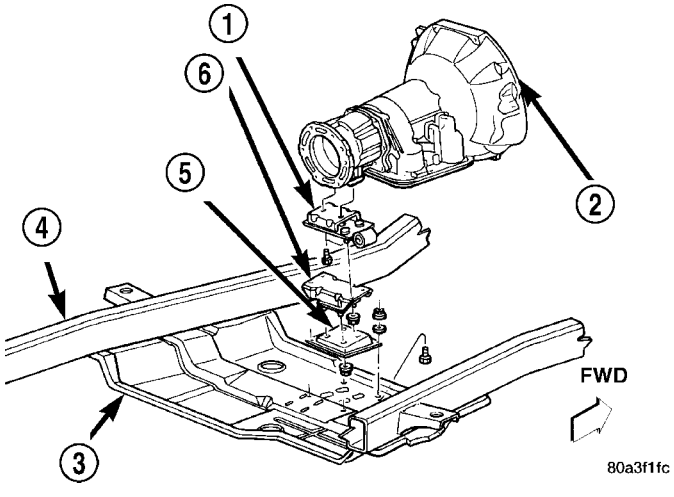

Fig. 16 Transmission Mount - Automatic Transmission

1 - TRANSMISSION SUPPORT BRACKET
2 - AUTOMATIC TRANSMISSION
3 - SKID PLATE
4 - FRAME
5 - TRANSMISSION MOUNT SUPPORT BRACKET
6 - CUSHION

AUTOMATIC TRANSMISSION - 42RLE (Continued)

& BUMPERS/FRAME/TRANSFER CASE SKID PLATE - REMOVAL)

(19) Disconnect transfer case vent hose.

(20) Remove transfer case. (Refer to 21 - TRANSMISSION/TRANSFER CASE - REMOVAL)

(21) Remove fill tube bracket bolts and pull tube out of transmission. Retain fill tube seal. Remove the bolt attaching transfer case vent tube to converter housing.

(22) Disconnect fluid cooler lines at transmission.

(23) Remove all converter housing bolts.

(24) Carefully work transmission and torque converter assembly rearward off engine block dowels.

(25) Hold torque converter in place during transmission removal.

(26) Lower transmission and remove assembly from under the vehicle.

(27) To remove torque converter, carefully slide torque converter out of the transmission.

DISASSEMBLY

NOTE: If the transmission is being reconditioned (clutch/seal replacement) or replaced, it is necessary to perform the Quick Learn Procedure using the DRBIII® Scan Tool (Refer to 8 - ELECTRICAL/ELECTRONIC CONTROL MODULES/TRANSMISSION CONTROL MODULE - STANDARD PROCEDURE).

Before disassembling transmission, move the shift lever clockwise as far as it will go and then remove the shift lever.

NOTE: Tag all clutch pack assemblies, as they are removed, for reassembly identification.

CAUTION: Do not intermix clutch discs or plates as the unit might then fail.

(1) Remove the torque converter from the transmission input shaft (Fig. 17).

(2) Measure input shaft end play using Tool 8266. Set up Tool 8266 and a dial indicator as shown in (Fig. 18). Move input shaft in and out to obtain end play reading. End play specifications are 0.13 to 0.64 mm (0.005 to 0.025 inch). Record indicator reading for reference when reassembling the transmission. If endplay exceeds the specified range, the #4 thrust plate needs to be inspected and changed if necessary.

NOTE: The four bolts along the bottom of the adapter housing have a sealing patch applied from the factory. Note the locations of these bolts and seperate these bolts for reuse.

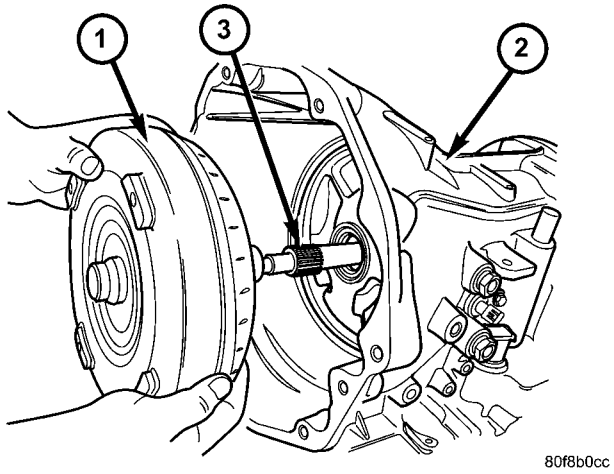

Fig. 17 Remove Torque Converter

1 - TORQUE CONVERTER
2 - TRANSMISSION
3 - INPUT SHAFT

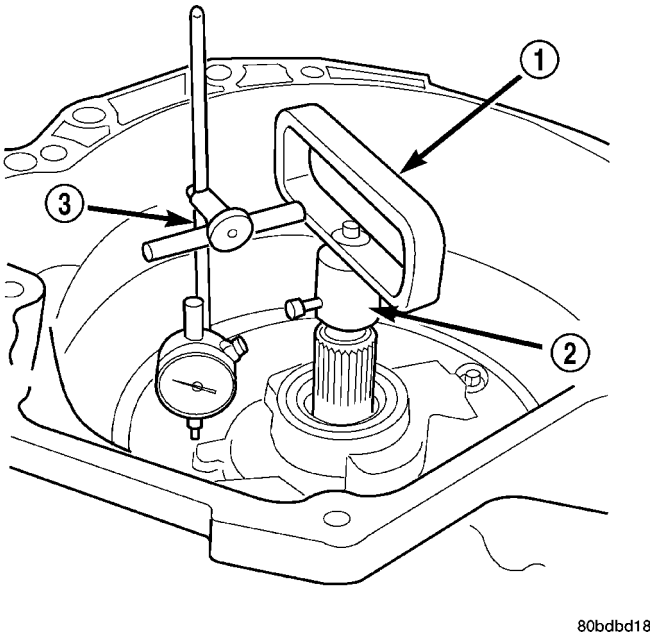

Fig. 18 Measure Input Shaft End Play Using Tool 8266 - Typical

1 - TOOL 8266-8
2 - TOOL 8266-2
3 - TOOL C-3339

AUTOMATIC TRANSMISSION - 42RLE (Continued)

(3) Remove the bolts (Fig. 19) that hold the adapter housing onto the transmission case.

(4) Remove the adapter (Fig. 21) housing from the transmission case. There are two pry slots (Fig. 20) located near the bottom corners of the housing for separating the housing from the transmission case.

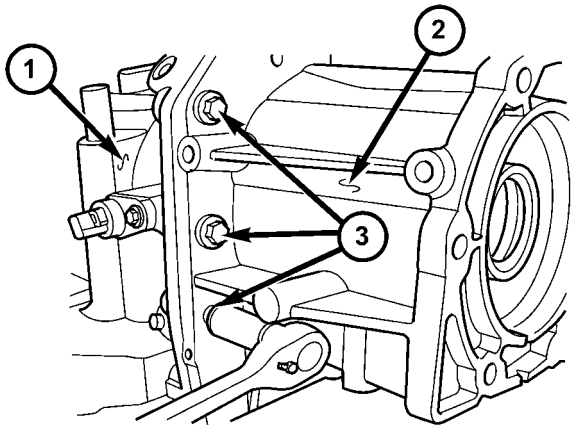

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Fig. 19 Remove Adapter Housing Bolts

1 - TRANSMISSION CASE
2 - ADAPTER HOUSING
3 - BOLTS

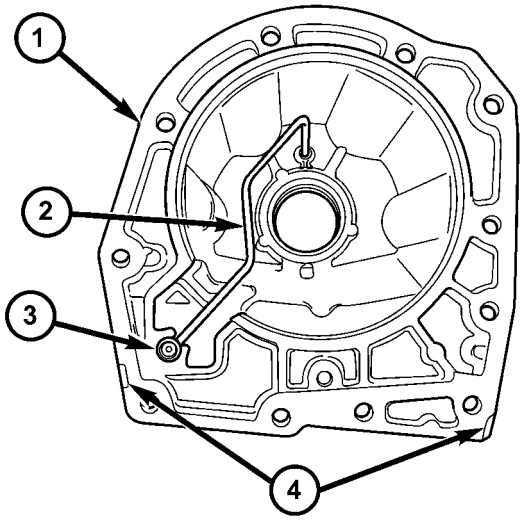

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Fig. 20 Lube Tube Grommet

1 - HOUSING
2 - LUBE TUBE
3 - GROMMET
4 - PRY SLOTS

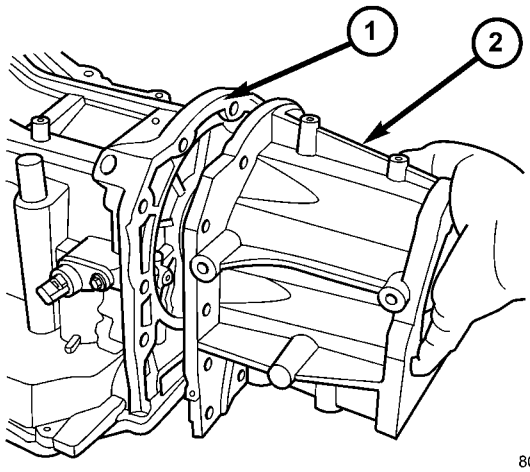

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Fig. 21 Remove Adapter Housing

1 - TRANSMISSION CASE
2 - ADAPTER HOUSING

(5) Inspect the lube tube grommet (Fig. 20) for damage. If the grommet lip is damaged, it will need to be replaced.

(6) Using a Slide Hammer C-3752 (Fig. 22), remove the 4X4 stub shaft (Fig. 23) from the transmission output shaft. Inspect the cir-clip on the shaft for damage and replace the clip if necessary.

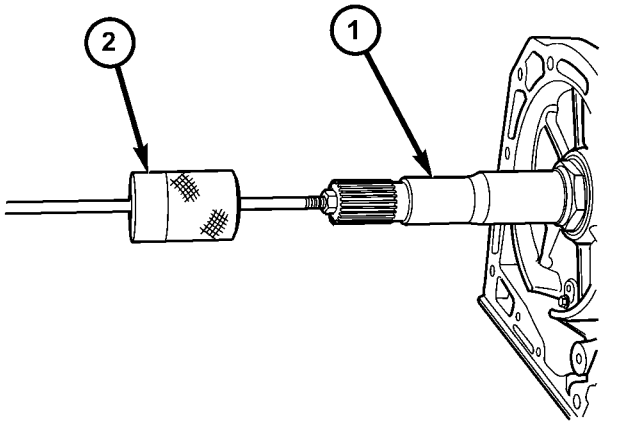

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Fig. 22 Remove the 4X4 Stub shaft Using C-3752

1 - 4X4 STUB SHAFT
2 - PULLER C-3752

AUTOMATIC TRANSMISSION - 42RLE (Continued)

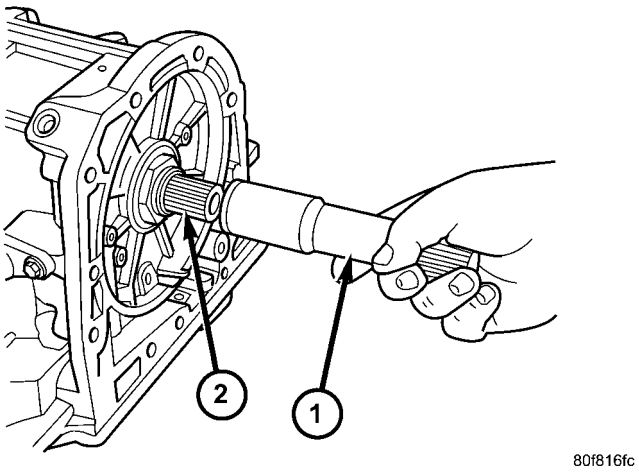


Fig. 23 Remove 4X4 Stub Shaft

1 - STUB SHAFT
2 - OUTPUT SHAFT

(7) Remove the input speed sensor bolt (Fig. 24).
(8) Remove the output speed sensor bolt (Fig. 25).

NOTE: The speed sensor bolts have a sealing patch applied from the factory. Seperate these bolts for reuse.

(9) Remove the input and output speed sensors (Fig. 26). Identify the speed sensors for re-installation since they are not interchangeable.

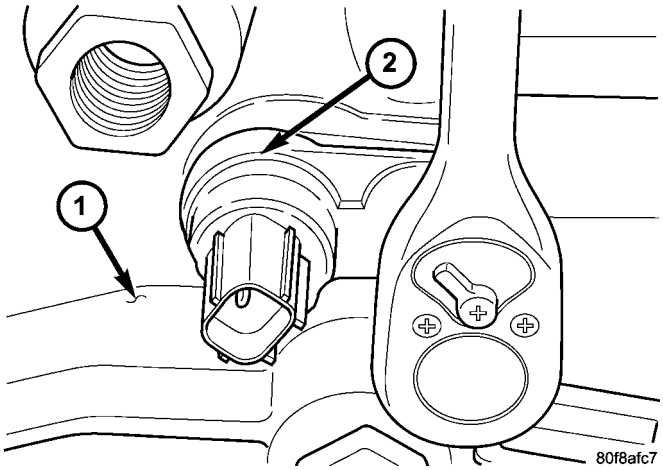


Fig. 24 Remove Input Speed Sensor Bolt

1 - INPUT SPEED SENSOR
2 - TRANSMISSION CASE

NOTE: One of the oil pan bolts has a sealing patch applied from the factory. Seperate this bolt for reuse.

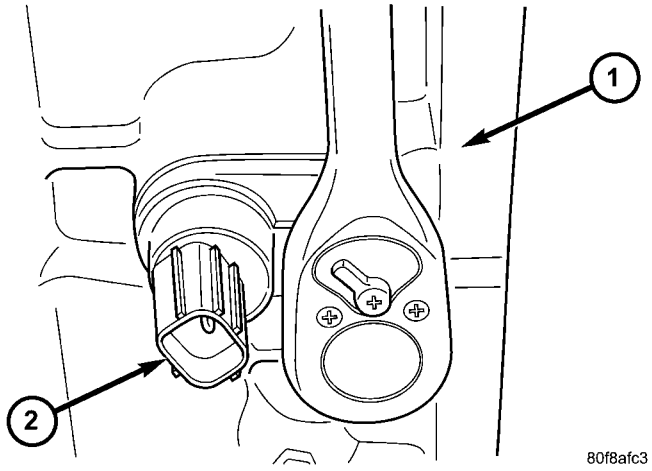


Fig. 25 Remove Output Speed Sensor Bolt

1 - OUTPUT SPEED SENSOR
2 - TRANSMISSION CASE

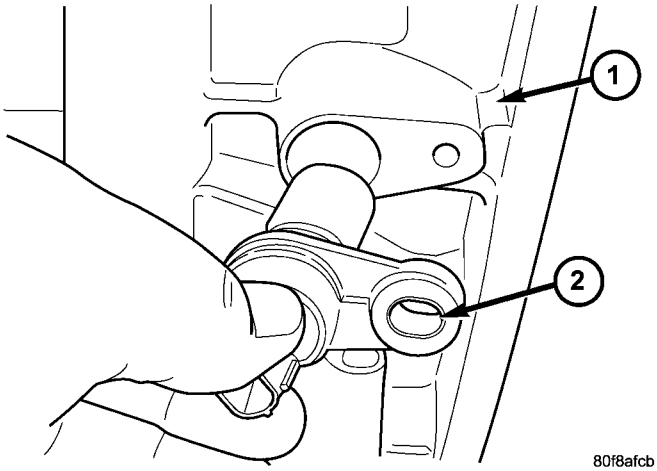


Fig. 26 Remove Output Speed Sensor

1 - OUTPUT SPEED SENSOR
2 - TRANSMISSION CASE

AUTOMATIC TRANSMISSION - 42RLE (Continued)

(10) Remove the transmission oil pan bolts (Fig. 27).

(11) Remove the transmission oil pan (Fig. 28).

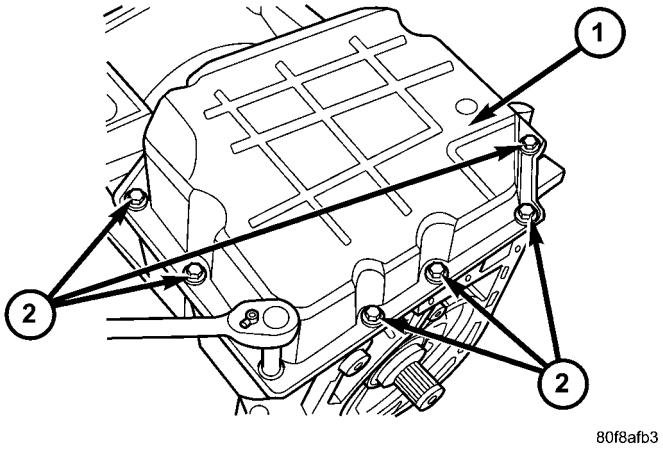


Fig. 27 Remove Transmission Oil Pan Bolts

1 - TRANSMISSION OIL PAN
2 - BOLTS

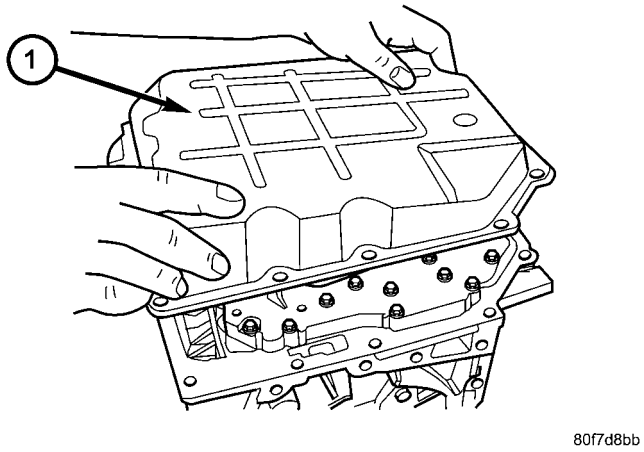


Fig. 28 Remove Transmission Oil Pan

1 - TRANSMISSION OIL PAN

(12) Remove the transmission oil filter screws (Fig. 29).

(13) Remove transmission oil filter (Fig. 30).

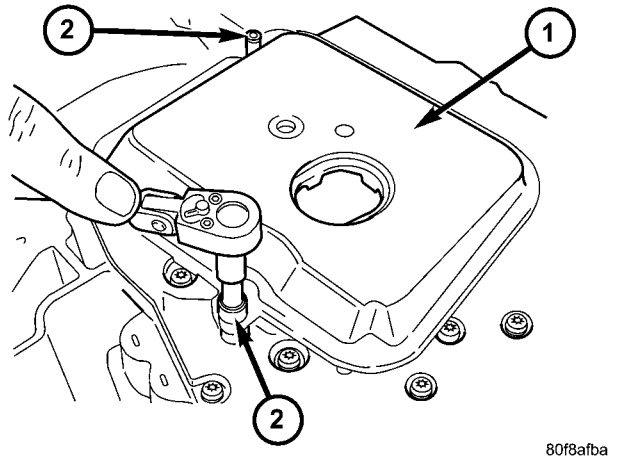


Fig. 29 Remove Oil Filter Screws

1 - OIL FILTER
2 - SCREWS

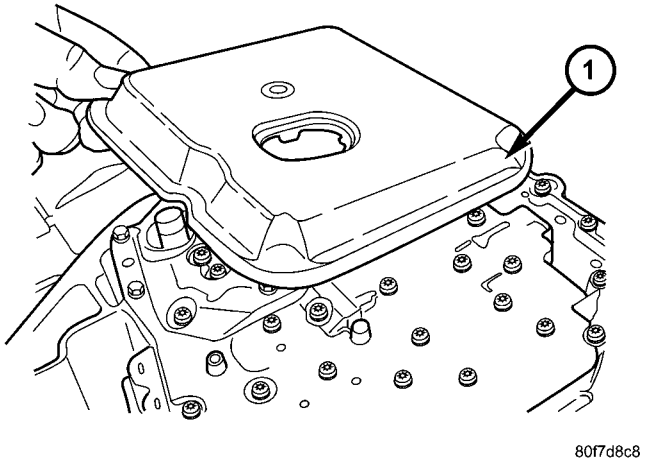


Fig. 30 Remove Transmission Filter

1 - TRANSMISSION FILTER

AUTOMATIC TRANSMISSION - 42RLE (Continued)

(14) Remove the oil filter o-ring from the valve body (Fig. 31).

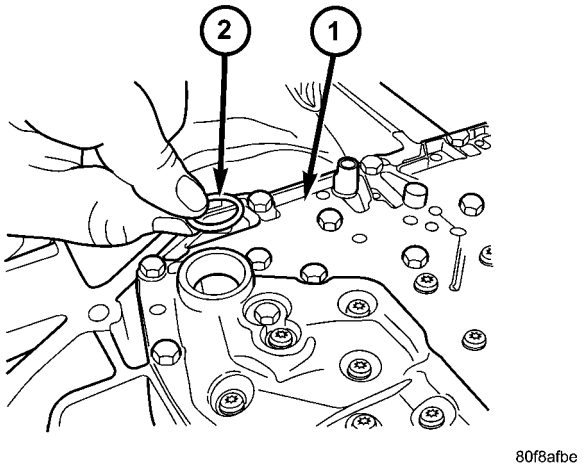


Fig. 31 Remove Oil Filter O-Ring

1 - VALVE BODY
2 - O-RING

(15) Remove valve body-to-case bolts (Fig. 32).

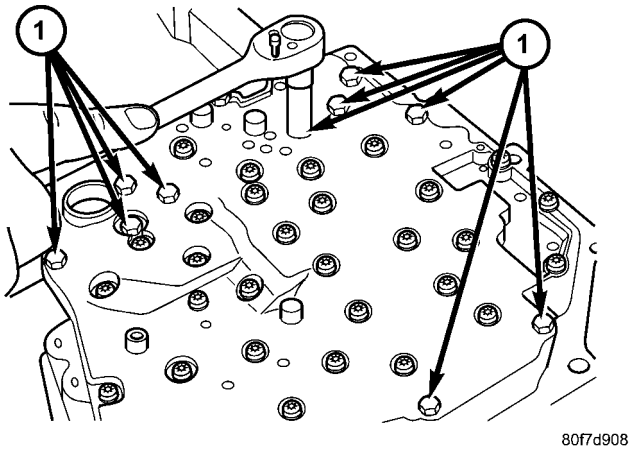


Fig. 32 Remove Valve Body Bolts

1 - BOLTS

CAUTION: Do not handle the valve body by the manual shaft. Damage could result.

(16) Remove valve body from transmission (Fig. 33).

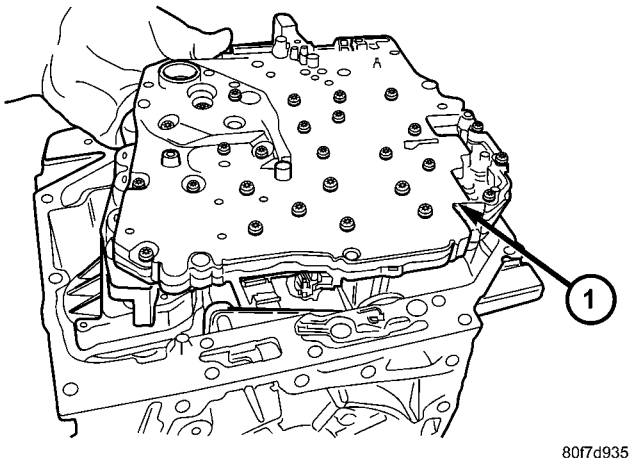


Fig. 33 Remove Valve Body From Transmission

1 - VALVE BODY

(17) Remove underdrive and overdrive accumulators (Fig. 34).

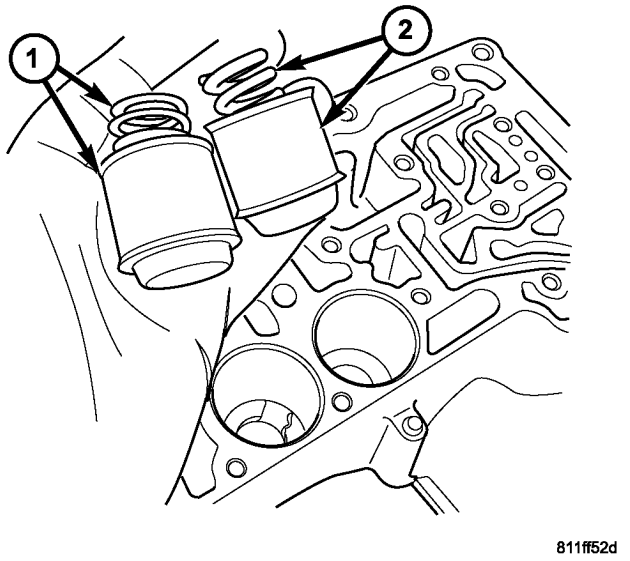


Fig. 34 Underdrive and Overdrive Accumulators

1 - OVERDRIVE PISTON AND SPRING
2 - UNDERDRIVE PISTON AND SPRING

AUTOMATIC TRANSMISSION - 42RLE (Continued)

(18) Remove the low/reverse accumulator snap ring (Fig. 35).

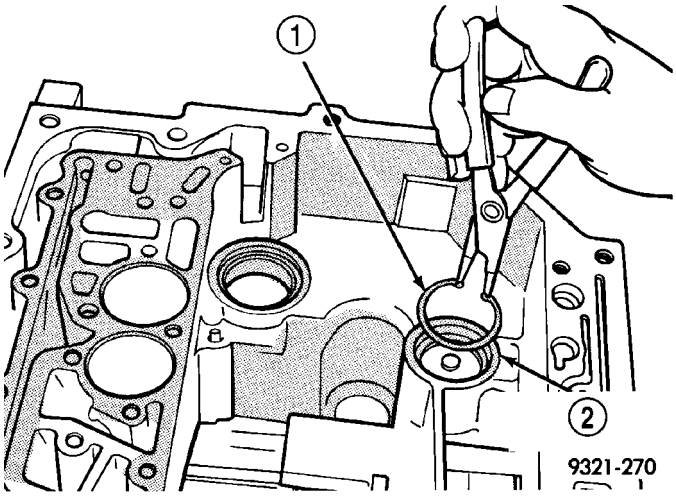


Fig. 35 Remove Low/Reverse Accumulator

1 - SNAP RING
2 - LOW/REVERSE ACCUMULATOR

(19) Remove the low/reverse accumulator plug (Fig. 36).

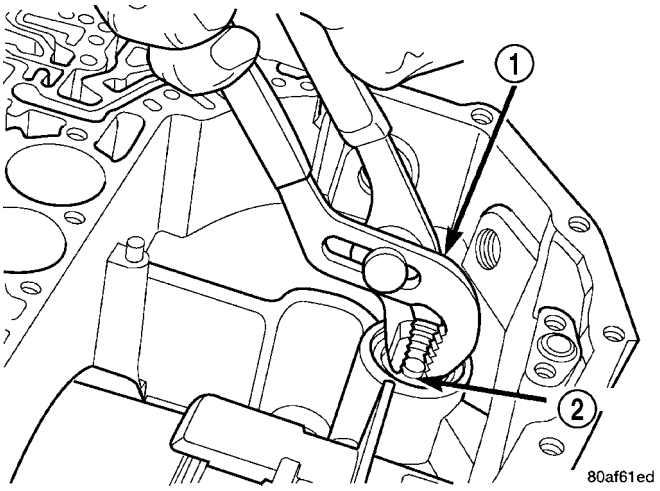


Fig. 36 Remove Low/Reverse Accumulator Plug

1 - ADJUSTABLE PLIERS
2 - PLUG

(20) Remove low/reverse accumulator piston using suitable pliers (Fig. 37). Remove piston and springs (Fig. 38).

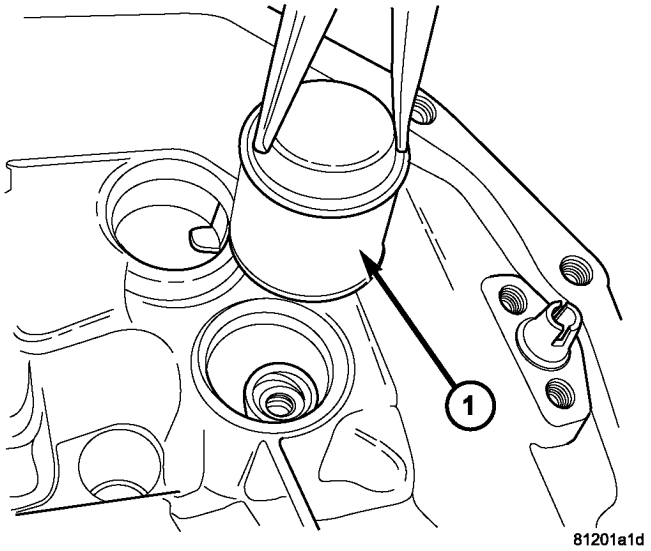


Fig. 37 Low/Reverse Accumulator Piston

1 - ACCUMULATOR PISTON

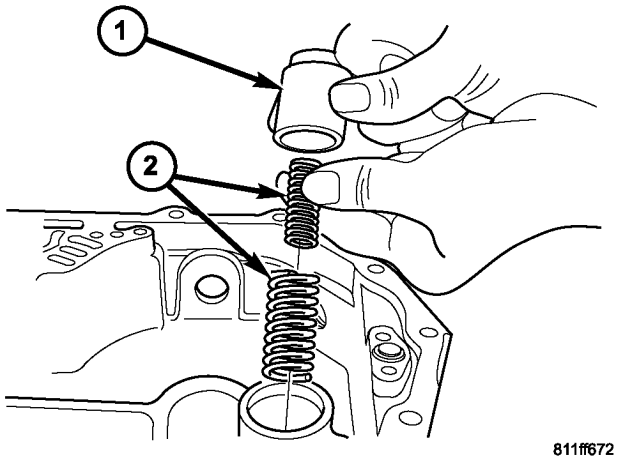


Fig. 38 Low/Reverse Accumulator

1 - PISTON
2 - RETURN SPRINGS

AUTOMATIC TRANSMISSION - 42RLE (Continued)

(21) Remove and discard the oil pump-to-case bolts (Fig. 39). The oil pump bolts are not to be reused.

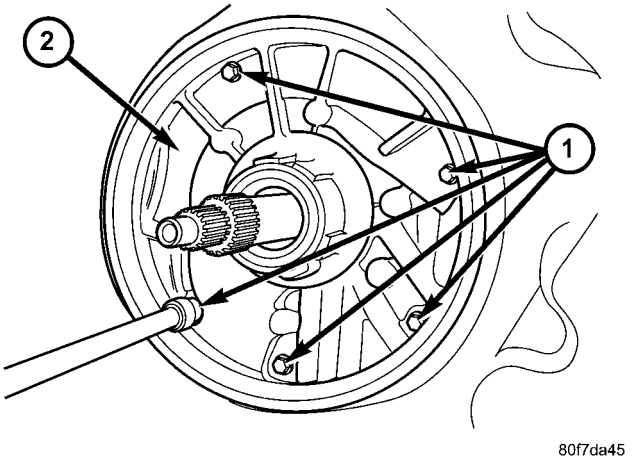


Fig. 39 Remove Oil Pump Attaching Bolts

1 - BOLTS
2 - OIL PUMP

(22) Remove oil pump using C-3752 Pullers (Fig. 40).

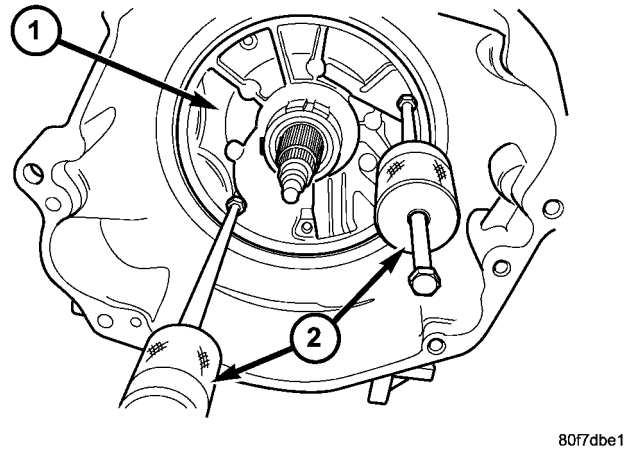


Fig. 40 Oil Pump Pullers

1 - OIL PUMP
2 - PULLERS

(23) Remove oil pump while pushing in on input shaft (Fig. 41).

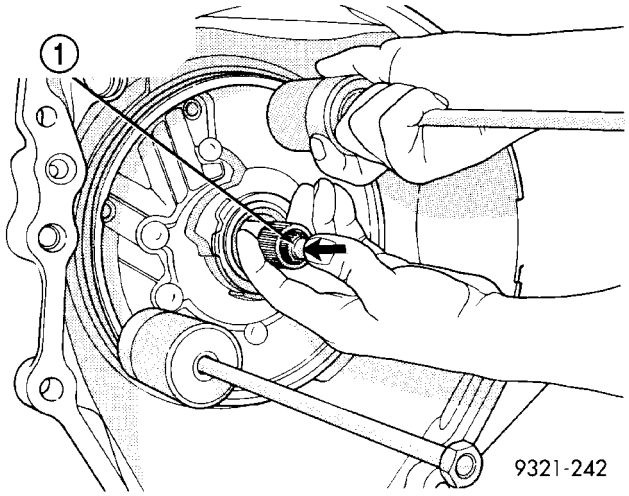


Fig. 41 Remove Oil Pump

1 - "PUSH IN" ON INPUT SHAFT WHILE REMOVING PUMP

(24) Remove oil pump gasket (Fig. 42).

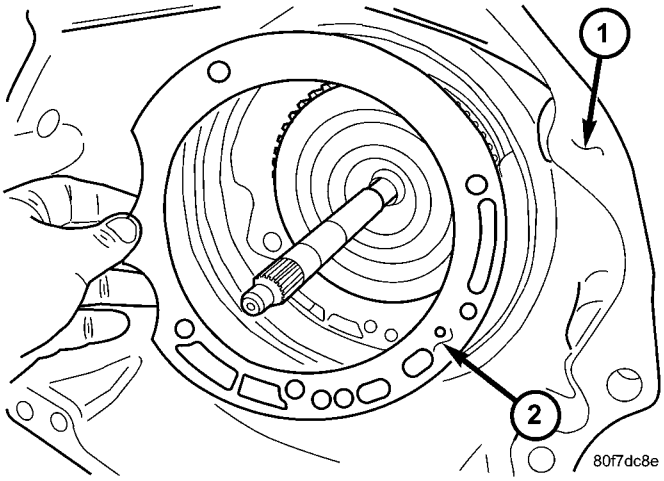


Fig. 42 Remove Oil Pump Gasket

1 - BELLHOUSING
2 - OIL PUMP GASKET

CAUTION: By-pass valve must be replaced if transmission failure occurs.

AUTOMATIC TRANSMISSION - 42RLE (Continued)

(25) Remove the cooler by-pass valve (Fig. 43).

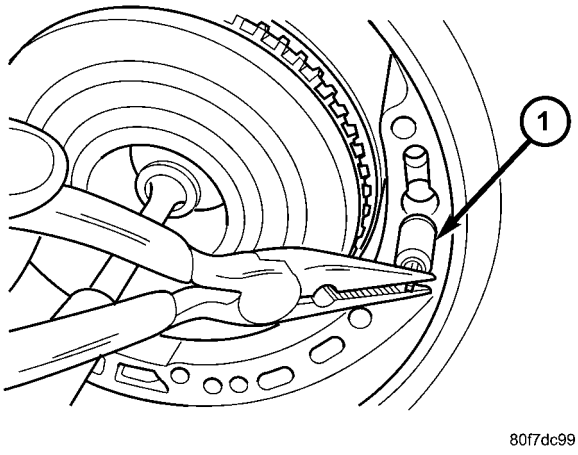


Fig. 43 Remove By-Pass Valve

1 - BYPASS VALVE

(26) Remove the #1 caged needle bearing (Fig. 44).

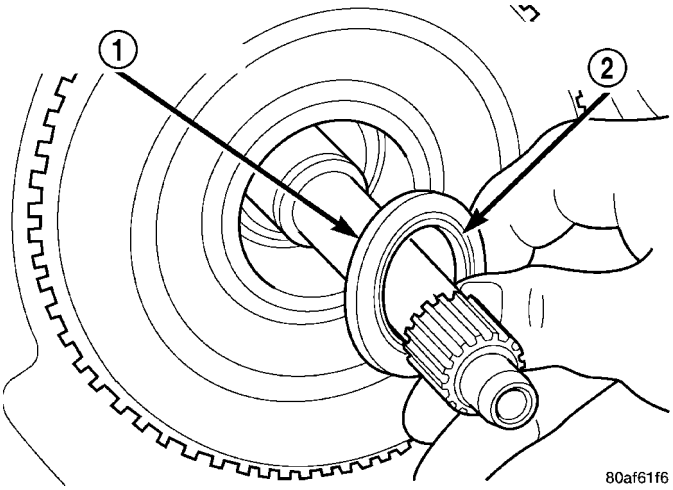


Fig. 44 Remove No. 1 Caged Needle Bearing

1 - #1 CAGED NEEDLE BEARING
2 - NOTE: TANGED SIDE OUT

(27) Remove the input clutch assembly (Fig. 45).

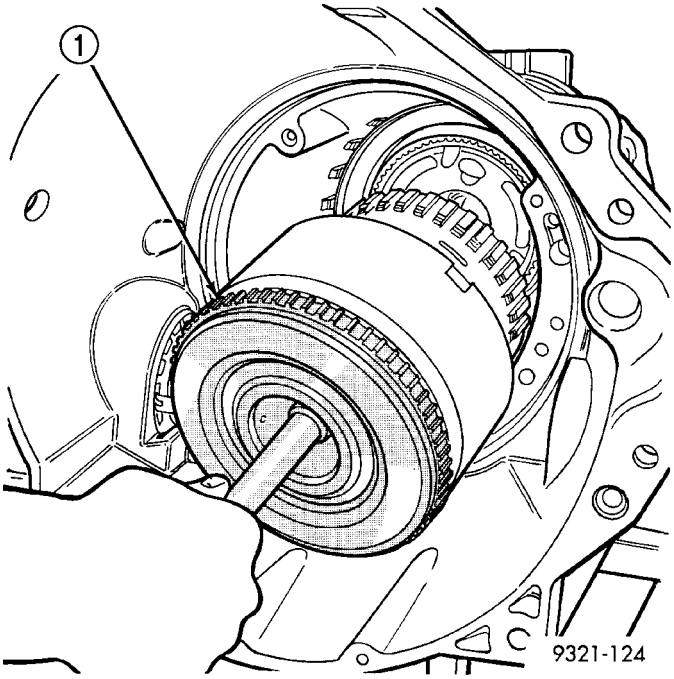


Fig. 45 Remove Input Clutch Assembly

1 - INPUT CLUTCH ASSEMBLY

(28) Remove the #4 thrust plate (Fig. 46).

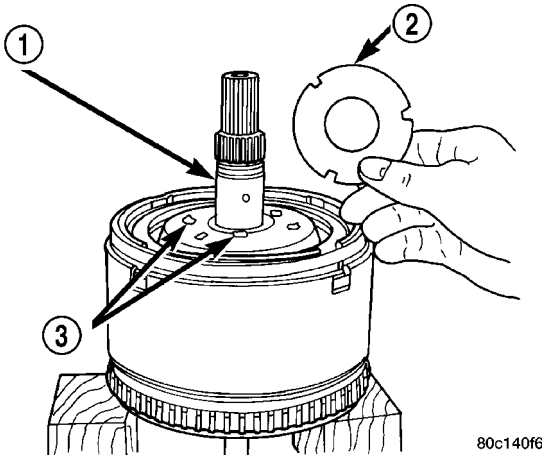


Fig. 46 Remove #4 Thrust Plate

1 - OVERDRIVE SHAFT ASSEMBLY
2 - #4 THRUST PLATE (SELECT)
3 - PETROLATUM FOR RETENTION

AUTOMATIC TRANSMISSION - 42RLE (Continued)

(29) Remove the front sun gear assembly and #4 thrust washer (if still in place) (Fig. 47).

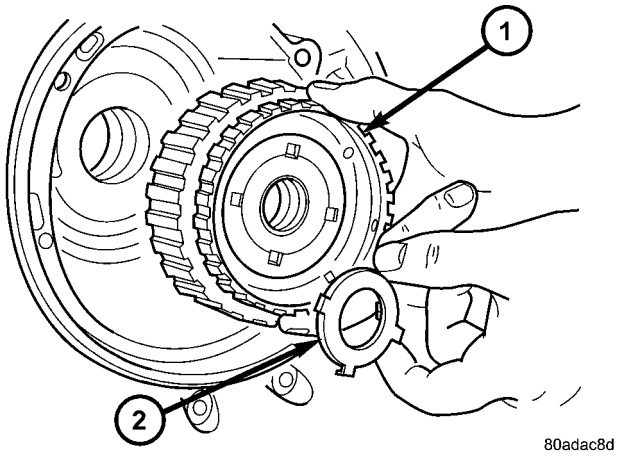


Fig. 47 Remove Front Sun Gear Assembly

1 - FRONT SUN GEAR ASSEMBLY
2 - #4 THRUST WASHER (FOUR TABS)

(30) Remove the front carrier/rear annulus and #6 needle bearing (Fig. 48).

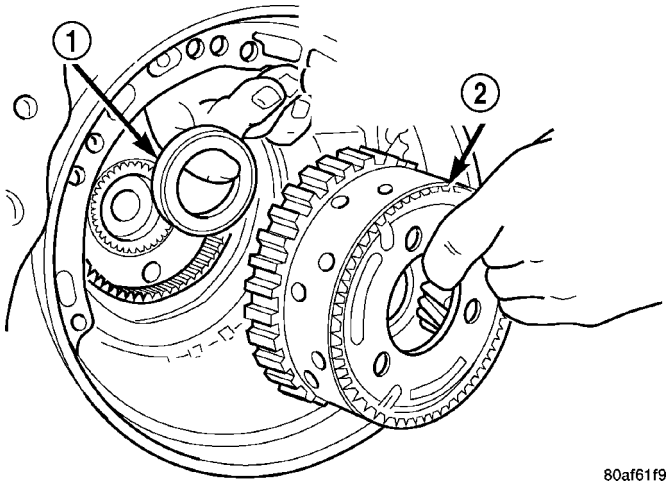


Fig. 48 Remove Front Carrier/Rear Annulus

1 - #6 NEEDLE BEARING
2 - FRONT CARRIER AND REAR ANNULUS ASSEMBLY (TWIST AND PULL OR PUSH TO REMOVE OR INSTALL).

(31) Remove the rear sun gear and #7 needle bearing (Fig. 49) and (Fig. 50).

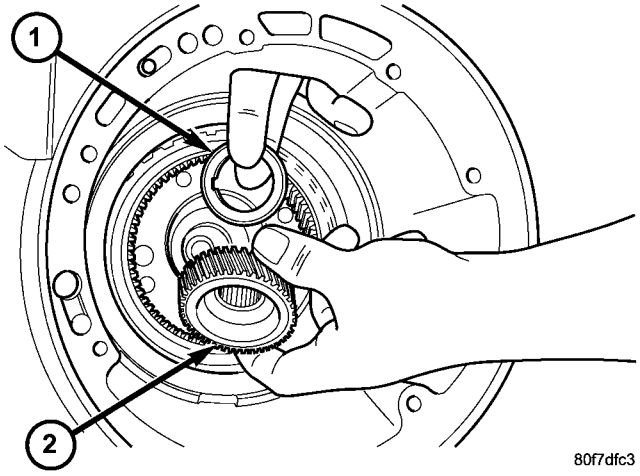


Fig. 49 Remove Rear Sun Gear

1 - #7 NEEDLE BEARING
2 - REAR SUN GEAR

NOTE: The number seven needle bearing has three antireversal tabs and is common with the number five and number two position. The orientation should allow the bearing to seat flat against the rear sun gear (Fig. 50).

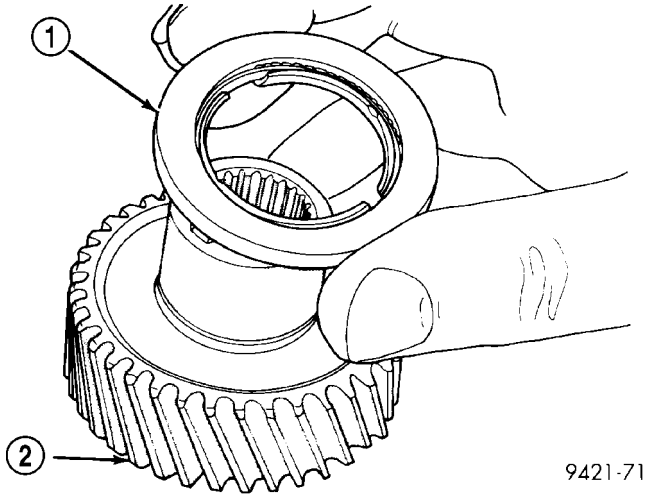


Fig. 50 Number 7 Bearing

1 - #7 BEARING
2 - REAR SUN GEAR

AUTOMATIC TRANSMISSION - 42RLE (Continued)

NOTE: Verify that Tool 5058A is centered properly over the 2/4 clutch retainer before compressing. If necessary, fasten the 5058A bar to the bellhousing flange with any combination of locking pliers and bolts to center the tool properly.

(32) Install and load Tool 5058A to remove the 2/4 clutch retainer snap ring (Fig. 51).

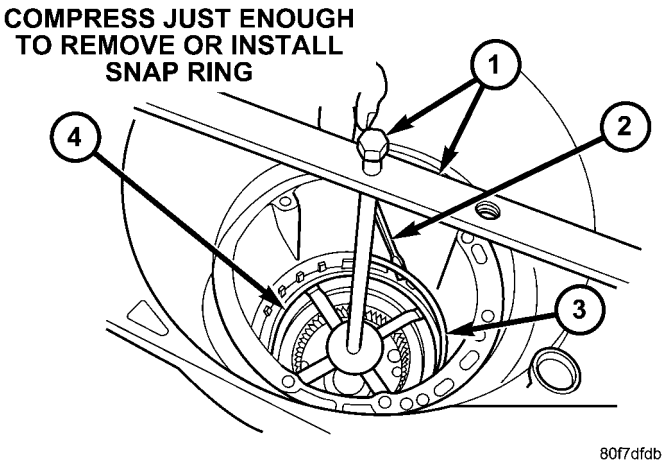


Fig. 51 Remove 2/4 Clutch Retainer Snap Ring

1 - TOOL 5058
2 - SCREWDRIVER
3 - SNAP RING
4 - 2/4 CLUTCH RETAINER

NOTE: The 2/4 Clutch Piston has bonded seals which are not individually serviceable. Seal replacement requires replacement of the piston assembly.

(33) Remove the 2/4 clutch retainer (Fig. 52) and (Fig. 53).

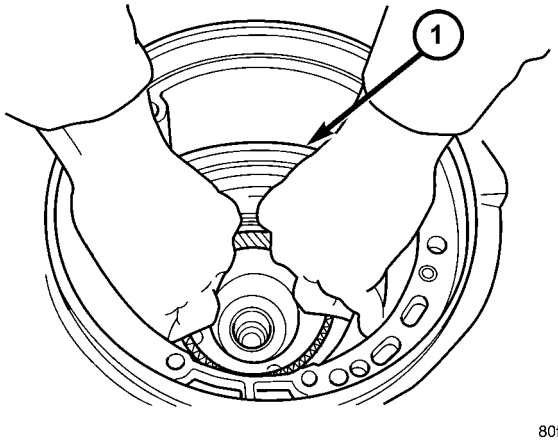


Fig. 52 Remove 2/4 Clutch Retainer

1 - 2/4 CLUTCH RETAINER

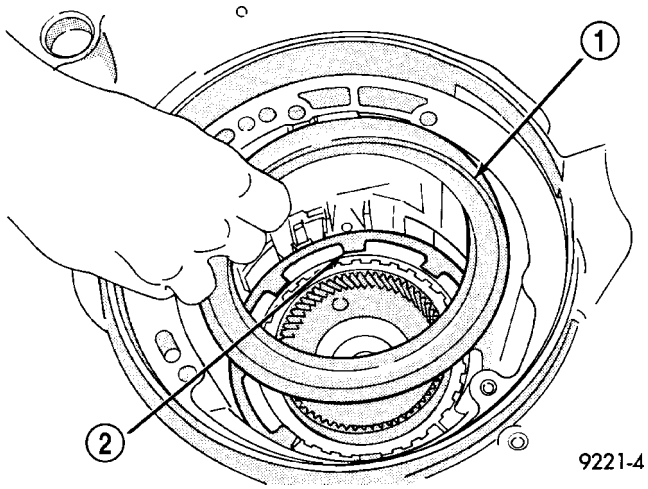


Fig. 53 2/4 Clutch Retainer

1 - 2/4 CLUTCH RETAINER
2 - 2/4 CLUTCH RETURN SPRING

(34) Remove the 2/4 clutch return spring (Fig. 54).

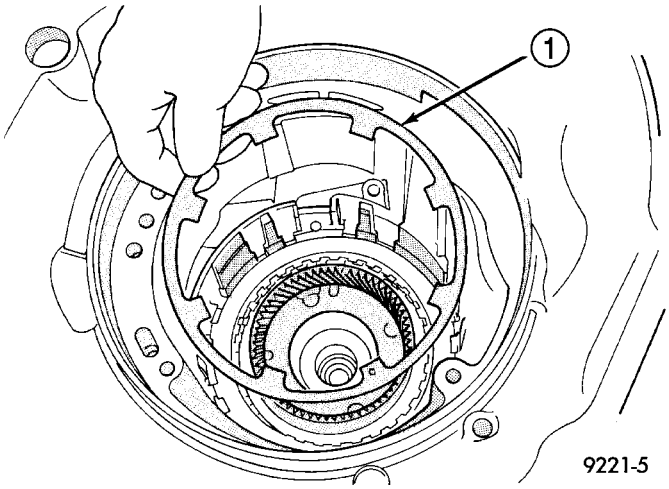


Fig. 54 Remove 2/4 Clutch Return Spring

1 - 2/4 CLUTCH RETURN SPRING

AUTOMATIC TRANSMISSION - 42RLE (Continued)

(35) Remove the 2/4 clutch pack (Fig. 55).

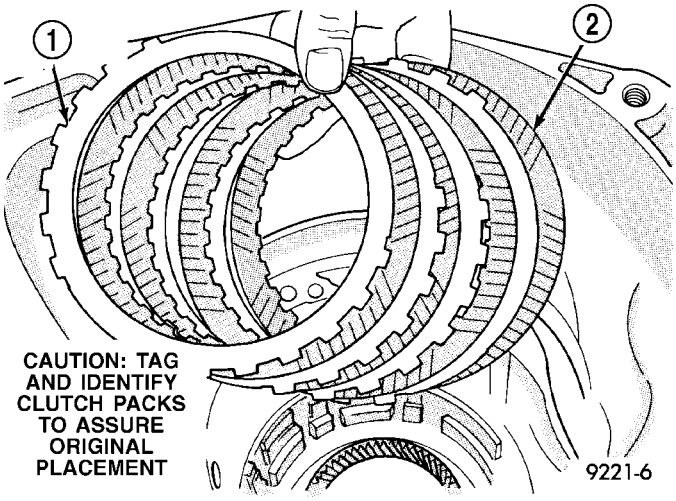


Fig. 55 Remove 2/4 Clutch Pack

1 - CLUTCH PLATE (4)
2 - CLUTCH DISC (4)

(36) Remove the tapered snap ring (Fig. 56).

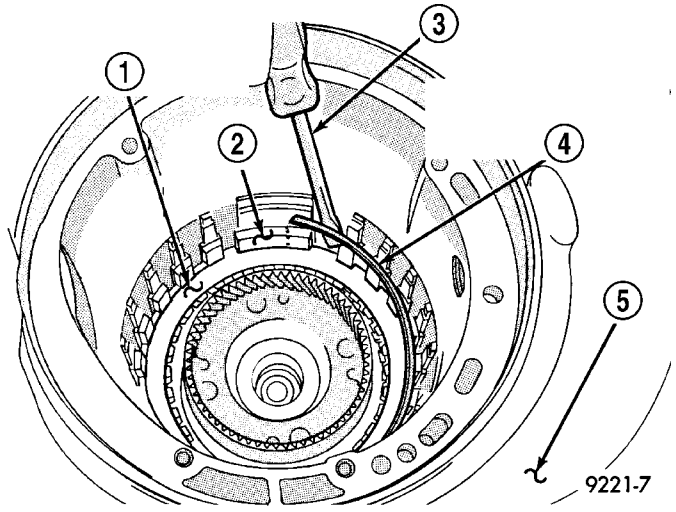


Fig. 56 Remove Tapered Snap Ring

1 - LOW/REVERSE CLUTCH REACTION PLATE
2 - LONG TAB
3 - SCREWDRIVER
4 - LOW/REVERSE TAPERED SNAP RING (TAPERED SIDE UP)
5 - OIL PAN FACE

(37) Remove the low/reverse reaction plate (Fig. 57).

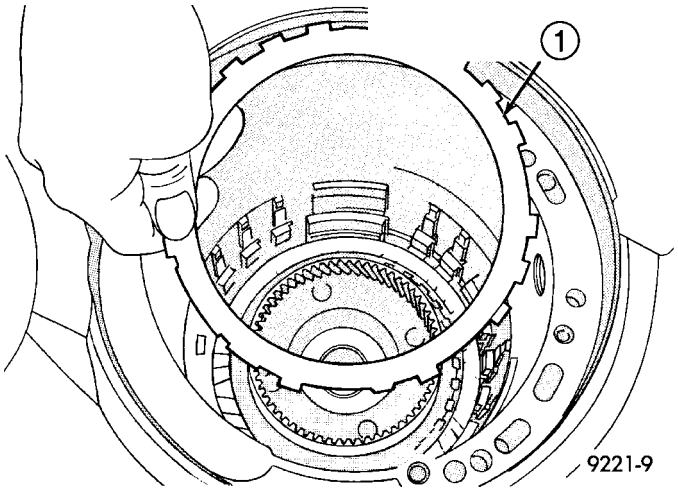


Fig. 57 Remove Low/Reverse Reaction Plate

1 - LOW/REVERSE REACTION PLATE (FLAT SIDE UP)

(38) Remove one (1) low/reverse clutch disc to facilitate snap ring removal (Fig. 58).

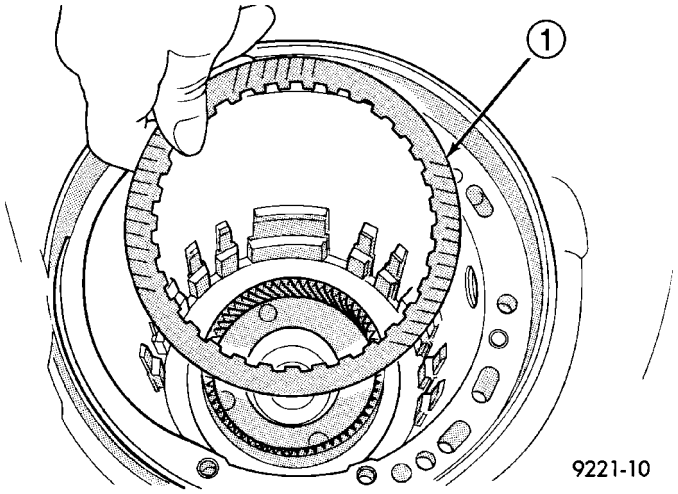


Fig. 58 Remove One Disc

1 - ONE DISC FROM LOW/REVERSE CLUTCH

(39) Remove the low/reverse reaction plate snap ring (Fig. 59).

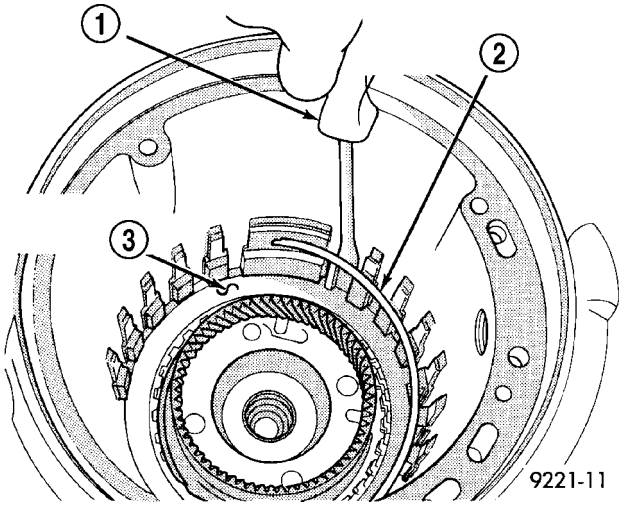


Fig. 59 Remove Low/Reverse Reaction Plate Snap Ring

1 - SCREWDRIVER
2 - LOW/REVERSE REACTION PLATE FLAT SNAP RING
3 - DO NOT SCRATCH CLUTCH PLATE

(40) Remove the low/reverse clutch pack (Fig. 60).

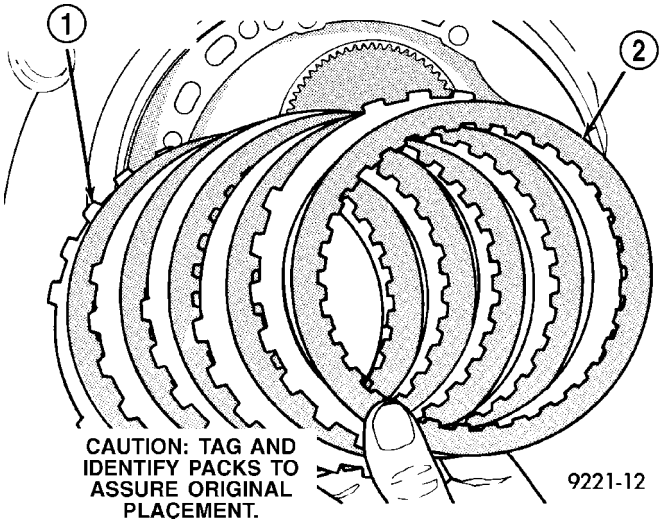


Fig. 60 Remove Low/Reverse Clutch Pack

1 - CLUTCH PLATES (5)
2 - CLUTCH DISCS (5)

CAUTION: Failure to grind and open stakes of the output shaft nut will result in thread damage to the shaft during nut removal.

WARNING: WEAR SAFETY GOGGLES WHILE GRINDING STAKE NUTS.

(41) Using a die grinder or equivalent, grind the stakes in the shoulder of the shaft nuts as shown in (Fig. 61) (Fig. 62). Do not grind all the way through the nut and into the shaft. There are two stakes on each nut.

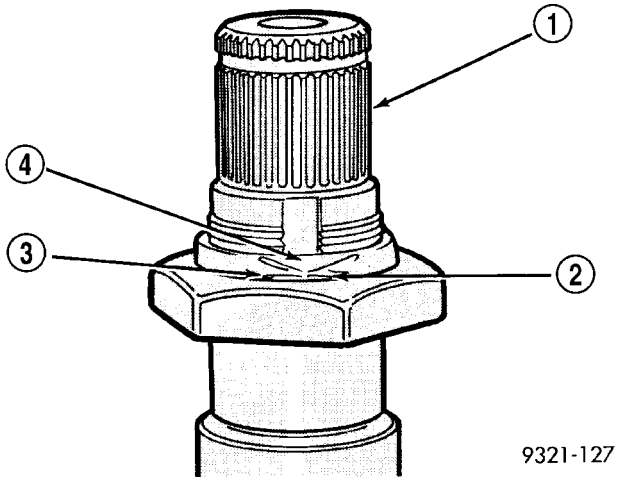


Fig. 61 Grinding Stakes

1 - TRANSFER SHAFT
2 - GRIND HERE
3 - GRIND HERE
4 - NUT STAKE

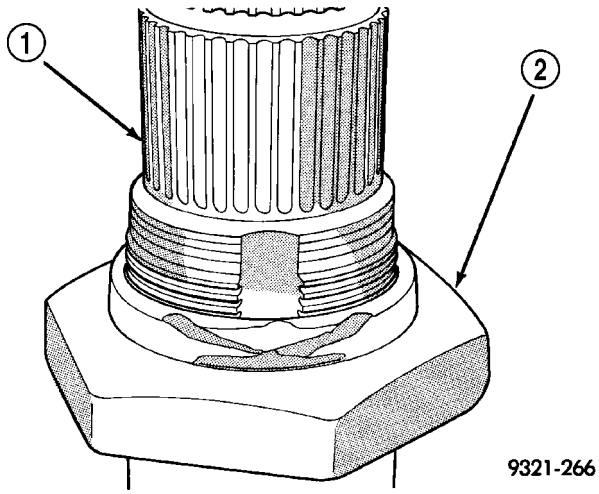


Fig. 62 Stake Grinding Pattern

1 - TRANSFER SHAFT
2 - TRANSFER SHAFT NUT

AUTOMATIC TRANSMISSION - 42RLE (Continued)

(42) Using a small chisel, carefully open the stakes on nut (Fig. 63).

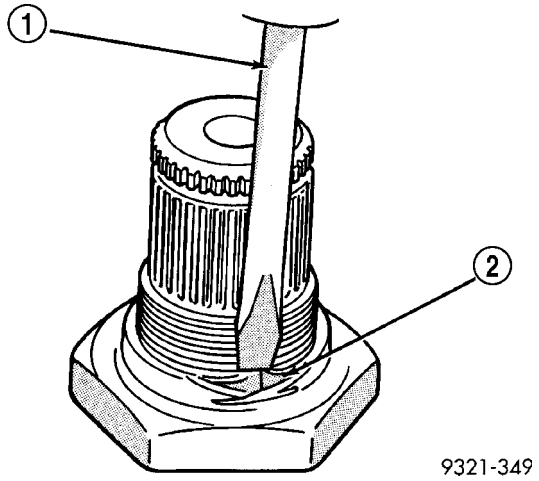


Fig. 63 Opening Nut Stakes

1 - CHISEL
2 - NUT STAKE

(43) Use special tool 6497 and 6498A to remove the transfer shaft nut or the output shaft nut (Fig. 64).

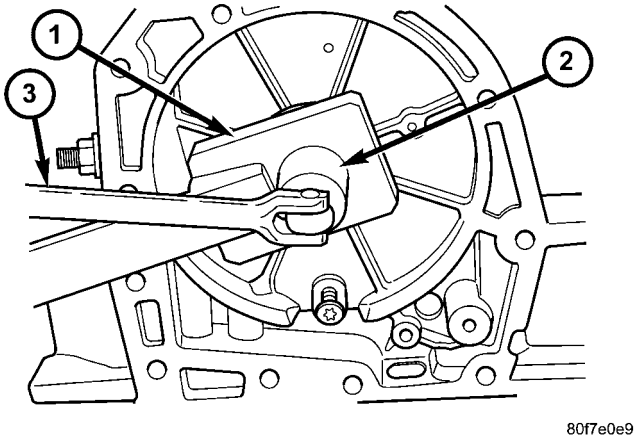


Fig. 64 Remove Output Shaft Nut

1 - SPECIAL TOOL 6497
2 - SPECIAL TOOL 6498A
3 - BREAKER BAR

(44) Remove the output shaft from case using a shop press (Fig. 65).

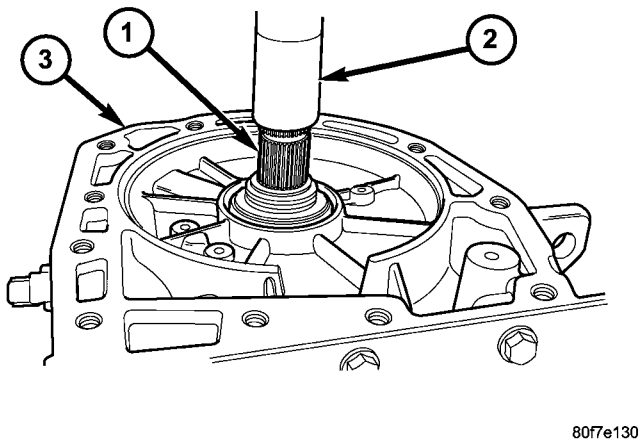


Fig. 65 Use Arbor Press to Remove Output Shaft from Case

1 - OUTPUT SHAFT
2 - ARBOR PRESS
3 - TRANSMISSION CASE

Use special tool 6596 with a shop press to remove the front output shaft bearing cup (Fig. 66).

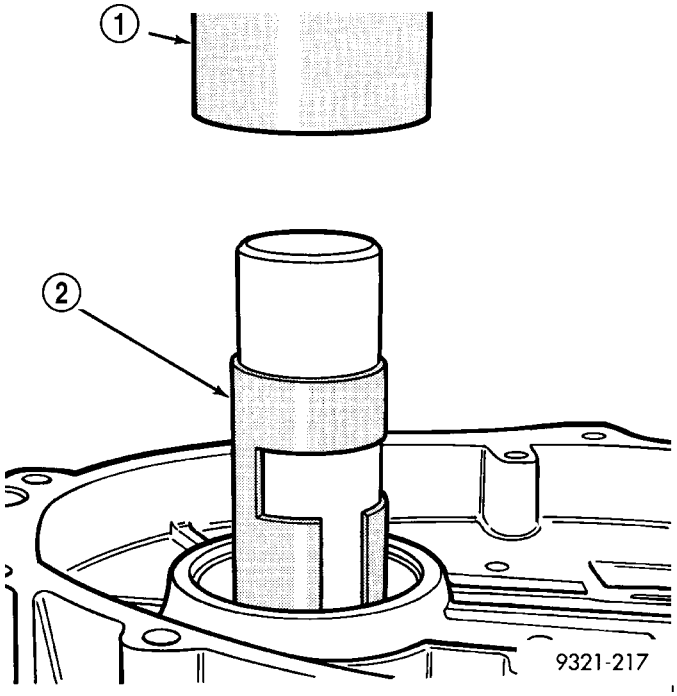


Fig. 66 Remove Front Bearing Cup - Typical

1 - ARBOR PRESS
2 - SPECIAL TOOL 6596

(45) Use special tool 6597 and handle C-4171 and C-4171-2 to press the rear output shaft bearing cup rearward (Fig. 67).

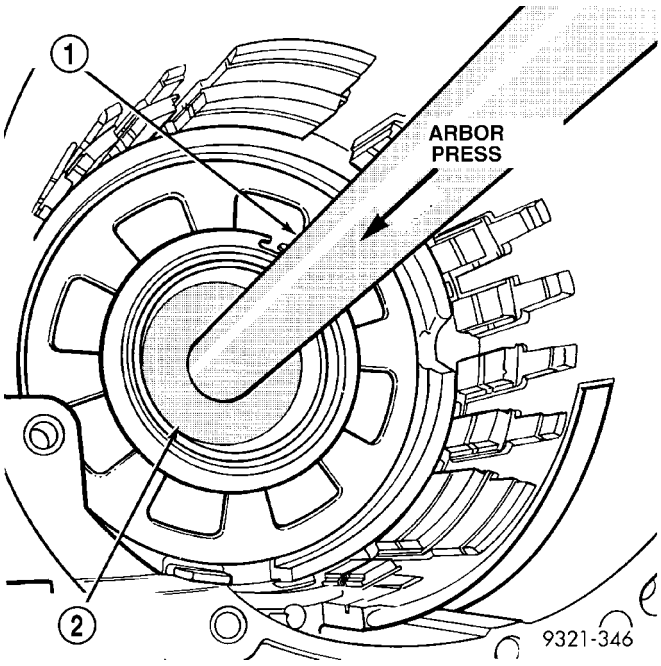


Fig. 67 Remove Rear Bearing Cup

1 - SPECIAL TOOL 4171 AND 4171-2
2 - SPECIAL TOOL 6597

(46) Remove the rear carrier front bearing cone (Fig. 68).

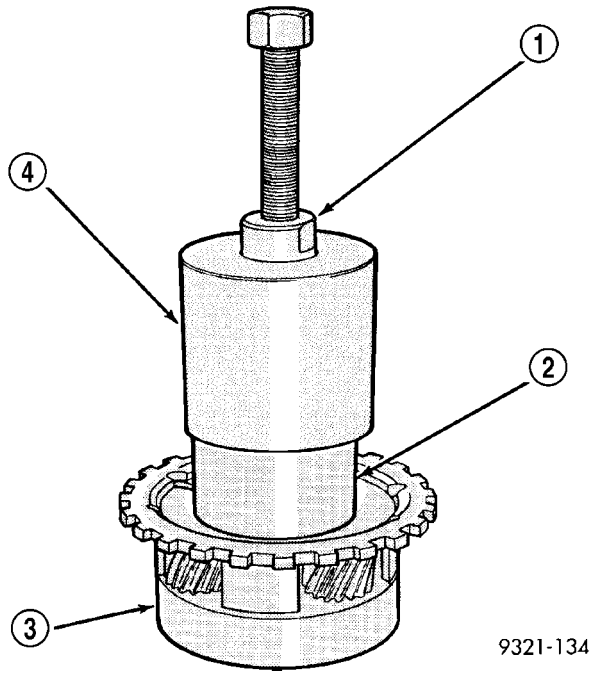


Fig. 68 Remove Rear Carrier Front Bearing Cone

1 - SPECIAL TOOL 5048-1
2 - SPECIAL TOOL 6545
3 - REAR CARRIER
4 - SPECIAL TOOL 5048

(47) Install and load compressor (Fig. 69) as shown in (Fig. 70).

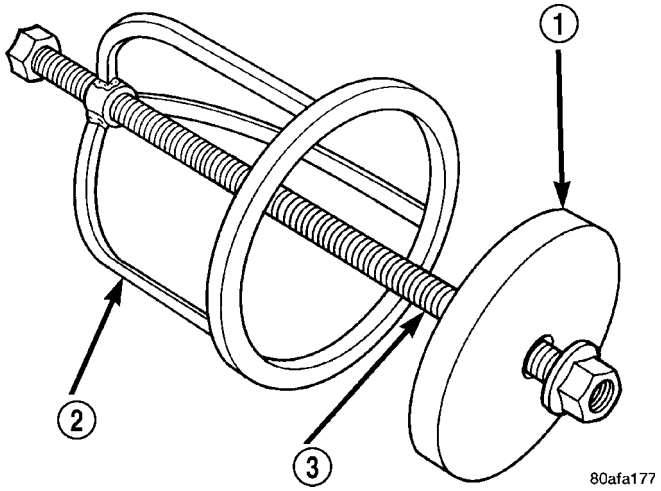


Fig. 69 Low/Reverse Spring Compressor Tool

1 - TOOL 6057
2 - TOOL 5059
3 - TOOL 5058-3

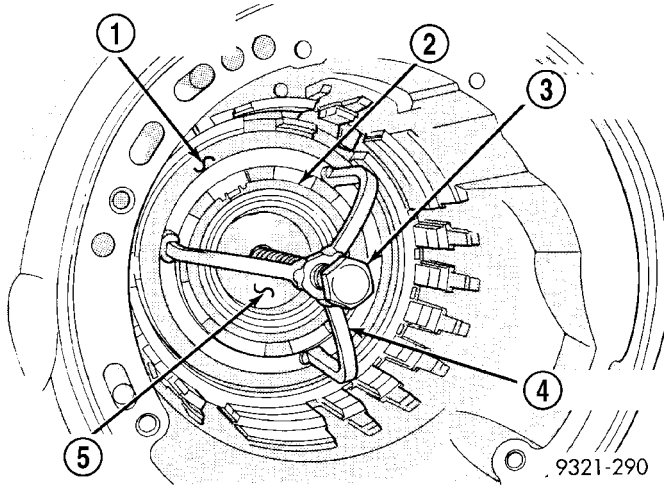


Fig. 70 Compressor Tool in Use

1 - LOW/REVERSE CLUTCH RETURN SPRING
2 - SNAP RING (INSTALL AS SHOWN)
3 - TOOL 5058A-3
4 - TOOL 5059A
5 - SPECIAL TOOL 6057

AUTOMATIC TRANSMISSION - 42RLE (Continued)

(48) Remove the low/reverse belleville spring snap ring (Fig. 71).

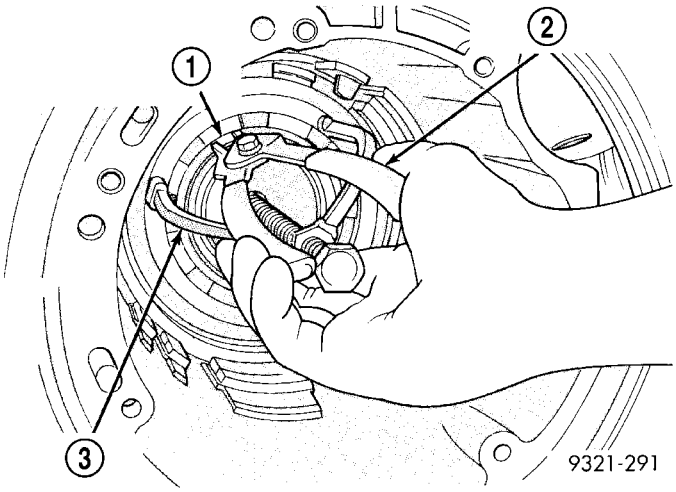


Fig. 71 Remove Snap Ring

1 - SNAP RING OPENING MUST BE BETWEEN SPRING LEVERS (AS SHOWN)
2 - SNAP RING PLIERS
3 - SPECIAL TOOL 5059A

(49) Remove the low/reverse piston belleville spring (Fig. 72).

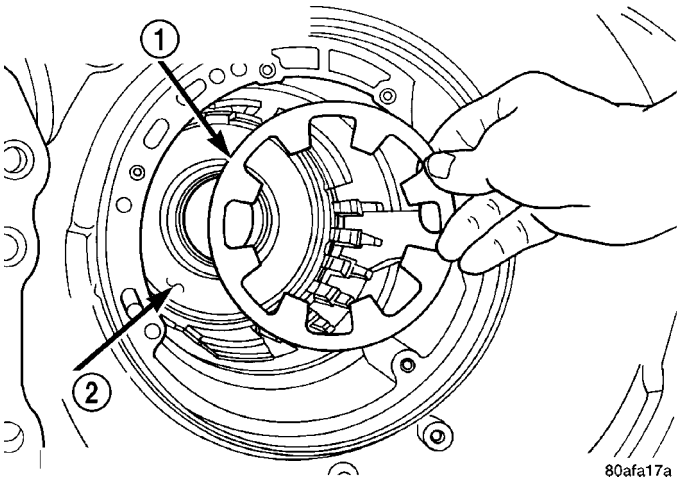


Fig. 72 Low/Reverse Piston Belleville Spring

1 - LOW/REVERSE PISTON RETURN SPRING
2 - PISTON

(50) Remove the park sprag pivot retaining screw.

(51) Drive out the anchor shaft using suitable punch (Fig. 73).

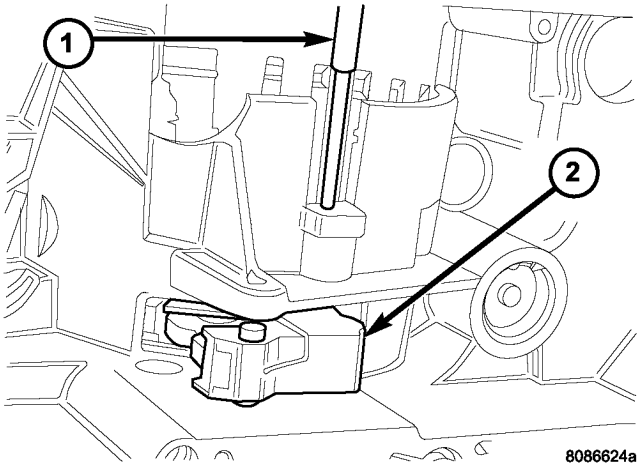


Fig. 73 Anchor Shaft Removal

1 - PIN PUNCH
2 - GUIDE BRACKET ASSEMBLY

(52) Remove the guide bracket pivot shaft (Fig. 74). Inspect all components (Fig. 75) for wear and replace if necessary.

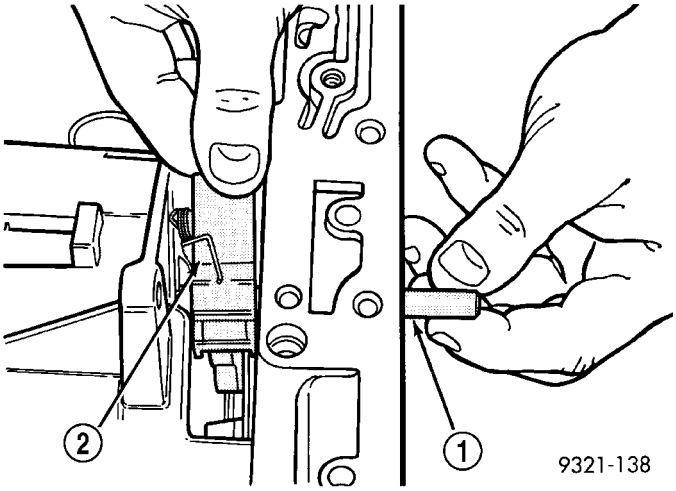


Fig. 74 Remove Guide Bracket Pivot Shaft

1 - PIVOT PIN
2 - GUIDE BRACKET ASSEMBLY

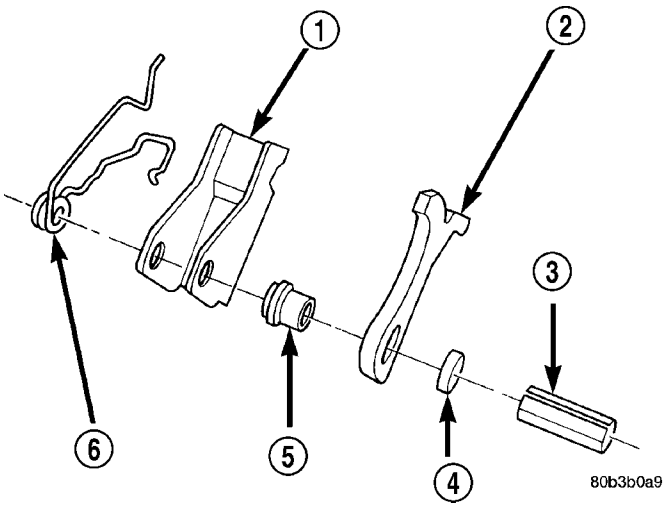


Fig. 75 Guide Bracket Disassembled

1 - GUIDE BRACKET
2 - PAWL
3 - SPLIT SLEEVE
4 - SPACER
5 - STEPPED SPACER
6 - ANTIRATCHET SPRING

NOTE: The Low/Reverse Clutch Piston has bonded seals which are not individually serviceable. Seal replacement requires replacement of the piston assembly.

(53) Remove the low/reverse clutch piston (Fig. 76).

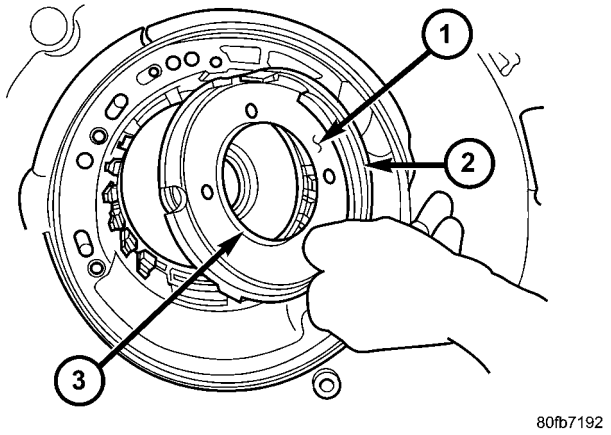


Fig. 76 Remove Low/Reverse Clutch Piston

1 - LOW/REVERSE CLUTCH PISTON
2 - BONDED SEAL
3 - BONDED SEAL

(54) Remove the low/reverse piston retainer screws.

(55) Remove low/reverse piston retainer (Fig. 77).

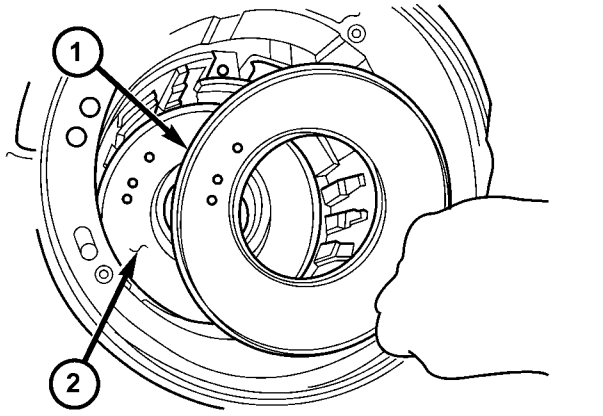


Fig. 77 Remove Piston Retainer

1 - LOW/REVERSE CLUTCH PISTON RETAINER
2 - GASKET

(56) Remove the low/reverse piston retainer gasket (Fig. 78).

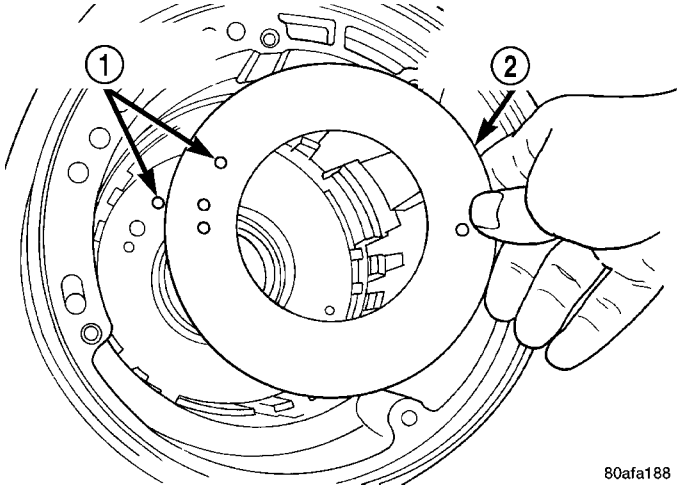


Fig. 78 Remove Piston Retainer Gasket

1 - GASKET HOLES MUST LINE UP
2 - LOW/REVERSE CLUTCH PISTON RETAINER GASKET

AUTOMATIC TRANSMISSION - 42RLE (Continued)

ASSEMBLY

NOTE: If the transmission assembly is being reconditioned (clutch/seal replacement) or replaced, it is necessary to perform the Quick Learn Procedure using the DRBIII® Scan Tool (Refer to 8 - ELECTRICAL/ELECTRONIC CONTROL MODULES/TRANSMISSION CONTROL MODULE - STANDARD PROCEDURE).

(1) Install the output bearing cups using Special Tool - 5050A (Fig. 79).

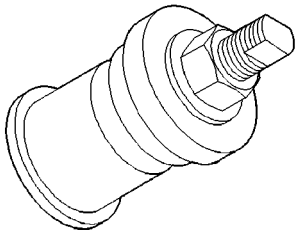


Fig. 79 Bearing Cup Installation Special Tool - 5050A

(2) Install low/reverse piston retainer gasket (Fig. 80).

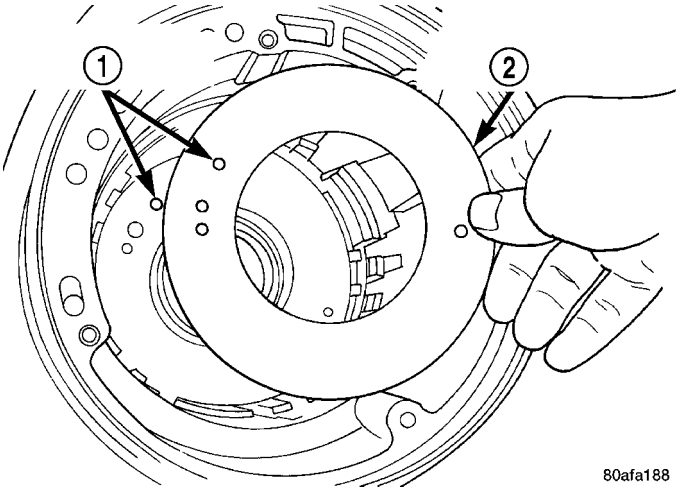


Fig. 80 Install Piston Retainer Gasket

1 - GASKET HOLES MUST LINE UP
2 - LOW/REVERSE CLUTCH PISTON RETAINER GASKET

(3) Install low/reverse piston retainer (Fig. 81).

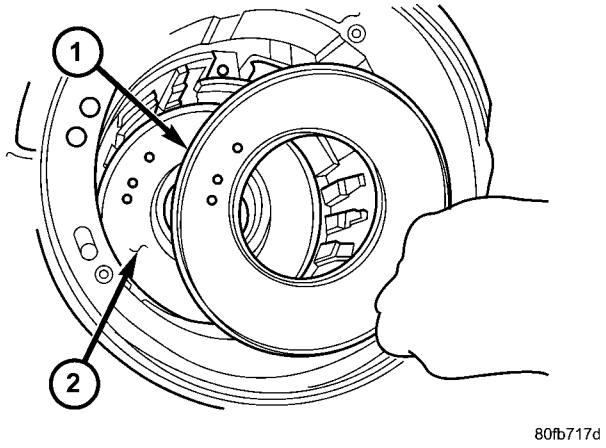


Fig. 81 Install Piston Retainer

1 - LOW/REVERSE CLUTCH PISTON RETAINER
2 - GASKET

(4) Install low/reverse piston retainer-to-case screws (Fig. 82) and torque to 5 N·m (45 in. lbs.).

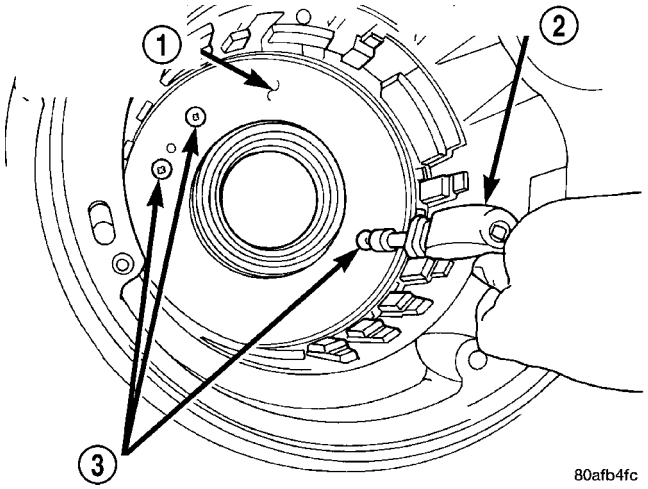


Fig. 82 Install Retainer Attaching Screws

1 - LOW/REVERSE CLUTCH PISTON RETAINER
2 - SCREWDRIVER
3 - TORX-LOC SCREWS

AUTOMATIC TRANSMISSION - 42RLE (Continued)

NOTE: The Low/Reverse Clutch Piston has bonded seals which are not individually serviceable. Seal replacement requires replacement of the piston assembly.

(5) Install low/reverse clutch piston (Fig. 83).

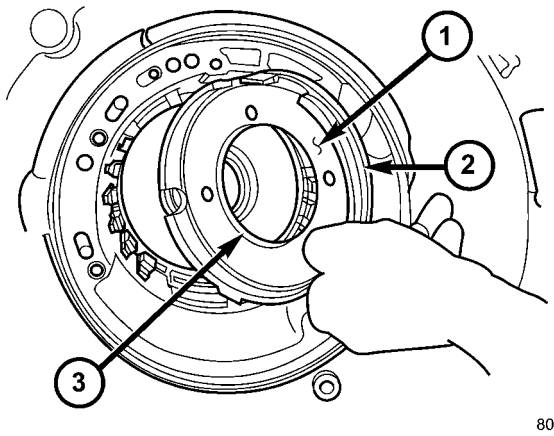


Fig. 83 Install Low/Reverse Clutch Piston

1 - LOW/REVERSE CLUTCH PISTON
2 - BONDED SEAL
3 - BONDED SEAL

(6) Assemble guide bracket assembly as shown in (Fig. 84) (Fig. 85).

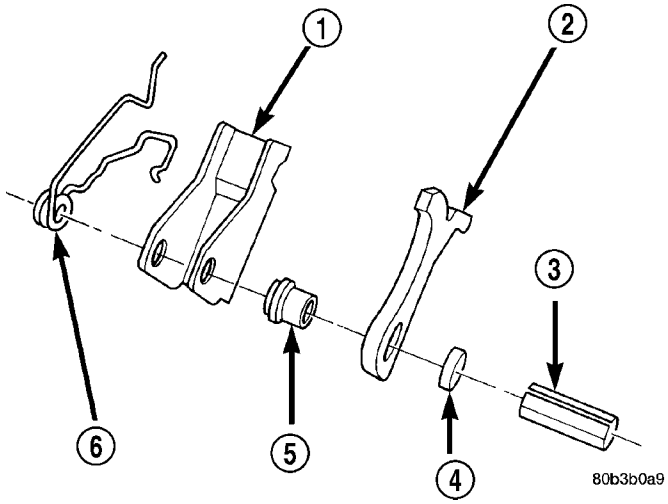


Fig. 84 Guide Bracket Assembly

1 - GUIDE BRACKET
2 - PAWL
3 - SPLIT SLEEVE
4 - SPACER
5 - STEPPED SPACER
6 - ANTIRATCHET SPRING

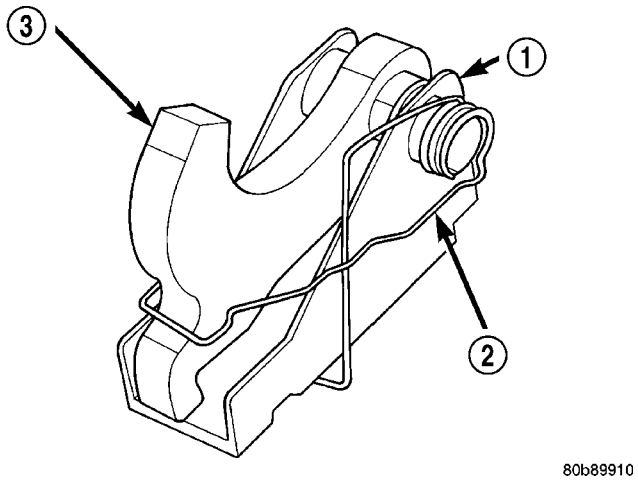


Fig. 85 Guide Bracket

1 - GUIDE BRACKET
2 - ANTIRATCHET SPRING (MUST BE ASSEMBLED AS SHOWN)
3 - PAWL

(7) Install guide bracket pivot shaft (Fig. 86).

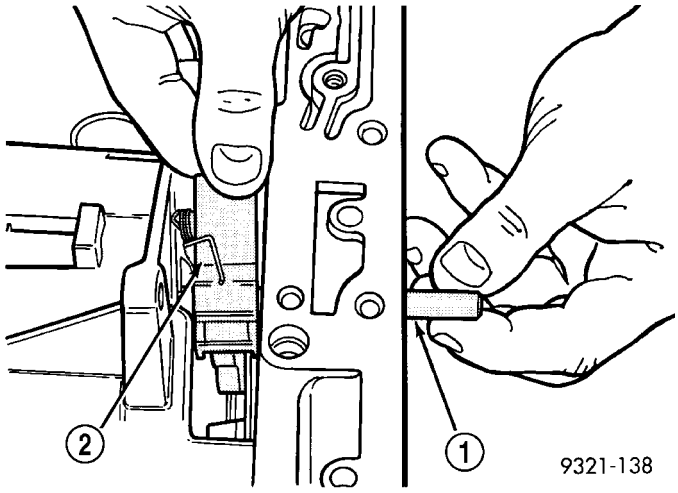


Fig. 86 Install Guide Bracket Pivot Pin

1 - PIVOT PIN
2 - GUIDE BRACKET ASSEMBLY

CAUTION: When installing, be sure guide bracket and split sleeve touch the rear of the transmission case.

(8) Install park sprag pivot retaining screw and torque to 4.5 N·m (40 in. lbs.).

AUTOMATIC TRANSMISSION - 42RLE (Continued)

(9) Install low/reverse piston belleville spring into position (Fig. 87).

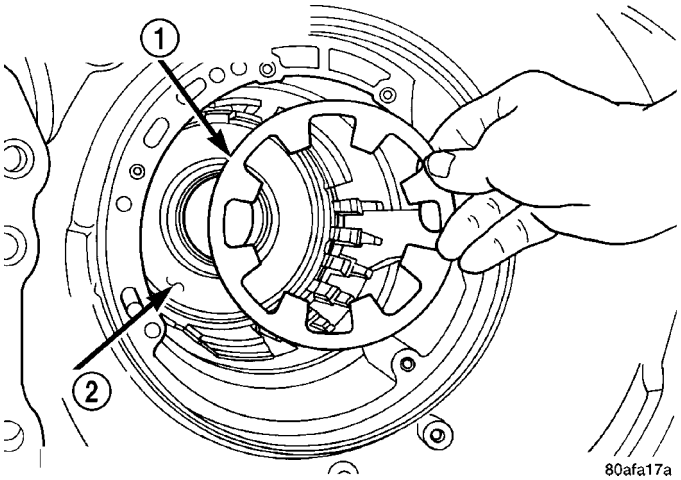


Fig. 87 Install Low/Reverse Piston Return Spring

1 - LOW/REVERSE PISTON RETURN SPRING
2 - PISTON

(10) Install and load low/reverse spring compressor tool as shown in (Fig. 88) (Fig. 89) to facilitate snap ring installation.

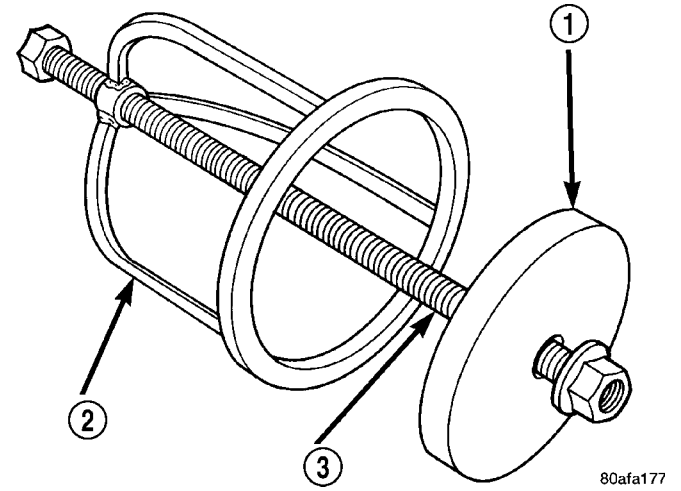


Fig. 88 Low/Reverse Spring Compressor Tool

1 - TOOL 6057
2 - TOOL 5059
3 - TOOL 5058-3

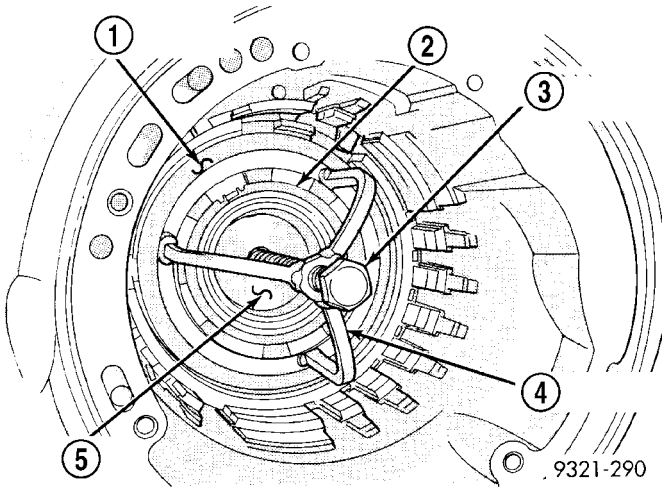


Fig. 89 Compressor Tool in Use

1 - LOW/REVERSE CLUTCH RETURN SPRING
2 - SNAP RING (INSTALL AS SHOWN)
3 - TOOL 5058A-3
4 - TOOL 5059A
5 - SPECIAL TOOL 6057

(11) Install snap ring and remove compressor tool (Fig. 90).

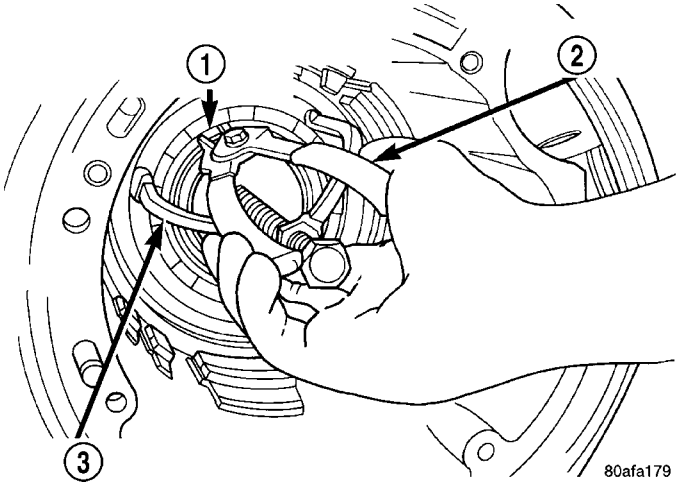


Fig. 90 Install Snap Ring

1 - SNAP RING OPENING MUST BE BETWEEN SPRING LEVERS (AS SHOWN)
2 - SNAP RING PLIERS
3 - TOOL 6057

AUTOMATIC TRANSMISSION - 42RLE (Continued)

(12) Install rear carrier front bearing cone (Fig. 91).

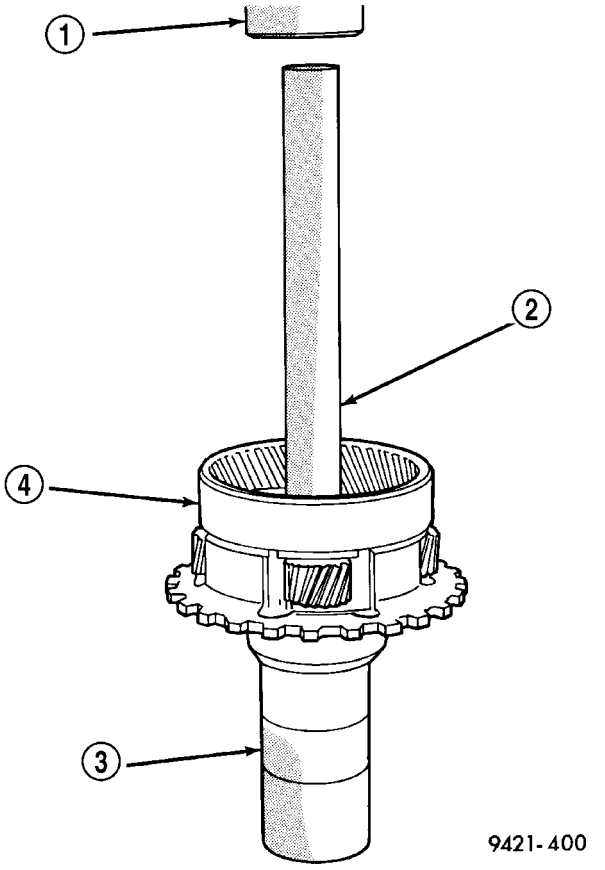


Fig. 91 Install Rear Carrier Front Bearing Cone

1 - ARBOR PRESS
2 - SPECIAL TOOL C-4171
3 - SPECIAL TOOL 6052
4 - REAR CARRIER

(13) Check output bearing preload. **Output bearing preload must be checked and/or adjusted if any of the following items have been replaced:**

- Output shaft (rear carrier assembly)
- Output shaft bearings
- Transmission case

(a) **PRELOAD CHECK/SHIM SELECTION:** Install rear output shaft bearing cone and special tool 6618A (Fig. 92).

(b) Install special tool 6618A (Fig. 93). Lightly tighten retaining screws. Screws should be below the plate surface, but do not snug screws.

(c) Turn case over on arbor press so that the plate is resting on the press base. **CAUTION: The output shaft will extend through the hole of tool 6618A. Ensure your press table has clearance for the output shaft.**

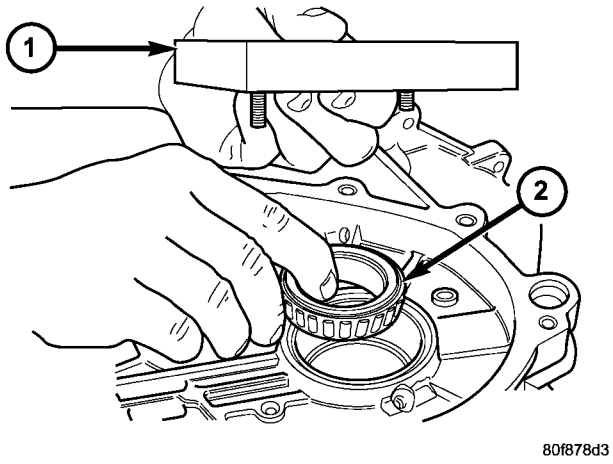


Fig. 92 Bearing Installation

1 - SPECIAL TOOL 6618-A
2 - REAR OUTPUT SHAFT BEARING

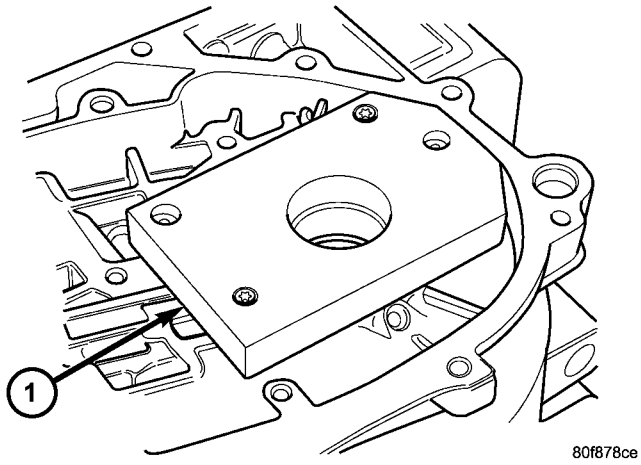


Fig. 93 Special Tool Installed

1 - SPECIAL TOOL 6618-A

AUTOMATIC TRANSMISSION - 42RLE (Continued)

(d) Install shim on output shaft (Fig. 94). Apply small amount of petrolatum onto the shim to hold it in place. Use the original shim as a starting point. If original shim is not available, use the thickest shim available.

(e) Install output shaft/rear carrier into rear bearing. The shaft must be pressed into position. Use special tool MD-998911 (Disc) and C- 4171 and C4171-2 (Handle) to press shaft into rear bearing (Fig. 95).

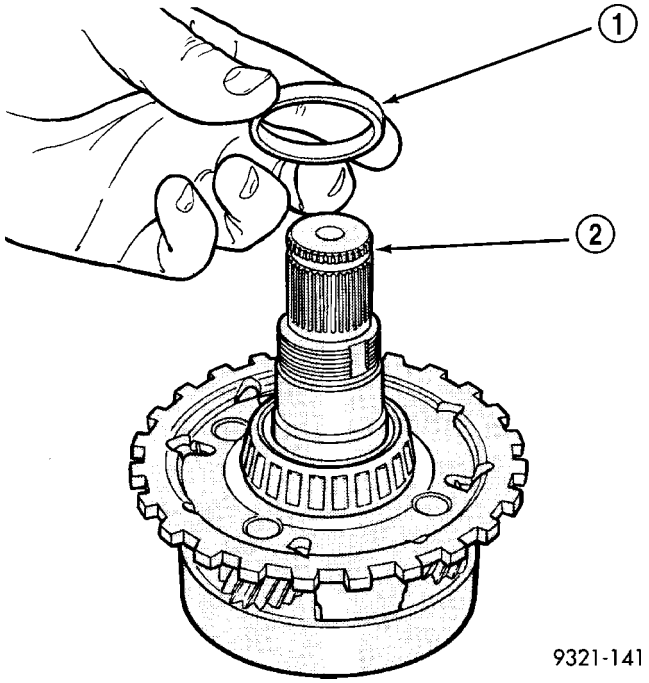


Fig. 94 Shim Installation

1 - SHIM
2 - OUTPUT SHAFT

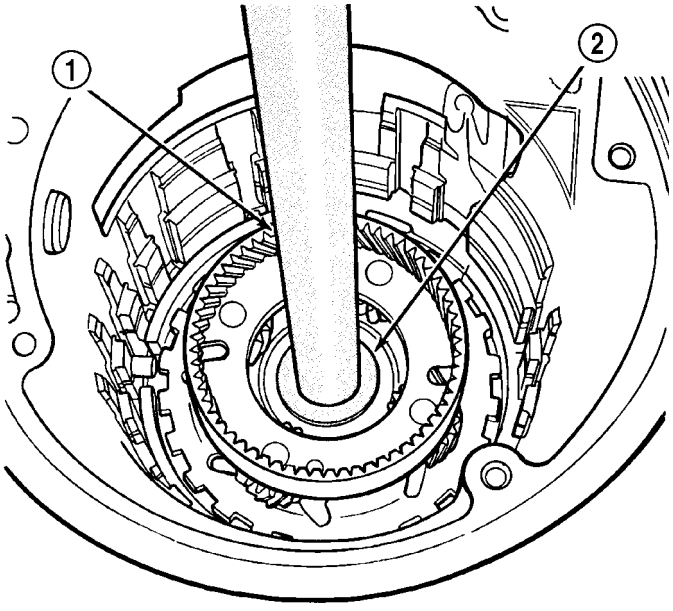


Fig. 95 Press Shaft Into Case

1 - SPECIAL TOOL C-4171 AND C-4171-2
2 - SPECIAL TOOL MD-998911

(f) **Do not re-use old output shaft nut because the removed stake weakens the nut flange.** Using special tools 6497 and 6498-A, install new output shaft nut. Do not reuse old output shaft nut. Tighten new output shaft nut to 271 N·m (200 ft. lbs.).

(g) Check the turning torque of the output shaft (Fig. 96). The shaft should have 1 to 8 in. lbs. of turning torque. If the turning torque is **higher than** 8 in. lbs., install a thicker shim. If turning torque is **less than** 1 in. lb., install a thinner shim. Make sure there is no end play.

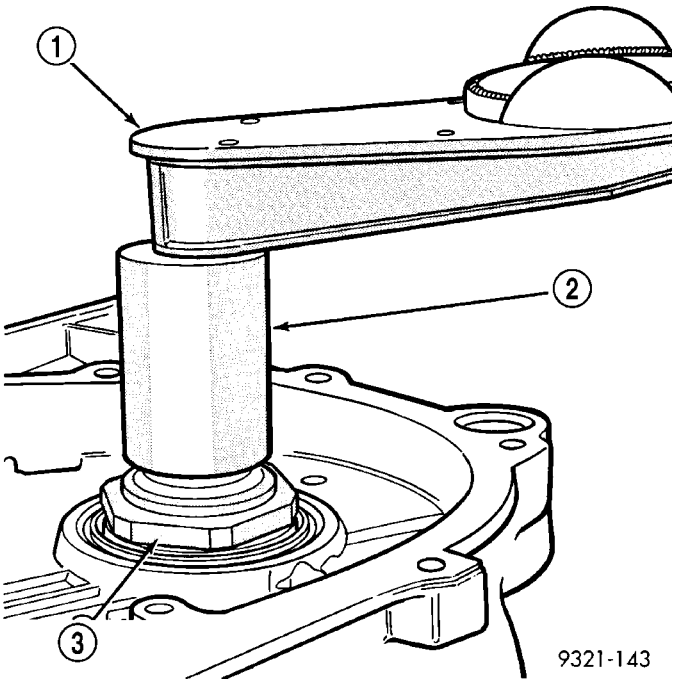


Fig. 96 Checking Turning Torque

1 - TORQUE WRENCH
2 - SPECIAL TOOL 6498-A
3 - OUTPUT SHAFT NUT

AUTOMATIC TRANSMISSION - 42RLE (Continued)

(h) The new nut must be staked after the correct turning torque is obtained (Fig. 97) (Fig. 98). Use special tool 6639 to stake output shaft nut. **CAUTION: Failure to stake nut could allow the nut to back-off during use.**

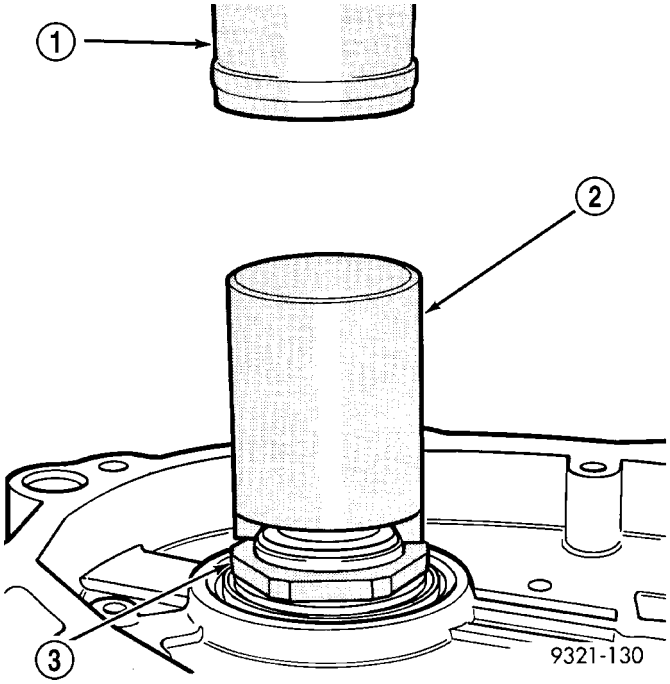


Fig. 97 Staking Output Shaft Nut - Typical

1 - ARBOR PRESS
2 - STAKING TOOL - 6639
3 - NEW NUT

(14) Install low/reverse clutch pack (Fig. 99). Leave uppermost disc out to facilitate snap ring installation.

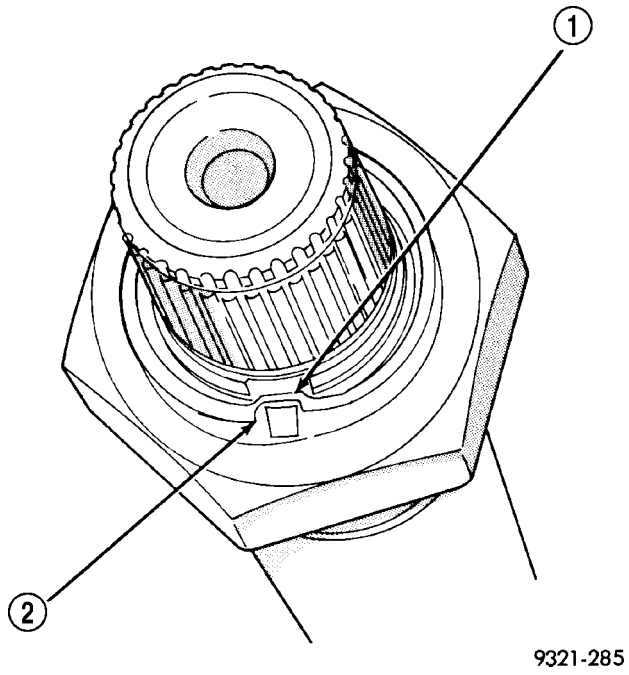


Fig. 98 Properly Staked Nut

1 - BOTTOMED IN SLOT
2 - CORRECTLY STAKED NUT

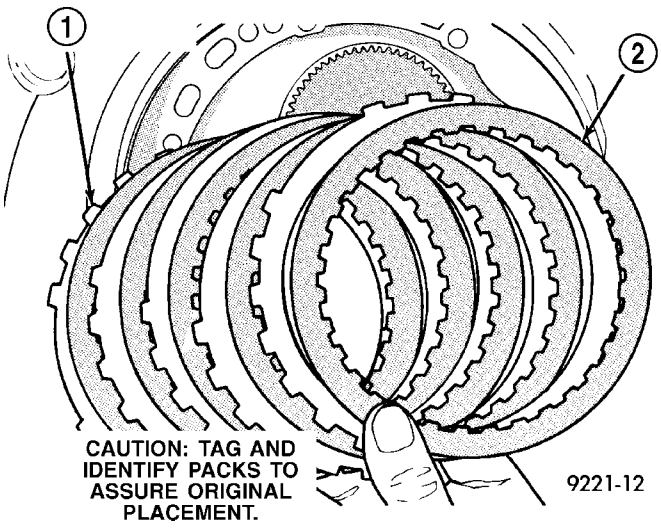


Fig. 99 Install Low/Reverse Clutch Pack

1 - CLUTCH PLATES (5)
2 - CLUTCH DISCS (5)

AUTOMATIC TRANSMISSION - 42RLE (Continued)

(15) Install low/reverse reaction plate snap ring (Fig. 100).

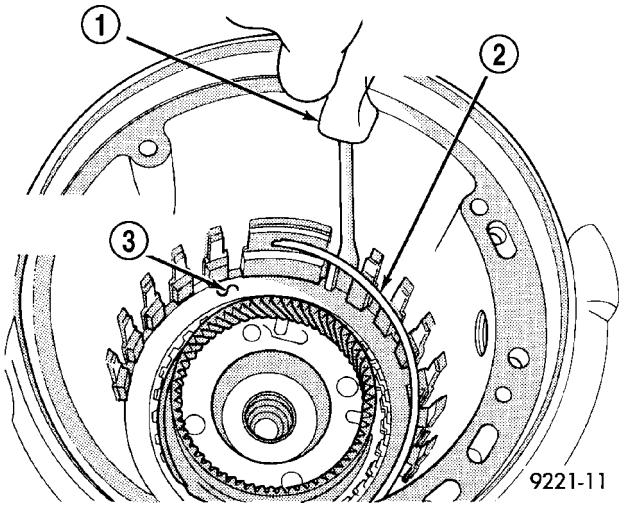


Fig. 100 Install Low/Reverse Reaction Plate Snap Ring

1 - SCREWDRIVER
2 - LOW/REVERSE REACTION PLATE FLAT SNAP RING
3 - DO NOT SCRATCH CLUTCH PLATE

(16) Install one low/reverse clutch disc (Fig. 101).

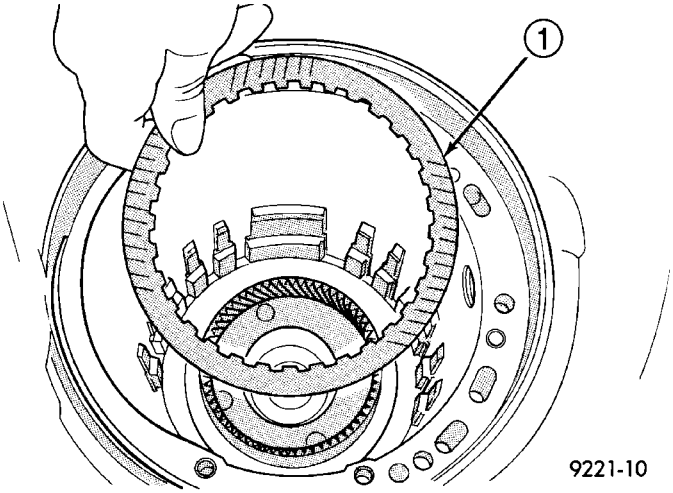


Fig. 101 Install One Disc

1 - ONE DISC FROM LOW/REVERSE CLUTCH

(17) Install low/reverse reaction plate with flat side up (Fig. 102).

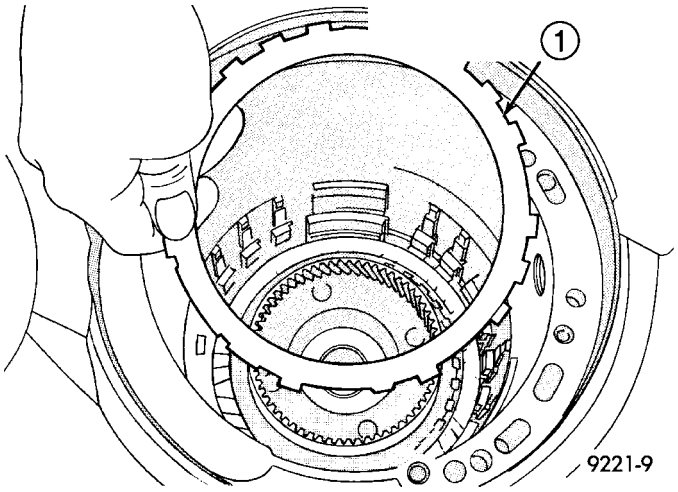


Fig. 102 Install Low/Reverse Reaction Plate

1 - LOW/REVERSE REACTION PLATE (FLAT SIDE UP)

(18) Install a new tapered snap ring (tapered side out) (Fig. 103). Make sure that the snap ring ends are oriented as shown (Fig. 104)

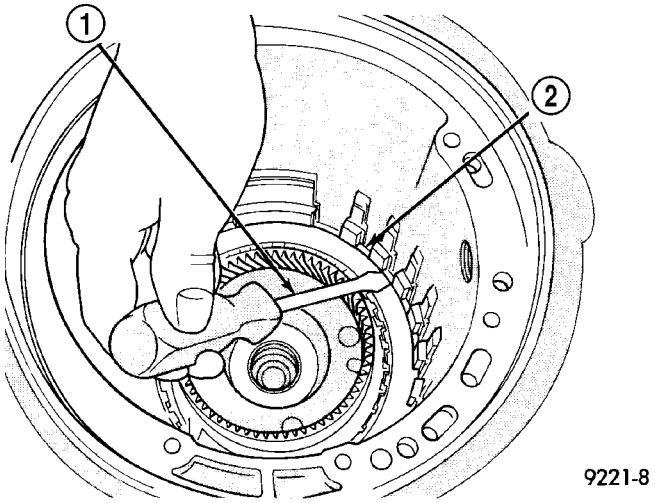


Fig. 103 Snap Ring Installed

1 - SCREWDRIVER
2 - TAPERED SNAP RING (INSTALL AS SHOWN)

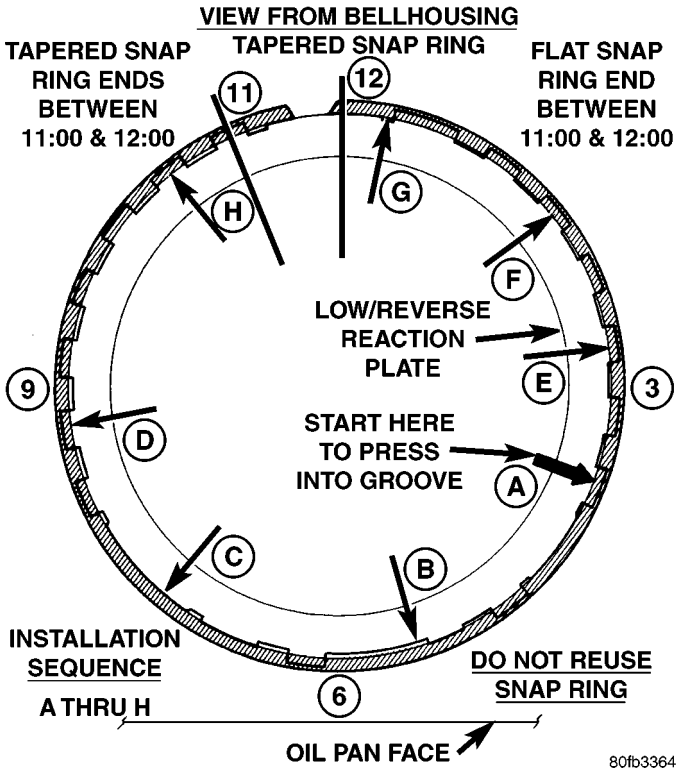


Fig. 104 Tapered Snap Ring Instructions

(19) Measure low/reverse clutch pack. Set up dial indicator as shown in (Fig. 105). Press down clutch pack with finger and zero dial indicator. Record measurement in four (4) places and take average reading. **Low/Reverse clutch pack clearance is 0.84 to 1.60 mm (0.033 to 0.063 inch).**

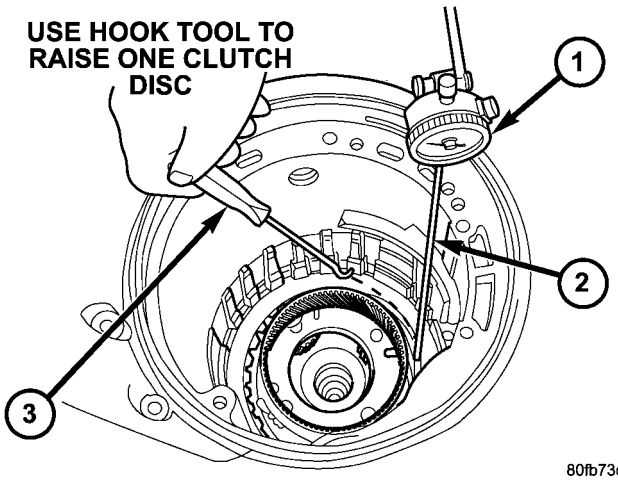


Fig. 105 Check Low/Reverse Clutch Clearance

1 - DIAL INDICATOR
2 - DIAL INDICATOR TIP TOOL 6268
3 - HOOK TOOL

(20) Select the proper low/reverse reaction plate to achieve specifications.

(21) Install 2/4 clutch pack (Fig. 106).

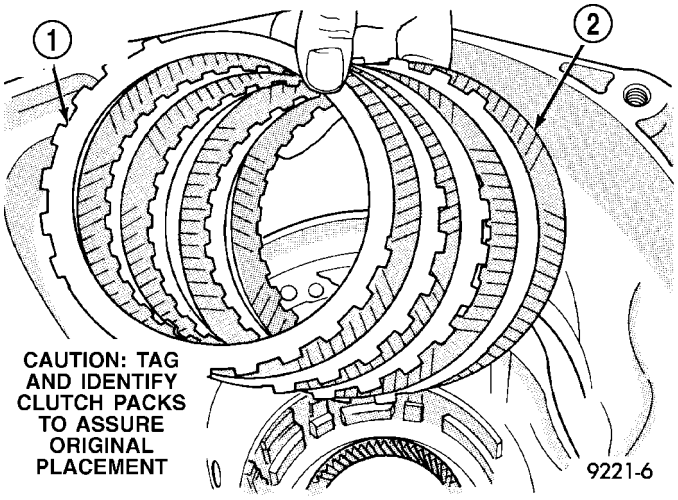


Fig. 106 Install 2/4 Clutch Pack

1 - CLUTCH PLATE (4)
2 - CLUTCH DISC (4)

NOTE: The 2/4 Clutch Piston has bonded seals which are not individually serviceable. Seal replacement requires replacement of the piston assembly.

(22) Install 2/4 clutch belleville spring (Fig. 107) (Fig. 108).

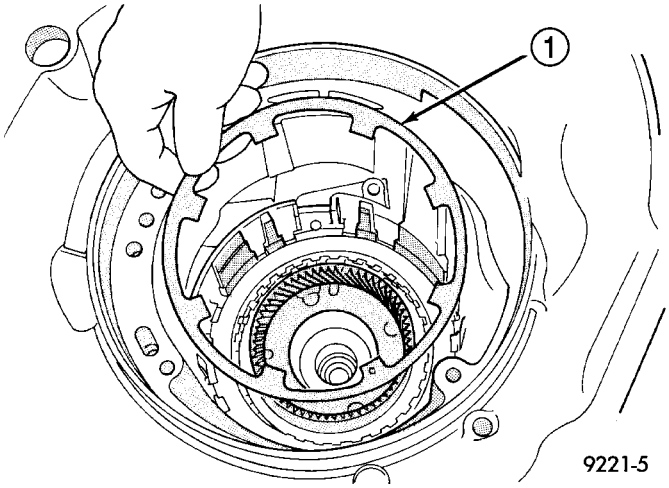


Fig. 107 Install 2/4 Clutch Return Spring

1 - 2/4 CLUTCH RETURN SPRING

AUTOMATIC TRANSMISSION - 42RLE (Continued)

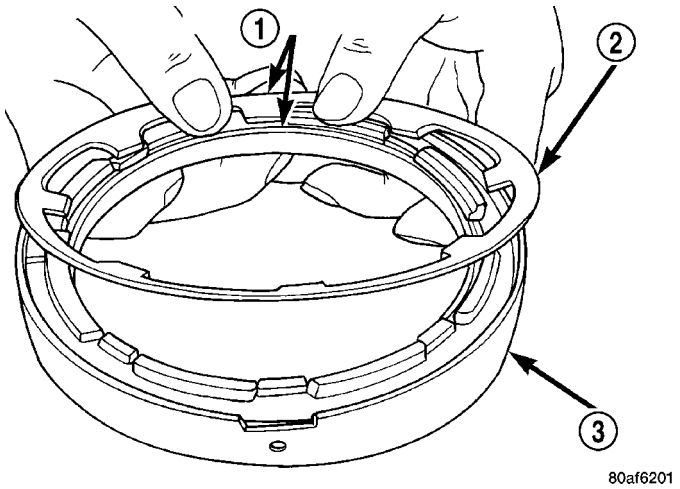


Fig. 108 Proper Orientation of 2/4 Clutch

1 - NOTE POSITION
2 - RETURN SPRING
3 - 2/4 CLUTCH RETAINER

(23) Install 2/4 clutch retainer (Fig. 109).

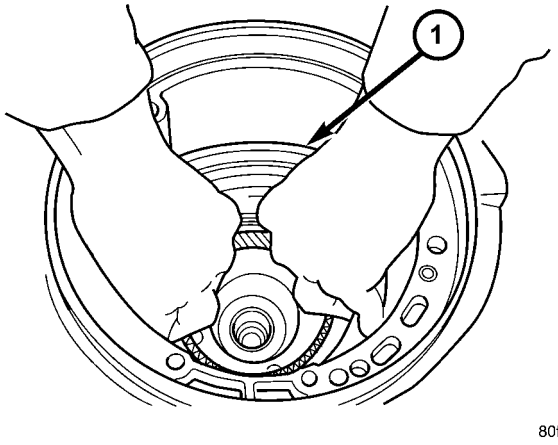


Fig. 109 Install 2/4 Clutch Retainer

1 - 2/4 CLUTCH RETAINER

NOTE: Verify that Tool 5058A is centered properly over the 2/4 clutch retainer before compressing. If necessary, fasten the 5058A bar to the bellhousing flange with any combination of locking pliers and bolts to center the tool properly.

(24) Set up Tool 5058 as shown in (Fig. 110). Compress 2/4 clutch just enough to facilitate snap ring installation.

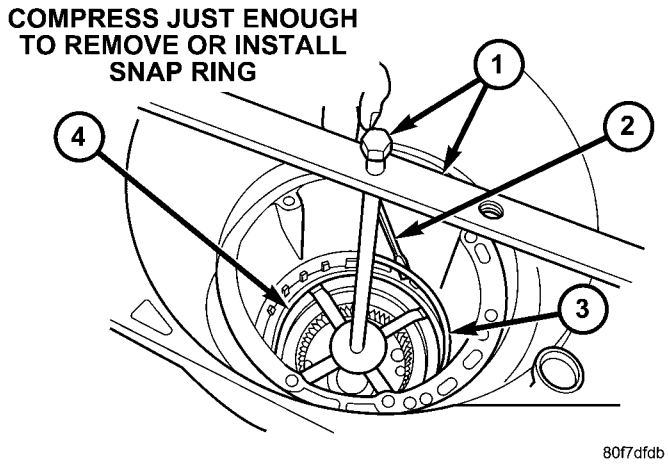


Fig. 110 Remove 2/4 Clutch Retainer Snap Ring

1 - TOOL 5058
2 - SCREWDRIVER
3 - SNAP RING
4 - 2/4 CLUTCH RETAINER

(25) **Measure 2/4 clutch clearance:** Set up dial indicator as shown in (Fig. 111). Press down clutch pack with finger and zero dial indicator. Record measurement in four (4) places and take average reading. **The 2/4 clutch pack clearance is 0.76 to 2.64 mm (0.030 to 0.104 inch).** If not within specifications, the clutch is not assembled properly or is excessively worn. **There is no adjustment for the 2/4 clutch clearance.**

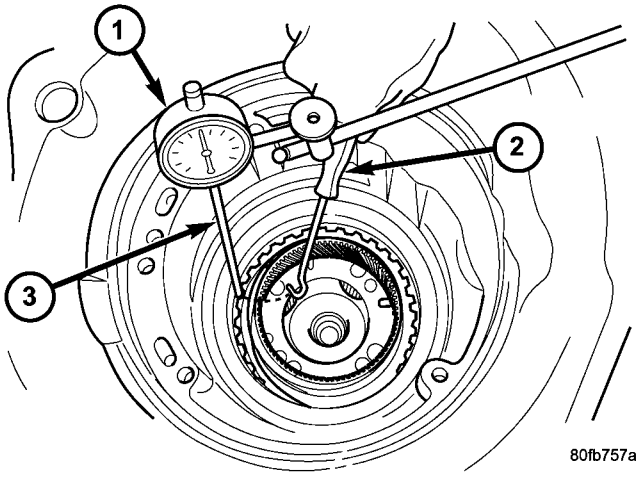


Fig. 111 Check 2/4 Clutch Clearance

1 - DIAL INDICATOR
2 - HOOK TOOL
3 - DIAL INDICATOR TIP TOOL 6268

(26) Install the #7 needle bearing to the rear sun gear (Fig. 112). **The number 7 needle bearing has three antireversal tabs and is common with the number 5 and number 2 position. The orientation should allow the bearing to seat flat against the rear sun gear. A small amount of petrolatum can be used to hold the bearing to the rear sun gear.**

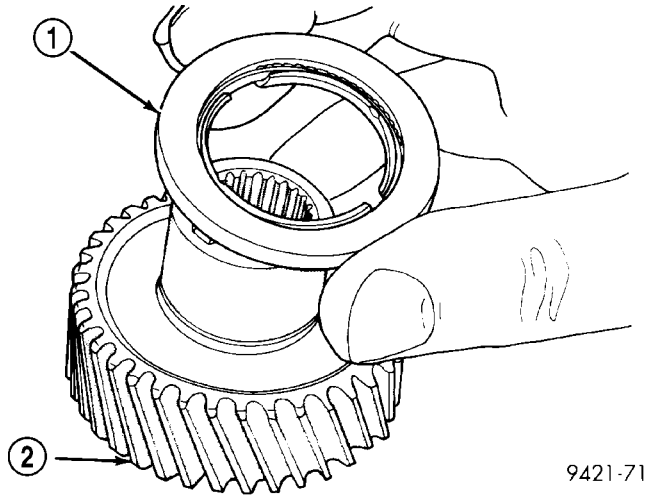


Fig. 112 Number 7 Bearing

1 - #7 BEARING
2 - REAR SUN GEAR

(27) Install rear sun gear and #7 needle bearing (Fig. 113).

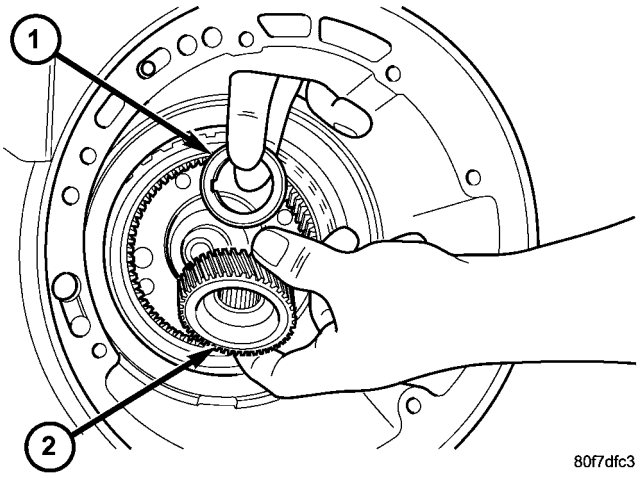


Fig. 113 Install Rear Sun Gear

1 - #7 NEEDLE BEARING
2 - REAR SUN GEAR

(28) Install front carrier/rear annulus assembly and #6 needle bearing (Fig. 114).

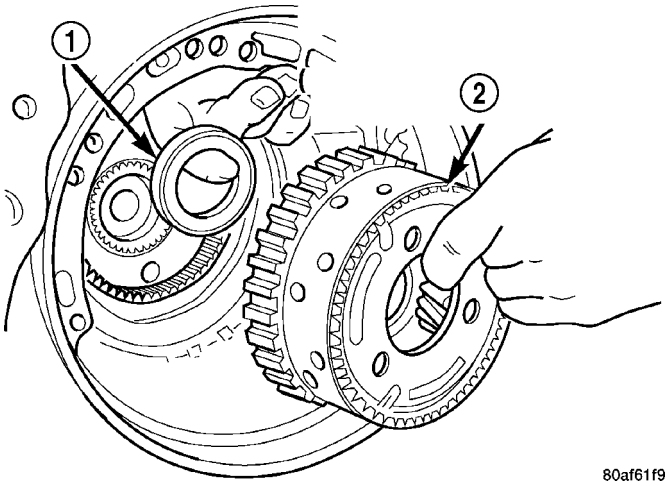


Fig. 114 Install Front Carrier/Rear Annulus

1 - #6 NEEDLE BEARING
2 - FRONT CARRIER AND REAR ANNULUS ASSEMBLY (TWIST AND PULL OR PUSH TO REMOVE OR INSTALL).

(29) Install front sun gear assembly and #4 thrust washer (Fig. 115).

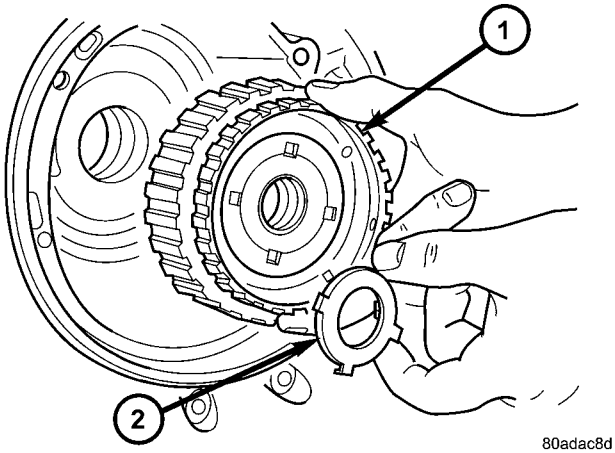


Fig. 115 Install Front Sun Gear Assembly

1 - FRONT SUN GEAR ASSEMBLY
2 - #4 THRUST WASHER (FOUR TABS)

AUTOMATIC TRANSMISSION - 42RLE (Continued)

(30) **Determine proper #4 thrust plate thickness.**

(a) Select the thinnest #4 thrust plate thickness.

(b) Install #4 thrust plate (Fig. 116) using petrolatum to hold into position.

(c) Install input clutch assembly. Ensure the input clutch assembly is completely seated by viewing position through input speed sensor hole. **If the speed sensor tone wheel is not centered in the opening, the input clutch assembly is not seated properly.**

(d) Remove the oil pump o-ring (Fig. 117) and install oil pump and gasket to transmission. Tighten the oil pump bolts to 30 N·m (265 in. lbs.). **Use screw-in dowels or phillips-head screwdrivers to align pump to case. Be sure to reinstall O-ring on oil pump after selecting the proper No. 4 thrust plate.**

(e) Measure the input shaft end play with the transmission in the vertical position. This will ensure that the measurement will be accurate.

(f) Set up and measure endplay using End Play Set 8266 and Dial Indicator Set C3339 as shown in (Fig. 118).

(g) Measure input shaft end play. **Input shaft end play must be 0.127 to 0.635 mm (0.005 to 0.025 inch).** For example, if end play reading is 0.055 inch, select No. 4 Thrust Plate which is 0.071 to 0.074 thick. This should provide an input shaft end play reading of 0.020 inch, which is within specifications.

(h) Remove oil pump, gasket, and input clutch assembly to gain access to and install proper #4 thrust plate.

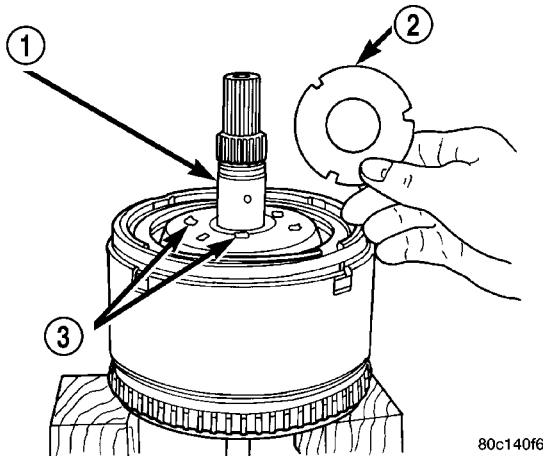


Fig. 116 Install #4 Thrust Plate

1 - OVERDRIVE SHAFT ASSEMBLY
2 - #4 THRUST PLATE (SELECT)
3 - PETROLATUM FOR RETENTION

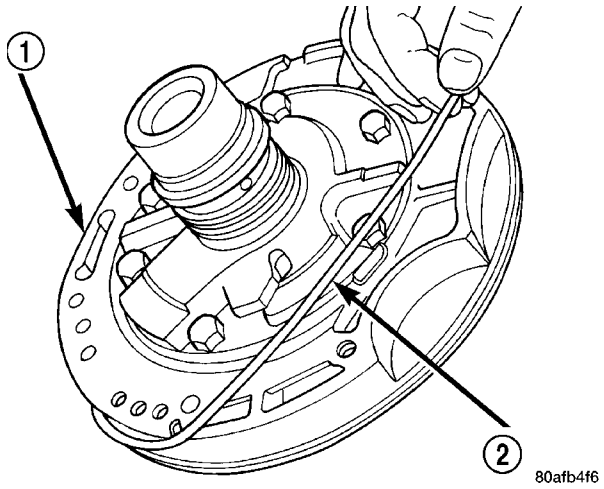


Fig. 117 Remove Oil Pump O-Ring

1 - OIL PUMP ASSEMBLY
2 - O-RING

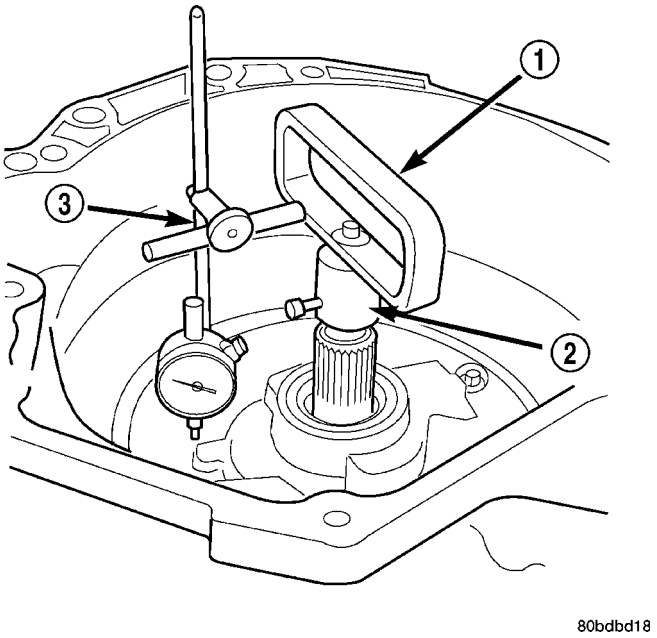


Fig. 118 Measure Input Shaft End Play Using Tool 8266 - Typical

1 - TOOL 8266-8
2 - TOOL 8266-2
3 - TOOL C-3339

(31) Install input clutch assembly with proper thrust plate (Fig. 119).

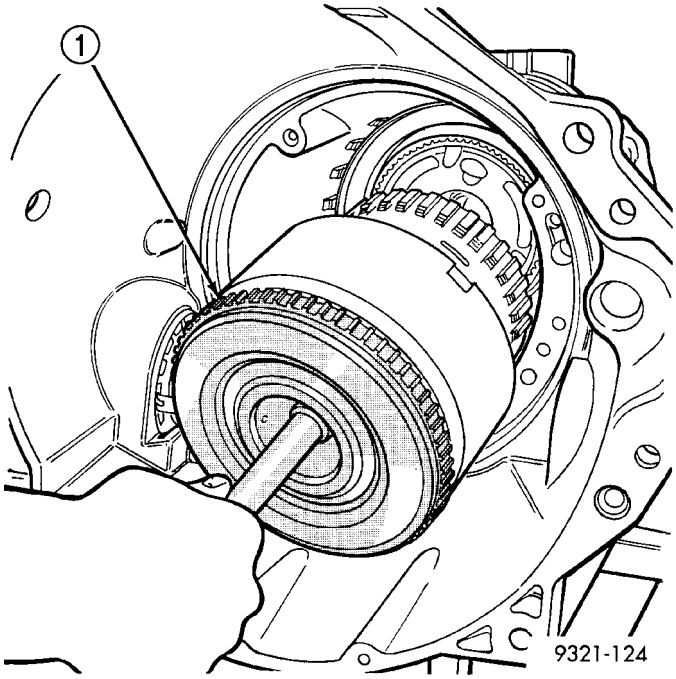


Fig. 119 Install Input Clutch Assembly

1 - INPUT CLUTCH ASSEMBLY

(32) Install #1 caged needle bearing (Fig. 120).

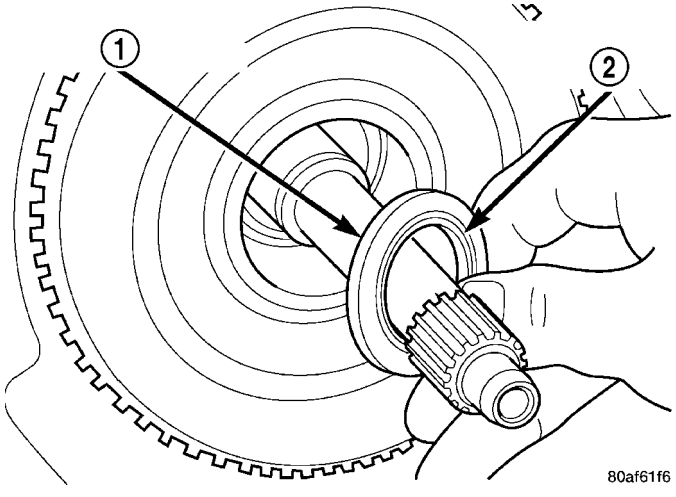


Fig. 120 Install No. 1 Caged Needle Bearing

1 - #1 CAGED NEEDLE BEARING
2 - NOTE: TANGED SIDE OUT

(33) Replace cooler by-pass valve if transmission failure has occurred (Fig. 121).

CAUTION: By-pass valve MUST be replaced if transmission failure occurs.

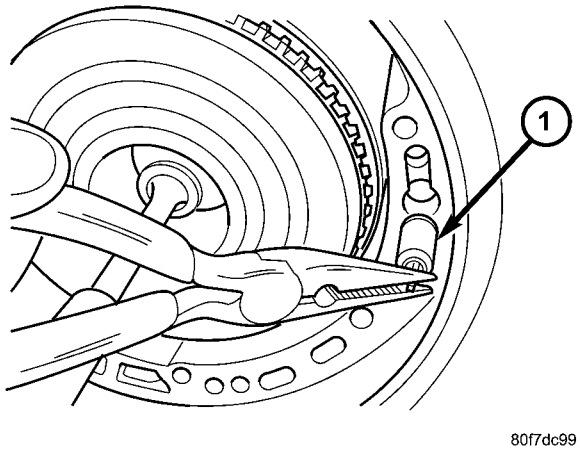


Fig. 121 Install By-Pass Valve

1 - BYPASS VALVE

NOTE: To align oil pump, gasket, and case during installation, use threaded dowels or phillips screwdrivers.

(34) Install oil pump gasket (Fig. 122).

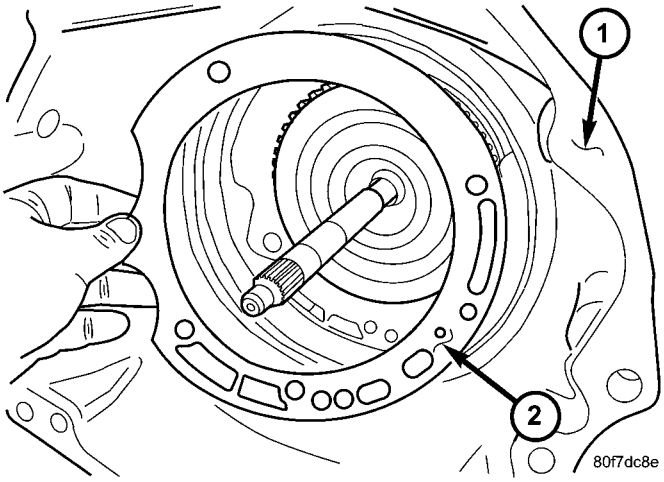


Fig. 122 Install Oil Pump Gasket

1 - BELLHOUSING
2 - OIL PUMP GASKET

AUTOMATIC TRANSMISSION - 42RLE (Continued)

(35) Install oil pump and torque the new oil pump-to-case bolts to 30 N·m (265 in. lbs.) (Fig. 123). Do not reuse original oil pump bolts.

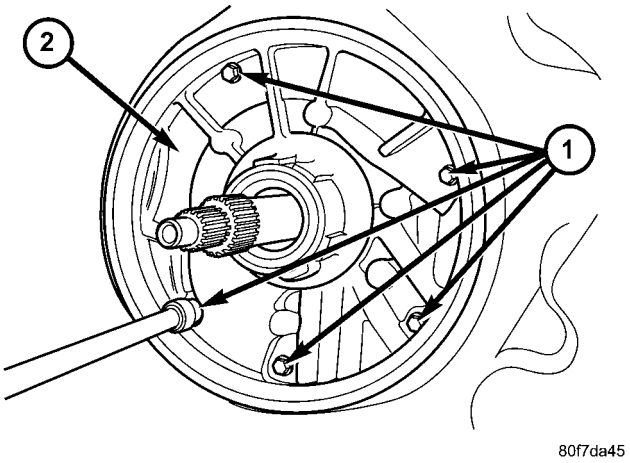


Fig. 123 Install Oil Pump Attaching Bolts

1 - BOLTS
2 - OIL PUMP

(36) Install low/reverse accumulator as shown in (Fig. 124).

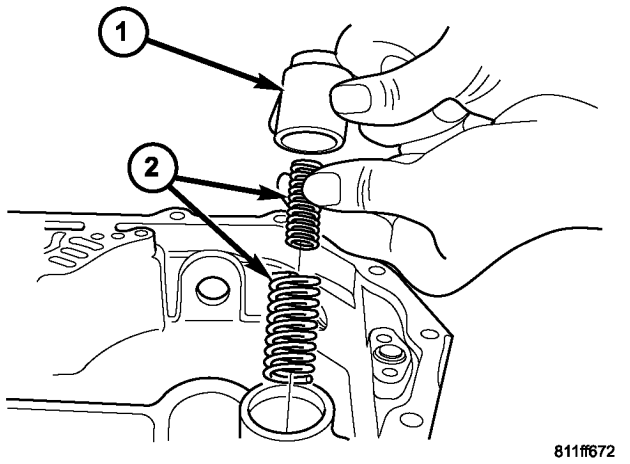


Fig. 124 Low/Reverse Accumulator

1 - PISTON
2 - RETURN SPRINGS

(37) Install low/reverse accumulator plug (Fig. 125).

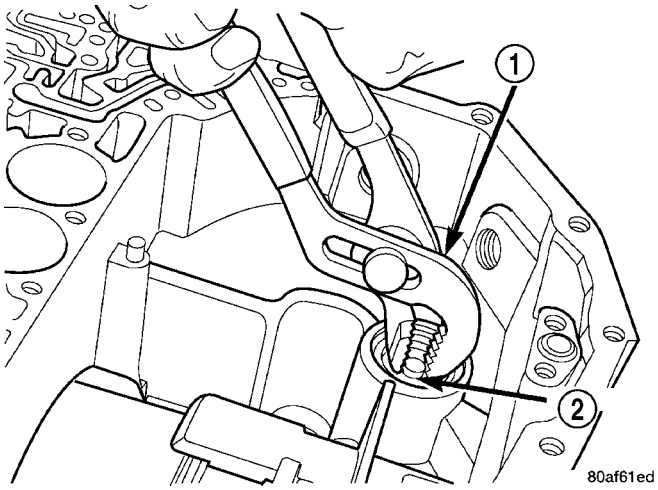


Fig. 125 Install Low/Reverse Accumulator Plug

1 - ADJUSTABLE PLIERS
2 - PLUG

(38) Install low/reverse accumulator snap ring (Fig. 126).

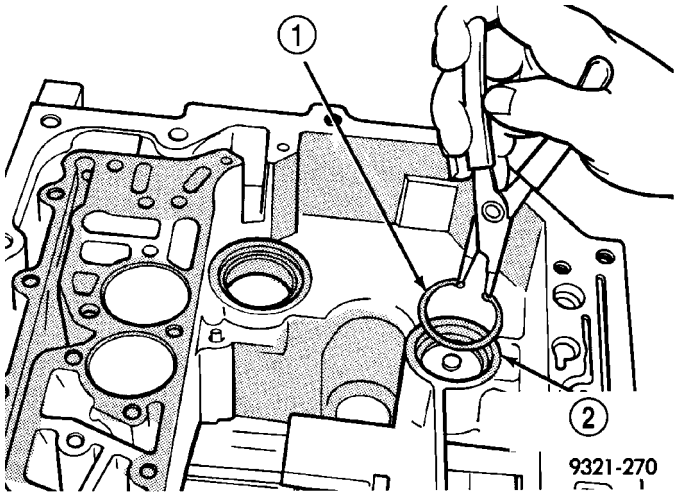


Fig. 126 Install Low/Reverse Accumulator Snap Ring

1 - SNAP RING
2 - LOW/REVERSE ACCUMULATOR

(39) Install underdrive and overdrive accumulators and springs (Fig. 127).

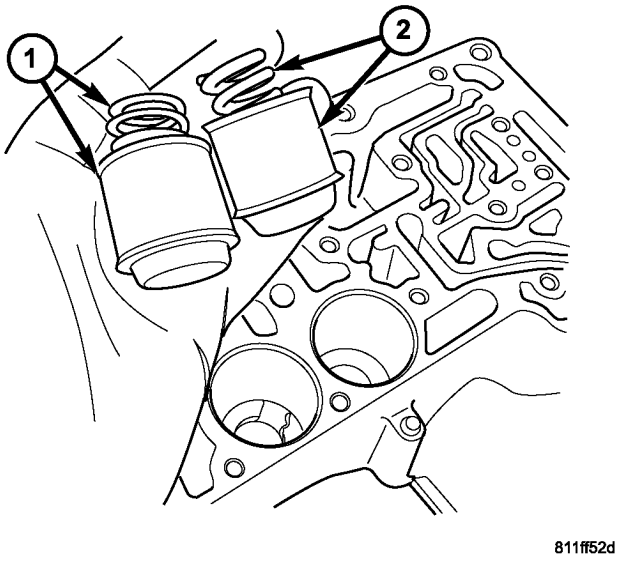


Fig. 127 Underdrive and Overdrive Accumulators

1 - OVERDRIVE PISTON AND SPRING
2 - UNDERDRIVE PISTON AND SPRING

CAUTION: Do not handle the valve body by the manual shaft. Damage could result.

(40) Install valve body into place as shown in (Fig. 128).

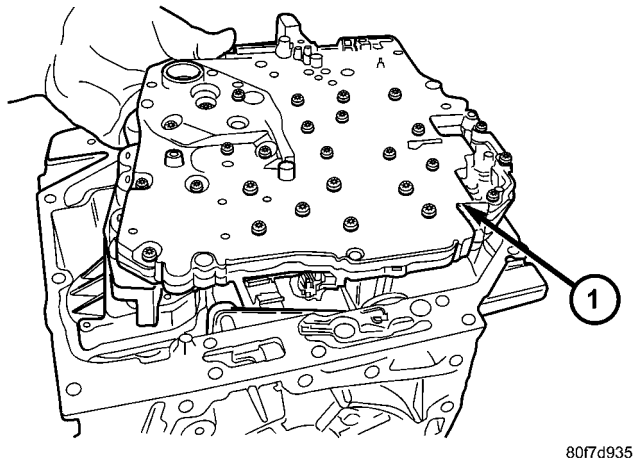


Fig. 128 Install Valve Body Onto Transmission

1 - VALVE BODY

(41) Install seven (7) valve body-to-case bolts (Fig. 129) and torque to 12 N·m (105 in. lbs.).

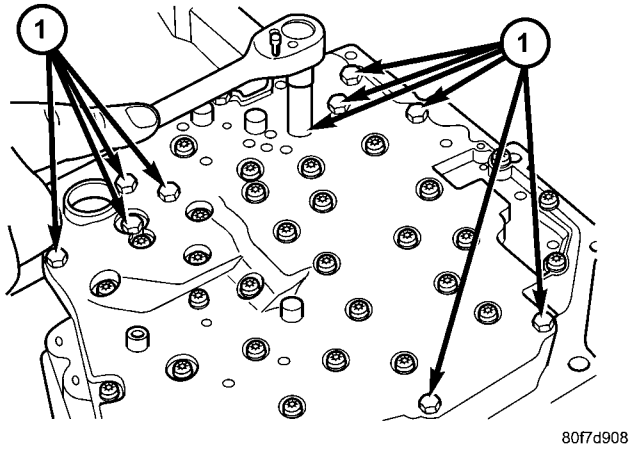


Fig. 129 Install Valve Body Bolts (7)

1 - BOLTS

(42) Install transmission oil filter (Fig. 130). Tighten the bolts to 5 N·m (45 in. lbs.)

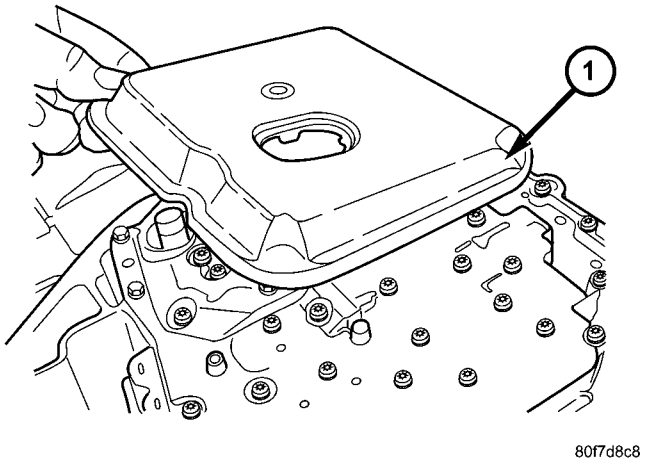


Fig. 130 Install Transmission Filter

1 - TRANSMISSION FILTER

AUTOMATIC TRANSMISSION - 42RLE (Continued)

NOTE: Before installing the oil pan bolt in the bolt hole located between the torque converter clutch on and U/D clutch pressure tap circuits (Fig. 131), it will be necessary to replentish the sealing patch on the bolt using Mopar® Lock & Seal Adhesive.

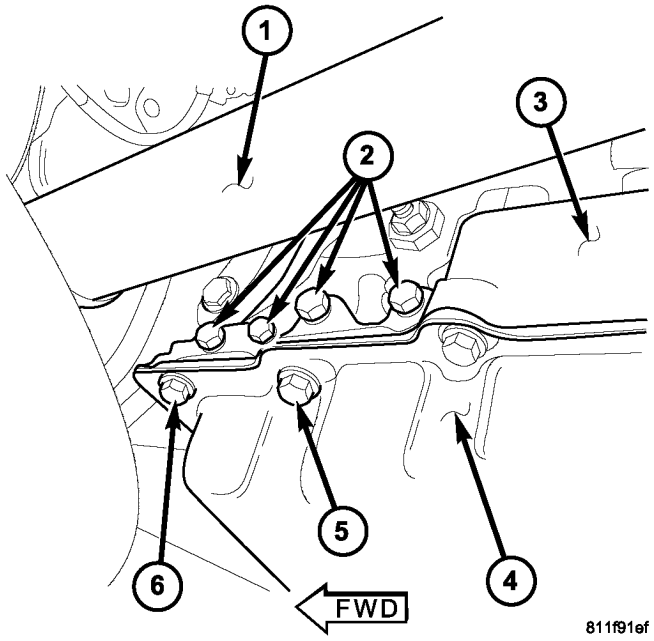


Fig. 131 Pan Fastener

1 - FRONT DRIVESHAFT
2 - PRESSURE PORTS
3 - TRANSMISSION CASE
4 - TRANSMISSION OIL PAN
5 - SECOND TRANSMISSION OIL PAN BOLT ON LEFT SIDE
6 - FIRST TRANSMISSION OIL PAN BOLT

(43) Install transmission oil pan (Fig. 132) with a bead of Mopar® ATF RTV. Torque oil pan-to-case bolts to 20 N·m (14.5 ft. lbs.).

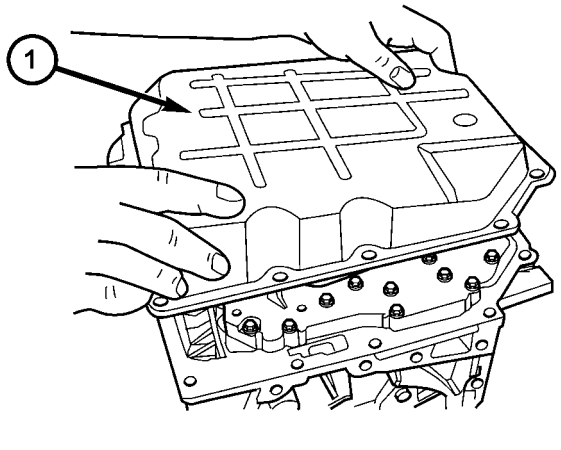


Fig. 132 Install Transmission Oil Pan

1 - TRANSMISSION OIL PAN

NOTE: Before installing either speed sensor bolt, it will be necessary to replentish the sealing patch on the bolt using Mopar® Lock & Seal Adhesive.

(44) Install both speed sensors into transmission case (Fig. 133). Torque the speed sensor bolts to 9 N·m (80 in. lbs.).

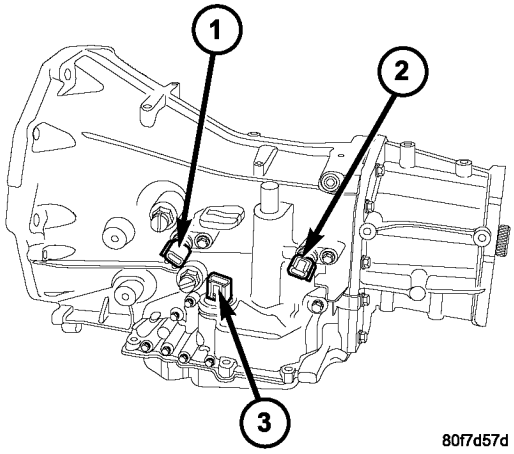


Fig. 133 Input and Output Speed Sensors and Transmission Range Sensor

1 - INPUT SPEED SENSOR
2 - OUTPUT SPEED SENSOR
3 - TRANSMISSION RANGE SENSOR

(45) As a final check of the transmission, measure the input shaft end play. This will indicate when a #4 thrust plate change is required. The #4 thrust plate is located behind the overdrive clutch hub. Attach a dial indicator to transmission bell housing with its plunger seated against end of input shaft (Fig. 134). Move input shaft in and out to obtain end play reading. **Input shaft end play must be 0.127 to 0.635 mm (0.005 to 0.025 inch).** If not within specifications, make the necessary thrust plate adjustment.

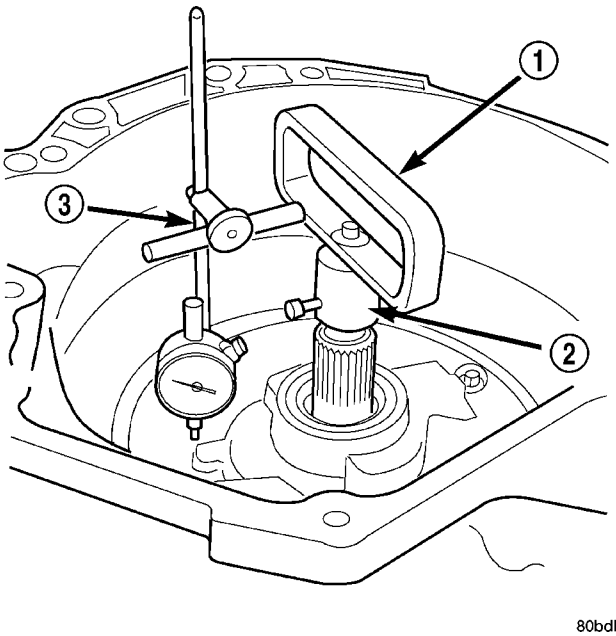


Fig. 134 Measure Input Shaft End Play Using Tool 8266 - Typical

1 - TOOL 8266-8
2 - TOOL 8266-2
3 - TOOL C-3339

(46) Inspect the lube tube grommet (Fig. 135) for damage. If the grommet lip is damaged, it will need to be replaced.

(47) Install the 4X4 stub shaft onto the transmission output shaft.

(48) Place a bead of Mopar® ATF RTV on the rear surface of the transmission case for the adapter housing.

(49) Install the adapter (Fig. 136) housing onto the transmission case.

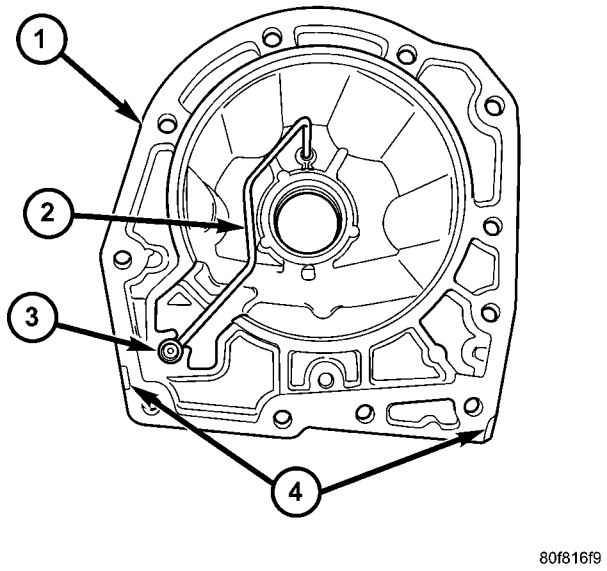


Fig. 135 Lube Tube Grommet

1 - HOUSING
2 - LUBE TUBE
3 - GROMMET
4 - PRY SLOTS

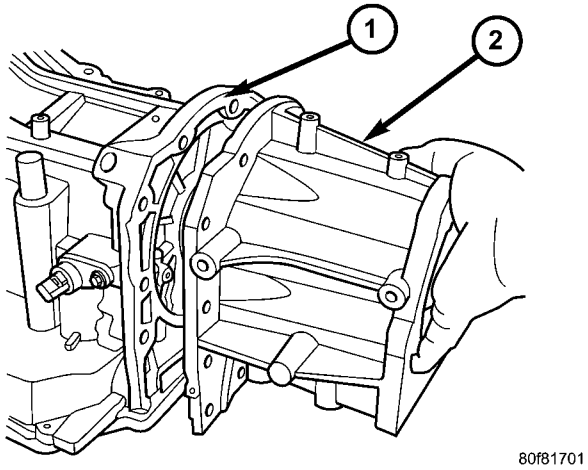


Fig. 136 Install Adapter Housing

1 - TRANSMISSION CASE
2 - ADAPTER HOUSING

AUTOMATIC TRANSMISSION - 42RLE (Continued)

NOTE: Before installing the lowermost four adapter housing bolts (Fig. 137), it will be necessary to replentish the sealing patch on the bolts using Mopar® Lock & Seal Adhesive.

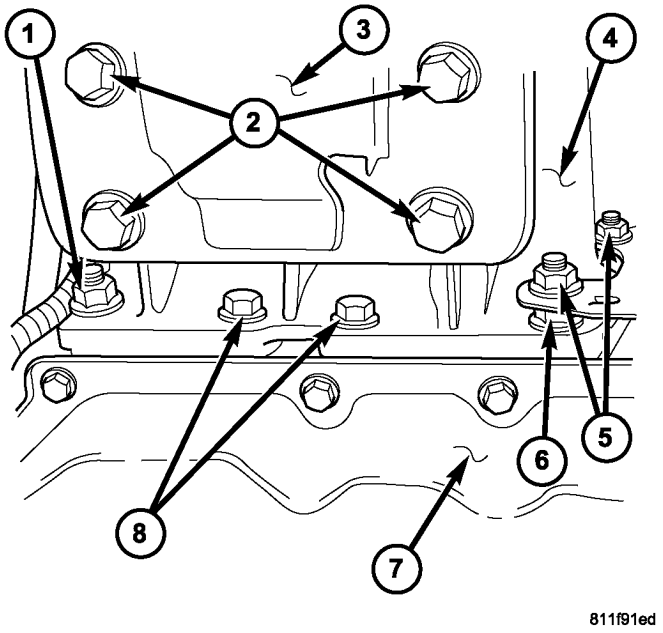


Fig. 137 Adapter Housing Fasteners

1 - STUD, ADAPTER/EXTENSION
2 - TRANSMISSION MOUNT FASTENERS (4)
3 - TRANSMISSION MOUNT
4 - TRANSMISSION CASE
5 - NUT, EXHAUST HANGER BRACKET (2)
6 - STUD, ADAPTER/EXTENSION
7 - TRANSMISSION OIL PAN
8 - BOLT, ADAPTER/EXTENSION (2)

(50) Install the bolts (Fig. 138) that hold the adapter housing onto the transmission case. Be sure to install any stud bolts to their original locations. Tighten the bolts to 54 N·m (40 ft.lbs.).

INSTALLATION

(1) Check torque converter hub and hub drive notches for sharp edges burrs, scratches, or nicks. Polish the hub and notches with 320/400 grit paper and crocus cloth if necessary. The hub must be smooth to avoid damaging pump seal at installation.

(2) Lubricate converter drive hub and oil pump seal lip with transmission fluid.

(3) Align converter and oil pump.

(4) Carefully insert converter in oil pump. Then rotate converter back and forth until fully seated in pump gears.

(5) Check converter seating with steel scale and straightedge (Fig. 139). Surface of converter lugs should be 1/2 in. to rear of straightedge when converter is fully seated.

(6) Temporarily secure converter with C-clamp.

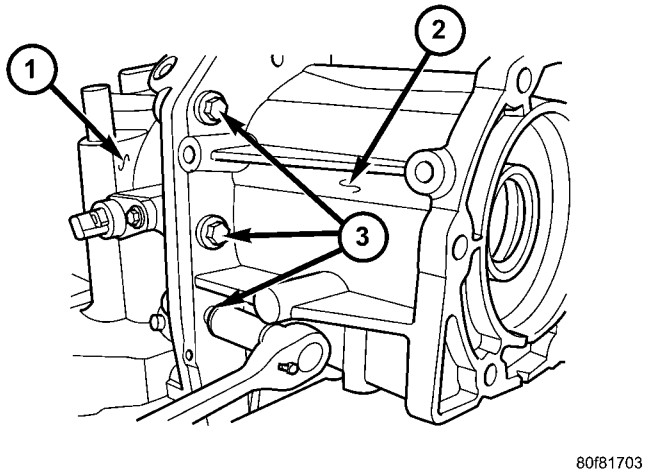


Fig. 138 Install Adapter Housing Bolts

1 - TRANSMISSION CASE
2 - ADAPTER HOUSING
3 - BOLTS

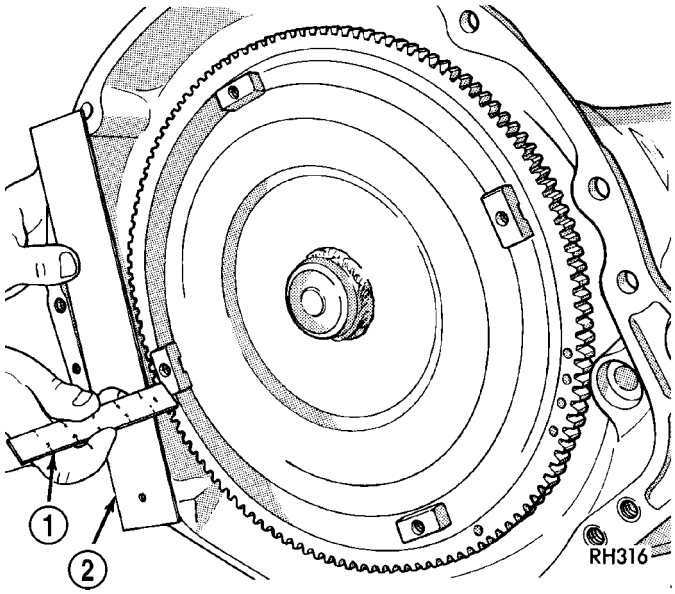


Fig. 139 Checking Converter Seating - Typical

1 - SCALE
2 - STRAIGHTEDGE

(7) Lightly grease crankshaft flange hole.

(8) Position transmission on jack and secure it with safety chains.

(9) Check condition of converter driveplate. Replace the plate if cracked, distorted or damaged. **Also be sure transmission dowel pins are seated in engine block and protrude far enough to hold transmission in alignment.**

(10) Raise transmission and align converter with drive plate and converter housing with engine block.

(11) Move transmission forward. Then raise, lower or tilt transmission to align converter housing with engine block dowels.

(12) Carefully work transmission forward and over engine block dowels until converter hub is seated in crankshaft.

(13) Install and tighten bolts that attach transmission converter housing to engine block.

CAUTION: Be sure the converter housing is fully seated on the engine block dowels before tightening any bolts.

(14) Install torque converter attaching bolts. Tighten bolts to 88 N·m (65 ft. lbs.).

(15) On 4.0L engine equipped vehicles, install the crankshaft position sensor (Fig. 140).

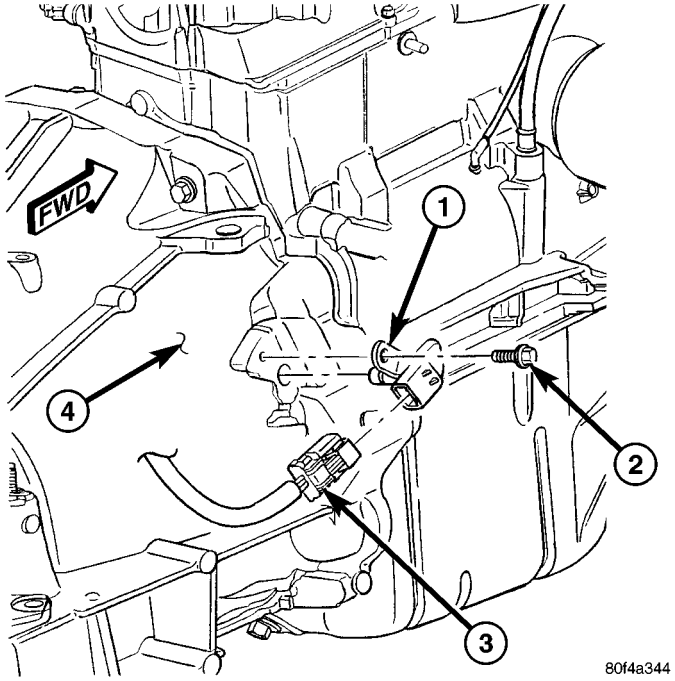


Fig. 140 Crankshaft Position Sensor

1 - CRANKSHAFT POSITION SENSOR
2 - MOUNTING BOLT
3 - ELECTRICAL CONNECTOR
4 - TRANSMISSION BELLHOUSING

(16) Install transmission fill tube and seal. Install new fill tube seal in transmission before installation.

(17) Connect transmission cooler lines to transmission.

(18) Install transfer case onto transmission. (Refer to 21 - TRANSMISSION/TRANSFER CASE - INSTALLATION)

(19) Install skid plate (Fig. 141) and attach transmission rear support to skid plate. (Refer to 13 - FRAME & BUMPERS/FRAME/TRANSFER CASE SKID PLATE - INSTALLATION)

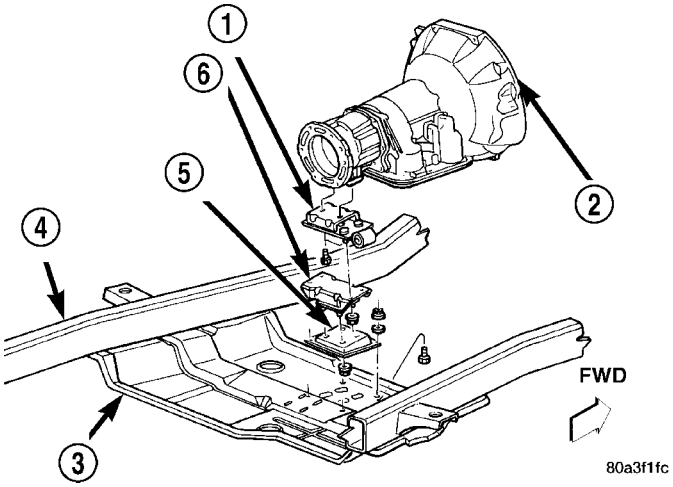


Fig. 141 Transmission Mount - Automatic Transmission

1 - TRANSMISSION SUPPORT BRACKET
2 - AUTOMATIC TRANSMISSION
3 - SKID PLATE
4 - FRAME
5 - TRANSMISSION MOUNT SUPPORT BRACKET
6 - CUSHION

(20) Remove engine support fixture.

(21) Remove transmission jack.

(22) Connect input and output speed sensor wires (Fig. 142).

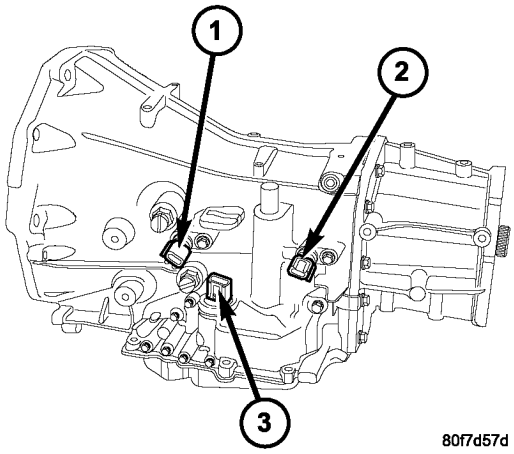


Fig. 142 Input and Output Speed Sensors and Transmission Range Sensor

1 - INPUT SPEED SENSOR
2 - OUTPUT SPEED SENSOR
3 - TRANSMISSION RANGE SENSOR

AUTOMATIC TRANSMISSION - 42RLE (Continued)

(23) Connect wires to the transmission range sensor (Fig. 142) and the solenoid/pressure switch assembly (Fig. 143).

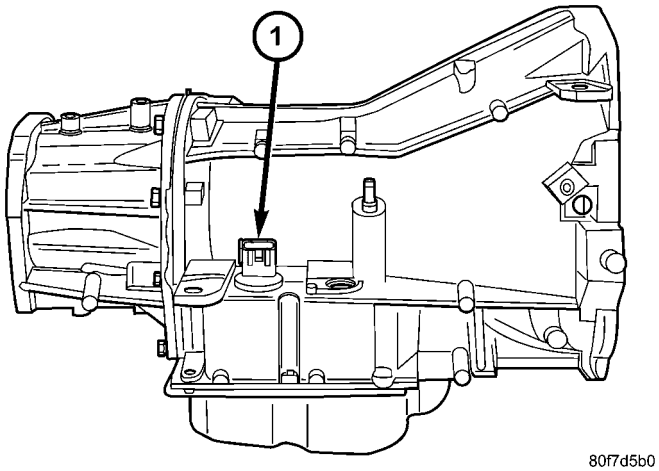


Fig. 143 Solenoid/Pressure Switch Assembly

1 - SOLENOID/PRESSURE SWITCH ASSEMBLY CONNECTOR

(24) Install converter housing access cover.

(25) Install exhaust pipes and support brackets, if removed.

(26) Install starter motor (Refer to 8 - ELECTRICAL/STARTING/STARTER MOTOR - INSTALLATION) and cooler line bracket.

(27) Install new plastic retainer grommet on any shift linkage rod or lever that was disconnected. Grommets should not be reused. Use pry tool to remove rod from grommet and cut away old grommet. Use pliers to snap new grommet into lever and to snap rod into grommet at assembly.

(28) Connect gearshift cable.

(29) Connect transfer case shift linkage.

(30) Adjust gearshift linkage, if necessary.

(31) Align and connect propeller shaft(s). (Refer to 3 - DIFFERENTIAL & DRIVELINE/PROPELLER SHAFT/PROPELLER SHAFT - INSTALLATION)

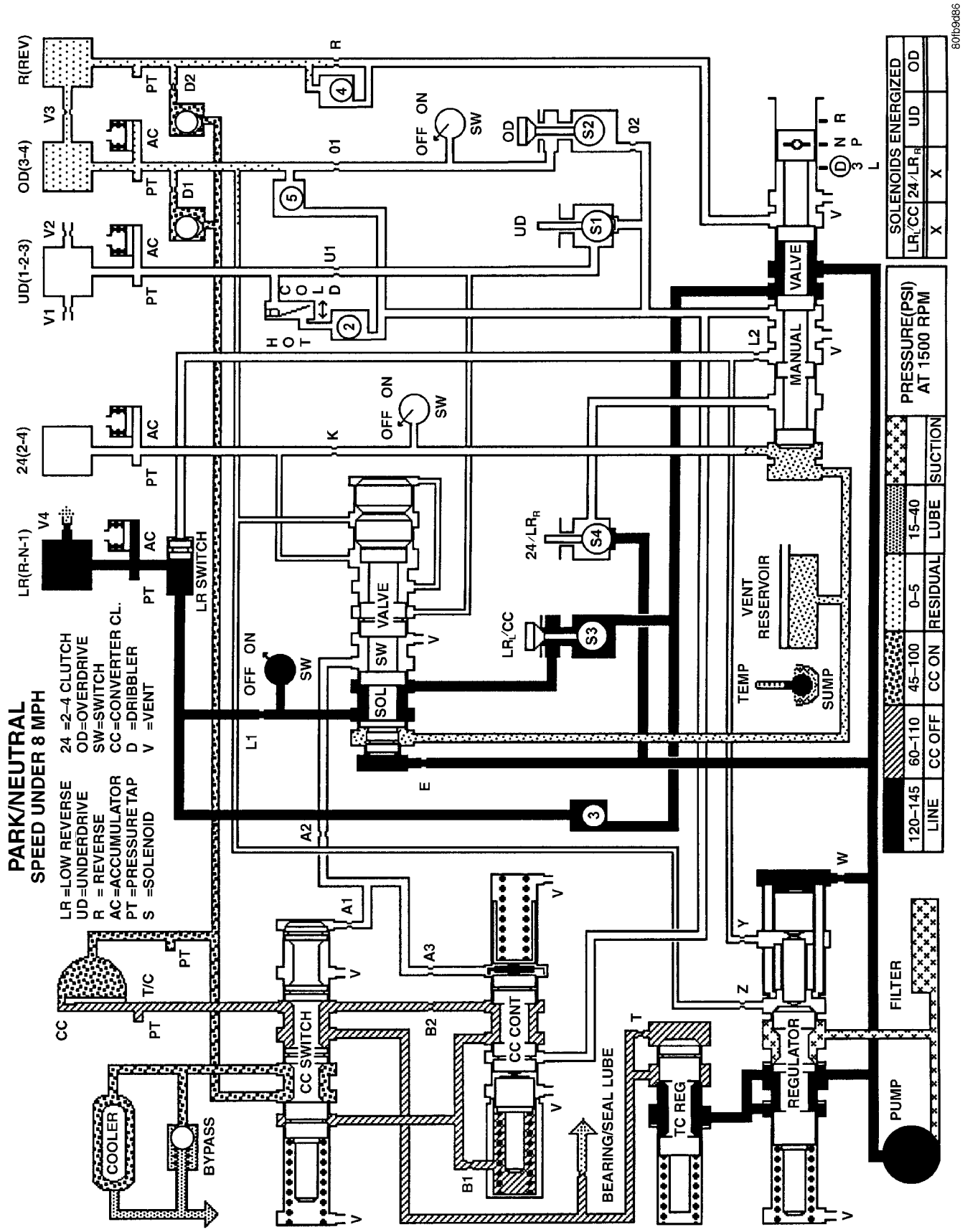
(32) Fill transfer case to bottom edge of fill plug hole.

(33) Lower vehicle and connect battery negative cable.

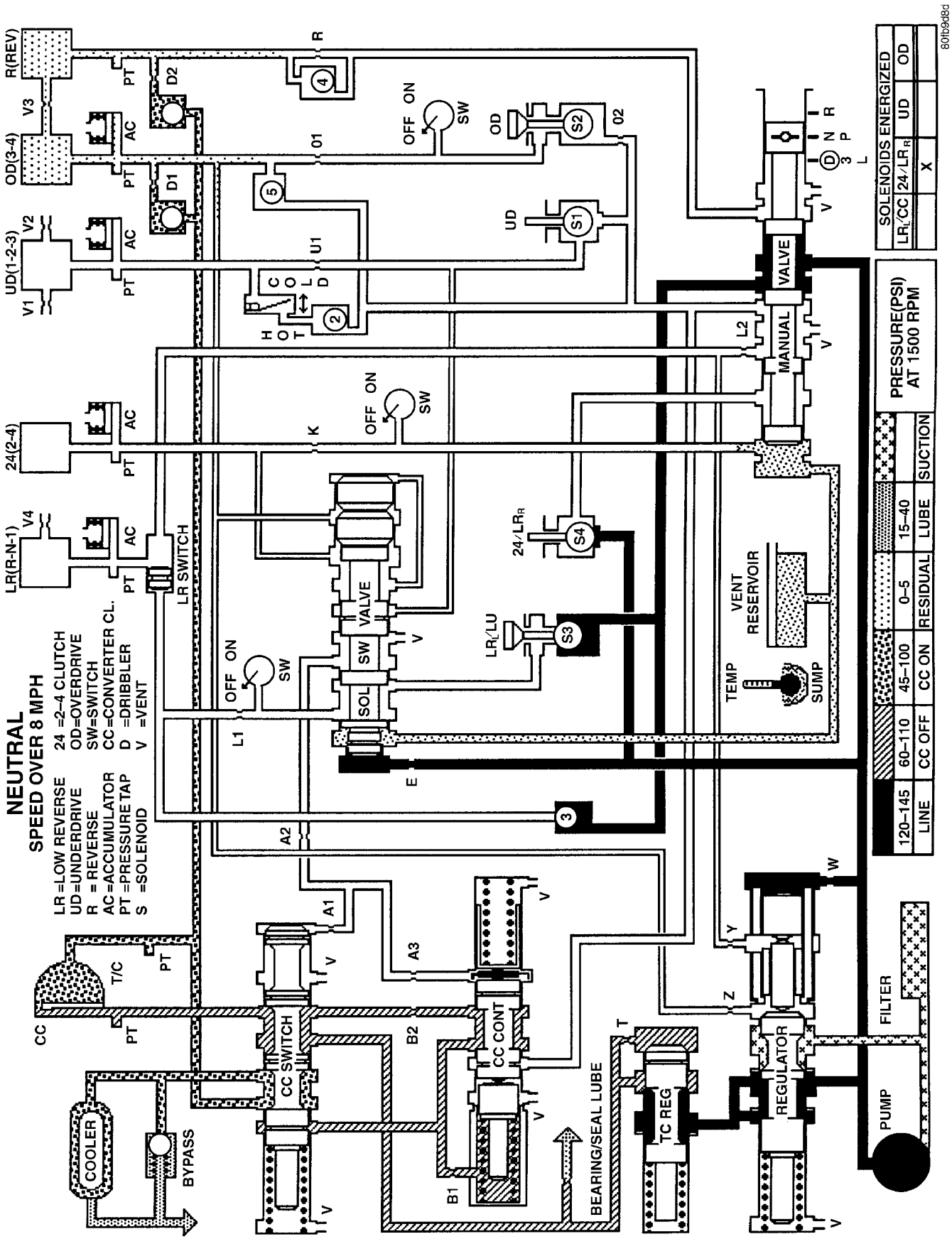
(34) Fill transmission to correct level with Mopar® ATF +4.

AUTOMATIC TRANSMISSION - 42RLE (Continued)

SCHEMATICS AND DIAGRAMS - 42RLE TRANSMISSION



AUTOMATIC TRANSMISSION - 42RLE (Continued)



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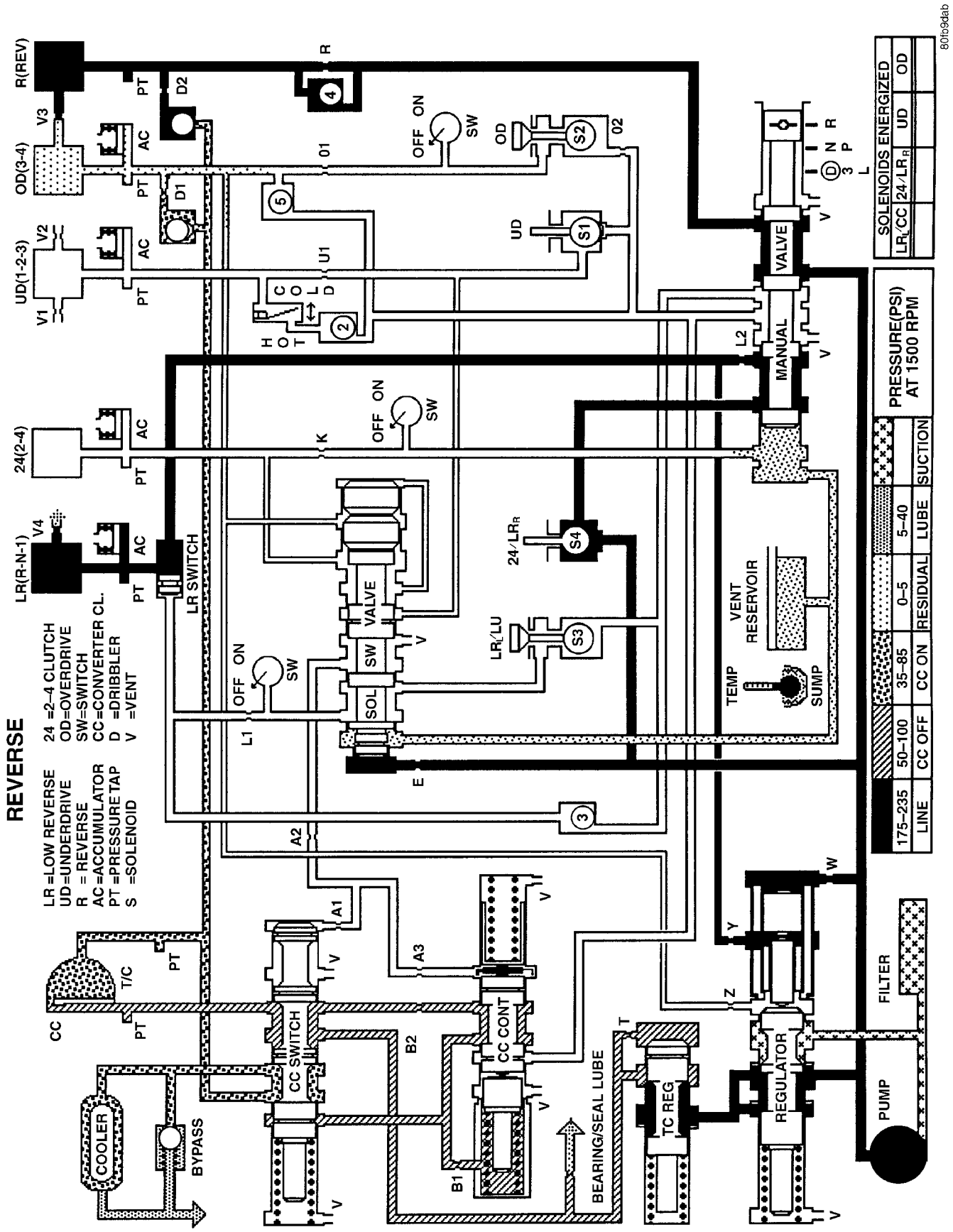

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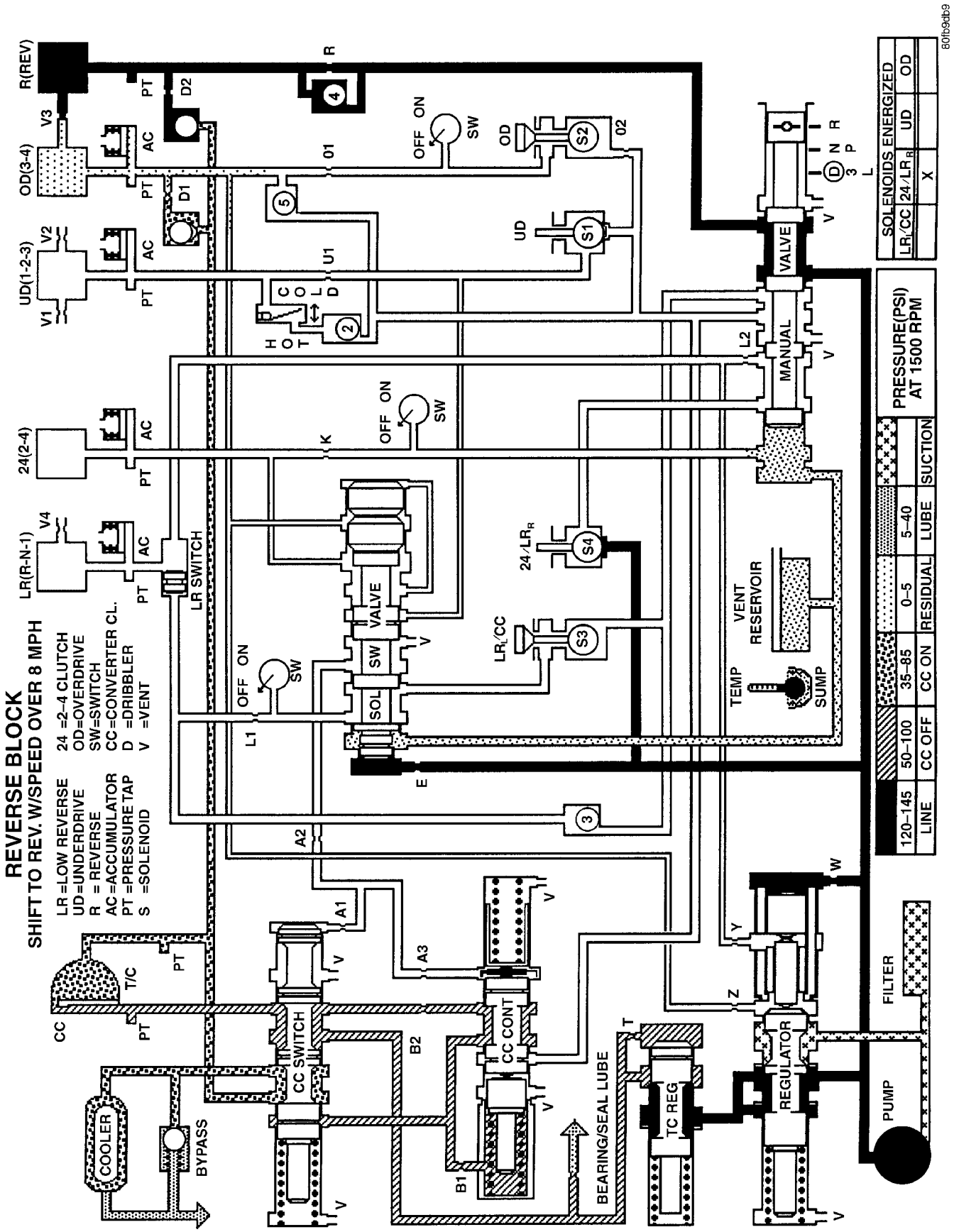
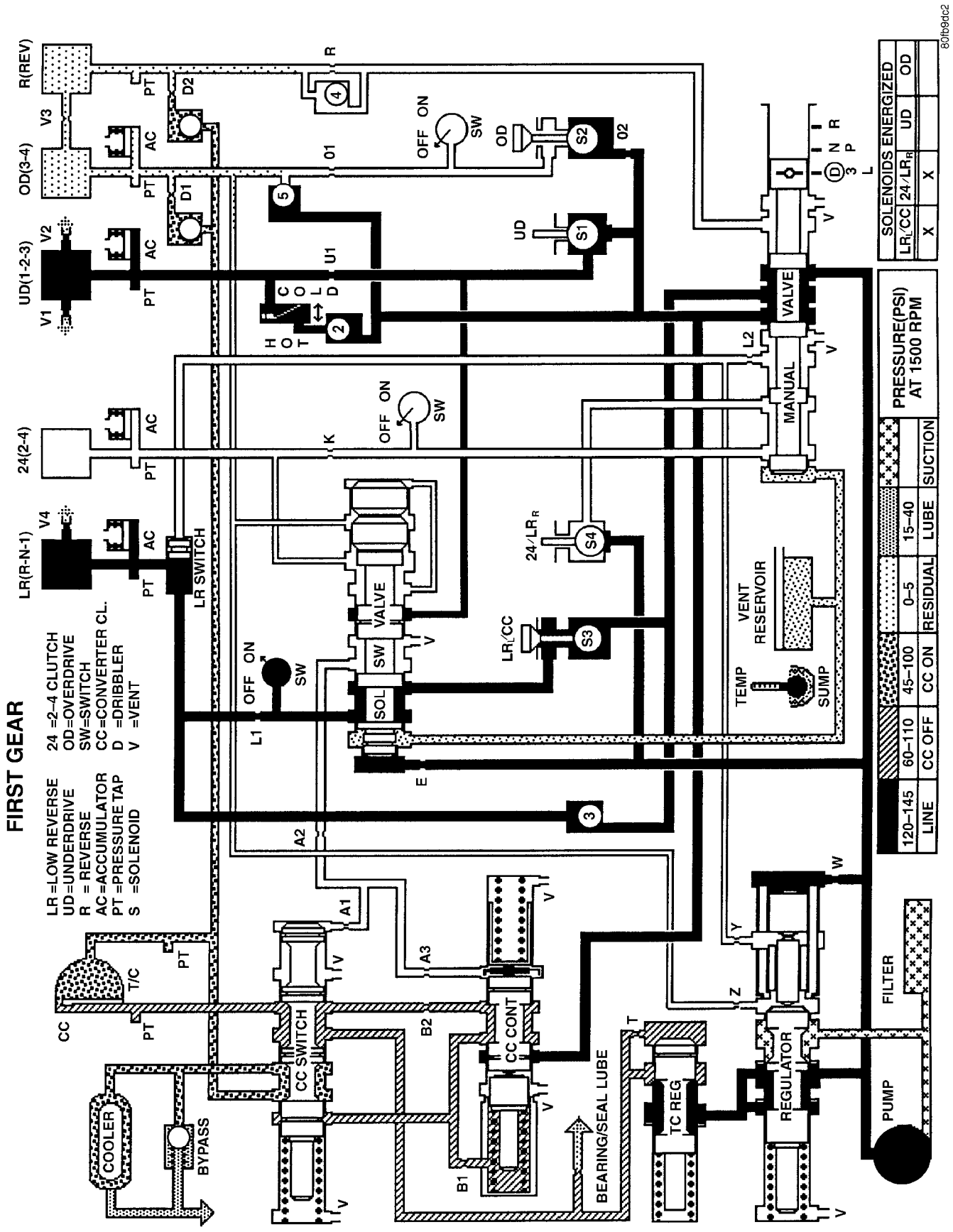


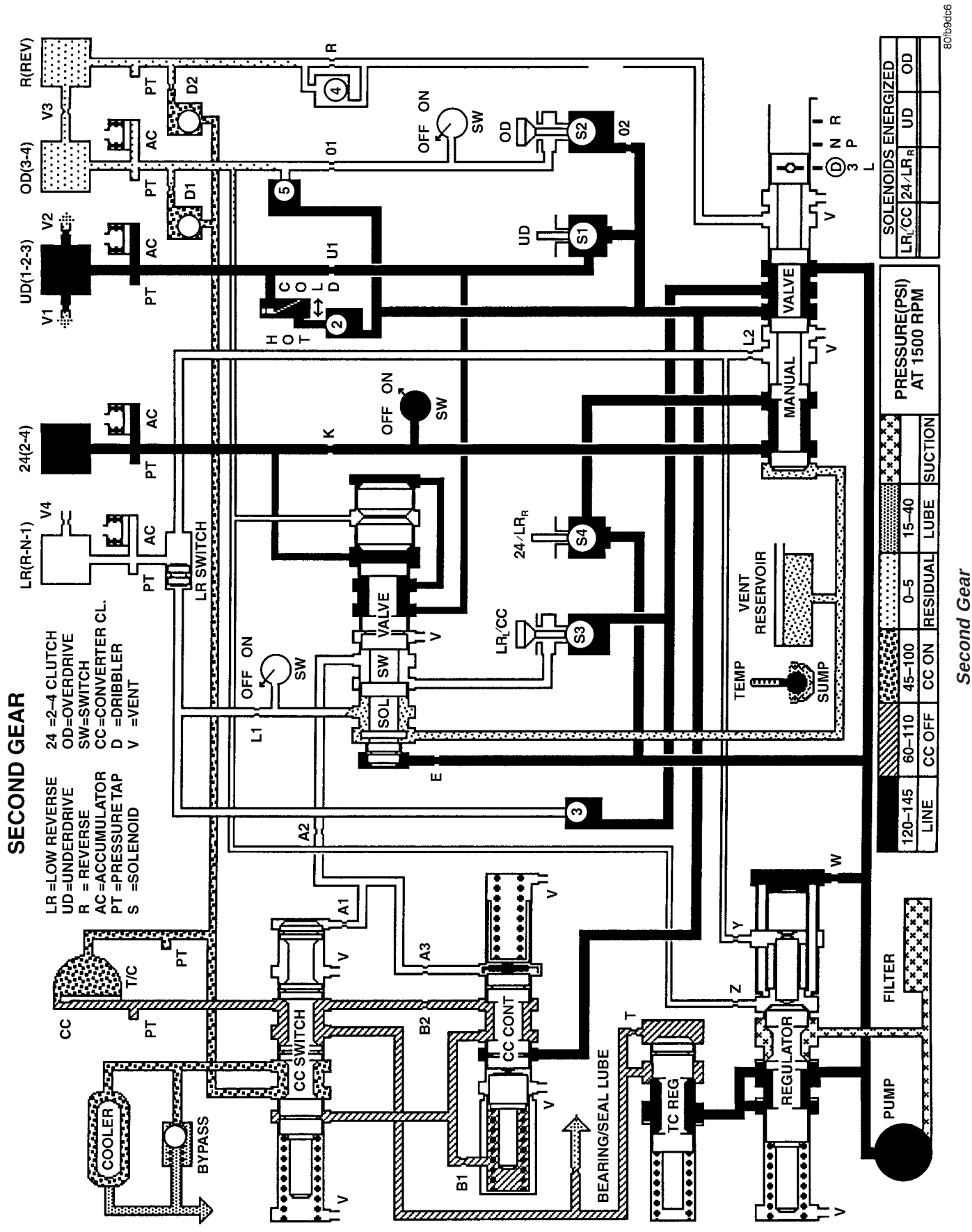
Reverse Block (Shift to Reverse w/Speed Over 8 mph)

AUTOMATIC TRANSMISSION - 42RLE (Continued)



First Gear

AUTOMATIC TRANSMISSION - 42RLE (Continued)



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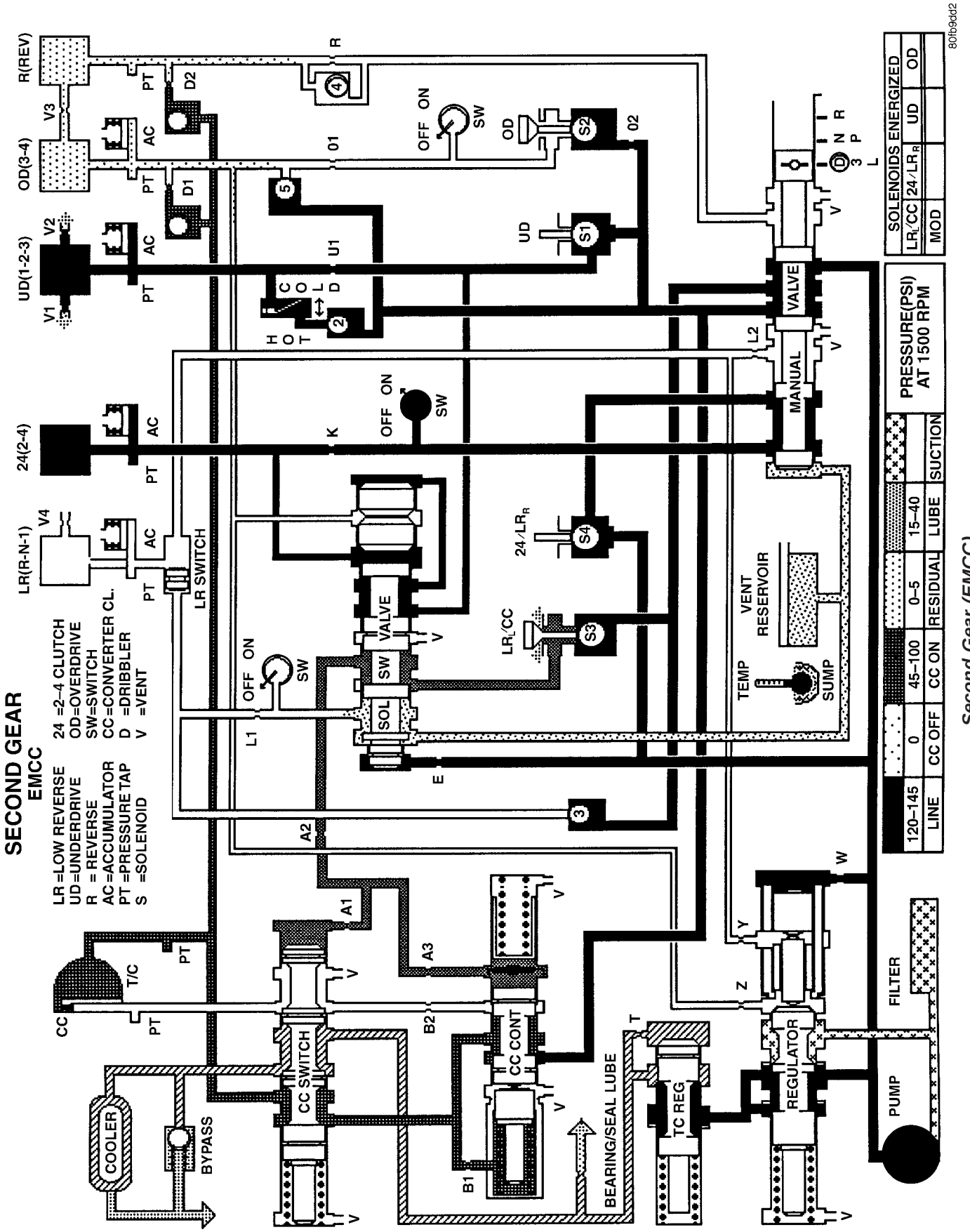

Second Gear (EMCC)

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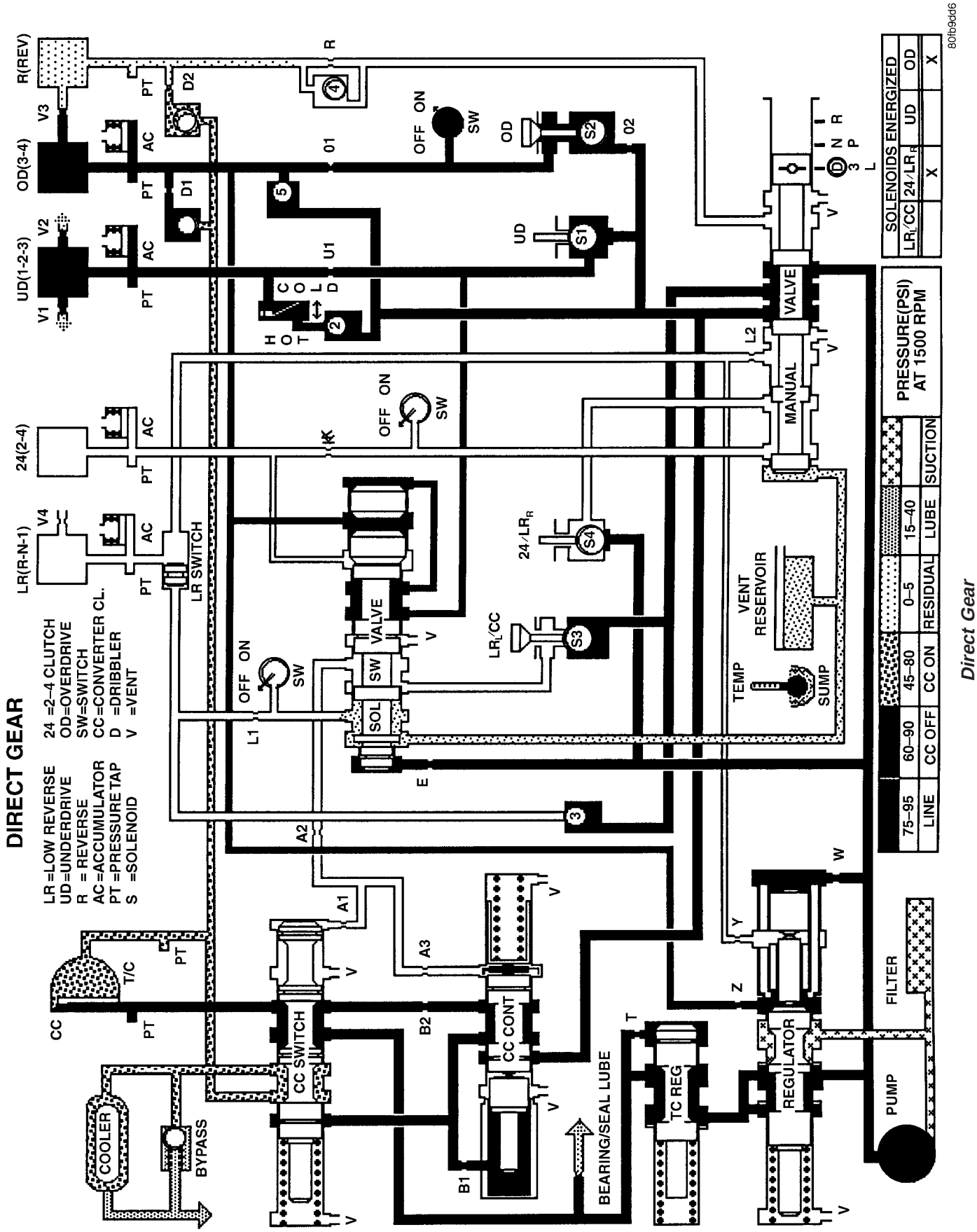

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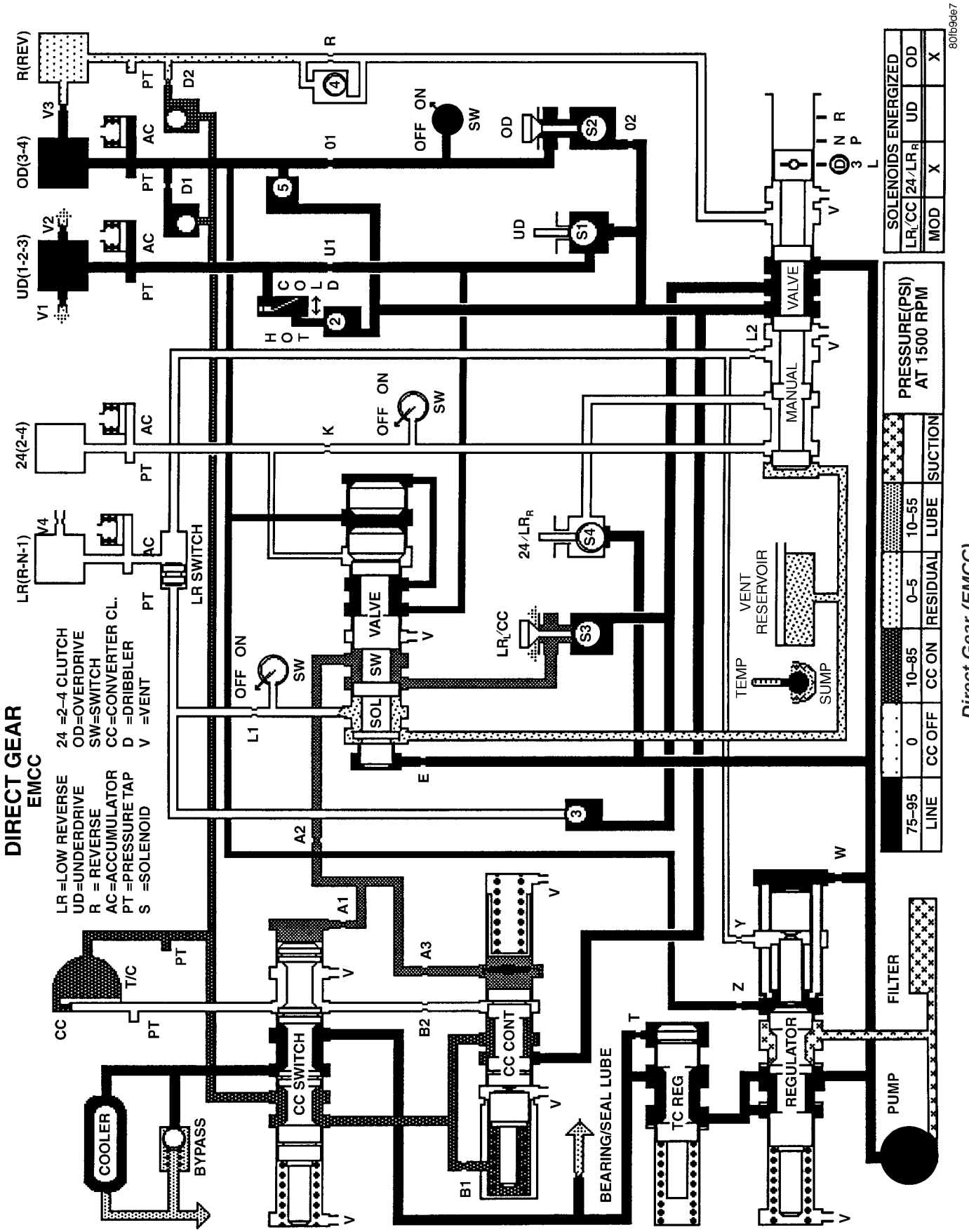
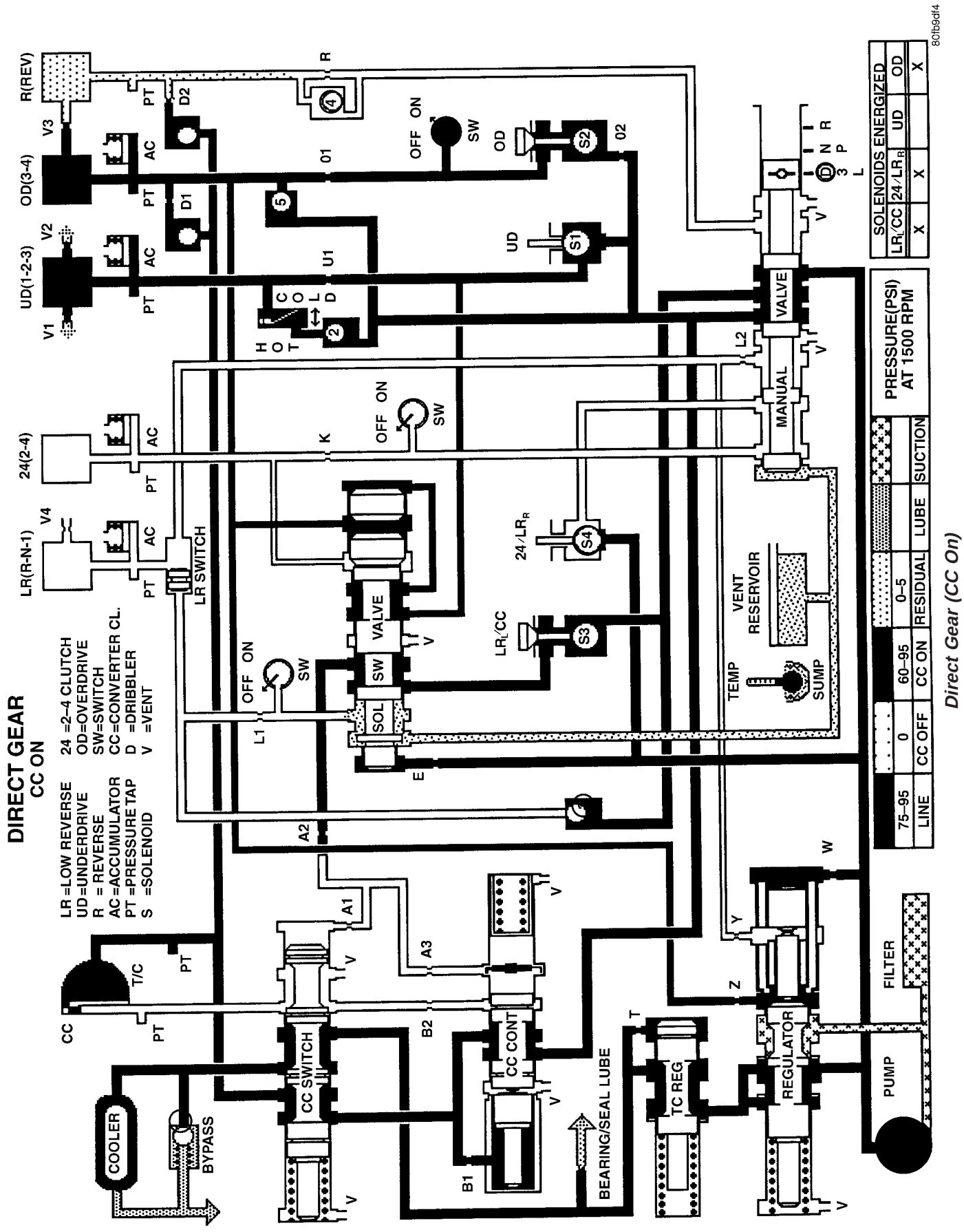


Direct Gear (EMCC)

AUTOMATIC TRANSMISSION - 42RLE (Continued)



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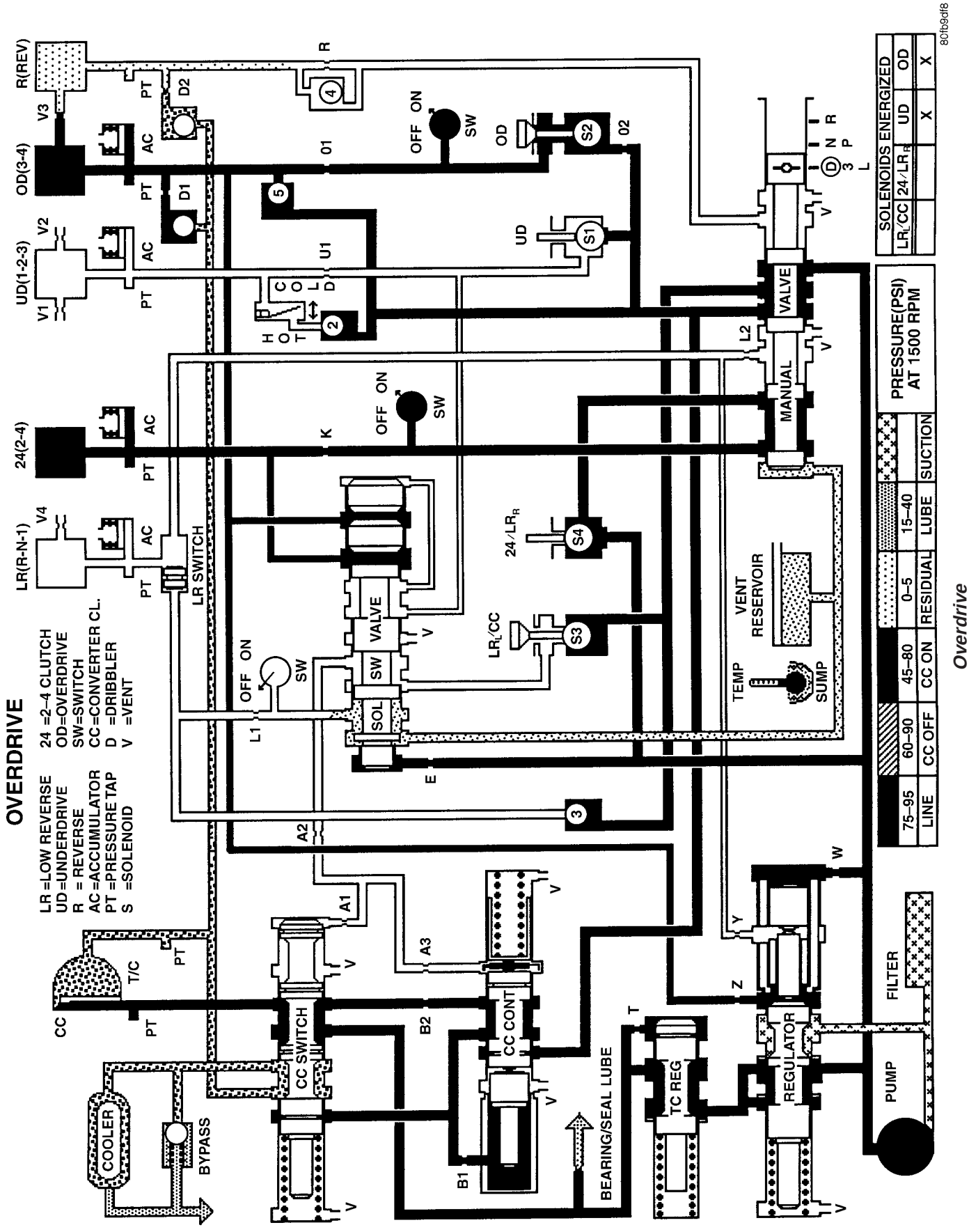

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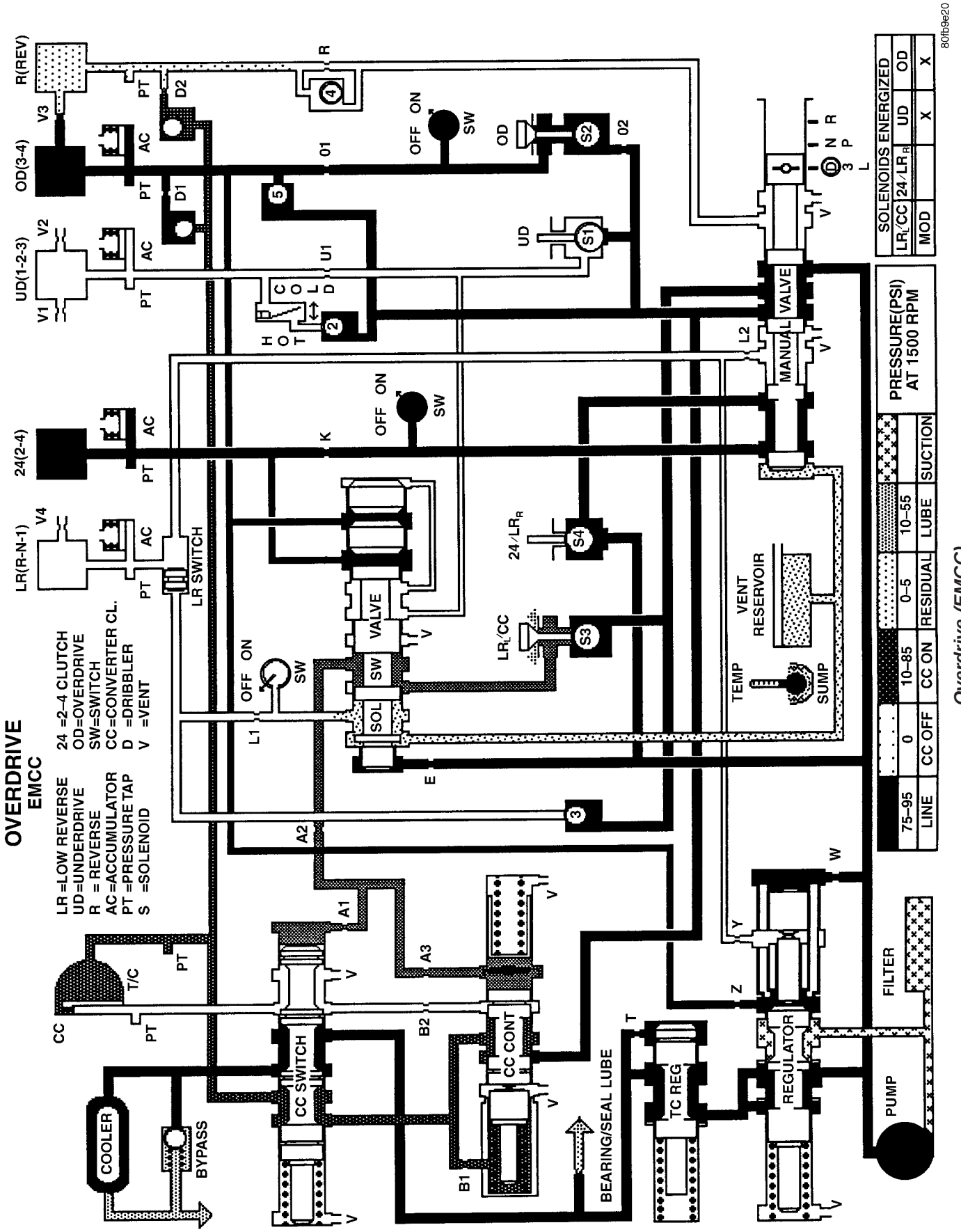

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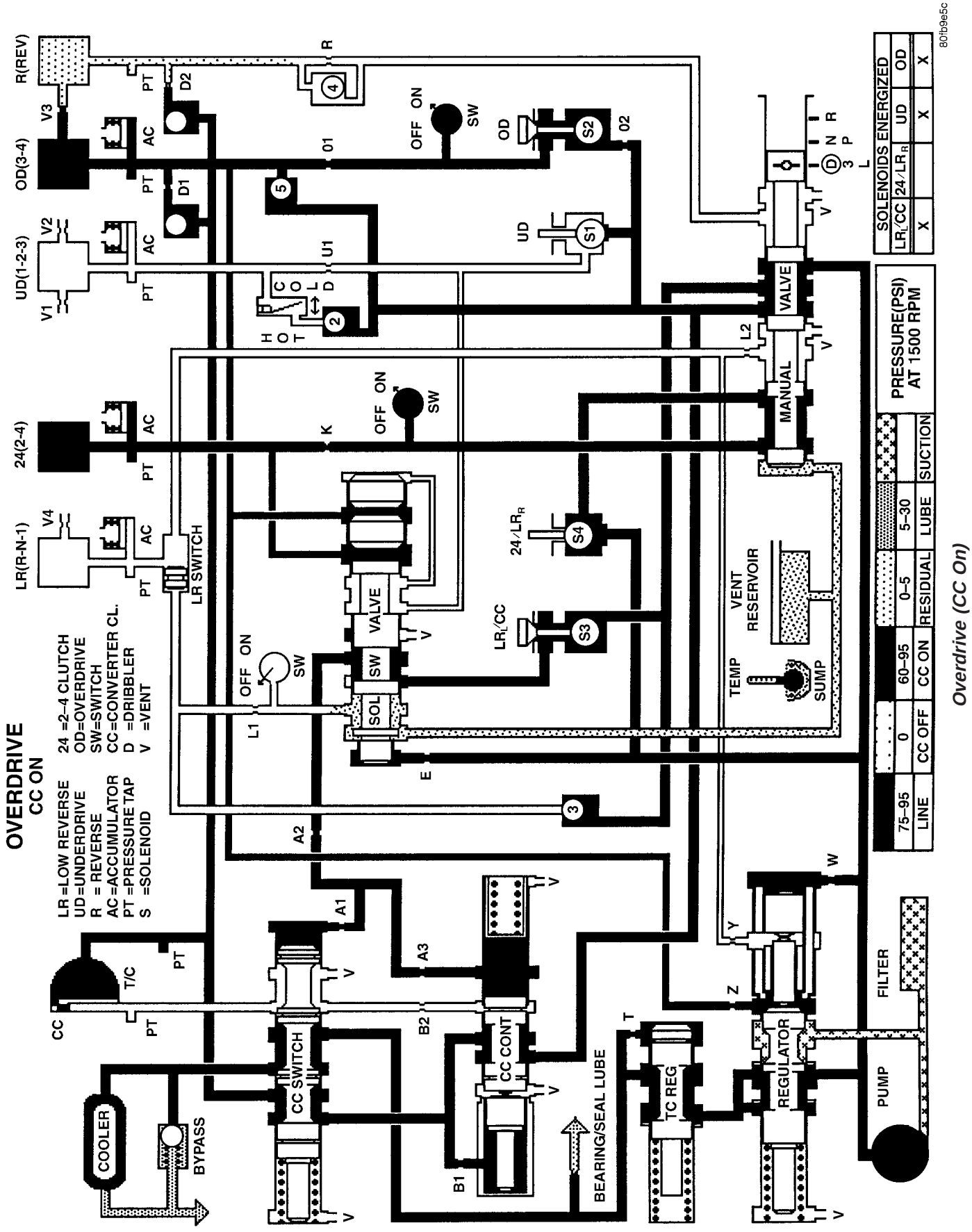


AUTOMATIC TRANSMISSION - 42RLE (Continued)

SPECIFICATIONS

42RLE AUTOMATIC TRANSMISSION

GENERAL SPECIFICATIONS

Transmission Type	Four-Speed Automatic, Electronically Controlled, Fully Adaptive, Electronically Modulated Torque Converter
Lubrication Method	Pump (internal - external gear-type)
Cooling Method	Water Heat Exchanger and/or Air-to-Oil Heat Exchanger

GEAR RATIOS

1st Gear	2.84:1
2nd Gear	1.57:1
3rd Gear (Direct)	1.00:1
4th Gear (Overdrive)	0.69:1
Reverse Gear	2.21:1

BEARING PRELOAD (DRAG TORQUE)

Description	Metric	Standard
Output Shaft	0.22-0.903 N·m	1-8 in. lbs.

CLUTCH PACK

Description	Metric	Standard
Low/Reverse Clutch (Select Reaction Plate)	0.84-1.60 mm	0.033-0.063 in.
Two/Four Clutch (No Select)	0.76-2.64 mm	0.030-0.104 in.
Reverse Clutch (Select Snap Ring)	0.89-1.37 mm	0.035-0.054 in.
Overdrive Clutch (No Select)	1.07-3.25 mm	0.042-0.128 in.
Underdrive Clutch (Select Reaction Plate)	0.94-1.50 mm	0.037-0.059 in.

INPUT SHAFT

Description	Metric	Standard
End Play	0.127-0.635 mm	0.005-0.025 in.

AUTOMATIC TRANSMISSION - 42RLE (Continued)

OIL PUMP CLEARANCES

DESCRIPTION	METRIC	STANDARD
Outer Gear-to-Crescent	0.060-0.298 mm	0.0023-0.0117 in.
Inner Gear-to-Crescent	0.093-0.385 mm	0.0036-0.0151 in.
Outer Gear-to-Pocket	0.089-0.202 mm	0.0035-0.0079 in.
Outer Gear Side Clearance	0.020-0.046 mm	0.0008-0.0018 in.
Inner Gear Side Clearance	0.020-0.046 mm	0.0008-0.0018 in.

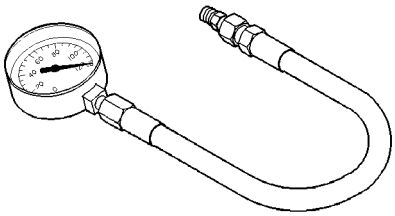
TORQUE SPECIFICATIONS

Description	N·m	Ft. Lbs.	In. Lbs.
Bolt, Converter-to-Driveplate	88	65	-
Bolt, Fluid Filter-to-Valve Body	5	-	45
Bolt, L/R Clutch Retainer-to-Case	5	-	45
Bolt, Adapter/Extension Housing	54	40	-
Bolt, Manual Valve Lever-to-Manual Valve	5	-	45
Bolt, Oil Pan-to-Case	20	14.5	-
Bolt, Oil Pump-to-Case	30	-	265
Bolt, Park Sprag Retainer	4.5	-	40
Bolt, Reaction Shaft Support Halves	28	-	250
Bolt, Solenoid/Pressure Switch Assy-to-Valve Body	5.5	-	50
Bolt, Valve Body-to-Case	12	-	105
Bolt, Valve Body-to-Transfer Plate	5	-	45
Fitting, Cooler Line	47.5	35	-
Nut, Output Shaft	271	200	-
Plug, Pressure Tap	5	-	45
Bolt, Input Speed-to-Case Sensor	9	-	80
Bolt, Output Speed-to-Case Sensor	9	-	80

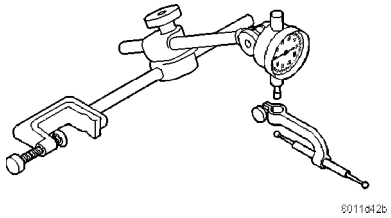
AUTOMATIC TRANSMISSION - 42RLE (Continued)

SPECIAL TOOLS

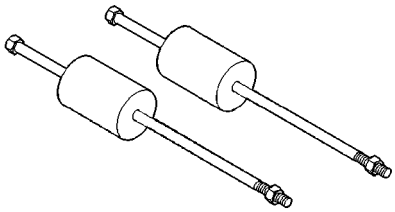
42RLE AUTOMATIC TRANSMISSION



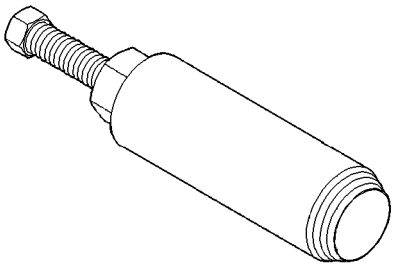
Pressure Gauge (High) C-3293SP



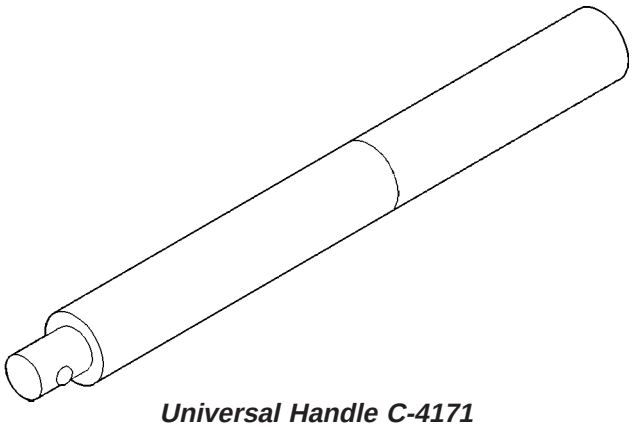
Dial Indicator C-3339



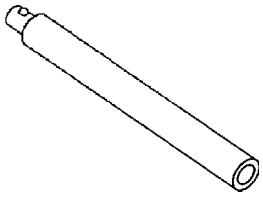
Hammer, Slide - C-3752



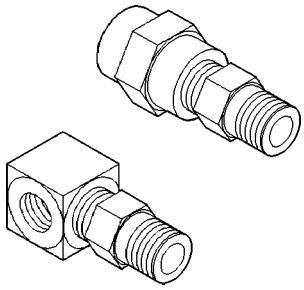
Puller, Seal - C-3981B



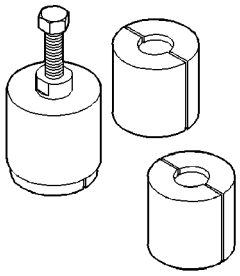
Universal Handle C-4171



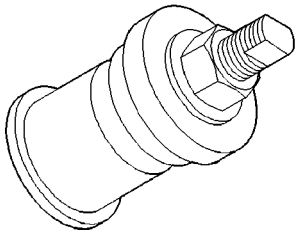
Extension, Handle - C-4171-2



Adapter Set - L-4559

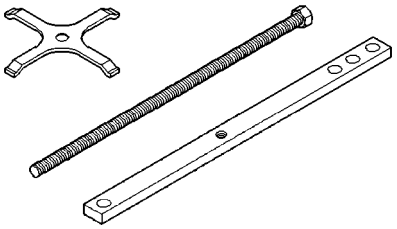


Puller Set 5048

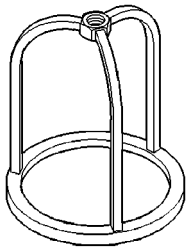


Installer - 5050A

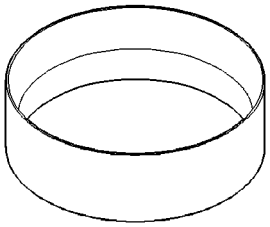
AUTOMATIC TRANSMISSION - 42RLE (Continued)



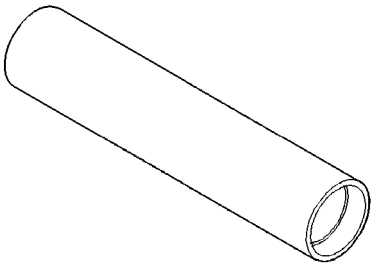
Compressor - 5058A



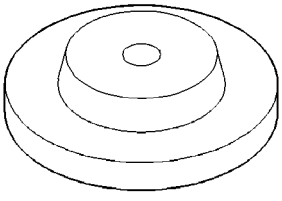
Compressor - 5059-A



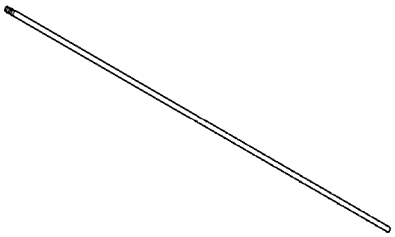
Installer - 5067



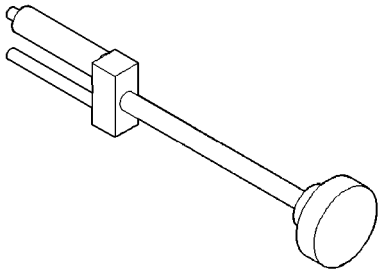
Installer - 6052



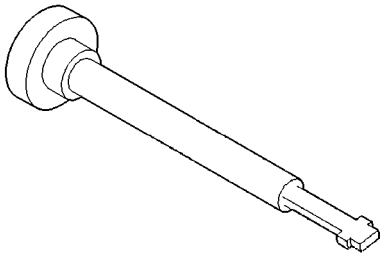
Disk - 6057



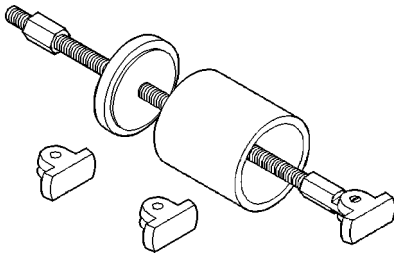
Tip - 6268



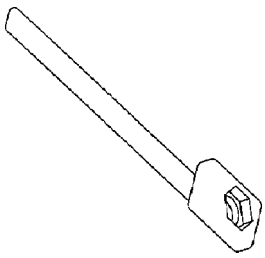
Remover/Installer - 6301



Remover/Installer - 6302

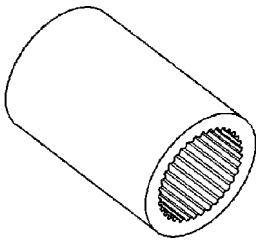


Remover 6310

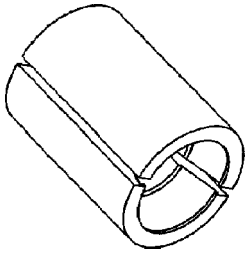


Wrench - 6497

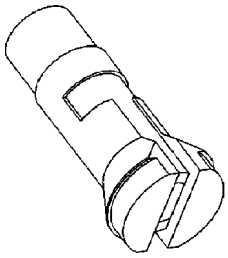
AUTOMATIC TRANSMISSION - 42RLE (Continued)



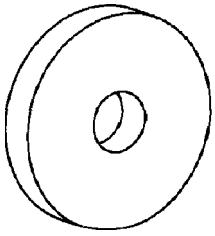
Wrench - 6498-A



Puller Jaws - 6545



Remover - 6596



Remover - 6597

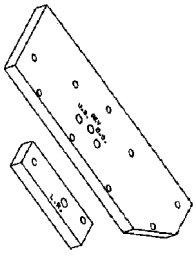
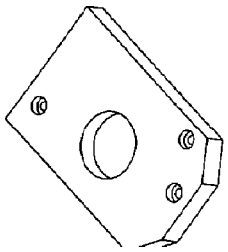
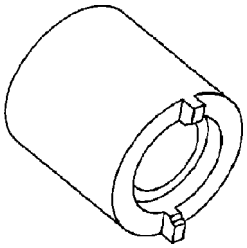


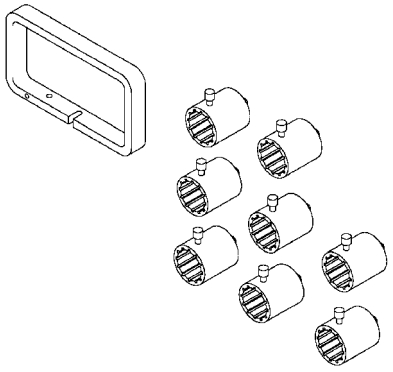
Plate Set - 6599



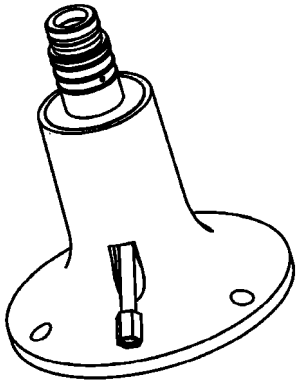
Plate, Support - 6618A



Tool, Staking - 6639



End Play Set - 8266



Fixture, Pressure - 8391

ACCUMULATOR

DESCRIPTION

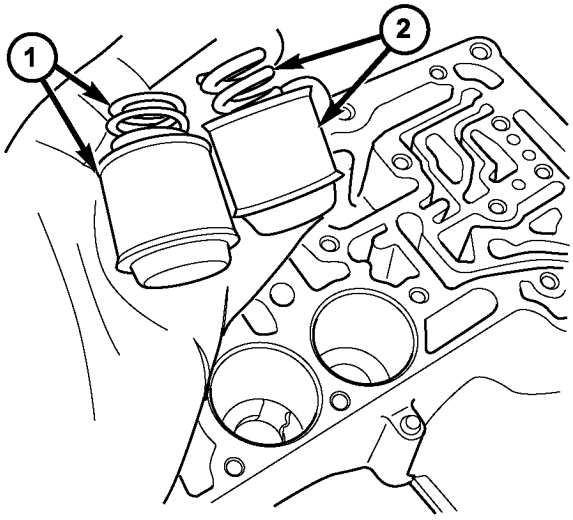


Fig. 144 Underdrive and Overdrive Accumulators

1 - OVERDRIVE PISTON AND SPRING
2 - UNDERDRIVE PISTON AND SPRING

The 42RLE underdrive, overdrive, low/reverse, and 2/4 clutch hydraulic circuits each contain an accumulator. An accumulator typically consists of a piston, return spring(s), and a cover or plug. The overdrive (1) and underdrive (2) accumulators are located within the transmission case, and are retained by the valve body (Fig. 144).

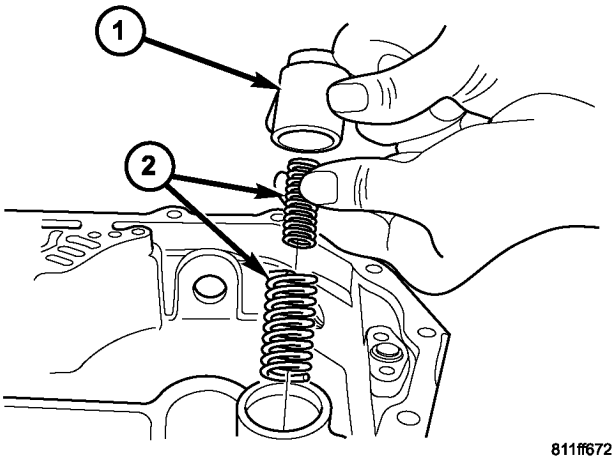


Fig. 145 Low/Reverse Accumulator

1 - PISTON
2 - RETURN SPRINGS

The low reverse (1) accumulator (Fig. 145) is also located within the transmission case, but the assembly is retained by a cover and a snap-ring.

The 2/4 accumulator (5) is located in the valve body. It is retained by a cover and retaining screws (Fig. 146).

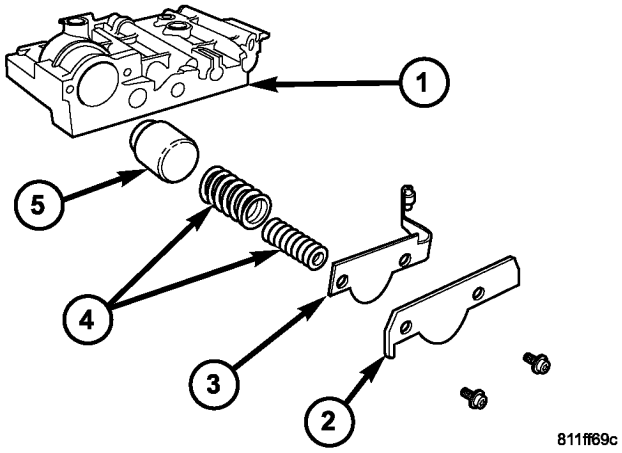


Fig. 146 2/4 Accumulator Assembly

1 - VALVE BODY
2 - RETAINER PLATE
3 - DETENT SPRING
4 - RETURN SPRINGS
5 - PISTON

OPERATION

The function of an accumulator is to cushion the application of a frictional clutch element. When pressurized fluid is applied to a clutch circuit, the application force is dampened by fluid collecting in the respective accumulator chamber against the piston and springs. The intended result is a smooth, firm clutch application.

ADAPTER HOUSING SEAL

REMOVAL

(1) Remove the transfer case (Refer to 21 - TRANSMISSION/TRANSFER CASE - REMOVAL).

(2) Using a screw mounted in a slide hammer, remove the adapter housing seal.

INSTALLATION

(1) Install a new adapter housing seal with Tool Handle C-4171 and Installer C-3860-A.

(2) Install the transfer case (Refer to 21 - TRANSMISSION/TRANSFER CASE - INSTALLATION).

BEARINGS

ADJUSTMENTS

BEARING ADJUSTMENT PROCEDURES

Take extreme care when removing and installing bearing cups and cones. **Use only an arbor press for installation,** as a hammer may not properly align the bearing cup or cone. Burrs or nicks on the

BEARINGS (Continued)

bearing seat will give a false end play reading, while gauging for proper shims. Improperly seated bearing cup and cones are subject to low-mileage failure.

Bearing cups and cones should be replaced if they show signs of pitting or heat distress.

If distress is seen on either the cup or bearing rollers, both cup and cone must be replaced.

NOTE: Bearing drag torque specifications must be maintained to avoid premature bearing failures.

Used (original) bearing may lose up to 50 percent of the original drag torque after break-in.

NOTE: All bearing adjustments must be made with no other component interference or gear intermesh.

Oil all bearings before checking turning torque.

BRAKE TRANSMISSION SHIFT INTERLOCK MECHANISM

DESCRIPTION

The Brake Transmission Shifter/Ignition Interlock (BTSI), is a cable and solenoid operated system. It interconnects the automatic transmission floor mounted shifter to the steering column ignition switch (Fig. 147).

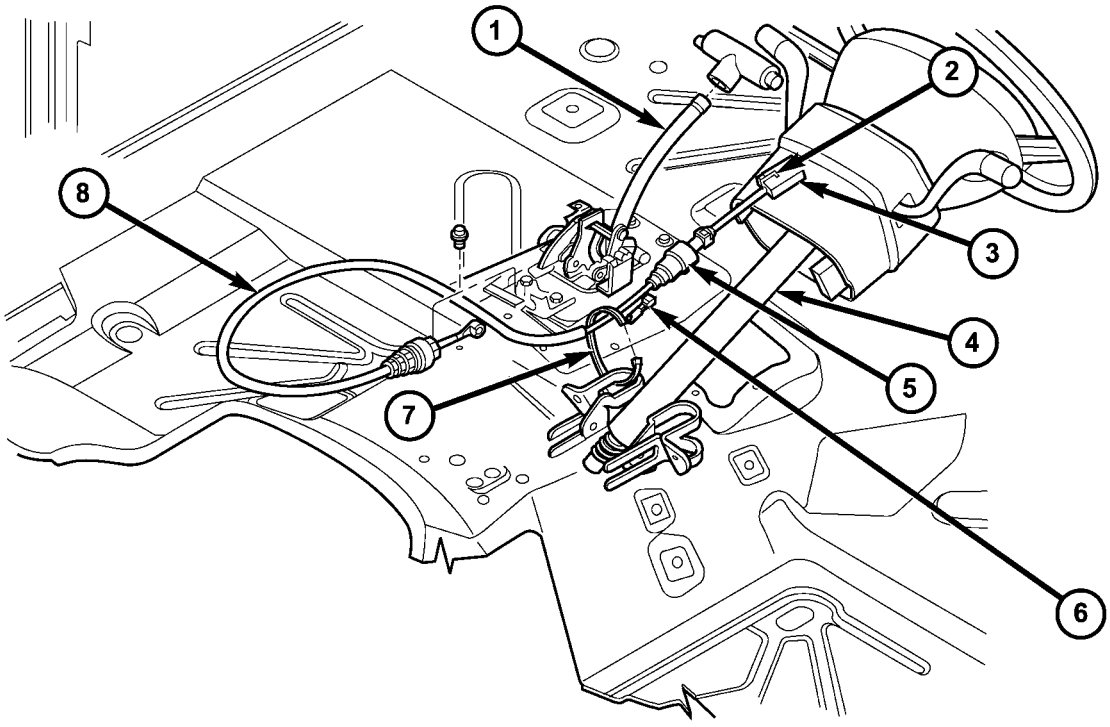


Fig. 147 Ignition Interlock Cable Routing

1 - SHIFT MECHANISM
2 - LOCK-TAB
3 - IGNITION LOCK INTERLOCK
4 - STEERING COLUMN
5 - SOLENOID
6 - WIRE CONNECTOR
7 - TIE STRAP
8 - PARK/BRAKE INTERLOCK CABLE

BRAKE TRANSMISSION SHIFT INTERLOCK MECHANISM (Continued)

OPERATION

The system locks the shifter into the PARK position. The Interlock system is engaged whenever the ignition switch is in the LOCK or ACCESSORY position. An additional electrically activated feature will prevent shifting out of the PARK position unless the brake pedal is depressed at least one-half an inch. A magnetic holding device in line with the park/brake interlock cable is energized when the ignition is in the RUN position. When the key is in the RUN position and the brake pedal is depressed, the shifter is unlocked and will move into any position. The interlock system also prevents the ignition switch from being turned to the LOCK or ACCESSORY position (Fig. 148) unless the shifter is fully locked into the PARK position.

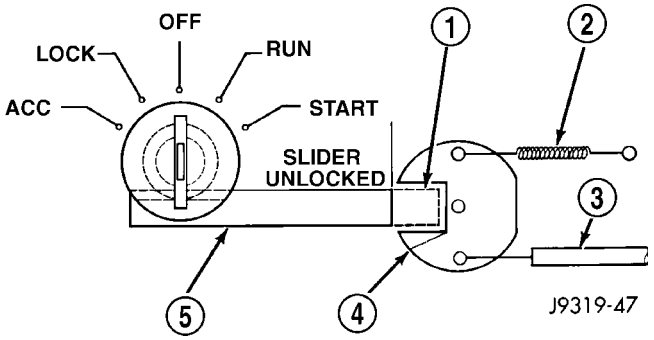


Fig. 148 Ignition Key Cylinder Actuation

1 - SLIDER LOCKED
2 - CAM RETURN SPRING
3 - INTERLOCK CABLE
4 - CAM
5 - SLIDER

REMOVAL

(1) Remove lower steering column cover. (Refer to 23 - BODY/INSTRUMENT PANEL/STEERING COLUMN OPENING COVER - REMOVAL)

(2) Remove lower steering column shroud.

(3) Remove tie strap near the solenoid retaining the brake transmission interlock cable to the steering column.

(4) Disengage wire connector from solenoid.

(5) With the ignition removed or in the unlocked position, disengage lock tab holding cable end to steering column (Fig. 149).

(6) Pull cable end from steering column.

(7) Remove the floor console and related trim. (Refer to 23 - BODY/INTERIOR/FLOOR CONSOLE - REMOVAL)

(8) Disconnect the cable from the bellcrank (Fig. 150).

(9) Disconnect and remove the cable from the shift bracket.

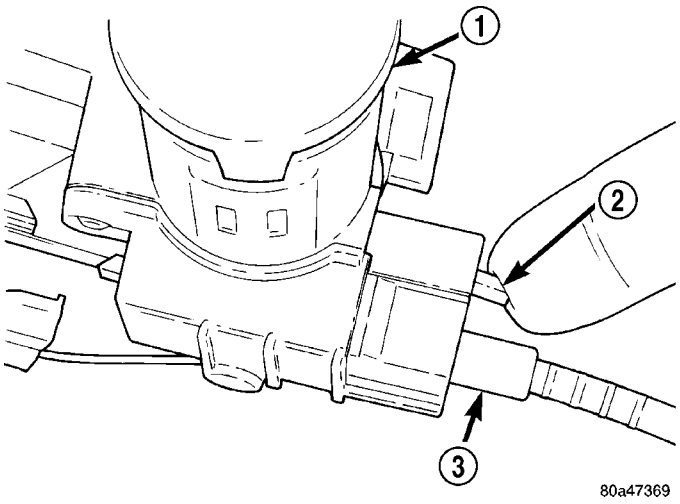


Fig. 149 Brake/Park Interlock Cable

1 - IGNITION LOCK
2 - LOCK TAB
3 - CABLE END

INSTALLATION

(1) Route replacement cable behind instrument panel and under floor console area to shift mechanism.

(2) Insert cable end into opening in steering column hub under ignition lock. Push cable inward until lock tab engages.

(3) Insert the cable end into the shifter bellcrank.

(4) Place gear selector in PARK.

(5) Push the spring-loaded cable adjuster forward and snap cable into bracket.

(6) Adjust the brake transmission shifter interlock cable.

(7) Verify that the cable adjuster lock clamp is pushed downward to the locked position.

(8) Test the park-lock cable operation.

(9) Install the floor console and related trim.

(10) Install tie strap to hold cable to base of steering column.

(11) Install lower steering column shroud and ignition lock.

(12) Install lower steering column cover.

ADJUSTMENTS

ADJUSTMENT - BRAKE TRANSMISSION SHIFT INTERLOCK CABLE

(1) Shift transmission into PARK.

(2) Remove shift lever bezel and console screws. Raise bezel and console for access to cable.

(3) Pull cable lock button up to release cable.

(4) Turn ignition switch to LOCK position.

(5) Use a spacer to create a one millimeter gap between the shifter pawl and top of the shift gate.

BRAKE TRANSMISSION SHIFT INTERLOCK MECHANISM (Continued)

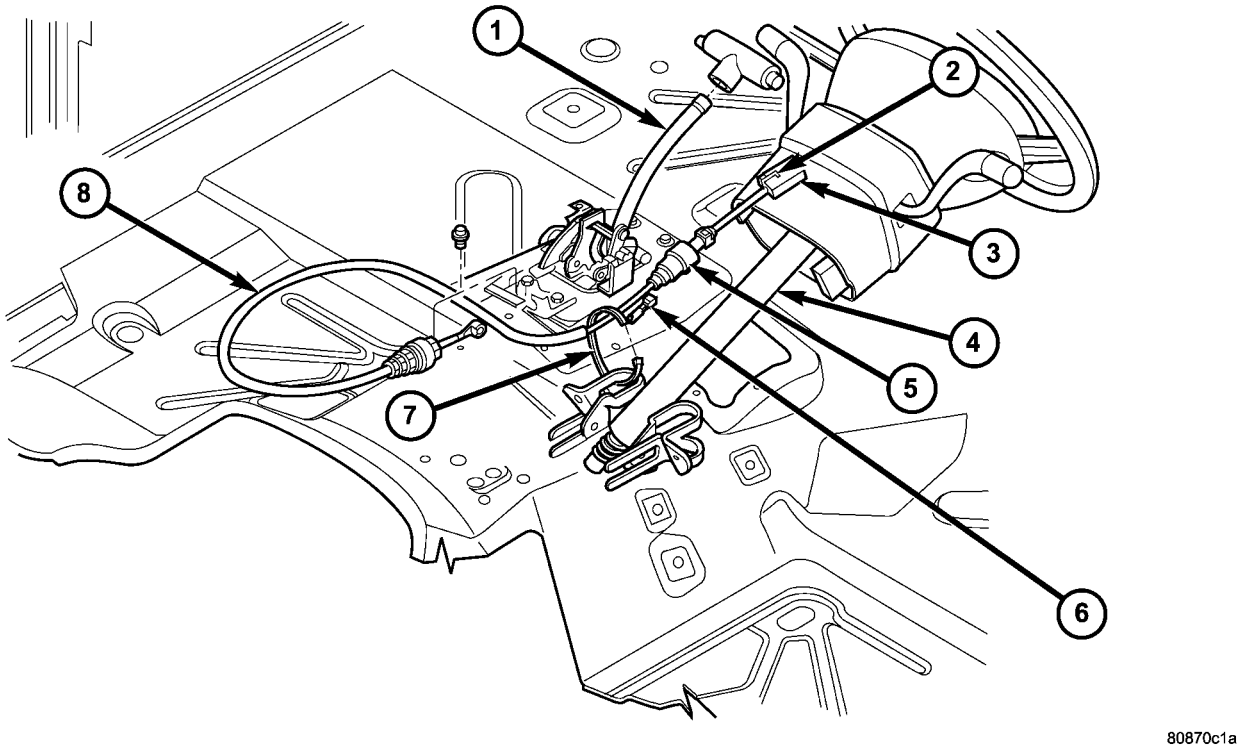


Fig. 150 Cable and Shifter

1 - SHIFT MECHANISM
2 - LOCK-TAB
3 - IGNITION LOCK INTERLOCK
4 - STEERING COLUMN
5 - SOLENOID
6 - WIRE CONNECTOR
7 - TIE STRAP
8 - PARK/BRAKE INTERLOCK CABLE

(6) Pull the cable forward and release.Ensure the cable end is seated in the bellcrank and press cable lock button down until it snaps in place.

(7) Check adjustment as follows:

(a) Check movement of release shift handle button (floor shift) or release lever (column shift). You should not be able to press button inward or move column lever.

(b) Turn ignition switch to RUN position.

(c) Shifting out of park should not be possible.

(d) Apply the brake and attempt to shift out of PARK. Shifting should be possible.

(e) While the transmission is shifted out of PARK, release the brake and attempt to shift through all gears. Release the shift button at least once during this procedure. The ignition key should not go to the LOCK position.

(f) Return transmission to the PARK position without applying the brake.

(8) Move shift lever back to PARK and check ignition switch operation. You should be able to turn switch to LOCK position and shift lever release button/lever should not move.

DRIVING CLUTCHES

DESCRIPTION

Three hydraulically applied input clutches are used to drive planetary components. The underdrive (2), overdrive (3), and reverse (4) clutches are considered input/driving clutches and are contained within the input clutch assembly (Fig. 151). The input clutch assembly also contains:

- Input shaft
- Input hub
- Clutch retainer
- Underdrive piston
- Overdrive/reverse piston
- Overdrive hub
- Underdrive hub

OPERATION

The three input clutches are responsible for driving different components of the planetary geartrain.

NOTE: (Refer to 21 - TRANSMISSION/AUTOMATIC - 42RLE - DIAGNOSIS AND TESTING) for a collective view of which clutch elements are applied at each position of the selector lever.

DRIVING CLUTCHES (Continued)

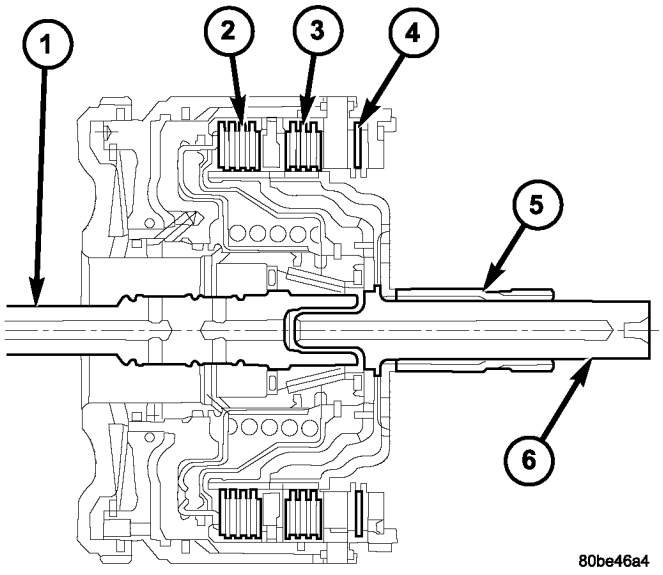


Fig. 151 Input Clutch Assembly

1 - INPUT SHAFT
2 - UNDERDRIVE CLUTCH
3 - OVERDRIVE CLUTCH
4 - REVERSE CLUTCH
5 - OVERDRIVE SHAFT
6 - UNDERDRIVE SHAFT

UNDERDRIVE CLUTCH

The underdrive clutch is hydraulically applied in first, second, and third (direct) gears by pressurized fluid against the underdrive piston. When the underdrive clutch is applied, the underdrive hub drives the rear sun gear.

OVERDRIVE CLUTCH

The overdrive clutch is hydraulically applied in third (direct) and overdrive gears by pressurized fluid against the overdrive/reverse piston. When the overdrive clutch is applied, the overdrive hub drives the front planet carrier.

REVERSE CLUTCH

The reverse clutch is hydraulically applied in reverse gear only by pressurized fluid against the overdrive/reverse piston. When the reverse clutch is applied, the front sun gear assembly is driven.

FLUID AND FILTER

DIAGNOSIS AND TESTING

DIAGNOSIS AND TESTING - CAUSES OF BURNT FLUID

Burnt, discolored fluid is a result of overheating which has two primary causes.

(1) A result of restricted fluid flow through the main and/or auxiliary cooler. This condition is usually the result of a damaged main cooler, or severe restrictions in the coolers and lines caused by debris or kinked lines.

(2) Heavy duty operation with a vehicle not properly equipped for this type of operation. Trailer towing or similar high load operation will overheat the transmission fluid if the vehicle is improperly equipped. Such vehicles should have an auxiliary transmission fluid cooler, a heavy duty cooling system, and the engine/axle ratio combination needed to handle heavy loads.

EFFECTS OF INCORRECT FLUID LEVEL

A low fluid level allows the pump to take in air along with the fluid. Air in the fluid will cause fluid pressures to be low and develop slower than normal. If the transmission is overfilled, the gears churn the fluid into foam. This aerates the fluid and causing the same conditions occurring with a low level. In either case, air bubbles cause fluid overheating, oxidation, and varnish buildup which interferes with valve and clutch operation. Foaming also causes fluid expansion which can result in fluid overflow from the transmission vent or fill tube. Fluid overflow can easily be mistaken for a leak if inspection is not careful.

FLUID CONTAMINATION

Transmission fluid contamination is generally a result of:

- adding incorrect fluid
- failure to clean dipstick and fill tube when checking level
- engine coolant entering the fluid
- internal failure that generates debris
- overheat that generates sludge (fluid breakdown)
- failure to replace contaminated converter after repair

The use of non-recommended fluids can result in transmission failure. The usual results are erratic shifts, slippage, abnormal wear and eventual failure due to fluid breakdown and sludge formation. Avoid this condition by using recommended fluids only.

The dipstick cap and fill tube should be wiped clean before checking fluid level. Dirt, grease and

FLUID AND FILTER (Continued)

other foreign material on the cap and tube could fall into the tube if not removed beforehand. Take the time to wipe the cap and tube clean before withdrawing the dipstick.

Engine coolant in the transmission fluid is generally caused by a cooler malfunction. The only remedy is to replace the radiator as the cooler in the radiator is not a serviceable part. If coolant has circulated through the transmission, an overhaul is necessary.

The torque converter should be replaced whenever a failure generates sludge and debris. This is necessary because normal converter flushing procedures will not remove all contaminants.

STANDARD PROCEDURE

STANDARD PROCEDURE - FLUID LEVEL CHECK

FLUID LEVEL CHECK

The transmission sump has a dipstick to check oil similar to most automatic transmissions. It is located on the left side of the engine. Be sure to wipe all dirt from dipstick handle before removing.

The torque converter fills in both the PARK and NEUTRAL positions. Place the selector lever in PARK to be sure that the fluid level check is accurate. **The engine should be running at idle speed for at least one minute, with the vehicle on level ground.** At normal operating temperature (approximately 82 C. or 180 F.), the fluid level is correct if it is in the HOT region (cross-hatched area) on the oil level indicator. The fluid level should be in COLD region at 70° F fluid temperature. Adjust fluid level as necessary. Use only Mopar® ATF+4, Automatic Transmission Fluid.

FLUID LEVEL CHECK USING DRB

NOTE: Engine and Transmission should be at normal operating temperature before performing this procedure.

(1) Start engine and apply parking brake.
(2) Connect DRBIII® scan tool and select transmission.
(3) Select sensors.
(4) Read the transmission temperature value.
(5) Compare the fluid temperature value with the chart.
(6) Adjust transmission fluid level shown on the dipstick according to the 42RLE Fluid Temperature Chart (Fig. 152). Use only Mopar® ATF+4, Automatic Transmission Fluid.
(7) Check transmission for leaks.

STANDARD PROCEDURE - FLUID/FILTER SERVICE

NOTE: Only fluids of the type labeled Mopar® ATF+4, Automatic Transmission Fluid, should be used in the transmission sump. A filter change should be made at the time of the transmission oil change. The magnet (on the inside of the oil pan) should also be cleaned with a clean, dry cloth.

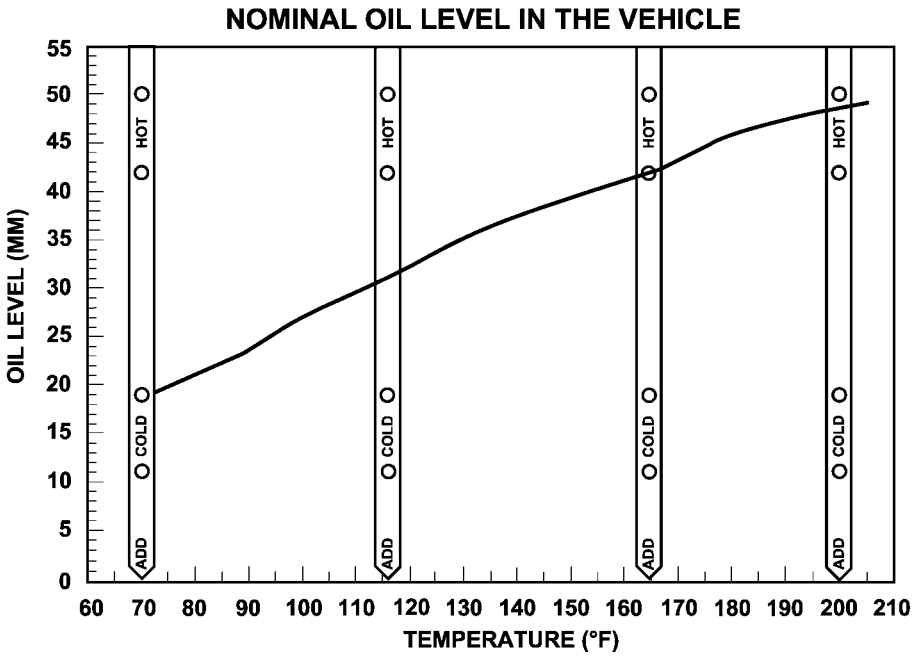


Fig. 152 42RLE Fluid Temperature Chart

FLUID AND FILTER (Continued)

NOTE: If the transmission is disassembled for any reason, the fluid and filter should be changed.

(1) Raise vehicle on a hoist. Place a drain container with a large opening, under transmission oil pan.

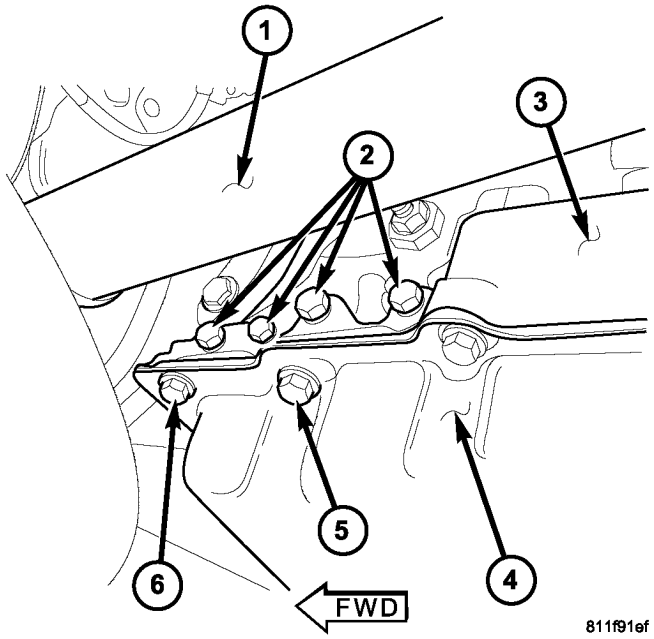


Fig. 153 Pan Fastener

1 - FRONT DRIVESHAFT
2 - PRESSURE PORTS
3 - TRANSMISSION CASE
4 - TRANSMISSION OIL PAN
5 - SECOND TRANSMISSION OIL PAN BOLT ON LEFT SIDE
6 - FIRST TRANSMISSION OIL PAN BOLT

NOTE: One of the oil pan bolts (5) has a sealing patch applied from the factory. Separate this bolt for reuse.

(2) Loosen pan bolts and tap the pan at one corner to break it loose allowing fluid to drain, then remove the oil pan.

(3) Install a new filter and o-ring on bottom of the valve body and tighten retaining screws to 5 N·m (40 in. lbs.).

NOTE: Before installing the oil pan bolt (5) in the bolt hole located between the torque converter clutch on and U/D clutch pressure tap circuits (Fig. 153), it will be necessary to replentish the sealing patch on the bolt using Mopar® Lock & Seal Adhesive.

(4) Clean the oil pan and magnet. Reinstall pan using new Mopar® Silicone Adhesive sealant. Tighten oil pan bolts to 19 N·m (165 in. lbs.).

(5) Pour four quarts of Mopar® ATF+4, Automatic Transmission Fluid, through the dipstick opening.

(6) Start engine and allow to idle for at least one minute. Then, with parking and service brakes applied, move selector lever momentarily to each position, ending in the park or neutral position.

(7) Check the transmission fluid level and add an appropriate amount to bring the transmission fluid level to 3mm (1/8 in.) below the lowest mark on the dipstick.

(8) Recheck the fluid level after the transmission has reached normal operating temperature (180°F.).

(9) To prevent dirt from entering transmission, make certain that dipstick is fully seated into the dipstick opening.

STANDARD PROCEDURE - TRANSMISSION FILL

To avoid overfilling transmission after a fluid change or overhaul, perform the following procedure:

(1) Remove dipstick and insert clean funnel in transmission fill tube.

(2) Add following initial quantity of Mopar® ATF +4, Automatic Transmission Fluid, to transmission:

(a) If only fluid and filter were changed, add **6 pints (3 quarts)** of ATF +4 to transmission.

(b) If transmission was completely overhauled, or torque converter was replaced or drained, add **10 pints (5 quarts)** of ATF +4 to transmission.

(3) Apply parking brakes.

(4) Start and run engine at normal curb idle speed.

(5) Apply service brakes, shift transmission through all gear ranges then back to NEUTRAL, set parking brake, and leave engine running at curb idle speed.

(6) Remove funnel, insert dipstick and check fluid level. If level is low, **add fluid to bring level to MIN mark on dipstick.** Check to see if the oil level is equal on both sides of the dipstick. If one side is noticably higher than the other, the dipstick has picked up some oil from the dipstick tube. Allow the oil to drain down the dipstick tube and re-check.

(7) Drive vehicle until transmission fluid is at normal operating temperature.

(8) With the engine running at curb idle speed, the gear selector in NEUTRAL, and the parking brake applied, check the transmission fluid level.

CAUTION: Do not overfill transmission, fluid foaming and shifting problems can result.

(9) Add fluid to bring level up to MAX arrow mark.

When fluid level is correct, shut engine off, release park brake, remove funnel, and install dipstick in fill tube.

GEARSHIFT CABLE

REMOVAL

(1) Shift transmission into PARK.
(2) Remove shift lever bezel and necessary console parts for access to shift lever assembly. (Refer to 23 - BODY/INTERIOR/FLOOR CONSOLE - REMOVAL)
(3) Disconnect cable at shift lever (Fig. 154) and feed cable through dash panel opening to underside of vehicle.
(4) Raise vehicle.
(5) Disengage cable eyelet at transmission shift lever and removel cable from the mounting bracket. Then remove old cable from vehicle.

INSTALLATION

(1) Route cable through hole in dash panel. Fully seat cable grommet into dash panel.
(2) Place the auto transmission manual shift control lever in “PARK” detent (rearmost) position and rotate prop shaft to ensure transmission is in PARK.
(3) Snap the cable into the transmission bracket so the retaining clip is engaged and connect cable end fitting onto the manual control lever ball stud.
(4) Place the floor shifter lever in PARK position.
(5) Connect shift cable to shifter mechanism by snapping cable retaining ears into shifter bracket and press cable end fitting onto lever ball stud.
(6) Snap the cable into the transmission bracket so the retaining ears are engaged and connect cable end fitting onto the manual control lever ball stud.
(7) Adjust the shift mechanism.
(8) Install any floor console components removed previously. (Refer to 23 - BODY/INTERIOR/FLOOR CONSOLE - INSTALLATION)

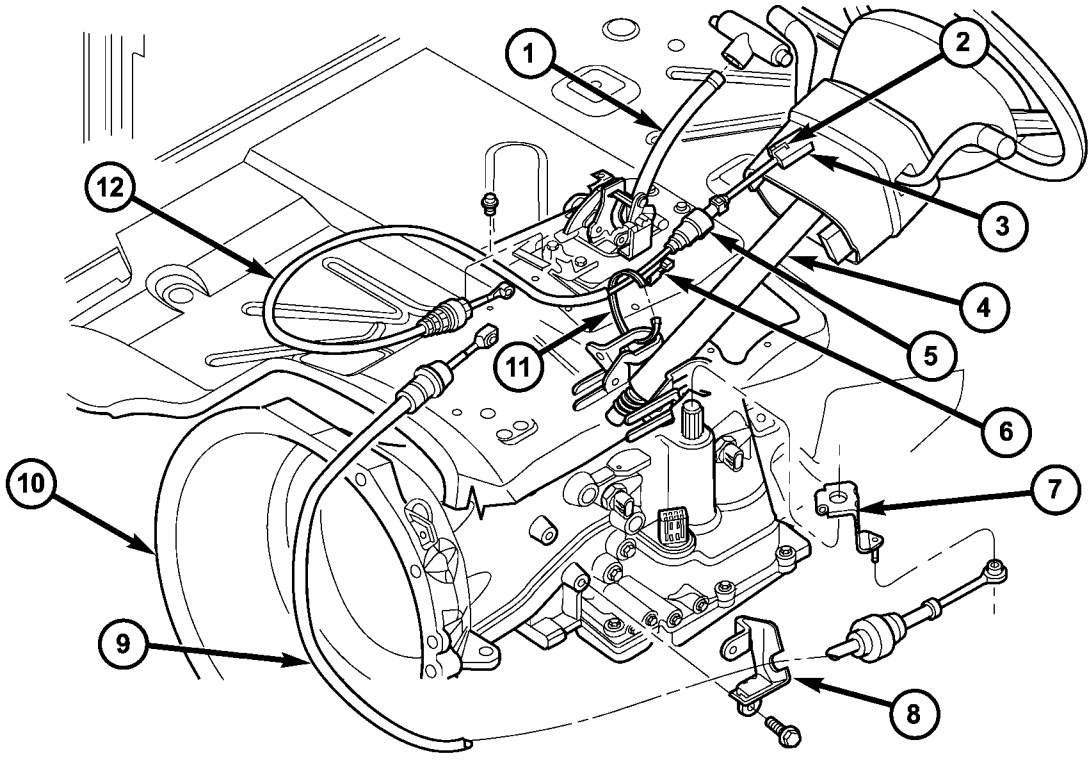


Fig. 154 Shifter Cable Routing

1 - SHIFT MECHANISM
2 - LOCK-TAB
3 - IGNITION LOCK INTERLOCK
4 - STEERING COLUMN
5 - SOLENOID
6 - WIRE CONNECTOR
7 - MANUAL LEVER
8 - CABLE BRACKET
9 - SHIFTER CABLE
10 - BELLHOUSING
11 - TIE STRAP
12 - PARK/BRAKE INTERLOCK CABLE

HOLDING CLUTCHES

DESCRIPTION

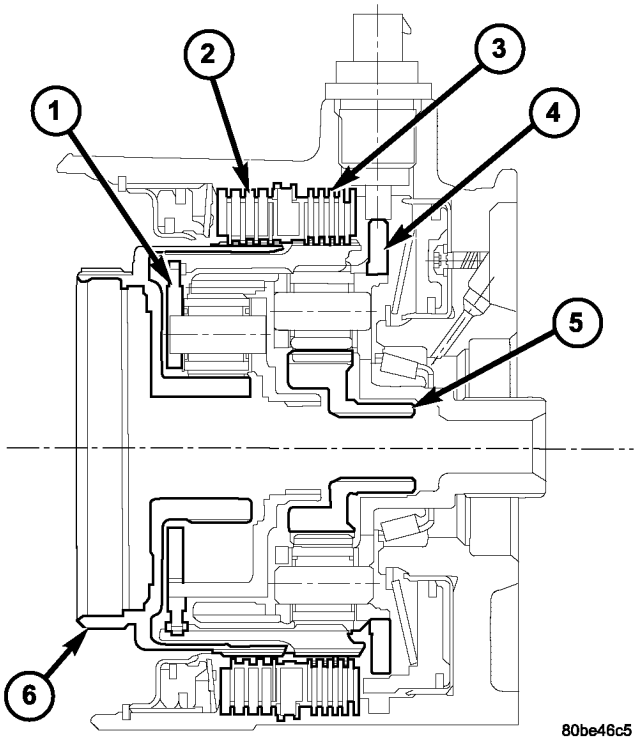


Fig. 155 2/4 and Low/Reverse Clutches

1 - FRONT PLANET CARRIER/REAR ANNULUS
2 - 2/4 CLUTCH
3 - L/R CLUTCH
4 - REAR PLANET CARRIER/FRONT ANNULUS
5 - REAR SUN GEAR
6 - FRONT SUN GEAR ASSEMBLY

Two hydraulically applied multi-disc clutches are used to hold planetary geartrain components stationary while the input clutches drive others. The 2/4 (2) and Low/Reverse (3) clutches are considered holding clutches and are contained at the rear of the transmission case (Fig. 155).

OPERATION

NOTE: (Refer to 21 - TRANSMISSION/AUTOMATIC - 42RLE - DIAGNOSIS AND TESTING) for a collective view of which clutch elements are applied at each position of the selector lever.

2/4 CLUTCH

The 2/4 clutch is hydraulically applied in second and fourth gears by pressurized fluid against the 2/4 clutch piston. When the 2/4 clutch is applied, the front sun gear assembly is held or grounded to the transmission case.

LOW/REVERSE CLUTCH

The Low/Reverse clutch is hydraulically applied in park, reverse, neutral, and first gears by pressurized fluid against the Low/Reverse clutch piston. When the Low/Reverse clutch is applied, the front planet carrier/rear annulus assembly is held or grounded to the transmission case.

INPUT CLUTCH ASSEMBLY

DISASSEMBLY

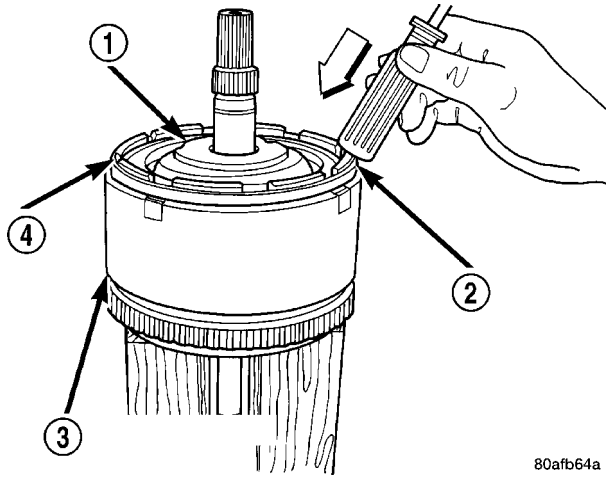


Fig. 156 Tapping Reaction Plate

1 - #4 THRUST PLATE (SELECT)
2 - TAP DOWN REVERSE CLUTCH REACTION PLATE TO REMOVE OR INSTALL SNAP RING
3 - INPUT CLUTCH RETAINER
4 - REVERSE CLUTCH REACTION PLATE

(1) Mount input clutch assembly to Input Clutch Pressure Fixture (Tool 8391).

(2) Tap down (2) reverse clutch reaction plate (4) to release pressure from snap ring (Fig. 156).

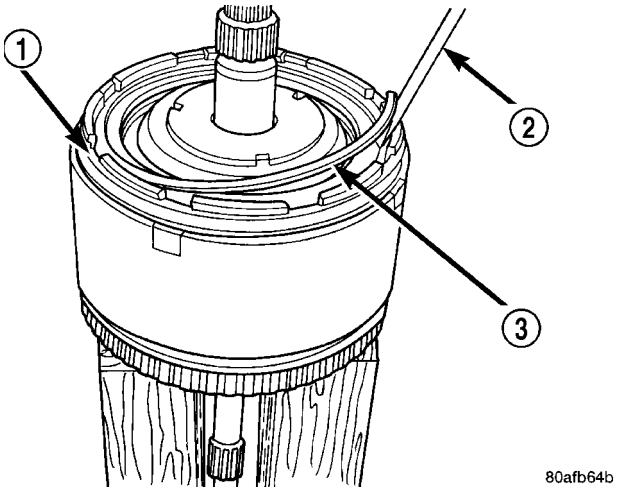


Fig. 157 Reverse Clutch Snap Ring

1 - REACTION PLATE
2 - SCREWDRIVER
3 - REVERSE CLUTCH SNAP RING (SELECT)

INPUT CLUTCH ASSEMBLY (Continued)

(3) Remove reverse clutch snap ring (3) (Fig. 157).

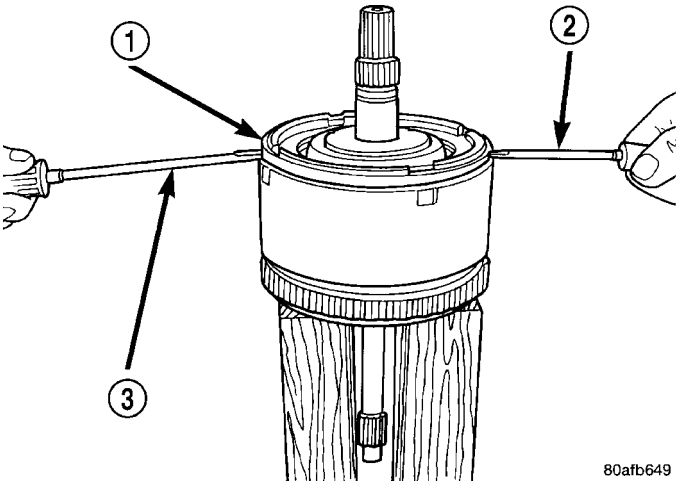


Fig. 158 Pry Reverse Clutch Reaction Plate

1 - REVERSE CLUTCH REACTION PLATE
2 - SCREWDRIVER
3 - SCREWDRIVER

(4) Pry up reverse clutch reaction plate (1) (Fig. 158).

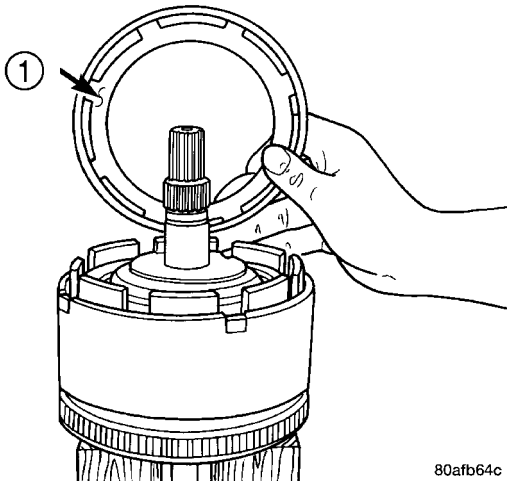


Fig. 159 Reverse Clutch Reaction Plate

1 - REVERSE CLUTCH REACTION PLATE (INSTALL FLAT SIDE DOWN)

(5) Remove reverse clutch reaction plate (1) (Fig. 159).

NOTE: Tag reverse clutch pack for reassembly identification.

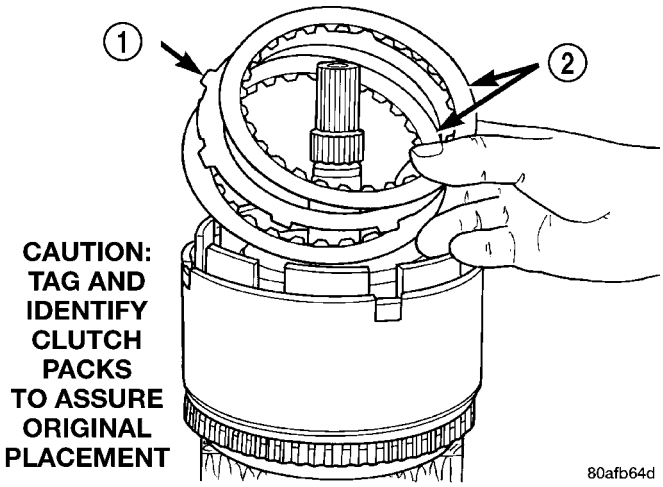


Fig. 160 Reverse Clutch Pack

1 - REVERSE CLUTCH PLATE
2 - REVERSE CLUTCH DISC

(6) Remove the reverse clutch pack (two fibers/one steel) (1, 2) (Fig. 160).

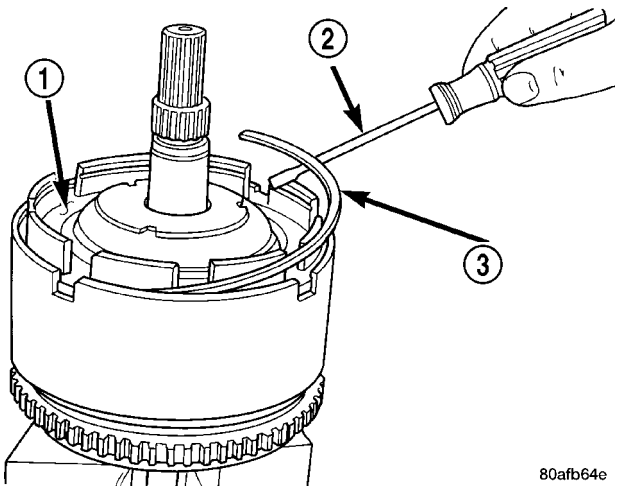


Fig. 161 OD/Reverse Pressure Plate Snap Ring

1 - OD/REVERSE PRESSURE PLATE
2 - SCREWDRIVER
3 - OD/REVERSE PRESSURE PLATE SNAP RING

(7) Remove the OD/Reverse reaction plate (1) snap ring (3) (Fig. 161).

INPUT CLUTCH ASSEMBLY (Continued)

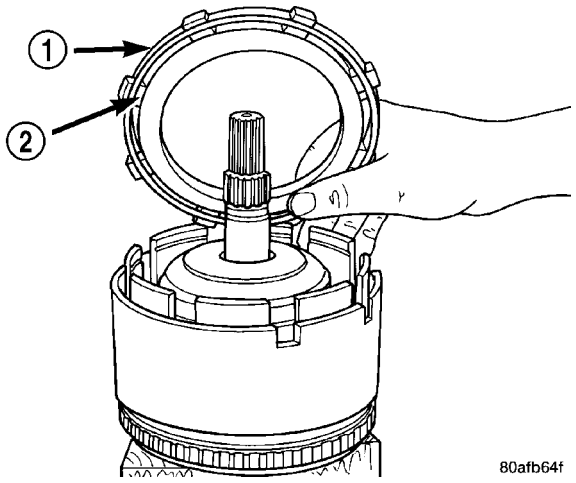


Fig. 162 OD/Reverse Reaction Plate

1 - OD/REVERSE PRESSURE PLATE (STEP SIDE DOWN)
2 - (STEP SIDE DOWN)

(8) Remove OD/Reverse pressure plate (1) (Fig. 162).

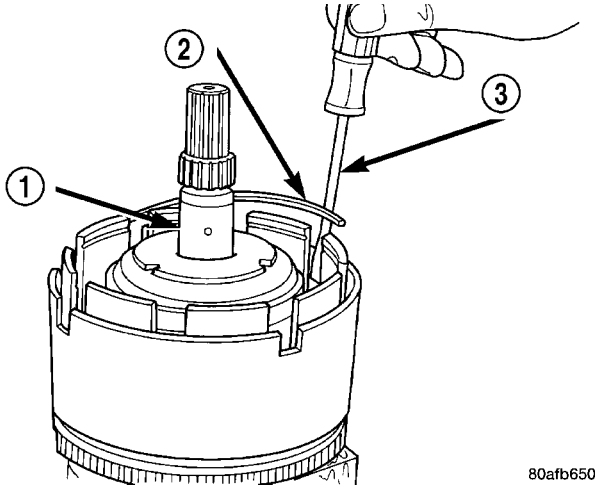


Fig. 163 Waved Snap Ring

1 - OVERDRIVE SHAFT ASSEMBLY
2 - OD/REVERSE CLUTCH WAVED SNAP RING
3 - SCREWDRIVER

(9) Remove OD/Reverse reaction plate wave snap ring (2) (Fig. 163).

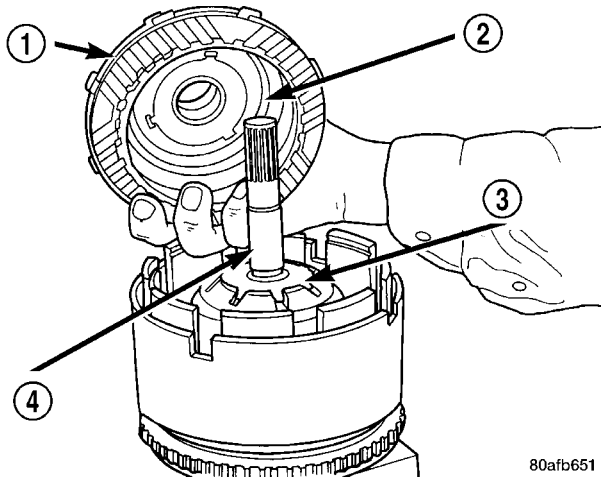


Fig. 164 Remove OD Clutch Pack

1 - OVERDRIVE SHAFT ASSEMBLY AND OD CLUTCH PACK
2 - #3 THRUST PLATE
3 - #3 THRUST WASHER
4 - UNDERDRIVE SHAFT ASSEMBLY

(10) Remove OD shaft/hub and OD clutch pack (1) (Fig. 164).

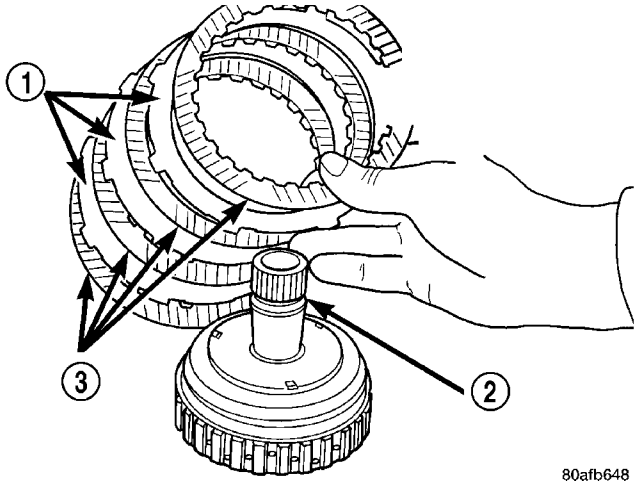


Fig. 165 Overdrive Clutch Pack

1 - OVERDRIVE CLUTCH PLATE
2 - OVERDRIVE SHAFT ASSEMBLY
3 - OVERDRIVE CLUTCH DISC

INPUT CLUTCH ASSEMBLY (Continued)

NOTE: Tag overdrive clutch pack for reassembly identification.

(11) Remove the overdrive clutch (1, 3) from the overdrive hub/shaft (2) (Fig. 165).

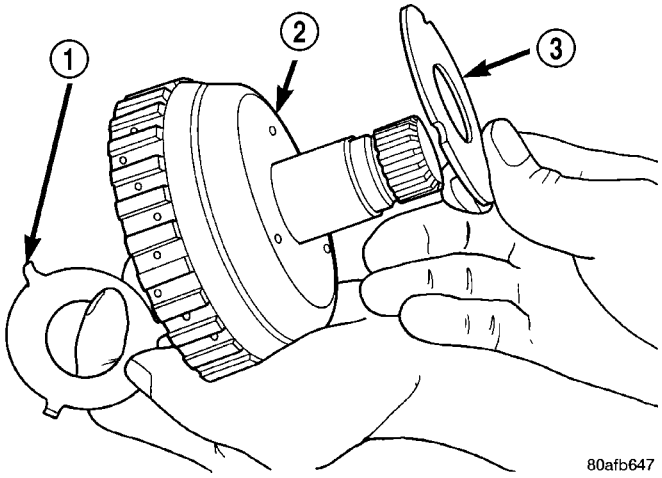


Fig. 166 #3 and #4 Thrust Washers

1 - #3 THRUST PLATE (3 TABS)
2 - OD SHAFT ASSEMBLY
3 - #4 THRUST PLATE (3 SLOTS)

(12) Remove and inspect number 3 and 4 thrust plates (1, 3) (Fig. 166).

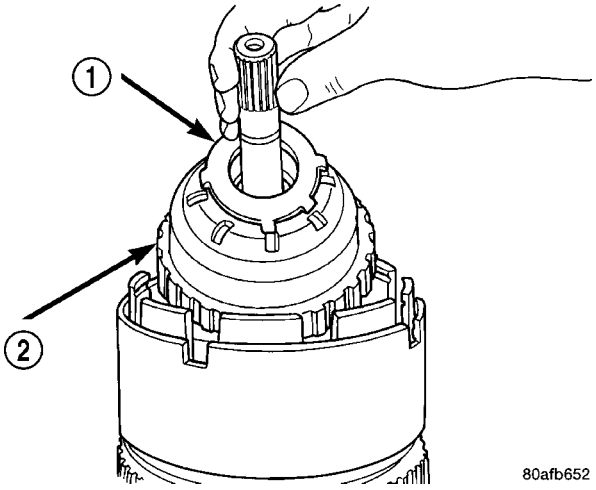


Fig. 167 Underdrive Shaft Assembly

1 - #3 THRUST WASHER (5 TABS)
2 - UNDERDRIVE SHAFT ASSEMBLY

(13) Remove the underdrive shaft assembly (2) (Fig. 167).

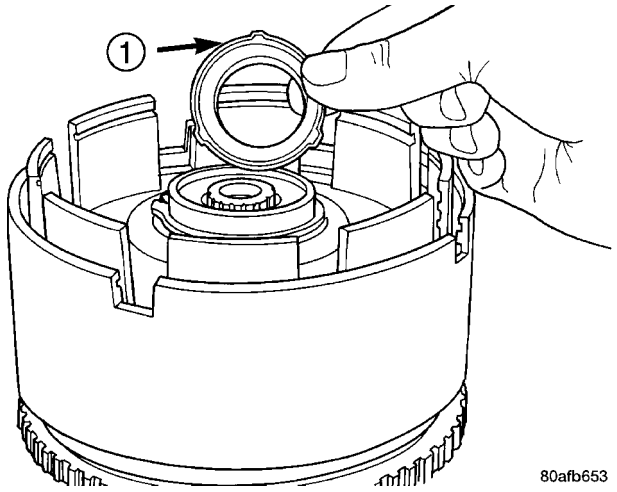


Fig. 168 No 2 Needle Bearing

1 - #2 NEEDLE BEARING (NOTE 3 TABS)

(14) Remove the number 2 needle bearing (1) (Fig. 168).

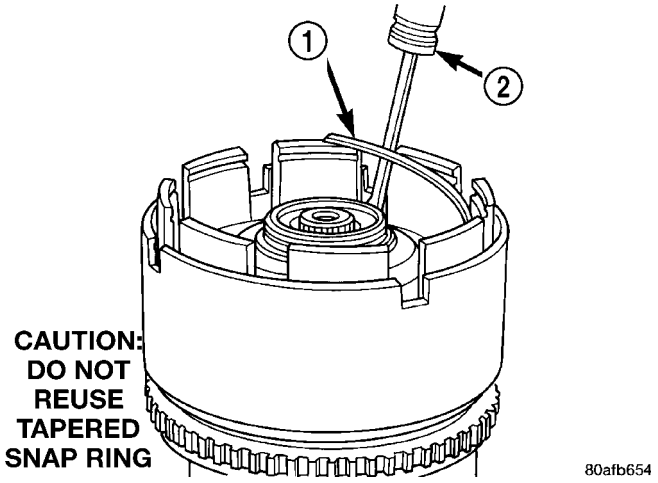


Fig. 169 Tapered Snap Ring

1 - OVERDRIVE/UNDERDRIVE CLUTCHES REACTION PLATE TAPERED SNAP RING
2 - SCREWDRIVER (DO NOT SCRATCH REACTION PLATE)

(15) Remove the OD/UD reaction plate tapered snap ring (1) (Fig. 169).

INPUT CLUTCH ASSEMBLY (Continued)

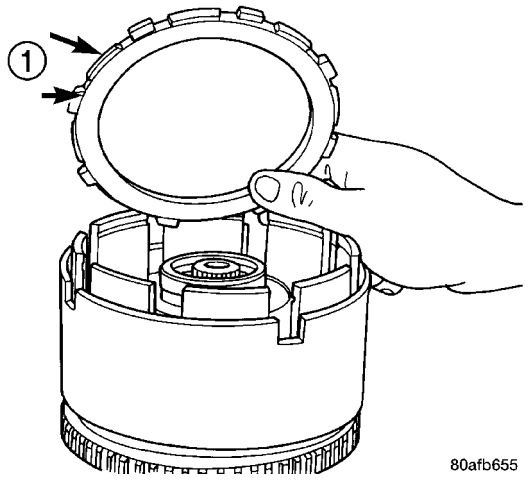


Fig. 170 OD/UD Reaction Plate

1 - OD/UD CLUTCH REACTION PLATE (STEP SIDE DOWN)

NOTE: The OD/UD clutch reaction plate has a step on both sides. The OD/UD clutches reaction plate goes tapered step side up.

(16) Remove the OD/UD reaction plate (1) (Fig. 170).

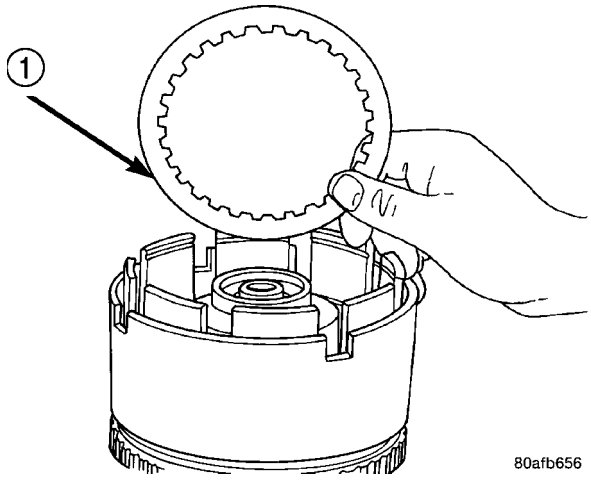


Fig. 171 Remove Last UD Clutch Disc

1 - ONE UNDERDRIVE CLUTCH DISC

(17) Remove the first UD clutch disc (1) (Fig. 171).

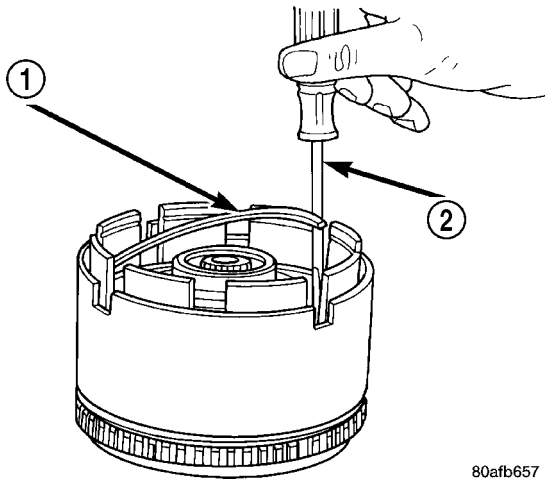


Fig. 172 UD Clutch Flat Snap Ring

1 - UNDERDRIVE CLUTCH REACTION PLATE FLAT SNAP RING
2 - SCREWDRIVER

(18) Remove the UD clutch flat snap ring (1) (Fig. 172).

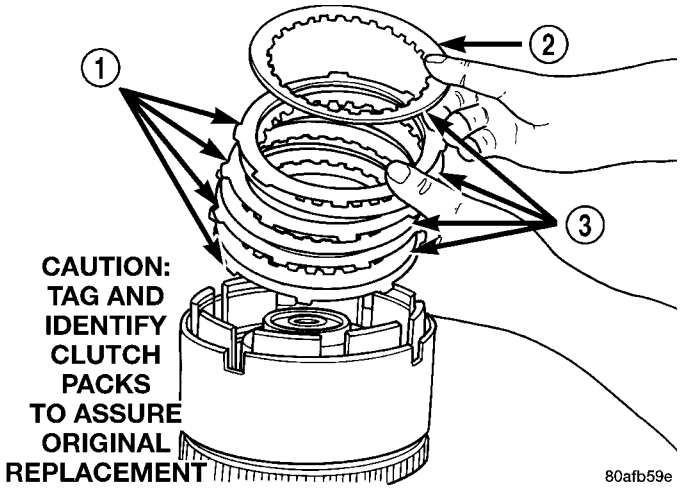


Fig. 173 Underdrive Clutch Pack

1 - CLUTCH PLATE
2 - ONE UD CLUTCH DISC
3 - CLUTCH DISC

NOTE: Tag underdrive clutch pack for reassembly identification.

(19) Remove the UD clutch pack (1, 3) (Fig. 173).

INPUT CLUTCH ASSEMBLY (Continued)

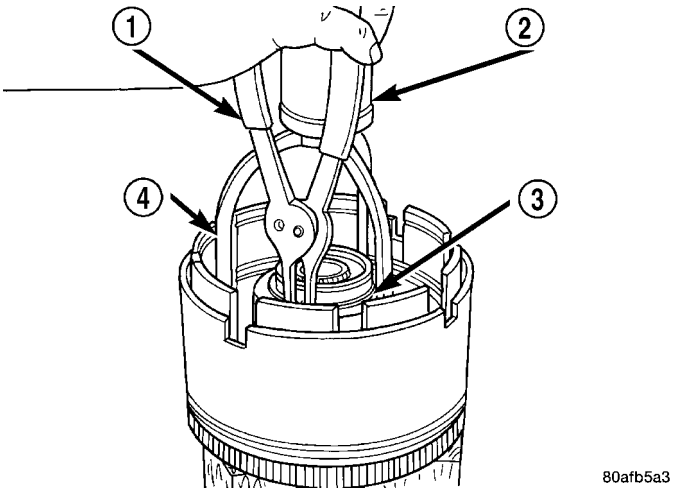


Fig. 174 UD Spring Retainer Snap Ring

1 - SNAP RING PLIERS
2 - ARBOR PRESS RAM
3 - SNAP RING
4 - SPECIAL TOOL 5059A

CAUTION: Compress return spring just enough to remove or install snap ring.

(20) Using Tool 5059A (4) and an arbor press (2), compress UD clutch piston enough to remove snap ring (3) (Fig. 174).

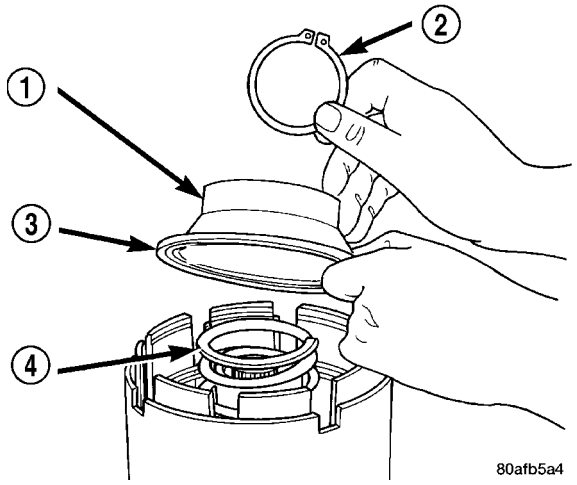


Fig. 175 UD Return Spring and Retainer

1 - UNDERDRIVE SPRING RETAINER
2 - SNAP RING
3 - SEAL
4 - PISTON RETURN SPRING

(21) Remove the underdrive spring retainer snap ring (2), spring retainer (1), and spring (4) (Fig. 175).

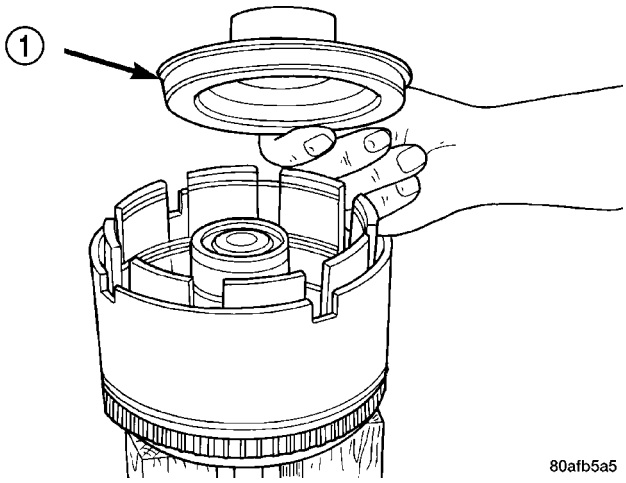


Fig. 176 Underdrive Clutch Piston

1 - PISTON

(22) Remove the UD clutch piston (1) (Fig. 176).

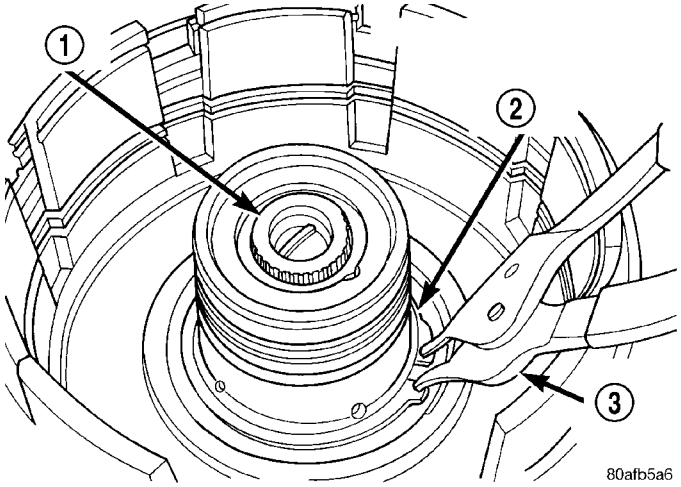


Fig. 177 Input Hub Tapered Snap Ring

1 - INPUT SHAFT
2 - INPUT HUB SNAP RING (TAPERED SIDE UP WITH TABS IN CAVITY)
3 - SNAP RING PLIERS

(23) Remove the input hub tapered snap ring (2) (Fig. 177).

INPUT CLUTCH ASSEMBLY (Continued)

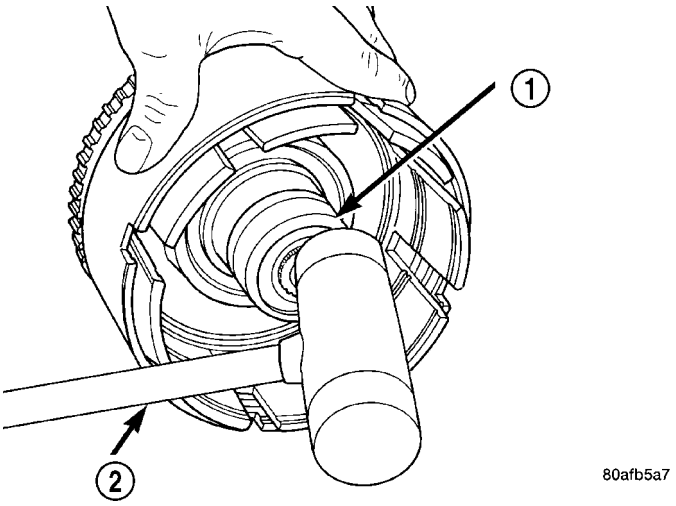


Fig. 178 Tap on Input Hub

1 - INPUT SHAFT AND HUB ASSEMBLY
2 - PLASTIC HAMMER

(24) Tap on input hub (1) with soft faced hammer (2) and separate input hub from OD/Reverse piston and clutch retainer (Fig. 178).

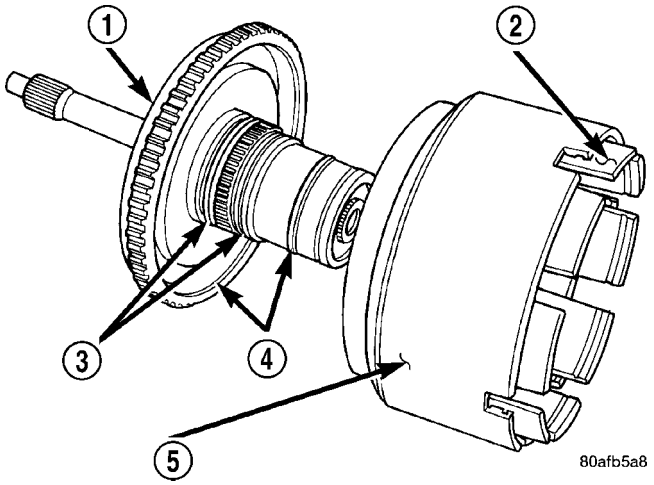


Fig. 179 Input Hub Removed

1 - INPUT SHAFT AND HUB ASSEMBLY
2 - INPUT CLUTCH RETAINER
3 - O-RING
4 - SEAL
5 - OVERDRIVE/REVERSE PISTON

(25) Separate the input hub from OD/Reverse piston (5) and clutch retainer (2) (Fig. 179).

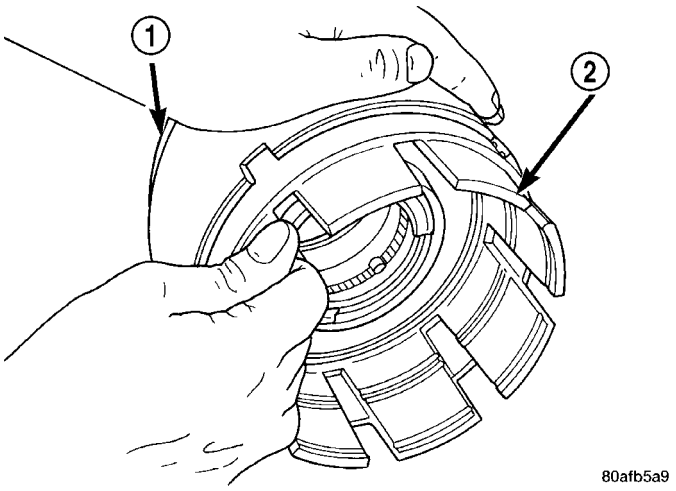


Fig. 180 Pull Retainer from Piston

1 - OVERDRIVE/REVERSE PISTON
2 - INPUT CLUTCH RETAINER

(26) Separate clutch retainer (2) from OD/Reverse piston (1) (Fig. 180).

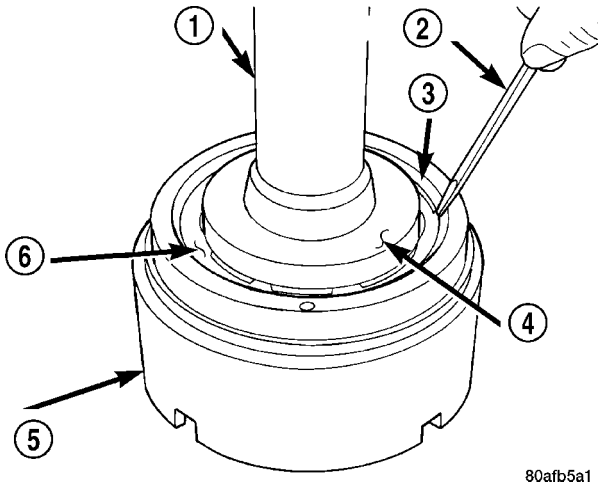


Fig. 181 Remove Snap Ring

1 - ARBOR PRESS RAM (COMPRESS RETURN SPRING JUST ENOUGH TO REMOVE OR INSTALL SNAP RING)
2 - SCREWDRIVER
3 - SNAP RING
4 - SPECIAL TOOL 6057
5 - OD/REVERSE PISTON
6 - RETURN SPRING

(27) Using Tool 6057 (4) and an arbor press(1), compress OD/Reverse piston (5) return spring just enough to remove snap ring (3) (Fig. 181).

INPUT CLUTCH ASSEMBLY (Continued)

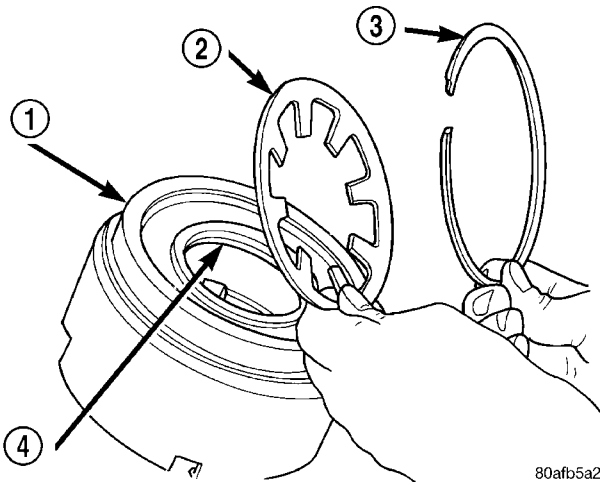


Fig. 182 Return Spring and Snap Ring

1 - OD/REVERSE PISTON
2 - RETURN SPRING
3 - SNAP RING
4 - O-RING

(28) Remove the OD/Reverse piston (5) return spring and snap ring (3) (Fig. 182).

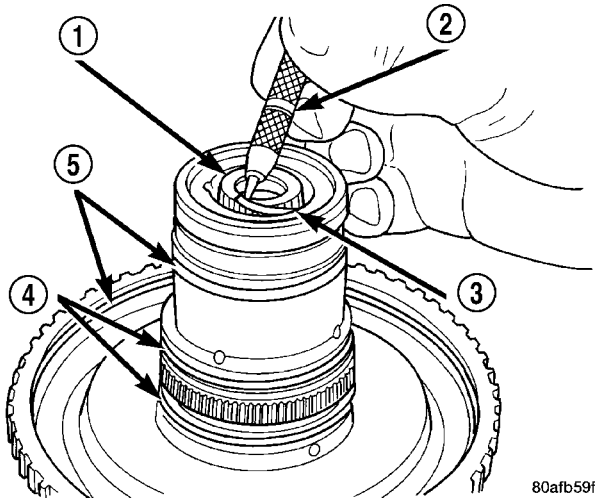


Fig. 183 Remove Input Shaft Snap Ring

1 - INPUT SHAFT
2 - SHARP-POINTED TOOL
3 - SNAP RING
4 - O-RINGS
5 - SEALS

(29) Remove input shaft (1) to input clutch hub snap ring (3) (Fig. 183).

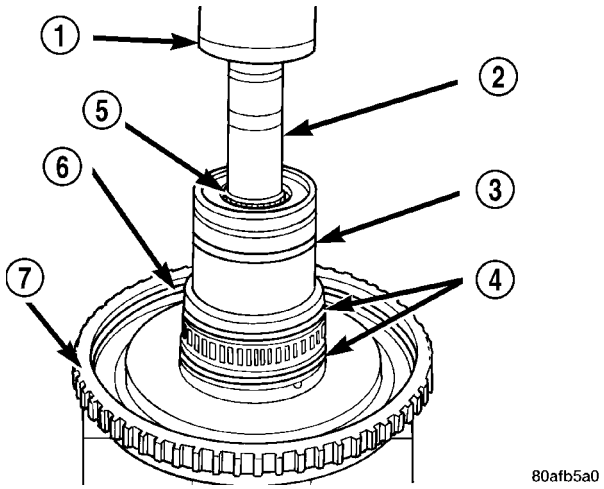


Fig. 184 Remove Input Shaft

1 - ARBOR PRESS RAM
2 - SOCKET
3 - SEAL
4 - O-RINGS
5 - INPUT SHAFT
6 - SEAL
7 - INPUT CLUTCH HUB

(30) Using a suitably sized socket (2) and an arbor press (1), remove input shaft (5) from input shaft hub (Fig. 184).

ASSEMBLY

Use petrolatum on all seals to ease assembly of components.

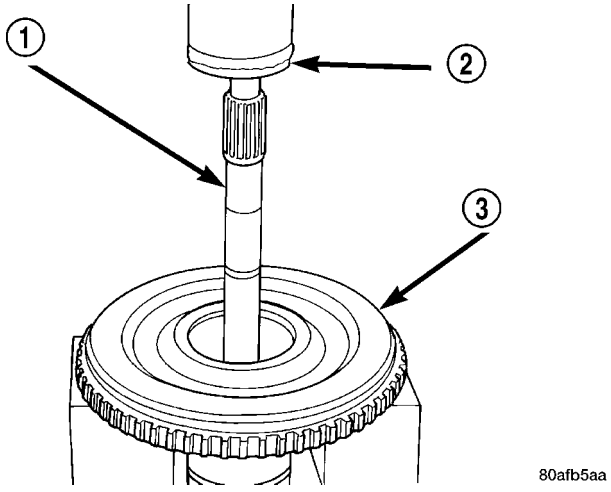


Fig. 185 Install Input Shaft

1 - INPUT SHAFT
2 - ARBOR PRESS RAM
3 - INPUT CLUTCH HUB

(1) Using an arbor press(2), install input shaft (1) to input shaft hub (2) (Fig. 185).

INPUT CLUTCH ASSEMBLY (Continued)

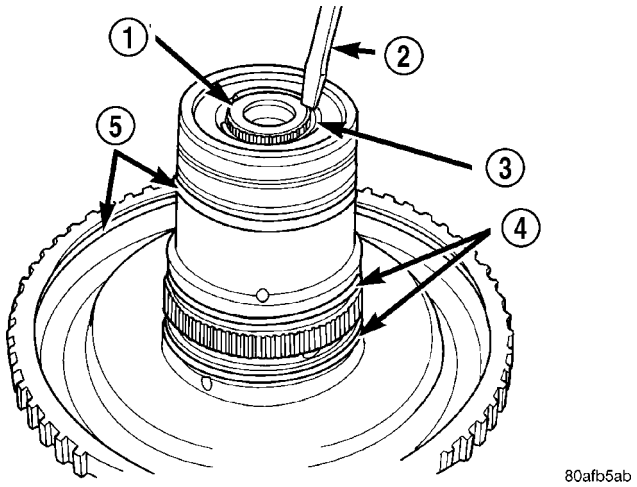


Fig. 186 Install Input Shaft Snap Ring

1 - INPUT SHAFT
2 - SCREWDRIVER (DO NOT SCRATCH BEARING SURFACE)
3 - SNAP RING
4 - O-RINGS
5 - SEALS

(2) Install input shaft snap ring (3) (Fig. 186).

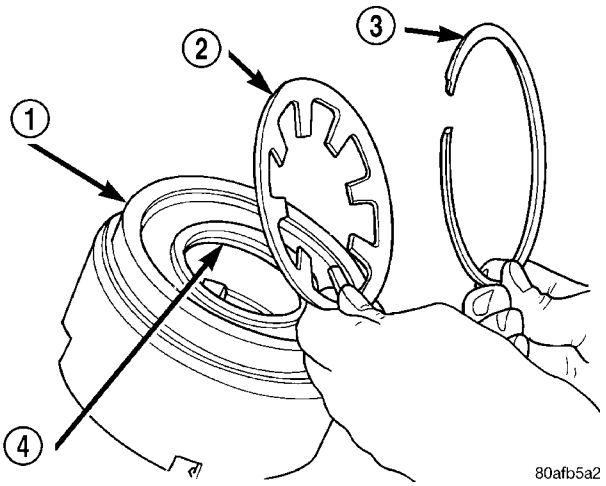


Fig. 187 Return Spring and Snap Ring

1 - OD/REVERSE PISTON
2 - RETURN SPRING
3 - SNAP RING
4 - O-RING

(3) Position the OD/Reverse piston return spring (2) and snap ring (3) onto the OD/Reverse piston (1) (Fig. 187).

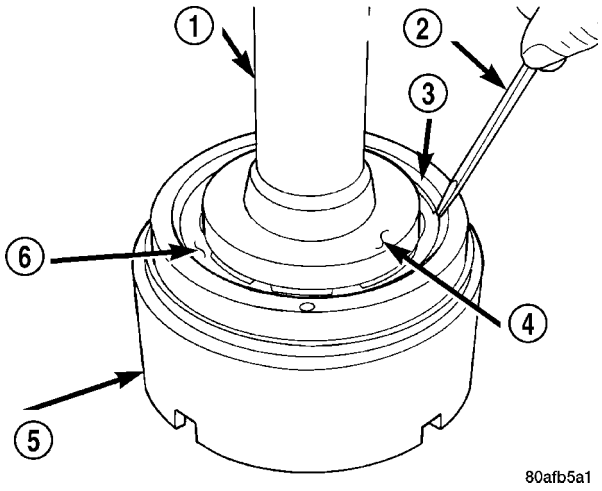


Fig. 188 Install Snap Ring

1 - ARBOR PRESS RAM (COMPRESS RETURN SPRING JUST ENOUGH TO REMOVE OR INSTALL SNAP RING)
2 - SCREWDRIVER
3 - SNAP RING
4 - SPECIAL TOOL 6057
5 - OD/REVERSE PISTON
6 - RETURN SPRING

(4) Using an arbor press (1) and Tool 6057 (4), install the OD/Reverse piston return spring (6) and snap ring (3) (Fig. 188).

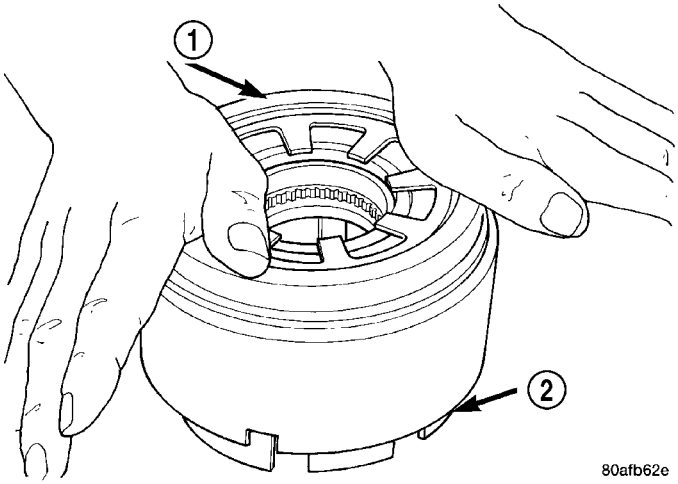


Fig. 189 Install OD/Reverse Piston

1 - PUSH DOWN TO INSTALL OVERDRIVE/REVERSE PISTON
2 - INPUT CLUTCHES RETAINER

(5) Install the OD/Reverse piston (1) assembly to the input clutch retainer (2) (Fig. 189).

INPUT CLUTCH ASSEMBLY (Continued)

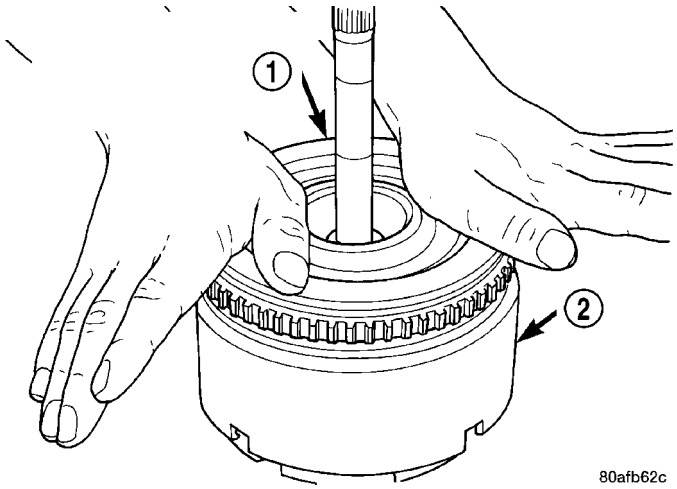


Fig. 190 Install Input Shaft Hub Assembly

1 - PUSH DOWN TO INSTALL INPUT SHAFT HUB ASSEMBLY (ROTATE TO ALIGN SPLINES)
2 - OD/REV. PISTON

(6) Install the input hub/shaft assembly (1) to the OD/Reverse piston/clutch retainer assembly (2) (Fig. 190).

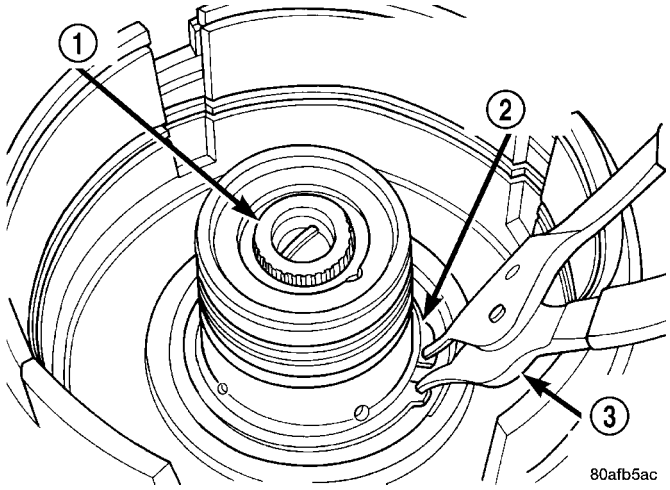


Fig. 191 Install Input Hub Tapered Snap Ring

1 - INPUT SHAFT
2 - INPUT HUB SNAP RING (TAPERED SIDE UP WITH TABS IN CAVITY)
3 - SNAP RING PLIERS

(7) Install input hub tapered snap ring (2) (Fig. 191). **Make sure snap ring is fully seated.**

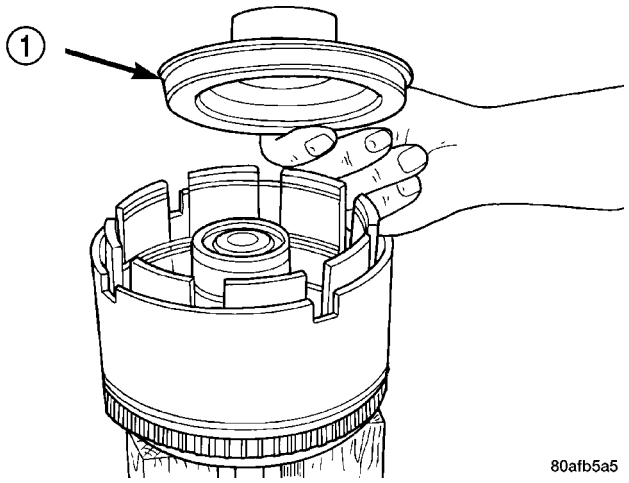


Fig. 192 Underdrive Clutch Piston

1 - PISTON

(8) Install the UD clutch piston (1) (Fig. 192).

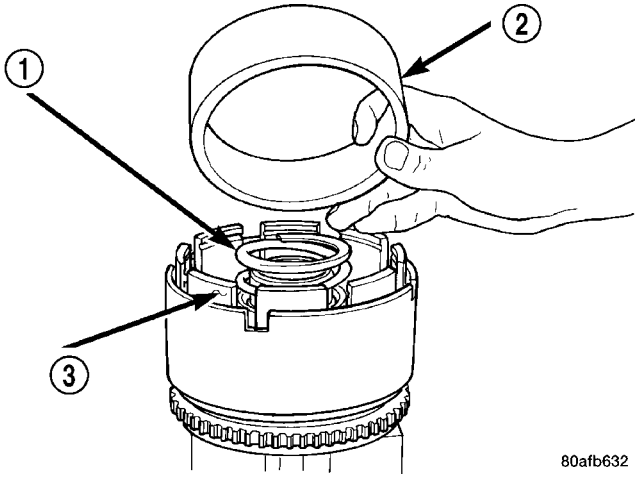


Fig. 193 Seal Compressor Special Tool 5067

1 - PISTON RETURN SPRING
2 - SPECIAL TOOL 5067
3 - INPUT CLUTCH RETAINER

(9) Install UD piston return spring (1)and Tool 5067 (2) (Fig. 193).

INPUT CLUTCH ASSEMBLY (Continued)

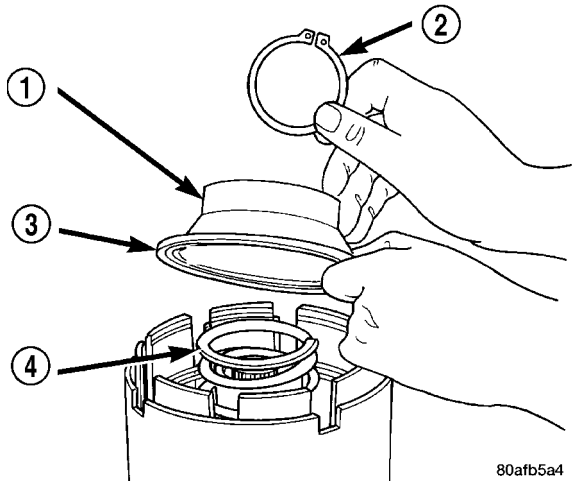


Fig. 194 UD Return Spring and Retainer

1 - UNDERDRIVE SPRING RETAINER
2 - SNAP RING
3 - SEAL
4 - PISTON RETURN SPRING

(10) Position the UD spring retainer (1) and snap ring (2) on the piston return spring (4) (Fig. 194).

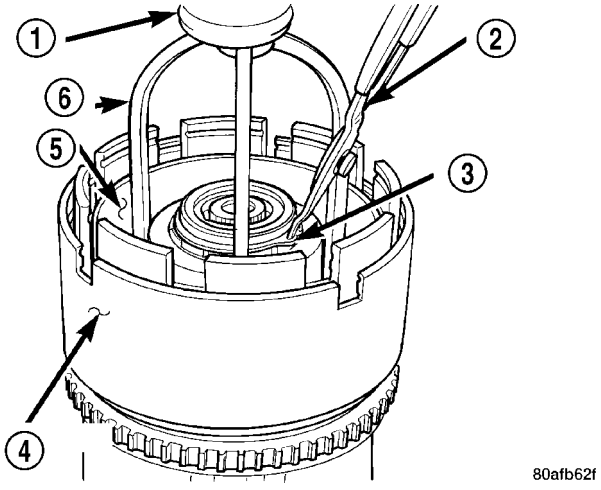


Fig. 195 Install UD Spring Retainer and Snap Ring

1 - ARBOR PRESS RAM
2 - SNAP RING PLIERS
3 - SNAP RING
4 - OD/REVERSE PISTON
5 - TOOL 5067
6 - TOOL 5059A

CAUTION: Compress return spring just enough to install snap ring.

(11) Using Tool 5059A (6) and an arbor press (1), install the UD spring retainer and snap ring (3) (Fig. 195).

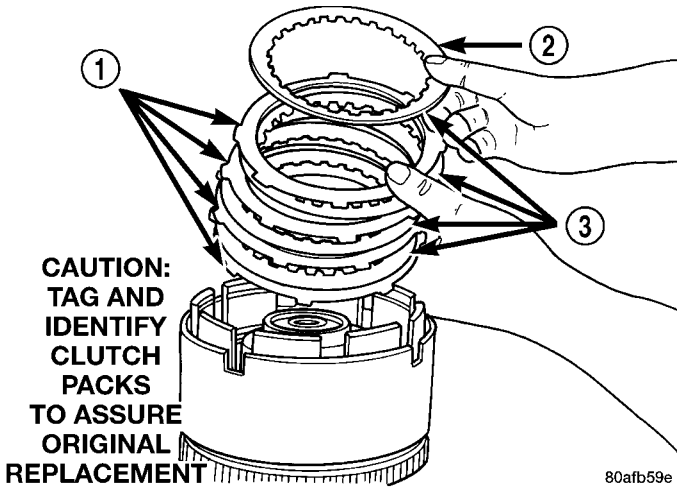


Fig. 196 Underdrive Clutch Pack

1 - CLUTCH PLATE
2 - ONE UD CLUTCH DISC
3 - CLUTCH DISC

(12) Install the UD clutch pack (four fibers/four steels) (1, 3) (Fig. 196). Leave the top disc (2) out until after the snap ring is installed.

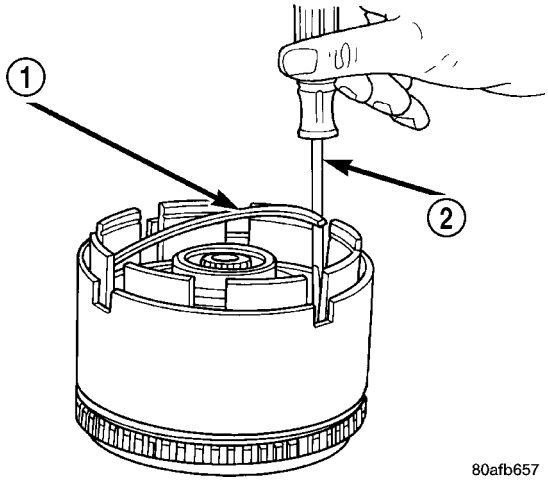


Fig. 197 UD Clutch Flat Snap Ring

1 - UNDERDRIVE CLUTCH REACTION PLATE FLAT SNAP RING
2 - SCREWDRIVER

(13) Install the UD clutch flat snap ring (1) (Fig. 197).

INPUT CLUTCH ASSEMBLY (Continued)

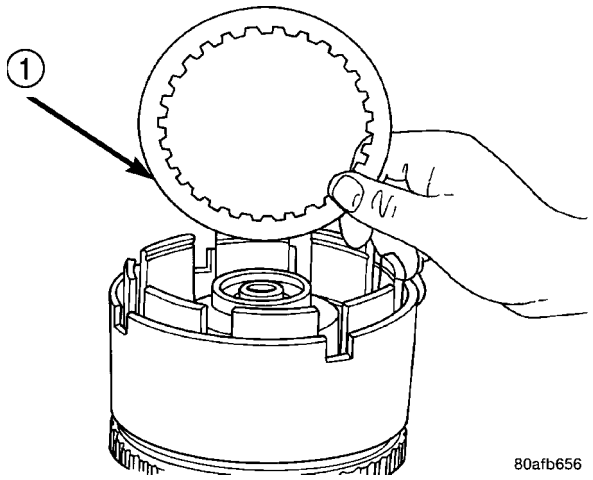


Fig. 198 Install Last UD Clutch Disc

1 - ONE UNDERDRIVE CLUTCH DISC

(14) Install the last UD clutch disc (1) (Fig. 198).

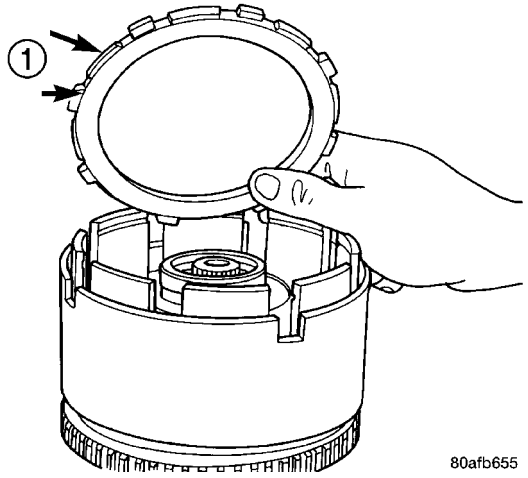


Fig. 199 OD/UD Reaction Plate

1 - OD/UD CLUTCH REACTION PLATE (STEP SIDE DOWN)

(15) Install the OD/UD clutch reaction plate (1) (Fig. 199). The OD/UD clutches reaction plate has a step on both sides. Install the OD/UD clutches reaction plate tapered step side up.

NOTE: Snap ring ends must be located within one finger of the input clutch hub. Be sure that snap ring is fully seated, by pushing with screwdriver, into snap ring groove all the way around.

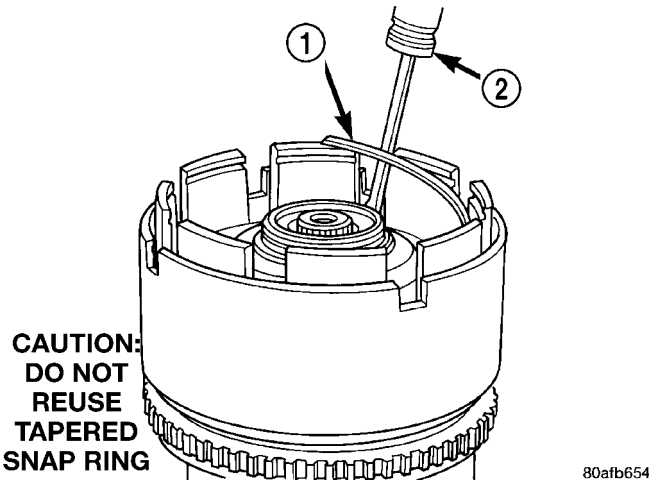


Fig. 200 Tapered Snap Ring

1 - OVERDRIVE/UNDERDRIVE CLUTCHES REACTION PLATE TAPERED SNAP RING
2 - SCREWDRIVER (DO NOT SCRATCH REACTION PLATE)

(16) Install the UD/OD tapered snap ring (1) (Fig. 200).

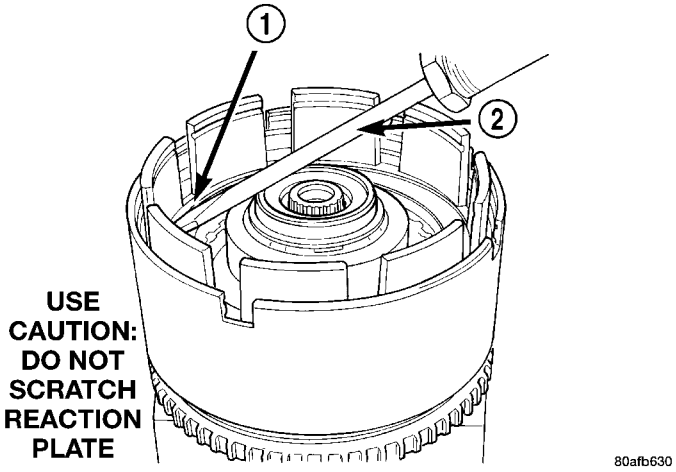


Fig. 201 Seating Tapered Snap Ring

1 - OVERDRIVE/UNDERDRIVE CLUTCHES REACTION PLATE TAPERED SNAP RING
2 - SCREWDRIVER

(17) Seat tapered snap ring (1) to ensure proper installation (Fig. 201).

INPUT CLUTCH ASSEMBLY (Continued)

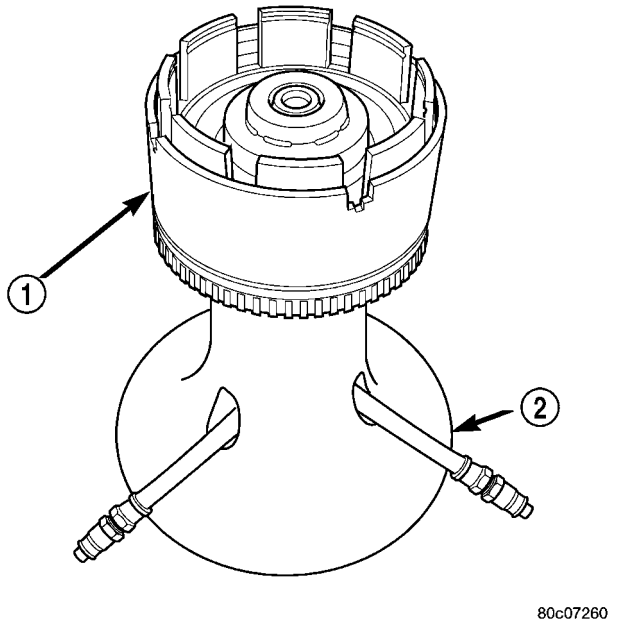


Fig. 202 Input Clutch Assembly on Pressure Fixture Tool - 8391

1 - INPUT CLUTCH ASSEMBLY
2 - INPUT CLUTCH PRESSURE FIXTURE - 8391

(18) Install input clutch assembly (1) to the Input Clutch Pressure Fixture 8391 (2) (Fig. 202).

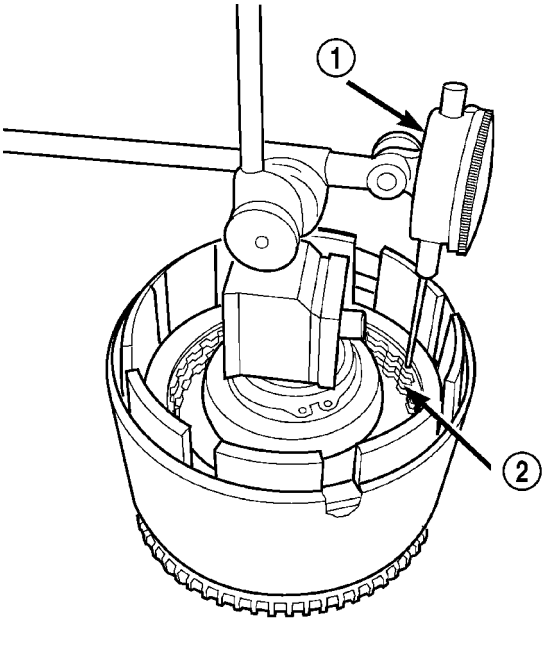


Fig. 203 Set Up Dial Indicator to Measure UD Clutch Clearance

1 - DIAL INDICATOR
2 - UNDERDRIVE CLUTCH

(19) Set up dial indicator (1) on the UD clutch pack (2) (Fig. 203).

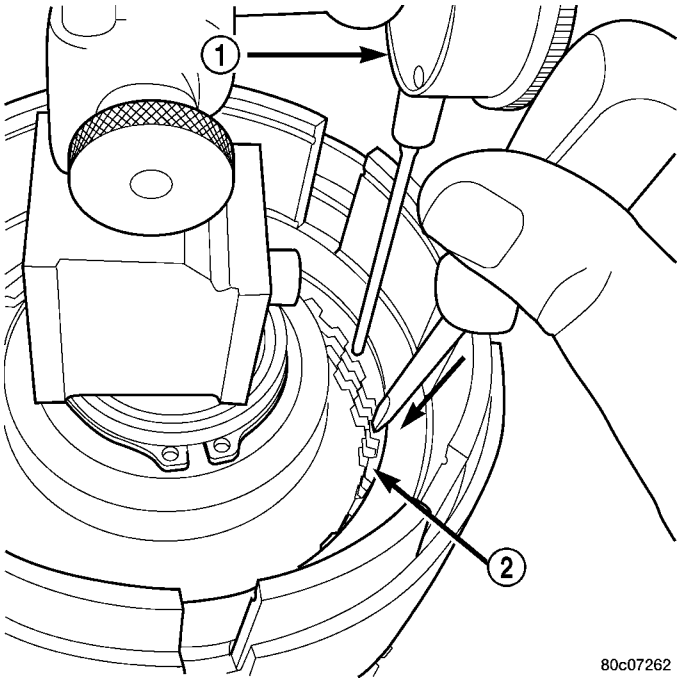


Fig. 204 Press Down on UD Clutch Pack and Zero Dial Indicator

1 - DIAL INDICATOR
2 - UNDERDRIVE CLUTCH

(20) Using moderate pressure, press down and hold (near indicator) the UD clutch pack (2) with screwdriver or suitable tool and zero dial indicator (1) (Fig. 204). When releasing pressure on clutch pack, indicator reading should advance 0.005-0.010 inches.

CAUTION: Do not apply more than 30 psi (206 kPa) to the underdrive clutch pack.

(21) Apply 30 psi (206 kPa) to the underdrive hose on Tool 8391 and measure UD clutch clearance. Measure and record UD clutch pack measurement in four (4) places, 90° apart.

(22) Take average of four measurements and compare with UD clutch pack clearance specification. **Underdrive clutch pack clearance must be 0.94-1.50 mm (0.037-0.059 in.).**

(23) If necessary, select the proper reaction plate to achieve specifications.

(24) Install the OD clutch pack (four fibers/three steels) (1) (Fig. 205).

(25) Install OD reaction plate waved snap ring (1) (Fig. 206).

(26) Install the OD/Reverse reaction plate (1) with large step down (towards OD clutch pack) (Fig. 207).

(27) Install OD reaction plate flat snap ring (3) (Fig. 208).

INPUT CLUTCH ASSEMBLY (Continued)

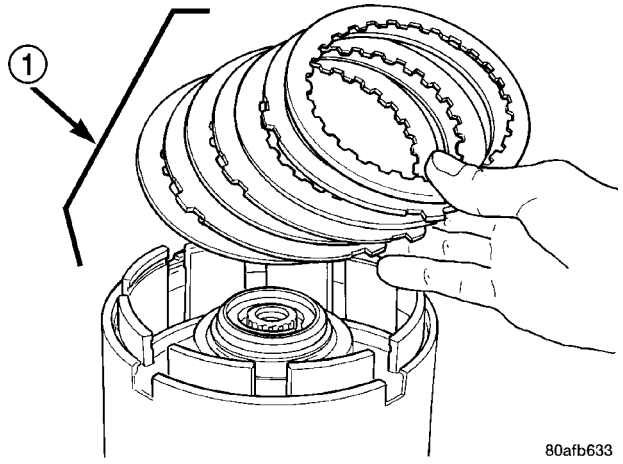


Fig. 205 Install OD Clutch Pack

1 - OVERDRIVE CLUTCH PACK

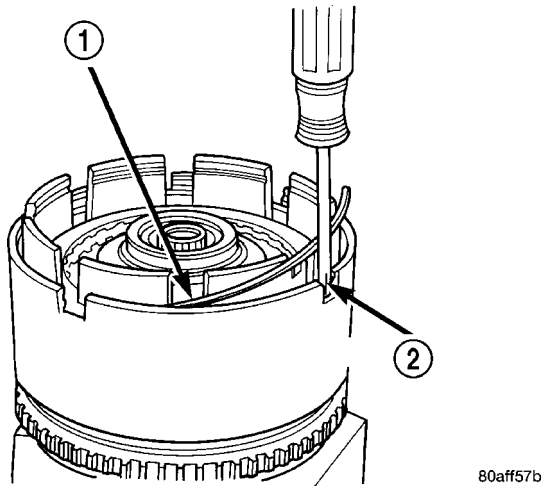


Fig. 206 Install Waved Snap Ring

1 - OVERDRIVE REACTION PLATE WAVED SNAP RING
2 - SCREWDRIVER

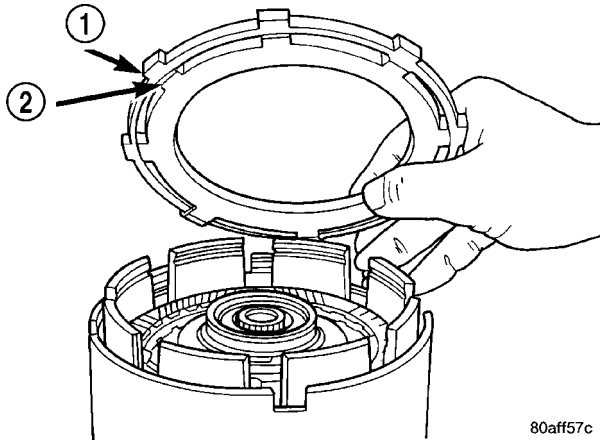


Fig. 207 OD/Reverse Reaction Plate

1 - OVERDRIVE/REVERSE PRESSURE PLATE
2 - (STEP SIDE DOWN)

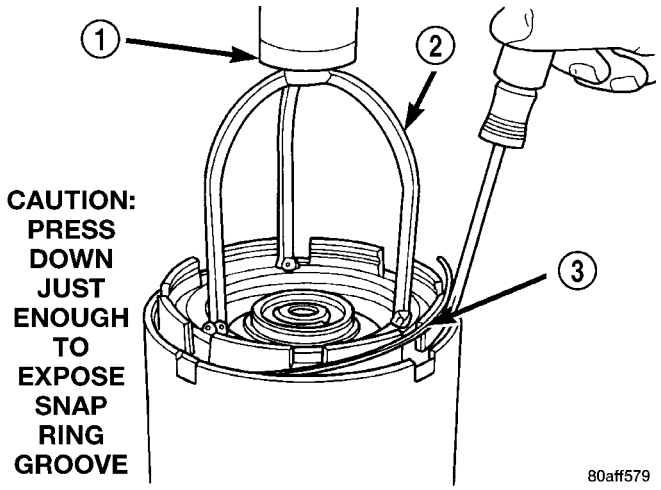


Fig. 208 Install Flat Snap Ring

1 - ARBOR PRESS RAM
2 - TOOL 5059A
3 - FLAT SNAP RING

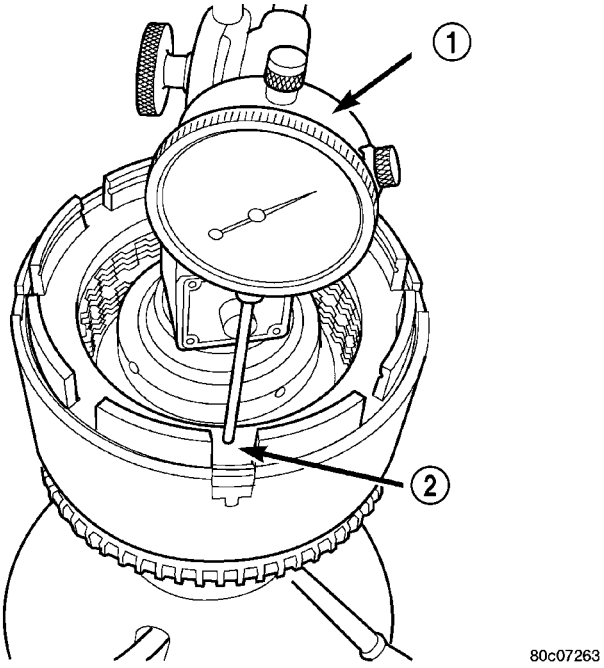


Fig. 209 Measure OD Clutch Pack Clearance

1 - DIAL INDICATOR
2 - OD/REVERSE PRESSURE PLATE

(28) Measure OD clutch pack clearance. Set up dial indicator (1) on top of the OD/Reverse reaction plate (2) (Fig. 209).

(29) Zero dial indicator and apply 30 psi (206 kPa) air pressure to the overdrive clutch hose on Tool 8391. Measure and record OD clutch pack measurement in four (4) places, 90° apart.

(30) Take average of four measurements and compare with OD clutch pack clearance specification. **The overdrive (OD) clutch pack clearance is 1.07-3.25 mm (0.042-0.128 in.).**

INPUT CLUTCH ASSEMBLY (Continued)

If not within specifications, the clutch is not assembled properly. There is no adjustment for the OD clutch clearance.

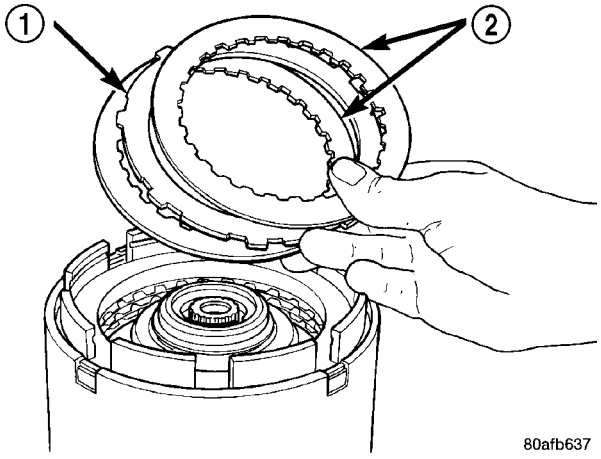


Fig. 210 Install Reverse Clutch Pack

1 - REVERSE CLUTCH PLATE
2 - REVERSE CLUTCH DISCS

(31) Install reverse clutch pack (two fibers/one steel) (1, 2) (Fig. 210).

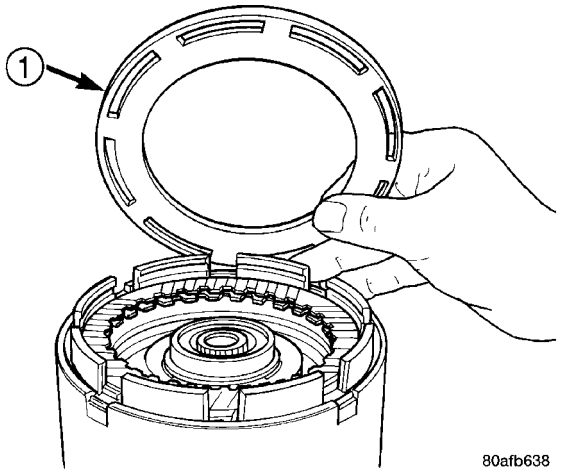


Fig. 211 Install Reaction Plate

1 - REVERSE CLUTCH REACTION PLATE (FLAT SIDE DOWN)

(32) Install reverse clutch reaction plate (1) with the flat side down towards reverse clutch (Fig. 211).

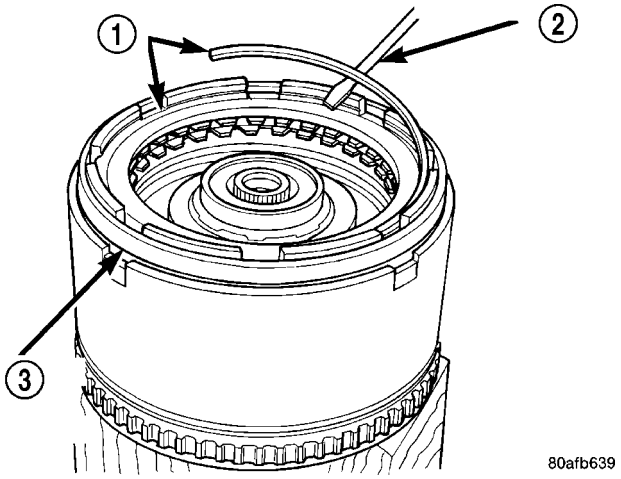


Fig. 212 Install Reverse Clutch Snap Ring

1 - REVERSE CLUTCH SNAP RING (SELECT)
2 - SCREWDRIVER
3 - REVERSE CLUTCH REACTION PLATE

(33) Tap reaction plate (3) down to allow installation of the reverse clutch snap ring (1). Install reverse clutch snap ring (1) (Fig. 212).

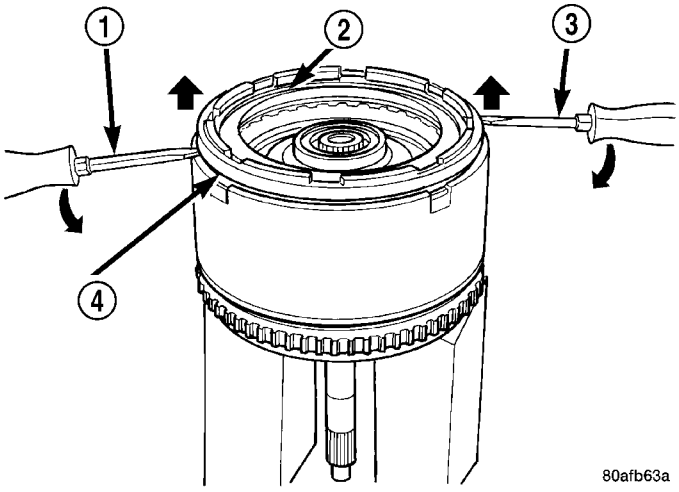


Fig. 213 Pry Up Reaction Plate

1 - SCREWDRIVER
2 - SNAP RING
3 - SCREWDRIVER
4 - MUST RAISE REVERSE REACTION PLATE TO RAISE SNAP RING

(34) Pry up reverse reaction plate (4) to seat against snap ring (2) (Fig. 213).

INPUT CLUTCH ASSEMBLY (Continued)

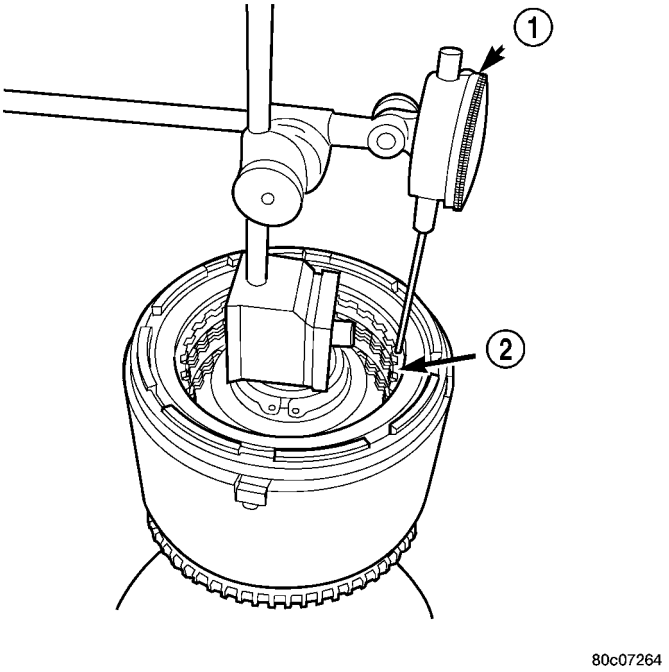


Fig. 214 Measure Reverse Clutch Pack Clearance

1 - DIAL INDICATOR
2 - REVERSE CLUTCH

(35) Set up a dial indicator (1) on the reverse clutch pack (2) (Fig. 214).

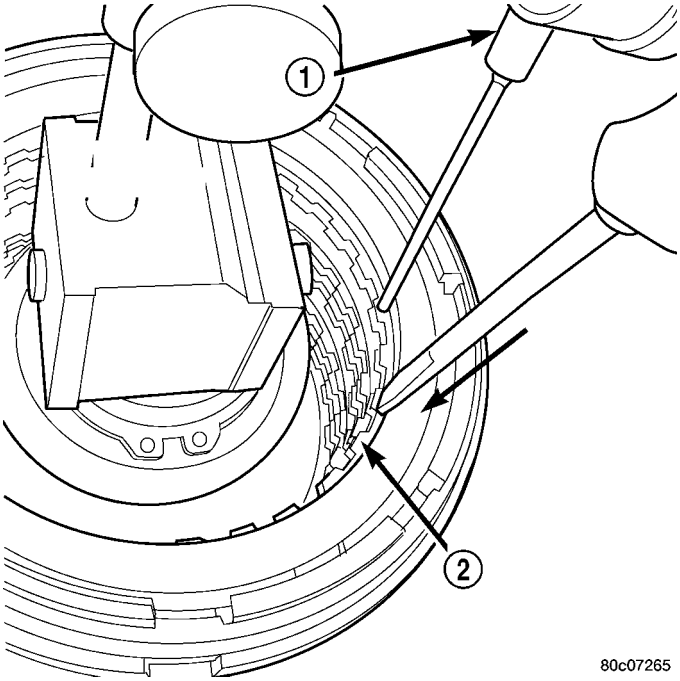


Fig. 215 Press Down on Reverse Clutch and Zero Indicator

1 - DIAL INDICATOR
2 - REVERSE CLUTCH

(36) Using moderate pressure, press down and hold (near indicator) reverse clutch disc (2) with screwdriver or suitable tool and zero dial indicator (1) (Fig. 215). When releasing pressure, indicator should advance 0.005-0.010 inches as clutch pack relaxes.

(37) Apply 30 psi (206 kPa) air pressure to the reverse clutch hose on Tool 8391. Measure and record reverse clutch pack measurement in four (4) places, 90° apart.

(38) Take average of four measurements and compare with reverse clutch pack clearance specification. **The reverse clutch pack clearance is 0.89-1.37 mm (0.035-0.054 in.).** Select the proper reverse clutch snap ring to achieve specifications.

(39) To complete the assembly, reverse clutch and overdrive clutch must be removed.

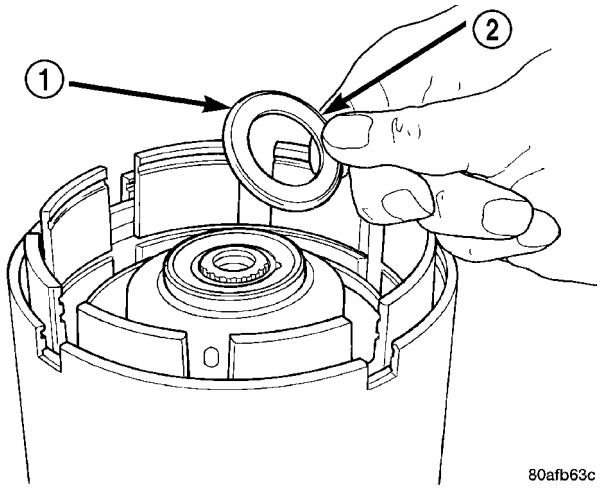


Fig. 216 Install No. 2 Needle Bearing

1 - #2 NEEDLE BEARING (NOTE 3 SMALL TABS)
2 - TABS UP

(40) Install the number 2 needle bearing (1) (Fig. 216).

INPUT CLUTCH ASSEMBLY (Continued)

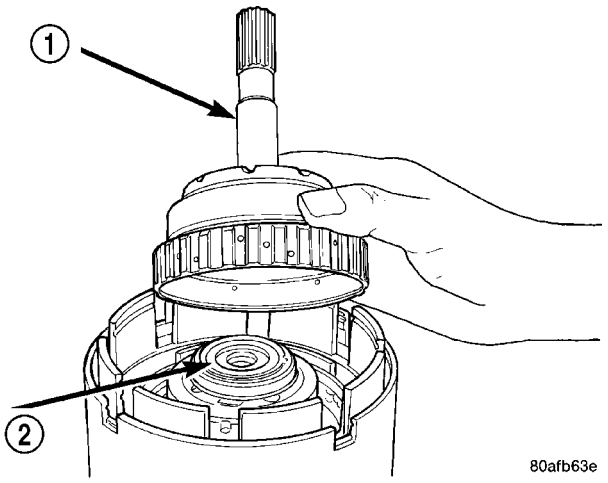


Fig. 217 Install Underdrive Shaft Assembly

1 - UNDERDRIVE SHAFT ASSEMBLY
2 - #2 NEEDLE BEARING

(41) Install the underdrive shaft assembly (1) (Fig. 217).

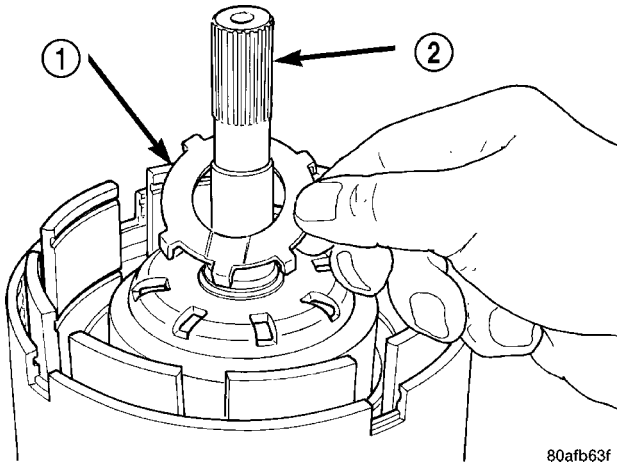


Fig. 218 Install No. 3 Thrust Washer

1 - #3 THRUST WASHER (NOTE 5 TABS)
2 - UNDERDRIVE SHAFT ASSEMBLY

(42) Install the number 3 thrust washer (1) (Fig. 218) to the underdrive shaft assembly (2). Be sure five tabs are seated properly.

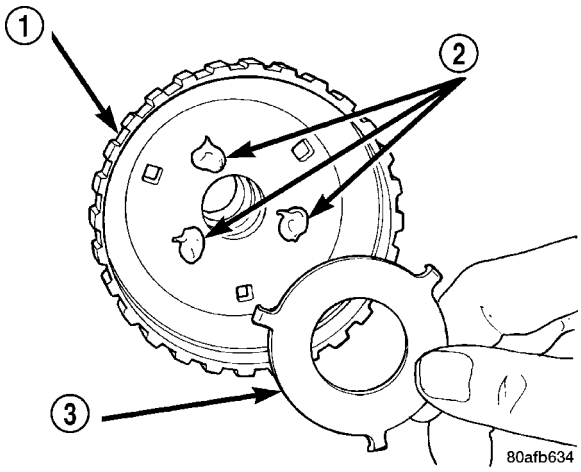


Fig. 219 Install No. 3 Thrust Plate

1 - OVERDRIVE SHAFT ASSEMBLY
2 - DABS OF PETROLATUM (FOR RETENTION)
3 - #3 THRUST PLATE (NOTE 3 TABS)

(43) Install the number 3 thrust plate (3) to the bottom of the overdrive shaft assembly (1). Retain with petrolatum or transmission assembly gel (2) (Fig. 219).

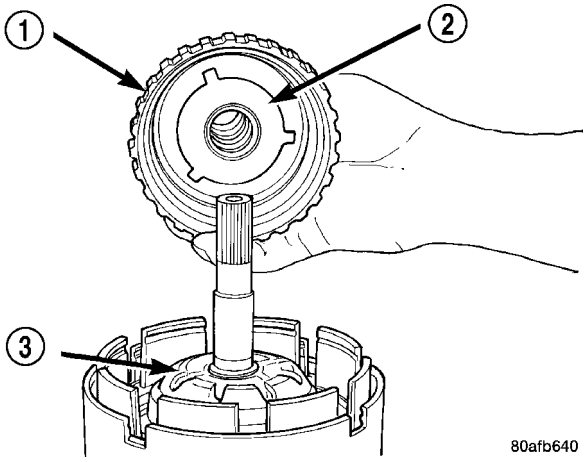


Fig. 220 Install Overdrive Shaft Assembly

1 - OVERDRIVE SHAFT ASSEMBLY
2 - #3 THRUST PLATE
3 - #3 THRUST WASHER

(44) Install the overdrive shaft assembly (1) (Fig. 220).

INPUT CLUTCH ASSEMBLY (Continued)

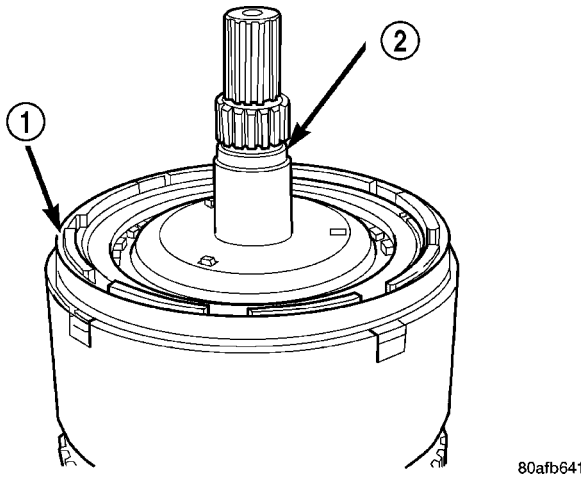


Fig. 221 Input Clutch Assembly

1 - INPUT CLUTCH ASSEMBLY
2 - OVERDRIVE SHAFT ASSEMBLY

(45) Reinstall overdrive and reverse clutch (Fig. 221). **Rechecking these clutch clearances is not necessary.**

INPUT SPEED SENSOR

DESCRIPTION

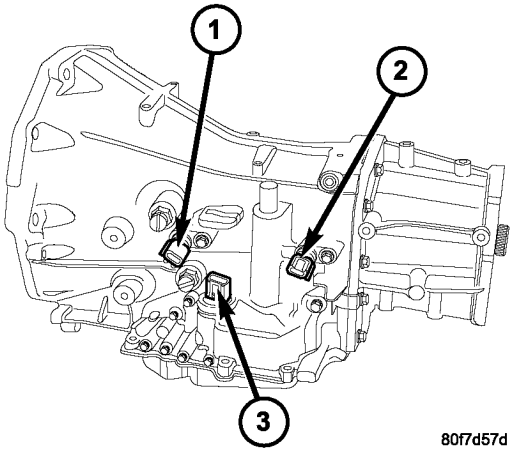


Fig. 222 Input and Output Speed Sensors and Transmission Range Sensor

1 - INPUT SPEED SENSOR
2 - OUTPUT SPEED SENSOR
3 - TRANSMISSION RANGE SENSOR

The Input and Output Speed Sensors (Fig. 222) are two-wire magnetic pickup devices that generate AC signals as rotation occurs. They are mounted in the left side of the transmission case and are considered primary inputs to the Transmission Control Module (TCM).

OPERATION

The Input Speed Sensor provides information on how fast the input shaft is rotating. As the teeth of the input clutch hub pass by the sensor coil, an AC voltage is generated and sent to the TCM. The TCM interprets this information as input shaft rpm.

The Output Speed Sensor generates an AC signal in a similar fashion, though its coil is excited by rotation of the rear planetary carrier lugs. The TCM interprets this information as output shaft rpm.

The TCM compares the input and output speed signals to determine the following:

- Transmission gear ratio
- Speed ratio error detection
- CVI calculation

The TCM also compares the input speed signal and the engine speed signal to determine the following:

- Torque converter clutch slippage
- Torque converter element speed ratio

REMOVAL

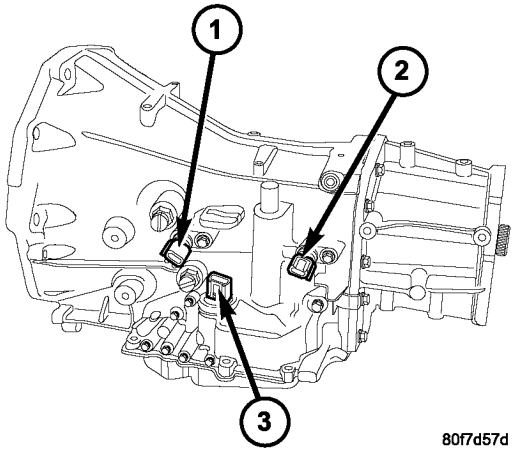


Fig. 223 Input Speed Sensor

1 - INPUT SPEED SENSOR
2 - OUTPUT SPEED SENSOR
3 - TRANSMISSION RANGE SENSOR

(1) Raise vehicle.

(2) Place a suitable fluid catch pan under the transmission.

(3) Remove the wiring connector from the input speed sensor (Fig. 223).

NOTE: The speed sensor bolt has a sealing patch applied from the factory. Be sure to reuse the same bolt.

(4) Remove the bolt holding the input speed sensor to the transmission case.

(5) Remove the input speed sensor from the transmission case.

INPUT SPEED SENSOR (Continued)

INSTALLATION

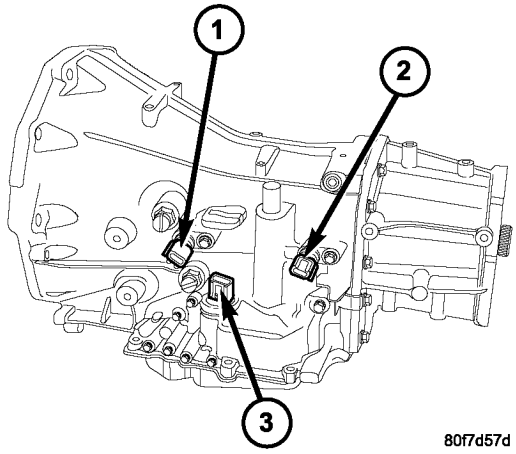


Fig. 224 Input Speed Sensor

1 - INPUT SPEED SENSOR
2 - OUTPUT SPEED SENSOR
3 - TRANSMISSION RANGE SENSOR

(1) Install the input speed sensor (1) (Fig. 224)into the transmission case.

NOTE: Before installing the speed sensor bolt, it will be necessary to replentish the sealing patch on the bolt using Mopar® Lock & Seal Adhesive.

(2) Install the bolt to hold the input speed sensor into the transmission case. Tighten the bolt to 9 N·m (80 in.lbs.).

(3) Install the wiring connector onto the input speed sensor

(4) Verify the transmission fluid level. Add fluid as necessary.

(5) Lower vehicle.

OIL PUMP

DESCRIPTION

The oil pump is located in the pump housing inside the bell housing of the transmission case. The oil pump assembly (Fig. 225) consists of an inner (3) and outer (2) gear, a housing (1), and a cover that also serves as the reaction shaft support (6).

OPERATION

As the torque converter rotates, the converter hub rotates the inner and outer gears. As the gears rotate, the clearance between the gear teeth increases in the crescent area, and creates a suction at the inlet side of the pump. This suction draws fluid through the pump inlet from the oil pan. As the clearance between the gear teeth in the crescent area decreases, it forces pressurized fluid into the pump outlet and to the valve body.

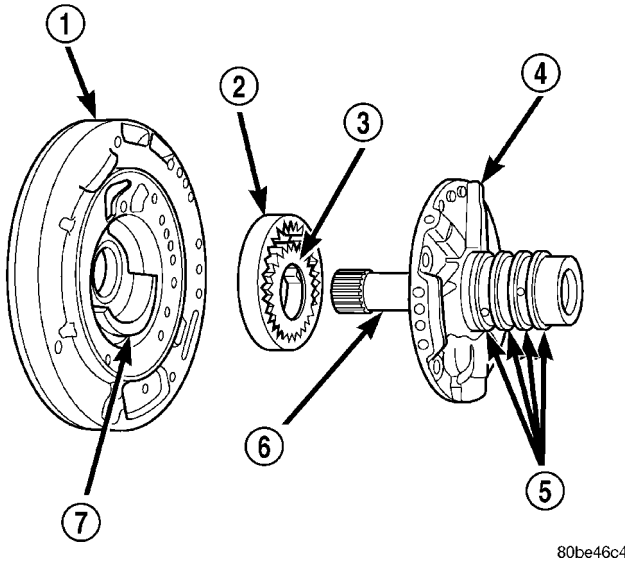


Fig. 225 Oil Pump Assembly

1 - PUMP HOUSING
2 - OUTER PUMP GEAR
3 - INNER PUMP GEAR
4 - REACTION SHAFT SUPPORT
5 - SEAL RINGS (4)
6 - REACTION SHAFT
7 - CRESCENT

DISASSEMBLY

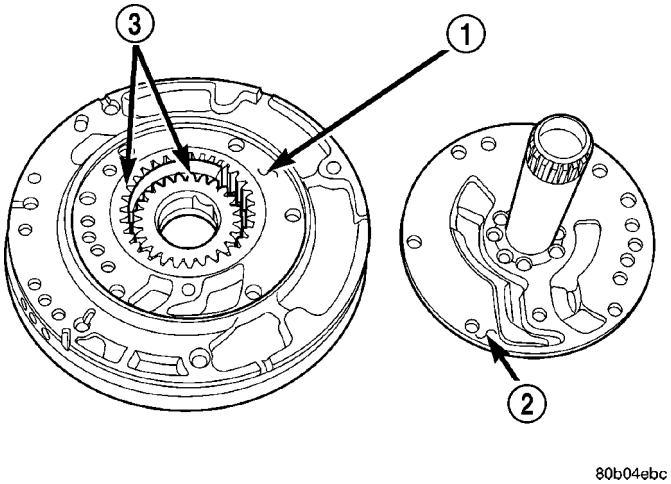


Fig. 226 Reaction Shaft Support

1 - PUMP HOUSING
2 - REACTION SHAFT SUPPORT
3 - PUMP GEARS

(1) Remove the reaction shaft support bolts.

(2) Remove the reaction shaft support(2) from the pump housing (1) (Fig. 226).

OIL PUMP (Continued)

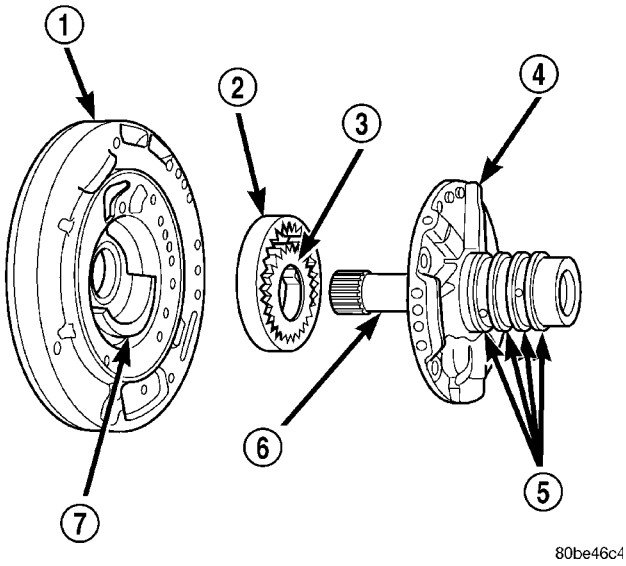


Fig. 227 Oil Pump Assembly

1 - PUMP HOUSING
2 - OUTER PUMP GEAR
3 - INNER PUMP GEAR
4 - REACTION SHAFT SUPPORT
5 - SEAL RINGS (4)
6 - REACTION SHAFT
7 - CRESCENT

(3) Remove the pump gears (2, 3) (Fig. 227) and check for wear and damage on pump housing (1) and gears (2, 3).

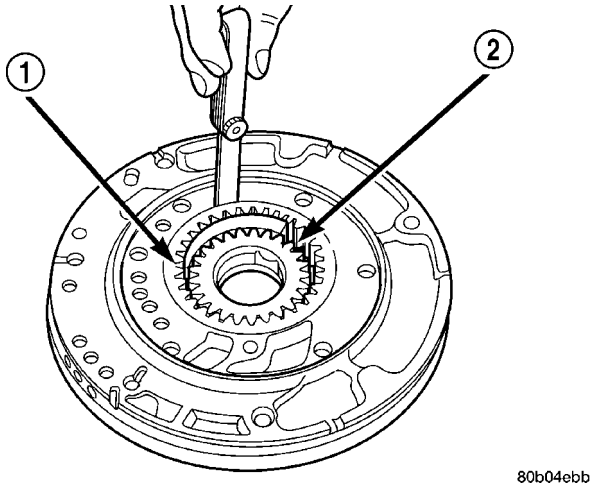


Fig. 228 Measure Outer Gear to Pocket

1 - OUTER GEAR
2 - POCKET

(4) Re-install the gears and check clearances.

(5) Measure the clearance between the outer gear (1) and the pump pocket (2) (Fig. 228). Clearance should be 0.089-0.202 mm (0.0035-0.0079 in.).

(6) Measure clearance between outer gear and crescent. Clearance should be 0.060-0.298 mm (0.0023-0.0117 in.).

(7) Measure clearance between inner gear and crescent. Clearance should be 0.093-0.385 mm (0.0036-0.0151 in.).

(8) Position an appropriate piece of Plastigage across both pump gears.

(9) Align the Plastigage to a flat area on the reaction shaft support housing.

(10) Install the reaction shaft to the pump housing. Tighten the bolts to 27 N·m (20 ft. lbs.).

(11) Remove bolts and carefully separate the housings. Measure the Plastigage following the instructions supplied.

(12) Clearance between outer gear side and the reaction shaft support should be 0.020-0.046 mm (0.0008-0.0018 in.). Clearance between inner gear side and the reaction shaft support should be 0.020-0.046 mm (0.0008-0.0018 in.).

ASSEMBLY

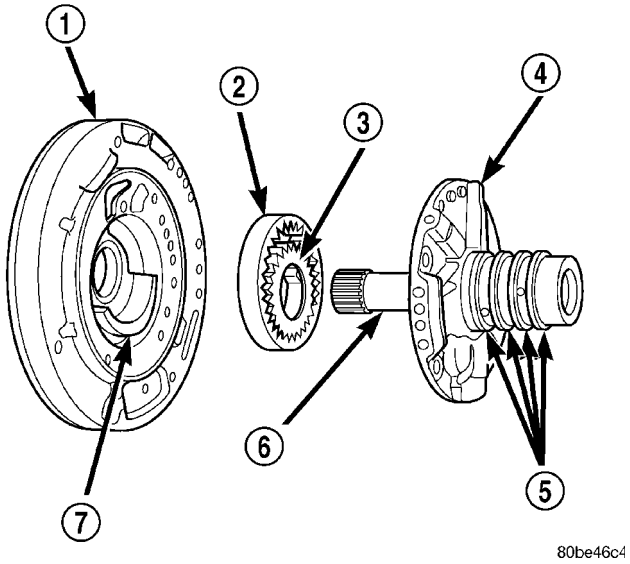


Fig. 229 Oil Pump Assembly

1 - PUMP HOUSING
2 - OUTER PUMP GEAR
3 - INNER PUMP GEAR
4 - REACTION SHAFT SUPPORT
5 - SEAL RINGS (4)
6 - REACTION SHAFT
7 - CRESCENT

(1) Assemble oil pump as shown (Fig. 229)

(2) Install and torque reaction shaft support-to-oil pump housing bolts to 28 N·m (20 ft. lbs.) torque.

OUTPUT SPEED SENSOR

DESCRIPTION

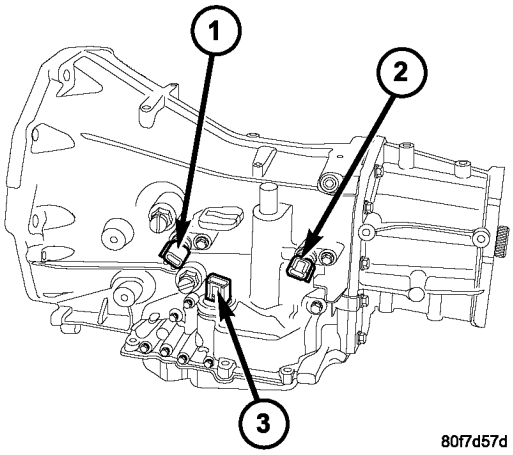


Fig. 230 Input and Output Speed Sensors and Transmission Range Sensor

1 - INPUT SPEED SENSOR
2 - OUTPUT SPEED SENSOR
3 - TRANSMISSION RANGE SENSOR

The Input and Output Speed Sensors (Fig. 230) are two-wire magnetic pickup devices that generate AC signals as rotation occurs. They are mounted in the left side of the transmission case and are considered primary inputs to the Transmission Control Module (TCM).

OPERATION

The Input Speed Sensor provides information on how fast the input shaft is rotating. As the teeth of the input clutch hub pass by the sensor coil, an AC voltage is generated and sent to the TCM. The TCM interprets this information as input shaft rpm.

The Output Speed Sensor generates an AC signal in a similar fashion, though its coil is excited by rotation of the rear planetary carrier lugs. The TCM interprets this information as output shaft rpm.

The TCM compares the input and output speed signals to determine the following:

- Transmission gear ratio
- Speed ratio error detection
- CVI calculation

The TCM also compares the input speed signal and the engine speed signal to determine the following:

- Torque converter clutch slippage
- Torque converter element speed ratio

REMOVAL

(1) Raise vehicle.

(2) Place a suitable fluid catch pan under the transmission.

(3) Remove the wiring connector from the output speed sensor (2) (Fig. 231).

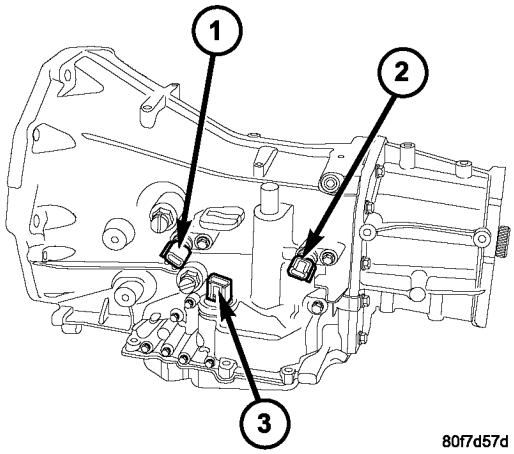


Fig. 231 Input and Output Speed Sensors and Transmission Range Sensor

1 - INPUT SPEED SENSOR
2 - OUTPUT SPEED SENSOR
3 - TRANSMISSION RANGE SENSOR

NOTE: The speed sensor bolt has a sealing patch applied from the factory. Be sure to reuse the same bolt.

(4) Remove the bolt holding the output speed sensor to the transmission case.

(5) Remove the output speed sensor (2) from the transmission case.

INSTALLATION

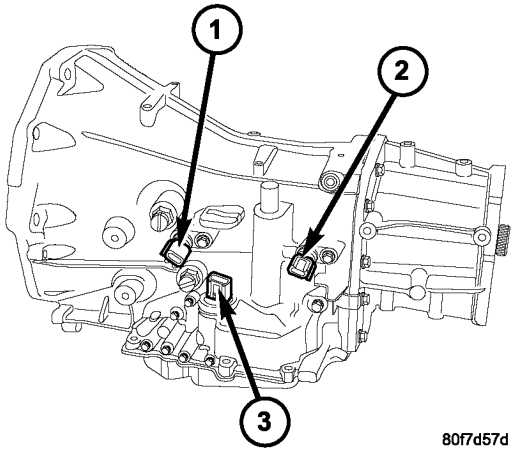


Fig. 232 Input and Output Speed Sensors and Transmission Range Sensor

1 - INPUT SPEED SENSOR
2 - OUTPUT SPEED SENSOR
3 - TRANSMISSION RANGE SENSOR

(1) Install the output speed sensor (2) (Fig. 232) into the transmission case.

OUTPUT SPEED SENSOR (Continued)

NOTE: Before installing the speed sensor bolt, it will be necessary to replentish the sealing patch on the bolt using Mopar® Lock & Seal Adhesive.

(2) Install the bolt to hold the output speed sensor into the transmission case. Tighten the bolt to 9 N·m (80 in.lbs.).

(3) Install the wiring connector onto the output speed sensor

(4) Verify the transmission fluid level. Add fluid as necessary.

(5) Lower vehicle.

OVERDRIVE SWITCH

DESCRIPTION

The overdrive OFF (control) switch is located in the center console (Fig. 233). The switch is a momentary contact device that signals the PCM to toggle current status of the overdrive function.

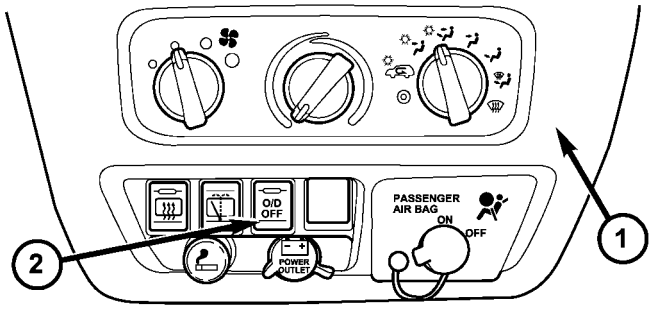


Fig. 233 Overdrive Off Switch

1 - ACCESSORY SWITCH BEZEL
2 - OVERDRIVE OFF SWITCH

OPERATION

At key-on, overdrive operation is allowed. Pressing the switch once causes the overdrive OFF mode to be entered and the overdrive OFF switch lamp to be illuminated. Pressing the switch a second time causes normal overdrive operation to be restored and the overdrive lamp to be turned off. The overdrive OFF mode defaults to ON after the ignition switch is cycled OFF and ON. The normal position for the control switch is the ON position. The switch must be in this position to energize the solenoid and allow a 3-4 upshift. The control switch indicator light illuminates only when the overdrive switch is turned to the OFF position, or when illuminated by the transmission control module.

REMOVAL

(1) Remove the accessory switch bezel (Fig. 234) (Refer to 23 - BODY/INSTRUMENT PANEL/ACCESSORY SWITCH BEZEL - REMOVAL).

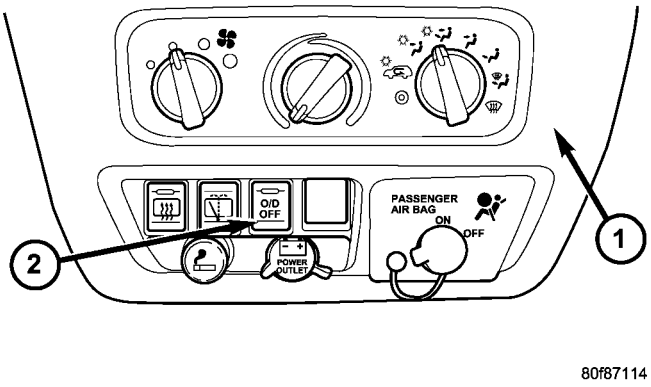


Fig. 234 Overdrive Off Switch

1 - ACCESSORY SWITCH BEZEL
2 - OVERDRIVE OFF SWITCH

(2) Remove the overdrive off switch from the accessory switch bezel.

INSTALLATION

(1) Install the overdrive off switch to the accessory switch bezel.

(2) Install the accesory switch bezel (Refer to 23 - BODY/INSTRUMENT PANEL/ACCESSORY SWITCH BEZEL - INSTALLATION).

PLANETARY GEARTRAIN

DESCRIPTION

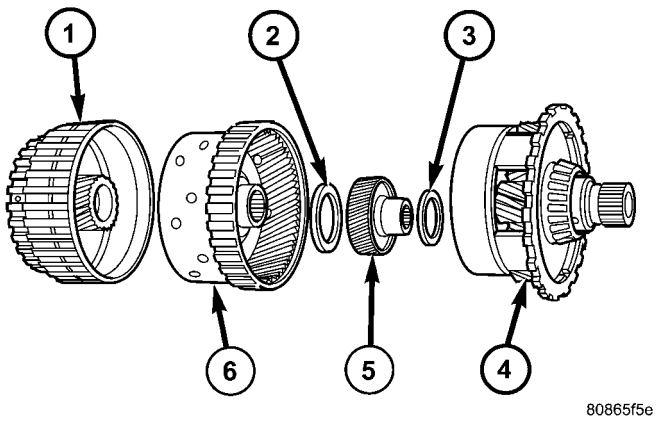


Fig. 235 Planetary Geartrain

1 - FRONT SUN GEAR ASSEMBLY
2 - #6 THRUST BEARING
3 - #7 THRUST BEARING
4 - REAR CARRIER FRONT ANNULUS ASSEMBLY
5 - REAR SUN GEAR
6 - FRONT CARRIER REAR ANNULUS ASSEMBLY

The planetary geartrain is located between the input clutch assembly and the rear of the transmission case. The planetary geartrain consists of two sun gears, two planetary carriers, two annulus (ring) gears, and one output shaft (Fig. 235).

OPERATION

The planetary geartrain utilizes two planetary gear sets that connect the transmission input shaft to the output shaft. Input and holding clutches drive or lock different planetary members to change output ratio or direction.

OIL PUMP SEAL

REMOVAL

(1) Remove the transmission from the vehicle (Refer to 21 - TRANSMISSION/AUTOMATIC - 42RLE - REMOVAL).

(2) Remove the torque converter from the transmission bellhousing.

(3) Use special tool C-3981B to remove oil pump seal.

INSTALLATION

(1) Clean and inspect oil pump seal seat. Then install seal using special tool C-4193-A.

(2) Clean and inspect torque converter hub. If nicks, scratches or hub wear are found, torque converter replacement will be required.

CAUTION: If the torque converter is being replaced, apply a light coating of grease to the crankshaft pilot hole. Also inspect the engine drive plate for cracks. If any cracks are found replace the drive plate. Do not attempt to repair a cracked drive plate. Always use new torque converter to drive plate bolts.

(3) Apply a light film of transmission oil to the torque converter hub and oil seal lips. Then install torque converter into transmission. Be sure that the hub lugs mesh with the front pump lugs when installing.

(4) Reinstall the transmission into the vehicle. (Refer to 21 - TRANSMISSION/TRANSAXLE/AUTOMATIC - 42RLE - INSTALLATION)

SHIFT MECHANISM

DESCRIPTION

The shift mechanism is cable operated and provides six shift positions. The shift indicator is located on the console next to the gear shift. The shift positions are:

- Park (P)
- Reverse (R)
- Neutral (N)
- Drive (D)
- Manual Second (2)
- Manual Low (1)

OPERATION

Manual low (1) range provides first gear only. Over run braking is also provided in this range. Manual second (2) range provides first and second gear only. Drive range provides first, second, and third gear ranges.

DIAGNOSIS AND TESTING - SHIFT MECHANISM

(1) The floor shifter lever and gate positions should be in alignment with all transmission PARK, NEUTRAL, and gear detent positions.

(2) Engine starts must be possible with floor shift lever in PARK or NEUTRAL gate positions only. Engine starts must not be possible in any other gear position.

(3) With floor shift lever handle push-button not depressed and lever in:

SHIFT MECHANISM (Continued)

(a) PARK position - Apply forward force on center of handle and remove pressure. Engine starts must be possible.

(b) PARK position - Apply rearward force on center of handle and remove pressure. Engine starts must be possible.

(c) NEUTRAL position - Normal position. Engine starts must be possible.

(d) NEUTRAL position - Engine running and brakes applied, apply forward force on center of shift handle. Transmission shall not be able to shift from NEUTRAL to REVERSE.

ADJUSTMENTS - SHIFT MECHANISM

Check adjustment by starting the engine in PARK and NEUTRAL. Adjustment is CORRECT if the engine starts only in these positions. Adjustment is INCORRECT if the engine starts in one but not both positions. If the engine starts in any position other than PARK or NEUTRAL, or if the engine will not start at all, the TRS may be faulty.

Gearshift Adjustment Procedure

(1) Shift transmission into PARK.

(2) Remove floor console as necessary for access to the shift cable adjustment. (Refer to 23 - BODY/INTERIOR/FLOOR CONSOLE - REMOVAL)

(3) Loosen the shift cable adjustment nut.

(4) Raise vehicle.

(5) Unsnap cable eyelet from transmission shift lever.

(6) Verify transmission shift lever is in PARK detent by moving lever fully rearward. Last rearward detent is PARK position.

(7) Verify positive engagement of transmission park lock by attempting to rotate propeller shaft. Shaft will not rotate when park lock is engaged.

(8) Snap cable eyelet onto transmission shift lever.

(9) Lower vehicle

(10) Tighten the shift cable adjustment screw to 12 N·m (105 in.lbs.).

(11) Verify correct operation.

(12) Install any floor console components removed for access. (Refer to 23 - BODY/INTERIOR/FLOOR CONSOLE - INSTALLATION)

SOLENOID

DESCRIPTION

The typical electrical solenoid used in automotive applications is a linear actuator. It is a device that produces motion in a straight line. This straight line motion can be either forward or backward in direction, and short or long distance.

A solenoid is an electromechanical device that uses a magnetic force to perform work. It consists of a coil of wire, wrapped around a magnetic core made from steel or iron, and a spring loaded, movable plunger, which performs the work, or straight line motion.

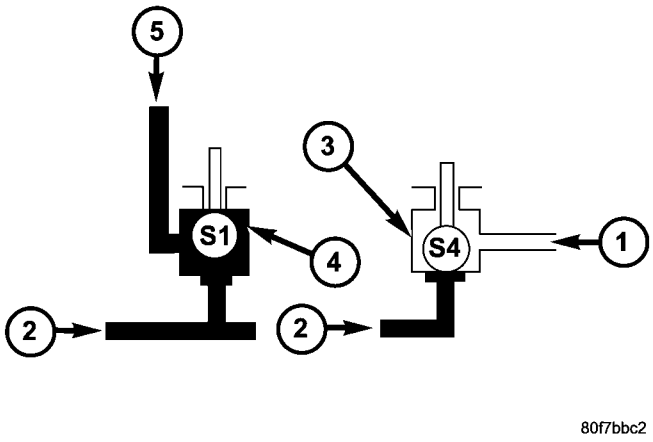


Fig. 236 2/4 - Low Reverse and Underdrive Solenoids

1 - MANUAL VALVE
2 - LINE PRESSURE
3 - 2/4 - LOW REVERSE SOLENOID ENERGIZED
4 - UNDERDRIVE SOLENOID DE-ENERGIZED
5 - UNDERDRIVE CLUTCH

The solenoids used in transmission applications are attached to valves which can be classified as **normally open (Fig. 236)** or **normally closed (Fig. 237)**. The **normally open** solenoid valve is defined as a valve which allows hydraulic flow when no current or voltage is applied to the solenoid. The **normally closed** solenoid valve is defined as a valve which does not allow hydraulic flow when no current or voltage is applied to the solenoid. These valves perform hydraulic control functions for the transmission and must therefore be durable and tolerant of dirt particles. For these reasons, the valves have hardened steel poppets and ball valves. The solenoids operate the valves directly, which means that the solenoids must have very high outputs to close the valves against the sizable flow areas and line pressures found in current transmissions. Fast response time is also necessary to ensure accurate control of the transmission.

The strength of the magnetic field is the primary force that determines the speed of operation in a particular solenoid design. A stronger magnetic field will cause the plunger to move at a greater speed than a weaker one. There are basically two ways to increase the force of the magnetic field:

1. Increase the amount of current applied to the coil or

2. Increase the number of turns of wire in the coil.

SOLENOID (Continued)

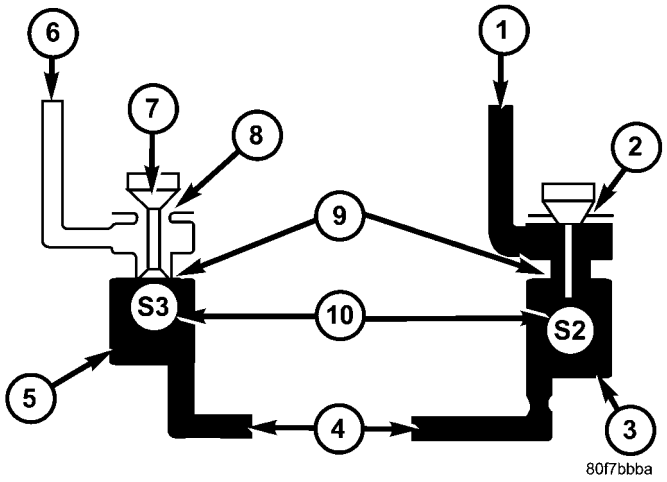


Fig. 237 Low Reverse/Converter Clutch and Overdrive Solenoids

1 - OVERDRIVE CLUTCH
2 - NO VENT
3 - OVERDRIVE SOLENOID ENERGIZED
4 - MANUAL VALVE
5 - LOW REVERSE/CONVERTER CLUTCH SOLENOID DE-ENERGIZED
6 - SOLENOID SWITCH VALVE
7 - TAPER
8 - VENT TO SUMP
9 - ORIFICE
10 - CHECK BALL

The most common practice is to increase the number of turns by using thin wire that can completely fill the available space within the solenoid housing. The strength of the spring and the length of the plunger also contribute to the response speed possible by a particular solenoid design.

A solenoid can also be described by the method by which it is controlled. Some of the possibilities include variable force, pulse-width modulated, constant ON, or duty cycle. The variable force and pulse-width modulated versions utilize similar methods to control the current flow through the solenoid to position the solenoid plunger at a desired position somewhere between full ON and full OFF. The constant ON and duty cycled versions control the voltage across the solenoid to allow either full flow or no flow through the solenoid's valve.

OPERATION

When an electrical current is applied to the solenoid coil, a magnetic field is created which produces an attraction to the plunger, causing the plunger to move and work against the spring pressure and the load applied by the fluid the valve is controlling. The plunger is normally directly attached to the valve which it is to operate. When the current is removed from the coil, the attraction is removed and the plunger will return to its original position due to spring pressure.

The plunger is made of a conductive material and accomplishes this movement by providing a path for the magnetic field to flow. By keeping the air gap between the plunger and the coil to the minimum necessary to allow free movement of the plunger, the magnetic field is maximized.

SOLENOID/PRESSURE SWITCH ASSY

DESCRIPTION

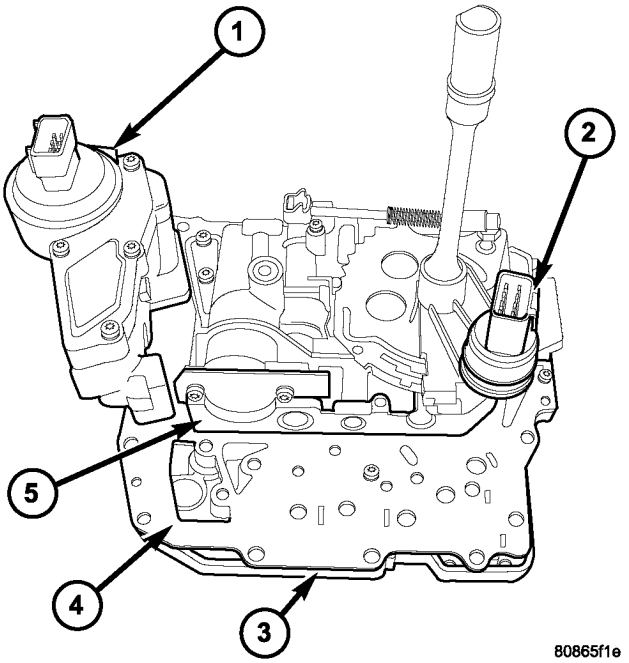


Fig. 238 Valve Body Assembly

1 - SOLENOID/PRESSURE SWITCH ASSEMBLY
2 - TRS
3 - TRANSFER PLATE
4 - SEPARATOR PLATE
5 - VALVE BODY

The Solenoid/Pressure Switch Assembly (1) (Fig. 238) is inside the transmission and mounted to the valve body assembly. The assembly consists of four solenoids that control hydraulic pressure to the L/R, 2/4, OD, and UD friction elements (transmission clutches), and the torque converter clutch. The reverse clutch is controlled by line pressure from the manual valve in the valve body. The solenoids are contained within the Solenoid/Pressure Switch Assembly, and can only be serviced by replacing the assembly.

The solenoid assembly also contains pressure switches that monitor and send hydraulic circuit information to the TCM. Likewise, the pressure switches can only be service by replacing the assembly.

SOLENOID/PRESSURE SWITCH ASSY (Continued)

OPERATION

SOLENOIDS

The solenoids receive electrical power from the Transmission Control Relay through a single wire. The TCM energizes or operates the solenoids individually by grounding the return wire of the solenoid needed. When a solenoid is energized, the solenoid valve shifts, and a fluid passage is opened or closed (vented or applied), depending on its default operating state. The result is an apply or release of a frictional element.

The 2/4 and UD solenoids are normally applied, which allows fluid to pass through in their relaxed or "off" state. By design, this allows transmission limp-in (P,R,N,2) in the event of an electrical failure.

The continuity of the solenoids and circuits are periodically tested. Each solenoid is turned on or off depending on its current state. An inductive spike should be detected by the TCM during this test. If no spike is detected, the circuit is tested again to verify the failure. In addition to the periodic testing, the solenoid circuits are tested if a speed ratio or pressure switch error occurs.

PRESSURE SWITCHES

The TCM relies on three pressure switches to monitor fluid pressure in the L/R, 2/4, and OD hydraulic circuits. The primary purpose of these switches is to help the TCM detect when clutch circuit hydraulic failures occur. The range for the pressure switch closing and opening points is 11-23 psi. Typically the switch opening point will be approximately one psi lower than the closing point. For example, a switch may close at 18 psi and open at 17 psi. The switches are continuously monitored by the TCM for the correct states (open or closed) in each gear as shown in the following chart:

PRESSURE SWITCH STATES

GEAR	L/R	2/4	OD
R	OP	OP	OP
P/N	CL	OP	OP
1st	CL	OP	OP
2nd	OP	CL	OP
D	OP	OP	CL
OD	OP	CL	CL

OP = OPEN

CL = CLOSED

A Diagnostic Trouble Code (DTC) will set if the TCM senses any switch open or closed at the wrong time in a given gear.

The TCM also tests the 2/4 and OD pressure switches when they are normally off (OD and 2/4 are tested in 1st gear, OD in 2nd gear, and 2/4 in 3rd gear). The test simply verifies that they are operational, by looking for a closed state when the corresponding element is applied. Immediately after a shift into 1st, 2nd, or 3rd gear with the engine speed above 1000 rpm, the TCM momentarily turns on element pressure to the 2/4 and/or OD clutch circuits to identify that the appropriate switch has closed. If it doesn't close, it is tested again. If the switch fails to close the second time, the appropriate Diagnostic Trouble Code (DTC) will set.

REMOVAL

NOTE: If the Solenoid/Pressure Switch Assembly is being replaced, the Quick Learn Procedure must be performed. (Refer to 8 - ELECTRICAL/ELECTRONIC CONTROL MODULES/TRANSMISSION CONTROL MODULE - STANDARD PROCEDURE)

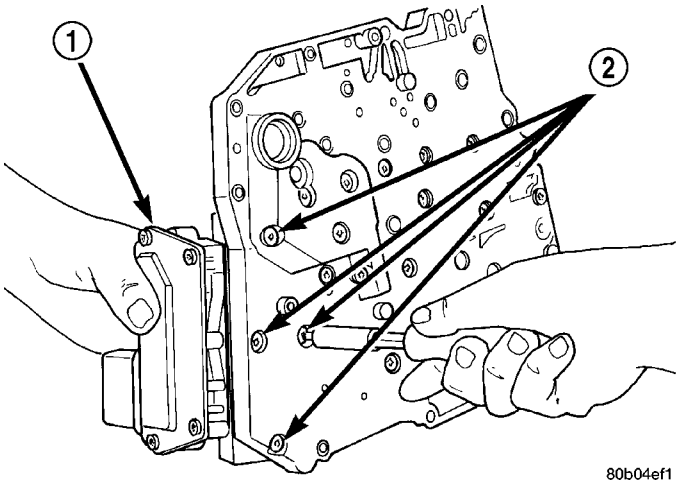


Fig. 239 Solenoid Retaining Screws

1 - SOLENOID/PRESSURE SWITCH ASSEMBLY
2 - RETAINING SCREWS

(1) Raise vehicle on hoist.

(2) Remove valve body assembly from transmission. (Refer to 21 - TRANSMISSION/AUTOMATIC - 42RLE/VALVE BODY - REMOVAL)

(3) Remove Solenoid/Pressure Switch Assembly retaining screws (2) from solenoid (Fig. 239).

SOLENOID/PRESSURE SWITCH ASSY (Continued)

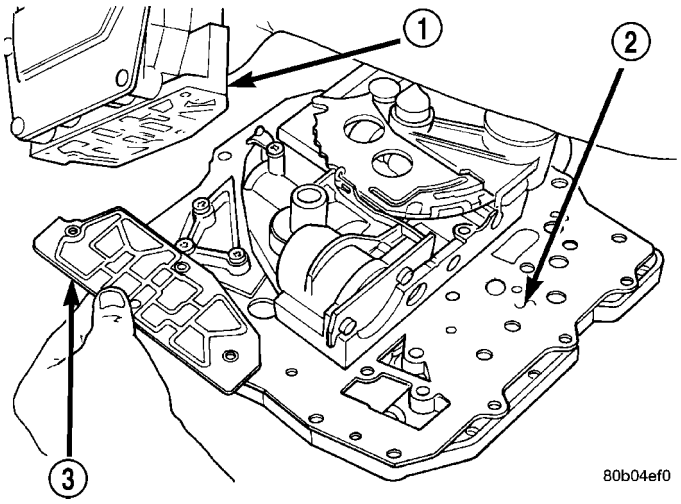


Fig. 240 Solenoid/Pressure Switch Assembly and Screen

1 - SOLENOID/PRESSURE SWITCH ASSEMBLY
2 - VALVE BODY
3 - SCREEN

(4) Remove Solenoid/Pressure Switch Assembly (1) and screen from valve body (2) (Fig. 240).

INSTALLATION

NOTE: If the Solenoid/Pressure Switch assembly is being replaced, the Quick Learn Procedure must be performed. (Refer to 8 - ELECTRICAL/ELECTRONIC CONTROL MODULES/TRANSMISSION CONTROL MODULE - STANDARD PROCEDURE)

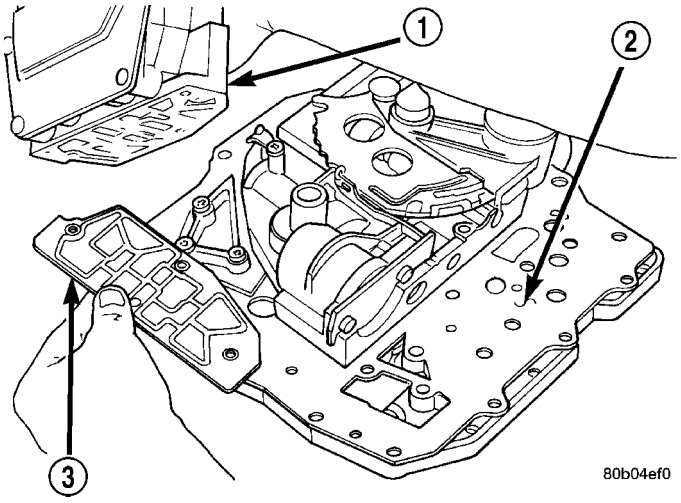


Fig. 241 Solenoid/Pressure Switch Assembly and Screen

1 - SOLENOID/PRESSURE SWITCH ASSEMBLY
2 - VALVE BODY
3 - SCREEN

(1) Install Solenoid/Pressure Switch Assembly (1) (Fig. 241) and screen (3) to the separator and transfer plates.

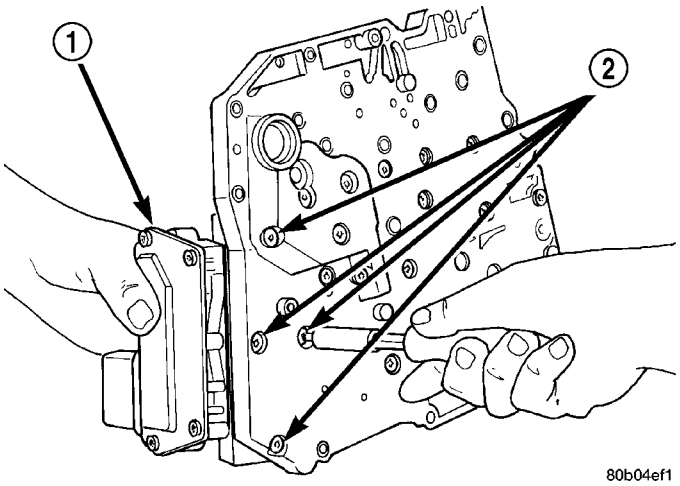


Fig. 242 Solenoid Retaining Screws

1 - SOLENOID/PRESSURE SWITCH ASSEMBLY
2 - RETAINING SCREWS

(2) Install and tighten retaining screws (2) (Fig. 242) to 5.5 N·m (50 in. lbs.) torque.

(3) Install valve body. (Refer to 21 - TRANSMISSION/AUTOMATIC - 42RLE/VALVE BODY - INSTALLATION)

TORQUE CONVERTER

DESCRIPTION

The torque converter (Fig. 243) is a hydraulic device that couples the engine crankshaft to the transmission. The torque converter consists of an outer shell with an internal turbine, a stator, an overrunning clutch, an impeller and an electronically applied converter clutch. The converter clutch provides reduced engine speed and greater fuel economy when engaged. Clutch engagement also provides reduced transmission fluid temperatures. The torque converter hub drives the transmission oil (fluid) pump.

The torque converter is a sealed, welded unit that is not repairable and is serviced as an assembly.

CAUTION: The torque converter must be replaced if a transmission failure resulted in large amounts of metal or fiber contamination in the fluid.

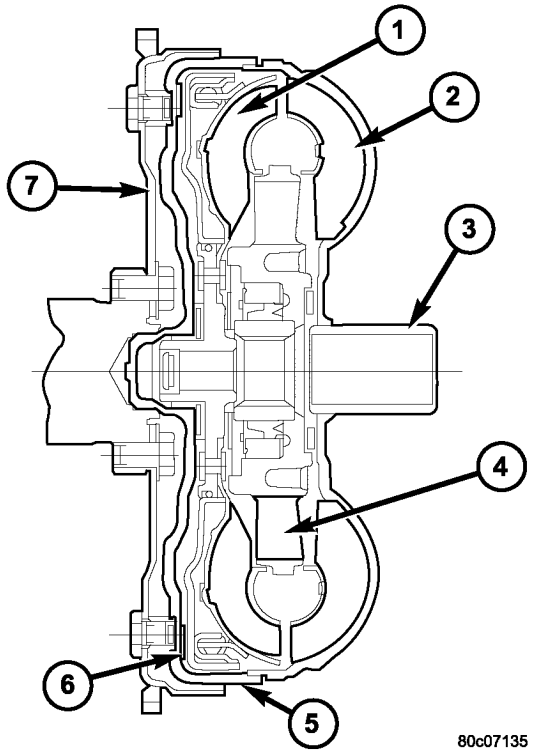


Fig. 243 Torque Converter Assembly

1 - TURBINE
2 - IMPELLER
3 - HUB
4 - STATOR
5 - FRONT COVER
6 - CONVERTER CLUTCH DISC
7 - DRIVE PLATE

TORQUE CONVERTER (Continued)

IMPELLER

The impeller (Fig. 244) is an integral part of the converter housing. The impeller consists of curved blades placed radially along the inside of the housing on the transmission side of the converter. As the converter housing is rotated by the engine, so is the impeller, because they are one and the same and are the driving members of the system.

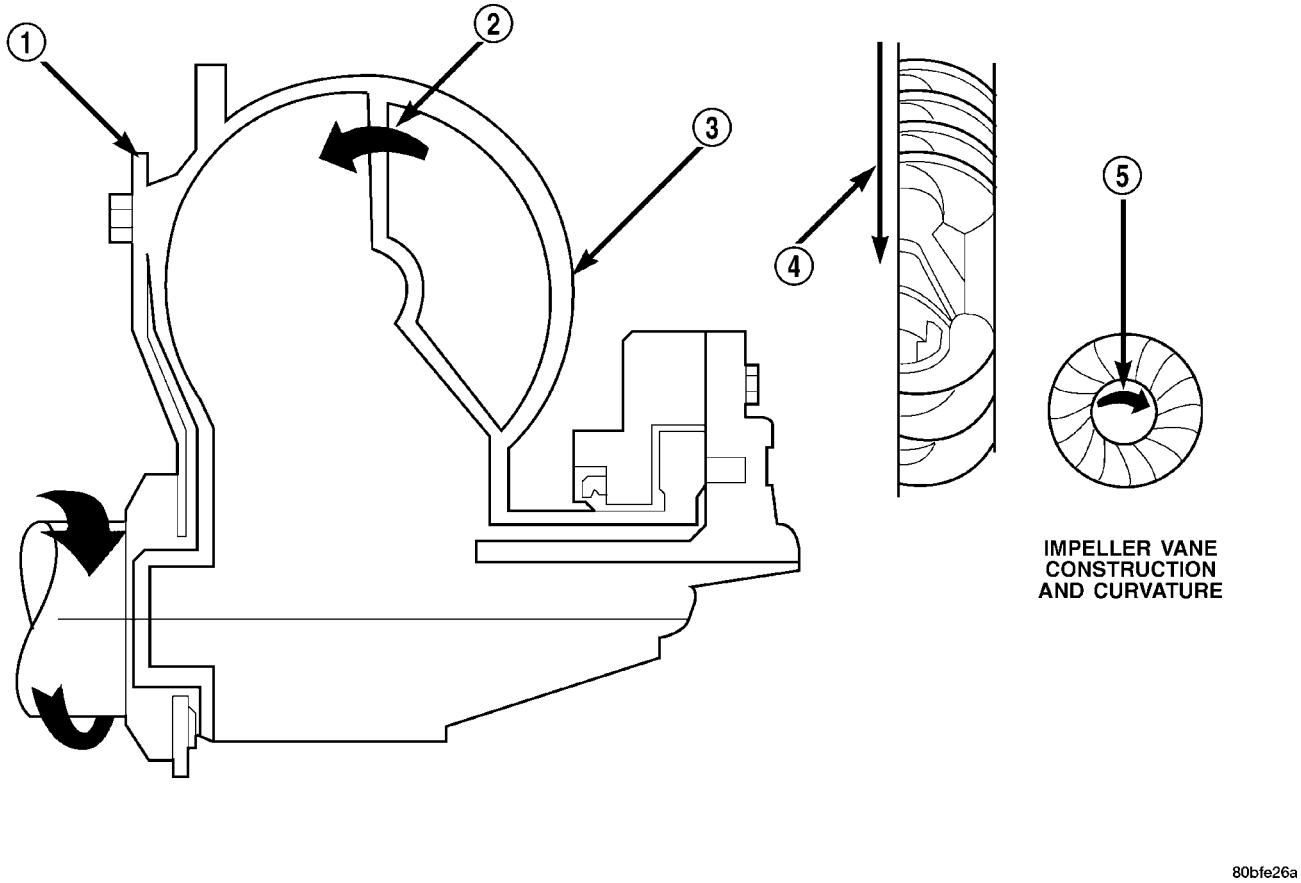


Fig. 244 Impeller

1 - ENGINE FLEXPLATE
2 - OIL FLOW FROM IMPELLER SECTION INTO TURBINE SECTION
3 - IMPELLER VANES AND COVER ARE INTEGRAL
4 - ENGINE ROTATION
5 - ENGINE ROTATION

TORQUE CONVERTER (Continued)

TURBINE

The turbine (Fig. 245) is the output, or driven, member of the converter. The turbine is mounted within the housing opposite the impeller, but is not attached to the housing. The input shaft is inserted through the center of the impeller and splined into the turbine. The design of the turbine is similar to the impeller, except the blades of the turbine are curved in the opposite direction.

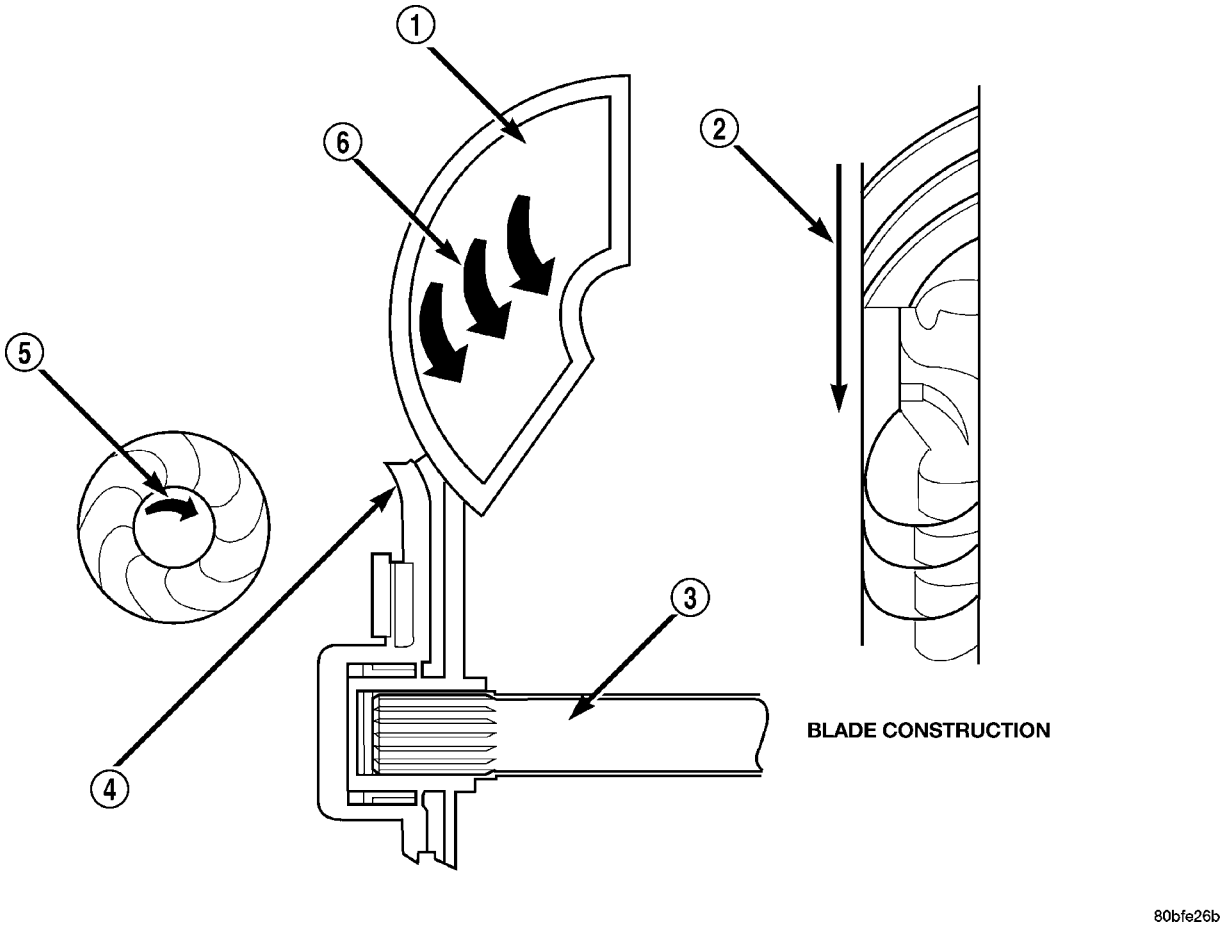


Fig. 245 Turbine

1 - TURBINE VANE
2 - ENGINE ROTATION
3 - INPUT SHAFT
4 - PORTION OF TORQUE CONVERTER COVER
5 - ENGINE ROTATION
6 - OIL FLOW WITHIN TURBINE SECTION

TORQUE CONVERTER (Continued)

STATOR

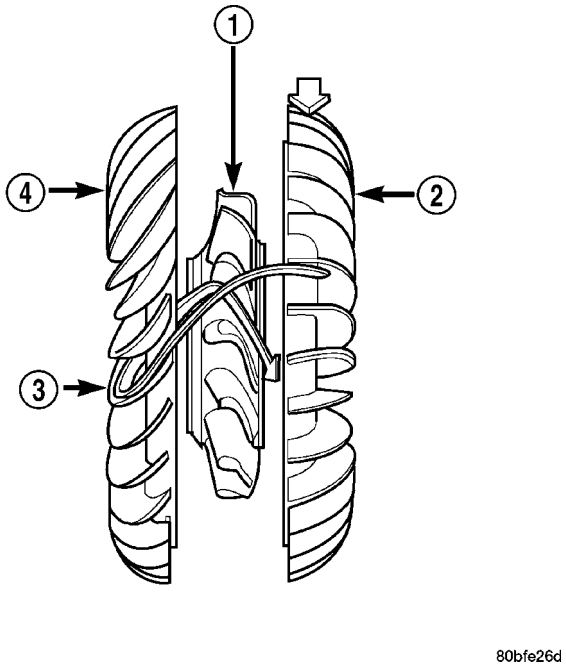


Fig. 246 Stator Location

1 - STATOR
2 - IMPELLER
3 - FLUID FLOW
4 - TURBINE

The stator assembly (Fig. 246) is mounted on a stationary shaft which is an integral part of the oil pump. The stator (1) is located between the impeller (2) and the turbine (4) within the torque converter case.

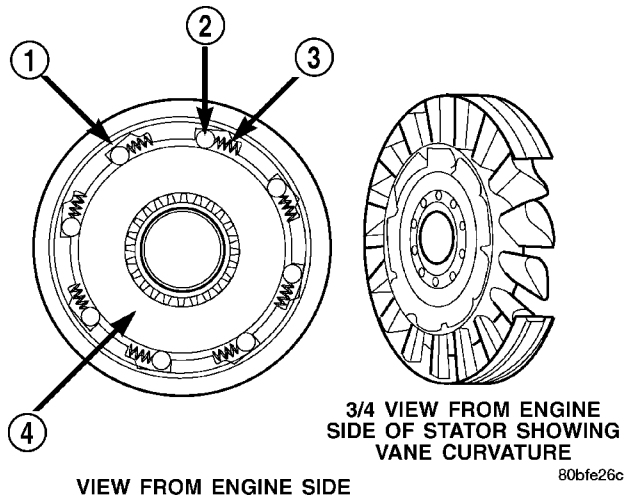


Fig. 247 Stator Components

1 - CAM (OUTER RACE)
2 - ROLLER
3 - SPRING
4 - INNER RACE

The stator contains an over-running clutch (1-4) (Fig. 247), which allows the stator to rotate only in a clockwise direction. When the stator is locked against the over-running clutch, the torque multiplication feature of the torque converter is operational.

TORQUE CONVERTER CLUTCH (TCC)

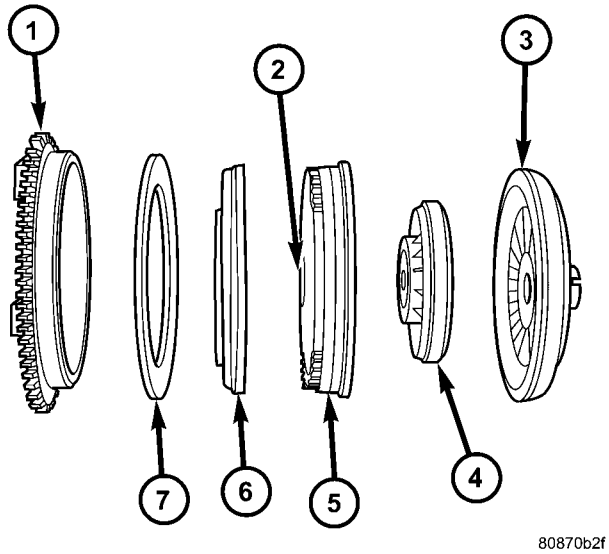


Fig. 248 Torque Converter Clutch (TCC)

1 - IMPELLER FRONT COVER
2 - THRUST WASHER ASSEMBLY
3 - IMPELLER
4 - STATOR
5 - TURBINE
6 - PISTON
7 - FRICTION DISC

The TCC (Fig. 248) was installed to improve the efficiency of the torque converter that is lost to the slippage of the fluid coupling. Although the fluid coupling provides smooth, shock-free power transfer, it is natural for all fluid couplings to slip. If the impeller (3) and turbine (5) were mechanically locked together, a zero slippage condition could be obtained. A hydraulic piston (6) with friction material (7) was added to the turbine assembly (5) to provide this mechanical lock-up.

In order to reduce heat build-up in the transmission and buffer the powertrain against torsional vibrations, the TCM can duty cycle the L/R-CC Solenoid to achieve a smooth application of the torque converter clutch. This function, referred to as Electronically Modulated Converter Clutch (EMCC) can occur at various times depending on the following variables:

- Shift lever position
- Current gear range
- Transmission fluid temperature
- Engine coolant temperature
- Input speed
- Throttle angle
- Engine speed

TORQUE CONVERTER (Continued)

OPERATION

The converter impeller (Fig. 249) (driving member), which is integral to the converter housing and bolted to the engine drive plate, rotates at engine speed. The converter turbine (driven member), which reacts from fluid pressure generated by the impeller, rotates and turns the transmission input shaft.

TURBINE

As the fluid that was put into motion by the impeller blades strikes the blades of the turbine, some of the energy and rotational force is transferred into the turbine and the input shaft. This causes both of them (turbine and input shaft) to rotate in a clockwise direction following the impeller. As the fluid is leaving the trailing edges of the turbine's blades it continues in a "hindering" direction back toward the impeller. If the fluid is not redirected before it strikes the impeller, it will strike the impeller in such a direction that it would tend to slow it down.

STATOR

Torque multiplication is achieved by locking the stator's over-running clutch to its shaft (Fig. 250). Under stall conditions (the turbine is stationary), the oil leaving the turbine blades strikes the face of the stator blades and tries to rotate them in a counterclockwise direction. When this happens the overrunning clutch of the stator locks and holds the stator from rotating. With the stator locked, the oil strikes the stator blades and is redirected into a "helping" direction before it enters the impeller. This circulation of oil from impeller to turbine, turbine to stator, and stator to impeller, can produce a maximum torque multiplication of about 2.4:1. As the turbine begins to match the speed of the impeller, the fluid that was hitting the stator in such as way as to cause it to lock-up is no longer doing so. In this condition of operation, the stator begins to free wheel and the converter acts as a fluid coupling.

TORQUE CONVERTER CLUTCH (TCC)

The torque converter clutch is hydraulically applied and is released when fluid is vented from the hydraulic circuit by the torque converter control (TCC) solenoid on the valve body. The torque converter clutch is controlled by the Powertrain Control Module (PCM). The torque converter clutch engages in fourth gear, and in third gear under various conditions, such as when the O/D switch is OFF, when the vehicle is cruising on a level surface after the vehicle has warmed up. The torque converter clutch will disengage momentarily when an increase in engine load is sensed by the PCM, such as when the vehicle begins to go uphill or the throttle pressure is increased.

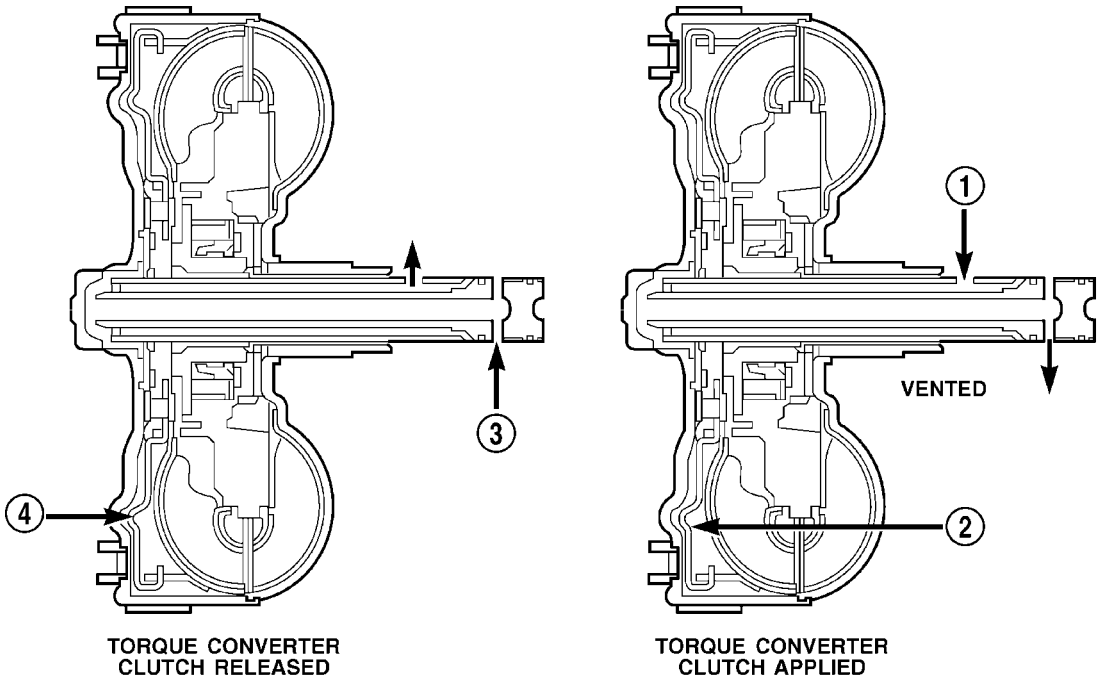


Fig. 249 Torque Converter Fluid Operation

1 - APPLY PRESSURE
2 - THE PISTON MOVES SLIGHTLY FORWARD
3 - RELEASE PRESSURE
4 - THE PISTON MOVES SLIGHTLY REARWARD

TORQUE CONVERTER (Continued)

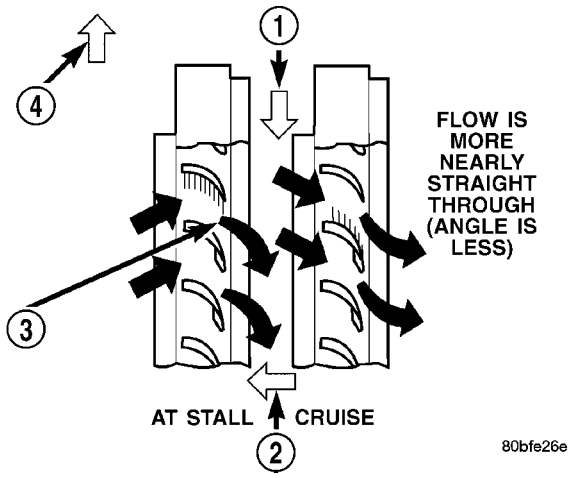


Fig. 250 Stator Operation

1 - DIRECTION STATOR WILL FREE WHEEL DUE TO OIL PUSHING ON BACKSIDE OF VANES
2 - FRONT OF ENGINE
3 - INCREASED ANGLE AS OIL STRIKES VANES
4 - DIRECTION STATOR IS LOCKED UP DUE TO OIL PUSHING AGAINST STATOR VANES

REMOVAL

(1) Remove transmission and torque converter from vehicle. (Refer to 21 - TRANSMISSION/AUTOMATIC - 45RFE/545RFE - REMOVAL)

(2) Place a suitable drain pan under the converter housing end of the transmission.

CAUTION: Verify that transmission is secure on the lifting device or work surface, the center of gravity of the transmission will shift when the torque converter is removed creating an unstable condition. The torque converter is a heavy unit. Use caution when separating the torque converter from the transmission.

(3) Pull the torque converter forward until the center hub clears the oil pump seal.

(4) Separate the torque converter from the transmission.

INSTALLATION

NOTE: Check converter hub and drive notches for sharp edges, burrs, scratches, or nicks. Polish the hub and notches with 320/400 grit paper or crocus cloth if necessary. The hub must be smooth to avoid damaging the pump seal at installation.

(1) Lubricate oil pump seal lip with transmission fluid.

(2) Place torque converter in position on transmission.

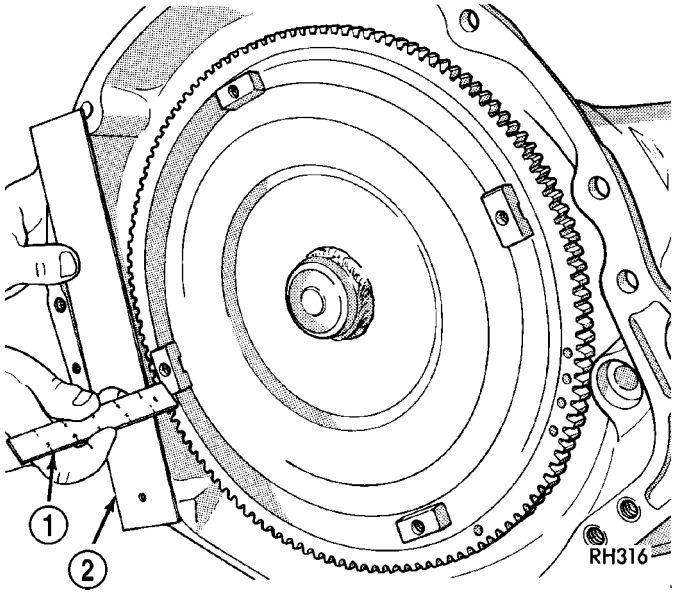


Fig. 251 Checking Torque Converter Seating - Typical

1 - SCALE
2 - STRAIGHTEDGE

CAUTION: Do not damage oil pump seal or bushing while inserting torque converter into the front of the transmission.

(3) Align torque converter to oil pump seal opening.

(4) Insert torque converter hub into oil pump.

(5) While pushing torque converter inward, rotate converter until converter is fully seated in the oil pump gears.

(6) Check converter seating with a scale (1) and straightedge (2) (Fig. 251). Surface of converter lugs should be 1/2 in. to rear of straightedge when converter is fully seated.

(7) If necessary, temporarily secure converter with C-clamp attached to the converter housing.

(8) Install the transmission in the vehicle.

(9) Fill the transmission with the recommended fluid.

TRANSMISSION CONTROL RELAY

DESCRIPTION

The relay is supplied fused B+ voltage, energized by the TCM, and is used to supply power to the solenoid pack when the transmission is in normal operating mode.

TRANSMISSION CONTROL RELAY (Continued)

OPERATION

When the relay is “off”, no power is supplied to the solenoid pack and the transmission is in “limp-in” mode. After a controller reset, the TCM energizes the relay. Prior to this, the TCM verifies that the contacts are open by checking for no voltage at the switched battery terminals. After this is verified, the voltage at the solenoid pack pressure switches is checked. After the relay is energized, the TCM monitors the terminals to verify that the voltage is greater than 3 volts.

TRANSMISSION RANGE SENSOR

DESCRIPTION

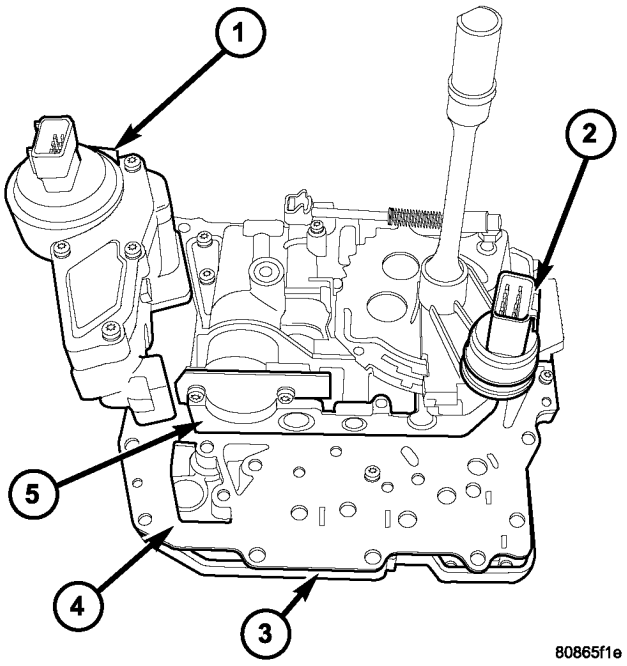


Fig. 252 Valve Body Assembly

1 - SOLENOID/PRESSURE SWITCH ASSEMBLY
2 - TRS
3 - TRANSFER PLATE
4 - SEPARATOR PLATE
5 - VALVE BODY

The Transmission Range Sensor (TRS) (2) (Fig. 252) is mounted to the top of the valve body inside the transmission and can only be serviced by removing the valve body assembly. The electrical connector extends through the transmission case.

The Transmission Range Sensor (TRS) has four switch contacts that monitor shift lever position and send the information to the PCM.

OPERATION

The Transmission Range Sensor (TRS) communicates shift lever position (SLP) to the TCM as a combination of open and closed switches. Each shift lever position has an assigned combination of switch states (open/closed) that the TCM receives from four sense circuits. The TCM interprets this information and determines the appropriate transmission gear position and shift schedule.

Since there are four switches, there are 16 possible combinations of open and closed switches (codes). Seven of these codes are related to gear position and three are recognized as “between gear” codes. This results in six codes which should never occur. These are called “invalid” codes. An invalid code will result in a DTC, and the TCM will then determine the shift lever position based on pressure switch data. This allows reasonably normal transmission operation with a TRS failure.

TRS SWITCH STATES

SLP	T42	T41	T3	T1
P	CL	CL	CL	OP
R	CL	OP	OP	OP
N	CL	CL	OP	CL
D	OP	OP	OP	CL
2	OP	OP	CL	OP
1	CL	OP	CL	CL

TRANSMISSION RANGE SENSOR (Continued)

REMOVAL

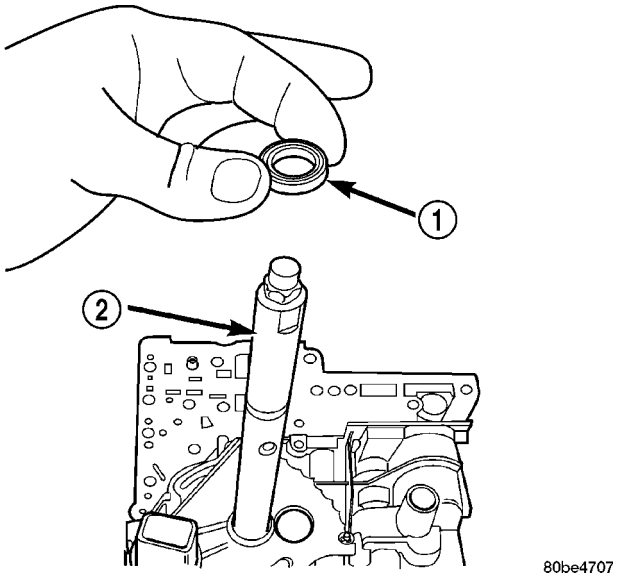


Fig. 253 Manual Shaft Seal - Typical

1 - SEAL
2 - MANUAL SHAFT

(1) Remove valve body assembly from vehicle. (Refer to 21 - TRANSMISSION/AUTOMATIC - 42RLE/VALVE BODY - REMOVAL)

(2) Remove the manual shaft seal (1) (Fig. 253).

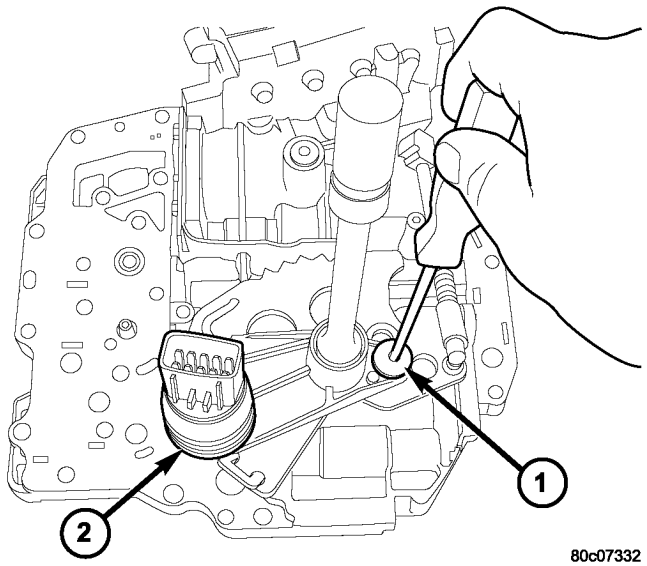


Fig. 254 Manual Shaft Retaining Screw

1 - SCREW
2 - TRS

(3) Remove manual shaft/TRS retaining screw (1) (Fig. 254).

(4) Slide TRS off of manual valve shaft.

INSTALLATION

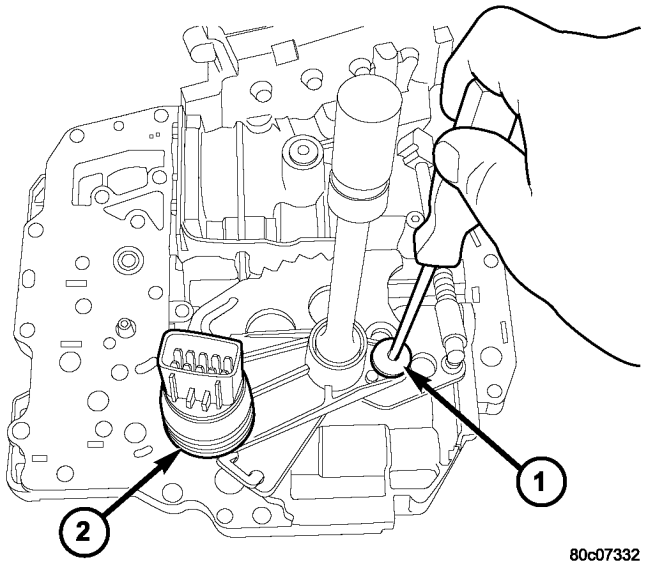


Fig. 255 Manual Shaft Retaining Screw

1 - SCREW
2 - TRS

(1) Install the TRS (2) to the manual shaft. Make sure TRS locating pin rests in manual valve bore slot.

(2) Install the TRS/manual shaft retaining screw (1) (Fig. 255) and torque to 5 N·m (45 in. lbs.) torque.

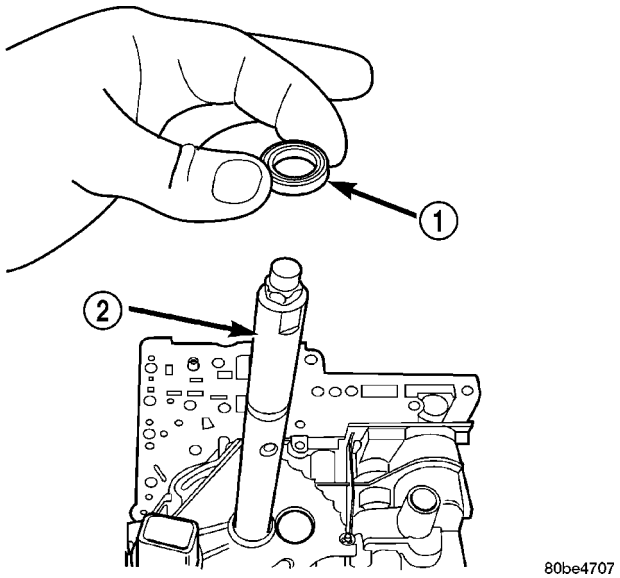


Fig. 256 Manual Shaft Seal - Typical

1 - SEAL
2 - MANUAL SHAFT

(3) Install the manual shaft seal (1) (Fig. 256).

(4) Install valve body to the transmission. (Refer to 21 - TRANSMISSION//AUTOMATIC - 42RLE/VALVE BODY - INSTALLATION)

TRANSMISSION TEMPERATURE SENSOR

DESCRIPTION

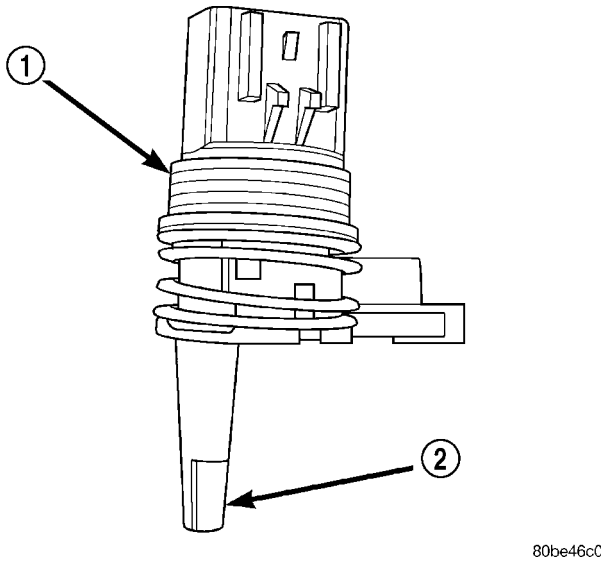


Fig. 257 Transmission Temperature Sensor

1 - TRANSMISSION RANGE SENSOR
2 - TEMPERATURE SENSOR

The transmission temperature sensor (2) (Fig. 257) is located in the transmission range sensor (1) and communicates transmission sump temperature to the TCM.

OPERATION

The transmission range sensor (TRS) has an integrated thermistor that the TCM uses to monitor the transmission's sump temperature. Since fluid temperature can affect transmission shift quality and convertor lock up, the TCM requires this information to determine which shift schedule to operate in. The TCM also monitors this temperature data so it can energize the vehicle cooling fan(s) when a transmission "overheat" condition exists. If the thermistor circuit fails, the TCM will revert to calculated oil temperature usage.

CALCULATED TEMPERATURE

A failure in the temperature sensor or circuit will result in calculated temperature being substituted for actual temperature. Calculated temperature is a predicted fluid temperature which is calculated from a combination of inputs:

- Battery (ambient) temperature
- Engine coolant temperature
- In-gear run time since start-up

VALVE BODY

DESCRIPTION

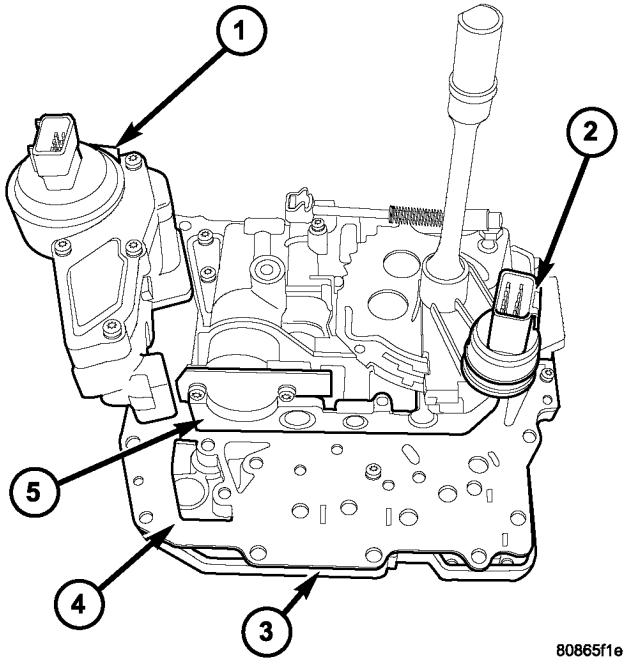


Fig. 258 Valve Body Assembly

1 - SOLENOID/PRESSURE SWITCH ASSEMBLY
2 - TRS
3 - TRANSFER PLATE
4 - SEPARATOR PLATE
5 - VALVE BODY

The valve body assembly (Fig. 258) consists of a cast aluminum valve body (5),separator plate (4), and transfer plate (3). The valve body contains valves and check balls that control fluid delivery to the torque converter clutch, solenoid/pressure switch assembly, and frictional clutches.

Also mounted to the valve body assembly are the solenoid/pressure switch assembly and the transmission range sensor (2) (Fig. 258).

The valves contained within the valve body (1) include the following (Fig. 259):

- Regulator valve(2)
- Solenoid switch valve(7)
- Manual valve(5)
- Converter clutch switch valve(6)
- Converter clutch control valve(4)
- Torque converter regulator valve(2)
- Low/Reverse switch valve(3)

In addition, the valve body also contains the thermal valve, #2, 3, 4 & 5 check balls and the 2/4 accumulator assembly.

VALVE BODY (Continued)

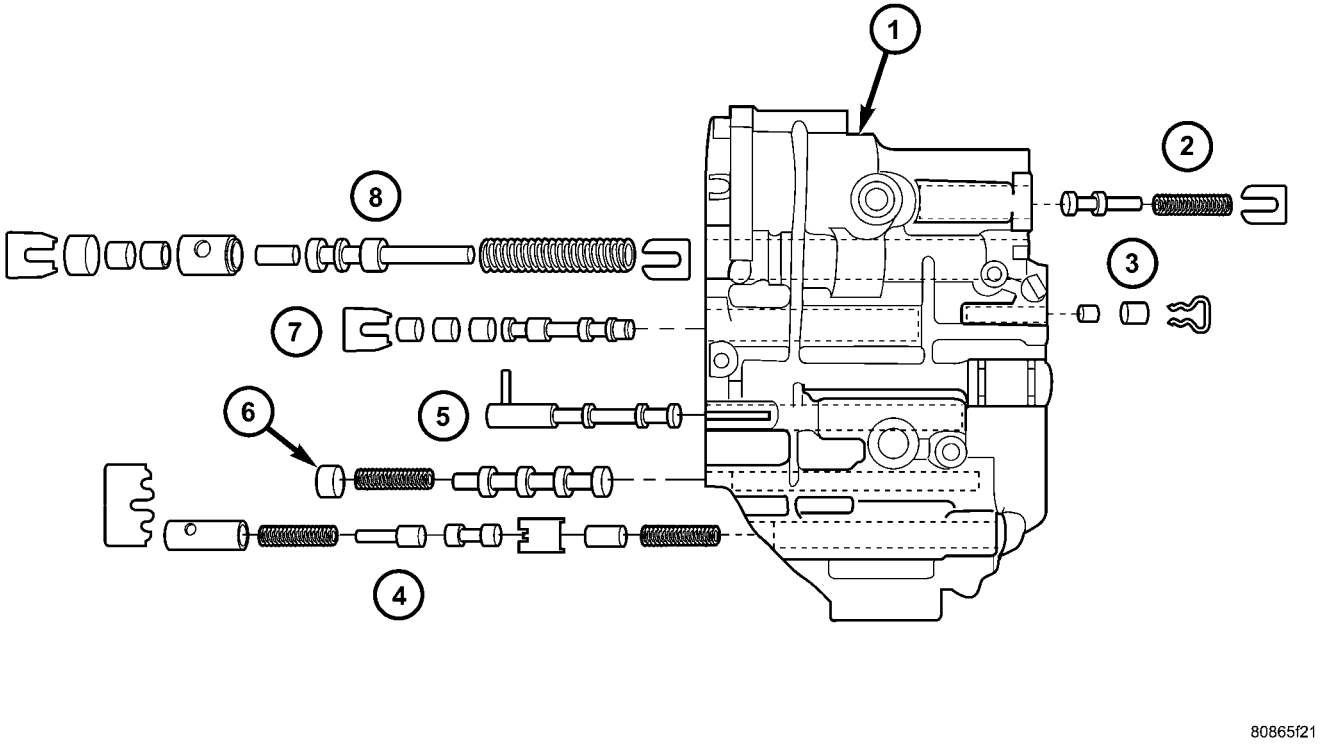


Fig. 259 Valve Body - Exploded

1 - VALVE BODY
2 - T/C REGULATOR VALVE
3 - L/R SWITCH VALVE
4 - CONVERTER CLUTCH CONTROL VALVE
5 - MANUAL VALVE
6 - CONVERTER CLUTCH SWITCH VALVE
7 - SOLENOID SWITCH VALVE
8 - REGULATOR VALVE

OPERATION

NOTE: (Refer to 21 - TRANSMISSION/AUTOMATIC - 42RLE - SCHEMATICS AND DIAGRAMS) for a visual aid in determining valve location, operation and design.

THERMAL VALVE

The thermal valve (1) (Fig. 260) is a bi-metallic shudder valve that helps control the venting rate of oil pressure in the underdrive clutch passage during release of the clutch. When the oil temperature is approximately 20 degrees Fahrenheit or less, the valve is fully open to assist in venting oil past the U1 orifice (2). At temperatures above 20 degrees, the valve starts to close and becomes fully closed at approximately 140 degrees. The thermal valve is located in the transfer plate of the valve body.

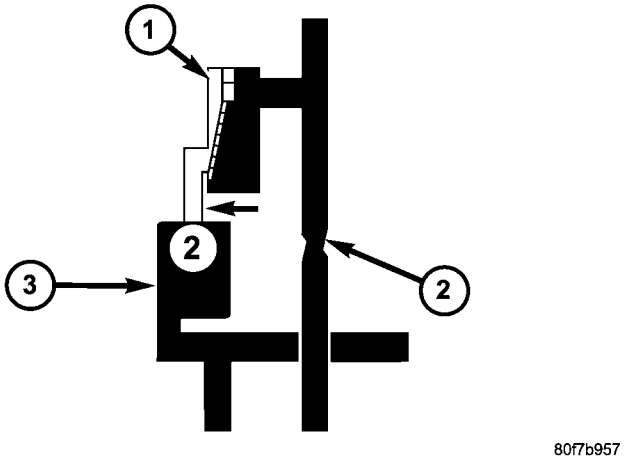


Fig. 260 Thermal Valve

1 - THERMAL VALVE
2 - U1 ORIFICE
3 - NUMBER 2 CHECK BALL

VALVE BODY (Continued)

REGULATOR VALVE

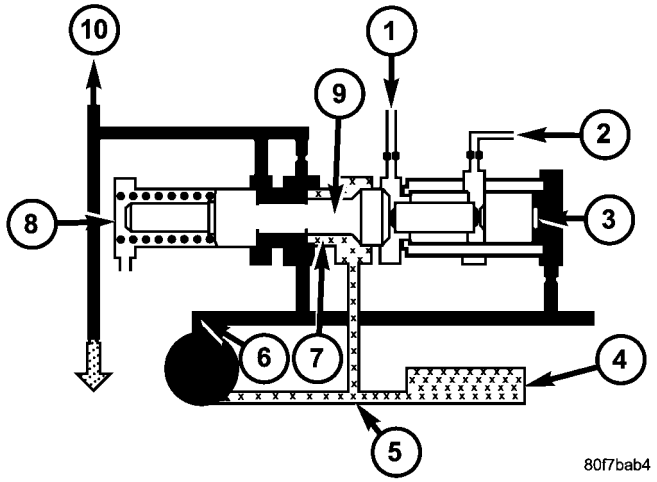


Fig. 261 Regulator Valve

1 - FROM OVERDRIVE CLUTCH CIRCUIT
2 - FROM MANUAL VALVE
3 - HYDRAULIC PRESSURE
4 - FILTER
5 - PUMP INLET
6 - PUMP OUTLET
7 - OIL PRESSURE REGULATED AT THIS POINT
8 - SPRING TENSION
9 - REGULATOR VALVE
10 - TORQUE CONVERTER CONTROL VALVE

The regulator valve (9) (Fig. 261) controls hydraulic pressure in the transmission. It receives unregulated pressure from the pump (6), which works against spring tension (8) to maintain oil at specific pressures. A system of sleeves and ports allows the regulator valve to work at one of three predetermined pressure levels. Regulated oil pressure is also referred to as “line pressure.”

SOLENOID SWITCH VALVE

The solenoid switch valve (Fig. 262) controls line pressure from the LR/CC solenoid (4). In one position, it allows the low/reverse clutch to be pressurized. In the other, it directs line pressure to the converter control and converter clutch valves (7).

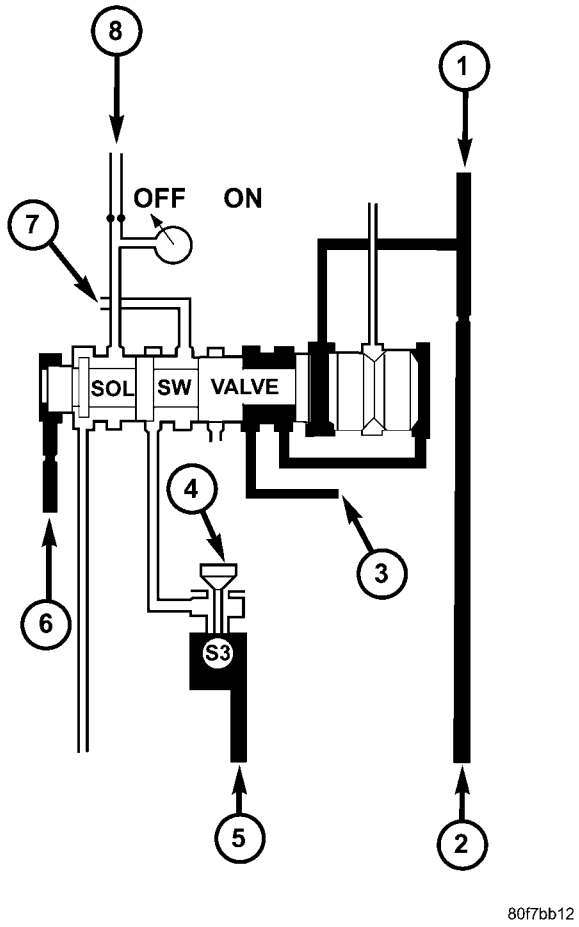


Fig. 262 Solenoid Switch Valve De-Energized

1 - 2/4 CLUTCH
2 - MANUAL VALVE
3 - UD CLUTCH
4 - LR/CC SOLENOID DE-ENERGIZED
5 - MANUAL VALVE
6 - LINE PRESSURE
7 - CONVERTER CLUTCH SWITCH AND CONTROL VALVES
8 - LR CLUTCH

VALVE BODY (Continued)

MANUAL VALVE

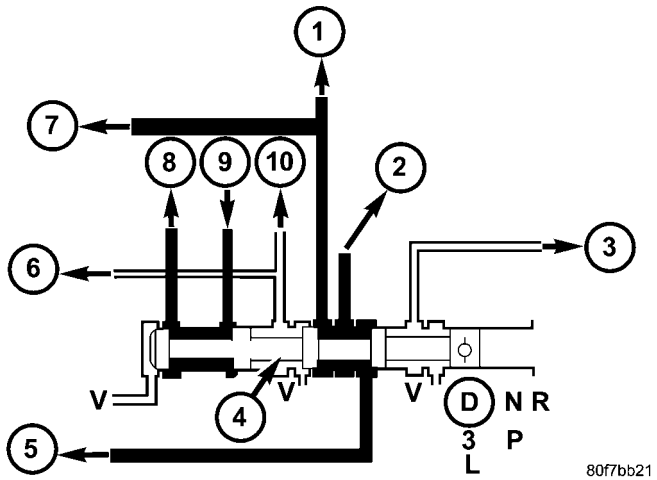


Fig. 263 Manual Valve

1 - UD CLUTCH
2 - LR/CC CLUTCH
3 - REVERSE CLUTCH
4 - MANUAL VALVE
5 - REGULATOR VALVE
6 - REGULATOR VALVE
7 - CONVERTER CLUTCH CONTROL VALVE
8 - 2/4 CLUTCH
9 - 2/4 - L/R SOLENOID
10 - L/R CLUTCH

The manual valve (4) (Fig. 263) is operated by the mechanical shift linkage. Its primary responsibility is to send line pressure to the appropriate hydraulic circuits and solenoids. The valve has three operating ranges or positions.

CONVERTER CLUTCH SWITCH VALVE

The main responsibility of the converter clutch switch valve (10) (Fig. 264) is to control hydraulic pressure applied to the front (off) side of the converter clutch piston. Line pressure from the regulator valve (5) is fed to the torque converter regulator valve (8). The pressure is then directed to the converter clutch switch valve (10) and to the front side of the converter clutch piston. This pressure pushes the piston back and disengages the converter clutch.

CONVERTER CLUTCH CONTROL VALVE

The converter clutch control valve (5) (Fig. 265) controls the back (on) side of the torque converter clutch (1). When the controller energizes or modulates the LR/CC solenoid to apply the converter clutch piston, both the converter clutch control valve (5) and the converter control valve move, allowing pressure to be applied to the back side of the clutch.

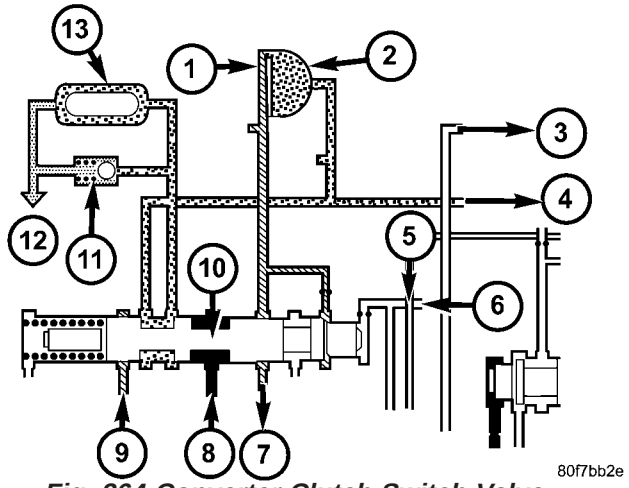


Fig. 264 Converter Clutch Switch Valve

1 - CONVERTER CLUTCH
2 - TORQUE CONVERTER
3 - LR CLUTCH
4 - DRIBBLERS
5 - REGULATOR VALVE
6 - SOLENOID SWITCH VALVE
7 - CONVERTER CLUTCH CONTROL VALVE
8 - TORQUE CONVERTER REGULATOR VALVE
9 - CONVERTER CLUTCH CONTROL VALVE
10 - CONVERTER CLUTCH SWITCH VALVE
11 - BYPASS VALVE
12 - LUBE
13 - COOLER

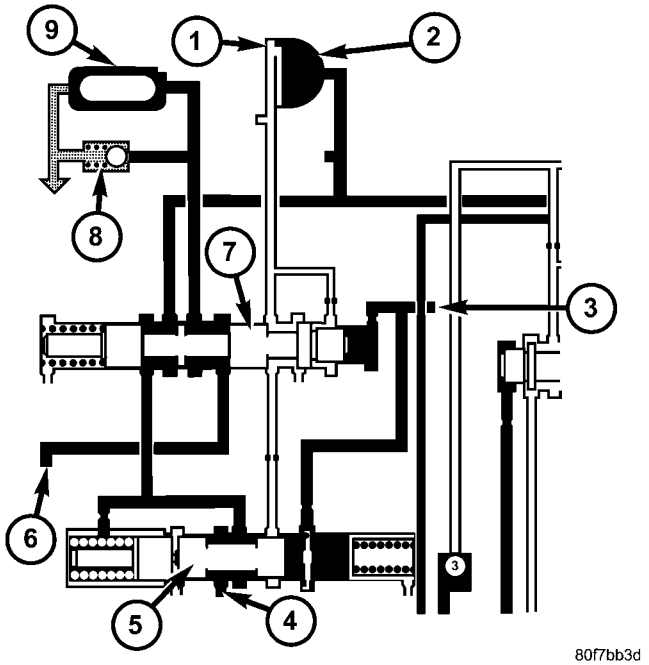


Fig. 265 Converter Clutch Control Valve

1 - CONVERTER CLUTCH
2 - TORQUE CONVERTER
3 - LR/CC SOLENOID
4 - FROM MANUAL VALVE
5 - CONVERTER CLUTCH CONTROL VALVE
6 - TORQUE CONVERTER REGULATOR VALVE
7 - CONVERTER CLUTCH SWITCH VALVE
8 - BYPASS VALVE
9 - COOLER

VALVE BODY (Continued)

T/C REGULATOR VALVE

The torque converter regulator valve slightly regulates the flow of fluid to the torque converter.

LOW/REVERSE SWITCH VALVE

The low/reverse clutch is applied from different sources, depending on whether low (1st) gear or reverse is selected. The low/reverse switch valve alternates positions depending on from which direction fluid pressure is applied. By design, when the valve is shifted by fluid pressure from one channel, the opposing channel is blocked. The switch valve alienates the possibility of a sticking ball check, thus providing consistent application of the low/reverse clutch under these operating conditions.

REMOVAL

NOTE: If valve body is being reconditioned or replaced, it is necessary to perform the Quick Learn Procedure. (Refer to 8 - ELECTRICAL/ELECTRONIC CONTROL MODULES/TRANSMISSION CONTROL MODULE - STANDARD PROCEDURE)

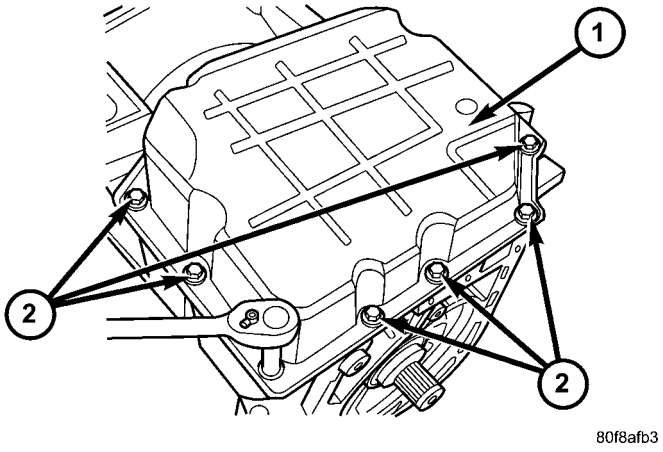


Fig. 266 Remove Transmission Oil Pan Bolts

1 - TRANSMISSION OIL PAN
2 - BOLTS

(1) Disconnect the TRS and solenoid wiring connectors.

(2) Disconnect the shift cable from the shift lever (at the transmission).

(3) Move the manual shift lever clockwise as far as it will go. This should be one position past the L position. Then remove the manual shift lever.

NOTE: One of the oil pan bolts has a sealing patch applied from the factory. Separate this bolts for reuse.

(4) Remove transmission pan bolts (2) (Fig. 266).

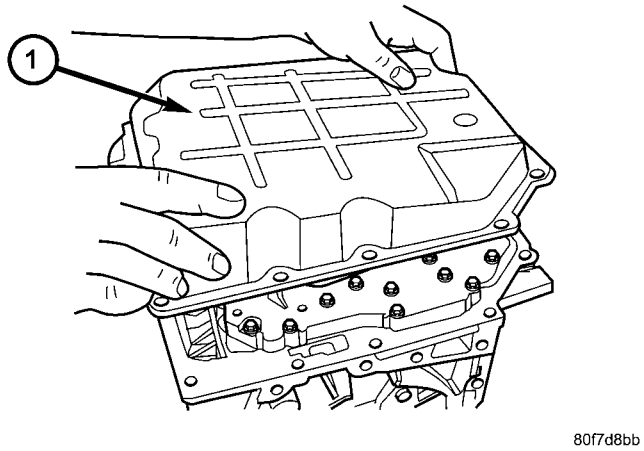


Fig. 267 Remove Transmission Oil Pan

1 - TRANSMISSION OIL PAN

(5) Remove transmission oil pan (1) (Fig. 267).

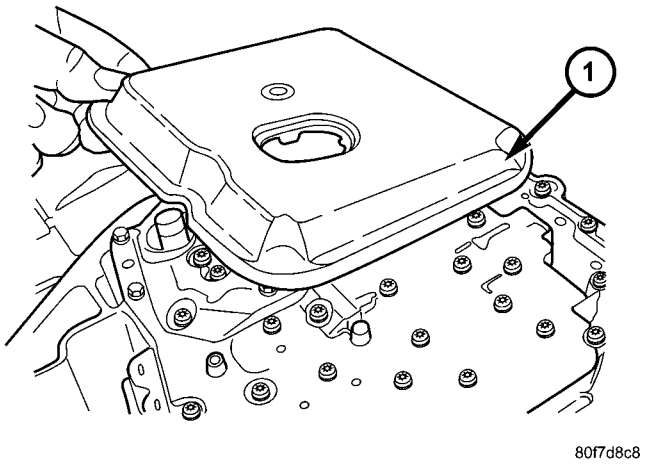


Fig. 268 Remove Transmission Filter

1 - TRANSMISSION FILTER

(6) Remove oil filter (1) (Fig. 268) from valve body. It is held in place by two screws.

VALVE BODY (Continued)

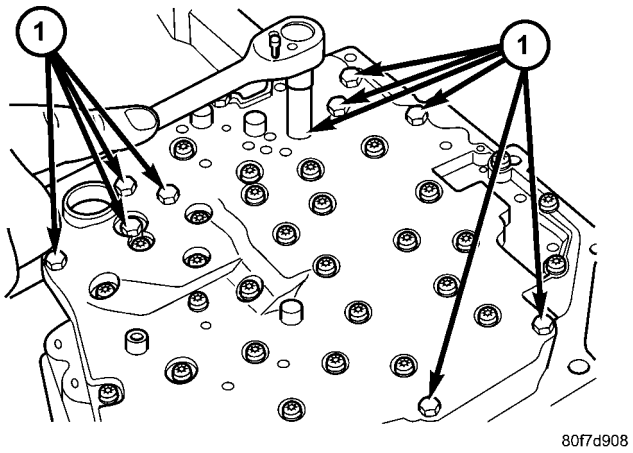


Fig. 269 Remove Valve Body Bolts

1 - BOLTS

(7) Remove valve body bolts-to-case (1) (Fig. 269).

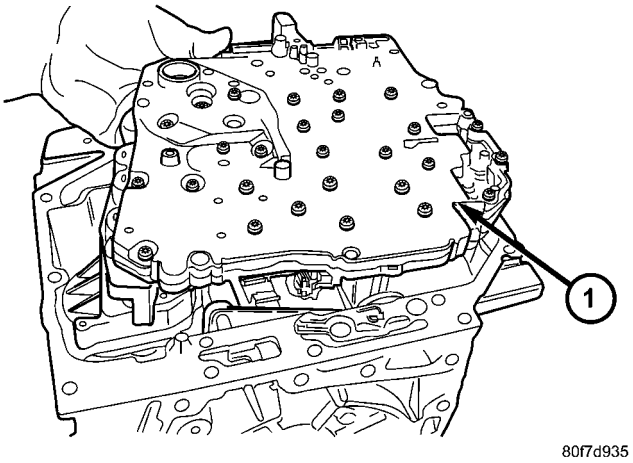


Fig. 270 Remove Valve Body From Transmission

1 - VALVE BODY

CAUTION: The overdrive and underdrive accumulators and springs may fall out when removing the valve body.

(8) Carefully remove valve body assembly (1) from the transmission (Fig. 270).

DISASSEMBLY

NOTE: If the valve body is being reconditioned or replaced, it is necessary to perform the Quick Learn Procedure using the DRBIII® Scan Tool (Refer to 8 - ELECTRICAL/ELECTRONIC CONTROL MODULES/ TRANSMISSION CONTROL MODULE - STANDARD PROCEDURE)

(1) Remove manual shaft seal.

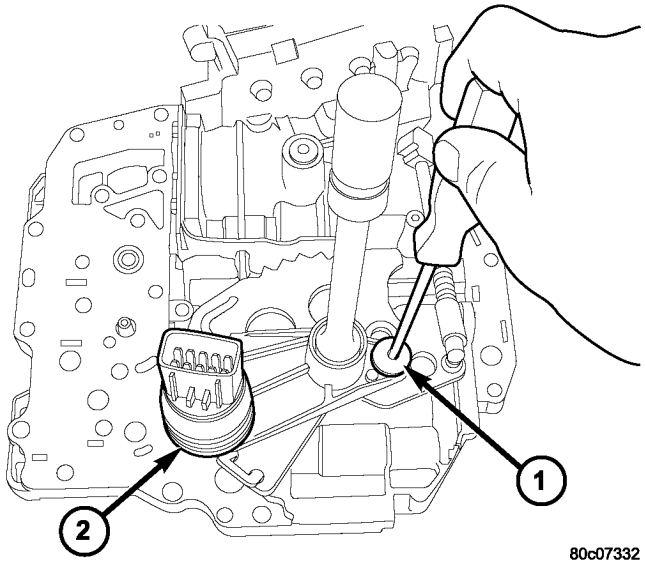


Fig. 271 Manual Shaft Retaining Screw

1 - SCREW
2 - TRS

(2) Remove manual shaft screw (1) (Fig. 271).

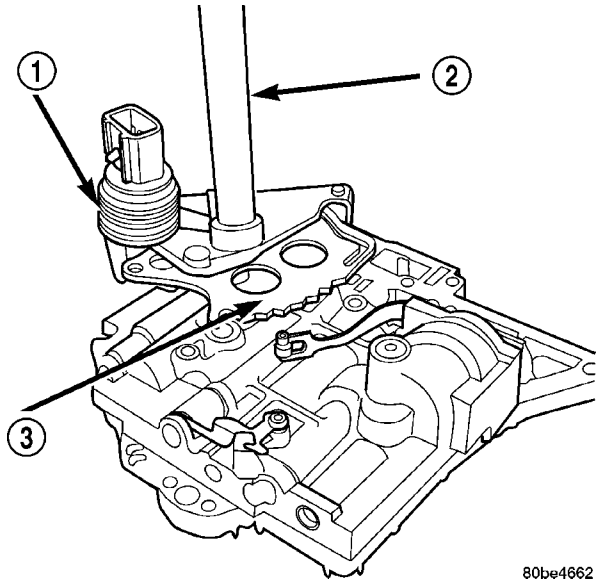


Fig. 272 Manual Shaft/Rooster Comb and Transmission Range Sensor

1 - TRANSMISSION RANGE SENSOR
2 - MANUAL SHAFT
3 - ROOSTER COMB

(3) Remove Transmission Range Sensor (TRS) (1) and manual shaft (2) (Fig. 272).

VALVE BODY (Continued)

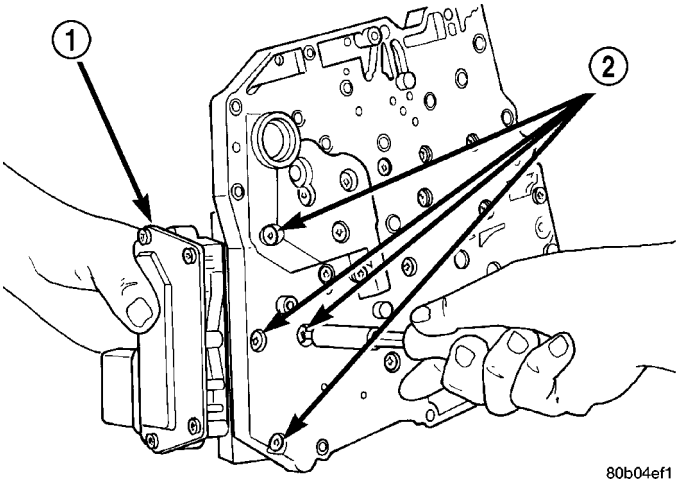


Fig. 273 Solenoid Retaining Screws

1 - SOLENOID/PRESSURE SWITCH ASSEMBLY
2 - RETAINING SCREWS

(4) Remove Solenoid/Pressure Switch Assembly (1) from valve body (Fig. 273).

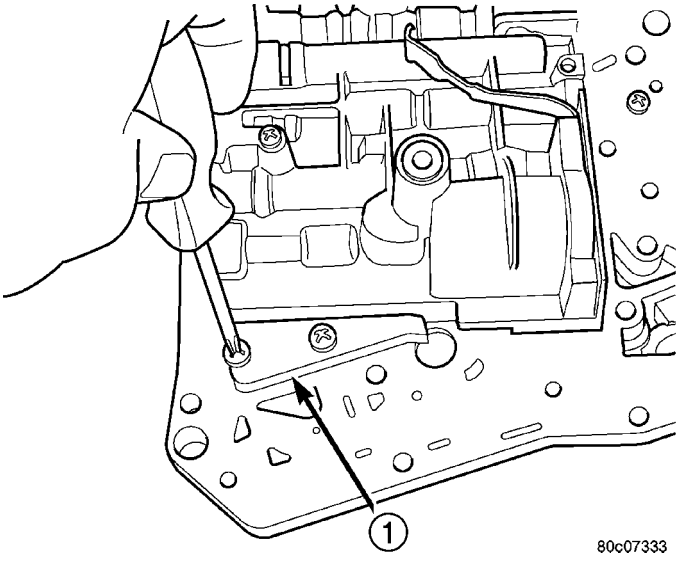


Fig. 274 Remove Stiffener Plate

1 - STIFFENER PLATE

(5) Remove valve body stiffener plate (1) (Fig. 274).

(6) Invert valve body assembly and remove transfer plate-to-valve body screws (1) (Fig. 275).

(7) Remove transfer/separator plate (1) from valve body (2) (Fig. 276)

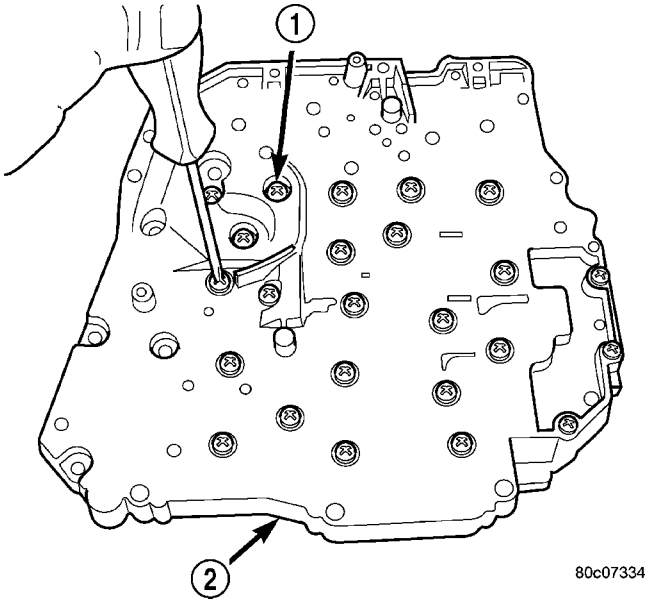


Fig. 275 Remove Transfer Plate-to-Valve Body Screws

1 - SCREW (24)
2 - TRANSFER PLATE

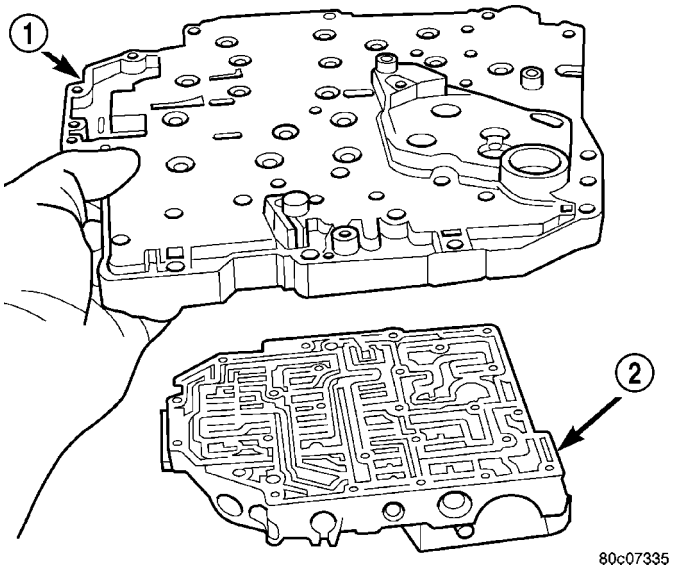


Fig. 276 Remove Transfer Plate to Valve Body

1 - TRANSFER PLATE
2 - VALVE BODY

VALVE BODY (Continued)

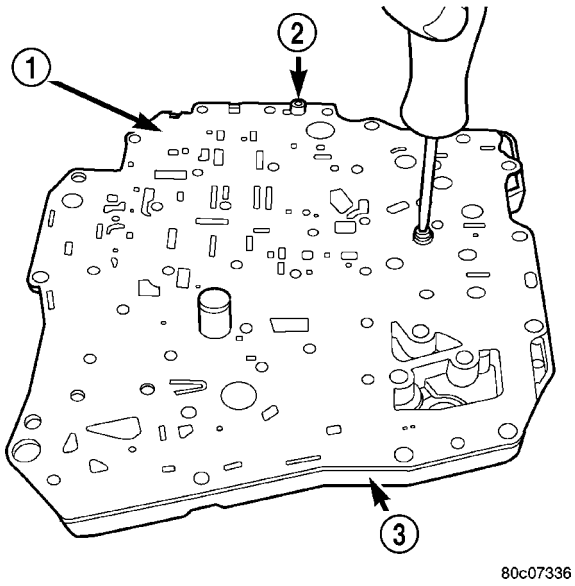


Fig. 277 Remove Separator Plate-to-Transfer Plate Screws

1 - SEPARATOR PLATE
2 - SCREW (2)
3 - TRANSFER PLATE

(8) Remove separator plate-to-transfer plate screws (2) (Fig. 277).

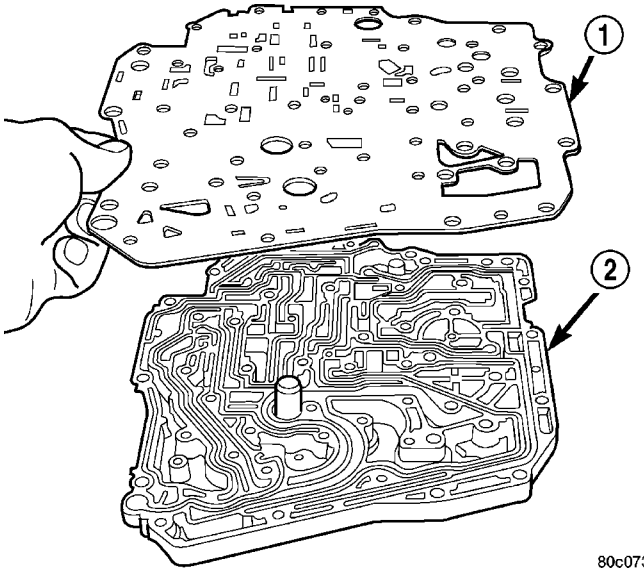


Fig. 278 Remove Separator Plate to Transfer Plate

1 - SEPARATOR PLATE
2 - TRANSFER PLATE

(9) Remove separator plate (1) from transfer plate (2) (Fig. 278).

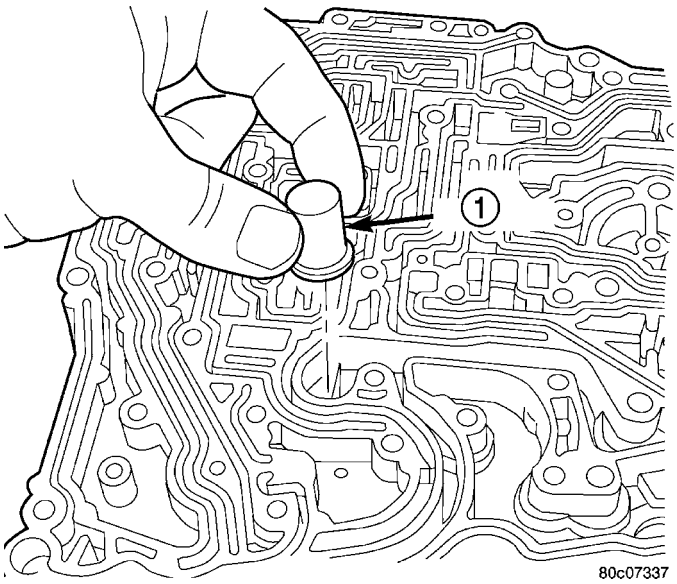


Fig. 279 Remove Oil Screen to Transfer Plate

1 - OIL SCREEN

(10) Remove the oil screen (1) from the transfer plate (Fig. 279).

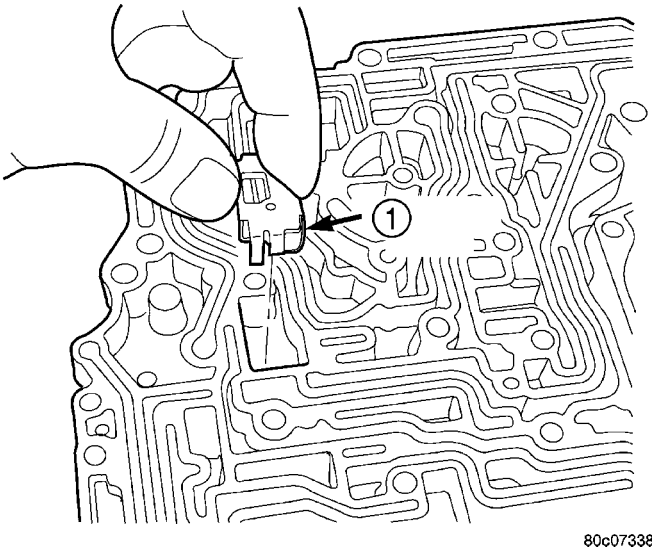


Fig. 280 Remove Thermal Valve to Transfer Plate

1 - THERMAL VALVE

(11) Remove thermal valve (1) (Fig. 280) from transfer plate.

VALVE BODY (Continued)

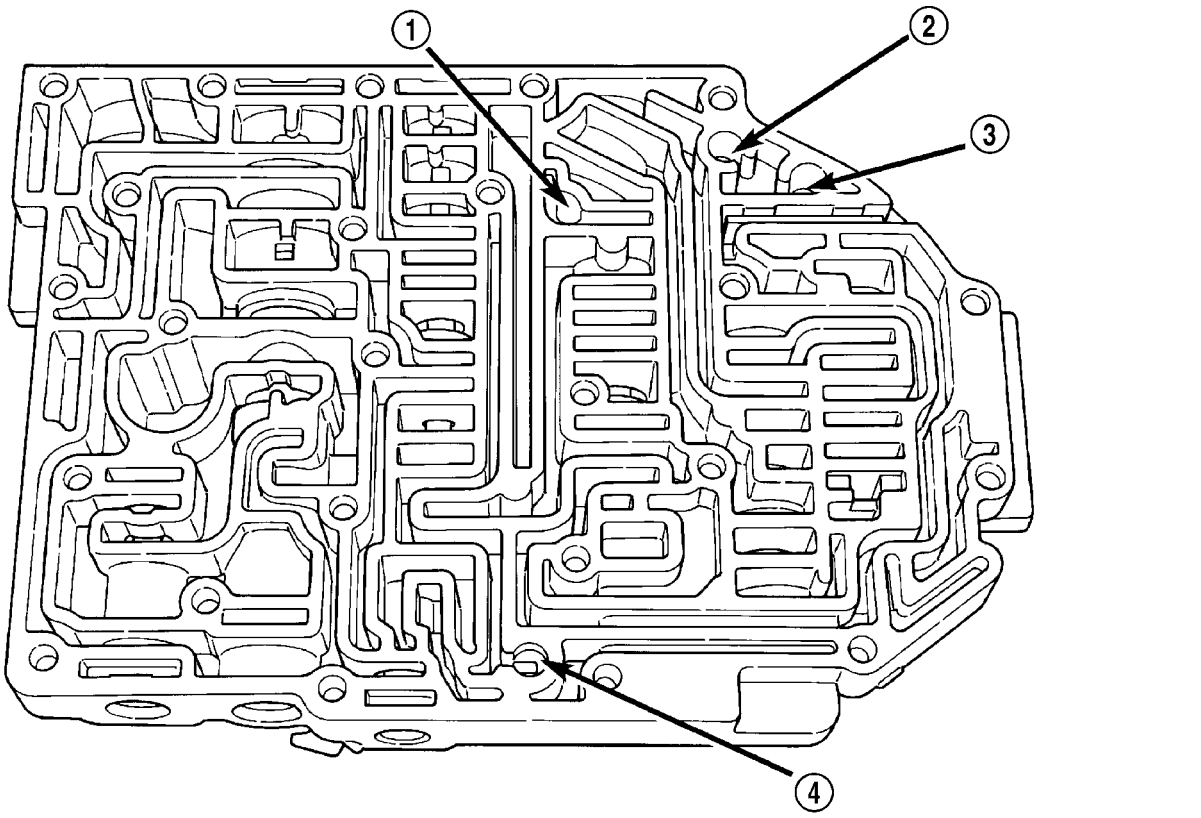


Fig. 281 Ball Check Location

1 - (#4) BALL CHECK LOCATION
2 - (#2) BALL CHECK LOCATION
3 - (#5) BALL CHECK LOCATION
4 - (#3) BALL CHECK LOCATION

(12) Remove valve body check balls (1-4). Note their location for assembly ease (Fig. 281).

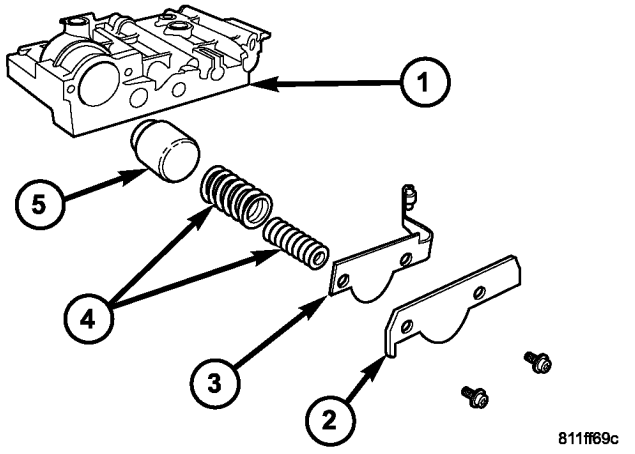


Fig. 282 2/4 Accumulator Assembly

1 - VALVE BODY
2 - RETAINER PLATE
3 - DETENT SPRING
4 - RETURN SPRINGS
5 - PISTON

(13) Remove 2/4 accumulator assembly (1-5) (Fig. 282).

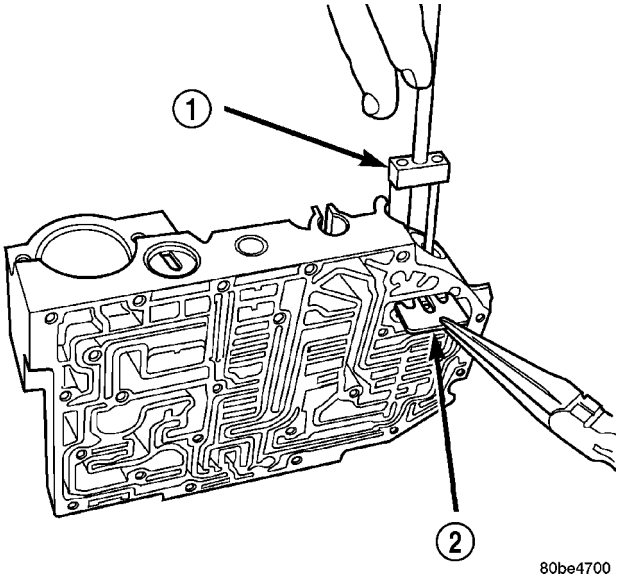


Fig. 283 Remove Dual Retainer Plate using Tool 6301

1 - TOOL 6301
2 - RETAINER

(14) Remove dual retainer plate (2) from valve body. Use special tool 6301 (1) to remove plate (2) (Fig. 283).

VALVE BODY (Continued)

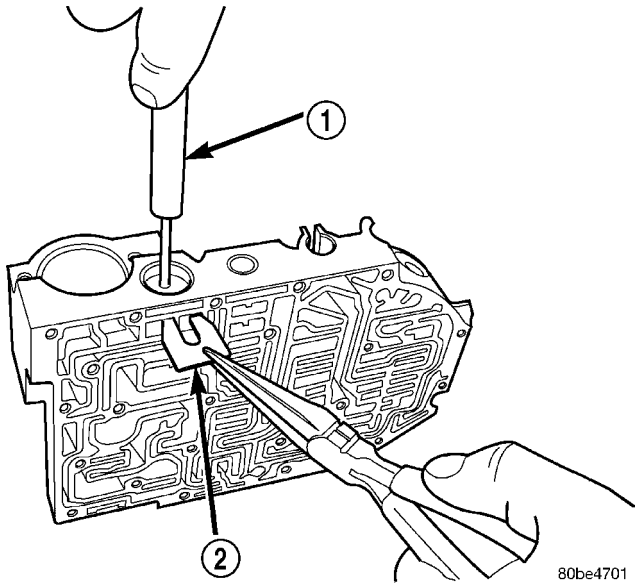


Fig. 284 Remove Regulator Valve Spring Retainer using Tool 6302

1 - TOOL 6302
2 - RETAINER

(15) Remove regulator valve spring retainer (2) (Fig. 284).

(16) Remove remaining retainers (1, 2) (Fig. 285).

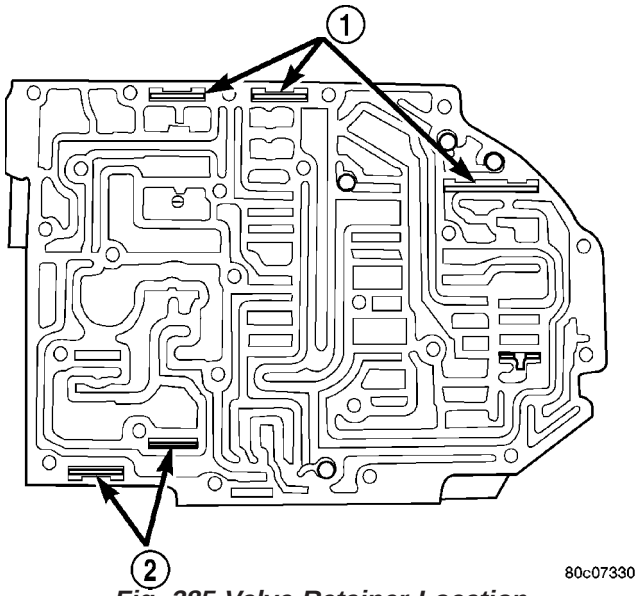


Fig. 285 Valve Retainer Location

1 - RETAINER
2 - RETAINER

(17) Remove all valves and springs (Fig. 286).

(18) Cleanliness through entire disassembly and assembly of the valve body cannot be overemphasized. When disassembling, each part should be washed in a suitable solvent, then dried by com-

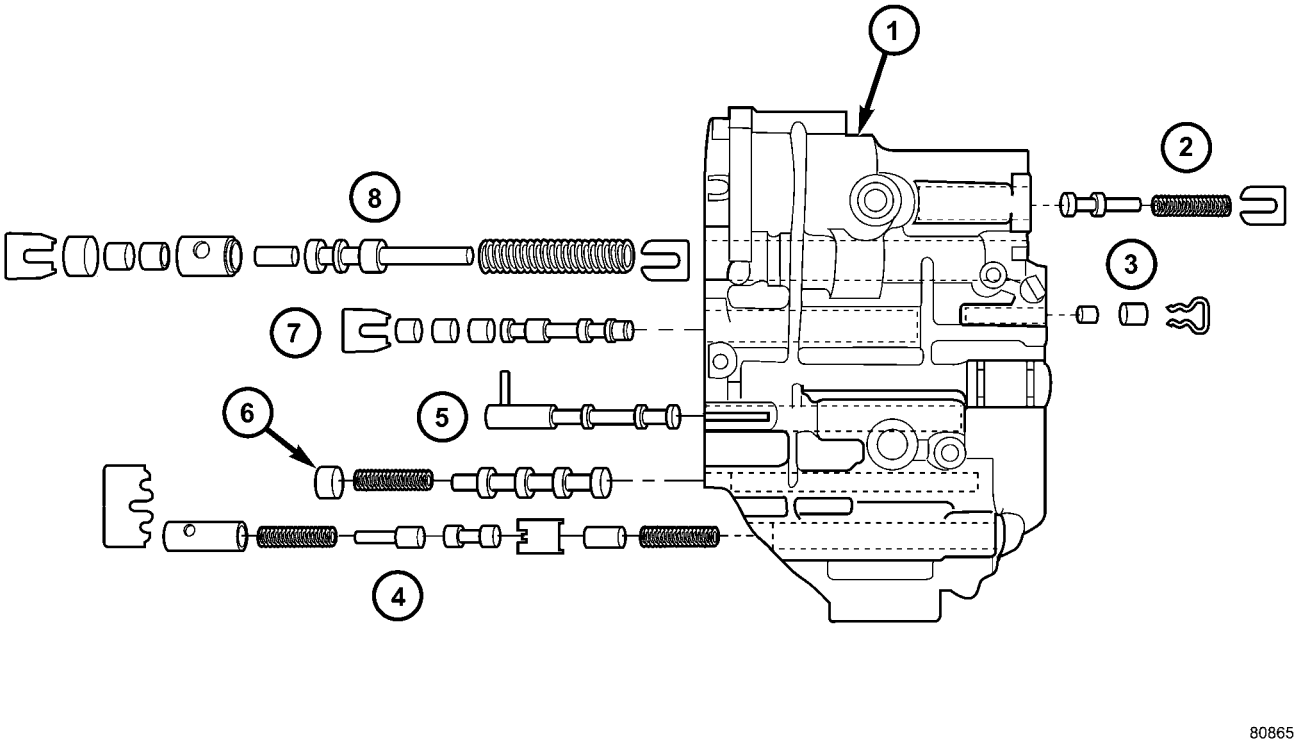


Fig. 286 Valve Body Assembly

1 - VALVE BODY
2 - T/C REGULATOR VALVE
3 - L/R SWITCH VALVE
4 - CONVERTER CLUTCH CONTROL VALVE
5 - MANUAL VALVE
6 - CONVERTER CLUTCH SWITCH VALVE
7 - SOLENOID SWITCH VALVE
8 - REGULATOR VALVE

VALVE BODY (Continued)

pressed air. **Do not wipe parts with shop towels.** All mating surfaces in the valve body are accurately machined; therefore, careful handling of all parts must be exercised to avoid nicks or burrs.

ASSEMBLY

NOTE: If the valve body assembly is being reconditioned or replaced, it is necessary to perform the Quick Learn Procedure using the DRBIII® Scan Tool. (Refer to 8 - ELECTRICAL/ELECTRONIC CONTROL MODULES/TRANSMISSION CONTROL MODULE - STANDARD PROCEDURE)

(1) Install all valves and springs as shown (Fig. 287).

(2) Install regulator valve spring retainer (2) (Fig. 288).

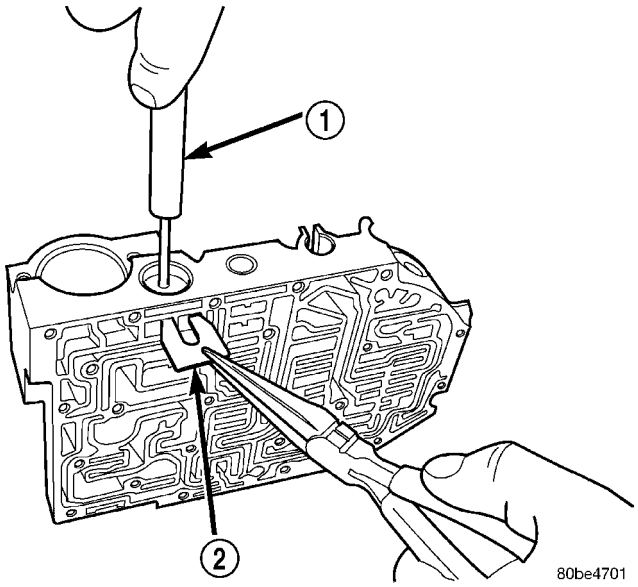


Fig. 288 Install Regulator Valve Spring Retainer using Tool 6302

1 - TOOL 6302
2 - RETAINER

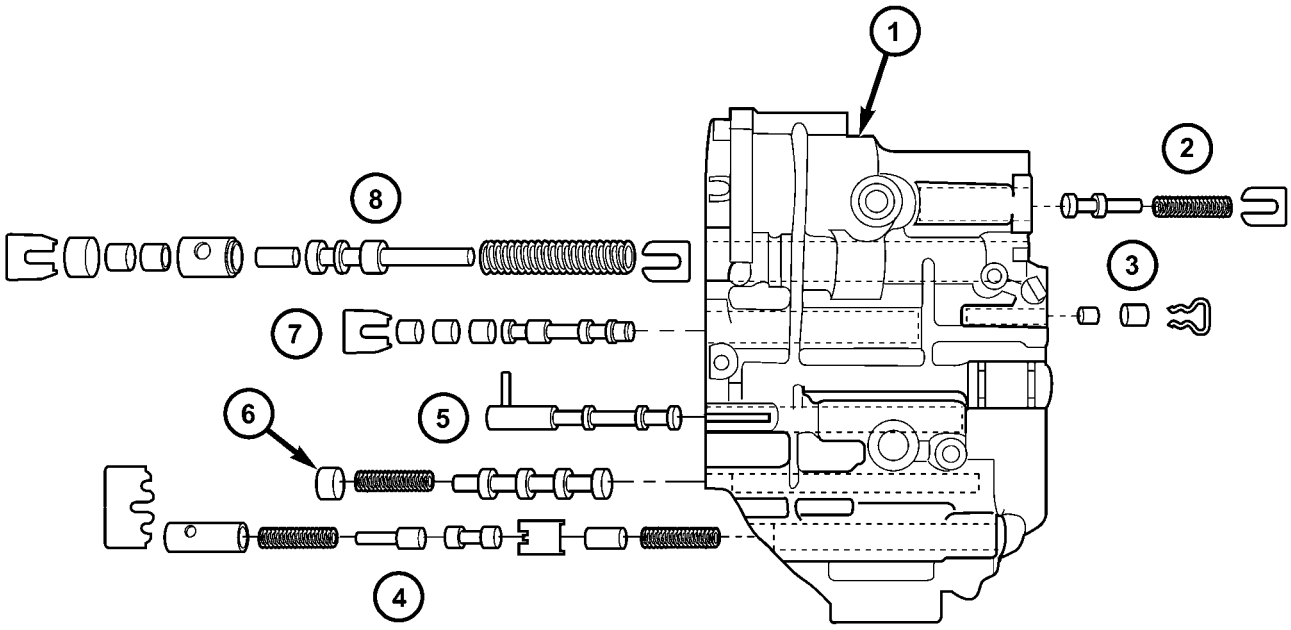


Fig. 287 Valve Body Assembly

1 - VALVE BODY
2 - T/C REGULATOR VALVE
3 - L/R SWITCH VALVE
4 - CONVERTER CLUTCH CONTROL VALVE
5 - MANUAL VALVE
6 - CONVERTER CLUTCH SWITCH VALVE
7 - SOLENOID SWITCH VALVE
8 - REGULATOR VALVE

VALVE BODY (Continued)

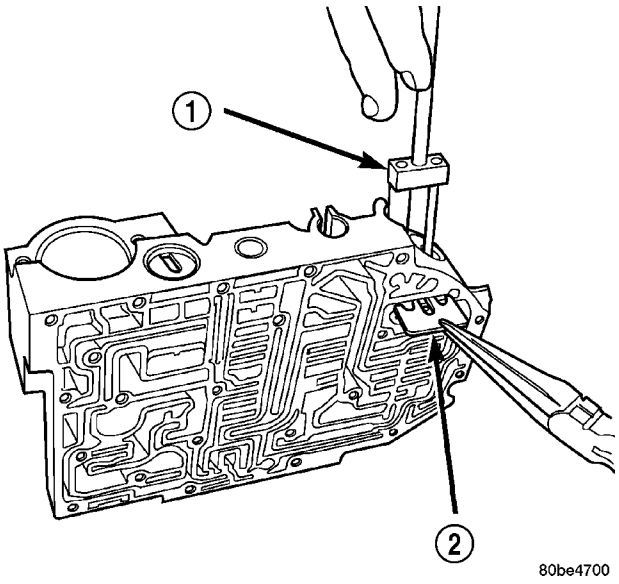


Fig. 289 Install Dual Retainer Plate using Tool 6301

1 - TOOL 6301
2 - RETAINER

(3) Install dual retainer plate (2) using Tool 6301 (1) (Fig. 289).

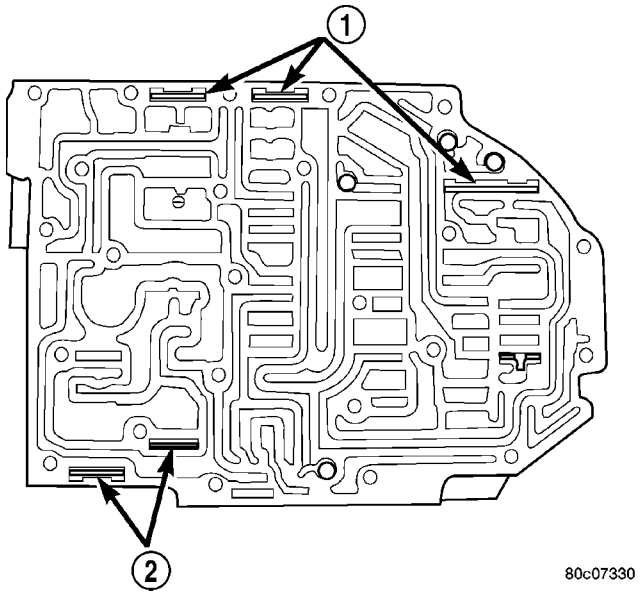


Fig. 290 Valve Retainer Location

1 - RETAINER
2 - RETAINER

(4) Verify that all retainers (1, 2) are installed as shown (Fig. 290). Retainers should be flush or below valve body surface.

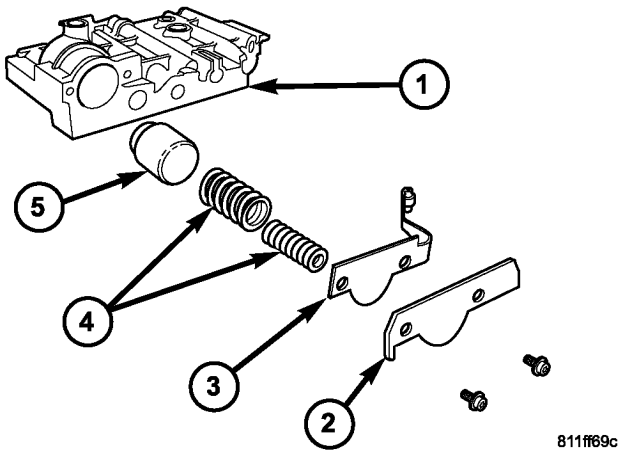


Fig. 291 2/4 Accumulator Assembly

1 - VALVE BODY
2 - RETAINER PLATE
3 - DETENT SPRING
4 - RETURN SPRINGS
5 - PISTON

(5) Install 2/4 Accumulator components (1-5) as shown (Fig. 291). Torque 2/4 Accumulator retainer plate to 5 N·m (45 in. lbs.).

(6) Install check balls into position as shown (Fig. 292). If necessary, secure them with petrolatum or transmission assembly gel for assembly ease.

(7) Install thermal valve (1) to the transfer plate (Fig. 293).

(8) Install the oil screen (1) to the transfer plate (Fig. 294).

VALVE BODY (Continued)

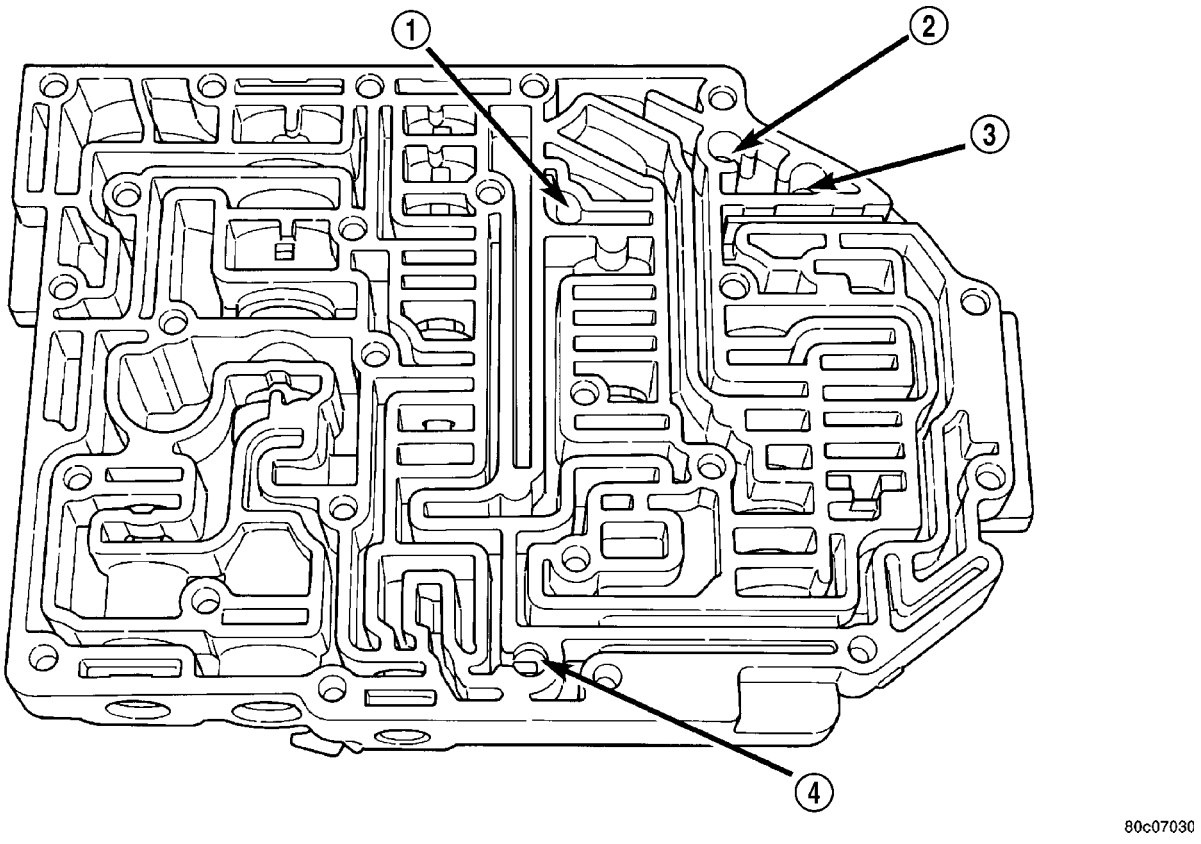


Fig. 292 Ball Check Location

1 - (#4) BALL CHECK LOCATION
2 - (#2) BALL CHECK LOCATION
3 - (#5) BALL CHECK LOCATION
4 - (#3) BALL CHECK LOCATION

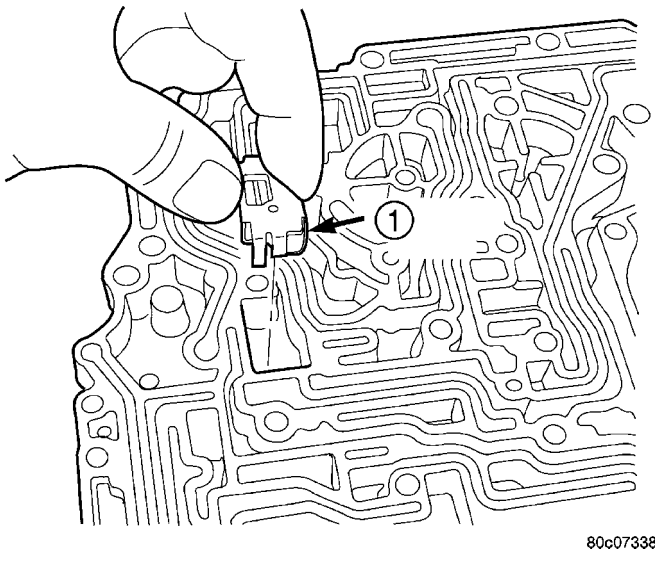


Fig. 293 Install Thermal Valve to Transfer Plate

1 - THERMAL VALVE

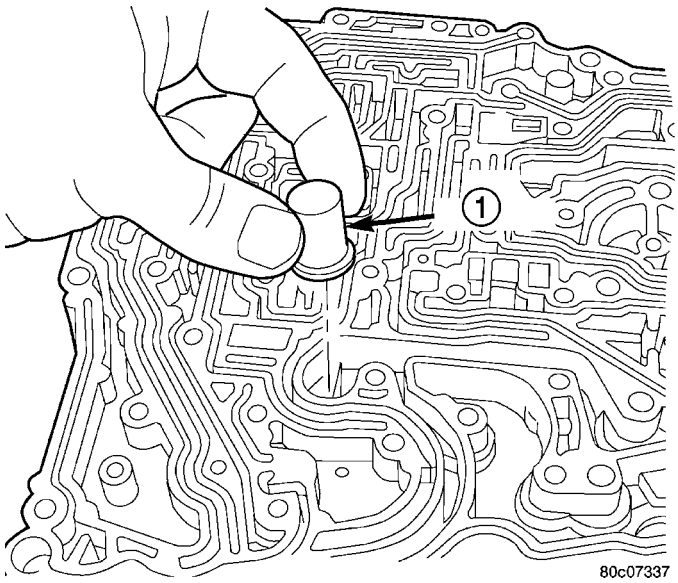


Fig. 294 Install Oil Screen to Transfer Plate

1 - OIL SCREEN

VALVE BODY (Continued)

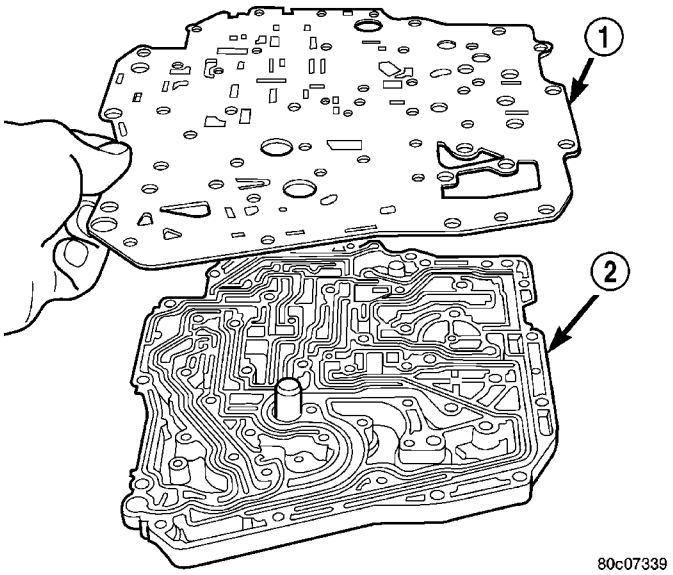


Fig. 295 Install Separator Plate to Transfer Plate

1 - SEPARATOR PLATE
2 - TRANSFER PLATE

(9) Install separator plate (1) to transfer plate (2) (Fig. 295).

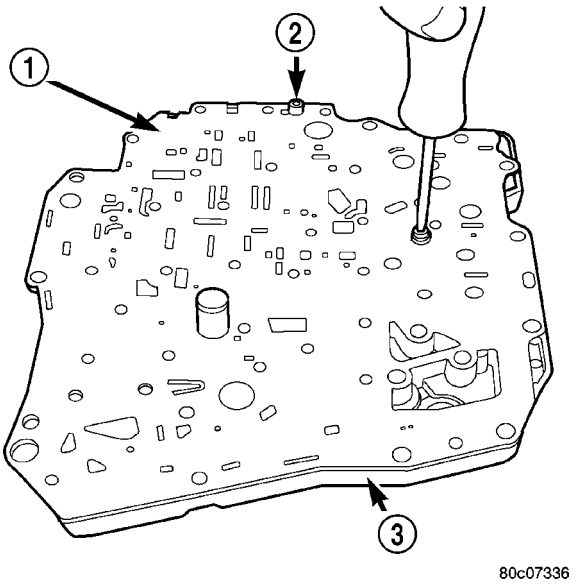


Fig. 296 Install Separator Plate-to-Transfer Plate Screws

1 - SEPARATOR PLATE
2 - SCREW (2)
3 - TRANSFER PLATE

(10) Install the two separator plate-to-transfer plate screws (2) (Fig. 296).

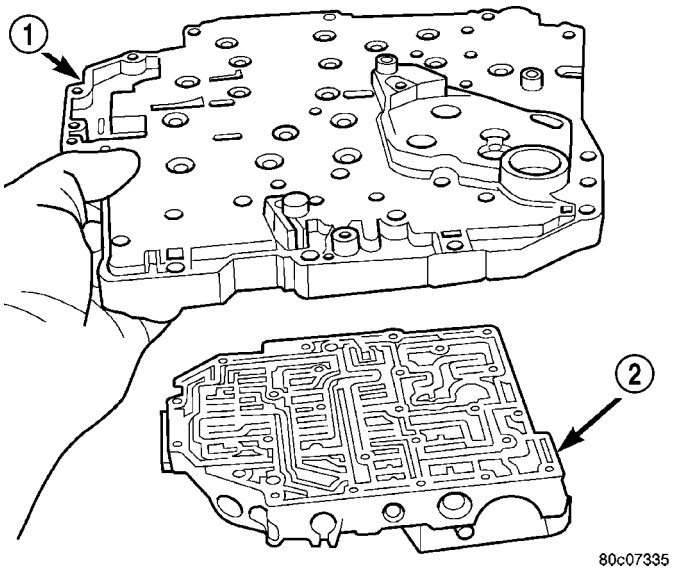


Fig. 297 Install Transfer Plate to Valve Body

1 - TRANSFER PLATE
2 - VALVE BODY

(11) Install the transfer plate (1) to the valve body (2) (Fig. 297).

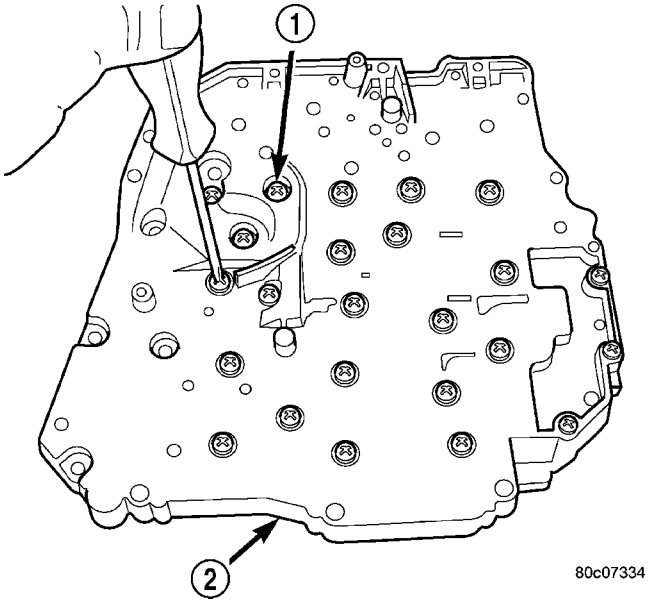


Fig. 298 Install Transfer Plate-to-Valve Body Screws

1 - SCREW (24)
2 - TRANSFER PLATE

(12) Install the transfer plate-to-valve body screws (1) (Fig. 298) and torque to 5 N·m (45 in. lbs.).

VALVE BODY (Continued)

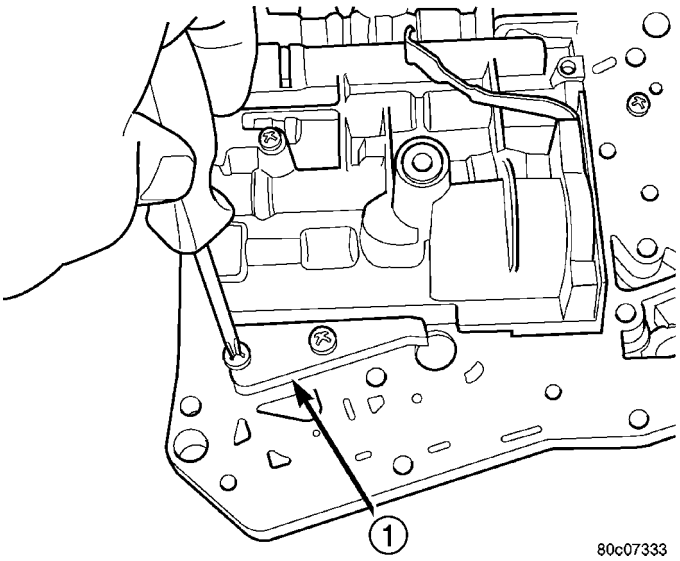


Fig. 299 Install Stiffener Plate

1 - STIFFENER PLATE

(13) Install the stiffener plate (1) (Fig. 299).

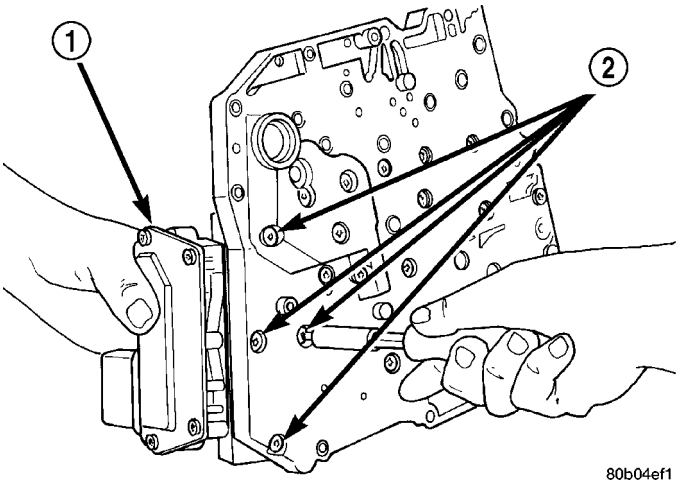


Fig. 300 Solenoid Retaining Screws

1 - SOLENOID/PRESSURE SWITCH ASSEMBLY
2 - RETAINING SCREWS

(14) Install the solenoid/pressure switch assembly (1) and screws to the transfer plate (Fig. 300) and torque to 5.5 N·m (50 in. lbs.).

(15) Install the manual shaft/rooster comb (3) and transmission range sensor (1) to the valve body (Fig. 301).

(16) Install the TRS/manual shaft retaining screw (1) (Fig. 302) and torque to 5 N·m (45 in. lbs.).

(17) Install manual shaft seal.

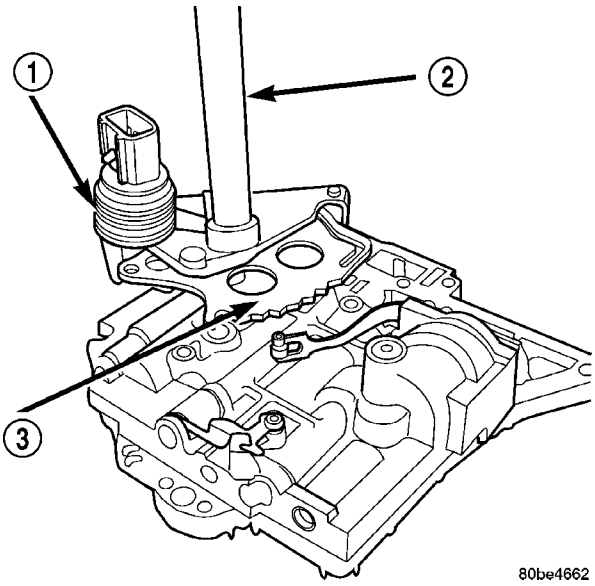


Fig. 301 Manual Shaft/Rooster Comb and Transmission Range Sensor

1 - TRANSMISSION RANGE SENSOR
2 - MANUAL SHAFT
3 - ROOSTER COMB

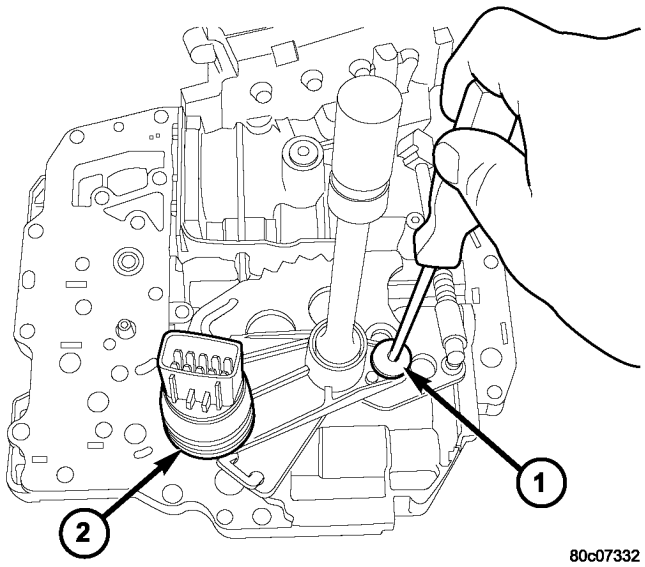


Fig. 302 Manual Shaft Retaining Screw

1 - SCREW
2 - TRS

VALVE BODY (Continued)

INSTALLATION

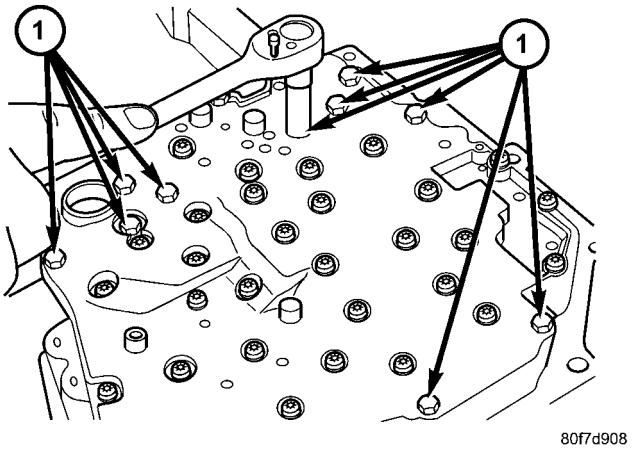


Fig. 303 Install Valve Body Bolts

1 - BOLTS

(1) Install valve body into position and start bolts (1). Torque valve body to transmission case bolts (1) (Fig. 303) to 12 N·m (105 in. lbs.) torque.

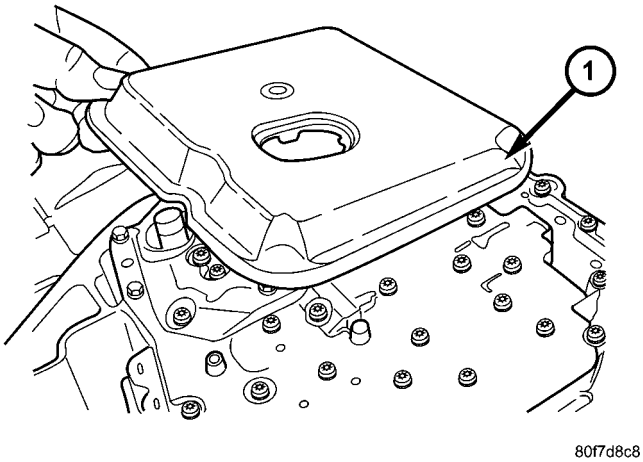


Fig. 304 Install Transmission Filter

1 - TRANSMISSION FILTER

(2) Install transmission oil filter (1) (Fig. 304).

NOTE: Before installing the oil pan bolt in the bolt hole (5) located between the torque converter clutch on and U/D clutch pressure tap circuits (Fig. 305), it will be necessary to replentish the sealing patch on the bolt using Mopar® Lock & Seal Adhesive.

(3) Make sure oil pan (1) and case rail are clean and dry. Install an 1/8" bead of RTV to the transmission oil pan and install to case. Tighten bolts (2) (Fig. 306) to 20 N·m (14.5 ft. lbs.).

(4) Lower vehicle and connect the TRS connector.

(5) Connect solenoid/pressure switch assembly connector.

(6) Lower vehicle.

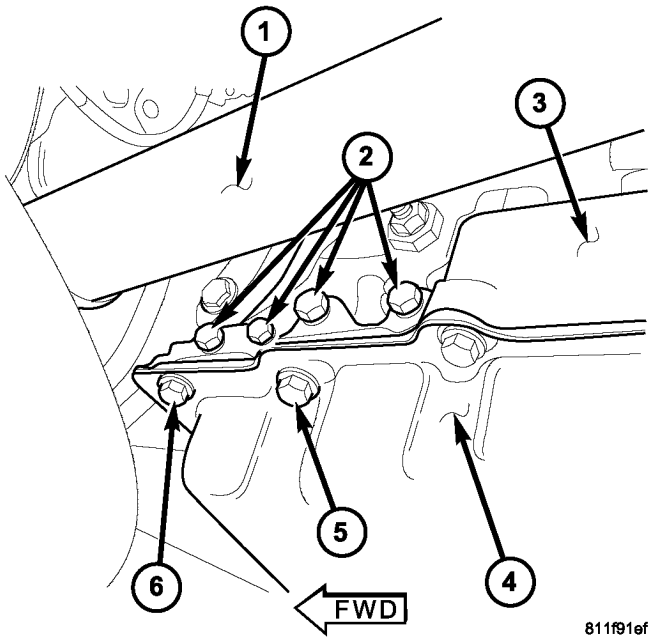


Fig. 305 Pan Fastener Location

1 - FRONT DRIVESHAFT
2 - PRESSURE PORTS
3 - TRANSMISSION CASE
4 - TRANSMISSION OIL PAN
5 - SECOND TRANSMISSION OIL PAN BOLT ON LEFT SIDE
6 - FIRST TRANSMISSION OIL PAN BOLT

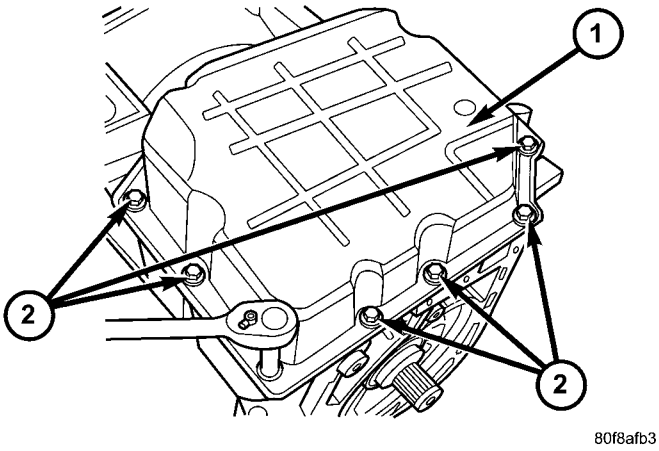


Fig. 306 Install Transmission Oil Pan Bolts

1 - TRANSMISSION OIL PAN
2 - BOLTS

(7) Fill transmission with ATF+4, Automatic Transmission Fluid. Verify proper fluid level. (Refer to 21 - TRANSMISSION/AUTOMATIC - 42RLE/ FLUID - STANDARD PROCEDURE)

NOTE: If the valve body has been reconditioned or replaced, it is necessary to perform the Quick Learn Procedure. (Refer to 8 - ELECTRICAL/ELECTRONIC CONTROL MODULES/TRANSMISSION CONTROL MODULE - STANDARD PROCEDURE)

TRANSFER CASE - NV231

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TRANSFER CASE - NV231

DESCRIPTION

The NV231 is a part-time transfer case with a low range reduction gear system. The NV231 has three operating ranges plus a NEUTRAL position. A low range system provides a reduction ratio for increased low speed torque capability.

The geartrain is mounted in two aluminum case halves attached with bolts. The mainshaft front and rear bearings are mounted in aluminum retainer housings bolted to the case halves.

OPERATING RANGES

Transfer case operating ranges are:

- 2WD (2-wheel drive)
- 4x4 (4-wheel drive)
- 4 Lo (4-wheel drive low range

The 2WD range is for use on any road surface at any time.

The 4x4 and 4 Lo ranges are for off road use only. They are not for use on hard surface roads. The only exception being when the road surface is wet or slippery or covered by ice and snow.

The low range reduction gear system is operative in 4 Lo range only. This range is for extra pulling power in off road situations. Low range reduction ratio is 2.72:1.

SHIFT MECHANISM

Operating ranges are selected with a floor mounted shift lever. The shift lever is connected to the transfer case range lever by an adjustable linkage rod. A straight line shift pattern is used. Range positions are marked on the shifter bezel cover plate.

TRANSFER CASE - NV231 (Continued)

IDENTIFICATION

A circular ID tag is attached to the rear case of each transfer case (Fig. 1). The ID tag provides the transfer case model number, assembly number, serial number, and low range ratio.

The transfer case serial number also represents the date of build.

OPERATION

The input gear is splined to the transmission output shaft. The input gear drives the mainshaft through the planetary assembly and range hub. The front output shaft is operated by a drive chain that connects the shaft to a drive sprocket on the mainshaft. The drive sprocket is engaged/disengaged by the mode fork, which operates the mode sleeve and hub. The sleeve and hub are not equipped with a synchronizer mechanism for shifting.

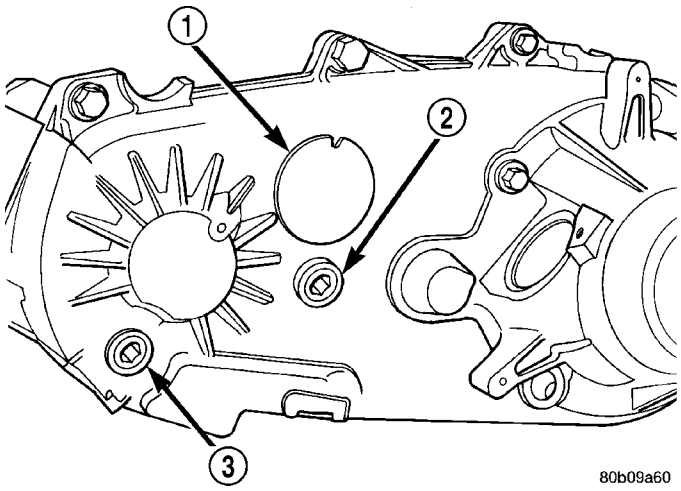

Fig. 1 Fill/Drain Plug And I.D. Tag Locations - Typical

1 - I.D. TAG
2 - FILL PLUG
3 - DRAIN PLUG

DIAGNOSIS AND TESTING - TRANSFER CASE - NV231

DIAGNOSIS CHART

Condition	Possible Cause	Correction
Transfer case difficult to shift or will not shift into desired range.	1) Vehicle speed to great to permit shifting.	1) Slow vehicle and shift into desired range.
	2) If vehicle was operated for an extended period in 4H mode on dry surface, driveline torque load may cause difficulty.	2) Stop vehicle and shift transfer case to Neutral position. Transfer case can then be shifted to the desired mode.
	3) Transfer case shift linkage binding.	3) Repair or replace linkage as necessary.
	4) Insufficient or incorrect lubricant.	4) Drain and refill transfer case with the correct type and quantity of lubricant.
	5) Internal transfer case components binding, worn, or damaged.	5) Repair or replace components as necessary.
Transfer case noisy in all drive modes.	1) Insufficient or incorrect lubricant.	1) Drain and refill transfer case with the correct type and quantity of lubricant.

TRANSFER CASE - NV231 (Continued)

Condition	Possible Cause	Correction
Transfer case noisy while in, or jumps out of, 4L mode.	1) Transfer case not completely engaged in 4L position.	1) Slow vehicle, shift transfer case to the Neutral position, and then shift into the 4L mode.
	2) Transfer case shift linkage out of adjustment.	2) Adjust linkage as necessary.
	3) Transfer case shift linkage loose or binding.	3) Repair, replace, or tighten linkage components as necessary.
	4) Range fork damaged, inserts worn, or fork is binding on the shift rail.	4) Repair or replace components as necessary.
	5) Low range gear worn or damaged.	5) Repair or replace components as necessary.
Lubricant leaking from transfer case seals or vent.	1) Transfer case overfilled.	1) Drain lubricant to the correct level.
	2) Transfer case vent closed or restricted.	2) Clean or replace vent as necessary.
	3) Transfer case seals damaged or installed incorrectly.	3) Replace suspect seal.
Abnormal tire wear.	1) Extended operation in 4H mode on dry surfaces,	1) Operate vehicle in 2H mode on dry surfaces.

REMOVAL

(1) Shift transfer case into NEUTRAL.
(2) Raise vehicle.
(3) Drain transfer case lubricant.
(4) Mark front and rear propeller shaft yokes for alignment reference.
(5) Support transmission with jack stand.
(6) Remove rear crossmember, or skid plate.
(7) Disconnect front/rear propeller shafts at transfer case.
(8) Disconnect vehicle speed sensor wires.
(9) Disconnect transfer case linkage rod from range lever.
(10) Disconnect transfer case vent hose (Fig. 2) and indicator switch harness, if necessary.
(11) Support transfer case with transmission jack.
(12) Secure transfer case to jack with chains.
(13) Remove nuts attaching transfer case to transmission.
(14) Pull transfer case and jack rearward to disengage transfer case.
(15) Remove transfer case from under vehicle.

DISASSEMBLY

Position transfer case on shallow drain pan. Remove drain plug and drain lubricant remaining in case.

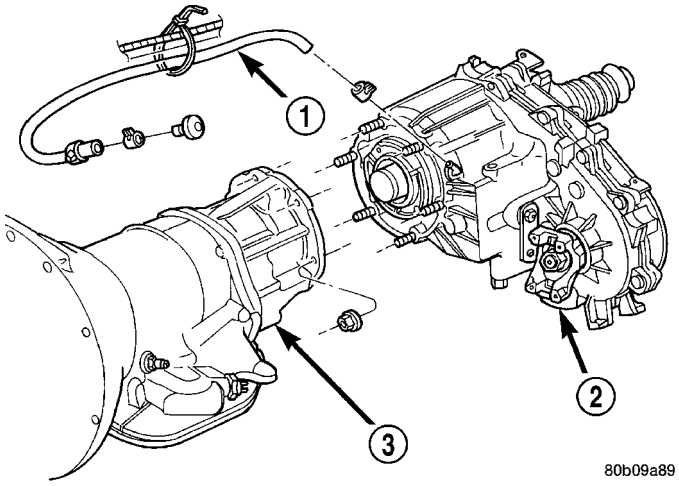

Fig. 2 Transfer Case Mounting

1 - VENT TUBE
2 - TRANSFER CASE
3 - TRANSMISSION

REAR RETAINER AND OIL PUMP

(1) Remove the speedometer adapter.
(2) Spread band clamp which holds output shaft boot to the output shaft slinger, or output shaft damper, with a suitable awl, or equivalent.

TRANSFER CASE - NV231 (Continued)

NOTE: Vehicles built with a 4.0L engine and a manual transmission use a damper weight on the transfer case output shaft. Be sure to identify the transfer case before proceeding.

(3) Remove output shaft boot from slinger, or output shaft damper, and output shaft.

(4) If the vehicle is not equipped with an output shaft damper, remove the output shaft rear slinger using Puller MD-998056-A (Fig. 3).

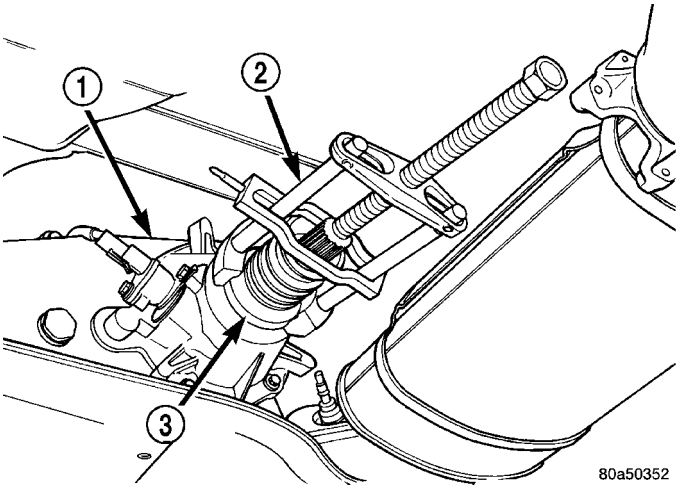

Fig. 3 Rear Slinger Removal

1 - TRANSFER CASE
2 - SPECIAL TOOL MD-998056-A
3 - SLINGER

(5) If the vehicle is equipped with an output shaft damper, use Screws 8421 and the puller yoke and forcing screw from a bolt-grip puller set, such as those used to remove steering wheels and harmonic balancers, to remove the transfer case output shaft damper.

(6) Use a suitable pry tool, or a slide hammer mounted screw, to remove the seal from the rear retainer (Fig. 4).

(7) Remove the rear output bearing I.D. retaining ring (Fig. 5).

(8) Remove the bolts holding the rear retainer to the rear case half.

(9) Tap rear retainer with rawhide or rubber mallet to loosen sealer bead.

(10) Remove rear retainer from rear case half (Fig. 6).

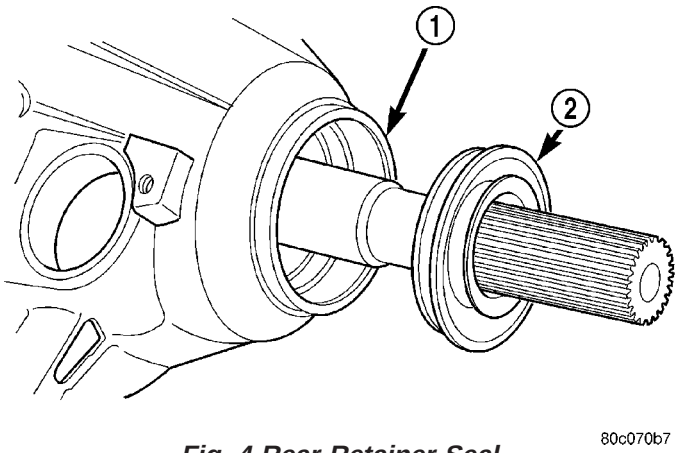

Fig. 4 Rear Retainer Seal

1 - REAR RETAINER
2 - OUTPUT SHAFT SEAL

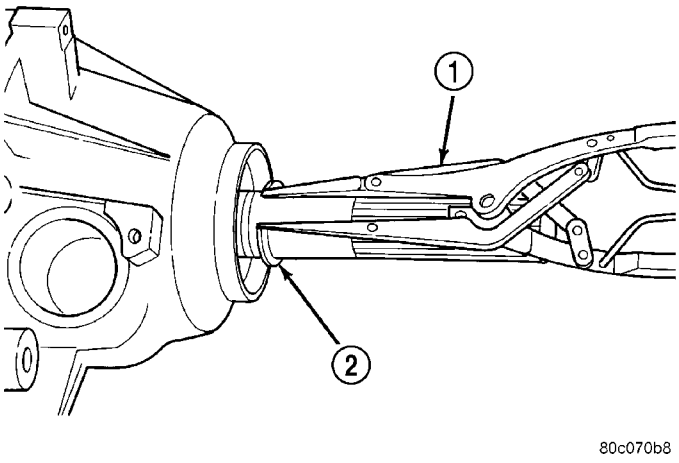

Fig. 5 Output Shaft Rear Bearing Retaining Ring

1 - SNAP-RING PLIERS
2 - REAR BEARING I.D. RETAINING RING

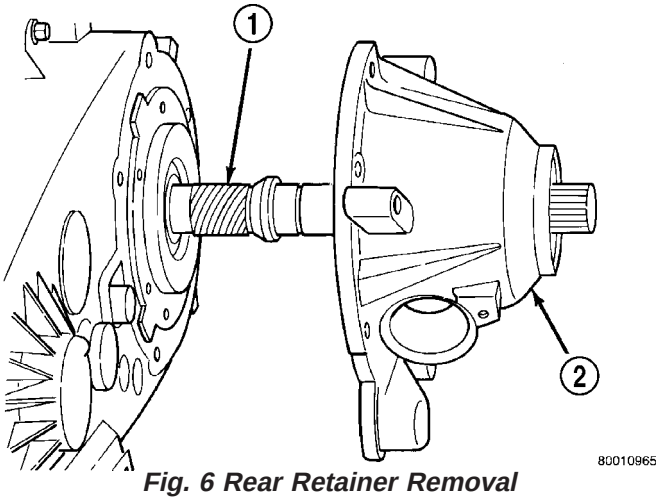

Fig. 6 Rear Retainer Removal

1 - MAINSHAFT
2 - REAR RETAINER

TRANSFER CASE - NV231 (Continued)

(11) Remove snap-ring holding oil pump in position on output shaft.

(12) Disengage oil pickup tube from oil pump and remove oil pump assembly. Remove oil pump by tilting the edge of the oil pump from under the edge of the rear case half and sliding the pump (Fig. 7).

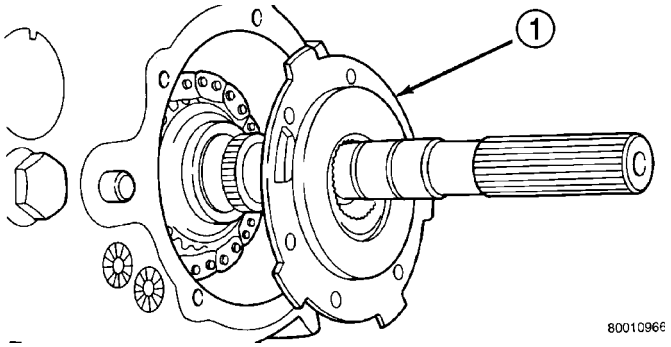

Fig. 7 Oil Pump Removal

1 - OIL PUMP

(13) Remove pick-up tube o-ring from oil pump (Fig. 8), if necessary. Do not disassemble the oil pump, it is not serviceable.

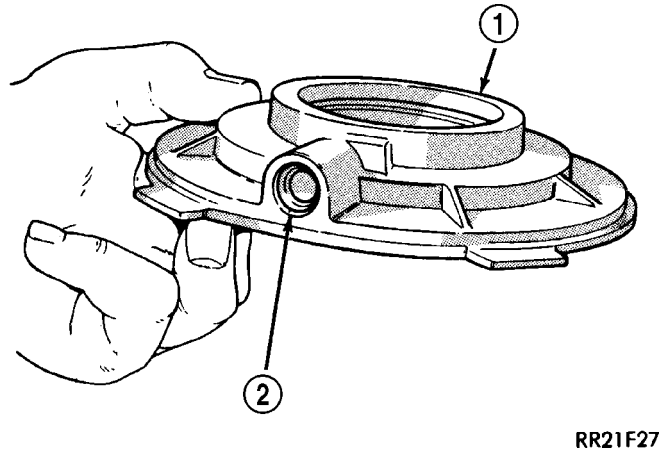

Fig. 8 Pick-up Tube O-ring Location

1 - OIL PUMP
2 - O-RING

YOKE AND RANGE LEVER

(1) Remove transfer case indicator switch.

(2) Remove front yoke nut as follows:

(a) Move range lever to 4L position.

(b) Then remove nut with socket and impact wrench (Fig. 9).

(3) Remove yoke. If yoke is difficult to remove by hand, remove it with bearing splitter, or with standard two jaw puller (Fig. 10). Be sure puller tool is positioned on yoke and not on slinger as slinger will be damaged.

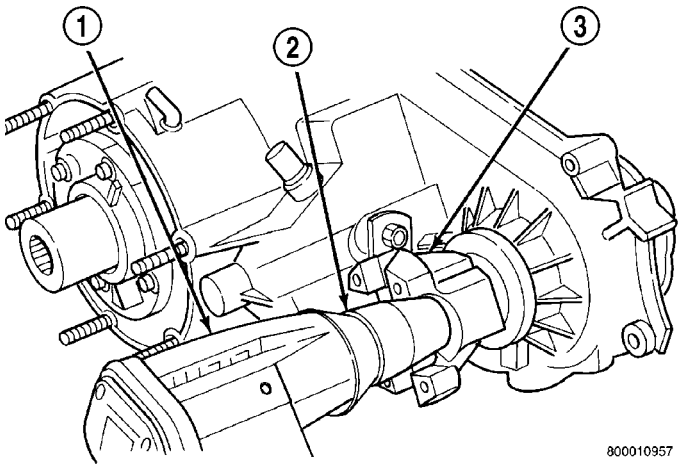

Fig. 9 Yoke Nut Removal

1 - IMPACT WRENCH
2 - SOCKET
3 - YOKE

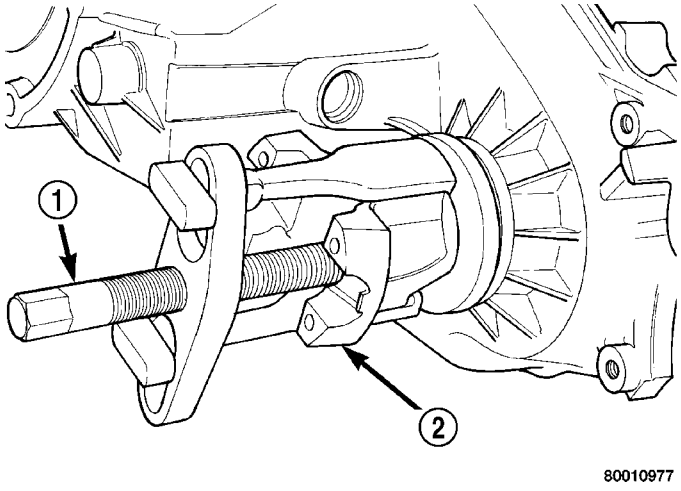

Fig. 10 Yoke Removal

1 - PULLER TOOL
2 - YOKE

(4) Remove seal washer from front output shaft. Discard washer as it should not be reused.

TRANSFER CASE - NV231 (Continued)

(5) Remove nut and washer that attach range lever to sector shaft. Then move sector to neutral position and remove range lever from shaft (Fig. 11).

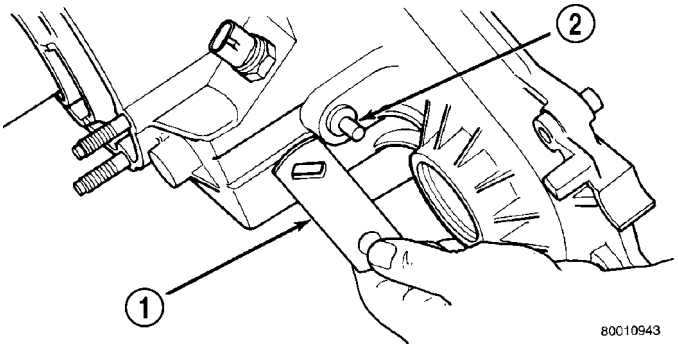

Fig. 11 Range Lever Removal

1 - RANGE LEVER
2 - SECTOR SHAFT

FRONT OUTPUT SHAFT AND DRIVE CHAIN

(1) Support transfer case so rear case is facing upward.

(2) Remove bolts holding front case to rear case. The case alignment bolts require flat washers (Fig. 12).

(3) Loosen rear case with flat blade screwdriver to break sealer bead. Insert pry tool blade only into notches provided at each end of case (Fig. 13).

(4) Remove rear case from front case.

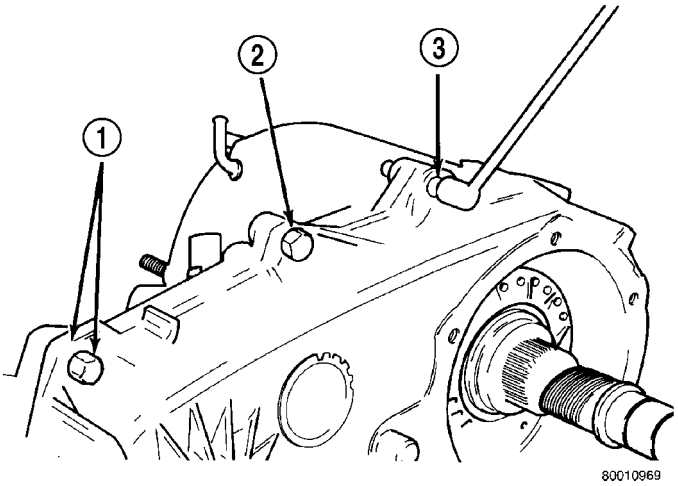

Fig. 12 Rear Case Alignment Bolt Locations

1 - DOWEL BOLT AND WASHER (2)
2 - CASE BOLT (5)
3 - SPLINE HEAD BOLT (1)

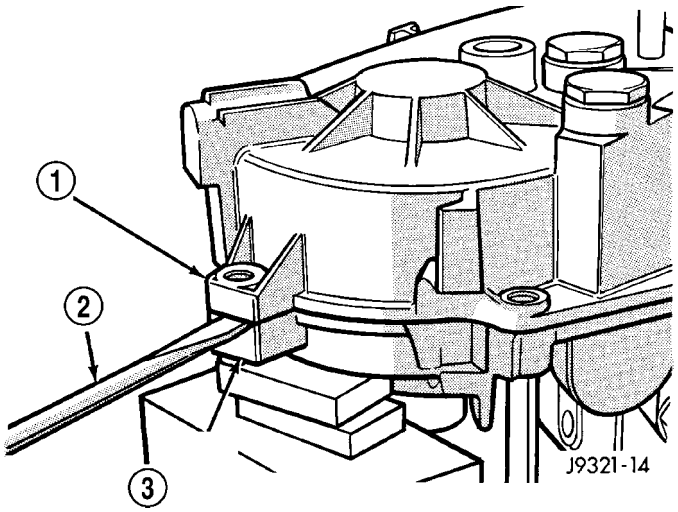

Fig. 13 Loosening Rear Case - Typical

1 - REAR CASE
2 - PRY TOOL (IN CASE SLOT)
3 - FRONT CASE

(5) Remove oil pickup tube from rear case (Fig. 14).

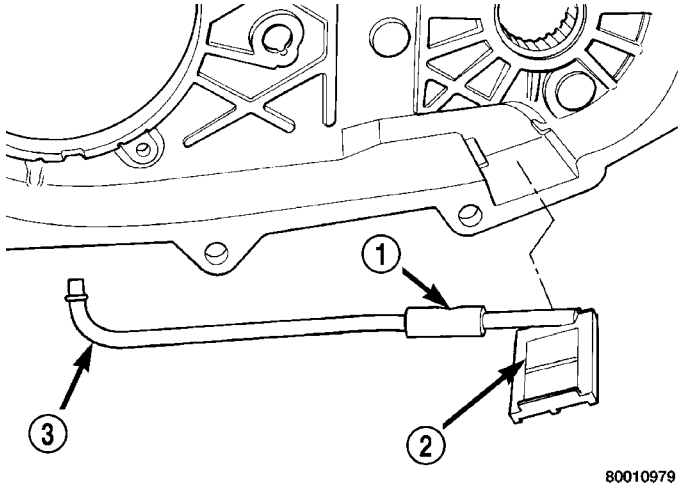

Fig. 14 Oil Pickup Tube Removal

1 - CONNECTING HOSE
2 - PICKUP SCREEN
3 - PICKUP TUBE

TRANSFER CASE - NV231 (Continued)

(6) Remove mode fork spring (Fig. 15).

(7) Pull front output shaft upward and out of front output shaft bearing (Fig. 16).

(8) Remove front output shaft and chain.

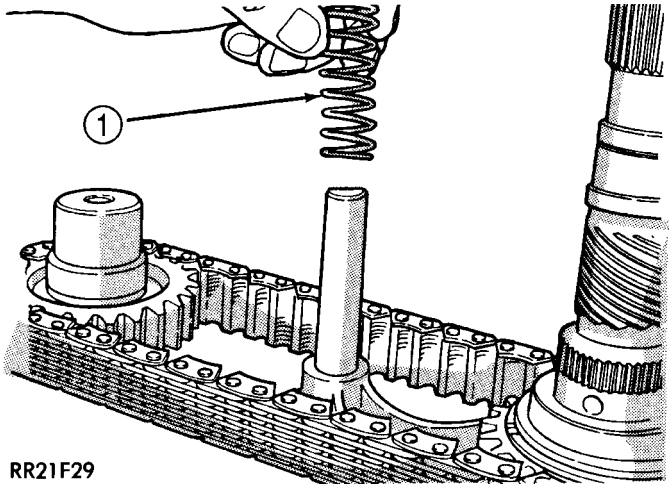

Fig. 15 Mode Fork Spring Removal

1 - MODE SPRING

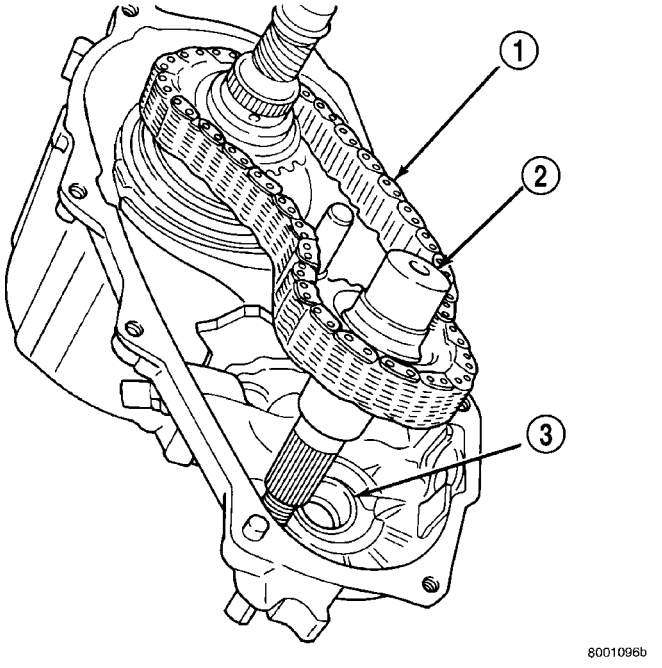

Fig. 16 Remove Front Output Shaft And Chain

1 - DRIVE CHAIN
2 - FRONT OUTPUT SHAFT
3 - SHAFT FRONT BEARING

SHIFT FORKS AND MAINSHAFT

(1) Remove detent plug, O-ring, detent spring and detent plunger (Fig. 17).

(2) Remove mainshaft from mode sleeve and input gear pilot bearing.

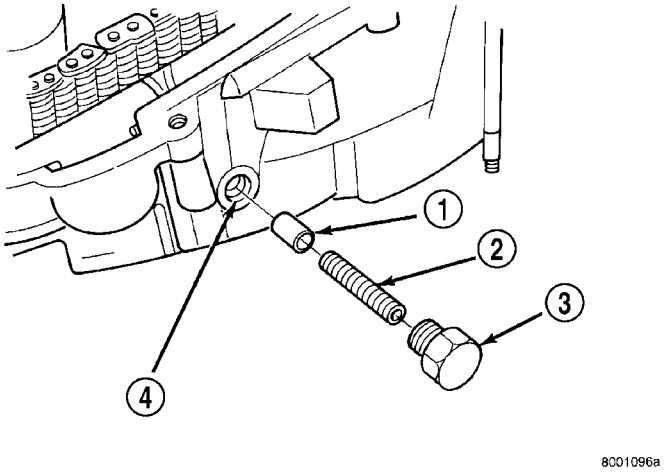

Fig. 17 Detent Plug, Spring And Plunger Removal

1 - POPPET
2 - SPRING
3 - SCREW
4 - POPPET BORE (IN CASE)

(3) Remove mode fork and sleeve as an assembly (Fig. 18). Note position of sleeve for assembly reference. The short side of the sleeve faces upward.

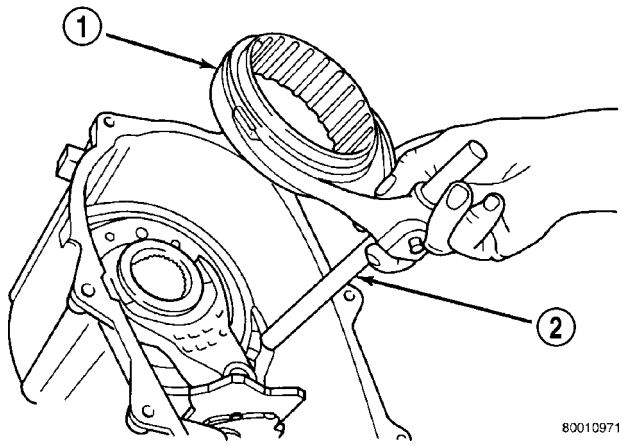

Fig. 18 Mode Fork And Sleeve Removal

1 - MODE SLEEVE
2 - MODE FORK AND RAIL

TRANSFER CASE - NV231 (Continued)

(4) Remove range fork and hub as an assembly (Fig. 19). Note fork position for installation reference.

(5) Remove shift sector from front case (Fig. 20).

(6) Remove shift sector bushing and O-ring (Fig. 21).

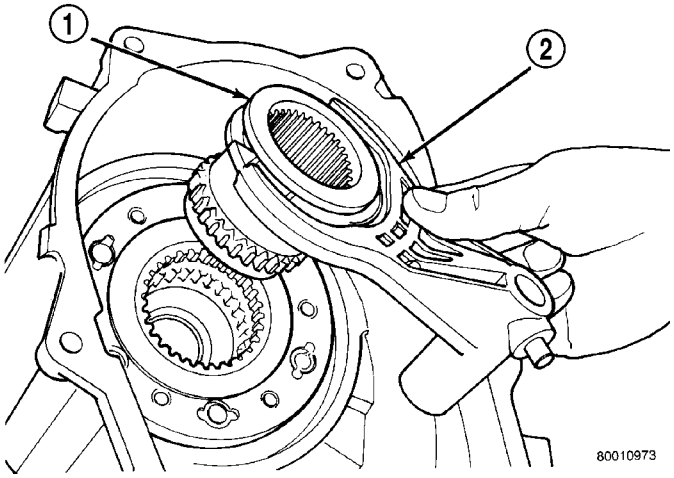

Fig. 19 Range Fork And Sleeve Removal

1 - RANGE HUB
2 - RANGE FORK

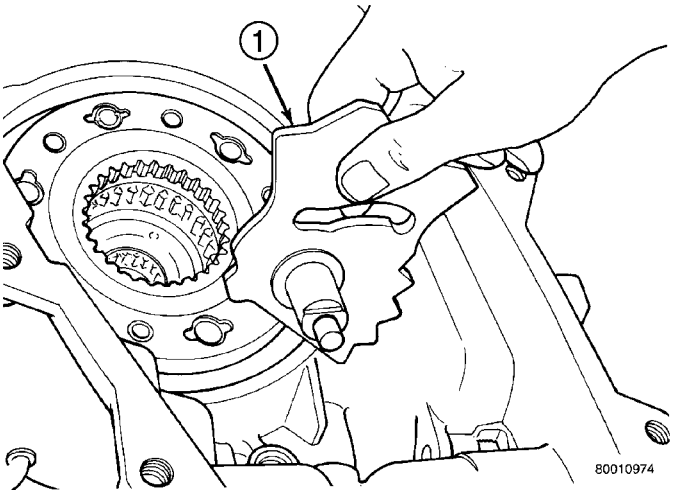

Fig. 20 Shift Sector Removal

1 - SHIFT SECTOR

MAINSHAFT

(1) Remove mode hub retaining ring with heavy duty snap-ring pliers (Fig. 22).

(2) Slide mode hub off mainshaft (Fig. 23).

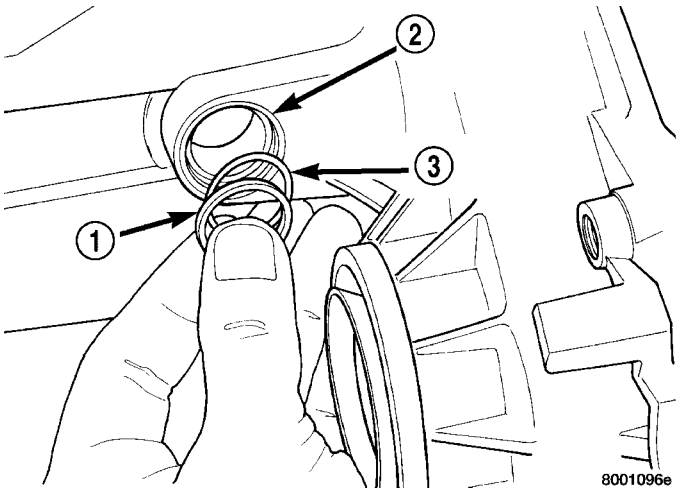

Fig. 21 Sector Bushing And O-Ring Removal

1 - SEAL RETAINER
2 - SECTOR SHAFT BORE
3 - O-RING SEAL

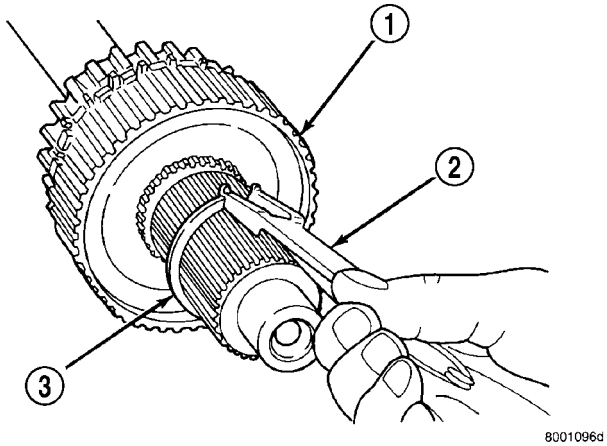

Fig. 22 Mode Hub Retaining Ring Removal

1 - MODE HUB
2 - SNAP-RING PLIERS (HEAVY DUTY)
3 - MODE HUB RETAINING RING

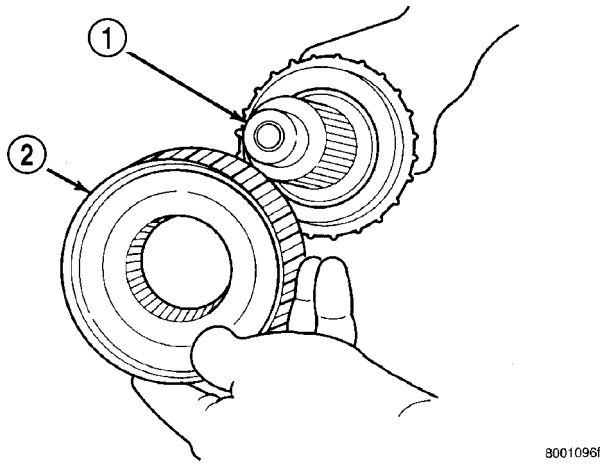

Fig. 23 Mode Hub Removal

1 - MAINSHAFT
2 - MODE HUB

TRANSFER CASE - NV231 (Continued)

(3) Slide drive sprocket off mainshaft (Fig. 24).

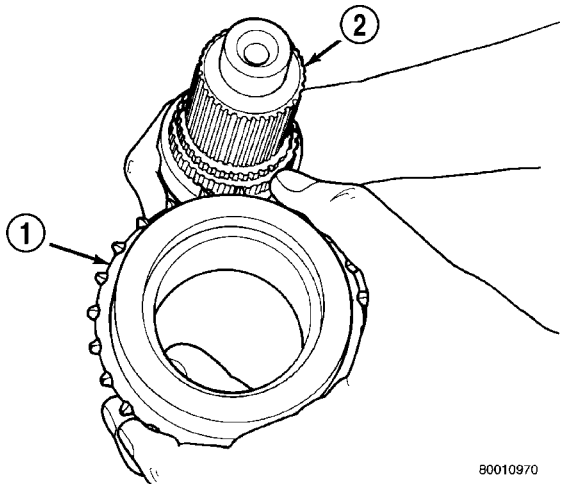

Fig. 24 Drive Sprocket Removal

1 - DRIVE SPROCKET
2 - MAINSHAFT

INPUT GEAR AND LOW RANGE GEAR

(1) Remove front bearing retainer attaching bolts (Fig. 25).

(2) Remove front bearing retainer. Pry retainer loose with pry tool positioned in slots at each end of retainer (Fig. 26).

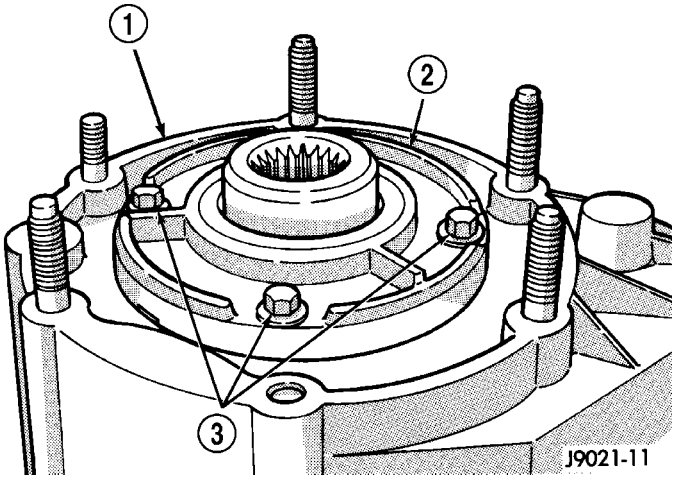

Fig. 25 Front Bearing Retainer Bolts

1 - FRONT CASE
2 - FRONT BEARING RETAINER
3 - RETAINER BOLTS

(3) Remove front bearing retainer seal. Tap seal out with drift and hammer.

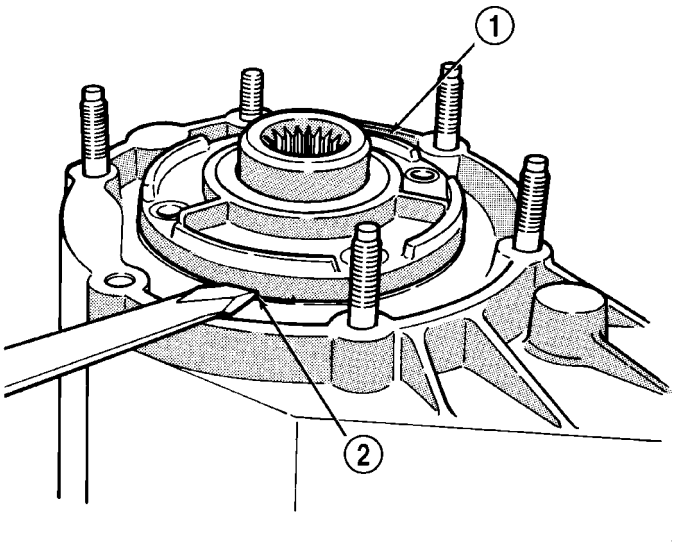

Fig. 26 Front Bearing Retainer Removal

1 - FRONT BEARING RETAINER
2 - RETAINER SLOT

(4) Remove input gear retaining ring with heavy duty snap-ring pliers (Fig. 27)

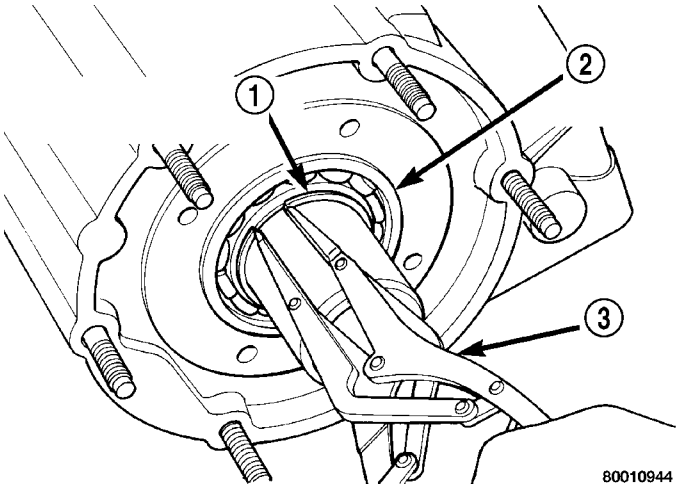

Fig. 27 Removing Input Gear Retaining Ring

1 - INPUT GEAR BEARING RETAINING RING
2 - INPUT GEAR BEARING
3 - SNAP-RING PLIERS

TRANSFER CASE - NV231 (Continued)

(5) Place front case in horizontal position. Then remove input gear and low range gear as an assembly (Fig. 28). Tap gear out of bearing with plastic mallet if necessary.

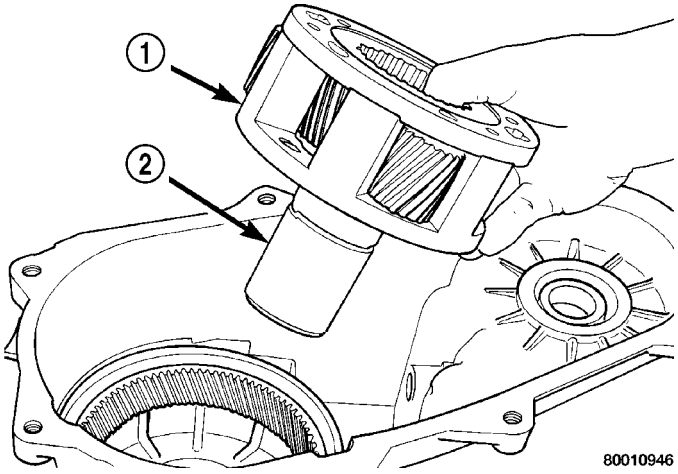

Fig. 28 Input Gear And Planetary Carrier Removal

1 - PLANETARY ASSEMBLY
2 - INPUT GEAR

INPUT AND LOW RANGE GEAR

(1) Remove snap-ring that retains input gear in low range gear (Fig. 29).
(2) Remove retainer (Fig. 30).
(3) Remove front tabbed thrust washer (Fig. 31).
(4) Remove input gear (Fig. 32).

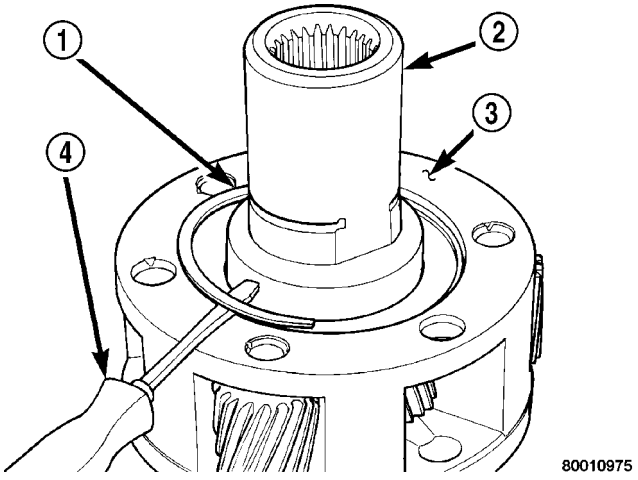

Fig. 29 Input Gear Snap-Ring Removal

1 - CARRIER LOCK RETAINING RING
2 - INPUT GEAR
3 - PLANETARY CARRIER
4 - SCREWDRIVER

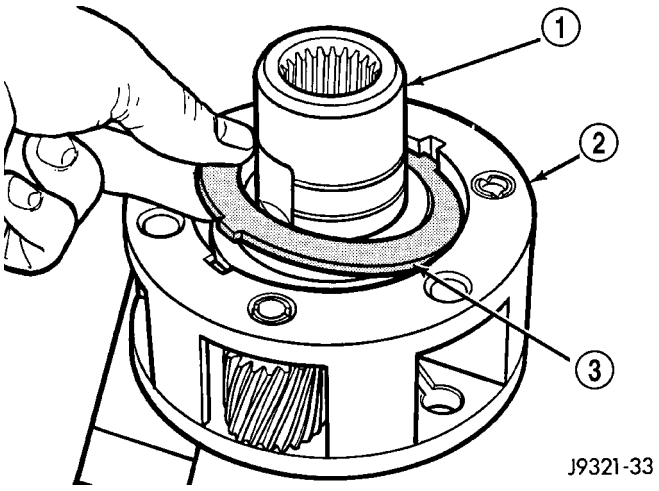

Fig. 30 Input Gear Retainer Removal

1 - INPUT GEAR
2 - LOW RANGE GEAR
3 - RETAINER

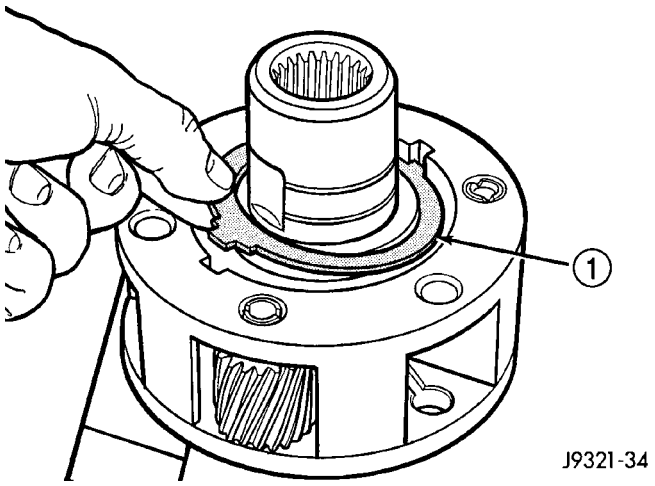

Fig. 31 Front Tabbed Thrust Washer Removal

1 - FRONT TABBED THRUST WASHER

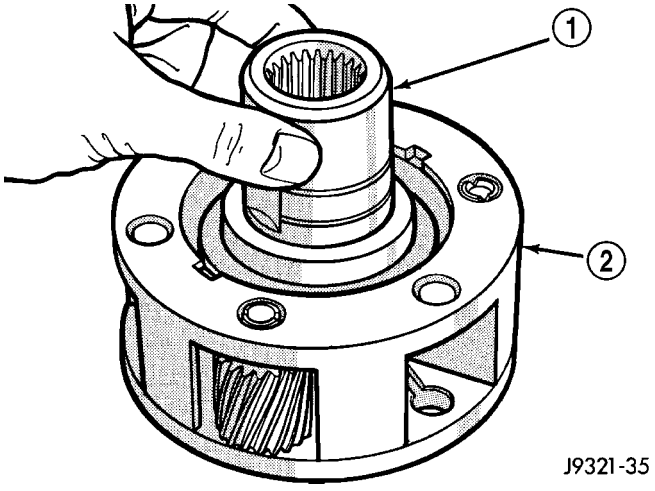

Fig. 32 Input Gear Removal

1 - INPUT GEAR
2 - LOW RANGE GEAR

(5) Remove rear tabbed thrust washer from low range gear (Fig. 33).

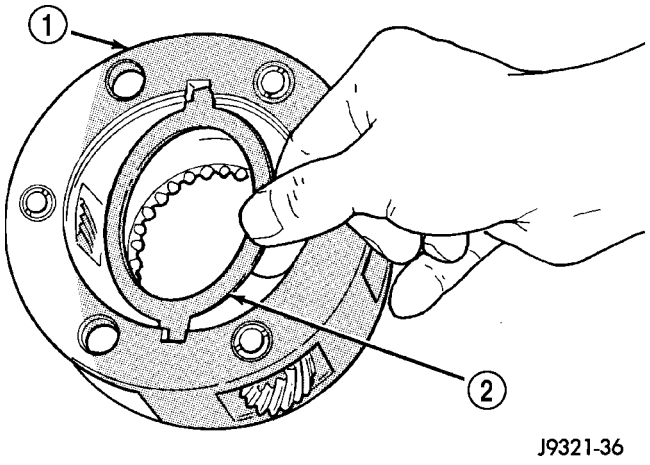

Fig. 33 Rear Tabbed Thrust Washer Removal

1 - LOW RANGE GEAR
2 - REAR TABBED THRUST WASHER

CLEANING

Clean the transfer case parts with a standard parts cleaning solvent. Remove all traces of sealer from the cases and retainers with a scraper and 3M™ all purpose cleaner. Use compressed air to remove solvent residue from oil feed passages in the case halves, retainers, gears, and shafts.

INSPECTION

MAINSHAFT/SPROCKET/HUB

Inspect the splines on the hub and shaft and the teeth on the sprocket (Fig. 34). Minor nicks and scratches can be smoothed with an oilstone. However, replace any part that is damaged.

Check the contact surfaces in the sprocket bore and on the mainshaft. Minor nicks and scratches can be smoothed with 320-400 grit emery cloth but do not try to salvage the shaft if nicks or wear is severe.

INPUT GEAR AND PLANETARY CARRIER

Check the teeth on the gear (Fig. 35). Minor nicks can be dressed off with an oilstone but replace the gear if any teeth are broken, cracked, or chipped. The bearing surface on the gear can be smoothed with 300-400 grit emery cloth if necessary.

Examine the carrier body and pinion gears for wear or damage. The carrier will have to be replaced as an assembly if the body, pinion pins, or pinion gears are damaged.

Check the lock ring and both thrust washers for wear or cracks. Replace them if necessary. Also replace the lock retaining ring if bent, distorted, or broken.

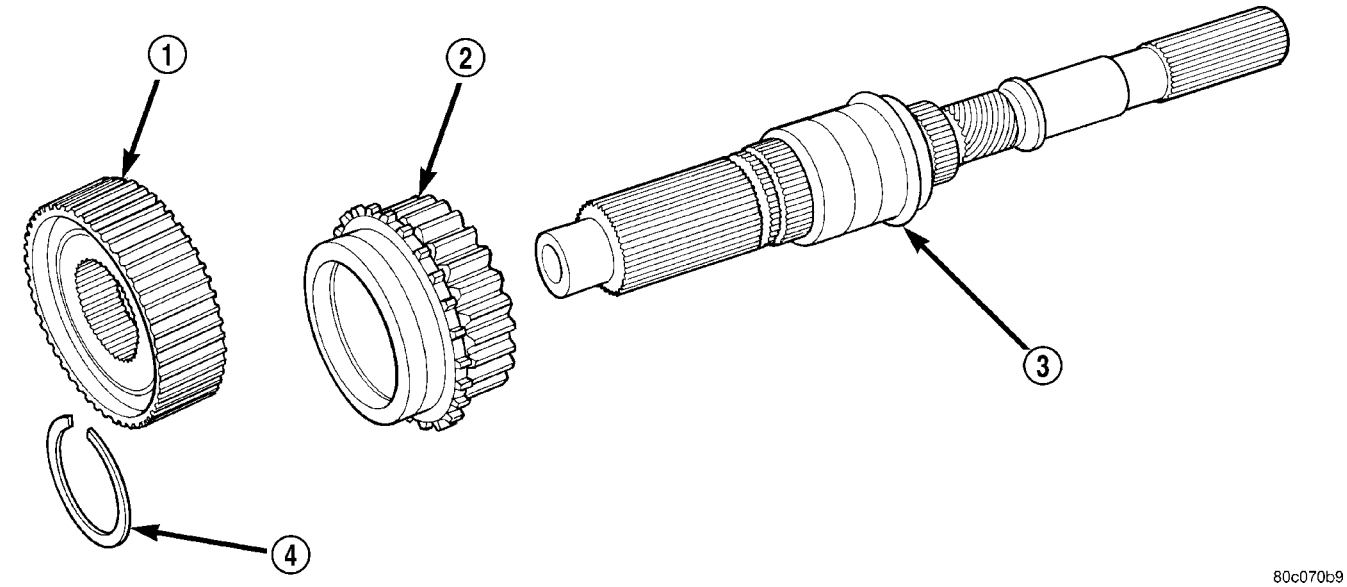

Fig. 34 Mainshaft, Mode Hub, And Drive Sprocket

1 - MODE HUB
2 - DRIVE SPROCKET
3 - MAINSHAFT
4 - MODE HUB RETAINING RING

TRANSFER CASE - NV231 (Continued)

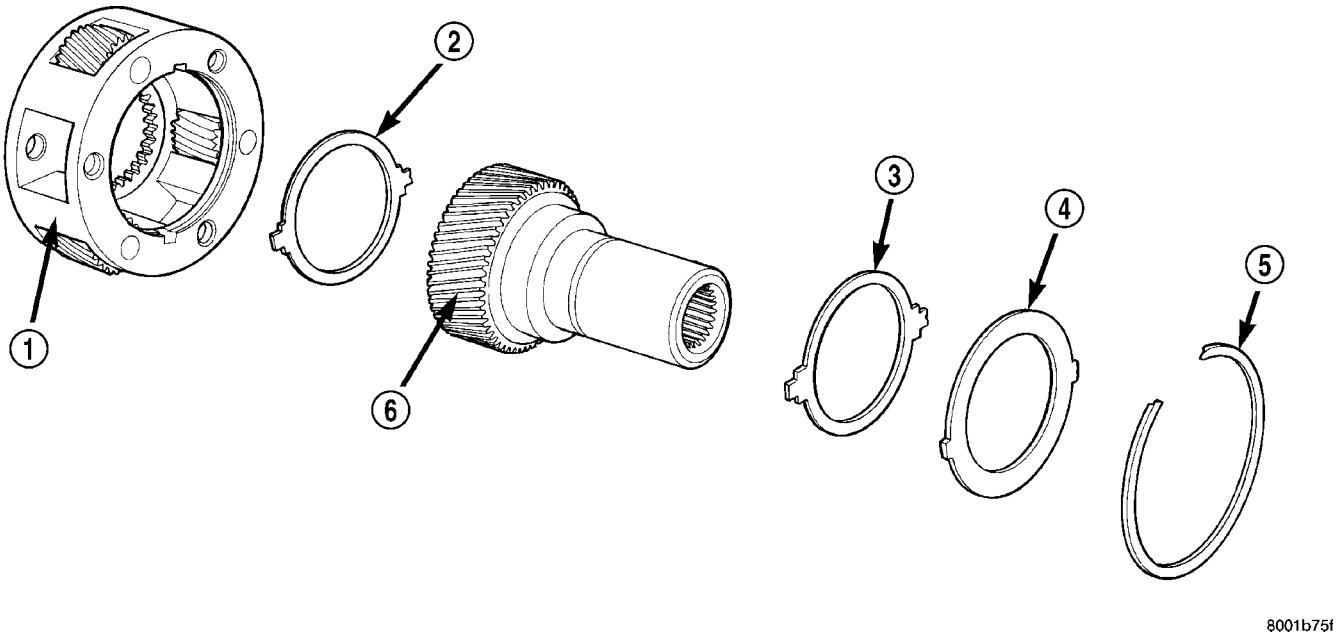

Fig. 35 Input Gear And Carrier Components

1 - PLANETARY CARRIER
2 - REAR THRUST WASHER
3 - FRONT THRUST WASHER
4 - CARRIER LOCK RING
5 - CARRIER LOCK RETAINING RING
6 - INPUT GEAR

SHIFT FORKS/HUBS/SLEEVES

Check condition of the shift forks and mode fork shift rail (Fig. 36). Minor nicks on the shift rail can be smoothed with 320-400 grit emery cloth.

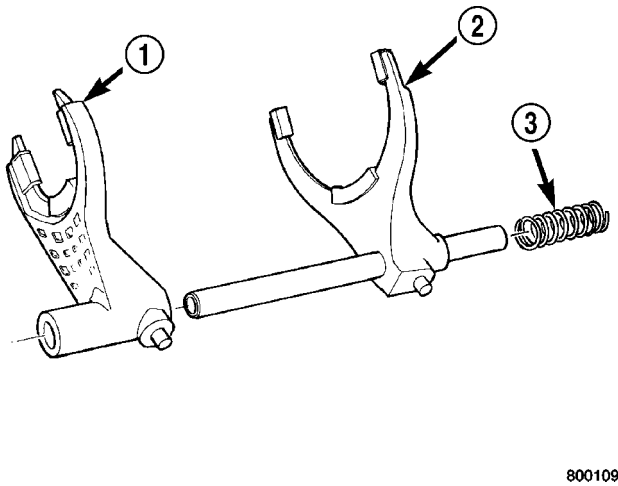

Fig. 36 Shift Forks

1 - RANGE FORK
2 - MODE FORK AND RAIL
3 - MODE SPRING

Inspect the shift fork wear pads (Fig. 37). The mode fork pads are serviceable and can be replaced if necessary. The range fork pads are not serviceable. The fork must be replaced as an assembly if the pads are worn or damaged.

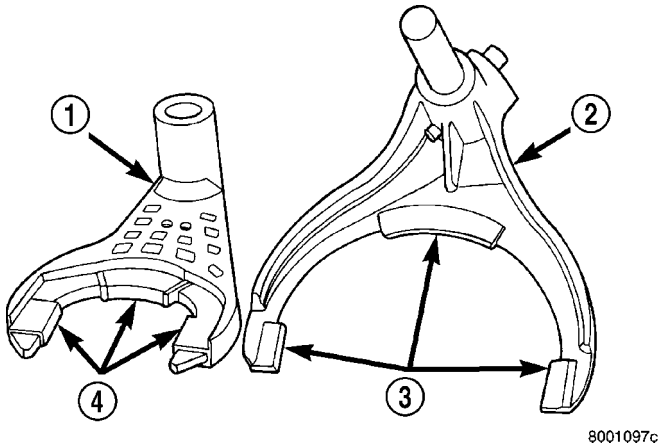

Fig. 37 Shift Fork And Wear Pad Locations

1 - RANGE FORK
2 - MODE FORK
3 - WEAR PADS (SERVICEABLE)
4 - WEAR PADS (NON-SERVICEABLE)

Check both of the sleeves for wear or damage, especially on the interior teeth. Replace the sleeves if wear or damage is evident.

TRANSFER CASE - NV231 (Continued)

REAR RETAINER/BEARING/ SEAL/SLINGER/BOOT

Inspect the retainer components (Fig. 38). Replace the bearing if rough or noisy. Check the retainer for cracks or wear in the bearing bore. Clean the retainer sealing surfaces with a scraper and 3M™ all purpose cleaner. This will ensure proper adhesion of the sealer during reassembly.

An output shaft slinger is used on some vehicles, while an output shaft damper is used on other vehicles. The output shaft slinger and seal should be replaced outright; do not reuse either part.

Replace any part if distorted, bent, or broken. Also replace the boot if cut or torn. Replace the boot band clamps, do not reuse them.

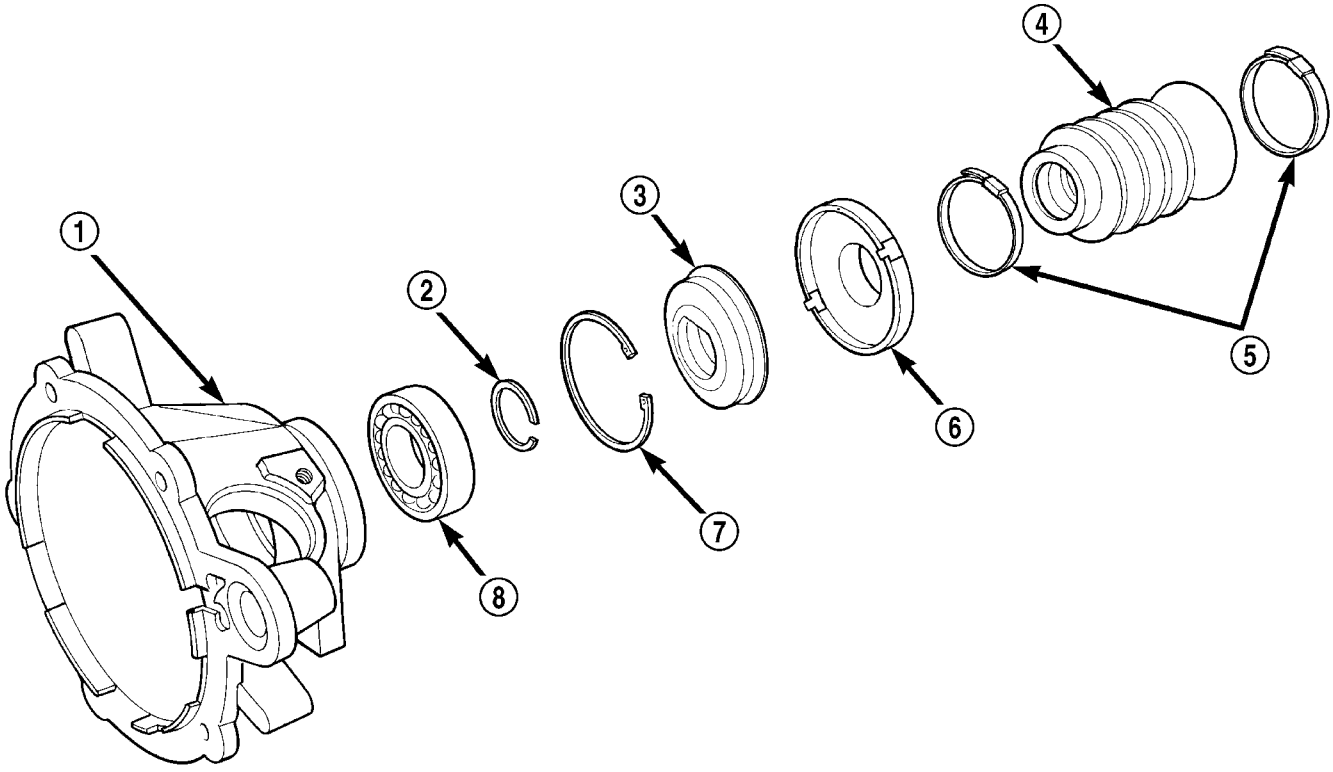

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Fig. 38 Rear Retainer Without Output Shaft Damper

1 - REAR RETAINER
2 - REAR BEARING I.D. MAINSHAFT RETAINING RING
3 - REAR SEAL
4 - BOOT
5 - BAND CLAMPS
6 - REAR SLINGER
7 - REAR BEARING O.D. RETAINING RING
8 - REAR BEARING

TRANSFER CASE - NV231 (Continued)

REAR OUTPUT SHAFT/YOKE/DRIVE CHAIN

Check condition of the seal contact surfaces of the yoke slinger (Fig. 39). This surface must be clean and smooth to ensure proper seal life. Replace the yoke nut and seal washer as neither part should be reused.

Inspect the shaft threads, sprocket teeth, and bearing surfaces. Minor nicks on the teeth can be smoothed with an oilstone. Use 320-400 grit emery to smooth minor scratches on the shaft bearing surfaces. Rough threads on the shaft can be chased if necessary. Replace the shaft if the threads are damaged, bearing surfaces are scored, or if any sprocket teeth are cracked or broken.

Examine the drive chain and shaft bearings. Replace the chain and both sprockets if the chain is stretched, distorted, or if any of the links bind. Replace the bearings if rough, or noisy.

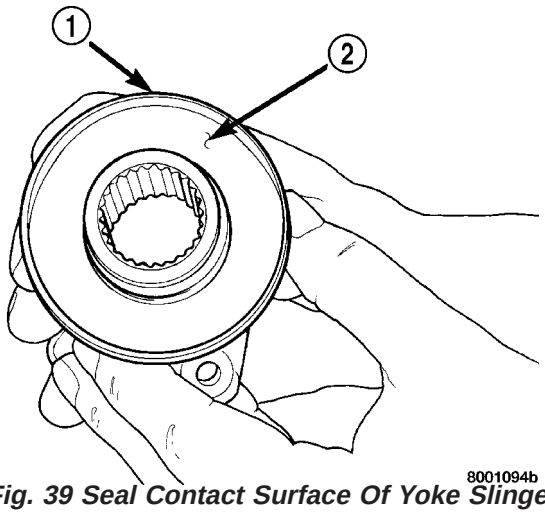


Fig. 39 Seal Contact Surface Of Yoke Slinger

1 - FRONT SLINGER (PART OF YOKE)
2 - SEAL CONTACT SURFACE MUST BE CLEAN AND SMOOTH

LOW RANGE ANNULUS GEAR

Inspect annulus gear condition carefully. The gear is only serviced as part of the front case. If the gear is damaged, it will be necessary to replace the gear and front case as an assembly. Do not attempt to remove the gear (Fig. 40)

FRONT/REAR CASES AND FRONT RETAINER

Inspect the cases and retainer for wear and damage. Clean the sealing surfaces with a scraper and

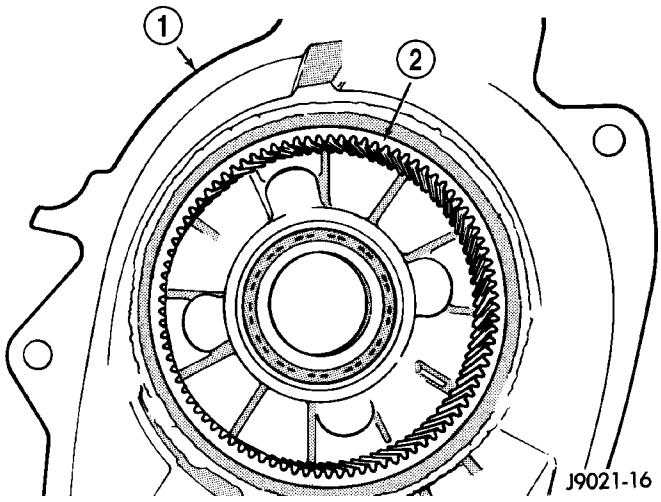


Fig. 40 Low Range Annulus Gear

1 - FRONT CASE
2 - LOW RANGE ANNULUS GEAR

3M™ all purpose cleaner. This will ensure proper sealer adhesion at assembly. Replace the input retainer seal; do not reuse it.

Check case condition. If leaks were a problem, look for gouges and severe scoring of case sealing surfaces. Also make sure the front case mounting studs are in good condition.

Check the front case mounting studs and vent tube. The tube can be secured with Loctite™ 271 or 680 if loose. The stud threads can be cleaned up with a die if necessary. Also check condition of the fill/drain plug threads in the rear case. The threads can be repaired with a thread chaser or tap if necessary. Or the threads can be repaired with Helicoil™ stainless steel inserts if required.

OIL PUMP/OIL PICKUP

Examine the oil pump pickup parts. Replace the pump if any part appears to be worn or damaged. Do not disassemble the pump as individual parts are not available. The pump is only available as a complete assembly. The pickup screen, hose, and tube are the only serviceable parts and are available separately.

TRANSFER CASE - NV231 (Continued)

ASSEMBLY

Lubricate transfer case components with Mopar® ATF +4, Automatic Transmission Fluid, or petroleum jelly (where indicated) during assembly.

BEARINGS AND SEALS

CAUTION: The bearing bores in various transfer case components contain oil feed holes. Make sure replacement bearings do not block the holes.

(1) Remove the front output shaft seal from case with pry tool (Fig. 41).

(2) Remove the front output shaft bearing retaining ring with screwdriver (Fig. 42).

(3) Remove bearing with Tool Handle C-4171 and Tool 5065 (Fig. 43).

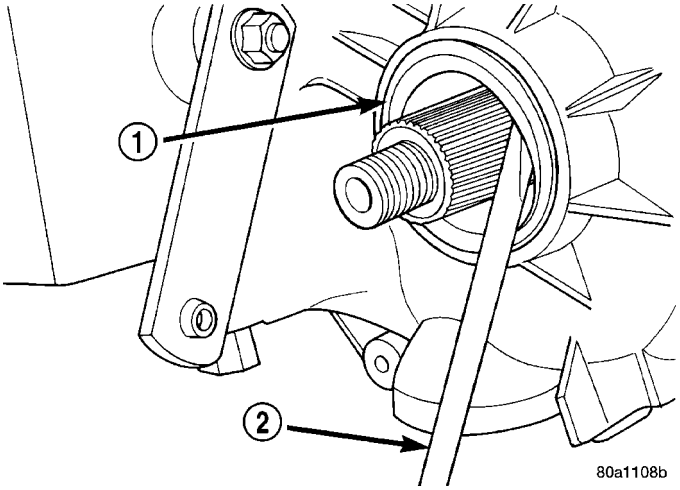


Fig. 41 Front Output Seal Removal - Typical

1 - OUTPUT SHAFT SEAL
2 - PRYBAR

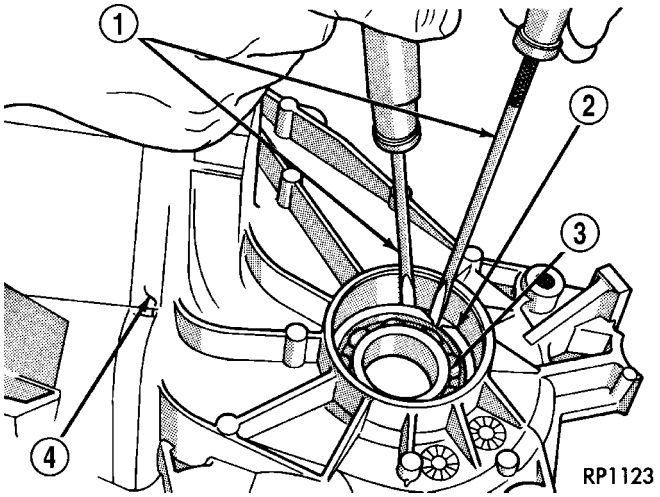


Fig. 42 Front Output Shaft Bearing Retaining Ring Removal

1 - SCREWDRIVERS
2 - SNAP-RING
3 - FRONT OUTPUT SHAFT BEARING
4 - FRONT CASE

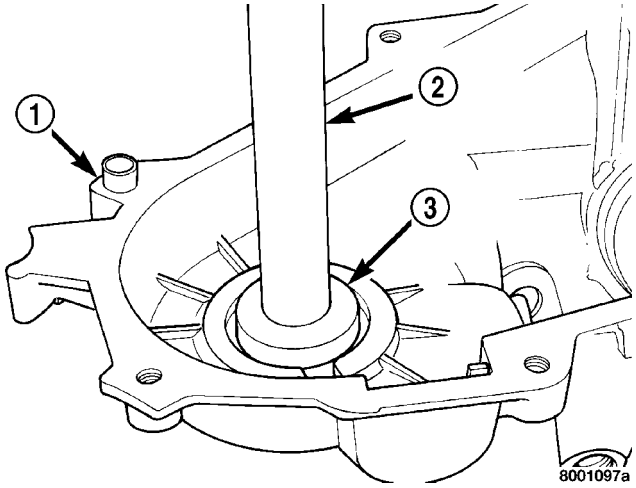


Fig. 43 Front Output Shaft Bearing Removal

1 - FRONT CASE
2 - SPECIAL TOOL C-4171
3 - SPECIAL TOOL 5065

TRANSFER CASE - NV231 (Continued)

(4) Install front output shaft front bearing in case with Tool Handle C-4171 and Installer 5064 (Fig. 44).

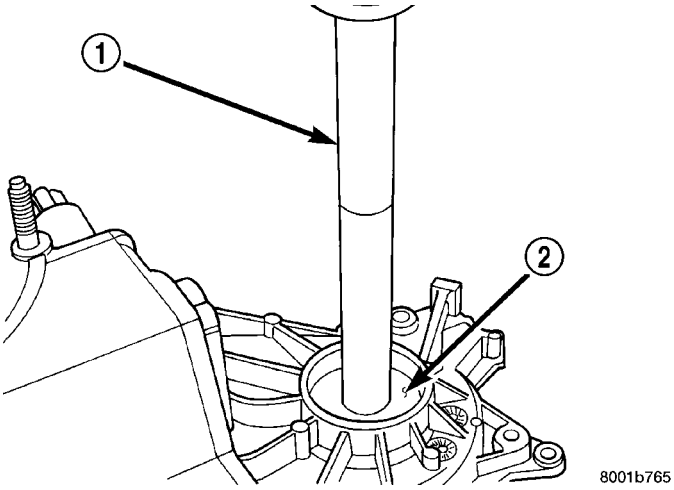


Fig. 44 Front Output Shaft Bearing Installation

1 - SPECIAL TOOL C-4171
2 - SPECIAL TOOL 5064

(5) Install output shaft front bearing retaining ring (Fig. 45). Start ring into place by hand. Then use small screwdriver to work ring into case groove. Be sure ring is fully seated before proceeding.

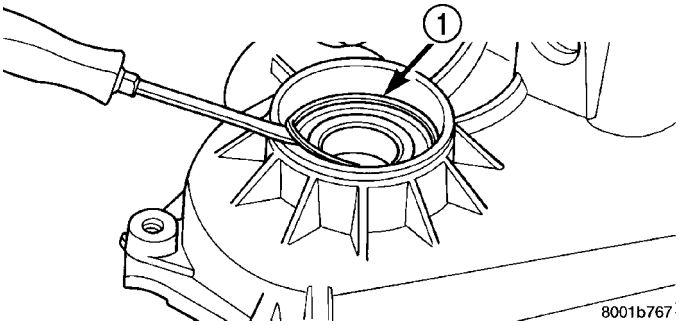


Fig. 45 Installing Output Shaft Front Bearing Retaining Ring

1 - WORK RETAINING RING INTO BORE GROOVE WITH SMALL SCREWDRIVER

(6) Install new front output seal in front case with Installer Tool 8143-A as follows:

(a) Place new seal on tool. **Garter spring on seal goes toward interior of case.**

(b) Start seal in bore with light taps from hammer (Fig. 46). Once seal is started, continue tapping seal into bore until installer tool bottoms against case.

(7) Remove the output shaft rear bearing with the screw and jaws from Remover L-4454 and Cup 8148 (Fig. 47).

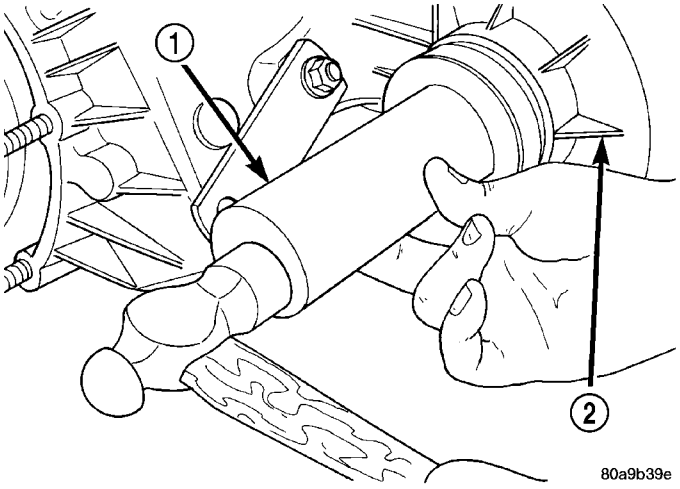


Fig. 46 Front Output Seal Installation - Typical

1 - INSTALLER 8143-A
2 - TRANSFER CASE

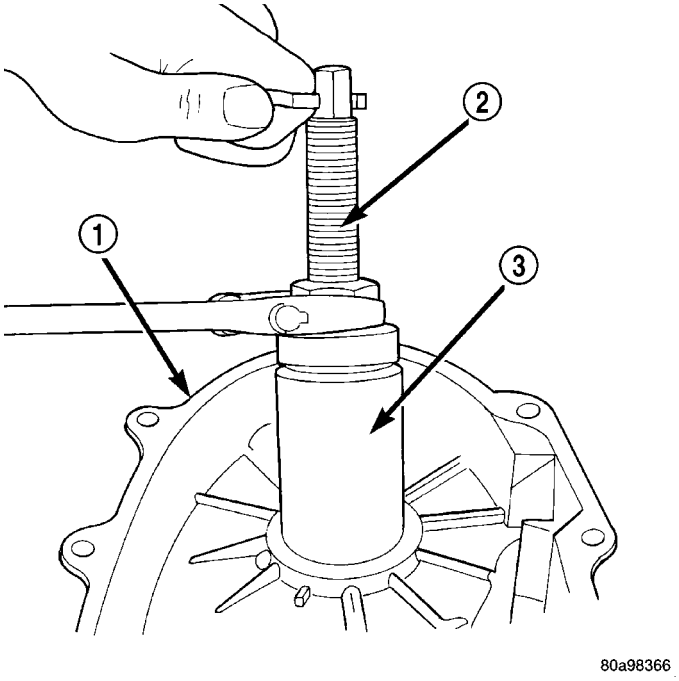


Fig. 47 Output Shaft Rear Bearing Removal

1 - REAR CASE
2 - SPECIAL TOOL L-4454-1 AND L-4454-3
3 - SPECIAL TOOL 8148

TRANSFER CASE - NV231 (Continued)

(8) Install new bearing with Tool Handle C-4171 and Installer 5066 (Fig. 48). The bearing bore is chamfered at the top. Install the bearing so it is flush with the lower edge of this chamfer (Fig. 49).

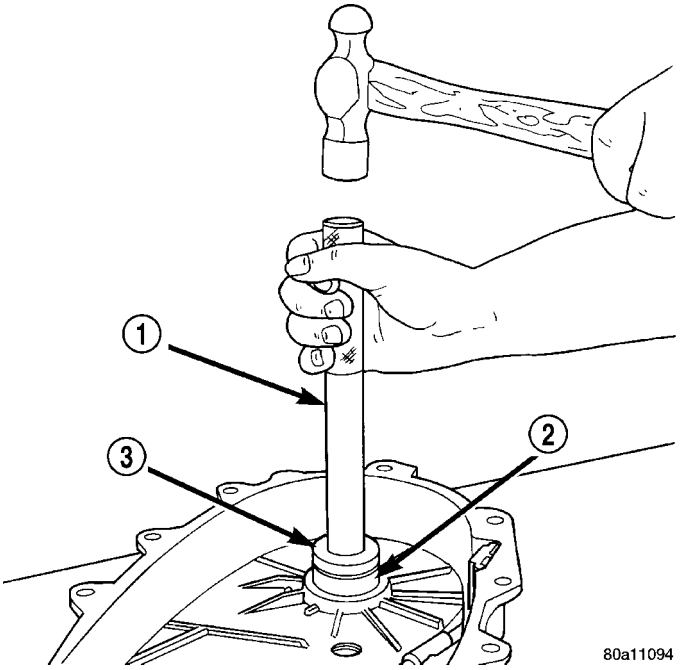


Fig. 48 Output Shaft Rear Bearing Installation

1 - HANDLE C-4171
2 - OUTPUT SHAFT INNER BEARING
3 - INSTALLER 5066

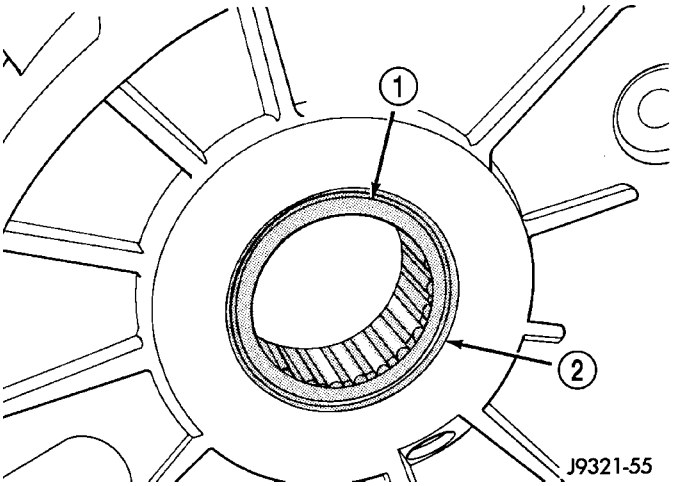


Fig. 49 Output Shaft Rear Bearing Installation Depth

1 - BEARING (SEATED) AT LOWER EDGE OF CHAMFER
2 - CHAMFER

(9) Using Remover C-4210 and Handle C-4171, drive input shaft bearing from inside the annulus gear opening in the case (Fig. 50).

(10) Install locating ring on new bearing.

(11) Position case so forward end is facing upward.

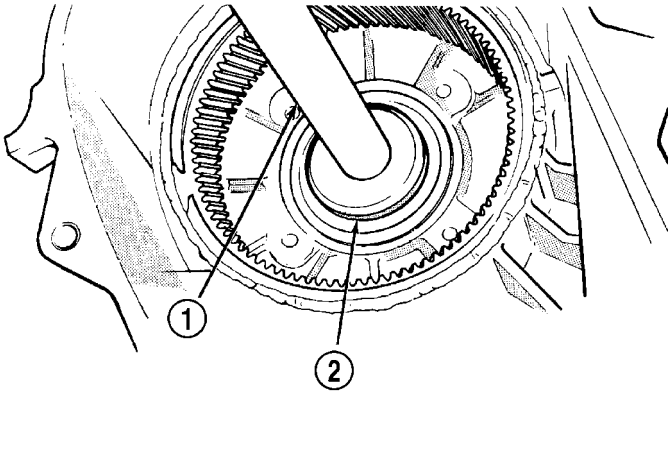


Fig. 50 Input Shaft Bearing Removal

1 - SPECIAL TOOL C-4171
2 - SPECIAL TOOL C-4210

(12) Using Remover C-4210 and Handle C-4171, drive input shaft bearing into case. The bearing locating ring must be fully seated against case surface (Fig. 51).

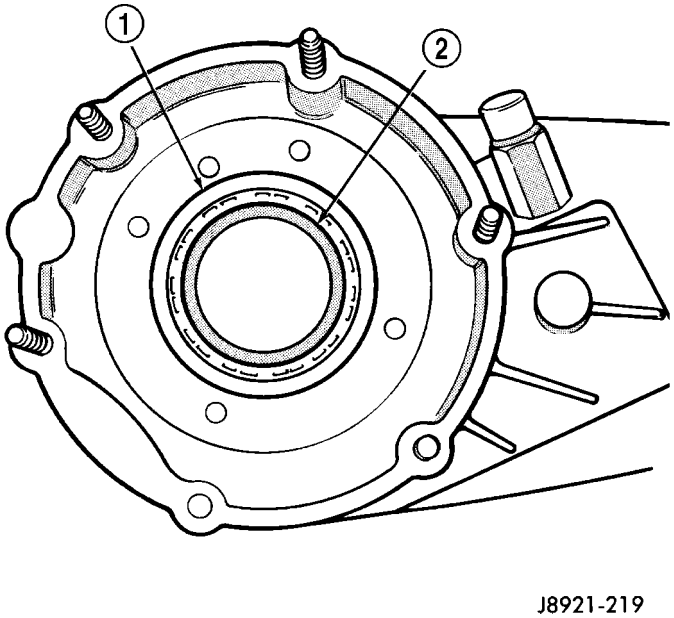


Fig. 51 Seating Input Shaft Bearing

1 - SNAP-RING
2 - INPUT SHAFT BEARING

TRANSFER CASE - NV231 (Continued)

(13) Remove input gear pilot bearing by inserting a suitably sized drift into the splined end of the input gear and driving the bearing out with the drift and a hammer (Fig. 52).

(14) Install new pilot bearing with Installer 5065 and Handle C-4171 (Fig. 53).

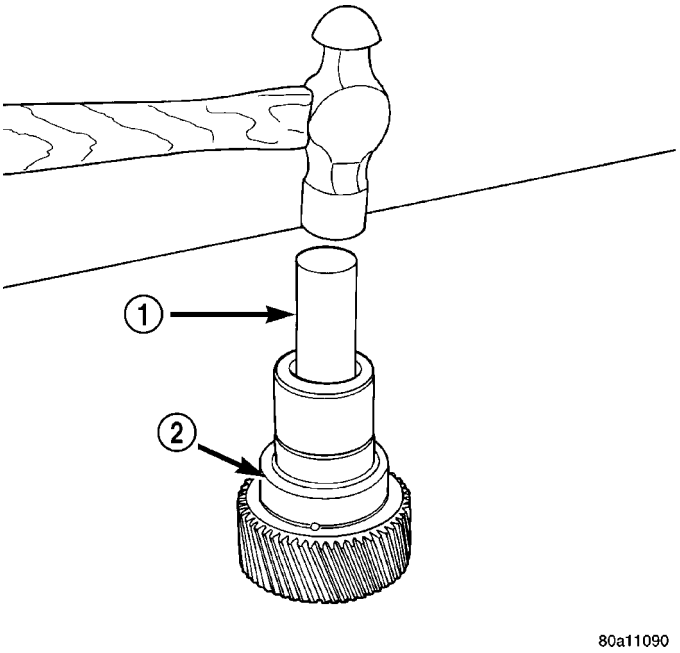


Fig. 52 Remove Input Gear Pilot Bearing

1 - DRIFT
2 - INPUT GEAR

(15) Remove front bearing retainer seal with suitable pry tool.

(16) Install new front bearing retainer seal with Installer 7884 (Fig. 54).

(17) Remove seal from oil pump housing with a suitable pry tool

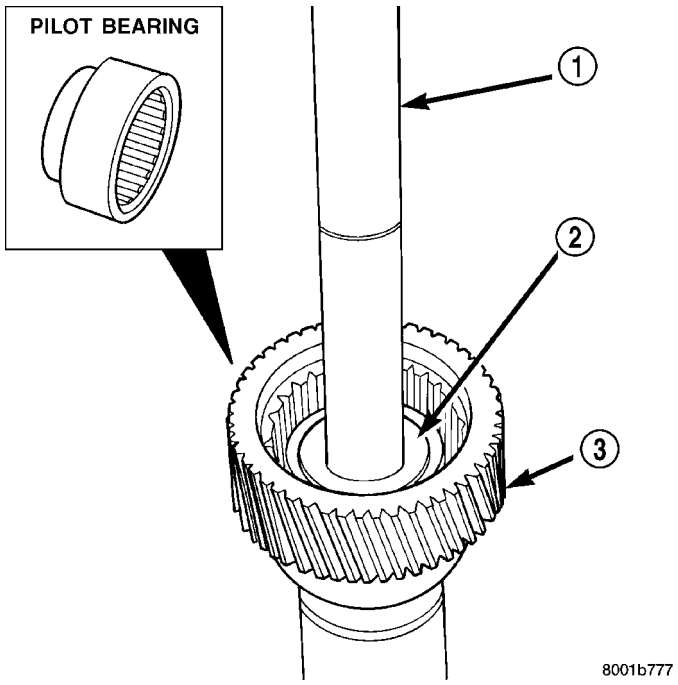


Fig. 53 Install Input Gear Pilot Bearing

1 - SPECIAL TOOL C-4171
2 - SPECIAL TOOL 5065
3 - INPUT GEAR

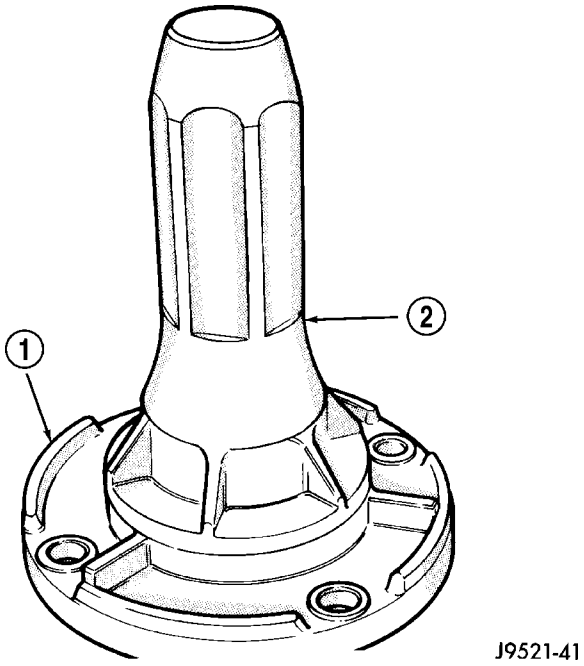


Fig. 54 Install Front Bearing Retainer Seal

1 - FRONT BEARING RETAINER
2 - SPECIAL TOOL 7884

TRANSFER CASE - NV231 (Continued)

(18) Install new seal in oil pump housing with Installer 7888 (Fig. 55).

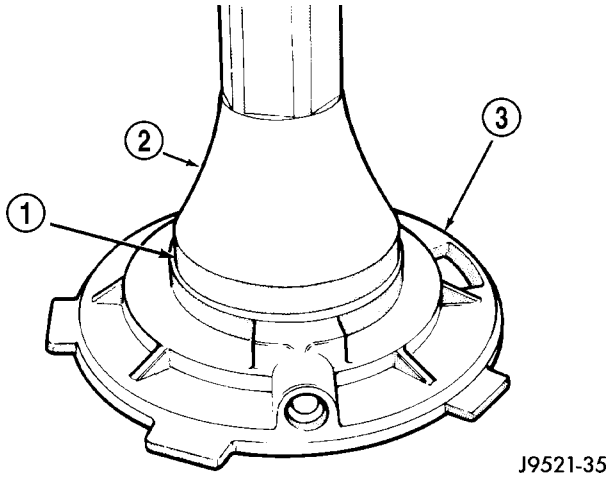


Fig. 55 Oil Pump Seal Installation

1 - HOUSING SEAL
2 - SPECIAL TOOL 7888
3 - OIL PUMP FEED HOUSING

(19) Remove rear retainer bearing with Installer 8128 and Handle C-4171.

(20) Install rear bearing in retainer with Handle C-4171 and Installer 5064 (Fig. 56).

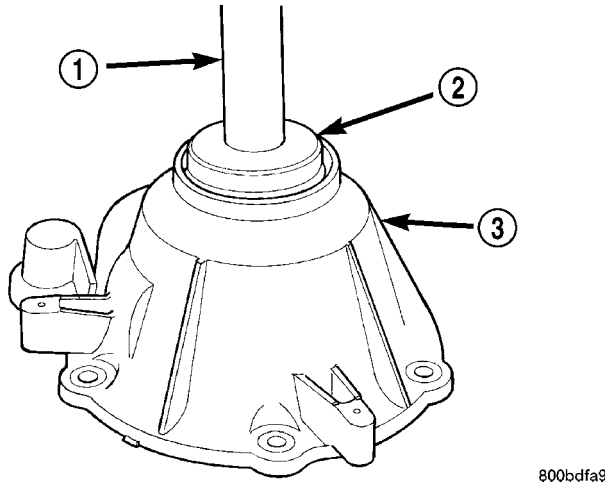


Fig. 56 Installing Rear Bearing In Retainer

1 - SPECIAL TOOL C-4171
2 - SPECIAL TOOL 5064
3 - REAR RETAINER

INPUT AND LOW RANGE GEAR

(1) Install first thrust washer in low range gear (Fig. 57). Be sure washer tabs are properly aligned in gear notches.

(2) Install input gear in low range gear. Be sure input gear is fully seated.

(3) Install remaining thrust washer in low range gear and on top of input gear. Be sure washer tabs are properly aligned in gear notches.

(4) Install retainer on input gear and install snap-ring.

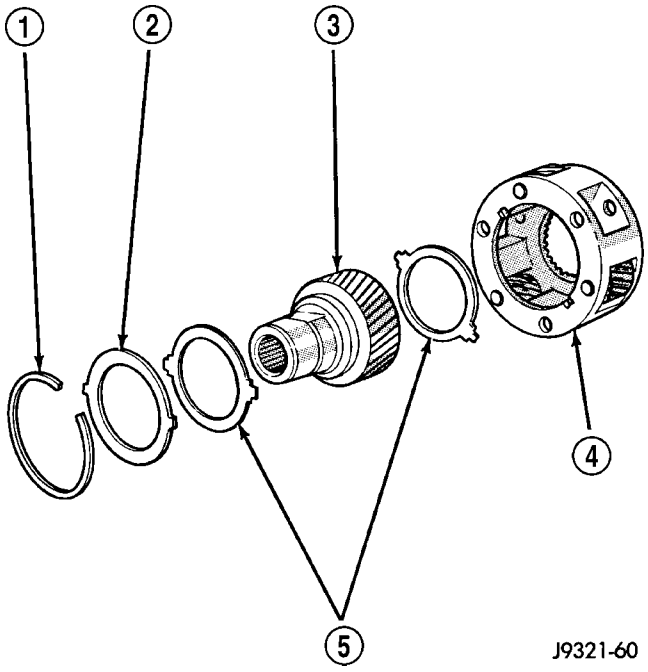


Fig. 57 Input/Low Range Gear Components

1 - SNAP-RING
2 - RETAINER PLATE
3 - INPUT GEAR
4 - LOW RANGE GEAR
5 - THRUST WASHERS

INPUT GEAR AND LOW RANGE GEAR

(1) Align and install low range/input gear assembly in front case (Fig. 58). Be sure low range gear pinions are engaged in annulus gear and that input gear shaft is fully seated in front bearing.

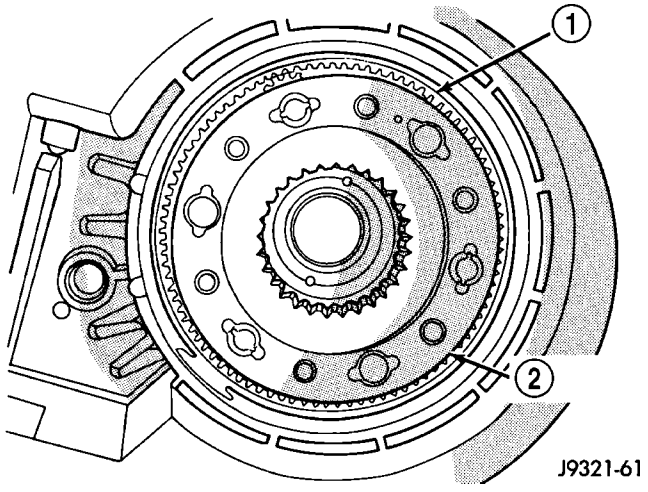


Fig. 58 Input/Low Range Gear Installation

1 - ANNULUS GEAR
2 - INPUT/LOW RANGE GEAR

TRANSFER CASE - NV231 (Continued)

(2) Install snap-ring to hold input/low range gear into front bearing (Fig. 59).

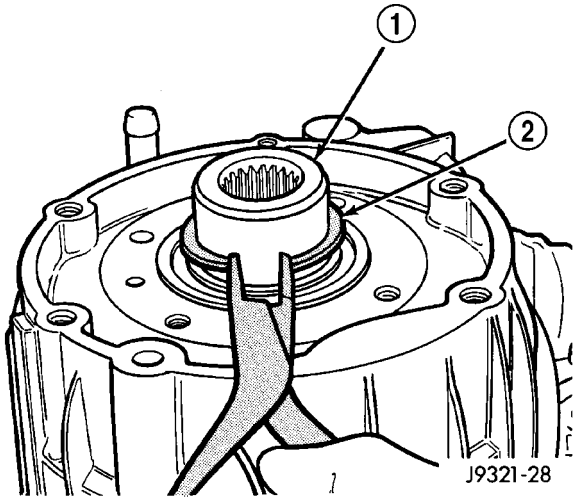


Fig. 59 Install Snap-Ring

1 - INPUT GEAR
2 - SNAP-RING

(3) Clean gasket sealer residue from retainer and inspect retainer for cracks or other damage.

(4) Apply a 3 mm (1/8 in.) bead of Mopar® gasket maker or silicone adhesive to sealing surface of retainer.

(5) Align cavity in seal retainer with fluid return hole in front of case.

CAUTION: Do not block fluid return cavity on sealing surface of retainer when applying Mopar® gasket maker or silicone adhesive sealer. Seal failure and fluid leak can result.

(6) Install bolts to hold retainer to transfer case (Fig. 60). Tighten to 21 N·m (16 ft. lbs.) of torque.

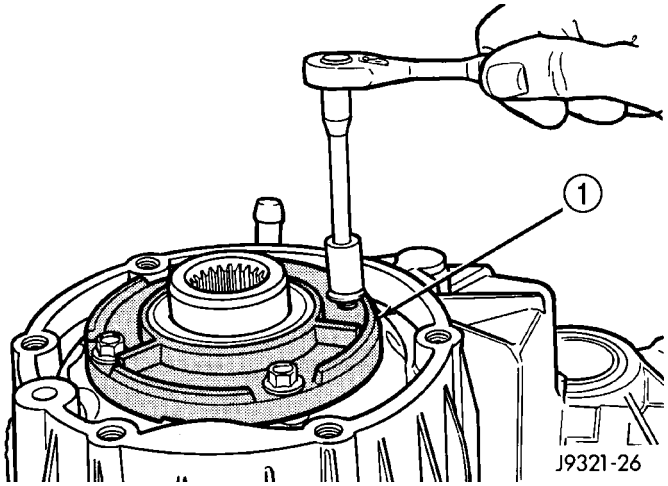


Fig. 60 Install Front Bearing Retainer

1 - FRONT BEARING RETAINER

MAINSHAFT

(1) Lubricate mainshaft splines with recommended transmission fluid.

(2) Slide drive sprocket onto mainshaft.

(3) Slide mode hub onto mainshaft.

(4) Install mode hub retaining ring. Verify that the retaining ring is fully seated in mainshaft groove.

SHIFT FORKS AND MAINSHAFT

(1) Install new sector shaft O-ring and bushing (Fig. 61).

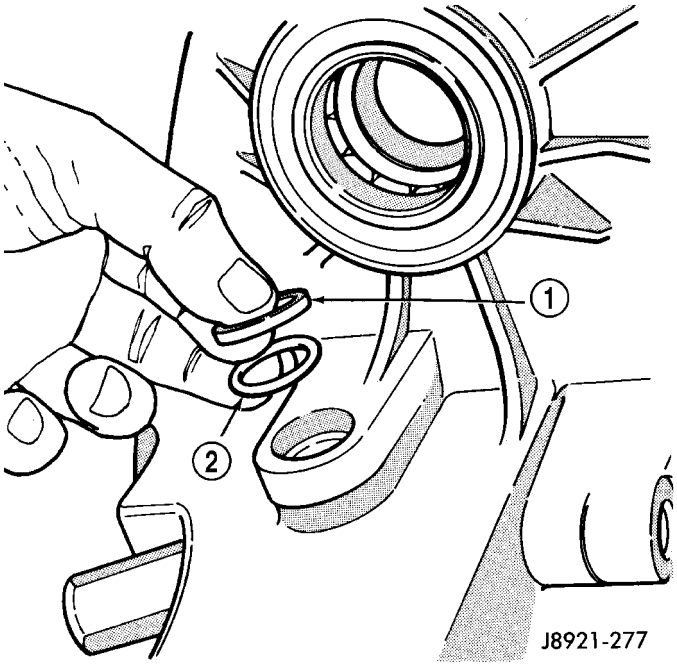


Fig. 61 Sector O-Ring And Bushing Installation

1 - SECTOR BUSHING
2 - O-RING

(2) Install shift sector in case (Fig. 62). Lubricate sector shaft with transmission fluid before installation.

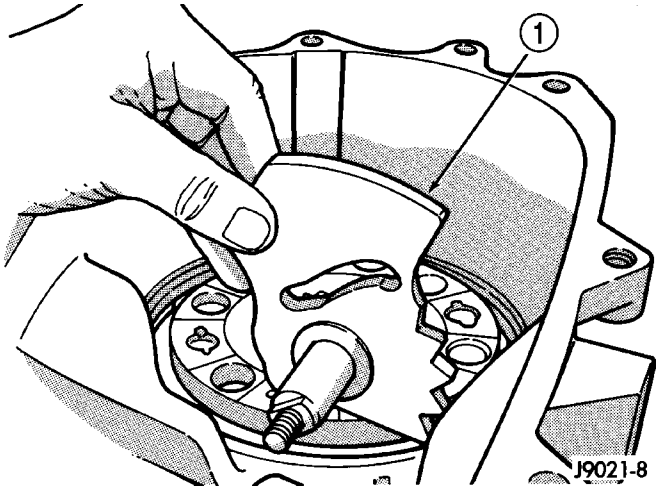


Fig. 62 Shift Sector Installation

1 - SHIFT SECTOR

TRANSFER CASE - NV231 (Continued)

(3) Install range lever, washer, and nut on sector shaft (Fig. 63). Tighten range lever nut to 27-34 N·m (20-25 ft. lbs.) torque.

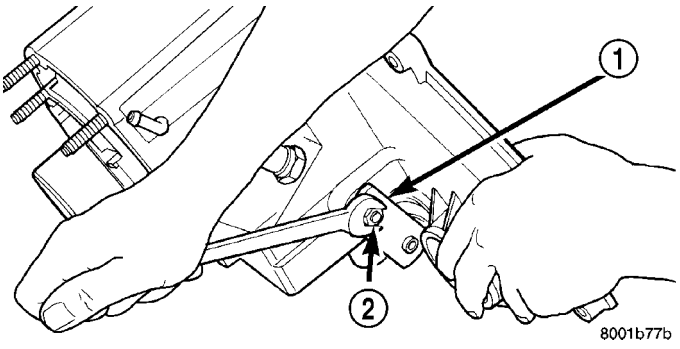


Fig. 63 Range Lever Installation

1 - RANGE LEVER
2 - LEVER NUT

(4) Assemble and install range fork and hub (Fig. 64). Be sure hub is properly seated in low range gear and engaged to the input gear.

(5) Align and insert range fork pin in shift sector slot.

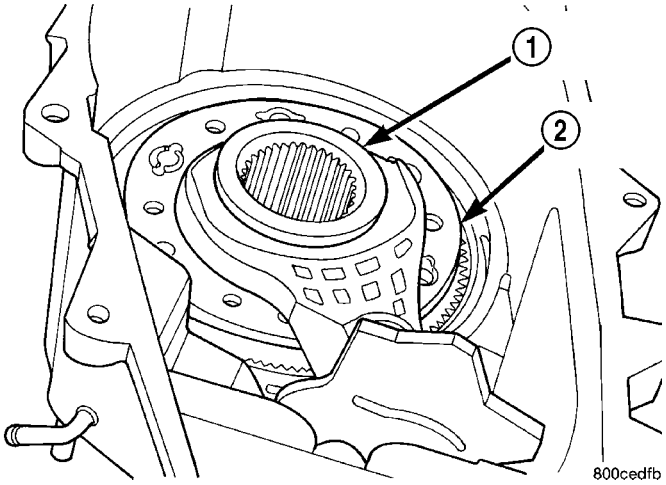


Fig. 64 Install Range Fork And Sleeve Assembly

1 - RANGE HUB
2 - RANGE FORK

(6) Install assembled mainshaft (Fig. 65). Be sure shaft is seated in pilot bearing and input gear.

(7) Install new pads on mode fork if necessary.

(8) Insert mode sleeve in mode fork mode fork. Be sure long side of sleeve is toward long end of shift rail (Fig. 66).

(9) Install assembled mode fork and sleeve (Fig. 67). Be sure fork rail goes through range fork and into case bore. Also be sure sleeve is aligned and seated on mainshaft hub.

(10) Rotate sector to NEUTRAL position.

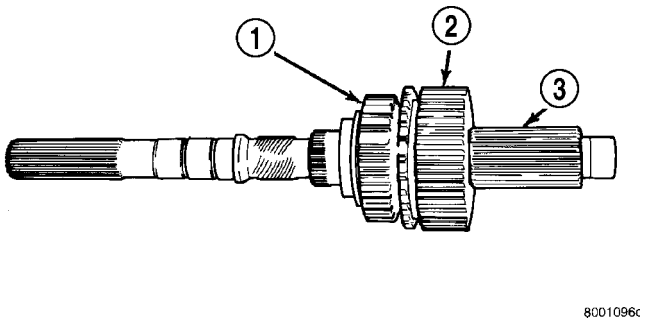


Fig. 65 Mainshaft Assembly Installation

1 - DRIVE SPROCKET
2 - MODE HUB
3 - MAINSHAFT

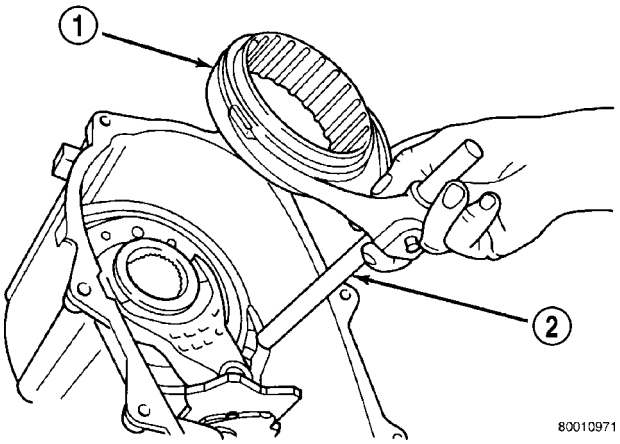


Fig. 66 Assembling Mode Fork And Sleeve

1 - MODE SLEEVE
2 - MODE FORK AND RAIL

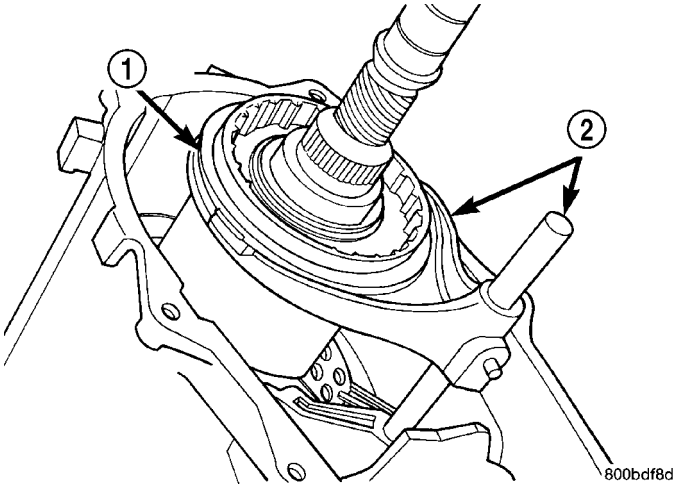


Fig. 67 Mode Fork And Sleeve Installation

1 - MODE SLEEVE
2 - MODE FORK AND RAIL

TRANSFER CASE - NV231 (Continued)

(11) Install new O-ring on detent plug (Fig. 68).

(12) Lubricate detent plunger with transmission fluid or light coat of petroleum jelly.

(13) Install detent plunger, spring and plug (Fig. 68).

(14) Verify that plunger is properly engaged in sector.

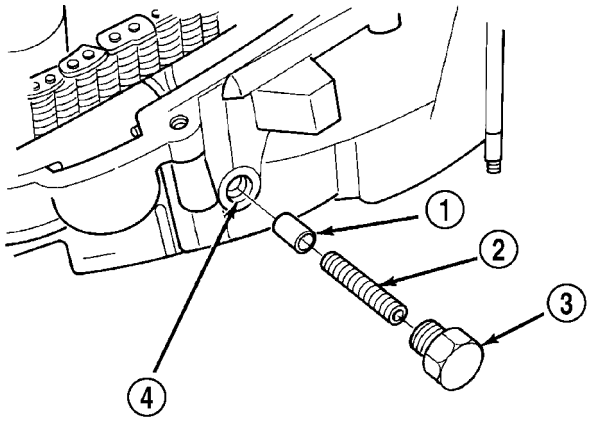


Fig. 68 Shift Detent Components

1 - POPPET
2 - SPRING
3 - SCREW
4 - POPPET BORE (IN CASE)

FRONT OUTPUT SHAFT AND DRIVE CHAIN

(1) Lubricate front output shaft-sprocket assembly, drive chain, and drive sprocket with transmission fluid.

(2) Assemble drive chain and front output shaft (Fig. 69).

(3) Start chain on mainshaft drive sprocket.

(4) Guide front shaft into bearing and drive sprocket onto mainshaft drive gear (Fig. 69).

(5) Install mode spring on upper end of mode fork shift rail (Fig. 70).

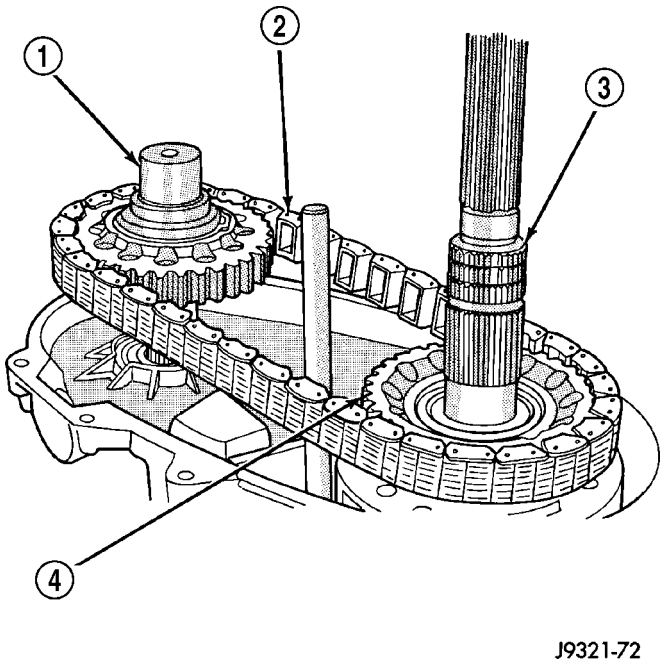


Fig. 69 Installing Drive Chain And Front Output Shaft

1 - FRONT OUTPUT SHAFT
2 - DRIVE CHAIN
3 - MAINSHAFT
4 - DRIVE SPROCKET

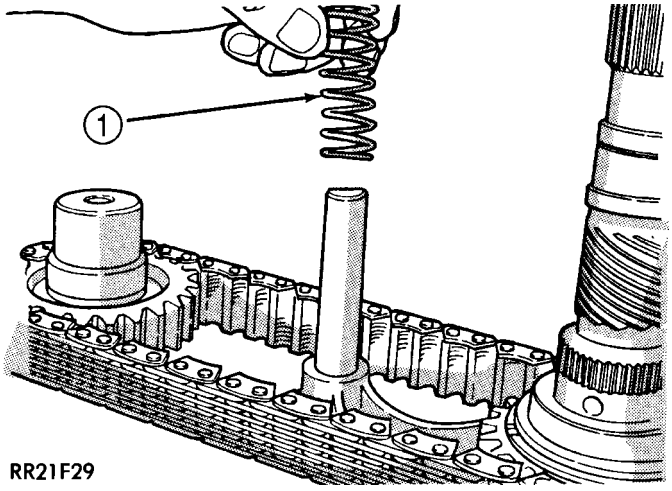


Fig. 70 Install Mode Fork Spring

1 - MODE SPRING

TRANSFER CASE - NV231 (Continued)

OIL PUMP AND REAR CASE

(1) Install magnet in front case pocket (Fig. 71).

(2) Assemble oil pickup screen, connecting hose, and tube.

(3) Install new pickup tube O-ring in oil pump (Fig. 72).

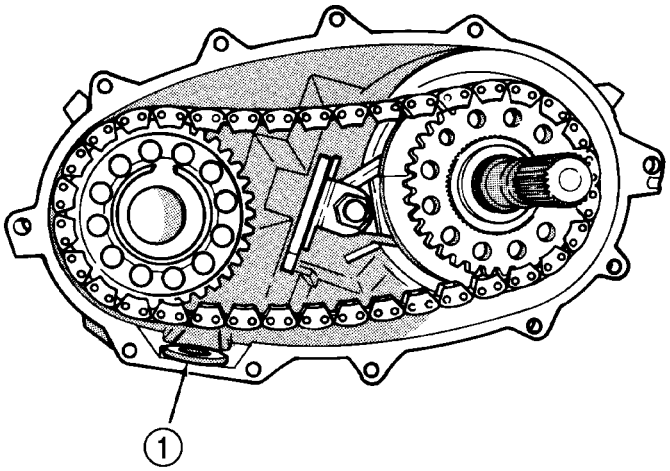


Fig. 71 Installing Case Magnet

1 - MAGNET

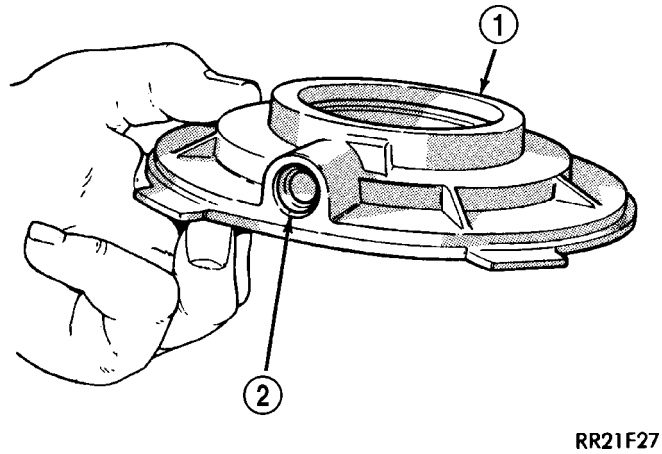


Fig. 72 Pickup Tube O-Ring Position

1 - OIL PUMP
2 - O-RING

(4) Insert oil pickup tube in oil pump inlet.

(5) Position assembled oil pump and pickup tube in rear case. Be sure pickup screen is securely seated in case slot. Also be sure oil pump locating tabs are outside rear case (Fig. 73).

(6) Apply 3 mm (1/8 in.) wide bead of Mopar® gasket maker or silicone adhesive sealer to mounting flange of front case. Work sealer bead around bolt holes.

(7) Lift rear case and oil pump and carefully position assembly on front case. Be sure case dowels are aligned and that mode fork rail extends through rear case before seating rear case on front case.

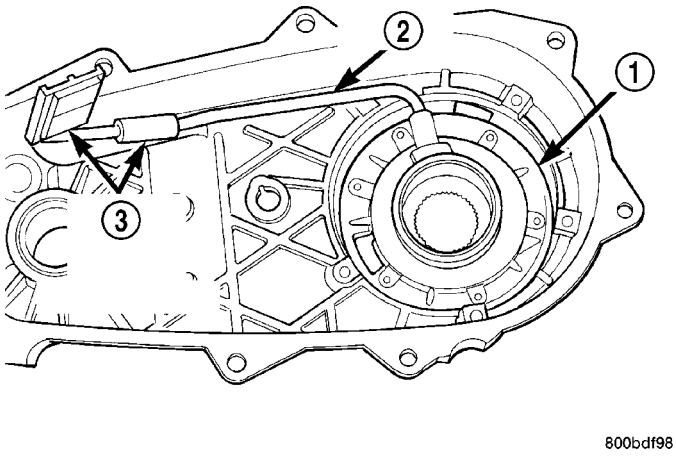


Fig. 73 Oil Pump And Pickup Tube Installation

1 - OIL PUMP
2 - PICKUP TUBE
3 - PICKUP SCREEN AND CONNECTOR

(8) Install case attaching bolts. Alignment bolts at each end of case are only ones requiring washers (Fig. 74).

(9) Tighten case bolts to 27-34 N·m (20-25 ft. lbs.) torque.

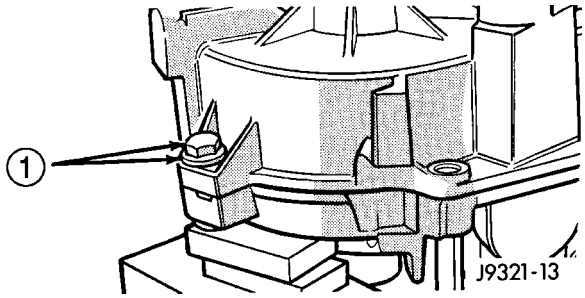


Fig. 74 Alignment Bolt Location

1 - ALIGNMENT BOLT AND WASHER (AT EACH END OF CASE)

YOKE AND RANGE LEVER

(1) Install indicator switch in front case. Tighten switch to 20-34 N·m (15-25 ft. lbs.) torque.

(2) Install range lever, washer and locknut on sector shaft (Fig. 75). Tighten locknut to 27-34 N·m (20-25 ft. lbs.) torque.

(3) Install new seal washer on front output shaft (Fig. 77).

(4) Lubricate yoke hub with transmission fluid and install yoke on front shaft.

(5) Install new seal washer on front shaft.

(6) Install yoke and new yoke nut on front output shaft (Fig. 76).

(7) Tighten yoke nut to 122-176 N·m (90-130 ft. lbs.) torque. Use Tool C-3281, or similar tool to hold yoke while tightening yoke nut.

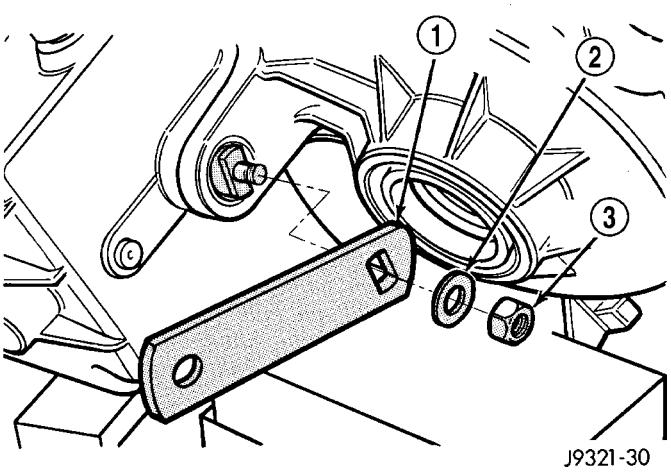


Fig. 75 Range Lever Installation

1 - RANGE LEVER
2 - WASHER
3 - LOCKNUT

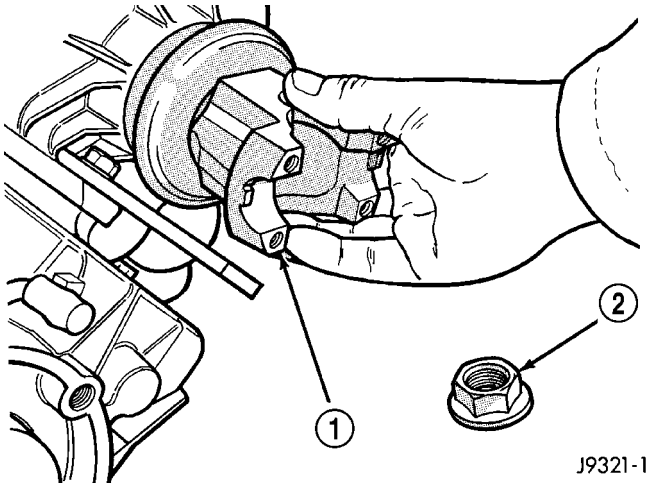


Fig. 76 Output Shaft Yoke Installation

1 - OUTPUT SHAFT YOKE
2 - YOKE NUT

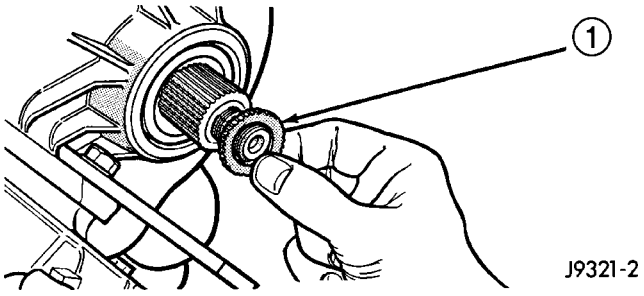


Fig. 77 Seal Washer Installation

1 - YOKE SEAL WASHER

REAR RETAINER

(1) Apply bead of Mopar® Sealer P/N 82300234, or Loctite™ Ultra Gray, to mating surface of rear retainer. Sealer bead should be a maximum of 3/16 inch.

(2) Install rear retainer on rear case. Tighten retainer bolts to 20-27 N·m (15-20 ft. lbs.) torque.

(3) Install rear bearing I.D. retaining ring and spacer on output shaft.

(4) Apply liberal quantity of petroleum jelly to new rear seal and to output shaft. Petroleum jelly is needed to protect seal lips during installation.

(5) Slide seal onto Seal Protector 6992 (Fig. 78). Slide seal protector and seal onto output shaft.

(6) Slide Installer C-4076-B onto seal protector with the recessed side of the tool toward the seal. Drive seal into rear bearing retainer with Installer C-4076-B and Handle MD-998323 (Fig. 79).

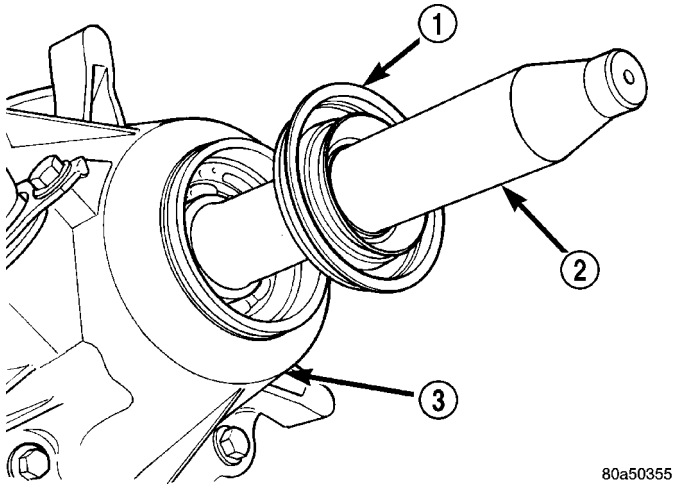


Fig. 78 Output Shaft Seal and Protector

1 - OUTPUT SHAFT SEAL
2 - SPECIAL TOOL 6992
3 - TRANSFER CASE

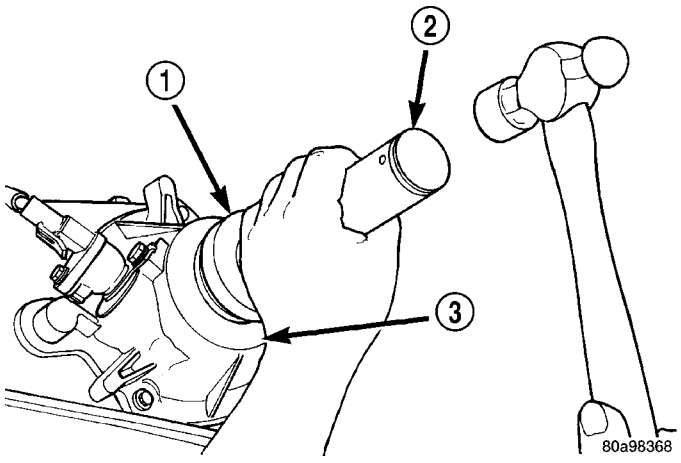


Fig. 79 Rear Seal Installation

1 - SPECIAL TOOL C-4076-B
2 - SPECIAL TOOL MD-998323
3 - TRANSFER CASE

TRANSFER CASE - NV231 (Continued)

NOTE: Vehicles built with a 4.0L engine and a manual transmission use a damper weight on the transfer case output shaft. Be sure to identify the transfer case before proceeding.

(7) Install a new output shaft rear slinger with Installer 8408, if the vehicle is not equipped with an output shaft damper.

(8) If the vehicle is equipped with an output shaft damper, install the output shaft damper as follows:

(a) Position the damper weight on the output shaft. Start the damper onto the output shaft chamfer, being careful to keep the weight square to the output shaft. (Fig. 80)

(b) Position the driver portion of Installer 8422 (Fig. 81) onto the damper, making sure the legs of the damper are positioned through the slots of the damper.

(c) Thread the puller screw of Installer 8422 into the output shaft by hand only. Make sure the screw is fully threaded into the output shaft.

(d) Using a wrench to hold the pulling screw stationary (Fig. 82), turn the pulling screw nut until the driver legs contact the rear face of the transfer case rear retainer. When the legs contact the retainer, the damper is properly positioned on the output shaft.

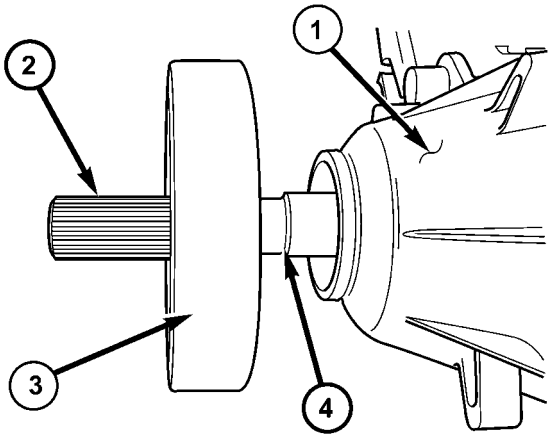


Fig. 80 Position Damper on Output Shaft

1 - Transfer Case
2 - Output Shaft
3 - Damper Weight
4 - Chamfer

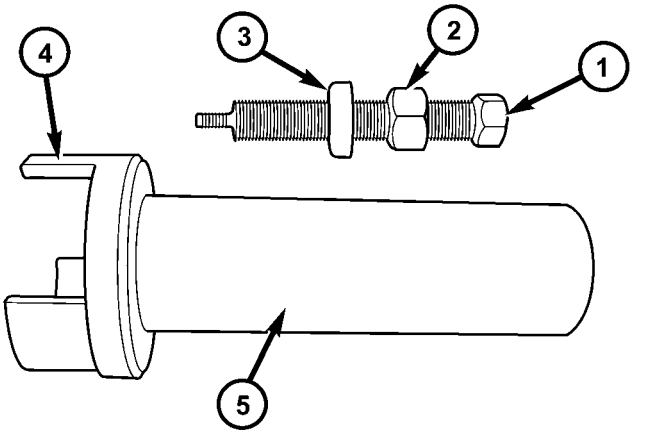


Fig. 81 Driver Installer 8422

1 - Pulling Screw
2 - Pulling Screw Nut
3 - Bearing
4 - Driver Legs
5 -Installer Driver

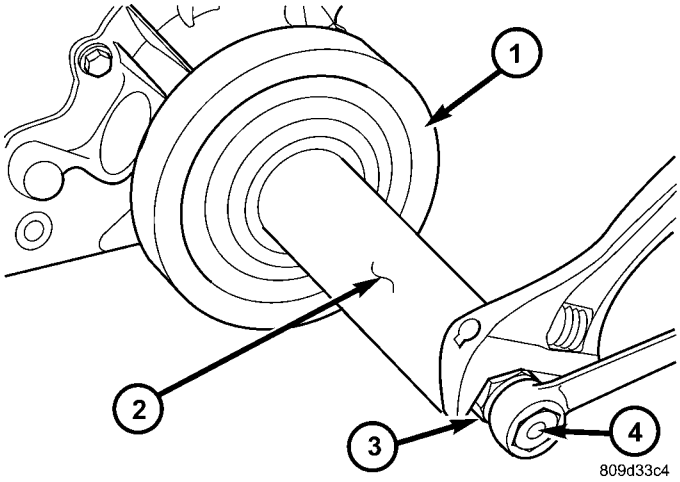


Fig. 82 Install Damper

1 - Damper
2 - Installer Driver
3 - Pulling Screw Nut
4 - Pulling Screw

TRANSFER CASE - NV231 (Continued)

(9) Install boot on output shaft slinger, or output shaft damper, and crimp retaining clamp with tool C-4975-A (Fig. 83).

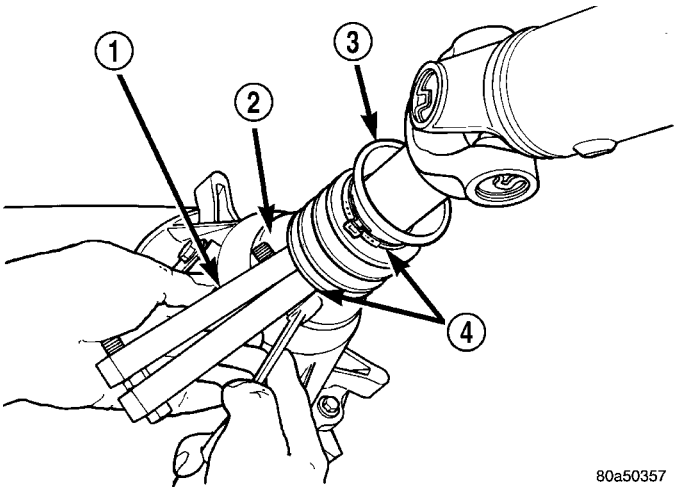


Fig. 83 Slinger Boot Installation - Typical

1 - SPECIAL TOOL C-4975–A
2 - SLINGER
3 - BOOT
4 - CLAMP

INSTALLATION

(1) Mount transfer case on a transmission jack.
(2) Secure transfer case to jack with chains.
(3) Position transfer case under vehicle.
(4) Align transfer case and transmission shafts and install transfer case on transmission.
(5) Install and tighten transfer case attaching nuts to 35 N·m (26 ft. lbs.) torque (Fig. 2).
(6) Connect vehicle speed sensor wires, and vent hose.
(7) Connect indicator switch harness to transfer case switch, if necessary. Secure wire harness to clips on transfer case.
(8) Align and connect propeller shafts. Refer to Differential and Driveline for proper procedures and specifications.
(9) Fill transfer case with correct fluid. Check transmission fluid level. Correct as necessary.
(10) Install rear crossmember, or skid plate. Tighten crossmember bolts to 41 N·m (30 ft. lbs.) torque.
(11) Remove transmission jack and support stand.
(12) Connect shift rod to transfer case range lever.
(13) Adjust transfer case shift linkage.
(14) Lower vehicle and verify transfer case shift operation.

SPECIFICATIONS

TRANSFER CASE - NV231

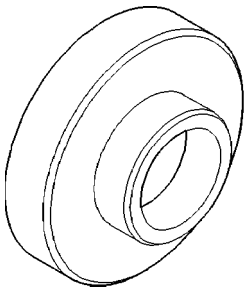
TORQUE

DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Plug, Detent	16-24	12-18	-
Plug, Drain/Fill	20-34	15-25	-
Bolt, Front Brg. Retainer	21	16	-
Bolt, Case Half	27-34	20-25	-
Nut, Front Yoke	122-176	90-130	-
Nut, Range Lever	27-34	20-25	-
Bolt, Rear Retainer	35-46	26-34	-
Nuts, Mounting	35-47	26-35	-
Switch, Indicator	20-34	15-25	-

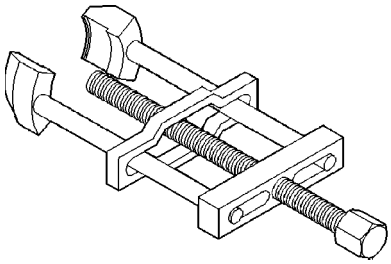
TRANSFER CASE - NV231 (Continued)

SPECIAL TOOLS

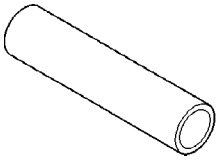
NV231



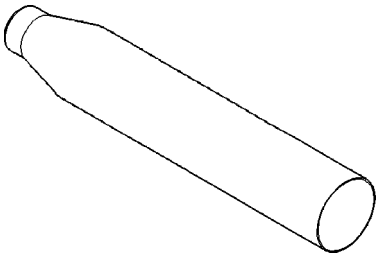
Installer - C-4076-B



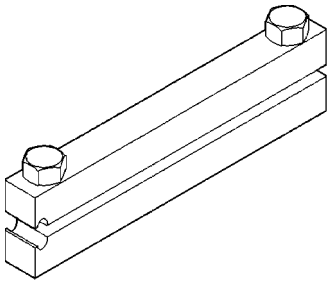
Puller, Slinger - MD-998056-A



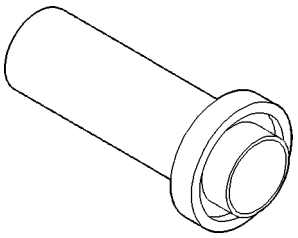
Installer - MD-998323



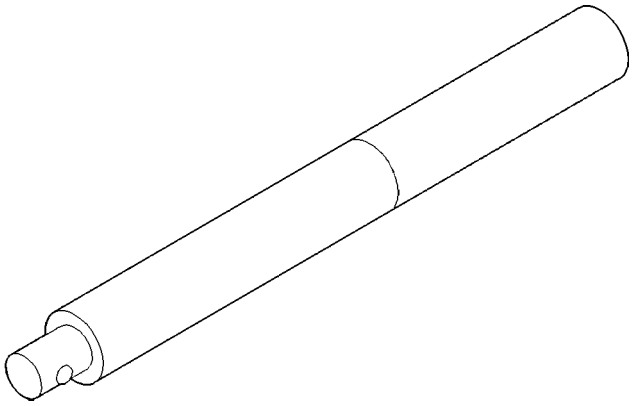
Protector, Seal - 8824



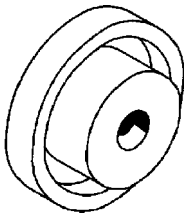
Installer, Boot Clamp - C-4975-A



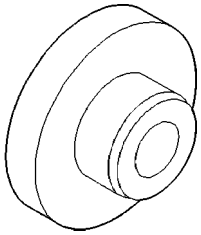
Installer, Seal - 8143-A



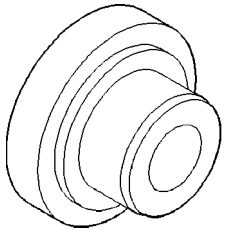
Handle, Universal - C-4171



Installer, Seal - C-4210

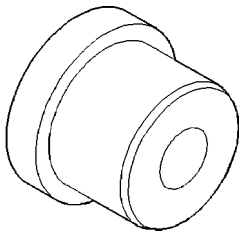


Installer, Bearing - 5052

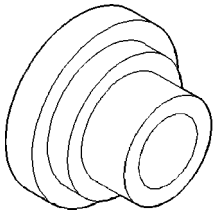


Installer, Bearing - 5065

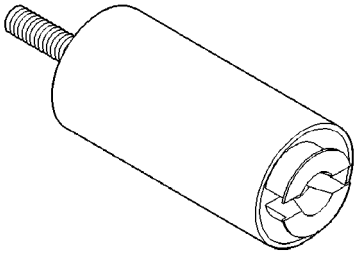
TRANSFER CASE - NV231 (Continued)



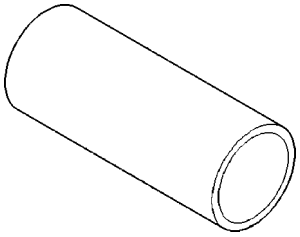
Installer, Bushing - 5066



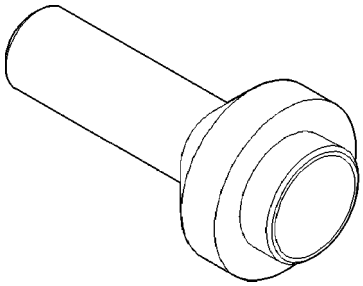
Installer, Bearing - 8128



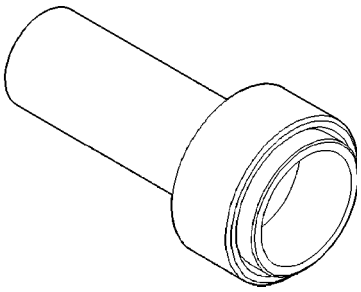
Remover - L-4454



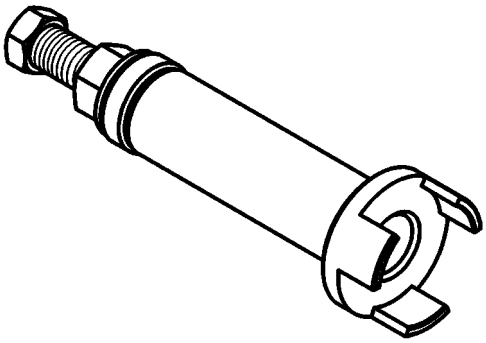
Cup - 8148



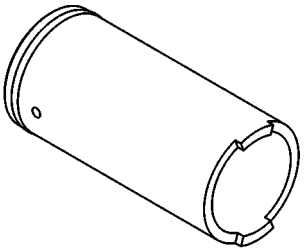
Installer, Seal - 7884



Installer, Pump Housing Seal - 7888



Installer, Transfer Case Damper Driver - 8422



Installer, Output Shaft Slinger - 8408

FLUID

STANDARD PROCEDURE - FLUID DRAIN AND FILL

The fill and drain plugs are both in the rear case (Fig. 84). Correct fill level is to the bottom edge of the fill plug hole. Be sure the vehicle is level to ensure an accurate fluid level check.

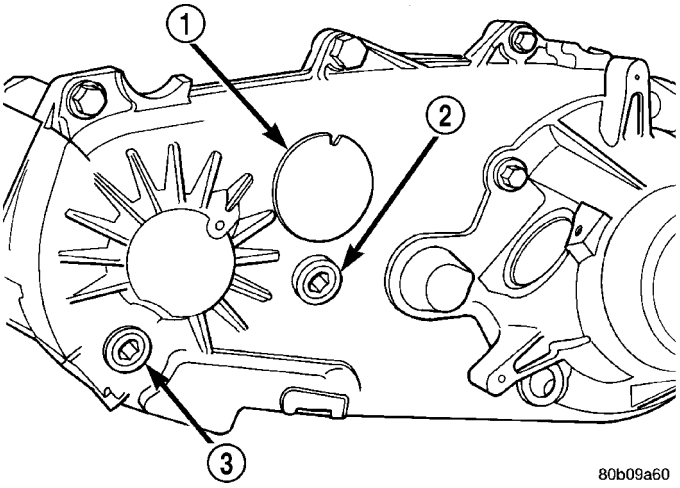


Fig. 84 Fill/Drain Plug And I.D. Tag Locations

1 - I.D. TAG
2 - FILL PLUG
3 - DRAIN PLUG

FRONT OUTPUT SHAFT SEAL

REMOVAL

(1) Raise vehicle.

(2) Remove front propeller shaft. (Refer to 3 - DIFFERENTIAL & DRIVELINE/PROPELLER SHAFT/PROPELLER SHAFT - REMOVAL)

(3) Remove front output shaft yoke.

(4) Remove seal from front case with pry tool (Fig. 85).

INSTALLATION

(1) Install new front output seal in front case with Installer Tool 8143-A as follows:

(a) Place new seal on tool. Garter spring on seal goes toward interior of case.

(b) Start seal in bore with light taps from hammer (Fig. 86). Once seal is started, continue tapping seal into bore until installer tool seats against case.

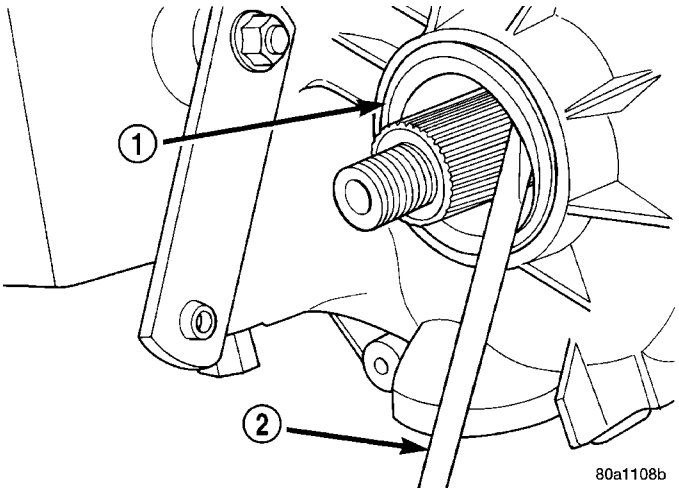


Fig. 85 Remove Front Output Shaft Seal - Typical

1 - OUTPUT SHAFT SEAL
2 - PRYBAR

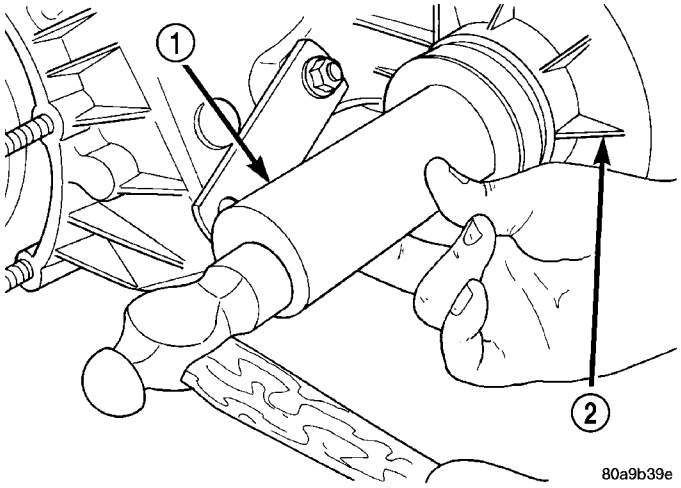


Fig. 86 Front Output Seal Installation - Typical

1 - INSTALLER 8143-A
2 - TRANSFER CASE

REAR OUTPUT SHAFT SEAL

REMOVAL

(1) Shift the transmission and transfer case into NEUTRAL.

(2) Raise and support vehicle.

(3) Mark a line across the pinion shaft and at each end of the propeller shaft for installation reference.

(4) Remove the U-joint strap bolts at the pinion shaft yoke.

(5) Pry open clamp holding the dust boot to propeller shaft yoke (Fig. 87).

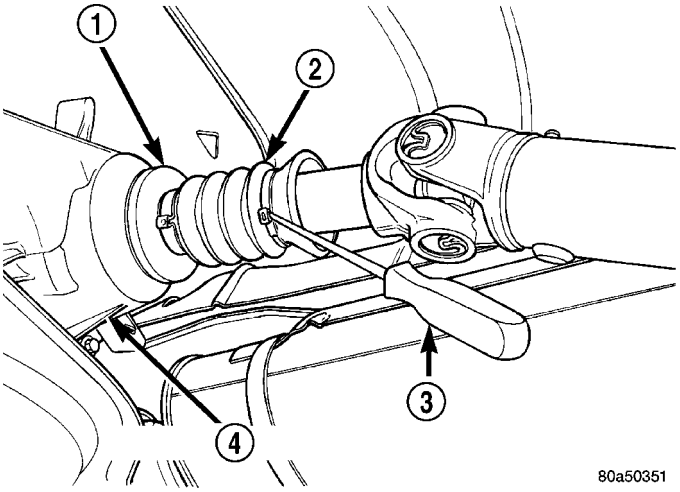


Fig. 87 Dust Boot Clamp

1 - SLINGER
2 - BOOT
3 - AWL
4 - TRANSFER CASE

(6) Slide the slip yoke off of the transmission/transfer case output shaft and remove the propeller shaft.

(7) Spread band clamp which holds output shaft boot to the output shaft slinger, or output shaft damper, with a suitable awl, or equivalent.

NOTE: Vehicles built with a 4.0L engine and a manual transmission use a damper weight on the transfer case output shaft. Be sure to identify the transfer case before proceeding.

(8) Remove output shaft boot from slinger, or output shaft damper, and output shaft.

(9) If the vehicle is not equipped with an output shaft damper, remove the output shaft rear slinger using Puller MD-998056-A (Fig. 88).

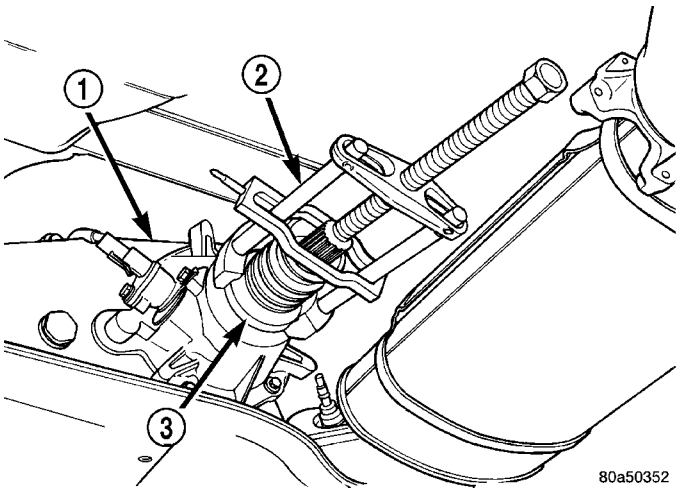


Fig. 88 Rear Slinger Removal

1 - TRANSFER CASE
2 - SPECIAL TOOL MD-998056-A
3 - SLINGER

(10) If the vehicle is equipped with an output shaft damper, use Screws 8421 and the puller yoke and forcing screw from a bolt-grip puller set, such as those used to remove steering wheels and harmonic balancers, to remove the transfer case output shaft damper.

(11) Use a suitable pry tool, or a slide hammer mounted screw, to remove the seal from the rear retainer (Fig. 89).

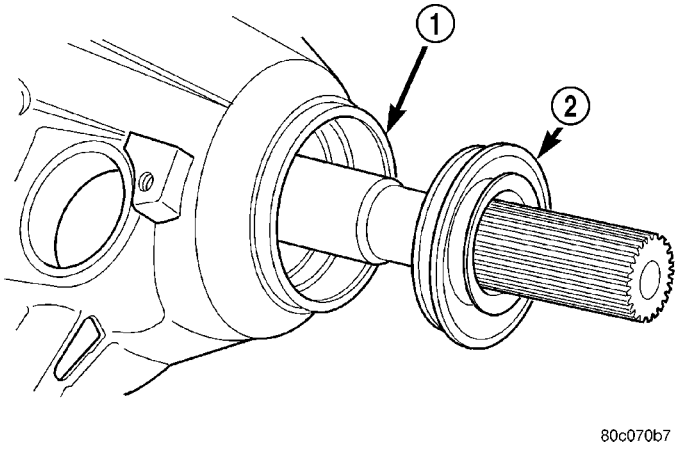


Fig. 89 Rear Output Shaft Seal

1 - REAR RETAINER
2 - OUTPUT SHAFT SEAL

REAR OUTPUT SHAFT SEAL (Continued)

INSTALLATION

(1) Apply liberal quantity of petroleum jelly to new rear seal and to output shaft. Petroleum jelly is needed to protect seal lips during installation.

(2) Slide seal onto Seal Protector 6992 (Fig. 90). Slide seal protector and seal onto output shaft.

(3) Slide Installer C-4076-B onto seal protector with the recessed side of the tool toward the seal. Drive seal into rear bearing retainer with Installer C-4076-B and Handle MD-998323 (Fig. 91).

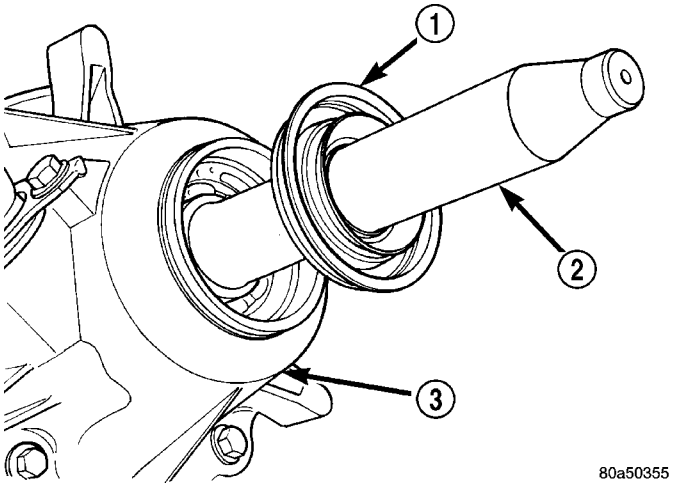


Fig. 90 Output Shaft Seal and Protector

1 - OUTPUT SHAFT SEAL
2 - SPECIAL TOOL 6992
3 - TRANSFER CASE

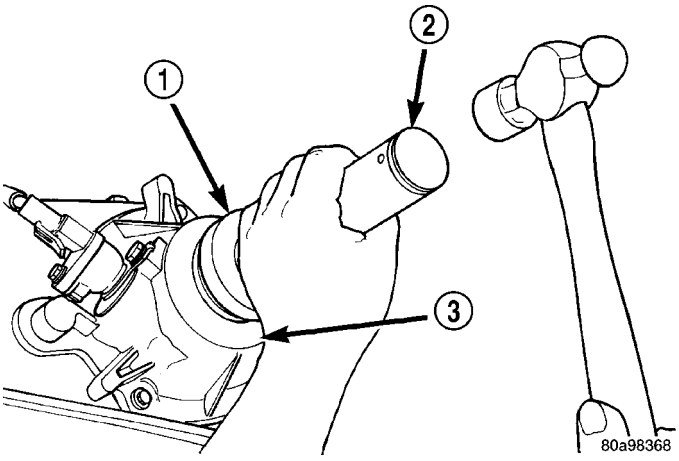


Fig. 91 Rear Seal Installation

1 - SPECIAL TOOL C-4076-B
2 - SPECIAL TOOL MD-998323
3 - TRANSFER CASE

NOTE: Vehicles built with a 4.0L engine and a manual transmission use a damper weight on the transfer case output shaft. Be sure to identify the transfer case before proceeding.

(4) Install a new output shaft rear slinger with Installer 8408, if the vehicle is not equipped with an output shaft damper.

(5) If the vehicle is equipped with an output shaft damper, install the output shaft damper as follows:

(a) Position the damper weight on the output shaft. Start the damper onto the output shaft chamfer, being careful to keep the weight square to the output shaft. (Fig. 92)

(b) Position the driver portion of Installer 8422 (Fig. 93) onto the damper, making sure the legs of the damper are positioned through the slots of the damper.

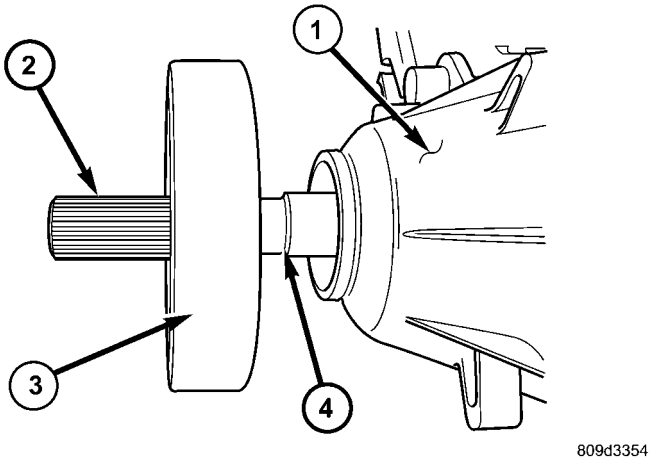


Fig. 92 Position Damper on Output Shaft

1 - Transfer Case
2 - Output Shaft
3 - Damper Weight
4 - Chamfer

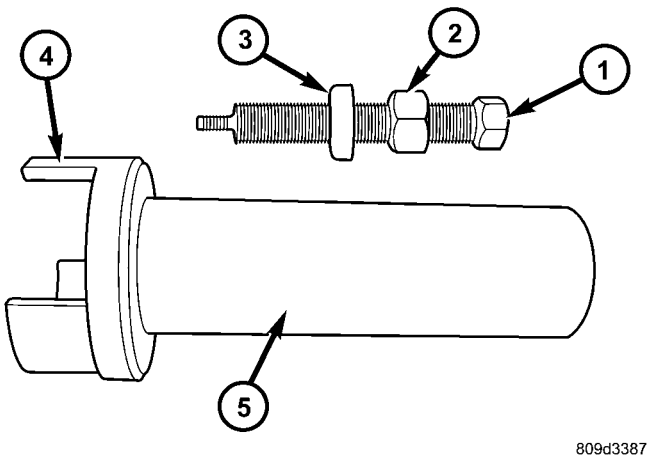


Fig. 93 Driver Installer 8422

1 - Pulling Screw
2 - Pulling Screw Nut
3 - Bearing
4 - Driver Legs
5 -Installer Driver

REAR OUTPUT SHAFT SEAL (Continued)

(c) Thread the puller screw of Installer 8422 into the output shaft by hand only. Make sure the screw is fully threaded into the output shaft.

(d) Using a wrench to hold the pulling screw stationary (Fig. 94), turn the pulling screw nut until the driver legs contact the rear face of the transfer case rear retainer. When the legs contact the retainer, the damper is properly positioned on the output shaft.

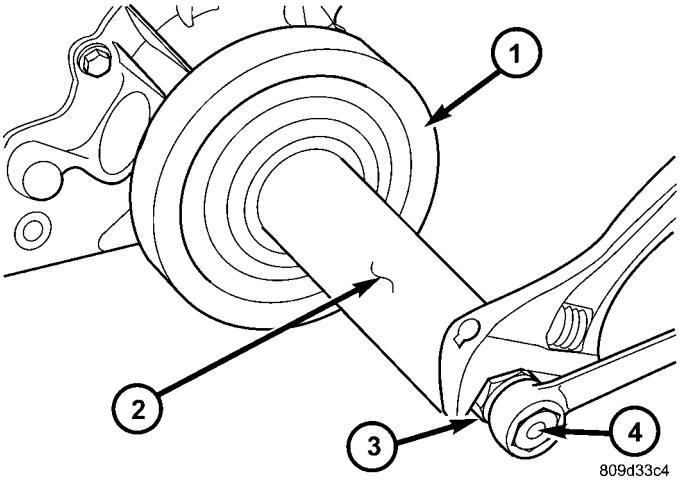


Fig. 94 Install Damper

1 - Damper
2 - Installer Driver
3 - Pulling Screw Nut
4 - Pulling Screw

(6) Install boot on output shaft slinger, or output shaft damper, and crimp retaining clamp with tool C-4975-A (Fig. 95).

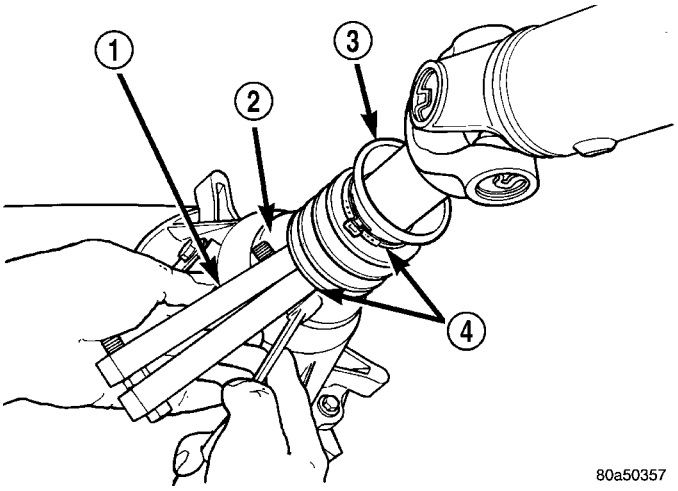


Fig. 95 Slinger Boot Installation - Typical

1 - SPECIAL TOOL C-4975–A
2 - SLINGER
3 - BOOT
4 - CLAMP

(7) Slide the slip yoke on the transmission/transfer case output shaft. Align installation reference marks at the axle yoke and install the propeller shaft.

(8) Tighten the U-joint strap/clamp bolts at the axle yoke to 19 N·m (14 ft. lbs.).

(9) Crimp clamp with Clamp Tool C-4975A to hold dust boot to propeller shaft yoke.

(10) Remove support and lower the vehicle.

SHIFT LEVER

REMOVAL

(1) Shift transfer case into 4L.

(2) Raise vehicle.

(3) Loosen adjusting trunnion locknut and slide shift rod out of trunnion (Fig. 96). If rod lacks enough travel to come out of trunnion, push trunnion out of torque shaft.

(4) Lower vehicle.

(5) Remove console. (Refer to 23 - BODY/INTERIOR/FLOOR CONSOLE - REMOVAL)

(6) Remove screws attaching lever assembly to floorpan and remove assembly and shift rod (if left attached).

INSTALLATION

(1) If shift rod was not removed from lever assembly, work rod down through floorpan opening. Then position lever assembly on floorpan and install assembly attaching screws.

(2) Install console. (Refer to 23 - BODY/INTERIOR/FLOOR CONSOLE - INSTALLATION)

(3) Raise vehicle.

(4) Connect trunnion to torque shaft arm. Or, slide shift rod into trunnion on range lever. Be sure shift rod slides freely in trunnion.

(5) Verify that range lever is in 4L position. Then tighten trunnion lock bolt.

(6) Lower vehicle and check transfer case shift operation.

SHIFT LEVER (Continued)

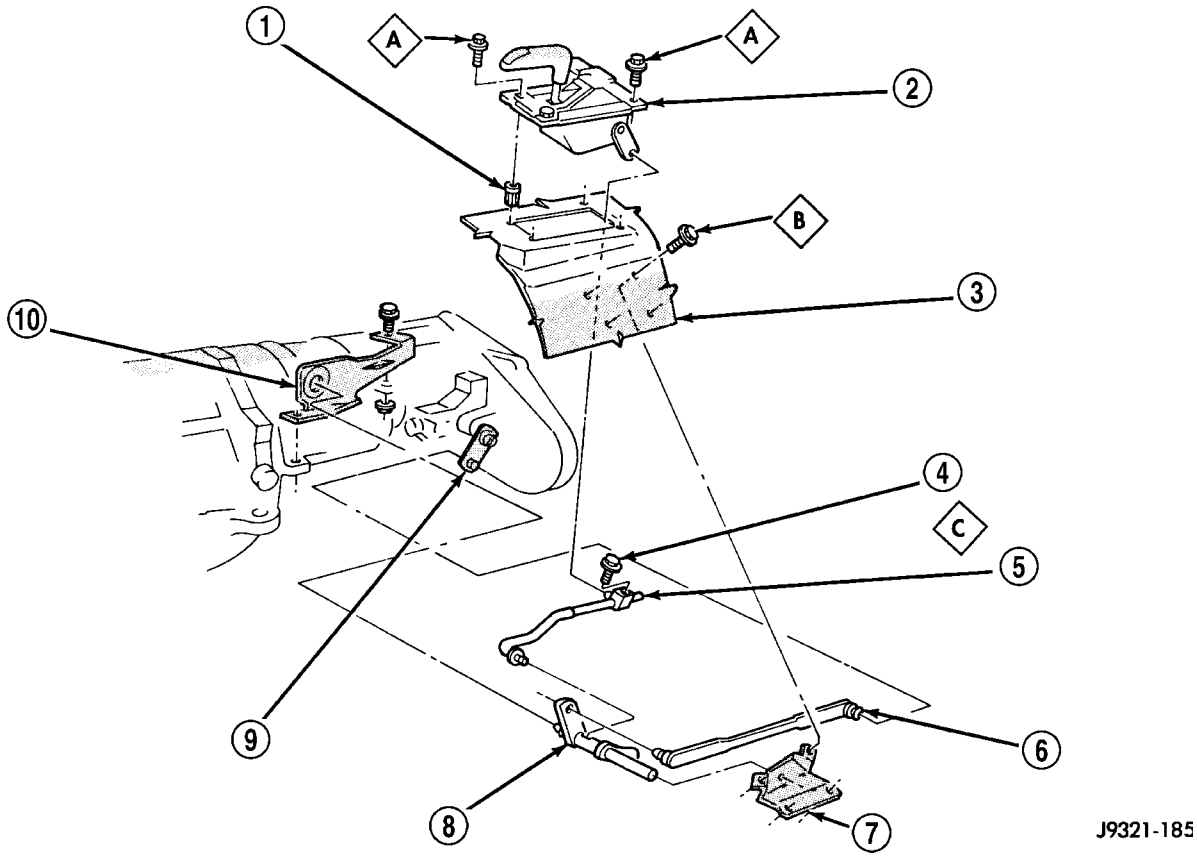


Fig. 96 Shift Linkage

1 - Rivnut (4)
2 - Shift Lever Assembly
3 - Floorpan
4 - Trunnion Lock Bolt
5 - Selector Rod and Trunnion
6 - Shift Lever Rod
7 - Torque Shaft Frame Bracket
8 - Torque Shaft
9 - Transfer Case Shift Lever
10 - Torque Shaft Transfer Case Bracket
A - 3-4 N·m (27-35 in. lbs.)
B - 11-14 N·m (97-123 in. lbs.)
C - 8-14 N·m (72-120 in. lbs.)

SHIFT LEVER (Continued)

ADJUSTMENTS - SHIFT LINKAGE

(1) Shift transfer case into 4L position.
(2) Raise vehicle.
(3) Loosen lock bolt on adjusting trunnion (Fig. 97).
(4) Be sure linkage rod slides freely in trunnion. Clean rod and apply spray lube if necessary.
(5) Verify that transfer case range lever is fully engaged in 4L position.
(6) Tighten adjusting trunnion lock bolt.
(7) Lower vehicle.

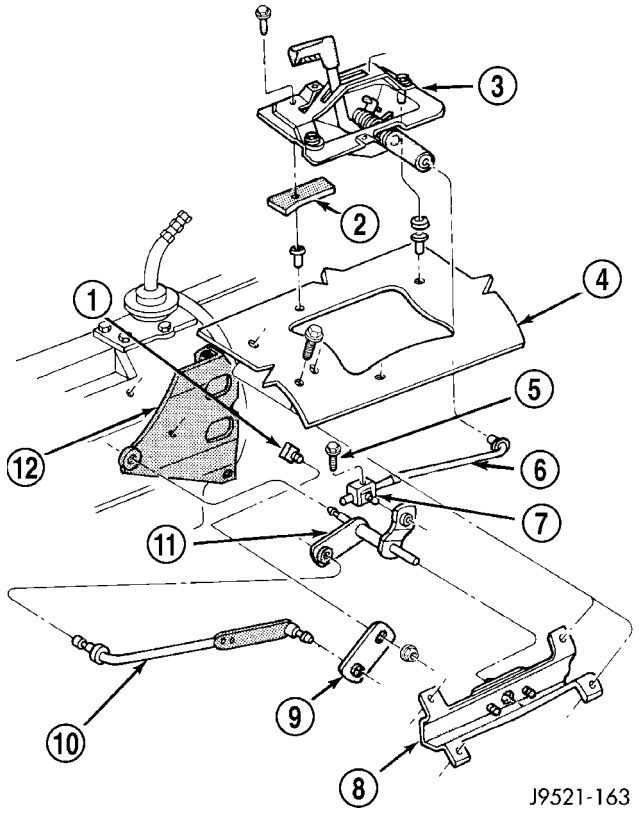


Fig. 97 Shift Linkage

1 - TRANSFER CASE SHIFT LEVER SHAFT
2 - SEAL
3 - TRANSFER CASE SHIFT LEVER ASSEMBLY
4 - FLOORPAN
5 - TRUNNION LOCK BOLT
6 - SHIFT ROD
7 - ADJUSTING TRUNNION
8 - TORQUE SHAFT BRACKET
9 - RANGE LEVER
10 - TORQUE SHAFT ROD
11 - TORQUE SHAFT
12 - LINKAGE BRACKET

SPEEDOMETER DRIVE ADAPTER

REMOVAL

(1) Raise vehicle.
(2) Disconnect wires from vehicle speed sensor.
(3) Remove adapter clamp and screw (Fig. 98).
(4) Remove speed sensor and speedometer adapter as an assembly.
(5) Remove speed sensor retaining screw and remove sensor from adapter.
(6) Remove speedometer pinion from adapter. Replace pinion if chipped, cracked, or worn.
(7) Inspect sensor and adapter O-rings (Fig. 98). Remove and discard O-rings if worn or damaged.
(8) Inspect terminal pins in speed sensor. Clean pins with Mopar® electrical spray cleaner if dirty or oxidized. Replace sensor if faulty, or if pins are loose, severely corroded, or damaged.

INSTALLATION

(1) Thoroughly clean adapter flange and adapter mounting surface in housing. Surfaces must be clean for proper adapter alignment and speedometer operation.
(2) Install new O-rings on speed sensor and speedometer adapter (Fig. 98), if necessary.
(3) Lubricate sensor and adapter O-rings with transmission fluid.
(4) Install vehicle speed sensor in speedometer adapter. Tighten sensor attaching screw to 2-3 N·m (15-27 in. lbs.) torque.
(5) Install speedometer pinion in adapter.
(6) Count number of teeth on speedometer pinion. Do this before installing assembly in housing. Then lubricate pinion teeth with transmission fluid.
(7) Note index numbers on adapter body (Fig. 99). These numbers will correspond to number of teeth on pinion.
(8) Install speedometer assembly in housing.
(9) Rotate adapter until required range numbers are at 6 o-clock position. Be sure range index numbers correspond to number of teeth on pinion gear.
(10) Install speedometer adapter clamp and retaining screw. Tighten clamp screw to 10-12 N·m (90-110 in. lbs.) torque.

SPEEDOMETER DRIVE ADAPTER (Continued)

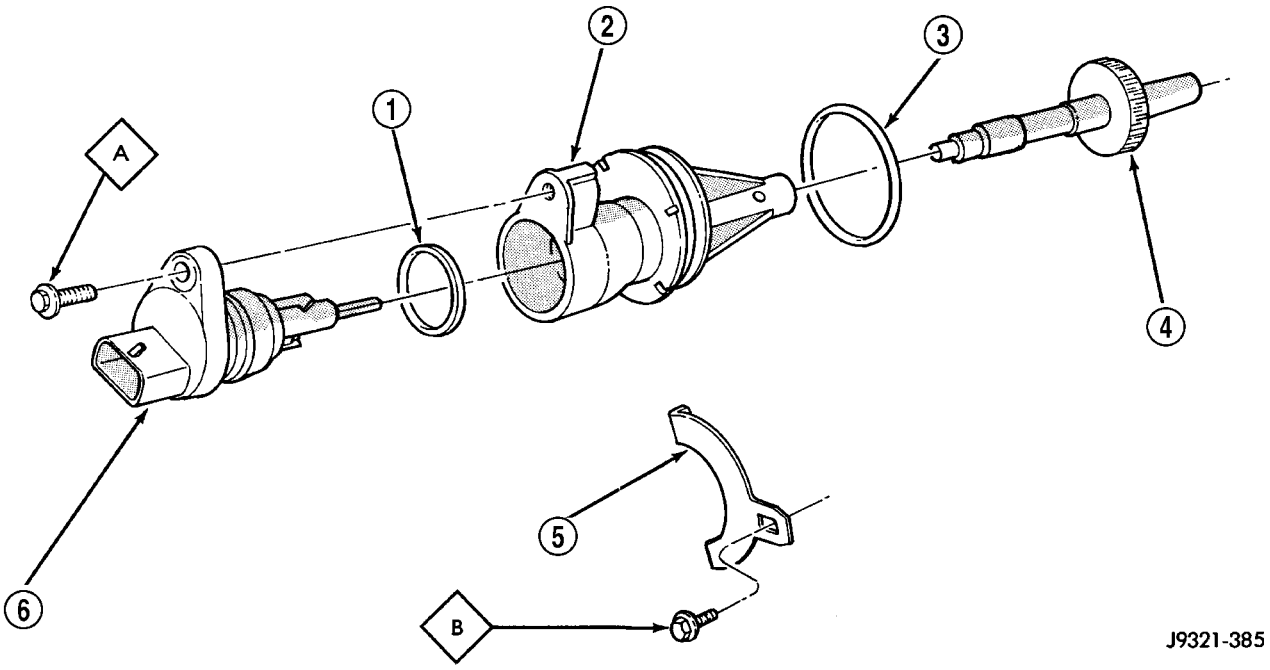


Fig. 98 Speedometer

1 - Sensor O-ring
2 - Speedometer Adapter
3 - Adapter O-ring
4 - Speedometer Pinion
5 - Adapter Clamp
6 - Vehicle Speed Sensor
A - 2-3 N·m (15-27 in. lbs.)
B - 10-12 N·m (90-110 in. lbs.)

(11) Connect wires to vehicle speed sensor.

(12) Lower vehicle and top off transmission fluid level if necessary.

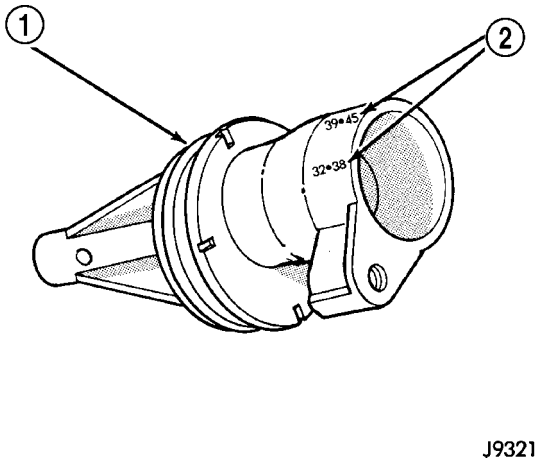


Fig. 99 Location Of Index Numbers On Speedometer Adapter

1 - SPEEDOMETER ADAPTER
2 - INDEX NUMBER LOCATION

VEHICLE SPEED SENSOR

DESCRIPTION

The 3-wire Vehicle Speed Sensor (VSS) is located on the speedometer pinion gear adapter. If equipped with 4WD, this adapter is located on the extension housing of the transfer case (drivers side). If equipped with 2WD, this adapter is located on the left side of the transmission extension housing.

OPERATION

The VSS is a 3-circuit (3–wire), magnetic, hall-effect sensor.

The 3 circuits are:

- A 5-volt power supply from the Powertrain Control Module (PCM).
- A ground is provided for the sensor though a low-noise sensor return circuit in the PCM.
- An input to the PCM is used to determine vehicle speed and distance traveled.

The speed sensor generates 8 pulses per sensor revolution. These signals, in conjunction with a closed throttle signal from the throttle position sensor, indicate a closed throttle deceleration to the PCM. When the vehicle is stopped at idle, a closed throttle signal is received by the PCM (but a speed sensor signal is not received).

Under deceleration conditions, the PCM adjusts the Idle Air Control (IAC) motor to maintain a desired MAP value. Under idle conditions, the PCM adjusts the IAC motor to maintain a desired engine speed.

VEHICLE SPEED SENSOR (Continued)

REMOVAL

The Vehicle Speed Sensor (VSS) is located on the speedometer pinion gear adapter. If equipped with 4WD, this adapter is located on the transfer case extension (left side) (Fig. 100). If equipped with 2WD, this adapter is located on the extension housing of the transmission (left side).

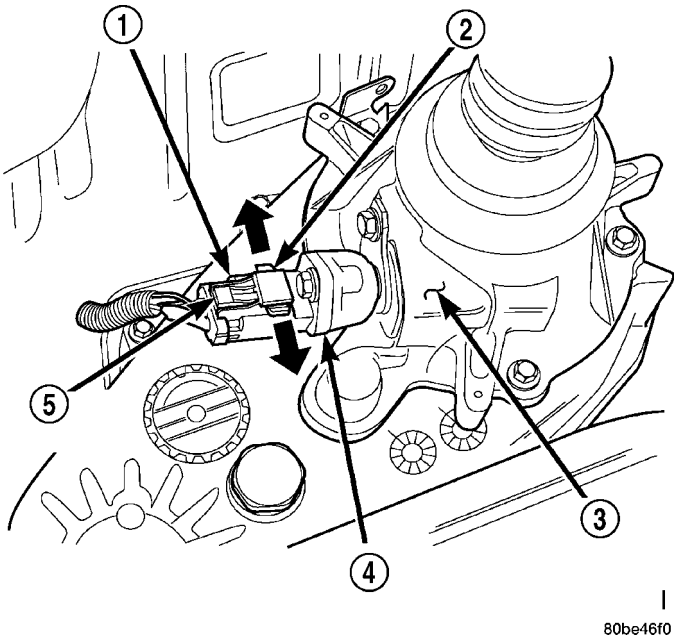


Fig. 100 VSS Location

1 - SENSOR ELECTRICAL CONNECTOR
2 - SLIDE TAB
3 - 4WD TRANSFER CASE EXTENSION
4 - VEHICLE SPEED SENSOR
5 - RELEASE LOCK

(1) Raise and support vehicle.

(2) Disconnect electrical connector from sensor by pushing slide tab (Fig. 100). After slide tab has been positioned, push in on secondary release lock (Fig. 100) on side of connector and pull connector from sensor.

(3) Remove sensor mounting bolt (Fig. 101).

(4) Remove sensor (pull straight out) from speedometer pinion gear adapter (Fig. 101). Do not remove gear adapter from transmission.

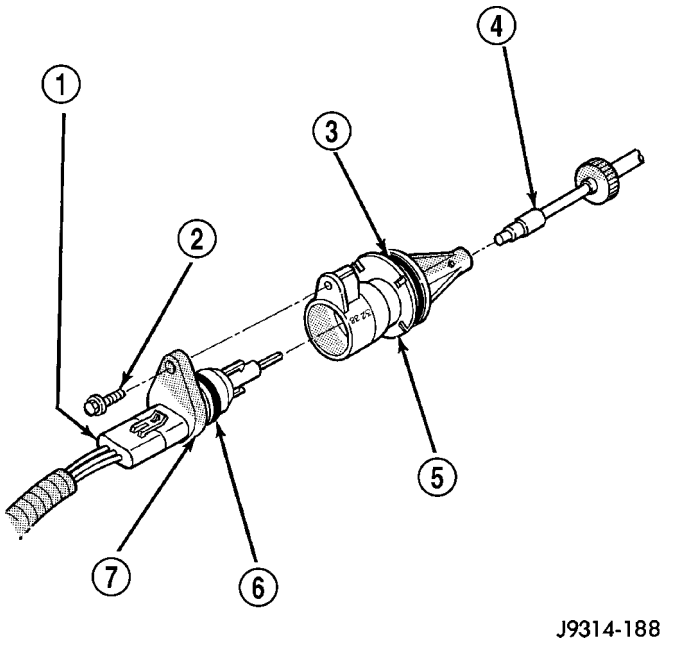


Fig. 101 VSS Removal/Installation

1 - ELECTRICAL CONNECTOR
2 - SENSOR MOUNTING BOLT
3 - O-RING
4 - SPEEDOMETER PINION GEAR
5 - SPEEDOMETER PINION GEAR ADAPTER
6 - O-RING
7 - VEHICLE SPEED SENSOR

INSTALLATION

(1) Clean inside of speedometer pinion gear adapter before installing speed sensor.

(2) Install sensor into speedometer gear adapter and install mounting bolt. Before tightening bolt, verify speed sensor is fully seated (mounted flush) to speedometer pinion gear adapter.

(3) Tighten sensor mounting bolt to 2.2 N·m (20 in. lbs.) torque.

(4) Connect electrical connector to sensor.

TRANSFER CASE - NV241

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TRANSFER CASE - NV241

DESCRIPTION

The NV241 transfer case (Fig. 1) is a part-time transfer case with a low-range gear system. It provides three operating ranges plus a NEUTRAL position. The low range position provides a gear reduction ratio of 4.0:1 for increased low speed torque capability.

The gear cases, retainer and extension are all of aluminum. Drive sprockets and an interconnecting drive chain are used to transmit engine torque to the front/rear propeller shafts. The mainshaft, input gear and front output shaft are supported by ball and needle bearings.

IDENTIFICATION

An identification tag (Fig. 2) is attached to the rear case of every transfer case. The tag provides the transfer case model number, assembly number, serial number, and low range ratio.

The transfer case serial number also represents the date of build.

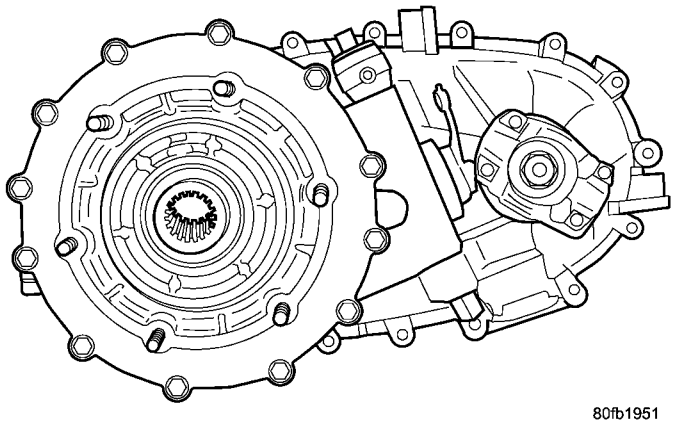


Fig. 1 NV241 - Front View

OPERATION

OPERATING RANGES

Transfer case operating ranges are:

- 2H (2-wheel drive)
- 4H (4-wheel drive)
- 4LO (4-wheel drive low range

The 2H range is for use on any road surface at any time.

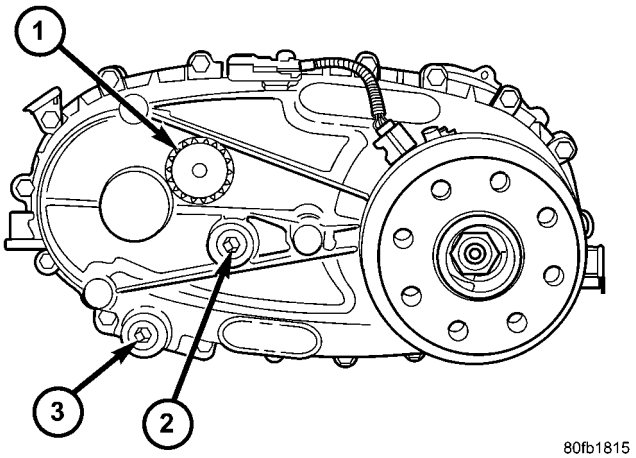


Fig. 2 Transfer Case Identification Tag and Fill/Drain Plugs

1 - IDENTIFICATION TAG
2 - FILL PLUG
3 - DRAIN PLUG

The 4H and 4LO ranges are for off road use only. They are not for use on hard surface roads. The only exception being when the road surface is covered by ice and snow.

The low range reduction gear system is operative in 4LO range only. This range is for extra pulling power in off road situations. Low range reduction ratio is 4.0:1.

SHIFT MECHANISM

The transfer case is operated by an adjustable floor mounted shift linkage. The transfer case shift lever is directly attached to the shift sector. The sector operates the range and mode forks within the transfer case.

A straight line shift pattern is used with a NEUTRAL detent. Lever range positions are imprinted in the shift knob.

SHIFTING

The vehicle must have the transmission placed in NEUTRAL, or the clutch depressed in the case of a manual transmission, and be moving less than 2-3 MPH when shifting into the 4L operating range.

DIAGNOSIS AND TESTING - TRANSFER CASE

Before beginning repair on a suspected transfer case malfunction, check all other driveline components beforehand.

The actual cause of a problem may be related to such items as: front hubs, axles, propeller shafts, wheels and tires, transmission, or clutch instead. If all other driveline components are in good condition and operating properly, refer to the Diagnosis Chart for further information.

DIAGNOSIS CHART

Condition	Possible Cause	Correction
Transfer Case difficult to shift or will not shift into desired range.	1) Vehicle speed too great to permit shifting.	1) Stop vehicle and shift into desired range. Or, reduce speed to below 3-4 km/h (2-3 mph) before attempting the shift.
	2) If vehicle was operated for an extended period in 4H on a dry paved surface, the driveline torque load may be causing a bind.	2) Stop vehicle and shift the transmission into neutral. Shift the transfer case to 2H and operate vehicle in 2H on dry paved surfaces.
	3) Transfer case external shift linkage binding.	3) Lubricate, repair, or replace linkage bushings, or tighten loose components as necessary.
	4) Insufficient or incorrect lubricant.	4) Drain and refill to edge of fill hole with Mopar® ATF +4, type 9602, Automatic Transmission fluid.
	5) Internal components binding, worn, or damaged.	5) Disassemble the transfer case and replace worn or damaged components as necessary.

TRANSFER CASE - NV241 (Continued)

Condition	Possible Cause	Correction
Transfer Case noisy in all operating ranges.	1) Insufficient or incorrect lubricant.	1) Drain and refill to edge of fill hole with Mopar® ATF +4, type 9602, Automatic Transmission fluid.
Noisy in, or jumps out of, four wheel drive low range.	1) Transfer case not completely engaged in 4L position.	1) With the transmission in NEUTRAL, or the clutch depressed in the case of a manual transmission and the vehicle moving under 3-4 km/h (2-3 mph), shift the transfer case to NEUTRAL and then shift into the 4L position.
	2) Shift linkage out of adjustment.	2) Adjust linkage.
	3) Shift linkage loose or binding.	3) Tighten, lubricate, or repair linkage as necessary.
	4) Range fork damaged, inserts worn, or fork is binding on the shift rail.	4) Disassemble unit and repair as necessary.
	5) Low range gear worn or damaged.	5) Disassemble unit and repair as necessary.
Lubricant leaking from output shaft seal or vent.	1) Transfer case overfilled.	1) Drain lubricant to the correct level.
	2) Vent closed or restricted.	2) Clear or replace vent as necessary.
	3) Output shaft seals damaged or installed incorrectly.	3) Replace seal as necessary. Check to ensure that another component, the propeller shaft slip yoke for example, is not causing damage to seal.
Abnormal tire wear.	1) Extended operation on hard, dry surfaces in the 4H position.	1) Operate vehicle in the 2H position on hard, dry surfaces.

REMOVAL

(1) Raise and support vehicle.

(2) Remove skid plate, if equipped. (Refer to 13 - FRAMES & BUMPERS/FRAME/TRANSFER CASE SKID PLATE - REMOVAL)

(3) Position drain oil container under transfer case.

(4) Remove transfer case drain plug and drain lubricant into container.

(5) Disconnect vent hose and vacuum harness at transfer case switch.

(6) Disconnect shift rod from grommet in transfer case shift lever, or from floor shift arm whichever provides easy access. Use channel lock style pliers to press rod out of lever grommet.

(7) Support transmission with jack stand.

(8) Remove rear crossmember.

(9) Mark front and rear propeller shafts for assembly reference.

(10) Remove front and rear propeller shafts. (Refer to 3 - DIFFERENTIAL & DRIVELINE/PROPELLER SHAFT/PROPELLER SHAFT - REMOVAL)

(11) Support transfer case with suitable jack. Secure transfer case to jack with safety chains.

(12) Remove nuts attaching transfer case to transmission.

(13) Move transfer case assembly rearward until free of transmission output shaft.

(14) Lower jack and move transfer case from under vehicle.

DISASSEMBLY

Position transfer case in a shallow drain pan. Remove drain plug and drain any remaining lubricant remaining in case.

TRANSFER CASE - NV241 (Continued)

REAR EXTENSION, RETAINER, AND REAR CASE

(1) Install bolts into two threaded holes in the rear propeller shaft companion flange 180° apart (Fig. 3).

(2) Place the Holder 6719 (Fig. 3) over the two bolts and use the Holder to remove the rear companion flange nut.

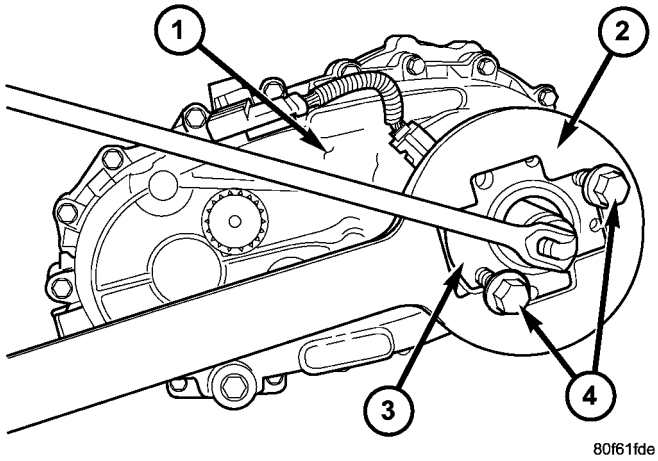


Fig. 3 Rear Companion Flange Nut Removal

1 - TRANSFER CASE
2 - HOLDER 6719
3 - COMPANION FLANGE
4 - BOLTS

(3) Remove the rear companion flange from the output shaft.

NOTE: The companion flange is a taper fit onto the output shaft. It may be necessary to use Puller 8992 to remove the companion flange.

(4) There is a seal slinger located on the rear of the companion flange that can be replaced if necessary. Use a suitable cold chisel to strike the slinger hub parallel to the centerline in several spots to expand the slinger hub. A new slinger can be driven onto the flange hub using a suitable pipe tool.

(5) Remove the speedometer sensor bolt (Fig. 4).

(6) Remove the speedometer sensor (Fig. 5) from the extension housing.

(7) Remove rear extension bolts (Fig. 6).

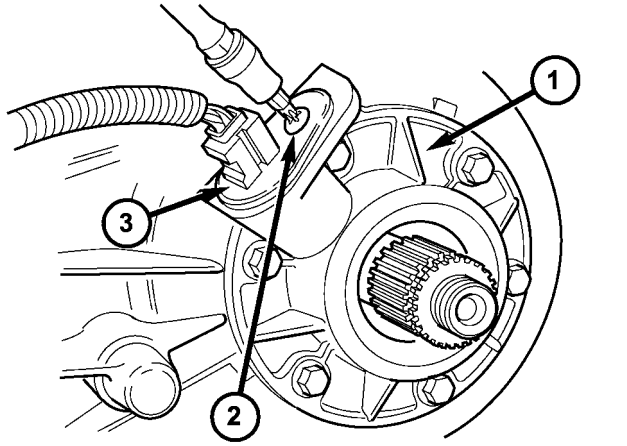


Fig. 4 Speedometer Sensor Bolt Removal

1 - EXTENSION HOUSING
2 - BOLT
3 - SPEEDOMETER SENSOR

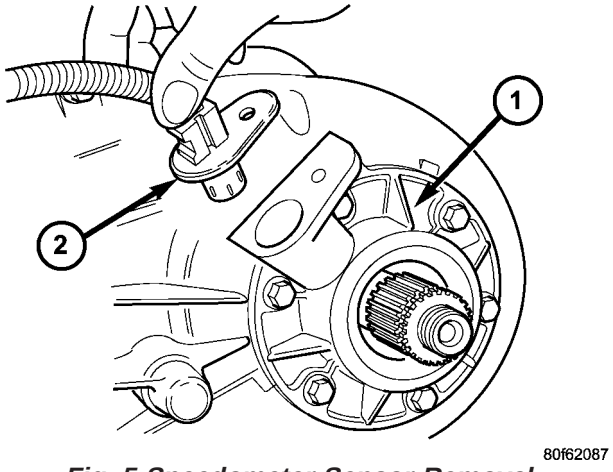


Fig. 5 Speedometer Sensor Removal

1 - EXTENSION HOUSING
2 - SPEEDOMETER SENSOR

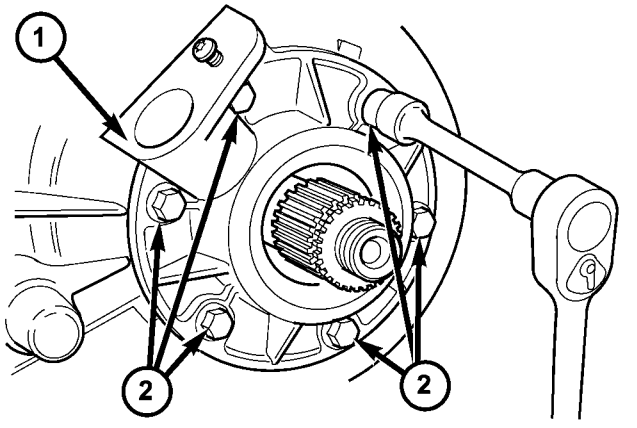


Fig. 6 Rear Extension Bolt Removal

1 - EXTENSION HOUSING
2 - BOLTS

TRANSFER CASE - NV241 (Continued)

(8) Remove rear extension housing (Fig. 7). Tap extension once or twice with a plastic mallet to break sealer bead and loosen it.

(9) Remove the speedometer sensor tone wheel (Fig. 8) from the output shaft.

(10) Remove the tone wheel dowel pin (Fig. 9) from the output shaft.

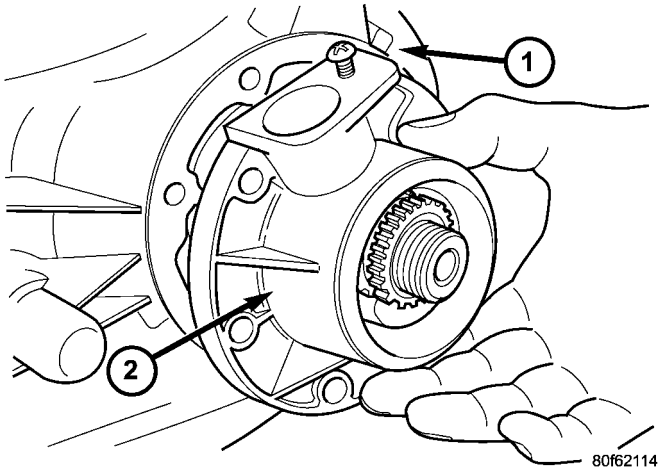


Fig. 7 Rear Extension Removal

1 - TRANSFER CASE
2 - EXTENSION HOUSING

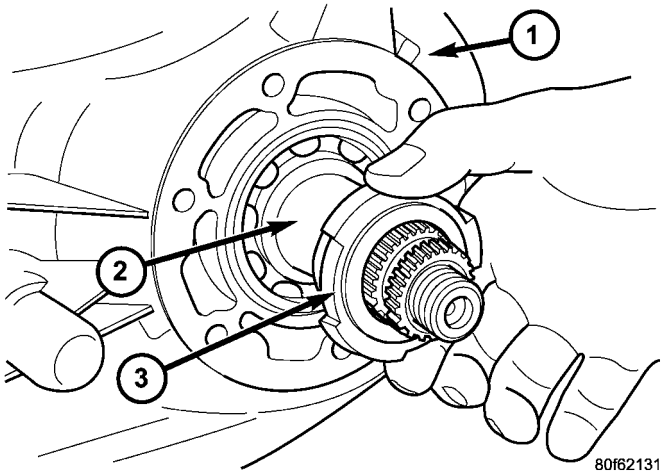


Fig. 8 Speedometer Sensor Tone Wheel Removal

1 - TRANSFER CASE
2 - OUPUT SHAFT
3 - SENSOR TONE WHEEL

YOKE AND SHIFT LEVER

(1) Shift transfer case into a 4WD mode.

(2) Use Holder 6958 to secure the yoke and remove the yoke nut (Fig. 10). Discard nut after removal. It is not reusable.

(3) Remove the yoke (Fig. 11) from the front output shaft.

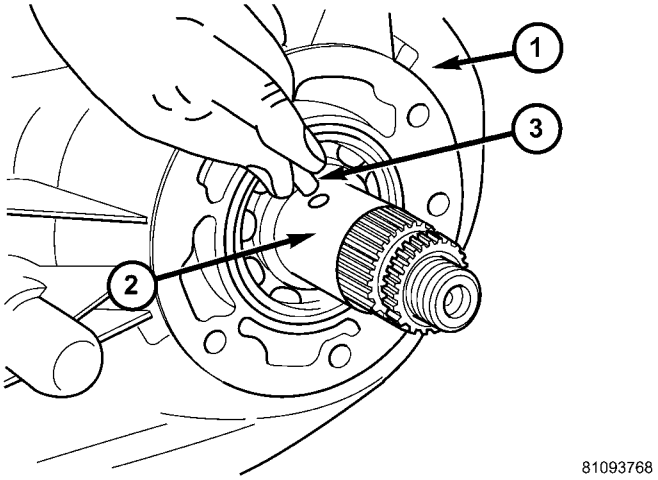


Fig. 9 Output Shaft Dowel Pin

1 - TRANSFER CASE
2 - OUPUT SHAFT
3 - DOWEL PIN

NOTE: The yoke is a taper fit onto the output shaft. It may be necessary to use a puller such as C-452 to remove the yoke.

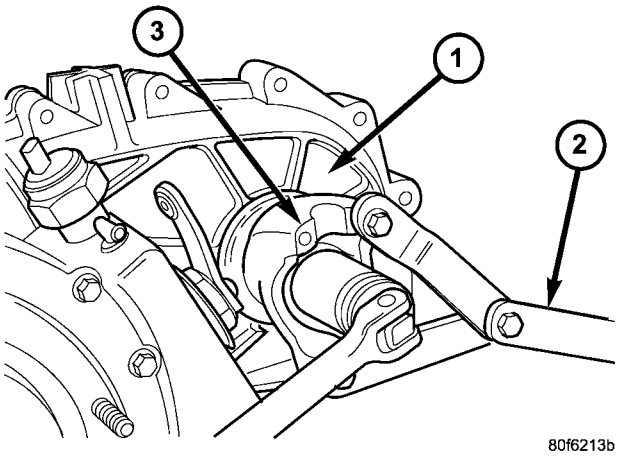


Fig. 10 Front Yoke Nut Removal

1 - TRANSFER CASE
2 - HOLDER 6958
3 - FRONT YOKE

(4) There is a seal slinger located on the rear of the yoke that can be replaced if necessary. Use a suitable cold chisel to strike the slinger hub parallel to the centerline in several spots to expand the slinger hub. A new slinger can be driven onto the yoke hub using a suitable pipe tool.

(5) Remove yoke rubber seal from front output shaft.

TRANSFER CASE - NV241 (Continued)

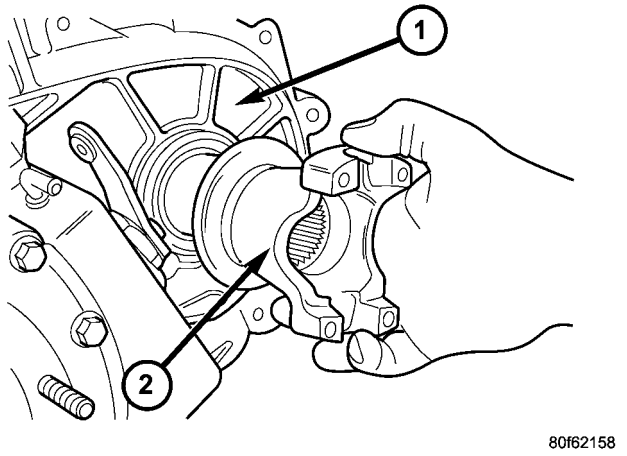


Fig. 11 Front Yoke Removal

1 - TRANSFER CASE
2 - FRONT YOKE

(6) Remove bolt (Fig. 12) that retains the shift lever to sector shaft.

(7) Remove the shift lever (Fig. 13) from the shift sector.

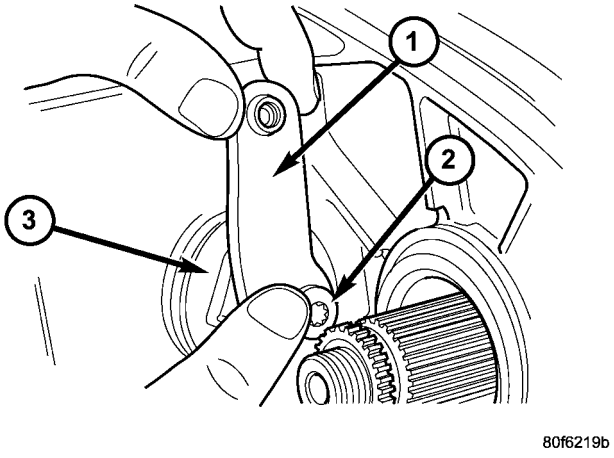


Fig. 12 Shift Lever Bolt Removal

1 - SHIFT LEVER
2 - BOLT
3 - SECTOR SUPPORT

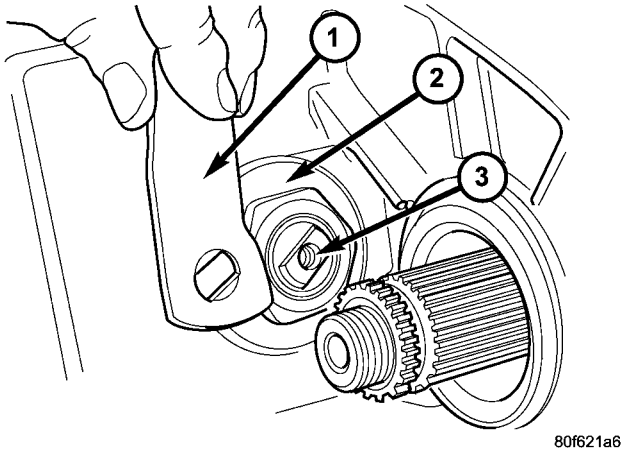


Fig. 13 Shift Lever Removal

1 - SHIFT LEVER
2 - SECTOR SUPPORT
3 - SHIFT SECTOR

(8) Remove the detent plug (Fig. 14).

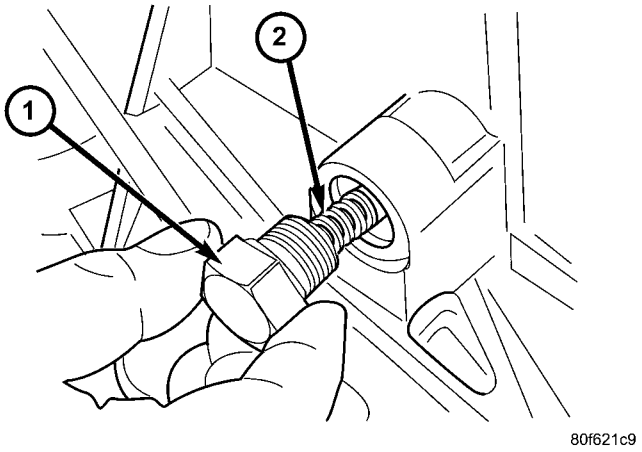


Fig. 14 Detent Plug Removal

1 - DETENT PLUG
2 - SPRING

TRANSFER CASE - NV241 (Continued)

(9) Remove the detent plug (Fig. 15), spring, and plunger from the transfer case.

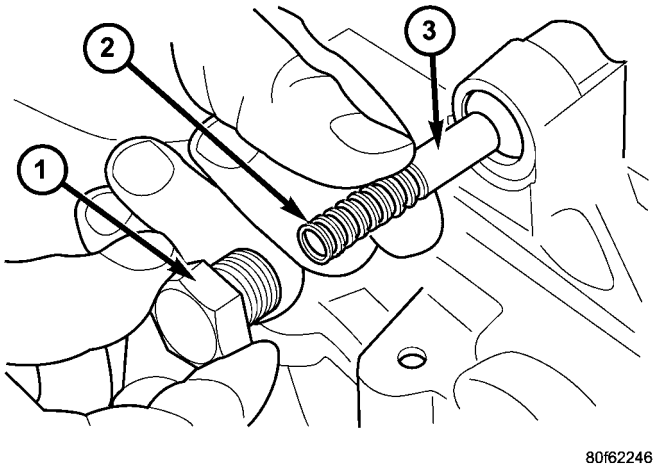


Fig. 15 Detent Plug, Spring, and Plunger Removal

1 - DETENT PLUG
2 - SPRING
3 - PLUNGER

(10) Remove the transfer case position sensor (Fig. 16).

(11) Remove the shift sector support (Fig. 17).

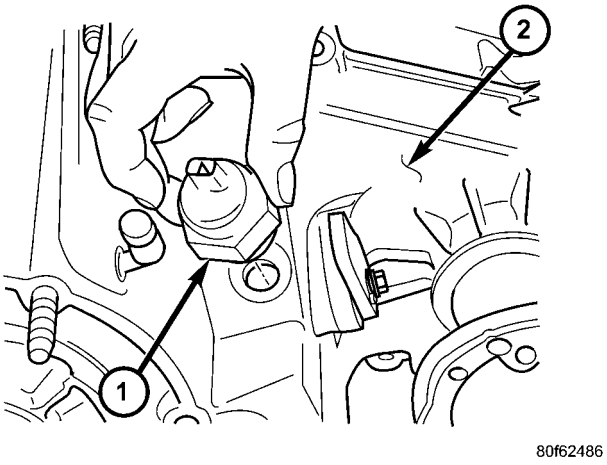


Fig. 16 Remove Transfer Case Position Sensor

1 - TRANSFER CASE POSITION SENSOR
2 - TRANSFER CASE

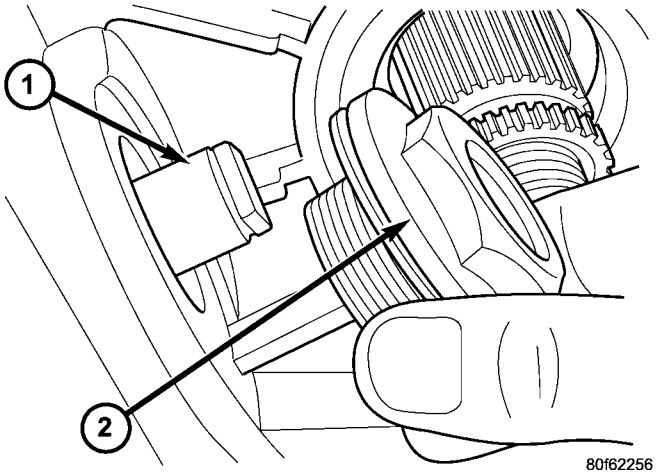


Fig. 17 Shift Sector Support Removal

1 - SHIFT SECTOR SHAFT
2 - SECTOR SUPPORT

MAINSHAFT, FRONT OUTPUT SHAFT, AND DRIVE CHAIN

(1) Note position of bolts that attach rear case to front case. Some bolts are unique. Mark position of these bolts with paint or scriber.

(2) Remove rear case-to-front case bolts (Fig. 18).

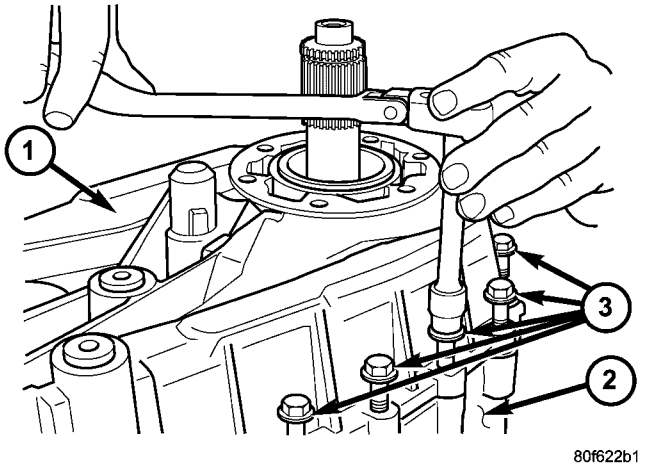


Fig. 18 Case Half Bolt Removal

1 - REAR CASE HALF
2 - FRONT CASE HALF
3 - BOLTS

(3) Loosen rear case with pry tool to break sealer bead.

TRANSFER CASE - NV241 (Continued)

(4) Unseat rear case from alignment dowels (Fig. 19).

(5) Remove rear case and oil pump assembly from front case (Fig. 20).

CAUTION: Do not remove the bolts holding the oil pump cover to the rear case half. The oil pump cover is aligned to the rear output shaft bearing inner race and will become mis-aligned if the bolts are loosened. If the transfer case failure has generated any debris which may have become trapped in the oil pump, the rear case and oil pump assembly MUST be replaced.

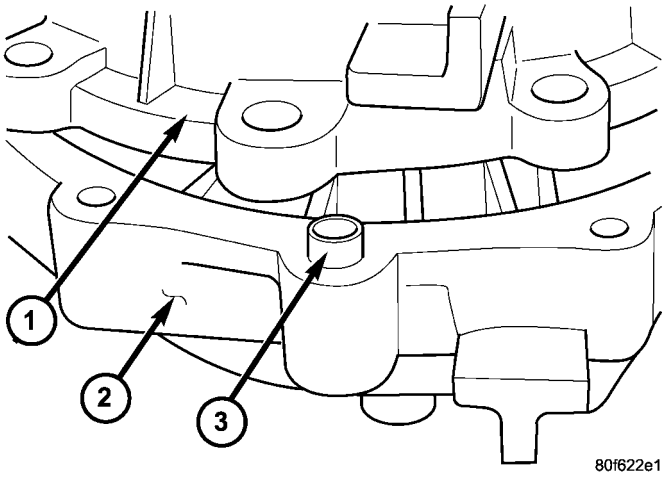


Fig. 19 Case Alignment Dowel

1 - REAR CASE HALF
2 - FRONT CASE HALF
3 - ALIGNMENT DOWEL

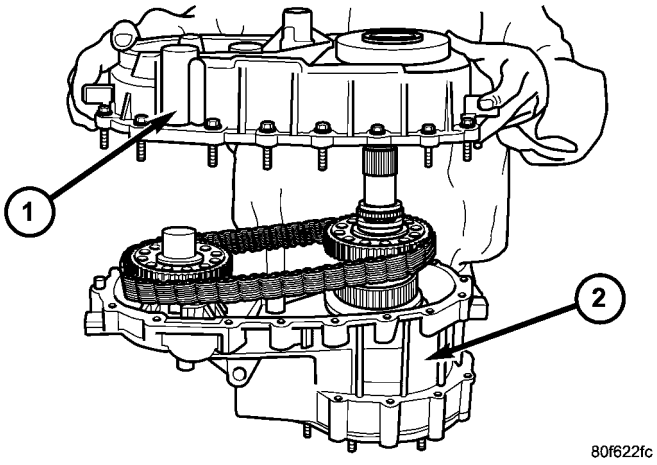


Fig. 20 Rear Case Half Removal

1 - REAR CASE HALF
2 - FRONT CASE HALF

(6) Remove shift rail cups and spring (Fig. 21).

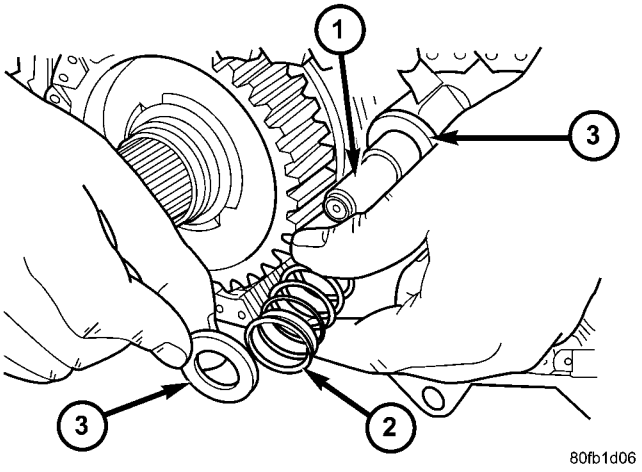


Fig. 21 Shift Rail Cup And Spring Removal

1 - SHIFT RAIL
2 - SPRING
3 - CUPS (2)

(7) Remove front drive sprocket retaining ring (Fig. 22).

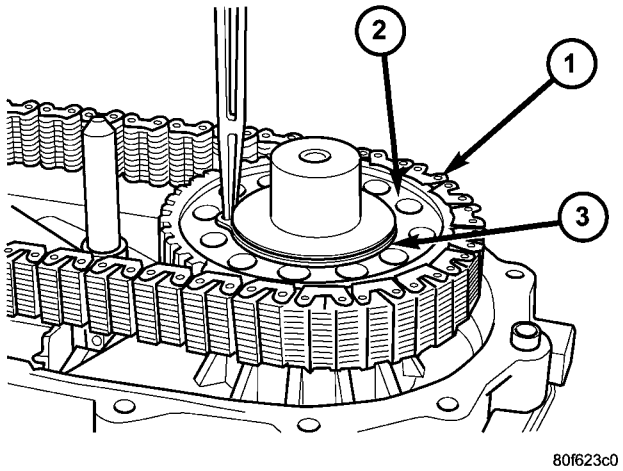


Fig. 22 Front Drive Sprocket Snap-Ring Removal

1 - DRIVE CHAIN
2 - FRONT DRIVE SPROCKET
3 - SNAP-RING

(8) Remove mainshaft, front sprocket, and chain as an assembly (Fig. 23).

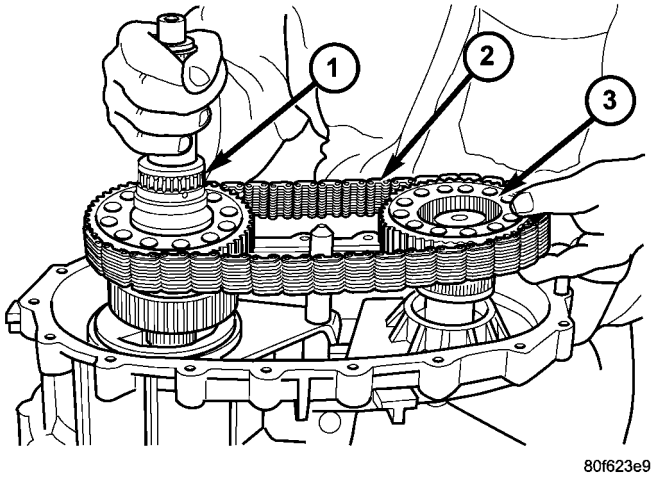


Fig. 23 Remove Mainshaft, Drive Chain, and Front Sprocket

1 - MAINSHAFT
2 - DRIVE CHAIN
3 - DRIVE SPROCKET

SHIFT FORKS

(1) Remove front output shaft from bearing in case (Fig. 24).

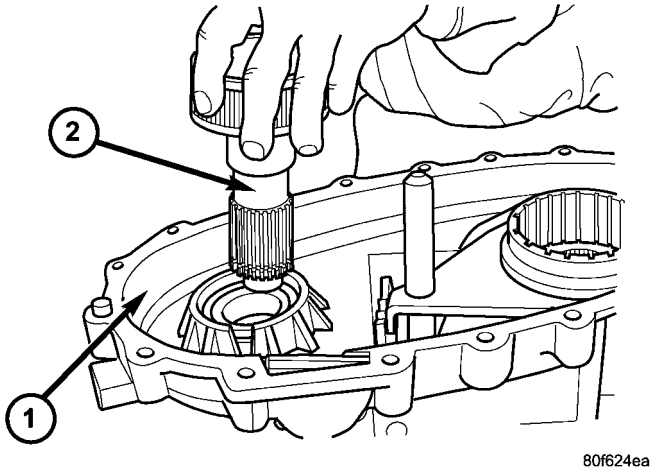


Fig. 24 Front Output Shaft Removal

1 - FRONT CASE HALF
2 - FRONT OUTPUT SHAFT

(2) Remove the shift rail (Fig. 25) from the shift forks.

(3) Remove mode fork, mode sleeve (Fig. 26). Note which way sleeve fits in fork (long side of sleeve goes to front).

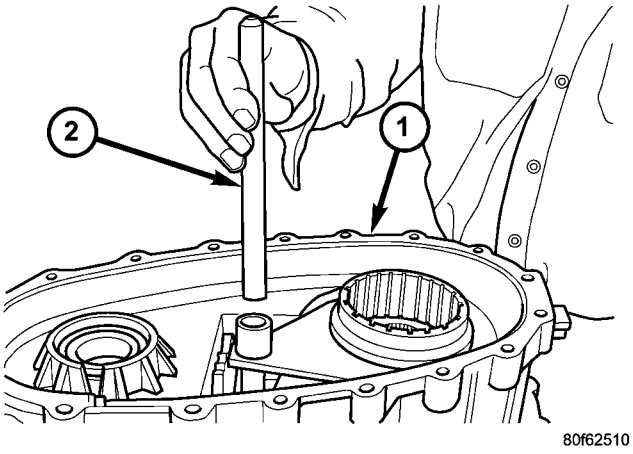


Fig. 25 Shift Fork Rail Removal

1 - FRONT CASE HALF
2 - SHIFT RAIL

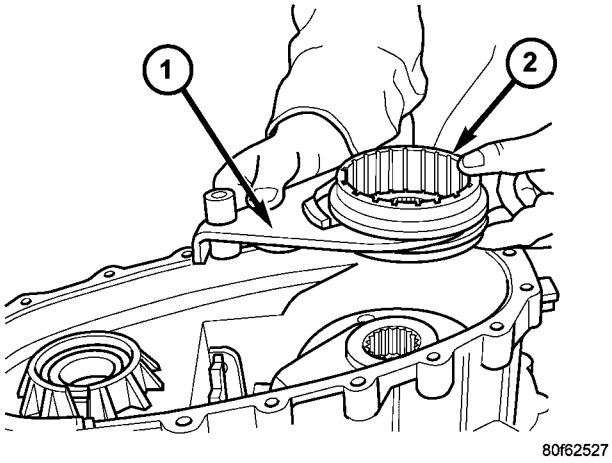


Fig. 26 Mode Fork and Sleeve Removal

1 - MODE FORK
2 - MODE SLEEVE

TRANSFER CASE - NV241 (Continued)

(4) Remove range fork and hub as an assembly (Fig. 27). Note fork position for installation reference.
(5) Remove shift sector (Fig. 28).

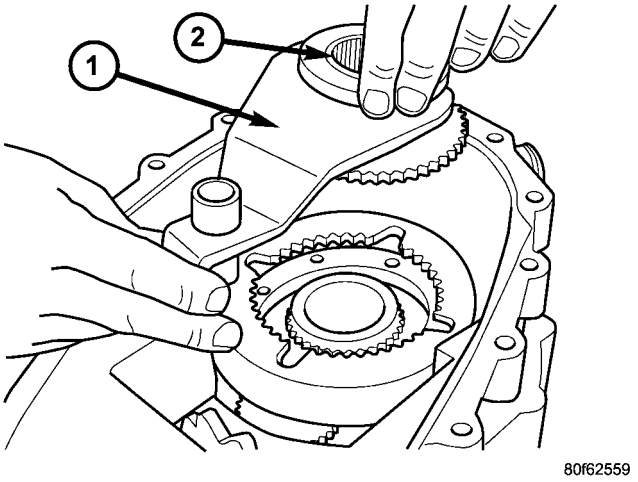


Fig. 27 Range Fork and Hub Removal

1 - RANGE FORK
2 - RANGE HUB

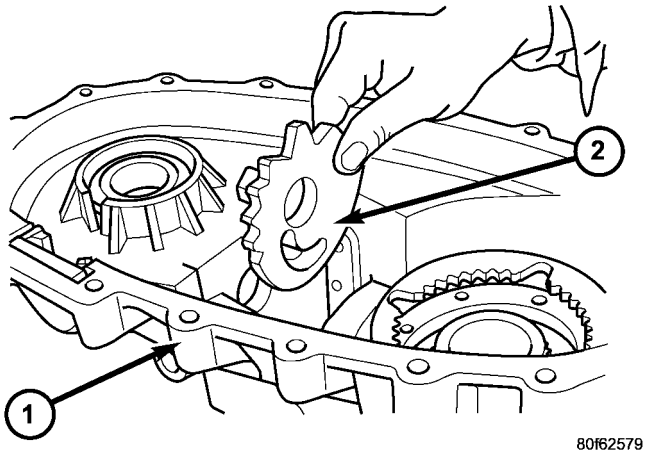


Fig. 28 Shift Sector Removal

1 - FRONT CASE HALF
2 - SHIFT SECTOR

MAINSHAFT

(1) Remove the snap-ring that secures the chain drive sprocket onto mainshaft (Fig. 29). Use standard (instead of parallel jaw) snap-ring pliers to remove this snap-ring.
(2) Remove drive sprocket (Fig. 30).

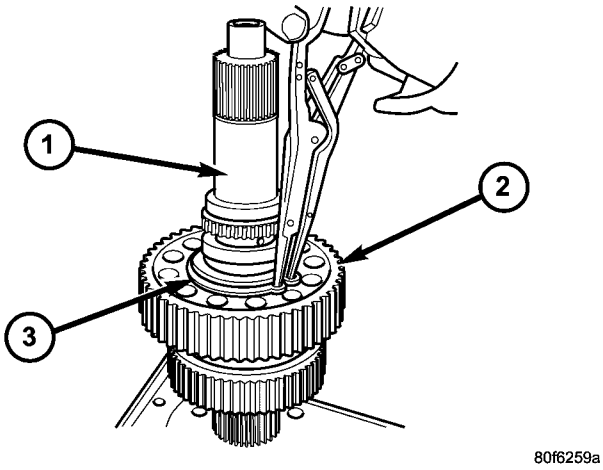


Fig. 29 Mainshaft Drive Sprocket Snap-Ring Removal

1 - MAINSHAFT
2 - DRIVE SPROCKET
3 - SNAP-RING

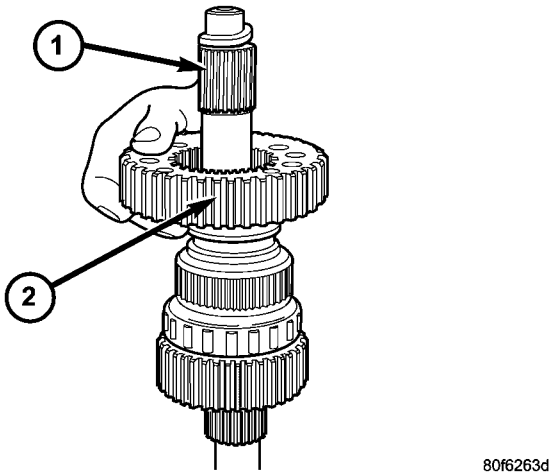


Fig. 30 Mainshaft Drive Sprocket Removal

1 - MAINSHAFT
2 - DRIVE SPROCKET

TRANSFER CASE - NV241 (Continued)

(3) Remove the sliding clutch gear (Fig. 31) from the mainshaft.

(4) Rotate the mainshaft 180° and remove the mode hub snap-ring (Fig. 32).

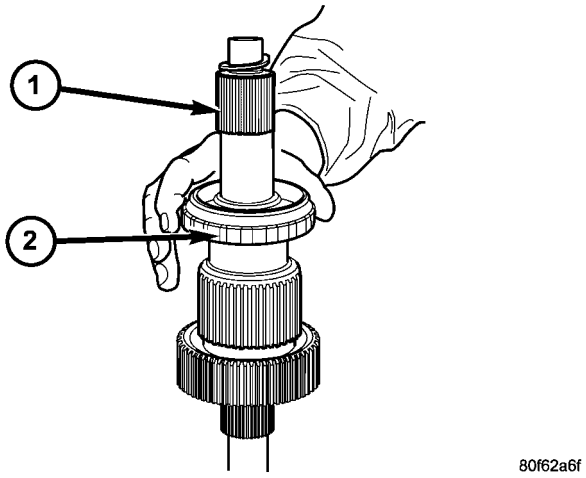


Fig. 31 Sliding Clutch Gear Removal

1 - MAINSHAFT
2 - SLIDING CLUTCH GEAR

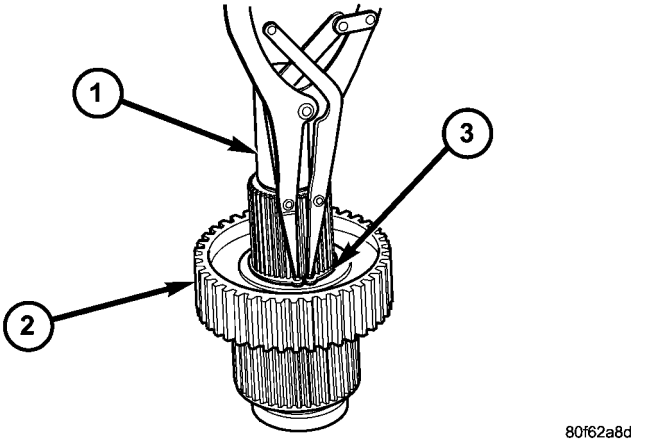


Fig. 32 Mode Hub Snap-Ring Removal

1 - MAINSHAFT
2 - MODE HUB
3 - SNAP-RING

(5) Remove the mode hub (Fig. 33) from the mainshaft.

(6) Remove the drive sprocket drive hub (Fig. 34) from the mainshaft.

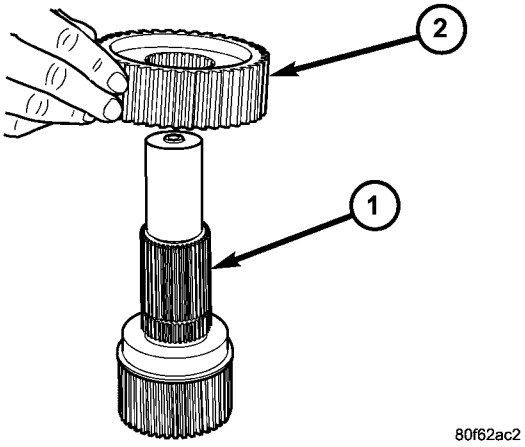


Fig. 33 Mode Hub Removal

1 - MAINSHAFT
2 - MODE HUB

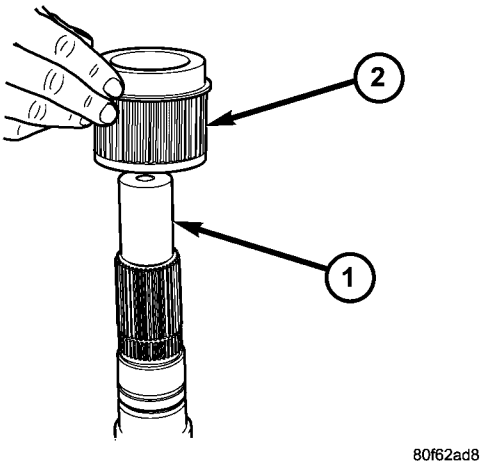


Fig. 34 Drive Sprocket Drive Hub Removal

1 - MAINSHAFT
2 - DRIVE SPROCKET DRIVE HUB

TRANSFER CASE - NV241 (Continued)

INPUT AND PLANETARY GEAR

(1) Remove front input retainer attaching bolts (Fig. 35).

(2) Remove the input gear seal with a suitable screw and slide hammer.

(3) Remove the input gear snap-ring (Fig. 36) from the input retainer.

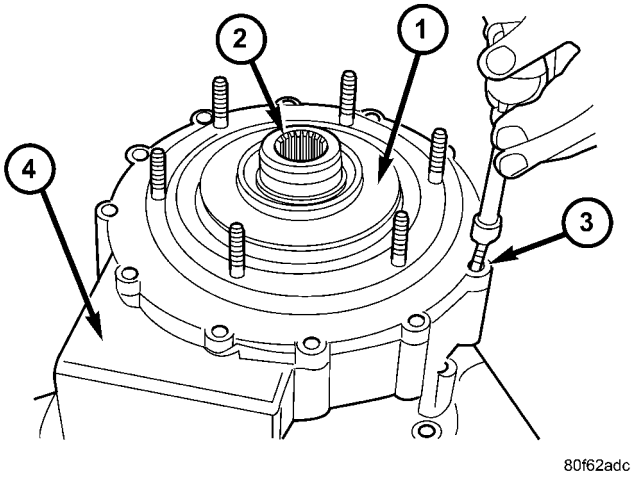


Fig. 35 Front Input Retainer Bolt Removal

1 - INPUT RETAINER
2 - INPUT GEAR
3 - BOLT
4 - FRONT CASE HALF

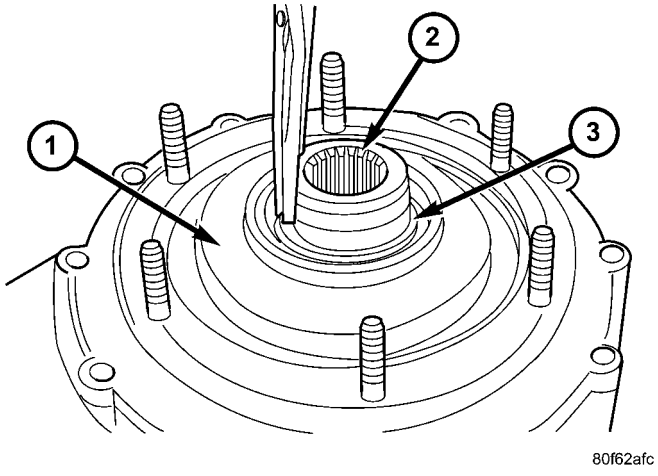


Fig. 36 Input Gear Snap-Ring Removal

1 - INPUT RETAINER
2 - INPUT GEAR
3 - SNAP-RING

(4) Remove the input gear (Fig. 37) from the input gear bearing with suitable hammer.

(5) Remove front input retainer (Fig. 38).

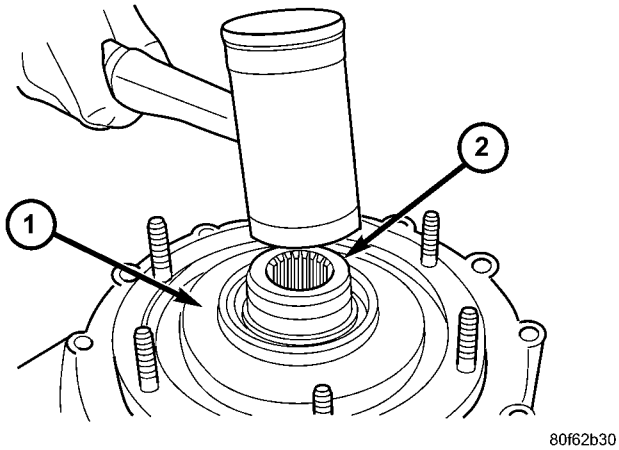


Fig. 37 Input Gear Removal

1 - INPUT RETAINER
2 - INPUT GEAR

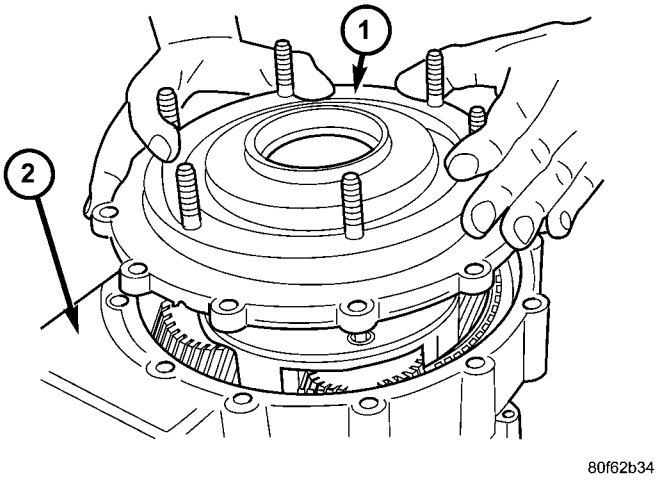


Fig. 38 Front Input Retainer Removal

1 - INPUT RETAINER
2 - FRONT CASE HALF

TRANSFER CASE - NV241 (Continued)

(6) Remove the low range planetary plastic spacer (Fig. 39) from the planetary carrier or back side of the input retainer.

(7) Remove the low range planetary gear set (Fig. 40) from the front case half.

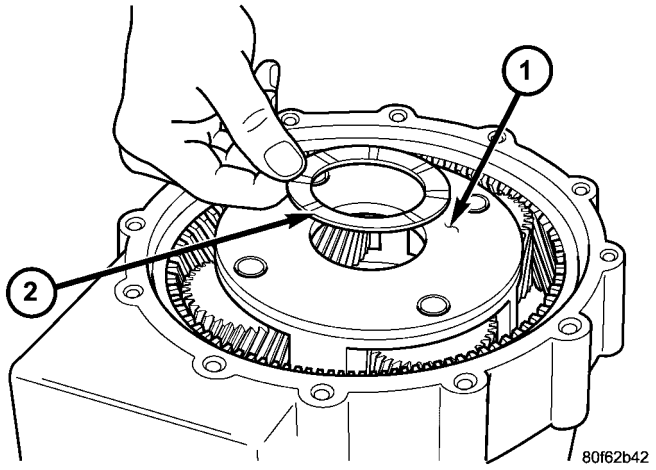


Fig. 39 Low Range Planetary Plastic Spacer

1 - LOW RANGE PLANETARY
2 - PLASTIC SPACER

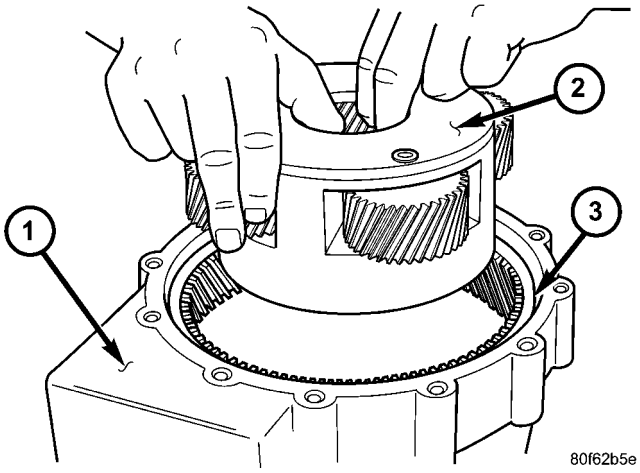


Fig. 40 Low Range Planetary Removal

1 - FRONT CASE HALF
2 - LOW RANGE PLANETARY
3 - ANNULUS GEAR

(8) Remove the annulus gear (Fig. 41) from the front case half.

(9) Support the input retainer on Cup 8148 as shown (Fig. 42).

(10) While using suitable snap-ring pliers to spread the input gear bearing snap-ring (Fig. 43), press downward on the retainer to remove the bearing.

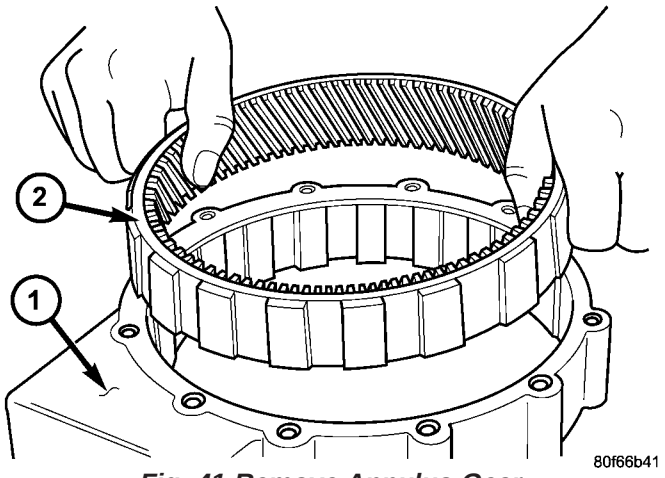


Fig. 41 Remove Annulus Gear

1 - FRONT CASE HALF
2 - ANNULUS GEAR

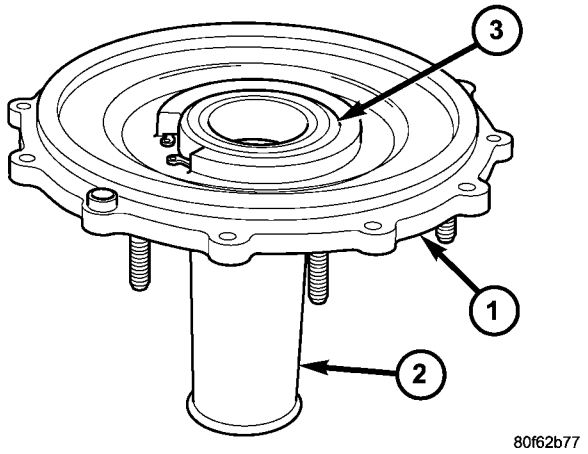


Fig. 42 Support Input Gear Bearing For Removal

1 - INPUT RETAINER
2 - CUP 8148
3 - INPUT GEAR BEARING

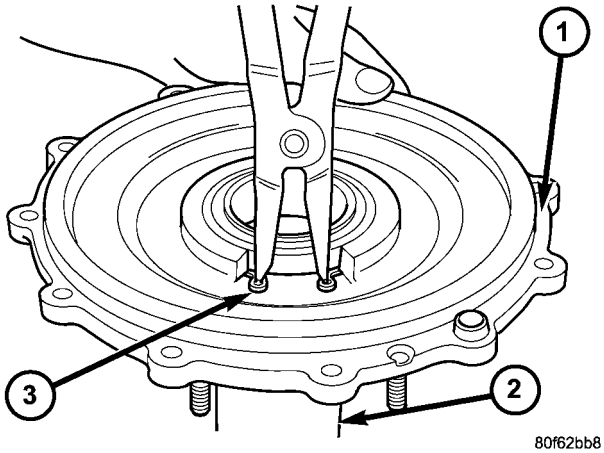


Fig. 43 Input Gear Bearing Removal

1 - INPUT RETAINER
2 - CUP 8148
3 - INPUT GEAR BEARING

TRANSFER CASE - NV241 (Continued)

(11) Remove the front output shaft bearing with Handle C-4171 and Installer 8239 (Fig. 44).

(12) Remove the rear output shaft bearing with Handle C-4171 and Installer 8239 (Fig. 45).

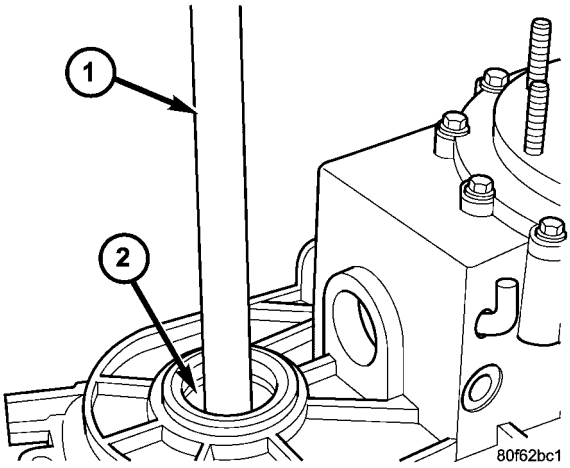


Fig. 44 Front Output Bearing Removal

1 - HANDLE C-4171
2 - INSTALLER 8239

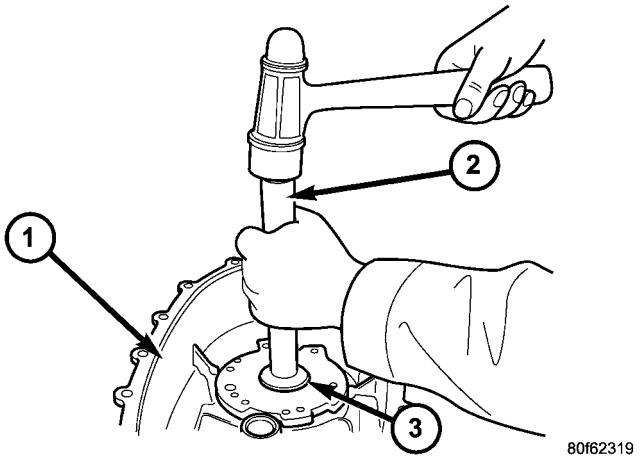


Fig. 45 Rear Output Bearing Removal

1 - REAR CASE HALF
2 - HANDLE C-4171
3 - INSTALLER 8239

CLEANING

Clean the transfer case parts with a standard parts cleaning solvent. Remove all traces of sealer from the cases and retainers with a scraper and 3M™ all purpose cleaner. Use compressed air to remove solvent residue from oil feed passages in the case halves, retainers, gears, and shafts.

INSPECTION

MAINSHAFT/SPROCKET/HUB

Inspect the splines on the hub and shaft and the teeth on the sprocket. Minor nicks and scratches can be smoothed with an oilstone, however, replace any part that is damaged.

Check the contact surfaces in the sprocket bore and on the mainshaft. Minor nicks and scratches can be smoothed with 320-400 grit emery cloth but do not try to salvage the shaft if nicks or wear is severe.

INPUT GEAR AND PLANETARY CARRIER

Check the teeth on the gear. Minor nicks can be dressed off with an oilstone but replace the gear if any teeth are broken, cracked, or chipped. The bearing surface on the gear can be smoothed with 300-400 grit emery cloth if necessary.

Examine the carrier body and pinion gears for wear or damage. The carrier will have to be replaced as an assembly if the body, pinion pins, or pinion gears are damaged. Inspect the plastic washer on the inside, bottom of the planetary carrier. The washer is trapped by the pinion gears and the complete planetary will need to be replaced if the washer is worn or damaged.

SHIFT FORKS/HUBS/SLEEVES

Check condition of the shift forks and mode fork shift rail. Minor nicks on the shift rail can be smoothed with 320-400 grit emery cloth.

Inspect the shift fork wear pads. The mode and range fork pads are serviceable and can be replaced if necessary.

Check both of the sleeves for wear or damage, especially on the interior teeth. Replace the sleeves if wear or damage is evident.

DRIVE CHAIN

Examine the drive chain and shaft bearings. replace the chain if stretched, distorted, or if any of the links bind. Replace the bearings if rough, or noisy.

LOW RANGE ANNULUS GEAR

Inspect annulus gear condition carefully. The gear can be serviced separately.

FRONT-REAR CASES

Inspect the cases for wear and damage. Replace the input retainer seal, do not reuse it.

Check case condition. If leaks were a problem, look for gouges and severe scoring of case sealing surfaces. Also make sure the front case mounting studs are in good condition.

TRANSFER CASE - NV241 (Continued)

Check the front case mounting studs and vent tube. The tube can be secured with Loctite™ 271 or 680 if loose. The stud threads can be cleaned up with a die if necessary. Also check condition of the fill/drain plug threads in the rear case. The threads can be repaired with a thread chaser or tap if necessary. Or the threads can be repaired with Helicoil™ stainless steel inserts if required.

OIL PUMP/OIL PICKUP

Examine the oil pump pickup parts. Replace the pump if any evidence of debris or damage exists. Do not disassemble the pump from the rear case half as individual parts are not available and it will not be possible to reassemble the pump cover to the rear case half. The pump is only available as part of the rear case half assembly. The pickup screen, hose, and tube are the only serviceable parts and are available separately.

ASSEMBLY

BEARINGS AND SEALS

(1) Position the input gear bearing in the input retainer bearing bore. Spread the snap-ring captured in the retainer with suitable snap-ring pliers and press the bearing into place. Verify that the snap-ring has engaged the groove in the bearing.

(2) Start front output shaft bearing in case (Fig. 46). Then seat bearing with Handle C-4171 and Installer 6953.

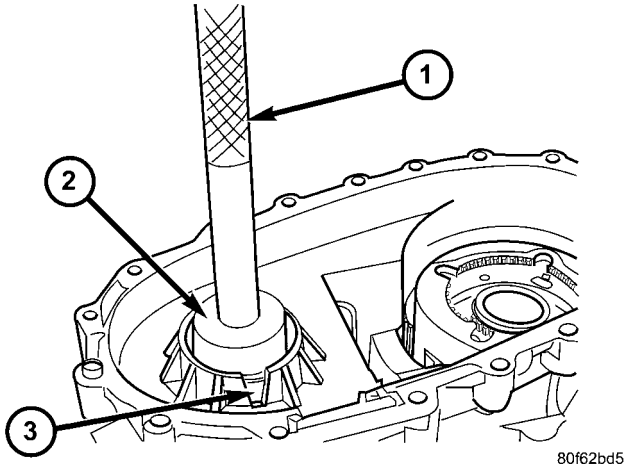


Fig. 46 Front Output Shaft Bearing Installation

1 - HANDLE C-4171
2 - INSTALLER 6953
3 - BEARING

(3) Install the front output shaft bearing retaining ring.

(4) Install the rear output shaft bearing into the rear case half with Installer 6953 and Handle C-4171 (Fig. 47).

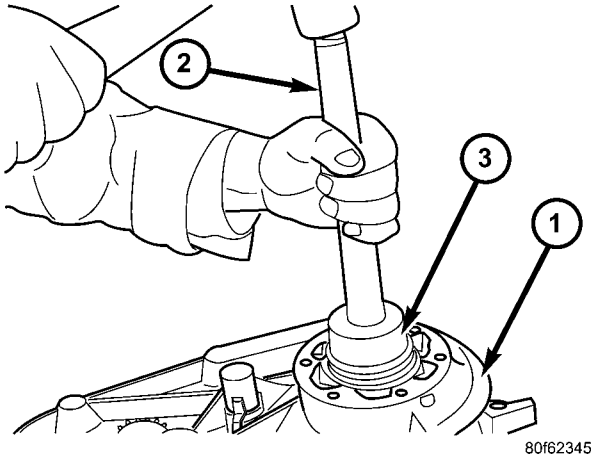


Fig. 47 Rear Output Bearing Installation

1 - REAR CASE HALF
2 - HANDLE C-4171
3 - INSTALLER 6953

(5) Remove the input gear pilot bearing using Puller Jaws 8240 and Cup 8148.

(6) Remove the input gear cup plug (Fig. 48) with a suitable drift and hammer inserted from the front of the input gear.

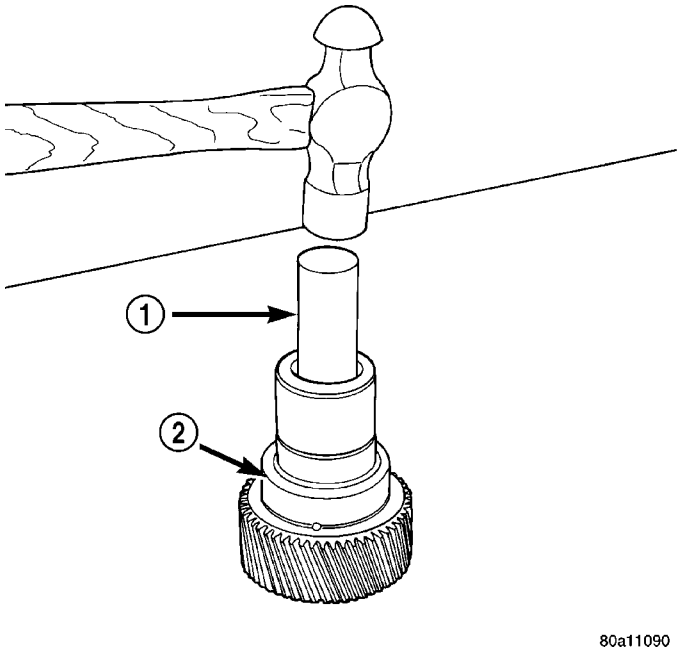


Fig. 48 Remove Input Gear Cup Plug

1 - DRIFT
2 - INPUT GEAR

(7) Install a new input gear cup plug with Installer 9045 and Handle C-4171.

TRANSFER CASE - NV241 (Continued)

(8) Install new input gear pilot bearing with Installer 9047 and Handle C-4171.

(9) Remove the front output shaft rear bearing with the screw and jaws from Remover L-4454 and Cup 8148 (Fig. 49).

(10) Install new bearing with Tool Handle C-4171 and Installer 5066 (Fig. 50). The bearing bore is chamfered at the top. Install the bearing so it is flush with the lower edge of this chamfer (Fig. 51).

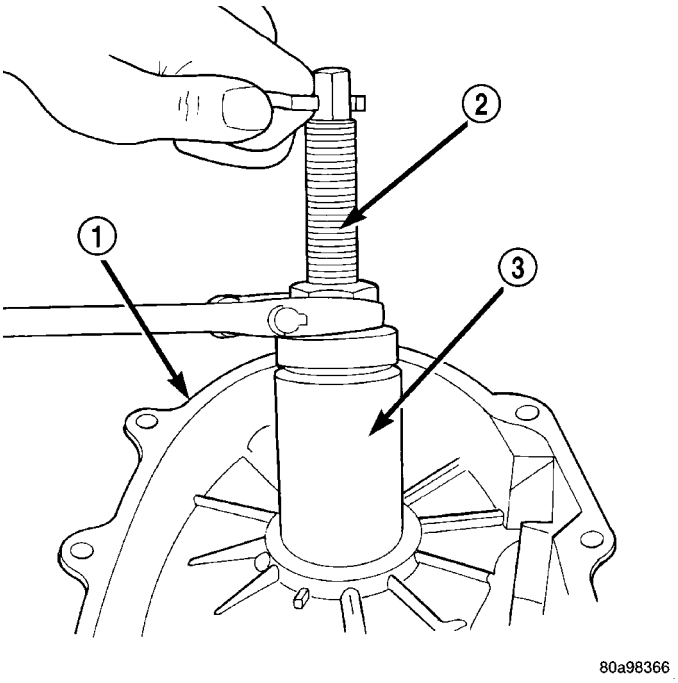


Fig. 49 Front Output Shaft Rear Bearing Removal

1 - REAR CASE
2 - SPECIAL TOOL L-4454-1 AND L-4454-3
3 - SPECIAL TOOL 8148

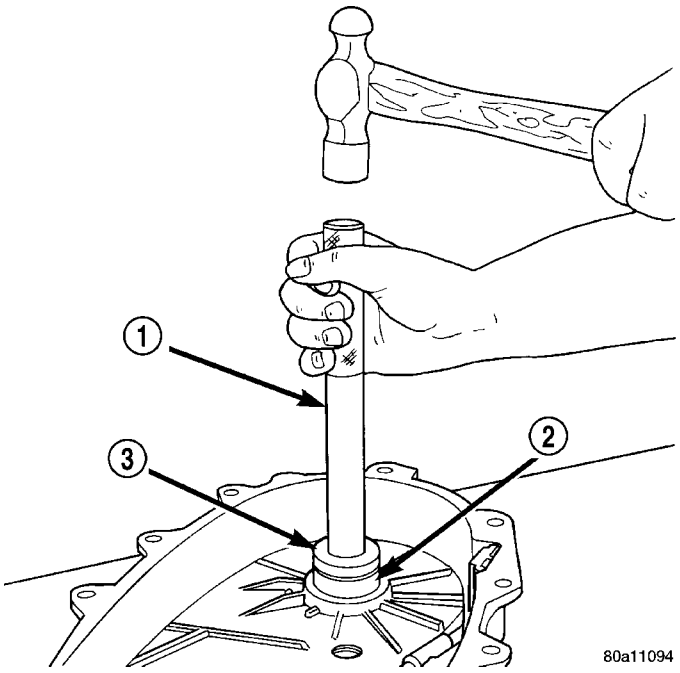


Fig. 50 Output Shaft Rear Bearing Installation

1 - HANDLE C-4171
2 - OUTPUT SHAFT INNER BEARING
3 - INSTALLER 5066

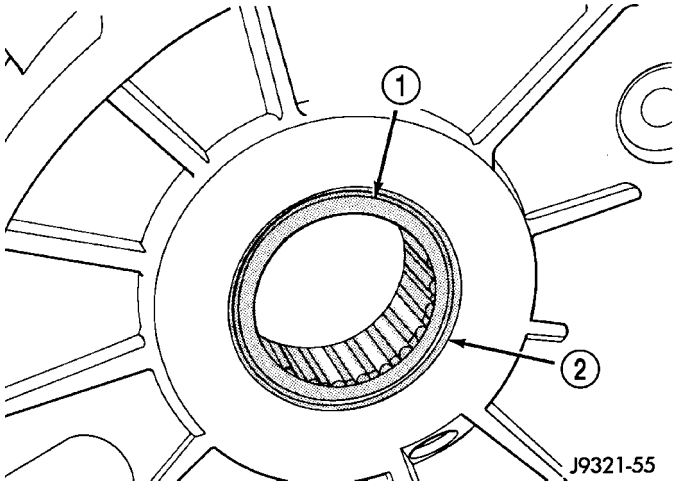


Fig. 51 Output Shaft Rear Bearing Installation Depth

1 - BEARING (SEATED) AT LOWER EDGE OF CHAMFER
2 - CHAMFER

TRANSFER CASE - NV241 (Continued)

INPUT AND PLANETARY GEAR

(1) Install the annulus gear (Fig. 52) into the front case half.

(2) Align and install low range planetary assembly into the front case half (Fig. 53). Be sure low range planetary pinions are engaged in annulus gear.

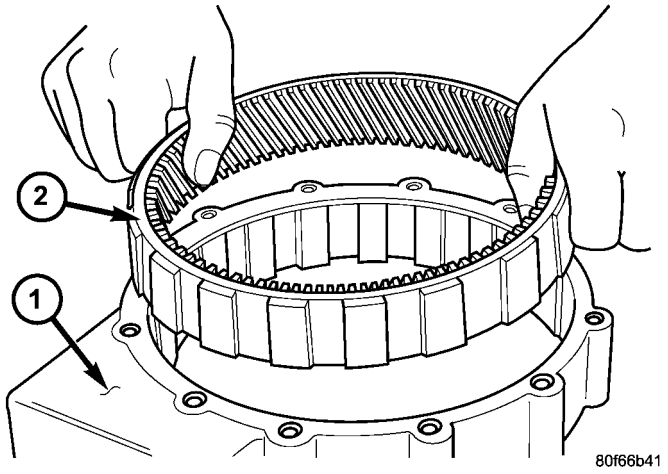


Fig. 52 Install Annulus Gear

1 - FRONT CASE HALF
2 - ANNULUS GEAR

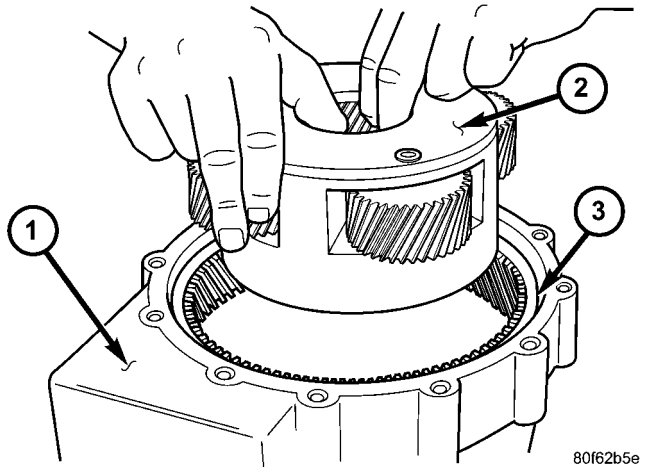


Fig. 53 Low Range Planetary Installation

1 - FRONT CASE HALF
2 - LOW RANGE PLANETARY
3 - ANNULUS GEAR

(3) Install the low range planetary plastic spacer (Fig. 54) onto the low range planetary carrier.

(4) Clean gasket sealer residue from retainer and inspect retainer for cracks or other damage.

(5) Apply a 3 mm (1/8 in.) bead of Mopar® Gasket Maker, or equivalent silicone adhesive, to sealing surface of retainer.

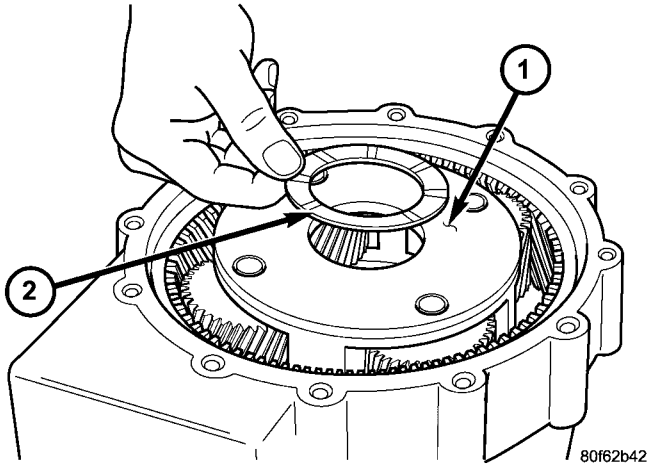


Fig. 54 Low Range Planetary Plastic Spacer

1 - LOW RANGE PLANETARY
2 - PLASTIC SPACER

(6) Install the front input retainer (Fig. 55) onto the front case half.

(7) Install bolts to hold retainer to transfer case (Fig. 56). Tighten to 21 N·m (16 ft. lbs.) of torque.

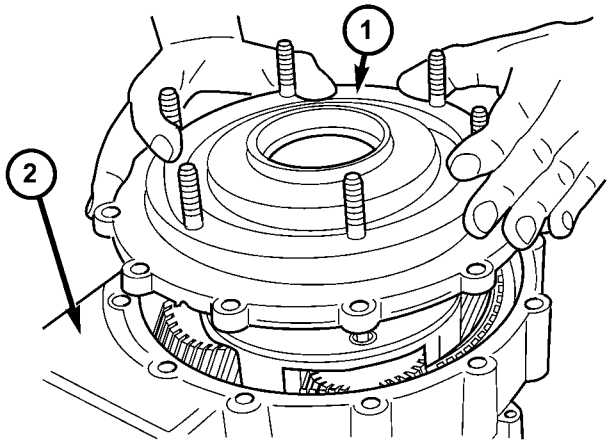


Fig. 55 Front Input Retainer Installation

1 - INPUT RETAINER
2 - FRONT CASE HALF

TRANSFER CASE - NV241 (Continued)

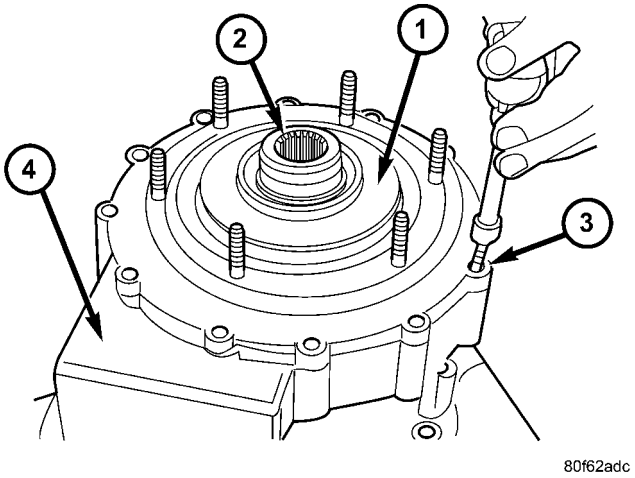


Fig. 56 Front Input Retainer Bolt Installation

1 - INPUT RETAINER
2 - INPUT GEAR
3 - BOLT
4 - FRONT CASE HALF

(8) Install the input gear to the low range planetary assembly and the input gear bearing.

(9) Install the input gear retaining snap-ring (Fig. 57) onto the input gear.

(10) Install new oil seal in front input retainer with Installer 8841 and Handle C-4171 (Fig. 58).

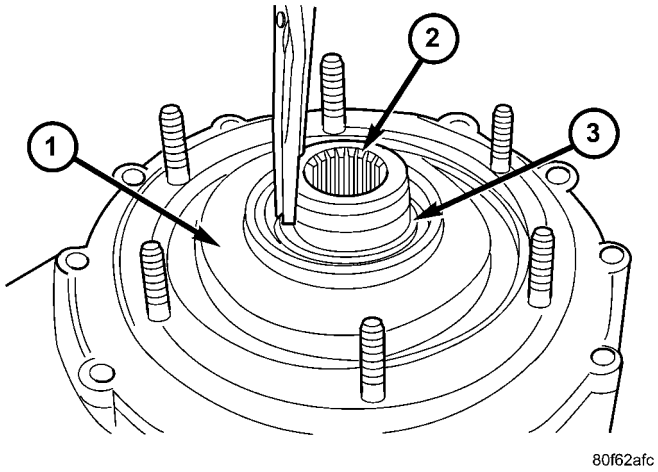


Fig. 57 Input Gear Snap-Ring Installation

1 - INPUT RETAINER
2 - INPUT GEAR
3 - SNAP-RING

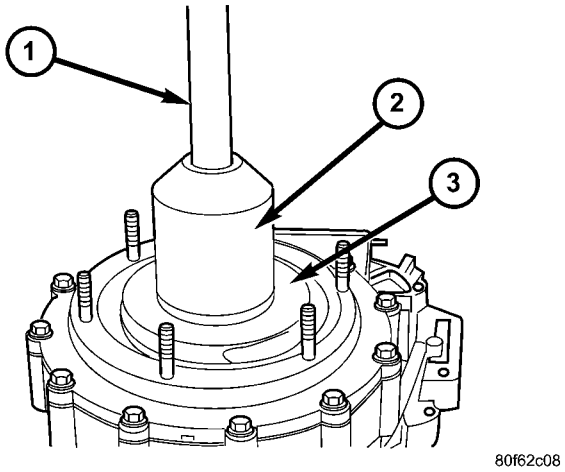


Fig. 58 Input Gear Seal Installation

1 - HANDLE C-4171
2 - INSTALLER 8841
3 - INPUT RETAINER

SHIFT FORKS AND MAINSHAFT

(1) Lubricate mainshaft splines with recommended transmission fluid.

(2) Install the drive sprocket drive hub onto the mainshaft (Fig. 59). Verify that the long shoulder is towards the front of the mainshaft.

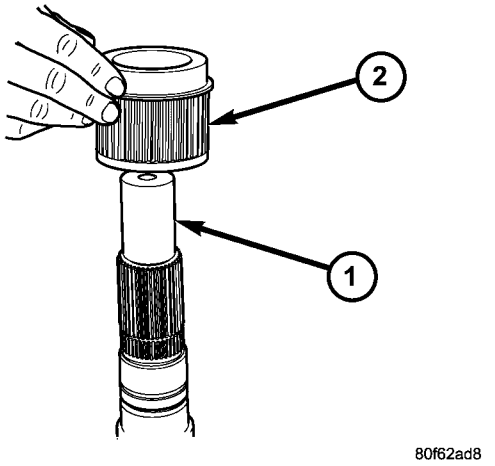


Fig. 59 Drive Sprocket Drive Hub Installation

1 - MAINSHAFT
2 - DRIVE SPROCKET DRIVE HUB

TRANSFER CASE - NV241 (Continued)

(3) Install the mode hub (Fig. 60) onto the mainshaft.

(4) Install the mode hub snap-ring (Fig. 61) onto the mainshaft.

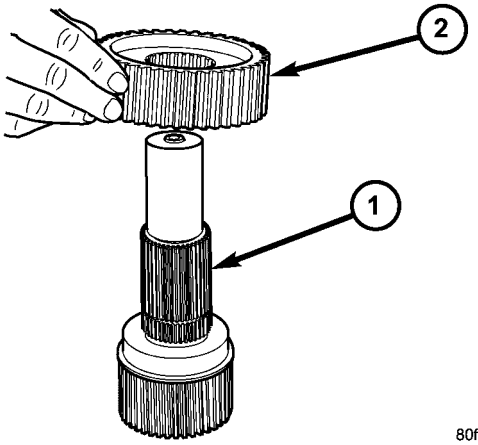


Fig. 60 Mode Hub Installation

1 - MAINSHAFT
2 - MODE HUB

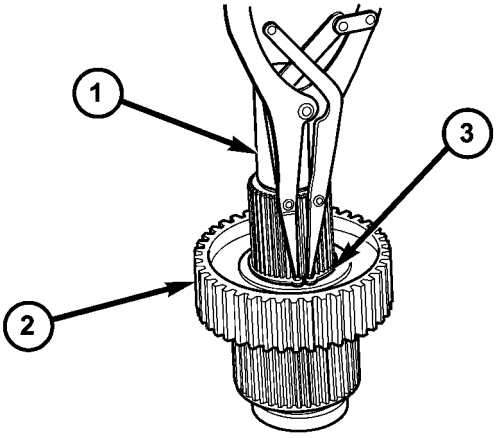


Fig. 61 Mode Hub Snap-Ring Installation

1 - MAINSHAFT
2 - MODE HUB
3 - SNAP-RING

(5) Rotate the mainshaft 180° and install the sliding clutch gear (Fig. 62) onto the drive sprocket drive hub.

(6) Install the mainshaft drive sprocket (Fig. 63) onto the drive hub.

(7) Install the mainshaft drive sprocket retaining ring (Fig. 64) onto the drive hub.

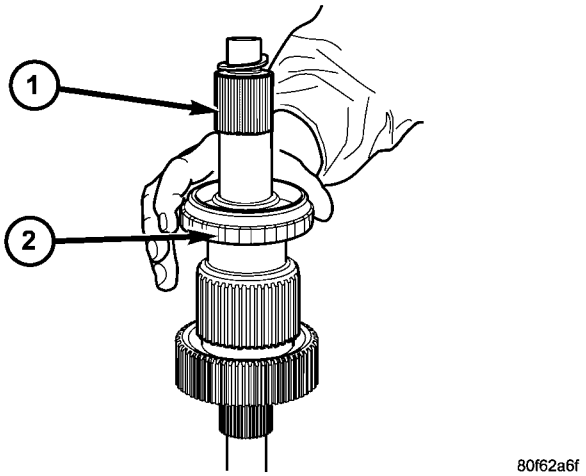


Fig. 62 Sliding Clutch Gear Installation

1 - MAINSHAFT
2 - SLIDING CLUTCH GEAR

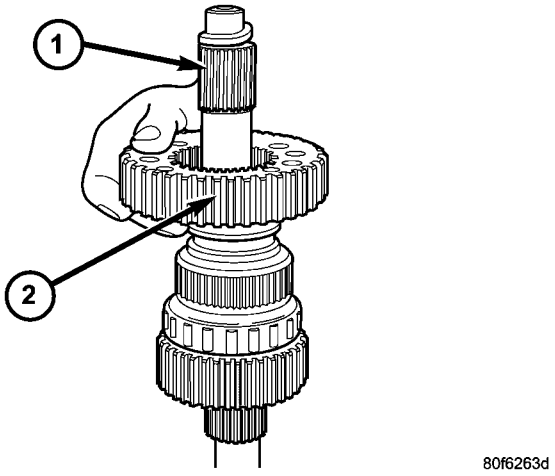


Fig. 63 Mainshaft Drive Sprocket Installation

1 - MAINSHAFT
2 - DRIVE SPROCKET

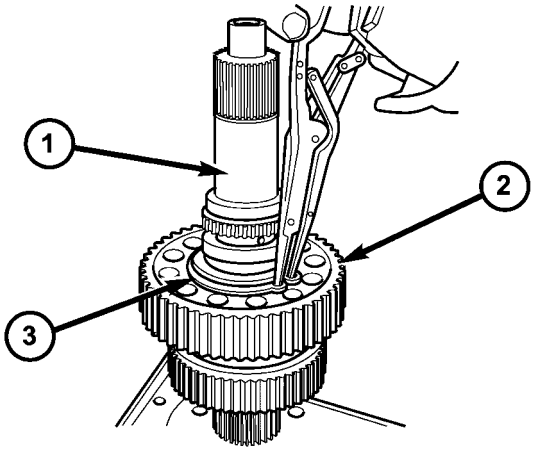


Fig. 64 Mainshaft Drive Sprocket Snap-Ring Installation

1 - MAINSHAFT
2 - DRIVE SPROCKET
3 - SNAP-RING

TRANSFER CASE - NV241 (Continued)

(8) Support front case on wood blocks so case interior is facing up. Place blocks between mounting studs on forward surface of case.

(9) Lubricate mainshaft components with transmission fluid.

(10) Lubricate sector shaft with transmission fluid and install shift sector in case (Fig. 65). Position slot in sector so it will be aligned with shift fork pin when shift forks are installed.

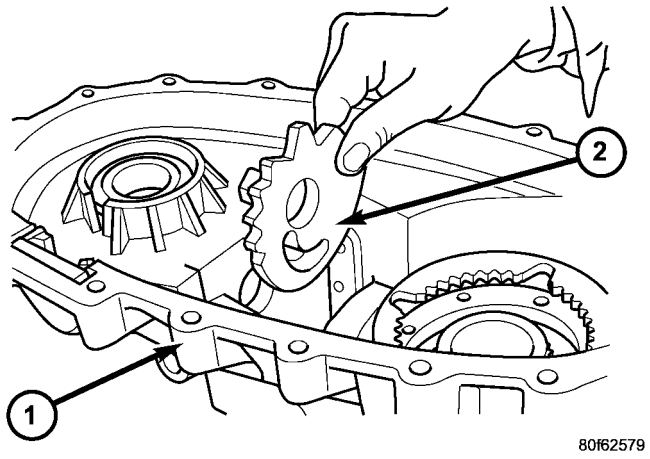


Fig. 65 Shift Sector Installation

1 - FRONT CASE HALF
2 - SHIFT SECTOR

(11) Assemble and install range fork and hub (Fig. 66). Be sure hub is properly seated in low range gear and engaged to the input gear.

(12) Align and insert range fork pin in shift sector slot.

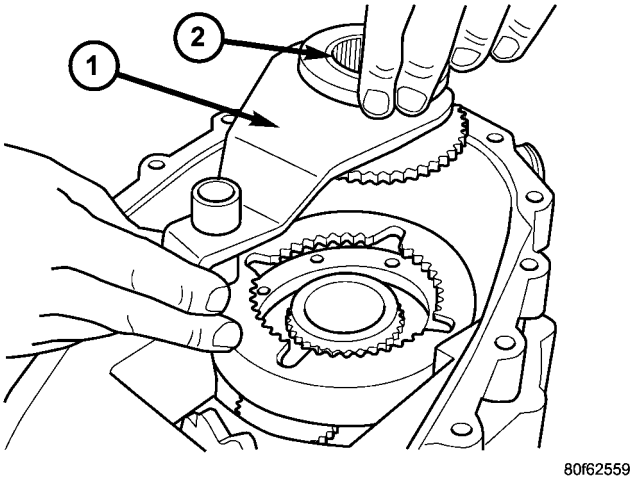


Fig. 66 Range Fork and Hub Installation

1 - RANGE FORK
2 - RANGE HUB

(13) Install mode fork and mode sleeve (Fig. 67). Verify that the long side of the mode sleeve is towards the front of the transfer case.

(14) Install the shift rail (Fig. 68) through the shift forks and into the shift rail pocket in the front case half.

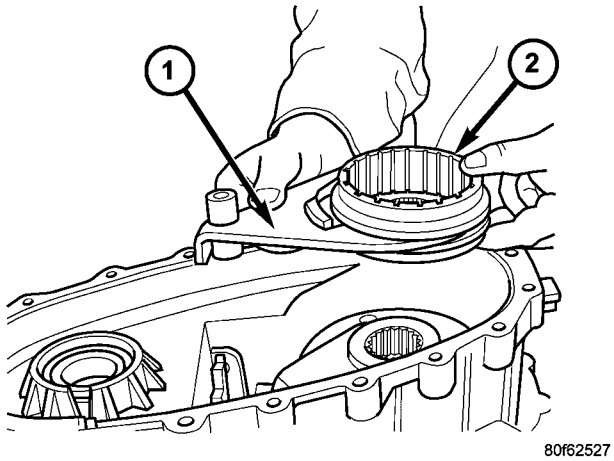


Fig. 67 Mode Fork and Sleeve Installation

1 - MODE FORK
2 - MODE SLEEVE

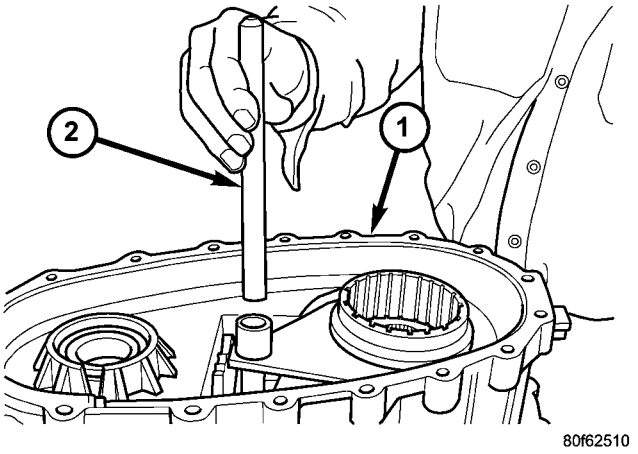


Fig. 68 Shift Fork Rail Installation

1 - FRONT CASE HALF
2 - SHIFT RAIL

TRANSFER CASE - NV241 (Continued)

FRONT OUTPUT SHAFT AND DRIVE CHAIN

(1) Install front output shaft into the font output shaft bearing (Fig. 69).

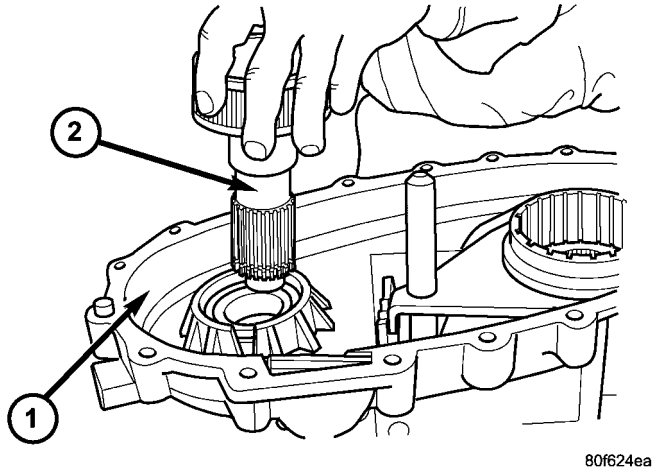


Fig. 69 Front Output Shaft Installation

1 - FRONT CASE HALF
2 - FRONT OUTPUT SHAFT

(2) Install the front output shaft snap-ring onto the front output shaft.

(3) Install new front output seal (Fig. 70) in front case with Installer 9041 and Handle C-4171.

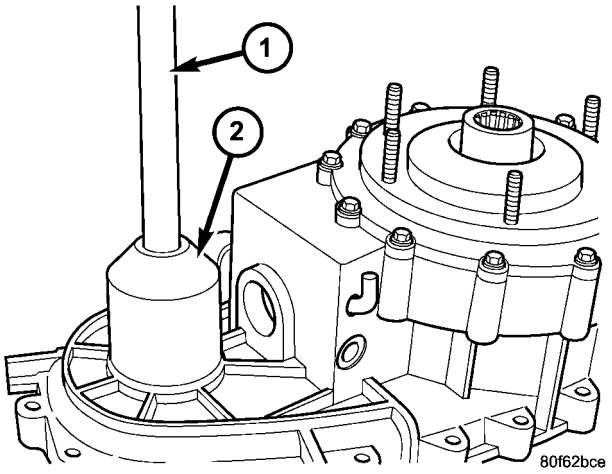


Fig. 70 Front Output Shaft Seal Installation

1 - HANDLE C-4171
2 - INSTALLER 9041

(4) Insert front sprocket in drive chain (Fig. 71).

(5) Install drive chain around mainshaft drive sprocket (Fig. 71). Then position front sprocket over front output shaft.

(6) Install the mainshaft into the range hub and mode sleeve and seat the front drive sprocket onto the front output shaft (Fig. 71).

(7) Install the front drive sprocket retaining ring (Fig. 72).

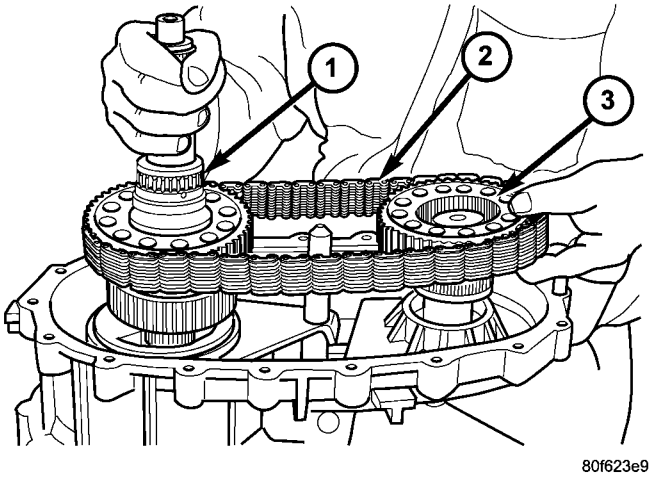


Fig. 71 Install Mainshaft, Drive Chain, and Front Sprocket

1 - MAINSHAFT
2 - DRIVE CHAIN
3 - DRIVE SPROCKET

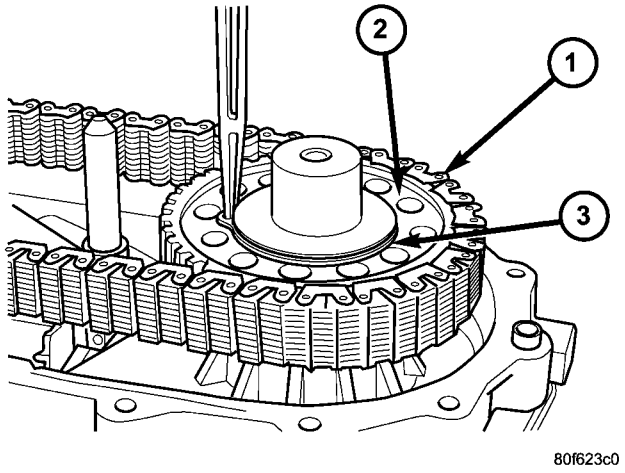


Fig. 72 Front Drive Sprocket Snap-Ring Installation

1 - DRIVE CHAIN
2 - FRONT DRIVE SPROCKET
3 - SNAP-RING

(8) Install spring and cups on shift rail (Fig. 73).

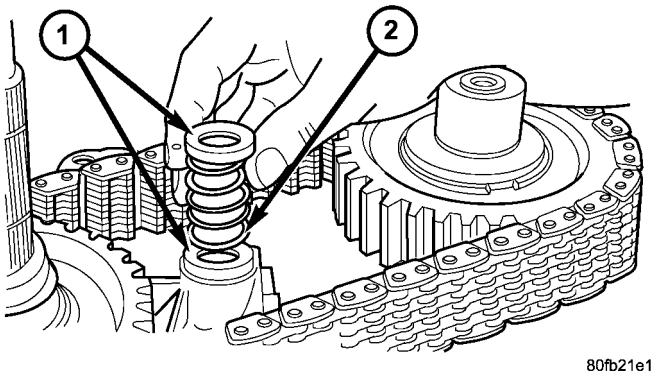


Fig. 73 Shift Rail Spring And Cups Installation

1 - CUPS (2)
2 - SPRING

(9) Insert magnet in front case pocket (Fig. 74).

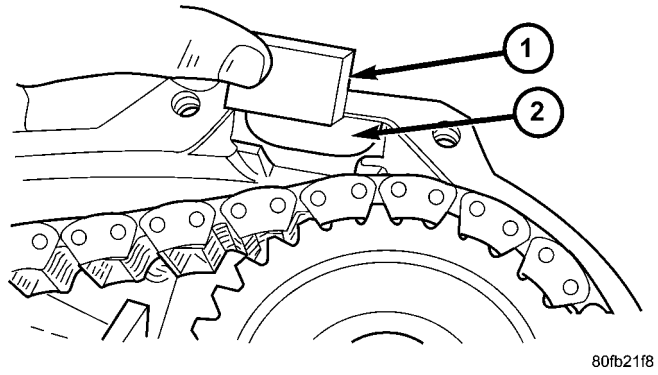


Fig. 74 Case Magnet Installation

1 - MAGNET
2 - CASE POCKET

OIL PUMP AND REAR CASE

CAUTION: Do not remove the bolts holding the oil pump cover to the rear case half. The oil pump cover is aligned to the rear output shaft inner bearing race and will become mis-aligned if the bolts are loosened. If the transfer case failure has generated any debris which may have become trapped in the oil pump. the rear case and oil pump assembly MUST be replaced.

Lubricate the oil pump components before installation. Prime the oil pickup tube by pouring a little oil into the tube before installation.

(1) Install new o-ring in pickup tube inlet of oil pump.

(2) Position oil pickup tube and filter in rear case. Be sure pickup filter is seated in case pocket and that pickup tube is aligned in case notches. Be sure hose that connects tube to filter is securely positioned.

(3) Install the transfer case position sensor (Fig. 75). Tighten the sensor to 20-34 N·m (16-25 ft.lbs.).

(4) Install the shift sector support (Fig. 76). Tighten the sector support to 27-42 N·m (20-30 ft.lbs.).

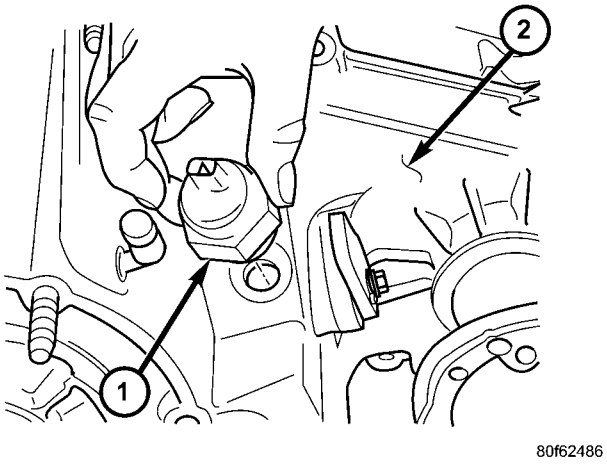


Fig. 75 Install Transfer Case Position Sensor

1 - TRANSFER CASE POSITION SENSOR
2 - TRANSFER CASE

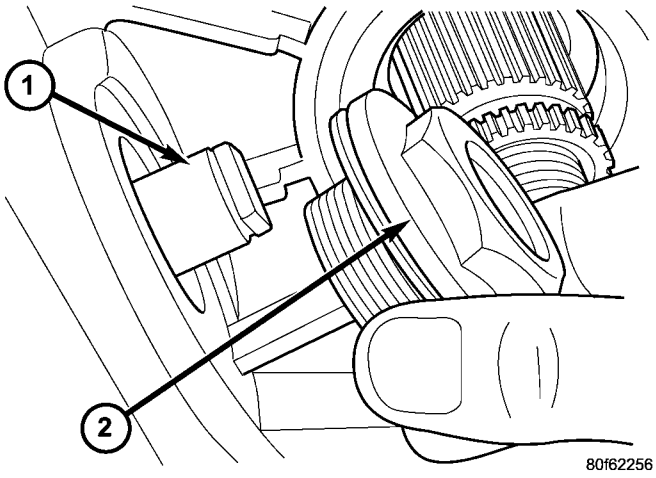


Fig. 76 Shift Sector Support Installation

1 - SHIFT SECTOR SHAFT
2 - SECTOR SUPPORT

TRANSFER CASE - NV241 (Continued)

(5) Install detent plunger and spring into the front case half. (Fig. 77).

(6) Install new o-ring on detent plug and install plug in front case (Fig. 77). Tighten plug to 16-25 N·m (12-18 ft. lbs.).

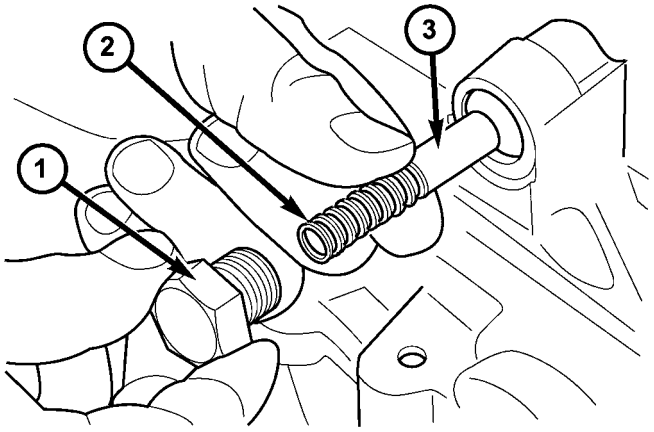


Fig. 77 Detent Plug, Spring, and Plunger Installation

1 - DETENT PLUG
2 - SPRING
3 - PLUNGER

(7) Install shift lever onto the sector shaft (Fig. 78).

(8) Install the shift lever bolt (Fig. 79). Tighten the bolt to 20-34 N·m (15-25 ft.lbs.)

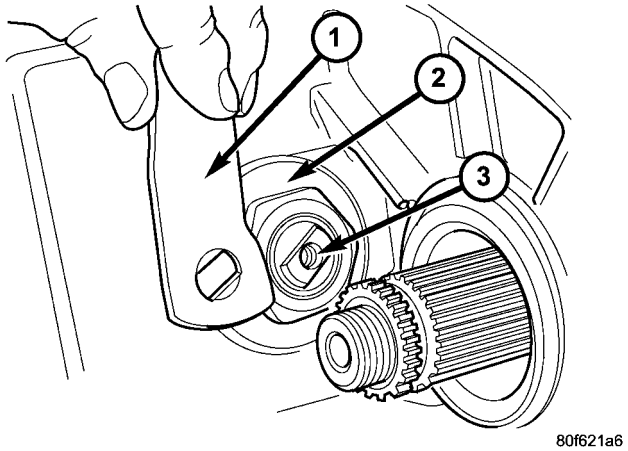


Fig. 78 Shift Lever Installation

1 - SHIFT LEVER
2 - SECTOR SUPPORT
3 - SHIFT SECTOR

(9) Shift the transfer case into a 4WD range. This will raise the mainshaft slightly and make installation of the rear case half much easier.

(10) Verify that the transfer case alignment dowels are properly installed.

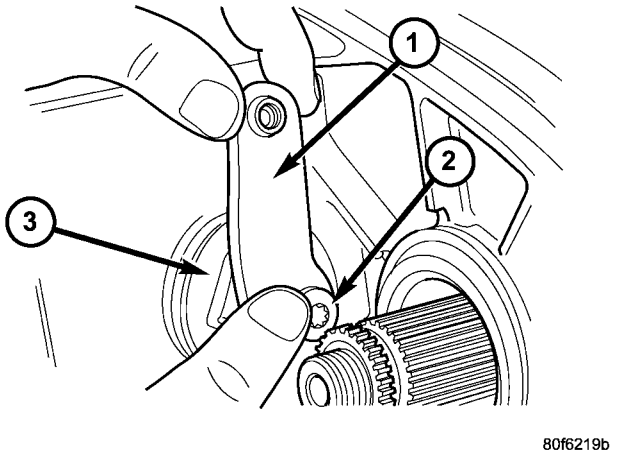


Fig. 79 Shift Lever Bolt Installation

1 - SHIFT LEVER
2 - BOLT
3 - SECTOR SUPPORT

(11) Apply bead of Mopar® Gasket Maker, or equivalent, to mating surface of front case. Keep sealer bead width to maximum of 3/16 inch. Do not use excessive amount of sealer as excess will be displaced into case interior.

(12) Align oil pump with mainshaft and align shift rail with bore in rear case. Then install rear case and oil pump assembly (Fig. 80). Be sure oil pickup tube remains in position during case installation.

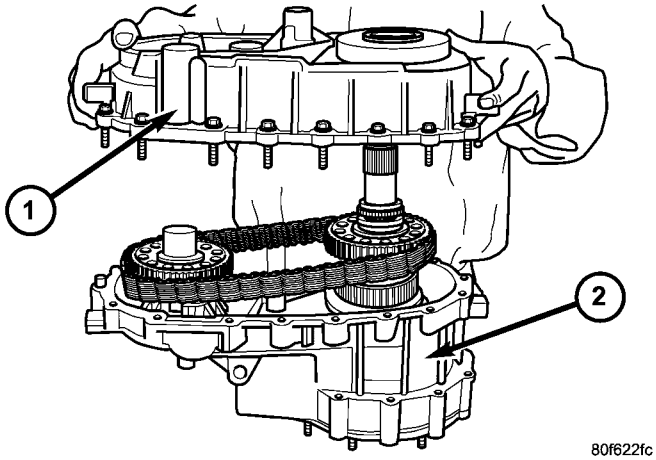


Fig. 80 Rear Case Half Installation

1 - REAR CASE HALF
2 - FRONT CASE HALF

(13) Install 4-5 rear case-to front case bolts to hold rear case in position. Tighten bolts snug but not to specified torque at this time.

TRANSFER CASE - NV241 (Continued)

CAUTION: Verify that shift rail, and case alignment dowels (Fig. 81) are seated before installing any bolts. Case could be cracked if shaft rail or dowels are misaligned.

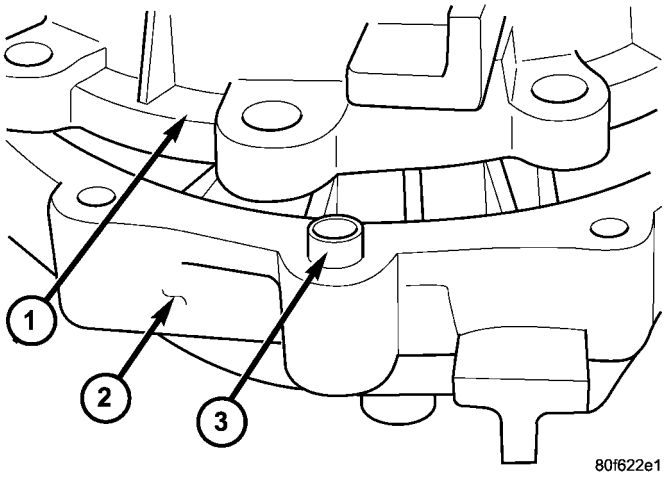


Fig. 81 Case Alignment Dowel

1 - REAR CASE HALF
2 - FRONT CASE HALF
3 - ALIGNMENT DOWEL

(14) Apply Loctite™ 242 to remainder of rear case-to-front case bolt threads and install bolts (Fig. 82). Tighten all bolts to 20-27 N·m (15-20 ft. lbs.).

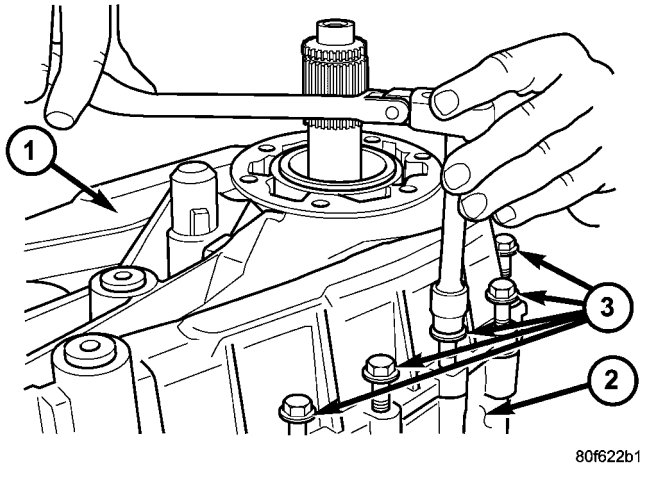


Fig. 82 Case Half Bolt Installation

1 - REAR CASE HALF
2 - FRONT CASE HALF
3 - BOLTS

YOKE

(1) Install yoke seal onto front shaft (Fig. 83).

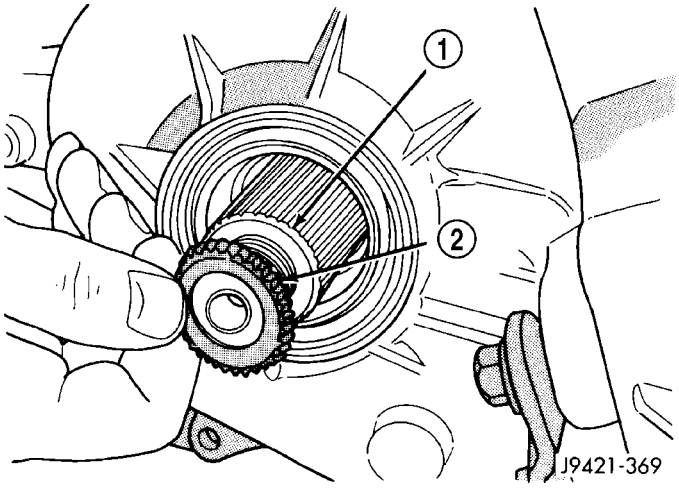


Fig. 83 Installing Flange Seal On Front Shaft

1 - FRONT OUTPUT SHAFT
2 - FLANGE SEAL

(2) Install yoke onto front shaft (Fig. 84). Then install and tighten a new yoke nut to 122-176 N·m (90-130 ft. lbs.) torque. Never re-use a yoke nut.

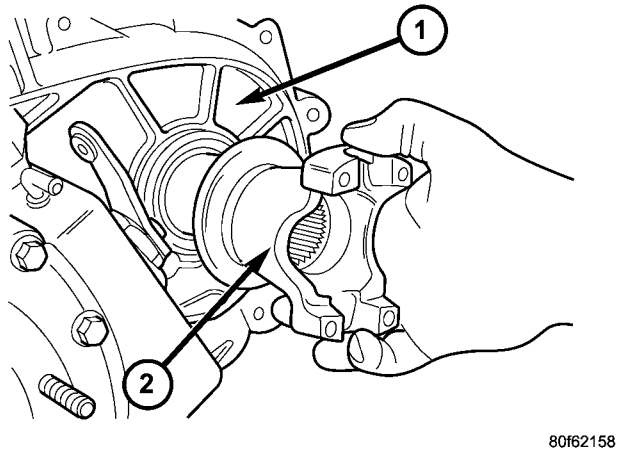


Fig. 84 Front Yoke Installation

1 - TRANSFER CASE
2 - FRONT YOKE

TRANSFER CASE - NV241 (Continued)

REAR EXTENSION

(1) Clean mating surfaces of transfer case housing and the rear retainer of any original gasket material.

(2) Apply bead of Mopar® Gasket Maker, or equivalent, to mating surface of rear case. Keep sealer bead width to maximum of 3/16 inch. Do not use excessive amount of sealer as excess will be displaced into case interior.

(3) Install the tone wheel dowel pin (Fig. 85) into the output shaft.

(4) Install the speedometer sensor tone wheel (Fig. 86) onto the output shaft with the notch in the tone wheel inner diameter toward the front of the transfer case. Make sure the notch fits over the dowel pin in the output shaft and the tone wheel is against the rear output shaft bearing inner race.

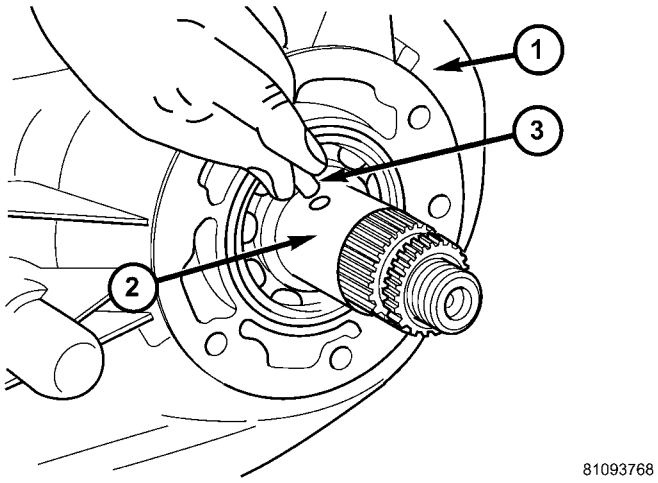


Fig. 85 Output Shaft Dowel Pin

1 - TRANSFER CASE
2 - OUPUT SHAFT
3 - DOWEL PIN

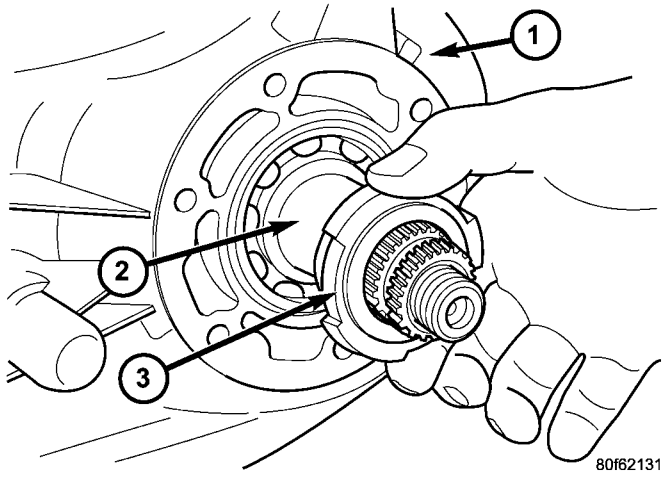


Fig. 86 Speedometer Sensor Tone Wheel Installation

1 - TRANSFER CASE
2 - OUPUT SHAFT
3 - SENSOR TONE WHEEL

(5) Align and install rear extension on rear case (Fig. 87).

(6) Apply Mopar® Silicone Sealer to threads of rear extension bolts. Then install extension bolts (Fig. 88). Tighten the extension bolts to 16-25 N·m (12-18 ft.lbs.).

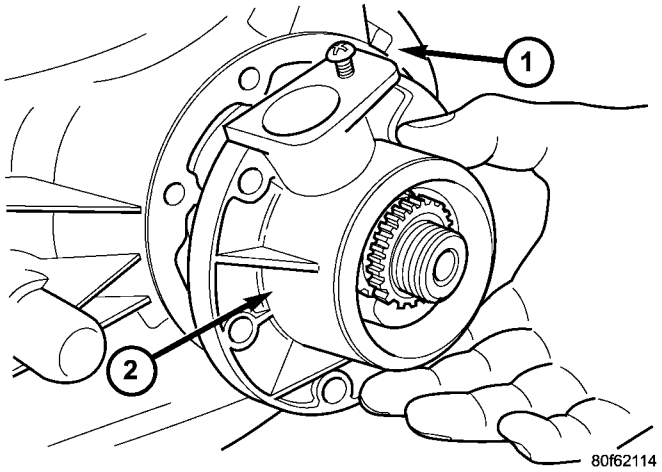


Fig. 87 Rear Extension Installation

1 - TRANSFER CASE
2 - EXTENSION HOUSING

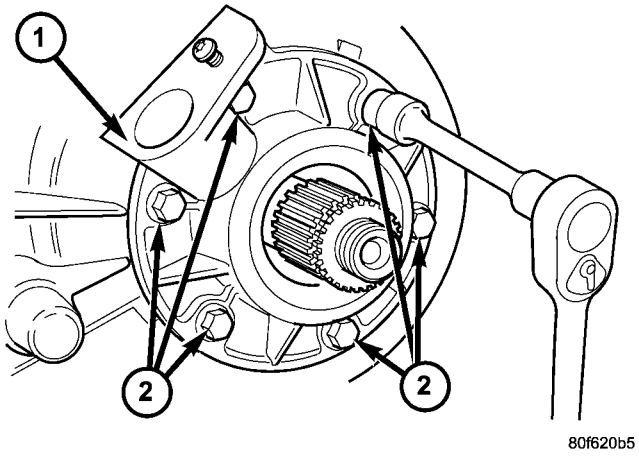


Fig. 88 Rear Extension Bolt Installation

1 - EXTENSION HOUSING
2 - BOLTS

TRANSFER CASE - NV241 (Continued)

(7) Install new seal in rear extension housing with Installer 9041 and Handle C-4171.

(8) Install the speedometer sensor (Fig. 89) into the extension housing.

(9) Install the speedometer sensor bolt (Fig. 90). Tighten the bolt to 11-16 N·m (8-12 ft.lbs.).

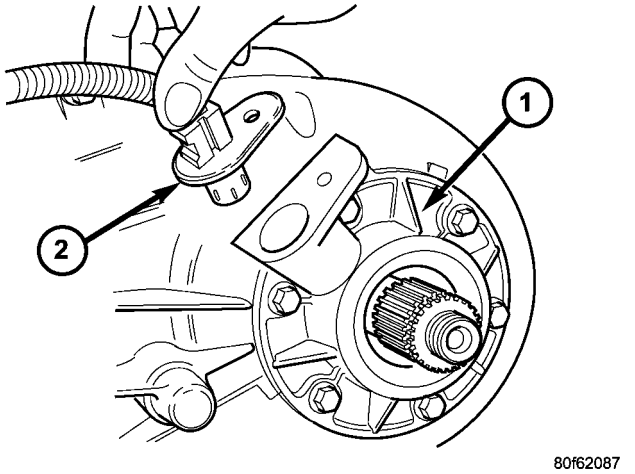


Fig. 89 Speedometer Sensor Installation

1 - EXTENSION HOUSING
2 - SPEEDOMETER SENSOR

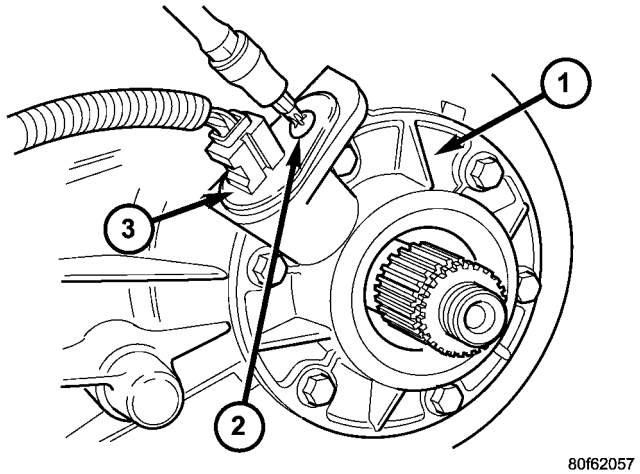


Fig. 90 Speedometer Sensor Bolt Installation

1 - EXTENSION HOUSING
2 - BOLT
3 - SPEEDOMETER SENSOR

(10) Install the rear companion flange onto the output shaft.

(11) Install bolts into two threaded holes in the rear propeller shaft companion flange 180° apart (Fig. 91).

(12) Place the Holder 6719 (Fig. 91) over the two bolts and use the Holder to install a new rear companion flange nut. Never re-use the companion flange nut once it has been installed. Tighten the companion flange nut to 122-176 N·m (90-130 ft.lbs.).

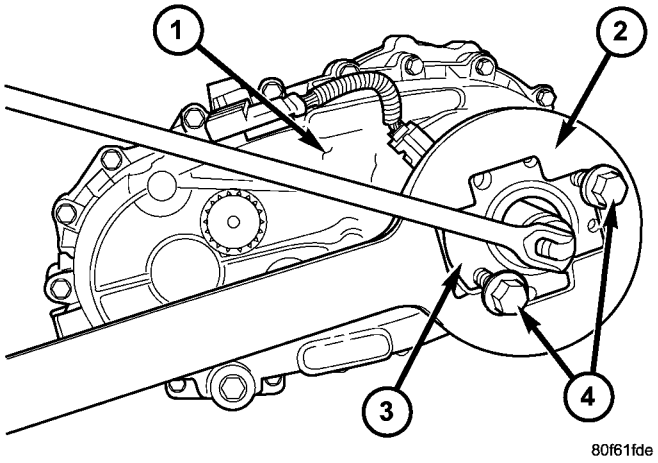


Fig. 91 Rear Companion Flange Nut Removal

1 - TRANSFER CASE
2 - HOLDER 6719
3 - COMPANION FLANGE
4 - BOLTS

INSTALLATION

(1) Align and seat transfer case on transmission. Be sure transfer case input gear splines are aligned with transmission output shaft. Align splines by rotating transfer case rear output shaft yoke if necessary. Do not install any transfer case attaching nuts until the transfer case is completely seated against the transmission.

(2) Install and tighten transfer case attaching nuts. Tighten nuts to 30-41 N·m (20-30 ft.lbs.).

(3) Install rear crossmember.

(4) Remove jack stand from under transmission.

(5) Align and connect propeller shafts. (Refer to 3 - DIFFERENTIAL & DRIVELINE/PROPELLER SHAFT/PROPELLER SHAFT - INSTALLATION)

(6) Connect vacuum harness and vent hose.

(7) Connect shift rod to transfer case lever or floor shift arm. Use channel lock style pliers to press rod back into lever grommet.

(8) Adjust shift linkage, if necessary.

(9) Fill transfer case with recommended transmission fluid and install fill plug.

(10) Install skid plate, if equipped. (Refer to 13 - FRAMES & BUMPERS/FRAME/TRANSFER CASE SKID PLATE - INSTALLATION)

(11) Lower vehicle

TRANSFER CASE - NV241 (Continued)

SPECIFICATIONS

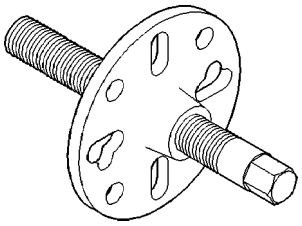
TRANSFER CASE - NV241

TORQUE SPECIFICATIONS

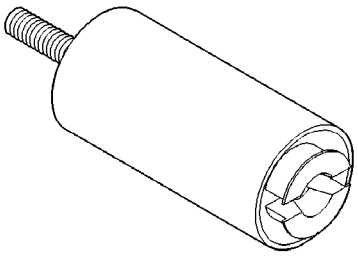
DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Plug, Detent	16-25	12-18	-
Plug, Drain/Fill	20-34	15-25	-
Bolt, Rear Extension	16-24	12-18	-
Bolt, Front Input Retainer	21	16	-
Bolt, Case Half	20-27	15-20	-
Nut, Front Yoke	122-176	90-130	-
Nut, Rear Companion Flange	258-312	190-230	-
Bolt, Shift Lever	20-34	15-25	-
Nuts, Mounting	30-41	20-30	-
Bolts, U-Joint	19	17	-
Support, Sector	27-42	20-30	-
Sensor, Transfer Case Position	20-34	16-25	-
Bolt, Speed Sensor	11-16	8-12	-

SPECIAL TOOLS

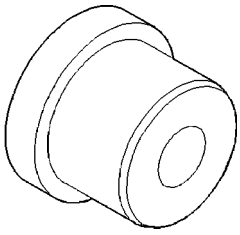
TRANSFER CASE - NV241



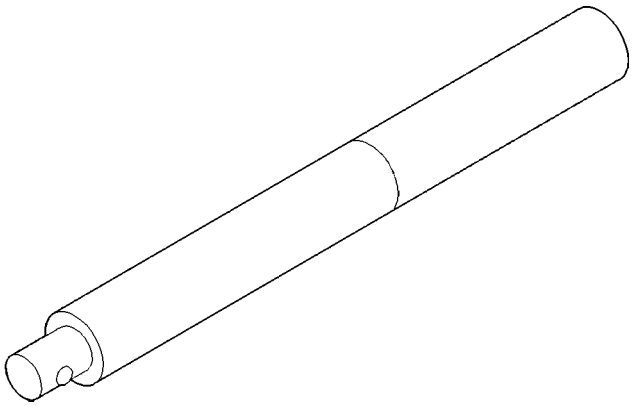
Puller - C-452



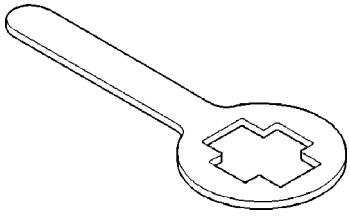
Remover - L-4454



Installer, Bushing - 5066

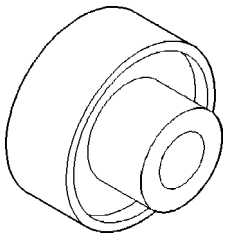


Handle, Universal - C-4171

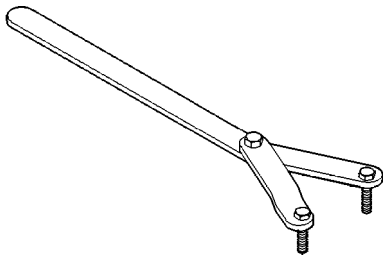


Holder, Yoke - 6719A

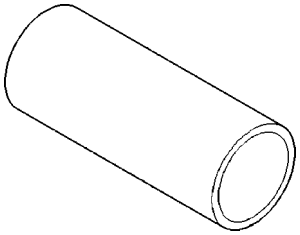
REAR EXTENSION SEAL (Continued)



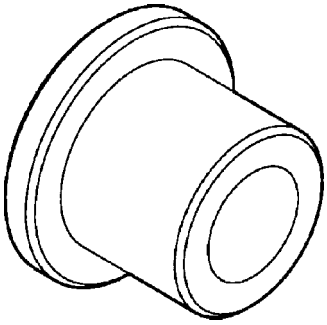
Installer, Bearing - 6953



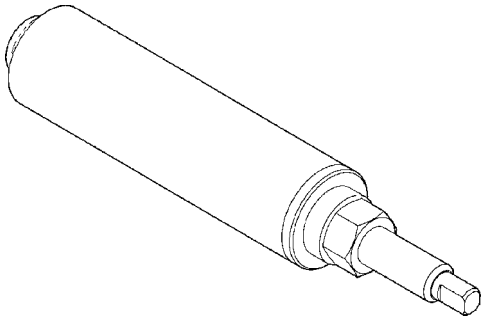
Spanner Wrench - 6958



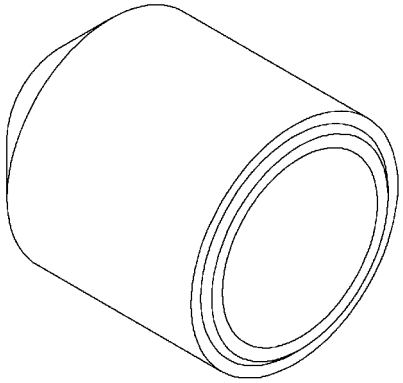
Cup - 8148



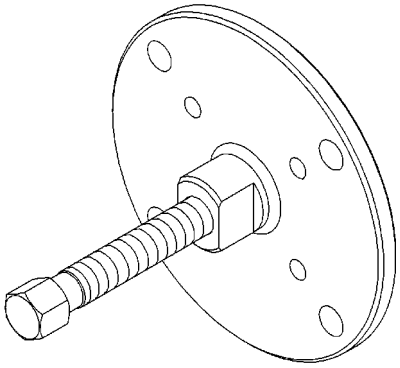
Installer, Bearing - 8239



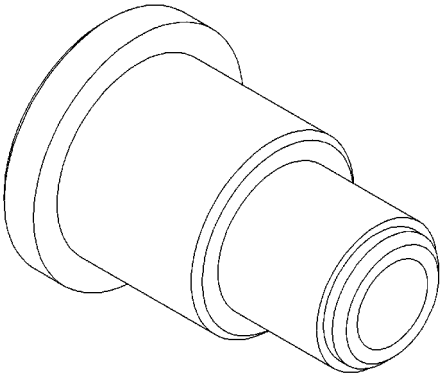
Remover - 8240



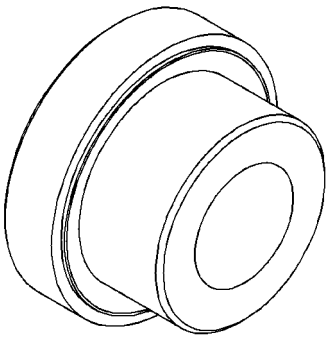
Installer, Seal - 8841



Flange Puller - 8992



Installer, Cup Seal - 9045



Installer, Bearing - 9047

REAR EXTENSION SEAL

REMOVAL

(1) Raise and support vehicle.
(2) Remove rear propeller shaft. (Refer to 3 - DIFFERENTIAL & DRIVELINE/PROPELLER SHAFT/PROPELLER SHAFT - REMOVAL)
(3) Remove the rear companion flange from the output shaft.
(4) Using a suitable pry tool or slide-hammer mounted screw, remove the rear extension seal.

INSTALLATION

(1) Clean fluid residue from sealing surface and inspect for defects.
(2) Using Installer 9041 and Handle C-4171, install seal in rear extension (Fig. 92).

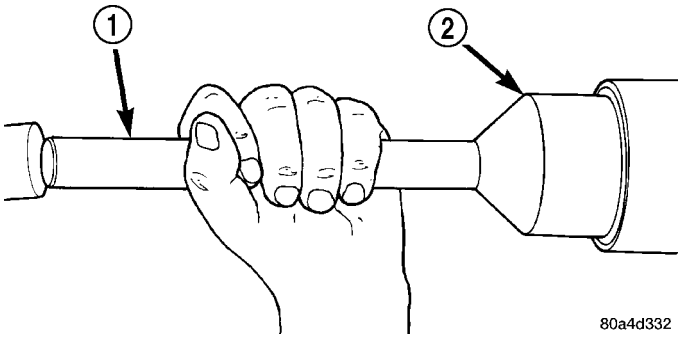


Fig. 92 Install Rear Extension Seal

1 - SPECIAL TOOL C-4171
2 - SPECIAL TOOL 9041

(3) Install the companion flange onto the output shaft.
(4) Install a new companion flange nut and tighten to 176-271 N·m (130-200 ft.lbs.).
(5) Install propeller shaft. (Refer to 3 - DIFFERENTIAL & DRIVELINE/PROPELLER SHAFT/PROPELLER SHAFT - INSTALLATION)
(6) Verify proper transfer case fluid level.
(7) Lower vehicle.

FLUID

STANDARD PROCEDURE - FLUID DRAIN AND FILL

The fill and drain plugs are both in the rear case (Fig. 93). Correct fill level is to the bottom edge of the fill plug hole. Be sure the vehicle is level to ensure an accurate fluid level check.

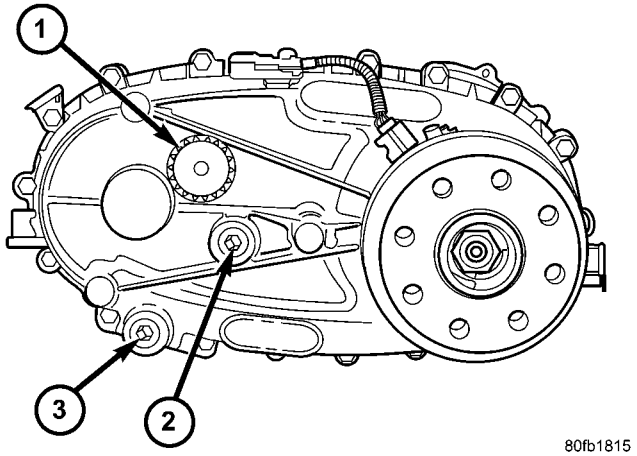


Fig. 93 Transfer Case Identification Tag and Fill/Drain Plugs

1 - IDENTIFICATION TAG
2 - FILL PLUG
3 - DRAIN PLUG

FRONT OUTPUT SHAFT SEAL

REMOVAL

(1) Shift transfer case into NEUTRAL.
(2) Raise vehicle.
(3) Remove front propeller shaft. (Refer to 3 - DIFFERENTIAL & DRIVELINE/PROPELLER SHAFT/PROPELLER SHAFT - REMOVAL)
(4) Using Spanner Wrench 6958, remove yoke nut (Fig. 94). Discard nut after removal. It is not reusable.

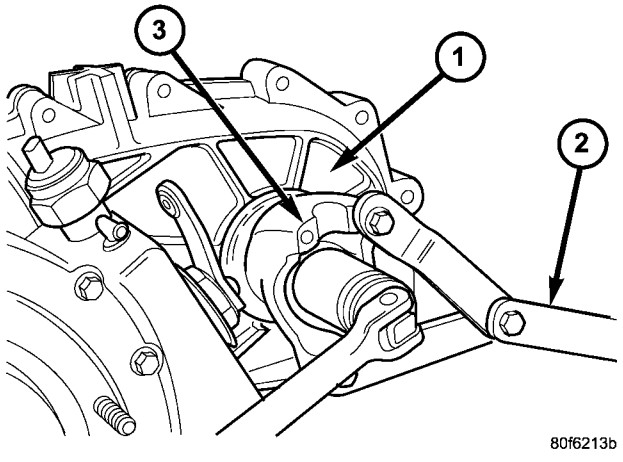


Fig. 94 Front Yoke Nut Removal

1 - TRANSFER CASE
2 - HOLDER 6958
3 - FRONT YOKE

FRONT OUTPUT SHAFT SEAL (Continued)

(5) Remove yoke from output shaft (Fig. 95). Use puller C-452 if flange can not be removed by hand.

(6) Remove yoke rubber seal from front output shaft (Fig. 96).

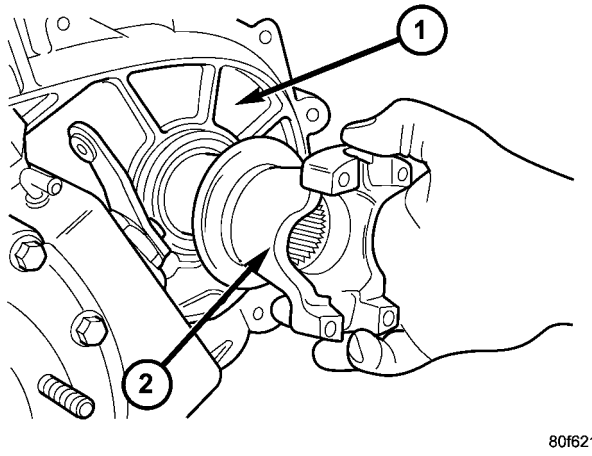


Fig. 95 Front Yoke Removal

1 - TRANSFER CASE
2 - FRONT YOKE

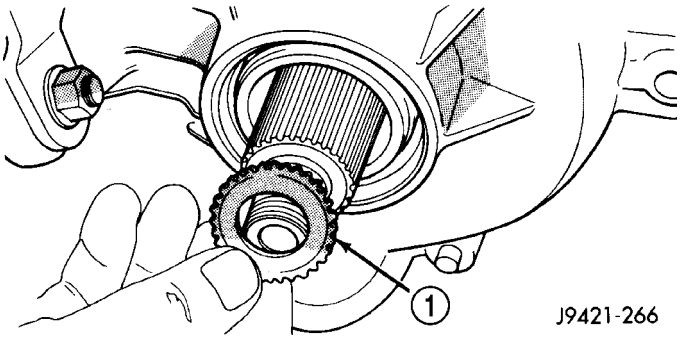


Fig. 96 Yoke Seal Removal

1 - FLANGE SEAL

(7) Remove front output shaft seal with suitable pry tool, or a slide hammer mounted screw.

INSTALLATION

(1) Install new front output seal in front case with Installer Tool 9041 and Handle C-4171 (Fig. 97).

(2) Install yoke seal on front output shaft (Fig. 98).

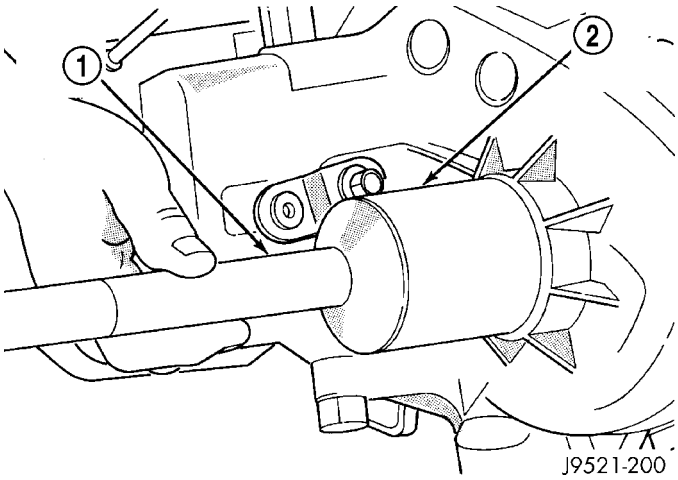


Fig. 97 Front Output Seal Installation - Typical

1 - SPECIAL TOOL C-4171
2 - SPECIAL TOOL 9041

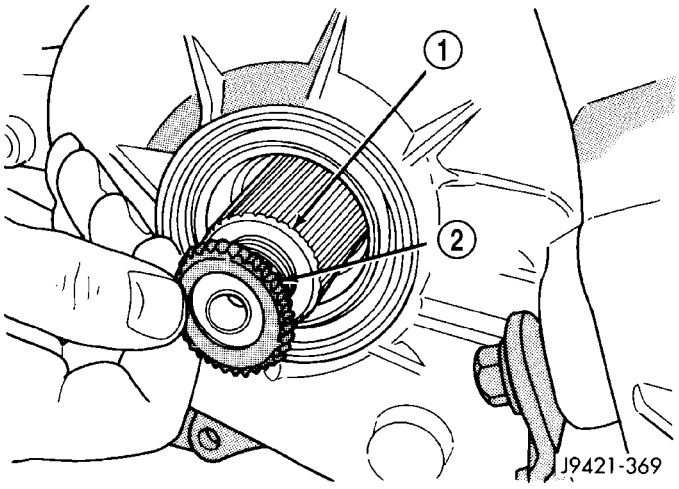


Fig. 98 Installing Flange Seal On Front Shaft

1 - FRONT OUTPUT SHAFT
2 - FLANGE SEAL

FRONT OUTPUT SHAFT SEAL (Continued)

(3) Install yoke on front output shaft (Fig. 99). Install a new yoke nut onto the front output shaft.

(4) Using Spanner Wrench 6958 (Fig. 100), tighten the yoke nut to 176-271 N·m (130-200 ft. lbs.) torque.

(5) Install propeller shaft. (Refer to 3 - DIFFERENTIAL & DRIVELINE/PROPELLER SHAFT/PROPELLER SHAFT - INSTALLATION)

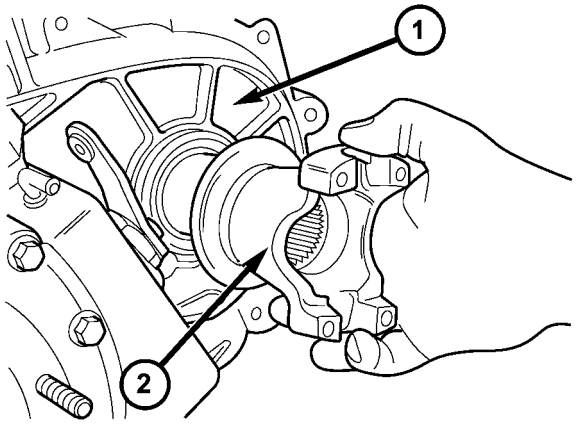


Fig. 99 Front Yoke Installation

1 - TRANSFER CASE
2 - FRONT YOKE

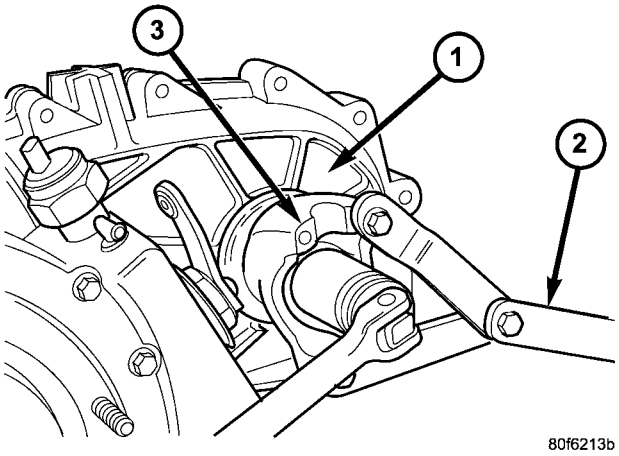


Fig. 100 Front Yoke Nut Installation

1 - TRANSFER CASE
2 - HOLDER 6958
3 - FRONT YOKE

POSITION SENSOR

DESCRIPTION

The transfer case position sensor is an electronic device whose output can be interpreted to indicate the transfer case's current operating mode. The sensor consists of a five position, resistive multiplexed circuit which returns a specific resistance value to the Powertrain Control Module (PCM) for each transfer case operating mode. The sensor is located on the top of the transfer case, just left of the transfer case centerline and rides against the sector plate roostercomb. The PCM supplies 5VDC (+/- 0.5V) to the sensor and monitors the return voltage to determine the sector plate, and therefore the transfer case, position.

OPERATION

OPERATING MODE VERSUS RESISTANCE

SENSOR POSITION	OPERATING MODE	SENSOR RESISTANCE (ohms)
1	2H	1124-1243
2	4H	650-719
3	NEUTRAL	389-431
4	4L	199-221
5	NOT USED	57-64

During normal vehicle operation, the Powertrain Control Module (PCM) monitors the transfer case position sensor return voltage to determine the operating mode of the transfer case. Refer to the Operating Mode Versus Resistance table for the correct resistance (1-6) (Fig. 101) for each position.

REMOVAL

(1) Raise and support the vehicle.

(2) Disengage the transfer case position sensor connector from the position sensor.

(3) Remove the position sensor from the transfer case.

INSTALLATION

(1) Inspect the o-ring seal on the transfer case position sensor. Replace the o-ring if necessary.

POSITION SENSOR (Continued)

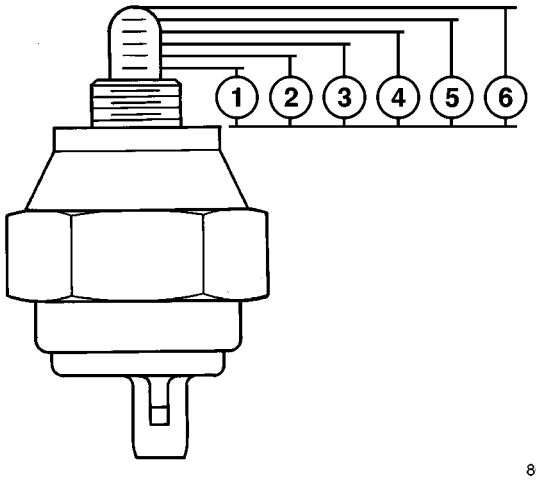


Fig. 101 Position Sensor Linear Movement

1 - POSITION 1 - 10mm ±0.5mm
2 - POSITION 2 - 12mm ±0.5mm
3 - POSITION 3 - 14mm ±0.5mm
4 - POSITION 4 - 16mm ±0.5mm
5 - POSITION 5 - 18mm ±0.5mm
6 - POSITION 6 - 20mm±0.5mm - FULL EXTENSION

(2) Install the transfer case position sensor into the transfer case. Torque the sensor to 20-34 N·m (15-25 ft.lbs.).

(3) Engage the transfer case position sensor connector to the position sensor.

(4) Lower vehicle.

(5) Verify proper sensor operation.

SHIFT LEVER

REMOVAL

(1) Shift transfer case into 4L.

(2) Raise vehicle.

(3) Loosen adjusting trunnion locknut and slide shift rod out of trunnion (Fig. 102). If rod lacks enough travel to come out of trunnion, push trunnion out of torque shaft.

(4) Lower vehicle.

(5) Remove console. (Refer to 23 - BODY/INTERIOR/FLOOR CONSOLE - REMOVAL)

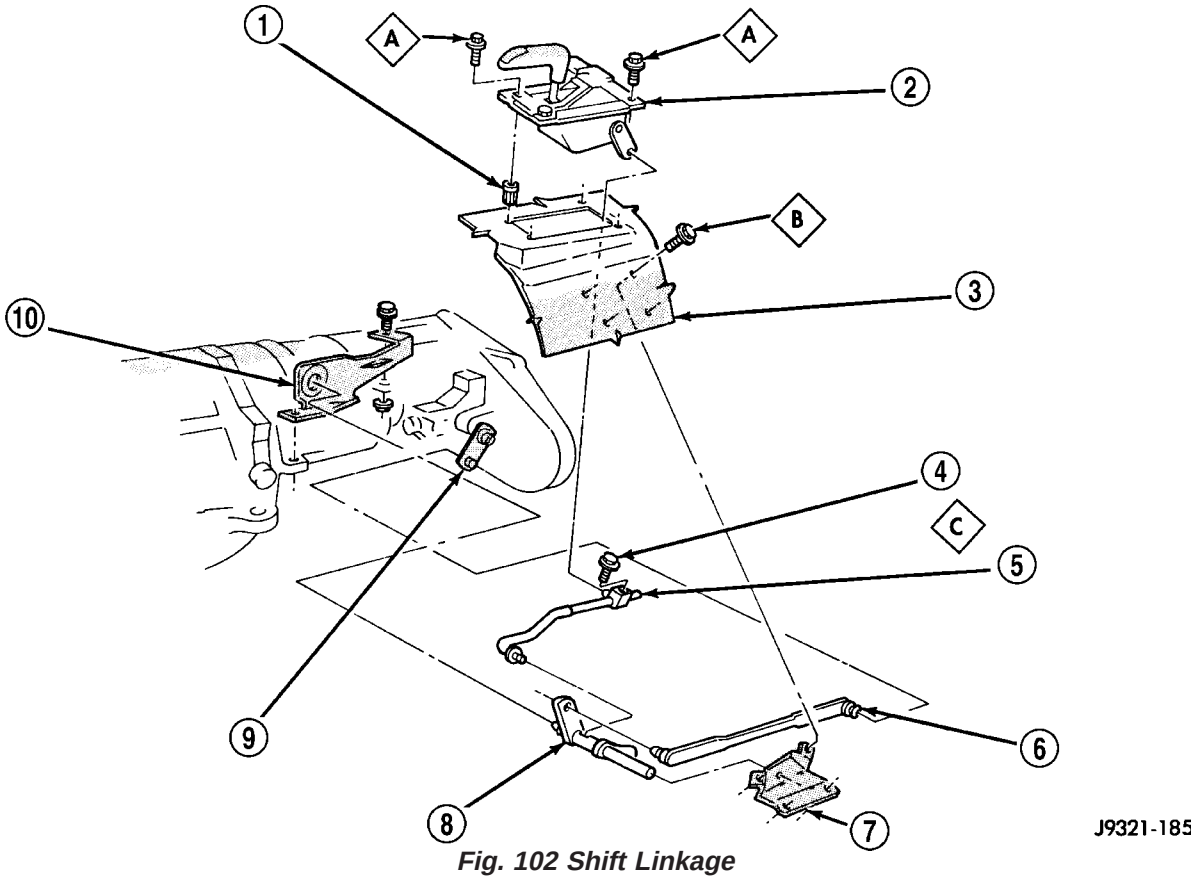


Fig. 102 Shift Linkage

1 - Rivnut (4)
2 - Shift Lever Assembly
3 - Floorpan
4 - Trunnion Lock Bolt
5 - Selector Rod and Trunnion
6 - Shift Lever Rod
7 - Torque Shaft Frame Bracket
8 - Torque Shaft
9 - Transfer Case Shift Lever
10 - Torque Shaft Transfer Case Bracket
A - 3-4 N·m (27-35 in. lbs.)
B - 11-14 N·m (97-123 in. lbs.)
C - 8-14 N·m (72-120 in. lbs.)

SHIFT LEVER (Continued)

(6) Remove screws attaching lever assembly to floorpan and remove assembly and shift rod (if left attached).

INSTALLATION

(1) If shift rod was not removed from lever assembly, work rod down through floorpan opening. Then position lever assembly on floorpan and install assembly attaching screws.

(2) Install console. (Refer to 23 - BODY/INTERIOR/FLOOR CONSOLE - INSTALLATION)

(3) Raise vehicle.

(4) Connect trunnion to torque shaft arm. Or, slide shift rod into trunnion on range lever. Be sure shift rod slides freely in trunnion.

(5) Verify that range lever is in 4L position. Then tighten trunnion lock bolt.

(6) Lower vehicle and check transfer case shift operation.

ADJUSTMENTS - SHIFT LINKAGE

(1) Shift transfer case into 4L position.

(2) Raise vehicle.

(3) Loosen lock bolt on adjusting trunnion (Fig. 103).

(4) Be sure linkage rod slides freely in trunnion. Clean rod and apply spray lube if necessary.

(5) Verify that transfer case range lever is fully engaged in 4L position.

(6) Tighten adjusting trunnion lock bolt.

(7) Lower vehicle.

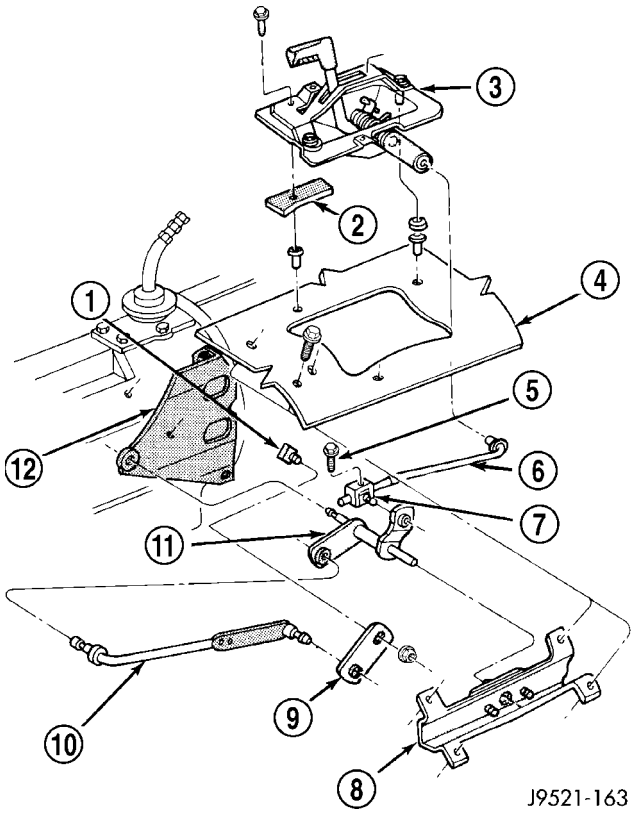


Fig. 103 Shift Linkage

1 - TRANSFER CASE SHIFT LEVER SHAFT
2 - SEAL
3 - TRANSFER CASE SHIFT LEVER ASSEMBLY
4 - FLOORPAN
5 - TRUNNION LOCK BOLT
6 - SHIFT ROD
7 - ADJUSTING TRUNNION
8 - TORQUE SHAFT BRACKET
9 - RANGE LEVER
10 - TORQUE SHAFT ROD
11 - TORQUE SHAFT
12 - LINKAGE BRACKET

VEHICLE SPEED SENSOR

DESCRIPTION

The 3-wire Vehicle Speed Sensor (VSS) (Fig. 104) is located in the transfer case rear extension.

OPERATION

The VSS is a 3-circuit (3–wire), magnetic, hall-effect sensor.

The 3 circuits are:

- A 5-volt power supply from the Powertrain Control Module (PCM).
- A ground is provided for the sensor though a low-noise sensor return circuit in the PCM.
- An input to the PCM is used to determine vehicle speed and distance traveled.

The speed sensor generates 8 pulses per sensor revolution. These signals, in conjunction with a closed throttle signal from the throttle position sensor, indicate a closed throttle deceleration to the PCM. When the vehicle is stopped at idle, a closed throttle signal is received by the PCM (but a speed sensor signal is not received).

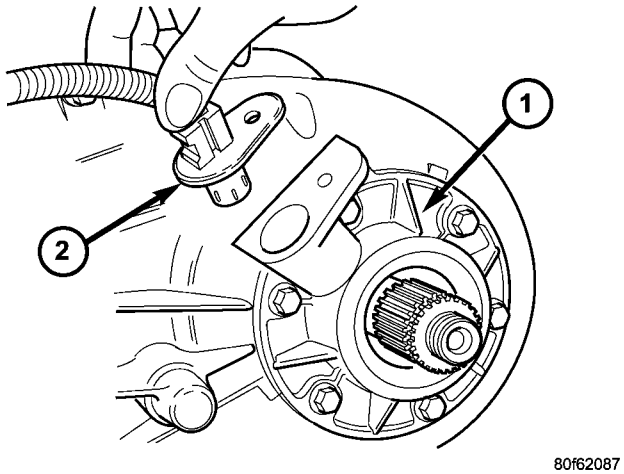


Fig. 104 Speedometer Sensor

1 - EXTENSION HOUSING
2 - SPEEDOMETER SENSOR

VEHICLE SPEED SENSOR (Continued)

Under deceleration conditions, the PCM adjusts the Idle Air Control (IAC) motor to maintain a desired MAP value. Under idle conditions, the PCM adjusts the IAC motor to maintain a desired engine speed.

REMOVAL

The Vehicle Speed Sensor (VSS) is located in the transfer case rear extension.

(1) Raise and support vehicle.

(2) Disconnect electrical connector from sensor by pushing slide tab. After slide tab has been positioned, push in on secondary release lock on side of connector and pull connector from sensor.

(3) Remove sensor mounting bolt.

(4) Remove sensor (pull straight out) from rear extension.

INSTALLATION

(1) Install sensor into transfer case rear extension and install mounting bolt. Before tightening bolt, verify speed sensor is fully seated (mounted flush) to rear extension.

(2) Tighten sensor mounting bolt to 11-16 N·m (8-12 ft. lbs.) torque.

(3) Connect electrical connector to sensor.

TIRES/WHEELS

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TIRES/WHEELS

DIAGNOSIS AND TESTING - TIRE AND WHEEL RUNOUT

Radial runout is the difference between the high and low points on the tire or wheel (Fig. 1).

Lateral runout is the **wobble** of the tire or wheel.

Radial runout of more than 1.5 mm (.060 inch) measured at the center line of the tread may cause the vehicle to shake.

Lateral runout of more than 2.0 mm (.080 inch) measured near the shoulder of the tire may cause the vehicle to shake.

Sometimes radial runout can be reduced. Relocate the wheel and tire assembly on the mounting studs (See Method 1). If this does not reduce runout to an acceptable level, the tire can be rotated on the wheel. (See Method 2).

METHOD 1 (RELOCATE WHEEL ON HUB)

(1) Drive vehicle a short distance to eliminate tire flat spotting from a parked position.

(2) Check wheel bearings and adjust if adjustable or replace if necessary.

(3) Check the wheel mounting surface.

(4) Relocate wheel on the mounting, two studs over from the original position.

(5) Tighten wheel nuts until all are properly torqued, to eliminate brake distortion.

(6) Check radial runout. If still excessive, mark tire sidewall, wheel, and stud at point of maximum runout and proceed to Method 2.

METHOD 2 (RELOCATE TIRE ON WHEEL)

NOTE: Rotating the tire on wheel is particularly effective when there is runout in both tire and wheel.

TIRES/WHEELS (Continued)

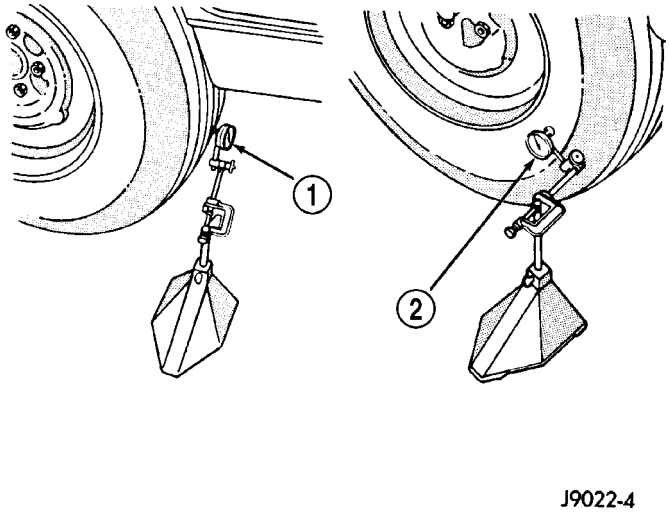

Fig. 1 Checking Tire/Wheel/Hub Runout

1 - RADIAL RUNOUT
2 - LATERAL RUNOUT

(1) Remove tire from wheel and mount wheel on service dynamic balance machine.

(2) Check wheel radial runout (Fig. 2) and lateral runout (Fig. 3).

- STEEL WHEELS: Radial runout 0.040 in., Lateral runout 0.045 in. (average-maximum)
- ALUMINUM WHEELS: Radial runout 0.030 in., Lateral runout 0.035 in. (average-maximum)

(3) If point of greatest wheel lateral runout is near original chalk mark, remount tire 180 degrees. Recheck runout or match mount, (Refer to 22 - TIRES/WHEELS - STANDARD PROCEDURE).

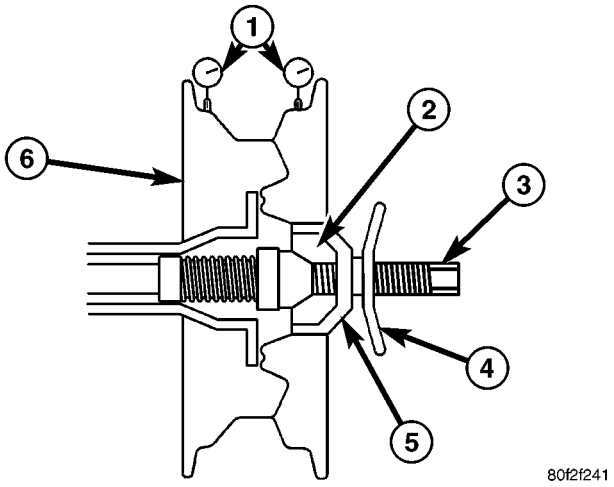

Fig. 2 RADIAL RUNOUT

1 - DIAL INDICATORS
2 - MOUNTING CONE
3 - SPINDLE SHAFT
4 - WING NUT
5 - PLASTIC CUP
6 - WHEEL

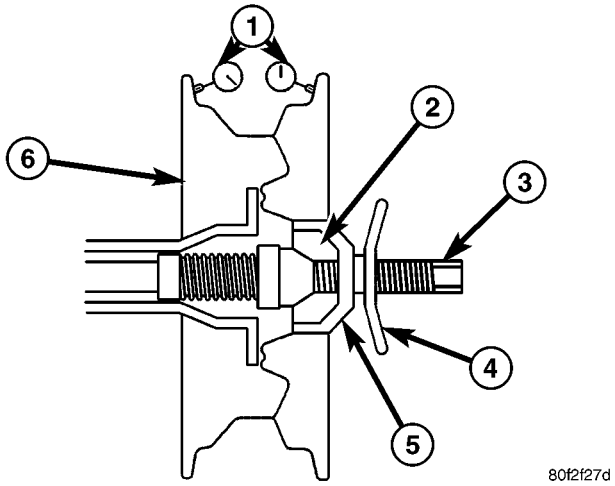

Fig. 3 LATERAL RUNOUT

1 - DIAL INDICATORS
2 - MOUNTING CONE
3 - SPINDLE SHAFT
4 - WING NUT
5 - PLASTIC CUP
6 - WHEEL

STANDARD PROCEDURE

STANDARD PROCEDURE - ROTATION

Tires on the front and rear operate at different loads and perform different steering, driving, and braking functions. For these reasons they wear at unequal rates and tend to develop irregular wear patterns. These effects can be reduced by rotating the tires at regular intervals. The benefits of tire rotation are:

- Increase tread life
- Maintain traction levels
- A smooth, quiet ride

The suggested method of tire rotation is (Fig. 4). Other rotation methods can be used, but they will not provide all the tire longevity benefits.

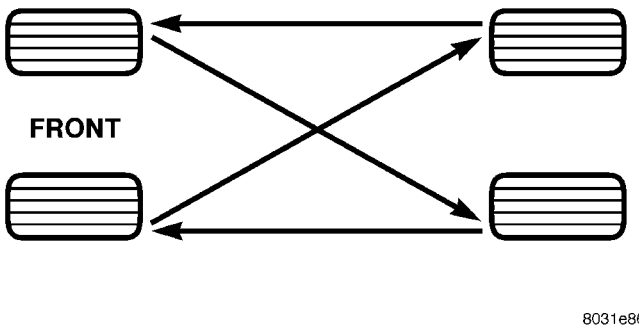

Fig. 4 Tire Rotation Pattern

TIRES/WHEELS (Continued)

STANDARD PROCEDURE - MATCH MOUNTING

Tires and wheels are currently match mounted at the factory. Match mounting is a technique used to reduce runout in the wheel/tire assembly. This means that the high spot of the tire is aligned with the low spot on the wheel rim. The high spot on the tire is marked with a paint mark or a bright colored adhesive label on the outboard sidewall. The low spot on the rim is identified with a label on the outside of the rim and a dot on the inside of the rim. If the outside label has been removed the tire will have to be removed to locate the dot on the inside of the rim.

Before dismounting a tire from its wheel, a reference mark should be placed on the tire at the valve stem location. This reference will ensure that it is remounted in the original position on the wheel.

(1) Use a dial indicator to locate the high spot of the tire on the center tread rib (Fig. 5). Record the indicator reading and mark the high spot on the tire. Place a mark on the tire at the valve stem location (Fig. 6).

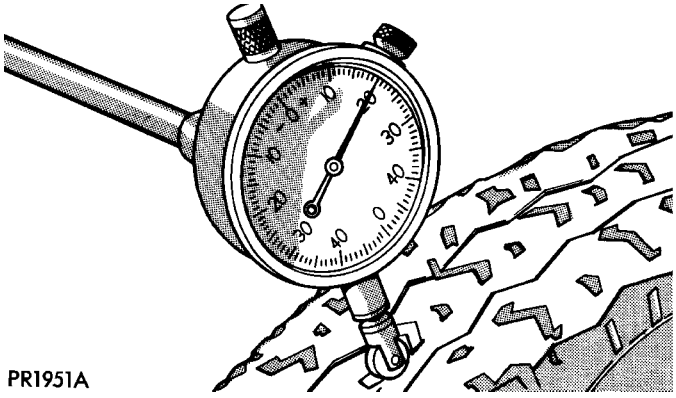

Fig. 5 Dial Indicator

(2) Break down the tire and remount it 180 degrees on the rim (Fig. 7).

(3) Measure the total runout again and mark the tire to indicate the high spot.

(4) If runout is still excessive use the following procedures.

(a) If the high spot is within 101.6 mm (4.0 in.) of the first spot and is still excessive, replace the tire.

(b) If the high spot is within 101.6 mm (4.0 in.) of the first spot on the wheel, the wheel may be out of specifications. (Refer to 22 - TIRES/WHEELS - DIAGNOSIS AND TESTING).

(c) If the high spot is NOT within 101.6 mm (4.0 in.) of either high spot, draw an arrow on the tread from second high spot to first. Break down the tire and remount it 90 degrees on rim in that direction (Fig. 8). This procedure will normally reduce the runout to an acceptable amount.

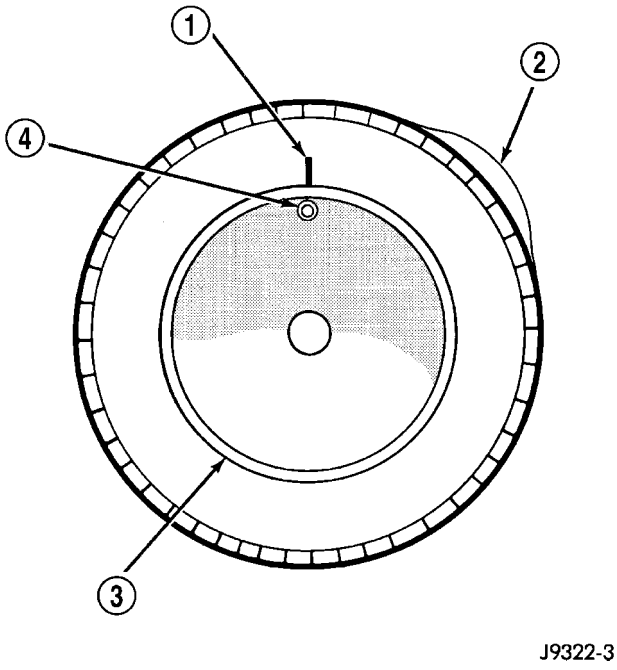

Fig. 6 First Measurement On Tire

1 - REFERENCE MARK
2 - 1ST MEASUREMENT
HIGH SPOT MARK TIRE AND RIM
3 - WHEEL
4 - VALVE STEM

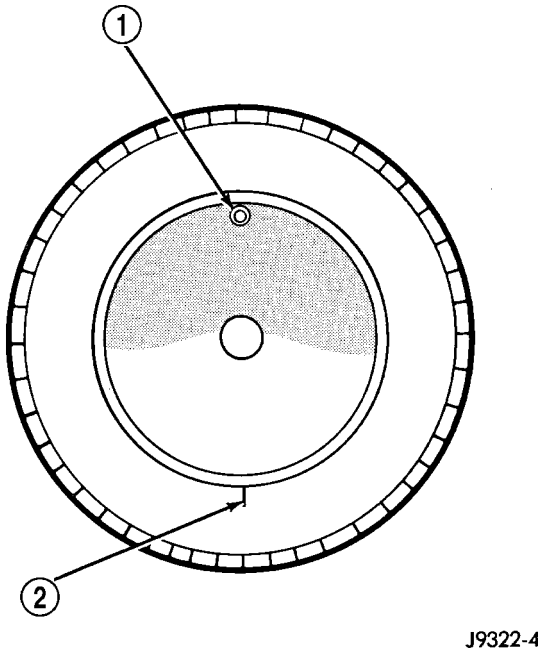

Fig. 7 Remount Tire 180 Degrees

1 - VALVE STEM
2 - REFERENCE MARK

TIRES/WHEELS (Continued)

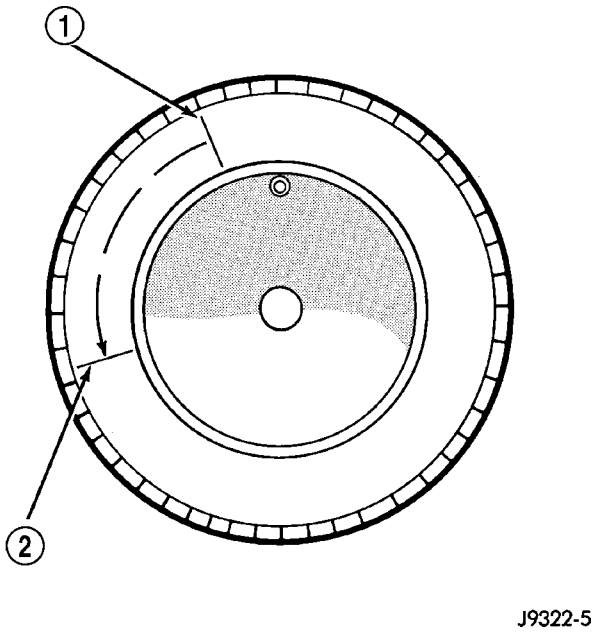

Fig. 8 Remount Tire 90 Degrees In Direction of Arrow

1 - 2ND HIGH SPOT ON TIRE
2 - 1ST HIGH SPOT ON TIRE

STANDARD PROCEDURE - TIRE AND WHEEL BALANCE

It is recommended that a two plane service dynamic balancer be used when a tire and wheel assembly require balancing. Refer to balancer operation instructions for proper cone mounting procedures. Typically use front cone mounting method for steel wheels. For aluminum wheel use back cone mounting method without cone spring.

NOTE: Static should be used only when a two plane balancer is not available.

NOTE: Cast aluminum and forged aluminum wheels require coated balance weights and special alignment equipment.

Wheel balancing can be accomplished with either on or off vehicle equipment. When using on-vehicle balancing equipment, remove the opposite wheel/tire. Off-vehicle balancing is recommended.

For static balancing, find location of heavy spot causing the imbalance. Counter balance wheel directly opposite the heavy spot. Determine weight required to counter balance the area of imbalance. Place half of this weight on the **inner** rim flange and the other half on the **outer** rim flange (Fig. 9).

For dynamic balancing, the balancing equipment is designed to locate the amount of weight to be applied to both the inner and outer rim flange (Fig. 10).

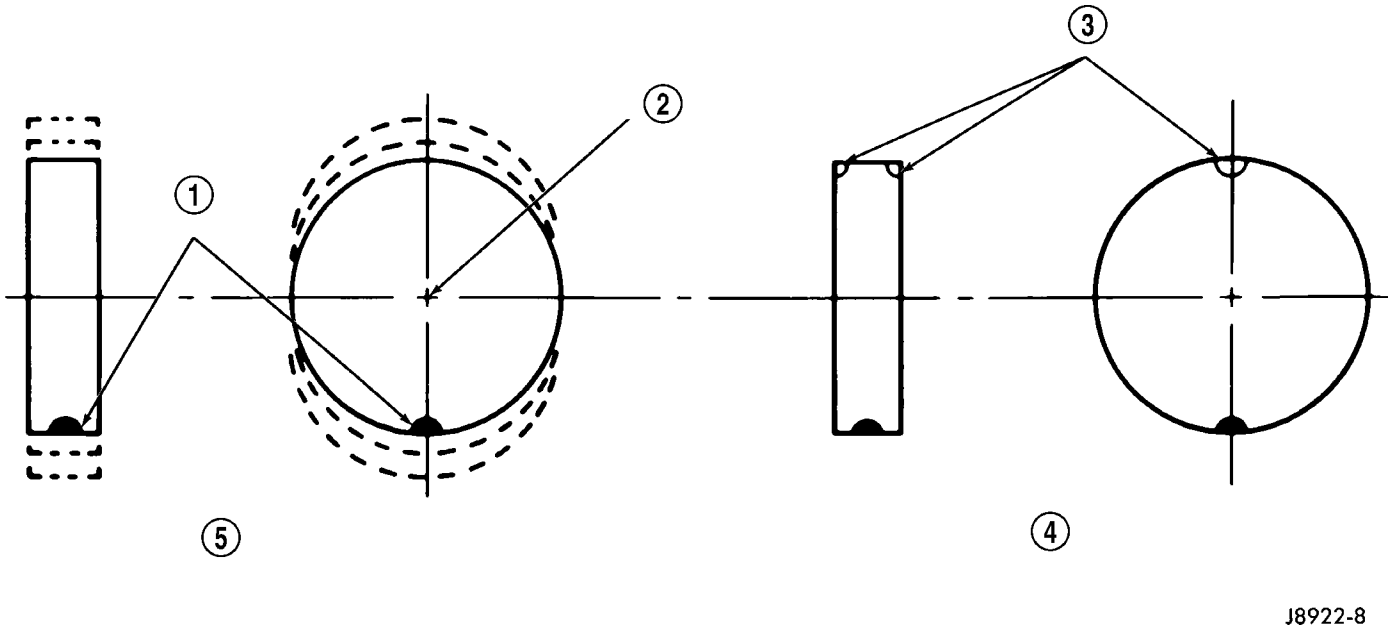

Fig. 9 Static Unbalance & Balance

1 - HEAVY SPOT
2 - CENTER LINE OF SPINDLE
3 - ADD BALANCE WEIGHTS HERE
4 - CORRECTIVE WEIGHT LOCATION
5 - TIRE OR WHEEL TRAMP, OR WHEEL HOP

TIRES/WHEELS (Continued)

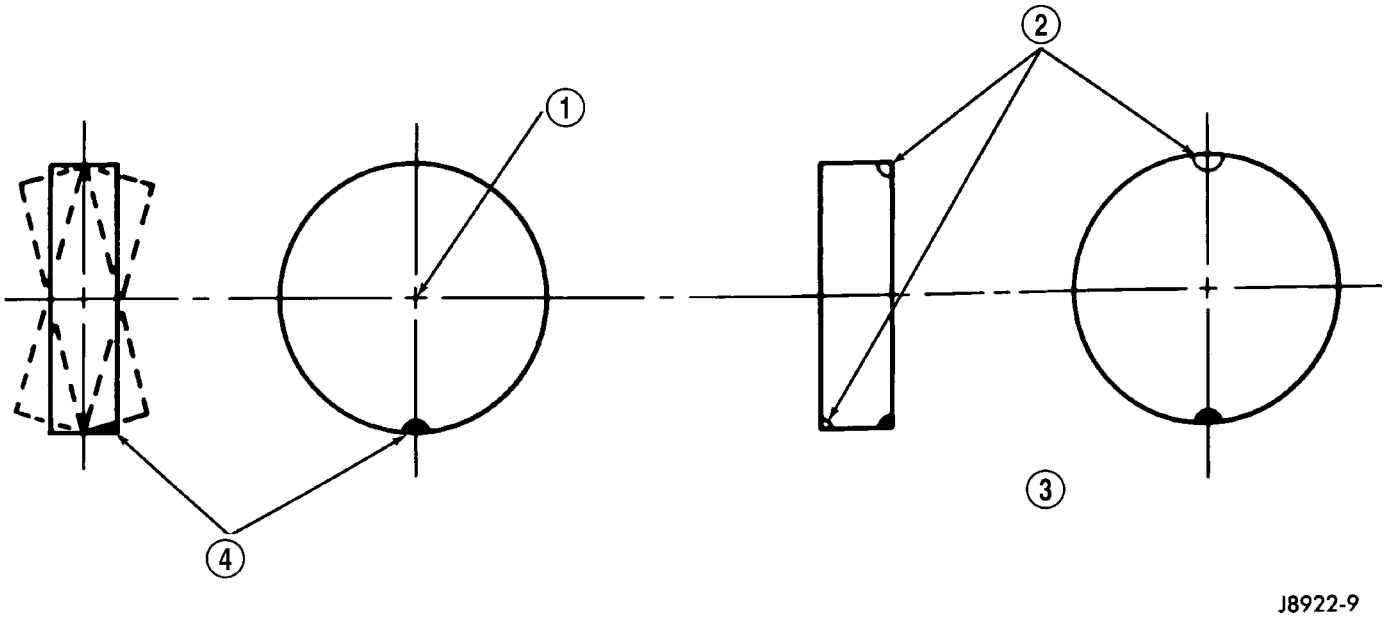

Fig. 10 Dynamic Unbalance & Balance

1 - CENTER LINE OF SPINDLE
2 - ADD BALANCE WEIGHTS HERE
3 - CORRECTIVE WEIGHT LOCATION
4 - HEAVY SPOT WHEEL SHIMMY AND VIBRATION

TIRES

DESCRIPTION

DESCRIPTION - TIRES

Tires are designed and engineered for each specific vehicle. They provide the best overall performance for normal operation. The ride and handling characteristics match the vehicle's requirements. With proper care they will give excellent reliability, traction, skid resistance, and tread life.

Driving habits have more effect on tire life than any other factor. Careful drivers will obtain in most cases, much greater mileage than severe use or careless drivers. A few of the driving habits which will shorten the life of any tire are:

- Rapid acceleration
- Severe brake applications
- High speed driving
- Excessive speeds on turns
- Striking curbs and other obstacles

Radial-ply tires are more prone to irregular tread wear. It is important to follow the tire rotation interval (Refer to 22 - TIRES/WHEELS - STANDARD PROCEDURE). This will help to achieve a greater tread life.

TIRE IDENTIFICATION

Tire type, size, aspect ratio and speed rating are encoded in the letters and numbers imprinted on the side wall of the tire. Refer to the chart to decipher the tire identification code (Fig. 11).

Performance tires have a speed rating letter after the aspect ratio number.

LETTER	SPEED RATING
P	150 km/h (93 mph)
S	180 km/h (112 mph)
T	190 km/h (118 mph)
U	200 km/h (124 mph)
H	210 km/h (130 mph)
V	240 km/h (149 mph)
W	270 km/h (168 mph)
Y	300 km/h (186 mph)

The speed rating is not always printed on the tire sidewall.

TIRE CHAINS

Tire snow chains may be used on **certain** models. Refer to the Owner's Manual for more information.

TIRES (Continued)

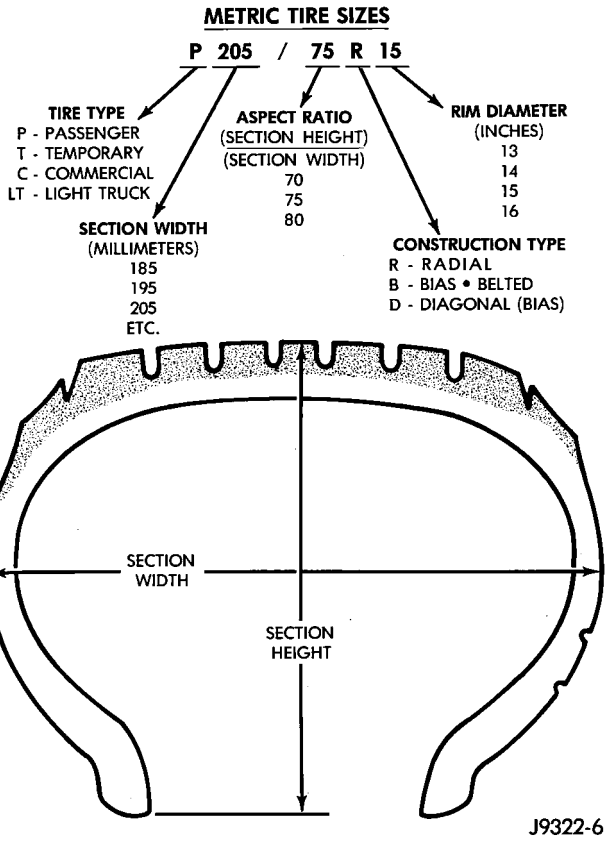

Fig. 11 Tire Identification

DESCRIPTION - RADIAL-PLY TIRES

Radial-ply tires improve handling, tread life and ride quality, and decrease rolling resistance.

Radial-ply tires must always be used in sets of four. Under no circumstances should they be used on the front only. They may be mixed with temporary spare tires when necessary. A maximum speed of 50 MPH is recommended while a temporary spare is in use.

Radial-ply tires have the same load-carrying capacity as other types of tires of the same size. They also use the same recommended inflation pressures.

The use of oversized tires, either in the front or rear of the vehicle, can cause vehicle drive train failure. This could also cause inaccurate wheel speed signals when the vehicle is equipped with Anti-Lock Brakes.

The use of tires from different manufactures on the same vehicle is NOT recommended. The proper tire pressure should be maintained on all four tires.

DESCRIPTION - TIRE INFLATION PRESSURES

Under inflation will cause rapid shoulder wear, tire flexing, and possible tire failure (Fig. 12).

Over inflation will cause rapid center wear and loss of the tire's ability to cushion shocks (Fig. 13).

Improper inflation can cause:

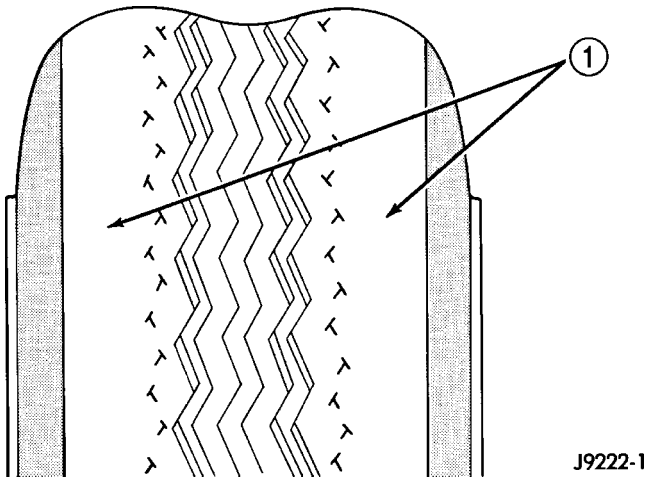

Fig. 12 Under Inflation Wear

1 - THIN TIRE THREAD AREAS

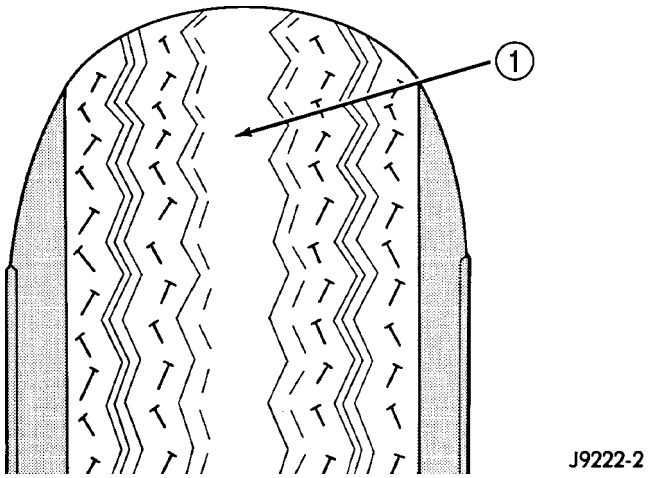

Fig. 13 Over Inflation Wear

1 - THIN TIRE THREAD AREA

- Uneven wear patterns
- Reduced tread life
- Reduced fuel economy
- Unsatisfactory ride
- Vehicle drift

For proper tire pressure specification refer to the Tire Inflation Pressure Chart provided with the vehicles Owners Manual. A Certification Label on the drivers side door pillar provides the minimum tire and rim size for the vehicle. The label also list the cold inflation pressure for these tires at full load operation

Tire pressures have been chosen to provide safe operation, vehicle stability, and a smooth ride. Tire pressure should be checked cold once a month. Tire pressure decreases as the ambient temperature drops. Check tire pressure frequently when ambient temperature varies widely.

TIRES (Continued)

Tire inflation pressures are cold inflation pressure. The vehicle must sit for at least 3 hours to obtain the correct cold inflation pressure reading. Or be driven less than one mile after sitting for 3 hours. Tire inflation pressures may increase from 2 to 6 pounds per square inch (psi) during operation. Do not reduce this normal pressure build-up.

WARNING: OVER OR UNDER INFLATED TIRES CAN AFFECT VEHICLE HANDLING AND TREAD WEAR. THIS MAY CAUSE THE TIRE TO FAIL SUDDENLY, RESULTING IN LOSS OF VEHICLE CONTROL.

DESCRIPTION - TIRE PRESSURE FOR HIGH SPEED

Where speed limits allow the vehicle to be driven at high speeds, correct tire inflation pressure is very important. For speeds up to and including 120 km/h (75 mph), tires must be inflated to the pressures shown on the tire placard. For continuous speeds in excess of 120 km/h (75 mph), tires must be inflated to the maximum pressure specified on the tire sidewall.

Vehicles loaded to the maximum capacity should not be driven at continuous speeds above 75 mph (120 km/h).

For emergency vehicles that are driven at speeds over 90 mph (144 km/h), special high speed tires must be used. Consult tire manufacturer for correct inflation pressure recommendations.

DESCRIPTION - REPLACEMENT TIRES

The original equipment tires provide a proper balance of many characteristics such as:

- Ride
- Noise
- Handling
- Durability
- Tread life
- Traction
- Rolling resistance
- Speed capability

It is recommended that tires equivalent to the original equipment tires be used when replacement is needed.

Failure to use equivalent replacement tires may adversely affect the safety and handling of the vehicle.

The use of oversize tires may cause interference with vehicle components. Under extremes of suspension and steering travel, interference with vehicle components may cause tire damage.

WARNING: FAILURE TO EQUIP THE VEHICLE WITH TIRES HAVING ADEQUATE SPEED CAPABILITY CAN RESULT IN SUDDEN TIRE FAILURE.

DIAGNOSIS AND TESTING

DIAGNOSIS AND TESTING - PRESSURE GAUGES

A quality air pressure gauge is recommended to check tire pressure. After checking the air pressure, replace valve cap finger tight.

DIAGNOSIS AND TESTING - TREAD WEAR INDICATORS

Tread wear indicators are molded into the bottom of the tread grooves. When tread depth is 1.6 mm (1/16 in.), the tread wear indicators will appear as a 13 mm (1/2 in.) band (Fig. 14).

Tire replacement is necessary when indicators appear in two or more grooves or if localized balding occurs.

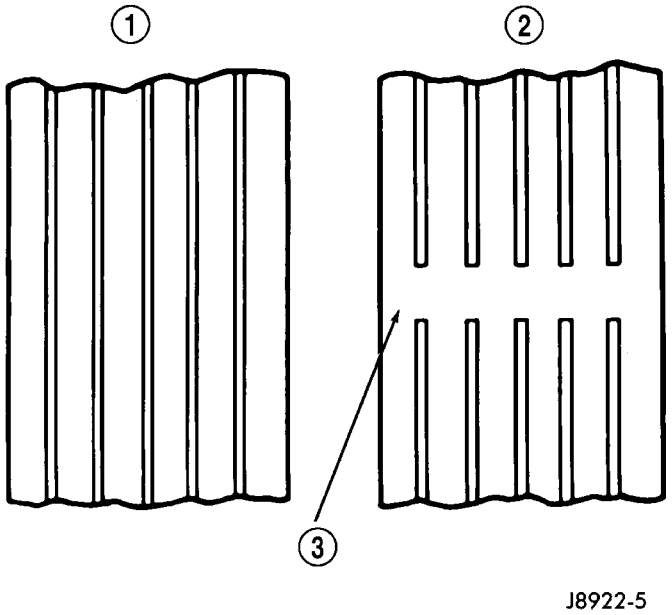

Fig. 14 Tread Wear Indicators

1 - TREAD ACCEPTABLE
2 - TREAD UNACCEPTABLE
3 - WEAR INDICATOR

DIAGNOSIS AND TESTING - TIRE WEAR PATTERNS

Under inflation will cause wear on the shoulders of tire. Over inflation will cause wear at the center of tire.

Excessive camber causes the tire to run at an angle to the road. One side of tread is then worn more than the other (Fig. 15).

TIRES (Continued)

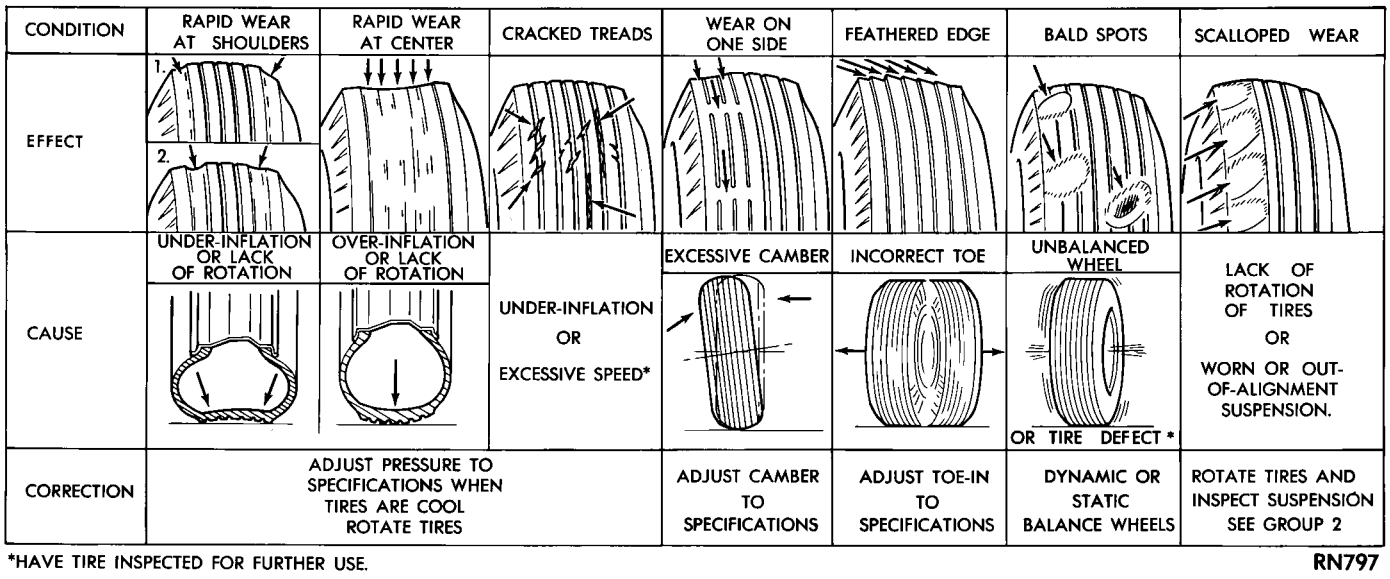

Fig. 15 Tire Wear Patterns

Excessive toe-in or toe-out causes wear on the tread edges and a feathered effect across the tread (Fig. 15).

DIAGNOSIS AND TESTING - TIRE NOISE OR VIBRATION

Radial-ply tires are sensitive to force impulses caused by improper mounting, vibration, wheel defects, or possibly tire imbalance.

To find out if tires are causing the noise or vibration, drive the vehicle over a smooth road at varying speeds. Note the noise level during acceleration and deceleration. The engine, differential and exhaust noises will change as speed varies, while the tire noise will usually remain constant.

DIAGNOSIS AND TESTING - TIRE/VEHICLE LEAD

Use the following Vehicle Lead Diagnosis And Correction Chart to diagnose and correct a vehicle lead or drift problem (Fig. 16).

TIRES (Continued)

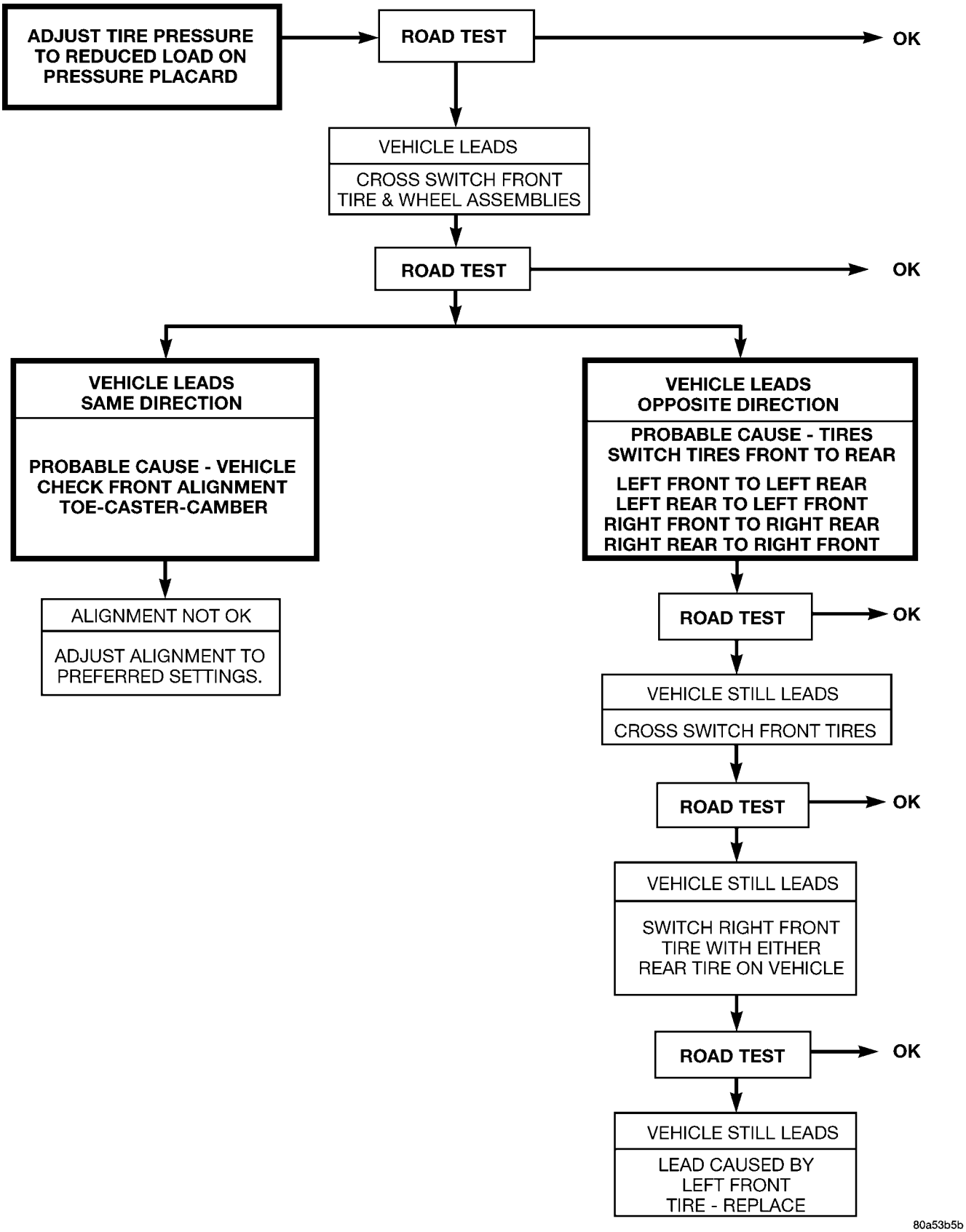

Fig. 16 VEHICLE LEAD DIAGNOSIS AND CORRECTION CHART

TIRES (Continued)

STANDARD PROCEDURE - REPAIRING LEAKS

For proper repairing, a radial tire must be removed from the wheel. Repairs should only be made if the defect, or puncture, is in the tread area (Fig. 17). The tire should be replaced if the puncture is located in the sidewall.

Deflate tire completely before removing the tire from the wheel. Use lubrication such as a mild soap solution when dismounting or mounting tire. Use tools free of burrs or sharp edges which could damage the tire or wheel rim.

Before mounting tire on wheel, make sure all rust is removed from the rim bead and repaint if necessary.

Install wheel on vehicle, and tighten to proper torque specification. (Refer to 22 - TIRES/WHEELS/ WHEELS - STANDARD PROCEDURE).

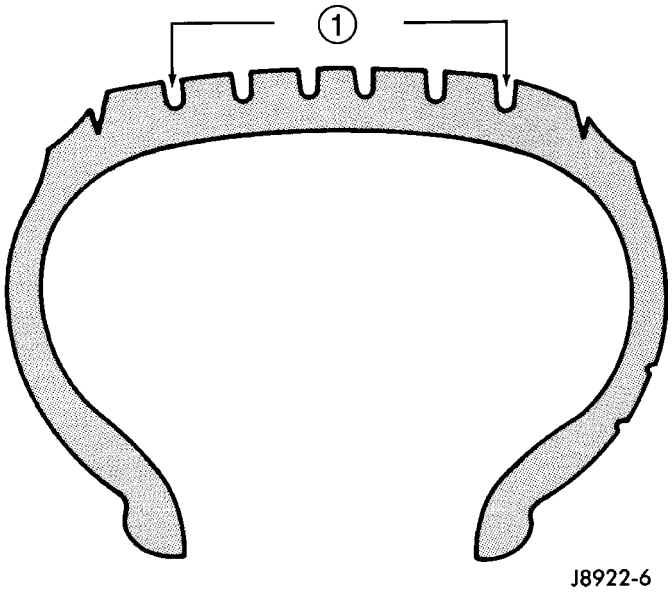

Fig. 17 Tire Repair Area

1 - REPAIRABLE AREA

CLEANING - TIRES

Remove the protective coating on the tires before delivery of a vehicle. This coating may cause deterioration of the tires.

To remove the protective coating, apply warm water and let it soak for a few minutes. Afterwards, scrub the coating away with a soft bristle brush. Steam cleaning may also be used to remove the coating.

NOTE: DO NOT use gasoline, mineral oil, oil-based solvent or a wire brush for cleaning.

SPECIFICATIONS

TIRE SIZE

DESCRIPTION	SPECIFICATION
TIRE	P215/75R15
TIRE	P225/75R15
TIRE	LT245/75R16
TIRE	30x9.50R15

SPARE TIRE

DESCRIPTION

DESCRIPTION - SPARE/TEMPORARY

The temporary spare tire is designed for emergency use only. The original tire should be repaired or replaced at the first opportunity, then reinstalled. Do not exceed speeds of 50 M.P.H. when using the temporary spare tire. Refer to Owner's Manual for complete details.

DESCRIPTION - FULL SIZE, SPARE WHEEL WITH MATCHING TIRE

The spare is a full usage wheel with a matching tire, It can be used within the (posted legal) speed limits or distance limitations as of the rest of the vehicles four tires. Refer to Owner's Manual for complete details.

SPARE TIRE CARRIER

REMOVAL

(1) Remove the spare tire from the wheel bracket (Fig. 18).

(2) Remove the bolts that attach the tire bracket to the tailgate (Fig. 19).

(3) Disconnect CHMSL.

(4) Remove the bracket and the gaskets from the tailgate.

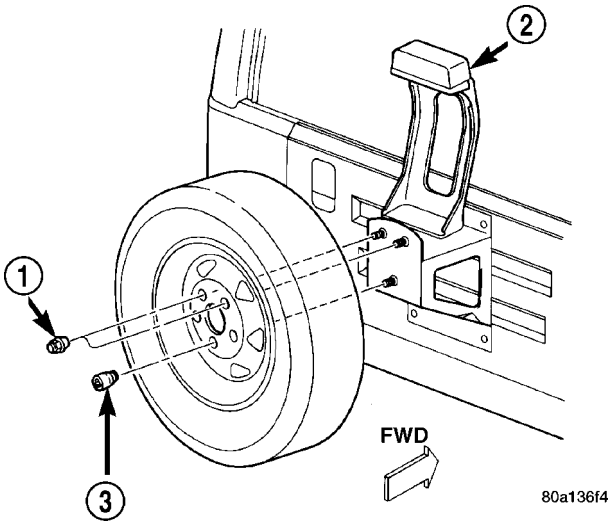

Fig. 18 Spare Tire

1 - WHEEL NUT
2 - CHMSL
3 - LOCK NUT

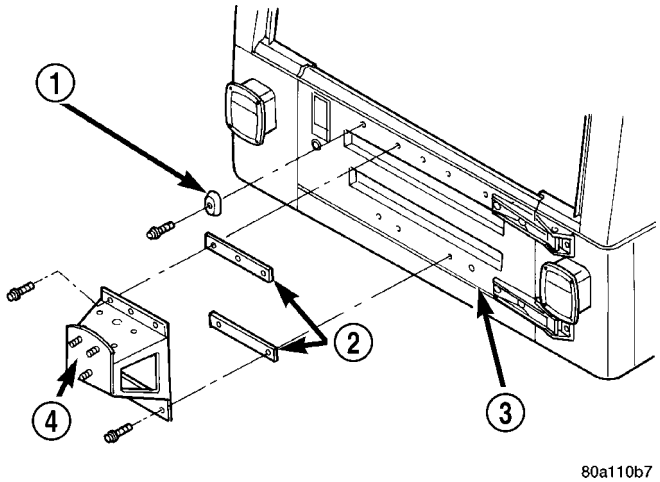

Fig. 19 Spare Tire Bracket

1 - BUMPER
2 - GASKET
3 - TAILGATE
4 - SPARE TIRE BRACKET

INSTALLATION

(1) Position the gaskets and the tire bracket on the tailgate and install the bolts. Tighten the bolts to 24 N·m (17 ft. lbs.) torque.

(2) Connect CHMSL connector.

(3) Install the spare tire on the tire bracket.

WHEELS

DESCRIPTION

The rim size is on the vehicle safety certification label located on the drivers door shut face. The size of the rim is determined by the drivetrain package. Original equipment wheels/rims are designed for operation up to the specified maximum vehicle capacity.

All models use stamped steel, cast aluminum or forged aluminum wheels. Every wheel has raised sections between the rim flanges and rim drop well called safety humps (Fig. 20).

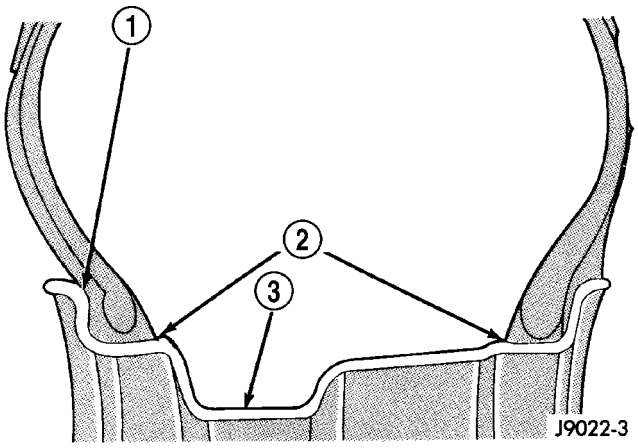

Fig. 20 Safety Rim

1 - FLANGE
2 - RIDGE
3 - WELL

Initial inflation of the tire forces the bead over these raised sections. In case of rapid loss of air pressure, the raised sections help hold the tire on the wheel.

The wheel studs and nuts are designed for specific applications. All aluminum and some steel wheels have wheel stud nuts with an enlarged nose. This enlarged nose is necessary to ensure proper retention of the wheels. Do not use replacement studs or nuts with a different design or lesser quality.

WHEELS (Continued)

DIAGNOSIS AND TESTING - WHEEL INSPECTION

Inspect wheels for:

- Excessive run out
- Dents or cracks
- Damaged wheel lug nut holes
- Air Leaks from any area or surface of the rim

NOTE: Do not attempt to repair a wheel by hammering, heating or welding.

If a wheel is damaged an original equipment replacement wheel should be used. When obtaining replacement wheels, they should be equivalent in load carrying capacity. The diameter, width, offset, pilot hole and bolt circle of the wheel should be the same as the original wheel.

WARNING: FAILURE TO USE EQUIVALENT REPLACEMENT WHEELS MAY ADVERSELY AFFECT THE SAFETY AND HANDLING OF THE VEHICLE. USED WHEELS ARE NOT RECOMMENDED. THE SERVICE HISTORY OF THE WHEEL MAY HAVE INCLUDED SEVERE TREATMENT OR VERY HIGH MILEAGE. THE RIM COULD FAIL WITHOUT WARNING.

STANDARD PROCEDURE

STANDARD PROCEDURE - WHEEL INSTALLATION

The wheel studs and nuts are designed for specific applications. They must be replaced with equivalent parts. Do not use replacement parts of lesser quality or a substitute design. All aluminum and some steel wheels have wheel stud nuts which feature an enlarged nose. This enlarged nose is necessary to ensure proper retention of the aluminum wheels.

Before installing the wheel, be sure to remove any build up of corrosion on the wheel mounting surfaces. Ensure wheels are installed with good metal-to-metal contact.

To install the wheel, first position it properly on the mounting surface. All wheel nuts should then be tightened just snug. Gradually tighten them in sequence to the proper torque specification (Fig. 21).

WARNING: NEVER USE OIL OR GREASE ON STUDS OR NUTS. INSTALLING WHEELS WITHOUT GOOD METAL-TO-METAL CONTACT OR USING CHROME PLATED LUG NUTS WITH CHROME PLATED WHEELS COULD CAUSE LOOSENING OF WHEEL NUTS. THIS COULD AFFECT THE SAFETY AND HANDLING OF THE VEHICLE.

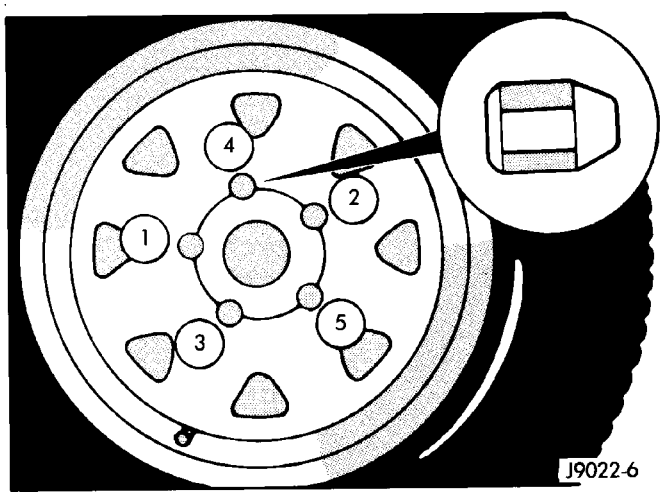

Fig. 21 Lug Nut Tightening Pattern

WHEEL REPLACEMENT

Wheels must be replaced if they have:

- Excessive runout
- Bent or dented
- Leak air through welds
- Have damaged bolt holes

Wheel repairs employing hammering, heating, or welding are not allowed.

Original equipment wheels are available through your dealer. Replacement wheels from any other source should be equivalent in:

- Load carrying capacity
- Diameter
- Width
- Offset
- Mounting configuration

Failure to use equivalent replacement wheels may affect the safety and handling of your vehicle. Replacement with **used** wheels is not recommended. Their service history may have included severe treatment.

STANDARD PROCEDURE - WHEEL REPLACEMENT

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WHEELS (Continued)

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- Load carrying capacity
- Diameter
- Width
- Offset
- Mounting configuration

Failure to use equivalent replacement wheels may affect the safety and handling of your vehicle. Replacement with **used** wheels is not recommended. Their service history may have included severe treatment.

SPECIFICATIONS

TORQUE CHART

TORQUE SPECIFICATIONS

DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Lug Nut 1/2 X 20 with 60° Cone	115-156	85-115	—

STUDS

REMOVAL

(1) Raise and support vehicle.
(2) Remove wheel and tire assembly.
(3) Remove brake caliper, caliper adapter and rotor, (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/ROTORS - REMOVAL).
(4) Remove stud from hub with Remover C-4150A (Fig. 22).

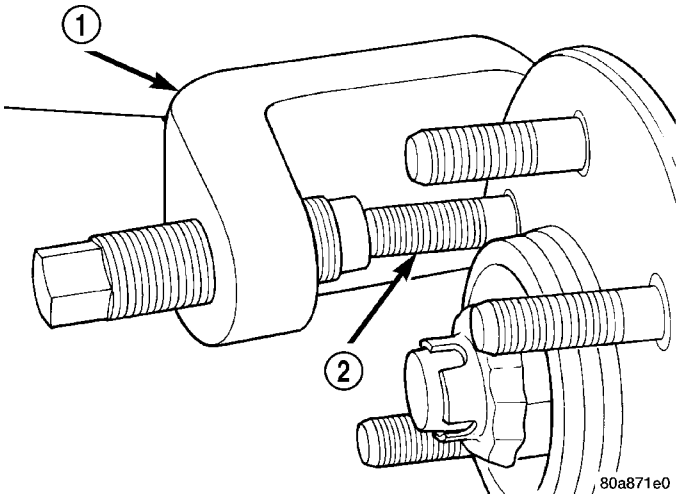

Fig. 22 Wheel Stud Removal

1 - REMOVER
2 - WHEEL STUD

INSTALLATION

(1) Install new stud into hub flange.
(2) Install three washers onto stud, then install lug nut with the flat side of the nut against the washers.
(3) Tighten lug nut until the stud is pulled into the hub flange. Verify that the stud is properly seated into the flange.
(4) Remove lug nut and washers.
(5) Install the brake rotor, caliper adapter, and caliper, (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/ROTORS - INSTALLATION).
(6) Install wheel and tire assembly, use new lug nut on stud or studs that were replaced. (Refer to 22 - TIRES/WHEELS/WHEELS - STANDARD PROCEDURE).
(7) Remove support and lower vehicle.

BODY

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BODY

WARNING

SAFETY PRECAUTIONS AND WARNINGS

WARNING: Use an osha approved breathing filter when spraying paint or solvents in a confined area. Personal injury can result.

- **Avoid prolonged skin contact with petroleum or alcohol – based cleaning solvents. Personal injury can result.**
- **Do not stand under a hoisted vehicle that is not properly supported on safety stands. Personal injury can result.**

CAUTION: When holes must be drilled or punched in an inner body panel, verify depth of space to the outer body panel, electrical wiring, or other components. Damage to vehicle can result.

- **Do not weld exterior panels unless combustible material on the interior of vehicle is removed from the repair area. Fire or hazardous conditions, can result.**
- **Always have a fire extinguisher ready for use when welding.**
- **Disconnect the negative (-) cable clamp from the battery when servicing electrical components that are live when the ignition is OFF. Damage to electrical system can result.**
- **Do not use abrasive chemicals or compounds on painted surfaces. Damage to finish can result.**
- **Do not use harsh alkaline based cleaning solvents on painted or upholstered surfaces. Damage to finish or color can result.**
- **Do not hammer or pound on plastic trim panel when servicing interior trim. Plastic panels can break.**

STANDARD PROCEDURE

STANDARD PROCEDURE - BODY LUBRICATION

All mechanisms and linkages should be lubricated when necessary. This will maintain ease of operation and provide protection against rust and excessive wear. The weatherstrip seals should be lubricated to prolong their life as well as to improve door sealing.

All applicable exterior and interior vehicle operating mechanisms should be inspected and cleaned. Pivot/sliding contact areas on the mechanisms should then be lubricated.

(1) When necessary, lubricate the operating mechanisms with the specified lubricants.

(2) Apply silicone lubricant to a cloth and wipe it on door seals to avoid over-spray that can soil passenger's clothing.

(3) Before applying lubricant, the component should be wiped clean. After lubrication, any excess lubricant should be removed.

(4) The hood latch, latch release mechanism, latch striker, and safety latch should be lubricated periodically.

(5) The door lock cylinders should be lubricated twice each year (preferably autumn and spring):

- Spray a small amount of lock cylinder lubricant directly into the lock cylinder.

• Apply a small amount to the key and insert it into the lock cylinder.
• Rotate it to the locked position and then back to the unlocked position several times.
• Remove the key. Wipe the lubricant from it with a clean cloth to avoid soiling of clothing.

Component	Fluid, Lubricant, or Genuine Part
Hinges:	
Door And Hood	Mopar® Engine Oil
Liftgate	Mopar® Multi-Purpose Lube NLGI Grade 2 EP, GC-LB
Latches:	
Door, Hood/Safety Catch, Liftgate	Mopar® Multi-Purpose Lube NLG Grade 2 EP, GC-LB
Seat Regulator & Track	Mopar® Multi-Purpose Lube NLGI Grade 2 EP, GC-LB
Window System Components	Mopar® Spray White Lube
Lock Cylinders	Mopar® Lock Cylinder Lube
Parking Brake Mechanism	Mopar® Wheel Brg. Grease NLGI Grade 1, GC-LBB
Soft Top	Mopar® Soft Top Zipper Cleaner & Lubricant

STANDARD PROCEDURE - PLASTIC BODY PANEL REPAIR

There are many different types of plastics used in today's automotive environment. We group plastics in three different categories: Rigid, Semi-Rigid, and Flexible. Any of these plastics may require the use of an adhesion promoter for repair. These types of plastic are used extensively on DaimlerChrysler Motors vehicles. Always follow repair material manufacturer's plastic identification and repair procedures.

Rigid Plastics:

Examples of rigid plastic use: Fascias, Hoods, Doors, and other Body Panels, which include SMC, ABS, and Polycarbonates.

Semi-Rigid Plastics:

Examples of semi-rigid plastic use: Interior Panels, Under Hood Panels, and other Body Trim Panels.

Flexible Plastics:

Examples of flexible plastic use: Fascias, Body Moldings, and upper and lower Fascia Covers.

Repair Procedure:

The repair procedure for all three categories of plastics is basically the same. The one difference is the material used for the repair. The materials must be specific for each substrate, rigid repair material for rigid plastic repair, semi-rigid repair material for semi-rigid plastic repair and flexible repair material for flexible plastic repair.

Adhesion Promoter/Surface Modifier:

Adhesion Promoters/Surface Modifiers are required for certain plastics. All three categories may have plastics that require the use of adhesion promoter/surface modifiers. Always follow repair material manufacturer's plastic identification and repair procedures.

SAFETY PRECAUTION AND WARNINGS

WARNING:

• EYE PROTECTION SHOULD BE USED WHEN SERVICING COMPONENTS. PERSONAL INJURY CAN RESULT.
• USE AN OSHA APPROVED BREATHING MASK WHEN MIXING EPOXY, GRINDING, AND SPRAYING PAINT OR SOLVENTS IN A CONFINED AREA. PERSONAL INJURY CAN RESULT.
• AVOID PROLONGED SKIN CONTACT WITH RESIN, PETROLEUM, OR ALCOHOL BASED SOLVENTS. PERSONAL INJURY CAN RESULT.
• DO NOT VENTURE UNDER A HOISTED VEHICLE THAT IS NOT PROPERLY SUPPORTED ON SAFETY STANDS. PERSONAL INJURY CAN RESULT.

NOTE:

• When holes must be drilled or cut in body panels, verify locations of internal body components and electrical wiring. Damage to vehicle can result.
• Do not use abrasive chemicals or compounds on undamaged painted surfaces around repair areas. Damage to finish can result.

BODY (Continued)

RIGID, SEMI-RIGID, AND FLEXIBLE PLASTIC PARTS TYPES

CODE	FAMILY NAME	COMMON TRADE NAME	TYPICAL APPLICATION
ASA	ACRYLONITRILE STYRENE ACRYLITE	LURANS	CONSOLES, GRILLES
ABS	ACRYLONITRILE BUTADIENE STYRENE	TERLURAN	″A″ PILLARS, CONSOLES, GRILLES
ABS/PC	ABS/PC ALLOY	PULSE, PROLOY, BAYBLEND	DOORS, INSTRUMENT PANELS
ABS/PVC	ABS/PV ALLOY	PROLOY, PULSE, LUSTRAN, CYCLOVIN	DOOR PANELS, GRILLES, TRIM
BMC	BULK MOLDING COMPOUND	BMC	FENDER EXTENSIONS
EMA	EHTYLENE METHYL ACRYLATE/IONOMER	SURLYN, EMA, IONOMER	BUMPER GUARDS, PADS
METTON	METTON	METTON	GRILLES, KICK PANELS, RUNNING BOARDS
MPPO	MODIFIED POLYPHENYLENE OXIDE	MPPO	SPOILER ASSEMBLY
PA	POLYAMID	ZYTEL, VYDYNE, PA, MINLON	FENDERS, QUARTER PANELS
PET	THERMOPLASTIC POLYESTER	RYNITE	TRIM
PBT/PPO	PBT/PPO ALLOY	GERMAX	CLADDINGS
PBTP	POLYBUTYLENE THEREPTHALATE	PBT, PBTP, POCAN, VALOX	WHEEL COVERS, FENDERS, GRILLES
PBTP/EEBC	POLYBUTYLENE THEREPTHALATE/EEBC ALLOY	BEXLOY, ″M″, PBTP/EEBC	FASCIAS, ROCKER PANEL, MOLDINGS
PC	POLYCARBONATE	LEXAN, MERLON, CALIBRE, MAKROLON PC	TAIL LIGHT LENSES, IP TRIM, VALANCE PANELS
PC/ABS	PC/ABS ALLOY	GERMAX, BAY BLENDS, PULSE	DOORS, INSTRUMENT PANELS
PPO	POLYPHENYLENE OXIDE	AZDEL, HOSTALEN, MARLEX, PRFAX, NORYL, GTX, PPO	INTERIOR TRIM, DOOR PANELS, SPLASH SHIELDS, STEERING COLUMN SHROUD
PPO/PA	POLYPHENYLENE/ POLYAMID	PPO/PA, GTX 910	FENDERS, QUARTER PANELS
PR/FV	FIBERGLASS REINFORCED PLASTIC	FIBERGLASS, FV, PR/FV	BODY PANELS
PS	POLYSTYRENE	LUSTREX, STYRON, PS	DOOR PANELS, DASH PANELS
RTM	RESIN TRANSFER MOLDING COMPOUND	RTM	BODY PANELS
SMC	SHEET MOLDED COMPOUND	SMC	BODY PANELS
TMC	TRANSFER MOLDING COMPOUND	TMC	GRILLES

BODY (Continued)

CODE	FAMILY NAME	COMMON TRADE NAME	TYPICAL APPLICATION
UP	UNSATURATED POLYESTER (THERMOSETTING)	SMC, BMC, TMC, ZMC, IMC, XSMC, UP	GRILLE OPENING PANEL, LIFTGATES, FLARESIDE FENDERS, FENDER EXTENSIONS
EEBC	ETHER/ESTER BLOCKED CO-POLYMER	EEBC	BUMPERS
EEBC/PBTP	EEBC/POLYBUTYLENE TEREPTHALATE	EEBC, PBTP, BEXLOY	BUMPER, ROCKER PANELS
EMPP	ETHYLENE MODIFIED POLYPROPYLENE	EMPP	BUMPER COVERS
EPDM	ETHYLENE/ PROPROPYLENE DIENE MONOMER	EPDM, NORDEL, VISTALON	BUMPERS
EPM	ETHYLENE/ PROPROPYLENE CO-POLYMER	EPM	FENDERS
MPU	FOAM POLYURETHANE	MPU	SPOILERS
PE	POLYETHYLENE	ALATHON, DYLAN, LUPOLEN, MARLEX	-
PP	POLYPROPYLENE (BLENDS)	NORYL, AZDEL, MARLOX, DYLON, PRAVEX	INNER FENDER, SPOILERS, KICK PANELS, A-PILLARS, DOOR PANELS, B-PILLARS, QUARTER PANELS, SPORT BAR TRIM, LIFTGATE TRIM, DECKLID TRIM SCUFF PLATES, KICK PANELS, CONSOLES
PP/EPDM	PP/EPDM ALLOY	PP/EPDM	SPOILERS, GRILLES
PUR	POLYURETHANE	COLONELS, PUR, PU	FASCIAS, BUMPERS
PUR/PC	PUR/PC ALLOY	TEXIN	BUMPERS
PVC	POLYVINYL CHLORIDE	APEX, GEON, VINYLITE	BODY MOLDINGS, WIRE INSULATION, STEERING WHEELS
RIM	REACTION INJECTED MOLDED POLYURETHANE	RIM, BAYFLEX	FRONT FASCIAS, MODULAR WINDOWS
RRIM	REINFORCED REACTION INJECTED MOLDED	PUR, RRIM	FASCIAS, BODY PANELS, BODY TRIMS
TPE	THERMO POLYETHYLENE	TPE, HYTREL, BEXLOY-V	FASCIAS, BUMPERS, CLADDINGS
TPO	THERMOPOLYOLEFIN	POLYTROPE, RENFLEX, SANTOPRENE, VISAFLEX, ETA, APEX, TPO, SHIELDS, CLADDINGS	BUMPERS, END CAPS, TELCAR, RUBBER, STRIPS, SIGHT, INTERIOR B POST
TPP	THERMO-POLYPROPYLENE	TPP	BUMPERS
TPU	THERMOPOLYURETHANE, POLYESTER	TPU, HYTREL, TEXIN, ESTANE	BUMPERS, BODY SIDE, MOLDINGS, FENDERS, FASCIAS

BODY (Continued)

PANEL SECTIONING

If it is required to section a large panel for a plastic repair, it will be necessary to reinforce the panel (Fig. 1). To bond two plastic panels together, a reinforcement must overlap both panels. The panels must be "V'd" at a 20 degree angle. The area to be reinforced should be washed, then sanded. Be sure to wipe off any excess soap and water when finished. Lightly sand or abrade the plastic with an abrasive pad or sandpaper. Blow off any dust with compressed air or wipe with a clean dry rag.

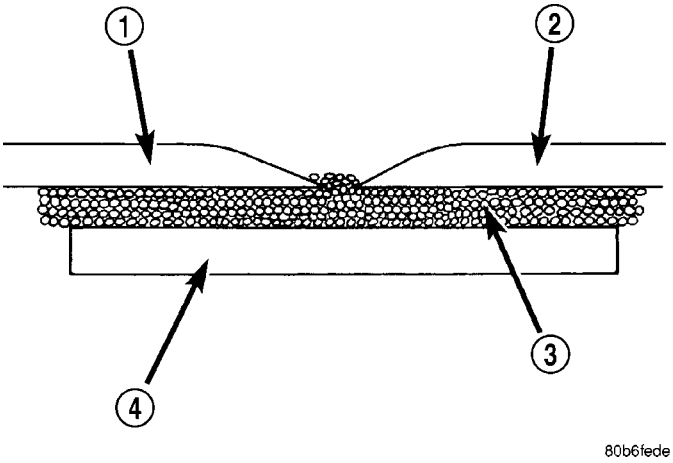


Fig. 1 PANEL SECTIONING

1 - EXISTING PANEL
2 - NEW PANEL
3 - PANEL ADHESIVE
4 - BONDING STRIP

When bonding plastic panels, Follow repair material manufacturers recommendations. Be sure that enough adhesive has been applied to allow squeeze out and to fill the full bond line. Once the pieces have been brought together, do not move them until the adhesive is cured. The assembly can be held together with clamps, rivets, etc. A faster cure can be obtained by heating with a heat lamp or heat gun. After the parts have been bonded and have had time to cure, rough sand the seam and apply the final adhesive filler to the area being repaired. Smooth the filler with a spreader, wooden tongue depressor, or squeegee. For fine texturing, a small amount of water can be applied to the filler surface while smoothing. The cured filler can be sanded as necessary and, as a final step, cleanup can be done with soapy water. Wipe the surface clean with a dry cloth allowing time for the panel to dry before moving on with the repair.

PANEL REINFORCEMENT

Structural repair procedures for rigid panels with large cracks and holes will require a reinforcement backing. Reinforcements can be made with several applications of glass cloth saturated with structural adhesive. Semi-rigid or flexible repair materials should be used for semi-rigid or flexible backing reinforcement (Fig. 2) and (Fig. 3). Open meshed fiberglass dry wall tape can be used to form a reinforcement. The dry wall tape allows the resin to penetrate through and make a good bond between the panel and the adhesive. Structurally, the more dry wall tape used, the stronger the repair.

Another kind of repair that can be done to repair large cracks and holes is to use a scrap piece of similar plastic and bond with structural adhesive. The reinforcement should cover the entire break and should have a generous amount of overlap on either side of the cracked or broken area.

When repairing plastic, the damaged area is first "V'd" out, or beveled. Large bonding areas are desirable when repairing plastic because small repairs are less likely to hold permanently. Beveling the area around a crack at a 20 degree angle will increase the bonding surface for a repair (Fig. 4). It is recommended that sharp edges be avoided because the joint may show through after the panel is refinished.

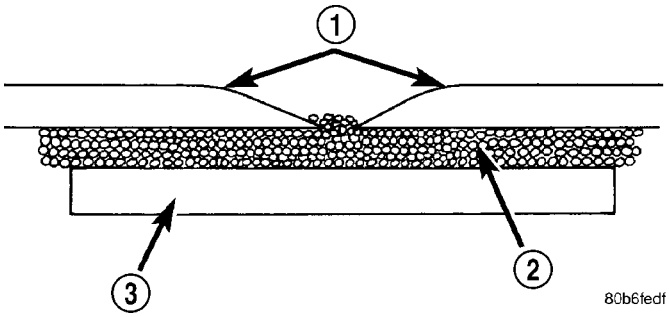


Fig. 2 SOFTENED EDGES

1 - SOFTENED EDGES
2 - PANEL ADHESIVE
3 - BONDING STRIP

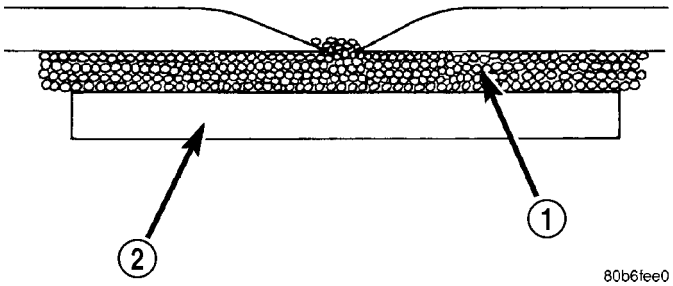


Fig. 3 PANEL REINFORCEMENT

1 - PANEL ADHESIVE
2 - REINFORCEMENT

- Panel repair for both flexible and rigid panels are basically the same. The primary difference between flexible panel repair and rigid panel repair is in the adhesive materials used (Fig. 5).

BODY (Continued)

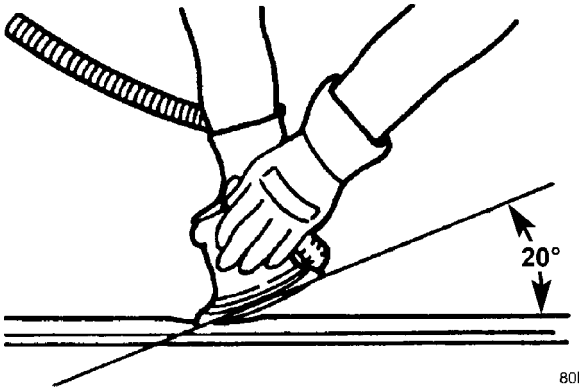


Fig. 4 BEVELING ANGLE - 20 DEGREE

- The technician should first decide what needs to be done when working on any type of body panel. One should determine if it is possible to return the damage part to its original strength and appearance without exceeding the value of the replacement part.
- When plastic repairs are required, it is recommended that the part be left on the vehicle when ever possible. That will save time, and the panel will remain stationary during the repair. Misalignment can cause stress in the repair areas and can result in future failure.

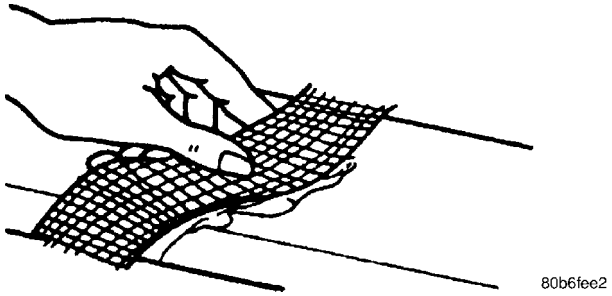


Fig. 5 FIBERGLASS TAPE

VISUAL INSPECTION

Composite materials can mask the severity of an accident. Adhesive bond lines, interior structure of the doors, and steel structures need to be inspected carefully to get a true damage assessment. Close inspection may require partial removal of interior trim or inner panels.

Identify the type of repair: Puncture or Crack - Damage that has penetrated completely through the panel. Damage is confined to one general area; a panel section is not required. However, a backer panel, open fiberglass tape, or matted material must be bonded from behind (Fig. 7) (Fig. 6).

PANEL SURFACE PREPARATION

If a body panel has been punctured, cracked, or crushed, the damaged area must be removed from the panel to achieve a successful repair. All spider web cracks leading away from a damaged area must be stopped or removed. To stop a running crack in a panel, drill a 6 mm (0.250 in.) hole at the end of the crack farthest away from the damage. If spider web cracks can not be stopped, the panel would require replacement. The surfaces around the damaged area should be stripped of paint and freed from wax and oil. Scuff surfaces around repair area with 360 grit wet/dry sandpaper, or equivalent, to assure adhesion of repair materials.

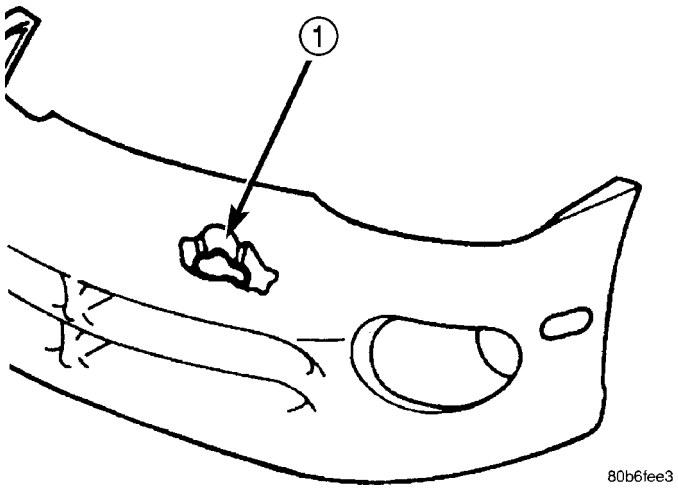


Fig. 6 DAMAGE COMPONENT

1 - PUNCTURE

PATCHING PANELS

An panel that has extensive puncture type damage can be repaired by cutting out the damaged material (Fig. 7). Use a suitable reciprocating saw or cut off wheel to remove the section of the panel that is damaged. The piece cut out can be used as a template to shape the new patch. It is not necessary to have access to the back of the panel to install a patch. Bevel edges of cutout at 20 degrees to expose a larger bonding area on the outer side. This will allow for an increased reinforcement areas.

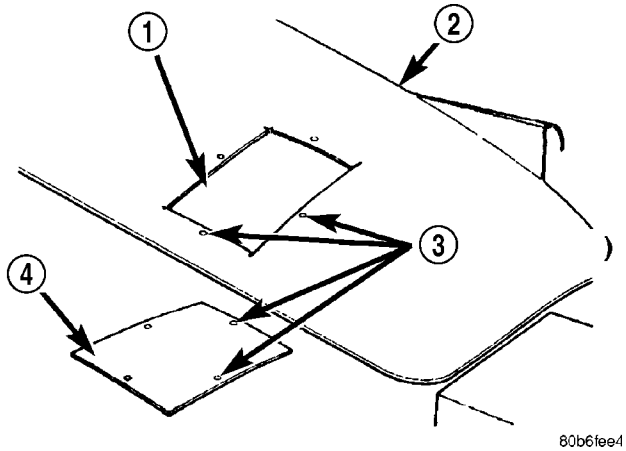


Fig. 7 DAMAGED PANEL CUTOUT AND PATCH

1 - CUTOUT
2 - DAMAGED BODY PANEL
3 - 4 MM (0.160 IN.) HOLES
4 - PATCH CUT TO SIZE

BODY (Continued)

PANEL PATCH FABRICATIONS

A patch can be fabricated from any rigid fiberglass panel that has comparable contour with the repair area. Lift gates and fenders can be used to supply patch material. If existing material is not available or compatible, a patch can be constructed with adhesive and reinforcement mesh (dry wall tape). Perform the following operation if required:

(1) Cover waxed paper or plastic with adhesive backed nylon mesh (dry wall tape) larger than the patch required (Fig. 8).

(2) Tape waxed paper or plastic sheet with mesh to a surface that has a compatible contour to the repair area.

(3) Apply a liberal coat of adhesive over the reinforcement mesh (Fig. 8). If necessary apply a second or third coat of adhesive and mesh after first coat has cured. The thickness of the patch should be the same as the repair area.

(4) After patch has cured, peel waxed paper or plastic from the back of the patch.

(5) If desired, a thin film coat of adhesive can be applied to the back of the patch to cover mesh for added strength.

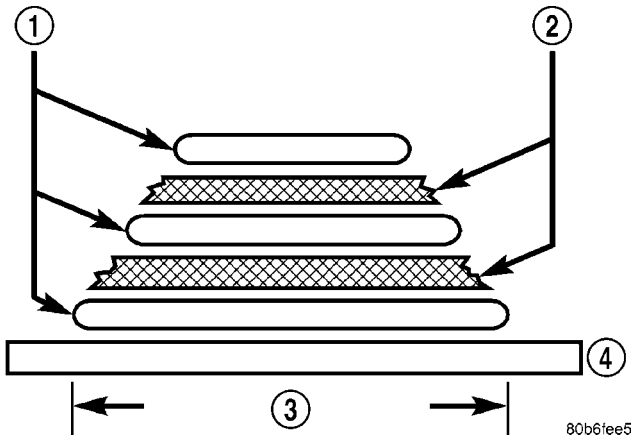


Fig. 8 FABRICATED PANEL

1 - STRUCTURAL ADHESIVE
2 - FIBERGLASS CLOTH OR FIBERGLASS MESH TAPE
3 - WIDTH OF V-GROOVE
4 - WAXED PAPER

PANEL PATCH INSTALLATION

(1) Make a paper or cardboard pattern the size and shape of the cutout hole in the panel.

(2) Trim 3 mm (0.125 in.) from edges of pattern so patch will have a gap between connecting surfaces.

(3) Using the pattern as a guide, cut the patch to size.

(4) Cut scrap pieces of patch material into 50 mm (2 in.) squares to use as patch supports to sustain the patch in the cutout.

(5) Drill 4 mm (0.160 in.) holes 13 mm (0.5 in.) in from edge of cutout hole (Fig. 7).

(6) Drill 4 mm (0.160 in.) holes 13 mm (0.5 in.) away from edge of patch across from holes drilled around cutout.

(7) Drill 3 mm (0.125 in.) holes in the support squares 13 mm (0.5 in.) from the edge in the center of one side.

(8) Scuff the backside of the body panel around the cutout hole with a scuff pad or sandpaper.

(9) Mix enough adhesive to cover one side of all support squares.

(10) Apply adhesive to cover one side of all support squares.

(11) Using number 8 sheet metal screws, secure support squares to back side of body panel with adhesive sandwiched between the panel and squares (Fig. 9).

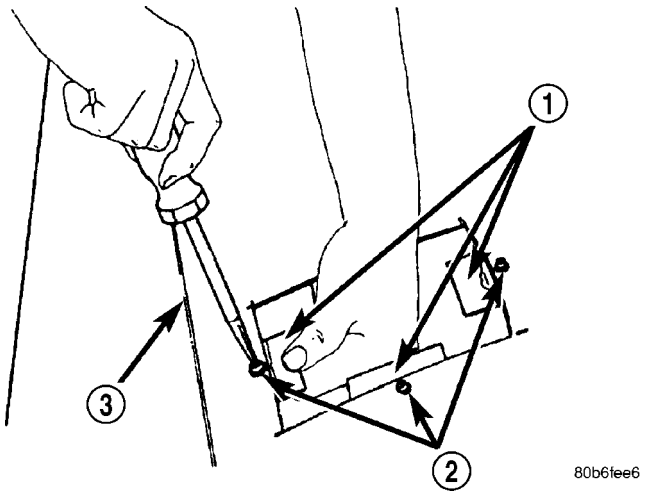


Fig. 9 SECURE SUPPORT SQUARES TO BODY PANEL

1 - SUPPORT SQUARES
2 - SCREWS
3 - DAMAGED BODY PANEL

BODY (Continued)

(12) Position patch in cutout against support squares and adjust patch until the gap is equal along all sides (Fig. 10).

(13) Drill 3 mm (0.125 in.) holes in the support squares through the pre-drilled holes in the patch.

(14) Apply a coat of adhesive to the exposed ends of the support squares (Fig. 11).

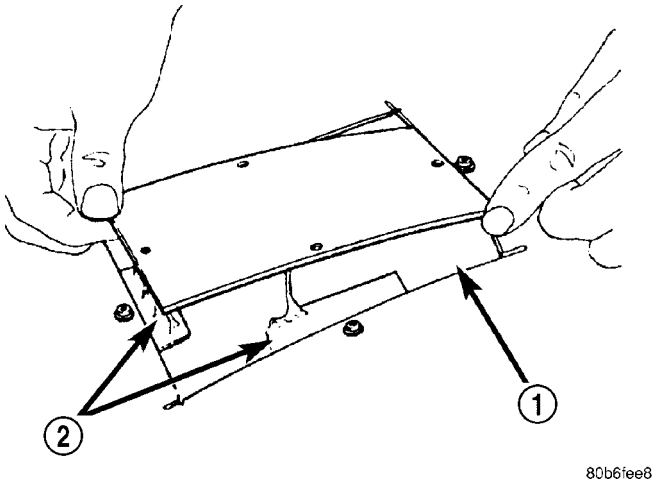


Fig. 10 POSITION PATCH IN CUTOUT AND ALIGN

1 - CUTOUT
2 - SUPPORT SQUARES

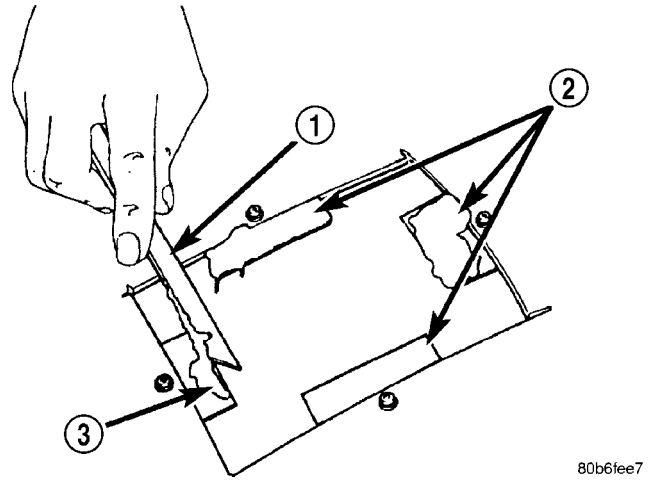


Fig. 11 APPLY ADHESIVE TO SUPPORT SQUARES

1 - APPLICATOR
2 - SUPPORT SQUARES
3 - ADHESIVE

(15) Install screws to hold the patch to support squares (Fig. 12). Tighten screws until patch surface is flush with panel surface.

(16) Allow adhesive to cure, and remove all screws.

(17) Using a 125 mm (5 in.) 24 grit disc grinder, grind a 50 mm (2 in.) to 75 mm (3 in.) wide and 2 mm (0.080 in.) deep path across the gaps around the patch (Fig. 13). With compressed air, blow dust from around patch.

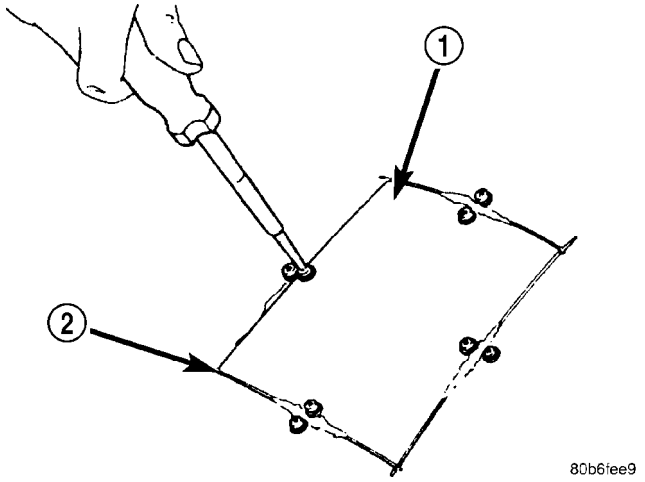


Fig. 12 INSTALL SCREWS

1 - PATCH
2 - GAP

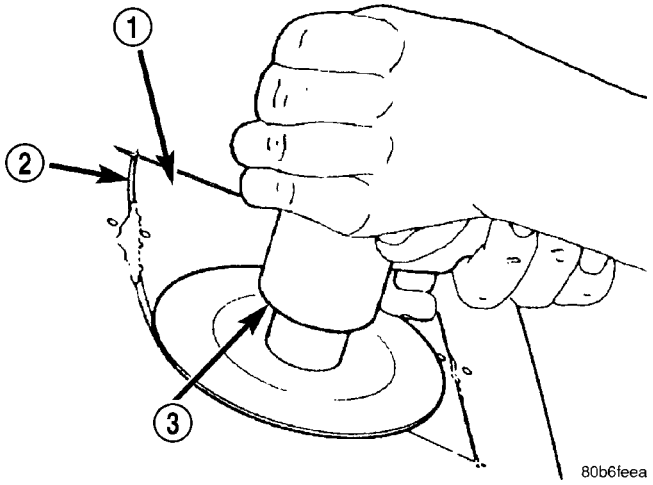


Fig. 13 GRIND SURFACE

1 - PATCH
2 - GAP
3 - DISC GRINDER

BODY (Continued)

(18) Apply adhesive backed nylon mesh (dry wall tape) over gaps around patch (Fig. 14).

(19) Mix enough adhesive to cover the entire patch area.

(20) Apply adhesive over the mesh around patch, and smooth epoxy with a wide spreader to reduce finish grinding. Use two to three layers of mesh and adhesive to create a stronger repair (Fig. 15).

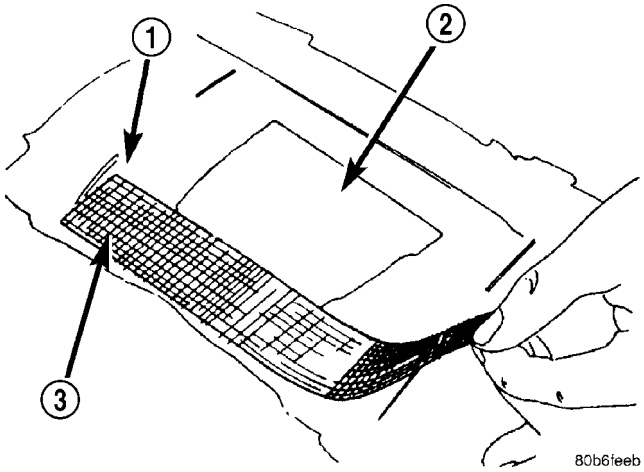


Fig. 14 COVER GAPS WITH MESH

1 - GROUND DOWN AREA
2 - PATCH
3 - MESH

PATCHED PANEL SURFACING

After patch panel is installed, the patch area can be finished using the same methods as finishing other types of body panels. If mesh material is exposed in the patched area, grind surface down, and apply a coat of high quality rigid plastic body filler. Prime, block sand, and paint as required.

BUZZ, SQUEAK & RATTLE

Buzz, Squeak & Rattles (BSR) may be caused by any one or more of the following and may be corrected as indicated:

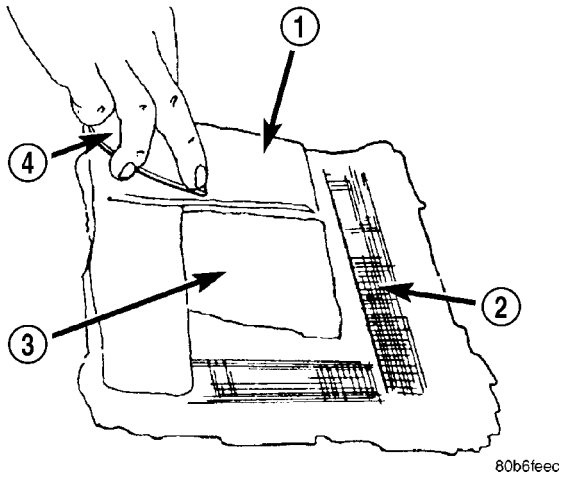


Fig. 15 COVER MESH WITH ADHESIVE

1 - ADHESIVE
2 - MESH
3 - PATCH
4 - SPREADER

- Loose fasteners should be tightened to specifications.
- Damaged or missing clips should be replaced.
- Damaged trim panels should be replaced.
- Incorrectly installed trim panels should be reinstalled properly.

Many BSR complaints such as loose trim, can be serviced using the Mopar® Parts BSR Noise Reduction Kit. This kit contains various tapes including foam, flock and anti-squeak used to eliminate noises caused by metal, plastic and vinyl components. Long life lubricants and greases can also be used on a variety of components. Refer to the Buzz, Squeak & Rattle Kit table for material contents and usage.

BODY (Continued)

BUZZ, SQUEAK & RATTLE KIT

ITEM	FEATURES	APPLICATIONS	SERVICE TEMP
Itch And Squeak Tape	An abrasion resistant material thin enough to conform to most irregular surfaces. Stops most itches and squeaks.	Between metal and metal, metal and plastic, metal and vinyl, vinyl and plastic. Interior. Examples: Trim panels and bezels.	-40° to 225° Fahrenheit (-40° to 107° Celsius)
Black Nylon Flock	Nylon Flock with an aggressive acrylic adhesive. Provides for cushioning and compression fit, also isolates components. Water-resistant.	Between metal and metal, metal and plastic, vinyl and plastic. Examples: Pull cups, bezels, clips, ducts, top cover to glass, cowl panel.	-40° to 180° Fahrenheit (-40° to 82° Celsius)
High Density Urethane Foam	Tear resistant, highly resilient and durable.	Between metal and metal, metal and plastic. Water-resistant. Examples: I/P, heavy metal rattles, isolating brackets.	-40° to 180° Fahrenheit (-40° to 82° Celsius)
Open Cell Foam Tape	Soft foam conforms to irregular surfaces.	Wire harness and connector wrap. Examples: Seals, gasket, wiring, heat ducts.	-40° to 180° Fahrenheit (-40° to 82° Celsius)
Closed Cell Low Density Foam Tape	Soft, conformable. Water-resistant.	Wherever bulk is needed. Prevents closing flutters and rattles when applied to door watershield. Examples: Door, I/P.	-40° to 180° Fahrenheit (-40° to 82° Celsius)
NYE® Grease 880	Long life.	Suspensions. Examples: Strut busings, sway bars.	-40° to 390° Fahrenheit (-40° to 200° Celsius)
Krytox® Oil	Long life. Will not dry out or harm plastics or rubber.	When access is not possible, oil will migrate to condition. Vinyl, rubber, plastic, metal. Examples: Convertible top bushings, pull cups trim panel inserts.	-30° to 400° Fahrenheit (-34° to 205° Celsius)
Krytox® Grease	Long life. Will not dry out or harm plastics or rubber.	Vinyl, rubber, plastic, metal, glass. Examples: Weather-strips, backlite and windshield moldings.	-30° to 400° Fahrenheit (-34° to 205° Celsius)

BODY (Continued)

SPECIFICATIONS - TORQUE

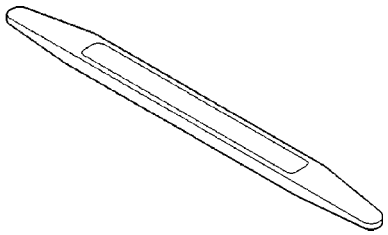
TORQUE SPECIFICATIONS

DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Body side guard/side bolts	5	—	45
Body side guard/support tab bolts	11	8	—
Hood assist spring bolt	3	—	25
Hood catch nuts	18	13	—
Hood catch bracket nuts	11	8	—
Hood hinge screws	24	18	—
Hood safety latch bolt	9	—	80
Bucket seat front anchor bolt	47	35	—
Bucket seat rear inboard anchor bolt	74	55	—
Bucket seat rear outboard anchor bolt	33	25	—
Liftgate glass ball stud nut	13	10	—
Liftgate glass hinge nut	6	—	53
Liftgate hinge to hardtop bolt	11	8	95
Front turning loop bolt	47	35	—
Front retractor bolt	47	35	—
Rear retractor bolt	47	35	—
Rear turning loop bolt	47	35	—
Rear belt anchor bolt	47	35	—
Rearview mirror set screw	1	—	9
Rear buckle anchor bolt	43	32	—
Side support bar to sport bar bolts	20	15	—
Side support sport bar to windshield frame bolts	32	24	—
Sport bar bracket bolt	68	50	—
Sport bar speaker pod bolts	68	50	—
Sport bar to wheelhouse bolts	40	30	—
Sport bar to cargo floor bolts	40	30	—
Tailgate hinge screws	23	17	—
Tailgate striker	71	52	—

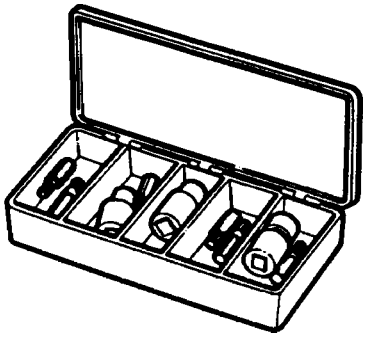
BODY (Continued)

SPECIAL TOOLS

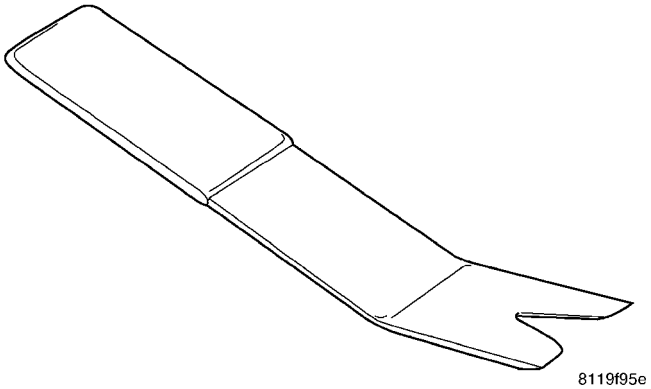
BODY



Trim Stick C-4755



TORX BIT SET C-4794-B



REMOVER, MOLDINGS C-4829-A

TAILGATE

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TAILGATE OUTSIDE HANDLE

REMOVAL

(1) Remove the latch from the tailgate. (Refer to 23 - BODY/TAILGATE/LATCH - REMOVAL)

(2) Remove the screws attaching the outside handle to the tailgate (Fig. 1).

(3) Separate the outside handle and seal from the tailgate.

INSTALLATION

(1) Position the seal and outside release handle on the tailgate. and install screws.

(2) Install the screws attaching the outside handle to the tailgate.

(3) Install the latch. (Refer to 23 - BODY/TAILGATE/LATCH - INSTALLATION)

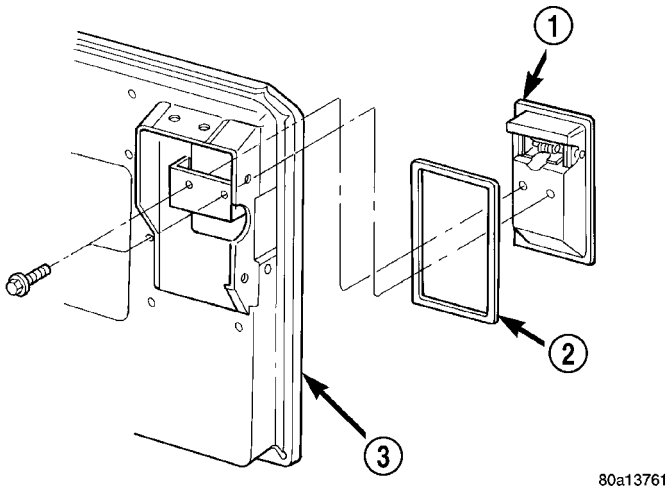

Fig. 1 TAILGATE OUTSIDE HANDLE

1 - OUTSIDE HANDLE
2 - SEAL
3 - TAILGATE

TAILGATE HINGE

REMOVAL

NOTE: Hinges may be serviced individually. If both are to be serviced, remove/install hinges one at a time.

(1) Using a grease pencil or equivalent, mark the position of the hinge on the body.
(2) Remove the screws attaching the hinge to the body and tailgate (Fig. 2).
(3) Separate the hinge from the tailgate.

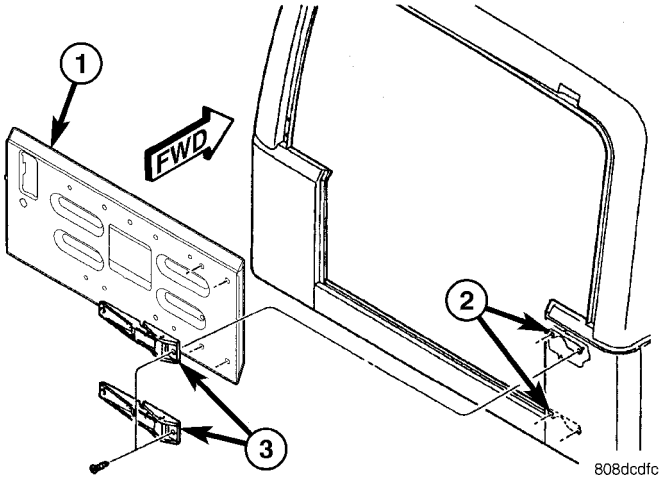

Fig. 2 TAILGATE HINGE

1 - TAILGATE
2 - NUT PLATE
3 - HINGE

INSTALLATION

NOTE: Hinges may be serviced individually. If both are to be serviced, remove/install hinges one at a time.

(1) Prepare and paint the replacement hinge to match the body paint color.
(2) Lubricate the hinge with spray lubricant.
(3) Align and position the hinge on the body and tailgate.
(4) Install the screws and tighten the screws to 23 N·m (200 in. lbs.).

TAILGATE LATCH

REMOVAL

(1) Open the tailgate and remove the latch trim cover (Fig. 3).
(2) Disconnect the outside handle to latch rod.
(3) Disconnect the lock cylinder to latch rod.
(4) Remove the screw attaching latch to tailgate.
(5) Separate the latch from the tailgate.

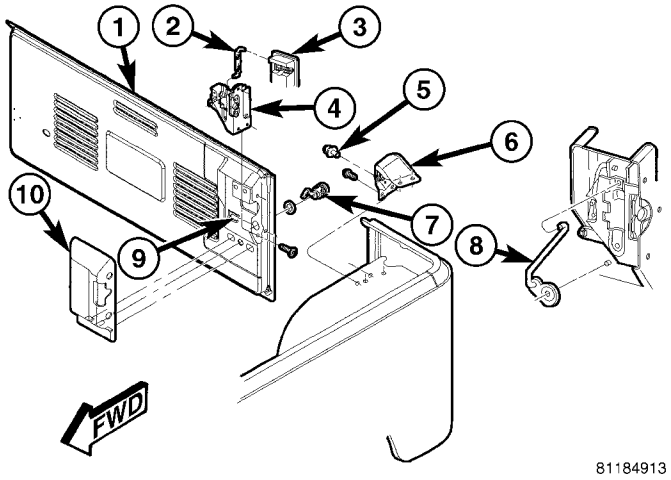

Fig. 3 TAILGATE LATCH

1 - TAILGATE
2 - OUTSIDE HANDLE ROD
3 - OUTSIDE HANDLE
4 - LATCH
5 - STRIKER
6 - STRIKER MOUNTING BRACKET
7 - LOCK CYLINDER
8 - LOCK CYLINDER ROD
9 - LOCK CYLINDER RETAINER CLIP
10 - LATCH COVER

INSTALLATION

(1) Position the latch in the tailgate.
(2) Install the screw attaching latch to tailgate. Do not tighten screw.
(3) Connect the lock cylinder to latch rod.
(4) Connect the outside handle to latch rod.
(5) Install the latch trim cover.

TAILGATE LATCH STRIKER

REMOVAL

(1) Remove the striker from the bracket.
(2) Remove the shim washers from the bracket.
(3) Remove the screws attaching the striker bracket to the body.

INSTALLATION

(1) Position the striker bracket on the body and install the screws.
(2) Position the striker and shim washers on the striker bracket.
(3) Install the striker in the bracket and tighten the striker to 71 N·m (52 ft. lbs.).

LIFTGATE GLASS HINGE

REMOVAL

(1) Open tailgate.
(2) Open and support liftgate glass.
(3) Remove wiper motor cover (right hinge only). (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS/REAR WIPER MOTOR TRIM COVER - REMOVAL)
(4) Remove the wiper blade. (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS/REAR WIPER BLADE - REMOVAL)
(5) Remove the nut attaching the liftgate hinge to the liftgate glass.
(6) Mark the position of the hinge in the top and remove the bolts attaching the hinge to the top (Fig. 4).

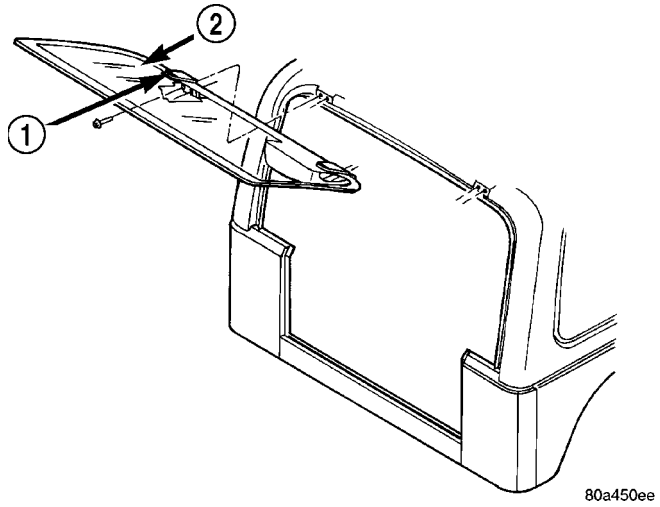

Fig. 4 LIFTGATE HINGE

1 - LIFTGATE HINGE
2 - LIFTGATE GLASS

INSTALLATION

(1) Align and position the hinge on the top and install the bolts.
(2) Install the nut attaching the liftgate hinge to the liftgate glass and tighten to 6 N·m (53 in. lbs.).
(3) Install wiper motor cover. (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS/REAR WIPER MOTOR TRIM COVER - INSTALLATION)
(4) Install the wiper blade. (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS/REAR WIPER BLADE - INSTALLATION)

LIFTGATE GLASS

REMOVAL

(1) Disconnect the rear defroster harness connectors.
(2) Disconnect the wiper motor harness connectors.

WARNING: DO NOT REMOVE THE LIFTGATE SUPPORT RODS WITH THE LIFTGATE CLOSED. THE SUPPORT ROD PISTONS ARE OPERATED BY HIGH PRESSURE GAS AND COULD CAUSE PERSONAL INJURY AND/OR VEHICLE DAMAGE IF THEY ARE REMOVED WITH THE PISTONS COMPRESSED (LIFTGATE CLOSED). ONCE REMOVED, DO NOT ATTEMPT TO DISASSEMBLE OR REPAIR THE SUPPORT RODS.

(3) Open the tailgate and liftgate.
(4) Remove support rod cylinders. (Refer to 23 - BODY/TAILGATE/SUPPORT CYLINDER - REMOVAL)
(5) Remove the bolts attaching the liftgate hinge to the hardtop (Fig. 5).
(6) Separate the liftgate glass from the hard top.

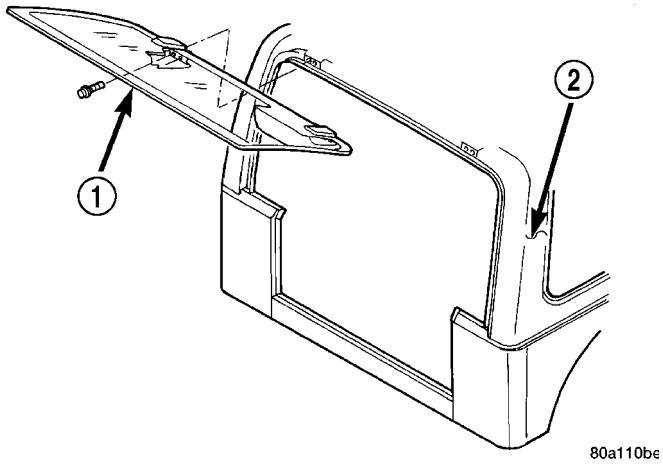

Fig. 5 LIFTGATE GLASS

1 - LIFTGATE REAR WINDOW
2 - HARD TOP

LIFTGATE GLASS (Continued)

INSTALLATION

Transfer all related components.

(1) If removed, install support rod ball studs and tighten the nut to 12 N·m (112 in. lbs.).

(2) Position the liftgate glass at the hard top.

(3) Install the bolts attaching the liftgate hinge to the hardtop and tighten to 10 N·m (95 in. lbs.).

(4) Install the support rod cylinders. (Refer to 23 - BODY/TAILGATE/SUPPORT CYLINDER - INSTALLATION)

LIFTGATE GLASS WEATHERSTRIP

DESCRIPTION

The liftgate glass weatherstrip is attached to the liftgate glass and is not serviceable. If the liftgate glass weatherstrip needs to be replaced, replace the liftgate glass. (Refer to 23 - BODY/LIFTGATE GLASS - REMOVAL)

TAILGATE LOCK CYLINDER

REMOVAL

(1) Open the tailgate.

(2) Remove the latch cover.

(3) Remove the lock cylinder retainer clip.

(4) Remove the lock cylinder from the tailgate opening.

INSTALLATION

(1) Position the lock cylinder in the tailgate opening.

(2) Connect the lock cylinder to latch rod.

(3) Install the lock cylinder retainer clip.

(4) Install the latch cover.

LIFTGATE GLASS SUPPORT CYLINDER

REMOVAL

WARNING: DO NOT REMOVE THE LIFTGATE SUPPORT RODS WITH THE LIFTGATE CLOSED. THE SUPPORT ROD PISTONS ARE OPERATED BY HIGH PRESSURE GAS AND COULD CAUSE PERSONAL INJURY AND/OR VEHICLE DAMAGE IF THEY ARE REMOVED WITH THE PISTONS COMPRESSED (LIFTGATE CLOSED). ONCE REMOVED, DO NOT ATTEMPT TO DISASSEMBLE OR REPAIR THE SUPPORT RODS.

(1) Open and support the liftgate glass.

(2) Release the support rod cylinder retaining clips at both ends of each support rod cylinder. (Fig. 6)

(3) Pull the support rods off the ball studs (Fig. 7).

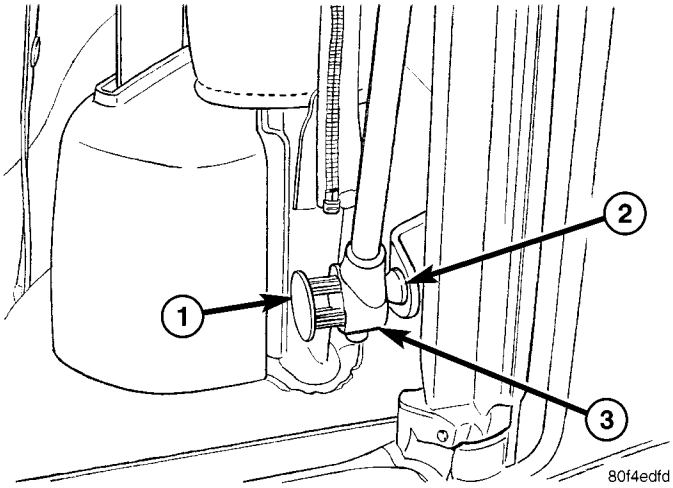

Fig. 6 SUPPORT ROD CYLINDER CONNECTION

1 - CLIP
2 - BALL STUD
3 - SUPPORT ROD END

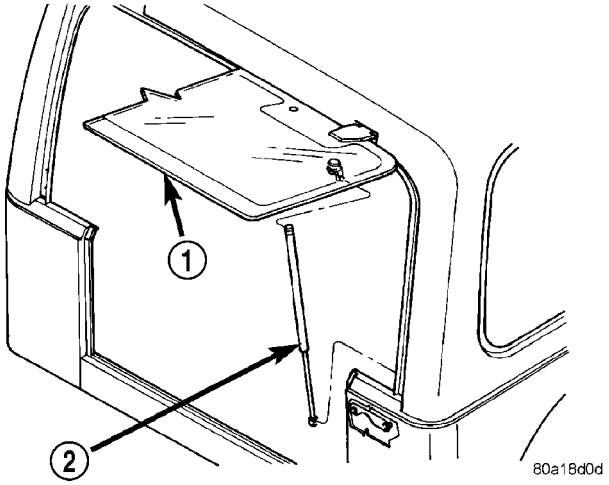

Fig. 7 SUPPORT ROD CYLINDER REMOVAL

1 - REAR WINDOW LIFT GLASS
2 - SUPPORT CYLINDER

INSTALLATION

(1) Position the support rod cylinders on the ball studs.

(2) Install the support rod cylinder retainer clips.

TAILGATE

REMOVAL

(1) Remove the spare tire.
(2) Open the tailgate and remove the center high mounted stop lamp (CHMSL) contact cover (Fig. 8).
(3) Disengage the CHMSL electrical connectors.
(4) Remove the screws that attach the tailgate hinge to the tailgate.
(5) Separate the tailgate from the vehicle.

INSTALLATION

NOTE: If necessary, transfer tailgate related components.

(1) Install the screws that attach the tailgate hinge to the tailgate.
(2) Engage the CHMSL electrical connectors.
(3) Install the CHMSL contact cover.
(4) Close the tailgate and install the spare tire.

ADJUSTMENTS

ADJUSTMENT

(1) Loosen the tailgate hinge-to-body screws.

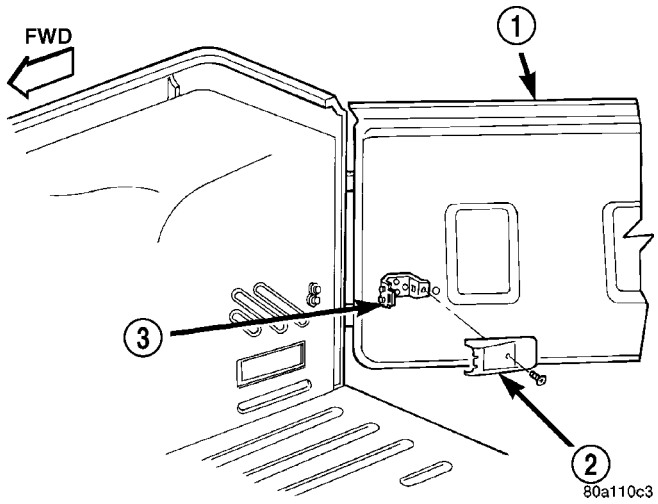

Fig. 8 CHMSL CONTACT COVER

1 - TAILGATE
2 - COVER
3 - CHMSL CONTACT COVER BRACKET

(2) Align the tailgate in the body opening and tighten the hinge screws. (Refer to 23 - BODY/BODY STRUCTURE/GAP AND FLUSH - SPECIFICATIONS)

FULL DOOR

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DOOR

REMOVAL

(1) Open the door.
(2) Disconnect the door restraint strap from the pin (Fig. 1).
(3) Remove the nuts at the door hinge pivots and lift the door from the body.

INSTALLATION

(1) Position the door in the hinge and install the nuts.
(2) Connect the door restraint strap at the pin.

ADJUSTMENTS

ADJUSTMENT

The doors are adjusted at the hinge attaching locations on either the body or the door. Enlarged holes are located in the body (lower hinge only) for fore, aft and tilt adjustments. Enlarged holes are also located in the door (upper and lower hinges) for up, down, fore, aft and tilt adjustments.

Prior to door adjustment or alignment, the door latch must be removed to allow the door to close freely and be properly aligned.

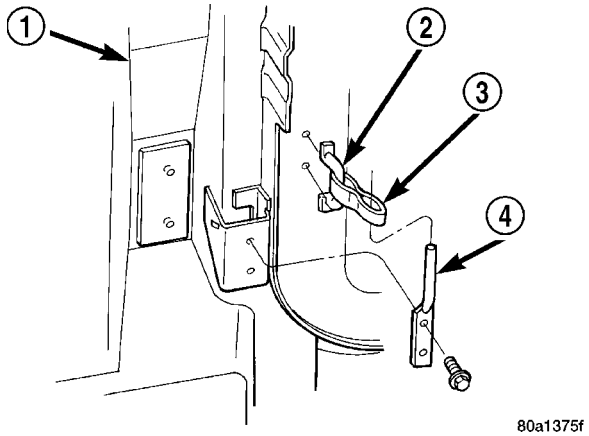

Fig. 1 RESTRAINT STRAP

1 - BODY
2 - FOOTMAN LOOP
3 - STRAP
4 - RESTRAINT PIN

The door latch striker should be adjusted in or out to allow the door latch to be fully engaged. The door should be flush with the adjacent body panels.

DOOR GLASS

REMOVAL

(1) Remove the door trim panel and the waterdam. (Refer to 23 - BODY/FULL DOOR/TRIM PANEL - REMOVAL)

(2) Pull the door glass run channel from the door sail.

(3) Roll glass fully downward.

(4) Using a trim stick C-4755 or equivalent, remove the screws and remove the door sail panel (Fig. 2) and (Fig. 3).

(5) Roll glass 1/4 upward to access regulator arm guide.

(6) Remove the screws that attach the regulator arm guide to the glass.

(7) Lift the glass upward while tilting inward and remove from the door.

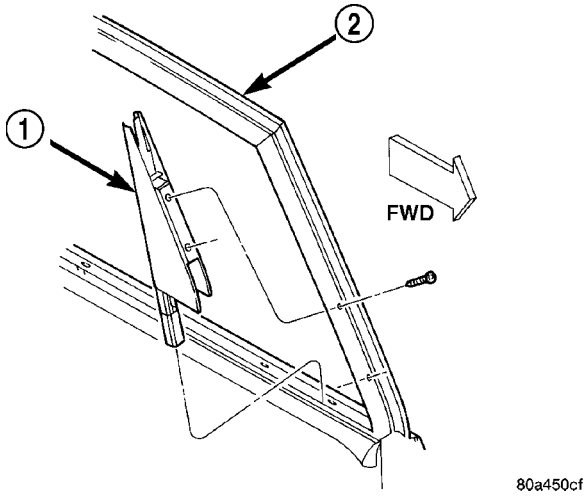

Fig. 2 DOOR SAIL SCREWS

1 - DOOR SAIL
2 - DOOR

INSTALLATION

(1) Position the glass in the door ensuring the glass is aligned in the glass run channel.

(2) Install the screws that attach the regulator arm guide to the glass.

(3) Install the door sail panel.

(4) Install the run channel in the door sail.

(5) Install the waterdam and the door trim panel. (Refer to 23 - BODY/FULL DOOR/TRIM PANEL - INSTALLATION)

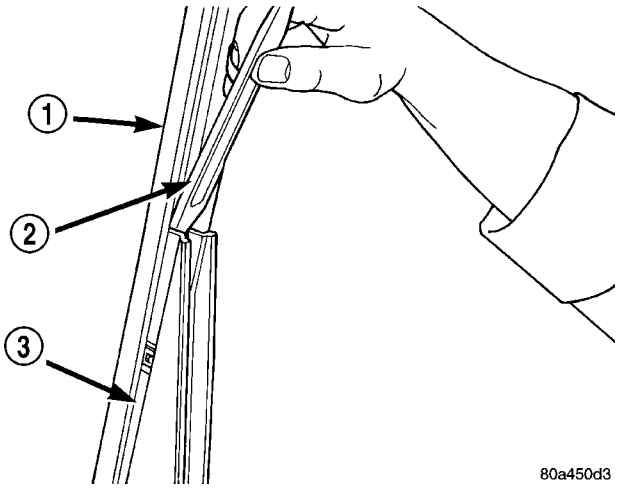

Fig. 3 DOOR SAIL REMOVAL

1 - DOOR FRAME
2 - TRIM STICK
3 - DOOR SAIL

WINDOW REGULATOR

REMOVAL

(1) Remove door glass. (Refer to 23 - BODY/FULL DOOR/DOOR GLASS - REMOVAL)

(2) Loosen the bolts in the slotted holes (Fig. 4).

(3) Remove the bolts attaching the regulator to the door inner panel.

(4) Lift the regulator upward to free it from the slotted holes in the door inner panel.

(5) Lower the regulator and remove it through the access hole in the door inner panel (Fig. 5).

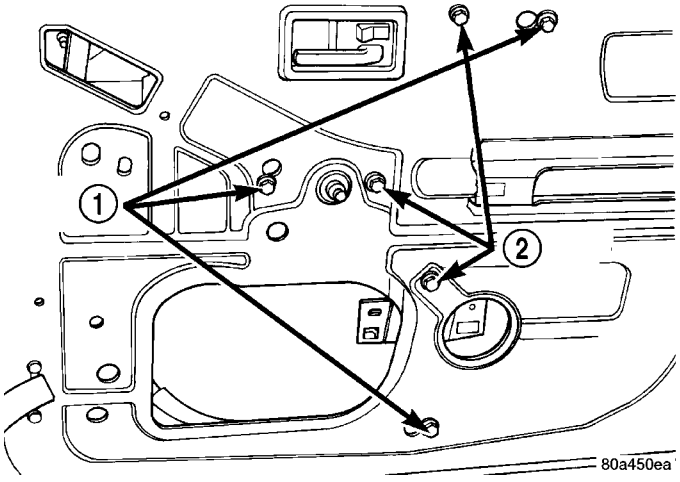

Fig. 4 WINDOW REGULATOR BOLTS

1 - LOOSEN BOLTS
2 - REMOVE BOLTS

WINDOW REGULATOR (Continued)

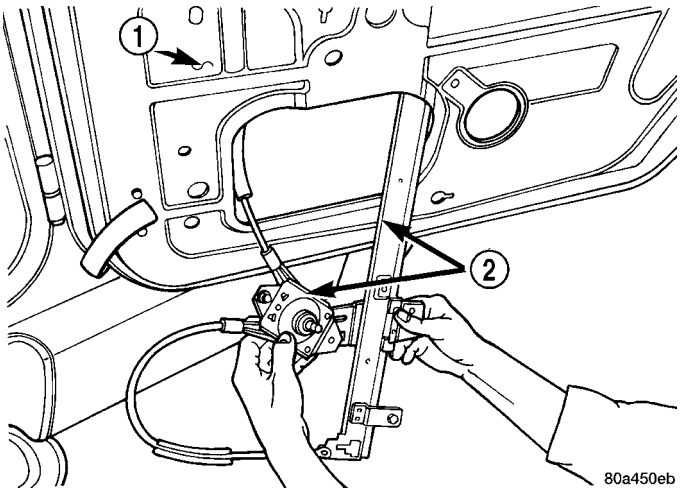

Fig. 5 REGULATOR REMOVAL

1 - DOOR
2 - REGULATOR ASSEMBLY

INSTALLATION

(1) Position the regulator in the door.
(2) Align regulator bolts into slotted holes.
(3) Install bolts attaching regulator to the inner door panel.
(4) Tighten the bolts in the slotted holes.
(5) Install door glass. (Refer to 23 - BODY/FULL DOOR/DOOR GLASS - INSTALLATION)

EXTERIOR HANDLE

REMOVAL

(1) Remove the door trim panel. (Refer to 23 - BODY/FULL DOOR/TRIM PANEL - REMOVAL)
(2) Position the window in the full upward position.
(3) Remove the grab handle support bracket (Fig. 6).
(4) Peel back the waterdam from the door inner panel to access the door latch.
(5) Disconnect from the latch, the inside lock knob to latch rod and, the outside release handle to latch rod.
(6) Disengage tail of retainer from handle keeper.
(7) Using a long flat blade, tap the handle keepers upward and remove from the door handle (Fig. 7).
(8) Remove the latch release rod from the door handle.
(9) Separate the handle and gasket from the door.

INSTALLATION

(1) Engage the latch release rod to the door handle.
(2) Position the gasket and handle in the door.

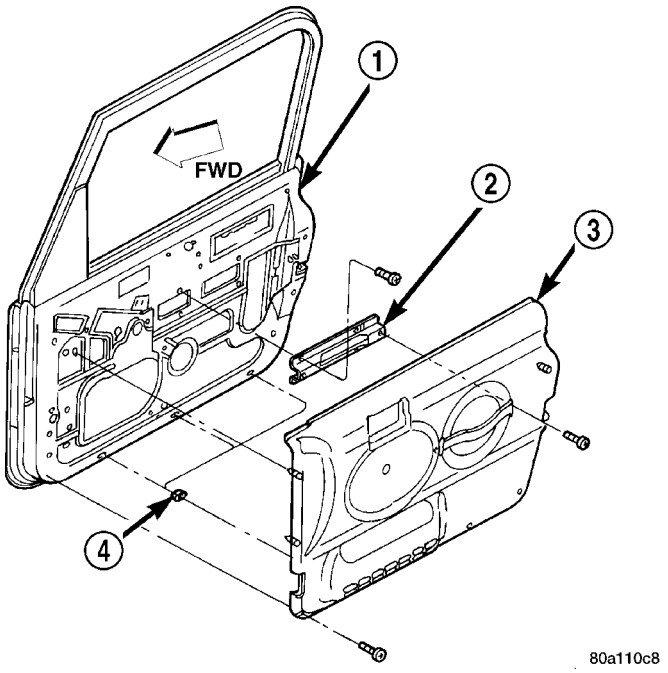

Fig. 6 GRAB HANDLE SUPPORT BRACKET

1 - FULL DOOR
2 - GRAB HANDLE BRACKET
3 - TRIM PANEL
4 - NUT

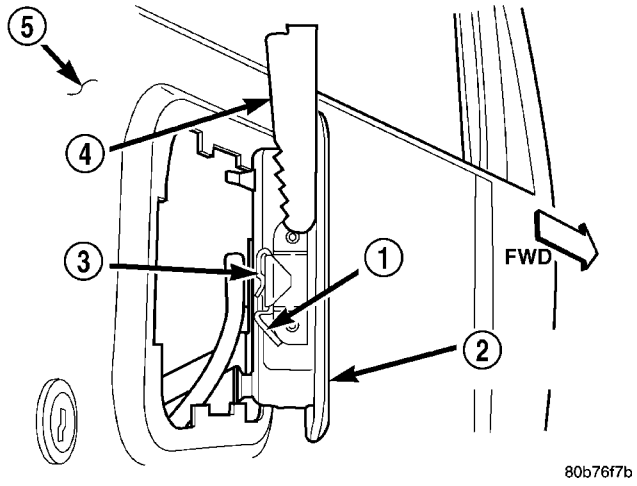

Fig. 7 OUTSIDE DOOR HANDLE REMOVAL

1 - TAIL
2 - OUTSIDE HANDLE
3 - RETAINER
4 - HANDLE KEEPER
5 - DOOR

(3) Slide the keepers into the door handle from the top.
(4) Lower the window.
(5) Using a long flat blade, lightly tap the handle keepers downward to secure the handle. The tail of the retainer must be positioned on the 2nd or 3rd step from the bottom on the handle keeper.
(6) Raise the window.

EXTERIOR HANDLE (Continued)

(7) Connect to the latch, the inside lock knob to latch rod and, the outside release handle to latch rod.
(8) Reposition the waterdam
(9) Install the grab handle support bracket.
(10) Install the door trim panel. (Refer to 23 - BODY/FULL DOOR/TRIM PANEL - INSTALLATION)

GLASS RUN CHANNEL

REMOVAL

(1) Lower the window.
(2) Using a trim stick C-4755 or equivalent, carefully pry the glass run channel weatherstrip from the window opening frame.
(3) Remove the door glass. (Refer to 23 - BODY/FULL DOOR/DOOR GLASS - REMOVAL)
(4) Grasp the glass run channel weatherstrip in the door (Fig. 8) and pull from the channel.

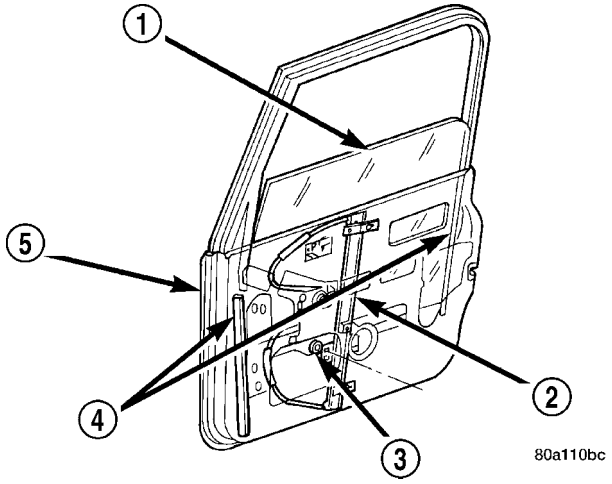

Fig. 8 FULL DOOR GLASS RUN CHANNEL WEATHERSTRIP

1 - WINDOW GLASS
2 - REGULATOR
3 - GROMMET
4 - GLASS RUN CHANNEL WEATHERSTRIP
5 - DOOR

INSTALLATION

NOTE: Applying a small amount of lubricant to the weatherstrip may ease the installation.

(1) Position the weatherstrip in the lower door channels and press into place.
(2) Install the door glass. (Refer to 23 - BODY/FULL DOOR/DOOR GLASS - INSTALLATION)
(3) Position the weatherstrip in the window opening frame and press into place.

NOTE: Ensure that the glass is seated properly. Improperly seated door glass will result in high glass roll-up/roll-down effort.

HINGE

REMOVAL

(1) Remove the door. (Refer to 23 - BODY/FULL DOOR/DOOR - REMOVAL)
(2) Using a grease pencil or equivalent, mark the outline of the existing hinge on the body and the door for installation alignment reference.
(3) Remove the nut from the upper hinge pin (Fig. 9).

NOTE: When removing the door or hinge DO NOT discard the plastic shims or the hinge pin.

(4) Remove the hinge-to-body screws and the hinge-to-door screws. Remove the hinge from the door and body. Support the door as necessary. The upper hinge is integrated with the windshield hinge. When removing it, support the windshield frame with an appropriate device prior to removal.

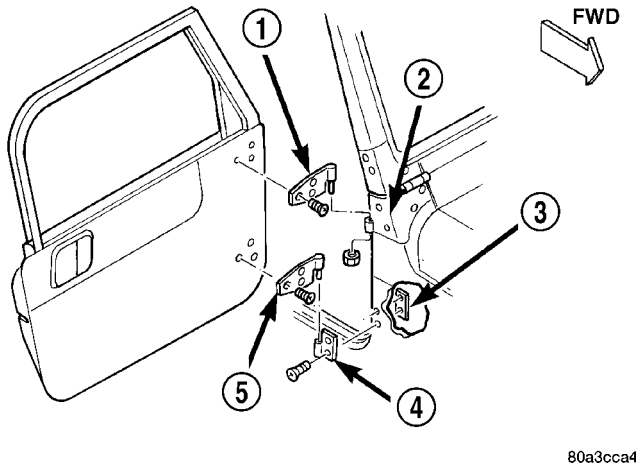

Fig. 9 FULL DOOR HINGE

1 - HINGE HALF
2 - HINGE HALF
3 - TAPPING PLATE
4 - HINGE HALF
5 - HINGE HALF

INSTALLATION

(1) Clean the replacement hinge with an appropriate solvent and dry it with compressed air.
(2) Paint the hinge to match the vehicle body.
(3) Lubricate the hinge with spray lubricant.
(4) Position the hinge on the door, align carefully with the wax pencil installation alignment reference marks, and install the screws.

HINGE (Continued)

(5) Position the hinge on the vehicle body. Align the wax pencil marks installation alignment reference marks. Install the screws.

(6) Install the door. (Refer to 23 - BODY/FULL DOOR/DOOR - INSTALLATION)

(7) Inspect the windshield alignment after hinge installation.

(8) Inspect the door alignment and adjust, if necessary. (Refer to 23 - BODY/FULL DOOR/DOOR - ADJUSTMENTS)

INSIDE HANDLE ACTUATOR

REMOVAL

(1) Remove the torx screw attaching the inside handle to the door.

(2) Carefully pull the handle from the door.

(3) Disconnect the latch rods from the handle (Fig. 10).

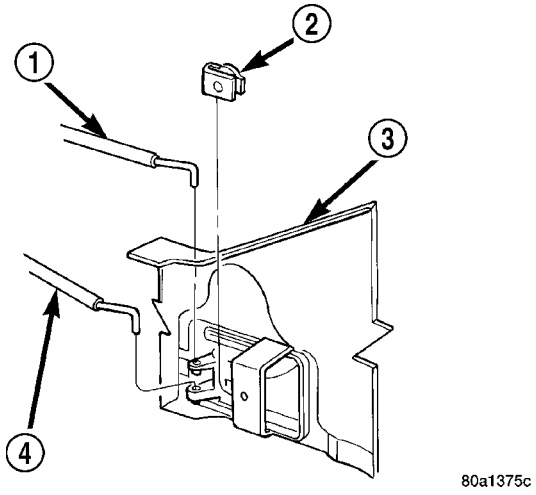

Fig. 10 INSIDE HANDLE ACTUATOR

1 - LOCK KNOB TO LATCH ROD
2 - U-NUT
3 - HALF DOOR TRIM PANEL
4 - INSIDE RELEASE HANDLE TO LATCH ROD

INSTALLATION

(1) Connect the latch rods to the handle.

(2) Position handle and seal in door.

(3) Install the torx screw attaching the inside handle the to door.

LATCH

REMOVAL

(1) Remove trim panel. (Refer to 23 - BODY/FULL DOOR/TRIM PANEL - REMOVAL)

(2) Roll window to full upward position.

(3) Disconnect the lock cylinder to latch rod. (Fig. 11)

(4) Disconnect the lock knob to latch rod.

(5) Disconnect the outside handle to latch release rod.

(6) Remove the screws attaching the latch to the door.

(7) Lower the latch in the door and disconnect the inside handle to latch rod.

(8) Remove the latch from the door.

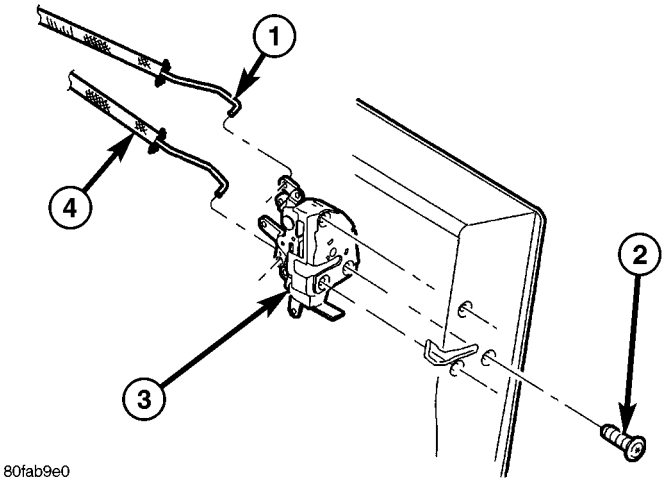

Fig. 11 DOOR LATCH ASSEMBLY

1 - RELEASE ROD
2 - SCREWS (3)
3 - LATCH ASSEMBLY
4 - LOCK ROD

INSTALLATION

(1) Position the latch in the door.

(2) Connect the inside handle to latch rod.

(3) Install the screws attaching the latch to the door.

(4) Position the door weatherstrip in place, apply adhesive as necessary.

(5) Connect the outside handle to latch rod.

(6) Connect the lock knob to latch rod.

(7) Connect the lock cylinder to latch rod.

(8) Install trim panel. (Refer to 23 - BODY/FULL DOOR/TRIM PANEL - INSTALLATION)

LATCH STRIKER

REMOVAL

(1) Remove the screws attaching the striker to the body.

(2) Separate the striker and the spacer from the body (Fig. 12).

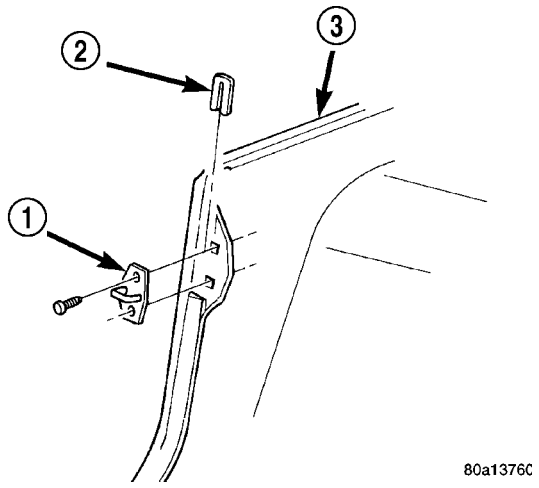

Fig. 12 LATCH STRIKER

1 - STRIKER
2 - SPACER
3 - BODY

INSTALLATION

(1) Position the striker and the spacer on the body.

(2) Install the screws attaching the striker and spacer to the body.

LOCK CYLINDER

REMOVAL

(1) Remove trim panel. (Refer to 23 - BODY/FULL DOOR/TRIM PANEL - REMOVAL)

(2) Peel back waterdam.

(3) Disconnect lock cylinder to latch rod.

(4) Remove lock cylinder retaining clip.

(5) Remove the lock cylinder from the door.

INSTALLATION

(1) Install the lock cylinder in the door.

(2) Install lock cylinder retaining clip.

(3) Connect lock cylinder to latch rod.

(4) Secure the waterdam to the door.

(5) Install trim panel. (Refer to 23 - BODY/FULL DOOR/LOCK CYLINDER - INSTALLATION)

TRIM PANEL

REMOVAL

(1) Lower the window.

(2) Remove the clip attaching the window glass regulator handle to the regulator. Remove the handle.

(3) Remove the screws attaching trim panel to door (Fig. 13).

(4) Using a special tool C-4829 or equivalent, remove push-in fasteners attaching trim panel to door.

(5) Lift the trim panel upward and separate the trim panel from the door.

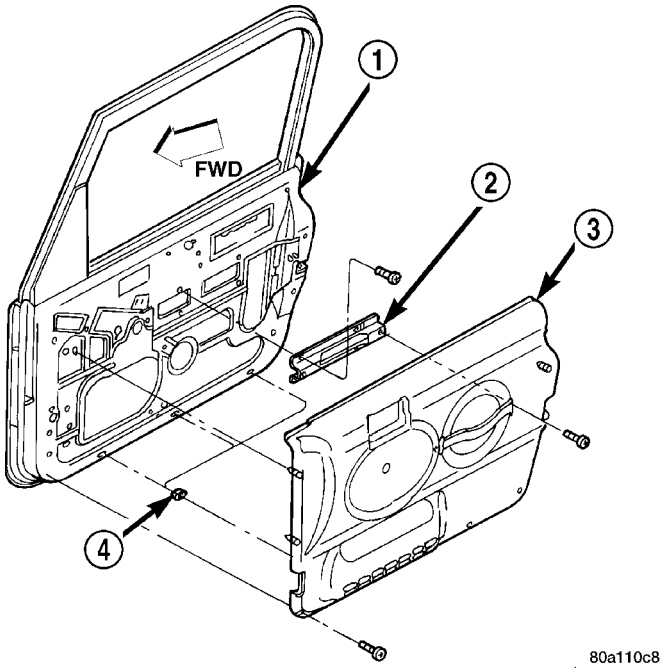

Fig. 13 FULL

1 - FULL DOOR
2 - GRAB HANDLE BRACKET
3 - TRIM PANEL
4 - NUT

INSTALLATION

(1) Position the trim panel on the door.

(2) Press the push-in fasteners attaching trim panel to door into place.

(3) Install the screws attaching trim panel to door.

(4) Position the clip on regulator handle and install the handle on the regulator.

HALF DOOR

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DOOR

REMOVAL

(1) Open the door.

(2) Disconnect the door restraint strap from the pin (Fig. 1).

(3) Remove the nuts at the door hinge pivots and lift the door from the body.

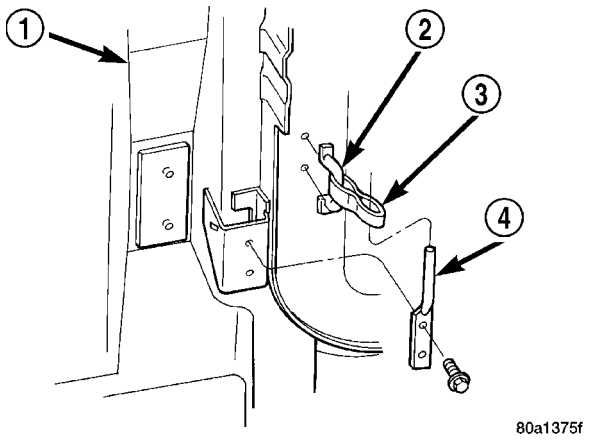

Fig. 1 RESTRAINT STRAP

1 - BODY
2 - FOOTMAN LOOP
3 - STRAP
4 - RESTRAINT PIN

INSTALLATION

(1) Position the door in the hinge and install the nuts.

(2) Connect the door restraint strap at the pin.

(3) Check for proper operation.

WINDOW

REMOVAL

(1) Open the door.

(2) Grasp the window at both front and rear edges and firmly lift upward (Fig. 2).

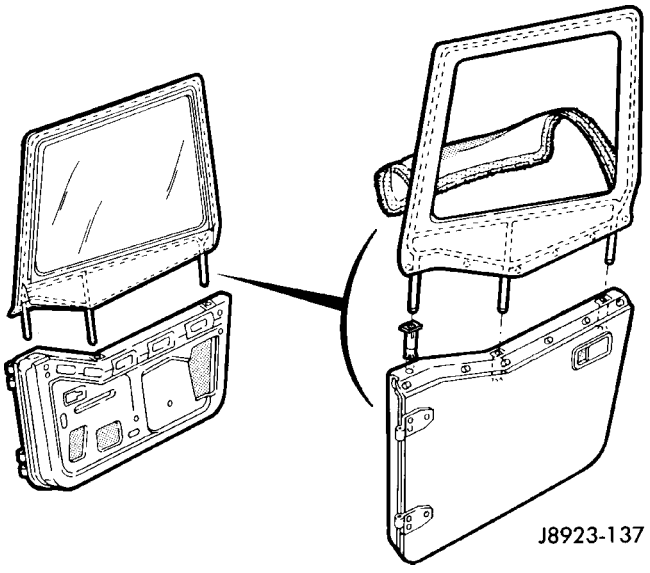

Fig. 2 HALF WINDOW

INSTALLATION

(1) Starting at the most forward alignment pin, position the window alignment pins into the restraint sleeves and push downward until seated.

EXTERIOR HANDLE

REMOVAL

(1) Remove trim panel. (Refer to 23 - BODY/HALF DOOR/TRIM PANEL - REMOVAL)
(2) Disconnect the outside handle to latch rod.
(3) Remove screws attaching the outside handle to the door.
(4) Separate the outside handle and seal from the door.

INSTALLATION

(1) Position the outside handle and seal in the door.
(2) Install screws attaching the outside handle to the door.
(3) Connect the outside handle to latch rod.
(4) Install trim panel. (Refer to 23 - BODY/HALF DOOR/TRIM PANEL - INSTALLATION)

HINGE

DESCRIPTION

The service procedures for the half door hinge are the same as the full door hinge. (Refer to 23 - BODY/FULL DOOR/HINGE - REMOVAL) and (Refer to 23 - BODY/FULL DOOR/HINGE - INSTALLATION).

INSIDE HANDLE ACTUATOR

REMOVAL

(1) Remove the torx screw attaching the inside handle to the door.
(2) Carefully pull handle from door.
(3) Disconnect the latch rods from the handle (Fig. 3).

INSTALLATION

(1) Connect the latch rods to the handle.
(2) Position handle and seal in door.
(3) Install the torx screw attaching the inside handle the to door.

LATCH

REMOVAL

(1) Remove trim panel. (Refer to 23 - BODY/HALF DOOR/TRIM PANEL - REMOVAL)
(2) Disconnect the lock cylinder to latch rod (Fig. 4).
(3) Disconnect the lock knob to latch rod.
(4) Disconnect the outside handle to latch rod.

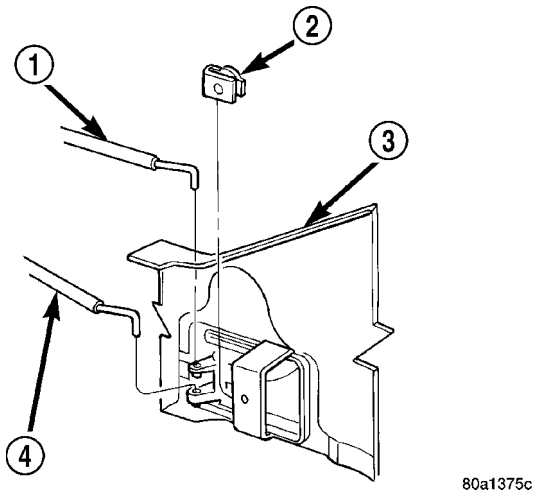

Fig. 3 INSIDE HANDLE ACTUATOR

1 - LOCK KNOB TO LATCH ROD
2 - U-NUT
3 - HALF DOOR TRIM PANEL
4 - INSIDE RELEASE HANDLE TO LATCH ROD

(5) Using a trim stick or equivalent, pry back the door weatherstrip at the latch to access the screw attaching the latch to the door.
(6) Remove the screws attaching the latch to the door (Fig. 5).
(7) Lower the latch in the door and disconnect the inside handle to latch rod.
(8) Remove the latch from the door.

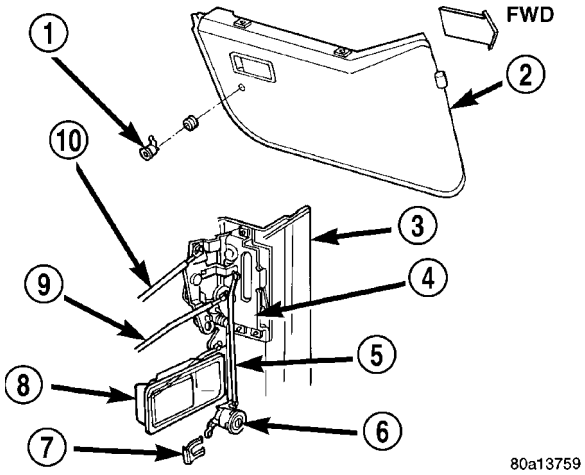

Fig. 4 HALF DOOR LATCH RODS

1 - LOCK CYLINDER
2 - HALF DOOR
3 - HALF DOOR
4 - LATCH
5 - LOCK CYLINDER TO LATCH ROD
6 - LOCK CYLINDER
7 - RETAINER
8 - OUTSIDE HANDLE
9 - INSIDE HANDLE TO LATCH ROD
10 - INSIDE LOCK TO LATCH ROD

LATCH (Continued)

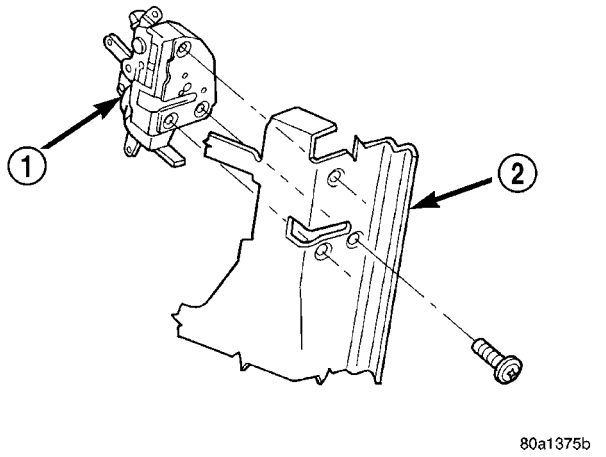

Fig. 5 DOOR LATCH

1 - LATCH
2 - HALF DOOR

INSTALLATION

(1) Position the latch in the door.
(2) Connect the inside handle to latch rod.
(3) Install the screws attaching the latch to the door.
(4) Position the door weatherstrip in place, apply adhesive as necessary.
(5) Connect the outside handle to latch rod.
(6) Connect the lock knob to latch rod.
(7) Connect the lock cylinder to latch rod (Fig. 4).
(8) Install trim panel. (Refer to 23 - BODY/HALF DOOR/TRIM PANEL - INSTALLATION)

LATCH STRIKER

REMOVAL

(1) Remove the screws attaching the striker to the body.
(2) Separate the striker and the spacer from the body (Fig. 6).

INSTALLATION

(1) Position the striker and the spacer on the body.
(2) Install the screws attaching the striker and spacer to the body.

LOCK CYLINDER

REMOVAL

(1) Remove trim panel. (Refer to 23 - BODY/HALF DOOR/TRIM PANEL - REMOVAL)
(2) Disconnect lock cylinder to latch rod (Fig. 4).
(3) Remove lock cylinder retaining clip.
(4) Remove the lock cylinder from the door.

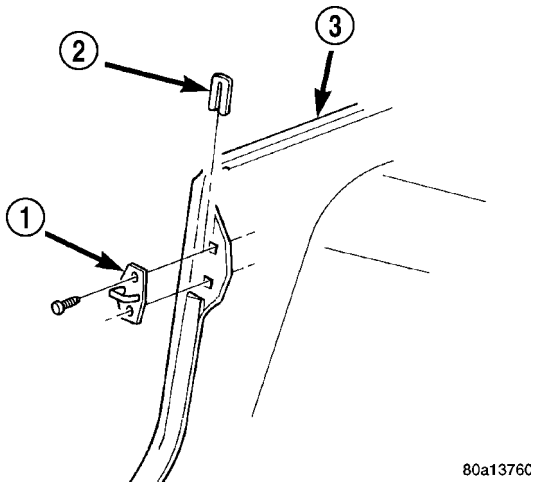

Fig. 6 LATCH STRIKER

1 - STRIKER
2 - SPACER
3 - BODY

INSTALLATION

(1) Install the lock cylinder in the door.
(2) Install lock cylinder retaining clip.
(3) Connect lock cylinder to latch rod.
(4) Install trim panel. (Refer to 23 - BODY/HALF DOOR/TRIM PANEL - INSTALLATION)

TRIM PANEL

REMOVAL

(1) Remove half door window. (Refer to 23 - BODY/HALF DOOR/WINDOW - REMOVAL)
(2) Rotate window retainer sleeves 90°. Using a trim stick C-4755 or equivalent, pry sleeve retainers from door.
(3) Remove the screws attaching trim panel to door.
(4) Using a trim stick C-4829 or equivalent, remove push-in fasteners attaching trim panel to door.
(5) Separate the trim panel from the door.

INSTALLATION

(1) Position the trim panel on the door.
(2) Press the push-in fasteners attaching trim panel to door into place.
(3) Install the screws attaching trim panel to door.
(4) Position retainer sleeves into door. Rotate retainer sleeves 90° to secure into place.
(5) Install half door window. (Refer to 23 - BODY/HALF DOOR/WINDOW - INSTALLATION)

EXTERIOR

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BODY DECALS

DESCRIPTION

TJ decals are durable tape decals with a adhesive backing.

To eliminate blisters and air bubbles in a decal, pierce them with a needle or pin. Force the trapped air out of the hole.

A heat gun can also be used to remove small wrinkles and irregularities in a decal.

REMOVAL

NOTE: The key to successful decal removal is to apply heat to area and slowly peel the decal from panel.

(1) Clean the surface as necessary.

(2) Place a piece of masking tape above or below the decal as a reference mark.

(3) Start at one end of the decal and apply heat with a heat gun. Slowly peel the decal from the panel by pulling it back. **Do not pull the decal outward from the panel.**

INSTALLATION

(1) The area that will be covered by the decal must be cleaned with an cleaning solution to remove any residue paint. Freshly painted surfaces must be thoroughly dry.

(2) Clean painted surface with a commercial wax and silicone removal solution. Wipe surface with a clean cloth and allow to dry.

(3) Position decal and carrier on panel and hold it in-place with pieces masking tape.

(4) Lift the bottom edge of the decal and carrier, use the tape sections as hinges, and reverse the position of the decal and carrier.

CAUTION: Always remove the carrier from the tape stripe/decal, never remove the tape stripe/decal from the carrier.

(5) Bend a corner of the carrier outward, separate the corner of the carrier from the decal.

(6) Using the masking tape on the body panel, align the decal.

(7) Separate the carrier from one end of the decal.

(8) Hold tape decal firmly against the panel surface while separating the carrier from the decal.

(9) Inspect tape decal with reflected light to check for defects that could have developed during the installation process. Remove all air and/or moisture bubbles.

COWL GRILLE AND SCREEN

REMOVAL

(1) Remove the wiper arms. (Refer to 8 - ELECTRICAL/WIPERS/WASHERS/WIPER ARMS - REMOVAL)

CAUTION: Use masking tape or equivalent on the wiper arm pivots to prevent damage to the cowl grille paint.

(2) Remove the four screws at the rear of the cowl grille.

(3) Open the hood and remove the screws that attach the cowl grille and screen to the cowl (Fig. 1).

(4) Remove the grille and screen from the cowl.

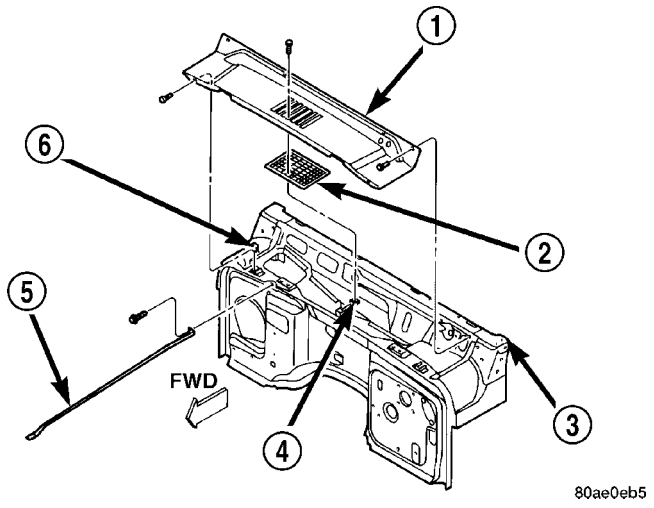

Fig. 1 COWL GRILLE AND SCREEN

1 - COWL GRILLE
2 - COWL GRILLE SCREEN
3 - COWL
4 - U NUT
5 - DASH PANEL TIE ROD
6 - U NUT

INSTALLATION

NOTE: When installing the cowl grille, ensure the snorkels on the cowl are positioned correctly and in good condition. Misaligned or damaged seals may allow water to enter the HVAC.

(1) Position the cowl screen and grille on the cowl.

(2) Install the screws that attach the grille and screen to the cowl.

(3) Remove the tape from the wiper pivots and install the wiper arms. (Refer to 8 - ELECTRICAL/WIPERS/WASHERS/WIPER ARMS - INSTALLATION)

RIGHT FRONT FENDER

REMOVAL

(1) Remove the battery tray. (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/TRAY - REMOVAL)

(2) Remove the air cleaner housing.

(3) Remove the bolts attaching the Power Distribution Center (PDC) to the fender.

(4) Disengage the PDC wire harness retainers on the battery tray and fender.

(5) Move and secure the PDC.

(6) Disengage the high pressure air conditioning line retainer on the fender.

(7) Disengage the front end lighting wire harness retainers on the fender.

(8) Disengage the battery temperature sensor connector.

(9) Disengage the vacuum line at the reservoir under the battery tray reinforcement bracket.

(10) Disengage the headlamp wire connector.

(11) Route the fog lamp (if equipped), park lamp and side marker wire harness through the access hole in the fender well.

(12) If equipped, remove the fender flare extension.

(13) Remove the bolts attaching the fender to the cowl (Fig. 2).

(14) Remove the bolts attaching the fender to the battery tray reinforcement bracket.

(15) Remove the bolts attaching the fender to the grille.

(16) Separate the fender from the vehicle.

INSTALLATION

Transfer all related components. Replace harness retainers if damaged.

(1) Position the fender on the vehicle.

(2) Install the bolts attaching the fender to the grille.

(3) Install the bolts attaching the fender to the battery tray reinforcement bracket.

(4) Install the bolts attaching the fender to the cowl.

(5) If equipped, install the fender flare extension and body side molding.

(6) Route the fog lamp (if equipped), park lamp and side marker wire harness through the access hole in the fender well. Seat the grommet.

(7) Engage the headlamp wire connector.

(8) Engage the battery temperature sensor connector.

(9) Engage the vacuum line at the reservoir under the battery tray reinforcement bracket.

(10) Position the front end lighting wire harness into the retainers on the fender. Engage the retainers to secure.

RIGHT FRONT FENDER (Continued)

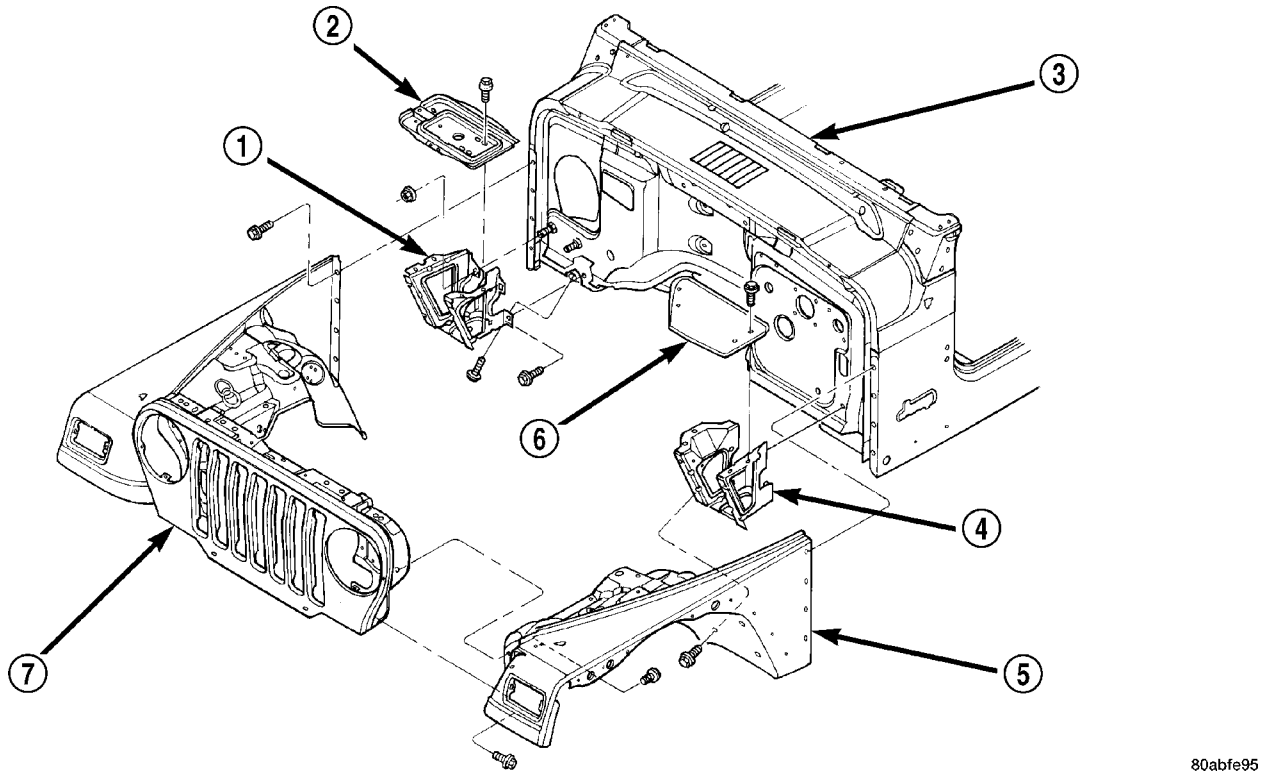

Fig. 2 FRONT FENDER

1 - REINFORCEMENT
2 - BATTERY TRAY
3 - BODY
4 - REINFORCEMENT
5 - FENDER
6 - TRAY
7 - RADIATOR GRILLE PANEL

(11) Position the high pressure air conditioning line into the retainer on the fender. Engage the retainer to secure.

(12) Position the PDC on the fender and install the bolts.

(13) Position the PDC wire harness into the retainers on the fender and battery tray. Engage the retainers to secure.

(14) Install the air cleaner housing.

(15) Install the battery tray. (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/TRAY - INSTALLATION)

LEFT FRONT FENDER

REMOVAL

(1) Disconnect the negative terminal on the battery. (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/CABLES - REMOVAL)

(2) Remove the windshield washer reservoir.

(3) Remove horns. (Refer to 8 - ELECTRICAL/HORN/HORN - REMOVAL)

(4) Remove EVAP canister.

(5) Remove the bolts attaching the ABS Hydraulic Control Unit (HCU) to the support tray.

(6) Secure the HCU.

(7) Remove the HCU tray.

(8) Disengage the front end lighting wire harness retainers on the fender.

(9) Disengage the headlamp wire connector.

(10) Route the fog lamp (if equipped), park lamp and side marker wire harness through the access hole in the fender well.

(11) Remove the bolts attaching the fender to the cowl (Fig. 3).

(12) Remove the bolts attaching the fender to the HCU tray reinforcement bracket.

(13) Remove the bolts attaching the fender to the grille.

(14) Separate the fender from the vehicle.

INSTALLATION

Transfer all related components. Replace harness retainers if damaged.

(1) Position the fender on the vehicle.

(2) Install the bolts attaching the fender to the grille.

LEFT FRONT FENDER (Continued)

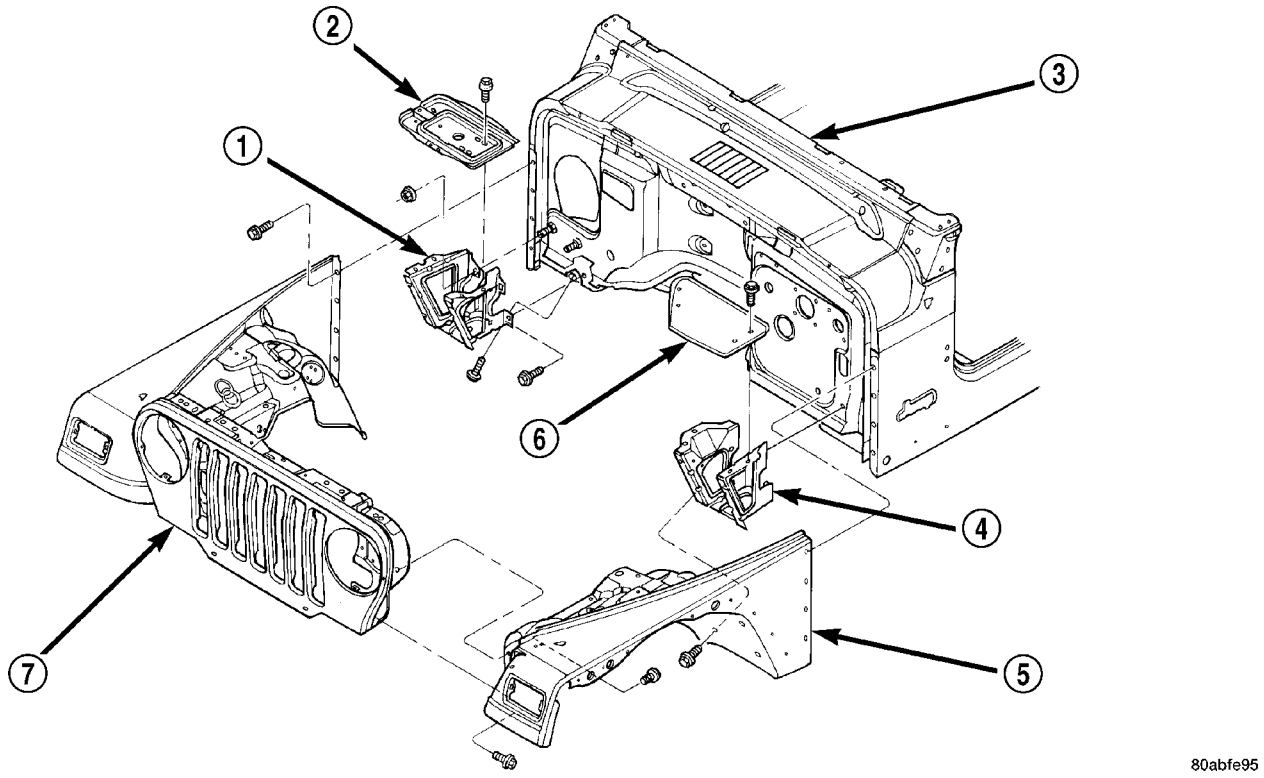

Fig. 3 FRONT FENDER

1 - REINFORCEMENT
2 - BATTERY TRAY
3 - BODY
4 - REINFORCEMENT
5 - FENDER
6 - TRAY
7 - RADIATOR GRILLE PANEL

(3) Position the front end lighting wire harness into the retainers on the fender. Engage the retainers to secure.

(4) Install the bolts attaching the fender to the HCU tray reinforcement bracket.

(5) Install the bolts attaching the fender to the cowl.

(6) If equipped, install the body side molding.

(7) Route the fog lamp (if equipped), park lamp and side marker wire harness through the access hole in the fender well. Seat the grommet.

(8) Engage the headlamp wire connector.

(9) Install the HCU tray.

(10) Position the HCU on the support tray and install the bolts.

(11) Install EVAP canister.

(12) Install horns. (Refer to 8 - ELECTRICAL/HORN/HORN - INSTALLATION)

(13) Engage horn wire connectors.

(14) Install the windshield washer reservoir.

(15) Connect the negative terminal on the battery. (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/CABLES - INSTALLATION)

RADIATOR GRILLE PANEL

REMOVAL

(1) Remove the front crossmember cover.

(2) Remove the crossmember valence cover.

(3) Remove the radiator overflow bottle.

(4) Remove the bolts that attach the radiator and shroud from the grille panel.

(5) If A/C equipped:

(a) Evacuate the system.

(b) Disconnect the high and low pressure lines at the quick disconnect couplings.

(c) Cover (cap) the lines to prevent contamination.

(6) Remove the bolts attaching the radiator support rods to the grille panel.

(7) Disconnect the head lamp, turn signal, marker lamp and horn wire harness connectors.

(8) Remove the bolts attaching the fenders to the grille panel.

(9) Remove the bolt attaching the grille to the frame mount.

(10) Separate the grille from the vehicle.

RADIATOR GRILLE PANEL (Continued)

INSTALLATION

Transfer all related components.

(1) Position the grille panel on the vehicle. Ensure the rubber support bumpers are aligned (Fig. 4).

(2) Install the bolt attaching the grille to the frame mount.

(3) Install the bolts attaching the fenders to the grille panel.

(4) Connect the head lamp, turn signal, marker lamp and horn wire harness connectors.

(5) Install the bolts attaching the radiator support rods to the grille panel.

(6) If A/C equipped:

(a) Connect the high and low pressure lines at the quick disconnect couplings.

(b) Evacuate and charge the system.

(7) Install the radiator and shroud to the grille panel.

(8) Install the radiator overflow bottle.

(9) Install the crossmember valence cover.

(10) Install the front crossmember cover.

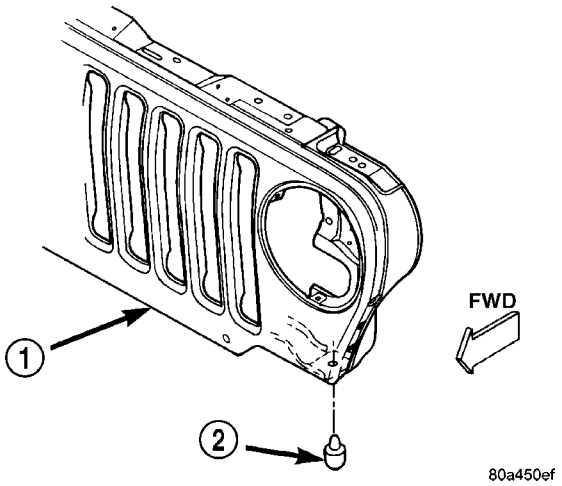

Fig. 4 GRILLE BUMPERS

1 - GRILLE PANEL
2 - BUMPER

WHEELHOUSE SPLASH SHIELD

REMOVAL

(1) Remove the plastic push pins that attach the splash shield to the wheelhouse (Fig. 5).

(2) Remove the push-in fasteners attaching the splash shield to the wheelhouse. (The push-in fasteners are molded into the splash shield.)

(3) Remove the splash shield from the wheelhouse.

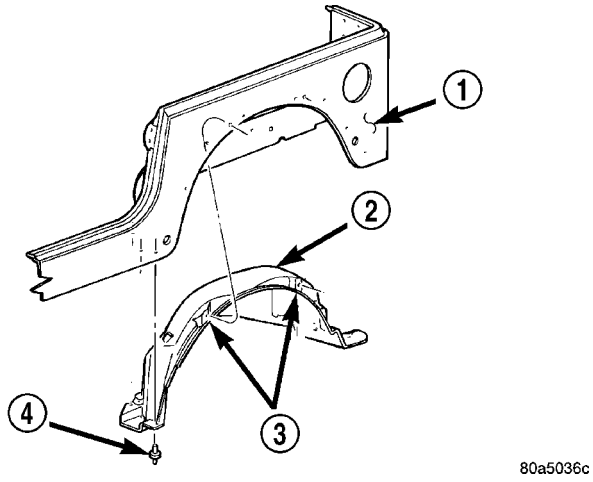

Fig. 5 Wheelhouse Splash Shield

1 - BODY
2 - WHEELHOUSE SPLASH SHIELD
3 - PUSH-IN FASTENERS
4 - RIVET

INSTALLATION

(1) Position the splash shield in the wheelhouse.

(2) Press the splash shield push-in fasteners into place.

(3) Attach the splash shield to the wheelhouse with push pins.

SIDE VIEW MIRROR

REMOVAL

(1) Remove the screws attaching the mirror to the door hinge (Fig. 6).
(2) Remove the mirror from the door hinge.
(3) Disconnect the electrical connector, if equipped.

INSTALLATION

(1) Clean the door hinge-mirror base contact surface.
(2) Position the mirror base at the door hinge.
(3) Install the three screws attaching the mirror base to the door hinge.

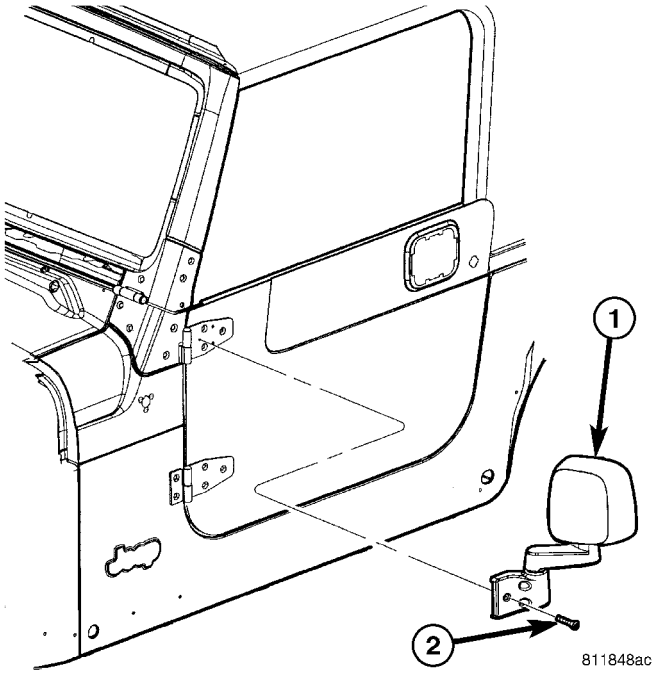

Fig. 6 SIDE VIEW MIRROR

1 - MIRROR ASSEMBLY
2 - SCREWS (3)

WHEEL OPENING FLARE MOLDING

REMOVAL

(1) Remove the side marker lamp. (Refer to 8 - ELECTRICAL/LAMPS/LIGHTING - EXTERIOR/ MARKER LAMP UNIT - REMOVAL)
(2) Remove the screws that attach the flare to the front fender or rear wheelhouse (Fig. 7).
(3) Separate the flare from the body.

INSTALLATION

(1) Clean the contact surface on the body.

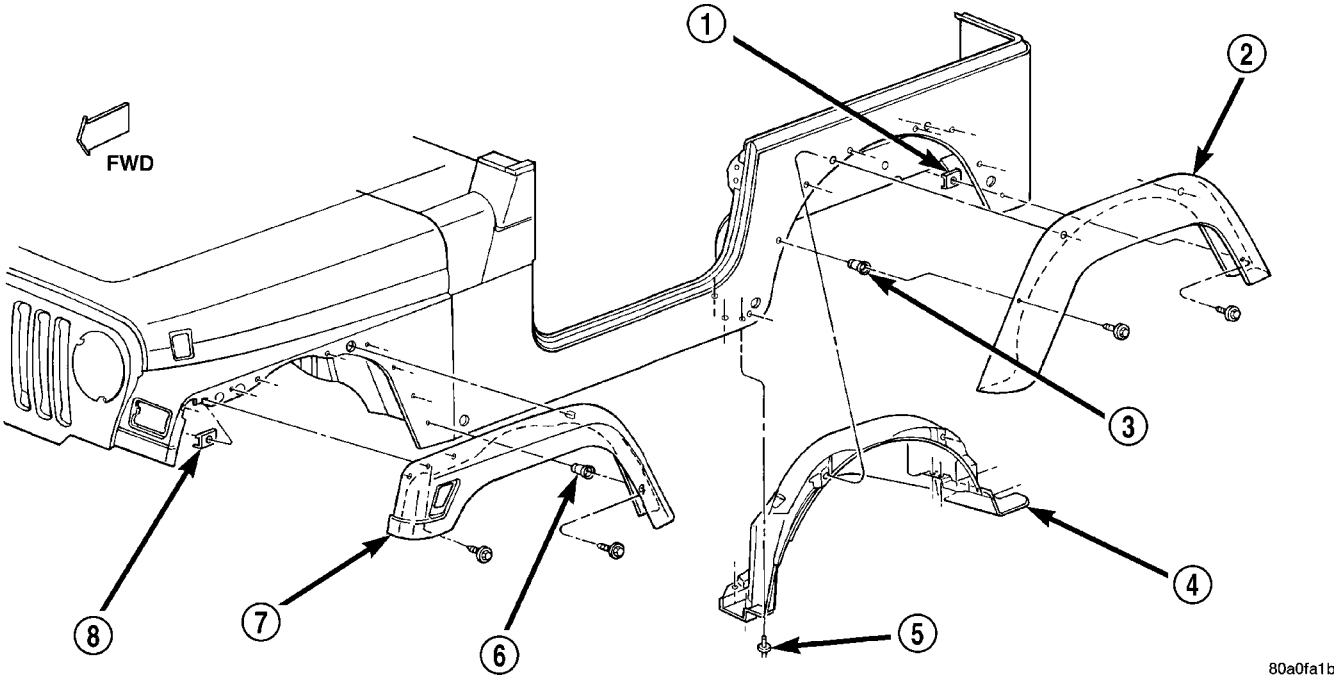

Fig. 7 FENDER FLARES

1 - U-NUT
2 - FENDER FLARE
3 - NUTSERT
4 - SPLASH SHIELD
5 - RIVET
6 - NUTSERT
7 - FENDER FLARE
8 - U-NUT

WHEEL OPENING FLARE MOLDING (Continued)

(2) Clean the contact surface on the flare and position it on the front fender or wheelhouse.

(3) Install the screws attaching the flares to the front fender or wheelhouse.

(4) If removed, install the side marker lamp.

EXHAUSTER

REMOVAL

(1) Remove the spare tire carrier. (Refer to 22 - TIRES/WHEELS/TIRES/SPARE TIRE CARRIER - REMOVAL)

(2) Using a trim stick C-4755 or equivalent, separate the exhauster from the tailgate. (Fig. 8)

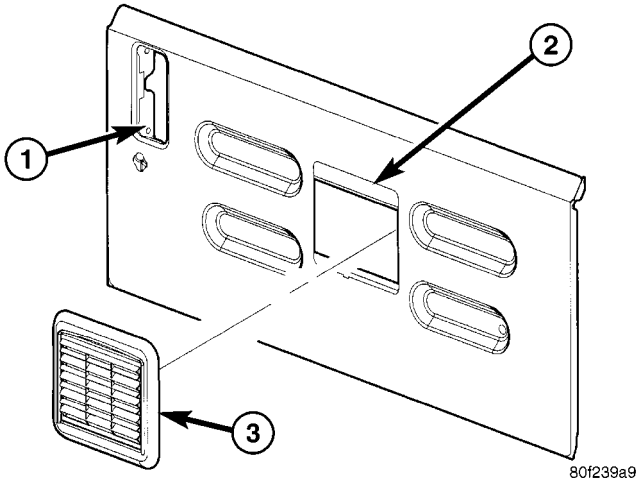

Fig. 8 BODY EXHAUSTER

1 - TAILGATE HANDLE
2 - EXHAUSTER OPENING
3 - EXHAUSTER

INSTALLATION

(1) Install the exhauster and seat fully.

(2) Install the spare tire carrier. (Refer to 22 - TIRES/WHEELS/TIRES/SPARE TIRE CARRIER - INSTALLATION)

BODY SIDE GUARD

REMOVAL

(1) Remove the bolts from the lower support tabs. (Fig. 9)

(2) Remove the side bolts and remove the side guard. (Fig. 10)

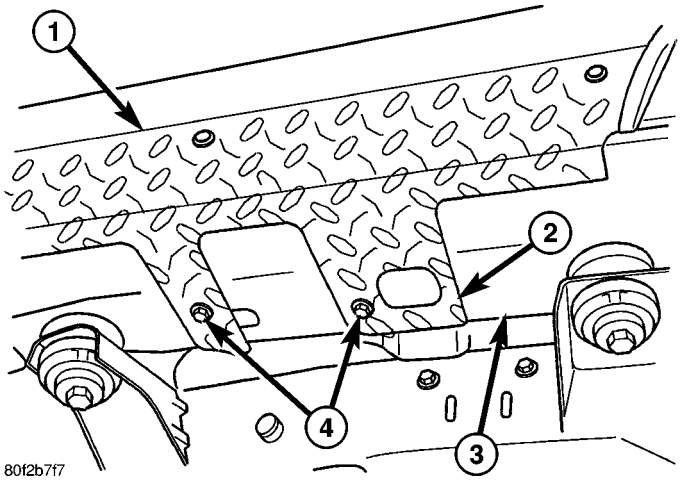

Fig. 9 SIDE GUARD - LOWER

1 - BODY SIDE GUARD
2 - SUPPORT TABS (5)
3 - BODY FLOOR SILL
4 - BOLTS (5)

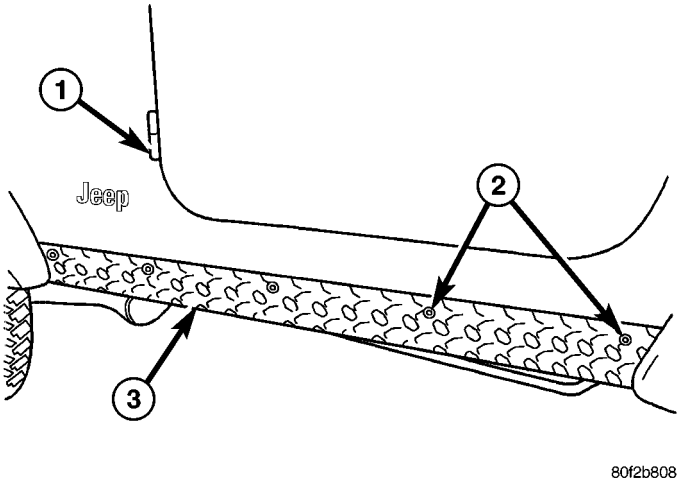

Fig. 10 SIDE GUARD - UPPER

1 - DOOR
2 - SIDE GUARD BOLTS (5)
3 - BODY SIDE GUARD

INSTALLATION

(1) Install the body side guard and install the side bolts.

(2) Tighten the bolts to 5 N·m (45 in. lbs.).

(3) Install the support tab bolts and tighten to 11 N·m (8 ft. lbs.).

SIDE STEP

REMOVAL

(1) Remove the bolts that attach the side step to the frame (Fig. 11).

(2) Tilt the step down and disengage the tabs from the frame slots.

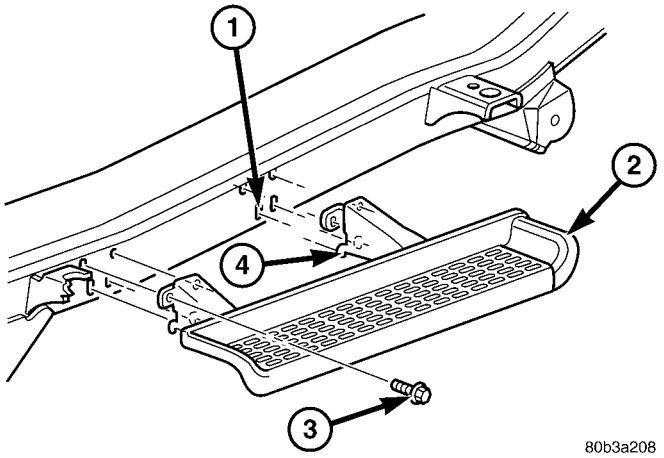

Fig. 11 SIDE STEP

1 - FRAME SLOTS
2 - SIDE STEP
3 - BOLTS
4 - STEP MOUNTING TABS

INSTALLATION

(1) Position the side step on the frame and engage the mounting tabs into the frame slots.

(2) Install the bolts that attach the side step to the frame and tighten to 21 N·m (15 ft. lbs.).

LICENSE PLATE BRACKET

REMOVAL

(1) If installed, remove the license plate.

(2) Remove the four screws attaching the license plate bracket to the body (Fig. 12).

(3) On export models remove the bolts and nuts. (Fig. 13)

(4) Separate the bracket from the body.

(5) Disconnect the electrical connectors, if equipped.

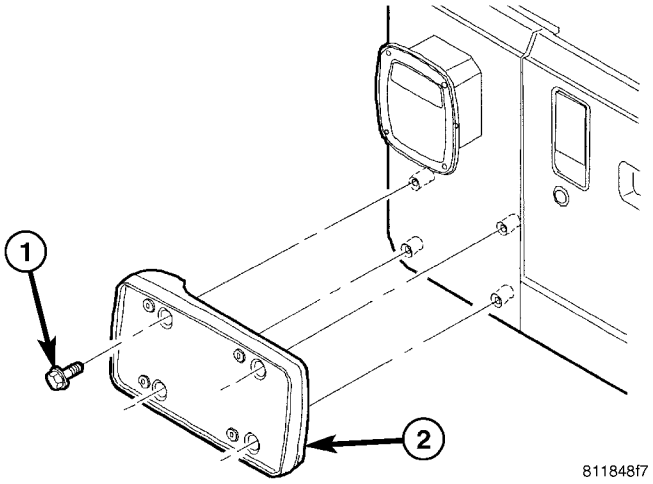

Fig. 12 LICENSE PLATE BRACKET

1 - BOLTS (4)
2 - BRACKET

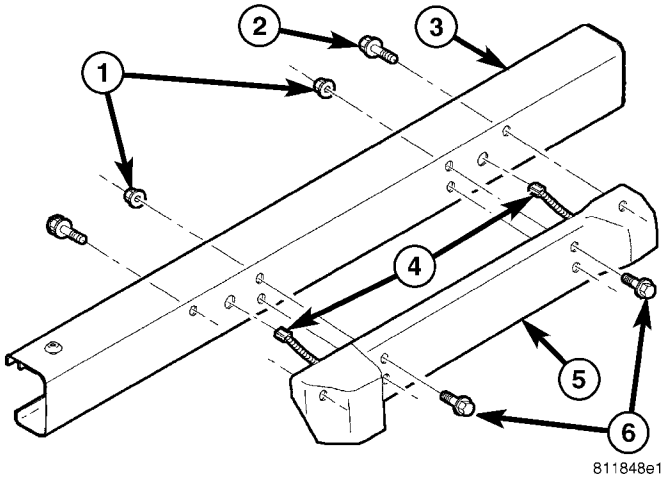

Fig. 13 LICENSE PLATE BRACKET - EXPORT

1 - NUTS (4)
2 - BOLTS (2)
3 - BUMPER
4 - ELECTRICAL CONNECTORS
5 - LICENSE PLATE BRACKET
6 - BOLTS (4)

INSTALLATION

(1) Connect the electrical connectors, if equipped.

(2) Position the bracket on the body.

(3) Install the four screws attaching the license plate bracket to the body.

(4) On export models, install the six bolts and four nuts.

(5) If removed, install the license plate.

HOOD

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HINGE

REMOVAL

(1) Remove the wiper arms. (Refer to 8 - ELECTRICAL/WIPERS/WASHERS/WIPER ARMS - REMOVAL)

(2) Remove the cowl panel and screen. (Refer to 23 - BODY/EXTERIOR/COWL GRILLE AND SCREEN - REMOVAL)

(3) Remove the bolts attaching the hinge to the cowl.

(4) Using a wax pencil, mark the position of the hinge on the hood for installation alignment reference.

(5) Remove the screws attaching the hinge to the hood (Fig. 1).

(6) Separate the hinge from the hood.

INSTALLATION

(1) Prepare and paint the replacement hinge to match the body paint color.

(2) Align the hinge with the installation reference marks on the hood

(3) Install the screws attaching the hinge to the hood and cowl and tighten the to 24 N·m (18 ft. lbs.).

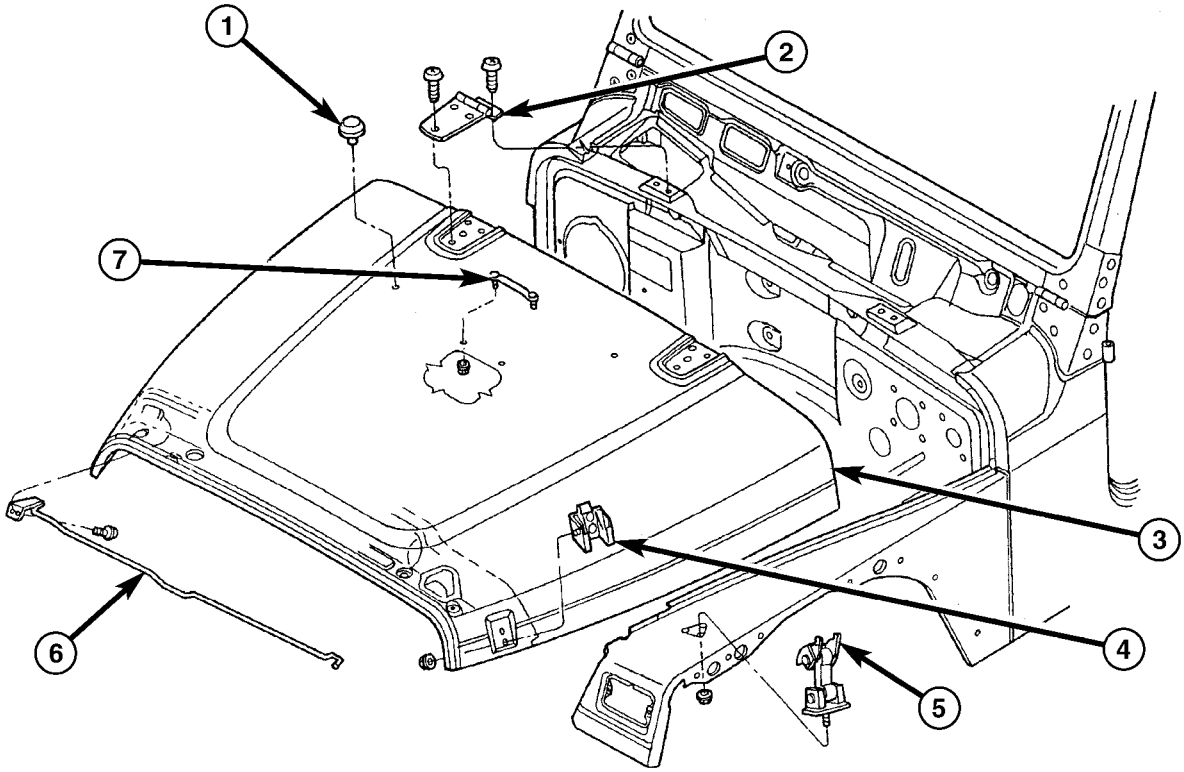

Fig. 1 HOOD COMPONENTS

1 - WINDSHIELD REST BUMPER
2 - HOOD HINGE
3 - HOOD
4 - HOOD CATCH BRACKET
5 - HOOD CATCH
6 - PROP ROD
7 - WINDSHIELD HOOD DOWN LOOP

HINGE (Continued)

(4) Install the bolts attaching the hinge to the cowl.

(5) Install the cowl panel and screen. (Refer to 23 - BODY/EXTERIOR/COWL GRILLE AND SCREEN - INSTALLATION)

(6) Install the wiper arms. (Refer to 8 - ELECTRICAL/WIPERS/WASHERS/WIPER ARMS - INSTALLATION)

HOOD

REMOVAL

(1) Raise and support the hood.

(2) Disconnect the underhood lamp wire harness connector.

(3) Disconnect the windshield washer nozzles.

(4) Disconnect the ground strap.

(5) Mark the position of the hinges on the hood for installation alignment reference.

(6) Remove the screws attaching the hood to the hinge and remove the hood (Fig. 1).

(7) If the hood must be replaced, remove and transfer the insulator panel, hinges, latches, bumpers, brackets, footman loop, hood lamp, support rod, and safety latch to the replacement hood (Fig. 1).

INSTALLATION

(1) Position the hood on the vehicle and install the screws attaching the hinge to the hood.

(2) Align the hinges with the installation reference marks on the hood and tighten the hinge screws to 24 N·m (18 ft. lbs.).

(3) Connect the underhood lamp wire harness connector.

(4) Connect the windshield washer nozzles.

(5) Connect the ground strap.

ADJUSTMENTS

ADJUSTMENT

The hood hinge screw holes are oversized to facilitate hood adjustment movement.

(1) Loosen the screws.

(2) Move the hood in the direction(s) required for correct alignment.

(3) Tighten the screws.

HOOD SAFETY LATCH

REMOVAL

(1) Raise and support the hood.

(2) Remove the bolt attaching the safety latch to the hood (Fig. 2).

(3) Remove the latch from the hood.

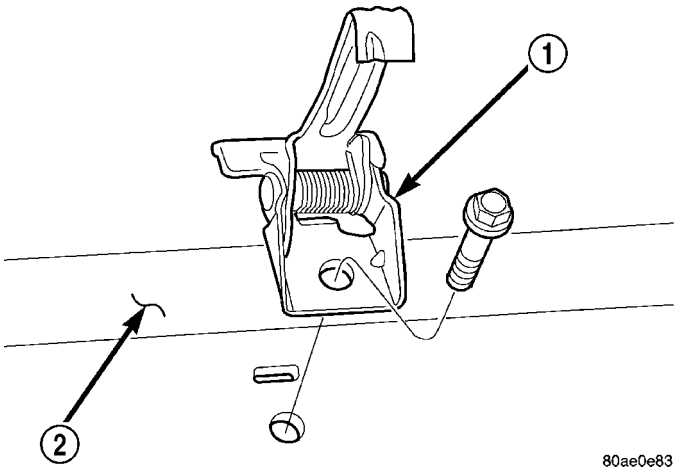

Fig. 2 HOOD SAFETY LATCH

1 - SAFETY LATCH
2 - HOOD

INSTALLATION

(1) Position the latch on the hood.

(2) Install the bolt attaching the safety latch to the hood and tighten to 9 N·m (80 in. lbs.).

(3) Remove the support rod and close the hood.

INSTRUMENT PANEL

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ACCESSORY SWITCH BEZEL

REMOVAL

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Disconnect and isolate the battery negative cable.

(2) Remove the center bezel from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/ CENTER BEZEL - REMOVAL).

(3) Remove the four screws that secure the accessory switch bezel to the instrument panel (Fig. 1).

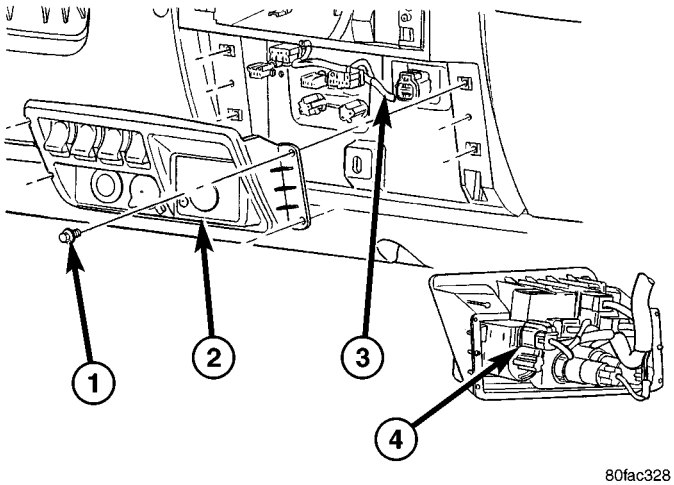

Fig. 1 ACCESSORY SWITCH BEZEL

1 - BEZEL SCREWS (4)
2 - BEZEL
3 - WIRE HARNESS
4 - ELECTRICAL CONNECTORS (7)

ACCESSORY SWITCH BEZEL (Continued)

(4) Pull the accessory switch bezel away from the instrument panel far enough to access the instrument panel wire harness connectors.

(5) Disconnect the instrument panel wire harness connectors from the connector receptacles for the accessory switches and the cigar lighter/power outlet on the back of the accessory switch bezel.

(6) Remove the accessory switch bezel from the instrument panel.

INSTALLATION

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Position the accessory switch bezel to the instrument panel.

(2) Reconnect the instrument panel wire harness connectors to the connector receptacles for the accessory switches and the cigar lighter/power outlet on the back of the accessory switch bezel.

(3) Position the accessory switch bezel onto the instrument panel.

(4) Install and tighten the four screws that secure the accessory switch bezel to the instrument panel. Tighten the screws to 2 N·m (20 in. lbs.).

(5) Reinstall the center bezel onto the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/ CENTER BEZEL - INSTALLATION).

(6) Reconnect the battery negative cable.

AXLE LOCK SWITCH

DESCRIPTION

If equipped, the Axle Lock switch is located in the Accessory Switch Bezel, below the HVAC controls. The switch will activate the Rear Axle locker and Rear/Front Axle locker. The switch is a double-action momentary rocker type. The Axle Lock Switch is secured by a snap fit within the bezel. The axle Lock Switch cannot be repaired and, if faulty or damaged, the entire switch must be replaced.

OPERATION

The Axle Lock switch receives battery current on a fused B(+) circuit from a fuse in the Power Distribution Center (PDC). The switch is grounded at all times. The Axle Lock switch is only operational when the instrument cluster sends an activation signal to the switch logic.

With the transfer case in the 4WD Low Range and the vehicle speed below 10 MPH (16kph), a momentary push to the lower portion of the rocker switch to Lock:

- 1st Push: Rear Axle Locked
- 2nd Push: Rear and Front Axle Locked

Momentary pushes, will toggle between Rear Axle Locked or Rear/Front Axle Locked. A momentary push to the upper portion of the rocker switch to “OFF”, (both axles unlocked).

DIAGNOSIS AND TESTING - AXLE LOCK SWITCH

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Check the fused B(+) fuse in the Power Distribution Center (PDC). If OK, go to step 2. If not, repair the shorted circuit or component required and replace the fuse.

(2) Check for battery voltage at the fused B(+) fuse in the PDC. If OK, go to step 3. If not, repair the open B(+) circuit to the fuse.

(3) Turn the ignition to the off position. Disconnect and isolate the negative battery cable. Remove the Axle Lock switch. Disconnect the switch harness connector. Check for continuity between the fuse cavity in the PDC and the Fused B(+) circuit in the switch harness connector. If OK got to step 4. If not OK, repair the open Fused B(+) circuit.

(4) Turn the ignition to the off position. Disconnect and isolate the negative battery cable. Remove the Axle Lock switch. disconnect the switch harness connector. check for continuity between the switch ground and a known good ground. If OK got to step 5. If not OK, repair open ground circuit.

AXLE LOCK SWITCH (Continued)

(5) Turn the ignition to the OFF position. Disconnect and isolate the negative battery cable. Remove the Instrument Cluster. Disconnect the cluster harness connectors C1 and C2. Check for continuity between each of the Enable circuits and each of the request circuits between the switch harness connector and the cluster harness connectors. If OK go to step 6. If not OK, repair the circuits as required.

(6) Turn the ignition to the OFF position. Disconnect and isolate the negative battery cable. Remove the axle Lock switch. Disconnect the switch harness connector. Check for continuity between each of the Request circuits, between the switch harness connector and each of the Locker relays in the PDC. If OK, refer to Axle Lock Indicator Problems in the Instrument Cluster section of the appropriate Body Manual or the Chassis Diagnostic Manual and perform the appropriate symptom. If not OK, repair the circuit(s) as required.

REMOVAL

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, DRIVER AIRBAG, PASSENGER AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Disconnect and isolate the battery negative cable.

(2) Remove the accessory switch bezel from the instrument panel center bezel from the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/ ACCESSORY SWITCH BEZEL - REMOVAL)

(3) Using a trim stick or another suitable wide flat-bladed tool, gently pry the latch tabs at the top and bottom of the axle lock switch on the back of the accessory switch bezel far enough to disengage the snap features on the switch housing then pull the switch out of the receptacle (Fig. 2).

INSTALLATION

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, DRIVER AIRBAG, PASSENGER AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

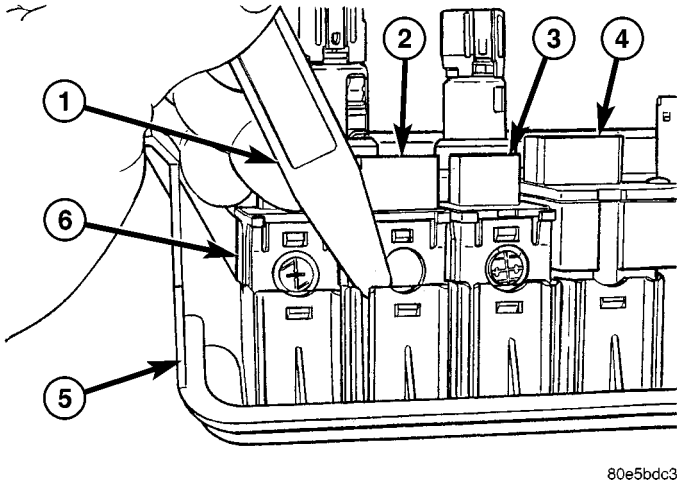

Fig. 2 ACCESSORY SWITCH REMOVE

1 - TRIM STICK
2 - REAR WIPER/WASHER SWITCH
3 - OVERDRIVE-OFF SWITCH
4 - AXLE LOCKER SWITCH
5 - ACCESSORY SWITCH BEZEL
6 - REAR WINDOW DEFOGGER SWITCH

(1) Position the axle lock switch over the proper receptacle on the back of the accessory switch bezel (Fig. 2).

(2) Evenly push the axle lock switch into the receptacle until the snap features on the top and bottom of the switch housing are fully engaged to the receptacle.

(3) Reinstall the accessory switch bezel onto the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/ACCESSORY SWITCH BEZEL - INSTALLATION).

(4) Reconnect the battery negative cable.

BASE TRIM

REMOVAL

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Disconnect and isolate the battery negative cable.

(2) Remove the instrument cluster from the instrument panel. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - REMOVAL).

(3) Remove the accessory switch bezel from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/ACCESSORY SWITCH BEZEL - REMOVAL).

(4) Remove the grab handle bezel from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/GRAB HANDLE BEZEL - REMOVAL).

(5) Remove the speakers from the instrument panel. (Refer to 8 - ELECTRICAL/AUDIO/SPEAKER - REMOVAL).

(6) Remove the radio from the instrument panel. (Refer to 8 - ELECTRICAL/AUDIO/RADIO - REMOVAL).

(7) Remove the heater-A/C control from the instrument panel. (Refer to 24 - HEATING & AIR CONDITIONING/CONTROLS/A/C-HEATER CONTROL - REMOVAL).

(8) Remove the outboard heater-A/C panel outlet barrels from the instrument panel. (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/PANEL OUTLET BARRELS - REMOVAL).

(9) Remove the instrument panel from the vehicle. (Refer to 23 - BODY/INSTRUMENT PANEL - REMOVAL).

(10) Place the instrument panel on a suitable work surface. Be certain to take the proper precautions to protect the instrument panel from any possible cosmetic damage.

(11) Remove the passenger airbag door from the instrument panel. (Refer to 8 - ELECTRICAL/RESTRAINTS/PASSENGER AIRBAG DOOR - REMOVAL).

(12) Remove the two screws that secure the 16-way data link connector to the instrument panel.

(13) Remove all of the screws around the perimeter of the instrument panel that secure the base trim to the structural support.

(14) Remove the base trim from the instrument panel structural support.

INSTALLATION

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Position the base trim onto the instrument panel structural support.

(2) Install and tighten all of the screws around the perimeter of the instrument panel that secure the base trim to the structural support. Tighten the screws to 2 N·m (20 in. lbs.).

(3) Install and tighten the two screws that secure the 16-way data link connector to the instrument panel. Tighten the screws to 2 N·m (20 in. lbs.).

(4) Reinstall the passenger airbag door into the instrument panel. (Refer to 8 - ELECTRICAL/RESTRAINTS/PASSENGER AIRBAG DOOR - INSTALLATION).

(5) Reinstall the instrument panel into the vehicle. (Refer to 23 - BODY/INSTRUMENT PANEL - INSTALLATION).

(6) Reinstall the outboard heater-A/C panel outlet barrels into the instrument panel. (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/PANEL OUTLET BARRELS - INSTALLATION).

(7) Reinstall the heater-A/C control into the instrument panel. (Refer to 24 - HEATING & AIR CONDITIONING/CONTROLS/A/C-HEATER CONTROL - INSTALLATION).

(8) Reinstall the radio into the instrument panel. (Refer to 8 - ELECTRICAL/AUDIO/RADIO - INSTALLATION).

(9) Reinstall the speakers into the instrument panel. (Refer to 8 - ELECTRICAL/AUDIO/SPEAKER - INSTALLATION).

(10) Reinstall the grab handle bezel onto the instrument panel. (Refer to 23 - BODY/INSTRU-

BASE TRIM (Continued)

MENT PANEL/GRAB HANDLE BEZEL - INSTALLATION).

(11) Reinstall the accessory switch bezel onto the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/ACCESSORY SWITCH BEZEL - INSTALLATION).

(12) Reinstall the instrument cluster into the instrument panel. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - INSTALLATION).

(13) Reconnect the battery negative cable.

CENTER BEZEL

REMOVAL

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Disconnect and isolate the battery negative cable.

(2) Remove the top cover from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/TOP COVER - REMOVAL).

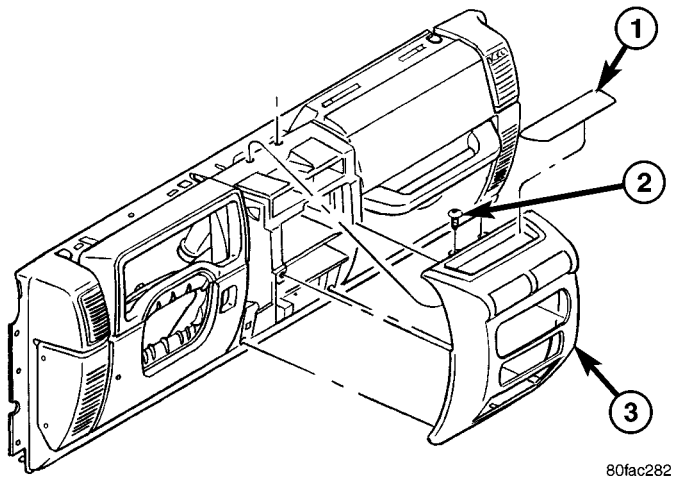

Fig. 3 CENTER BEZEL

1 - BEZEL MAT
2 - TOP SCREWS (2)
3 - CENTER BEZEL

(3) Remove the two screws that secure the top of the center bezel to the top of the instrument panel structural support. (Fig. 3)

(4) Using a trim stick or another suitable wide flat-bladed tool, gently pry the lower edge of the center bezel away from the instrument panel.

(5) Pull the lower edge of the center bezel away from the instrument panel far enough to disengage the four snap clip retainers that secure it to the receptacles in the instrument panel base trim.

(6) Remove the center bezel from the instrument panel.

INSTALLATION

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(1) Position the center bezel onto the instrument panel.

(2) Align the snap clips on the center bezel with the receptacles in the instrument panel base trim.

(3) Using hand pressure, press firmly on the center bezel over each of the snap clip locations until each of the snap clips is fully engaged in its receptacle in the instrument panel base trim.

(4) Install and tighten the two screws that secure the top of the center bezel to the top of the instrument panel structural support. Tighten the screws to 2 N·m (20 in. lbs.).

(5) Reinstall the top cover onto the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/TOP COVER - INSTALLATION).

(6) Reconnect the battery negative cable.

CLUSTER BEZEL

REMOVAL

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(1) Disconnect and isolate the battery negative cable.

(2) Remove the steering column opening cover from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/STEERING COLUMN OPENING COVER - REMOVAL).

(3) Remove the top cover from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/TOP COVER - REMOVAL).

(4) Remove the two screws that secure the lower mounting tabs of the cluster bezel to the instrument panel (Fig. 4).

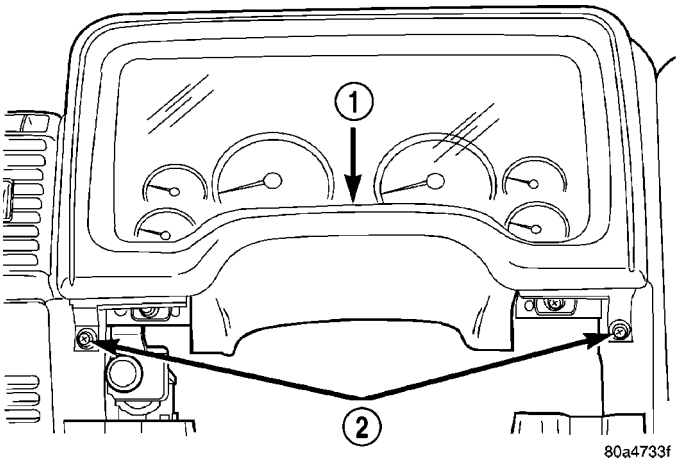

Fig. 4 CLUSTER BEZEL LOWER SCREWS REMOVE/INSTALL

1 - CLUSTER BEZEL
2 - SCREW (2)

(5) Remove the three screws that secure the upper mounting flange of the cluster bezel to the top of the instrument panel (Fig. 5).

(6) Remove the cluster bezel from the instrument panel.

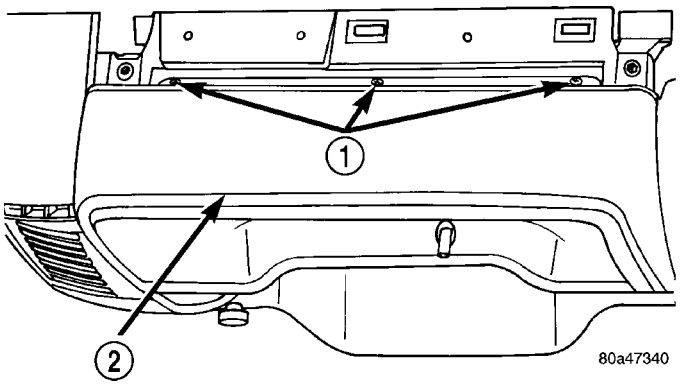

Fig. 5 CLUSTER BEZEL UPPER SCREWS REMOVE/INSTALL

1 - SCREW (3)
2 - CLUSTER BEZEL

INSTALLATION

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(1) Position the cluster bezel onto the instrument panel.

(2) Install and tighten the three screws that secure the upper mounting flange of the cluster bezel to the top of the instrument panel (Fig. 5). Tighten the screws to 2 N·m (20 in. lbs.).

(3) Install and tighten the two screws that secure the lower mounting tabs of the cluster bezel to the instrument panel (Fig. 4). Tighten the screws to 2 N·m (20 in. lbs.).

(4) Reinstall the top cover onto the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/TOP COVER - INSTALLATION).

(5) Reinstall the steering column opening cover onto the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/STEERING COLUMN OPENING COVER - INSTALLATION).

(6) Reconnect the battery negative cable.

GLOVE BOX

REMOVAL

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(1) Disconnect and isolate the battery negative cable.

(2) Release the glove box latch and open the glove box door.

(3) While supporting the glove box door with one hand, grasp the check strap as close to the glove box door as possible and slide the rolled end of the check strap out of the slot in the edge of the door (Fig. 6).

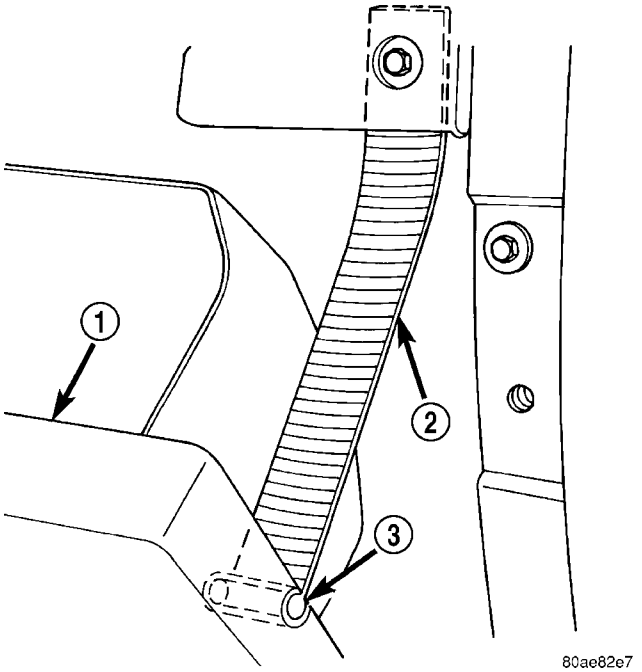

Fig. 6 GLOVE BOX REMOVE/INSTALL

1 - GLOVE BOX DOOR
2 - CHECK STRAP
3 - SLOT

(4) Lower the glove box door far enough to disengage the hinge hook formations on the lower edge of the door from the hinge pins on the lower edge of the instrument panel.

(5) Remove the glove box from the instrument panel.

DISASSEMBLY

Some of the components of the glove box used in this vehicle are serviced individually. The serviced components include the glove box latch and handle unit, and the glove box lock cylinder (Fig. 7). Following are the procedures for disassembling these components from the glove box.

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

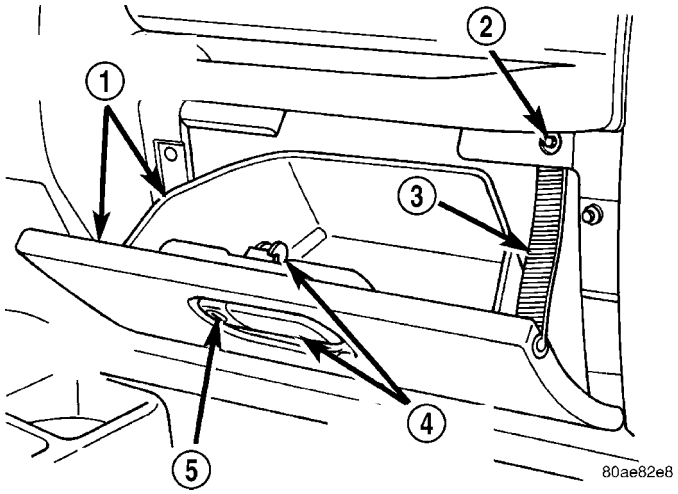

Fig. 7 GLOVE BOX COMPONENTS

1 - GLOVE BOX DOOR AND BIN
2 - SCREW
3 - CHECK STRAP
4 - LATCH AND HANDLE
5 - LOCK CYLINDER

GLOVE BOX LATCH AND HANDLE

(1) Disconnect and isolate the battery negative cable.

(2) Remove the glove box from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/GLOVE BOX - REMOVAL).

GLOVE BOX (Continued)

(3) Remove the four screws that secure the glove box latch and handle to the glove box door from the inside of the glove box.

(4) Remove the latch and handle from the inside of the glove box door.

GLOVE BOX LOCK CYLINDER

(1) Remove the glove box latch and handle unit from the glove box. Refer to GLOVE BOX LATCH AND HANDLE.

(2) Insert the key into the glove box lock cylinder.

(3) Insert a small screwdriver into the retaining tumbler release slot and depress the retaining tumbler (Fig. 8).

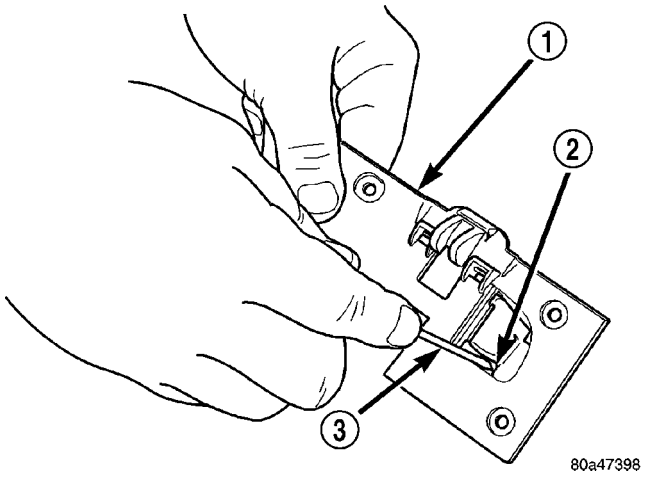

Fig. 8 GLOVE BOX LOCK CYLINDER REMOVE/ INSTALL

1 - GLOVE BOX LATCH
2 - RELEASE SLOT
3 - SMALL SCREWDRIVER

(4) Using a gentle twisting and pulling action with the key, pull the lock cylinder out of the glove box latch and handle unit.

ASSEMBLY

Some of the components of the glove box used in this vehicle are serviced individually. The serviced components include the glove box latch and handle unit, and the glove box lock cylinder (Fig. 7). Following are the procedures for assembling these components onto the glove box.

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

GLOVE BOX LATCH AND HANDLE

(1) Position the latch and handle onto the inside of the glove box door.

(2) Install and tighten the four screws that secure the glove box latch and handle to the glove box door from the inside of the glove box. Tighten the screws to 2 N·m (20 in. lbs.).

(3) Reinstall the glove box onto the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/ GLOVE BOX - INSTALLATION).

(4) Reconnect the battery negative cable.

GLOVE BOX LOCK CYLINDER

(1) Insert the key into the glove box lock cylinder.

(2) Using a gentle twisting and pushing action on the key, push the lock cylinder into the glove box latch and handle unit.

(3) Reinstall the glove box latch and handle onto the glove box. Refer to GLOVE BOX LATCH AND HANDLE.

INSTALLATION

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(1) Position the glove box to the instrument panel.

(2) Engage the hinge hook formations on the lower edge of the glove box door with the hinge pins on the lower edge of the instrument panel.

(3) Tilt the upper edge of the glove box door up toward the instrument panel far enough to engage the check strap with the door (Fig. 6).

(4) While supporting the glove box door with one hand, grasp the check strap as close to the glove box door as possible and slide the rolled end of the check strap into the slot in the edge of the door.

(5) Close and latch the glove box door.

(6) Reconnect the battery negative cable.

GLOVE BOX CHECK STRAP

REMOVAL

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(1) Disconnect and isolate the battery negative cable.

(2) Remove the glove box from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/ GLOVE BOX - REMOVAL).

(3) Remove the screw that secures the glove box check strap to the instrument panel on the upper glove box opening reinforcement (Fig. 9).

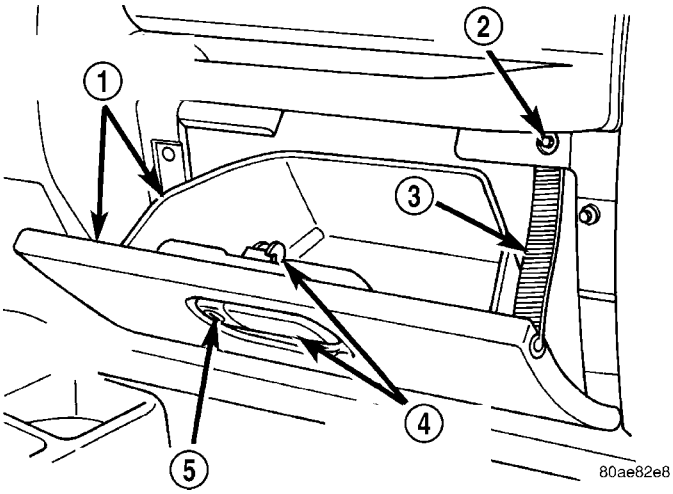

Fig. 9 GLOVE BOX CHECK STRAP

1 - GLOVE BOX DOOR AND BIN
2 - SCREW
3 - CHECK STRAP
4 - LATCH AND HANDLE
5 - LOCK CYLINDER

(4) Remove the check strap from the upper glove box opening reinforcement.

INSTALLATION

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(1) Position the check strap to the upper glove box opening reinforcement (Fig. 9).

(2) Install and tighten the screw that secures the glove box check strap to the upper glove box opening reinforcement. Tighten the screw to 2 N·m (20 in. lbs.).

(3) Reinstall the glove box onto the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/ GLOVE BOX - INSTALLATION).

(4) Reconnect the battery negative cable.

GLOVE BOX LATCH STRIKER

REMOVAL

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(1) Disconnect and isolate the battery negative cable.

(2) Remove the glove box from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/ GLOVE BOX - REMOVAL).

GLOVE BOX LATCH STRIKER (Continued)

(3) Remove the two screws that secure the latch striker to the upper glove box opening reinforcement (Fig. 10).

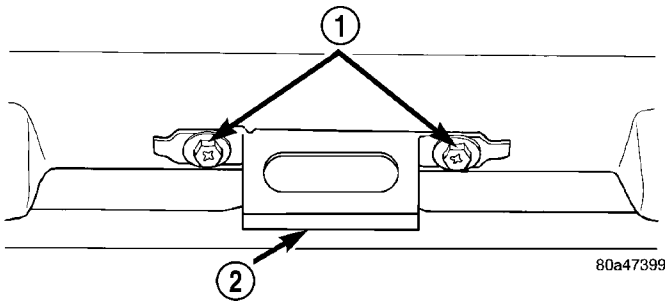

Fig. 10 GLOVE BOX LATCH STRIKER REMOVE/ INSTALL

1 - SCREW (2)
2 - LATCH STRIKER

(4) Remove the latch striker from the upper glove box opening reinforcement.

INSTALLATION

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(1) Position the latch striker onto the upper glove box opening reinforcement (Fig. 10).

(2) Install and tighten the two screws that secure the latch striker to the upper glove box opening reinforcement. Tighten the screws to 2 N·m (20 in. lbs.).

(3) Reinstall the glove box onto the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/ GLOVE BOX - INSTALLATION).

(4) Reconnect the battery negative cable.

GRAB HANDLE

REMOVAL

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(1) Disconnect and isolate the battery negative cable.

(2) Remove the glove box from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/ GLOVE BOX - REMOVAL).

(3) Reach through and above the glove box opening to access and remove the two nuts that secure the stud on each end of the grab handle to the instrument panel structural support (Fig. 11). Discard the used grab handle mounting nuts.

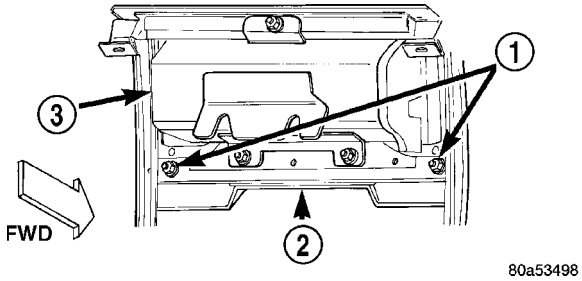

Fig. 11 GRAB HANDLE REMOVE/INSTALL

1 - NUT (2)
2 - GLOVE BOX OPENING
3 - PASSENGER AIRBAG

(4) Remove the grab handle from the face of the instrument panel.

INSTALLATION

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GRAB HANDLE (Continued)

(1) Position the grab handle onto the face of the instrument panel.

NOTE: Always use new fasteners to install the grab handle. The removed grab handle fasteners should be discarded.

(2) Reach through and above the glove box opening to install and tighten two new nuts to secure the stud on each end of the grab handle to the instrument panel structural support (Fig. 11). Tighten the nuts to 6 N·m (50 in. lbs.).

(3) Reinstall the glove box into the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/GLOVE BOX - INSTALLATION).

(4) Reconnect the battery negative cable.

GRAB HANDLE BEZEL

REMOVAL

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(1) Disconnect and isolate the battery negative cable.

(2) Remove the grab handle from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/GRAB HANDLE - REMOVAL).

(3) Remove the glove box latch striker from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/GLOVE BOX LATCH STRIKER - REMOVAL).

(4) Remove the two screws that secure the grab handle bezel to the upper glove box opening reinforcement (Fig. 12).

(5) Remove the grab handle bezel from the instrument panel.

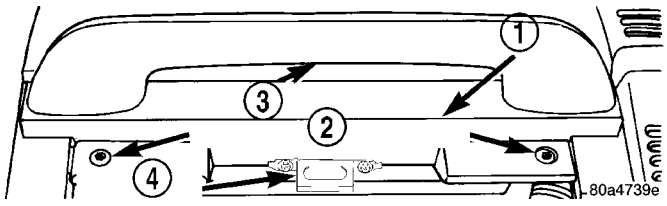

Fig. 12 GRAB HANDLE BEZEL REMOVE/INSTALL

1 - GRAB HANDLE BEZEL
2 - SCREW (2)
3 - GRAB HANDLE
4 - LATCH STRIKER

INSTALLATION

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(1) Position the grab handle bezel onto the instrument panel (Fig. 12).

(2) Install and tighten the two screws that secure the grab handle bezel to the upper glove box opening reinforcement. Tighten the screws to 2 N·m (20 in. lbs.).

(3) Reinstall the glove box latch striker onto the glove box opening upper reinforcement (Refer to 23 - BODY/INSTRUMENT PANEL/GLOVE BOX LATCH STRIKER - INSTALLATION).

(4) Reinstall the grab handle onto the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/GRAB HANDLE - INSTALLATION).

(5) Reconnect the battery negative cable.

INSTRUMENT PANEL ASSEMBLY

REMOVAL

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NOTE: Before starting this procedure, be certain to turn the steering wheel until the front wheels are in the straight-ahead position.

(1) Disconnect and isolate the battery negative cable.

(2) Remove the steering column opening cover from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/STEERING COLUMN OPENING COVER - REMOVAL).

(3) Remove the steering column from the vehicle, but do not remove the driver airbag, the steering wheel, or the switches from the steering column. Be certain that the steering wheel is locked and secured from rotation to prevent the loss of clockspring centering. (Refer to 19 - STEERING/COLUMN - REMOVAL).

(4) From beneath the driver side end of the instrument panel, perform the following:

(a) Disconnect the instrument panel wire harness connectors from the 100-way cross body wire harness connector near the left cowl side inner panel.

(b) Disconnect the driver side window demister hose at the heater-A/C housing demister/defroster duct.

(5) Remove the glove box from the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/GLOVE BOX - REMOVAL).

(6) Reach through the instrument panel glove box opening to perform the following:

(a) Disconnect the heater-A/C control vacuum harness connector from the heater-A/C housing vacuum harness connector.

(b) Disconnect the instrument panel wire harness connector from the heater-A/C housing wire harness connector.

(c) Disconnect the cross body wire harness connector from the passenger airbag pigtail wire connector.

(d) Disconnect the passenger side window demister hose at the heater-A/C housing demister/defroster duct.

(e) Disconnect the two halves of the radio antenna coaxial cable connector.

(f) Remove the two nuts that secure the passenger airbag lower bracket to the studs on the dash panel (Fig. 13).

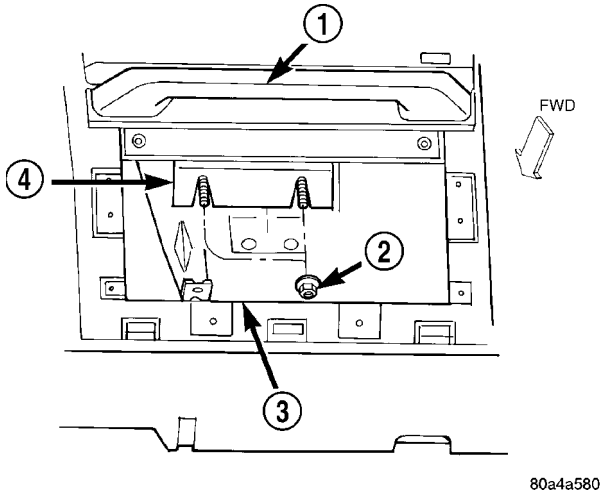

Fig. 13 PASSENGER AIRBAG LOWER BRACKET NUTS REMOVE/INSTALL

1 - GRAB HANDLE
2 - NUT (2)
3 - GLOVE BOX OPENING
4 - PASSENGER AIRBAG LOWER BRACKET

(7) Remove the top cover from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/TOP COVER - REMOVAL).

(8) Remove the three screws that secure each end of the instrument panel structural support to the cowl side inner panels at the front of each door opening (Fig. 14).

(9) Remove the four nuts that secure the top of the instrument panel structural support to the studs on the top of the dash panel.

(10) With the aid of an assistant, lift the instrument panel off of the dash panel studs and remove it from the vehicle.

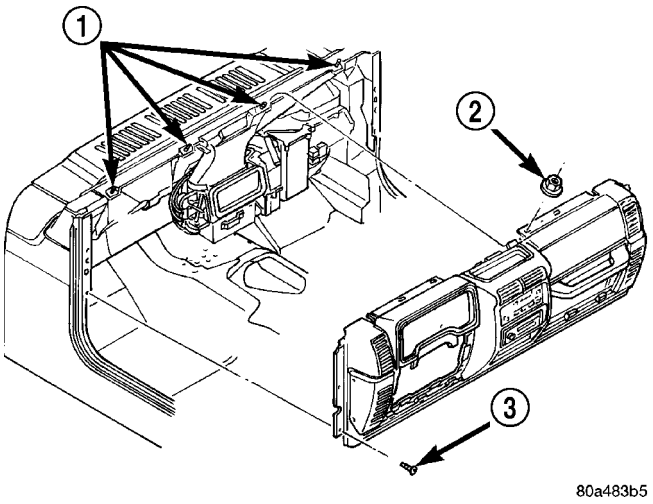

Fig. 14 INSTRUMENT PANEL REMOVE/INSTALL

1 - STUD (4)
2 - NUT (4)
3 - SCREW (6)

INSTALLATION

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) With the aid of an assistant, position the instrument panel into the vehicle and onto the dash panel studs (Fig. 14).

(2) Install and tighten the four nuts that secure the top of the instrument panel structural support to the studs on the top of the dash panel. Tighten the nuts to 12 N·m (105 in. lbs.).

(3) Install and tighten the three screws that secure each end of the instrument panel structural support to the cowl side inner panels at the front of each door opening. Tighten the screws to 12 N·m (105 in. lbs.).

(4) Reinstall the top cover onto the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/TOP COVER - INSTALLATION).

(5) Reach through the instrument panel glove box opening to perform the following:

(a) Install and tighten the two nuts that secure the lower passenger airbag bracket to the studs on the dash panel (Fig. 13). Tighten the nuts to 28 N·m (21 ft. lbs.).

(b) Reconnect the two halves of the radio antenna coaxial cable connector.

(c) Reconnect the passenger side window demister hose to the heater-A/C housing demister/defroster duct.

(d) Reconnect the cross body wire harness connector to the passenger airbag pigtail wire connector.

(e) Reconnect the instrument panel wire harness connector to the heater-A/C housing wire harness connector.

(f) Reconnect the heater-A/C control vacuum harness connector to the heater-A/C housing vacuum harness connector.

(6) Reinstall the glove box onto the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/GLOVE BOX - INSTALLATION).

(7) From beneath the driver side of the instrument panel, perform the following:

(a) Reconnect the driver side window demister hose to the heater-A/C housing demister/defroster duct.

(b) Reconnect the instrument panel wire harness connectors to the 100-way cross body wire harness connector near the left cowl side inner panel.

(8) Reinstall the steering column into the vehicle. Be certain that the steering wheel is locked and secured from rotation to prevent the loss of clockspring centering. (Refer to 19 - STEERING/COLUMN - INSTALLATION).

(9) Reinstall the steering column opening cover onto the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/STEERING COLUMN OPENING COVER - INSTALLATION).

(10) Reconnect the battery negative cable.

STEERING COLUMN OPENING COVER

REMOVAL

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(1) Disconnect and isolate the battery negative cable.

(2) If the vehicle is so equipped, move the tilt steering column to the fully raised position.

(3) Remove the two screws that secure the steering column opening cover to the instrument panel structural support (Fig. 15).

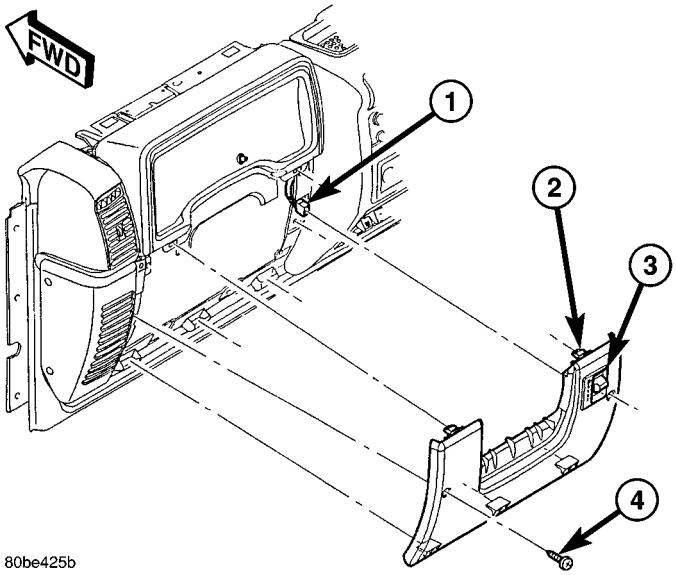

Fig. 15 STEERING COLUMN OPENING COVER REMOVE/INSTALL

1 - WIRE HARNESS CONNECTOR
2 - STEERING COLUMN OPENING COVER
3 - HEADLAMP LEVELING SWITCH
4 - SCREW (2)

(4) Using a trim stick or another suitable wide flat-bladed tool, gently pry the top of the steering column opening cover away from the instrument panel on each side of the steering column far enough to disengage the two snap clip retainers from their receptacles in the instrument panel base trim.

(5) If the vehicle is so equipped, roll the top of the steering column opening cover downward far enough to access and disconnect the wire harness connector for the headlamp leveling switch from the switch connector receptacle.

(6) Pull the top of the steering column opening cover rearward far enough to disengage the hinge hook formations on the lower edge of the cover from the hinge pins on the lower edge of the instrument panel.

(7) Remove the steering column opening cover from the instrument panel.

INSTALLATION

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(1) Position the steering column opening cover to the instrument panel.

(2) Engage the hinge hook formations on the lower edge of the steering column opening cover with the hinge pins on the lower edge of the instrument panel (Fig. 15).

(3) If the vehicle is so equipped, roll the top of the steering column opening cover upward far enough to reconnect the wire harness connector for the headlamp leveling switch to the switch connector receptacle.

(4) Tilt the upper edge of the steering column opening cover up and align the two snap clip retainers on the cover with their receptacles in the instrument panel base trim.

(5) Using hand pressure, press firmly on the steering column opening cover over each of the snap clip locations until each of the snap clips is fully engaged in its receptacle in the instrument panel base trim.

(6) Install and tighten the two screws that secure the steering column opening cover to the instrument panel structural support. Tighten the screws to 2 N·m (20 in. lbs.).

(7) Reconnect the battery negative cable.

TOP COVER

REMOVAL

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Disconnect and isolate the battery negative cable.

(2) Using a trim stick or another suitable wide flat-bladed tool, gently pry the instrument panel top cover up and away from the instrument panel far enough to disengage the five snap clip retainers from their receptacles in the instrument panel structural support (Fig. 16).

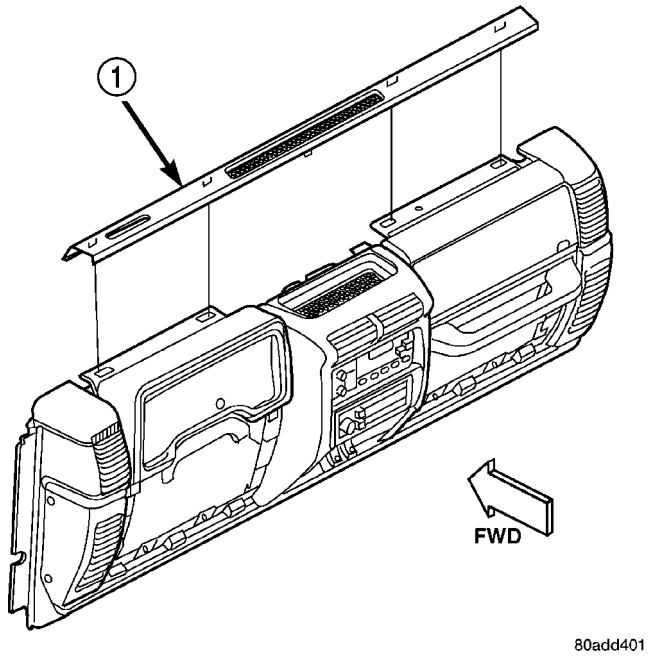

Fig. 16 TOP COVER REMOVE/INSTALL

1 - TOP COVER

(3) Remove the top cover from the instrument panel.

INSTALLATION

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(1) Position the top cover onto the instrument panel (Fig. 16).

(2) Align the snap clips on the top cover with the snap clip receptacles in the instrument panel structural support.

(3) Using hand pressure, press firmly downward on the top cover over each of the snap clip locations until each of the snap clips is fully seated in its receptacle in the instrument panel structural support.

(4) Reconnect the battery negative cable.

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ADD-A-TRUNK

REMOVAL

(1) Release latches under trunk panel and lift panel up.

(2) Remove bolts attaching trunk to inner body panel (Fig. 1).

(3) Separate trunk from vehicle.

INSTALLATION

(1) Position the trunk in the cargo space.

(2) Install the bolts.

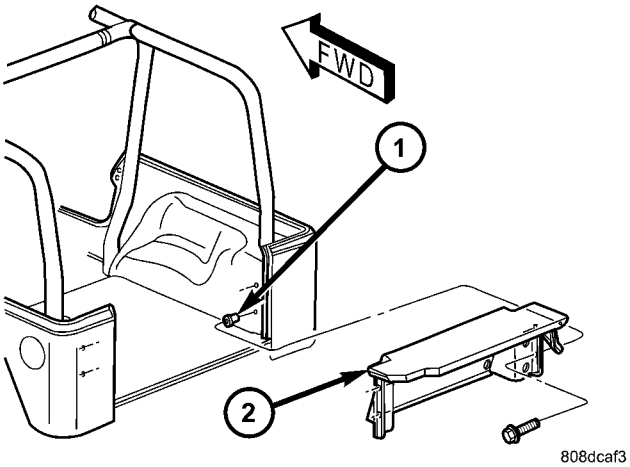

Fig. 1 ADD-A-TRUNK

1 - SCREW
2 - ADD-A-TRUNK PANEL

FRONT CARPET

REMOVAL

(1) If necessary, remove the center console. (Refer to 23 - BODY/INTERIOR/CENTER CONSOLE - REMOVAL)

(2) Remove the retainers attaching the carpet to the dash panel (Fig. 2).

(3) Disengage the snaps under front seats.

(4) Remove carpet from the vehicle

INSTALLATION

(1) Position the carpet in the vehicle

(2) Engage the snaps around under front seats.

(3) Install the retainers attaching the carpet to the dash panel.

(4) If previously removed, install the center console. (Refer to 23 - BODY/INTERIOR/CENTER CONSOLE - INSTALLATION)

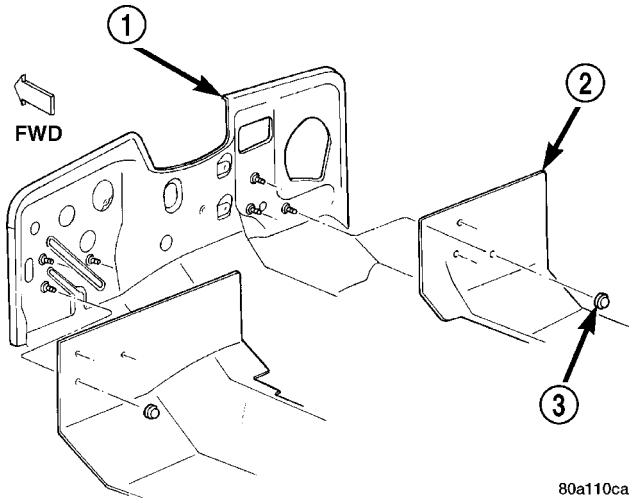

Fig. 2 FRONT CARPET

1 - DASH PANEL
2 - CARPET
3 - PUSH ON RETAINER

CENTER CARPET

REMOVAL

(1) Disengage the snaps under front seats.
(2) Remove the carpet.

INSTALLATION

(1) Position the carpet in the vehicle.
(2) Engage the snaps under front seats.

CARGO AREA CARPET

REMOVAL

(1) Position the rear seat in the fold and tumbled position.
(2) Pull the carpet from under the rear seat.
(3) Remove the Add-A-Trunk, if equipped. (Refer to 23 - BODY/INTERIOR/ADD-A-TRUNK - REMOVAL)
(4) Route the rear seat belt buckles through the cargo area carpet.
(5) Separate the carpet from the vehicle (Fig. 3).

INSTALLATION

(1) Position the carpet in the vehicle.
(2) Route the rear seat belt buckles through the cargo area carpet.
(3) Install the Add-A-Trunk, if equipped. (Refer to 23 - BODY/INTERIOR/ADD-A-TRUNK - INSTALLATION)
(4) Return the rear seat to the full rearward position.

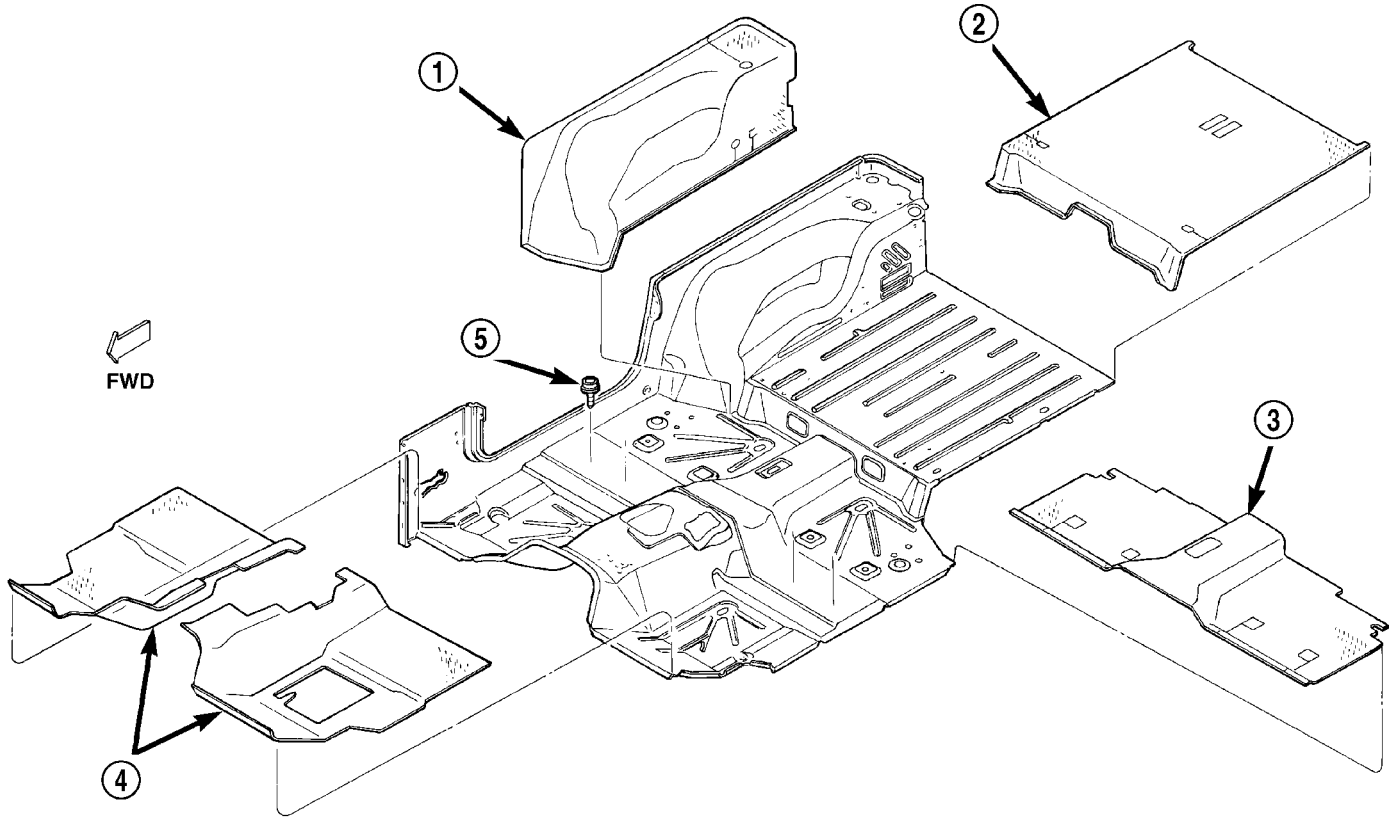

Fig. 3 VEHICLE CARPET

1 - WHEEL HOUSE CARPET
2 - CARGO AREA CARPET
3 - CENTER FLOOR CARPET
4 - FRONT FLOOR CARPET
5 - SNAP

WHEELHOUSE CARPET

REMOVAL

(1) Position the rear seat in the full forward position.
(2) Remove the Add-A-trunk, if equipped. (Refer to 23 - BODY/INTERIOR/ADD-A-TRUNK - REMOVAL)
(3) Grasp wheelhouse carpet and remove from vehicle (Fig. 3).

INSTALLATION

(1) Position wheelhouse carpet in vehicle and adjust as necessary.
(2) Install the Add-A-trunk, if equipped. (Refer to 23 - BODY/INTERIOR/ADD-A-TRUNK - INSTALLATION)
(3) Return the rear seat to the full rearward position.

REAR VIEW MIRROR

REMOVAL

(1) Disconnect the electrical connector, if equipped. (Fig. 4)
(2) Loosen the mirror set screw.
(3) Slide the mirror up and off the support button (bracket).

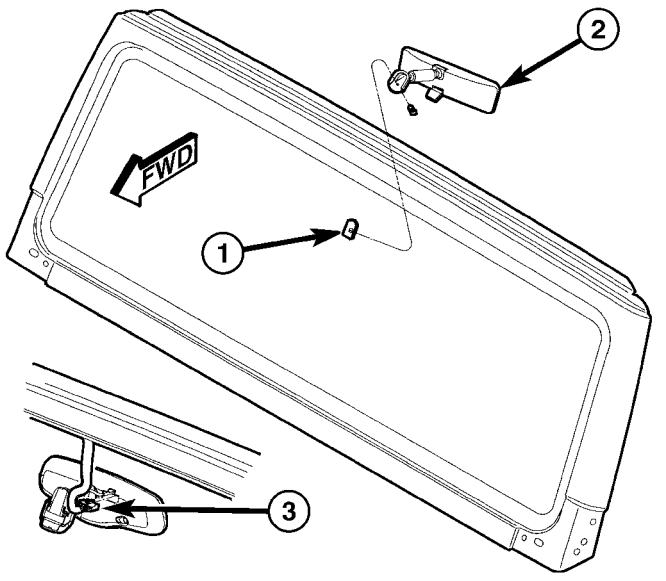

Fig. 4 REAR VIEW MIRROR

1 - SUPPORT BRACKET
2 - MIRROR ASSEMBLY
3 - ELECTRICAL CONNECTOR

INSTALLATION

(1) Slide the mirror onto the support button (bracket).

CAUTION: Do not over-tighten the setscrew because glass chipping and/or breakage could result.

(2) Tighten the mirror set screw to 1 N·m (9 in. lbs.).
(3) Connect the electrical connector, if equipped.

REAR VIEW MIRROR SUPPORT BRACKET

INSTALLATION

(1) Mark the position for the mirror bracket on the outside of the windshield glass with a wax pencil.
(2) Clean the bracket contact area on the glass. Use a mild powdered cleanser on a cloth saturated with isopropyl (rubbing) alcohol. Finally, clean the glass with a paper towel dampened with alcohol.
(3) Sand the surface on the support bracket with fine grit-sandpaper. Wipe the bracket surface clean with a paper towel.
(4) Apply accelerator to the surface on the bracket according to the following instructions:

- Crush the vial to saturate the felt applicator.
- Remove the paper sleeve.
- Apply accelerator to the contact surface on the bracket.
- Allow the accelerator to dry for five minutes.
- Do not touch the bracket contact surface after the accelerator has been applied.

(5) Apply adhesive accelerator to the bracket contact surface on the windshield glass. Allow the accelerator to dry for one minute. Do not touch the glass contact surface after the accelerator has been applied.
(6) Install the bracket according to the following instructions:

- Apply one drop of adhesive at the center of the bracket contact-surface on the windshield glass.
- Apply an even coat of adhesive to the contact surface on the bracket.
- Align the bracket with the marked position on the windshield glass.
- Press and hold the bracket in place for at least one minute.

NOTE: Verify that the mirror support bracket is correctly aligned, because the adhesive will cure rapidly.

REAR VIEW MIRROR SUPPORT BRACKET (Continued)

(7) Allow the adhesive to cure for 8-10 minutes. Remove any excess adhesive with an alcohol-dampened cloth.

(8) Allow the adhesive to cure for an additional 8-10 minutes before installing the mirror.

SUNVISOR

REMOVAL

(1) Remove the screws that attach the sunvisor arm support brackets to the windshield frame (Fig. 5).

(2) Remove the sunvisor from the windshield frame.

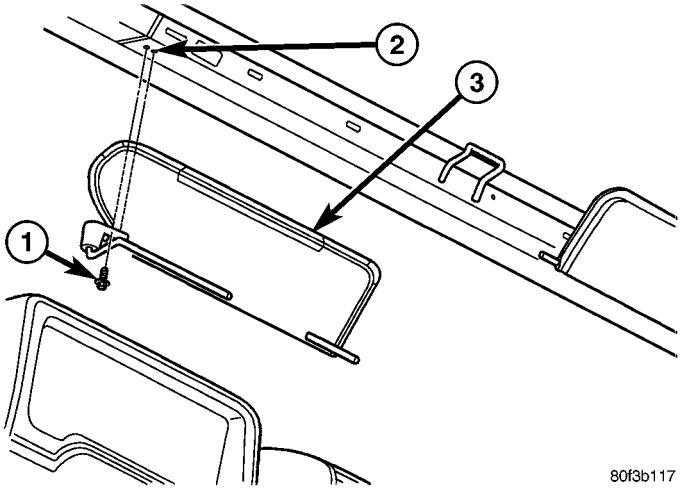

Fig. 5 SUNVISOR

1 - SCREWS (2)
2 - WINDSHIELD FRAME
3 - VISOR

INSTALLATION

(1) Position the sunvisor on the windshield frame and align the arm support bracket holes with the frame.

(2) Install the screws that attach the sunvisor arm support brackets to the frame. Tighten the screws securely.

SHIFT BEZEL - AUTOMATIC TRANSMISSION

REMOVAL

(1) Pull shift lever handle off of shift lever (Fig. 6).

(2) Using a trim stick C-4755 or equivalent, remove the shift bezel.

(3) Twist and remove the bezel light.

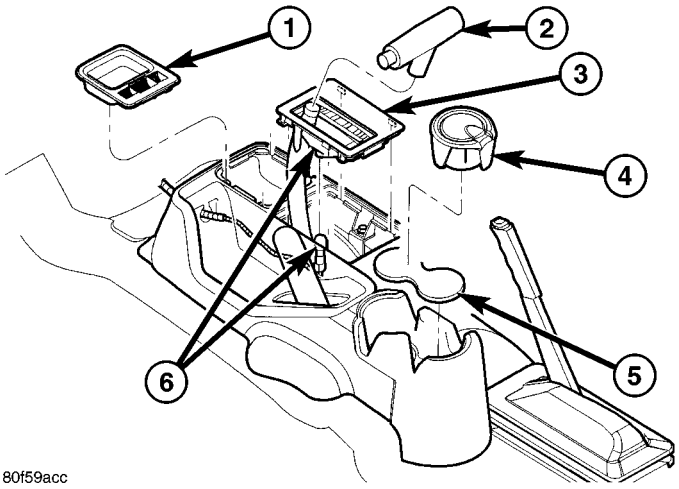

Fig. 6 SHIFT BEZEL

1 - COIN TRAY
2 - TRANSMISSION SHIFT LEVER HANDLE
3 - SHIFT BEZEL
4 - ASH RECEIVER (IF EQUIPPED)
5 - CUP HOLDER MAT
6 - SHIFT BEZEL LIGHT/SOCKET

INSTALLATION

(1) Connect the shift bezel light by twisting into socket on the bezel.

(2) Position the shift bezel over the shift lever and install onto the console.

(3) Align the shifter handle keyways and push handle onto the shifter until fully seated.

SHIFT BOOT

REMOVAL

(1) Using a trim stick C-4755 or equivalent, pry the shift boot from the bezel.

(2) Using a small flat blade, pry the shift pattern insert from the shift knob.

(3) Remove the nut attaching the shift knob to the shift lever (Fig. 7).

(4) Remove the knob and slide the shift boot from the shift lever.

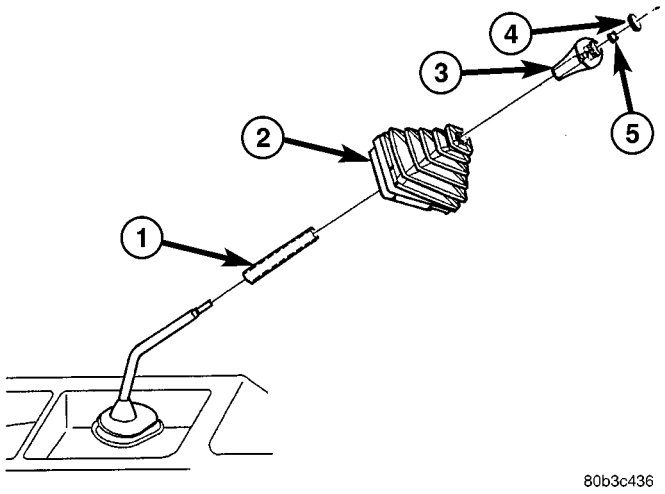

Fig. 7 SHIFT BOOT

1 - BOOT SUPPORT TUBE
2 - SHIFT BOOT
3 - SHIFT KNOB
4 - SHIFT PATTERN CAP
5 - NUT

INSTALLATION

(1) Slide the shift boot over the shift lever.

(2) Snap the shift boot into place in the center console.

(3) Position the shift knob on the lever, install the nut and tighten to 34 N·m (25 ft. lbs.).

(4) Position the shift pattern insert on the knob and press into place.

CENTER CONSOLE

REMOVAL

(1) Remove the cup holder, if equipped. (Refer to 23 - BODY/INTERIOR/CENTER CONSOLE CUPHOLDER ASSEMBLY - REMOVAL)

(2) On full console models remove the trim disc from the bottom of the cup holder (Fig. 10).

(3) Remove the automatic transmission shift bezel, if equipped. (Refer to 23 - BODY/INTERIOR/SHIFT BEZEL - REMOVAL)

(4) Using a trim stick C-4755 or equivalent, remove the air bag deactivation switch and disconnect the electrical connector, if equipped.

(5) Remove the bolt(s) attaching the console to the floor pan (Fig. 8) and (Fig. 9).

(6) Shift transfer case to four low position.

(7) Shift transmission to L (2nd gear for manual transmission) and remove the console assembly.

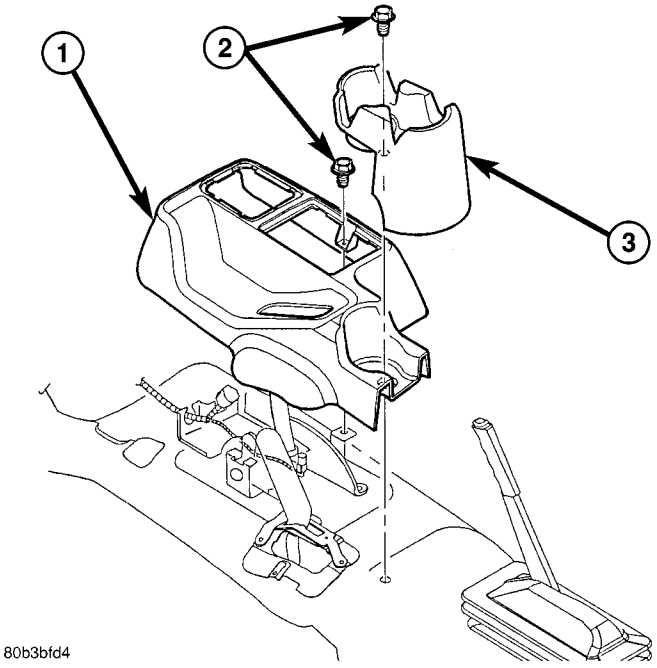

Fig. 8 MINI CONSOLE

1 - MINI CONSOLE
2 - BOLTS
3 - CUPHOLDER

CENTER CONSOLE (Continued)

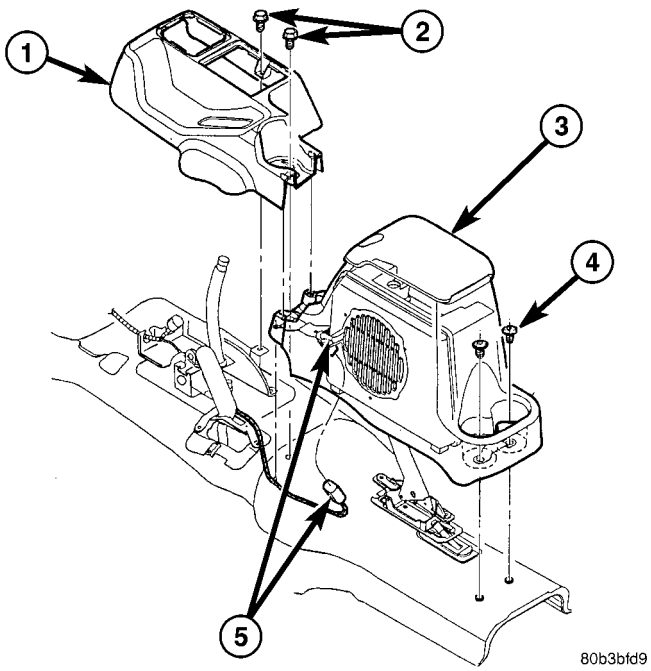

Fig. 9 FULL CONSOLE

1 - FRONT CONSOLE
2 - BOLTS
3 - REAR TOWER
4 - BOLTS (2)
5 - ELECTRICAL CONNECTOR

INSTALLATION

(1) Connect the electrical connector and install console tower assembly over the park brake lever, if equipped.

(2) Install the tower bolts.

(3) Install the mini console assembly over the transmission and transfer case shifters.

(4) Install the console bolts.

(5) Connect the air bag deactivation switch electrical connector and install the switch, if equipped.

(6) Connect the air bag deactivation switch electrical connector and install the switch, if equipped.

(7) Install the auto transmission shift bezel, if equipped. (Refer to 23 - BODY/INTERIOR/SHIFT BEZEL - INSTALLATION)

(8) Install the manual transmission shift boot, if equipped. (Refer to 23 - BODY/INTERIOR/SHIFT BOOT - INSTALLATION)

(9) Install the cup holder and trim disc, if equipped. (Refer to 23 - BODY/INTERIOR/CENTER CONSOLE CUPHOLDER ASSEMBLY - INSTALLATION)

CENTER CONSOLE CUP HOLDER

REMOVAL

(1) Using a trim stick C-4755 or equivalent, remove the cup holder insert. (Fig. 10) and (Fig. 8)

(2) On mini console equipped models remove the bolt and remove the cup holder. (Fig. 8)

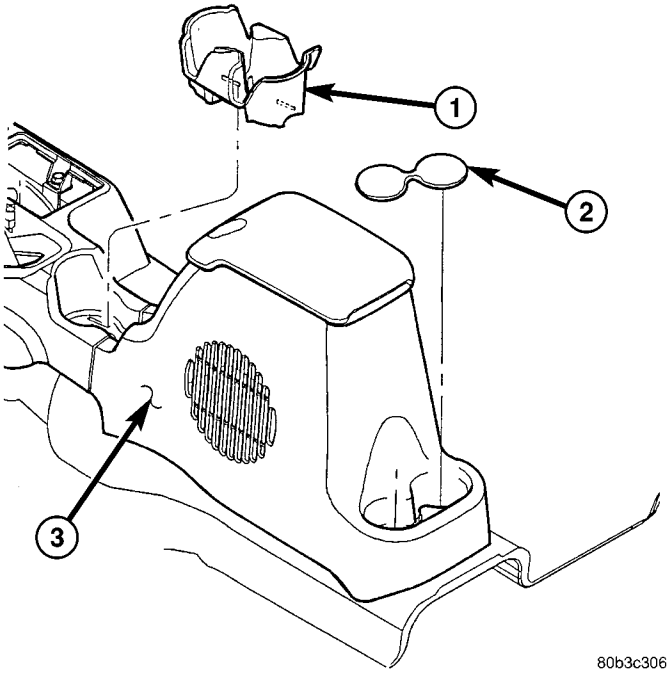

Fig. 10 FULL CONSOLE CUP HOLDER

1 - FULL CONSOLE CUP HOLDER
2 - TRIM PIECE
3 - CONSOLE TOWER

INSTALLATION

(1) On mini console equipped models install the cup holder and install the bolt.

(2) Tighten the bolt to 4 N·m (35 in lbs.).

(3) Install the cup holder insert.

CONSOLE LOCK CYLINDER

REMOVAL

(1) Open the console cover.
(2) Remove the screw that attaches the retainer to the lock and then remove the retainer from the lock (Fig. 11).
(3) Remove the lock cylinder from the console cover.

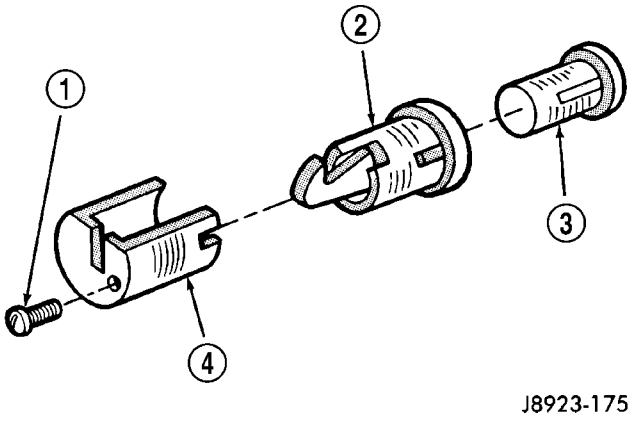

Fig. 11 CONSOLE LOCK CYLINDER

1 - ATTACHING SCREW
2 - LOCK
3 - LOCK CYLINDER
4 - RETAINER

INSTALLATION

(1) Insert the assembled lock in the console cover hole and position the retainer on the lock and install the screw.

SPORT BAR

REMOVAL

(1) Remove hard top and/or soft top. (Refer to 23 - BODY/REMOVEABLE TOP/HARD TOP - REMOVAL) or (Refer to 23 - BODY/REMOVEABLE TOP/SOFT TOP - REMOVAL).
(2) Remove the door opening frames. (Refer to 23 - BODY/INTERIOR/DOOR OPENING FRAME - REMOVAL)
(3) Remove the sunvisors. (Refer to 23 - BODY/INTERIOR/SUN VISOR - REMOVAL)
(4) Remove the a-pillar weatherstrips.
(5) Remove the speaker pods. (Refer to 23 - BODY/INTERIOR/SPORT BAR - SPEAKER POD - REMOVAL)
(6) Disengage center support bar cover zipper.
(7) Remove the bolts attaching the sport bars to the center support bar (Fig. 12).
(8) Remove the bolts attaching the side support bars to the windshield frame.
(9) Remove the bolts attaching the side support bars to the sport bars.
(10) Separate the side support bars from the vehicle.
(11) Pull back the center section of the carpet and remove the bolts attaching the sport bar to the cargo floor panel.
(12) Remove the bolt attaching the sport bar to the brackets behind the door strikers.
(13) Lower the rear seat and lift rear seat to the full forward position.
(14) Pull back wheelhouse carpet and remove bolts attaching the sport bars to the wheelhouse.
(15) Remove the bolts attaching the seatbelt anchors to the wheelhouse.
(16) Carefully lift the sport bar upward and remove it from the vehicle.
(17) If necessary, remove the pads and covers from the sport bar.

INSTALLATION

(1) If necessary, transfer all attached components.
(2) Clean the base plate contact surface areas on the floor and wheelhouse panels.
(3) Apply epoxy chromate primer to the attaching hole edges for protection against corrosion.
(4) Position the sport bar base plates on the floor and wheelhouse panels with the holes aligned.

NOTE: To prevent water seepage, apply 3M Drip-Chek Sealant (or an equivalent product) to the underside of the sport bar base flanges and all the bolt heads before installation.

(5) Position the center support bar into the sport bars correctly with the large horizontal slots facing rearward and hole/slot facing downward.
(6) Install the bolts attaching the seatbelt anchors to the wheelhouse.
(7) Install the bolts attaching the sport bars to the wheelhouse and install the wheelhouse carpet. Tighten the bolts to 40 N·m (30 ft. lbs.).
(8) Return seat back to upright position.
(9) Install the bolts attaching the sport bar to the cargo floor panel and install the center carpet. Tighten the bolts to 40 N·m (30 ft. lbs.).
(10) Install the bolts attaching the sport bar to the brackets behind the door strikers and tighten to 68 N·m (50 ft. lbs.).
(11) Position side supports at the windshield and install the bolts attaching the side support bars to the windshield frame. Tighten the bolts to 32 N·m (24 ft. lbs.).
(12) Install the bolts attaching the side support bars to the sport bars and tighten to 20 N·m (15 ft. lbs.).

SPORT BAR (Continued)

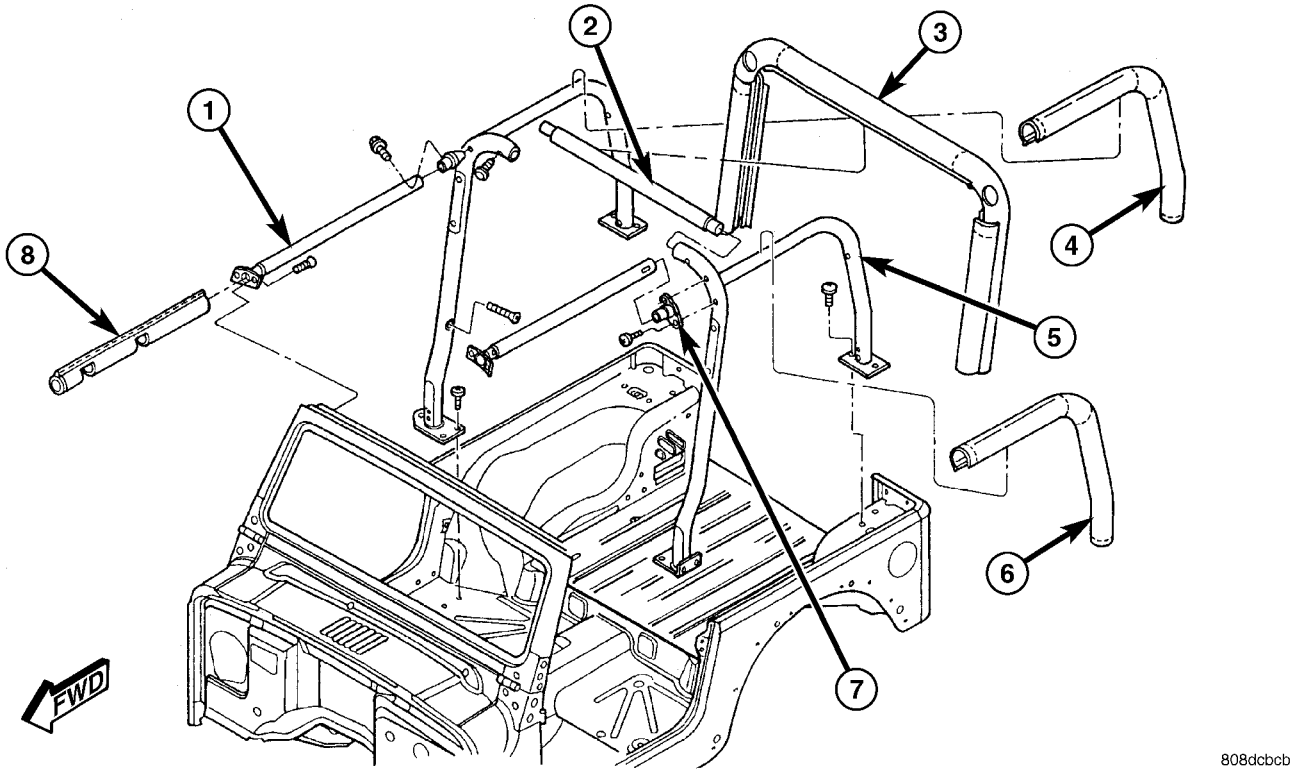

Fig. 12 SPORT BAR

1 - SIDE SUPPORT BAR
2 - CENTER SUPPORT BAR
3 - COVER
4 - COVER
5 - SPORT BAR
6 - COVER
7 - BRACKET
8 - COVER

(13) Install the bolts attaching the center support bar to the sport bar and tighten to 68 N·m (50 ft. lbs.).

(14) Engage center support bar cover zipper.

(15) Install the speaker pods. (Refer to 23 - BODY/INTERIOR/SPORT BAR - SPEAKER POD - INSTALLATION)

(16) Install the a-pillar weatherstrips.

(17) Install the sun visors (Refer to 23 - BODY/INTERIOR/SUN VISOR - INSTALLATION)

(18) Install the door opening frames. (Refer to 23 - BODY/INTERIOR/DOOR OPENING FRAME - INSTALLATION)

(19) Install hard top and/or soft top. (Refer to 23 - BODY/REMOVEABLE TOP/HARD TOP - INSTALLATION) or (Refer to 23 - BODY/REMOVEABLE TOP/SOFT TOP - INSTALLATION).

A-PILLAR TRIM

REMOVAL

(1) Remove the a-pillar trim screws. (Fig. 13)

(2) Starting at the bottom, pull off the trim and remove.

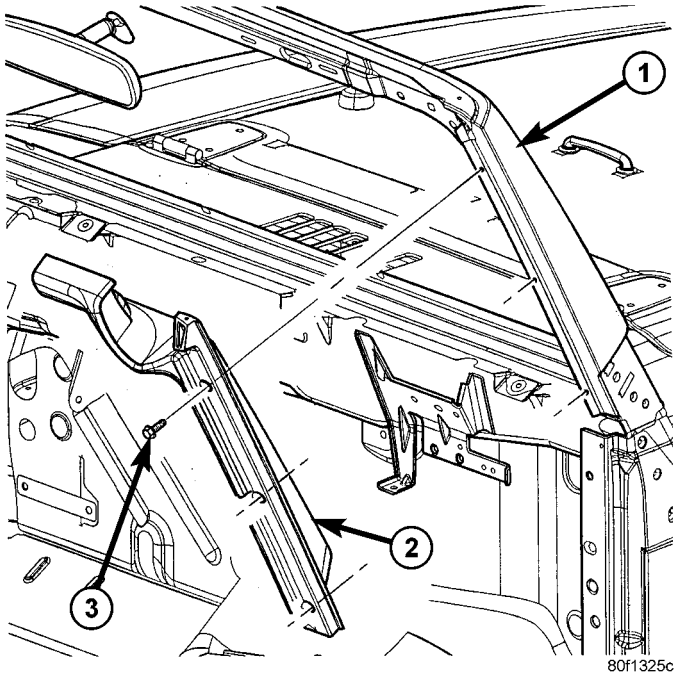

Fig. 13 A-PILLAR TRIM

1 - WINDSHIELD FRAME
2 - A-PILLAR TRIM
3 - SCREWS (3)

INSTALLATION

(1) Install the trim over the sports bar and position into the top.

(2) Press the bottom into place and install the screws.

REAR HEADER TRIM

REMOVAL

(1) Disconnect the rear window defogger electrical connectors, if equipped.

(2) Remove the wiper motor cover and disconnect the wiper motor electrical connector and washer hose. (Fig. 14)

(3) Using a trim stick C-4755 or equivalent, separate the header trim and retainer clips from the top. (Fig. 15)

(4) Pull the wire harnesses and washer hose through the trim panel and remove.

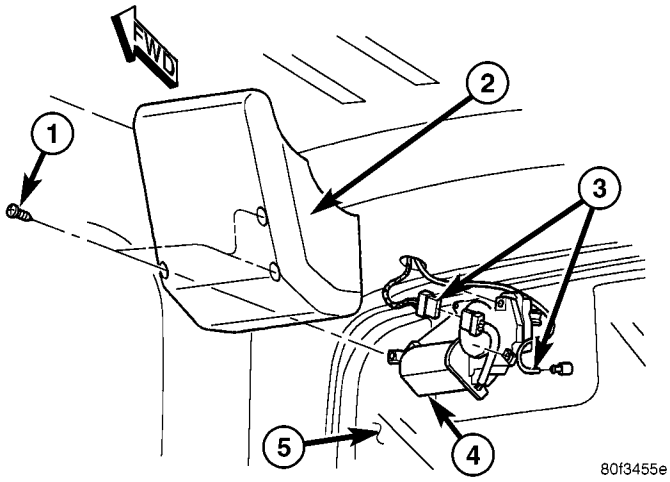

Fig. 14 REAR WIPER MOTOR TRIM COVER REMOVE/INSTALL

1 - SCREW (3)
2 - TRIM COVER
3 - ELECTRICAL & WASHER HOSE CONNECTIONS
4 - WIPER MOTOR
5 - LIFTGATE GLASS

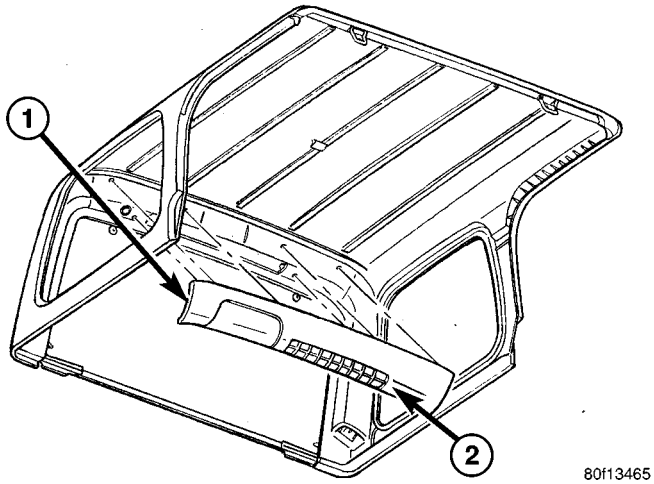

Fig. 15 REAR HEADER TRIM

1 - REAR HEADER TRIM
2 - WIRE OPENINGS (2)

INSTALLATION

(1) Route the wire harnesses and washer hose through the trim panel.

(2) Install the header trim and seat the retaining clips fully.

(3) Connect the wiper motor electrical and the washer hose connection.

(4) Connect the rear window defogger electrical connectors, if equipped.

WINDSHIELD HEADER

REMOVAL

(1) Position both sun visors out.

(2) Using a trim stick C-4755 or equivalent, separate the retaining clips and remove the trim from behind the a-pillar trim panels. (Fig. 16)

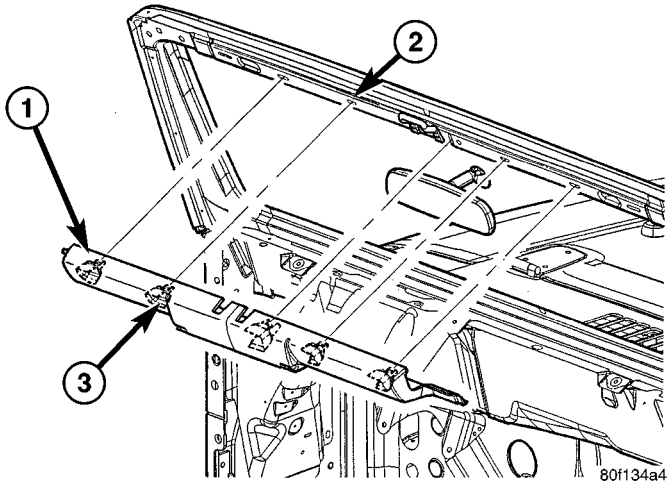

Fig. 16 WINDSHIELD HEADER MOLDING

1 - WINDSHIELD HEADER TRIM
2 - WINDSHIELD FRAME
3 - SPRING CLIPS (4)

INSTALLATION

(1) Install the windshield header trim and position the end tabs behind the a-pillar trim.

(2) Seat the retaining clips fully.

SPORT BAR - SPEAKER POD

REMOVAL

(1) On vehicles equipped with a soft top, unzip the side panel next to the pod.

(2) On vehicles equipped with a hard top, unclip the windshield clamps.

(3) Remove the bolts on side of the top next to the pod and loosen the opposite side bolts. (Refer to 23 - BODY/REMOVEABLE TOP/HARD TOP - REMOVAL)

(4) Lift up the side of the top to gain access to the pod bolts and support with a block of wood or similar.

(5) Remove the two bolts securing the pod to the sport bar.

(6) Separate the pod guide pin from the sport bar and disconnect the electrical connector.

INSTALLATION

(1) Connect the pod electrical connector and install the pod onto the guide pin.

(2) Install the bolts and tighten to 68 N·m (50 ft. lbs.).

(3) On vehicles equipped with a hard top, install the hard top. (Refer to 23 - BODY/REMOVABLE TOP/HARD TOP - INSTALLATION)

(4) On vehicles equipped with a soft top, zip up the side panel.

PAINT

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PAINT

SPECIFICATIONS - PAINT CODES

NOTE: Because of late model changes to the available paint colors (Refer to VEHICLE DATA/VEHICLE INFORMATION/VEHICLE CERTIFICATION LABEL - DESCRIPTION) or (Refer to VEHICLE DATA/VEHICLE INFORMATION/BODY CODE PLATE - DESCRIPTION) for the correct paint codes for each vehicle. (Refer to 23 - BODY/PAINT/PAINT CODE - DESCRIPTION)

EXTERIOR COLORS

EXTERIOR COLOR	DAIMLERCHRYSLER CODE
Flame Red Clearcoat	PR4
Impact Orange Clearcoat	BV5
Light Khaki Pearlcoat	AJC
Electric Lime Green Pearlcoat	BGL
Deep Beryl Green Pearlcoat	CGV
Moss Green Pearlcoat	RJN
Patriot Blue Pearlcoat	WBT
Bright Silver Metallic Clearcoat	WSB
Black Clearcoat	DX8
Stone White	SW1

INTERIOR COLORS

INTERIOR COLOR	DAIMLERCHRYSLER CODE
Dark Slate Gray	DV
Khaki	J3

EXTERIOR HARD AND SOFT TOP COLORS

EXTERIOR HARD TOP COLORS	DAIMLERCHRYSLER CODE
Medium Gray	BDL
Dark Khaki	ZJ8

PAINT CODE

DESCRIPTION

Exterior vehicle body colors are identified on the Vehicle Certification Label (Refer to VEHICLE DATA/VEHICLE INFORMATION/VEHICLE CERTIFICATION LABEL - DESCRIPTION) or the Body Code Plate (Refer to VEHICLE DATA/VEHICLE INFORMATION/BODY CODE PLATE - DESCRIPTION). The first digit of the paint code listed on the vehicle indicates the sequence of application, i.e.: P = primary coat, Q = secondary coat. The color names provided in the Paint and Trim Code Description chart are the color names used on most repair product containers. (Refer to 23 - BODY/PAINT - SPECIFICATIONS)

BASECOAT/CLEARCOAT FINISH

DESCRIPTION

On most vehicles a two-stage paint application (basecoat/clearcoat) is used. Color that is applied to primer is called basecoat. The clearcoat protects the basecoat from ultraviolet light and provides a durable high-gloss finish.

PAINT TOUCH-UP

STANDARD PROCEDURE - PAINT TOUCH-UP

When a painted metal surface has been scratched or chipped, it should be touched-up as soon as possible to avoid corrosion. For best results, use Mopar® Scratch Filler/Primer, Touch-Up Paints and Clear Topcoat. Refer to Introduction group of this manual for Body Code Plate information.

WARNING: USE AN OSHA APPROVED RESPIRATOR AND SAFETY GLASSES WHEN SPRAYING PAINT OR SOLVENTS IN A CONFINED AREA. PERSONAL INJURY CAN RESULT.

(1) Scrape loose paint and corrosion from inside scratch or chip.

(2) Clean affected area with Mopar® Tar/Road Oil Remover, and allow to dry.

(3) Fill the inside of the scratch or chip with a coat of filler/primer. Do not overlap primer onto good surface finish. The applicator brush should be wet enough to puddle-fill the scratch or chip without running. Do not stroke brush applicator on body surface. Allow the filler/primer to dry hard.

(4) Cover the filler/primer with color touch-up paint. Do not overlap touch-up color onto the original color coat around the scratch or chip. Butt the new color to the original color, if possible. Do not stroke applicator brush on body surface. Allow touch-up paint to dry hard.

(5) On vehicles without clear coat, the touch-up color can be lightly finesse sanded (1500 grit) and polished with rubbing compound.

(6) On vehicles with clearcoat, apply clear topcoat to touch-up paint with the same technique as described in Step 4. Allow clear topcoat to dry hard. If desired, Step 5 can be performed on clear topcoat.

WARNING: AVOID PROLONGED SKIN CONTACT WITH PETROLEUM OR ALCOHOL – BASED CLEANING SOLVENTS. PERSONAL INJURY CAN RESULT. AVOID PROLONGED SKIN CONTACT WITH PETROLEUM OR ALCOHOL – BASED CLEANING SOLVENTS. PERSONAL INJURY CAN RESULT.

FINESSE SANDING, BUFFING & POLISHING

DESCRIPTION

CAUTION: Do not remove more than .5 mils of clearcoat finish, if equipped. Basecoat paint must retain clearcoat for durability.
Use a Paint Thickness Gauge #PR-ETG-2X or equivalent to determine film thickness before and after the repair.

Minor acid etching, orange peel, or smudging in clearcoat or single-stage finishes can be reduced with light finesse sanding, hand buffing, and polishing. **If the finish has been finesse sanded in the past, it cannot be repeated. Finesse sanding operation should be performed by a trained automotive paint technician.**

REMOVEABLE TOP

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REMOVEABLE TOP

STANDARD PROCEDURE

HARD TOP REPAIR

NOTE: The hard top is manufactured from a sheet molded compound (SMC).
SMC is constructed using short fiberglass strands usually less than 5.08 cm (2 inches) long. Sheet stock of glass impregnated resin matting is placed into the mold and pressed under heat to flow material throughout the mold. Tooling is shear edge designed to mold to net (i.e. no trimming at periphery required).
SMC is repairable in accordance with established procedures described within this manual. (Refer to 23 - BODY - STANDARD PROCEDURE - PLASTIC BODY PANEL REPAIR)

Hard Top Locators

There are two hard top locator devices positioned approximately 150 mm (6 in.) from the most forward attachment holes. If the locator device is missing or damaged it must be replaced, as it will assist in placing the hard top back into the correct position for top re-attachment.

If the locator device is missing just make sure that the attachment holes are clear and simply insert the new locator (part is orientated by hole size and inserts from the bottom of the rail before the hard top is re-attached to the vehicle).

If the locator device is damaged, cut off the locking tabs, remove and replace as described above.

HARD TOP REPAIR

If it has been determined that the hard top can be repaired. The following required materials and procedures are to be used.

(1) Use a grinder to remove the paint and outline the damaged area. Use a grade 150 grit disc for paint removal.

(2) Grind the outlined surface area again with a 240 grit disc to prevent coarse scratches from appearing in the final finish.

(3) If cracks extend from the hole, it will be necessary to stop-drill the crack(s) with a 3-mm (1/8-in) diameter drill bit. This will prevent the crack from spreading and allow repair material to be effective.

(4) Position a fiberglass mat or cloth on the repair surface area. Cut the mat 2.5 cm (1 in.) larger than repair area but smaller than the outline area. Make the outline area larger if required.

(5) Clean the outlined area.

(6) Place the fiberglass cloth on aluminum foil.

(7) Pour the fiberglass resin into a clean container.

(8) Mix the appropriate amount of hardener and resin. Follow the manufacturers instructions.

(9) Apply the hardener/resin mixture to both sides of the fiberglass cloth.

(10) Place the fiberglass cloth over the repair area. Next, place the aluminum foil over the cloth using a plastic spreader to smooth-out the cloth and resin. Use firm pressure to remove air bubbles and to smooth-out the cloth. Remove the aluminum foil.

(11) Allow the resin to cure.

(12) Smooth-out the surface area to the contour of the hard top with a 150-grit disc.

(13) Apply plastic filler to complete the repair. Finish smoothing the surface area with 240-320 grit paper.

REMOVEABLE TOP (Continued)

(14) Repeat the previous steps on the inside area of the hard top.
(15) Featheredge the repaired surface area.
(16) Prime the repaired surface area with PPG® Epoxy Primer, or an equivalent product.
(17) Apply surface primer to the surface area.
(18) Prime the surface area for the color coat.
(19) Apply color coat to the repaired surface area.

HARD TOP FRACTURE REPAIR

If it has been determined that the hard top can be repaired. The following required materials and procedures are to be used.
(1) Use a grinder to remove the paint and outline the damaged area. Use a grade 150 grit disc for paint removal.
(2) Grind the outlined surface area again with a 240 grit disc to prevent coarse scratches from appearing in the final finish.
(3) If cracks extend from the damaged area, it will be necessary to stop drill the cracks with a 3-mm (1/8-in) diameter drill bit. This will prevent the crack and/or fracture from spreading and allow repair material to be effective.
(4) Bevel the edges of the crack/fracture on both sides with a rotary file.

NOTE: The edges should be beveled on the inside and outside of the top to ensure sufficient surface area for good bonding.

(5) Complete the repairs with fiberglass cloth and resin as described in the hard top hole repair procedure. (Refer to 23 - BODY/REMOVABLE TOP - STANDARD PROCEDURE)

HARD TOP SAG REPAIR

Inspect the roof/hard top assembly for the presence of sag, or if the customer complains of water accumulation on the roof. Perform the following procedure to repair.
(1) Disengage latches at windshield frame (Fig. 1).

NOTE: The two forward bolts/nuts are not captured. Do not lose the nuts when removing.

(2) Remove the six bolts that attach the hard top to the body (Fig. 2).
(3) Depress tab on rear wiper motor connector and pull downward to disengage (Fig. 3).
(4) Make two marks on the inside of the hard top 6 cm (2.5 in.) on both sides of the center roof rib, directly above the sport bar.
(5) Move the hard top back about 10 cm (4 in.) to expose the inner rib section.
(6) Clean the roof area with isopropyl alcohol.
(7) Install the foam repair blocks by removing the protective backing tape from the adhesive side. The foam blocks must be centered above the sport bar pad.
(8) Move the hard top assembly back into position and centered at the windshield frame.
(9) Inspect the hard top seals for damage and replace, if necessary.
(10) Inspect the windshield header seal and make sure it is clean, undamaged, and free of debris.
(11) Carefully position the hard top assembly on the vehicle making sure that the latches are not pinched between the windshield frame and top.

CAUTION: Do not over tighten the top bolts. Over tightening can cause cracking of the hardtop assembly.

(12) Loosely install the six bolts. Ensure that the top is centered on the vehicle and locators are in place in the body side slots and tighten the bolts to 17 N·m (150 in. lbs.).
(13) Connect the wire wiper motor harness connector.
(14) Connect the rear washer fluid hose.
(15) Engage the latches at windshield frame and snap the clamps home.

HARD TOP

REMOVAL

(1) Disengage latches at windshield frame (Fig. 1).

NOTE: The two forward bolts/nuts are not captured. Do not lose the nuts when removing.

(2) Remove the six bolts that attach the hard top to the body (Fig. 2).

(3) Depress tab on rear wiper motor connector and pull downward to disengage (Fig. 3).

(4) Disconnect the rear washer fluid hose. Cap the hose to prevent washer fluid leakage (Fig. 4).

CAUTION: Protect the lower edges of the top from potential damage, cracks, paint chips, gouges and scratches that are not covered under warranty.

(5) Carefully remove the hard top assembly from the vehicle.

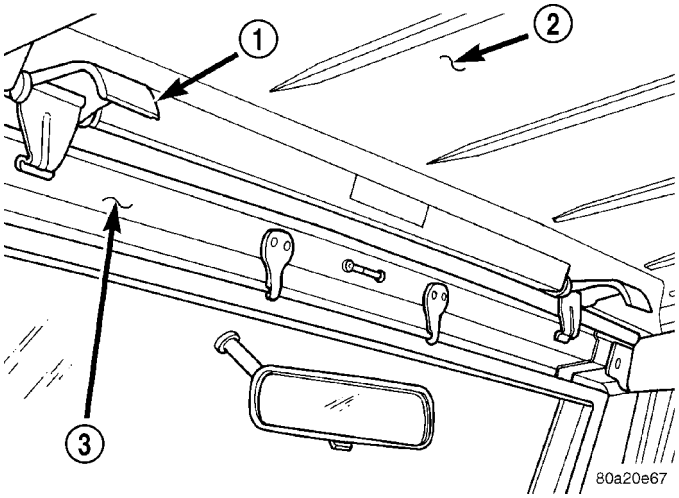

Fig. 1 HARD TOP LATCH

1 - LATCH
2 - HARD TOP
3 - WINDSHIELD FRAME

INSTALLATION

(1) Inspect the hard top seals for damage and replace, if necessary.

(2) Inspect the windshield header seal and make sure it is clean, undamaged, and free of debris.

(3) Carefully position the hard top assembly on the vehicle making sure that the latches are not pinched between the windshield frame and top.

CAUTION: Do not over tighten the top bolts. Over tightening can cause cracking of the hardtop assembly.

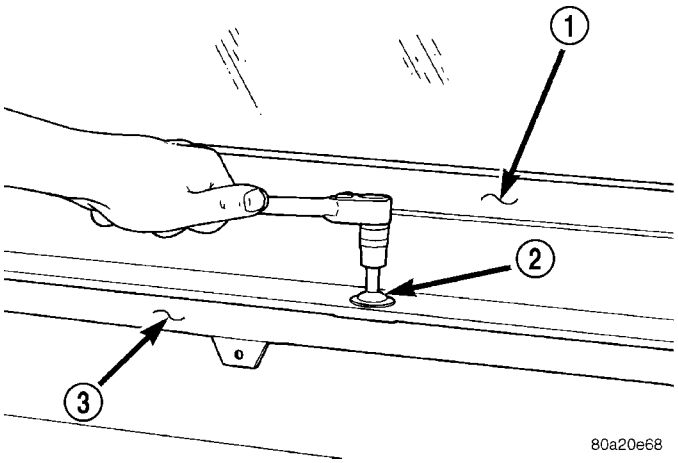

Fig. 2 HARD TOP REMOVAL

1 - HARD TOP
2 - HOLD DOWN BOLT
3 - BODY RAIL

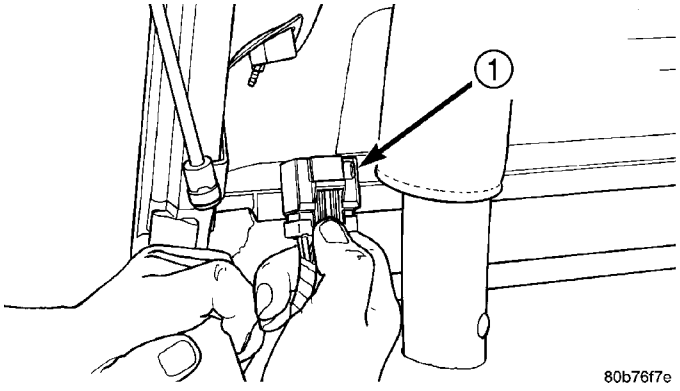

Fig. 3 REAR WIPER WIRE HARNESS CONNECTOR

1 - WIPER MOTOR CONNECTOR

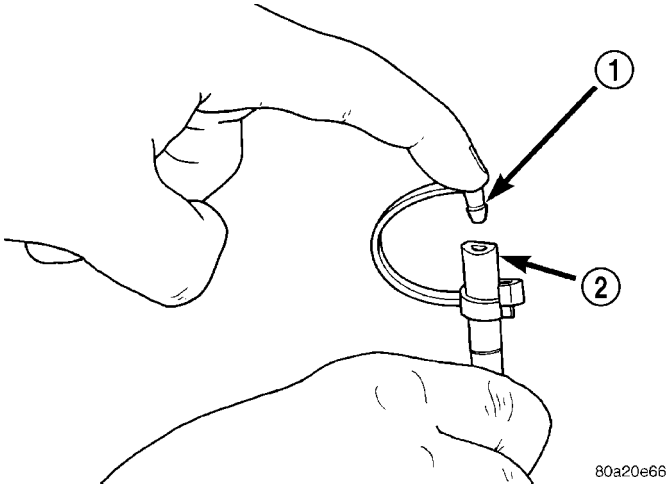

Fig. 4 REAR WASHER FLUID TUBE

1 - WASHER HOSE CAP
2 - REAR WASHER SUPPLY HOSE - BODY HALF

HARD TOP (Continued)

(4) Loosely install the six bolts. Ensure that the top is centered on the vehicle and tighten the bolts to 17 N·m (150 in. lbs.).

(5) Connect the wire wiper motor harness connector.

(6) Connect the rear washer fluid hose.

(7) Engage the latches at windshield frame and snap the clamps home.

SOFT TOP

REMOVAL

(1) Disengage the retainers attaching the rear window to the body.

(2) Remove rear window, unzipping from right to left.

(3) Disengage J-straps at soft top rear corners (Fig. 5).

(4) Unzip quarter windows, disengage J-strap and remove quarter windows.

(5) Starting at the rear of the upper door opening frame and working forward, disengage drip rail retainers attaching the soft top to the door opening frame.

(6) Unlatch top at windshield frame.

(7) Lower the top to the rearward position.

(8) Remove the screws attaching the roof bows to the pivot bracket (Fig. 6).

(9) Lift up bows at pivot bracket to disengage from pivot bracket.

(10) Remove the top (Fig. 7).

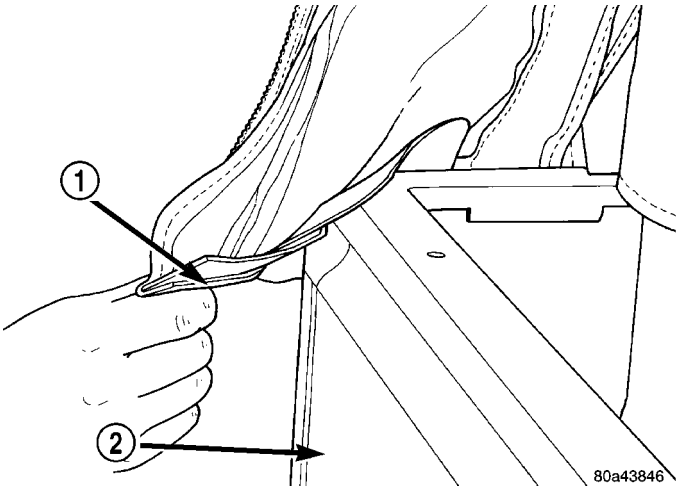

Fig. 5 SOFT TOP J-STRAPS

1 - SOFT TOP CORNER J-STRAP
2 - QUARTER PANEL

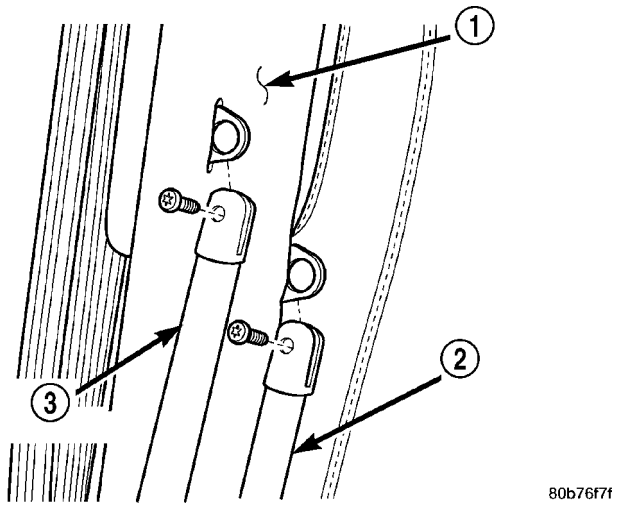

Fig. 6 ROOF BOW REMOVAL

1 - SPORT BAR
2 - REAR ROOF BOW
3 - SIDE ROOF BOW

INSTALLATION

(1) Position the top on the vehicle.

(2) Install the screws attaching the roof bows to the pivot bracket. (The front bow is attached to the pivot bracket on the upper outward location).

(3) Raise the top.

(4) Position latch in windshield frame.

(5) Install the quarter windows.

(6) Working from front to rear, engage the J-straps attaching the quarter window to the body.

(7) Install rear window.

(8) Engage drip rail retainers above door opening frame.

SOFT TOP (Continued)

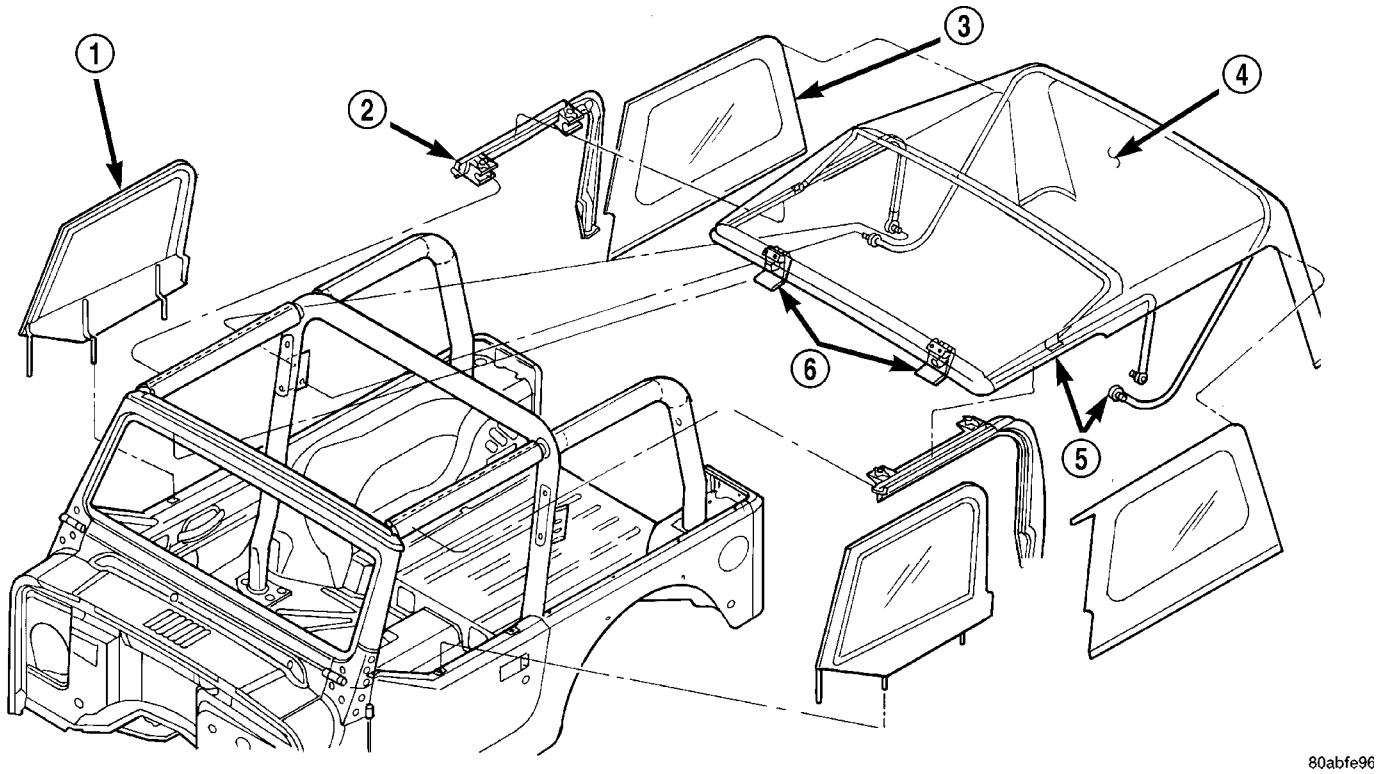

Fig. 7 SOFT TOP

1 - HALF DOOR WINDOW
2 - DOOR OPENING FRAME
3 - QUARTER WINDOW
4 - SOFT TOP
5 - ROOF BOW
6 - LATCH REINFORCEMENT

(9) Working from front to rear, engage J-straps at soft top rear corners.

(10) Engage the retainers attaching the rear window to the body.

(11) Close latch at windshield frame.

SOFT TOP FABRIC

REMOVAL

(1) Disengage the snaps attaching the soft top fabric to the rear roof bow.

(2) Disengage the hook and loop fastener attaching soft top fabric to the center roof bow.

(3) Lower the soft top.

(4) Remove the screws attaching the soft top fabric to the front roof bow and fold back fabric.

(5) Separate the soft top fabric from the frame.

INSTALLATION

(1) Position the soft top fabric on the frame.

(2) Install the screws attaching the soft top fabric to the front roof bow.

(3) Engage the hook and loop fastener attaching soft top fabric to the center roof bow.

(4) Engage the snaps attaching the soft top fabric to the rear roof bow.

(5) Raise and secure the soft top.

DOOR OPENING FRAME

REMOVAL

Vehicles equipped with a soft top require a door opening frame to complete the seal for the soft top door assembly.

(1) Lower the top to the rearward position.

(2) Turn the knobs located on top of the door opening frame counter clockwise and remove completely (Fig. 8).

(3) Pull door opening frame outward and up. Separate from vehicle.

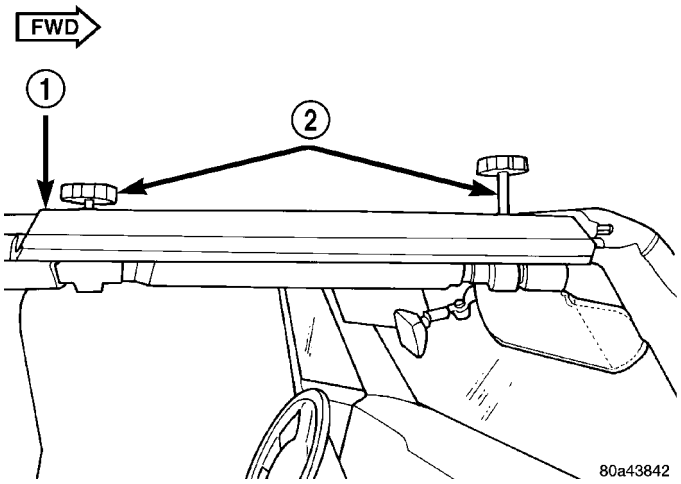

Fig. 8 DOOR OPENING FRAME

1 - DOOR OPENING FRAME
2 - KNOBS

INSTALLATION

Vehicles equipped with a soft top require a door opening frame to complete the seal for the soft top door assembly.

(1) Install the alignment pin at the base of the door opening frame into the hole at the top of the quarter panel.

(2) Position the door opening frame on the side support bar and install the knobs.

(3) Raise and secure the top.

HARD/SOFT TOP LATCH

REMOVAL

(1) Unlatch the top (Fig. 9).

(2) Using a grease pencil or equivalent, mark the position of the latch on the top.

(3) Remove the screws attaching the latch to the top.

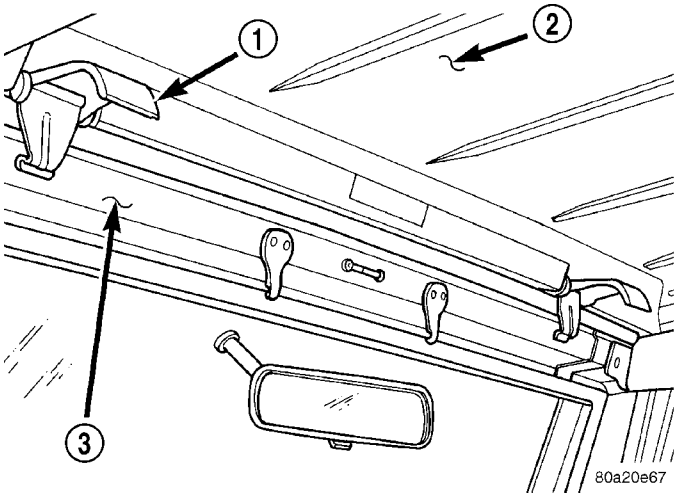

Fig. 9 HARD/SOFT TOP LATCH

1 - LATCH
2 - HARD TOP
3 - WINDSHIELD FRAME

INSTALLATION

(1) Position the latch on the top and install the screws.

SEATS

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FRONT SEAT

REMOVAL

(1) Disengage seat belt electrical connector (Fig. 1).
(2) Remove the bolts attaching the seat adjuster to the floor panel (Fig. 2).
(3) Remove the seat from the vehicle.

INSTALLATION

(1) Position the seat in the vehicle.
(2) Install the bolts attaching the rear of seat frame to the floor panel. Tighten outboard bolt to 33 N·m (25 ft. lbs.). Tighten inboard bolt to 74 N·m (55 ft. lbs.).

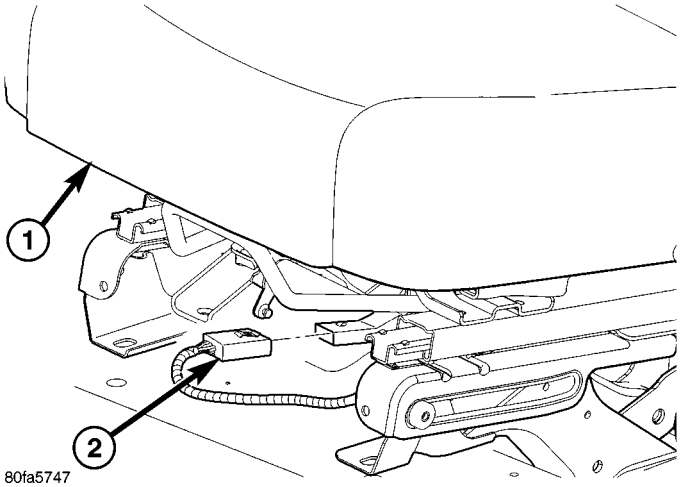

Fig. 1 SEAT BELT ELECTRICAL

1 - FRONT SEAT
2 - SEAT BELT ELECTRICAL CONNECTOR

FRONT SEAT (Continued)

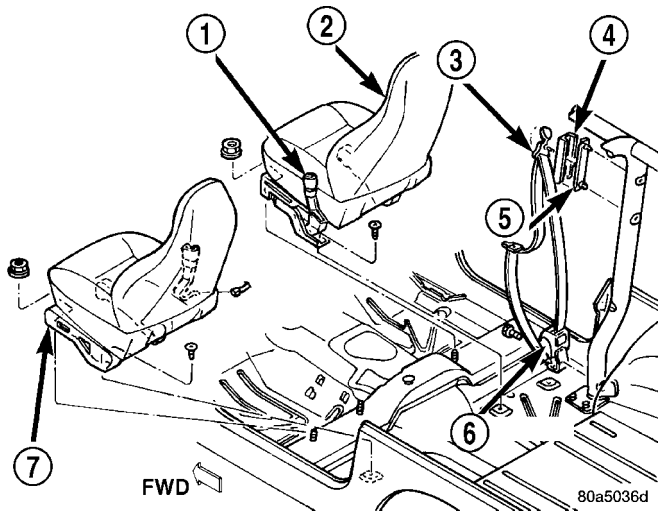

Fig. 2 BUCKET SEAT REMOVAL

1 - BUCKLE
2 - SEAT
3 - TURNING LOOP
4 - COVER
5 - ADJUSTER
6 - RETRACTOR
7 - MOUNTING BRACKET

(3) Install the bolts attaching the front of seat frame to the floor panel and tighten bolts to 33 N·m (25 ft. lbs.).

(4) Engage seat belt electrical connector.

FRONT SEAT BACK

REMOVAL

(1) Remove seat. (Refer to 23 - BODY/SEATS/ FRONT SEAT - REMOVAL)

(2) Disengage the j-straps at the lower edge of the seat back and position the seat back cover aside.

(3) Using a small flat bladed tool, release the hinge cover retaining tab and remove. (Fig. 3)

(4) Disconnect the recliner cables. (Fig. 4)

(5) Remove the upper recliner bolts and remove the seat back

INSTALLATION

(1) Position the seatback on the seat cushion.

(2) Passenger seat and driver dumping seat:

(a) Engage seat dump cable to clip on recliner.

(3) Install the bolts attaching the recliner to the seat cushion frame.

(4) Engage the seat cushion corner cover j-retainers.

(5) Engage the retainers attaching the cushion cover to the outboard seat cushion frame.

(6) Install seat. (Refer to 23 - BODY/SEATS/ FRONT SEAT - INSTALLATION)

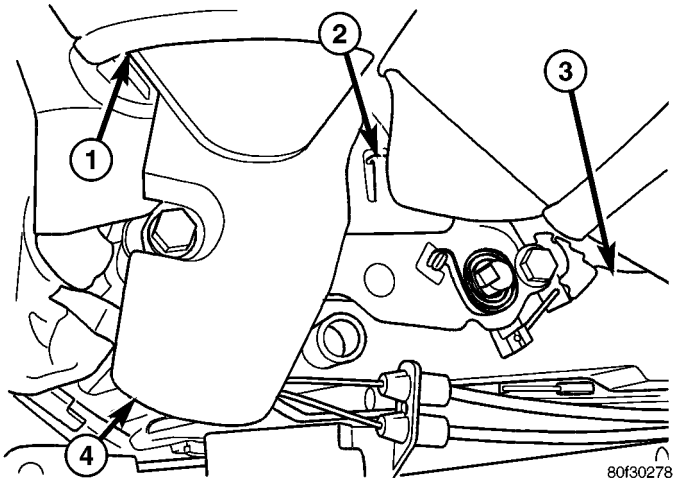

Fig. 3 SEAT BACK HINGE COVER

1 - SEAT BACK COVER
2 - RECLINER COVER RETAINING TAB
3 - SEAT CUSHION FRAME
4 - RECLINER COVER

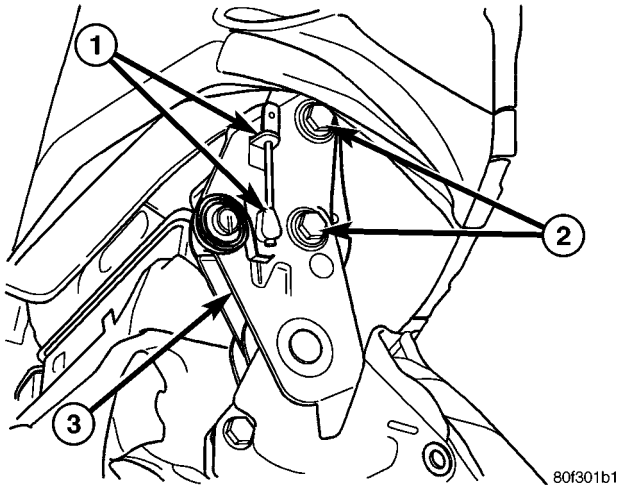

Fig. 4 SEAT BACK HINGE

1 - DUMP CABLE CONNECTIONS
2 - SEAT BACK BOLTS (4)
3 - SEAT BACK RECLINER

FRONT SEAT BACK COVER

REMOVAL

(1) Remove the seat. (Refer to 23 - BODY/SEATS/ SEAT - REMOVAL)
(2) Disconnect the j-straps at the bottom of the seat back.
(3) Remove the bottom hog rings.
(4) Remove the upper hog rings.
(5) Remove the support rods from the seat back.
(6) Using a trim stick C-4755 or equivalent, remove and discard the seat release handle knob. (Fig. 5)
(7) Remove the screw and remove the release handle bezel.
(8) Remove the seat cover and cushion.

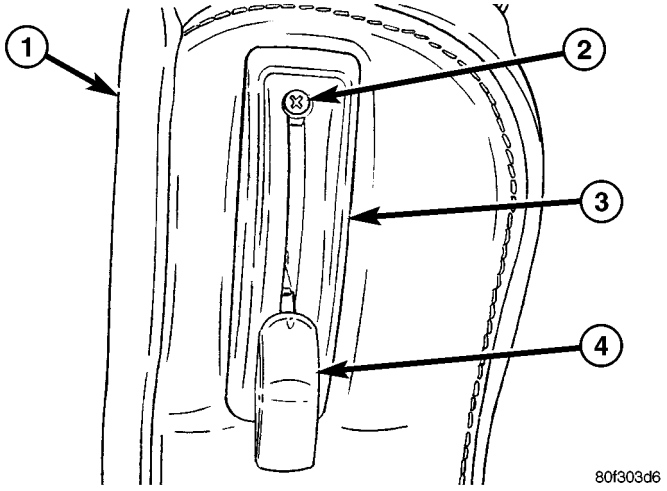

Fig. 5 FORWARD FOLDING SEAT RELEASE HANDLE

1 - SEAT BACK
2 - BEZEL SCREW
3 - RELEASE HANDLE BEZEL
4 - RELEASE HANDLE KNOB

INSTALLATION

(1) Position cover and cushion on seatback.
(2) Install the release handle bezel and install the screw.
(3) Install a new release handle knob.
(4) Install the support rods.
(5) Install the upper hog rings.
(6) Install the bottom hog rings.
(7) Connect the j-straps at the bottom of the seat back.
(8) Install seat back. (Refer to 23 - BODY/SEATS/ FRONT SEAT BACK - INSTALLATION)

FRONT SEAT CUSHION COVER

REMOVAL

(1) Remove seat back. (Refer to 23 - BODY/SEATS/ FRONT SEAT BACK - REMOVAL)
(2) Disengage inboard j-strap.
(3) Disengage front j-strap.
(4) Disengage rear j-strap
(5) Roll cover up to access hog rings.
(6) Disengage inboard and outboard hog rings.
(7) Separate cover from cushion.

INSTALLATION

(1) Position cover on cushion and align seams.
(2) Engage inboard and outboard hog rings.
(3) Roll cover over cushion edges.
(4) Engage inboard j-strap.
(5) Engage front j-strap.
(6) Engage rear j-strap
(7) Install seat back. (Refer to 23 - BODY/SEATS/ FRONT SEAT BACK - INSTALLATION)

FRONT SEAT BACK RECLINER/ COVER

REMOVAL

(1) Remove the seat. (Refer to 23 - BODY/SEATS/ FRONT SEAT - REMOVAL)
(2) Disconnect the j-straps at the lower edge of the seat back and position aside.
(3) Disconnect the rear j-straps of the cushion and position aside.
(4) Using a small flat bladed tool, release the recliner cover retaining tab and remove. (Fig. 3)
(5) Disconnect the seat back dump cables. (Fig. 4)
(6) Remove the two seat back bolts.
(7) Remove the two seat cushion frame bolts.
(8) Disconnect the recliner cable.
(9) For the inboard recliner, disconnect the forward folding cable assembly, if equipped.

INSTALLATION

(1) Connect the forward folding cable assembly, if equipped.
(2) Connect the recliner cable.
(3) Install the two seat cushion frame bolts.
(4) Install the two seat back bolts.
(5) Connect the seat back dump cables.
(6) Install the recliner cover.
(7) Connect the rear j-straps of the cushion.
(8) Connect the j-straps at the lower edge of the seat back.
(9) Install the seat. (Refer to 23 - BODY/SEATS/ SEAT - INSTALLATION)

FRONT SEAT BACK RECLINER CABLE

REMOVAL

(1) Remove seat cushion cover. (Refer to 23 - BODY/SEATS/SEAT CUSHION COVER - REMOVAL)

(2) Disengage cable from recliner release handle. (Fig. 6) and (Fig. 7)

(3) Route cable through seat cushion pad.

(4) Disengage cable from mounting bracket.

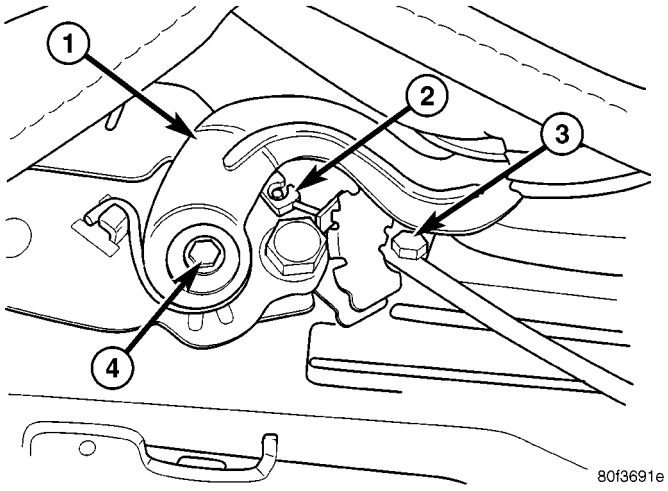

Fig. 6 RECLINER HANDLE/CABLE

1 - RECLINER HANDLE
2 - RECLINER CABLE END
3 - RECLINER CABLE
4 - RECLINER HANDLE BOLT

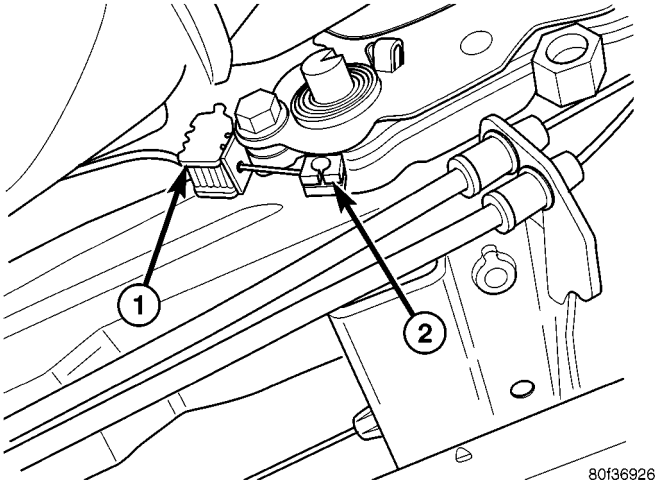

Fig. 7 RECLINER CABLE END

1 - RECLINER CABLE
2 - RECLINER CABLE END

INSTALLATION

(1) Engage cable to mounting bracket.

(2) Route cable through seat cushion pad.

(3) Engage cable to recliner release.

(4) Install seat back cover. (Refer to 23 - BODY/SEATS/FRONT SEAT BACK COVER - INSTALLATION)

FRONT SEAT FOLDING RELEASE CABLE

REMOVAL

(1) Remove the seat. (Refer to 23 - BODY/SEATS/SEAT - REMOVAL)

(2) Disconnect the release cable folding end. (Fig. 8)

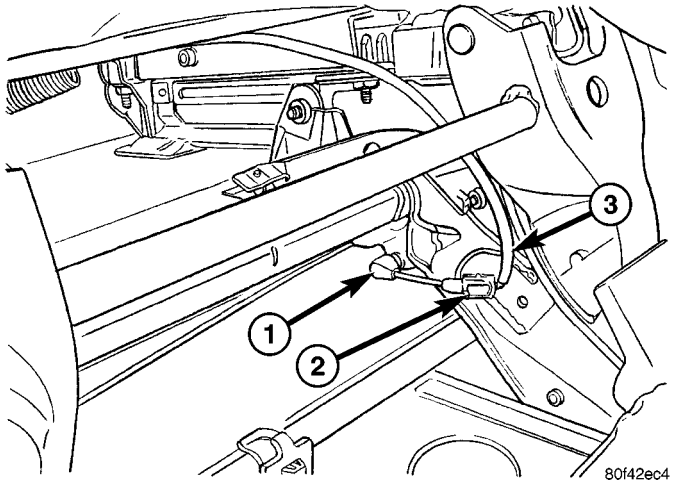

Fig. 8 RELEASE CABLE/FOLDING END

1 - CABLE END CLIP
2 - CABLE HOUSING
3 - CABLE ASSEMBLY

FRONT SEAT FOLDING RELEASE CABLE (Continued)

(3) Remove the inboard recliner. (Refer to 23 - BODY/SEATS/SEAT BACK RECLINER - REMOVAL)

(4) Disconnect the release cable assembly at the inboard recliner, from the support bracket. (Fig. 9)

(5) Disconnect the cable assembly at the slider and remove. (Fig. 10)

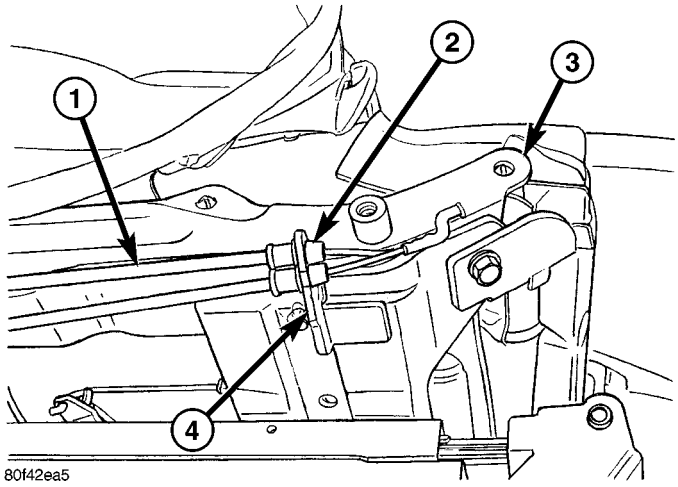

Fig. 9 RELEASE CABLE/RECLINER END

1 - CABLE ASSEMBLY
2 - CABLE HOUSING
3 - SEAT CUSHION FRAME
4 - CABLE SUPPORT BRACKET

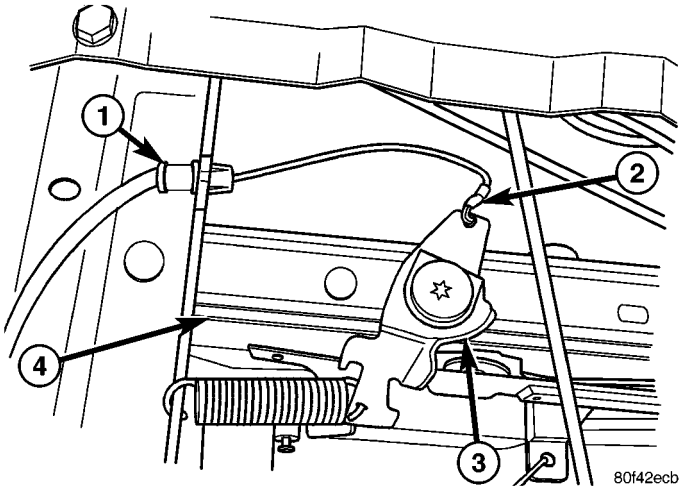

Fig. 10 RELEASE CABLE/SLIDER END

1 - CABLE HOUSING
2 - CABLE END
3 - SLIDER
4 - SEAT CUSHION FRAME

INSTALLATION

(1) Connect the cable assembly at the slider.

(2) Connect the inboard release cable assembly to the support bracket at the recliner.

(3) Install the inboard recliner. (Refer to 23 - BODY/SEATS/SEAT BACK RECLINER - INSTALLATION)

(4) Connect the folding end of the release cable assembly.

(5) Install the seat. (Refer to 23 - BODY/SEATS/SEAT - INSTALLATION)

FRONT SEAT BACK DUMP CABLE

REMOVAL

(1) Remove the seat back cover. (Refer to 23 - BODY/SEATS/FRONT SEAT BACK COVER - REMOVAL)

(2) Disconnect the dump cables from the recliners. (Fig. 4)

(3) Remove the two dump handle assembly screws and remove the release cable assembly.

INSTALLATION

(1) Install the dump handle assembly and install the two screws.

(2) Connect the dump cables at the recliners.

(3) Install the seat back cover. (Refer to 23 - BODY/SEATS/SEAT BACK COVER - INSTALLATION)

FRONT SEAT TRACK ADJUSTER

REMOVAL

(1) Remove the seat. (Refer to 23 - BODY/SEATS/SEAT - REMOVAL)

(2) Position aside the seat back cover. (Refer to 23 - BODY/SEATS/SEAT BACK COVER - REMOVAL)

(3) Remove the recliner covers. (Refer to 23 - BODY/SEATS/SEAT BACK RECLINER - REMOVAL)

(4) Position aside the seat cushion cover. (Refer to 23 - BODY/SEATS/SEAT CUSHION COVER - REMOVAL)

(5) Disconnect the recliner cable ends. (Fig. 11) and (Fig. 12)

(6) Remove the seat cushion to hinge bolts. (Fig. 11) and (Fig. 12)

(7) Disconnect the folding release cable from the seat back recliner, if equipped. (Fig. 11)

(8) Remove the seat back assembly.

(9) Remove the two bolts on either side of the seat cushion and remove the seat track assembly. (Fig. 13)

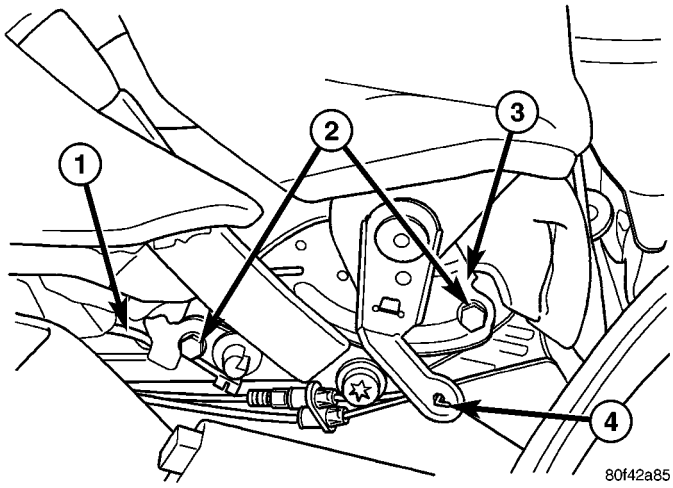

Fig. 11 INBOARD SEAT CUSHION AND HINGE

1 - RECLINER CABLE
2 - SEAT CUSHION FRAME BOLTS
3 - SEAT CUSHION FRAME
4 - FOLDING RELEASE CABLE ASSEMBLY END

INSTALLATION

(1) Install the seat track assembly and install the two front bolts.

(2) Install the seat back assembly.

(3) Connect the folding release cable assembly end to the seat back recliner.

(4) Install the seat back assembly bolts.

(5) Tighten the seat track bolts to 50 N·m (37 ft. lbs.)

(6) Connect the recliner cables.

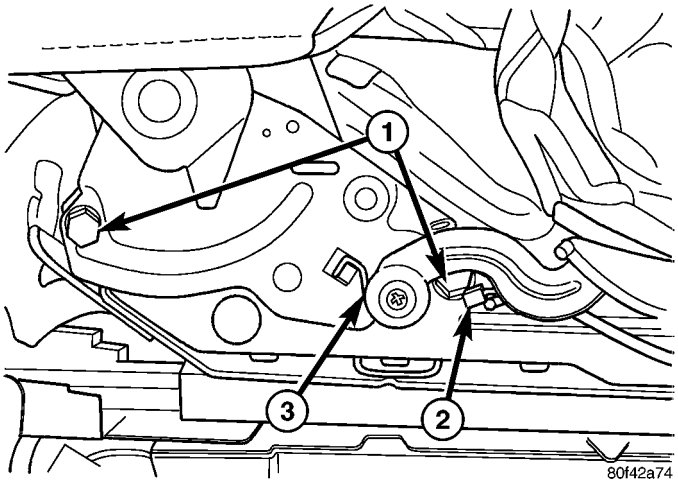

Fig. 12 SEAT CUSHION AND HINGE

1 - SEAT CUSHION FRAME BOLTS
2 - RECLINER CABLE END
3 - RECLINER HANDLE

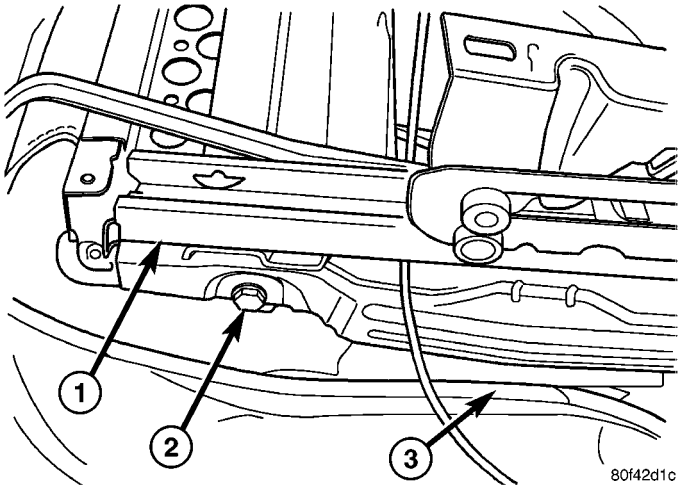

Fig. 13 FRONT SEAT CUSHION BOLTS

1 - SEAT TRACK
2 - SEAT CUSHION FRAME BOLT
3 - SEAT CUSHION

(7) Install the seat cushion cover. (Refer to 23 - BODY/SEATS/SEAT CUSHION COVER - INSTALLATION)

(8) Install the recliner covers. (Refer to 23 - BODY/SEATS/SEAT BACK RECLINER - INSTALLATION)

(9) Install the seat back cover. (Refer to 23 - BODY/SEATS/SEAT BACK COVER - INSTALLATION)

(10) Install the seat. (Refer to 23 - BODY/SEATS/SEAT - INSTALLATION)

REAR SEAT

REMOVAL

(1) Remove the add-a-trunk, if equipped. (Refer to 23 - BODY/INTERIOR/ADD-A-TRUNK - REMOVAL)

(2) Using the latch handle, fold the seat back down and tip the seat assembly forward. (Fig. 14)

(3) Squeeze the release bar and disengage the front latches. (Fig. 15)

(4) Remove the seat through tailgate opening.

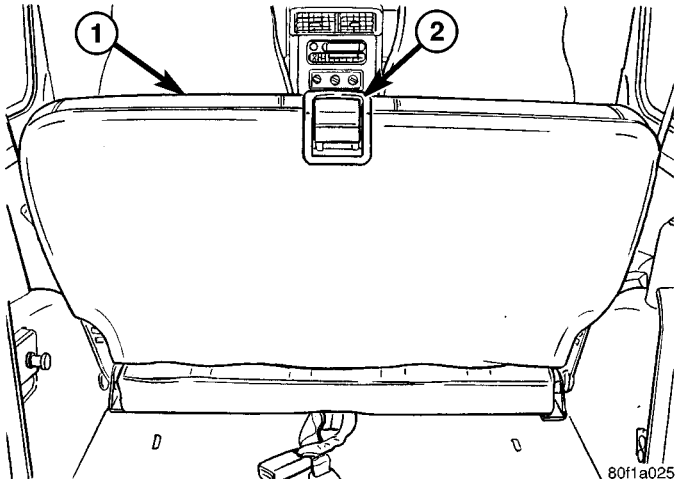

Fig. 14 REAR SEAT BACK

1 - REAR SEAT BACK
2 - RELEASE HANDLE

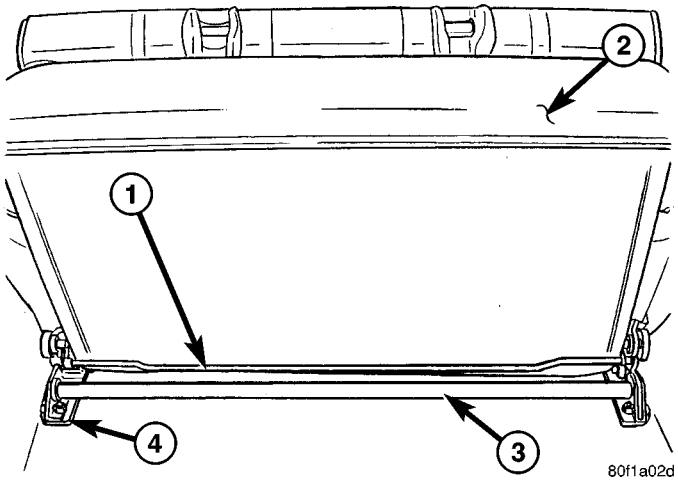

Fig. 15 REAR SEAT RELEASE LATCH ROD

1 - RELEASE LATCH ROD
2 - REAR SEAT CUSHION
3 - REAR SEAT LATCH ASSEMBLY
4 - REAR SEAT LATCH STRIKER (4)

INSTALLATION

(1) Position the seat on the rear floor panel and engage the seat frame release latch with the front strikers.

(2) Roll seat back and engage the rear latches with the rear strikers.

(3) Return the seat back to the up position.

(4) Install the add-a-trunk, if equipped. (Refer to 23 - BODY/INTERIOR/ADD-A-TRUNK - INSTALLATION)

REAR SEAT BACK

REMOVAL

(1) Remove rear seat. (Refer to 23 - BODY/SEATS/REAR SEAT - REMOVAL)

(2) Remove the bolts attaching the seat back assembly to the seat cushion and remove the seat back. (Fig. 16)

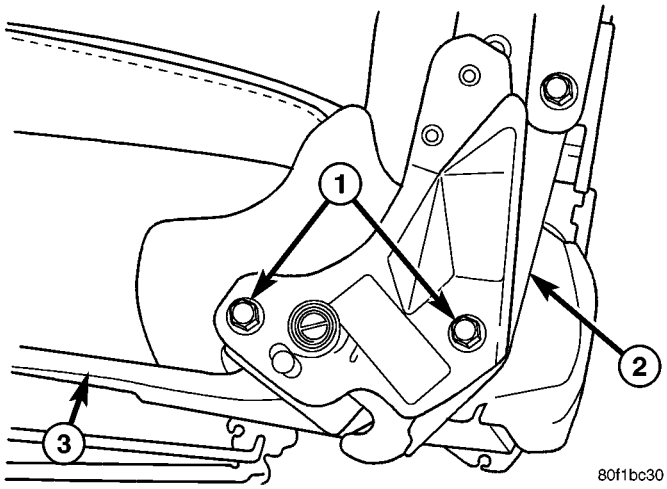

Fig. 16 REAR SEAT BACK HINGE

1 - SEAT CUSHION BOLTS
2 - SEAT BACK HINGE
3 - SEAT CUSHION

INSTALLATION

(1) Position the seat back on the seat cushion and install the bolts.

(2) Tighten the bolts to 50 N·m (37 ft. lbs.).

(3) Install rear seat. (Refer to 23 - BODY/SEATS/REAR SEAT - INSTALLATION)

REAR SEAT BACK HINGE

REMOVAL

(1) Remove the seat. (Refer to 23 - BODY/SEATS/ REAR SEAT - REMOVAL)
(2) Remove the lower hinge bolts. (Fig. 16)
(3) Disconnect the lower j-straps of the seat back and position aside the seat back cover.
(4) Disconnect the release cable. (Fig. 17)
(5) Remove the bolts and remove the hinge.

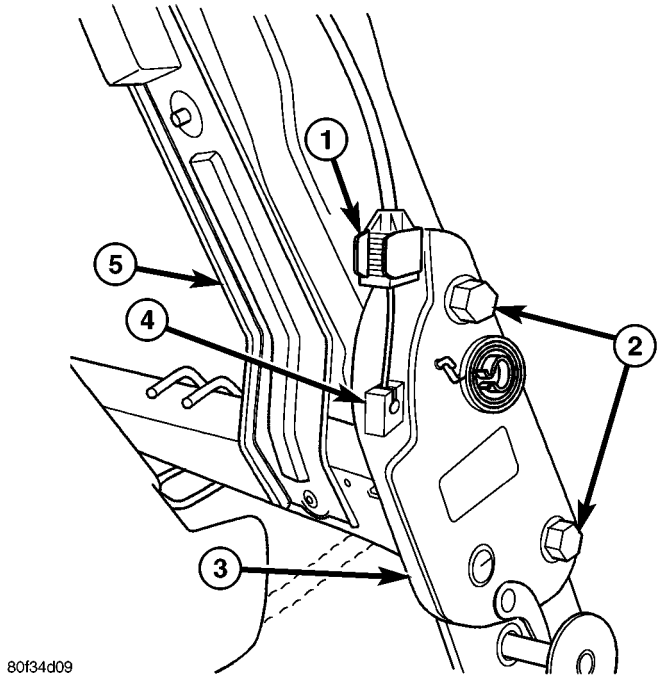

Fig. 17 REAR SEAT BACK HINGE

1 - RELEASE CABLE GUIDE
2 - BOLTS
3 - SEAT BACK HINGE
4 - RELEASE CABLE CONNECTION
5 - SEAT BACK FRAME

INSTALLATION

(1) Install the hinge and install the bolts.
(2) Tighten the bolts to 50 N·m (37 ft. lbs.).
(3) Connect the release cable.
(4) Connect the lower j-straps of the seat back.
(5) Install the lower hinge bolts and tighten to 50 N·m (37 ft. lbs.).
(6) Install the seat. (Refer to 23 - BODY/SEATS/ REAR SEAT - INSTALLATION)

REAR SEAT BACK COVER

REMOVAL

(1) Remove the seat. (Refer to 23 - BODY/SEATS/ SEAT - REMOVAL)
(2) Remove the screws and remove the seat back release handle bezel. (Fig. 18)
(3) Release the j-clips at the bottom of the seat back and remove the seat cover.
(4) Remove the hog rings and remove the seat cushion. (Fig. 19)

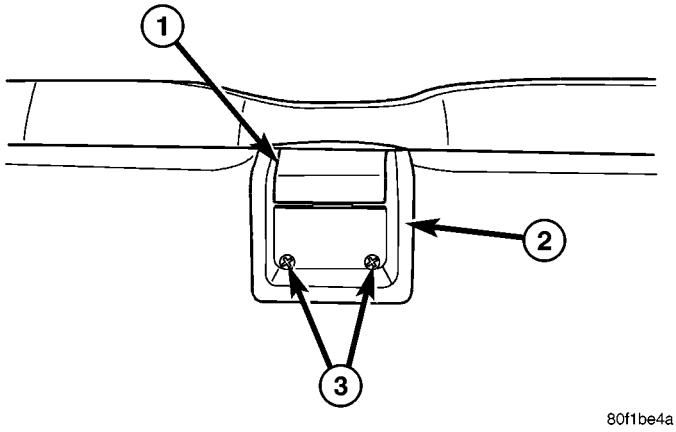

Fig. 18 RELEASE HANDLE

1 - RELEASE HANDLE
2 - BEZEL
3 - SCREWS (2)

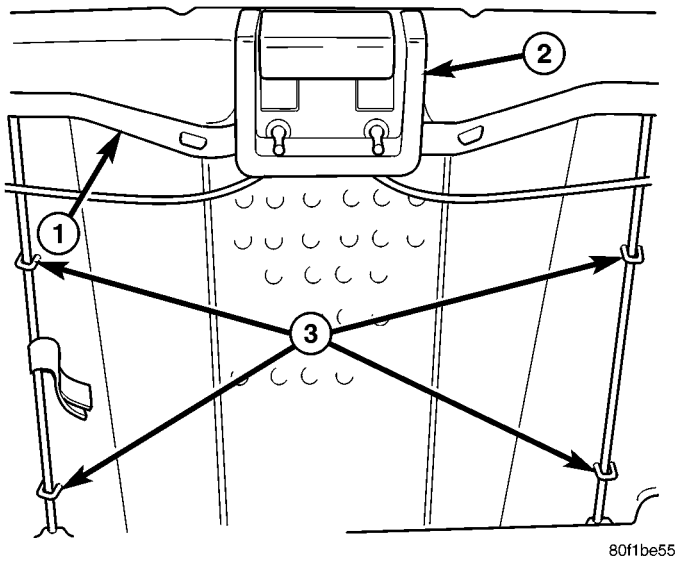

Fig. 19 SEAT BACK CUSHION

1 - SEAT BACK FRAME
2 - SEAT BACK RELEASE HANDLE
3 - HOG RINGS (4)

INSTALLATION

(1) Position the cushion onto the seat back frame and install new hog ring fasteners.
(2) Install the cover over the seat back frame and cushion.

REAR SEAT BACK COVER (Continued)

(3) Connect the bottom j-clips.
(4) Install the release handle bezel and install the screws.
(5) Install the rear seat back. (Refer to 23 - BODY/SEATS/REAR SEAT BACK - INSTALLATION)

REAR SEAT CUSHION COVER

REMOVAL

(1) Remove the rear seat back. (Refer to 23 - BODY/SEATS/REAR SEAT BACK - REMOVAL)
(2) Un-zip the cover. (Fig. 20)
(3) Release the J-clips. (Fig. 21)
(4) Roll the cover from seat cushion and separate from the seat cushion frame.
(5) Remove the hogrings from the foam to cover.

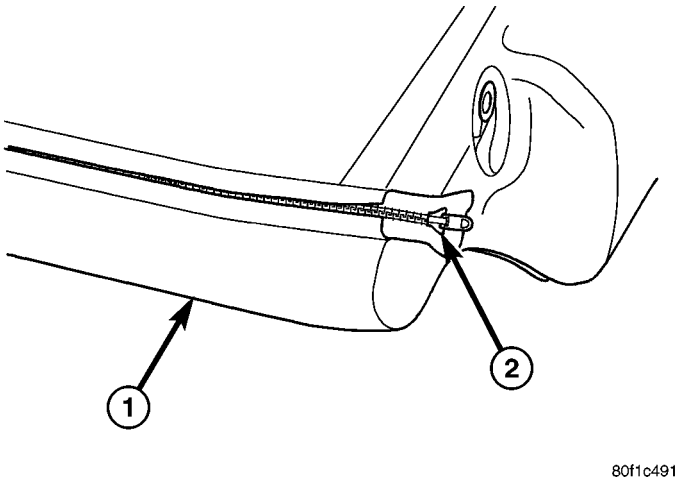

Fig. 20 SEAT CUSHION COVER

1 - SEAT CUSHION
2 - COVER ZIPPER

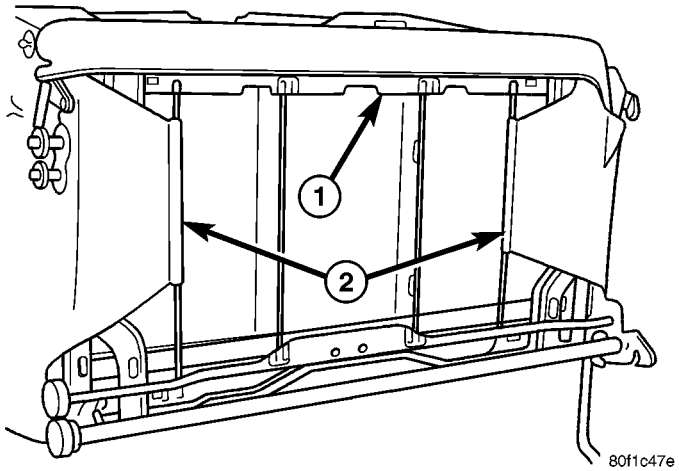

Fig. 21 SEAT CUSHION

1 - SEAT CUSHION FRAME
2 - J-STRAPS

INSTALLATION

(1) Position the cover on the cushion, install new hogrings to foam, and roll cover downward over the corners.
(2) Connect the J-clips to the seat cushion frame.
(3) Close the cover zipper and tuck the zipper ends into the cover.
(4) Install the rear seat back. (Refer to 23 - BODY/SEATS/REAR SEAT BACK - INSTALLATION)

REAR SEAT RELEASE CABLE

REMOVAL

(1) Remove the seat. (Refer to 23 - BODY/SEATS/REAR SEAT - REMOVAL)
(2) Remove the screws and remove the seat back release handle bezel. (Fig. 18)
(3) Release the j-clips at the bottom of the seat back and remove the seat cover.
(4) Disconnect the cables at the seat back hinges. (Fig. 17)
(5) Position the seat back cushion aside and remove the screws. (Fig. 22)
(6) Remove the seat back release cable assembly.

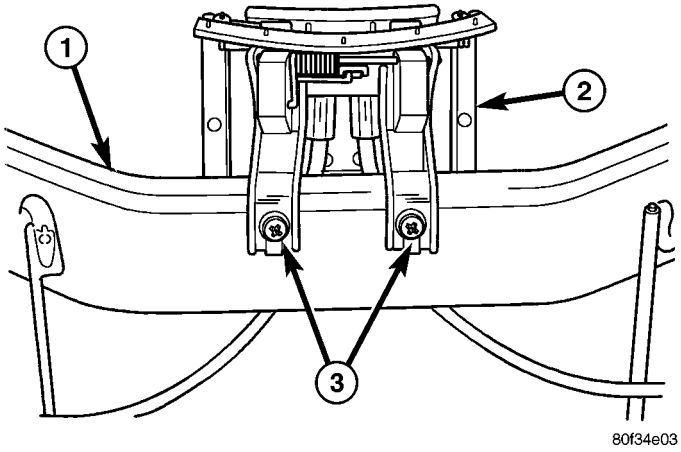

Fig. 22 RELEASE CABLE ASSEMBLY

1 - SEAT BACK FRAME
2 - RELEASE CABLE ASSEMBLY HANDLE
3 - SCREWS

INSTALLATION

(1) Install the seat back release cable assembly and install the screws.
(2) Connect the cables at the seat back hinges.
(3) Install the seat cover and engage the j-clips at the bottom of the seat back.
(4) Install the handle bezel and install the screws.
(5) Install the seat. (Refer to 23 - BODY/SEATS/REAR SEAT - INSTALLATION)

REAR SEAT STRIKERS

REMOVAL

Rear Strikers

(1) Release the seat back and fold seat assembly forward.

(2) Remove the bolts and remove the striker. (Fig. 23)

Front Strikers

(1) Remove the seat. (Refer to 23 - BODY/SEATS/ REAR SEAT - REMOVAL)

(2) Remove the bolts and remove the striker. (Fig. 23)

INSTALLATION

Front Strikers

(1) Install the striker and install the bolts.

(2) Tighten the bolts to 61 N·m (45 ft. lbs.).

(3) Install the seat. (Refer to 23 - BODY/SEATS/ REAR SEAT - INSTALLATION)

Rear Strikers

(1) Install the striker and install the bolts.

(2) Tighten the bolts to 61 N·m (45 ft. lbs.).

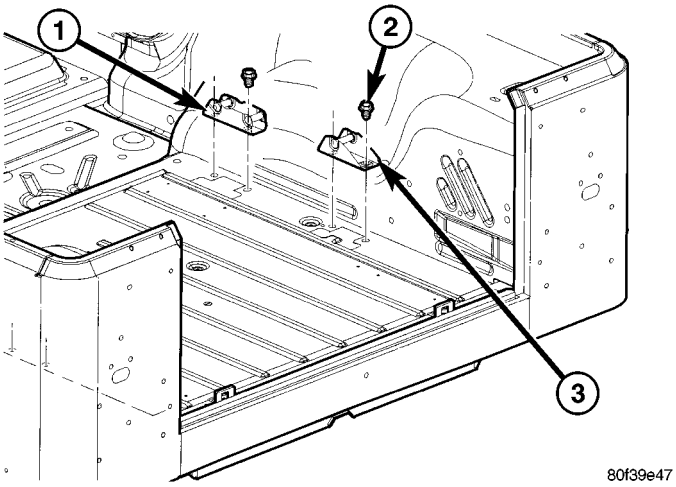

Fig. 23 REAR SEAT STRIKERS

1 - FRONT STRIKER
2 - BOLTS (2 PER STRIKER)
3 - REAR STRIKER

(3) Return the seat to the down position.

STATIONARY GLASS

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STATIONARY GLASS

DESCRIPTION

WINDSHIELD SAFETY PRECAUTIONS

DESCRIPTION

WARNING: DO NOT OPERATE THE VEHICLE WITHIN 24 HOURS OF WINDSHIELD INSTALLATION. IT TAKES AT LEAST 24 HOURS FOR URETHANE ADHESIVE TO CURE. IF IT IS NOT CURED, THE WINDSHIELD MAY NOT PERFORM PROPERLY IN AN ACCIDENT.

- **URETHANE ADHESIVES ARE APPLIED AS A SYSTEM. USE GLASS CLEANER, GLASS PREP SOLVENT, GLASS PRIMER, PVC (VINYL) PRIMER AND PINCH WELD (FENCE) PRIMER PROVIDED BY THE ADHESIVE MANUFACTURER. IF NOT, STRUCTURAL INTEGRITY COULD BE COMPROMISED.**
- **DAIMLERCHRYSLER DOES NOT RECOMMEND GLASS ADHESIVE BY BRAND. TECHNICIANS SHOULD REVIEW PRODUCT LABELS AND TECHNICAL DATA SHEETS, AND USE ONLY ADHESIVES THAT THEIR MANUFACTURES WARRANT WILL RESTORE A VEHICLE TO THE REQUIREMENTS OF FMVSS 212. TECHNICIANS SHOULD ALSO INSURE THAT PRIMERS AND CLEANERS ARE COMPATIBLE WITH THE PARTICULAR ADHESIVE USED.**
- **BE SURE TO REFER TO THE URETHANE MANUFACTURER'S DIRECTIONS FOR CURING TIME SPECIFICATIONS, AND DO NOT USE ADHESIVE AFTER ITS EXPIRATION DATE.**
- **VAPORS THAT ARE EMITTED FROM THE URETHANE ADHESIVE OR PRIMER COULD CAUSE PERSONAL INJURY. USE THEM IN A WELL-VENTILATED AREA.**
- **SKIN CONTACT WITH URETHANE ADHESIVE SHOULD BE AVOIDED. PERSONAL INJURY MAY RESULT.**
- **ALWAYS WEAR EYE AND HAND PROTECTION WHEN WORKING WITH GLASS.**

CAUTION: Protect all painted and trimmed surfaces from coming in contact with urethane or primers. Be careful not to damage painted surfaces when removing moldings or cutting urethane around windshield.

OPERATION

The windshield is attached to the window frame with urethane adhesive. The urethane adhesive is applied cold and seals the surface area between the window opening and the glass. The primer adheres the urethane adhesive to the windshield.

It is difficult to salvage a windshield during the removal operation. The windshield is part of the structural support for the roof. The urethane bonding used to secure the windshield to the fence is difficult to cut or clean from any surface. If the moldings are set in urethane, it would also be unlikely they could be salvaged. Before removing the windshield, check the availability of the windshield and moldings from the parts supplier.

QUARTER GLASS

REMOVAL

(1) Cover surface areas with protective covering to avoid paint damage and extra clean-up time.

(2) Using a razor knife, slide the blade between the quarter glass and the inboard edge of the reveal molding.

(3) Cut around the interior perimeter of the reveal molding and sever the cap of the reveal molding.

(4) Using a cold knife, cut the urethane around the perimeter of the quarter glass.

(5) Remove the quarter glass from the opening (Fig. 1).

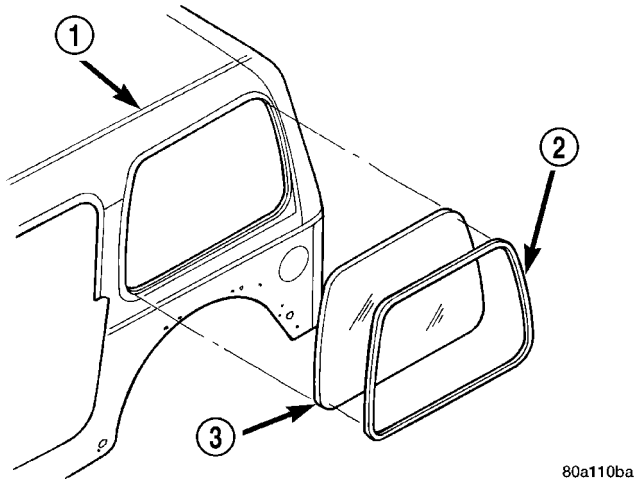

Fig. 1 HARD TOP QUARTER GLASS

1 - HARD TOP
2 - QUARTER GLASS REVEAL MOLDING
3 - QUARTER GLASS

INSTALLATION

(1) Trim the urethane from the quarter glass opening fence. Leave a 3 mm (0.1 in.) level base of urethane on the quarter glass opening fence.

(2) Place replacement quarter glass into quarter glass opening and position glass in the center of the opening against fence.

(3) Verify the glass lays evenly against the fence at the sides, top and bottom of the replacement quarter glass. Next, make alignment marks on glass and top with a grease pencil.

(4) Remove replacement quarter glass from opening.

(5) Position the quarter glass inside up on a suitable work surface.

WARNING: DO NOT USE SOLVENT BASED GLASS CLEANER TO CLEAN QUARTER GLASS BEFORE APPLYING GLASS PREP AND PRIMER. POOR ADHESION CAN RESULT.

(6) Clean inside of quarter glass with ammonia based glass cleaner and lint-free cloth.

(7) Clean the outer edge of the window glass with naphtha or a similar product.

(8) Apply molding to perimeter of quarter glass. The butt weld of the molding should be centered at the bottom edge of the quarter glass.

(9) Apply Glass Prep adhesion promoter 25 mm (1 in.) wide around perimeter of the quarter glass and wipe with clean/dry lint-free cloth until no streaks are visible.

(10) Apply Glass Primer 25 mm (1 in.) wide around perimeter of quarter glass. Allow at least three minutes drying time.

(11) Apply Pinchweld primer 15 mm (0.75 in.) wide around the quarter glass fence. Allow at least three minutes drying time.

(12) Apply a 10 mm (0.4 in.) diameter bead of urethane to the center of the quarter glass fence surface area.

CAUTION: Be prepared to install the quarter glass immediately after applying the adhesive. The adhesive begins to cure within 10-15 minutes.

(13) Align the quarter glass with the grease pencil marks and position quarter glass on fence.

(14) Push the quarter glass inward until the reveal molding is seated on the hardtop. Use care to avoid excessive squeeze-out of adhesive.

(15) Open windows and liftgate to prevent pressure build-up while the urethane is curing.

(16) Apply 150 mm (6 in.) lengths of 50 mm (2 in.) masking tape spaced 250 mm (10 in.) apart to hold quarter glass in place until urethane cures.

(17) After urethane has cured, remove tape strips and water test quarter glass to verify repair.

WINDSHIELD

REMOVAL

The windshield is positioned in the reveal molding and is bonded to the windshield frame with urethane adhesive. The windshield interior trim molding is positioned onto the inner windshield frame pinchweld.

(1) Cover body surface areas with protective covering to avoid paint damage and extra clean-up time.

(2) Remove the windshield wiper arms. (Refer to 8 - ELECTRICAL/WIPERS/WASHERS/WIPER ARMS - REMOVAL)

(3) Remove the rear view mirror. (Refer to 23 - BODY/INTERIOR/REAR VIEW MIRROR - REMOVAL)

(4) Using a razor knife, slide the blade between the windshield glass and the inboard edge of the reveal molding.

(5) Cut around the interior perimeter of the reveal molding and sever the cap of the reveal molding.

(6) Using a cold knife, cut the urethane around the perimeter of the windshield.

(7) Remove the windshield glass from the frame (Fig. 2).

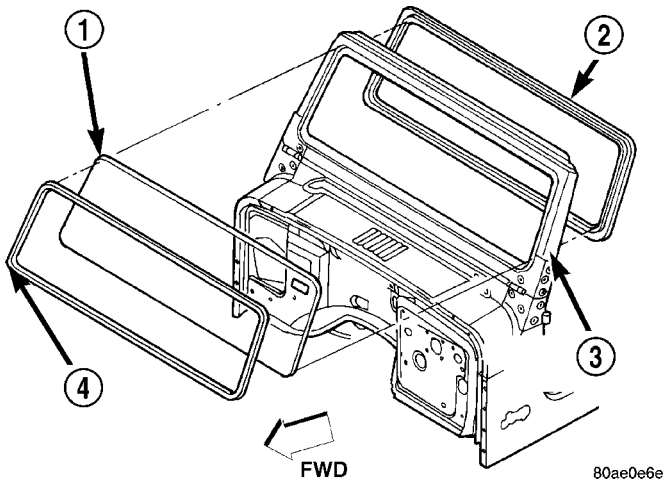

Fig. 2 WINDSHIELD

1 - WINDSHIELD
2 - INNER TRIM MOLDING
3 - WINDSHIELD FRAME
4 - WINDSHIELD MOLDING

INSTALLATION

(1) Trim the urethane from the pinchweld flanges. Leave a 3 mm (0.1 in.) level base of urethane on the pinchweld flanges.

(2) Place replacement windshield into windshield opening and position glass in the center of the opening against pinchweld flange.

(3) Verify the glass lays evenly against the pinch weld fence at the sides, top and bottom of the replacement windshield. If not, the pinchweld flange must be formed to the shape of the new glass. Next, make alignment marks on glass and body with a grease pencil.

(4) Remove replacement windshield from windshield opening.

(5) Position the windshield inside up on a suitable work surface with two padded, wood 10 cm by 10 cm by 50 cm (4 in. by 4 in. by 20 in.) blocks, placed parallel 75 cm (2.5 ft.) apart (Fig. 3).

WARNING: DO NOT USE SOLVENT BASED GLASS CLEANER TO CLEAN WINDSHIELD BEFORE APPLYING GLASS PREP AND PRIMER. POOR ADHESION CAN RESULT.

(6) Clean inside of windshield with ammonia based glass cleaner and lint-free cloth.

(7) Apply molding to perimeter of windshield. The butt weld of the molding should be centered at the bottom edge of the windshield.

(8) Apply Glass Prep adhesion promoter 25 mm (1 in.) wide around perimeter of windshield and wipe with clean/dry lint-free cloth until no streaks are visible.

(9) Apply Glass Primer 25 mm (1 in.) wide around perimeter of windshield. Allow at least three minutes drying time.

(10) Apply Pinchweld primer 15 mm (.75 in.) wide around the windshield fence. Allow at least three minutes drying time.

(11) Apply a urethane bead (Fig. 4) on the pinchweld flange surface area 6 mm (.25 in.) from the outboard edge.

CAUTION: Be prepared to install the windshield glass immediately after applying the adhesive. The adhesive begins to cure within 10-15 minutes.

(12) Align the windshield with the grease pencil marks and position windshield on pinchweld flanges.

(13) Push the windshield glass inward until the reveal molding is seated on the windshield frame. Use care to avoid excessive squeeze-out of adhesive.

(14) Open windows and liftgate to prevent pressure build-up while the urethane is curing.

(15) Starting in each corner, apply 150 mm (6 in.) lengths of 50 mm (2 in.) masking tape spaced 250 mm (10 in.) apart to hold windshield in place until urethane cures.

(16) Install the rear view mirror support bracket. (Refer to 23 - BODY/INTERIOR/REAR VIEW MIRROR SUPPORT BRACKET - INSTALLATION)

(17) Install the rear view mirror. (Refer to 23 - BODY/INTERIOR/REAR VIEW MIRROR - INSTALLATION)

WINDSHIELD (Continued)

(18) Install the wiper arms. (Refer to 8 - ELECTRICAL/WIPERS/WASHERS/WIPER ARMS - INSTALLATION)

(19) After urethane has cured, remove tape strips and water test windshield to verify repair.

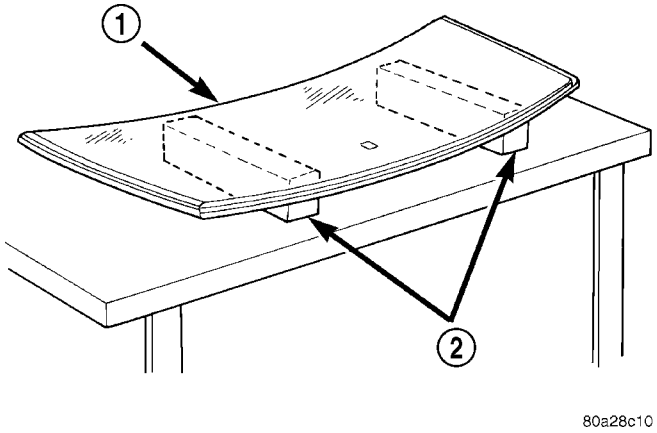

Fig. 3 WORK SURFACE SET UP AND MOLDING INSTALLATION

1 - WINDSHIELD AND MOLDINGS
2 - BLOCKS

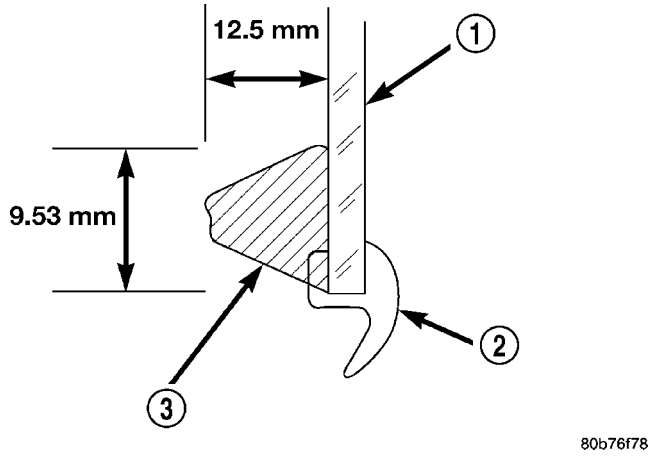

Fig. 4 URETHANE BEAD

1 - WINDSHIELD
2 - MOLDING
3 - URETHANE BEAD

WINDSHIELD HINGE

REMOVAL

NOTE: If both hinges are to be replaced, the windshield must be tilted to the full forward position. (Refer to 23 - BODY/STATIONARY GLASS/WINDSHIELD - REMOVAL)

(1) Remove door. (Refer to 23 - BODY/DOOR - FRONT/DOOR - REMOVAL)

(2) Remove the bolts attaching the hinge to the cowl (Fig. 5).

(3) Remove the bolts attaching the hinge to the windshield frame.

(4) Separate the hinge from the vehicle.

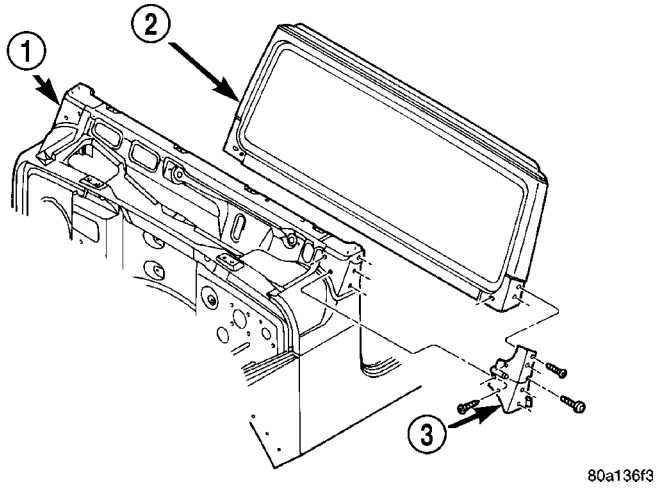

Fig. 5 WINDSHIELD HINGE

1 - COWL
2 - WINDSHIELD
3 - WINDSHIELD HINGE

INSTALLATION

(1) Paint as required.

(2) Clean the contact surface of the hinge and cowl with isopropyl alcohol or equivalent.

(3) Apply a 4 mm bead of Mopar® Vinyl Acrylic Sealant or equivalent around the perimeter of the hinge contact surface. The bead should be 10 mm inboard of the edge.

(4) Position the hinge on the vehicle.

(5) Install the bolts attaching the hinge to the windshield frame.

(6) Install the bolts attaching the hinge to the cowl.

(7) Ensure that the sealant provides complete coverage. Wipe away excess sealant.

(8) Install door. (Refer to 23 - BODY/DOOR - FRONT/DOOR - INSTALLATION)

WINDSHIELD FRAME

REMOVAL

(1) Unlatch top.
(2) Remove the bolts attaching the sport bar to the windshield frame.
(3) Remove the windshield wiper arms. (Refer to 8 - ELECTRICAL/WIPERS/WASHERS/WIPER ARMS - REMOVAL)
(4) Remove the torx screw closest to the hinge pivot point and tilt the windshield forward.
(5) Remove the torx screws attaching the windshield hinge to the windshield frame (Fig. 6).
(6) Separate the windshield frame from the vehicle.

INSTALLATION

NOTE: Inspect windshield to cowl seal for damage and confirm correct position.

(1) Position the windshield frame on the vehicle.
(2) Install the torx screws attaching the windshield hinge to the windshield frame.
(3) Tilt the windshield rearward.
(4) Install the torx screw closest to the hinge pivot point and lock the windshield in the upright position.

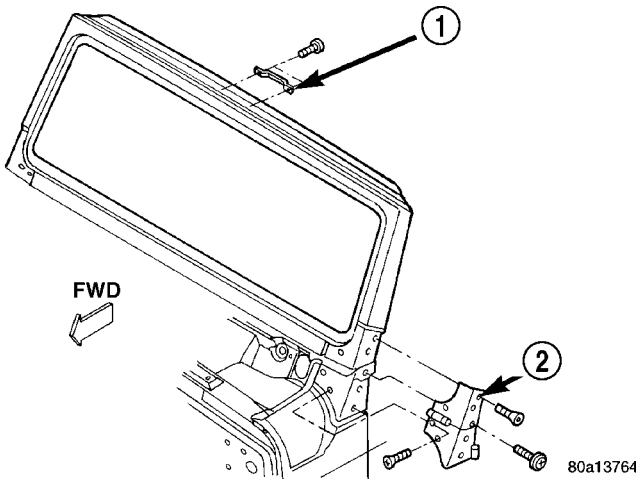

Fig. 6 WINDSHIELD FRAME

1 - WINDSHIELD HOLD DOWN LOOP
2 - WINDSHIELD HINGE

(5) Install the windshield wiper arms. (Refer to 8 - ELECTRICAL/WIPERS/WASHERS/WIPER ARMS - INSTALLATION)
(6) Install the bolts attaching the sport bar to the windshield frame.
(7) Latch top.

WEATHERSTRIP/SEALS

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HOOD TO COWL WEATHERSTRIP

REMOVAL

(1) Carefully separate the weatherstrip from the cowl flange (Fig. 1).

INSTALLATION

(1) Position the weatherstrip on the cowl flange and press it into place.

FULL DOOR INNER BELT WEATHERSTRIP

DESCRIPTION - FULL DOOR INNER BELT WEATHERSTRIP

The inner belt weatherstrip is attached to the door trim panel and is not serviceable. If the inner belt weatherstrip needs to be replaced, replace the door trim panel.

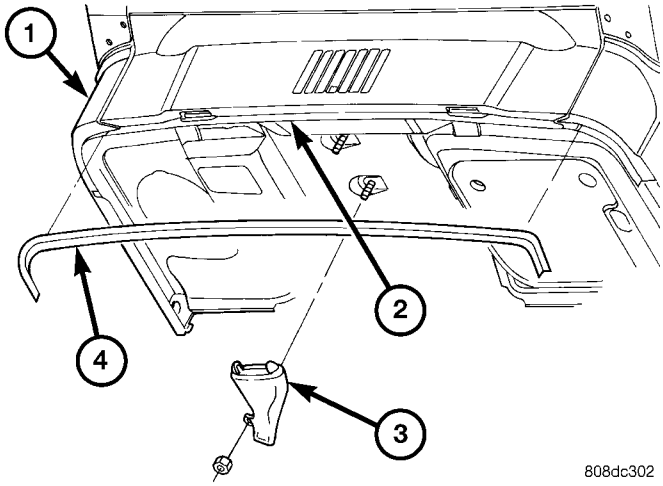

Fig. 1 HOOD TO COWL WEATHERSTRIP

1 - COWL
2 - COWL FLANGE
3 - COWL PLENUM DRAIN
4 - HOOD TO COWL WEATHERSTRIP

FULL DOOR OUTER BELT SEAL

REMOVAL

(1) Remove the door sail panel.
(2) Disengage the clips attaching the outer belt seal to the door (Fig. 2).
(3) Separate the seal from the door.

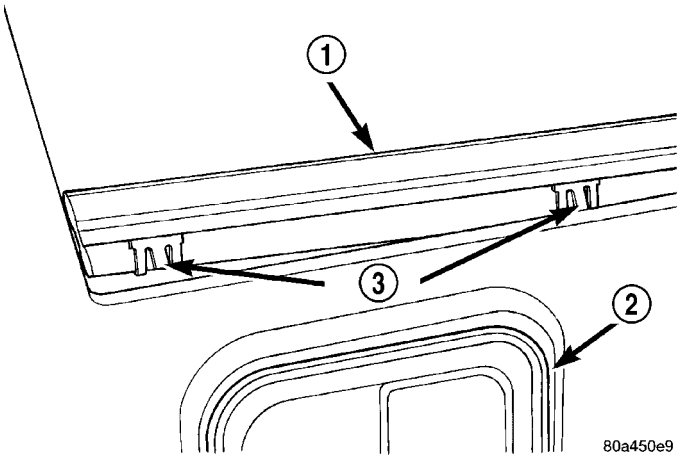

Fig. 2 FULL DOOR OUTER BELT SEAL

1 - OUTER BELTLINE WEATHERSTRIP
2 - OUTSIDE DOOR HANDLE
3 - CLIPS

INSTALLATION

(1) Position the seal on the door.
(2) Engage the clips attaching the outer belt seal to the door.
(3) Install the door sail panel.

FULL DOOR WEATHERSTRIP

REMOVAL

NOTE: The upper portion of the weatherstrip is seated into a channel around the window opening frame. The lower portion of the weatherstrip is attached to the door with push-in fasteners.

(1) Peel the weatherstrip from the channel.
(2) Remove the push-in fasteners attaching the weatherstrip to the door (Fig. 3).

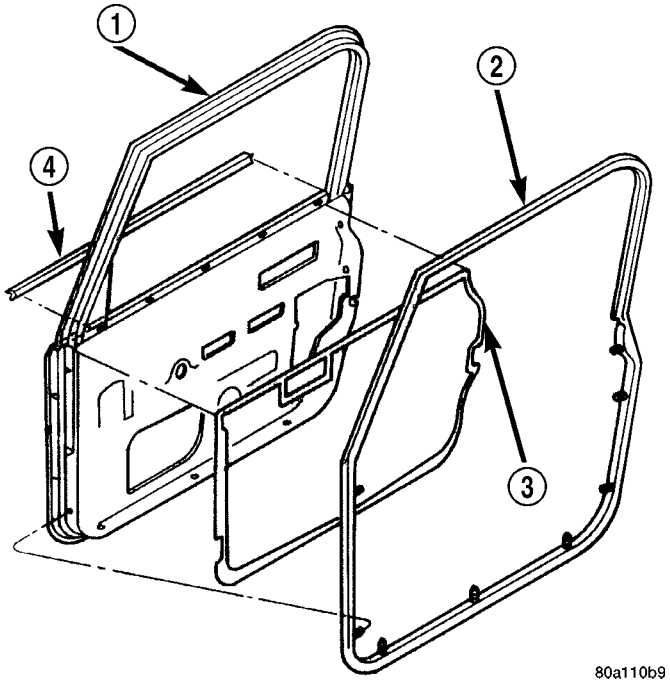

Fig. 3 FULL DOOR WEATHERSTRIP

1 - FULL DOOR
2 - WEATHERSTRIP
3 - WATERDAM
4 - OUTER DOOR BELT SEAL

INSTALLATION

(1) Position the lower part of the weatherstrip and seat the push-in fasteners fully.
(2) Install the weatherstrip in the upper weatherstrip channel seat fully.

HALF DOOR WEATHERSTRIP

REMOVAL

NOTE: The weatherstrip is attached to the door with push-in fasteners.

(1) Remove trim panel. (Refer to 23 - BODY/HALF DOOR/TRIM PANEL - REMOVAL)
(2) Remove window retaining sleeve.
(3) Remove the push-in fasteners attaching the weatherstrip to the door. (Fig. 4)

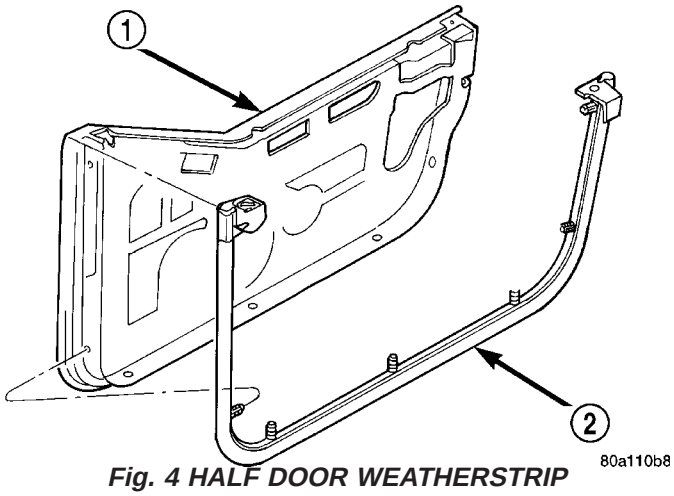

Fig. 4 HALF DOOR WEATHERSTRIP

1 - HALF DOOR
2 - WEATHERSTRIP

INSTALLATION

NOTE: The weatherstrip is attached to the door with push-in fasteners.

(1) Position the weatherstrip onto the door and seat the push-in fasteners fully.
(2) Install window retaining sleeve.
(3) Install trim panel. (Refer to 23 - BODY/HALF DOOR/TRIM PANEL - INSTALLATION)

TAILGATE WEATHERSTRIP AND CHANNEL

REMOVAL

(1) Open the tailgate.
(2) Remove the push-in fasteners attaching the weatherstrip to the top corners of the tailgate (Fig. 5).
(3) Peel the weatherstrip from the upper tailgate corners.
(4) Slide the weatherstrip out of the tailgate channel.
(5) If the weatherstrip channel requires replacement, peel the weatherstrip channel from the tailgate.

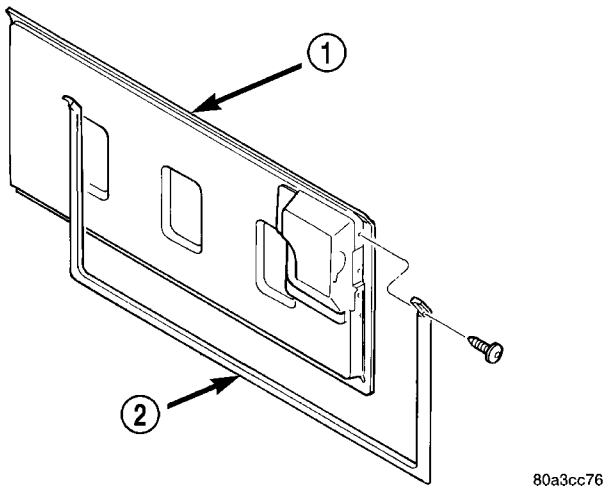

Fig. 5 TAILGATE WEATHERSTRIP

1 - TAILGATE
2 - WEATHERSTRIP

INSTALLATION

(1) If the weatherstrip channel is being replaced;
(a) Clean the channel contact surface on the tailgate with isopropyl alcohol, or equivalent.
(b) Peel the paper backing from the weatherstrip channel.
(c) Install the push pin fasteners attaching the weatherstrip to the tailgate.
(d) Position weatherstrip channel to the tailgate and press into place.
(e) Use hand pressure or a roller to wet out the tape adhesive holding the weatherstrip channel to the tailgate.
(2) Install the push pin fasteners attaching the weatherstrip to the tailgate.
(3) Slide the weatherstrip into the weatherstrip channel.

WINDSHIELD FRAME WEATHERSTRIP

REMOVAL

UPPER (Header)

(1) Disconnect the top from the windshield frame.
(2) Disengage the push-in fasteners attaching the weatherstrip to the windshield frame.
(3) Peel the weatherstrip from the frame.

LOWER

NOTE: The lower windshield frame weatherstrip can be removed with the frame tilted forward to the full horizontal position.

(1) Remove the wiper arms. (Refer to 8 - ELECTRICAL/WIPERS/WASHERS/WIPER ARMS - REMOVAL)
(2) Disconnect the top from the windshield frame.
(3) Remove the cowl grille. (Refer to 23 - BODY/EXTERIOR/COWL GRILLE AND SCREEN - REMOVAL)
(4) Remove the torx screws on each side of the windshield frame allowing the windshield frame to tilt to the full horizontal position.
(5) Disengage the outboard push-in fasteners at the top of cowl on each hinge pillar (Fig. 6).
(6) Remove the weatherstrip from the cowl.

INSTALLATION

UPPER (Header)

(1) Clean the seal contact surface on the windshield frame with isopropyl alcohol or equivalent.

NOTE: Ensure that the contact surface is dry and free from any residue, poor adhesion will result.

(2) Position the weatherstrip on the windshield frame, align the push-in fasteners and press it into place (Fig. 6).
(3) Remove adhesive backing from the bottom of the weatherstrip.
(4) Using forceful hand pressure, seat the adhesive on the contact surface.

NOTE: If tape surface becomes contaminated, it will not adhere to the windshield frame.

(5) Connect the top to the windshield frame.

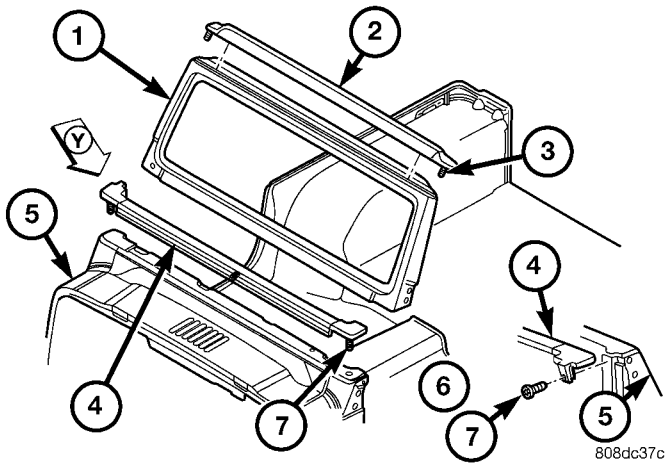

Fig. 6 WINDSHIELD FRAME WEATHERSTRIP

1 - WINDSHIELD FRAME
2 - UPPER HEADER WEATHERSTRIP
3 - PUSH-IN FASTENER
4 - WINDSHIELD TO COWL WEATHERSTRIP
5 - COWL
6 - VIEW OF ARROW Y
7 - PUSH-IN FASTENERS (2 per end)

LOWER

(1) Position the weatherstrip on the cowl.
(2) Align the outer push-in fasteners and press them into place.
(3) Tilt the windshield frame rearward to the full vertical position.

NOTE: Ensure weatherstrip outer detail mates with the a-pillar trim. Outer detail must be flush with trim surface for proper sealing performance.

(4) Install the torx screws on each side of the windshield securing the windshield frame.
(5) Connect the top to the windshield frame.
(6) Install cowl grille. (Refer to 23 - BODY/EXTERIOR/COWL GRILLE AND SCREEN - INSTALLATION)
(7) Install the wiper arms. (Refer to 8 - ELECTRICAL/WIPERS/WASHERS/WIPER ARMS - INSTALLATION)

BODY STRUCTURE

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GAP AND FLUSH

SPECIFICATIONS - GAP AND FLUSH

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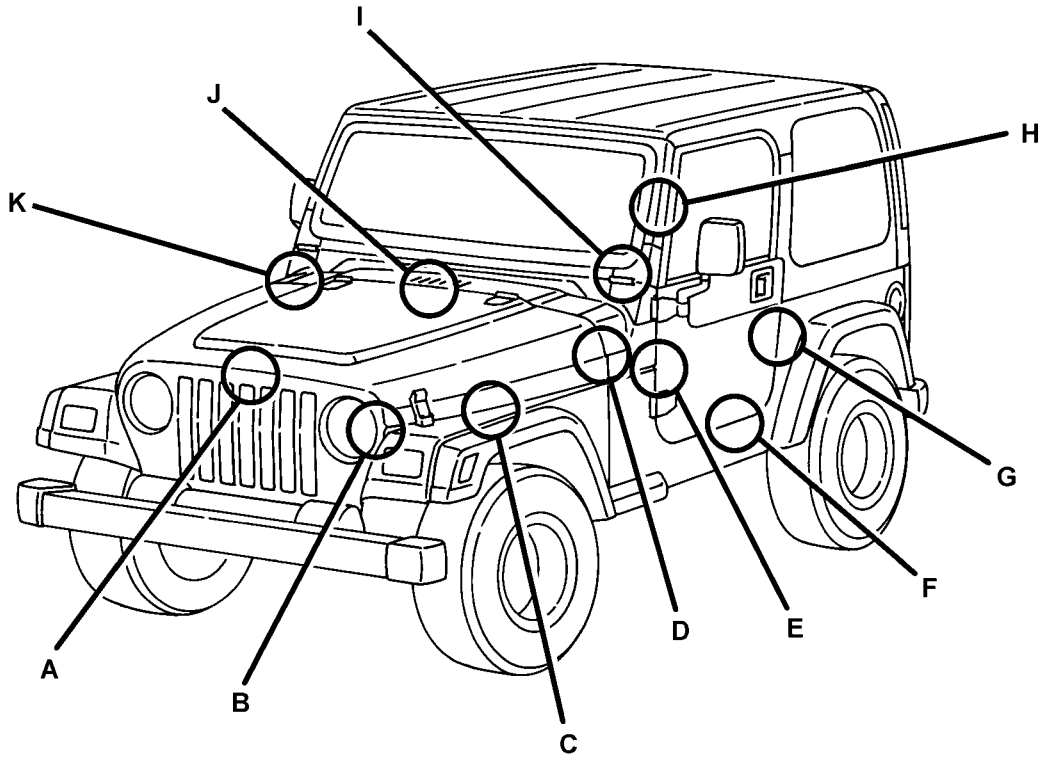
GAP AND FLUSH (Continued)



	DESCRIPTION	GAP	FLUSH
A	Hood to Radiator Guard	7.0 +/- 1.5	N/A
B	Hood to Lamp Bezel	4.0 +/- 2.0	N/A
C	Hood to Fender	7.0 +/- 1.5	N/A
D	Hood to Cowl at Side	7.0 +/- 1.5	0.0 +/- 1.5
E	Door to Cowl	5.0 +/- 1.5	0.0 +/- 1.5
F	Door to Bodyside at Bottom	5.0 +/- 1.5	0.0 +/- 1.5
G	Door to Bodyside at Rear	5.0 +/- 1.5	0.0 +/- 1.5
H	Door to Windshield Frame	5.0 +/- 1.5	0.0 +/- 2.0
I	Windshield Hinge to Windshield Frame	N/A	0.0 +/- 1.0
J	Hood to Cowl	7.0 +/- 1.5	0.0 +/- 1.5
K	Cowl Top Grille to Cowl Top End	5.0 +/- 1.5	0.0 +/- 1.5

NOTE: ALL MEASUREMENTS ARE IN MM.

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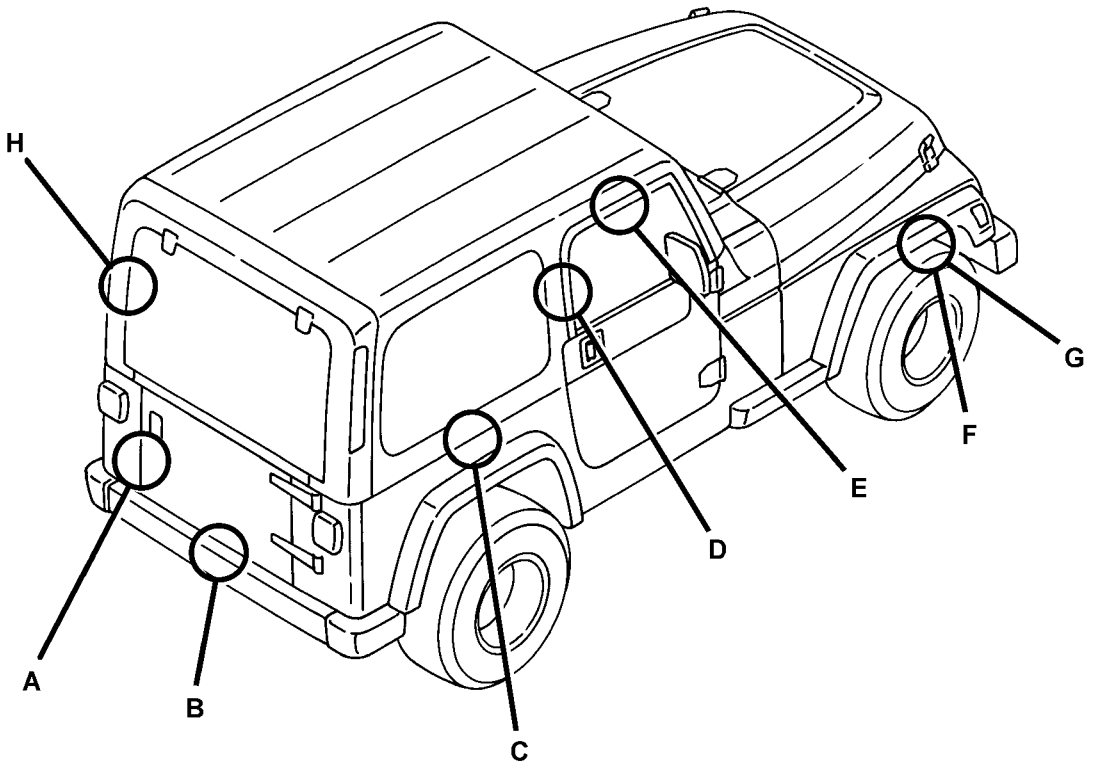
Fig. 1 FRONT/SIDE DIMENSIONS

GAP AND FLUSH (Continued)



	DESCRIPTION	GAP	FLUSH
A	Tailgate to Rear Corner Panel	5.0 +/- 1.5	0.0 +/- 1.5
B	Tailgate to Rear Center Lower	5.0 +/- 1.5	0.0 +/- 1.5
C	Hardtop to Bodyside	N/A	2.0 +/- 1.5
D	Door Header to Hardtop at Rear	4.5 +/- 1.5	0.0 +/- 2.0
E	Door to Top at Top	5.0 +/- 1.5	0.0 +/- 2.0
F	Wheelflare to Fender (Highline)	N/A	3.0 +/- 1.5
G	Wheelflare to Fender (Base Level)	N/A	14.0 +/- 1.5
H	Liftglass to Hardtop	4.5 +/- 2.0	4.0 +/- 2.0

NOTE: ALL MEASUREMENTS ARE IN MM.

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Fig. 2 REAR/SIDE DIMENSIONS

OPENING DIMENSIONS

SPECIFICATIONS

BODY OPENING DIMENSIONS

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DOOR OPENING	4
QUARTER WINDOW OPENING	5
ENGINE COMPARTMENT	6
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- A & B. Center of radius at bottom to center of radius top.

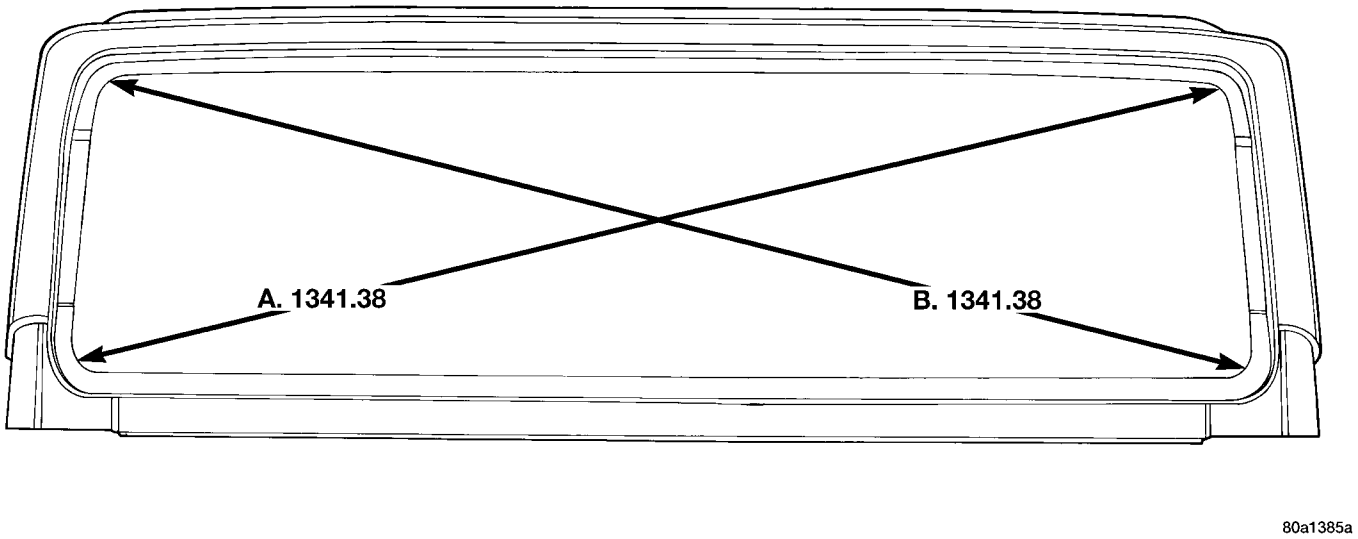

Fig. 3 WINDSHIELD OPENING

OPENING DIMENSIONS (Continued)

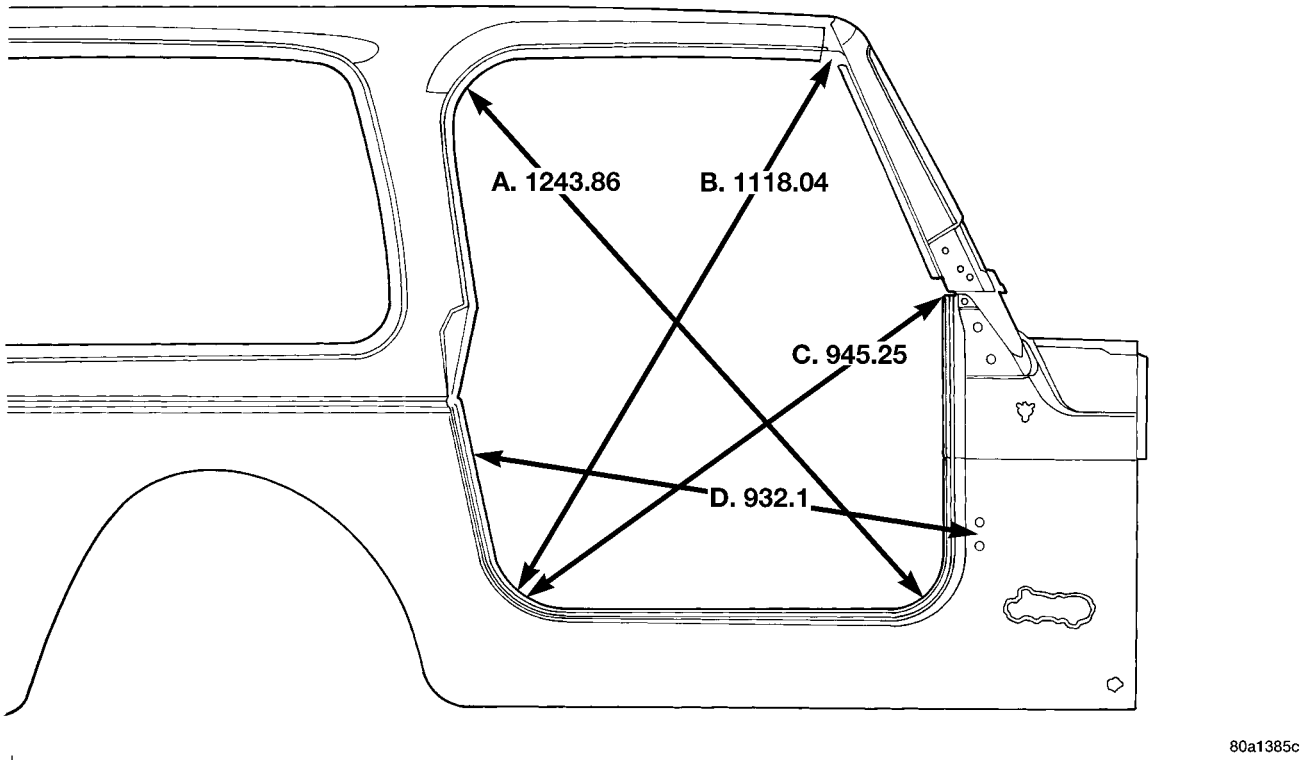

Fig. 4 DOOR OPENING

OPENING DIMENSIONS (Continued)

- A. Center of radius at bottom front to center of radius at top rear.
- B. Center of door lower rear corner to center of top of windshield frame.
- C. Center of door lower rear corner to top of cowl.
- D. Center of door hinge mount to center of door striker mount.

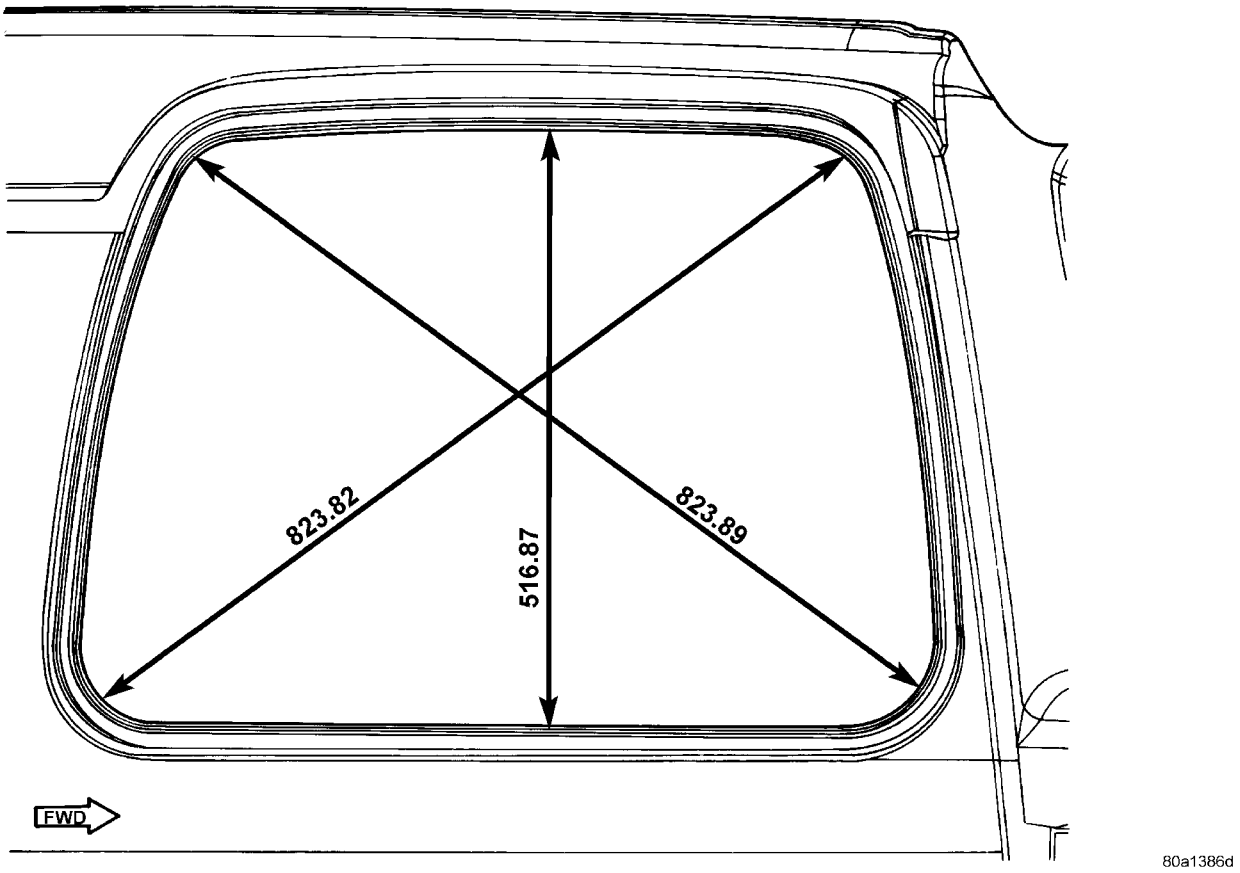

Fig. 5 QUARTER WINDOW OPENING

OPENING DIMENSIONS (Continued)

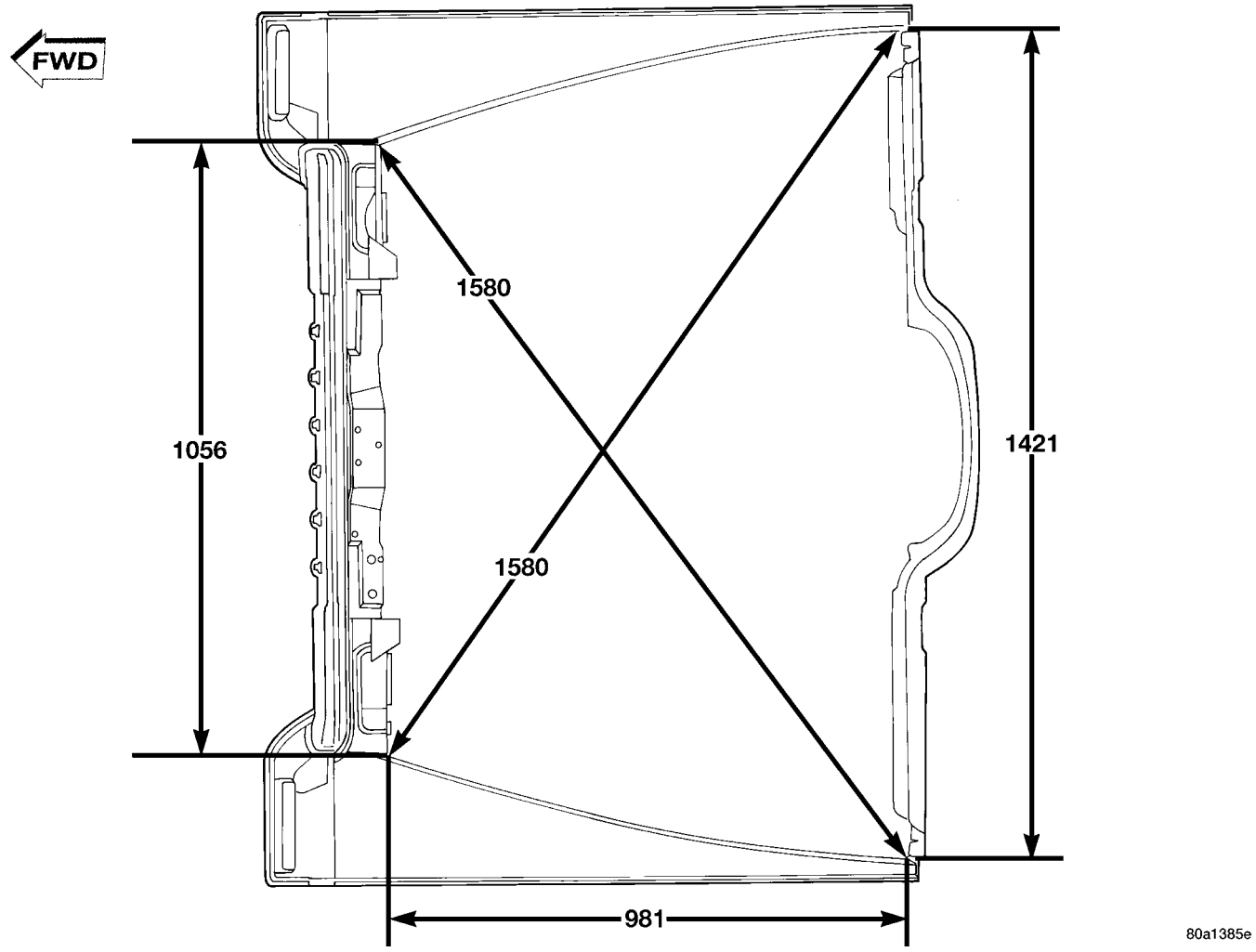

Fig. 6 ENGINE COMPARTMENT

OPENING DIMENSIONS (Continued)

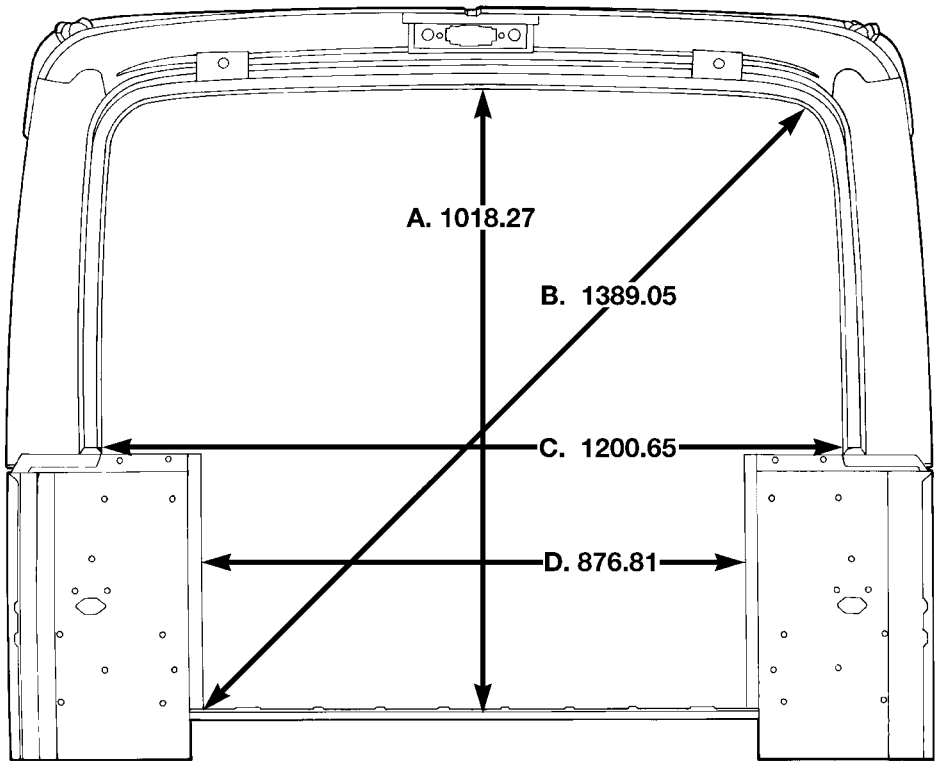

Fig. 7 TAILGATE AND LIFTGATE OPENING

- A. Center of liftgate opening to floor.
- B. Center of radius upper corner to center of body and floor corner.
- C. Liftgate opening distance.
- D. Tailgate opening distance.

SEALER LOCATIONS

SPECIFICATIONS

SEALER LOCATIONS

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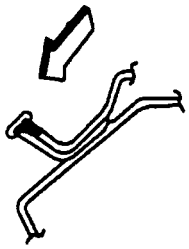
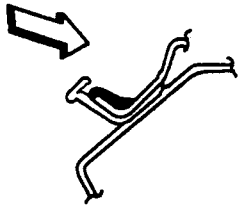
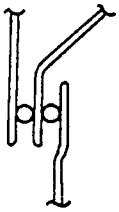
SEALER LOCATIONS (Continued)



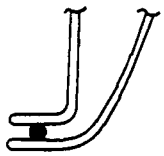
HOLD GUN NOZZLE IN DIRECTION OF ARROW IN ORDER TO EFFECTIVELY SEAL METAL JOINTS.



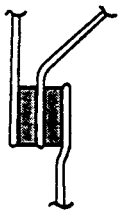
DO NOT HOLD GUN NOZZLE IN DIRECTION OF ARROW. SEALER APPLIED AS SHOWN IS INEFFECTIVE.



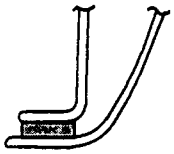
3 METAL THICKNESS



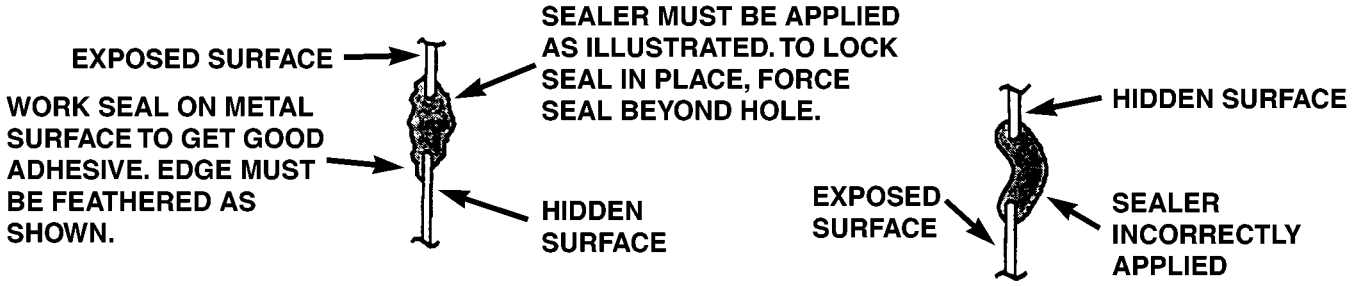
2 METAL THICKNESS



3 METAL THICKNESS



2 METAL THICKNESS



SYMBOLS

- THUMBGRADEABLE SEALER
- EXTRUDABLE THERMOPLASTIC
- EXPOSED THERMOPLASTIC SEALANT
- zzzzzzzz HIDDEN SEALANT

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Fig. 8 APPLICATION METHODS

SEALER LOCATIONS (Continued)

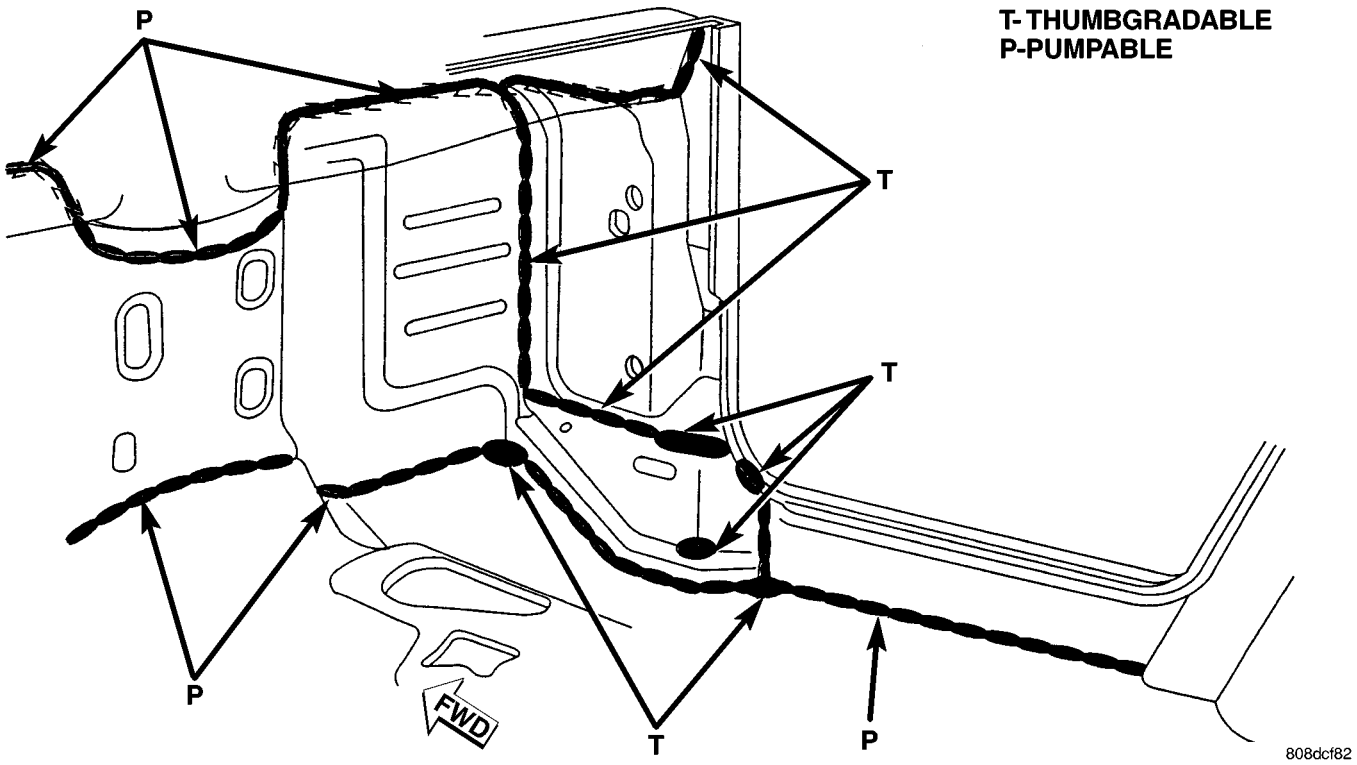

Fig. 9 COWL AND PLENUM

SEALER LOCATIONS (Continued)

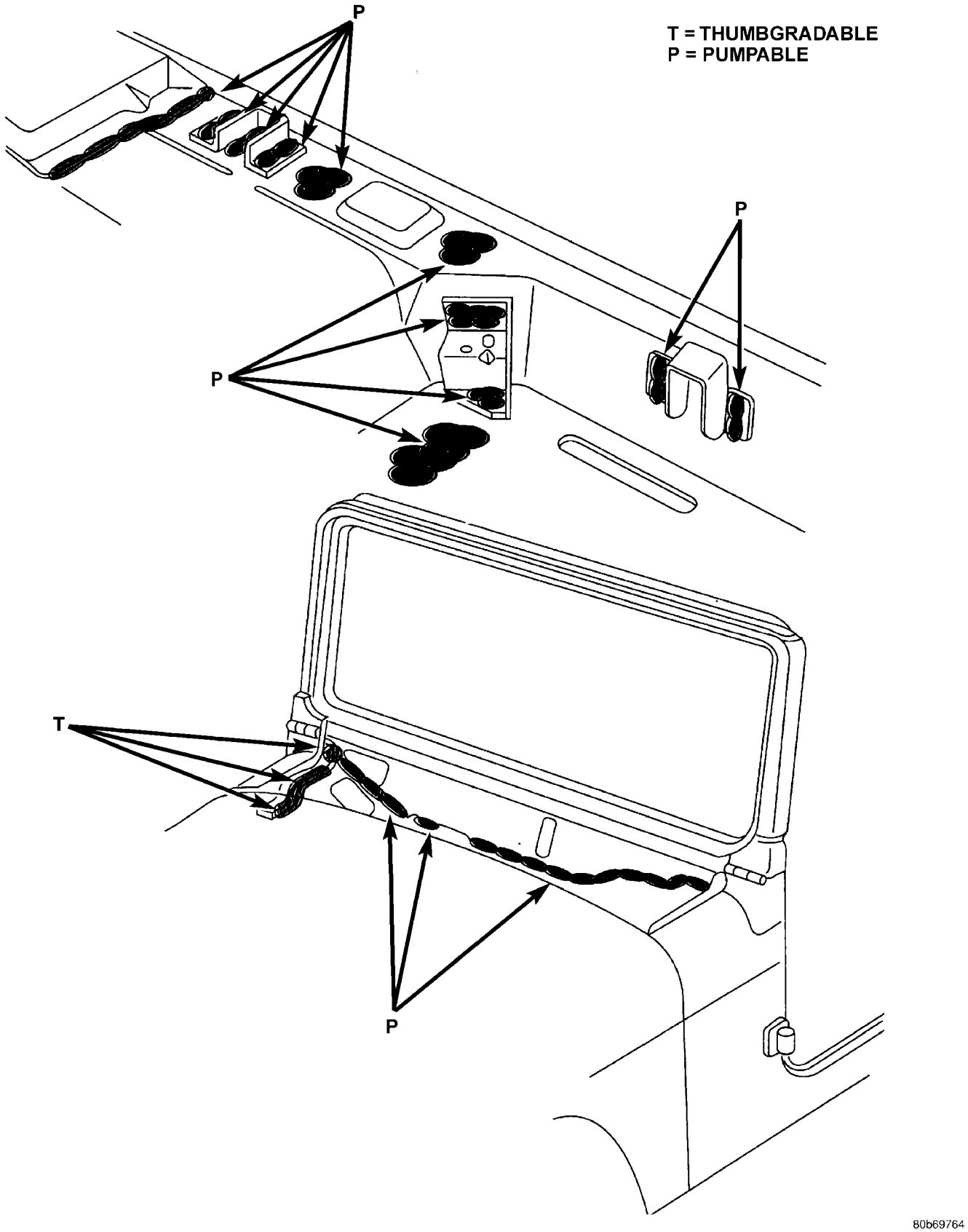

Fig. 10 COWL TOP END AND PLENUM

SEALER LOCATIONS (Continued)

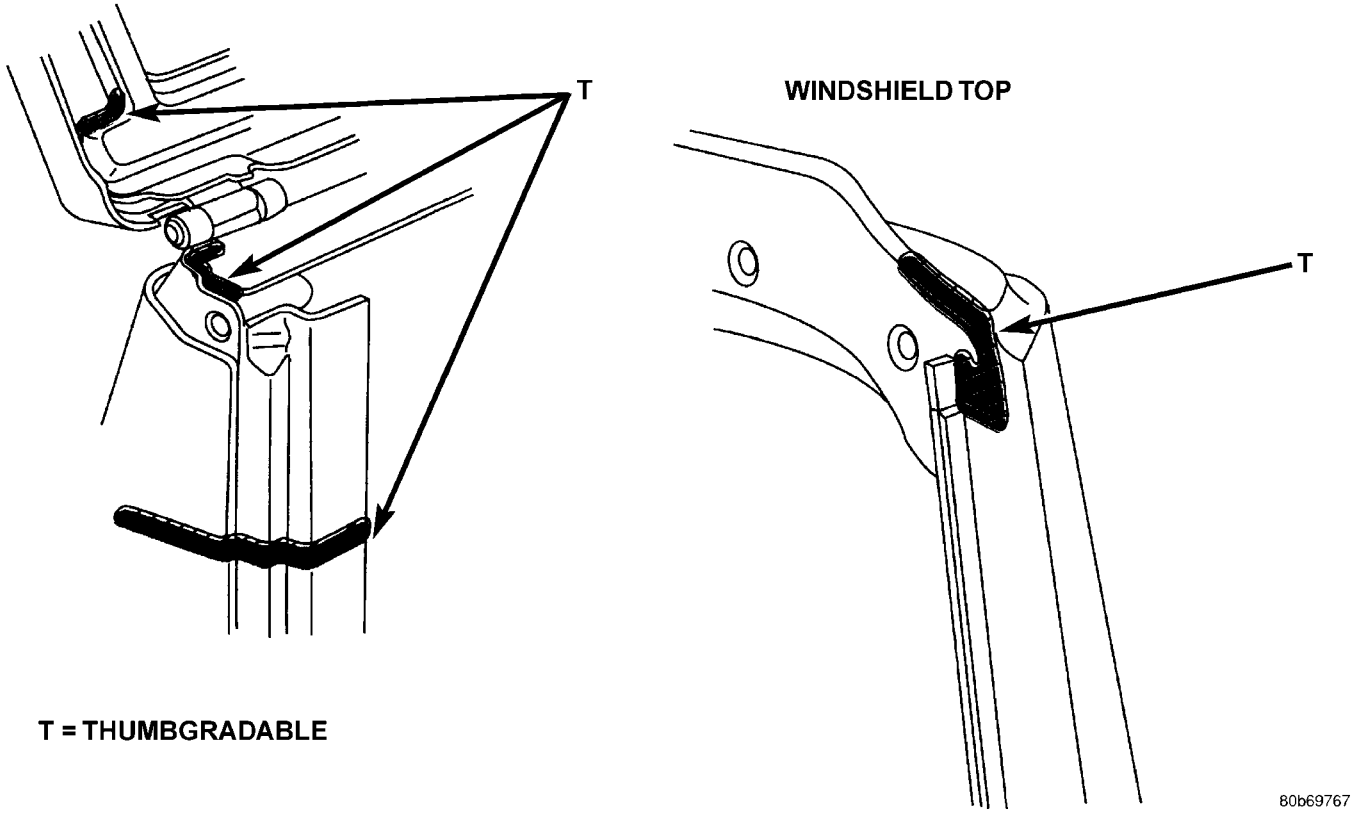

Fig. 11 COWL AND WINDSHIELD OUTER PANEL

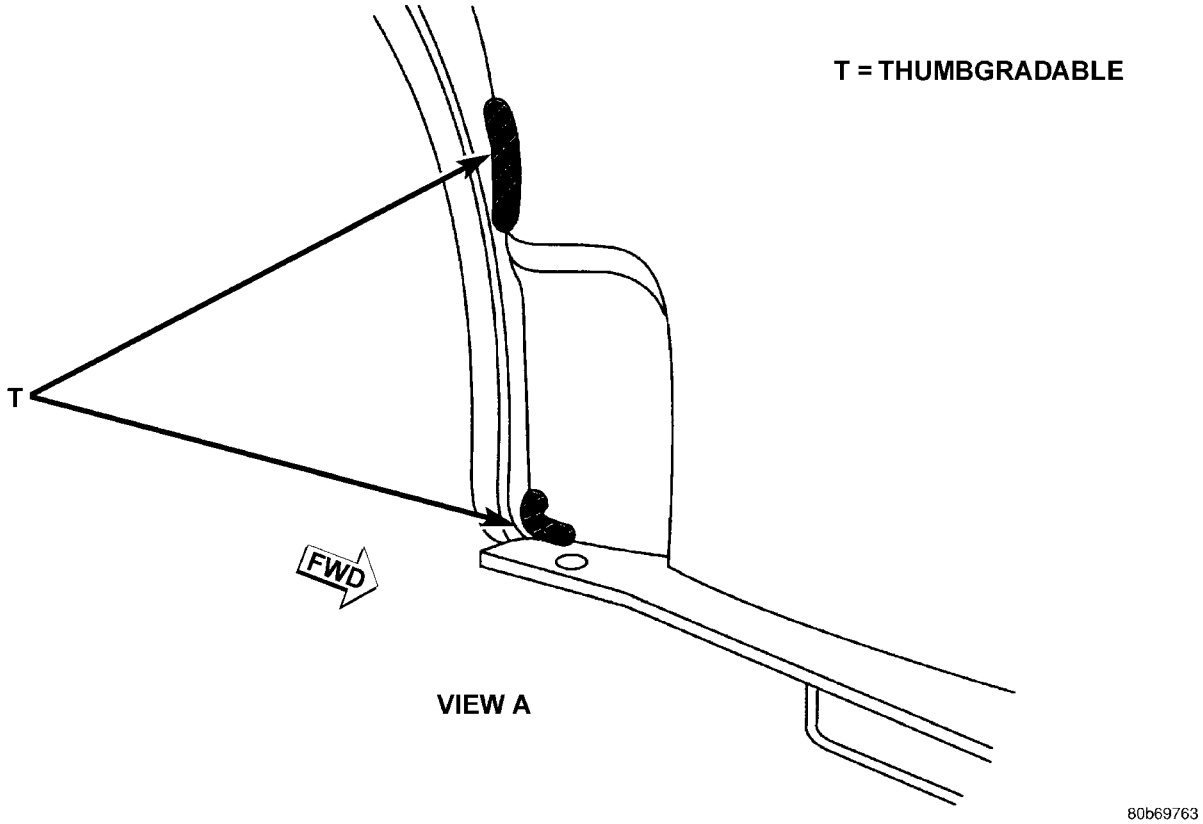

Fig. 12 INNER WHEELHOUSE

SEALER LOCATIONS (Continued)

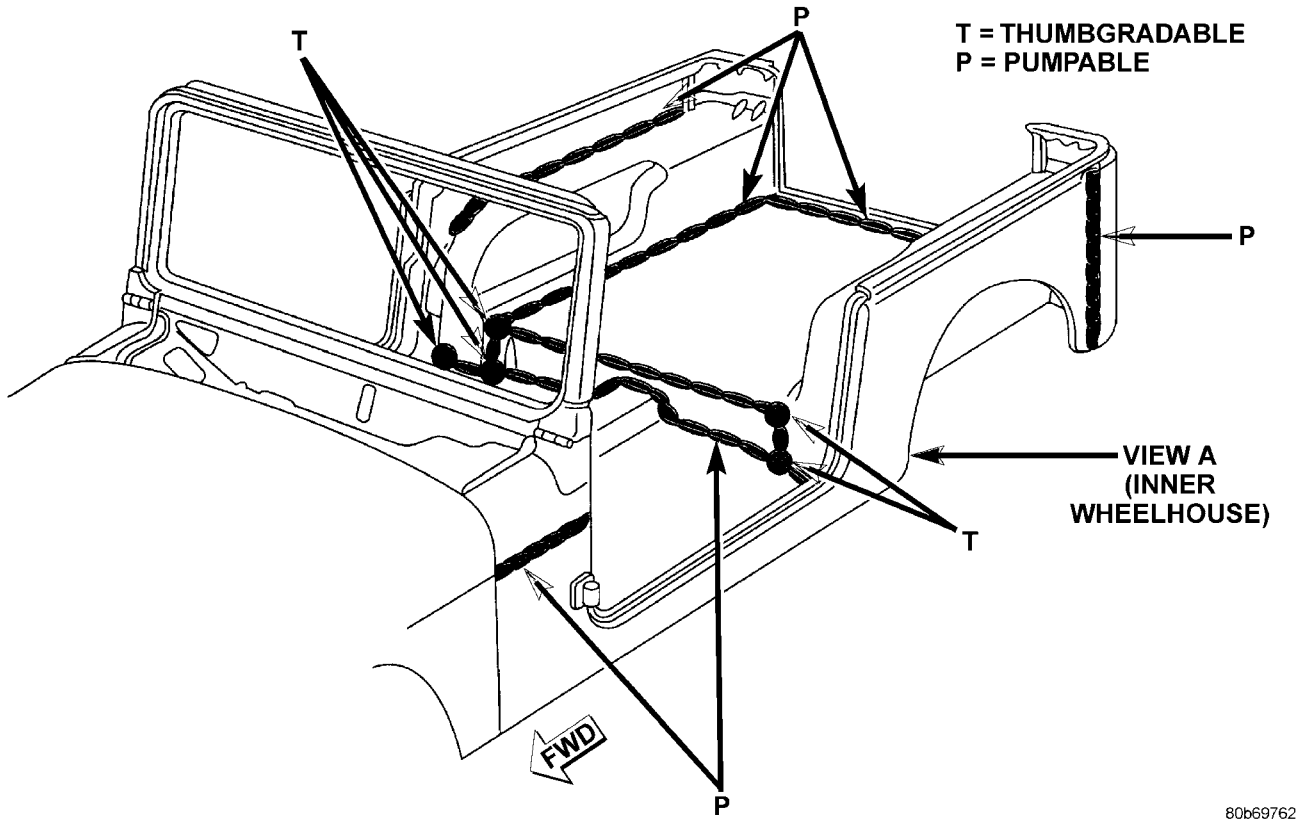

Fig. 13 REAR FLOOR RISER AND OUTER BODY SEAMS

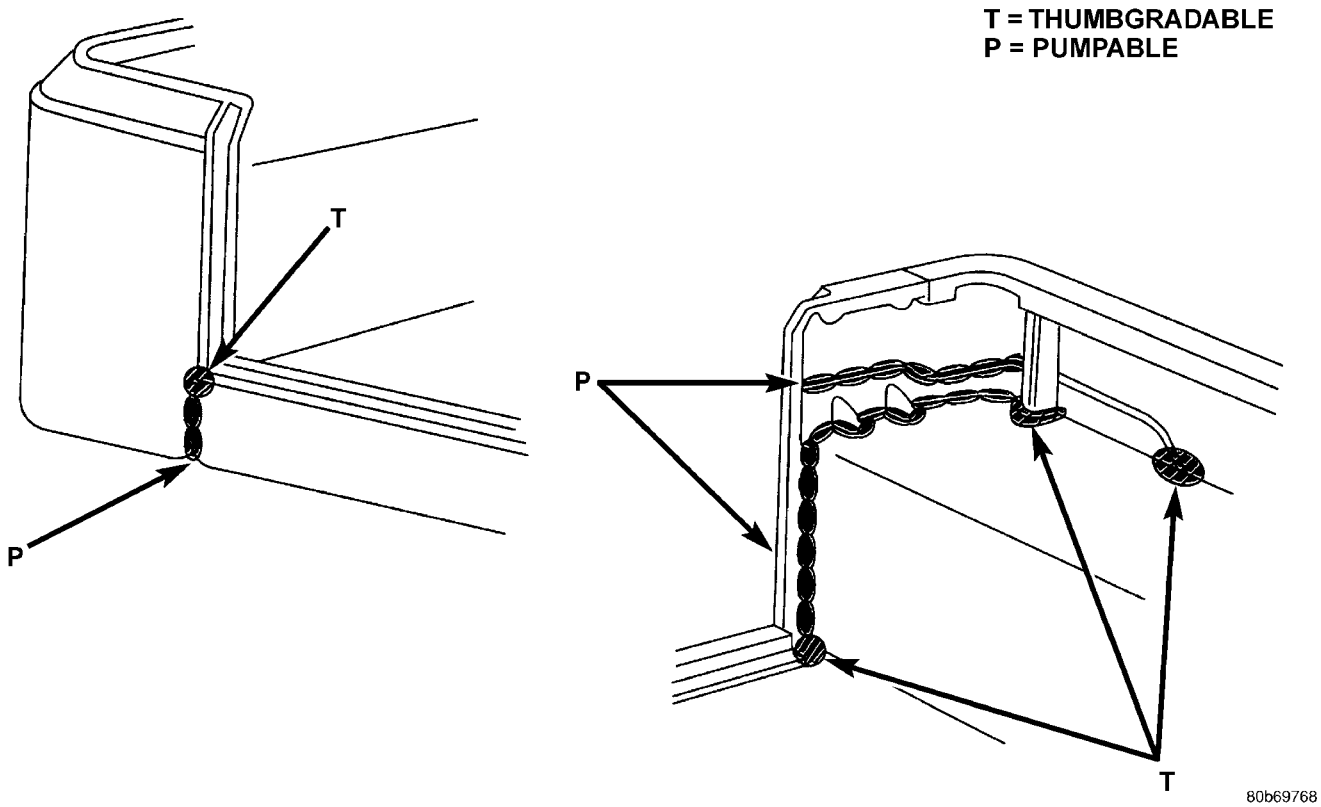

Fig. 14 REAR CORNER PANEL

SEALER LOCATIONS (Continued)

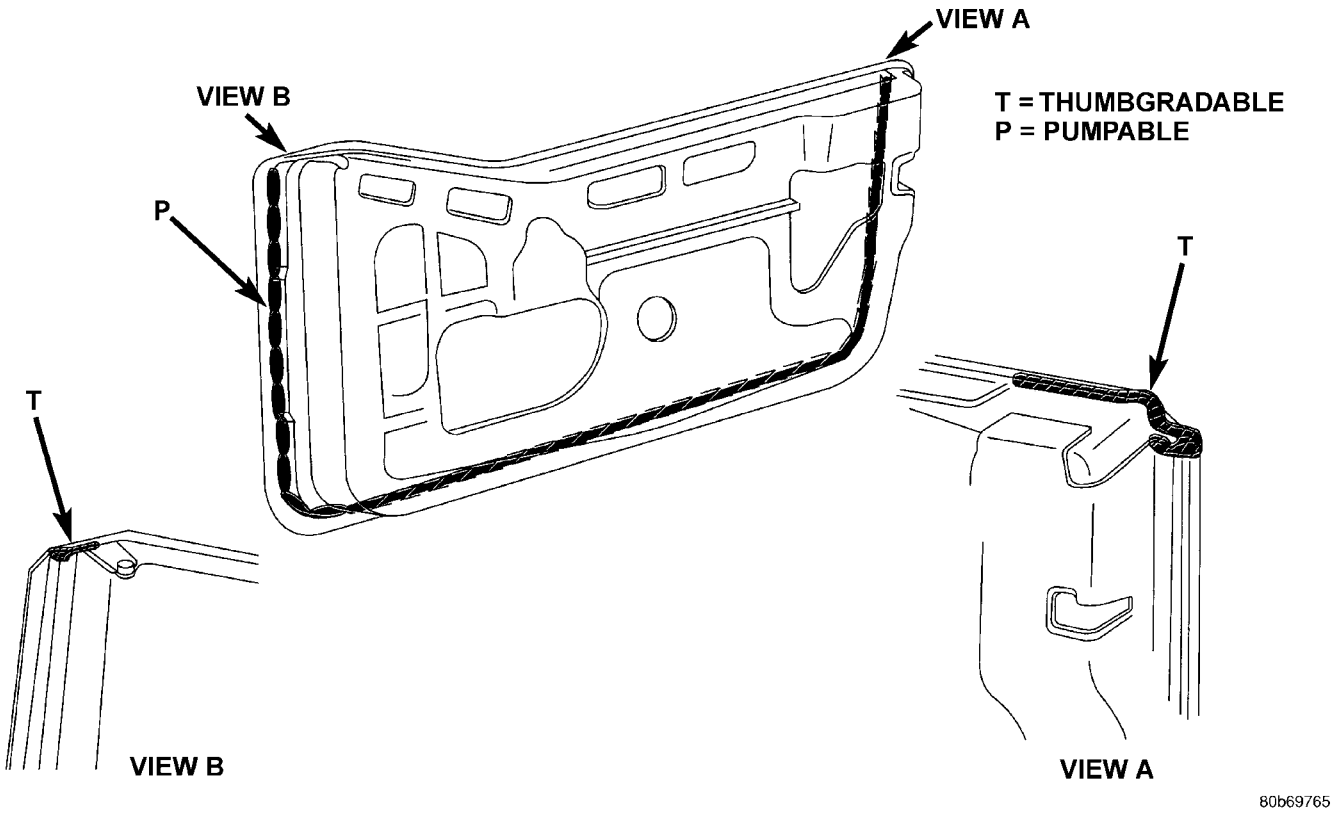

Fig. 15 HALF DOOR

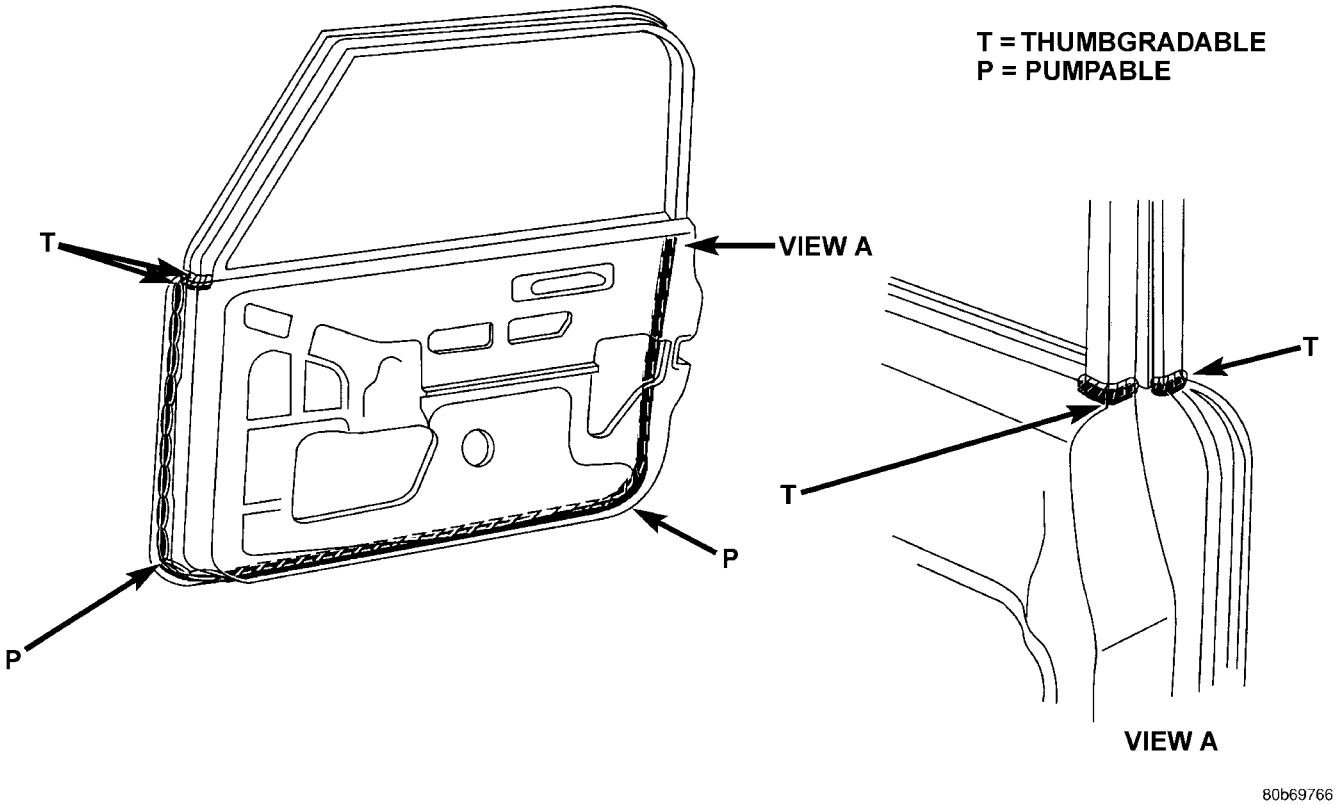

Fig. 16 FULL DOOR

SEALER LOCATIONS (Continued)

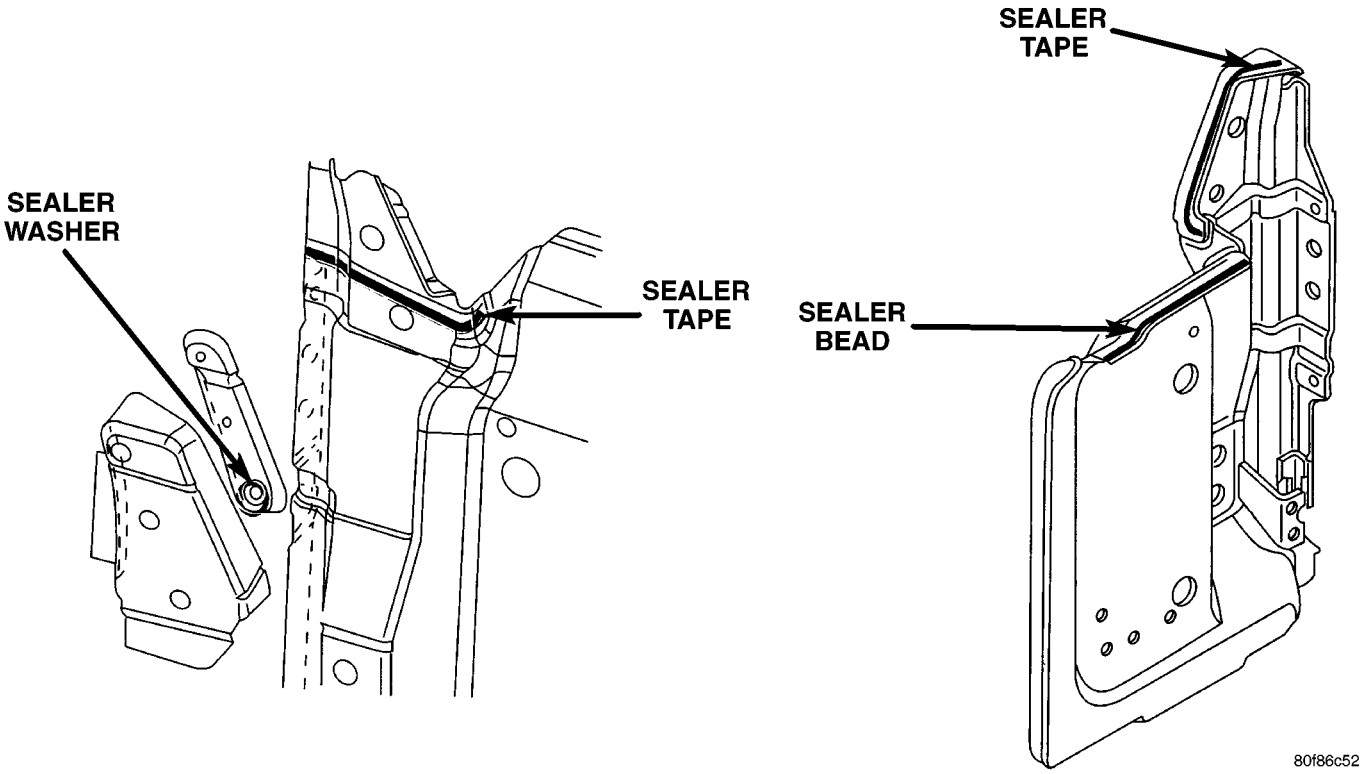

Fig. 17 DASH COWL AND PLENUM

SEALER LOCATIONS (Continued)

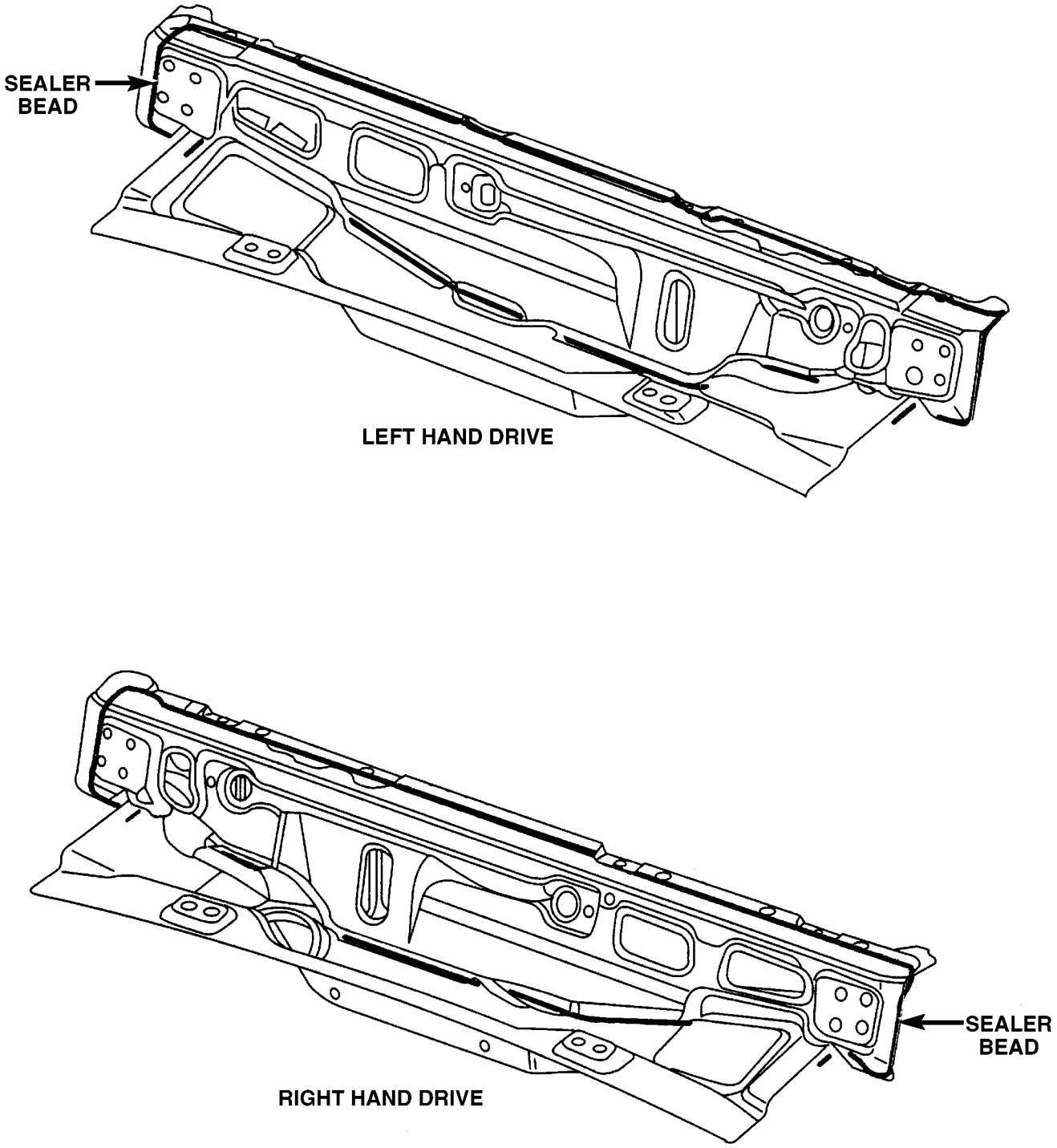

Fig. 18 DASH, COWL AND PLENUM

SEALER LOCATIONS (Continued)

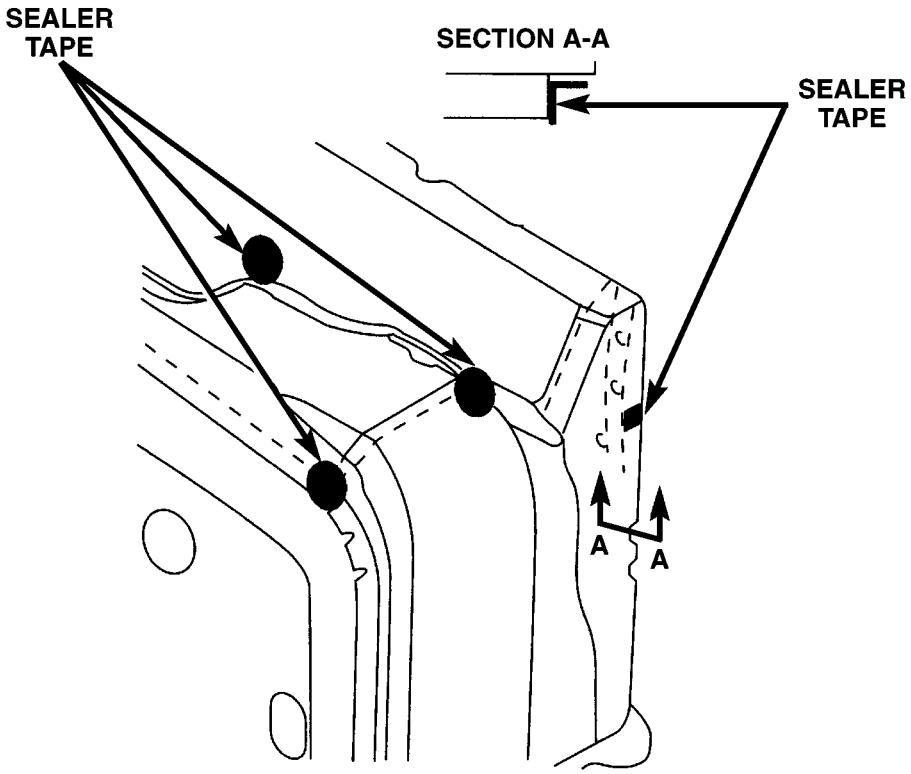

Fig. 19 DASH, COWL AND PLENUM

SEALER LOCATIONS (Continued)

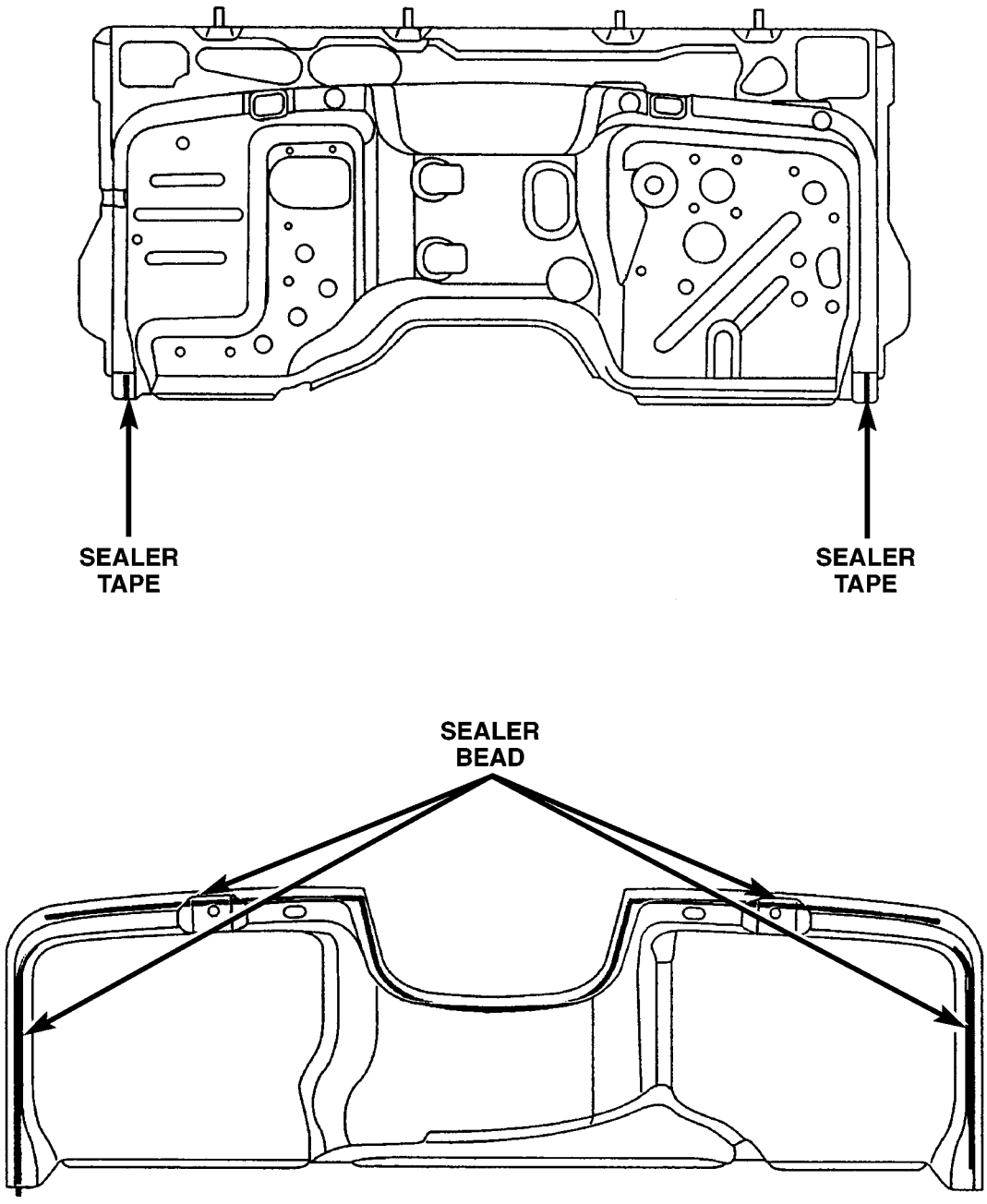

Fig. 20 DASH, COWL AND PLENUM

SEALER LOCATIONS (Continued)

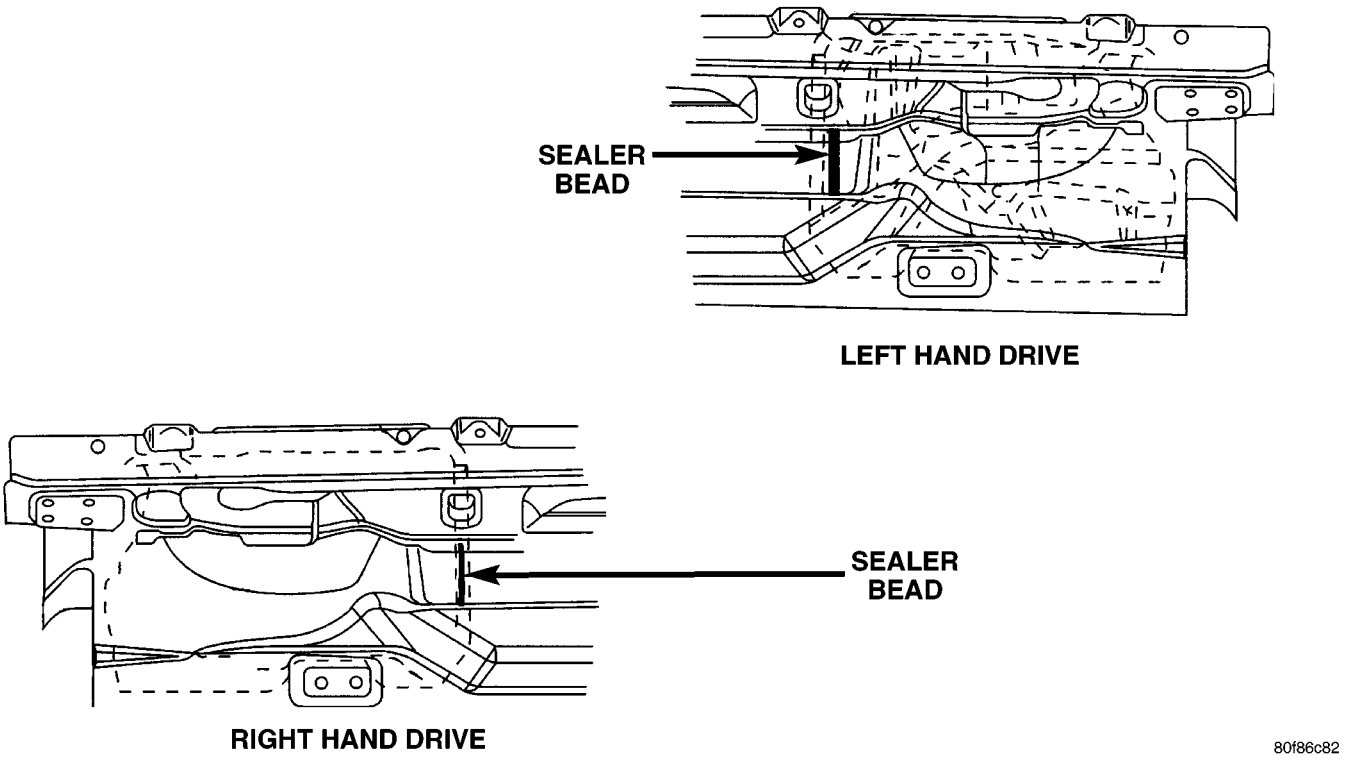

Fig. 21 DASH, COWL AND PLENUM SUPPORTS AND REINFORCEMENTS

SEALER LOCATIONS (Continued)

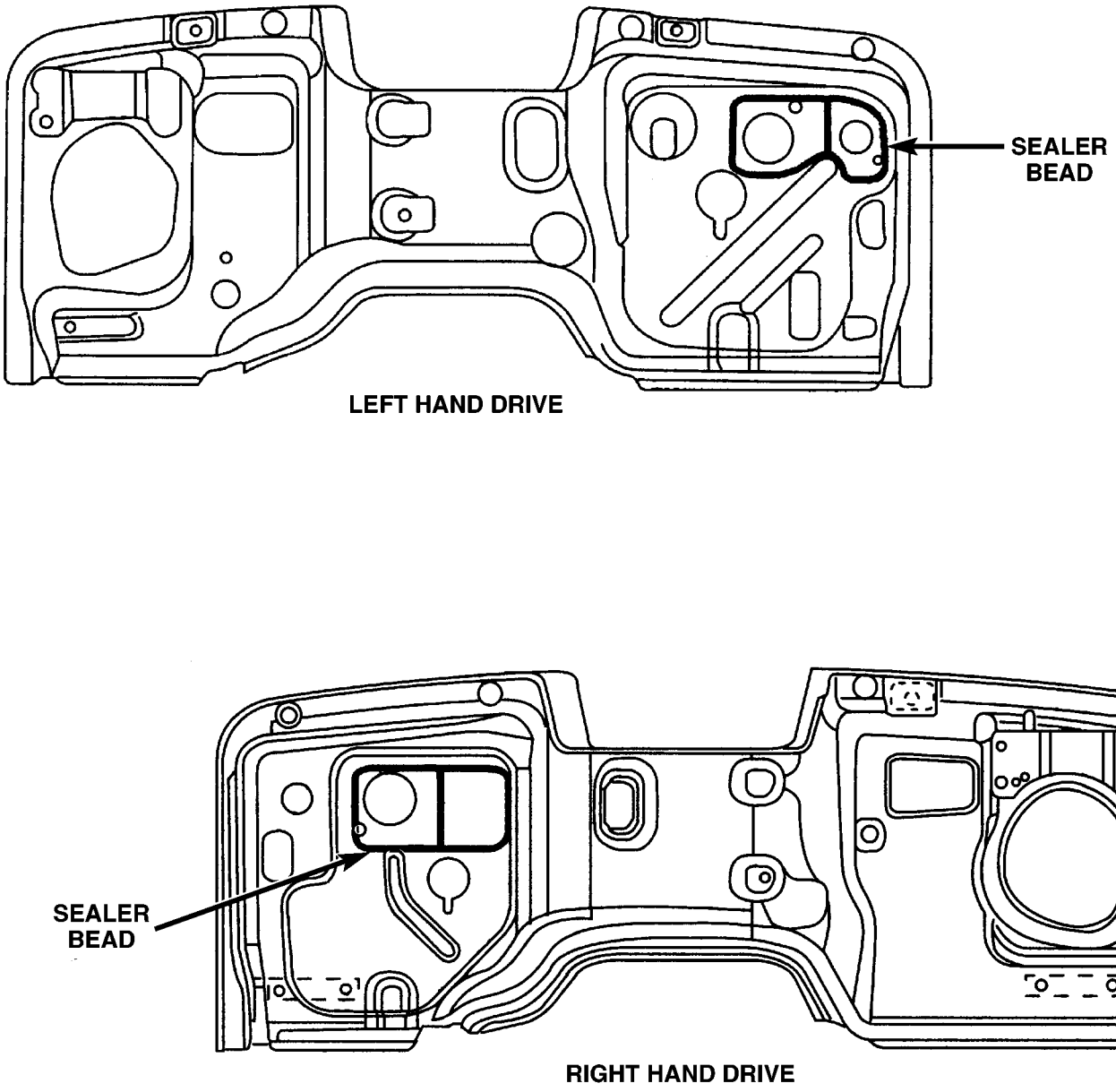

Fig. 22 DASH, COWL AND PLENUM SUPPORTS AND REINFORCEMENTS

SEALER LOCATIONS (Continued)

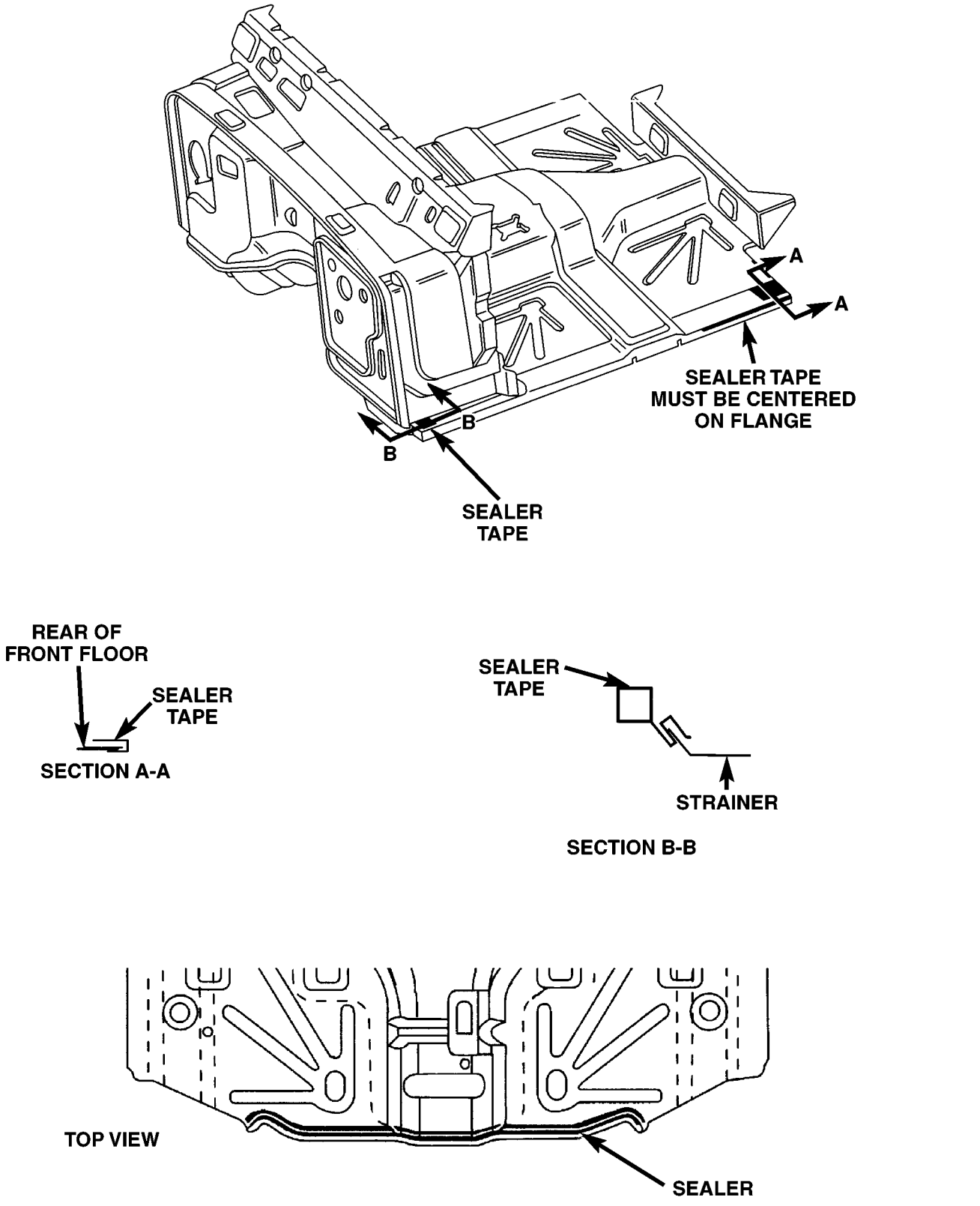

Fig. 23 FRONT FLOOR, STRAINER, REINFORCEMENT AND COWL SIDE LOWER

SEALER LOCATIONS (Continued)

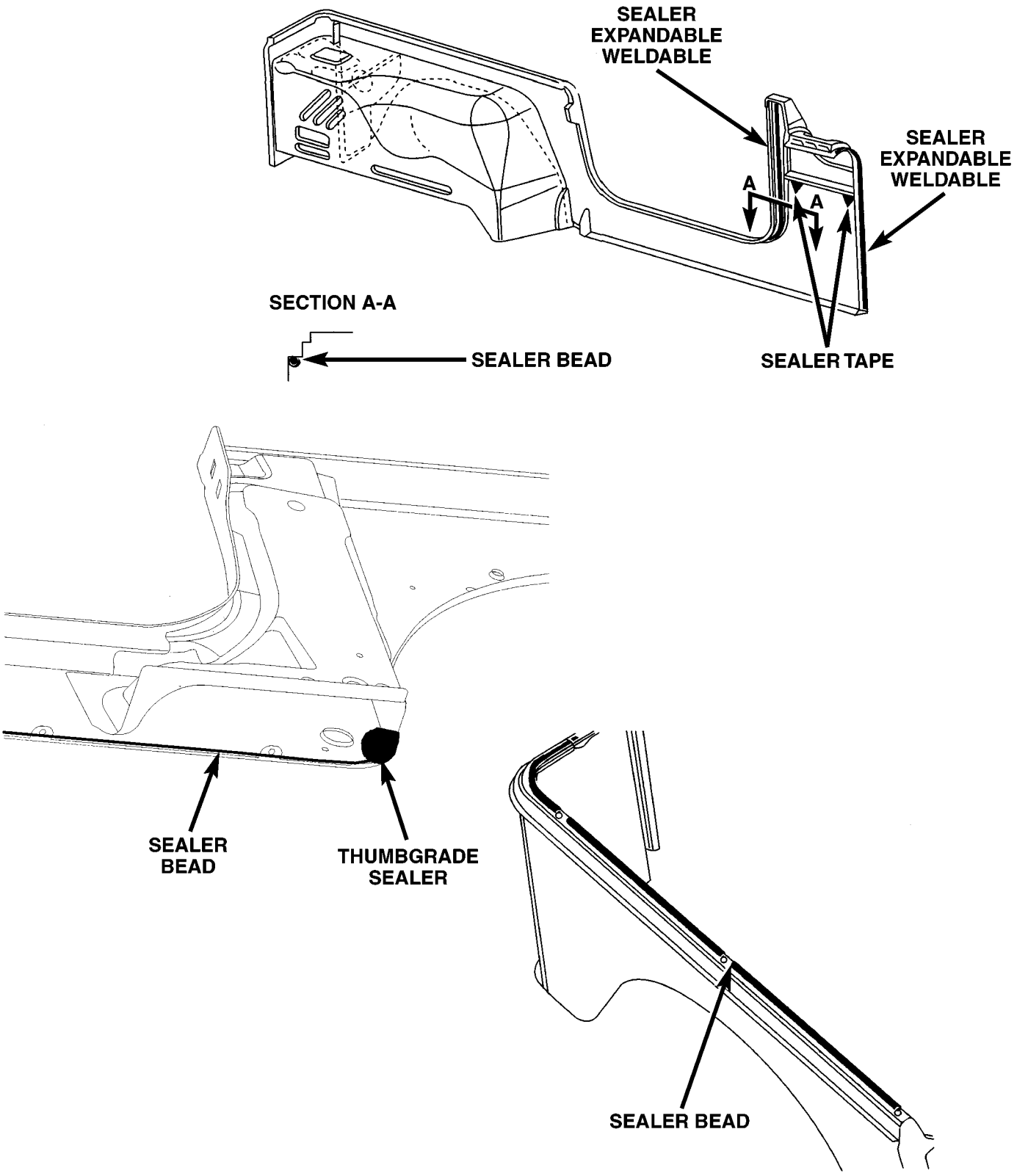

Fig. 24 BODY SIDE APERTURE

SEALER LOCATIONS (Continued)

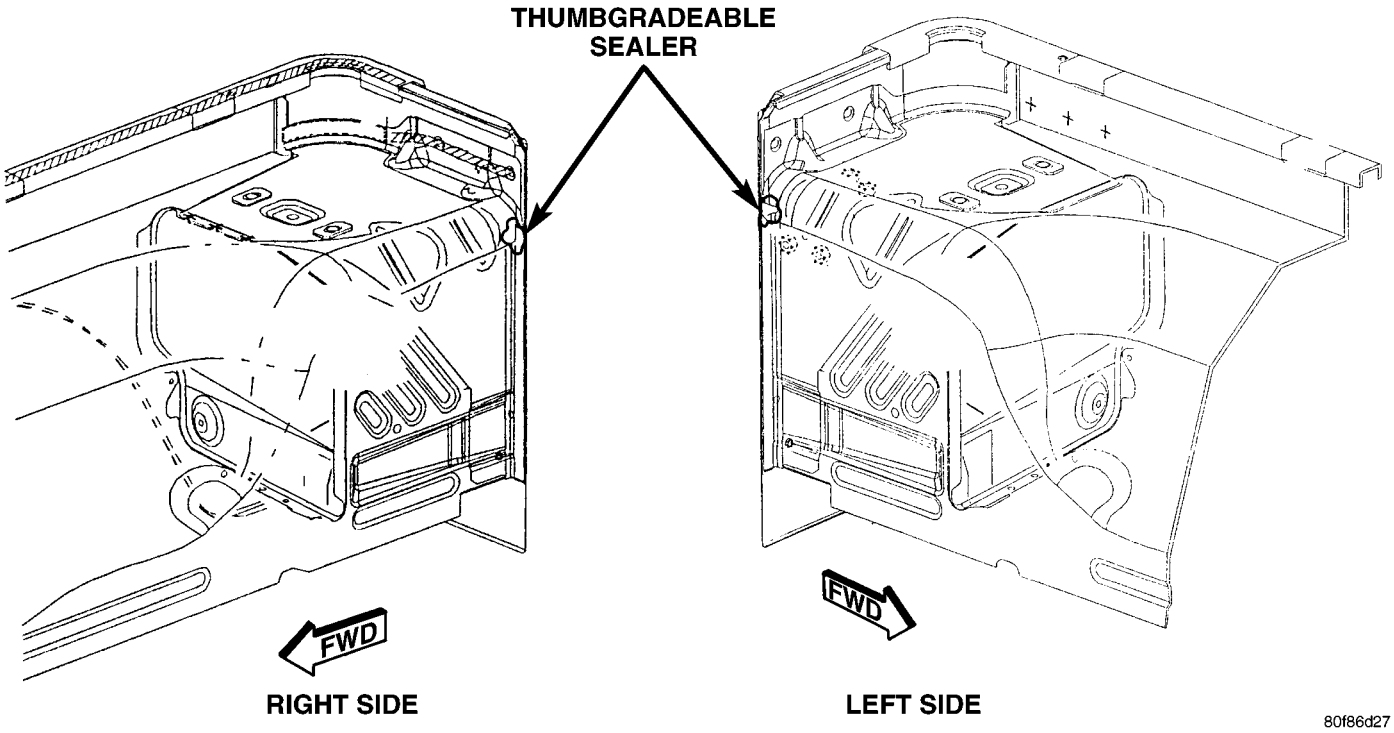

Fig. 25 WHEELHOUSE

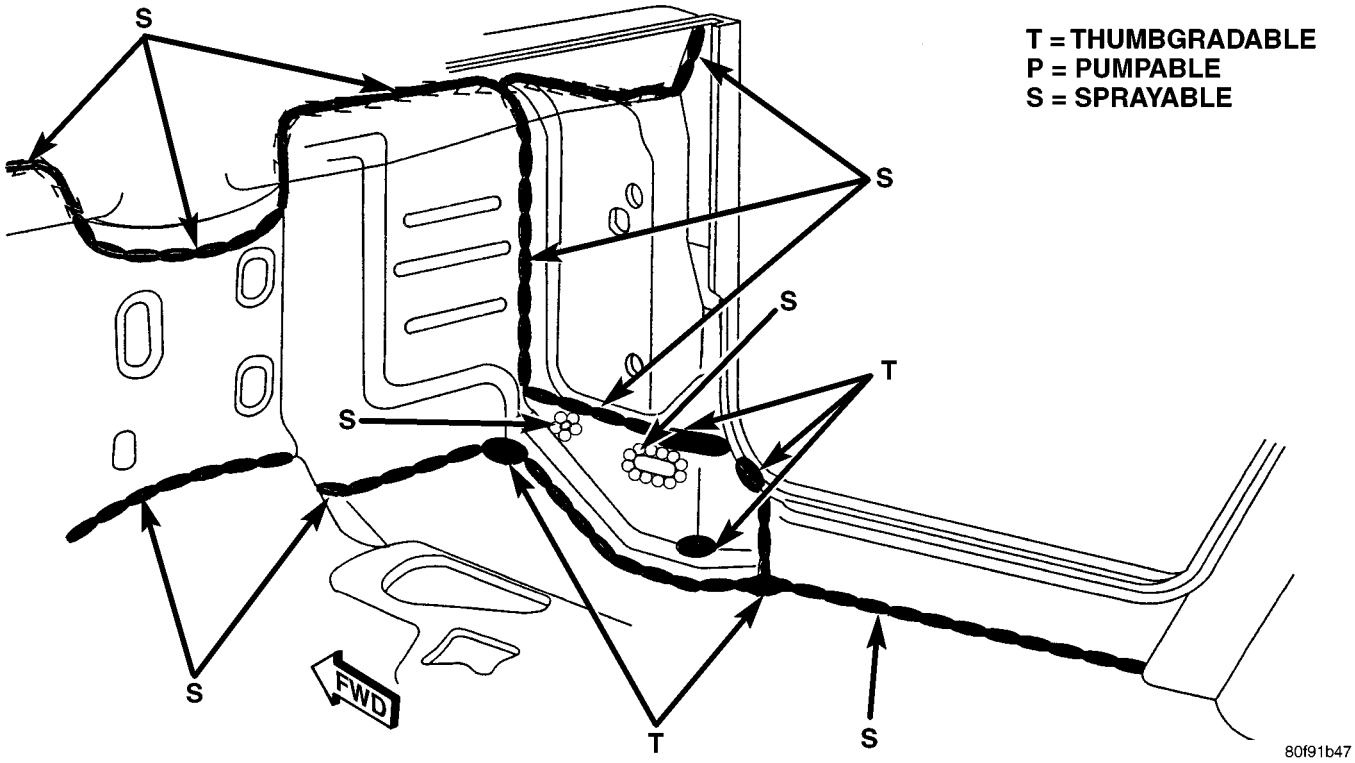

Fig. 26 COWL AND PLENUM

SEALER LOCATIONS (Continued)

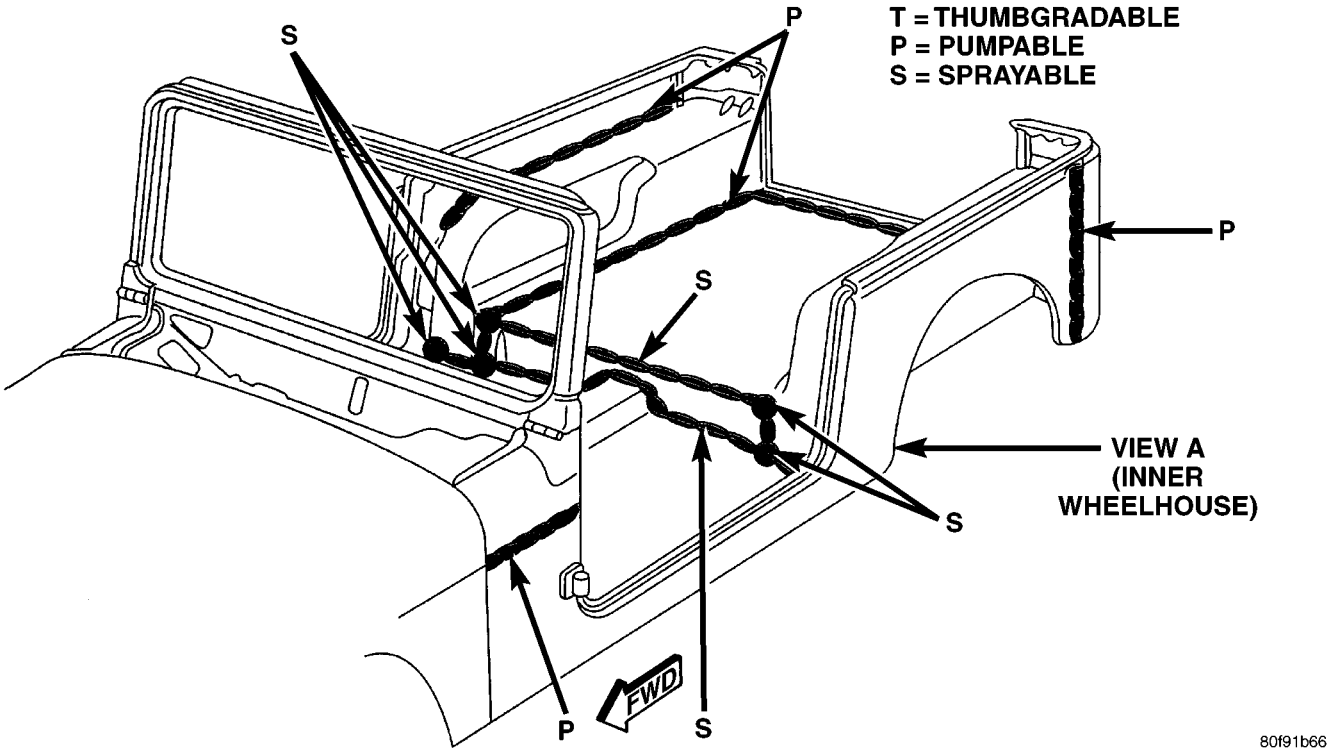

Fig. 27 REAR FLOOR AND OUTER BODY SEAMS

STRUCTURAL ADHESIVE LOCATIONS

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BODY COMPLETE LWB	38

STRUCTURAL ADHESIVE LOCATIONS (Continued)

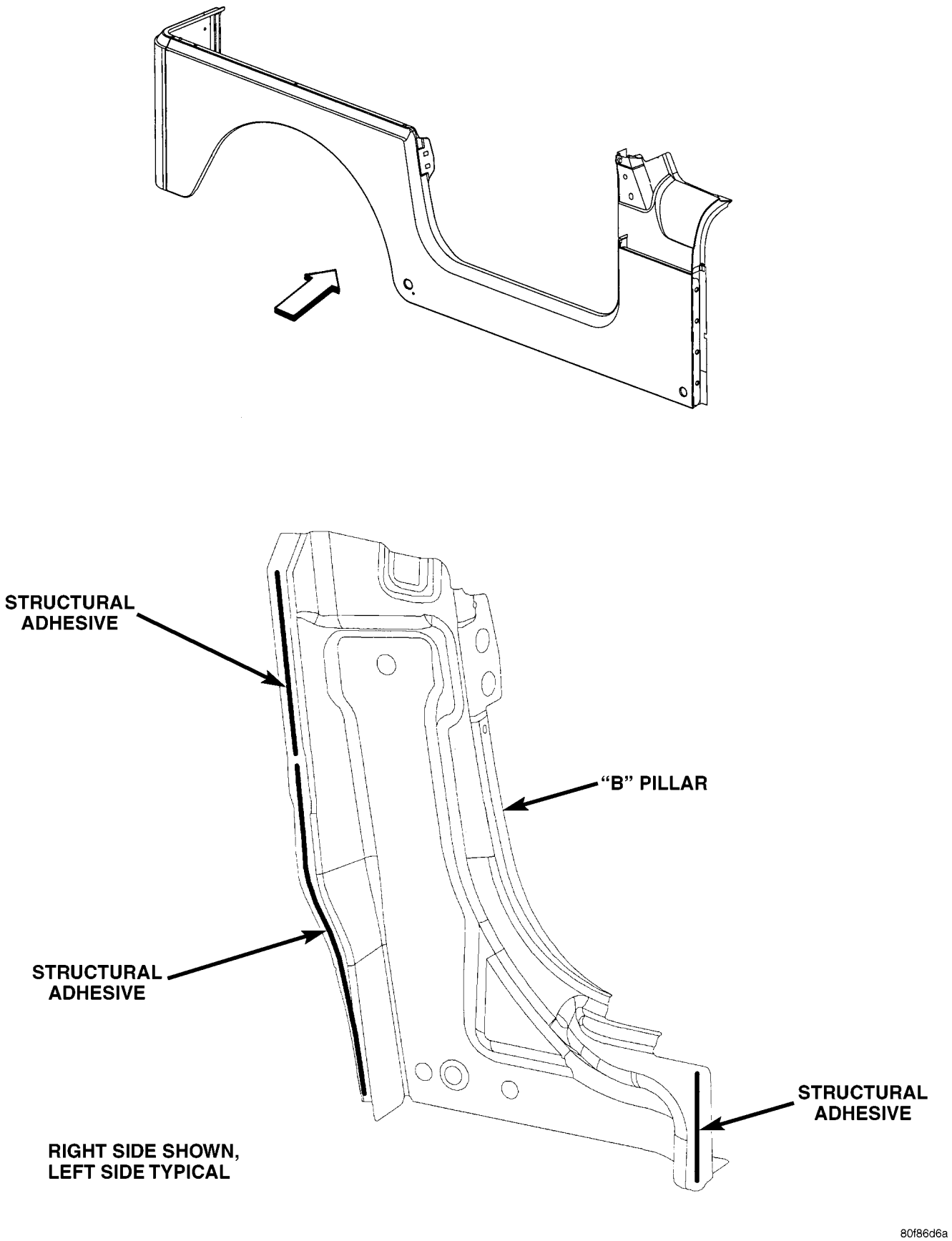

Fig. 28 BODY SIDE APERTURE

STRUCTURAL ADHESIVE LOCATIONS (Continued)

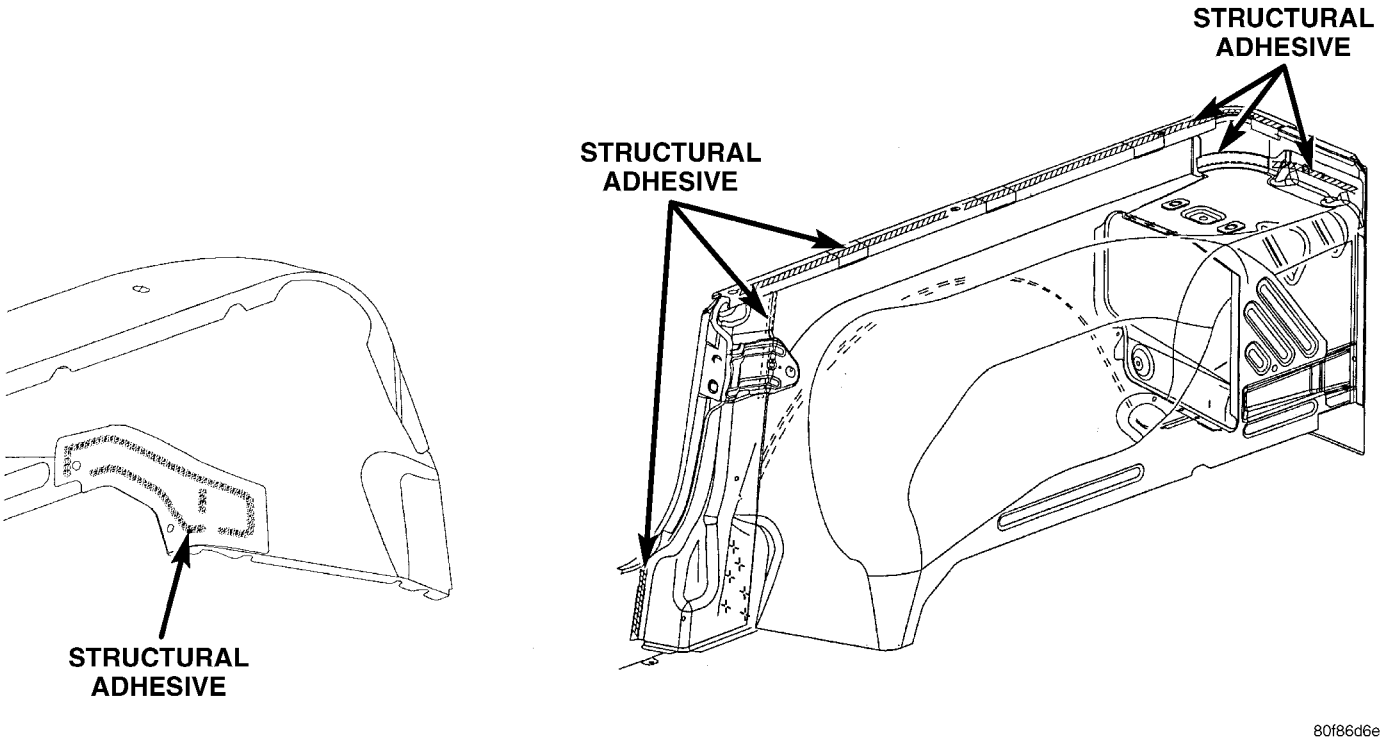

Fig. 29 WHEELHOUSE

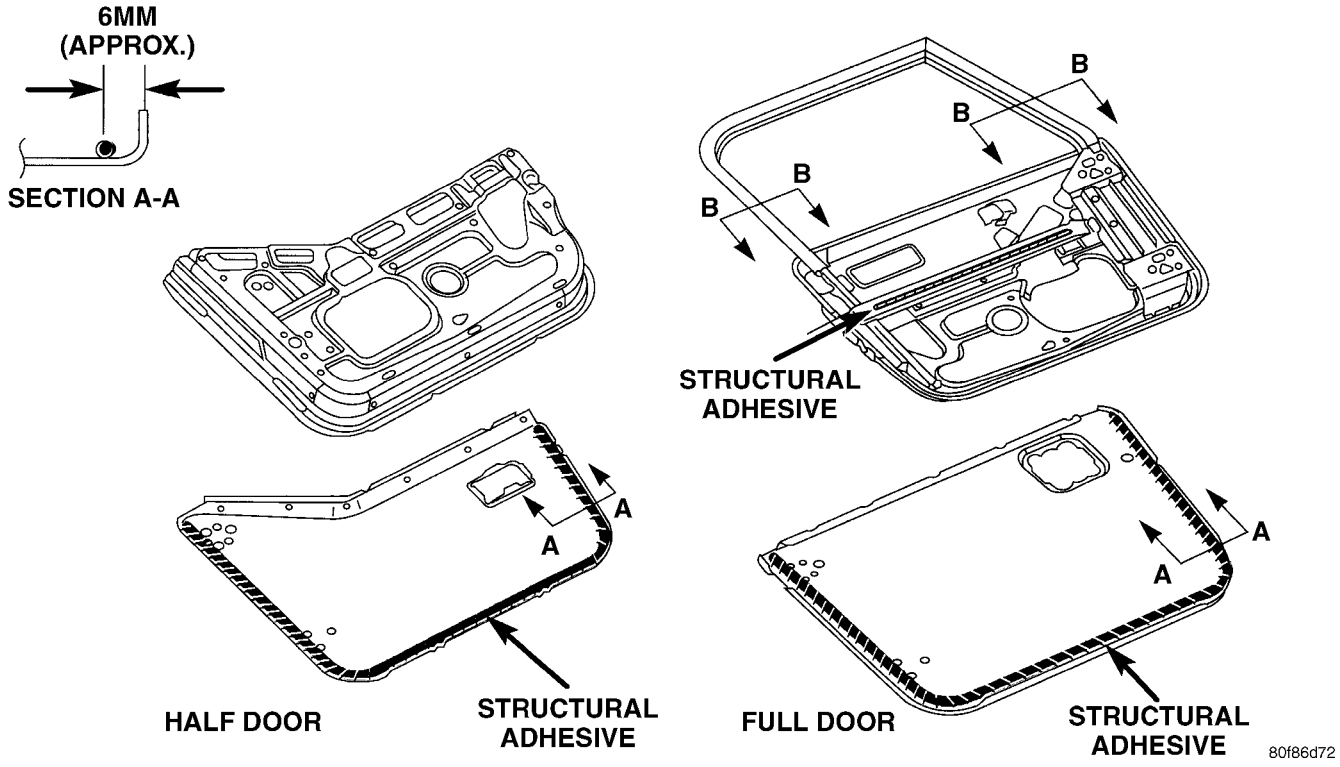

Fig. 30 DOOR OUTER PANELS

STRUCTURAL ADHESIVE LOCATIONS (Continued)

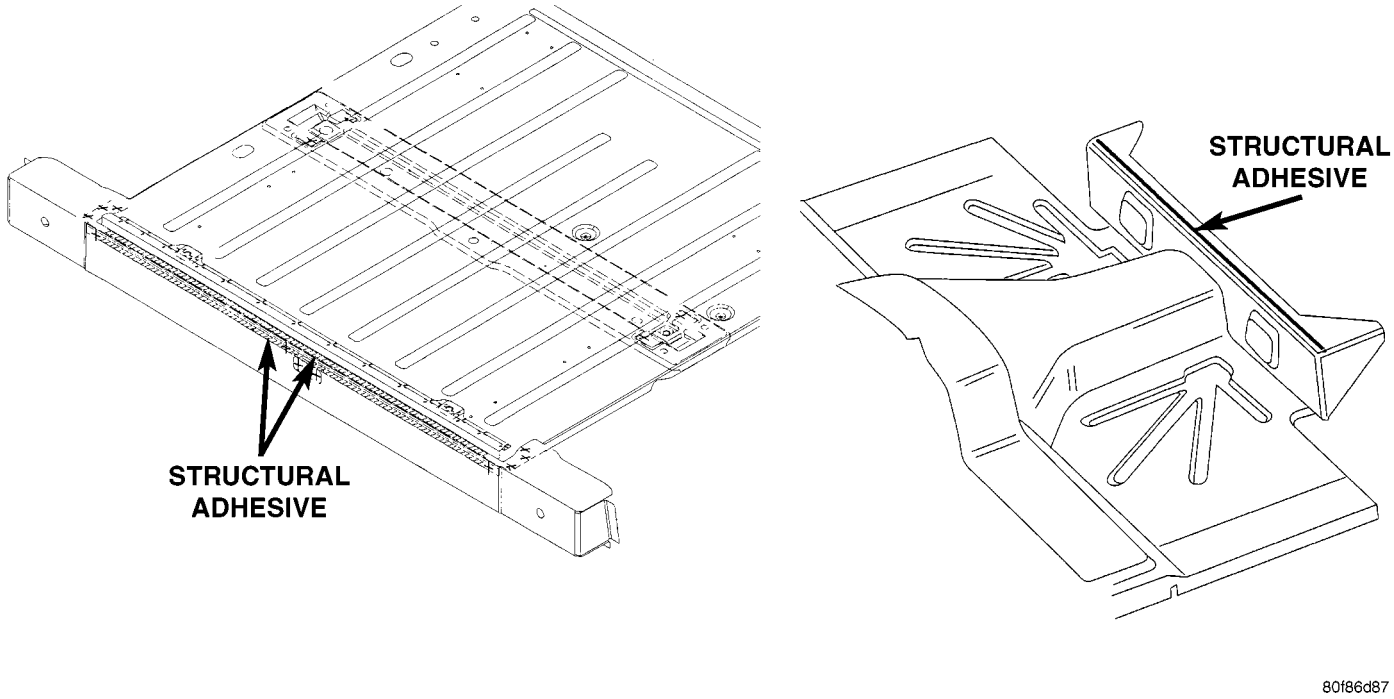

Fig. 31 REAR FLOOR PAN

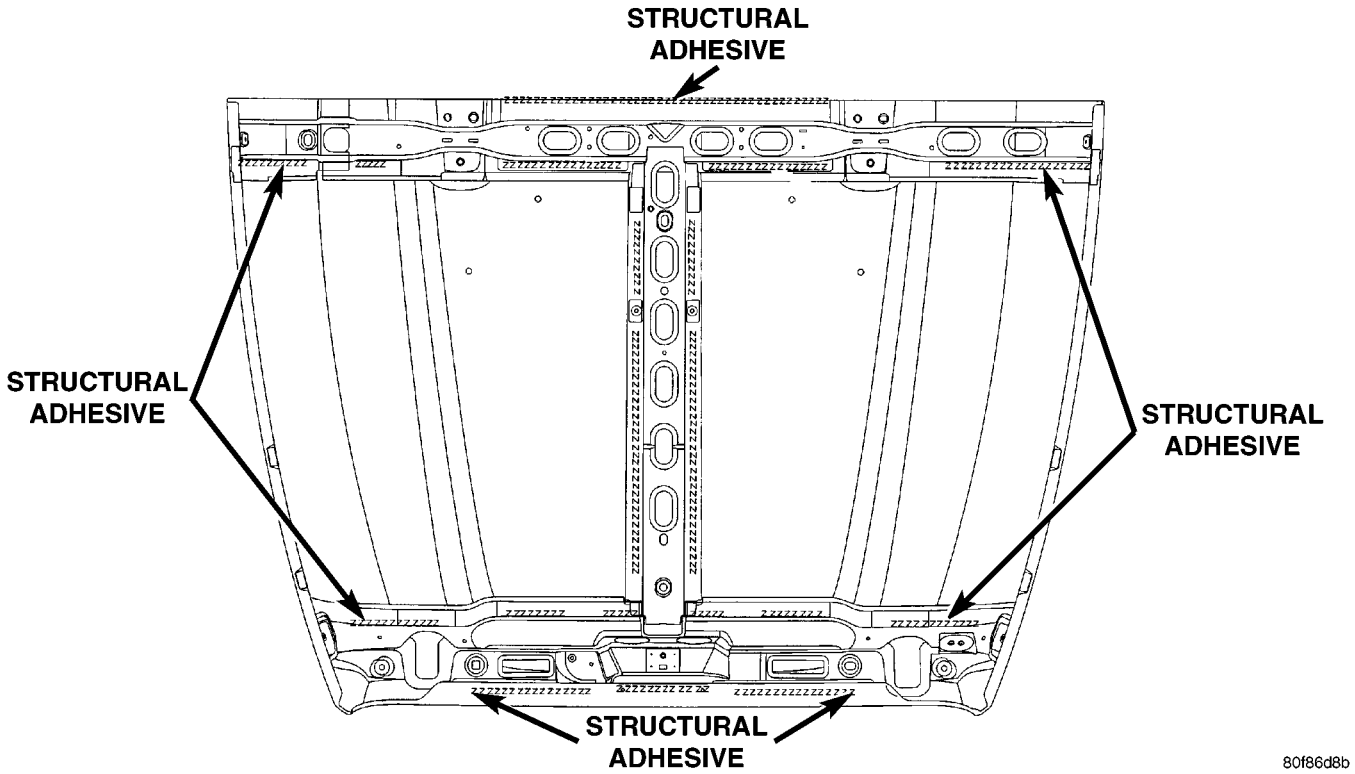

Fig. 32 HOOD ASSEMBLY

STRUCTURAL ADHESIVE LOCATIONS (Continued)

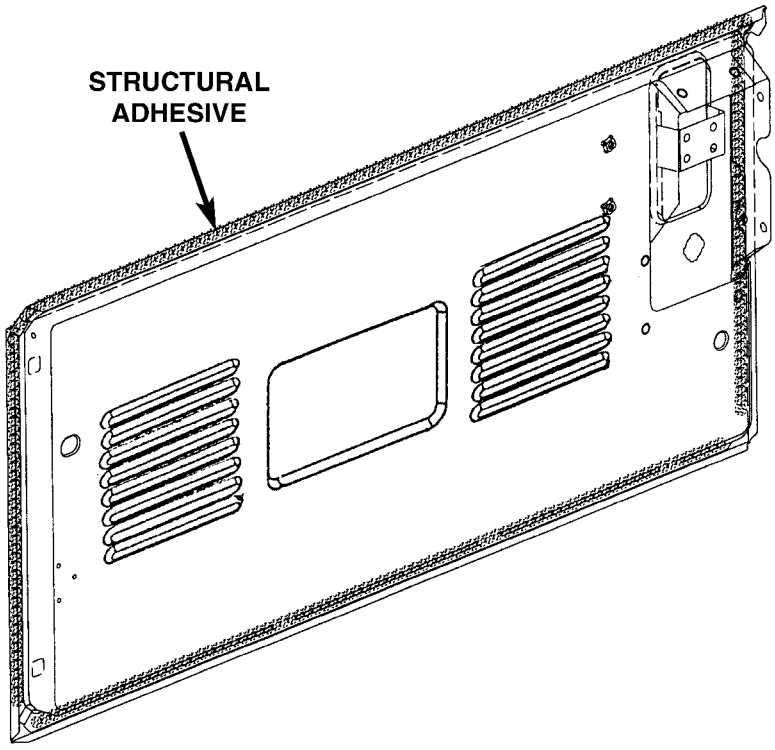

Fig. 33 TAIL GATE

STRUCTURAL ADHESIVE LOCATIONS (Continued)

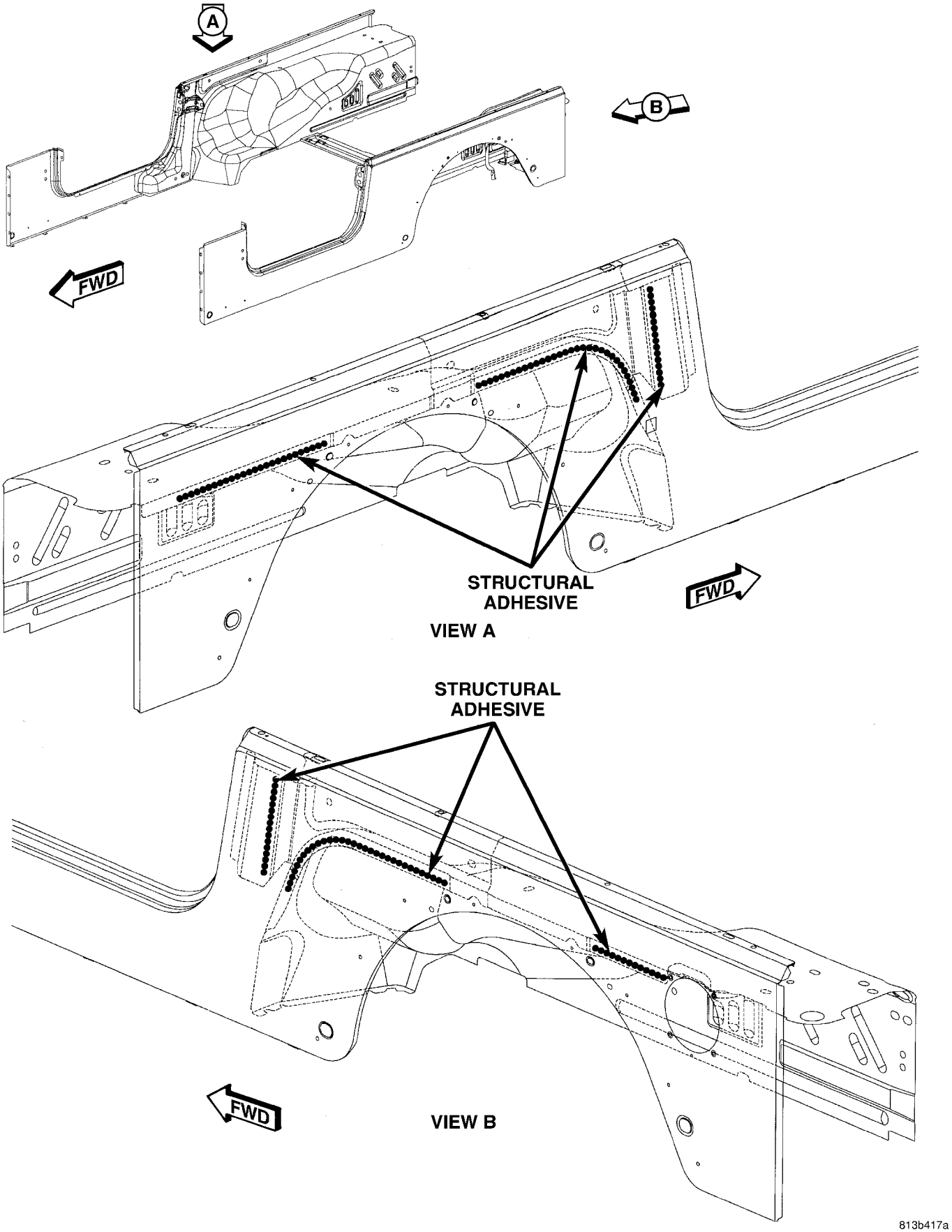

Fig. 34 BODY SIDE/WHEELHOUSE LONG WHEEL BASE (LWB)

STRUCTURAL ADHESIVE LOCATIONS (Continued)

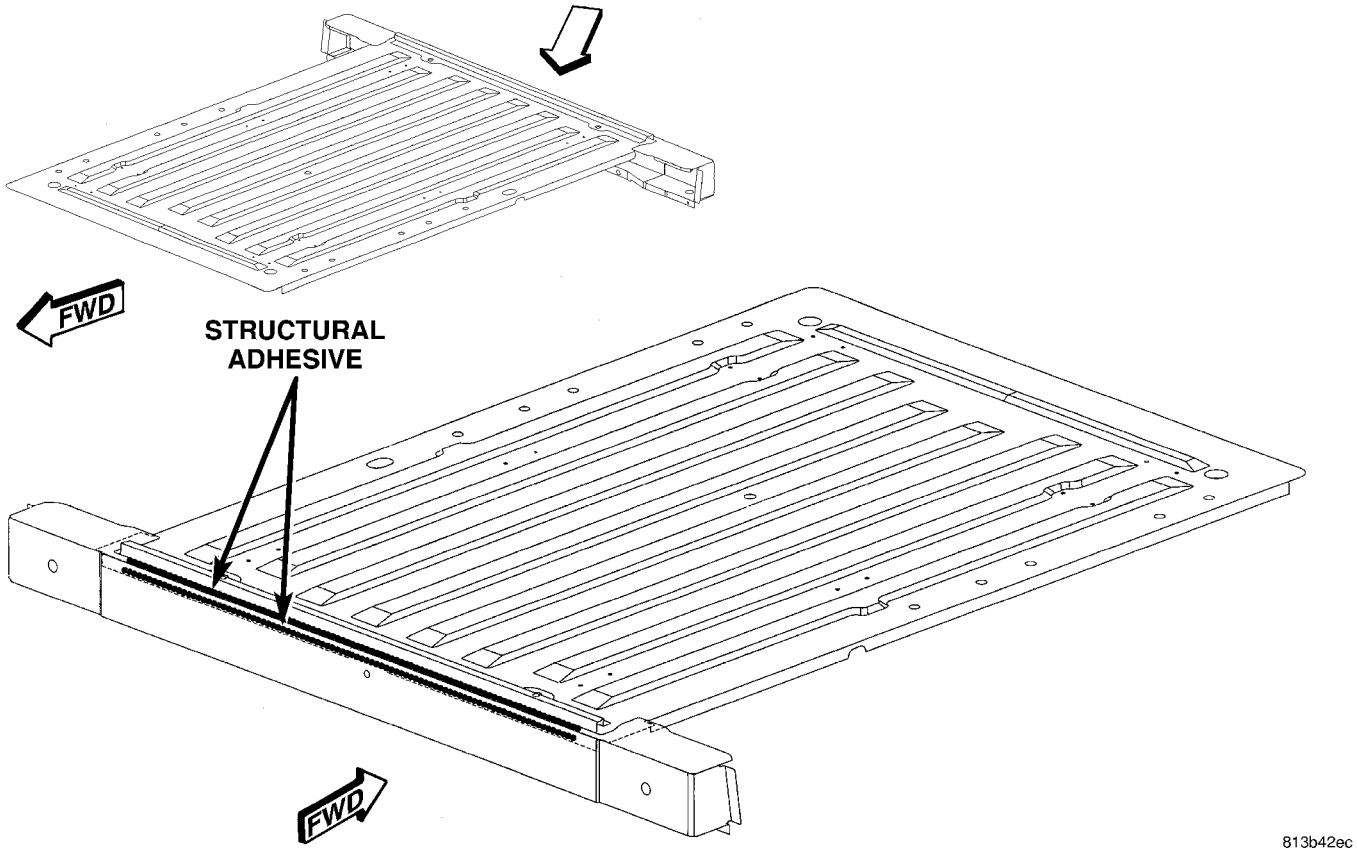

Fig. 35 REAR FLOOR LWB

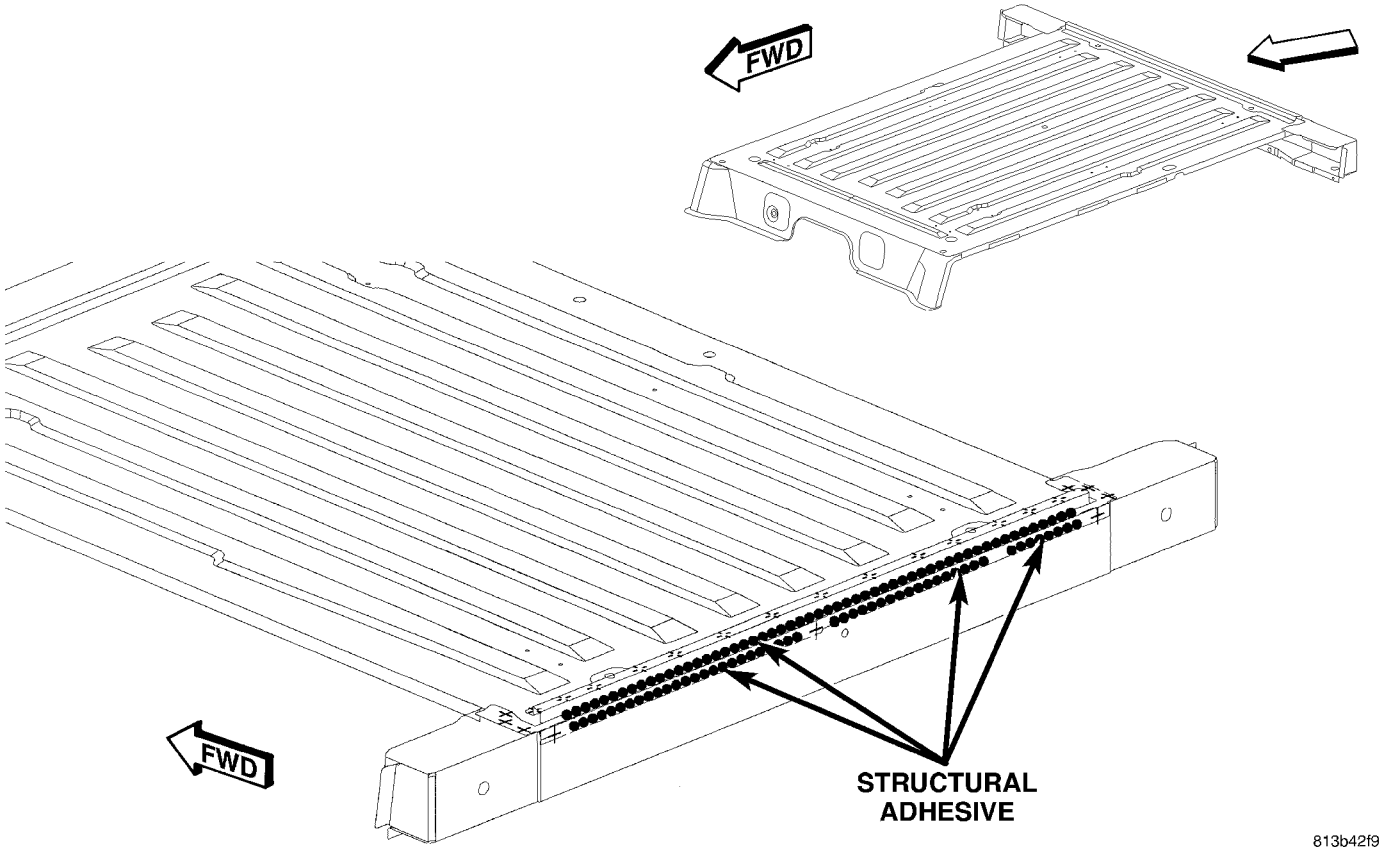

Fig. 36 UNDERBODY LWB

STRUCTURAL ADHESIVE LOCATIONS (Continued)

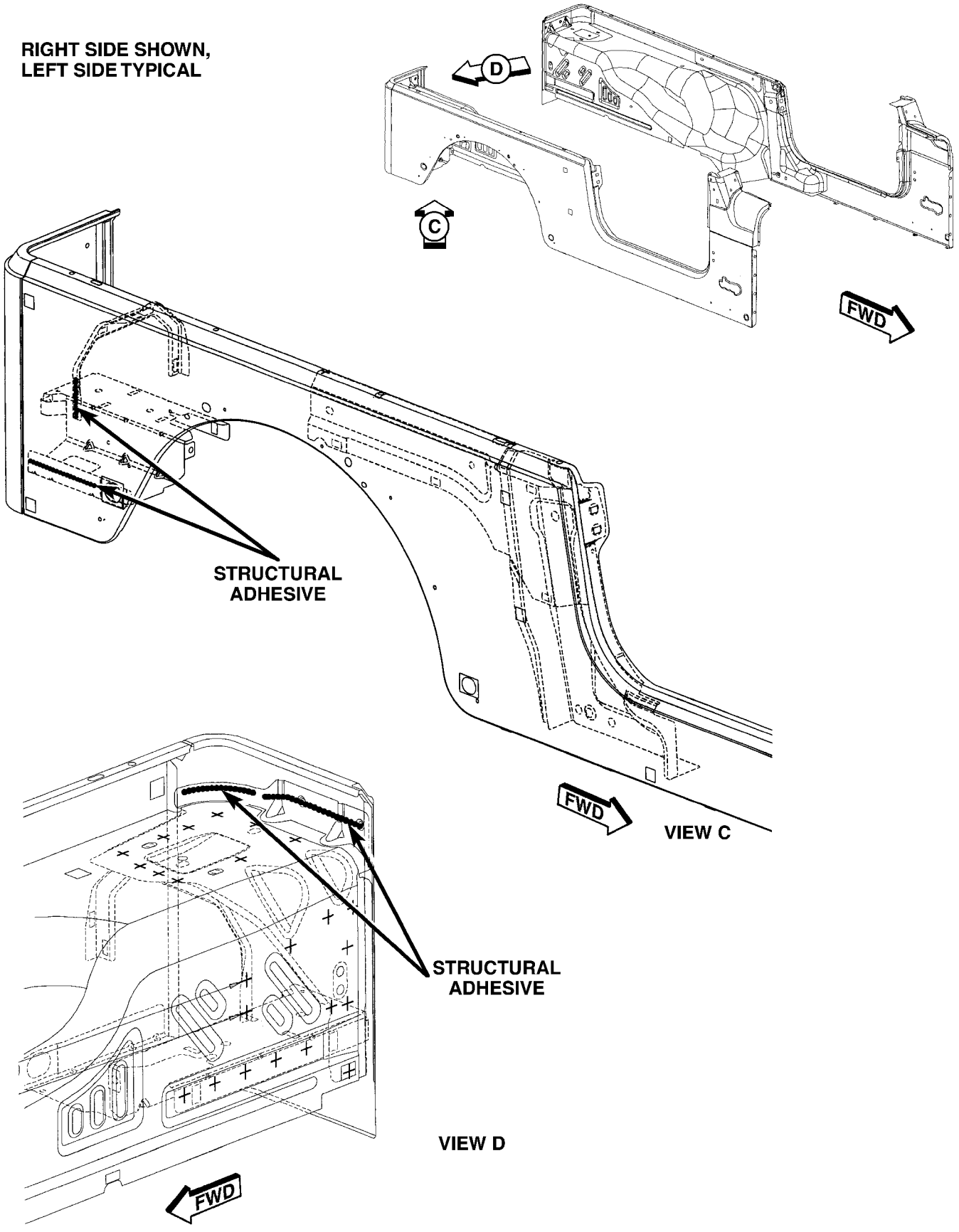

Fig. 37 BODY SIDE COMPLETE LWB

STRUCTURAL ADHESIVE LOCATIONS (Continued)

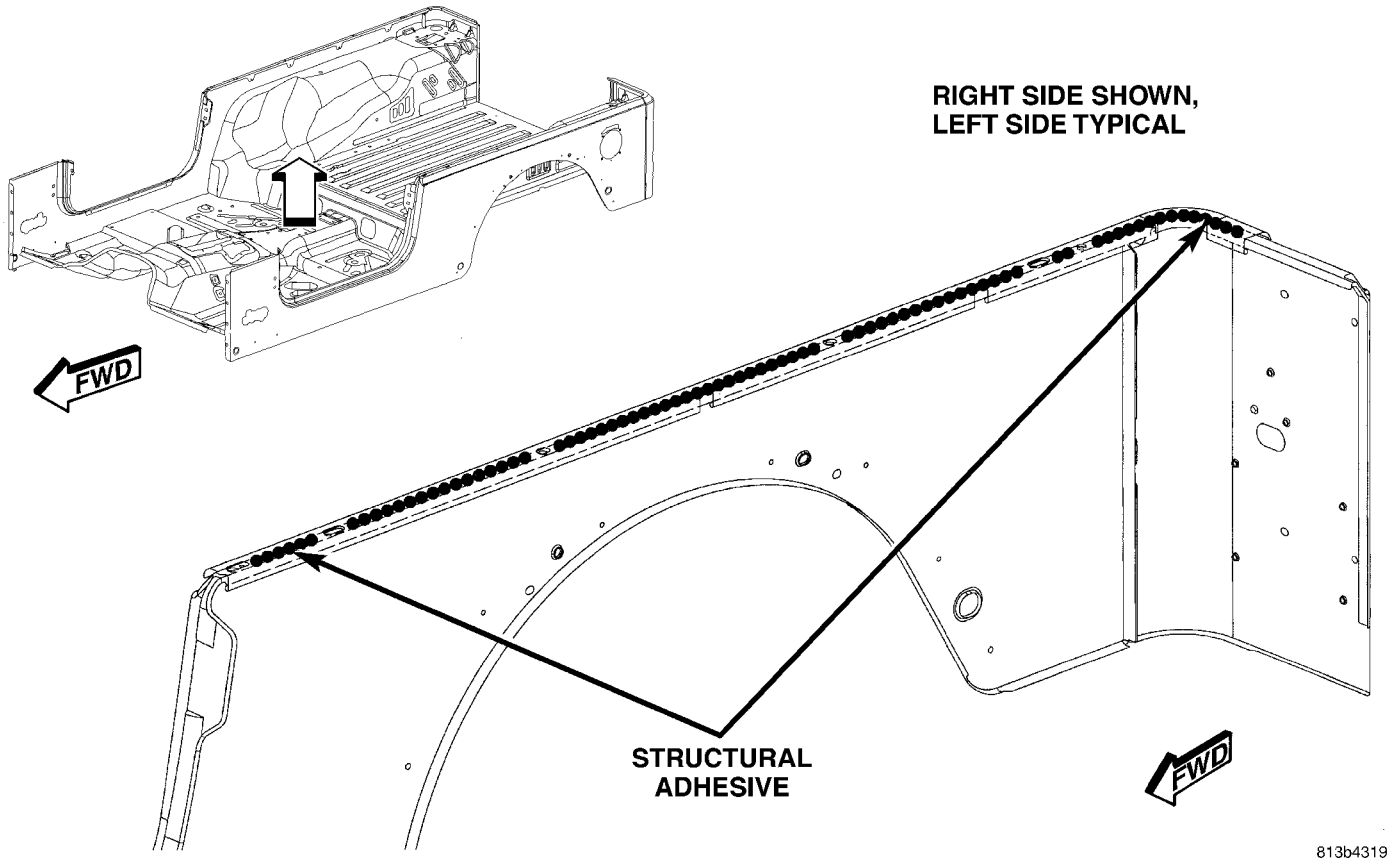

Fig. 38 BODY COMPLETE LWB

WELD LOCATIONS

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WELD LOCATIONS

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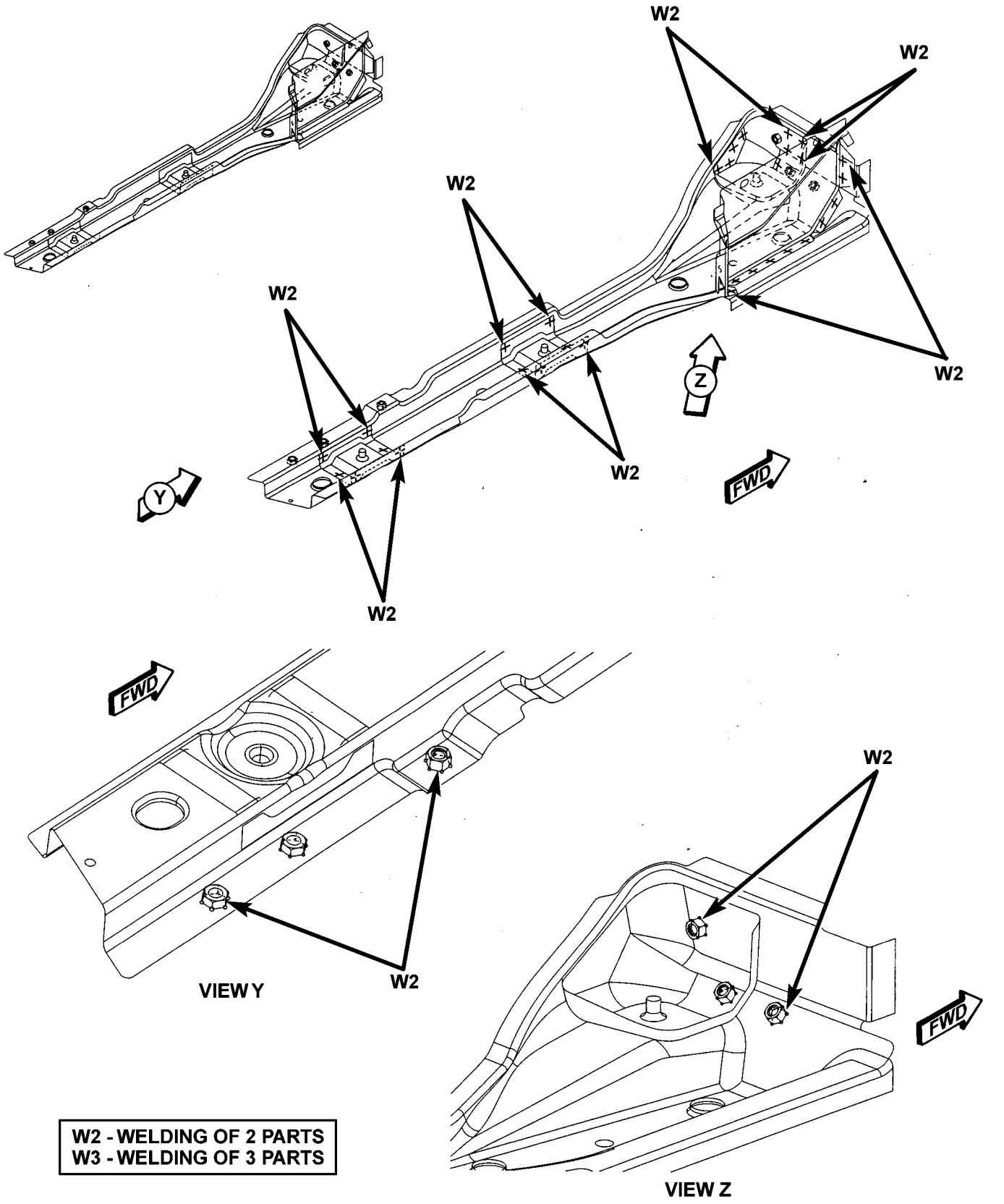
WELD LOCATIONS (Continued)



BOTH SIDES TYPICAL

80f8586e

Fig. 39 CROSSMEMBERS/STRAINERS

WELD LOCATIONS (Continued)

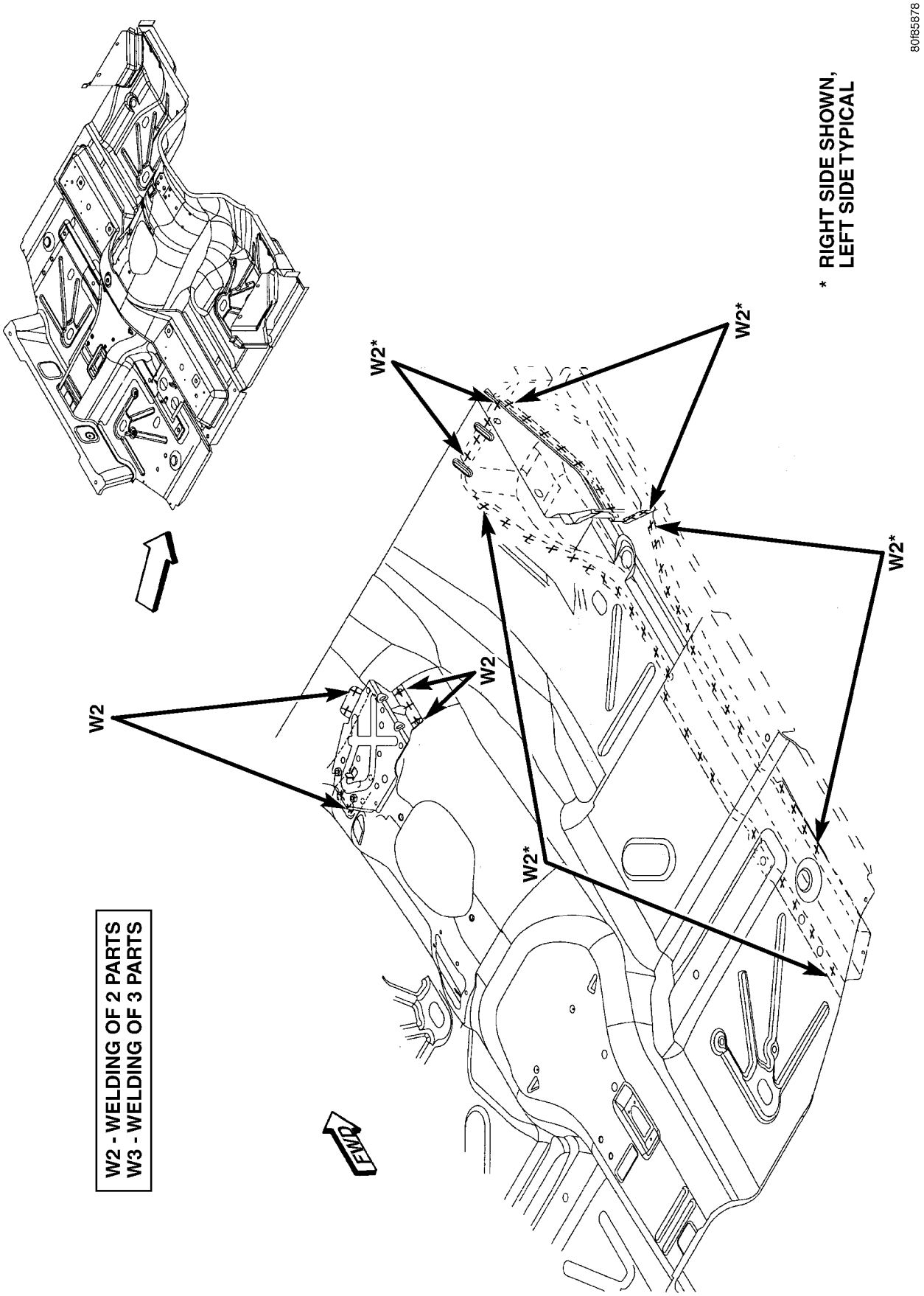


Fig. 40 FRONT FLOOR PAN (1 OF 3)

WELD LOCATIONS (Continued)

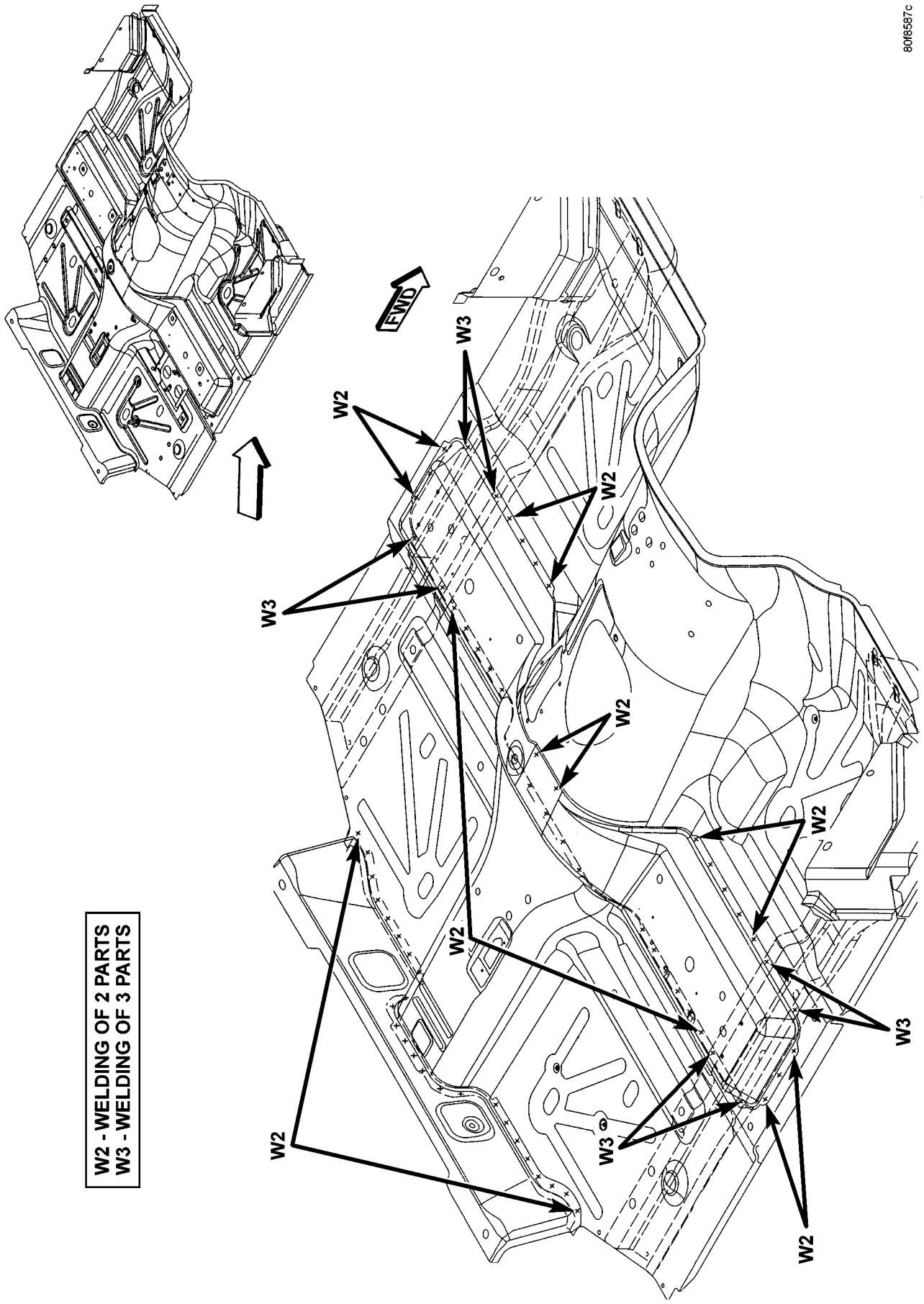


Fig. 41 FRONT FLOOR PAN (2 OF 3)

WELD LOCATIONS (Continued)

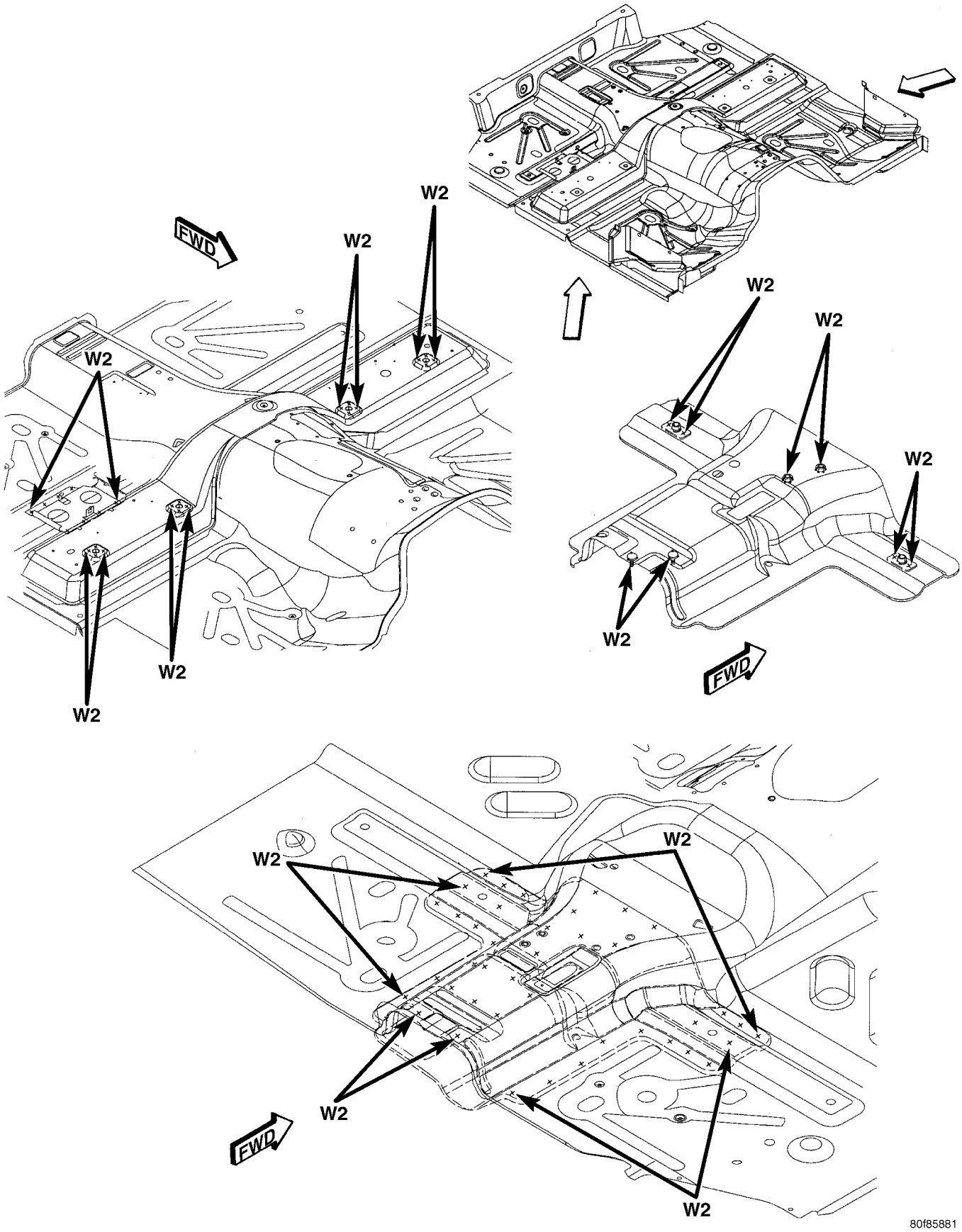


Fig. 42 FRONT FLOOR PAN (3 OF 3)

WELD LOCATIONS (Continued)

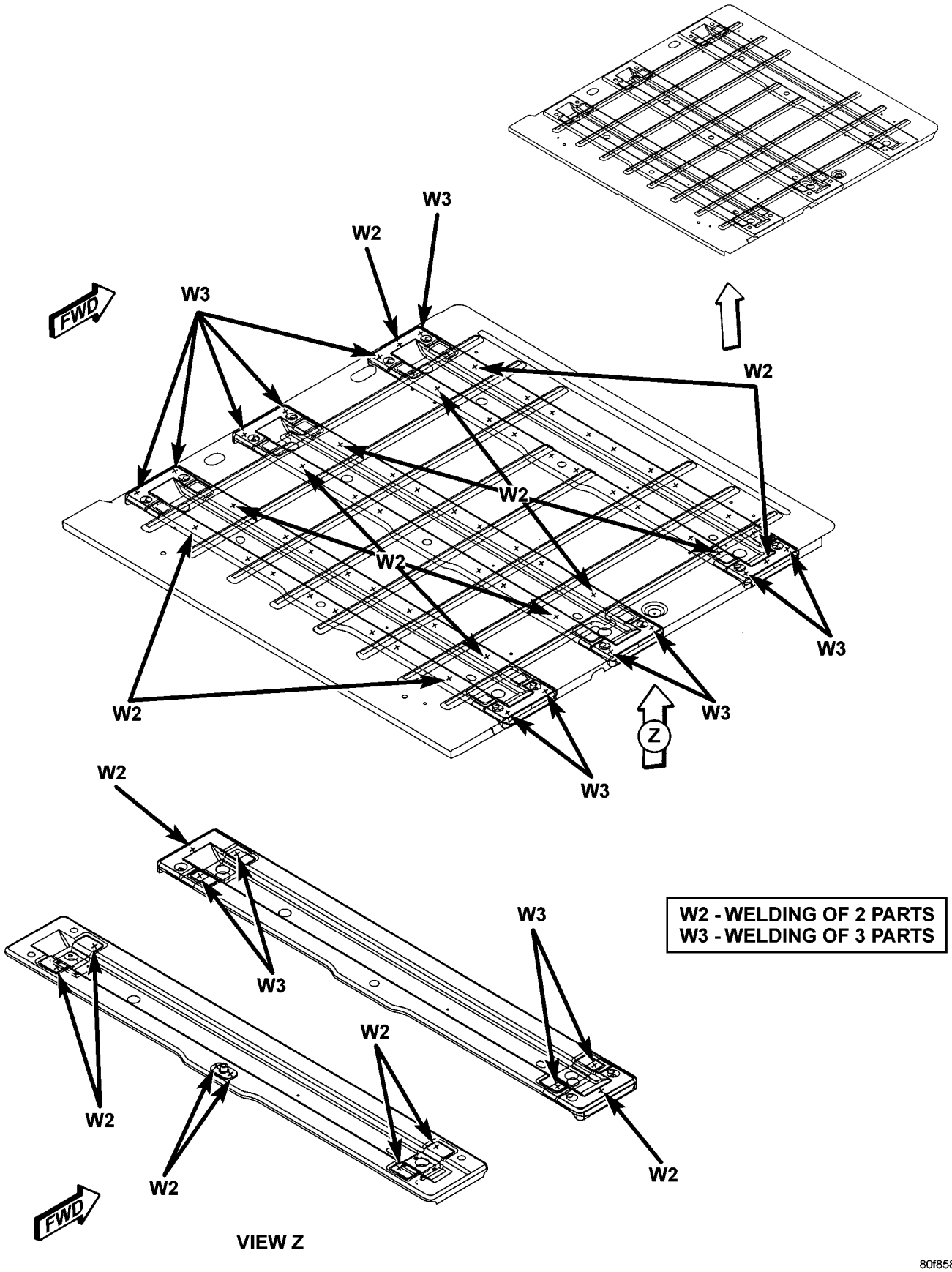


Fig. 43 REAR FLOOR PAN (1 OF 2)

WELD LOCATIONS (Continued)

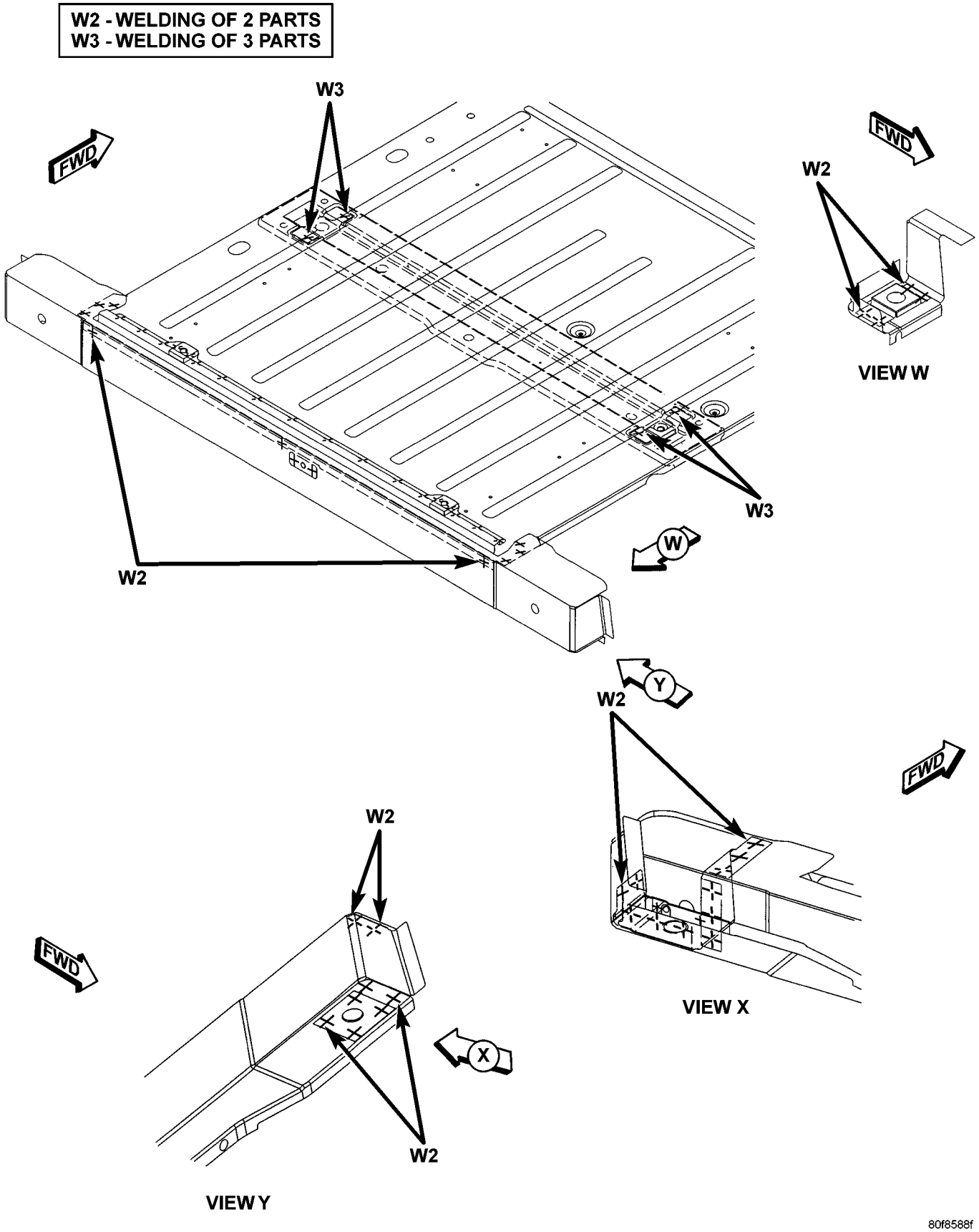


Fig. 44 REAR FLOOR PAN (2 OF 2)

WELD LOCATIONS (Continued)

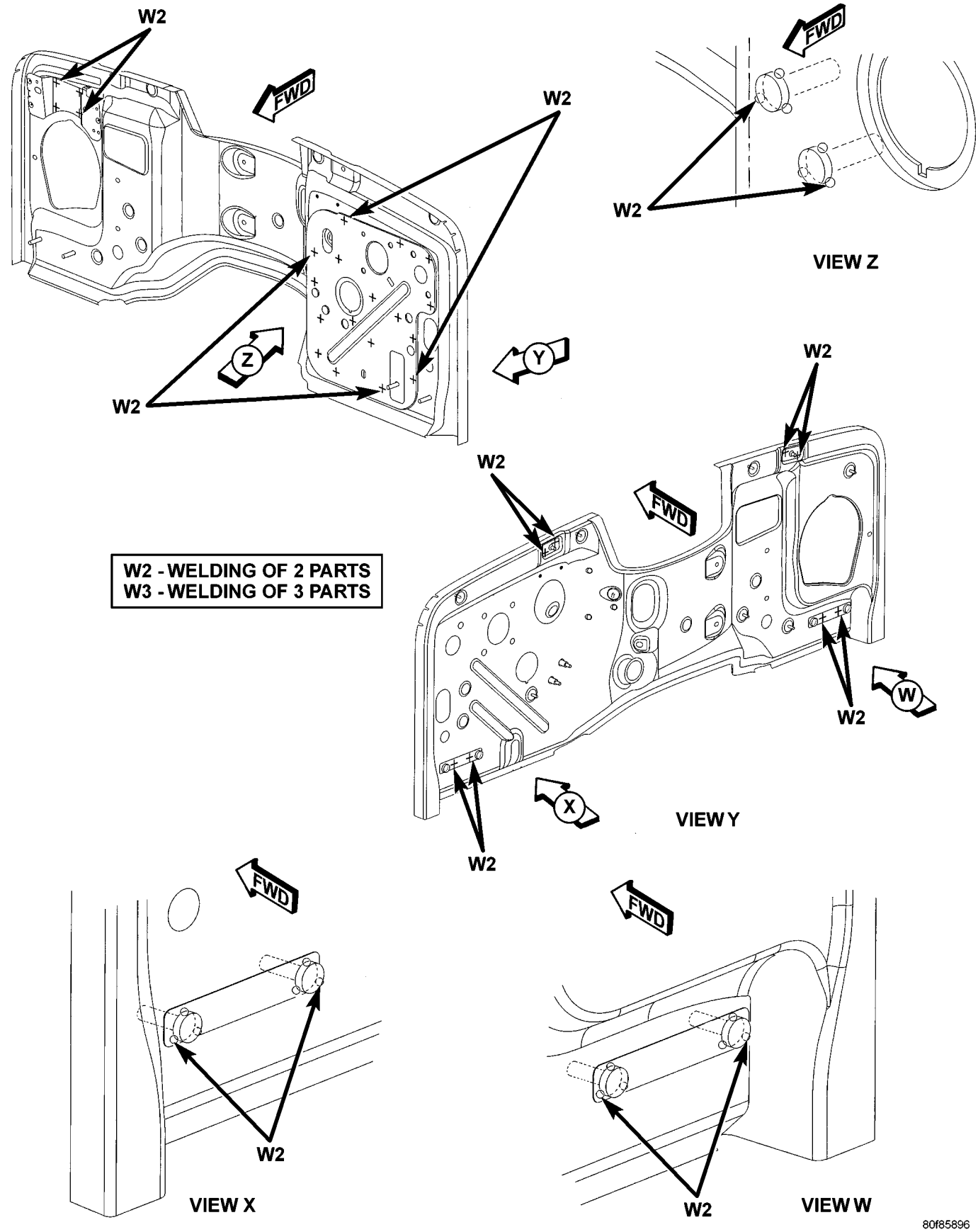


Fig. 45 DASH PANEL - LEFT HAND DRIVE

WELD LOCATIONS (Continued)

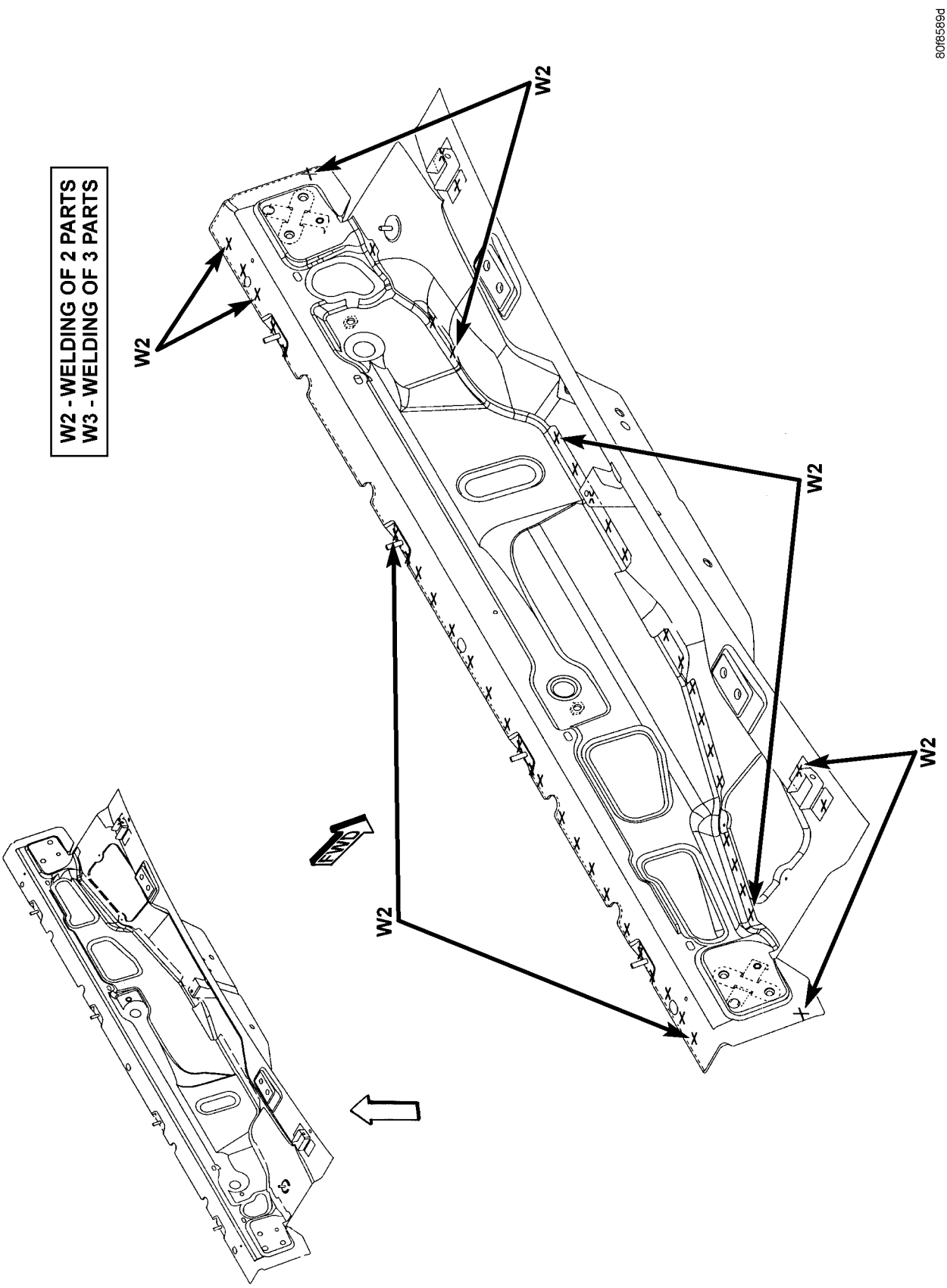


Fig. 46 COWL ASSEMBLY - LEFT HAND DRIVE (1 OF 3)

WELD LOCATIONS (Continued)

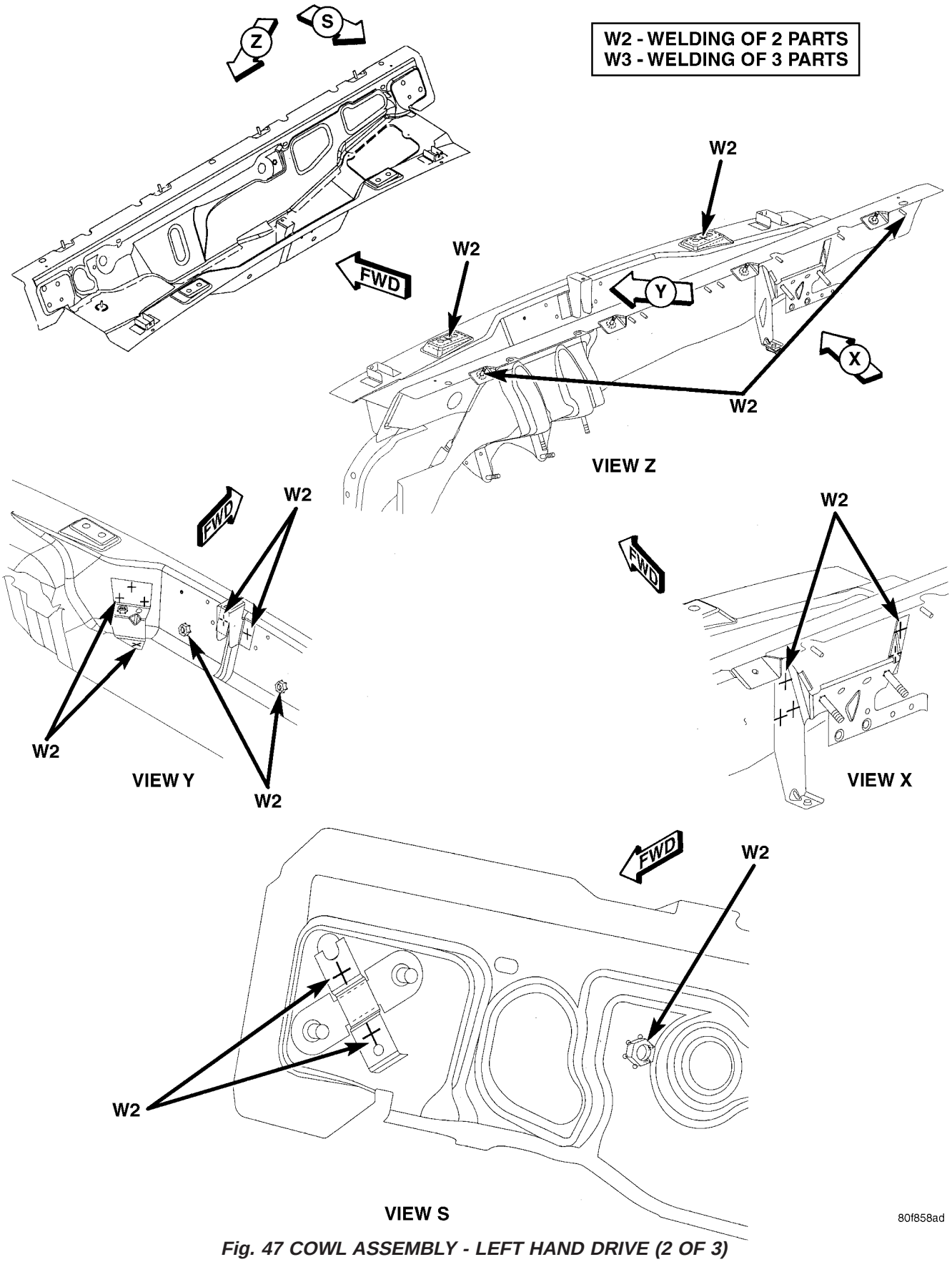


Fig. 47 COWL ASSEMBLY - LEFT HAND DRIVE (2 OF 3)

WELD LOCATIONS (Continued)

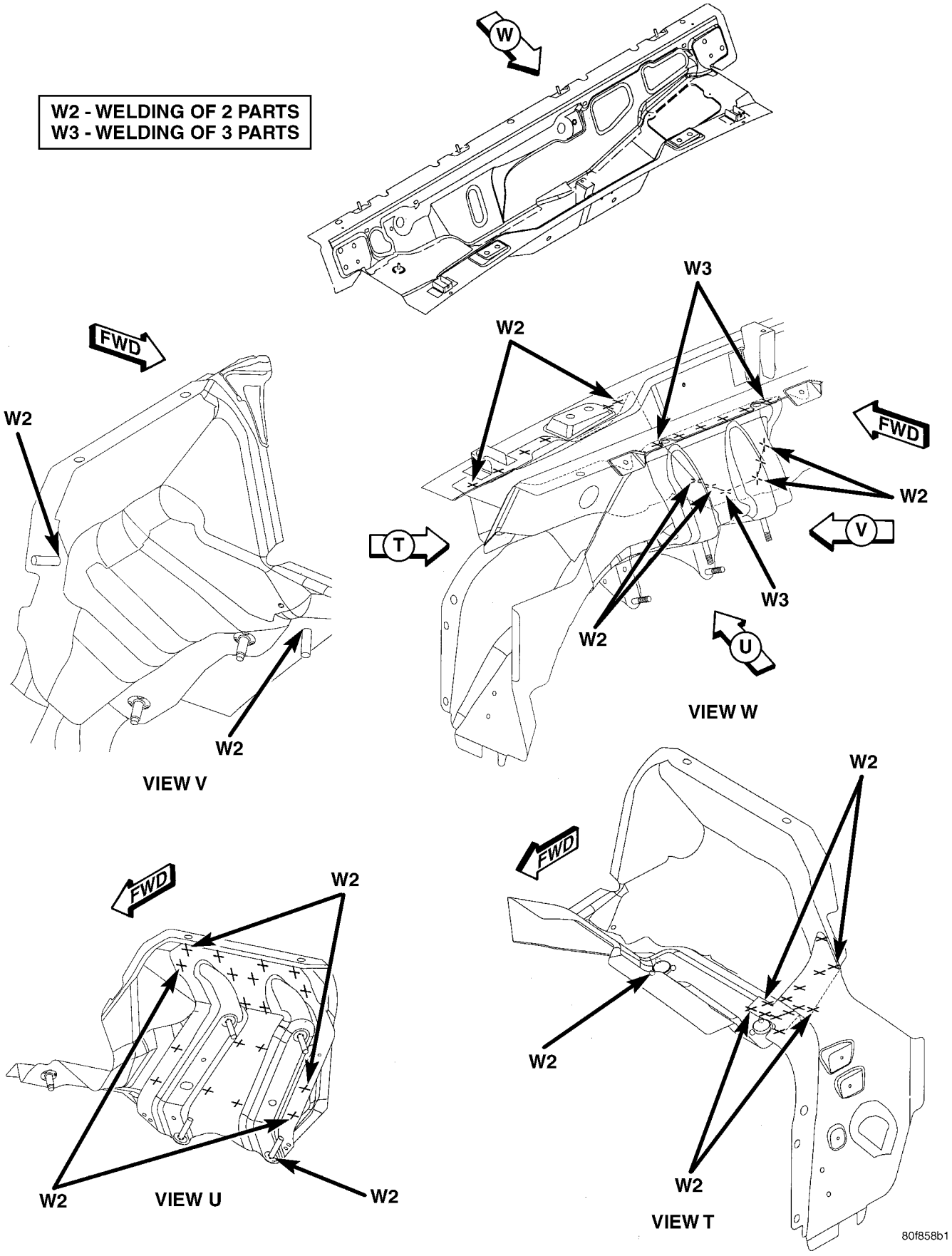


Fig. 48 COWL ASSEMBLY - LEFT HAND DRIVE (3 OF 3)

WELD LOCATIONS (Continued)

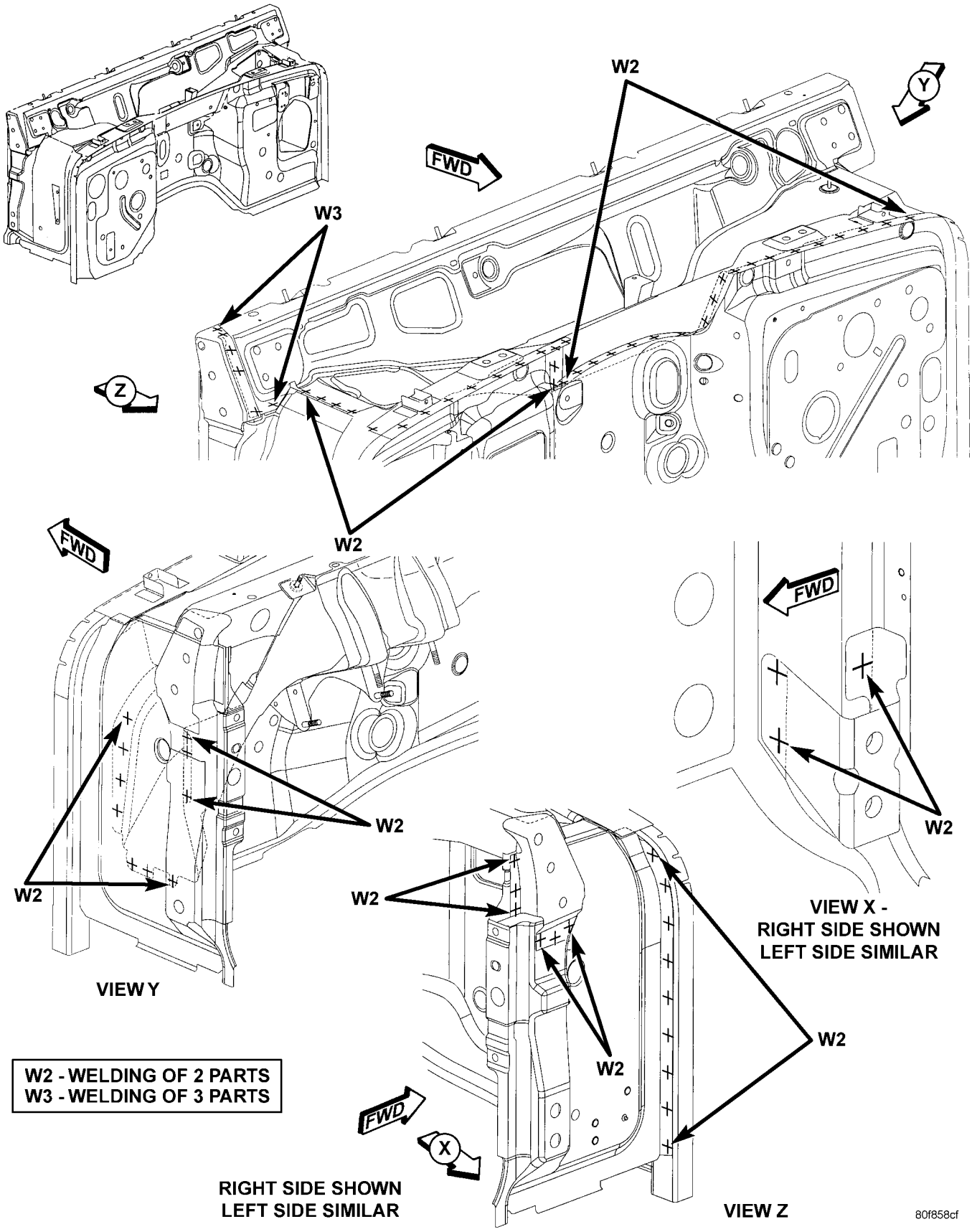


Fig. 49 DASH COWL ASSEMBLY - LEFT HAND DRIVE

WELD LOCATIONS (Continued)

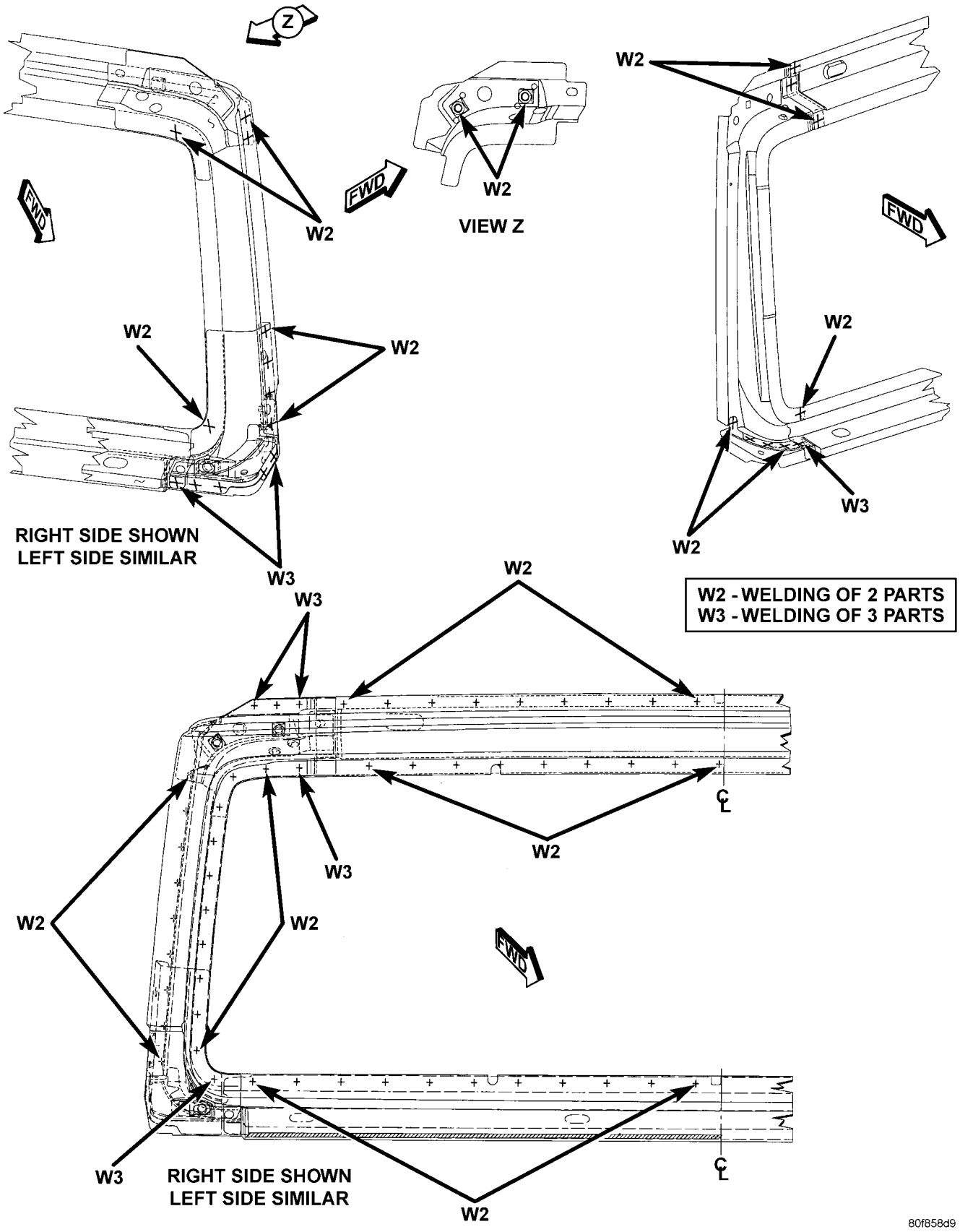


Fig. 50 WINDSHIELD FRAME

WELD LOCATIONS (Continued)

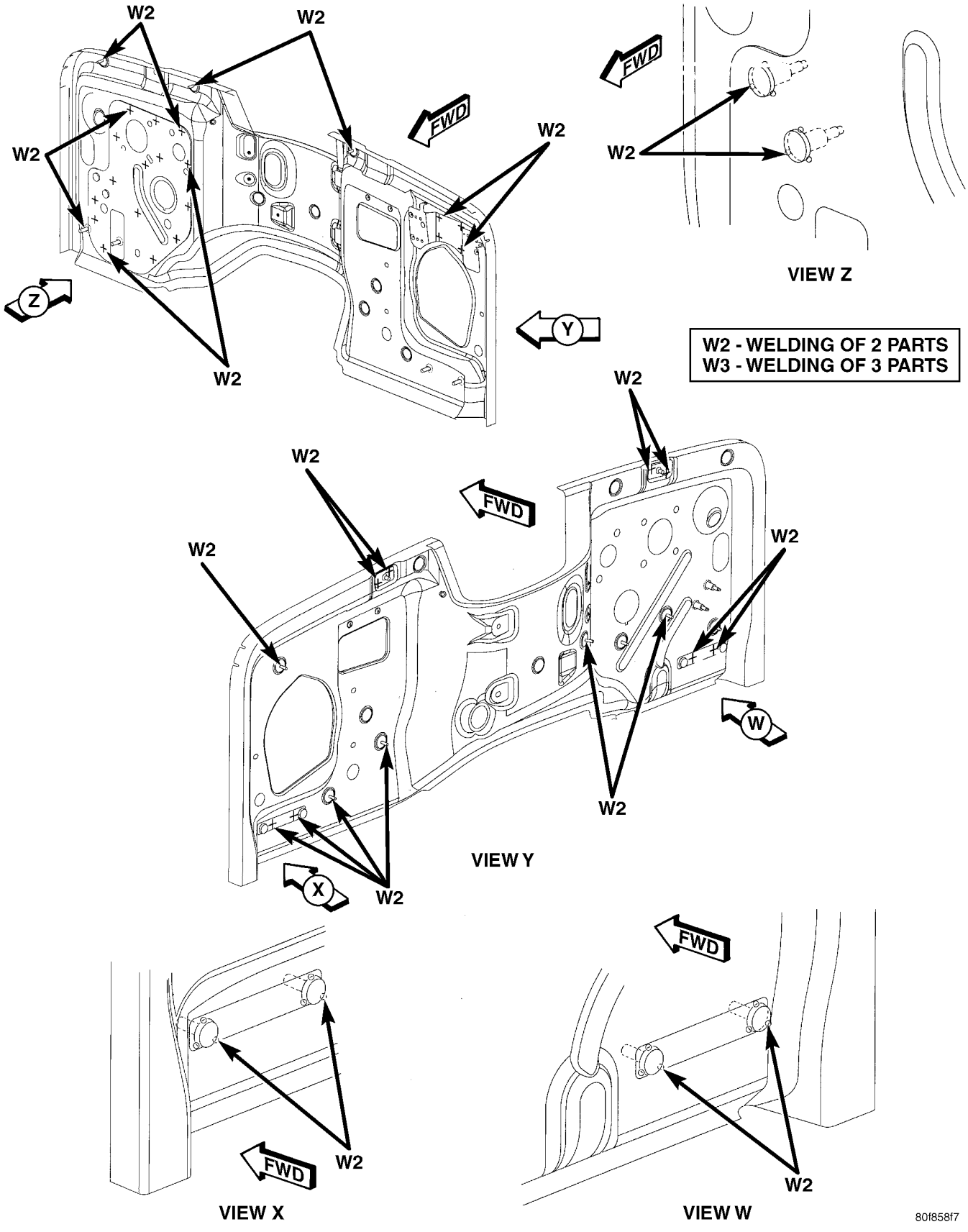


Fig. 51 DASH PANEL - RIGHT HAND DRIVE

WELD LOCATIONS (Continued)

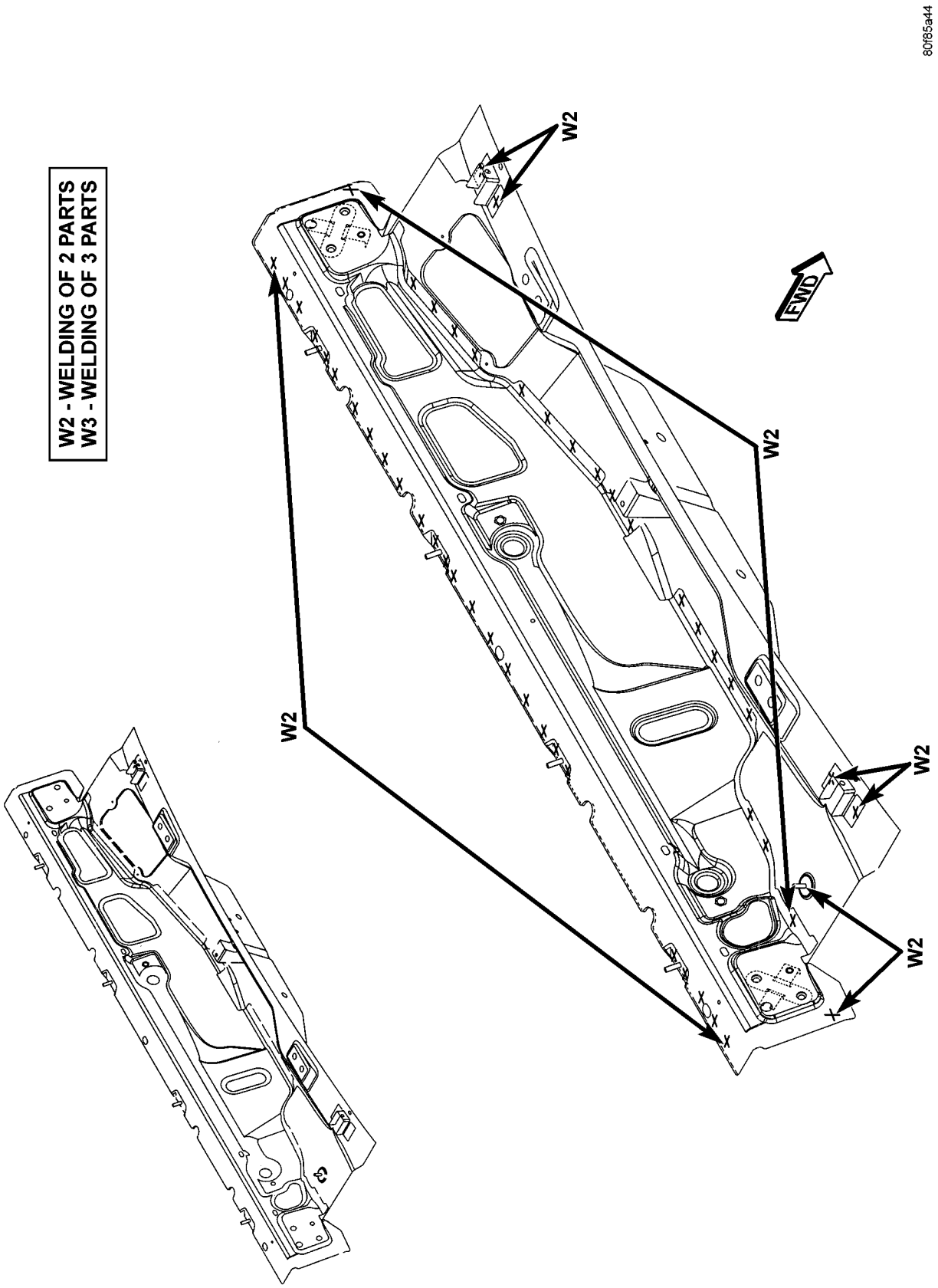


Fig. 52 COWL ASSEMBLY - RIGHT HAND DRIVE (1 OF 3)

WELD LOCATIONS (Continued)

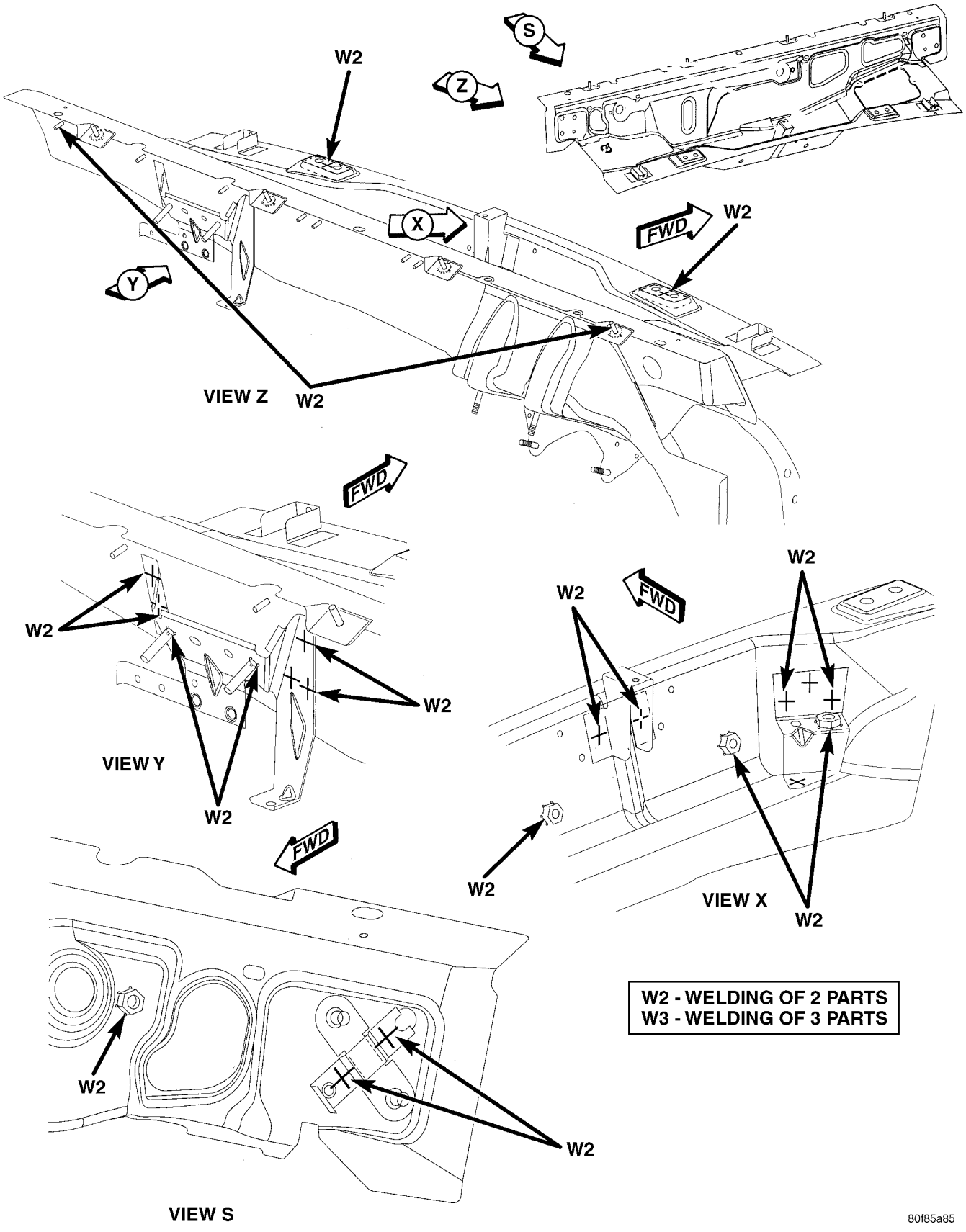


Fig. 53 COWL ASSEMBLY - RIGHT HAND DRIVE (2 OF 3)

WELD LOCATIONS (Continued)

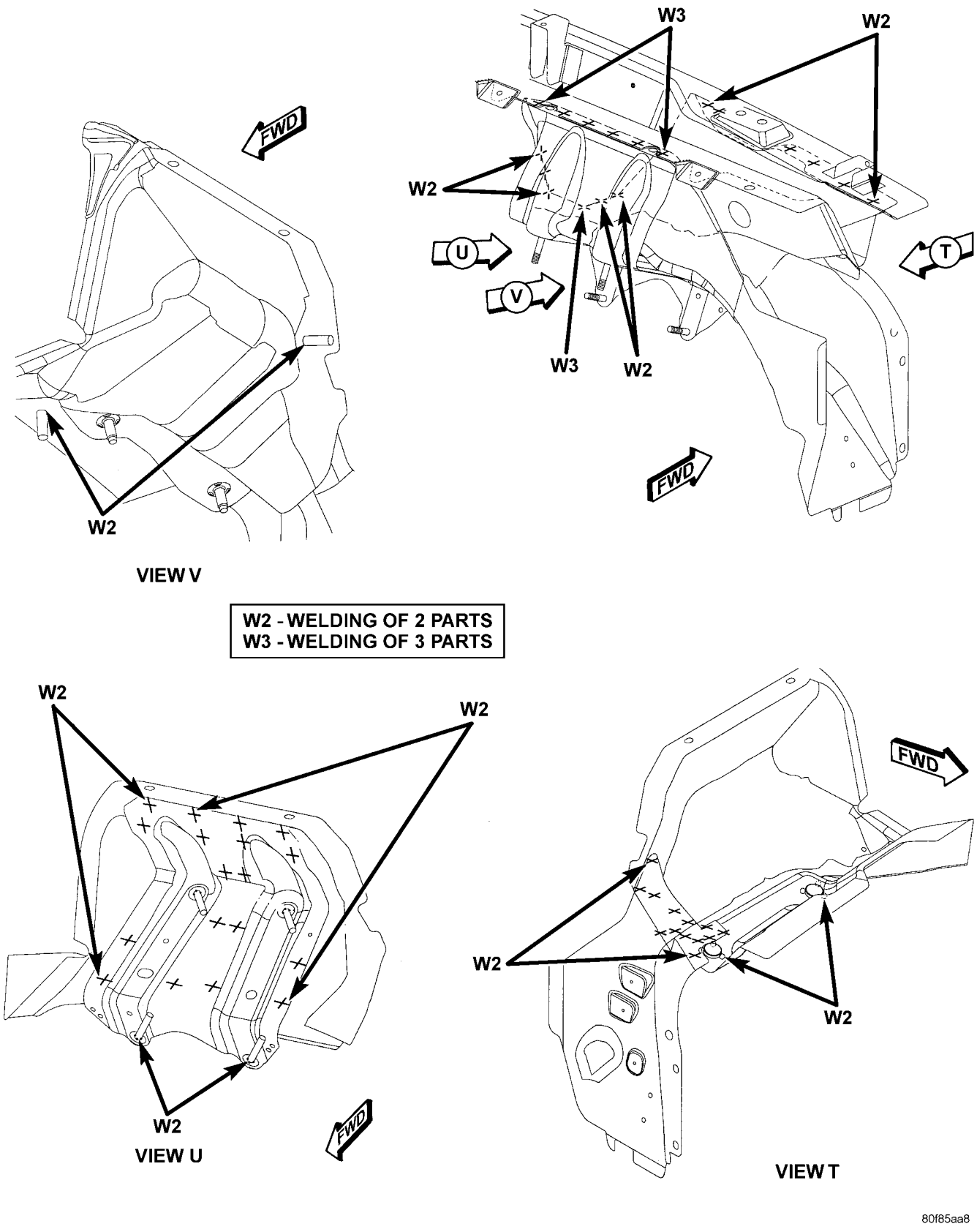


Fig. 54 COWL ASSEMBLY - RIGHT HAND DRIVE (3 OF 3)

WELD LOCATIONS (Continued)

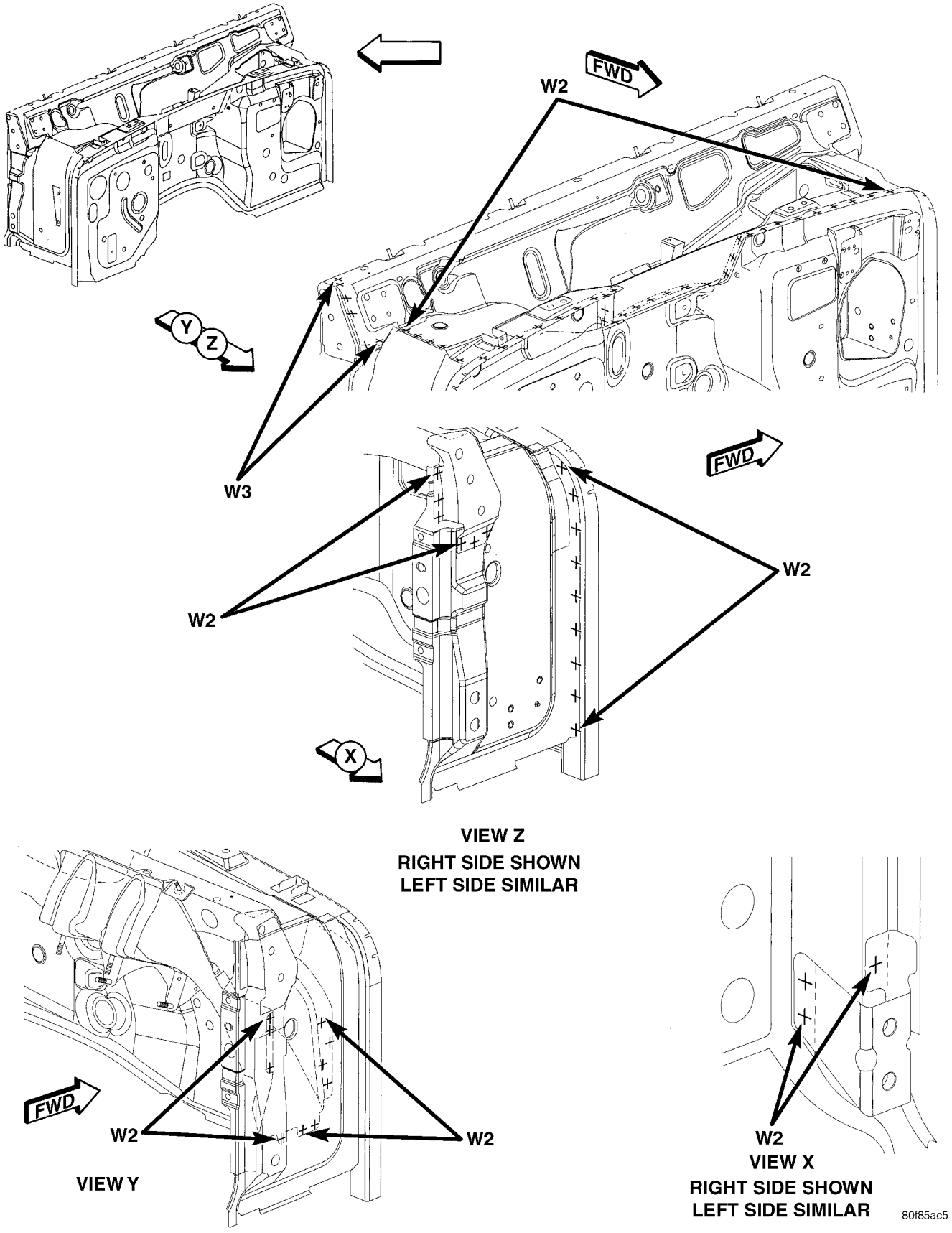


Fig. 55 DASH COWL ASSEMBLY - RIGHT HAND DRIVE

WELD LOCATIONS (Continued)

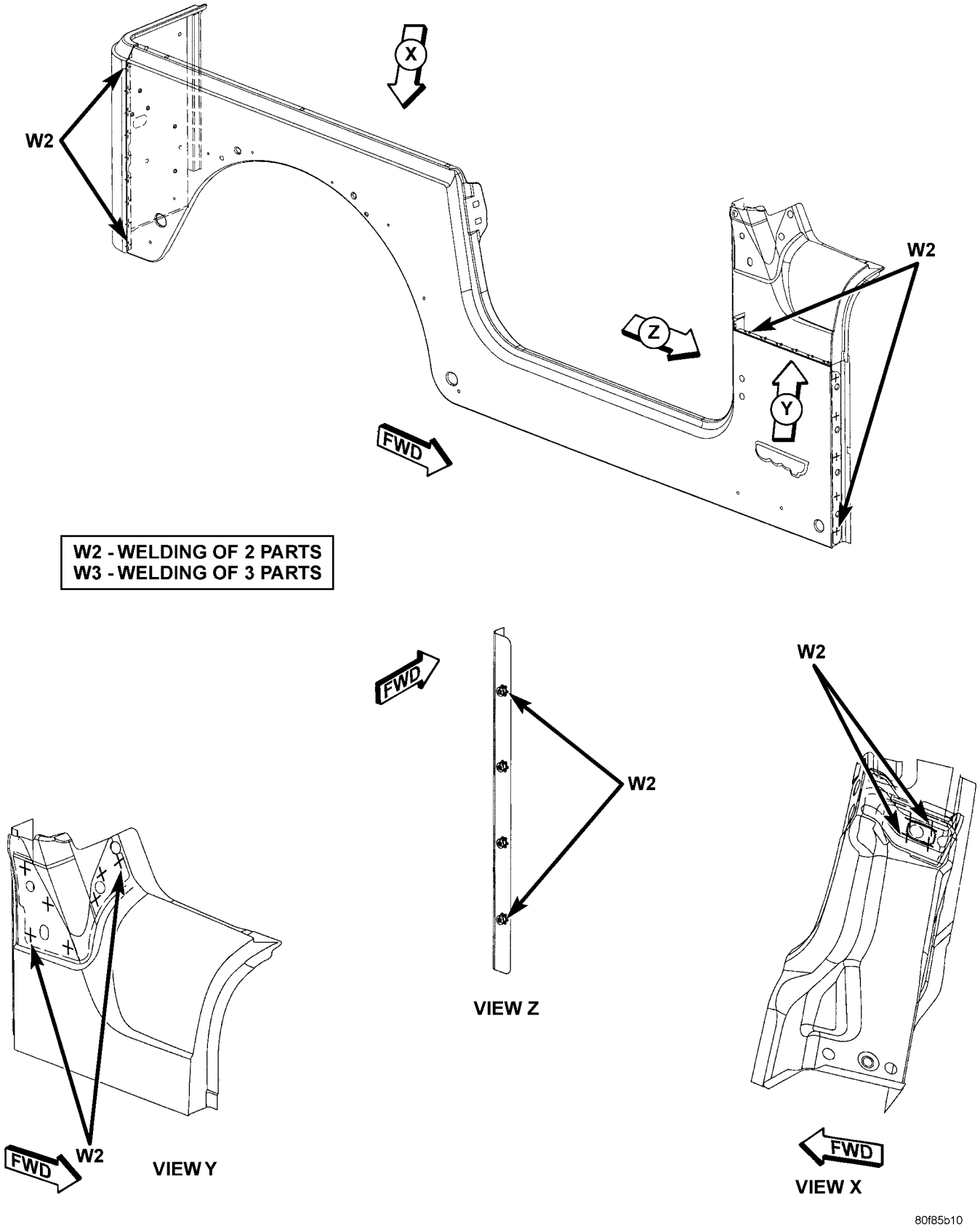
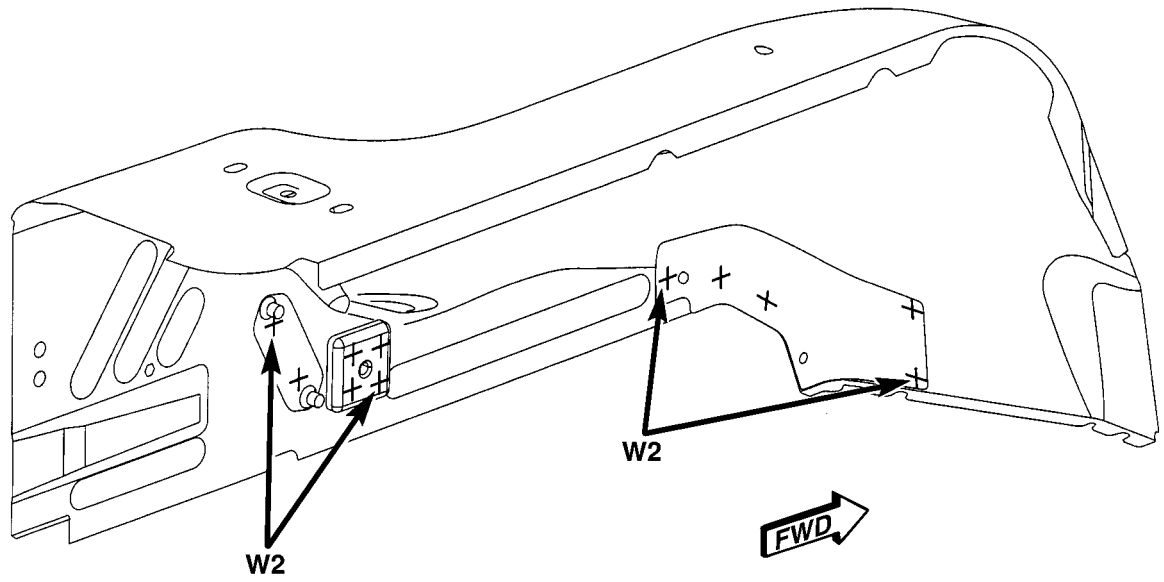


Fig. 56 BODY SIDE ASSEMBLY

WELD LOCATIONS (Continued)



W2 - WELDING OF 2 PARTS
W3 - WELDING OF 3 PARTS

RIGHT SIDE SHOWN,
LEFT SIDE SIMILAR

80f85b86

Fig. 57 WHEELHOUSE ASSEMBLY

WELD LOCATIONS (Continued)

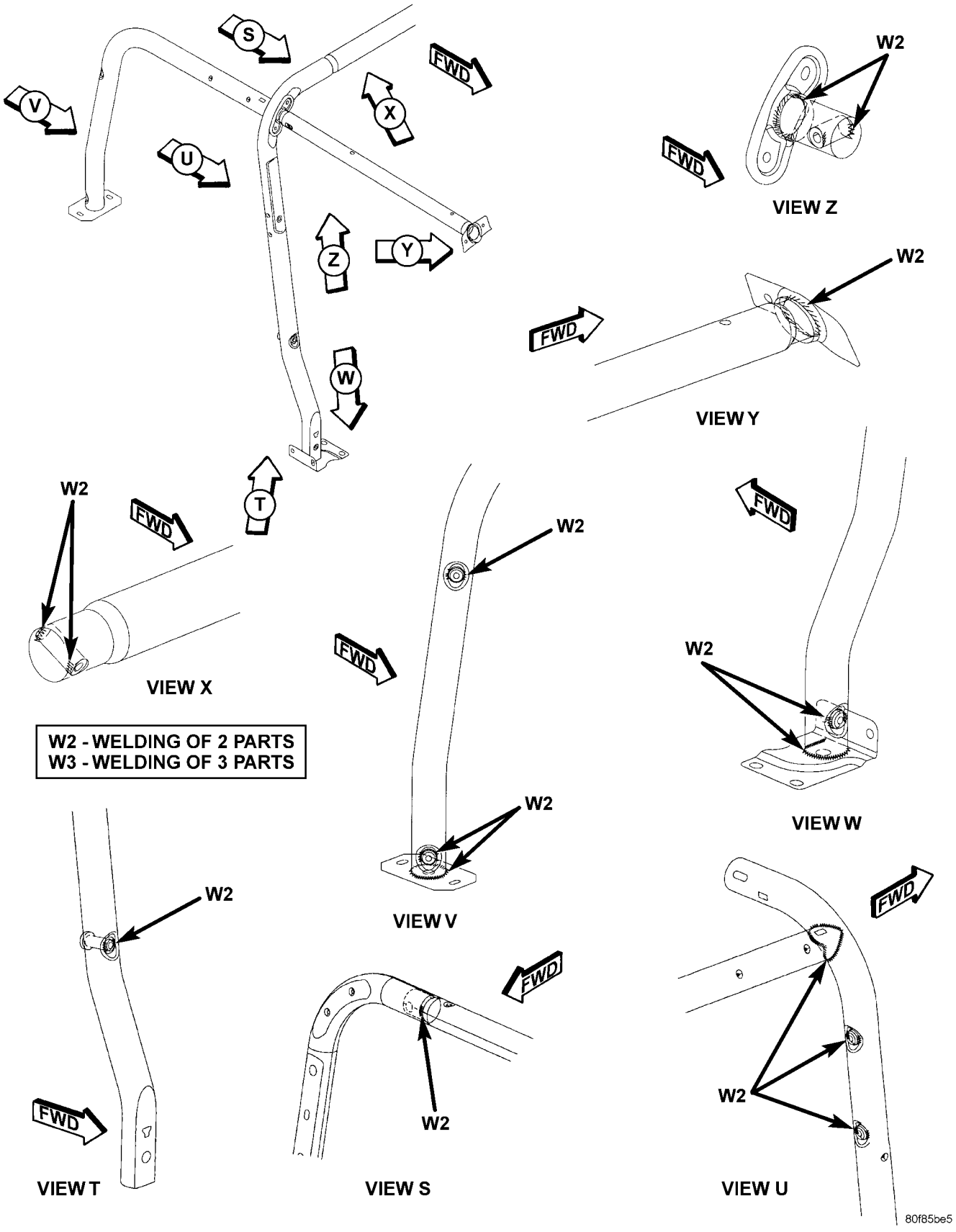


Fig. 58 SPORT BAR ASSEMBLY

WELD LOCATIONS (Continued)

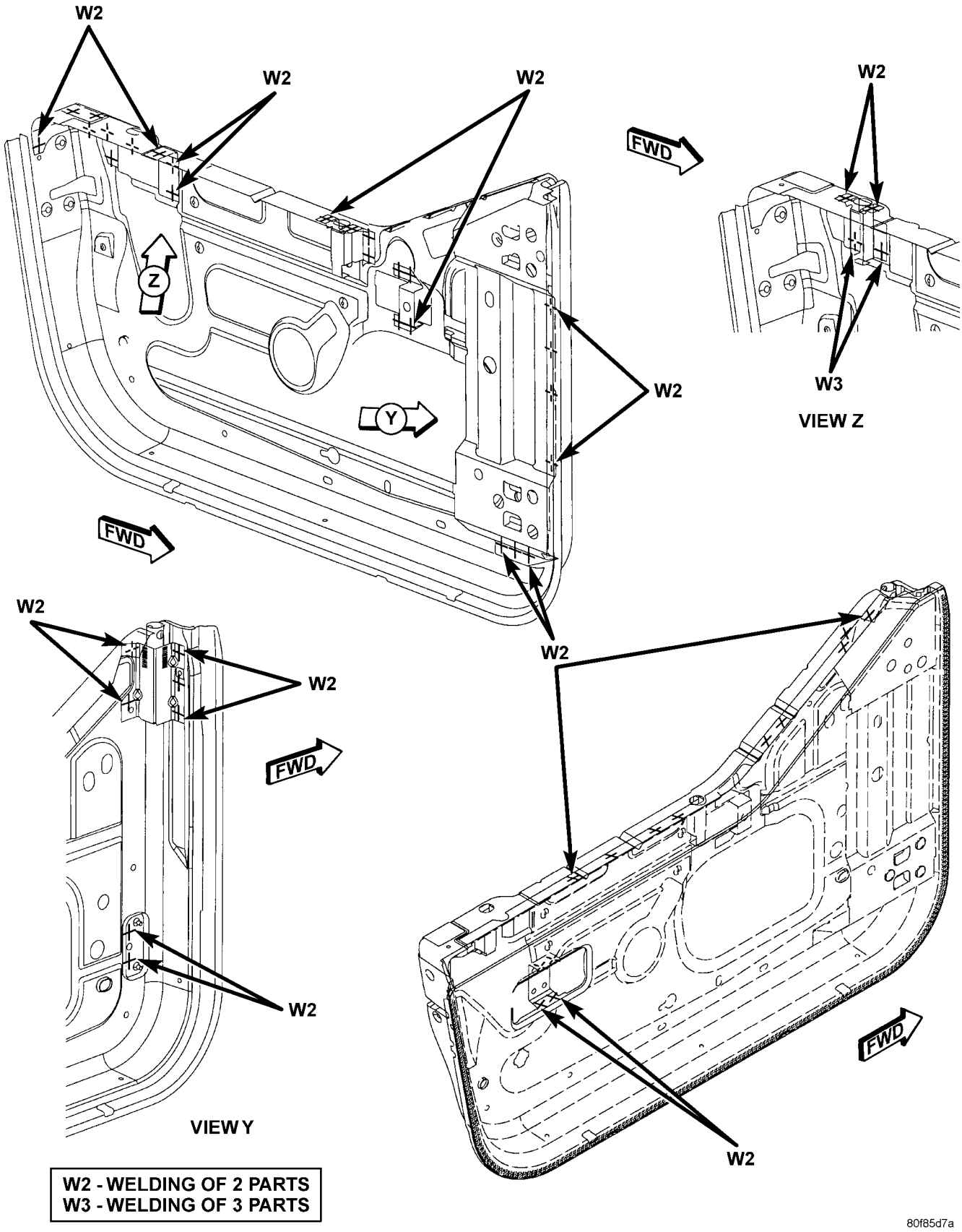


Fig. 59 HALF DOOR ASSEMBLY

WELD LOCATIONS (Continued)

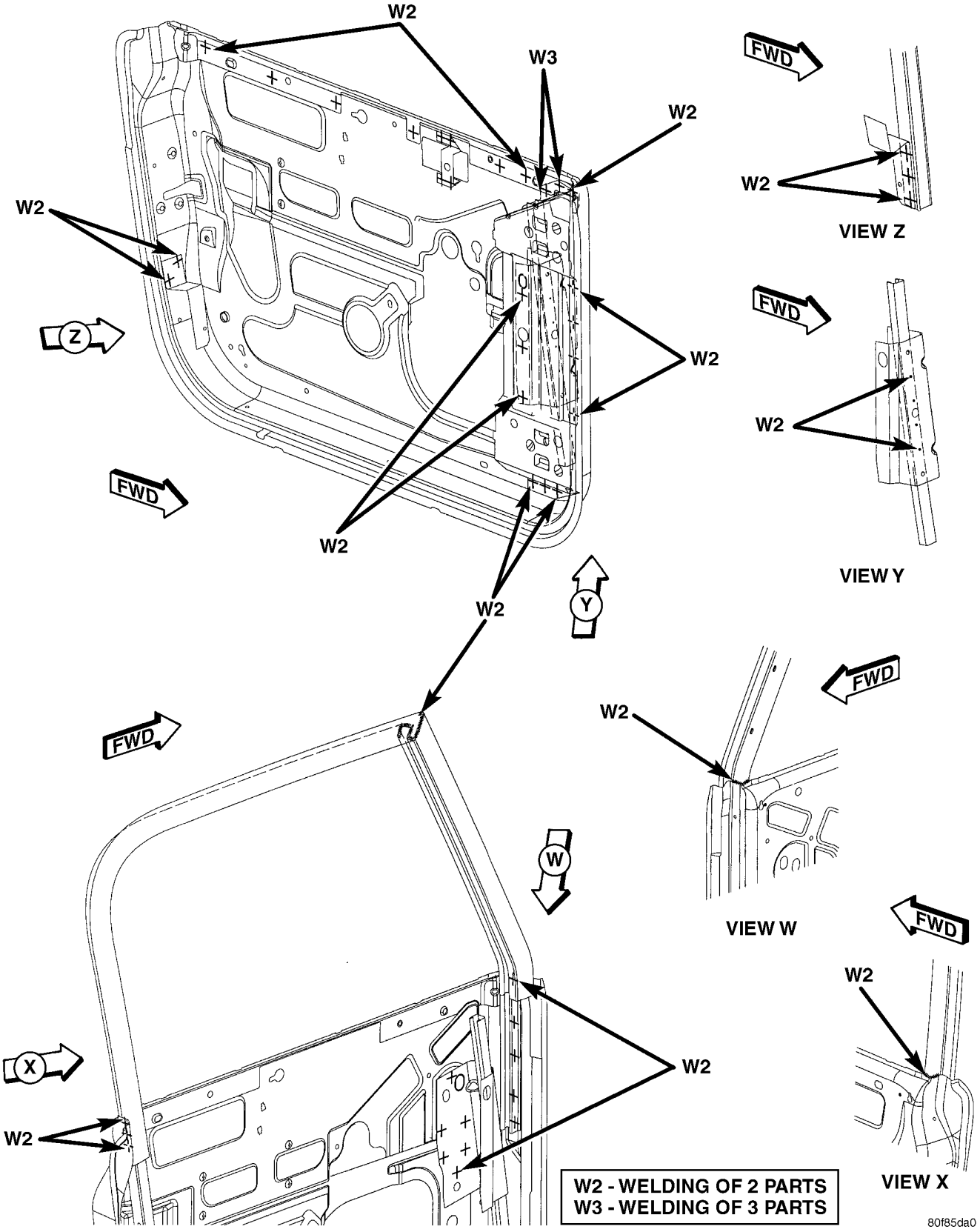


Fig. 60 FULL DOOR ASSEMBLY (1 OF 2)

WELD LOCATIONS (Continued)

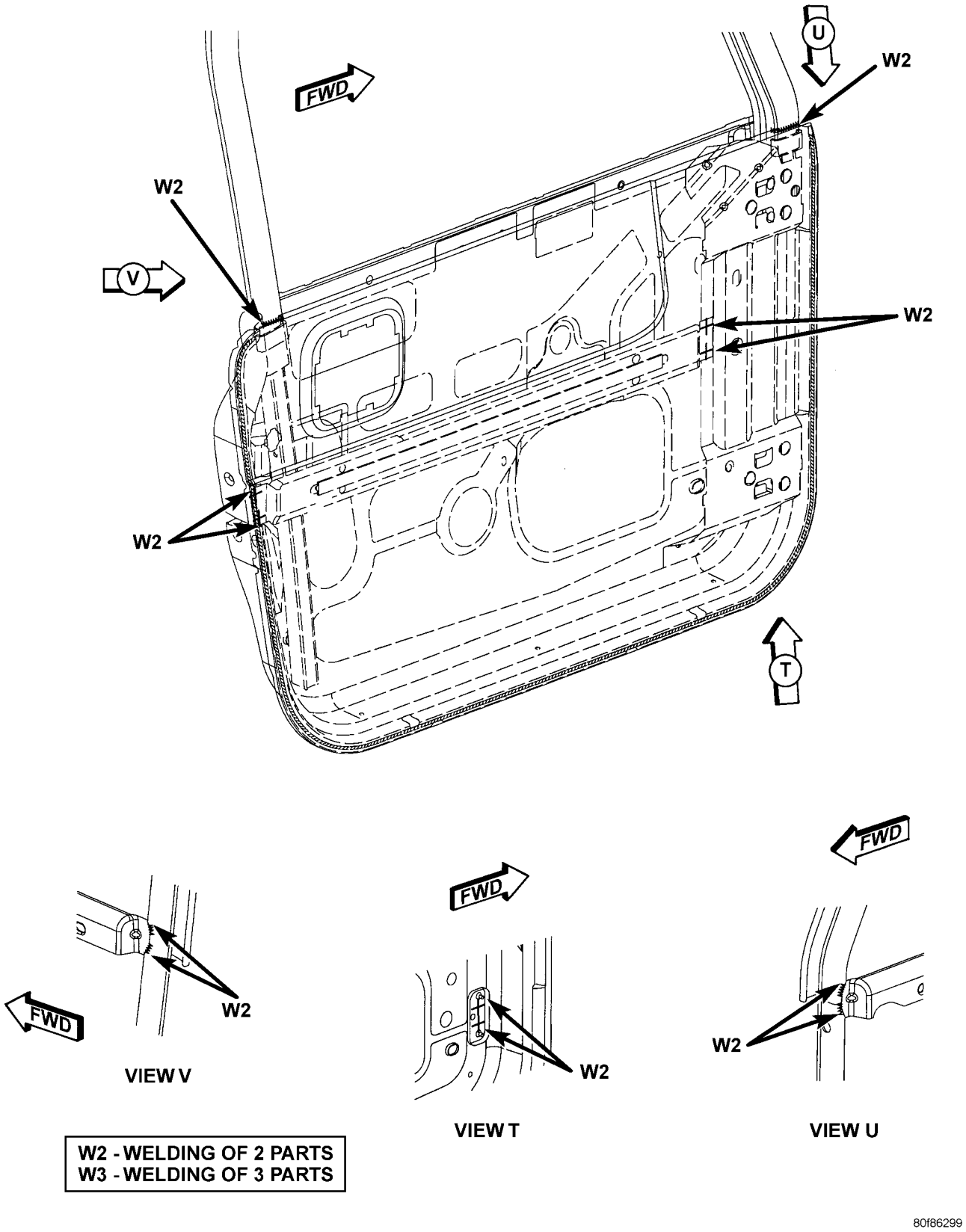


Fig. 61 FULL DOOR ASSEMBLY (2 OF 2)

WELD LOCATIONS (Continued)

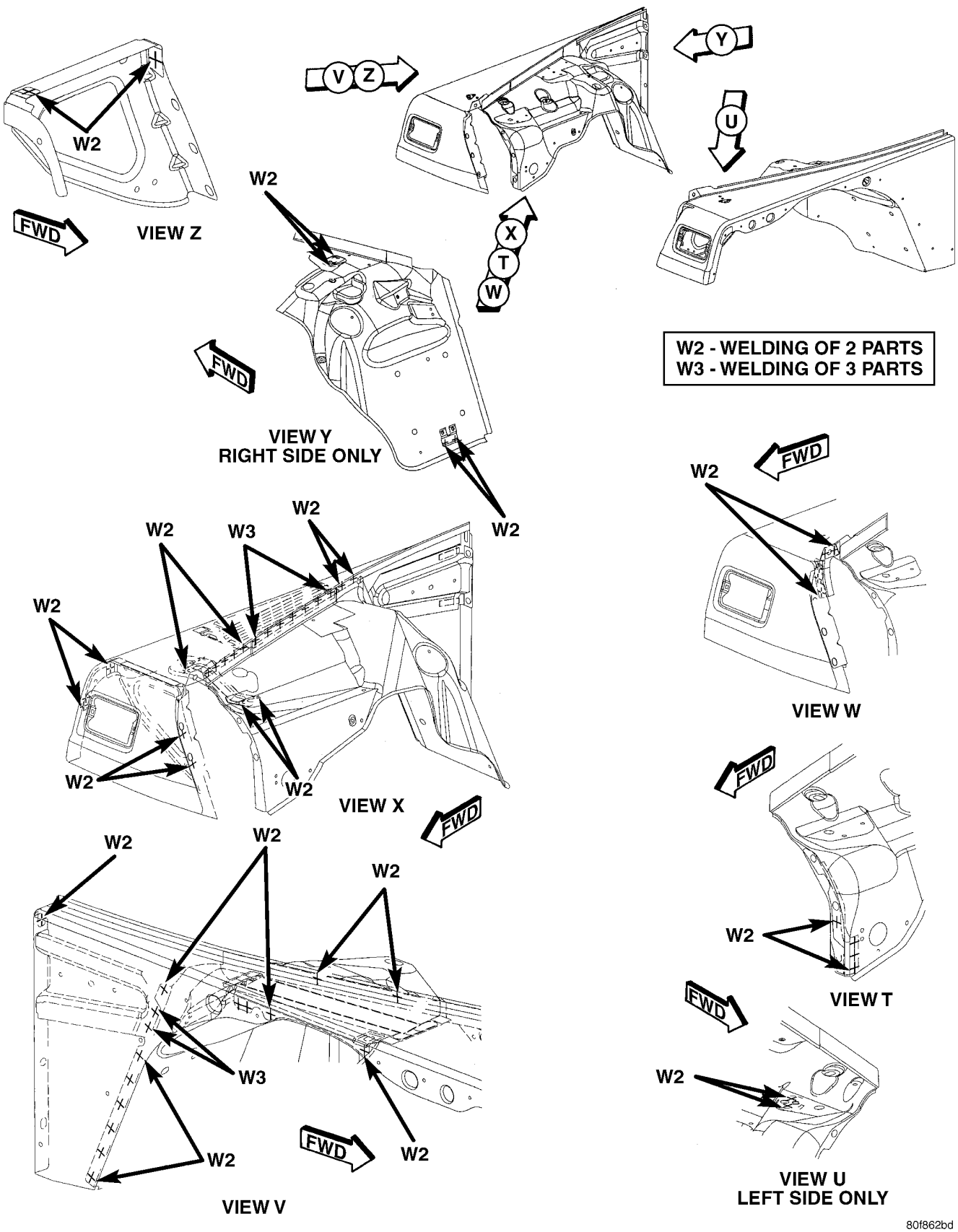


Fig. 62 FRONT FENDER ASSEMBLY

WELD LOCATIONS (Continued)

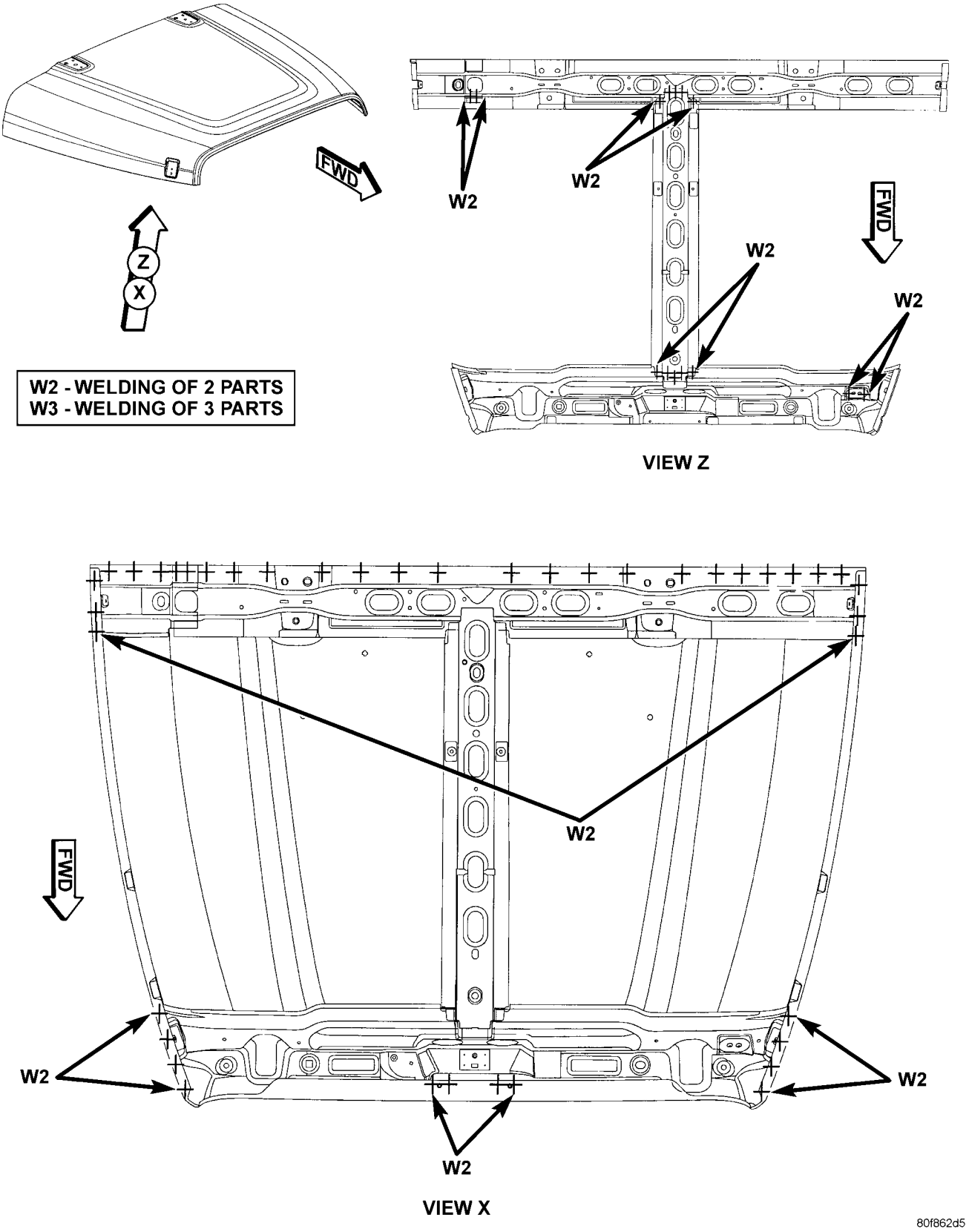


Fig. 63 HOOD ASSEMBLY

WELD LOCATIONS (Continued)

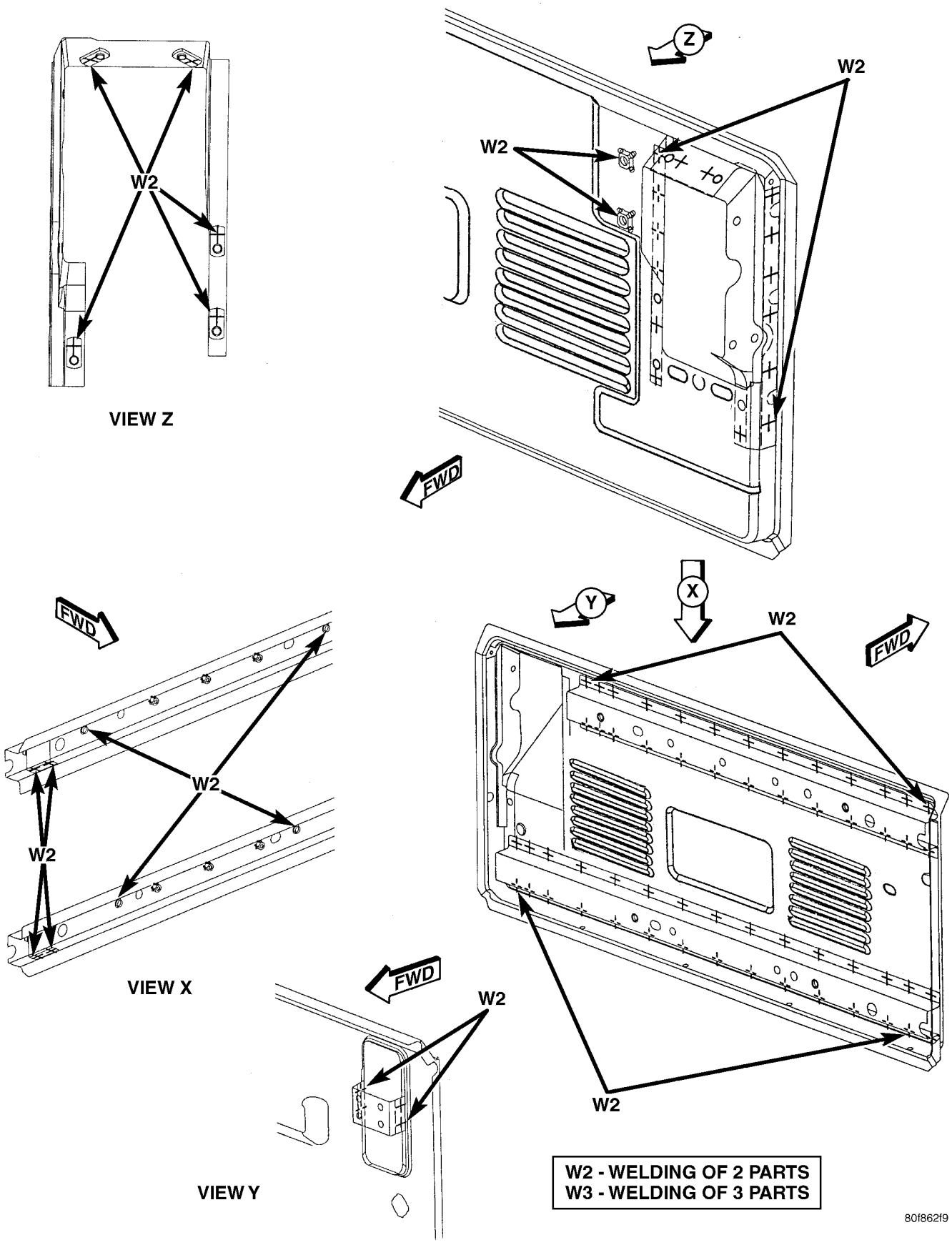


Fig. 64 TAILGATE ASSEMBLY

WELD LOCATIONS (Continued)

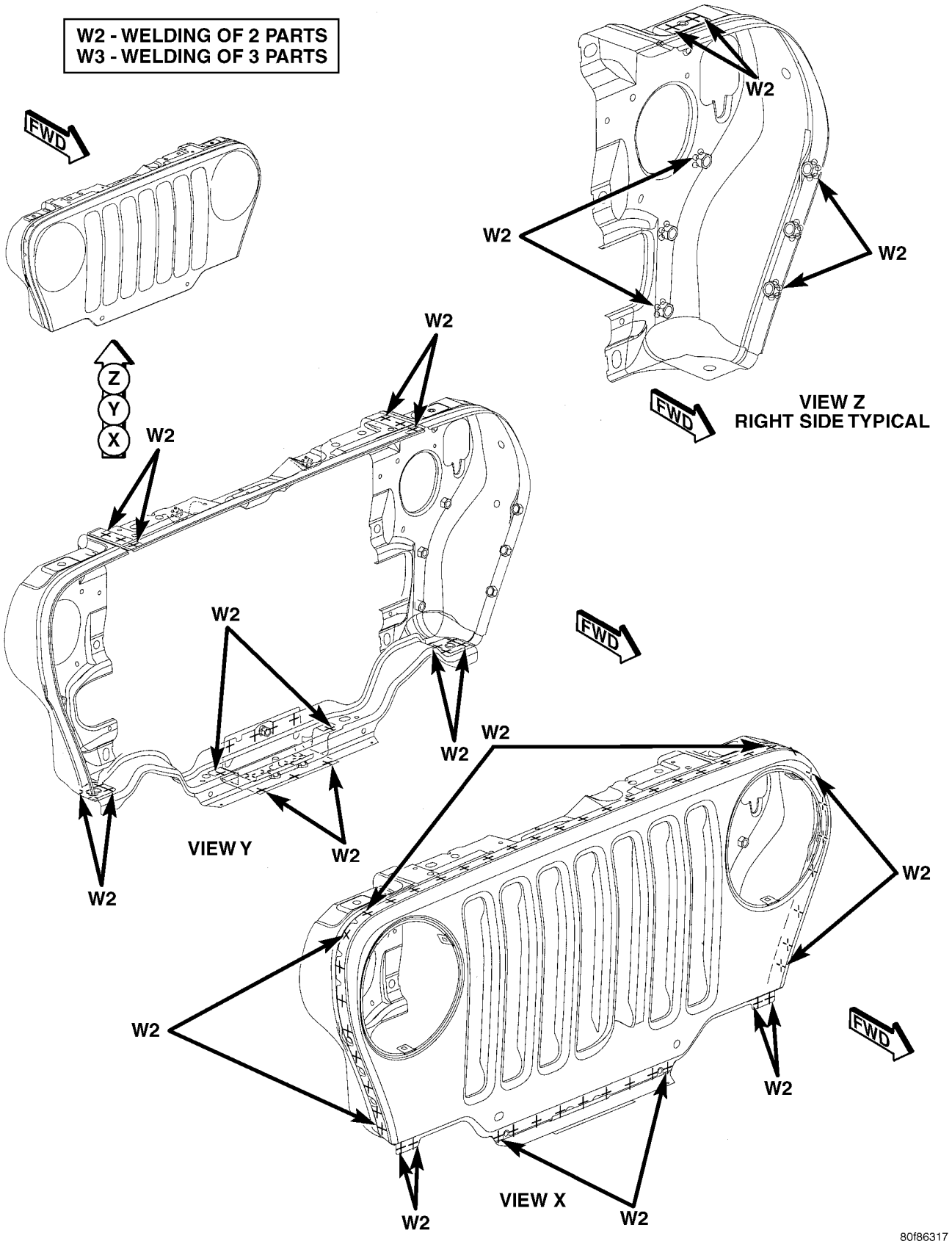


Fig. 65 RADIATOR GUARD ASSEMBLY

WELD LOCATIONS (Continued)

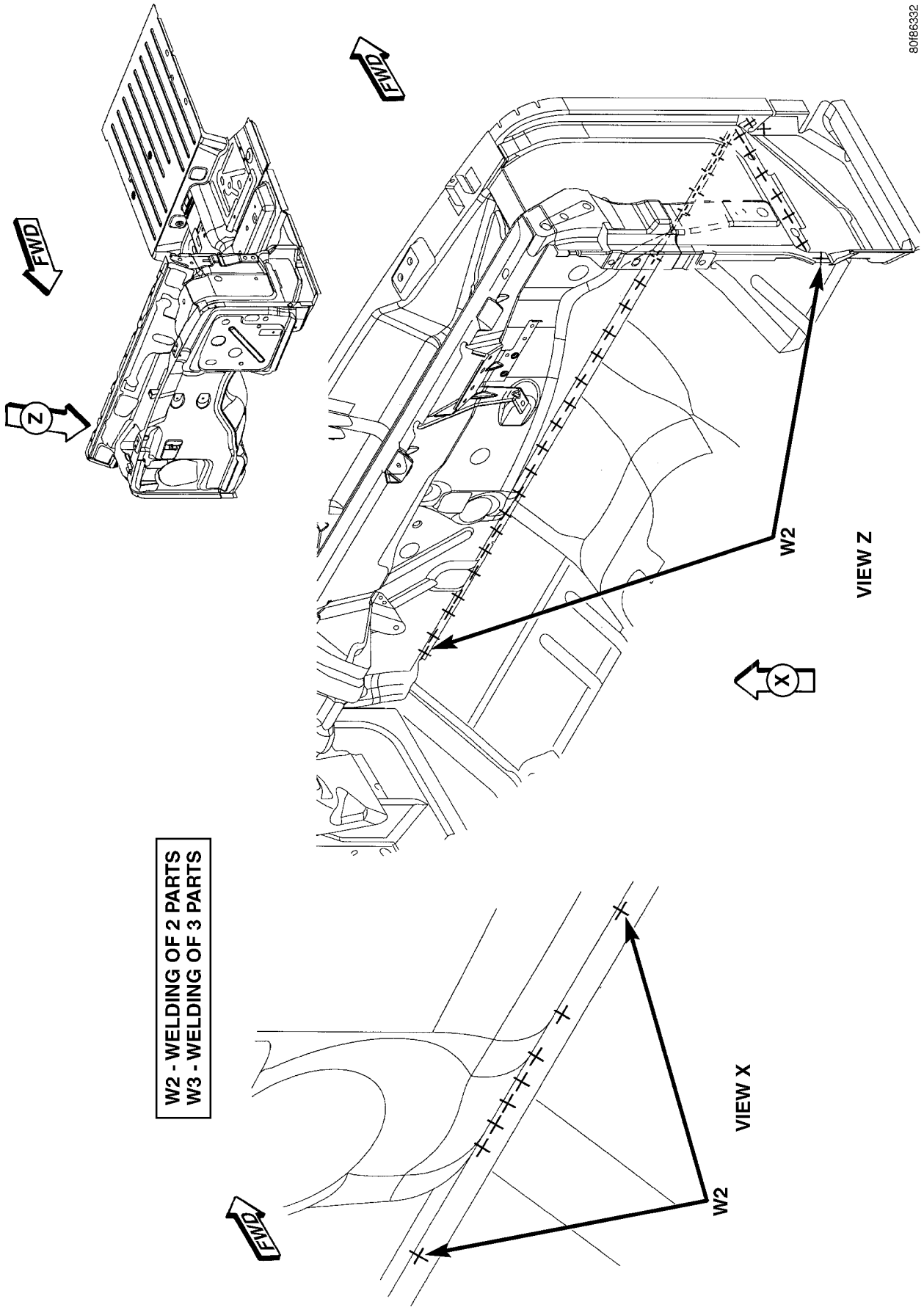


Fig. 66 DASH, COWL AND FLOOR ASSEMBLY - LEFT HAND DRIVE

WELD LOCATIONS (Continued)

Fig. 67 DASH, COWL AND FLOOR ASSEMBLY - RIGHT HAND DRIVE

WELD LOCATIONS (Continued)

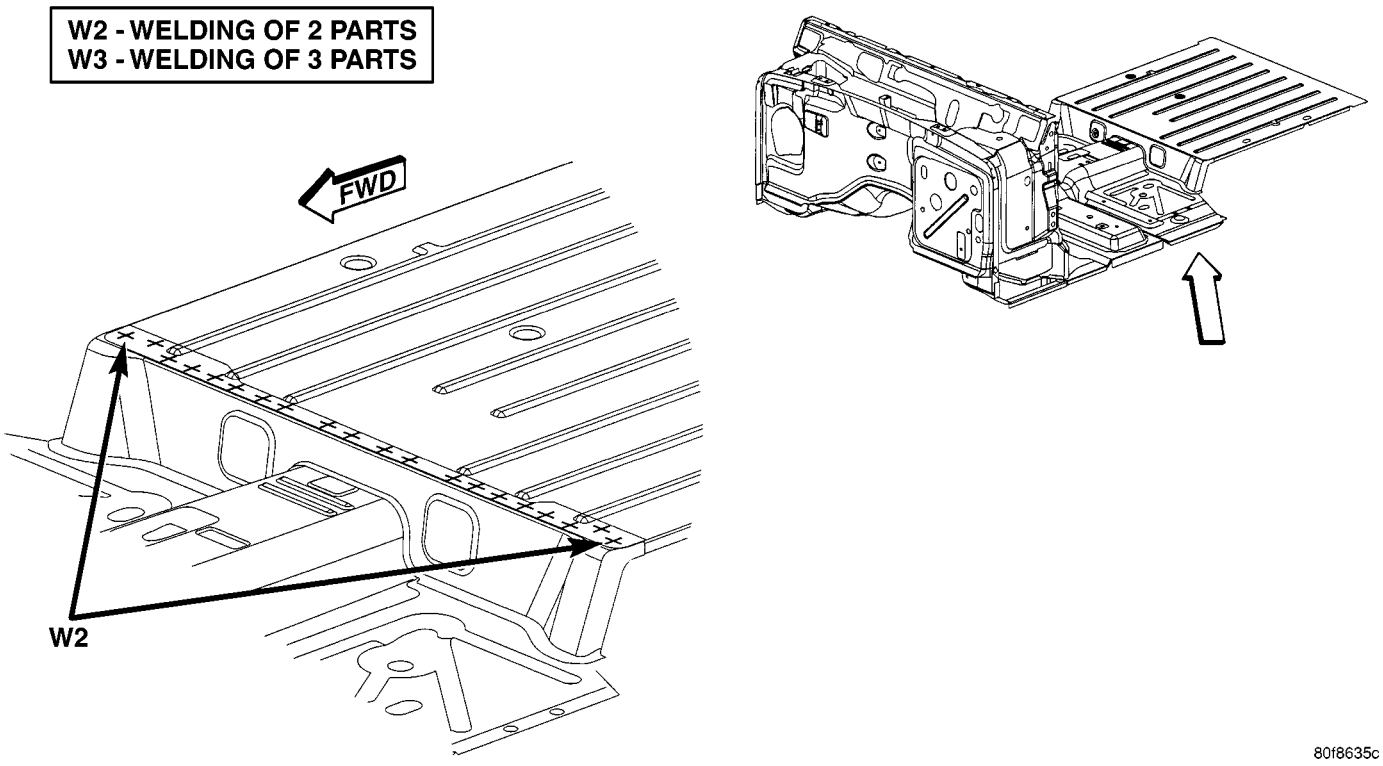


Fig. 68 FRONT AND REAR FLOOR PAN

WELD LOCATIONS (Continued)

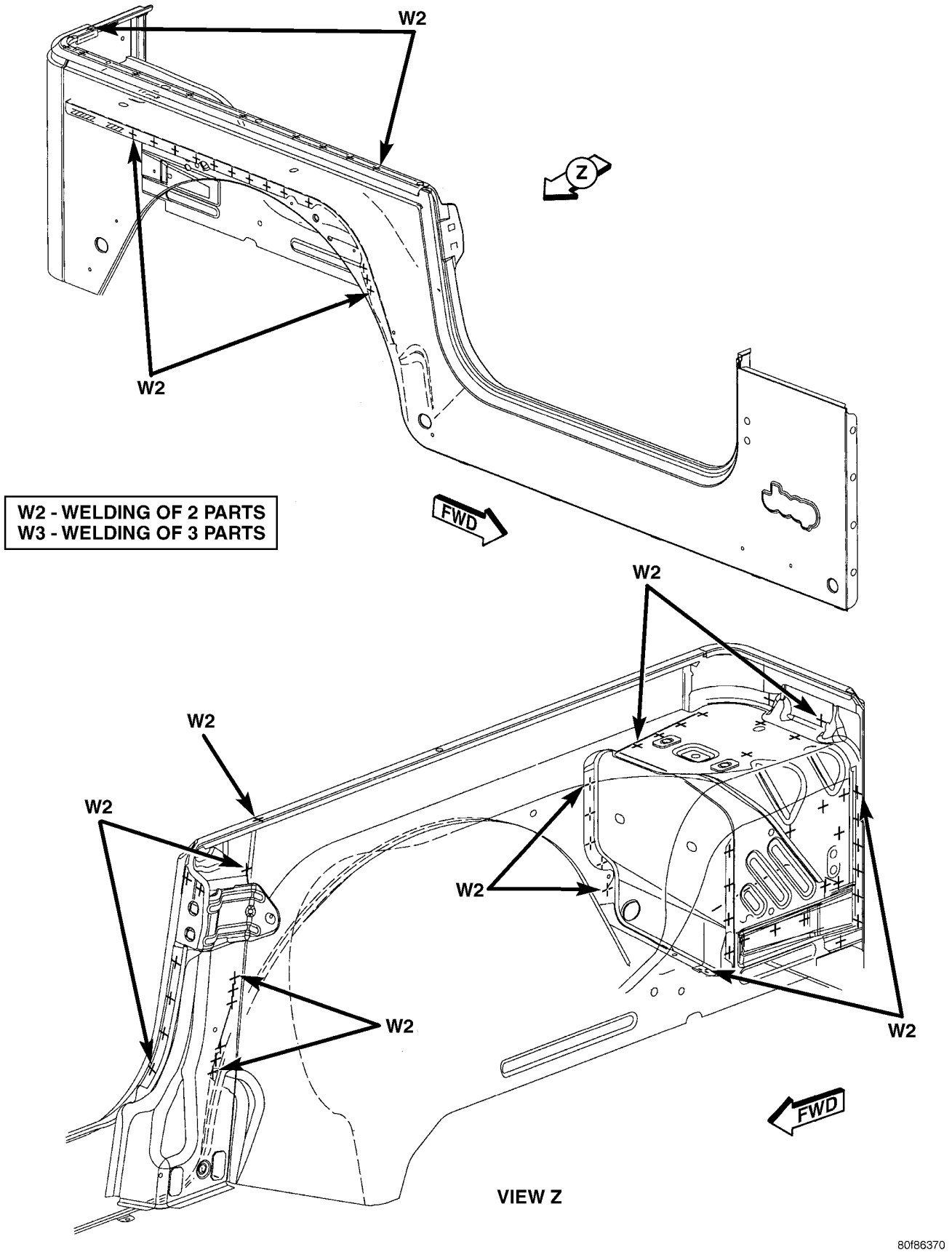


Fig. 69 BODY SIDE (1 OF 2)

WELD LOCATIONS (Continued)

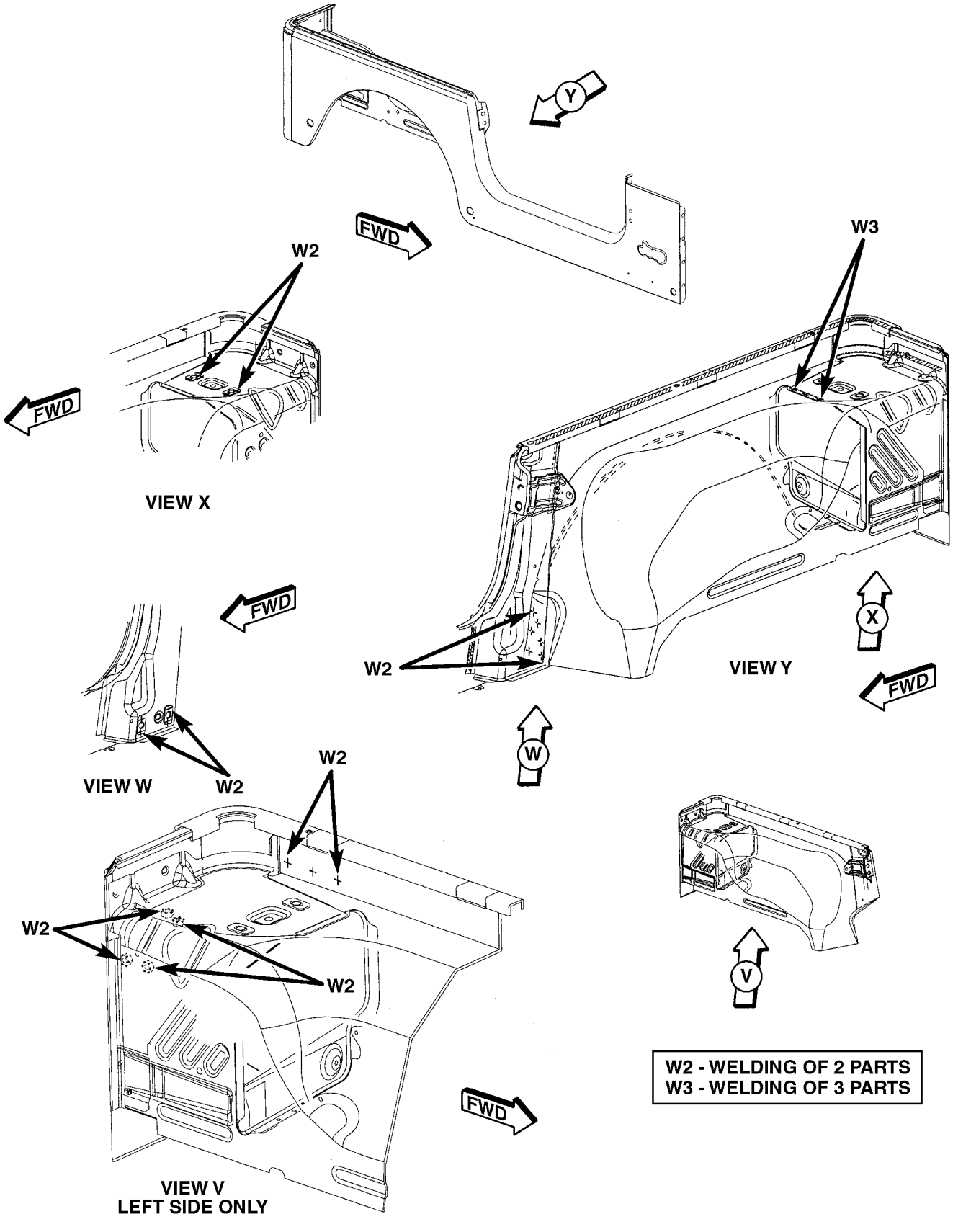


Fig. 70 BODY SIDE (2 OF 2)

WELD LOCATIONS (Continued)

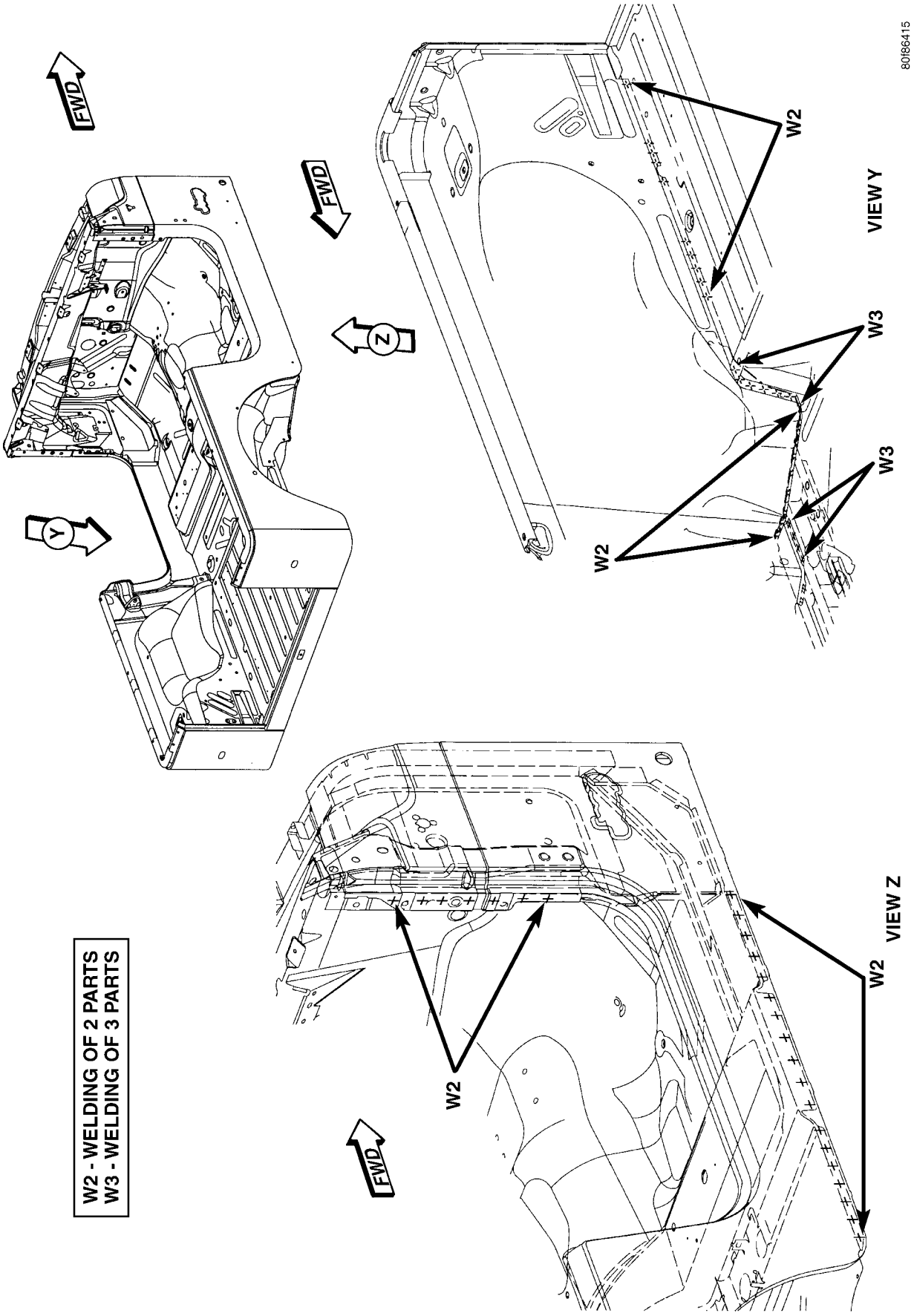


Fig. 71 BODY COMPLETE (1 OF 4)

WELD LOCATIONS (Continued)

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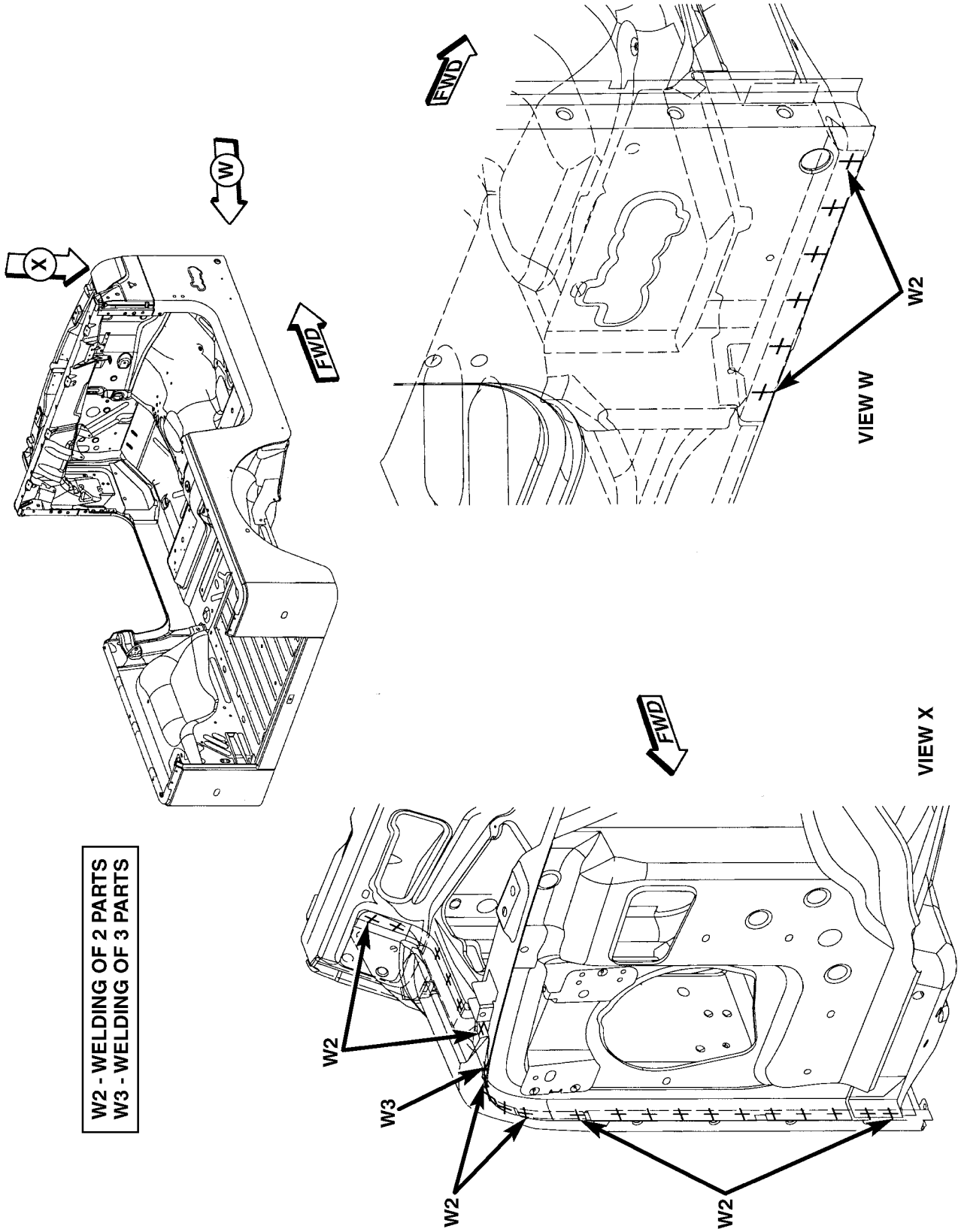

Fig. 72 BODY COMPLETE (2 OF 4)

WELD LOCATIONS (Continued)

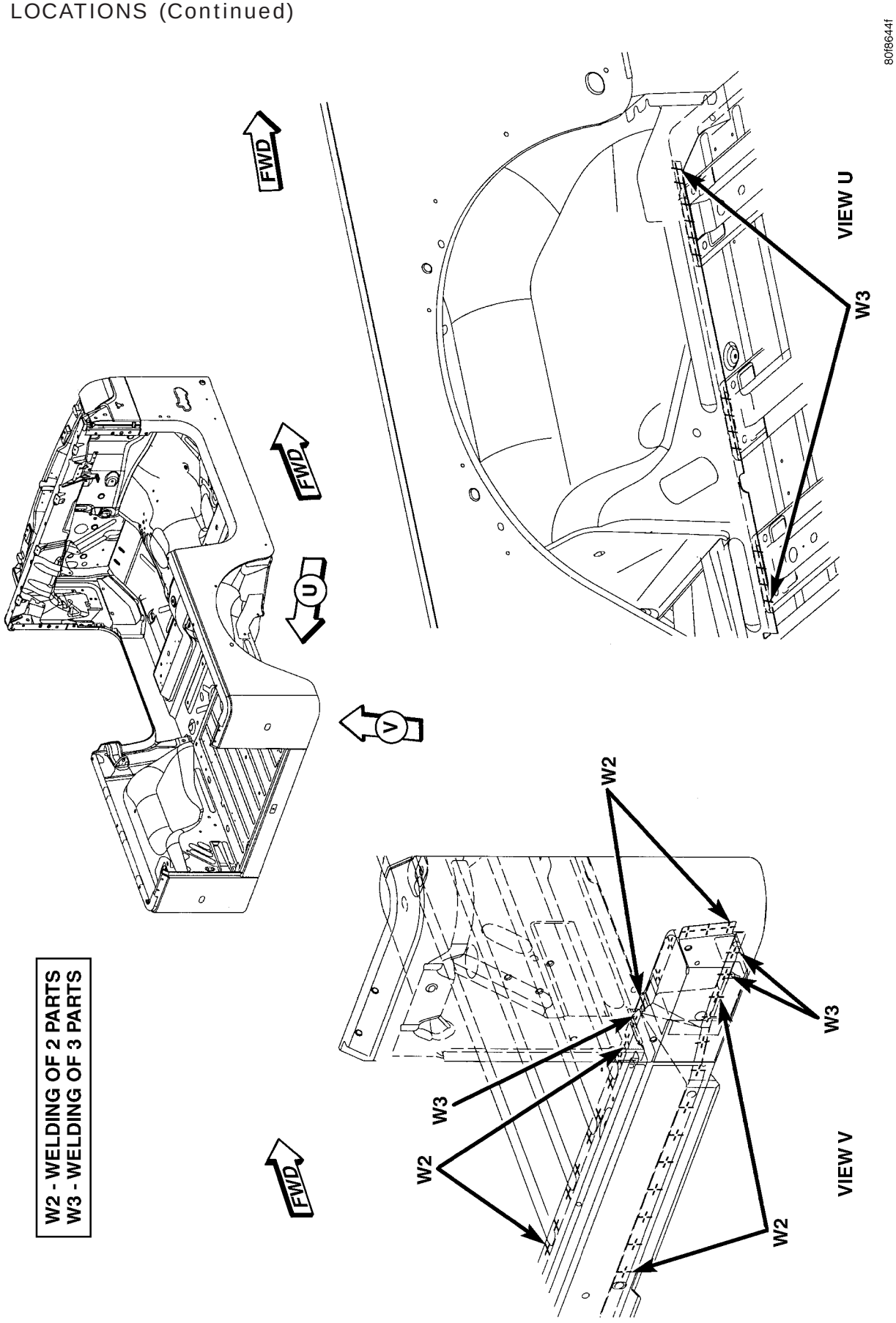

Fig. 73 BODY COMPLETE (3 OF 4)

WELD LOCATIONS (Continued)

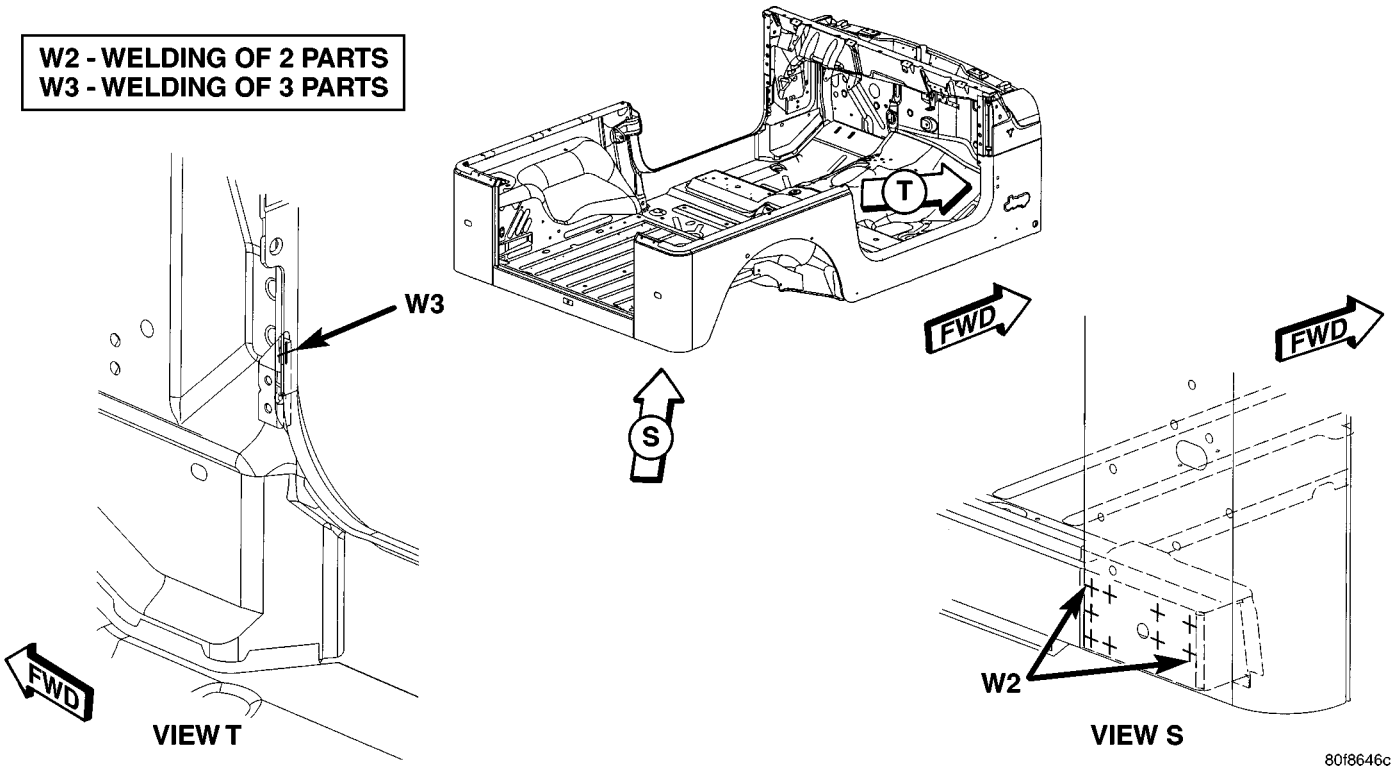

Fig. 74 BODY COMPLETE (4 OF 4)

WELD LOCATIONS (Continued)

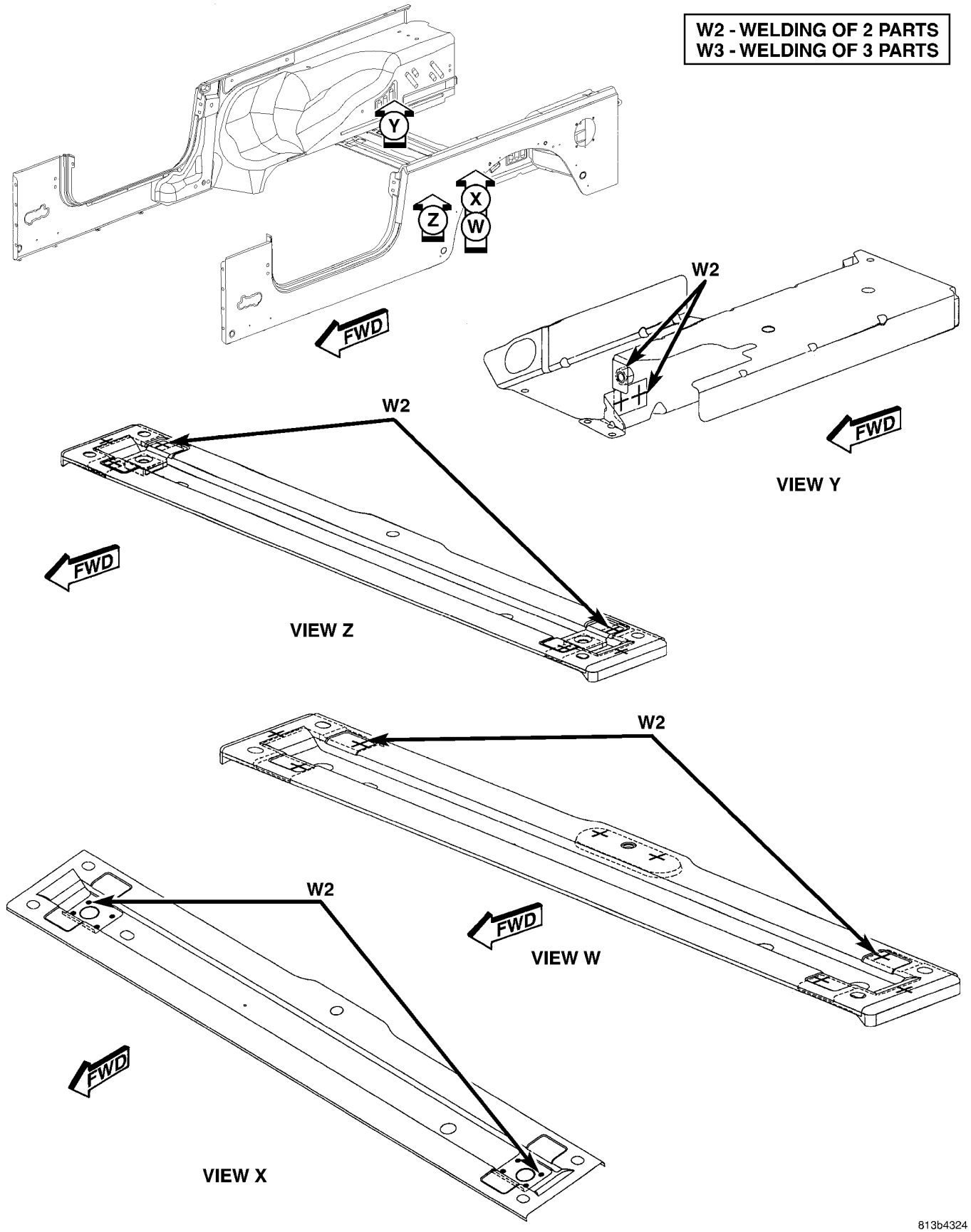

Fig. 75 BODY COMPLETE LONG WHEEL BASE (LWB) (1 OF 3)

WELD LOCATIONS (Continued)

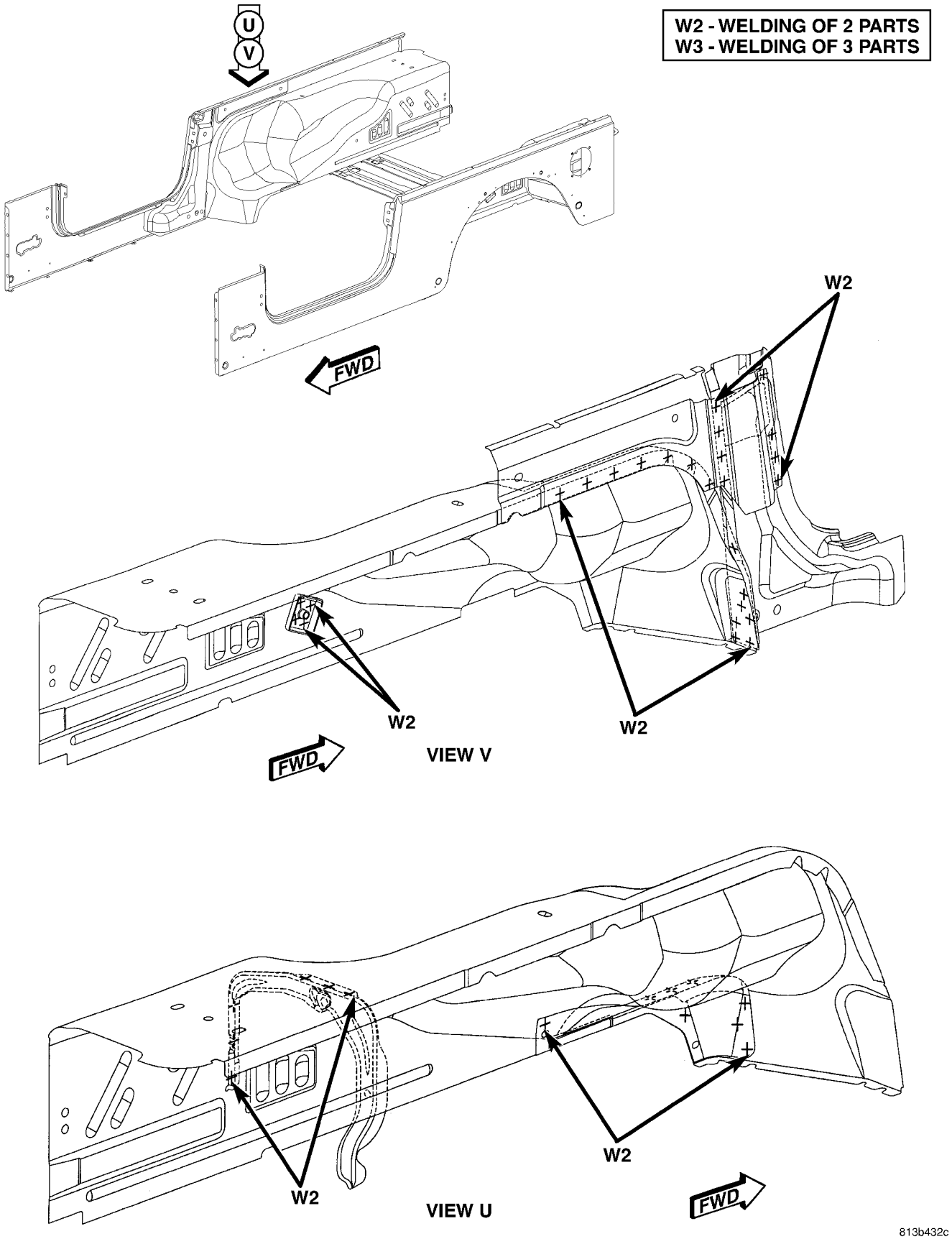

Fig. 76 BODY COMPLETE LWB (2 OF 3)

WELD LOCATIONS (Continued)

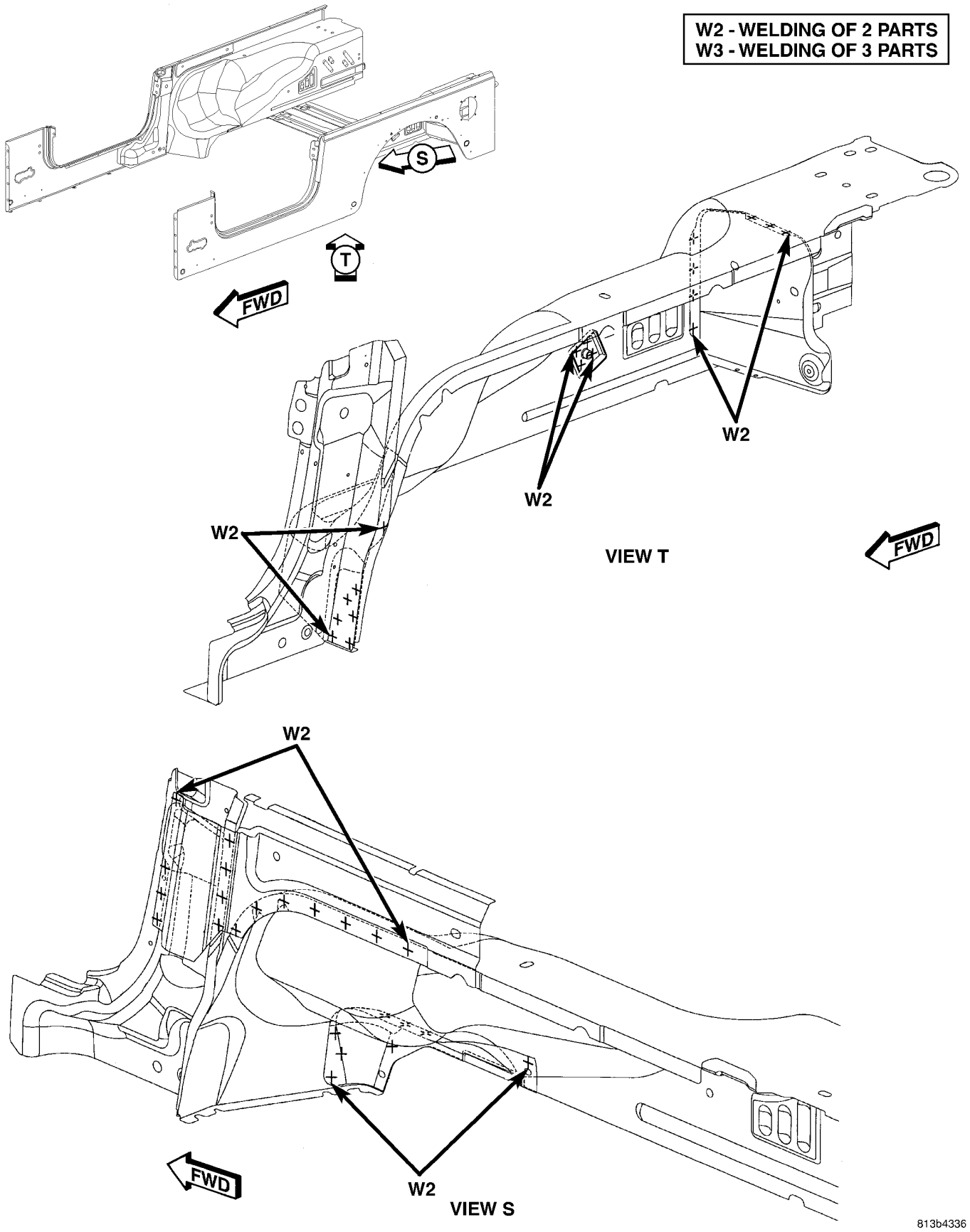

Fig. 77 BODY COMPLETE LWB (3 OF 3)

WELD LOCATIONS (Continued)

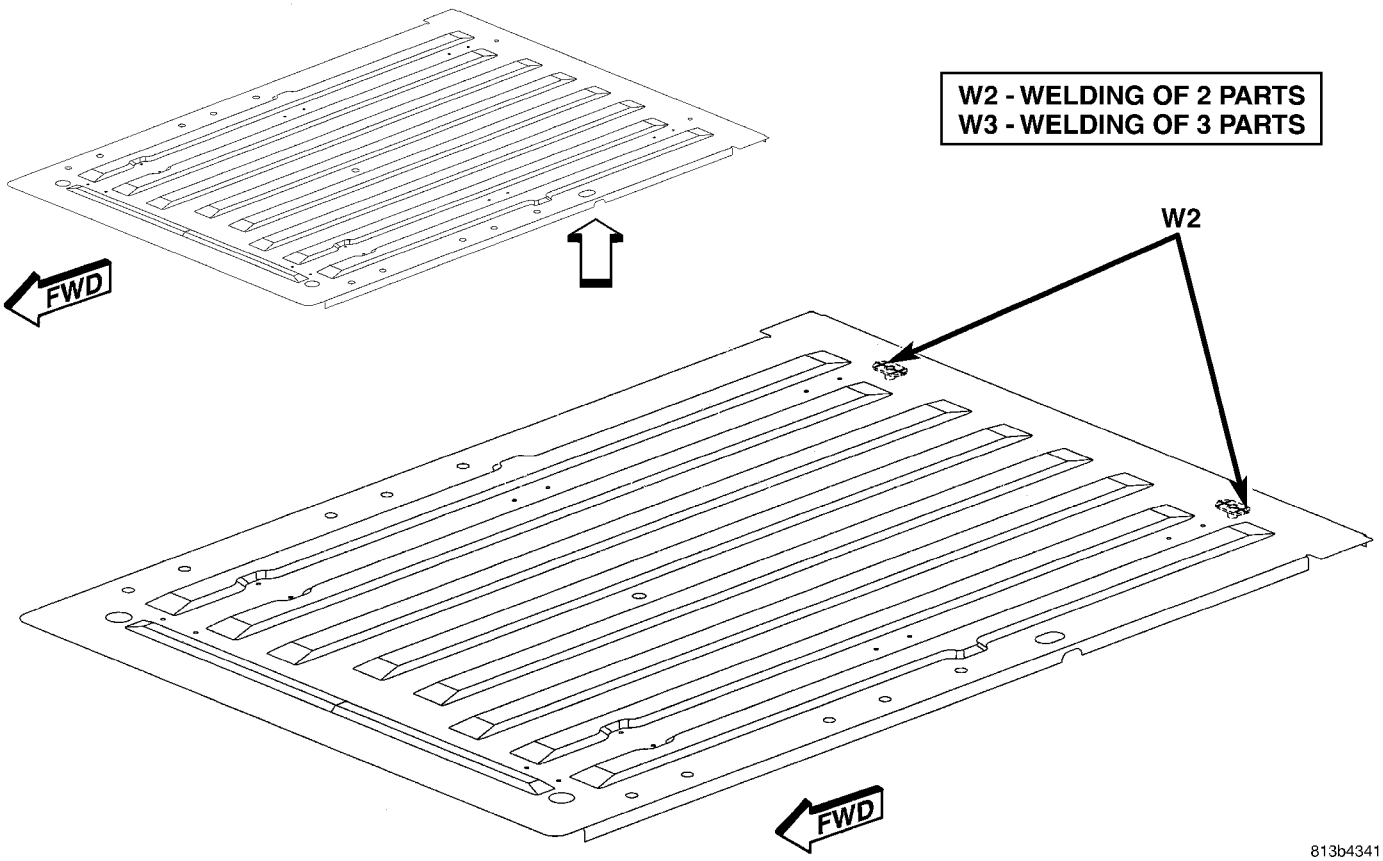

Fig. 78 REAR FLOOR LWB (1 OF 3)

WELD LOCATIONS (Continued)

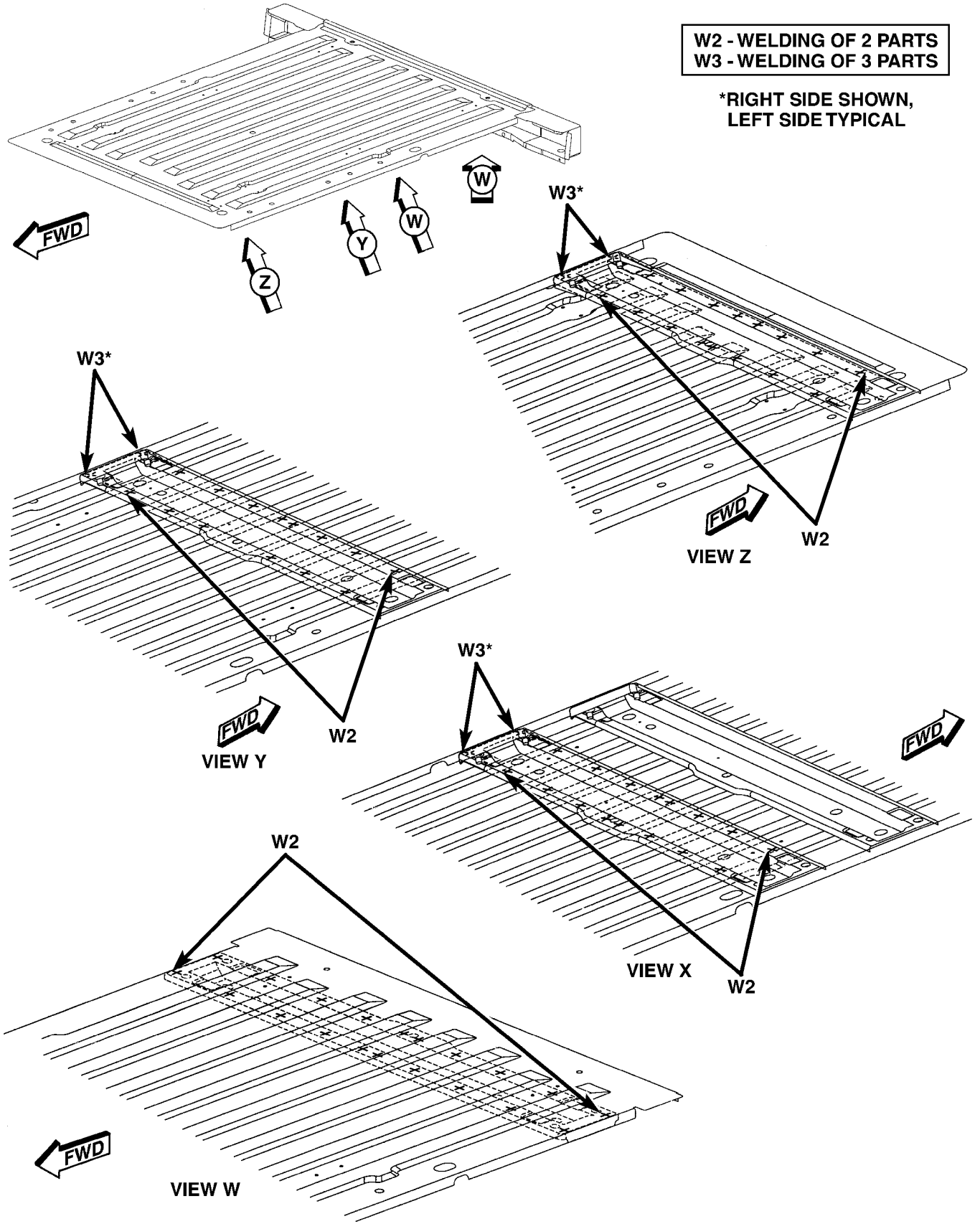

Fig. 79 REAR FLOOR LWB (2 OF 3)

WELD LOCATIONS (Continued)

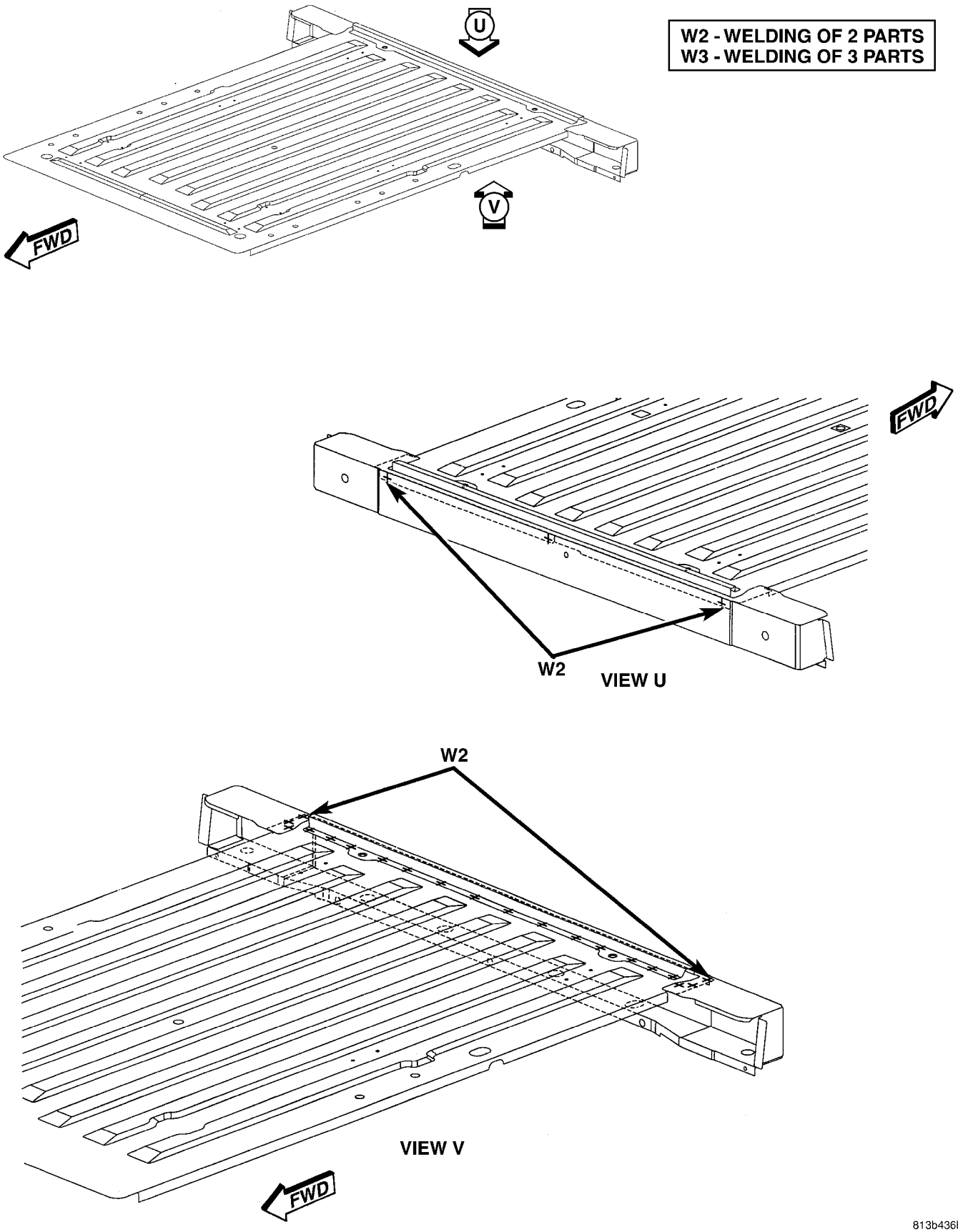

Fig. 80 REAR FLOOR LWB (3 OF 3)

WELD LOCATIONS (Continued)

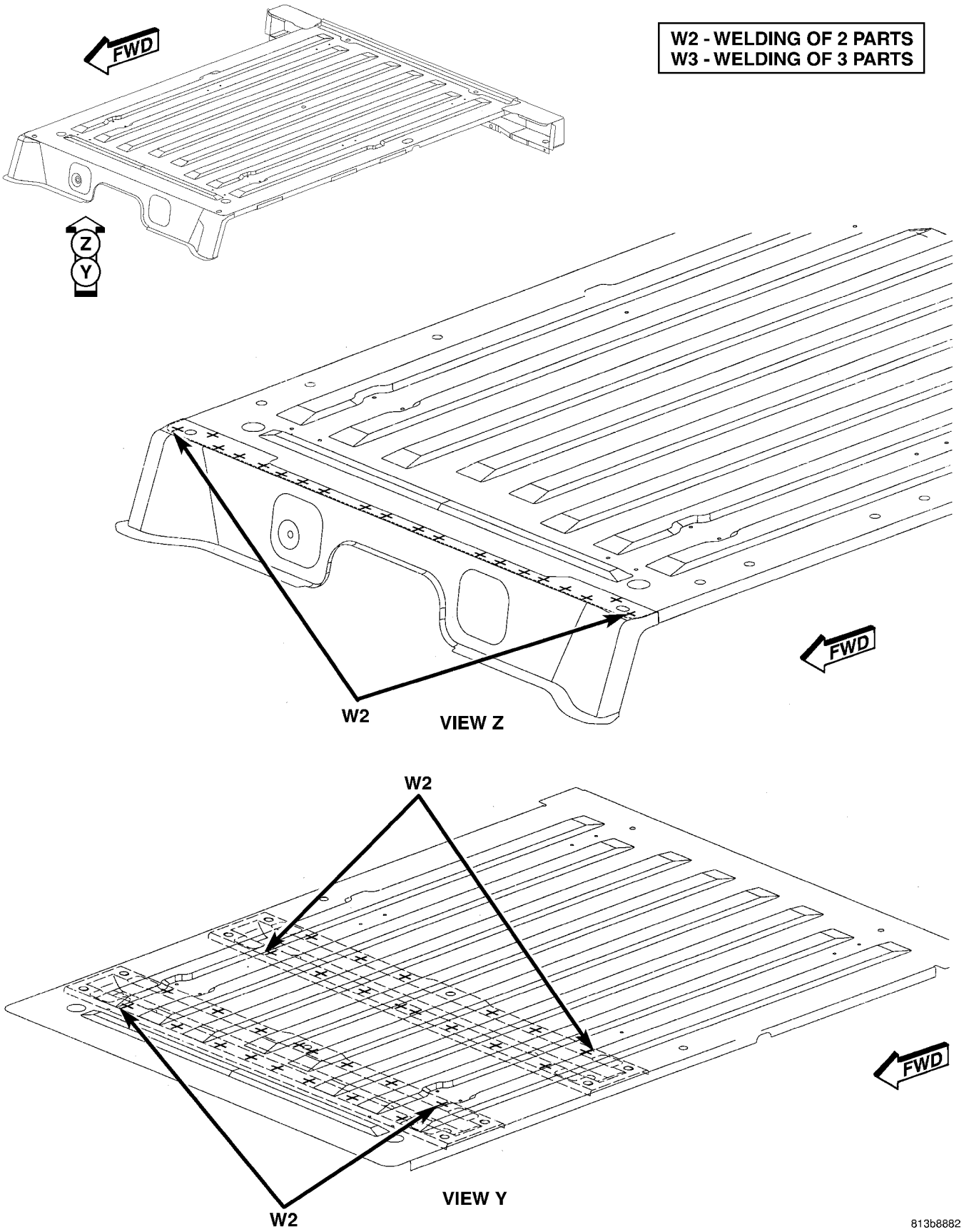

Fig. 81 UNDERBODY COMPLETE LWB (1 OF 3)

WELD LOCATIONS (Continued)

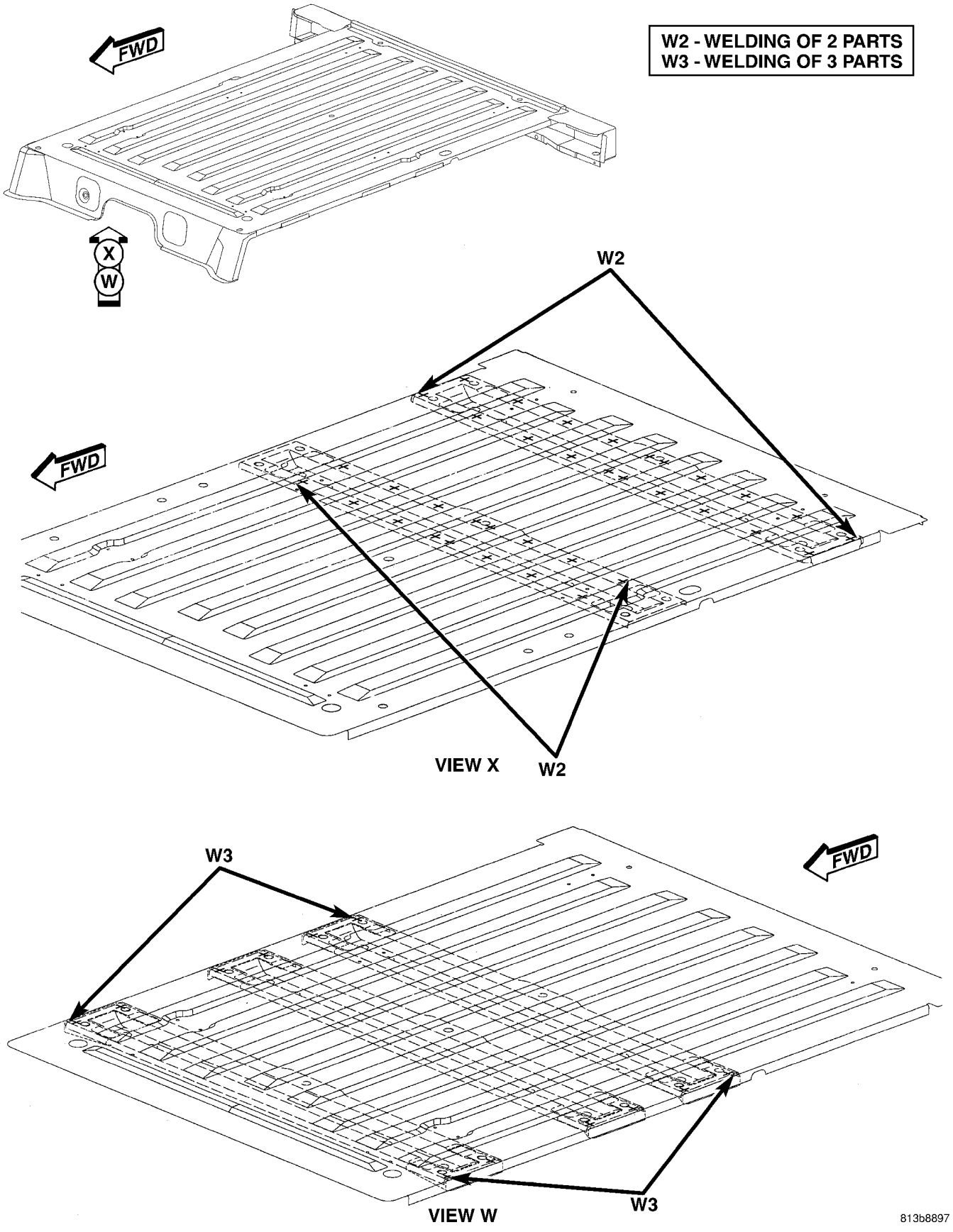

Fig. 82 UNDERBODY COMPLETE LWB (2 OF 3)

WELD LOCATIONS (Continued)

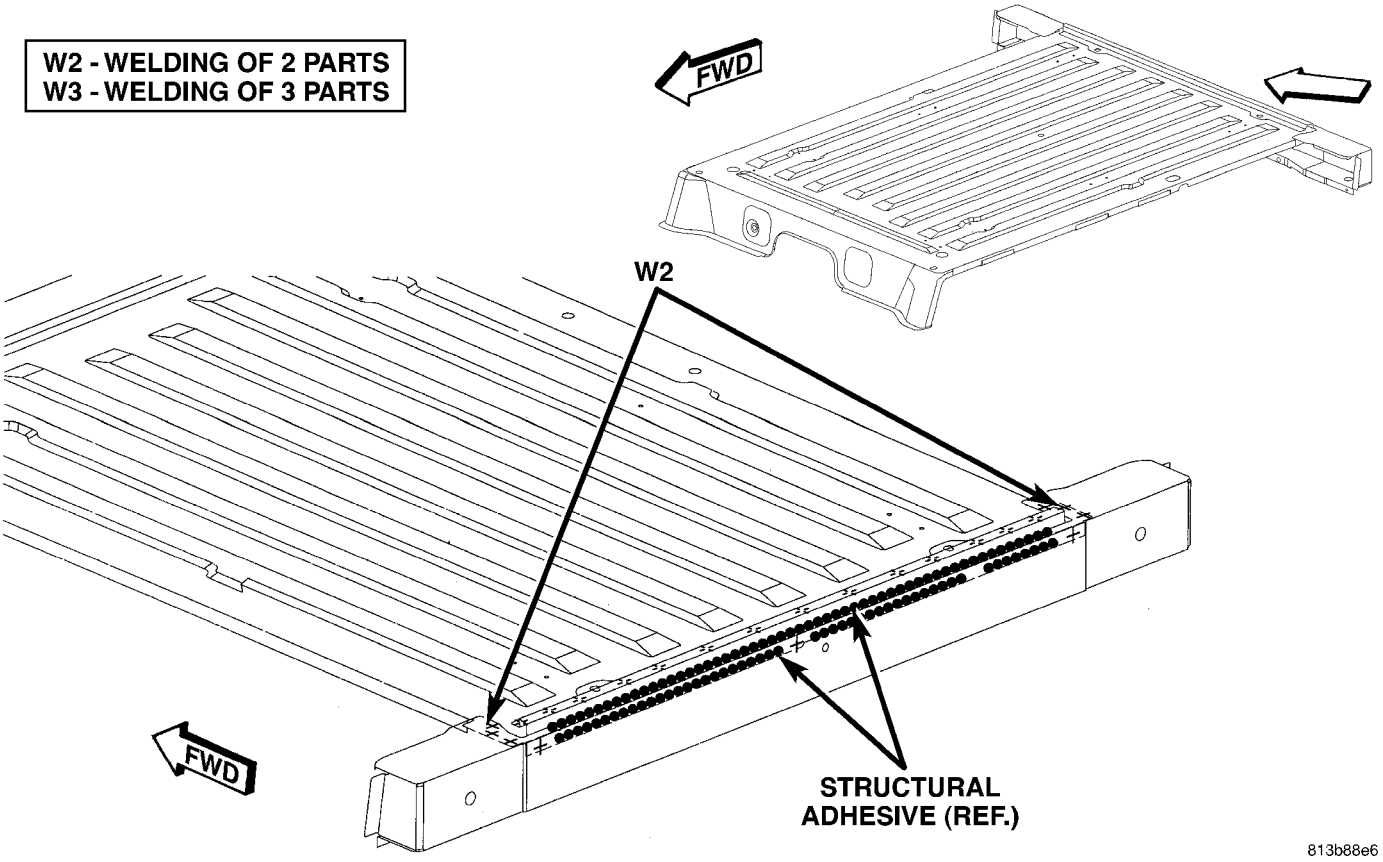

Fig. 83 UNDERBODY COMPLETE LWB (3 OF 3)

WELD LOCATIONS (Continued)

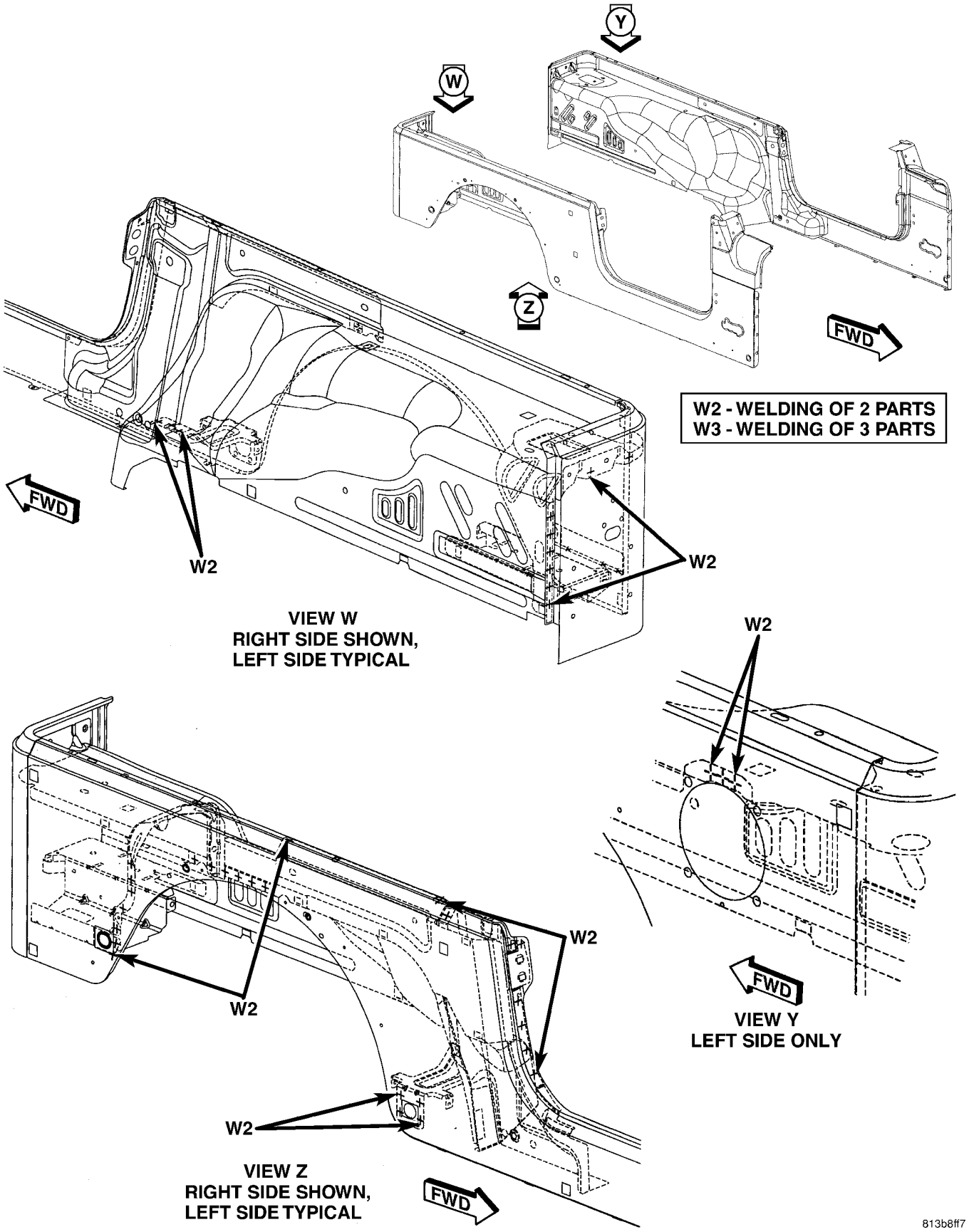

Fig. 84 BODY SIDE COMPLETE LWB (1 OF 2)

WELD LOCATIONS (Continued)

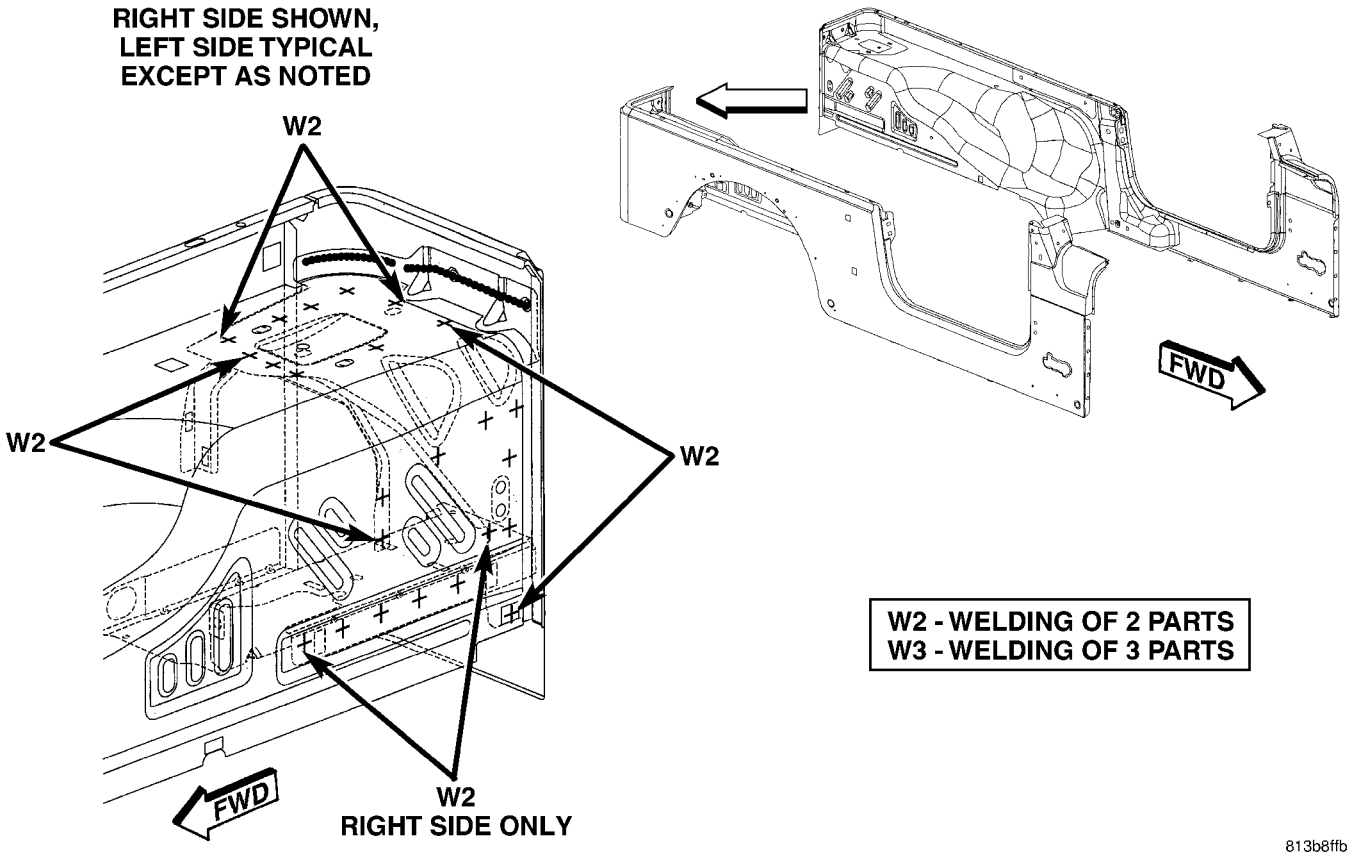

Fig. 85 BODY SIDE COMPLETE LWB (2 OF 2)

WELD LOCATIONS (Continued)

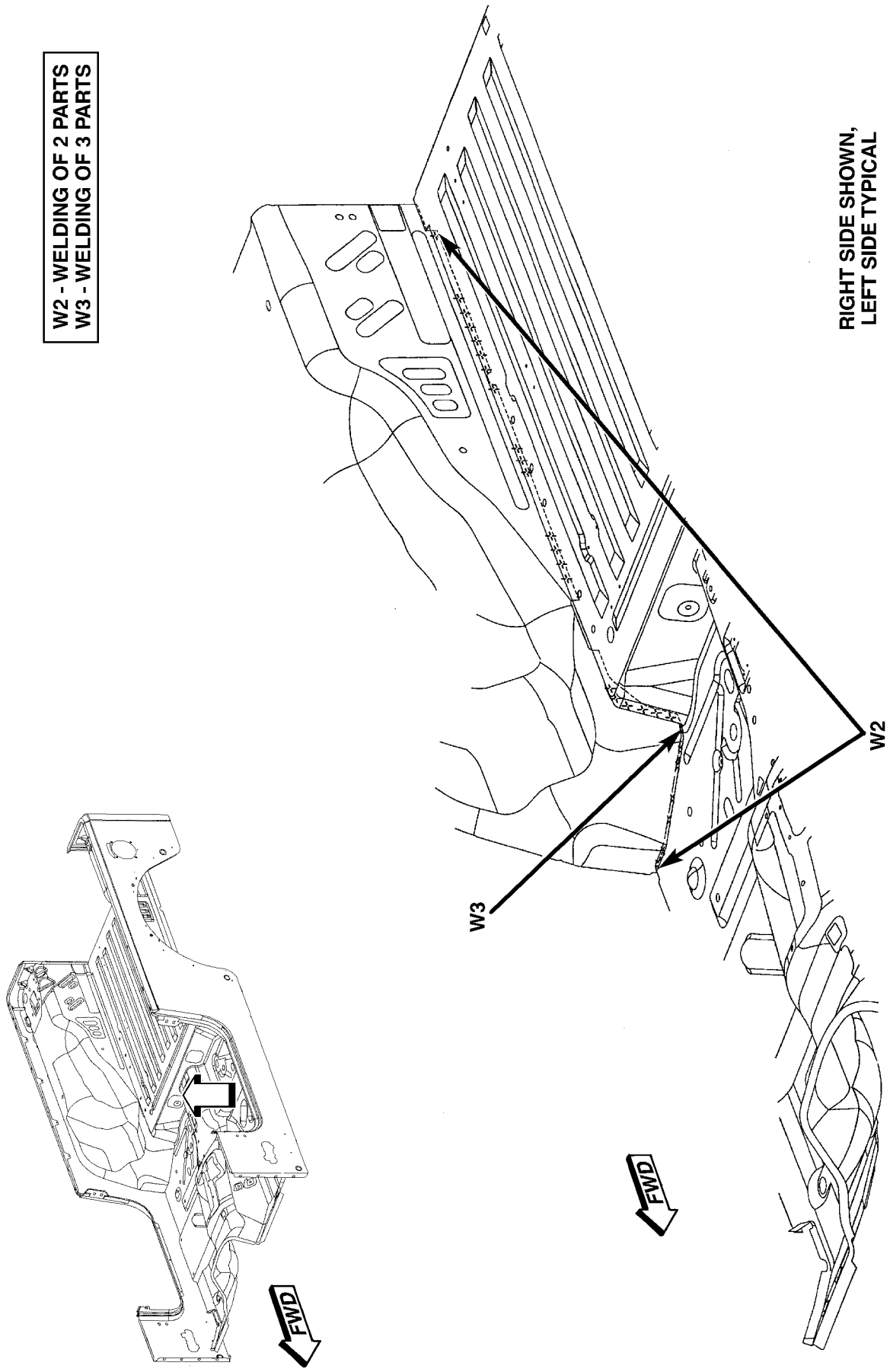

Fig. 86 BODY COMPLETE LWB (1 OF 4)

WELD LOCATIONS (Continued)

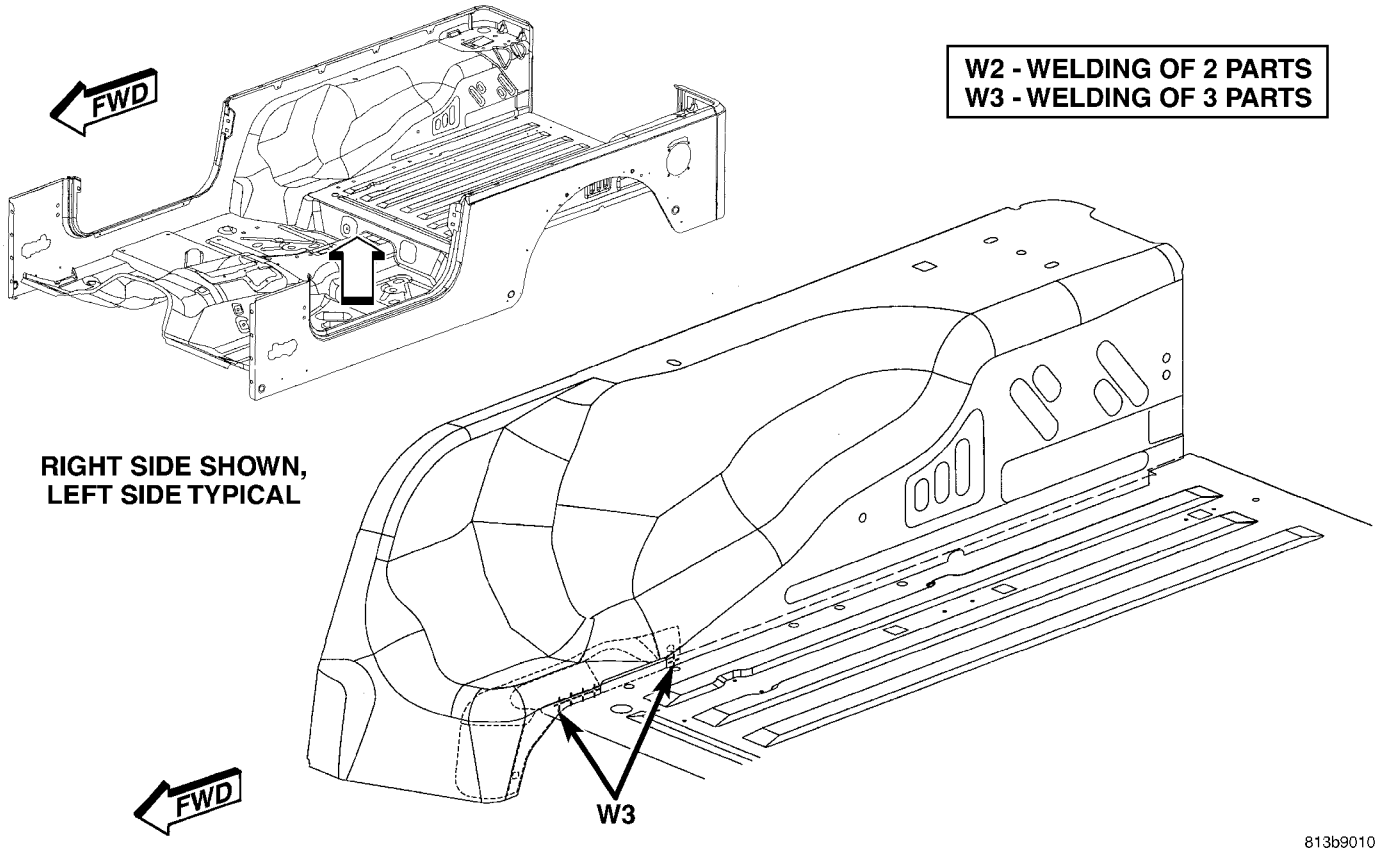

Fig. 87 BODY COMPLETE LWB (2 OF 4)

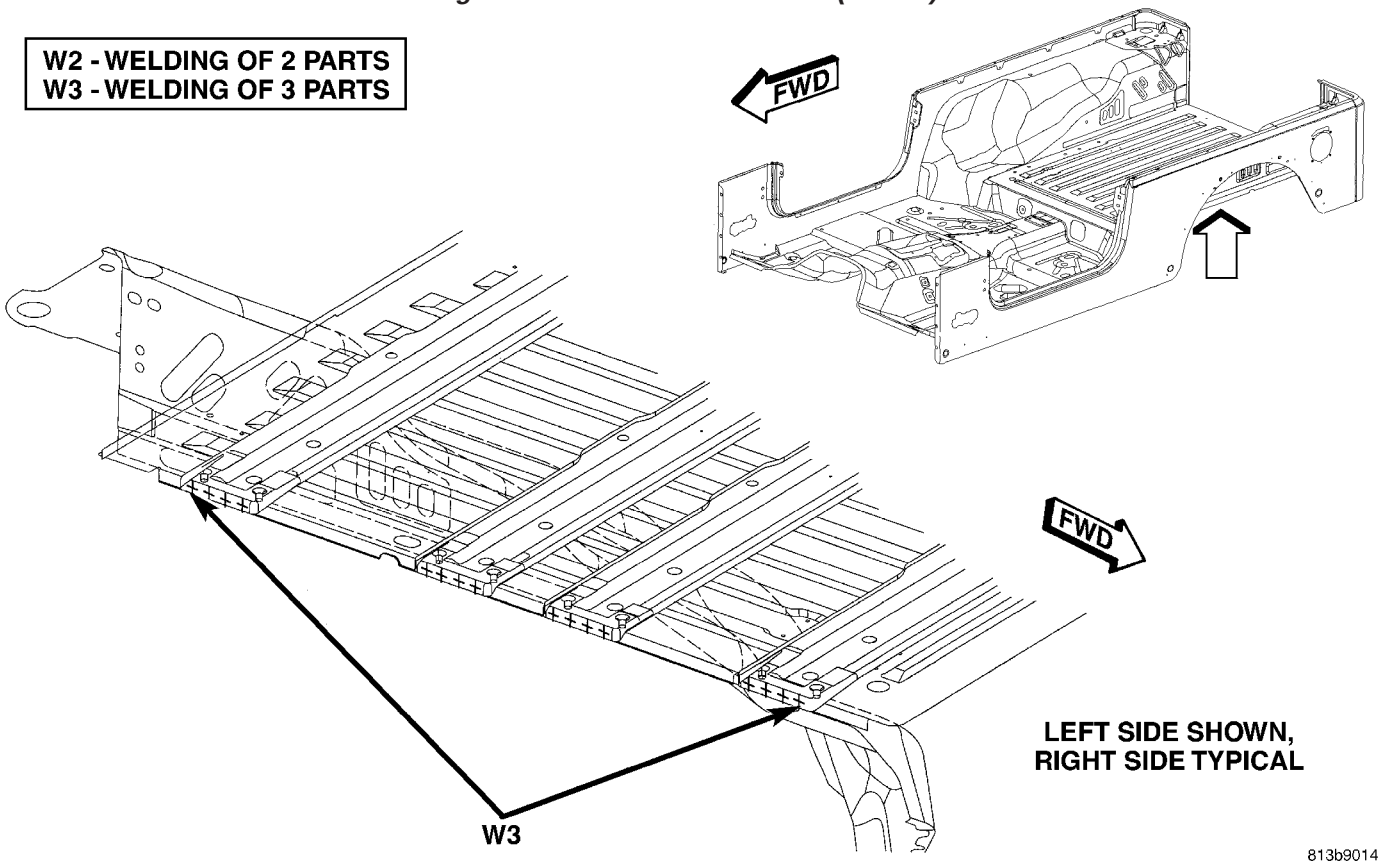

Fig. 88 BODY COMPLETE LWB (3 OF 4)

WELD LOCATIONS (Continued)

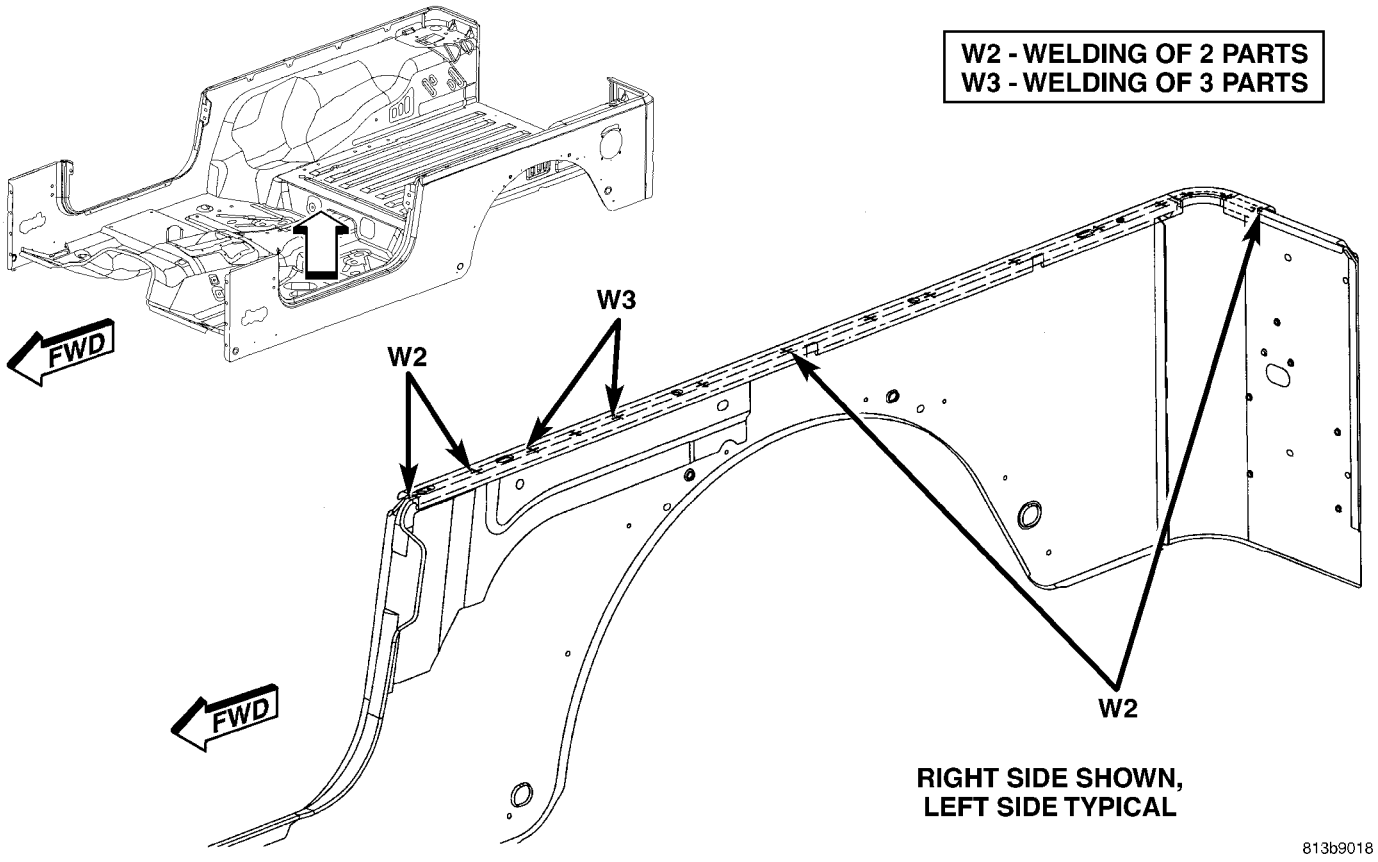

Fig. 89 BODY COMPLETE LWB (4 OF 4)

HEATING & AIR CONDITIONING

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HEATING & AIR CONDITIONING

DESCRIPTION

A manually controlled single zone type heating-air conditioning system or a manually controlled heater-only system is available on this model (depending on market).

To maintain the performance level of the heating, ventilation and air conditioning (HVAC) system, the engine cooling system must be properly maintained. The use of a bug screen is not recommended. Any obstructions in front of the radiator or A/C condenser will reduce the performance of the A/C and engine cooling systems.

The engine cooling system includes the radiator, thermostat, radiator hoses and the engine coolant pump. Refer to 7 - Cooling for more information before opening or attempting any service to the engine cooling system.

All vehicles are equipped with a common heater, ventilation and air conditioning (HVAC) housing (Fig. 1). The system combines air conditioning, heating and ventilation capabilities in a single unit mounted within the passenger compartment under the instrument panel. The HVAC housing includes:

- Blower motor
- Blower motor relay
- Blower motor resistor
- Recirculation-air door and actuator (A/C system only)
- Heater core
- A/C evaporator (A/C system only)
- Blend-air door and actuator
- Floor/defrost and panel-air doors and actuators

On heater-only systems, the A/C evaporator and recirculation-air door are omitted from the housing.

Based upon the system and mode selected, conditioned air can exit the heater-only or heater-A/C HVAC housing through one or a combination of the three main housing outlets: defrost, panel or floor.

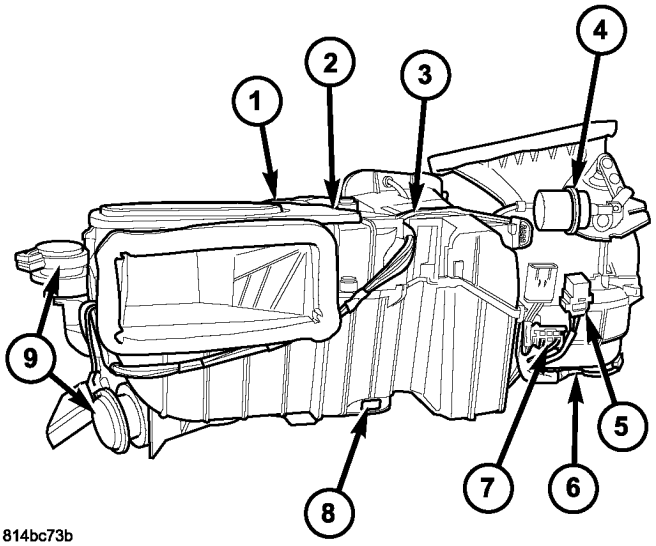

Fig. 1 HVAC Housing - LHD shown, RHD similar

1 - HVAC HOUSING
2 - HEATER CORE
3 - A/C EVAPORATOR
4 - RECIRCULATION DOOR ACTUATOR
5 - BLOWER MOTOR RELAY
6 - BLOWER MOTOR
7 - BLOWER MOTOR RESISTOR
8 - BLEND DOOR ACTUATOR
9 - MODE DOOR ACTUATORS

The defrost outlet is located on the top of the housing, the panel outlet is located on the face of the housing and the floor outlet is located on the bottom of the housing. Once the conditioned air exits the HVAC housing, it is further directed through molded plastic ducts to the various outlets within the vehicle interior. These outlets and their locations are as follows:

- **Defroster Outlet** - A single large defroster outlet is located in the center of the instrument panel top cover, near the base of the windshield.
- **Side Window Demister Outlets** - There are two side window demister outlets, one is located at

each outboard end of the instrument panel, above the panel outlets.

- **Panel Outlets** - There are four panel outlets in the instrument panel, one located near each outboard end of the instrument panel facing the rear of the vehicle and two located near the top of the instrument panel center bezel.
- **Front Floor Outlets** - There are two front floor outlets, one located above each side of the floor panel center tunnel near the dash panel.

OPERATION

Both the single zone manual temperature control (MTC) heating-A/C system and the heater-only system are blend-air type systems. In a blend-air heating-A/C system (Fig. 2), a blend-air door controls the amount of conditioned air that is allowed to flow through, or around, the heater core. The temperature control determines the discharge air temperature by operating the blend door actuator, which move the blend-air door. This design allows almost immediate control of output air temperatures.

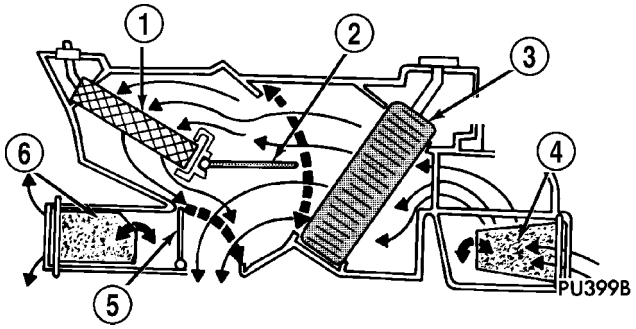

Fig. 2 Typical Blend-Air HVAC System

1 - HEATER CORE
2 - BLEND-AIR DOOR
3 - A/C EVAPORATOR
4 - RECIRCULATION-AIR DOOR
5 - FLOOR MODE-AIR DOOR
6 - PANEL/DEFROST MODE-AIR DOOR

The heating and A/C systems pulls outside (ambient) air through the cowl opening at the base of the windshield, then into the air inlet housing above the heating, ventilation and air conditioning (HVAC) housing. On models equipped with A/C, the air passes through the A/C evaporator. Air flow can be directed either through or around the heater core. This is done by adjusting the blend door with the temperature control knob on the A/C-heater control in the instrument panel. The air flow can then be directed from the panel, floor and defrost outlets in various combinations using the mode control located on the A/C-heater control. Air flow velocity can be adjusted with the blower fan speed control located on the A/C-heater control.

On models equipped with A/C, the outside air intake can be shut off by selecting the Recirculation Mode. This will operate a vacuum actuated recirculating-air door that closes off the fresh air intake and recirculates the air that is already inside the vehicle.

The A/C compressor can be engaged by turning the mode control clockwise from the Off position. It can also be engaged by placing the mode control in the mix to defrost positions. This will remove heat and humidity from the air before it is directed through or around the heater core. The mode control on the A/C-heater control is used to also direct the conditioned air to the selected system outlets. The mode control uses engine vacuum to control the mode-air doors.

The defroster outlet receives airflow from the HVAC housing through the molded plastic defroster duct, which is connected to the HVAC housing defroster outlet. The airflow from the defroster outlet is directed by fixed vanes in the defroster grille and cannot be adjusted.

The side window demisters receive airflow from the HVAC housing through the molded plastic defroster duct, two molded plastic demister ducts and two molded plastic demister outlets. The airflow from the side window demister outlets is directed by fixed vanes in the demister grilles and cannot be adjusted. The demisters direct air from the HVAC housing through the outlets located on the top corners of the instrument panel. The demisters operate when the mode control is positioned in the floor-defrost and defrost-only settings. Some air may be noticeable from the demister outlets when the mode control is in the bi-level to floor positions.

The four instrument panel outlets receive airflow from the HVAC housing through a single molded plastic instrument panel duct. The single instrument panel duct directs airflow to all of the instrument panel outlets. Each of these outlets can be individually adjusted to direct the flow of air.

The floor outlets receive airflow from the HVAC housing through the floor distribution duct. The front floor outlets are integral to the molded plastic floor distribution duct, which is secured to the bottom of the HVAC housing. The floor outlets cannot be adjusted.

HEATING & AIR CONDITIONING (Continued)

NOTE: It is important to keep the air intake opening clear of debris. Leaf particles and other debris that is small enough to pass through the cowl opening screen can accumulate within the HVAC housing. The closed, warm, damp and dark environment created within the housing is ideal for the growth of certain molds, mildews and other fungi. Any accumulation of decaying plant matter provides an additional food source for fungal spores, which enter the housing with the fresh intake-air. Excess debris, as well as objectionable odors created by decaying plant matter and growing fungi can be discharged into the passenger compartment during heater-A/C operation if the air intake opening is not kept clear of debris.

The A/C system on models so equipped is designed for the use of non-CFC, R-134a refrigerant and uses an A/C fixed orifice tube located in the liquid line to meter the flow of refrigerant to the A/C evaporator. The A/C evaporator cools and dehumidifies the incoming air prior to blending it with the heated air. To maintain minimum evaporator temperature and prevent evaporator freezing, the A/C clutch is cycled on and off by the A/C low pressure switch mounted on the accumulator.

DIAGNOSIS AND TESTING

A/C PERFORMANCE

The A/C system is designed to provide the passenger compartment with low temperature and low humidity air. The A/C evaporator, located in the HVAC housing is cooled to temperatures near the freezing point. As warm damp air passes over the fins of the A/C evaporator, the air transfers its heat to the refrigerant in the evaporator coils and the moisture in the air condenses on the evaporator fins. During periods of high heat and humidity, an A/C system will be more effective in the Recirculation mode (max-A/C). With the system in the Recirculation mode, only air from the passenger compartment passes through the A/C evaporator. As the passenger compartment air dehumidifies, the A/C system performance levels rise.

Humidity has an important bearing on the temperature of the air delivered to the interior of the vehicle. It is important to understand the effect that humidity has on the performance of the A/C system. When humidity is high, the A/C evaporator has to perform a double duty. It must lower the air temperature, and it must lower the temperature of the moisture in the air that condenses on the evaporator fins. Condensing the moisture in the air transfers heat energy into the evaporator fins and coils. This reduces the amount of heat the A/C evaporator can absorb from the air. High humidity greatly reduces the ability of the A/C evaporator to lower the temperature of the air.

However, evaporator capacity used to reduce the amount of moisture in the air is not wasted. Wringing some of the moisture out of the air entering the vehicle adds to the comfort of the passengers. Although, an owner may expect too much from their A/C system on humid days. A performance test is the best way to determine whether the system is performing up to design standards. This test also provides valuable clues as to the possible cause of trouble with the A/C system. The ambient air temperature in the location where the vehicle will be tested must be a minimum of 21° C (70° F) for this test.

A/C PERFORMANCE TEST

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

NOTE: When connecting the service equipment coupling to the line fitting, verify that the valve of the coupling is fully closed. This will reduce the amount of effort required to make the connection.

HEATING & AIR CONDITIONING (Continued)

(1) Connect a tachometer and a manifold gauge set or an A/C recycling/charging station.

(2) Set the mode control to the Recirculation mode, the temperature control to the full cool position, and the blower fan to the highest speed position.

(3) Start the engine and hold at 1,000 rpm with the A/C clutch engaged.

(4) The engine should be warmed up to operating temperature with the doors closed and windows open.

(5) Insert a thermometer in the driver side center panel air outlet and operate the engine for five minutes.

(6) The A/C clutch may cycle, depending upon the ambient temperature and humidity. If the A/C clutch cycles, unplug the wire harness connector from the A/C low pressure switch located on the accumulator (Fig. 3), then place a jumper wire across the terminals of the wire harness connector for the switch.

(7) With the A/C clutch engaged, record the air outlet discharge temperature, the discharge pressure (high side service port) and the suction pressure (low side service port). Compare the air temperature at the center panel outlet and the A/C compressor discharge pressure (high side) to the A/C Performance Temperature and Pressure chart.

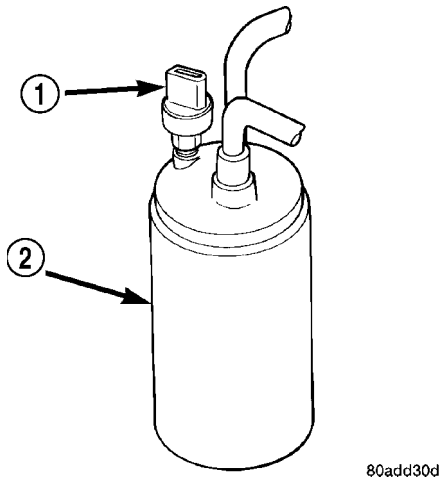

Fig. 3 A/C Low Pressure Switch - Typical

1 - A/C LOW PRESSURE SWITCH
2 - ACCUMULATOR

A/C PERFORMANCE TEMPERATURE AND PRESSURE

Ambient Air Temperature	21° C (70° F)	27° C (80° F)	32° C (90° F)	38° C (100° F)	43° C (110° F)
Air Temperature at Center Panel Outlet	-3 to 3° C (27 to 38° F)	1 to 7° C (33 to 44° F)	3 to 9° C (37 to 48° F)	6 to 13° C (43 to 55° F)	10 to 18° C (50 to 64° F)
Evaporator Inlet Pressure at Charge Port	179 to 241 kPa (26 to 35 psi)	221 to 283 kPa (32 to 41 psi)	262 to 324 kPa (38 to 47 psi)	303 to 365 kPa (44 to 53 psi)	345 to 414 kPa (50 to 60 psi)
Compressor Discharge Pressure	1240 to 1655 kPa (180 to 240 psi)	1380 to 1790 kPa (200 to 260 psi)	1720 to 2070 kPa (250 to 300 psi)	1860 to 2345 kPa (270 to 340 psi)	2070 to 2690 kPa (300 to 390 psi)

(8) If the air outlet temperature fails to meet the specifications in the A/C Performance Temperature and Pressure chart, or if the A/C compressor discharge pressure is high, refer to the A/C Pressure Diagnosis chart.

HEATING & AIR CONDITIONING (Continued)

A/C PRESSURE DIAGNOSIS

Condition	Possible Causes	Correction
Rapid A/C clutch cycling (ten or more cycles per minute).	1. Low refrigerant system charge.	1. See Refrigerant System Leaks in this group. Test the refrigerant system for leaks. Repair, evacuate and charge the refrigerant system, if required.
Equal pressures, but the A/C clutch does not engage.	1. No refrigerant in the refrigerant system.	1. See Refrigerant System Leaks in this group. Test the refrigerant system for leaks. Repair, evacuate and charge the refrigerant system, if required.
	2. Faulty fuse.	2. Check the fuses in the fuse block and the power distribution center (PDC). Repair the shorted circuit or component and replace the fuses, if required. Refer to Group 8.
	3. Faulty A/C clutch field coil.	3. See A/C Compressor Clutch in this group. Test the A/C clutch field coil and replace, if required.
	4. Faulty A/C clutch relay.	4. See A/C Clutch Relay in this group. Test the A/C clutch relay and relay circuits. Repair the circuits or replace the relay, if required.
	5. Improperly installed or faulty A/C low pressure switch.	5. See A/C Low Pressure Switch in this group. Test the A/C low pressure switch and tighten or replace, if required.
	6. Faulty A/C pressure transducer (2.4L engine).	6. See A/C Pressure Transducer in this group. Test the A/C pressure transducer and replace, if required.
	6. Faulty A/C high pressure switch (4.0L engine).	6. See A/C High Pressure Switch in this group. Test the A/C high pressure switch and replace, if required.
	7. Faulty powertrain control module (PCM).	7. Refer to the proper Diagnostic Procedures manual for testing of the PCM. Test the PCM and replace, if required.
Normal pressures, but A/C Performance Test air temperatures at center panel outlet are too high.	1. Excessive refrigerant oil in system.	1. See Refrigerant Oil Level in this group. Recover the refrigerant from the refrigerant system and inspect the refrigerant oil content. Restore the refrigerant oil to the proper level, if required.
	2. Blend door actuator improperly installed or faulty.	2. See Blend Door Actuator in this group. Inspect the actuator for proper operation and replace, if required.
	3. Blend-air door inoperative or sealing improperly.	3. See HVAC Housing in this group. Inspect the blend-air door for proper operation and sealing and correct, if required.
The low side pressure is normal or slightly low, and the high side pressure is too low.	1. Low refrigerant system charge.	1. See Refrigerant System Leaks in this group. Test the refrigerant system for leaks. Repair, evacuate and charge the refrigerant system, if required.
	2. Refrigerant flow through the accumulator is restricted.	2. See Accumulator in this group. Replace the restricted accumulator, if required.

HEATING & AIR CONDITIONING (Continued)

Condition	Possible Causes	Correction
	3. Refrigerant flow through the A/C evaporator is restricted.	3. See A/C Evaporator in this group. Replace the restricted A/C evaporator, if required.
	4. Faulty A/C compressor.	4. See A/C Compressor in this group. Replace the A/C compressor, if required.
The low side pressure is normal or slightly high, and the high side pressure is too high.	1. A/C condenser air flow restricted.	1. Check the A/C condenser for damaged fins, foreign objects obstructing air flow through the condenser fins, and missing or improperly installed air seals. Clean, repair, or replace components as required.
	2. Inoperative radiator cooling fan.	2. Test the radiator cooling fan and replace, if required. Refer to Group 7.
	3. Refrigerant system overcharged.	3. See Refrigerant System Charge in this group. Recover the refrigerant from the refrigerant system. Charge the refrigerant system to the proper level, if required.
	4. Air in the refrigerant system.	4. See Refrigerant System Leaks in this group. Test the refrigerant system for leaks. Repair, evacuate and charge the refrigerant system, if required.
	5. Engine overheating.	5. Test the engine cooling system and repair, if required. Refer to Group 7.
The low side pressure is too high, and the high side pressure is too low.	1. Accessory drive belt slipping.	1. Inspect the accessory drive belt condition and tension. Tighten or replace the accessory drive belt, if required. Refer to Group 7.
	2. Faulty A/C orifice tube.	2. See A/C Orifice Tube in this group. Replace the liquid line, if required.
	3. Faulty A/C compressor.	3. See A/C Compressor in this group. Replace the A/C compressor, if required.
The low side pressure is too low, and the high side pressure is too high.	1. Restricted refrigerant flow through the refrigerant lines.	1. See Liquid Line and Suction and Discharge Line in this group. Inspect the refrigerant lines for kinks, tight bends or improper routing. Correct the routing or replace the refrigerant line, if required.
	2. Restricted refrigerant flow through the A/C orifice tube.	2. See A/C Orifice Tube in this group. Replace the liquid line, if required.
	3. Restricted refrigerant flow through the A/C condenser.	3. See A/C Condenser in this group. Replace the restricted condenser, if required.

HEATING & AIR CONDITIONING (Continued)

HEATER PERFORMANCE

WARNING: Review safety precautions and warnings in this group before performing this procedure (Refer to 24 - HEATING & AIR CONDITIONING/ PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

Check the coolant level, drive belt tension, radiator air flow and fan operation. Start engine and allow to warm up to normal operating temperature.

WARNING: Do not remove radiator cap when engine is hot, personal injury can result.

If vehicle has been run recently, wait 15 minutes before removing the radiator cap. Place a rag over the cap and turn it to the first safety stop. Allow pressure to escape through the overflow tube. When the system pressure stabilizes, remove the cap completely.

MAXIMUM HEATER OUTPUT: TEST AND ACTION

Engine coolant is provided to the heater system by two heater hoses. With the engine idling at normal operating temperature, set the temperature control to maximum heat, the mode control to the floor position, and the blower in the highest speed position. Using a test thermometer, check the temperature of the air being discharged from the floor outlets. Compare the test thermometer reading to the Heater Temperature Reference chart.

HEATER TEMPERATURE REFERENCE

Ambient Air Temperature	15.5° C (60° F)	21.1° C (70° F)	26.6° C (80° F)	32.2° C (90° F)
Minimum Air Temperature at Floor Outlet	62.2° C (144° F)	63.8° C (147° F)	65.5° C (150° F)	67.2° C (153° F)

If the floor outlet air temperature is insufficient, check that the cooling system is operating to specifications (Refer to 7 - COOLING/ENGINE - DIAGNOSIS AND TESTING). Both heater hoses should be HOT to the touch (the coolant return hose should be slightly cooler than the supply hose). If the coolant return hose is much cooler than the supply hose, locate and repair the engine coolant flow obstruction in heater system.

OBSTRUCTED COOLANT FLOW

Possible locations or causes of obstructed coolant flow are as follows:

- Pinched or kinked heater hoses.
- Improper heater hose routing.
- Plugged heater hoses or supply and return ports at the cooling system connections.
- Plugged heater core.

If proper coolant flow through the cooling system is verified, and heater outlet air temperature is insufficient, a mechanical problem may exist.

MECHANICAL PROBLEMS

Possible causes of insufficient heat due to mechanical problems are as follows:

- Faulty engine thermostat.
- Faulty blend door actuator.
- Obstructed fresh air intake.
- Obstructed heater system outlets.
- Faulty blend-air door.

TEMPERATURE CONTROL

If the heater outlet air temperature cannot be adjusted with the temperature control on the A/C-heater control panel, the following could require service:

- Faulty blend door actuator.
- Faulty A/C-heater control.
- Faulty related wiring harness or connectors.
- Faulty blend-air door.

HEATING & AIR CONDITIONING (Continued)

SPECIFICATIONS

HEATING-A/C SYSTEM

A/C SYSTEM

Item	Description	Notes
A/C Compressor	Denso 10PA17	ND-8 PAG Oil
Freeze–up Control	A/C low pressure switch	Input to PCM, accumulator mounted - cycles A/C clutch off below 172 kPa (25 psi) - cycles back on above 276 kPa (40 psi)
Low psi Control	A/C low pressure switch	Input to PCM, opens below 172 kPa (25 psi) - resets above 276 kPa (40 psi)
High psi Control	A/C high pressure switch (4.0L engine)	Input to PCM, A/C discarge line mounted - opens at discharge pressure above 3100 - 3375 kPa (450 - 490 psi) - resets at 1860 - 2275 kPa (270 - 330 psi)
	A/C pressure transducer (2,4L engine)	Input to PCM, A/C discarge line mounted - opens at discharge pressure above 2971 kPa (431 psi) and below 206 kPa (30 psi)

Item	Description	Notes
Refrigerant Charge Capacity	Refer to the A/C Underhood Specification Label located in the engine compartment	R-134a refrigerant
A/C Clutch Coil Draw	2.0 - 3.7 amps	@ 12V ± 0.5V @ 21° C (70° F)
A/C Clutch Air Gap	0.35 - 0.65 mm (0.014 - 0.026 in)	

HEATING & AIR CONDITIONING (Continued)

FASTENER TORQUE

Description	N·m	Ft. Lbs.	In. Lbs.
All Screws NOT Listed Below	2	–	17
A/C Compressor Shaft Bolt	13	–	115
A/C Compressor Mounting Bolts (4.0L)	45-65	35-50	–
A/C Compressor Rear Brace Bolts (4.0L)	40-55	30-40	–
A/C Compressor Mounting Bolts (2.4L)	28	21	–
A/C Condenser Mounting Bolts	5	–	44
Accumulator Retaining Band Bolt	4.5	–	40
Discharge Line to Compressor Nut	25	18	–
Discharge Line to Condenser Nut	12	–	105
HVAC Housing Stud Nuts	6	–	55
Liquid Line to Condenser Nut	12	–	105
Refrigerant Line Bracket Nut (4.0L)	5	–	44
Refrigerant Line to Accumulator Nut	9	–	80
Radiator Shroud Bolts	8	–	72
Suction Line to Compressor Nut	25	18	–

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A/C COMPRESSOR CLUTCH

DESCRIPTION

The A/C compressor clutch assembly consists of a stationary electromagnetic A/C clutch field coil, a pulley bearing and pulley assembly, and a clutch plate (Fig. 1). These components provide the means to engage and disengage the A/C compressor from the engine accessory drive belt.

The A/C clutch field coil and the pulley bearing and pulley assembly are both retained on the nose of the A/C compressor with snap rings. The clutch plate is splined to the compressor shaft and secured with a bolt.

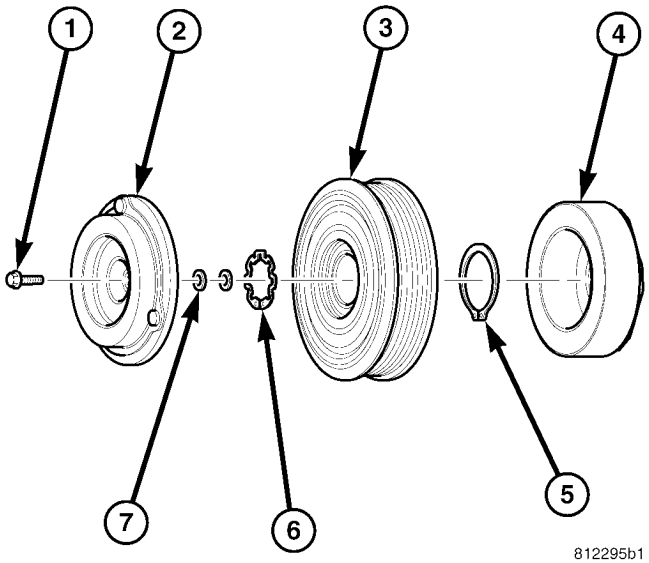

Fig. 1 A/C Compressor Clutch

1 - BOLT
2 - CLUTCH PLATE
3 - PULLEY AND BEARING
4 - FIELD COIL
5 - SNAP RING
6 - SNAP RING
7 - SHIMS

OPERATION

The A/C compressor clutch components provide the means to engage and disengage the A/C compressor from the engine accessory drive belt. When the electromagnetic A/C clutch field coil is energized, it magnetically draws the clutch plate into contact with the clutch pulley and drives the compressor shaft. When the coil is not energized, the pulley freewheels on the clutch hub bearing, which is part of the pulley.

A/C clutch engagement is controlled by the following:

- A/C-heater mode control
- A/C clutch relay
- A/C high pressure switch
- A/C low pressure switch
- Powertrain control module (PCM)

The A/C compressor clutch components cannot be repaired and, if faulty or damaged, they must be replaced.

DIAGNOSIS AND TESTING

A/C COMPRESSOR CLUTCH COIL

The A/C compressor clutch coil electrical circuit is controlled by the powertrain control module (PCM) through the A/C clutch relay. Begin testing of a suspected compressor clutch coil problem by performing the preliminary checks.

PRELIMINARY CHECKS

(1) If no diagnostic trouble codes (DTCs) are found in the powertrain control module (PCM), go to Step 2. If any DTCs are found, repair as required.

(2) If the A/C compressor clutch still will not engage, verify the refrigerant charge level (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - DIAGNOSIS AND TESTING - REFRIGERANT SYSTEM LEAKS). If the refrigerant charge level is OK, go to Step 3. If the refrigerant charge level is not OK, adjust the refrigerant charge as required.

(3) On models equipped with the 2.4L engine, if the A/C compressor clutch still will not engage, disconnect the wire harness connector for the A/C pressure transducer and check for battery current at the connector with the engine running and the A/C-heater control set to the A/C mode. If OK, go to TESTS. If not OK, refer to the Body Diagnostic Procedures to perform further diagnosis. On models equipped with the 4.0L engine, go to TESTS.

TESTS

(1) Verify the battery state of charge (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/BATTERY - DIAGNOSIS AND TESTING).

(2) Connect an ammeter (0 to 10 ampere scale selected) in series with the clutch coil feed terminal. Connect a voltmeter (0 to 20 volt scale selected) to measure voltage across the battery and the clutch coil.

(3) With the A/C-heater control in the A/C mode and the blower motor at low speed, start the engine and allow it to run at a normal idle speed.

(4) The A/C compressor clutch should engage immediately, and the clutch coil supply voltage should be within two volts of the battery voltage. If the coil supply voltage is OK, go to Step 5. If the coil supply voltage is not within two volts of battery voltage, test the clutch coil feed circuit for excessive voltage drop and repair as necessary. If there is no voltage reading at the clutch coil, use a DRB III ® scan tool and refer to appropriate diagnostic informa-

A/C COMPRESSOR CLUTCH (Continued)

tion for testing of the compressor clutch circuit and PCM control. The following components must be checked and repaired as required before you can complete testing of the clutch coil:

- Fuses in the fuse block and the power distribution center (PDC)
- A/C-heater mode control
- A/C clutch relay
- A/C High pressure switch (4.0L engine)
- A/C Low pressure switch
- Powertrain control module (PCM)

(5) The A/C compressor clutch coil is acceptable if the current draw is 2.0 to 3.7 amperes. Specifications apply for a work area temperature of 21° C (70° F). If voltage is more than 12.5 volts, add electrical loads by turning on electrical accessories until voltage reads below 12.5 volts.

(a) If the A/C clutch coil current reading is zero, the coil is open and must be replaced.

(b) If the A/C clutch coil current reading is above specifications, the coil is shorted and must be replaced.

STANDARD PROCEDURE

DIODE REPLACEMENT

(1) Disconnect and isolate the negative battery cable.

(2) Locate the diode in the A/C clutch wire harness and remove the protective covering.

(3) Remove the diode from the wire harness while paying attention to the current flow direction (Fig. 2).

(4) Remove the insulation from the wires in the harness. Only remove enough insulation to solder in the new diode.

(5) Install the new diode in the harness, making sure current flow is correct. If necessary refer to the appropriate wiring diagram for current flow.

CAUTION: To prevent damage to the diode and wiring, DO NOT use acid core solder.

(6) Solder the connections together using rosin core type solder only.

(7) Tape the diode to the harness using electrical tape making, sure the diode is completely sealed from the elements.

(8) Reconnect the negative battery cable, and test A/C clutch operation.

A/C CLUTCH INSPECTION

NOTE: The A/C clutch can be serviced in the vehicle. The refrigerant system can remain fully-charged during compressor clutch, pulley and bearing assembly, or coil replacement.

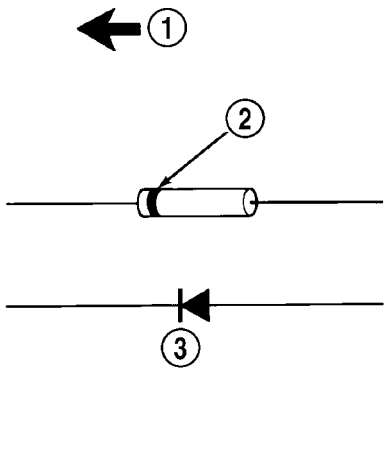

Fig. 2 Diode Identification

1 - CURRENT FLOW
2 - BAND INDICATES CURRENT FLOW
3 - DIODE AS SHOWN IN WIRE DIAGRAMS

Examine the friction surfaces of the clutch pulley and the clutch plate for wear (Fig. 3). The pulley and plate should be replaced if there is excessive wear or scoring.

If the friction surfaces are oily, inspect the shaft and nose area of the A/C compressor for refrigerant oil. If refrigerant oil is found, the compressor shaft seal is leaking and the A/C compressor must be replaced.

Check the clutch pulley bearing for roughness or excessive leakage of grease. Replace the pulley and bearing assembly, if required.

A/C CLUTCH BREAK-IN

After a new A/C compressor clutch has been installed, cycle the compressor clutch approximately twenty times (five seconds on, then five seconds off). During this procedure, set the A/C-heater controls to the A/C Recirculation Mode, the blower motor in the highest speed position, and the engine speed at 1500 to 2000 rpm. This procedure (burnishing) will seat the opposing friction surfaces and provide a higher compressor clutch torque capability.

REMOVAL

NOTE: The compressor clutch can be serviced in the vehicle. The refrigerant system can remain fully-charged during compressor clutch, pulley, or coil replacement.

(1) Disconnect and isolate the battery negative cable.

(2) Remove the serpentine drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - REMOVAL) and (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - REMOVAL).

A/C COMPRESSOR CLUTCH (Continued)

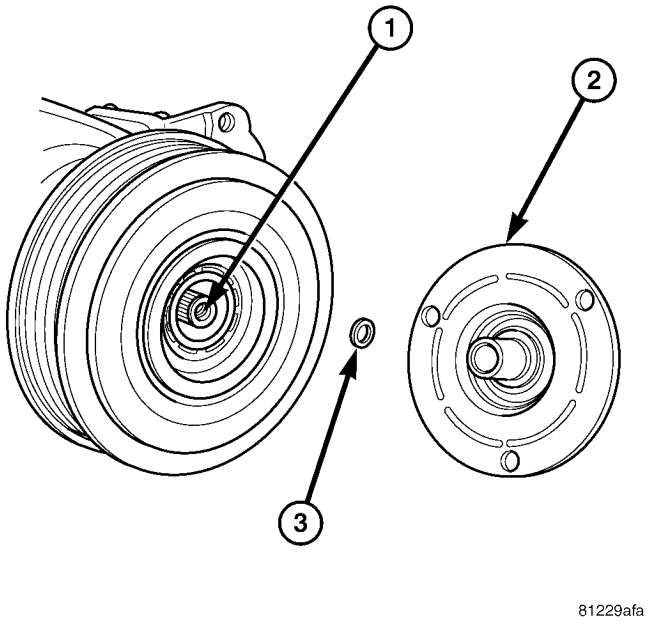

Fig. 3 Clutch Plate and Shim(s)

1 - A/C COMPRESSOR SHAFT
2 - CLUTCH PLATE
3 - SHIM

(3) Remove the bolt that secures the compressor clutch to the compressor shaft (Fig. 4). A band-type oil filter wrench may be used to secure the clutch during bolt removal.

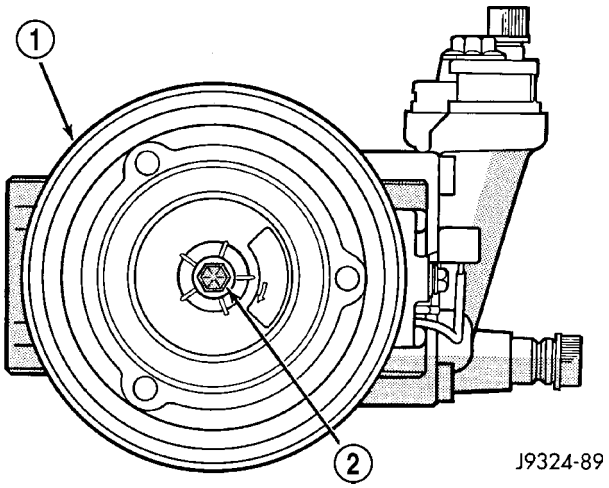

Fig. 4 A/C Compressor Shaft Bolt

1 - COMPRESSOR CLUTCH PLATE
2 - COMPRESSOR SHAFT BOLT

(4) Tap the clutch plate with a plastic mallet to release it from the splines on the compressor shaft. Remove the clutch plate and shim(s) from the compressor shaft (Fig. 5).

NOTE: Use care not to lose any of the shim(s).

CAUTION: Do not use screwdrivers between the clutch plate assembly and pulley to remove front plate as this may damage the front plate assembly.

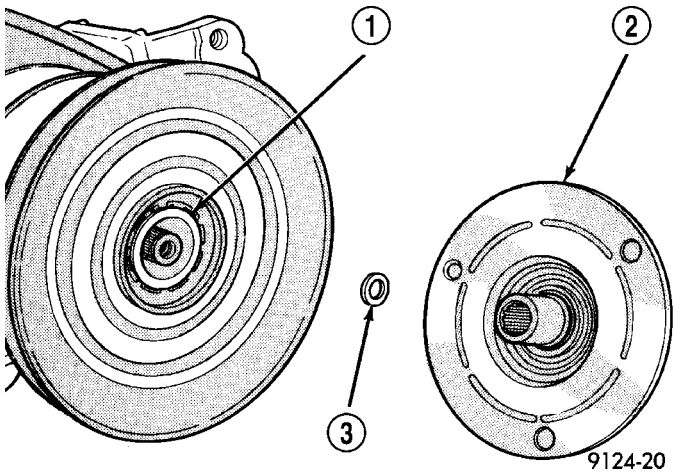

Fig. 5 Clutch Plate and Shim(s)

1 - COMPRESSOR SHAFT
2 - CLUTCH PLATE
3 - CLUTCH PLATE SHIM

(5) Remove the external snap ring that secures the compressor clutch pulley to the nose of the compressor front housing with snap ring pliers (Special Tool C-4574) or equivalent and slide the pulley assembly off of the compressor (Fig. 6).

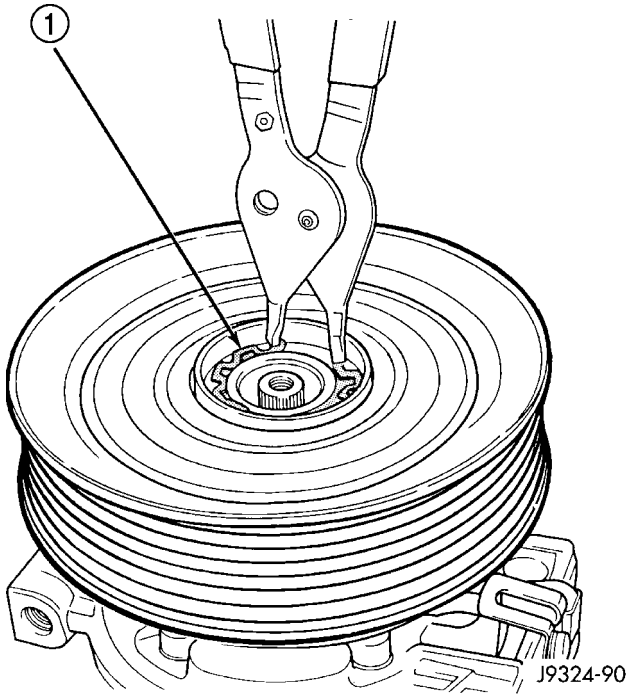

Fig. 6 Pulley Snap Ring

1 - SNAP RING

A/C COMPRESSOR CLUTCH (Continued)

(6) Remove the screw and retainer from the clutch coil lead wire harness on the compressor front housing.

(7) Remove the external snap ring that secures the compressor clutch coil to the nose of the compressor front housing with snap ring pliers and slide the coil assembly off of the compressor (Fig. 7).

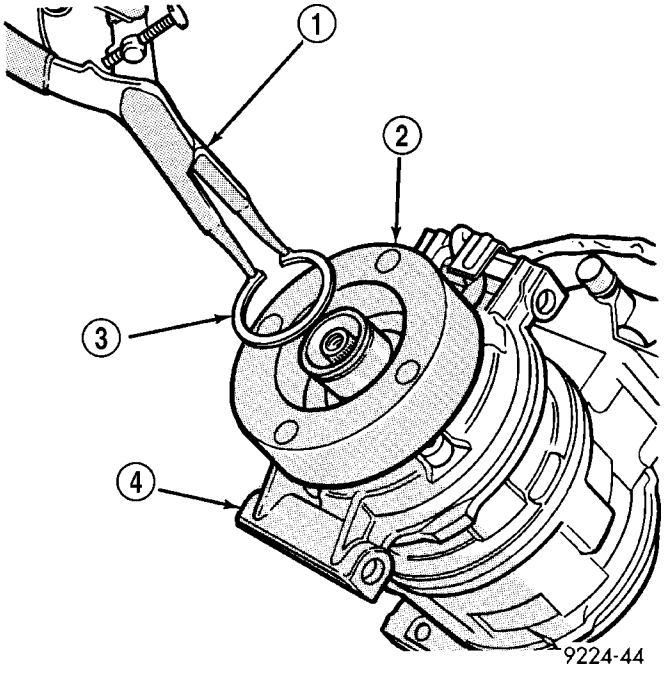

Fig. 7 Clutch Coil Snap Ring

1 - SNAP RING PLIERS
2 - CLUTCH COIL
3 - SNAP RING
4 - COMPRESSOR

INSTALLATION

(1) Align the dowel pin on the back of the clutch field coil with the hole in the compressor front housing and press the field coil into place over the nose of the compressor.

(2) Install the clutch coil lead wire harness retaining clip on the compressor front housing and tighten the retaining screw.

(3) Install the clutch field coil and snap ring with snap ring pliers (Special Tool C-4574). The bevel side of the snap ring must be facing outward. Also, both eyelets of the snap ring must be to the right or left of the pin on the compressor. Press in on the snap ring to be certain that it is properly seated in the groove.

CAUTION: If the snap ring is not fully seated in the groove it will vibrate out, resulting in a clutch failure and severe damage to the front housing of the compressor.

(4) Install the pulley assembly onto the compressor. If necessary, place a block of wood on the friction surface and tap gently with a hammer (Fig. 8).

CAUTION: Do not mar the pulley friction surface.

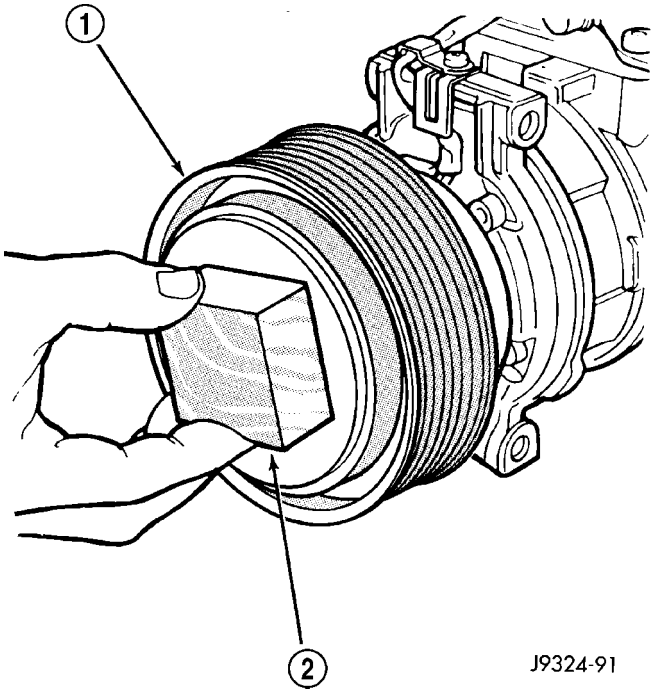

Fig. 8 Pulley Assembly - Install

1 - PULLEY ASSEMBLY
2 - WOOD BLOCK

(5) Install the pulley assembly retaining snap ring (bevel side outward) with snap ring pliers (Special Tool C-4574). Press in on the snap ring to be certain that it is properly seated in the groove.

(6) If the original clutch plate assembly and pulley assembly are to be reused, the old shim(s) can be used. If not, place a stack of shim(s) equal to the old shim(s) on the shaft against the shoulder.

(7) Install the clutch plate assembly onto the shaft.

NOTE: The shims may compress after tightening the shaft bolt. Check the air gap in four or more places to verify the air gap is still correct. Spin the pulley before performing a final check of the air gap.

(8) With the clutch plate assembly tight against the shim(s), measure the air gap between the clutch plate and the pulley face with feeler gauges. The air gap should be between 0.35 to 0.65 mm (0.014 to 0.026 in.). If the proper air gap is not obtained, add or subtract shims as needed until the desired air gap is obtained.

A/C COMPRESSOR CLUTCH (Continued)

(9) Install the compressor shaft bolt. Tighten the bolt to 13 N·m (115 in. lbs.).

(10) Install the accessory drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - INSTALLATION).

(11) Reconnect the negative battery cable.

A/C CLUTCH RELAY

DESCRIPTION

The A/C clutch relay (Fig. 9) is a International Standards Organization (ISO) micro-relay. Relays conforming to the ISO specifications have common physical dimensions, current capacities, terminal functions and patterns. The ISO micro-relay terminal functions are the same as a conventional ISO relay. However, the ISO micro-relay terminal pattern (or footprint) is different, the current capacity is lower, and the physical dimensions are smaller than those of the conventional ISO relay.

The A/C clutch relay is located in the power distribution center (PDC) in the engine compartment.

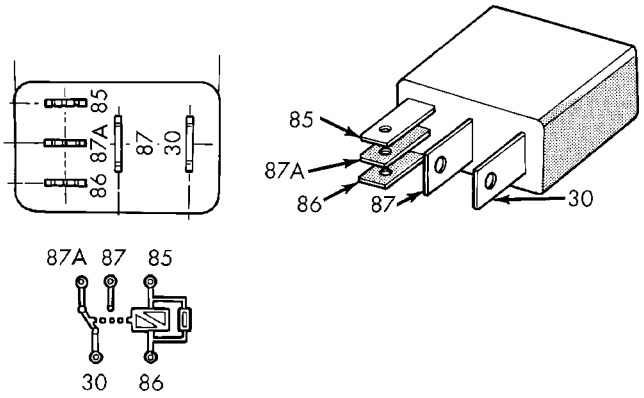

Fig. 9 A/C Compressor Clutch Relay

30 - COMMON FEED
85 - COIL GROUND
86 - COIL BATTERY
87 - NORMALLY OPEN
87A - NORMALLY CLOSED

OPERATION

The A/C clutch relay is an electromechanical switch that uses a low current input controlled by the powertrain control module (PCM) to control the high current output to the A/C clutch field coil. The movable, common feed relay contact is held against the fixed, normally closed relay contact by spring pressure. When the electromagnetic relay coil is energized, it draws the movable common feed relay contact away from the fixed, normally closed relay contact and, holds it against the fixed, normally open relay contact. This action allows high current to flow to the A/C clutch field coil.

When the relay coil is de-energized, spring pressure returns the movable relay contact back against the fixed, normally closed contact point. The resistor or diode is connected in parallel with the relay coil, and helps to dissipate voltage spikes and electromagnetic interference that can be generated as the electromagnetic field of the relay coil collapses.

The A/C clutch relay terminals are connected to the vehicle electrical system through a receptacle in the power distribution center (PDC). The inputs and outputs of the A/C clutch relay include:

- The common feed terminal (A10) receives battery current through a fuse in the PDC at all times.
- The coil ground terminal (A8) receives a ground input from the PCM through the A/C clutch relay control circuit only when the PCM electronically pulls the control circuit to ground.
- The coil battery terminal (A6) receives battery current through a fuse in the fuse block only when the ignition switch is in the Run or Start position.
- The normally open terminal (A9) provides battery current to the A/C clutch coil through the A/C clutch relay only when the A/C clutch relay coil is energized.
- The normally closed terminal (87A) is not connected to any circuit in this application, but provides a battery current output only when the A/C clutch relay coil is de-energized.

The A/C clutch relay cannot be repaired and, if faulty or damaged, it must be replaced. Refer to the appropriate wiring information for diagnosis and testing of the ISO-standard micro-relay and for complete HVAC wiring diagrams.

A/C CLUTCH RELAY (Continued)

REMOVAL

(1) Disconnect and isolate the battery negative cable.

(2) Remove the cover from the power distribution center (PDC) (Fig. 10) located in the engine compartment.

NOTE: Refer to the fuse and relay layout map on the inside of the PDC cover for A/C clutch relay location.

(3) Remove the A/C clutch relay from the PDC.

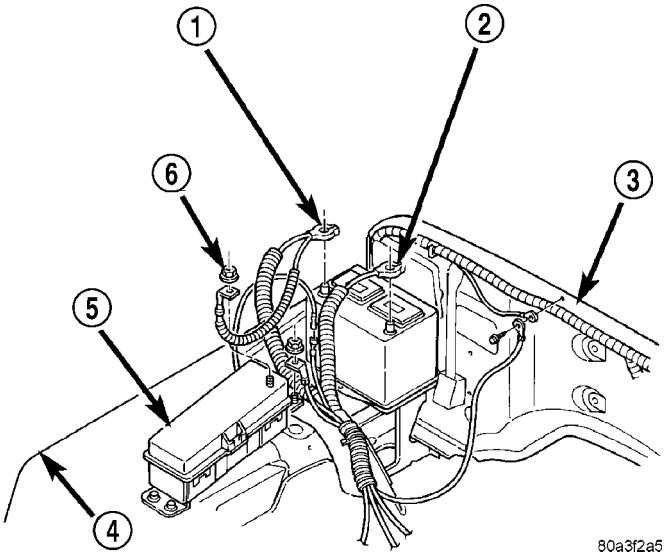

Fig. 10 Power Distribution Center - LHD shown, RHD similar

1 - BATTERY POSITIVE CABLE
2 - BATTERY NEGATIVE CABLE
3 - DASH PANEL
4 - INNER FENDER
5 - POWER DISTRIBUTION CENTER (PDC)
6 - NUT

INSTALLATION

NOTE: Refer to the fuse and relay layout map on the inside of the power distribution center (PDC) cover for A/C clutch relay location.

(1) Position the A/C clutch relay into the receptacle of the PDC.

(2) Align the A/C clutch relay terminals with the terminal cavities in the PDC receptacle and push down firmly on the relay until the terminals are fully seated.

(3) Close the cover of the PDC.

(4) Reconnect the negative battery cable.

A/C HEATER CONTROL

DESCRIPTION

The A/C-heater control (Fig. 11) for both the manual temperature control (MTC) single zone heating-only and the heating-A/C system allows one temperature setting for the entire vehicle. All controls are identified by ISO graphic symbols.

The A/C-heater control is located in the instrument panel and contains:

- a rotary control knob for blower fan speed selection.
- a rotary control knob for recirculation and mode control of the discharged air.
- a rotary control knob for temperature control of the discharged air and turning the heating-A/C system off.

The A/C-heater control cannot be repaired and, if faulty or damaged, it must be replaced. The blower motor and mode control switch, control knobs and illumination lamps are available for service replacement.

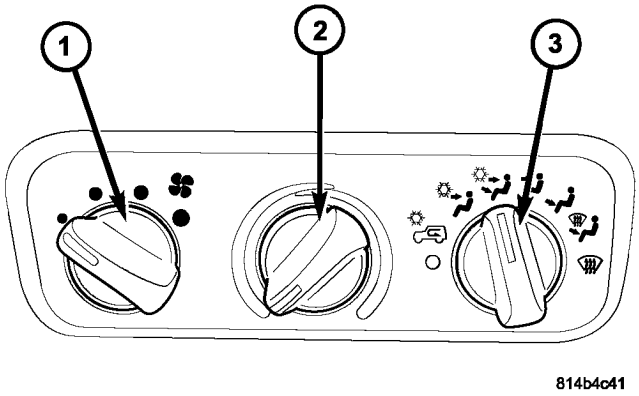

Fig. 11 A/C-Heater Control - Typical

1 - BLOWER SPEED CONTROL
2 - TEMPERATURE SELECT COPNTROL
3 - MODE SELECT CONTROL

A/C HEATER CONTROL (Continued)

DIAGNOSIS AND TESTING

VACUUM CONTROL SYSTEM

Vacuum control is used to operate the mode-air doors within the HVAC housing. Testing the operation of the heating-only and heating-A/C system mode control will determine if the vacuum and mechanical controls are functioning. However, it is possible that a vacuum control system that operates correctly at engine idle (high engine vacuum) may not function properly at high engine speeds or loads (low engine vacuum). This can be caused by leaks within the vacuum system, or a faulty vacuum check valve.

A vacuum system test will help to identify the source of poor vacuum system performance or vacuum system leaks. Before starting this test, stop the engine and make certain that the problem is not a disconnected vacuum supply line at the engine intake manifold vacuum tap or at the vacuum reservoir.

Use an adjustable vacuum test set (Special Tool C-3707-B) and a suitable vacuum pump to test the HVAC vacuum control system. With a finger placed over the end of the vacuum test hose probe (Fig. 12), adjust the bleed valve on the test set gauge to obtain a vacuum of exactly 27 kPa (8 in. Hg.). Release and block the end of the probe several times to verify that the vacuum reading returns to the exact 27 kPa (8 in. Hg.) setting. Otherwise, a false reading will be obtained during testing.

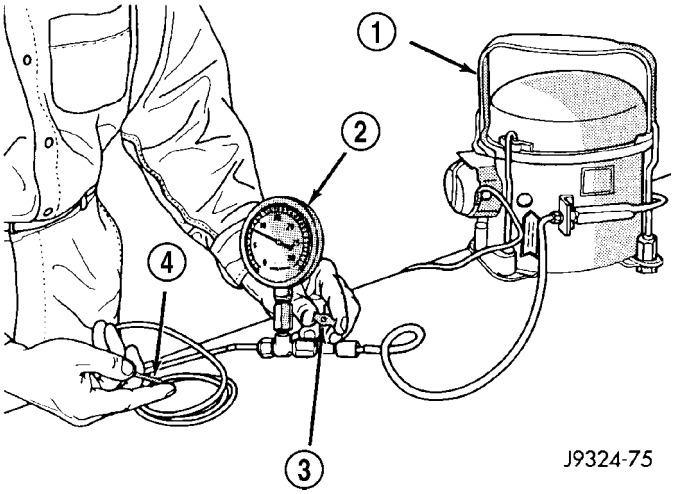

Fig. 12 Adjust Vacuum Test Bleed Valve

1 - VACUUM PUMP TOOL C-4289
2 - VACUUM TEST SET C-3707
3 - BLEED VALVE
4 - PROBE

VACUUM CHECK VALVE

(1) Remove the vacuum check valve. The valve is located in the (black) vacuum supply tube at the intake manifold vacuum tap.

(2) Connect the test set vacuum supply hose to the A/C-heater control side of the vacuum check valve. When connected to this side of the check valve, no vacuum should pass and the test set gauge should return to the 27 kPa (8 in. Hg.) setting. If OK, go to Step 3. If not OK, replace the faulty vacuum check valve.

(3) Connect the test set vacuum supply hose to the engine vacuum side of the vacuum check valve. When connected to this side of the check valve, vacuum should flow through the check valve without restriction. If not OK, replace the faulty vacuum check valve.

A/C HEATER CONTROLS

(1) Connect the test set vacuum probe to the heater-A/C vacuum supply (black) tube in the engine compartment. Position the test set gauge so that it can be viewed from the passenger compartment.

(2) Place the A/C-heater mode control in each mode position, one position at a time, and pause after each selection. The test set gauge should return to the 27 kPa (8 in. Hg.) setting shortly after each selection is made. If not OK, a component or vacuum line in the vacuum circuit of the selected mode has a leak. See the procedure in Locating Vacuum Leaks.

LOCATING VACUUM LEAKS

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the battery negative (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. This is the only sure way to disable the airbag system. Failure to take the proper precautions could result in accidental airbag deployment and possible personal injury or death.

CAUTION: Do not use lubricant on the switch ports or in the holes of the connector, as lubricant will ruin the vacuum valve in the switch. A drop of clean water in the connector holes will help the connector slide onto the switch ports.

A/C HEATER CONTROL (Continued)

(1) Disconnect the vacuum harness connector at the back of the A/C-heater control (Refer to 24 - HEATING & AIR CONDITIONING/CONTROLS/A/C HEATER CONTROL - REMOVAL).

(2) Connect the test set vacuum hose probe to each port in the vacuum harness connector, one port at a time, and pause after each connection (Fig. 13). The test set gauge should return to the 27 kPa (8 in. Hg.) setting shortly after each connection is made. If OK, replace the faulty mode control switch. If not OK, go to Step 3.

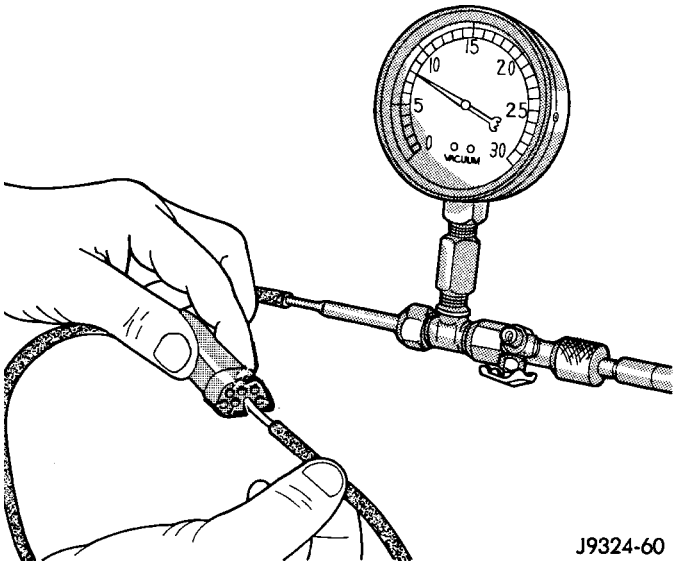

Fig. 13 Vacuum Circuit Test

(3) Determine the vacuum line color of the vacuum circuit that is leaking. To determine the vacuum line colors, refer to the Vacuum Circuit charts (Fig. 14) and (Fig. 15).

(4) Disconnect and plug the vacuum line from the component (fitting, actuator, valve, switch, or reservoir) on the other end of the leaking circuit. Instrument panel disassembly or removal may be necessary to gain access to some components. See the appropriate service procedures.

(5) Connect the test set hose or probe to the open end of the leaking circuit. The test set gauge should return to the 27 kPa (8 in. Hg.) setting shortly after each connection is made. If OK, replace the faulty disconnected component. If not OK, go to Step 6.

(6) To locate a leak in a vacuum line, leave one end of the line plugged and connect the test set hose or probe to the other end of the line. Run your fingers slowly along the line while watching the test set gauge. The vacuum reading will fluctuate when your fingers contact the source of the leak. To repair the vacuum line, cut out the leaking section of the line. Then, insert the loose ends of the line into a suitable length of 3 millimeter (0.125 inch) inside diameter rubber hose.

A/C HEATER CONTROL (Continued)

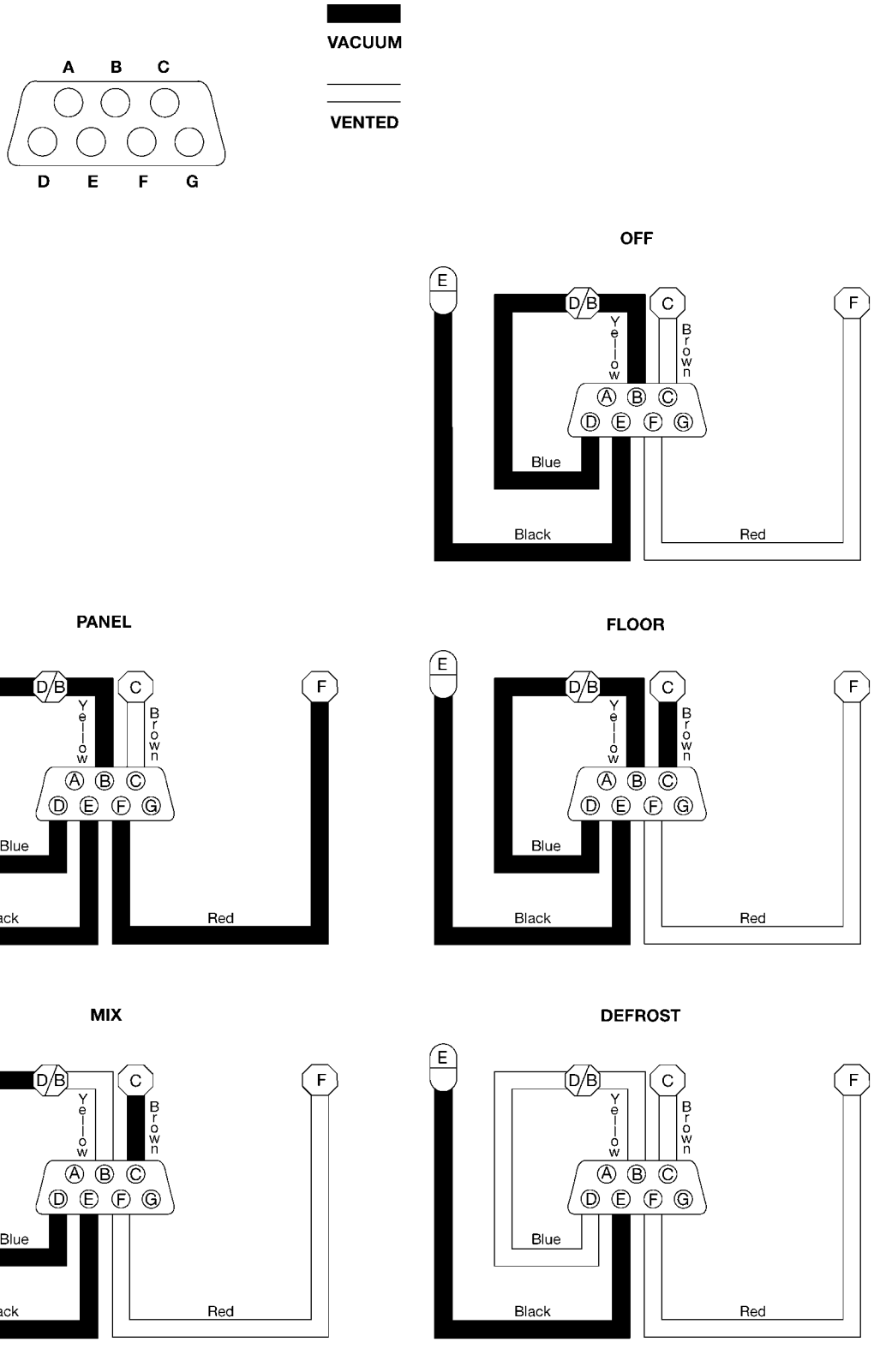

Fig. 14 Vacuum Circuits - Heater Only

A - NOT USED
B - DEFROST ACTUATOR, FULL POSITION (YELLOW)
C - FLOOR ACTUATOR (BROWN)
D - DEFROST ACTUATOR, MID-POSITION (BLUE)
E - VACUUM SUPPLY, RESERVOIR (BLACK)
F - PANEL ACTUATOR (RED)
G - NOT USED

A/C HEATER CONTROL (Continued)

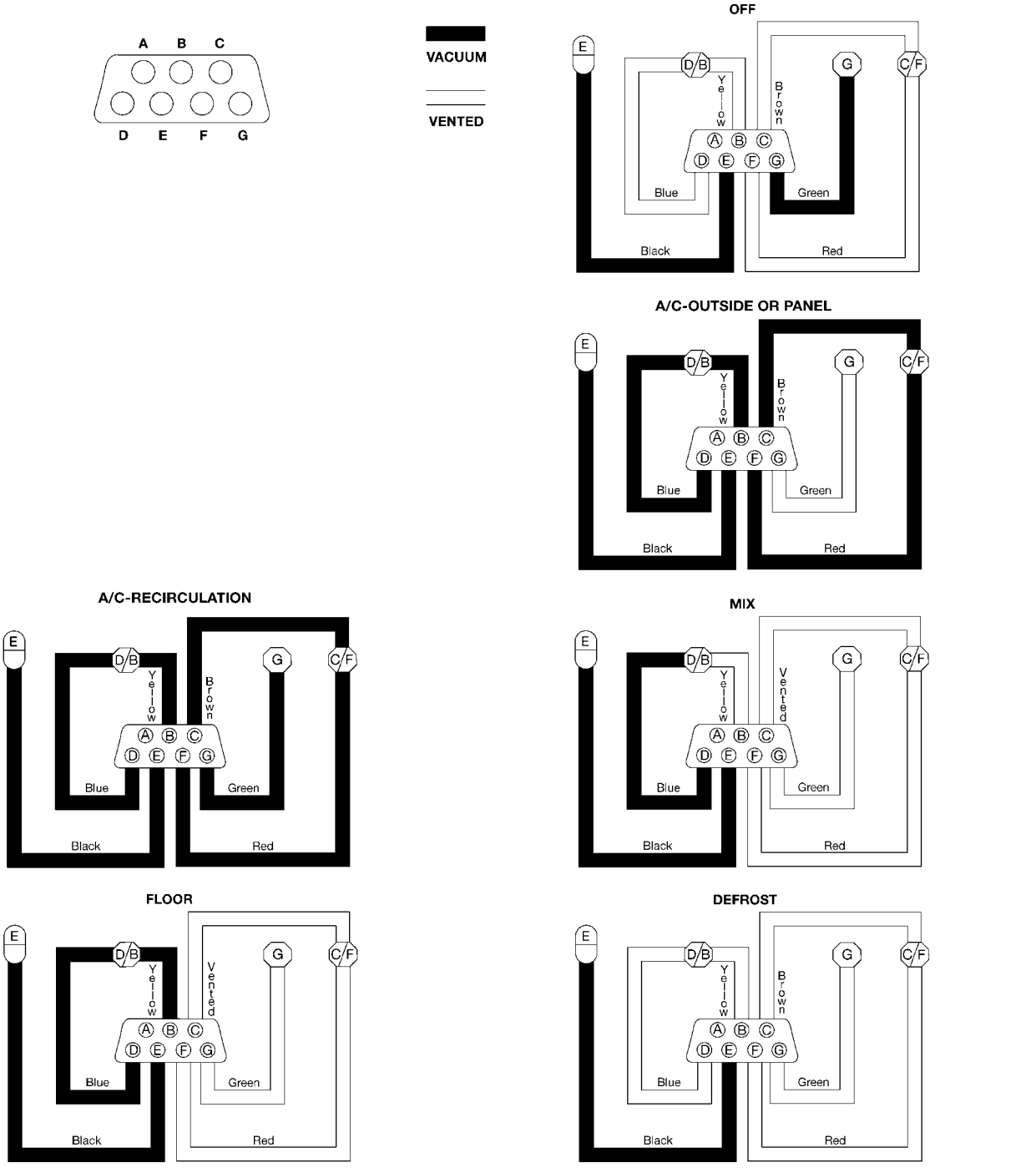

Fig. 15 Vacuum Circuits - Heater-A/C

A - NOT USED
B - DEFROST ACTUATOR (YELLOW)
C - FLOOR ACTUATOR (BROWN)
D - DEFROST ACTUATOR, MID-POSITION (BLUE)
E - VACUUM SUPPLY, RESERVOIR (BLACK)
F - PANEL ACTUATOR (RED)
G - RECIRCULATION ACTUATOR (GREEN)

A/C HEATER CONTROL (Continued)

REMOVAL

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the negative battery (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. This is the only sure way to disable the airbag system. Failure to take the proper precautions could result in accidental airbag deployment and possible personal injury or death.

(1) Disconnect and isolate the battery negative cable.

(2) Remove the center bezel from the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/ INSTRUMENT PANEL CENTER BEZEL - REMOVAL).

(3) Remove the glove box from the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/ GLOVE BOX - REMOVAL).

(4) Reach through the glove box opening and disconnect the two halves of the A/C-heater control vacuum harness connector (Fig. 16).

(5) Remove the two screws that secure the A/C-heater control to the instrument panel.

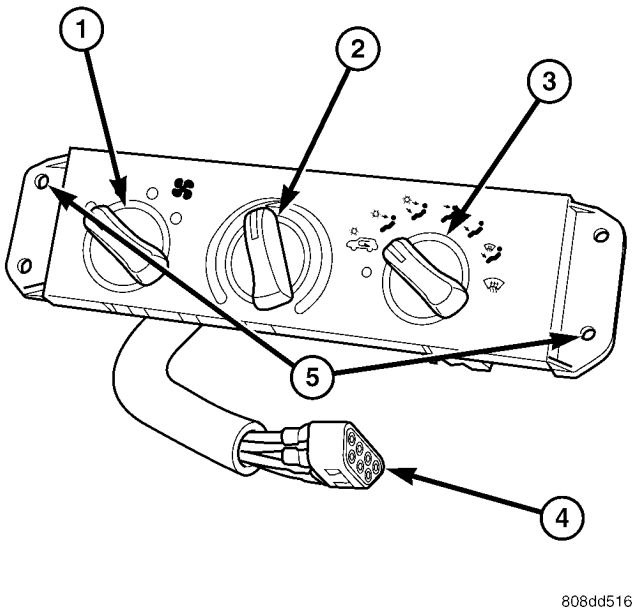

Fig. 16 A/C-Heater Control - Typical

1 - BLOWER MOTOR CONTROL
2 - TEMPERATURE CONTROL
3 - MODE CONTROL
4 - VACUUM HARNESS CONNECTOR
5 - MOUNTING SCREW LOCATIONS

(6) Pull the A/C-heater control away from the instrument panel to access the connections on the back of the control.

(7) Disconnect the three wire harness connectors from the back of the A/C-heater control (Fig. 17) and remove the control from the vehicle.

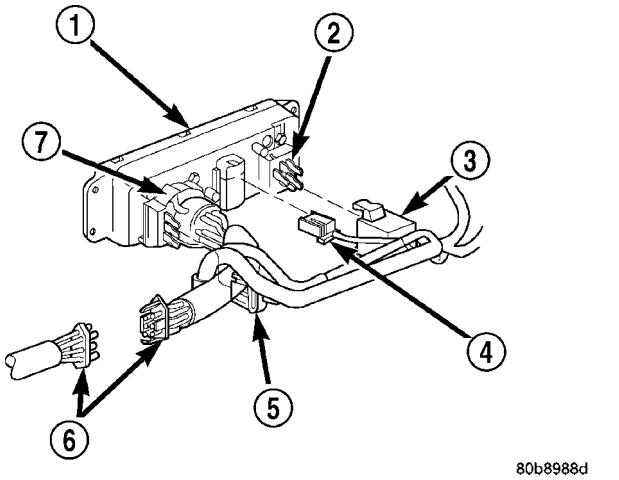

Fig. 17 A/C-Heater Control Connections - Typical

1 - A/C-HEATER CONTROL
2 - BLOWER MOTOR SWITCH
3 - BLOWER MOTOR SWITCH CONNECTOR
4 - TEMPERATURE/ILLUMINATION CONNECTOR
5 - MODE SWITCH CONNECTOR
6 - VACUUM HARNESS
7 - MODE SWITCH

INSTALLATION

(1) Connect the three wire harness connectors to the back of the A/C-heater control.

(2) Position the A/C-heater control onto the instrument panel and install the two retaining screws. Tighten the screws to 2 N·m (17 in. lbs.).

(3) Reach through the glove box opening and connect the A/C-heater vacuum harness connector.

(4) Install the glove box (Refer to 23 - BODY/INSTRUMENT PANEL/GLOVE BOX - INSTALLATION).

(5) Install the center bezel (Refer to 23 - BODY/ INSTRUMENT PANEL/INSTRUMENT PANEL CENTER BEZEL - INSTALLATION).

(6) Reconnect the negative battery cable.

A/C HIGH PRESSURE SWITCH

DESCRIPTION

The A/C high pressure switch controls A/C compressor clutch engagement and disengagement when equipped with the 4.0L engine. The A/C high pressure switch is mounted on a fitting located on the A/C discharge line near the A/C compressor. The A/C high pressure switch turns off the A/C compressor if

A/C HIGH PRESSURE SWITCH (Continued)

the refrigerant system pressure exceeds 3100 - 3375 kPa (450 - 490 psi).

The fitting for the A/C high pressure switch on the A/C discharge line is equipped with an O-ring seal and contains a Schrader-type valve, which allows the switch to be serviced without discharging the refrigerant system.

OPERATION

The A/C high pressure switch is electrically connected in series with the A/C low pressure switch between ground and the powertrain control module (PCM). The contacts in the A/C high pressure switch open and close causing the PCM to turn the A/C compressor clutch on and off. This prevents A/C compressor operation when the discharge line pressure approaches high levels, and also reduces electrical surging from A/C clutch engagement.

The A/C high pressure switch contacts open when the A/C discharge line pressure rises above 3100 to 3375 kPa (450 to 490 psi). The A/C high pressure switch contacts close when the A/C discharge line pressure drops to 1860 to 2275 kPa (270 to 330 psi).

The A/C high pressure switch is factory-calibrated and cannot be adjusted or repaired and, if faulty or damaged, it must be replaced.

DIAGNOSIS AND TESTING

A/C HIGH PRESSURE SWITCH

Before performing diagnosis of the A/C high pressure switch, verify that the refrigerant system has the correct refrigerant charge (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM CHARGE)

For circuit descriptions and diagrams, refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, further details on wire harness routing and retention, as well as pin-out and location views for the various wire harness connectors, splices and grounds.

(1) Disconnect and isolate the negative battery cable.

(2) Disconnect the wire harness connector from the A/C high pressure switch.

(3) Check for continuity between both terminals of the A/C high pressure switch. There should be continuity. If there is not continuity, replace the faulty A/C high pressure switch. If there is continuity, test the A/C high pressure switch sense circuit (Refer to 8 - ELECTRICAL/WIRING DIAGRAM INFORMATION - DIAGNOSIS AND TESTING). Repair as required.

REMOVAL

NOTE: It is not necessary to discharge the refrigerant system to replace the A/C high pressure switch.

(1) Disconnect and isolate the negative battery cable.

(2) Disconnect the wire harness connector from the A/C high pressure switch (Fig. 18).

(3) Remove the A/C high pressure switch from the A/C discharge line and remove and discard the O-ring seal.

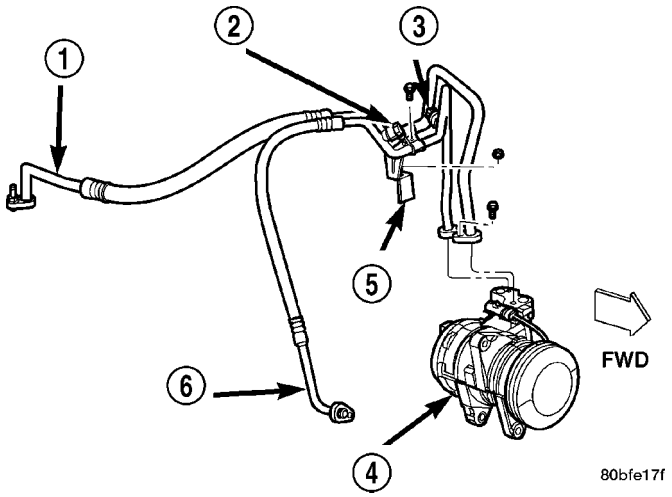

Fig. 18 A/C High Pressure Switch - Typical

1 - A/C SUCTION LINE
2 - SERVICE PORT
3 - A/C HIGH PRESSURE SWITCH
4 - A/C COMPRESSOR
5 - MOUNTING BRACKET
6 - A/C DISCHARGE LINE

INSTALLATION

NOTE: Use only the specified O-ring as it is made of special material for R-134a. Use only refrigerant oil of the type required for the A/C compressor.

(1) Lubricate a new rubber O-ring seal with clean refrigerant oil and install it on the A/C high pressure switch fitting.

(2) Install the A/C high pressure switch onto the A/C discharge line. Hand-tightened the switch securely.

(3) Connect the wire harness connector to the A/C high pressure switch.

(4) Reconnect the negative battery cable.

A/C LOW PRESSURE SWITCH

DESCRIPTION

The A/C low pressure switch is a single pole, single throw, pressure actuated switch that is used to control evaporator freeze up.

The A/C low pressure switch is mounted on a fitting located on the top of the A/C accumulator. The A/C low pressure switch fitting on the accumulator is equipped with an O-ring and contains a Schrader-type valve, which allows the switch to be serviced without discharging the refrigerant system.

OPERATION

The A/C low pressure switch monitors the pressure of the refrigerant leaving the accumulator to the A/C compressor. When equipped with the 4.0L engine, the A/C low pressure switch is electrically connected in series with the A/C high pressure switch, between ground and the powertrain control module (PCM).

The contacts in the A/C low pressure switch open or close the path to ground, signaling the PCM to turn the A/C compressor clutch on and off. This regulates the refrigerant system pressure and controls A/C evaporator temperature. Controlling the evaporator temperature prevents condensate water on the evaporator fins from freezing and obstructing A/C system air flow.

The A/C low pressure switch contacts are open when the A/C suction line pressure is approximately 172 kPa (25 psi) or lower. The A/C low pressure switch contacts will close when the A/C suction line pressure rises to approximately 276 kPa (40 psi) or above. Lower ambient temperatures, below approximately -1° C (30° F), will also cause the A/C low pressure switch contacts to open. This is due to the pressure/temperature relationship of the refrigerant in the system.

The A/C low pressure switch is factory-calibrated and cannot be adjusted or repaired. If the A/C low pressure switch is faulty or damaged, it must be replaced.

DIAGNOSIS AND TESTING

A/C LOW PRESSURE SWITCH

NOTE: Lower ambient temperatures, below about -1° C (30° F), during cold weather will open the A/C low pressure switch contacts and prevent A/C compressor operation due to the pressure/temperature relationship of the refrigerant.

Before performing diagnosis of the A/C low pressure switch, verify that the refrigerant system has the correct refrigerant charge (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM CHARGE) and that the A/C low pressure switch is properly installed on the accumulator fitting. If the A/C low pressure switch is not properly installed, it may not open the Schrader-type valve in the accumulator fitting, which will prevent the switch from correctly monitoring the refrigerant system pressure.

For circuit descriptions and diagrams, refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, further details on wire harness routing and retention, as well as pin-out and location views for the various wire harness connectors, splices and grounds.

(1) With gear selector in Park or Neutral and with park brake set, start the engine and allow it to idle.

(2) Place the A/C-heater mode control in any A/C position.

(3) Disconnect the wire harness connector from the A/C low pressure switch.

(4) Using a suitable jumper wire, install the jumper wire between the two terminals of the wire harness connector for the A/C low pressure switch. The A/C compressor clutch should engage. If the A/C clutch does engage, replace the faulty A/C low pressure switch. If the A/C clutch does not engage, the A/C clutch, A/C clutch relay, A/C-heater control , A/C high pressure switch (4.0L engine application), A/C pressure transducer (2.4L engine application), PCM, fuses or related wiring circuits may be defective (Refer to 8 - ELECTRICAL/WIRING DIAGRAM INFORMATION - DIAGNOSIS AND TESTING). Repair as required.

A/C LOW PRESSURE SWITCH (Continued)

REMOVAL

NOTE: Note: It is not necessary to discharge the refrigerant system to replace the A/C low pressure switch.

(1) Disconnect and isolate the negative battery cable.
(2) Disconnect the wire harness connector from the A/C low pressure switch (Fig. 19).
(3) Remove the A/C low pressure switch from the fitting on the top of the accumulator and remove and discard the O-ring seal.

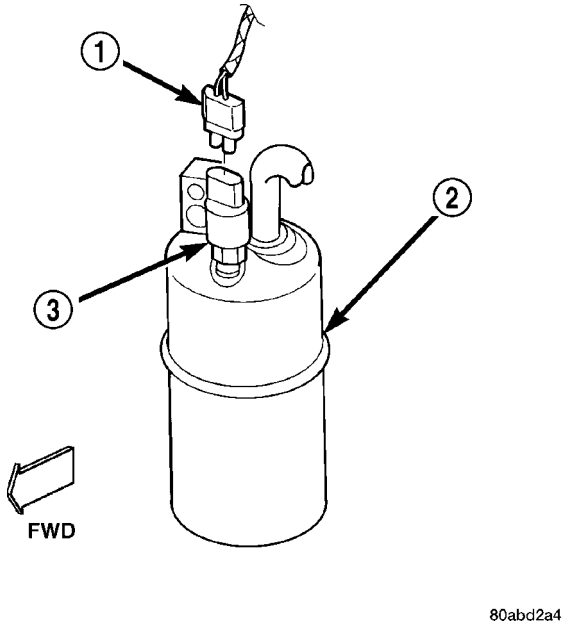

Fig. 19 A/C Low Pressure Switch - Typical

1 - WIRE HARNESS CONNECTOR
2 - ACCUMULATOR
3 - A/C LOW PRESSURE SWITCH

INSTALLATION

NOTE: Use only the specified O-ring as it is made of special material for R-134a. Use only refrigerant oil of the type required for the A/C compressor.

(1) Lubricate a new rubber O-ring seal with clean refrigerant oil and install it on the A/C low pressure switch fitting.
(2) Install the A/C low pressure switch on the accumulator. Hand-tightened the switch securely.
(3) Connect the wire harness connector to the A/C low pressure switch.
(4) Reconnect the negative battery cable.

A/C PRESSURE TRANSDUCER

DESCRIPTION

The A/C pressure transducer (Fig. 20) is a switch that is installed on a fitting located on the A/C discharge line when equipped with the 2.4L gasoline engine. An internally threaded fitting on the A/C pressure transducer connect it to the externally threaded Schrader-type fitting on the A/C discharge line. A rubber O-ring seals the connection between the A/C pressure transducer and the discharge line fitting. The A/C pressure transducer is connected to the vehicle electrical system by a molded plastic connector with three terminals.

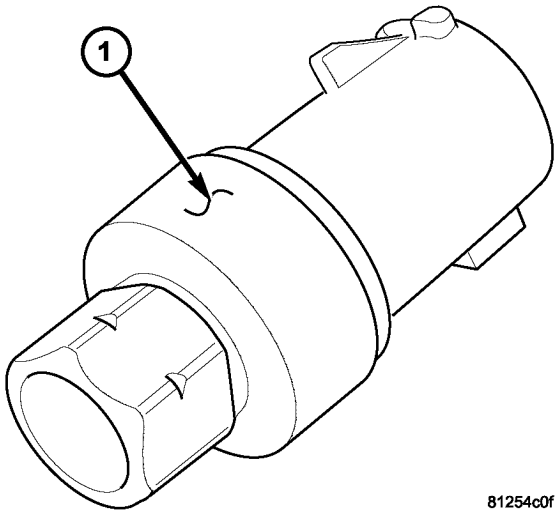

Fig. 20 A/C Pressure Transducer

1 - A/C PRESSURE TRANSDUCER

OPERATION

The A/C pressure transducer monitors the pressures in the high side of the refrigerant system through its connection to a fitting on the A/C discharge line. The A/C pressure transducer will change its internal resistance in response to the pressures it monitors. A Schrader-type valve in the discharge line fitting permits the A/C pressure transducer to be removed or installed without disturbing the refrigerant in the A/C system.

The powertrain control module (PCM) provides a five volt reference signal and a sensor ground to the A/C pressure transducer, then monitors the output voltage of the A/C pressure transducer on a sensor return circuit to determine refrigerant pressure. The PCM is programmed to respond to the A/C pressure transducer and other sensor inputs by controlling the operation of the A/C compressor clutch and the radiator cooling fan to help optimize A/C system perfor-

mance and to protect the system components from damage PCM will disengage the A/C compressor clutch when high side pressure rises above 2971 kPa (431 psi) or fall below 206 kPa (30 psi). The A/C pressure transducer input to the PCM will also prevent the A/C compressor clutch from engaging when ambient temperatures are below about 1.0° C (50° F) due to the pressure/temperature relationship of the refrigerant.

The A/C pressure transducer is diagnosed using a DRBIII® scan tool. Refer to the appropriate diagnostic information.

The A/C pressure transducer cannot be adjusted or repaired and, if faulty or damaged, it must be replaced.

DIAGNOSIS AND TESTING

A/C PRESSURE TRANSDUCER

The A/C pressure transducer is tested using a DRBIII® scan tool. Refer to the appropriate diagnostic information. Before testing the A/C pressure transducer, be certain that the transducer wire harness connection is clean of corrosion and properly connected. For the A/C to operate, an A/C pressure transducer voltage reading between 0.451 and 4.519 volts is required. Voltages outside this range indicate a low or high refrigerant system pressure condition to the powertrain control module (PCM). The PCM is programmed to respond to a low or high refrigerant system pressure by suppressing operation of the A/C compressor. Refer to the A/C Pressure Transducer Voltage chart for the possible conditions indicated by the transducer voltage reading.

A/C PRESSURE TRANSDUCER VOLTAGE

Voltage	Possible Indication
0.0	1. No sensor supply voltage from PCM. 2. Shorted sensor circuit. 3. Faulty transducer.
0.150 TO 0.450	1. Ambient temperature below 10° C (50° F). 2. Low refrigerant system pressure.
0.451 TO 4.519	1. Normal refrigerant system pressure.
4.520 TO 4.850	1. High refrigerant system pressure.
5.0	1. Open sensor circuit. 2. Faulty transducer.

REMOVAL

NOTE: Note: It is not necessary to discharge the refrigerant system to replace the A/C pressure transducer.

(1) Disconnect and isolate the negative battery cable.

(2) Disconnect the wire harness connector from the A/C pressure transducer (Fig. 21).

(3) Remove the A/C pressure transducer from the A/C discharge line and remove and discard the O-ring seal.

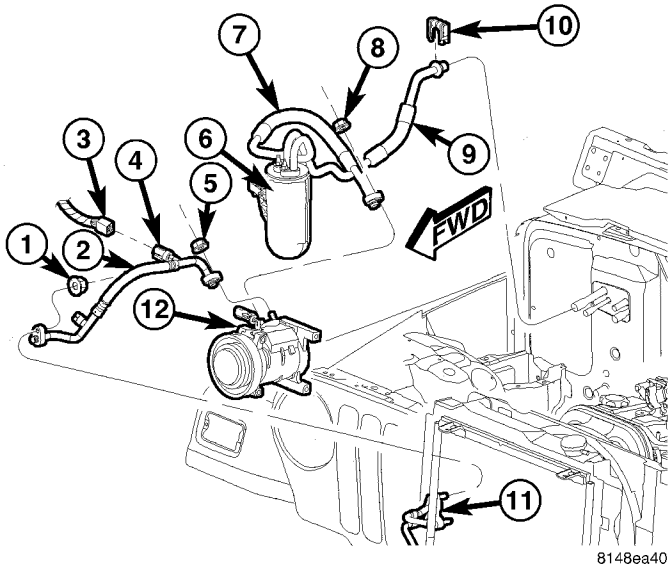

Fig. 21 A/C Pressure Transducer - 2.4L Engine

1 - NUT
2 - A/C DISCHARGE LINE
3 - WIRE HARNESS CONNECTOR
4 - A/C PRESSURE TRANSDUCER
5 - NUT
6 - ACCUMULATOR
7 - A/C SUCTION LINE (FRONT SECTION)
8 - NUT (4)
9 - A/C SUCTION LINE (REAR SECTION)
10 - SECONDARY RETAINING CLIP
11 - A/C CONDENSER
12 - A/C COMPRESSOR

INSTALLATION

NOTE: Use only the specified O-ring as it is made of special material for R-134a. Use only refrigerant oil of the type required for the A/C compressor.

(1) Lubricate a new rubber O-ring seal with clean refrigerant oil and install it on the A/C pressure transducer fitting.

(2) Install the A/C pressure transducer onto the A/C discharge line. Hand-tighten the transducer securely.

A/C PRESSURE TRANSDUCER (Continued)

(3) Connect the wire harness connector to the A/C pressure transducer.

(4) Reconnect the negative battery cable.

BLEND DOOR ACTUATOR

DESCRIPTION

The blend door actuator is a reversible, 12-volt direct current (DC), servo motor which mechanically positions the blend-air door (Fig. 22). The blend door actuator is located on the bottom of the HVAC housing.

The blend door actuator is contained within a black molded plastic housing with an integral wire connector receptacle. The actuator also has an output shaft that connects it to the mode door linkage and three integral mounting tabs that allow it to be secured to the HVAC housing. The blend door actuator requires mechanical indexing to the blend door linkage.

The blend door actuator can be accessed for service from under the instrument panel.

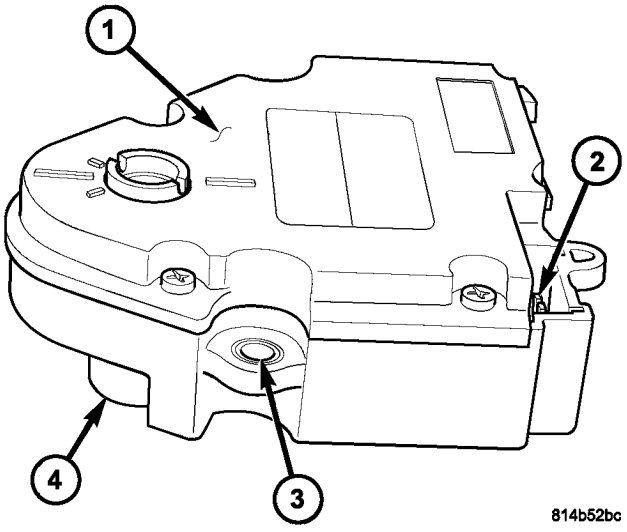

Fig. 22 Blend Door Actuator

1 - BLEND DOOR ACTUATOR
2 - WIRE CONNECTOR RECEPTACLE
3 - MOUNTING TAB (3)
4 - ACTUATOR OUTPUT SHAFT

OPERATION

The blend door actuator is connected to the A/C-heater control through the vehicle electrical system by a dedicated three-wire lead and connector of the HVAC wire harness. The blend door actuator can move the blend-air door in two directions. A potentiometer within the actuator allows the A/C-heater control to know the exact position of the blend-air door at all times.

The blend door actuator can be diagnosed using a DRBIII® scan tool. Refer to Body Diagnostic Procedures for more information.

The blend door actuator cannot be adjusted or repaired and, if faulty or damaged, it must be replaced.

REMOVAL

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the negative battery (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. This is the only sure way to disable the airbag system. Failure to take the proper precautions could result in accidental airbag deployment and possible personal injury or death.

NOTE: Prior to performing this procedure, turn the ignition key to the On position and set the A/C-heater control in the mid-temperature position and wait 10 seconds.

(1) Disconnect and isolate the negative battery cable.

(2) Disconnect the wire harness connector from the blend door actuator located at the bottom of the HVAC housing (Fig. 23).

(3) Remove the three screws that secure the blend door actuator to the HVAC housing and remove the actuator.

BLEND DOOR ACTUATOR (Continued)

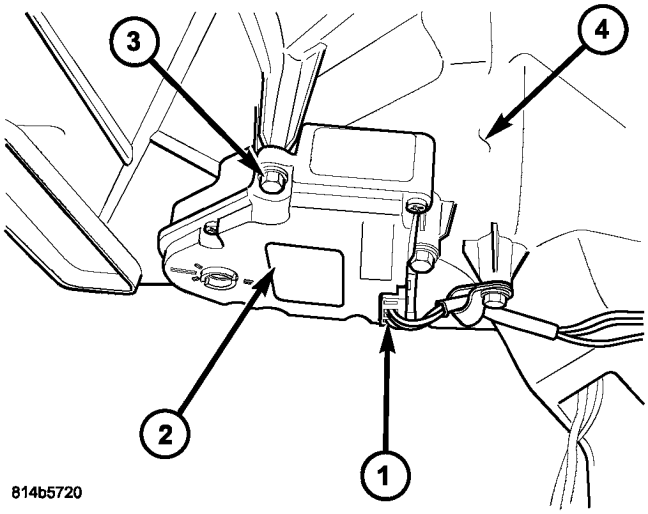

Fig. 23 Blend Door Actuator

1 - WIRE HARNESS CONNECTOR
2 - BLEND DOOR ACTUATOR
3 - SCREWS (3)
4 - HVAC HOUSING

INSTALLATION

(1) Position the blend door actuator onto the bottom of the HVAC housing. If necessary, rotate the actuator slightly to align the actuator output shaft with the blend door linkage.

(2) Install the three screws that secure the blend door actuator to the HVAC housing. Tighten the screws to 2 N·m (17 in. lbs.).

(3) Connect the HVAC wire harness connector to the blend door actuator.

(4) Reconnect the battery negative cable.

BLOWER MOTOR RELAY

DESCRIPTION

The blower motor relay (Fig. 24) is a International Standards Organization (ISO)-type relay. Relays conforming to the ISO specifications have common physical dimensions, current capacities, terminal functions and patterns. The blower motor relay is a electromechanical device that switches battery current from a fuse in the power distribution center (PDC) directly to the blower motor. The blower motor relay is energized when the relay coil is provided a voltage signal by the ignition switch.

The blower motor relay is mounted to the HVAC housing, directly behind the glove box.

OPERATION

The ISO-standard blower motor relay is an electromechanical switch that uses a low current input from the ignition switch to control the high current output to the blower motor. The movable, common feed relay contact is held against the fixed, normally closed relay contact by spring pressure. When the electromagnetic relay coil is energized, it draws the movable common feed relay contact away from the fixed, normally closed relay contact and, holds it against the fixed, normally open relay contact. This action allows high current to flow to the blower motor.

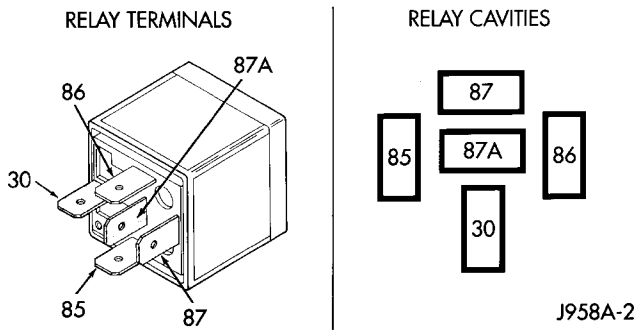

Fig. 24 Blower Motor Relay

30 - COMMON FEED
85 - COIL GROUND
86 - COIL BATTERY
87 - NORMALLY OPEN
87A - NORMALLY CLOSED

When the relay coil is de-energized, spring pressure returns the movable relay contact back against the fixed, normally closed contact point. The resistor or diode is connected in parallel with the relay coil, and helps to dissipate voltage spikes and electromagnetic interference that can be generated as the electromagnetic field of the relay coil collapses.

The blower motor relay terminals are connected to the vehicle electrical system through a connector receptacle mounted to the HVAC housing. The inputs and outputs of the blower motor relay include:

- The common feed terminal (4) receives battery current from a fuse in the PDC at all times.
- The coil ground terminal (2) is connected to ground at all times.
- The coil battery terminal (5) receives battery current from a fuse in the fuse block through a fused ignition switch output (run) circuit only when the ignition switch is in the On position.
- The normally open terminal (1) provides battery current to the blower motor through the blower motor relay only when the blower motor relay coil is energized.
- The normally closed terminal (87A) is not connected to any circuit in this application, but provides

BLOWER MOTOR RELAY (Continued)

a battery current output only when the blower motor relay coil is de-energized.

The blower motor relay cannot be repaired and, if faulty or damaged, it must be replaced. Refer to the appropriate wiring information for diagnosis and testing of the ISO-standard relay and for complete HVAC wiring diagrams.

REMOVAL

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the battery negative (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. This is the only sure way to disable the airbag system. Failure to take the proper precautions could result in accidental airbag deployment and possible personal injury or death.

(1) Disconnect and isolate the negative battery cable.

(2) Remove the glove box from the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/GLOVE BOX - REMOVAL).

(3) Locate the blower motor relay through the instrument panel glove box opening. The relay is mounted upright and to the right of the instrument panel harness (Fig. 25).

(4) Remove the blower motor relay from its wire harness connector.

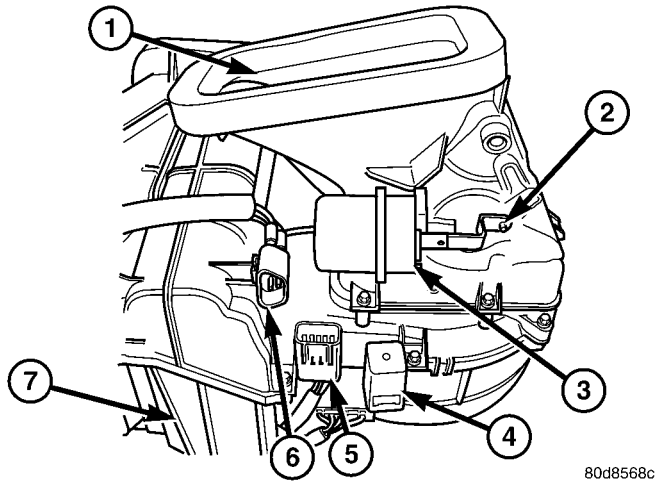

Fig. 25 Blower Motor Relay - LHD shown, RHD similar

1 - AIR INTAKE
2 - RECIRCULATION DOOR LINKAGE
3 - RECIRCULATION DOOR ACTUATOR
4 - BLOWER MOTOR RELAY
5 - HVAC ELECTRICAL CONNECTOR
6 - HVAC VACUUM CONNECTOR
7 - HVAC HOUSING

INSTALLATION

(1) Align the blower motor relay terminals with the terminal cavities in the wire harness connector.

(2) Push down firmly on the blower motor relay until the terminals are fully seated in the connector receptacles.

(3) Install the glove box (Refer to 23 - BODY/INSTRUMENT PANEL/GLOVE BOX - INSTALLATION).

(4) Reconnect the negative battery cable.

BLOWER MOTOR RESISTOR

DESCRIPTION

The blower motor resistor (Fig. 26) is mounted to the rear of the HVAC housing, near the glove box. The blower motor resistor consists of a molded plastic mounting plate with an integral wire connector receptacle. Concealed behind the mounting plate is the resistor circuit board.

The blower motor resistor can be accessed for service from under the instrument panel.

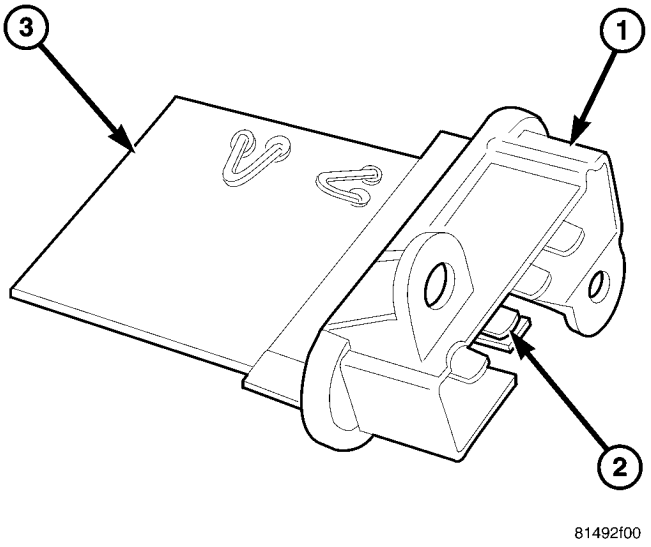

Fig. 26 Blower Motor Resistor

1 - MOUNTING PLATE
2 - WIRE HARNESS CONNECTOR
3 - RESISTOR CIRCUIT BOARD

OPERATION

The blower motor resistor is connected to the vehicle electrical system through a dedicated wire lead and connector of the HVAC wire harness. The blower motor resistor has multiple resistor circuits, each of which will reduce the current flow through the blower motor to change the blower motor speed.

The blower motor switch in the MTC heating-A/C system directs the ground path for the blower motor

through the correct resistor circuit to obtain the selected speed. With the blower motor control in the lowest speed position, the ground path for the blower motor is applied through all of the resistor circuits. Each higher speed selected with the blower motor control applies the blower motor ground path through fewer of the resistor circuits, increasing the blower motor speed. When the blower motor control is in the highest speed position, the blower motor resistor is bypassed and the blower motor receives a direct path to ground.

The blower motor resistor cannot be adjusted or repaired and, if faulty or damaged, it must be replaced.

DIAGNOSIS AND TESTING

BLOWER MOTOR RESISTOR

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the negative battery (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. This is the only sure way to disable the airbag system. Failure to take the proper precautions could result in accidental airbag deployment and possible personal injury or death.

NOTE: For circuit descriptions and diagrams, refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, further details on wire harness routing and retention, as well as pinout and location views for the various wire harness connectors, splices and grounds.

(1) Disconnect and isolate the negative battery cable.

(2) Disconnect the wire harness connector from the blower motor resistor (Refer to 24 - HEATING & AIR CONDITIONING/CONTROLS/BLOWER MOTOR RESISTOR - REMOVAL).

(3) Using an ohmmeter, check for continuity between all of the blower motor resistor terminals. In each case there should be continuity. If OK, repair the wire harness circuits between the blower motor switch and the blower motor resistor or blower motor as required. If not OK, replace the faulty blower motor resistor.

REMOVAL

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the negative battery (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. This is the only sure way to disable the airbag system. Failure to take the proper precautions could result in accidental airbag deployment and possible personal injury or death.

WARNING: The blower motor resistor may get very hot during normal operation. If the blower motor was turned on prior to servicing the blower motor resistor, wait five minutes to allow the blower motor resistors to cool before performing diagnosis or service. Failure to take this precaution can result in possible personal injury.

(1) Disconnect and isolate the negative battery cable.

(2) Remove the glove box (Refer to 23 - BODY/INSTRUMENT PANEL/GLOVE BOX - REMOVAL).

(3) Unlock the blower motor resistor wire harness connector and disconnect the connector from the blower motor resistor (Fig. 27).

(4) Reach through the glove box opening and remove the upper screw that secures the blower motor resistor to the HVAC housing.

(5) From under the instrument panel, remove the lower screw that secures the blower motor resistor to the HVAC housing and remove the resistor.

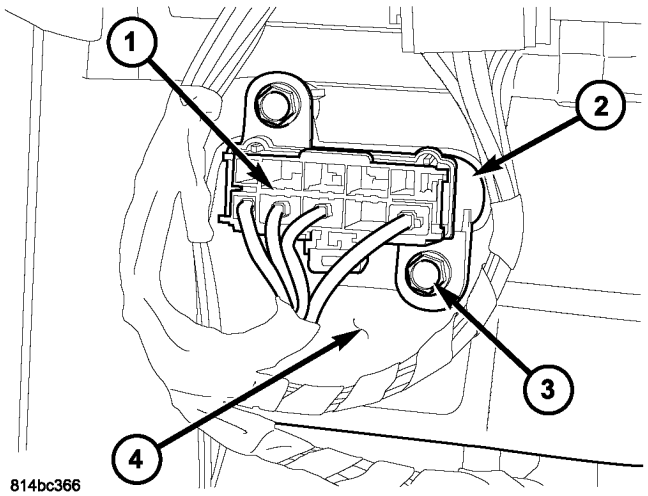

Fig. 27 Blower Motor Resistor-R and I

1 - WIRE HARNESS CONNECTOR
2 - BLOWER MOTOR RESISTOR
3 - SCREW (2)
4 - HVAC HOUSING

BLOWER MOTOR RESISTOR (Continued)

INSTALLATION

(1) From under the instrument panel, position the blower motor resistor into the HVAC housing.

(2) From under the instrument panel, install the lower screw that secures the blower motor resistor to the HVAC housing. Tighten the screw to 2 N·m (17 in. lbs.).

(3) Reach through the glove box opening and install the upper screw that secures the blower motor resistor to the HVAC housing. Tighten the screw to 2 N·m (17 in. lbs.).

(4) Connect the wire harness connector to the blower motor resistor and engage the connector lock.

(5) Install the glove box (Refer to 23 - BODY/INSTRUMENT PANEL/GLOVE BOX - INSTALLATION).

(6) Reconnect the negative battery cable.

BLOWER MOTOR SWITCH

DESCRIPTION

The blower motor for the heater-only or heater-A/C system is controlled by a rotary-type blower motor switch, mounted in the A/C-heater control (Fig. 28). The blower motor switch allows the selection of four different blower motor speeds and includes an Off position.

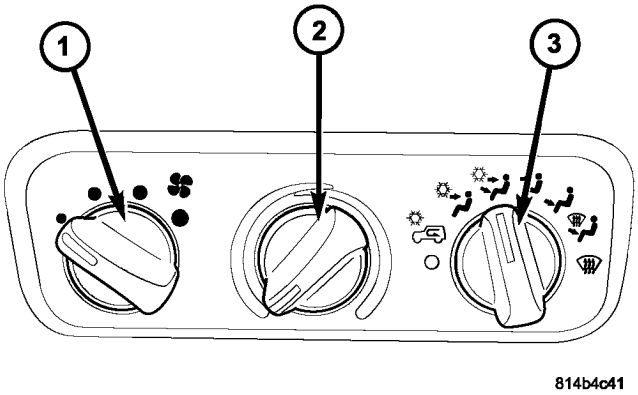

Fig. 28 A/C-Heater Control - Typical

1 - BLOWER SPEED CONTROL
2 - TEMPERATURE SELECT COPNTROL
3 - MODE SELECT CONTROL

OPERATION

The blower motor switch provides a ground path for the blower motor through the A/C-heater mode control. The blower motor switch directs this ground path through or around the blower motor resistor wires, as required to achieve the selected blower motor speed.

The blower motor switch cannot be repaired and, if faulty or damaged, it must be replaced. The blower motor switch knob is available for service replacement.

DIAGNOSIS AND TESTING

BLOWER MOTOR SWITCH

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the battery negative (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. This is the only sure way to disable the airbag system. Failure to take the proper precautions could result in accidental airbag deployment and possible personal injury or death.

For circuit descriptions and diagrams, refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, further details on wire harness routing and retention, as well as pin-out and location views for the various wire harness connectors, splices and grounds.

(1) Check for battery voltage at the fuse in the power distribution center (PDC). If OK, go to Step 2. If not OK, repair the shorted circuit or component as required and replace the faulty fuse.

(2) Turn the ignition switch to the Off position. Disconnect and isolate the negative battery cable. Remove the A/C-heater control from the instrument panel (Refer to 24 - HEATING & AIR CONDITIONING/CONTROLS/A/C HEATER CONTROL - REMOVAL). Check for continuity between the ground circuit cavity of the A/C-heater control wire harness connector and a good ground. There should be continuity. If OK, go to Step 3. If not OK, repair the open circuit to ground as required.

(3) With the A/C-heater control wire harness connector disconnected, place the A/C-heater mode control in any position except Off. Check for continuity between the ground circuit terminal and each of the blower motor driver circuit terminals of the A/C-heater control as you move the blower motor switch to each of the four speed positions. There should be continuity at each driver circuit terminal in only one blower motor switch speed position. If OK, test and repair the blower driver circuits between the A/C-heater control and the blower motor resistor as required. If not OK, replace the faulty blower motor switch.

BLOWER MOTOR SWITCH (Continued)

REMOVAL

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the battery negative (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. This is the only sure way to disable the airbag system. Failure to take the proper precautions could result in accidental airbag deployment and possible personal injury or death.

(1) Disconnect and isolate the negative battery cable.

(2) Using a trim stick C-4755 or equivalent, gently pry the blower motor switch knob off of the switch stem from the front of the A/C-heater control.

(3) Remove the A/C-heater control from the instrument panel (Refer to 24 - HEATING & AIR CONDITIONING/CONTROLS/A/C HEATER CONTROL - REMOVAL).

(4) Remove the screw that secures the blower motor switch to the rear of the A/C-heater control (Fig. 29).

(5) Remove the blower motor switch from the A/C-heater control.

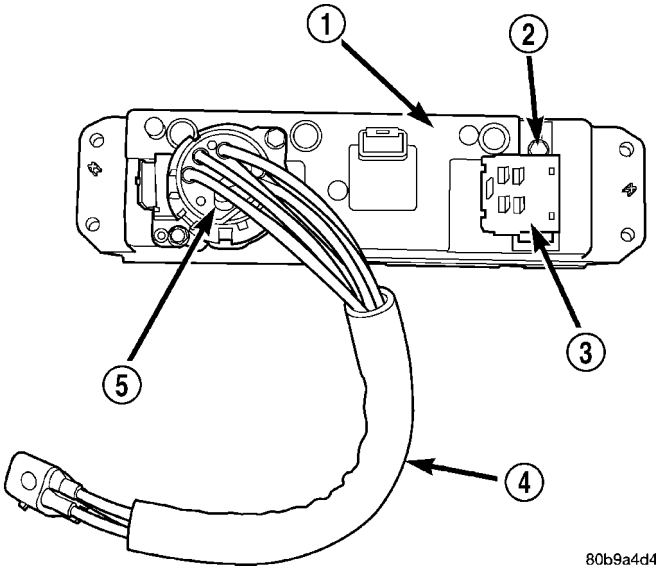

Fig. 29 Blower Motor Switch

1 - A/C-HEATER CONTROL
2 - SCREW
3 - BLOWER MOTOR SWITCH
4 - VACUUM HARNESS
5 - MODE CONTROL SWITCH

INSTALLATION

(1) Position the blower motor switch onto the rear of the A/C-heater control.

(2) Install the screw that secures the blower motor switch to the A/C-heater control. Tighten the screw to 2 N·m (17 in. lbs.).

(3) Install the A/C-heater control (Refer to 24 - HEATING & AIR CONDITIONING/CONTROLS/A/C HEATER CONTROL - INSTALLATION).

(4) Install the blower motor switch knob onto the switch stem on the front of the A/C-heater control.

(5) Reconnect the negative battery cable.

MODE DOOR ACTUATORS

DESCRIPTION

Both the heating-only system and the heating-A/C system use two vacuum operated mode door actuators to control the movement of the mode-air doors within the HVAC housing (Fig. 30).

When vacuum is supplied by the A/C-heater control to one side of the defrost/floor actuator or the panel door actuator, the actuator rod is pulled into the actuator, which moves the mode door lever and mode-air door in one direction. When vacuum is supplied by the A/C-heater control to the other side of the actuator, the rod moves the mode-air door in the other direction.

The mode door actuators can be accessed for service from under the left side of the instrument panel.

The mode door actuators cannot be adjusted or repaired and, if faulty or damaged, they must be replaced.

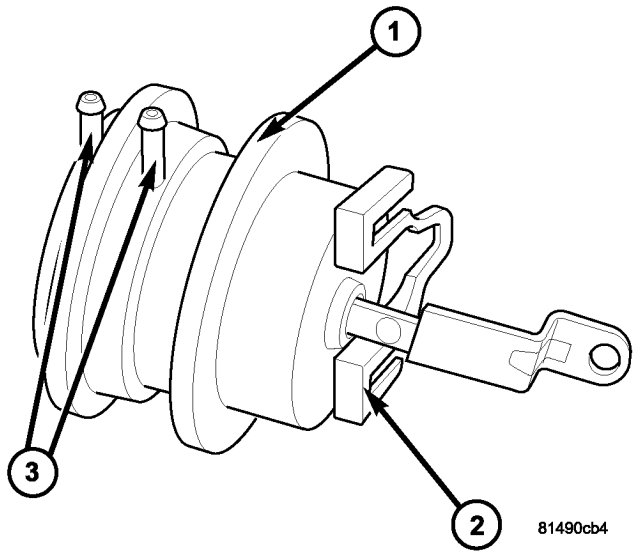

Fig. 30 Vacuum Operated Mode Door Actuator - Typical

1 - MODE DOOR ACTUATOR
2 - MOUNTING PROVISION
3 - VACUUM LINE CONNECTORS

MODE DOOR ACTUATORS (Continued)

REMOVAL

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the battery negative (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. This is the only sure way to disable the airbag system. Failure to take the proper precautions could result in accidental airbag deployment and possible personal injury or death.

FLOOR/DEFROST DOOR ACTUATOR

(1) Disconnect and isolate the negative battery cable.

(2) Remove the knee blocker from the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/KNEE BLOCKER - REMOVAL).

(3) Disconnect the vacuum harness from the floor/defrost door actuator (Fig. 31).

(4) Remove the push-nut that secures the actuator rod to the floor/defrost door lever.

(5) Using a small flat bladed tool, gently pry the actuator latch while pulling firmly outwards on the floor/defrost door actuator. Rotate and tilt the actuator as required to remove the actuator from the HVAC housing mount.

(6) Disengage the actuator rod from the floor/defrost door lever and remove the actuator.

PANEL DOOR ACTUATOR

(1) Remove the airbag control module (Refer to 8 - ELECTRICAL/RESTRAINTS/AIRBAG CONTROL MODULE - REMOVAL).

(2) Remove the floor distribution duct (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/FLOOR DISTRIBUTION DUCT - REMOVAL).

(3) Disconnect the vacuum harness from the panel door actuator (Fig. 32).

(4) Using a small flat bladed tool, gently pry the actuator latch while pulling firmly outwards on the panel door actuator. Rotate and tilt the actuator as required to remove the actuator from the HVAC housing mount.

(5) Using needle nose pliers, disengage the actuator rod from the push-pin type retainer located on the panel door lever and remove the actuator.

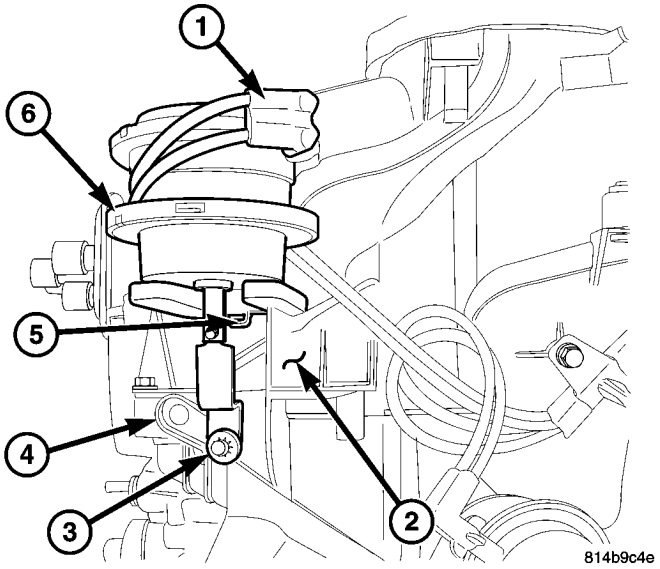

Fig. 31 Floor/Defrost Door Actuator

1 - VACUUM HARNESS
2 - HVAC HOUSING
3 - PUSH-NUT
4 - FLOOR/DEFROST DOOR LEVER
5 - ACTUATOR LATCH
6 - FLOOR/DEFROST (MODE) DOOR ACTUATOR

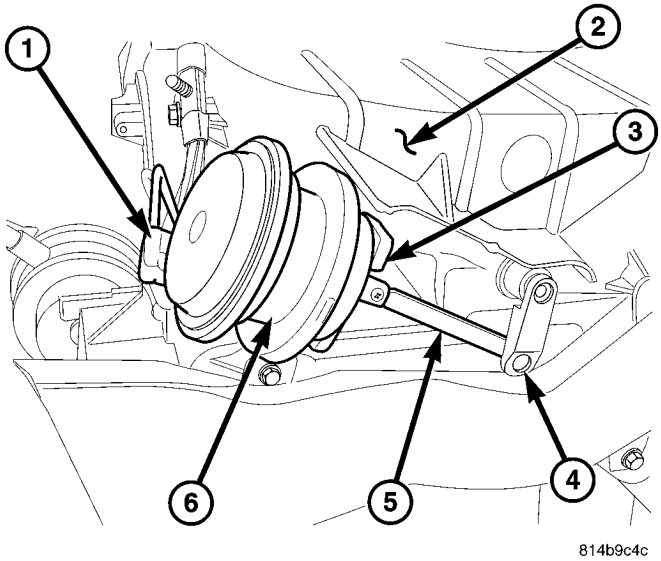

Fig. 32 Panel Door Actuator

1 - VACUUM HARNESS
2 - HVAC HOUSING
3 - ACTUATOR LATCH
4 - PANEL DOOR LEVER
5 - ACTUATOR ROD
6 - PANEL (MODE) DOOR ACTUATOR

INSTALLATION

FLOOR/DEFROST DOOR ACTUATOR

(1) Rotate and tilt the floor/defrost door actuator as required to install the actuator mount onto the HVAC housing mounting boss. Push the actuator onto the mount until it snaps into the locked position.

(2) Engage the actuator rod to the floor/defrost door lever and install the push-nut.

MODE DOOR ACTUATORS (Continued)

(3) Connect the vacuum harness to the floor/defrost door actuator.

(4) Install the knee blocker onto the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/KNEE BLOCKER - INSTALLATION).

(5) Reconnect the negative battery cable.

PANEL DOOR ACTUATOR

(1) Using needle nose pliers, engage the actuator rod to the push-pin type retainer located on the panel door lever.

(2) Rotate and tilt the panel door actuator as required to install the actuator mount onto the HVAC housing mounting boss. Push the actuator onto the mount until it snaps into the locked position.

(3) Connect the vacuum harness to the panel door actuator.

(4) Install the floor distribution duct (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/FLOOR DISTRIBUTION DUCT - INSTALLATION).

(5) Install the airbag control module (Refer to 8 - ELECTRICAL/RESTRAINTS/AIRBAG CONTROL MODULE - INSTALLATION).

RECIRCULATION DOOR ACTUATOR

DESCRIPTION

When equipped with A/C, a vacuum operated recirculation door actuator is used to control the movement of the recirculation-air door within the HVAC housing (Fig. 33).

When vacuum is supplied by the A/C-heater control to the recirculation door actuator, the actuator linkage is pulled into the actuator, which moves the recirculation door lever and recirculation door to two Recirculation position. The recirculation door actuator is spring loaded, so when vacuum is released from the actuator, the linkage moves the recirculation-air door back to its static position (fresh air).

The recirculation door actuator can be accessed for service through the glove box opening in the instrument panel.

The recirculation door actuator cannot be adjusted or repaired and, if faulty or damaged, it must be replaced.

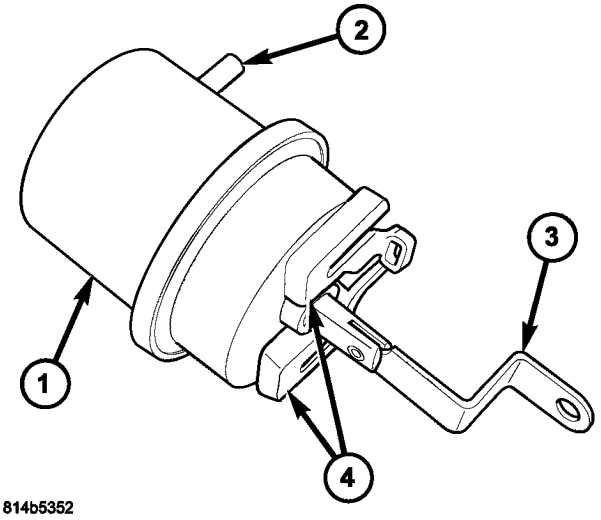

Fig. 33 Recirculation Door Actuator

1 - RECIRCULATION DOOR ACTUATOR
2 - VACUUM CONNECTOR RECEPTACLE
3 - ACTUATOR ROD
4 - ACTUATOR MOUNT

REMOVAL

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the battery negative (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. This is the only sure way to disable the airbag system. Failure to take the proper precautions could result in accidental airbag deployment and possible personal injury or death.

NOTE: The recirculation-air door and actuator are used only on models equipped with A/C.

RECIRCULATION DOOR ACTUATOR (Continued)

(1) Disconnect and isolate the negative battery cable.

(2) Remove the glove box (Refer to 23 - BODY/INSTRUMENT PANEL/GLOVE BOX - REMOVAL).

(3) Remove the right front speaker bezel and speaker (Refer to 8 - ELECTRICAL/AUDIO/SPEAKER - REMOVAL).

(4) Using a small flat bladed tool, gently pry the actuator latch while pulling firmly outwards on the actuator (Fig. 34). Rotate and tilt the actuator as required to remove the actuator from the HVAC air inlet housing mount.

(5) Disengage the actuator rod from the push-pin type retainer located on the end of the recirculation door actuator lever.

(6) Disconnect the vacuum harness from the recirculation door actuator and remove the actuator from the vehicle.

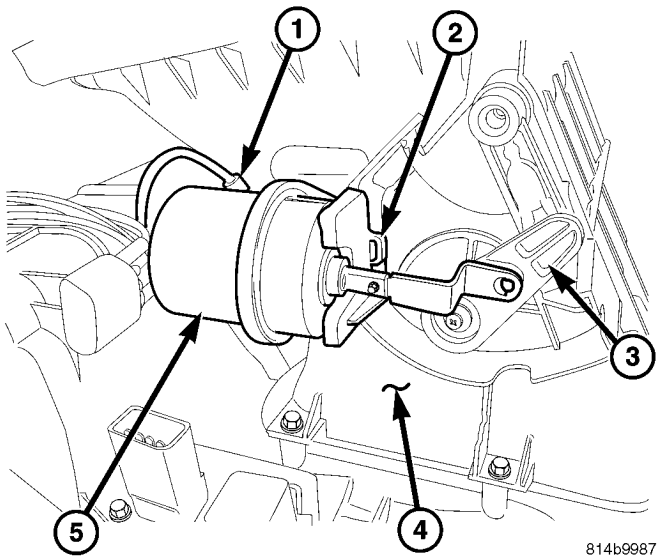

Fig. 34 Recirculation Door Actuator

1 - VACUUM HARNESS
2 - ACTUATOR LATCH
3 - RECIRCULATION DOOR ACTUATOR LEVER
4 - HVAC AIR INLET HOUSING
5 - RECIRCULATION DOOR ACTUATOR

INSTALLATION

NOTE: The recirculation-air door and actuator are used only on models equipped with A/C.

(1) Connect the vacuum harness to the recirculation door actuator.

(2) Engage the actuator rod to the push-pin type retainer located on the end of the recirculation door actuator lever.

(3) Rotate and tilt the recirculation door actuator as required to install the actuator mount onto the HVAC air inlet housing mounting boss. Push the actuator onto the mount until it snaps into the locked position.

(4) Install the right front speaker and speaker bezel (Refer to 8 - ELECTRICAL/AUDIO/SPEAKER - INSTALLATION).

(5) Install the glove box (Refer to 23 - BODY/INSTRUMENT PANEL/GLOVE BOX - INSTALLATION).

(6) Reconnect the negative battery cable.

VACUUM CHECK VALVE

DESCRIPTION

Two vacuum check valves (Fig. 35) are installed in the accessory vacuum supply system. One is on the accessory vacuum supply line in the engine compartment, near the vacuum tap on the engine intake manifold. A second vacuum check valve is located at the HVAC system takeout. The vacuum check valves are designed to allow vacuum to flow in only one direction through the vacuum supply circuits.

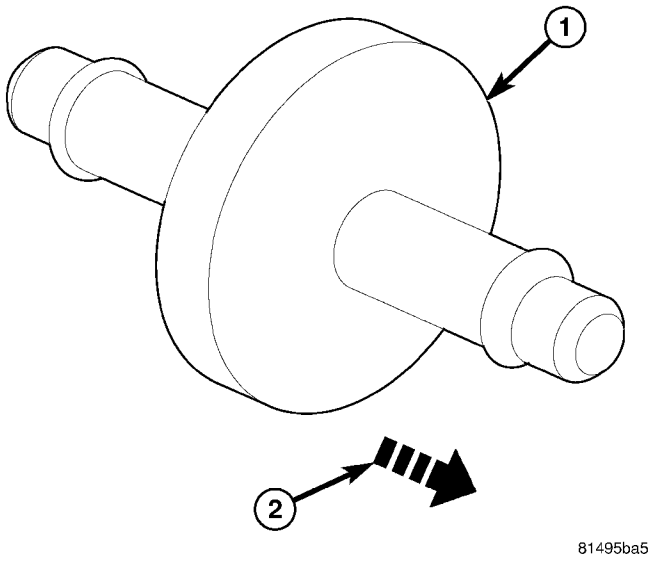

Fig. 35 Vacuum Check Valve - Typical

1 - VACUUM CHECK VALVE
2 - DIRECTION OF AIR FLOW

OPERATION

The use of the two vacuum check valves help to maintain the system vacuum needed to retain the selected A/C-heater mode settings. The check valves prevent the engine from bleeding down system vacuum through the intake manifold during extended heavy engine load (low engine vacuum) operation.

The vacuum check valves cannot be repaired and, if faulty or damaged, must be replaced.

VACUUM CHECK VALVE (Continued)

REMOVAL

NOTE: Note the orientation of the check valves in the vacuum supply line for correct reinstallation.

(1) Disconnect the vacuum check valve from the fitting at the engine intake manifold and the accessory vacuum supply line (Fig. 36) and remove the valve.

(2) Disconnect the vacuum check valve from the HVAC vacuum supply line fitting and the accessory vacuum supply line and remove the valve.

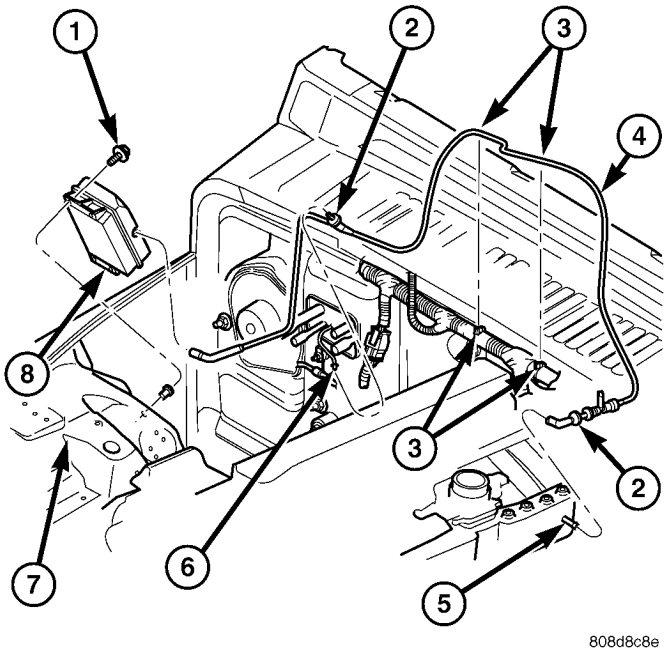

Fig. 36 Vacuum Check Valves - Typical

1 - SCREW
2 - VACUUM CHECK VALVE (2)
3 - VACUUM LINE RETAINER (2)
4 - ACCESSORY VACUUM SUPPLY LINE
5 - ENGINE VACUUM FITTING
6 - HVAC VACUUM SUPPLY LINE
7 - INNER FENDER
8 - VACUUM RESERVOIR

INSTALLATION

(1) Using the orientation noted during Removal, connect the vacuum check valve to the HVAC vacuum supply line fitting and the accessory vacuum supply line.

(2) Using the orientation noted during Removal, connect the vacuum check valve to the fitting at the engine intake manifold and the accessory vacuum supply line.

VACUUM RESERVOIR

DESCRIPTION

The vacuum reservoir (Fig. 37) is mounted to the rear of the right front inner fender in the engine compartment, under the battery tray.

The vacuum reservoir can be accessed for service by removing the battery and battery tray.

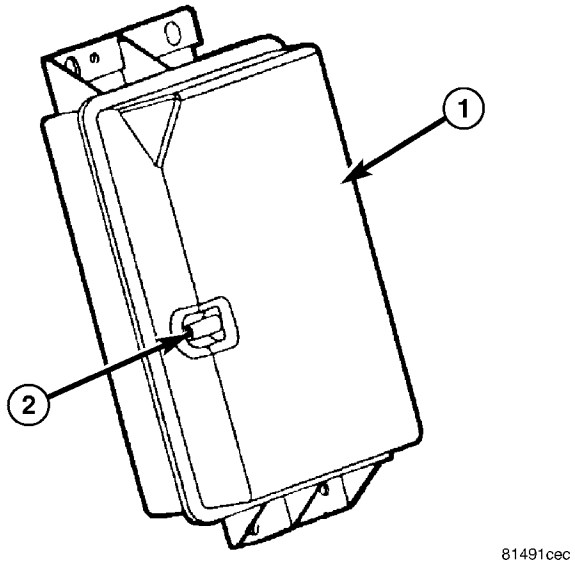

Fig. 37 Vacuum Reservoir - Typical

1 - VACUUM RESERVOIR
2 - VACUUN LINE CONNECTION

OPERATION

Engine vacuum is stored in the vacuum reservoir. The stored vacuum is used to operate the vacuum-controlled vehicle accessories during periods of low engine vacuum such as when the vehicle is climbing a steep grade, or under other high engine load operating conditions.

The vacuum reservoir cannot be repaired and, if faulty or damaged, it must be replaced.

VACUUM RESERVOIR (Continued)

REMOVAL

(1) Remove the battery and battery tray from the engine compartment (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/BATTERY - REMOVAL) and (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/TRAY - REMOVAL).

(2) Disconnect the vacuum supply line from the vacuum reservoir (Fig. 38).

(3) Remove the screw that secures the vacuum reservoir to the inner fender.

(4) Remove the vacuum reservoir from the engine compartment.

INSTALLATION

(1) Position the vacuum reservoir into the engine compartment.

(2) Install the screw that secures the vacuum reservoir to the inner fender panel. Tighten the screw to 2 N·m (17 in. lbs.).

(3) Connect the vacuum supply line to the vacuum reservoir.

(4) Install the battery tray and battery (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/TRAY - INSTALLATION) and (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/BATTERY - INSTALLATION).

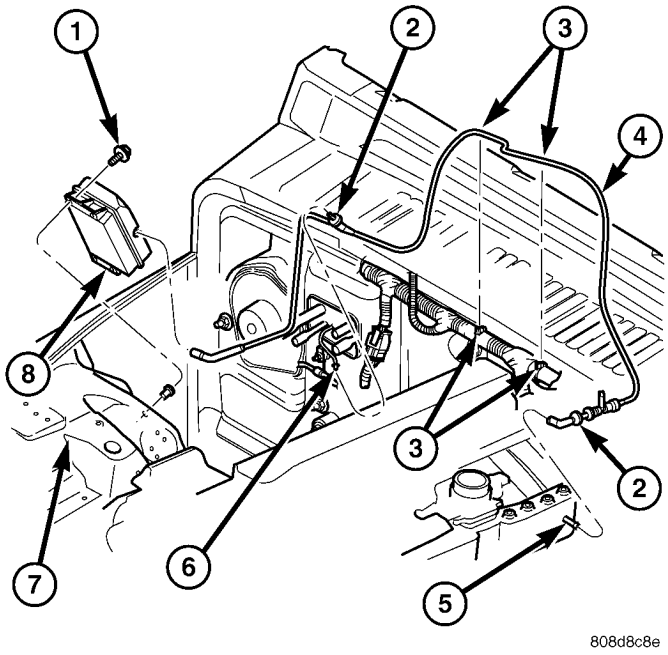

Fig. 38 Vacuum Reservoir - Typical

1 - SCREW
2 - VACUUM CHECK VALVE (2)
3 - VACUUM LINE RETAINER (2)
4 - ACCESSORY VACUUM SUPPLY LINE
5 - ENGINE VACUUM FITTING
6 - HVAC VACUUM SUPPLY LINE
7 - INNER FENDER
8 - VACUUM RESERVOIR

DISTRIBUTION

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AIR OUTLETS

DESCRIPTION

There is one defroster air outlet located an the top of the instrument panel (Fig. 1). The airflow from the defroster outlet is directed by fixed vanes in the defroster grille and cannot be adjusted. The defroster outlet and grille are not serviceable from the instrument panel top cover (Refer to 23 - BODY/INSTRUMENT PANEL/INSTRUMENT PANEL TOP COVER - REMOVAL).

There are four instrument panel air outlets. One air outlet is located near each outboard end of the instrument panel facing the rear of the vehicle, and two located near the top of the instrument panel center bezel. Each of the instrument panel air outlets contain a serviceable grille with movable vanes that are used to direct or shut off the flow of the conditioned air leaving the instrument panel outlets. The instrument panel air outlets can be removed individually for service replacement.

REMOVAL

NOTE: The instrument panel air outlets are retained to the pivots within the air outlet housings by a snap fit.

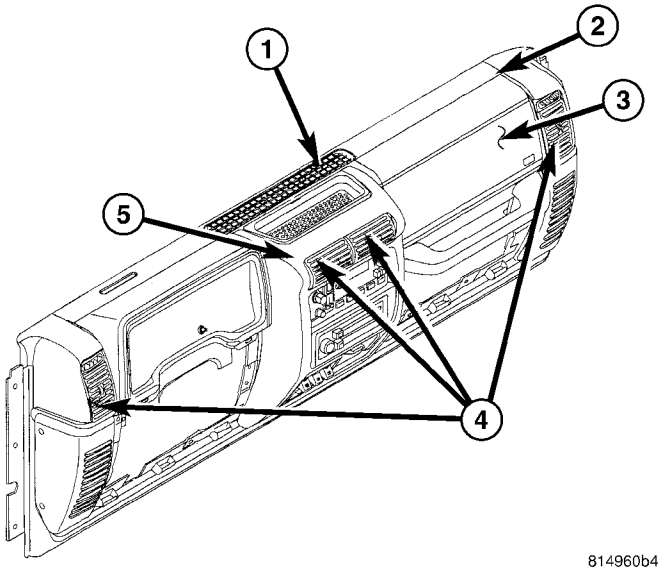

Fig. 1 Air Outlets - LHD shown, RHD similar

1 - DEFROSTER AIR OUTLET
2 - INSTRUMENT PANEL TOP COVER
3 - INSTRUMENT PANEL
4 - INSTRUMENT PANEL AIR OUTLETS (4)
5 - INSTRUMENT PANEL CENTER BEZEL

(1) Using a trim stick C-4755 or equivalent (Fig. 2), gently pry the panel air outlet out of the panel outlet housing as required.

AIR OUTLETS (Continued)

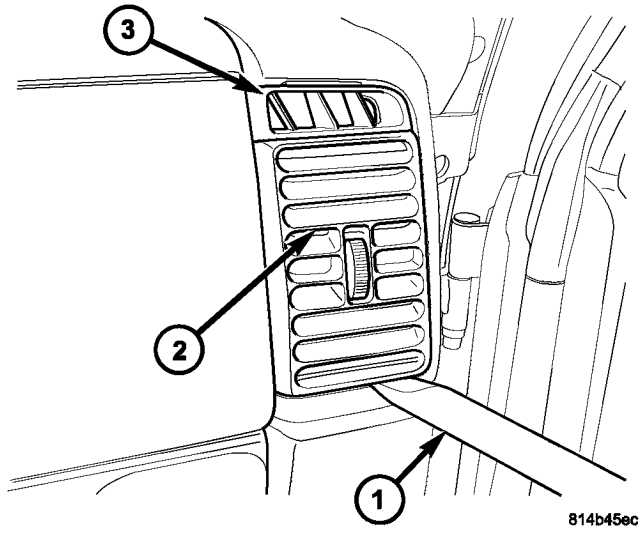

Fig. 2 Instrument Panel Air Outlets

1 - TRIM STICK
2 - PANEL AIR OUTLETS (4)
3 - PANEL OUTLET HOUSING

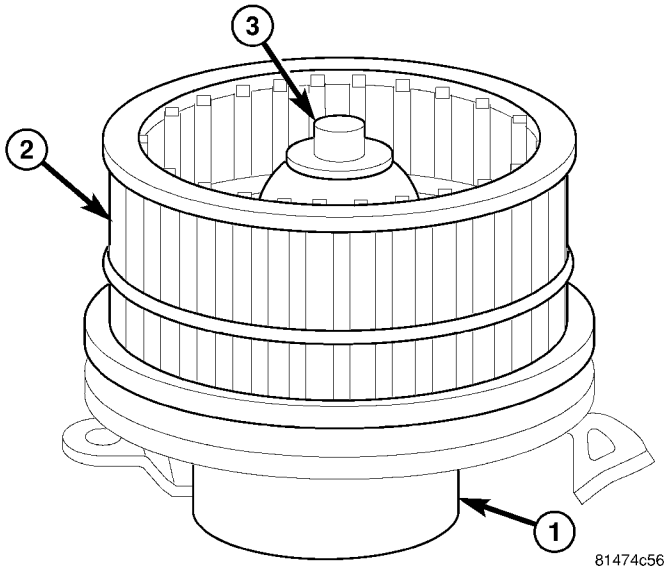

Fig. 3 Blower Motor - Typical

1 - BLOWER MOTOR
2 - BLOWER WHEEL
3 - MOTOR SHAFT

INSTALLATION

CAUTION: The instrument panel air outlets can only be installed in one direction. If an air outlet is not properly oriented prior to installation, damage to the outlet will occur.

NOTE: Each air outlet is retained into the outlet opening by two pivot shafts. One pivot shaft is larger than the other. The air outlet must be first installed onto the larger pivot shaft, then snapped into place over the smaller pivot shaft.

(1) Install the air outlet onto the larger pivot shaft located in the instrument panel outlet opening.

(2) Gently push the air outlet onto the smaller pivot shaft until it snaps into position.

BLOWER MOTOR

DESCRIPTION

The blower motor (Fig. 3)is a 12-volt, direct current (DC) motor mounted within a plastic housing with an integral wire connector and squirrel cage-type blower wheel that is secured to the blower motor shaft. The blower motor wheel is located in the HVAC air inlet housing which is mounted to the passenger side end of the HVAC housing.

The blower motor can be accessed for service from underneath the instrument panel.

OPERATION

The blower motor is used to control the velocity of air moving through the HVAC housing by spinning the blower wheel within the HVAC air inlet housing at the selected speed.

The blower motor will operate whenever the ignition switch is in the On position and the blower motor control in any position except Off. Blower motor speed is controlled by regulating the path to ground through the blower motor resistor and through the blower motor control located within the A/C-heater control.

The blower motor and blower motor wheel are factory balanced and cannot be adjusted or repaired. If faulty or damaged, the blower motor and blower wheel must be replaced as an assembly.

DIAGNOSIS AND TESTING

BLOWER MOTOR

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the negative battery (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. Failure to take the proper precautions could result in accidental airbag deployment and possible personal injury or death.

BLOWER MOTOR (Continued)

NOTE: For circuit descriptions and diagrams, refer to Air Conditioning/Heater in Group 8W - Wiring Diagrams.

OPERATION

Possible causes of an inoperative blower motor or blower motor not operating in all speeds include:

- Faulty fuse
- Faulty blower motor resistor
- Faulty blower motor switch
- Faulty blower motor
- Faulty blower motor circuit wiring or wire harness connectors

VIBRATION

Possible causes of blower motor vibration include:

- Improper blower motor mounting
- Improper blower wheel mounting
- Deformed blower wheel
- Out of balance blower wheel due to foreign material in the wheel
- Faulty blower motor

NOISE

To determine if the blower motor is the source of the noise, simply switch the blower motor from Off to On. To verify that the blower motor is the source of the noise, unplug the blower motor wire harness connector and operate the heating-A/C system. If the noise goes away, possible causes include:

- Foreign material in the HVAC air inlet housing
- Improper blower motor mounting
- Improper blower wheel mounting
- Faulty blower motor

REMOVAL

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the battery negative (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. This is the only sure way to disable the airbag system. Failure to take the proper precautions could result in an accidental airbag deployment and possible personal injury or death.

NOTE: The blower motor is located on the passenger side of the vehicle under the instrument panel. The blower motor can be removed without having to remove the instrument panel or the HVAC housing.

(1) Disconnect and isolate the negative battery cable.

(2) Disconnect the wire harness connector from the blower motor (Fig. 4).

(3) Release the locking tab that secures the blower motor to the HVAC housing and rotate blower motor counterclockwise.

(4) Rotate and tilt the blower motor as needed for clearance to remove the blower motor and wheel from the HVAC housing.

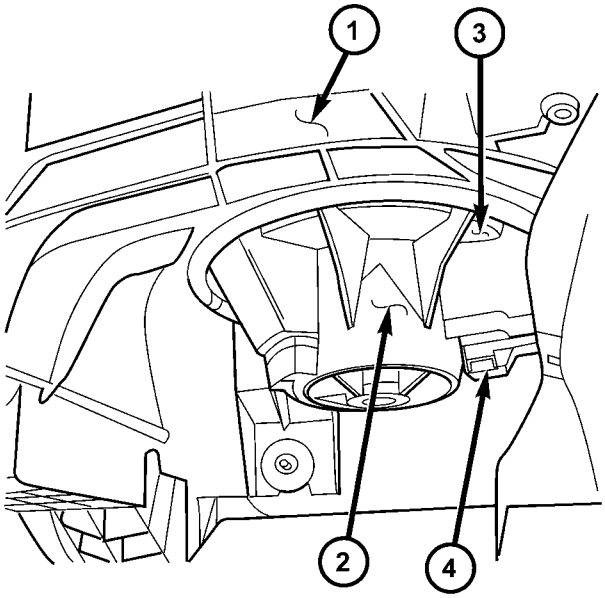

Fig. 4 Blower Motor - LHD shown, RHD similar

1 - HVAC HOUSING
2 - BLOWER MOTOR
3 - RETAINER-LOCKING TAB
4 - WIRE HARNESS CONNECTOR

INSTALLATION

NOTE: Failure to install the blower motor assembly correctly could result in an air leak or the blower motor assembly becoming completely disengaged from the HVAC housing.

(1) Align and install the blower motor into the HVAC housing.

(2) Rotate the blower motor until all of the locking tabs have secured the blower motor assembly to the HVAC housing.

(3) Connect the wire harness connector to the blower motor.

(4) Reconnect the battery negative cable.

(5) Test the blower motor for proper installation by turning the blower motor speed to its fastest position and checking around the outer edges of the blower assembly for air leaks. If any air leaks are found, remove and reinstall the blower motor.

DEFROSTER DUCT

REMOVAL

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the battery negative (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. This is the only sure way to disable the airbag system. Failure to take the proper precautions could result in accidental airbag deployment and possible personal injury or death.

(1) Remove the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/INSTRUMENT PANEL ASSEMBLY - REMOVAL).

(2) Remove the two nuts that secure the defroster duct to the dash panel (Fig. 5).

(3) Remove the defroster duct from the dash panel and disconnect it from the HVAC housing.

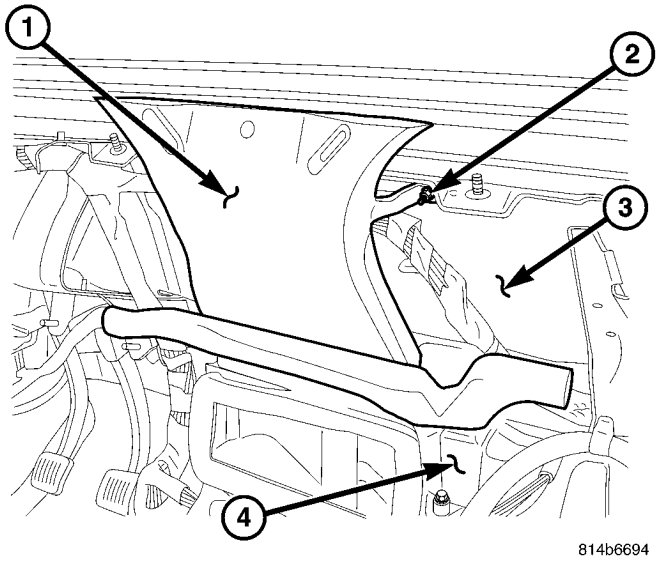

Fig. 5 Defroster Duct

1 - DEFROSTER DUCT
2 - NUT (2)
3 - DASH PANEL
4 - HVAC HOUSING

INSTALLATION

(1) Install the defroster duct onto the top of the HVAC housing and onto the studs located on the dash panel.

(2) Install the two push-nuts that secure the defroster duct to the dash panel.

(3) Install the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/INSTRUMENT PANEL ASSEMBLY - INSTALLATION).

FLOOR DISTRIBUTION DUCT

REMOVAL

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the battery negative (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. This is the only sure way to disable the airbag system. Failure to take the proper precautions could result in accidental airbag deployment and possible personal injury or death.

(1) Remove the airbag control module (Refer to 8 - ELECTRICAL/RESTRAINTS/AIRBAG CONTROL MODULE - REMOVAL).

(2) Remove the three screws that secure the floor distribution duct to the bottom of the HVAC housing and remove the duct (Fig. 6).

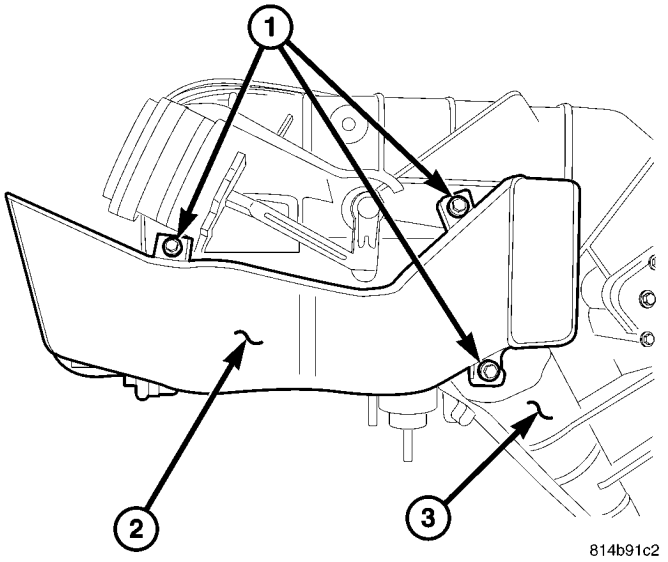

Fig. 6 Floor Distribution Duct

1 - SCREW (3)
2 - FLOOR DISTRIBUTION DUCT
3 - HVAC HOUSING

INSTALLATION

(1) Position the floor distribution duct onto the bottom of the HVAC housing.

FLOOR DISTRIBUTION DUCT (Continued)

(2) Install the three screws that secure the floor distribution duct to HVAC housing. Tighten the screws to 2 N·m (17 in. lbs.).

(3) Install the airbag control module (Refer to 8 - ELECTRICAL/RESTRAINTS/AIRBAG CONTROL MODULE - INSTALLATION).

HVAC HOUSING

DESCRIPTION

Both the single zone manual temperature control (MTC) heating-A/C system and the heater-only system are equipped with a common HVAC housing assembly that combines A/C and heating capabilities into a single unit located within the passenger compartment (Fig. 7). The HVAC housing assembly consists of the following:

- **HVAC housing** — The HVAC housing consists of an upper housing and a lower housing that are attached together by screws and clips and is mounted to the dash panel behind the instrument panel. The HVAC housing contains the heater core , A/C evaporator , blend-air and mode-air doors, and has mounting provisions for the air-door actuators, floor distribution duct, blower motor, blower motor resistor, blower motor relay and the HVAC wire harness and vacuum harness.
- **Air inlet housing** — The air inlet housing is mounted to the right end of the HVAC housing and contains the recirculation-air door and has mounting provisions for the recirculation door actuator and its linkage.

The heating-A/C system is a blend-air type system. The blend-air door controls the amount of conditioned air that is allowed to flow through, or around, the heater core.

The A/C system is designed for the use of a non-CFC, R-134a refrigerant and uses an A/C evaporator to cool and dehumidify the incoming air prior to blending it with the heated air. A temperature control determines the discharge air temperature by operating the blend door actuator, which moves the blend-air door. This allows an almost immediate control of the output air temperature of the system. The two mode door actuators operate the mode-air doors which direct the flow of the conditioned air out the various air outlets, depending on the mode selected. The recirculation door actuator operates the recirculation-air door which closes off the fresh air intake and recirculates the air already inside the vehicle. The electric recirculation door actuator is connected to the vehicle electrical system by the HVAC wire harness and, the recirculation door actuator and mode door actuators are connected to the A/C-heater control by the HVAC vacuum harness. The blower motor controls the velocity of air flowing through the HVAC housing assembly by spinning the blower wheel within the HVAC housing at the selected speed by use of the blower motor resistor.

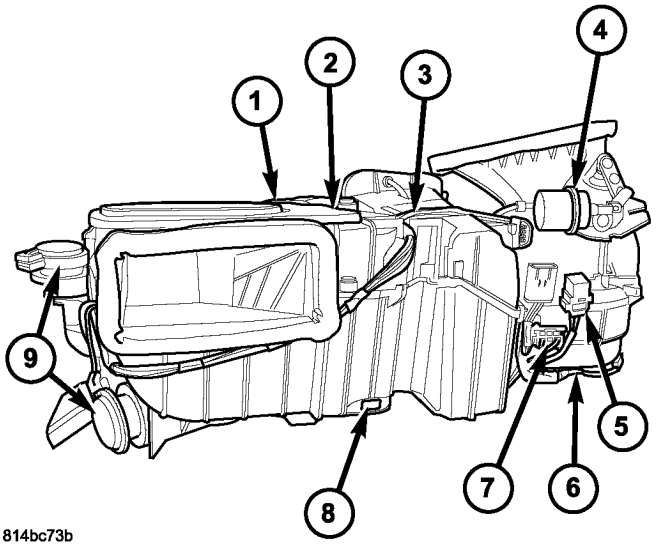

Fig. 7 HVAC Housing - LHD shown, RHD similar

1 - HVAC HOUSING
2 - HEATER CORE
3 - A/C EVAPORATOR
4 - RECIRCULATION DOOR ACTUATOR
5 - BLOWER MOTOR RELAY
6 - BLOWER MOTOR
7 - BLOWER MOTOR RESISTOR
8 - BLEND DOOR ACTUATOR
9 - MODE DOOR ACTUATORS

The HVAC housing must be removed from the vehicle and disassembled for service of the heater core, A/C evaporator, blend-air and mode-air doors. The air inlet housing must be removed from HVAC housing and disassembled for service of the recirculation-air door.

REMOVAL

HVAC HOUSING

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the battery negative (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. This is the only sure way to disable the airbag system. Failure to take the proper precautions could result in an accidental airbag deployment and possible personal injury or death.

HVAC HOUSING (Continued)

NOTE: The HVAC housing must be removed from the vehicle and the two halves of the housing separated for service of the heater core, A/C evaporator, blend door and each of the various mode doors.

(1) Disconnect and isolate the negative battery cable.
(2) Remove the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/INSTRUMENT PANEL ASSEMBLY - REMOVAL).
(3) If the vehicle is not equipped with A/C, go to Step 6. If the vehicle is equipped with A/C, recover the refrigerant from the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM RECOVERY).
(4) Disconnect the A/C liquid and suction lines from the A/C evaporator and remove and discard the O-ring seals (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/REFRIGERANT LINE COUPLER - REMOVAL).
(5) Install plugs in, or tape over the opened suction and liquid line fittings and both evaporator tubes.
(6) Drain the engine cooling system (Refer to 7 - COOLING - STANDARD PROCEDURE).
(7) Disconnect the heater hoses from the heater core tubes. Install plugs in, or tape over the opened heater core tubes.
(8) Disconnect the HVAC system vacuum supply line connector from the tee fitting near the heater core tubes.
(9) Remove the five nuts that secure the HVAC housing to the dash panel in the engine compartment (Fig. 8). If necessary, loosen the battery hold-downs and reposition the battery for additional access (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/BATTERY - REMOVAL).
(10) Remove the two push-nuts that secure the defroster duct to the dash panel in the passenger compartment and remove the duct.
(11) Remove the bolt that secures the HVAC housing to the bracket located on the dash panel in the passenger compartment.
(12) Remove the HVAC housing from inside the vehicle taking care not to allow any remaining coolant to drain on the vehicles interior.

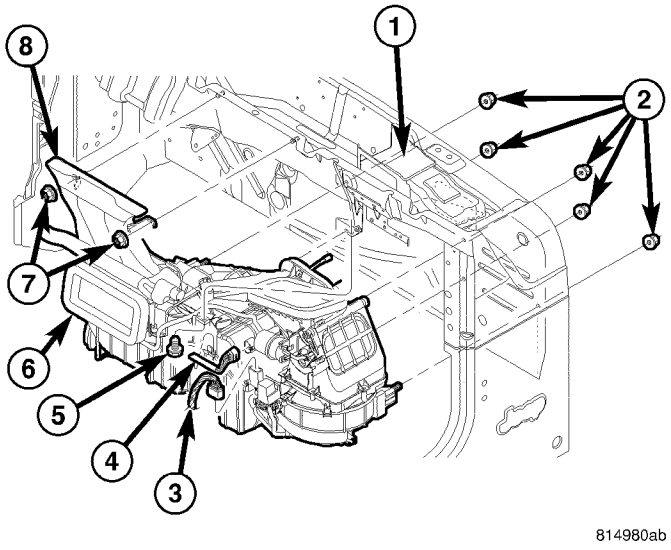

Fig. 8 HVAC Housing - LHD shown, RHD similar

1 - DASH PANEL
2 - NUT (5)
3 - INSTRUMENT PANEL WIRE HARNESS
4 - INSTRUMENT PANEL VACUUM HARHESS
5 - BOLT
6 - HVAC HOUSING
7 - NUT (2)
8 - DEFROSTER DUCT

AIR INLET HOUSING

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the negative battery (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. This is the only sure way to disable the airbag system. Failure to take the proper precautions could result in accidental airbag deployment and possible personal injury or death.

NOTE: The recirculation-air door and actuator are used only on models equipped with A/C.

NOTE: The air inlet housing must be removed from HVAC housing and disassembled for service of the recirculation-air door.

(1) Remove the HVAC housing (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/HVAC HOUSING - REMOVAL).
(2) If equipped, disconnect the HVAC vacuum harness connector from the recirculation door actuator.

HVAC HOUSING (Continued)

(3) Remove the five screws that secure the air inlet housing to the top of the HVAC housing and remove the air inlet housing (Fig. 9).

NOTE: If the foam seal on the air inlet housing is deformed or damaged, it must be replaced.

(4) If required, disassemble the air inlet housing (Refer to 24 - HEATING & AIR CONDITIONING/ DISTRIBUTION/HVAC HOUSING - AIR INLET HOUSING - DISASSEMBLY).

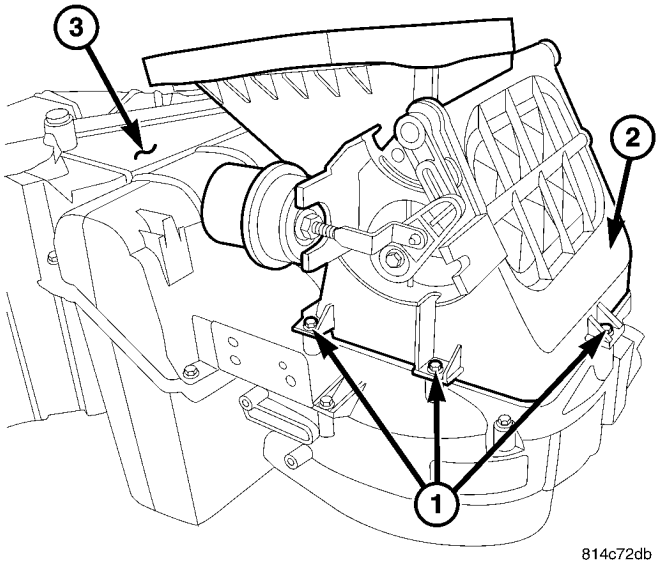

Fig. 9 HVAC Air Inlet Housing - LHD shown, RHD similar

1 - SCREWS (5)
2 - AIR INLET HOUSING
3 - HVAC HOUSING

DISASSEMBLY

HVAC HOUSING

NOTE: The HVAC housing must be removed from the vehicle and disassembled for service of the A/C evaporator, heater core, blend-air and mode-air doors.

(1) Remove the HVAC housing assembly and place it on a workbench (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/HVAC HOUSING - REMOVAL).

(2) Disconnect the HVAC vacuum harness connectors from the mode door actuators and the recirculation door actuator, if equipped (Fig. 10).

(3) Disengage the vacuum harness from any retaining clips located on the HVAC housing and carefully pull the vacuum supply line through the foam seal located at the front of the housing and remove the vacuum harness.

(4) Remove the mode door actuators from the HVAC housing (Refer to 24 - HEATING & AIR CONDITIONING/CONTROLS/MODE DOOR ACTUATORS - REMOVAL).

(5) Remove the blend door actuator from the HVAC housing (Refer to 24 - HEATING & AIR CONDITIONING/CONTROLS/BLEND DOOR ACTUATOR - REMOVAL).

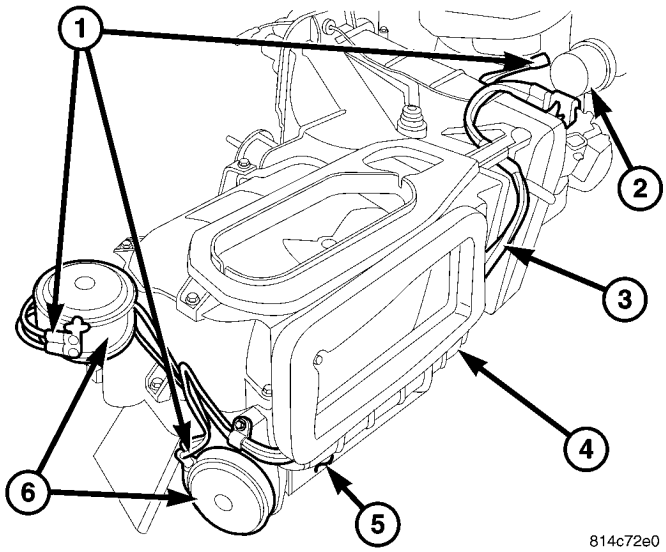

Fig. 10 Vacuum Harness and Actuators - LHD shown, RHD similar

1 - VACUUM HARNESS CONNECTORS
2 - RECIRCULATION DOOR ACTUATOR
3 - HVAC VACUUM HARNESS
4 - BLEND DOOR ACTUATOR
5 - HVAC HOUSING
6 - MODE DOOR ACTUATORS

HVAC HOUSING (Continued)

(6) Disconnect the wire harness connectors from the blower motor resistor and the blower motor (Fig. 11).

(7) Disengage the push-in retainers for the HVAC wire harness connector and the blower motor relay from the HVAC housing.

(8) Remove the three screws that secure the HVAC wire harness to the HVAC housing and remove the wire harness.

(9) Remove the two screws that secure the blower motor resistor to the HVAC housing and remove the resistor.

(10) Disengage the blower motor retaining tab and turn the blower motor counter-clockwise and remove the blower motor from the HVAC housing.

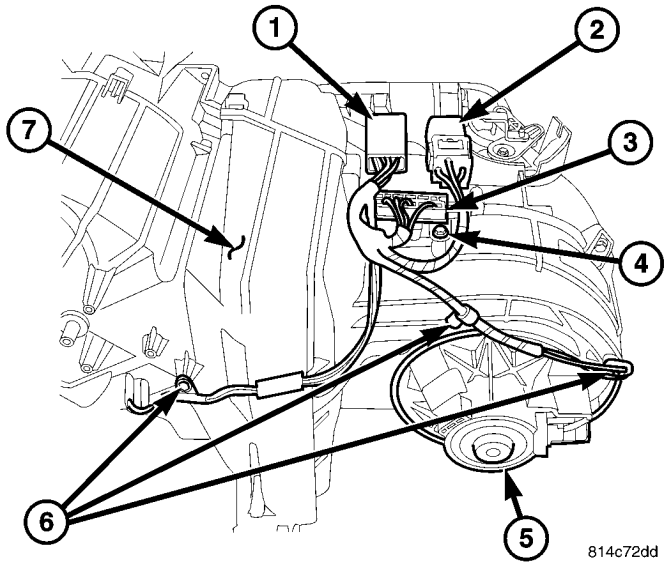

Fig. 11 HVAC Wire Harness - LHD shown, RHD similar

1 - HVAC WIRE HARNESS
2 - BLOWER MOTOR RELAY
3 - BLOWER MOTOR RESISTOR
4 - SCREW (2)
5 - BLOWER MOTOR
6 - SCREW (3)
7 - HVAC HOUSING

(11) Remove the five screws that secure the air inlet housing to the top of the HVAC housing and remove the inlet housing (Fig. 12).

(12) Remove the three screws that secure the floor distribution duct to the bottom of the HVAC housing and remove the duct (Fig. 13).

(13) Remove the one screw that secures the upper and lower HVAC housing halves together at the bottom of the HVAC housing.

NOTE: If the foam seal for the defrost/demister outlet cover or the panel outlet is deformed or damaged, the seal must be replaced.

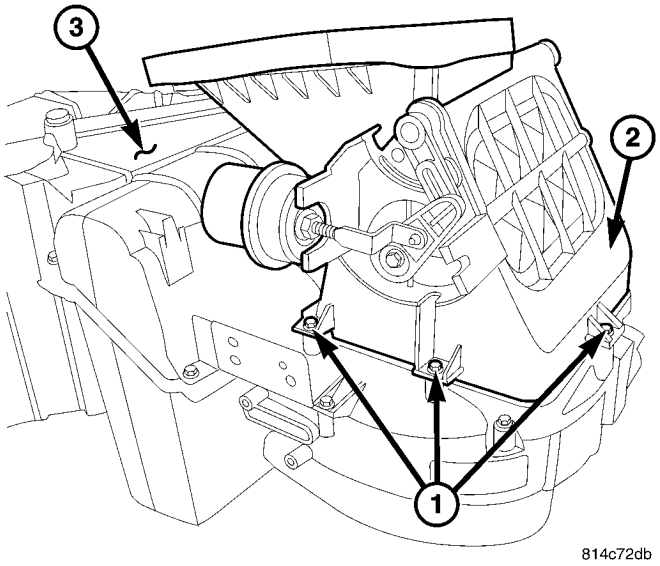

Fig. 12 HVAC Air Inlet Housing - LHD shown, RHD similar

1 - SCREWS (5)
2 - AIR INLET HOUSING
3 - HVAC HOUSING

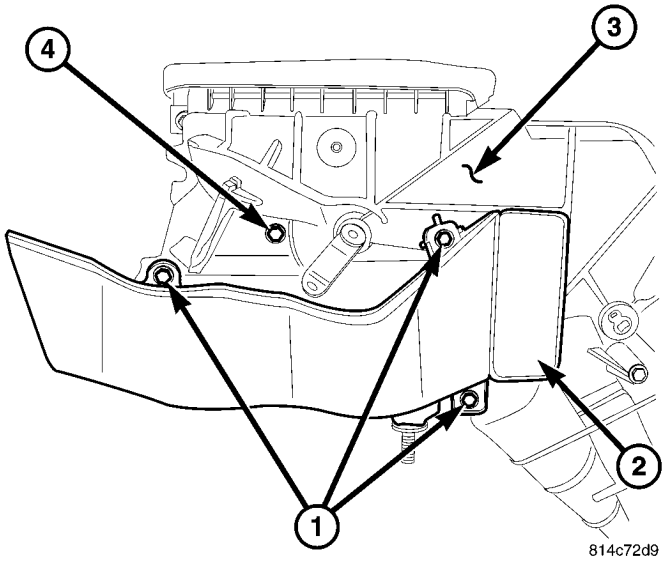

Fig. 13 Floor Distribution Duct and Screw - LHD shown, RHD similar

1 - SCREW (3)
2 - FLOOR DISTRIBUTION DUCT
3 - HVAC HOUSING (LOWER HALF)
4 - SCREW

(14) Carefully cut the foam seal for the defrost/demister outlet cover along the two parting lines of the outlet cover (Fig. 14).

(15) Carefully cut the foam seal for the panel outlet along the parting line of the two HVAC housing halves.

HVAC HOUSING (Continued)

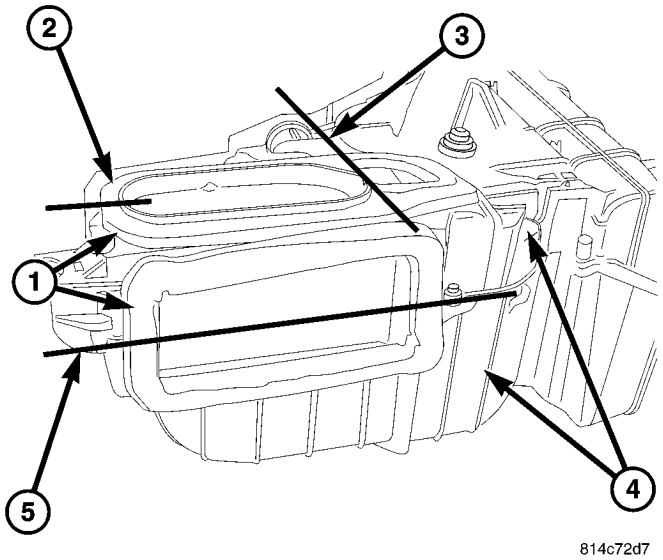

Fig. 14 Foam Seals and Parting Lines - Typical

1 - FOAM SEAL (2)
2 - PARTING LINE
3 - PARTING LINE
4 - HVAC HOUSING HALVES
5 - PARTING LINE

NOTE: If the foam seal for the defrost/demister outlet cover is deformed or damaged, the seal must be replaced.

(16) Remove the five screws that secure the defrost/demister outlet cover to the top of the HVAC housing and remove the cover (Fig. 15).

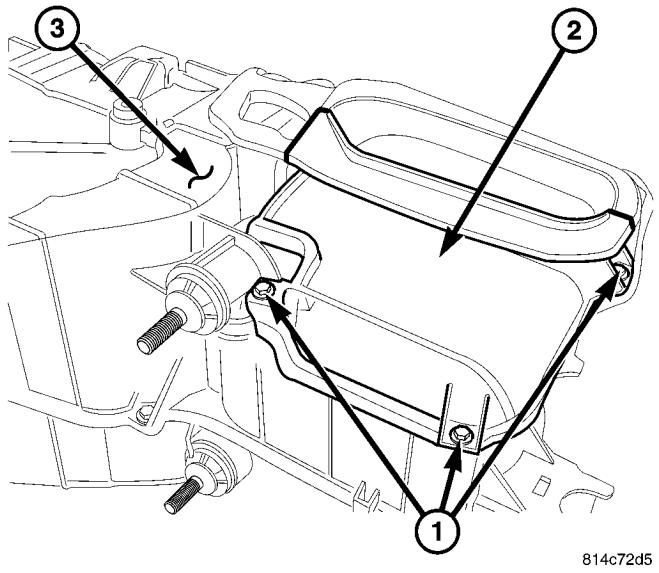

Fig. 15 Defrost/Demister Outlet Cover - LHD shown, RHD similar

1 - SCREW (5)
2 - DEFROST/DEMISTER OUTLET COVER
3 - HVAC HOUSING

NOTE: If the foam seal for the evaporator and heater core tubes is deformed or damaged, the seal must be replaced.

(17) Carefully cut the foam seal for the evaporator and heater core tubes along the parting line of the two HVAC housing halves (Fig. 16).

NOTE: If the foam seal for the evaporator condensate drain tube is deformed or damaged, the seal must be replaced.

(18) Remove the foam seal from the evaporator condensate drain tube.

(19) Remove the fourteen screws and two clips that secure the upper and lower HVAC housing halves together.

(20) Carefully separate the two HVAC housing halves.

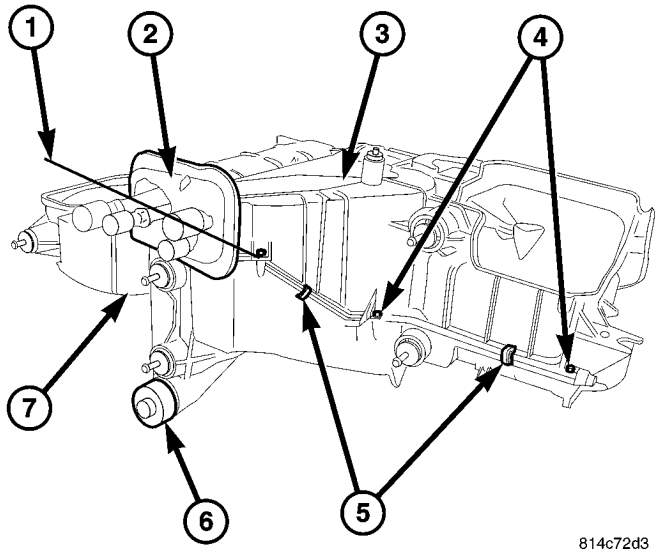

Fig. 16 Foam Seal and HVAC Housing Screws - Typical

1 - PARTING LINE
2 - FOAM SEAL
3 - HVAC HOUSING (UPPER HALF)
4 - SCREWS (14)
5 - CLIP (2)
6 - FOAM SEAL
7 - HVAC HOUSING (LOWER HALF)

HVAC HOUSING (Continued)

NOTE: If either foam seal on the panel-air door is deformed or damaged, the air door must be replaced.

(21) Using a pick or similar tool, release the retaining tab that secures panel door lever to the panel-air door and remove the lever (Fig. 17).

(22) Remove the panel-air door from the lower half of the HVAC housing.

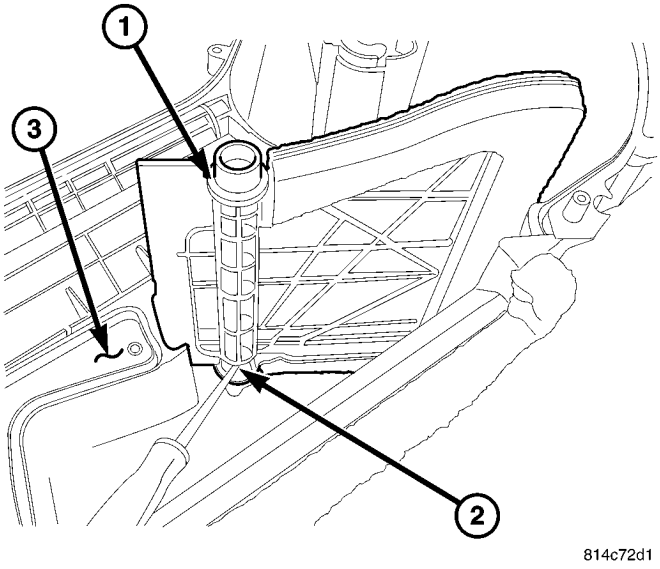

Fig. 17 Panel-Air Door - LHD shown, RHD similar

1 - PANEL-AIR DOOR
2 - PANEL DOOR LEVER RETAINING TAB
3 - HVAC HOUSING (LOWER HALF)

NOTE: If either foam seal on the air doors is deformed or damaged, the air door must be replaced.

(23) Remove the floor/defrost-air door and the blend-air door from the lower half of the HVAC housing (Fig. 18).

NOTE: If the foam insulator around the heater core is deformed or damaged, the insulator must be replaced.

(24) Carefully lift the heater core and foam insulator out of the lower half of the HVAC housing (Fig. 19).

NOTE: If the foam insulator around the A/C evaporator is deformed or damaged, the insulator must be replaced.

(25) If equipped with A/C, disengage the rubber seal from around the tubes of the A/C evaporator and carefully lift the A/C evaporator and foam insulator out of the lower half of the HVAC housing.

NOTE: If the rubber seal for the A/C evaporator tubes is deformed or damaged, the seal must be replaced.

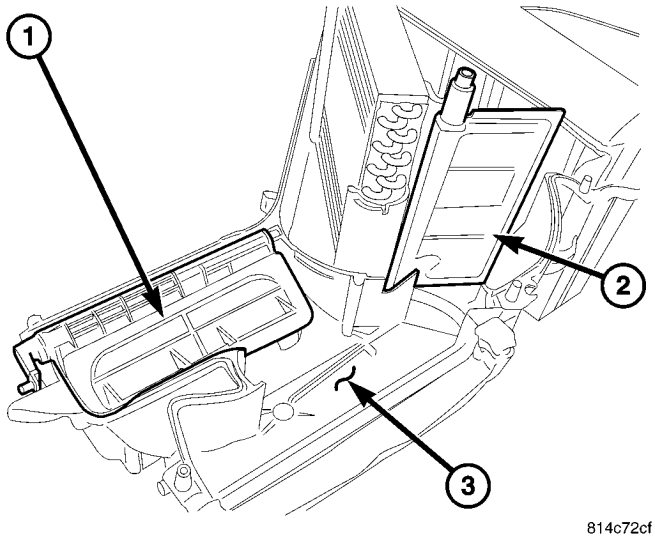

Fig. 18 Floor/Defrost and Blend-Air Doors - LHD shown, RHD similar

1 - FLOOR/DEFROST-AIR DOOR
2 - BLEND-AIR DOOR
3 - HVAC HOUSING (LOWER HALF)

(26) Remove the rubber seal for the evaporator tubes from the lower half of the HVAC housing, if equipped.

(27) If required, remove the screw that secures the evaporator tube clamp to the evaporator tubes and remove the clamp.

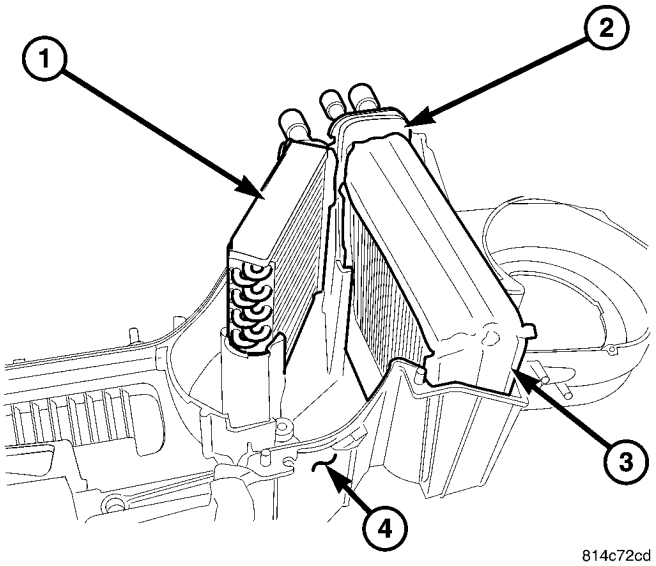

Fig. 19 Heater Core and A/C Evaporator - Typical

1 - HEATER CORE
2 - RUBBER SEAL
3 - A/C EVAPORATOR (IF EQUIPPED)
4 - HVAC HOUSING (LOWER HALF)

HVAC HOUSING (Continued)

AIR INLET HOUSING

NOTE: The recirculation-air door and actuator are used only on models equipped with A/C.

NOTE: The air inlet housing must be removed from HVAC housing and disassembled for service of the recirculation-air door.

(1) Remove the air inlet housing from the HVAC housing and place it on a workbench (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/ HVAC HOUSING - AIR INLET HOUSING - REMOVAL).

(2) Using a small flat bladed tool, gently pry the recirculation door actuator latch while pulling firmly outwards on the actuator and remove the actuator from the HVAC air inlet housing mount (Fig. 20).

(3) Disengage the actuator rod from the push-pin type retainer located on the end of the recirculation door actuator lever.

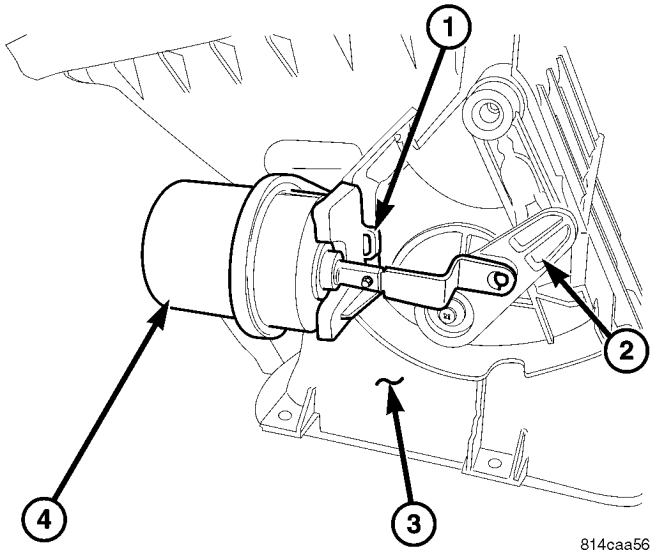

Fig. 20 Recirculation Door Actuator

1 - ACTUATOR LATCH
2 - RECIRCULATION DOOR ACTUATOR LEVER
3 - HVAC AIR INLET HOUSING
4 - RECIRCULATION DOOR ACTUATOR

(4) Remove the screw that secures the recirculation door actuator lever to the air inlet housing and remove the lever (Fig. 21).

(5) Using a pick or similar tool, release the retaining tab of the recirculation door lever located on the inside surface of the air inlet housing and pull the lever out of the pivot shaft of the recirculation-air door (Fig. 22).

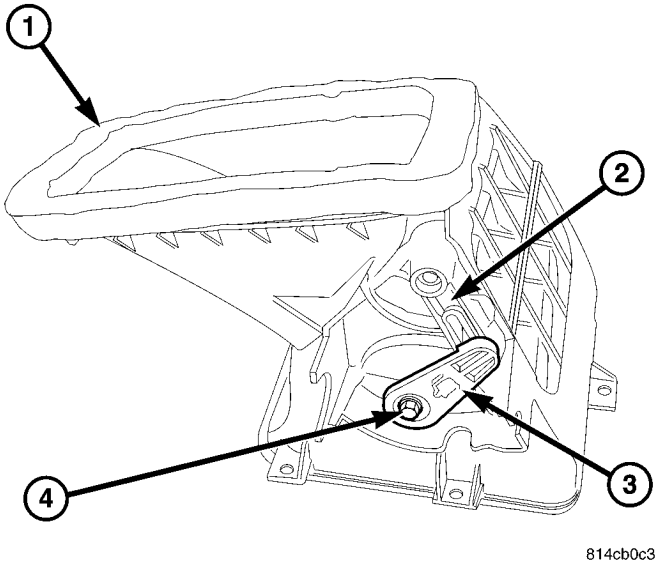

Fig. 21 Recirculation Door Actuator Lever

1 - HVAC AIR INLET HOUSING
2 - RECIRCULATION DOOR LEVER
3 - RECIRCUALTION DOOR ACTUATOR LEVER
4 - SCREW

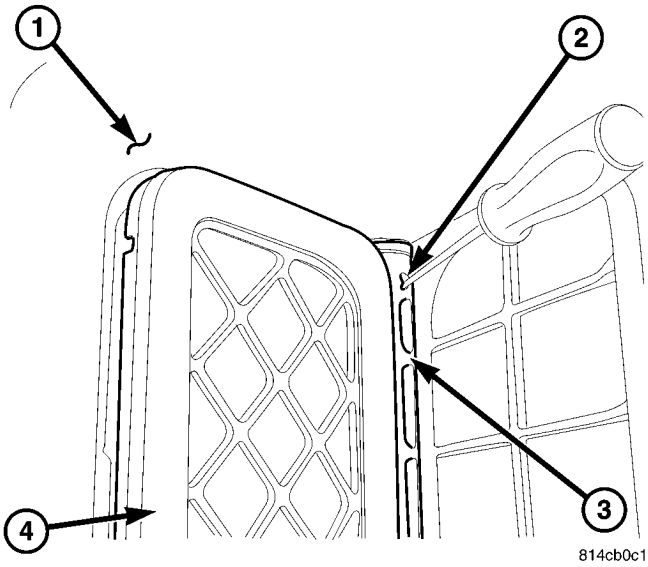

Fig. 22 Recirculation Door Lever

1 - HVAC AIR INLET HOUSING (INSIDE VIEW)
2 - RECIRCULATION DOOR LEVER RETAINING TAB
3 - RECIRCULATION DOOR PIVOT SHAFT
4 - RECIRCULATION-AIR DOOR

HVAC HOUSING (Continued)

(6) Using a pick or similar tool, release the retaining tab of the recirculation door retainer located on the inside surface of the air inlet housing and pull the retainer out of the pivot shaft of the recirculation-air door (Fig. 23).

NOTE: If a foam seal on the recirculation-air door is deformed or damaged, the air door must be replaced.

(7) Remove the recirculation-air door from the air inlet housing.

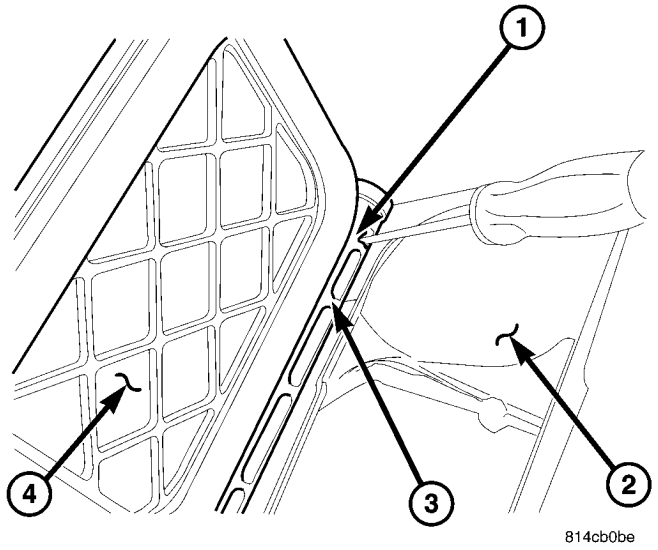

Fig. 23 Recirculation Door Retainer

1 - RETAINING TAB
2 - HVAC AIR INLET HOUSING (INSIDE VIEW)
3 - RECIRCULATION DOOR PIVOT SHAFT
4 - RECIRCULATION-AIR DOOR

ASSEMBLY

HVAC HOUSING

NOTE: The HVAC housing must be removed from the vehicle and disassembled for service of the A/C evaporator, heater core, blend-air and mode-air doors.

NOTE: If any foam or rubber seal on the HVAC housing or an air-door is deformed or damaged, it must be replaced.

(1) Carefully install the A/C evaporator, foam insulator and the rubber seal into the lower half of the HVAC housing. If removed, install the evaporator tube clamp (Fig. 24).

(2) Carefully install the heater core and foam insulator into the lower half of the HVAC housing.

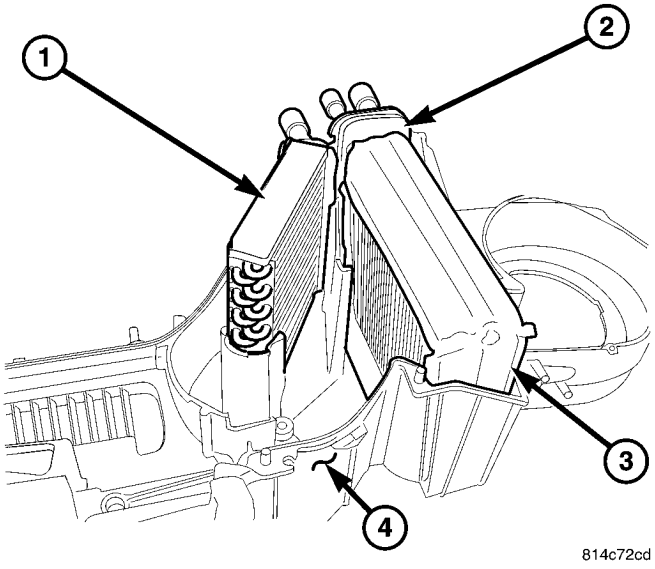

Fig. 24 Heater Core and A/C Evaporator - Typical

1 - HEATER CORE
2 - RUBBER SEAL
3 - A/C EVAPORATOR (IF EQUIPPED)
4 - HVAC HOUSING (LOWER HALF)

(3) Install the floor/defroster-air door and the blend-air door into their pivot holes in the lower half of the HVAC housing (Fig. 25).

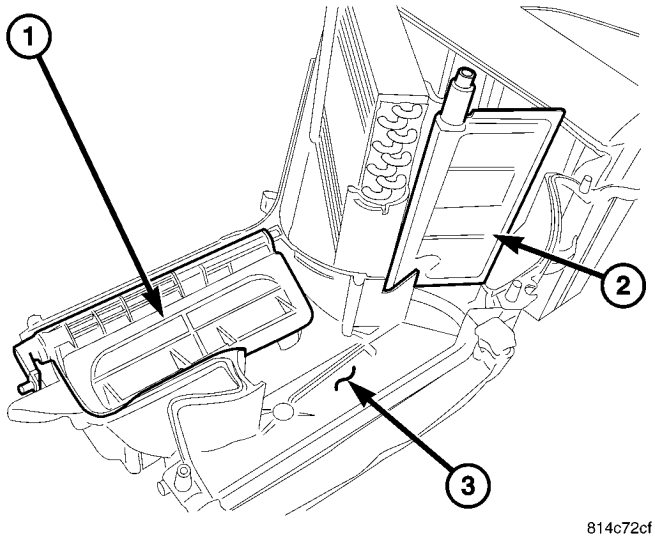

Fig. 25 Floor/Defrost and Blend-Air Doors - LHD shown, RHD similar

1 - FLOOR/DEFROST-AIR DOOR
2 - BLEND-AIR DOOR
3 - HVAC HOUSING (LOWER HALF)

HVAC HOUSING (Continued)

(4) Position the panel-air door into the pivot hole located in the lower half of the HVAC housing (Fig. 26).

(5) Align the panel door lever with the panel-air door pivot shaft and install the lever into the shaft. Be sure that the lever retaining tab fully engages the pivot shaft.

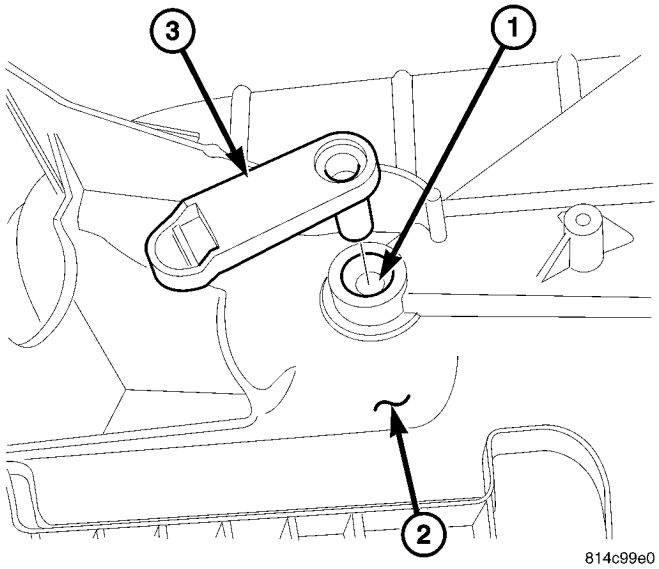

Fig. 26 Panel Door Lever - Typical

1 - PANEL-AIR DOOR PIVOT SHAFT
2 - HVAC HOUSING (LOWER HALF)
3 - PANEL DOOR LEVER

(6) Position the two halves of the HVAC housing together. Be certain that each of the air-door pivot shafts are properly engaged in its pivot hole and, if equipped with A/C, that the evaporator tube rubber seal is properly positioned (Fig. 27).

(7) Install the fourteen screws and two clips that secure the two housing halves together. Tighten the screws to 2 N·m (17 in. lbs.).

(8) Install the foam seal onto the evaporator condensate drain tube.

(9) Position the defrost/demister outlet cover onto the top of the HVAC housing (Fig. 28).

(10) Install the five screws that secure the defrost/demister outlet cover to the HVAC housing. Tighten the screws to 2 N·m (17 in. lbs.).

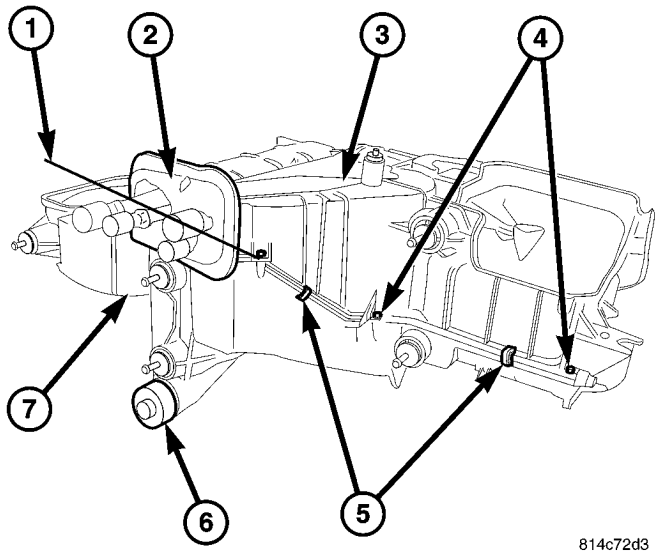

Fig. 27 Foam Seal and HVAC Housing Screws - Typical

1 - PARTING LINE
2 - FOAM SEAL
3 - HVAC HOUSING (UPPER HALF)
4 - SCREWS (14)
5 - CLIP (2)
6 - FOAM SEAL
7 - HVAC HOUSING (LOWER HALF)

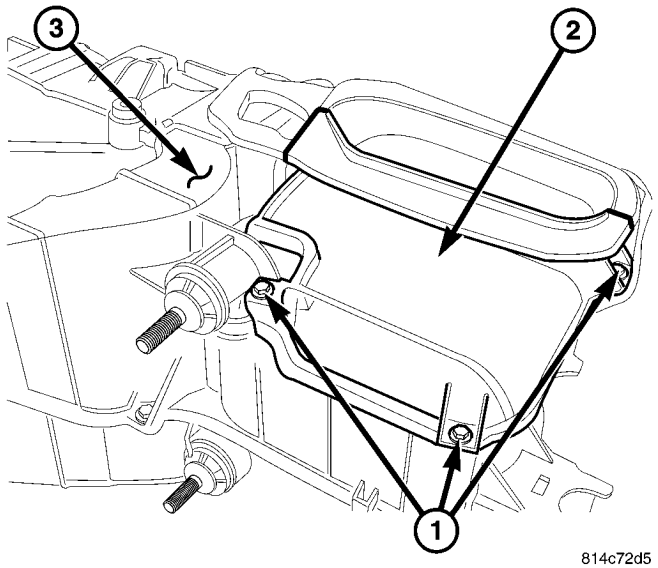

Fig. 28 Defrost/Demister Outlet Cover - LHD shown, RHD similar

1 - SCREW (5)
2 - DEFROST/DEMISTER OUTLET COVER
3 - HVAC HOUSING

HVAC HOUSING (Continued)

(11) Install the one screw that secures the two housing halves together at the bottom of the HVAC housing. Tighten the screw to 2 N·m (17 in. lbs.) (Fig. 29).

(12) Position the floor distribution duct onto the bottom of the HVAC housing.

(13) Install the three screws that secure the floor distribution duct to the HVAC housing. Tighten the screw to 2 N·m (17 in. lbs.).

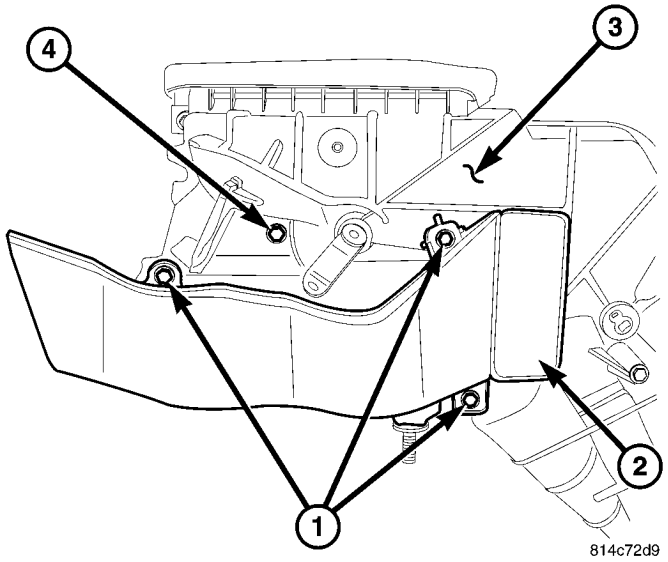

Fig. 29 Floor Distribution Duct and Screw - LHD shown, RHD similar

1 - SCREW (3)
2 - FLOOR DISTRIBUTION DUCT
3 - HVAC HOUSING (LOWER HALF)
4 - SCREW

(14) Position the air inlet housing onto the top of the HVAC housing (Fig. 30).

(15) Install the five screws that secure the air inlet housing to the HVAC housing. Tighten the screw to 2 N·m (17 in. lbs.).

(16) Install the HVAC wire harness onto the HVAC housing. Make sure the wires are routed through all wiring retainers (Fig. 31).

(17) Engage the push-in retainers for the HVAC wire harness connector and the blower motor relay to the HVAC housing.

(18) Install the three screws that secure the HVAC wire harness to the HVAC housing. Tighten the screws to 2 N·m (17 in. lbs.).

(19) Position the blower motor into the HVAC housing and turn it clockwise until the retaining tab is fully engaged.

(20) Position the blower motor resistor into the HVAC housing and install the two retaining screws. Tighten the screws to 2 N·m (17 in. lbs.).

(21) Connect the HVAC wire harness connectors to the blower motor and the blower motor resistor.

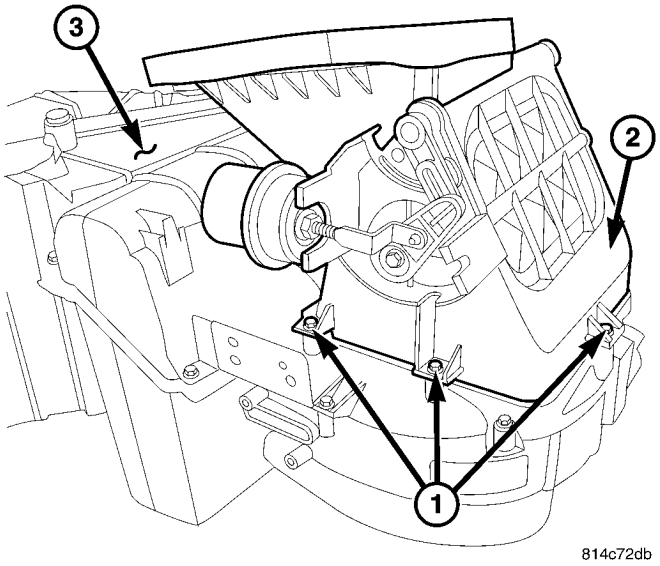

Fig. 30 HVAC Air Inlet Housing - LHD shown, RHD similar

1 - SCREWS (5)
2 - AIR INLET HOUSING
3 - HVAC HOUSING

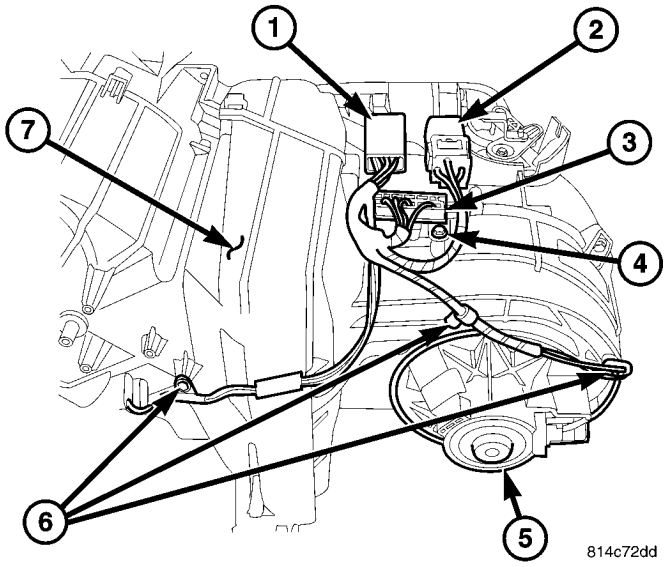

Fig. 31 HVAC Wire Harness - LHD shown, RHD similar

1 - HVAC WIRE HARNESS
2 - BLOWER MOTOR RELAY
3 - BLOWER MOTOR RESISTOR
4 - SCREW (2)
5 - BLOWER MOTOR
6 - SCREW (3)
7 - HVAC HOUSING

(22) Install the mode door actuators (Fig. 32) (Refer to 24 - HEATING & AIR CONDITIONING/ CONTROLS/MODE DOOR ACTUATORS - INSTALLATION).

HVAC HOUSING (Continued)

(23) Install the blend door actuator (Refer to 24 - HEATING & AIR CONDITIONING/CONTROLS/ BLEND DOOR ACTUATOR - INSTALLATION).

(24) Install the HVAC vacuum harness onto the HVAC housing. Make sure the harness is routed through all the retaining clips located on the HVAC housing.

(25) Insert the vacuum supply line through the foam seal located at the front of the HVAC housing and connect the vacuum harness connectors to the mode door actuators and the recirculation door actuator, if equipped.

(26) Install the HVAC housing assembly (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/HVAC HOUSING - INSTALLATION).

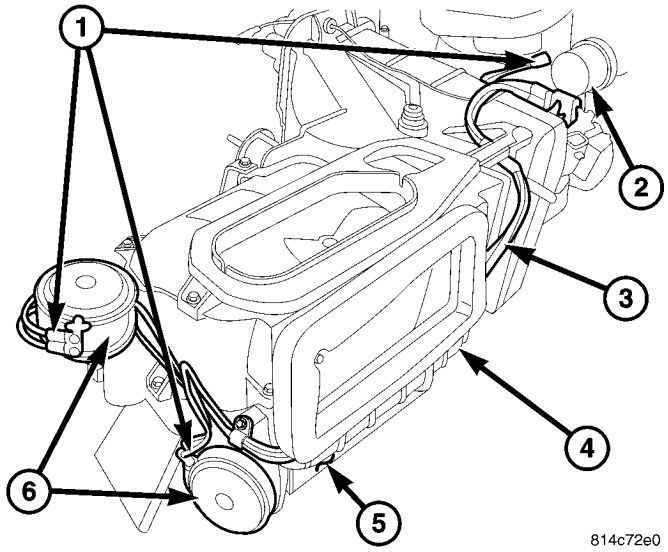

Fig. 32 Vacuum Harness and Actuators - LHD shown, RHD similar

1 - VACUUM HARNESS CONNECTORS
2 - RECIRCULATION DOOR ACTUATOR
3 - HVAC VACUUM HARNESS
4 - BLEND DOOR ACTUATOR
5 - HVAC HOUSING
6 - MODE DOOR ACTUATORS

AIR INLET HOUSING

NOTE: The recirculation-air door and actuator are used only on models equipped with A/C.

NOTE: The air inlet housing must be removed from HVAC housing and disassembled for service of the recirculation-air door.

NOTE: If a foam seal on the recirculation-air door is deformed or damaged, the air door must be replaced.

(1) Position the recirculation-air door into the air inlet housing.

CAUTION: Make sure that the recirculation-air door pivot shaft is properly seated in the pivot seats located in the air inlet housing.

(2) Align the recirculation door lever with the recirculation-air door pivot shaft and install the lever until the retaining tab is fully engaged to the pivot shaft (Fig. 33).

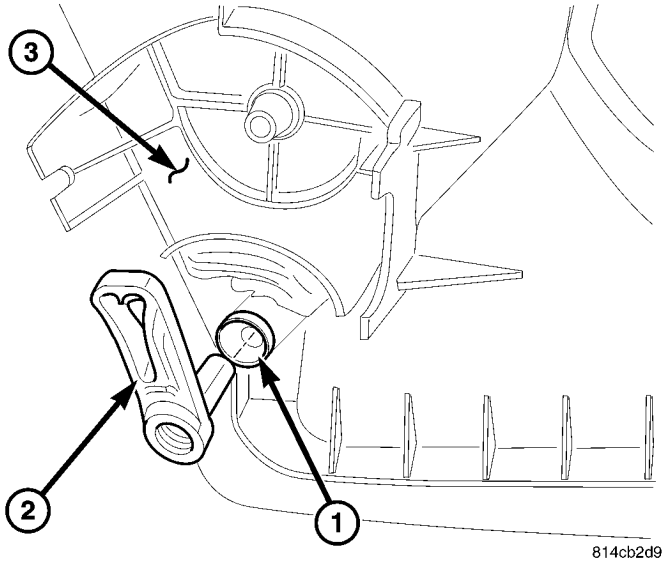

Fig. 33 Recirculation Door Lever Installation

1 - RECIRCULATION DOOR PIVOT SHAFT
2 - RECIRCULATION DOOR LEVER
3 - HVAC AIR INLET HOUSING

HVAC HOUSING (Continued)

(3) Align the recirculation door retainer with the recirculation-air door pivot shaft and install the retainer until the retaining tab is fully engaged to the pivot shaft (Fig. 34).

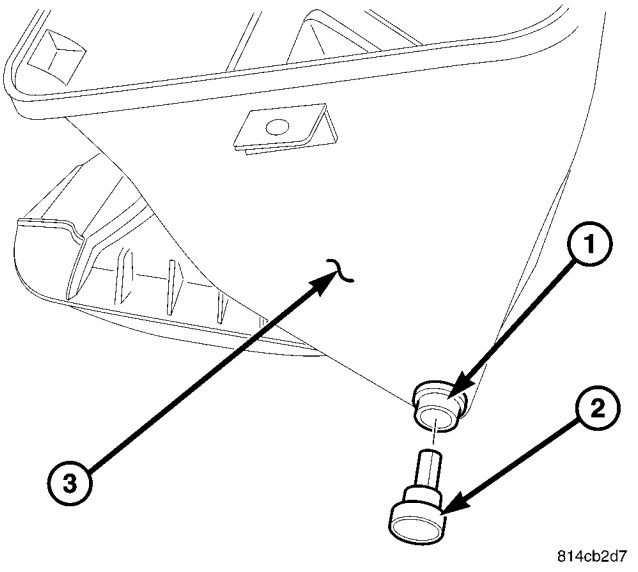

Fig. 34 Recirculation Door Retainer Installation

1 - RECIRCULATION DOOR PIVOT SHAFT
2 - RECIRCULATION DOOR RETAINER
3 - HVAC AIR INLET HOUSING

(4) Install the recirculation door actuator lever onto the pivot shaft on the air distribution housing (Fig. 35).

NOTE: Align the guide pin on the recirculation door actuator lever with the guides on the recirculation door lever during installation of the actuator lever.

(5) Connect the recirculation door actuator lever to the recirculation door lever.

(6) Install the screw that secures the recirculation door actuator lever to the air distribution housing. Tighten the screw to 2 N·m (17 in. lbs.).

(7) Install the recirculation door actuator mount onto the air inlet housing mounting boss (Fig. 36). Push the actuator onto the mount until it snaps into the locked position.

(8) Install the air inlet housing onto the HVAC housing (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/HVAC HOUSING - AIR INLET HOUSING - INSTALLATION).

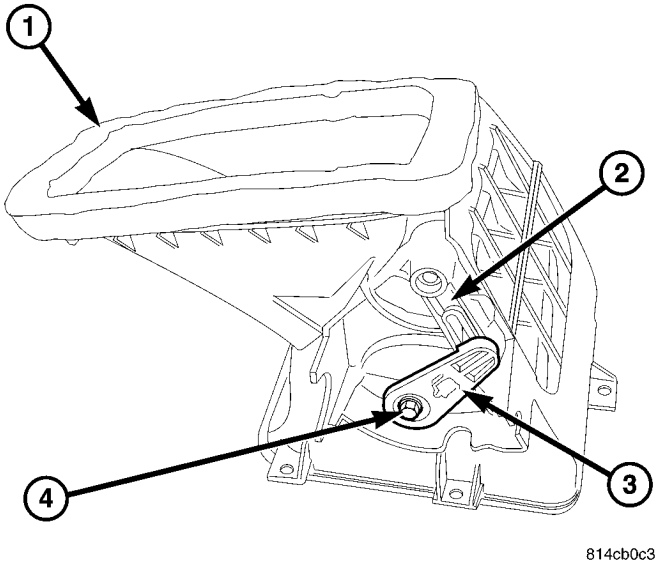

Fig. 35 Recirculation Door Actuator Lever

1 - HVAC AIR INLET HOUSING
2 - RECIRCULATION DOOR LEVER
3 - RECIRCUALTION DOOR ACTUATOR LEVER
4 - SCREW

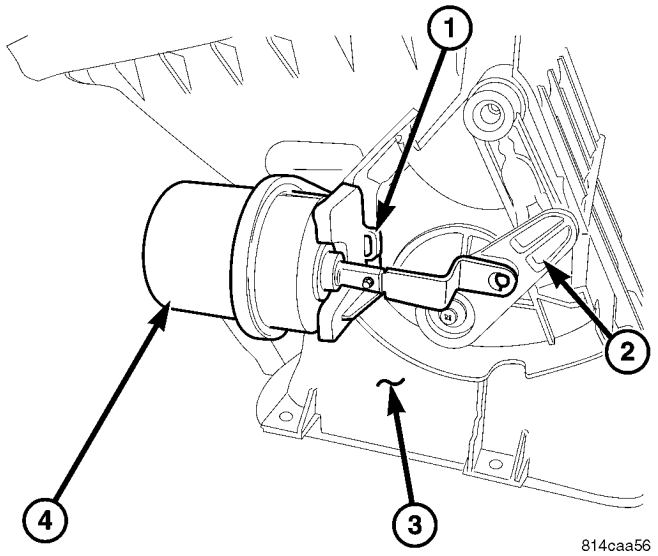

Fig. 36 Recirculation Door Actuator

1 - ACTUATOR LATCH
2 - RECIRCULATION DOOR ACTUATOR LEVER
3 - HVAC AIR INLET HOUSING
4 - RECIRCULATION DOOR ACTUATOR

INSTALLATION

HVAC HOUSING

NOTE: Use only the specified O-rings as they are made of special material for R-134a. Use only refrigerant oil of the type required for the A/C compressor in the vehicle.

(1) Position the HVAC housing to the dash panel. Be certain that the evaporator condensate drain tube and the housing mounting studs are inserted into their correct mounting holes.

(2) Install the bolt that secures the HVAC housing to the bracket located on the dash panel in the pas-

senger compartment. Tighten the bolt to 3 N·m (27 in. lbs.).

(3) Install the defroster duct onto the top of the HVAC housing.

(4) Install the two push-nuts that secure the defroster duct to the dash panel in the passenger compartment.

(5) Install the five nuts that secure the HVAC housing to the dash panel in the engine compartment. Tighten the nuts to 6 N·m (53 in. lbs.).

(6) If the battery was repositioned during the removal procedure, position the battery and tighten the hold-downs (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/BATTERY - INSTALLATION).

(7) Connect the HVAC system vacuum supply line connector to the tee fitting near the heater core tubes.

(8) Unplug or remove the tape from the heater core tubes and connect the heater hoses to the heater core tubes.

(9) If the vehicle is not equipped with A/C, go to Step 15. If the vehicle is equipped with A/C, unplug or remove the tape from the liquid and suction line fittings and both evaporator tubes.

(10) Lubricate new rubber O-ring seals with clean refrigerant oil and install them on the liquid and suction line fittings.

(11) Connect the A/C liquid and suction lines to the A/C evaporator (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/REFRIGERANT LINE COUPLER - INSTALLATION).

(12) Install the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/INSTRUMENT PANEL ASSEMBLY - INSTALLATION).

(13) Evacuate the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM EVACUATE).

(14) Charge the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM CHARGE).

(15) Reconnect the negative battery cable.

(16) If the heater core is being replaced, flush the cooling system (Refer to 7 - COOLING - STANDARD PROCEDURE - COOLING SYSTEM CLEANING/ REVERSE FLUSHING).

(17) Refill the engine cooling system (Refer to 7 - COOLING - STANDARD PROCEDURE - COOLING SYSTEM REFILL).

AIR INLET HOUSING

NOTE: The recirculation-air door and actuator are used only on models equipped with A/C.

NOTE: If the foam seal on the air inlet housing is deformed or damaged, it must be replaced.

(1) Position the air inlet housing onto the top of the HVAC housing.

(2) Install the five screws that secure the air inlet housing to the HVAC housing. Tighten the screws to 2 N·m (17 in. lbs.).

(3) If equipped, connect the HVAC vacuum harness connector to the recirculation door actuator.

(4) Install the HVAC housing (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/HVAC HOUSING - INSTALLATION).

INSTRUMENT PANEL DEMISTER DUCTS

REMOVAL

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the battery negative (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. This is the only sure way to disable the airbag system. Failure to take the proper precautions could result in accidental airbag deployment and possible personal injury or death.

NOTE: The two demister ducts can be serviced without removing the instrument panel. However, if either of the demister outlets require servicing, the instrument panel must be removed.

DEMISTER DUCTS

(1) Disconnect and isolate the negative battery cable.

(2) To gain access to the passenger side demister duct, remove the glove box from the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/ GLOVE BOX - REMOVAL).

(3) To gain access to the driver side demister duct remove the top cover, knee blocker and instrument cluster from the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/INSTRUMENT PANEL TOP COVER - REMOVAL), (Refer to 23 - BODY/INSTRUMENT PANEL/KNEE BLOCKER - REMOVAL) and (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - REMOVAL).

(4) Reach through the glove box opening and/or the cluster opening of the instrument panel and dis-

INSTRUMENT PANEL DEMISTER DUCTS (Continued)

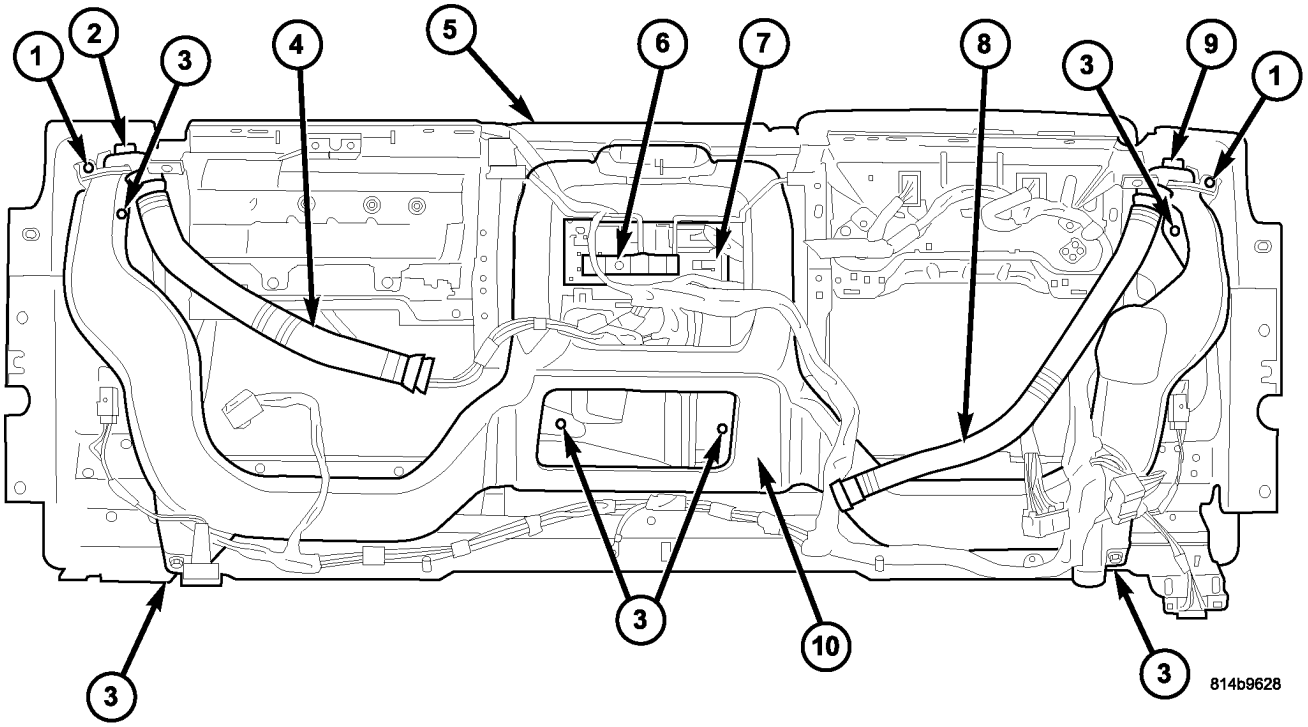

Fig. 37 Demister Ducts - IP Removed for Clarity

1 - SCREW (2)
2 - RH DEMISTER OUTLET
3 - SCREW (10)
4 - RH DEMISTER DUCT
5 - INSTRUMENT PANEL
6 - RADIO REAR SUPPORT BRACKET
7 - RADIO
8 - LH DEMISTER DUCT
9 - LH DEMISTER OUTLET
10 - INSTRUMENT PANEL DUCT

connect the demister ducts from the demister outlets and the defroster duct as necessary (Fig. 37).

DEMISTER OUTLETS

NOTE: Take the proper precautions to protect the front face of the instrument panel from cosmetic damage while performing this procedure.

(1) Remove the instrument panel assembly and place it on a workbench (Refer to 23 - BODY/INSTRUMENT PANEL/INSTRUMENT PANEL ASSEMBLY - REMOVAL)
(2) Disconnect the demister ducts from the demister outlets as necessary (Fig. 38).
(3) Remove the screw that secures each demister outlet to the instrument panel duct as necessary.
(4) Remove the demister outlets from the instrument panel as necessary.

INSTALLATION

NOTE: The two demister ducts can be serviced without removing the instrument panel. However, if either of the demister outlets require servicing, the instrument panel must be removed.

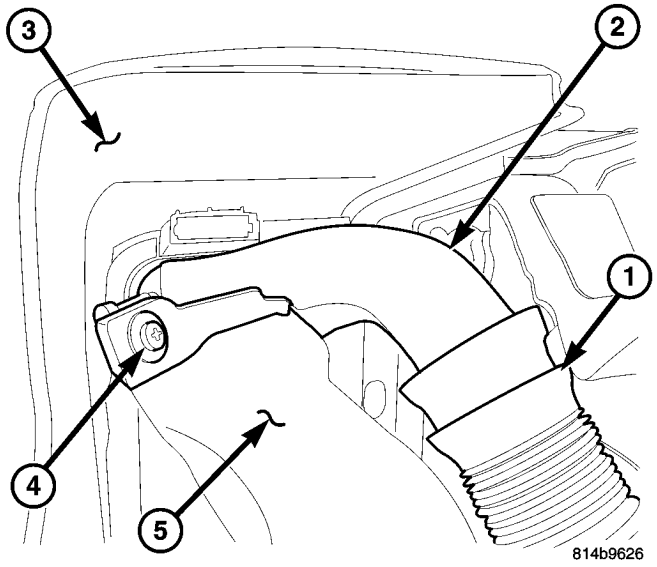

Fig. 38 Demister Outlet - Right Side shown, Left Side similar

1 - DEMISTER HOSE (2)
2 - DEMISTER OUTLET (2)
3 - INSTRUMENT PANEL
4 - SCREW (2)
5 - INSTRUMENT PANEL DUCT

DEMISTER DUCTS

(1) Reach through the glove box opening and/or the cluster opening of the instrument panel to connect the demister ducts to the demister outlets and the defroster duct as necessary.

(2) If removed, install the top cover, knee blocker and instrument cluster (Refer to 23 - BODY/INSTRUMENT PANEL/INSTRUMENT PANEL TOP COVER - INSTALLATION), (Refer to 23 - BODY/INSTRUMENT PANEL/KNEE BLOCKER - INSTALLATION) and (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - INSTALLATION).

(3) If removed, install the glove box (Refer to 23 - BODY/INSTRUMENT PANEL/GLOVE BOX - INSTALLATION).

(4) Reconnect the negative battery cable.

DEMISTER OUTLETS

(1) Position the demister outlets into the instrument panel as necessary.

(2) Install the screw that secures each demister outlet to the instrument panel duct as necessary.

(3) Connect the demister ducts to the demister outlets as necessary.

(4) Install the instrument panel assembly (Refer to 23 - BODY/INSTRUMENT PANEL/INSTRUMENT PANEL ASSEMBLY - INSTALLATION).

INSTRUMENT PANEL DUCTS

REMOVAL

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the battery negative (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. This is the only sure way to disable the airbag system. Failure to take the proper precautions could result in accidental airbag deployment and possible personal injury or death.

NOTE: The air ducts for the instrument panel outlets are integral to each other and cannot be serviced separately.

NOTE: Take the proper precautions to protect the front face of the instrument panel from cosmetic damage while performing this procedure.

(1) Disconnect and isolate the negative battery cable.

(2) Remove the instrument panel and place it on a workbench (Refer to 23 - BODY/INSTRUMENT PANEL/INSTRUMENT PANEL ASSEMBLY - REMOVAL).

(3) Remove the center bezel from the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/INSTRUMENT PANEL CENTER BEZEL - REMOVAL).

(4) Remove the accessory switch bezel, A/C-heater control, radio and radio rear support bracket from the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/ACCESSORY SWITCH BEZEL - REMOVAL), (Refer to 24 - HEATING & AIR CONDITIONING/CONTROLS/A/C HEATER CONTROL - REMOVAL) and (Fig. 39) (Refer to 8 - ELECTRICAL/AUDIO/RADIO - REMOVAL).

(5) Remove the demister ducts and the demister outlets (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/INSTRUMENT PANEL DEMISTER DUCTS - REMOVAL).

(6) Disconnect the instrument panel wire harness connectors from the left and right side courtesy lamps located at the bottom of the instrument panel.

(7) Disengage the instrument panel wire harness retainers from the instrument panel duct and position the wire harness out of the way.

(8) Remove the ten screws that secure the instrument panel duct to the instrument panel and remove the duct.

INSTRUMENT PANEL DUCTS (Continued)

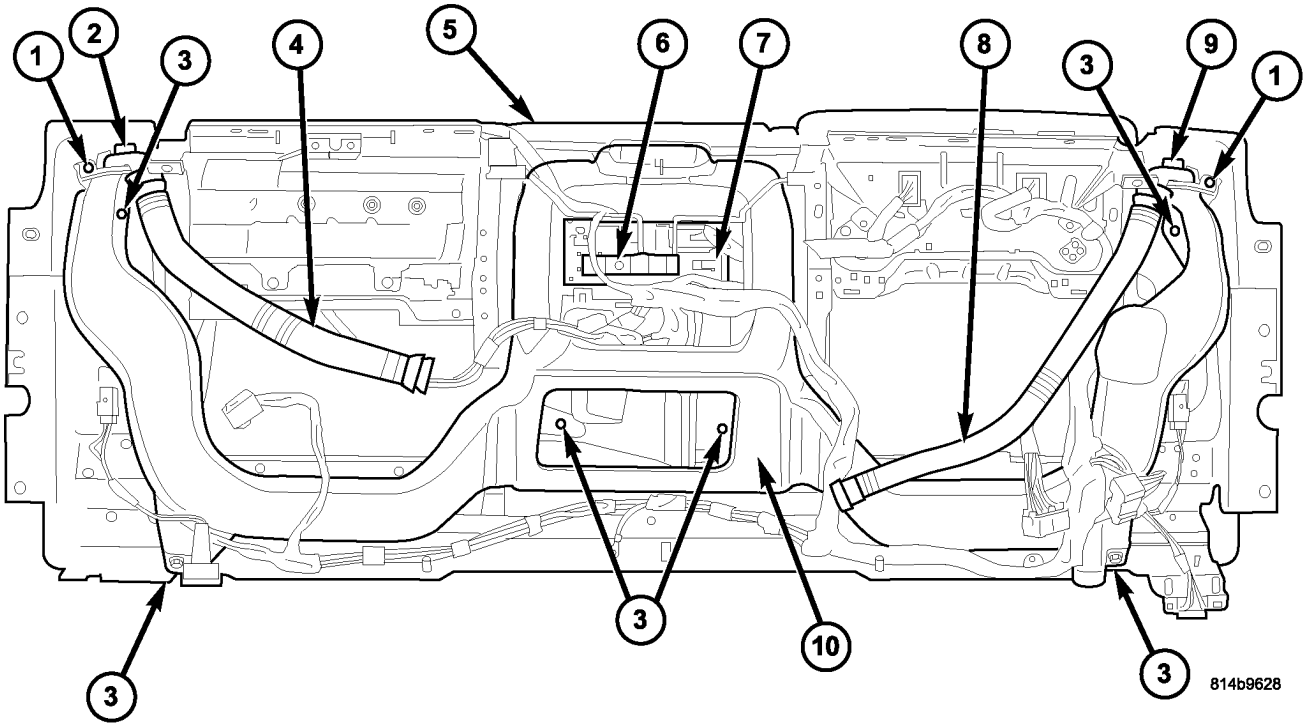

Fig. 39 Instrument Panel Duct - LHD shown, RHD typical

1 - SCREW (2)
2 - RH DEMISTER OUTLET
3 - SCREW (10)
4 - RH DEMISTER DUCT
5 - INSTRUMENT PANEL
6 - RADIO REAR SUPPORT BRACKET
7 - RADIO
8 - LH DEMISTER DUCT
9 - LH DEMISTER OUTLET
10 - INSTRUMENT PANEL DUCT

INSTALLATION

(1) Position the instrument panel duct into the instrument panel.

(2) Install the ten screws that secure the instrument panel duct to the instrument panel. Tighten the screws to 2 N·m (17 in. lbs.).

(3) Engage the instrument panel wire harness retainers to the instrument panel duct and connect the wire harness to the left and right courtesy lamps.

(4) Install the demister outlets and the demister ducts (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/INSTRUMENT PANEL DEMISTER DUCTS - INSTALLATION.

(5) Install the accessory switch bezel, A/C-heater control and the radio and the radio rear support bracket into the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/ACCESSORY SWITCH BEZEL - INSTALLATION), (Refer to 24 - HEATING & AIR CONDITIONING/CONTROLS/A/C HEATER CONTROL - INSTALLATION) and (Refer to 8 - ELECTRICAL/AUDIO/RADIO - INSTALLATION).

(6) Install the center bezel onto the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/INSTRUMENT PANEL CENTER BEZEL - INSTALLATION).

(7) Install the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/INSTRUMENT PANEL ASSEMBLY - INSTALLATION).

(8) Reconnect the negative battery cable.

PLUMBING

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PLUMBING

DESCRIPTION

The A/C refrigerant lines and hoses are used to carry the refrigerant between the various A/C system components. The refrigerant lines and hoses for the R-134a system on this vehicle consist of a barrier-hose design with a nylon tube sandwiched between rubber layers. The nylon tube helps to contain the R-134a refrigerant, which has a smaller molecular structure than R-12 refrigerant. The ends of the refrigerant lines are made from lightweight aluminum or steel, and commonly use braze-less fittings.

Any kinks or sharp bends in the refrigerant lines and hoses will reduce the capacity of the entire A/C system and can reduce the flow of refrigerant in the system. The radius of all bends in the flexible hose refrigerant lines should be at least ten times the diameter of the hose and the refrigerant lines should be routed so they are at least 80 millimeters (3 inches) away from the exhaust manifold(s) and exhaust pipe(s).

OPERATION

High pressures are produced in the refrigerant system when the A/C compressor is operating. Extreme care must be exercised to make sure that each of the refrigerant system connections is pressure-tight and leak free. It is a good practice to inspect all flexible hose refrigerant lines at least once a year to make sure they are in good condition and properly routed.

The refrigerant lines and hoses are coupled to other A/C system components with block-type fittings. An O-ring seal, or a flat steel gasket with an integral O-ring (dual plane seal), is used to mate the refrigerant line fittings with A/C system components to ensure the integrity of the refrigerant system.

The refrigerant lines and hoses cannot be repaired and, if faulty or damaged, they must be replaced.

WARNINGS

WARNING: The A/C system contains refrigerant under high pressure. Repairs should only be performed by qualified service personnel. Severe personal injury or death may result from improper service procedures.

WARNING: Avoid breathing the refrigerant and refrigerant oil vapor or mist. Exposure may irritate the eyes, nose, and/or throat. Wear eye protection when servicing the A/C refrigerant system. Serious eye injury can result from direct contact with the refrigerant. If eye contact occurs, seek medical attention immediately.

WARNING: Do not expose the refrigerant to open flame. Poisonous gas is created when refrigerant is burned. An electronic leak detector is recommended. Severe personal injury or death may result from improper service procedures.

WARNING: If accidental system discharge occurs, ventilate the work area before resuming service. Large amounts of refrigerant released in a closed work area will displace the oxygen and cause suffocation and death.

WARNING: The evaporation rate of R-134a refrigerant at average temperature and altitude is extremely high. As a result, anything that comes in contact with the refrigerant will freeze. Always protect the skin or delicate objects from direct contact with the refrigerant.

WARNING: The R-134a service equipment or the vehicle refrigerant system should not be pressure tested or leak tested with compressed air. Some mixtures of air and R-134a have been shown to be combustible at elevated pressures. These mixtures are potentially dangerous, and may result in fire or explosion causing property damage, personal injury or death.

WARNING: The engine cooling system is designed to develop internal pressures of 97 to 123 kilopascals (14 to 18 pounds per square inch). Do not remove or loosen the coolant pressure cap, cylinder block drain plugs, radiator drain, radiator hoses, heater hoses, or hose clamps while the engine cooling system is hot and under pressure. Allow the vehicle to cool for a minimum of 15 minutes before opening the cooling system for service. Failure to observe this warning can result in serious burns from the heated engine coolant.

CAUTIONS

CAUTION: Never add R-12 to a refrigerant system designed to use R-134a. Do not use R-12 equipment or parts on a R-134a A/C system. These refrigerants are not compatible and damage to the A/C system will result.

CAUTION: Never use R-12 refrigerant oil in a A/C system designed to use R-134a refrigerant oil. These refrigerant oils are not compatible and damage to the A/C system will result.

PLUMBING (Continued)

CAUTION: The use of A/C system sealers may result in damage to A/C refrigerant recovery/evacuation/recharging equipment and/or A/C system. Many federal, state/provincial and local regulations prohibit the recharge of A/C systems with known leaks. DaimlerChrysler recommends the detection of A/C system leaks through the use of approved leak detectors and fluorescent leak detection dyes. Vehicles found with A/C system sealers should be treated as contaminated and replacement of the entire A/C refrigerant system is recommended. A/C systems found to be contaminated with A/C system sealers, A/C stop-leak products or seal conditioners voids the warranty for the A/C system.

CAUTION: Recover the refrigerant before opening any fitting or connection. Open the fittings with caution, even after the system has been discharged. Never open or loosen a connection before recovering the refrigerant.

CAUTION: If equipped, do not remove the secondary retention clip from any spring-lock coupler connection while the refrigerant system is under pressure. Recover the refrigerant before removing the secondary retention clip. Open the fittings with caution, even after the system has been discharged. Never open or loosen a connection before recovering the refrigerant.

CAUTION: The internal parts of the A/C system will remain stable as long as moisture-free refrigerant and refrigerant oil is used. Abnormal amounts of dirt, moisture or air can upset the chemical stability. This may cause operational troubles or even serious damage if present in more than very small quantities. Before disconnecting a component, clean the outside of the fittings thoroughly to prevent contamination from entering the refrigerant system. Keep service tools and the work area clean. Do not open the refrigerant system or uncap a replacement component until you are ready to service the system. Immediately after disconnecting a component from the refrigerant system, seal the open fittings with a cap or plug. This will prevent contamination from entering the A/C system.

CAUTION: Refrigerant oil will absorb moisture from the atmosphere if left uncapped. Do not open a container of refrigerant oil until you are ready to use it. Replace the cap on the oil container immediately after using. Store refrigerant oil only in a clean, airtight, and moisture-free container.

CAUTION: Do not overcharge the refrigerant system. Overcharging will cause excessive compressor head pressure and can cause compressor noise and A/C system failure.

DIAGNOSIS AND TESTING

REFRIGERANT SYSTEM LEAKS

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

If the A/C system is not cooling properly, determine if the refrigerant system is fully-charged (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - DIAGNOSIS AND TESTING - REFRIGERANT SYSTEM CHARGE LEVEL).

If the refrigerant system is low or empty; a leak at a refrigerant line, connector fitting, component, or component seal is likely. While an oily residue on or near refrigerant system lines, connector fittings, components, or component seals can indicate the general location of a possible refrigerant leak, the exact leak location should be confirmed with an electronic leak detector prior to component repair or replacement. An electronic leak detector designed for R-134a refrigerant is recommended for locating and confirming refrigerant system leaks. See the operating instructions supplied by the equipment manufacturer for proper care and use of this equipment.

To detect a leak in the refrigerant system, perform one of the following procedures as indicated by the results of the refrigerant system charge level test.

SYSTEM EMPTY

(1) Evacuate the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM EVACUATE).

(2) Connect and dispense 0.283 kilograms (0.625 pounds or 10 ounces) of R-134a refrigerant into the evacuated refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM CHARGE).

(3) Proceed to the SYSTEM LOW procedures.

PLUMBING (Continued)

SYSTEM LOW

(1) Position the vehicle in a wind-free work area. This will aid in detecting small leaks.

(2) Bring the refrigerant system up to operating temperature and pressure. This is done by allowing the engine to run at idle under the following conditions for five minutes, then turning the engine off.

(a) Front windows open.

(b) Transmision in Park or Neutral with the park brake set.

(c) A/C-heater controls set to fresh air, panel mode, high blower speed and A/C compressor engaged.

(3) Wait five to seven minutes and, with the engine not running, use a electronic R-134a leak detector and search for leaks. Because R-134a refrigerant is heavier than air, the leak detector probe should be moved slowly along the bottom side of all refrigerant lines, connector fittings and components.

(4) To inspect an A/C evaporator for leaks, insert the electronic leak detector probe into a floor outlet or the recirculation air intake.

STANDARD PROCEDURE

REFRIGERANT SYSTEM SERVICE EQUIPMENT

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

WARNING: Eye protection must be worn when servicing an air conditioning refrigerant system. Turn off (rotate clockwise) all valves on the equipment being used, before connecting to or disconnecting from the refrigerant system. Failure to observe these warnings may result in personal injury or death.

RECOVERY/RECYCLING STATION

When servicing the A/C system, a R-134a refrigerant recovery/recycling/charging station that meets SAE Standard J2210 must be used (Fig. 1). Contact an automotive service equipment supplier for refrigerant recovery/recycling/charging equipment. See the operating instructions supplied by the equipment manufacturer for proper care and use of this equipment.

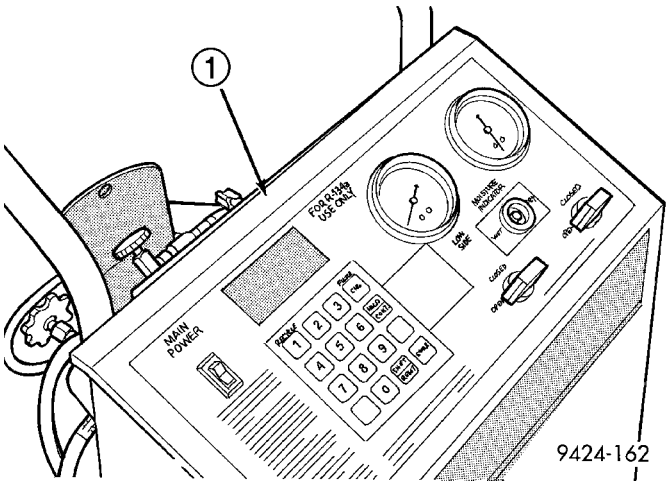

Fig. 1 Refrigerant Recovery/Recycling Station - Typical

1 - R-134a REFRIGERANT STATION

MANIFOLD GAUGE SET

A manifold gauge set may be needed with some recovery/recycling/charging equipment (Fig. 2). The service hoses on the gauge set being used should have manual (turn wheel), or automatic back-flow valves at the service port connector ends. This will prevent refrigerant from being released into the atmosphere.

MANIFOLD GAUGE SET CONNECTIONS

CAUTION: Do not use an R-12 manifold gauge set on an R-134a system. The refrigerants are not compatible and system damage will result.

- **LOW PRESSURE GAUGE HOSE** - The low pressure hose (Blue with Black stripe) attaches to the suction (low-side) service port. This port is located on the A/C liquid line.
- **HIGH PRESSURE GAUGE HOSE** - The high pressure hose (Red with Black stripe) attaches to the discharge (high-side) service port. This port is located on the A/C discharge line.
- **RECOVERY, RECYCLING, EVACUATION AND CHARGING HOSE** - The center manifold hose (Yellow, or White, with Black stripe) is used to recover, evacuate, and charge the refrigerant system. When the low or high pressure valves on the manifold gauge set are opened, the refrigerant in the system will escape through this hose.

PLUMBING (Continued)

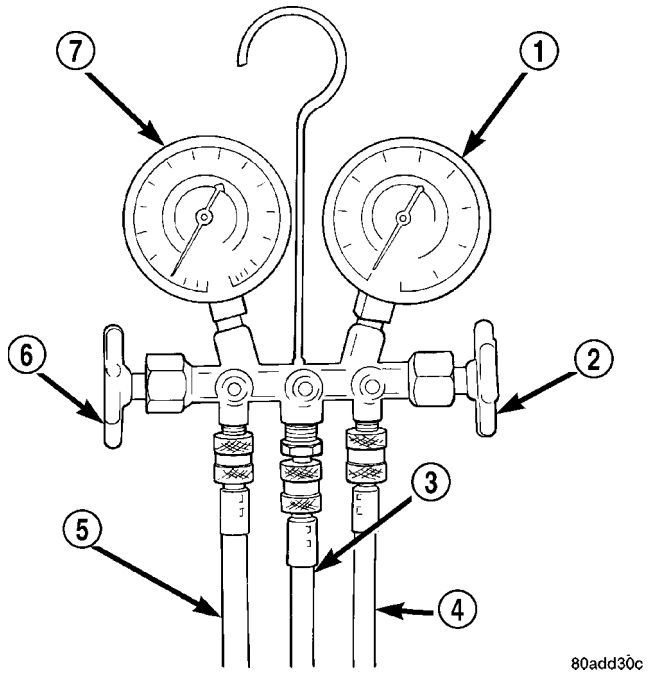

Fig. 2 Manifold Gauge Set - Typical

1 - HIGH PRESSURE GAUGE
2 - VALVE
3 - VACUUM/REFRIGERANT HOSE (YELLOW W/BLACK STRIPE)
4 - HIGH PRESSURE HOSE (RED W/BLACK STRIPE)
5 - LOW PRESSURE HOSE (BLUE W/BLACK STRIPE)
6 - VALVE
7 - LOW PRESSURE GAUGE

REFRIGERANT SYSTEM CHARGE LEVEL

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

NOTE: Always refer to the A/C Underhood Specification Label for the refrigerant fill capacity of the vehicle being serviced.

The following procedure should be used to determine whether the A/C refrigerant system contains the proper refrigerant charge. Symptoms of an improper refrigerant charge (low) include: poor A/C performance, fog emitted from the air outlets, a hissing sound from the A/C orifice tube/liquid line area. There are two different methods with which the refrigerant charge level may be tested:

1. Using a DRBIII® scan tool, a thermocouple and the Charge Determination Chart (Fig. 3). Refer to the appropriate diagnostic information.

2. Using a manifold gauge set, a thermocouple and the Charge Determination Chart (Fig. 3).

A temperature probe is required to measure liquid line temperature. The clamp-on, Type K thermocouple temperature probe used in this procedure is available through the DaimlerChrysler Professional Service Equipment (PSE) program. This probe (PSE #66-324-0014 or #80PK-1A) is compatible with temperature-measuring instruments that accept Type K thermocouples, and have a miniature connector input. Other temperature probes are available through aftermarket sources; however, all references in this procedure will reflect the use of the probe made available through the PSE program.

In order to use the temperature probe, a digital thermometer will also be required. If a digital thermometer is not available, an adapter is available through the PSE program that will convert any standard digital multimeter into a digital thermometer. This adapter is designed to accept any standard Type K thermocouple. If a digital multimeter is not available, this tool is also available through the PSE program.

NOTE: When connecting the service equipment couplings to the refrigerant system service ports, be certain that the valve of each coupling is fully closed. This will reduce the amount of effort required to make the connection.

(1) Remove the caps from the refrigerant system service ports and attach a manifold gauge set or a R-134a refrigerant recovery/recycling/charging station that meets SAE Standard J2210 to the refrigerant system.

(2) Attach a clamp-on thermocouple to the A/C liquid line. The thermocouple must be placed as close as possible to the inlet end of the A/C orifice tube to accurately observe liquid line temperature.

(3) Bring the refrigerant system up to operating temperature and pressure. This is done by allowing the engine to run at idle under the following conditions for five minutes.

(a) Front windows are open.

(b) Transmission in Park or Neutral with the parking brake set.

(c) A/C-heater controls set to fresh air, full cool, panel outlet mode, high blower speed and the A/C compressor engaged.

(4) Raise the liquid line (discharge) pressure to about 1793 kPa (260 psi) by placing a piece of cardboard over the front of the A/C condenser. To place the cardboard properly, remove the upper radiator sight shield (if equipped) and cover only enough of the A/C condenser to raise and maintain the liquid line pressure at the previously specified level.

PLUMBING (Continued)

(5) Observe the liquid line (discharge) pressure and liquid line temperature. Using the Charge Determination Chart (Fig. 3), determine whether the refrigerant system is operating within the proper charge range.

(a) If the refrigerant system is operating in the undercharged area of the chart, add 0.057 kilogram (0.125 pound or 2 ounces) of R-134a refrigerant to the system.

(b) If the refrigerant system is operating in the overcharged area of the chart, reclaim 0.057 kilogram (0.125 pound or 2 ounces) of refrigerant from the system.

(6) Recheck the refrigerant system charge level following each refrigerant level adjustment. Continue this process until the refrigerant system readings are in the proper charge range on the Charge Determination Chart.

REFRIGERANT SYSTEM RECOVERY

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

A R-134a refrigerant recovery/recycling/charging station that meets SAE Standard J2210 must be used to recover the refrigerant from an R-134a refrigerant system. Refer to the operating instructions supplied by the equipment manufacturer for the proper care and use of this equipment.

REFRIGERANT SYSTEM EVACUATE

NOTE: Special effort must be used to prevent moisture from entering the A/C system oil. Moisture in the oil is very difficult to remove and will cause a reliability problem with the A/C compressor.

If an A/C compressor designed to use R-134a refrigerant is left open to the atmosphere for an extended period of time. It is recommended that the refrigerant oil be drained and replaced with new oil or a new A/C compressor be used. This will eliminate the possibility of contaminating the refrigerant system.

If the refrigerant system has been open to the atmosphere, it must be evacuated before the refrigerant system can be filled. Moisture and air mixed with the refrigerant system will raise the compressor head pressure above acceptable operating levels. This

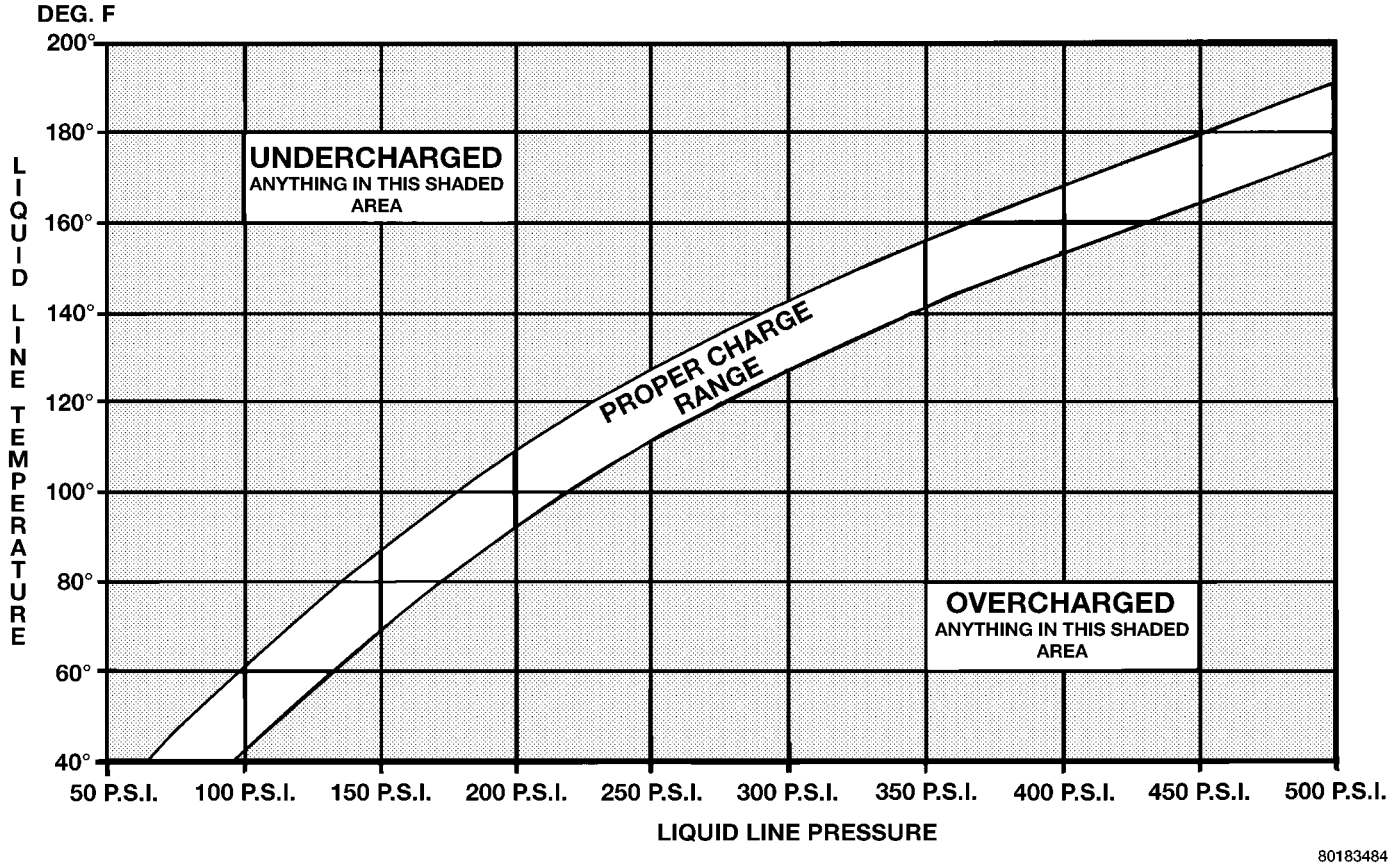

Fig. 3 Charge Determination Chart (Ambient Test Condition 85° F)

PLUMBING (Continued)

will reduce the performance of the A/C system and damage the A/C compressor. Moisture will boil at near room temperature when exposed to vacuum. A R-134a refrigerant recovery/recycling/charging station that meets SAE Standard J2210 must be used to evacuate the refrigerant system. See the operating instructions supplied by the equipment manufacturer for proper care and use of this equipment. To evacuate the refrigerant system, use the following procedure:

NOTE: When connecting the service equipment couplings to the refrigerant system service ports, be certain that the valve of each coupling is fully closed. This will reduce the amount of effort required to make the connection.

(1) Connect a suitable charging station, refrigerant recovery machine or a manifold gauge set with vacuum pump and refrigerant recovery equipment (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM SERVICE EQUIPMENT).

(2) Recover the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM RECOVERY).

(3) Open the suction and discharge valves and start the vacuum pump. The vacuum pump should run a minimum of 45 minutes prior to charge to eliminate all moisture within the refrigerant system. When the suction gauge reads -88 kPa (-26 in. Hg) vacuum or greater for 30 minutes, close all valves and turn off vacuum pump. If the refrigerant system fails to reach the specified vacuum, the refrigerant system likely has a leak that must be corrected. If the refrigerant system maintains specified vacuum for at least 30 minutes, start the vacuum pump, open the suction and discharge valves. Then allow the system to evacuate an additional 10 minutes.

(4) Close all valves. Turn off and disconnect the vacuum pump.

(5) Charge the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM CHARGE).

REFRIGERANT SYSTEM CHARGE

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

NOTE: Always refer to the A/C Underhood Specification Label for the refrigerant fill capacity of the vehicle being serviced.

After all refrigerant system leaks have been repaired and the refrigerant system has been evacuated, a refrigerant charge can be injected into the system. For the proper amount of the refrigerant charge, always refer to the A/C Underhood Specification Label located in the engine compartment. An R-134a refrigerant recovery/recycling/charging station that meets SAE Standard J2210 must be used to charge the refrigerant system with R-134a refrigerant. See the operating instructions supplied by the equipment manufacturer for proper care and use of this equipment.

CHARGING PROCEDURE

CAUTION: A small amount of refrigerant oil is removed from the A/C system each time the refrigerant system is recovered and evacuated. Before charging the A/C system, you MUST replenish any oil lost during the recovery process. Refer the equipment manufacturer instructions for more information.

(1) Evacuate the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM EVACUATE).

(2) A manifold gauge set and a R-134a refrigerant recovery/recycling/charging station that meets SAE standard J2210 should still be connected to the A/C refrigerant system.

(3) Measure the proper amount of R-134a refrigerant and heat it to 52° C (125° F) with the charging station. See the operating instructions supplied by the equipment manufacturer for proper use of this equipment.

(4) Open both the suction and discharge valves, then open the charge valve to allow the heated refrigerant to flow into the system.

(5) When the transfer of refrigerant has stopped, close both the suction and discharge valves.

(6) If all of the refrigerant charge did not transfer from the dispensing device, open all of the windows in the vehicle and set the A/C-heater controls so that the A/C compressor is engaged and the blower motor is operating at its lowest speed setting. Run the engine at a steady high idle (about 1400 rpm). If the A/C compressor will not engage, test the compressor clutch control circuit and repair as required.

(7) Open the suction valve to allow the remaining refrigerant to transfer to the refrigerant system.

PLUMBING (Continued)

WARNING: Take care not to open the discharge (high pressure) valve at this time.

(8) Close the suction valve and test the A/C system performance (Refer to 24 - HEATING & AIR CONDITIONING - STANDARD PROCEDURE - A/C PERFORMANCE TEST).

(9) Disconnect the charging station and manifold gauge set from the refrigerant system service ports.

(10) Install the protective caps onto the refrigerant system service ports.

A/C COMPRESSOR

DESCRIPTION

A/C COMPRESSOR

The A/C system uses a Denso 10PA17 seven cylinder, reciprocating wobble plate-type A/C compressor on all models. This A/C compressor has a fixed displacement of 150 cubic centimeters (9.375 cubic inches), and has both the suction and discharge ports located on the cylinder head.

A label identifying the use of R-134a refrigerant is located on the A/C compressor.

HIGH PRESSURE RELIEF VALVE

A high pressure relief valve is located on the compressor manifold. This mechanical valve is designed to vent refrigerant from the A/C system to protect against damage to the A/C compressor and other A/C system components, caused by a restriction in air flow through the A/C condenser or, an overcharge of refrigerant.

OPERATION

A/C COMPRESSOR

The A/C compressor is driven by the engine through an electric clutch, drive pulley and belt arrangement. The A/C compressor is lubricated by refrigerant oil that is circulated throughout the refrigerant system with the refrigerant.

The A/C compressor draws in low-pressure refrigerant vapor from the A/C evaporator through its suction port. It then compresses the refrigerant into a high-pressure, high-temperature refrigerant vapor, which is then pumped to the A/C condenser through the compressor discharge port.

The A/C compressor cannot be repaired and, if faulty or damaged, it must be replaced. The compressor clutch, pulley and bearing assembly, and clutch field coil are available for service.

HIGH PRESSURE RELIEF VALVE

The high pressure relief valve vents the A/C system when a discharge pressure of 3445 to 4135 kPa (500 to 600 psi) or above is reached. The high pressure relief valve closes when a minimum discharge pressure of 2756 kPa (400 psi) is reached.

The high pressure relief valve vents only enough refrigerant to reduce the A/C system pressure, and then re-seats itself. The majority of the refrigerant is conserved in the A/C system. If the valve vents refrigerant, it does not mean that the valve is faulty.

The high pressure relief valve is factory-calibrated. The high pressure relief valve cannot be adjusted or repaired, and must not be removed or otherwise disturbed. The high pressure relief valve is only serviced as a part of the A/C compressor.

DIAGNOSIS AND TESTING

A/C COMPRESSOR NOISE

When investigating an A/C system related noise, you must first know the conditions under which the noise occurs. These conditions include: weather, vehicle speed, transmission in gear or neutral, engine speed, engine temperature, and any other special conditions. Noises that develop during A/C operation can often be misleading. For example: What sounds like a failed front engine bearing or connecting rod, may be caused by loose bolts, nuts, mounting brackets or a loose compressor clutch assembly.

Drive belts are speed sensitive. At different engine speeds and depending upon drive belt tension, drive belts can develop noises that are mistaken for an A/C compressor noise. Improper drive belt tension can cause a misleading noise when the compressor clutch is engaged, which may not occur when the compressor clutch is disengaged. Check the accessory drive belt condition and tension as described in Cooling before beginning this procedure.

(1) Select a quiet area for testing. Duplicate the complaint conditions as much as possible. Switch the compressor on and off several times to clearly identify the compressor noise. Listen to the compressor while the clutch is engaged and disengaged. Probe the compressor with an engine stethoscope or a long screwdriver with the handle held to your ear to better localize the source of the noise.

(2) Loosen all of the compressor mounting hardware and retighten. Tighten the compressor clutch mounting nut. Be certain that the clutch coil is mounted securely to the compressor, and that the clutch plate and pulley are properly aligned and have the correct air gap (Refer to 24 - HEATING & AIR CONDITIONING/CONTROLS/A/C COMPRESSOR CLUTCH - INSTALLATION).

A/C COMPRESSOR (Continued)

(3) To duplicate a high-ambient temperature condition (high head pressure), restrict the air flow through the condenser. Install a manifold gauge set to be certain that the discharge pressure does not exceed 2760 kPa (400 psi).

(4) Check the refrigerant system plumbing for incorrect routing, rubbing or interference, which can cause unusual noises. Also check the refrigerant lines for kinks or sharp bends that will restrict refrigerant flow, which can cause noises (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - DESCRIPTION - REFRIGERANT LINES).

(5) If the noise is from opening and closing of the high pressure relief valve, recover, evacuate and recharge the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM RECOVERY), (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM EVACUATE) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM CHARGE).

(6) If the noise is from liquid slugging on the suction line, replace the accumulator (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/ACCUMULATOR - REMOVAL) and check the refrigerant oil level and the refrigerant system charge (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/REFRIGERANT OIL - STANDARD PROCEDURE - REFRIGERANT SYSTEM OIL LEVEL).

(7) If the liquid slugging condition continues following accumulator replacement, replace the A/C compressor and repeat Step 1.

REMOVAL

2.4L ENGINE

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

NOTE: The A/C compressor may be removed and repositioned without disconnecting the refrigerant lines or discharging the refrigerant system. Discharging is not necessary if servicing the A/C clutch, clutch field coil or the engine.

(1) Recover the refrigerant from the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT RECOVERY).

(2) Disconnect and isolate the negative battery cable.

(3) Remove the accessory drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - REMOVAL).

(4) Disconnect the wire harness connector from the A/C clutch field coil.

(5) Remove the nuts that secure the A/C suction and discharge lines to the A/C compressor and remove and discard the O-ring seals (Fig. 4).

(6) Install plugs in, or tape over all of the opened refrigerant line fittings and compressor ports.

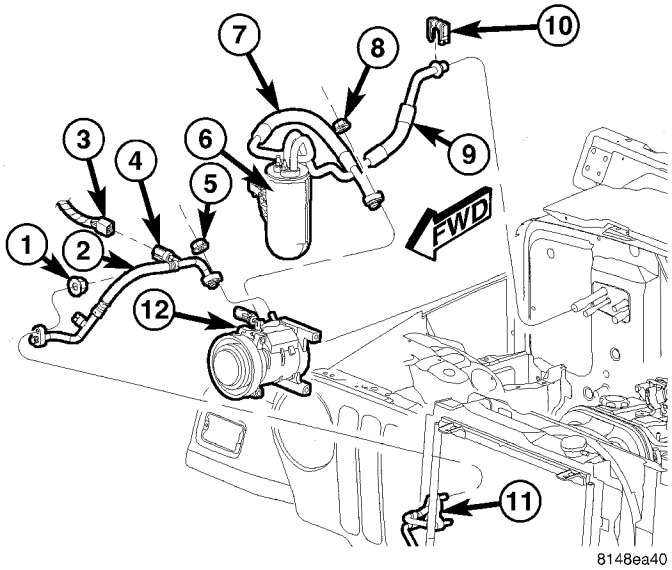

Fig. 4 A/C Compressor Refrigerant Lines - 2.4L Engine

1 - NUT
2 - A/C DISCHARGE LINE
3 - WIRE HARNESS CONNECTOR
4 - A/C PRESSURE TRANSDUCER
5 - NUT
6 - ACCUMULATOR
7 - A/C SUCTION LINE (FRONT SECTION)
8 - NUT (4)
9 - A/C SUCTION LINE (REAR SECTION)
10 - SECONDARY RETAINING CLIP
11 - A/C CONDENSER
12 - A/C COMPRESSOR

A/C COMPRESSOR (Continued)

(7) Remove the bolts and nuts that secure the A/C compressor to the engine (Fig. 5)

(8) Remove the A/C compressor from the engine compartment.

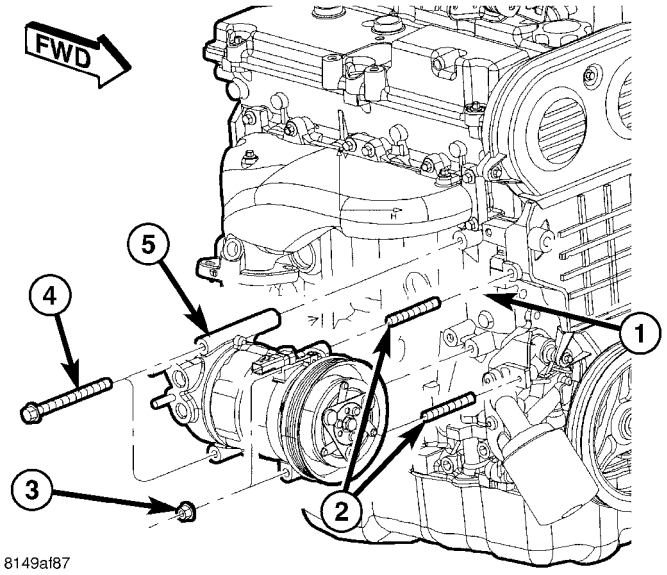

Fig. 5 A/C Compressor - 2.4L Engine

1 - CYLINDER BLOCK
2 - STUDS
3 - NUT (2)
4 - BOLT (2)
5 - A/C COMPRESSOR

4.0L ENGINE

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

NOTE: The A/C compressor may be removed and repositioned without disconnecting the refrigerant lines or discharging the refrigerant system. Discharging is not necessary if servicing the A/C clutch, clutch field coil or the engine.

(1) Recover the refrigerant from the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT RECOVERY).

(2) Disconnect and isolate the negative battery cable.

(3) Remove the accessory drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - REMOVAL).

(4) Disconnect the wire harness connector from the A/C clutch field coil.

(5) Remove the nuts that secure the A/C suction and discharge lines to the A/C compressor and remove and discard the O-ring seals (Fig. 6).

(6) Install plugs in, or tape over all of the opened refrigerant line fittings and compressor ports.

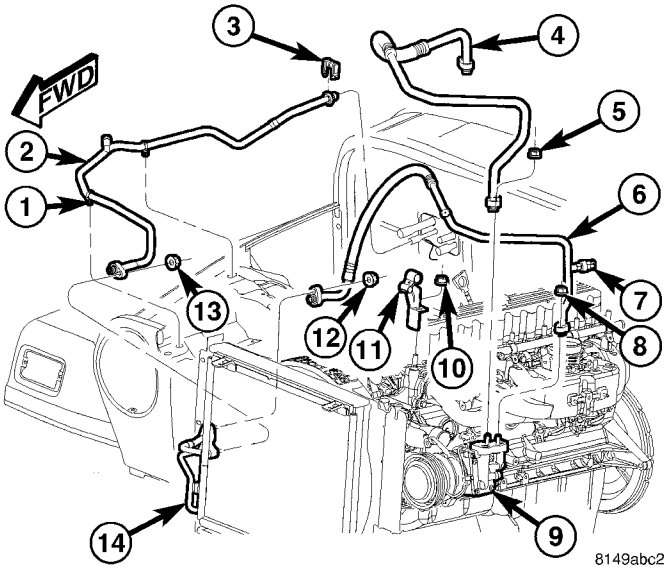

Fig. 6 A/C Compressor Refrigerant Lines - 4.0L LHD shown, RHD similar

1 - REFRIGERANT LINE RETAINING CLIP (2)
2 - A/C LIQUID LINE
3 - SECONDARY RETAINING CLIP
4 - A/C SUCTION LINE (FRONT SECTION)
5 - NUT
6 - A/C DISCHARGE LINE
7 - A/C HIGH PRESSURE SWITCH
8 - NUT
9 - A/C COMPRESSOR
10 - NUT
11 - REFRIGERANT LINE BRACKET
12 - NUT
13 - NUT
14 - A/C CONDENSER

A/C COMPRESSOR (Continued)

(7) Remove the bolts and nuts that secure the A/C compressor to the engine (Fig. 7).

(8) Remove the A/C compressor from the engine compartment.

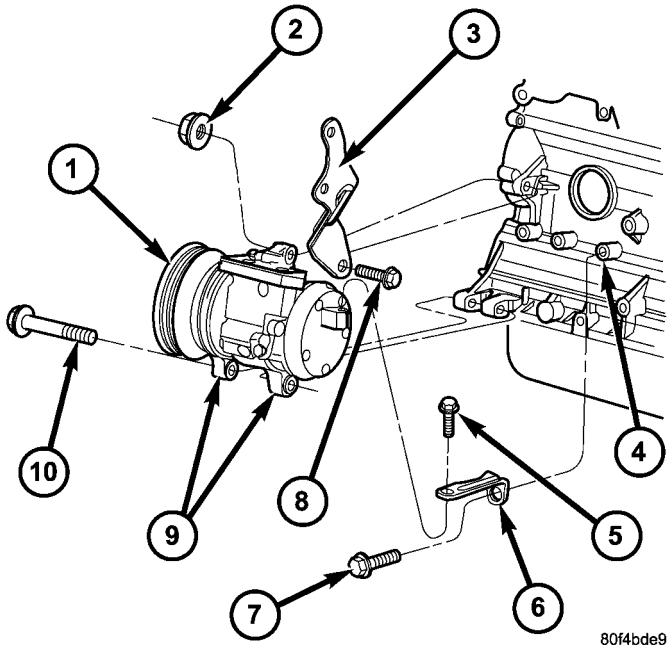

Fig. 7 A/C Compressor - 4.0L Engine

1 - A/C COMPRESSOR
2 - NUT
3 - UPPER MOUNTING BRACKET
4 - CYLINDER BLOCK
5 - BOLT
6 - LOWER MOUNTING BRACKET
7 - BOLT
8 - BOLT
9 - COMPRESSOR LOWER MOUNTING POINT
10 - BOLT

INSTALLATION

2.4L ENGINE

NOTE: If a replacement A/C compressor is being installed, be certain to check the refrigerant oil level. See Refrigerant Oil Level in this group for the procedures. Use only refrigerant oil of the type recommended for the A/C compressor in the vehicle.

(1) Position the A/C compressor into the engine compartment.

(2) Install the bolts and nuts that secure the A/C compressor to the engine. Tighten the bolts and nuts to 28 N·m (21 ft. lbs.).

(3) Remove the tape or plugs from the opened refrigerant line fittings and compressor ports.

(4) Lubricate new rubber O-ring seals with clean refrigerant oil and install them on the discharge and liquid line fittings. Use only the specified O-rings as they are made of a special material for the R-134a system. Use only refrigerant oil of the type recommended for the A/C compressor in the vehicle.

(5) Install the A/C suction and discharge lines onto the A/C compressor.

(6) Install the nuts that secure the A/C suction and discharge lines to the A/C compressor. Tighten the nuts to 25 N·m (18 ft. lbs.).

(7) Connect the wire harness connector to the A/C clutch field coil.

(8) Install the accessory drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - INSTALLATION).

(9) Reconnect the negative battery cable.

(10) Evacuate the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM EVACUATE).

(11) Charge the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM CHARGE).

4.0L ENGINE

NOTE: If a replacement A/C compressor is being installed, be certain to check the refrigerant oil level. See Refrigerant Oil Level in this group for the procedures. Use only refrigerant oil of the type recommended for the A/C compressor in the vehicle.

(1) Position the A/C compressor into the engine compartment.

(2) Install the bolts and nut that secure the A/C compressor to the engine.

- Tighten the bolts and nut fastening the compressor to the cylinder block to 45-65 N·m (35-50 ft. lbs.).
- Tighten the bolts fastening the rear brace to the compressor and cylinder block to 40-55 N·m (30-40 ft. lbs.).

(3) Remove the tape or plugs from the opened refrigerant line fittings and compressor ports.

(4) Lubricate new rubber O-ring seals with clean refrigerant oil and install them on the discharge and liquid line fittings. Use only the specified O-rings as they are made of a special material for the R-134a system. Use only refrigerant oil of the type recommended for the A/C compressor in the vehicle.

(5) Install the A/C suction and discharge lines onto the A/C compressor.

(6) Install the nuts that secure the A/C suction and discharge lines to the A/C compressor. Tighten the nuts to 25 N·m (18 ft. lbs.).

(7) Connect the wire harness connector to the A/C clutch field coil.

A/C COMPRESSOR (Continued)

(8) Install the accessory drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - INSTALLATION).

(9) Reconnect the negative battery cable.

(10) Evacuate the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM EVACUATE).

(11) Charge the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM CHARGE).

A/C CONDENSER

DESCRIPTION

The A/C condenser is located in the front of the engine compartment behind the grille (Fig. 8). The A/C condenser is a heat exchanger that allows the high-pressure refrigerant gas being discharged by the A/C compressor to give up its heat to the air passing over the condenser fins, which causes the refrigerant to cool and change to a liquid state.

The A/C condenser is equipped with mounting provisions and a tapping block for the A/C refrigerant lines.

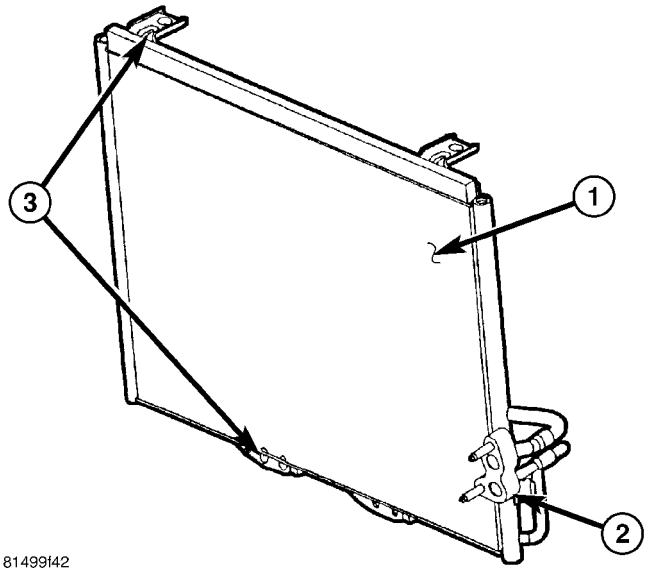

Fig. 8 A/C Condenser - Typical

1 - A/C CONDENSER
2 - MOUNTING BRACKET
3 - REFRIGERANT LINE TAPPING BLOCK

OPERATION

When air passes through the fins of the A/C condenser, the high-pressure refrigerant gas within the A/C condenser gives up its heat. The refrigerant then condenses as it leaves the A/C condenser and becomes a high-pressure liquid. The volume of air flowing over the condenser fins is critical to the proper cooling performance of the A/C system. Therefore, it is important that there are no objects placed in front of the radiator grille openings at the front of the vehicle or foreign material on the condenser fins that might obstruct proper air flow. Also, any factory-installed air seals or shrouds must be properly reinstalled following radiator or A/C condenser service.

The A/C condenser cannot be repaired and, if faulty or damaged, it must be replaced.

REMOVAL

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

CAUTION: Before removing the A/C condenser, note the location of each of the radiator/condenser air seals. These air seals are used to direct air through the A/C condenser and radiator. The air seals must be reinstalled in their proper locations in order for the A/C and engine cooling systems to perform as designed.

(1) Disconnect and isolate the negative battery cable.

(2) Recover the refrigerant from the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM RECOVERY).

(3) Partially drain the engine cooling system (Refer to 7 - COOLING - STANDARD PROCEDURE - COOLING SYSTEM DRAINING).

(4) Remove the cooling recovery container (Refer to 7 - COOLING/ENGINE/COOLANT RECOVERY CONTAINER - REMOVAL.

(5) Disconnect the upper radiator hose from the radiator and position the hose out of the way (Refer to 7 - COOLING - OPERATION - HOSE CLAMPS).

(6) Remove the radiator cooling fan and fan shroud (Refer to 7 - COOLING/ENGINE/FAN DRIVE VISCOUS CLUTCH - REMOVAL).

(7) Raise and support the vehicle.

(8) Position the lower condenser/radiator air seal out of the way and remove the two bolts that secure the lower condenser mounting bracket to the bottom of the grille panel.

(9) Lower the vehicle.

A/C CONDENSER (Continued)

(10) If equipped, remove the upper condenser/radiator air seal (Fig. 9).

(11) Remove the nuts that secure the discharge and liquid lines to the condenser tapping block.

(12) Disconnect the discharge and liquid lines from the condenser tapping block and remove and discard the O-ring seals.

(13) Install plugs in, or tape over the opened refrigerant line fittings and the condenser ports.

(14) Remove the two bolts that secure the upper condenser mounting brackets to the top of the grille panel.

(15) Remove the three bolts on each side of the radiator that secure the radiator mounting brackets to the grille panel.

(16) Tilt the radiator back towards the engine. Use care to prevent damaging the radiator fins.

(17) Carefully remove the A/C condenser from the engine compartment.

INSTALLATION

NOTE: If the A/C condenser is being replaced, add 22 milliliters (0.75 fluid ounce) of refrigerant oil to the refrigerant system. Use only refrigerant oil of the type recommended for the A/C compressor in the vehicle.

(1) Carefully position the A/C condenser into the engine compartment.

(2) Install the two bolts that secure the upper condenser mounting brackets to the top of the grille panel. Tighten the bolts to 5 N·m (44 in. lbs.).

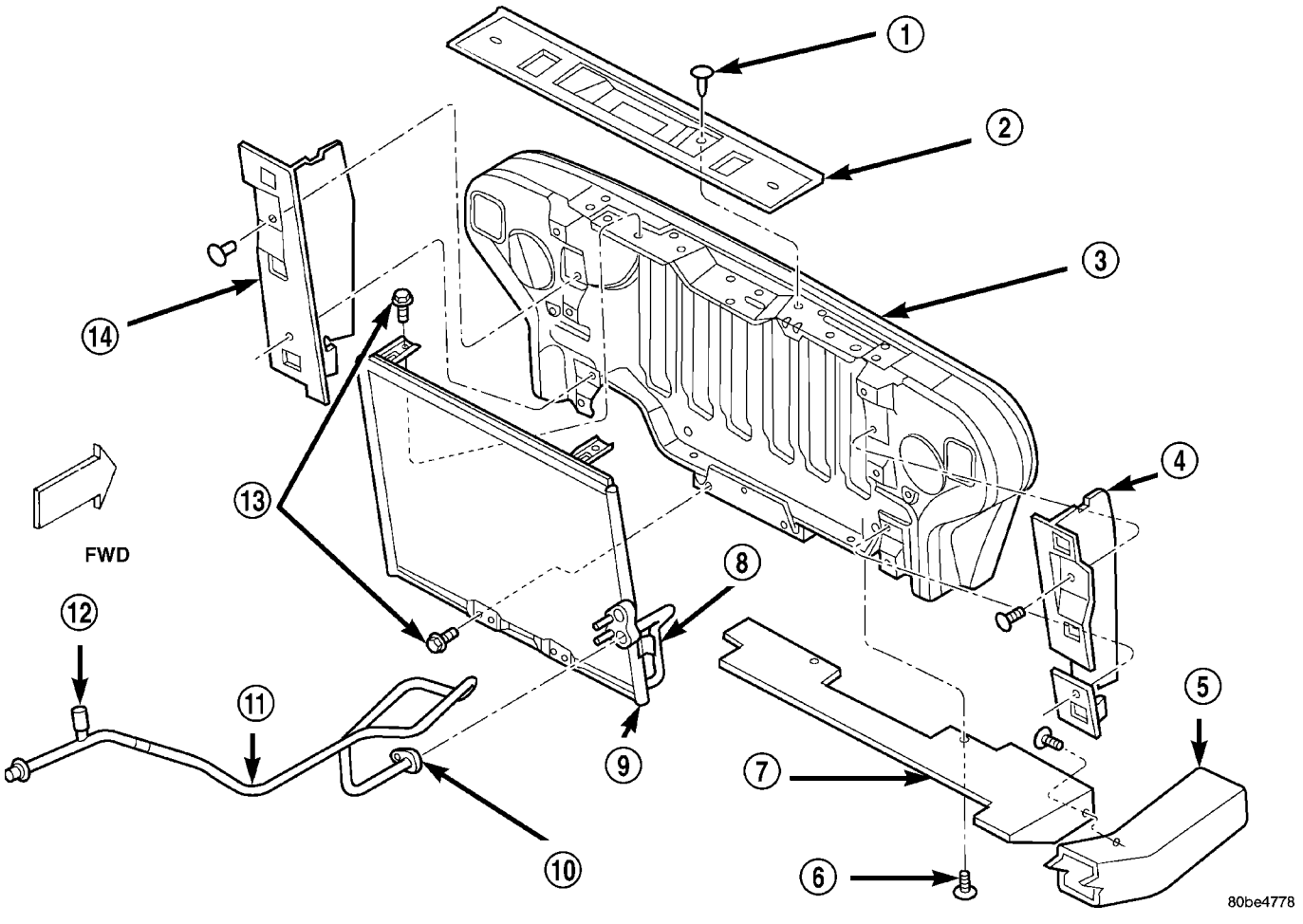

Fig. 9 A/C Condenser

1 - PUSH-PIN RETAINER
2 - UPPER AIR SEAL
3 - GRILLE PANEL
4 - AIR SEAL
5 - FRAME RAIL
6 - PUSH-PIN RETAINER
7 - LOWER AIR SEAL
8 - CONDENSER TAPPING BLOCK
9 - A/C CONDENSER
10 - REFRIGERANT LINE FITTING
11 - LIQUID LINE
12 - SERVICE PORT
13 - BOLTS
14 - AIR SEAL

A/C CONDENSER (Continued)

(3) Align the radiator mounting brackets to the grille panel and install the six bolts that secure the radiator to the grille panel. Tighten the bolts to 8 N·m (72 in. lbs.).

(4) Raise and support the vehicle.

(5) Position the lower condenser/radiator air seal out of the way and install the two bolts that secure the lower condenser mounting bracket to the bottom of the grille panel. Tighten the bolts to 5 N·m (44 in. lbs.).

(6) Lower the vehicle.

(7) Remove the tape or plugs from the discharge and liquid line fittings and the condenser ports.

(8) Lubricate new rubber O-ring seals with clean refrigerant oil and install them onto the discharge and liquid line fittings. Use only the specified O-rings as they are made of a special material for the R-134a system. Use only refrigerant oil of the type recommended for the A/C compressor in the vehicle.

(9) Install the discharge and liquid lines onto the condenser tapping block.

(10) Install the nuts that secures the discharge and liquid lines to the condenser tapping block. Tighten the nuts to 12 N·m (105 in. lbs.).

(11) If equipped, install the upper condenser/radiator air seal.

(12) Install the radiator cooling fan and fan shroud (Refer to 7 - COOLING/ENGINE/FAN DRIVE VISCOUS CLUTCH - INSTALLATION).

(13) Connect the upper radiator hose to the radiator and install the hose clamp (Refer to 7 - COOLING - OPERATION - HOSE CLAMPS).

(14) Install the coolant recovery container (Refer to 7 - COOLING/ENGINE/COOLANT RECOVERY CONTAINER - INSTALLATION).

(15) Reconnect the battery negative cable.

(16) Fill the engine cooling system (Refer to 7 - COOLING - STANDARD PROCEDURE - COOLING SYSTEM REFILLING)

(17) Evacuate the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM EVACUATE).

(18) Charge the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM CHARGE).

A/C DISCHARGE LINE

DESCRIPTION

The A/C discharge line is the refrigerant line that carries refrigerant from the A/C compressor to the A/C condenser. The A/C discharge line includes the high-side service port.

The A/C discharge line on models equipped with the 2.4L engine include a fitting for the A/C pressure transducer. For more information refer to 24 - HEATING & AIR CONDITIONING/CONTROLS/A/C PRESSURE TRANSDUCER - DESCRIPTION.

The A/C discharge line on models equipped with the 4.0L engine include a fitting for the A/C high pressure switch. For more information refer to 24 - HEATING & AIR CONDITIONING/CONTROLS/A/C HIGH PRESSURE SWITCH - DESCRIPTION.

CAUTION: Use only seals specified for the vehicle. Failure to use the correct seals will cause the refrigerant system connections to leak.

The A/C discharge line has no serviceable parts except for the O-ring seals and high-side service port valve and it's protective cap.

If the A/C discharge line is found to be leaking or is damaged, it must be replaced.

REMOVAL

2.4L ENGINE

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

(1) Disconnect and isolate the negative battery cable.

(2) Recover the refrigerant from the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM RECOVERY).

(3) Disconnect the wire harness connector from the A/C pressure transducer located on the A/C discharge line (Fig. 10).

(4) Remove the nut that secures the A/C discharge line to the A/C compressor.

(5) Disconnect the A/C discharge line from the A/C compressor and remove and discard the O-ring seal.

(6) Install plug in, or tape over the opened discharge line fitting and the compressor port.

(7) Remove the nut that secures theA/C discharge line to the A/C condenser.

(8) Disconnect the A/C discharge line from the A/C condenser and remove discard the O-ring seal.

(9) Install plug in, or tape over the opened discharge line fitting and the condenser port.

A/C DISCHARGE LINE (Continued)

(10) Remove the A/C discharge line from the engine compartment and if required, remove the A/C pressure transducer from the A/C discharge line.

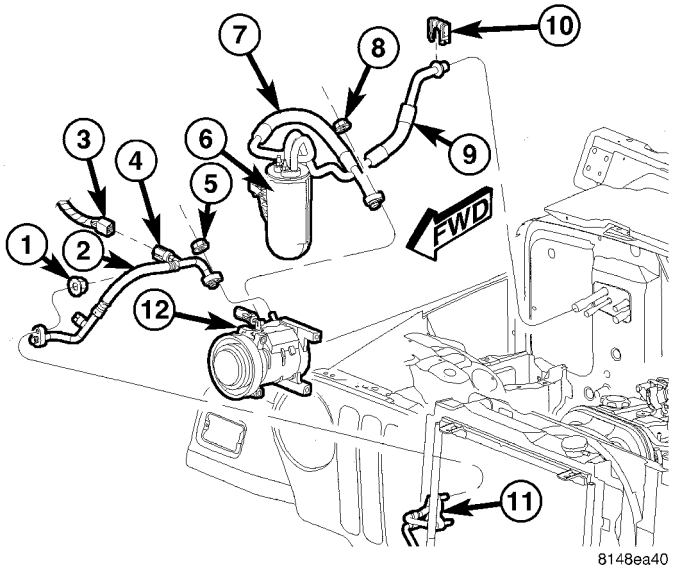

Fig. 10 A/C Discharge Line - 2.4L Engine

1 - NUT
2 - A/C DISCHARGE LINE
3 - WIRE HARNESS CONNECTOR
4 - A/C PRESSURE TRANSDUCER
5 - NUT
6 - ACCUMULATOR
7 - A/C SUCTION LINE (FRONT SECTION)
8 - NUT (4)
9 - A/C SUCTION LINE (REAR SECTION)
10 - SECONDARY RETAINING CLIP
11 - A/C CONDENSER
12 - A/C COMPRESSOR

4.0L ENGINE

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

(1) Disconnect and isolate the negative battery cable.

(2) Recover the refrigerant from the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM RECOVERY).

(3) Disconnect the wire harness connector from the A/C high pressure switch located on the A/C discharge line (Fig. 11).

(4) Remove the nut that secures the refrigerant line support bracket to the top of the engine and remove the discharge line from the bracket.

(5) Remove the nut that secures the A/C discharge line to the A/C compressor.

(6) Disconnect the A/C discharge line from the A/C compressor and remove and discard the O-ring seal.

(7) Install plug in, or tape over the opened discharge line fitting and the compressor port.

(8) Remove the nut that secures the A/C discharge line to the A/C condenser.

(9) Disconnect the A/C discharge line from theA/C condenser and remove discard the O-ring seal.

(10) Install plug in, or tape over the opened discharge line fitting and the condenser port.

(11) Remove the A/C discharge line from the engine compartment and if required, remove the A/C high pressure switch from the A/C discharge line.

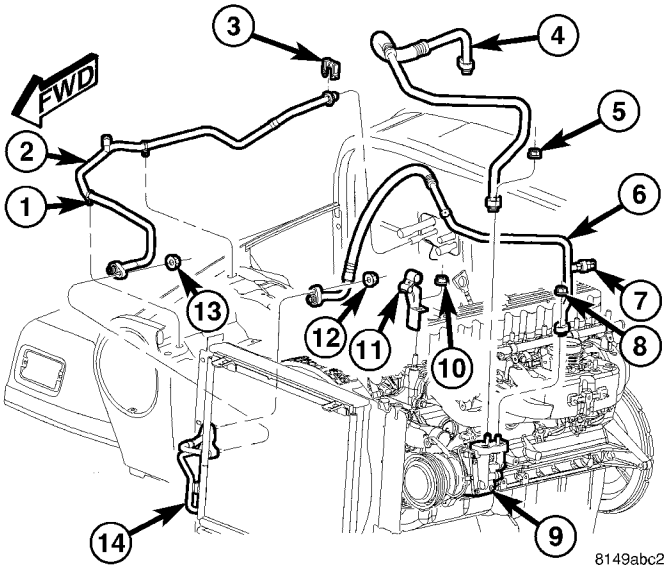

Fig. 11 A/C Discharge Line - 4.0L LHD shown, RHD similar

1 - REFRIGERANT LINE RETAINING CLIP (2)
2 - A/C LIQUID LINE
3 - SECONDARY RETAINING CLIP
4 - A/C SUCTION LINE (FRONT SECTION)
5 - NUT
6 - A/C DISCHARGE LINE
7 - A/C HIGH PRESSURE SWITCH
8 - NUT
9 - A/C COMPRESSOR
10 - NUT
11 - REFRIGERANT LINE BRACKET
12 - NUT
13 - NUT
14 - A/C CONDENSER

INSTALLATION

2.4L ENGINE

NOTE: Replacement of the refrigerant line O-ring seals is required anytime a refrigerant line is opened. Failure to replace the rubber O-ring seals could result in a refrigerant system leak.

A/C DISCHARGE LINE (Continued)

(1) If removed, install the A/C pressure transducer onto the A/C discharge line. Hand-tighten the transducer securely.
(2) Position the A/C discharge line into the engine compartment.
(3) Remove the tape or plugs from the condenser port and the discharge line fitting.
(4) Lubricate a new rubber O-ring seal with clean refrigerant oil and install it on the discharge line fitting. Use only the specified O-ring as it is made of a special material for the R-134a system. Use only refrigerant oil of the type recommended for the A/C compressor in the vehicle.
(5) Connect the A/C discharge line to the A/C condenser.
(6) Install the nut that secures the A/C discharge line to the A/C condenser. Tighten the nut to 12 N·m (105 in. lbs.).
(7) Remove the tape or plugs from the compressor port and the discharge line fitting.
(8) Lubricate a new rubber O-ring seal with clean refrigerant oil and install it on the discharge line fitting. Use only the specified O-ring as it is made of a special material for the R-134a system. Use only refrigerant oil of the type recommended for the A/C compressor in the vehicle.
(9) Connect the A/C discharge line to the A/C compressor.
(10) Install the bolt that secures the A/C discharge line to the A/C compressor. Tighten the bolt to 25 N·m (18 ft. lbs.).
(11) Connect the wire harness connector to the A/C pressure transducer.
(12) Reconnect the negative battery cable.
(13) Evacuate the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM EVACUATE).
(14) Charge the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM CHARGE).

4.0L ENGINE

NOTE: Replacement of the refrigerant line O-ring seals is required anytime a refrigerant line is opened. Failure to replace the rubber O-ring seals could result in a refrigerant system leak.

(1) If removed, install the A/C high pressure switch onto the A/C discharge line. Hand-tighten the switch securely.
(2) Position the A/C discharge line into the engine compartment.
(3) Remove the tape or plugs from the condenser port and the discharge line fitting.
(4) Lubricate a new rubber O-ring seal with clean refrigerant oil and install it on the discharge line fitting. Use only the specified O-ring as it is made of a special material for the R-134a system. Use only refrigerant oil of the type recommended for the A/C compressor in the vehicle.
(5) Connect the A/C discharge line to the A/C condenser.
(6) Install the nut that secures the A/C discharge line to the A/C condenser. Tighten the nut to 12 N·m (105 in. lbs.).
(7) Remove the tape or plugs from the compressor port and the discharge line fitting.
(8) Lubricate a new rubber O-ring seal with clean refrigerant oil and install it on the discharge line fitting. Use only the specified O-ring as it is made of a special material for the R-134a system. Use only refrigerant oil of the type recommended for the A/C compressor in the vehicle.
(9) Connect the A/C discharge line to the A/C compressor.
(10) Install the bolt that secures the A/C discharge line to the A/C compressor. Tighten the bolt to 25 N·m (18 ft. lbs.).
(11) Install the A/C discharge line into the refrigerant line bracket and install the retaining nut. Tighten the nut to 5 N·m (44 in. lbs.).
(12) Connect the wire harness connector to the A/C high pressure switch.
(13) Reconnect the negative battery cable.
(14) Evacuate the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM EVACUATE).
(15) Charge the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM CHARGE).

A/C EVAPORATOR

DESCRIPTION

The A/C evaporator (Fig. 12) for the heating-A/C system is located within the HVAC housing, behind the instrument panel. The A/C evaporator is positioned in the HVAC housing so that all air entering the housing must pass over the evaporator fins before it is distributed through the heating-A/C system ducts and outlets. However, air passing over the evaporator fins will only be conditioned when the A/C compressor is engaged and circulating refrigerant through the A/C evaporator.

The A/C evaporator tubes are connected and sealed to the A/C liquid and suction lines by use of rubber O-rings and quick-connect type fittings

A/C EVAPORATOR (Continued)

The A/C evaporator can only be serviced by removing and disassembling the HVAC housing.

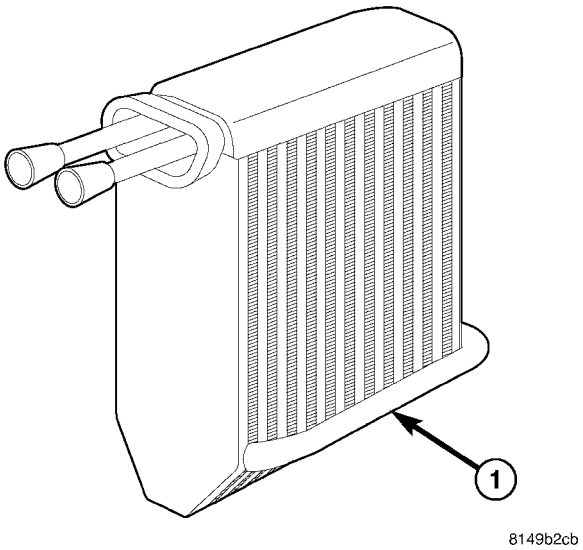

Fig. 12 A/C Evaporator - Typical

1 - A/C EVAPORATOR

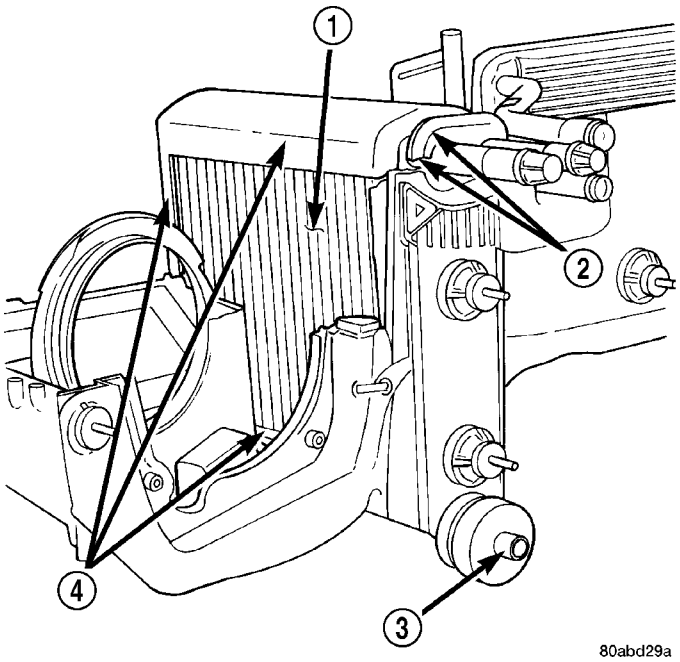

Fig. 13 A/C Evaporator - Typical

1 - A/C EVAPORATOR
2 - RUBBER GROMMET
3 - CONDENSATE DRAIN TUBE
4 - FOAM INSULATOR

OPERATION

Refrigerant enters the A/C evaporator from the A/C orifice tube as a low-temperature, low-pressure mixture of liquid and gas. As air flows over the fins of the A/C evaporator, the humidity in the air condenses on the fins, and the heat from the air is absorbed by the refrigerant. Heat absorption causes the refrigerant to boil and vaporize. The refrigerant becomes a low-pressure gas when it leaves the A/C evaporator.

The A/C evaporator cannot be repaired and, if faulty or damaged, it must be replaced.

REMOVAL

(1) Remove the HVAC housing (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/HVAC HOUSING - REMOVAL).

(2) Disassemble the HVAC housing as necessary to access the A/C evaporator (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/HVAC HOUSING - DISASSEMBLY).

(3) Carefully lift the A/C evaporator and insulator out of the lower half of the HVAC housing (Fig. 13).

(4) If required, remove the rubber grommet and the foam insulator from the A/C evaporator. If the insulator is deformed or damaged, it must be replaced.

INSTALLATION

NOTE: If the A/C evaporator is being replaced, add 45 milliliters (1.5 fluid ounces) of refrigerant oil to the refrigerant system. Use only refrigerant oil of the type recommended for the A/C compressor in the vehicle.

(1) If removed, install the foam insulator over the A/C evaporator. If the insulator is deformed or damaged, it must be replaced.

(2) Install the A/C evaporator into the lower half of the HVAC housing. Make sure that the evaporator drain within the HVAC housing is clean and unrestricted and that the foam insulator and rubber grommet are properly installed.

(3) Assemble the HVAC housing (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/HVAC HOUSING - ASSEMBLY).

(4) Install the HVAC housing (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/HVAC HOUSING - INSTALLATION).

A/C ORIFICE TUBE

DESCRIPTION

The fixed A/C orifice tube is installed in the section of the A/C liquid line closest to the A/C condenser and provides a restriction in the liquid refrigerant line between the A/C condenser and the A/C evaporator. This restriction established the pressure differential between the high and low-pressure sides of the A/C system.

The inlet end of the fixed A/C orifice tube has a nylon mesh filter screen, which filters the refrigerant

A/C ORIFICE TUBE (Continued)

and helps to reduce the potential for blockage of the metering orifice by refrigerant system contaminants (Fig. 14). The outlet end of the tube has a nylon mesh diffuser screen. The O-rings on the plastic body of the tube seal it to the inside of the liquid line and prevent the refrigerant from bypassing the fixed metering orifice.

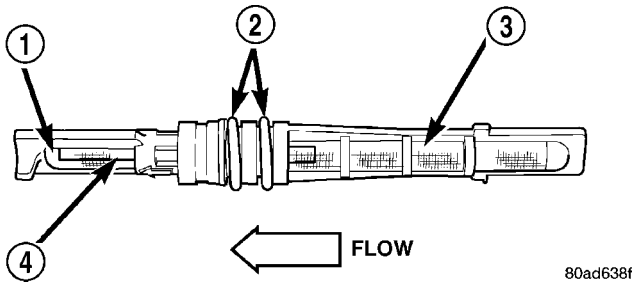

Fig. 14 Fixed A/C Orifice Tube - Typical

1 - DIFFUSER SCREEN
2 - O-RING SEALS
3 - INLET FILTER SCREEN
4 - ORIFICE

OPERATION

The fixed A/C orifice tube is used to meter the flow of liquid refrigerant into the A/C evaporator. The high-pressure liquid refrigerant from the A/C condenser expands into a low-pressure liquid as it passes through the metering orifice and diffuser screen of the A/C orifice tube.

The A/C orifice tube is not serviceable. It cannot be repaired, and if faulty or plugged, it must be replaced as part of the A/C liquid line (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/LIQUID LINE - REMOVAL).

DIAGNOSIS AND TESTING

A/C ORIFICE TUBE

WARNING: The A/C liquid line between the A/C condenser and the A/C orifice tube can become hot enough to burn the skin. Use extreme caution when performing the following test to prevent possible personal injury.

NOTE: The A/C orifice tube can be checked for proper operation using the following procedure. However, the A/C orifice tube is only serviced as a part of the A/C liquid line. If the results of this test indicate that the A/C orifice tube is obstructed or missing, the A/C liquid line must be replaced.

(1) Confirm that the refrigerant system is properly charged (Refer to 24 - HEATING & AIR CONDITIONING - DIAGNOSIS AND TESTING - A/C PERFORMANCE TEST).

(2) Start the engine. Turn on the A/C system and confirm that the compressor clutch is engaged.

(3) Allow the A/C system to operate for five minutes.

(4) Lightly and cautiously touch the A/C liquid line near the condenser outlet at the front of the engine compartment. The liquid line should be hot to the touch.

(5) Touch the A/C liquid line near the evaporator inlet at the rear of the engine compartment. The liquid line should be cold to the touch.

(6) If there is a distinct temperature differential between the two ends of the A/C liquid line, the A/C orifice tube is in good condition. If there is little or no detectable temperature differential between the two ends of the A/C liquid line, the A/C orifice tube is obstructed or missing and the A/C liquid line must be replaced (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/A/C LIQUID LINE - REMOVAL).

ACCUMULATOR

DESCRIPTION

The A/C accumulator (Fig. 15) is mounted in the engine compartment between the A/C evaporator and the A/C compressor. An integral mounting bracket is used to secure the A/C accumulator to the frame rail. The A/C low pressure switch is mounted on a fitting located on the top of the A/C accumulator.

The A/C accumulator cannot be repaired and, if faulty or damaged, it must be replaced. The rubber O-rings used on accumulator connections and the A/C low pressure switch are available for service replacement.

OPERATION

Refrigerant enters the A/C accumulator mostly as a low pressure vapor through the inlet tube. Any liquid, oil-laden refrigerant falls to the bottom of the canister, which acts as a separator. A desiccant bag is mounted inside the accumulator canister to absorb any moisture which may have entered and become trapped within the refrigerant system A filter is also mounted inside the canister to trap any foreign material that may have entered the refrigerant system during assembly. The low pressure vapor exits the A/C accumulator through the outlet tube.

The A/C accumulator cannot be repaired. If the A/C accumulator is faulty or damaged, or if the refrigerant system has been contaminated or left open to the atmosphere for an indeterminable period or if the A/C compressor has failed, it must be replaced.

ACCUMULATOR (Continued)

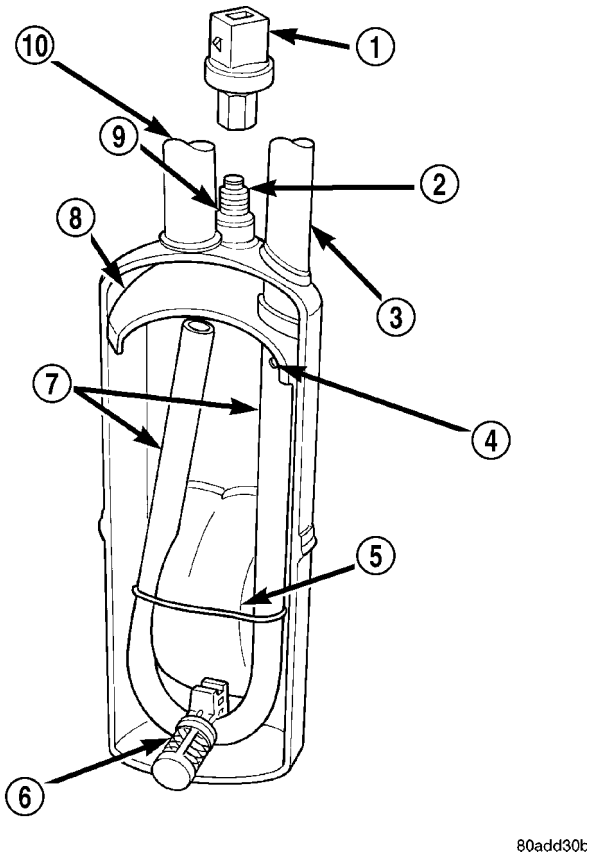

Fig. 15 Accumulator - Typical

1 - A/C LOW PRESSURE SWITCH
2 - PRESSURE SWITCH FITTING
3 - OUTLET TO COMPRESSOR
4 - ANTI-SIPHON HOLE
5 - DESICCANT BAG
6 - OIL RETURN ORIFICE FILTER
7 - VAPOR RETURN TUBE
8 - ACCUMULATOR DOME
9 - O-RING SEAL
10 - INLET FROM EVAPORATOR

REMOVAL

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

(1) Disconnect and isolate the negative battery cable.

(2) Recover the refrigerant from the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM RECOVERY).

(3) Disconnect the wire harness connector from the A/C low pressure switch.

(4) If required, remove the A/C low pressure switch from the accumulator and remove and discard the O-ring seal.

(5) Remove the plastic retaining clip that secures the suction line to the liquid line near the accumulator.

(6) Loosen the bolt that secures the accumulator retaining band to the support bracket located on the dash panel.

(7) Remove the nuts that secure the refrigerant lines to the accumulator (Fig. 16).

(8) Disconnect the refrigerant lines from the accumulator and remove and discard the O-ring seals.

(9) Install plugs in, or tape over the opened refrigerant line fittings and accumulator ports.

(10) Pull the accumulator and retaining band forward until the bolt in the band is clear of the slotted hole in the support bracket.

(11) Remove the accumulator from the engine compartment and if necessary, remove the retaining band from the accumulator.

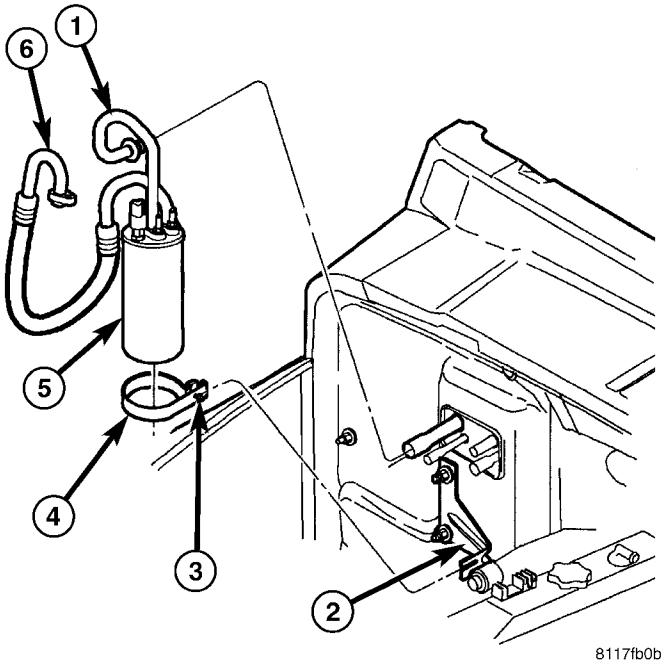

Fig. 16 A/C Accumulator - Typical

1 - SUCTION LINE (EVAPORATOR SIDE)
2 - SUPPORT BRACKET
3 - BOLT
4 - RETAINING BAND
5 - ACCUMULATOR
6 - SUCTION LINE (COMPRESSOR SIDE)

ACCUMULATOR (Continued)

INSTALLATION

NOTE: If the A/C accumulator is being replaced, add 90 milliliters (3 fluid ounces) of refrigerant oil to the refrigerant system. Use only refrigerant oil of the type recommended for the A/C compressor in the vehicle.

(1) If removed, position the accumulator into the retaining band and loosely install the bolt.

(2) Position the accumulator and retaining band into the slotted hole of the support bracket on the dash panel.

(3) Remove the tape or plugs from the opened refrigerant line fittings and the accumulator ports.

(4) Lubricate new rubber O-ring seals with clean refrigerant oil and install them on the refrigerant line fittings. Use only the specified O-rings as they are made of a special material for the R-134a system. Use only refrigerant oil of the type recommended for the A/C compressor in the vehicle.

(5) Connect the refrigerant lines to the accumulator.

(6) Install the nuts that secure the refrigerant lines to the accumulator. Tighten the nuts to 9 N·m (80 in. lbs.).

(7) Tighten the bolt that secures the accumulator retaining band to the support bracket. Tighten the bolt to 4.5 N·m (40 in. lbs.).

(8) Install the plastic retaining clip that secures the A/C suction line to the liquid line near the accumulator.

(9) If removed, install the A/C low pressure switch onto the accumulator using a new O-ring seal. Hand-tighten the switch securely.

(10) Connect the wire harness connector to the A/C low pressure switch.

(11) Reconnect the negative battery cable.

(12) Evacuate the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM EVACUATE).

(13) Charge the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM CHARGE).

(14) Charge the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE).

HEATER CORE

DESCRIPTION

The heater core (Fig. 17) is mounted in the HVAC housing, located behind the instrument panel. The heater core is a heat exchanger made of rows of tubes and fins. The heater core inlet and outlet tubes are integral to the heater core and cannot be serviced separately.

The heater core can be serviced by removing the HVAC housing from the vehicle.

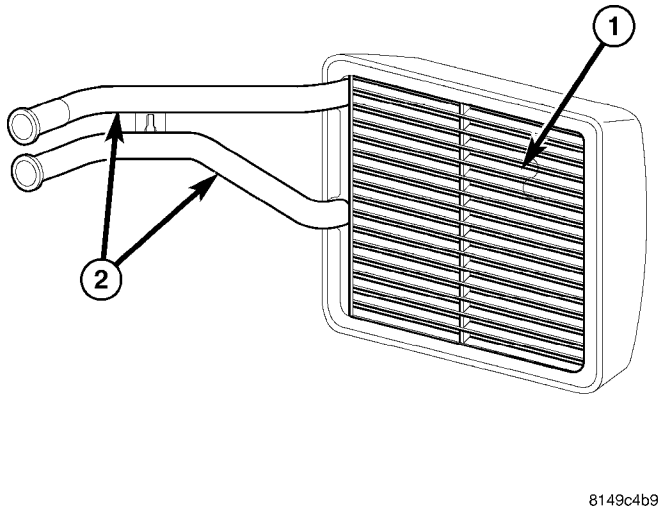

Fig. 17 Heater Core - Typical

1 - HEATER CORE
2 - INLET AND OUTLET TUBES

OPERATION

Engine coolant is circulated through the heater hoses and heater core at all times. As the coolant flows through the heater core, heat removed from the engine is transferred to the heater core fins and tubes. Air directed through the heater core picks up the heat from the heater core fins. The blend door allows control of the heater output air temperature by regulating the amount of air that is flowing through the heater core within the HVAC housing. The blower motor speed controls the volume of air flowing through the HVAC housing.

The heater core cannot be repaired and, if faulty or damaged, it must be replaced.

HEATER CORE (Continued)

REMOVAL

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the negative battery (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. This is the only sure way to disable the airbag system. Failure to take the proper precautions could result in accidental airbag deployment and possible personal injury or death.

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

(1) Remove the HVAC housing (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/HVAC HOUSING - REMOVAL).

(2) Disassemble the HVAC housing as necessary to access the heater core (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/HVAC HOUSING - DISASSEMBLY).

(3) Lift the heater core out of the lower half of the HVAC housing (Fig. 18).

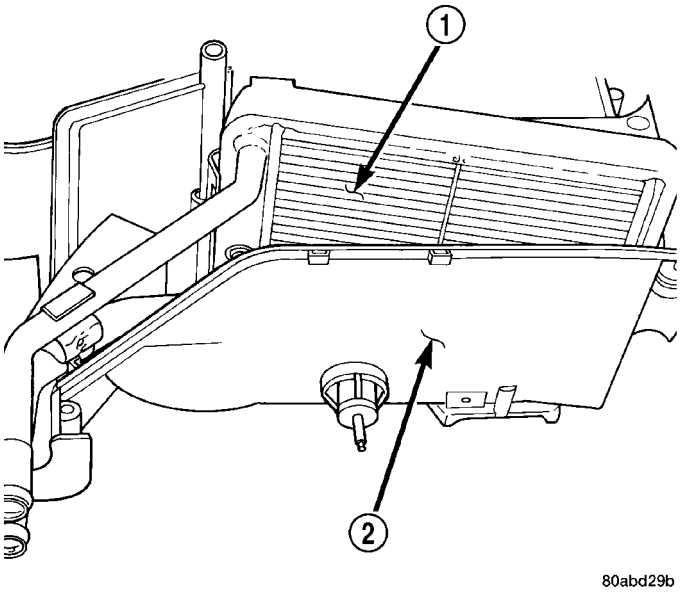

Fig. 18 Heater Core - LHD shown, RHD similar

1 - HEATER CORE
2 - HVAC HOUSING

INSTALLATION

(1) Install the heater core into the bottom half of the HVAC housing.

(2) Assemble the HVAC housing (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/HVAC HOUSING - ASSEMBLY).

NOTE: If the heater core is being replaced, flush the cooling system (Refer to 7 - COOLING - STANDARD PROCEDURE - COOLING SYSTEM CLEANING/REVERSE FLUSHING).

(3) Install the HVAC housing (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/HVAC HOUSING - INSTALLATION).

LIQUID LINE

DESCRIPTION

The A/C liquid line is the refrigerant line that carries refrigerant from the A/C condenser to the A/C evaporator. The A/C liquid contains the fixed A/C orifice tube and the low-side service port.

CAUTION: Use only O-ring seals specified for the vehicle. Failure to use the correct O-ring seals will cause the refrigerant system connections to leak.

The A/C liquid line has no serviceable parts except for the O-ring seals and low-side service port valve and it's protective cap.

If the A/C liquid line is found to be leaking or damaged, it must be replaced.

REMOVAL

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

(1) Disconnect and isolate the negative battery cable.

(2) Recover the refrigerant from the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM RECOVERY).

(3) Depending on application, remove the engine air cleaner housing (Refer to 9 - ENGINE/AIR INTAKE SYSTEM/AIR CLEANER HOUSING - REMOVAL).

LIQUID LINE (Continued)

(4) Remove the secondary retaining clip from the spring-lock coupler that secures the liquid line to the A/C evaporator (Fig. 19) or (Fig. 20).

(5) Using the proper A/C line disconnect tool, disconnect the liquid line from the A/C evaporator and remove and discard the O-ring seal (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/REFRIGERANT LINE COUPLER - REMOVAL).

(6) Install plugs in, or tape over the opened liquid line fitting and the evaporator port.

(7) Remove the liquid line from the retaining clips located on the right inner fender.

(8) Remove the nut that secures the liquid line to the A/C condenser.

(9) Disconnect the liquid line from the A/C condenser and remove and discard the O-ring seal.

(10) Install plugs in, or tape over the opened liquid line fitting and the condenser port.

(11) On LHD models, remove the liquid line from the engine compartment. On RHD models, proceed to Step 12.

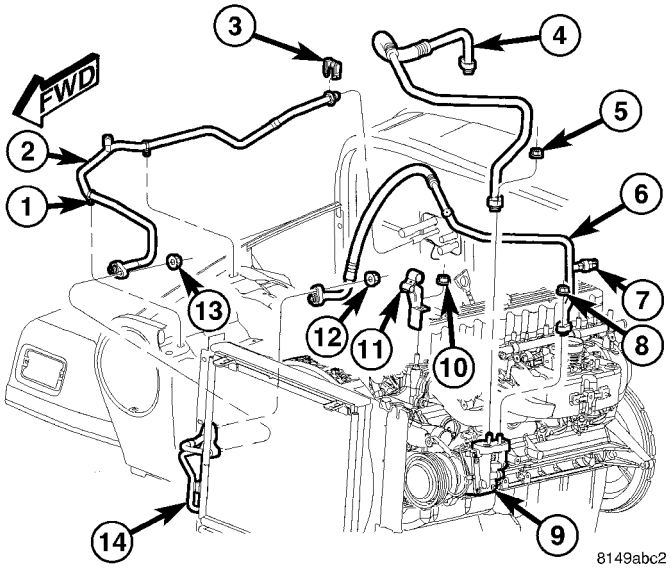

Fig. 19 A/C Liquid Line - 4.0L LHD shown, 2.4L similar

1 - REFRIGERANT LINE RETAINING CLIP (2)
2 - A/C LIQUID LINE
3 - SECONDARY RETAINING CLIP
4 - A/C SUCTION LINE (FRONT SECTION)
5 - NUT
6 - A/C DISCHARGE LINE
7 - A/C HIGH PRESSURE SWITCH
8 - NUT
9 - A/C COMPRESSOR
10 - NUT
11 - REFRIGERANT LINE BRACKET
12 - NUT
13 - NUT
14 - A/C CONDENSER

(12) On RHD models, remove the rear section of the liquid line from the retaining clips located on the dash panel (Fig. 20).

(13) Remove the secondary retaining clip from the spring-lock coupler that secures the rear section of the liquid line to the front section of the liquid line.

(14) Using the proper A/C line disconnect tool, disconnect the rear section of the liquid line from the front section of the liquid line and remove and discard the O-ring seal (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/REFRIGERANT LINE COUPLER - REMOVAL).

(15) Install plugs in, or tape over the two opened liquid line fittings.

(16) Remove the front and rear section of the liquid line from the engine compartment.

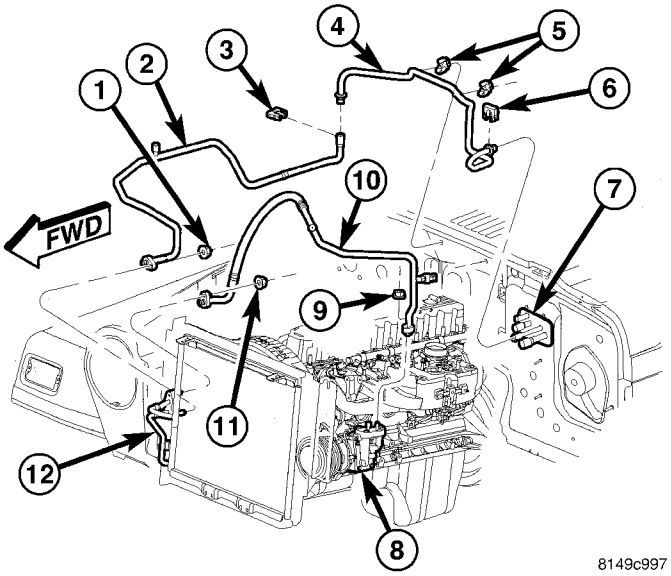

Fig. 20 A/C Liquid Line - 4.0L RHD

1 - NUT
2 - A/C LIQUID LINE (FRONT SECTION)
3 - SECONDARY RETAINING CLIP
4 - A/C LIQUID LINE (REART SECTION)
5 - REFRIGERANT LINE RETAINING CLIPS
6 - SECONDARY RETAINING CLIP
7 - A/C EVAPORATOR
8 - A/C COMPRESSOR
9 - NUT
10 - A/C DISCHARGE L;INE
11 - NUT
12 - A/C CONDENSER

INSTALLATION

NOTE: Replacement of the refrigerant line O-ring seals is required anytime a refrigerant line is opened. Failure to replace the rubber O-ring seals could result in a refrigerant system leak.

(1) Position the liquid line into the engine compartment.

LIQUID LINE (Continued)

(2) Remove the tape or plugs from the liquid line fitting and the condenser port.

(3) Lubricate a new rubber O-ring seal with clean refrigerant oil and install it on the liquid line fitting. Use only the specified O-rings as they are made of a special material for the R-134a system. Use only refrigerant oil of the type recommended for the compressor in the vehicle.

(4) Connect the liquid line to the A/C condenser.

(5) Install the nut that secures the liquid line to the A/C condenser. Tighten the nut to 12 N·m (105 in. lbs.).

(6) Install the liquid line into the plastic retaining clips located on the right inner fender.

(7) Remove the tape or plugs from the liquid line fitting and the evaporator port.

(8) Lubricate a new rubber O-ring seal with clean refrigerant oil and install it on the liquid line quick-connect fitting. Use only the specified O-rings as they are made of a special material for the R-134a system. Use only refrigerant oil of the type recommended for the compressor in the vehicle.

(9) Connect the liquid line to the A/C evaporator (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/REFRIGERANT LINE COUPLER - INSTALLATION).

(10) Install the secondary retaining clip onto the spring-lock coupler that secures the liquid line to the A/C evaporator.

(11) On LHD models, go to Step 17. On RHD models, proceed to Step 12.

(12) On RHD models, remove the tape or plugs from the two liquid line fittings.

(13) Lubricate a new rubber O-ring seal with clean refrigerant oil and install it on the liquid line quick-connect fitting. Use only the specified O-rings as they are made of a special material for the R-134a system. Use only refrigerant oil of the type recommended for the compressor in the vehicle.

(14) Connect the rear section of the liquid line to the front section of the liquid line (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/REFRIGERANT LINE COUPLER - INSTALLATION).

(15) Install the secondary retaining clip onto the spring-lock coupler that secures the rear section of the liquid line to the front section of the liquid line.

(16) Install the rear section of the liquid line into the retaining clips located on the dash panel.

(17) If removed, install the engine air cleaner housing (Refer to 9 - ENGINE/AIR INTAKE SYSTEM/AIR CLEANER HOUSING - INSTALLATION).

(18) Reconnect the negative battery cable.

(19) Evacuate the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM EVACUATE).

(20) Charge the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM CHARGE).

REFRIGERANT

DESCRIPTION

The refrigerant used in this air conditioning system is a HydroFluoroCarbon (HFC), type R-134a. Unlike R-12, which is a ChloroFluoroCarbon (CFC), R-134a refrigerant does not contain ozone-depleting chlorine. R-134a refrigerant is a non-toxic, non-flammable, clear, and colorless liquefied gas.

Even though R-134a does not contain chlorine, it must be reclaimed and recycled just like CFC-type refrigerants. This is because R-134a is a greenhouse gas and can contribute to global warming.

OPERATION

R-134a refrigerant is not compatible with R-12 refrigerant in an air conditioning system. Even a small amount of R-12 added to an R-134a refrigerant system will cause compressor failure, refrigerant oil sludge or poor air conditioning system performance. In addition, the PolyAlkylene Glycol (PAG) synthetic refrigerant oils used in an R-134a refrigerant system are not compatible with the mineral-based refrigerant oils used in an R-12 refrigerant system.

R-134a refrigerant system service ports, service tool couplers and refrigerant dispensing bottles have all been designed with unique fittings to ensure that an R-134a refrigerant system is not accidentally contaminated with the wrong refrigerant (R-12). There are also labels posted in the engine compartment of the vehicle and on the A/C compressor to identify that the A/C system is equipped with R-134a refrigerant.

REFRIGERANT LINE COUPLER

DESCRIPTION

Spring-lock type refrigerant line couplers are used to connect the A/C liquid line to the A/C evaporator and, on RHD models the front section of the liquid line to the rear section of the liquid line. Secondary retaining clips are installed over the connected couplers for added protection.

The spring-lock refrigerant line couplers require special disconnect tools for disengaging the two coupler halves.

REFRIGERANT LINE COUPLER (Continued)

OPERATION

The spring-lock coupler is held together by a garter spring inside a circular cage on the male half of the fitting (Fig. 21). When the two coupler halves are connected, the flared end of the female fitting slips behind the garter spring inside the cage on the male fitting. The garter spring and cage prevent the flared end of the female fitting from pulling out of the cage.

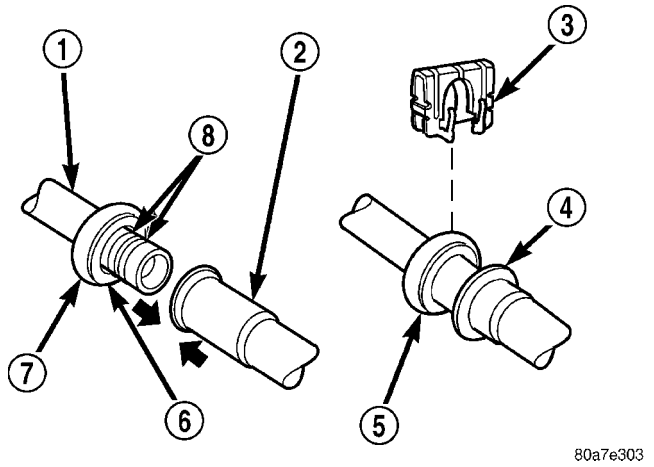

Fig. 21 Spring-Lock Coupler - Typical

1 - MALE HALF SPRING-LOCK COUPLER
2 - FEMALE HALF SPRING-LOCK COUPLER
3 - SECONDARY CLIP
4 - CONNECTION INDICATOR RING
5 - COUPLER CAGE
6 - GARTER SPRING
7 - COUPLER CAGE
8 - O-RING SEALS

Two O-rings on the male half of the fitting are used to seal the connection. These O-rings are compatible with R-134a refrigerant and must be replaced with O-rings made of the same material.

Secondary clips are installed over the two connected coupler halves at the factory for added blowoff protection. In addition, some models have a plastic ring that is used at the factory as a visual indicator to confirm that these couplers are connected. After the coupler is connected, the plastic indicator ring is no longer needed; however, it will remain on the refrigerant line near the coupler cage.

REMOVAL

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

(1) Recover the refrigerant from the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM RECOVERY).

(2) Remove the secondary retaining clip from the spring-lock coupler.

(3) Fit the proper size A/C line disconnect tool (Special Tool Kit 7193) over the spring-lock coupler cage (Fig. 22).

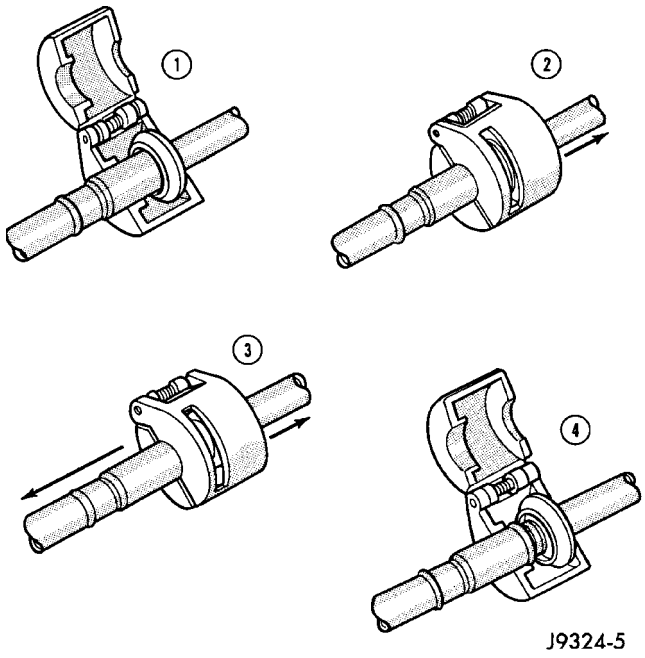

Fig. 22 Refrigerant Line Spring-Lock Coupler Disconnect

(4) Close the two halves of the A/C line disconnect tool around the spring-lock coupler.

(5) Push the A/C line disconnect tool into the open side of the coupler cage to expand the garter spring. Once the garter spring is expanded and while still pushing the disconnect tool into the open side of the coupler cage, pull on the refrigerant line attached to the female half of the coupler fitting until the flange on the female fitting is separated from the garter spring and cage on the male fitting within the disconnect tool.

NOTE: The garter spring may not release if the A/C line disconnect tool is cocked while pushing it into the coupler cage opening.

(6) Open and remove the A/C line disconnect tool from the disconnected spring-lock coupler.

(7) Complete the separation of the two halves of the coupler fitting.

REFRIGERANT LINE COUPLER (Continued)

INSTALLATION

(1) Check to ensure that the garter spring is located within the cage of the male coupler fitting, and that the garter spring is not damaged.

(a) If the garter spring is missing, install a new spring by pushing it into the coupler cage opening.

(b) If the garter spring is damaged, remove it from the coupler cage with a small wire hook (DO NOT use a screwdriver) and install a new garter spring.

(2) Clean any dirt or foreign material from both halves of the coupler fitting.

(3) Install new O-rings on the male half of the coupler fitting.

CAUTION: Use only the specified O-rings as they are made of a special material for the R-134a system. The use of any other O-rings may allow the connection to leak intermittently during vehicle operation.

(4) Lubricate the male fitting and O-rings, and the inside of the female fitting with clean R-134a refrigerant oil. Use only refrigerant oil of the type recommended for the compressor in the vehicle.

(5) Fit the female half of the coupler fitting over the male half of the fitting.

(6) Push together firmly on the two halves of the coupler fitting until the garter spring in the cage on the male half of the fitting snaps over the flanged end on the female half of the fitting.

(7) Ensure that the spring-lock coupler is fully engaged by trying to separate the two coupler halves. This is done by pulling the refrigerant lines on either side of the coupler away from each other.

(8) Reinstall the secondary retaining clip over the spring-lock coupler cage.

(9) Evacuate the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM EVACUATE).

(10) Charge the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM CHARGE).

REFRIGERANT OIL

DESCRIPTION

The refrigerant oil used in R-134a refrigerant systems is a synthetic-based, PolyAlkylene Glycol (PAG), wax-free lubricant. Mineral-based R-12 refrigerant oils are not compatible with PAG oils, and should never be introduced to an R-134a refrigerant system.

There are different PAG oils available, and each contains a different additive package. The 10PA17 compressor used in this vehicle is designed to use an ND-8 PAG refrigerant oil. Use only refrigerant oil of this same type to service the refrigerant system.

OPERATION

After performing any refrigerant recovery or recycling operation, always replenish the refrigerant system with the same amount of the recommended refrigerant oil as was removed. Too little refrigerant oil can cause A/C compressor damage, and too much can reduce A/C system performance.

PAG refrigerant oil is more hygroscopic than mineral oil, and will absorb any moisture it comes into contact with, even moisture in the air. The PAG oil container should always be kept tightly capped until it is ready to be used. After use, recap the oil container immediately to prevent moisture contamination.

STANDARD PROCEDURE

REFRIGERANT OIL LEVEL

WARNING: Review safety precautions and warnings in this group before performing this procedure (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

CAUTION: The oil used in the A/C compressor is ND-8 PAG R-134a refrigerant oil. Only refrigerant oil of the same type should be used to service the A/C system. Do not use any other refrigerant oil. The oil container should be kept tightly capped until it is ready for use and then tightly capped after use to prevent contamination from dirt and moisture. Refrigerant oil will quickly absorb any moisture it comes in contact with, therefore, special effort must be used to keep all R-134a system components moisture-free. Moisture in the refrigerant oil is very difficult to remove and will cause a reliability problem with the A/C compressor.

NOTE: Most reclaim/recycling equipment will measure the lubricant being removed during recovery. This amount of lubricant should be added back into the system. Refer to the reclaim/recycling equipment manufacturers instructions.

REFRIGERANT OIL (Continued)

It will not be necessary to check the oil level in the A/C compressor or to add oil, unless there has been an oil loss. An oil loss may occur due to a rupture or leak from a refrigerant line, a connector fitting, a component, or a component seal. If a leak occurs, add 30 milliliters (1 fluid ounce) of refrigerant oil to the refrigerant system after the repair has been made. Refrigerant oil loss will be evident at the leak point by the presence of a wet, shiny surface around the leak.

Refrigerant oil must be added when an accumulator, A/C evaporator or A/C condenser is replaced. See the Refrigerant Oil Capacities chart. When an A/C compressor is replaced, the refrigerant oil must be drained from the old compressor and measured. Drain all of the refrigerant oil from the new A/C compressor, then fill the new compressor with the same amount of refrigerant oil that was drained out of the old compressor.

REFRIGERANT OIL CAPACITIES

Component	ml	oz
Total System Fill	180	6.1
Accumulator	90	3.0
A/C Condenser	22	0.75
A/C Evaporator	45	1.5
A/C Compressor	Drain and measure the oil from the old compressor - see text.	

SERVICE PORT VALVE CORE

DESCRIPTION

The A/C service port valve cores are serviceable items. The high side valve is located on the located on the manifold directly over the discharge port of the A/C compressor. The low side valve is located on the liquid line near the evaporator inlet tube at the rear of the engine compartment.

Refrigerant system service ports are used to recover, recycle, evacuate, charge and test the A/C refrigerant system. Unique sizes are used on the two service ports for the R-134a refrigerant system to ensure the system is not accidentally contaminated with R-12 refrigerant or by service equipment used for R-12 refrigerant.

The high-side service port is located on the A/C discharge. The low-side service port is located on the A/C liquid line. Both the high-side and low-side A/C service port valve cores are serviceable.

NOTE: The protective cap aids in service port sealing and helps protects the refrigerant system from contamination. Remember to always reinstall the protective cap onto the service port when refrigerant system service is complete.

Each of the service ports has a threaded plastic protective cap installed over it from the factory. The service port caps are serviceable items.

REMOVAL

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

(1) Remove the protective cap from the service port (Fig. 23).

(2) Recover the refrigerant from the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM RECOVERY).

(3) Using a Schrader-type valve core tool, remove the valve core from the service port.

(4) Install a plug in or tape over the opened service port(s).

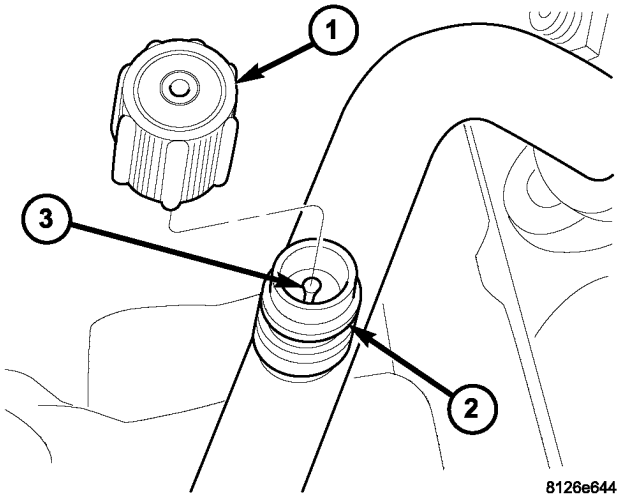

Fig. 23 A/C Service Port-Typical

1 - PROTECTIVE CAP
2 - SERVICE PORT
3 - VALVE CORE

INSTALLATION

(1) Lubricate the valve core with clean refrigerant oil prior to installation. Use only refrigerant oil of the type recommended for the A/C compressor in the vehicle.

SERVICE PORT VALVE CORE (Continued)

(2) Remove the tape or plug from the service port.

CAUTION: A valve core that is not fully seated in the A/C service port can result in damage to the valve during refrigerant system evacuation and charge. Such damage may result in a loss of system refrigerant while uncoupling the charge adapters.

(3) Using a Schrader-type valve core tool, install and tighten the valve core into the service port(s).

(4) Evacuate the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM EVACUATE).

(5) Charge the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM CHARGE).

NOTE: The protective cap helps aid in service port sealing and helps protects the refrigerant system from contamination. Remember to always reinstall the protective cap onto the service port when refrigerant system service is complete.

(6) Install the protective cap onto the service port.

SUCTION LINE

DESCRIPTION

The A/C suction line is the refrigerant line that goes from the A/C evaporator to the A/C accumulator, then to the A/C compressor. The A/C suction line is serviced in two sections.

CAUTION: Use only O-ring seals specified for the vehicle. Failure to use the correct O-ring seal will cause the refrigerant system connections to leak.

The A/C suction lines have no serviceable parts except for the rubber O-ring seals.

If an A/C suction line is found to be leaking or is damaged, it must be replaced.

REMOVAL

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

(1) Disconnect and isolate the negative battery cable.

(2) Recover the refrigerant from the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM RECOVERY).

(3) Remove the suction line from the retaining clip located near the accumulator.

(4) Remove the secondary retaining clip from the spring-lock coupler that secures the suction line to the A/C evaporator

(5) Using the proper A/C line disconnect tool, disconnect the suction line from the A/C evaporator and remove and discard the O-ring seal (Fig. 24) (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/REFRIGERANT LINE COUPLER - REMOVAL).

(6) Install plugs in, or tape over the opened suction line fitting and the evaporator port.

(7) Remove the nuts that secure the suction lines to the accumulator and to the A/C compressor.

(8) Disconnect the suction lines from the accumulator and the A/C compressor.

(9) Install plugs in, or tape over the opened suction line fittings and the accumulator and compressor ports.

(10) Remove the suction lines from the vehicle.

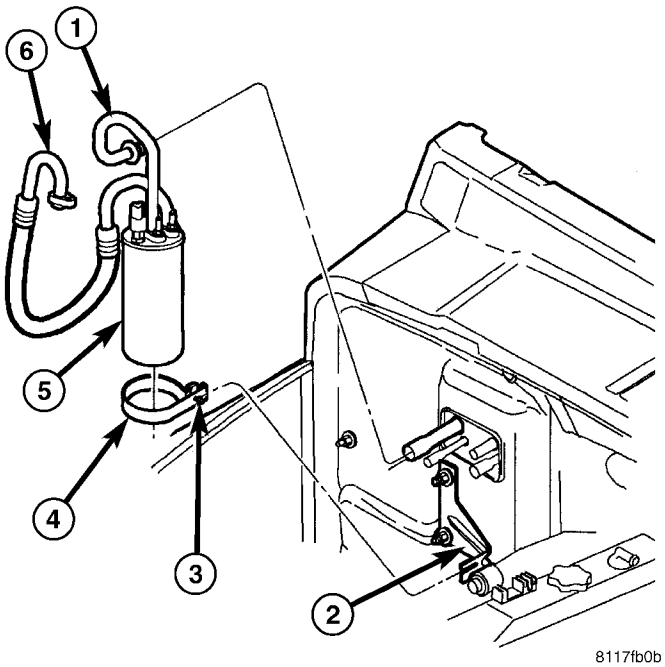

Fig. 24 Suction Lines - 4.0L LHD shown, 2.4L and RHD similar

1 - SUCTION LINE (EVAPORATOR SIDE)
2 - SUPPORT BRACKET
3 - BOLT
4 - RETAINING BAND
5 - ACCUMULATOR
6 - SUCTION LINE (COMPRESSOR SIDE)

SUCTION LINE (Continued)

INSTALLATION

NOTE: Replacement of the refrigerant line O-ring seals is required anytime a refrigerant line is opened. Failure to replace the rubber O-ring seals could result in a refrigerant system leak.

(1) Position the A/C suction line into the engine compartment, as required.

(2) Remove the tape or plugs from the opened suction line fittings and the A/C accumulator, A/C compressor and A/C evaporator as required.

(3) Lubricate new rubber O-ring seals with clean refrigerant oil and install them on the suction line fittings. Use only the specified O-ring seals as they are made of a special material for the R-134a system. Use only refrigerant oil of the type recommended for the A/C compressor in the vehicle.

(4) Connect the suction lines onto the A/C accumulator and A/C compressor, as required.

(5) Install the nuts that secure the suction lines to the accumulator. Tighten the nuts to 9 N·m (80 in. lbs.).

(6) Install the nut that secures the suction line to the A/C compressor. Tighten the nut to 25 N·m (18 ft. lbs.).

(7) Connect the suction line spring-lock coupler to the A/C evaporator (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/REFRIGERANT LINE COUPLER - INSTALLATION).

(8) Install the secondary retaining clip onto the spring-lock coupler that secures the suction line to the A/C evaporator.

(9) Install the suction line into the retaining clip located near the accumulator.

(10) Reconnect the negative battery cable.

(11) Evacuate the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM EVACUATE).

(12) Charge the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM CHARGE).

EMISSIONS CONTROL

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EMISSIONS CONTROL

DESCRIPTION

DESCRIPTION - EMISSION CONTROL SYSTEM

The Powertrain Control Module (PCM) monitors many different circuits in the fuel injection, ignition, emission and engine systems. If the PCM senses a problem with a monitored circuit often enough to indicate an actual problem, it stores a Diagnostic Trouble Code (DTC) in the PCM's memory. If the code applies to a non-emissions related component or system, and the problem is repaired or ceases to exist, the PCM cancels the code after 40 warm-up cycles. Diagnostic trouble codes that affect vehicle emissions illuminate the Malfunction Indicator Lamp (MIL). The MIL is displayed as an engine icon on the instrument panel. Refer to Malfunction Indicator Lamp (MIL) in this section.

Certain criteria must be met before the PCM stores a DTC in memory. The criteria may be a specific range of engine RPM, engine temperature, and/or input voltage to the PCM.

The PCM might not store a DTC for a monitored circuit even though a malfunction has occurred. This may happen because one of the DTC criteria for the circuit has not been met. **For example** , assume the diagnostic trouble code criteria requires the PCM to monitor the circuit only when the engine operates between 750 and 2000 RPM. Suppose the sensor's output circuit shorts to ground when engine operates above 2400 RPM (resulting in 0 volt input to the PCM). Because the condition happens at an engine speed above the maximum threshold (2000 rpm), the PCM will not store a DTC.

There are several operating conditions for which the PCM monitors and sets DTC's. Refer to Monitored Systems, Components, and Non-Monitored Circuits in this section.

Technicians must retrieve stored DTC's by connecting the DRB scan tool (or an equivalent scan tool) to the 16–way data link connector (Fig. 1).

NOTE: Various diagnostic procedures may actually cause a diagnostic monitor to set a DTC. For instance, pulling a spark plug wire to perform a spark test may set the misfire code. When a repair is completed and verified, connect the DRB scan tool to the 16–way data link connector to erase all DTC's and extinguish the MIL.

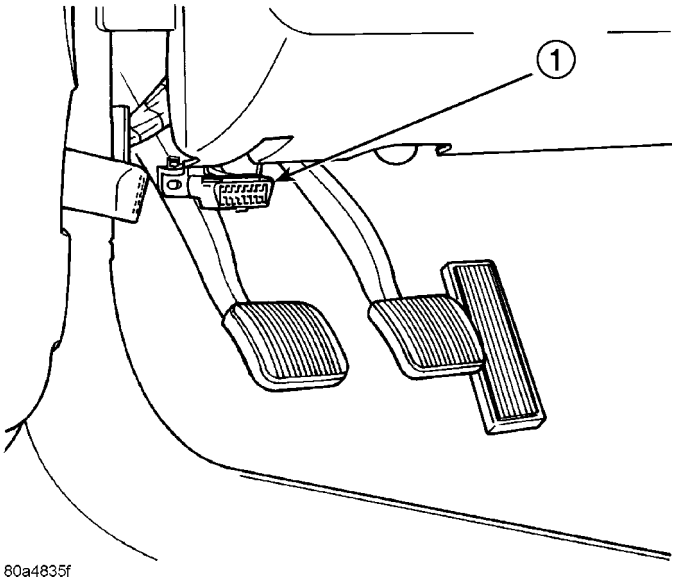

Fig. 1 DATA LINK (DIAGNOSTIC) CONNECTOR LOCATION

1 - 16–WAY DATA LINK CONNECTOR

EMISSIONS CONTROL (Continued)

DESCRIPTION - STATE DISPLAY TEST MODE

The switch inputs to the Powertrain Control Module (PCM) have two recognized states; HIGH and LOW. For this reason, the PCM cannot recognize the difference between a selected switch position versus an open circuit, a short circuit, or a defective switch. If the State Display screen shows the change from HIGH to LOW or LOW to HIGH, assume the entire switch circuit to the PCM functions properly. Connect the DRB scan tool to the data link connector and access the state display screen. Then access either State Display Inputs and Outputs or State Display Sensors.

DESCRIPTION - CIRCUIT ACTUATION TEST MODE

The Circuit Actuation Test Mode checks for proper operation of output circuits or devices the Powertrain Control Module (PCM) may not internally recognize. The PCM attempts to activate these outputs and allow an observer to verify proper operation. Most of the tests provide an audible or visual indication of device operation (click of relay contacts, fuel spray, etc.). Except for intermittent conditions, if a device functions properly during testing, assume the device, its associated wiring, and driver circuit work correctly. Connect the DRB scan tool to the data link connector and access the Actuators screen.

DESCRIPTION - DIAGNOSTIC TROUBLE CODES

A Diagnostic Trouble Code (DTC) indicates the PCM has recognized an abnormal condition in the system.

Remember that DTC's are the results of a system or circuit failure, but do not directly identify the failed component or components.

BULB CHECK

Each time the ignition key is turned to the ON position, the malfunction indicator (check engine) lamp on the instrument panel should illuminate for approximately 2 seconds then go out. This is done for a bulb check.

OBTAINING DTC'S USING DRB SCAN TOOL

(1) Obtain the applicable Powertrain Diagnostic Manual.

(2) Obtain the DRB Scan Tool.

(3) Connect the DRB Scan Tool to the data link (diagnostic) connector. This connector is located in the passenger compartment; at the lower edge of instrument panel; near the steering column.

(4) Turn the ignition switch on and access the "Read Fault" screen.

(5) Record all the DTC's and "freeze frame" information shown on the DRB scan tool.

(6) To erase DTC's, use the "Erase Trouble Code" data screen on the DRB scan tool. **Do not erase any DTC's until problems have been investigated and repairs have been performed.**

DESCRIPTION - TASK MANAGER

The PCM is responsible for efficiently coordinating the operation of all the emissions-related components. The PCM is also responsible for determining if the diagnostic systems are operating properly. The software designed to carry out these responsibilities is referred to as the 'Task Manager'.

DESCRIPTION - MONITORED SYSTEMS

There are new electronic circuit monitors that check fuel, emission, engine and ignition performance. These monitors use information from various sensor circuits to indicate the overall operation of the fuel, engine, ignition and emission systems and thus the emissions performance of the vehicle.

The fuel, engine, ignition and emission systems monitors do not indicate a specific component problem. They do indicate that there is an implied problem within one of the systems and that a specific problem must be diagnosed.

If any of these monitors detect a problem affecting vehicle emissions, the Malfunction Indicator Lamp (MIL) will be illuminated. These monitors generate Diagnostic Trouble Codes that can be displayed with the MIL or a scan tool.

The following is a list of the system monitors:

- Misfire Monitor
- Fuel System Monitor
- Oxygen Sensor Monitor
- Oxygen Sensor Heater Monitor
- Catalyst Monitor
- Leak Detection Pump Monitor (if equipped)

All these system monitors require two consecutive trips with the malfunction present to set a fault.

Refer to the appropriate Powertrain Diagnostics Procedures manual for diagnostic procedures.

The following is an operation and description of each system monitor:

OXYGEN SENSOR (O2S) MONITOR

Effective control of exhaust emissions is achieved by an oxygen feedback system. The most important element of the feedback system is the O2S. The O2S is located in the exhaust path. Once it reaches operating temperature 300° to 350°C (572° to 662°F), the sensor generates a voltage that is inversely proportional to the amount of oxygen in the exhaust. The information obtained by the sensor is used to calculate the fuel injector pulse width. This maintains a 14.7 to 1 Air Fuel (A/F) ratio. At this mixture ratio,

EMISSIONS CONTROL (Continued)

the catalyst works best to remove hydrocarbons (HC), carbon monoxide (CO) and nitrogen oxide (NOx) from the exhaust.

The O2S is also the main sensing element for the Catalyst and Fuel Monitors.

The O2S can fail in any or all of the following manners:

- slow response rate
- reduced output voltage
- dynamic shift
- shorted or open circuits

Response rate is the time required for the sensor to switch from lean to rich once it is exposed to a richer than optimum A/F mixture or vice versa. As the sensor starts malfunctioning, it could take longer to detect the changes in the oxygen content of the exhaust gas.

The output voltage of the O2S ranges from 0 to 1 volt. A good sensor can easily generate any output voltage in this range as it is exposed to different concentrations of oxygen. To detect a shift in the A/F mixture (lean or rich), the output voltage has to change beyond a threshold value. A malfunctioning sensor could have difficulty changing beyond the threshold value.

OXYGEN SENSOR HEATER MONITOR

If there is an oxygen sensor (O2S) shorted to voltage DTC, as well as a O2S heater DTC, the O2S fault MUST be repaired first. Before checking the O2S fault, verify that the heater circuit is operating correctly.

Effective control of exhaust emissions is achieved by an oxygen feedback system. The most important element of the feedback system is the O2S. The O2S is located in the exhaust path. Once it reaches operating temperature 300° to 350°C (572 ° to 662°F), the sensor generates a voltage that is inversely proportional to the amount of oxygen in the exhaust. The information obtained by the sensor is used to calculate the fuel injector pulse width. This maintains a 14.7 to 1 Air Fuel (A/F) ratio. At this mixture ratio, the catalyst works best to remove hydrocarbons (HC), carbon monoxide (CO) and nitrogen oxide (NOx) from the exhaust.

The voltage readings taken from the O2S sensor are very temperature sensitive. The readings are not accurate below 300°C. Heating of the O2S sensor is done to allow the engine controller to shift to closed loop control as soon as possible. The heating element used to heat the O2S sensor must be tested to ensure that it is heating the sensor properly.

The O2S sensor circuit is monitored for a drop in voltage. The sensor output is used to test the heater by isolating the effect of the heater element on the O2S sensor output voltage from the other effects.

LEAK DETECTION PUMP MONITOR (IF EQUIPPED)

The leak detection assembly incorporates two primary functions: it must detect a leak in the evaporative system and seal the evaporative system so the leak detection test can be run.

The primary components within the assembly are: A three port solenoid that activates both of the functions listed above; a pump which contains a switch, two check valves and a spring/diaphragm, a canister vent valve (CVV) seal which contains a spring loaded vent seal valve.

Immediately after a cold start, between predetermined temperature thresholds limits, the three port solenoid is briefly energized. This initializes the pump by drawing air into the pump cavity and also closes the vent seal. During non test conditions the vent seal is held open by the pump diaphragm assembly which pushes it open at the full travel position. The vent seal will remain closed while the pump is cycling due to the reed switch triggering of the three port solenoid that prevents the diaphragm assembly from reaching full travel. After the brief initialization period, the solenoid is de-energized allowing atmospheric pressure to enter the pump cavity, thus permitting the spring to drive the diaphragm which forces air out of the pump cavity and into the vent system. When the solenoid is energized and de energized, the cycle is repeated creating flow in typical diaphragm pump fashion. The pump is controlled in 2 modes:

Pump Mode: The pump is cycled at a fixed rate to achieve a rapid pressure build in order to shorten the overall test length.

Test Mode: The solenoid is energized with a fixed duration pulse. Subsequent fixed pulses occur when the diaphragm reaches the Switch closure point.

The spring in the pump is set so that the system will achieve an equalized pressure of about 7.5" water. The cycle rate of pump strokes is quite rapid as the system begins to pump up to this pressure. As the pressure increases, the cycle rate starts to drop off. If there is no leak in the system, the pump would eventually stop pumping at the equalized pressure. If there is a leak, it will continue to pump at a rate representative of the flow characteristic of the size of the leak. From this information we can determine if the leak is larger than the required detection limit (currently set at .040" orifice by CARB). If a leak is revealed during the leak test portion of the test, the test is terminated at the end of the test mode and no further system checks will be performed.

After passing the leak detection phase of the test, system pressure is maintained by turning on the LDP's solenoid until the purge system is activated. Purge activation in effect creates a leak. The cycle rate is again interrogated and when it increases due

EMISSIONS CONTROL (Continued)

to the flow through the purge system, the leak check portion of the diagnostic is complete.

The canister vent valve will unseal the system after completion of the test sequence as the pump diaphragm assembly moves to the full travel position.

Evaporative system functionality will be verified by using the stricter evap purge flow monitor. At an appropriate warm idle the LDP will be energized to seal the canister vent. The purge flow will be clocked up from some small value in an attempt to see a shift in the 02 control system. If fuel vapor, indicated by a shift in the 02 control, is present the test is passed. If not, it is assumed that the purge system is not functioning in some respect. The LDP is again turned off and the test is ended.

MISFIRE MONITOR

Excessive engine misfire results in increased catalyst temperature and causes an increase in HC emissions. Severe misfires could cause catalyst damage. To prevent catalytic convertor damage, the PCM monitors engine misfire.

The Powertrain Control Module (PCM) monitors for misfire during most engine operating conditions (positive torque) by looking at changes in the crankshaft speed. If a misfire occurs the speed of the crankshaft will vary more than normal.

FUEL SYSTEM MONITOR

To comply with clean air regulations, vehicles are equipped with catalytic converters. These converters reduce the emission of hydrocarbons, oxides of nitrogen and carbon monoxide. The catalyst works best when the Air Fuel (A/F) ratio is at or near the optimum of 14.7 to 1.

The PCM is programmed to maintain the optimum air/fuel ratio of 14.7 to 1. This is done by making short term corrections in the fuel injector pulse width based on the O2S sensor output. The programmed memory acts as a self calibration tool that the engine controller uses to compensate for variations in engine specifications, sensor tolerances and engine fatigue over the life span of the engine. By monitoring the actual fuel-air ratio with the O2S sensor (short term) and multiplying that with the program long-term (adaptive) memory and comparing that to the limit, it can be determined whether it will pass an emissions test. If a malfunction occurs such that the PCM cannot maintain the optimum A/F ratio, then the MIL will be illuminated.

CATALYST MONITOR

To comply with clean air regulations, vehicles are equipped with catalytic converters. These converters reduce the emission of hydrocarbons, oxides of nitrogen and carbon monoxide.

Normal vehicle miles or engine misfire can cause a catalyst to decay. This can increase vehicle emissions and deteriorate engine performance, driveability and fuel economy.

The catalyst monitor uses dual oxygen sensors (O2S's) to monitor the efficiency of the converter. The dual O2S's sensor strategy is based on the fact that as a catalyst deteriorates, its oxygen storage capacity and its efficiency are both reduced. By monitoring the oxygen storage capacity of a catalyst, its efficiency can be indirectly calculated. The upstream O2S is used to detect the amount of oxygen in the exhaust gas before the gas enters the catalytic converter. The PCM calculates the A/F mixture from the output of the O2S. A low voltage indicates high oxygen content (lean mixture). A high voltage indicates a low content of oxygen (rich mixture).

When the upstream O2S detects a lean condition, there is an abundance of oxygen in the exhaust gas. A functioning converter would store this oxygen so it can use it for the oxidation of HC and CO. As the converter absorbs the oxygen, there will be a lack of oxygen downstream of the converter. The output of the downstream O2S will indicate limited activity in this condition.

As the converter loses the ability to store oxygen, the condition can be detected from the behavior of the downstream O2S. When the efficiency drops, no chemical reaction takes place. This means the concentration of oxygen will be the same downstream as upstream. The output voltage of the downstream O2S copies the voltage of the upstream sensor. The only difference is a time lag (seen by the PCM) between the switching of the O2S's.

To monitor the system, the number of lean-to-rich switches of upstream and downstream O2S's is counted. The ratio of downstream switches to upstream switches is used to determine whether the catalyst is operating properly. An effective catalyst will have fewer downstream switches than it has upstream switches i.e., a ratio closer to zero. For a totally ineffective catalyst, this ratio will be one-to-one, indicating that no oxidation occurs in the device.

The system must be monitored so that when catalyst efficiency deteriorates and exhaust emissions increase to over the legal limit, the MIL will be illuminated.

DESCRIPTION - TRIP DEFINITION

The term “Trip” has different meanings depending on what the circumstances are. If the MIL (Malfunction Indicator Lamp) is OFF, a Trip is defined as when the Oxygen Sensor Monitor and the Catalyst Monitor have been completed in the same drive cycle.

When any Emission DTC is set, the MIL on the dash is turned ON. When the MIL is ON, it takes 3

EMISSIONS CONTROL (Continued)

good trips to turn the MIL OFF. In this case, it depends on what type of DTC is set to know what a "Trip" is.

For the Fuel Monitor or Mis-Fire Monitor (continuous monitor), the vehicle must be operated in the "Similar Condition Window" for a specified amount of time to be considered a Good Trip.

If a Non-Contiuous OBDII Monitor fails twice in a row and turns ON the MIL, re-running that monitor which previously failed, on the next start-up and passing the monitor, is considered to be a Good Trip. These will include the following:

- Oxygen Sensor
- Catalyst Monitor
- Purge Flow Monitor
- Leak Detection Pump Monitor (if equipped)
- EGR Monitor (if equipped)
- Oxygen Sensor Heater Monitor

If any other Emission DTC is set (not an OBDII Monitor), a Good Trip is considered to be when the Oxygen Sensor Monitor and Catalyst Monitor have been completed; or 2 Minutes of engine run time if the Oxygen Sensor Monitor or Catalyst Monitor have been stopped from running.

It can take up to 2 Failures in a row to turn on the MIL. After the MIL is ON, it takes 3 Good Trips to turn the MIL OFF. After the MIL is OFF, the PCM will self-erase the DTC after 40 Warm-up cycles. A Warm-up cycle is counted when the ECT (Engine Coolant Temperature Sensor) has crossed 160°F and has risen by at least 40°F since the engine has been started.

DESCRIPTION - COMPONENT MONITORS

There are several components that will affect vehicle emissions if they malfunction. If one of these components malfunctions the Malfunction Indicator Lamp (MIL) will illuminate.

Some of the component monitors are checking for proper operation of the part. Electrically operated components now have input (rationality) and output (functionality) checks. Previously, a component like the Throttle Position sensor (TPS) was checked by the PCM for an open or shorted circuit. If one of these conditions occurred, a DTC was set. Now there is a check to ensure that the component is working. This is done by watching for a TPS indication of a greater or lesser throttle opening than MAP and engine rpm indicate. In the case of the TPS, if engine vacuum is high and engine rpm is 1600 or greater and the TPS indicates a large throttle opening, a DTC will be set. The same applies to low vacuum if the TPS indicates a small throttle opening.

All open/short circuit checks or any component that has an associated limp in will set a fault after 1 trip with the malfunction present. Components without an associated limp in will take two trips to illuminate the MIL.

Refer to the Diagnostic Trouble Codes Description Charts in this section and the appropriate Powertrain Diagnostic Procedure Manual for diagnostic procedures.

DESCRIPTION - NON-MONITORED CIRCUITS

The PCM does not monitor the following circuits, systems and conditions that could have malfunctions causing driveability problems. The PCM might not store diagnostic trouble codes for these conditions. However, problems with these systems may cause the PCM to store diagnostic trouble codes for other systems or components. For example, a fuel pressure problem will not register a fault directly, but could cause a rich/lean condition or misfire. This could cause the PCM to store an oxygen sensor or misfire diagnostic trouble code

FUEL PRESSURE

The fuel pressure regulator controls fuel system pressure. The PCM cannot detect a clogged fuel pump inlet filter, clogged in-line fuel filter, or a pinched fuel supply or return line. However, these could result in a rich or lean condition causing the PCM to store an oxygen sensor or fuel system diagnostic trouble code.

SECONDARY IGNITION CIRCUIT

The PCM cannot detect an inoperative ignition coil, fouled or worn spark plugs, ignition cross firing, or open spark plug cables.

CYLINDER COMPRESSION

The PCM cannot detect uneven, low, or high engine cylinder compression.

EXHAUST SYSTEM

The PCM cannot detect a plugged, restricted or leaking exhaust system, although it may set a fuel system fault.

FUEL INJECTOR MECHANICAL MALFUNCTIONS

The PCM cannot determine if a fuel injector is clogged, the needle is sticking or if the wrong injector is installed. However, these could result in a rich or lean condition causing the PCM to store a diagnostic trouble code for either misfire, an oxygen sensor, or the fuel system.

EXCESSIVE OIL CONSUMPTION

Although the PCM monitors engine exhaust oxygen content when the system is in closed loop, it cannot determine excessive oil consumption.

EMISSIONS CONTROL (Continued)

THROTTLE BODY AIRFLOW

The PCM cannot detect a clogged or restricted air cleaner inlet or filter element.

VACUUM ASSIST

The PCM cannot detect leaks or restrictions in the vacuum circuits of vacuum assisted engine control system devices. However, these could cause the PCM to store a MAP sensor diagnostic trouble code and cause a high idle condition.

PCM SYSTEM GROUND

The PCM cannot determine a poor system ground. However, one or more diagnostic trouble codes may be generated as a result of this condition. The module should be mounted to the body at all times, also during diagnostic.

PCM CONNECTOR ENGAGEMENT

The PCM may not be able to determine spread or damaged connector pins. However, it might store diagnostic trouble codes as a result of spread connector pins.

DESCRIPTION - HIGH AND LOW LIMITS

The PCM compares input signal voltages from each input device with established high and low limits for the device. If the input voltage is not within limits and other criteria are met, the PCM stores a diagnostic trouble code in memory. Other diagnostic trouble code criteria might include engine RPM limits or input voltages from other sensors or switches that must be present before verifying a diagnostic trouble code condition.

DESCRIPTION - LOAD VALUE

ENGINE	IDLE/NEUTRAL	2500 RPM/NEUTRAL
All Engines	2% to 8% of Maximum Load	9% to 17% of Maximum Load

OPERATION - TASK MANAGER

The Task Manager determines which tests happen when and which functions occur when. Many of the diagnostic steps required by OBD II must be performed under specific operating conditions. The Task Manager software organizes and prioritizes the diagnostic procedures. The job of the Task Manager is to determine if conditions are appropriate for tests to be run, monitor the parameters for a trip for each test, and record the results of the test. Following are the responsibilities of the Task Manager software:

- Test Sequence
- MIL Illumination
- Diagnostic Trouble Codes (DTCs)
- Trip Indicator
- Freeze Frame Data Storage
- Similar Conditions Window

Test Sequence

In many instances, emissions systems must fail diagnostic tests more than once before the PCM illuminates the MIL. These tests are know as 'two trip monitors.' Other tests that turn the MIL lamp on after a single failure are known as 'one trip monitors.' A trip is defined as 'start the vehicle and operate it to meet the criteria necessary to run the given monitor.'

Many of the diagnostic tests must be performed under certain operating conditions. However, there are times when tests cannot be run because another test is in progress (conflict), another test has failed (pending) or the Task Manager has set a fault that may cause a failure of the test (suspend).

- Pending

Under some situations the Task Manager will not run a monitor if the MIL is illuminated and a fault is stored from another monitor. In these situations, the Task Manager postpones monitors **pending** resolution of the original fault. The Task Manager does not run the test until the problem is remedied.

For example, when the MIL is illuminated for an Oxygen Sensor fault, the Task Manager does not run the Catalyst Monitor until the Oxygen Sensor fault is remedied. Since the Catalyst Monitor is based on signals from the Oxygen Sensor, running the test would produce inaccurate results.

- Conflict

There are situations when the Task Manager does not run a test if another monitor is in progress. In these situations, the effects of another monitor running could result in an erroneous failure. If this **conflict** is present, the monitor is not run until the conflicting condition passes. Most likely the monitor will run later after the conflicting monitor has passed.

For example, if the Fuel System Monitor is in progress, the Task Manager does not run the EGR Monitor. Since both tests monitor changes in air/fuel ratio and adaptive fuel compensation, the monitors will conflict with each other.

- Suspend

Occasionally the Task Manager may not allow a two trip fault to mature. The Task Manager will **sus-**

EMISSIONS CONTROL (Continued)

pend the maturing of a fault if a condition exists that may induce an erroneous failure. This prevents illuminating the MIL for the wrong fault and allows more precis diagnosis.
For example, if the PCM is storing a one trip fault for the Oxygen Sensor and the EGR monitor, the Task Manager may still run the EGR Monitor but will suspend the results until the Oxygen Sensor Monitor either passes or fails. At that point the Task Manager can determine if the EGR system is actually failing or if an Oxygen Sensor is failing.

MIL Illumination

The PCM Task Manager carries out the illumination of the MIL. The Task Manager triggers MIL illumination upon test failure, depending on monitor failure criteria.

The Task Manager Screen shows both a Requested MIL state and an Actual MIL state. When the MIL is illuminated upon completion of a test for a third trip, the Requested MIL state changes to OFF. However, the MIL remains illuminated until the next key cycle. (On some vehicles, the MIL will actually turn OFF during the third key cycle) During the key cycle for the third good trip, the Requested MIL state is OFF, while the Actual MIL state is ON. After the next key cycle, the MIL is not illuminated and both MIL states read OFF.

Diagnostic Trouble Codes (DTCs)

With OBD II, different DTC faults have different priorities according to regulations. As a result, the priorities determine MIL illumination and DTC erasure. DTCs are entered according to individual priority. DTCs with a higher priority overwrite lower priority DTCs.

Priorities

- Priority 0 —Non-emissions related trouble codes
- Priority 1 — One trip failure of a two trip fault for non-fuel system and non-misfire.
- Priority 2 — One trip failure of a two trip fault for fuel system (rich/lean) or misfire.
- Priority 3 — Two trip failure for a non-fuel system and non-misfire or matured one trip comprehensive component fault.
- Priority 4 — Two trip failure or matured fault for fuel system (rich/lean) and misfire or one trip catalyst damaging misfire.

Non-emissions related failures have no priority. One trip failures of two trip faults have low priority. Two trip failures or matured faults have higher priority. One and two trip failures of fuel system and misfire monitor take precedence over non-fuel system and non-misfire failures.

DTC Self Erasure

With one trip components or systems, the MIL is illuminated upon test failure and DTCs are stored.

Two trip monitors are components requiring failure in two consecutive trips for MIL illumination. Upon failure of the first test, the Task Manager enters a maturing code. If the component fails the test for a second time the code matures and a DTC is set.

After three good trips the MIL is extinguished and the Task Manager automatically switches the trip counter to a warm-up cycle counter. DTCs are automatically erased following 40 warm-up cycles if the component does not fail again.

For misfire and fuel system monitors, the component must pass the test under a Similar Conditions Window in order to record a good trip. A Similar Conditions Window is when engine RPM is within ±375 RPM and load is within ±10% of when the fault occurred.

NOTE: It is important to understand that a component does not have to fail under a similar window of operation to mature. It must pass the test under a Similar Conditions Window when it failed to record a Good Trip for DTC erasure for misfire and fuel system monitors.

DTCs can be erased anytime with a DRB III. Erasing the DTC with the DRB III erases all OBD II information. The DRB III automatically displays a warning that erasing the DTC will also erase all OBD II monitor data. This includes all counter information for warm-up cycles, trips and Freeze Frame.

Trip Indicator

The **Trip** is essential for running monitors and extinguishing the MIL. In OBD II terms, a trip is a set of vehicle operating conditions that must be met for a specific monitor to run. All trips begin with a key cycle.

Good Trip

The Good Trip counters are as follows:

- Specific Good Trip
- Fuel System Good Trip
- Misfire Good Trip
- Alternate Good Trip (appears as a Global Good Trip on DRB III)
  - Comprehensive Components
  - Major Monitor
- Warm-Up Cycles

Specific Good Trip

The term Good Trip has different meanings depending on the circumstances:

- If the MIL is OFF, a trip is defined as when the Oxygen Sensor Monitor and the Catalyst Monitor have been completed in the same drive cycle.

EMISSIONS CONTROL (Continued)

- If the MIL is ON and a DTC was set by the Fuel Monitor or Misfire Monitor (both continuous monitors), the vehicle must be operated in the Similar Condition Window for a specified amount of time.
- If the MIL is ON and a DTC was set by a Task Manager commanded once-per-trip monitor (such as the Oxygen Sensor Monitor, Catalyst Monitor, Purge Flow Monitor, Leak Detection Pump Monitor, EGR Monitor or Oxygen Sensor Heater Monitor), a good trip is when the monitor is passed on the next start-up.
- If the MIL is ON and any other emissions DTC was set (not an OBD II monitor), a good trip occurs when the Oxygen Sensor Monitor and Catalyst Monitor have been completed, or two minutes of engine run time if the Oxygen Sensor Monitor and Catalyst Monitor have been stopped from running.

Fuel System Good Trip

To count a good trip (three required) and turn off the MIL, the following conditions must occur:

- Engine in closed loop
- Operating in Similar Conditions Window
- Short Term multiplied by Long Term less than threshold
- Less than threshold for a predetermined time

If all of the previous criteria are met, the PCM will count a good trip (three required) and turn off the MIL.

Misfire Good Trip

If the following conditions are met the PCM will count one good trip (three required) in order to turn off the MIL:

- Operating in Similar Condition Window
- 1000 engine revolutions with no misfire

Warm-Up Cycles

Once the MIL has been extinguished by the Good Trip Counter, the PCM automatically switches to a Warm-Up Cycle Counter that can be viewed on the DRB III. Warm-Up Cycles are used to erase DTCs and Freeze Frames. Forty Warm-Up cycles must occur in order for the PCM to self-erase a DTC and Freeze Frame. A Warm-Up Cycle is defined as follows:

- Engine coolant temperature must start below and rise above 160° F
- Engine coolant temperature must rise by 40° F
- No further faults occur

Freeze Frame Data Storage

Once a failure occurs, the Task Manager records several engine operating conditions and stores it in a Freeze Frame. The Freeze Frame is considered one frame of information taken by an on-board data recorder. When a fault occurs, the PCM stores the input data from various sensors so that technicians can determine under what vehicle operating conditions the failure occurred.

The data stored in Freeze Frame is usually recorded when a system fails the first time for two trip faults. Freeze Frame data will only be overwritten by a different fault with a higher priority.

CAUTION: Erasing DTCs, either with the DRB III or by disconnecting the battery, also clears all Freeze Frame data.

Similar Conditions Window

The Similar Conditions Window displays information about engine operation during a monitor. Absolute MAP (engine load) and Engine RPM are stored in this window when a failure occurs. There are two different Similar conditions Windows: Fuel System and Misfire.

FUEL SYSTEM

- **Fuel System Similar Conditions Window** — An indicator that 'Absolute MAP When Fuel Sys Fail' and 'RPM When Fuel Sys Failed' are all in the same range when the failure occurred. Indicated by switching from 'NO' to 'YES'.
- **Absolute MAP When Fuel Sys Fail** — The stored MAP reading at the time of failure. Informs the user at what engine load the failure occurred.
- **Absolute MAP** — A live reading of engine load to aid the user in accessing the Similar Conditions Window.
- **RPM When Fuel Sys Fail** — The stored RPM reading at the time of failure. Informs the user at what engine RPM the failure occurred.
- **Engine RPM** — A live reading of engine RPM to aid the user in accessing the Similar Conditions Window.
- **Adaptive Memory Factor** — The PCM utilizes both Short Term Compensation and Long Term Adaptive to calculate the Adaptive Memory Factor for total fuel correction.
- **Upstream O2S Volts** — A live reading of the Oxygen Sensor to indicate its performance. For example, stuck lean, stuck rich, etc.
- **SCW Time in Window (Similar Conditions Window Time in Window)** — A timer used by the PCM that indicates that, after all Similar Conditions have been met, if there has been enough good engine running time in the SCW without failure detected. This timer is used to increment a Good Trip.
- **Fuel System Good Trip Counter** — A Trip Counter used to turn OFF the MIL for Fuel System DTCs. To increment a Fuel System Good Trip, the engine must be in the Similar Conditions Window, Adaptive Memory Factor must be less than calibrated threshold and the Adaptive Memory Factor

EMISSIONS CONTROL (Continued)

must stay below that threshold for a calibrated amount of time.

- **Test Done This Trip** — Indicates that the monitor has already been run and completed during the current trip.

MISFIRE

- **Same Misfire Warm-Up State** — Indicates if the misfire occurred when the engine was warmed up (above 160° F).
- **In Similar Misfire Window** — An indicator that 'Absolute MAP When Misfire Occurred' and 'RPM When Misfire Occurred' are all in the same range when the failure occurred. Indicated by switching from 'NO' to 'YES'.
- **Absolute MAP When Misfire Occurred** — The stored MAP reading at the time of failure. Informs the user at what engine load the failure occurred.
- **Absolute MAP** — A live reading of engine load to aid the user in accessing the Similar Conditions Window.
- **RPM When Misfire Occurred** — The stored RPM reading at the time of failure. Informs the user at what engine RPM the failure occurred.
- **Engine RPM** — A live reading of engine RPM to aid the user in accessing the Similar Conditions Window.
- **Adaptive Memory Factor** — The PCM utilizes both Short Term Compensation and Long Term Adaptive to calculate the Adaptive Memory Factor for total fuel correction.
- **200 Rev Counter** — Counts 0–100 720 degree cycles.
- **SCW Cat 200 Rev Counter** — Counts when in similar conditions.
- **SCW FTP 1000 Rev Counter** — Counts 0–4 when in similar conditions.
- **Misfire Good Trip Counter** — Counts up to three to turn OFF the MIL.
- **Misfire Data**— Data collected during test.
- **Test Done This Trip**— Indicates YES when the test is done.

EVAPORATIVE EMISSIONS

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EVAPORATIVE EMISSIONS

DESCRIPTION - EVAPORATION CONTROL SYSTEM

The evaporation control system prevents the emission of fuel tank vapors into the atmosphere. When fuel evaporates in the fuel tank, the vapors pass through vent hoses or tubes to a charcoal filled evaporative canister. The canister temporarily holds the vapors. The Powertrain Control Module (PCM) allows intake manifold vacuum to draw vapors into the combustion chambers during certain operating conditions.

All engines use a duty-cycle purge system. The PCM controls vapor flow by operating the duty-cycle EVAP purge solenoid. Refer to Duty-Cycle EVAP Canister Purge Solenoid.

An ORVR system is used. This system is part of the EVAP Control System. Refer to On-Board Refueling Vapor Recovery (ORVR) for information.

When equipped with certain emissions packages, a Leak Detection Pump (LDP) will be used as part of the evaporative system for OBD II requirements. Also refer to Leak Detection Pump.

NOTE: The evaporative system uses specially manufactured lines/hoses. If replacement becomes necessary, only use fuel resistant hose.

EVAPORATIVE EMISSIONS (Continued)

SPECIFICATIONS

TORQUE - EMISSIONS

DESCRIPTION	N-m	Ft. Lbs.	In. Lbs.
EVAP Canister Mounting Nuts	9	-	80
Leak Detection Pump Filter Mounting Bolt	7	-	65
LDP Pump Mounting Bolts	1	-	11
EVAP Canister/LDP Mounting Bracket-to-Body Nuts	-	-	250
EVAP Canister/LDP Mounting Bracket-to-Body Bolts	-	-	212

CCV SYSTEM

DESCRIPTION - CCV SYSTEM

The 4.0L 6-cylinder engine is equipped with a Closed Crankcase Ventilation (CCV) system and a fixed orifice valve (Fig. 1). The 2.4L 4-cylinder uses a PCV valve. Refer to PCV Valve for 2.4L information.

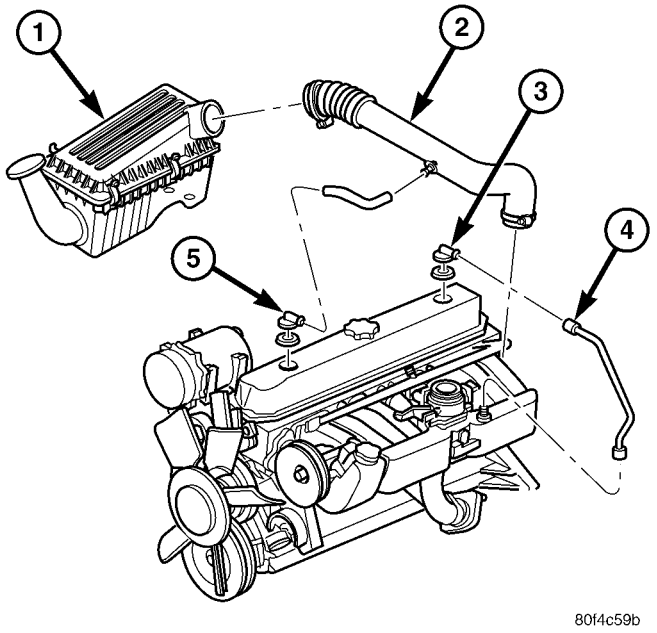

Fig. 1 CRANKCASE VENTILATION (CCV) FITTING - 4.0L

1 - AIR CLEANER ASSEMBLY
2 - AIR INTAKE TUBE
3 - FIXED ORIFICE FITTING
4 - VACUUM LINE
5 - AIR INLET FITTING

OPERATION - CCV SYSTEM

The CCV system performs the same function as a conventional PCV system, but does not use a vacuum controlled valve.

On 4.0L 6-cylinder engines, a molded vacuum tube connects manifold vacuum to the top of the cylinder head (valve) cover. The vacuum fitting contains a fixed orifice of a calibrated size. It meters the amount of crankcase vapors drawn out of the engine.

When the engine is operating, fresh air enters the engine and mixes with crankcase vapors. Manifold vacuum draws the vapor/air mixture through the fixed orifice and into the intake manifold. The vapors are then consumed during combustion.

DIAGNOSIS AND TESTING - CCV SYSTEM

TESTING/CLEANING

The Crankcase Ventilation (CCV) system performs the same function as a conventional PCV system, but does not use a vacuum controlled valve. A vacuum fitting containing a fixed orifice of a calibrated size is used. It meters the amount of crankcase vapors drawn out of the engine.

(1) Check each CCV system tube (line) for leaks, cracks, kinks or bends. Replace as necessary

(2) Disconnect each CCV tube.

(3) Remove fixed orifice fitting from valve cover.

(4) Blow compressed air through each tube, and the fixed orifice fitting. Check for blockage or restrictions. If cleaning is necessary, spray a soapy-type all-purpose cleaner into each component and blow out. After restriction is cleared, rinse out component with clear water. Blow water from component and install to vehicle. **To prevent damage to plastic components, never spray carburetor-type cleaner into**

CCV SYSTEM (Continued)

any of the plastic tubes or the fixed orifice fitting. Never attempt to clean the fixed orifice fitting with a metal object as calibration could be affected. If fixed fitting cannot be cleared, replace it.

PCV VALVE

DESCRIPTION

2.4L

The 2.4L 4-cylinder engine is equipped with a closed crankcase ventilation system and a Positive Crankcase Ventilation (PCV) valve.

This system consists of:

- a PCV valve attached to the left/front side of the valve cover (Fig. 2). It is secured with 1 bolt. An o-ring is used to seal valve to valve cover (Fig. 3). Another type of threaded PCV valve may be used (Fig. 4).
- the air cleaner housing
- tubes and hoses to connect the system components.

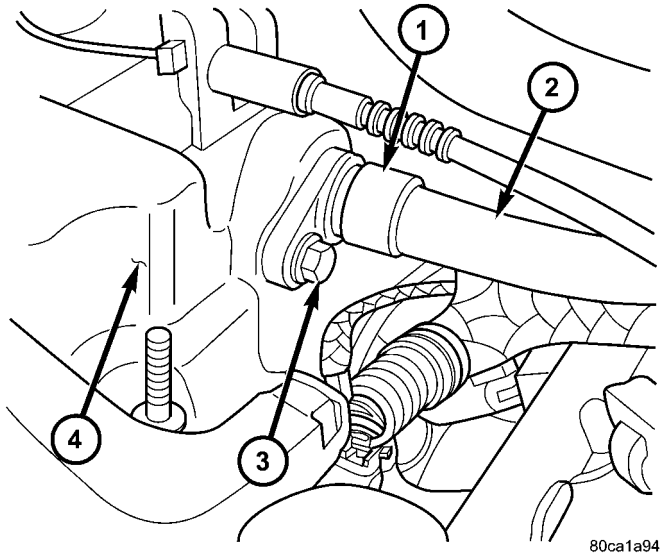

Fig. 2 PCV VALVE LOCATION - 2.4L

1 - PCV VALVE
2 - HOSE
3 - MOUNTING BOLT
4 - VALVE COVER (LEFT SIDE)

OPERATION

2.4L

The PCV system operates by engine intake manifold vacuum. Filtered air is routed into the crankcase through the air cleaner hose and crankcase breather(s) (if used). The metered air, along with crankcase vapors, are drawn through the PCV valve and into a passage in the intake manifold. The PCV system manages crankcase pressure and meters blow-by gases to the intake system, reducing engine sludge formation.

The PCV valve contains a spring loaded plunger. This plunger meters the amount of crankcase vapors

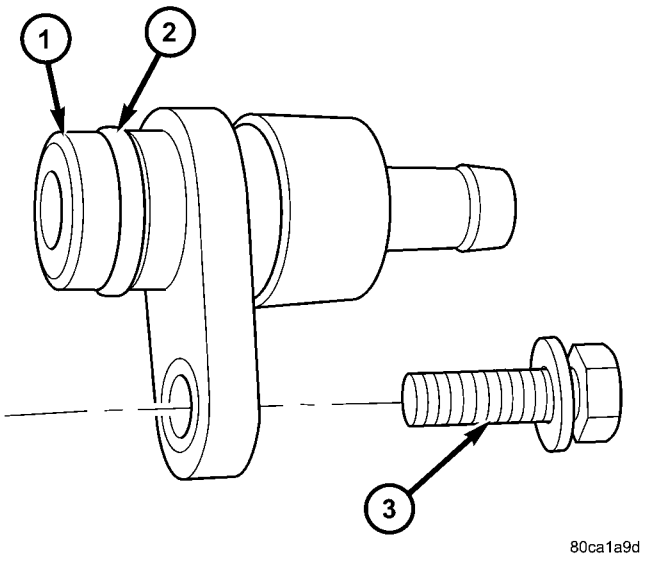

Fig. 3 PCV VALVE AND O-RING - 2.4L

1 - PCV VALVE
2 - O-RING
3 - MOUNTING BOLT

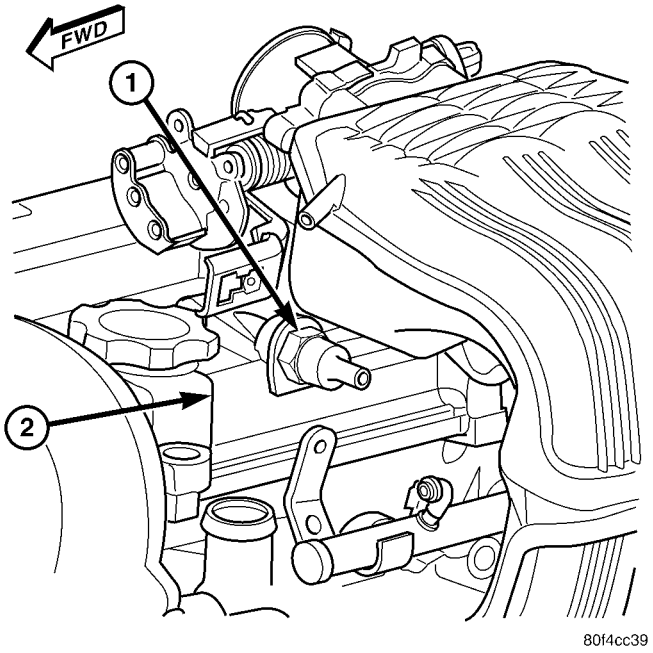

Fig. 4 PCV VALVE - 2.4L - THREADED

1 - PCV VALVE (THREADED)
2 - VALVE COVER (LEFT SIDE)

PCV VALVE (Continued)

routed into the combustion chamber based on intake manifold vacuum.

TYPICAL PCV valves are shown in (Fig. 5), (Fig. 6) and (Fig. 7).

When the engine is not operating, or during an engine pop-back, the spring forces the plunger back against the seat (Fig. 5). This will prevent vapors from flowing through the valve.

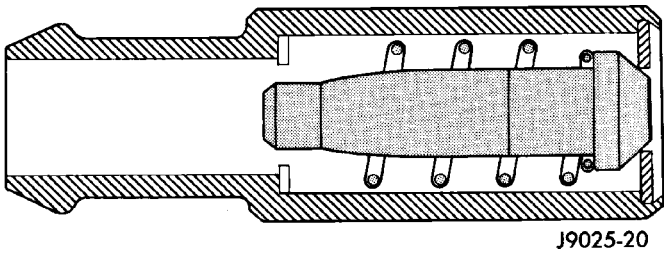

Fig. 5 ENGINE OFF OR ENGINE BACKFIRE - NO VAPOR FLOW

During periods of high manifold vacuum, such as idle or cruising speeds, vacuum is sufficient to completely compress spring. It will then pull the plunger to the top of the valve (Fig. 6). In this position there is minimal vapor flow through the valve.

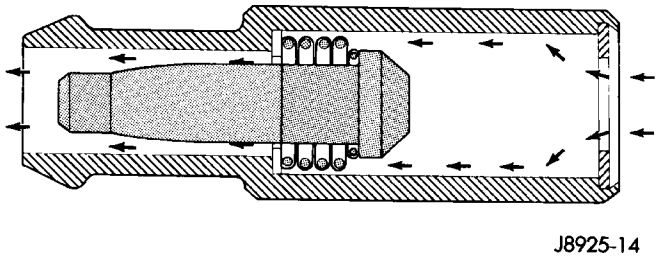

Fig. 6 HIGH INTAKE MANIFOLD VACUUM - MINIMAL VAPOR FLOW

During periods of moderate manifold vacuum, the plunger is only pulled part way back from inlet. This results in maximum vapor flow through the valve (Fig. 7).

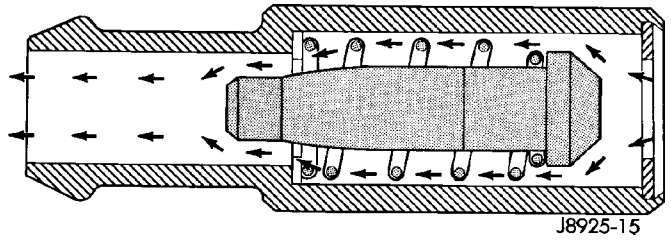

Fig. 7 MODERATE INTAKE MANIFOLD VACUUM - MAXIMUM VAPOR FLOW

REMOVAL

2.4L

The PCV valve is attached to the left/front side of the valve cover (Fig. 2). It is secured with 1 bolt. An o-ring is used to seal valve to valve cover (Fig. 3). Another type of threaded PCV valve may be used (Fig. 4).

(1) Remove hose from valve. Check condition of hose.

(2) Bolted PCV Valve:
  (a) Remove 1 bolt.
  (b) Remove PCV valve from valve cover.
  (c) Check condition of valve o-ring.

(3) Threaded PCV Valve:
  (a) Unscrew valve from valve cover.

EVAP/PURGE SOLENOID

DESCRIPTION

The duty cycle EVAP canister purge solenoid (DCP) regulates the rate of vapor flow from the EVAP canister to the intake manifold. The Powertrain Control Module (PCM) operates the solenoid.

OPERATION

During the cold start warm-up period and the hot start time delay, the PCM does not energize the solenoid. When de-energized, no vapors are purged. The PCM de-energizes the solenoid during open loop operation.

The engine enters closed loop operation after it reaches a specified temperature and the time delay ends. During closed loop operation, the PCM cycles (energizes and de-energizes) the solenoid 5 or 10 times per second, depending upon operating conditions. The PCM varies the vapor flow rate by changing solenoid pulse width. Pulse width is the amount of time that the solenoid is energized. The PCM adjusts solenoid pulse width based on engine operating condition.

REMOVAL

The duty cycle EVAP canister purge solenoid is located in the engine compartment near the front of the brake master cylinder (Fig. 8). It is attached to a slip-on type bracket.

The EVAP System Test Port is located near the EVAP/Purge Solenoid (Fig. 8).

(1) Disconnect electrical wiring connector at solenoid.

(2) Disconnect vacuum lines/hoses at solenoid.

(3) Lift solenoid and rubber solenoid support from mounting bracket.

EVAP/PURGE SOLENOID (Continued)

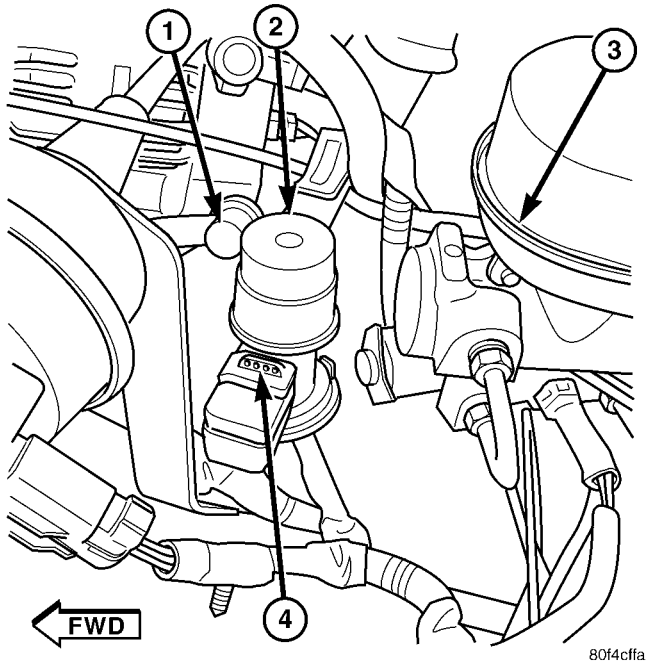

Fig. 8 EVAP/PURGE SOLENOID

1 - EVAP SYSTEM TEST PORT
2 - EVAP/PURGE SOLENOID
3 - FRONT OF BRAKE MASTER CYLINDER
4 - SOLENOID MOUNTING BRACKET

INSTALLATION

(1) Install purge solenoid and rubber support to its mounting bracket.

(2) Connect vacuum harness and wiring connector.

FUEL FILLER CAP

DESCRIPTION

The plastic fuel tank filler tube cap is threaded onto the end of the fuel fill tube. Certain models are equipped with a 1/4 turn cap.

OPERATION

The loss of any fuel or vapor out of fuel filler tube is prevented by the use of a pressure-vacuum fuel fill cap. Relief valves inside the cap will release fuel tank pressure at predetermined pressures. Fuel tank vacuum will also be released at predetermined values. This cap must be replaced by a similar unit if replacement is necessary. This is in order for the system to remain effective.

CAUTION: Remove fill cap before servicing any fuel system component to relieve tank pressure. If equipped with a California emissions package and a Leak Detection Pump (LDP), the cap must be tightened securely. If cap is left loose, a Diagnostic Trouble Code (DTC) may be set.

REMOVAL

If replacement of the 1/4 turn fuel tank filler tube cap is necessary, it must be replaced with an identical cap to be sure of correct system operation.

CAUTION: Remove the fuel tank filler tube cap to relieve fuel tank pressure. The cap must be removed prior to disconnecting any fuel system component or before draining the fuel tank.

ORVR

DESCRIPTION

The ORVR (On-Board Refueling Vapor Recovery) system consists of a unique fuel tank, flow management valve (5) (Fig. 9) , fluid control valve, one-way check valve and EVAP (vapor) canister (1) (Fig. 9).

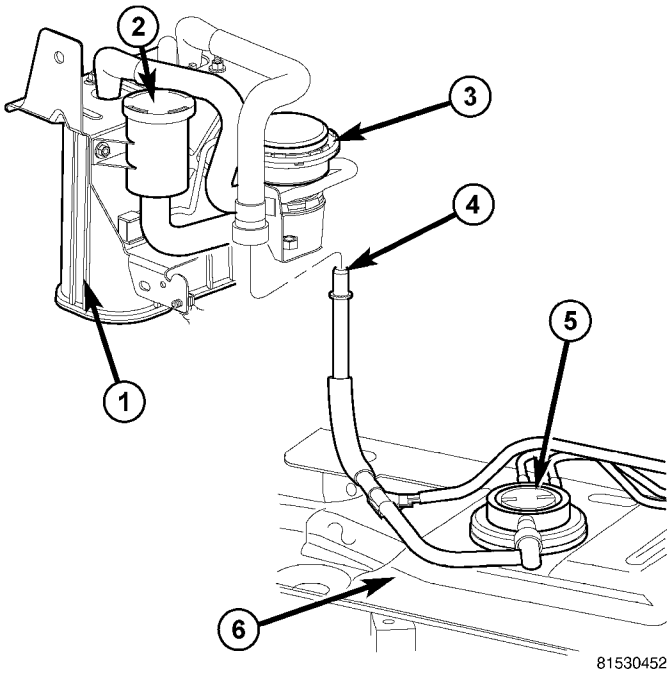

Fig. 9 ORVR VAPOR LINE

1 - EVAP CANISTER
2 - NVLD FILTER
3 - NVLD PUMP
4 - VAPOR LINES
5 - FLOW VALVE
6 - TOP OF FUEL TANK)

A vapor line (4) and vent line (9) (Fig. 10) are also used to connect the fuel tank into the ORVR system.

ORVR (Continued)

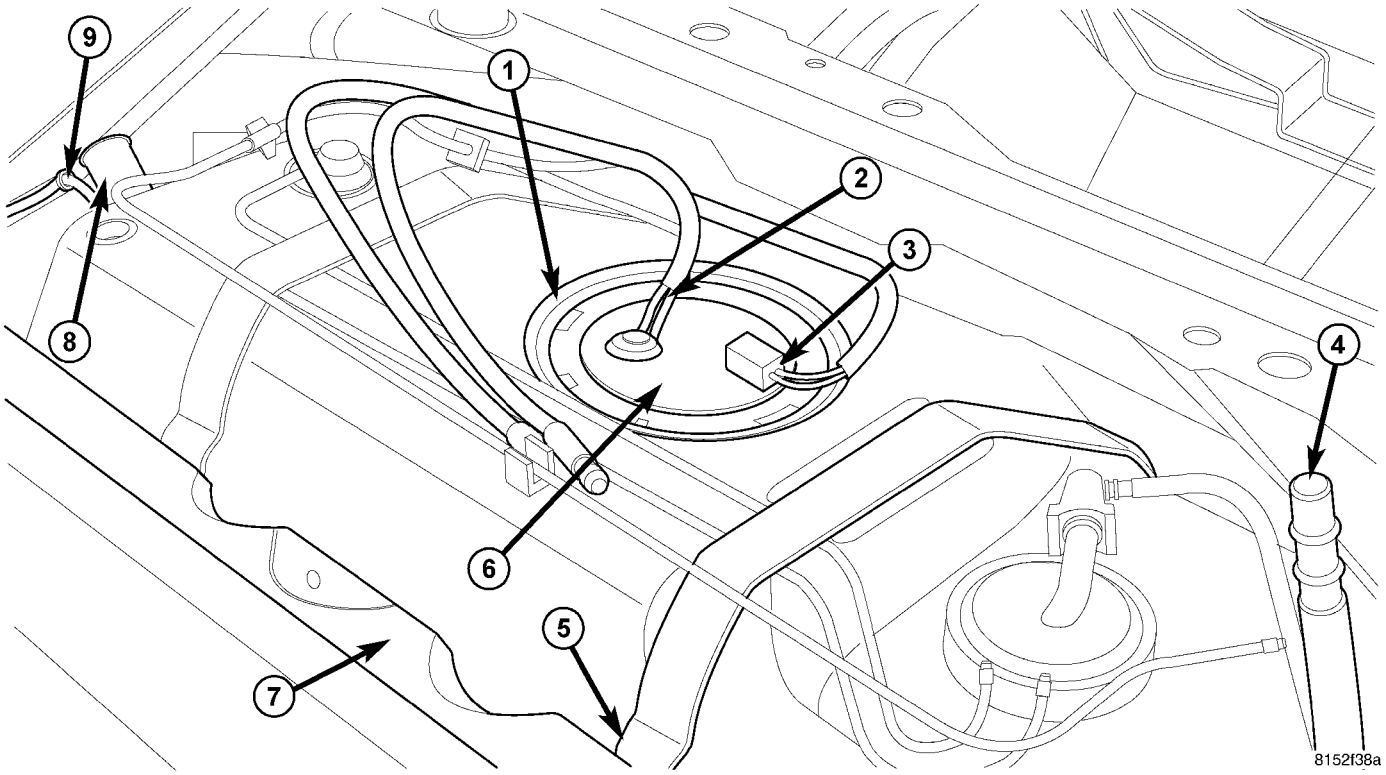

Fig. 10 TOP OF FUEL TANK

1 - FUEL PUMP MODULE LOCKRING
2 - FUEL SUPPLY LINE
3 - ELECTRICAL CONNECTOR
4 - VAPOR LINE
5 - FUEL TANK MOUNTING STRAPS (2)
6 - FUEL PUMP MODULE ASSEMBLY
7 - SKID PLATE
8 - FUEL FILL FITTING
9 - FUEL VENT FITTING

OPERATION

The ORVR (On-Board Refueling Vapor Recovery) system is used to remove excess fuel tank vapors. This is done while the vehicle is being refueled.

Fuel flowing into the fuel filler tube (approx. 1" I.D.) creates an aspiration effect drawing air into the fuel fill tube. During refueling, the fuel tank is vented to the EVAP canister to capture escaping vapors. With air flowing into the filler tube, there are no fuel vapors escaping to the atmosphere. Once the refueling vapors are captured by the EVAP canister, the vehicle's computer controlled purge system draws vapor out of the canister for the engine to burn. The vapor flow is metered by the purge solenoid so that there is no, or minimal impact on driveability or tailpipe emissions.

As fuel starts to flow through the fuel fill tube, it opens the normally closed check valve and enters the fuel tank. Vapor or air is expelled from the tank through the control valve and on to the vapor canister. Vapor is absorbed in the EVAP canister until vapor flow in the lines stops. This stoppage occurs following fuel shut-off, or by having the fuel level in the tank rise high enough to close the control valve. This control valve contains a float that rises to seal the large diameter vent path to the EVAP canister. At this point in the refueling process, fuel tank pressure increases, the check valve closes (preventing liquid fuel from spiting back at the operator), and fuel then rises up the fuel filler tube to shut off the dispensing nozzle.

NATURAL VACUUM LEAK DETECTION ASSEMBLY

DESCRIPTION

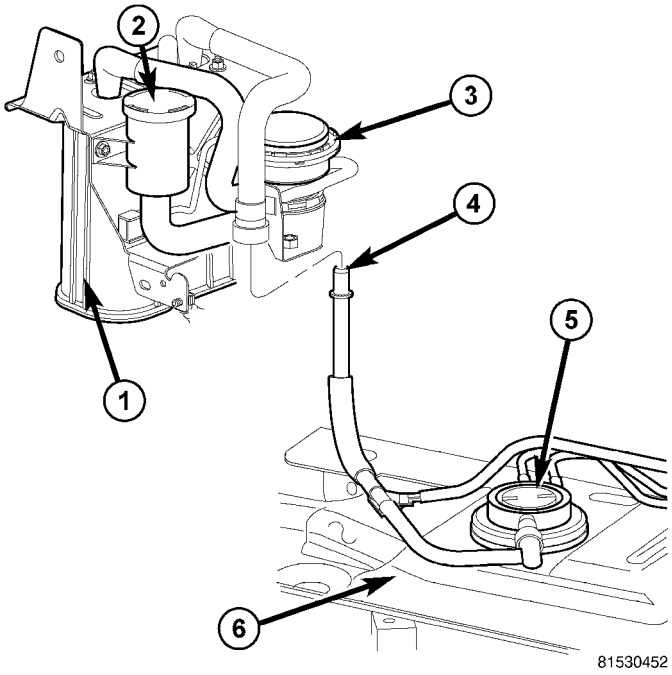

Fig. 11 ORVR VAPOR LINE

1 - EVAP CANISTER
2 - NVLD FILTER
3 - NVLD PUMP
4 - VAPOR LINES
5 - FLOW VALVE
6 - TOP OF FUEL TANK)

The Natural Vacuum Leak Detection (NVLD) pump (3) (Fig. 11) is located behind the right-rear wheel and system.

OPERATION

The Natural Vacuum Leak Detection (NVLD) system is the next generation evaporative leak detection system that will first be used on vehicles equipped with the Next Generation Controller (NGC). This new system replaces the leak detection pump as the method of evaporative system leak detection. This is to detect a leak equivalent to a 0.020″ (0.5 mm) hole. This system has the capability to detect holes of this size very dependably.

The basic leak detection theory employed with NVLD is the ″Gas Law″. This is to say that the pressure in a sealed vessel will change if the temperature of the gas in the vessel changes. The vessel will only see this effect if it is indeed sealed. Even small leaks will allow the pressure in the vessel to come to equilibrium with the ambient pressure. In addition to the detection of very small leaks, this system has the capability of detecting medium as well as large evaporative system leaks.

A vent valve seals the canister vent during engine off conditions. If the vapor system has a leak of less than the failure threshold, the evaporative system will be pulled into a vacuum, either due to the cool down from operating temperature or diurnal ambient temperature cycling. The diurnal effect is considered one of the primary contributors to the leak determination by this diagnostic. When the vacuum in the system exceeds about 1″ H2O (0.25 KPA), a vacuum switch closes. The switch closure sends a signal to the NGC. The NGC, via appropriate logic strategies, utilizes the switch signal, or lack thereof, to make a determination of whether a leak is present.

The NVLD device is designed with a normally open vacuum switch, a normally closed solenoid, and a seal, which is actuated by both the solenoid and a diaphragm. The NVLD is located on the atmospheric vent side of the canister. The NVLD assembly may be mounted on top of the canister outlet, or in-line between the canister and atmospheric vent filter. The normally open vacuum switch will close with about 1″ H2O (0.25 KPA) vacuum in the evaporative system. The diaphragm actuates the switch. This is above the opening point of the fuel inlet check valve in the fill tube so cap off leaks can be detected. Submerged fill systems must have recirculation lines that do not have the in-line normally closed check valve that protects the system from failed nozzle liquid ingestion, in order to detect cap off conditions.

The normally closed valve in the NVLD is intended to maintain the seal on the evaporative system during the engine off condition. If vacuum in the evaporative system exceeds 3″ to 6″ H2O (0.75 to 1.5 KPA), the valve will be pulled off the seat, opening the seal. This will protect the system from excessive vacuum as well as allowing sufficient purge flow in the event that the solenoid was to become inoperative.

The solenoid actuates the valve to unseal the canister vent while the engine is running. It also will be used to close the vent during the medium and large leak tests and during the purge flow check. This solenoid requires an initial 1.5 amps of current to pull the valve open, but after 100 mili-seconds, will be duty cycled down to an average of about 150 mA for the remainder of the drive cycle.

Another feature in the device is a diaphragm that will open the seal in the NVLD with pressure in the evaporative system. The device will ″blow off″ at about 0.5″ H2O (0.12 KPA) pressure to permit the venting of vapors during refueling. An added benefit to this is that it will also allow the tank to ″breathe″ during increasing temperatures, thus limiting the pressure in the tank to this low level. This is beneficial because the induced vacuum during a subse-

NATURAL VACUUM LEAK DETECTION ASSEMBLY (Continued)

quent declining temperature will achieve the switch closed (pass threshold) sooner than if the tank had to decay from a built up pressure.

The device itself has 3 wires: Switch sense, solenoid driver and ground. It also includes a resistor to protect the switch from a short to battery or a short to ground. The NGC utilizes a high-side driver to energize and duty-cycle the solenoid.

REMOVAL

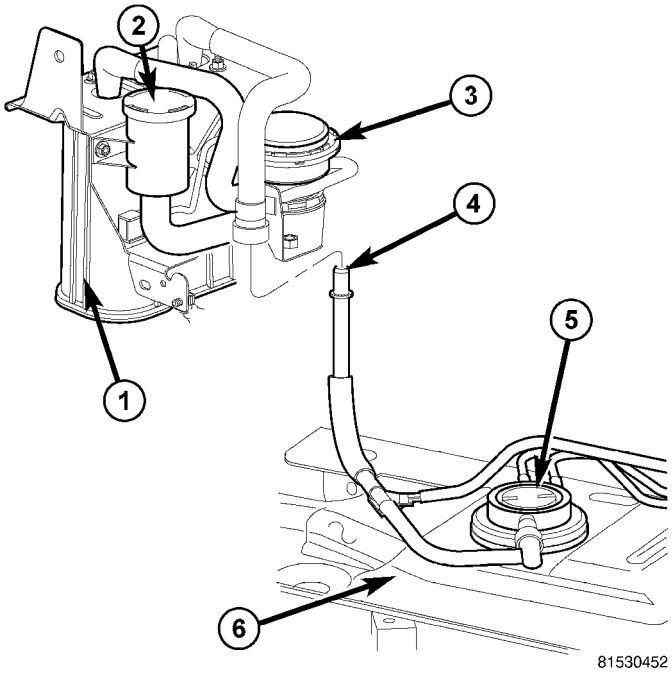

Fig. 12 ORVR VAPOR LINE

1 - EVAP CANISTER
2 - NVLD FILTER
3 - NVLD PUMP
4 - VAPOR LINES
5 - FLOW VALVE
6 - TOP OF FUEL TANK)

The Natural Vacuum Leak Detection (NVLD) pump (3) (Fig. 12) and EVAP canister (1) are attached to a common support bracket. This support assembly is located to the rear of the right / rear tire under the plastic wheelhouse liner. The NVLD pump filter is also located near the EVAP canister. The pump and filter are replaced (serviced) as one unit.

(1) Remove right/rear tire/wheel.

(2) Remove wheelhouse liner at right/rear wheel.

(3) Remove vertical support bracket (2) (Fig. 13) to gain access to ORVR vapor lines.

(4) A vapor line (4) (Fig. 14) connects the fuel tank to the EVAP canister and NVLD pump. This connection is made near the right/rear corner of the fuel tank. Carefully disconnect this line. Be very careful not to bend or kink the vapor lines. If lines leak, a Diagnostic Trouble Code (DTC) will be set.

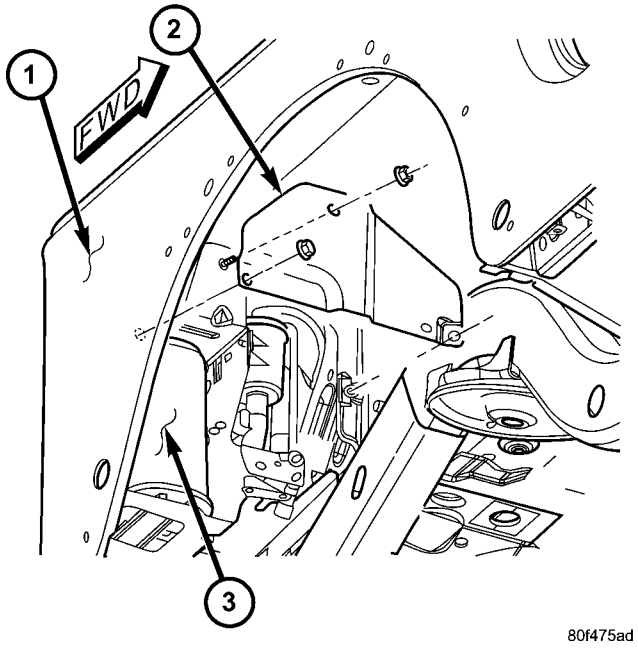

Fig. 13 EVAP CANISTER / LDP LOCATION

1 - RIGHT / REAR FENDER (WHEELHOUSE)
2 - VERTICAL SUPPORT BRACKET
3 - EVAP CANISTER

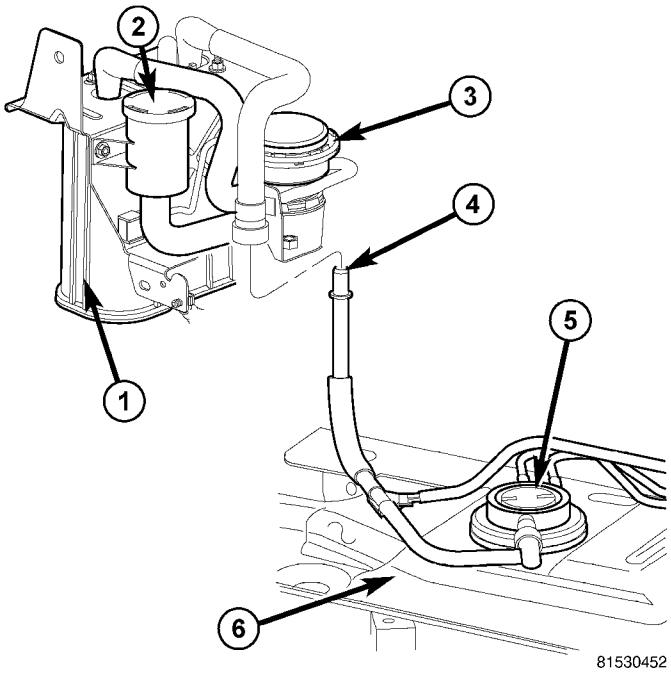

Fig. 14 ORVR VAPOR LINE

1 - EVAP CANISTER
2 - NVLD FILTER
3 - NVLD PUMP
4 - VAPOR LINES
5 - FLOW VALVE
6 - TOP OF FUEL TANK)

NATURAL VACUUM LEAK DETECTION ASSEMBLY (Continued)

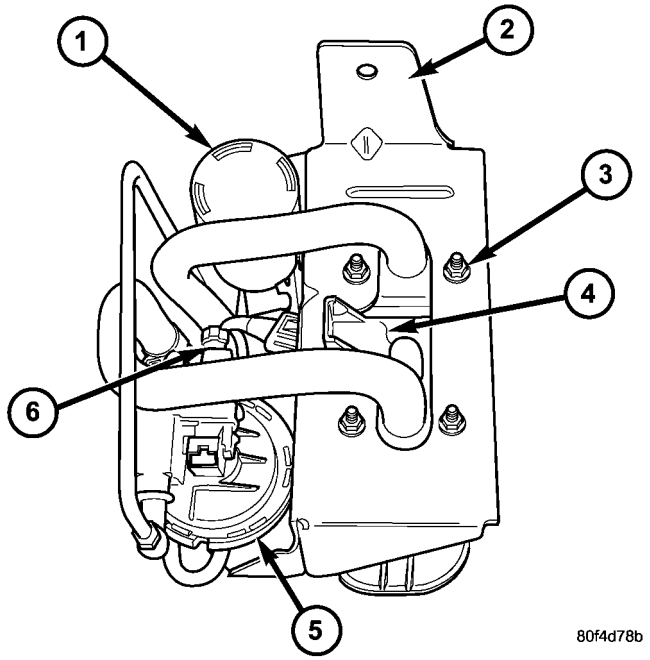

Fig. 15 COMMON SUPPORT BRACKET ASSEMBLY

1 - LDP FILTER
2 - COMMON SUPPORT BRACKET
3 - EVAP CANISTER NUTS (4)
4 - EVAP CANISTER
5 - LDP
6 - LDP ELEC. CONNECT.

(5) Remove common support bracket assembly (2) (Fig. 15) containing pump, pump filter and EVAP canister from vehicle. While lowering this assembly, disconnect pump electrical connector (6) (Fig. 15) at LDP pump. **Do not allow the assembly to hang from the wire harness.**

(6) Carefully separate hose at bottom of pump filter.

(7) Carefully remove vapor/vacuum lines at pump.

(8) If necessary, remove pump filter mounting bolt and remove filter.

(9) Remove EVAP canister from common support bracket by removing four nuts (3) (Fig. 15).

(10) Remove three pump mounting bolts (2) (Fig. 16) and remove pump from support bracket.

INSTALLATION

(1) Install pump and three mounting bolts (3) (Fig. 17) to common support bracket. Tighten bolts to 1 N·m (11 in. lbs.) torque.

(2) If necessary, install pump filter to common support bracket. Tighten bolt to 7 N·m (65 in. lbs.) torque.

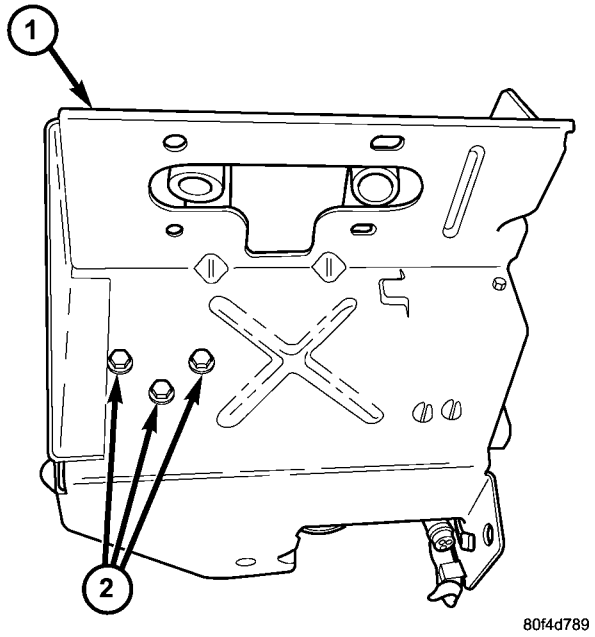

Fig. 16 LDP MOUNTING BOLTS

1 - COMMON SUPPORT BRACKET
2 - LDP MOUNTING BOLTS (3)

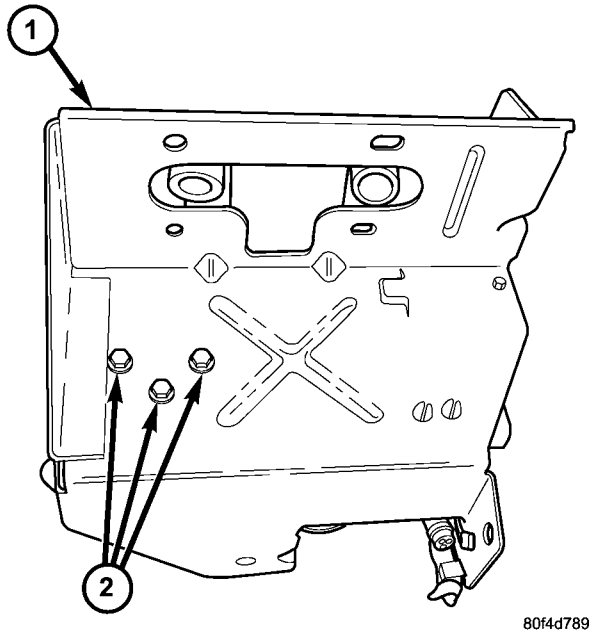

Fig. 17 LDP MOUNTING BOLTS

1 - COMMON SUPPORT BRACKET
2 - LDP MOUNTING BOLTS (3)

NATURAL VACUUM LEAK DETECTION ASSEMBLY (Continued)

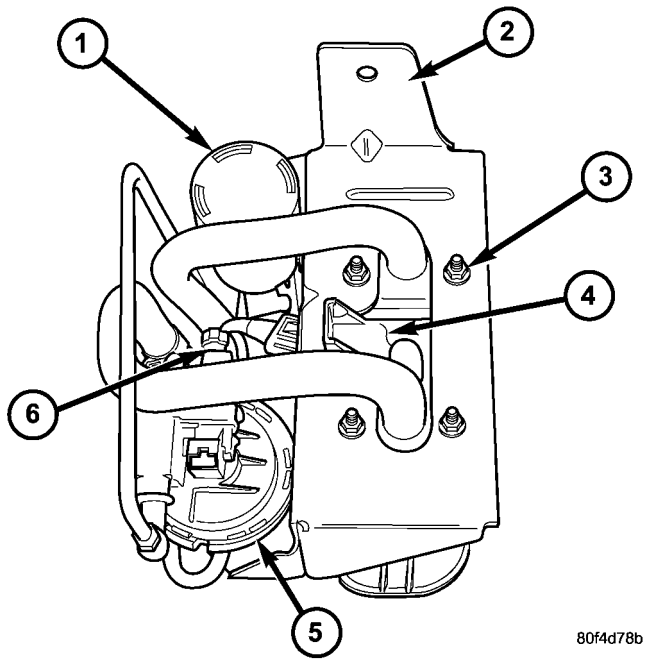

Fig. 18 COMMON SUPPORT BRACKET ASSEMBLY

1 - LDP FILTER
2 - COMMON SUPPORT BRACKET
3 - EVAP CANISTER NUTS (4)
4 - EVAP CANISTER
5 - LDP
6 - LDP ELEC. CONNECT.

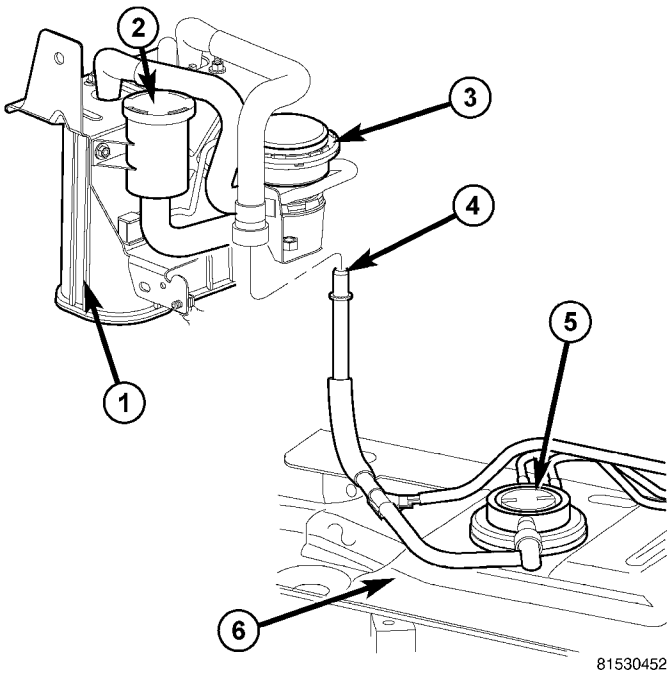

Fig. 19 ORVR VAPOR LINE

1 - EVAP CANISTER
2 - NVLD FILTER
3 - NVLD PUMP
4 - VAPOR LINES
5 - FLOW VALVE
6 - TOP OF FUEL TANK)

(3) Install EVAP canister to common support bracket and install four mounting nuts (3) (Fig. 18).

(4) Carefully install vapor/vacuum lines to pump, filter and EVAP canister. **The vapor/vacuum lines and hoses must be firmly connected. Check the vapor/vacuum lines at the LDP, LDP filter and EVAP canister purge solenoid for damage or leaks. If a leak is present, a Diagnostic Trouble Code (DTC) will be set.**

(5) Raise assembly into position while connecting electrical connector to pump. Do not allow this assembly to hang from the wire harness. While assembly is still being supported, connect vapor line (4) (Fig. 19) to fuel tank.

(6) Position common support bracket assembly to body. Install and tighten bolts.

(7) Install vertical support bracket.

(8) Install wheelhouse liner at right/rear wheel.

(9) Install right/rear tire/wheel.

VACUUM LINES

DESCRIPTION

A vacuum schematic for emission related items can be found on the VECI label. Refer to Vehicle Emission Control Information (VECI) Label for label location.

VAPOR CANISTER

DESCRIPTION

The EVAP (vapor) canister (1), Natural Vacuum Leak Detection (NVLD) pump (3) and pump filter are attached to a common support bracket. This support assembly is located to the rear of the right / rear tire under the plastic wheelhouse liner (Fig. 20).

VAPOR CANISTER (Continued)

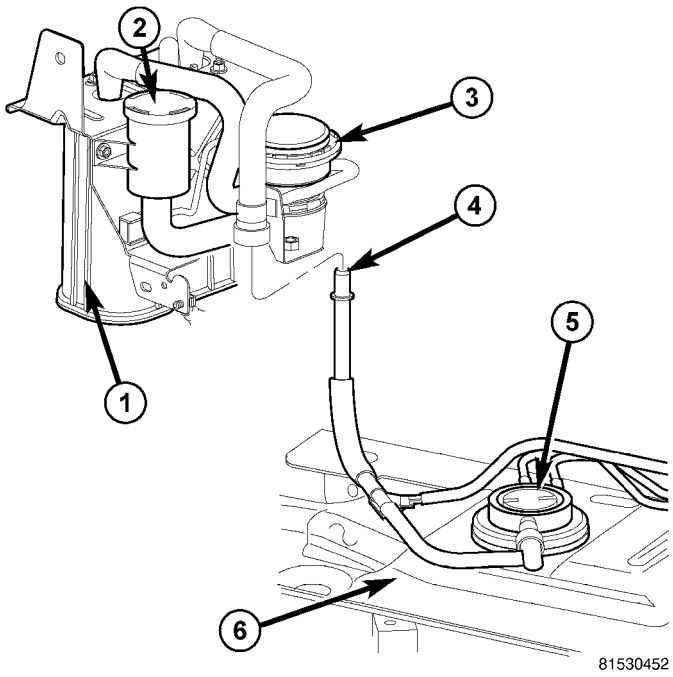

Fig. 20 ORVR VAPOR LINE

1 - EVAP CANISTER
2 - NVLD FILTER
3 - NVLD PUMP
4 - VAPOR LINES
5 - FLOW VALVE
6 - TOP OF FUEL TANK)

OPERATION

A maintenance free, EVAP canister is used on all vehicles. The EVAP canister is filled with granules of an activated carbon mixture. Fuel vapors entering the EVAP canister are absorbed by the charcoal granules.

Fuel tank pressure vents into the EVAP canister. Fuel vapors are temporarily held in the canister until they can be drawn into the intake manifold. The duty cycle EVAP canister purge solenoid allows the EVAP canister to be purged at predetermined times and at certain engine operating conditions.

The EVAP canister is also to be considered as part of the ORVR system.

REMOVAL

The Natural Vacuum Leak Detection (NVLD) pump, pump filter, and EVAP canister are attached to a common support bracket. This support assembly is located to the rear of the right / rear tire under the plastic wheelhouse liner.

For EVAP canister removal procedures, refer to Natural Vacuum Leak Detection (NVLD) pump Removal/Installation.

INSTALLATION

The Natural Vacuum Leak Detection (NVLD) pump, pump filter, and EVAP canister are attached to a common support bracket. This support assembly is located to the rear of the right / rear tire under the plastic wheelhouse liner.

For EVAP canister removal procedures, refer to Natural Vacuum Leak Detection (NVLD) pump Removal/Installation.

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SERVICE MANUAL COMMENTS

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